Modern Algebraic Geometry

— *EYNTKA* Series —

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Contents

Algebraic Geometry originated as the study of the zero-sets of polynomials, namely it was (and in many ways behind all the abstraction predominantly still is) the solution to a system of polynomials. By explicitly working with the zero sets of polynomials, we can discover a great multitude of geometric results within algebra, however as the field progressed it was discovered zero sets are in many ways insufficient to capture all the geometric data. For some quick examples:

1. it was discovered in the mid 20th century that results in number theory seem eerily similar to the study of 1 dimensional curves.
2. The galois group of a field share many of the same properties of as the fundamental group of a topological space.
3. The fact that a polynomial may have a root of high multiplicity is lost in the classical geometric picture $V(f)$, but this is often key information.
4. Number theorists do not work over algebraically closed fields as they are looking for integer (or integral) solutions to polynomial equations. However, these do not have the same nice geometric correspondence within classical algebraic geometry as the Nullstellensatz works well when the field is algebraically closed.
5. All rings can be represented as the quotient of $\mathbb{Z}[x_i, \mid i \in \mathcal{I}]$ or $k[x_i, \mid i \in \mathcal{I}]$ for some indexing set I, hence all ring (not just reduced finitely generated rings over algebraically closed fields) have some notion of being a function with evaluation
6. When studying a family of varieties (ex. $x^2 + ty^2 = 1$ as t varies) the resulting object will often have "bad" singularities which varieties are poorly equipped to work with.

Such realizations lead to the need to ground algebraic geometry with a new object that encapsulates all of these ideas. this new object is the *scheme*. As compared to varieties that requires an ambient space and working over polynomial rings over fields, schemes are *intrinsic* in that they don't require any ambient space to be defined in, and they generalize classical algebraic geometry to be able to capture the geometry of both Diophantine equations (for number theory) as well as the geometry of (quasi-projective) varieties and other more novel. Perhaps key in their success over other variations is how well they also remember "multiplicity" information of roots of polynomials. This data shall be paramount in many results we shall introduce[1].

This book shall be written with graduate or well-read undergraduates in mind. The reason is to show how inter-connected schemes are to various other fields of mathematics. For example:

1. The geometry of schemes draws many parallels to complex geometry and some results from complex analysis (ex. Hartog's lemma from Complex analysis over several variables shall have a strong appearance when discussing normal schemes).
2. A sheaf draws many similarities to the bundles that are put on manifolds; in fact a manifold can be defined equivalently by using a sheaf of differentiable functions instead of an atlas. Some results such as Poincaré duality shall be proven for schemes (see 6.5)
3. Chapter 6 shall strongly rely on the readers intuitions from algebraic topology and use modern homological algebra.

[1]We shall see for Elliptic Curves that the finite-generation of certain torsion groups shall rely on Schemes in a key way to remember multiplicity

4. Many of the properties in commutative algebra shall become interesting geometric properties in algebraic geometry.

Essentially, anything non-optional material from EYNTKA Undergraduate Algebra ([20]), EYNTKA Differential Topology ([53]), and EYNTKA Algebraic Topology ([51]) will be considered as assumed prerequisites. Some results from EYNTKA complex analysis ([52]) will be drawn on as well given the strong connection between complex geometry and the geometry of schemes.

All essential category theory will be assumed; it can be reviewed in EYNTKA Undergraduate Algebra.

Throughout this book, we shall use the following terminology:

- Let k be a field, $\overline{k}$ the algebraic closure of k
- Let R an arbitrary ring, $M_n(R)$ the matrix ring over R, $R_\mathfrak{p}$ the localization of R by $R \setminus \mathfrak{p}$, R_f the localization of R by $\{f, f^2, ...\}$, $\text{Frac}(R)$ the field of fractions of R is R is an integral domain
- A is an arbitrary R-algebra.

see [20].

adding new citation: [47]

The material covered in the preliminary chapter is at a break-neck pace and is only there for expository purposes for the reader to see if they have the right intuitions to cover algebraic geometry in its greater generality. The first main Part, Scheme Theory, focuses heavily on building up all the necessary theoretical framework to have a comfortable grasp of the material and is heavily inspired by the many famous texts covering the material (ex. Hartshorne, Shafarevich, Vakil, Serre, Eisenbud, and so forth).

Preliminary Chapter

In this chapter, we shall quickly review elementary concepts linking algebra and geometry without doing any proofs, that is we are going to explore some classical algebraic geometry. For all the important details, see [20].

0.1 Recall Classical Algebraic Geometry

If A is an R-algebra, then by the universal property of free R-algebras we have if S is a set of generators for A,

$$\begin{tikzcd} R[S] \arrow[r, "\varphi"] & A \\ S \arrow[u, "\iota"] \arrow[ur, "\iota"'] & \end{tikzcd}$$

If S is finite, then $R[S] \cong R[x_1, x_2, \ldots, x_n]$ by relabelling S. Since every ring is a $\mathbb{Z}$-algebra, we see that polynomial rings are central to the study of general rings. It is thus fruitful to realize that polynomial rings can naturally be interpreted as functions, that is:

$$f \in R[S], \; R^S \to R[S]^\vee \qquad f \mapsto f(s_i)_{i \in S}$$

In the finite case:

$$f \in R[x_1, x_2, \ldots, x_n], \qquad R^n \to R[x_1, x_2, \ldots, x_n]^\vee \qquad (a_1, a_2, \ldots, a_n) \mapsto (f \mapsto f(a_1, a_2, \ldots, a_n))$$

We don't want to consider R^S as having any structure besides being an affine space, hence we shall label it as $\mathbb{A}_R^S$ or in the finite case $\mathbb{A}_R^n$. If the ring is evident in context, we shall write $\mathbb{A}^n$.

0.1.1 Conventions in Number Theory In a number-theoretic context, $\mathbb{A}^n$ often implicitly means $\mathbb{A}_{\mathbb{C}}^n$ (or $\mathbb{A}_{\mathbb{Q}}^n$), with $\mathbb{A}^n(R)$ denoting the R-points. This convention arises because polynomials over $\mathbb{C}$ always have solutions, which is not guaranteed over $\mathbb{R}$ or $\mathbb{Q}$.

For now, we shall stick to the finitely generated case. Note that in $\mathbb{A}^n_R$, the point $(0, \ldots, 0)$ has no special property, as we are treating this space as simply a collection of points. To emphasize this geometric interpretation, any $p \in \mathbb{A}^n_R$ will be called a *point*, and for

$$p = (r_1, r_2, \ldots, r_n)$$

each r_i is called a *coordinate* of p. Hence, we may view any $f \in R[x_1, x_2, \ldots, x_n]$ as a function:

$$f : \mathbb{A}^n_R \to R \qquad (r_1, r_2, \ldots, r_n) \mapsto f(r_1, r_2, \ldots, r_n)$$

Algebraic Sets and the Zariski Topology

The zeros of polynomials form many interesting algebraic shapes. A prototypical example is $V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{R}}$, the unit circle.

Definition 0.1.2: Zero Set

For $S \subseteq R[x_1, \ldots, x_n]$, the *zero set* (or *vanishing locus*) of S is

$$\mathcal{Z}(S) = \{p \in \mathbb{A}^n_R \mid f(p) = 0, \ \forall f \in S\}$$

Definition 0.1.3: Algebraic Set

A subset $X \subseteq \mathbb{A}^n_R$ is called an *algebraic set* (or *closed set*) if there exists $S \subseteq R[x_1, \ldots, x_n]$ such that $\mathcal{Z}(S) = X$.

The collection of algebraic sets satisfies the axioms for closed sets of a topology:

1. $\mathbb{A}^n_R = \mathcal{Z}(0)$ and $\emptyset = \mathcal{Z}(1)$ are algebraic.
2. Finite unions: $\mathcal{Z}(S_1) \cup \mathcal{Z}(S_2) = \mathcal{Z}(S_1 \cdot S_2)$ where $S_1 \cdot S_2 = \{fg : f \in S_1, g \in S_2\}$.
3. Arbitrary intersections: $\bigcap_\alpha \mathcal{Z}(S_\alpha) = \mathcal{Z}\big(\bigcup_\alpha S_\alpha\big)$.

Definition 0.1.4: Zariski Topology

The *Zariski topology* on $\mathbb{A}^n_R$ is the topology whose closed sets are precisely the algebraic sets.

0.1.5 Properties of the Zariski Topology The Zariski topology is quite coarse compared to the Euclidean topology on $\mathbb{R}^n$ or $\mathbb{C}^n$:

1. It is generally *not* Hausdorff: in $\mathbb{A}^1_k$, the only proper closed subsets are finite sets of points, so any two nonempty open sets intersect.
2. Every nonempty open set is dense.
3. Despite this, the Zariski topology is well-suited to algebraic geometry because its closed sets are precisely the sets defined by polynomial equations.

The Ideal–Variety Correspondence

Conversely, given a subset $S \subseteq \mathbb{A}^n_R$, we can ask which polynomials vanish on S:

Definition 0.1.6: Ideal of a Set

For $S \subseteq \mathbb{A}^n_R$, the *ideal of* S is

$$\mathcal{I}(S) = \{f \in R[x_1, \dots, x_n] \mid f(p) = 0, \ \forall p \in S\}$$

One readily checks that $\mathcal{I}(S)$ is always an ideal in $R[x_1, \dots, x_n]$. Moreover, $\mathcal{I}(S)$ is always a *radical* ideal, i.e., $f^n \in \mathcal{I}(S)$ implies $f \in \mathcal{I}(S)$. This is immediate: if f^n vanishes on S, then f vanishes on S.

The operators $\mathcal{Z}$ and $\mathcal{I}$ form an order-reversing correspondence:

$$S_1 \subseteq S_2 \implies \mathcal{I}(S_1) \supseteq \mathcal{I}(S_2), \qquad I_1 \subseteq I_2 \implies \mathcal{Z}(I_1) \supseteq \mathcal{Z}(I_2)$$

However, this correspondence is not bijective in general. The precise relationship requires the notion of radical ideals and irreducibility.

Definition 0.1.7: Radical Ideal

An ideal $I \subseteq R$ is *radical* if $f^n \in I$ implies $f \in I$ for all $f \in R$ and $n \geq 1$. Equivalently, $I = \sqrt{I}$ where $\sqrt{I} = \{f \in R \mid \exists n \geq 1, \ f^n \in I\}$.

Definition 0.1.8: Irreducible Set

An algebraic set V is *irreducible* if whenever $V = V_1 \cup V_2$ with V_1, V_2 closed, then $V = V_1$ or $V = V_2$. Equivalently, V is irreducible if and only if any two nonempty open subsets of V intersect.

Definition 0.1.9: Variety (Classical)

An algebraic set $V \subseteq \mathbb{A}^n_k$ is an (affine) *variety* if V is irreducible.

The key algebraic characterization of irreducibility is:

Proposition 0.1.10: Irreducibility and Prime Ideals

An algebraic set V is irreducible if and only if $\mathcal{I}(V)$ is a prime ideal.

***Proof*:**
See [20].

Example 0.1.11: Reducible vs. Irreducible

Consider $V(xy) \subseteq \mathbb{A}^2_k$. This is the union of the two coordinate axes:

$$V(xy) = V(x) \cup V(y)$$

The ideal (xy) is radical but not prime (since $xy \in (xy)$ but $x \notin (xy)$ and $y \notin (xy)$). Hence $V(xy)$

is an algebraic set but *not* a variety.

In contrast, $V(y - x^2) \subseteq \mathbb{A}^2_k$ (a parabola) is irreducible: the ideal $(y - x^2)$ is prime.

We can now state the Nullstellensatz precisely. From here on, we assume k is an algebraically closed field.

Theorem 0.1.12: Hilbert's Nullstellensatz

Let k be algebraically closed. Then:

1. **(Weak form)** If $I \subsetneq k[x_1, \dots, x_n]$ is a proper ideal, then $\mathcal{Z}(I) \neq \emptyset$.
2. **(Strong form)** For any ideal $I \subseteq k[x_1, \dots, x_n]$, we have $\mathcal{I}(\mathcal{Z}(I)) = \sqrt{I}$.
3. **(Maximal ideal form)** The maximal ideals of $k[x_1, \dots, x_n]$ are precisely those of the form $\mathfrak{m}_p = (x_1 - a_1, \dots, x_n - a_n)$ for points $p = (a_1, \dots, a_n) \in \mathbb{A}^n_k$.

***Proof*:**
See [20].

The strong form immediately gives:

Corollary 0.1.13: The Ideal–Variety Correspondence

Let k be algebraically closed. The maps $\mathcal{Z}$ and $\mathcal{I}$ induce inclusion-reversing bijections:

$$\{\text{radical ideals in } k[x_1, \dots, x_n]\} \overset{\mathcal{Z}}{\longleftrightarrow} \{\text{algebraic sets in } \mathbb{A}^n_k\}$$

Under this correspondence:

Ideals	$\longleftrightarrow$	**Algebraic Sets**
radical ideals	$\longleftrightarrow$	algebraic sets
prime ideals	$\longleftrightarrow$	varieties (irreducible)
maximal ideals	$\longleftrightarrow$	points

The Coordinate Ring

The passage from geometry to algebra is achieved through the coordinate ring:

Definition 0.1.14: Coordinate Ring

Let $V \subseteq \mathbb{A}^n_k$ be an algebraic set. The *coordinate ring* (or *ring of regular functions*) of V is

$$k[V] := k[x_1, \dots, x_n]/\mathcal{I}(V)$$

The elements of $k[V]$ are equivalence classes of polynomials, where two polynomials are equivalent if they agree on V. Thus $k[V]$ can be interpreted as the ring of *polynomial functions* $V \to k$.

Proposition 0.1.15: Properties of the Coordinate Ring

Let $V \subseteq \mathbb{A}^n_k$ be an algebraic set with coordinate ring $k[V]$. Then:

1. $k[V]$ is a finitely generated k-algebra.
2. $k[V]$ is reduced (i.e., has no nilpotents) if and only if $\mathcal{I}(V)$ is radical.
3. $k[V]$ is an integral domain if and only if V is irreducible.
4. Points $p \in V$ correspond to maximal ideals $\mathfrak{m}_p \subseteq k[V]$.

***Proof*:**
See [20].

Example 0.1.16: Coordinate Rings

1. $k[\mathbb{A}^n] = k[x_1, \ldots, x_n]$, since $\mathcal{I}(\mathbb{A}^n) = (0)$.
2. For the parabola $V = V(y - x^2) \subseteq \mathbb{A}^2$, we have $k[V] = k[x, y]/(y - x^2) \cong k[x]$, reflecting that $V \cong \mathbb{A}^1$.
3. For the circle $V = V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{C}}$, we have $k[V] = \mathbb{C}[x, y]/(x^2 + y^2 - 1)$. This is an integral domain since $(x^2 + y^2 - 1)$ is prime over $\mathbb{C}$.
4. For the nodal cubic $V = V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2$, the coordinate ring is $k[V] = k[x, y]/(y^2 - x^3 - x^2)$. This is an integral domain (the curve is irreducible), but V is singular at the origin.

Morphisms of Affine Varieties

To form a category of varieties, we need morphisms.

Definition 0.1.17: Regular Map / Morphism

Let $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$ be algebraic sets. A *regular map* (or *morphism*) $\varphi : V \to W$ is a map of the form

$$\varphi(p) = (f_1(p), f_2(p), \ldots, f_m(p))$$

where each $f_i \in k[V]$. That is, φ is given by polynomial functions in coordinates.

Proposition 0.1.18: Morphisms and Algebra Homomorphisms

A regular map $\varphi : V \to W$ induces a k-algebra homomorphism

$$\varphi^* : k[W] \to k[V], \qquad g \mapsto g \circ \varphi$$

called the *pullback*. Conversely, every k-algebra homomorphism $k[W] \to k[V]$ arises as φ^* for a unique regular map $\varphi : V \to W$.

***Proof*:**
See [20].

Note the contravariance: a map $V \to W$ gives a map $k[W] \to k[V]$ in the opposite direction. This leads to the fundamental equivalence:

Theorem 0.1.19: Equivalence of Categories

Let k be algebraically closed. There is a contravariant equivalence of categories:

$$\text{Aff-Var}_k \underset{\text{mSpec}}{\overset{V \mapsto k[V]}{\rightleftarrows}} \left(\text{Alg}_k^{\text{f.g., red}}\right)^{\text{op}}$$

where Aff-Var_k is the category of affine varieties over k with regular maps, and $\text{Alg}_k^{\text{f.g., red}}$ is the category of finitely generated reduced k-algebras with k-algebra homomorphisms.

***Proof*:**
The functor $V \mapsto k[V]$ sends varieties to their coordinate rings and sends morphisms $\varphi : V \to W$ to pullbacks $\varphi^* : k[W] \to k[V]$. The quasi-inverse sends a finitely generated reduced k-algebra A to its maximal spectrum $\text{mSpec}(A)$, the set of maximal ideals with the Zariski topology. The Nullstellensatz guarantees that for $A = k[V]$, the points of V are in bijection with maximal ideals of A. See [20] for details.

0.1.20 Why Reduced Algebras? The restriction to reduced algebras (no nilpotents) corresponds to the fact that classical varieties are determined by their point-sets. Modern algebraic geometry (via schemes) removes this restriction, allowing nilpotent elements to encode infinitesimal data – this is one of the key generalizations.

Local Rings and Regularity

To study the local behavior of a variety at a point, we first introduce the algebraic machinery of localization. This process allows us to systematically make elements of a ring invertible, essentially allowing us to study rings "locally" by focusing on fractions whose denominators do not vanish at the point of interest.

Definition 0.1.21: Localization of a Ring

Let R be a commutative ring. A subset $S \subseteq R$ is *multiplicatively closed* if $1 \in S$ and for all $a, b \in S$, we have $ab \in S$. The *localization* of R at S, denoted $S^{-1}R$, is the set of equivalence classes of fractions $\frac{r}{s}$ with $r \in R$ and $s \in S$, modulo the equivalence relation:

$$\frac{r}{s} \sim \frac{r'}{s'} \iff \text{there exists } t \in S \text{ such that } t(rs' - r's) = 0$$

This becomes a ring under the standard addition and multiplication of fractions.

Localization is completely characterized by the following universal property, which states that $S^{-1}R$ is the "most efficient" way to force the elements of S to become units.

Proposition 0.1.22: Universal Property of Localization

Let S be a multiplicative set of a ring R. The natural map $\iota : R \to S^{-1}R$ given by $\iota(r) = \frac{r}{1}$ is a ring homomorphism that maps elements of S to units in $S^{-1}R$. Furthermore, if A is any ring and $g : R \to A$ is a ring homomorphism such that $g(S) \subseteq A^\times$ (the group of units of A), then there exists a unique ring homomorphism $h : S^{-1}R \to A$ such that $h \circ \iota = g$.

***Proof*:**

We define $h\left(\frac{r}{s}\right) = g(r)g(s)^{-1}$. To see that h is well-defined, suppose $\frac{r}{s} = \frac{r'}{s'}$. Then there is some $t \in S$ such that $t(rs' - r's) = 0$. Applying g gives $g(t)(g(r)g(s') - g(r')g(s)) = 0$. Since $t \in S$, $g(t)$ is a unit in A, so we can multiply by $g(t)^{-1}$ to obtain $g(r)g(s') = g(r')g(s)$. Multiplying by $g(s)^{-1}g(s')^{-1}$ yields $g(r)g(s)^{-1} = g(r')g(s')^{-1}$, confirming h is well-defined.

It is straightforward to verify that h is a ring homomorphism. For uniqueness, if $h' : S^{-1}R \to A$ satisfies $h' \circ \iota = g$, then $h'\left(\frac{r}{1}\right) = g(r)$ and $h'\left(\frac{1}{s}\right) = h'\left(\frac{s}{1}\right)^{-1} = g(s)^{-1}$. Thus, $h'\left(\frac{r}{s}\right) = g(r)g(s)^{-1} = h\left(\frac{r}{s}\right)$.

An important consequence of this universal property is that localization is a functor.

Proposition 0.1.23: Functoriality of Localization

Let R, R' be rings with multiplicative sets S, S' respectively. If $\varphi : R \to R'$ is a ring homomorphism such that $\varphi(S) \subseteq S'$, then there is a unique induced ring homomorphism $\widetilde{\varphi} : S^{-1}R \to (S')^{-1}R'$ making the following diagram commute:

$$\widetilde{\varphi} \circ \iota = \iota' \circ \varphi$$

where ι and ι' are the natural localization maps.

***Proof*:**

Consider the composition $\iota' \circ \varphi : R \to R' \to (S')^{-1}R'$. For any $s \in S$, we have $\varphi(s) \in S'$. Therefore, $\iota'(\varphi(s))$ is a unit in $(S')^{-1}R'$. By the universal property of $S^{-1}R$ applied to the homomorphism $\iota' \circ \varphi$, there exists a unique ring homomorphism $\widetilde{\varphi} : S^{-1}R \to (S')^{-1}R'$ satisfying

$\widetilde{\varphi} \circ \iota = \iota' \circ \varphi$. Explicitly, this map is given by $\widetilde{\varphi}\left(\frac{r}{s}\right) = \frac{\varphi(r)}{\varphi(s)}$.

We now apply this general algebraic construction to the coordinate ring of a variety to study its local geometry.

Definition 0.1.24: Local Ring at a Point

Let V be an affine variety and $p \in V$ a point with corresponding maximal ideal $\mathfrak{m}_p \subseteq k[V]$. The set $S = k[V] \setminus \mathfrak{m}_p$ is multiplicatively closed. The *local ring of V at p* is the localization of $k[V]$ at S:

$$\mathcal{O}_{V,p} := k[V]_{\mathfrak{m}_p} = \left\{ \frac{f}{g} \;\middle|\; f, g \in k[V],\ g(p) \neq 0 \right\}$$

This is a local ring with unique maximal ideal $\mathfrak{m}_{V,p} = \mathfrak{m}_p \cdot \mathcal{O}_{V,p}$.

The local ring $\mathcal{O}_{V,p}$ captures the "germs" of regular functions near p: two functions are identified if they agree on some neighborhood of p.

Definition 0.1.25: Zariski Tangent Space

The *Zariski tangent space* to V at p is the k-vector space

$$T_pV := \left(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2\right)^\vee = \mathrm{Hom}_k\left(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2, k\right)$$

The *Zariski cotangent space* is $T_p^*V := \mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2$.

0.1.26 Tangent Space via Derivations Equivalently, T_pV can be defined as the space of k-derivations $\mathcal{O}_{V,p} \to k$ at p, or as the space of "ε-points": morphisms $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \to V$ centered at p. This latter perspective generalizes naturally to schemes.

Definition 0.1.27: Smooth Point, Singular Point

Let V be a variety of dimension d (see below). A point $p \in V$ is:

- *smooth* (or *regular*) if $\dim_k T_pV = d$;
- *singular* if $\dim_k T_pV > d$.

A variety is *smooth* (or *nonsingular*) if every point is smooth.

Example 0.1.28: Singularities

1. The parabola $V(y - x^2)$ is smooth everywhere. At any point, $\dim T_pV = 1 = \dim V$.
2. The cuspidal cubic $V(y^2 - x^3)$ has a singularity (cusp) at the origin. One computes $\dim T_0V = 2 > 1 = \dim V$.
3. The nodal cubic $V(y^2 - x^2(x+1))$ has a node at the origin where two branches cross.

Proposition 0.1.29: Regularity and Local Rings

A point $p \in V$ is smooth if and only if $\mathcal{O}_{V,p}$ is a *regular local ring*, i.e., $\dim_k(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2) = \dim \mathcal{O}_{V,p}$ (Krull dimension).

***Proof*:**
See [20].

Rational Functions and Dimension

For an irreducible variety, we can form the field of fractions of its coordinate ring:

Definition 0.1.30: Function Field

Let V be an affine variety (irreducible). The *function field* of V is

$$k(V) := \mathrm{Frac}(k[V])$$

Elements of $k(V)$ are called *rational functions* on V. A rational function f/g is defined on the open set $\{p \in V : g(p) \neq 0\}$.

Definition 0.1.31: Dimension

The *dimension* of an affine variety V is

$$\dim V := \mathrm{tr.deg}_k\ k(V)$$

the transcendence degree of the function field over k. Equivalently, $\dim V = \dim k[V]$, the Krull dimension of the coordinate ring.

Example 0.1.32: Dimensions

1. $\dim \mathbb{A}^n = n$, since $k(\mathbb{A}^n) = k(x_1, \ldots, x_n)$ has transcendence degree n.
2. Any plane curve $V(f) \subseteq \mathbb{A}^2$ with f irreducible has dimension 1.
3. A hypersurface $V(f) \subseteq \mathbb{A}^n$ with f irreducible has dimension $n - 1$.

Proposition 0.1.33: Dimension and Chains of Primes

For an affine variety V, we have $\dim V = \dim k[V]$, where the Krull dimension of $k[V]$ is the supremum of lengths of chains of prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_d$$

in $k[V]$.

***Proof*:**
See [20].

0.2 Recall Projective Spaces

Projective space is the natural setting for algebraic geometry. Many phenomena that are awkward or incomplete in affine space become clean and uniform in projective space: intersection theory works properly (Bézout's theorem), curves have well-defined degrees, and there are no "missing points at infinity." This section reviews the essential constructions.

Definition and Homogeneous Coordinates

Definition 0.2.1: Projective Space

Let k be a field. *Projective n-space* over k is

$$\mathbb{P}^n_k := \left(k^{n+1} \setminus \{0\}\right) / \sim$$

where $(a_0, a_1, \ldots, a_n) \sim (\lambda a_0, \lambda a_1, \ldots, \lambda a_n)$ for all $\lambda \in k^\times$. The equivalence class of $(a_0, \ldots, a_n)$ is denoted $[a_0 : a_1 : \cdots : a_n]$ and called a *point* in $\mathbb{P}^n$. The a_i are *homogeneous coordinates.*

0.2.2 Projective Space as Lines Equivalently, $\mathbb{P}^n_k$ is the set of 1-dimensional linear subspaces (lines through the origin) in k^{n+1}. This is sometimes denoted $\mathbb{P}(k^{n+1})$ or $\mathrm{Proj}(k^{n+1})$.

Example 0.2.3: Low-Dimensional Projective Spaces

1. $\mathbb{P}^0$ is a single point.
2. $\mathbb{P}^1$ is the *projective line.* Points are $[a_0 : a_1]$ with $(a_0, a_1) \neq (0, 0)$. If $a_1 \neq 0$, we can normalize to $[t : 1]$ for $t = a_0/a_1 \in k$. The remaining point $[1 : 0]$ is the "point at infinity." Thus $\mathbb{P}^1 \cong \mathbb{A}^1 \sqcup \{\infty\}$.
3. $\mathbb{P}^2$ is the *projective plane*, the natural home for plane curves.

Affine Charts

Projective space is covered by affine open sets:

Definition 0.2.4: Standard Affine Charts

For each $i \in \{0, 1, \dots, n\}$, define the *standard affine open*

$$U_i := \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid a_i \neq 0\}$$

There is an isomorphism $\varphi_i : U_i \xrightarrow{\sim} \mathbb{A}^n$ given by

$$\varphi_i([a_0 : \cdots : a_n]) = \left(\frac{a_0}{a_i}, \dots, \frac{\widehat{a_i}}{a_i}, \dots, \frac{a_n}{a_i}\right)$$

where the hat denotes omission.

Proposition 0.2.5: Affine Cover of Projective Space

The standard affine opens cover $\mathbb{P}^n$:

$$\mathbb{P}^n = U_0 \cup U_1 \cup \cdots \cup U_n$$

The complement of U_i is the hyperplane $H_i = \{[a_0 : \cdots : a_n] : a_i = 0\} \cong \mathbb{P}^{n-1}$.

Example 0.2.6: The Projective Line Revisited

For $\mathbb{P}^1$, we have two charts:

- $U_0 = \{[a_0 : a_1] : a_0 \neq 0\} \cong \mathbb{A}^1$ via $[a_0 : a_1] \mapsto a_1/a_0$
- $U_1 = \{[a_0 : a_1] : a_1 \neq 0\} \cong \mathbb{A}^1$ via $[a_0 : a_1] \mapsto a_0/a_1$

On the overlap $U_0 \cap U_1$, if $t = a_1/a_0$ is the coordinate on U_0 and $s = a_0/a_1$ on U_1, then $s = 1/t$. This is the standard gluing of two copies of $\mathbb{A}^1$ to form $\mathbb{P}^1$.

Homogeneous Polynomials and Graded Rings

Polynomials do not define functions on $\mathbb{P}^n$ directly, since $f(\lambda a_0, \dots, \lambda a_n) \neq f(a_0, \dots, a_n)$ in general. However, the *vanishing* of homogeneous polynomials is well-defined.

Definition 0.2.7: Homogeneous Polynomial

A polynomial $f \in k[x_0, x_1, \dots, x_n]$ is *homogeneous of degree* d if every monomial in f has total degree d. Equivalently, $f(\lambda x_0, \dots, \lambda x_n) = \lambda^d f(x_0, \dots, x_n)$ for all $\lambda \in k$.

Definition 0.2.8: Graded Ring

The polynomial ring $S = k[x_0, \dots, x_n]$ has a *grading* $S = \bigoplus_{d \geq 0} S_d$, where S_d is the k-vector space of homogeneous polynomials of degree d (including 0). We have $S_d \cdot S_e \subseteq S_{d+e}$.

Definition 0.2.9: Homogeneous Ideal

An ideal $I \subseteq k[x_0, \dots, x_n]$ is *homogeneous* if it is generated by homogeneous polynomials. Equivalently, $I = \bigoplus_{d \geq 0}(I \cap S_d)$.

Proposition 0.2.10: Characterization of Homogeneous Ideals

An ideal I is homogeneous if and only if for every $f \in I$, all homogeneous components of f are also in I.

***Proof*:**
See [20].

Projective Varieties

Definition 0.2.11: Projective Algebraic Set

For a set T of homogeneous polynomials in $k[x_0, \dots, x_n]$, the *projective zero set* is

$$\mathcal{Z}_+(T) := \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid f(a_0, \dots, a_n) = 0, \ \forall f \in T\}$$

This is well-defined since f homogeneous implies $f(\lambda \mathbf{a}) = \lambda^d f(\mathbf{a})$, so $f(\mathbf{a}) = 0 \iff f(\lambda \mathbf{a}) = 0$. A subset $V \subseteq \mathbb{P}^n$ is a *projective algebraic set* if $V = \mathcal{Z}_+(T)$ for some set T of homogeneous polynomials.

Definition 0.2.12: Homogeneous Ideal of a Set

For $V \subseteq \mathbb{P}^n$, define

$$\mathcal{I}_+(V) := \text{ideal generated by } \{f \in k[x_0, \dots, x_n] \text{ homogeneous} \mid f(p) = 0, \ \forall p \in V\}$$

This is always a homogeneous radical ideal.

The projective algebraic sets form the closed sets of the *Zariski topology on* $\mathbb{P}^n$.

Definition 0.2.13: Projective Variety

A projective algebraic set $V \subseteq \mathbb{P}^n$ is a *projective variety* if it is irreducible in the Zariski topology.

0.2.14 The Irrelevant Ideal The ideal $\mathfrak{m}_+ = (x_0, x_1, \dots, x_n) \subseteq k[x_0, \dots, x_n]$ is called the *irrelevant ideal*. It satisfies $\mathcal{Z}_+(\mathfrak{m}_+) = \emptyset$, since the only common zero of all x_i in k^{n+1} is the origin, which does not correspond to a point in $\mathbb{P}^n$. This is an important subtlety in the projective Nullstellensatz.

Theorem 0.2.15: Projective Nullstellensatz

Let k be algebraically closed and let $I \subseteq k[x_0, \ldots, x_n]$ be a homogeneous ideal. Then:

1. $\mathcal{Z}_+(I) = \emptyset$ if and only if $\sqrt{I} \supseteq \mathfrak{m}_+^N$ for some $N \geq 1$ (equivalently, $\sqrt{I} = \mathfrak{m}_+$ or $\sqrt{I} = (1)$).
2. If $\mathcal{Z}_+(I) \neq \emptyset$, then $\mathcal{I}_+(\mathcal{Z}_+(I)) = \sqrt{I}$.

Hence $\mathcal{Z}_+$ and $\mathcal{I}_+$ give a bijection between:

$$\left\{\text{homogeneous radical ideals } I \text{ with } \sqrt{I} \neq \mathfrak{m}_+\right\} \longleftrightarrow \{\text{projective algebraic sets in } \mathbb{P}^n\}$$

***Proof*:**
See [20].

Homogeneous Coordinate Ring and the Proj Construction

Definition 0.2.16: Homogeneous Coordinate Ring

For a projective variety $V = \mathcal{Z}_+(I) \subseteq \mathbb{P}^n$, the *homogeneous coordinate ring* is the graded ring

$$S(V) := k[x_0, \ldots, x_n]/I$$

with the induced grading $S(V) = \bigoplus_{d \geq 0} S(V)_d$.

0.2.17 Homogeneous Coordinate Ring is Not Intrinsic Unlike the affine case, the homogeneous coordinate ring $S(V)$ depends on the embedding $V \hookrightarrow \mathbb{P}^n$, not just on V as an abstract variety. Different embeddings of the same variety can give non-isomorphic graded rings.

Definition 0.2.18: The Proj Construction

For a graded ring $S = \bigoplus_{d \geq 0} S_d$, we define

$$\operatorname{Proj}(S) := \{\mathfrak{p} \in \operatorname{Spec}(S) \mid \mathfrak{p} \text{ is homogeneous and } \mathfrak{p} \not\supseteq S_+\}$$

where $S_+ = \bigoplus_{d>0} S_d$ is the irrelevant ideal. This set is equipped with a natural topology and structure sheaf making it a scheme. For $S = k[x_0, \ldots, x_n]$, we recover $\operatorname{Proj}(S) = \mathbb{P}^n_k$.

Relation Between Affine and Projective

The affine charts give a way to study projective varieties locally:

Definition 0.2.19: Affine Cone

For a projective variety $V \subseteq \mathbb{P}^n$, the *affine cone* over V is

$$C(V) := \mathcal{Z}(\mathcal{I}_+(V)) \subseteq \mathbb{A}^{n+1}$$

This is an affine variety in $\mathbb{A}^{n+1}$, consisting of all lines through the origin that correspond to points of V, together with the origin itself.

Proposition 0.2.20: Affine Pieces of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ be a projective variety and $U_i \cong \mathbb{A}^n$ a standard affine chart. Then $V \cap U_i$ is an affine variety in $\mathbb{A}^n$.
Explicitly, if $V = \mathcal{Z}_+(f_1, \ldots, f_r)$ with each f_j homogeneous of degree d_j, then $V \cap U_i = \mathcal{Z}(f_1^{(i)}, \ldots, f_r^{(i)})$ where

$$f_j^{(i)}(y_0, \ldots, \widehat{y_i}, \ldots, y_n) := f_j(y_0, \ldots, y_{i-1}, 1, y_{i+1}, \ldots, y_n)$$

is the *dehomogenization* of f_j with respect to x_i.

Definition 0.2.21: Homogenization

Conversely, given $g \in k[y_1, \ldots, y_n]$ of degree d, its *homogenization* with respect to x_0 is

$$g^h(x_0, x_1, \ldots, x_n) := x_0^d \cdot g\left(\frac{x_1}{x_0}, \ldots, \frac{x_n}{x_0}\right)$$

This is homogeneous of degree d.

Example 0.2.22: Projective Closure

The parabola $V(y - x^2) \subseteq \mathbb{A}^2$ has homogenization $yz - x^2 \in k[x, y, z]$. Its projective closure is $\overline{V} = \mathcal{Z}_+(yz - x^2) \subseteq \mathbb{P}^2$. The “new” point $[0 : 1 : 0]$ (where $z = 0$ and $x = 0$) is the point at infinity where the parabola “closes up.”

Quasi-Projective Varieties

Definition 0.2.23: Quasi-Projective Variety

A *quasi-projective variety* is a locally closed subset of $\mathbb{P}^n$, i.e., an open subset of a projective variety. Equivalently, it is the intersection of an open set and a closed set in $\mathbb{P}^n$.

Proposition 0.2.24: Affine and Projective are Quasi-Projective

Both affine varieties and projective varieties are quasi-projective:

1. A projective variety $V \subseteq \mathbb{P}^n$ is quasi-projective (it is closed, hence locally closed).
2. An affine variety $V \subseteq \mathbb{A}^n \cong U_0 \subseteq \mathbb{P}^n$ is quasi-projective (it is closed in the open set U_0).

The category of quasi-projective varieties with regular maps is the classical category of "algebraic varieties" in much of the literature.

Morphisms of Projective Varieties

Defining morphisms of projective varieties requires care, since homogeneous polynomials do not define functions.

Definition 0.2.25: Morphism of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ and $W \subseteq \mathbb{P}^m$ be projective varieties. A *morphism* $\varphi : V \to W$ is a map that is locally given by homogeneous polynomials of the same degree. That is, for each $p \in V$, there exist homogeneous polynomials $f_0, \ldots, f_m \in k[x_0, \ldots, x_n]$ of the same degree d such that:

1. $f_0, \ldots, f_m$ do not simultaneously vanish on a neighborhood of p in V.
2. $\varphi(q) = [f_0(q) : \cdots : f_m(q)]$ for all q in this neighborhood.

0.2.26 Local Nature of Morphisms Different homogeneous polynomials may be needed on different open subsets of V. On overlaps, they must agree as maps to $\mathbb{P}^m$. This local definition is necessary because no single tuple of homogeneous polynomials may work globally.

Example 0.2.27: The Veronese Embedding

The *Veronese embedding* of degree d is

$$\nu_d : \mathbb{P}^n \to \mathbb{P}^N, \qquad [x_0 : \cdots : x_n] \mapsto [\cdots : x_0^{i_0} \cdots x_n^{i_n} : \cdots]$$

where the coordinates on $\mathbb{P}^N$ are indexed by monomials of degree d, so $N = \binom{n+d}{d} - 1$.

For example, $\nu_2 : \mathbb{P}^1 \to \mathbb{P}^2$ is given by $[s : t] \mapsto [s^2 : st : t^2]$. The image is the conic $\mathcal{Z}_+(xz - y^2)$, showing that $\mathbb{P}^1$ is isomorphic to this conic.

Example 0.2.28: The Segre Embedding

The *Segre embedding* is

$$\sigma : \mathbb{P}^n \times \mathbb{P}^m \to \mathbb{P}^{(n+1)(m+1)-1}, \qquad ([x_i], [y_j]) \mapsto [x_i y_j]_{0 \le i \le n,\, 0 \le j \le m}$$

For example, $\sigma : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$ is given by

$$([s : t], [u : v]) \mapsto [su : sv : tu : tv]$$

The image is the quadric surface $\mathcal{Z}_+(xw - yz) \subseteq \mathbb{P}^3$.

The Segre embedding shows that $\mathbb{P}^n \times \mathbb{P}^m$ can be given the structure of a projective variety.

Products of Projective Varieties

Proposition 0.2.29: Products of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ and $W \subseteq \mathbb{P}^m$ be projective varieties. Then $V \times W$, embedded via the Segre embedding $\mathbb{P}^n \times \mathbb{P}^m \hookrightarrow \mathbb{P}^{(n+1)(m+1)-1}$, is a projective variety.

***Proof*:**
See [20].

0.2.30 Contrast with Affine Products For affine varieties, products are straightforward: $V \times W \subseteq \mathbb{A}^n \times \mathbb{A}^m = \mathbb{A}^{n+m}$, and $k[V \times W] \cong k[V] \otimes_k k[W]$. The projective case requires the Segre embedding because $\mathbb{P}^n \times \mathbb{P}^m \ncong \mathbb{P}^{n+m}$.

Completeness and Properness

Projective varieties enjoy a crucial property not shared by affine varieties:

Definition 0.2.31: Complete Variety

A variety V is *complete* if for every variety W, the projection $\pi : V \times W \to W$ is a closed map (sends closed sets to closed sets).

Theorem 0.2.32: Projective Varieties are Complete

Every projective variety is complete.

***Proof*:**
See [20].

0.2.33 Completeness as Compactness Completeness is the algebraic analogue of compactness. Over $\mathbb{C}$, a projective variety is compact in the classical (Euclidean) topology. Affine varieties of positive dimension are never complete: for example, the projection $\mathbb{A}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$ does not send the closed hyperbola $\mathcal{Z}(xy - 1)$ to a closed set.

Proposition 0.2.34: Affine and Complete Implies Finite

If V is both affine and complete, then V is a finite set of points.

***Proof*:**
See [20].

0.3 Noetherian Induction or Well-founded Induction

This is something that will come up in later proofs (for example, see lemma 5.1.14). This is a form of induction that generalizes the usual induction where the well-ordered set is $\mathbb{N}$. Recall that a binary relation R on S is a subset of $S \times S$. For $(a, b) \in R$, we may write $a \preceq b$ or $b \succeq a$. Then the symbol $\preceq$ is sometimes called the binary relation. A binary relation is a strict partial order if it is:

1. Irreflexive: $\forall x \in A, \neg(x < x)$
2. Transitive: $\forall x, y, z \in A, (x < y \wedge y < z) \implies x < z$
3. Asymmetric: $\forall x, y \in A, x < y \implies \neg(y < x)$

Given a set S with a strict partial order $\preceq$, a descending chain is a sequence $a_1, a_2, a_3, \ldots$ of elements in A such that $a_1 \succeq a_2 \succeq a_3 \succeq \ldots$

Definition 0.3.1: Well-founded Relation

A strict partial order $\preceq$ on a set A is well-founded if there exist no infinite descending chains in A with respect to $\preceq$.

Definition 0.3.2: Minimal Element

Given a subset S of a well-founded set A, an element $m \in S$ is minimal if there is no element $x \in S$ such that $x \preceq m$.

Theorem 0.3.3: Noetherian Induction

Let $(A, \preceq)$ be a set with a well-founded relation $\preceq$, and let P be a property defined on A. If for all $x \in A : [\forall y \in A, (y \preceq x \implies P(y))] \implies P(x)$ then $P(x)$ holds for all $x \in A$.

***Proof*:**
see cite:HERE.

The following may be an insightful alternative:

Theorem 0.3.4: Noetherian Induction - Alternative Form

Let $(A, \preceq)$ be a set with a well-founded relation $\preceq$. If $S \subseteq A$ is a subset such that for all minimal elements m of $A \backslash S$, we have $m \in S$, then $S = A$.

***Proof*:**
see cite:HERE.

The connection between these forms lies in taking S to be the set of elements where P holds, and showing that if S isn't all of A, then any minimal element of $A \backslash S$ must actually be in S, giving a contradiction.

0.4 Categorical Reminder

Category theory provides the language and organizational framework for modern algebraic geometry. Schemes are defined as certain locally ringed spaces, morphisms are defined via universal properties, and many constructions (fiber products, sheafification, etc.) are instances of general categorical machinery. This section collects the essential concepts.

Categories, Functors, and Natural Transformations

Definition 0.4.1: Category

A *category* $\mathcal{C}$ consists of:

1. A class $\mathrm{Ob}(\mathcal{C})$ of *objects*.
2. For each pair of objects X, Y, a set of *morphisms* from X to Y denoted $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ or $\mathrm{Mor}_{\mathbb{C}}(X, Y)$.
3. For each triple X, Y, Z, a *composition* map $\mathrm{Hom}(Y, Z) \times \mathrm{Hom}(X, Y) \to \mathrm{Hom}(X, Z)$.
4. For each object X, an *identity* morphism $\mathrm{id}_X \in \mathrm{Hom}(X, X)$.

These satisfy associativity of composition and the identity laws.

Example 0.4.2: Categories in Algebra and Geometry

1. **Set**: sets and functions.
2. **Grp**, **Ab**, **Ring**, **CRing**: groups, abelian groups, rings, commutative rings.
3. $\mathbf{Mod}_R$: R-modules and R-linear maps.
4. **Top**: topological spaces and continuous maps.
5. $\mathbf{Aff}_k$: affine varieties over k and regular maps.

Definition 0.4.3: Functor

A *(covariant) functor* $F : \mathcal{C} \to \mathcal{D}$ consists of:

1. A map $F : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$.
2. For each $X, Y \in \mathcal{C}$, a map $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$.

These must satisfy $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ and $F(g \circ f) = F(g) \circ F(f)$.
A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ reverses arrows: $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(Y), F(X))$ with $F(g \circ f) = F(f) \circ F(g)$. Equivalently, it is a covariant functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$.

Example 0.4.4: Functors in Algebraic Geometry

1. The coordinate ring functor $V \mapsto k[V]$ is a contravariant functor $\mathbf{Aff}_k \to \mathbf{CRing}$.
2. Forgetful functors: $\mathbf{Grp} \to \mathbf{Set}$, etc.
3. We shall define the spectrum functor $A \mapsto \mathrm{Spec}\,(A)$ is a contravariant functor $\mathbf{CRing} \to \mathbf{Sch}$.
4. We shall define the global sections functor $X \mapsto \Gamma(X, \mathcal{O}_X)$ is a contravariant functor $\mathbf{Sch} \to \mathbf{CRing}$.

Definition 0.4.5: Natural Transformation

Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation* $\eta : F \Rightarrow G$ is a family of morphisms $\eta_X : F(X) \to G(X)$ for each $X \in \mathcal{C}$, such that for every morphism $f : X \to Y$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{tikzcd} F(X) \arrow[r, "\eta_X"] \arrow[d, "F(f)"'] & G(X) \arrow[d, "G(f)"] \\ F(Y) \arrow[r, "\eta_Y"'] & G(Y) \end{tikzcd}$$

A *natural isomorphism* is a natural transformation where each η_X is an isomorphism.

Definition 0.4.6: Equivalence of Categories

An *equivalence of categories* between $\mathcal{C}$ and $\mathcal{D}$ consists of functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ together with natural isomorphisms $G \circ F \cong \mathrm{id}_{\mathcal{C}}$ and $F \circ G \cong \mathrm{id}_{\mathcal{D}}$.
A functor F is an equivalence if and only if it is:

- *Fully faithful*: $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ is a bijection for all X, Y.
- *Essentially surjective*: every object of $\mathcal{D}$ is isomorphic to $F(X)$ for some $X \in \mathcal{C}$.

Example 0.4.7: The Classical Equivalence

The equivalence $\mathbf{Aff}_k \simeq (\mathbf{Alg}_k^{\text{f.g., red}})^{\mathrm{op}}$ from section 0.1 is an equivalence of categories: the coordinate ring functor is fully faithful and essentially surjective.

Universal Properties: Limits and Colimits

Many constructions in mathematics are characterized by universal properties. Limits and colimits provide a unified framework.

Definition 0.4.8: Diagram and Cone

Let J be a small category (the "index category"). A *diagram* of shape J in $\mathcal{C}$ is a functor $D : J \to \mathcal{C}$.
A *cone* over D with vertex $X \in \mathcal{C}$ is a collection of morphisms $\pi_j : X \to D(j)$ for each $j \in J$, such that for every morphism $\alpha : j \to j'$ in J, we have $D(\alpha) \circ \pi_j = \pi_{j'}$.

Definition 0.4.9: Limit

The *limit* of a diagram $D : J \to \mathcal{C}$, denoted $\varprojlim D$ or $\lim_J D$, is a cone $(\pi_j : L \to D(j))_{j \in J}$ that is universal: for any other cone $(f_j : X \to D(j))$, there exists a unique morphism $f : X \to L$ such that $\pi_j \circ f = f_j$ for all j.

$$\begin{tikzcd} X \arrow[d, dashed, "\exists! f"'] \arrow[dr, "f_j"] & \\ L \arrow[r, "\pi_j"'] & D(j) \end{tikzcd}$$

Definition 0.4.10: Colimit

Dually, a *cocone* under D with vertex X is a collection $\iota_j : D(j) \to X$ compatible with the diagram. The *colimit* $\varinjlim D$ is the universal cocone.

Example 0.4.11: Specific Limits and Colimits

Shape J	**Limit**	**Colimit**
Discrete (no arrows)	Product $\prod_i X_i$	Coproduct $\coprod_i X_i$
$\bullet \rightrightarrows \bullet$	Equalizer	Coequalizer
$\bullet \to \bullet \leftarrow \bullet$	Pullback (fiber product)	—
$\bullet \leftarrow \bullet \to \bullet$	—	Pushout
Empty	Terminal object $*$	Initial object $\emptyset$
Directed poset	Directed (inverse) limit	Directed (direct) limit

Products and Coproducts

Definition 0.4.12: Product

The *product* of objects $\{X_i\}_{i \in I}$ is an object $\prod_i X_i$ together with projection morphisms $\pi_j : \prod_i X_i \to X_j$ such that for any object Y with morphisms $f_j : Y \to X_j$, there exists a unique $f : Y \to \prod_i X_i$ with $\pi_j \circ f = f_j$.

Definition 0.4.13: Coproduct

Dually, the *coproduct* $\coprod_i X_i$ has inclusion morphisms $\iota_j : X_j \to \coprod_i X_i$ universal for morphisms out of the X_j.

Example 0.4.14: Products and Coproducts in Various Categories

Category	Product	Coproduct
Set	Cartesian product	Disjoint union
Grp	Direct product	Free product
Ab, $\mathbf{Mod}_R$	Direct product	Direct sum
CRing	Direct product	Tensor product $A \otimes_{\mathbb{Z}} B$
Top	Product topology	Disjoint union topology
Sch	Fiber product over $\operatorname{Spec} \mathbb{Z}$	Disjoint union

Fiber products are ubiquitous in algebraic geometry: base change, intersections, and fiber constructions all use them.

Definition 0.4.15: Fiber Product / Pullback

Given morphisms $f : X \to S$ and $g : Y \to S$, the *fiber product* (or *pullback*) is an object $X \times_S Y$ with morphisms $p : X \times_S Y \to X$ and $q : X \times_S Y \to Y$ such that $f \circ p = g \circ q$, universal with this property:

$$\begin{tikzcd} X \times_S Y \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, phantom, "\square"] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & S \end{tikzcd}$$

For any Z with morphisms to X and Y making the square commute, there is a unique morphism $Z \to X \times_S Y$.

Example 0.4.16: Fiber Products in Sets

In **Set**, the fiber product is

$$X \times_S Y = \{(x, y) \in X \times Y : f(x) = g(y)\}$$

If $S = *$ is a point, this is the ordinary Cartesian product.

Adjoint Functors

Adjunctions capture many important relationships between categories.

Definition 0.4.17: Adjoint Pair

Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be functors. We say F is *left adjoint* to G (and G is *right adjoint* to F), written $F \dashv G$, if there is a natural bijection

$$\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$$

for all $X \in \mathcal{C}$ and $Y \in \mathcal{D}$.

Proposition 0.4.18: Characterizations of Adjunctions

The following are equivalent to $F \dashv G$:

1. Natural bijection $\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$.
2. Existence of a *unit* $\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow G \circ F$ and *counit* $\varepsilon : F \circ G \Rightarrow \mathrm{id}_{\mathcal{D}}$ satisfying the triangle identities.
3. F preserves colimits and satisfies a solution set condition (Freyd's adjoint functor theorem).

***Proof*:**
See any standard reference on category theory, e.g., Mac Lane [9].

Example 0.4.19: Adjunctions in Algebra and Geometry

1. **Free-Forgetful**: The free group functor $F : \mathbf{Set} \to \mathbf{Grp}$ is left adjoint to the forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$:

$$\mathrm{Hom}_{\mathbf{Grp}}(F(S), G) \cong \mathrm{Hom}_{\mathbf{Set}}(S, U(G))$$

2. **Tensor-Hom**: For R-modules, $(-) \otimes_R M \dashv \mathrm{Hom}_R(M, -)$:

$$\mathrm{Hom}_R(N \otimes_R M, P) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, P))$$

3. **Inverse and Direct Image**: For a continuous map $f : X \to Y$, the inverse image functor f^{-1} on sheaves is left adjoint to the direct image f_*.

Two adjunctions we shall encounter are:

1. **Spec-Global Sections**: The functor $\mathrm{Spec} : \mathbf{CRing}^{\mathrm{op}} \to \mathbf{Sch}$ is left adjoint to $\Gamma : \mathbf{Sch} \to \mathbf{CRing}^{\mathrm{op}}$:

$$\mathrm{Hom}_{\mathbf{Sch}}(\mathrm{Spec}\,(A), X) \cong \mathrm{Hom}_{\mathbf{CRing}}(\Gamma(X, \mathcal{O}_X), A)$$

2. **Sheafification**: The sheafification functor is left adjoint to the inclusion of sheaves into presheaves.

Proposition 0.4.20: Adjoints and Limits

1. Left adjoints preserve colimits.
2. Right adjoints preserve limits.

In particular, if $F \dashv G$, then F preserves coproducts, coequalizers, and pushouts, while G preserves products, equalizers, and pullbacks.

***Proof*:**
See [9].

Representable Functors and the Yoneda Lemma

The Yoneda perspective is fundamental to modern algebraic geometry: schemes can be understood through their "functor of points."

Definition 0.4.21: Representable Functor

A functor $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ is *representable* if there exists an object $X \in \mathcal{C}$ and a natural isomorphism

$$F \cong \mathrm{Hom}_{\mathcal{C}}(-, X) =: h_X$$

The object X is called the *representing object*. We say "X represents F" or "F is represented by X."

Theorem 0.4.22: Yoneda Lemma

Let $\mathcal{C}$ be a category, $X \in \mathcal{C}$, and $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ a functor. There is a natural bijection

$$\mathrm{Nat}(h_X, F) \cong F(X)$$

where $\mathrm{Nat}(h_X, F)$ is the set of natural transformations from h_X to F. The bijection sends $\eta : h_X \Rightarrow F$ to $\eta_X(\mathrm{id}_X) \in F(X)$.

***Proof*:**
See [9].

Corollary 0.4.23: Yoneda Embedding

The functor $h : \mathcal{C} \to \mathbf{Fun}(\mathcal{C}^{\mathrm{op}}, \mathbf{Set})$ given by $X \mapsto h_X = \mathrm{Hom}(-, X)$ is fully faithful. In particular:

1. If $h_X \cong h_Y$, then $X \cong Y$ (representing objects are unique up to unique isomorphism).
2. $\mathrm{Hom}_{\mathcal{C}}(X, Y) \cong \mathrm{Nat}(h_X, h_Y)$.

Example 0.4.24: Moduli Problems as Functors
Many moduli problems are phrased as functors $F : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$:

- $F(S) = \{\text{families of elliptic curves over } S\}$
- $F(S) = \{\text{closed subschemes of } \mathbb{P}^n_S \text{ flat over } S\}$ (Hilbert functor)

If F is representable by a scheme M, then M is the *moduli space*. Often F is not representable by a scheme but by an algebraic stack.

Filtered Colimits

Filtered colimits arise naturally when taking stalks of sheaves and germs of functions.

Definition 0.4.25: Filtered Category

A category J is *filtered* if:

1. J is nonempty.
2. For any $i, j \in J$, there exists $k \in J$ with morphisms $i \to k$ and $j \to k$.
3. For any parallel morphisms $f, g : i \to j$, there exists $h : j \to k$ with $h \circ f = h \circ g$.

A *directed set* (poset where every pair has an upper bound) is filtered when viewed as a category.

Definition 0.4.26: Filtered Colimit

A *filtered colimit* (or *direct limit*) is a colimit over a filtered index category.

Example 0.4.27: Localization as Filtered Colimit
For a ring A and multiplicative set S, the localization is

$$S^{-1}A = \varinjlim_{s \in S} A_s$$

where $A_s = A[1/s]$ and the colimit is over S ordered by divisibility.

Proposition 0.4.28: Filtered Colimits Commute with Finite Limits

In **Set** (and more generally in any *finitely presentable* category), filtered colimits commute with finite limits. In particular, filtered colimits are exact in module categories.

***Proof*:**
See [9].

Abelian Categories

For homological algebra and cohomology, we need abelian categories.

Definition 0.4.29: Abelian Category

A category $\mathcal{A}$ is *abelian* if:

1. $\mathcal{A}$ has a zero object (both initial and terminal).
2. $\mathcal{A}$ has all binary products and coproducts (which coincide: biproducts).
3. Every morphism has a kernel and cokernel.
4. Every monomorphism is a kernel; every epimorphism is a cokernel.

Example 0.4.30: Abelian Categories

1. $\mathbf{Ab}$, $\mathbf{Mod}_R$ for any ring R.
2. $\mathbf{Sh}(X, \mathbf{Ab})$: sheaves of abelian groups on X.
3. $\mathbf{QCoh}(X)$: quasi-coherent sheaves on a scheme X.
4. $\mathbf{Coh}(X)$: coherent sheaves on a Noetherian scheme X.

0.4.31 Homological Algebra in Abelian Categories In an abelian category, one can define:

- Exact sequences: $A \to B \to C$ is exact at B if $\ker(B \to C) = \operatorname{im}(A \to B)$.
- Chain complexes and homology.
- Derived functors (if there are enough injectives or projectives).

Sheaf cohomology $H^i(X, \mathcal{F})$ is defined as the right derived functors of the global sections functor $\Gamma(X, -)$.

0.4.1 Summary: Categorical Tools for Schemes

Concept	**Use in Algebraic Geometry**
Equivalence of categories	Classical: $\mathbf{Aff}_k \simeq (\mathbf{Alg}_k^{\text{f.g.,red}})^{\text{op}}$
Fiber products	Base change, fibers, intersections
Adjunctions	$\operatorname{Spec} \dashv \Gamma$; $f^{-1} \dashv f_*$; sheafification
Representable functors	Schemes via functor of points
Yoneda lemma	Universal properties; moduli problems
Filtered colimits	Stalks, localizations
Abelian categories	Sheaf cohomology, derived categories

0.5 *Key Results from Differential Geometry

Many constructions in algebraic geometry—tangent bundles, differential forms, cohomology, metrics—have their origins in differential geometry. This section reviews the essential concepts from the smooth category that will inform and motivate their algebraic counterparts.

Throughout this section, "smooth" means C^∞ (infinitely differentiable).

Smooth Manifolds

Definition 0.5.1: Smooth Manifold

A *topological manifold* of dimension n is a Hausdorff, second-countable topological space M that is locally homeomorphic to $\mathbb{R}^n$. A *chart* is a pair (U, φ) where $U \subseteq M$ is open and $\varphi : U \xrightarrow{\sim} \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism.
A *smooth structure* on M is a maximal atlas $\{(U_\alpha, \varphi_\alpha)\}$ such that all transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

are smooth (as maps between open subsets of $\mathbb{R}^n$).
A *smooth manifold* is a topological manifold equipped with a smooth structure.

Example 0.5.2: Examples of Smooth Manifolds

1. $\mathbb{R}^n$ with the standard atlas (single chart $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$).
2. Open subsets of $\mathbb{R}^n$ (or of any smooth manifold).
3. The n-sphere $S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}$.
4. Real and complex projective spaces $\mathbb{RP}^n$, $\mathbb{CP}^n$.
5. Lie groups: $\mathrm{GL}_n(\mathbb{R})$, $\mathrm{O}(n)$, $\mathrm{U}(n)$, $\mathrm{SL}_n(\mathbb{R})$, etc.
6. Any smooth algebraic variety over $\mathbb{R}$ or $\mathbb{C}$ (with its classical topology).

Definition 0.5.3: Smooth Map

A map $f : M \to N$ between smooth manifolds is *smooth* if for every $p \in M$ and charts (U, φ) around p and (V, ψ) around $f(p)$, the composition

$$\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \to \psi(V)$$

is smooth as a map between open subsets of Euclidean spaces.
A *diffeomorphism* is a smooth bijection with smooth inverse.

Definition 0.5.4: Tangent Space (via Derivations)

Let M be a smooth manifold and $p \in M$. A *derivation at* p is an $\mathbb{R}$-linear map $v : C^\infty(M) \to \mathbb{R}$ satisfying the Leibniz rule:

$$v(fg) = f(p) \cdot v(g) + g(p) \cdot v(f)$$

The *tangent space* T_pM is the vector space of all derivations at p.

Proposition 0.5.5: Dimension of Tangent Space

If $\dim M = n$, then $\dim T_pM = n$ for all $p \in M$. In local coordinates $(x^1, \ldots, x^n)$ centered at p, a basis for T_pM is given by the partial derivative operators:

$$\left\{ \frac{\partial}{\partial x^1}\bigg|_p, \ldots, \frac{\partial}{\partial x^n}\bigg|_p \right\}$$

Definition 0.5.6: Differential / Pushforward

Let $f : M \to N$ be smooth and $p \in M$. The *differential* (or *pushforward*) of f at p is the linear map

$$df_p = f_{*,p} : T_pM \to T_{f(p)}N, \qquad (df_p(v))(g) := v(g \circ f)$$

for $v \in T_pM$ and $g \in C^\infty(N)$.

Definition 0.5.7: Tangent Bundle

The *tangent bundle* of M is the disjoint union

$$TM := \bigsqcup_{p \in M} T_pM = \{(p, v) : p \in M,\ v \in T_pM\}$$

equipped with the natural projection $\pi : TM \to M$, $(p, v) \mapsto p$. This is a smooth manifold of dimension $2n$ and a vector bundle of rank n over M.

Definition 0.5.8: Vector Field

A *vector field* on M is a smooth section of the tangent bundle, i.e., a smooth map $X : M \to TM$ such that $\pi \circ X = \mathrm{id}_M$. Equivalently, X assigns to each $p \in M$ a tangent vector $X_p \in T_pM$, varying smoothly with p.
The space of vector fields is denoted $\mathfrak{X}(M)$ or $\Gamma(TM)$.

Proposition 0.5.9: Vector Fields as Derivations

A vector field $X \in \mathfrak{X}(M)$ defines a derivation $X : C^\infty(M) \to C^\infty(M)$ by $(Xf)(p) := X_p(f)$. This gives an isomorphism between $\mathfrak{X}(M)$ and the $C^\infty(M)$-module of derivations of $C^\infty(M)$.

Definition 0.5.10: Lie Bracket

The *Lie bracket* of vector fields $X, Y \in \mathfrak{X}(M)$ is the vector field $[X, Y]$ defined by

$$[X, Y](f) := X(Y(f)) - Y(X(f))$$

This makes $\mathfrak{X}(M)$ into a Lie algebra over $\mathbb{R}$.

Cotangent Spaces and Differential Forms

Definition 0.5.11: Cotangent Space

The *cotangent space* at $p \in M$ is the dual vector space

$$T_p^*M := (T_pM)^* = \mathrm{Hom}_{\mathbb{R}}(T_pM, \mathbb{R})$$

Elements of T_p^*M are called *cotangent vectors* or *covectors*.

Definition 0.5.12: Differential of a Function

For $f \in C^\infty(M)$ and $p \in M$, the *differential* $df_p \in T_p^*M$ is defined by

$$df_p(v) := v(f) \quad \text{for } v \in T_pM$$

In local coordinates, $df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i$, where $\{dx^1, \ldots, dx^n\}$ is the dual basis to $\{\partial/\partial x^1, \ldots, \partial/\partial x^n\}$.

Definition 0.5.13: Cotangent Bundle

The *cotangent bundle* is $T^*M := \bigsqcup_{p \in M} T_p^*M$, a vector bundle of rank n over M.

Definition 0.5.14: Differential k-Forms

A *differential k-form* (or simply *k-form*) on M is a smooth section of $\bigwedge^k T^*M$. That is, a k-form ω assigns to each $p \in M$ an alternating k-linear map

$$\omega_p : \underbrace{T_pM \times \cdots \times T_pM}_{k \text{ times}} \to \mathbb{R}$$

varying smoothly with p.
The space of k-forms is denoted $\Omega^k(M)$. We set $\Omega^0(M) = C^\infty(M)$.

Example 0.5.15: Differential Forms in Coordinates
In local coordinates $(x^1, \dots, x^n)$, every k-form can be written as

$$\omega = \sum_{i_1 < \cdots < i_k} f_{i_1 \dots i_k} \, dx^{i_1} \wedge \cdots \wedge dx^{i_k}$$

where $f_{i_1 \dots i_k} \in C^\infty(M)$ and $\wedge$ denotes the wedge product.

Definition 0.5.16: Wedge Product

The *wedge product* $\wedge : \Omega^k(M) \times \Omega^\ell(M) \to \Omega^{k+\ell}(M)$ is the unique bilinear, associative operation satisfying:

1. $df \wedge dg = -dg \wedge df$ for $f, g \in C^\infty(M)$.
2. $(f\omega) \wedge \eta = f(\omega \wedge \eta) = \omega \wedge (f\eta)$ for $f \in C^\infty(M)$.
3. $\omega \wedge \eta = (-1)^{k\ell} \eta \wedge \omega$ for $\omega \in \Omega^k$, $\eta \in \Omega^\ell$.

This makes $\Omega^*(M) := \bigoplus_{k \geq 0} \Omega^k(M)$ into a graded-commutative algebra.

Definition 0.5.17: Pullback of Forms

Let $f : M \to N$ be smooth. The *pullback* $f^* : \Omega^k(N) \to \Omega^k(M)$ is defined by

$$(f^*\omega)_p(v_1, \dots, v_k) := \omega_{f(p)}(df_p(v_1), \dots, df_p(v_k))$$

The pullback satisfies $f^*(\omega \wedge \eta) = f^*\omega \wedge f^*\eta$ and $(g \circ f)^* = f^* \circ g^*$.

Exterior Derivative and de Rham Cohomology

Definition 0.5.18: Exterior Derivative

The *exterior derivative* is the unique family of $\mathbb{R}$-linear maps $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying:

1. On $\Omega^0(M) = C^\infty(M)$, d is the differential: $(df)(X) = X(f)$.
2. $d \circ d = 0$.
3. $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ for $\omega \in \Omega^k$ (graded Leibniz rule).

Example 0.5.19: Exterior Derivative in Coordinates
For $\omega = \sum_I f_I \, dx^I$ (where I is a multi-index), we have

$$d\omega = \sum_I \sum_{j=1}^n \frac{\partial f_I}{\partial x^j} dx^j \wedge dx^I$$

For instance, if $\omega = f\,dx + g\,dy$ on $\mathbb{R}^2$, then

$$d\omega = \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) dx \wedge dy$$

Proposition 0.5.20: Naturality of d

The exterior derivative commutes with pullback: for $f : M \to N$ smooth,

$$f^* \circ d = d \circ f^*$$

That is, $f^*(d\omega) = d(f^*\omega)$ for all $\omega \in \Omega^*(N)$.

Definition 0.5.21: de Rham Complex

The *de Rham complex* of M is the cochain complex

$$0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(M) \to 0$$

A form ω is *closed* if $d\omega = 0$, and *exact* if $\omega = d\eta$ for some η. Since $d^2 = 0$, every exact form is closed.

Definition 0.5.22: de Rham Cohomology

The *k-th de Rham cohomology* of M is

$$H^k_{\mathrm{dR}}(M) := \frac{\ker(d : \Omega^k \to \Omega^{k+1})}{\operatorname{im}(d : \Omega^{k-1} \to \Omega^k)} = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}}$$

This is a real vector space measuring the “failure of closed forms to be exact.”

Theorem 0.5.23: de Rham's Theorem

For a smooth manifold M, there is a natural isomorphism

$$H^k_{\mathrm{dR}}(M) \cong H^k(M; \mathbb{R})$$

where $H^k(M;\mathbb{R})$ is the singular cohomology of M with real coefficients. The isomorphism is given by integration of forms over cycles.

***Proof*:**
See [53]

Example 0.5.24: de Rham Cohomology Examples

1. $H^k_{\mathrm{dR}}(\mathbb{R}^n) = 0$ for $k > 0$, and $H^0_{\mathrm{dR}}(\mathbb{R}^n) = \mathbb{R}$ (Poincaré lemma).
2. $H^0_{\mathrm{dR}}(M) = \mathbb{R}^{\pi_0(M)}$, where $\pi_0(M)$ is the number of connected components.

3. $H^1_{\mathrm{dR}}(S^1) = \mathbb{R}$, generated by $[d\theta]$.
4. $H^k_{\mathrm{dR}}(S^n) = \mathbb{R}$ for $k = 0, n$, and 0 otherwise.

0.5.25 Algebraic de Rham Cohomology For a smooth algebraic variety X over a field k of characteristic zero, one can define *algebraic de Rham cohomology* using the complex of algebraic differential forms $\Omega^*_{X/k}$. Over $\mathbb{C}$, this agrees with singular cohomology by Grothendieck's algebraic de Rham theorem.

Vector Bundles

Definition 0.5.26: Vector Bundle

A *(real) vector bundle of rank* r over a smooth manifold M is a smooth manifold E together with:

1. A smooth surjection $\pi : E \to M$.
2. For each $p \in M$, the fiber $E_p := \pi^{-1}(p)$ has the structure of an r-dimensional real vector space.
3. Local triviality: each $p \in M$ has a neighborhood U with a diffeomorphism $\varphi : \pi^{-1}(U) \xrightarrow{\sim} U \times \mathbb{R}^r$ such that φ is linear on each fiber.

Example 0.5.27: Vector Bundle Examples

1. The *trivial bundle* $M \times \mathbb{R}^r \to M$.
2. The tangent bundle TM and cotangent bundle T^*M.
3. The *tautological line bundle* $\mathcal{O}(-1)$ over $\mathbb{RP}^n$ or $\mathbb{CP}^n$: the fiber over a line ℓ is ℓ itself.
4. Exterior powers $\bigwedge^k T^*M$ (whose sections are k-forms).
5. The *canonical bundle* $K_M := \bigwedge^n T^*M$ (top exterior power).

Definition 0.5.28: Section of a Vector Bundle

A *section* of $\pi : E \to M$ is a smooth map $s : M \to E$ with $\pi \circ s = \mathrm{id}_M$. The space of sections is denoted $\Gamma(E)$ or $\Gamma(M, E)$. This is a $C^\infty(M)$-module.

Definition 0.5.29: Operations on Vector Bundles

Given vector bundles E, F over M:

- *Direct sum*: $(E \oplus F)_p = E_p \oplus F_p$.
- *Tensor product*: $(E \otimes F)_p = E_p \otimes_{\mathbb{R}} F_p$.
- *Dual*: $E_p^* = (E_p)^*$.
- *Exterior powers*: $(\bigwedge^k E)_p = \bigwedge^k E_p$.
- *Pullback*: for $f : N \to M$, $(f^*E)_q = E_{f(q)}$.

Definition 0.5.30: Transition Functions

If $\{U_\alpha\}$ is an open cover of M with local trivializations $\varphi_\alpha : E|_{U_\alpha} \xrightarrow{\sim} U_\alpha \times \mathbb{R}^r$, the *transition functions* are

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}_r(\mathbb{R}), \qquad \varphi_\alpha \circ \varphi_\beta^{-1}(p, v) = (p, g_{\alpha\beta}(p) \cdot v)$$

These satisfy the cocycle condition: $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps.

0.5.31 Line Bundles and Divisors A *line bundle* is a vector bundle of rank 1. Line bundles are classified by $H^1(M; C^\infty(M, \mathbb{R}^\times))$ via transition functions. In algebraic geometry, line bundles on a variety X correspond to Cartier divisors and form the Picard group $\mathrm{Pic}(X)$.

Connections and Curvature

Definition 0.5.32: Connection on a Vector Bundle

A *connection* (or *covariant derivative*) on a vector bundle $E \to M$ is an $\mathbb{R}$-linear map

$$\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E) = \Omega^1(M, E)$$

satisfying the Leibniz rule: for $f \in C^\infty(M)$ and $s \in \Gamma(E)$,

$$\nabla(fs) = df \otimes s + f\nabla s$$

For a vector field X, we write $\nabla_X s := \iota_X(\nabla s) \in \Gamma(E)$ for the covariant derivative of s in direction X.

0.5.33 Connections Exist Every vector bundle over a paracompact manifold admits a connection. Connections form an affine space over $\Omega^1(M, \mathrm{End}(E))$: if ∇ is a connection and $A \in \Omega^1(M, \mathrm{End}(E))$, then $\nabla + A$ is also a connection.

Definition 0.5.34: Curvature

The *curvature* of a connection ∇ is the $\mathrm{End}(E)$-valued 2-form $F_\nabla \in \Omega^2(M, \mathrm{End}(E))$ defined by

$$F_\nabla(X, Y) := \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]}$$

for vector fields X, Y. A connection is *flat* if $F_\nabla = 0$.

Proposition 0.5.35: Curvature as Obstruction

A connection ∇ is flat if and only if parallel transport is path-independent (depends only on the homotopy class of the path). Flat connections on a trivial bundle correspond to representations $\pi_1(M) \to \mathrm{GL}_r(\mathbb{R})$.

Riemannian Metrics

Definition 0.5.36: Riemannian Metric

A *Riemannian metric* on M is a smooth assignment of an inner product $g_p : T_pM \times T_pM \to \mathbb{R}$ to each $p \in M$. In local coordinates, $g = \sum_{i,j} g_{ij}\, dx^i \otimes dx^j$ where $(g_{ij}(p))$ is a positive definite symmetric matrix for each p.
A *Riemannian manifold* is a pair (M, g).

Example 0.5.37: Riemannian Metrics

1. The standard metric on $\mathbb{R}^n$: $g = dx^1 \otimes dx^1 + \cdots + dx^n \otimes dx^n$.
2. The round metric on S^n (induced from $\mathbb{R}^{n+1}$).
3. The hyperbolic metric on the upper half-plane: $g = \frac{dx^2+dy^2}{y^2}$.
4. The Fubini–Study metric on $\mathbb{CP}^n$.

Proposition 0.5.38: Existence of Riemannian Metrics

Every smooth manifold admits a Riemannian metric. (This uses partitions of unity.)

Definition 0.5.39: Levi-Civita Connection

On a Riemannian manifold (M, g), there is a unique connection ∇ on TM that is:

1. *Metric-compatible*: $X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.
2. *Torsion-free*: $\nabla_X Y - \nabla_Y X = [X, Y]$.

This is called the *Levi-Civita connection*.

Definition 0.5.40: Geodesics

A *geodesic* is a curve $\gamma : I \to M$ satisfying $\nabla_{\dot\gamma}\dot\gamma = 0$. Geodesics are locally length-minimizing curves.

Integration and Orientation

Definition 0.5.41: Orientation

An *orientation* on an n-manifold M is a choice of smoothly varying orientation on each tangent space T_pM. Equivalently, it is a nowhere-vanishing n-form $\omega \in \Omega^n(M)$, or a trivialization of the canonical bundle $K_M = \bigwedge^n T^*M$.
A manifold is *orientable* if it admits an orientation, and *oriented* if one has been chosen.

Example 0.5.42: Orientability

1. $\mathbb{R}^n$, S^n, and $\mathbb{CP}^n$ are orientable.
2. $\mathbb{RP}^n$ is orientable if and only if n is odd.
3. The Möbius strip is not orientable.

Definition 0.5.43: Integration of Top Forms

Let M be an oriented n-manifold and $\omega \in \Omega^n_c(M)$ a compactly supported n-form. The *integral* $\int_M \omega$ is defined by:

1. In a single oriented chart (U, φ) with $\operatorname{supp}(\omega) \subseteq U$: write $\omega = f\,dx^1 \wedge \cdots \wedge dx^n$ and set
$$\int_M \omega := \int_{\varphi(U)} f(\varphi^{-1}(x))\,dx^1 \cdots dx^n$$
2. For general ω, use a partition of unity to reduce to the local case.

Theorem 0.5.44: Stokes' Theorem

Let M be an oriented n-manifold with boundary ∂M (given the induced orientation), and let $\omega \in \Omega^{n-1}_c(M)$. Then
$$\int_M d\omega = \int_{\partial M} \omega$$
In particular, if M is closed (compact without boundary), then $\int_M d\omega = 0$.

***Proof*:**
See [53]

0.5.45 Classical Theorems as Special Cases Stokes' theorem generalizes:

- The fundamental theorem of calculus ($n = 1$).
- Green's theorem ($n = 2$, planar region).
- The classical Stokes theorem ($n = 3$, surface in $\mathbb{R}^3$).
- The divergence theorem ($n = 3$, solid region in $\mathbb{R}^3$).

Partitions of Unity

Definition 0.5.46: Partition of Unity

Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of M. A *partition of unity subordinate to* $\{U_\alpha\}$ is a collection of smooth functions $\{\rho_\alpha : M \to [0,1]\}_{\alpha \in A}$ such that:

1. $\operatorname{supp}(\rho_\alpha) \subseteq U_\alpha$ for each α.
2. The collection $\{\operatorname{supp}(\rho_\alpha)\}$ is locally finite.
3. $\sum_\alpha \rho_\alpha(p) = 1$ for all $p \in M$.

Theorem 0.5.47: Existence of Partitions of Unity

Every open cover of a smooth manifold admits a subordinate partition of unity.

***Proof*:**
See [53].

0.5.48 Partitions of Unity in Algebraic Geometry Partitions of unity are a crucial tool in differential geometry, enabling:

- Existence of Riemannian metrics.
- Existence of connections.
- Extension of local constructions to global ones.
- Proof that smooth manifolds are paracompact.

Importantly, partitions of unity do not exist in algebraic geometry. Polynomial or algebraic functions cannot be "bump functions." This is one reason why sheaf cohomology becomes essential in algebraic geometry: it replaces the role of partitions of unity in extending local to global.

0.5.1 Summary: Differential vs. Algebraic Geometry

Concept	Differential Geometry	Algebraic Geometry
Local model	$\mathbb{R}^n$ or $\mathbb{C}^n$	Spec (A)
Functions	$C^\infty(M)$	$\mathcal{O}_X(X)$ (regular functions)
Tangent space	T_pM (derivations)	$(\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$
Cotangent space	T_p^*M	$\mathfrak{m}_p/\mathfrak{m}_p^2$
Vector bundle	Smooth fiber bundle	Locally free sheaf
Differential forms	$\Omega^k(M)$	$\Omega^k_{X/k}$ (Kähler differentials)
Cohomology	de Rham $H^*_{\mathrm{dR}}(M)$	Sheaf cohomology $H^*(X, \mathcal{F})$
Partitions of unity	Exist	Do not exist
Gluing	Smooth	Via sheaf axioms

The absence of partitions of unity in the algebraic world makes sheaf-theoretic methods indispensable. Many constructions that are "automatic" in differential geometry (e.g., existence of metrics) require careful sheaf-cohomological arguments in algebraic geometry.

0.6 *Key Results from Complex Geometry

Complex geometry sits at the intersection of differential geometry and algebraic geometry. Over $\mathbb{C}$, algebraic varieties carry both a Zariski topology (algebraic) and a classical topology (analytic), and Serre's GAGA theorems establish connections between these perspectives. This section reviews the essential concepts from complex analytic geometry that inform and enrich algebraic geometry over $\mathbb{C}$.

Complex Manifolds

Definition 0.6.1: Complex Manifold

A *complex manifold* of dimension n is a smooth manifold M of real dimension $2n$ equipped with an atlas $\{(U_\alpha, \varphi_\alpha)\}$ where each $\varphi_\alpha : U_\alpha \xrightarrow{\sim} \varphi_\alpha(U_\alpha) \subseteq \mathbb{C}^n$ is a homeomorphism onto an open subset of $\mathbb{C}^n$, and all transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

are *holomorphic* (i.e., complex analytic).

Definition 0.6.2: Holomorphic Function

A function $f : U \to \mathbb{C}$ on an open set $U \subseteq \mathbb{C}^n$ is *holomorphic* if it satisfies the Cauchy–Riemann equations in each variable, or equivalently, if it is complex differentiable at every point. In coordinates $z_j = x_j + iy_j$, this means

$$\frac{\partial f}{\partial \bar{z}_j} = 0 \quad \text{for all } j$$

where $\frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right)$.

Definition 0.6.3: Holomorphic Map

A map $f : M \to N$ between complex manifolds is *holomorphic* if it is holomorphic in local coordinates. A *biholomorphism* is a holomorphic bijection with holomorphic inverse.

Example 0.6.4: Complex Manifolds

1. $\mathbb{C}^n$ with the standard complex structure.
2. Open subsets of $\mathbb{C}^n$ (or of any complex manifold).
3. Complex projective space $\mathbb{CP}^n$: covered by $n+1$ charts $U_i = \{[z_0 : \cdots : z_n] : z_i \neq 0\} \cong \mathbb{C}^n$.
4. Smooth projective varieties over $\mathbb{C}$ (with the classical topology).
5. Complex tori $\mathbb{C}^n/\Lambda$ where $\Lambda \cong \mathbb{Z}^{2n}$ is a lattice.
6. Riemann surfaces: complex manifolds of dimension 1.

0.6.5 Smooth vs. Complex vs. Algebraic We have strict inclusions of categories:

$$\{\text{smooth projective varieties}/\mathbb{C}\} \subsetneq \{\text{compact Kähler manifolds}\} \subsetneq \{\text{compact complex manifolds}\}$$

Not every complex manifold is algebraic (e.g., generic complex tori of dimension ≥ 2), and not every compact complex manifold is Kähler (e.g., most Hopf surfaces).

Almost Complex and Complex Structures

Definition 0.6.6: Almost Complex Structure

An *almost complex structure* on a smooth manifold M is a smooth bundle endomorphism $J : TM \to TM$ satisfying $J^2 = -\mathrm{id}$. This makes each tangent space T_pM into a complex vector space via $(a + bi) \cdot v := av + bJ(v)$.

Proposition 0.6.7: Complex Manifolds Have Almost Complex Structures

Every complex manifold has a natural almost complex structure: in holomorphic coordinates $(z_1, \ldots, z_n)$ with $z_j = x_j + iy_j$,

$$J\left(\frac{\partial}{\partial x_j}\right) = \frac{\partial}{\partial y_j}, \qquad J\left(\frac{\partial}{\partial y_j}\right) = -\frac{\partial}{\partial x_j}$$

The converse is not automatic:

Definition 0.6.8: Integrable Almost Complex Structure

An almost complex structure J is *integrable* if it arises from a complex manifold structure. The *Nijenhuis tensor* N_J measures the obstruction to integrability.

Theorem 0.6.9: Newlander–Nirenberg Theorem

An almost complex structure J is integrable if and only if the Nijenhuis tensor vanishes: $N_J = 0$.

***Proof*:**
see [14]

Complexified Tangent Bundle and (p, q)-Forms

Definition 0.6.10: Complexified Tangent Bundle

For a complex manifold M, the *complexified tangent bundle* is $T_{\mathbb{C}}M := TM \otimes_{\mathbb{R}} \mathbb{C}$. The almost complex structure J extends $\mathbb{C}$-linearly and has eigenvalues $\pm i$. This gives a decomposition

$$T_{\mathbb{C}}M = T^{1,0}M \oplus T^{0,1}M$$

where $T^{1,0}M$ is the $+i$ eigenspace and $T^{0,1}M$ is the $-i$ eigenspace.
In local coordinates, $T^{1,0}M$ is spanned by $\frac{\partial}{\partial z_j} := \frac{1}{2}\left(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\right)$ and $T^{0,1}M$ by $\frac{\partial}{\partial \bar{z}_j} := \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right)$.

Definition 0.6.11: (p,q)-Forms

The complexified cotangent bundle similarly decomposes: $T^*_{\mathbb{C}}M = (T^{1,0}M)^* \oplus (T^{0,1}M)^*$. A *differential form of type* (p,q) is a section of

$$\bigwedge^{p,q} T^*_{\mathbb{C}}M := \bigwedge^{p}(T^{1,0}M)^* \otimes \bigwedge^{q}(T^{0,1}M)^*$$

The space of (p,q)-forms is denoted $\Omega^{p,q}(M)$ or $\mathcal{A}^{p,q}(M)$.

Proposition 0.6.12: Decomposition of Forms

The space of complex k-forms decomposes as

$$\Omega^k(M,\mathbb{C}) = \bigoplus_{p+q=k} \Omega^{p,q}(M)$$

In local coordinates, a (p,q)-form can be written as

$$\omega = \sum_{|I|=p,|J|=q} f_{I,J}\, dz^I \wedge d\bar{z}^J$$

where $dz^I = dz_{i_1} \wedge \cdots \wedge dz_{i_p}$ and $d\bar{z}^J = d\bar{z}_{j_1} \wedge \cdots \wedge d\bar{z}_{j_q}$.

The Dolbeault Operators

Definition 0.6.13: Dolbeault Operators

The exterior derivative $d : \Omega^k(M,\mathbb{C}) \to \Omega^{k+1}(M,\mathbb{C})$ decomposes as $d = \partial + \bar{\partial}$ where:

$$\partial : \Omega^{p,q}(M) \to \Omega^{p+1,q}(M)$$
$$\bar{\partial} : \Omega^{p,q}(M) \to \Omega^{p,q+1}(M)$$

These satisfy $\partial^2 = 0$, $\bar{\partial}^2 = 0$, and $\partial\bar{\partial} + \bar{\partial}\partial = 0$.

Example 0.6.14: Dolbeault Operators in Coordinates

For $f \in C^\infty(M,\mathbb{C}) = \Omega^{0,0}(M)$:

$$\partial f = \sum_{j=1}^{n} \frac{\partial f}{\partial z_j} dz_j, \qquad \bar{\partial} f = \sum_{j=1}^{n} \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j$$

A function is holomorphic if and only if $\bar{\partial} f = 0$.

Definition 0.6.15: Dolbeault Complex

For each p, the *Dolbeault complex* is

$$0 \to \Omega^{p,0}(M) \xrightarrow{\bar{\partial}} \Omega^{p,1}(M) \xrightarrow{\bar{\partial}} \Omega^{p,2}(M) \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \Omega^{p,n}(M) \to 0$$

Definition 0.6.16: Dolbeault Cohomology

The *Dolbeault cohomology groups* are

$$H^{p,q}_{\bar{\partial}}(M) := \frac{\ker(\bar{\partial} : \Omega^{p,q} \to \Omega^{p,q+1})}{\operatorname{im}(\bar{\partial} : \Omega^{p,q-1} \to \Omega^{p,q})}$$

These are complex vector spaces measuring the failure of the $\bar{\partial}$-Poincaré lemma.

Theorem 0.6.17: Dolbeault's Theorem

For a complex manifold M, there is a natural isomorphism

$$H^{p,q}_{\bar{\partial}}(M) \cong H^q(M, \Omega^p_M)$$

where Ω^p_M is the sheaf of holomorphic p-forms.

***Proof*:**
See [14]

Hermitian and Kähler Metrics

Definition 0.6.18: Hermitian Metric

A *Hermitian metric* on a complex manifold M is a smoothly varying Hermitian inner product h_p on each $T^{1,0}_p M$. Equivalently, it is a Riemannian metric g such that $g(JX, JY) = g(X, Y)$ for all tangent vectors X, Y.
In local coordinates, $h = \sum_{j,k} h_{j\bar{k}}\, dz_j \otimes d\bar{z}_k$ with $(h_{j\bar{k}})$ a positive definite Hermitian matrix.

Definition 0.6.19: Fundamental Form

The *fundamental form* (or *associated* $(1,1)$*-form*) of a Hermitian metric h is

$$\omega := -\operatorname{Im}(h) = \frac{i}{2} \sum_{j,k} h_{j\bar{k}}\, dz_j \wedge d\bar{z}_k$$

This is a real $(1,1)$-form satisfying $\omega(X, JX) > 0$ for nonzero X.

Definition 0.6.20: Kähler Manifold

A Hermitian metric h is *Kähler* if its fundamental form ω is closed: $d\omega = 0$. A *Kähler manifold* is a complex manifold equipped with a Kähler metric. The form ω is then called the *Kähler form*.

Proposition 0.6.21: Equivalent Characterizations of Kähler

The following are equivalent for a Hermitian manifold (M, h):

1. $d\omega = 0$ (the Kähler condition).
2. The Levi-Civita connection ∇ of $g = \mathrm{Re}(h)$ satisfies $\nabla J = 0$.
3. Locally, $\omega = i\partial\bar{\partial}\varphi$ for some real function φ (a *Kähler potential*).
4. The holonomy of g is contained in $\mathrm{U}(n) \subset \mathrm{SO}(2n)$.

***Proof*:**
See [14].

Example 0.6.22: Kähler Manifolds

1. $\mathbb{C}^n$ with the standard metric $\omega = \frac{i}{2}\sum_j dz_j \wedge d\bar{z}_j$.
2. $\mathbb{CP}^n$ with the *Fubini–Study metric*:

$$\omega_{\mathrm{FS}} = \frac{i}{2\pi}\partial\bar{\partial}\log\left(|z_0|^2 + \cdots + |z_n|^2\right)$$

3. Any smooth projective variety over $\mathbb{C}$ (inherits Kähler structure from $\mathbb{CP}^n$).
4. Complex submanifolds of Kähler manifolds are Kähler.
5. Complex tori $\mathbb{C}^n/\Lambda$ with the flat metric.

Example 0.6.23: Non-Kähler Manifolds

1. The *Hopf surface* $(\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z\rangle \cong S^1 \times S^3$ is a compact complex surface with $b_2 = 0$, hence not Kähler (Kähler manifolds have $b_2 \geq 1$ by the existence of $[\omega]$).
2. Most compact complex manifolds are not Kähler.

Hodge Theory

Hodge theory provides a bridge between the topology, complex structure, and Riemannian geometry of a Kähler manifold.

Definition 0.6.24: Hodge Star and Laplacian

On a Riemannian manifold (M, g) of dimension n, the *Hodge star* $* : \Omega^k(M) \to \Omega^{n-k}(M)$ is defined by

$$\alpha \wedge *\beta = g(\alpha, \beta)\,\mathrm{vol}_g$$

The *Hodge Laplacian* is $\Delta := dd^* + d^*d$ where $d^* := (-1)^{n(k+1)+1} * d*$ is the formal adjoint of d.
A form is *harmonic* if $\Delta\omega = 0$.

Theorem 0.6.25: Hodge Theorem (Riemannian)

On a compact oriented Riemannian manifold M:

1. Every de Rham cohomology class contains a unique harmonic representative.
2. $H^k_{\mathrm{dR}}(M) \cong \mathcal{H}^k(M) := \{\omega \in \Omega^k(M) : \Delta\omega = 0\}$.

***Proof*:**
See [14]

On a Kähler manifold, there are additional Laplacians:

Definition 0.6.26: Dolbeault Laplacians

On a Hermitian manifold, define

$$\Delta_\partial := \partial\partial^* + \partial^*\partial, \qquad \Delta_{\bar\partial} := \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$$

Theorem 0.6.27: Kähler Identities

On a Kähler manifold:

$$\Delta = 2\Delta_\partial = 2\Delta_{\bar\partial}$$

In particular, a form is harmonic if and only if it is ∂-harmonic if and only if it is $\bar\partial$-harmonic.

***Proof*:**
See [14]

Theorem 0.6.28: Hodge Decomposition

On a compact Kähler manifold M:

1. There is a decomposition
$$H^k(M, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(M)$$
where $H^{p,q}(M) \cong H^{p,q}_{\bar{\partial}}(M) \cong H^q(M, \Omega^p_M)$.
2. Complex conjugation gives $\overline{H^{p,q}(M)} = H^{q,p}(M)$.
3. The decomposition is independent of the choice of Kähler metric.

***Proof*:**
See [14]

Definition 0.6.29: Hodge Numbers

The *Hodge numbers* of a compact Kähler manifold are
$$h^{p,q}(M) := \dim_{\mathbb{C}} H^{p,q}(M)$$
These satisfy:

1. $h^{p,q} = h^{q,p}$ (complex conjugation).
2. $h^{p,q} = h^{n-p,n-q}$ (Serre duality).
3. $b_k = \sum_{p+q=k} h^{p,q}$ (Betti numbers).

The Hodge numbers are often displayed in the *Hodge diamond*.

Example 0.6.30: Hodge Diamond of $\mathbb{CP}^n$
For $\mathbb{CP}^n$, the only nonzero Hodge numbers are $h^{p,p} = 1$ for $0 \leq p \leq n$:
$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 1 & & 0 \\ & \cdots & & \cdots & \end{array}$$
This reflects $H^{2k}(\mathbb{CP}^n, \mathbb{C}) = \mathbb{C}$ and $H^{2k+1}(\mathbb{CP}^n, \mathbb{C}) = 0$.

Example 0.6.31: Hodge Diamond of a Curve

For a smooth projective curve C of genus g:

$$\begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}$$

Here $h^{1,0} = h^{0,1} = g = \dim H^0(C, \Omega^1_C) = \dim H^1(C, \mathcal{O}_C)$.

Line Bundles and Divisors

Definition 0.6.32: Divisor (Complex Manifold)

A *divisor* on a complex manifold M is a formal $\mathbb{Z}$-linear combination $D = \sum_i n_i V_i$ where each V_i is an irreducible analytic hypersurface (codimension 1) and $n_i \in \mathbb{Z}$. The group of divisors is denoted $\mathrm{Div}(M)$.

Definition 0.6.33: Line Bundle Associated to a Divisor

To a divisor D, one associates a holomorphic line bundle $\mathcal{O}(D)$ (or $\mathcal{L}(D)$) whose sections over U are meromorphic functions f with $(f) + D|_U \geq 0$, i.e., f may have poles along negative components of D and must vanish along positive components.

Proposition 0.6.34: Divisors and Line Bundles

On a compact complex manifold M:

1. Every holomorphic line bundle L is isomorphic to $\mathcal{O}(D)$ for some divisor D.
2. $\mathcal{O}(D_1) \cong \mathcal{O}(D_2)$ if and only if $D_1 - D_2 = (f)$ for some meromorphic function f (linear equivalence).
3. The *Picard group* $\mathrm{Pic}(M) = \{\text{line bundles}\}/\cong$ satisfies $\mathrm{Pic}(M) \cong \mathrm{Div}(M)/\sim$.

Definition 0.6.35: First Chern Class

The *first Chern class* of a line bundle L is a cohomology class $c_1(L) \in H^2(M, \mathbb{Z})$ (or $H^{1,1}(M) \cap H^2(M, \mathbb{Z})$ for Kähler manifolds). It can be computed as:

$$c_1(L) = \frac{i}{2\pi}[F_\nabla]$$

where F_∇ is the curvature of any connection on L.

Proposition 0.6.36: Chern Class Properties

1. $c_1(L_1 \otimes L_2) = c_1(L_1) + c_1(L_2)$.
2. $c_1(L^*) = -c_1(L)$.
3. $c_1(\mathcal{O}(D)) = [D]$ (the cohomology class of D).
4. On $\mathbb{CP}^n$: $c_1(\mathcal{O}(1)) = [\omega_{\mathrm{FS}}]$, the class of a hyperplane.

0.6.1 Summary: Complex Geometry and Algebraic Geometry

This table has more than what was covered for the purpose of reference:

Concept	Complex Analytic	Algebraic
Space	Complex manifold	Smooth variety / scheme
Functions	Holomorphic	Regular (polynomial)
Topology	Classical (Euclidean)	Zariski
Local model	$\mathbb{C}^n$	Spec (A)
Differential forms	(p, q)-forms, $\bar{\partial}$	Kähler differentials $\Omega^p_{X/k}$
Cohomology	Dolbeault $H^{p,q}_{\bar{\partial}}$	Sheaf cohomology $H^q(X, \Omega^p)$
Vector bundles	Holomorphic bundles	Locally free sheaves
Divisors	Analytic hypersurfaces	Weil/Cartier divisors
Line bundles	Pic(M)	Pic(X)
Metrics	Hermitian, Kähler	(No direct analogue)
Key theorem	Hodge decomposition	Serre duality, GAGA

0.6.37 Foreshadowing GAGA For projective varieties over $\mathbb{C}$, one can freely pass between algebraic and analytic viewpoints via GAGA:

- **Algebraic methods**: Work over any field, use scheme theory, étale cohomology.
- **Analytic methods**: Use Hodge theory, L^2-methods, Kähler geometry, transcendental techniques.

Many results (e.g., Hodge conjecture, Torelli theorems) involve both perspectives essentially.

Part I

Scheme Theory

I believe there will be more than just basic alg geo. This part will go over the core theory, Hartshorne, Vakil, Eisenbud, and so forth. Other parts will surely be specializations.

(I want to mention somewhere that for all the concepts we shall generalize from geometry, the integral shall be interesting as it will be recovers using cohomology by using the pushforward and trace maps).

1

Sheaves

Roughly speaking, Sheaf Theory is the theory of augmenting a topological space[1] that adds information to each open set (adds “local” information) which can be “glued” to get information in larger open sets, and can be “restricted” to get information on smaller sets. The prototypical example of a sheaf is the set of smooth functions on a manifold. Smooth functions on a manifold can be restricted to open subsets of a manifold which are homeomorphic to $U \subseteq \mathbb{R}^n$ and if a smooth function on each part of the manifold is defined in a compatible way (i.e. they agree on intersections), they can be glued back together to form a function on the entire manifold. As an open covering of M can always be chosen so that each U is simply connected connected, the non-trivial information is in the ability or choice of gluing of functions.

A key motivator behind sheaves is that they can hold the “geometric” information of a space. The reader may recall that a vector-space and subspaces can be characterized via their dual spaces, that is all functions from the vector space to $\mathbb{F}$. A similar result holds for Varieties with the coordinate ring uniquely characterizing a Variety. This duality was found to exist everywhere in mathematics, to name three more examples:

- A classical example that started the duality between algebra and geometry is that of $C(X)$ and X where X is a compact space and $C(X)$ all continuous function on X. By taking the stone-Cech compactification of a space X, the space can be achieved for any space (see [37, p.144] for details)
- Similarly, a radon measure's can be characterized using the duals of $C_c(X)$ where X is a locally compact space (classical result can be found in EYTNKA at [56])
- Spin manifolds can be characterized by spectral triples (see [21])

[1]more generally, a topoi, which concretely translates in our case to a category whose objects are open sets and morphisms are inclusions

- Sets with set-function can be thought of as the most "abstract" geometric object (they only have information about containment). There is an appropriate dual called the *complete atomic Boolean algebras*. The reader may be interested in knowing that Boolean algebra's characterize logical operations, which gives an interesting geometric interpretation to logic[2].

There is a long list of such examples (the reader may be curious to checkout this link). This gave rise to the idea that a geometric space can be understood by studying some subset of function on this space with certain. In algebraic cases, there are relatively few functions that are defined globally on a space, but many that are defined on local components (the prototypical example of this would be $\mathbb{C}$[3]).[4]

Why do functions capture this geometric information? One can think of functions as tools for representing relations. If our function is of the form $f : X \to \mathbb{R}$, then this is relating X to numbers, i.e. quantification, usually X being a topological space and f being continuous for some notion of closeness. If we have a continuous function of the form $f : X \to \mathbb{R}^n$ (or $\mathbb{C}^n$), with X a topological space, then we may pullback a lot of geometric results from $\mathbb{R}^n$ (or $\mathbb{C}^n$) onto X, depending on how f is defined (ex. it can be shown that f is a diffeomorphism if and only if it pulls back the differential structure of $\mathbb{R}^n$). Very often, it is not possible to define a diffeomorphism $f : X \to \mathbb{R}^n$, however we may define it on components. Functions have the property that we may restrict them or construct them by defining what happening at each point (or in the continuous case, on each open set where they agree on intersections[5]). But you may say that morphisms in a category already capture this intuition. This is indeed the case, what makes functions special is that you can construct larger morphisms from smaller morphisms with an appropriate notion of *gluing* and *restricting*. In particular, the closeness given by the notion of topological spaces and the locality properties inspired by functions shall be the two main properties of sheaves. As is the course of mathematics, it may be asked if these properties of functions can be generalized, and we may ask if we can take special categories which have these features; this is called a *site*. For an introduction to algebraic geometry, limiting to the sites defined on topological spaces is plenty sufficient and shall be the focus of this book.

A particularly important realization will be that commutative rings will in many ways behave like functions on a space, that is if R is a ring, there is a sense in which the elements of R will be the "functions" on a special space (namely, the space $\operatorname{Spec}(R)$). This shall be elaborated on in chapter 2 and was commented on in the front-matter. At this stage it is motivating to know that there is a reason why we may care about sheaves other than the sheaf of functions, namely the interpolation between commutative rings and functions.

Motivating Example from Vakil

Before exploring sheaves in generality, it is useful to have the prototypical example of a sheaf in mind in more detail:

[2] This connection is further refined using Topoi

[3] entire holomorphic are necessarily power series, while holomorphic on (not necessarily simply connected) open sets are analytic and may even have non-trivial Laurent expansion around holes

[4] see this link for more details

[5] by the gluing lemma

Example 1.0.1: Sheaf Of Differentiable Functions On Manifold

Let M be a smooth manifold. Then the we shall constuct the sheaf of differentiable function on M. For each open subset $U \subseteq M$, we shall assocaite to it the set of smooth function $C^\infty(U)$. Denote this collection by $\mathcal{O}(U)$ (this will be common notation for sheaves). Then if we have an open subset of U, $V \subseteq U$, we may restrict the function from U to function on V, namely we have a map $\text{res}_{U,V} : \mathcal{O}(U) \to \mathcal{O}(V)$ mapping $f \mapsto f|_V$. Naturally, if we have $W \hookrightarrow V \hookrightarrow U$, then the following commutative diagram.

$$\begin{tikzcd} & V \arrow[dr, "\text{res}_{V,W}"] & \\ U \arrow[ur, "\text{res}_{U,V}"] \arrow[rr, "\text{res}_{U,W}"'] & & W \end{tikzcd}$$

This collection of maps satisfy two important properties. In the following let and $U = \bigcup_i U_i$.

1. *Defining Locally*: Let's say $f_1, f_2 \in \mathcal{O}(U)$, Then if f_1 and f_2 agree on all restriction, that is $\text{res}_{U,U_i} f_1 = \text{res}_{U,U_j} f_2$, then $f_1 = f_2$

2. *Gluing*: If there exists a collection $f_i \in \mathcal{O}(U_i)$ such that f_i and f_j agree on intersections, that is $\text{res}_{U_i, U_i \cap U_j} f_i = \text{res}_{U_j, U_i \cap U_j} f_j$, then there is a $f \in \mathcal{O}(U)$ such that $\text{res}_{U,U_i} f = f_i$ for all i.

Then a sheaf will respect these properties. Note that in fact, we may define a manifold M to be $(M, \mathcal{O}_M)$ where M Is a topological space and $\mathcal{O}_M$ is a sheaf of smooth functions on M. If instead $\mathcal{O}(U)$ was the collection of continuous functions, we would get a topological manifold. Even more generally, we may let the functions be set functions, for set functions work well under restrictions and we may define set functions locally and construct set function via gluing.

Note that Diffeology can be thought of as a generalization of this result or of a special case of sheaf theory, and we shall explore it briefly at the end of this chapter.

Example 1.0.2: Sheaf On Set Functions

Take $\text{Hom}_{\textbf{Set}}(X, Y)$ to be the collection of arbitrary set functions (or equivalently, let X, Y have the discrete topology so that all set functions are continuous). Then given an arbitrary collection of sets $\{U_i\}_{i \in I}$ of X with functions $f_i : U_I \to Y$, if the collection of functions agree on all pair-wise intersections $U_i \cap U_j$, $i, j \in I$, then there exists a unique function f on $\bigcup_{i \in I} U_i$ that restricts to all these functions. Notice the lack of essentially any structure. This shows that the heavy-lifter insuring the gluability of local data is the *associativity* axiom functions must satisfy.

So if even set functions can form a sheaf, then why did I write "nice" function earlier? As it turns out, what we shall do is abstract the notion of function! In particular, we shall look at systems where we may have the behavior of localizing and gluing "functions" on a topological space. Hence, in the following abstraction, we will consider $\mathcal{O}_M$ to be the data that stores a collection of objects that *behave like functions.*

A key property that distinguishes functions from general morphisms is the notion of evaluation (taking $f(x)$ for some element x and function f). Since we are abstracting the notion of function, we shall need a notion of *evaluation.* To motivate this generalization, let's come back to $(M, \mathcal{O}_M)$ and consider some $\mathcal{O}(U) = C^\infty(U)$ for some open $U \subseteq M$ and take $p \in U$. Recall that $\mathcal{O}(U)$ is a ring and that the *germ* at p is collection of equivalence relations:

$$[f] = \{(f, U) \mid p \in U,\ f \in \mathcal{O}(U)\}$$
$$(f, U) \sim (g, V) \ \Leftrightarrow W \subseteq U, W \subseteq V, p \in W,\ f|_W = g|_W$$

that is, $[f] = [g]$ if and only if there is an open neighborhood $W \subseteq U, V$ containing p ($p \in W$) such that $f|_W = g|_W$ (in the language of sheaf theory, $\mathrm{res}_{U,W} f = \mathrm{res}_{V,W} g$). Denote this collection $\mathcal{O}(U)_p$. The key is that the germ of $\mathcal{O}(U)$ at p is a *local ring* with maximal ideal $\mathfrak{m}_p$ of germs such that $[f](p) = 0$ (or simply $f(p) = 0$ if unambiguous). Then:

$$0 \to \mathfrak{m}_p \to \mathcal{O}(U)_p \xrightarrow{f \mapsto f(p)} \mathbb{R} \to 0$$

and all elements $\mathcal{O}(U) \setminus \mathfrak{m}_p$ are invertible. Hence, we shall see that we may thing of "evalutation" of f at some point $p \in U$ as taking first the germ $\mathcal{O}(U)_p$ and quotient it by the maximal ideal $\mathfrak{m}_p$, $\mathcal{O}(U)_p/\mathfrak{m}_p$, and then taking $\overline{f} \in \mathcal{O}(U)_p/\mathfrak{m}_p$ to be the value of f. Note that the above construction required that $\mathcal{O}(U)$ is a *local ring*. This is evidently the case when $\mathcal{O}(U) = C^\infty(U)$, however in general sheaf theory this ring will not in general be local. In fact, $\mathcal{O}(U)$ will not even need to be a ring! When each $\mathcal{O}(U)$ is a ring, then we shall say that X is a *ringed space*, and when all the germs are local rings, then we shall say that X is a *locally ringed space*.

A final word on generalization: we have been working over a topological space, however we may abstract further and work over a general category; in this way, we would define the notion of a "topology" on a category, known as a *Grothendeick topology* (a category with this structure is called a *site*), and develop sheaf theory accordingly. Though this is fruitful, this generlization shall be ommited for now to concentrate on the geoemtric properties at hand when using a topological space.

1.1 Definitions and Examples

We start by formalizing the above intuitions.

Definition 1.1.1: Presheaf Axiomatic Definition

Let X be a topological space. Then a *presheaf* over a category $\mathbf{C}$ on X, denoted $\mathscr{F}_X$ contains the following data:

1. for each open set $U \subseteq X$, there is an object c of $\mathbf{C}$ and denote it by $\mathscr{F}(U)$. If C is a category whose objects are sets with additioanl structure, then the elements of $\mathscr{F}(U)$ are called the *sections* of $\mathscr{F}$ over U. If we say that s is a section of $\mathscr{F}$, then we mean that s is a section of $\mathscr{F}$ over X.

2. For each inclusion map $U \hookrightarrow V$, there is a morphism $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ called the *restriction* morphism (or *restriction map*) such that

   (a) $\mathrm{res}_{U,U} = \mathrm{id}_U$

   (b) If $W \hookrightarrow V \hookrightarrow U$, then the following diagram commutes

$$\begin{tikzcd} & \mathscr{F}(V) \arrow[dr, "\mathrm{res}_{V,W}"] & \\ \mathscr{F}(U) \arrow[ur, "\mathrm{res}_{U,V}"] \arrow[rr, "\mathrm{res}_{U,W}"'] & & \mathscr{F}(W) \end{tikzcd}$$

The collection of all presheaves on a topological space X is denoted $\mathbf{PSh}_{\mathbf{C}}(X)$ (or $\mathbf{PSh}(X)$ if unambiguous). If U is clear from context, we shall often write $\mathrm{res}_{U,V} f$ as $f|_V$.

The nomenclature "section" will be made clear in proposition 1.1.13. When it is no ambiguous the image of the restriction morphism for $s \in \mathscr{F}(U)$ shall be denoted:

$$\mathrm{res}_{U,V}(s) = s|_V$$

This notation can be ambiguous as it is not explicit which $U \subseteq X$ is taken from which $s \in \mathscr{F}(U)$, however it will be unambiguous in-context.

When constructing sections of non-affine scheme's in chapter 2, it shall be important to have internalized the following consequence of the restriction maps. Let $U = U_1 \cup U_2$ and $W = U_1 \cap U_2$. Then $W \subseteq U_1 \subseteq U$ and $W \subseteq U_2 \subseteq U$. Then by definition of the restriction map, the following diagram commutes:

$$\begin{tikzcd} & \mathscr{F}(U_1) \arrow[dr, "\mathrm{res}_{U_1,W}"] & \\ F(U) \arrow[ur, "\mathrm{res}_{U,U_1}"] \arrow[rr, "\mathrm{res}_{U,W}"'] \arrow[dr, "\mathrm{res}_{U,U_2}"'] & & F(W) \\ & \mathscr{F}(U_2) \arrow[ur, "\mathrm{res}_{U_2,W}"'] & \end{tikzcd}$$

Importantly:

$$f|_W = (f|_{U_1})|_W = (f|_{U_2})|_W$$

The key take-away is that a larger section on U can be restricted to different sections on an open cover of U, but these sections map to the same restriction on intersections. In this way, presheaves preserve elements "going down". We shall see example of how "going up" fails after introducing some more basic terminology.

For practical computational purposes, it shall be important to re-interpret a presheaf in more categorical language, Notice that it is equivalent to define a presheaf as a *contravariant functor*. Namely

Definition 1.1.2: Presheaf Functorial Definition

Let X be a topological space and let $O(X)$ be the category whose objects are the open sets of X and whose morphism are $\subseteq$ (i.e. the category induced by the partial order given by $\subseteq$). Then a *presheaf* over $\mathbf{C}$ is a contravariant functor from $O(X)$ to $\mathbf{C}$, that is a functor $\mathscr{F} : O(X)^{op} \to \mathbf{C}$.

Usually, our presheaves shall be over **Set**, **Ab**, Mod_k or **CRing**[6].

An important notion for geometric objects is their behavior locally. The extreme (or limit) of the notation of locality is the behavior around a point p. This idea originated when studying germs of differential functions in Differential Geometry. These are well-defined on presheaves;

Definition 1.1.3: Stalk as Equivalence Relation

let $(X, \mathcal{O}_X)$ be a presheaf. Then the *stalks* at $p \in X$ is the collection of equivalence class:

$$[f] = \{(f, U) \mid p \in U,\ f \in \mathcal{O}(U)\}$$
$$(f, U) \sim (g, V) \qquad \Leftrightarrow \qquad W \subseteq U, W \subseteq V, p \in W,\ \text{res}_{U,W} f = \text{res}_{V,W} g$$

The stalks at p is usually denoted $\mathcal{O}_{X,p}$ or $\mathcal{O}_p$ if unambiguous in context.

If $\mathbf{C}$ is a category whose objects are augmented sets[a], then if $p \in U$ and $f \in \mathscr{F}(U)$, $[f] \in \mathscr{F}_p$ is called the *germ* of f at p.

[a] ex. **Set**, **Ring**, **Grp**, etc.

Note that in general, the notion of "evaluation" of a stalk/germ as it is usually defined on differential geometry won't make sense; though the symbol f is written the open sets need not contain functions or objects that act like function. We shall later define nice presheaves in which we may recover the notion of evaluation and treat the elements like functions. There is again a categorical way of interpreting stalks that is useful to us:

Definition 1.1.4: Stalks As Colimits

Let $\mathscr{F} \in \mathbf{PSh}_{\mathbf{C}}(X)$ be a presheaf where $\mathbf{C}$ is a category with small filtered colimits. Then a *stalk* at $p \in X$ is:

$$\mathscr{F}_p := \varinjlim \mathscr{F}(U)$$

that is $\mathscr{F}_p$ is the colimit over all $\mathscr{F}(U)$ where $p \in U$.

The adjectives "small filtered colimit" are only there to guarantee that the colimit exists. Most base category that will be worked over shall have the stronger property of bieng *cocompelte* (having all colimits) and often also being *complete* (having all limits). Categories that are complete and cocomplete include:

[6] Technically, **Ab** is $\mathbf{Mod}_{\mathbb{Z}}$, however distinguishing these two is useful as we usually mean k to be a field

1. **Set** (Category of Sets)
2. **Grp** (Category of Groups)
3. **Ab** (Category of Abelian Groups)
4. **Ring** (Category of Rings)
5. R-**Mod** (Category of modules over a ring R)
6. **Top** (Category of Topological Spaces)

Hence, this shows us that as long as the colimit exist in our category, we may define the notion of stalks for presheaves over that category without needing to reference the notion of elements. Note that it is the colimit and not the limit since we need $\mathscr{F}_p$ to "represent" the behaviour around p rather than "embed" germs.

One of the most important properties of stalks is their ability to be defined locally. First, if $U \subseteq X$ is an open set, we can define the sub-presheaf on U by letting $\mathcal{F}|_U(V) = \mathcal{F}(V \cap U)$. The following proposition will be really useful later when we need to work with stalks between larger and smaller (pre)sheaves, and mirrors the property in differential geometry (recall [53, chapter 2.1])

Proposition 1.1.5: Stalk Isomorphism

Let X be a presheaf of sets and $U \subseteq X$ an open set. Then for any $x \in U$:

$$\mathscr{F}_{X,x} \cong_{\mathbf{C}} \mathscr{F}_{U,x}$$

***Proof*:**
We shall prove this in the case where $\mathbf{C} = \mathbf{Set}$. The general case is left as an exercise.

Let us start by defining the map $\varphi : \mathscr{F}_{X,x} \to \mathscr{F}_{U,x}$. Take $[(f, V)] \in \mathcal{F}_{X,x}$ such that $x \in V$. As $x \in U$, $x \in U \cap V$. Hence, notice that:

$$(f, V) \sim (f|_{U\cap V}, U \cap V) \qquad \text{as} \qquad \text{res}_{U,U\cap V} f = \text{res}_{U\cap V,U\cap V} f|_{U\cap V} = f|_{U\cap V}$$

Then define the map between stalk via:

$$\varphi : (f, V) \sim (f|_{U\cap V}, U \cap V) \mapsto (f|_{U\cap V}, U \cap V)$$

This map is well-defined. Take $(f, V_1) \sim (g, V_2)$ where $V_1, V_2 \subseteq X$ so that we have to show that $(f|_{U\cap V_1}, U \cap V_1) \sim (g|_{U\cap V_2}, U \cap V_2)$. As $(f, V_1) \sim (g, V_2)$ there exists $W \subseteq V_1, V_2$ such that

$$f|_W = g|_W$$

Take $U \cap W \subseteq W \subseteq V_1, V_2$. Then by the restriction axioms we have that:

$$f|_{U\cap W} = g|_{U\cap W}$$

So that $(f|_{U\cap W}, U \cap W) \sim (g|_{U\cap W}, U \cap W)$. Then as:

$$\text{res}_{U\cap V_1,U\cap W}(f|_{U\cap V_1}) = \text{res}_{U\cap W,U\cap W}(f|_{U\cap W})$$

(and similarly for V_2), we get:

$$(f|_{U\cap V_1}, U\cap V_1) \sim (f|_{U\cap W}, U\cap W) \sim (g|_{U\cap W}, U\cap W) \sim (g|_{U\cap V_2}, U\cap V_2)$$

and hence the map is well-defined. This map has a natural inverse, namely

$$\varphi^{-1} : \mathscr{F}_{U,x} \to \mathscr{F}_{X,x} \qquad (f,V) \mapsto (f,V)$$

which can be shown to be well-defined and $\varphi \circ \varphi^{-1} = \varphi^{-1} \circ \varphi = \mathrm{id}$, hence these sets are bijective, which are the isomorphisms in set. To generalize to categories that are augmented sets, take advantage of the restriction maps being morphisms.

Now, presheaves allow for the notion of restricting and localizing, that is there is structure preserved "going down". But there are no condition on any structure being preserved "going up". In particular, not all presheaves are amenable to gluing local information to create more global information (think of bounded holomorphic functions on open subsets of $\mathbb{C}$). We must add this condition axiomatically:

Definition 1.1.6: Sheaf using Axioms

Let $\mathscr{F} \in \mathbf{PSh}_C(X)$ be a presheaf over a category $\mathbf{C}$ whose objects are sets possibly with additional structure. Then $\mathscr{F}$ is also a sheaf if it satisfies the following two axioms;

1. *Gluing Axiom (Existence)*: Let $U = \bigcup_i U_i$. If there exists a collection $f_i \in \mathscr{F}(U_i)$ such that $\mathrm{res}_{U_i, U_i\cap U_j} f_i = \mathrm{res}_{U_j, U_i\cap U_j} f_j$, then there is a $f \in \mathscr{F}(U)$ such that $\mathrm{res}_{U,U_i} f = f_i$ for all i.

2. *Locality Axiom (Uniqueness)*. Let $U = \bigcup_i U_i$. If $f_1, f_2 \in \mathscr{F}(U)$, then if $\mathrm{res}_{U,U_i} f_1 = \mathrm{res}_{U,U_i} f_2$ for all i, then $f_1 = f_2$

3. $\mathscr{F}(\emptyset)$ is associated with the terminal object (if we're working over set, $\mathscr{F}(\emptyset)$ is a singleton).

If only the locality axiom is satisfied, then our presheaf is called a *separated presheaf.* The collection of all sheaves on a topological space X is denoted $\mathbf{Sh}_C(X)$ (or $\mathbf{Sh}(X)$ if unambiguous)

We may think of the locality axiom telling us there is *at most* one way to glue, while the gluing axiom is telling us there is *at least* one way to glue. A separated presheaf is so named as it allows sections to "separate" points (there a cannot be two sections identical on restrictions of an open covering).

1.1.7 Separation and Evaluation Later, the notion of "evaluation" shall be defined for sections where the objects of $\mathbf{C}$ behave like sets of functions. Being separated *does not* imply a function is defined by the evaluation at every point as the intuition of separated algebra for the Stone-Weierstrass Theorem would imply. This is because evaluation on locally ringed spaces will carry more data that needs to be taken into account (ex. see lemma 2.2.6).

The following is sometimes (redundantly) included in the definition of a presheaf:

Lemma 1.1.8: Sheaf and Zero Section

Let **C** be a category with the zero object 0 and $\mathscr{F}$ a sheaf (or particularly, a separated presheaf) on X. Let $\bigcup_{i\in I} U_i = X$ be an open cover and $s \in \mathscr{F}(X)$. Then if $s|_{U_i} = 0$ for all i, then $s = 0$.

***Proof*:**
exercise.

Notice that we are taking elements of each $\mathscr{F}(U)$ because $\mathscr{F}(U)$ is an object in a category whose objects are sets with possibly additional structure. This is a deviation from the generalization presented in the definition of presheaves as **C** may be more general. There is in fact a way to define a sheaf for general categories:

Proposition 1.1.9: Sheaf Categorical Interpretation

Let $\mathscr{F} \in \mathbf{PSh}(X)$ be a presheaf, and let $\cdot$ be a terminal object (in **Set**, an arbitrary 1-point set). Let $U = \bigcup_i U_i$, and consider:

$$\cdot \to \mathscr{F}(U) \to \prod_{i\in I} \mathscr{F}(U_i) \rightrightarrows \prod_{i,j\in I} \mathscr{F}(U_i \cap U_j)$$

for any indexing set I. Then if the sequence is exact at $\mathscr{F}(U)$, the locality axiom holds, and if it is exact at $\prod \mathscr{F}(U_i)$, the gluing axiom holds.

***Proof*:**
This shall be proven when $\mathbf{C} = \mathbf{Ab}$, the general result can be proven by the reader interested in practicing their category theory.

Assume $\mathscr{F}(U) \xrightarrow{\alpha} \prod \mathscr{F}(U_i)$ is injective so that

$$s \hookrightarrow (s|_{U\cap U_i})_i$$

Then by definition of injectivity any s_a, s_b mapping to this collection of restricted sections is equal, $s_a = s_b$, giving the locality.

Now, let $s_i = \mathscr{F}(U_i)$. Take the equalizer:

$$\prod_{i\in I} \mathscr{F}(U_i) \xrightarrow{\beta,\gamma} \prod_{j\in J} \mathscr{F}(U_i \cap U_j)$$

where:

$$\beta((s_i)_i) = \big(s_i|_{U_i\cap U_j}\big)_{i,j\in I} \qquad \text{and} \qquad \gamma((s_i)_i) = \big(s_j|_{U_i\cap U_j}\big)_{i,j\in I}$$

For example Let $I = \{1, 2\}$ and $U_1 \cup U_2 = U$. Let us not put in the "redundant" elements for the pairs $(1, 1), (2, 2)$, and $(2, 1)$; the last pair is not automatically excluded in the definition to avoid putting an unnecessary ordering on I, and the former two are not excluded for the simplicity of notation. As we are working with abelian groups, we can turn the equalizer into:

$$\cdot \to \mathscr{F}(U) \to \prod_{i\in I} \mathscr{F}(U_i) \xrightarrow{s_i|_{U_i\cap U_j} - s_j|_{U_i\cap U_j}} \prod_{i,j\in I} \mathscr{F}(U_i \cap U_j)$$

Thus the kernel of the equalize is when these two are equal, namely:

$$\beta(s_1, s_2) = (s_1|_{U_1 \cap U_2}) \qquad \text{and} \qquad \gamma(s_1, s_2) = (s_2|_{U_1 \cap U_2})$$

are equal:

$$(s_1|_{U_1 \cap U_2}) = \beta(s_1, s_2) = \gamma(s_1, s_2) = (s_2|_{U_1 \cap U_2})$$

Giving the gluing condition. In general, let us denote the kernel of the equalizer $\mathrm{Eq}(\beta, \gamma)$. By exactness we get that:

$$\mathrm{im}(\alpha) = \mathrm{Eq}(\beta, \gamma)$$

Thus, every element in the equalizer comes from a section in U. If α is also a monomorphism, it is unique, as we sought to show.

1.1.10 Foreshadowing Čech Cohomology

What if we extend this to have further equalizers? Sheaves that satisfy higher-order intersection-exactness conditions are a proper subset of sheaves (over non-trivial categories and topological spaces). Such sheaves can be thought of as never having any gluing obstruction, and hence have trivial cohomology. We shall return to these ideas when exploring Čech cohomology in section 6.1.1.

Example 1.1.11: Presheaves failing exactness (i.e. gluability and locality)

Let X be any topological space and define:

$$\mathcal{F}(X) = \mathbb{Z} \qquad \mathcal{F}(U) = 0 \qquad (\text{for } U \subsetneq X)$$

Then this sheaf is gluable, but not separated. If $U \subsetneq X$ is covered, then it is clear that gluing must hold. If $U = X = \cup_i U_i$, then that restrictions holds as any $z \in \mathbb{Z}$ has $z|_{U_i} = 0$ and so they certainly agree on all intersections. Hence, we see that each $z \in \mathbb{Z}$ is glued by a bunch of 0-elements!

Example 1.1.12: Presheaves And Sheaves

Most of these examples will be on sheaves of sets.

1. Verify that the sheaf of differentiable functions over M, namely $\mathcal{O}_M$, is indeed a sheaf. More generally, if X is any topological space, then the data defined by $\mathscr{F}(U) = \mathrm{Hom}_{\mathbf{C}}(U, \mathbb{R})$ where $\mathbf{C}$ is either **Top** or **Set** is a sheaf.

   Note how in a sheaf, we are "forced" to have interesting global behaviour if there are local section on non-intersecting sets. Consider $U_1 = (0, 1)$ and $U_2 = (3, 4)$. Then $U_1 \cap U_2 = 0$, so $\mathscr{F}(U_1 \cap U_2) = \{*\}$ is the terminal object (a singleton). If there is a function $f \in \mathscr{F}(U_1)$ and $g \in \mathscr{F}(U_2)$, then the restriction map

   $$\mathrm{res}_{U_1,\emptyset} f = * = \mathrm{res}_{U_2,\emptyset}$$

   Meaning they *have* to agree on intersection. Thus, by the gluability axiom, there must exist a function F on $U_1 \cup U_2$ that restricts to f, g. For functions, this is obvious as it is the function defined by:

$$F(x) = \begin{cases} f(x) & x \in (0,1) \\ g(x) & x \in (3,4) \end{cases}$$

This becomes more interesting when the category in question is not obviously interpret able as functions on a space (ex. some sheaves represent logical proposition, see ref:HERE).

2. (*Restriction of Sheaves*) Let $\mathscr{F} \in \mathbf{Sh}(X)$ and $U \subseteq X$ an open subset of X. Then the *restriction* of $\mathscr{F}$ to U, denoted $\mathscr{F}|_U$ has as data for each open $V \subseteq U$ $\mathscr{F}|_U(V) = \mathscr{F}(V)$ (note hos it is important that U is open for well-definedness). We shall later see how we may define restrictions of sheaves to arbitrary sets[a]

3. (*Skyscraper Sheaf*) Let X be a topological space, $p \in X$, and S be any set. Let $\iota_p : \{p\} \to X$ be the inclusion map. Then define the sheaf of sets $(\iota_p)_* S$ on X (not on S!) to be:

$$(\iota_p)_* S(U) = \begin{cases} S & p \in U \\ \{e\} & p \notin U \end{cases}$$

This sheaf is called the *skyscraper sheaf.* The name comes from the fact that the informal picture for this sheaf looks like a skyscraper at p. For sheaves of Abelian groups, $\{e\}$ is the trivial group. More generally, $(\iota_p)_*(U)$ would be the final object if $p \notin U$.

4. (*Constant Presheaf*) Let X be a topological space and S be any set. Define $\underline{S}_{\text{pre}}(U) = S$ for all open $U \subseteq X$ Then it should be verified that $\underline{S}_{\text{pre}}$ forms a presheaf on X, called the *constant presheaf associated to* S. This set is not in general a sheaf if $|S| > 1$: take $X = U_1 \coprod U_2$ to be the disjoint union of two (open) sets, take the constant presheaf so that all restriction maps are constant maps. Chose $s_1 \in S$ and $s_2 \in S$ which are distinct: $s_1 \neq s_2$. Then by the gluing axiom, there would have to exist a $s \in S$ such that $s|_{U_1} = s_1$ and $s|_{U_2} = s_2$ and their intersection is empty and hence tautologically follow the gluing axiom, however as we have the identity map we have $s_1 = s = s_2$ however $s_1 \neq s_2$.

   This is fixed by a process called *sheafification* which will add the appropriate section equivalent to (s_1, s_2) that would restrict to both of these elements.

   What makes the sky-scrapper sheaf special is the "distinguishing" of a point making it act like a generalization of an indicator function. In the above example, one of the two sets will not have the point p.

5. (*Constant Sheaf*) Again let X be a topological space and S be any set. For every open $U \subseteq X$, let $\mathscr{F}(U)$ be the collection of all *locally constant* continuous functions from U to S, that is if $f \in \mathscr{F}(U)$, for each $p \in U$ there is an open neighborhood $V \subseteq U$ containing p, $p \in V \subseteq U$, such that $f|_V : V \to S$ is constant (on euclidean space, this is equivalent to having a constant function on each connected component). Then this sheaf is called the *constant sheaf associated to* S, and is denoted $\underline{S}$.

   Another way of describing this sheaf is if we endow S with the discrete topology and let $\mathscr{F}(U)$ be the continuous maps $U \to S$.

6. A great example of a pre-sheaf on $\mathbb{C}$ that is not a sheaf is the sheaf of bounded holomorphic functions on $\mathbb{C}$. Recall that bounded entire holomorphic functions (defined on all of $\mathbb{C}$) must be constant by Louisville's Theorem, however there do exist bounded functions on open subsets of $\mathbb{C}$. For example , on each disk of radius r, $\mathbb{D}_r^2$, the holomorphic function z^2 is

bounded but $\mathbb{C} = \cup_r \mathbb{D}_r^2$ has no bounded holomorphic function for which the restriction to any disk $\mathbb{D}_r^2$ is z^2; hence the gluing axiom fails.

Another example would be the sheaf composed of holomorphic functions admitting holomorphic square roots, as there is no global square-root function on $\mathbb{C}$ due to non-trivial monodromy.

7. (*Sheaf of continuous maps*) Let X and Y be topological spaces. Define $\mathscr{F}(U) = \mathrm{Hom}_{\mathbf{Top}}(U, Y)$. This sheaf is a generalization sheaf of differentiable functions and the constant sheaf, namely

   - The sheaf of differentiable functions has $Y = \mathbb{R}$
   - the constant sheaf has $Y = S$ with the discrete topology.

   If Y is also a topological group, then the continuous maps to Y form a sheaf of groups.

8. (*Sheaf of section of a maps*) This is an important special case of the above sheaf. Let X, Y be topological spaces and let $\mu : Y \to X$ be a continuous map. Then the sections of μ (in the sense of the existence of a map $s : X \to Y$ such that $\mu \circ s = \mathrm{id}$) form a sheaf. In particular, for each open $U \subseteq X$, $\mathscr{F}(U)$ is the set of section maps $s : U \to Y$ (continuous maps $s : U \to Y$ such that $\mu \circ s = \mathrm{id}|_U$). Vector bundles and vector fields are a good example of such a sheaf.

9. The following examples shows how different sheaves can be put on the same object to characterize different type of information. Throughout this example, let $X = \mathbb{A}_k^1$ be the (classical) affine line (see definition 0.1.9). The open subsets in the Zariski topology are all cofinite subsets. If $U \subseteq \mathbb{A}_k^1$ is open let $a_1, ..., a_n$ be the points that are omitted from U.

   (a) (Sheaf of regular function) Then letting $f_i = (x - a_i)$ and $F = \{f_1, ..., f_n\}$, define $\mathscr{F}(U) = k[x]_F$ i.e. $\mathscr{F}(U)$ is the ring generated by $k[x]$ and $1/f_i$ for each i. The reader should verify that this is indeed a pre-sheaf, and even a sheaf.

   (b) (Sheaf of invertible functions) Let X be as above, and let $\mathscr{F}(U) = (k[x]_F)^*$, that is consider only the invertible functions. Then X is again a sheaf.

   (c) (sheaf of roots of unit) Again consider X as above. Then let $\mathscr{F}(U)$ be the subgroup of $k[x]_F^*$ consisting of all roots of unity present in $\mathscr{F}(U)$. This forms a sheaf

   (d) (Sheaf of differential forms) If the reader is familiar with differential geometry, let $k = \mathbb{R}$ and let $\mathscr{F}(U)$ be the collection of all closed forms on U (or more generally, all smooth forms on U). Note that only $\mathscr{F}(\mathbb{A}_{\mathbb{R}}^1)$ are all the closed forms exact.

[a] see ref:HERE

If $\mathscr{F} \in \mathbf{PSh}(U)$ and $f \in \mathscr{F}(U)$ for some open $U \subseteq X$, then f is called a section. The following proposition motivates this idea

Proposition 1.1.13: Space of Sections of (Pre)Sheaf: étale spaces

Let $\mathscr{F}$ be a presheaf of sets. Then there exists a topological space F and a local homeomorphism $\pi : F \to X$ so that $\mathscr{F}$ can be represented as the sheaf of sections from F to X. The space F is called the *étalé space*. In particular, F is the *étalé space* of $\mathscr{F}$.

Furthermore, there is an equivalence between the category $\mathbf{Sh}(X)$ and the category of étalé spaces over X (with bundle morphisms as morphisms).

The idea is that (pre)sheaves correspond to a space F that is it is locally homeomorphic to X. Thus, geometrically sheaves could be thought of as local homeomorphisms, though the total space F may be rather ugly as shall be shown in example 1.1.14. Note that local homeomorphisms need not be surjective; this is a common misconceptions as many examples of surjective local homeomorphisms are covering spaces, but examples such as the sky-scrapper sheaf should show why this need not be the case.

***Proof*:**
Let F be the disjoint union of the stalks on F.

$$F = \coprod_{x \in X} \mathscr{F}_x$$

Then $\pi : F \to X$ is naturally a set map. Define a basis for F as follows: for each $s \in \mathscr{F}(U)$, take the collection

$$\{[s]_x \in \mathscr{F}_x \mid x \in U\}$$

These form a basis for F (note that each stalk has the discrete topology). With the setup, it is now easy to see the equivalence of categories.

The étale space may seem like it ought to be pretty similar to X being locally homeomorphic, however this can be far from the case:

Example 1.1.14: Non-Hausdorff Étale Space

Take $X = \mathbb{R}$ be the real numbers and take $\mathcal{F}$ to be the smooth functions on $\mathbb{R}$. Take the constant germ around 0 as well as the germ which can be represented by the function:

$$g(x) = \begin{cases} x\sin(1/x) & x\sin(1/x) > 0 \\ 0 & x\sin(1/x) \leq 0 \\ 0 & x < 0 \end{cases}$$

In other words, g is zero when x is negative and when the function $x\sin(1/x)$ would be negative; we want to keep when $x\sin(1/x)$ is positive. Then these two are different points in the étale space, but *cannot* be separated by open sets.

Note that if we were working with $X = \mathbb{C}$ and took analytic or holomorphic functions, then this cannot happen as we have a non-isolated 0 without all of g being zero. This turns out to be the main impingement for the étale space being Hausdorff. In general sheaves of analytic functions shall form Hausdorff étale spaces.

1.1.1 Stalks and Sheaves

As mentioned earlier, presheaves induce stalks. It was mentioned that sheaves but *not* presheaves can be characterized on the level of stalks, meaning the stalks of a sheaf shall give us all the information about the morphism a sheaf. This in fact almost characterizes sheaves from presheaves, as it in fact characterizes *separated presheaves* (recall they satisfy the locality but not gluing axiom). The following shows the value of working with separated presheaves:

Lemma 1.1.15: Sections Determined By Germs

Let $\mathscr{F}$ be a sheaf (more specifically, a separated presheaf). Then the map:

$$\mathscr{F}(U) \to \prod_{p \in U} (\mathscr{F}_p)$$

is injective

This map being injective justifies the term "separated", namely the points are separated by the sections.

***Proof*:**
Let f be the map and consider $f(s) = (s_x)_{x \in U}$ Say:

$$(s_x)_{x \in U} = (t_x)_{x \in U} \qquad \text{equiv.} \qquad [(s, V_x)]_{x \in U} = [(t, W_x)]_{x \in U}$$

Then by definition there exists a $C_x \subseteq V_x \cap W_x$ such that

$$s|_{C_x} = t|_{C_x}$$

Now as $x \in C_x$ for each $x \in U$ we have $U = \bigcup_{x \in U} C_x$ and furthermore for for any pair C_x, C_y in the open cover we have:

$$\text{res}_{C_x, C_x \cap C_y} s = \text{res}_{C_y, C_x \cap C_y} t$$

hence, by the gluing axiom $s = t$, as we sought to show.

Note that we can show the locality axiom form this argument:

Corollary 1.1.16: Global Section Over Local Sections

Let $\mathscr{F}$ be a sheaf on X and $U_i \subseteq X$ is an open cover. Then:

$$\mathscr{F}(X) \to \prod_i \mathscr{F}(U_i)$$

is injective

***Proof*:**
Map $s \mapsto \left(s|_{U_i}\right)_i$. Consider

$$\left(s|_{U_i}\right)_i = \left(t|_{U_i}\right)_i$$

then s and t agree on intersections of U_i, hence there exists a unique element in $\mathcal{F}(X)$ for both these elements, namely $s = t$

In example 1.1.11, the sheaf $\mathcal{F}(X) = \mathbb{Z}$ and $\mathcal{F}(U) = 0$ for $U \subsetneq X$ was not a separated presheaf, and certainly $\mathcal{F}(X) \to \prod_{x\in X} \mathcal{F}_x$ is *not* injective, as each $\mathcal{F}_x$ consist of a single equivalence class representing the zero element (and hence $\prod_{x\in X} \mathcal{F}_x \equiv \{[0]\}$, and $\mathcal{F}(X) = \mathbb{Z}$. This example shows that we can somehow construct a nonzero object from a bunch of zero-objects. This can be thought of as being the key-blocker, especially in the case of abelian categories where most of the time the pre-image of 0 is only 0. This motivate the following:

Definition 1.1.17: Support of Sections

Let $\mathscr{F}$ be a sheaf of an abelian group on X, and s a global section of $\mathscr{F}$. Then the *support* of s, denoted $\operatorname{Supp}(s)$, is the points $p \in X$ where s has a nonzero germ:

$$\operatorname{Supp}(s) := \{p \in X \mid s_p \neq 0 \text{ in } \mathscr{F}_p\}$$

This notion differs a bit from the usual notion of support in analysis, in particular notice that it is the germ itself that is nonzero and not that the evaluation of the germ s_p at the point p that is nonzero.

Lemma 1.1.18: Support is Closed

Let $\mathscr{F}$ be a sheaf of abelian groups on X and $s \in \mathscr{F}(X)$. Then $\operatorname{Supp}(s)$ is closed

***Proof*:**
We'll show $\operatorname{Supp}(s)^c$ is open, that is the set of all points $p \in X$ such that $s_p = 0$ is open. If $s_p = 0$ in the stalk, there exists an open subset $W_p \subseteq X$ such that $s|_{W_p} = 0$. Taking the union of arbitrary open sets is open, and hence taking the union of all W_p cover $\operatorname{Supp}(s)^c$ making $\operatorname{Supp}(s)$ closed, completing the proof.

The equivalent in analysis is the definition of the *closed support* of a function. Due to this, the compliment of the support of an element s is the 0-section. The section can also be defined on presheaves, however there are examples of presheaves where there are nonzero sections that "live nowhere" because it is 0 in every stalk[7]

Let us now take a closer look at $\prod_{p\in U}(\mathscr{F}_p)$. This set is much bigger than sections of a sheaf. We thus take the special sub-collection that associates with $\mathscr{F}(U)$ (essentially characterizing the image of $\mathscr{F}(U)$:

[7]Take $\mathscr{F}$ to be the sheaf of holomorphic functions on $\mathbb{C}$, $\mathscr{G}$ the sheaf of bounded holomorphic functions. Then take the sheaf $\mathcal{H} = \mathscr{F}/\mathscr{G}$. Then show that $\mathcal{H}_x = 0$, however $\mathcal{H}(\mathbb{C}) = \{\text{entire functions}\}/\mathbb{C}$

Definition 1.1.19: Compatible Germs

Let $\mathscr{F}$ be a sheaf on X and $U \subseteq X$ an open set. Then we say an elements $(s_p)_{p\in U} \in \prod_{p\in U} \mathscr{F}_p$ consists of *compatible germs* if there exists some representative open sets U_p for s_p:

$$U_p \subseteq U \text{ open}, \ \tilde{s}_p \in \mathscr{F}(U_p)$$

such that the germ of $\tilde{s}_p$ for all $q \in U_p$ is s_q. In other words, if $\{U_i\}$ is an open cover of U where $p \in U_i$ for all i, then s_p is the germ of f_i at p.

Hence, a compatible germ is the collection of germs that are "compatible" when taking subsets, that is *restrictions*. Using the gluing axiom, it is a good exercise to show that any choice of compatible germs for a sheaf $\mathscr{F}$ over U is the image of a section of $\mathscr{F}$ over U, that is we have successfully identified the right hand side of the equation in lemma 1.1.15. Note how important it is that we use sheaves (or at least separated presheaves) for the proof. The proof also motivate the agricultural terminology of "sheaf", for in agriculture a sheaf is a collection of "Stalks" (which is the main stem of a plant).

1.1.20 Remark When working with sheaves of sets or some structures on sets, the elements of the section can literally be thought of as functions using the above definition, namely function of the form $s : U \to \coprod_{p\in U} \mathscr{F}_p$. The generalization presented above allows to generalize the notion of "open set U" so that the topological space X is replaced special category called a *site*, see ref:HERE.

1.1.2 Sheaf from a Topological Base

When defining a differential or topological structure, one of the strategies that is taken is to define pieces of the manifold in a compatible way, or said differently define it on some basis of the manifold. This section endeavours to do this for sheaves.

Let X be a topological space with a sheaf $\mathscr{F}$. Take a basis $\{B_i\}$ for X and take the subset of $\mathscr{F}$ consisting of $\mathscr{F}(B_i)$ along with all the appropriate restriction maps. Then there is enough Stalk information to recover all of $\mathscr{F}$ (show this), and hence we my want to start with such a collection to generate a sheaf. This collection alone will generate a presheaf, we will require the basis to satisfy the locality and gluability axioms:

Definition 1.1.21: Sheaf On A Base

Let X be a topological space and $\mathcal{B}$ be a topological base. Then this collection is caleld a *sheaf on a base* if it satisfies:

1. **Base locality axiom**: If $B = \cup_i B_i$, then if $f, g \in \mathscr{F}(B)$ are such $\text{res}_{B,B_i} f = \text{res}_{B,B_i} g$ for all i, then $f = g$.
2. **Base Gluability**: If $B = \cup_i B_i$, and $f_i \in \mathscr{F}(B_i)$ such that f_i agrees with f_j on any basic open set contained in $B_i \cap B_j$, then there exists $f \in \mathscr{F}(B)$ such that $\text{res}_{B,B_i} f = f_i$ for all i

Theorem 1.1.22: Define Sheaf From Base

Let $\{B_i\}$ be a base for the topological space X and let F be a of sets on the base. Then there is a sheaf $\mathscr{F}$ extending F with isomorphisms $\mathscr{F}(B_i) \cong F(B_i)$ agreeing with restrictions. Furthermore, this sheaf is unique up to unique isomorhpism

***Proof*:**
The sheaf $\mathscr{F}$ will be defined as the sheaf of compatible germs of F. To this end, define the stalk of a presheaf F on a base at $p \in X$ by taking:

$$F_p = \varinjlim F(B_i)$$

Define:

$$\mathscr{F}(U) =$$
$$\{(f_p \in F_p)_{p\in U} \mid \forall p \in U, \exists B \text{ with } p \in B \subseteq U,\ s \in F(B) \text{ that is compatible: } s_q = f_q\ \forall q \in B\}$$

Then the map $F(B) \to \mathscr{F}(B)$ is an isomorphism.

Lots more to cover here that is worth going over

1.1.3 Gluing Sheaves on an Open Cover

Just as we can construct a sheaf from data on a topological base, we can construct a sheaf from sheaves defined on the pieces of an open cover, provided the local sheaves are compatible on overlaps. This result is foundational: it will be used in chapter 2 to construct the structure sheaf when gluing schemes.

Theorem 1.1.23: Gluing Sheaves On An Open Cover

Let X be a topological space and $\{U_i\}_{i\in I}$ an open cover. Suppose we are given:

1. a sheaf $\mathscr{F}_i$ on each U_i,
2. isomorphisms of sheaves $\varphi_{ij} : \mathscr{F}_i|_{U_i\cap U_j} \xrightarrow{\sim} \mathscr{F}_j|_{U_i\cap U_j}$ for all i, j

satisfying the cocycle condition on triple overlaps:

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik} \quad \text{on } U_i \cap U_j \cap U_k$$

Then there exists a sheaf $\mathscr{F}$ on X, unique up to unique isomorphism, together with isomorphisms $\psi_i : \mathscr{F}|_{U_i} \xrightarrow{\sim} \mathscr{F}_i$ such that $\varphi_{ij} \circ \psi_i = \psi_j$ on $U_i \cap U_j$.

***Proof*:**
For each open set $V \subseteq X$, define:

$$\mathscr{F}(V) = \left\{(s_i)_{i\in I} \;\middle|\; s_i \in \mathscr{F}_i(V \cap U_i), \quad \varphi_{ij}\left(s_i|_{V\cap U_i\cap U_j}\right) = s_j|_{V\cap U_i\cap U_j}\ \forall i, j\right\}$$

with restriction maps defined componentwise: if $W \subseteq V$ then $\mathrm{res}_{V,W}((s_i)_i) = (s_i|_{W\cap U_i})_i$.

$\mathscr{F}$ is a presheaf. The restriction of a compatible tuple is compatible (as φ_{ij} commutes with restriction, being a sheaf morphism), and the identity and composition axioms are inherited componentwise.

Locality axiom. Let $V = \bigcup_\alpha V_\alpha$ and suppose $(s_i), (t_i) \in \mathscr{F}(V)$ satisfy $(s_i)|_{V_\alpha} = (t_i)|_{V_\alpha}$ for all α. Then for each i, we have $s_i|_{V_\alpha \cap U_i} = t_i|_{V_\alpha \cap U_i}$ for all α. Since $\{V_\alpha \cap U_i\}_\alpha$ is an open cover of $V \cap U_i$ and $\mathscr{F}_i$ is a sheaf (satisfying locality), $s_i = t_i$ for all i.

Gluing axiom. Let $V = \bigcup_\alpha V_\alpha$ and suppose $(s_i^{(\alpha)})_i \in \mathscr{F}(V_\alpha)$ agree on overlaps, i.e. $s_i^{(\alpha)}|_{V_\alpha \cap V_\beta \cap U_i} = s_i^{(\beta)}|_{V_\alpha \cap V_\beta \cap U_i}$ for all α, β, i. For each fixed i, the sections $s_i^{(\alpha)} \in \mathscr{F}_i(V_\alpha \cap U_i)$ agree on overlaps $V_\alpha \cap V_\beta \cap U_i$. As $\mathscr{F}_i$ is a sheaf, there exists a unique $s_i \in \mathscr{F}_i(V \cap U_i)$ with $s_i|_{V_\alpha \cap U_i} = s_i^{(\alpha)}$ for all α.

It remains to check the compatibility condition for the glued tuple $(s_i)_i$: we need $\varphi_{ij}(s_i|_{V \cap U_i \cap U_j}) = s_j|_{V \cap U_i \cap U_j}$. Both sides are sections of $\mathscr{F}_j(V \cap U_i \cap U_j)$ that agree on each $V_\alpha \cap U_i \cap U_j$ (since $(s_i^{(\alpha)})_i \in \mathscr{F}(V_\alpha)$ satisfies the compatibility condition). By the locality axiom for $\mathscr{F}_j$, they are equal.

Hence $\mathscr{F}$ is a sheaf. The isomorphisms $\psi_i : \mathscr{F}|_{U_i} \to \mathscr{F}_i$ are given by projection onto the i-th component: $(s_j)_j \mapsto s_i$, which is an isomorphism because the compatibility conditions determine all other components on U_i via the φ_{ij}. Uniqueness of $\mathscr{F}$ follows from corollary 1.3.4 applied to the open cover.

1.2 Morphisms and Exactness

Now that we have defined (pre)sheaves, it is natural to define morphism between sheaves to show that $\mathbf{PSh}(X)$ and $\mathbf{Sh}(X)$ form a category. Sheaf morphisms shall be over the same object, and hence can be thought of as the equivalent of looking at local homeomorphisms between two étale spaces:

Definition 1.2.1: Morphism Of Sheaves

Let $\mathscr{F}, \mathscr{G} \in \mathbf{PSh}(X)$ (note that $\mathbf{Sh}(X) \subseteq \mathbf{PSh}(X)$). Then a *morphism of (pre)sheaves* (over the chosen category) $\varphi : \mathscr{F} \to \mathscr{G}$ is a natural transformation between the two functors. Concretely, it contains the data $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ such that if $V \subseteq U$ then the following diagram commutes:

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[r, "\varphi(U)"] \arrow[d, "\mathrm{res}_{U,V}"'] & \mathscr{G}(U) \arrow[d, "\mathrm{res}_{U,V}"] \\
\mathscr{F}(V) \arrow[r, "\varphi(V)"'] & \mathscr{G}(V)
\end{tikzcd}$$

that is

$$\mathrm{res}_{U,V}(\varphi(U)(-)) = \varphi(V)(\mathrm{res}_{U,V}(-))$$

The collection of morphisms between two sheaves is denoted $\mathrm{Hom}_{\mathbf{Sh}(X)}(\mathscr{F}, \mathscr{G})$ or $\mathrm{Mor}_{\mathbf{Sh}(X)}(\mathscr{F}, \mathscr{G})$

To emphasize when the domain and codomain of a presheaf morphism is a sheaf, the morphism shall be called a *sheaf morphism.* A useful geometric intuition of sheaf morphisms is that they apply a transformation to each map (ex. $\exp(f)(z) = e^{f(z)}$). With the morphisms and objects defined, it is easy to show that they form a category. We shall denote the category of sheaves over a category $\mathbf{C}$ on a space X as $\mathbf{C}_X$. Thus:

$$\mathbf{Set}_X \qquad \mathbf{Ab}_X \qquad \mathbf{Ring}_X$$

are the categories of sheaves over sets, abelian groups, and rings respectively. Since the morphisms are the same, the category $\mathbf{Sh}(X)$ is a full subcategory of $\mathbf{PSh}(X)$. For presheaves, we shall denote the category with an extra superscript:

$$\mathbf{Set}_X^{\text{pre}} \qquad \mathbf{Ab}_X^{\text{pre}} \qquad \mathbf{Ring}_X^{\text{pre}}$$

Example 1.2.2: Morphism of Presheaves and Sheaves

(If the reader is familiar with Riemann surface) Let X be a Riemann surface and $\mathcal{O}_X(-P)$ be the collection of meromorphic function $f \in K^*(X)$ such that $\operatorname{div}(f) + (-p) \geq 0$ ($P \in X$ a point). Then if $\mathcal{O}_X$ is the sheaf of holomorphic functions on X there are sheaf morphisms:

$$0 \to \mathcal{O}_K(-P) \to \mathcal{O}_X \to \kappa_P \to 0$$

where κ_P is the sky-scrapper sheaf at P, namely:

$$\kappa_P = \begin{cases} (0) & x \neq P \\ \mathbb{C} & x = P \end{cases}$$

This is even an exact sequence of sheaves, but we need to build-up to show how to properly define exactness in $\mathbf{C}_X$.

From now on, we shall be working with *abelian categories* unless stated otherwise: they all have 0 objects, they are closed under finite products and coproducts, and their kernel/cokernels behave well with epimorphisms and monomorphisms; see section 0.4 for a refresher or [9] for more details. We shall build-up to showing that $\mathbf{C}_X$ and $\mathbf{C}_X^{\text{pre}}$ are abelian categories.

As a presheaf morphism is a natural transformation, it important to emphasize that it does not make sense to say that a sheaf morphism is injective or surjective. Instead, it can be a *monomorphism* or an *epimorphism* and have corresponding kernel, cokernels, and images. Let us find objects in $\mathbf{PSh}(X)$ and $\mathbf{Sh}(X)$ that can represent these kernel, cokernel, and image. Starting with the kernel of a presheaf morphism $\varphi : \mathscr{F} \to \mathscr{G}$ which is defined to be be object $\mathcal{K}$ with the morphism $K : \mathcal{K} \to \mathcal{F}$ such that $\varphi \circ K = 0$ universally.

Proposition 1.2.3: Kernels in Presheaf Category

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then $\ker\varphi$ can be defined to be the kernel of each $\varphi(U)$:

$$\mathcal{K}(U) = \ker(\varphi(U))$$

The resulting presheaf is denoted $\ker_{\text{pre}}(\varphi)$. Furthermore, if $\mathscr{F}$ is a sheaf and $\mathscr{G}$ satisfies at least the locality axiom (in particular if it is also a sheaf), then $\mathcal{K}$ is a sheaf.

***Proof*:**
chase the diagram, and use a similar separated/locality condition in the case of a sheaf as was done before.

Note that as the representation of the kernel for sheaves is the same as the characterization of presheaves, we can just write $\ker(\varphi)$ without any ambiguity. As the monomorphism condition can be checked on open sets, the following usual definition of injectivity is recovered on the level of open sets:

Definition 1.2.4: Injective Presheaf and Sheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between shaves or presheaves. Then φ is *injective* if $\ker \varphi(U) = 0$ for all open $U \subseteq X$.

The image of a presheaf morphism $\varphi : \mathscr{F} \to \mathscr{G}$ is defined to be be object $\mathscr{I}$ with the monomorphism $I : \mathscr{G} \to \mathscr{I}$ and a morphism $E : \mathscr{F} \to \mathscr{I}$ such that $\varphi = E \circ I$ universally, namely:

$$\begin{tikzcd}
\mathscr{F} \arrow[rr, "\varphi"] \arrow[dr, "E"] \arrow[ddr, bend right, "E'"'] & & \mathscr{G} \\
& \mathscr{I} \arrow[ur, "I"] \arrow[d, dashed, "\exists! v"] & \\
& \mathscr{I}' \arrow[uur, bend right, "I'"'] &
\end{tikzcd}$$

The reader familiar with category may be more familiar with images in abelian categories defined as:

$$\operatorname{im}(\varphi) = \ker(\operatorname{coker}(\varphi))$$

As we are working with abelian categories, $\ker(\operatorname{coker}(\varphi)) \cong \operatorname{coker}(\ker(\varphi))$, and hence there is no need of defining the coimage.

Proposition 1.2.5: Images in Presheaf and Sheaf Category

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then $\operatorname{im}_{\text{pre}}\varphi$ can be defined to be the image of each $\varphi(U)$:

$$\operatorname{im}_{\text{pre}}\varphi(U) = \operatorname{im}(\varphi(U))$$

and similarly for cokernels:

$$\operatorname{coker}_{\text{pre}}(\varphi(U)) = \frac{\mathscr{G}(U)}{\operatorname{im}(\varphi(U))}$$

The resulting $\operatorname{im}_{\text{pre}}(\varphi)$ and $\operatorname{coker}_{\text{pre}}(\varphi)$ is called the *image presheaf* and *cokernel presheaf* respectively.

This is *not* necessarily the case if $\mathscr{F}$ is a sheaf. Images and cokernel in $\mathbf{C}_X$ will require a bit more work to be characterized.

***Proof*:**
chase the diagram.

Definition 1.2.6: Surjective Presheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then φ is *surjective* if $\operatorname{im}(\varphi(U)) = \mathscr{G}(U)$ for all open $U \subseteq X$.

As noted earlier, images of sheaves need not be sheaves:

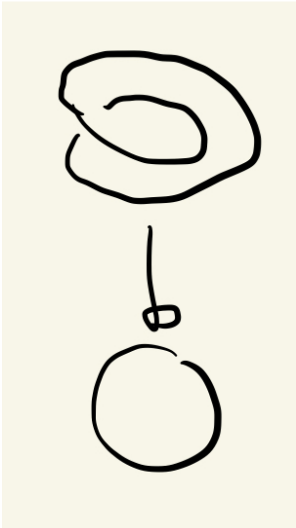
Example 1.2.7: Sheaves: difficulty With Cokernels and Images

1. Let $X = \mathbb{C}$ with the usual topology, $\underline{\mathbb{Z}}$ be the constant sheaf on X, $\mathcal{O}_X$ the sheaf of holomorphic functions, and $\mathscr{F}$ the *presheaf* of functions admitting a holomorphic logarithm. Then there is an exact sequence of presheaves on X:
$$0 \to \underline{\mathbb{Z}} \to \mathcal{O}_X \to \mathscr{F} \to 0$$
where $\underline{\mathbb{Z}} \to \mathcal{O}_X$ is the natural inclusion on each open set, and $\mathcal{O}_X \to \mathscr{F}$ is given by $f \mapsto \exp(2\pi i f)$ (again, over each open set). This is indeed a short exact sequence, for if $U \subseteq V \subseteq \mathbb{C}$, then:
$$\begin{tikzcd} 0 \arrow[r] & \underline{\mathbb{Z}}(V) \arrow[r] \arrow[d, "\text{res}"] & \mathcal{O}_X(V) \arrow[r] \arrow[d, "\text{res}"] & \mathscr{F}(V) \arrow[r] \arrow[d, "\text{res}"] & 0 \\ 0 \arrow[r] & \underline{\mathbb{Z}}(U) \arrow[r] & \mathcal{O}_X(U) \arrow[r] & \mathscr{F}(U) \arrow[r] & 0 \end{tikzcd}$$
commutes. Certainly each $\underline{\mathbb{Z}}(V) \to \mathcal{O}_X(V)$ is an inclusion, and $\mathcal{O}_X(V) \to \mathscr{F}(V)$ is surjective (by definition) with kernel being the constant functions $\mathbb{Z}$ since if f admits a logarithmic inverse, then $f = \log(g)$, so $g \mapsto \exp(2\pi i \log g) = f$.

   Then it should be clear that $\mathscr{F}$ is *not* a sheaf: take $U = \mathbb{C}^*$, cover it with $U_1 = \mathbb{C} \setminus \mathbb{R}_{\geq 0}$ and $U_2 = \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Then $\mathscr{F}(U)$, $\mathscr{F}(U_1)$, and $\mathscr{F}(U_2)$ each have multiple non-zero functions admitting logarithms (see [52]). If $\mathscr{F}$ were a sheaf, then as z admits logarithmic inverse on U_1 and U_2 and tautologically $z|_{U_1 \cap U_2} = z|_{U_1 \cap U_2}$ there should be a function defined on U restricting to these two, that restricts to both of these, but no such function exists (consider that such a function would have to be continuous, but going around the origin introduces non-trivial monodromy).

2. Consider the following similar but subtly different example. Consider $\mathcal{O}_{\mathbb{C}}^*$ the sheaf of all non-vanishing holomorphic functions (this sheaf is strictly "bigger" than the above $\mathscr{F}$, ex. $\mathscr{F}(\mathbb{C}^*) \subsetneq \mathcal{O}_{\mathbb{C}}^*(\mathbb{C}^*)$). Consider the following short exact sequence of *sheaves*:
$$0 \to \underline{\mathbb{Z}} \xrightarrow{\iota} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \xrightarrow{!} 0$$
Notice that on $U = \mathbb{C}^*$, z does not vanish and so $z \in \mathcal{O}_{\mathbb{C}}^*(U)$. However, z does not admit a logarithmic inverse (again see [52]). This would imply that $\exp(\mathbb{C}^*)$ is *not* surjective and so exp is not surjective on all open sets. And yet it is indeed the case that there exists a morphism $\xrightarrow{!}$, namely exp is an *epimorphism* on sheaves. This will be an example of an epimorphism that is *not* characterize by surjectivity on open sets, as shall be explained by defining sheafification after these examples.

3. For the reader less comfortable with complex analysis, here is a more rudimentary example: take the sheaf of continuous sections of the circle S^1. For our first sheaf of sections, let $\pi : S^1 \to S^1$ be the identity (the trivial bundle), and let $\mathcal{F}_1$ be the sheaf of sections of this bundle. The second sheaf of section will with the projection from the twisted circle, which can be visualized as:

   Label this shape S^1_2. The second sheaf $\mathcal{F}_2$ is all the continuous sections of all the open subsets of the codomain for the map $\pi : S^1_2 \to S^1$ with a reasonable definition of projection map mirroring the above image (over the complex numbers, we can take $z \mapsto z^2$). Note that $\mathcal{F}_2(S^2) = 0$, i.e. it has no global sections. Then there is certainly a surjective continuous map φ satisfying the commutative diagram:

$$\begin{tikzcd} S^1_2 \arrow[rr, "\varphi"] \arrow[dr] & & S^1 \arrow[dl] \\ & S^1 & \end{tikzcd}$$

   However, there is no surjective maps:

$$\mathcal{F}_2(X) \to \mathcal{F}_1(X)$$

   namely since there no continuous section from S^1 to S^1_2 that would satisfy the projection condition.

In the above, the first example shows that exactness has no issue with pre-sheaves, and in the second example we claimed the short exact sequence of sheaves was still exact in the category of sheaves. Notice that the image of exp in $\mathcal{O}^*_{\mathbb{C}}$ was the presheaf $\mathscr{F}$ in the first example. What we shall show is that this image uniquely determines $\mathcal{O}^*_{\mathbb{C}}$. In the same vein in which we can from an abelian group via a abelian semi-group using groupification, or a normal subgroup by normalizing a subgroup, we can from a sheaf from a presheaf via sheafification. As is usual, this is done in a universal way that is adjoint to the forgetful functor. This sheafification shall be the solution to defining right notion of surjection for sheaves:

Theorem 1.2.8: Sheafification

Let $\mathscr{F}$ be a presheaf on X. Then there exists a sheaf $\mathscr{F}^{\text{sh}}$ and a map $\text{sh} : \mathscr{F} \to \mathscr{F}^{\text{sh}}$ such that for any sheaf $\mathscr{G}$ and any presheaf morphism $g : \mathscr{F} \to \mathscr{G}$, there exists a unique morphism of sheaves $f : \mathscr{F}^{\text{sh}} \to \mathscr{G}$ such that the following diagram commutes:

$$\begin{tikzcd} & \mathscr{F}^{\text{sh}} \arrow[d, "{g,\ \text{sheaf mor.}}"] \\ \mathscr{F} \arrow[ur, "\text{sh}"] \arrow[r, "f", "\text{presheaf mor.}"'] & \mathscr{G} \end{tikzcd}$$

The sheafification of the presheaf $\mathscr{F}$ is often denoted $\mathscr{F}^+$.

The idea will be to remove all section that violate locality, and add sections to respect gluability. Another good take is the following post here.

***Proof*:**

Let $\mathscr{F}$ be the presheaf in the question. Define $\mathscr{F}^{\text{sh}}$ as the set of compatible germs of the presheaf $\mathcal{F}$ over U, namely:

$$\mathscr{F}^{\text{sh}}(U) := \{(f_p \in \mathscr{F}_p)_{p \in U} \mid \forall p \in U, \exists p \in V \subseteq U \text{ open and } s \in \mathscr{F}(V) \text{ such that } s_q = f_q, \forall q \in V\}$$

Note this resemble to the étale space. To see that $\mathscr{F}^{\text{sh}}$ is a sheaf, first note that restriction is defined by restricting the domain of the tuple of germs. For the gluing axiom, if we have sections $s^{(i)} \in \mathscr{F}^{\text{sh}}(U_i)$ agreeing on overlaps, we can define a global tuple s on $U = \bigcup U_i$ by setting $s_p = s_p^{(i)}$ for $p \in U_i$. This tuple belongs to $\mathscr{F}^{\text{sh}}(U)$ because the condition of being "locally represented by a section of $\mathscr{F}$" is satisfied locally by the witnesses for each $s^{(i)}$.

We define the map $\text{sh} : \mathscr{F} \to \mathscr{F}^{\text{sh}}$ by sending a section $\sigma \in \mathscr{F}(U)$ to the tuple of its germs $((\sigma)_p)_{p \in U}$, which is clearly a presheaf morphism.

To prove the universal property, let $\mathscr{G}$ be a sheaf and $g : \mathscr{F} \to \mathscr{G}$ a presheaf morphism. We wish to construct a unique $f : \mathscr{F}^{\text{sh}} \to \mathscr{G}$. Let $s \in \mathscr{F}^{\text{sh}}(U)$. By definition, there exists an open cover $\{V_i\}$ of U and sections $\sigma_i \in \mathscr{F}(V_i)$ such that $s|_{V_i} = \text{sh}(\sigma_i)$ (i.e., the germs of s match the germs of σ_i on V_i). We propose to define $f(s)$ by gluing the sections $g_{V_i}(\sigma_i) \in \mathscr{G}(V_i)$.

To do this, we must verify that $g(\sigma_i)$ and $g(\sigma_j)$ agree on $V_i \cap V_j$. For any point $p \in V_i \cap V_j$, we have $(\sigma_i)_p = s_p = (\sigma_j)_p$. By the definition of germs, there exists a small neighborhood $W_p \subseteq V_i \cap V_j$ such that $\sigma_i|_{W_p} = \sigma_j|_{W_p}$. Since g is a presheaf morphism, it respects restriction, so $g(\sigma_i)|_{W_p} = g(\sigma_j)|_{W_p}$. Because $\mathscr{G}$ is a sheaf (specifically, separated), and the sections $g(\sigma_i)$ and $g(\sigma_j)$ agree on a neighborhood of every point in the intersection, they must be equal on the entire intersection $V_i \cap V_j$. Thus, by the gluing axiom of $\mathscr{G}$, there exists a unique section $T \in \mathscr{G}(U)$ such that $T|_{V_i} = g(\sigma_i)$. We define $f_U(s) = T$. This map is unique because any morphism f must respect restriction, forcing it to agree with this local construction.

The "failure" of a presheaf being a sheaf can be measured as the lack of exactness of at the $\prod_i F(U_i)$ in proposition 1.1.9.

Example 1.2.9: Sheafification

To gain intuition, show the following sheafifications:

1. Show that $(\underline{S}_{\text{pre}})^{\text{sh}} = \underline{S}$, that is the presheaf of constant function sheafifies to the sheaf of constant functions.
2. Let $\mathscr{F}$ on $\mathbb{R}$ be all bounded continuous functions on each open set. Show that $\mathscr{F}^{\text{sh}}$ is *all* continuous functions on each open subset.
3. Show that $\mathcal{H}$ on $\mathbb{C}$ of non-vanishing holomorphic functions that admit a logarithm sheafify to all non-vanishing holomorphic functions.
4. Let $\mathscr{F}$ be a non-trivial presheaf where all stalks are trivial (see exercise ref:HERE). Show that $\mathscr{F}^{\text{sh}}$ is trivial.

Corollary 1.2.10: Sheafification and Stalks

Let $\mathcal{F} \in \mathbf{PSh}(X)$ be a presheaf. Then for all $p \in X$

$$\mathcal{F}_p \cong \mathcal{F}_p^{\text{sh}}$$

***Proof*:**

Use something similar given in the proof of proposition 1.1.5

Let us now put together the universal properties we've shown so far to get that the category of sheaves also have cokernels:

Proposition 1.2.11: Sheaf Morphisms Have Cokernels

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a sheaf morphism. Then it has a cokernel and it is the sheafification of the cokernel-presheaf.

***Proof*:**

Recall that the universal property of the cokernel can be seen as the colimit of the following diagram:

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{G} \\ 0 & \end{tikzcd}$$

As we know, the cokernel forms a presheaf $\mathcal{H}_{\text{pre}}$. By sheafification we have a unique map

$\mathrm{sh}: \mathcal{H}_{\mathrm{pre}} \to \mathcal{H}$. Now let $\mathcal{A}$ be any sheaf and consider the following diagram:

$$\begin{tikzcd}
\mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{G} \arrow[d] \arrow[rrdd, bend left] & & \\
0 \arrow[r] \arrow[rrrd] & \mathcal{H}_{\mathrm{pre}} \arrow[r, "\mathrm{sh}"] & \mathcal{H} \arrow[rd, dashed, "\exists!"] & \\
& & & \mathcal{A}
\end{tikzcd}$$

By the universal property of cokernels for presheaves, we have a unique map $\mathcal{H}_{\mathrm{pre}} \to \mathcal{A}$. By the universal property of sheafification, we will get a unique map composed from the two unique universal maps giving us $\mathcal{H} \to \mathcal{A}$, as we sought to show.

Corollary 1.2.12: Image and Sheafification

Let $\varphi \in \mathbf{Sh}(X)$ be a morphism of sheaves over an abelian category. Then the image of φ is the sheafification of the image presheaf.

***Proof*:**
Recall that in an abelian category, the image of a morphism is defined as the kernel of its cokernel:

$$\mathrm{Im}(\varphi) \cong \mathrm{Ker}(\mathcal{G} \to \mathrm{coker}(\varphi))$$

Let $\mathcal{K}_{\mathrm{pre}}$ be the presheaf cokernel of φ. We have, by definition, a short exact sequence of presheaves:

$$0 \to \mathrm{Im}_{\mathrm{pre}}(\varphi) \to \mathcal{G} \to \mathcal{K}_{\mathrm{pre}} \to 0$$

Since the sheafification functor is exact (it preserves exact sequences and finite limits), applying it to the sequence above yields an exact sequence of sheaves:

$$0 \to (\mathrm{Im}_{\mathrm{pre}}(\varphi))^{\mathrm{sh}} \to \mathcal{G}^{\mathrm{sh}} \to (\mathcal{K}_{\mathrm{pre}})^{\mathrm{sh}} \to 0$$

Since $\mathcal{G}$ is already a sheaf, $\mathcal{G}^{\mathrm{sh}} \cong \mathcal{G}$. By proposition 1.2.11, $(\mathcal{K}_{\mathrm{pre}})^{\mathrm{sh}}$ is the sheaf cokernel of φ. Consequently, the object $(\mathrm{Im}_{\mathrm{pre}}(\varphi))^{\mathrm{sh}}$ is the kernel of the morphism $\mathcal{G} \to \mathrm{coker}(\varphi)$, which is precisely the definition of the image sheaf.

Definition 1.2.13: Surjective Sheaf Morphism

Let $\varphi: \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves and $\mathrm{im}\varphi$ the image presheaf. Then the cokernel and image of φ is the sheafification of the cokernel-presheaf and image-presheaf respectively. The morphism φ is *surjective* if

$$(\mathrm{im}_{\mathrm{pre}}\varphi)^{\mathrm{sh}} = \mathscr{G}$$

Using this, we see that the map $\mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^*$ is *surjective* in the category of sheaves, even though it is not surjective in the category of presheaves. Hence, exactness in presheaves and sheaves is *different*. The following example using simpler categories should demonstrate why this is not so strange:

Example 1.2.14: Ab Vs TF-Ab

Let **Ab** be the category of abelian groups and TF-Ab be the category of torsion-free abelian groups. Consider the morphism $\varphi : \mathbb{Z} \to \mathbb{Z}$, $\varphi(n) = 2n$. This morphism exists in both categories. The cokernel of φ in **Ab** is $\mathbb{Z}/2\mathbb{Z}$, which is torsion. To find the cokernel in TF-AB, the "best" torsion-free group must be found so that its composition with φ is universal. It isn't hard to see this is 0, so:

$$\operatorname{coker}_{\mathbf{Ab}}(\varphi) = \mathbb{Z}/2\mathbb{Z} \neq 0 = \operatorname{coker}_{\text{TF-Ab}}(\varphi)$$

Thus, though the morphism in each category are the same, the epimorphism differ.

Thus, a theoretical understanding of epimorphisms in $\mathbf{C}_X$ is done. However, it is computationally intractable to always sheafify the image sheaf to check for surjectivity. As sheaves are created to be "local", we shall be able to shall be able to check their condition on *stalks*. Proving this shall fit in better into the broader framework on when the functor taking $\mathscr{F} \mapsto \mathscr{F}(U)$ and $\mathscr{F} \mapsto \mathscr{F}_p$ is exact.

Functors from (pre)sheaves: simplifying checking exactness

As presheaves and sheaves store all local information, we may want that there are some natural functors that preserve some exactness to extract that information. The two natural candidates for functors to would be $-(U)$ taking a (pre)sheaf to its sections over an open set U, and $(-)_p$ the functor taking a (pre)sheaf to it's stalk at p.

It is clear that $-(U) : \mathbf{C}_X \to \mathbf{C}$ is a functor for:

$$\mathscr{F} \mapsto \mathscr{F}(U) \qquad (\varphi : \mathscr{F} \to \mathscr{G}) \mapsto (\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U))$$

preserves composition's an identities. Let us show $(-)_p$ is also a functor:

Proposition 1.2.15: Morphism Of (Pre)Sheaves Induces Morphism On Stalks

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of (pre)sheaves over any category $\mathbf{C}$ having colimits (so that stalks are well-defined). Then there is an induced morphism on stalks $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$, that is, it induced a functor $\mathbf{C}_X \to \mathbf{C}$.

***Proof*:**

This is definition pushing: For each open $U \subseteq X$ we have the morphism $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$. Define the map:

$$[(f, U)]_p \mapsto [(\varphi(U)(f), U]_p$$

For this map to be well-defined if $(f, U) \sim (f', V)$ then we need:

$$[(f', V)]_p \mapsto [(\varphi(V)(f'), V]_p = [(\varphi(U)(f), U]_p$$

Hence we need that $((\varphi(V)(f'), V) \sim ((\varphi(U)(f), U)$. As $(f, U) \sim (f', V)$ we have there exists $W \subseteq U, V$ such that

$$\operatorname{res}_{U,W} f = \operatorname{res}_{V,W} f' = g$$

Then as φ is a pre-sheaf morphism we have:

$$\begin{tikzcd} \varphi(U) : f \arrow[r, mapsto] \arrow[d, "\mathrm{res}_{U,W}"] & \varphi(U)(f) \arrow[d, "\mathrm{res}_{U,W}"] \\ \varphi(W) : \mathrm{res}_{U,W}(f) = g \arrow[r, mapsto] & \varphi(W)(g) \end{tikzcd} \qquad \begin{tikzcd} \varphi(V) : f' \arrow[r, mapsto] \arrow[d, "\mathrm{res}_{V,W}"] & \varphi(U)(f') \arrow[d, "\mathrm{res}_{V,W}"] \\ \varphi(W) : \mathrm{res}_{V,W}(f) = g \arrow[r, mapsto] & \varphi(W)(g) \end{tikzcd}$$

which implies

$$\mathrm{res}_{U,W}\varphi(U)(f) = \mathrm{res}_{V,W}\varphi(V)(f')$$

But that means they are similar, $((\varphi(V)(f'), V) \sim ((\varphi(U)(f), U)$, as we sought to show

Let us now see how these functors preserve monomorphisms/epimorphisms (i.e. injectivity and surjectivity on opens/stalks). Starting with open subsets:

Proposition 1.2.16: On Presheaf: Open Set Functor is Exact

The functor $-(U) : \mathbf{PSh}_{\mathbf{C}}(X) \to \mathbf{C}$ (where $\mathbf{C}$ is abelian) is an exact functor

On $\mathbf{Sh}$, the functor $-(U)$ is left-exact, and hence preserves injective morphisms.

***Proof*:**
Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a monomorphism of presheaves between presheaves. By definition, for any presheaf $\mathcal{H}$ and any morphisms $f, g : \mathcal{H} \to \mathcal{F}$, the relation $\varphi \circ f = \varphi \circ g$ implies $f = g$. Since morphisms of presheaves are natural transformations defined component-wise, the equality of morphisms holds if and only if it holds for every open set U. Thus, the condition implies:

$$\varphi(U) \circ f(U) = \varphi(U) \circ g(U) \implies f(U) = g(U)$$

This is precisely the definition of a monomorphism in the target abelian category $\mathbf{C}$, giving left-exactness and preservation of injectivity. A symmetric argument holds for epimorphisms: if $\psi : \mathcal{G} \to \mathcal{Q}$ is an epimorphism, then $\alpha \circ \psi = \beta \circ \psi \implies \alpha = \beta$, which translates component-wise to $\psi(U)$ being an epimorphism and preserving surjectivity in $\mathbf{C}$. Since the functor $-(U)$ preserves both monomorphisms (kernels) and epimorphisms (cokernels), it is exact on presheaves.

Turning to the category of sheaves $\mathbf{Sh}(X)$, the functor $-(U)$ remains left-exact. This is because the inclusion $\mathbf{Sh}(X) \hookrightarrow \mathbf{PSh}(X)$ preserves limits: the sheafification function $(-)^{\mathrm{sh}} : \mathbf{PSh}(X) \to \mathbf{Sh}(X)$ is *left adjoint* to the inclusion $\iota : \mathbf{PSh}(X) \to \mathbf{Sh}(X)$, hence the inclusion is a *right adjoint* so by ref:HERE (put it in section 0.4) it preserves limits. As kernels are limits they are preserved. Consequently, if φ is a monomorphism of sheaves, it is also a monomorphism of presheaves, and as shown above, $\varphi(U)$ is a monomorphism (and thus injective) in $\mathbf{C}$. However, $-(U)$ is not right-exact on sheaves. While epimorphisms in presheaves imply component-wise surjectivity, there are not enough sheaves for component-wise surjectivity of sheaves and sheafification is required. As seen in example 1.2.7, a map can be an epimorphism of sheaves (locally surjective) without the component map $\varphi(U)$ being surjective, failing exactness on the right (ex. take the global section functor Γ which is will not perserve exactness of example 2).

1.2.17 Remark In Vakil's proof of this result, [64], he uses ""indicator sheaves" to verify these cancellations. In categorical terms, to test the value of a section $s \in \mathcal{F}(U)$, one uses the Yoneda Lemma. The indicator sheaf mentioned corresponds to the representable presheaf h_U (which is the sheafification of the constant section on subsets of U and empty elsewhere). Mapping out of h_U allows one to probe the specific algebraic properties of $\mathcal{F}(U)$ essentially element-by-element.

Proposition 1.2.18: Exactness of Presheaves On The Level of Open Sets

An sequence of presheaves

$$0 \to \mathscr{F}_1 \to \mathscr{F}_2 \to \cdots \to \mathscr{F}_n \to 0$$

is exact if and only if the sequence

$$0 \to \mathscr{F}_1(U) \to \mathscr{F}_2(U) \to \cdots \to \mathscr{F}_n(U) \to 0$$

is exact for all U.

***Proof*:**
The only if direction is induction on proposition 1.2.16. Conversely, if $0 \to \mathscr{F}_1(U) \to \mathscr{F}_2(U) \to \cdots \to \mathscr{F}_n(U) \to 0$ is exact for each open $U \subseteq X$, then as we already have morphism of presheaves we have these commute with restrictions and $\mathrm{im}\mathscr{F}_i = \ker \mathscr{F}_i$, completing the proof.

Corollary 1.2.19: Exactness of Stalk For Presheaves

Let $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ be an exact sequence of presheaves. Then if filtered colimits exists (so that stalks exist), then

$$0 \to \mathscr{F}_p \to \mathscr{G}_g \to \mathscr{H}_p \to 0$$

is exact.

***Proof*:**
Use that each functor $-(U)$ is exact for each U; so we can take their colimit.

Note that for presheaves, exactness on all stalks does not imply exactness on presheaves as example 1.2.7(2) shows.

Theorem 1.2.20: Presheaves Over Ab Category Form Ab Category

The category $\mathbf{PSh}_{\mathbf{C}}(X)$ over an abelian category C is an abelian category

***Proof*:**
The main thing that wasn't checked was how to define addition of morphism. By the above work, this can be done on each open set: if $\varphi, \psi : \mathcal{F} \to \mathcal{G}$ then $\varphi + \psi$ defined via open sets as:

$$\varphi + \psi(U) := \varphi(U) + \psi(U)$$

is a map of presheaf (it respects restriction by the commutativity of both natural transformation), and so the hom-sets form an abelian group. Then as we also have a 0-object, and we have finite products (again, something the reader could verify), we have an additive category.

To show kernels and cokernels exist, by proposition 1.2.18 it suffices to check on the level of open sets. Then we define:

$$(\ker_{\mathrm{pre}} \varphi)(U) = \ker \varphi(U)$$

The key is to consider the following diagram where $U \hookrightarrow V$:

$$\begin{tikzcd}
0 \arrow[r] & \ker_{\mathrm{pre}}(\varphi(V)) \arrow[r] \arrow[d, dashed, "\exists!"] & \mathcal{F}(V) \arrow[r] \arrow[d, "\mathrm{res}_{V,U}"] & \mathcal{G}(V) \arrow[d, "\mathrm{res}_{V,U}"] \\
0 \arrow[r] & (\ker_{\mathrm{pre}} \varphi)(U) \arrow[r] & \mathcal{F}(U) \arrow[r] & \mathcal{G}(U)
\end{tikzcd}$$

which shows that this is indeed well-defined. The reader can further check that this does indeed satisfy the universal property of the kernel. By dualizing the argument, we can do the same for the cokernel, and hence the additive category of presheaves has kernels and cokernels, making it abelian.

To keep track of all morphisms between open sets, we box the following definition:

Definition 1.2.21: Sheaf Hom

Let $\mathscr{F}, \mathscr{G} \in \mathbf{Sh}_{\mathbf{C}}(X)$. Then the *sheaf hom* is defined to be:

$$\mathcal{H}om(\mathscr{F}, \mathscr{G})(U) := \mathrm{Mor}(\mathscr{F}_U, \mathscr{G}|_U)$$

where $\mathrm{Mor}(\mathscr{F}|_U, \mathscr{G}|_U)$ are all morphism of (pre)sheaves. The dual of $\mathscr{F}$ is:

$$\mathscr{F}^\vee = \mathrm{Hom}_{\mathrm{Mod}_{\mathcal{O}_X}}(\mathscr{F}, \mathcal{O}_X)$$

Just like Hom, $\mathcal{H}om$ is a (co/contra)variant functor, depending on which position you choose.

The situation is not so clean for $(-)_p$, namely the functor on $\mathbf{Sh}_{\mathbf{C}}(X)$ is not always exact for *any* abelian category; it need not be the case that filtered colimits preserve exactness, and hence that $(-)_p$ preserve exactness (think of the category of finite abelian groups which doesn't have $\varinjlim_p \mathbb{Z}/p^n\mathbb{Z}$). Grothendieck axiomatically added this condition to abelian categories (such categories are called *Grothendieck categories*). Fortunately for us, this will never be an issue as all the categories of interest to us shall have this property, namely:

1. abelian groups, more generally R-Mod
2. Sheaves of abelian groups (more generally $\mathcal{O}_X$-modules which are defined later)
3. the product of Grothendieck categories

4. In ref:HERE, it shall be important to have internal Hom's: given a small category $\mathbf{C}$ and a Grothendieck category $\mathbf{G}$, all functors from $\mathbf{C}$ to $\mathbf{G}$ (with morphism being natural transformation) form Grothendieck categories.

Proposition 1.2.22: Sheaves and exactness For Stalk

Let $\mathbf{Sh}(X)$ be the category of sheaves over any R-Mod. Then $(-)_p : \mathbf{Sh}(X) \to R$-Mod is exact.

***Proof*:**
Recall that the stalk of a sheaf $\mathcal{F}$ at a point p is defined as the colimit over the filtered system of open neighborhoods of p:

$$\mathcal{F}_p = \varinjlim_{p \in U} \mathcal{F}(U)$$

By the general properties of abelian categories, the colimit functor is always right-exact (it preserves cokernels). The non-trivial property required is left-exactness. In the category of R-modules (unlike the category of finite abelian groups mentioned above), filtered colimits commute with finite limits. Consequently, the filtered colimit commutes with kernels.

Therefore, if $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ is an exact sequence of sheaves, applying the stalk functor yields a sequence of filtered colimits. Since exactness is preserved by filtered colimits in R-Mod, the resulting sequence of stalks

$$0 \to \mathcal{F}_p \to \mathcal{G}_p \to \mathcal{H}_p \to 0$$

is exact.

With the right notion of compatible stalks defined, we may show how to determine morphisms of sheaves at the level of stalks, namely a scheme morphisms is determined by stalks, and a scheme morphism that is an isomorphism on stalks induces an inverse scheme morphism:

Proposition 1.2.23: Morphisms Determined By Stalks

Let φ_1, φ_2 be morphisms from a presheaf of sets $\mathscr{F}$ to a *sheaf* of sets $\mathscr{G}$ that induce the same maps on each stalk. Then $\varphi_1 = \varphi_2$

***Proof*:**
For every open U, $\varphi_1(U) = \varphi_2(U)$. Let $s \in \mathscr{F}(U)$. Since $\mathscr{G}$ is a sheaf, the section $t = \varphi_1(U)(s)$ is uniquely determined by its germs $(t_p)_{p \in U}$.

For any $p \in U$, the germ of the image is the image of the germ:

$$(\varphi_1(s))_p = (\varphi_1)_p(s_p)$$

By assumption, $(\varphi_1)_p = (\varphi_2)_p$, so:

$$(\varphi_1)_p(s_p) = (\varphi_2)_p(s_p) = (\varphi_2(s))_p$$

Since $\varphi_1(s)$ and $\varphi_2(s)$ have identical germs at every point in U, and $\mathscr{G}$ is a sheaf (satisfying the identity axiom), the sections must be identical. Thus $\varphi_1(U)(s) = \varphi_2(U)(s)$.

The above proof can be summarized by the following diagram:

$$\begin{tikzcd} \mathscr{F}(U) \arrow[r] \arrow[d, hook] & \mathscr{G}(U) \arrow[d, hook] \\ \prod_{p\in U}\mathscr{F}_p \arrow[r] & \prod_{p\in U}\mathscr{G}_p \end{tikzcd}$$

Proposition 1.2.24: Stalks-Local Isomorphism induces Isomorphism

Let φ be a morphism of sheaves. Then φ is an isomorphism if and only if it induces an isomorphism for all stalks.

***Proof*:**
Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a sheaf morphism. If φ is an isomorphism, then certainly each φ_p is an isomorphism, hence say each φ_p is an isomorphism. Then it must be shown that each $\varphi(U)$ is an isomorphism. For injectivity, say for $s \in \mathscr{F}(U)$ $\varphi(s) = 0 \in \mathscr{G}(U)$. Then for each $p \in U$, $\varphi(s)_p = 0 \in \mathscr{G}_p$. As φ_p is injective at each p, we have $s_p = 0 \in \mathscr{F}_p$. Using the definition of stalk, we have that s is zero on some open set, namely $s|_{W_p} = 0$ for $W_p \subseteq U$. As U is covered by such neighborhoods, by the gluing axiom we have $s = 0$

For surjectivity, let $t \in \mathscr{G}(U)$. Consider $t_p \in \mathscr{G}_p$. As φ_p is surjective, we have $s_p \in \mathscr{F}_F$ such that $\varphi_p(s_p) = t_p$. Now, let $(s(p), V_p)$ be a representative of s_p on the set V_p containing the point p. Then $\varphi(s(p))$ and $t|_{V_p}$ are two elements of $\mathscr{G}(V_p)$ whose germs at p are the same. Thus, replacing V_p by a smaller neighborhood if necessary, we have $\varphi(s(p)) = t|_{V_p} \in \mathscr{G}(V_p)$.

Now, U is covered by open sets V_p where on each V_p we have $s(p) \in \mathscr{F}(V_p)$ and $\varphi(s(p)) = t|_{V_p}$. If p, q are two points, then $s(p)|_{V_p\cap V_q}$ and $s(q)|_{V_p\cap V_q}$ are two sections of $\mathscr{F}(V_p \cap V_q)$ which are both sent to $t|_{V_p\cap V_q}$ via φ. As φ was shown to be injective, they are equal. then by the gluing axiom, there exists an $s \in \mathscr{F}(U)$ such that

$$s|_{V_p} = s(p) \qquad \forall p \in U$$

Finally, to show that $\varphi(s) = t$, notice that $\varphi(s)$ and t are two section of $\mathscr{F}(U)$ where for each p $\varphi(s)|_{V_p} = t|_{V_p}$, hence they glue to create $\varphi(s) = t$, as we sought to show.

This shows that a *morphism* $\varphi : \mathscr{F} \to \mathscr{G}$ can be determined on the level of stalks. This does not imply that if two *sheaves* have isomorphic stalks, then they are isomorphic! You need to already have a sheaf homomorphism which in turn induces the isomorphic stalks. Conceptually, this is saying that being locally isomorphic does not imply it is globally isomorphic; there are plenty concrete example form differential geometry to choose from.

Example 1.2.25: Isomorphic Stalks, Not Isomorphic Sheaves

1. Let X be a Hausdorff space. Consider the constant sheaf $\underline{\mathbb{Z}}$ and the sky-scrapper sheaf $\iota_*(\mathbb{Z})$. Show that the stalks at each point is $\mathbb{Z}$, however these two sheaves are not isomorphic.

2. The following will rely on the reader's intuitions of varieties (see [20]). Take the variety associated to the coordinate ring:

$$A = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$$

which is the 2-sphere in $\mathbb{R}^3$. Take the tangent bundle, i.e. A-module, of all derivations $E = \mathrm{Der}_{\mathbb{R}}(A)$. Then E is a rank 2 locally trivial module since for any open subset or distinguished open subset, $E_x \cong \mathbb{R}^2$, showing each stalk is isomorphic. However, $E \ncong A^2$.

3. Another famous example would be the trivial line bundle on S^1 vs. the möbius bundle.

As stalks define maps on some open neighbourhood, we can use this condition to find the closest condition that allows to check if a map between topological spaces is sheaf map on the level of open sets:

Proposition 1.2.26: Characterizing Stalks Inducing Sheaf Morphism

Let X be a topological space with two sheaves $\mathscr{F}, \mathscr{G}$. Let $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$ be a collection of stalk morphisms for all $p \in X$. Then this gives a map of sheaves $\Phi : \mathscr{F} \to \mathscr{G}$ if and only if for all open subsets $U \subseteq X$, for all sections $s \in \mathscr{F}(U)$, there is an open cover $U = \cup_i U_i$ and sections $t_i \in \mathscr{G}(U_i)$ such that

$$\varphi_p(s_p) = (t_i)_p$$

for all $p \in U_i$ and all i.

***Proof*:**
($\Rightarrow$) Assume $\Phi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves. Then for any open U and section $s \in \mathscr{F}(U)$, we can simply take the trivial cover $U_1 = U$. Define $t_1 = \Phi(U)(s) \in \mathscr{G}(U)$. By the definition of the induced map on stalks (proposition 1.2.15), for any $p \in U$, the map φ_p sends the germ of s to the germ of $\Phi(U)(s)$. Thus, $\varphi_p(s_p) = (t_1)_p$, satisfying the condition.

($\Leftarrow$) Conversely, assume the local condition holds. We must construct a morphism $\Phi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ for every open set U. Let $s \in \mathscr{F}(U)$. By hypothesis, there exists a cover $\{U_i\}$ and sections $t_i \in \mathscr{G}(U_i)$ such that for every $p \in U_i$, the germ of t_i at p is $\varphi_p(s_p)$.

We first show that these local sections t_i glue together. Consider the intersection $U_{ij} = U_i \cap U_j$. For any point $p \in U_{ij}$, we have:

$$(t_i)_p = \varphi_p(s_p) = (t_j)_p$$

Since $\mathscr{G}$ is a sheaf (specifically, a separated presheaf), sections are determined by their germs (lemma 1.1.15). Because t_i and t_j have identical germs at every point in U_{ij}, they must be equal as sections:

$$\mathrm{res}_{U_i,U_{ij}}(t_i) = \mathrm{res}_{U_j,U_{ij}}(t_j)$$

Since the sections agree on overlaps, the Gluing Axiom for $\mathscr{G}$ implies there exists a unique section $t \in \mathscr{G}(U)$ such that $t|_{U_i} = t_i$. We define $\Phi(U)(s) = t$.

To ensure Φ is well-defined, suppose there was another cover $\{V_j\}$ and sections $\{r_j\}$ satisfying the condition, yielding a glued section t'. For any $p \in U$, $t_p = \varphi_p(s_p)$ (by the construction on the cover U_i) and $t'_p = \varphi_p(s_p)$ (by the construction on V_j). Thus t and t' have the same germs everywhere, and since $\mathscr{G}$ is a sheaf, $t = t'$.

Finally, Φ commutes with restrictions naturally because the construction is defined pointwise via germs: for $V \subseteq U$, the germs of $\Phi(V)(s|_V)$ are $\varphi_p((s|_V)_p) = \varphi_p(s_p)$, which matches the germs of $(\Phi(U)(s))|_V$.

Summarizing the results for morphisms of sheaves:

Theorem 1.2.27: Equivalence of Mono- and Epimorphisms

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves on X (over **Set** or an abelian category).

The following conditions are equivalent:

1. φ is a monomorphism in the category of sheaves.
2. φ is injective on the level of stalks: $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is injective for all $p \in X$.
3. φ is injective on the level of open sets: $\varphi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ is injective for all open $U \subseteq X$.

The following conditions are equivalent:

1. φ is an epimorphism in the category of sheaves.
2. φ is surjective on the level of stalks: $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is surjective for all $p \in X$.

If $\varphi(U)$ is surjective for all open sets U, then φ is an epimorphism (surjectivity on sections implies surjectivity on stalks). The converse is *false*: an epimorphism of sheaves need not be surjective on global sections (as seen in the exponential sequence).

***Proof*:**
This is combining the results from earlier. In particular, apply the results just proven about stalks to the proof of proposition 1.2.16.

Theorem 1.2.28: Sheaves Over Ab Category Form Ab Category

The category $\mathbf{Sh_C}(X)$ over an abelian category C is an abelian category

***Proof*:**
exercise.

Morphisms and Stalks

Since the stalk functor $(-)_p$ is exact by proposition 1.2.22, it preserves all finite limits and colimits. This yields the following immediate corollaries.

Corollary 1.2.29: Commuting Kernel and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the kernel is the kernel of the stalk:

$$(\ker(\varphi))_p \cong \ker(\varphi_p)$$

***Proof*:**
Since $(-)_p$ is an exact functor, it is left-exact. Left-exact functors preserve kernels.

Corollary 1.2.30: Commuting Cokernel and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the cokernel is the cokernel of the stalk:

$$(\mathrm{coker}(\varphi))_p \cong \mathrm{coker}(\varphi_p)$$

***Proof*:**
Since $(-)_p$ is an exact functor, it is right-exact. Right-exact functors preserve cokernels.

Due to sheafification, the surjectivity of a sheaf morphism is no longer conditional on the surjectivity of sections on all open sets. This leads to the following counter-intuitive examples.

Example 1.2.31: Surjective on Stalks $\neq$ Surjective on Global Sections

1. (if you know Complex Analysis): Let $X = \mathbb{C} \setminus \{0\}$. Let $\mathcal{O}_X$ be the sheaf of holomorphic functions and Ω^1_X be the sheaf of holomorphic differential 1-forms. Define $\varphi : \mathcal{O}_X \to \Omega^1_X$ by $f \mapsto df$.
   - *Locally:* On any simply connected disk U, every closed form is exact. Since holomorphic forms are closed, $\omega = df$ has a solution. Thus φ_p is surjective.
   - *Globally:* The form $\omega = \frac{dz}{z}$ is a global section of $\Omega^1_X(X)$. However, its primitive is $\log(z)$, which is not a function on $\mathbb{C} \setminus \{0\}$. Thus $\frac{dz}{z} \notin \mathrm{im}(\varphi(X))$.
2. Let $X = \mathbb{R}$, $\mathscr{F} = \underline{\mathbb{Z}}$ (constant sheaf), and $\mathscr{G} = i_{p*}(\mathbb{Z}) \oplus i_{q*}(\mathbb{Z})$ (skyscrapers at distinct points p, q). Define $\varphi : \mathscr{F} \to \mathscr{G}$ by restriction.
   - *Stalks:* At p, $\mathscr{F}_p = \mathbb{Z}$ and $\mathscr{G}_p = \mathbb{Z} \oplus 0 \cong \mathbb{Z}$. The map is identity (surjective). Similarly at q. Elsewhere, both are 0.
   - *Globally:* $\mathscr{F}(X) = \mathbb{Z}$ (constant functions). $\mathscr{G}(X) = \mathbb{Z} \oplus \mathbb{Z}$. The map sends $n \mapsto (n, n)$. The image is the diagonal, which is not the whole plane $\mathbb{Z} \oplus \mathbb{Z}$.

Corollary 1.2.32: Commuting Image and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the image is the image of the stalk:

$$(\mathrm{im}(\varphi))_p \cong \mathrm{im}(\varphi_p)$$

***Proof*:**
In an abelian category, $\mathrm{im}(\varphi) = \ker(\mathrm{coker}(\varphi))$. Since $(-)_p$ commutes with both kernels and cokernels, it commutes with images.

1.2.33 Summary

- **Presheaves** form an abelian category because computation is done on the level of open sets.
- **Sheaves** form an abelian category because computation is done on the level of stalks.

Exercise 1.2.1

1. Show that $\varphi : \mathscr{F} \to \mathscr{G}$ is surjective if and only if for every $s \in \mathscr{G}(U)$, there exists an open covering $\bigcup_i U_i = U$ and elements $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s|_{U_i}$ (hint: recall φ is surjective if and only if it is surjective on stalks).

   This exercise will come back in proposition 5.1.16

2. Let $(X, \mathcal{O}_X)$ be a ringed space and $U \subseteq X$ an open subset. Show that we can define non-vanishing distinguished open subset by:

$$\begin{tikzcd} U_f \arrow[r] \arrow[d] & \mathcal{O}_X^* \arrow[d] \\ U \arrow[r, "f"] & \mathcal{O}_X \end{tikzcd}$$

   that is, $U_f = f^{-1}(\mathcal{O}_X^*) = \left\{x \in U \;\middle|\; f(x) \in \mathcal{O}_{X,x}^*\right\}$. Show these form a sheaf.

3. Define a *local ringed space* to be a ringed space $(X, \mathcal{O}_X)$ such that for all open $U \subseteq X$ and all $f \in \mathcal{O}_X(U)$, $U = U_f \cup U_{1-f}$, namely for all $x \in U$, f or $1 - f$ is invertible.

   a) Show that if X, $\operatorname{mSpec} \mathcal{O}_X(X)$ is a singleton.

   b) Let $C(X)$ be all continuous real function on X. Show that $(X, \mathbb{C}(X))$ is local.

4. Let $(X, \mathcal{O}_X)$ is a local ringed space, $U \subseteq X$ open, and $f \in \mathcal{O}_X(U)$. Then if $f_1 + \cdots + f_r = 1$, then $\bigcup U_i = U$.

5. Let $\mathscr{F}$ be a sheaf of sets, show that $\mathcal{H}om(\underline{\{p\}}, \mathscr{F}) \cong \mathscr{F}$

6. Let $\mathscr{F}$ be a sheaf of groups, show that $\mathcal{H}om(\underline{\mathbb{Z}}, \mathscr{F}) \cong \mathscr{F}$

7. It is tempting to then think that $\mathcal{H}om(\mathscr{F}, \mathscr{G})_p \cong \operatorname{Hom}(\mathscr{F}_p, \mathscr{G}_p)$. However this is *not* the case: think of a counter-example. There is a map from one of these sets to another, what is it?

1.3 Grothendieck's Functors: Pushforward, Pullback, and more

We next endeavor to move sheaves around from one topological space to another given a continuous map $f : X \to Y$. Grothendieck identified 4 key ways of doing so that come in adjoint pairs:

1. The pushforward and pullback: f_* and f^*
2. the proper direct image and proper inverse image: $f_!$ and $f^!$

These are some of the key functors in Grothendieck *6 functor formalism* where the adjoint pair Hom and $\otimes$ is missing. These functor's are more naturally introduced in a cohomological setting and will be present in chapter 6.

1.3.1 Pushforward and Pullback

The easiest of these is the pushforward. If $U \subseteq V \subseteq Y$ and $f : X \to Y$, then $f^{-1}(U) \subseteq f^{-1}(V)$. Hence, f^{-1} can be thought of as a contravariant functor preserving $\subseteq$. In fact, more is true: f^{-1} also preserves the properties in proposition 1.1.9, and hence works on sheaves as well:

Definition 1.3.1: Pushforward Of (Pre)sheaf

Let $f : X \to Y$ be a continuous map and $\mathscr{F} \in \mathbf{PSh}(X)$. Then the *pushforward* or *direct image* of $\mathscr{F}$ through f is defined on every open set $U \subseteq Y$ to be

$$f_*\mathscr{F}(U) = \mathscr{F}(f^{-1}(U))$$

and induces a (per)sheaf on Y. The pushforward defines a [covariant] functor $f_* : \mathbf{Sh}(X) \to \mathbf{Sh}(Y)$ sending:

$$f_* : \mathscr{F} \to f_*(\mathscr{F})$$

If working with a particular category, for example **Set**, we can write $f_* : \mathbf{Set}_X \to \mathbf{Set}_Y$.

The term "pushforward" comes from the fact that we are taking a sheaf on X and defining a sheaf on Y through the map $f : X \to Y$. The pullback shall do the same thing in reverse.

It is perhaps unsurprising that f^{-1} preserves pre-sheaves, however it is a little less obvious that it preserves sheaves as well; it is worth thinking this through. To me, this shows that sheaves are a very natural concept, as the usually topological function f^{-1} naturally preserves them. This also motivates the Grothendieck topology that will be introduced much later. We have already seen an example of a pushforward: the skyscraper sheaf is induced that pushing forward along the inclusion map $\iota_p : \{p\} \to X$ the constant sheaf $\underline{S}$.

Proposition 1.3.2: Pushforward Induces Map On Stalks

Let $f : X \to Y$ be a continuous map and $\mathscr{F} \in \mathbf{Sh}(X)$. Then for each $\pi(p) = q$, there exists a map

$$(f_*\mathscr{F})_q \to \mathscr{F}_p$$

between the stalks of $f_*\mathscr{F}$ and $\mathscr{F}$.

***Proof*:**
exercise. This can be verified using the equivalence class definition or the universal property. Doing it using equivalence classes, define the map

$$[s]_q \mapsto [f^{-1}(s)]_p$$

Proposition 1.3.3: Sheaf Morphisms Determined by a Basis

Let X be a topological space and $\mathcal{B}$ a basis for the topology. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. Suppose that for every basis set $B \in \mathcal{B}$, we have a morphism $\varphi_B : \mathcal{F}(B) \to \mathcal{G}(B)$ such that for any inclusion of basis sets $V \subseteq U$ (where $U, V \in \mathcal{B}$), the maps commute with restriction:

$$\varphi_V(s|_V) = (\varphi_U(s))|_V$$

Then there exists a unique sheaf morphism $\Phi : \mathcal{F} \to \mathcal{G}$ extending this collection.

***Proof*:**
Let $U \subseteq X$ be an arbitrary open set. Since $\mathcal{B}$ is a basis, there exists a cover $U = \bigcup_{i \in I} B_i$ where $B_i \in \mathcal{B}$. We wish to construct a map $\Phi_U : \mathcal{F}(U) \to \mathcal{G}(U)$.

Take a section $s \in \mathcal{F}(U)$. We restrict this section to the basis elements to get $s_i = s|_{B_i} \in \mathcal{F}(B_i)$. Using our hypothesis, we can map these to the target sheaf:

$$t_i := \varphi_{B_i}(s_i) \in \mathcal{G}(B_i)$$

We now claim that the collection $\{t_i\}$ glues together to form a global section $t \in \mathcal{G}(U)$. By the sheaf axioms for $\mathcal{G}$, it suffices to show that t_i and t_j agree on the overlap $B_i \cap B_j$.

Unlike the specific case of affine schemes, the intersection $B_i \cap B_j$ might not be a single basis element. However, by the definition of a topology basis, we can cover the intersection by smaller basis elements:

$$B_i \cap B_j = \bigcup_k V_k \quad \text{where } V_k \in \mathcal{B}$$

Since $V_k \subseteq B_i$, our compatibility hypothesis tells us that restricting the mapped section is the same as mapping the restricted section:

$$t_i|_{V_k} = (\varphi_{B_i}(s|_{B_i}))|_{V_k} = \varphi_{V_k}(s|_{V_k})$$

By the exact same logic for j, we have $t_j|_{V_k} = \varphi_{V_k}(s|_{V_k})$. Thus, t_i and t_j agree on every V_k covering the intersection. Since $\mathcal{G}$ is a sheaf (specifically, using the identity axiom locally), this implies t_i and t_j agree on the entire intersection $B_i \cap B_j$.

Since the sections are compatible, there exists a unique $t \in \mathcal{G}(U)$ such that $t|_{B_i} = t_i$. We define $\Phi_U(s) = t$.

To ensure this is well-defined, one must check that choosing a different basis cover for U yields the same result. This follows from standard refinement arguments (taking the union of two covers and applying the uniqueness of gluing on the common refinement). Uniqueness of Φ is guaranteed because two sheaf morphisms agreeing on a basis must agree everywhere.

Corollary 1.3.4: Sheaf Isomorphism Determined By A Basis

Let X be a topological space and $\mathcal{B}$ a basis for the topology. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. Suppose that for every basis set $B \in \mathcal{B}$, we have an isomorphism $\varphi_B : \mathcal{F}(B) \to \mathcal{G}(B)$ such that for any inclusion of basis sets $V \subseteq U$ (where $U, V \in \mathcal{B}$), the maps commute with restriction:

$$\varphi_V(s|_V) = (\varphi_U(s))|_V$$

Then there exists a unique sheaf isomorphism $\Phi : \mathcal{F} \to \mathcal{G}$ extending this collection.

***Proof*:**
As each φ_B is an isomorphism, they each have an inverse. It is easy to show that they satisfy the same restriction compatibility condition, hence by proposition 1.3.3 there exists a global inverse morphism, and hence φ is an isomorphism.

Pullback

So far, most of the discussion of sheaves where with the assumption of a topological X. In example 1.1.12, there were examples of sheaves being pushed forward (Definition 1.3.1) via a continuous map $f : X \to Y$. Namely the sheaf $\mathscr{F}$ maps to the sheaf $f_*\mathscr{F}$ on Y defined via:

$$\pi_*\mathscr{F}(U) = \mathscr{F}\left(f^{-1}(U)\right)$$

There is an adjoint, the *pullback*, called the inverse image sheaf. It is called a pullback as it takes a sheaf $\mathscr{F}$ and Y and defines a sheaf $f^*(\mathscr{F})$ on X. It is not as clear as the case for the push-forward as the image of an open set need not be open, hence the existence is more subtle (similar to the existence of cokernels for groups). The adjoint will be important not only for having more than one perspective on sheaves, but for definition morphisms of schemes.

Definition 1.3.5: Inverse Image Sheaf

Let $f : X \to Y$ be a function function. and $\mathscr{F}, \mathscr{G}$ be sheaves on X and Y respectively. Then the *inverse image sheaf* or *pullback* $f^*\mathscr{G}$ on X is the sheaf associated to the presheaf:

$$f^*_{\text{pre}}\mathscr{G}(U) = \varinjlim_{V \supseteq f(U)} \mathscr{G}(V)$$

or more concisely

$$f^*\mathscr{G} = (f^*_{\text{pre}}\mathscr{G})^{\text{sh}}$$

Giving us a map $f^* : \mathbf{Sh}(Y) \to \mathbf{Sh}(X)$.

1.3.6 On Vakil's Notation Vakil uses the f^{-1} notation intend of f^* as he wants to reserve the f^* notation for pulling back $\mathcal{O}_X$-modules:

$$f^*\mathcal{G} := f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$$

This is because we still want pullback as defined in this section to apply to $\mathcal{O}_X$-modules. We shall instead keep track of which category we are performing the operations.

The construction of the sheaf is done in the usual quotient way: the elements of $f^*_{\text{pre}}\mathscr{F}(U)$ are equivalent classes $(s, U), (t, V)$ where $(s, U) \sim (t, V)$ if there exists a $W \subseteq U \cap V$ where $f(U) \subseteq W$ and the restriction of s, t are equal on W.

The sheafification is crucial:

Example 1.3.7: Presheaf Inverse Image

Let $f : Y \to X$ where $Y = X \coprod X$ and f is the natural projection map and a sheaf $\mathscr{G}$ defined on X. Then by definition, we would get $f^*_{\text{pre}}\mathscr{G}(U) = \mathscr{G}(U)$, however the sheafification will get:

$$f^*\mathscr{G}(U) = \mathscr{G}(U) \times \mathscr{G}(U)$$

showing the two are in general not equal.

Proposition 1.3.8: push-forward Sheaf Adjoint

Let $f : X \to Y$ be a continuous functions and $\mathscr{F}, \mathscr{G}$ be sheaves on X and Y respectfully. Then there is a bijection:

$$\text{Mor}_X(f^*\mathscr{G}, \mathscr{G}) \leftrightarrow \text{Mor}_Y(\mathscr{G}, f_*\mathscr{F})$$

That is, they are adjoints

***Proof*:**
exercise.

One advantage of the inverse image sheaf is the computation of stalk

Proposition 1.3.9: Stalks Via Inverse Image Sheaf

Let $f : X \to Y$ be a continuous function with $\mathscr{G}$ a sheaf on Y. Then:

$$(f^*\mathscr{G})_p = \mathscr{F}_{f(p)}$$

***Proof*:**
exercise, follows more naturally via the definition.

In fact, the adjoint of this map is the *skyscraper sheaf*, namely given $f : \{p\} \to X$ where $p \in X$, then $f^*\mathscr{G} = \mathscr{G}_p$. Conversely, if we define the sheaf $\mathscr{F}(\{p\}) = S$, then $f_*\mathscr{F}(U)$ is the skyscraper sheaf.

will get back to this when it is more needed

1.3.2 Proper Direct and Inverse Image

While the standard pushforward f_* and pullback f^* are sufficient for many geometric operations, they lack certain "clean" properties regarding the supports of sections:

Example 1.3.10: Smeared Section

Consider $X = \mathbb{R} - \{0\}$ with the euclidean topology and take the inclusion $j : X \hookrightarrow \mathbb{R}$. Then let $\mathscr{F}$ on X be the $\mathbb{Z}$-constant sheaf. Then what is $j_*\mathscr{F}$ at 0? Consider that on any open neighbourhood $(-\epsilon, \epsilon)$, the pushforward is $(-\epsilon, 0) \cup (0, \epsilon)$. We can choose two arbitrary choices of constant functions. Hence:

$$j_*(\mathscr{F})_0 \cong \mathbb{Z} \oplus \mathbb{Z}$$

it is *not* 0!

For those who already encountered the Zariski topology on $\mathbb{A}^1_k$ (k infinite), consider $X = \mathbb{A}^1_k \setminus \{0\}$ and the inclusion $j : X \hookrightarrow \mathbb{A}^1_k$. Then as the open sets of $\mathbb{A}^1_k$ are cofinite, hence any open subset U containing 0 is cofinite; it is easy to show that $X \cap U$ is always *connected*. Hence, if we put the $\mathbb{Z}$-constant sheaf on X, $j_*(\mathscr{F})_0 = \mathbb{Z}$. The functor $j_!$ that will be defined soon will be characterized by $j_!(\mathscr{F})_x = 0$ if $x \notin X$.

To address this, Grothendieck introduced the "shriek" functors (denoted with an exclamation mark in the subscript, ex. $j_!$, or in Hartshorne with a tick in the subscript, ex. $j_{\prime}$), which handle supports more strictly. These functors behave differently depending on whether the map is an open immersion or a closed immersion.

Extension by Zero (Open Immersion)

Let $j : U \hookrightarrow X$ be an inclusion of an open set. The standard pushforward j_* "smears" the information from U up to the boundary in X. Often, we wish to extend a sheaf from U to X such that it is exactly zero outside of U.

Definition 1.3.11: Extension by Zero

Let $j : U \to X$ be an open immersion and $\mathscr{F}$ a sheaf on U. The *extension by zero*, denoted $j_!\mathscr{F}$, is the sheaf on X associated to the presheaf:

$$j_!(V) = \begin{cases} \mathscr{F}(V) & \text{if } V \subseteq U \\ 0 & \text{otherwise} \end{cases}$$

This functor $j_! : \mathbf{Sh}(U) \to \mathbf{Sh}(X)$ is also called the *proper direct image* or *lower shriek*.

The behavior of this functor is best understood at the level of stalks. Unlike j_*, which may have non-zero stalks on the boundary ∂U, $j_!$ is "clean":

$$(j_!\mathscr{F})_p = \begin{cases} \mathscr{F}_p & \text{if } p \in U \\ 0 & \text{if } p \notin U \end{cases}$$

Just as f^* is the left adjoint to f_*, the extension by zero provides a *left* adjoint to the restriction.

Proposition 1.3.12: Adjunction of Extension by Zero

Let $j : U \to X$ be an open immersion. Then $j_!$ is the left adjoint to the pullback (restriction) j^*. That is, for $\mathscr{F} \in \mathbf{Sh}(U)$ and $\mathscr{G} \in \mathbf{Sh}(X)$:

$$\mathrm{Mor}_X(j_!\mathscr{F}, \mathscr{G}) \cong \mathrm{Mor}_U(\mathscr{F}, j^*\mathscr{G})$$

***Proof*:**
exercise. (Hint: A map from zero is unique, so the morphisms on X are entirely determined by their behavior on the open set U).

Sections with Support (Closed Immersion)

Conversely, let $i : Z \hookrightarrow X$ be a closed immersion. The standard pullback i^* restricts sections to Z, effectively "forgetting" everything outside. The right adjoint to the pushforward i_* is the functor that finds sections supported *only* on Z.

Definition 1.3.13: Exceptional Inverse Image

Let $i : Z \to X$ be a closed immersion. The *exceptional inverse image*, denoted $i^!\mathscr{F}$ (or sometimes $\mathcal{H}^0_Z(\mathscr{F})$), is the functor $i^! : \mathbf{Sh}(X) \to \mathbf{Sh}(Z)$ defined by the subgroup of sections:

$$\Gamma(V, i^!\mathscr{F}) = \{s \in \Gamma(V, i^*\mathscr{F}) \big| \mathrm{supp}(s) \subseteq Z\}$$

It serves as the right adjoint to the pushforward i_*.

1.3.3 Hom and Tensor

here (I think this is a good place for it, to round off the 6-functor formalism.)

Summary of the Six Operations

It is common to group these functors into a "Six Operation" formalism. While defined generally for derived categories, for sheaves on topological spaces, they exhibit a specific symmetry between open and closed maps.

1. **For an Open Immersion j:** The exceptional inverse image collapses to the standard pullback ($j^! \cong j^*$), leaving us with the triple $(j_!, j^*, j_*)$.
2. **For a Closed Immersion i:** The proper direct image collapses to the standard pushforward ($i_! \cong i_*$), leaving us with the triple $(i^*, i_*, i^!)$.

These functors allow for the decomposition of a sheaf $\mathscr{F}$ on X into its parts on an open set U and the closed complement Z via the fundamental exact sequence:

$$0 \to j_!j^*\mathscr{F} \to \mathscr{F} \to i_*i^*\mathscr{F} \to 0$$

Exercise 1.3.1

1. Let $U \subseteq Y$ open and $\iota : U \hookrightarrow Y$ be the inclusion. Show that $\iota^*\mathscr{G} \cong \mathscr{G}|_U$.
2. Show that f^* is an exact functor (naturally on abelian categories, however is just as good to show it for any ${}_R\text{Mod}$)
3. Let $U \subseteq X$ be an open subset of a topological space so that $V = X \setminus U$ is closed. Let $\mathscr{F}$ be a sheaf on V, and $\iota : V \hookrightarrow X$ the inclusion.
   a) Show that $(\iota_*\mathscr{F})_p \cong \mathscr{F}_p$ if $p \in V$ and 0 otherwise.
   b) Let $j_!(\mathscr{F})$ be a sheaf on X given by
   $$W \mapsto \mathscr{F}(W) \qquad W \mapsto 0$$
   where the first happens when $W \subseteq V$, and the second in all other cases. Show that $(j_!\mathscr{F})_p \cong \mathscr{F}_p$ if $p \in U$ and 0 otherwise. .
   c) conclude there is a short exact sequence $0 \to j_!(\mathscr{F}|_U) \to \mathscr{F} \to \iota_*(\mathscr{F}|_V) \to 0$.

1.4 Ringed Space and Locally Ringed Space

Having defined the morphisms in $\mathbf{Sh}(X)$, an important class of sheaves to distinguish are sheaves over rings, as one of the ultimate goals in this Part of the textbook is the generalize the Nullstellensatz to arbitrary (commutative) rings:

Theorem 1.4.1: (Commutative) Ringed Objects in Sheaf

let $\mathbf{Sh}_{\mathbf{C}}(X)$ be the category of sheaves over a category $\mathbf{C}$ that has finite products and a terminal object. Then commutative ringed object exists, and the subcategory $\mathcal{R} \subseteq \mathbf{Sh}_{\mathbf{C}}(X)$ of commutative ringed object is equivalent to the category of sheaves $\mathbf{Sh}_{\mathscr{R}(C)}(X)$ on the *commutative ringed objects* of $\mathbf{C}$.

***Proof*:**
This is a lot of diagram chasing. With some of the proofs later in this section, this result can be proved more easily on the level of stalks.

We shall almost always work in the case of $\mathbf{C} = \mathbf{Set}$ so that $\mathscr{R}(\mathbf{C}) = \mathbf{Ring}$:

Definition 1.4.2: Ringed Space And Structure Sheaf

Let $\mathbf{Sh}_{\mathbf{Ring}}(X)$ be the category of sheaves over $\mathbf{Ring}$ on X. Then given a sheaf of [commutative] rings over X, denote it $\mathcal{O}_X$, the tuple $(X, \mathcal{O}_X)$ is called a *ringed space*. The sheaf $\mathcal{O}_X$ is called the *structure sheaf*, and the sections of the structure sheaf $\mathcal{O}_X$ over an open subset U are called *[regular] functions on U*.

If it is unambiguous, the $\mathcal{O}_X$ is dropped and X is simply called the ringed space.

1.4.3 Warning Though these are called functions, they are elements of an arbitrary [commutative] ring. The reader should have regular function from [57, chapter 22.3] in mind.

Before defining morphisms formally, it is useful to understand why the sheaf morphism appears to go in the "opposite" direction. Geometrically, if we have a continuous map $f : X \to Y$ and a "function" φ on an open set $V \subseteq Y$ (i.e., $\varphi \in \mathcal{O}_Y(V)$), we naturally obtain a function on X by *pulling back* φ via composition: $\varphi \circ f$. The domain of this new function is $f^{-1}(V)$.

In the language of sheaves, this operation assigns to a section in $\mathcal{O}_Y(V)$ a section in $\mathcal{O}_X(f^{-1}(V))$. Recall that by definition of the pushforward sheaf, $(f_*\mathcal{O}_X)(V) := \mathcal{O}_X(f^{-1}(V))$. Therefore, the geometric notion of pulling back functions induces an algebraic map $\mathcal{O}_Y(V) \to (f_*\mathcal{O}_X)(V)$ for every open set V, or globally, a sheaf morphism $\mathcal{O}_Y \to f_*\mathcal{O}_X$.

Definition 1.4.4: Ringed Space Morphism

A map $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_X)$ is a *ringed space morphism* if $f : X \to Y$ is a continuous map and $f^\# : \mathcal{O}_Y \to f_*(\mathcal{O}_X)$ is a sheaf morphism.

It is easy to see that these form a category and shall often be denoted **RingedSp**.

Example 1.4.5: Ringed Space

1. The simplest and prototypical ringed space is again the sheaf of [real/complex] continuous/smooth/holomorphic functions on a manifold/variety with component-wise addition and multiplication.
2. Schemes in the next part shall form a large class of ringed spaces
3. vector fields on a manifold do not form a ringed space as they are not closed under multiplication (they are closed under the lie bracket operation). However, they shall form a $\mathcal{O}_X$-module over manifold, as shall be explored in chapter 5.

An important sheaf that shall often be used is the sheaf of units:

Definition 1.4.6: Unit Sheaf

Let $(X, \mathcal{O}_X)$ be a ringed space. Then $\mathcal{O}_X^*$ is the sheaf given by the fibered product:

$$\begin{tikzcd} \mathcal{O}_X^* \arrow[r] \arrow[d] & \{1\} \arrow[d] \\ \mathcal{O}_X \times \mathcal{O}_X \arrow[r, "\cdot"] & \mathcal{O}_X \end{tikzcd}$$

We shall almost never refer to the categories interpretations of the sheaf of rings in this Part of the book (it shall be important in ref:HERE), however ringed spaces are the central object of study in algebraic geometry.

As elements the sections in $\mathcal{O}_X(U)$ are called functions, there should be a notion of evaluation. When working over coordinate rings, evaluation of f at a point a is equivalent to modding f by the (maximal) ideal $(x - a)$. This works over coordinate rings since by the Nullstellensatz the maximal ideals correspond to points. For a general (commutative) ring, there can be many more pathological behaviours, namely it is not clear how to associate to each point $x \in X$ a maximal ideal $\mathfrak{m} \subseteq R$.

The first natural place to consider is the behaviour at the stalk, namely each point x is associated to the ring $\mathcal{O}_{X,x}$. Is there a natural choice of ideal in this ring? In this case, a natural choice would mean there is only *one* choice of ideal, in which case any $f \in \mathcal{O}_X(U)$ where $x \in U$ maps into $\mathcal{O}_{X,x}$ and can be modded by this maximal ideal. In general the answer is no:

Example 1.4.7: Stalk that is not a Local Ring

Take $X = \{p, q\}$ with the topology given by $\mathcal{T}_X = \{\emptyset, \{p\}, X\}$. Let:

$$\mathcal{O}_X(\emptyset) = \mathbf{0} \qquad \mathcal{O}_X(\{p\}) = k \qquad \mathcal{O}_X(X) = k \times k$$

where k is a field with the only non-trivial restriction map being

$$\mathcal{O}_X(X) \to \mathcal{O}_X(\{p\}) \qquad (a, b) \mapsto a$$

Then the stalk at p and q are:

$$\mathcal{O}_{X,p} = k \qquad \text{and} \qquad \mathcal{O}_{X,q} = k \times k$$

showing that the stalk at q has more than one maximal ideal (namely $k \times \{0\}$ and $\{0\} \times k$).

When working over coordinate rings of varieties, all stalks will be local rings (exercise 1.4.1(1.4.1 (1))). In fact, the same holds for (smooth, complex, topological) manifolds [53, section 2.1]. This motivate focusing on ringed spaces whose stalks are local rings:

Definition 1.4.8: Locally Ringed Space

Let X be a ringed space. Then X is a *locally ringed space* if its stalks are local rings.

Definition 1.4.9: Evaluation on Locally Ringed Space

Let $(X, \mathcal{O}_X)$ be a locally ringed space. Then as every stalk is a local ring, it has a unique maximal ideal, hence a unique field

$$\mathcal{O}_{X,x}/\mathfrak{m}_x = \kappa(x)$$

called the *residue field*. Define *evaluation* of an element $f \in \mathcal{O}_X(U)$ where $x \in U$ by defining:

$$f(x) := \overline{f} \bmod \mathfrak{m}_x$$

where $\overline{f}$ is the image of f in $\mathcal{O}_X(U) \to \mathcal{O}_{X,x}$.

In general, ringed space morphisms *do not* preserve the local ring structure of the stalks (an example is worked through at the beginning of chapter 3, see example 3.1.1). This condition must be imposed:

Definition 1.4.10: Locally Ringed Space Morphism

Let X, Y be locally ringed spaces. Then a morphism of locally ringed spaces $\left(\pi, \pi^{\#}\right) : X \to Y$ is a morphism of ringed spaces such that the induced map of stalks $\pi^{\sharp} : \mathcal{O}_{Y,q} \to \pi_* \mathcal{O}_{X,p}$ (where $\pi(p) = q$) are local ring homomorphisms (the pre-image of the maximal ideal is contained in the maximal ideal).

It can be checked that locally ringed spaces with locally ringed morphisms form a category which shall often be denoted **LocRingedSp** or **LRS**.

Example 1.4.11: Manifolds and Locally ringed sheaves

The two basic examples are varieties and manifolds. Abstracting, topological spaces with continuous function form locally ringed spaces. Complex analytic space (which generalize varieties by taking the vanishing set of holomorphic functions) also form locally ringed spaces. p-adic spaces (central to number theory and modern representation theory) form locally ringed spaces. More generally, Algebraic Spaces and Stacks (defined in Part ref:HERE) form locally ringed spaces.

Overall, locally ringed spaces are the "right" over-arching category when considering the duality between algebra and geometry.

Definition 1.4.12: Isomorphism Of Ringed Spaces

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be two ringed spaces. Then an isomorphism or ringed space is a map $\pi : X \to Y$ where:

1. π is a homeomorphism
2. There is an isomorphism of sheaves Between $\mathcal{O}_X$ and $\mathcal{O}_Y$ where two two sheaves are given via the pushforward π_* (i.e. $\mathcal{O}_Y \cong \pi_* \mathcal{O}_X$ on Y)[a]

[a] By proposition 1.2.24, it is enough to check this isomorphism on stalks

Exercise 1.4.1

1. Show that the stalks of coordinate rings at maximal ideals are all local rings

1.5 *Diffeology

(would love to add a word here)

Pontificating about Points

(I think it is worth writing here my thoughts on what really constitutes a point. Points are fundamental in geometry, you can say there are the necessary indivisible object, and Treating points as secial collections of functions *I think* makes sense because of the result on locally compact spaces withe the upsisde down pi isomorphism, and the more elementary result on compact groups)

(Also, This may veer into "functor of points" as well as "Yoneda's lemma" as well as Distributions as well as Mitchell's theorem were that any abelian category can be seen as an R-module, and as R-modules have elements, we get something reminiscent to recovering points)

2

Schemes I: Basics

Recall the following correspondence:

$$\left\{\text{Affine Varieties}\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\begin{array}{c}\text{Finitely Generated,}\\ \text{Reduced rings}\\ \text{over algebraically closed fields}\end{array}\right\} \tag{2.1}$$

This setups an algebraic-geometric correspondence, where the algebraic object can be thought of as functions on the geometric object, or conversely that the algebraic objects can induce a new geometric space with a notion of "points" (or more generally local information).

finish this above!!

The question naturally arises then if this duality can be extended to general commutative rings. In particular, does there exist some topological space such that:

$$\left\{\begin{array}{c}\\ ???\\ \\ \end{array}\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\begin{array}{c}\\ \text{Commutative Ring}\\ \\ \end{array}\right\}$$

The necessity of commutativity stems from the need of prime ideals being well-defined[1]. For a more geometric motivation, smooth/holomorphic functions over a real/complex manifold $\mathbb{C}^\infty(M)/\mathcal{H}(M)$ form a commutative ring since they map into $\mathbb{R}$ or $\mathbb{C}$, and hence we copy this behaviour.

Why do this? This abstraction is not simply for abstractions-sake. It is because the dictionary between algebra and geometry that has been built up to is incredibly useful, and will fix some of the problems of algebraic varieties, for example:

[1]The theory of "non-commutative geometry" using non-commutative rings can be explored in *Spectral Theory* in a class on functional analysis (see [54])

1. There are many algebras over more general rings that have geometric interpretations. For a rather trivial but illustrative example, the coordinate ring of a line whose points are in some arbitrary (commutative) ring R (the coordinate ring being $R[x]$) currently does not have a good representation since the Nullstellensatz requires k to be an algebraically closed fields. Another example that may at first seem "un-geometric" but turns out to have many strong geometric properties are rings such as $\mathbb{Z}$, $\mathbb{Z}[i]$ (the Gaussian integers), and more generally *algebraic number rings* (see [8]). Mathematicians have found that these corresponds naturally to looking for integer solutions (or integral solutions when working over an algebraic ring) of a curves (see [8]). In particular, the element of $\mathbb{Z}[i] = \mathbb{Z}[x]/(x^2+1)$ should be functions that "evaluate" at points that would find information mod $\mathfrak{p}$ to solutions of x^2+1.

2. As we saw in front-matter, localization gives us powerful tools to explore the behavior of a variety at a point. In the theoretical framework given by the Nullstellensatz, we cannot consider such rings, for example, if we take the coordinate ring of an affine line, $k[x]$, takes the closure $k(x)$, or study $k[x]_{(x)}$, we loose finite generation. However, as we saw these certainly still have interpretations as functions on a geometric space (the former being the rational functions on a variety, the latter being the "germs"[2] around $0 \in \mathbb{A}^1_k$).

3. Varieties loose multiplicity information. Take for example two varieties over $\mathbb{C}$ defined by the equations
$$y = x^2 \qquad \text{and} \qquad y = 0$$
then if we take their intersection, we get a point $(0,0)$. The corresponding affine variety is given by taking the union of quotient of the two coordinate rings, namely:
$$\frac{k[x,y]}{(y, x^2 - y)} = \frac{k[x]}{(x^2)}$$
Usually, we would omit the x^2 term and just use $k[x]/(x)$ for the coordinate ring, however this eliminate the multiplicity information, namely that the variety $y = x^2$ "doubly intersects" the variety $y = 0$. The ring $k[x]/(x^2)$ keeps track that we have a "double-point", while $k[x]/(x)$ does not. This is a achieved by the fact that the ring has nilpotent elements.

4. (If you know differential geometry) A geometrical object that has many algebraic parallels is the vector bundle. Most vector bundles are very algebraic, however many are not projective (cannot be embedded into $\mathbb{P}^n_k$, or even $\mathbb{P}^n_R$ for some ring R). We thus need need to generalize better our notion of projective space. This shall lead to different sheaves on spectra and will be studied in chapter 5.

This build-up will introduce to us the *affine scheme*. Many familiar geometric objects shall look like affine schemes, but not all familiar geometric objects shall look like them (notably, projective space). We will need to generalize a bit more and define a *scheme* which is locally an affine scheme. This shall complete the liaison that algebraic geometry promises between algebra and geometry.

We shall then focus on how what properties of commutative rings translate to geometric properties on a scheme, and conversely when imposing desirable geometric properties how does this translate as algebraic properties. Properties such as being an integral domain, PID, UFD, Noetherian, reduced, and so forth will have geometric interpretations, and conversely notions like dimension, smoothness, open/closed immersions, and so forth shall have algebraic counter-parts.

[2]Somewhat surprisingly, germs are also a "purely algebraic" notion given by nilpotent elements and localization, and hence appear as schemes

2.1 Defining Spec

Recall that the maximal ideals of $k[x_1, x_2, \cdots, x_n]/V$ correspond to points in a variety V (theorem 0.1.12). It may then be natural to say that the collection of maximal ideals should be considered the "points" of an arbitrary ring R. These collection were called the spectrum (plural spectra). For such an association to work, there would need to be a functor translating maps between rings to maps between the collection of maximal ideals. In particular, we would want that the functor translating a ring homomorphism $\varphi : R \to S$ to the associated continuous function mapping the maximal ideal $\mathfrak{m} \subseteq S$ to a maximal ideal $\varphi^{-1}(\mathfrak{m})$ (recall that regular function corresponds contra-variantly to k-algebra homomorphisms), However, the pre-image of a maximal ideal need not be maximal breaking functoriality (ex. take $\mathbb{Z} \hookrightarrow \mathbb{Q}$. Then (0) is maximal in $\mathbb{Q}$, but certainly not maximal in $\mathbb{Z}$).

When the theory was originally being developed within spectral theory from functional analysis (see [54]), the spaces were compact Hausdorff spaces, in which case it would have sufficed to work with the maximal ideals as within this setting the pre-image of maximal ideals (of the ring of continuous functions) is usually maximal. However, as demonstrated in the example above this doesn't hold for general rings.

Fortunately, if we loosen our requirement a bit and consider prime ideals, then it is indeed the case that $\varphi^{-1}(P)$ is prime, making it the more natural choice of "points" for functorial purposes, even though it shall produce some more "complicated" points (see example 2.1.3). Working with primes also brings the added benefit of incurring a lot more structure which shall turn out to be very insightful and make the addition of primes not only necessary for the definition to work, but to explain a lot of phenomena[3].

Definition 2.1.1: Spectrum

Let R be an [arbitrary] commutative ring. Then $\operatorname{Spec} R$ is the collection of prime ideals of R. Let $\operatorname{mSpec}(R)$ represent the collection of maximal ideals of R.

We call an element of $\operatorname{Spec} R$ a *point*, and we call an element of R a *regular functions*

In the special case where R is a coordinate ring $k[x_1, ..., x_n]/I(V)$, $\operatorname{Spec} R$ corresponds to points of an affine variety as well as "extra" points associated to primes that are not maximal (ex. $(y - x^2) \subseteq k[x, y]$) along with the zero ideal which is prime in integral coordinate rings; the intuition behind having these extra points shall be elucidated throughout this section.

To avoid confusion between points of $\operatorname{Spec} R$ and ideals of R, we shall write points in square brackets, for example $[\mathfrak{p}] \in \operatorname{Spec} R$ for $\mathfrak{p} \subseteq R$. There is some merit in writing it in the same notation as the equivalence relation notation, as points on in the topology are defined up to the radical of an ideal, for example for $(x^2) \subseteq \mathbb{C}[x]$, $V(\{(x^2)\}) = [(x)]$ (the $V(-)$ function will be defined in section 2.1.1, or see [20]). As we shall see, once a structure sheaf is introduced, the sets $\operatorname{Spec} \mathbb{C}[x]/(x^2)$ and $\operatorname{Spec} \mathbb{C}[x]/(x)$ which each have one element shall not be have the same structure sheaf added to them. We shall see that in most cases, we can interpret the maximal ideals as points and can write an ideal of the form $[(x_1 - a_1, ..., x_n - a_n)]$ as $(a_1, ..., a_n)$ without ambiguity, however this is not always the case (we shall define non-trivial schemes that shall have no such point[4]). In example 2.1.3, we shall

[3]For example, recall that if k was not algebraically closed, there exists regular functions that are not polynomials. Another is the distinct characterization of divisors in number theory which unifies describing the same cohomological properties of schemes

[4]In particular, there shall be no closed points

see that we can write any point associated to a maximal ideal from a coordinate ring in this way.

Another advantage of working with spectra is that there is no need to reference an ambient space for the algebraic subsets, the information is "within" the space! In particular, every quasi-projective variety in [20] was a subset of $\mathbb{P}^n_k$, while a spectra need not be a subset of $\mathbb{P}^n_k$! In fact, there shall be schemes that are not embed-able in projective space, though they are few and far between[5]. We shall later see under which condition we can embed a scheme into $\mathbb{P}^n_R$ for a ring R, and naturally this shall always be be the case for schemes associated to quasi-projective varieties (covered in section 5.3.3).

When working with coordinate rings R of a variety V, we may consider R to be the set of functions on V, since we may have a notion of "evaluation" for each element of R. We can do the same for a general commutative ring R, motivating the notation f for a general element of a commutative ring R ($f \in R$) and for calling such elements *regular functions*. Notice that if $p(x) \in k[x]$, then we may find the value of the evaluation $p(a)$ for some $a \in k$ by doing

$$\frac{p(x)}{(x-a)} = p(a) \bmod (x-a)$$

To represent the above more formally, every element $f \in k[x]$ (for some algebraically closed field k) is associated to an element of $\mathrm{Hom}_{\mathbf{Set}}\left(\mathrm{Spec}\,(k[x]), \coprod_{(x-a)\in \mathrm{Spec}\,(k[x])} k[x]/(x-a)\right)$, in particular:

$$k[x] \to \mathrm{Hom}_{\mathbf{Set}}\left(\mathrm{Spec}\,(k[x]), \coprod_{(x-a)\in \mathrm{Spec}\,(k[x])} \frac{k[x]}{(x-a)}\right)$$

$$f \mapsto ((x-a) \mapsto f \mod(x-a))$$

Stare at the above until it makes sense; this is essentially a result of the Nullstellensatz ([20]). Notably, we can think of a function as holding local information (indeed, the "most local" as it gives point-wise information rather than open-set local, component local, germ local, and so forth) at each point of the spectrum[6]. We extract this point-wise information by evaluating, or in the case of spectra by modding.

Recall that if we take subsets of the classical affine space, we can often define more regular functions[7], for example by taking the (classical) affine line minus the origin, $\mathbb{A}^{\setminus}_k\{0\}$, we have $f(x) = 1/x$ is a well-defined regular function in the classical sense. We shall thus want to focus on all possible functions from $\mathrm{Frac}(R)$[8] that are regular at a point $\mathfrak{p} \in \mathrm{Spec}\, R$; we saw in [10] that this collection is given by localizing at the prime: $R_\mathfrak{p}$ (for $R = k[x]$ with $\mathfrak{p} = (x-a)$ we had $k[x]_{(x-a)}$). This collection is called the *stalk* at the point $\mathfrak{p}$. Notice that the codomain of the hom-set can be thought of as the disjoint union of field of fractions of the germs $k[x]_{(x-a)}$ as

$$\frac{k[x]_{(x-a)}}{\mathfrak{m}_{(x-a)}} \cong \mathrm{Frac}\left(\frac{k[x]}{(x-a)}\right) \cong \mathrm{Frac}(k) = k$$

which show that we recover the same notion of evaluation when working over the localization. This greater generality shall be important when we define the sheaf on $\mathrm{Spec}\, R$ namely since the ring $R_\mathfrak{p}$

[5]We shall see in ref:HERE that all proper scheme are covered by a projective scheme, hence we shall require some "awkward blow-ups" to forcefully construct a scheme that cannot be embedded in projective space

[6]or more generally, at each point of a geometric space, like a manifold

[7]though not always, for example the punctured plane has the same algebraic functions as the plane, see example 2.2.51

[8]Technically, it is required that R be an integral domain as this corresponds to irreducible subsets. Each irreducible subsets will have its own $\mathrm{Frac}(R_i)$ of possible regular functions

shall be a *local ring* that has only a single maximal ideal $\mathfrak{m}_\mathfrak{p}$ making it unambiguous which ideal to mod by (as explored in section 1.4), and so we shall use it for our definition:

Definition 2.1.2: Value Of Regular Function

Let $f \in R$. Then the *value* of f at the point $\mathfrak{p} \in \operatorname{Spec}(R)$ is:

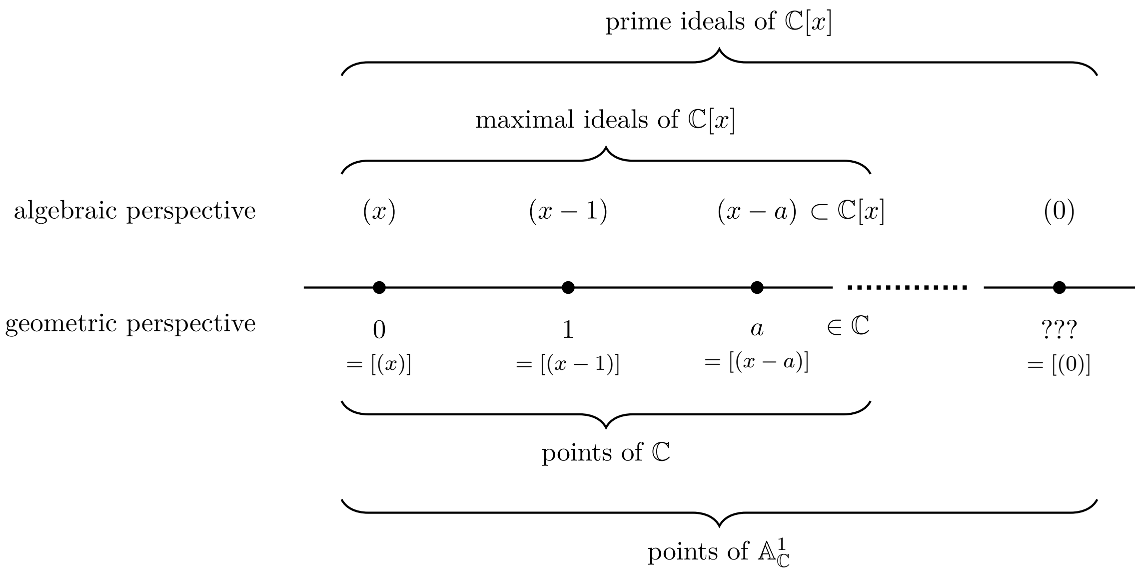
$$f(\mathfrak{p}) := \overline{f} \bmod \mathfrak{m}_\mathfrak{p} \in R_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}$$

Let us now give multiple examples to get some grasp on spec and the functions on Spec

Example 2.1.3: Spectra and [regular] functions

This is a long set of examples, the reader should cover as many as they need to feel comfortable with the notion of spectrum; they may return here later for further elucidation on spectra.

1. Take $\operatorname{Spec}(\mathbb{C}[x])$ which consists of all prime (in fact maximal) ideals $(x-a)$ for $a \in \mathbb{C}$ and the prime ideal (0). This spectrum is denoted $\mathbb{A}^1_\mathbb{C}$; more generally the spectrum of $R[x_1, \ldots, x_n]$ is denoted $\mathbb{A}^n_R$. As $\mathbb{C}[x]$ is a finitely generated reduced ring over an algebraically closed field, the maximal ideals correspond exactly to the classical points of $\mathbb{A}^1_\mathbb{C}$ by the Nullstellensatz, with the exception of the special point (0) called the *generic point* which we shall return to soon. To visualize this set, imagine a real line (to remember that it is a 1-dimensional object over $\mathbb{C}$):



The generic point was located "away" from the line, but that is not in fact accurate. The problem is that it is a single point that is near every other point, while still being a single point. What it means for points to be near shall be defined shortly when the Zariski topology on spectra is introduced, but for now you can imagine that this causes issues when trying to draw such a point on a Euclidean piece of paper. There are many ways to visualize it: some write it as a cloud of points in the corner, others at some edge of the visual. Personally, I find it fruitful to think of it as a "higher-dimensional" point that lives over all other points of $\mathbb{A}^1_{\mathbb{C}}$. This perspective shall be corroborated in section **??**.

Let us evaluate some regular functions $f \in \mathbb{C}[x]$. Let's say $f = x^2 + 2x + 1$ and input $[(x-3)]$. Then the answer is

$$16 \bmod (x-3) = f(3) \bmod (x-3)$$

which can be thought of as our usual answer $f(3)$! Notice that $\mathbb{C}[x]/(x-a) \cong \mathbb{C}$, and hence the codomain when evaluating at each point is the same. We shall see this is not the case soon. We can think of the fact that all the codomains of evaluation coincide as being a special symmetry for varieties over algebraically closed fields (that the evaluation information locally can all be embedded into a single global function-space).

Let us now plug in the generic point (0) into our function f (i.e. the point that isn't in our usual visualization of the complex line). We get:

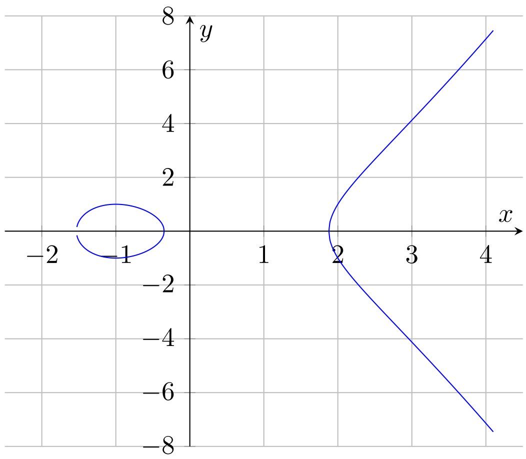
$$f \bmod 0 \in \mathbb{C}[x]$$

that is, it gives back the function itself, and is the only point for which the codomain of f is not isomorphic to $\mathbb{C}$. You can think of (0) as a "1"-[complex]-dimensional point. When we have spectra of higher dimensional (irreducible) schemes, say dimension n, then the generic point will be "n"-dimensional. We can't yet define dimension as we are still working generically with sets; this shall be rectified in section 2.6.

More generally, this entire discussion works just as well over any algebraically closed field $k = \overline{k}$, as the completeness of $\mathbb{C}$ did not factor into the above results.

Often when working with $\overline{k}$-polynomials over several variables, we shall write the maximal ideals as tuples of points, e.g. $[(x-2, y-3)] \in \mathbb{A}^2_k$ will be represented as $(2,3) \in \mathbb{A}^2_k$. We shall often then denote the generic point $[(0)]$ as η to distinguish it from the point 0 which is associated to the ideal (x).

2. Take $y^2 - x^3 + 3x - 1 \in \mathbb{C}[x,y]$ and consider Spec $\frac{\mathbb{C}[x,y]}{(y^2-x^3+3x-1)}$. This spectrum corresponds to the elliptic curve $y^2 = x^3 - 3x + 1$. The spectrum will contain all the maximal ideals of the ring $\frac{\mathbb{C}[x,y]}{(y^2-x^3+3x-1)}$ as well as a generic point η corresponding to $[(0)]$ representing the entire space, and hence can be visualized as:



More generally, all the points of an affine algebraic set correspond to the maximal ideals of a spectrum, and so a subset of the geometry of spectra includes the classical geometry of curves, surfaces, and so forth!

3. Here is a Spectra that has no corresponding algebraic set in the classical sense: $\mathrm{Spec}(\mathbb{Z}) = \{(2),(5),(7),\cdots,(p),...,(0)\}$. In other words, $\mathrm{Spec}(\mathbb{Z}) = \{(p)\big|p \text{ is prime}\} \cup \{0\}$ and $\mathrm{mSpec}(\mathbb{Z}) = \mathrm{Spec}(\mathbb{Z}) \backslash \{(0)\}$.

   Let us evaluate some of the regular function $n \in \mathbb{Z}$ on the points $\mathrm{Spec}\,(\mathbb{Z})$. The regular function 15 at the point $[(7)] \in \mathrm{Spec}\,(\mathbb{Z})$ is $15 \bmod (7) = 1 \bmod (7)$. The regular function 9 evaluated at $[(3)]$ is $0 \bmod (3)$. We may even say it has a double-zero at 3.

   Hence, functions on $\mathrm{Spec}\,(\mathbb{Z})$ return some information about divisibility by primes.

4. For any field k, $\mathrm{Spec}\, k = \mathrm{mSpec}\, k = \{(0)\}$ (or $\mathrm{Spec}\, k = \{\eta\}$). For example, $\mathrm{Spec}\,\mathbb{Z}/3\mathbb{Z} = \{(0)\}$. Thus, fields are like "points" within algebraic geometry. An advantage of adding a structure sheaf will be to remember the original field k: we shall see that as schemes $\mathrm{Spec}\, K \ncong \mathrm{Spec}\, L$ if $K \ncong L$, as their sheaves will differ. Though the geometric picture as spectra is trivial, they shall still hold valuable non-trivial information.

   The evaluation of each function at the single point returns the function modulo (0), namely the function itself as an element of k:

$$3 \bmod 0 = 3 \in k$$

5. Take $\mathrm{Spec}\ \mathbb{R}[x]$. This consists of $[(0)]$, all prime (even maximal) ideals of the form $[(x-a)]$, and all prime (even maximal) ideals that can be represented by an irreducible 2nd degree polynomial $\left[\left(x^2+bx+c\right)\right]$, for example $\left[\left(x^2+1\right)\right]$. Evaluating at any of these points (besides $[(0)]$) will give a field: if evaluating points of the form $(x-a)$ the field will be isomorphic to $\mathbb{R}$, and it evaluating points of the form $\left(x^2+ax+b\right)$, the field will be isomorphic to $\mathbb{C}$. As every irreducible quadratic splits into conjugate pairs of linear terms over $\mathbb{C}$, we can think of $\mathrm{Spec}(\mathbb{R}[x])$ as being a copy of $\mathbb{C}$ folded over itself attaching the conjugate pairs (don't continue reading until this is clear)
   Let us take the function x^3-1 and evaluate it as some point. At the point $(x-2)$ we get $7 \bmod (x-2)$, or in our classical interpretation, $f(2)$. At the point $\left(x^2+1\right)$, we get:

$$x^3 - 1 \equiv -x - 1 \bmod \left(x^2+1\right)$$

   which we can think of as being $-i-1$. Hence, the Spectra of polynomials over fields that are not algebraically closed will behave more like algebraically closed fields as the non-linear maximal primes shall play the role of the required "additional points"! Hence, one piece of the puzzle for the "additional points" in a spectra is that they allow us to think of the spectrum as being "algebraically closed"!

6. Generalizing the above result, show that the points of $\mathrm{Spec}\,\mathbb{Q}[x]$ (minus the generic point 0) are in bijective correspondence with the points of $\overline{\mathbb{Q}}$, the algebraic closure of $\mathbb{Q}$, where the points of $\overline{\mathbb{Q}}$ are glue to their galois-conjugates: $\mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$.

7. Consider $\mathrm{Spec}\ \mathbb{Z}[x]$. Unlike $\mathbb{Q}[x]$, the ring $\mathbb{Z}[x]$ is not a PID and has Krull dimension 2, As it is not a PID, it has prime ideals of the form $(p, f(x))$ where $f(x)$ and $f(x) \bmod p$ is irreducible. This is the first example where the irreducible functions of a polynomial ring

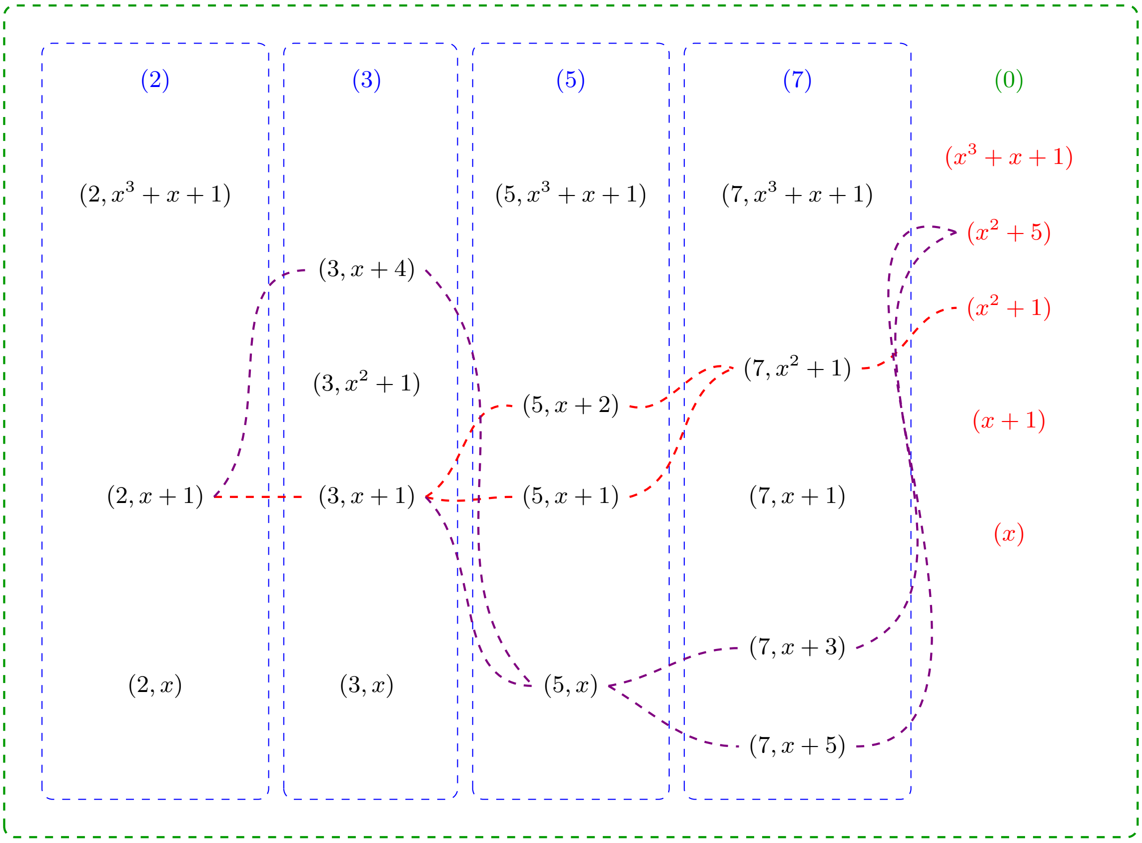
are not in one-to-one correspondence with the points, there are more points than there are irreducible functions[a]! The maximal ideals are of the form

$$(p, f(x))$$

while the prime ideals are of the form:

$$(p) \quad (f(x)) \quad (p, f(x))$$

There are now non-maximal prime ideals, namely the principal ideals $(\ p\ )$ for prime p and principal ideals $(f(x))$ where $f(x)$ is irreducible over $\mathbb{Z}$ (or equivalently by Gauss's lemma over $\mathbb{Q}$). To understand this spectrum better, the following is a part of a visual of Spec $\mathbb{Z}[x]$:



Let us parse this image: the maximal points are all of the form $(p, f(x))$ and they are the black points. The prime (p) are contained in each maximal ideal $(p, f(x))$. They are like a more specialized generic point, where they are not near all the points, but near every point for which the ideal representing the point contains (p). Due to this, each column is surround by dotted blue lines that indicating the generic location of each prime $(\ p\ )$. The entire image is surround by a green dotted line which represents the generic point $[(0)]$ and how it is close to every point. Notice that there are gaps in the columns exactly where the irreducible polynomial $f(x)$ is reducible mod$p$. For example $x^2+1 \bmod 2$ is reducible with solution $x=1$, and so $\left(2, x^2+1\right)$ is not prime, namely given $\mathbb{Z}[x]/\left(2, x^2+1\right) = (\mathbb{Z}/2\mathbb{Z})[\omega]$ we have

$$(\omega+1)(\omega+1)=\omega^2+2\omega+1=1+1=0$$

showing that $\left(2, x^2+1\right)$ is not prime, and hence not a point.
Next, each point $(f(x))$ is also a generic point, however their "shape" is not a simple as the points (p). The simplest case is when a point $(p, g(x))$ contains $f(x)$, namely $f(x)=g(x)$. In the image, we see that $\left[\left(3, x^2+1\right)\right]$ is a point and hence $\left[\left(x^2+1\right)\right]$ is near this point. On the other hand, both $(5, x+2),(5, x+1)$ are contained in $\left(x^2+1\right)$, precisely because x^2+1 factors into a product of those two linear terms mod 5 . Hence $\left[\left(x^2+1\right)\right]$ is near $[(5, x+2)]$ and $[(5, x+1)]$. This closeness is shown via a red-line passing near these points and represents the fact that $\left[\left(x^2+1\right)\right]$ is near each of these points. Essentially, this represents the quadratic residues of x^2+1 !
The same thing has been done for the point $\left[\left(x^2+5\right)\right]$. Some extra points have been added to the image to demonstrate more clearly the branching behavior. Notice that $\left[\left(x^2+5\right)\right]$ and $\left[\left(x^2+1\right)\right]$ "intersect", for example at $[(2, x+1)]$ and at $[(13, x+5)]$. These give us some number-theoretic information about these equations!
The way I like to interpret these "fuzzy" points corresponding to the irreducible polynomials from some intuition given by complex analysis. This is not unreasonable as the spectrum in many ways mirror the behavior of algebraically closed fields and complex analysis is the analysis of functions that are locally infinite polynomials, and indeed these two fields are very closed linked, something we shall cover in ref:HERE.
For now, consider the following: if we have a "germ" around a single point of a holomorphic function (i.e. the information given by its power-series representation around an arbitrarily small neighborhood around the point), we in fact have information about the entire holomorphic function. From this perspective, it doesn't matter at which point we choose to take the power-series representation, it will return equivalent information. Thus, the information does not live at any point point in particular but "generically" at all these points. If we choose to "specialize" and specify which data we want, we shall evaluate further to a single maximal ideal.
Notice that $\mathbb{Z}[x]$ is seen on a two-dimensional grid. We shall later see that $\operatorname{Spec} \mathbb{Z}[x]$ is naturally seen as a 2 -dimensional scheme, while $\operatorname{Spec} \mathbb{Q}[x]$ shall be 1-dimensional. Furthermore, the red-lines and blue dotted lines will be seen as 1-dimensional points, the generic point will be seen as a 2 -dimensional point, while all the maximal ideals will be 0 -dimensional points.

Finally, let us consider evaluation of functions $f \in \mathbb{Z}[x]$ at points. At maximal points, we shall get values in the field $\mathbb{F}_p(\omega)$[b] where ω is a root of the irreducible polynomial. For example, take $\left[\left(7, x^3+x+1\right)\right] \in \operatorname{Spec} \mathbb{Z}[x]$ and $x^2-4 \in \mathbb{Z}[x]$. Then the value will be in $(\mathbb{Z}/7\mathbb{Z})(\omega)$ where ω is an element such that $\omega^3+\omega+1=0$. Then we would get that the value of x^2-4 is $\omega^2+3 \in \mathbb{F}_7(\omega)$.

Another example would be evaluating at the point $\left[\left(3, x^2+1\right)\right]$ the function x^4-1. Working in $\mathbb{Z}[x]/\left(3, x^2+1\right) \cong \mathbb{F}_3[x]/\left(x^2+1\right) \cong \mathbb{F}_9$, we use $x^2 \equiv -1$ to compute:

$$x^4-1 \equiv\left(x^2\right)^2-1 \equiv(-1)^2-1=1-1=0 \pmod{3,\, x^2+1}$$

In other words, $f\left(\left[\left(3, x^2+1\right)\right]\right)=0 \in \mathbb{F}_3(i)$. As we are evaluating at maximal ideals, we shall always get a "number" as the output; in particular, since we can always embed finite fields into an algebraic closure, the output of a function at a maximal point always lies within an algebraically closed field.

Let us now evaluate at point that corresponds to a non-maximal prime ideal, say at 3. The output of each function $f \in \mathbb{Z}[x]$ will then be:

$$\bar{f} \in (\mathbb{Z}/3\mathbb{Z})[x] = \mathbb{F}_3[x]$$

The way I like to think about these output is that we are asking for some of the information stored in f, namely it shall return enough information about f to determine the generic behavior of that function around $[(3)]$. If we wanted the generic behavior around a different generic point, for example $\left[\left(x^2+1\right)\right]$, then we would get:

$$f(i) \in \mathbb{Z}[i]$$

Namely, if we choose any irreducible polynomial, we are choosing which new element of an integral extension we are adding that we then evaluate the function f at. We may then extract further information by modding by different primes. It shall often be the case that evaluating at a generic point corresponds to reducing the information of a function down to a smaller geometric object or to modding information.

8. If you know some number theory, $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]$ is the quintessential example of a ring that has primes that are *not* principle, for example $(2, 1+\sqrt{-5})$ is prime but is provably not principal (see [8]). Evaluating at (2) and $(1+\sqrt{-5})$ will return 0, however the zero set of both of these function's shall be larger than $(2, 1+\sqrt{-5})$; we shall require the vanishing set to contain both of these.

9. The following is another example of a spectrum with a single element. Take $\operatorname{Spec}\left(k[x]/(x^2)\right)$, the spectrum of a ring with nilpotent elements. This ring can be thought of as an extension of k by an element $\epsilon = x \bmod x^2$ that is thought to be "infinitely small". It has the interesting property that $\epsilon^2 = 0$, even though ϵ itself is nonzero. As the spectrum only contains one point[c], namely $[(\epsilon)]$, this can be thought of as a "thicker" point (as shall be explored in greater detail in section 2.2.2). The ring $k[x]/(x^2)$ is called the *dual numbers* and shall be a recurring ring to study the behaviour of nilpotent elements. This ring has the interesting property of having the element $\epsilon \in k[\epsilon]$ which always evaluates to 0 even though it is not the zero element! This gives us our first example of a function[d] that is not defined by what it does at every point, namely because the functions 0 and ϵ have the same outputs.

   More generally, $k[x]/(x^n)$ has a single element in its spectrum. As k and $k[x]/(x^n)$ have the same underlying topological space, what shall distinguish them will be the sheaf structure associated to these two spectra (they shall not be isomorphic as schemes).

10. The above ring only had one point in its spectrum. There are many spectrum's with only two points: a generic point and a non-generic point. A large class of such rings shall be *local rings*. For example $k[x]_{(x-a)}$ will have (0) and $(x-a)$ as the two points of the spectrum. Another example example of a spectrum with only two points is $k[[x]]$ which has (x) and (0) as its two points. As spectra, $k[x]_{(x)}$ and $k[[x]]$ are indistinguishable. What shall distinguish these two spectra shall be the sheaf-structure associated to these two spectra.

11. Let us take a look at the number theoretic equivalent of localization: take $\mathbb{Z}_{(p)}$, the localization of $\mathbb{Z}$ at $\mathbb{Z}\backslash(p)$. This spectrum contains two points, $[(0)]$ and $[(p)]$. In particular, $\operatorname{Spec}\mathbb{Z}_{(p)}$ is not a single point like $\operatorname{Spec}\mathbb{F}_p$, it somehow has some more "fuzz" around it's single point $[(p)]$. This can be thought of as $\operatorname{Spec}\mathbb{Z}_{(p)}$ "remembering" that there is some arbitrarily small neighborhood around the point $[(p)]$ that it must keep track of. Once we introduce a topology on spectra's, we'll see that the closure of $[(p)]$ is itself (it is a closed point), while

the closure of $[(0)]$ is $\{[(0)], [(p)]\}$, motivating the idea that $[(0)]$ "remembers" that there is some more information outside of just the single point! This shall be even more clear once we introduce the notion of dimension: the point $[(p)]$ will be zero-dimensional, while $[(0)]$ will be one-dimensional!

12. Let $R = k[x] \times k[x]$. What is $\operatorname{Spec} R$? Notice that $\operatorname{Spec} R$ does not contain just one generic point, but two! Importantly, $[(0)]$ is not prime as $(1,0)\cdot(0,1) \in (0)$ but $(1,0),(0,1) \notin (0)$.

    In general, the primes of a product ring $A \times B$ are all of the form $\mathfrak{p} \times B$ or $A \times \mathfrak{q}$ where $\mathfrak{p} \in \operatorname{Spec}(A)$ and $\mathfrak{q} \in \operatorname{Spec}(B)$ (exercise). Hence $\operatorname{Spec}(A \times B) \cong \operatorname{Spec}(A) \sqcup \operatorname{Spec}(B)$: the spectrum of a product is the disjoint union of the spectra. In our case, $\operatorname{Spec}(k[x] \times k[x]) \cong \mathbb{A}^1_k \sqcup \mathbb{A}^1_k$, two disjoint copies of the affine line, each with its own generic point.

13. Take $\operatorname{Spec} \mathbb{F}_p[x] = \mathbb{A}^1_{\mathbb{F}_p}$. As it is a Euclidean domain, the points are (0) and $(f(x))$ for all irreducible polynomials over $\mathbb{F}_p$. One of the interesting facets of $\mathbb{A}^1_{\mathbb{F}_p}$ is that its closed points are far richer than the p elements of $\mathbb{F}_p$: for each $n \geq 1$, each irreducible polynomial of degree n over $\mathbb{F}_p$ gives a closed point whose residue field is $\mathbb{F}_{p^n}$. Since there are irreducible polynomials of every degree, $\mathbb{A}^1_{\mathbb{F}_p}$ has infinitely many closed points despite $\mathbb{F}_p$ being finite. This richness in points will allow us to distinguish functions on $\mathbb{F}_p$ by their evaluation: recall that a polynomial is not distinguished by evaluation at all points in $\mathbb{F}_p$ (e.g. $x^p - x$ vanishes everywhere on $\mathbb{F}_p$ but is nonzero), however a polynomial *will* be distinguished by its evaluation at all points in $\operatorname{Spec}(\mathbb{F}_p[x])$, as we shall show using proposition 2.1.4.

14. Take $\operatorname{Spec}(\overline{k}[x,y]) = \mathbb{A}^2_{\overline{k}}$ (e.g. $\overline{k} = \mathbb{C}$). Then the points are:

    $$(0) \qquad (x-a, y-b) \qquad (f(x,y))$$

    where $f(x,y)$ is an irreducible polynomial in $\overline{k}[x,y]$. Visually, all the primes of the form $(x-a, y-b)$ correspond to the usual points in $\mathbb{C}^2$. The points $(f(x,y))$ can be thought of as tracing out their curve in $\mathbb{C}^2$; for example $(y - x^2)$ is the point that is the curve $y = x^2$. Notice how the maximal ideals correspond to "0"-dimensional points, while the ideal $(y - x^2)$ corresponds to a "1"-dimensional point. Similarly, (0) is a "2"-dimensional point. We shall come back to this through Krull dimensions in section 2.6.

    More generally, by the Nullstellensatz the maximal points of $\overline{k}[x_1, \ldots, x_n]$ are the ideals of the form:

    $$(x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n)$$

    The prime ideals of this ring don't lend themselves to a nice classification. Each prime ideal corresponds to an irreducible variety of a certain dimension, and hence the classification of the prime ideals is equivalent to the classification of irreducible varieties, which is a rather difficult task that is carried out in special cases, and more generally leads to the theory of moduli spaces. Some context on this classification in a classical setting was covered in [20].

15. Find the points of $\operatorname{Spec}(k[x_1, \ldots, x_n])$ where k is not algebraically closed, ex. $\mathbb{Q}$; generalize example 5[e].

[a]This sort of exploration on the correspondence between objects that "principal" (or locally principal) and how much they are not will be the focus of section 5.4

[b]Recall that $k[\omega] = k(\omega)$, or re-prove this

[c]Prove that $[(0)]$ is not a prime ideal: $\epsilon \cdot \epsilon = 0 \in (0)$ but $\epsilon \notin (0)$

[d]granted, a generalization of the notion of function

[e]If you solved it, note that $(\sqrt{2}, \sqrt{2})$ is glued to $(-\sqrt{2}, -\sqrt{2})$ but *not* to $(\sqrt{2}, -\sqrt{2})$

2.1.1 Zariski Topology

Having defined the set that represents the space, the next thing to do is give a topology to this set. This topology would have to be a generalization of the Zariski topology defined on $k[x_1, ..., x_n]$ using the definition of regular function that was just introduced:

Definition 2.1.4: Zariski Topology

Let R be a commutative ring and $\text{Spec}(R)$ the collection of prime ideals of R. Then define the *closed sets* for the Zariski topology on $\text{Spec}(R)$ to be:

$$V(S) = \{x \in \text{Spec}(R) \mid f(x) = 0,\ \forall f \in S\} = \{[\mathfrak{p}] \in \text{Spec}(R) \mid S \subseteq \mathfrak{p}\}$$

Take a moment to see how this parallels the condition that all the functions in the coordinate ring must equal zero when evaluated at $\mathfrak{p}$. the notation $f(x) = 0$ will eventually be correctly re-interpreted when we show the sheaf naturally induced by R is a *locally ringed space*, and hence evaluation is well-defined for any ring element. It is an exercise in commutative algebra to check that this is indeed a topology, namely:

Lemma 2.1.5: Zariski-closed Sets Form Topology

Let $\text{Spec}(R)$ be the collection of primes and let:

$$\mathcal{T} = \{V(S) \mid S \subseteq R\}$$

Then:

1. $\mathcal{T}$ contain the empty set and $\text{Spec}(R)$ (these are both the vanishing set of single elements)
2. $\mathcal{T}$ is closed under arbitrary unions and finite intersections, namely:

$$\cap_i V(I_i) = V\left(\sum_i I_i\right)$$

for arbitrary indexing set, and that

$$V(I_1) \cup V(I_2) = V(I_1 I_2)$$

Furthermore, for any radical $I \subseteq R$, $V(I) \cong \text{Spec}\, R/I$ as as topological spaces[a].

[a] and as sheaves, which we'll see later

***Proof*:**
see [20].

Note that the vanishing set $V(-)$ defined for the spectra of rings usually have more points than their counterpart for coordinate rings in classical algebraic geometry as they will contains primes that are not maximal ideals (particularly for rings of "dimension" ≥ 2 as we have briefly discussed), and these primes will be within $V(S)$ for appropriate S.

The reader should check how this topology compares to the topology for classical algebraic sets. For example, as there exist generic points in most spectra we have that this space is no longer T_1 (it shall soon be shown that generic points are not closed). If R is Noetherian, then $\operatorname{Spec} R$ is a Noetherian topological space; see exercise 2.1.1 and section 2.1.3 for more comparisons.

Just like for varieties, we may define the inverse function $I(-)$, that is for [nonempty] $S \subseteq \operatorname{Spec}(R)$,

$$I(S) \subseteq R \qquad I(S) = \bigcap_{[\mathfrak{p}] \in S} \mathfrak{p}$$

Hence $I(S)$ is a radical ideal (recall the intersection of prime ideals is a radical ideal, think $2\mathbb{Z} \cap 3\mathbb{Z} = 6\mathbb{Z}$). Just like in classical algebraic geometry:

Proposition 2.1.6: Properties of $I(-)$ and $V(-)$

1. $I(S)$ is an ideal of R, and furthermore it is always *radical*.
2. $I(-)$ is inclusion-reversing
3. $I(A \cup B) = I(A) \cap I(B)$, and $I(A \cap B) = \sqrt{I(A) + I(B)}$.
4. $I(\overline{S}) = I(S)$
5. $V(I(S)) = \overline{S}$ and $I(V(J)) = \sqrt{J}$.
6. If S is closed, then $V(I(S)) = S$. If I is radical, then $I(V(J)) = J$.
7. $I(-)$ and $V(-)$ are an inclusion-reversing bijection between closed subsets of $\operatorname{Spec}(R)$ and radical ideals of R. This is called a *galois correspondence*.

Hence:

$$\text{closed subset} \subseteq \operatorname{Spec}(R) \qquad \Leftrightarrow \qquad \text{radical ideal of } R$$

***Proof*:**
exercise.

Thus, we get the (surjective) correspondence:

$$\{\text{Radical ideals of } R\} \underset{V(-)}{\overset{I(-)}{\rightleftarrows}} \{\text{closed sets of } \operatorname{Spec}(R)\} \tag{2.2}$$

Notice that prime ideals are radical and hence are part of this association. This correspondence does not extend to all ideals until a sheaf, the structure sheaf, is introduced (see section 2.2.2).

As we see, the Zariski topology on $\operatorname{Spec}(R)$ is a generalization of the Zariski topology on $k[x_1, ..., x_n]$. The closed sets of $\operatorname{Spec}(R)$ will contain the "traditional" points as well as the new points, for example $V(xy, yz)$ will contain all the "traditional" points where $y = 0$ or $x = z = 0$, as well as the generic point that vanish on $V(xy)$ and $V(yz)$ (i.e. $[(y)]$ vanishes on this set). Just like for affine varieties, $\operatorname{Spec}(R)$ is almost never Hausdorff. This is even clearer in $\operatorname{Spec}(R)$, for if there are primes that are not maximal, these are *not* closed points (in a Hausdorff or T_1 space, every point is closed)

Example 2.1.7: Non-closed Points

Take $\mathbb{A}^2_{\mathbb{C}}$ with $[(y^2-x)]$ and $[(x-2,y-4)]$. The point $[(y^2-x)]$ is not closed. Notice that:

$$[(y-4,x-2)] \in V(I([(y^2-x)])) = V(y^2-x) \overset{\text{pr. 2.1.6}}{=} \overline{\{[(y^2-x)]\}}$$

Showing that the point is *not* closed, namely points associated to non-maximal prime ideals are not closed. However, notice the set $V(I([(y^2-x)]))$ is closed by definition. In a moment, the point $[(y^2-x)]$ will be seen as the generic point of the above closed set.

In fact, due to Hilbert's Nullstellensatz, we know that in $\mathbb{A}^n_k$ the maximal ideals correspond to the "traditional points", that is the closed points. From here on, we shall call the "traditional points" (more generally points associated to maximal ideals) the *closed points*, and the "extra" or "bonus" points that were being referred to shall be called the *non-closed points* or *generic points*.

With a topology, we can now link points like $[(x-2,y-4)]$ to points like $[(y-x^2)]$. As these are point, it doesn't make sense to say one point contains another. However we we be able to link these points using closure:

Definition 2.1.8: Specialization And Generization

Let $x, y \in X$ be two points in a topological space. Then we say that x is a *specialization* of y and y a *generization* of x if

$$x \in \overline{\{y\}}$$

Algebraically, $[\mathfrak{q}]$ is a specialization of $[\mathfrak{p}]$ if and only if $\mathfrak{p} \subseteq \mathfrak{q}$, hence $V(\mathfrak{p}) = \overline{\{[\mathfrak{p}]\}}$.

We can now define generic point more precisely:

Definition 2.1.9: Generic Point

Let X be a topological space. Then $p \in X$ is a *generic point* for a closed subset K iff

$$\overline{\{p\}} = K$$

By this definition, closed points are technically generic points associated the closet set of a maximal idea; we shall only use the term generic point for closed sets that are not given by a maximal ideal. With generic points and the Zariski topology defined, the idea that generic points of a closed set (including all of $\text{Spec}(R)$) are close to all the points in the closed subset (it is contained in all the open subsets (or all the open subsets of $V(S) \cap U$).

Let us end with upgrading proposition proposition 2.2.7

Proposition 2.1.10: Nilpotent Spectrum Homeomorphism

Let R be a ring and I an ideal of nilpotent elements. Then

$$\text{Spec}\, R/I \to \text{Spec}\, R$$

is a homeomorphism.

***Proof*:**
The map is a bijective continuous map. Show the inverse is continuous.

Non-closed points shall have their use in maps such as $k[x,y] \to k[x](y)$ as well as in intersection theory where the intersection or "creases" of schemes shall often be generic points. We shall see that there is a "local-ring" associated to both closed and non-closed points alike and their reduction behaviour is the same (though naturally being in different dimensions means that the behaviour may differ in exactitude's).

2.1.2 (Topological) Basis: Distinguished Open Sets

As the topology of $|\mathrm{Spec}\,(R)|$ is generated by the closed sets $V(S)$, and these sets are the intersection of:

$$V(S) = \bigcap_{f \in S} V(f)$$

the sets $V(f)$ generate the topology of $\mathrm{Spec}\,(R)$. We may want to call these sets "distinguished closed sets", however we shall favor their open counterpart due to their simpler behaviour when introducing sheaf theory to spectra[9]:

Definition 2.1.11: Distinguished Open Sets

The sets $D(f)$ are called the *distinguished open sets*, *basic open sets*, or *principal open sets*, and are defined to be

$$D(f) = X_f = |\mathrm{Spec}\,(R)| \setminus V(f) = \{\mathfrak{p} \in \mathrm{Spec}\,(R) \mid f \notin \mathfrak{p}\}$$

that is, it is the collection of prime ideals of R that do not contain f. Any open set U can be written as:

$$U = \mathrm{Spec}\,(R) \setminus V(S) = \mathrm{Spec}\,(R) \setminus \bigcap_{f \in S} V(f) = \bigcup_{f \in S} D(f)$$

Writing it as $D(f)$ may be helpful; Prof. Ravi Vakil calls these "doesn't-vanish sets" which may serve as a helpful mnemonic. These sets are usually not closed (especially if $\mathrm{Spec}\,(R)$ is connected, in which case unless $D(f) = \emptyset$ it cannot be), and hence they are not the vanishing set for any functions in R. However, they are still isomorphic to a spectrum:

[9] There shall be a natural unique open subscheme structure up to isomorphism, but infinitely many natural closed subscheme structures up to isomorphism due to the possibility of nilpotents.

Proposition 2.1.12: Distinguished Open Sets and Localization

Let $D(f) \subseteq \operatorname{Spec}(R)$ be a distinguished open set. Then:

$$D(f) \cong_{\mathbf{Top}} \operatorname{Spec}(R_f)$$

justifying the X_f notation. Furthermore, if $g = f_1 f_2 \cdots f_n$, then

$$\bigcap_{i=1,\cdots,n} (\operatorname{Spec} R)_{f_i} = (\operatorname{Spec} R)_g$$

or as distinguished open sets if $D(f_1), \dots, D(f_n) \subseteq \operatorname{Spec}(R)$, then their intersection is the distinguished open

$$D(f_1 \cdots f_n)$$

***Proof*:**
Recall that $R_f = R[f^{-1}]$, and that $\mathfrak{p} \subseteq R[f^{-1}]$ iff "$f \notin \mathfrak{p}$ and $\mathfrak{p} \in R$". This is exactly the points of $D(f)$; it is left to check that the map is bi-continuous. The second part is left as an exercise.

Corollary 2.1.13: Distinguished Open Sets Form Topological Basis

The collection of distinguished open sets $\{D(f)\}_{f \in R}$ from a topological basis for $\operatorname{Spec} R$ with the Zariski topology

***Proof*:**
Distinguished opens cover $\operatorname{Spec} R$ by definition, and proposition 2.1.12 show they are closed under intersections.

A mistake the reader may make is to believe that each $D(f)$ can be thought of as the spectrum minus some points, but this need not be the case:

Example 2.1.14: Spectrum and Distinguished Open Sets
Take $X = \operatorname{Spec}(k[x, y])$ and take $U = X \setminus \{[(x, y)]\}$, i.e. X minus the origin (which is a closed set as it is a maximal ideal). Show that $U \ncong D(f)$ for any f (hint: notice that (x, y) is not principal). This shows an important distinction between points and functions.

Show further that U can be represented as the union of two distinguished open sets.

Another result we may want to conclude is that each open set $U \subseteq \operatorname{Spec}(R)$ corresponds to $\operatorname{Spec}(R_S)$ for an appropriate S, i.e. $U \cong_{\mathbf{Top}} \operatorname{Spec}(R_S)$. This shall be shown to be *false*; the curious reader may jump to example 2.2.51 for a counter-example.

The following result, while elementary, is used repeatedly in gluing arguments and in the proof of the Affine Communication Lemma. It says that a distinguished open inside a distinguished open is itself distinguished in the ambient spectrum.

Proposition 2.1.15: Distinguished Open of a Distinguished Open

Let $D(f) \subseteq \operatorname{Spec}(R)$ be a distinguished open, so that $D(f) \cong \operatorname{Spec}(R_f)$. Let $g' \in R_f$ and consider the distinguished open $D(g') \subseteq \operatorname{Spec}(R_f)$. Then $D(g')$, viewed as a subset of $\operatorname{Spec}(R)$, is itself a distinguished open of $\operatorname{Spec}(R)$.
More precisely, if $g' = h/f^n$ with $h \in R$ and $n \geq 0$, then:

$$\operatorname{Spec}((R_f)_{g'}) \;=\; \operatorname{Spec}(R_{fh}).$$

In particular, $D(g') = D(fh)$ as subsets of $\operatorname{Spec}(R)$.

***Proof*:**
Since f is already a unit in R_f, inverting $g' = h/f^n$ in R_f is the same as inverting h in R_f:

$$(R_f)_{h/f^n} = (R_f)_h.$$

Now $(R_f)_h$ is the localization of R at the multiplicative set generated by f and h, which is the same as localizing at fh:

$$(R_f)_h = R_{fh}.$$

(Formally: $(R_f)_h = R[f^{-1}][h^{-1}] = R[(fh)^{-1}] = R_{fh}$, since inverting fh inverts both f and h, and conversely.) The corresponding open set is $D(fh) = D(f) \cap D(h) \subseteq \operatorname{Spec}(R)$.

2.1.16 Remark The proposition says that the collection of distinguished open sets is closed under "refinement": passing to a smaller distinguished open inside a distinguished open stays within the class. This is what makes distinguished opens a good basis for defining sheaves—one never needs to leave the class when restricting.

Let us record two further useful consequences.

Corollary 2.1.17: Unit Ideal and Distinguished Cover

Elements $f_1, \ldots, f_n \in R$ generate the unit ideal $(f_1, \ldots, f_n) = R$ if and only if $\operatorname{Spec}(R) = \bigcup_{i=1}^n D(f_i)$.

***Proof*:**
A prime $\mathfrak{p}$ is not in $\bigcup D(f_i)$ if and only if $f_i \in \mathfrak{p}$ for all i, if and only if $(f_1, \ldots, f_n) \subseteq \mathfrak{p}$. Thus $\bigcup D(f_i) = \operatorname{Spec}(R)$ if and only if no prime contains all the f_i, if and only if $(f_1, \ldots, f_n) = R$.

Corollary 2.1.18: Powers Generate the Unit Ideal

If $f_1, \ldots, f_n$ generate the unit ideal in R, then for any positive integers $k_1, \ldots, k_n$, the powers $f_1^{k_1}, \ldots, f_n^{k_n}$ also generate the unit ideal.

***Proof*:**
We have $D(f_i) = D(f_i^{k_i})$ (a prime contains f_i if and only if it contains $f_i^{k_i}$), so $\bigcup D(f_i^{k_i}) = \bigcup D(f_i) = \operatorname{Spec}(R)$, which by the previous corollary means $(f_1^{k_1}, \ldots, f_n^{k_n}) = R$.

2.1.3 Topological Properties

Let us see what classical topological properties applies to $\operatorname{Spec} R$. An important consequence is that for any open cover of $\operatorname{Spec}(R)$ given by distinguished open sets, there exists a finite open sub-cover. To see this, notice that $\cup_{i\in I} D(f_i) = \operatorname{Spec}(R)$ For some indexing set I if and only if $(\{f_i\}_{i\in I}) = R$ that is the ideal generated by the set $\{f_i\}_{i\in I}$. Then 1 must be inside that ideal, and must be a finite linear combination, and the result follows by the algebraic/geometric correspondence. This shows that $\operatorname{Spec}(R)$ is always compact in the topological sense. We may wish that this implies that $\operatorname{Spec}(R)$ has the usual nice properties of compactness, but this is very much not the case (notice this implies $\mathbb{A}^n_{\mathbb{C}}$ is compact). This turns out to be a consequence of lacking Hausdroffness. For this reason, a different name is given to compactness:

Definition 2.1.19: Quasicompact

Let X be a topological space. Then if X is compact and non-Hausdorff, X is *quasicompact*

As an exercise, show that if R is Noetherian, every subset of $\operatorname{Spec}(R)$ is quasicompact. In section 3.5.4, we shall show how the notion of proper maps from analysis is the right way of defining the notion of compactness which recovers many of the nice results expected from compactness.

The companion property of [quasi]compactness is connectedness. Recall a set is connected if it cannot be written as a disjoint of two nonempty open sets. Disconnectedness of Spec is equivalent to a ring representable as the product of two rings, $R = R_1 \times R_2$, or if the reader remembers the brief discussion in Representation theory(see [18]), if R has idempotent elements

Proposition 2.1.20: Spectrum Connectedness And Idempotent Elements

IF $R = R_1 \times \cdots \times R_n = \prod_i^n R_i$ is a ring, then show that

$$\operatorname{Spec}\left(\prod_i^n R_i\right) = \coprod_i \operatorname{Spec}(R_i)$$

More generally, show that $\operatorname{Spec}(R)$ is not connected if R is isomorphic to the product of two zero-rings, that is there exists $a_1, a_2 \in R$ st $a_1^2 = a_2, a_2^2 = a_2, a_1 + a_2 = 1$.

***Proof*:**
good exercise.

Note that for the homeomorphism, the product must be finite, if the product is infinite then $\operatorname{Spec}(\prod_i R_i)$ is larger than the right hand side. In fact, an infinite disjoint union of $\operatorname{Spec}(R_i)$ cannot be represented as $\operatorname{Spec}(R)$ for an appropriate R; it can be seen as the "simplest" example of a scheme that is not an affine scheme (see example 2.2.12(6)).

A notion similar to connectedness is *irreducibility*. Recall that a topological space is said to be *irreducible* if it is nonempty and it is not the union of two proper closed subsets. Written differently, X is irreducible if whenever $X = Y \cup Z$ with Y, Z closed in X, then $Y = X$ or $Z = X$[10]. Note that Y, Z may have nonempty intersection. In the euclidean topology, this notion is much less important (think $\mathbb{C}$), however in the Zariski topology we shall see that many interesting sets are irreducible. The intuition stems from traditional varieties where irreducibility corresponds to irreducible polynomials, and indeed this intuition when properly generalized translates over to irreducible [closed] sets in the Zariski topology of Spectra.

Let $X = Y \cup Z$ for closed sets Y, Z. Then there are radical ideals J_X, J_Y, J_Z such that $V(J_X) = V(J_Y) \cup V(J_Z) = V(J_Y J_Z) = V(J_Y \cap J_Z)$. As these are all radical ideals, $J_X = J_Y \cap J_Z$. Then if J_X is prime, it is an exercise in commutative algebra to show that either $J_Y \subseteq J_Z$ or $J_Z \subseteq J_Y$. Hence, irreducibility is in correspondence with prime ideals:

$$\{\text{prime ideals of } R\} \underset{V(-)}{\overset{I(-)}{\rightleftarrows}} \{\text{closed irreducible sets of Spec}\,(R)\} \tag{2.3}$$

If we look at irreducible closed sets, then $V(-)$ and $I(-)$ gives a bijection between the irreducible closed subsets of Spec$\,(R)$ and the prime ideals of R, in other words there is a bijection between the *points* of Spec$\,(R)$ and the irreducible closed subsets of Spec$\,(R)$. Since it is a bijection, every irreducible closed subset of Spec$\,(R)$ has *exactly* one generic point associated to it.

As usual, we define the Noetherian property on Spec$\,(R)$ to mean that the descending chain condition for closed sets is satisfied (to mirror the ascending chain of ideals in R). In the Noetherian case, all closed sets can be decomposed into irreducibles:

Proposition 2.1.21: Noetherian Decomposition

Let X be a Noetherian topological space. Then every nonempty closed subset Z can be expressed uniquely as a finite union

$$Z = Z_1 \cup \cdots \cup Z_n$$

of irreducible closed subsets, none contained in another.That is , any closed subset Z has a finite number of pieces.

***Proof*:**
see [20].

The reader should check the equivalent result for prime ideals

2.1.4 Ring to Top Functor: Induced Spectra Maps

As spectra are given by rings, we shall want to translate over ring homomorphisms to maps between spectra:

[10]The finiteness is important, for there may be irreducible sets that contain within them reducible sets of lower dimension

Definition 2.1.22: Induced Spectra Map

let $Y = \operatorname{Spec}(S)$ and $X = \operatorname{Spec}(R)$ be two spectrum. Then given the ring homomorphism: $\varphi : R \to S$, the induced *spectrum map* $\varphi^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ is given by:

$$\varphi^a(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$$

The reader ought to verify that if $\varphi : R \to S$ is a ring homomorphism, then $\varphi^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ sending $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ is functorial (recall or reprove that the pre-image of a prime ideal is prime). We now have two functions happening: φ^a given by ring homomorphisms, and regular functions given by ring elements. Often, φ^a is called a map, while the elements of R are called functions (following a similar convention from differential geometry).

Example 2.1.23: Maps between Spectra: φ^a

1. Show that the surjective map $R \to R/I$ induce an injective map $\operatorname{Spec}(R/I) \to \operatorname{Spec}(R)$. Algebraically, the surjective map adds relations. Geometrically, we are taking the points with the extra algebraic relations given by I and associating them to the points in R.

2. Show that the injective map $R \to S^{-1}R$ induce a injective map $\operatorname{Spec}(S^{-1}R) \to \operatorname{Spec}(R)$. Algebraically, the map $R \to S^{-1}R$ takes (integral) elements[a] and maps it to units.

3. Here is a concrete example generalizing the usual notion of a regular morphism as introduced in[20]: Take $\mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec}\mathbb{C}[x]$ which by Nullstellensatz we can think of as the traditional points $\mathbb{C}$ along with the generic point (0). Consider the ring homomorphism between these rings defined by:
$$\varphi : \mathbb{C}[y] \to \mathbb{C}[x] \qquad y \mapsto x^2$$
Note this is not the squaring map, as that is not a ring morphism, but rather the map that maps the variable y to x^2 and then extends linearly. Then the map on the sets Spec's is given by:
$$\varphi^a : \operatorname{Spec}(\mathbb{C}[x]) \to \operatorname{Spec}(\mathbb{C}[y])$$
$$\begin{aligned}\varphi^a(\mathfrak{p}) &= \varphi^{-1}((x-a)) \\ &= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; f(x^2) \in (x-a) \right\} \\ &= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; x-a \middle| f(x^2) \right\} \\ &= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; f(x^2) = (x-a)q(x) \right\} \\ &= (y-a^2)\end{aligned}$$
In one line:
$$\varphi^a([x-a]) = [(y-a^2)]$$
verify that
$$(\varphi^a)^{-1}((y-a)) = \{(x-\sqrt{a}), (x+\sqrt{a})\}$$
Perhaps more intuitively, this is the Spec-equivalent of the more usual map $\mathbb{C} \to \mathbb{C}$ given by $z \to z^2$ which can be thought of as the double-covering of $\mathbb{C}$ with branching point 0. Note the importance of visualizing schemes using complex numbers, or more generally an algebraically closed field, even if we replaced $\mathbb{C}$ with $\mathbb{R}$; for example the image of (x^2+1) is

$(y+1)$ (i.e. $-i^2 = -1$) and $(\varphi^a)^{-1}\left((y^2+1)\right) = \{(x^2+1)\}$ where the information about the 2 points is captured in the degree of the (irreducible) polynomial in the pre-image (recall they are glued together in $\operatorname{Spec}\mathbb{R}[x]$).

4. More generally, any collection of regular functions can produce a spectra map, a "regular map" (in particular, this mirror's regular maps between affine varieties). Let k be any field and take polynomial $f_1, ..., f_n \in k[x_1, ..., x_m]$. Then define:

$$\varphi : k[y_1, ..., y_n] \to k[x_1, ..., x_m] \qquad y_i \to f_i$$

Let A represent the domain and B the codomain. Then show that for any ideal $I \subseteq A$, $J \subseteq B$ such that $\varphi(I) \subseteq J$, φ induces a map of sets:

$$\operatorname{Spec}\left(k[x_1, ..., x_m]/J\right) \to \operatorname{Spec}\left(k[y_1, ..., y_n]/I\right)$$

More particularly, show that this maps sends the point $[(x_1 - a_1, ..., x_m - a_m)]$ (traditionally, $(a_1, ..., a_m) \in k^m$ to the point traditionally represented as:

$$(f_1(a_1, ..., a_m), ..., f_n(a_1, ..., a_m)) \in k^n$$

From now on, we shall sometimes take liberties in spectra maps between spectra resembling varieties by defining polynomial maps on the maximal ideals (this shall pose no problems with extending the definition to the prime ideals or to the later sheaf structure that shall be added by theorem 3.7.1).

5. The map $\mathbb{A}^1_k \to \mathbb{A}^1_k$ given by $a \mapsto p(a)$ can always be represented as a map to the graph:

$$\mathbb{A}^1_k \to \mathbb{A}^2_k \qquad a \mapsto (a, p(a))$$

Furthermore, by the projection map $\pi : \mathbb{A}^2_k \to \mathbb{A}^1_k$ given by $(a, b) \mapsto a$, we shall see that this maps $\mathbb{A}^1_k$ isomorphic to all of its graphs (this works even for higher dimensions). Note that this does not extend if we have more variable where more than one term is not the identity, for example:

$$\mathbb{A}^1_k \to \mathbb{A}^3_k \qquad t \mapsto (t, t^2, t^3)$$

maps to the point of $\mathbb{A}^1_k$ to the associated to the variety $\mathcal{Z}(x - y^2, x - z^3)$ which in [20] was shown to be rational but not isomorphic. Another example that isn't even rational is a map from $\mathbb{P}^1_k$ to an elliptic curve E.

6. This example gives a way of thinking of $\mathbb{A}^n_{\mathbb{Z}} = \operatorname{Spec}\left(\mathbb{Z}[x_1, ..., x_n]\right)$ as an "$\mathbb{A}^n$-bundle" on $\operatorname{Spec}(\mathbb{Z})$ and is an important example for number theory. Define the map $\pi : \mathbb{A}^n_{\mathbb{Z}} \to \operatorname{Spec}(\mathbb{Z})$ given by the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[x_1, ..., x_n]$. Then show there is a bijection between $\pi^{-1}([(p)])$ where $(p) \neq (0)$and $\mathbb{A}^n_{\mathbb{F}_p}$! If $(p) = (0)$ then the pre-image is $\mathbb{A}^n_k$. The picture of this bundle can be seen as follows:

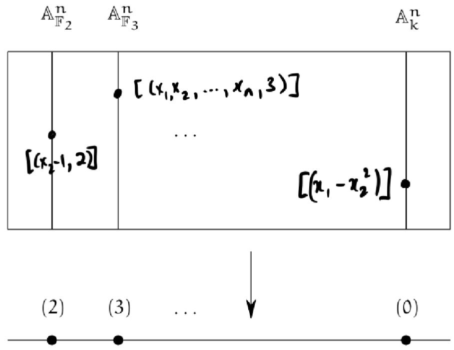


Figure 2.1: Bundle on Spec $\mathbb{Z}$

7. Consider $\varphi : \mathbb{Z} \hookrightarrow k[x]$. Then $\varphi^a : \operatorname{Spec} \mathbb{C}[x] \to \operatorname{Spec} \mathbb{Z}$ is the zero map, namely $\varphi^a(\operatorname{Spec} k[x]) = \{[(0)]\}$. From this perspective, we can see that classical field theory has trivial number theory as it has no interesting fibers over Spec $\mathbb{Z}$.

8. Let $\varphi : R \to S$ be an integral extension. Show that $\operatorname{Spec}(S) \to \operatorname{Spec}(R)$ is surjective (hint: Lying Over). This results also motivates the terminology "lying over" geometrically. Recall that is R or S is a field, the other must be too. Show that S being a field and the map being surjective can be used to conclude R is a field. What is the geometric argument if R was a field?

9. Let us see why it is important to work with primes instead of maximal ideals. Consider Spec $k(x)[y] \to$ Spec $k[x]$ induced by $k[x] \to k(x)[y]$ where $k[x]$ maps to itself (notably, all elements become units). Then as $k(x)[y]$ is a PID, all of its primes are maximal and are of the form $(y - f(x))$ for some $f(x) \in k(x)$. Then the pre-image of this ideal under the inclusion map must be the ideal (0), which is certainly not maximal (in fact it is the generic point). Hence the map Spec $k(x)[y] \to$ Spec $k[x]$ will map $[(y - f(x))] \mapsto \eta$, notably a 0 -dimensional point mapped to a 1-dimensional point!

10. Let R be a ring with nilpotence and $I = \sqrt{(0)}$ the nilradical. Take $\varphi : R \to R/I$ to be the usual projection. Then φ^a is a bijection; the key is to recall that the nilradical intersection of all prime ideals, and hence any prime $\overline{\mathfrak{p}} \subseteq R/I$ which is of the for $\mathfrak{p} + I$ must in fact equal itself: $\mathfrak{p} + I = \mathfrak{p}$ (as $I \subseteq \mathfrak{p}$). The example to keep in mind should be $R = \mathbb{C}[x]/(x^2)$ with $I = (x)$.

[a] Note that if an element is nilpotent, localizing it will map it to zero

Lemma 2.1.24: Induced Sepctra Map Is Continuous

Let $\varphi : R \to S$ be a ring homomorphism. Then φ^a is a continuous map

***Proof*:**
It suffices to show the pre-image of a closed set is closed, so take $V(S) \subseteq \operatorname{Spec}(R)$ and consider $(\varphi^a)^{-1}(V(I))$. Show that this is equal to $V(\varphi(I))$, showing it is closed.

The map φ^a given by the natural inverting function is a naturally continuous map, that is φ^a is a continuous function between the Spec of two rings with the Zariski topology. Hence Spec is a [contravariant] functor from **CRing** to **Top**. Furthermore, continuity can be shown on distinguished open sets, namely as they form a basis we can show that:

$$(\varphi^a)^{-1}(D([\mathfrak{p}]) = D(\varphi([\mathfrak{p}]))$$

Note that there will be continuous maps between the spectrum of rings that are not φ^a,

Example 2.1.25: Continuous Not Induced By Ring Homomorphism
The map from $\mathbb{A}^1_{\mathbb{C}}$ to itself that is the identity on all points except $[(x)]$ and $[(x-1)]$ (which will be represented with 0 and 1) where it swaps these two values is *not* a polynomial and hence cannot have been induced by a ring homomorphism, however it *is* continuous. If $0, 1 \notin V(S)$, then $\varphi^{-1}(V(S)) = V(S)$. Similarly if both values are inside. If 0 or 1 is in $V(S)$, say $0 \in V(S)$, then $\varphi^{-1}(V(S)) = V(S) \setminus \{0\} \cup \{1\}$ which is closed as all finite subsets of $\mathbb{A}^1_{\mathbb{C}}$ are closed.

Can you now find two ring homomorphisms that induce the same Spectra map? See exercise 2.1.1(2.1.1 (13))

Exercise 2.1.1(2.1.1 (14)) goes over some of the properties preserved by continuous maps. Most of the properties that we shall be want for schemes to have (ex. separatedness, properness, smoothness, multiplicity information, finiteness conditions) shall require more than continuity.

The correspondence is terse as Spec does not carry enough information to fully and faithfully correspond to ring homomorphism. In chapter 3 it shall be shown that scheme morphisms are the right object to carry this information (namely, they shall insure that around every point, that is each stalk, the spectra map behaves like a polynomial).

2.1.5 *Subsets of Spectra via quotient and Localization

Two common constructions that came up in a few example is the process of taking the quotient and the localization of a ring R. These form important subsets of $\operatorname{Spec}(R)$, namely by the 4th isomorphism theorem and the prime-correspondence theorem for prime ideals in localization (see [18] and [10]). By these theorems, $\operatorname{Spec}(R/I), \operatorname{Spec}(S^{-1}R) \subseteq \operatorname{Spec}(R)$. If $I = \mathfrak{p}$ and $S = R \setminus \mathfrak{p}$,the first corresponds to all prime ideals containing $\mathfrak{p}$, while the second corresponds to all prime ideals that are contained in the prime ideal $\mathfrak{p}$[11] These two sets need not union to the entire $\operatorname{Spec}(R)$ (take $\mathbb{Z}$ and 5), however they do correspond to two “opposite” ways of carving out a part of $\operatorname{Spec}(R)$. In the following visual is a very common illustration:

[11] More generally it is all prime ideals not having trivial intersection with the multiplicative set.

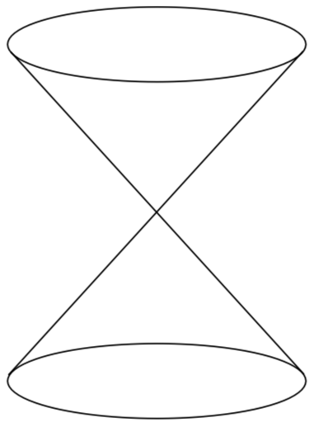


Figure 2.2: partial decomposition of Spec (R)

Geometrically, Spec (R/I) can be thought of as "carving out" a part of Spec (R). For example, take $R = \mathbb{C}[x, y, z]$ and $\mathfrak{p} = (x^2 + y^2 - z)$. If we were working with affine complex varieties, then we would say this is the 3D-parabola in $\mathbb{C}^3$, however as we are on Spec $(R/\mathfrak{p})$ this would come with some extra points (which ones?). For Spec $(S^{-1}R)$, we may think it as the opposite (we shall cover the special case of $R_{\mathfrak{p}}$ afterwards). For example let's say $S = \{1, f, f^2, ...\}$ and take $S^{-1}R$. Then the primes of Spec $(S^{-1}R)$ would be all primes that do not containing f, that is all points where f doesn't vanish! As an illustration, the set Spec $(\mathbb{C}[x, y]_{y-x^2})$ are the points in the affine plane, minus all the points on the parabola $y - x^2$, namely all the singletons (a, a^2) for $a \in \mathbb{C}$ (so $[(x - a, y - a^2)]$), as well as the point $[(y - x^2)]$.

Proposition 2.1.26: Closed And Open Subsets

Let R be a commutative ring, $I \subseteq R$ an ideal, $f \in R$, $\mathfrak{p} \subseteq R$ prime, and S a multiplicative set. Then:

1. Spec $(R/I) \subseteq$ Spec (R) is a closed subset
2. Spec (R_f) is an open subset
3. Spec $\left(S^{-1}R\right) \subseteq$ Spec(R) need not be open or closed in general. In particular Spec $(R_{\mathfrak{p}}) \subseteq$ Spec(R) need not be open or closed. However, it is an embedding, namely $\iota : \operatorname{Spec} S^{-1}R \hookrightarrow \operatorname{Spec} R$ is injective, continuous, and closed.

***Proof*:**

1. Think of $V(I)$ (induce the topology on Spec (R/I) via the inclusion map)
2. Think Spec $(R) \setminus V(f)$
3. think of Spec $(\mathbb{Q}) \subseteq$ Spec $(\mathbb{Z})$ and Spec $(\mathbb{Z}_{\mathfrak{p}}) \to$ Spec $(\mathbb{Z})$ induced by $\mathbb{Z} \to \mathbb{Z}_{\mathfrak{p}}$ (this shows

that Spec $(R_{\mathfrak{p}})$ need not be open.

For the case of $S = R - \mathfrak{p}$, we can think of Spec $(R_{\mathfrak{p}})$ as keeping near the $\mathfrak{p}$, namely since the prime ideals away from $\mathfrak{p}$ will be turned into units, the primes intersecting $R \setminus \mathfrak{p}$ will be eliminated. For example, if $R = \mathbb{C}[x, y]$ and $\mathfrak{p} = (x, y)$, then we keep all the points corresponding to being "near" the origin, that is the point $[(x, y)]$, i.e. $(0, 0)$, and the 1-dimensional points $[(f(x, y))]$ where $f(0, 0) = 0$ (i.e. where $f \equiv 0 \mod (x, y)$) that is irreducible curves that pass through the origin. In the special case where $R = k[x]$, then the localization will always only have two points, usually called the classical point and the generic point. These shall eventually be seen as geometric interpretations of DVR.

Using these geometric results, they can lend intuition towards algebraic manipulation:

Example 2.1.27: Illustration Of Quotient And Localization

Using the above intuitions, we can quickly come up with some intuition isomorphisms. Take $R = \mathbb{C}[x, y]/(xy)$. Visually, this is the union of the y and x-axis. If we localize this at x, then by our above intuition this should eliminated all points where the "function" x vanishes. The function x vanishes at the y-axis, and so we must be left with the x-axis minus the origin. This happens to be Spec $\mathbb{C}[x]_x$. In fact, these two rings are isomorphic!

$$\left(\frac{\mathbb{C}[x, y]}{(xy)}\right)_x \cong \mathbb{C}[x]_x$$

which the reader should check.

Exercise 2.1.1

1. Show that $D(f) \cap D(g) = D(fg)$.
2. $D(f) \subseteq D(g)$ iff $f^n \in (g)$ iff g is invertible in R_f. How would you prove this via "geometric explanation"? For an algebraic explanation, change to $V(f) \supseteq V(g)$ and take $I(-)$ to inverse and conclude.
3. $D(f) = \emptyset$ iff f is in the Nilradical.
4. Show that any connected component is the union of irreducible components (think of the $+$ shape, i.e. Spec $(\mathbb{C}[x, y]/(xy))$).
5. Show that $\mathbb{A}^1_{\mathbb{C}}$ is irreducible.
6. Show that in an irreducible topological space, any nonempty open set is dense.
7. If X is a topological and Z an irreducible subspace, then $\overline{Z}$ is irreducible as well
8. Show that an irreducible space is connected.
9. If A is a integral domain, Spec (A) is irreducible.
10. Show that Spec $(\mathbb{C}[x, y]/(xy))$ is connected but reducible.
11. Show that there is a natural bijection irreducible components (maximal irreducible subsets) of Spec (R) and the *minimal* prime ideals of R.
12. these next two exercises show us how the theory of classical varieties persists with Schemes

a) Let A be a finitely generated k-algebra. Then show that the closed points of $\operatorname{Spec}(A)$ are dense; show that if $f \in A$ and $D(f)$ is nonempty, then $D(f)$ contains a closed point of $\operatorname{Spec}(A)$[12]

b) Let $A = \overline{k}[x_1, ..., x_n]/I$ where the Nilradical is trivial, and let $X = \operatorname{Spec}(A) \subseteq \mathbb{A}^n_{\overline{k}}$. Show that [regular] functions on X are determined by their valued on the closed points. Hint: if f, g are different [regular] functions on X, then $f - g$ is nowhere zero on an open subset of X. Use the above result.

13. Find two ring homomorphism that induce the same spectra map.

14. Let $f : X \to Y$ be a continuous map.

a) Show that irreducible sets map to irreducible sets to irreducible sets.

b) Show that if X is Noetherian, then the image of f is Noetherian

c) Show that the dimension of X is greater than or equal to the dimension of the image of Y (also see 2.6).

2.2 Schemes

The spectrum of a ring with the Zariski Topology does not hold all data provided by a commutative ring, for example nilpotence is forgotten. The *structure sheaf* will be added to $\operatorname{Spec} R$ in order to add this extra information (namely, we shall have the collection of regular function defined on each open set as part of the data we are tracking). The resulting object is called the *affine scheme*.

After defining the affine scheme, we shall require one more crucial abstraction that links it to differential geometry: we shall want our object to the glued-together open sets. In differential geometry, these would all be open subsets of euclidean space, usually simply connected and all isomorphic. A key distinguishing feature in algebraic geometry is that the sets that can be glued can be very different, namely $\operatorname{Spec} R$ for all commutative rings R will be the sets from which we can select, and as the reader has already seen these sets can have very different topologies.

2.2.1 The Structure Sheaf for Affine Schemes

When working with geometric objects (varieties, real and complex manifolds, etc), we study the local behaviour of the space, namely by studying some given information on open sets or germs/stalks. In the theory of differential manifold, the information at each local was the "linear" equation (recall that one way of defining the tangent bundle is by taking all the derivations of smooth functions, which can be shown to "preserve" the linear part of each smooth function). For an affine scheme, we are interested in the functions functions are *regular* on open sets. By section 2.1.2, it is enough to define the regular functions on distinguished open sets. For those motivated by analytic parallels, Complex manifold have a sheaf of meromorphic functions which are defined on the open subsets of the complex manifold, and we usually have holomorphic functions on the entire manifold.

To motivate the following sheaf, recall that on $\mathbb{C}^n$, if we take open subsets there are many more holomorphic functions that could be defined on that open subset, namely that rational functions are

[12]Hint: A_f is also a finitely generated k-algebra use generalized Nullstellensatz. Show the closed points of the Spec of a finitely generated k-algebra are those for which the residue field is a finite extension of k

well-defined when the roots of the denominator (which are not cancel by the numerator) are outside the open set. For example, on $\mathbb{C}^2 \setminus \{x\text{-axis} \cup y\text{-axis}\}$, the function

$$\frac{x^2+4x+y+1}{x^5y^4}$$

is well-defined. Moving this to Spec, in $\mathbb{A}^2_{\mathbb{C}}$, we ought to have the above function to be well-defined on $D(xy)$. Furthermore, we want the collection of global sections, $\mathcal{O}_X$, to be R, mirroring the "global section" on $\mathbb{C}^2$ being the (non-rational) holomorphic functions in two variables.

As we saw in theorem 1.1.22, it suffice to define a sheaf on a base of $\operatorname{Spec}(R)$, namely on the distinguished open sets. Thus, we may define on $D(f) \subseteq \operatorname{Spec}(R)$ the ring that is the localization of R at the multiplicative subset of all function that do not vanish on $V(f)$ (note how this is subtly different than functions that do not vanish on f, what if f is nilpotent?). This can be represented as:

$$\mathcal{O}_X(D(f)) = \mathcal{O}_X(X_f) \cong R_f$$

First off, this is a presheaf: if $D(f) \supseteq D(g)$, by exercise 2.1.1 there is a minimal $n \in \mathbb{N}_{>0}$ such that $g^n \in (f)$ so $rf = g^n$. Then as $D(g^n) = D(g)$, we may define the restriction map:

$$\operatorname{res}_{X_f, X_g} : R_f \to R_g \qquad \frac{s}{f^m} \mapsto \frac{r^m s}{g^{nm}}$$

which shows we have defined a presheaf on a base. We next must show that it satisfies the sheaf on base axioms.

Proposition 2.2.1: Distinguished Opens form Sheaf-Basis for $\operatorname{Spec} R$

Let $X = \operatorname{Spec} R$, and suppose that $D(f)$ is covered by distinguished open sets $D(f_a) \subseteq D(f)$. Then:

1. If $g, h \in R_f$ are equal in R_{f_a} when passing through the restriction map for every a, then $g = h$.
2. If for each a, there exists a $g_a \in R_{f_a}$ such that for each pair a, b, the images of g_a and g_b in $R_{f_a f_b}$ are equal, then there is an element $g \in R_f$ whose image in R_{f_a} is g_a.

In other words, $\mathcal{O}_X$ defines a $\mathcal{B}$-sheaf on the distinguished base, and by theorem 1.1.22, $\mathcal{O}_X$ extends to a sheaf on X.

***Proof*:**
Let $\operatorname{Spec}(R) = \bigcup_{i \in I} D(f_i)$, which is equivalent to saying the f_i generate the unit ideal: $(f_i)_{i \in I} = R$. We show the locality and gluability axioms, or in categorical terms the exactness of:

$$0 \longrightarrow R \longrightarrow \prod_{i \in I} R_{f_i} \rightrightarrows \prod_{i \neq j \in I} R_{f_i f_j}$$

Starting with locality. As $s_1 = s_2$ is equivalent to $s_1 - s_2 = 0$, it suffices to show the result in the case where $s = 0$. By quasicompactness we may pass to a finite subcover:

$$\operatorname{Spec}(R) = \bigcup_{i=1}^{n} D(f_i) \qquad (f_1, \ldots, f_n) = R$$

Suppose that for some $s \in R$, $\operatorname{res}_{\operatorname{Spec}(R),D(f_i)} s = 0$ for all i. We must show $s = 0$. By definition of R_{f_i}:

$$s|_{D(f_i)} = 0 \quad \text{if and only if} \quad f_i^{m_i} s = 0 \quad \text{for some } m_i \geq 0$$

Taking $m = \max_i(m_i)$, we have $f_i^m s = 0$ for all i. Now, as $D(f_i) = D(f_i^m)$, the cover $\bigcup D(f_i^m) = \operatorname{Spec}(R)$ still holds, so $(f_1^m, \ldots, f_n^m) = R$. Thus, for appropriate choices $r_i \in R$ with $\sum_{i=1}^n r_i f_i^m = 1$, we get:

$$s = 1 \cdot s = \left(\sum_i r_i f_i^m \right) s = \sum_i r_i (f_i^m s) = 0$$

For general $D(f)$, repeat the above argument replacing R with R_f (quasi-compactness of $D(f)$ ensures the cover can still be taken to be finite, and the same algebraic manipulations apply in R_f).

We next show gluability[a]. Starting with the case $\operatorname{Spec}(R) = \bigcup_i D(f_i)$, suppose there are elements in each R_{f_i} that agree on overlaps $R_{f_i f_j}$ (note that intersections of distinguished open sets are distinguished open sets). As there may be infinitely many sets on which there are overlaps, it cannot be assumed that I is finite. However, the finite case is a good start, say $I = \{1, \ldots, n\}$. Then we have elements $\frac{a_i}{f_i^{l_i}} \in R_{f_i}$ agreeing on the overlaps $R_{f_i f_j}$, meaning:

$$(f_i^{l_i} f_j^{l_j})^{m_{ij}} (f_j^{l_j} a_i - f_i^{l_i} a_j) = 0$$

Simplifying notation by writing $g_i = f_i^{l_i}$:

$$(g_i g_j)^{m_{ij}} (g_j a_i - g_i a_j) = 0$$

As there are finitely many m_{ij}, take $m = \max_{i,j}(m_{ij})$:

$$(g_i g_j)^m (g_j a_i - g_i a_j) = 0$$

Next, convert to a different set of distinguished opens: take $b_i = a_i g_i^m$ and $h_i = g_i^{m+1}$ so that $D(h_i) = D(g_i)$. Then on each $D(h_i)$, we have the section b_i/h_i, and the overlap condition simplifies to:

$$h_j b_i = h_i b_j$$

Noting that $\bigcup_i D(h_i) = \operatorname{Spec}(R)$ gives $1 = \sum_{i=1}^n r_i h_i$ for appropriate $r_i \in R$, define:

$$r = \sum_i r_i b_i$$

Then r is the element of R that restricts to b_j/h_j on each $D(h_j)$:

$$r h_j = \sum_i r_i b_i h_j = \sum_i r_i h_i b_j = \left(\sum_i r_i h_i \right) b_j = b_j$$

showing gluability in the finite case. For the infinite case, we use the previously shown base-locality axiom. By quasi-compactness, we can choose a finite subcover $D(f_1), \ldots, D(f_n)$ and construct r as above. For any $z \in I \setminus \{1, \ldots, n\}$, repeat the argument on $\{1, \ldots, n, z\}$ to get $r' \in R$ which restricts to a_i for $i \in \{1, \ldots, n, z\}$. By base locality, $r' = r$. Hence r restricts to $a_z/f_z^{l_z}$ for all z.

To finish off the proof, we must handle the case of an arbitrary $D(f)$ rather than all of $\operatorname{Spec}(R)$. This follows by applying the same argument with R replaced by R_f throughout.

[a] Vakil pointed out that Serre called this a "partition of unity argument," which is the local-to-global principle physically manifested in differential geometry – a rather good perspective on what is about to happen.

It may be tempting to say that $\mathcal{O}_{\operatorname{Spec}(R)}(U)$ for any open subset U is a localization of R at some multiplicatively closed subset S, however this is not true:

Example 2.2.2: Subtle Local Section Of Affine Scheme

Let $R = k[x,y,z,w]/(xw-yz)$; $\operatorname{Spec} R$ is called the *quadric cone*. Note that R is an integral domain as $(xw-yz)$ is prime[a]. Let $U = D(y) \cup D(w)$. Then a section $s \in \mathcal{O}(U)$ is determined by a choice of $f \in R_y$ and $g \in R_w$ that agree on R_{yw}, namely $\mathcal{O}(U)$ is determined by the left exact sequence:

$$0 \to \mathcal{O}(U) \hookrightarrow R_y \oplus R_w \xrightarrow{(f,g)\mapsto f-g} R_{yw}$$

The kernel of the difference map is $\mathcal{O}(U)$. Take $(x/y,\ z/w)$ and consider:

$$\frac{x}{y} = \frac{z}{w} \qquad \text{if and only if} \qquad xw - zy = 0$$

where there is no need for a factor of $(yw)^n$ in front, as there are no zero-divisors. In R, this is exactly the defining relation of the quotient, hence this equality holds. Therefore there is an element $t \in \mathcal{O}(U)$ such that:

$$t \mapsto \left(\frac{x}{y}, \frac{z}{w}\right)$$

Next, for any $r \in R$, certainly $r \in R_y, R_w$ and $r - r = 0$, so $r \in \mathcal{O}(U)$. Is it possible that $t = r$ for some $r \in R$? As the map $\mathcal{O}(U) \hookrightarrow R_y \oplus R_w$ is injective, if $t = r$ then since $r \mapsto (r,r)$ we would need $r = x/y = z/w$, but y and w are not invertible in R, and hence $t \notin R$. Thus the ring $R[t] \neq R$. Certainly $R[t] \subseteq \mathcal{O}(U)$ and any R-polynomial $f(t) \in R[t]$ maps to:

$$f(t) \mapsto \left(f\left(\frac{x}{y}\right), f\left(\frac{z}{w}\right)\right)$$

Let us show that $yt - x = 0$ (equivalently $wt - z = 0$). By definition of a sheaf, this can be checked on the open cover $\{D(y), D(w)\}$. By construction (recall that restriction maps are ring homomorphisms and that $r \mapsto r$ under these maps):

$$(yt - x)|_{D(y)} = y \cdot \frac{x}{y} - x = x - x = 0$$

and

$$(yt - x)|_{D(w)} = y \cdot \frac{z}{w} - x = \frac{yz - xw}{w} = \frac{0}{w} = 0$$

giving $yt - x = 0$. Thus:

$$R[t] \cong \frac{R[T]}{(yT - x)} \cong R\left[\frac{x}{y}\right]$$

Note that y is *not* a unit in $R[x/y]$. Let us show $\mathcal{O}(U) \cong R[t]$. If $s \in \mathcal{O}(U)$,

$$s|_{D(y)} = \frac{a}{y^n} \qquad \text{and} \qquad s|_{D(w)} = \frac{b}{w^m}$$

for some $a, b \in R$ and $n, m \in \mathbb{N}$. They must agree on $D(yw)$:

$$\left.\frac{a}{y^n}\right|_{D(yw)} = \frac{aw^n}{(yw)^n} = \frac{by^m}{(yw)^m} = \left.\frac{b}{w^m}\right|_{D(yw)}$$

Since R is an integral domain, this simplifies to $aw^m = by^n$ in R.

To show that $s \in R[t]$, we use the key relations $x = ty$ and $z = tw$ (which hold in $R[t]$ since $yt = x$ and $wt = z$). In R_y, the element a/y^n can be rewritten by substituting $x = ty$ and $z = tw$: every monomial in a involving x or z can be expressed using t, y, and w. More precisely, any element of R_y can be written as a polynomial in t with coefficients in $k[y, y^{-1}, w]$ (after eliminating x and z using the relations). Similarly, any element of R_w is a polynomial in t with coefficients in $k[y, w, w^{-1}]$. The overlap condition $aw^m = by^n$ ensures that the two expressions, when restricted to R_{yw}, give the same polynomial in t. By the locality axiom (already proved), this polynomial is unique, and it lies in $R[t]$ (no negative powers of y or w are needed, as one can verify by comparing degrees). Hence $s \in R[t]$, and so $\mathcal{O}_X(U) \cong R[x/y]$.

Now, $R[x/y]$ is *not* a localization of R. Indeed, $R[x/y] \ncong R$ (since $x/y \notin R$), yet $R[x/y]$ has the same units as R (namely k^*): no element becomes invertible. Since any nontrivial localization R_S contains units not present in R, and isomorphisms must preserve units, no localization of R is isomorphic to $R[x/y]$.

More generally, given an arbitrary open subset U, we can cover it by affine opens $\{U_i\}_{i\in I}$ and cover the intersections $U_i \cap U_j$ by affine opens $\{U_{ijk}\}_{k\in K_{ij}}$. Then the sheaf condition gives that $\mathcal{O}_X(U)$ is the equalizer of:

$$\prod_{i\in I} \mathcal{O}_X(U_i) \rightrightarrows \prod_{i,j\in I,\, k\in K_{ij}} \mathcal{O}_X(U_{ijk})$$

The general idea is that $\mathcal{O}_X(U)$ may contain rational functions f/g where g is nonzero on all of U. In practice, we shall often only require the local sections on distinguished open sets or on affine open sets, and many computations reduce to these two cases.

Note that even the following weaker hope fails: given $U \subseteq \operatorname{Spec} R$, is $\mathcal{O}_X(U)$ isomorphic to the localization of R at the multiplicative set S of all elements that do not vanish at any point of U? Consider $U = D(x) \cup D(y) \subseteq \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$. We will show in example 2.2.51 that U is not affine. However, $\mathcal{O}_{\mathbb{A}^2_k}(D(x) \cup D(y)) \cong k[x, y]$ (a Hartogs-type phenomenon: regular functions extend across a codimension-2 locus). This is just $k[x, y]$ itself, yet $D(x) \cup D(y) \ncong \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$.

This shows that the relationship between local geometry and algebra is more nuanced than the affine case suggests.

[a] As $k[x, y, z, w]$ is a UFD, a principal ideal (p) where p is irreducible is prime. It is tedious but rudimentary to show that $xw - yz$ is irreducible.

Keeping track of the structure sheaf $\mathcal{O}_{\operatorname{Spec} R}$ along with $\operatorname{Spec}(R)$ gives us our first definition we need:

Definition 2.2.3: Affine Scheme

Let $X = \operatorname{Spec}(R)$ be the spectrum of the ring R with the Zariski topology, and let $\mathcal{O}_{\operatorname{Spec}(R)}$ be the structure sheaf. Then $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ is called an *affine scheme*.

In some textbooks, an affine scheme is defined as a locally ringed space (recall definition 1.4.2) $(X, \mathcal{O}_X)$ that is an *affine scheme* if it is locally-ringed-space isomorphic to $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ for appropriate ring R.

With the sheaf defined, we have now "justified" thinking of elements on an arbitrary ring R as functions, as

$$\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec} R) = \mathcal{O}_{\operatorname{Spec}(R)}(D(1)) = R_1 = R$$

Even more is true, due to the section's being constructed through localization of R, they may be ratio's of functions, and we give elements of section a particular name:

Definition 2.2.4: Regular Functions

Let $X = \operatorname{Spec} R$ be an and $U \subseteq X$ an open subset. Then the set $\mathcal{O}_X(U)$ is called the set of *regular functions* on U.

Note that like complex manifolds (and unlike general smooth manifolds), there are in general relatively few functions in the global section of an as compared to the global section (in the global section of an affine scheme, all rational functions are regular, that is they are R). On the other hand, $\mathcal{O}_X(U)$ may have many interesting rational functions. In fact, in many cases all open subsets of X are dense meaning the local behavior can give a lot of fruitful information on the entire scheme (as shall be the case for abstract varieties, see ref:HERE)

Example 2.2.5: Affine Schemes

All spectra can be equipped with a structure sheaf by definition and so they are all affine schemes. For less obvious examples:

1. If $k \ncong k'$ are distinct fields fields, then as schemes $\operatorname{Spec}(k) \ncong \operatorname{Spec}(k')$ as their sheaves are different. Since there is only one point $[(0)]$ an all such Spectra, the functions (elements of the sheaf) will all return themselves, which the reader can think of as there only being constant functions, where each sheaf have different constant functions.

2. Let $R = k[x]_{(x)}$. Then $\operatorname{Spec} R$ contains two points, $[(x)]$ and $[(0)]$ and has 3 open sets, namely

$$\emptyset \quad \{[(0)]\} \quad \{[(0)], [(x)]\} = \operatorname{Spec} R$$

Note that U and $\emptyset$ are distinguished open sets (since $\{(0)\} = X_x$). The corresponding structure sheaf is:

$$\mathcal{O}_X(X) = R = k[x]_{(x)}$$
$$\mathcal{O}_X(U) = k(x)$$

The restriction map is the natural inclusion. This schemes is an example of a *local scheme*.

3. Let $R = k[[x]]$, the completion of $k[x]$ at x. The only points of Spec R are again $[(0)]$ and $[(x)]$, and so it has the same open sets. The corresponding structure sheaf is:

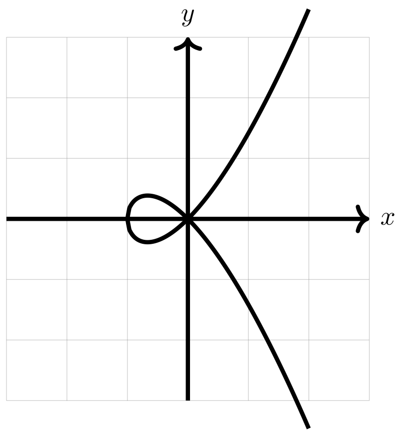
$$\mathcal{O}_X(X) = R = k[[x]]$$
$$\mathcal{O}_X(U) = \operatorname{Frac}(k[[x]])$$

Thus, we see that there can be schemes with very similar sheaves and identical spectra, but are distinct.

This scheme is considered to be *even more local* than localization. Consider $\mathbb{A}^2_k$. Then the sequence of maps:

$$k[x, y] \hookrightarrow k[x, y]_{(x,y)} \hookrightarrow k[[x, y]]$$

induces a sequence of inclusion spectra maps $Y \to X \to \mathbb{A}_k^2$, showing how the completion at the point (x, y) is "smaller" than the localization at that point. To demonstrate this, assume k does not have characteristic 2 or 3 and consider the affine scheme given by $y^2 = x^2 + x^3$:



Note that $y^2 - x^2 - x^3$ is irreducible, and localizing at (x, y) keeps the corresponding $(\frac{k[x,y]}{(y^2-x^2-x^3)})_{(x,y)}$ with only two points. However, in

$$\frac{k[[x, y]]}{(y^2 - x^2 - x^3)}$$

we have the access to power series expansion around $(0, 0)$, and we have that the power series:

$$x + \frac{1}{2}x^2 - \frac{1}{8}x^3 + \cdots$$

belongs to this ring, and notably this power series is the square-root of $x^2 + x^3$. Thus, we may write:

$$y^2 - x^3 - x^2 = (y - \sqrt{x^3 + x^2})(y + \sqrt{x^3 + x^2})$$

and so we get that $\frac{k[[x,y]]}{(y^2-x^3-x^2)}$ is *not* an integral domain, and in particular we shall see that Spec $\frac{k[[x,y]]}{(y^2-x^3-x^2)}$ has *two* irreducible components corresponding to the two branches at $(0, 0)$.

4. The reader may recall that most power series do not satisfy an algebraic equation. For this reason, the above ring is often considered too big, and instead we take an intermediary ring H which contains all power series that satisfy algebraic equations over $\mathrm{Frac}(R)$. This is called the *Henselization* of $\operatorname{Spec} R$ at a point $\mathfrak{p}$.

5. If X, Y are then the product of X and Y are:

$$\operatorname{Spec}(R) \times \operatorname{Spec}(S) = \operatorname{Spec}(R \otimes_{\mathbb{Z}} S) = \operatorname{Spec}(R \otimes S)$$

where the tensor product with no base ring by default assume it is over the initial ring. The generalization of this concept shall bee shown in section 3.2.

2.2.2 Ideal Correspondence in Affine Schemes

So far, the dictionary between certain types of ideals of R and the geometric equivalent in $\operatorname{Spec}(R)$ has been establish, namely through $V(-)$ and $I(-)$:

$$\begin{aligned}
\text{maximal ideal of } R &\quad\Leftrightarrow\quad \text{closed points of } \operatorname{Spec}(R)\\
\text{minimal prime ideals of } R &\quad\Leftrightarrow\quad \text{irreducible components of } \operatorname{Spec}(R)\\
\text{prime ideal of } R &\quad\Leftrightarrow\quad \text{irreducible closed subset of } \operatorname{Spec}(R)\\
\text{radical ideals of } R &\quad\Leftrightarrow\quad \text{closed subsets of } \operatorname{Spec}(R)
\end{aligned}$$

What of a general ideal $J \subseteq R$? In section 2.1.1, this information was "lost" as $I(V(J)) = \sqrt{J}$. However, a sheaf would capture this information!

Let's start with the simplest case of $R = \mathbb{C}[x]/(x^2)$ with $X = \operatorname{Spec}(R)$ that was mentioned in example 2.1.3(6). If we consider the set $\operatorname{Spec}(R)$, then this is just the origin, i.e. $[(x)]$. Namely, by proposition 2.1.10[13],

$$\operatorname{Spec}(\mathbb{C}[x]/(x^2)) \cong_{\mathbf{Top}} \operatorname{Spec}(\mathbb{C}[x]/(x)) \cong_{\mathbf{Top}} \{\overline{(x)}\} \equiv \{0\}$$

Note that (0) is no longer prime since $x^2 \in (0)$ but $x \notin (0)$, and thus $[(0)] \notin \operatorname{Spec}(\mathbb{C}[x]/(x^2))$. To show that the $\left(\operatorname{Spec}\left(\mathbb{C}[x]/\left(x^2\right)\right), \mathcal{O}_{\operatorname{Spec}(\mathbb{C}[x]/(x^2))}\right)$ is different from $\left(\operatorname{Spec}(\mathbb{C}[x]/(x)), \mathcal{O}_{\operatorname{Spec}(\mathbb{C}[x]/(x))}\right)$, let us first label:

$$(X, \mathcal{O}_X) = \left(\operatorname{Spec}\frac{\mathbb{C}[x]}{(x^2)}, \mathcal{O}_{\operatorname{Spec}\mathbb{C}[x]/(x^2)}\right)$$

Consider the image of a polynomial $f(x)$ (i.e. an element in the global section) in $\mathbb{C}[x]/\left(x^2\right)$. For example if $\overline{f(x)} = \overline{(x-a)^2} \in \mathbb{C}[x]/\left(x^2\right)$, then:

$$\bar{f} = \overline{(x-a)^2} = \overline{x^2 - 2ax + a^2} = \overline{-2ax + a^2} \in \mathbb{C}[x]/\left(x^2\right) = \mathcal{O}_X(X)$$

Evaluating at the single point $[(x)] = \epsilon$ we get

$$\bar{f}(\epsilon) = a^2$$

showing that evaluating at ϵ acts like plugging in zero. as x is a function, we get:

$$x(\epsilon) = 0$$

even though x is *not* the zero-function. In fact, one very important algebraic fact that is "obvious" for traditional functions but is not necessarily the case for ring with nilpotents is that a function that is zero everywhere will have to be zero:

Lemma 2.2.6: Nilpotence and Nonzero "Zero" Functions

Let R be a ring with nilpotence (i.e. a nonreduced ring). Then $\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R))$ contains a function that evaluate to 0 everywhere but is not zero.

[13]Note that this is a topological isomorphism, these are not scheme isomorphic as we shall see in section 3.1.1!

***Proof*:**
Let $f \in R$ be a nilpotence element so that $f \neq 0$ but $f^n = 0$. Then f is contained in each prime ideal, for $ff^{n-1} = f^n = 0 \in \mathfrak{p}$, and so either $f, f^{n-1} \in \mathfrak{p}$, which we can then reduce to having $f \in \mathfrak{p}$. Hence it is in the Nilradical, and so as a function

$$f(\mathfrak{p}) = 0$$

However $f \neq 0$, as we sought to show.

For completeness, we register this result whose proof was outlined in example 2.1.23(8). As it turns out, the failure of a regular function to be determined by its evaluation at points can completely be put at the feet of nilpotents:

Proposition 2.2.7: Spectrum And Nilpotent Elements

Let R be a ring and I an ideal of nilpotent elements. Then

$$\operatorname{Spec} R/I \to \operatorname{Spec} R$$

is a homeomorphism. As a consequence, two regular functions (elements of R) have the same value if they differ by a nilpotent element.

Thus, as topological spaces, nilpotents have no effect. Their presence is felt within the sheaf we put on spectra (see section 2.2.1)

***Proof*:**
Hint: recall that the nilradical is equal to the intersection of all prime elements (see [10]).

Hence if the nilradical is zero, $\sqrt{(0)} = (0)$, functions (in global sections of affine schemes) are determined by their points[14]! In the new language we developed, this say that a function vanishes at every point of an affine scheme if and only if it is nilpotent! Note how this contrasts to the case of (classical) varieties where we required k to be algebraically closed for the same result, showing the greater generality achieved by having the extra prime points!

How to then visualize the Spec (R) along with its structure sheaf? It is usually done by picturing a "cloud" or "fuzz" around the point:

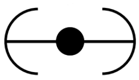


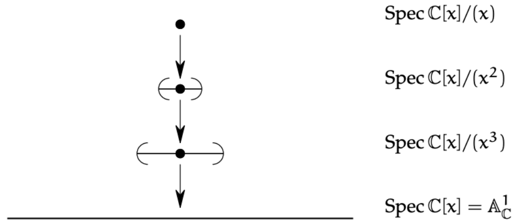
Figure 2.3: Picture of $(X, \mathcal{O}_X)$, not just $X = \operatorname{Spec}(R)$

This can be thought of as the "first degree germ" information around the point (recall that in $\mathbb{C}[\epsilon]$, $f(a+\epsilon) = f(a) + f'(a)\epsilon$, showing that the output of f at a plus the "fudge" of ϵ gives $f(a)$ plus an infinitesimal amount in the direction $f'(a)$. The sequence of restrictions $\mathbb{C}[x] \to \mathbb{C}[x]/(x^2) \to \mathbb{C}[x]/(x)$ should be thought of as inching closer to $f(0)$:

[14] Proposition 2.4.5 shall push this result to its maximal generalization on quasicompact schemes

$$
\begin{tikzcd}
\mathbb{C}[x] \arrow[r, two heads] & \mathbb{C}[x]/(x^2) \arrow[r, two heads] & \mathbb{C}[x]/(x) \\
f(x) \arrow[rr, mapsto] & & f(0),
\end{tikzcd}
$$

The reason the point is visualized to have the fuzz is that it is the point that creates this "effect" when a function evaluates at this point, in other words remembering the functions behaviour changes the geometric interpretation of the point. If we take, $\mathbb{C}[x]/(x^3)$, then then the spectrum will again be a single point, and when finding an element from the global section, the value, 1st derivative, and 2nd derivative will be remembered. This can either be thought of a larger fuzz or a fuzz that is "bending" with more degree's of precision. We can have the following chain of inclusions:



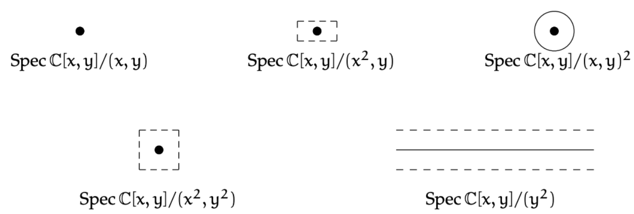
Going to higher dimensions and considering:

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(x,y)^2}\right)$$

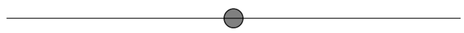
this can be thought of as remembering the derivative in all directions, and can be represented by a circle.

Example 2.2.8: Geometric Interpretation of Nilpotence

1. The reader should stare at this image to make sure it makes sense:



2. consider,
$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(y^2,xy)}\right)$$
Then this can be thought of as the intersection of a thickened x-axis and the "+" created by xy, that is this can be thought of as the x-axis as well as the first-order differential information around the origin.



Once we introduce stalks in section 2.2.5, these points will be more detectable. Notice that $[(x)]$ is *no longer* a generic point since $y^2 = 0 \in \overline{(x)}$ but $y \notin \overline{(x)}$, and hence the ideal is not prime, thus no longer a point. On the other hand, the function y when evaluating at the points $[\overline{(x-a, y-b)}] = [\overline{(x-a,y)}]$ is 0 at each point

3. Consider
$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(y^2)}\right)$$
By proposition 2.1.10, this is homeomorphic to $\mathbb{A}^1_{\mathbb{C}}$. As this is a rather simple example, this can be seen concretely: if $\mathfrak{p} \subseteq \mathbb{C}[x,y]/(y^2)$ is prime and contains y^2, then by primality $y \in \mathfrak{p}$, and hence the prime ideals correspond to those of $\mathbb{C}[x,y]/(y^2,y) \cong \mathbb{C}[x]$, i.e. those of $\mathbb{A}^1_{\mathbb{C}}$.

Now, in $\mathbb{C}[x,y]/(y^2)$, there cannot be y-terms of degree higher than 1: if $F(x,y) \in \mathbb{C}[x,y]/(y^2)$, then:
$$F(x,y) = f(x) + g(x)\,y$$
This can be thought of as $F(x,y)$ "remembering" some first-order information in the y-direction, as though it were the restriction of a function on $\mathbb{A}^2_{\mathbb{C}}$ to an infinitesimal neighborhood of the x-axis. In this decomposition, $f(x)$ is the value on the x-axis and $g(x)$ plays the role of $\partial F/\partial y|_{y=0}$ – a tangent direction at each point. Algebraic operations respect this: for instance,
$$(f_1 + g_1 y)(f_2 + g_2 y) = f_1 f_2 + (f_1 g_2 + f_2 g_1)\,y$$
which mirrors the product rule for derivatives.

Another behaviour that may seem strange is the addition of new units! Consider $1 + xy$. Since y is nilpotent ($y^2 = 0$), the element xy is also nilpotent: $(xy)^2 = x^2y^2 = 0$. We can verify directly that $1 + xy$ is a unit:
$$(1+xy)(1-xy) = 1 - (xy)^2 = 1 - 0 = 1$$
More generally, an element $f(x) + g(x)\,y$ is a unit in $\mathbb{C}[x,y]/(y^2)$ if and only if $f(x)$ is a unit in $\mathbb{C}[x]$, i.e. $f(x) \in \mathbb{C}^*$ is a nonzero constant. This is because the nilpotent part $g(x)\,y$ can always be "absorbed" by the geometric series trick $(1+n)^{-1} = 1 - n$ when $n^2 = 0$, but the non-nilpotent part $f(x)$ must already be invertible.

Geometrically, this means the unit group of $\mathbb{C}[x,y]/(y^2)$ is *much larger* than that of $\mathbb{C}[x]$. For $\mathbb{C}[x]$, the only units are the nonzero constants $\mathbb{C}^*$. For the "fuzzy line" $\mathbb{C}[x,y]/(y^2)$, every element $a + g(x)\,y$ with $a \in \mathbb{C}^*$ is a unit – regardless of the choice of $g(x)$. The unit group is thus $\mathbb{C}^* \times \mathbb{C}[x]$, an infinite-dimensional group. All of these units evaluate to the constant a at every point (since y maps to 0 in every residue field), yet they are genuinely distinct elements of the ring, distinguished by their "tangent data" $g(x)$. In this sense, $1+g(x)\,y$ is an invertible function that is infinitesimally close to 1: it equals 1 on the underlying topological space but carries a nontrivial first-order perturbation. This is the multiplicative counterpart of the earlier observation that y is a nonzero function that evaluates to 0 everywhere – here, $1+g(x)\,y$ is a non-constant unit that evaluates to the *same* constant everywhere.

Observe that $D(x)$ is the "punctured fuzzy line": it removes the point $[(x)] \in \operatorname{Spec}(\mathbb{C}[x,y]/(y^2))$ but retains the nilpotent thickening at every remaining point. On the other hand, $D(y)=\emptyset$, since every prime ideal contains y (as $y^2 = 0 \in \mathfrak{p}$ implies $y \in \mathfrak{p}$ by primality). Hence the underlying topological space $|\operatorname{Spec} R|$ does not detect the nilpotent "germ" information carried by y – but the structure sheaf does, as y is a nonzero section of $\mathcal{O}_X$ that vanishes at every point.

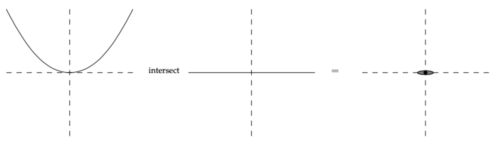
Intersection Theory and Family of Schemes

Intersection theory is covered extensively in *Intersection Theory* [16]. The intersection theory of curves as covered in [57, chapter 21.6] gives insightful ways in which the discussion above has practical applications. Recall that the multiplicity information of a curve is defined as an equivalence class of elements of $k[x,y]$ up to scaling. This was to distinguish f and f^2 in $k[x,y]$, something varieties cannot do in classical algebraic geometry. Schemes naturally capture multiplicity information.

For example, consider the following:

$$\begin{aligned}\operatorname{Spec}(\mathbb{C}[x,y]/(y-x^2)) \cap \operatorname{Spec}(\mathbb{C}[x,y]/(y)) &= \operatorname{Spec}(\mathbb{C}[x,y]/(y-x^2,y))\\ &= \operatorname{Spec}(\mathbb{C}[x,y]/(y,x^2))\end{aligned}$$

That is, intersecting the parabola $y=x^2$ with the x-axis produces a fuzzy point that remembers the multiplicity:



Let us push this idea further and identify the local subscheme associated to a curve passing through the origin with a given tangent direction. Let $R = k[\epsilon]/(\epsilon^2)$ be the dual numbers, and consider a surjection:

$$\varphi : k[x,y] \to R$$

that sends the origin to the unique point of $\operatorname{Spec} R$, i.e. $\varphi(x), \varphi(y) \in (\epsilon)$. The most general such map sends $x \mapsto b\epsilon$ and $y \mapsto -a\epsilon$ for some $a,b \in k$ not both zero (different choices of (a,b) up to scaling give different tangent directions). As $\epsilon^2 = 0$, the map φ vanishes on $(x,y)^2 = (x^2, xy, y^2)$, and hence φ factors through:

$$\overline{\varphi} : \frac{k[x,y]}{(x^2,xy,y^2)} \to R$$

As R is 2-dimensional while $k[x,y]/(x^2, xy, y^2)$ is 3-dimensional (with basis $\{1, x, y\}$), the map $\overline{\varphi}$ has a 1-dimensional kernel. One can check that this kernel is generated by $ax + by$ (the linear form vanishing in the chosen tangent direction), and hence we get the isomorphism:

$$X_{a,b} = \operatorname{Spec} \frac{k[x,y]}{(x^2, xy, y^2, ax + by)} \cong \operatorname{Spec} R$$

How can we interpret this scheme?

- Algebraically, we can interpret $X_{a,b} \subseteq \mathbb{A}^2_k$ as the subscheme associated to the ideal containing all functions $f \in k[x,y]$ that vanish at the origin *and* have partial derivatives satisfying:

  $$b \left.\frac{\partial f}{\partial x}\right|_0 - a \left.\frac{\partial f}{\partial y}\right|_0 = 0$$

  This says that the gradient of f at the origin is proportional to (a, b), or equivalently, $f = c(ax + by) + g(x, y)$ where $g(x,y) \in (x,y)^2$ consists of higher-order terms.

- Geometrically, $X_{a,b}$ is the first-order neighborhood of the origin along the line $ax + by = 0$. It arises as the composite

  $$\operatorname{Spec} \frac{k[t]}{(t^2)} \hookrightarrow \mathbb{A}^1_k \xrightarrow{t \mapsto (bt, -at)} \mathbb{A}^2_k$$

  which embeds a "fuzzy point" into $\mathbb{A}^2$ along the tangent direction $(b, -a)$.

We interpret $X_{a,b}$ as containing the point $(0,0)$ together with the infinitesimally near information in the direction specified by the line $ax + by = 0$ (the "tangent-line" information!). Such a scheme arises naturally when intersecting two curves non-transversally (the example with the parabola and the line is the simplest such case).

Another place where such schemes naturally arise is when looking at the *limit of families of schemes*. To understand this, take the case of two points $(0,0), (a,b) \in \mathbb{A}^2_k$. These two points together form a scheme:

$$X_1 = \operatorname{Spec} R = \operatorname{Spec} \frac{k[x,y]}{(x,y) \cap (x-a,\, y-b)} = \operatorname{Spec} \frac{k[x,y]}{(x^2 - ax,\, xy - bx,\, xy - ay,\, y^2 - by)}$$

It should not be hard to see (e.g. via the Chinese Remainder Theorem) that $R \cong k \times k$.

Now, suppose (a,b) moves towards $(0,0)$ along a polynomial curve. In particular, say there are two polynomial functions $a(t), b(t)$ such that $(a(0), b(0)) = (0,0)$. Let $X_t = \{(0,0),\, (a(t), b(t))\}$. Then what scheme would we expect to get when $t \to 0$? The advantage of using schemes for intersection theory, as outlined above, suggests that this limit should remember what direction the points came from – and indeed it does. Even more than this, the limit shall still be a length-2 scheme: two points that have "collided" but retained a memory of their approach direction.

Let us elucidate the details. Take

$$I_t = (x,y) \cap (x - a(t),\, y - b(t)) = (x^2 - a(t)x,\ xy - b(t)x,\ xy - a(t)y,\ y^2 - b(t)y)$$

As $t \to 0$, the ideal I_0 must contain x^2, xy, and y^2. Furthermore, I_t always contains a linear form:

$$a(t)y - b(t)x = (xy - b(t)x) - (xy - a(t)y)$$

Writing $a(t) = a_1 t + a_2 t^2 + \cdots$ and $b(t) = b_1 t + b_2 t^2 + \cdots$ (since $a(0) = b(0) = 0$), we can divide by t to obtain an element of I_t for $t \neq 0$:

$$\frac{a(t)y - b(t)x}{t} = a_1 y - b_1 x + t \cdot (a_2 y - b_2 x + \cdots)$$

Taking $t \to 0$, the limit ideal contains:

$$(x^2,\, xy,\, y^2,\, a_1 y - b_1 x) \subseteq I_0$$

These are in fact equal. To see this, note that the quotient $k[x,y]/I_t$ is a k-vector space of dimension 2 for each $t \neq 0$ (since I_t cuts out two distinct points). By a continuity argument (in the sense of flat limits), the limiting quotient $k[x,y]/I_0$ also has dimension 2[15]. Since the quotient by the left-hand side $(x^2, xy, y^2, a_1 y - b_1 x)$ already has dimension 2 (with basis $\{1, b_1 x\}$ if $b_1 \neq 0$, or $\{1, a_1 y\}$ if $a_1 \neq 0$), the inclusion must be an equality. Labelling $\alpha = a_1$, $\beta = b_1$ and $X_0 = X_{\alpha,\beta}$, we may write:

$$\lim_{t \to 0} X_t = X_{\alpha,\beta}$$

In higher dimensions, things get more complicated, and when our schemes are no longer local even more so. We shall see in section 6.8 that what we have been examining is a *flat family* of schemes. Flatness provides the correct conditions for a parameterized family of schemes to have well-behaved limits, generalizing the "continuity" argument used above.

Exercise 2.2.1

1. Show that for any polynomial $f \in k[x]$, if $a + b\epsilon \in \mathbb{C}(\epsilon)$, then

   $$f(a + b\epsilon) = f(a) + bf'(a)\epsilon$$

2. let k be algebraically closed, R be a local k-algebra of dimension 2 with maximal ideal $\mathfrak{m}$. Show that $R \cong k[x]/(x^2)$ (hint: use some dimension argument and Nakayama's lemma to conclude $\mathfrak{m}^2 = 0$).

3. Let k be algebraically closed. Find two non-isomorphic local k-algebras of dimension 3. Geometrically, we can think of this as representing the extra flexibility in directions or local schemes can capture.

4. Let k be algebraically closed and $\operatorname{Spec} R = \operatorname{Spec} k[x_1, ..., x_n]/I \subseteq \mathbb{A}^n_k$ be a local subscheme of dimension 0 and degree 3 (R has dimension 3 over k) supported at the origin. Then $\operatorname{Spec} R$ is either isomorphic to $\operatorname{Spec} k[x]/(x^3)$ or $\operatorname{Spec} k[x,y]/(x^2, xy, y^2)$, and these two schemes are not isomorphic to each other.

5. Let R be a ring that is not a field. Show that the restriction map on the structure sheaf is almost never surjective. Hence, there are more local functions than global function.

6. Let R be a reduced ring (contains no nilpotent elements). Show that the restriction maps are injective, that is each function on a larger open set can only restrict to one function. Give a counter-example for a ring with nilpotent elements.

[15]This argument is more subtle without the finite-dimensionality hypothesis, which shall be addressed when we cover this in fuller generality in section 6.8.

2.2.3 Schemes

The last step in the definition of schemes is to observe that many geometric objects are not the spectrum of a ring, but are *locally* the spectrum of a ring. The prototypical examples are projective spaces, their closed subsets, and sequences of blow ups—none of which are in general affine.

Definition 2.2.9: Scheme

A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ that is locally affine: there exists an open cover $X = \bigcup_i U_i$ such that for each i, there is a ring R_i and an isomorphism of locally ringed spaces

$$(U_i,\ \mathcal{O}_X|_{U_i}) \cong_{\mathbf{LRS}} (\operatorname{Spec}(R_i),\ \mathcal{O}_{\operatorname{Spec}(R_i)}).$$

Such a cover is called an *affine open cover* of X.

The requirement that $(X, \mathcal{O}_X)$ be a locally ringed space—that is, that each stalk $\mathcal{O}_{X,x}$ be a local ring—is essential. Without it, the "evaluation" of a section at a point (via the residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$) would not be well-defined, and the connection to geometry would be lost. We saw in section 1.4 that locally ringed spaces are the correct generalization of the spaces that arise in both differential and algebraic geometry. Since the stalk of $\mathcal{O}_X$ at a point $x \in X$ is a local ring (by the locally ringed space condition), we denote it $\mathcal{O}_{X,x}$, its maximal ideal $\mathfrak{m}_x$, and its residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$.

The topology of affine schemes is generated by distinguished open sets. Lemma 2.2.29 shall show that given $\operatorname{Spec} R, \operatorname{Spec} S \subseteq X$ where their intersection is non-empty, the intersection can be covered by distinguished open sets that are distinguished on both $\operatorname{Spec} R$ and $\operatorname{Spec} S$. Hence the topology of schemes is also generated by distinguished open sets. See section 2.8 for a deeper discussion about the purely topological properties of schemes.

The way to interpret the agreement on intersections is that locally a scheme respects the algebraic relations imposed by a ring R_i, and the compatibility on $U_i \cap U_j$ (when nonempty) corresponds to the algebraic relations between R_i and R_j being consistent. A detailed example is presented in example 2.2.55.

2.2.10 Lack of $V(-)$ and $I(-)$? The topology of an affine scheme is characterized by the ideals of its global ring. General schemes are no longer tethered to the global sections (as shall be momentarily demonstrated). However, there is a generalization of $V(-)$ and $I(-)$ that works with the (quasi-coherent) *ideal sheaves* of a scheme X; the details are covered in section 5.1.4. From this perspective, the structure sheaf $\mathcal{O}_X$ can be thought of as generalizing the notion of a ring to a "local" object, remembering information that no single ring can capture.

Definition 2.2.11: Morphism of Schemes

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be schemes. Then a *morphism* of schemes is a morphism of locally ringed spaces $(\varphi, \varphi^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$.

A *isomorphism of schemes* is an isomorphism of locally ringed spaces: a homeomorphism $f : X \to Y$ together with a sheaf isomorphism $f^\# : \mathcal{O}_Y \xrightarrow{\sim} f_*\mathcal{O}_X$.

Not every morphism of ringed spaces preserves local rings (see example 3.1.1). The full exploration of the nuances shall be covered in section 3.1.1.

Example 2.2.12: Schemes

1. All affine schemes are schemes, covered by a single open set.
2. The key non-affine scheme is projective space $\mathbb{P}^n_k$. We shall construct the general projective scheme $\mathbb{P}^n_R$ in section 2.3; for now, the reader can verify that the usual projective space as defined in [20] is a scheme by exhibiting the standard affine charts $U_i = \{x_i \neq 0\} \cong \mathbb{A}^n_k$. Any closed subvariety of $\mathbb{P}^n_k$ (with the appropriate structure sheaf) is also a scheme, typically not affine.
3. The following is the smallest non-affine scheme. Take $X = \{p, q_1, q_2\}$ with the topology generated by $X_1 = \{p, q_1\}$ and $X_2 = \{p, q_2\}$ as a sub-base. Define the sheaf of rings by
$$\mathcal{O}(X) = \mathcal{O}(X_1) = \mathcal{O}(X_2) = K[x]_{(x)}, \qquad \mathcal{O}(\{p\}) = K(x),$$
with restriction maps $\mathcal{O}(X) \to \mathcal{O}(X_i)$ the identity and $\mathcal{O}(X_i) \to \mathcal{O}(\{p\})$ the natural localization. This presheaf is a sheaf, and $(X, \mathcal{O})$ is a scheme but *not* an affine scheme: the ring $K[x]_{(x)}$ has two points in its spectrum, but X has three. Geometrically, this is the "germ of a doubled point"—the generic point of $\mathbb{A}^1$ has been split into two.
4. By proposition 2.1.20, a finite disjoint union of schemes is a scheme. An arbitrary (infinite) disjoint union of affine schemes is a scheme but not an affine scheme, as it fails to be quasi-compact.
5. All affine schemes are "separated" (a property to be defined in section 3.5.1). There exist non-separated schemes; the simplest is the "line with doubled origin," constructed by gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$.
6. If the reader has already encountered blow-ups, the schemes associated to a blow-up are typically not affine.
7. The open subset $D(x) \cup D(y) \subseteq \mathbb{A}^2_k$ is a scheme that is quasi-compact, separated, and a subset of an affine scheme, yet is not itself affine. The details are in Example 2.2.51.
8. Let $X = \{*\}$ be a singleton and $\mathcal{O}_X(X) = R$ for a commutative ring R. Is $(X, \mathcal{O}_X)$ a scheme? First, it must be affine so $X = \operatorname{Spec}(R)$. Next, $\operatorname{Spec} R$ must have a single prime ideal. This must be a 0-dimensional is a local ring. For example:
$$R \in \{k, \mathbb{Z}/p^n\mathbb{Z}, k[x]/(x^m)\}$$
Note there are local rings with more than one point, for example $k[[x]]$.

It is not always immediate whether a scheme is affine. For example, $\mathbb{A}^1_{\mathbb{C}} \setminus \{[(x)]\}$ may seem non-affine since it is not an affine variety in the classical sense, but it is isomorphic to $\operatorname{Spec}(\mathbb{C}[x, x^{-1}])$. On the other hand, $\mathbb{A}^2_{\mathbb{C}} \setminus \{[(x, y)]\}$ is genuinely not affine (Example 2.2.51). We shall establish tools for detecting affineness shortly.

Given a scheme X, we write $|X|$ for the underlying topological space. The elements of $\mathcal{O}_X(U)$ are called *sections* over U, and sections of $\mathcal{O}_X(X)$ are called *global sections*. These are variously denoted

$$\mathcal{O}_X(U), \qquad \Gamma(U, \mathcal{O}_X), \qquad H^0(U, \mathcal{O}_X),$$

depending on context: the first is convenient when only the structure sheaf is present; the second will be useful when multiple sheaves on X are in play; and the third hints at the fact that the global section functor $\Gamma(X, -)$ is left-exact but generally not exact, with its right-derived functors $H^i(X, -)$ forming the cohomology theory developed in chapter 6.

Recovering the ring from an affine scheme

A natural question arises: if a scheme X happens to be affine, can we recover the ring R from the locally ringed space $(X, \mathcal{O}_X)$? The answer is yes, and the ring is simply the global sections.

Proposition 2.2.13: Distinguished Open Sets and Schemes

Let R be a ring and $f \in R$. Then:

$$(D(f),\ \mathcal{O}_{\operatorname{Spec}(R)}|_{D(f)}) \;\cong\; (\operatorname{Spec}(R_f),\ \mathcal{O}_{\operatorname{Spec}(R_f)}).$$

In particular, taking $f = 1$: if $(X, \mathcal{O}_X)$ is an affine scheme, meaning $(X, \mathcal{O}_X) \cong (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ for some ring R, then R is recovered as the global sections:

$$\mathcal{O}_X(X) \;\cong\; R.$$

***Proof*:**
The first statement is the scheme-theoretic upgrade of Proposition 2.1.12: the homeomorphism $D(f) \cong \operatorname{Spec}(R_f)$ extends to an isomorphism of locally ringed spaces because the structure sheaf on $D(f)$ is defined by the same localization data as $\mathcal{O}_{\operatorname{Spec}(R_f)}$.

For the recovery of R: since X is affine, $X = D(1) = \operatorname{Spec}(R)$, and

$$\mathcal{O}_X(X) = \mathcal{O}_X(D(1)) = R_1 \cong R,$$

where the middle equality is the definition of the structure sheaf on $\operatorname{Spec}(R)$.

The motivation behind this proposition is that for affine schemes, the global sections completely determine the scheme. This is emphatically *not* the case for general schemes: the global sections of $\mathbb{P}^n_k$ are just k (for $n \geq 1$), which carries no information about the geometry of projective space. The structure sheaf, not the global sections, is the fundamental invariant.

Proposition 2.2.14: Characterization of Affine Schemes

Let $(X, \mathcal{O}_X)$ be a locally ringed space with $R = \mathcal{O}_X(X)$. Then $(X, \mathcal{O}_X) \cong (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ if and only if:

1. For every $f \in R$, $\mathcal{O}_X(U_f) \cong R_f$, where $U_f = \{x \in X : f \notin \mathfrak{m}_x\}$.
2. The natural map $X \to |\operatorname{Spec}(R)|$ (sending x to the prime $\mathfrak{m}_x \cap R$) is a homeomorphism.

***Proof*:**
Exercise. (The content is that conditions (1) and (2) together give an isomorphism of locally ringed spaces, not of topological spaces or of ringed spaces.)

Theorem 2.2.15: Ring Homomorphisms and Affine Scheme Isomorphisms

Let R and S be rings. An isomorphism of locally ringed spaces $\pi : \mathrm{Spec}\,(R) \to \mathrm{Spec}\,(S)$ induces an isomorphism $\pi^{\#} : S \xrightarrow{\sim} R$ on global sections. Conversely, an isomorphism $\varphi : S \xrightarrow{\sim} R$ induces an isomorphism $\mathrm{Spec}\,(\varphi) : \mathrm{Spec}\,(R) \xrightarrow{\sim} \mathrm{Spec}\,(S)$, and these two constructions are inverse to each other.

***Proof*:**
Corollary of theorem 3.1.4.

Given a scheme X, the collection of distinguished open sets from any affine cover forms a basis for the topology on X. This is important because the intersection of two affine opens need not itself be affine (see section 3.5.1), but distinguished opens within each affine chart suffice.

Proposition 2.2.16: Distinguished Open Sets Form a Basis

Let X be a scheme and $\{U_i = \mathrm{Spec}\,(R_i)\}$ an affine open cover. Then the collection of all distinguished open sets $D(f) \subseteq U_i$, as i ranges over the index set and f ranges over R_i, forms a basis for the topology on X.

***Proof*:**
Every open subset of X restricts to an open subset of each U_i, which is a union of distinguished opens in U_i (since distinguished opens form a basis for the Zariski topology on $\mathrm{Spec}\,(R_i)$). The union over all i covers the original open set.

Finiteness conditions and base ring

In algebra, every ring is a $\mathbb{Z}$-algebra, and additional structure arises by specifying a base ring. The scheme-theoretic counterpart is the following.

Definition 2.2.17: R-scheme and Finite Type Schemes

An *R-scheme* (or a *scheme over R*) is a scheme X together with a morphism $X \to \mathrm{Spec}\,(R)$, called the *structure morphism*. Equivalently, the sections of $\mathcal{O}_X$ are R-algebras and all restriction maps are R-algebra homomorphisms.
An R-scheme X is *locally of finite type over R* if it can be covered by affine opens $\mathrm{Spec}\,(S_i)$ where each S_i is a finitely generated R-algebra. If X is also quasi-compact, then X is *of finite type over R*.

In number theory, integral extensions S/R (where S is a finitely generated R-module) play a central role. The scheme-theoretic analogue is:

Definition 2.2.18: Finite R-Schemes

An R-scheme X is *locally finite over* R if it can be covered by affine opens $\operatorname{Spec}(S_i)$ where each S_i is a finitely generated R-module. If X is quasi-compact, then X is *finite over* R.

If $R = k$ is a field (or more generally an Artinian ring), then a finite R-scheme is a finite collection of "fat points"—spectra of Artinian local k-algebras.

2.2.19 Terminology Warning Do not confuse *finite type* with *finite*. A scheme is locally of finite type if its coordinate rings are finitely generated R-*algebras* (finitely many generators as a ring); it is locally finite if its coordinate rings are finitely generated R-*modules* (finitely many generators as a module). The latter is a much stronger condition: $k[x]$ is a finitely generated k-algebra but not a finitely generated k-module.

Example 2.2.20: Finite Type R-schemes

1. Affine n-space $\mathbb{A}^n_k = \operatorname{Spec}(k[x_1, \ldots, x_n])$ is of finite type over k.
2. For polynomials $f_1, \ldots, f_m \in R[x_1, \ldots, x_n]$, the scheme $\operatorname{Spec}(R[x_1, \ldots, x_n]/(f_1, \ldots, f_m))$ is of finite type over R. Most classical intuitions from algebraic sets concern $\overline{k}$-schemes of finite type over $\overline{k}$ (abstract varieties shall require further properties; see section 3.7).
3. If R is the localization of a polynomial ring over k at a prime, then $\operatorname{Spec}(R)$ is generally not of finite type over k. Intuitively, local rings live in a "different dimension" from the global picture.
4. If the fibers of $X \to \operatorname{Spec}(R)$ are infinite-dimensional, then X is not of finite type over R.

2.2.4 Comparing Schemes and Real/Complex Manifolds

Before proceeding to further explore properties of schemes, it is worth taking a moment to catalogue how schemes and real smooth manifolds have some fundamentally different geometric properties:

1. The connected scheme given by $V(x) \cup V(yz) \subseteq \mathbb{A}^3_k$ has two irreducible components of different dimensions (as we shall more rigorously show in section 2.6). A connected manifold (smooth or even topological) is of a single, well-defined dimension at every point.
2. An abstract manifold cannot self-intersect[16], while schemes (even affine schemes, even varieties) can: consider $V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2_k$, which has a node at the origin.
3. The patches of a manifold are very regular, being diffeomorphic to $\mathbb{R}^n$; in this sense manifold patches all carry "linear smooth" information, and there is essentially no local diversity from one patch to another. The affine patches of a scheme, by contrast, may be "pathological"—for example, the local rings can be non-reduced or non-Noetherian (in the latter case there may even be uncountably many connected components). This reflects how affine schemes represent

[16] An *immersion* of a manifold may of course have self-intersections in the target, but the source manifold itself is not singular at such points.

"local algebraic information," which is far richer than the "local smooth linear information" that real manifolds encode. Some of the conditions and properties we shall introduce later (e.g. Noetherianity, regularity) will restore some order.

4. In Manifold Theory, the notion of irreducible component is trivial, since in any Hausdorff space the only irreducible subsets are singletons. From an analytic perspective, this existence of non-trivial irreducible components in schemes can be thought of as a consequence of the failure of the Zariski topology to separate points by functions; part of this phenomenon is the existence of "large" (i.e. non-closed) points for schemes. More generally, the correspondence between irreducible closed sets and non-closed points (in at least Noetherian schemes) shows that schemes encode much more "global" information about codimension subsets than real smooth manifolds do.

5. The topology of $\mathbb{R}^n$ (and hence of $\mathbb{C}^n$ as a real manifold) is the product topology: the sets

$$\prod_{i=1}^{n}(a_i, b_i), \qquad a_i, b_i \in \mathbb{R},\ a_i < b_i,$$

form a topological basis for $\mathbb{R}^n$. This is far from the case for $\mathbb{A}^n_k$: the Zariski topology on $\mathbb{A}^n_k$ is *not* the product of the Zariski topologies on $\mathbb{A}^1_k$ (indeed, the product would be the cofinite topology, whose closed sets are always finite unions of points, whereas $\mathbb{A}^2_k$ has closed curves). However, there will still be a meaningful way to construct the scheme $\mathbb{A}^n_k$ from $\mathbb{A}^1_k$ via the *fibered product* (see section 3.2).

6. If a scheme X is irreducible, all its nonempty open sets are dense. Many of the schemes we shall work with will be unions of irreducible subschemes, where for each irreducible subscheme every nonempty affine open subset is dense in that subscheme. Dense open subsets are far from typical for manifolds (e.g. a disjoint union of two manifolds has open subsets contained in a single component, which are not dense).

7. More generally, the behaviour of schemes matches more closely with the behaviour of complex analytic geometry than with smooth real geometry. We shall draw such analogies often: for example, *Hartogs's Extension Lemma* ([52, chapter 8]) will have parallels with normal schemes, the existence of certain holomorphic/meromorphic functions will tie into divisor theory, compact complex Riemann surfaces will correspond to projective curves, and so forth.

8. (embedding properties) Any smooth real manifold M can be embedded into $\mathbb{R}^N$ for appropriate N (Whitney Embedding Theorem). For classical affine (resp. projective) varieties of dimension n, they essentially by definition embed into $\mathbb{A}^N_k$ (resp. $\mathbb{P}^N_k$) for some N. For complex manifolds, it is *not* the case that they all embed in $\mathbb{C}^N$[17], and in particular no compact complex manifold of positive dimension embeds into $\mathbb{C}^N$. This is because complex manifolds possess much more rigidity than real manifolds [14][18]. Schemes exhibit this same type of rigidity. The only types of schemes that have a universal embedding theorem into one of the classical spaces $\mathbb{A}^N_k$ or $\mathbb{P}^N_k$ are curves (see **??**) which embed into $\mathbb{P}^3$. the category of all surfaces already fail to have such a universal embedding theorem (by example **??**).

[17] If a compact complex manifold does embed holomorphically, it must reduce to a point, since any global holomorphic function on a compact complex manifold is constant. Non-compact complex manifolds that embed holomorphically in $\mathbb{C}^N$ are called *Stein manifolds* and share many properties of quasi-projective varieties.

[18] For instance, the complex torus $\mathbb{C}/\Lambda$ depends on the lattice Λ up to biholomorphism, giving a one-complex-parameter family of non-isomorphic structures.

By the Kodaira Embedding Theorem [14], a compact Kähler manifold with a positive line bundle can be embedded into projective space. Proper schemes (the algebraic analogue of compact complex manifolds) with an *ample line bundle* (the algebraic analogue of a positive line bundle) similarly embed into projective space. In fact, this means that compact Kähler manifolds with positive line bundle are (as complex analytic spaces) isomorphic to a smooth proper scheme with an ample bundle. This is the closest to a general embedding theorem that is widely used, and will be proven in section 5.3.3.

For a closed immersion $X \hookrightarrow \mathbb{A}^n_R$ so that $X \cong V \subseteq \mathbb{A}^n_R$ where V is a closed subscheme, X is topologically a sub-variety, though it may have a more complicated sheaf structure (for more details see section 3.7).

For an open embedding $X \hookrightarrow \mathbb{A}^n_R$ so that $X \cong U \subseteq \mathbb{A}^n_R$ where U is open, X is a *quasi-affine subscheme*, which by definition is an open subscheme of affine space. Their characterization is more subtle: the map $X \to \operatorname{Spec}(\mathcal{O}_X(X))$ must be an open embedding and, as will be discussed in section 5.1.1, all quasi-coherent sheaves of $\mathcal{O}_X$-modules must be generated by global sections. Verifying that every $\mathcal{O}_X$-module is generated by global sections is difficult; there is a simpler characterization given by the Goodman–Hartshorne Theorem [34], but it is still a nontrivial condition, illustrating the greater complexities that arise when working with non-proper schemes.

More generally, much later in ref:HERE, there shall be other interesting ambient spaces into which certain types of schemes may be embedded (e.g. toric varieties, Grassmannians, flag varieties, etc.), each serving different purposes depending on the moduli problem under study.

9. Schemes can have "doubled points": recall that the line with two origins is a standard counterexample for manifolds, as it fails to be Hausdorff at the origin. In scheme theory the analogous phenomenon is permitted; indeed, many moduli spaces that are schemes are naturally non-separated schemes (see section 3.3.6).

10. A scheme remembers multiplicity. For example, on a classical point there is little room for interesting functions, but scheme-theoretic functions (i.e. regular functions on $\operatorname{Spec} k[x]/(x^2)$) can encode multiplicity and infinitesimal information. A smooth manifold $(M, \mathcal{A})$ with its maximal atlas does not retain such data.

11. There can be multiple structure sheaves on the same underlying topological space, defining genuinely different schemes (think of the different fields that can serve as the structure sheaf over a single point). In contrast, manifolds of dimension ≤ 3 have unique differential structures up to diffeomorphism (by work of Radó in dimension 2 and Moise in dimension 3); in fact, the categories of piecewise-linear, topological, and smooth manifolds coincide in these dimensions, showing how "linear" manifold theory is in low dimensions. In section 3.2 we shall introduce the process of *base change*, by which one can change the structure sheaf without changing the underlying topological space.

12. Though we have not defined an atlas for schemes, Theorem 2.2.54 will show that schemes can be constructed by gluing affine patches, just as a manifold is built from its atlas, see section 2.2.7. (want to say why not as useful here).

13. The natural additional structure on a smooth manifold is a *tangent bundle*, whose transition functions take the form $D(\mathbf{y} \circ \mathbf{x}^{-1})$; that is, they are not just smooth maps between open subsets but must respect a $\mathrm{GL}_n(\mathbb{R})$-action given by the Jacobian. We shall define the algebraic analogue for schemes (the sheaf of Kähler differentials and the tangent sheaf) in chapter 5,

section 5.5, though more care is required since schemes are not "locally linear" everywhere in the way smooth manifolds are. Much of the intuition from Poincaré duality (see [51, section 4.3]) will inform the generalization to Serre duality.

14. Schemes can better capture different levels of *local* behaviour. For example, a germ of a smooth manifold at a point is not itself a smooth manifold, but the analogous object for a scheme—the spectrum of a local ring—is again a scheme, called a *local scheme*. This is because commutative rings faithfully encode this local information, and (affine) schemes are the geometric realization of commutative rings.

15. Scheme functions are more "rigid": on a smooth manifold, partitions of unity exist thanks to the flexibility of the Euclidean topology, but no exact analogue holds for schemes. This rigidity, however, also adds structure and sometimes simplifies definitions (e.g. closed subschemes, introduced in section 3.1.3, are a rather "natural" concept compared to the work that goes into verifying that closed submanifolds are locally diffeomorphic to $\mathbb{R}^k \subseteq \mathbb{R}^n$).

    As smooth functions on each $U \subseteq M$ form a sheaf, sheaves that admit a "partition of unity" are called *fine sheaves*. A weaker notion of *soft sheaf* will be introduced in section 6.1; this is the closest scheme-theoretic analogue of the partition-of-unity property.

16. Schemes unify "arithmetic" and "geometry" in a way that manifold theory is not equipped for. As examples, there are no manifold analogues of $\operatorname{Spec}\mathbb{Z}$, $\operatorname{Spec}\mathcal{O}_K$ (where $\mathcal{O}_K$ is a ring of integers of a number field), or $\operatorname{Spec}\mathbb{Z}[t]$. In this way, scheme theory gives geometric meaning to objects that are not tractable in differential geometry. A consequence is that different points of a single scheme can have residue fields of different characteristics (e.g. on Spec $\mathbb{Z}$, the residue field at the prime (p) is $\mathbb{F}_p$ while the residue field at the generic point is $\mathbb{Q}$)—a phenomenon with no manifold analogue. To work in a more "purely geometric" setting, one often restricts to $\overline{k}$-schemes where $\overline{k}$ denotes the algebraic closure of a field of characteristic 0. In fact, when defining Z-valued points in section 3.3.2 for a scheme Z, if $Z = \operatorname{Spec}\overline{k}$ then such points are called *geometric points*. Working over a non-algebraically closed field often signals that the scheme carries arithmetic information.

17. Schemes have non-closed ("generic") points. Such points are essential: for instance, the ring map $k[x, y] \to k(x)[y]$ (localizing away from the irreducible polynomials in x) induces a morphism Spec $k(x)[y] \to$ Spec $k[x, y]$ whose image consists of points lying over the generic point of Spec $k[x]$. We shall later see that the image of such a morphism is *dense*.

18. Though the above list highlights many differences, many of the usual geometric tools remain applicable in the study of schemes. For example, $\mathbb{C}[x, y]/(x^2 - y^3)$ is not a normal ring. This can be hard to verify algebraically, but viewing Spec $\mathbb{C}[x, y]/(x^2 - y^3)$ as a curve in $\mathbb{C}^2$ (in the Euclidean topology), the fact that it has a cusp at the origin—and hence is singular there, at a codimension-0 point of the curve—is precisely why it fails to be normal. (Normality for varieties is equivalent to regularity in codimension 1, also known as Serre's condition R_1 together with S_2.) This type of geometric reasoning can be powerful: one can study the smoothness (or more specifically the normality) of a variety to determine the integral closure of its coordinate ring.

Though this is a long list consisting mostly of differences, very often in practice (especially in the setting of k-schemes, or when studying curves, surfaces, moduli spaces, etc.) we work with schemes that are proper (the correct generalization of compact manifolds) and have many methods to "resolve" singularities (e.g. eliminate nodes, cusps, etc.) and make the resulting scheme regular[19]. In fact, as we shall see in section 6.9, a result known as *GAGA* (Géométrie Algébrique et Géométrie Analytique) shows that proper schemes over $\mathbb{C}$ with appropriate coherent sheaves are equivalent, as categories, to compact complex analytic spaces with their coherent analytic sheaves. Naturally, arithmetic schemes will not be part of this manifold-scheme correspondence.

2.2.5 Evaluation and Stalks on Schemes

In differential geometry, the stalks of a manifold can be used define the evaluation of a function. This same trick is endeavored for scheme:

Proposition 2.2.21: Stalks of Schemes

Let $\operatorname{Spec}(R)$ be a scheme. Then the stalk at $\mathcal{O}_{\operatorname{Spec}(R)}$ at the point $[\mathfrak{p}]$ is isomorphic to the local ring $R_{\mathfrak{p}}$.

$$\mathcal{O}_{\operatorname{Spec}(R),\mathfrak{p}} \cong R_{\mathfrak{p}}$$

Furthermore, for any $[\mathfrak{p}] \in X$, as $[\mathfrak{p}] \in \operatorname{Spec}(R_i)$ for appropriate R_i, we have

$$\mathcal{O}_{X,[\mathfrak{p}]} \cong \mathcal{O}_{\operatorname{Spec}(R_i),[\mathfrak{p}]}$$

***Proof*:**
For the first part, prove that $R_{\mathfrak{p}}$ is the colimit of R_f where $f \notin \mathfrak{p}$. The second part comes from proposition 1.1.5.

Definition 2.2.22: Local Scheme

Let X be a scheme. Then X is a *local scheme* if it has a unique closed points.

The scheme $(\operatorname{Spec} \mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}}, \mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}})$ is called a *local scheme* at the point $\mathfrak{p}$.

The reader should show that the local scheme at the point $\mathfrak{p}$ is the intersection of all open subsets containing $\mathfrak{p}$. Hence, local schemes are the germs of schemes. Note that as there are "points within points", namely as there are non-closed points in schemes, then there is also stalks that are subsets of stalks, in particular if $\mathfrak{p}$ and $\mathfrak{m}$ is a prime and maximal ideal respectively where $\mathfrak{p} \subsetneq \mathfrak{m} \in \operatorname{Spec} R$ so that $[\mathfrak{m}] \in \overline{[\mathfrak{p}]}$, then $\mathcal{O}_{X,\mathfrak{p}} \cong R_{\mathfrak{p}}$ is a localization of $\mathcal{O}_{X,\mathfrak{m}} \cong R_{\mathfrak{m}}$. Due to this, notice that the local schemes in higher dimensions (section 2.6) can have local schemes contained within them. For example, the scheme $\operatorname{Spec} k[x,y]_{(x,y)}$ is the local scheme at the origin, it contain a single closed and a non-closed point for every irreducible curve in $\mathbb{A}^2_k$. Notice too how the local schemes are still schemes, as compared to the collection of germs at a point along with the point is not considered to be a manifold.[20]

[19] Though as we shall see, there is no universal resolution of singularities in all characteristics and dimensions. Resolution has been completely proved in characteristic 0 (Hironaka) and in low dimensions, but in positive characteristic the general case remains open. Moreover, the singularity information itself is sometimes important and should not always be discarded.

[20] In fact, germs not being manifolds can be seen as a lacking property in the category of manifolds as this shows manifolds are not closed under filtered colimits. Synthetic differential geometry is an attempt to rectify this.

The main idea is that the single maximal ideal represents that there is one point at which we can evaluate for each germ. As all local rings have a maximal ideal, this motivates remembering the following notion form commutative algebra:

Definition 2.2.23: Residue Field

Let R be a commutative ring and $\mathfrak{p} \subseteq R$ a prime ideal. Then the ring of fractions of the integral domain domain $R/\mathfrak{p}$ is called the *residue field*, i.e. $\operatorname{Frac}(R/\mathfrak{p})$ is the residue field of $\mathfrak{p}$ (with respect to R). This field is often denoted $\kappa(\mathfrak{p})$.

The use of kappa in $\kappa(\mathfrak{p})$ is due to the fact that there can be many different residue fields at different points. For example in $\operatorname{Spec} \mathbb{R}[x]$, all residue fields are isomorphic to either $\mathbb{R}$ or $\mathbb{C}$ (depending if the localization is being taken at a non-maximal or maximal prime ideal).

Using residue fields, we can generalize the evaluation of a regular function (i.e. the evaluation of any element in the sections of $\mathcal{O}_X$). Recall that:

$$R_\mathfrak{p}/\mathfrak{m}_\mathfrak{p} \cong \operatorname{Frac}(R/\mathfrak{p})$$

as localization and taking the quotient commutes (see [10]). This makes it much easier to evaluate a function if we can determine R. Then as the valuation of a function is usually in $R/\mathfrak{p}$, as rational functions are now permitted on (proper) open subsets of X, the evaluation will be in $\operatorname{Frac}(R/\mathfrak{p})$. For a general point $x \in X$ in a scheme, we may diminish to the to define evaluation.

Definition 2.2.24: Evaluation of Function [on a Scheme]

Let X be a scheme, $U \subseteq X$ an open subset, and $f \in \mathcal{O}_X(U)$ a section. Then the value of f at the point $[\mathfrak{p}] \in U$ is the element:

$$f(\mathfrak{p}) := \bar{f} \in \mathcal{O}_{X,\mathfrak{p}}/\mathfrak{m}_\mathfrak{p} \cong \operatorname{Frac}(R/\mathfrak{p}) = \kappa(\mathfrak{p})$$

where $\operatorname{Spec}(R)$ is some affine local neighborhood of $\mathfrak{p}$.

As the quotient of any ring by a prime ideal is an integral domain, by taking its field of fraction we are forcing the codomain of each function to be a disjoint union of fields:

$$f : U \to \coprod_{\mathfrak{p}} \mathcal{O}_{U,\mathfrak{p}} \quad (\text{ and in affine case }) \quad f : U \to \coprod_{\mathfrak{p} \in U} \operatorname{Frac}(R/\mathfrak{p})$$

If $\mathfrak{p}$ is maximal, then $\operatorname{Frac}(R/\mathfrak{p}) = R/\mathfrak{m}$. For certain rings, it suffices to evaluate sections only over maximal points (namely Jacobson Schemes, see corollary 3.7.6). In the case of affine Jacobson schemes, we can consider:

$$f : U \to \coprod_{\mathfrak{p} \in U} R/\mathfrak{m}$$

Varieties will be an example of Jacobson schemes. If $R/\mathfrak{m}_i \cong R/\mathfrak{m}_j$ for all maximal ideals (which shall be shown to be the case for varieties), then we can simply write:

$$f : U \to R/\mathfrak{m} \cong k$$

which recovers the notion of evaluation from classical algebraic geometry!

2.2.25 Foreshadowing: What Type of Function? As f is defined above, it is a set function between a relatively nice space U and a bad codomain. Can this function be characterized, or represented, by a known nicer function? In fact, it shall be representable with relatively simple scheme morphisms, recovering how in classical algebraic geometry sections are regular functions; details in proposition 3.3.9.

Example 2.2.26: Evaluation On Affine Schemes

The key idea is that the stalk $R_{\mathfrak{p}}$ gives all function's that will not vanish at $\mathfrak{p}$, and hence can divide by the function at that point, and then taking the quotient by the maximal ideal evaluate the function at that point, equivalently considering the value of the rational function in $\operatorname{Frac}(R/\mathfrak{p})$:

1. Take $X = \mathbb{C}[x]$. Then each $f/g \in \mathcal{O}_X(U)$ is a rational polynomial where $g(\mathfrak{p}) \neq 0$ for all $\mathfrak{p} \in U$. For example if $U = D(x)$, then $h = \frac{x+1}{x} \in \mathcal{O}_X(D(x))$ and:

$$h(3) = \frac{3+1}{3} = \frac{4}{3} \in \mathbb{C} = \frac{\mathbb{C}[x]}{(x-3)} \cong \mathcal{O}_{X,3}/(x-3)$$
$$h(-1) = \frac{-1+1}{-1} = 0 \in \mathbb{C} = \frac{\mathbb{C}[x]}{(x+1)} \cong \mathcal{O}_{X,-1}(x+1)$$

   As the points of $\mathbb{C}[x]$ correspond to the variety, we see that the evaluation will always be in $\mathbb{C}$, and there is no need to take the field of fractions as $\operatorname{Frac}(\mathbb{C}) = \mathbb{C}$.
   Another important example, take (want the vanishing at 0 , but something feels off)

2. Take $\mathbb{A}^2_k$ for a field k where char $\neq 2$. Then in the open set U away from the y-axis and the curve $y^2 - x^5$ has in it's local section $\mathcal{O}_{\mathbb{A}^2_k}(U)$ the rational function $\left(x^2+y^2\right)/x\left(y^2-x^5\right) \in \mathcal{O}_{\mathbb{A}^2_k}(U)$. It's value at the point $(2,4)$ (i.e. $[(x-2,y-4)]$) give a values in $\operatorname{Frac}(R/(x-2,y-4)) = \operatorname{Frac}(k[x,y]/(x-2,y-4)) \cong R_{(x-2,y-4)}/\overline{(x-2,y-4)}$. Then:

$$\frac{x^2+y^2}{x\left(y^2-x^5\right)} \bmod (x-2,y-4) = \frac{2^2+4^2}{2\left(4^2-2^5\right)} \in \mathbb{C} = \frac{\mathbb{C}[x,y]}{(x-2,y-4)} \cong \mathcal{O}_{\mathbb{A}^2_k,(2,4)}$$

   The value at the point $[(y)]$, which is the generic point of the x axis, is

$$\frac{x^2+y^2}{x\left(y^2-x^5\right)} \bmod (y) = \frac{x^2}{-x^6} = -1\frac{1}{x^4} \in \mathbb{C}(x) \cong \operatorname{Frac}(\mathbb{C}[x,y]/(y)) \cong \mathcal{O}_{\mathbb{A}^2_k,[(y)]}$$

3. Take $X = \operatorname{Spec}(\mathbb{Z})$ and take the open set $U = \operatorname{Spec}(\mathbb{Z})\backslash\{(2),(7)\}$. This open set has many rational function, but let's take $27/4$. Then evaluating this function at different points of U we get:

$$(5) : 2/(-1) \equiv -2 \equiv 3 \bmod 5$$
$$(0) : 27/4$$
$$(3) : 0/1 = 0 \bmod 3$$

Notice that at 3, the function became 0. A (rational) function whose output is 0 is said to *vanish* at that point. If $\mathfrak{m}_\mathfrak{p}$ represents the maximal ideal of $\mathcal{O}_{X,\mathfrak{p}}$, then functions on an open subset U (i.e. section in $\mathscr{F}(U)$) have values at each point of U whose value lies in $\kappa(\mathfrak{p})$. A function *vanishes* at $\mathfrak{p}$ if its value at $\mathfrak{p}$ is 0. This idea is not well-defined on ringed spaces in general! There is furthermore the following properties mirroring manifold theory:

Proposition 2.2.27: Properties of Functions on Locally Ringed Spaces

Let f be function on a locally ringed space X. Then:

1. The subset of X where f vanishes is closed
2. if f vanishes nowhere, then f is invertible

Note that the property "if f vanishes everywhere, then $f = 0$" is not one of the properties. This will be addressed in proposition 2.4 .10 .

***Proof*:**

1. the subset of X where the germ of f is invertible is open
2. If $f \not\equiv 0 \bmod \mathfrak{p}$, then in particular $f \not\equiv 0 \bmod \mathfrak{m}$ for all maximal ideal, meaning f is contained in no maximal ideal. But then f must be a unit. If f where not a unit, then there would exist an $\mathfrak{m}$ that f belongs to, namely the maximal ideal containing f. But only units are not contained in any maximal ideal (as the ring is non-trivial)

Note how this is is *not* true on a variety defined in the classical sense:

Example 2.2.28: On Variety: Vanishes Nowhere, Not Invertible

Take

$$\overline{x-100} \in \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)}$$

The reader should check that this function vanishes nowhere on $V(x^2+y^2-1)$ as a variety, but it is *not* in invertible (i.e it is not a unit). However, $\overline{x-100}$ *does* vanish on Spec $\mathbb{R}[x,y]/(x^2+y^2-1)$; can the reader find what prime ideal it vanishes on?

This ties in with the intuition that the spectrum of $\mathbb{R}[x,y]/(x^2+y^2-1)$ would be some "gluing" of $\mathbb{C}[x,y]/(x^2+y^2-1)$ via galois conjugate.

Finally, a moment should be taken to interpret evaluating at generic points. Let us consider $X = \mathbb{A}^1_k$ and let η be the generic point. Then looking at the fiber at η, we see that:

$$\mathcal{O}_{\mathbb{A}^1_k,\eta} = (k[x])_{(0)} = k(x)$$

that is, everything is inverted. The earlier intuition about $R_\mathfrak{p}$ is that we only evaluate functions *near* the point $\mathfrak{p}$ and every other $\mathfrak{q} \in \operatorname{Spec} R$ can be inverted, so this means that η is near *no* points as we can invert every point! This contrasts with the intuition that $\overline{\eta} = \mathbb{A}^1_k$ which seems to suggest that it is *close* to every point. This is where the nomenclature *generic* can be of use: it doesn't "see" what global sections capture, and can be thought of as "ignoring" any constraints given by points. This also explains the evaluation at the generic point, it returns the function itself motivating how it is not "particularly" extracting any information.

2.2.6 Independence of Affine Covering

A scheme X can be covered by affine open subsets in many different ways. Unlike the situation for smooth manifolds in low dimensions—where Moise's theorem [48] guarantees a unique smooth structure on topological manifolds of dimension ≤ 3—schemes admit no such rigidity. Even in dimension zero, the topological space underlying a field is always a single closed point, yet different fields give non-isomorphic affine schemes. It is therefore essential to know when a property of a scheme can be checked on *any* affine covering, rather than depending on a particular choice.

The tool for this is the *Affine Communication Lemma*. It identifies a class of properties—those that are stable under localization and can be recovered from a generating cover—for which checking on a single affine cover suffices. Such properties are called *affine-local*, and they include most of the "geometric" properties of schemes (reducedness, Noetherianness, integrality, normality, etc.). Properties that depend on "arithmetic" or global information—such as the number of connected components—are typically not affine-local.

The key preliminary observation is that the intersection of two affine opens, while generally not affine, can always be covered by open sets that are simultaneously distinguished in both.

Lemma 2.2.29: Build-up Lemma

Let $\operatorname{Spec}(R)$ and $\operatorname{Spec}(S)$ be affine open subschemes of a scheme X. Then $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$ can be covered by open sets that are simultaneously distinguished open in $\operatorname{Spec}(R)$ and distinguished open in $\operatorname{Spec}(S)$.

***Proof*:**
Let $[\mathfrak{p}] \in \operatorname{Spec}(R) \cap \operatorname{Spec}(S)$. Since $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$ is open in $\operatorname{Spec}(R)$, and the distinguished open sets form a basis for the Zariski topology (section 2.1.2), there exists $f \in R$ with $[\mathfrak{p}] \in \operatorname{Spec}(R_f) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S)$. Now, as

$$\operatorname{Spec}(R_f) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S) \subseteq \operatorname{Spec}(S)$$

$\operatorname{Spec}(R_f)$ is an open subset of $\operatorname{Spec}(S)$. Thus, by construction of the topology on $\operatorname{Spec}(S)$ using distinguished opens (given by S), there exists $g \in S$ with

$$[\mathfrak{p}] \in \operatorname{Spec}(S_g) \subseteq \operatorname{Spec}(R_f) \cap \operatorname{Spec}(S) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S).$$

In particular, $\operatorname{Spec}(S_g) \subseteq \operatorname{Spec}(R_f)$.

Now, $g \in \mathcal{O}_X(\operatorname{Spec}(S))$ restricts to an element $g' \in \mathcal{O}_X(\operatorname{Spec}(R_f)) = R_f$, and the locus in $\operatorname{Spec}(R_f)$ where g is nonvanishing is exactly $\operatorname{Spec}(S_g)$, so:

$$\operatorname{Spec}(S_g) = \operatorname{Spec}((R_f)_{g'}).$$

By definition of R_f, write $g' = h/f^n$ for some $n \geq 0$ and $h \in R$. By Proposition 2.1.15, a distinguished open of a distinguished open is distinguished:

$$(R_f)_{g'} = (R_f)_{h/f^n} = R_{fh}$$

where the last equality holds since f is already invertible in R_f. Hence:

$$\operatorname{Spec}(S_g) = \operatorname{Spec}(R_{fh}) = D(fh) \subseteq \operatorname{Spec}(R).$$

This is a distinguished open in $\operatorname{Spec}(R)$, and it is simultaneously equal to $\operatorname{Spec}(S_g) = D(g) \subseteq \operatorname{Spec}(S)$, a distinguished open in $\operatorname{Spec}(S)$. Since $[\mathfrak{p}]$ was arbitrary, these sets cover $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$.

With this in hand, we can state and prove the main tool.

Theorem 2.2.30: Affine Communication Lemma

Let X be a scheme and let Q be a property enjoyed by some affine open subschemes of X. Suppose Q satisfies:

1. *(Localization)* If $\operatorname{Spec}(A) \subseteq X$ satisfies Q and $f \in A$, then $\operatorname{Spec}(A_f)$ satisfies Q.
2. *(Gluing)* If $f_1, \ldots, f_n \in A$ generate the unit ideal $(f_1, \ldots, f_n) = A$ and each $\operatorname{Spec}(A_{f_i})$ satisfies Q, then $\operatorname{Spec}(A)$ satisfies Q.

If there exists an affine open cover $X = \bigcup_i \operatorname{Spec}(R^{(i)})$ such that each $\operatorname{Spec}(R^{(i)})$ satisfies Q, then *every* affine open subscheme of X satisfies Q.

***Proof*:**
Let $\operatorname{Spec}(A) \subseteq X$ be an arbitrary affine open subscheme. We must show $\operatorname{Spec}(A)$ satisfies Q. For each i, the intersection $\operatorname{Spec}(A) \cap \operatorname{Spec}(R^{(i)})$ is an open subset of both $\operatorname{Spec}(A)$ and $\operatorname{Spec}(R^{(i)})$. By the Build-up Lemma (lemma 2.2.29), this intersection can be covered by open sets that are simultaneously distinguished in $\operatorname{Spec}(A)$ and distinguished in $\operatorname{Spec}(R^{(i)})$.

Since $\operatorname{Spec}(R^{(i)})$ satisfies Q, condition (1) implies that each of these distinguished opens (viewed as localizations of $R^{(i)}$) also satisfies Q. But these same opens are also distinguished opens of $\operatorname{Spec}(A)$, say of the form $\operatorname{Spec}(A_{g_j})$ for various $g_j \in A$.

Now, $\operatorname{Spec}(A)$ is quasi-compact, and the union $\bigcup_i (\operatorname{Spec}(A) \cap \operatorname{Spec}(R^{(i)})) = \operatorname{Spec}(A)$ covers it. Extracting a finite subcover from among the $\operatorname{Spec}(A_{g_j})$, we obtain elements $g_1, \ldots, g_m \in A$ such that:

(a) each $\operatorname{Spec}(A_{g_j})$ satisfies Q (inherited from the $R^{(i)}$ via localization), and

(b) $\operatorname{Spec}(A) = \bigcup_{j=1}^m \operatorname{Spec}(A_{g_j})$, which means $g_1, \ldots, g_m$ generate the unit ideal in A.

By condition (2), $\operatorname{Spec}(A)$ satisfies Q.

The power of the lemma is that once the two conditions (localization and gluing) are verified for a property Q—which is typically a purely algebraic statement about rings—the conclusion is that Q holds for *every* affine open, not just those in the chosen cover. This means the property can be attributed to the scheme X itself, independent of the presentation.

The Affine Communication Lemma gives a uniform method for proving that properties of schemes are well-defined. We collect some important instances;

Example 2.2.31: Affine-local Properties

1. *Reducedness*: Let Q be the property "A is a reduced ring" (i.e., A has no nonzero nilpotents). We verify the two conditions:

(a) *Localization:* If A is reduced, then A_f is reduced. Indeed, if $(a/f^n)^m = 0$ in A_f, then $f^k a^m = 0$ in A for some k, hence $(f^j a)^m = 0$ for j large enough, so $f^j a = 0$, hence $a/f^n = 0$ in A_f.

(b) *Gluing:* If $(f_1, \ldots, f_n) = A$ and each A_{f_i} is reduced, then A is reduced. Indeed, if $a^m = 0$ in A, then $a^m = 0$ in each A_{f_i}, so $a = 0$ in each A_{f_i} (as A_{f_i} is reduced), meaning $f_i^{k_i} a = 0$ in A for some k_i. Since the f_i generate the unit ideal, so do the $f_i^{k_i}$ (see section 2.1.2), hence $a = 0$.

By the Affine Communication Lemma, a scheme X is reduced if and only if some (equivalently, every) affine open cover consists of reduced rings. This shall be the foundation for definition given in section 2.4.1.

2. *Noetherian*: Let Q be the property "A is a Noetherian ring."

(a) *Localization:* If A is Noetherian, then A_f is Noetherian (a standard result in commutative algebra: every ideal of A_f is an extension of an ideal of A, and extensions of finitely generated ideals are finitely generated).

(b) *Gluing:* Suppose $(f_1, \ldots, f_n) = A$ and each A_{f_i} is Noetherian. Let $I \subseteq A$ be an ideal. Each extension $I \cdot A_{f_i}$ is finitely generated, say by elements $a_{i,1}/f_i^{k_1}, \ldots, a_{i,r_i}/f_i^{k_{r_i}}$. Clearing denominators, we obtain finitely many elements of I whose images generate $I \cdot A_{f_i}$ for each i. Since the f_i generate the unit ideal, these elements generate I itself.

A scheme is *locally Noetherian* if it admits an affine cover by spectra of Noetherian rings. By the Affine Communication Lemma, this is equivalent to every affine open being the spectrum of a Noetherian ring.

3. *Normality*: Let Q be the property "A is a normal ring" (integrally closed in its total ring of fractions). This can be checked via the Affine Communication Lemma: if A is normal then A_f is normal, and normality can be recovered from a generating cover. Thus, a scheme is *normal* if and only if every affine open is the spectrum of a normal ring.

Local Properties and the Affine Communication Lemma

As more properties of schemes are defined, we must be precise about what it means for a scheme to "have" a property and how to check for it. There at least two natural levels at which a property can be detected, and they form a hierarchy.

Definition 2.2.32: Property of Schemes

A *property* P of schemes is a class of schemes closed under isomorphism: if X satisfies P and $Y \cong X$, then Y satisfies P.

Definition 2.2.33: Affine-local and Stalk-local Properties

Let P be a property of schemes.

1. P is *affine-local* if for any scheme X and any affine open cover $X = \bigcup U_i$, the scheme X satisfies P if and only if every U_i satisfies P.
2. P is *Zariski-local* if for any scheme X and any (Zariski) open cover $X = \bigcup U_i$, the scheme X satisfies P if and only if every U_i satisfies P.
3. P is *stalk-local* if for any scheme X, X satisfies P if and only if for every point $x \in X$, the affine scheme $\operatorname{Spec} \mathcal{O}_{X,x}$ satisfies P.

The Affine Communication Lemma provides a practical test for affine-locality: it suffices to verify the localization and gluing conditions for the corresponding ring property. As sheaf maps can always be verified stalk-locally, stalk-locality is another natural condition to consider. We shall see that morphism properties are not always verified affine-locally; those that are give rise to *affine morphisms* (see definition 3.4.5).

Note that $\operatorname{Spec} \mathcal{O}_{X,x}$ *need not* be open in X, hence $\{\operatorname{Spec} \mathcal{O}_{X,x}\}_{x\in X}$ need not be an affine open cover. For example $X = \mathbb{A}^1_{\mathbb{C}}$ and $\mathcal{O}_{X,2} \cong \operatorname{Spec} k[x]_{(x-2)}$ so that $\operatorname{Spec} \mathcal{O}_{X,2}$ has only 2 points but any open subscheme of X has cofinitely many points plus the generic point. Thus, a stalk-local is not the condition that it suffices to check the specific "stalk-cover".

Zariski-local properties are another natural choice, but as any scheme has a topological basis of distinguished open sets, these are identical to affine-locals:

Proposition 2.2.34: Affine-local Iff Zariski-local

Let P be a property of schemes. Then P is affine-local if and only if it is Zariski-local.

***Proof*:**
exercise.

It is natural to ask whether stalk-locality is a stronger or weaker condition than affine-locality. The answer is that it is strictly stronger.

Proposition 2.2.35: Stalk-local implies Affine-local

If a property of a scheme is stalk-local, then it is affine-local.

***Proof*:**
Let P be stalk-local and let $X = \bigcup U_i$ be an affine open cover.

($\Rightarrow$) Suppose X satisfies P. For any $x \in U_i$, the stalk $\mathcal{O}_{U_i,x} \cong \mathcal{O}_{X,x}$ by proposition 1.1.5, and $\operatorname{Spec} \mathcal{O}_{X,x}$ satisfies property P since X satisfies P. As this holds for all $x \in U_i$, the scheme U_i satisfies P by stalk-locality.

($\Leftarrow$) Suppose every U_i satisfies P. For any $x \in X$, choose some U_i containing x. Then $\mathcal{O}_{X,x} \cong \mathcal{O}_{U_i,x}$ satisfies the local-ring property since U_i satisfies P. As x was arbitrary, X satisfies P.

Common examples are:

1. "X is reduced" is stalk-local with respect to "$\mathcal{O}_{X,x}$ has no nonzero nilpotents"
2. "X is regular" is stalk-local with respect to "$\mathcal{O}_{X,x}$ is a regular local ring"
3. "X is normal" is stalk-local with respect to "$\mathcal{O}_{X,x}$ is an integrally closed domain."

The converse is false: there exist affine-local properties that cannot be detected at the level of stalks. The issue is that stalks capture "infinitely close" behavior but lose information about finiteness conditions that involve the global structure of a neighborhood.

Example 2.2.36: Affine-local Does Not Imply Stalk-local

Consider the property "locally Noetherian." As shown in example 2.2.31, this is affine-local. We show it is not stalk-local.

Let $R = \prod_{i=1}^{\infty} \mathbb{F}_2$, the countably infinite product of copies of $\mathbb{F}_2$. The ring R is not Noetherian: the ideals $I_n = \{(a_1, a_2, \dots) : a_1 = \cdots = a_n = 0\}$ form a strictly ascending chain. Thus $\operatorname{Spec}(R)$ is not locally Noetherian.

However, every stalk of $\mathcal{O}_{\operatorname{Spec}(R)}$ is Noetherian. For any prime ideal $\mathfrak{p} \subseteq R$, the localization $R_{\mathfrak{p}}$ is isomorphic to $\mathbb{F}_2$. To see this, recall that R is a Boolean ring ($a^2 = a$ for all a), so every prime ideal is maximal with residue field $\mathbb{F}_2$. Since R is zero-dimensional and Boolean, the localization at any prime is a field.

If "locally Noetherian" were stalk-local, the fact that all stalks are Noetherian fields would force R to be Noetherian—a contradiction. Thus stalk-locality is strictly stronger than affine-locality.

The upshot is a hierarchy of "locality" conditions for scheme properties:

$$\begin{tikzcd}
 & & \text{closed under open restrictions} \\
\text{stalk-local} \arrow[r, Rightarrow] & \text{affine-local} \arrow[ur, Rightarrow] \arrow[dr, Rightarrow] & \\
 & & \text{closed under gluing over open covers}
\end{tikzcd}$$

The first implication is strict, as the example above shows. In practice, most "algebraic" properties (reduced, normal, regular, Cohen–Macaulay) are stalk-local, while "finiteness" properties (Noetherian, finite type) are affine-local but not stalk-local.

2.2.37 Advanced Foreshadowing: Other Covers Notice how the collection of maps $\operatorname{Spec} \mathcal{O}_{X,x} \to X$ for all $x \in X$ can still be seen as a cover which is neither open nor closed. Such a cover still has important properties; in this case it fpqc (fidèlement plate et quasi-compacte). We shall not address such covers in the textbook, however the idea of replacing open sets with covering maps with "desirable" properties will be a key intuition in defining the *étale topology* using *étale coverings* in section ref:HERE.
Readers interested in Descent Theory [67] may be interested to know that the above hierarchy is often extended to the following:

$$\text{fpqc-local} \implies \text{fppf-local} \implies \text{étale-local} \implies \text{Zariski-local}$$

where stalk-local becomes a part of fpqc-local.

2.2.7 *Affine Atlases

By definition, a scheme is a locally ringed space that admits some affine open cover. The structure sheaf $\mathcal{O}_X$ is part of the data: it is given, and the affine cover witnesses that it is "locally affine." But there is an equivalent formulation—closer in spirit to the definition of a smooth manifold—in which the structure sheaf is not given a priori, but is *recovered* from an atlas. This perspective clarifies the sense in which a scheme is "independent of coordinates."

Recall that a smooth manifold of dimension n can be defined in two equivalent ways:

1. A topological space M equipped with a sheaf C^∞_M of smooth functions such that (M, C^∞_M) is locally isomorphic to $(\mathbb{R}^n, C^\infty_{\mathbb{R}^n})$.
2. A topological space M equipped with a *maximal smooth atlas*: a maximal collection of charts (U_i, φ_i) where $\varphi_i : U_i \xrightarrow{\sim} V_i \subseteq \mathbb{R}^n$ is a homeomorphism, and the transition maps $\varphi_j \circ \varphi_i^{-1}$ are smooth (see section 0.5)

The two definitions are equivalent because the sheaf C^∞_M can be recovered from the atlas: a function $f : U \to \mathbb{R}$ is smooth if and only if $f \circ \varphi_i^{-1}$ is smooth for every chart (U_i, φ_i) with $U_i \subseteq U$. Conversely, the atlas is recoverable from the sheaf: a chart is precisely a local isomorphism of the sheaf with the standard model.

In the manifold setting, most textbooks adopt definition (2) because the transition maps are concrete and checkable. In scheme theory, definition (1) has historically been preferred because the structure sheaf interacts well with the algebraic machinery of localization, sheaf cohomology, and base change. Nevertheless, the atlas formulation works equally well and is illuminating.

Affine atlases for schemes

Definition 2.2.38: Affine Chart

Let X be a topological space. An *affine chart* on X is a pair (U, φ) where $U \subseteq X$ is open and $\varphi : U \xrightarrow{\sim} |\operatorname{Spec}(R)|$ is a homeomorphism for some commutative ring R. The ring R is called the *coordinate ring* of the chart.

Two charts (U_i, φ_i) with coordinate ring R_i and (U_j, φ_j) with coordinate ring R_j must be compatible on their overlap. In the manifold case, compatibility means the transition map is smooth. For schemes, it means the transition map is "algebraic" in the following sense.

Definition 2.2.39: Compatibility of Charts

Two affine charts (U_i, φ_i) and (U_j, φ_j) on a topological space X are *compatible* if for every point $x \in U_i \cap U_j$, there exist elements $f \in R_i$ and $g \in R_j$ such that:

1. $x \in \varphi_i^{-1}(D(f)) \cap \varphi_j^{-1}(D(g))$,
2. $\varphi_i^{-1}(D(f)) = \varphi_j^{-1}(D(g))$ as subsets of X, and
3. The composite homeomorphism

$$D(f) \xrightarrow{\varphi_i^{-1}} \varphi_i^{-1}(D(f)) \;=\; \varphi_j^{-1}(D(g)) \xrightarrow{\varphi_j} D(g)$$

is induced by a ring isomorphism $(R_j)_g \xrightarrow{\sim} (R_i)_f$.

Condition (3) is the key algebraic content: the transition "map" between two charts must not be a homeomorphism of topological spaces, but must arise from an isomorphism of the corresponding localized rings. This is the condition that fails for arbitrary homeomorphisms between spectra—and it is exactly what the structure sheaf encodes in the locally ringed space definition.

Definition 2.2.40: Affine Atlas

An *affine atlas* on a topological space X is a collection $\mathcal{A} = \{(U_i, \varphi_i)\}_{i \in I}$ of pairwise compatible affine charts such that $X = \bigcup_{i \in I} U_i$.

Definition 2.2.41: Maximal Affine Atlas

An affine atlas $\mathcal{A}$ is *maximal* if it contains every affine chart on X that is compatible with all charts already in $\mathcal{A}$. That is, if (V, ψ) is an affine chart compatible with every $(U_i, \varphi_i) \in \mathcal{A}$, then $(V, \psi) \in \mathcal{A}$.

Definition 2.2.42: Scheme via Atlas

A *scheme* is a pair $(X, \mathcal{A})$ where X is a topological space and $\mathcal{A}$ is a maximal affine atlas on X.

This definition makes no reference to sheaves, locally ringed spaces, or stalks. The algebraic data is encoded entirely in the transition isomorphisms between charts: the atlas definition is equivalent to definition 2.2.9. The equivalence rests on two constructions: recovering the structure sheaf from an atlas, and recovering the maximal atlas from a locally ringed space.

Proposition 2.2.43: From Atlas to Structure Sheaf

Let $(X, \mathcal{A})$ be a scheme in the sense of definition 2.2.42. Then there exists a unique sheaf of rings $\mathcal{O}_X$ on X such that $(X, \mathcal{O}_X)$ is a scheme in the sense of definition 2.2.9, and for every chart $(U, \varphi) \in \mathcal{A}$ with coordinate ring R:

$$(U,\ \mathcal{O}_X|_U) \ \cong\ (\operatorname{Spec}(R),\ \mathcal{O}_{\operatorname{Spec}(R)}).$$

***Proof*:**
Define $\mathcal{O}_X$ on the basis of opens $\{U : (U, \varphi) \in \mathcal{A} \text{ for some } \varphi\}$ by setting $\mathcal{O}_X(U) = R$ whenever $\varphi : U \xrightarrow{\sim} \operatorname{Spec}(R)$. For a distinguished open $D(f) \subseteq \operatorname{Spec}(R)$, the chart $(\varphi^{-1}(D(f)),\ \varphi|_{D(f)})$ is in $\mathcal{A}$ (by maximality) with coordinate ring R_f, so we set $\mathcal{O}_X(\varphi^{-1}(D(f))) = R_f$. The restriction maps are the natural localization maps $R \to R_f$.

The compatibility condition on charts ensures that this is well-defined on overlaps: if two charts (U_i, φ_i) and (U_j, φ_j) both contain a point x, the transition isomorphism $(R_j)_g \cong (R_i)_f$ identifies the sections. By the sheaf axiom for the structure sheaf of $\operatorname{Spec}(R)$, this basis assignment extends uniquely to a sheaf on all of X.

The stalks of $\mathcal{O}_X$ are local rings: at a point $x \in U_i$, the stalk $\mathcal{O}_{X,x}$ is the localization $(R_i)_{\mathfrak{p}}$ where $\mathfrak{p} = \varphi_i(x)$, which is local. Hence $(X, \mathcal{O}_X)$ is a locally ringed space, and each chart gives an affine open, so it is a scheme.

Proposition 2.2.44: From Locally Ringed Space to Maximal Atlas

Let $(X, \mathcal{O}_X)$ be a scheme in the sense of definition 2.2.9. Then the collection

$$\mathcal{A}_{\max} = \left\{ (U, \varphi) : \begin{array}{l} U \subseteq X \text{ is open, } \varphi : (U, \mathcal{O}_X|_U) \xrightarrow{\sim} (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)}) \\ \text{is an isomorphism of locally ringed spaces, for some } R \end{array} \right\}$$

is a maximal affine atlas on X. The atlas $\mathcal{A}_{\max}$ depends only on $(X, \mathcal{O}_X)$, not on any particular affine open cover.

***Proof*:**
The charts in $\mathcal{A}_{\max}$ are pairwise compatible: if (U_i, φ_i) and (U_j, φ_j) are in $\mathcal{A}_{\max}$, then on $U_i \cap U_j$ the isomorphisms φ_i and φ_j are both isomorphisms of locally ringed spaces, and the transition map is an isomorphism of the appropriate localizations (this is the content of the Build-up Lemma, lemma 2.2.29). The atlas covers X because the definition of a scheme guarantees the existence of at least one affine cover. It is maximal by construction: every affine open is included.

These two constructions are inverse to each other: starting from an atlas, building the sheaf, and taking the maximal atlas of the resulting scheme recovers the original atlas; starting from a scheme, taking $\mathcal{A}_{\max}$, and building the sheaf from it recovers the original structure sheaf.

Just as for manifolds, the practical consequence is that one need not specify a maximal atlas. Any atlas suffices.

Proposition 2.2.45: Uniqueness of Maximal Atlas

Let $\mathcal{A}$ be any affine atlas on a topological space X. Then there exists a unique maximal affine atlas $\mathcal{A}_{\max}$ containing $\mathcal{A}$. Explicitly,

$$\mathcal{A}_{\max} = \{(V, \psi) : (V, \psi) \text{ is compatible with every } (U, \varphi) \in \mathcal{A}\}.$$

***Proof*:**
One must verify that any two charts in $\mathcal{A}_{\max}$ are compatible with each other (not just with the charts in $\mathcal{A}$). Let (V_1, ψ_1) and (V_2, ψ_2) be in $\mathcal{A}_{\max}$, and let $x \in V_1 \cap V_2$. Choose a chart $(U, \varphi) \in \mathcal{A}$ with $x \in U$. By compatibility of (V_1, ψ_1) with (U, φ), there exist distinguished opens around x where the transition is an isomorphism of localizations. By compatibility of (V_2, ψ_2) with (U, φ), the same holds. Composing (and using that distinguished opens of distinguished opens are distinguished, by proposition 2.1.15), we obtain a transition isomorphism between localizations of V_1 and V_2.

Uniqueness is immediate: any maximal atlas containing $\mathcal{A}$ must contain every compatible chart, hence must equal $\mathcal{A}_{\max}$.

The upshot is that defining a scheme requires only specifying a topological space together with *any* affine atlas—the maximal atlas is then uniquely determined. The Affine Communication Lemma (theorem 2.2.30) is the tool that ensures properties of the scheme can be checked on the given atlas rather than on the full maximal atlas: its localization and gluing conditions propagate a property from any covering atlas to all affine opens.

The atlas viewpoint also gives an explicit construction of schemes from purely algebraic data, analogous to constructing a manifold by specifying charts and transition maps. Suppose we are given:

1. Rings $R_1, \ldots, R_n$.
2. For each pair (i, j), elements $f_{ij} \in R_i$ specifying open subsets $U_{ij} = D(f_{ij}) \subseteq \operatorname{Spec}(R_i)$.
3. Ring isomorphisms $\varphi_{ij} : (R_i)_{f_{ij}} \xrightarrow{\sim} (R_j)_{f_{ji}}$.

Subject to the *cocycle condition*: for each triple (i, j, k), the composition $\varphi_{jk} \circ \varphi_{ij}$ agrees with φ_{ik} on the appropriate localizations.

These data determine a unique scheme X (up to unique isomorphism) by the gluing construction of section 2.2.9: the underlying set is the disjoint union $\coprod_i |\operatorname{Spec}(R_i)|$ modulo the identifications given by the φ_{ij}, the topology is the quotient topology, and the affine atlas consists of the images of the $\operatorname{Spec}(R_i)$. Conversely, every scheme with a given finite affine cover arises in this way: the rings are the coordinate rings of the charts, and the transition isomorphisms are read off from the structure sheaf on overlaps.

2.2.8 Subschemes: Open

We seek the natural notions of subschemes induced by the parent scheme. As a scheme has a topology, a natural starting point is to look for open or closed subsets with a sheaf naturally induced by the parent scheme[21]. To define the topology of open and closed subschemes is easy. The difficult part is to insure the subscheme has a sheaf structure that is in a natural way compatible with the parent scheme. This shall require a deeper discussion on monomorphic scheme morphisms as the sheaf structure of subschemes is “non-trivial”, and hence this discussion is mostly delayed till section 3.1.3. There is one type of subscheme that shall be used in this chapter and requires no use of scheme morphism since it is directly compatible with sheaves: *open subschemes*.

Proposition 2.2.46: Open Subscheme

Let X be a scheme and $U \subseteq X$ an open subset. Then $(U, \mathcal{O}_X|_U)$ is a scheme.

***Proof*:**
U is certainly a subspace and a subsheaf with local rings as stalks. To show it is locally affine, for any $x \in U$ choose affine open $\operatorname{Spec} R \cong V \subseteq X$ where $x \in U \cap V$. As $D(f)$ forms a (topological) basis for $\operatorname{Spec} R$, there exists an f_0 such that $x \in D(f_0) \subseteq U \cap V \subseteq U$, showing that U is covered in affine, completing the proof.

Definition 2.2.47: Open Subscheme

Let X be a scheme. Then $(U, \mathcal{O}_X|_U)$ is called an *open subscheme*. If U is also an affine scheme, then U is often said to be an *affine open subscheme* or *affine open subset*.

If a scheme is isomorphic to an open subscheme, we shall say that the isomorphism map (onto the image) is an open embedding:

Definition 2.2.48: Open Embedding

A map of schemes $\pi : X \to Y$ is an open embedding if:

$$(X, \mathcal{O}_X) \xrightarrow{\sim} (U, \mathcal{O}_Y|_U) \hookrightarrow (Y, \mathcal{O}_Y)$$

2.2.49 Remark about Hartshorne’s Definition of Open Subscheme Note that in Hartshorne, an open subscheme was technically not defined as $(U, \mathcal{O}_X|_U)$ but as $(U, [\mathcal{O}_X|_U])$ where $[\mathcal{O}_X|_U]$ is the equivalence class of all isomorphic sheaves. This is in fact not correct: isomorphic sheaves shall give technically different open subschemes (which are of course isomorphic). Hence, Hartshorne claims that an open subscheme has a unique subscheme structure, not unique up to isomorphism.

Corollary 2.2.50: Subspace and Zariski Topology On Open Subscheme

Let $U = \operatorname{Spec}(R) \subseteq X$ be an affine open subset and take $(U, \mathcal{O}_X|_U)$. Then the Zariski topology on U is the subspace topology on U with respect to X

[21] Categorically, a natural starting point is to look for subobjects, but surprisingly these are not so useful in **Sch**; see section 3.1.3

***Proof*:**
exercise.

The prototypical example of an open subscheme would be the open sets $D_+(x_i) \subseteq \mathbb{P}^n_k$ of projective space or $D(f) \subseteq \operatorname{Spec} R$ of affine space. Note that an open sub-scheme of an affine sub-scheme need not be affine:

Example 2.2.51: Open Sub-Scheme need not be Affine

Let $R = k[x, y]$, $X = \mathbb{A}^2_k = \operatorname{Spec}(k[x, y])$, and consider $U = \mathbb{A}^2_k \setminus \{(x, y)\}$ which may be considered the plane without the origin. Before seeing why this is a scheme, try to show that U is indeed an open subset (it is the compliment of intersection of certain closed subsets, equivalently the union of certain open sets, there is more than one way of creating it).

Now, there does not exist an R such that $\operatorname{Spec}(R) \cong U$. It may be tempting to say that there ought to be a distinguished open set of $\mathbb{A}^2_k$ that produces U, but the reader should think through why this is not the case (hint: (x, y) is not principle). However, U can certainly be described as the union of two distinguished open sets, namely:

$$U = D(x) \cup D(y)$$

What is the global section on the open sub-scheme U? This would be all functions in $D(x) \cup D(y)$ that agree on the intersection, $D(x) \cap D(y) = D(xy) = R_{xy} = k[x, y]_{xy}$. Computing the local sections, as $D(x)$ is all points in $\mathbb{A}^2_k$ that do not vanish in the function x, and $D(y)$ is all the points that do not vanish in the function y, we get:

$$\mathcal{O}_U(D(x)) = R_x = k\left[x, y, \frac{1}{x}\right] \qquad \mathcal{O}_U(D(y)) = R_y = k\left[x, y, \frac{1}{y}\right]$$

Now consider the functions that agree on the intersection: $R_x \cap R_y \subseteq R_{xy}$. These are all rationals functions with only powers of x in the denominator, and only powers of y in the denominator (as it must be in the intersection). But this leaves only the elements of R: $R_x \cap R_y = R$ (where R is identified with it's canonical embedding in R_{xy}). Thus:

$$\mathcal{O}_{\mathbb{A}^2_k}(U) = \mathcal{O}_U(U) = k[x, y]$$

in other words, the global section of U is identical to the global section of $\mathbb{A}^2_k$. Now, if U was affine, then by proposition 2.2.13, we would have that the is determined by the global ring, that is:

$$U \cong \mathbb{A}^2_k$$

that is, this is a scheme isomorphism, and hence also a homeomorphic map between prime ideals that must respect $V(-)$ and $I(-)$. But in U we have:

$$V(x) \cap V(y) = \emptyset$$

while this is certainly not the case in $\mathbb{A}^2_k$ (as it gives $[(x - 0, y - 0)]$). Hence U is *not* an affine scheme!

The main idea is that $D(f)$ misses a codimension 1 subset if R is Noetherian integral domain (see exercise 5.3.1 (1)), where Noetherian is to avoid infinite pathologies and integral domain is to insure there is only one irreducible component (see section 2.4). More generally, given R is Noetherian and an integral domain, $\operatorname{Spec} R_S$ for some $S \subseteq R$ not containing units misses a (pure) codimension 1 subset (as it is open). The reason that higher codimensional sets "don't matter" when considering global sections very reminiscent of complex geometry, namely this is an algebraic version of *Hartog's Lemma* (see [52, chapter 8]).

It is possible to have an affine open subscheme of an affine scheme, and hence by dualizing we may look for algebraic conditions on the ring homomorphisms. We state the result for those familiar with flatness (see [10]) but leave the proof till flatness is introduced in ref:HERE:

Theorem 2.2.52: Characterizing Affine Open Immersions

$\mathrm{Spec}\,(S) \to \mathrm{Spec}\,(R)$ is an open immersion if and only $R \to S$ is flat, of finite presentation, and an epimorphism.

***Proof*:**
see https://mathoverflow.net/questions/20782/ring-theoretic-characterization-of-open-affines.

2.2.53 For the expert: Cover Condition The reader well-versed in descent theory may have noticed this is almost the condition of an fppf covering, with faithful (i.e. surjective) the epimorphism condition. This motivates how fppf covers are the local open covers and appropriately generalize Zariski covers, see ref:HERE.

2.2.9 Gluing (Coproduct) of Schemes

In Differential Geometry, one of the most common way of constructing a manifold, and indeed what is at the heart of geometry, is the process of gluing pieces together with some amount of consistency (see [53, chapter 1]). More generally, gluing manifolds, and gluing schemes, is a special case of a pushout (or fibered coproduct) $X \cup_Z Y$:

$$\begin{tikzcd}
& Z \arrow[dl] \arrow[dr] & \\
X \arrow[dr] \arrow[ddr, dashed, bend right] & & Y \arrow[dl] \arrow[ddl, dashed, bend left] \\
& X \cup_Z Y \arrow[d, dashed, "\exists!"] & \\
& Q &
\end{tikzcd}$$

The fibered product $X \times_Z Y$ can be thought of as taking a product of spaces $X \times Y$ and restricting to where the maps agree (i.e., carving out a subspace), while the fibered coproduct can be thought of as taking the disjoint union of spaces and doing some gluing. The fibered product shall be more nuanced as defining $X \times Y$ does not sense and hence is left for section 3.2.

Just like for manifolds, we can't glue schemes any way we want and get another scheme, we must be a bit more careful. Example 3.2.7 shall give an example of "bad" gluing.

The key idea for coproducts of schemes is to ensure that the gluing preserves the local affine structure (note that for a manifold it also has to preserve Hausdorffness). Hence, it is marginally easier to construct schemes via gluing:

Theorem 2.2.54: Coproduct (Pushout) of Schemes

Let $\{X_i\}_{i \in I}$ be a collection of schemes. Then if there exists:

1. A collection of open subschemes, not necessarily an open cover, $X_{ij} \subseteq X_i$ (where $X_{ii} = X_i$)
2. a collection of isomorphisms $f_{ij} : X_{ij} \to X_{ji}$ (where f_{ii} would be the identity) where

$$f_{ij}(X_{ij} \cap X_{ik}) \subseteq X_{ji} \cap X_{jk}$$

such that the isomorphisms agree on triple intersection[a]:

$$f_{ik}|_{X_{ij} \cap X_{ik}} = f_{jk}|_{X_{ji} \cap X_{jk}} \circ f_{ij}|_{X_{ij} \cap X_{ik}}$$

then there is a unique scheme X (up to unique isomorphism) with open subsets isomorphic to X_i respecting the above gluing conditions.

[a]this is called the *cocycle condition*

This is a translation of the techniques from differential geometry (this condition may look more familiar if written as $\mathbf{y} \circ \mathbf{x}^{-1} : \mathbf{x}(U \cap V) \to \mathbf{y}(U \cap V)$). The cocycle condition can be visualized like so:

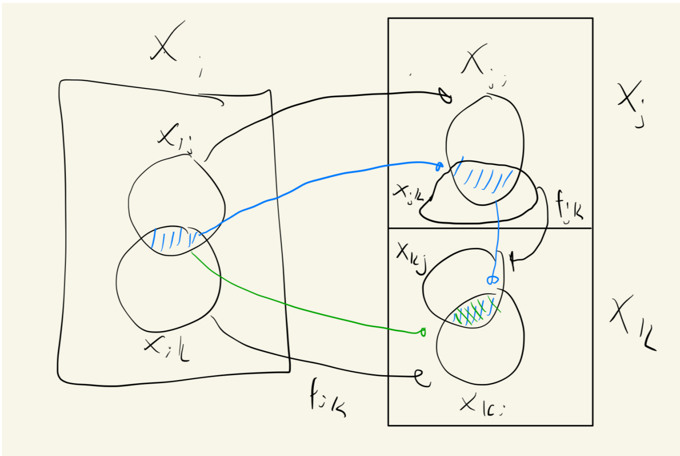


Figure 2.4: Green map must match blue map

Note the cocycle condition forces:

1. when $i = j = k$ that $\varphi_{ii} = \mathrm{id}$
2. when $i = k$ that $\varphi_{ij} = \varphi_{jk}^{-1}$

***Proof*:**
Define X as the quotient of the disjoint union of the schemes X_i:

$$X = \frac{\coprod_{i \in I} X_i}{(X_{ij} \ni x \sim f_{ij}(x) \in X_{ji})}$$

where the equivalence relation is defined by identifying points via the gluing maps: $x \sim y$ if $x \in X_{ij}$, $y \in X_{ji}$, and $f_{ij}(x) = y$. The cocycle condition ensures that this relation is transitive (and symmetry follows from $f_{ji} = f_{ij}^{-1}$, which is a consequence of the cocycle condition with $k = i$), so this is a valid equivalence relation. We endow X with the quotient topology relative to the natural projection map $\pi : \coprod X_i \to X$.

Let $\iota_i : X_i \to X$ denote the restriction of π to the component X_i, and let $U_i = \iota_i(X_i)$ be its image in X. Because the X_{ij} are open sets, one can verify that ι_i is a homeomorphism onto the open set U_i. The sets U_i form an open cover of X, where each U_i is a topological copy of X_i.

With the topological space established, we define the structure sheaf $\mathcal{O}_X$. For any open set $V \subseteq X$, we define a section $s \in \mathcal{O}_X(V)$ to be a collection of sections $\{s_i\}_{i \in I}$ where $s_i \in \mathcal{O}_{X_i}(\iota_i^{-1}(V))$, satisfying the compatibility condition on overlaps:

$$s_i\big|_{\iota_i^{-1}(V) \cap X_{ij}} = f_{ij}^{\#}\Big(s_j\big|_{\iota_j^{-1}(V) \cap X_{ji}}\Big)$$

This defines a valid sheaf of rings by theorem 1.1.23, applied to the open cover $\{U_i\}$ with sheaves $(\iota_i)_* \mathcal{O}_{X_i}$ and transition isomorphisms induced by $f_{ij}^{\#}$.

Finally, we must verify that the ringed space $(X, \mathcal{O}_X)$ is indeed a scheme. It suffices to check this locally. Consider the open set U_i equipped with the restricted sheaf $\mathcal{O}_X|_{U_i}$. By our construction of the global sections, the sheaf $\mathcal{O}_X|_{U_i}$ is canonically isomorphic to the sheaf $\mathcal{O}_{X_i}$ via the map ι_i. Thus, we have an isomorphism of ringed spaces $(U_i, \mathcal{O}_X|_{U_i}) \cong (X_i, \mathcal{O}_{X_i})$. Since each X_i is a scheme by hypothesis, each open set U_i is a scheme. As X is covered by open subsets that are schemes (and hence covered by affine schemes), X is a scheme as we sought to show.

What would the resulting global sections of a glued scheme look like? Consider the case of two schemes X_1, X_2 glued along $X_{12} \subseteq X_1$ and $X_{21} \subseteq X_2$ via an isomorphism $f_{12} : X_{12} \xrightarrow{\sim} X_{21}$, with inclusion maps $\iota_i : X_i \to X$. Then:

$$\mathcal{O}_X(U) =$$
$$\Big\{(s_1, s_2) \;\Big|\; s_1 \in \mathcal{O}_{X_1}(\iota_1^{-1}(U)),\ s_2 \in \mathcal{O}_{X_2}(\iota_2^{-1}(U)) \text{ such that } s_1\big|_{\iota_1^{-1}(U) \cap X_{12}} = f_{12}^{\#}\Big(s_2\big|_{\iota_2^{-1}(U) \cap X_{21}}\Big)\Big\}$$

This formula is common enough to be given a name: we call it the *gluing formula*.

For three schemes X_1, X_2, X_3, the formula extends naturally: a global section is a triple $(s_1, s_2, s_3) \in \mathcal{O}_{X_1}(X_1) \times \mathcal{O}_{X_2}(X_2) \times \mathcal{O}_{X_3}(X_3)$ satisfying the pairwise compatibility conditions on all overlaps simultaneously:

$$s_i\big|_{X_{ij}} = f_{ij}^{\#}\Big(s_j\big|_{X_{ji}}\Big) \qquad \text{for all } i, j \in \{1, 2, 3\}$$

Note that one does not glue two schemes first and then attach the third; the cocycle condition ensures that all pairwise constraints are mutually consistent, so all three (or finitely many) pieces are handled at once. The pattern for n schemes is identical: global sections are n-tuples satisfying $\binom{n}{2}$ compatibility conditions.

When thinking about global sections of glued schemes, it is helpful to view the gluing maps as imposing additional *relations*: the global sections of the glued scheme are the elements of the product $\prod_i \mathcal{O}_{X_i}(X_i)$ that are compatible under all the transition isomorphisms. The more gluing constraints there are, the smaller the ring of global sections becomes.

Example 2.2.55: Gluing [affine] Schemes

Let $X = \operatorname{Spec}(k[x])$ and $Y = \operatorname{Spec}(k[y])$. Take $D(x) \subseteq X$ and $D(y) \subseteq Y$, which is the collection of all prime ideals that do not contain x and y respectively. As shown in proposition ref:HERE, these subsets are isomorphic to:

$$D(x) \cong \operatorname{Spec}\left(k\left[x, \frac{1}{x}\right]\right) \qquad D(y) \cong \operatorname{Spec}\left(k\left[y, \frac{1}{y}\right]\right)$$

As a diagram, we have:

$$\begin{tikzcd} \operatorname{Spec}(k[x]) & & \operatorname{Spec}(k[y]) \\ & \operatorname{Spec}(k[x,1/x]) \arrow[ul] \arrow[ur] & \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} k[x] \arrow[dr] & & k[y] \arrow[dl] \\ & k[x,1/x] & \end{tikzcd}$$

or it may be more helpful to see this as:

$$\begin{tikzcd} \operatorname{Spec} k[x] & & \operatorname{Spec} k[y] \\ & \operatorname{Spec} k[x,1/x] \cong \operatorname{Spec} k[y,1/y] \arrow[ul] \arrow[ur] & \end{tikzcd}$$

and

$$\begin{tikzcd} k[x] \arrow[dr] & & k[y] \arrow[dl] \\ & k[x,1/x] \cong k[y,1/y] & \end{tikzcd}$$

where the cocycle condition must be satisfied by the isomorphisms. Notice how these are similar to restriction conditions: we are "manually" figuring out how the restrictions should work so that by uniqueness of gluing local sections on a sheaf we can get the global sections. These two open subsets can be glued in at least two different ways to produce two non-affine schemes that will be familiar to the reader:

1. (Line with double origin). Define the isomorphism $D(x) \cong D(y)$ via the ring isomorphism $\varphi^\# : k[y, 1/y] \to k[x, 1/x]$ given by $y \mapsto x$. This is a scheme by theorem 2.2.54. Label this scheme S_1. We claim S_1 is *not* affine, even though its global sections coincide with those of $\mathbb{A}^1_k$.

   By the gluing formula, a global section of $\mathcal{O}_{S_1}$ is a pair $(f(x), g(y)) \in k[x] \times k[y]$ satisfying the compatibility condition:

   $$f(x) = \varphi^\#(g(y)) = g(x) \qquad \text{in } k[x, 1/x]$$

Since $k[x] \hookrightarrow k[x, 1/x]$ is injective, this simply says $f(x) = g(x)$ as polynomials (after identifying the variable names). In particular, any polynomial satisfies this condition, and so:

$$\mathcal{O}_{S_1}(S_1) \cong k[x]$$

By proposition 2.2.13, an affine scheme is determined by its global sections, so if S_1 were affine it would be isomorphic to $\operatorname{Spec}(k[x]) = \mathbb{A}^1_k$. But this is impossible: S_1 has two distinct closed points (the two origins) that are not separated by any open set, while $\mathbb{A}^1_k$ does not. Hence S_1 is not affine. The same construction works in higher dimensions (e.g. $\mathbb{A}^n_k$ with double origin).

In differential geometry, such spaces are examples of non-Hausdorff manifolds, however all non-trivial schemes already fail to be Hausdorff. The relevant condition for schemes is *separatedness*: spaces with double-origins fail to be separated (as a hint: show that the intersection of the two copies of $\mathbb{A}^1_k$ inside S_1 is $D(x)$, which is not affine), while $\mathbb{A}^n_k$ is separated.

2. (Projective space). Define the isomorphism $D(x) \cong D(y)$ via the ring isomorphism $\varphi^{\#} : k[y, 1/y] \to k[x, 1/x]$ defined by $y \mapsto 1/x$ (equivalently, $1/y \mapsto x$). The reader will recognize the resulting scheme as $\mathbb{P}^1_k$. When $k \in \{\mathbb{R}, \mathbb{C}\}$, this is isomorphic (as a manifold) to a circle and a sphere respectively, while in dimension n, $\mathbb{P}^n_k$ cannot be embedded in $\mathbb{A}^n_k$. For schemes, one must account for the generic point (0) and the non-closed points given by irreducible polynomials (recall that linear factors that are Galois conjugates are identified).

   Let us show that $\mathbb{P}^1_k$ is not affine by computing $\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$. A global section is a pair $(f(x), g(y)) \in k[x] \times k[y]$ satisfying:

$$f(x) = \varphi^{\#}(g(y)) = g(1/x) \qquad \text{in } k[x, 1/x]$$

   Write $f(x) = \sum_{n=0}^{N} a_n x^n$ and $g(y) = \sum_{m=0}^{M} b_m y^m$, so that:

$$g(1/x) = \sum_{m=0}^{M} b_m \, x^{-m}$$

   In $k[x, 1/x]$, the Laurent monomials $\{x^n\}_{n \in \mathbb{Z}}$ are linearly independent over k. The left-hand side $f(x)$ involves only non-negative powers of x, while the right-hand side $g(1/x)$ involves only non-positive powers. For these to be equal, both must be constant:

$$a_n = 0 \text{ for } n \geq 1, \qquad b_m = 0 \text{ for } m \geq 1, \qquad a_0 = b_0$$

   Hence f and g are both the same constant, giving:

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = k$$

   By proposition 2.2.13, if $\mathbb{P}^1$ were affine then $\mathbb{P}^1 \cong \operatorname{Spec}(\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)) = \operatorname{Spec}(k)$, which is a single point. But $\mathbb{P}^1_k$ has more than one point, so $\mathbb{P}^1_k$ is not affine[a].

[a] Those familiar with complex analysis may recall that the only holomorphic functions on $\mathbb{P}^1_{\mathbb{C}}$ (the Riemann sphere) are constants, which is a manifestation of the same phenomenon.

2.2.10 *Gluing Affine Schemes and Fiber Products of Rings

Let us now specialize to the case where the pieces being glued are affine. Recall that the functors $\operatorname{Spec}(-)$ and $\Gamma(-)$ are contravariant and form an adjunction that exchanges limits and colimits. Hence, the cofibered product (pushout) of schemes dualizes to the fiber product (pullback) of rings. If $X_1 = \operatorname{Spec}(A)$, $X_2 = \operatorname{Spec}(B)$, and the overlap maps factor through a ring C, then the gluing formula for global sections becomes:

$$\mathcal{O}_X(X) = A \times_C B := \{(a, b) \in A \times B \mid \text{images of } a \text{ and } b \text{ agree in } C\}$$

This is dual to the familiar fact that the fibered product of affine schemes corresponds to the tensor product of rings:

$$\operatorname{Spec} A \times_{\operatorname{Spec} C} \operatorname{Spec} B \cong \operatorname{Spec}(A \otimes_C B)$$

In both cases, the contravariance of Spec exchanges limits and colimits, and hence exchanges tensor products (colimits in **CRing**) with fiber products (limits in **CRing**). More generally, for an arbitrary number of affine pieces, the global sections are the limit of the diagram of rings formed by all the overlap maps.

It is natural to ask: since **CRing** is both complete and cocomplete, does $\operatorname{Spec}(A \times_C B)$ give the pushout of $\operatorname{Spec}(A)$ and $\operatorname{Spec}(B)$ over $\operatorname{Spec}(C)$ in the category of schemes? The answer is *no* in general. The subtlety is that the universal property must be tested against different classes of objects depending on the ambient category:

- *In **Aff***: the object $\operatorname{Spec}(A \times_C B)$ satisfies the universal property of the pushout with respect to maps into affine schemes, and this pushout always exists since **CRing** is complete (so fiber products of rings always exist).
- *In **Sch***: the pushout must satisfy the universal property for maps into *all* schemes. A morphism from an affine scheme to a non-affine target Q is not determined by a ring homomorphism out of $\Gamma(Q, \mathcal{O}_Q)$ (since Q is not affine, the adjunction $\operatorname{Spec} \dashv \Gamma$ does not apply), so the affine pushout may fail the universal property.

In categorical language, the inclusion **Aff** $\hookrightarrow$ **Sch** does not preserve pushouts. The pushout computed in **Sch** can be genuinely non-affine: the line with doubled origin and the projective line from example 2.2.55 are both pushouts of affine schemes in **Sch** that are not affine. In each case, $\operatorname{Spec}(A \times_C B)$ exists as an affine scheme, but it is *not* isomorphic to the glued scheme X – it is only the "best affine approximation" to the pushout.

The preceding discussion raises a natural question: if $\operatorname{Spec}(A \times_C B)$ is not the pushout in **Sch**, what *is* it? The answer is given by the adjunction between global sections and Spec.

Proposition 2.2.56: $\Gamma \dashv \operatorname{Spec}$ Adjunction

The global sections functor $\Gamma : \mathbf{Sch} \to \mathbf{CRing}^{op}$ is left adjoint to $\operatorname{Spec} : \mathbf{CRing}^{op} \to \mathbf{Sch}$:

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}(R)) \cong \operatorname{Hom}_{\mathbf{CRing}}(R, \Gamma(X, \mathcal{O}_X))$$

for any scheme X and any ring R.

***Proof*:**
See proposition 3.1.12

Since left adjoints preserve colimits and right adjoints preserve limits, this single adjunction encodes both dualities between schemes and rings:

- Γ (left adjoint) sends pushouts in **Sch** to colimits in $\mathbf{CRing}^{op}$, i.e. limits (fiber products) in **CRing**:
$$\Gamma(X_1 \cup_Z X_2) \cong \Gamma(X_1) \times_{\Gamma(Z)} \Gamma(X_2)$$
This is why the gluing formula computes global sections as a fiber product of rings – it is a formal consequence of the adjunction.
- Spec (right adjoint) sends limits in $\mathbf{CRing}^{op}$ (i.e. colimits in **CRing**, such as tensor products) to limits (fiber products) in **Sch**:
$$\operatorname{Spec}(A \otimes_C B) \cong \operatorname{Spec}(A) \times_{\operatorname{Spec}(C)} \operatorname{Spec}(B)$$
- Spec , being a right adjoint, does *not* preserve colimits in general. The fiber product $A \times_C B$ is a limit in **CRing**, hence a colimit in $\mathbf{CRing}^{op}$, and there is no reason for Spec to preserve it.

Definition 2.2.57: Affinization

Let X be a scheme. The *affinization* of X is the affine scheme $X^{\mathbf{Aff}} := \operatorname{Spec}(\Gamma(X, \mathcal{O}_X))$, together with the canonical morphism
$$\eta_X : X \to \operatorname{Spec}(\Gamma(X, \mathcal{O}_X))$$
given by the unit of the $\Gamma \dashv \operatorname{Spec}$ adjunction.

By the universal property of adjunctions, η_X is the universal map from X to an affine scheme: any morphism $X \to \operatorname{Spec}(R)$ factors uniquely through η_X. For the pushout $X = X_1 \cup_Z X_2$ of affine schemes, since $\Gamma(X) \cong A \times_C B$, this gives:
$$X_1 \cup_Z X_2 \xrightarrow{\eta} \operatorname{Spec}(A \times_C B)$$
Hence $\operatorname{Spec}(A \times_C B)$ is not the pushout in **Sch**, but rather the *best affine approximation* to the pushout – the closest affine scheme that the glued scheme maps to.

Example 2.2.58: Affinization of Glued Schemes
Returning to the two gluings from example 2.2.55:

1. *Line with doubled origin.* Here $A \times_C B \cong k[x]$, so the affinization is $\mathbb{A}^1_k$. The map $\eta : S_1 \to \mathbb{A}^1_k$ identifies the two origins to a single point, collapsing exactly the non-affine pathology.
2. *Projective line.* Here $A \times_C B \cong k$, so the affinization is $\operatorname{Spec}(k)$, a single point. The map $\eta : \mathbb{P}^1_k \to \operatorname{Spec}(k)$ collapses the entire projective line to a point, reflecting the fact that $\mathbb{P}^1_k$ has only constant global functions.

In both cases, the affinization map η "kills" precisely the geometric information that cannot be captured by global sections.

Exercise 2.2.2

1. Let X be the line with two origins, $Y = \mathbb{A}^1_k$ the affine line, and $\pi : X \to Y$ the natural projection. Compute $f_*(\mathcal{O}_X)_0$, the talk of the pushforward at 0.
2. Let X be a scheme. Show that the affine open subsets form a base.
3. A subset Z of a scheme X is closed if and only if there exists an open cover of X by affine subschemes, $\{U_i = \mathrm{Spec}(A_i)\}_{i\in I}$, such that for every $i \in I$, the intersection $Z \cap U_i$ is a closed subset of the affine scheme U_i. In other words, a closed subset of X is characterized as being locally closed on an affine open covering.

2.3 Projective Schemes

Recall projective space from [53]. In [20] a few key properties of projective space $\mathbb{P}^n_{\mathbb{C}}$ were proved, namely:

1. Projective space was constructed by gluing together affine spaces. The cocycle condition is straightforward to verify over any ring (the transition maps are rational functions that do not require algebraic closure), though understanding the closed points is simpler when $k = \overline{k}$. We shall construct Proj in a more "natural" way without any choice of coordinates.
2. There shall be a homogeneous coordinate ring S from which we can construct the scheme $\mathrm{Proj}\, S$. Just like in the classical setting, S shall not be the ring of regular functions on $\mathrm{Proj}\, S$: the global sections of $\mathrm{Proj}\, S$ shall be constant, and $\mathrm{Proj}\, S$ shall behave like the "compact" sets in algebraic geometry (namely the proper sets, as mentioned in [20]).

   We shall show that the algebraic object S will be associated to a different sheaf over $\mathrm{Proj}\, S$ that is not the structure sheaf, see section 5.3.
3. The properties of Bézout's theorem shall turn out to be the special case of the theory of *divisors* on projective space. See section **??** for more details.

Classically, n-dimensional projective space $\mathbb{R}\mathrm{P}^n$ or $\mathbb{C}\mathrm{P}^n$ is defined on $\mathbb{R}^{n+1} \setminus \{0\}$ or $\mathbb{C}^{n+1} \setminus \{0\}$ by taking an equivalence class:

$$(x_0, \ldots, x_n) \sim (y_0, \ldots, y_n) \qquad \Leftrightarrow \qquad (x_0, \ldots, x_n) = (\lambda y_0, \ldots, \lambda y_n) \qquad \text{for some } \lambda \neq 0$$

This produces an n-dimensional smooth manifold with the usual patches defined as:

$$U_i = D(x_i) = \{[x_0 : \cdots : x_n] \in \mathbb{P}^n \mid x_i \neq 0\}$$

where, using the notation $x_{k,j} = x_k/x_j$ for the affine coordinates on U_j, the transition maps were defined as (see [53, chapter 1]):

$$\varphi_i \circ \varphi_j^{-1}(x_{0,j}, \ldots, \widehat{x_{j,j}}, \ldots, x_{n,j}) = \left(\frac{x_{0,j}}{x_{i,j}}, \ldots, \frac{1}{x_{i,j}}, \ldots, \frac{\widehat{x_{i,j}}}{x_{i,j}}, \ldots, \frac{x_{n,j}}{x_{i,j}}\right)$$

In [20], we saw we can do the same for any field k, namely since inverses exist. Let us do the same but with a general ring R and the graded polynomial ring $S = R[x_0, x_1, \ldots, x_n]$ with the standard grading ($\deg x_i = 1$, R in degree 0), and $S_+ = \bigoplus_{d>0} S_d$ being the irrelevant ideal. Though S is naturally $\mathbb{N}$-graded, we shall work in the setting of $\mathbb{Z}$-graded rings since localizations introduce negative degrees[22].

As a set, define:

Definition 2.3.1: Projective Space – as a Set

$$\operatorname{Proj} S = \{[\mathfrak{p}] \mid \mathfrak{p} \subseteq S \text{ is a homogeneous prime with } S_+ \not\subseteq \mathfrak{p}\}$$

The condition $S_+ \not\subseteq \mathfrak{p}$ excludes the "irrelevant" prime consisting of all positive-degree elements, which would correspond to the non-existent point $[0 : \cdots : 0]$ in projective space. With the set defined, $\operatorname{Proj} S$ has the Zariski topology induced by homogeneous elements of positive degree:

$$V_+(T) = \{[\mathfrak{p}] \in \operatorname{Proj} S \mid T \subseteq \mathfrak{p}\}$$

The basis for the topology can be given by the positive-degree homogeneous elements:

$$D_+(f) = \{[\mathfrak{p}] \in \operatorname{Proj} S \mid f \notin \mathfrak{p}\}$$

If $f = x_i$, we would like this to correspond to the usual affine space $\mathbb{A}^n_R$. In the classical case, we got this isomorphism by doing:

$$\begin{aligned}
\{[x_0 : \cdots : x_n] \mid x_i \neq 0\} &= \left\{ \left[\frac{x_0}{x_i} : \cdots : 1 : \cdots : \frac{x_n}{x_i} \right] \,\middle|\, x_i \neq 0 \right\} \\
&\cong k\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \\
&= k[x_{0,i}, \ldots, x_{n,i}] && \text{relabeling} \\
&= \mathbb{A}^n_k
\end{aligned}$$

Notice how these are all degree 0 in the grading of S. These sets are not quite the same as $\operatorname{Spec} S_f$; however it turns out that taking $D_+(f)$ and working with the degree-0 part of the localization gives us exactly the desired result:

Lemma 2.3.2: Characterizing Basic Open Sets For Projective Space

Let $f \in S$ be a homogeneous element of positive degree, and define $S_{(f)} := (S_f)_0$ to be the degree-0 part of the localization. Then:

$$D_+(f) \cong \operatorname{Spec}\left(S_{(f)}\right)$$

***Proof*:**
We construct explicit inverse maps. Define the forward map $\Phi : D_+(f) \to \operatorname{Spec}(S_{(f)})$ by:

$$\Phi(\mathfrak{p}) = \mathfrak{p} \cdot S_f \cap S_{(f)}$$

Since $f \notin \mathfrak{p}$, the extension $\mathfrak{p} \cdot S_f$ is a proper homogeneous prime of S_f, and its intersection with

[22] More general gradings are sometimes useful, however we shall not need this either in this book or for a while

the subring $S_{(f)}$ is a prime ideal of $S_{(f)}$.

For the inverse, let $e = \deg f$ and define $\Psi : \operatorname{Spec}(S_{(f)}) \to D_+(f)$ by:

$$\Psi(\mathfrak{q}) = \bigoplus_{d \geq 0} \left\{ a \in S_d \;\middle|\; \frac{a^e}{f^d} \in \mathfrak{q} \right\}$$

Note that a^e has degree de and f^d has degree de, so $a^e/f^d \in S_{(f)}$ and the condition makes sense.

$f \notin \Psi(\mathfrak{q})$: If $f \in \Psi(\mathfrak{q})$, then $f^e/f^e = 1 \in \mathfrak{q}$, contradicting $\mathfrak{q}$ being proper.

Closure under addition: Let $a, b \in S_d$ with $a^e/f^d, b^e/f^d \in \mathfrak{q}$. Consider a binomial term $a^j b^{e-j}$ with $0 \leq j \leq e$. Observe:

$$\left(\frac{a^j b^{e-j}}{f^d}\right)^e = \left(\frac{a^e}{f^d}\right)^j \cdot \left(\frac{b^e}{f^d}\right)^{e-j} \in \mathfrak{q}$$

Since $\mathfrak{q}$ is prime (hence radical: $x^n \in \mathfrak{q} \Rightarrow x \in \mathfrak{q}$), we get $a^j b^{e-j}/f^d \in \mathfrak{q}$ for each j. Summing over all binomial terms gives $(a+b)^e/f^d \in \mathfrak{q}$.

Closure under S-multiplication: If $a \in \Psi(\mathfrak{q})_d$ and $c \in S_k$, then $(ca)^e/f^{d+k} = (c^e/f^k)(a^e/f^d)$. The second factor lies in $\mathfrak{q}$ and the first in $S_{(f)}$, so the product lies in $\mathfrak{q}$.

Primality: If $a \in S_d, b \in S_k$ with $ab \in \Psi(\mathfrak{q})$, then $(ab)^e/f^{d+k} = (a^e/f^d)(b^e/f^k) \in \mathfrak{q}$. Since $\mathfrak{q}$ is prime, either $a^e/f^d \in \mathfrak{q}$ or $b^e/f^k \in \mathfrak{q}$, giving $a \in \Psi(\mathfrak{q})$ or $b \in \Psi(\mathfrak{q})$.

One can verify that Φ and Ψ are inverse to each other: the key step is that $a \in \mathfrak{p}_d$ implies $a^e/f^d \in \mathfrak{p}S_f \cap S_{(f)}$, and conversely if $a^e/f^d \in \mathfrak{p}S_f$ then $a \in \mathfrak{p}$ by primality (since $f \notin \mathfrak{p}$). For the homeomorphism, the preimage of a basic open $D(g^e/f^{\deg g}) \subseteq \operatorname{Spec}(S_{(f)})$ is $D_+(fg) \cap D_+(f) = D_+(fg)$, so both sides have matching bases.

Let us apply this to $f = x_i$:

$$\begin{aligned} D_+(x_i) &= \{[\mathfrak{p}] \in \operatorname{Proj} S \mid x_i \notin \mathfrak{p}\} \\ &\cong \operatorname{Spec}\left(S_{(x_i)}\right) \\ &= \operatorname{Spec}\left(\text{degree-0 elements of } R[x_0, \ldots, x_n, 1/x_i]\right) \\ &\overset{!}{=} \operatorname{Spec} R\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \\ &\cong \operatorname{Spec} \frac{R[x_{0,i}, \ldots, x_{n,i}]}{(x_{i,i} - 1)} \\ &\cong \mathbb{A}^n_R \end{aligned}$$

where $\overset{!}{=}$ comes from the fact that $1/x_i$ is now a unit of degree -1, hence each x_j/x_i is of degree 0, and these elements generate $S_{(x_i)}$ over R. Relabeling x_j/x_i as $x_{j,i}$ and "omitting" $x_{i,i}$ (which equals one, equivalently we quotient by $(x_{i,i} - 1)$, a form of dehomogenizing), we see that it is isomorphic to the usual $\mathbb{A}^n_R$, and hence we recover the usual affine cover! Note the importance of taking the degree-0 component, for

$$\operatorname{Spec} R[x_0, \ldots, x_n, 1/x_i] \cong \mathbb{A}^{n+1}_R \setminus \{x_i = 0\}$$

Let us now construct the structure sheaf on $\operatorname{Proj} S$. We shall see that $\operatorname{Proj} S$ is not affine (we hope not, as the classical projective variety is not affine!), and so it is not determined by its global sections in the way that affine schemes are. Nonetheless, the construction below endows $\operatorname{Proj} S$ with a natural scheme structure. As we just saw, each $D_+(f)$ is affine and hence we certainly have an affine cover. Let us ensure that the gluing satisfies the cocycle condition from theorem 2.2.54, namely that all intersections glue appropriately to form a scheme.

Theorem 2.3.3: Structure Sheaf On Proj

Take the Zariski topology on $\operatorname{Proj} S$ given by the basis $D_+(f) \cong \operatorname{Spec}(S_{(f)})$. Then $\operatorname{Proj} S$ is a scheme.

***Proof*:**
Let $f, g \in S$ be homogeneous elements of positive degree and take $D_+(f), D_+(g)$ with non-trivial overlap. It is not difficult to show that $D_+(f) \cap D_+(g) = D_+(fg)$. We verify the cocycle condition from theorem 2.2.54. Consider:

$$D(g^{\deg f}/f^{\deg g}) \subseteq D_+(f) \qquad \text{and} \qquad D(f^{\deg g}/g^{\deg f}) \subseteq D_+(g)$$

We shall show that:

$$S_{(f)}\left[\frac{1}{g^{\deg f}/f^{\deg g}}\right] \cong S_{(fg)} \cong S_{(g)}\left[\frac{1}{f^{\deg g}/g^{\deg f}}\right]$$

This will induce the desired isomorphism on spectra. There is a natural map:

$$\varphi : S_{(f)} \to S_{(fg)}$$

mapping $h/f^k \mapsto hg^k/(fg)^k$. Taking $U = \left\{(g^{\deg f}/f^{\deg g})^k \;\middle|\; k \in \mathbb{N}\right\}$, notice that $\varphi(U)$ consists of units in $S_{(fg)}$, thus there is a natural map:

$$\widetilde{\varphi} : S_{(f)}\left[\frac{1}{g^{\deg f}/f^{\deg g}}\right] \to S_{(fg)}$$

It is straightforward to verify that this map is in fact an isomorphism (injectivity is easy, and matching degrees gives surjectivity). The same argument works in the other direction, giving the desired result. Thus, we have established the desired ring isomorphism, which induces the required isomorphism of schemes, completing the proof.

Definition 2.3.4: Projective Scheme

Let $S_\bullet$ be a $\mathbb{Z}$-graded ring with $S_0 = R$. Then $\operatorname{Proj} S_\bullet$ is called the projective scheme associated to $S_\bullet$. If $S = R[x_0, \ldots, x_n]$ with the standard grading, then the above construction defines the classical projective n-scheme or prototypical projective scheme over R, and is denoted $\mathbb{P}^n_R$ or $\mathbb{P}^n$ if unambiguous.

We can naturally recover the notion of quasiprojective as open subschemes of projective schemes:

Definition 2.3.5: Quasiprojective Scheme

A *quasiprojective R-scheme* is an open subscheme of a projective R-scheme.

Notice that the scalars can be defined for any ring, which generalizes the result from classical algebraic geometry. If $R = k$ is an algebraically closed field, the closed points of $\mathbb{P}^n_k$ will behave like the traditional points of projective space defined in classical algebraic geometry.

Proposition 2.3.6: Global Sections of $\operatorname{Proj} S_\bullet$

Let $S_\bullet$ be an $\mathbb{N}$-graded ring generated by S_1 as an S_0-algebra (e.g. $S = R[x_0, \dots, x_n]$ with the standard grading), and let $X = \operatorname{Proj} S_\bullet$. Then:

$$\mathcal{O}_X(X) = S_0$$

Hence $\mathbb{P}^n_R$ is not an affine scheme (the case $n = 1$ was given in example 2.2.55).

***Proof*:**
exercise

Recall from [20] that homogeneous polynomials of $k[x_0, \dots, x_n]$ cut out closed sets from $\mathbb{P}^n_k$. This is because if $a \in k^{n+1}$ is a root of a homogeneous polynomial, so are all scalar multiples of a. In modern algebraic geometry, the global sections $\mathcal{O}_X(X)$ are regarded as the "functions" on a scheme X. However, $\mathcal{O}_{\mathbb{P}^n_R}(\mathbb{P}^n_R) = R$ by proposition 2.3.6, so the graded ring S that *generates* $\operatorname{Proj} S$ does not manifest as the structure sheaf.

How, then, should we interpret S? The answer is that each graded piece S_d gives rise to a sheaf $\mathcal{O}(d)$ on $\mathbb{P}^n_R$ called a *twisting sheaf* (or *Serre twist*). These are examples of *line bundles* (invertible sheaves), and the structure sheaf $\mathcal{O}_{\mathbb{P}^n_R}$ is the special case $\mathcal{O}(0)$. We shall show in proposition 5.3.10 that the graded ring S can be recovered from these sheaves:

$$S_d \cong \Gamma(\mathbb{P}^n_R, \mathcal{O}(d))$$

This is significant because line bundles and their global sections control the geometry of morphisms *from* projective space: a morphism $\mathbb{P}^n_R \to \mathbb{P}^m_R$ is determined by a choice of $m+1$ global sections of a line bundle on $\mathbb{P}^n_R$ (with no common vanishing), a correspondence that will be made precise in section 5.3.3. In this way, the graded ring S – despite not being the ring of global functions – encodes the "embedding data" of projective space.

The reader ought to remember that if $f(x_{0,i}, \dots, x_{n,i})$ is a homogeneous polynomial of degree d on $D(x_i)$, then when converting coordinates to $D(x_j)$ we get:

$$f(x_{0,i}, \dots, x_{n,i}) = x_{j,i}^{\deg f}\, f(x_{0,j}, \dots, x_{n,j})$$

Example 2.3.7: Projective Scheme Homogeneous Polynomial Cut

There are many examples in [20], and hence we shall just put a few as a refresher.

1. Take $C = V(x^2 + y^2 - z^2) \subseteq \mathbb{P}^2_k$ for an algebraically closed field k. Then if $z = 0$, we would get $[(0,0,0)] \in C$, however this point doesn't exist in $\mathbb{P}^n_k$. Hence consider $z \neq 0$. Then for each solution $[(a,b,c)] = [(\lambda a, \lambda b, \lambda c)] \in C$. In k^3, the solutions form a cone of points.

Loosely visualizing $\mathbb{P}^2_k$ as a sphere (we should glue the antipodal points), we see that this resembles the image:

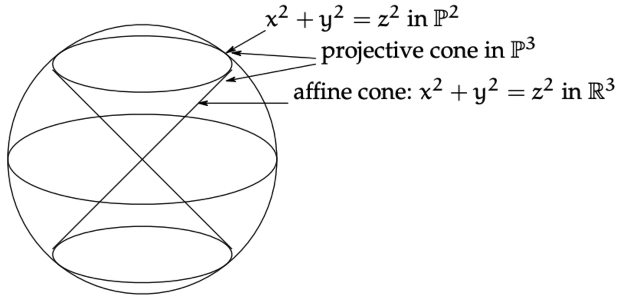


Figure 2.5: Visualization of $V(x^2 + y^2 - z^2)$

Such vanishing sets are called *vanishing cones.*

2. $\mathbb{P}^0_R = \operatorname{Proj} R[x_0] \cong \operatorname{Spec} R$, hence all 0-dimensional projective schemes are affine. Note then that $\mathbb{P}^0_k = \{\overline{0}\}$ since $\operatorname{Spec} k$ is a singleton; or this can be verified directly by noticing that the only non-irrelevant homogeneous prime ideal of $k[x]$ is (0), for all other homogeneous prime ideals are of the form (x^n) and hence contain S_+.

3. For an example that is not a hypersurface, take $V(wz - xy, x^2 - wy, y^2 - xz)$. This is called the *twisted cubic curve.* This ideal is prime, and the variety is isomorphic to $\mathbb{P}^1_k$ (hint: consider $k[w, x, y, z] \to k[s, t]$, $(w, x, y, z) \mapsto (s^3, s^2t, st^2, t^3)$).

4. When $R = k$ is a field, the definition of $\mathbb{P}^n_k$ as a scheme matches the classical definition of projective space as a quotient of $k^{n+1} \setminus \{0\}$ by $k^\times$. The points in the scheme correspond exactly to lines through the origin (plus the generic point). However, when R is not a field, $\mathbb{P}^n_R$ behaves like a "compact" object in a very powerful way. Consider the case where R is a Discrete Valuation Ring (DVR), for example $R = k[[t]]$ or $\mathbb{Z}_{(p)}$, with fraction field K. Consider a point $P \in \mathbb{P}^n_K$ (i.e., a point in projective space over the fraction field). We can write this point in coordinates $[f_0 : \cdots : f_n]$ where $f_i \in K$. We might ask: does this point "extend" to a point in $\mathbb{P}^n_R$? In other words, if we have a curve defined for $t \neq 0$, does it have a unique limit as $t \to 0$?

   The answer is yes. Since we are in projective space, we can scale coordinates by any non-zero element.

   (a) Let $v(f_i)$ be the valuation of each coordinate.
   (b) Multiply the entire tuple by t^{-m}, where $m = \min_i(v(f_i))$.
   (c) Now all coordinates are in R, and at least one coordinate is a unit (valuation 0).
   (d) We can now safely set $t = 0$ to find the unique limit point in the special fiber $\mathbb{P}^n_k \subset \mathbb{P}^n_R$.

   For example, if $R = k[[t]]$ and $P = [1/t^2 : 1/t] \in \mathbb{P}^1_K$, we scale by t^2 to get $[1 : t]$. Setting $t = 0$ gives the limit point $[1 : 0]$. This property – that any map from the punctured curve $\operatorname{Spec} K$ extends uniquely to the whole curve $\operatorname{Spec} R$ – is called the *Valuative Criterion for Properness.* It essentially states that projective schemes are "compact" (curves don't run off to infinity without hitting a limit). This implies that $\mathbb{P}^n_R$ is connected in a way affine schemes are not (an affine line with a missing origin cannot be patched!). See section 3.5.4 for more details.

5. In an affine scheme, two polynomials over a field cut out the same vanishing set if they differ by a scalar. This is not the case for homogeneous polynomials cutting out parts of $\mathbb{P}^n$: consider $x^2y = 0$ and $xy^2 = 0$. The problem is that these two are homogenizations of different polynomials that produce the same set. However, as we know from section 2.2.2, schemes remember multiplicity. This behaviour shall later be properly taken into account by saying that two polynomials shall cut out different *closed subschemes* (rather than the same *closed subset*). With this addendum, it shall once again be the case that two polynomials cut out the same closed subscheme if and only if they differ by a scalar multiple, as closed subschemes shall essentially correspond to polynomial cut-outs (as sets) that have sheaves remembering multiplicity information.

Proposition 2.3.8: Irreducibility of classical projective scheme

Let R be a domain. Then the scheme $\mathbb{P}^n_R$ is irreducible and is of finite type over R.

As irreducible spaces are connected, $\mathbb{P}^n_k$ is connected.

***Proof*:**
Since R is a domain, $S = R[x_0, \dots, x_n]$ is a domain, so the zero ideal (0) is a homogeneous prime not containing S_+. Hence $(0) \in \operatorname{Proj} S$ is a point. For each i, $x_i \notin (0)$, so $(0) \in D_+(x_i)$ for every i. Since $\{D_+(x_i)\}_{i=0}^n$ covers $\operatorname{Proj} S$, the generic point (0) lies in every basic open of the standard cover. As any nonempty open set contains some $D_+(x_i)$, it contains (0), so $\{(0)\}$ is dense and $\mathbb{P}^n_R$ is irreducible. Finite type over R follows from each chart $D_+(x_i) \cong \mathbb{A}^n_R = \operatorname{Spec} R[x_1, \dots, x_n]$ being of finite type over R, with finitely many charts.

To conclude this section, let us formally define the affine and projective cone:

Definition 2.3.9: Affine Cone

Let $S_\bullet$ be a graded ring. Then the affine cone of $\operatorname{Proj} S_\bullet$ is $\operatorname{Spec}(S_\bullet)$.

Note that $\operatorname{Spec}(S_\bullet)$ depends on $S_\bullet$, not just $\operatorname{Proj}(S_\bullet)$. Think of the affine cone as eliminating the points at infinity and adding the origin. In fact, given a field k, there is a natural map $\operatorname{Spec} S_\bullet \setminus V(S_+) \to \operatorname{Proj} S_\bullet$ (more generally, if $S_\bullet$ is a finitely generated graded ring over a base ring R, there is a natural morphism).

Definition 2.3.10: Projective Cone

Take $\operatorname{Spec} S_\bullet$. Then the projective cone or projective completion is $\operatorname{Proj}(S_\bullet[t])$ for some indeterminate t of degree 1.

Take for example $\operatorname{Proj}(k[x, y, z]/(z^2 - x^2 - y^2))$. Then the projective cone is $\operatorname{Proj}(k[x, y, z, t]/(z^2 - x^2 - y^2))$. The vanishing locus $V_+(t)$ returns the original projective scheme, and taking the complement (the distinguished open set $D_+(t)$) gives $\operatorname{Spec}(S_\bullet)$. This construction can be thought of as the opposite of the affine cone, where we attach the points at infinity of a conic curve.

The Non-Uniqueness of Graded Rings

Before moving on, we must highlight a fundamental difference between the Spec and Proj constructions. The spectrum functor provides a perfect anti-equivalence between commutative rings and affine schemes; if $\operatorname{Spec} A \cong \operatorname{Spec} B$ as schemes, then $A \cong B$ as rings.

The Proj construction completely fails this property. Two wildly different graded rings can produce the exact same projective scheme. For example, let $S_\bullet = k[x, y]$ and let $T_\bullet = k[x^2, xy, y^2] \cong k[u, v, w]/(uw - v^2)$. As standard rings, these are not isomorphic: $S_\bullet$ is a Unique Factorization Domain, while $T_\bullet$ is not (since $uw = v^2$ represents two distinct factorizations into irreducibles). Geometrically, their affine cones are drastically different: $\operatorname{Spec} S_\bullet$ is the smooth plane $\mathbb{A}^2_k$, whereas $\operatorname{Spec} T_\bullet$ is a quadric cone which is singular at the origin.

However, because Proj effectively removes the origin (by restricting to homogeneous prime ideals not containing S_+) and takes a quotient by the grading, these differences vanish: $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} T_\bullet \cong \mathbb{P}^1_k$ (proven in theorem 3.1.35).

In general, the Proj construction only determines a graded ring "up to truncation at finite degrees". If two graded rings $S_\bullet$ and $T_\bullet$ share the exact same graded pieces for all degrees $d \geq N$ for some large integer N, they will yield isomorphic projective schemes. We will formalize exactly how to recover the "canonical" coordinate ring of a projective scheme using twisted sheaves in section 5.3.

Proj-like Functors for other schemes?

here

2.4 Properties of Schemes

In this section, we shall cover various different properties a scheme may have. In the process, we shall see if these properties can be checked affine-locally or stalk-locally Many of the topological properties inquired about affine schemes can be asked about for general schemes. For example, general schemes need not be quasicompact, while all affine schemes are quasicompact. Properties that have been explored already are:

1. (dis)connected: In the affine case, this corresponds to the existence of non-trivial idempotent elements, for if $e \in R$ is such an element then $R \cong eR \times (1 - e)R$, and so the spectrum is a disjoint union. For the general case of schemes, it is usually given by the disjoint union of known schemes.

2. (ir)reducible: In the affine case, this corresponds to the ring having *no zero divisors*, i.e. an integral domain. Proposition 2.4.19 shall characterize irreducible schemes[23].

   Recall that an equivalent conditions to irreducibility is that every non-empty open subset is dense (as the reader should verify).

3. quasicompact: Essentially all schemes we shall work with will be quasicompact (with the "trivial" exception being the infinite disjoint union of schemes). Most of the time, non-quasicompact schemes will come from some infinite gluing.

[23] with the reducedness condition insuring no "fuzz" in the sheaf

These are global topological properties, hence they cannot be checked affine- or stalk-locally (as the reader should verify with some counter-examples).

Finiteness Conditions

The most common types of rings in commutative algebra are Noetherian rings. The geometric equivalent is the following:

Definition 2.4.1: Noetherian Scheme

Let X be a scheme. If X can be covered by affine open sets where each ring R_i of $\mathrm{Spec}\,(R_i)$ is Noetherian, then X is said to be a *locally Noetherian Scheme*. If X is also quasicompact, then X is said to be a *Noetherian Scheme*.

All coordinate rings are Noetherian, and hence we may think of varieties as examples of Noetherian schemes[24]. We can think of the Noetherian condition as insuring that the open affine don't have "difficult" infinite behavior. Most examples which involve non-Noetherian rings involve infinitely many variables (ex. $k[x_1, x_2, ...]$, $k[x, y, y/x, y/x^2, ...]$ and so forth). In part I; examples of non-Noetherian topological spaces we shall encounter will be the fibered product of Noetherian rings (see ref:HERE), and the special product topology put on the ring of adèles. In general, we shall see that there are many results that do not require finite generation, and hence it is worth keeping track of it when this condition is necessary.

One important consequence of being Noetherian is the existence of finitely many irreducible components:

Lemma 2.4.2: Finitely Many Components

Let X be a Noetherian scheme. Then X has finitely many irreducible components.

***Proof*:**
As X is Noetherian, it is locally Noetherian, so let it be covered be Noetherian affine open schemes. As it is quasicompact, we may take a finite subcover so $X = \bigcup_i^n \mathrm{Spec}\, R_i$. As each R_i is Noetherian, it can have only finitely many irreducible components (check this), and hence their union can have only finitely many irreducible components, completing the proof.

This is not true if the ring is not Noetherian: the spectrum of the ring

$$\varinjlim_{n\to\infty} \frac{k[x_i \,|\, 1 \leq i \leq n]}{\prod_i^n x_i}$$

has infinitely many components, but it is a quasicompact scheme (as it is affine).

Let us next turn to an important property of quasicompact schemes. When working with affine schemes, we always have that there exists closed points corresponding to the maximal ideals of the ring. When working over general schemes, there may be no closed points[25]. Fortunately, all quasicompact scheme have closed points, and even better:

[24] Once we introduce theorem 3.7.1 this will be made precise
[25] see https://math.stanford.edu/~vakil/files/schwede03.pdf

Proposition 2.4.3: Quasicompact Schemes and Closed Points

Let X be a quasicompact scheme. Then X contains closed points. Furthermore, the closure of any non-closed point contains closed points.

***Proof*:**
First, if X is affine, it certainly contains closed points as $\mathcal{O}_X(X) = R$ contain at least one maximal ideal. Furthermore, the closure of any non-closed point would contain maximal ideal as any prime ideal is contained in a maximal ideal.

Now, let X be some quasicompact scheme. Let $X = \bigcup_i^n U_i$ be a finite covering of affine open subsets. Take any closed point $\mathfrak{p}_1 \in U_1$. If it is closed in X we're done. If not, take its closure $\overline{\mathfrak{p}_1}$ in X. As $\mathfrak{p}_1$ is not closed in X, its closure contains more than $\mathfrak{p}_1$, call this point $\mathfrak{p}_2$, and without loss of generality say it is in U_2. If $\mathfrak{p}_2$ is closed in X, we are done. If not, take $\overline{\mathfrak{p}_2}$ in X. Its closure must contain a point $\mathfrak{p}_3$ that is not in U_2 or U_1 (as $\mathfrak{p}_3$ would be in the closure of $\mathfrak{p}_1$ or $\mathfrak{p}_2$ in U_1 and U_2 respectively). Continue this process until a point is closed, or if we get to $\mathfrak{p}_n \in U_n$, then the closure of this point can't contain any other point in $U_1, \ldots, U_n$, making $\mathfrak{p}_n$ closed and completing the proof.
Note that $\mathfrak{p}_n$ is in the closure of $\mathfrak{p}_1$ in X : by a topological argument it is easy to show that:

$$\overline{\mathfrak{p}_1} \supseteq \overline{\mathfrak{p}_2} \supseteq \cdots \supseteq \overline{\mathfrak{p}_n} = \mathfrak{p}_n$$

and hence the closure of any non-closed point contains a closed point.

Note that this does not mean that the closed points are dense, that is it does not imply that the closure of the collection of closed point is the entire space. Such schemes are called Jacobson schemes and are related to Jacobson rings and Jacobson radicals (see [10]). Once we introduce (abstract) varieties, we shall see that the closed points are dense for varieties. The following gives a good set of (affine) schemes whose closed points are dense:

Lemma 2.4.4: Dense Closed Points

Let A be a finitely generated k-algebra. Then the closed points of $\operatorname{Spec} A$ are dense

***Proof*:**
It suffices show that for each $f \in A$ where $D(f) \neq \emptyset$ (i.e. f is not nilpotent, see exercise ref:HERE), $D(f)$ $\operatorname{Spec}(A_f) \cong D(f)$, and A_f certainly contains maximal ideals, there is a maximal ideal $\mathfrak{p}_f \subseteq A_f$ which corresponds to a (at least) prime ideal $\mathfrak{p} \subseteq A$ not containing f. If we show that $A/\mathfrak{p}$ is a field, we're done. By Zariski's lemma, F is a finitely generated k-algebra if it is a finitely generated k-module, hence it suffice to show that $A/\mathfrak{p}$ is a finitely generated k-vector space. As $(A/\mathfrak{p})_f \cong A_f/\mathfrak{p}_f$, if we show that $A_f/\mathfrak{p}_f$ is a finite dimensional k-vector space, we can lift to a finite basis for $A/\mathfrak{p}$.

Now, as $A_f = A[1/f]$, it is a finitely generated k-algebra and $\mathfrak{p}_f$ is maximal, $A_f/\mathfrak{p}_f$ is a finite dimensional k-vector space by Zariski. But then we get the desired conclusion

We can also show it in the following more "straightforward" with a more subtle application of Nullstellensatz. For the sake of contradiction, suppose $D(f)$ has no maximal ideals. Then for each maximal ideal $\mathfrak{m} \subseteq A$, $f \in \mathfrak{m}$. Thus:

$$f \in \bigcap_{\mathfrak{m} \subseteq A} \mathfrak{m} \stackrel{!}{=} \sqrt{(0)}$$

where $\stackrel{!}{=}$ comes as a consequence of the Nullstellensatz (key here is the finitely generation as a k-algebra). But then f is nilpotent, and so $D(f) = \emptyset$. Hence, any nonempty set contains a closed point, completing the proof.

Recall in proposition 2.3.18 it was mentioned that invertibility can be determined if a function is non-zero at every point, however a function being zero at all points requires more care to study. Part of this is because this is true for quasicompact schemes

Proposition 2.4.5: Quasicompact Schemes and Vanishing Function

Let X be a quasicompact scheme. Then if $f \in \mathcal{O}_X(X)$ vanishes at all points in X, then there exists an n such that $f^n = 0$.

***Proof*:**
Say $f([\mathfrak{p}]) = 0$ so that $f \in \mathfrak{p}$ for all $\mathfrak{p}$. Then on $U_i \cong \operatorname{Spec} R_i \subseteq X, f \in \sqrt{R_i}$, implying $f^{n_i} = 0$ for some n. Cover X with these open sets, and by Quasicompactness we only require finitely many such sets, where in each set we have an f_i and n_i such that $f_i^{n_i} = 0$. As X is a scheme, on intersection $U_i \cap U_j$ we have $g_{i,j} = f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ and as ring homomorphisms must map 0 to 0, $g_{i,j}^{\max n_i, n_j} = 0$. Take $n = \max_i n_i$ so that for each intersection $U_i \cap U_j$:

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j} = g_{i,j} \quad g_{i,j}^n = 0$$

Now, as these agree on intersections and cover X, by definition there must exist a function $f \in \mathcal{O}_X(X)$ such that $f|_{U_i} = f_i$. Then as $\operatorname{res}_{X,U_i}(f^n) = \operatorname{res}_{X,U_i}(f)^n = f_i^n f_i^{n_i n'} = 0 f^{n'} = 0$, we have:

$$(f^n)|_{U_i} = \left(f|_{U_i}\right)^n = 0$$

which implies $(f^n)|_{U_i} = 0 \in R_i$. Then by guability, it must be that $f^n = 0$, completing the proof.

This is not necessarily true for a general scheme, namely since we can have nilpotence of every degree:

Example 2.4.6: Non-vanishing Function On Scheme
Take

$$X = \coprod_{n \geq 1} \operatorname{Spec}\left(k[x]/\left(x^n\right)\right)$$

Then the global function ϵ is zero every-where, namely there is only one point to evaluate to, but essentially by definition there cannot exist an n such that $\epsilon^n = 0$.

Lemma 2.4.7: Projective Scheme is Noetherian

Let $X = \mathbb{P}^n_{\mathbb{Z}}$ or $\mathbb{P}^n_k$. Then X is Noetherian.

***Proof*:**
exercise.

To conclude this section, we shall define one more type of scheme here due to the fact that all affine and projective schemes have this property, however it shall not be of great use to us until chapter ref:HERE. The following establishes some finiteness conditions on the intersection of important open subsets:

Definition 2.4.8: Quasi-separated

Let X be a topological space. Then X is *quasiseparated* if the intersection of any two quasicompact open subsets is quasicompact.

Proposition 2.4.9: Characterizing Quasi-separated Schemes

Let X be scheme. Then X is quasi-separated if and only if the intersection of any two affine open subsets is a finite union of affine open subsets

***Proof*:**
This is very similar to a compactness argument and is left as an exercise

Proposition 2.4.10: Affine Scheme and Quasiseparatedned

Let $X = \operatorname{Spec} R$ be an affine scheme. Then X is quasiseparated.

***Proof*:**
The intersection of affine open subsets is affine (example ref:HERE shows that $U \cap V \cong \operatorname{Spec}(S \otimes_R S')$ for $U, V \subseteq \operatorname{Spec} R$ and $U \cong \operatorname{Spec} S$, $V \cong \operatorname{Spec} S'$. Then by quasi-compactness this can be covered by finitely many distinguished open subsets.

From this, we can reduce to the case of affine opens and the rest follows quickly.

A large family of schemes that are quasi-separated and quasicompact are projective R-schemes, and as we shall see many schemes we shall be working with can be embedded into a projective R-scheme, and hence these two finiteness properties will be very common. There do exist non quasi-separated schemes:

Example 2.4.11: Non-quasiseparated Scheme
Let $X = \operatorname{Spec}(k[x_1, x_2, \dots])$ and let $U = X \backslash [\mathfrak{m}]$ where $\mathfrak{m}$ is the maximal ideal $(x_1, x_2, \dots)$. Take two copies of X and glue along U to create a double-origin scheme, namely the infinite version of the double-origin scheme seen in example 2.15. Show this is not quasiseparated.

Lemma 2.4.12: Projective Scheme Quasicompact And Quasiseperated

Let X be a projective R-scheme. Then X is quasicompact and quasiseparated

Note that R need not be Noetherian, hence any projective $\mathbb{P}^n_R$ scheme is quasicompact.

***Proof*:**
hint: affine communication lemma.

Exercise 2.4.1

1. Let X be a scheme. Show that X is a locally Noetherian scheme if and only if every open affine subset corresponds to a Noetherian ring.
2. Show that $\prod_i^\infty \operatorname{Spec} R_i$ properly embeds in $\operatorname{Spec}\left(\prod_i^\infty R_i\right)$. More precisely, the right hand side is (significantly) larger than the left hand side.

2.4.1 Reducedness and Integrality

The next important property that general rings have that coordinate rings for varieties don't have is the presence of zero-divisors and nilpotence. A ring with zero-divisors can be decomposed into a product to two rings and corresponds to a function vanishing on separate irreducible components (there must be more than 1), while a ring with nilpotence have effect on the sheaf of a ring as discussed in section 2.2.2. A ring with no nilpotent elements is said to be *reduced* (note that it may have zero-divisors[26]). A scheme that has this property of being reduced for every ring of sections is given a name:

Definition 2.4.13: Reduced Scheme

Let X be a scheme. Then X is a *reduced scheme* if $\mathcal{O}_X(U)$ is reduced for every open set U of X

Naturally, on a reduced scheme by proposition 2.4.5 a global section that evaluates to zero at every point will be the zero-section.

Proposition 2.4.14: Reducedness Is Stalk-local

Let X be a scheme. Then X is reduced if and only if non of the stalks have nonzero nilpotents. As a consequence, if $f, g \in \mathcal{O}_X(U)$ are two functions and agree on all points, then $f = g$

***Proof*:**
if X is a reduced scheme, then the localization will preserve reducedness, and hence each stalk is reduced. For the converse, we can show the contrapositive and assume there exists a $f \in \mathcal{O}_X(U)$ such that $f^n = 0$. Note that f cannot be a unit (as the reader could verify). Then some affine open subset $\operatorname{Spec} R \subseteq U$ we have $f|_U^n = f^n|_U = 0|_U$, showing that f is still nilpotent in $\mathcal{O}_X(\operatorname{Spec} R)$. Then it is a non-unit element in $\operatorname{Spec} R$, and hence belongs in some maximal ideal

[26] if D is the set of all zero-divisors of a reduced ring, then it is the union of all minimal ideals

$f \in \mathfrak{m}$. Then localizing at $[\mathfrak{m}]$, it is easy to see that f remains to be a nilpotent element, showing that not all stalks are reduced.

For the next assertion, this will be a simpler version of proposition 2.4.5 as we do not need to keep track of any exponents, and hence the Quasicompactness condition is not needed! Assume $f, g \in \mathcal{O}_X(U)$ such that $f(x) = g(x)$ for all $x \in U$. We want to show that $f = g$. Take $h = f - g$. Now, by definition $h(x) = f(x) - g(x) = 0$, so h must be contained in every prime ideal on every affine open subset of U. As $\mathcal{O}_X(U)$ is reduced, $\sqrt{(0)} = (0)$ on each affine open, and so $h|_{\operatorname{Spec} R} = 0$ for each element in the affine open covering, which by the gluging gives us $h = 0$, completing the proof.

Lemma 2.4.15: Reducedness For Affine And Projective Schemes

Let R be a reduced ring. Then $\operatorname{Spec}(R)$ is reduced. Hence, $\mathbb{A}^n$ and $\mathbb{P}^n$ are reduced.

***Proof*:**
exercise.

Proposition 2.4.16: Reducedness On Quasicompact Schemes

Let X be a quasicompact scheme. Then X is reduced if and only if it is reduced at closed points

***Proof*:**
Assume it is reduced at all closed points. As X is quasi-compact, the closure of each non-closed point contains a closed point; say $Y \in X$ is a non-closed and $x \in X$ is a closed point contained in its closure. Then $\mathcal{O}_{X,Y}$ is the localization of $\mathcal{O}_{X,x}$. As localization preserves reducedness, $\mathcal{O}_{X,Y}$ must be reduced, completing the proof.

The above showed that reducedness for quasicompact schemes is a “closed-point” local condition. It is not an open set local condition. Furthermore, it is not enough to check the global sections:

Example 2.4.17: Reducedness Edge Case

1. The scheme given by:

$$X = \operatorname{Spec} \frac{\mathbb{C}[x, y_1, y_2, \dots]}{(y_1^2, y_2^2, \dots, (x-1)y - 1, (x-2)y_2, \dots)}$$

   is nonreduced at the closed points corresponding to the positive integers. The complement of this set is not Zariski open.
   The key is that if X is also locally Noetherian, then reducedness is a locally open condition.

2. Another mistake that is easy to make: let X be scheme where the global sections is reduced. This doesn't imply X is reduced (take the scheme cut out by $x^2 = 0$ in $\mathbb{P}^2_k$; note that the global section will be k, but X will be non-reduced).

One of the most important type of reduced ring is an integral domain. If a scheme is an integral domain for each section, it is given a name:

Definition 2.4.18: Integral Scheme

Let X be a scheme. Then X is *integral* if it is nonempty and $\mathcal{O}_X(U)$ is an integral domain for every nonempty open subset $U \subseteq X$.

Recall that the product of integral domain is not an integral domain, or geometrically, the disjoint union of integral scheme need not be integral. This global behavior cannot be captured through stalks, as the reader should verify, and hence integrality is not stalk-local. Integral schemes can be thought of as schemes that are in "one piece" and have "no fuzz":

Proposition 2.4.19: Integral, then Irreducible and Reduced

Let X be a scheme. Then if X is an integral scheme, it is irreducible and reduced

It is in fact an if and only if, which we shall show in proposition 2.4.25.

***Proof*:**
If X is integral, it is certainly reduced as each ring of local sections is integral (and hence reduced). To show it is irreducible, assume $X = X_1 \cup X_2$, X_1, X_2 closed. For the sake of contradiction, let's say these sets don't contain each other. Then there exists open subsets $U \subseteq X_1, V \subseteq X_2$ such that $U \cap V = \emptyset$. Indeed, given $X_1 \not\subseteq X_2$, then here exists a point $p \in X_1 \setminus X_2$. Let

$$U = (X \setminus X_2) \cap X_1$$

Then U is open in X_1, and hence by construction in X. Similarly, define $V = (X \setminus X_1) \cap X_2$. Then $U \cap V = \emptyset$.

Then, by exercise ref:HERE, we have by an almost direct consequence of the sheaf axioms that:

$$\mathcal{O}_X(U \coprod V) \cong \mathcal{O}_X(U) \times \mathcal{O}_X(V)$$

But then $\mathcal{O}_X(U \coprod V)$ certainly contains zero-divisors, and hence cannot be integral.

An important property that integral domains is the ability to localize to obtain a fractional field. Localizing at an integral schemes generic point shall give us a "global" function field in which all local sections can embed, similarity to varieties! We first prove that an integral scheme has a unique generic point. Note the proof can be done if we replace integrality with irreducibility, that is we don't need every ring of local sections to be integral, see exercise ref:HERE.

Lemma 2.4.20: Irreducible Schemes and Generic Points

Let X be an integral scheme. Then for every irreducible closed subset $Z \subseteq X$, there exists a unique generic point η as $\overline{\eta} = Z$. In particular, as X is irreducible, then there exists a point η such that $\overline{\eta} = X$.

***Proof*:**
Let $Z \subseteq X$ be an irreducible subscheme. Then $Z = \bigcup_i U_i$ is an affine open cover. As $U_i = \operatorname{Spec} R_i$ is open and Z is irreducible, $\overline{U_i} = Z$ (exercise in topology). Now, as X is integral, $[(0_i)]$ is the generic point of U_i as (0_i) is prime. Let $\eta_i = [(0_i)]$. Then:

$$\overline{\eta_i} = Z$$

This shows a generic point exists, let's shows the generic point is unique. Say η_1, η_2 are generic points of Z so that $\overline{\eta_1} = \overline{\eta_2} = Z$. Take any affine open $U_i \subseteq Z$. Then we must have $\eta_1, \eta_2 \in U_i$. If they weren't, then $\eta_i \in Z \setminus U_i$ would be a closed and hence there exists a smaller closed set containing η_i which would imply $\overline{\eta_i} = Z \setminus U_i$, a contradiction. Thus, we get $\eta_1, \eta_2 \in U_i$. Hence, in the subspace topology

$$\overline{\eta_1} = \overline{\eta_2} = U_i$$

Showing that these are also the generic points of U_i. But as U_i is affine and integral, there can only exist a single generic point, hence $\eta_1 = \eta_2$.

In the special case where X is itself irreducible, we can conclude it has a unique generic point η, as we sought to show.

This is a purely topological fact, and hence can also be proven using only irreducibility, see exercise ref:HERE.

The existence was given by reducing to an affine open and "using" its generic point. Using this, we can quickly calculate the stalk at the generic point of an integral scheme. We give this stalk a name:

Definition 2.4.21: Rational Field

Let X be an integral scheme. Then the function field is the collection

$$\mathcal{O}_{X,\eta} = R_{(0)} = \operatorname{Frac}(R)$$

for any R given by $\operatorname{Spec}(R) \subseteq X$, and it is often denoted $\kappa(X)$. If X is a k-scheme, then it is denoted $k(X)$.

Thus, integral schemes have a similar property to that of classical affine and projective varieties, namely that their sections over different open sets can be considered as subsets of a single ring! Gluing is easy as we can see that two elements are glued if and only if they are the same element of $\kappa(X)$.

Proposition 2.4.22: Integral Scheme: Restriction Maps and Function Field

Let X be a scheme that is irreducible and reduced. If $U \subseteq V$ where $V \neq \emptyset$, then

$$\operatorname{res}_{V,U} : \mathcal{O}_X(V) \hookrightarrow \mathcal{O}_X(U)$$

is an inclusion.

***Proof*:**
As X is reduced, its sections are all determined by their value on the points, and hence can be treated as function (see example 2.4.23 on why this proof fails otherwise). Take $U \subseteq V \subseteq X$, $f, g \in \mathcal{O}_X(U)$ and assume $f|_U = g|_U$. We want to show that $f|_V = g|_V$. Define:

$$Z = \{x \in V \mid f(x) = g(x)\}$$

Z is closed in V as it is the zero set of $f - g$. By definition $U \subseteq Z$, $Z \subseteq V$, and U is open in V. As X is integral, the open subset V is irreducible. But then U is *dense* in V. As Z is closed in V and contains U, it must be that $Z = V$, but then

$$f|_V = g|_V$$

as we sought to show

Note that we require *both* the irreducibility and the reducedness, having *only* irreducibility is not enough

Example 2.4.23: Non-injective Non-reduced Scheme

Let $R = \frac{k[x,y]}{(xy,y^2)}$ so that $\operatorname{Spec} R$ is the x axis there is a bit of fluff in direction of the y-axis at the point 0. Then given $D(x) \subseteq \operatorname{Spec} R$, the map $\operatorname{res} : \mathcal{O}_{\operatorname{Spec} R}(\operatorname{Spec} R) \to \mathcal{O}_{\operatorname{Spec} R}(D(x))$ maps $y \mapsto 0$ as

$$y \mapsto \frac{y}{1} = \frac{xy}{x} = \frac{0}{x} = 0$$

Corollary 2.4.24: Integral Scheme: Generic Point

Let X be an integral scheme. Then there exists a generic point η such that given any choice of $\operatorname{Spec}(R) \subseteq X$, there is always an inclusion:

$$\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_{X,\eta} \cong \kappa(R)$$

***Proof*:**
By definition:

$$\mathcal{O}_{X,\eta} = \varinjlim_{\eta \in U} \mathcal{O}_X(U)$$

and as we saw lemma 2.4.20 the generic point is in each open set. By proposition 2.4.22, each map in the commutative diagram are injective, hence the map $\mathcal{O}_X(U)$ is injective into $\mathcal{O}_{X,\eta}$.

Proposition 2.4.25: Irreducible and Reduced, then Integral

Let X be a scheme. If X is irreducible and reduced, it is integral

***Proof*:**
Let X be irreducible and reduced. Then any open subscheme is irreducible and reduced (the irreducibility is proved topologically, and the reducedness can be deduced by the stalks). In

particular, any affine open subscheme $\operatorname{Spec} R$ must be irreducible and reduced. Certainly R must then be reduced, hence to show that $\operatorname{Spec} R$ is integral consider $xy \in R$ such that $xy = 0$. Then

$$\operatorname{Spec} R = V(0) = V(xy) = V(x) \cup V(y)$$

But that contradicts irreducedness. Hence, all ring of sections of affine open subsets are integral. For general open $U \subseteq X$, we must have an affine open $\operatorname{Spec} R \subseteq U$, and hence a restriction $\mathcal{O}_X(U) \to \mathcal{O}_X(\operatorname{Spec} R)$. By proposition 2.4.22, this map is injective, and hence $\mathcal{O}_X(U)$ is a subring of an integral domain, making it an integral domain. Hence all rings of local sections are integral domains, completing the proof.

If we have the added condition of being Noetherian, then we only need to check that it is connected and that all its stalks are integral domains:

Proposition 2.4.26: Characterizing Integrality in Noetherian Schemes

Let X be a Noetherian Scheme. Then X is integral if and only if X is nonempty, connected, and all stalks $\mathcal{O}_{X,\mathfrak{p}}$ are integral domains.

Intuitively, the stalk at the intersection of irreducible components will "remember" the generic points of the respective irreducible component.

***Proof*:**
By proposition 2.4.19 and proposition 2.4.25, X is integral if and only if it is reduced and irreducible. Then X is certainly non-empty, connected, and the stalks are integral (the localization of integral domains is integral).

Conversely, assume X is nonempty, connected, with integral stalks. First of all, as the stalks are integral, they are reduced. We must show that X is irreducible. As X is Noetherian, by lemma 2.4.2 X can be decomposed into finitely many irreducible components:

$$X = X_1 \cup X_2 \cup \cdots \cup X_n$$

As X is connected, for each i, j we must have $X_i \cap X_j \neq \emptyset$. Take an affine open subset $\operatorname{Spec} R \subseteq X$ and:

$$\operatorname{Spec} R = \bigcup_i^n (X_i \cap \operatorname{Spec} R)$$

Then as each X_i is irreducible, it is a topological argument to see that the sets $(X_i \cap \operatorname{Spec} R)$ form an irreducible decomposition of $\operatorname{Spec} R$. Hence, they must each have a generic point that corresponds to a minimal prime ideal. Hence, if the X_i are distinct, R_i must contains at least two distinct minimal prime ideals. Then if $[\mathfrak{p}] \in X_i \cap X_j$, $\mathcal{O}_{X,[\mathfrak{p}]} = (R_i)_{\mathfrak{p}}$ contains two distinct minimal prime ideals. But integral domains could only contain a single minimal prime ideal (namely the one associated to the generic point $[(0)]$), and so this stalk cannot be integral. Thus, it must be that $X_i = X_j$ for all i, j, showing that X is irreducible and completing the proof.

Exercise 2.4.2

1. Show that every point $x \in X$ in a topological space must be contains in an irreducible component (you will use Zorn's Lemma).

2. Show that connected components (maximal connected subset) are closed. Show that the connected components of $\mathrm{Spec}\left(\prod_i^\infty \mathbb{F}_2\right)$ are not open.
3. Show that $D(f)$ is empty if and only if f is nilpotent. Show that if X is an integral scheme, then $D(f)$ is always nonempty and contains the generic point.
4. If $f \in R$ is nilpotent, then R_f is the zero-ring.
5. Let X be an irreducible scheme. Show that there is a unique generic point η such that $\overline{\eta} = X$ (note that lack of reducedness). More generally, show that if $Z \subseteq X$ is a (nonempty) irreducible subset, there is a unique generic point η such that $\overline{\eta} = Z$
6. Let X be a reduced scheme and $Y \hookrightarrow X$ a closed immersion where $\iota(Y) = X$. Then ι is an isomorphism.

2.4.2 Normality and Factoriality

Recall that a normal ring R has the property that it an integral domain that is is integrally closed in it's field of fractions, $\overline{R}^{\mathrm{Frac}(R)} = R$. Due to their importance in number theory, we shall take the time to develop the theory of Schemes for which all the rings of sections are normal. These shall correspond to schemes that have some nicer behaviour for their singularities. It can be thought of as a condition weaker than smoothness (smoothness shall be called regular in section REF:HERE) as normality shall still allow for singularities, however these singularities shall be more tame. In particular, the reader can intuitively have the following geometric picture in mind for subschemes of Normal Schemes:

1. In codimension 1, all singularities are resolved, meaning it is “smooth” in codim 1
2. In codimension ≥ 2, being normal means being nice enough so that rational functions defined on the compliment of the singularities can be extended to the singularity (i.e. Algebraic Hartogs's lemma applies)

Definition 2.4.27: Normal Scheme

Let X be a scheme. Then X is said to be *normal* if every stalk $\mathcal{O}_{X,\mathfrak{p}}$ is normal domains.

Proposition 2.4.28: Normal Scheme Stalk Local

Let X be a scheme. Then X is a normal scheme if and only if $\mathcal{O}_{X,\mathfrak{p}}$ is normal for each $\mathfrak{p} \in X$

***Proof*:**
Recall (or reprove) that normality is preserved when localizing (hint: recall the proof that a UFD is normal)

Note that reducedness was stalk-local, and so it is immediate that normal schemes are reduced. As being integrally closed is well-behaved under localization (recall the proof that a UFD is normal), for an if R is normal then $\mathrm{Spec}\,(R)$ is a normal scheme. If each stalk is a UFD, then it is immediately a Normal scheme. Let's give a name to such a scheme

Definition 2.4.29: Factorial Scheme

Let X be a scheme. Then X is said to be *normal* if every stalk $\mathcal{O}_{X,\mathfrak{p}}$ is a UFD.

If R is a UFD, then $\operatorname{Spec} R$ is factorial (by the above reasoning), however the converse need be true! Namely, we may have factorial schemes where the stalks are all UFD's but the global sections do not form a UFD. A quintessential example will be the spectra of elliptic curves.

Example 2.4.30: Normal Schemes

1. The schemes $\mathbb{A}_k^n$, $\mathbb{P}_k^n$ and $\operatorname{Spec} \mathbb{Z}$ are normal. If R is normal, by definition $\operatorname{Spec} R$ is normal.
2. Starting on p.164 section 5.4.8, lots and lots of good examples

Exercise 2.4.3

1. Find the structure for the following rings:
   a) $\operatorname{Spec} \mathbb{C}[x]/(x^3 - x^2)$
   b) $\operatorname{Spec} \mathbb{C}[x]/(x^2 - d)$
   c) $\operatorname{Spec} \mathbb{C}[x]/(x^2 + 1)$
2. Let X be a quasicompact scheme. Show that the closure of each point contains a closed point (this is not true for non quasicompact schemes in general)
3. Let X be a quasicompact scheme. Suppose f vanishes at all point of X. Then there exists some N st $f^n = 0$. Show that this fails when X is not quasicompact (hint: take infinite disjoint unions of $\operatorname{Spec}(R_n)$ where $R_n := k[x]/(x^n)$)
4. Let X be a quasicompact scheme. Then it suffices to cheek reducedness at closed points (show that any nonreduced point has a nonreduced closed point in its closure)
5. Show that reducedness is not in general an open condition by considering:
$$X = \operatorname{Spec}\left(\frac{\mathbb{C}[x, y_1, y_2, ...,}{(y_1^2, y_2^2, y_3^3, ..., (x-1)y_1, (x-2)y_2, (x-3)y_3), ...}\right)$$
and $Y = \operatorname{Spec}[x]$. Show the nonreduced points of X are the closed points corresponding to the positive integers. The compliment of this set is *not* Zariski open.
6. Let X be an integral scheme. Show that if $\emptyset \neq V \subseteq U$, then the map $\operatorname{res}_{U,V} : \mathcal{O}_X(U) \to \mathcal{O}_X(V)$ is an inclusion. very important, see p.156 under exercise
7. Let X be an integral scheme, γ the generic point, and $\operatorname{Spec}(R) \cong U \subseteq X$ is any nonempty affine open subset of X. Then $\mathcal{O}_X(U) \to \mathcal{O}_{X,\gamma} = K(A)$ is an inclusion. Conclude that gluing is easy: every f_i in U_i for an open cover for U glue together if they are same element in $\operatorname{Frac}(X)$ (recall that $\operatorname{Frac}(X)$ is defined to be $\operatorname{Frac}(R)$ where R is given by $\mathcal{O}_X(\operatorname{Spec}(R))$)
8. Show that integrality is not stalk local via the example $\operatorname{Spec}(A) \coprod \operatorname{Spec}(B) = \operatorname{Spec}(A \times B)$.
9. Show that reducedness is an affine local property.

10. Let X be a locally Noetherian Scheme. Then X is quasiseparated.
11. Let X be a Noetherian Scheme. Then it has a finite number of connected components, each a finite union of irreducible components.
12. Show that the ring of sections of a Noetherian Scheme need not be Noetherian.
13. Let X be a scheme of locally finite type over $\mathbb{Z}$ or k. Then X is locally Noetherian.
14. A scheme that is of finite type over any Noetherian ring is Neotherian.
15. Let X be a quasiprojective R-scheme. Then if R is Noetherian, X is a Noetherian scheme, and hence has a finite number of irreducible components.
16. Let X be a locally Noetherian scheme. Then the reduced locus is open. (Vakil p.156)
17. Let $U \subseteq X$ be an open subscheme of a projective R-scheme X. Show that U is locally of finite type R
18. Find a counter-example showing that normal schemes are not in general integral schemes (hint: think of the disjoint union of two normal scheme. For more hint, Vakil p. 162).
19. Let X be a Noetherian scheme. Then X is normal if and only if if is a finite disjoint union of integral normal schemes.

2.5 Associated Points

Associated points capture the "component structure" of a scheme, including both the obvious irreducible components and more subtle "embedded" components arising from nilpotent structure. For a reduced scheme, the associated points are simply the generic points of the irreducible components. For non-reduced schemes, there can be additional associated points that "remember" how components come together or where nilpotent thickening occurs.

The algebraic foundation is the theory of associated primes of modules, which we briefly recall before giving the scheme-theoretic definition.

2.5.1 Associated Primes of Modules

Definition 2.5.1: Associated Prime

Let R be a ring and M an R-module. A prime ideal $\mathfrak{p} \in \operatorname{Spec} R$ is an *associated prime* of M if $\mathfrak{p} = \operatorname{Ann}(m)$ for some $m \in M$. The set of associated primes of M is denoted $\operatorname{Ass}_R(M)$ or simply $\operatorname{Ass}(M)$.

Equivalently, $\mathfrak{p} \in \operatorname{Ass}(M)$ if and only if $R/\mathfrak{p}$ embeds into M as an R-submodule (via $1 \mapsto m$). The condition $\mathfrak{p} = \operatorname{Ann}(m)$ is quite stringent: it is not enough for $\mathfrak{p}$ to annihilate some element; it must be the *full* annihilator.

Example 2.5.2: Associated Primes

1. Let $R = k[x]$ and $M = R$. The only associated prime is (0): if $\mathrm{Ann}(f) = \mathfrak{p}$ for some $f \neq 0$, then $\mathfrak{p} \cdot f = 0$, but $k[x]$ is a domain so $\mathfrak{p} = (0)$.

2. Let $R = k[x, y]$ and $M = R/(xy)$. The associated primes are (x) and (y): we have $\mathrm{Ann}(\bar{x}) = (y)$ and $\mathrm{Ann}(\bar{y}) = (x)$ in the quotient.

3. Let $R = k[x, y]$ and $M = R/(y^2, xy)$. The associated primes are (y) and (x, y): we have $\mathrm{Ann}(\bar{x}) = (y)$, and $\mathrm{Ann}(\bar{y}) = (x, y)$. The prime (x, y) is an *embedded* associated prime (see below).

4. Let $R = \mathbb{Z}$ and $M = \mathbb{Z}/12\mathbb{Z}$. Then $\mathrm{Ass}(M) = \{(2), (3)\}$: we have $\mathrm{Ann}(\bar{4}) = (3)$ and $\mathrm{Ann}(\bar{3}) = (4) \supsetneq (2)$ ... but $\mathrm{Ann}(\bar{6}) = (2)$. Note (2) and (3) are the primes appearing in the primary decomposition $12 = 4 \cdot 3 = 2^2 \cdot 3$.

The key properties of associated primes in the Noetherian setting are:

Proposition 2.5.3: Properties of Associated Primes

Let R be a Noetherian ring and M a finitely generated R-module. Then:

1. $\mathrm{Ass}(M) \neq \emptyset$ if and only if $M \neq 0$.

2. $\mathrm{Ass}(M)$ is finite.

3. the set of zero divisors on M is given by:

$$\bigcup_{\mathfrak{p} \in \mathrm{Ass}(M)} \mathfrak{p} = \{r \in R \mid rm = 0 \text{ for some } m \neq 0\}$$

,

4. The minimal elements of $\mathrm{Ass}(M)$ are exactly the minimal primes of $\mathrm{Supp}(M)$ (equivalently, the minimal primes of $\mathrm{Ann}(M)$).

***Proof*:**
See [10] for the full proof. We sketch the key ideas:

1. If $M \neq 0$, choose $m \neq 0$. The set of ideals $\{\mathrm{Ann}(m') : m' \in Rm,\ m' \neq 0\}$ is nonempty and has a maximal element $\mathrm{Ann}(m_0)$ by the Noetherian condition. One checks this maximal annihilator is prime.

2. Primary decomposition: $0 = \bigcap_{i=1}^{n} Q_i$ in M gives $\mathrm{Ass}(M) \subseteq \{\sqrt{Q_1}, \ldots, \sqrt{Q_n}\}$. Equality holds for the minimal primes; the others may or may not be associated.

3. If $r \in \mathfrak{p} \in \mathrm{Ass}(M)$, write $\mathfrak{p} = \mathrm{Ann}(m)$; then $rm = 0$ with $m \neq 0$. Conversely, if $rm = 0$ for $m \neq 0$, then $r \in \mathrm{Ann}(m) \subseteq \mathfrak{p}$ for some associated prime $\mathfrak{p}$ containing $\mathrm{Ann}(m)$.

4. Follows from the relationship between associated primes and primary decomposition.

Property (3) is particularly important: it says that in a Noetherian ring, the zero divisors are precisely the elements lying in some associated prime. This will translate to the geometric statement that a function is a zero divisor if and only if it vanishes at an associated point.

Definition 2.5.4: Minimal and Embedded Primes

Let R be a Noetherian ring and M a finitely generated R-module. An associated prime $\mathfrak{p} \in \mathrm{Ass}(M)$ is:

- *minimal* (or *isolated*) if it is minimal among associated primes, i.e., there is no $\mathfrak{q} \in \mathrm{Ass}(M)$ with $\mathfrak{q} \subsetneq \mathfrak{p}$;
- *embedded* if it is not minimal, i.e., there exists $\mathfrak{q} \in \mathrm{Ass}(M)$ with $\mathfrak{q} \subsetneq \mathfrak{p}$.

Minimal associated primes correspond to the irreducible components of $\mathrm{Supp}(M)$. Embedded associated primes correspond to "hidden" structure not visible in the support as a set, but detected by the module structure. Geometrically, embedded points arise from nilpotent thickening or non-transverse intersections.

2.5.2 Associated Points of Schemes

Definition 2.5.5: Associated Point of a Scheme

Let X be a scheme. A point $x \in X$ is an *associated point* of X if, for some (equivalently, every) affine open $U = \mathrm{Spec}\, R$ containing x with x corresponding to $\mathfrak{p} \in \mathrm{Spec}\, R$, we have:

$$\mathfrak{p} \in \mathrm{Ass}_A(A)$$

The set of associated points of X is denoted $\mathrm{Ass}(X)$.

The "some/every" equivalence follows from the fact that associated primes behave well under localization and thus by the affine communication lemma: if $\mathfrak{p} \in \mathrm{Ass}(R)$ and $S \subseteq A$ is a multiplicative set with $S \cap \mathfrak{p} = \emptyset$, then $S^{-1}\mathfrak{p} \in \mathrm{Ass}(S^{-1}R)$.

Proposition 2.5.6: Characterization of Associated Points

Let X be a locally Noetherian scheme. A point $x \in X$ is an associated point if and only if the maximal ideal $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ is an associated prime of $\mathcal{O}_{X,x}$ (as a module over itself). Equivalently, $x \in \mathrm{Ass}(X)$ if and only if there exists $f \in \mathcal{O}_{X,x}$ with $\mathrm{Ann}(f) = \mathfrak{m}_x$.

***Proof*:**
Take an affine open $U = \mathrm{Spec}\, A$ containing x, with x corresponding to $\mathfrak{p}$. Then $\mathcal{O}_{X,x} = A_\mathfrak{p}$ and $\mathfrak{m}_x = \mathfrak{p}A_\mathfrak{p}$. We have $\mathfrak{p} \in \mathrm{Ass}_A(A)$ if and only if $\mathfrak{p}A_\mathfrak{p} \in \mathrm{Ass}_{A_\mathfrak{p}}(A_\mathfrak{p})$ by the localization property of associated primes.

Proposition 2.5.7: Associated Points of Locally Noetherian Schemes

Let X be a locally Noetherian scheme. Then:

1. $\mathrm{Ass}(X)$ is a discrete subset of X: every associated point has an open neighborhood containing no other associated points.
2. If X is quasi-compact (e.g., Noetherian), then $\mathrm{Ass}(X)$ is finite.
3. If X is reduced, then $\mathrm{Ass}(X)$ equals the set of generic points of the irreducible components of X.
4. The minimal associated points are exactly the generic points of the irreducible components of X.
5. X has embedded points if and only if X is not reduced "at" those points (more precisely: X has no embedded points if and only if X is reduced).

***Proof*:**

1. Each associated point x lies in an affine open $\operatorname{Spec} A$ where it corresponds to $\mathfrak{p} \in \mathrm{Ass}(A)$. Shrinking to $\operatorname{Spec} A_{\mathfrak{p}}$, the only associated prime is $\mathfrak{p}A_{\mathfrak{p}}$ by the localization property.
2. Cover X by finitely many affine opens $\operatorname{Spec} A_i$; each has finitely many associated primes, and associated points of X lying in $\operatorname{Spec} A_i$ correspond to associated primes of A_i.
3. If X is reduced, then for any affine open $\operatorname{Spec} A$, we have A reduced (no nilpotents). A reduced Noetherian ring has $\mathrm{Ass}(A) = \mathrm{Min}(A)$ (the minimal primes), and these correspond to the generic points of irreducible components.
4. The minimal primes of A are both the minimal elements of $\mathrm{Ass}(A)$ and the generic points of the irreducible components of $\operatorname{Spec} A$.
5. If X is reduced, (3) shows there are no embedded points. Conversely, if X has no embedded points and $\mathfrak{p}$ is a minimal prime, then $A_{\mathfrak{p}}$ has unique associated prime $\mathfrak{p}A_{\mathfrak{p}}$, which forces $A_{\mathfrak{p}}$ to be a field (it's a local ring with no zero divisors of depth 0), so A is generically reduced. A bit more work shows A is reduced.

Example 2.5.8: Associated Points of Integral Schemes

1. Let R be an integral domain. Then the generic point (0) is the unique associated point of $\operatorname{Spec} R$. Indeed, $\mathrm{Ass}(R) = \{(0)\}$ since R has no zero divisors: if $\mathrm{Ann}(r) = \mathfrak{p}$ for some $r \neq 0$, then $\mathfrak{p} \cdot r = 0$ forces $\mathfrak{p} = (0)$.

   More generally, if X is an integral scheme, then $\mathrm{Ass}(X) = \{\eta\}$ where η is the generic point.
2. Let $X = \operatorname{Spec}(k[x,y]/(xy))$. This is two lines (the x-axis and y-axis) meeting at the origin. The associated primes of $k[x,y]/(xy)$ are (x) and (y):

$$\mathrm{Ann}(\bar{x}) = (y), \qquad \mathrm{Ann}(\bar{y}) = (x)$$

These are both minimal, corresponding to the generic points of the two irreducible components. The origin (x, y) is *not* an associated point: one can check that $\mathrm{Ann}(\bar{f}) \neq (x, y)$ for any $\bar{f} \in k[x, y]/(xy)$.

Geometrically: the two lines meet transversally at the origin, and transverse intersections do not create embedded points.

3. Let $X = \mathrm{Spec}\,(k[x, y]/(y^2, xy))$. This scheme has the same underlying topological space as $\mathrm{Spec}\,(k[x, y]/(y)) = \mathbb{A}^1_k$ (the x-axis), but with nilpotent "fuzz" at the origin.

   The associated primes are (y) and (x, y):

   - $\mathrm{Ann}(\bar{x}) = (y)$: if $f \cdot \bar{x} = 0$ in $k[x, y]/(y^2, xy)$, then $fx \in (y^2, xy)$, forcing $f \in (y)$.
   - $\mathrm{Ann}(\bar{y}) = (x, y)$: we have $x \cdot \bar{y} = \overline{xy} = 0$ and $y \cdot \bar{y} = \overline{y^2} = 0$, so $(x, y) \subseteq \mathrm{Ann}(\bar{y})$. Maximality is clear since $\bar{y} \neq 0$.

   Thus $\mathrm{Ass}(X) = \{[(y)], [(x, y)]\}$: the generic point of the x-axis (minimal) and the origin (embedded). The embedded point at the origin "remembers" that X is non-reduced there, even though the underlying set is just a line.

4. Continuing the previous example, let $X = \mathrm{Spec}\,(k[x, y]/(y^2, xy))$ and consider the function $\bar{y} \in \mathcal{O}_X(X)$.

   Where does $\bar{y}$ vanish? At a point $\mathfrak{p} \in X$, the function $\bar{y}$ vanishes if and only if its image in the residue field $\kappa(\mathfrak{p})$ is zero, which happens if and only if $y \in \mathfrak{p}$. Thus $\bar{y}$ vanishes at every point of X (since every prime of $k[x, y]/(y^2, xy)$ contains y).

   What is $\mathrm{Supp}(\bar{y})$*?* The support of $\bar{y}$ as a section is the set of points where $\bar{y}$ is not zero in the stalk:

   $$\mathrm{Supp}(\bar{y}) = \{\mathfrak{p} \in X \mid \bar{y} \neq 0 \in \mathcal{O}_{X,\mathfrak{p}}\}$$

   At the generic point $[(y)]$: the stalk is $(k[x, y]/(y^2, xy))_{(y)} = k(x)[y]/(y^2)$ (localize away from (y), but y remains nilpotent). Here $\bar{y} \neq 0$.

   At the origin $[(x, y)]$: the stalk is $(k[x, y]/(y^2, xy))_{(x,y)}$, and $\bar{y} \neq 0$ there as well.

   At any other closed point $[(x - a, y)]$ for $a \neq 0$: the stalk is $k[x, y]_{(x-a,y)}/(y^2, xy)$. Since $x - a$ is a unit in this localization, and $xy = 0$ implies $y = (x - a)^{-1} \cdot 0 = 0$ after multiplying by the unit... actually, let's be more careful. We have $xy = 0$, and $x = (x - a) + a$, so $((x - a) + a)y = 0$, giving $(x - a)y = -ay$. Since $x - a$ is a unit, $y = -a^{-1}(x - a)y$... hmm, this gives $y = -a^{-1}(x - a)y$, so $y(1 + a^{-1}(x - a)) = 0$. Since $a \neq 0$ and $x - a$ is in the maximal ideal, $1 + a^{-1}(x - a)$ is a unit, so $y = 0$ in this stalk.

   Thus $\mathrm{Supp}(\bar{y}) = \{[(y)], [(x, y)]\}$, which is exactly $\mathrm{Ass}(X)$. The function $\bar{y}$ is a *zero divisor* (since $(x) \cdot \bar{y} = 0$), and its support is concentrated at the associated points.

Proposition 2.5.9: Zero Divisors and Associated Points

Let X be a locally Noetherian scheme and $f \in \mathcal{O}_X(X)$ a global section. Then f is a zero divisor if and only if f vanishes at some associated point of X. Equivalently:

$$f \text{ is a non-zero-divisor} \quad \Longleftrightarrow \quad f|_\xi \neq 0 \text{ for all } \xi \in \mathrm{Ass}(X)$$

***Proof*:**
The question is local, so we may assume $X = \operatorname{Spec} A$ for a Noetherian ring A. Then f is a zero divisor in A if and only if $f \in \bigcup_{\mathfrak{p} \in \operatorname{Ass}(A)} \mathfrak{p}$ by proposition 2.5.3(3), which happens if and only if $f \in \mathfrak{p}$ for some associated prime $\mathfrak{p}$, i.e., f vanishes at the corresponding associated point.

This proposition is extremely useful: to check whether a function is a non-zero-divisor, it suffices to check non-vanishing at the finitely many associated points, rather than at every point.

Example 2.5.10: Double Line

Let $X = \operatorname{Spec}(k[x, y]/(y^2))$, the "double line" or x-axis with nilpotent thickening. The only prime ideal of $k[x, y]/(y^2)$ containing (y^2) is (y) (since (y) is radical and $(y^2) \subseteq (y)$, and any prime containing y^2 must contain y).

Is (y) an associated prime? We need $\operatorname{Ann}(f) = (y)$ for some f. Take $f = \bar{y}$: then $g \cdot \bar{y} = 0$ iff $gy \in (y^2)$ iff $g \in (y)$. So $\operatorname{Ann}(\bar{y}) = (y)$.

Thus $\operatorname{Ass}(X) = \{[(y)]\}$: just the generic point. There is no embedded point! Even though X is non-reduced, the nilpotent thickening is "uniform" along the entire line, not concentrated at any particular point.

Compare this to $k[x, y]/(y^2, xy)$, where the xy relation creates an embedded point at the origin by making y "more nilpotent" there.

Example 2.5.11: Associated Points of $\operatorname{Spec} \mathbb{Z}$

Let $X = \operatorname{Spec} \mathbb{Z}$. Since $\mathbb{Z}$ is an integral domain, the unique associated point is the generic point (0). The closed points (p) for primes p are *not* associated.

Now let $M = \mathbb{Z}/n\mathbb{Z}$ for some $n > 1$, and consider $\widetilde{M}$ as a quasi-coherent sheaf on $\operatorname{Spec} \mathbb{Z}$. The associated primes of M are the primes dividing n. For $n = 12 = 2^2 \cdot 3$, we have $\operatorname{Ass}(M) = \{(2), (3)\}$, corresponding to the points where the "support" of the module is concentrated.

Associated points play a key role in many constructions:

Proposition 2.5.12: Sections Determined by Associated Points

Let X be a locally Noetherian scheme and $\mathscr{F}$ a quasi-coherent sheaf on X. A section $s \in \mathscr{F}(X)$ is zero if and only if $s_\xi = 0$ in the stalk $\mathscr{F}_\xi$ for every associated point $\xi \in \operatorname{Ass}(X)$.

***Proof*:**
This reduces to the affine case: if $X = \operatorname{Spec} A$ and $\mathscr{F} = \widetilde{M}$ for an A-module M, then $s \in M$ is zero iff $s = 0$ in $M_\mathfrak{p}$ for all $\mathfrak{p} \in \operatorname{Ass}(M)$. This follows from the fact that the natural map $M \to \prod_{\mathfrak{p} \in \operatorname{Ass}(M)} M_\mathfrak{p}$ is injective for finitely generated modules over Noetherian rings.

2.5.13 Associated Points and Depth For a locally Noetherian scheme X, the associated points are exactly the points where the depth of the local ring is zero:

$$\mathrm{Ass}(X) = \{x \in X \mid \mathrm{depth}(\mathcal{O}_{X,x}) = 0\}$$

Recall that the depth of a local ring $(R, \mathfrak{m})$ is the length of a maximal regular sequence in $\mathfrak{m}$. A point x has depth 0 if and only if $\mathfrak{m}_x$ consists entirely of zero divisors (plus nilpotents), which happens precisely when $\mathfrak{m}_x \in \mathrm{Ass}(\mathcal{O}_{X,x})$.
This perspective connects associated points to Cohen–Macaulay and Gorenstein conditions: a Noetherian scheme is Cohen–Macaulay if and only if at every point, the depth equals the dimension, which imposes strong restrictions on the associated points.

Exercise 2.5.1

1. Let $A = k[x, y, z]/(xy, xz)$. Find $\mathrm{Ass}(A)$ and describe the associated points of $\operatorname{Spec} A$ geometrically. Which are minimal and which are embedded?
2. Let $A = k[x, y]/(x^2, xy)$. Show that $\mathrm{Ass}(A) = \{(x), (x, y)\}$. Interpret this geometrically: what is the underlying space, and where is the embedded point?
3. Prove that a Noetherian scheme X is reduced if and only if it has no embedded associated points.
4. Let X be a Noetherian scheme and $U \subseteq X$ an open subset. Show that $\mathrm{Ass}(U) = \mathrm{Ass}(X) \cap U$. (Hint: use the localization property of associated primes.)
5. Let $f : X \to Y$ be a closed immersion of Noetherian schemes. Show that $\mathrm{Ass}(X) \subseteq f^{-1}(\mathrm{Ass}(Y))$ does NOT hold in general. Give a counterexample. (Hint: consider $\operatorname{Spec}(k) \hookrightarrow \operatorname{Spec}(k[x]/(x^2))$.)
6. Let X be an integral scheme with generic point η. Show that a rational function $f \in k(X)$ is regular (i.e., lies in $\mathcal{O}_X(U)$ for some open U) if and only if it is regular at η. (This is a version of the "associated points control regularity" principle for integral schemes.)
7. (*Primary decomposition and associated points*) Let $I \subseteq R$ be an ideal in a Noetherian ring with primary decomposition $I = Q_1 \cap \cdots \cap Q_n$. Show that $\mathrm{Ass}(R/I) = \{\sqrt{Q_1}, \ldots, \sqrt{Q_n}\}$, and identify which are minimal vs. embedded.

2.6 Dimension

The reader may want to review [10] for an in-depth look at the algebraic counterpart. Recall that the *Krull dimension* of a commutative ring R is the supremum of the lengths of proper chains of prime ideals:

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

with indexing starting at 0. Geometrically, for $\operatorname{Spec} R$ and topological spaces more generally (including schemes), the *dimension* is the supremum of lengths of proper chains of closed irreducible subsets with indexing starting at 0. Since the closed irreducible subsets of $\operatorname{Spec} R$ are exactly the sets $V(\mathfrak{p})$

for $\mathfrak{p}$ prime, we have $\dim \operatorname{Spec} R = \operatorname{kdim} R$. Furthermore:

$$\dim \operatorname{Spec} R = \dim \operatorname{Spec} (R/\sqrt{0})$$

since $\operatorname{Spec} R$ and $\operatorname{Spec}(R/\sqrt{0})$ are homeomorphic (quotienting by the nilradical does not change the prime ideals).

If the space is not irreducible, then the dimension per component can vary (for example $\mathbb{A}^1 \cup \mathbb{A}^2 \subseteq \mathbb{A}^3$, interpreted as the x-axis union the yz-plane). Hence we usually work with irreducible schemes. If necessary, we say a scheme has *pure dimension n* (or is *equidimensional*) if all of its irreducible components have the same dimension n. Dimension respects inclusions: if $X \subseteq Y$ then $\dim X \leq \dim Y$.

Dimension	**Name**
0	point (or collection of points)
1	curve
2	surface
$n \geq 3$	n-fold

Lemma 2.6.1: Dimension of Open Subset

Let $U \subseteq X$ be an open subset of a topological space. Then there is a bijection between irreducible closed subsets of U and the irreducible closed subsets of X that meet U, given by $Z \mapsto \overline{Z}$ (closure in X) with inverse $W \mapsto W \cap U$.

***Proof*:**
If $Z \subseteq U$ is irreducible and closed in U, then $\overline{Z}$ is irreducible and closed in X with $\overline{Z} \cap U = Z$ (since Z is closed in U, it equals the intersection of its closure with U). Conversely, if $W \subseteq X$ is irreducible closed with $W \cap U \neq \emptyset$, then $W \cap U$ is irreducible (an open subset of an irreducible set is irreducible) and closed in U, with $\overline{W \cap U} = W$ (since $W \cap U$ is dense in W). These operations are mutually inverse.

Proposition 2.6.2: Dimension of Scheme

Let X be a scheme. Then $\dim X = n$ if and only if X admits an open cover by affine open subsets of dimension at most n, where equality is achieved by at least one affine open.

***Proof*:**
If $\dim X = n$, then each open subset in an affine cover has dimension $\leq n$ (since irreducible closed subsets of an open set correspond to those of X meeting it, by lemma 2.6.1).

Conversely, suppose every affine open in some cover has dimension $\leq n$ and at least one has dimension exactly n. We must show $\dim X = n$. The dimension is at least n since some affine open has dimension n, and a chain in U_i lifts to a chain in X via lemma 2.6.1. For the reverse inequality, suppose $\dim X > n$. Then there exists a chain of irreducible closed subsets in X:

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_{n+1}$$

Since Z_0 is nonempty, it meets some affine open U_i in the cover. Then each Z_j also meets U_i

(since $Z_0 \subseteq Z_j$ and $Z_0 \cap U_i \neq \emptyset$). By lemma 2.6.1, the chain:

$$Z_0 \cap U_i \subsetneq Z_1 \cap U_i \subsetneq \cdots \subsetneq Z_{n+1} \cap U_i$$

consists of distinct irreducible closed subsets of U_i (since the closure map is a bijection). This gives $\dim U_i \geq n+1 > n$, contradicting our assumption.

Lemma 2.6.3: Integral Extensions and Dimension

Let $f : \operatorname{Spec} R \to \operatorname{Spec} S$ correspond to an integral extension $S \hookrightarrow R$. Then:

$$\dim \operatorname{Spec} R = \dim \operatorname{Spec} S$$

***Proof*:**
The going-up theorem gives $\dim \operatorname{Spec} R \geq \dim \operatorname{Spec} S$ (any chain in S lifts to a chain in R), and the incomparability theorem gives $\dim \operatorname{Spec} R \leq \dim \operatorname{Spec} S$ (primes lying over the same prime cannot be comparable). See [10] for the details.

2.6.4 A word on Noetherian Schemes It is *not* the case that a Noetherian scheme must be finite-dimensional: Nagata constructed a Noetherian ring of infinite Krull dimension (see Vakil, chapter 11).
However, Noetherian *local* rings are always finite-dimensional (by [10]), and so points of locally Noetherian schemes always have finite codimension.

2.6.1 Codimension

We are often concerned with subspaces that are a few dimensions lower than the ambient scheme X. Since schemes can be unions of irreducible components of different dimension, we define codimension only for irreducible subschemes:

Definition 2.6.5: Codimension

Let X be a topological space and $Y \subseteq X$ an irreducible closed subset. Then:

$$\begin{aligned}&\operatorname{codim}_X Y \\ &= \sup\{n : \text{there exists a chain } Y = Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \text{ of irreducible closed subsets of } X\}\end{aligned}$$

For a scheme X and a point $x \in X$, the codimension of $\overline{\{x\}}$ in X equals the Krull dimension of the local ring $\mathcal{O}_{X,x}$.

For the affine case: if $\mathfrak{p} \in \operatorname{Spec} R$, then $\operatorname{codim}_{\operatorname{Spec} R} V(\mathfrak{p})$ equals the *height* of $\mathfrak{p}$, i.e. the supremum of lengths of chains:

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_h = \mathfrak{p}$$

and this equals $\operatorname{kdim} R_{\mathfrak{p}}$. For example, in $\operatorname{Spec} \mathbb{Z}$, the generic point $\eta = [(0)]$ has codimension 0, while any closed point $[(p)]$ has codimension 1.

If Y has codimension 0 in X, then Y must be an irreducible component of X.

Definition 2.6.6: Hypersurface

Let $Y \subseteq X$ be an irreducible closed subscheme. Then Y is called a *hypersurface* if $\operatorname{codim}_X Y = 1$.

Lemma 2.6.7: Sum of Dimension and Codimension

Let X be a topological space and $Y \subseteq X$ an irreducible closed subset. Then:

$$\operatorname{codim}_X Y + \dim Y \leq \dim X$$

***Proof*:**
A chain of length c descending from Y and a chain of length d ascending from Y can be concatenated to give a chain of length $c + d$ in X. Taking suprema gives $\operatorname{codim}_X Y + \dim Y \leq \dim X$.

Equality holds for irreducible varieties of finite type over a field, but *not* in general:

Example 2.6.8: Codimension Pathology
Consider the scheme $X = \operatorname{Spec}(k[x]_{(x)}[t])$, which can be thought of as a surface with a line attached at one point. Then $\dim X = 2$, but the closed point $p = (x, t)$ has dimension 0 and codimension 1 (not 2!), since the maximal chain from p is:

$$(0) \subsetneq (x) \subsetneq (x, t)$$

but (x) already has codimension 1 and dimension 1 (the "line" component), while the "surface" component (0) has dimension 2. Note too that this scheme has open subsets of *different dimension* – a phenomenon that does *not* occur for open subsets of n-manifolds.

The pathology above is excluded by the following class of rings:

Definition 2.6.9: Catenary Ring

A commutative ring R is *catenary* if for every pair of nested prime ideals $\mathfrak{q} \subseteq \mathfrak{p}$, all maximal chains of prime ideals between $\mathfrak{q}$ and $\mathfrak{p}$ have the same length.

Many rings are catenary: localizations of finitely generated $\mathbb{Z}$-algebras, complete Noetherian local rings, Dedekind domains, Cohen–Macaulay rings, and in particular all localizations of finitely generated k-algebras (see exercise 2.6.1).

2.6.2 Transcendence Dimension

Working with Krull dimension directly can be difficult. For finitely generated algebras over a field, we have a more computable invariant:

Definition 2.6.10: Transcendence Dimension

Let A be an integral domain that is a k-algebra. Then the *transcendence dimension* of A over k is:

$$\operatorname{trdeg}_k A := \operatorname{trdeg}_k \operatorname{Frac}(A)$$

i.e. the transcendence degree of the fraction field of A over k. Transcendence bases, the transcendence degree of a field extension, and its additivity in towers are developed in *Core Algebra* [18]; only the translation to algebras is made here.

Theorem 2.6.11: Transcendence and Krull Dimension

Let A be a finitely generated integral domain over a field k. Then:

$$\dim \operatorname{Spec} A = \operatorname{trdeg}_k \operatorname{Frac}(A)$$

Thus, if X is an irreducible k-variety with function field $k(X)$, then $\dim X = \operatorname{trdeg}_k k(X)$.

***Proof*:**
See [10] for the algebraic proof. The key steps are: Noether normalization gives a finite (hence integral) extension $k[y_1, \dots, y_d] \hookrightarrow A$ with $d = \operatorname{trdeg}_k \operatorname{Frac}(A)$; by lemma 2.6.3, $\dim \operatorname{Spec} A = \dim \mathbb{A}^d_k = d$.

For catenary rings, dimension and codimension add up:

Theorem 2.6.12: Dimension Formula for Finite Type k-Schemes

Let X be an irreducible scheme locally of finite type over a field k, $Y \subseteq X$ an irreducible closed subset, and η the generic point of Y. Then:

$$\operatorname{codim}_X Y + \dim Y = \dim X$$

equivalently:

$$\dim \mathcal{O}_{X,\eta} + \dim Y = \dim X$$

***Proof*:**
The question is local, so we may reduce to $X = \operatorname{Spec} A$ for a finitely generated k-domain A. Let $\mathfrak{p}$ be the prime corresponding to η. Then $\operatorname{codim}_X Y = \operatorname{kdim} A_{\mathfrak{p}}$ and $\dim Y = \operatorname{kdim} A/\mathfrak{p}$, so we must show $\operatorname{kdim} A_{\mathfrak{p}} + \operatorname{kdim} A/\mathfrak{p} = \operatorname{kdim} A$. This follows from the fact that finitely generated k-algebras are catenary and equidimensional (see [10] or Vakil, chapter 11 for a guided exercise).

2.6.3 Krull's Height Theorem and Algebraic Hartogs' Lemma

We now turn to two fundamental results about codimension-1 phenomena. Krull's theorem gives an upper bound on codimension in terms of the number of defining equations, while the algebraic Hartogs' lemma characterizes regular functions on normal schemes in terms of codimension-1 behavior. Both will be used extensively in the theory of divisors (section 5.4).

Krull's Height Theorem

The following theorem, proved in [10], is one of the most important tools for computing codimension:

Theorem 2.6.13: Krull's Height Theorem (Hauptidealsatz)

Let R be a Noetherian ring. If $\mathfrak{a} = (f_1, \dots, f_n)$ is a proper ideal generated by n elements, then every minimal prime $\mathfrak{p}$ over $\mathfrak{a}$ has:

$$\text{height}(\mathfrak{p}) \le n$$

In particular, if $f \in R$ is neither a zero-divisor nor a unit, then every minimal prime over (f) has height exactly 1.

***Proof*:**
See [10]. The proof uses the dimension theorem for Noetherian local rings: localizing at $\mathfrak{p}$, the ideal $\mathfrak{a}_{\mathfrak{p}}$ is $\mathfrak{p}_{\mathfrak{p}}$-primary and generated by n elements, so $\text{kdim} R_{\mathfrak{p}} \le n$ by the characterization of dimension as the minimal number of generators of a parameter ideal.

Geometrically, Krull's theorem says: in a Noetherian scheme, the vanishing locus of n functions has all irreducible components of codimension $\le n$. In particular:

Corollary 2.6.14: Principal Ideal Theorem – Geometric Form

Let X be a locally Noetherian scheme and $f \in \mathcal{O}_X(X)$ a regular function that is not a zero-divisor on any irreducible component. Then every irreducible component of $V(f)$ has codimension exactly 1 in X.

Corollary 2.6.15: Codimension Bound for Complete Intersections

Let X be a locally Noetherian scheme and $Z = V(f_1, \dots, f_n) \subseteq X$. Then every irreducible component of Z has codimension $\le n$. If equality holds for every component, Z is called a *(set-theoretic) complete intersection*.

Example 2.6.16: Krull's Theorem in Practice

1. In $\mathbb{A}^3_k = \operatorname{Spec} k[x, y, z]$, the ideal (xy, xz) defines the vanishing locus $V(x) \cup V(y, z)$, which is the yz-plane $\cup$ the x-axis. The yz-plane has codimension 1, while the x-axis has codimension 2. Krull's theorem guarantees codimension ≤ 2, and both bounds are achieved.

2. The ideal $(x^2 + y^2 + z^2, x + y + z) \subseteq k[x, y, z]$ defines a curve in $\mathbb{A}^3_k$. By Krull's theorem, each component has codimension ≤ 2. In this case, the curve has codimension exactly 2, so it is a complete intersection.

3. The twisted cubic curve $C \subseteq \mathbb{A}^3_k$ (the image of $t \mapsto (t, t^2, t^3)$) has codimension 2 but ideal $I(C) = (y - x^2, z - x^3)$, which requires 2 generators. However, in $\mathbb{P}^3_k$ the twisted cubic requires 3 generators of its homogeneous ideal (it is *not* a complete intersection).

The converse direction of Krull's theorem also holds and is equally useful:

Proposition 2.6.17: Converse to Krull's Theorem

Let R be a Noetherian ring and $\mathfrak{p}$ a prime of height n. Then there exist $f_1, \ldots, f_n \in \mathfrak{p}$ such that $\mathfrak{p}$ is a minimal prime over $(f_1, \ldots, f_n)$.

***Proof*:**
See [10]. One constructs the f_i inductively using prime avoidance: at each step, choose f_i in $\mathfrak{p}$ but outside all minimal primes of $(f_1, \ldots, f_{i-1})$ of height $< i$.

Algebraic Hartogs' Lemma

In complex analysis, Hartogs' extension theorem states that a holomorphic function defined on the complement of a codimension-≥ 2 set extends uniquely to the ambient space. Amazingly, there is an algebraic analogue for normal schemes:

Theorem 2.6.18: Algebraic Hartogs' Lemma

Let A be a Noetherian normal domain. Then:

$$A = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \operatorname{height}(\mathfrak{p})=1}} A_{\mathfrak{p}}$$

where the intersection takes place inside $\operatorname{Frac}(A)$.

In geometric language: a rational function on a Noetherian normal scheme with no poles along any codimension-1 subvariety is in fact a regular function. Equivalently, regular functions on a normal scheme are determined by their behavior at codimension-1 points – you cannot have a function that is "only irregular in codimension ≥ 2."

***Proof*:**
The inclusion $A \subseteq \bigcap_{\operatorname{ht}(\mathfrak{p})=1} A_{\mathfrak{p}}$ is clear (any element of A lies in every localization). For the reverse, suppose $x \in \operatorname{Frac}(A)$ lies in $A_{\mathfrak{p}}$ for every height-1 prime $\mathfrak{p}$. Let $I = \{a \in A \mid ax \in A\}$ be the ideal of "denominators" of x. If $x \notin A$, then $I \neq A$. Let $\mathfrak{q}$ be a minimal prime over I.

Claim: $\operatorname{height}(\mathfrak{q}) \leq 1$.

If $\operatorname{height}(\mathfrak{q}) = 0$, this is fine. If $\operatorname{height}(\mathfrak{q}) \geq 2$, then $x \in A_{\mathfrak{p}}$ for all height-1 primes $\mathfrak{p} \subseteq \mathfrak{q}$, and since $A_{\mathfrak{q}}$ is a normal local domain, $A_{\mathfrak{q}} = \bigcap_{\operatorname{ht}(\mathfrak{p})=1,\, \mathfrak{p} \subseteq \mathfrak{q}} (A_{\mathfrak{q}})_{\mathfrak{p}A_{\mathfrak{q}}}$ (applying the result in the local case, where it holds by the characterization of normal domains as S_2 and R_1). This gives $x \in A_{\mathfrak{q}}$, contradicting $I \subseteq \mathfrak{q}$. So $\operatorname{height}(\mathfrak{q}) = 1$.

But then $x \in A_{\mathfrak{q}}$ by hypothesis, so there exist $a \in A$, $s \notin \mathfrak{q}$ with $x = a/s$, giving $sx = a \in A$ and hence $s \in I$. But $I \subseteq \mathfrak{q}$ and $s \notin \mathfrak{q}$, a contradiction. Therefore $I = A$ and $x \in A$.

Example 2.6.19: Algebraic Hartogs in Practice

1. Consider $\mathbb{A}^2_k = \operatorname{Spec} k[x,y]$ and the open set $U = D(x) \cup D(y) = \mathbb{A}^2 \setminus \{(0,0)\}$. We showed in example 2.2.2 that $\mathcal{O}_{\mathbb{A}^2}(U) = k[x,y]$. This is now immediate from algebraic Hartogs: the complement of U is $\{(0,0)\}$, which has codimension 2, and $k[x,y]$ is a normal domain. Hence any regular function on U extends to all of $\mathbb{A}^2$.

2. The same argument fails for $\mathbb{A}^1$: removing a point from $\mathbb{A}^1$ removes a codimension-1 subset, and indeed $\mathcal{O}_{\mathbb{A}^1}(D(x)) = k[x, 1/x] \neq k[x]$.

3. Let $A = k[x,y,z]/(x^2+y^2-z^2)$, the quadric cone, which is a normal domain (exercise). The singular point at the origin has codimension 2, so algebraic Hartogs implies that regular functions on the smooth locus extend to all of $\operatorname{Spec} A$.

4. The algebraic Hartogs' lemma *fails* without normality. Consider the nodal cubic $A = k[x,y]/(y^2 - x^2(x+1))$, which is not normal at the origin. The element y/x in the fraction field is regular at all codimension-1 points away from the node, yet $y/x \notin A$.

The connection between these two results is fascinating: Krull's theorem says that the zero locus of a single function has codimension ≤ 1, while Hartogs' lemma says that on normal schemes, a rational function is determined by its poles, which also occur in codimension 1. Both point toward the centrality of codimension-1 phenomena, which will be formalized in the theory of *Weil and Cartier divisors* in section 5.4.

Exercise 2.6.1

1. Show that a Noetherian scheme of dimension 0 has finitely many points.

2. Let $f : X \to Y$ be an integral morphism. Show that $\dim f^{-1}(y) = 0$ for all $y \in Y$ (i.e., the fibers are zero-dimensional).

3. If $\nu : \widetilde{X} \to X$ is the normalization of a scheme, show that $\dim \widetilde{X} = \dim X$.

4. If K/k is an algebraic field extension and X is a k-scheme of finite type, show that $\dim X = \dim X_K$ where $X_K = X \times_k K$ is the base change.

5. Generalize the above result: if K/k is an arbitrary field extension and X is an irreducible k-scheme of finite type, show that $\dim X_K = \dim X$.

6. Show the following rings are catenary: localizations of finitely generated $\mathbb{Z}$-algebras, complete Noetherian local rings, Dedekind domains.

7. Let X be an integral scheme locally of finite type over a field k, and let $U \subseteq X$ be a nonempty open subset. Show that $\dim U = \dim X$.

8. (*Hartogs' lemma for schemes*) Let X be a normal Noetherian integral scheme, and let $U \subseteq X$ be an open subset whose complement has codimension ≥ 2. Show that the restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X(U)$ is an isomorphism. (Hint: reduce to the affine case and apply theorem 2.6.18.)

9. Show that the quadric cone $\operatorname{Spec} k[x,y,z]/(x^2+y^2-z^2)$ is a normal domain ($\operatorname{char} k \neq 2$). (Hint: show it is R_1 and S_2, or use the Jacobian criterion to show regularity in codimension 1

and then invoke Serre's criterion.)

2.7 Regular and Smooth Schemes

The notion of smoothness is central to both differential and algebraic geometry. In differential geometry, a manifold is smooth if it locally looks like $\mathbb{R}^n$. In algebraic geometry, the analogous condition is that the local ring at each point behaves like a polynomial ring – this is the notion of *regularity*. Over perfect fields (in particular, algebraically closed fields of any characteristic), regularity and smoothness coincide; over imperfect fields, smoothness is the stronger and more geometrically natural condition.

The algebraic foundations for this section – Hilbert functions, the dimension theorem, regular local rings, and their characterization via homological methods – were developed in [10, 15]. We recall the key definitions and build on them.

Definition 2.7.1: Tangent Space

Let X be a scheme and $x \in X$ a point. Then the *(Zariski) cotangent space* at x is the $\kappa(x)$-vector space:

$$\mathfrak{m}_x/\mathfrak{m}_x^2$$

where $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ is the maximal ideal of the stalk at x, and $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$ is the residue field. The *(Zariski) tangent space* is its dual:

$$T_xX := (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee = \mathrm{Hom}_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2,\, \kappa(x))$$

Definition 2.7.2: Regular Local Ring

Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then R is *regular* if:

$$\mathrm{kdim} R = \dim_k(\mathfrak{m}/\mathfrak{m}^2)$$

By [10], we always have $\mathrm{kdim} R \le \dim_k(\mathfrak{m}/\mathfrak{m}^2)$, so regularity asks for equality.

Proposition 2.7.3: Regular Local Rings are Domains

Every regular local ring is an integral domain.

***Proof*:**
By [15], if $G_\mathfrak{m}(R) \cong k[t_1, \ldots, t_d]$ is a domain and $\bigcap_n \mathfrak{m}^n = 0$ (which holds by the Krull intersection theorem for Noetherian local rings), then R is a domain.

Definition 2.7.4: Regular Scheme

A scheme X is *regular* if $\mathcal{O}_{X,x}$ is a regular local ring for every point $x \in X$. A scheme is *singular* at a point x if $\mathcal{O}_{X,x}$ is not regular.

Example 2.7.5: Regular and Singular Schemes

1. $\mathbb{A}_k^2$ *is regular.* For a closed point $\mathfrak{m} = (x-a, y-b)$, first observe that $\kappa(\mathfrak{m}) = k$ (since k is algebraically closed). The cotangent space $\mathfrak{m}/\mathfrak{m}^2$ is spanned by the images of $x-a$ and $y-b$, so $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 2$. The local ring $k[x,y]_\mathfrak{m}$ has a maximal chain $0 \subsetneq (x-a) \subsetneq (x-a, y-b)$, giving kdim $= 2$.

   Since we are working over schemes, we also have higher-dimensional points. Consider $[\mathfrak{p}] = [(x^2-y)] \in \mathbb{A}_k^2$. The local ring $k[x,y]_\mathfrak{p}$ has dimension 1 (one can show the chain $0 \subsetneq (x^2-y)$ is maximal), and $\mathfrak{m}_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}^2$ is a 1-dimensional $\kappa(\mathfrak{p})$-vector space. Hence this point is regular.

   An important observation: $[(x^3-y^2)] \in \mathbb{A}_k^2$ is *also* a regular point of $\mathbb{A}_k^2$, even though x^3-y^2 defines a singular curve! This is because regularity at a generic point of $\mathbb{A}_k^2$ concerns the local ring of the *ambient* scheme, not the subscheme $V(x^3-y^2)$. The point $[(x^3-y^2)]$ is "generically regular" in $\mathbb{A}_k^2$ – further motivating the name *generic* point.

2. *The cuspidal cubic is singular.* Let $X = \operatorname{Spec} k[x,y]/(x^3-y^2)$. Then X is not regular at the origin: the local ring $\mathcal{O}_{X,(x,y)}$ has kdim $= 1$ (since X is an irreducible curve), but the cotangent space $\mathfrak{m}/\mathfrak{m}^2$ at the origin has $\dim_k = 2$, since neither $\overline{x}$ nor $\overline{y}$ is redundant modulo $\mathfrak{m}^2$ (the relation $x^3 = y^2$ lies in $\mathfrak{m}^3$, not $\mathfrak{m}^2$). On the other hand, X is regular at every other closed point (exercise).

3. *Normal does not imply regular.* If X is a normal scheme, then its stalks are integrally closed, and all codimension-1 points are regular (by [10]). However, normality does *not* imply regularity in dimension ≥ 2. For example, $\operatorname{Spec} k[x,y,z]/(x^2+y^2-z^2)$ is normal but not regular at the origin (exercise: show $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 3$ while kdim $= 2$).

   However, for a 1-dimensional connected Noetherian scheme, normality *does* imply regularity: the codimension-1 points are exactly the closed points, and the generic point is always regular. In fact, a 1-dimensional normal local domain is a DVR by [10], which is a regular local ring of dimension 1.

The singular locus of a scheme is typically "small." For varieties over algebraically closed fields, this was shown in [20]: the set of singular points forms a proper closed subset.

For schemes of finite type over a field, regularity can be checked via partial derivatives, generalizing the classical criterion from [20]:

Proposition 2.7.6: Jacobian Criterion for Schemes

Let k be a field and $X = \operatorname{Spec}(k[x_1, \ldots, x_n]/(f_1, \ldots, f_m))$ a closed subscheme of $\mathbb{A}^n_k$. Let $x \in X$ be a closed point with residue field $\kappa(x)$. Then X is regular at x if and only if:

$$\operatorname{rank}\left(\frac{\partial f_i}{\partial x_j}(x)\right) = n - \dim \mathcal{O}_{X,x}$$

where the partial derivatives are evaluated in $\kappa(x)$.

***Proof*:**
This is the scheme-theoretic translation of [20]. The key computation is that the cotangent space satisfies:

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \cong \frac{\mathfrak{m}_{\mathbb{A}^n,x}}{(\mathfrak{m}^2_{\mathbb{A}^n,x} + (f_1, \ldots, f_m))}$$

and so $\dim_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2) = n - \operatorname{rank}(J_x)$. Regularity is the condition that this equals $\dim \mathcal{O}_{X,x}$, which is equivalent to $\operatorname{rank}(J_x) = n - \dim \mathcal{O}_{X,x}$.

Example 2.7.7: Jacobian Criterion in Practice

1. The cuspidal cubic $X = V(y^2 - x^3) \subseteq \mathbb{A}^2_k$. The Jacobian is $(-3x^2, 2y)$. At the origin, this is $(0,0)$, which has rank $0 < 2 - 1 = 1$. At any other closed point (a,b) with $b \neq 0$, the rank is $1 = n - \dim X$. Hence X is singular only at the origin.

2. The nodal cubic $X = V(y^2 - x^2(x+1)) \subseteq \mathbb{A}^2_k$. The Jacobian is $(-3x^2 - 2x, 2y)$. At the origin, this is $(0,0)$, so the origin is again singular. At any other point on X where $y \neq 0$, the rank is 1.

3. The quadric surface $X = V(x^2 + y^2 - z^2) \subseteq \mathbb{A}^3_k$ ($\operatorname{char} k \neq 2$). The Jacobian is $(2x, 2y, -2z)$, which has rank 0 only at the origin. Hence X is regular away from the origin.

Smooth Morphisms and Smooth Schemes

Regularity is a property of a scheme in isolation, while *smoothness* is a relative notion: it concerns a morphism $X \to S$ and asks whether the fibers are "non-singular" in a way that varies well over the base. The two notions coincide over perfect fields.

Definition 2.7.8: Smooth Morphism (Preliminary)

A morphism $f : X \to S$ of schemes locally of finite type is *smooth of relative dimension* n at a point $x \in X$ if there exists an open neighborhood U of x and a factorization:

$$\begin{tikzcd} U \arrow[r, hook, "\text{open}"] \arrow[dr, "f|_U"'] & \operatorname{Spec} \frac{\mathcal{O}_S(V)[x_1,\dots,x_{n+r}]}{(f_1,\dots,f_r)} \arrow[d] \\ & V \end{tikzcd}$$

where V is an open neighborhood of $f(x)$ in S, and the Jacobian matrix $(\partial f_i/\partial x_j)$ has rank r at x.
The morphism f is *smooth* if it is smooth at every point. It is *étale* if it is smooth of relative dimension 0.

The definition is local and built from a factorization, so the two stability properties one wants follow by manipulating factorizations rather than by any new geometry.

Proposition 2.7.9: Stability of Smooth Morphisms

Let $f\colon X \to S$ be smooth of relative dimension n.

1. *Base change.* For any morphism $S' \to S$, the projection $X \times_S S' \to S'$ is smooth of relative dimension n.
2. *Composition.* If $g\colon S \to T$ is smooth of relative dimension m, then $g \circ f$ is smooth of relative dimension $m + n$.
3. *Products.* If X and Y are smooth over a field k of dimensions m and n, then $X \times_k Y$ is smooth over k of dimension $m + n$.

***Proof*:**
(1). Smoothness is local on source and target, so work with a factorization as in definition 2.7.8: U open in X mapping to $\operatorname{Spec} \mathcal{O}_S(V)[x_1, \dots, x_{n+r}]/(f_1, \dots, f_r)$ over V, with the Jacobian of rank r. Base change along $S' \to S$ replaces $\mathcal{O}_S(V)$ by $\mathcal{O}_{S'}(V')$ and leaves the polynomials f_i and hence the Jacobian matrix formally unchanged, so its rank at a point of the fibre product is again r. The relative dimension $(n + r) - r = n$ is unchanged.

(2). Compose the two factorizations: over a suitable open, X is cut out in an affine space of relative dimension $n+r$ over S by r equations, and S in one of relative dimension $m+s$ over T by s equations. Pulling the first set of equations up and adjoining the second exhibits X inside an affine space of relative dimension $(m+s)+(n+r)$ over T cut out by $r+s$ equations, and the Jacobian is block triangular with the two original blocks on the diagonal, hence of rank $r+s$. The relative dimension is $(m+s+n+r) - (r+s) = m+n$.

(3). The projection $X \times_k Y \to Y$ is the base change of $X \to \operatorname{Spec} k$ along $Y \to \operatorname{Spec} k$, so it is smooth of relative dimension m by (1). Composing with $Y \to \operatorname{Spec} k$, smooth of relative dimension n, gives $X \times_k Y \to \operatorname{Spec} k$ smooth of relative dimension $m+n$ by (2), completing the proof.

Smoothness is a condition on a Jacobian, and flatness is a condition on modules; that the first implies the second is the bridge every downstream use of π^* crosses. The bridge is the local criterion for flatness, which the series already owns in the homological form that the argument below actually needs [15], so the result is proved here rather than cited.

Proposition 2.7.10: Smooth Morphisms Are Flat

A morphism $f : X \to S$ smooth of relative dimension n (definition 2.7.8) with S locally Noetherian is flat, and all of its nonempty fibres have pure dimension n.

***Proof*:**
Both conclusions are local on source and target, so fix $x \in X$ and take the presentation of definition 2.7.8: an affine open $V = \operatorname{Spec} A$ of S around $f(x)$, with A Noetherian, and an open neighbourhood of x realized as $\operatorname{Spec} B$ for

$$B \;=\; C/(f_1, \ldots, f_r), \qquad C \;=\; A[x_1, \ldots, x_N], \qquad N = n + r$$

the Jacobian $(\partial f_i/\partial x_j)$ having rank r at x. Write $\mathfrak{p} \subseteq A$ for the image of x, $\kappa = \kappa(\mathfrak{p})$, and $\mathfrak{q} \subseteq B$ for x itself. Reduction modulo $\mathfrak{p}$ identifies $C \otimes_A \kappa$ with the polynomial ring $\kappa[x_1, \ldots, x_N]$, inside which the images $\bar{f}_1, \ldots, \bar{f}_r$ cut out the fibre through x. Let $\mathfrak{n}$ be the prime of $\kappa[x]$ corresponding to x and set $R = \kappa[x_1, \ldots, x_N]_{\mathfrak{n}}$.

Step 1: R is regular. Iterating the formula $\operatorname{gldim} R'[t] = \operatorname{gldim} R' + 1$ of [15] down to the field κ, whose global dimension is 0, gives $\operatorname{gldim} \kappa[x_1, \ldots, x_N] = N$. Localization is exact, so localizing a projective resolution of length $\le N$ resolves the localized module, and every R-module is the localization of itself viewed over $\kappa[x]$; hence $\operatorname{gldim} R \le N$ is finite. The Auslander–Buchsbaum–Serre theorem of [15] then says exactly that R is a regular local ring.

Step 2: $\bar{f}_1, \ldots, \bar{f}_r$ is a regular sequence on R. First, their classes in $\mathfrak{n}R/\mathfrak{n}^2R$ are linearly independent over $\kappa(x)$. Indeed the universal κ-derivation sends $g \mapsto dg = \sum_j (\partial g/\partial x_j)\, dx_j$ into the free $\kappa(x)$-module on $dx_1, \ldots, dx_N$, and it kills $\mathfrak{n}^2$ after reduction because $d(gh) = g\, dh + h\, dg$ vanishes at x when both factors do. So a relation $\sum_i a_i \bar{f}_i \in \mathfrak{n}^2R$ with $a_i \in R$ yields, using $\bar{f}_i \in \mathfrak{n}R$, the relation $\sum_i a_i(x)\, d\bar{f}_i = 0$; since the Jacobian has rank r at x the vectors $d\bar{f}_i$ are independent, so every $a_i(x) = 0$, that is $a_i \in \mathfrak{n}R$. This is the asserted independence.

Now induct, writing $d = \dim R$: suppose $R_m = R/(\bar{f}_1, \ldots, \bar{f}_m)$ is regular of dimension $d - m$ for some $m < r$, the case $m = 0$ being Step 1. Independence in the cotangent space says $\bar{f}_{m+1}$ lies in the maximal ideal of R_m but not its square, so it is nonzero; and R_m is a domain by proposition 2.7.3, so $\bar{f}_{m+1}$ is a non-zero divisor on R_m. Killing a non-zero divisor drops the dimension by exactly one [10], while killing an element outside the square of the maximal ideal drops the cotangent dimension by exactly one; both were $d - m$, so R_{m+1} is regular of dimension $d - m - 1$. The induction runs to $m = r$, proving both that the sequence is regular and that the local ring R_r of the fibre at x is regular of dimension $d - r$.

Step 3: flatness. Write $B_m = C/(f_1, \ldots, f_m)$ and localize everything at x; each $(B_m)_{\mathfrak{q}}$ is a Noetherian local ring receiving a local homomorphism from $A_{\mathfrak{p}}$, and is finitely generated as a module over itself. We show by induction that $(B_m)_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$. For $m = 0$, C is a polynomial ring over A, hence free, hence flat, and flatness survives localization. Given flatness at stage m, note that $(B_m)_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} \kappa = R_m$, on which $\bar{f}_{m+1}$ is a non-zero divisor by Step 2. The Fibrewise Nonzerodivisor Criterion of [15] therefore makes f_{m+1} a non-zero divisor on $(B_m)_{\mathfrak{q}}$ and $(B_{m+1})_{\mathfrak{q}} = (B_m)_{\mathfrak{q}}/f_{m+1}(B_m)_{\mathfrak{q}}$ flat over $A_{\mathfrak{p}}$. At $m = r$ this says $B_{\mathfrak{q}}$ is flat over $A_{\mathfrak{p}}$, and x

was arbitrary, so f is flat.

Step 4: fibre dimension. Steps 1 and 2 apply at every point of every nonempty fibre, so each fibre is a regular scheme of finite type over the residue field of its image, and is therefore a disjoint union of integral components (proposition 2.7.21). Take a closed point x of such a component Z. Then $\mathfrak{n}$ is a maximal ideal of $\kappa[x_1, \dots, x_N]$, so $d = \dim R = N$, and Step 2 gives $\dim \mathcal{O}_{Z,x} = N - r = n$. For an integral scheme of finite type over a field the local ring at a closed point has dimension equal to that of the whole [10], so $\dim Z = n$. Every component has dimension n, which is pure dimension n.

Definition 2.7.11: Smooth Scheme Over a Field

Let k be a field. A k-scheme X is *smooth over* k (or k*-smooth*) if the structure morphism $X \to \operatorname{Spec} k$ is smooth.

The relationship between smoothness and regularity is as follows:

Proposition 2.7.12: Smooth Implies Regular

Let k be a field and X a k-scheme locally of finite type. If X is smooth over k, then X is regular.

***Proof*:**
Smoothness over k implies that the cotangent space $\mathfrak{m}_x/\mathfrak{m}_x^2$ has dimension equal to the local dimension of X at x for every point x, which is exactly regularity. The key point is that smoothness gives the Jacobian rank condition over every residue field, not just over k itself.

The converse holds over perfect fields:

Theorem 2.7.13: Regularity and Smoothness Over Perfect Fields

Let k be a perfect field (e.g., any field of characteristic 0, or any finite field) and X a k-scheme locally of finite type. Then X is smooth over k if and only if X is regular.

***Proof*:**
The forward direction is proposition 2.7.12. For the converse, regularity gives the Jacobian rank condition at all k-rational points. The perfection of k ensures that this extends to all points: if k is perfect, then every finitely generated field extension $\kappa(x)/k$ is separably generated, and the cotangent space computation over $\kappa(x)$ agrees with the one over k. The details require the theory of Kähler differentials, which will be developed in section 5.5.

Example 2.7.14: Regularity $\neq$ Smoothness Over Imperfect Fields
Let $k = \mathbb{F}_p(t)$ (an imperfect field) and consider $X = \operatorname{Spec} k[x]/(x^p - t)$. Then X is a single closed point with residue field $k(\alpha)$ where $\alpha^p = t$. The local ring is $k(\alpha)$, which is a field and hence a regular local ring of dimension 0.

However, X is *not* smooth over k: the polynomial $f(x) = x^p - t$ has $f'(x) = px^{p-1} = 0$, so the Jacobian criterion fails. Geometrically, the extension $k(\alpha)/k$ is purely inseparable, and the fiber over $\operatorname{Spec} k$ is "infinitesimally thickened" from the perspective of k. This is the prototypical example of a regular but non-smooth scheme.

The Singular Locus

Definition 2.7.15: Singular Locus

Let X be a locally Noetherian scheme. The *singular locus* of X is the set:

$$\operatorname{Sing}(X) = \{x \in X \mid \mathcal{O}_{X,x} \text{ is not a regular local ring}\}$$

Proposition 2.7.16: Singular Locus is Closed

Let X be a scheme locally of finite type over a field k. Then $\operatorname{Sing}(X)$ is a closed subset of X.

***Proof*:**
The question is local, so we may assume $X = \operatorname{Spec}(k[x_1, \ldots, x_n]/(f_1, \ldots, f_m))$. By the Jacobian criterion, the singular locus is cut out by the vanishing of all $(n - \dim X) \times (n - \dim X)$ minors of the Jacobian matrix, which defines a closed set.

For irreducible varieties, the singular locus is not just closed but *proper*:

Theorem 2.7.17: Generic Smoothness

Let X be an irreducible variety over an algebraically closed field k. Then $\operatorname{Sing}(X)$ is a proper closed subset of X. In particular, the smooth locus is a dense open subset.

***Proof*:**
See [20]. The key idea is that if X is a hypersurface $V(f)$ and $\operatorname{Sing}(X) = X$, then all partial derivatives of f vanish identically on X. In characteristic 0, this forces f to be constant. In characteristic $p > 0$, it forces $f = g^p$ for some polynomial g (using algebraic closure to take p-th roots), contradicting the irreducibility of f. The general case reduces to the hypersurface case via birational equivalence.

Over non-algebraically-closed fields, the correct generalization replaces "regular" with "smooth":

Corollary 2.7.18: Generic Smoothness Over Perfect Fields

Let X be a geometrically integral[a] k-scheme of finite type over a perfect field k. Then the smooth locus of $X \to \operatorname{Spec} k$ is a dense open subset.

[a] geometrically integral means change the base ring k to $\overline{k}$; see ref:HERE

2.7.1 Properties of Regular Schemes

Beyond being domains, regular local rings enjoy two stronger finiteness properties:

Proposition 2.7.19: Regular Local Rings are Normal

Every regular local ring is integrally closed (normal).

***Proof*:**
See [15]. This follows from the fact that the associated graded ring of a regular local ring is a polynomial ring (hence a UFD), and one can lift the integral closure property from the associated graded ring.

Proposition 2.7.20: Regular Local Rings are UFDs

Every regular local ring is a unique factorization domain.

***Proof*:**
This is a theorem of Auslander–Buchsbaum, using the characterization of regular local rings via finite global dimension (Serre's theorem, [15]). The key step is showing that every height-1 prime in a regular local ring is principal, and then invoking the Nagata criterion for UFDs.

The chain of implications for Noetherian local domains is:

$$\text{regular} \implies \text{UFD} \implies \text{normal} \implies \text{regular in codimension 1}$$

None of these implications reverse in general (except in dimension 1, where regular, normal, and DVR are equivalent by [10]).

Proposition 2.7.21: Regular Scheme and Irreducible Components

A regular scheme is a disjoint union of its irreducible components (each of which is integral).

***Proof*:**
Since each stalk is a domain, every point lies on a unique irreducible component. Hence distinct components cannot meet, and the scheme decomposes as a disjoint union. Each component is integral (irreducible and reduced) since regularity implies the stalks are domains.

2.7.2 Dimension and Regularity for Schemes Over $\mathbb{Z}$

An important feature of the scheme-theoretic framework is that the notion of regularity applies uniformly to schemes over any base, including $\operatorname{Spec} \mathbb{Z}$. This allows number-theoretic objects to be treated geometrically:

Example 2.7.22: Regularity of Arithmetic Schemes

1. $\operatorname{Spec}\mathbb{Z}$ is a regular scheme of dimension 1: for each prime (p), the local ring $\mathbb{Z}_{(p)}$ is a DVR (hence a regular local ring of dimension 1), and the generic point has local ring $\mathbb{Q}$, a field.
2. $\operatorname{Spec}\mathbb{Z}[i]$ is a regular scheme. At each maximal ideal $\mathfrak{p}$, the local ring $\mathbb{Z}[i]_{\mathfrak{p}}$ is a DVR (since $\mathbb{Z}[i]$ is a Dedekind domain).
3. $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]$ is *not* regular at the prime $\mathfrak{p} = (2, 1+\sqrt{-5})$: the local ring $\mathbb{Z}[\sqrt{-5}]_{\mathfrak{p}}$ is a 1-dimensional local domain that is not a DVR (since $\mathfrak{p}$ is not principal). See [8] for the failure of unique factorization in this ring. The normalization (integral closure in the fraction field) produces the ring of integers $\mathbb{Z}\left[\frac{1+\sqrt{-5}}{2}\right]$, which *is* a Dedekind domain and hence regular.
4. More generally, if K is a number field and $\mathcal{O}_K$ its ring of integers, then $\operatorname{Spec}\mathcal{O}_K$ is a regular scheme if and only if $\mathcal{O}_K$ is a principal ideal domain (equivalently, the class number of K is 1). This follows from the equivalence of regularity and DVR in dimension 1 (by [10]).

Exercise 2.7.1

1. Show that $\mathbb{A}^n_k$ is a regular scheme for any field k.
2. Let $X = \operatorname{Spec} k[x,y]/(y^2 - x^3 + x)$. Show that X is a regular scheme (assuming $\operatorname{char} k \neq 2$).
3. Show that the generic point of any integral scheme is a regular point.
4. Let $X = \operatorname{Spec} k[x,y,z,w]/(xw - yz)$ be the quadric cone (from example 2.2.2). Find $\operatorname{Sing}(X)$ and show it consists of a single point.
5. Let k be a perfect field and $f \in k[x_1, \dots, x_n]$ irreducible. Show that $V(f)$ is smooth over k if and only if f and its partial derivatives have no common zero.
6. Show that the product of regular schemes is regular (hint: use that regularity is detected at stalks, and compute the stalks of a product).
7. Let R be a regular local ring and $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. Show that $R/(x)$ is again a regular local ring (of dimension one less).

2.8 *How Much Does the Topology Know?

Throughout this chapter, we have constructed schemes as locally ringed spaces: a topological space X together with a structure sheaf $\mathcal{O}_X$. The topological space carries the Zariski topology, which for affine schemes $\operatorname{Spec}(R)$ is entirely determined by the lattice of prime ideals of R. A natural question arises: how much of the scheme-theoretic structure is already visible at the level of topology?

One might initially expect a clean dichotomy: that the topological spaces arising from affine schemes form a special class, and that the topology of a non-affine scheme would look fundamentally "different"—perhaps failing some property that all affine topologies share. This turns out to be false, and understanding *why* it is false sharpens our appreciation for the role of the structure sheaf.

Spectral spaces

To state the result precisely, we need to isolate the topological properties that characterize the spaces $|\mathrm{Spec}\,(R)|$.

Definition 2.8.1: Spectral Space

A topological space X is a *spectral space* (or *coherent space*) if it satisfies the following four conditions:

1. X is quasi-compact.
2. X is sober: every irreducible closed subset has a unique generic point.
3. The quasi-compact open subsets of X are closed under finite intersection (i.e. X is quasi-separated, see definition 3.5.6) and form a basis for the topology.
4. Every non-empty irreducible closed subset of X has a generic point.

Condition (2) already implies (4), so the definition can be stated with just (1)–(3), but we include (4) for emphasis as it is the condition most often violated by "pathological" topological spaces. Note that condition (3) asks for two things at once: the quasi-compact opens form a basis (so there are enough of them), and a finite intersection of quasi-compact opens is again quasi-compact (a condition that can fail even for well-behaved spaces—it fails, for instance, for the Zariski topology on an infinite product of affine lines, though not for any Noetherian space).

The reader should recognize that every affine scheme satisfies these conditions. In $\mathrm{Spec}\,(R)$: the space is quasi-compact (section 2.1.3); it is sober by construction (generic points of irreducible closed sets correspond to prime ideals); and the distinguished open sets $D(f)$, which are quasi-compact, form a basis that is closed under finite intersection ($D(f) \cap D(g) = D(fg)$).

The fact, proved by Hochster in 1969, is that these conditions are not just necessary but *sufficient*.

Theorem 2.8.2: Hochster's Theorem

A topological space X is homeomorphic to $\mathrm{Spec}\,(R)$ for some commutative ring R if and only if X is a spectral space.

***Proof*:**
see [44]

The "only if" direction is the easy half: we have just verified that every $\mathrm{Spec}\,(R)$ is spectral. The "if" direction is: given any spectral space X, Hochster explicitly constructs a commutative ring R such that $|\mathrm{Spec}\,(R)| \cong X$. The construction uses the lattice of quasi-compact open subsets of X to build R as a quotient of a polynomial ring, with generators corresponding to the quasi-compact opens and relations encoding their intersection and covering behavior.

We do not reproduce Hochster's construction here, but the theorem has an immediate and perhaps surprising consequence.

Non-affine schemes with affine topology

Consider projective space $\mathbb{P}^n_k$ over a field k, with $n \geq 1$. This is emphatically not an affine scheme: its ring of global sections is just k, and $\operatorname{Spec}(k)$ is a single point, while $\mathbb{P}^n_k$ has many points. Yet the underlying topological space $|\mathbb{P}^n_k|$ is:

1. quasi-compact (it is covered by finitely many affine charts $U_i \cong \mathbb{A}^n$),
2. sober (it is a scheme, and every scheme is sober),
3. equipped with a basis of quasi-compact opens closed under finite intersection (the distinguished opens in each affine chart, together with their finite intersections, which are again quasi-compact by proposition 2.1.15).

Hence $|\mathbb{P}^n_k|$ is a spectral space, and by Hochster's theorem, there exists a commutative ring R with $|\mathbb{P}^n_k| \cong |\operatorname{Spec}(R)|$ as topological spaces. The topology of projective space is indistinguishable from that of an affine scheme.

This is not an isolated phenomenon. Any quasi-compact and quasi-separated scheme has an underlying spectral space, and hence is homeomorphic to $\operatorname{Spec}(R)$ for some ring R. This includes virtually all schemes encountered in practice: all Noetherian schemes, all schemes of finite type over a field or over $\mathbb{Z}$, all separated schemes.

Proposition 2.8.3: Topology of Quasi-Compact Quasi-Separated Schemes

Let X be a quasi-compact and quasi-separated scheme. Then $|X|$ is a spectral space, and hence there exists a commutative ring R such that $|X| \cong |\operatorname{Spec}(R)|$ as topological spaces.

***Proof*:**
Quasi-compactness of X gives condition (1). Every scheme is sober (a point $x \in X$ has closure $\overline{\{x\}}$ which is irreducible, and if Z is irreducible closed then its generic point is the unique point η with $\overline{\{\eta\}} = Z$; this follows from the corresponding fact for affine schemes, which glues). For condition (3): the quasi-compact opens of X include all affine opens and their finite unions, which form a basis. The quasi-separated hypothesis says exactly that the intersection of two quasi-compact opens is quasi-compact, giving closure under finite intersection.

To make this concrete, here are some examples.

Example 2.8.4: Non-Affine Schemes with Affine Topology

1. *Projective space.* As discussed above, $|\mathbb{P}^n_k|$ is spectral. Over $k = \overline{k}$ algebraically closed, the closed points of $\mathbb{P}^1_k$ are in bijection with $\overline{k} \cup \{\infty\}$, and the topology is the cofinite topology on this set together with a generic point whose closure is the whole space. This is homeomorphic to $|\operatorname{Spec}(k[x])|$ (which also has the cofinite topology on its closed points, plus a generic point). In particular, $|\mathbb{P}^1_k| \cong |\mathbb{A}^1_k|$ as topological spaces, even though $\mathbb{P}^1_k$ and $\mathbb{A}^1_k$ are not isomorphic as schemes.
2. *The line with doubled origin.* The non-separated scheme obtained by gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$ is quasi-compact and quasi-separated, so its topology is spectral. This scheme is not affine (it is not even separated), yet topologically it is homeomorphic to $\operatorname{Spec}(R)$ for

some ring R.

3. *The punctured plane.* The scheme $U = \mathbb{A}^2_k \setminus \{(x, y)\}$ from Example 2.2.51 is quasi-compact and quasi-separated (it is separated, being an open subscheme of an affine scheme), hence spectral. It is not affine, yet its topology is that of some $\operatorname{Spec}(R)$.

The constructible topology and Stone spaces

Hochster's theorem tells us that a topological space is the spectrum of a ring if and only if it is spectral. But the definition of a spectral space involves a somewhat asymmetric collection of conditions. There is a cleaner perspective that explains *what structure a spectral space actually carries*, and it comes from a classical duality in topology and logic.

Given a spectral space X, define the *constructible topology* (also called the *patch topology*) on the underlying set of X by declaring both the quasi-compact open subsets *and their complements* to be open. Equivalently, the constructible topology is the coarsest topology in which every quasi-compact open of X is clopen (both open and closed).

Proposition 2.8.5: The Constructible Topology is a Stone Space

Let X be a spectral space. Then X equipped with the constructible topology is a *Stone space*: a topological space that is compact, Hausdorff, and totally disconnected.

***Proof*:**
(Sketch) The constructible topology is Hausdorff because any two distinct points of a spectral space can be separated by a quasi-compact open (one point lies in it, the other does not), which is clopen in the constructible topology. It is compact because the constructible topology is finer than the original spectral topology (which is already quasi-compact), and one can verify that any cover by constructible opens has a finite subcover using the quasi-compactness of the spectral topology together with the finiteness of intersections. Total disconnectedness follows because the clopen sets (which include all quasi-compact opens and their complements) separate points.

We denote the set X equipped with the constructible topology by X^{cons}. The identity map $X^{\text{cons}} \to X$ is continuous (the constructible topology is finer), but not a homeomorphism: the constructible topology is strictly finer than the spectral topology whenever X has a non-closed point (i.e., whenever X is not discrete).

The name "constructible topology" is literal. A subset $C \subseteq X$ of a topological space is *constructible* if it belongs to the Boolean algebra generated by the open sets—equivalently, if C is a finite union of locally closed subsets ($U \cap Z$ where U is open and Z is closed). In a spectral space, the constructible subsets have a clean characterization:

Proposition 2.8.6: Constructible Sets are Clopens

Let X be a spectral space. A subset $C \subseteq X$ is constructible (in the classical sense: a finite Boolean combination of quasi-compact opens) if and only if C is clopen in the constructible topology.

This connects directly to Chevalley's theorem (**??**): if $f : X \to Y$ is a morphism of finite type between Noetherian schemes, then the image of a constructible set is constructible. In the language of the constructible topology, this says that f induces a continuous map $X^{\text{cons}} \to Y^{\text{cons}}$ between the associated Stone spaces, and continuous maps between Stone spaces preserve clopens. From this perspective, Chevalley's theorem is a statement about the *continuity of scheme morphisms with respect to the constructible topology*—a viewpoint that is both conceptually cleaner and technically powerful.

Stone duality and the lattice of opens

The appearance of Stone spaces is not accidental; it reflects a duality from mathematical logic and lattice theory.

Stone duality (1936) establishes an anti-equivalence of categories between Boolean algebras and Stone spaces: every Boolean algebra B is the algebra of clopen subsets of a unique Stone space $S(B)$, and conversely, every Stone space arises in this way. Restricting to distributive lattices (which need not be Boolean), one obtains *spectral spaces*: a spectral space is precisely a Stone space together with a distinguished sublattice of its Boolean algebra of clopens.

More precisely, let X^{cons} be a Stone space. A sublattice $L \subseteq \text{Clopen}(X^{\text{cons}})$ that forms a basis for some topology on the underlying set, and is closed under finite intersection and finite union, determines a spectral topology on X: the topology generated by L. Different choices of L give different spectral topologies on the same underlying Stone space.

This gives a clean factorization of the data in a spectral space:

$$\text{Spectral space} \;=\; \text{Stone space} \;+\; \text{Choice of lattice.}$$

The Stone space is the "combinatorial skeleton"—it knows which subsets are constructible (i.e., definable by first-order formulas in model-theoretic language). The lattice records which constructible sets are "open," i.e., which direction the specialization order runs.

Example 2.8.7: Different Spectral Structures on the Same Stone Space

Consider the Cantor set $C \subseteq [0,1]$, which is a Stone space (compact, Hausdorff, totally disconnected). Every countable Boolean algebra has C as its Stone space. In particular, the spaces $|\text{Spec}(\mathbb{Z})|$ and $|\text{Spec}(k[x])|$ (for k algebraically closed) both have the same constructible topology—they are both countable spectral spaces whose Stone space is the Cantor set—but they carry different spectral topologies (different lattices of opens) corresponding to the different lattice structures of their prime ideals.

More concretely, both spaces consist of a generic point η together with countably many closed points, and the topology in both cases has $\{\eta\}$ dense and the closed points discrete in the complement. But the "arithmetic" of which opens contain which closed points differs: in $\text{Spec}(\mathbb{Z})$, the closed point (p) lies in $D(q)$ for $q \neq p$, while in $\text{Spec}(k[x])$, the closed point $(x-a)$ lies in $D(x-b)$ for $b \neq a$. These are different lattices on the same Stone space.

What does the topology actually store?

The data of a scheme $(X, \mathcal{O}_X)$ decomposes into three layers:

1. *The Stone space* X^{cons}: the combinatorial skeleton. It knows the set of points, the constructible subsets, and (via Stone duality) the Boolean algebra they generate. This is the coarsest invariant—it is shared by many different schemes.

2. *The spectral topology*: a choice of sublattice L of the clopens of X^{cons}, determining which constructible sets are "open." This encodes the specialization order (which points are limits of which others) and hence the notion of "generic" and "closed" points. Different spectral topologies on the same Stone space give genuinely different spaces (e.g., $\operatorname{Spec}(\mathbb{Z})$ vs. $\operatorname{Spec}(k[x])$), but by Hochster's theorem, both are realizable as $\operatorname{Spec}(R)$ for some ring.

3. *The structure sheaf* $\mathcal{O}_X$: the algebraic data. It determines which continuous maps are "algebraic," what the global sections and cohomology are, and whether the scheme is affine, projective, smooth, etc. This is the finest invariant, and it is where the geometry lives.

The passage from (3) to (2) forgets the algebra; the passage from (2) to (1) forgets the specialization order. Neither passage is reversible: many non-isomorphic schemes share the same topology (as we saw with $\mathbb{P}^1_k$ and $\mathbb{A}^1_k$), and many non-homeomorphic spectral spaces share the same Stone space (as with $\operatorname{Spec}(\mathbb{Z})$ and $\operatorname{Spec}(k[x])$).

The conclusion is that the Zariski topology, while important for defining the notion of "local" and for the combinatorics of specialization and generization, carries far less information than one might expect. The structure sheaf is the essential datum: it determines which continuous maps are "algebraic," which open subsets are "distinguished," and what "functions" look like. Two schemes can share the same topological space—or even the same constructible topology—while having completely different algebraic geometry.

3

Schemes II: Morphisms and Constructions

Scheme morphisms are inherently more *algebraic* than *smooth* in nature: they are in some ways more stringent than holomorphic maps (only "polynomial-like" functions are allowed), but they also permit singularities. In this chapter, we shall properly explore morphisms of schemes, classify many important types, and interpret them geometrically. In the process, we will be able to "recover" the correct analogues of Hausdorffness and compactness for schemes in the form of *separatedness* and *properness*. The condition of being proper will turn out to be closely related to being projective: by Chow's lemma, every proper scheme over a Noetherian base admits a surjective birational morphism from a projective scheme, showing that projective schemes are the natural "compact schemes" to consider. The chapter concludes by defining an abstract variety and showing that $\mathbf{Var}_k$ is a fully faithful subcategory of $\mathbf{Sch}_k$.

Algebraically, smooth and holomorphic morphisms share an important feature with scheme morphisms: a continuous map $f : M \to N$ between smooth manifolds is a *smooth map* if and only if $f^*(C^\infty(N)) \subseteq C^\infty(M)$ where

$$f^* : C^\infty(N) \to C^\infty(M) \qquad f^*(\varphi) = \varphi \circ f$$

and in particular if f is a homeomorphism, it is a diffeomorphism if and only if f^* is a ring isomorphism. In essence, f "preserves" and is "compatible with" all the relations on N given by smooth functions. As schemes are defined by their algebraic relations, this pullback approach generalizes naturally[1].

With this in mind, we shall require the following properties from scheme morphisms:

1. A scheme morphism $f : X \to Y$ must preserve the structure sheaf (which encodes all algebraic

[1] In the smooth case, the "if and only if" makes this an equivalent definition of smooth map; since there is only one type of "affine patch" (open subsets of $\mathbb{R}^n$), one can then recover the usual definition. Proposition 3.1.10 establishes the analogous result for schemes.

information). Concretely, for every open set $V \subseteq Y$, there should be a ring homomorphism $f^{\#}(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$ that pulls back local sections on Y to local sections on X. The intuition is that we treat sections as genuine (scheme) functions, and we require their composition with f to remain a scheme function. Thus, a scheme morphism must be a *morphism of ringed spaces* (a continuous map together with a compatible sheaf map).

This additional structure is genuinely necessary: as we saw in example 2.1.25, not all continuous maps between spectra are induced by ring homomorphisms, and there is no natural way to recover a ring homomorphism from a mere continuous map. A simple instance is that $\operatorname{Spec} k$ and $\operatorname{Spec} k'$ for two non-isomorphic fields k, k' have homeomorphic underlying spaces (both are single points) yet are not isomorphic as schemes.

2. The pullback of functions must be compatible with evaluation: if $s \in \mathcal{O}_Y(V)$ vanishes at $f(x) \in Y$, then $f^{\#}(s)$ must vanish at $x \in X$. This means the pullback must respect the local ring structure at each point, making f a morphism of *locally* ringed spaces.

3. Ring homomorphisms between global sections and scheme morphisms between affine schemes must be dual to each other (generalizing the fact that a map between k-varieties is regular if and only if the contravariant map between coordinate rings is a k-algebra homomorphism).

This final condition means we want an equivalence of categories:

$$\mathbf{Sch}^{\mathbf{Aff}} \underset{\operatorname{Spec}(-)}{\overset{\Gamma(-)}{\rightleftarrows}} \mathbf{CRing}^{\mathrm{op}}$$

In section ref:HERE, this equivalence is generalized from affine schemes to all schemes by replacing $\mathbf{CRing}^{\mathrm{op}}$ with the appropriate category of sheaves of rings on a site (the functor-of-points perspective).

For those familiar with differential geometry [53], let us take a moment to compare and contrast smooth maps with scheme morphisms.

In differential geometry, continuous maps between manifolds can be arbitrarily well approximated by smooth maps (as a consequence of *Whitney's Approximation Theorem*). Hence, smooth maps can be thought of as "smoothing out" continuous maps – the smooth and continuous worlds are not far apart. This is *not at all* the case in the Zariski topology. Consider, for example, any permutation $\sigma : \overline{k} \to \overline{k}$ that preserves finite subsets (where $\overline{k}$ is an algebraically closed field). Such a σ induces a homeomorphism of $\mathbb{A}^1_{\overline{k}}$ in the Zariski topology – since the closed sets are exactly the finite sets and the whole space, any bijection preserving finiteness is continuous. Yet a "wild" permutation (one not given by a polynomial) will not be a scheme morphism[2]. Thus, there is no hope of approximating arbitrary continuous maps by regular maps: scheme morphisms are fundamentally more rigid than smooth maps.

In fact, regular maps behave much more like *holomorphic* maps. Both are determined by their local algebraic/analytic expressions, both satisfy analogues of the identity theorem (a regular map vanishing on a dense open must vanish everywhere), and both have fibers of controlled dimension. This similarity becomes an exact correspondence for the right categories: the GAGA theorems of Serre establish an equivalence between algebraic and analytic coherent sheaves on projective varieties over $\mathbb{C}$ (see section 6.9). Hence, when building geometric intuition for scheme morphisms via [53]

[2]Over $\mathbb{C}$, any discontinuous field automorphism $\sigma \in \operatorname{Aut}(\mathbb{C}/\mathbb{Q})$ gives such an example; these exist in abundance (assuming the axiom of choice) but none are $\mathbb{C}$-algebra maps.

and [52], it is holomorphic maps, not smooth maps, that provide the better mental model. A key difference is that scheme morphisms freely allow singularities and have unique algebraic-geometric features absent from the complex-analytic or smooth settings (for example, the Frobenius morphism from example 3.1.11, or the behavior of morphisms between schemes over $\operatorname{Spec}\mathbb{Z}$).

3.1 Elementary Concepts

Some of the discussion here shall overlap with the content in section 1.4; this was done as the self-completeness of the material in this section is a priority.

Recall (definition 1.4.4) that a morphism of ringed spaces $\pi : X \to Y$ is a continuous map of topological spaces together with a map $\pi^{\#} : \mathcal{O}_Y \to \pi_* \mathcal{O}_X$ that replicates composing any function in $\mathcal{O}_Y$ with π[3]. However, this notion is not restrictive enough for scheme theory for two reasons:

(i) Not all *continuous* maps between spectra are induced by ring homomorphisms (see Example 2.1.25), so specifying a sheaf map is genuinely additional data beyond the continuous map. Again, to abuse the same example, $\operatorname{Spec}(k)$ and $\operatorname{Spec}(k')$ for non-isomorphic fields k, k' are homeomorphic (both are single points) but carry different structure sheaves.

(ii) Even among ringed space morphisms, the global sections functor $\Gamma(-)$ is *not injective* on morphism sets: distinct ringed space morphisms can induce the same ring homomorphism on global sections. The following example demonstrates this.

Example 3.1.1: Two Ringed Space Morphisms for One Ring Homomorphism

Take $R = \mathbb{Z}_{(p)}$ and $S = \mathbb{Q}$ so that $\operatorname{Spec}(R) = \{[(p)], [(0)]\}$ and $\operatorname{Spec}(S) = \{[(0)]\}$. Let us drop the parentheses and square brackets to declutter notation. The open sets of $\operatorname{Spec}(R)$ are:

$$\emptyset, \quad \{0\}, \quad \{0, p\} = \operatorname{Spec}(R)$$

The stalks of $\operatorname{Spec}(R)$ are $\mathbb{Z}_{(p)}$ at p and $\mathbb{Q}$ at 0. The stalk of $\operatorname{Spec}(S)$ is $\mathbb{Q}$ at 0.

There is only one ring homomorphism $\psi : R \to S$, namely the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

In the category **Sch**, this should correspond to a unique map $\iota^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $\psi^{-1}((0)) = (0)$. However, in the category **RingedSp** we can define two morphisms between these affine schemes:

1. *Map 1 (The "correct" map):* $\pi_1 : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $0 \mapsto 0$. The sheaf map is determined by the stalk map at the unique point $0 \in \operatorname{Spec}(S)$:

$$(\pi_1)_0^{\#} : \mathcal{O}_{R,0} \to \mathcal{O}_{S,0} \qquad \Longrightarrow \qquad \mathbb{Q} \xrightarrow{\text{id}} \mathbb{Q}$$

This is a local ring homomorphism (both sides are fields, with maximal ideal (0) mapping to (0)). On global sections, it induces the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

2. *Map 2 (The "phantom" map):* $\pi_2 : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $0 \mapsto p$. Since $\{p\}$ is closed, this is a continuous map. The sheaf map is determined by the stalk map at $0 \in \operatorname{Spec}(S)$:

$$(\pi_2)_0^{\#} : \mathcal{O}_{R,p} \to \mathcal{O}_{S,0} \qquad \Longrightarrow \qquad \mathbb{Z}_{(p)} \xrightarrow{\text{inclusion}} \mathbb{Q}$$

[3] Or equivalently by adjointness, the map $\pi^{-1}(\mathcal{O}_Y) \to \mathcal{O}_X$.

This is a valid ring homomorphism, so π_2 is a valid morphism of ringed spaces. On global sections, this *also* induces the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

Thus, we have found two distinct ringed space morphisms $\pi_1 \neq \pi_2$ that induce the *same* ring homomorphism on global sections. We reject π_2 because the stalk map $\mathbb{Z}_{(p)} \to \mathbb{Q}$ is *not* a local ring homomorphism: the maximal ideal $(p) \subseteq \mathbb{Z}_{(p)}$ maps to the unit ideal in $\mathbb{Q}$ (since p is invertible in $\mathbb{Q}$), violating the condition $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$. The definition of locally ringed space morphism correctly filters out this non-geometric map.

Thus, if $\mathbf{RingedSp}^{\mathbf{Aff}}$ denotes the category whose objects are $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ and whose morphisms are ringed space morphisms, the functor:

$$\Gamma(-) : \mathbf{RingedSp}^{\mathbf{Aff}} \to \mathbf{CRing}^{\mathrm{op}}$$

is surjective on hom-sets (every ring homomorphism $R \to S$ induces at least one ringed space morphism via the standard contraction-of-primes construction), but not injective (as the example above demonstrates).

The key realization is that ring homomorphisms induce local structure that must be preserved. By proposition 2.2.21, the stalks of schemes are all *local rings*. A map between local rings need not preserve maximal ideals (for instance, the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$ sends the maximal ideal (p) to the unit ideal). We thus single out maps that do preserve this structure:

Definition 3.1.2: Local Ring Homomorphism

Let $\varphi : (R, \mathfrak{m}_R) \to (S, \mathfrak{m}_S)$ be a ring homomorphism between local rings with maximal ideals $\mathfrak{m}_R$ and $\mathfrak{m}_S$ respectively. Then φ is a *local ring homomorphism* if $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$, or equivalently $\varphi^{-1}(\mathfrak{m}_S) = \mathfrak{m}_R$.

A ringed space morphism that is a local ring homomorphism at each stalk is given a name:

Definition 3.1.3: Locally Ringed Space Morphism

Let X, Y be locally ringed spaces. A *morphism of locally ringed spaces* $(\pi, \pi^\#) : X \to Y$ is a morphism of ringed spaces such that the induced stalk map $\pi_p^\# : \mathcal{O}_{Y,\pi(p)} \to \mathcal{O}_{X,p}$ is a local ring homomorphism for every $p \in X$.

Observe that ring homomorphisms *automatically* induce local maps on localizations: if $\varphi : R \to S$ is a ring homomorphism and $\mathfrak{p} \subseteq S$ is prime, then the induced map $\varphi_\mathfrak{p} : R_{\varphi^{-1}(\mathfrak{p})} \to S_\mathfrak{p}$ given by $r/t \mapsto \varphi(r)/\varphi(t)$ satisfies $\varphi_\mathfrak{p}(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$. Indeed, if $r/t \in \mathfrak{m}_R$ then $r \in \varphi^{-1}(\mathfrak{p})$, so $\varphi(r) \in \mathfrak{p}$, giving $\varphi(r)/\varphi(t) \in \mathfrak{p}S_\mathfrak{p} = \mathfrak{m}_S$. Equivalently, $\mathfrak{m}_R = \varphi_\mathfrak{p}^{-1}(\mathfrak{m}_S)$[4]. Example 3.1.1 shows that this locality condition need *not* hold for arbitrary ringed space morphisms, even between locally ringed spaces. The local ring condition forces the morphism to “respect” not just the lattice structure of opens but also the evaluation of functions at each point.

With this added structure, we recover a bijection:

[4]Note this does not imply $\varphi(\mathfrak{m}_R) = \mathfrak{m}_S$: the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Z}_{(p)}[1/q]$ for primes $p \neq q$ is a local ring homomorphism where $\varphi(\mathfrak{m}_R) \subsetneq \mathfrak{m}_S$.

Theorem 3.1.4: Ring Homomorphisms and Locally Ringed Space Morphisms

Let R, S be commutative rings. Then:

1. Every ring homomorphism $\varphi : R \to S$ induces a natural morphism of locally ringed spaces:
$$(f, f^{\#}) : (\mathrm{Spec}\,(S), \mathcal{O}_{\mathrm{Spec}\,(S)}) \to (\mathrm{Spec}\,(R), \mathcal{O}_{\mathrm{Spec}\,(R)})$$
2. Every morphism of locally ringed spaces $(\mathrm{Spec}\,(S), \mathcal{O}_{\mathrm{Spec}\,(S)}) \to (\mathrm{Spec}\,(R), \mathcal{O}_{\mathrm{Spec}\,(R)})$ induces a ring homomorphism $\varphi : R \to S$ (via global sections).
3. These operations are inverse to each other. In particular, the locally ringed space morphism is uniquely determined by the ring homomorphism on global sections:
$$\mathrm{Hom}_{\mathbf{LRS}}(\mathrm{Spec}\,(S), \mathrm{Spec}\,(R)) \cong \mathrm{Hom}_{\mathbf{CRing}}(R, S)$$

***Proof*:**
(1) Ring homomorphism $\Rightarrow$ locally ringed space morphism. Let $\varphi : R \to S$ be a ring homomorphism. Define the continuous map $f : \mathrm{Spec}\,(S) \to \mathrm{Spec}\,(R)$ by the standard contraction $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$.

We construct the sheaf morphism $f^{\#} : \mathcal{O}_{\mathrm{Spec}\,(R)} \to f_*\mathcal{O}_{\mathrm{Spec}\,(S)}$ on the basis of distinguished open sets, using proposition 1.3.3. For each $g \in R$, we must define a map

$$f^{\#}(D(g)) : \mathcal{O}_{\mathrm{Spec}\,(R)}(D(g)) \longrightarrow (f_*\mathcal{O}_{\mathrm{Spec}\,(S)})(D(g)) = \mathcal{O}_{\mathrm{Spec}\,(S)}(f^{-1}(D(g))).$$

The preimage of a distinguished open under f is again distinguished:

$$\begin{aligned} f^{-1}(D(g)) &= \left\{[\mathfrak{p}] \in \mathrm{Spec}\,(S) \;\middle|\; \varphi^{-1}(\mathfrak{p}) \in D(g)\right\} \\ &= \left\{[\mathfrak{p}] \in \mathrm{Spec}\,(S) \;\middle|\; g \notin \varphi^{-1}(\mathfrak{p})\right\} \\ &= \{[\mathfrak{p}] \in \mathrm{Spec}\,(S) \mid \varphi(g) \notin \mathfrak{p}\} \\ &= D(\varphi(g)) \end{aligned}$$

This is a special property of the affine setting: the pullback of a "basic" geometric object is again basic. Using the identifications $\mathcal{O}_{\mathrm{Spec}\,(R)}(D(g)) \cong R_g$ and $\mathcal{O}_{\mathrm{Spec}\,(S)}(D(\varphi(g))) \cong S_{\varphi(g)}$, the global map φ induces a map on localizations. Since $\varphi(g)$ is a unit in $S_{\varphi(g)}$, the universal property of localization provides a unique ring homomorphism:

$$\Phi_g : R_g \longrightarrow S_{\varphi(g)}, \qquad \frac{a}{g^n} \longmapsto \frac{\varphi(a)}{\varphi(g)^n}.$$

These maps are compatible with restriction: if $D(h) \subseteq D(g)$ (so $h \in \sqrt{(g)}$), the diagram

$$\begin{tikzcd} R_g \arrow[r, "\Phi_g"] \arrow[d, "\mathrm{loc}"] & S_{\varphi(g)} \arrow[d, "\mathrm{loc}"] \\ R_h \arrow[r, "\Phi_h"] & S_{\varphi(h)} \end{tikzcd}$$

commutes because both paths send a/g^n to $\varphi(a)/\varphi(g)^n$ viewed in $S_{\varphi(h)}$. By proposition 1.3.3, the maps Φ_g glue to a unique sheaf morphism $f^{\#}$.

At each stalk, the induced map $f_{\mathfrak{p}}^{\#} : R_{\varphi^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ is the localization of φ at $\mathfrak{p}$, which is a local ring homomorphism as shown above. Hence $(f, f^{\#})$ is a morphism of locally ringed spaces.

(2) Locally ringed space morphism ⇒ ring homomorphism. Given a morphism of locally ringed spaces $(f, f^{\#}) : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$, evaluate the sheaf map on global sections:

$$f^{\#}(\operatorname{Spec}(R)) : \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)) \to \mathcal{O}_{\operatorname{Spec}(S)}(f^{-1}(\operatorname{Spec}(R)))$$

Since $\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)) \cong R$ and $\mathcal{O}_{\operatorname{Spec}(S)}(\operatorname{Spec}(S)) \cong S$, this yields a ring homomorphism $\varphi : R \to S$.

(3) These operations are inverse. Starting with $\varphi : R \to S$, constructing $(f, f^{\#})$, and taking global sections recovers φ by construction.

Conversely, starting with a locally ringed space morphism $(f, f^{\#})$ and letting $\varphi = f^{\#}(\operatorname{Spec}(R)) : R \to S$, and g be the morphism defined in step (1) given by $g(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$, we must show that $(f, f^{\#})$ equals the morphism $(g, g^{\#})$ constructed in step (1).

The continuous maps agree: Let $\mathfrak{p} \in \operatorname{Spec}(S)$. Consider the commutative diagram relating global sections to stalks:

$$\begin{tikzcd} R \arrow[r, "\varphi"] \arrow[d, "\mathrm{loc}"] & S \arrow[d, "\mathrm{loc}"] \\ R_{f(\mathfrak{p})} \arrow[r, "f_{\mathfrak{p}}^{\#}"] & S_{\mathfrak{p}} \end{tikzcd}$$

We compute $\varphi^{-1}(\mathfrak{p}) \subseteq R$ by tracing the preimage of $\mathfrak{m}_{S_{\mathfrak{p}}} \subseteq S_{\mathfrak{p}}$ along two paths:

- *Algebraic path (right then down):* The localization $S \to S_{\mathfrak{p}}$ pulls $\mathfrak{m}_{S_{\mathfrak{p}}}$ back to $\mathfrak{p}$. Then φ pulls $\mathfrak{p}$ back to $\varphi^{-1}(\mathfrak{p})$.
- *Geometric path (down then right):* Since $(f, f^{\#})$ is a morphism of *locally* ringed spaces, $f_{\mathfrak{p}}^{\#}$ is a local ring homomorphism, so $(f_{\mathfrak{p}}^{\#})^{-1}(\mathfrak{m}_{S_{\mathfrak{p}}}) = \mathfrak{m}_{R_{f(\mathfrak{p})}}$. The localization $R \to R_{f(\mathfrak{p})}$ pulls this maximal ideal back to $f(\mathfrak{p})$.

By commutativity, both paths yield the same ideal: $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p}) = g(\mathfrak{p})$. Hence $f = g$ as continuous maps.

The sheaf maps agree: Since $f = g$, both $(f, f^{\#})$ and $(g, g^{\#})$ are sheaf morphisms over the same continuous map. To show $f^{\#} = g^{\#}$, it suffices to check equality on stalks ($\mathcal{O}_{\operatorname{Spec}(R)}$ is a sheaf, so morphisms out of it are determined by stalks). At each $\mathfrak{p} \in \operatorname{Spec}(S)$, both stalk maps are local ring homomorphisms $R_{\varphi^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ making the above diagram commute. Since the localization map $R \to R_{\varphi^{-1}(\mathfrak{p})}$ is an epimorphism in the category of rings (every element of $R_{\varphi^{-1}(\mathfrak{p})}$ is a fraction r/s with $r, s \in R$), any ring homomorphism out of $R_{\varphi^{-1}(\mathfrak{p})}$ that agrees with φ when precomposed with the localization map must be the unique localization $\varphi_{\mathfrak{p}}$. Hence $f_{\mathfrak{p}}^{\#} = g_{\mathfrak{p}}^{\#}$ for all $\mathfrak{p}$, giving $f^{\#} = g^{\#}$.

Affine schemes form the correct category to be the opposite category of commutative rings. We can rephrase what we just proved as:

Corollary 3.1.5: Equivalence of Rings and Affine Schemes

The category of affine schemes with locally ringed space morphisms is equivalent to the opposite category of commutative rings:

$$\mathbf{Sch}^{\mathbf{Aff}} \underset{\mathrm{Spec}\,(-)}{\overset{\Gamma(-)}{\rightleftarrows}} \mathbf{CRing}^{\mathrm{op}}$$

where Spec and Γ are inverse equivalences. More generally, for any locally ringed space X and any ring R, this equivalence extends to an adjunction on all of **LRS**:

$$\mathrm{Hom}_{\mathbf{CRing}}(R, \Gamma(X)) \cong \mathrm{Hom}_{\mathbf{LRS}}(X, \mathrm{Spec}\,(R))$$

where $\Gamma(-)$ is the left adjoint and Spec $(-)$ is the right adjoint. On the subcategory of affine schemes, this adjunction restricts to an equivalence because both the unit $(X \to \mathrm{Spec}\,\Gamma(X))$ and the counit $(\Gamma(\mathrm{Spec}\,(R)) \to R)$ are isomorphisms.

Having shown that locally ringed morphisms are the right choice, let us take a moment to understand better how evaluation at a point is "preserved" by locally ringed space morphisms. Given a morphism of locally ringed spaces $\pi : X \to Y$, a point $q \in Y$ and a point $p \in X$ with $\pi(p) = q$, how do we interpret the local ring homomorphism between stalks? Say there is a section $g \in \mathcal{O}_Y(V)$ on an open subset $V \subseteq Y$ that vanishes at q, i.e., $g(q) = 0$ (equivalently, $g \in \mathfrak{m}_q$ in the stalk $\mathcal{O}_{Y,q}$). We want the pullback $\pi^\#(g)$ to vanish at p:

$$g(\pi(p)) = 0 \quad \Rightarrow \quad (\pi^\# g)(p) = 0.$$

The local ring condition ensures exactly this: since $\pi_p^\# : \mathcal{O}_{Y,q} \to \mathcal{O}_{X,p}$ satisfies $\pi_p^\#(\mathfrak{m}_q) \subseteq \mathfrak{m}_p$, the vanishing $g \in \mathfrak{m}_q$ implies $\pi^\#(g) \in \mathfrak{m}_p$, which means $\pi^\#(g)$ vanishes at p.

$$\begin{tikzcd}
\mathcal{O}_{Y,q} \arrow[r, "\pi_p^\#"] & \mathcal{O}_{X,p} \\
g \in \mathfrak{m}_q \arrow[r, mapsto] & \pi^\#(g) \in \mathfrak{m}_p \\
Y \supseteq V & X \supseteq \pi^{-1}(V) \\
\in & \in \\
q = \pi(p) & p \arrow[l, "\pi"']
\end{tikzcd}$$

The top row encodes the algebraic condition (the stalk map sends the maximal ideal into the maximal ideal); the bottom row encodes the geometric picture (the point p maps to q under π). The locally ringed space condition is the requirement that these two rows are compatible: vanishing is preserved under pullback. π is not injective, for each $p \in \pi^{-1}(q)$, there would be a local map.

This is "obvious" when g and π are our usual functions since $g(\pi(p)) = (g \circ \pi)(p)$. However, the phantom map from Example 3.1.1 shows that this can fail for general ringed space morphisms.

Consider for the last time:

$$\pi_2 : \mathrm{Spec}\,(\mathbb{Q}) \to \mathrm{Spec}\,(\mathbb{Z}_{(p)}), \qquad \pi_2([(0)]) = [(p)].$$

The function $p \in \mathcal{O}_Y(Y) = \mathbb{Z}_{(p)}$ vanishes at the closed point $[(p)]$: its value in the residue field $\kappa([(p)]) = \mathbb{Z}_{(p)}/(p) \cong \mathbb{F}_p$ is $p \bmod p = 0$. However, the pullback $\pi_2^{\#}(p)$ is $p \in \mathbb{Q}$ (via the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$), and the value of p at the unique point $[(0)] \in \mathrm{Spec}\,(\mathbb{Q})$ is $p \bmod (0) = p \neq 0$ in $\kappa([(0)]) = \mathbb{Q}$. The vanishing was not preserved.

3.1.6 Remark: the generic point of $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$ The reader may find it surprising that $\{[(0)]\}$ is an open subset of $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$: it is the distinguished open $D(p)$, the complement of the closed point $V(p) = \{[(p)]\}$. This is *not* typical behaviour for generic points. In $\mathrm{Spec}\,(\mathbb{Z})$ or $\mathrm{Spec}\,(k[x,y])$, the generic point is *not* open—its closure is the entire space. The peculiarity here is that $\mathbb{Z}_{(p)}$ is a local domain with exactly two prime ideals, so the space has only two points, and removing a closed point from a two-point space leaves an open point. This makes $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$ the simplest non-trivial example for testing morphism conditions, precisely because the topology is so small that pathological maps are easy to construct explicitly.

3.1.7 Sections as Functions: Three Perspectives The reader may be troubled by a philosophical point: in differential geometry, a section $f \in C^\infty(U)$ *is* a function $f : U \to \mathbb{R}$, and evaluation at a point x is simply $f(x)$. For schemes, a section $f \in \mathcal{O}_X(U)$ is an element of a ring, and evaluation at x is the composite

$$\mathcal{O}_X(U) \longrightarrow \mathcal{O}_{X,x} \longrightarrow \kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x,$$

which is an additional construction. Is the datum of evaluation "missing" from the definition? The answer is no—it is encoded in the locally ringed space condition, but implicitly. There are three equivalent ways to see this.

(a) Hartshorne's approach. Define a section over U to be a function $f : U \to \coprod_{x\in U} \mathcal{O}_{X,x}$ such that $f(x) \in \mathcal{O}_{X,x}$ for each x, and such that f is "locally a quotient of ring elements." This makes sections into genuine functions and evaluation into literal function evaluation. The cost is that the codomain $\coprod \mathcal{O}_{X,x}$ varies from point to point and the definition requires a local coherence condition.

(b) The Modern approach. Keep sections as abstract ring elements. The locally ringed space axiom guarantees that each stalk $\mathcal{O}_{X,x}$ is local with a unique maximal ideal $\mathfrak{m}_x$, hence a unique residue field $\kappa(x)$, hence a unique evaluation map. The datum of evaluation is not "added"—it is a *consequence* of the locally ringed space structure. If the stalks were not local (i.e., if there were multiple maximal ideals), there would be multiple residue fields and no canonical notion of "the value of f at x." The LRS axiom is precisely the condition that evaluation is well-defined.

(c) The functor-of-points approach. By the representability of the global section functor (Proposition 3.3.9), a section $f \in \mathcal{O}_X(U)$ corresponds to a morphism $U \to \mathbb{A}^1_k$. In this picture, sections *are* morphisms, and evaluation at a point x is the composite $\mathrm{Spec}\,(\kappa(x)) \to U \to \mathbb{A}^1_k$, which is genuinely a function value.

All three perspectives carry exactly the same information. Approach (a) is most concrete; (b) is most algebraically efficient; (c) is most categorically natural. This text primarily uses (b), with (c) developed in section 3.3.

3.1.1 Scheme Morphisms

Inspired by the algebraic-geometric properties of locally ringed space morphisms, we define morphisms between schemes as locally ringed space morphisms:

Definition 3.1.8: Morphism of Schemes

Let X, Y be schemes. A morphism of locally ringed spaces $\pi : X \to Y$ is called a *morphism of schemes*. Schemes together with scheme morphisms form the category **Sch**.

One of the most important ways to control scheme morphisms is to reduce them to an affine cover. The following result is slightly more general but implies the desired result:

Lemma 3.1.9: Gluing Morphisms of Ringed Spaces

Let X, Y be ringed spaces and $X = \bigcup_i U_i$ an open cover. Suppose there exist ringed space morphisms $f_i : U_i \to Y$ that agree on overlaps:

$$f_i\big|_{U_i \cap U_j} = f_j\big|_{U_i \cap U_j}$$

Then there is a unique morphism of ringed spaces $f : X \to Y$ such that $f\big|_{U_i} = f_i$ for all i. Furthermore, if each f_i is a morphism of locally ringed spaces, then so is f.

***Proof*:**

Continuity. By the topological gluing lemma, the maps f_i glue to a unique continuous map $f : X \to Y$ with $f|_{U_i} = f_i$.

Sheaf morphism. We must construct $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$. For any open $V \subseteq Y$, we need a ring homomorphism $f^\#(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$.

The open set $f^{-1}(V) \subseteq X$ is covered by $\{f^{-1}(V) \cap U_i\}_i$. Since $f|_{U_i} = f_i$, we have $f_i^{-1}(V) = f^{-1}(V) \cap U_i$. The sheaf morphism $f_i^\# : \mathcal{O}_Y \to (f_i)_*\mathcal{O}_{U_i}$ gives, for each i, a ring homomorphism:

$$f_i^\#(V) : \mathcal{O}_Y(V) \longrightarrow \mathcal{O}_{U_i}(f^{-1}(V) \cap U_i)$$

On overlaps $f^{-1}(V) \cap U_i \cap U_j$, the agreement condition $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ implies $f_i^\#|_{U_i \cap U_j} = f_j^\#|_{U_i \cap U_j}$, so:

$$f_i^\#(V)(s)\big|_{f^{-1}(V) \cap U_i \cap U_j} = f_j^\#(V)(s)\big|_{f^{-1}(V) \cap U_i \cap U_j} \qquad \forall s \in \mathcal{O}_Y(V)$$

The sets $\{f^{-1}(V) \cap U_i\}_i$ cover $f^{-1}(V)$, and the local images $f_i^\#(V)(s)$ are sections of $\mathcal{O}_X$ that agree on overlaps. By the gluability axiom of $\mathcal{O}_X$, there exists a unique section $t \in \mathcal{O}_X(f^{-1}(V))$ restricting to $f_i^\#(V)(s)$ on each $f^{-1}(V) \cap U_i$. Define $f^\#(V)(s) := t$. This is a ring homomorphism since each $f_i^\#(V)$ is, and it commutes with restriction by construction. Uniqueness of $f^\#$ follows from the locality axiom of $\mathcal{O}_X$.

Locally ringed space condition. If each f_i is an LRS morphism, then for any $p \in X$, p lies in some U_i and the stalk map $(f^\#)_p = (f_i^\#)_p$ is a local ring homomorphism by assumption. Hence f is an LRS morphism.

By lemma 3.1.9, we can control morphisms of schemes by looking at ring homomorphisms. We can even ensure the codomain is an affine scheme:

Proposition 3.1.10: Scheme Morphisms are Locally Affine

A morphism $\pi : X \to Y$ between schemes is a morphism of schemes if and only if it is *locally affine*: every point $x \in X$ has affine open neighborhoods $U = \operatorname{Spec} R \ni x$ and $V = \operatorname{Spec} S \ni \pi(x)$ with $\pi(U) \subseteq V$, such that $\pi|_U : U \to V$ is induced by a ring homomorphism $\varphi : S \to R$.

***Proof*:**
($\Rightarrow$) Scheme morphisms are locally affine. Let $\pi : X \to Y$ be a morphism of schemes and $x \in X$. Since Y is a scheme, there exists an affine open $V = \operatorname{Spec} S$ containing $\pi(x)$. By continuity, $\pi^{-1}(V)$ is an open neighborhood of x in X; since X is a scheme, it can be covered by affine opens, so there exists an affine open $U = \operatorname{Spec} R \ni x$ with $U \subseteq \pi^{-1}(V)$, hence $\pi(U) \subseteq V$.

The restricted map $\pi|_U : \operatorname{Spec} R \to \operatorname{Spec} S$ is a morphism of locally ringed spaces between affine schemes. By theorem 3.1.4, it is uniquely induced by the ring homomorphism on global sections:

$$\varphi := \Gamma(\pi|_U) : S \cong \mathcal{O}_Y(V) \xrightarrow{\pi|_U^{\#}(V)} \mathcal{O}_X(U) \cong R$$

($\Leftarrow$) Locally affine morphisms are scheme morphisms. Suppose $\pi : X \to Y$ is an LRS morphism and every $x \in X$ admits affine neighborhoods as above. Let $\{U_i = \operatorname{Spec} R_i\}$ be an open cover of X with corresponding affine opens $\{V_i = \operatorname{Spec} S_i\}$ in Y such that $\pi(U_i) \subseteq V_i$ and $\pi|_{U_i}$ is induced by a ring homomorphism $\varphi_i : S_i \to R_i$.

Each $\pi|_{U_i}$ is an LRS morphism (by theorem 3.1.4), so the stalk map at every $p \in U_i$ is a local ring homomorphism. Since $\{U_i\}$ covers X, the stalk map at every $p \in X$ is local. Hence π is an LRS morphism between schemes, i.e., a morphism of schemes.

Example 3.1.11: Morphisms Of Schemes

1. Recall the ring map $\varphi : \mathbb{C}[y] \to \mathbb{C}[x]$ given by $y \mapsto x^2$. Geometrically, this corresponds to the map $f : \mathbb{A}^1_x \to \mathbb{A}^1_y$ sending $a \mapsto a^2$. Let us verify the sheaf map on a specific open set. Consider the section $s = 3/(y-4) \in \mathcal{O}_{\mathbb{A}^1_y}(D(y-4))$, defined on the open set $V = D(y-4) = \mathbb{A}^1 \setminus \{4\}$. The preimage of this set is:

$$f^{-1}(V) = \{a \in \mathbb{C} \mid a^2 \neq 4\} = \mathbb{A}^1 \setminus \{-2, 2\}$$

   The induced map on sections $f^{\#}_V : \mathcal{O}_{\mathbb{A}^1_y}(V) \to \mathcal{O}_{\mathbb{A}^1_x}(f^{-1}(V))$ sends s to:

$$f^{\#}_V(s) = \frac{3}{x^2 - 4}$$

   which is indeed a regular function on $f^{-1}(V) = D(x-2) \cap D(x+2)$.

2. Let $U \subseteq Y$ be an open subset of a scheme Y. There is a natural morphism of ringed spaces:

$$\iota : (U, \mathcal{O}_Y|_U) \to (Y, \mathcal{O}_Y)$$

This is the open embedding as seen in definition 2.2.48. It is important to note the behavior of the stalks. For any point $p \in U$, the stalk map $\iota_p^\# : \mathcal{O}_{Y,\iota(p)} \to \mathcal{O}_{U,p}$ is an *isomorphism* (specifically, the identity map). This distinguishes open embeddings from other types of immersions.

3. *Tautological Projection (Affine Cone):* Consider the projection from the punctured affine space to projective space:

$$\pi : \mathbb{A}_k^{n+1} \setminus \{0\} \to \mathbb{P}_k^n \qquad (x_0, \ldots, x_n) \mapsto [x_0 : \cdots : x_n]$$

This is a morphism of schemes. To see the sheaf map, we check it on the standard affine cover of $\mathbb{P}^n$. Let $U_i = D_+(x_i) \cong \mathbb{A}_k^n$ be the standard open charts of projective space. The preimage is the distinguished open set $\pi^{-1}(U_i) = D(x_i) \subseteq \mathbb{A}^{n+1} \setminus \{0\}$. The rings of sections are:

$$\mathcal{O}_{\mathbb{P}^n}(U_i) \cong k\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \qquad \text{and} \qquad \mathcal{O}_{\mathbb{A}^{n+1}\setminus\{0\}}(D(x_i)) \cong k[x_0, \ldots, x_n]_{x_i}$$

The sheaf map $\pi^\#(U_i)$ is the natural inclusion of the degree-0 subring into the full localization:

$$k\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \hookrightarrow k[x_0, \ldots, x_n]_{x_i}$$

Since these inclusions commute with further localization (restriction), they glue to form a valid sheaf morphism. Furthermore, the topological map π restricted to $D(x_i) \to U_i$ is exactly the map of spectra induced by this ring homomorphism (given by pulling back prime ideals $\mathfrak{p} \mapsto \mathfrak{p} \cap \mathcal{O}_{\mathbb{P}^n}(U_i)$), confirming this matched pair forms a valid morphism of schemes.

4. *Structure Map of Proj:* Let $S_\bullet$ be a finitely generated graded R-algebra (where $S_0 = R$). There is a natural structure morphism:

$$\pi : \operatorname{Proj} S_\bullet \to \operatorname{Spec}(R)$$

This is induced locally. On the basic open set $D_+(f) \subseteq \operatorname{Proj} S_\bullet$ (for homogeneous f), the ring of sections is $S_{(f)}$. The map $R \to S_\bullet$ induces a map $R \to S_{(f)}$ via $r \mapsto r/1$. These local maps glue to give the global morphism π.

5. Let $\varphi : S_\bullet \to R_\bullet$ be a *surjective* homomorphism of graded rings. This induces a *closed embedding* of schemes in the opposite direction:

$$f : \operatorname{Proj} R_\bullet \hookrightarrow \operatorname{Proj} S_\bullet$$

For example, the projection $k[x, y, z] \to k[x, y, z]/(x^2 + y^2 - z^2)$ induces the inclusion of the conic into $\mathbb{P}^2$.

6. Consider the projection $\pi : \operatorname{Spec}(\mathbb{C}[x, y]) \to \operatorname{Spec}(\mathbb{C}[x])$ induced by the inclusion $\mathbb{C}[x] \hookrightarrow \mathbb{C}[x, y]$.

   - *Closed Points:* A point $(x - 3, y - a)$ maps to the contraction $(x - 3)$. Geometrically, $(3, a) \mapsto 3$.
   - *Generic Points:* Consider the generic point of the parabola, $\eta = [(y - x^2)]$. The image is the contraction $(y - x^2) \cap \mathbb{C}[x]$. Since no nonzero polynomial in $\mathbb{C}[x]$ alone lies in $(y - x^2)$, the intersection is (0). Thus, the parabola maps to the generic point of the affine line!

7. Let $p \in X$ be a point of a scheme. We can form a scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$. There is a canonical morphism $\operatorname{Spec}(\mathcal{O}_{X,p}) \to X$ induced by the localization maps $\mathcal{O}_X(U) \to \mathcal{O}_{X,p}$ for open neighborhoods U of p. For $X = \mathbb{A}^1_k$ and $p = (x)$, this is the map $\operatorname{Spec}(k[x]_{(x)}) \to \operatorname{Spec}(k[x])$.

   The scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$ can be thought of as the "intersection of all open neighborhoods of p," but it is *not* an open embedding: $\operatorname{Spec}(\mathcal{O}_{X,p})$ has only the points of X that specialize to p (i.e., the prime ideals contained in $\mathfrak{m}_p$), which typically includes the generic point but not other closed points. The key property is that the stalk map at p is an isomorphism $\mathcal{O}_{X,p} \xrightarrow{\sim} \mathcal{O}_{\operatorname{Spec}(\mathcal{O}_{X,p}),\,\mathfrak{m}_p}$, since localizing a local ring at its maximal ideal has no effect.

8. *The Frobenius Morphism (Non-isomorphism bijection):* Let $X = \operatorname{Spec}(\mathbb{F}_p[t])$. Consider the ring homomorphism $\varphi : \mathbb{F}_p[t] \to \mathbb{F}_p[t]$ given by $t \mapsto t^p$. This induces a scheme morphism $F : X \to X$.

   - *Topologically:* The map on points is $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$. For a closed point $(t - a)$ where $a \in \overline{\mathbb{F}_p}$, we compute:

     $$\varphi^{-1}((t-a)) = \{f \in \mathbb{F}_p[t] \mid f(t^p) \in (t-a)\} = \{f \mid f(a^p) = 0\} = (t - a^p)$$

     Since $a^p = a$ in $\mathbb{F}_p$ (by Fermat's little theorem), we get $\varphi^{-1}((t-a)) = (t-a)$. Similarly, $\varphi^{-1}((0)) = (0)$. Thus, F is the *identity* on the underlying topological space.
   - *Sheaf-theoretically:* The map on sections is $t \mapsto t^p$. This is *not* an isomorphism: it is not surjective, as t is not in the image.

   Thus, F is a homeomorphism that is not an isomorphism of schemes. This distinction is invisible to the Zariski topology but detected by the structure sheaf.

   To give some insight on how we can extract information from this geometric perspective: once we define differentials in section 5.5, we shall see that $d(t^p) = pt^{p-1}\,dt = 0$, and hence the tangent map between tangent spaces will be the *zero map* at each point. We can think of this as "infinitesimal shrinking" at each point, while globally it is the identity!

If the codomain of a scheme morphism is an affine scheme, then we can completely determine the behaviour of the scheme morphisms by looking at the ring morphism between the global sections:

Proposition 3.1.12: Adjunction of Affine Schemes

Let $\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}(R))$ be the collection of morphism of scheme from a scheme X to an $\operatorname{Spec}(R)$. Then it is in natural bijection to the ring homomorphisms $\operatorname{Hom}_{\text{Ring}}(R, \mathcal{O}_X(X))$:

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}(R)) \cong \operatorname{Hom}_{\mathbf{CRing}}(R, \mathcal{O}_X(X))$$

In particular, The map $\theta : X \to \operatorname{Spec}(\Gamma(X, \mathcal{O}_X))$ is an injection. Furthermore, this is an isomorphism if and only if X is an affine scheme.

Note that the ring $\mathcal{O}_X(X)$ can be very complex, even in nice cases where X is a finite-type k-scheme, for example it need not be finitely generated.

***Proof*:**
If $X = \operatorname{Spec}(S)$, then theorem 3.1.4 gives the bijection. Next, map $\pi : X \to \operatorname{Spec}(R)$ certainly induces a ring homomorphism $\varphi : R \to \mathcal{O}_X(X)$, we shall show that a ring homomorphism $\varphi : R \to \mathcal{O}_X(X)$ induces a unique scheme map $\pi : X \to \operatorname{Spec}(R)$ and that this map is the inverse of the above map. By definition of a scheme:

$$\pi : \bigcup_i \operatorname{Spec}(S_i) \to \operatorname{Spec}(R)$$

where we get a restriction map $\pi_i : \operatorname{Spec}(S_i) \to \operatorname{Spec}(R)$ along with sheaf morphisms, for which the global section map is $\varphi_i : R \to \mathcal{O}_X(\operatorname{Spec}(S_i))$. As $\operatorname{Spec} S_i \subseteq X$ is an open subset, there is a natural restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X(\operatorname{Spec} S_i)$. We thus get a composition:

$$R \to \mathcal{O}_X(X) \to \mathcal{O}_X(\operatorname{Spec} S_i)$$

The map $R \to \mathcal{O}_X(\operatorname{Spec} S_i)$ is uniquely determined by the ring homomorphism $S_i \to R$, and as X is covered by the $\operatorname{Spec} S_i$, by lemma 3.1 .6 there is a unique way to glue the maps of $\mathcal{O}_X(\operatorname{Spec} S_i)$ to $\mathcal{O}_X(X)$, completing the proof.

The converse is not true:

Example 3.1.13: Maps From Affine To Projective

Take $\operatorname{Hom}_{\mathrm{Sch}}\left(\operatorname{Spec}\mathbb{Q}[x], \mathbb{P}^1_{\mathbb{Q}}\right)$ and $\operatorname{Hom}_{\mathbf{Ring}}(\mathbb{Q}, \mathbb{Q}[x])$ (recall that $\mathcal{O}_X\left(\mathbb{P}^1_{\mathbb{Q}}\right) \cong \mathbb{Q})$). Then

$$\operatorname{Hom}_{\mathrm{Ring}}(\mathbb{Q}, \mathbb{Q}[x]) = \{\mathrm{id}\}$$

namely since since the units of $\mathbb{Q}$ must map to the units of $\mathbb{Q}[x]$, which are all contains in $\mathbb{Q}$, and as it must be that $1 \mapsto 1$, we are forced to have the identity. However, there are multiple maps from $\operatorname{Spec}\mathbb{Q}[x]$ to $\mathbb{P}^1_k$; simply map $\operatorname{Spec}\mathbb{Q}[x]$ into the open subset $U_0 \cong k[y] \subseteq \mathbb{P}^1_k$ by taking $x \mapsto p(y)$ for some polynomial $p(y) \in k[y]$. More generally, we can let choose a field k[a]

The intuition is that $\mathcal{O}_X(X)$ in general contains less information than X as the global sections on a general scheme need to restrict to local sections on affine open subsets, and hence there may be in general less of them.

Show that the canonical morphism $X \to \operatorname{Spec}(\mathcal{O}_X(X))$ is an isomorphism if and only if X is a affine.

[a]Note that an [injective] ring homomorphism $k \to k$ need not be surjective, consider $F(x) \to F(x)$ given by $x \mapsto x^2$, hence the argument requires a bit more work

Using proposition 3.1.12, we can get a few more interesting results:

Example 3.1.14: Morphisms to Affine and Projective Space

1. *Maps to Affine Space are Tuples of Functions:* Let X be *any* scheme (not necessarily affine). What is a morphism $X \to \mathbb{A}^n_k$? Using the Adjunction property (proposition 3.1.12), we have:

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \mathbb{A}^n_k) \cong \operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec} k[t_1, \ldots, t_n])$$

$$\cong \operatorname{Hom}_{\mathbf{Ring}}(k[t_1, \ldots, t_n], \mathcal{O}_X(X))$$

A ring homomorphism from a polynomial ring is uniquely determined by where it sends the variables t_i. Let f_i be the image of t_i in $\Gamma(X, \mathcal{O}_X)$. Thus, a morphism to affine space is exactly equivalent to choosing n global functions $(f_1, \ldots, f_n)$ on X. This generalizes the classical fact that a map to $\mathbb{R}^n$ is given by n coordinate functions.

Notice how this also shows that there is no non-trivial map $\mathbb{P}^n_k \to \mathbb{A}^m_k$ (exercise 3.1.2(3.1.2 (4))).

s

2. *Maps to Projective Space (The need for Line Bundles):* One might hope to define a map $X \to \mathbb{P}^n_k$ similarly by choosing $n+1$ global functions $f_0, \ldots, f_n$ to serve as homogeneous coordinates $[f_0, \ldots, f_n]$. This naturally fails when using the structure sheaf since global functions on projective space are just constants, which would make such maps trivial (as illustrated in example 3.1.13). Instead, the coordinate functions x_i on $\mathbb{P}^n$ are sections of a twisted sheaf, not the structure sheaf. Consequently, to define a scheme morphism to projective space, one requires a *Line Bundle* (an invertible sheaf $\mathcal{L}$) on X and $n+1$ global sections $s_i \in \Gamma(X, \mathcal{L})$ that do not vanish simultaneously. This distinction – that maps to affine space use functions, while maps to projective space use bundles – is the primary motivation for the study of *Invertible Sheaves* in section 5.3.3.

Our last result in this section is a special case where gluing affine schemes *does* produce an affine scheme:

Corollary 3.1.15: Characterizing Affine Schemes Locally

Let X be a scheme. Then X is an affine scheme if and only if there is a finite subset of elements $f_1, \ldots, f_n \in \mathcal{O}_X(X)$ such that X_f is affine and $(f_1, \ldots, f_n) = \mathcal{O}_X(X)$

***Proof*:**
If X is affine, take $f_1 = 1$. Conversely, assume there exist $f_1, \ldots, f_n \in \mathcal{O}_X(X)$ such that each X_{f_i} is affine and $(f_1, \ldots, f_n) = \mathcal{O}_X(X)$. This ideal condition implies there exist $a_i \in \mathcal{O}_X(X)$ such that $\sum_{i=1}^n a_i f_i = 1$. Let $R = \mathcal{O}_X(X)$. We claim that $X \cong \operatorname{Spec} R$.

By proposition 3.1.12, the identity homomorphism on R induces a canonical morphism of schemes:

$$\theta : X \longrightarrow \operatorname{Spec} R$$

To show θ is an isomorphism, it suffices to verify it locally on an open cover of the target $\operatorname{Spec} R$. The condition $\sum_{i=1}^n a_i f_i = 1$ ensures that the distinguished open sets $D(f_1), \ldots, D(f_n)$ form an open cover of $\operatorname{Spec} R$.

For each i, the preimage of $D(f_i)$ under θ is precisely the locus of points in X where the global section f_i does not vanish, which is by definition X_{f_i}. Since the $D(f_i)$ cover $\operatorname{Spec} R$, their preimages X_{f_i} necessarily cover X.

Consider the restriction of θ to one of these patches:

$$\theta|_{X_{f_i}} : X_{f_i} \longrightarrow D(f_i) \cong \operatorname{Spec}(R_{f_i})$$

By our hypothesis, X_{f_i} is an affine scheme. Because the global sections of a scheme on a distinguished open set defined by a global function f_i are given by the localization of the global sections ring, we have

$$\mathcal{O}_X(X_{f_i}) \cong R_{f_i}$$

Thus, by the Adjunction property (corollary 3.1.5), the morphism $\theta|_{X_{f_i}}$ corresponds exactly to the identity ring isomorphism $R_{f_i} \xrightarrow{\sim} R_{f_i}$. Hence, each restriction $\theta|_{X_{f_i}}$ is an isomorphism of schemes.

Since θ is a morphism of schemes that restricts to an isomorphism on the sets of an open cover $\{X_{f_i}\}$ mapping to an open cover $\{D(f_i)\}$, we can apply the gluing lemma for morphisms (lemma 3.1.9). The local inverses glue uniquely to form a global inverse, proving that θ is an isomorphism and $X \cong \operatorname{Spec} R$.

3.1.2 S-Schemes

Recall that all abelian rings can be seen as $\mathbb{Z}$-algebras, and that $\mathbb{Z}$ is initial in the category of commutative rings. From this perspective, all ring homomorphism $\varphi : R \to S$ are in bijective correspondence with morphisms that make the following diagram commute:

$$\begin{tikzcd} R \arrow[rr] & & S \\ & \mathbb{Z} \arrow[ul] \arrow[ur] & \end{tikzcd}$$

Then we may generalize the notion of abelian groups to R-algebras by replacing $\mathbb{Z}$ with a ring R. We may further augment the restriction morphisms to R-algebra homomorphisms as we are working with commutative rings; this was done in definition definition 2.2.17. Now, this structure was given by a scheme morphism $X \to \operatorname{Spec} R$, and an R-scheme morphism is a scheme morphism between R-schemes $X \to Y$ where the following diagram commutes:

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & Y \arrow[dl] \\ & \operatorname{Spec}(R) & \end{tikzcd}$$

This is the commutative diagram of fiber-preserving morphism, the same requirement that is imposed of fiber-bundle morphisms. And indeed, R-schemes behave very much like bundles over $\operatorname{Spec} R$ (as the examples bellow will demonstrates), and when the morphism $X \to \operatorname{Spec} R$ is nice enough (ex. it is étale, see ref:HERE), then it shall even be locally trivial.

Thus, it is natural to consider the base space to be an arbitrary scheme:

Definition 3.1.16: S-Scheme

Let $\mathbf{Sch}_S$ be the category who objects are scheme morphisms:

$$X \to S$$

and whose morphisms are commutative diagrams:

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

As compositions, $f \circ \pi_X = \pi_Y$. The objects of this category or called *S-scheme*. The morphism of schemes $X \to S$ is called the *structure morphism*.

To see what extra structure is added, let us start the case of $S = \operatorname{Spec}(R)$. Then each open subset of X has the structure of an R-algebra, and these generalize definition 2.2.17. If $S = \operatorname{Spec}(R)$, then we would often say that the S-scheme (i.e. $\operatorname{Spec}(R)$-scheme) is an R-scheme

Proposition 3.1.17: Final Object Of Scheme

In **Sch**, the scheme $\operatorname{Spec}\mathbb{Z}$ is the final object. This makes the category of schemes isomorphic to the category of $\mathbb{Z}$-schemes.

If k is a field, $\operatorname{Spec} k$ is the final object in the category of k-schemes.

***Proof*:**
Recall that $\mathbb{Z}$ is initial in Ring, and thus it is easy to show it is initial in Comm-Ring, then as Schemes (resp. k-schemes) are the co-categories to rings (resp. k-algebras), the result is immediate.

Working over S-schemes shall restrict to much "simpler" cases than over general schemes. For example, if studying complex analysis and complex varieties, we would expect Spec($\mathbb{C}$) to have trivial automorphisms, and we may want to interpret $\operatorname{Spec}\left(\mathbb{R}[x]/\left(x^2+1\right)\right)$ without too much "gluing" done by the automorphism ring, namely we want $\mathbb{C}$ be be trivial.

Example 3.1.18: The Category of S-schemes

An *S-scheme* is a scheme X equipped with a morphism $\pi : X \to S$, called the structure morphism. A morphism of S-schemes $f : X \to Y$ is a morphism of schemes such that the triangle with the base S commutes .

1. *Schemes over k ("Geometry" Schemes)*: If $S = \operatorname{Spec}(k)$ for a field k, an S-scheme closely resembles an algebraic set (over k). By proposition 3.1.12, the structure morphism $X \to \operatorname{Spec}(k)$ is equivalent to the data of a k-algebra homomorphism $k \to \Gamma(X, \mathcal{O}_X)$. This fixes the field of scalars, ensuring that the functions on X are k-algebras, not just rings.

   When working with polynomials, we are often interested in k-linear homomorphisms $k[x] \to k[x]$ where k is fixed (e.g., coordinate changes). If we only looked at ring homomorphisms, include field automorphisms (like complex conjugation or the Frobenius). We enforce linearity by requiring the algebra diagram to commute:

$$\begin{tikzcd} k[x] \arrow[rr, "\varphi"] & & k[x] \\ & k \arrow[ul, "\iota"] \arrow[ur, "\iota"'] & \end{tikzcd}$$

In the language of schemes, this corresponds to a morphism of S-schemes (where $S = \mathrm{Spec}(k)$):

$$\begin{tikzcd} \mathrm{Spec}(k[x]) \arrow[rr, "f"] \arrow[dr, "\pi_X"'] & & \mathrm{Spec}(k[x]) \arrow[dl, "\pi_Y"] \\ & \mathrm{Spec}(k) & \end{tikzcd}$$

The condition $f \circ \pi_X = \pi_Y$ ensures that the morphism preserves the underlying field structure.

2. *Projective Schemes:* Projective schemes over a field, $\mathbb{P}^n_k = \mathrm{Proj}(k[x_0, \ldots, x_n])$, are naturally k-schemes. The structure map $\mathbb{P}^n_k \to \mathrm{Spec}(k)$ is induced by the inclusion of the degree 0 scalars $k \hookrightarrow k[x_0, \ldots, x_n]$.

3. *Field Extensions as Coverings:* Let L/K be a finite field extension. Then Spec(L) and Spec(K) are both single points topologically. However, the inclusion $K \hookrightarrow L$ induces a morphism of schemes $\pi : \mathrm{Spec}(L) \to \mathrm{Spec}(K)$. We don't want to include morphisms permuting K, hence we limit to k-scheme morphisms While topologically trivial, this morphism encodes the arithmetic complexity of the extension.

3.1.19 Foreshadowing Étale Fundamental Group Later, we shall see that Spec(L) acts as a "covering space" for Spec(K). This analogy can be made more precise K, L be perfect fields and let L/K be a finite extension. Then $L = k[\alpha]$ for a root of the appropriate irreducible polynomial $m_{\alpha,L/K}$. Then taking advantage of the fact that the category of K-field extensions is *isomorphic* (not just equivalent) to the category of rings of integers with base ring $\mathcal{O}_K$ (exercise if the reader forgot) The morphism $\mathrm{Spec}\,(L) \to \mathrm{Spec}\,(K)$ is associated to a (dense) morphism $\mathrm{Spec}\,(\mathcal{O}_L) \to \mathrm{Spec}\,(\mathcal{O}_K)$ (as $\mathcal{O}_K \to \mathcal{O}_L$ is injective). In section 3.5.1, we shall see morphism between affine schemes are uniquely determined by their value on dense sets. As $\mathcal{O}_L$ and $\mathcal{O}_K$ are both dimension 1 rings, these are *covering of curves*. This covering hints at the generalization of covering spaces from algebraic topology, leading to the *Étale Fundamental Group*, which generalizes the Galois group Gal(L/K) to schemes, allowing us to study loops and homotopy in an arithmetic setting.

Lemma 3.1.20: Stalks of k-schemes

Let X be a k-scheme and $x \in X$ so that $x = [\mathfrak{p}]$ for some prime $\mathfrak{p} \subseteq R_i = k\left[u_i\right]_i$. Then

$$\mathcal{O}_{X,\mathfrak{p}} \cong \left(k\left[u_i\right]_i\right)_{\mathfrak{p}}$$

In particular, we have that the quotient by $\mathfrak{m}_x$ is a k-algebra:

$$\kappa(x) = k(x)$$

***Proof*:**
As X is a k-scheme, each $\mathcal{O}_X(U)$ is a k-algebra, and the colimit of k-algebra's is a k-algebra (they are closed under products and equalizers, see [20]). The rest comes from proposition 1.1.5.

Note that the residue may be other fields, ex. $\mathbb{R}(x) = \mathbb{C}$ (take $\operatorname{Spec} \mathbb{R}[t]$ where and choose a degree > 1 point).

One of the most important first impositions for many algebraic manipulations are finiteness conditions:

Definition 3.1.21: Locally Finite Type And Finite Type

Let X, Y be S-schemes. Then a morphism $f : X \to Y$ is locally of finite type if there exists an affine open cover $V_j = \operatorname{Spec}\left(S_j\right)$ of Y where $f^{-1}\left(V_i\right)$ can be covered by affine open subsets $U_{ij} = \operatorname{Spec}\left(A_{ij}\right)$ where A_{ij} is a finitely generated S_i-algebra, that is

$$f^{\#}\left(V_i\right) : S_j = \mathcal{O}_Y\left(V_j\right) \to \mathcal{O}_X\left(f^{-1}\left(V_i\right)\right) \xrightarrow{\text{res}} \mathcal{O}_Y\left(U_{ij}\right) = R_{ij}$$

induces a finitely generated S_j-algebra on each R_{ij}. If furthermore each $f^{-1}\left(V_j\right)$ can be covered by finite number of affine subsets, f is said to be of finite type.

An S-scheme X is said to be of locally finite type (resp. finite type) over S if $f : X \to S$ is of locally finite type (resp. finite type).

3.1.3 Subschemes: Closed, Locally Closed, and More

Open subschemes (section 2.2.8) are the simplest type of subscheme: given an open subset $U \subseteq X$, the restriction $(U, \mathcal{O}_X|_U)$ carries a unique scheme structure compatible with X. The situation for closed subsets is fundamentally different: a closed subset can carry *multiple* scheme structures, distinguished by the amount of "infinitesimal data" they retain. Beyond the open and closed cases, there are subschemes that are locally closed (closed inside an open subset), and there are natural "local" subobjects that do not correspond to any topological subset at all.

The reader may remark that sub-objects usually correspond to monomorphisms. While the definition is formally definable the resulting schemes for **Sch**

Closed subschemes and closed immersions

Given a surjection $R \twoheadrightarrow R/I$, by proposition 2.1.26 the image $\operatorname{Spec}(R/I) \subseteq \operatorname{Spec}(R)$ is a closed subset, and the inclusion is a morphism of locally ringed spaces. For a general scheme, this condition generalizes to requiring that the sheaf map be surjective.

The multiplicity issue deserves emphasis. Consider the closed point $\{0\} \subseteq \mathbb{A}^1_k$. As a topological subspace, $\{0\}$ is just a point. But there are infinitely many scheme structures on this point compatible with the ambient scheme: $\operatorname{Spec}(k[x]/(x))$, $\operatorname{Spec}(k[x]/(x^2))$, $\operatorname{Spec}(k[x]/(x^3))$, and so on. These differ in how much nilpotent (infinitesimal) structure they retain. The underlying reason is elementary: $0^n = 0$ for all n, so the vanishing locus of x and x^n are identical as subsets, but $k[x]/(x^n)$ remembers the first $n-1$ derivatives. In contrast, open subschemes have a *unique* compatible scheme structure: the restriction of $\mathcal{O}_X$ to the open set.

Definition 3.1.22: Closed Immersion and Closed Subscheme

Let $f : Y \to X$ be a morphism of schemes. Then f is a *closed immersion* if:

1. f induces a homeomorphism from $|Y|$ onto a closed subset of $|X|$, and
2. the sheaf map $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective (equivalently, surjective on stalks).

A *closed subscheme* of X is a closed subset $Z \subseteq |X|$ together with a scheme structure on Z such that the inclusion $\iota : Z \hookrightarrow X$ is a closed immersion.

The distinction between a closed immersion and a closed subscheme is that a closed immersion $f : Y \to X$ is a morphism (which might come with an isomorphism $Y \cong Z$ for some closed subscheme Z), while a closed subscheme is always given with the literal inclusion as the map. In practice, the two are used interchangeably.

When define closed subsets of affine schemes, we used $V(-)$. This construction can also be used to construct closed subschemes of an affine scheme, but it shall forget the sheaf-theorem information (ex. nilpotence). This shall be recovered in definition 5.1.36 by inputting *ideal shaves* in $V(-)$.

Note that the complement of a closed subscheme is a unique open subscheme (with its canonical structure), while the complement of an open subscheme admits *multiple* closed subscheme structures—usually infinitely many, corresponding to different choices of nilpotent thickening. This asymmetry between open and closed is a recurring theme. There is a natural choice of closd subscheme if nilpotent information is removed; this is a *reduced subscheme*, see exercise ref:HERE.

Example 3.1.23: Closed Subschemes

1. *A point with its reduced structure.* The inclusion $\{0\} \hookrightarrow \mathbb{A}^1_k$ corresponds to the surjection $k[x] \twoheadrightarrow k[x]/(x) \cong k$. The image is closed, the sheaf map is surjective (and remains so on all localizations), so this is a closed immersion.
2. *A fat point.* The inclusion $\operatorname{Spec}(k[x]/(x^2)) \hookrightarrow \mathbb{A}^1_k$ corresponds to $k[x] \twoheadrightarrow k[x]/(x^2)$. The image is the same closed point $\{0\}$, but the scheme structure is different: the coordinate ring $k[x]/(x^2) \cong k[\epsilon]/(\epsilon^2)$ remembers one infinitesimal direction. More generally, $\operatorname{Spec}(k[x]/(x^n)) \hookrightarrow \mathbb{A}^1_k$ is a closed immersion for every $n \geq 1$, giving countably many closed subscheme structures on the single point $\{0\}$.

   This example also shows that the structure sheaf of a closed subscheme is *not* a subsheaf of

$\mathcal{O}_X$: the quotient $k[x]/(x^2)$ is not a subring of $k[x]$.

3. *The general affine case.* For any ideal $I \subseteq R$, the surjection $R \twoheadrightarrow R/I$ induces a closed immersion $\mathrm{Spec}\,(R/I) \hookrightarrow \mathrm{Spec}\,(R)$ with image $V(I)$.

4. *Projective subspaces.* The inclusion $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^2_k$ given by $[s:t] \mapsto [s:t:0]$ is a closed immersion. On the standard affine chart $U_0 = \{x_0 \neq 0\}$, it corresponds to the surjection $k[y_1, y_2] \twoheadrightarrow k[y_1, y_2]/(y_2) \cong k[y_1]$.

It is worth noting that neither of the two conditions in the definition is redundant. The surjectivity of the sheaf map is not enough. The open immersion $D(x) \cup D(y) \hookrightarrow \mathbb{A}^2_k$ has a surjective sheaf map (in fact an isomorphism on each open), but the image is open, not closed. More exotic examples arise from non-separated schemes.

Similarly, topological closedness is not enough. The normalization of the cuspidal cubic provides an example. Let $Y = \mathrm{Spec}\,(k[x,y]/(y^2 - x^3))$ and consider the map $\nu : \mathbb{A}^1_k \to Y$ given by $t \mapsto (t^2, t^3)$, corresponding to the ring map $k[x,y]/(y^2 - x^3) \to k[t]$ with $x \mapsto t^2$, $y \mapsto t^3$. This map is a homeomorphism onto its image (which is all of $|Y|$) and $|Y|$ is closed in itself, but the ring map is *not* surjective: the element $t \in k[t]$ is not in the image (even though $y/x = t^3/t^2 = t$ is a rational function on Y, it is not regular). The failure of surjectivity reflects the singularity of Y at the origin—or algebraically, the failure of $k[t^2, t^3]$ to be integrally closed in $k[t]$.

Proposition 3.1.24: Properties of Closed Immersions

1. A morphism $f : Y \to X$ is a closed immersion if and only if for every affine open $\mathrm{Spec}\,(S) \subseteq X$, the preimage $f^{-1}(\mathrm{Spec}\,(S)) = \mathrm{Spec}\,(R)$ is affine and the induced map $S \to R$ is surjective.

2. Every surjective ring homomorphism $S \twoheadrightarrow R$ induces a closed immersion $\mathrm{Spec}\,(R) \hookrightarrow \mathrm{Spec}\,(S)$.

3. The composition of two closed immersions is a closed immersion.

***Proof*:**

1. One direction is immediate: if f is a closed immersion, then on each affine open $\mathrm{Spec}\,(S)$ the preimage is the closed subscheme $\mathrm{Spec}\,(S/I)$ for the appropriate ideal I, and $S \to S/I$ is surjective. The converse uses that surjectivity of a sheaf map can be checked on an affine cover of the codomain: if $f^\#|_{\mathrm{Spec}\,(S)}$ is surjective for each member of an affine cover, then $f^\#$ is surjective. A proof using ideal sheaves is given in section 5.1.4.

2. The surjection $S \twoheadrightarrow R = S/I$ induces a homeomorphism $\mathrm{Spec}\,(R) \cong V(I) \subseteq \mathrm{Spec}\,(S)$ onto a closed subset, and the sheaf map $\mathcal{O}_{\mathrm{Spec}\,(S)} \to f_*\mathcal{O}_{\mathrm{Spec}\,(R)}$ is surjective on each distinguished open (it corresponds to $S_g \twoheadrightarrow (S/I)_{\overline{g}}$, which is surjective).

3. Let $Z \xrightarrow{g} Y \xrightarrow{f} X$ be closed immersions. The composition $f \circ g$ is a homeomorphism onto a closed subset (a closed subset of a closed subset is closed). On each affine open $\mathrm{Spec}\,(S) \subseteq X$, by part (1), $f^{-1}(\mathrm{Spec}\,(S)) = \mathrm{Spec}\,(S/I)$ and $g^{-1}(\mathrm{Spec}\,(S/I)) = \mathrm{Spec}\,(S/J)$ for ideals $I \subseteq J \subseteq S$. The composite corresponds to $S \twoheadrightarrow S/J$, which is surjective.

As closed subschemes are determined by ideals on each open subset, it is natural to ask whether they can be described globally by an "ideal subsheaf" of $\mathcal{O}_X$. This is indeed the case, and the theory is developed in section 5.1.4 once we have the language of quasi-coherent sheaves.

Locally closed subschemes: immersions

If $X = \operatorname{Spec} R$ is an affine scheme and $U \subseteq X$ is an open subset, for any $S \subseteq \mathcal{O}_X(U)$, $V(S)$ is usually *not* a closed subset but a *locally closed subset*. These are the natural local solutions to regular functions, and thus deserve some attention:

Definition 3.1.25: Locally Closed Immersion

A morphism $f : Y \to X$ is a *locally closed immersion* (or simply an *immersion*) if f factors as

$$Y \xrightarrow{i} U \xhookrightarrow{\text{open}} X$$

where $U \subseteq X$ is an open subscheme and $i : Y \hookrightarrow U$ is a closed immersion. Equivalently, f induces a homeomorphism from $|Y|$ onto a locally closed subset of $|X|$ (i.e., the intersection $C \cap U$ of a closed set C and an open set U), with $f^\#$ surjective onto the structure sheaf of Y. A *locally closed subscheme* of X is a locally closed subset with the scheme structure making the inclusion an immersion.

Proposition 3.1.26: Open and Closed Immersions are Immersions

Every open immersion and every closed immersion is a locally closed immersion.

***Proof*:**

An open immersion $U \hookrightarrow X$ factors as $U \xrightarrow{\text{id}} U \hookrightarrow X$, where id is (trivially) a closed immersion. A closed immersion $Y \hookrightarrow X$ factors as $Y \hookrightarrow X \xrightarrow{\text{id}} X$, taking $U = X$.

Closedness of a subset $C \subseteq X$ can be verified on an open cover: C is closed if and only if $C \cap U_i$ is closed in U_i for every member of an open cover $\{U_i\}$ of X. Local closedness is a strictly weaker condition: C must be the intersection of a closed set with a *single* open set, not locally closed in each chart.

Locally closed subsets appear naturally in at least two important contexts. First, images of morphisms of varieties are typically locally closed (or finite unions thereof); this is the content of Chevalley's theorem (**??**). Second, the diagonal morphism $\Delta : X \to X \times_S X$ is always a locally closed immersion, and the condition that Δ be a *closed* immersion is precisely the definition of separatedness (section 3.5.1).

Local schemes: beyond topological subsets

The open, closed, and locally closed subschemes above all correspond to topological subsets of X equipped with a scheme structure. There is a fourth type of "subobject" that arises naturally but does *not* correspond to any topological subset: the *local scheme* at a point.

Let X be a scheme and $x \in X$ a point. The stalk $\mathcal{O}_{X,x}$ is a local ring, and we can form the scheme $\operatorname{Spec}(\mathcal{O}_{X,x})$. There is a canonical morphism

$$\iota_x : \operatorname{Spec}(\mathcal{O}_{X,x}) \longrightarrow X$$

defined as follows. Choose any affine open $\operatorname{Spec}(R) \subseteq X$ containing x, with x corresponding to a prime $\mathfrak{p} \subseteq R$. The localization map $R \to R_{\mathfrak{p}} = \mathcal{O}_{X,x}$ induces $\operatorname{Spec}(\mathcal{O}_{X,x}) \to \operatorname{Spec}(R) \hookrightarrow X$. This morphism is independent of the choice of affine open (the stalk is a colimit over all open neighborhoods, so any two choices yield the same map).

In what sense is $\operatorname{Spec}(\mathcal{O}_{X,x})$ a "subobject" of X? The morphism ι_x is *not* an immersion in any topological sense:

Proposition 3.1.27: The Local Scheme is Not a Topological Subscheme

Let X be a scheme of finite type over a field k and let $x \in X$ be a non-isolated point. Then $\iota_x : \operatorname{Spec}(\mathcal{O}_{X,x}) \to X$ is not an open immersion, not a closed immersion, and not a locally closed immersion. In particular, the image of ι_x is not a locally closed subset of $|X|$.

***Proof*:**
The image of ι_x consists of those primes $\mathfrak{q} \subseteq R$ with $\mathfrak{q} \subseteq \mathfrak{p}$ (these are exactly the primes that survive the localization $R \to R_{\mathfrak{p}}$). This is the set of *generizations* of x: the points $y \in X$ with $x \in \overline{\{y\}}$. In particular, the image contains the generic point of every irreducible component passing through x.

This set is neither open nor closed in general. It is not closed because it contains the generic point η of X (if X is irreducible), and $\overline{\{\eta\}} = X \neq \operatorname{im}(\iota_x)$. It is not open because removing the image from X does not yield a closed set (the closure of $\{x\}$ is not contained in $X \setminus \operatorname{im}(\iota_x)$). It is not locally closed because a locally closed set in a Noetherian space is a finite union of locally closed irreducible sets, and the set of generizations of x does not have this form when x has positive codimension.

Example 3.1.28: Local Schemes

1. *The generic point of* $\mathbb{A}^1_k$. Let $X = \mathbb{A}^1_k = \operatorname{Spec}(k[x])$ and let $\eta = [(0)]$ be the generic point. Then $\mathcal{O}_{X,\eta} = k(x)$, the function field, and $\operatorname{Spec}(\mathcal{O}_{X,\eta}) = \operatorname{Spec}(k(x))$ is a single point. The morphism $\iota_\eta : \operatorname{Spec}(k(x)) \to \mathbb{A}^1_k$ sends the unique point to η. This is the inclusion of the generic point, which is neither open nor closed in $\mathbb{A}^1_k$ (its closure is all of $\mathbb{A}^1_k$, but it is not itself closed).

   Actually, for $\mathbb{A}^1_k$ the generic point *is* a locally closed subset (it is the intersection of the open set X with the closure $\overline{\{\eta\}} = X$, so it is a dense locally closed subset). But the map ι_η is still not a locally closed immersion, because $\operatorname{Spec}(k(x))$ and $\{\eta\}$ (with its induced scheme structure as a subspace of $\mathbb{A}^1_k$) are not isomorphic as locally ringed spaces: the stalk of $\mathcal{O}_{\mathbb{A}^1}$ at η is $k(x)$, but the induced structure sheaf on $\{\eta\}$ as a subspace is $\mathcal{O}_{\mathbb{A}^1}(\mathbb{A}^1) = k[x]$, not $k(x)$.

2. *A closed point of* $\mathbb{A}^2_k$. Let $X = \mathbb{A}^2_k$ and $x = [(x,y)]$, the origin. Then $\mathcal{O}_{X,x} = k[x,y]_{(x,y)}$, and $\operatorname{Spec}(\mathcal{O}_{X,x})$ has two points: the closed point (the maximal ideal (x,y)) and the generic point (the zero ideal). The image of ι_x is $\{[(x,y)], [(0)]\}$—the origin together with the generic

point of $\mathbb{A}^2$. This is not open, not closed, and not locally closed in $\mathbb{A}^2_k$.

3. *A point on a curve in* $\mathbb{A}^2_k$. Let $C = V(y - x^2) \subseteq \mathbb{A}^2_k$ and let $x = [(x, y)]$ be the origin of C. Then $\mathcal{O}_{C,x} = (k[x, y]/(y - x^2))_{(x,y)} \cong k[x]_{(x)}$, which has two primes. The image of ι_x consists of the origin and the generic point of C—but not the generic point of $\mathbb{A}^2$ (which does not lie on C).

Despite not being a topological subscheme, the local scheme $\operatorname{Spec}(\mathcal{O}_{X,x})$ is arguably the most important "subobject" associated to a point. It captures the local algebra of X at x—the ring $\mathcal{O}_{X,x}$ encodes the germs of regular functions near x—and many properties of X at x (regularity, normality, Cohen–Macaulay-ness) are properties of this single local ring. The morphism ι_x is flat (since localization is an exact functor), which makes it well-behaved from a homological perspective even though it is topologically ill-behaved.

We can place this in context with the other types of subobjects:

Type	*Topological condition*	*Sheaf condition*	*Uniqueness*
Open subscheme	image is open	$f^\#$ is an isomorphism	unique
Closed subscheme	image is closed	$f^\#$ is surjective	not unique
Locally closed	image is loc. closed	$f^\#$ surjective onto image	not unique
Local scheme at x	image = generizations of x	$f^\#$ is localization	unique

The first three are topological subobjects: the image is a subset with a specific topological property, and the scheme structure is constrained by the sheaf map. The fourth is algebraic rather than topological: it is determined by the localization at a prime, and the image has no clean topological description. Yet it is the one that controls the local geometry.

3.1.29 Immersions as finite-type monomorphisms There is a clean characterization that explains the role of locally closed immersions. Among monomorphisms of schemes, the immersions are (essentially) the ones that are *locally of finite type*: they impose only finitely many algebraic conditions locally. All three of our topological types—open immersions, closed immersions, and locally closed immersions—are locally of finite type. The local scheme map $\operatorname{Spec}(\mathcal{O}_{X,x}) \to X$ is a monomorphism but is *not* locally of finite type (the local ring $\mathcal{O}_{X,x}$ is generally not a finitely generated algebra over the coordinate ring of any affine chart).
This gives a clean heuristic: schemes are locally affine, and the sub-objects of an affine scheme $\operatorname{Spec}(R)$ that are "of finite type" are exactly the closed subschemes $\operatorname{Spec}(R/I)$. The locally closed immersions are what you get by globalizing this: locally on X, a finite-type sub-object is a closed subscheme of an affine chart, which is a locally closed subscheme of X.

Formal local schemes: the analytic germ

One can zoom in even further than the local scheme by taking the completion. Let $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ be the maximal ideal. We can form the $\mathfrak{m}_x$-adic completion of the local ring, $\widehat{\mathcal{O}}_{X,x}$, and consider the affine scheme $\operatorname{Spec}(\widehat{\mathcal{O}}_{X,x})$.

The natural map $\mathcal{O}_{X,x} \to \widehat{\mathcal{O}}_{X,x}$ induces a canonical morphism

$$\widehat{\iota}_x : \operatorname{Spec}(\widehat{\mathcal{O}}_{X,x}) \longrightarrow \operatorname{Spec}(\mathcal{O}_{X,x}) \longrightarrow X.$$

For Noetherian schemes, the completion map $\mathcal{O}_{X,x} \to \widehat{\mathcal{O}}_{X,x}$ is faithfully flat. Consequently, the induced map on spectra is surjective, meaning the image of $\widehat{\iota}_x$ in X is exactly the same as the image of the local scheme: the set of generizations of x.

If the image is the same, how is this a "smaller" neighborhood? While the local scheme captures arbitrarily small *Zariski* neighborhoods, the completion captures arbitrarily small *formal* (or analytic) neighborhoods. The Zariski topology is famously coarse, and the completion detects geometric features that the local scheme cannot.

Example 3.1.30: Formal Branches of a Node

Let $X = \operatorname{Spec}(k[x,y]/(y^2 - x^2(x+1)))$ be a nodal cubic curve over an algebraically closed field of characteristic $\neq 2$, and let $p = [(x,y)]$ be the node at the origin.

The local ring $\mathcal{O}_{X,p} = (k[x,y]/(y^2 - x^2(x+1)))_{(x,y)}$ is an integral domain; in the Zariski topology, any open neighborhood of the origin is irreducible, so the local scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$ is an integral scheme.

However, analytically, the curve consists of two branches crossing at the origin. If we pass to the completion $\widehat{\mathcal{O}}_{X,p} \cong k[\![x,y]\!]/(y^2 - x^2(x+1))$, the element $x+1$ has a square root (by Hensel's Lemma, or the binomial series). Thus, the defining equation factors:

$$y^2 - x^2(x+1) = (y - x\sqrt{x+1})(y + x\sqrt{x+1}).$$

The complete local ring is not a domain! The formal local scheme $\operatorname{Spec}(\widehat{\mathcal{O}}_{X,p})$ has two irreducible components, perfectly capturing the two analytic branches of the curve passing through the node.

Geometrically, passing to the formal local scheme feels like taking a smaller, more precise subspace. Categorically, however, this marks a hard boundary: *the morphism $\widehat{\iota}_x$ is not a monomorphism.* For a map of affine schemes to be a monomorphism, the underlying ring map must be an epimorphism (meaning the multiplication map $S \otimes_R S \to S$ is an isomorphism). However, there is a fundamental algebraic fact: *any faithfully flat epimorphism of rings is an isomorphism.* Because the completion of a Noetherian local ring is faithfully flat, if it were an epimorphism, the local ring would already have to be complete!

Because $\widehat{\iota}_x$ is not a monomorphism, the formal local scheme does not belong in our hierarchy of subobjects (immersions and local schemes). It is not "injective on points" functorially, and its diagonal map is very far from being an isomorphism. Instead of strictly restricting the space of functions, completion conceptually "thickens" the local ring by adding infinite limits (Taylor series). This breaks categorical injectivity, but it is exactly the mechanism required to reveal finer analytic structure. Geometrically, we can think of the fact that the two branches of a node our now two separate points as the analytic scheme "adding" more points, and thus failing the required injectivity.

In modern algebraic geometry, one often prefers to treat the completion as a *formal scheme*, denoted $\operatorname{Spf}(\widehat{\mathcal{O}}_{X,x})$. As a formal scheme, its underlying topological space is just the single point $\{x\}$, but its sheaf of topological rings remembers the "infinitesimal thickenings" of the point. See section 5.6 for more details.

Quick Aside: Dominant morphisms

Closed immersions of affine schemes correspond to surjective ring maps $R \twoheadrightarrow R/I$. The opposite extreme—injective ring maps—does not yield a surjective scheme morphism, but it yields the next best thing.

Definition 3.1.31: Dominant Morphism

A scheme morphism $f : X \to Y$ is *dominant* if the image of f is dense in Y, i.e., $\overline{f(X)} = Y$.

Proposition 3.1.32: Injective Ring Maps and Dominant Morphisms

Let $\varphi : R \hookrightarrow S$ be an injective ring homomorphism. Then the corresponding scheme morphism $f : \mathrm{Spec}\,(S) \to \mathrm{Spec}\,(R)$ is dominant.

***Proof*:**
The image $f(\mathrm{Spec}\,(S))$ is the set $\{\varphi^{-1}(\mathfrak{q}) : \mathfrak{q} \in \mathrm{Spec}\,(S)\}$. Its closure in $\mathrm{Spec}\,(R)$ is $V(I)$ where $I = \bigcap_{\mathfrak{q}\in\mathrm{Spec}\,(S)} \varphi^{-1}(\mathfrak{q}) = \varphi^{-1}\left(\bigcap_{\mathfrak{q}\in\mathrm{Spec}\,(S)} \mathfrak{q}\right) = \varphi^{-1}(\mathrm{Nil}(S))$, the preimage of the nilradical of S. Since φ is injective, $\varphi^{-1}(\mathrm{Nil}(S)) \subseteq \mathrm{Nil}(R)$. Thus $V(I) \supseteq V(\mathrm{Nil}(R)) = \mathrm{Spec}\,(R)$, giving $\overline{f(\mathrm{Spec}\,(S))} = \mathrm{Spec}\,(R)$.

Note that such a map is *not* necessarily injective, and in fact often fails to be injective (ex. on curves it fails at the non-branching points). Thus, we do not consider such maps subschemes; dominant maps are recorded here as it is likely the reader would be curious to know how to think of injective ring homomorphisms at this early stage. A fuller treatment of dominant morphisms and rational maps is given in **??**. In particular, we will see that for morphisms of finite type, dominance can be characterized by the density of the image, and Chevalley's theorem (theorem 5.2.7) gives a precise description of the image (it is a constructible set).

Exercise 3.1.1

1. Let X be a scheme and $Z \subseteq |X|$ a closed subset. Define an ideal sheaf $\mathcal{I}$ on X by

$$\mathcal{I}(U) = \{f \in \mathcal{O}_X(U) \mid f(x) = 0 \text{ for all } x \in Z \cap U\}$$

where $f(x) = 0$ means the image of f in $\kappa(x)$ vanishes. Suppose $\mathrm{Spec}\,(R) \subseteq X$ is an affine open subset such that $Z \cap \mathrm{Spec}\,(R) = V(I)$ for some ideal $I \subseteq R$. Show that the ideal of R corresponding to $\mathcal{I}(\mathrm{Spec}\,(R))$ is $\sqrt{I}$.

2. Use the previous result to conclude that the closed subscheme Z_{red} defined by $\mathcal{I}$ is a scheme whose underlying topological space is exactly Z, and whose structure sheaf has no nilpotents. Argue why Z_{red} is the unique closed subscheme structure on Z with this property.

3. Consider the closed subset $Z = \{0\}$ in the affine line $\mathbb{A}^1_k = \mathrm{Spec}\,(k[x])$. Describe the reduced subscheme Z_{red} and compare it to the closed subschemes defined by the ideals (x^n) for $n \geq 2$.

4. The *reduction* of a scheme X, denoted X_{red}, is the reduced closed subscheme supported on the entire underlying space $|X|$. Show that the inclusion $X_{\mathrm{red}} \hookrightarrow X$ satisfies the following

universal property: for any reduced scheme Y and any morphism $f : Y \to X$, f factors uniquely through X_{red}.

3.1.4 More on Projective Schemes

In section 2.3, we defined the prototypical projective scheme $\mathbb{P}^n_R = \operatorname{Proj} R[x_0, \ldots, x_n]$. To do geometry, we need to understand its closed subschemes.

As we will prove rigorously in the next chapter using ideal sheaves (specifically corollary 5.3.15), closed subschemes of a projective scheme are governed by homogeneous ideals. For now, we take this as our primary geometric construction: given a graded ring $S_\bullet$ and a homogeneous ideal $I \subseteq S_\bullet$, the canonical surjective graded ring homomorphism $S_\bullet \twoheadrightarrow S_\bullet/I$ induces a closed immersion:

$$\operatorname{Proj}(S_\bullet/I) \hookrightarrow \operatorname{Proj} S_\bullet$$

Conceptually, this is the exact generalization of the affine case. If $S_\bullet$ is a finitely generated graded R-algebra generated in degree 1, we can choose $n+1$ degree 1 generators to form a surjective graded map $R[x_0, \ldots, x_n] \twoheadrightarrow S_\bullet$. This implies $S_\bullet \cong R[x_0, \ldots, x_n]/I$, meaning $\operatorname{Proj} S_\bullet$ naturally embeds as a closed subscheme of $\mathbb{P}^n_R$. Choosing these generators is geometrically equivalent to choosing projective coordinates.

We can classify the simplest closed subschemes of $\mathbb{P}^n_k$ by the generators of their ideals:

Definition 3.1.33: Projective Hypersurface and Linear Spaces

A *hypersurface* of $\mathbb{P}^n_k$ is a closed subscheme $\operatorname{Proj} k[x_0, \ldots, x_n]/(f)$ given by a single homogeneous polynomial f. The *degree* of the hypersurface is the degree of f.

A hypersurface of degree 1 is called a *hyperplane*. A hypersurface of degree $2, 3, 4, 5 \ldots$ is called a *quadric*, *cubic*, *quartic*, *quintic*, and so forth.

If $n = 2$ (so we are in $\mathbb{P}^2_k$), a hypersurface is called a *curve*. A hypersurface of degree $1, 2, 3$ is called a *line*, *conic curve* (or *conic*), and *cubic curve* respectively.

A *linear space* is a closed subscheme cut out by a collection of degree 1 homogeneous polynomials (an intersection of hyperplanes). Any injective linear transformation $V \to W$ of vector spaces induces a closed immersion $\mathbb{P}(V) \hookrightarrow \mathbb{P}(W)$ whose image is a linear space.

Recall that $\mathbb{P}^n_k$ has only constant global functions, so these subschemes are cut out by regular functions defined only on open affine patches. Let us look at some classic examples.

Example 3.1.34: Closed Projective Subschemes

1. *Twisted Cubic*: Take the homogeneous ideal $I = (xy - zw, x^2 - wy, y^2 - xz)$ and look at the closed subscheme $\operatorname{Proj} k[w, x, y, z]/I \subseteq \mathbb{P}^3_k$. This shape is called the *twisted cubic*. It is a *curve*, and it is isomorphic to $\mathbb{P}^1_k$ via the map induced by the graded ring homomorphism:

$$k[w, x, y, z] \to k[s, t] \qquad (w, x, y, z) \mapsto (s^3, s^2t, st^2, t^3)$$

To see this is an isomorphism of schemes, one can take the standard affine open sets and show it gives closed immersions on the charts. For example, on the chart $D_+(w)$ where

$w \neq 0$ and $s \neq 0$, the map of degree 0 localizations is:

$$k\left[\frac{x}{w}, \frac{y}{w}, \frac{z}{w}\right] \to k\left[\frac{t}{s}\right] \qquad \frac{x}{w} \mapsto \frac{t}{s}, \ \frac{y}{w} \mapsto \frac{t^2}{s^2}, \ \frac{z}{w} \mapsto \frac{t^3}{s^3}$$

2. *Lines on a Quadric Surface*: The following shows a way to parameterize certain projective varieties. Take the quadric surface $V_+(wz - xy) \subseteq \mathbb{P}^3_k$ (with projective coordinates w, x, y, z). For any two elements $\alpha, \beta \in k$ that are not both zero, notice that as $[c, d] \in \mathbb{P}^1_k$ varies, $[\alpha c, \beta c, \alpha d, \beta d]$ forms a line completely contained on the surface. Show there is one other parameterization of this surface via lines, and show it must be the case that these are the only two families (hint: one parameterization contains $V_+(w, x)$ and the other $V_+(w, y)$).

Another important construction arises from taking a different grading of the polynomial ring, denoted $S_{d\bullet}$. This allows us to "linearize" high-degree equations. First, we must establish that altering the grading in this specific way does not change the underlying abstract scheme.

Theorem 3.1.35: Isomorphism of Veronese Subrings

Let $S_\bullet = R[x_0, \ldots, x_n]$ be the standard graded polynomial ring. For any integer $d > 0$, let $S_{d\bullet} = \bigoplus_{m=0}^{\infty} S_{md}$ be the subring consisting of degrees that are multiples of d. Then there is a canonical isomorphism of schemes:

$$\operatorname{Proj} S_{d\bullet} \cong \operatorname{Proj} S_\bullet$$

***Proof*:**
We construct this isomorphism by comparing the standard affine open covers. By lemma 2.3.2, the scheme $\operatorname{Proj} S_\bullet$ is covered by the distinguished open sets $D_+(x_i) \cong \operatorname{Spec}((S_{x_i})_0)$ for $i = 0, \ldots, n$.

For the subring $S_{d\bullet}$, the elements x_i^d have degree d and therefore reside in $S_{d\bullet}$. The corresponding distinguished open sets $D_+(x_i^d)$ cover $\operatorname{Proj} S_{d\bullet}$, and applying lemma 2.3.2 again, each is isomorphic to $\operatorname{Spec}(((S_{d\bullet})_{x_i^d})_0)$.

We claim the rings $(S_{x_i})_0$ and $((S_{d\bullet})_{x_i^d})_0$ are identically the same. An arbitrary element in $(S_{x_i})_0$ is of the form a/x_i^k, where $a \in S_k$ is a homogeneous polynomial of degree k. We can rewrite this fraction by multiplying the numerator and denominator by x_i^{kd-k}:

$$\frac{a}{x_i^k} = \frac{a x_i^{kd-k}}{x_i^{kd}} = \frac{a x_i^{kd-k}}{(x_i^d)^k}$$

The new numerator $a x_i^{kd-k}$ has degree $k + (kd - k) = kd$, meaning it is an element of S_{kd}, which is a valid component of $S_{d\bullet}$. The denominator is a power of x_i^d. Therefore, this fraction is naturally an element of $((S_{d\bullet})_{x_i^d})_0$. Since this containment goes both ways, the rings are equal. These local isomorphisms canonically glue together to yield the global isomorphism $\operatorname{Proj} S_{d\bullet} \cong \operatorname{Proj} S_\bullet$.

Definition 3.1.36: Veronese Embedding

Let $S_\bullet = k[x_0, \dots, x_n]$ and let $S_{d\bullet}$ be the d-tuple Veronese subring. By theorem 3.1.35, $\operatorname{Proj} S_{d\bullet} \cong \mathbb{P}^n_k$. We can view $S_{d\bullet}$ as being generated in "degree 1" by the monomials of degree d from the original ring. There are $N+1 = \binom{n+d}{d}$ such monomials. Mapping a new polynomial ring $k[y_0, \dots, y_N] \twoheadrightarrow S_{d\bullet}$ by sending the variables y_i to these monomials induces a closed immersion:

$$\mathbb{P}^n_k \cong \operatorname{Proj} S_{d\bullet} \hookrightarrow \mathbb{P}^N_k$$

This map is called the *d-tuple Veronese Embedding.*

The geometric utility of the Veronese embedding is that a hypersurface of degree d in $\mathbb{P}^n_k$ gets mapped exactly to a hyperplane intersection (degree 1) in $\mathbb{P}^N_k$.

3.1.37 Foreshadowing: Sheaves and Morphisms In the next chapter, we will show that morphisms to projective space are governed by invertible sheaves. The Veronese embedding is precisely the morphism $X \to \mathbb{P}^N_k$ generated by the global sections of the twisting sheaf $\mathcal{O}_X(d)$.

As noted earlier, choosing a set of degree 1 generators is equivalent to choosing an embedding into a specific ambient projective space $\mathbb{P}^N_R$. Therefore, it is necessary to distinguish the intrinsic geometry of the abstract scheme from the extrinsic geometry of its embedding.

Consider the scheme $X = \mathbb{P}^1_k$. We can embed it in several ways:

1. *Degree 1*: Use $S_\bullet = k[x, y]$. It is generated in degree 1 by 2 elements. This gives the identity embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^1_k$ as a straight line.
2. *Degree 2*: Consider the 2-tuple Veronese subring $S_{2\bullet} = k[x^2, xy, y^2]$. As an abstract scheme, $\operatorname{Proj} S_{2\bullet} \cong \mathbb{P}^1_k$. To write $S_{2\bullet}$ as a ring generated in degree 1, we require 3 generators: let $u = x^2$, $v = xy$, $w = y^2$. They satisfy $uw = v^2$, yielding an embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^2_k$. The image is a conic curve.
3. *Degree 3*: The 3-tuple Veronese subring $S_{3\bullet} = k[x^3, x^2y, xy^2, y^3]$ requires 4 generators, yielding an embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^3_k$. The image is the twisted cubic.

The line in $\mathbb{P}^1_k$, the conic in $\mathbb{P}^2_k$, and the twisted cubic in $\mathbb{P}^3_k$ are isomorphic as abstract schemes. Extrinsically, however, they are embedded differently, have different geometric degrees, and are cut out by different homogeneous ideals.

One might ask why we do not always restrict to degree 1 generators. Certain equations become unnecessarily complicated if forced into a standard degree 1 embedding, but simplify if we assign different *weights* (degrees) to the variables. For example, embedding certain curves or moduli spaces into standard $\mathbb{P}^N$ via Veronese embeddings increases the ambient dimension N significantly and requires many equations to cut out. This motivates the following construction.

Definition 3.1.38: Weighted Projective Space

Let $S_\bullet = k[a_1, \dots, a_n]$ be a graded ring where each variable a_i is assigned a specific degree $d_i \in \mathbb{Z}_{>0}$. Then $\operatorname{Proj} k[a_1, \dots, a_n]$ is called a *weighted projective space* and is denoted $\mathbb{P}(d_1, \dots, d_n)$.

Geometrically, over a field, assigning different weights corresponds to taking the quotient of $\mathbb{A}^n_k \setminus \{0\}$ by the action of the multiplicative group $\mathbb{G}_m$ (i.e. $(\mathbb{A}^*_k, \cdot)$) where the scaling acts as $\lambda \cdot (x_1, \dots, x_n) = (\lambda^{d_1} x_1, \dots, \lambda^{d_n} x_n)$.

Example 3.1.39: Weighted Projective Space Examples

1. *Show that $\mathbb{P}(m, n) \cong \mathbb{P}^1_k$ for any positive integers m, n.*

   Let $S_\bullet = k[x, y]$ with $\deg(x) = m$ and $\deg(y) = n$. The scheme is covered by the two open sets $D_+(x)$ and $D_+(y)$. For $D_+(x) \cong \operatorname{Spec}((S_x)_0)$, the degree 0 elements are generated by $u = y^m/x^n$. Thus, $(S_x)_0 \cong k[u]$, which is isomorphic to $\mathbb{A}^1_k$. Similarly, for $D_+(y) \cong \operatorname{Spec}((S_y)_0)$, the degree 0 elements are generated by $v = x^n/y^m$. Thus, $(S_y)_0 \cong k[v]$, which is also isomorphic to $\mathbb{A}^1_k$. On the intersection $D_+(xy)$, the variable v is exactly u^{-1}. This translates to gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$ via the map $u \mapsto u^{-1}$, which is the precise definition of $\mathbb{P}^1_k$.

2. *Show that $\mathbb{P}(1, 1, 2) \cong \operatorname{Proj} k[u, v, w, z]/(uw - v^2)$.*

   Let $S_\bullet = k[x, y, t]$ with $\deg(x) = 1, \deg(y) = 1$, and $\deg(t) = 2$. By theorem 3.1.35, $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} S_{2\bullet}$. The subring $S_{2\bullet}$ is generated by all degree 2 monomials from $S_\bullet$. These monomials are exactly x^2, xy, y^2, and t. Let us map a standard polynomial ring into $S_{2\bullet}$ by setting $u = x^2$, $v = xy$, $w = y^2$, and $z = t$. The only algebraic relation between these generators is $uw = x^2y^2 = (xy)^2 = v^2$. The generator z is algebraically independent of the others. Therefore, $S_{2\bullet} \cong k[u, v, w, z]/(uw - v^2)$. Since the spaces are isomorphic, it follows that $\mathbb{P}(1, 1, 2)$ can be realized as the quadric cone $\operatorname{Proj} k[u, v, w, z]/(uw - v^2)$ in standard $\mathbb{P}^3_k$.

3. *Smooth Compactifications of Hyperelliptic Curves.*

   Weighted projective spaces are highly practical for compactifying curves without introducing artificial singularities. Consider the affine hyperelliptic curve $y^2 = x^5 + 1$. If we attempt to compactify this by homogenizing it into standard $\mathbb{P}^2_k$ (where all variables have degree 1), we obtain the curve $y^2z^3 = x^5 + z^5$. Calculating the Jacobian reveals that this curve has a singularity at the point at infinity $[0 : 1 : 0]$.

   However, if we embed the curve into the weighted projective space $\mathbb{P}(1, 1, 3)$ with coordinates (x, z, y) having degrees $1, 1$, and 3 respectively, we can form a homogeneous equation of total degree 6:
   $$y^2 = x^5 z + z^6$$
   In this weighted space, the curve remains smooth at infinity.

3.1.5 The Category of Schemes

Having defined schemes, morphisms, subschemes, and projective schemes, we have formally constructed the category of schemes, denoted **Sch** (or $\mathbf{Sch}_S$ for S-schemes). Before moving on, it is valuable to look at the categorical properties of **Sch** as a whole.

How well-behaved is this category? Does it have limits and colimits? The short answer is that the local affine structure is well-behaved, but the global structure is much more complex.

The Affine Subcategory

Let $\mathbf{Sch}^{\mathbf{Aff}}$ be the full subcategory of affine schemes. By the definition of the spectrum functor, we have an anti-equivalence of categories:

$$\mathbf{Sch}^{\mathbf{Aff}} \simeq \mathbf{CRing}^{op}$$

Because the category of commutative rings is both complete (has all limits) and cocomplete (has all colimits), the category $\mathbf{Sch}^{\mathbf{Aff}}$ is also completely well-behaved: it has all limits and colimits. As discussed in section 2.2.10, operations here simply mirror the algebraic operations in rings (for example, the categorical coproduct of affine schemes corresponds to the direct product of their rings).

For instance, the initial object in $\mathbf{CRing}$ is $\mathbb{Z}$, which means the terminal object in $\mathbf{Sch}^{\mathbf{Aff}}$ is $\operatorname{Spec}\mathbb{Z}$. In fact, $\operatorname{Spec}\mathbb{Z}$ is the terminal object in the entirety of $\mathbf{Sch}$. To prove this quickly, let X be any scheme, and let X be covered by affine open sets $\operatorname{Spec} A_i$. A morphism $X \to \operatorname{Spec}\mathbb{Z}$ restricts to local morphisms $\operatorname{Spec} A_i \to \operatorname{Spec}\mathbb{Z}$, which correspond exactly to ring homomorphisms $\mathbb{Z} \to A_i$. Since $\mathbb{Z}$ is the initial object in the category of rings, there exists exactly one such ring homomorphism for each A_i. Because these local morphisms are completely unique, they trivially agree on all overlaps and glue together to form a unique global morphism $X \to \operatorname{Spec}\mathbb{Z}$.

The Projective Subcategory

On the opposite end of the spectrum is the subcategory of projective schemes over a fixed base ring. While affine schemes are determined entirely by their global functions, connected projective schemes have only constant global functions.

Categorically, the projective subcategory is highly rigid and lacks the completeness of $\mathbf{Sch}^{\mathbf{Aff}}$. It does not admit infinite limits, and finite colimits (such as general pushouts) often fail to produce a scheme that remains projective. However, what the projective subcategory lacks in categorical completeness, it makes up for in geometric properness. Because projective schemes satisfy the valuative criterion for properness (as we will see later), curves do not escape to infinity without a limit. This makes the projective subcategory the ideal setting for intersection theory and global counting problems, whereas the affine category remains the setting for local algebraic calculations.

Subobjects and Immersions

In standard category theory, a "subobject" is defined as an equivalence class of monomorphisms. If we apply this strict categorical definition to $\mathbf{Sch}$, we encounter non-geometric behavior. A morphism of schemes $X \to Y$ being a categorical monomorphism simply means it is left-cancellative.

To see why this is geometrically problematic, consider the affine subcategory $\mathbf{Sch}^{\mathbf{Aff}}$. Because of the anti-equivalence to $\mathbf{CRing}$, a monomorphism of affine schemes corresponds exactly to an *epimorphism* of commutative rings. Ring epimorphisms certainly include surjective maps $A \to A/I$ (which yield closed immersions) and principal localizations $A \to A_f$ (which yield basic open immersions). However, ring epimorphisms also include total localizations like $\mathbb{Z} \to \mathbb{Q}$. Therefore, the map $\operatorname{Spec}\mathbb{Q} \to \operatorname{Spec}\mathbb{Z}$ is a strict categorical subobject in $\mathbf{Sch}^{\mathbf{Aff}}$, even though geometrically $\operatorname{Spec}\mathbb{Q}$ does not resemble a "piece" of $\operatorname{Spec}\mathbb{Z}$ (it is neither an open nor a closed subset).

For this reason, algebraic geometry rarely focuses on bare categorical monomorphisms. Instead, we restrict our attention to specific geometric embeddings: *open immersions* and *closed immersions* (and their composition, locally closed immersions). These immersions ensure that the subscheme maps homeomorphically onto a topological subspace of the target and that the structure sheaves behave compatibly.

In $\mathbf{Sch}^{\mathbf{Aff}}$, these geometric subobjects are strictly generated by ideal quotients and localizations. In the projective subcategory, the geometric subobjects are almost exclusively closed immersions; because projective schemes are proper (compact), any open immersion that is also a projective scheme must actually be a closed set (a union of connected components). Thus, geometric "subschemes" form a much narrower and better-behaved class of morphisms than general categorical subobjects.

Limits in Sch

When we glue affine schemes together to form general schemes, we lose some of the categorical completeness found in $\mathbf{Sch}^{\mathbf{Aff}}$.

1. *Finite limits exist:* As proved above, the category **Sch** contains a terminal object $\operatorname{Spec}\mathbb{Z}$. Furthermore, as we will see in section 3.2, the category of schemes also admits pullbacks (fibered products). A standard result in category theory dictates that any category with a terminal object and pullbacks automatically has all finite limits (including equalizers and finite products).
2. *Infinite limits generally do not exist:* Unlike affine schemes, general schemes are not closed under infinite products. For example, the infinite product $\prod_{i=1}^{\infty} \mathbb{P}^1_k$ does not exist in the category of schemes. The intuitive reason for this failure is that the Zariski topology behaves poorly under infinite Cartesian products, making it impossible to find a valid affine open cover for the resulting space (for a formal proof involving the failure of the representability of its functor of points, see ref:HERE).
3. *Cofiltered limits exist:* The category of schemes does have cofiltered limits (often called inverse limits), provided the transition maps are affine. This is easily justified: because the spectrum functor is contravariant, taking the inverse limit of these affine schemes translates algebraically to taking the direct limit (filtered colimit) of their corresponding commutative rings, which is a well-behaved algebraic operation. A classic example is the scheme of p-adic integers, which is the limit of affine schemes: $\operatorname{Spec}\mathbb{Z}_p = \varprojlim \operatorname{Spec}(\mathbb{Z}/p^n\mathbb{Z})$.

Colimits in Sch

1. *Arbitrary coproducts exist:* One of the simplest and most useful colimits in **Sch** is the coproduct. Any arbitrary collection of schemes $\{X_i\}$ has a coproduct given by their disjoint union $\coprod X_i$. As we established in section 2.2.9, this holds true for both finite and infinite collections.
2. *Finite colimits generally do not exist:* Outside of disjoint unions, the category of schemes does not have general pushouts or coequalizers. A standard counter-example involves trying to glue a scheme to itself along a generic point. For instance, the pushout of $\operatorname{Spec}(K) \rightrightarrows \mathbb{A}^1_K$ (where we attempt to glue the generic point to something else) fails to be a scheme. We will see this explicitly in example 3.2.7.
3. *Pushouts of open immersions exist:* There is one important exception to the failure of pushouts. If $f : X \to Z$ and $g : Y \to Z$ are both open immersions, their pushout always exists. This is exactly the categorical formulation of gluing schemes together, which we covered in section 2.2.9.

4. *Filtered colimits exist:* The category does admit filtered colimits (often called direct limits). A geometric example is infinite-dimensional projective space, $\mathbb{P}^\infty_k = \varinjlim \mathbb{P}^n_k$, which exists as a scheme (specifically $\operatorname{Proj} k[x_0, x_1, \dots]$), though it loses the property of being locally Noetherian.

Schemes as Colimits of Affine Schemes

We have seen that the category **Sch** contains $\mathbf{Aff} = \mathbf{CRing}^{\mathrm{op}}$ as a full subcategory, and that general schemes are obtained by "gluing" affine pieces. We can now make the gluing process categorically precise: every scheme is a colimit of affine schemes, and the colimit has a particularly clean form.

The coequalizer picture. Let X be a scheme admitting a finite affine open cover $X = U_1 \cup \cdots \cup U_n$. Form the disjoint union

$$U = U_1 \sqcup \cdots \sqcup U_n,$$

which is itself an affine scheme: if $U_i = \operatorname{Spec}(A_i)$, then $U = \operatorname{Spec}(A_1 \times \cdots \times A_n)$. The natural map $\pi : U \to X$ (the inclusion of each U_i into X) is surjective but not injective: a point lying in $U_i \cap U_j$ appears twice in U, once in the U_i-copy and once in the U_j-copy.

To recover X from U, we must identify these redundant copies. Define the "overlap scheme"

$$R = U \times_X U = \coprod_{i,j} (U_i \cap U_j).$$

By the Build-up Lemma (lemma 2.2.29), each intersection $U_i \cap U_j$ can be covered by open sets that are simultaneously distinguished in U_i and U_j. If X is separated, the intersections $U_i \cap U_j$ are themselves affine (section 3.5.1), and R is affine as well; in general, R can be covered by affines, and the construction still works by refining the cover.

The scheme R comes equipped with two natural morphisms to U: the projections

$$s, t : R \rightrightarrows U$$

where s includes $U_i \cap U_j$ into its U_i-copy and t includes it into its U_j-copy. The scheme X is then recovered as the *coequalizer* of s and t in the category of locally ringed spaces:

$$R \underset{t}{\overset{s}{\rightrightarrows}} U \xrightarrow{\pi} X.$$

That is, X is the universal locally ringed space receiving a map from U that identifies $s(r)$ with $t(r)$ for every point $r \in R$.

Let us form an equivalence relation between the glued points categorically. We shall look at the *groupoid object* in the category of schemes (or, when everything is affine, in **Aff**). The additional structure maps are:

1. *Identity:* $e : U \to R$, the diagonal, sending a point of U_i to the "trivial overlap" of U_i with itself.
2. *Composition:* $c : R \times_{s,U,t} R \to R$, expressing transitivity: if a point of $U_i \cap U_j$ and a point of $U_j \cap U_k$ agree on U_j, they compose to a point of $U_i \cap U_k$.
3. *Inverse:* $\iota : R \to R$, the swap: a point of $U_i \cap U_j$ is also a point of $U_j \cap U_i$.

These satisfy the usual groupoid axioms (associativity, unit, inverse), which encode the fact that the relation "x and y represent the same point of X" is an equivalence relation. On the ring side, dualizing: if $U = \operatorname{Spec}(A)$ and $R = \operatorname{Spec}(B)$, the two ring homomorphisms $s^{\#}, t^{\#} : A \rightrightarrows B$ together with the identity, composition, and inverse maps make (A, B) into a *cogroupoid object* in **CRing**.

Conversely, one can *start* with affine groupoid data and produce a scheme. Given rings A and B with homomorphisms $s^{\#}, t^{\#} : A \rightrightarrows B$ and maps $e^{\#} : B \to A$, $c^{\#} : B \to B \otimes_A B$, $\iota^{\#} : B \to B$ satisfying the cogroupoid axioms, together with the condition that s and t are open immersions (so that the "gluing maps" identify open subsets), the coequalizer of $\operatorname{Spec}(B) \rightrightarrows \operatorname{Spec}(A)$ in the category of locally ringed spaces is a scheme.

This gives a precise sense in which the category of schemes is built from the category of affine schemes: **Sch** is obtained from **Aff** by freely adjoining coequalizers of affine groupoid presentations whose structure maps are open immersions. Every scheme admits such a presentation (by choosing an affine cover), and the result is independent of the choice (by the Affine Communication Lemma and the uniqueness of colimits).

The reader should keep this in mind when going through the functorial perspective developed in section 3.3.7, where schemes are characterized as Zariski sheaves with affine open covers. The two viewpoints are dual: the functorial approach defines schemes as certain *limits* (sheaves are defined by equalizer conditions), while the groupoid approach constructs schemes as certain *colimits* (coequalizers of equivalence relations). The equivalence of these two perspectives is a manifestation of a general principle: in algebraic geometry, limit-based and colimit-based descriptions of the same object coexist, and switching between them is sometimes the key to a proof.

Exercise 3.1.2

1. Show that the map $\operatorname{Spec} k(x) \to \operatorname{Spec} k[y]_{(y)}$ sending the unique (generic) point η to the closed point of the codomain is a valid morphism of ringed spaces, but *not* a morphism of locally ringed spaces. (This highlights why the local ring condition is necessary for scheme morphisms).
2. Let $p \in X$ be a point in a scheme. Show there is a canonical morphism of schemes $\operatorname{Spec}(\mathcal{O}_{X,p}) \to X$.
3. Using the above, define a canonical morphism $\operatorname{Spec}(\kappa(p)) \to X$. This is the formal scheme-theoretic morphism corresponding to the inclusion of a point.
4. Show there is no non-trivial map $\mathbb{P}^n_k \to \mathbb{A}^m_k$.

3.2 (Fiber) Product and Fibers

https://stacks.math.columbia.edu/tag/01JW

One of the most common tools in constructing new algebraic and geometric objects are limits and colimits. Given a category that is complete and cocomplete (having all limits and colimits), then:

1. limits are a combination of products and equalizers (i.e. cutting)
2. colimits are a combination of coproducts and coequalizers (i.e. gluing)

The category $\mathbf{Sch}_S$ is closed under coproducts (disjoint union), but not necessarily colimits as we require non-trivial gluing conditions (recall theorem 2.2.54). However, it *is* closed under products and equalizes (equalizer of an affine scheme R usually becomes a closed subscheme). To do this, we must show it is closed under products. Starting with the affine case, by corollary 3.1.5 it is equivalence for R-algebra to be closed under coproducts, and indeed the coproduct of A, B is $A \otimes_R B$ (see [20]).

Theorem 3.2.1: Fibered Product Of Schemes

Let X, Y be schemes over S. Then then the fibered product of X and Y over S exists, that is the limit of the diagram exists:

$$\begin{tikzcd} X \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

The fibered product is denoted $X \times_S Y$

More commonly, we would say that the fibered product satisfies the universal property that if $Z \to X$ and $Z \to Y$ commutes in the diagram, then there exists a unique morphism $Z \to X \times_S Y$ such that

$$\begin{tikzcd} & Z \arrow[d] \arrow[ddl, bend right] \arrow[ddr, bend left] & \\ & X \times_S Y \arrow[dl] \arrow[dr] & \\ X \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

commutes. If S is the final object, or more concisely it is empty or "doesn't matter", then we have the usual product. If S is not specified, the product shall be defined as $X \times Y := X \times_{\mathrm{Spec}(\mathbb{Z})} Y$

***Proof*:**
We shall construct the products of s, and then glue them together.

Let $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, and $S = \mathrm{Spec}(R)$ be affine. Then A, B are R-algebras, and the product of X and Y over S is:

$$\mathrm{Spec}(A \otimes_R B)$$

This comes form the universal property of tensor products along with the property of adjoints. Indeed, if $Z \to \mathrm{Spec}(A \otimes_R B)$ is a scheme morphism, then this gives a ring homomorphism $A \otimes_R B$ into $\mathcal{O}_Z(Z)$. By the universal property of tensor products, a morphism of $A \otimes_R B$ into is the same as a morphism from A and B to that ring. Then by proposition 3.1.12, this gives a morphism of Z into $\mathrm{Spec}(A \otimes_R B)$, and this morphism will satisfy the universal property of fibered products given by the universal property of tensor products.

Let us now construct the more general case via gluing. Let X, Y be schemes. To do this, we shall reduce to a covering of $X \times_S Y$. For any open affine $U \subseteq X$, if the product existed then $\pi_1^{-1}(U) \cong U \times_S Y$ (check this[a]). Now if $\{X_i\}$ is an open covering for x, then I claim that if $X_i \times_S X$ exists, so does $X \times_S Y$. Indeed, for each $U_{ij} = \pi_1^{-1}(X_i) \subseteq X_i \times_S Y$, by the uniqueness of the product there are unique isomorphisms: $\varphi_{ij} : U_{ij} \to U_{ji}$ for each i, j for compatible projections. Furthermore, these isomorphisms are compatible with the gluing condition (as you should check if you are not convinced!). As a remind, the condition was:

$$\varphi_{ik}|_{X_{ij} \cap X_{ik}} = \varphi_{jk} \circ \varphi_{ij}|_{X_{ij} \cap X_{ik}}$$

Thus, we get that we may glue them together by lemma 3.1.9 we have that this glued scheme is isomorphic to $X \times_S Y$, and through this isomorphism we can check that the universal property is satisfied.

By interchanging X and Y, we can now conclude from the proof on affine schemes that if X and Y are any schemes, and S is an affine, then $X \times_S Y$ exists. To show that S can be an arbitrary scheme, let $\{S_i\}$ be an open affine cover of S. Let $\alpha : X \to S$ and $\beta : Y \to S$ be two scheme morphisms, and let $X_i = \alpha^{-1}(S_i)$, $Y_i = \beta^{-1}(S_i)$. Then $X_i \times_{S_i} Y_i$ is well-defined.

Now, what's key is that this scheme is the same as the scheme $X_i \times_S Y$. To see this, we must invoke the uniqueness of the universal property, and indeed if $f : Z \to X_i$ and $g : Z \to Y$ are scheme morphisms over S, by the properties of the commutative diagram we must have that the image of v must land inside Y_i. But by universality they must be equal. Thus $X_i \times_S Y$ is well-defined for each i, and so by the above argument it glues to create $X \times_S Y$, as we sought to show.

[a] If $f : Z \to U$, compose with the inclusion map and invoke the appropriate universal properties

Example 3.2.2: Fibered Products

1. Show that $\mathbb{A}_k^n \times_k \mathbb{A}_k^m \cong \mathbb{A}_k^{n+m}$, which is exactly what we would hope from a product. Notably, prove $k[x] \otimes_k k[y] \cong k[x, y]$.

2. Similarly, if k is a characteristic 0 field show that $\mathbb{Z}[x] \otimes_{\mathbb{Z}} k[y] \cong k[x, y]$. Hence $\mathbb{A}_{\mathbb{Z}}^1 \times \mathbb{A}_k^1 \cong \mathbb{A}_k^2$.

3. Let $X = \operatorname{Spec} \mathbb{C}[x, y]/(x^2 - y)$ and $Y = \mathbb{C}[z, w]/(z^3 - w^4)$. Then:

$$X \times_{\mathbb{C}} Y \cong \operatorname{Spec} \frac{\mathbb{C}[x, y, z, w]}{(x^2 - y, z^3 - w^3)}$$

   The reader should stare at this until they understand why it is called a fibered product.

4. There is new behaviour when considering schemes being tensored over non-algebraically closed fields. Take $\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C}$. As these are affine, this is $\operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C})$. As $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$ (exercise),

$$\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C} \cong \operatorname{Spec}(\mathbb{C}) \amalg \operatorname{Spec}(\mathbb{C})$$

   Notice here how the choice of sheaf completely changed the answer, notably $\operatorname{Spec} \mathbb{C}$ and $\operatorname{Spec} \mathbb{R}$ are schemes over a single point, hence the fibered product diagram when only considering points is an identity map on points:

$$\begin{tikzcd} \bullet \arrow[dr] & & \bullet \arrow[dl] \\ & \bullet & \end{tikzcd}$$

and it is easy to see that in **Top** this results is a single point. However, the resulting scheme is *two* points. Thus, the fibered product here somehow "unravelled" the fact that $\operatorname{Spec}\mathbb{R}$ should be thought of as "gluing" conjugates together (think of the intuition for $\operatorname{Spec}\mathbb{R}[x]$ as a $\operatorname{Spec}\mathbb{C}[x]/\mathrm{Gal}_{\mathbb{R}}(\mathbb{C})$, the real line then don't glue to another points, but their residue field is $\mathbb{R}$ instead of $\mathbb{C}$, and hence it codimension 2 with respect to $\mathbb{C}$).

More generally show that if L/K is a finite galois extension, then

$$L \otimes_K L \cong \prod_{|\mathrm{Aut}(L/K)|} K$$

is isomorphic to $L^{|G|}$($|G|$ copies of L), hint: this is the linear independence of characters proof in geometric form. Hence, over fields fibered products capture some idea properties of galois groups.

5. Recall that the intersection of rings is a ring. This is a special case of the fibered product in the category of commutative rings, namely that if $S \to R$ and $S' \to R$ exists, then:

$$\begin{tikzcd} & S \cap S' \arrow[dl] \arrow[dr] & \\ S \arrow[dr] & & S' \arrow[dl] \\ & R & \end{tikzcd}$$

Let us see how this translates to schemes and affine schemes. Recall that the product and coproduct of affine schemes is an affine scheme:

$$\operatorname{Spec} S \times \operatorname{Spec} S' = \operatorname{Spec} S \otimes_{\mathbb{Z}} S'$$
$$\operatorname{Spec} S \amalg \operatorname{Spec} S' = \operatorname{Spec} S \times S'$$

If $U, U' \subseteq \operatorname{Spec} R$ are open affine, $U = \operatorname{Spec} S$ and $U' = \operatorname{Spec} S'$, Then as $\operatorname{Spec}(-)$ is contravariant we need that the intersection $(\operatorname{Spec}(S)) \cap (\operatorname{Spec}(S'))$ becomes the "union" of rings. Naturally, the union of rings need not be a ring, and as both S, S' contain units then $\langle S, S' \rangle = R$. Instead, the natural dual is:

$$\operatorname{Spec} S \cap \operatorname{Spec} S' = \operatorname{Spec} S \otimes_R S'$$

As we saw, the union of affine need not be affine (see example 2.2.55).

One of the most important uses of the fibered is to change the base scheme:

Definition 3.2.3: Base Change (Scheme-theoretic Pullback)

Let X be an S-scheme, and $T \to S$ a scheme morphism. Then the *base change* of X to T is the T-scheme:

$$X_T = T \times_S X$$

If $f : X \to Y$ is an S-scheme morphism, and $T \to S$ a scheme morphism, then the *base change* of f is the map $f' : X_T \to Y_T$ where $f' = \mathrm{id}_T \times_{\mathrm{id}_S} f$.

Example 3.2.4: Base Change

1. For any R-scheme X and morphism $\operatorname{Spec} S \to \operatorname{Spec} R$ (i.e. ring homomorphism $R \to S$), we get the S-scheme:
$$X' = S \times_R X$$
The category of R-schemes maps to the category of S schemes via base change.
2. If $X = R[x_1, ..., x_m]/I$, is an R-scheme and $S \to R$ is a ring homomorphism, then:
$$X_S = \frac{S[x_1, ..., x_m]}{IS}$$
that is, the set of constant change.
3. As the map $S \to R$ need not be surjective, we can get something like $X = \operatorname{Spec} \mathbb{C}$ as a $\mathbb{C}$-scheme and base change $\mathbb{R} \hookrightarrow \mathbb{C}$.
4.

As the tensor product is a very versatile operation, the fibered product is as well. Let us demonstrate by translating some important results:

Proposition 3.2.5: Base Change Properties

The following properties hold for fibered products:

1. Let $\iota : U \to Z$ be an open immersion and $\varphi : Y \to Z$ be any scheme morphism. Then $U \times_Z Y$ is isomorphic to the open subscheme of Y given by the preimage of U:
$$U \times_Z Y \cong \left(\varphi^{-1}(U), \mathcal{O}_Y|_{\varphi^{-1}(U)}\right)$$
2. *Stability of Closed Immersions:* If $X \to Z$ is a closed immersion, then for any map $Y \to Z$, the base change $X \times_Z Y \to Y$ is a closed immersion.
3. *Stability of Open Immersions:* If $X \to Z$ is an open immersion, then for any map $Y \to Z$, the base change $X \times_Z Y \to Y$ is an open immersion.

***Proof*:**
Reduction to the Affine Case (for 2 and 3): The properties of being a closed immersion or an open immersion are *local on the target.* That is, a morphism $f : W \to Y$ is a closed (resp. open) immersion if and only if there exists an open cover $\{V_i\}$ of Y such that each restriction $f^{-1}(V_i) \to V_i$ is a closed (resp. open) immersion.

Let $X \to Z$ be the immersion we are base changing. Let $\{Z_\alpha = \operatorname{Spec}(R_\alpha)\}$ be an affine open cover of Z. Since $X \to Z$ is an immersion, the restriction $X_\alpha = X \times_Z Z_\alpha \to Z_\alpha$ corresponds to a map of affine schemes (either a quotient or a localization). Now, let $\{Y_{ij} = \operatorname{Spec}(S_{ij})\}$ be an affine open cover of Y such that each Y_{ij} maps into some Z_α. Since fibered products commute with open restriction, the base change restricted to Y_{ij} is:

$$(X \times_Z Y) \times_Y Y_{ij} \cong X \times_Z Y_{ij} \cong (X \times_Z Z_\alpha) \times_{Z_\alpha} Y_{ij}$$

This reduces the problem to showing that the base change of an affine immersion by an affine morphism is an immersion. If we prove this for rings, the global property follows by gluing.

1. The fibered product is the limit in the category of schemes. Maps $W \to U \times_Z Y$ correspond to pairs of maps $W \to U$ and $W \to Y$ that agree in Z. Since $U \hookrightarrow Z$ is an inclusion, a map to U is just a map to Z whose image lies in U. Thus, a map to the product is a map $W \to Y$ such that the composition $W \to Y \xrightarrow{\varphi} Z$ factors through U. This is exactly the definition of a map into the open subscheme $\varphi^{-1}(U) \subseteq Y$.

2. *Affine Case:* Let $Z = \operatorname{Spec}(R)$, $X = \operatorname{Spec}(R/I)$, and $Y = \operatorname{Spec}(S)$. The closed immersion corresponds to the surjection $\pi : R \twoheadrightarrow R/I$. We compute the base change by tensoring with S over R. Since the tensor product is *right-exact*, applying $S \otimes_R -$ to the surjection preserves surjectivity:
$$S \cong S \otimes_R R \xrightarrow{1 \otimes \pi} S \otimes_R (R/I) \longrightarrow 0$$
Thus, the ring map corresponding to the base change $X \times_Z Y \to Y$ is surjective. Since surjective ring maps correspond to closed immersions of affine schemes, and we have established the property is local, the general base change is a closed immersion.

3. *Affine Case:* Let $Z = \operatorname{Spec}(R)$ and let $X \to Z$ correspond to a localization $R \to R_f$ for some $f \in R$ (a basic open immersion). Let $Y = \operatorname{Spec}(S)$ with map $\varphi : R \to S$. We compute the base change by tensoring:
$$S \otimes_R R_f = S \otimes_R R[f^{-1}] \cong S[\varphi(f)^{-1}] = S_{\varphi(f)}$$
The resulting map $S \to S_{\varphi(f)}$ is a localization. Since any open immersion is locally isomorphic to a basic open immersion (localization), and we have established the property is local, the general base change is an open immersion.

This motivates an important definition generalizing affine and projective schemes (idk if this should be here)

Definition 3.2.6: Affine Schemes over a Scheme

Let X be a scheme. Then:

$$\mathbb{A}^n_X = X \times_{\mathbb{Z}} \mathbb{Z}[x_1, \ldots, x_n]$$

Example 3.2.7: Pushout Need Not Exist

Consider two copies of $\mathbb{A}^1_{\mathbb{C}}$. Consider:

$$\begin{tikzcd} & \operatorname{Spec}\mathbb{C} \arrow[dl] \arrow[dr] & \\ \mathbb{A}^1_{\mathbb{C}} & & \mathbb{A}^1_{\mathbb{C}} \end{tikzcd}$$

where $[(0)] \mapsto [(0)]$ in both cases. This is like the double-origin example, except every point is a double origin. This shall be shown to not be a scheme as it is "too non-separated".
We cannot fix this by limiting to the category of affine schemes. Consider for example:

$$\begin{tikzcd} & \operatorname{Spec} k[t]_t \arrow[dl] \arrow[dr] & \\ \operatorname{Spec} k[t] & & \operatorname{Spec} k[t] \end{tikzcd}$$

where $\operatorname{Spec} k[t]_t = \mathbb{A}^1_k \backslash \{0\}$, and the maps are the maps given by $t \mapsto t$ and $t \mapsto 1/t$. Then we saw in example 2.15 that the colimit of this diagram is projective space, $\mathbb{P}^1_k$. Hence affine schemes are not in general closed under pushout. Note that the corresponding fibered product does exist in CRing, namely it will have to be:

$$\begin{tikzcd} & k \arrow[dl, dashed] \arrow[dr, dashed] & \\ k[t] \arrow[dr] & & k[t] \arrow[dl] \\ & k[t]_t & \end{tikzcd}$$

However, applying Spec to this will not return the pushout, namely the pair of adjoint functors (Spec, Γ) does not preserve pushouts.

Note however that colimits do exist in the category of locally ringed spaces. See this link (which link??) for more information.

(words here)

Definition 3.2.8: Geometrically Integral

Let S be a k-scheme. Then S is *geometrically integral* if it is integral over the algebraic closure$\overline{k}$.

The term "geometrically PROPERTY" is a common term to specify that a PROPERTY occurs for the k-scheme when taking the algebraic closure of k.

(want to include this somewhere: *Tensor Product Interpretation (Base Change):* The map constructed above is canonically isomorphic to the base change of the ring R_r to S. Using the tensor product:

$$S \otimes_R R_r \cong S \otimes_R (R[x]/\langle xr - 1\rangle) \cong S[x]/\langle x\varphi(r) - 1\rangle \cong S_{\varphi(r)}$$

This interpretation is powerful because it links the *geometric pullback* of sheaves to the *algebraic tensor product.*)

3.2.1 Fibers and family of Schemes

This gives a huge class of new object's that we may define. Let us start with defining fibers:

Definition 3.2.9: Fibers

Let $f : X \to Y$ be a morphism of scheme. Let $y \in Y$, and $k(y)$ be the residue field of y. Let $\operatorname{Spec}(k(y)) \to Y$ be the natural morphism[a]. Then the *fiber* of the morphism f over the point y is the scheme

$$X_y = X \times_Y \operatorname{Spec}(\kappa(y))$$

In the case where $X = \operatorname{Spec}(R), Y = \operatorname{Spec}(S)$ are affine, then for a point $\mathfrak{p} \in Y$:

$$f^{-1}(\mathfrak{p}) = \operatorname{Spec} R \times_S \operatorname{Spec} \kappa(\mathfrak{p}) = \operatorname{Spec}\left(R \otimes_S S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}}\right)$$

[a] exercise ref:HERE

Proposition 3.2.10: Topological Space of Fibers

Let $f : X \to Y$ be a morphism of schemes, and let $y \in Y$. The natural map $|X_y| \to f^{-1}(y)$ is a homeomorphism.

Furthermore, $X_y \to X$ is a *closed immersion* identified with the ideal $f^{\#}(\mathfrak{m}_y)$.

***Proof*:**
The question is affine local on source and target (proving bijectivity can be reduced to compatible open covers), so let's assume $X = \operatorname{Spec}(R)$ and $Y = \operatorname{Spec}(S)$ are affine schemes. Then $f : \operatorname{Spec} R \to \operatorname{Spec} S$ corresponds to a ring homomorphism $f^{\#} : S \to R$. Let y correspond to the prime ideal $\mathfrak{p}$ in S. Then:

$$X_y = X \times_Y \operatorname{Spec}(\kappa(y)) = \operatorname{Spec}(R \otimes_S S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}})$$

We shall take advantage of example **??**(1) where points are associated to maps to $\kappa(\mathfrak{q})$. Suppose $(f^{\#})^{-1}(\mathfrak{q}) = \mathfrak{p}$ where $\mathfrak{q} \in \operatorname{Spec}(R)$ (i.e. $f(\mathfrak{q}) = \mathfrak{p}$). Then Let g represent the composition $S \to R \to R_{\mathfrak{q}}$. Then $g : S \to R_{\mathfrak{q}}$ factors uniquely through $\overline{g} : S_{\mathfrak{p}} \to R_{\mathfrak{q}}$. We can extend this map by tensoring with $R \otimes_S (-)$ so that:

$$\Phi : R \otimes_S S_{\mathfrak{p}} \to R_{\mathfrak{q}} \qquad (r, s/t) = r\overline{g}(s/t)$$

Note that any element in $\mathfrak{p}S_\mathfrak{p}$ maps to $\mathfrak{q}R_\mathfrak{q}$, hence have have a unique map:

$$R \otimes_S S_\mathfrak{p}/\mathfrak{p}S_\mathfrak{p} \twoheadrightarrow R_\mathfrak{q}/\mathfrak{q}R_\mathfrak{q} = \kappa(\mathfrak{q})$$

As this map is unique, the correspondence between $k(\mathfrak{q})$ and the points of the fiber is unique, and certainly it surjective. Thus, for each point $\operatorname{Spec}(R_\mathfrak{q}/\mathfrak{q}R_\mathfrak{q})$, it is associated to a unique point in $f^{-1}(\mathfrak{p})$.

Finally, it is clear that $f^{-1}(\{y\})$ satisfies the (topological) pullback

$$\begin{tikzcd} f^{-1}(y) \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec}(k(y)) \arrow[r] & Y \end{tikzcd}$$

Taking:

$$\begin{tikzcd} X_y \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec}(k(y)) \arrow[r] & Y \end{tikzcd}$$

The maps are all continuous and follow from properties of fiber product, and the algebra above shows it is a bijection. It is easy to see it is a homeomorphism. For the closed

In some ways, fibers of schemes are more "regular" than fibers of sets, as the extra information of schemes will provide some more possible pre-images:

Example 3.2.11: Fibers

1. Take the familiar projection of the parabola onto the x axis induced by the ring map $\mathbb{Q}[x] \to \mathbb{Q}[y]$ mapping $x \mapsto y^2$. We can think of this as the map $\mathbb{Q}[x] \to \mathbb{Q}[y^2]$ where the codomain is isomorphic to:
$$\mathbb{Q}[y^2] \cong \frac{\mathbb{Q}[x,y]}{x-y^2}$$
This is sometimes called the parameterization of the map. Then the pre-image of 1 is:
$$\begin{aligned} \operatorname{Spec} \mathbb{Q}[x,y]/\left(y^2-x\right) \otimes_{\mathbb{Q}[x]} \mathbb{Q}[x]/(x-1) &\cong \operatorname{Spec} \mathbb{Q}[x,y]/\left(y^2-x, x-1\right) \\ &\cong \operatorname{Spec}\left(\mathbb{Q}[y]/\left(y^2-1\right)\right) \\ &\cong \operatorname{Spec}(\mathbb{Q}[y]/(y-1)(y+1)) \\ &\overset{\text{C.R.T}}{\cong} \operatorname{Spec}(\mathbb{Q}[y]/(y-1) \times \mathbb{Q}[y]/(y+1)) \\ &\cong \operatorname{Spec}(\mathbb{Q}[y]/(y-1)) \amalg \operatorname{Spec}(\mathbb{Q}[y]/(y+1)) \end{aligned}$$
That is, it is the points ± 1. The pre-image of 0 is:
$$\operatorname{Spec} \mathbb{Q}[x,y]/(y^2-x, x) \cong \operatorname{Spec} \mathbb{Q}[y]/(y^2)$$
Notice that the pre-image remembers multiplicity information!
As the spectrum of a polynomial ring over rationals is the quotient by the galois action on

the polynomial ring over $\bar{k}$, the pre-image of -1 exists! It is:

$$\operatorname{Spec}\mathbb{Q}[x,y]/(y^2-x,x+1)\cong\operatorname{Spec}\mathbb{Q}[y]/(y^2+1)\cong\operatorname{Spec}\mathbb{Q}[i]=\operatorname{Spec}\mathbb{Q}(i)$$

This can be thought of as "size 2" point over the base field. The pre-image of the generic point is a reduced point, and can be thought of as "size 2" over the residue field:

$$\operatorname{Spec}\mathbb{Q}[x,y]/(y^2-x)\otimes_{\mathbb{Q}[x]}\mathbb{Q}(x)\cong\operatorname{Spec}\mathbb{Q}[y]\otimes_{\mathbb{Q}[y]}\mathbb{Q}(y^2)$$

Try working through the pre-image of other points that are not commonly associated to $\mathbb{Q}$. Notice that the pre-image is the two points, or you get nonreduced behaviour that is "of degree 2" or you get a field extension of degree 2.

2. Let $X=\operatorname{Spec}(\mathbb{Z}[i])$ (the Gaussian integers) and $Y=\operatorname{Spec}(\mathbb{Z})$. The inclusion $\mathbb{Z}\hookrightarrow\mathbb{Z}[i]$ induces a morphism $\pi:X\to Y$. Let us analyze the fibers of this map (the pre-images of points).
   - *Ramification:* Consider the point $(2)\in\operatorname{Spec}(\mathbb{Z})$. The fiber is $\operatorname{Spec}(\mathbb{Z}[i]\otimes_{\mathbb{Z}}\mathbb{Z}/(2))\cong\operatorname{Spec}(\mathbb{F}_2[x]/(x^2+1))=\operatorname{Spec}(\mathbb{F}_2[x]/(x+1)^2)$. Topologically, there is one point, but the ring has a nilpotent element. The prime 2 "ramifies."
   - *Splitting:* Consider $(5)\in\operatorname{Spec}(\mathbb{Z})$. In $\mathbb{Z}[i]$, $5=(2+i)(2-i)$. The fiber corresponds to $\mathbb{F}_5[x]/(x^2+1)$. Since x^2+1 has roots $2,3$ in $\mathbb{F}_5$, this ring is $\mathbb{F}_5\times\mathbb{F}_5$. Geometrically, the point (5) pulls back to *two* distinct points.
   - *Inert:* Consider $(3)\in\operatorname{Spec}(\mathbb{Z})$. x^2+1 is irreducible in $\mathbb{F}_3$. The fiber is $\operatorname{Spec}(\mathbb{F}_9)$, a single point with a larger residue field.

   This example demonstrates that scheme morphisms provide the correct geometric framework for Algebraic Number Theory.

3. As $X\to\operatorname{Spec}(\mathbb{Z})$ is always well-defined and unique, there is always fibers over $\operatorname{Spec}(\mathbb{Z})$ and hence they are usually given a name: The fiber over (0) is often labeled $X_{\mathbb{Q}}$ (over $\mathbb{Q}$), and the fibers for all the primes are X_p (over $\mathbb{F}_p$). A common terminology is that X_p *arises by reduction* mod p of the scheme X.

4. One way of characterizing surjectivity $\operatorname{Spec}(B)\to\operatorname{Spec}(A)$ ring theoretically is to demand that for each $\mathfrak{p}\in A$, $B\otimes_{\mathbb{Z}}\kappa(\mathfrak{p})\neq 0$ (i.e. the preimage is nonempty). In section ref:HERE.

Proposition 3.2.12: Pre-image of Generic (Spreading out)

Let $f:X\to Y$ be a morphism of finite type and Y Noetherian. Then if $f^{-1}(\eta)=\emptyset$ for a generic point $\eta\in Y$, there exists an open subset $U\subseteq Y$ containing η such that $f^{-1}(U)=\emptyset$.

***Proof*:**
Since the problem is local on Y around η, we may assume $Y=\operatorname{Spec}(A)$ is affine. Since Y has a generic point η, Y is irreducible. As Y is Noetherian, we can treat A as an integral domain (modulo nilpotents, which do not affect the topology or emptiness of schemes). Let $K=\operatorname{Frac}(A)$ be the residue field of η.

Since f is of finite type, X is quasi-compact. We can cover X by a finite number of affine open subsets $X = \bigcup_{i=1}^n \operatorname{Spec}(B_i)$, where each B_i is a finitely generated A-algebra.

The hypothesis $f^{-1}(\eta) = \emptyset$ means the fiber product $X \times_Y \operatorname{Spec}(K)$ is the empty scheme. For each affine patch $\operatorname{Spec}(B_i)$, this corresponds to the ring:

$$B_i \otimes_A K = 0$$

Note that $B_i \otimes_A K$ is isomorphic to the localization of B_i by the multiplicative set $S = A \setminus \{0\}$. If a localization is the zero ring, then $1 = 0$ in that ring. By the definition of localization, this implies there exists an element $s_i \in S$ (i.e., a non-zero element $s_i \in A$) such that:

$$s_i \cdot 1_{B_i} = 0 \quad \text{in } B_i$$

Now, consider the basic open set $U_i = D(s_i) \subseteq Y$. Since $s_i \neq 0$, $\eta \in U_i$. The preimage of this open set in the chart $\operatorname{Spec}(B_i)$ is $\operatorname{Spec}((B_i)_{s_i})$. However, since s_i annihilates the unity in B_i, the localized ring is zero:

$$1 = \frac{s_i}{s_i} = \frac{1}{s_i} \cdot (s_i \cdot 1_{B_i}) = 0$$

Thus, $\operatorname{Spec}((B_i)_{s_i}) = \emptyset$.

To handle the entire scheme X, let $s = s_1 s_2 \cdots s_n$. Since A is a domain, $s \neq 0$, so $U = D(s)$ is a non-empty open set containing η. Furthermore, $U \subseteq U_i$ for all i. Consequently, the preimage $f^{-1}(U)$ is the union of the preimages restricted to each chart. Since s is a multiple of s_i, s acts as zero on every B_i, making the localization $(B_i)_s = 0$ for all i. Therefore, $f^{-1}(U) = \emptyset$.

On p. 260 of Vakil he gives more on general fibers, generic fibers, and generically finite morphisms

don't know where to put these examples yet

Example 3.2.13: Parameterization of Curves using Schemes

Take

$$X = \operatorname{Spec} \frac{k[x,y,t]}{(ty - x^2)}$$
$$Y = \operatorname{Spec} k[t]$$

Let $f : X \to Y$ be given by the natural ring homomorphism $k[t] \to \frac{k[x,y,t]}{(ty-x^2)}$. Note both these schemes are integral schemes of finite type over k.

Now, identify the closed points of Y with k. For $a \neq 0 \in k$, we get the fibers correspond to the parabola's $ay = x^2$ or $y = x^2/a$. All of these are irreducible, reduced curves. At $a = 0$, we get $x^2 = 0$, which is a reduced scheme.

More generally, S-schemes can be geometrically interpreted as schemes paramaterized by the points of S whose maps are fiber-bundle morphisms. Hence, the scheme $\operatorname{Spec} k[t]$ *paramaterized* the curve.

A quick note must be done about general products and why it may be preferable to take the fibered product over a field:

Example 3.2.14: Strange Fibered Products

1. Show that $\operatorname{Spec}(\mathbb{Z}/m\mathbb{Z}) \times_{\operatorname{Spec}(\mathbb{Z})} \operatorname{Spec}(\mathbb{Z}/n\mathbb{Z}) = \emptyset$.
2. Another product of taking the "absolute" fibered product is that if $X = \operatorname{Spec}\mathbb{Z}[x]$ and $Y = \operatorname{Spec}\mathbb{Z}[y]$ then:

$$\dim X \times Y = \dim(X) + \dim(Y) - \dim(\operatorname{Spec} Z) = \dim X + \dim Y - 1$$

If we restrict to k-schemes for a field k, these two pathologies do not occur.

Put an appropriate definition here for the following

Hence, given a surjective map $Y \to X$, Y can be regarded as a collection of schemes paramaterized by Y. Conversely, we may want to study a scheme X_0, and see if there are some deformations we may want to do. Then a common way of defining a *family of algebraic deformation* is by a scheme morphisms $f : X \to Y$ where one of the fibers of f is isomorphic to X_0.

Some more here on representable Functors

Exercise 3.2.1

1. Let $\varphi : X \to Y$ be a morphism of S-scheme of finite type. Show that if $S \to S'$ is any base extension, $\varphi' : X' \to Y'$ is an S'-scheme morphism preserves finite type.
2. Let $\varphi : X \to Y$ be a scheme morphism of integral schemes. Show the fibers needn't be irreducible or reduced.
3. Let $\mathfrak{p} \in X$ be a point in a scheme. Show there is a canonical map

$$\operatorname{Spec}(\mathcal{O}_{X,\mathfrak{p}}) \to X$$

4. Using the above, define a canonical morphism $\operatorname{Spec}(\kappa(\mathfrak{p})) \to X$.
5. Perhaps put here some base-change exercises relating the tensor product exercises you had in [20] and now give them their geometric counter-part.
6. Show if $U, V \subseteq X$ are open subschemes then $U \times_X V \cong U \cap V$.
7. Let $\varphi : B \to A$ be a ring morphism, $S \subseteq B$ a multiplicative subset, so $\varphi(S)$ is a multiplicative subset of A. Show that $\varphi(S)^{-1}A \cong A \otimes_B (S^{-1}B)$. What does this imply for the relation between localization and the fibered product?
8. Let X be a k-scheme. Show that if L/k is a field extension, $X \times_{\operatorname{Spec}(k)} \operatorname{Spec}(L)$ is an L-scheme. Show that this in fact gives a functor (define the appropriate maps, and verify).

3.2.2 *Monomorphisms in Sch

Using fiber products, we can better characterize subschemes categorically. It is natural to ask: what are the *categorical* sub-objects of a scheme? In any category, the correct notion of "sub-object" is a monomorphism: a morphism $f : Y \to X$ such that for any two morphisms $g, h : Z \rightrightarrows Y$, the equality $f \circ g = f \circ h$ implies $g = h$. In the category of sets, monomorphisms are injective functions. In the category of groups, they are injective homomorphisms. In the category of schemes, the answer is more interesting.

Proposition 3.2.15: Monomorphisms of Schemes

A morphism $f : Y \to X$ of schemes is a monomorphism if and only if the diagonal morphism $\Delta : Y \to Y \times_X Y$ is an isomorphism.

***Proof*:**
By definition, f is a monomorphism if and only if for every scheme Z, the map $f_* : \operatorname{Hom}(Z, Y) \to \operatorname{Hom}(Z, X)$ is injective. By the universal property of the fiber product, $\operatorname{Hom}(Z, Y \times_X Y) \cong \operatorname{Hom}(Z, Y) \times_{\operatorname{Hom}(Z,X)} \operatorname{Hom}(Z, Y)$, so f_* is injective for all Z if and only if the two projections $Y \times_X Y \rightrightarrows Y$ are equal, which holds if and only if Δ is an isomorphism.

For affine schemes, the characterization is purely algebraic. A morphism $\operatorname{Spec}(S) \to \operatorname{Spec}(R)$ is a monomorphism if and only if the corresponding ring map $\varphi : R \to S$ is an *epimorphism of commutative rings*: a map such that for any two ring maps $\alpha, \beta : S \rightrightarrows T$, the equality $\alpha \circ \varphi = \beta \circ \varphi$ implies $\alpha = \beta$. Equivalently, φ is an epimorphism if and only if the multiplication map $S \otimes_R S \to S$ is an isomorphism.

Note that monomorphissm of schemes are at least always injective. It is easy to see this reduces to affine patches. THen for a ring map $f : R \to S$ to be an epimorphism in the category of commutative rings, it must satisfy a strict tensor condition: the multiplication map $S \otimes_R S \to S$ must be an isomorphism. Geometrically, this injective on points, and it doesn't ramify or fold back on itself; see exercise ref:HERE.

The crucial observation is that epimorphisms of rings are *much more general* than surjections. The three main classes are:

Example 3.2.16: Epimorphisms in CRing

1. *Surjections* $R \twoheadrightarrow R/I$. These are epimorphisms because any map out of R/I is determined by its values on the images of elements of R. The corresponding monomorphisms of affine schemes are the *closed immersions*.

2. *Localizations* $R \to S^{-1}R$. These are epimorphisms because any map out of $S^{-1}R$ is determined by its values on elements of R (the images of elements of S must be invertible, which leaves no freedom). In general these maps are neither open nor closed (recall proposition 2.1.26). This includes both $R \to R_f$ (giving *open immersions* to distinguished opens) and $R \to R_{\mathfrak{p}}$ (giving the *local scheme* morphisms $\operatorname{Spec}(\mathcal{O}_{X,x}) \to X$).

   The most famous such example is $\mathbb{Z} \to \mathbb{Q}$ which is the localization away from the prime (0).

3. *Gabriel Localizations.* Most localization is done at either a single element or the compliment of a prime ideal. However, we can localize ideals. More generally, we can localize on *filters of ideals* (Gabriel localization) corresponding to an arbitrary intersection of open sets. Naturally, open sets are not closed under arbitrary intersection! These can result in pathological "fractal" sets that are still categorically a subscheme.

All four types of subobjects from the summary table are monomorphisms of schemes, but the converse is false: there exist monomorphisms that do not fall into any of the four classes. For instance, the map $\mathrm{Spec}\,(\mathbb{Q}) \to \mathrm{Spec}\,(\mathbb{Z})$ is a monomorphism (the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism of rings—it is the localization at the multiplicative set $\mathbb{Z} \setminus \{0\}$), but the image is the single point $\{[(0)]\}$, which is the generic point of $\mathrm{Spec}\,(\mathbb{Z})$. This is neither open, nor closed, nor locally closed: it is a local scheme map.

3.3 The Functor of Points

Throughout the development of scheme theory, we have built our geometric objects from locally ringed spaces: a topological space equipped with a structure sheaf, glued from affine pieces. This is a powerful framework, but it carries a certain asymmetry: the points of a scheme and the morphisms between schemes are defined by quite different-looking data. The goal of this section is to develop a perspective—the *functor of points*—in which schemes are understood entirely through their morphisms. The result is not a rephrasing; it reveals structure that the locally ringed space definition obscures, and it provides the language in which most modern moduli problems are formulated.

The idea has a simple origin. Recall from section 2.2.5 that a global section $f \in \Gamma(X, \mathcal{O}_X)$ of a scheme X assigns to each point $x \in X$ an element $f(x)$ in the residue field $\kappa(x)$, but that these residue fields vary from point to point. A global section is therefore *not* a function in the classical sense. Let us trace how this observation leads, step by step, to a functorial description of schemes.

3.3.1 From Evaluation to Morphisms

When the base field is algebraically closed, the situation is familiar. Over an algebraically closed field $k = \overline{k}$, a polynomial $f \in \overline{k}[x_1, \ldots, x_n]$ determines a genuine set-theoretic function:

$$\overline{k}[x_1, \ldots, x_n] \longrightarrow \mathrm{Hom}_{\mathbf{Set}}(\overline{k}^n, \overline{k}), \qquad f \longmapsto \big((a_1, \ldots, a_n) \mapsto f(a_1, \ldots, a_n)\big). \tag{3.1}$$

Here $\overline{k}^n$ can be identified with $\mathrm{mSpec}\,(\overline{k}[x_1, \ldots, x_n])$ via the Nullstellensatz, and each evaluation $f(a_1, \ldots, a_n)$ lands in $\overline{k} \cong \kappa(\mathfrak{m}_{(a_1,\ldots,a_n)})$. The map in equation (3.1) is injective precisely because an algebraically closed field has "enough points" to separate polynomials.

Over a general field k, this breaks down: the maximal ideals of $k[x_1, \ldots, x_n]$ no longer correspond to k-rational tuples. As discussed in example 2.1.3, $\mathrm{mSpec}\,(k[x_1, \ldots, x_n])$ contains orbits of geometric points under the absolute Galois group. One can partially recover the situation by mapping not into k, but into the affine line $\mathbb{A}^1_k$, which itself carries these additional closed points. Concretely, a polynomial $f \in k[x_1, \ldots, x_n]$ induces a morphism of k-algebras

$$k[t] \longrightarrow k[x_1, \ldots, x_n], \qquad t \longmapsto f,$$

which, by the equivalence of affine schemes with rings, corresponds to a scheme morphism $\mathbb{A}^n_k \to \mathbb{A}^1_k$. Evaluating at a closed point $\mathfrak{m}$ of $\mathbb{A}^n_k$ yields the image of f in $\kappa(\mathfrak{m})$, and this image determines a closed point of $\mathbb{A}^1_k$. Thus, we arrive at a map

$$k[x_1, \ldots, x_n] \longrightarrow \mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{mSpec}\, k[x_1, ..., x_n],\, \mathrm{mSpec}\, k[t]), \qquad f \longmapsto \text{the morphism induced by } t \mapsto f. \tag{3.2}$$

This is already an improvement: it works over any field, and the target is a set of *scheme morphisms* rather than set-theoretic functions. However, it still only detects the behavior of f at closed points.

Recall that such schemes are called *Jacobson schemes* (see definition 3.7.4). For general schemes this is insufficient. For instance, on $\operatorname{Spec}(k[[x]])$ or $\operatorname{Spec}(\mathbb{Z}_{(p)})$, a function is not determined by its values at closed points alone.

The natural fix is to allow *all* prime ideals as test points. For any k-algebra A, a scheme morphism $\operatorname{Spec}(A) \to \mathbb{A}^1_k$ is the same data as a k-algebra map $k[t] \to A$, which since this is a k-algebra homomorphism is the same as choosing an element of A for t to map to. This gives:

$$\Gamma(\operatorname{Spec}(A), \mathcal{O}_{\operatorname{Spec}(A)}) = A \;\cong\; \operatorname{Hom}_{\mathbf{Sch}_k}(\operatorname{Spec}(A), \mathbb{A}^1_k). \tag{3.3}$$

The left-hand side is the ring of global sections; the right-hand side is the set of scheme morphisms to the affine line. Each element $f \in A$ corresponds to the unique morphism $\pi_f : \operatorname{Spec}(A) \to \mathbb{A}^1_k$ for which $\pi_f^{\#}(t) = f$. The "value" of f at a prime $\mathfrak{p}$ is recovered by following $t \mapsto f \mapsto f \bmod \mathfrak{p} \in \kappa(\mathfrak{p})$, which is precisely the scheme-theoretic notion of evaluation from section 2.2.5.

Two observations are now in order. First, the construction (3.3) is not specific to affine schemes: it will extend to arbitrary k-schemes X once we establish the representability of the global section functor (Proposition 3.3.9 below). Second, and more importantly, there is nothing special about $\mathbb{A}^1_k$ as a target apart from the fact that it *represents* the functor of global sections. One can ask: what happens if we replace $\mathbb{A}^1_k$ by another scheme T?

The set $\operatorname{Hom}_{\mathbf{Sch}_k}(X, T)$ makes perfect sense for any T, and different choices of T probe different aspects of X. Taking $T = \mathbb{A}^1_k$ recovers global sections; taking $T = \mathbb{G}_m$ recovers invertible global sections; taking $T = \mathbb{P}^n_k$ recovers line bundles generated by $n+1$ global sections (cf. section 5.3.3). The test object T acts as a "measuring device" that detects a particular piece of the geometry of X. The fact—a consequence of the Yoneda Lemma applied to schemes—is that if we keep track of $\operatorname{Hom}(-, X)$ for *all* possible test objects simultaneously, we recover X entirely.

3.3.2 Z-valued Points

Let us make the above discussion concrete. Consider the scheme

$$X = \operatorname{Spec} \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)},$$

whose underlying topological space consists of $\operatorname{Gal}(\mathbb{C}/\mathbb{R})$-conjugate pairs of maximal ideals together with a generic point. The closed points do not individually look like points on a circle—many have residue field $\mathbb{C}$, not $\mathbb{R}$—and so the "usual" picture of the unit circle is not immediately visible.

If we want to recover that picture, we should look for $\mathbb{R}$-valued points: ring homomorphisms

$$\mathrm{ev}_{a,b} : \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \longrightarrow \mathbb{R}$$

sending $x \mapsto a$ and $y \mapsto b$. The quotient relation forces $a^2+b^2=1$, so the collection of all such homomorphisms is in bijection with

$$X(\mathbb{R}) \;\simeq\; \left\{(a,b) \in \mathbb{R}^2 \;\middle|\; a^2+b^2=1\right\},$$

which is exactly the unit circle. Equivalently, each such homomorphism factors through the evaluation map $\mathbb{R}[x,y] \to \mathbb{R}$ as a map that kills the ideal (x^2+y^2-1). Via the contravariant equivalence of affine schemes and rings, these ring homomorphisms correspond to scheme morphisms $\mathrm{Spec}\,(\mathbb{R}) \to X$. The sheaf morphism on global sections is:

$$\mathrm{ev}_{a,b}^{\#}(X): \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \longrightarrow \mathbb{R}.$$

Now suppose we change the target ring. Over $\mathbb{Z}$, the equation $a^2+b^2=1$ has only the four solutions $(\pm 1, 0)$ and $(0, \pm 1)$. Over $\mathbb{Q}$, a Pythagorean-triple parametrization yields infinitely many solutions such as $\left(\frac{3}{5}, \frac{4}{5}\right)$. Over $\mathbb{C}$, the circle becomes a rational curve, and the solution set is much larger still. Each choice of "coefficient ring" yields a different collection of evaluation maps, probing different arithmetic and geometric features of the same scheme.

There is one subtlety to address regarding the base. For the evaluation map $\mathrm{ev}_{a,b}$ to exist as a map of $\mathbb{R}$-algebras, the target ring must receive a structure map from $\mathbb{R}$. There is no nontrivial ring homomorphism $\mathbb{R} \to \mathbb{Q}$, so asking for $\mathbb{Q}$-valued points of the $\mathbb{R}$-scheme X is not well-posed. To allow all these evaluations simultaneously, it is natural to work over the universal base $\mathbb{Z}$. Set $X_{\mathbb{Z}} = \mathrm{Spec}\ \mathbb{Z}[x,y]/(x^2+y^2-1)$. Then for any commutative ring R, the R-valued points of $X_{\mathbb{Z}}$ are the ring homomorphisms $\mathbb{Z}[x,y]/(x^2+y^2-1) \to R$, or equivalently the solutions of $a^2+b^2=1$ in R. If $A \to B$ is a ring homomorphism and X_A, X_B denote the base changes, then $X_A(B) = X_B(B)$: the B-valued points see only the base-changed scheme.

It is worth pausing to understand the $\mathbb{Z}$-points of $X_{\mathbb{Z}}$ scheme-theoretically.

A $\mathbb{Z}$-point of $X_{\mathbb{Z}}$ is a morphism $\mathrm{Spec}\,(\mathbb{Z}) \to X_{\mathbb{Z}}$, which is a section of the structure map $X_{\mathbb{Z}} \to \mathrm{Spec}\,(\mathbb{Z})$. Geometrically, one can visualize $X_{\mathbb{Z}}$ as an arithmetic surface fibered over $\mathrm{Spec}\,(\mathbb{Z})$, with each fiber X_p being the circle $x^2+y^2=1$ reduced modulo the prime p. A $\mathbb{Z}$-point is then a "horizontal" section of this fibration: it passes through every fiber simultaneously.

These $\mathbb{Z}$-points are *not* closed points of $X_{\mathbb{Z}}$. Consider the point $P=(1,0)$, corresponding to the homomorphism $\varphi: \mathbb{Z}[x,y]/(x^2+y^2-1) \to \mathbb{Z}$ defined by $x \mapsto 1$, $y \mapsto 0$. The kernel is the prime ideal $\mathfrak{p} = (x-1, y)$, and the quotient $\mathbb{Z}[x,y]/\mathfrak{p} \cong \mathbb{Z}$. Since $\mathbb{Z}$ is not a field, $\mathfrak{p}$ is prime but not maximal, and the image of P is a closed subscheme isomorphic to $\mathrm{Spec}\,(\mathbb{Z})$—it has dimension 1 and residue field $\mathbb{Q}$ at its generic point. In contrast, a closed point of $X_{\mathbb{Z}}$ has dimension 0 and a finite residue field $\mathbb{F}_q$.

The four $\mathbb{Z}$-points interact with the fibers in instructive ways. Modulo 2, since $1 \equiv -1$ in $\mathbb{F}_2$, the points $(1,0)$ and $(-1,0)$ collapse to the same point of X_2, as do $(0,1)$ and $(0,-1)$. Modulo an odd prime $p \geq 3$, all four points remain distinct in $X_p(\mathbb{F}_p)$. These four "horizontal sections" are the only integer solutions on the circle; any other rational point, such as $\left(\frac{3}{5}, \frac{4}{5}\right)$, gives a $\mathbb{Q}$-point but not a $\mathbb{Z}$-point.

More generally, there is no reason to restrict the "value scheme" to be affine. One might wish to evaluate points on a torus with values in a curve, or to study morphisms between projective schemes. The evaluation interpretation as a global ring homomorphism is then lost, but this is precisely what the sheaf was built to handle: evaluation can be performed locally. Geometrically, a morphism $Z \to X$ between arbitrary schemes is the datum of compatible local evaluations. This motivates the following definition.

Definition 3.3.1: Z-valued Points

Let X be a scheme. The *Z-valued points* of X are the morphisms $Z \to X$ in the category of schemes. The set of all such morphisms is denoted

$$X(Z) := \mathrm{Hom}_{\mathbf{Sch}}(Z, X).$$

When $Z = \mathrm{Spec}\,(R)$ is affine, this is written $X(R)$.

The reader should note a potential source of confusion: a "Z-valued point" is the *morphism* $Z \to X$ itself, not any particular point in Z or in the image of Z. This terminology is justified by the fact that a morphism of schemes is not determined by the underlying continuous map of topological spaces—the sheaf map $\varphi^{\#}$ carries additional information. The full morphism is the "point".

the fibered product of two schemes need not be the schemes of these two fibered products. In a concrete sense, this is because we must be more specific with what types of points we are dealing with. If we specify the points, we in fact recover the usual "sets"[5].

Proposition 3.3.2: Fibered Product With Z-valued Points

Let X, Y be schemes and take Z-valued points on each scheme i.e. morphisms in $\mathrm{Hom}(Z, X)$ and $\mathrm{Hom}(Z, Y)$. Then show that the fibered product will be:

$$\mathrm{Hom}(W, X \otimes_W Y) = \mathrm{Hom}(Z, X) \times_{\mathrm{Hom}(Z,W)} \mathrm{Hom}(Z, Y)$$

***Proof*:**
exercise.

Example 3.3.3: Non-trivial Projective Line: Severi–Brauer Variety

A Severi–Brauer Variety over k is a variety P that becomes isomorphic to $\mathbb{P}^n$ for appropriate n after base-changing to an algebraic closure[a]. Take the conic:

$$C : x^2 + y^2 + z^2 = 0 \subseteq \mathbb{P}^2_{\mathbb{R}}$$

This connect is simply Proj $\frac{\mathbb{R}[x,y,z]}{(x^2+y^2+z^2)}$. Taking the standard chart $D_+(x)$ and letting $u = y/x$, $v = z/x$ we see that:

$$C \cap D_+(x) = \mathrm{Spec}\left(\frac{R[u, v]}{(1 + u^2 + v^2)}\right)$$

It is easy to see that in this chart that $(v, u^2 + 1)$ is a closed point, since:

$$\frac{\mathbb{R}[u, v]/(1 + u^2 + v^2)}{(v, u^2 + 1)} \cong \frac{\mathbb{R}[u]}{(u^2 + 1)} \cong \mathbb{C}$$

In C itself, this becomes the homogeneous principle ideal $(z, x^2 + y^2)$. When base-changing to $\mathbb{C}$, this becomes the two point $[1 : i : 0], [1 : -i : 0] \in C_{\mathbb{C}}$.

However, there are no $\mathbb{R}$-points: there is no triple $a, b, c \in \mathbb{R} \setminus \{0\}$ such that $a^2 + b^2 + c^2 = 0$, hence $C(\mathbb{R}) = \emptyset$.

[5] The topological product can also be thought of as being over the right valued points

In cite:HERE, we shall see this corresponds to its Brauer class being the quaternion algebra $\left(\frac{-1,-1}{\mathbb{R}}\right)$, and that $\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$.

[a]For those who know projective bundles, it's the function-field analogue of a projective bundle with no vector-bundle lift

3.3.3 The Yoneda Embedding for Schemes

Having defined Z-valued points, we can now ask: how much of a scheme X is captured by the totality of its valued points? The answer, given by the Yoneda Lemma, is: *everything*.

Proposition 3.3.4: Scheme Morphisms from Valued Points

Let X and Y be schemes. A morphism of schemes $\varphi : X \to Y$ is equivalent to a natural transformation $\eta : h_X \to h_Y$ between the associated functors of points. That is, giving a collection of maps of sets

$$\eta_Z : X(Z) \longrightarrow Y(Z)$$

for every scheme Z, natural in Z, uniquely determines a scheme morphism $\varphi : X \to Y$, and conversely.

***Proof*:**
Every scheme X defines a contravariant functor $h_X : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ by

$$h_X(Z) = \mathrm{Hom}_{\mathbf{Sch}}(Z, X) = X(Z).$$

A morphism $\varphi : X \to Y$ induces a natural transformation $\varphi_* : h_X \to h_Y$ by post-composition: for each Z and each $f \in X(Z)$, set $(\varphi_*)_Z(f) = \varphi \circ f$. Naturality follows from the associativity of composition.

Conversely, given a natural transformation $\eta : h_X \to h_Y$, we evaluate at $Z = X$. The set $X(X) = \mathrm{Hom}(X, X)$ contains the identity id_X, and we define

$$\varphi := \eta_X(\mathrm{id}_X) \in Y(X) = \mathrm{Hom}(X, Y).$$

By the Yoneda Lemma (section 0.4),

$$\mathrm{Nat}(h_X, h_Y) \;\cong\; h_Y(X) = \mathrm{Hom}(X, Y),$$

and this correspondence is bijective. Thus, the assignment $X \mapsto h_X$ defines a fully faithful embedding $\mathbf{Sch} \hookrightarrow \mathrm{Fun}(\mathbf{Sch}^{\mathrm{op}}, \mathbf{Set})$: the category of schemes embeds into the category of presheaves on $\mathbf{Sch}$.

The proof is short, but the content is not trivial. It says that a scheme is *completely determined* by how all other schemes map into it. No information is lost by passing from the locally ringed space X to the functor h_X.

A useful analogy organizes this perspective. In linear algebra, a vector $v \in V$ is determined by the collection of values $\ell(v)$ for all linear functionals $\ell \in V^*$: the dual space "separates points." Choosing a basis for V^* gives coordinates on V. The Yoneda embedding plays the same role for schemes. The functor h_X is the "dual description" of X: the scheme is determined by the collection of sets $X(Z)$ for all test schemes Z. Each choice of test scheme is analogous to choosing a basis vector in the dual—it extracts one piece of information about X.

Different test objects give genuinely different perspectives on the same scheme:

Test object Z	$X(Z)$	*What it detects*
$\mathrm{Spec}\,(\overline{k})$	geometric points	the classical variety
$\mathrm{Spec}\,(k)$	rational points	arithmetic information
$\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))$	tangent vectors	infinitesimal geometry (order 1)
$\mathrm{Spec}\,(k[t]/(t^n))$	$(n{-}1)$-jets	higher infinitesimal geometry
$\mathrm{Spec}\,(k[[t]])$	formal arcs	formal local behavior
$\mathrm{Spec}\,(K),\ K = k(C)$	rational curves	birational geometry
$\mathrm{Spec}\,(\mathcal{O}_{X,x})^\dagger$	the germ at x	local scheme structure
$\mathrm{Spec}\,(\hat{\mathcal{O}}_{X,x})^\dagger$	formal neighborhood	formal geometry at x

† The last two entries differ in character from the others: they depend on the scheme X itself, rather than being universal test objects. Their role is not to "probe" X from outside, but to isolate the local structure at a point $x \in X$.

The local ring $\mathcal{O}_{X,x}$ is the stalk of the structure sheaf, and $\mathrm{Spec}\,(\mathcal{O}_{X,x})$ is the "germ" of X at x: the limit of all open neighborhoods of x, retaining only the information visible in an arbitrarily small neighborhood. There is a canonical morphism $\mathrm{Spec}\,(\mathcal{O}_{X,x}) \to X$ (induced by the localization maps $\mathcal{O}_X(U) \to \mathcal{O}_{X,x}$), and any morphism $f : Y \to X$ sending a point $y \in Y$ to x factors through $\mathrm{Spec}\,(\mathcal{O}_{X,x})$ at the level of germs:

$$\mathrm{Spec}\,(\mathcal{O}_{Y,y}) \longrightarrow \mathrm{Spec}\,(\mathcal{O}_{X,x}) \longrightarrow X.$$

In this sense, the local ring "sees" every morphism that passes through x. This is why properties of morphisms (flatness, smoothness, étaleness) can often be checked at the level of local rings.

The completion $\hat{\mathcal{O}}_{X,x}$ captures even finer information: the formal power series behavior at x, analogous to the Taylor expansion of a smooth function. The formal arc entry $\mathrm{Spec}\,(k[[t]])$ in the table is in fact the special case $\mathrm{Spec}\,(\hat{\mathcal{O}}_{\mathbb{A}^1,0})$—the formal neighborhood of the origin in the affine line. For a general point x on a smooth n-dimensional variety, the completion is $\hat{\mathcal{O}}_{X,x} \cong k[[t_1, \ldots, t_n]]$, and the formal neighborhood is a "formal polydisk" of dimension n.

The passage from one test object to another is analogous to a change of basis. A morphism $Z' \to Z$ induces, by precomposition, a map $X(Z) \to X(Z')$ that relates the two perspectives. For instance, the closed immersion $\mathrm{Spec}\,(k) \hookrightarrow \mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))$, dual to the quotient map $k[\epsilon]/(\epsilon^2) \twoheadrightarrow k$, sends a tangent vector to its basepoint. This is the "forgetful" map from infinitesimal data to point data. The inclusions among the infinitesimal test objects form a tower:

$$\mathrm{Spec}\,(k) \hookrightarrow \mathrm{Spec}\,(k[\epsilon]/(\epsilon^2)) \hookrightarrow \mathrm{Spec}\,(k[t]/(t^3)) \hookrightarrow \cdots \hookrightarrow \mathrm{Spec}\,(k[[t]])$$

where each step remembers one more order of infinitesimal information, and the formal arc $\mathrm{Spec}\,(k[[t]])$ is the limit, retaining all orders simultaneously.

3.3.4 Key Examples

The definition of Z-valued points subsumes many classical constructions. We now develop the most important instances in detail.

Example 3.3.5: Geometric Points

Let X be a scheme locally of finite type over a field k, and let $\overline{k}$ be an algebraic closure. A $\overline{k}$-valued point is a k-morphism

$$\varphi : \operatorname{Spec}(\overline{k}) \longrightarrow X.$$

Since $\operatorname{Spec}(\overline{k})$ has a unique point η, the morphism φ is determined by the image $y = \varphi(\eta) \in X$ together with the stalk map $\varphi_y^\# : \mathcal{O}_{X,y} \to \overline{k}$. Passing to the residue field, we obtain a k-embedding $\kappa(y) \hookrightarrow \overline{k}$.

We claim that y must be a closed point. Indeed, if y were not closed, then $\kappa(y)$ would contain an element transcendental over k; say t. In particular, as $\varphi_y(t) \in \overline{k}$ must be algebraic, there exists a k-polynomial p such that $\varphi_y(p(t)) = p(\varphi_y(t)) = 0$. As $\varphi_y(0) = 0$ and t is transcendental, $p(t) \neq 0$, demonstrating the failure of injectivity—a contradiction, since field homomorphisms are injective. Conversely, if y is closed, then $\kappa(y)/k$ is a finite extension (by Zariski's Lemma, since X is locally of finite type), and any finite extension of k embeds into $\overline{k}$. Thus, y must be closed, and every closed point arises in this way.

However, there may be *more* $\overline{k}$-valued points than closed points. For example, the $\mathbb{R}$-scheme $\operatorname{Spec}(\mathbb{R}[x]/(x^2+1)) \cong \operatorname{Spec}(\mathbb{C})$ has a single closed point, but two $\mathbb{C}$-valued points: these correspond to the two $\mathbb{R}$-algebra homomorphisms $\mathbb{C} \to \mathbb{C}$ (namely the identity and complex conjugation). In general, the number of $\overline{k}$-valued points lying over a closed point y equals the number of k-embeddings $\kappa(y) \hookrightarrow \overline{k}$, which is $[\kappa(y) : k]$ when $\kappa(y)/k$ is separable.

To obtain a bijection, one can either base-change to $\overline{k}$ (replacing X by $X_{\overline{k}} = X \times_k \operatorname{Spec}(\overline{k})$, which identifies all the Galois conjugates; see section 3.2), or work directly over an algebraically closed field. In the latter case, $\kappa(y) = \overline{k}$ for every closed point, and the correspondence

$$\left\{\operatorname{Spec}(\overline{k}) \to X\right\} \xrightarrow{\sim} \{y \in X : y \text{ is closed}\}$$

is bijective. This recovers the classical variety associated to X: the "geometric points" are exactly the closed points over an algebraically closed field.

Example 3.3.6: Generic Points and Rational Maps

Let C be a curve over $\mathbb{C}$ with function field $K = \mathbb{C}(C)$. A K-valued point of a $\mathbb{C}$-scheme X is a $\mathbb{C}$-morphism $\operatorname{Spec}(K) \to X$. Since $\operatorname{Spec}(K)$ is the generic point of C, such a morphism amounts to a rational map $C \dashrightarrow X$: it is defined on some dense open subset of C, but may not extend to all of C.

More concretely, taking $K = \mathbb{C}(t)$ (the function field of $\mathbb{A}^1_{\mathbb{C}}$), any $\mathbb{C}(t)$-valued point fits into the commutative diagram:

$$\begin{tikzcd} \operatorname{Spec} \mathbb{C}(t) \arrow[rr] \arrow[dr] & & X \arrow[dl] \\ & \operatorname{Spec} \mathbb{C} & \end{tikzcd}$$

This is the algebro-geometric analogue of a "generic curve" in X: rather than picking out a single closed point, it describes a one-parameter family of points parametrized by the generic point of the affine line. Whether such a rational map extends to a regular morphism $\mathbb{A}^1_{\mathbb{C}} \to X$ (or $C \to X$ in general) is precisely the content of the valuative criteria discussed in section 3.5.1 and section 3.5.4.

Example 3.3.7: Tangent Vectors and the Dual Numbers

Let $D = k[\epsilon]/(\epsilon^2)$ be the ring of dual numbers. The scheme $\operatorname{Spec}(D)$ is a "fat point": it has a unique closed point $\overline{o}$ (corresponding to the maximal ideal (ϵ)), but its structure sheaf remembers one infinitesimal direction. The canonical surjection $D \twoheadrightarrow k$ corresponds to a closed immersion $\iota : \operatorname{Spec}(k) \hookrightarrow \operatorname{Spec}(D)$, and we write $o = \iota^{-1}(\overline{o})$.

Let X be a k-scheme of finite type and let $x \in X$ be a closed point (so $\kappa(x) = k$). We classify all D-valued points of X based at x, namely all morphisms $\varphi : \operatorname{Spec}(D) \to X$ such that $\varphi \circ \iota$ sends o to x. Choose an affine open neighborhood $x \in \operatorname{Spec}(R) \subseteq X$; then $\operatorname{Hom}_x(\operatorname{Spec}(D), X) \cong \operatorname{Hom}_x(\operatorname{Spec}(D), \operatorname{Spec}(R))$, where the subscript x indicates that the closed point maps to x. By the equivalence of affine schemes and rings, these are in bijection with k-algebra homomorphisms $f : R \to D$ satisfying $f(\mathfrak{m}_x) \subseteq (\epsilon)$.

Now R decomposes as a k-vector space as $R = k \oplus \mathfrak{m}_x$. Indeed, as x is closed,

$$0 \to \mathfrak{m}_x \hookrightarrow R \twoheadrightarrow R/\mathfrak{m}_x \cong k \to 0$$

and as k is free this splits. Then f is the identity on k, so f is determined by its restriction to $\mathfrak{m}_x$. Since $\epsilon^2 = 0$ in D, we have $f(\mathfrak{m}_x^2) = 0$, and so f descends to a k-linear functional on $\mathfrak{m}_x/\mathfrak{m}_x^2$. Conversely, any linear functional $\ell : \mathfrak{m}_x/\mathfrak{m}_x^2 \to k$, extended by zero on $\mathfrak{m}_x^2$ and by the identity on k, determines a k-algebra map $R \to D$ sending $\mathfrak{m}_x$ into (ϵ). We conclude:

$$\operatorname{Hom}_x(\operatorname{Spec}(D),\, X) \;\cong\; (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee = \Theta_{X,x},$$

the Zariski tangent space at x. This is a satisfying realization: the tangent space, originally defined via the maximal ideal and its square (cf. section 2.7), is recovered purely by probing X with the fat point $\operatorname{Spec}(D)$.

Moreover, if $f : X \to Y$ is a morphism of k-schemes, then post-composition with f sends a D-valued point based at x to one based at $f(x)$:

$$\mathrm{d}_x f :\; \Theta_{X,x} \longrightarrow \Theta_{Y,f(x)}, \qquad v \longmapsto f \circ v.$$

This is the differential of f, now defined without any reference to local rings or Kähler differentials—it is simply the functoriality of D-valued points.

Example 3.3.8: Group Schemes via Valued Points

By Proposition 3.3.4, a group scheme G over k can equivalently be described as a scheme such that for every k-scheme Z, the set $G(Z)$ carries a group structure, and all maps $G(Z) \to G(Z')$ induced by morphisms $Z' \to Z$ are group homomorphisms. In other words, G is a "group object" in the functor-of-points sense: the functor $h_G : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ factors through the forgetful functor $\mathbf{Grp} \to \mathbf{Set}$.

For a concrete instance, consider the multiplicative group scheme $\mathbb{G}_m = \operatorname{Spec}(k[t, t^{-1}])$. For any k-algebra A:

$$\mathbb{G}_m(A) = \operatorname{Hom}_{k\text{-}\mathbf{Alg}}(k[t, t^{-1}],\, A) \;\cong\; A^\times.$$

A k-algebra homomorphism out of $k[t, t^{-1}]$ is determined by the image of t, and invertibility of t in the source forces the image to be a unit in A. Since the units of any ring form a group under multiplication, and ring homomorphisms preserve units, $\mathbb{G}_m(A)$ is a group for every A, functorially in A.

Similarly, the additive group scheme $\mathbb{G}_a = \operatorname{Spec}(k[t]) = \mathbb{A}^1_k$ satisfies $\mathbb{G}_a(A) = A$ with its additive group structure; and the general linear group $\mathrm{GL}_{n,k} = \operatorname{Spec}(k[x_{ij}, \det^{-1}])$ satisfies $\mathrm{GL}_{n,k}(A) \cong \mathrm{GL}_n(A)$ for any k-algebra A. These examples demonstrate a recurring theme: when working over a non-algebraically-closed base, or with non-reduced schemes, the functor of points often gives a cleaner description than the underlying topological space. One *defines* the algebraic group by specifying what it does on all test rings, and the scheme structure is a consequence.

3.3.5 Representable Functors in Algebraic Geometry

The examples of the previous subsection probe a fixed scheme X by varying the test object Z. We now reverse the perspective: fix the type of geometric data we wish to extract, and ask whether there exists a "universal" scheme that classifies it. Formally, given a contravariant functor $\mathcal{F} : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$, we say $\mathcal{F}$ is *representable* if there exists a scheme M and a natural isomorphism $\mathcal{F} \cong h_M$. The scheme M is then called a *fine moduli space* for the problem encoded by $\mathcal{F}$, and by the Yoneda Lemma it is unique up to unique isomorphism.

We have already encountered the simplest instance of this idea.

Proposition 3.3.9: Global Sections and the Affine Line

Let X be a k-scheme. There is a natural bijection

$$\Gamma(X, \mathcal{O}_X) \;\cong\; \operatorname{Hom}_{\mathbf{Sch}_k}(X,\, \mathbb{A}^1_k).$$

Concretely, a global section $f \in \Gamma(X, \mathcal{O}_X)$ corresponds to the unique morphism $\pi_f : X \to \mathbb{A}^1_k$ such that $\pi_f^\#(t) = f$, where t is the coordinate on $\mathbb{A}^1_k = \operatorname{Spec}(k[t])$. In categorical language, the global section functor $\Gamma(-, \mathcal{O})$ is representable by $\mathbb{A}^1_k$.

***Proof*:**
By the adjunction between Spec and Γ (proposition 3.1.12):

$$\operatorname{Hom}_{\mathbf{Sch}_k}(X,\, \operatorname{Spec} k[t]) \;\cong\; \operatorname{Hom}_{k\text{-}\mathbf{Alg}}(k[t],\, \Gamma(X, \mathcal{O}_X)).$$

The polynomial ring $k[t]$ is the free k-algebra on one generator: a k-algebra homomorphism $k[t] \to R$ is uniquely determined by the image of t. Thus:

$$\operatorname{Hom}_{k\text{-}\mathbf{Alg}}(k[t],\, \Gamma(X, \mathcal{O}_X)) \;\cong\; \Gamma(X, \mathcal{O}_X).$$

Let us verify that the bijection matches our intuition of evaluation. Let $x \in X$ and let $\pi_f : X \to \mathbb{A}^1_k$ be the morphism corresponding to $f \in \Gamma(X, \mathcal{O}_X)$. The definition of a scheme morphism gives a commutative diagram of local rings:

$$\begin{tikzcd} k[t] \arrow[r, "\psi"] \arrow[d] & \Gamma(X, \mathcal{O}_X) \arrow[r] & \mathcal{O}_{X,x} \arrow[d, "\text{residue}"] \\ \mathcal{O}_{\mathbb{A}^1, \pi_f(x)} \arrow[rr] & & \kappa(x) \end{tikzcd}$$

The composition across the top sends $t \mapsto f \mapsto f(x) \in \kappa(x)$, and the image point $\pi_f(x)$ is the prime ideal $\mathfrak{q} = \{P(t) \in k[t] : P(f(x)) = 0 \text{ in } \kappa(x)\}$. This is exactly the scheme-theoretic "value" of f at x.

3.3.10 Remark: no field hypothesis is needed The above proof uses only that $k[t]$ is free on one generator; it did not use that k is a field. For any base scheme S, the relative affine line $\mathbb{A}^1_S$ represents the global section functor on $\mathbf{Sch}_S$. The same remark applies to several of the examples that follow.

The representability of $\Gamma(-, \mathcal{O})$ by $\mathbb{A}^1_k$ is the first instance of a broad pattern. Let us now survey the most important representable functors in algebraic geometry. In each case, a piece of geometric data attached to a scheme X is captured by morphisms from X into a single "classifying" scheme.

The multiplicative group. Replacing the additive structure of $\mathbb{A}^1_k$ with the multiplicative structure of $\mathbb{G}_m = \operatorname{Spec}(k[t, t^{-1}])$ gives:

$$\operatorname{Hom}_{\mathbf{Sch}_k}(X, \mathbb{G}_m) \;\cong\; \Gamma(X, \mathcal{O}_X)^{\times}.$$

The proof follows the same pattern: a k-algebra map $k[t, t^{-1}] \to R$ is determined by the image of t, and invertibility of t in the source forces the image to be a unit. The $\mathbb{G}_m$-valued points of X are precisely the invertible global sections.

Projective space and line bundles. The representability of projective space is quite interesting. Define the functor

$$\mathcal{F}(X) = \{(\mathcal{L}, s_0, \dots, s_n) \mid \mathcal{L} \in \operatorname{Pic}(X),\ s_i \in \Gamma(X, \mathcal{L}),\ \text{the } s_i \text{ generate } \mathcal{L}\} /\cong$$

where two tuples $(\mathcal{L}, s_0, \dots, s_n)$ and $(\mathcal{L}', s'_0, \dots, s'_n)$ are identified if there is an isomorphism $\mathcal{L} \xrightarrow{\sim} \mathcal{L}'$ carrying s_i to s'_i for each i. Then there is a natural bijection

$$\operatorname{Hom}_{\mathbf{Sch}_k}(X, \mathbb{P}^n_k) \;\cong\; \mathcal{F}(X).$$

A morphism to $\mathbb{P}^n$ is the same data as a line bundle equipped with $n+1$ globally generating sections; these sections serve as "homogeneous coordinates" for the map. This is the content of the theory developed in section 5.3.3. When $n = 1$, a map $X \to \mathbb{P}^1$ corresponds to a line bundle with two generating sections—the algebro-geometric analogue of a meromorphic function.

Grassmannians. The Grassmannian $\operatorname{Gr}(r, n)$ generalizes projective space from lines to r-dimensional subspaces. It represents the functor

$$\mathcal{F}(X) = \left\{\mathcal{O}_X^{\oplus n} \twoheadrightarrow \mathcal{E} \;\middle|\; \mathcal{E} \text{ is a locally free sheaf of rank } r\right\} /\cong.$$

A morphism $X \to \operatorname{Gr}(r, n)$ classifies a rank-r quotient bundle of the trivial bundle $\mathcal{O}_X^{\oplus n}$, or equivalently, a sub-bundle of rank $n - r$. When $r = 1$, this recovers the description of projective space above.

The Grassmannian is itself a smooth projective variety; its functor-of-points description gives a uniform construction that works over any base ring, without the need to choose coordinates or Plücker embeddings.

The general linear group. The group scheme $\mathrm{GL}_{n,k} = \mathrm{Spec}\,(k[x_{11}, \ldots, x_{nn}, \det^{-1}])$ represents the functor

$$\mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{Spec}\,(A), \mathrm{GL}_{n,k}) \;\cong\; \mathrm{GL}_n(A)$$

for any k-algebra A. A map into $\mathrm{GL}_{n,k}$ is the same as giving an invertible $n \times n$ matrix with entries in A. Combined with the group scheme structure from Example 3.3.8, this gives a definition of the general linear group that works uniformly over all base rings, including rings with nilpotents or mixed characteristic.

Hilbert schemes. Perhaps the most application of representability is given by the Hilbert scheme, which we develop in the next subsection.

The common thread in each of the above examples is the Yoneda philosophy: a geometric invariant of X—global sections, line bundles, quotient bundles, invertible matrices—is captured by morphisms from X into a single classifying scheme. The Yoneda Lemma ensures that the classifying scheme, when it exists, is unique up to unique isomorphism.

3.3.6 Hilbert Schemes

The representable functors considered so far classify data that is "linear" in nature: sections of a sheaf, quotients of a free module, invertible matrices. The Hilbert scheme is of a fundamentally different character: it parametrizes *subschemes* of a given projective scheme. That such a moduli problem admits a scheme as its solution is one of the most important results in the functorial approach to algebraic geometry, and it provides a rich source of examples and applications.

Flatness as the continuity condition

Before stating the result, we must address a foundational question: what does it mean for a collection of subschemes to "vary continuously" over a parameter space? Consider a family of closed subschemes $\{Z_s\}_{s \in S}$ of a fixed projective scheme $Y \subseteq \mathbb{P}^n_k$, parametrized by a scheme S. Algebraically, this family is encoded by a closed subscheme $Z \subseteq Y \times_k S$ such that each fiber $Z_s = Z \times_S \{s\}$ is the corresponding member of the family.

Not every such Z should be considered a "nice" family. For instance, one could construct a family where a curve degenerates to a union of a curve and an isolated point, causing the Hilbert polynomial to jump. The condition that prevents such pathologies is *flatness*: we require that Z be flat over S. Flatness ensures that the Hilbert polynomial $P_{Z_s}(m) = \chi(Z_s, \mathcal{O}_{Z_s}(m))$ is constant as s varies over connected components of S. This is the correct algebraic analogue of "continuous variation": the members of the family all share the same discrete numerical invariants.

The reader who has not yet encountered flatness in detail should think of it as a homological condition: a morphism $\pi : Z \to S$ is flat if the fibers Z_s do not "jump" in any cohomological sense. A thorough treatment will be given in section 6.8.

The Hilbert functor and Grothendieck's theorem

Fix a projective scheme $Y \subseteq \mathbb{P}^n_k$ and a numerical polynomial $p \in \mathbb{Q}[t]$ (recall that the Hilbert polynomial of any coherent sheaf on projective space is a polynomial with rational coefficients). The *Hilbert functor* is the contravariant functor $\mathcal{H}ilb^p_Y : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ defined by

$$\mathcal{H}ilb^p_Y(S) = \left\{ Z \subseteq Y \times_k S \;\middle|\; \begin{array}{l} Z \text{ is a closed subscheme, flat over } S, \\ \text{with Hilbert polynomial } p \text{ on each fiber} \end{array} \right\}.$$

For a morphism $f : S' \to S$, the induced map $\mathcal{H}ilb^p_Y(S) \to \mathcal{H}ilb^p_Y(S')$ sends a family $Z \subseteq Y \times S$ to its pullback $Z \times_S S' \subseteq Y \times S'$.

Theorem 3.3.11: Grothendieck's Existence Theorem

Let $Y \subseteq \mathbb{P}^n_k$ be a projective scheme over a Noetherian ring k, and let $p \in \mathbb{Q}[t]$ be a numerical polynomial. Then the Hilbert functor $\mathcal{H}ilb^p_Y$ is representable by a projective scheme Hilb^p_Y over k:

$$\mathrm{Hom}_{\mathbf{Sch}_k}(S, \mathrm{Hilb}^p_Y) \;\cong\; \mathcal{H}ilb^p_Y(S)$$

naturally in S.

The proof is beyond our scope (it relies on a careful analysis of Castelnuovo–Mumford regularity and the Grassmannian embedding), but the statement is worth absorbing. It says that the totality of all subschemes of Y with a given Hilbert polynomial is itself a geometric object—a projective scheme—on which one can do geometry. There is a *universal family*: a closed subscheme $\mathcal{Z} \subseteq Y \times \mathrm{Hilb}^p_Y$, flat over Hilb^p_Y, such that every flat family $Z \subseteq Y \times S$ with Hilbert polynomial p is the pullback of $\mathcal{Z}$ along a unique morphism $S \to \mathrm{Hilb}^p_Y$.

Example: the Hilbert scheme of points

The simplest interesting case arises when p is the constant polynomial $p(m) = n$ for a positive integer n. Then a closed subscheme $Z \subseteq Y$ with Hilbert polynomial $p(m) = n$ is a zero-dimensional subscheme of length n: it is supported on at most n points of Y, possibly with multiplicity (i.e., with nilpotent structure). The corresponding Hilbert scheme is denoted $\mathrm{Hilb}^n(Y)$ and is called the *Hilbert scheme of n points* on Y.

When Y is a smooth surface, $\mathrm{Hilb}^n(Y)$ turns out to be a smooth variety of dimension $2n$. This is a nontrivial theorem (due to Fogarty) and stands in contrast to the naive expectation: the "symmetric product" $\mathrm{Sym}^n(Y) = Y^n/S_n$, which parametrizes unordered n-tuples of points, is singular along the diagonals where points collide. The Hilbert scheme resolves these singularities by remembering the infinitesimal directions in which colliding points approach each other.

To see this concretely, consider $Y = \mathbb{A}^2_k$ and $n = 2$. A length-2 subscheme $Z \subseteq \mathbb{A}^2$ is one of two types:

1. Two distinct points $\{p, q\}$, corresponding to the ideal $I_p \cap I_q \subseteq k[x, y]$ where I_p and I_q are the maximal ideals of p and q. The quotient $k[x, y]/I$ has dimension 2 as a k-vector space.
2. A single point p with a tangent direction $v \in T_p\mathbb{A}^2$. If $p = (0, 0)$ and v points in the x-direction, the corresponding ideal is (y, x^2), and $k[x, y]/(y, x^2) \cong k[x]/(x^2)$ has dimension 2. The extra nilpotent direction records "how the two points came together."

Thus $\mathrm{Hilb}^2(\mathbb{A}^2)$ parametrizes pairs of distinct points together with a "blow-up" of the diagonal: at each point of the diagonal, the fiber is a $\mathbb{P}^1$ recording the tangent direction of approach. This is precisely the blow-up of $\mathrm{Sym}^2(\mathbb{A}^2)$ along its singular locus.

Tangent spaces to the Hilbert scheme

The functor-of-points perspective makes the computation of tangent spaces to moduli spaces almost mechanical. By Example 3.3.7, a tangent vector to Hilb^p_Y at a point $[Z]$ (corresponding to a subscheme $Z \subseteq Y$) is a morphism $\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2)) \to \mathrm{Hilb}^p_Y$ sending the closed point to $[Z]$. By the universal property, this is the same as a flat family of subschemes of Y over $\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))$ whose special fiber is Z. One can show that such a family corresponds to a first-order deformation of Z inside Y, and that:

$$\Theta_{\mathrm{Hilb}^p_Y,\,[Z]} \;\cong\; \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Z,\, \mathcal{O}_Z)$$

where $\mathcal{I}_Z$ is the ideal sheaf of Z in Y. This identification—tangent vectors to the Hilbert scheme are the same as homomorphisms from the ideal sheaf to the structure sheaf of the subscheme—is a prototype for the general principle that tangent spaces to moduli spaces are computed by Hom or Ext groups.

3.3.7 Schemes as Functors

The reader may have noticed an apparent circularity in the functor-of-points perspective. We define the functor $h_X : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ using the category **Sch** of schemes, but if the functor of points is to serve as a *definition* of schemes, we seem to need the category **Sch** already in hand.

The resolution, due to Grothendieck and developed systematically by Demazure and Gabriel, proceeds in stages and is worth spelling out.

Step 1: Start with rings. The category **CRing** of commutative rings is purely algebraic. We define the category of *affine schemes* as $\mathbf{Aff} = \mathbf{CRing}^{\mathrm{op}}$, the formal opposite category. No geometry has been invoked.

Step 2: Embed into functors. Any functor $F : \mathbf{CRing} \to \mathbf{Set}$ (equivalently, a presheaf on **Aff**) can be thought of as a "generalized space." An affine scheme $\mathrm{Spec}\,(A)$ defines such a functor by $h_{\mathrm{Spec}\,(A)}(R) = \mathrm{Hom}_{\mathbf{CRing}}(A, R)$. By the Yoneda Lemma, the assignment $\mathrm{Spec}\,(A) \mapsto h_{\mathrm{Spec}\,(A)}$ is a fully faithful embedding $\mathbf{Aff} \hookrightarrow \mathrm{Fun}(\mathbf{CRing}, \mathbf{Set})$.

Step 3: Impose a sheaf condition. Not every functor $\mathbf{CRing} \to \mathbf{Set}$ deserves to be called geometric. We require that F be a *sheaf* for the Zariski topology. Concretely, if $f_1, \ldots, f_n$ generate the unit ideal in a ring A (so that $\mathrm{Spec}\,(A) = \bigcup_i D(f_i)$ is a cover by distinguished open sets), then the diagram

$$F(A) \longrightarrow \prod_i F(A_{f_i}) \rightrightarrows \prod_{i,j} F(A_{f_i f_j})$$

should be an equalizer. This is the local-to-global property that geometric objects possess: compatible data on an open cover should glue uniquely. A sheaf satisfying this property is called a *Zariski sheaf.*

Step 4: Require an open cover by affines. A *scheme* is a Zariski sheaf F that admits an open cover by representable subfunctors. That is, there exist rings A_i and natural transformations $h_{\mathrm{Spec}\,(A_i)} \hookrightarrow F$ that are "open immersions" (in the sense defined below) and that jointly cover F.

This definition recovers the usual category of schemes without circularity: we begin with rings, pass to representable functors (affine schemes), impose a sheaf condition (gluing), and characterize schemes as those sheaves admitting an affine open cover. The result is a full subcategory of $\mathrm{Fun}(\mathbf{CRing}, \mathbf{Set})$ defined by intrinsic conditions.

The practical payoff is significant. To define a scheme, it suffices to specify a functor $\mathcal{F} : \mathbf{CRing} \to \mathbf{Set}$ and verify the Zariski sheaf condition and the existence of an affine open cover. This is often considerably easier than constructing a locally ringed space by hand, especially for moduli problems: the Hilbert scheme was originally constructed by Grothendieck precisely by first defining the functor $\mathcal{H}ilb^p_Y$ and then proving it satisfies the above conditions. The same strategy applies to Picard schemes, moduli of curves, and many other classifying spaces that arise in algebraic geometry.

Open subfunctors

Step 4 above relies on a notion of "open immersion" for functors, which we now make precise. Let $F : \mathbf{CRing} \to \mathbf{Set}$ be a Zariski sheaf. A subfunctor $G \subseteq F$ (meaning $G(A) \subseteq F(A)$ for every ring A, compatibly with restriction maps) is called an *open subfunctor* if the following condition holds: for every ring A and every element $\xi \in F(A)$, there exist elements $f_1, \ldots, f_n \in A$ generating the unit ideal such that, for each i, the image of ξ in $F(A_{f_i})$ lies in $G(A_{f_i})$.

Informally, G is open in F if membership in G can be "checked locally." To see why this is the right definition, consider the representable case. If $F = h_{\mathrm{Spec}\,(B)}$ and G corresponds to an open subscheme $U \subseteq \mathrm{Spec}\,(B)$, then a morphism $\xi : \mathrm{Spec}\,(A) \to \mathrm{Spec}\,(B)$ factors through U if and only if its image lands in U. The preimage $\xi^{-1}(U)$ is an open subset of $\mathrm{Spec}\,(A)$, and the condition that $f_1, \ldots, f_n$ generate the unit ideal with $\xi|_{D(f_i)}$ landing in U says exactly that $\xi^{-1}(U)$ covers $\mathrm{Spec}\,(A)$—i.e., ξ factors through U.

A collection of open subfunctors $\{G_i \subseteq F\}$ is said to *cover* F if for every field K, the sets $G_i(K)$ jointly cover $F(K)$. The condition in Step 4 can now be restated: a Zariski sheaf F is a scheme if it admits a cover by open subfunctors, each of which is representable by an affine scheme.

Tangent spaces to functors

One of the practical advantages of the functorial viewpoint is that many geometric constructions can be performed directly on functors, without first knowing that the functor is representable. We have already seen in Example 3.3.7 that the tangent space to a scheme X at a k-rational point x is recovered by probing with the dual numbers $D = k[\epsilon]/(\epsilon^2)$. This construction generalizes immediately to arbitrary functors.

Definition 3.3.12: Tangent Space of a Functor

Let $F : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ be a functor and let $\xi_0 \in F(\mathrm{Spec}\,(k))$ be a k-rational point of F. The *tangent space* of F at ξ_0 is

$$T_{\xi_0} F := \left\{ \xi \in F(\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))) \;\middle|\; \xi \text{ maps to } \xi_0 \text{ under } F(\mathrm{Spec}\,(k) \hookrightarrow \mathrm{Spec}\,(D)) \right\}.$$

When $F = h_X$ for a scheme X and ξ_0 corresponds to a closed point $x \in X$ with $\kappa(x) = k$, this recovers the Zariski tangent space $\Theta_{X,x}$ by Example 3.3.7. The key point is that the definition makes sense even when F is not representable.

For the tangent space to carry a k-vector space structure, one needs the functor F to satisfy a mild condition: $T_{\xi_0}F$ should be compatible with the k-module structure on square-zero extensions of k. This is automatic for representable functors, and holds for most functors arising in practice (including the Hilbert functor and the Picard functor).

As an application, we computed the tangent space of the Hilbert functor at the end of section 3.3.6: for a subscheme $Z \subseteq Y$ with ideal sheaf $\mathcal{I}_Z$, the tangent space $T_{[Z]} \operatorname{Hilb}_Y^p \cong \operatorname{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Z, \mathcal{O}_Z)$. This computation was carried out purely at the level of the functor $\mathcal{H}ilb_Y^p$; the fact that Hilb_Y^p is actually a scheme (Theorem 3.3.11) was not needed for the computation itself, only for the conclusion that this tangent space has geometric meaning.

3.3.8 Moduli Problems and the Limits of Representability

The Hilbert scheme demonstrates the power of the functorial approach: define a moduli functor, prove it is representable, and obtain a geometric parameter space for free. A natural question is how far this method extends. Can every reasonable classification problem in algebraic geometry be solved by a representing scheme?

The answer is no, and understanding *why* it fails is as instructive as the successes.

The moduli problem for elliptic curves

An *elliptic curve* over a field k is a smooth projective curve of genus 1 equipped with a distinguished rational point (the reader unfamiliar with this notion may consult section 7.1.7 and return here afterward). Two elliptic curves are isomorphic if and only if they have the same j-invariant, an element of k.

We would like a "moduli space of elliptic curves": a scheme M whose points correspond to isomorphism classes of elliptic curves, and such that families of elliptic curves over a base S correspond to morphisms $S \to M$. The functor to be represented is:

$$\mathcal{M}_{1,1}(S) = \left\{ (E \to S,\, \sigma : S \to E) \;\middle|\; \begin{array}{l} E \to S \text{ is a smooth proper morphism} \\ \text{with genus-1 geometric fibers,} \\ \sigma \text{ is a section} \end{array} \right\} /{\cong_S}$$

where $\cong_S$ denotes isomorphism of families over S. If this functor were representable by a scheme M, there would exist a *universal family*: an elliptic curve $\mathcal{E} \to M$ such that every family $E \to S$ is the pullback of $\mathcal{E}$ along a unique morphism $S \to M$.

Over an algebraically closed field, the isomorphism classes of elliptic curves are classified by the j-invariant, so the underlying set of M should be $\mathbb{A}^1$. Indeed, there is a scheme—the j-line $\mathbb{A}^1_k$—that serves as a *coarse moduli space*: its closed points are in bijection with isomorphism classes, and for any family $E \to S$ there exists a unique morphism $S \to \mathbb{A}^1$ whose fiber over a point s has j-invariant equal to the j-invariant of E_s. But the j-line is *not* a fine moduli space. The functor $\mathcal{M}_{1,1}$ is not representable.

The failure is easy to spot: every elliptic curve E over a field of characteristic $\neq 2$ has a nontrivial automorphism, namely it has at least the inversion map $\iota : E \to E$ sending $P \mapsto -P$ (with respect to the group law). For the curves with $j = 0$ and $j = 1728$, the automorphism group is even larger ($\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$, respectively). These automorphisms are the obstruction to representability.

To see why concretely, suppose a fine moduli space M existed with universal family $\mathcal{E} \to M$. Consider the trivial family $E \times S \to S$ for a fixed elliptic curve E with nontrivial automorphism α. The identity map $\mathrm{id} : E \times S \to E \times S$ and the automorphism $\alpha \times \mathrm{id}_S : E \times S \to E \times S$ are *different* isomorphisms of the family, but they give rise to the *same* classifying morphism $S \to M$ (since they classify the same family up to isomorphism). This violates the requirement that the classifying morphism be unique.

More precisely, the issue is that the functor $\mathcal{M}_{1,1}$ does not satisfy the sheaf condition in sufficient generality: it fails descent for étale covers. A family that is locally trivial in the étale topology need not be globally trivial, and the automorphisms of the fibers obstruct the gluing.

Coarse moduli spaces

The j-line is an example of a *coarse moduli space*: a scheme M equipped with a natural transformation $\mathcal{M}_{1,1} \to h_M$ that is universal among maps to representable functors, and that induces a bijection $\mathcal{M}_{1,1}(\mathrm{Spec}\,(\overline{k})) \xrightarrow{\sim} M(\overline{k})$ on geometric points. A coarse moduli space classifies objects up to isomorphism at the level of points, but does not carry a universal family and does not faithfully track families over general bases.

Beyond schemes: stacks

The correct solution is to enlarge the category of geometric objects. Rather than forcing $\mathcal{M}_{1,1}(S)$ to be a *set* (of isomorphism classes), one retains it as a *groupoid*: the category whose objects are families $E \to S$ and whose morphisms are isomorphisms of families. The resulting structure is an *algebraic stack*, often denoted $\mathcal{M}_{1,1}$ as well. The stack remembers the automorphisms of each object, and this additional data resolves the descent obstruction.

An intermediate notion is that of an *algebraic space*. These arise when the moduli problem has no nontrivial automorphisms but the functor still fails to be representable by a scheme—for instance, because the "correct" topology for gluing is the étale topology rather than the Zariski topology. Algebraic spaces are Zariski-locally schemes but may not be globally so; they are to schemes what manifolds are to open subsets of $\mathbb{R}^n$.

The hierarchy, in increasing generality, is:

$$\text{Schemes} \subset \text{Algebraic Spaces} \subset \text{Algebraic Stacks} \subset \text{Higher Stacks},$$

where each step relaxes a condition on the moduli functor. Schemes require representability by a Zariski sheaf with an affine cover. Algebraic spaces replace the Zariski topology with the étale topology. Algebraic stacks allow the functor to take values in groupoids rather than sets. The development of this hierarchy is beyond our scope, but the reader should be aware that the functor-of-points perspective developed in this section is the entry point to this hierarchy: every step in it is formulated in terms of functors and their properties.

3.3.9 *The Fourier–Mukai Transform

The functor-of-points philosophy associates to a scheme X the collection of all morphisms into X. One can push this idea further by asking: what information is carried not by morphisms, but by *sheaves* on products? This line of thought leads to the Fourier–Mukai transform, which can be understood as a "categorified" version of the change-of-basis perspective developed above. This subsection is intended as a brief orientation for the interested reader; the full theory belongs to the realm of derived categories and will not be developed in detail here.

From graphs to correspondences

Given a morphism $f : X \to Y$, its *graph* $\Gamma_f = \{(x, f(x)) \mid x \in X\}$ is a closed subscheme of $X \times Y$. The morphism f is entirely recoverable from Γ_f: project to X (an isomorphism) and then to Y.

A natural generalization is to drop the requirement that the projection to X be an isomorphism. Any closed subscheme $Z \subseteq X \times Y$—or, more generally, any sheaf $\mathcal{P}$ on $X \times Y$—defines a "correspondence" between X and Y. For a point $x \in X$, the fiber $Z_x = Z \cap (\{x\} \times Y)$ is a subscheme of Y, and as x varies, one obtains a family of subschemes of Y parametrized by X. This is precisely the kind of data that Hilbert schemes classify.

A word on derived categories

To formulate the Fourier–Mukai transform precisely, one works not with the abelian category $\mathbf{Coh}(X)$ of coherent sheaves, but with its *bounded derived category* $D^b(X)$. The motivation is that many natural operations on sheaves—tensor products, pushforwards, global sections—are not exact, and their "corrections" (the higher derived functors Tor, $R^i f_*$, H^i) carry essential geometric information. The derived category is a framework in which a sheaf is replaced by a *complex* of sheaves, and these higher derived functors become part of a single, coherent operation.

An object of $D^b(X)$ is a bounded complex

$$\cdots \to \mathcal{F}^{i-1} \to \mathcal{F}^i \to \mathcal{F}^{i+1} \to \cdots$$

of coherent sheaves, considered up to *quasi-isomorphism*: two complexes are identified if they have isomorphic cohomology sheaves. A single sheaf $\mathcal{F}$ is regarded as a complex concentrated in degree zero. The derived category retains the information of the abelian category (via the cohomology functor H^i) but also remembers the "process" of resolution, which is essential for the functoriality that follows.

The reader familiar with the construction of Ext groups via projective or injective resolutions will recognize the underlying idea: in the derived category, one works with the resolutions themselves rather than discarding them after taking cohomology.

The transform

Let X and Y be smooth projective varieties over a field k, and let $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ denote the projection morphisms. Given a "kernel" $\mathcal{P} \in D^b(X \times Y)$, the *Fourier–Mukai transform* with kernel $\mathcal{P}$ is the functor

$$\Phi_{\mathcal{P}} : D^b(X) \longrightarrow D^b(Y), \qquad \mathcal{F} \longmapsto R\pi_{Y,*}(\mathcal{P} \otimes^L \pi_X^* \mathcal{F}),$$

where π_X^* is the (exact) pullback, $\otimes^L$ is the derived tensor product, and $R\pi_{Y,*}$ is the derived pushforward. In words: pull the sheaf $\mathcal{F}$ back from X to the product, "multiply" by the kernel $\mathcal{P}$, and "integrate" (push forward) along Y.

The analogy with the classical Fourier transform on $\mathbb{R}^n$ is worth making explicit. Recall that the classical transform sends a function f on $\mathbb{R}^n$ to a function $\hat{f}$ on the dual space $(\mathbb{R}^n)^*$ via

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)\, e^{2\pi i\langle x,\xi\rangle}\, dx.$$

The exponential $e^{2\pi i\langle x,\xi\rangle}$ is a "kernel" on the product $\mathbb{R}^n \times (\mathbb{R}^n)^*$; the multiplication by f is the analogue of tensoring with $\pi_X^*\mathcal{F}$; and the integration over $\mathbb{R}^n$ is the analogue of the derived pushforward $R\pi_{Y,*}$. The Fourier–Mukai transform is, in this precise sense, the algebraic geometer's Fourier transform, with coherent sheaves playing the role of functions and the derived pushforward replacing integration.

From delta-points to character-points

The classical Fourier transform admits a particularly illuminating description in terms of two distinguished "bases" for the space of functions on a locally compact abelian group G. The first basis consists of the *delta functions* δ_x for $x \in G$: any function can be written as $f = \int_G f(x)\, \delta_x\, dx$, expressing f in terms of where it is "supported." The second basis consists of the *characters* χ_ξ for $\xi \in \hat{G}$ (the Pontryagin dual group): the Fourier transform rewrites f as $\hat{f} = \int_G f(x)\, \chi_\xi(x)\, dx$, expressing f in terms of its "frequency content." The Fourier transform is, at its core, a change of basis from delta-points to character-points:

$$\widehat{\delta_x} = \chi_x, \qquad \widehat{\chi_\xi} = \delta_\xi.$$

The Fourier–Mukai transform makes this change of basis precise in algebraic geometry, and the natural setting is that of *abelian varieties*. Let A be an abelian variety of dimension g over k (a smooth projective group variety—the higher-dimensional generalization of an elliptic curve). Its *dual abelian variety* $\hat{A} = \mathrm{Pic}^0(A)$ parametrizes the degree-zero line bundles on A: a point $\xi \in \hat{A}$ corresponds to a line bundle L_ξ on A that is algebraically equivalent to zero. The dual $\hat{A}$ is the algebro-geometric analogue of the Pontryagin dual $\hat{G}$.

On the product $A \times \hat{A}$ there is a canonical line bundle $\mathcal{P}$, the *Poincaré bundle*, characterized by the universal property: for each $\xi \in \hat{A}$, the restriction $\mathcal{P}|_{A\times\{\xi\}} \cong L_\xi$. The Poincaré bundle is the algebro-geometric analogue of the exponential kernel $e^{2\pi i\langle x,\xi\rangle}$: it "pairs" a point of A with a character from $\hat{A}$.

The Fourier–Mukai transform $\Phi_\mathcal{P} : D^b(A) \to D^b(\hat{A})$ with kernel $\mathcal{P}$ now enacts the change of basis between delta-points and character-points. The two fundamental computations are:

Deltas map to characters. Let $a \in A$ be a closed point and let $k(a)$ denote the skyscraper sheaf at a (the sheaf supported at the single point a with stalk k). The skyscraper $k(a)$ is the algebro-geometric δ-function: it is "concentrated at a" and detects nothing elsewhere. Applying the transform:

$$\Phi_\mathcal{P}(k(a)) = R\pi_{\hat{A},*}(\mathcal{P} \otimes^L \pi_A^* k(a)) = R\pi_{\hat{A},*}(\mathcal{P}|_{\{a\}\times\hat{A}}) = \mathcal{P}|_{\{a\}\times\hat{A}}.$$

The restriction $\mathcal{P}|_{\{a\}\times\hat{A}}$ is a line bundle on $\hat{A}$. Under the double-duality identification $A \cong \hat{\hat{A}}$, this is precisely the "character" on $\hat{A}$ associated to the point a—just as the classical Fourier transform sends δ_x to the character χ_x.

Characters map to deltas. Let $\xi \in \hat{A}$ and let $L_\xi = \mathcal{P}|_{A\times\{\xi\}}$ be the corresponding degree-zero line bundle on A. The line bundle L_ξ is the algebro-geometric "character": tensoring with L_ξ defines a "twist" of sheaves on A, analogous to multiplication by χ_ξ. One computes:

$$\Phi_{\mathcal{P}}(L_\xi) \cong k(\xi)[-g]$$

where $k(\xi)$ is the skyscraper sheaf at the point $\xi \in \hat{A}$ and $[-g]$ denotes a cohomological shift by $g = \dim A$. Up to this shift (which accounts for the difference between underived and derived pushforward on a g-dimensional variety), the character L_ξ is sent to the delta-function $k(\xi)$ on the dual—exactly as $\widehat{\chi_\xi} = \delta_\xi$ in the classical setting.

The cohomological shift $[-g]$ is the price of working in the derived category. It appears because the higher direct images $R^i\pi_{\hat{A},*}$ of L_ξ vanish for $i \neq g$ (a consequence of the fact that L_ξ has no global sections for $\xi \neq 0$, and its cohomology is concentrated in degree g). The shift is harmless: it is an automorphism of $D^b(\hat{A})$, and Mukai's theorem accounts for it by asserting that the *composition*

$$D^b(A) \xrightarrow{\Phi_{\mathcal{P}}} D^b(\hat{A}) \xrightarrow{\Phi_{\mathcal{P}^\vee}} D^b(A)$$

is isomorphic to the shift functor $(-1)_A^*[-g]$, where $(-1)_A$ is the inversion on A. This is the precise analogue of the Fourier inversion formula $\hat{\hat{f}}(x) = f(-x)$.

Mukai's theorem and Orlov representability

The computations above are instances of a fundamental result due to Mukai: the Fourier–Mukai transform $\Phi_{\mathcal{P}} : D^b(A) \to D^b(\hat{A})$ is an *equivalence of categories*. This is a strong statement: it says that the derived category of coherent sheaves on an abelian variety is equivalent to that of its dual. Two geometrically distinct varieties—A and $\hat{A}$ need not be isomorphic—are "the same" from the perspective of their sheaf theory.

More broadly, Orlov's representability theorem asserts that for smooth projective varieties, every equivalence $D^b(X) \xrightarrow{\sim} D^b(Y)$ is isomorphic to a Fourier–Mukai transform $\Phi_{\mathcal{P}}$ for some kernel $\mathcal{P} \in D^b(X \times Y)$. This is the categorified Yoneda Lemma: just as a natural transformation $h_X \to h_Y$ is determined by a single element of $\mathrm{Hom}(X, Y)$, a sufficiently nice functor $D^b(X) \to D^b(Y)$ is determined by a single object of $D^b(X \times Y)$.

The analogy with the change-of-basis perspective from section 3.3.3 is clean. The Yoneda embedding probes X by mapping test objects *into* X; the Fourier–Mukai transform probes X by its category of sheaves. The kernel $\mathcal{P}$ on $X \times Y$ mediates between these two descriptions, playing the role of a "change-of-basis matrix" between two categorical coordinate systems. On abelian varieties, this change of basis is literally the passage from delta-points to character-points—the oldest and most fundamental duality in harmonic analysis, now transplanted into algebraic geometry.

3.4 Finiteness and Affine Conditions

As schemes are rather general notions, so to scheme morphisms can be very general. The following notions are guiding principles presented by Vakil on what a good type of morphism should have. This terminology will be very useful in labeling many of the following lemmas an propositions. Let's say that C is a collection of morphisms between schemes. Then:

1. it is *local on target* if

   (a) $\pi : X \to Y$ is in the class, then for any open subset $V \subseteq Y$, the restricted homomorphism $\pi^{-1}(V) \to V$ is in the class
   (b) $\pi : X \to Y$ is a morphism, $\cup_i V_i = Y$ is an open cover where each $\pi^{-1}(V_i) \to V_i$ is in this class, then π is in this class

   Usually we consider affine opens, but it need not be the case.

2. is *local on source* if there is
   (a) $\pi : X \to Y$ is in the class, then for any open subset $U \subseteq X$, the restricted homomorphism $U \to \pi(U)$ is in the class
   (b) $\pi : X \to Y$ is a morphism, $\cup_i U_i = X$ is an open cover where each $U_i \to \pi(U_i)$ is in this class, then π is in this class
3. it is closed under composition
4. it is closed under fibered products. As a special case, is closed under change of basis from an S-scheme to a T-scheme

Finiteness Conditions

Definition 3.4.1: Quasicompact Morphism

Let $\pi : X \to Y$ be a morphism of scheme. Then π is *quasicompact* if every open affine subset $U \subseteq Y$ has quasicompact pre-image, that is $\pi^{-1}(U)$ is quasicompact.

Naturally, Quasicompactness is is closed under composition, and is local on target.

Example 3.4.2: Quasicompact Morphism

We shall in this section show that other types of morphism shall be quasicompact (finite, embeddings, proper, etc). At the moment, we shall give concrete examples:

1. Any scheme morphism of affine schemes is quasicompact, this relies on the use of the distinguished open base: Let $f : \operatorname{Spec} S \to \operatorname{Spec} R$ and $U \subseteq \operatorname{Spec} R$ be quasi-compact. Then $U = \bigcup_i^n D(g_i)$ for appropriate distinguished open sets. Then since $f^{-1}(D(g_i)) = D(f^\#(g_i))$, we have that
$$f^{-1}(U) = \bigcup_i^n D(f^\#(g_i))$$
showing the pre-image is quasi-compact.
2. Let $X = \coprod_i^\infty \operatorname{Spec} k$ and $Y = \operatorname{Spec} k$. Then the natural projection $\pi : X \to Y$ is certainly not quasicompact, the pre-image of $\operatorname{Spec} k$ not being quasicompact.
3. Let X be Noetherian schemes and $f : X \to Y$ a scheme morphism. Then f is quasicompact. This follows from the following topological property: If X is a Noetherian topological space, every subset is Noetherian.

   Hence, any non-quasicompact map *must* have X as a non-Noetherian scheme, and so in practice we shall not often see non-quasicompact schemes.

4. Let $X = \operatorname{Spec} k[x]$ and $\tilde{X}$ be the blow up at every point. Then the natural map $\varphi : \tilde{X} \to X$ is not quasicompact.

For future reference, the following definition is put down (see ref:HERE for its application):

Definition 3.4.3: Quasiseparated Morphism

Let $\pi : X \to Y$ be a morphism of scheme. Then π is *quasiseparated* if every open affine subset $U \subseteq Y$ has quasicompact pre-image.

Example 3.4.4: Quasiseparated Morphism
We shall see examples of such morphisms later.

The next important type of morphism that will always have affine open pre-images:

Definition 3.4.5: Affine Morphism

Let $\pi : X \to Y$ be a morphism of schemes. Then π is *affine* if for every affine open subset $U \subseteq Y$, $\pi^{-1}(U)$ is an affine scheme (interpreted as an open subscheme of X)

Affine morphisms can be thought of as being the closest types of morphisms between schemes that "behave" like morphisms of affine schemes, namely since the pre-image of every affine is affine we can just cover in affine sets and study those without needing to "think" about the gluing structure. Proposition 3.4.10 shall show we only need to check a morphism is affine on a single affine cover, which is not immediately obvious as it is not *a priori* clear that we cannot "gerrymander" a partition to force affine pre-images on the open subsets of the cover.

3.4.6 Advanced Intuition This is for the reader already familiar with all this material. Let $f : X \to Y$ is a morphism. Then f is an affine morphisms if and only if there exists a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras $\mathcal{A}$ such that $X \cong \operatorname{Spec}_Y(\mathcal{A})$ where $\operatorname{Spec}_Y(\mathcal{A})$ is the relative spectrum with respect to S (this is a generalization of X as an S-algebra)

We shall give examples of affine morphisms after we prove some nice results, one important such example shall be that when the domain and codomain are affine, the morphism must be affine. Let us start by showing that if the domain/codomain are not affine, the morphism may not be affine:

Example 3.4.7: Non Affine Morphisms
The prototypical example of a non affine morphism is $\mathbb{P}^1_k \to \operatorname{Spec} k$ (i.e. the structure morphism). Then the pre-image is *not* an (as shown in proposition 2.3.6). Example of a non-affine scheme where the codomain is not affine is the embedding $\mathbb{A}^n_k \to \mathbb{P}^n_k$ for $\mathbb{A}^n_k$ embeds in a choice of U_i ($n = 1$ it is enough to just do $n = 1$ to see why).

Certainly affine morphisms are stable under composition, and are stable under base change. This condition would be nice to be affine-local on target. It takes some build-up:

Lemma 3.4.8: Affine Morphisms and Qcqs

Let $\pi : X \to Y$ be an affine morphism. Then it is quasicompact and quasiseparated.

***Proof*:**
The key is that *affine schemes* are quasi-separated (and quasicompact). By the definition of an affine morphism, for every affine open subset $V \subseteq Y$, the preimage $U = \pi^{-1}(V)$ is an affine scheme, say $U \cong \operatorname{Spec} A$.

1. *Quasicompactness:* Every affine scheme is quasicompact. Since the preimage of every affine open $V \subseteq Y$ is quasicompact, the morphism π is quasicompact.

2. *Quasiseparatedness:* A morphism is quasiseparated if the preimage of every affine open in Y is a quasiseparated scheme. Since $U = \operatorname{Spec} A$ is an affine scheme, it is separated (the diagonal $\Delta : U \to U \times_{\mathbb{Z}} U$ is a closed immersion), and every separated scheme is quasiseparated. Therefore, π is quasiseparated.

Lemma 3.4.9: Qcqs Lemma

Let X be a quasicompact and quasiseparated scheme and $s \in \mathcal{O}_X(X)$. Then

$$\mathcal{O}_X(X)_s \cong \mathcal{O}_X(X_s)$$

This theorem can be thought of as the generalization of the result where $X = \operatorname{Spec}(R)$, we we already know that:

$$\mathcal{O}_{\operatorname{Spec}(R)}\left(\operatorname{Spec}(R)\right)_s \cong, \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)_s) = \mathcal{O}_{\operatorname{Spec}(R)}(D(s))$$

***Proof*:**
As X is quasicompact, cover it with finitely many affine open sets $U_i = \operatorname{Spec} R_i$. take $U_{ij} = U_i \cap U_j$. Then the following is exact

$$0 \to \mathcal{O}_X(X) \to \prod_i R_i \to \prod_{i,j} \mathcal{O}_X(U_{ij})$$

As X is quasiseparated, each U_{ij} is covered by finitely many affine open sets $U_{ijk} = \operatorname{Spec}(R_{ij})$ so that the following is exact:

$$0 \to \mathcal{O}_X(X) \to \prod_i R_i \to \prod_{(i,j),k} R_{ijk}$$

Now, localization (being an exact functor) gives:

$$0 \to \mathcal{O}_X(X)_s \to \left(\prod_i R_i\right)_s \to \left(\prod_{(i,j),k} R_{ijk}\right)_s$$

which is still exact. As localization commutes with finite products (importance on finiteness), we get for appropriates values:

$$0 \to \mathcal{O}_X(X)_s \to \prod_i (R_i)_{s_i} \to \prod_{i,j,k} (R_{ijk})_{s_{ijk}}$$

which again is still exact, where the s_i and s_{ijk} is is the restriction to the rings R_i and R_{ijk}. Now, the scheme X_s can be covered by the affine open sets $\text{Spec}\,(R_i)_{s_i}$ and $\text{Spec}\,(R_i)_{s_i} \cap \text{Spec}\,(R_{ijk})_{s_{ijk}}$ giving almost the same exact sequence:

$$0 \to \mathcal{O}_X(X_s) \to \prod_i (R_i)_{s_i} \to \prod_{i,j,k} (R_{ijk})_{s_{ijk}}$$

Hence, the kernel of the map

$$\mathcal{O}_X(X)_s \to \mathcal{O}_X(X_s)$$

is the same, and we get an isomorphism.

Proposition 3.4.10: Affine Morphisms Are Affine-local On Target

Let $\pi : X \to Y$ be a scheme morphism. Then π is an affine morphism if and only if there is a affine open cover $\cup_i U_i = Y$ such that $\pi^{-1}(U_i)$ is affine.

***Proof*:**
We shall want to apply the affine communication lemma. For that, we must show the conditions are satisfied. First, suppose $\pi : X \to Y$ is affine over $\text{Spec}\,(S)$ so that $\pi^{-1}(\text{Spec}\,(S)) = \text{Spec}\,(R)$. Then notice that for any $s \in S$:

$$\pi^{-1}(\text{Spec}\,(S_s)) = \text{Spec}\,(R)_{\pi^\# s}$$

showing the first condition. For the next, suppose that $\pi : X \to \text{Spec}\,(S)$ and $(s_1, ..., s_n)$ with $X_{\pi^\# s_i} = \text{Spec}\,(R_i)$. We'll show that X is affine as well. To that end, let $R = \mathcal{O}_X(X)$. Then $X \to \text{Spec}\,(S)$ factors through $\alpha : X \to \text{Spec}\,(R)$ (given by the isomorphism $R \to \mathcal{O}_X(X)$), giving us:

$$\begin{tikzcd}
\bigcup_i X_{\pi^\# s_i} \arrow[rr, "\alpha"] \arrow[dr, "\pi"'] & & \text{Spec}\,(R) \arrow[dl, "\beta"] \\
& \bigcup_i D(s_i) = \text{Spec}\,(S) &
\end{tikzcd}$$

Then, as we are working with scheme morphisms, the pre-image of a distinguished open set is a distinguished open set and so

$$\beta^{-1}(D(s_i)) = D(\beta^\#(s_i)) \cong \text{Spec}\, R_{\beta^\# s_i}$$

and $\pi^{-1}(D(s_i)) = \text{Spec}\,(R_i)$. Now, as X is quasicompact and quasiseparated, by the Qcqs lemma we have the isomorphism:

$$A_{\beta^\# s_i} = \mathcal{O}_X(X)_{\beta^\# s_i} = \mathcal{O}_X(X_{\beta^\# s_i}) = R_i$$

Hence, α induces an isomorphism $\text{Spec}\,(R_i) \cong \text{Spec}\,(R_{\beta^\# s_i})$, and so α is an isomorphism over each $\text{Spec}\,(R_{\beta^\# s_i})$, which covers $\text{Spec}\,(R)$, and hence α is an isomorphism, and so $X \cong \text{Spec}\,(R)$, completing the proof.

From this, we can conclude that if the domain/codomain are affine, the morphisms must be affine

Corollary 3.4.11: Morphism Between Affine is Affine

Let $f : \operatorname{Spec} R \to \operatorname{Spec} S$ be a morphism of schemes. Then f is an affine morphism

Proof:
By proposition 3.4.10, it suffices to prove it on an affine open cover of the codomain. Then naturally, take an open cover by distinguished open sets, $D(g_i)$ whose pre-image $D(f^{\#}(g_i))$ which is an affine scheme.

Corollary 3.4.12: Hypersurfaces and Affine Morphissm

Let $X \subseteq \operatorname{Spec}(R)$ be a closed subset which is locally cut out by one equation (Z is covered by open subsets which are each given by the quotient of one equation). Then the complement of Z, Y, is affine

Note that in the global case, $Y = \operatorname{Spec}(R_f)$, but Y need not be of this form for the local case.

Proof:
exercise.

Example 3.4.13: Affine Morphisms

1. Take $\mathbb{A}^2_k \to \mathbb{P}^1_k$ given by $(a, b) \mapsto [a : b]$ which was shown to be a scheme morphism in example 3.1.11 (uhm, this is obviously not a morphism, where does the origin map too... Should probably take the blow-up). This is an affine morphism: take the usual open cover U_0, U_1. Then:

$$f^{-1}(U_0) \cong \operatorname{Spec} k[x, y]_x \qquad \text{and} \qquad f^{-1}(U_1) \cong \operatorname{Spec} k[x, y]_y$$

   hence by proposition 3.4.10 it is affine.

2. More generally let $\iota : X \to Y$ be either an open or closed immersion. Show that ι is an affine morphism.

The next two types of morphisms are affine morphisms where the pre-image has some extra properties. The next one can be thought of as the generalization of finite integral extensions (or more simply a finite extension or finite ring homomorphism):

Definition 3.4.14: Finite Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *finite* if for every affine open set $\operatorname{Spec}(S) \subseteq Y$, $\pi^{-1}(\operatorname{Spec}(S))$ is the spectrum of a S-algebra that is finitely generated as a S-module, i.e. it is a finite S-algebra.

Note how being a finitely generated S-module is a rather stronger condition than being a finitely generated S-algebra: it forces each element to be integral and "globally bounded" (since R is a finitely generated S-module, there can't be elements of arbitrary order). The reader should see that the composition of two finite morphisms is finite (see exercise ref:HERE)

Lemma 3.4.15: Algebra Build-up For Local On Target

Let $R \to S$ be a ring map, and $(f_1, ..., f_n) = R$ for appropriate $f_i \in R$. Then:

1. If $R_{f_i} \to S_{f_i}$ is integral, $R \to S$ is integral
2. If $R_{f_i} \to S_{f_i}$ is finite, $R \to S$ is finite

***Proof*:**

1. Take $s \in S$, and consider the ideal

$$(p \in R[x] \,|\, p(s) = 0) \subseteq R[x]$$

   Then take the subset $J \subseteq R$ of the leading coefficients of the elements in I. This is in fact an ideal, as the reader could verify or see [10]. By clearing denominators, we get that:

$$f_i^{n_i} \in J$$

   Hence, as shown in [18], $1 \in J$

2. The above shows that $R \to S$ is integral, for the finiteness note that we may choose a maximal N_i as the degree's of the generators are globally bounded.

Proposition 3.4.16: Finite Morphism Affine Local On Target

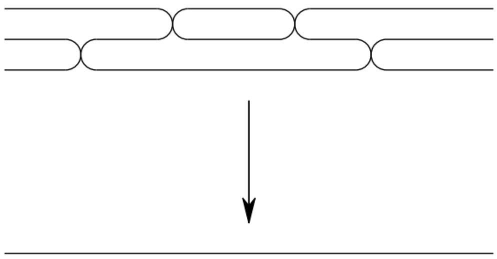
Let $f : X \to Y$ be a scheme morphism. Then it is finite if there is an affine cover $\bigcup_i \mathrm{Spec}\,(S_i) = Y$ such that $f^{-1}(\mathrm{Spec}\,(S_i))$ is the spectrum of a finite S_i-algebra.

***Proof*:**

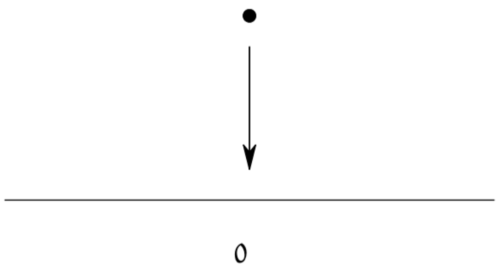
If f is finite, the result is *a fortiori* true, for the converse take $\mathrm{Spec}\, S_i \subseteq Y$, cover it in distinguished open sets, and apply corollary 3.1.15 and lemma 3.4.15.

Example 3.4.17: Finite Morphism

1. If L/K is a finite extension, $\mathrm{Spec}\,(K) \to \mathrm{Spec}\,(L)$ is a finite morphism.
2. Take the map $\mathrm{Spec}\, k[t] \to \mathrm{Spec}\, k[u]$ given by $u \mapsto p(t)$ for some degree n polynomial $p(t) \in k[t]$. Then this map is finite: notice that $k[t]$ is generated as a $k[u]$-module by $1, t, ..., t^{n-1}$. This can be visualized as:



3. Let $I \subseteq R$ be an ideal and take the natural schemes and map $\operatorname{Spec}(R/I) \to \operatorname{Spec}(R)$. Then this is a finite morphism, as R/I is generated as an R-module by $1 \in R/I$. This can be visualized as:



4. More generally, all closed immersions are finite morphisms.

5. Let $f \in R[x]$ be a polynomial whose leading coefficient is *not* a unit. Show that

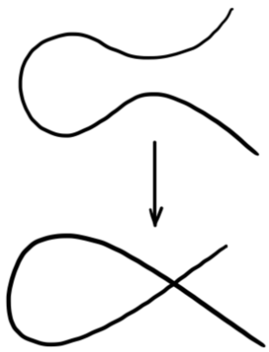
$$\operatorname{Spec} \frac{R[x]}{(f)} \to \operatorname{Spec} R$$

is *not* a finite morphism. Show that it *is* a finite morphism if the leading coefficient is a unit, making this an iff.

6. Take $\operatorname{Spec} k[t] \to \operatorname{Spec} \frac{k[x,y]}{(y^2-x^2-x^3)}$ corresponding to the ring map given by:

$$x \mapsto t^2 - 1 \qquad y \to t^3 - t$$

Namely, this shows this curve is rational. This is a finite morphism, since $k[t]$ is generated as a $\frac{k[x,y]}{(y^2-x^2-x^3)}$-module by 1 and t. This can be visualized as:



As the reader may have noticed, there is only one point that is 2-to-1. The reader should check that this will induce an isomorphism between $D(t^2 - 1)$ and $D(x)$, i.e. if the node is removed there is an isomorphism. This shall come up many times as an example of normalizing a curve.

7. Let $X \to \operatorname{Spec} k$ be a scheme morphism. Show that if the morphism if finite, X is the finite union of points with the discrete topology, each points having as residue field a finite extension of k. This can be visualized as:

Notice that all the fibers were finite. This is no happenstance of the chosen examples:

Proposition 3.4.18: Finite Morphism is Finite-to-one

Let $\pi : X \to Y$ be a finite morphism. Then it's fibers are finite

***Proof*:**
Since the property of the fiber is local on the target, we may assume $Y = \operatorname{Spec} B$ is affine. By the definition of a finite morphism, X is affine, say $X = \operatorname{Spec} A$, and the induced ring map $B \to A$ makes A a *finite* B-module (finitely generated as a module).

Let $y \in Y$ be a point corresponding to the prime ideal $\mathfrak{p} \subseteq B$. The fiber over y, denoted X_y, is given by the spectrum of the fiber ring:

$$A_y = A \otimes_B \kappa(y)$$

where $\kappa(y)$ is the residue field at $\mathfrak{p}$. Since A is generated by finitely many elements as a B-module, the tensor product A_y is generated by the images of those elements as a vector space over $\kappa(y)$. Thus, A_y is a finite-dimensional $\kappa(y)$-algebra.

Any finite-dimensional algebra over a field is an Artinian ring. A basic property of Artinian rings is that they have finitely many prime ideals (in fact, finitely many points, all of which are closed). Therefore, $\operatorname{Spec}(A_y)$ consists of finitely many points.

The converse is not the case

Example 3.4.19: Finite Fibers But Not Finite Morphisms

The open embedding $\mathbb{A}^2_k \setminus \{(0,0)\} \hookrightarrow \mathbb{A}^2_k$ has finite fibered (they are one-to-one), but it is not affine (as $\mathbb{A}^2_k \setminus \{(0,0)\}$ is not affine).

It also possible to have finite fibers and affine, but not finite: Take $\mathbb{A}^1_{\mathbb{C}} \setminus \{0\} \hookrightarrow \mathbb{A}^1_{\mathbb{C}}$ (intuitively, localizing is "changing perspectives" to the infinitesimal).

It looks important to add 7.3.I to get a grasp on finite morphisms to affine spectrums relation to projective scheme

Definition 3.4.20: Integral Morphism

Let $\pi : X \to Y$ be a scheme morphism. The nit is an *integral morphism* if π is affine, and for every affine open subset $\operatorname{Spec}(S) \subseteq Y$, where $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$, the induced map $S \to R$ is an integral extension.

As integral extensions must be finitely generated over the source ring, an finite morphism is a integral morphism, however an integral morphism need not be finite (think $\operatorname{Spec} \overline{\mathbb{Q}} \to \operatorname{Spec} \mathbb{Q}$). It is further easy to verify that the composition of integral maps is integral.

Proposition 3.4.21: Integral Morphisms Affine-local On Target

Let $\pi : X \to Y$ be an scheme morphism. Then π is an integral morphism if and only if there is a affine open cover $\cup_i U_i = Y$ such that we get integral extension for each pre-image.

***Proof*:**
exercise.

Proposition 3.4.22: Integral Morphism Are Closed Maps

Let $\pi : X \to Y$ be a Integral morphism. Then it is a closed map: it maps closed subsets to closed subsets

As a consequence, finite morphisms are closed

***Proof*:**
idea: reduce first to ring map., and consider the induced map. Now take $I \subseteq R$ and $J = (\pi^{\#})^{-1}(I)$ and consider the integral extension $S/J \subseteq A/I$. Apply the Lying Over theorem.

The definition above invites the question of which properties survive the passage to $\overline{k}$, and connectedness is the first case worth settling. It can fail: $\operatorname{Spec}\mathbb{C}$ is a single point over $\mathbb{R}$, while its base change

$$\operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}) \cong \operatorname{Spec}(\mathbb{C} \times \mathbb{C})$$

is a pair of points. The failure is visible in the residue field, which is $\mathbb{C}$, strictly larger than $\mathbb{R}$, and that extra room is what splits. A point whose residue field is exactly k leaves no such room. Call a morphism $\operatorname{Spec} k \to X$ of k-schemes a *k-rational point* of X; its image is a point $x_0 \in X$ with $\kappa(x_0) = k$, since the composite $\operatorname{Spec} k \to X \to \operatorname{Spec} k$ is the identity and so the induced k-algebra map $\kappa(x_0) \to k$ is an isomorphism. Write $X(k)$ for the set of k-rational points.

The topological input is recorded first, and it is here that the finiteness hypothesis on X gets spent.

Lemma 3.4.23: Base Change to the Algebraic Closure is Open and Closed

Let k be a field and let X be a scheme locally of finite type over k. Then the projection $\pi \colon X_{\overline{k}} \to X$ is surjective, closed, and open.

***Proof*:**
Step 1: π is integral, hence surjective and closed. By definition 3.1.21 we may cover X by affine opens $\operatorname{Spec} A$ with A a finitely generated k-algebra, and by proposition 3.2.5(1) together with example 3.2.4 we have $\pi^{-1}(\operatorname{Spec} A) = \operatorname{Spec}(A \otimes_k \overline{k})$. As an A-algebra, $A \otimes_k \overline{k}$ is generated by the elements $1 \otimes \lambda$ for $\lambda \in \overline{k}$, because $a \otimes \lambda = (a \otimes 1)(1 \otimes \lambda)$. Each λ is algebraic over k, so $1 \otimes \lambda$ is a root of the monic minimal polynomial of λ, read in $A[T]$. An algebra generated by integral elements is integral over its base [10], so π is an integral morphism (definition 3.4.20) and hence a closed map by proposition 3.4.22.

For surjectivity, fix $\mathfrak{p} \in \operatorname{Spec} A$. Localizing at $\mathfrak{p}$ and killing $\mathfrak{p}$ shows that the primes of $A \otimes_k \overline{k}$ contracting to $\mathfrak{p}$ are the primes of

$$(A \otimes_k \overline{k}) \otimes_A \kappa(\mathfrak{p}) \cong \kappa(\mathfrak{p}) \otimes_k \overline{k}$$

This ring is a tensor product of nonzero k-vector spaces, hence nonzero, hence has a prime.
Step 2: openness reduces to a finite subextension. Openness may be tested on the cover: for $U \subseteq X_{\overline{k}}$ open one has $\pi(U) \cap \operatorname{Spec} A = \pi(U \cap \pi^{-1}(\operatorname{Spec} A))$, and every open of $\operatorname{Spec}(A \otimes_k \overline{k})$ is a union of distinguished opens, so it is enough to show that $\pi(D(f))$ is open for $f \in A \otimes_k \overline{k}$. Write $f = \sum_{i=1}^{n} a_i \otimes \lambda_i$ and set $k' = k(\lambda_1, \ldots, \lambda_n)$, a finite extension of k since each λ_i is algebraic. Put $B = A \otimes_k k'$ and $C = A \otimes_k \overline{k} \cong B \otimes_{k'} \overline{k}$, and let $f' \in B$ be the element $\sum_i a_i \otimes \lambda_i$, which maps to f. Factor π as $\pi' \circ \rho$ with $\rho \colon \operatorname{Spec} C \to \operatorname{Spec} B$ and $\pi' \colon \operatorname{Spec} B \to \operatorname{Spec} A$. Since $\overline{k}$ is an algebraic

closure of k' and B is a finitely generated k'-algebra, Step 1 applies to ρ and makes it surjective; as $D(f) = \rho^{-1}(D(f'))$ this gives $\rho(D(f)) = D(f')$ and therefore $\pi(D(f)) = \pi'(D(f'))$.
Step 3: the finite case. Choose a k-basis $\lambda_1, \dots, \lambda_d$ of k', where $d = [k' : k]$. Then $1 \otimes \lambda_1, \dots, 1 \otimes \lambda_d$ is an A-basis of B, so B is a free A-module of rank d. For $b \in B$ let

$$\chi_b(T) = T^d + c_{d-1}(b)T^{d-1} + \cdots + c_0(b) \in A[T]$$

be the characteristic polynomial of multiplication by b on B. We claim $\pi'(D(b)) = \bigcup_{i=0}^{d-1} D(c_i(b))$, which is open.

Fix $\mathfrak{p} \in \operatorname{Spec} A$. By the correspondence of Step 1, the primes of B over $\mathfrak{p}$ are the primes of $\overline{B} = B \otimes_A \kappa(\mathfrak{p})$, a $\kappa(\mathfrak{p})$-algebra free of rank d on the reduced basis, and the matrix of multiplication by the image $\overline{b}$ is the reduction of the matrix over A; so $\chi_{\overline{b}} = \chi_b \bmod \mathfrak{p}$. Now $\mathfrak{p} \notin \pi'(D(b))$ says that $\overline{b}$ lies in every prime of $\overline{B}$, that is, $\overline{b}$ is nilpotent. If $\overline{b}$ is nilpotent then multiplication by $\overline{b}$ is a nilpotent endomorphism of a finite dimensional vector space, so $\chi_{\overline{b}} = T^d$ and every $c_i(b)$ lies in $\mathfrak{p}$. Conversely, if every $c_i(b)$ lies in $\mathfrak{p}$ then $\chi_{\overline{b}} = T^d$, so Cayley-Hamilton makes multiplication by $\overline{b}$ nilpotent of order d, whence $\overline{b}^d = 0$. The two conditions agree, which proves the claim and shows π' is open. Combining with Step 2, π carries distinguished opens to opens and is therefore open, completing the proof.

The finiteness hypothesis on X enters through Step 2, where a single element of $A \otimes_k \overline{k}$ is seen to involve only finitely many elements of $\overline{k}$; this is what keeps the infinite extension $\overline{k}/k$ out of the topological argument below.

Proposition 3.4.24: Connected Plus a Rational Point Gives Geometric Connectedness

Let k be a field and let X be a connected scheme locally of finite type over k with $X(k) \neq \emptyset$. Then X is geometrically connected, that is, $X_{\overline{k}}$ is connected.

***Proof*:**
Step 1: the fiber over a rational point is a single point. Fix a k-rational point with image $x_0 \in X$, so $\kappa(x_0) = k$. Choose an affine open $\operatorname{Spec} A \subseteq X$ containing x_0 with A finitely generated over k, and let $\mathfrak{p} \subseteq A$ be the prime of x_0. By the correspondence in Step 1 of lemma 3.4.23, the points of $X_{\overline{k}}$ over x_0 are the primes of

$$\kappa(\mathfrak{p}) \otimes_k \overline{k} = k \otimes_k \overline{k} \cong \overline{k}$$

a field, so there is exactly one. Write $\tilde{x}_0$ for it.
Step 2: every connected component of $X_{\overline{k}}$ contains $\tilde{x}_0$. The scheme $X_{\overline{k}}$ is covered by spectra of finitely generated $\overline{k}$-algebras, which are Noetherian, so each such affine has finitely many irreducible components by lemma 2.4.2 and hence finitely many connected components, each of them open. So $X_{\overline{k}}$ is locally connected and its connected components are open as well as closed. Let Z be one of them. By lemma 3.4.23 the image $\pi(Z)$ is then open and closed in X, and it is nonempty, so $\pi(Z) = X$ because X is connected. In particular $x_0 \in \pi(Z)$, so Z meets the fiber over x_0, which by Step 1 forces $\tilde{x}_0 \in Z$.
Step 3: conclusion. The scheme $X_{\overline{k}}$ is nonempty, since it contains $\tilde{x}_0$. Distinct connected components are disjoint, so Step 2 leaves exactly one component, and $X_{\overline{k}}$ is connected, as we sought to show.

The classical phrasing of the same argument uses the action of $\mathrm{Aut}(\overline{k}/k)$ on $X_{\overline{k}}$ over X: the group permutes the connected components and fixes $\tilde{x}_0$, since that point is alone in its fiber, so the component through $\tilde{x}_0$ is stable. Step 2 above trades the transitivity of that action on fibers for the cheaper statement that π is open and closed. Dropping the rational point returns the failure that opened the discussion: $(\operatorname{Spec}\mathbb{C})(\mathbb{R}) = \emptyset$, and $\operatorname{Spec}\mathbb{C}$ is a connected $\mathbb{R}$-scheme of finite type that is not geometrically connected.

Definition 3.4.25: Locally Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *locally of finite type* if for every affine open subset $\operatorname{Spec}(S) \subseteq Y$, and every affine open subset $\operatorname{Spec}(R) \subseteq \pi^{-1}(\operatorname{Spec}(R))$, the induced morphism $S \to R$ makes R a finitely generated S-algebra. Equivalently, $\pi^{-1}(\operatorname{Spec}(S))$ can be covered by affine open subsets $\operatorname{Spec}(R_i)$ where each R_i is a finitely generated S-algebra.

Definition 3.4.26: Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *of finite type* if it is locally of finite type and quasicompact.

Proposition 3.4.27: Locally Finite Type Morphism Affine-local On Target

Let $\pi : X \to Y$ be an scheme morphism. Then π is locally of finite type (resp. of finite type) if there is a cover of Y by affine open sets $\operatorname{Spec}(S_i)$ such that $\pi^{-1}(\operatorname{Spec}(S_i)$ is locally finite type over S_i.

***Proof*:**
exercise.

Example 3.4.28: Finite Type Morphisms

1. All Schemes that are finite type over k are example of finite type scheme morphisms $X \to \operatorname{Spec} k$
2. The map $\mathbb{P}^n_R \to \operatorname{Spec} R$ is of finite type ($\mathbb{P}^n_R$ is covered by $n+1$ open sets)

Proposition 3.4.29: Integral and Finite Type iff Finite

1. finite morphisms are of finite type
2. A morphism is finite if it is integral and of finite type

***Proof*:**
exercise.

Definition 3.4.30: Quasifinite Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *quasifinite* if it is of finite type, and for each $\mathfrak{q} \in Y$, $\pi^{-1}(\mathfrak{q})$ is a finite set.

By proposition proposition 3.4.18 and proposition 3.4.29(1), finite morphisms are quasifinite. The converse is not true (think of $\mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{A}^2_k$. The finite type condition is necessary, there exists morphisms with finite fibers that are not quasifinite

Example 3.4.31: Finite Fibers, But Not Quasifinite

The morphism $\operatorname{Spec}(\mathbb{C}(t)) \to \operatorname{Spec}(\mathbb{C})$ has finite fibers, but is not quasifinite. Similarly to $\operatorname{Spec}(\overline{\mathbb{Q}}) \to \operatorname{Spec}(\mathbb{Q})$.

The failure of the converse in the example above is the model for the general situation: the morphism $\mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{A}^2_k$ is quasifinite but not finite, and the only thing standing between it and finiteness is that it has been restricted to an open subset. Zariski's Main Theorem says this is always the obstruction, and it is what makes quasifiniteness a useful hypothesis.

Theorem 3.4.32: Zariski's Main Theorem

Let Y be a Noetherian scheme and let $\pi\colon X \to Y$ be a separated, quasifinite morphism of finite type. Then π factors as

$$X \xrightarrow{j} Z \xrightarrow{g} Y$$

where j is an open immersion and g is finite.

***Proof Key ideas*:**

The full proof is outside the scope of this book. It rests either on the theorem on formal functions together with Stein factorization, neither of which is developed here, or on a dévissage reducing the general case to that of a finitely generated algebra over a Noetherian ring; the argument is long and its details are not illuminating at this stage. See EGA IV_3, 8.12.6 [27], or [63, Tag 05K0] for a self-contained modern account.

The shape of the argument is worth recording, because the substance of the theorem is a statement about rings. Reduce to the case $Y = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$ affine, with B a finitely generated A-algebra. The local assertion is that if $\mathfrak{q} \in \operatorname{Spec} B$ is isolated in its fibre over A, then there are an element $b \in B \setminus \mathfrak{q}$ and a finite A-subalgebra $A' \subseteq B$ with $B_b = A'_b$; that is, after inverting a single element the algebra becomes finite over A. The distinguished open $D(b)$ is the piece of X on which π already looks finite, and Z is assembled by gluing the finite algebras A' obtained this way over a cover of Y.

The theorem separates the two mechanisms by which a morphism can have finite fibres. A finite morphism has them because it is proper and affine; a quasifinite morphism has them because its fibres are discrete, and it may have been cut down to an open subset along the way. Theorem 3.4.32 says that these are the only two mechanisms, and that the second sits inside the first as an open immersion.

The form in which the theorem is usually applied adds a normality hypothesis on the target, which collapses the finite part to an isomorphism and leaves only the open immersion.

Corollary 3.4.33: Birational Form of Zariski's Main Theorem

Let $\pi\colon X \to Y$ be a separated, quasifinite morphism of finite type between integral Noetherian schemes, and suppose that Y is normal and that π is dominant and induces an isomorphism $K(Y) \to K(X)$ of function fields (such a morphism is called *birational*; see definition 3.6.2). Then π is an open immersion onto an open subscheme of Y.

***Proof*:**
Step 1: factor, and cut the middle term down to an integral scheme. By theorem 3.4.32 there is a factorization $\pi = g \circ j$ with $j\colon X \to Z$ an open immersion and $g\colon Z \to Y$ finite. Let Z' be the closure of $j(X)$ in Z, taken with its reduced induced structure. Then Z' is integral, being the closure of the integral scheme $j(X) \cong X$, and $j(X)$ is an open subscheme of Z': it is open in Z and contained in Z', and its two induced structures agree because both are reduced with the same underlying space. The restriction $g' = g|_{Z'}\colon Z' \to Y$ is finite, since a closed subscheme of a scheme finite over Y is again finite over Y.

Step 2: g' is surjective and changes no function field. A finite morphism is proper (proposition 3.5.22) and hence closed, so $g'(Z')$ is closed in Y; it contains $\pi(X)$, which is dense because π is dominant, so g' is surjective. The generic point of Z' is the image of the generic point of X, so $K(Z') = K(X)$, and the hypothesis on π gives $K(Z') = K(Y)$.

Step 3: normality forces g' to be an isomorphism. The claim is local on Y, so let $\operatorname{Spec} A \subseteq Y$ be affine, with A a normal domain with fraction field $K(Y)$. A finite morphism is affine, so $g'^{-1}(\operatorname{Spec} A) = \operatorname{Spec} B$ with B a finite A-module, and B is a domain with fraction field $K(Z') = K(Y)$. The map $A \to B$ is injective because g' is dominant between integral schemes, so $A \subseteq B$ is an inclusion of subrings of $K(Y)$. Every element of B is integral over A, being an element of a finite A-module, and A is integrally closed in $K(Y)$ by normality; hence $B \subseteq A$ and so $B = A$. Therefore g' is an isomorphism.

Step 4: conclusion. The morphism π is the composite of the open immersion $X \to Z'$ with the isomorphism g', hence is an open immersion, as we sought to show.

Normality of the target is not a technical convenience; without it the corollary is false, and the standard counterexample is the one every treatment of normalization begins with.

Example 3.4.34: Normalization of the Cuspidal Cubic

Let k be algebraically closed and let $C = \operatorname{Spec} k[u,v]/(v^2 - u^3)$ be the cuspidal cubic. The morphism $\nu\colon \mathbb{A}^1_k \to C$ determined by $u \mapsto t^2$ and $v \mapsto t^3$ is finite, hence separated and quasifinite, and it is bijective on points; it is birational, being an isomorphism over $C \setminus \{(0,0)\}$.

It is not an open immersion. Being surjective, an open immersion would make it an isomorphism, and it is not: at the origin the maximal ideal of $\mathcal{O}_{C,(0,0)}$ is generated by the images of u and v, and neither lies in its square, since the only relation $v^2 = u^3$ has all its terms of degree at least

2. So $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2$ while $\dim \mathcal{O}_{C,(0,0)} = 1$, and the local ring is not regular, whereas $\mathcal{O}_{\mathbb{A}^1_k,0}$ is. The hypothesis of corollary 3.4.33 that fails is normality of C; the morphism ν is precisely its normalization.

The example names a construction that has not yet been made in general. It exists for every variety, and it is finite, which is what makes it usable: normalizing does not leave the category of schemes of finite type, and it does not disturb properness.

Theorem 3.4.35: Normalization of a Variety

Let k be a field and let X be an integral scheme of finite type over k. Then there exist an integral normal scheme $\widetilde{X}$ of finite type over k and a morphism $\nu\colon \widetilde{X} \to X$ such that

1. ν is finite and surjective;
2. ν is birational, inducing an isomorphism $K(X) \to K(\widetilde{X})$ on function fields, and an isomorphism over the normal locus of X;
3. $(\widetilde{X}, \nu)$ is universal: every dominant morphism $Z \to X$ from an integral normal scheme factors uniquely through ν.

The pair $(\widetilde{X}, \nu)$ is unique up to unique isomorphism over X, and is called the *normalization* of X. If X is proper over k, so is $\widetilde{X}$.

***Proof*:**
Step 1: the affine case. Let $X = \operatorname{Spec} A$ with A a domain finitely generated over k, and let $\overline{A}$ be the integral closure of A in $K = \operatorname{Frac}(A)$. By the finiteness of integral closure for finitely generated algebras over a field, proved in *Commutative Algebra* [10], the ring $\overline{A}$ is a finite A-module, so it is again a finitely generated k-algebra, and it is a normal domain with fraction field K. Set $\widetilde{X} = \operatorname{Spec} \overline{A}$. The inclusion $A \subseteq \overline{A}$ is an integral ring extension with $\overline{A}$ finite over A, so $\nu\colon \widetilde{X} \to X$ is finite by proposition 3.4.29, and it is surjective because an integral extension induces a surjection on spectra. It is birational since the two rings share the fraction field K, and it is an isomorphism over the locus where A is already integrally closed.

Step 2: gluing. For a distinguished open $\operatorname{Spec} A_f \subseteq X$, integral closure commutes with localization (*Commutative Algebra* [10]), giving $\overline{A_f} = (\overline{A})_f$, so the affine normalizations agree on overlaps and glue: covering a general X by affine opens $\operatorname{Spec} A_i$ and normalizing each, the resulting schemes patch to a scheme $\widetilde{X}$ with a finite morphism to X, since finiteness is affine-local on the target (proposition 3.4.16). The glued $\widetilde{X}$ is integral with function field $K(X)$ and is normal, normality being a stalk condition.

Step 3: the universal property and uniqueness. Let $g\colon Z \to X$ be dominant with Z integral and normal. Dominance gives an inclusion of function fields $K(X) \subseteq K(Z)$. Working over an affine $\operatorname{Spec} A \subseteq X$ with preimage covered by affines $\operatorname{Spec} C \subseteq Z$, every element of $\overline{A}$ is integral over A, hence over C, and lies in $K(Z)$; as C is integrally closed in $K(Z)$, it lies in C. So $A \to C$ extends uniquely to $\overline{A} \to C$, and these glue to the required factorization. Uniqueness up to unique isomorphism is the usual consequence of a universal property.

> *Step 4: properness.* A finite morphism is proper (proposition 3.5.22), so if $X \to \operatorname{Spec} k$ is proper then the composite $\widetilde{X} \to X \to \operatorname{Spec} k$ is proper, as we sought to show.

Two features of the normalization matter for what follows. It changes nothing in codimension zero, since it is birational; and by corollary 3.4.33 it is an isomorphism exactly where X was already normal, so it is a repair restricted to the non-normal locus. What it gives is that $\widetilde{X}$ is regular in codimension 1, because a normal Noetherian scheme is, and therefore that the divisor theory of chapter 5 applies to it.

Vakil promises exercise 9.3.H shows why this would naturally arise

Exercise 3.4.1

1. Let $\pi : X \to Y$ be an open embedding. Then if Y is locally Noetherian, X is locally Noetherian. If Y is Noetherian, X is Noetherian.
2. Let $\pi : X \to Y$ be an open embedding. Show that if Y is quasicompact, X need not be (there is an affine counter-example)
3. Show that the composition of two finite morphism is a finite morphism.
4. Show that being an open embedding is not local on the source.
5. Let B/A be an integral extension. Then show that any ring homomorphism $B \to C$ induces an integral extensions $C \otimes_B A/C$. Conclude that integrality is preserved by base change (see scalar extensions in [20] if you are unsure what it means to change base).
6. Show that every open embedding is locally of finite type. Hence, every quasicompact open embedding is of finite type.
7. Every open embedding into a locally Noetherian scheme is of finite type
8. The composition of two morphisms locally of finite type is locally of finite type.
9. Let $\pi : X \to Y$ be locally of finite type and Y locally Noetherian. Then X is locally Noetherian. IF $\pi : X \to Y$ is of finite type and Y is Noetherian, then X is Noetherian,
10. Show that quasifinite morphisms to $\operatorname{Spec}(k)$ are finite)
11. Let $k = \overline{\mathbb{F}_p}$. Show that the morphism $F : \mathbb{A}^n_k \to \mathbb{A}^n_k$ of k-schemes corresponding to the bijection $x_i \mapsto x_i^p$ is a bijection, and no power of F is the identity (as sets)
12. (Generalizing the above) Let X be an $\mathbb{F}_p$-scheme. Define an endomorphism $F : X \to X$ such that for each affine open subset $\operatorname{Spec}(R) \subseteq R<\ F$ corresponds to the map $R \to R$ given by $f \mapsto f^p$. The morphism F is called the *absolute Frobenius morphism.*

3.5 Separated and Proper Schemes

The last two major types of schemes we must cover are *separated schemes* that recover the notion of Hausdroffness for schemes, and *proper schemes* which recover a notion of compactness for schemes. As is the theme in this section, these two properties can be defined purely in terms of scheme morphisms.

From an algebraic-geometric perspective, these conditions allow for better behaviour of closed sets, whether the intersection of two closed sets is closed or whether the image of closed sets is closed is governed by separatedness and properness respectively. Both separatedness and properness can be defined in terms of morphisms, and this is in fact the more practical approach. As closed sets are locally closed set of affine scheme (i.e. vanishing sets of global sections of $\operatorname{Spec} R_i$), the preservation of closedness of these sets under various kinds of mappings is of key interest in scheme theory.

3.5.1 Separated Morphisms

Recall that if $f : X \to Y$ is a continuous map and $K \subseteq X$ is compact, then $f(K)$ is compact. In euclidean topology, this implies closed and bounded maps to closed and bounded, importantly this is a special case of when closed sets maps to closed sets. As affine schemes are quasi-compact (recall we use the term quasi-compact instead of compact due to the differing behaviour without Hausdorffness), if $K \subseteq X$ is closed it is quasi-compact, and hence its image is quasi-compact if $f : X \to Y$ is a scheme morphism (which is continuous). However, such a map *need not be closed*, namely it need not map closed sets to closed sets (recall example ref:HERE). In particular a quasicompact set need not be closed (see [25, section 3.2]).

Another example that shows the failure of separatedness is the following: Take X to be the line with double origin (recall example 2.2.55). Then the two natural inclusions $\iota_1, \iota_2 : \mathbb{A}^1_k \to X$ do not intersect on a closet set!

To recover the good properties of Hausdroffness, the notion of *separatedness* is defined. The idea is that we want to avoid cases where there can be "double points" or even more points that are intuitively arbitrarily close to each other; such points cannot be distinguished, or *separated* by (regular) functions.

In the topological setting, we saw X is Hausdorff if and only if $\Delta \subseteq X \times X$ is closed. In the case of scheme every "diagonal" (when working with fibered products) is a locally closed immersion:

Proposition 3.5.1: Diagonal Locally closed immersion

Let $f : X \to S$ be a morphism of schemes so that X is an S-scheme. Then the diagonal morphism $\delta : X \to X \times_S X$ is a locally closed immersion.

Proof:

We will describe a union of open subsets of $X \times_S X$ covering the image of X, such that the image of X is a closed immersion in this union.

Say S is covered with affine open sets V_i and X is covered with affine open sets U_{ij}, with $f : U_{ij} \to V_i$. Take $W = \bigcup_{i,j} U_{ij} \times_{V_i} U_{ij}$. This is an affine open subscheme of the product $X \times_S X$ (recall this is how the product was constructed). Then the diagonal is covered by these affine open subsets $U_{ij} \times_{V_i} U_{ij}$. Any point $p \in X$ lies in some U_{ij}; then $\delta(p) \in U_{ij} \times_{V_i} U_{ij}$. Note that $\delta^{-1}(U_{ij} \times_{V_i} U_{ij}) = U_{ij}$ since certainly $U_{ij} \subset \delta^{-1}(U_{ij} \times_{V_i} U_{ij})$, and as $\pi_1 \circ \delta = \mathrm{id}_X$ (as π_1 is the first projection), $\delta^{-1}(U_{ij} \times_{V_i} U_{ij}) \subset U_{ij}$. Finally, to check that $U_{ij} \to U_{ij} \times_{V_i} U_{ij}$ is a closed immersion, say $V_i = \operatorname{Spec} B$ and $U_{ij} = \operatorname{Spec} A$. Then this corresponds to the natural ring map $A \otimes_B A \to A$ ($a_1 \otimes a_2 \mapsto a_1 a_2$), which is certainly surjective, completing the proof.

The key is that these open subsets need not cover $X \times_S X$, hence this does not imply the embedding is closed.

In our case, a subset is closed if it is locally closed in an affine covering, this means that the diagonal can be expressed (locally) as vanishing sets. Furthermore, the product is not well-defined for schemes, as it must always be fibered over some other scheme. This leads to the following:

Definition 3.5.2: Separated Scheme

Let X be a scheme. Then X is *separated* if the map $\delta : X \to X \times_{\mathbb{Z}} X$ is a closed immersion.

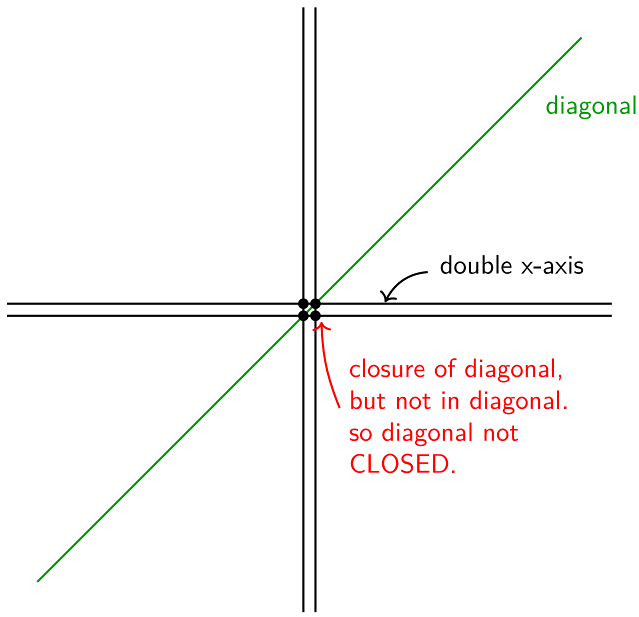
More generally, a map $X \to S$ between schemes is *separated* if the map $\delta : X \to X \times_S X$ is a closed immersion. Denote the image of δ as $\Delta(X)$ or Δ_X.

As a historical footnote, the term scheme used to mean separated scheme back before the 1960s, and the term pre-scheme was used for the modern term scheme.

Note that the first condition is equivalent to saying that the morphism $X \to \operatorname{Spec}(\mathbb{Z})$ is separated. The fact that separatedness is given by the purely topological condition of $\Delta(X)$ being closed shows its closeness to the original topological condition of Hausdorffness.

Example 3.5.3: Separated Morphisms

1. Take the line with two origins over k, label it X. Then when considering $X \times_{\operatorname{Spec} k} X$, we can think to it as in image bellow. The double origin will correspond to 4 points, but the diagonal will only have 2 points. Taking the closure of the diagonal will include all 4 points (show that any open set containing either $(0_1, 0_2)$ or $(0_2, 0_1)$ must intersect $\Delta(X)$, and hence is in the closure; see [25, section 1.5]), showing that it is *not* separated.



Note that $X \to X \times_k X$ is still a locally closed immersion. To show that it is not closed, we can take an open cover that covers *all* of $X \times_k X$ and show that for the affine open W_i containing $P_1 = (0_1, 0_2)$, $\Delta(X) \cap W_i$ is not closed. Indeed, As $P_1 \notin \Delta(X)$, $P_1 \notin \Delta(X) \cap W_i$.

It is easy to see that P_1 is in the closure of this set, $P_1 \in \overline{\Delta(X) \cap W_i}$, we see that $\Delta(X) \cap W_i$ is *not* closed.

Note that if we take $X \to X$ to be the identity map, then this *is* a separated morphism, since:

$$X \cong X \times_X X$$

showing the relativeness of separatedness when picking different base schemes.

2. Let $X = \operatorname{Spec}(A) \to \operatorname{Spec}(B) = Y$ be any morphism of schemes between affine schemes. Then this is *always* a separated morphism. Unwind the definitions: when converting into rings we get the map:

$$A \otimes_B A \to A$$

which is surjective, and hence we get:

$$A \cong A \otimes_B A/\mathfrak{a}$$

that is, we are taking the quotient by an ideal, which we know will give us a closed subscheme of $X \times_Y X$ isomorphic to X.

3. More generally, if $X \to Y$ is an affine morphism, it is separated (follows from proposition 3.5.1).

4. The following scheme is a famous example of non-separatedness appearing naturally. Consider $\mathbb{C}^* \curvearrowright \mathbb{A}^2_{\mathbb{C}}$ via $t \cdot (x, y) = (tx, t^{-1}x)$; this is usually sated as "the algebraic torus acts on the affine plane with weights $(1, -1)$". Then taking $(\mathbb{A}^2_{\mathbb{C}} \setminus \{0\})/\mathbb{C}^*$ where the origin is removed as the action is non-free, the resulting space is a non-separated scheme.

   In many cases, non-separated scheme appear naturally in moduli theory.

An important example worth singling out is that of projective scheme:

Lemma 3.5.4: Projective Scheme is Separated

The scheme $\mathbb{P}^1_{\mathbb{Z}}$ is separated.

***Proof*:**
Let $X = \mathbb{P}^1_{\mathbb{Z}}$ and take a standard affine cover $U_0 = \operatorname{Spec} \mathbb{Z}[t]$ and $U_1 = \operatorname{Spec} \mathbb{Z}[u]$, glued via $u = 1/t$. To prove separatedness, we must show $\Delta(X)$ is closed in $X \times_{\mathbb{Z}} X$. It suffices to show the intersection of $\Delta(X)$ with the members of an open cover of the *product* is closed. The product is covered by the four affine patches $U_{ij} = U_i \times U_j$.

We analyze the corresponding ring maps (dual maps) $\Delta^\#$. Recall that a morphism of affine schemes is a closed immersion if and only if its ring dual map is surjective. If surjective, the image corresponds to $V(\ker(\Delta^\#))$.

- *Case 1: The standard charts* (e.g., $W_{00} = U_0 \times U_0$).
  Algebraically, $W_{00} \cong \operatorname{Spec}(\mathbb{Z}[t] \otimes_{\mathbb{Z}} \mathbb{Z}[t]) \cong \operatorname{Spec} \mathbb{Z}[t_1, t_2]$. The diagonal restricted to this chart maps U_0 into $U_0 \times U_0$. The corresponding ring dual map $\Delta^\#$ is:

$$\varphi : \mathbb{Z}[t_1, t_2] \to \mathbb{Z}[t], \quad \text{defined by } t_1 \mapsto t, \; t_2 \mapsto t.$$

This map is clearly surjective. By the First Isomorphism Theorem, $\mathbb{Z}[t] \cong \mathbb{Z}[t_1, t_2]/\ker(\varphi)$. The kernel is the ideal generated by $(t_1 - t_2)$. Consequently, the diagonal corresponds to the closed subscheme defined by this ideal, $V(t_1 - t_2)$, representing the line $y = x$.

- *Case 2: The mixed charts* (e.g., $W_{01} = U_0 \times U_1$).
  Algebraically, the target is $W_{01} \cong \operatorname{Spec} \mathbb{Z}[t] \times \operatorname{Spec} \mathbb{Z}[u] \cong \operatorname{Spec} \mathbb{Z}[t, u]$. The source (the part of the diagonal mapping into this chart) is the intersection $U_0 \cap U_1$. This intersection is the locus where $t \neq 0$, which is affine: $\operatorname{Spec} \mathbb{Z}[t, t^{-1}]$. The ring dual map $\Delta^\#$ maps the coordinates of the product to their values on the intersection (determined by the gluing $u = 1/t$):
  $$\psi : \mathbb{Z}[t, u] \to \mathbb{Z}[t, t^{-1}].$$
  Defined by $t \mapsto t$ and $u \mapsto t^{-1}$. Since t and t^{-1} are in the image, ψ is surjective. This implies the geometric map is a closed immersion. The algebraic geometry dictionary tells us the image is defined by the kernel of ψ:
  $$\operatorname{Spec} \mathbb{Z}[t, t^{-1}] \cong \operatorname{Spec} \left(\mathbb{Z}[t, u]/\ker(\psi) \right).$$
  Since $t(t^{-1}) - 1 = 0$ in the codomain, the element $(tu - 1)$ is in the kernel. In fact, $\ker(\psi) = (tu - 1)$. Thus, the image of the diagonal in this chart is the closed subset $V(tu - 1) \subseteq \mathbb{A}^2_{\mathbb{Z}}$, which describes the hyperbola $u = 1/t$.

Since the image is closed in every chart of an open cover of the target $X \times_{\mathbb{Z}} X$, the diagonal map is a closed immersion.

The details for the case of $\mathbb{P}^n_R$ are a bit laborious, so until I feel comfortable with them I am putting this corollary here as a refresher for my future self:

Corollary 3.5.5: Projective Space is Separated

For any ring R, the projective space $\mathbb{P}^n_R$ is separated.

***Proof*:**
Let $X = \mathbb{P}^n_R$. We use the standard affine open cover $\{U_i\}_{i=0}^n$, where $U_i = \{[x_0 : \cdots : x_n] \big| x_i \neq 0\}$. The affine ring for U_i is $R[y_0^{(i)}, \ldots, y_n^{(i)}]$ subject to $y_i^{(i)} = 1$, where the coordinates represent $y_k^{(i)} = x_k/x_i$.

To show the diagonal $\Delta : X \to X \times_R X$ is a closed immersion, we check that its image is closed in the open affine cover of the product given by $\{U_i \times U_j\}_{0 \le i,j \le n}$.

Consider a specific chart $W_{ij} = U_i \times U_j$.

- **Algebraic Setup:** The ring corresponding to W_{ij} is the tensor product of the rings for U_i and U_j:
  $$A_{ij} = R[y_0^{(i)}, \ldots, y_n^{(i)}] \otimes_R R[z_0^{(j)}, \ldots, z_n^{(j)}]$$
  where variables y come from the first factor (U_i) and z from the second (U_j), with $y_i^{(i)} = 1$ and $z_j^{(j)} = 1$.

- **The Diagonal Map:** A point lies in the image of the diagonal in this chart if it is in

$U_i \cap U_j$ in both factors. The map on rings $\varphi : A_{ij} \to \mathcal{O}_X(U_i \cap U_j)$ is induced by the coordinate change formulas:

$$z_k^{(j)} = \frac{x_k}{x_j} = \frac{x_k/x_i}{x_j/x_i} = \frac{y_k^{(i)}}{y_j^{(i)}}$$

Thus, the map is defined by $y_k^{(i)} \mapsto y_k^{(i)}$ and $z_k^{(j)} \mapsto y_k^{(i)} \cdot (y_j^{(i)})^{-1}$.

- **The Closed Relations:** We can find the generators of the kernel by clearing denominators. The relations are:
$$z_k^{(j)} y_j^{(i)} - y_k^{(i)} = 0 \quad \text{for all } k = 0, \ldots, n.$$
Note specifically that when $k = i$, since $y_i^{(i)} = 1$, we get the relation $z_i^{(j)} y_j^{(i)} - 1 = 0$. This is the generalized "hyperbola" $(x_{i/j} \cdot x_{j/i} = 1)$.

  The ideal generated by these polynomials $J = \langle z_k^{(j)} y_j^{(i)} - y_k^{(i)} \rangle_k$ defines a closed subset of W_{ij}. Since φ is surjective onto the localization of the polynomial ring (image is $U_i \cap U_j$), this closed set is exactly the image of the diagonal.

As the image is closed in every member of an open cover of the product, $\mathbb{P}_R^n$ is separated.

The next concept will not be too useful to us for awhile, and shall come back up when we see algebraic stacks and algebraic spaces, but is useful to define:

Definition 3.5.6: Quasi-separated

Let X be a scheme. Then X is quasi-separated if the image of $X \to X \times X$ is quasi-compact. More generally, a map $X \to S$ between schemes is quasi-separated if the image map of $X \to X \times_S X$ is quasi-compact.

Proposition 3.5.7: Separated Implies Quasi-separated

Let X be a separated scheme. Then X is quasi-separated.

***Proof*:**
By definition, X is separated if the diagonal morphism $\Delta : X \to X \times X$ is a closed immersion. By definition, X is quasi-separated if Δ is a quasi-compact morphism. Thus, it suffices to prove that any closed immersion is quasi-compact.

Let $f : Z \to Y$ be a closed immersion. To prove f is quasi-compact, we check the condition on an affine open cover. Let $V \subseteq Y$ be an affine open subset, say $V = \operatorname{Spec}(B)$. Because f is a closed immersion, the restriction $f|_V : f^{-1}(V) \to V$ corresponds to a surjective ring homomorphism $B \to A$ (specifically $A \cong B/I$ for the ideal I cutting out Z). Therefore, the preimage $f^{-1}(V)$ is isomorphic to $\operatorname{Spec}(A)$. Since every affine scheme is quasi-compact, $f^{-1}(V)$ is quasi-compact.

As the preimage of any affine open is quasi-compact, the map f is quasi-compact. Applying this to Δ, we conclude X is quasi-separated.

Separatedness gives us nicer behaviour on closed sets. The following results outline some important examples:

Proposition 3.5.8: Intersection of Separated affine schemes

Let $X \to S$ be a separated scheme morphism where S is affine. Then if $U, V \subseteq X$ are open affine subsets, their intersection

$$U \cap V$$

is an affine open subset.

***Proof*:**
Let $S = \operatorname{Spec} R$, $U = \operatorname{Spec} A$, and $V = \operatorname{Spec} B$.

1. **The product of affines is affine.** Recall that the product $X \times_S X$ is constructed by gluing affine charts. Specifically, the open subset $U \times_S V \subseteq X \times_S X$ is isomorphic to the spectrum of the tensor product:

$$U \times_S V \cong \operatorname{Spec}(A \otimes_R B).$$

   Thus, $U \times_S V$ is an affine open subscheme of the product.

2. **The intersection is a preimage.** Consider the diagonal morphism $\Delta : X \to X \times_S X$. We observe that the intersection of the open sets is exactly the pullback of their product by the diagonal:

$$\Delta^{-1}(U \times_S V) = \{x \in X \big| \Delta(x) \in U \times_S V\} = \{x \in X \big| x \in U \text{ and } x \in V\} = U \cap V.$$

3. **Separatedness implies affine.** By the definition of separatedness, Δ is a closed immersion. This means that Δ induces a closed immersion of $U \cap V$ into the open subset $U \times_S V$:

$$U \cap V \hookrightarrow U \times_S V.$$

   A closed immersion into an affine scheme corresponds to a surjective ring map. Since $U \times_S V$ is affine ($\operatorname{Spec}(A \otimes_R B)$), $U \cap V$ must be isomorphic to $\operatorname{Spec}((A \otimes_R B)/I)$ for some ideal I. Since the spectrum of a quotient ring is an affine scheme, $U \cap V$ is affine.

For a counter-example, taking affine *plane* with double origin we get that taking two affine lines each containing a different origin will intersect to produce the plane without the origin, which in example 2.2.51 is not affine.

Proposition 3.5.9: Graphs Are Closed

Let X be a separated S-schemes and Y any S-scheme. Then the graph of the S-morphism $f : X \to Y$:

$$\Gamma_f = \operatorname{im}((\mathrm{id}_Y, f)Y \to Y \times_S X)$$

is closed

***Proof*:**

1. Show the graph is constructed via the pullback of the diagonal $\Delta_X \subseteq X \times_S X$ along the map $id_Y \times g : Y \times_S X \to X \times_S X$.

2. pullbacks preserve closed immersions (proposition 3.2.5(2)).

For the case where it is not separated as usual let X be the line with two origins (not separated). Let Y be the affine line $\mathbb{A}^1$. Consider the morphism $g : Y \setminus \{0\} \to X$ which is the natural inclusion into the "un-doubled" part of X. What is the graph of this morphism? It's a subset of $(Y \setminus \{0\}) \times_k X$. The key is that this map cannot be extended to the origin of Y in a unique way. Indeed, $0 \in Y$ could map to either $0_1 \in X$ or $0_2 \in X$. This ambiguity means the graph of any potential extension is *not* closed; the point $(0, 0_1)$ and $(0, 0_2)$ are "limit points" of the graph but there is no natural choice of which one is in the graph. The graph is not well-behaved at the boundary. This intuition will in fact be a central way of interpreting separatedness, as shall be explored in the next section.

Proposition 3.5.10: Morphism Determined By Dense Open Subset

Let X be a *separated* S-scheme, let Z be a *reduced* S-scheme, and let $U \subseteq Z$ be a dense open subset. Then if two S-morphisms $f, g : Z \to X$ agree on U, they agree everywhere:

$$f|_U = g|_U \qquad \Rightarrow \qquad f = g$$

***Proof*:**
Consider the locus in Z where f and g agree. The key is that this can be described as the preimage of the diagonal of X, namely, take $(f, g) : Z \to X \times_S X$ and set

$$E = (f, g)^{-1}(\Delta_X)$$

Because X is separated over S, the diagonal Δ_X is a closed subscheme of $X \times_S X$ by definition 3.5.2, so E is a closed subscheme of Z. Its underlying closed set contains U, which is dense, so that closed set is all of Z.

It remains to upgrade this equality of *sets* to an equality of *morphisms*, and this is exactly where reducedness of the source enters. The scheme E is a closed subscheme of Z whose underlying space is all of Z, so its ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_Z$ has vanishing stalks modulo nilpotents; as Z is reduced, $\mathcal{I} = 0$ and $E = Z$ as schemes. Therefore (f, g) factors through the diagonal, which says precisely that $f = g$, as we sought to show.

Both hypotheses are needed, and each fails in a way worth seeing once. Dropping separatedness, let X be the line with two origins of the discussion above, let $Z = \mathbb{A}^1$ and $U = \mathbb{A}^1 \setminus \{0\}$, and let f and g be the two evident inclusions: they agree on the dense open U and differ at the origin. Dropping reducedness, let $Z = \operatorname{Spec} k[x, \epsilon]/(\epsilon^2, x\epsilon)$ with $U = D(x)$, and let $f, g : Z \to \mathbb{A}^1$ be given by $y \mapsto 0$ and $y \mapsto \epsilon$. Since $x\epsilon = 0$, the function ϵ restricts to 0 on U, so f and g agree there even though they differ on Z; here the target is as well behaved as one could ask, being both reduced and separated, and it is the nilpotent on the source that defeats the conclusion.

3.5.2 Valuation Criterion for Separatedness

We shall next characterize separated schemes and morphisms in using valuation rings. This will be analogous to the fact that in a Hausdorff space, the limit of a sequence has at most 1 point. From this result, think of a continuous map $(0,1) \to X$. If we want to extend this to a map $[0,1] \to X$, if X is Hausdorff, there is only one possible extension. If X wasn't, say it had two origins, then there can be more than one map that can extend (the new 0 point will have two possible places to map to).

Let us now extend this idea to a map $X \to Y$, and we have an embedding $(0,1) \to X$ an identity map $(0,1) \to [0,1]$ and an embedding $(0,1] \to Y$ where we are only adding one point. Then there should only be one lift. Note that this lift is not dependent on X and Y. Let's say Y was the double-origin scheme and X was the double y-axis scheme. Then the embedding $(0,1)$ can be extended to two way into Y, but that will correspond to a unique lift.

With this intuition in mind, let us now look at what is the equivalent of the open interval for schemes. This would be a Discrete valuation ring (DVR) R. It has only two points. We can think of as $\operatorname{Spec}(\operatorname{Frac}(R))$ as the open interval and $\operatorname{Spec}(R)$ as adding the closed 0 (notice that the new point in $\operatorname{Spec}(R)$ is also a closed point, while the other point is "open" and is the generic point that can be thought of was closed to everywhere else). Then we have the following situation:

$$\begin{tikzcd}
\operatorname{Spec}(\operatorname{Frac}(R)) \arrow[r] \arrow[d] & Y \arrow[d] \\
\operatorname{Spec}(R) \arrow[r] \arrow[ur, dashed] & X
\end{tikzcd}$$

With the hope that having at most one lift corresponding to the map being separable. This is *sometimes* the case. To prove this, we require two technical lemmas regarding valuation rings and specialization.

Lemma 3.5.11: Lifts from Valuation Rings

Let R be a valuation ring of a field K. Let $T = \operatorname{Spec} R$ and $U = \operatorname{Spec} K$.

1. A morphism $\operatorname{Spec} K \to X$ is equivalent to giving a point $x_1 \in X$ and an inclusion of fields $\kappa(x_1) \subseteq K$.
2. A morphism $\operatorname{Spec} R \to X$ is equivalent to
   - giving two points $x_0, x_1 \in X$, with x_0 a specialization of x_1 (i.e., $x_0 \in \overline{\{x_1\}}$), and
   - an inclusion $\kappa(x_1) \subseteq K$, such that R dominates the local ring $\mathcal{O}_{x_0,Z}$ where $Z = \overline{\{x_1\}}$ with its reduced induced structure.

Proof:

1. U is a single point scheme equipped with the structure sheaf K. To give a morphism $U \to X$ is to give a continuous map (picking a point $x_1 \in X$) and a morphism of sheaves $\mathcal{O}_{X,x_1} \to K$. This is a local homomorphism from a local ring to a field, which is equivalent to an inclusion of the residue field $\kappa(x_1) \subseteq K$.
2. Let $t_0 = \mathfrak{m}_R$ be the closed point of T and $t_1 = (0)$ be the generic point. Given a morphism $T \to X$, let x_0, x_1 be the images of t_0, t_1. Since the closure of t_1 contains t_0, by continuity

the closure of x_1 must contain x_0, so x_0 is a specialization of x_1. Since T is reduced, the morphism factors through the reduced induced scheme structure on the closure $Z = \overline{\{x_1\}}$. Algebraically, $\kappa(x_1)$ is the function field of Z. The morphism gives a local homomorphism $\mathcal{O}_{x_0,Z} \to R$. Since the map on function fields is compatible with the inclusion $\kappa(x_1) \subseteq K$, this exactly means that the valuation ring R dominates the local ring $\mathcal{O}_{x_0,Z}$.

Conversely, given the data of x_0, x_1 and the domination $\mathcal{O}_{x_0,Z} \to R$, we construct the map. The map of topological spaces is determined by the points. The map of sheaves is determined by the ring map $\mathcal{O}_{x_0,Z} \to R$ (since T is local, mapping to the global sections of T is sufficient), which composes with the natural map $\operatorname{Spec} \mathcal{O} \to X$ to give the desired morphism.

To make the conditions of lemma 3.5.11 concrete, let us look at a geometric example where T is a curve germ and X is the affine plane.

Example 3.5.12: Visualizing Valuation Lifts

1. Let $X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$. Let $R = k[t]_{(t)}$ be the discrete valuation ring at the origin of the affine line. Then $K = k(t)$. Let $T = \operatorname{Spec} R = \{t_0, t_1\}$ where $t_0 = (t)$ is the closed point and $t_1 = (0)$ is the generic point.

   Consider the morphism $T \to X$ defined by the ring homomorphism:

   $$\varphi^\# : k[x, y] \to k[t]_{(t)}, \quad x \mapsto t, \quad y \mapsto t^2$$

   Geometrically, this represents a curve parameterized by t approaching the origin along the parabola $y = x^2$.

   (a) *Where the generic point maps ($U \to X$):* The generic point t_1 corresponds to the prime ideal $(0) \subset R$. Its image $x_1 \in X$ is the prime ideal $\mathfrak{p} = (\varphi^\#)^{-1}((0))$. Since $y - x^2 \mapsto t^2 - t^2 = 0$, the ideal $(y - x^2)$ is in the kernel. In fact, the kernel is exactly $(y - x^2)$. Thus, x_1 is the *generic point of the parabola* $V(y - x^2)$. The inclusion of fields is the map on function fields:

   $$\kappa(x_1) = \operatorname{Frac}\left(\frac{k[x, y]}{(y - x^2)}\right) \xrightarrow{\cong} k(t) = K$$

   (b) *Where the closed point maps:* The closed point t_0 corresponds to the maximal ideal $(t) \subset R$. Its image $x_0 \in X$ is $(\varphi^\#)^{-1}((t)) = (x, y)$, which is the origin.

   (c) *Specialization:* We see clearly that x_0 (the origin) is in the closure of x_1 (the whole parabola). This satisfies the condition $x_0 \in \overline{\{x_1\}}$.

   (d) *Domination:* Let $Z = \overline{\{x_1\}}$ be the parabola. The local ring $\mathcal{O}_{x_0,Z}$ is the ring of functions on the parabola regular at the origin, which is isomorphic to $k[t]_{(t)}$. The map induced by $\varphi^\#$ is the identity $k[t]_{(t)} \to k[t]_{(t)}$, which is a local homomorphism (maps maximal ideal to maximal ideal), satisfying the domination condition.

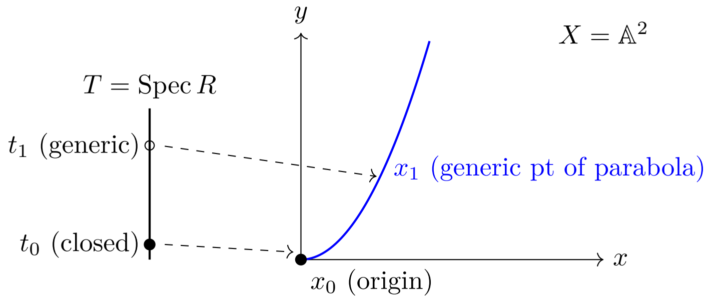


Figure 3.1: The generic point of the DVR maps to the generic point of the curve, while the closed point of the DVR maps to the limit point on the curve.

2. Let us look at a case where the generic point of T maps to the *generic point* of $X = \mathbb{A}^2_k$, rather than a curve inside it. This requires constructing a valuation on the function field $k(x, y)$.

   Let $K = k(x, y)$. We define a valuation $v : K^\times \to \mathbb{Z}$ by the "order of vanishing along the y-axis". Any rational function $f \in k(x, y)$ can be written as $f = x^n \frac{g}{h}$ where x does not divide g or h. We define $v(f) = n$.

   Let R be the valuation ring for v:

   $$R = \{f \in k(x, y) \big| v(f) \geq 0\} = k(x, y)_{(x)}$$

   This is the local ring of the affine plane localized at the prime ideal (x) (the generic point of the y-axis).

   Let us analyze the map $T = \operatorname{Spec} R \to X = \operatorname{Spec} k[x, y]$ induced by the natural inclusion $k[x, y] \subset R$.

   (a) *Generic Point:* The generic point t_1 of T (ideal (0) in R) maps to the ideal (0) in $k[x, y]$. Thus, $t_1 \mapsto \eta$, the generic point of the plane.

   (b) *Closed Point:* The closed point t_0 of T (maximal ideal xR) maps to the ideal $(x) \subset k[x, y]$. This is the *generic point of the line* $x = 0$.

   (c) *Visualization:*

   $$\begin{tikzcd} U = \operatorname{Spec} k(x, y) \arrow[r] \arrow[d] & \eta \text{ (generic pt of } \mathbb{A}^2) \arrow[d, dashed, "specializes"] \\ T = \operatorname{Spec} k(x, y)_{(x)} \arrow[r] & \xi \text{ (generic pt of } y\text{-axis)} \end{tikzcd}$$

   Even though t_0 is a "closed point" in the topological space T, it maps to a non-closed point in X.

   (d) *Domination:* The stalk of X at the image point ξ is $\mathcal{O}_{X,\xi} = k[x, y]_{(x)}$. In this specific case, the valuation ring R is *isomorphic* to the stalk. This happens because we approached a codimension 1 point. If we approached the origin (codimension 2), R would be a further localization/quotient of the stalk.

Lemma 3.5.13: Closure and Specialization

Let $f : X \to Y$ be a quasicompact morphism of schemes (e.g., if X, Y are Noetherian). Then the subset $f(X) \subseteq Y$ is closed if and only if it is stable under specialization.

***Proof*:**
One direction is topological: closed sets are always stable under specialization (if a point is in the set, its closure is in the set). For the converse, assume $f(X)$ is stable under specialization. We want to show $\overline{f(X)} \subseteq f(X)$. We may assume X and Y are reduced, and replace Y with the reduced induced structure on $\overline{f(X)}$ so we want to show f is surjective onto Y. Let $y \in Y$. We can replace Y with an affine neighborhood of y, so assume $Y = \operatorname{Spec} B$. Since f is quasicompact, X is a finite union of affine opens $X_i = \operatorname{Spec} A_i$. Since y is in the closure of the image, $y \in \overline{f(X_i)}$ for some i. Let $Y_i = \overline{f(X_i)}$ with the reduced structure. Then $X_i \to Y_i$ corresponds to an injective ring map $B \to A_i$ (injective because the map is dominant). The point y corresponds to a prime $\mathfrak{p} \subseteq B$. Let $\mathfrak{p}' \subseteq \mathfrak{p}$ be a *minimal* prime ideal of B contained in $\mathfrak{p}$, such a prime exists by Zorn's lemma. This prime $\mathfrak{p}'$ corresponds to a point y' which specializes to y. I claim $y' \in f(X_i)$. Consider the localization $B_{\mathfrak{p}'}$. This is a field since $\mathfrak{p}'$ is minimal in a reduced ring. The map $B_{\mathfrak{p}'} \to A_i \otimes_B B_{\mathfrak{p}'}$ is an inclusion of a field into a non-zero ring as $B \to A_i$ is injective. Thus, the target ring is not the zero ring, so it has a prime ideal $\mathfrak{q}'$. This prime contracts to $\mathfrak{p}'$. Thus, y' is in the image. Since $f(X)$ is stable under specialization and $y' \rightsquigarrow y$, $y \in f(X)$ as we sought to show.

Theorem 3.5.14: Valuation Criterion for Separatedness

Let $f : X \to Y$ be a morphism of schemes, and assume that X is Noetherian. Then f is separated if and only if for any field K, and for any valuation ring R with quotient field K, $T = \operatorname{Spec} R, U = \operatorname{Spec} K$, and the morphism $i : U \to T$ is the morphism induced by the inclusion $R \subseteq K$, given a morphism of T to Y, and given a morphism of U to X, the following diagram commutes:

$$\begin{tikzcd} U \arrow[r] \arrow[d, "\iota"] & X \arrow[d, "f"] \\ T \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

and there is *at most one* morphism of T to X making the whole diagram commutative.

***Proof*:**
First, suppose f is separated. Let $h, h' : T \to X$ be two morphisms making the diagram commute. We obtain a morphism $L = (h, h') : T \to X \times_Y X$. Since the restrictions of h and h' to U are the same (given by the diagram), the generic point t_1 of T maps to the diagonal $\Delta(X) \subseteq X \times_Y X$. Since f is separated, the diagonal $\Delta(X)$ is a closed immersion, so $\Delta(X)$ is a closed set. The image of T is the closure of the image of the generic point (as $T = \operatorname{Spec} R$ has a generic point whose closure includes the closed point). Since the generic point lands in the closed set $\Delta(X)$, the entire image $L(T)$ must lie in $\Delta(X)$. Therefore, h and h' coincide everywhere on T, so $h = h'$. The equality of the sheaf maps follows from lemma 3.5.11 as the inclusions into K are identical.

Conversely, suppose the valuation condition holds. To show f is separated, we must show that the diagonal $\Delta : X \to X \times_Y X$ is a closed immersion. Since Δ is a locally closed immersion, it suffices to show the image $\Delta(X)$ is a closed subset of $X \times_Y X$. Since X is Noetherian, the product $X \times_Y X$ is Noetherian (or at least the morphism Δ is quasicompact). By lemma 3.5.13, it suffices to show that $\Delta(X)$ is stable under specialization.

Let $\xi_1 \in \Delta(X)$ and let $\xi_1 \rightsquigarrow \xi_0$ be a specialization in $X \times_Y X$. We must show $\xi_0 \in \Delta(X)$. Let $K = \kappa(\xi_1)$ and let $\mathcal{O}$ be the local ring of ξ_0 on the subscheme $\overline{\{\xi_1\}}$ (with reduced induced structure). By domination properties of valuation rings, there exists a valuation ring R of K which dominates $\mathcal{O}$. Let $T = \operatorname{Spec} R$ and $U = \operatorname{Spec} K$. By lemma 3.5.11, this data induces a morphism $T \to X \times_Y X$ mapping the generic point to ξ_1 and the closed point to ξ_0. Composing this with the two projections $p_1, p_2 : X \times_Y X \to X$, we get two morphisms $h, h' : T \to X$. Since $\xi_1 \in \Delta(X)$, we have $p_1(\xi_1) = p_2(\xi_1)$, so h and h' agree on the generic point U. Furthermore, they both map to Y identically (as they factor through $X \times_Y X$). By the hypothesis of the theorem, there is at most one such lift, so $h = h'$. Consequently, the map $T \to X \times_Y X$ must factor through the diagonal, implying $\xi_0 = L(t_0) \in \Delta(X)$. Thus $\Delta(X)$ is stable under specialization and is closed.

Example 3.5.15: Valuation Criterion on $\mathbb{A}^1$

Let us demonstrate the mechanics of the valuation criterion by proving that the affine line $X = \mathbb{A}^1_k = \operatorname{Spec} k[t]$ is separated over $Y = \operatorname{Spec} k$.

Suppose we are given a valuation ring R with fraction field K, and the standard diagram:

$$\begin{tikzcd}
U = \operatorname{Spec} K \arrow[r, "\alpha"] \arrow[d, "\iota"'] & \operatorname{Spec} k[t] \arrow[d] \\
T = \operatorname{Spec} R \arrow[r] \arrow[ur, dashed, "\exists ? \tilde{\alpha}"] & \operatorname{Spec} k
\end{tikzcd}$$

We translate this geometric data into algebraic data (reversing arrows):

1. The morphism $\alpha : \operatorname{Spec} K \to \operatorname{Spec} k[t]$ corresponds to a k-algebra homomorphism $\alpha^\# : k[t] \to K$. This homomorphism is completely determined by the image of the generator t. Let $\alpha^\#(t) = \lambda \in K$.

2. The inclusion $\iota : \operatorname{Spec} K \to \operatorname{Spec} R$ corresponds to the inclusion of the valuation ring into its fraction field $R \hookrightarrow K$.

3. A lift $\tilde{\alpha} : \operatorname{Spec} R \to \operatorname{Spec} k[t]$ corresponds to a homomorphism $\tilde{\alpha}^\# : k[t] \to R$. Let $\tilde{\alpha}^\#(t) = r \in R$.

For the diagram to commute, we require that the composition $k[t] \xrightarrow{\tilde{\alpha}^\#} R \hookrightarrow K$ equals the original map $\alpha^\#$. Evaluating at t, this means we must have $r = \lambda$ inside K.

Now we analyze the possibilities based on the element $\lambda \in K$:

- *Case 1:* $\lambda \in R$. Then we must define $\tilde{\alpha}^\#(t) = \lambda$. Since the map is determined on generators, this is the *unique* homomorphism making the diagram commute.

- *Case 2:* $\lambda \notin R$. Then there is no element $r \in R$ such that $r = \lambda$. Thus, no lift exists.

In either case, the number of lifts is *at most one* (it is either 1 or 0). Therefore, $\mathbb{A}^1_k$ is separated.

Contrast this with the "Line with two origins" (X). If we map $\operatorname{Spec} K$ to the generic point of X, and R is a DVR, we essentially have a map sending t to 0 in the residue field. In the algebraic formulation, the gluing data allows for two distinct maps $T \to X$ (one to each origin) that become identical when restricted to the generic point U. This violates the uniqueness condition.

Corollary 3.5.16: Properties of Separated Morphisms

Assume that all schemes are Noetherian in the following statements.

1. Open and closed immersions are separated.
2. A composition of two separated morphisms is separated.
3. Separated morphisms are stable under base extension.
4. If $f : X \to Y$ and $f' : X' \to Y'$ are separated morphisms of schemes over a base scheme S, then the product morphism $f \times f' : X \times_S X' \to Y \times_S Y'$ is also separated.
5. If $f : X \to Y$ and $g : Y \to Z$ are two morphisms and if $g \circ f$ is separated, then f is separated.
6. A morphism $f : X \to Y$ is separated if and only if Y can be covered by open subsets V_i such that $f^{-1}(V_i) \to V_i$ is separated for each i.

***Proof*:**
These statements all follow immediately from the condition of the theorem (Valuation Criterion, theorem 3.5.14). We follow Hartshorne [33] and give the proof of (3) to illustrate the method.

Let $f : X \to Y$ be a separated morphism, let $Y' \to Y$ be any morphism, and let $X' = X \times_Y Y'$ be obtained by base extension. We must show that the induced map $f' : X' \to Y'$ is separated. So suppose we are given a valuation ring R with field of fractions K, let $T = \operatorname{Spec} R$ and $U = \operatorname{Spec} K$. Suppose we are given morphisms of T to Y' and U to X' as in the theorem, and two morphisms of T to X' making the diagram

$$\begin{tikzcd}
U \arrow[r] \arrow[d, "i"'] & X' \arrow[r] \arrow[d, "f'"] & X \arrow[d, "f"] \\
T \arrow[r] \arrow[ur, shift left] \arrow[ur, shift right] & Y' \arrow[r] & Y
\end{tikzcd}$$

commutative (where the double arrows represent the two potential lifts $T \to X'$). Composing with the map $X' \to X$, we obtain two morphisms of T to X. Since the outer diagram commutes with respect to f (which is separated), and the maps agree on the dense open set U, these two morphisms $T \to X$ must be the same by theorem 3.5.14.

We also have two morphisms $T \to X' \to Y'$. Since the diagram commutes, both of these must equal the given map $T \to Y'$. Thus, we have two maps $T \to X'$ such that their compositions to X are equal and their compositions to Y' are equal. But X' is the fibered product of X and Y' over Y, so by the universal property of the fibered product, a map to X' is uniquely determined by its projections to X and Y'. Therefore, the two maps of T to X' are the same. Hence f' is separated.

3.5.17 Note on Noetherian Condition You have probably noticed that in order to apply the theorem, it is not necessary to assume that all the schemes mentioned in the corollary are Noetherian. In fact, even in the theorem itself, you can get by with assuming something less than X Noetherian (see EGA I, new ed., 5.5.4).
What Hartshorne says is that if a Noetherian hypothesis will make statements and proofs substantially simpler, then he takes the hypothesis, even though it may not be necessary. This is reasonable as most cases where the valuation citation is useful is when working with stacks which will satisfy these conditions.

Rational maps to Separated Schemes

here

Exercise 3.5.1

1. add exercise 4.5 of Hartshorne: important for Weil divisors to show characterization of irred. closed subscheme of (separated noethean integral) reg. in codim 1 scheme
2. Show that $X \cong \Delta(X)$. Show that in general, $\Delta(X)$ is a locally closed immersion.
3. Show that a scheme X over S is separated if and only if it is separated over $\mathbb{Z}$.
4. If $\varphi : A \to B$ is a ring homomorphism, then the corresponding morphism $\mathrm{Spec}\,(B) \to \mathrm{Spec}\,(A)$ is dominant if and only if $\ker \varphi$ is contained in the nilradical of A.

3.5.3 Universally Closed Morphisms

We now build up to the notion of properness. Proper schemes and morphisms capture the idea of compactness and "compact-preserving" maps. In topology, a map is often called *proper* if the preimage of any compact set is compact. In algebraic geometry, because the Zariski topology is so coarse (and affine schemes are almost always quasicompact), this definition is not strong enough.

Instead, we look for a property that ensures the *image* of a closed set is closed, and remains closed after any base change.

Definition 3.5.18: Universally Closed

Let $f : X \to Y$ be a morphism of schemes. Then f is *universally closed* if it is a closed map (images of closed sets are closed), and for any morphism $Y' \to Y$, the base change $f' : X \times_Y Y' \to Y'$ is also a closed map.

The intuition is that we want to avoid "missing points" at the boundary. Recall the example of $\mathbb{A}^1_k \to \mathrm{Spec}\, k$. This is closed, but base changing to $\mathbb{A}^1_k$ (projection of the plane to the line) maps the hyperbola $xy = 1$ to the open set $\mathbb{A}^1 \setminus \{0\}$. The "missing point" at infinity prevents the map from being universally closed.

Just as Separatedness was characterized by the *uniqueness* of lifts in the valuation criterion, Universally Closed morphisms are characterized by the *existence* of lifts.

Theorem 3.5.19: Valuation Criterion for Universally Closed Morphisms

Let $f : X \to Y$ be a quasicompact morphism of schemes. Then f is universally closed if and only if for any valuation ring R with fraction field K, and any diagram:

$$\begin{tikzcd} U = \operatorname{Spec} K \arrow[r] \arrow[d, "\iota"'] & X \arrow[d, "f"] \\ T = \operatorname{Spec} R \arrow[r] \arrow[ur, dashed, "\exists"] & Y \end{tikzcd}$$

there exists **at least one** morphism $T \to X$ making the diagram commute.

From the perspective of functors of points, the valuative criterion is naturally a statement about extending Spec (K)-valued points to Spec (R)-valued points.

Proof:
(⇒) Existence implies Universally Closed: Suppose the lifting property holds. We want to show f is universally closed. Let $Y' \to Y$ be any base change. We must show $f' : X' \to Y'$ is closed. Since f is quasicompact, f' is quasicompact. By lemma 3.5.13, it suffices to show that the image $f'(X')$ is stable under specialization. Let $z_1 \in X'$ and let $y_1 = f'(z_1)$. Suppose $y_1 \rightsquigarrow y_0$ is a specialization in Y'. We need to find a $z_0 \in X'$ such that $f'(z_0) = y_0$.

Let $K = \kappa(z_1)$. We have a map $\operatorname{Spec} K \to X'$. Let $\mathcal{O}$ be the local ring of y_0 in the closure $\overline{\{y_1\}}$ (with reduced structure). By domination, there exists a valuation ring R of K dominating $\mathcal{O}$. This gives us maps $U \to X'$ and $T \to Y'$. Composing with the projection $X' \to X$, we get maps $U \to X$ and $T \to Y$ fitting the hypothesis diagram. By assumption, there exists a lift $T \to X$. By the universal property of the fibered product X', the pair of maps $T \to X$ and $T \to Y'$ induce a map $T \to X'$. Let z_0 be the image of the closed point of T in X'. Since the map commutes, $f'(z_0)$ must be the image of the closed point of T in Y', which is y_0. Thus y_0 is in the image.

(⇐) Universally Closed implies Existence: Suppose f is universally closed. Consider the diagram with $U \to X$ and $T \to Y$. We base change by $T \to Y$ to get $X_T = X \times_Y T$. The map $U \to X$ and $U \to T$ induce a map $U \to X_T$. Let ξ_1 be the image of the unique point of U in X_T. Let $Z = \overline{\{\xi_1\}}$ with reduced structure. Since f is universally closed, the projection $X_T \to T$ is closed. Thus the image of Z is closed in T. The generic point of T is in the image (it is the image of ξ_1). Since $T = \operatorname{Spec} R$, the closure of the generic point is all of T. Thus, the image of Z is all of T. Therefore, there exists a point $\xi_0 \in Z$ that maps to the closed point $t_0 \in T$.

This gives us a local homomorphism of local rings $R \to \mathcal{O}_{\xi_0, Z}$. Since R is a valuation ring of $K = \kappa(\xi_1)$, and R dominates the local ring (by construction of the map $T \to Y$), R must actually be the local ring (or dominate it via isomorphism). By lemma 3.5.11, this domination data provides a morphism $T \to X_T$ sending $t_0 \to \xi_0$ and $t_1 \to \xi_1$. Composing $T \to X_T \to X$ gives the desired lift.

3.5.4 Proper Morphisms

We now combine the concepts. We want a scheme (or morphism) that behaves like a compact Hausdorff space.

- Hausdorff iff Separated iff Uniqueness of limits.
- Compact iff Universally Closed + Finite Type iff Existence of limits.

Definition 3.5.20: Proper Morphism

A morphism $f : X \to Y$ is called *proper* if it satisfies three conditions:

1. Separated (Hausdorff-like).
2. Universally Closed (Image of closed is closed).
3. Finite Type (Finite covering / geometrically bounded).

A scheme X is said to be *proper* if the morphism $X \to \operatorname{Spec}(\mathbb{Z})$ is proper, and an R-scheme is proper if the map $X \to \operatorname{Spec}(R)$ is proper.

Proposition 3.5.21: Properties of Proper Morphisms

Assume all schemes are Noetherian.

1. A closed immersion is proper.
2. A composition of proper morphisms is proper.
3. Proper morphisms are stable under base change.
4. Products of proper morphisms are proper.
5. If $f : X \to Y$ and $g : Y \to Z$ are morphisms, g is separated, and $g \circ f$ is proper, then f is proper.
6. Properness is local on the base (target).

***Proof*:**

1. A closed immersion is separated (diagonal is iso), finite type (surjection of rings), and universally closed (closed sets in a subspace are closed in the ambient space).
2. Let $f : X \to Y$ and $g : Y \to Z$ be proper. Separatedness and finite type are preserved under composition. For universally closed: Let $Z' \to Z$. Base change $X \to Z$ by Z' is the composition of base changing g then f. Since pullback preserves closed maps, the composition is closed.
3. Finite type and separatedness are stable. Universal closedness is stable by definition.
4. Follows formally from Base Change and Composition.

5. This is best proved using the Valuation Criterion (theorem 3.5.25). Let R be a valuation ring with field K. Suppose we have a diagram:

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

We want a unique lift. Compose with $g : Y \to Z$ to get a map $\operatorname{Spec} K \to X$ and $\operatorname{Spec} R \to Z$. Since $g \circ f$ is proper, there exists a unique lift $\operatorname{Spec} R \to X$. This lift makes the big diagram commute. Does it make the triangle with Y commute? We have two maps $\operatorname{Spec} R \to Y$: one given, and one via the lift through X. Both map the generic point to the same place, and both compose with g to give the same map to Z. Since g is *separated*, by the valuation criterion for separatedness, these two maps must be equal. Thus the lift works for f.

The following two propositions give a large class of useful proper morphisms, and consequence proper schemes over Y:

Proposition 3.5.22: Finite Morphisms are Proper

Let $f : X \to Y$ be a finite morphism. Then f is proper.

***Proof*:**
A finite morphism is affine, hence separated. It is finite type (actually finite presentation if Noetherian). We need only show it is universally closed. Since properness is local on the target, assume $Y = \operatorname{Spec} R$. Then $X = \operatorname{Spec} S$ where S is a finite R-module. Let $Y' \to Y$ be a base change. The base change of a finite morphism is finite ($S \otimes_R R'$ is a finite R'-module). Thus, it suffices to show that any finite morphism is a closed map. Let $\varphi : A \to B$ be a finite ring map. Let $V(I) \subseteq \operatorname{Spec} B$ be a closed set. The image is $V(\varphi^{-1}(I))$ inside $\operatorname{Spec} A$. This follows from the *Going Up Theorem* [10] (or the Cohen-Seidenberg Theorem). Since B is integral over A, the map $\operatorname{Spec} B \to \operatorname{Spec} A$ is closed.

If the focus is put on morphisms between affine schemes (or slightly more generally affine morphisms), then properness reduces to finiteness:

Proposition 3.5.23: Affine Proper Morphisms are Finite

Let $f : X \to Y$ be an affine morphism. Then f is proper if and only if f is finite.

***Proof*:**
($\Leftarrow$) This follows immediately from proposition 3.5.22. A finite morphism is always affine and proper.

($\Rightarrow$) Suppose f is affine and proper. We must show it is finite. Since finiteness is local on the base, we may assume $Y = \operatorname{Spec} A$ is affine. Because f is affine, $X = \operatorname{Spec} B$ can also be assumed to be affine, and so f corresponds to a ring homomorphism $\varphi : A \to B$.

Since f is proper, it is of *finite type*. This means B is a finitely generated A-algebra.

To show finiteness, we must show that φ is *integral*. We use the Valuative Criterion for Universally Closed Morphisms (theorem 3.5.19). Let K be the fraction field of B (or any field containing B) and let $R \subset K$ be any valuation ring containing A (more precisely, containing $\varphi(A)$). Consider the diagram:

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & \operatorname{Spec} B \arrow[d] \\ \operatorname{Spec} R \arrow[r] & \operatorname{Spec} A \end{tikzcd}$$

Since f is universally closed, a lift $\operatorname{Spec} R \to \operatorname{Spec} B$ exists. Algebraically, this means the map $B \to K$ factors through R, i.e., $B \subseteq R$.

Thus, B is contained in every valuation ring of K that contains A. A standard theorem in commutative algebra states that the integral closure of A in K is the intersection of all valuation rings of K containing A, see lemma 5.4.13 for a slightly stronger proof. Therefore, B is integral over A.

We now have that B is a finitely generated A-algebra that is integral over A. A standard result in commutative algebra states that such a ring is a finitely generated A-module (see [10]). Since B is a finite A-module, f is a finite morphism, completing the proof.

Example 3.5.24: Non-Proper Morphisms

1. $\mathbb{A}^1_k \to \operatorname{Spec} k$ is not proper. It is not universally closed (base change to $\mathbb{A}^1$ maps the hyperbola to an open set).
2. $\mathbb{A}^1_k \setminus \{0\} \to \mathbb{A}^1_k$ (open immersion). This is separated and finite type, but not closed (image is open).
3. The "Line with two origins" is not proper because it is not separated.

Theorem 3.5.25: Valuation Criterion for Properness

Let $f : X \to Y$ be a morphism of finite type, with X Noetherian. Then f is proper if and only if for every valuation ring R with fraction field K, and every diagram:

$$\begin{tikzcd} U \arrow[r] \arrow[d, "\iota"'] & X \arrow[d, "f"] \\ T \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

there exists a *unique* lifting $T \to X$ making the diagram commute.

***Proof*:**
See derivation in previous section.

Projective Schemes and Consequences

Theorem 3.5.26: Projective Morphisms are Proper

A projective morphism between Noetherian schemes is proper.

***Proof*:**
(See previous section for proof via valuation criterion).

One of the most powerful consequences of properness is the behavior of global functions. For compact complex manifolds, Liouville's theorem states that holomorphic functions on a compact connected manifold are constant. We have the algebraic analogue:

Theorem 3.5.27: Global Sections of Proper Schemes

Let k be a field. Let X be a proper k-scheme that is *geometrically* reduced and *geometrically* connected. Then:

$$\Gamma(X, \mathcal{O}_X) = k$$

(i.e., the only global regular functions are constants). Over an algebraically closed field the two hypotheses are just reducedness and connectedness.

The adverb is not decoration. Take $k = \mathbb{R}$ and $X = \operatorname{Spec}\mathbb{C}$: this is proper over $\mathbb{R}$ (it is finite), reduced, and connected, yet $\Gamma(X, \mathcal{O}_X) = \mathbb{C} \neq \mathbb{R}$. What fails is that X has a point whose residue field is larger than k, and $X_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C})$ is disconnected. Geometric connectedness is exactly the hypothesis that rules this out, by proposition 3.4.24 and the discussion around it.

***Proof*:**
Let $f \in \Gamma(X, \mathcal{O}_X)$ be a global section. By proposition 3.3.9, we get a scheme morphism (that we shall also denote as f)

$$f : X \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$$

Consider the open immersion $\iota : \mathbb{A}^1_k \hookrightarrow \mathbb{P}^1_k$. Compose them to get $\varphi : X \to \mathbb{P}^1_k$. Since X is proper over k and $\mathbb{P}^1_k$ is separated over k, the image $\varphi(X)$ must be a *closed* subset of $\mathbb{P}^1_k$. However, f maps into the affine line, so the image $\varphi(X)$ does not contain the point at infinity, ∞. Thus, $\varphi(X)$ is a closed subset of $\mathbb{P}^1_k$ contained entirely in $\mathbb{A}^1_k$. The closed subsets of $\mathbb{P}^1_k$ are finite sets of points or the whole space. Since it misses ∞, it must be a finite set of points. Since X is connected, the image is a single closed point $\mathfrak{p} \in \mathbb{A}^1_k$.

It remains to see that the residue field at $\mathfrak{p}$ is k, and this is where the geometric hypotheses enter; without them $\mathfrak{p}$ may be a closed point of positive degree, as the example after the statement shows. Base change to $\overline{k}$. Formation of global sections commutes with the flat base change $\operatorname{Spec}\overline{k} \to \operatorname{Spec} k$, so $\Gamma(X, \mathcal{O}_X) \otimes_k \overline{k} = \Gamma(X_{\overline{k}}, \mathcal{O}_{X_{\overline{k}}})$, and it suffices to prove the claim for $X_{\overline{k}}$. That scheme is proper over $\overline{k}$, and reduced and connected because X was geometrically so. Running the argument above over $\overline{k}$, the image of f is a single closed point of $\mathbb{A}^1_{\overline{k}}$, whose residue field is $\overline{k}$ because $\overline{k}$ is algebraically closed. So f is the constant $\lambda \in \overline{k}$, giving $\Gamma(X_{\overline{k}}, \mathcal{O}) = \overline{k}$ and therefore $\dim_k \Gamma(X, \mathcal{O}_X) = 1$. As $k \subseteq \Gamma(X, \mathcal{O}_X)$, equality holds, completing the proof.

While Projective implies Proper, the converse is not true (though examples are pathological, like Hironaka's example). However, Chow's Lemma states that proper schemes are "birationally" projective.

Theorem 3.5.28: Chow's Lemma

Let $f : X \to S$ be a proper morphism of finite type over a Noetherian base S. Then there exists a scheme X' and a morphism $\pi : X' \to X$ such that:

1. π is a surjective projective morphism.
2. The composite $X' \to S$ is a projective morphism.
3. There exists a dense open $U \subseteq X$ such that $\pi^{-1}(U) \to U$ is an isomorphism.

***Proof*:**
(Sketch). The proof involves taking a finite cover of X by quasi-projective schemes U_i. We form the disjoint union $U = \coprod U_i$ and a map $U \to X$. We then take the closure of the graph of this map inside $X \times_S \mathbb{P}^N$ (for sufficiently large N). The resulting scheme X' dominates X and sits inside a projective space over S, making it projective. The dense open isomorphism comes from the fact that we started with a cover. See [33, Hartshorne II.4.10] for the complete derivation.

Exercise 3.5.2

1. Show that two copies of $\operatorname{Spec}(k[x_1, x_2, ...])$ minus the origin, glued at the origin, is not quasi-separated.
2. Prove that the intersection of two proper subschemes is proper.

3.6 Rational Maps

When working with [quasiprojective] varieties, we had the weaker notion of a rational function that was defined almost everywhere. A rational function was more flexible, allowing for some important classification results, the definition of Kodaira dimension, study singularities, more invariances by looking at the birational automorphisms, and some "local to global" principles. The following brings these concepts over to schemes

Definition 3.6.1: Rational Map

Let X, Y be schemes. Then a *rational map* $\pi : X \dashrightarrow Y$, is an equivalence relation:

$$(\alpha U \to Y) \sim (\beta : V \to Y) \qquad \text{if and only if} \qquad Z \subseteq U \cap V \text{ s.t. } \alpha|_Z = \beta|_Z$$

where U, V, Z are all dense open subsets[a].

If X, Y are S-schemes, then rational maps of S-schemes are defined similarly.

If the image of one of the representatives is dense, then the rational map is said to be *dominant* or *dominating*.

[a] When Y is separated, we will let $Z = U \cap V$

Note that a rational map is is an equivalence relation, not necessarily defined on a single domain.

Definition 3.6.2: Birational Map

A rational map $\pi : X \dashrightarrow Y$ is *birational* if it is dominant, and it has a dominant rational inverse. Two schemes X, Y are *birationally equivalent* if there is a birational map $\pi : X \dashrightarrow Y$.

Example 3.6.3: Rational Map

1. A scheme morphisms always gives a rational map, similarily to how all regular functions are special rational functions
2. The projection $\mathbb{P}^n_R \dashrightarrow \mathbb{P}^{n-1}_R$ projecting down a dimension is a rational map (recall the same reasoning for rational functions in [57, chapter 22.3]

Lemma 3.6.4: Dominating Rational Map on Irreducible Schemes

Let $\pi : X \dashrightarrow Y$ be a rational map between irreducible schemes. Then π is dominating iff it sends the generic point of X to the generic point of Y.

***Proof*:**
recall the properties of generic points in irreducible schemes.

We may naturally want to have a result that is similar to the correspondence of rational functions and field extensions as given in [20]. The full correspondence requires irreducible varieties (integral finite type k-schemes), however we may at least always induce a map of function fields:

Proposition 3.6.5: Rational Maps of Integral Schemes induces Function Field Map

The collection of integrals schemes with dominant maps induces morphisms of function fields in the opposite direction.

***Proof*:**
First show that the form a category (importantly, composition works). Then think of how to create this correspondence.

Corollary 3.6.6: Birational Map on Irreducible Schemes

Let $\pi : X \dashrightarrow Y$ be a birational map between irreducible schemes. Then the induced map of function fields is an isomorphism.

For the converse failing

Example 3.6.7: Function Field Not Inducing Rational Map
Take $\mathrm{Spec}\,(k[x])$ and $\mathrm{Spec}\,(k(X)$ which have the same function field $k(x)$. Then certainly there is no corresponding rational map

$$\mathrm{Spec}\,k[x] \dashrightarrow \mathrm{Spec}\,k(x)$$

of k-schemes, as such a morphism would correspond to a morphism from $U \subseteq \mathrm{Spec}\,k[x]$ (ex. $U = \mathrm{Spec}\,k[x, 1/f(x)]$) to $\mathrm{Spec}\,k(x)$, but not map of rings $k(X) \to k[x, 1/f(x)]$ satisfies the right properties for any $f(x)$.

Proposition 3.6.8: Reduced Schemes and Birational Maps

Let X, Y be reduced schemes. Then X and nY are birational if and only if there exists a dense open subscheme $U \subseteq X$ and $V \subseteq Y$ such that

$$U \cong V$$

***Proof*:**
Generalize the proof from [20] (If stuck, look at Vakil p. 190)

Vakil looks here at rational maps of irreducible varieties that look like translating the results over from varieties, so I will omit for now

I like his translation of the Pythagorean problem into a scheme problem

non-rational complex curve, cool presentation

3.7 Abstract Varieties

Let us use the tools we've developed to our usual notion of variety. We shall see that defining varieties abstractly (other common adverbs being inherently, integrally, intrinsically) shall give us many of the same benefits as defining schemes as spectrum. Most results about quasi-projectile varieties from [20] readily translate to the abstract setting with a few exceptions that we shall go over.

There is a natural way in which we can embed (quasi-projective) varieties into the category $\mathbf{Sch}_{\bar{k}}$:

Theorem 3.7.1: Varieties and Schemes: Fully Faithful Functor

Let $\bar{k}$ be an algebraically closed field. Then there exists a fully faithful functor $\mathbf{Var}_{\bar{k}} \to \mathbf{Sch}_{\bar{k}}$ where $\mathbf{Var}_{\overline{k}}$ is the usual category of varieties (over $\overline{k}$) as defined in [20].

Recall that a fully faithful means the hom-sets are in bijection, and it implies injectivity on objects (but not necessarily surjective, for example Ab $\to$ Grp is a fully faithful functor). We shall also stick with Hartshorne's notation for the functor, which is the letter t, it shall come up a few times later in the book.

***Proof*:**
For any (quasi-projective) variety, we shall first find the "points" that shall be the points of the scheme by working with the irreducible closed subsets. We shall then find the structure sheaf to put on it by finding all the regular functions, which shall be enough to define the functor.

First off, for any topological space X, and let $t(X)$ be the set of (nonempty) irreducible closed subsets of X. If Y is a closed subset of X, then $t(Y) \subseteq t(X), t(Y_1 \cup Y_2) = t(Y_1) \cup t(Y_2)$, and $t\left(\bigcap Y_i\right) = \bigcap t(Y_i)$. Hence, we can define a topology $t(X)$ by taking as closed sets the subsets of the form $t(Y)$, where Y is a closed subset of X. If $f: X_1 \to X_2$ is a continuous map, then we obtain a map $t(f): t(X_1) \to t(X_2)$ by sending an irreducible closed subset to the closure of its image. Thus t is a functor on topological spaces. Furthermore, define a continuous map $\alpha: X \to t(X)$ by $\alpha(P) = \{P\}^-$. Note that α induces a bijection between the set of open subsets of X and the set of open subsets of $t(X)$.

Next, let k be an algebraically closed field, V be a variety over k, and let $\mathcal{O}_V$ be its sheaf of regular functions. I claim that $(t(V), \alpha_*(\mathcal{O}_V))$ is a scheme over k. Since any variety can be covered by open affine sub-varieties, it will be sufficient to show that if V is affine, then $(t(V), \alpha_*(\mathcal{O}_V))$ is a scheme. Hence, let V be an affine variety with affine coordinate ring A. Define a morphism of locally ringed spaces

$$\beta: (V, \mathcal{O}_V) \to X = \operatorname{Spec} A$$

as follows: for each point $P \in V$, let $\beta(P) = \mathfrak{m}_P$, the ideal of A consisting of all regular functions which vanish at P. Then β is a bijection of V onto the set of closed points of X. It is easy to see that β is a homeomorphism onto its image. Now for any open set $U \subseteq X$, we will define a homomorphism of rings $\mathcal{O}_X(U) \to \beta_*(\mathcal{O}_V)(U) = \mathcal{O}_V\left(\beta^{-1}U\right)$. Given a section $s \in \mathcal{O}_X(U)$, and given a point $P \in \beta^{-1}(U)$, we define $s(P)$ by taking the image of s in the stalk $\mathcal{O}_{X,\beta(P)}$, which is isomorphic to the local ring $A_{\mathfrak{m}_P}$, and then passing to the quotient ring $A_{\mathfrak{m}_P}/\mathfrak{m}_P$ which is isomorphic to the field k. Thus s gives a function from $\beta^{-1}(U)$ to k. It is easy to see that this is a regular function, and that this map gives an isomorphism $\mathcal{O}_X(U) \cong \mathcal{O}_V\left(\beta^{-1}U\right)$. Finally, since the prime ideals of A are in 1-1 correspondence with the irreducible closed subsets of V, these remarks show that $(X, \mathcal{O}_X)$ is isomorphic to $(t(V), \alpha_*(\mathcal{O}_V))$, so the latter is indeed an affine scheme, completing the proof.

We have defined upon enough properties to characterize the image of this embedding:

Definition 3.7.2: k-Variety

Let X be a k-scheme. Then if X is reduced, irreducible, separated, and of finite type over k, then X is said to be a *variety over* k or a k*-variety*. If X is affine/(quasi)projective, then it is an affine/(quasi)projective variety over k respectively.

It is important to note that in [20], a variety was an affine algebraic set that was irreducible, which motivates adding irreducible in the definition. However some authors drop the condition and "variety" would be "algebraic set". Note too that we have defined k-varieties and not varieties in general. Furthermore, some authors require the field $k = \overline{k}$ be algebraically closed as part of the definition to more closely reflect the geometric results of the Nullstellensatz, and also to avoid the following problem:

Example 3.7.3: Variety Not Closed Under Base Change

Take the $\mathbb{R}$-variety $\mathbb{C} \cong \mathbb{R}[x]/(x^2+1) = V$ which is a single point, hence irreducible. Then $C \times_{\mathbb{R}} V = \operatorname{Spec} \mathbb{C}[x](x^2+1)$ which now has two points, hence it is not irreducible and so *not* a variety as we have defined it.

While the abstract definition of a scheme allows for elegant categorical constructions, it introduces points that are not closed (the generic points corresponding to non-maximal prime ideals). Notice in the proof of theorem 3.7.1 that the map β is a bijection of the classical variety V onto *only* the closed points of $X = \operatorname{Spec} A$. One might naturally worry: do we lose algebraic information by only evaluating functions at the closed points?

This is where the concept of a *Jacobson scheme* is useful, serving as the exact geometric justification for why our classical intuition of evaluating polynomials at points perfectly captures the algebraic structure. Indeed, it is fair to say that Jacobson schemes are the "most general" setting in which the Nullstellensatz holds.

Definition 3.7.4: Jacobson Scheme

A topological space X is *Jacobson* if its closed points are dense in every closed subset. A scheme X is a *Jacobson scheme* if its Zariski topology is Jacobson.

Recall ([10]) that a *Jacobson ring* is a ring where the intersection of every maximal ideal of R containing $I \subseteq R$ is equal to the radical $\sqrt{I}$.

Proposition 3.7.5: Characterizing Jacobson Affine Schemes

An affine scheme $X = \operatorname{Spec} R$ is Jacobson if and only if R is *Jacobson*.

***Proof*:**
Let $X = \operatorname{Spec} R$. Any closed subset of X is of the form $V(I)$ for some ideal $I \subseteq R$. The closed points of X are exactly the maximal ideals $\operatorname{mSpec}(R)$. Therefore, the closed points residing inside $V(I)$ are exactly the maximal ideals containing I.

The closure of these points inside X is given by taking the vanishing locus of their intersection:

$$\overline{V(I) \cap \mathrm{mSpec}(R)} = V\left(\bigcap_{\substack{\mathfrak{m} \in \mathrm{mSpec}(R) \\ I \subseteq \mathfrak{m}}} \mathfrak{m}\right)$$

If R is a Jacobson ring, this intersection of maximal ideals is exactly $\sqrt{I}$. Thus, the closure becomes $V(\sqrt{I})$, which is equal to $V(I)$. This shows the closed points in $V(I)$ are dense in $V(I)$, making X a Jacobson space.

Conversely, if X is a Jacobson space, the closed points are dense in $V(I)$, so $V\left(\bigcap_{\mathfrak{m} \supset I} \mathfrak{m}\right) = V(I)$. Taking the radical of the defining ideals gives $\bigcap_{\mathfrak{m} \supset I} \mathfrak{m} = \sqrt{I}$, proving R is a Jacobson ring.

Let us formally prove that our geometric intuition of "density" translates exactly to this algebraic property of the nilradical, which ensures we don't lose information.

Corollary 3.7.6: Density of Closed Points and the Nilradical

Let R be a commutative ring and $X = \operatorname{Spec} R$. If the closed points of X are dense in X, then the intersection of all maximal ideals in R is exactly the nilradical $\mathrm{Nil}(R)$.

***Proof*:**
This is a direct application of the logic from proposition 3.7.5 using the zero ideal $I = (0)$.

The entire space X is the closed subset $V(0)$. The condition that the closed points are dense in X means that the closure of the maximal ideals is the whole space:

$$V\left(\bigcap_{\mathfrak{m} \in \mathrm{mSpec}(R)} \mathfrak{m}\right) = V(0)$$

Taking the radical of the defining ideals on both sides yields:

$$\bigcap_{\mathfrak{m} \in \mathrm{mSpec}(R)} \mathfrak{m} = \sqrt{(0)} = \mathrm{Nil}(R)$$

(Note that the left side is its own radical because an intersection of prime ideals is already a radical ideal).

Because the nilradical is zero, the geometry of the maximal ideals is dense enough to completely determine the algebra. This classically justifies viewing polynomials as functions to the base field k. Furthermore, as we saw early in the chapter in section 3.3 (Functor of Points), this classical intuition is perfectly mirrored in scheme theory, where a global section $f \in \Gamma(X, \mathcal{O}_X)$ is canonically equivalent to a scheme morphism $X \to \mathbb{A}^1_k$.

Example 3.7.7: Non-Jacobson Schemes
A scheme that is not Jacobson must have a prime ideal that is not the intersection of maximal ideals. This represents a higher dimensional point that is "not covered" by closed points. Schemes

likes $\operatorname{Spec} k[[x]]$ and $\operatorname{Spec} \mathbb{Z}_{(p)}$. The scheme $\operatorname{Spec} k[x]/(x^n)$ *is* Jacobson, having only (x) as its prime and maximal ideal.

Note that:

$$k[[x]] \cong \varprojlim_n \frac{k[x]}{(x^n)}$$

giving an example of how different the geometry and the algebra can behave under limits. We shall address this in section 5.6.

Having reconciled the abstract algebraic structure with our geometric intuition of points, let us look at the standard ways to construct these varieties.

Proposition 3.7.8: Affine and Projective Varieties

1. $\mathbb{A}^n_k/I$ is an affine k-variety if and only if $I \subseteq k[x_1, ..., x_n]$ is a radical ideal
2. Let $I \subseteq k[x_1, ..., x_n]$ be a radical graded ideal. Then $\operatorname{Proj} k[x_0, ..., x_n]/I$ is a projective k-variety [a]

[a] I need not be radical, ex. $I = (x_0^2, x_0x_1, ..., x_0x_n)$

***Proof*:**
exercise.

An important class of varieties are *proper varieties*. For historical reasons, namely the 19th century Italian school, the following is a common name for proper varieties;

Definition 3.7.9: Complete Variety

Let X be a k-variety. Then it is said to be *complete* if it is proper.

The adjective "complete" for a proper variety comes from the fact that we are in some way thinking of a complete variety as an extension into projective space. It can be shown (see theorem 3.5.28[6]) that if C is complete, then there exists a morphism $C \to \mathbb{P}^n$ to a projective scheme that is isomorphic to an open dense subset. There do exist complete non-projective varieties, but they are hard to come by, partly due to how bad the singularity must be[7].

Before concluding this section, it is crucial to see how finite type schemes (and thus varieties) behave under categorical products, as standard geometric products naturally map to fibered products in the category of schemes.

Proposition 3.7.10: Fibered Products of Finite Type Schemes

Let S be a k-scheme of finite type. If X and Y are k-schemes of finite type over S, then their fibered product $X \times_S Y$ is a finite type k-scheme. Consequently, the product $X \times_k Y$ of two k-varieties over an algebraically closed field is a k-variety.

[6] see Borcherds scheme video abstract and projective varieties

[7] again see the same Borcherds video for an overview. The example he gave also allows for producing algebraic spaces that are not schemes

***Proof*:**
Because $X \to S$ and $Y \to S$ are morphisms of finite type, their base changes are also of finite type. Specifically, the projection $X \times_S Y \to Y$ is a base change of $X \to S$, so it is of finite type. Composing this with the finite type structure morphism $Y \to \operatorname{Spec} k$, we find that $X \times_S Y \to \operatorname{Spec} k$ is of finite type.

In the specific case where $S = \operatorname{Spec} k$ and X, Y are k-varieties over an algebraically closed field k, we know $X \times_k Y$ is of finite type. Furthermore, varieties are separated by definition, and the product of separated schemes is separated. Since k is algebraically closed, the tensor product of two integral k-algebras over k is again an integral domain. Because this property holds locally on affine charts, $X \times_k Y$ is reduced and irreducible. Having satisfied all conditions, $X \times_k Y$ is a k-variety.

One question about varieties has gone untouched so far, and it comes up every time a statement about an arbitrary variety is to be reduced to a statement about curves: given two closed points of a variety of positive dimension, is there an irreducible curve inside the variety through both of them? The strategy for producing one is the strategy classical geometry suggests. Embed the variety in projective space, cut it by a hypersurface through the two points, and repeat until the dimension has come down to one. Each cut must do two things: drop the dimension by exactly one, and leave behind something irreducible. The first is dimension theory and is settled by Krull's theorem; the second is Bertini's theorem, and it is the one ingredient below that this book quotes rather than proves.

Lemma 3.7.11: Hypersurface Sections Drop the Dimension by One

Let k be a field, let $X \subseteq \mathbb{P}^n_k$ be an irreducible closed subvariety with $\dim X = d \geq 1$, and let $F \in k[x_0, \dots, x_n]$ be homogeneous of degree $m \geq 1$ with $X \not\subseteq V_+(F)$. Then every irreducible component of $X \cap V_+(F)$ has dimension $d - 1$.

***Proof*:**
Let W be an irreducible component of $X \cap V_+(F)$, so W is nonempty, and choose an index j with $W \cap D_+(x_j) \neq \emptyset$. Write $U = X \cap D_+(x_j)$, a nonempty affine open subscheme of X. Being irreducible and reduced, X is integral by proposition 2.4.25, so $\mathcal{O}_X(U)$ is a domain by definition 2.4.18.

The quotient $f = F/x_j^m$ is a regular function on $D_+(x_j)$, since $\mathcal{O}_{\mathbb{P}^n_k}(D_+(x_j)) = k[x_0/x_j, \dots, x_n/x_j]$ and F is homogeneous of degree m; let f also denote its restriction to U. If f vanished on U then U would lie in the closed set $V_+(F)$, and U is dense in the irreducible X, so $X \subseteq V_+(F)$, against the hypothesis. Hence f is a nonzero element of a domain, so it is a zero divisor on no irreducible component of U, and U has exactly one such component. Corollary 2.6.14 applies and gives that every irreducible component of

$$V(f) = (X \cap V_+(F)) \cap U$$

has codimension 1 in U.

By lemma 2.6.1 the assignment $Z \mapsto \overline{Z}$ is an inclusion-preserving bijection from the irreducible closed subsets of an open subset onto those of the ambient space meeting it, with inverse

$W \mapsto W \cap U$. Applied to the open subset $(X \cap V_+(F)) \cap U$ of $X \cap V_+(F)$, it shows that $W \cap U$ is an irreducible component of $V(f)$, so $\operatorname{codim}_U(W \cap U) = 1$; applied to the open subset U of X, it turns chains descending from W in X into chains descending from $W \cap U$ in U and back, so $\operatorname{codim}_X W = 1$ as well. Since X is irreducible and of finite type over k, theorem 2.6.12 gives

$$\dim W = \dim X - \operatorname{codim}_X W = d - 1$$

as we sought to show.

The lemma says nothing about irreducibility, and for good reason: the section it produces is usually reducible. What is worse for the problem at hand is that the reducibility can be forced by the two points one wants to keep, so that no amount of generality in the choice of hypersurface removes it. The following is the standard warning example, and it is the reason the sections taken below have degree larger than one.

Example 3.7.12: Hyperplanes Through Two Points Need Not Cut Irreducibly

Let $Q = V_+(wz - xy) \subseteq \mathbb{P}^3_k$ be the quadric surface, which carries two families of lines by example 3.1.34, and let

$$q_0 = [0, 0, 1, 0], \qquad q = [0, 0, 0, 1]$$

be two closed points of the line $\ell = V_+(w, x) \subseteq Q$. A hyperplane $V_+(aw + bx + cy + dz)$ containing q_0 and q has $c = d = 0$, so it is $H = V_+(aw + bx)$ and contains ℓ. For $b \neq 0$ one may substitute $x = -(a/b)w$ into the equation of Q, which becomes $w(bz + ay)/b$ on H, so

$$Q \cap H = V_+(w, x) \cup V_+(aw + bx,\ ay + bz)$$

and for $b = 0$ one gets $Q \cap V_+(w) = V_+(w, x) \cup V_+(w, y)$. In every case the section is a union of two distinct lines, one of them ℓ. So the hyperplanes through q_0 and q cut Q reducibly, all of them, no matter how they are chosen. The lemma this section is aiming at survives the example, since ℓ itself is an irreducible curve through q_0 and q; what does not survive is the naive induction by hyperplane sections.

The obstruction in the example is that the two points span a line lying on the surface, and every hyperplane through two points of a line contains that line. Sections of degree larger than the number of points to be joined are not subject to it, because forms of degree m vanishing at a finite set S still separate the points outside S once m exceeds the number of points of S. By the Veronese embedding (definition 3.1.36) such sections are hyperplane sections in a re-embedding, so nothing is lost by taking them, and the curve produced at the end of the induction still sits inside the original variety. With that adjustment the classical Bertini theorem applies. For a finite set S of closed points of $\mathbb{P}^n_k$ and an integer $m \geq 1$ write

$$W_m(S) = \{F \in k[x_0, \ldots, x_n]_m \mid F(s) = 0 \text{ for every } s \in S\}$$

for the space of degree-m forms vanishing on S, and let $\Lambda_m(S)$ denote the projective space of hypersurfaces (definition 3.1.33) whose defining form lies in $W_m(S)$.

Theorem 3.7.13: Bertini Irreducibility for Sections Through a Finite Set

Let k be algebraically closed, let $X \subseteq \mathbb{P}^n_k$ be an irreducible closed subvariety with $\dim X \geq 2$, let $S \subseteq X$ be a nonempty finite set of closed points, and let $m > |S|$. Then there is a dense open subset $\Omega \subseteq \Lambda_m(S)$ such that for every $H \in \Omega$ the closed set $X \cap H$ is irreducible of dimension $\dim X - 1$. Every such H contains S, so every such section does too.

***Proof Key ideas*:**
The dimension statement and the openness are elementary and are proved here; the irreducibility is the classical Bertini theorem for linear systems and is quoted.

Step 1: the members that swallow X form a proper closed subset. For $s \in S$ the linear forms vanishing at s cut out $\{s\}$, so if all of them vanished on X then $X \subseteq \{s\}$ and $\dim X = 0$, which is excluded; choose therefore a linear form L_s with $L_s(s) = 0$ and $L_s \notin I(X)$, and a linear form $L_1 \notin I(X)$, which exists because X is nonempty. The homogeneous ideal $I(X)$ is prime because X is irreducible, so

$$F_0 = \Big(\prod_{s \in S} L_s\Big) \cdot L_1^{m-|S|}$$

lies in $W_m(S) \backslash I(X)$, and $X \not\subseteq V_+(F_0)$. Consequently $W_m(S) \cap I(X)$ is a proper linear subspace of $W_m(S)$, so the members of $\Lambda_m(S)$ containing X form a proper closed subset. On its complement the section $X \cap H$ is nonempty, since it contains S, and every irreducible component of it has dimension $\dim X - 1$ by lemma 3.7.11.

Step 2: what the system does away from S. Let $p \neq q$ be closed points of $\mathbb{P}^n_k$ outside S. Choosing linear forms L_s with $L_s(s) = 0 \neq L_s(q)$, then L' with $L'(p) = 0 \neq L'(q)$, and L'' with $L''(q) \neq 0$, the form

$$\Big(\prod_{s \in S} L_s\Big) \cdot L' \cdot L''^{\,m-|S|-1}$$

lies in $W_m(S)$, vanishes at p and not at q; here $m > |S|$ is what makes the exponent nonnegative. Dropping the factor L' from the same product gives a member of $W_m(S)$ not vanishing at q, so the common zero locus of $W_m(S)$ is exactly S. The construction with L' chosen to have nonzero differential along a prescribed tangent direction at p separates tangent vectors as well. The base locus S is finite and so of codimension at least 2 in X, and the rational map $\varphi\colon X \dashrightarrow \mathbb{P}(W_m(S)^\vee)$ attached to the system is injective and unramified on $X \setminus S$. It is therefore birational onto its image, and that image has dimension $\dim X \geq 2$.

Step 3: the input that is not available here. What remains is the assertion that a linear system on an irreducible projective variety whose base locus has codimension at least 2 and whose associated rational map is birational onto an image of dimension at least 2 has irreducible general member. The standard proof passes to the incidence variety

$$Z = \{(p, H) \in (X \setminus S) \times \Lambda_m(S) \mid p \in H\}$$

which is irreducible because its projection to $X \setminus S$ is a projective-space bundle, and then compares the number of components of a general fiber of $Z \to \Lambda_m(S)$ with the degree of the finite part of the Stein factorization of that projection. Stein factorization rests on the theorem on formal functions, and neither is developed in this book; it is the same input that forced theorem 3.4.32 to be quoted rather than proved. See [63] for a modern account of the Bertini theorems. The characteristic-zero route through the smoothness form of Bertini's theorem, [33, p. II.8.18], together with connectedness of the general section, is not enough for what follows, since the applications need all characteristics.

The theorem is the only step below that is imported. Everything else is dimension bookkeeping, and the payoff is the statement the section opened with. Note that the two points in it are allowed to coincide, in which case S is a single point; note also that the variety is not assumed complete, and that the curve produced is closed in the variety but generally not complete.

Lemma 3.7.14: Two Closed Points Lie on an Irreducible Curve

Let k be algebraically closed and let V be a quasiprojective k-variety with $\dim V \geq 1$. Given closed points $v_0, v \in V$, there is a closed subvariety $C \subseteq V$, irreducible of dimension 1, with $v_0, v \in C$.

***Proof*:**
Step 1: pass to the projective closure. By definition 3.7.2 and definition 2.3.5 the variety V is an open subscheme of a projective k-scheme, hence a locally closed subscheme of some $\mathbb{P}^n_k$; fix such an immersion and identify V with its image. Let X be the closure of V in $\mathbb{P}^n_k$, with its reduced induced structure. Then X is an irreducible closed subvariety of $\mathbb{P}^n_k$, integral by proposition 2.4.25, and V is open in X: writing $V = Y \cap U$ with Y closed and U open in $\mathbb{P}^n_k$ gives $X \subseteq Y$ and $V = X \cap U$. The generic point η of X (lemma 2.4.20) lies in V, since otherwise the closed set $X \setminus V$ would contain $\overline{\{\eta\}} = X$ and V would be empty. So V and X have the same local ring at η, hence the same function field by corollary 2.4.24, and theorem 2.6.11 gives $\dim X = \dim V =: d$.

Step 2: cut the dimension down inside X. The points v_0 and v are closed in X and not just in V. Indeed, let Z be the closure of v_0 in X, so that $Z \cap V = \{v_0\}$; by lemma 2.6.1 the closure operation matches chains descending from $\{v_0\}$ in V with chains descending from Z in X, so $\operatorname{codim}_X Z = \operatorname{codim}_V \{v_0\} = d$ by theorem 2.6.12 applied to V, and a second application to X gives $\dim Z = 0$. Since Z is irreducible with generic point v_0, a second point of Z would produce a chain of length one, so $Z = \{v_0\}$. Being closed in X, which is closed in $\mathbb{P}^n_k$, the two points are closed points of $\mathbb{P}^n_k$.

Put $S = \{v_0, v\}$, a set of one or two closed points of X, and fix $m = 3$, so that $m > |S|$. Set $X_0 = X$ and suppose an irreducible closed subvariety $X_i \subseteq \mathbb{P}^n_k$ of dimension $d - i \geq 2$ containing S has been produced. Theorem 3.7.13 applied to X_i supplies a hypersurface $H \in \Lambda_3(S)$ with $X_i \cap H$ irreducible of dimension $d - i - 1$; let X_{i+1} be this closed set with its reduced induced structure. It contains S, since $S \subseteq X_i$ and $S \subseteq H$. After $d - 1$ steps this yields an irreducible closed subvariety $C_0 = X_{d-1} \subseteq X$ of dimension 1 containing v_0 and v. If $d = 1$ no step is performed and $C_0 = X$.

Step 3: intersect back into V. Let $C = C_0 \cap V$ with its reduced induced structure. Since C_0 is closed in X and V is an open subscheme of X, the subscheme C is closed in V and open in C_0. It contains v_0 and v, so it is nonempty, and a nonempty open subscheme of an irreducible scheme is irreducible and dense; so C is irreducible and shares the generic point and function field of C_0 by lemma 2.4.20 and corollary 2.4.24, whence $\dim C = \dim C_0 = 1$ by theorem 2.6.11. Finally C is reduced, being open in a reduced scheme, and separated and of finite type over k, being a locally closed subscheme of $\mathbb{P}^n_k$; so C is a k-variety in the sense of definition 3.7.2, irreducible of dimension 1 and closed in V, containing both points, completing the proof.

Step 3 is where the hypothesis of quasiprojectivity earns its place. The cutting in Step 2 has to happen in the projective closure, because that is where hypersurface sections live, and the curve it produces need not lie in V; but it meets V in a nonempty open subset of itself, and a nonempty open subset of an irreducible curve is again an irreducible curve, closed in V because the ambient curve was closed in X. The same device settles the complete case, where the variety at hand need not be projective at all. There the projective model is supplied by Chow's lemma, and the curve is pushed forward rather than intersected back.

Corollary 3.7.15: Curves Joining Two Points of a Complete Variety

Let k be algebraically closed, let V be a complete k-variety (definition 3.7.9) with $\dim V \geq 1$, and let $v_0 \neq v$ be closed points of V. Then there is a closed subvariety $C \subseteq V$, irreducible of dimension 1, with $v_0, v \in C$.

***Proof*:**
Step 1: a projective model. The structure morphism $V \to \operatorname{Spec} k$ is proper and of finite type, and the base $\operatorname{Spec} k$ is Noetherian, so theorem 3.5.28 provides a projective k-scheme X', a surjective morphism $\pi\colon X' \to V$, and a dense open $U \subseteq V$ with $\pi^{-1}(U) \to U$ an isomorphism. Let X be the closure of $\pi^{-1}(U)$ in X' with its reduced induced structure. As U is irreducible, so are $\pi^{-1}(U)$ and X; thus X is a projective k-variety. It contains the dense open subscheme $\pi^{-1}(U) \cong U$, so $\dim X = \dim U = \dim V \geq 1$ by lemma 2.4.20, corollary 2.4.24 and theorem 2.6.11. The restriction $\pi|_X$ factors as the closed immersion $X \hookrightarrow X'$ followed by π, so it is proper by theorem 3.5.26 and proposition 3.5.21; its image is therefore closed and contains the dense subset U, which makes $\pi|_X$ surjective onto V.

Step 2: lift the two points. The fibers $\pi|_X^{-1}(v_0)$ and $\pi|_X^{-1}(v)$ are nonempty closed subsets of X, and X is a closed subspace of some $\mathbb{P}^N_k$, which is Noetherian by lemma 2.4.7; so the closed subsets of X satisfy the descending chain condition and each fiber contains a minimal nonempty closed subset T. By minimality every point of T has closure equal to T, so T is irreducible and each of its points is a generic point of it; the generic point of an irreducible closed subset of an integral scheme is unique by lemma 2.4.20, so T is a single point, closed in X. Choose such points $x_0 \in \pi|_X^{-1}(v_0)$ and $x \in \pi|_X^{-1}(v)$.

Step 3: produce the curve upstairs and push it down. A projective scheme is an open subscheme of itself, so X is a quasiprojective k-variety and lemma 3.7.14 supplies an irreducible closed subvariety $C' \subseteq X$ of dimension 1 with $x_0, x \in C'$. Let $C = \pi(C')$ with its reduced induced structure. It is closed, since $\pi|_X$ is proper and C' is closed in X, and it is irreducible, being the continuous image of an irreducible space; and it contains v_0 and v. It is a k-variety, being a reduced irreducible closed subscheme of the separated finite-type scheme V.

It remains to compute $\dim C$. The generic point η' of C' maps to the generic point of C, because the closure of its image contains $\pi(C') = C$; so π induces a homomorphism $k(C) \to k(C')$ of function fields (corollary 2.4.24), which is injective because its source is a field. Theorem 2.6.11 therefore gives $\dim C \leq \dim C' = 1$. The dimension is not 0: if $\dim C = 0$ then theorem 2.6.12 applied to the irreducible closed subset $\{v_0\} \subseteq C$ forces $\operatorname{codim}_C\{v_0\} = 0$ and hence $\{v_0\} = C$, contradicting $v \in C$ and $v \neq v_0$. So $\dim C = 1$, completing the proof.

The restriction to distinct points is what the argument gives and not an artifact of bookkeeping: the image of a curve under π can be a single point, and the only thing ruling that out above is that C has to contain two distinct points. The two statements together cover the varieties that arise in practice, and they are what licenses the reduction of a question about an arbitrary variety to the same question about a curve, which is then normalized and compactified. That reduction is the engine behind the rigidity lemma and the theorem of the cube in *Abelian Varieties* [6]. One gap is left open by the pair of statements: a variety that is separated and of finite type but neither quasiprojective nor complete is reached only by first compactifying it, and Nagata's compactification theorem is not developed in this book.

3.8 Bertini Theorems

The construction of a curve through two prescribed points quoted its one serious input, the irreducibility of a general hypersurface section, from outside the book. That debt is now paid, and paid in a stronger currency. This section proves two Bertini theorems from scratch: general hyperplane sections of a smooth variety are smooth, and the preimage of a general hyperplane under a map to projective space with two-dimensional image is geometrically irreducible. The second is the theorem theorem 3.7.13 was reaching for, though it is not the same statement, and the difference is the subject of box 3.8.11.

The section closes with a quantitative refinement, and that refinement is the third of its theorems. For the applications it is not enough to know that a general member of a family behaves; one wants to know how large the bad locus is inside the parameter space, because the members one is allowed to use are often confined to a subfamily. The refinement measures the bad locus by its codimension in a Grassmannian of linear systems, and the measurement is sharp enough that a subfamily of codimension one still meets the good locus. Its proof is an induction that removes one hypersurface at a time, and the mechanism which makes a single cut suffice, a count of the linear subspaces lying inside a fixed closed set, is proved here as well.

Throughout, $(\mathbb{P}^N_k)^\vee$ denotes the dual projective space, whose points parametrize hyperplanes: a point $[a] = [a_0 : \cdots : a_N]$ corresponds to the hyperplane

$$H_a = V_+(a_0X_0 + \cdots + a_NX_N) \subseteq \mathbb{P}^N_k$$

and this is a bijection between the k-rational points of $(\mathbb{P}^N_k)^\vee$ and the hyperplanes of $\mathbb{P}^N_k$ defined over k. A property of hyperplanes holds for a *general* hyperplane if there is a dense open subset $\Omega \subseteq (\mathbb{P}^N_k)^\vee$ such that every H_a with $[a] \in \Omega$ a k-rational point has it. The same language is used for members of any family parametrized by a variety.

Over a finite field this convention can be vacuous, since a dense open subset of a projective space over $\mathbb{F}_q$ may have no $\mathbb{F}_q$-rational point at all. That is why an infinite ground field appears in the hypotheses of every irreducibility statement below. Bertini theorems over finite fields are a separate subject, opened up by Charles and Poonen [5]; nothing proved here says anything about that case.

Lemma 3.8.1: Rational Points of a Nonempty Open Set over an Infinite Field

Let k be an infinite field and $\Omega \subseteq \mathbb{P}^M_k$ a nonempty open subscheme. Then Ω has a k-rational point.

***Proof*:**
The standard charts $D_+(X_i) \cong \mathbb{A}^M_k$ cover $\mathbb{P}^M_k$, so Ω meets one of them; replacing Ω by $\Omega \cap D_+(X_i)$ reduces to $\Omega \subseteq \mathbb{A}^M_k = \operatorname{Spec} k[y_1, \ldots, y_M]$. A nonempty open subscheme of an affine scheme contains a nonempty basic open $D(f)$, and $D(f) \neq \emptyset$ forces $f \neq 0$.

It remains to produce $c \in k^M$ with $f(c) \neq 0$, since such a c is a k-rational point of $\mathbb{A}^M_k$ lying in $D(f)$. Induct on M. For $M = 1$ a nonzero polynomial in one variable has at most $\deg f$ roots and k is infinite. For $M > 1$ write $f = \sum_i f_i(y_1, \ldots, y_{M-1}) y_M^i$ and choose i_0 with $f_{i_0} \neq 0$; by induction there is $c' \in k^{M-1}$ with $f_{i_0}(c') \neq 0$, so $f(c', y_M)$ is a nonzero polynomial in one variable and the case $M = 1$ supplies c_M with $f(c', c_M) \neq 0$, completing the proof.

Two further preliminaries recur. The first bounds the size of preimages under a dominant morphism and produces a dense open subset of the target that is genuinely hit. It is proved here rather than quoted because the general theory of fiber dimension is developed later in the book.

Lemma 3.8.2: Generic Fiber Dimension

Let k be an algebraically closed field and $\pi\colon U \to W$ a dominant morphism of irreducible k-varieties. Put $D = \dim U$, $w = \dim W$ and $e = D - w$. Then there are dense open subschemes $U^\circ \subseteq U$ and $W^\circ \subseteq W$ with $\pi(U^\circ) \subseteq W^\circ$ and a finite surjective morphism

$$\psi\colon U^\circ \longrightarrow W^\circ \times_k \mathbb{A}^e_k$$

whose composite with the first projection is $\pi|_{U^\circ}$. In particular $\pi(U^\circ) = W^\circ$, and for every closed subset $T \subseteq W$,

$$\dim\left(\pi^{-1}(T) \cap U^\circ\right) \leq \dim(T \cap W^\circ) + e$$

whenever the left side is nonempty.

***Proof*:**
Choose a nonempty affine open $W_1 \subseteq W$ and then a nonempty affine open $U_1 \subseteq \pi^{-1}(W_1)$; both are dense, being nonempty opens of irreducible schemes, and $\pi^{-1}(W_1)$ is nonempty because π is dominant. Write $A = \mathcal{O}_W(W_1)$ and $B = \mathcal{O}_U(U_1)$. Both are domains (proposition 2.4.25 and definition 2.4.18), and π induces $A \to B$, injective because $\pi|_{U_1}$ is dominant: a nonzero element of the kernel would confine $\pi(U_1)$ to a proper closed subset of W_1.

Let $K = \operatorname{Frac} A$ and $L = \operatorname{Frac} B$. Theorem 2.6.11 gives $\operatorname{trdeg}_k K = w$ and $\operatorname{trdeg}_k L = D$, hence $\operatorname{trdeg}_K L = e$. Choose $u_1, \ldots, u_e \in B$ forming a transcendence basis of L over K, which is possible because B generates L as a field over K and a transcendence basis may be extracted from any generating set. Then L is algebraic over $K(u_1, \ldots, u_e)$.

Write $B = A[z_1, \dots, z_q]$. Each z_j is algebraic over $K(u_1, \dots, u_e)$, hence satisfies a polynomial relation with coefficients in $A[u_1, \dots, u_e]$; clearing denominators and dividing by the leading coefficient makes z_j integral over $A_{a_j}[u_1, \dots, u_e]$ for a suitable nonzero $a_j \in A$. Put $a = a_1 \cdots a_q$. Then every z_j is integral over $R := A_a[u_1, \dots, u_e]$, so B_a is a finitely generated R-algebra generated by integral elements, hence a finite R-module.

Set $W^\circ = W_1 \cap D(a)$ and $U^\circ = \operatorname{Spec} B_a$, the open subscheme $U_1 \cap \pi^{-1}(D(a))$ of U; it is nonempty because a is a nonzero element of the domain B, hence dense. The inclusion $R \hookrightarrow B_a$ is finite and injective, so $\psi \colon U^\circ \to \operatorname{Spec} R = W^\circ \times \mathbb{A}^e$ is finite, and it is surjective because an injective integral ring extension satisfies lying over. Composing with the projection $\operatorname{Spec} R \to W^\circ$ recovers $\pi|_{U^\circ}$, and surjectivity of ψ together with surjectivity of the projection gives $\pi(U^\circ) = W^\circ$.

For the dimension bound let $T \subseteq W$ be closed and put $T' = T \cap W^\circ$. Then $\pi^{-1}(T) \cap U^\circ = \psi^{-1}(T' \times \mathbb{A}^e)$, and ψ restricts to a finite morphism from this scheme onto a closed subscheme of $T' \times \mathbb{A}^e$. A finite morphism corresponds to an integral ring extension on affine pieces, so lemma 2.6.3 gives

$$\dim \left(\pi^{-1}(T) \cap U^\circ\right) \leq \dim(T' \times_k \mathbb{A}^e_k)$$

Finally $\dim(T' \times \mathbb{A}^e) = \dim T' + e$: it suffices to check this for T' irreducible and reduced, and then $T' \times \mathbb{A}^e$ is an irreducible k-variety by proposition 3.7.10 whose function field is $k(T')(t_1, \dots, t_e)$, of transcendence degree $\dim T' + e$ by theorem 2.6.11, as we sought to show.

The second preliminary transports a statement proved over $\overline{k}$ back to k-rational parameters. It replaces a Galois-descent argument by a linear-algebra one and needs no separability hypothesis.

Lemma 3.8.3: Descent of a Dense Open Locus

Let k be a field with algebraic closure $\overline{k}$ and let $V \subseteq \mathbb{P}^M_{\overline{k}}$ be a dense open subscheme. Then there is a dense open subscheme $\Omega \subseteq \mathbb{P}^M_k$ such that every k-rational point of Ω, viewed as an $\overline{k}$-rational point of $\mathbb{P}^M_{\overline{k}}$, lies in V.

***Proof*:**
The complement $F = \mathbb{P}^M_{\overline{k}} \setminus V$ is a proper closed subset, so $F = V_+(g_1, \dots, g_s)$ for homogeneous $g_1, \dots, g_s \in \overline{k}[X_0, \dots, X_M]$, not all zero. Choose $\theta_1, \dots, \theta_h \in \overline{k}$, linearly independent over k, whose k-span contains every coefficient of every g_j, and write

$$g_j = \sum_{l=1}^{h} \theta_l \, g_{jl}, \qquad g_{jl} \in k[X_0, \dots, X_M]$$

with each g_{jl} homogeneous of the same degree as g_j.

For $c \in k^{M+1}$ the values $g_{jl}(c)$ lie in k, so linear independence of the θ_l gives $g_j(c) = 0$ for all j if and only if $g_{jl}(c) = 0$ for all j and l. Hence the k-rational points of $\mathbb{P}^M_k$ whose associated $\overline{k}$-rational points lie in F are exactly the k-rational points of the closed subscheme $F_0 = V_+(\{g_{jl}\}) \subseteq \mathbb{P}^M_k$. Some g_j is nonzero, so some g_{jl} is a nonzero polynomial and $F_0 \neq \mathbb{P}^M_k$. Take $\Omega = \mathbb{P}^M_k \setminus F_0$, a nonempty open subscheme of an irreducible scheme and therefore dense, completing the proof.

Smoothness of a general hyperplane section

The first Bertini theorem is a statement about tangent spaces. A hyperplane section of a smooth variety fails to be smooth at a point exactly when the hyperplane contains the embedded tangent space there, and the hyperplanes through a fixed d-dimensional linear subspace of $\mathbb{P}^N$ form a linear subspace of $(\mathbb{P}^N)^\vee$ of dimension $N-d-1$. Sweeping this over the d-dimensional X produces a family of dimension $N-1$, one short of $(\mathbb{P}^N)^\vee$, and the theorem follows. The content is that the tangent-space condition is algebraic in the point, which the Jacobian criterion supplies.

Theorem 3.8.4: Bertini's Smoothness Theorem

Let k be an algebraically closed field and $X \subseteq \mathbb{P}^N_k$ a smooth quasiprojective k-variety of dimension d. Then there is a dense open subset $\Omega \subseteq (\mathbb{P}^N_k)^\vee$ such that for every $[a] \in \Omega$ the scheme $X \cap H_a$ is smooth over k, and of pure dimension $d-1$ wherever it is nonempty.

***Proof*:**
If $d = N$ then X is a nonempty open subscheme of $\mathbb{P}^N_k$, no hyperplane contains it, and $X \cap H_a$ is an open subscheme of $H_a \cong \mathbb{P}^{N-1}_k$, hence smooth of pure dimension $N-1$; take $\Omega = (\mathbb{P}^N_k)^\vee$. Assume from now on that $d < N$.

Step 1: when a section is singular at a point. Fix a closed point $x \in X$ and an index i_0 with $x_{i_0} \neq 0$, and define a k-linear map

$$\epsilon_x \colon k^{N+1} \longrightarrow \mathcal{O}_{X,x}/\mathfrak{m}_x^2, \qquad a \longmapsto \Big[\sum_{i=0}^{N} a_i\, X_i/X_{i_0}\Big]$$

Replacing i_0 by another index multiplies ϵ_x by a unit of $\mathcal{O}_{X,x}$ and does not change its kernel, written W_x. The local ring $\mathcal{O}_{\mathbb{P}^N,x}$ is generated as a k-algebra by the fractions X_j/X_{i_0}, so $\mathcal{O}_{\mathbb{P}^N,x}/\mathfrak{m}^2$ is spanned over k by the classes of 1 and of the X_j/X_{i_0}, all of which lie in the image of the corresponding map for $\mathbb{P}^N$; and $\mathcal{O}_{X,x}/\mathfrak{m}_x^2$ is a quotient of $\mathcal{O}_{\mathbb{P}^N,x}/\mathfrak{m}^2$. Hence ϵ_x is surjective. Since X is smooth over the algebraically closed k it is regular (proposition 2.7.12), and $\kappa(x) = k$, so $\dim_k \mathfrak{m}_x/\mathfrak{m}_x^2 = \dim \mathcal{O}_{X,x} = d$ by definition 2.7.2 and theorem 2.6.12. Therefore

$$\dim_k W_x = (N+1) - (d+1) = N - d$$

Suppose $X \not\subseteq H_a$ and set $\ell = \sum_i a_i X_i/X_{i_0}$, a regular function on $X \cap D_+(X_{i_0})$. If $a \in W_x$ then $\ell \in \mathfrak{m}_x^2 \subseteq \mathfrak{m}_x$, so $x \in H_a$; conversely $x \in H_a$ says $\ell \in \mathfrak{m}_x$. The local ring of the section is $\mathcal{O}_{X\cap H_a,x} = \mathcal{O}_{X,x}/(\ell)$. Now $\ell \neq 0$ in $\mathcal{O}_{X,x}$: otherwise ℓ would vanish on a nonempty open subset of the irreducible scheme $X \cap D_+(X_{i_0})$, hence on all of it, hence on X by density, against $X \not\subseteq H_a$. As $\mathcal{O}_{X,x}$ is a domain (proposition 2.7.3), every minimal prime $\mathfrak{p}$ over (ℓ) has height exactly 1: at most 1 by theorem 2.6.13, and at least 1 because $\mathfrak{p} \neq 0$. Applying theorem 2.6.12 to X and to the integral closed subscheme cut out by $\mathfrak{p}$ gives $\dim \mathcal{O}_{X,x}/\mathfrak{p} = d-1$, so $\dim \mathcal{O}_{X\cap H_a,x} = d-1$. The cotangent space of the section is $\mathfrak{m}_x/(\mathfrak{m}_x^2 + (\ell))$, of dimension d if $\ell \in \mathfrak{m}_x^2$ and $d-1$ otherwise.

Comparing the two computations,

$$X \cap H_a \text{ is regular of dimension } d-1 \text{ at } x \iff a \notin W_x$$

and when $a \in W_x$ the section is singular at x.

Step 2: the bad hyperplanes sweep out at most $N-1$ dimensions. Every point of the locally closed subscheme $X \subseteq \mathbb{P}^N_k$ has an affine open neighbourhood in X of the form

$$U = \operatorname{Spec} \frac{k[y_1, \dots, y_N, z]}{(f_1, \dots, f_s,\ zg-1)}$$

where $y_1, \dots, y_N$ are the affine coordinates X_j/X_{i_0} $(j \neq i_0)$ of a chart $D_+(X_{i_0})$ and X is closed in the principal open $D(g)$ of that chart. Since X is Noetherian (lemma 2.4.7 and lemma 2.6.1), finitely many such U cover it. Fix one.

By the cotangent computation inside proposition 2.7.6, the cotangent space of X at a closed point $x \in U$ is the quotient of that of the ambient chart by the differentials of the f_i, so for ℓ as above with $\ell(x) = 0$,

$$\ell \in \mathfrak{m}_x^2 \iff (a_j)_{j \neq i_0} \in \operatorname{Row} J(x), \qquad J(x) = \left(\frac{\partial f_i}{\partial y_j}(x)\right)_{i,j}$$

because the linear part of ℓ at x has coefficient a_j in y_j. Smoothness of X and proposition 2.7.6 give $\operatorname{rank} J(x) = N - d$ at every closed point of U. Consequently

$$W_x = \left\{ a \in k^{N+1} \;\middle|\; a_{i_0} + \textstyle\sum_{j \neq i_0} a_j y_j(x) = 0 \text{ and } (a_j)_{j \neq i_0} \in \operatorname{Row} J(x) \right\}$$

the first condition being the statement $\ell(x) = 0$.

For each choice σ of $N-d$ rows and $N-d$ columns of J, let $U_\sigma \subseteq U$ be the open subscheme where the corresponding minor is invertible. These cover U, the rank being $N-d$ everywhere, and on U_σ the $N-d$ rows of J indexed by σ are linearly independent, hence a basis of $\operatorname{Row} J(x)$. Define

$$\Phi_\sigma \colon U_\sigma \times_k \mathbb{P}^{N-d-1}_k \longrightarrow (\mathbb{P}^N_k)^\vee, \qquad (x, [c]) \longmapsto [a]$$

by $(a_j)_{j \neq i_0} = \sum_{t=1}^{N-d} c_t J_{\sigma_t}(x)$ and $a_{i_0} = -\sum_{j \neq i_0} a_j y_j(x)$. Each coordinate of a is a regular function on U_σ times a linear form in c, and $a \neq 0$ when $c \neq 0$ because the chosen rows are independent, so Φ_σ is a morphism; by the description of W_x its image is exactly the set of $[a]$ for which some closed point $x \in U_\sigma$ has $a \in W_x$.

The source of Φ_σ is an irreducible k-variety (proposition 3.7.10) with function field $k(U_\sigma)(t_1, \dots, t_{N-d-1})$, hence of dimension $d + (N-d-1) = N-1$ by theorem 2.6.11. The closure of its image is irreducible and the induced map of function fields is injective, so theorem 2.6.11 bounds $\dim \overline{\operatorname{im}\Phi_\sigma} \leq N-1$. Let Σ be the union of the closures $\overline{\operatorname{im}\Phi_\sigma}$ over the finitely many pairs (U, σ): a closed subset of $(\mathbb{P}^N_k)^\vee$ of dimension at most $N-1$, hence proper.

Step 3: conclusion. Let $\Lambda \subseteq (\mathbb{P}^N_k)^\vee$ be the set of $[a]$ with $X \subseteq H_a$, the projectivization of the space of linear forms vanishing on X; it is closed, and proper because a point of X lies outside some hyperplane. Put $\Omega = (\mathbb{P}^N_k)^\vee \setminus (\Sigma \cup \Lambda)$, a dense open subset.

Fix $[a] \in \Omega$ with $X \cap H_a \neq \emptyset$. Every closed point x of $X \cap H_a$ has $a \notin W_x$, since otherwise $[a]$ would lie in Σ, so by Step 1 the section is regular of dimension $d-1$ at x. The singular locus of $X \cap H_a$ is closed (proposition 2.7.16); if it were nonempty it would contain a closed point of $X \cap H_a$, because closed points are dense in any affine open of a scheme of finite type over a field (lemma 2.4.4) and a closed point of a closed subscheme of $X \cap H_a$ is a closed point of $X \cap H_a$. Hence $X \cap H_a$ is regular, therefore smooth over the perfect field k by theorem 2.7.13, and of pure dimension $d-1$, completing the proof.

The theorem says nothing about the section being nonempty and nothing about its irreducibility, and both fail routinely. A smooth conic in $\mathbb{P}^2$ meets a general line in two points, a smooth zero-dimensional scheme that is not irreducible, and a general line in $\mathbb{P}^3$ misses a fixed line altogether. What the theorem delivers is the inductive step for cutting a smooth variety down to a smooth curve, provided irreducibility is supplied separately.

Example 3.8.5: Hyperplane Sections of the Twisted Cubic

Let k be algebraically closed and let $X \subseteq \mathbb{P}^3_k$ be the twisted cubic, the image of $\mathbb{P}^1_k \to \mathbb{P}^3_k$, $[s:t] \mapsto [s^3 : s^2t : st^2 : t^3]$. It is a smooth curve, so $d = 1$ and $N = 3$, and the incidence family constructed in the proof has dimension $N - 1 = 2$ inside the three-dimensional $(\mathbb{P}^3)^\vee$. Concretely, the embedded tangent line at the parameter $[s:t]$ is spanned by $[s^3 : s^2t : st^2 : t^3]$ and $[3s^2 : 2st : t^2 : 0]$, and the planes containing that line form a line in $(\mathbb{P}^3)^\vee$; sweeping over the one-dimensional X gives a surface in $(\mathbb{P}^3)^\vee$, the dual surface. A plane H off that surface meets X in three distinct points, a smooth scheme of pure dimension 0; a plane tangent to X meets it in a length-two point together with a further point, and the section is singular there. The count $2 < 3$ is the inequality that makes the general plane work.

Irreducibility of a general hyperplane preimage

Irreducibility is not a local condition and cannot be read off tangent spaces, so the second Bertini theorem needs a different mechanism. The mechanism is a change of viewpoint: instead of examining the members of a family one at a time, one assembles them into a single morphism and asks about its generic fiber. Irreducibility of the generic fiber is a statement about a function field, and the point of the next lemmas is that it can be certified by a single rational point and that it then propagates to almost every closed fiber.

Geometric irreducibility, not just irreducibility, is what is required. The double cover $\mathbb{A}^1 \to \mathbb{A}^1$, $u \mapsto u^2$, has irreducible generic fiber $\operatorname{Spec} k(t)[u]/(u^2 - t)$ while every closed fiber away from the origin is a pair of points; the generic fiber is not geometrically irreducible, and that is the only thing separating the two behaviours.

Lemma 3.8.6: A Smooth Rational Point Forces Geometric Irreducibility

Let F be a field and Z an irreducible F-scheme of finite type having an F-rational point at which Z is smooth over F. Then Z is geometrically irreducible, that is, $Z_{\overline{F}}$ is irreducible.

***Proof*:**
Let Z^{sm} be the locus where $Z \to \operatorname{Spec} F$ is smooth. It is open, contains the given point, and is therefore a nonempty open subscheme of the irreducible Z, hence dense, irreducible and connected, with an F-rational point. By proposition 3.4.24 the base change $Z^{\mathrm{sm}}_{\overline{F}}$ is connected. Smoothness is preserved by base change, so $Z^{\mathrm{sm}}_{\overline{F}}$ is smooth over $\overline{F}$, hence regular by proposition 2.7.12, hence the disjoint union of its irreducible components by proposition 2.7.21. A connected scheme with such a decomposition has exactly one component, so $Z^{\mathrm{sm}}_{\overline{F}}$ is irreducible.

It remains to pass from the open subscheme to $Z_{\overline{F}}$. The scheme Z is integral (proposition 2.4.25); work on an affine open $\operatorname{Spec} B \subseteq Z$ meeting Z^{sm}, so B is a domain and $Z^{\mathrm{sm}} \cap \operatorname{Spec} B \supseteq D(f)$ for some nonzero $f \in B$. The ring $B \otimes_F \overline{F}$ is free, hence flat, over B, so the nonzerodivisor f remains a nonzerodivisor in it, and a nonzerodivisor lies in no associated prime, in particular in no minimal prime (proposition 2.5.3). Hence every irreducible component of $\operatorname{Spec}(B \otimes_F \overline{F})$ meets $D(f)_{\overline{F}}$. Running this over an affine cover shows that every irreducible component of $Z_{\overline{F}}$ meets $Z^{\mathrm{sm}}_{\overline{F}}$, in distinct components of it. Since $Z^{\mathrm{sm}}_{\overline{F}}$ is irreducible, $Z_{\overline{F}}$ has exactly one component, as we sought to show.

The propagation lemma needs one input from field theory which the book does not develop, isolated here so that its use is visible.

Lemma 3.8.7: Separably Generated Function Fields from a Smooth Point

Let F be a field and Z an integral F-scheme of finite type which is smooth over F at some point. Then $F(Z)$ is separably generated over F.

***Proof Key ideas*:**
The argument is short but ends in a field-theoretic fact this book does not prove, so that fact is quoted rather than developed: for a finite extension of fields L/F_0, separability of L over F_0 is equivalent to the vanishing of Ω_{L/F_0} and equivalent to the vanishing of every F_0-derivation of L into L. The implication used below is that a finite extension admitting no nonzero derivation over its base is separable, and the equivalence is proved in *Core Algebra* [18].

By definition 2.7.8 the smooth point has a neighbourhood of the form $\operatorname{Spec} F[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$ on which the Jacobian matrix $(\partial f_i / \partial x_j)$ has rank r, with $n - r = \dim Z$. After renumbering, the minor on the first r columns is invertible at the point, hence on a neighbourhood. Write $F_0 = F(x_{r+1}, \ldots, x_n)$, a purely transcendental subextension of $L = F(Z)$ of transcendence degree $n - r = \dim Z$, so that L/F_0 is finite. An F_0-derivation δ of L kills $x_{r+1}, \ldots, x_n$ and satisfies $\sum_{j \le r} (\partial f_i / \partial x_j)\, \delta(x_j) = 0$ for each i; invertibility of the matrix forces $\delta = 0$. By the quoted fact L/F_0 is separable, so $x_{r+1}, \ldots, x_n$ is a separating transcendence basis, completing the account.

Lemma 3.8.8: Irreducibility of the General Fiber

Let k be an algebraically closed field, let Z and T be irreducible k-varieties, and let $f\colon Z \to T$ be a dominant morphism with generic fiber Z_η over the generic point η of T. Assume Z_η has a $\kappa(\eta)$-rational point at which it is smooth over $\kappa(\eta)$. Then there is a dense open subset $T^\circ \subseteq T$ such that $f^{-1}(\lambda)$ is nonempty and irreducible for every closed point $\lambda \in T^\circ$.

***Proof*:**
Write $D = \dim Z$, $t = \dim T$, $e = D - t$, $K = \kappa(\eta) = k(T)$ and $L = k(Z)$.

Step 1: the generic fiber is geometrically irreducible. The points of Z lying over η, with the subspace topology, are the underlying space of Z_η, and its irreducible closed subsets are the traces of the irreducible closed subsets of Z dominating T. Since Z is irreducible it is the only maximal such, so Z_η is irreducible with function field L. By hypothesis it has a smooth K-rational point, so lemma 3.8.6 makes it geometrically irreducible and lemma 3.8.7 makes L/K separably generated.

Step 2: shrinking Z is harmless. Shrink T to its smooth locus, dense by theorem 2.7.17. For a closed point $\lambda \in T$ the maximal ideal $\mathfrak{m}_\lambda$ is then generated by t elements, so $f^{-1}(\lambda)$ is locally cut out in Z by t functions and theorem 2.6.13 bounds the codimension of each of its components by t; with theorem 2.6.12 every component of every nonempty fiber has dimension at least e.

Now let $F \subsetneq Z$ be closed. There is a dense open $T_F \subseteq T$ with $\dim(F \cap f^{-1}(\lambda)) \le e - 1$ for every closed $\lambda \in T_F$. Induct on $\dim F$, treating the irreducible components separately. For a component F' with $\overline{f(F')} \neq T$, delete $\overline{f(F')}$ from T. Otherwise $f|_{F'}$ is dominant with $\dim F' \le D - 1$, so lemma 3.8.2 supplies dense opens $F'^\circ \subseteq F'$ and $T'^\circ \subseteq T$ with $\dim(F'^\circ \cap f^{-1}(\lambda)) \le \dim F' - t \le e - 1$ for $\lambda \in T'^\circ$, while $F' \setminus F'^\circ$ has smaller dimension and is handled by induction. Combining, for λ in a dense open subset every irreducible component of $f^{-1}(\lambda)$ meets any prescribed dense open $Z^\circ \subseteq Z$, so the components of $f^{-1}(\lambda)$ correspond bijectively to those of $f^{-1}(\lambda) \cap Z^\circ$.

Step 3: a hypersurface model. Fix a separating transcendence basis $u_1, \ldots, u_e$ of L over K, so that $L/K(u)$ is finite separable, and let θ be a primitive element, scaled to be integral over $K[u]$, with minimal polynomial $P(\Theta) \in K(u)[\Theta]$, monic of degree δ and separable. Choose a nonempty affine open $T_1 \subseteq T$ and put $A = \mathcal{O}_T(T_1)$, so that $K = \mathrm{Frac} A$. Only finitely many elements of K occur among the coefficients of P, so there is a nonzero $\alpha \in A$ with $P \in A_\alpha[u][\Theta]$. Put

$$B' = \frac{A_\alpha[u_1, \ldots, u_e][\Theta]}{(P)}$$

Being monic, P is primitive over the unique factorization domain $A_\alpha[u]$ and irreducible over $K(u)$, so Gauss's lemma makes it irreducible in $A_\alpha[u][\Theta]$ and B' a domain with fraction field L.

Both B' and an affine model B'' of a dense open subscheme of Z over T_1 are finitely generated A-algebras with fraction field L, so they agree after inverting one element: writing each generator of one as a fraction of elements of the other and clearing the finitely many denominators produces nonzero h' and h'' with $(B')_{h'} \cong (B'')_{h''}$. Let $Z^\circ \subseteq Z$ be the corresponding dense open subscheme. By Step 2 it suffices to prove that the fiber of $\operatorname{Spec}(B')_{h'}$ over a general closed point λ is nonempty and irreducible.

Step 4: specializing the equation. For a closed point $\lambda \in T_1$ with $\alpha(\lambda) \neq 0$, write $P_\lambda \in k[u_1, \ldots, u_e][\Theta]$ for the reduction of P, monic of degree δ in Θ; the fiber of $\operatorname{Spec} B'$ over λ is $\operatorname{Spec} k[u][\Theta]/(P_\lambda)$. Separability of P makes its discriminant with respect to Θ a nonzero element of $A_\alpha[u]$, so for λ outside a proper closed subset P_λ is squarefree, and then the fiber is irreducible precisely when P_λ is irreducible in $k[u][\Theta]$.

The generic fiber of $\operatorname{Spec} B'$ over T is $\operatorname{Spec} K[u][\Theta]/(P)$, an open subscheme of Z_η under the identification of Step 3 and therefore geometrically irreducible by Step 1; so P is a power of an irreducible polynomial over $\overline{K}$, and being squarefree it is irreducible over $\overline{K}$.

Fix i with $0 < i < \delta$ and let $\mathcal{M}_i$ be the affine space of polynomials in (u, Θ) monic of degree i in Θ and of total degree at most $\deg P$; a factorization of the monic P_λ into two factors of positive Θ-degree can always be normalized so that both factors are monic in Θ, and their total degrees are then bounded by $\deg P$. Let $\mathcal{Y}_i$ be the closed subscheme of $T_1 \times \mathcal{M}_i \times \mathcal{M}_{\delta-i}$ cut out by the coefficientwise equations $P_\lambda = GH$. If the projection of $\mathcal{Y}_i$ to T_1 had dense image, some irreducible component of $\mathcal{Y}_i$ would have dense image and the generic point ζ of that component would map to η; then P would factor into two factors of positive Θ-degree over the field $\kappa(\zeta) \supseteq K$. Choosing an algebraic closure Ξ of $\kappa(\zeta)$, which contains an algebraic closure of K, the polynomial P would be reducible over Ξ. But $\overline{K}[u][\Theta]/(P)$ is a domain, and a finitely generated domain over an algebraically closed field remains a domain after any extension of the base field, so $\Xi[u][\Theta]/(P)$ is a domain and P is irreducible over Ξ. Hence the projection of each $\mathcal{Y}_i$ to T_1 has image inside a proper closed subset, and P_λ is irreducible for λ outside a proper closed subset of T.

Step 5: conclusion. Delete from T the proper closed subsets produced in Steps 2 and 4, the zero locus of α, the zero locus of the discriminant, and the complement of the dense open image supplied by lemma 3.8.2 applied to $\operatorname{Spec}(B')_{h'} \to T$; what remains is a dense open T°. For $\lambda \in T^\circ$ the fiber of $\operatorname{Spec}(B')_{h'}$ is a nonempty open subscheme of the irreducible $\operatorname{Spec} k[u][\Theta]/(P_\lambda)$, hence irreducible, and Step 2 transfers this to $f^{-1}(\lambda)$, completing the proof.

One case of the theorem below resists every argument of this kind. If π is inseparable, for instance the Frobenius $\mathbb{P}^n \to \mathbb{P}^n$ in characteristic p, then the differential of π vanishes identically, no preimage of a hyperplane is smooth anywhere, and no rational point of any generic fiber is a smooth point. The repair is to twist the target by a power of Frobenius until the map becomes generically smooth, which changes neither the topology of the source nor the family of hyperplanes on the target. The field theory behind the twist is isolated next.

Lemma 3.8.9: Generic Smoothness after a Frobenius Twist

Let k be an algebraically closed field of characteristic $p > 0$, let U be an irreducible k-variety, let $\pi\colon U \to \mathbb{P}^r_k$ be a morphism whose image is dense in an irreducible closed subvariety W, and for $s \geq 0$ let $F^s\colon \mathbb{P}^r_k \to \mathbb{P}^r_k$ be the morphism $[X_i] \mapsto [X_i^{p^s}]$. Write $U_{(s)}$ for the reduced closed subscheme underlying $U \times_{\mathbb{P}^r_k, F^s} \mathbb{P}^r_k$. Then $U_{(s)} \to U$ is a homeomorphism for every s, and for all sufficiently large s the induced morphism $U_{(s)} \to \mathbb{P}^r_k$ is smooth at some point.

***Proof Key ideas*:**
The homeomorphism statement is elementary. The morphism F^s is bijective on points, k being perfect, and it is closed, so it is a homeomorphism; its base change along π is again bijective and closed, and passing to the reduced subscheme does not change the underlying space.

The second statement is a field-theoretic assertion this book does not develop, and it is quoted for that reason: for a finitely generated extension L/K in characteristic p, the compositum $L \cdot K^{1/p^s}$ is separably generated over K^{1/p^s} once s is large. Here $K = k(W)$ and $L = k(U)$, and the function field of the twisted target is K^{1/p^s} because k is perfect. The proof compares the modules of Kähler differentials Ω_{L_s/K_s} as s grows: they are generated by the differentials of a fixed finite generating set, the transition maps are surjective, and over the perfect closure the dimension equals the transcendence degree, so the dimensions stabilize there. Kähler differentials are not available at this point in the book, and the limit statement is not developed in it at all; the assertion in this geometric form is [40, Sec. 5], completing the account.

Kollár states the theorem for a map drawn as a dashed arrow but called a morphism. The two are not interchangeable, and for a rational map the symbol $\pi^{-1}(H)$ has to be given a meaning; the statement below fixes a domain of definition and takes preimages there, which is what the proof manipulates. It also weakens dominance to the condition the proof consumes, that the image has dimension at least two.

Theorem 3.8.10: Bertini's Irreducibility Theorem

Let k be an infinite field, let X be a geometrically irreducible k-variety, and let $\pi\colon X \dashrightarrow \mathbb{P}^r_k$ be a rational map represented by a morphism $\pi_U\colon U \to \mathbb{P}^r_k$ on a dense open subscheme $U \subseteq X$, and suppose the closure of the image of π_U has dimension at least 2. Then there is a dense open subset $\Omega \subseteq (\mathbb{P}^r_k)^\vee$ such that $\pi_U^{-1}(H_a)$ is geometrically irreducible for every k-rational point $[a] \in \Omega$. Such $[a]$ exist by lemma 3.8.1.

***Proof*:**
Step 1: reduction to an algebraically closed field. The scheme $\pi_U^{-1}(H_a)$ is geometrically irreducible exactly when its base change to $\overline{k}$ is irreducible, and that base change is $(\pi_U)_{\overline{k}}^{-1}\big((H_a)_{\overline{k}}\big)$, preimages commuting with base change. The scheme $X_{\overline{k}}$ is irreducible by hypothesis, $U_{\overline{k}}$ is a dense open subscheme of it, and the closure of the image still has dimension at least 2. Granting the theorem over $\overline{k}$, lemma 3.8.3 converts the dense open subset of $(\mathbb{P}^r_{\overline{k}})^\vee$ it produces into a dense open subset of $(\mathbb{P}^r_k)^\vee$ whose k-rational points work. So assume $k = \overline{k}$, replace X by U, and produce a dense open subset of $(\mathbb{P}^r_k)^\vee$ over whose closed points the preimage is irreducible.

Step 2: normal form. Let $W = \overline{\pi(U)}$ with the reduced induced structure, an irreducible closed subvariety of $\mathbb{P}^r_k$ of dimension $w \geq 2$, and regard π as a dominant morphism $U \to W$; note $r \geq w \geq 2$. Put $D = \dim U$ and $e = D - w$. For a hyperplane H one has $\pi^{-1}(H) = \pi^{-1}(H \cap W)$, and $\pi^{-1}(H)$ is cut out in U by one equation which is not identically zero, so by theorem 2.6.13 and theorem 2.6.12 it is either empty or of pure dimension $D - 1$.

Suppose first that π is smooth at some point of U, so that the smooth locus U^{sm} of π is a dense open subscheme; the other case is Step 6. Shrinking W to its smooth locus (theorem 2.7.17) and applying lemma 3.8.2 to $\pi|_{U^{\mathrm{sm}}}$, fix dense opens $U^\circ \subseteq U^{\mathrm{sm}}$ and $W^\circ \subseteq W$ with $\pi(U^\circ) = W^\circ$, with W° inside the smooth locus of W, and with a finite surjective $\psi \colon U^\circ \to W^\circ \times \mathbb{A}^e$ over W°.

Discarding $U \setminus U^\circ$ costs only a proper closed subset of hyperplanes. Indeed, let $F_1, \dots, F_c$ be the irreducible components of $U \setminus U^\circ$ of dimension $D - 1$; the others cannot be components of a fiber, by the purity just recorded. A component F_i is contained in $\pi^{-1}(H_a)$ exactly when $\pi(F_i) \subseteq H_a$, and the $[a]$ with $\pi(F_i) \subseteq H_a$ form a proper linear subspace of $(\mathbb{P}^r_k)^\vee$, since F_i is nonempty. Deleting those finitely many subspaces, every component of $\pi^{-1}(H_a)$ meets U°, so $\pi^{-1}(H_a)$ is irreducible if and only if $\pi^{-1}(H_a) \cap U^\circ$ is irreducible and nonempty.

Step 3: the incidence variety and its diagonal section. Put

$$\mathcal{Q} = \{(x, [a]) \in U^\circ \times (\mathbb{P}^r_k)^\vee \mid \pi(x) \in H_a\}, \qquad \mathcal{J} = \{(y, x, [a]) \in U^\circ \times \mathcal{Q} \mid \pi(y) \in H_a\}$$

both closed in the ambient products. The projection $\mathcal{Q} \to U^\circ$ has all fibers equal to the hyperplane of $(\mathbb{P}^r_k)^\vee$ cut out by the single nonzero linear condition $\sum_i a_i \pi_i(x) = 0$, and on the open subscheme of U° where a fixed homogeneous coordinate of π is invertible one may solve for the corresponding a_i, exhibiting $\mathcal{Q}$ there as a product with $\mathbb{P}^{r-1}$. These opens cover U° and pairwise meet, so $\mathcal{Q}$ is irreducible of dimension $D + r - 1$. The fiber of $\mathcal{J}$ over a point $(x, [a])$ of $\mathcal{Q}$ is $\pi^{-1}(H_a) \cap U^\circ$, which is what the theorem is about, and

$$\varsigma \colon \mathcal{Q} \longrightarrow \mathcal{J}, \qquad (x, [a]) \longmapsto (x, x, [a])$$

is a section of $\mathcal{J} \to \mathcal{Q}$, because $\pi(x) \in H_a$ by the definition of $\mathcal{Q}$. This tautological section replaces the base point of a pencil in Kollár's argument, and it is available over the whole of $\mathcal{Q}$ rather than over a single line in $(\mathbb{P}^r_k)^\vee$.

Step 4: $\mathcal{J}$ has a unique component dominating $\mathcal{Q}$. Let $\mathcal{J}^{\neq}$ be the open subscheme of $\mathcal{J}$ where $\pi(y) \neq \pi(x)$ and $\mathcal{J}^{=}$ its closed complement. The projection of $\mathcal{J}^{\neq}$ to $U^\circ \times U^\circ$ lands in the open subscheme where $\pi(y) \neq \pi(x)$, which is nonempty because $\pi(U^\circ) = W^\circ$ has dimension $w \geq 1$, and which is irreducible of dimension $2D$ by proposition 3.7.10. Its fibers are the linear subspaces of $(\mathbb{P}^r_k)^\vee$ cut out by the two independent conditions $\sum_i a_i \pi_i(y) = \sum_i a_i \pi_i(x) = 0$, that is, copies of $\mathbb{P}^{r-2}$, and the same solving-for-coordinates argument trivializes them over an open cover. Hence $\mathcal{J}^{\neq}$ is irreducible of dimension $2D + r - 2$.

For $\mathcal{J}^{=}$, the locus $\{(y, x) \in U^\circ \times U^\circ \mid \pi(y) = \pi(x)\}$ is $U^\circ \times_{W^\circ} U^\circ$, which maps finitely to $(W^\circ \times \mathbb{A}^e) \times_{W^\circ} (W^\circ \times \mathbb{A}^e) = W^\circ \times \mathbb{A}^{2e}$ under $\psi \times \psi$; so lemma 2.6.3 bounds its dimension by $w + 2e = 2D - w$. Its fibers in $\mathcal{J}^{=}$ are copies of $\mathbb{P}^{r-1}$, whence

$$\dim \mathcal{J}^{=} \leq 2D - w + r - 1 \leq 2D + r - 3 < \dim \mathcal{J}^{\neq}$$

the middle inequality being where the hypothesis $w \geq 2$ is spent. For $w = 1$ the two dimensions agree and the theorem is false, a general hyperplane then meeting W in $\deg W$ points and $\pi^{-1}(H)$ being a union of that many fibers.

Let $\mathcal{J}_0 = \overline{\mathcal{J}^{\neq}}$, irreducible of dimension $2D + r - 2$. For $(x, [a])$ in $\mathcal{Q}$ outside the proper closed subset over which some component of $\mathcal{J}^{=}$ not contained in $\mathcal{J}_0$ lies, the fiber $\pi^{-1}(H_a) \cap U^\circ$ is pure of dimension $D - 1$ and meets $\pi^{-1}(\pi(x))$ in dimension at most $e \leq D - 2$ by lemma 3.8.2, so the locus $\pi(y) \neq \pi(x)$ is dense in that fiber and the whole fiber lies in $\mathcal{J}_0$. A component of $\mathcal{J}$ dominating $\mathcal{Q}$ therefore has a dense subset inside the closed set $\mathcal{J}_0$, hence is contained in it and equals it. Consequently the generic fiber of $\mathcal{J} \to \mathcal{Q}$ is irreducible and coincides with that of $\mathcal{J}_0 \to \mathcal{Q}$; and ς maps a dense subset of $\mathcal{Q}$, hence all of $\mathcal{Q}$, into the closed set $\mathcal{J}_0$.

Step 5: the section point is smooth, and conclusion. The scheme U° is smooth over k, being smooth over the smooth W°. Fix a closed point $x \in U^\circ$, an index i_0 with $\pi_{i_0}(x) \neq 0$, and repeat the computation of Step 1 of the proof of theorem 3.8.4 with the regular functions π_i/π_{i_0} in place of the coordinates X_i/X_{i_0}: the map

$$\epsilon_x \colon k^{r+1} \longrightarrow \mathcal{O}_{U^\circ,x}/\mathfrak{m}_x^2, \qquad a \longmapsto \Big[\sum_i a_i\, \pi_i/\pi_{i_0}\Big]$$

is k-linear, and $\pi^{-1}(H_a) \cap U^\circ$ is singular at x exactly when a lies in $W_x := \ker \epsilon_x$, by the same comparison of the cotangent space with the dimension. Here ϵ_x factors through $\mathcal{O}_{W,\pi(x)}/\mathfrak{m}_{\pi(x)}^2$, of dimension $w + 1$, and the induced map on that quotient is injective because π is smooth at x; so the image of ϵ_x has dimension $w + 1$ and $\dim_k W_x = r - w$. The hyperplanes through $\pi(x)$ form a linear subspace of $(\mathbb{P}^r_k)^\vee$ of dimension $r - 1$, whereas $\mathbb{P}(W_x)$ has dimension $r - w - 1 < r - 1$ since $w \geq 2$; so for each fixed x the singular condition is proper on the fiber of $\mathcal{Q}$ over x.

The condition $a \in W_x$ is closed on $\mathcal{Q}$, being the vanishing of the minors of a Jacobian matrix augmented by the row a, exactly as in Step 2 of the proof of theorem 3.8.4 applied to a local presentation of U° together with the functions π_i/π_{i_0}. Being closed and containing no fiber of $\mathcal{Q} \to U^\circ$, it is a proper closed subset, so its complement is a nonempty open subset of the irreducible $\mathcal{Q}$ and contains the generic point θ. Hence the generic fiber of $\mathcal{J}_0 \to \mathcal{Q}$ has the $\kappa(\theta)$-rational point $\varsigma(\theta)$ and is smooth there over $\kappa(\theta)$.

Lemma 3.8.8 now applies to $\mathcal{J}_0 \to \mathcal{Q}$ and yields a dense open $\mathcal{Q}^\circ \subseteq \mathcal{Q}$ over whose closed points the fiber is irreducible; intersecting with the dense open of Step 4 on which the fiber of $\mathcal{J}_0$ agrees with that of $\mathcal{J}$, the scheme $\pi^{-1}(H_a) \cap U^\circ$ is irreducible for every closed point $(x, [a]) \in \mathcal{Q}^\circ$. The projection $\mathcal{Q} \to (\mathbb{P}^r_k)^\vee$ is dominant: a hyperplane meets the projective variety W of positive dimension, because the affine cone over W has dimension $w + 1 \geq 3$ and meets the hyperplane $\sum_i a_i X_i = 0$ in dimension at least $w \geq 2$ by theorem 2.6.13, hence in more than the origin; and the $[a]$ for which $H_a \cap W$ avoids W° lie in the finitely many proper linear subspaces of hyperplanes containing a fixed $(w-1)$-dimensional component of $W \setminus W^\circ$. Applying lemma 3.8.2 to $\mathcal{Q}^\circ \to (\mathbb{P}^r_k)^\vee$ produces a dense open subset of $(\mathbb{P}^r_k)^\vee$ contained in the image, and by Step 2 the preimage $\pi^{-1}(H_a)$ itself is irreducible over that subset.

Step 6: the inseparable case. Suppose π is smooth at no point of U, so $\operatorname{char} k = p > 0$. Lemma 3.8.9 supplies s with $\pi_{(s)} \colon U_{(s)} \to \mathbb{P}^r_k$ smooth at a point, where $U_{(s)}$ is the reduced scheme underlying $U \times_{\mathbb{P}^r_k, F^s} \mathbb{P}^r_k$; the morphism $U_{(s)} \to U$ is a homeomorphism, so $U_{(s)}$ is an irreducible k-variety and a closed subset of U is irreducible exactly when its preimage is. For a hyperplane H_a one has $(F^s)^{-1}(H_a) = H_{a'}$ as sets, with $a'_i = a_i^{p^{-s}}$, again a hyperplane, and $[a] \mapsto [a']$ is a homeomorphism of $(\mathbb{P}^r_k)^\vee$ onto itself, inverse to the Frobenius of the dual space. Under the homeomorphism $U_{(s)} \to U$ the closed set $\pi_{(s)}^{-1}(H_{a'})$ corresponds to $\pi^{-1}(H_a)$. The closure of the image of $\pi_{(s)}$ is $(F^s)^{-1}(W)$, of the same dimension $w \geq 2$. Steps 2 to 5 applied to $\pi_{(s)}$ give a dense open subset of $(\mathbb{P}^r_k)^\vee$, whose preimage under $[a] \mapsto [a']$ is again dense open and serves as Ω. Irreducibility over the algebraically closed k is geometric irreducibility, and Step 1 transports the conclusion back to the original ground field, completing the proof.

3.8.11 Warning: Two Different Bertini Irreducibility Theorems Theorem 3.7.13 and theorem 3.8.10 are different statements, and neither implies the other. The first fixes a subvariety $X \subseteq \mathbb{P}^n_k$ and a finite set S of closed points of it, and asserts that a general hypersurface of degree $m > |S|$ *through* S cuts X irreducibly; the divisor moves in the incomplete system $\Lambda_m(S)$ of degree-m forms vanishing on S, and the content is that the constraint imposed by S does not destroy irreducibility.

The second fixes a map π to a projective space and lets the hyperplane be entirely unconstrained. The variety being cut is the source of π, which need not be embedded in the target at all, and the divisor cutting it is the preimage $\pi_U^{-1}(H)$, which can be far from a hypersurface section of anything.

Neither direction of implication holds. Taking π to be a closed immersion $X \hookrightarrow \mathbb{P}^n_k$ and $m = 1$, theorem 3.8.10 gives irreducibility of $X \cap H$ for a general hyperplane, which theorem 3.7.13 cannot deliver, its hypothesis $m > |S| \geq 1$ excluding hyperplanes; and example 3.7.12 shows the exclusion is not an artifact. Conversely theorem 3.8.10 says nothing about members of a proper subsystem such as $\Lambda_m(S)$: a dense open subset of $(\mathbb{P}^n)^\vee$ can miss a fixed linear subspace of it entirely, so generality in the full system carries no information about generality in a subsystem. That failure is not a defect of the statement but the reason theorem 3.8.20 below is phrased with codimension bounds rather than with a single dense open subset. A reader who reaches for one theorem where the other is needed will find the hypotheses do not match.

Iterating theorem 3.8.10 together with theorem 3.8.4 cuts a variety down to a curve.

Corollary 3.8.12: General Linear Sections of a Smooth Projective Variety

Let k be an infinite field and $Y \subseteq \mathbb{P}^M_k$ a geometrically irreducible projective k-variety of dimension $n \geq 1$ which is smooth over k. Then for general hyperplanes $H_1, \ldots, H_{n-1}$ the scheme $Y \cap H_1 \cap \cdots \cap H_{n-1}$ is a geometrically irreducible curve, smooth over k.

***Proof*:**
Induct on n, the case $n = 1$ being the hypothesis. Let $n \geq 2$ and apply theorem 3.8.10 to the closed immersion $Y \hookrightarrow \mathbb{P}^M_k$, whose image has dimension $n \geq 2$, and theorem 3.8.4 to $Y_{\overline{k}}$. Intersecting the two dense open subsets produced, after passing the second through lemma 3.8.3, gives a dense open subset of $(\mathbb{P}^M_k)^\vee$ over whose k-rational points $Y \cap H_1$ is geometrically irreducible and smooth of dimension $n-1$. Lemma 3.8.1 supplies such an H_1 over the infinite

field k, and the induction hypothesis applies to $Y \cap H_1$, completing the proof.

Codimension bounds in the Grassmannian

The corollary just proved is not strong enough for the applications that motivate the section. There the linear sections are not free to be fully general: they are required to pass through prescribed points, or they arise as the fibers of a fixed linear projection, and in both cases they range over a proper subvariety of the parameter space. A statement of the form “the bad locus is contained in a proper closed subset” is then useless, since the subvariety one is confined to may lie inside the bad locus. What is needed is a bound on the codimension of the bad locus, which survives intersection with subvarieties of small enough codimension.

The parameter space is a Grassmannian. Fix a projective variety $Y \subseteq \mathbb{P}^M_k$ of dimension $n \geq 2$ and an integer $m \geq 1$, and let

$$V_m = \frac{k[X_0, \ldots, X_M]_m}{k[X_0, \ldots, X_M]_m \cap I(Y)}$$

be the space of degree-m forms restricted to Y; its projectivization $|H| = \mathbb{P}(V_m)$ is the linear system of degree-m hypersurface sections of Y. Write $\mathrm{Grass}(j, V_m)$ for the Grassmannian of j-dimensional linear subspaces of the vector space V_m; its construction and projectivity are in *Classical Algebraic Geometry* [20], which also supplies the standard affine charts used below. The relevant subspaces have *vector* dimension $n-1$, that is, projective dimension $n-2$ inside $|H|$: a subspace of vector dimension $n-1$ is spanned by $n-1$ hypersurface sections, whose common zero locus on the n-dimensional Y is a curve. Subspaces of projective dimension $n-1$ would cut out a finite set instead.

Definition 3.8.13: The Universal Family of Curve Sections

With Y, m and V_m as above, the *universal family of curve sections* is

$$\mathbf{C} = \{(y, L) \in Y \times \mathrm{Grass}(n-1, V_m) \mid G(y) = 0 \text{ for every } G \in L\}$$

a closed subscheme of $Y \times \mathrm{Grass}(n-1, V_m)$. Its fiber over a point L is written C_L; as a set it is the intersection of all the members of the linear system $\mathbb{P}(L) \subseteq |H|$.

One general fact about Grassmannians is isolated first, because it is what makes the codimension-two bound work in every dimension. The argument below cuts Y by one hypersurface at a time and has to control the subspaces L none of whose members is good. Those L are exactly the linear subspaces of $|H|$ lying inside a fixed proper closed subset, and the following count says that such subspaces are rare, with a codimension that grows with the dimension of the subspace. Nothing about linear systems enters.

Lemma 3.8.14: Linear Subspaces Contained in a Closed Set

Let k be an algebraically closed field, let $Z \subsetneq \mathbb{P}^M_k$ be a closed subset and let $r \geq 1$. Write $\mathrm{Grass}(r, \mathbb{P}^M_k)$ for the Grassmannian of r-dimensional linear subspaces of $\mathbb{P}^M_k$, of dimension $(r+1)(M-r)$, and put

$$\mathbb{F}_r(Z) = \left\{\Lambda \in \mathrm{Grass}(r, \mathbb{P}^M_k) \;\middle|\; \Lambda \subseteq Z\right\}$$

Then $\mathbb{F}_r(Z)$ is closed and

$$\mathrm{codim}\mathbb{F}_r(Z) \geq (r+1)\mathrm{codim}Z \geq r+1$$

with $\mathrm{codim}Z = M - \dim Z$ and the convention that the empty set has infinite codimension. The bound is attained when Z is a hyperplane.

***Proof*:**
Step 0: closedness. Let $\mathcal{A} = \{(x, \Lambda) \mid x \in \Lambda\}$ be the tautological incidence set, closed in $\mathbb{P}^M_k \times \mathrm{Grass}(r, \mathbb{P}^M_k)$. Over each standard chart of the Grassmannian [20] the projection $\mathcal{A} \to \mathrm{Grass}(r, \mathbb{P}^M_k)$ is the projection of a trivial $\mathbb{P}^r$-bundle, hence an open map. The complement of $\mathbb{F}_r(Z)$ is the image of the open subset of $\mathcal{A}$ where $x \notin Z$, so it is open.

Step 1: reduction to Z irreducible. Let $Z_1, \ldots, Z_s$ be the irreducible components of Z. A linear subspace is irreducible, so $\Lambda \subseteq Z$ forces $\Lambda \subseteq Z_i$ for some i, and $\mathbb{F}_r(Z) = \bigcup_i \mathbb{F}_r(Z_i)$ while $\mathrm{codim}Z = \min_i \mathrm{codim}Z_i$. It therefore suffices to treat Z irreducible, taken with its reduced structure.

Step 2: induction on $\dim Z$. Induct on $\dim Z$. If $\dim Z < r$ then $\mathbb{F}_r(Z) = \emptyset$ and there is nothing to prove; since $r \geq 1$ this covers every Z of dimension at most 0 and starts the induction. Assume now $\dim Z = M - c$ with $1 \leq c \leq M - r$, let Z^{sm} be the smooth locus of Z, a dense open subset by theorem 2.7.17, and let $\mathrm{Sing}\, Z = Z \setminus Z^{\mathrm{sm}}$, closed (proposition 2.7.16) of dimension strictly less than $\dim Z$. Split

$$\mathbb{F}_r(Z) = \mathbb{F}' \cup \mathbb{F}_r(\mathrm{Sing}\, Z), \qquad \mathbb{F}' = \{\Lambda \subseteq Z \mid \Lambda \cap Z^{\mathrm{sm}} \neq \emptyset\}$$

The second piece is handled by the inductive hypothesis: it has codimension at least $(r+1)\big(M - \dim \mathrm{Sing}\, Z\big) \geq (r+1)(c+1) > (r+1)c$.

Step 3: a subspace through a smooth point lies in the tangent space. For $x \in Z^{\mathrm{sm}}$ let $\mathbb{T}_x Z \subseteq \mathbb{P}^M_k$ be the embedded tangent space: choosing affine coordinates centred at x, it is the common zero locus of the degree-one parts at x of the dehomogenizations of the forms in $I(Z)$. It is a linear subspace, and proposition 2.7.6 gives $\dim \mathbb{T}_x Z = \dim Z = M - c$ because x is a smooth point.

Let $\Lambda \in \mathbb{F}'$ and $x \in \Lambda \cap Z^{\mathrm{sm}}$. Every line $\ell \subseteq \Lambda$ through x lies in Z, so each $F \in I(Z)$ vanishes identically on ℓ; in the coordinates above the degree-one part of F at x then vanishes on the direction of ℓ, that is, $\ell \subseteq \mathbb{T}_x Z$. Since Λ is the union of the lines through x that it contains, $\Lambda \subseteq \mathbb{T}_x Z$.

Step 4: the incidence count. Put

$$\mathcal{I} = \{(x, \Lambda) \in Z^{\text{sm}} \times \mathbb{F}' \mid x \in \Lambda\}$$

a locally closed subset of $\mathbb{P}^M_k \times \text{Grass}(r, \mathbb{P}^M_k)$ and so a k-variety with finitely many irreducible components. By Step 3 the fiber of $\mathcal{I}$ over a point $x \in Z^{\text{sm}}$ is contained in the set of r-dimensional subspaces of $\mathbb{T}_x Z \cong \mathbb{P}^{M-c}_k$ passing through x; those correspond to the $(r-1)$-dimensional subspaces of the quotient $\mathbb{P}^{M-c-1}_k$, so the fiber has dimension at most $r(M-c-r)$. Applying lemma 3.8.2 to each irreducible component of $\mathcal{I}$ over the closure of its image in Z^{sm},

$$\dim \mathcal{I} \leq (M-c) + r(M-c-r)$$

In the other direction let $\mathbb{F}'_0$ be an irreducible component of the locally closed set $\mathbb{F}'$. The restriction of $\mathcal{A}$ to $\mathbb{F}'_0$ is a $\mathbb{P}^r$-bundle over $\mathbb{F}'_0$, hence irreducible of dimension $\dim \mathbb{F}'_0 + r$, and $\mathcal{I}$ meets it in the open subset where $x \in Z^{\text{sm}}$, which is nonempty by the definition of $\mathbb{F}'$ and therefore dense. Hence $\dim \mathbb{F}'_0 + r \leq \dim \mathcal{I}$, and

$$\dim \mathbb{F}' \leq (M-c) + r(M-c-r) - r$$

Step 5: conclusion. Subtracting from $\dim \text{Grass}(r, \mathbb{P}^M_k) = (r+1)(M-r)$ and expanding,

$$\text{codim}\mathbb{F}' \geq (r+1)(M-r) - (M-c) - r(M-c-r) + r = c(r+1)$$

Combining with the bound of Step 2 for $\mathbb{F}_r(\text{Sing}\, Z)$ gives $\text{codim}\mathbb{F}_r(Z) \geq c(r+1) = (r+1)\text{codim}Z$, and $c \geq 1$ gives the second inequality.

For sharpness take Z a hyperplane, so that $c = 1$ and $\mathbb{F}_r(Z) = \text{Grass}(r, \mathbb{P}^{M-1}_k)$, of dimension $(r+1)(M-1-r)$ and hence of codimension exactly $r+1$, completing the proof.

The case $r = 1$, the statement that the lines inside a proper closed subset form a family of codimension at least twice the codimension of that subset, is classical: it goes back to Segre in 1948 and is proved in this form by Rogora, *Manuscripta Mathematica* **82** (1994). The general case is proved above because the induction below needs it for $r = n-2$, and the growth of the bound with r is exactly what a one-shot reduction to surfaces cannot supply.

A piece of dimension bookkeeping comes next, because it is what the applications consume. Put $\nu = \dim_k V_m$, so that $\text{Grass}(n-1, V_m)$ has dimension $(n-1)(\nu - n + 1)$. If p and q are distinct closed points of Y and $V_m(-p-q) \subseteq V_m$ is the subspace of forms vanishing at both, then $V_m(-p-q)$ has codimension 2 as soon as the degree-m forms separate p from q, and the subvariety

$$\text{Grass}\left(n-1, V_m(-p-q)\right) \subseteq \text{Grass}(n-1, V_m)$$

of subspaces all of whose members pass through p and q has codimension exactly $2(n-1)$. This is the locus the two-point application is confined to, and $2(n-1) \geq 2$ for $n \geq 2$. A codimension bound of 2, or even of 1, in the ambient Grassmannian therefore says nothing about whether the bad locus avoids this subvariety, and the two-point statement has to be obtained by applying the theorem to the constrained system instead; corollary 3.8.24 does that.

Four further inputs are needed before the theorem, and it is worth being explicit about which are proved here and which are quoted.

The first is that the bad locus is constructible. Every bound below is obtained by exhibiting a bad set as the image of an incidence variety and comparing dimensions, and that comparison is valid for constructible sets and false for arbitrary ones. The failure is not hypothetical: a subset of $\mathbb{P}^2$ meeting every line in exactly two points exists, by transfinite induction, and is Zariski dense, so knowing that a set meets every member of a covering family in a small set says nothing about its dimension. The statement below is the one place in this section where a result from the theory of constructible sets is imported, and it is imported explicitly.

Lemma 3.8.15: Constructibility of the Geometrically Irreducible Locus

Let $f\colon X \to T$ be a morphism of finite type between Noetherian schemes. Then

$$\{t \in T \mid X_t \text{ is geometrically irreducible}\}$$

is a constructible subset of T, and so are the corresponding loci for "geometrically reduced" and for "of dimension at most d" for each fixed d.

***Proof Key ideas*:**
The three statements are proved together in EGA by a limit argument that reduces to the case of a finitely presented morphism over a Noetherian base and then constructs a stratification of T over which the geometry of the fibers is constant; the reference is [27, IV_3, 9.7.7], and [63] has a modern account. The constructibility of the dimension loci is the semicontinuity of fiber dimension. None of this is developed in this book, and the statement is used only through the following consequence: if $E \subseteq T$ is constructible then $\dim \overline{E} = \dim E$, and the image of a constructible set under a morphism is constructible, so the fiber-dimension bound of lemma 3.8.2 may be applied to E by passing to the irreducible components of $\overline{E}$, on each of which E contains a dense open subset.

The second input is connectedness. A reducible member of a linear system on a surface is only useful if its two halves meet, and that is what turns reducibility into a singularity and so into a codimension count. Kollár's Corollary 24 attributes the step to Bertini's irreducibility theorem, which does not give it: Bertini controls a general member, while the claim quantifies over every reducible member. Connectedness is the input the step consumes, and it is proved here from a pencil.

Lemma 3.8.16: Connectedness of the Fibers of a Family over a Curve

Let k be an algebraically closed field and let $q\colon \Gamma \to T$ be a proper surjective morphism from an irreducible k-variety onto a smooth irreducible k-curve. Suppose $q^{-1}(t)$ is irreducible for every closed point t in some dense open subset of T. Then $q^{-1}(t)$ is connected for every closed point $t \in T$.

***Proof Key ideas*:**
This is Zariski's connectedness theorem. What is quoted is the existence of the Stein factorization of q, a factorization $\Gamma \to T' \to T$ with $T' \to T$ finite and $\Gamma \to T'$ surjective with connected fibers; see [63, Tag 03H0]. It rests on the coherence of $q_*\mathcal{O}_\Gamma$ and on the theorem on formal

functions, neither of which is available at this point in the book, and it is the same input that forced theorem 3.4.32 to be quoted rather than proved.

Granting it, the rest is short and is recorded so that the size of the import is visible. The scheme T' is integral because Γ is and $\Gamma \to T'$ is surjective with connected fibers, and the number of connected components of $q^{-1}(t)$ equals the number of points of T' over t. If the general fiber of q is irreducible, then $T' \to T$ is injective over a dense open subset of T, hence purely inseparable, hence a homeomorphism; when it has degree one it is an isomorphism by corollary 3.4.33, the target T being a smooth curve and therefore normal. Either way $T' \to T$ is a bijection on points, so every fiber of q is connected, completing the account.

Lemma 3.8.17: Members of a Very Ample System Are Connected

Let k be an algebraically closed field and let $S \subseteq \mathbb{P}^N_k$ be an integral projective k-variety with $\dim S \geq 2$. Then $S \cap V_+(F)$ is connected for every homogeneous form F that does not vanish identically on S.

***Proof*:**
Write $d = \dim S$, let $m = \deg F$ and let $C = S \cap V_+(F)$, of pure dimension $d-1$ by lemma 3.7.11. Under the m-th Veronese re-embedding (example 0.2.27) the degree-m hypersurface sections of S become hyperplane sections, so theorem 3.8.10, applied to the resulting closed immersion of S, whose image has dimension $d \geq 2$, supplies a dense open set of degree-m forms G with $S \cap V_+(G)$ irreducible. Shrinking it further, require also that $V_+(G)$ contain no irreducible component of C, which excludes finitely many proper linear subspaces, and that G be independent of F modulo $I(S)$. If no such G exists then F itself already has $S \cap V_+(F)$ irreducible and there is nothing to prove; fix one otherwise.

Step 1: the total space of the pencil. Put

$$\Gamma = \left\{(x, [\lambda : \mu]) \in S \times \mathbb{P}^1_k \;\middle|\; \lambda F(x) + \mu G(x) = 0\right\}$$

with its reduced structure, a closed subscheme of $S \times \mathbb{P}^1_k$, and let $q \colon \Gamma \to \mathbb{P}^1_k$ be the second projection. The fiber of q over $[\lambda : \mu]$ is $S \cap V_+(\lambda F + \mu G)$, so q is surjective, and q is proper because Γ is closed in $S \times \mathbb{P}^1_k$ and S is proper over k (proposition 3.5.21 and theorem 3.5.26). The base locus $B = C \cap V_+(G)$ has dimension $d-2$, since $V_+(G)$ contains no component of the pure $(d-1)$-dimensional C, and over $S \setminus B$ the projection $\Gamma \to S$ is bijective, exactly one ratio $[\lambda : \mu]$ solving the equation.

The scheme Γ is cut out in the $(d+1)$-dimensional $S \times \mathbb{P}^1_k$ by a single equation which is not identically zero, so every irreducible component of it has dimension d by theorem 2.6.13 and theorem 2.6.12. The closed subset $B \times \mathbb{P}^1_k$ has dimension $d-1$, so no component of Γ lies inside it; every component therefore meets $\Gamma \setminus (B \times \mathbb{P}^1_k) \cong S \setminus B$, which is irreducible, and Γ is irreducible.

Step 2: conclusion. The map $[\lambda : \mu] \mapsto [\lambda F + \mu G]$ identifies $\mathbb{P}^1_k$ with the line in the space of degree-m hypersurface sections spanned by F and G, and that line meets the dense open subset of irreducible members in the point $[G]$, hence in a dense open subset of $\mathbb{P}^1_k$. So the fiber of q

over a general closed point is irreducible, and lemma 3.8.16 applies: every fiber of q is connected. Taking the fiber over $[1:0]$ gives that C is connected, completing the proof.

The third input is what makes the induction close. Cutting Y by one member D of the system produces a variety of one dimension less, and the whole argument depends on the linear system on that variety being the quotient of V_m by the line spanned by D, with nothing else collapsing. That is the content of the next lemma, and it is where regularity in codimension one is spent.

Lemma 3.8.18: One Cut Removes Exactly One Dimension

Let k be an algebraically closed field and let $Y \subseteq \mathbb{P}^M_k$ be an integral projective k-variety of dimension $n \geq 2$ which is regular in codimension one. Let F be a degree-m form not vanishing on Y and suppose the scheme $Y \cap V_+(F)$ is irreducible and generically reduced; write Y_F for it with its reduced structure. Then the restriction map

$$V_m(Y) \longrightarrow V_m(Y_F)$$

is surjective with kernel the line spanned by the class of F. In particular $\dim_k V_m(Y_F) = \dim_k V_m(Y) - 1$.

***Proof*:**
Both spaces are quotients of the space of degree-m forms on $\mathbb{P}^M_k$, so the map is surjective, and F vanishes on Y_F, so the class of F lies in the kernel; it is nonzero because F does not vanish on Y.

For the converse let G be a degree-m form vanishing on Y_F. Since Y is regular in codimension one, the local ring at every codimension-one point y of Y is a discrete valuation ring, and after trivializing degree-m forms near y by a form not vanishing at y one obtains order functions ord_y. Let y_0 be the generic point of Y_F. The scheme $Y \cap V_+(F)$ is irreducible, so Y_F is the only codimension-one subvariety of Y on which F vanishes, and it is generically reduced, so $\mathcal{O}_{Y,y_0}/(F)$ is a field and F generates the maximal ideal of the discrete valuation ring $\mathcal{O}_{Y,y_0}$. Hence $\mathrm{ord}_{y_0}(F) = 1$ and $\mathrm{ord}_y(F) = 0$ for every other codimension-one point y, while $\mathrm{ord}_{y_0}(G) \geq 1$ and $\mathrm{ord}_y(G) \geq 0$ throughout. The ratio $\varphi = G/F$ is therefore a rational function on Y with $\mathrm{ord}_y(\varphi) \geq 0$ at every codimension-one point.

Let $\pi\colon \widetilde{Y} \to Y$ be the normalization (theorem 3.4.35), a finite birational morphism from an integral normal projective variety which is an isomorphism over the normal locus of Y. Regularity in codimension one puts every codimension-one point of Y in that locus, and a finite morphism preserves dimensions, so the codimension-one points of $\widetilde{Y}$ correspond bijectively to those of Y with matching order functions. Hence $\pi^*\varphi$ has nonnegative order at every codimension-one point of the normal $\widetilde{Y}$, and a normal domain is the intersection of its localizations at the height-one primes (*Commutative Algebra* [10]), so $\pi^*\varphi$ is a global regular function on $\widetilde{Y}$. That variety is integral and proper over the algebraically closed k, so theorem 3.5.27 gives $\pi^*\varphi = c \in k$, and π is birational, so $\varphi = c$. Then $G - cF$ vanishes at the generic point of Y, hence on Y, and the class of G in $V_m(Y)$ is c times that of F, completing the proof.

The fourth and last input is the dichotomy at a smooth point. On a surface, a member of the system that is singular at a prescribed smooth point s costs three conditions, and only a two-dimensional family of choices of s is available to pay for them; the surplus of one is what makes reducibility a codimension-two condition rather than a codimension-one one, provided a general such member is still irreducible. That proviso fails exactly when the system of members singular at s maps the surface to a curve, and the following lemma says how rarely that happens.

Lemma 3.8.19: The Second-Order Projection Has Two-Dimensional Image

Let k be an algebraically closed field, let $S \subseteq \mathbb{P}^N_k$ be an integral projective surface and let $m \geq 2$. Write V_m for the space of degree-m forms restricted to S and $\nu = \dim_k V_m$; for a smooth closed point $s \in S$ let $V_m(-2s) \subseteq V_m$ be the subspace of forms whose restriction lies in $\mathfrak{m}_s^2$, and let $\psi_s \colon S \dashrightarrow \mathbb{P}\big(V_m(-2s)^\vee\big)$ be the associated rational map. If the closure of the image of ψ_s is a curve for every s in a dense open subset of the smooth locus of S, then $\nu = 6$ and $m = 2$.

***Proof*:**
Step 1: the system of singular members. Fix a smooth closed point s. The local ring $\mathcal{O}_{S,s}$ is regular of dimension 2, so $\mathcal{O}_{S,s}/\mathfrak{m}_s^3$ has dimension $1+2+3=6$ over k. Restriction of linear forms surjects onto $\mathfrak{m}_s/\mathfrak{m}_s^2$, the closed immersion $S \subseteq \mathbb{P}^N_k$ making $\mathcal{O}_{\mathbb{P}^N,s} \to \mathcal{O}_{S,s}$ surjective, so products of j linear forms vanishing at s span $\mathfrak{m}_s^j/\mathfrak{m}_s^{j+1}$. Multiplying such products by a power of a linear form not vanishing at s shows that $V_m \to \mathcal{O}_{S,s}/\mathfrak{m}_s^3$ is surjective for $m \geq 2$. Hence $\nu \geq 6$, the subspace $V_m(-2s)$ has codimension 3, and the members of $\mathbb{P}(V_m(-2s))$ whose restriction lies in $\mathfrak{m}_s^3$ form a proper subspace, so a general member has multiplicity exactly 2 at s. Here the multiplicity $\operatorname{mult}_s M$ of an effective divisor M on S at s is the largest j with the local equation of M in $\mathfrak{m}_s^j$.

The base locus of $\mathbb{P}(V_m(-2s))$ is $\{s\}$: given a closed point $y \neq s$, choose linear forms σ, τ vanishing at s and not at y and a degree-$(m-2)$ form not vanishing at y; the product lies in $V_m(-2s)$ and does not vanish at y, which uses $m \geq 2$. So ψ_s is defined on $S \setminus \{s\}$ and has no fixed component.

Step 2: every fiber of ψ_s passes through s. Assume the closure Γ_s of the image of ψ_s is a curve, and let E be the closure of a fiber, an irreducible curve in S. If $s \notin E$, choose a hyperplane of $\mathbb{P}(V_m(-2s)^\vee)$ missing the point of Γ_s below E; the corresponding member M of the system satisfies $M \cap E \subseteq \{s\}$ and hence $M \cap E = \emptyset$. But M is cut on S by a degree-m form, and a projective variety of positive dimension meets every hypersurface by lemma 3.7.11. So $s \in E$.

Step 3: the degree of Γ_s is at most 2. Let $\delta = \deg \Gamma_s$ and let H be a general hyperplane, meeting the irreducible curve Γ_s in δ distinct points [20]. The member M of $\mathbb{P}(V_m(-2s))$ corresponding to H contains the closures $E_1, \ldots, E_\delta$ of the δ fibers over those points, pairwise distinct irreducible curves, each passing through s by Step 2. The local ring $\mathcal{O}_{S,s}$ is regular, hence a unique factorization domain (*Commutative Algebra* [10]), so each E_i has a prime local equation $p_i \in \mathfrak{m}_s$, the p_i are pairwise non-associate, and the local equation of M is divisible by $p_1 \cdots p_\delta \in \mathfrak{m}_s^\delta$. Taking M general, $\operatorname{mult}_s M = 2$ by Step 1, so $\delta \leq 2$.

Step 4: the numerical conclusion. The image Γ_s is nondegenerate in $\mathbb{P}(V_m(-2s)^\vee)$, a projective space of dimension $\nu - 4$: a hyperplane containing the image would be a member of the system containing S, and no nonzero form vanishes on S by construction of V_m. An irreducible nondegenerate curve in $\mathbb{P}^d_k$ has degree at least d [20], so $\nu - 4 \leq \delta \leq 2$ and $\nu \leq 6$. With $\nu \geq 6$ from Step 1 this gives $\nu = 6$.

Step 5: $m \geq 3$ is excluded. Suppose $m \geq 3$. Choose a linear form ℓ with $\ell(s) = 0$ that does not vanish on S, possible because the linear forms vanishing at s cut out $\{s\}$ and S is not a point, and a linear form ℓ' not vanishing on S. The form $\ell^3 \ell'^{m-3}$ has degree m and does not vanish on S, the ideal of S being prime, so it is a nonzero element of V_m; and its restriction lies in $\mathfrak{m}_s^3$, since ℓ restricts into $\mathfrak{m}_s$. Hence $V_m \to \mathcal{O}_{S,s}/\mathfrak{m}_s^3$ has nonzero kernel and $\nu \geq 7$, contradicting Step 4. Therefore $m = 2$, completing the proof.

The route through Steps 3 and 4 is chosen deliberately.[8]

The statement below is Kollár's Corollary 24, restated. Several corrections to the source are built in. The integer n, undefined there, is $\dim Y$, as his own proof forces. The singular-locus condition is imposed on Y, not on an unrelated X. The Grassmannian parametrizes subspaces of vector dimension $n-1$, since subspaces of vector dimension n would cut out a finite set rather than a curve. The degree bound is $m \geq 3$ throughout, which removes every exception at once and matches what the two-point corollary needs. And the conclusion of the codimension-two part is irreducibility, not integrality, for a reason recorded after the proof.

Theorem 3.8.20: Bertini's Irreducibility Theorem II

Let k be an infinite field and let $Y \subseteq \mathbb{P}^M_k$ be a closed subscheme of dimension $n \geq 2$ which is geometrically irreducible and generically reduced and whose non-regular locus $\operatorname{Sing} Y$ has codimension at least 2 in Y. Let $m \geq 3$ and let V_m, $\operatorname{Grass}(n-1, V_m)$ and C_L be as above. Then:

1. there is a closed subset $Z_1 \subseteq \operatorname{Grass}(n-1, V_m)$ of codimension at least 1 such that for every k-rational $L \notin Z_1$ the scheme C_L is a geometrically integral curve, smooth over k;
2. there is a closed subset $Z_2 \subseteq \operatorname{Grass}(n-1, V_m)$ of codimension at least 2 such that for every k-rational $L \notin Z_2$ the set C_L is a geometrically irreducible curve.

There is no exceptional case.

***Proof*:**
By lemma 3.8.3, applied to the Grassmannian in a projective embedding [20], it suffices to construct the Z_i after base change to $\overline{k}$. Assume therefore $k = \overline{k}$.

Step 0: only the reduced structure enters. The space V_m was defined by the forms vanishing on Y, which are the forms vanishing on the underlying reduced subscheme, and part (2) is a statement about the underlying set of C_L. If Y is regular at a point then its local ring there is a domain (proposition 2.7.3) and in particular reduced, so Y and Y_{red} agree near that point and Y_{red} is regular there too; hence $\operatorname{Sing} Y_{\text{red}} \subseteq \operatorname{Sing} Y$ and the hypotheses pass to Y_{red}. Replacing Y

[8]The classical way to identify a surface all of whose second-order projections collapse is the Kronecker-Castelnuovo theorem on irreducible plane curve systems. That theorem has no known analogue in positive characteristic, as Ji records in Remark 2.14 of [36]; a reader who shortens the argument above by invoking it would silently restrict the whole section to characteristic zero. The count used here is characteristic-free.

by Y_{red} changes neither V_m nor any C_L as a set, so assume from now on that Y is integral. Note that irreducible and generically reduced is what the hypothesis says, not integral; the reason is recorded after the proof.

Step 0': from subspaces to bases. Let $\mathcal{F} \subseteq (V_m)^{n-1}$ be the open subscheme of linearly independent tuples $(G_1, \dots, G_{n-1})$ and let $\rho\colon \mathcal{F} \to \mathrm{Grass}(n-1, V_m)$ send a tuple to its span. In each standard chart of the Grassmannian [20] a subspace carries a canonical basis, so ρ is there the projection of a trivial bundle with fiber GL_{n-1}; it is surjective and open. A dense open subset of $\mathcal{F}$ that is a union of fibers of ρ therefore has dense open image.

Step 1: the good members. Under the m-th Veronese re-embedding (example 0.2.27) the degree-m hypersurface sections of Y become the hyperplane sections of Y inside $\mathbb{P}(V_m^\vee)$, and $\mathbb{P}(V_m)$ is the corresponding dual projective space. Let $U \subseteq \mathbb{P}(V_m)$ be the intersection of the following three dense open subsets: the set of $[F]$ with $Y \cap V_+(F)$ irreducible of dimension $n-1$, dense open by theorem 3.8.10 applied to the closed immersion of Y, whose image has dimension $n \geq 2$; the set of $[F]$ with $Y^{\mathrm{sm}} \cap V_+(F)$ smooth over k, dense open by theorem 3.8.4 applied to the smooth quasiprojective Y^{sm} in the same re-embedding; and the complement of the finitely many proper linear subspaces consisting of the $[F]$ whose $V_+(F)$ contains an irreducible component of $\operatorname{Sing} Y$.

For $[F] \in U$ write Y_F for $Y \cap V_+(F)$ with its reduced structure, an integral projective variety of dimension $n-1$. Off $\operatorname{Sing} Y$ the scheme $Y \cap V_+(F)$ is smooth, hence reduced, hence equal to Y_F there, so $\operatorname{Sing} Y_F \subseteq \operatorname{Sing} Y \cap V_+(F)$, of dimension at most $n-3$ because no component of $\operatorname{Sing} Y$ lies in $V_+(F)$. Thus $\operatorname{Sing} Y_F$ has codimension at least 2 in Y_F, and Y_F satisfies the hypotheses of the theorem in dimension $n-1$. Moreover $\dim(\operatorname{Sing} Y \cap V_+(F)) < n-1$, so the generic point of $Y \cap V_+(F)$ is a smooth point and $Y \cap V_+(F)$ is generically reduced; lemma 3.8.18 therefore applies and

$$V_m/\langle F \rangle \xrightarrow{\sim} V_m(Y_F), \qquad \dim_k V_m(Y_F) = \nu - 1$$

Two consequences are recorded for use below. First, for $L \in \mathrm{Grass}(n-1, V_m)$ containing F the image $\overline{L}$ of L in $V_m(Y_F)$ has vector dimension $n-2$, and $L \mapsto \overline{L}$ is an isomorphism from the set of such L onto $\mathrm{Grass}\big(n-2, V_m(Y_F)\big)$. Second, since $F \in L$ forces $C_L \subseteq Y \cap V_+(F)$, and since two forms differing by a multiple of F have the same zero set on Y_F,

$$C_L = C_{\overline{L}} \quad \text{as subsets of } Y$$

the right-hand side computed on Y_F. This is the recursion, and it is exact.

Step 2: part (1). Let $\mathcal{F}^\circ \subseteq \mathcal{F}$ be the set of tuples $(G_1, \dots, G_{n-1})$ such that $[G_1] \in U$, then $[\overline{G_2}]$ lies in the corresponding good open subset for Y_{G_1}, and so on down to a curve. Each condition is dense open given the previous ones, so $\mathcal{F}^\circ$ is dense open, and C_L depends only on the span, so $\mathcal{F}^\circ$ may be enlarged to a union of fibers of ρ without leaving the good locus. Shrink it once more so that no cut contains an irreducible component of what is left of $\operatorname{Sing} Y$ at that stage, again finitely many proper linear conditions. After $n-1$ such cuts the curve C_L is irreducible, and it misses $\operatorname{Sing} Y$ altogether: that set has dimension at most $n-2$ and each cut drops the dimension of what remains of it by one, so after $n-1$ cuts nothing is left. Hence $C_L \subseteq Y^{\mathrm{sm}}$ and C_L is smooth over k by theorem 3.8.4, therefore reduced and integral. Take $Z_1 = \mathrm{Grass}(n-1, V_m) \setminus \rho(\mathcal{F}^\circ)$, closed of codimension at least 1 by Step 0'.

Step 3: the bad locus, and the base case $n = 2$. Let

$$\mathcal{B} = \{L \in \mathrm{Grass}(n-1, V_m) \mid C_L \text{ is not a geometrically irreducible curve}\}$$

By lemma 3.8.15, applied to the universal family $\mathbf{C} \to \mathrm{Grass}(n-1, V_m)$, the set $\mathcal{B}$ is constructible. This is the point at which the dimension counts below become legitimate.

Suppose $n = 2$, so that $\mathrm{Grass}(1, V_m) = \mathbb{P}(V_m)$ and C_L is a single member of the system, of pure dimension 1 by lemma 3.7.11. By lemma 3.8.17 every member is connected. If a member C is reducible, two distinct irreducible components of it meet at a point s; if $s \notin \operatorname{Sing} Y$ then $\mathcal{O}_{C,s}$ is not a domain, hence not regular by proposition 2.7.3, so C is singular at s. Every reducible member therefore meets $\operatorname{Sing} Y$, a finite set since it has codimension 2 in the surface Y, or is singular at a smooth point of Y.

Take first $s \in \operatorname{Sing} Y$ and let $V_m(-s) \subseteq V_m$ be the forms vanishing at s, of codimension 1 because the system separates points. The base locus of $\mathbb{P}(V_m(-s))$ is $\{s\}$ and the associated rational map is the linear projection of Y from s in the embedding by $|H|$. If the image of that projection has dimension 2, then theorem 3.8.10 makes a general member of $\mathbb{P}(V_m(-s))$ irreducible, so $\mathcal{B} \cap \mathbb{P}(V_m(-s))$ lies in a proper closed subset and has dimension at most $\nu - 3$. If instead the image is a curve, then for a general $y \in Y$ the line joining s to y meets Y in a curve and hence lies in Y, so Y is a cone with vertex s whose rulings are lines; but the hyperplane class of that embedding is m times the hyperplane class of $Y \subseteq \mathbb{P}^M_k$, which has degree at least 1 on any curve, so a ruling would have degree at least $m \geq 3$, while a line has degree 1. That case does not occur. Since $\operatorname{Sing} Y$ is finite, the union of these loci has dimension at most $\nu - 3$.

Take now s a smooth point. By Step 1 of the proof of lemma 3.8.19 the subspace $V_m(-2s)$ has codimension 3, so $\mathbb{P}(V_m(-2s))$ has dimension $\nu - 4$. Let

$$\mathcal{I} = \{(s, [C]) \in Y^{\mathrm{sm}} \times \mathcal{B} \mid \operatorname{mult}_s C \geq 2\}$$

a constructible set, since the multiplicity condition is closed, and let T be the set of $s \in Y^{\mathrm{sm}}$ for which $\mathcal{B} \cap \mathbb{P}(V_m(-2s))$ is dense in $\mathbb{P}(V_m(-2s))$; the set T is constructible, being a fiber-dimension locus of the constructible $\mathcal{I}$. For $s \notin T$ the fiber of $\mathcal{I}$ has dimension at most $\nu - 5$. For $s \in T$ the intersection $\mathcal{B} \cap \mathbb{P}(V_m(-2s))$ is constructible and dense, hence contains a dense open subset, so a general member of $\mathbb{P}(V_m(-2s))$ is reducible and theorem 3.8.10 forbids the associated map ψ_s from having two-dimensional image; its image is a curve. Were T dense in Y it would, being constructible, contain a dense open subset, and lemma 3.8.19 would force $m = 2$, against $m \geq 3$. So T is not dense and $\dim T \leq 1$. Consequently

$$\dim \mathcal{I} \leq \max\left\{2 + (\nu - 5),\ 1 + (\nu - 4)\right\} = \nu - 3$$

and the projection of $\mathcal{I}$ to $\mathbb{P}(V_m)$, which contains every reducible member singular at a smooth point, has dimension at most $\nu - 3$.

Combining the two, $\dim \mathcal{B} \leq \nu - 3 = \dim \mathbb{P}(V_m) - 2$, and $Z_2 = \overline{\mathcal{B}}$ is a closed subset of codimension at least 2. This proves part (2) for $n = 2$.

Step 4: the one-cut induction. Let $n \geq 3$ and assume part (2) for all varieties of dimension $n - 1$

satisfying the hypotheses, with the same m. Put $Z = \mathbb{P}(V_m) \setminus U$, a closed subset of codimension at least 1, and split $\mathcal{B}$ according to whether the linear subspace $\mathbb{P}(L) \subseteq \mathbb{P}(V_m)$, of projective dimension $n-2$, meets U.

If $\mathbb{P}(L) \cap U = \emptyset$ then $\mathbb{P}(L) \subseteq Z$, so L lies in $\mathbb{F}_{n-2}(Z)$ in the notation of lemma 3.8.14, applied with $M = \nu - 1$ and $r = n - 2 \geq 1$. That lemma gives

$$\operatorname{codim}\mathbb{F}_{n-2}(Z) \geq (n-1)\operatorname{codim}Z \geq n-1 \geq 2$$

so this half of the bad locus is contained in a closed subset of codimension at least 2. This is the half that a one-shot reduction to a surface cannot see, and it costs nothing.

For the other half consider the incidence variety

$$\mathcal{J} = \{([F], L) \in U \times \operatorname{Grass}(n-1, V_m) \mid F \in L\}$$

locally closed in $\mathbb{P}(V_m) \times \operatorname{Grass}(n-1, V_m)$, and write π_1 and π_2 for its two projections. Its fiber over $[F] \in U$ is the set of L containing F, isomorphic by Step 1 to $\operatorname{Grass}\big(n-2, V_m(Y_F)\big)$, of dimension $(n-2)(\nu - n + 1)$; hence

$$\dim \mathcal{J} = (\nu - 1) + (n-2)(\nu - n + 1)$$

Its fiber over L is $\mathbb{P}(L) \cap U$, a nonempty open subset of $\mathbb{P}(L) \cong \mathbb{P}^{n-2}_k$ precisely when L belongs to the second half, so that fiber has dimension $n-2$.

Put $\mathcal{J}_{\mathrm{bad}} = \mathcal{J} \cap \pi_2^{-1}(\mathcal{B})$, constructible because $\mathcal{B}$ is. Fix $[F] \in U$. Under the isomorphism of Step 1 the fiber of $\mathcal{J}_{\mathrm{bad}}$ over $[F]$ is exactly the bad locus of Y_F inside $\operatorname{Grass}(n-2, V_m(Y_F))$, because $C_L = C_{\overline{L}}$ as sets. The inductive hypothesis, applied to the $(n-1)$-dimensional Y_F, puts that locus inside a closed subset of codimension at least 2, so the fiber has dimension at most $(n-2)(\nu - n + 1) - 2$. Applying lemma 3.8.2 to the irreducible components of the closure of $\mathcal{J}_{\mathrm{bad}}$ over U,

$$\dim \mathcal{J}_{\mathrm{bad}} \leq (\nu - 1) + (n-2)(\nu - n + 1) - 2$$

Every fiber of $\mathcal{J}_{\mathrm{bad}} \to \operatorname{Grass}(n-1, V_m)$ over a point of the second half has dimension $n-2$, so the same lemma, applied to the components of the closure, bounds the image by

$$\dim \pi_2(\mathcal{J}_{\mathrm{bad}}) \leq (n-2)(\nu - n + 1) + \nu - n - 1$$

while $\dim \operatorname{Grass}(n-1, V_m) = (n-2)(\nu - n + 1) + (\nu - n + 1)$. The difference is 2, so the second half of the bad locus also has codimension at least 2.

Taking $Z_2 = \mathbb{F}_{n-2}(Z) \cup \overline{\pi_2(\mathcal{J}_{\mathrm{bad}})}$, a closed subset of codimension at least 2 containing $\mathcal{B}$, completes the induction and the proof.

The hypothesis is that Y be geometrically irreducible and generically reduced, not that it be geometrically integral, and the conclusion of part (2) is irreducibility, not integrality. Neither is pedantry. Both are forced by the induction, and the mechanism is depth. Cutting a variety by a general member of a very ample system drops the depth at every point by one, so a point of Y where the local ring has depth 1 becomes an embedded point of $Y \cap V_+(F)$ for every F through it. A variety that is only regular in codimension one may have a positive-dimensional locus of such points, since they are confined to $\operatorname{Sing} Y$ but nothing bounds their depth; and every hypersurface meets a positive-dimensional set. For such a Y no member of the system cuts an integral divisor at all, and an induction phrased with an integrality hypothesis would not close after a single step. The sharp form of the obstruction is due to Benoist, *Le théorème de Bertini en famille*, Bulletin de la Société Mathématique de France **139** (2011), Theorem 0.5(iii): if Y has a closed point of depth 1 then the locus of members with non-integral section has codimension exactly 1. A codimension-two claim about integrality is therefore false, while the irreducible and generically reduced version above is insensitive to depth.

3.8.21 Warning: What Part (2) Does Not Say Part (2) of theorem 3.8.20 asserts that C_L is irreducible, and says nothing about C_L being reduced. Part (1) does assert integrality, and the two are consistent because part (1) only speaks about the complement of a codimension-one set. Sharpening part (2) to integrality would be a genuine error, not a missed opportunity: by the theorem of Benoist quoted above, a variety with a closed point of depth 1 has a codimension-one family of members cutting a non-reduced divisor, and no codimension-two subset can contain that family.

The statement of part (2) is corroborated independently. It is Corollary 2.15 of Ji's *The Noether-Lefschetz theorem in arbitrary characteristic* [36], where it appears in the same form and with the same codimension. That is corroboration, not the proof used here: her argument runs on the theory of Flenner, O'Carroll and Vogel, on SGA 2 XIII 2.1 and on the Enriques-Severi-Zariski vanishing theorem, none of which this series develops. The proof above uses only the two Bertini theorems proved earlier in this section, the Grassmannian count of lemma 3.8.14, and the two quoted inputs isolated in lemma 3.8.15 and lemma 3.8.16.

3.8.22 Warning: Why the Degree Bound Is $m \geq 3$ Theorem 3.8.20 is stated with $m \geq 3$, and part (2) of it is false at $m = 2$. Take $Y = \mathbb{P}^2$ embedded as a plane and $m = 2$, so $|H|$ is the five-dimensional system of conics and $\mathrm{Grass}(1, V_2)$ is $|H|$ itself, a $\mathbb{P}^5$; a subspace of vector dimension $n - 1 = 1$ is a single conic, and C_L is that conic. A conic is irreducible unless it is a pair of lines or a double line, and those form the discriminant hypersurface, of codimension exactly 1 in $\mathbb{P}^5$. So a Z_1 exists and no Z_2 can.

The mechanism is visible in lemma 3.8.19. What makes reducibility a codimension-two rather than a codimension-one condition is that a member singular at a prescribed smooth point s is still usually irreducible: the singular point costs three conditions, and only the two-dimensional family of choices of s is available to pay for them, leaving a surplus of one. For conics the surplus is spent, because a conic with a point of multiplicity two meets every line through that point in a scheme of length 2 concentrated there, so any such line meeting the conic elsewhere is a component and the conic splits. Raising m to 3 restores the general nodal cubic, and Step 5 of the proof of lemma 3.8.19 is exactly the observation that the form $\ell^3 \ell'^{\,m-3}$ becomes available at $m = 3$.

The plane with its conics is the only exception. Lemma 3.8.19 forces $\nu = 6$ and $m = 2$ in the uncontrolled case, and the classical inequality of Hopf and Riemann, that finite-dimensional spaces W_1, W_2 of functions on an integral variety over an algebraically closed field satisfy $\dim(W_1 W_2) \geq \dim W_1 + \dim W_2 - 1$, then gives $\nu \geq mN + 1$ for a surface nondegenerate in $\mathbb{P}^N$; so $6 \geq 2N + 1$, forcing $N = 2$ and $Y = \mathbb{P}^2$. That inequality is quoted rather than proved, and it is used only for this classification. The theorem itself does not need it.

The bound $m \geq 3$ is what theorem 3.8.20 assumes, and it disposes of the exception in every dimension at once, so nothing below depends on the classification just given.

Two ways of reaching a subfamily

The applications never use a fully general L. They reach the Grassmannian along a constrained family, and the two families that occur behave in opposite ways, so it is worth separating them.

The first is the two-point family. As recorded before theorem 3.8.20, the subspaces all of whose members pass through two prescribed points form $\mathrm{Grass}(n-1, V_m(-p-q))$, of codimension exactly $2(n-1)$ in $\mathrm{Grass}(n-1, V_m)$. This is a *special* subvariety, chosen for its geometry, and there is no reason for the bad locus to meet it in the expected codimension; a codimension-two bound in the ambient Grassmannian carries no information about it at all. The remedy is not to transfer but to re-run the argument inside the constrained system, which is what corollary 3.8.24 does.

The second is the family that a general linear projection produces: the fibers of a projection are cut by the hyperplanes of a general n-dimensional subspace $V_0 \subseteq V_m$. Here the subvariety is *general*, and generality is exactly what makes the ambient bound transfer with no loss. This is the one place where a codimension bound in the full Grassmannian is genuinely what is required, and it is why theorem 3.8.20 is proved in the ambient space rather than only in subsystems.

Lemma 3.8.23: Transfer to a General Subsystem

Let k be an algebraically closed field, let W be a k-vector space of finite dimension ν and let $n \geq 2$. For an n-dimensional subspace $W_0 \subseteq W$ let

$$\mathbb{P}(W_0^\vee) = \{L \in \mathrm{Grass}(n-1, W) \mid L \subseteq W_0\}$$

a closed subvariety of $\mathrm{Grass}(n-1, W)$ isomorphic to $\mathbb{P}_k^{n-1}$. Let $Z \subseteq \mathrm{Grass}(n-1, W)$ be closed and irreducible of codimension c. Then there is a dense open subset of $\mathrm{Grass}(n, W)$ over whose points

$$Z \cap \mathbb{P}(W_0^\vee) \text{ is empty, or of dimension exactly } n-1-c$$

and it is empty for every W_0 in a dense open subset as soon as $c \geq n$.

***Proof*:**
Put

$$\mathcal{K} = \{(L, W_0) \in Z \times \mathrm{Grass}(n, W) \mid L \subseteq W_0\}$$

a closed subset of the product of two projective varieties. Its projection to Z has fiber over L the set of n-dimensional subspaces containing L, that is, the lines of W/L, a projective space of dimension $\nu - n$; over each standard chart this is a trivial bundle, so $\mathcal{K}$ is irreducible of dimension

$$\dim \mathcal{K} = (n-1)(\nu - n + 1) - c + (\nu - n)$$

Since $\dim \mathrm{Grass}(n, W) = n(\nu - n)$, expanding gives

$$\dim \mathcal{K} - \dim \mathrm{Grass}(n, W) = n - 1 - c$$

The projection $\pi\colon \mathcal{K} \to \mathrm{Grass}(n, W)$ is a morphism of projective varieties, so its image is closed, and its fiber over W_0 is $Z \cap \mathbb{P}(W_0^\vee)$. If π is not dominant its image is a proper closed subset and every W_0 in the complement has empty intersection. If π is dominant, lemma 3.8.2 supplies a dense open subset of $\mathrm{Grass}(n, W)$ over whose points the fiber has dimension exactly $\dim \mathcal{K} - \dim \mathrm{Grass}(n, W) = n - 1 - c$. Finally, if $c \geq n$ then $\dim \mathcal{K} < \dim \mathrm{Grass}(n, W)$, so π cannot be dominant and the first alternative holds, completing the proof.

Applied to a component of the bad locus of theorem 3.8.20, which has $c \geq 2$, the lemma gives $\dim(Z \cap \mathbb{P}(V_0^\vee)) \leq n - 3$ for a general n-dimensional $V_0 \subseteq V_m$, so a general hyperplane of a general V_0 is good, and for $c \geq n$ the bad locus is missed outright. The ambient bound and the constrained bound agree, which is the whole point: no re-running of the argument inside a subsystem is possible here, because the subsystem is not given in advance.

The last statement is the form the two-point application needs, and it is the opposite case. The remedy there is to run the cutting argument inside the constrained system from the start.

Corollary 3.8.24: Curves Through Two Prescribed Points

Let k be an infinite field, let $Y \subseteq \mathbb{P}_k^M$ be a geometrically irreducible and generically reduced closed subscheme of dimension $n \geq 2$, and let $p \neq q$ be k-rational points of Y. Let $m \geq 3$ and let $V_m(-p-q) \subseteq V_m$ be the subspace of degree-m forms vanishing at p and at q, of codimension 2. Then there is a closed subset

$$Z \subseteq \operatorname{Grass}\big(n-1, V_m(-p-q)\big), \qquad \operatorname{codim} Z \geq 1$$

such that for every k-rational $L \notin Z$ the set C_L is a geometrically irreducible curve containing p and q. Such L exist by lemma 3.8.1.

Proof:
Every member of L vanishes at p and at q, so $p, q \in C_L$ for every L in this Grassmannian, which is the point of the constraint. It remains to run the cutting argument inside the constrained system, and by Step 0 of the proof of theorem 3.8.20 it suffices to work with Y integral.

The system $\mathbb{P}(V_m(-p-q))$ has base locus $\{p, q\}$ and separates 2-jets at every $y \notin \{p, q\}$: choosing degree-1 forms σ_p, σ_q vanishing at p respectively q and not at y, the products $\sigma_p \sigma_q G$ with G of degree $m-2 \geq 1$ lie in $V_m(-p-q)$ and already realize every 2-jet at y, since degree-1 forms surject onto $\mathcal{O}_{Y,y}/\mathfrak{m}_y^2$. So off $\{p, q\}$ the associated rational map is a closed immersion, and its image has dimension $n \geq 2$.

Theorem 3.8.10 applied to that map makes a general member of $\mathbb{P}(V_m(-p-q))$ cut Y irreducibly. Iterating $n-1$ times as in corollary 3.8.12, and passing from tuples to spans by the bundle ρ of Step 0' of the proof of theorem 3.8.20 applied to $V_m(-p-q)$ in place of V_m, gives a dense open subset of $\operatorname{Grass}(n-1, V_m(-p-q))$ over which C_L is an irreducible curve. Its complement is the required Z, and lemma 3.8.3 together with lemma 3.8.1 produces a k-rational point outside it, completing the proof.

3.9 *The Category of Affine Schemes as Ringed Spaces

Having established that scheme morphisms must be morphisms of *locally* ringed spaces to properly reflect geometric evaluation, it is natural to ask: just how wildly does the equivalence $\mathbf{Sch}^{\mathbf{Aff}} \cong \mathbf{CRing}^{\mathrm{op}}$ fail if we drop the "locally" condition? What does the category of affine schemes look like if we restrict our morphisms to ringed space morphisms?

As we saw with the "phantom map" in example 3.1.1, multiple distinct ringed space morphisms can induce the exact same ring homomorphism on global sections. The topological map f becomes somehow decoupled from the algebraic pullback $f^\#$. Fortunately, this decoupling is not entire; it is classified by the topology of the Zariski spectrum, specifically:

Proposition 3.9.1: Classification of Ringed Space Morphisms

Let $X = \operatorname{Spec}(S)$ and $Y = \operatorname{Spec}(R)$ be affine schemes. A pair consisting of a continuous map $f : X \to Y$ and a ring homomorphism $\varphi : R \to S$ defines a valid morphism of ringed spaces $(f, f^{\#}) : X \to Y$ (where $f^{\#}$ on global sections is φ) if and only if:

$$\varphi^{-1}(\mathfrak{p}) \subseteq f(\mathfrak{p}) \qquad \text{for all } \mathfrak{p} \in \operatorname{Spec}(S).$$

Furthermore, when this condition holds, the sheaf map $f^{\#}$ is uniquely determined.

***Proof*:**
Let $g : X \to Y$ be the canonical "true" continuous map defined by $g(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$.

For $(f, f^{\#})$ to be a valid morphism of ringed spaces, the global ring homomorphism $\varphi : R \to S$ must induce a well-defined local map on the stalks for every $\mathfrak{p} \in X$. Specifically, the composite map $R \to S \to S_{\mathfrak{p}}$ must factor uniquely through the localization of R at $f(\mathfrak{p})$:

$$\begin{tikzcd}
R \arrow[r, "\varphi"] \arrow[d, "\text{loc}"] & S \arrow[d, "\text{loc}"] \\
R_{f(\mathfrak{p})} \arrow[r, dashed, "\exists!\, f^{\#}_{\mathfrak{p}}"] & S_{\mathfrak{p}}
\end{tikzcd}$$

By the universal property of localization, this factorization exists if and only if every element in the denominator of $R_{f(\mathfrak{p})}$ maps to a unit in $S_{\mathfrak{p}}$.

- The denominators of $R_{f(\mathfrak{p})}$ are exactly the elements of the set $R \setminus f(\mathfrak{p})$.
- The units of $S_{\mathfrak{p}}$ are exactly the fractions with numerators in S and denominators in $S \setminus \mathfrak{p}$, meaning the image of the element in S must lie in $S \setminus \mathfrak{p}$.

Thus, we require:

$$\varphi(R \setminus f(\mathfrak{p})) \subseteq S \setminus \mathfrak{p}$$

Taking the set-theoretic complement of this condition in R, we see that an element $r \in R$ avoids the left-hand side if and only if it avoids the right-hand side. This gives:

$$\varphi^{-1}(\mathfrak{p}) \subseteq f(\mathfrak{p})$$

Since localization maps are epimorphisms in the category of commutative rings, if this factorization exists, it is unique, uniquely determining the sheaf map $f^{\#}$ on all stalks, and hence on all of X.

Recall that in the Zariski topology, the containment of prime ideals $\mathfrak{q} \subseteq \mathfrak{q}'$ means exactly that $\mathfrak{q}'$ is a specialization of $\mathfrak{q}$ (i.e., $\mathfrak{q}'$ lies in the closure of the point $\mathfrak{q}$, so $\mathfrak{q}' \in \overline{\{\mathfrak{q}\}}$). The proposition reveals a geometric reality about the category of ringed spaces: the algebraic condition $\varphi^{-1}(\mathfrak{p}) \subseteq f(\mathfrak{p})$ translates to:

$$f(\mathfrak{p}) \in \overline{\{g(\mathfrak{p})\}}$$

A ringed space morphism allows the topological map f to take a point $\mathfrak{p}$ and map it to *any specialization* (any point in the closure) of the "correct" algebraic target $g(\mathfrak{p})$, provided the resulting assignment f remains globally continuous.

Example 3.9.2: Revisiting the Phantom Map

Consider again $R = \mathbb{Z}_{(p)}$, $S = \mathbb{Q}$, and φ the standard inclusion. We have $\operatorname{Spec}(\mathbb{Q}) = \{[(0)]\}$ and $\operatorname{Spec}(\mathbb{Z}_{(p)}) = \{[(0)], [(p)]\}$.

The "true" algebraic target for the unique point in $\operatorname{Spec}(\mathbb{Q})$ is $g([(0)]) = \varphi^{-1}((0)) = [(0)]$.

What are the specializations of $[(0)]$ in $\operatorname{Spec}(\mathbb{Z}_{(p)})$? The closure of the generic point $[(0)]$ is the entire space, since the prime ideal (0) is contained in the maximal ideal (p). Therefore, $[(p)]$ is a valid specialization of $[(0)]$.

By proposition 3.9.1, mapping $f([(0)]) = [(p)]$ satisfies the condition $(0) \subseteq (p)$. Thus, it forms a valid ringed space morphism, giving us the phantom map π_2.

4

Schemes II.5: Group Schemes

This chapter can comfortably skipped if the reader is not interested in studying algebraic groups or abelian varieties in the near future. The chapter is placed here since it is a natural follow up to all the constructions we have presented in the previous chapter, and (I think) does not require much in the way of quasi-coherent or coherent sheaves, and similar for sheaf cohomology.

In classical algebraic geometry, an *algebraic group* is a variety that is simultaneously a group, with the multiplication and inversion maps being morphisms of varieties. The theory of algebraic groups over algebraically closed fields is rich and well-developed, encompassing the classification of reductive groups, the structure theory of Borel and torus subgroups, and the representation theory that underpins much of modern mathematics.

Over a general base scheme, however, the correct notion is that of a *group scheme*: a scheme G equipped with morphisms encoding multiplication, identity, and inversion, subject to the usual group axioms expressed diagrammatically. This generalization is essential for arithmetic applications (where the base may be $\operatorname{Spec}(\mathbb{Z})$ or $\operatorname{Spec}(\mathbb{Z}_p)$), for the study of families of groups (where the group structure may degenerate or change character from fiber to fiber), and for capturing phenomena invisible over algebraically closed fields—such as infinitesimal group schemes in characteristic p.

This section develops the foundations of group schemes. The reader should be comfortable with fiber products (section 3.2) throughout.

4.1 Definition and First Examples

Recall that in any category $\mathcal{C}$ with finite products and a terminal object e, a *group object* is an object G equipped with morphisms

$$\mu : G \times G \to G, \qquad \epsilon : e \to G, \qquad \iota : G \to G$$

satisfying the usual group axioms—associativity, identity, and inverse—expressed as commutative diagrams. In the category **Set**, a group object is an ordinary group. In the category of smooth manifolds, a group object is a Lie group. In the category of schemes, a group object is a group scheme.

Definition 4.1.1: Group Scheme

Let S be a scheme. A *group scheme over S* is an S-scheme G together with S-morphisms

$$\mu : G \times_S G \longrightarrow G, \qquad \epsilon : S \longrightarrow G, \qquad \iota : G \longrightarrow G$$

called *multiplication*, *identity*, and *inversion*, such that the following diagrams commute:
Associativity:

$$\begin{tikzcd}
G \times_S G \times_S G \arrow[r, "\mu\times\mathrm{id}"] \arrow[d, "\mathrm{id}\times\mu"'] & G \times_S G \arrow[d, "\mu"] \\
G \times_S G \arrow[r, "\mu"'] & G
\end{tikzcd}$$

Identity:

$$\begin{tikzcd}
S \times_S G \arrow[r, "\epsilon\times\mathrm{id}"] \arrow[dr, "\sim"'] & G \times_S G \arrow[d, "\mu"] & G \times_S S \arrow[l, "\mathrm{id}\times\epsilon"'] \arrow[dl, "\sim"] \\
& G &
\end{tikzcd}$$

Inverse:

$$\begin{tikzcd}
G \arrow[r, "(\iota,\mathrm{id})"] \arrow[d, "(\mathrm{id},\iota)"'] & G \arrow[d, "\mu"] & \\
G \times_S G \arrow[r, "\mu"'] & G & S \arrow[l, "\epsilon"'] \arrow[ul, "\epsilon\circ\pi"'] \arrow[ull, bend right, "\epsilon\circ\pi"']
\end{tikzcd}$$

where $\pi : G \to S$ is the structure morphism.
A group scheme is *commutative* (or *abelian*) if additionally $\mu \circ \sigma = \mu$, where $\sigma : G \times_S G \to G \times_S G$ is the swap morphism.

The identity morphism $\epsilon : S \to G$ is a section of the structure map $\pi : G \to S$. Its image is a closed subscheme of G when G is separated over S, but in general it need not be a closed point or even a closed subscheme—it is an S-valued point of G in the sense of section 3.3.2. Over a field k, however, $\epsilon : \mathrm{Spec}\,(k) \to G$ picks out a k-rational point, which is closed if G is of finite type over k.

Proposition 4.1.2: Group Schemes over a Field are Separated

Let G be a group scheme of finite type over a field k. Then G is separated over k.

We shall assume a proposition from proven later, the intuition that translated directly from topology is that separatedness is the equivalent of Hausdorffness, and a space is Hausdorff is the diagonal is closed.

***Proof*:**
We must show the diagonal $\Delta : G \to G \times_k G$ is a closed immersion (section 3.5.1). Consider the morphism $\varphi : G \times_k G \to G$ defined by $\varphi = \mu \circ (\mathrm{id} \times \iota)$, i.e., $(g, h) \mapsto g \cdot h^{-1}$. The diagonal $\Delta(G)$ is the fiber $\varphi^{-1}(\epsilon(\mathrm{Spec}\,(k)))$. Since ϵ is a section of a morphism of finite type (and k is a field), the image of ϵ is a closed point of G. The fiber of a morphism over a closed point is a closed subscheme of the source. Hence $\Delta(G)$ is closed.

For a group scheme the rational point is supplied for free by the identity, and smoothness upgrades the conclusion from connectedness to integrality.

Corollary 4.1.3: Group Varieties Are Geometrically Integral

Let k be a field and let G be a smooth connected group scheme of finite type over k (definition 4.1.1). Then G is geometrically integral, and it is separated over k.

***Proof*:**
Step 1: geometric connectedness. The identity section $\epsilon\colon \mathrm{Spec}\, k \to G$ is a morphism of k-schemes, so $G(k) \neq \emptyset$. Being of finite type over k, the scheme G is locally of finite type, and it is connected by hypothesis, so proposition 3.4.24 makes $G_{\overline{k}}$ connected.
Step 2: $G_{\overline{k}}$ is regular. Smoothness survives extension of scalars. Locally G admits a factorization as in definition 2.7.8 through $\mathrm{Spec}\,(k[x_1, \ldots, x_{n+r}]/(f_1, \ldots, f_r))$ along an open U, and the rank condition says that U misses the closed set $V(J)$ cut out by the ideal J of $r \times r$ minors of the Jacobian. Extending scalars to $\overline{k}$ changes neither the f_i nor the minors, and $V(J\overline{k}[x_1, \ldots, x_{n+r}])$ is the preimage of $V(J)$, so $U_{\overline{k}}$ misses it too. Hence $G_{\overline{k}}$ is smooth over $\overline{k}$, and proposition 2.7.12 makes it regular.
Step 3: regular plus connected is integral. By proposition 2.7.21 a regular scheme is the disjoint union of its irreducible components, and $G_{\overline{k}}$ is Noetherian, so by lemma 2.4.2 there are finitely many of them and each is open and closed. Step 1 leaves exactly one, so $G_{\overline{k}}$ is irreducible. Its local rings are regular, hence domains by proposition 2.7.3, and in particular reduced, so $G_{\overline{k}}$ is reduced by proposition 2.4.14. Irreducible and reduced is integral by proposition 2.4.25, so $G_{\overline{k}}$ is integral and G is geometrically integral in the sense of definition 3.2.8. Separatedness over k is proposition 4.1.2, completing the proof.

Smoothness enters only through regularity after base change, and that is the exact strengthening of "regular" the geometric statement wants: over an imperfect field a regular k-scheme need not stay reduced over $\overline{k}$, so the corollary fails with "smooth" weakened to "regular". Finite type is used twice, once for proposition 3.4.24 and once for the finiteness of the component set in Step 3. Separatedness is not an extra assumption on G, since every group scheme of finite type over a field has it; it is recorded here because geometric integrality is usually invoked alongside it.

Let us now examine the fundamental examples.

Example 4.1.4: The Additive Group Scheme
The *additive group scheme* $\mathbb{G}_{a,S} = \mathbb{A}^1_S$ has underlying scheme $\mathrm{Spec}_S(\mathcal{O}_S[t])$, with multiplication given by the ring map

$$\mathcal{O}_S[t] \longrightarrow \mathcal{O}_S[t] \otimes_{\mathcal{O}_S} \mathcal{O}_S[t], \qquad t \longmapsto t \otimes 1 + 1 \otimes t,$$

identity $t \mapsto 0$, and inversion $t \mapsto -t$. These encode, respectively, addition, zero, and negation: for any S-scheme T,

$$\mathbb{G}_a(T) = \Gamma(T, \mathcal{O}_T)$$

with its additive group structure. Over a field k, this is simply the additive group of the coordinate ring.

Example 4.1.5: The Multiplicative Group Scheme

The *multiplicative group scheme* $\mathbb{G}_{m,S} = \operatorname{Spec}_S(\mathcal{O}_S[t, t^{-1}])$ has multiplication

$$t \longmapsto t \otimes t,$$

identity $t \mapsto 1$, and inversion $t \mapsto t^{-1}$. For any S-scheme T:

$$\mathbb{G}_m(T) = \Gamma(T, \mathcal{O}_T)^\times,$$

the group of invertible global sections under multiplication.

Example 4.1.6: The General Linear Group Scheme

For a positive integer n, the *general linear group scheme* is

$$\mathrm{GL}_{n,S} = \operatorname{Spec}_S\big(\mathcal{O}_S[x_{11}, x_{12}, \ldots, x_{nn},\, d]/(d \cdot \det(x_{ij}) - 1)\big)\,.$$

The variable d is the formal inverse of $\det(x_{ij})$, so the underlying scheme parametrizes $n \times n$ matrices with invertible determinant. The multiplication map is given by matrix multiplication: the ring map sends x_{ij} to $\sum_k x_{ik} \otimes x_{kj}$, and the identity sends $x_{ij} \mapsto \delta_{ij}$. For any S-scheme T:

$$\mathrm{GL}_n(T) \cong \mathrm{GL}_n(\Gamma(T, \mathcal{O}_T)),$$

the group of invertible $n \times n$ matrices with entries in $\Gamma(T, \mathcal{O}_T)$. When $n = 1$, we recover $\mathrm{GL}_1 \cong \mathbb{G}_m$.

Every example so far was produced by writing down a coordinate ring and reading off a comultiplication. A second route names a functor from k-algebras to groups and then exhibits it as the functor of points of a scheme (section 3.3), and the projective linear group is the first case where that route is the easier one. Its classical description as the quotient $\mathrm{GL}_n/\mathbb{G}_m$ is unavailable here, since quotients of group schemes are not constructed until section 4.4 and even there require faithfully flat descent rather than a pointwise construction. The way around this is to present the same object as a symmetry group: PGL_n is what acts on the matrix algebra M_n.

Definition 4.1.7: The Projective Linear Group Scheme

Let k be a commutative ring and let $n \geq 1$. For a k-algebra R let $M_n(R)$ denote the R-algebra of $n \times n$ matrices, free as an R-module on the matrix units e_{ij}. The *projective linear group* $\mathrm{PGL}_{n,k}$ is the group functor

$$\mathrm{PGL}_n \colon k\text{-}\mathbf{Alg} \longrightarrow \mathbf{Grp}, \qquad \mathrm{PGL}_n(R) := \mathrm{Aut}_{R\text{-alg}}(M_n(R))$$

assigning to R the group of R-algebra automorphisms of $M_n(R)$ under composition, and to a map $R \to R'$ the base change $\varphi \mapsto \varphi \otimes_R R'$.
Write $\mathrm{GL}(M_n)$ for the group scheme of R-linear automorphisms of the free module M_n, so that the basis $\{e_{ij}\}$ gives $\mathrm{GL}(M_n) \cong \mathrm{GL}_{n^2,k}$ as in example 4.1.6. Then $\mathrm{PGL}_{n,k}$ is the closed subscheme of $\mathrm{GL}_{n^2,k}$ cut out by the conditions

$$\varphi(e_{ij})\,\varphi(e_{kl}) = \delta_{jk}\,\varphi(e_{il}), \qquad \sum_{i=1}^{n} \varphi(e_{ii}) = 1$$

imposed for all indices, with composition of automorphisms as group law. In particular $\mathrm{PGL}_{n,k}$ is an affine group scheme and a closed subgroup scheme of $\mathrm{GL}_{n^2,k}$.

Unwinding the second display: an R-point of GL_{n^2} is an R-linear automorphism φ of $M_n(R)$, recorded by the coefficients $a_{(ij),(kl)}$ in $\varphi(e_{kl}) = \sum_{i,j} a_{(ij),(kl)} e_{ij}$. Multiplication on $M_n(R)$ is R-bilinear, so such a φ is multiplicative as soon as it is multiplicative on matrix units and unital as soon as it fixes $\sum_i e_{ii}$; expanding the two conditions in the $a_{(ij),(kl)}$ therefore yields a finite list of quadratic and linear polynomials with integer coefficients whose vanishing is equivalent to φ being an algebra map. Their zero locus is closed under composition and inversion, so $\mathrm{PGL}_{n,k} = \mathrm{Spec}\,(A/\mathfrak{a})$ is a closed subgroup scheme, where A is the coordinate ring of $\mathrm{GL}_{n^2,k}$ and $\mathfrak{a}$ is the ideal they generate; affineness is then proposition 4.1.9. The integer coefficients make the construction compatible with base change, so the same equations define $\mathrm{PGL}_{n,S}$ over an arbitrary base scheme S.

It remains to reconcile this with the classical picture. Each $g \in \mathrm{GL}_n(R)$ gives the automorphism $c_g \colon x \mapsto gxg^{-1}$ of $M_n(R)$, and $c_g = c_h$ exactly when gh^{-1} is central in $M_n(R)$, that is, scalar. Conjugation is thus a homomorphism $\mathrm{GL}_n \to \mathrm{PGL}_n$ whose kernel is the copy of $\mathbb{G}_m$ sitting in GL_n as the scalar matrices (example 4.1.5). Over a field it is surjective on points, which is the Skolem-Noether theorem, and the argument is three lines. Let k be a field and $\varphi \in \mathrm{PGL}_n(k)$; give $V = k^n$ the twisted structure $a \cdot v := \varphi(a)v$ and call the result V_φ. The ring $M_n(k)$ is simple with V as its unique simple module up to isomorphism and every module a direct sum of copies of V, so the n-dimensional module V_φ is isomorphic to V; such an isomorphism is a k-linear bijection g with $g\varphi(a) = ag$ for all a, that is, $\varphi = c_{g^{-1}}$. With the kernel computation this gives

$$\mathrm{PGL}_n(k) = \mathrm{GL}_n(k)/k^\times$$

so on points over a field the automorphism functor returns the naive quotient.

The equality concerns fields and does not survive verbatim over an arbitrary ring. For a k-algebra R the map $\mathrm{GL}_n(R)/R^\times \to \mathrm{PGL}_n(R)$ is injective but need not be surjective: an automorphism φ twists the standard module R^n, and under the Morita equivalence between $M_n(R)$-modules and R-modules the twisted module corresponds to an invertible R-module L_φ, with φ of the form c_g precisely when L_φ is free. The obstruction lies in $\mathrm{Pic}\,(R)$ and vanishes for fields and local rings; see [68] and [45]. This is also why $R \mapsto \mathrm{GL}_n(R)/R^\times$ cannot serve as the definition: it is a group functor, but it fails to be a sheaf and is not representable.

Example 4.1.8: The Case $n = 2$
Take $n = 2$, so M_2 is free of rank 4 on $e_{11}, e_{12}, e_{21}, e_{22}$ and PGL_2 is a closed subgroup scheme of GL_4. The scalars can be watched disappearing on the diagonal torus: for $t \in R^\times$ and $g = \mathrm{diag}(t, 1)$,

$$c_g(e_{11}) = e_{11}, \qquad c_g(e_{12}) = t\, e_{12}, \qquad c_g(e_{21}) = t^{-1} e_{21}, \qquad c_g(e_{22}) = e_{22}$$

so in the ordered basis $(e_{11}, e_{12}, e_{21}, e_{22})$ the automorphism c_g is $\mathrm{diag}(1, t, t^{-1}, 1) \in \mathrm{GL}_4(R)$. A general diagonal $g = \mathrm{diag}(t_1, t_2)$ produces $\mathrm{diag}(1, t_1 t_2^{-1}, t_1^{-1} t_2, 1)$, depending only on the ratio $t_1 t_2^{-1}$. The diagonal torus $\mathbb{G}_m^2 \subseteq \mathrm{GL}_2$ therefore maps onto a one-dimensional torus in PGL_2, the scalars $t_1 = t_2$ collapsing to the identity, as the kernel computation predicts. Over a field, $\mathrm{PGL}_2(k) = \mathrm{GL}_2(k)/k^\times$ acts faithfully on $\mathbb{P}^1_k$ by fractional linear transformations.

The definition claimed affineness on the strength of PGL_n being closed inside GL_{n^2}. That step is worth isolating, since it is invoked again for every kernel and every stabilizer met later in this chapter, none of which arrive with a coordinate ring already in hand.

Proposition 4.1.9: Closed Subgroup Schemes of an Affine Group Scheme Are Affine

Let S be a scheme and let G be an S-group scheme (definition 4.1.1) whose structure morphism $G \to S$ is affine. If $\iota \colon H \to G$ is a closed subgroup scheme, then the structure morphism $H \to S$ is affine. In particular, if $S = \mathrm{Spec}\,(k)$ and G is an affine scheme, then H is an affine scheme.

***Proof*:**
Only the underlying schemes are at issue; the group structure plays no role. Let $U \subseteq S$ be an affine open subset and let G_U denote its preimage in G, an affine scheme because $G \to S$ is affine (definition 3.4.5). Both conditions defining a closed immersion (definition 3.1.22) are local on the target, so the restriction $\iota^{-1}(G_U) \to G_U$ is again a closed immersion. A closed subscheme of an affine scheme $\mathrm{Spec}\, A$ is $\mathrm{Spec}\,(A/\mathfrak{a})$ for an ideal $\mathfrak{a} \subseteq A$ by proposition 3.1.24(1), hence affine, and $\iota^{-1}(G_U)$ is precisely the preimage of U under $H \to S$. So every affine open of S has affine preimage in H, which is the definition of an affine morphism, completing the proof.

Short as it is, this keeps the affine theory closed under the constructions of the present chapter. Kernels (definition 4.2.2) and stabilizers (definition 4.3.3) are built as fiber products over a section and are therefore closed subgroup schemes, so an affine group scheme has affine kernels and affine stabilizers, each of which is again the spectrum of a Hopf algebra by proposition 4.1.14. The pattern fails for non-affine group schemes: an abelian variety (example 4.1.12) is proper over the base, so every closed subgroup scheme of it is proper as well, and among proper schemes the affine ones are exactly the finite ones.

Example 4.1.10: Roots of Unity
For a positive integer n, the *group scheme of n-th roots of unity* is

$$\mu_n = \mathrm{Spec}\,(\mathbb{Z}[t]/(t^n - 1)).$$

The group structure is inherited from $\mathbb{G}_m$: the comultiplication is $t \mapsto t \otimes t$ (the same as for $\mathbb{G}_m$, but now $t^n = 1$). For any ring A:

$$\mu_n(A) = \{a \in A \mid a^n = 1\} \subseteq A^\times.$$

The behavior of μ_n depends dramatically on the characteristic of the base. Over a field k with $\mathrm{char}(k) \nmid n$, the polynomial $t^n - 1$ has n distinct roots in $\overline{k}$, and $\mu_{n,k}$ is a reduced scheme of degree n over k—an étale group scheme. Over $\overline{k}$, it is isomorphic (as a scheme) to the disjoint union of n copies of $\operatorname{Spec}(\overline{k})$, and $\mu_n(\overline{k}) \cong \mathbb{Z}/n\mathbb{Z}$.

In characteristic $p > 0$ with $p \big| n$, the situation changes. Taking $n = p$, we have $t^p - 1 = (t-1)^p$ in $\mathbb{F}_p[t]$, so

$$\mu_{p,\mathbb{F}_p} = \operatorname{Spec}(\mathbb{F}_p[t]/((t-1)^p)) = \operatorname{Spec}(\mathbb{F}_p[s]/(s^p))$$

where $s = t - 1$. This is a *non-reduced* scheme: a single fat point at $t = 1$ with a p-dimensional tangent space. The group $\mu_p(\mathbb{F}_p) = \{1\}$ is trivial, yet the scheme μ_p is non-trivial—it carries nilpotent structure that remembers the "ghost" of the p-th roots of unity that have collapsed to 1.

Example 4.1.11: Constant Group Schemes

Let Γ be a finite (abstract) group. The *constant group scheme* associated to Γ over a base S is the scheme

$$\Gamma_S = \coprod_{\gamma \in \Gamma} S,$$

a disjoint union of copies of S indexed by elements of Γ. The group structure is defined by declaring that the component indexed by γ times the component indexed by γ' lands in the component indexed by $\gamma\gamma'$. For any connected S-scheme T:

$$\Gamma_S(T) = \Gamma.$$

The constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ over a field k is always étale (ref:HERE) and reduced, regardless of the characteristic. It should be carefully distinguished from μ_n: over $\mathbb{F}_p$, the scheme $(\mathbb{Z}/p\mathbb{Z})_{\mathbb{F}_p}$ consists of p disjoint points, while $\mu_{p,\mathbb{F}_p}$ is a single fat point.

Example 4.1.12: Elliptic Curves as Group Schemes

An elliptic curve E over a field k (see section 7.1.7) is a smooth projective curve of genus 1 equipped with a k-rational point $O \in E(k)$. The chord-and-tangent construction endows E with a group scheme structure: the multiplication is the addition law on points, $\epsilon = O$, and ι is the map $P \mapsto -P$. Unlike the previous examples, E is *not* affine—it is a projective group scheme. Commutative projective group schemes (of finite type over a field) are called *abelian varieties*; elliptic curves are abelian varieties of dimension 1.

4.1.1 The Hopf Algebra Perspective

The examples above were all defined by specifying ring maps encoding multiplication, identity, and inversion. For affine group schemes, this algebraic data has a clean formalization: the coordinate ring carries the structure of a *Hopf algebra*. This perspective is particularly useful for explicit computations and for connecting group schemes to representation theory.

Let $G = \operatorname{Spec}(A)$ be an affine group scheme over a field k. The group operations on G dualize to k-algebra maps on A:

$$\begin{aligned}
\mu : G \times_k G \to G \quad &\rightsquigarrow \quad \Delta : A \to A \otimes_k A \qquad &\text{(comultiplication)}\\
\epsilon : \operatorname{Spec}(k) \to G \quad &\rightsquigarrow \quad \varepsilon : A \to k \qquad &\text{(counit)}\\
\iota : G \to G \quad &\rightsquigarrow \quad S : A \to A \qquad &\text{(antipode)}
\end{aligned}$$

The group axioms translate directly:

Definition 4.1.13: Hopf Algebra

A *commutative Hopf algebra* over k is a commutative k-algebra A equipped with k-algebra homomorphisms $\Delta : A \to A \otimes_k A$ (comultiplication) and $\varepsilon : A \to k$ (counit), and a k-algebra map $S : A \to A$ (antipode), satisfying:

1. *Coassociativity:* $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$.
2. *Counit:* $(\varepsilon \otimes \mathrm{id}) \circ \Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon) \circ \Delta$.
3. *Antipode:* $\mu_A \circ (S \otimes \mathrm{id}) \circ \Delta = \varepsilon \cdot 1_A = \mu_A \circ (\mathrm{id} \otimes S) \circ \Delta$, where $\mu_A : A \otimes_k A \to A$ is the multiplication of A.

The coassociativity and counit axioms say that (A, Δ, ε) is a *coalgebra*; the antipode axiom says that S "undoes" the comultiplication in the same way that inversion undoes multiplication in a group. The key constraint is that Δ and ε must be *algebra* homomorphisms—this encodes the fact that multiplication $\mu : G \times G \to G$ is a morphism of schemes, not a set-theoretic map.

Proposition 4.1.14: Affine Group Schemes and Hopf Algebras

The functor $G \mapsto \mathcal{O}(G)$ is an anti-equivalence of categories:

$$\{\text{Affine group schemes over } k\}^{\mathrm{op}} \xrightarrow{\sim} \{\text{Commutative Hopf algebras over } k\}.$$

***Proof*:**
This follows from the anti-equivalence between affine k-schemes and commutative k-algebras. An affine group scheme is a group object in the category of affine k-schemes; dualizing, its coordinate ring becomes a cogroup object in k-**Alg**. The axioms of a cogroup object in a category of commutative algebras are exactly the axioms of a commutative Hopf algebra.

Let us revisit the earlier examples through this lens.

Example 4.1.15: Hopf Algebras

1. *The additive group.* The Hopf algebra of $\mathbb{G}_a = \operatorname{Spec}(k[t])$ is $k[t]$ with $\Delta(t) = t \otimes 1 + 1 \otimes t$, $\varepsilon(t) = 0$, and $S(t) = -t$. In the language of coalgebras, t is a *primitive* element: $\Delta(t) = t \otimes 1 + 1 \otimes t$.
2. *The multiplicative group.* The Hopf algebra of $\mathbb{G}_m = \operatorname{Spec}(k[t, t^{-1}])$ is $k[t, t^{-1}]$ with $\Delta(t) = t \otimes t$, $\varepsilon(t) = 1$, and $S(t) = t^{-1}$. Here t is a *group-like* element: $\Delta(t) = t \otimes t$.

3. *The general linear group.* The Hopf algebra of GL_n is $k[x_{ij}, \det^{-1}]$ with $\Delta(x_{ij}) = \sum_k x_{ik} \otimes x_{kj}$ (matrix multiplication), $\varepsilon(x_{ij}) = \delta_{ij}$ (the identity matrix), and $S(x_{ij})$ given by the (i,j)-cofactor of (x_{ij}) divided by det—this is the Cramer's rule formula for the entries of the inverse matrix.

4. *Roots of unity.* The Hopf algebra of μ_n is $k[t]/(t^n - 1)$ with the same comultiplication as $\mathbb{G}_m$. The quotient by the ideal $(t^n - 1)$ is compatible with the Hopf structure because $\Delta(t^n - 1) = t^n \otimes t^n - 1 \otimes 1 = (t^n - 1) \otimes t^n + 1 \otimes (t^n - 1)$, which lies in the ideal generated by $t^n - 1$ in $A \otimes A$. In Hopf algebra language, $(t^n - 1)$ is a *Hopf ideal*.

The Hopf algebra viewpoint makes certain constructions transparent. For instance, a *homomorphism* of group schemes $\varphi : G \to H$ corresponds to a homomorphism of Hopf algebras $\varphi^* : \mathcal{O}(H) \to \mathcal{O}(G)$ (respecting Δ, ε, and S). The kernel of φ is the group scheme whose Hopf algebra is $\mathcal{O}(G)/(\ker \varepsilon_H \cdot \mathcal{O}(G))$; we will formalize this in section 4.2.

4.1.16 Remark: non-commutative Hopf algebras If we drop the requirement that A be commutative, we obtain the notion of a *non-commutative Hopf algebra*, which corresponds to a "quantum group" rather than a group scheme. The theory of quantum groups, developed by Drinfeld and Jimbo, is a rich subject in its own right but lies outside our scope. In this text, all Hopf algebras are commutative (corresponding to the fact that the coordinate ring of an affine scheme is always commutative), though the group schemes themselves need not be.

4.1.2 Finite and Infinitesimal Group Schemes

Over an algebraically closed field of characteristic zero, the theory of affine algebraic groups is largely controlled by Lie theory: connected algebraic groups are smooth, their structure is determined by their Lie algebras, and finite group schemes are just finite groups viewed as zero-dimensional varieties. In characteristic $p > 0$, the situation is fundamentally richer: there exist group schemes that are "purely infinitesimal"—non-reduced, supported at a single point, yet carrying nontrivial group structure encoded in the nilpotent elements of their coordinate ring.

Definition 4.1.17: Finite Group Scheme

A group scheme G over a field k is *finite* if it is of the form $G = \operatorname{Spec}(A)$ where A is a finite-dimensional k-algebra. The *order* of G is $\dim_k A$.

A finite group scheme of order n has at most n geometric points, but may have fewer if A has nilpotents. The extreme cases are:

Étale group schemes. A finite group scheme is *étale* if its coordinate ring A is a product of separable field extensions of k. Over an algebraically closed field, an étale group scheme of order n is simply a disjoint union of n copies of $\operatorname{Spec}(k)$, and the group structure is that of an ordinary finite group. The constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ is the prototypical example.

Infinitesimal (or connected) group schemes. A finite group scheme $G = \operatorname{Spec}(A)$ is *infinitesimal* if A is a local ring, i.e., G is supported at a single point. The group structure is then entirely encoded in the nilpotent elements of A. These exist only in positive characteristic (or over non-reduced bases).

We have already seen the simplest examples of both types.

Example 4.1.18: μ_p in Characteristic p

As computed in Example 4.1.10, over $\mathbb{F}_p$ the group scheme $\mu_p = \operatorname{Spec}(\mathbb{F}_p[t]/(t^p - 1)) = \operatorname{Spec}(\mathbb{F}_p[s]/(s^p))$ (where $s = t-1$) is infinitesimal of order p. Its Hopf algebra structure (expressed in terms of s) is:

$$\Delta(s) = s \otimes 1 + 1 \otimes s + s \otimes s, \qquad \varepsilon(s) = 0, \qquad S(s) = \sum_{i=1}^{p-1} (-1)^i s^i.$$

The comultiplication is neither purely primitive ($s \otimes 1 + 1 \otimes s$) nor purely group-like ($s \otimes s$)—it is a mixture, reflecting the fact that μ_p is a "twisted" infinitesimal group.

Example 4.1.19: The Frobenius Kernel α_p

The group scheme $\alpha_p = \operatorname{Spec}(k[t]/(t^p))$, defined over a field k of characteristic $p > 0$, is the kernel of the Frobenius endomorphism on $\mathbb{G}_a$. Its Hopf algebra is $k[t]/(t^p)$ with

$$\Delta(t) = t \otimes 1 + 1 \otimes t, \qquad \varepsilon(t) = 0, \qquad S(t) = -t.$$

Note that t is primitive, just as in $\mathbb{G}_a$, but the relation $t^p = 0$ (rather than no relation) makes α_p a finite group scheme of order p. For any k-algebra A:

$$\alpha_p(A) = \{a \in A \mid a^p = 0\},$$

which is a subgroup of $(A, +)$. Over a perfect field, α_p has only the trivial k-point, yet it is a non-trivial group scheme: for instance, $\alpha_p(k[\epsilon]/(\epsilon^p)) = k \cdot \epsilon$ is a p-element group.

The schemes α_p and μ_p are both infinitesimal group schemes of order p, but they are *not* isomorphic: α_p is connected and has no non-trivial characters (homomorphisms to $\mathbb{G}_m$), while μ_p is the Cartier dual of $\mathbb{Z}/p\mathbb{Z}$ and detects p-torsion in the multiplicative group.

The contrast between α_p and μ_p illustrates a general structural result. Over a perfect field k of characteristic $p > 0$, every finite group scheme G admits a canonical filtration

$$0 \subseteq G^0 \subseteq G$$

where G^0 is the *connected component of the identity* (the largest connected subgroup scheme) and G/G^0 is étale. This is the *connected-étale sequence*:

$$1 \longrightarrow G^0 \longrightarrow G \longrightarrow G^{\text{ét}} \longrightarrow 1,$$

and over a perfect field this sequence splits (non-canonically), so $G \cong G^0 \times G^{\text{ét}}$. The connected part G^0 is an infinitesimal group scheme (its coordinate ring is local), and the étale part $G^{\text{ét}}$ is determined by the action of $\operatorname{Gal}(\overline{k}/k)$ on the geometric points $G(\overline{k})$.

This decomposition has no analogue in characteristic zero, where every finite group scheme over a field is étale. It is a manifestation of the richer arithmetic of positive characteristic, and plays a fundamental role in the theory of abelian varieties (where the p-torsion $A[p]$ of an abelian variety splits into connected and étale parts whose relative sizes determine the p-rank).

The Frobenius and Verschiebung

The characteristic-p phenomena above are organized by two fundamental endomorphisms. Let k be a perfect field of characteristic p, and let G be a group scheme over k.

The *absolute Frobenius* $F_{\text{abs}} : G \to G$ is the identity on the underlying topological space and acts as $a \mapsto a^p$ on the structure sheaf. This is a morphism of schemes but *not* a morphism of k-schemes (it does not commute with the structure map unless $k = \mathbb{F}_p$). To obtain a k-morphism, one passes to the *relative Frobenius* $F : G \to G^{(p)}$, where $G^{(p)}$ is the base change of G along the Frobenius $\operatorname{Spec}(k) \to \operatorname{Spec}(k)$. The relative Frobenius is a homomorphism of group schemes.

For a commutative group scheme, there is a dual morphism $V : G^{(p)} \to G$ called the *Verschiebung* (German for "shift"). It satisfies $V \circ F = [p]_G$ and $F \circ V = [p]_{G^{(p)}}$, where $[p]$ denotes multiplication by p in the group law. The kernels $\ker(F)$ and $\ker(V)$ are finite group schemes that capture, respectively, the "infinitesimal" and "étale" parts of the p-torsion.

For example, $\ker(F : \mathbb{G}_a \to \mathbb{G}_a^{(p)}) = \alpha_p$ (since F acts as $t \mapsto t^p$ on $k[t]$, and $\ker = \operatorname{Spec}(k[t]/(t^p))$), while $\ker(F : \mathbb{G}_m \to \mathbb{G}_m^{(p)}) = \mu_p$.

4.2 Homomorphisms, Kernels, and Exact Sequences

Definition 4.2.1: Homomorphism of Group Schemes

A *homomorphism* of S-group schemes $\varphi : G \to H$ is a morphism of S-schemes that is compatible with the group structures:

$$\varphi \circ \mu_G = \mu_H \circ (\varphi \times \varphi)$$

, $\varphi \circ \epsilon_G = \epsilon_H$, and $\varphi \circ \iota_G = \iota_H \circ \varphi$.

In the functor-of-points language, φ is a homomorphism if and only if for every S-scheme T, the induced map $\varphi(T) : G(T) \to H(T)$ is a group homomorphism.

Definition 4.2.2: Kernel of a Homomorphism

Let $\varphi : G \to H$ be a homomorphism of S-group schemes. The *kernel* of φ is the fiber product

$$\ker(\varphi) = G \times_H S$$

where the two maps to H are $\varphi : G \to H$ and $\epsilon_H : S \to H$.

By the universal property of fiber products, for any S-scheme T:

$$\ker(\varphi)(T) = \{g \in G(T) \mid \varphi(T)(g) = e_H \in H(T)\},$$

which is exactly the kernel of the group homomorphism $\varphi(T) : G(T) \to H(T)$. The kernel $\ker(\varphi)$ is a closed subgroup scheme of G (being a fiber of a morphism over a section).

Example 4.2.3: Classical Exact Sequences

1. *The n-th power map on $\mathbb{G}_m$.* The homomorphism $[n] : \mathbb{G}_m \to \mathbb{G}_m$ defined by $t \mapsto t^n$ has kernel μ_n. This gives a short exact sequence of group schemes:
$$1 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \xrightarrow{(\,)^n} \mathbb{G}_m \longrightarrow 1.$$
This is the *Kummer sequence*. Over a field k with $\mathrm{char}(k) \nmid n$, passing to $\overline{k}$-points recovers the familiar exact sequence $1 \to \mu_n(\overline{k}) \to \overline{k}^\times \xrightarrow{n} \overline{k}^\times \to 1$.

2. *The determinant.* The determinant $\det : \mathrm{GL}_n \to \mathbb{G}_m$ is a homomorphism of group schemes with kernel SL_n:
$$1 \longrightarrow \mathrm{SL}_n \longrightarrow \mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m \longrightarrow 1.$$

3. *The Frobenius on $\mathbb{G}_a$ in characteristic p.* The Frobenius $F : \mathbb{G}_a \to \mathbb{G}_a$ defined by $t \mapsto t^p$ has kernel α_p:
$$0 \longrightarrow \alpha_p \longrightarrow \mathbb{G}_a \xrightarrow{F} \mathbb{G}_a \longrightarrow 0.$$
This is the *Artin–Schreier sequence* (in its scheme-theoretic form).

A word of caution on exactness is in order. A sequence $1 \to N \xrightarrow{i} G \xrightarrow{\varphi} H \to 1$ of group schemes is said to be *exact* if: i identifies N with $\ker(\varphi)$ as a closed subgroup scheme of G, and φ is faithfully flat. The surjectivity of φ is *not* the naive statement that $\varphi(T) : G(T) \to H(T)$ is surjective for all T; this is often false even when the sequence is exact. For instance, in the Kummer sequence over $\mathbb{Q}$, the map $\mathbb{Q}^\times \xrightarrow{n} \mathbb{Q}^\times$ is not surjective (not every rational number is an n-th power), but the map of schemes $\mathbb{G}_m \xrightarrow{n} \mathbb{G}_m$ is faithfully flat. The correct notion of surjectivity for group schemes is faithfully flat descent, not surjectivity on points.

4.3 Actions of Group Schemes on Schemes

Definition 4.3.1: Action of a Group Scheme

Let G be an S-group scheme and X an S-scheme. An *action* of G on X is an S-morphism
$$\sigma : G \times_S X \longrightarrow X$$
satisfying the usual axioms: the diagram
$$\begin{tikzcd} G \times_S G \times_S X \arrow[r, "\mathrm{id}\times\sigma"] \arrow[d, "\mu\times\mathrm{id}"'] & G \times_S X \arrow[d, "\sigma"] \\ G \times_S X \arrow[r, "\sigma"'] & X \end{tikzcd}$$
commutes (associativity), and $\sigma \circ (\epsilon \times \mathrm{id}_X) = \mathrm{id}_X$ (the identity acts trivially).

In the functor-of-points language, this means: for every S-scheme T, the group $G(T)$ acts on the set

$X(T)$, and these actions are compatible with base change.

Example 4.3.2: Standard Actions

1. *Translation.* Any group scheme G acts on itself by left multiplication: $\sigma(g, h) = g \cdot h$. This is the *left regular action*. Each element $g \in G(T)$ defines an automorphism $\lambda_g : G_T \to G_T$ called *left translation by g*.
2. *Conjugation.* G acts on itself by conjugation: $\sigma(g, h) = ghg^{-1}$. This is the *adjoint action* of G on itself. Its derivative at the identity gives the adjoint representation of $\mathfrak{g}$ (the Lie algebra of G) on itself; see section 4.5.
3. *Linear representations.* An action of G on $\mathbb{A}^n_S$ that is $\mathcal{O}_S$-linear corresponds to a homomorphism $G \to \mathrm{GL}_{n,S}$. This is the scheme-theoretic notion of a *linear representation*.
4. $\mathbb{G}_m$*-actions and gradings.* An action of $\mathbb{G}_m$ on an affine scheme $\mathrm{Spec}\,(A)$ corresponds to a $\mathbb{Z}$-grading on A: $A = \bigoplus_{d\in\mathbb{Z}} A_d$. The $\mathbb{G}_m$-equivariant morphisms are the graded ring homomorphisms. This is the algebraic origin of the weight decompositions ubiquitous in representation theory.

Definition 4.3.3: Orbits and Stabilizers

Let G act on X and let $x \in X(T)$ be a T-valued point. The *orbit map* is

$$\sigma_x : G_T \longrightarrow X_T, \qquad g \longmapsto g \cdot x.$$

The *stabilizer* (or *isotropy group*) of x is the fiber product

$$G_x = G_T \times_{X_T} T$$

where the two maps to X_T are $\sigma_x : G_T \to X_T$ and $x : T \to X_T$. For any T-scheme T':

$$G_x(T') = \{g \in G(T') \mid g \cdot x = x \in X(T')\}.$$

The stabilizer is a closed subgroup scheme of G_T.

4.4 Quotients by Group Actions

Having defined actions, we now ask: given an action $\sigma : G \times_S X \to X$, does the "quotient" X/G exist as a scheme? This question is considerably more subtle than its set-theoretic counterpart, and its resolution leads to geometric invariant theory (GIT).

The categorical quotient

The most basic notion is that of a *categorical quotient*. We want a scheme Y and a G-invariant morphism $\pi : X \to Y$ (meaning $\pi \circ \sigma = \pi \circ \mathrm{pr}_2$, where $\mathrm{pr}_2 : G \times X \to X$ is the second projection) that is universal: any G-invariant morphism $X \to Z$ factors uniquely through π.

Definition 4.4.1: Categorical and Geometric Quotients

Let G act on X over S. A *categorical quotient* is an S-scheme Y with a G-invariant morphism $\pi : X \to Y$ such that:

1. (Universal property) For every G-invariant morphism $f : X \to Z$, there exists a unique morphism $\overline{f} : Y \to Z$ with $f = \overline{f} \circ \pi$.
2. (Coinvariant ring, in the affine case) If $X = \operatorname{Spec}(A)$ and G acts on A by ring automorphisms, then $Y = \operatorname{Spec}(A^G)$ where A^G denotes the subring of G-invariants.

A categorical quotient $\pi : X \to Y$ is a *geometric quotient* if additionally:

(3) π is surjective,

(4) the fibers of π are exactly the G-orbits, and

(5) π is submersive (a subset $U \subseteq Y$ is open if and only if $\pi^{-1}(U)$ is open in X).

A geometric quotient, when it exists, is the best possible outcome: the scheme Y faithfully represents the orbit space. However, geometric quotients frequently fail to exist—the simplest example being the action of $\mathbb{G}_m$ on $\mathbb{A}^2$ by scalar multiplication, where the origin is a “bad” orbit (it is the only closed orbit of dimension less than 1).

The GIT approach

Geometric invariant theory, developed by Mumford, provides a systematic method for constructing quotients by reductive group actions on projective or quasi-projective varieties. The central idea is to work not with the full quotient, but with a *good quotient* obtained by restricting to “semistable” points.

The word *reductive* is used here without being defined. It belongs to the structure theory of linear algebraic groups, which this book does not develop, and it is established where it is actually needed, in the geometric-invariant-theory chapter of *Moduli, Deformation Theory, and Stacks* [17]. For the paragraphs below it is enough to read it as naming the groups already met in section 4.1: the general and special linear groups GL_n and SL_n, the tori $\mathbb{G}_m^r$, and PGL_n. What the hypothesis gives is finite generation of the ring of invariants, without which the construction below does not produce a scheme at all. Over a field of characteristic 0 that property is equivalent to every finite-dimensional linear representation of G decomposing into irreducible subrepresentations; in characteristic p the equivalence fails, and the weaker notion of geometric reductivity is what the finite-generation theorem actually requires.

Let G be a reductive group acting on a projective variety $X \subseteq \mathbb{P}^n$ via a linearization (an action that lifts to the homogeneous coordinate ring). The *semistable locus* X^{ss} consists of those points $x \in X$ for which there exists a G-invariant section $f \in \Gamma(X, \mathcal{O}_X(d))^G$ (for some $d > 0$) with $f(x) \neq 0$. The *stable locus* $X^s \subseteq X^{ss}$ consists of those semistable points with finite stabilizer and closed orbit in X^{ss}.

The fundamental theorem of GIT asserts that the categorical quotient $X^{ss} \to X^{ss}/\!/G$ exists as a projective variety, and its restriction to the stable locus $X^s \to X^s/G$ is a geometric quotient. The notation “$/\!/$” (double slash) is reserved for the GIT quotient to distinguish it from the naive set-theoretic quotient.

A thorough treatment of GIT lies outside the scope of this text, but we record the most important special that we use in theorem 7.1.40 to bound the number of automorphisms of most curves:

Proposition 4.4.2: Quotient by a Finite Group

Let G be a finite group acting on an affine scheme $X = \mathrm{Spec}\,(A)$ by ring automorphisms. Then the categorical quotient $X/G = \mathrm{Spec}\,(A^G)$ exists, and the morphism $\pi : X \to X/G$ is finite and surjective. If $|G|$ is invertible in A, then π is a geometric quotient and $X \to X/G$ is étale over the locus of points with trivial stabilizer.

***Proof*:**
The ring of invariants A^G is a subring of A, and A is integral over A^G by the theorem on integral extensions (each $a \in A$ satisfies the monic polynomial $\prod_{g\in G}(T - g\cdot a) \in A^G[T]$). Hence $\pi : \mathrm{Spec}\,(A) \to \mathrm{Spec}\,(A^G)$ is finite and surjective. The universal property of $\mathrm{Spec}\,(A^G)$ as a categorical quotient follows from the fact that G-invariant maps $A \to B$ factor uniquely through A^G. When $|G|$ is invertible, the Reynolds operator $R = \frac{1}{|G|}\sum_{g\in G} g$ is a projection $A \to A^G$, and the étaleness statement follows from the Jacobian criterion applied to the map π away from the ramification locus.

4.5 The Lie Algebra of a Group Scheme

In the theory of Lie groups, the tangent space at the identity carries the structure of a Lie algebra, and this infinitesimal invariant determines the local structure of the group. The same construction works for group schemes, using the Zariski tangent space from section 2.7. (In the following section, we will see that this tangent space admits a clean reformulation via dual-number valued points; see Example 3.3.7.)

Let G be a group scheme of finite type over a field k, with identity element $e \in G(k)$. The Zariski tangent space at e is:

$$T_eG = (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$$

where $\mathfrak{m}_e$ is the maximal ideal of $\mathcal{O}_{G,e}$.

Definition 4.5.1: Lie Algebra of a Group Scheme

The *Lie algebra* of G is the tangent space $\mathfrak{g} = T_eG$, equipped with a k-linear bracket $[\,,\,] : \mathfrak{g} \otimes_k \mathfrak{g} \to \mathfrak{g}$ defined via the adjoint action (specified below). We write $\mathfrak{g} = \mathrm{Lie}(G)$.

To define the bracket, observe that the group structure on G induces a group structure on the set of dual-number points:

$$G(k[\epsilon]/(\epsilon^2))$$

is a group via the multiplication μ. The kernel of the natural map $G(k[\epsilon]/(\epsilon^2)) \to G(k)$ (induced by $\epsilon \mapsto 0$) is exactly $\mathfrak{g}$, and it sits in a split short exact sequence of groups:

$$0 \longrightarrow \mathfrak{g} \longrightarrow G(k[\epsilon]/(\epsilon^2)) \longrightarrow G(k) \longrightarrow 1.$$

In this sequence, $\mathfrak{g}$ is an *abelian* subgroup (the product of two tangent vectors $v, w \in \mathfrak{g}$ under the group law of $G(k[\epsilon]/(\epsilon^2))$ is $v + w$, since the quadratic terms in ϵ vanish). However, the *commutator* of elements in $G(k[\epsilon_1, \epsilon_2]/(\epsilon_1^2, \epsilon_2^2, \epsilon_1\epsilon_2))$ is well-defined and lands in $\mathfrak{g}$; this commutator defines the Lie bracket $[v, w]$.

More concretely, the adjoint action $\mathrm{ad} : G \to \mathrm{Aut}(\mathfrak{g})$ is defined by conjugation: for $g \in G(T)$ and $v \in \mathfrak{g} \otimes_k \mathcal{O}(T)$, set $\mathrm{ad}(g)(v) = gvg^{-1}$ (computed in $G(T[\epsilon]/(\epsilon^2))$). Differentiating ad at the identity gives a map $\mathfrak{g} \to \mathrm{End}_k(\mathfrak{g})$, and the Lie bracket is $[v, w] = (\mathrm{d}_e\mathrm{ad})(v)(w)$.

Proposition 4.5.2: Lie Algebra as Primitive Elements

Let $G = \mathrm{Spec}\,(A)$ be an affine group scheme over k with Hopf algebra (A, Δ, ε) and augmentation ideal $I = \ker(\varepsilon)$. Then:

$$\mathfrak{g} = (I/I^2)^\vee$$

as a k-vector space, and the Lie bracket corresponds to the commutator in the convolution algebra $\mathrm{Hom}_k(A, k)$. Equivalently, $\mathfrak{g}$ is the space of *primitive elements* of the dual coalgebra: those $\delta \in A^\vee$ satisfying $\delta(ab) = \varepsilon(a)\delta(b) + \delta(a)\varepsilon(b)$ for all $a, b \in A$ (i.e., δ is a derivation $A \to k$ at ε).

***Proof*:**
By the Hopf algebra structure, the augmentation ideal $I = \ker(\varepsilon)$ is the maximal ideal at the identity, so $(I/I^2)^\vee$ is the Zariski tangent space at e as expected. A k-linear functional $\delta : A \to k$ vanishing on $k \cdot 1$ corresponds to an element of $I^\vee$; it descends to $(I/I^2)^\vee$ if and only if $\delta(I^2) = 0$, which is equivalent to the derivation (Leibniz) rule $\delta(ab) = \varepsilon(a)\delta(b) + \delta(a)\varepsilon(b)$.

Example 4.5.3: Lie Algebras of Classical Group Schemes

1. $\mathbb{G}_a$*:* The augmentation ideal of $k[t]$ (with $\varepsilon(t) = 0$) is (t), and $(t)/(t^2) \cong k$. Thus $\mathrm{Lie}(\mathbb{G}_a) = k$ with the zero bracket—the Lie algebra is one-dimensional and abelian.

2. $\mathbb{G}_m$*:* The augmentation ideal of $k[t, t^{-1}]$ (with $\varepsilon(t) = 1$) is $(t-1)$. Setting $s = t-1$, we have $(s)/(s^2) \cong k$, so $\mathrm{Lie}(\mathbb{G}_m) = k$ with the zero bracket—again one-dimensional and abelian.

3. GL_n*:* The augmentation ideal of $k[x_{ij}, \det^{-1}]$ (with $\varepsilon(x_{ij}) = \delta_{ij}$) is generated by the elements $e_{ij} := x_{ij} - \delta_{ij}$. The quotient I/I^2 is freely generated by the images of the e_{ij}, so $\mathrm{Lie}(\mathrm{GL}_n) \cong M_n(k)$, the space of all $n \times n$ matrices. The Lie bracket is the matrix commutator $[A, B] = AB - BA$, which recovers the classical Lie algebra $\mathfrak{gl}_n$.

4. SL_n*:* The additional relation $\det(x_{ij}) = 1$ imposes the linear condition $\mathrm{tr}(e_{ij}) = 0$ at the level of I/I^2. Hence $\mathrm{Lie}(\mathrm{SL}_n) = \{A \in M_n(k) : \mathrm{tr}(A) = 0\} = \mathfrak{sl}_n$.

5. μ_n *over a field with* $\mathrm{char}(k) \nmid n$*:* The augmentation ideal of $k[t]/(t^n - 1)$ (with $\varepsilon(t) = 1$) is $I = (t-1)$. Writing $t^n - 1 = (t-1)(t^{n-1} + \cdots + 1)$, and noting that $t^{n-1} + \cdots + 1$ evaluates to $n \neq 0$ at $t = 1$ (hence is coprime to $t - 1$), we find that $(t-1)^2$ and $t^n - 1$ together generate $(t-1)$, so $I/I^2 = 0$. Thus $\mathrm{Lie}(\mu_n) = 0$—the Lie algebra is trivial. This reflects the fact that μ_n is étale: it has no infinitesimal structure.

6. μ_p *in characteristic* p*:* Now $t^p - 1 = (t-1)^p$, so the coordinate ring is $k[s]/(s^p)$ with

$s = t - 1$, and $I/I^2 = (s)/(s^2) \cong k$. Hence $\mathrm{Lie}(\mu_p) = k$ is one-dimensional. The infinitesimal group scheme μ_p carries tangent information at its unique point, unlike its étale counterpart.

The contrast between the last two examples is worth emphasizing: μ_n for $\mathrm{char}(k) \nmid n$ is étale with trivial Lie algebra, while μ_p in characteristic p is infinitesimal with a one-dimensional Lie algebra. The Lie algebra "sees" the infinitesimal structure of the group scheme, and in characteristic p this structure is genuinely richer.

This observation foreshadows a general theme: in characteristic zero, the Lie algebra functor $G \mapsto \mathrm{Lie}(G)$ is an equivalence between connected algebraic groups and their Lie algebras (this is the scheme-theoretic incarnation of Lie's third theorem). In characteristic p, the Lie algebra loses information—it cannot distinguish between α_p and $\mathbb{G}_a$, for instance, since both have $\mathrm{Lie} = k$. The correct replacement is the theory of *p-Lie algebras* (or *restricted Lie algebras*), which equips $\mathfrak{g}$ with an additional "p-th power" operation $x \mapsto x^{[p]}$ that captures the Frobenius structure. This theory, while important for the classification of algebraic groups in positive characteristic, lies beyond our present scope.

4.6 Cartier Duality

For finite abelian groups (or in functional analysis, a locally compact abelian group like $\mathbb{R}$ with the euclidean topology), Pontryagin duality states that a group A is isomorphic to its double dual $\mathrm{Hom}(\mathrm{Hom}(A, S^1), S^1)$. In the realm of finite commutative group schemes over a field k, there is a perfect geometric analogue where the circle group S^1 is replaced by the multiplicative group scheme $\mathbb{G}_m$.

Definition 4.6.1: Cartier Dual

Let G be a finite commutative group scheme over a field k. The *Cartier dual* of G, denoted G^D, is the group scheme representing the functor of homomorphisms from G to $\mathbb{G}_m$. That is, for any k-algebra R:

$$G^D(R) = \mathrm{Hom}_{\text{gp-sch}/R}(G_R, \mathbb{G}_{m,R}).$$

At first glance, it is not obvious that this functor is representable by a finite commutative group scheme. However, the Hopf algebra perspective makes the construction transparent and purely algebraic.

Let $G = \mathrm{Spec}\,(A)$. Since G is finite, A is a finite-dimensional k-vector space. We can form the dual vector space $A^\vee = \mathrm{Hom}_k(A, k)$. Because G is a commutative group scheme, its Hopf algebra A is both commutative and cocommutative (meaning $\sigma \circ \Delta = \Delta$, where σ swaps the tensor factors in $A \otimes_k A$).

Dualizing the structural maps of A perfectly swaps the algebra and coalgebra structures:

- The multiplication $A \otimes_k A \to A$ dualizes to a comultiplication $A^\vee \to A^\vee \otimes_k A^\vee$.
- The comultiplication $\Delta : A \to A \otimes_k A$ dualizes to a multiplication $A^\vee \otimes_k A^\vee \to A^\vee$.
- The unit $k \to A$ dualizes to a counit $A^\vee \to k$.
- The counit $\varepsilon : A \to k$ dualizes to a unit $k \to A^\vee$.

- The antipode $S : A \to A$ dualizes to an antipode $S^\vee : A^\vee \to A^\vee$.

Since A is commutative and cocommutative, $A^\vee$ is also a commutative and cocommutative Hopf algebra. Thus, it defines a new finite commutative affine group scheme.

Proposition 4.6.2: Cartier Duality Theorem

Let $G = \operatorname{Spec}(A)$ be a finite commutative group scheme over k. Then the Cartier dual G^D is represented by the affine group scheme $\operatorname{Spec}(A^\vee)$. Furthermore, the natural evaluation map gives an isomorphism:

$$G \xrightarrow{\sim} (G^D)^D.$$

Example 4.6.3: Cartier Duals of Core Examples

1. *Roots of unity and constant groups.* The most classical pairing is between the constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ and the group of roots of unity μ_n. A homomorphism from $\mathbb{Z}/n\mathbb{Z}$ to $\mathbb{G}_m$ is completely determined by where it sends the generator $1 \in \mathbb{Z}/n\mathbb{Z}$, and this image must be an n-th root of unity. Hence, for any k-algebra R:

$$\operatorname{Hom}((\mathbb{Z}/n\mathbb{Z})_R, \mathbb{G}_{m,R}) \cong \mu_n(R).$$

   Thus, $((\mathbb{Z}/n\mathbb{Z})_k)^D \cong \mu_n$. By the duality theorem, we also have $\mu_n^D \cong (\mathbb{Z}/n\mathbb{Z})_k$.

2. *The Frobenius kernel α_p.* Let k be a perfect field of characteristic $p > 0$, and recall $\alpha_p = \operatorname{Spec}(k[t]/(t^p))$ with $\Delta(t) = t \otimes 1 + 1 \otimes t$. To find a homomorphism from α_p to $\mathbb{G}_m = \operatorname{Spec}(k[u, u^{-1}])$, we must specify a group-like element in $k[t]/(t^p)$. The truncated exponential series:

$$E(t) = \sum_{i=0}^{p-1} \frac{t^i}{i!}$$

   is well-defined in characteristic p (since we stop before dividing by $p!$) and satisfies $E(t+s) = E(t)E(s)$ in $k[t,s]/(t^p, s^p)$. This gives a Hopf algebra map $u \mapsto E(t)$, which defines an isomorphism $\alpha_p^D \cong \alpha_p$. Thus, unlike μ_p, the infinitesimal group scheme α_p is self-dual.

4.7 Rational Maps into Group Schemes

Definition 3.6.1 introduced a rational map as an equivalence class of morphisms defined on dense open subsets, and said nothing about where such a map is actually defined. When the target carries a group law the answer is as good as it could be: the map is defined everywhere it has any right to be. This is Weil's extension theorem, and it is the mechanism behind the construction of Néron models and behind the structure theory of algebraic groups that *Abelian Varieties* [6] takes up.

Two independent obstructions block extension in general. The target may fail to be proper: for a proper open subscheme $W \subsetneq V$, the identity of W is a rational map $V \dashrightarrow W$ with no extension, and no hypothesis on V repairs it. The source may fail to be normal: the parametrization of the cuspidal cubic is an isomorphism away from the cusp, and its inverse does not extend across it (example 4.7.5).

Removing both obstructions is still not enough for a general target, as the blow-up of the plane at a point shows (example 4.7.12). The group law is what closes the remaining gap.

Definition 4.7.1: Domain of Definition

Let X be an integral scheme, let Y be a separated scheme, and let $\varphi\colon X \dashrightarrow Y$ be a rational map. Then φ is *defined at* a point $x \in X$ if some representative (U, α) of φ has $x \in U$. The set of such points is the *domain of definition* $\operatorname{dom}(\varphi)$, and its complement $X \setminus \operatorname{dom}(\varphi)$ is the *locus of indeterminacy* of φ.

The definition is only useful once the representatives are known to fit together, so that $\operatorname{dom}(\varphi)$ carries a single morphism rather than a compatible family of them. Separatedness of the target is exactly what makes this work.

Proposition 4.7.2: The Domain of Definition Carries a Canonical Morphism

Let X be an integral scheme, Y a separated scheme, and $\varphi\colon X \dashrightarrow Y$ a rational map. Then $\operatorname{dom}(\varphi)$ is a dense open subset of X, and the representatives of φ glue to a single morphism $\operatorname{dom}(\varphi) \to Y$ representing φ.

***Proof*:**
The domain of definition is a union of the open sets underlying representatives, hence open, and it contains at least one nonempty such set, hence is dense because X is irreducible.

Let (U, α) and (V, β) be representatives. Both U and V are nonempty open subsets of an irreducible space, so $U \cap V$ is nonempty, and by definition 3.6.1 there is a dense open $Z \subseteq U \cap V$ with $\alpha|_Z = \beta|_Z$. The locus in $U \cap V$ where α and β agree is the preimage of the diagonal $\Delta_Y \subseteq Y \times Y$ under (α, β), and this is a closed subscheme of $U \cap V$ because Y is separated. Its underlying space contains Z and is closed, hence is all of $U \cap V$; and $U \cap V$ is reduced because X is integral, so a closed subscheme carried by the whole space is the whole scheme. Thus α and β agree on $U \cap V$, and the representatives glue, as we sought to show.

The gluing argument is used repeatedly below in the following form: two morphisms out of an integral scheme into a separated scheme that agree at the generic point are equal, since the locus of agreement is closed and contains a dense point. The next lemma is the other piece of routine machinery the section needs. It says that a morphism defined on the spectrum of a local ring, an object with no room in it, already comes from a genuine neighborhood.

Lemma 4.7.3: Spreading Out a Morphism from a Local Ring

Let k be a field, let X be an integral k-scheme, let $x \in X$, and let Y be a k-scheme locally of finite type. Every k-morphism $h\colon \operatorname{Spec} \mathcal{O}_{X,x} \to Y$ factors as

$$\operatorname{Spec} \mathcal{O}_{X,x} \longrightarrow V \xrightarrow{h_V} Y$$

for some open neighborhood V of x and some k-morphism h_V.

***Proof*:**
Step 1: the morphism lands in an affine chart. Let $\mathfrak{m}$ denote the closed point of $\operatorname{Spec}\mathcal{O}_{X,x}$ and put $y = h(\mathfrak{m})$. Choose an affine open $\operatorname{Spec} B \subseteq Y$ containing y, with B a finitely generated k-algebra. The preimage $h^{-1}(\operatorname{Spec} B)$ is open in $\operatorname{Spec}\mathcal{O}_{X,x}$ and contains $\mathfrak{m}$. An open subset of the spectrum of a local ring containing the closed point is the whole space: its complement is $V(\mathfrak{a})$ for some ideal $\mathfrak{a}$ with $\mathfrak{a} \not\subseteq \mathfrak{m}$, which forces $\mathfrak{a} = \mathcal{O}_{X,x}$. Hence h factors through $\operatorname{Spec} B$ and corresponds to a k-algebra homomorphism $\theta\colon B \to \mathcal{O}_{X,x}$.

Step 2: the homomorphism is defined on a neighborhood. Write $B = k[T_1, \ldots, T_n]/J$ and let $\theta_i \in \mathcal{O}_{X,x}$ be the image of T_i. Each θ_i is a germ at x, so there is an open neighborhood V of x on which all n of them are regular, say $\theta_i = s_i|_x$ with $s_i \in \mathcal{O}_X(V)$. Because X is integral, both $\mathcal{O}_X(V)$ and $\mathcal{O}_{X,x}$ are subrings of the function field $K(X)$ and the restriction $\mathcal{O}_X(V) \to \mathcal{O}_{X,x}$ is injective. For $f \in J$ the section $f(s_1, \ldots, s_n) \in \mathcal{O}_X(V)$ restricts to $f(\theta_1, \ldots, \theta_n) = 0$, so it is itself zero. Therefore $T_i \mapsto s_i$ descends to a k-algebra homomorphism $B \to \mathcal{O}_X(V)$.

Step 3: conclusion. The homomorphism $B \to \mathcal{O}_X(V)$ is a morphism $h_V : V \to \operatorname{Spec} B \subseteq Y$, and composing it with $\operatorname{Spec}\mathcal{O}_{X,x} \to V$ returns the homomorphism θ, hence returns h, completing the proof.

The first extension theorem needs no group law at all. It isolates what properness of the target gives on its own: indeterminacy cannot occur along a divisor, because a divisor is seen by a discrete valuation and a proper target has nowhere for a valuation to escape to.

Theorem 4.7.4: Rational Maps to a Proper Target

Let k be a field, let X be a normal integral k-scheme of finite type, and let Y be a proper k-scheme. Then the locus of indeterminacy of any rational map $\varphi\colon X \dashrightarrow Y$ has codimension at least 2 in X.

***Proof*:**
Write $N = X \setminus \operatorname{dom}(\varphi)$. No irreducible component of N has codimension 0, since $\operatorname{dom}(\varphi)$ is dense by proposition 4.7.2. Suppose some component W has codimension 1, and let ξ be its generic point.

Step 1: the local ring at ξ is a valuation ring. The local ring $\mathcal{O}_{X,\xi}$ is Noetherian of dimension $\operatorname{codim} W = 1$, and it is normal because X is. A one-dimensional Noetherian normal local domain is a discrete valuation ring [10], and its fraction field is the function field $K(X)$.

Step 2: the valuative criterion extends the map. The generic point of X lies in every nonempty open subset, so composing $\operatorname{Spec} K(X) \to \operatorname{dom}(\varphi)$ with the morphism of proposition 4.7.2 gives a k-morphism $\operatorname{Spec} K(X) \to Y$. Together with the structure morphism $\operatorname{Spec}\mathcal{O}_{X,\xi} \to \operatorname{Spec} k$, this is a lifting problem for $Y \to \operatorname{Spec} k$ against the valuation ring $\mathcal{O}_{X,\xi}$. The morphism $Y \to \operatorname{Spec} k$ is proper and of finite type and Y is Noetherian, so theorem 3.5.25 supplies a k-morphism $h\colon \operatorname{Spec}\mathcal{O}_{X,\xi} \to Y$ restricting to it on the generic point.

Step 3: descending to a neighborhood. By lemma 4.7.3 there are an open neighborhood V of ξ

and a morphism $h_V : V \to Y$ through which h factors. The set $V \cap \mathrm{dom}(\varphi)$ is a nonempty open subset of the irreducible scheme X, and on it h_V and the morphism of proposition 4.7.2 agree at the generic point by construction, hence agree, because Y is separated. So (V, h_V) represents φ and $\xi \in \mathrm{dom}(\varphi)$, contradicting $\xi \in N$. Hence no component of N has codimension 1, completing the proof.

Example 4.7.5: The Cuspidal Cubic

Let k be algebraically closed and let $C = \operatorname{Spec} k[u,v]/(v^2 - u^3)$ be the cuspidal cubic. The morphism $\nu\colon \mathbb{A}^1_k \to C$ determined by $u \mapsto t^2$ and $v \mapsto t^3$ identifies the coordinate ring of C with the subring $k[t^2, t^3] \subseteq k[t]$ of polynomials with no linear term, and restricts to an isomorphism $\mathbb{A}^1_k \setminus \{0\} \to C \setminus \{(0,0)\}$ with inverse $(u,v) \mapsto v/u$. Let $\varphi\colon C \dashrightarrow \mathbb{A}^1_k$ be the rational map $t = v/u$, so that $\varphi \circ \nu$ is the identity wherever it is defined.

The map φ is not defined at the cusp. Otherwise $t \in \mathcal{O}_{C,(0,0)}$, so $t = a/b$ with $a, b \in k[t^2, t^3]$ and $b(0) \neq 0$. Then $tb = a$ in $k[t]$, and the coefficient of t in tb is $b(0) \neq 0$, while a has no linear term. Thus the locus of indeterminacy is the single closed point $(0,0)$, of codimension 1 in the curve C.

Composing with $\mathbb{A}^1_k \subseteq \mathbb{P}^1_k$ replaces the target by a proper one without changing the conclusion: if the resulting map extended to a morphism $\psi\colon C \to \mathbb{P}^1_k$, then $\psi \circ \nu$ and the identity would agree on a dense open subset of the reduced scheme $\mathbb{A}^1_k$, hence everywhere, giving $\psi((0,0)) = 0$. Then ψ would carry a neighborhood of the cusp into $\mathbb{A}^1_k$ and so exhibit t as an element of $\mathcal{O}_{C,(0,0)}$, which was just excluded. The hypothesis of theorem 4.7.4 that fails is normality of C, and the cusp is precisely where normality fails.

Passing from codimension 2 to codimension infinity is where the group law enters, and it does so through a statement about rational functions. The following lemma is the divisor-theoretic input, proved here from unique factorization rather than from the theory of divisors developed in chapter 5, which is not yet available.

Lemma 4.7.6: Poles Are Detected by the Denominator

Let R be a unique factorization domain with fraction field K, and let $f \in K$ be written as $f = a/b$ with $a, b \in R$, $b \neq 0$, and a and b having no common irreducible factor. Then for every prime ideal $\mathfrak{p} \subseteq R$,

$$f \in R_{\mathfrak{p}} \qquad \text{if and only if} \qquad b \notin \mathfrak{p}$$

***Proof*:**

If $b \notin \mathfrak{p}$, then $f = a/b$ lies in $R_{\mathfrak{p}}$ by the definition of the localization. Conversely, suppose $b \in \mathfrak{p}$ and $f \in R_{\mathfrak{p}}$, say $f = c/s$ with $c \in R$ and $s \notin \mathfrak{p}$. Then $as = cb$ in R. The element b is a nonzero nonunit, since $b \in \mathfrak{p}$, so it has an irreducible factorization, and $\mathfrak{p}$ being prime forces some irreducible factor π of b to lie in $\mathfrak{p}$. Now π divides $cb = as$, and π does not divide a because a and b have no common irreducible factor, so π divides s. Then $s \in \mathfrak{p}$, contrary to the choice of s, completing the proof.

Lemma 4.7.7: The Polar Locus is Pure of Codimension One

Let X be a regular integral Noetherian scheme and let $f \in K(X)$ be a nonzero rational function. Then the *polar locus*

$$P(f) = \{x \in X \mid f \notin \mathcal{O}_{X,x}\}$$

is closed, and every irreducible component of $P(f)$ has codimension 1 in X.

***Proof*:**
Step 1: the polar locus is closed. If $f \in \mathcal{O}_{X,x}$, then f is represented by a section of $\mathcal{O}_X$ over some open neighborhood of x, and hence lies in $\mathcal{O}_{X,x'}$ for every x' in that neighborhood. The complement of $P(f)$ is therefore open.

Step 2: a local factorization. Let W be an irreducible component of $P(f)$ with generic point w, and put $R = \mathcal{O}_{X,w}$. This is a regular local ring, hence a unique factorization domain by proposition 2.7.20, and its fraction field is $K(X)$. Write $f = a/b$ with $a, b \in R$, $b \neq 0$, and a and b having no common irreducible factor. Since $w \in P(f)$ the element f does not lie in R, so b lies in the maximal ideal of R by lemma 4.7.6; in particular b is a nonzero nonunit.

Step 3: Krull's theorem produces a codimension-one pole. Being a nonzero nonunit of a domain, b is neither a unit nor a zero-divisor, so every minimal prime over (b) has height exactly 1 by theorem 2.6.13. Fix such a prime $\mathfrak{q}$ and let $w' \in X$ be the corresponding point, so that $\mathcal{O}_{X,w'} = R_{\mathfrak{q}}$ and $\operatorname{codim}\overline{\{w'\}} = \operatorname{height}(\mathfrak{q}) = 1$. Since $b \in \mathfrak{q}$, lemma 4.7.6 gives $f \notin \mathcal{O}_{X,w'}$, so $w' \in P(f)$ and therefore $\overline{\{w'\}} \subseteq P(f)$.

Step 4: conclusion. The primes of R correspond to the generizations of w in X, so $w \in \overline{\{w'\}}$. Thus $\overline{\{w'\}}$ is an irreducible closed subset of $P(f)$ containing $W = \overline{\{w\}}$, and maximality of the component W forces $W = \overline{\{w'\}}$. Hence $\operatorname{codim} W = 1$, as we sought to show.

Everything is now in place for the central lemma. The idea is to trade the rational map φ for the rational map $(x, y) \mapsto \varphi(x)\varphi(y)^{-1}$ on $X \times_k X$, which takes the constant value e along the diagonal. A pole of a function pulled back from a neighborhood of e therefore misses the diagonal generically, and lemma 4.7.7 converts that into a codimension-one statement back on X.

Lemma 4.7.8: Indeterminacy of a Rational Map to a Group Scheme

Let k be an algebraically closed field, let X be a smooth integral k-scheme of finite type, and let G be a group scheme over k that is separated and of finite type. Then the locus of indeterminacy of a rational map $\varphi \colon X \dashrightarrow G$ is either empty or of pure codimension 1 in X.

***Proof*:**
Write $U = \operatorname{dom}(\varphi)$ and $N = X \setminus U$, and assume $N \neq \emptyset$. Since k is algebraically closed, X is geometrically integral, so $X \times_k X$ is integral; it is smooth over k and therefore regular, and it is Noetherian.

Step 1: the comparison map. Let $\Phi\colon X \times_k X \dashrightarrow G$ be the rational map represented on $U \times_k U$ by the composite

$$U \times_k U \xrightarrow{\varphi\times\varphi} G \times_k G \xrightarrow{\mathrm{id}\times\iota} G \times_k G \xrightarrow{\mu} G$$

informally $(x, y) \mapsto \varphi(x)\varphi(y)^{-1}$. If φ is defined at a point x, then $(x, x) \in U \times_k U$ and Φ is defined at (x, x).

Step 2: Φ is trivial along the diagonal. Let $\Delta\colon X \to X \times_k X$ be the diagonal, a closed immersion since X is separated. The composite $\Phi \circ \Delta|_U$ is $\mu \circ (\mathrm{id}, \iota) \circ \varphi|_U$, which is the constant morphism e by the inverse axiom of definition 4.1.1. In particular $\Delta(U) \subseteq \mathrm{dom}(\Phi)$ and Φ takes the value e there.

Step 3: indeterminacy of φ at a closed point forces indeterminacy of Φ. Let $x \in N$ be a closed point and suppose Φ were defined at (x, x). Because k is algebraically closed and x is closed, $\kappa(x) = k$ and the fibre of the first projection over x is a copy of X. Its intersection with $\mathrm{dom}(\Phi)$ is open and nonempty, as it contains (x, x), so its intersection with the dense open U is a nonempty open subset of X and hence contains a closed point b, which is k-rational. Then $(x, b) \in \mathrm{dom}(\Phi)$ with $b \in U(k)$. Let $\sigma_b\colon X \to X \times_k X$ be the section $y \mapsto (y, b)$ of the first projection. On U the composite $\Phi \circ \sigma_b$ is $y \mapsto \varphi(y)\varphi(b)^{-1}$, so

$$\varphi = \mu \circ \big(\Phi \circ \sigma_b,\ \varphi(b)\big)$$

as rational maps, the second entry being a constant k-point of G. The right-hand side is a morphism on $\sigma_b^{-1}(\mathrm{dom}(\Phi))$, which is an open set containing x, so φ is defined at x, contradicting $x \in N$. Hence Φ is not defined at (x, x).

Step 4: a pole near the identity. Keep $x \in N$ closed and put $p = (x, x)$. Pullback along Φ makes sense on functions regular near e: by Step 2 the map Φ carries the nonempty open set $\Delta(U)$ to e, so for $f \in \mathcal{O}_{G,e}$ the composite $f \circ \Phi$ is regular on a nonempty, hence dense, open subset of the integral scheme $X \times_k X$, and therefore defines an element $\Phi^* f \in K(X \times_k X)$.

Suppose every $f \in \mathcal{O}_{G,e}$ satisfied $\Phi^* f \in \mathcal{O}_{X\times_k X,p}$. Choosing an affine open $\operatorname{Spec} B \subseteq G$ containing e, the composite $B \to \mathcal{O}_{G,e} \to \mathcal{O}_{X\times_k X,p}$ would be a morphism $\operatorname{Spec} \mathcal{O}_{X\times_k X,p} \to \operatorname{Spec} B \subseteq G$, which lemma 4.7.3 descends to a morphism on a neighborhood of p. That morphism agrees with Φ at the generic point by construction, hence agrees with it wherever both are defined because G is separated, so it represents Φ near p and makes Φ defined at p, contradicting Step 3. Hence some $f \in \mathcal{O}_{G,e}$ has $g = \Phi^* f \notin \mathcal{O}_{X\times_k X,p}$; in particular $g \neq 0$.

Step 5: restricting the pole to the diagonal. Put $R = \mathcal{O}_{X\times_k X,p}$, a regular local ring, and write $g = a/b$ with $a, b \in R$ having no common irreducible factor, as in lemma 4.7.7; by Step 4 the element b is a nonzero nonunit. Let $\delta = \Delta(\eta)$, where η is the generic point of X. Then δ is the generic point of $\Delta(X)$ and a generization of p, so it corresponds to a prime $\mathfrak{p}_\delta \subseteq R$, and $\mathfrak{p}_\delta$ is the ideal cutting out the integral closed subscheme $\Delta(X)$ near p; thus $R/\mathfrak{p}_\delta = \mathcal{O}_{X,x}$. Since $\eta \in U$, Step 2 puts δ in $\mathrm{dom}(\Phi)$ with $\Phi(\delta) = e$, so g is regular at δ, that is $g \in R_{\mathfrak{p}_\delta}$. By lemma 4.7.6, $b \notin \mathfrak{p}_\delta$. Writing $S = R/\mathfrak{p}_\delta = \mathcal{O}_{X,x}$, the image $\overline{b} \in S$ is therefore nonzero, and it is a nonunit because b lies in the maximal ideal of R. The ring S is a regular local Noetherian domain, so theorem 2.6.13 gives a minimal prime over $(\overline{b})$ of height 1, say corresponding to a point $x' \in X$ with $\operatorname{codim}\overline{\{x'\}} = 1$.

Step 6: conclusion. The minimal prime produced in Step 5 is the image of a prime $\tilde{\mathfrak{q}} \subseteq R$ that contains both $\mathfrak{p}_\delta$ and b, and $\tilde{\mathfrak{q}}$ is the prime of R at the point $\Delta(x') = (x', x')$. Since $b \in \tilde{\mathfrak{q}}$, lemma 4.7.6 gives $g \notin \mathcal{O}_{X \times_k X,(x',x')}$, and hence Φ is not defined at (x', x'). By the contrapositive of the last sentence of Step 1, φ is not defined at x', so $\overline{\{x'\}} \subseteq N$ is an irreducible closed subset of codimension 1 containing x.

Now let W be an irreducible component of N. The complement in W of the remaining components is a nonempty open subset of a scheme of finite type over k, so it contains a point x that is closed in X and lies on no other component. The previous paragraph produces an irreducible closed $\overline{\{x'\}} \subseteq N$ of codimension 1 with $x \in \overline{\{x'\}}$; being irreducible and contained in N, it lies in a single component, which must be W. Hence $\operatorname{codim} W \leq \operatorname{codim}\overline{\{x'\}} = 1$, and $\operatorname{codim} W \geq 1$ because U is dense, completing the proof.

Theorem 4.7.9: Weil's Extension Theorem

Let k be an algebraically closed field, let X be a smooth integral k-scheme of finite type, and let G be a group scheme over k that is separated and of finite type. If a rational map $\varphi\colon X \dashrightarrow G$ is defined at every point of X of codimension 1, then φ is a morphism $X \to G$.

***Proof*:**
By lemma 4.7.8 the locus of indeterminacy is empty or of pure codimension 1. In the second case it has an irreducible component of codimension 1, whose generic point is a point of X of codimension 1 at which φ is not defined, contrary to hypothesis. So the locus is empty and proposition 4.7.2 delivers the morphism, as we sought to show.

Corollary 4.7.10: Rational Maps to a Proper Group Scheme

Let k be an algebraically closed field, let X be a smooth integral k-scheme of finite type, and let G be a group scheme over k that is proper over k. Then every rational map $\varphi\colon X \dashrightarrow G$ is a morphism.

***Proof*:**
A regular local ring is a unique factorization domain by proposition 2.7.20, and a unique factorization domain is integrally closed [10], so the smooth scheme X is normal. A proper k-scheme is separated and of finite type, so theorem 4.7.4 applies and the locus of indeterminacy has codimension at least 2. In particular φ is defined at every point of codimension 1, and theorem 4.7.9 makes it a morphism, completing the proof.

Corollary 4.7.10 is the form in which the theorem is consumed downstream. A proper group scheme over a field that is in addition smooth and geometrically connected is an abelian variety, and the corollary then says that every rational map to one from a smooth source is a morphism. That is what makes the Néron mapping property reasonable to ask for, and it is why a rational map from a smooth variety to an abelian variety may be manipulated as though it were a morphism from the start.

4.7.11 Note: The Hypothesis on the Ground Field Algebraic closure is used at exactly one place in the proof of lemma 4.7.8: Step 3 produces a k-rational point b in a nonempty open set and uses the section $y \mapsto (y, b)$ of the first projection to convert a statement about Φ into one about φ. Over an arbitrary field such a point need not exist, and the section is replaced by faithfully flat descent along the projection. The theorem holds in that generality, and indeed over an arbitrary base scheme for a smooth source and a smooth separated group scheme; see Bosch, Lütkebohmert and Raynaud [3], 4.4/1.

Example 4.7.12: The Blow-Up of the Plane at a Point

Let k be algebraically closed, let $P = (1:0:0) \in \mathbb{P}^2_k$, and let

$$Y = \big\{\big((x_0 : x_1 : x_2),\, (y_1 : y_2)\big) \in \mathbb{P}^2_k \times \mathbb{P}^1_k \;\big|\; x_1 y_2 = x_2 y_1\big\}$$

be the blow-up of $\mathbb{P}^2_k$ at P, with projection $\pi\colon Y \to \mathbb{P}^2_k$. Then π restricts to an isomorphism over $\mathbb{P}^2_k \setminus \{P\}$, while $\pi^{-1}(P)$ is a copy of $\mathbb{P}^1_k$. Both Y and $\mathbb{P}^2_k$ are smooth, integral and proper.

The inverse of that isomorphism is a rational map $\rho\colon \mathbb{P}^2_k \dashrightarrow Y$, and it is not a morphism. If it were, then $\pi \circ \rho$ and the identity would agree on a dense open subset of the reduced scheme $\mathbb{P}^2_k$, hence everywhere, so ρ would be a section of π and therefore a closed immersion. Its image would be a closed subscheme of the irreducible surface Y isomorphic to $\mathbb{P}^2_k$, hence of dimension 2, hence all of Y; so ρ would be an isomorphism and π its inverse, contradicting the fact that π contracts a curve to a point.

The locus of indeterminacy of ρ is thus the single point P, of codimension 2. This shows that the bound in theorem 4.7.4 is attained and cannot be improved, and that the group hypothesis in lemma 4.7.8 is not decorative: source and target here are smooth and proper, and the indeterminacy locus is nonempty of codimension 2, so it is not of pure codimension 1. What Y lacks is a group law.

Theorem 4.7.9 is a statement about the domain of definition of a rational map into a group scheme. The group law on the *target* converts a criterion in codimension 1 into an extension, and nothing is asked about a group law on the source, which need not have one. A second mechanism becomes available when the source is itself a group and the rational map respects that group law generically. Multiplicativity on a dense open set forces the extension on its own, with no criterion in codimension 1, no smoothness of the source, and no hypothesis on the target beyond separatedness. The two statements are not comparable and neither implies the other, so a reader who runs them together will look for the wrong hypothesis: Weil's theorem sees only the group law on the target, the theorem below sees the group laws on both.

What replaces the divisor theory is homogeneity. A group is swept out by the translates of any nonempty open subset, provided the translating points are drawn from a dense enough supply, and that is what turns a formula valid near one point into a morphism defined everywhere.

Lemma 4.7.13: Translates of an Open Set Along a Dense Set of Points Cover the Group

Let k be an algebraically closed field and let G be a group scheme over k whose underlying scheme is integral and of finite type over k. For $a \in G(k)$ write $\lambda_a \colon G \to G$ for left translation by a (example 4.3.2), and write $aB = \lambda_a(B)$ for a subset $B \subseteq G$. Call a subset $S \subseteq G(k)$ dense when the set of points of G it determines is dense. Let $B \subseteq G$ be a nonempty open subset.

1. If $S \subseteq G(k)$ is dense, then every $g \in G(k)$ factors as $g = ab$ with $a \in S$ and $b \in B(k)$, and the open sets aB for $a \in S$ cover G.

2. If $A \subseteq G$ is a nonempty open subset, then $A(k)$ is dense. Consequently $A(k)B(k) = G(k)$, and the open sets aB for $a \in A(k)$ cover G.

***Proof*:**
Two facts are used throughout. Since G is irreducible, a nonempty open subset of G is dense and any two nonempty open subsets meet. And a nonempty open subset U of G has $U(k) \neq \emptyset$: being a nonempty open subset of a k-scheme of finite type it contains a closed point, and such a point is k-rational because k is algebraically closed. A k-point of G whose image lies in an open subscheme U is a k-point of U, and this identification is used without further comment.

Left translation is recalled next. For $a \in G(k)$ the morphism λ_a sends $x \mapsto ax$ on T-valued points, for every k-scheme T, with a read in $G(T)$ through the group homomorphism $G(k) \to G(T)$. So $\lambda_a \circ \lambda_{a^{-1}}$ and $\lambda_{a^{-1}} \circ \lambda_a$ induce the identity on T-valued points for every T and are therefore both id_G, and the same computation gives $\iota \circ \iota = \mathrm{id}_G$. Thus λ_a, with inverse $\lambda_{a^{-1}}$, and the inversion ι are automorphisms of the scheme G; each carries an open subscheme isomorphically onto its image and so carries k-points bijectively onto k-points.

1. Fix $g \in G(k)$ and put $C = (\lambda_g \circ \iota)(B)$, a nonempty open subset of G whose k-points are exactly the gb^{-1} for $b \in B(k)$. The set S is dense and C is nonempty open, so some $a \in S$ has its image in C; as a k-point of C it is $a = gb^{-1}$ for a unique $b \in B(k)$, whence $b = a^{-1}g$ and $g = ab$.

   Put $V = \bigcup_{a \in S} aB$, an open subset of G. The factorization just produced gives $g = \lambda_a(b) \in aB$, so V contains the image of every k-point of G. Were the closed complement $Z = G \setminus V$ nonempty, then Z with its reduced induced structure would be a nonempty k-scheme of finite type and would contain a closed point, again k-rational; that point would be a k-point of G outside V. Hence $Z = \emptyset$ and $V = G$.

2. Let $D \subseteq G$ be a nonempty open subset. Then $A \cap D$ is a nonempty open subset of G, so $(A \cap D)(k) \neq \emptyset$, and $A(k)$ meets D. Thus $A(k)$ is dense, and part (1) with $S = A(k)$ covers G by the sets aB and gives $G(k) \subseteq A(k)B(k)$; the reverse inclusion is immediate, completing the proof.

The lemma is why no hypothesis in codimension 1 is needed below. Weil's theorem has to work inward from the divisors of X because a general X offers no way of moving one point to another;

a group offers exactly that, and the price is that the rational map has to be compatible with the motion.

Stating that compatibility takes one piece of bookkeeping. A rational map φ out of a group G has a domain of definition Ω, and the equation $\varphi(xy) = \varphi(x)\varphi(y)$ is not even well formed at a pair (x, y) unless all three of x, y and xy lie in Ω. The pairs for which they do form an open subset of $G \times_k G$, and the hypothesis is that the two sides agree on a dense open subset of it.

Theorem 4.7.14: A Rational Homomorphism Is a Homomorphism

Let k be an algebraically closed field, let G be a group scheme over k whose underlying scheme is integral and of finite type over k, and let H be a separated group scheme over k. Let $\varphi\colon G \dashrightarrow H$ be a rational map, write $\Omega = \mathrm{dom}(\varphi)$, and write φ also for the morphism $\Omega \to H$ representing it (definition 4.7.1 and proposition 4.7.2). Let $p_1, p_2\colon G \times_k G \to G$ be the two projections and let

$$\Omega_2 = p_1^{-1}(\Omega) \cap p_2^{-1}(\Omega) \cap \mu_G^{-1}(\Omega) \subseteq G \times_k G$$

be the open subset over which x, y and xy all lie in Ω. On Ω_2 put

$$\alpha = \varphi \circ \mu_G|_{\Omega_2}, \qquad \beta = \mu_H \circ \big(\varphi \circ p_1|_{\Omega_2},\ \varphi \circ p_2|_{\Omega_2}\big)$$

informally $\alpha(x, y) = \varphi(xy)$ and $\beta(x, y) = \varphi(x)\varphi(y)$. Suppose there is a dense open subset W of $G \times_k G$ with $W \subseteq \Omega_2$ and $\alpha|_W = \beta|_W$. Then φ is represented by a unique morphism $\widetilde{\varphi}\colon G \to H$, and $\widetilde{\varphi}$ is a homomorphism of group schemes.

***Proof*:**
Since k is algebraically closed, G is geometrically integral (definition 3.2.8), so $G \times_k G$ is integral; it is of finite type over k because G is. Every closed point of G or of $G \times_k G$ is therefore k-rational, and a k-point of $G \times_k G$ is a pair of k-points of G. Since H is separated, proposition 4.7.2 applies and Ω is a dense open subset of G carrying the morphism φ.

One principle is used four times below, and it is worth naming once: if Y is an integral scheme and $f, g\colon Y \to H$ are morphisms agreeing on a dense open subscheme of Y, then $f = g$. This is proposition 3.5.10, applicable because an integral scheme is reduced and H is separated by hypothesis. Call it the *agreement principle* below.

Step 1: a dense supply of translating points. For $a \in G(k)$ let $\sigma_a = (a \circ \pi_G,\ \mathrm{id}_G)\colon G \to G \times_k G$ be the section $y \mapsto (a, y)$ of p_2, where $\pi_G\colon G \to \operatorname{Spec} k$ is the structure morphism, and put $W_a = \sigma_a^{-1}(W)$, an open subset of G. Let

$$P = \{a \in G(k) \mid a \in \Omega \text{ and } W_a \neq \emptyset\}$$

Then P is dense. Let $D \subseteq G$ be a nonempty open subset. Since G is irreducible, $D \cap \Omega$ is nonempty, so $p_1^{-1}(D \cap \Omega)$ is a nonempty open subset of $G \times_k G$; it meets the nonempty open set W because $G \times_k G$ is irreducible, and their intersection, a nonempty open subset of a k-scheme of finite type, contains a closed point, which is k-rational. Write that point as (a, b) with $a, b \in G(k)$. Then $a \in (D \cap \Omega)(k)$, and $\sigma_a \circ b = (a, b)$ has its image in W, so $b \in W_a$. Hence $a \in P \cap D$.

Step 2: the translated maps agree with φ. Fix $a \in P$. The k-point a factors through Ω, so composing it with φ gives a k-point $\varphi(a) \in H(k)$ and hence an automorphism $\lambda_{\varphi(a)}$ of H; λ_a is an automorphism of G. Put $a\Omega = \lambda_a(\Omega)$, a dense open subset of G, and

$$\rho_a = \lambda_{\varphi(a)} \circ \varphi \circ \lambda_{a^{-1}}|_{a\Omega} \colon a\Omega \longrightarrow H$$

informally $\rho_a(x) = \varphi(a)\,\varphi(a^{-1}x)$; this is a morphism because $\lambda_{a^{-1}}$ carries $a\Omega$ isomorphically onto Ω.

Three inclusions are needed before the two maps can be compared. By construction $\sigma_a(W_a) \subseteq W \subseteq \Omega_2$. Since $p_2 \circ \sigma_a = \mathrm{id}_G$ and $\Omega_2 \subseteq p_2^{-1}(\Omega)$, this gives $W_a = p_2(\sigma_a(W_a)) \subseteq \Omega$. Since $\mu_G \circ \sigma_a = \lambda_a$, which on T-valued points is the identity $\mu_G(a, y) = ay$, and $\mu_G(\Omega_2) \subseteq \Omega$, it also gives $aW_a = \mu_G(\sigma_a(W_a)) \subseteq \Omega$. So both W_a and aW_a lie in Ω, and consequently $aW_a \subseteq a\Omega \cap \Omega$.

Now restrict α and β along $\sigma_a|_{W_a} \colon W_a \to W$. Using $p_1 \circ \sigma_a = a \circ \pi_G$ and $p_2 \circ \sigma_a = \mathrm{id}_G$, together with the inclusions just established,

$$\alpha \circ \sigma_a|_{W_a} = \varphi \circ \mu_G \circ \sigma_a|_{W_a} = \varphi \circ \lambda_a|_{W_a}, \qquad \beta \circ \sigma_a|_{W_a} = \mu_H \circ \big(\varphi(a) \circ \pi_G|_{W_a},\ \varphi|_{W_a}\big) = \lambda_{\varphi(a)} \circ \varphi|_{W_a}$$

the second equality on the right holding because μ_H with a constant first entry is left translation by that constant. The hypothesis $\alpha|_W = \beta|_W$ therefore reads

$$\varphi \circ \lambda_a|_{W_a} = \lambda_{\varphi(a)} \circ \varphi|_{W_a}$$

and precomposing with the isomorphism $\lambda_{a^{-1}} \colon aW_a \to W_a$ turns the left side into $\varphi|_{aW_a}$ and the right side into $\rho_a|_{aW_a}$. So φ and ρ_a agree on aW_a, which is nonempty because $a \in P$ and λ_a is a homeomorphism.

Both φ and ρ_a are morphisms on $a\Omega \cap \Omega$, a nonempty open subscheme of the integral scheme G and hence integral, and aW_a is a nonempty open subset of it, hence dense in it. The agreement principle gives

$$\varphi = \rho_a \quad \text{on } a\Omega \cap \Omega$$

Step 3: the translated maps agree with one another. Let $a, b \in P$. The sets $a\Omega$, $b\Omega$ and Ω are dense open subsets of the irreducible G, so $a\Omega \cap b\Omega \cap \Omega$ is a nonempty open subset of the integral scheme $a\Omega \cap b\Omega$ and hence dense in it. On it Step 2 gives $\rho_a = \varphi = \rho_b$, and the agreement principle upgrades this to $\rho_a = \rho_b$ on $a\Omega \cap b\Omega$. This compatibility is what licenses the next step.

Step 4: gluing. By Step 1 the set P is a dense set of k-points of G, so lemma 4.7.13(1), applied with $S = P$ and $B = \Omega$, says the open sets $a\Omega$ for $a \in P$ cover G. By Step 3 the ρ_a agree on overlaps, so they glue to a morphism $\widetilde{\varphi} \colon G \to H$. Step 2 gives $\widetilde{\varphi} = \varphi$ on $a\Omega \cap \Omega$ for every $a \in P$, and those sets cover Ω, so $\widetilde{\varphi}|_\Omega = \varphi$ and $\widetilde{\varphi}$ represents the rational map.

Step 5: uniqueness. If $\psi \colon G \to H$ is a morphism representing φ, then ψ and $\widetilde{\varphi}$ agree on a dense open subset of G by definition 3.6.1, so $\psi = \widetilde{\varphi}$ by the agreement principle.

Step 6: multiplicativity. Consider the morphisms $\widetilde{\varphi} \circ \mu_G$ and $\mu_H \circ (\widetilde{\varphi} \times \widetilde{\varphi})$ from $G \times_k G$ to H.

On W one has $\mu_G(W) \subseteq \Omega$ and $p_i(W) \subseteq \Omega$ for $i = 1, 2$, and $\widetilde{\varphi}|_\Omega = \varphi$, so the two restrict on W to $\alpha|_W$ and $\beta|_W$, which agree by hypothesis. As W is a dense open subset of the integral scheme $G \times_k G$, the agreement principle gives

$$\widetilde{\varphi} \circ \mu_G = \mu_H \circ (\widetilde{\varphi} \times \widetilde{\varphi})$$

Step 7: the remaining axioms. For a k-scheme T the identification $(G \times_k G)(T) = G(T) \times G(T)$ turns μ_G and μ_H into the group laws on $G(T)$ and $H(T)$, so Step 6 says exactly that $\widetilde{\varphi}(T)\colon G(T) \to H(T)$ is multiplicative, that is, a homomorphism of abstract groups. A homomorphism of groups carries the identity to the identity and inverses to inverses, so $\widetilde{\varphi} \circ \epsilon_G$ and ϵ_H induce the same map on T-valued points for every T, as do $\widetilde{\varphi} \circ \iota_G$ and $\iota_H \circ \widetilde{\varphi}$. By the Yoneda lemma the corresponding identities of morphisms hold, so $\widetilde{\varphi}$ satisfies the three conditions of definition 4.2.1, completing the proof.

Since $\widetilde{\varphi}$ is defined on all of G and represents φ, the conclusion contains the assertion that $\operatorname{dom}(\varphi) = G$. The confinement of W to Ω_2 is not a technicality that could be dropped: outside Ω_2 one of $\varphi(x)$, $\varphi(y)$, $\varphi(xy)$ is undefined and there is no equation to impose. Inside Ω_2, on the other hand, the choice of W carries no information. Both α and β are morphisms on all of Ω_2, which is a nonempty open subscheme of the integral $G \times_k G$ and hence integral, so agreement on a dense open subset propagates to Ω_2 because H is separated. The theorem could have been stated with $W = \Omega_2$; it is stated with a general W because that is the shape in which the hypothesis is usually met, as an identity of rational functions valid away from some divisor (example 4.7.15).

Both hypotheses on the two group schemes are weaker than they look. Integrality of G holds for every connected group variety, since a smooth connected group scheme of finite type over a field is geometrically integral (corollary 4.1.3); smoothness is used for nothing else, and the density arguments see only integrality. Separatedness of H is automatic as soon as H is of finite type over k (proposition 4.1.2), and is stated in place of finite type because nothing in the proof bounds H.

Example 4.7.15: Additive Rational Functions in One Variable

Let k be algebraically closed and let $f \in k(x)$ be a rational function satisfying

$$f(x + y) = f(x) + f(y)$$

as an identity in $k(x, y)$. Write $f = g/h$ with $g, h \in k[x]$ having no common irreducible factor. The additive group $\mathbb{G}_a = \mathbb{A}^1_k$ (example 4.1.4) is a group scheme over k whose underlying scheme is integral and of finite type, and $k[x]$ is a unique factorization domain, so lemma 4.7.6 identifies the domain of definition of the rational map $\varphi\colon \mathbb{G}_a \dashrightarrow \mathbb{G}_a$ determined by f as $\Omega = \mathbb{A}^1_k \setminus V(h)$.

Multiplication on $\mathbb{G}_a$ is $(x, y) \mapsto x + y$, so the set Ω_2 of theorem 4.7.14 is the complement in $\mathbb{A}^2_k$ of the three closed sets cut out by $h(x)$, $h(y)$ and $h(x + y)$, each a proper closed subset; hence Ω_2 is a dense open subset of $\mathbb{A}^2_k$. On it both $(x, y) \mapsto f(x + y)$ and $(x, y) \mapsto f(x) + f(y)$ are regular, and a morphism to $\mathbb{A}^1_k$ is the same thing as a regular function; since $\mathbb{A}^2_k$ is integral, the regular functions on Ω_2 inject into $k(x, y)$, so the displayed identity says precisely that the two morphisms agree on Ω_2. Taking $W = \Omega_2$, theorem 4.7.14 makes φ a morphism $\mathbb{A}^1_k \to \mathbb{A}^1_k$, that is, a global section of $\mathcal{O}_{\mathbb{A}^1_k}$. Thus $f \in k[x]$: an additive rational function has no poles at all.

Which polynomials survive is a coefficient comparison. Write $f = \sum_{n \geq 0} c_n x^n$. Substituting

$x = y = 0$ gives $c_0 = 0$, and for $n \geq 2$ the coefficient of xy^{n-1} in $f(x+y) - f(x) - f(y)$ is nc_n, since $f(x)$ and $f(y)$ contribute no mixed monomials. In characteristic 0 this forces $c_n = 0$ for all $n \geq 2$, so the additive rational functions are exactly the $c_1 x$ with $c_1 \in k$. In characteristic p the relation $nc_n = 0$ is vacuous whenever $p \big| n$, and there are more of them: $(x+y)^p = x^p + y^p$, so $x \mapsto x^p$ is additive, as is every k-linear combination of the x^{p^e}.

4.7.16 Remark: Why PGL_n Needs This Theorem Corollary 4.7.10 requires a proper target and so says nothing about $\mathrm{PGL}_{n,k}$, which is a closed subgroup scheme of $\mathrm{GL}_{n^2,k}$ (definition 4.1.7) and therefore affine and of finite type over k. For $n \geq 2$ it is not proper over k. An affine morphism is proper exactly when it is finite (proposition 3.5.23), and a finite k-scheme is $\operatorname{Spec} A$ for a finite-dimensional k-algebra A; distinct maximal ideals of A are comaximal, so A surjects onto the product of the corresponding residue fields and there are at most $\dim_k A$ of them. A k-point of $\operatorname{Spec} A$ is a k-algebra map $A \to k$ and is determined by its kernel, a maximal ideal, so a finite k-scheme has finitely many k-points. But $\mathrm{PGL}_n(k) = \mathrm{Aut}_{k\text{-alg}}(M_n(k))$ is infinite for $n \geq 2$: conjugation by $\operatorname{diag}(t, 1, \ldots, 1)$ fixes e_{11} and sends e_{12} to $t\,e_{12}$, so these automorphisms are pairwise distinct as t runs over $k^\times$, which is infinite because an algebraically closed field is (the case $n = 2$ is the computation in example 4.1.8).

Theorem 4.7.14 is what is left, and it suffices. Being a group scheme of finite type over a field, $\mathrm{PGL}_{n,k}$ is separated (proposition 4.1.2), so a rational map into $\mathrm{PGL}_{n,k}$ from a group scheme whose underlying scheme is integral and of finite type over k, multiplicative on a dense open set, is a morphism and a homomorphism of group schemes.

Exercise 4.7.1

1. Let k be a field and let $\varphi\colon \mathbb{P}^2_k \dashrightarrow \mathbb{P}^1_k$ be the rational map $(x_0 : x_1 : x_2) \mapsto (x_0 : x_1)$. Determine the locus of indeterminacy of φ and verify that it is consistent with theorem 4.7.4.
2. Let k be a field, let X be a normal integral k-scheme of finite type of dimension 1, and let Y be a proper k-scheme. Show that every rational map $X \dashrightarrow Y$ is a morphism.
3. Let X be a regular integral Noetherian scheme and let $f \in K(X)$ be nonzero with $P(f) = \emptyset$. Show that $f \in \mathcal{O}_X(X)$. (*Hint:* for an integral scheme, $\mathcal{O}_X(X)$ is the intersection of the $\mathcal{O}_{X,x}$ inside $K(X)$.)
4. The target $\mathbb{A}^1_k = \mathbb{G}_a$ of example 4.7.5 is a group scheme, and the locus of indeterminacy there is a single closed point. Explain why this is not a counterexample to lemma 4.7.8, and identify the hypothesis of that lemma which the cuspidal cubic fails.
5. Deduce from example 4.7.12 that the surface Y appearing there carries no group scheme structure over k, and identify the step in the proof of lemma 4.7.8 at which the group law is used.

5

Schemes III: Sheaves of Modules, Divisors, and Differentials

So far, we have worked exclusively with the structure sheaf $\mathcal{O}_X$ of a scheme X. But just as the geometry of a smooth manifold M is enriched by studying vector bundles, principal bundles[1], and their sections, the geometry of a scheme is enriched by studying additional sheaves that carry an $\mathcal{O}_X$-module structure. Every vector bundle on M gives rise to a sheaf of sections that is a module over the ring of smooth functions $C^\infty(M)$; in the same way, every "bundle-like" object on a scheme gives rise to a sheaf of $\mathcal{O}_X$-modules.

The most important such sheaves are those that are *locally* isomorphic to the sheaf associated to a module: on each affine open $U_i = \operatorname{Spec} R_i$, the sheaf restricts to $\widetilde{M_i}$ for some R_i-module M_i. These are the *quasi-coherent sheaves*, and when the modules M_i satisfy a finite presentation condition they are called *coherent sheaves*[2].

Quasi-coherent sheaves can be understood as a natural extension of vector bundles. *Locally free sheaves* (the algebraic incarnation of vector bundles) are not closed under kernels and cokernels — the category of vector bundles is not abelian. The category $\mathbf{QCoh}(X)$ is obtained by adding precisely those kernels and cokernels; it is the smallest abelian subcategory of $\mathcal{O}_X$-modules containing the locally free sheaves, just as the category of all R-modules is the abelian completion of the category of free R-modules. This perspective is developed in section 5.1.3.

By working with quasi-coherent sheaves, we gain access to a wealth of new geometric objects: ideal sheaves (encoding closed subschemes with scheme structure), twisted sheaves on projective space (from which one can recover the graded ring $S_\bullet$ of $\operatorname{Proj} S_\bullet$), skyscraper-like sheaves remembering

[1] more generally, gerbes and n-gerbes on manifolds, see ref:HERE

[2] The term "coherent" originates with Serre [62], who required sheaves to "cohere" (stick together) in a controlled way. The prefix "quasi-" signals that finite generation is dropped, at the cost of losing some finiteness results — for instance, quasi-coherent sheaves are not in general preserved under pushforward.

multiplicities (generalizing example 1.1.12), and sheaves of differentials (from which we recover differential forms, tangent bundles, and smoothness criteria). The study of these objects occupies the present chapter and sets the stage for sheaf cohomology in chapter 6.

There shall be sheaves on schemes that are *not* bundle-like (not quasi-coherent), such as the sheaf given by taking all units at each local section. Therefore, we start with actions of abelian sheave on ringed spaces.

5.1 Definition and Basic Properties

We begin by sheafifying the notion of a module over a ring. If $(X, \mathcal{O}_X)$ is a ringed space, an $\mathcal{O}_X$-module is a sheaf of abelian groups on which $\mathcal{O}_X$ acts section-wise, compatibly with restriction.

Definition 5.1.1: $\mathcal{O}_X$-Module

Let $(X, \mathcal{O}_X)$ be a ringed space. An *$\mathcal{O}_X$-module* is a sheaf $\mathscr{F} \in \mathbf{Sh}_{\mathbf{Ab}}(X)$ equipped with the following data:

1. For each open $U \subseteq X$, the abelian group $\mathscr{F}(U)$ carries the structure of an $\mathcal{O}_X(U)$-module.
2. For each inclusion $U \subseteq V$ of open sets, the restriction map $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ is compatible with the module structure:

$$\begin{tikzcd} \mathcal{O}_X(V) \times \mathscr{F}(V) \arrow[r, "\curvearrowright"] \arrow[d, "\mathrm{res}_{V,U} \times \mathrm{res}_{V,U}"'] & \mathscr{F}(V) \arrow[d, "\mathrm{res}_{V,U}"] \\ \mathcal{O}_X(U) \times \mathscr{F}(U) \arrow[r, "\curvearrowright"] & \mathscr{F}(U) \end{tikzcd}$$

A *morphism of $\mathcal{O}_X$-modules* $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves of abelian groups such that for each open $U \subseteq X$, the map $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ is an $\mathcal{O}_X(U)$-module homomorphism. The collection of all such morphisms is denoted $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$, or simply $\mathrm{Hom}_X(\mathscr{F}, \mathscr{G})$ when the structure sheaf is understood.

The reader should think of $\mathcal{O}_X$-modules as the scheme-theoretic analogue of (sections of) fibre bundles: the structure sheaf plays the role of smooth functions, and the module structure encodes how sections transform under multiplication by functions.

Given a morphism of ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$, there are natural operations that transport $\mathcal{O}_X$-modules to $\mathcal{O}_Y$-modules and vice versa:

Definition 5.1.2: $\mathcal{O}_X$-module Pushforward and Pullback

Let $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces.

1. *(Direct image.)* If $\mathscr{F}$ is an $\mathcal{O}_X$-module, then the pushforward sheaf $f_*\mathscr{F}$ is naturally an $f_*\mathcal{O}_X$-module on Y. Via the sheaf map $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$, this acquires the structure of an $\mathcal{O}_Y$-module: for $V \subseteq Y$ open and $g \in \mathcal{O}_Y(V)$, $s \in f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(V))$, the action is $g \cdot s := f^\#(V)(g) \cdot s$.

2. *(Inverse image.)* If $\mathscr{G}$ is a $\mathcal{O}_Y$-module, then the sheaf $f^{-1}\mathscr{G}$ is an $f^{-1}\mathcal{O}_Y$-module on X. The *pullback* (or *inverse image $\mathcal{O}_X$-module*) is
$$f^*\mathscr{G} \ := \ f^{-1}\mathscr{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$
The tensor product re-bases the module structure from $\mathcal{O}_Y$ to $\mathcal{O}_X$.

These operations form an adjoint pair: there is a natural isomorphism $\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathscr{G}, \mathscr{F}) \cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathscr{G}, f_*\mathscr{F})$.

Example 5.1.3: General $\mathcal{O}_X$-Module Constructions

Fix a ringed space $(X, \mathcal{O}_X)$.

1. *The structure sheaf.* The sheaf $\mathcal{O}_X$ is an $\mathcal{O}_X$-module with the action given by ring multiplication. Since each restriction map is a ring homomorphism, compatibility with the module structure is automatic.

2. *Abelian groups as modules.* Just as every abelian group is a $\mathbb{Z}$-module, every sheaf of abelian groups $\mathscr{G}$ on X is a module over the constant sheaf $\underline{\mathbb{Z}}$. The category of $\mathcal{O}_X$-modules therefore generalizes $\mathbf{Sh}_{\mathbf{Ab}}(X)$.

3. *Kernels, images, cokernels.* If $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of $\mathcal{O}_X$-modules, then $\ker\varphi$, $\mathrm{im}\varphi$, and $\mathrm{coker}\varphi$ are all $\mathcal{O}_X$-modules (by the same arguments as for sheaves of abelian groups, since the module structure is inherited section-wise). The category of $\mathcal{O}_X$-modules is abelian, with products and coproducts computed section-wise.

4. *Sheaf Hom.* For $\mathcal{O}_X$-modules $\mathscr{F}, \mathscr{G}$, the assignment
$$U \ \longmapsto \ \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathscr{F}|_U, \mathscr{G}|_U)$$
defines a sheaf of $\mathcal{O}_X$-modules, the *sheaf Hom*, denoted $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$.

5. *Tensor product.* The *tensor product* $\mathscr{F} \otimes_{\mathcal{O}_X} \mathscr{G}$ is the sheafification of the presheaf $U \mapsto \mathscr{F}(U) \otimes_{\mathcal{O}_X(U)} \mathscr{G}(U)$. Sheafification is essential here: the naive presheaf need not satisfy the gluing axiom, and the resulting sections can differ dramatically from the section-wise tensor product. See example 5.3.18 for a instance of this phenomenon.

The following result is a useful way

Proposition 5.1.4: Evaluation via Skyscraper Sheaves

Let X be a scheme and $x \in X$ a point. Let $\mathcal{O}_x$ denote the skyscraper sheaf at x with stalk $\kappa(x)$ (formally, the pushforward of $\mathcal{O}_{\mathrm{Spec}\,(\kappa(x))}$ along the inclusion $i : \mathrm{Spec}\,(\kappa(x)) \hookrightarrow X$). Then there is a natural isomorphism:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_x) \;\cong\; \kappa(x).$$

***Proof*:**
For any sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, morphisms from the structure sheaf $\mathcal{O}_X$ to $\mathcal{F}$ are naturally identified with the global sections of $\mathcal{F}$. Explicitly, a morphism $\varphi : \mathcal{O}_X \to \mathcal{F}$ is uniquely determined by the global section $\varphi(X)(1) \in \Gamma(X, \mathcal{F})$, giving an isomorphism:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F}) \;\cong\; \Gamma(X, \mathcal{F})$$

Applying this general fact to the skyscraper sheaf $\mathcal{F} = \mathcal{O}_x$, we obtain:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_x) \;\cong\; \Gamma(X, \mathcal{O}_x)$$

By the definition of the skyscraper sheaf, its global sections evaluate exactly to the stalk at the point x, which is the residue field $\kappa(x)$. Therefore:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_x) \;\cong\; \kappa(x)$$

This proposition demonstrates that taking the global section of a function $f \in \Gamma(X, \mathcal{O}_X) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_X)$ and composing it with the natural projection $\mathcal{O}_X \to \mathcal{O}_x$ precisely yields the evaluation $f(x) \in \kappa(x)$.

5.1.1 Quasi-Coherent and Coherent Sheaves

A natural hope is that, for an affine scheme $X = \mathrm{Spec}\, R$, the category of $\mathcal{O}_X$-modules is equivalent to R-Mod. This turns out to be false: the category of $\mathcal{O}_X$-modules is too large.

Example 5.1.5: Too Many $\mathcal{O}_X$-modules Over Affine Schemes

For any such equivalence to work, the sheaf $\widetilde{M}$ associated to an R-module M would need to satisfy $\widetilde{M}_{\mathfrak{p}} \cong M_{\mathfrak{p}}$ at each prime $\mathfrak{p}$. We show that not every $\mathcal{O}_X$-module has this form.

Let $X = \mathrm{Spec}\, \mathbb{Z}_{(2)}$, which has two points: the closed point (2) and the generic point $\eta = (0)$. The nontrivial open sets are $\{(2), \eta\} = X$ and $\{\eta\}$. The structure sheaf satisfies $\mathcal{O}_X(X) = \mathbb{Z}_{(2)}$ and $\mathcal{O}_X(\{\eta\}) = \mathbb{Q}$.

To specify an $\mathcal{O}_X$-module $\mathscr{F}$, we must give a $\mathbb{Z}_{(2)}$-module $\mathscr{F}(X)$, a $\mathbb{Q}$-module $\mathscr{F}(\{\eta\})$, and a restriction (module) homomorphism $\mathscr{F}(X) \to \mathscr{F}(\{\eta\})$. Consider the choice

$$\mathscr{F}(X) = \mathbb{Z}_{(2)}, \qquad \mathscr{F}(\{\eta\}) = 0, \qquad \mathbb{Z}_{(2)} \twoheadrightarrow 0.$$

This defines a valid $\mathcal{O}_X$-module. However, if $\mathscr{F} = \widetilde{M}$ for some $\mathbb{Z}_{(2)}$-module M, then $M = \Gamma(X, \widetilde{M}) = \mathbb{Z}_{(2)}$ and $\widetilde{M}(\{\eta\}) = M_{(0)} = M[1/2] = \mathbb{Q} \neq 0$. This contradicts $\mathscr{F}(\{\eta\}) = 0$, so $\mathscr{F}$ does not arise from any module.

The problem is that a general $\mathcal{O}_X$-module need not preserve the localization structure in compatible with the affine structure. The remedy is to restrict attention to $\mathcal{O}_X$-modules that are locally of the form $\widetilde{M}$. We first define the "affine building blocks."

Definition 5.1.6: Sheaf Associated to M

Let R be a ring and M an R-module. The *sheaf associated to* M on $\operatorname{Spec} R$ is the $\mathcal{O}_{\operatorname{Spec} R}$-module $\widetilde{M}$ defined on the basis of distinguished open sets by

$$\widetilde{M}(D(f)) \;=\; M_f \qquad \text{(the localization of } M \text{ at } f\text{)},$$

with the natural restriction maps induced by further localization. One checks (by the same argument as for the structure sheaf) that this defines a sheaf on $\operatorname{Spec} R$.

Since we now have multiple sheaves on a scheme, we adopt the standard notation $\Gamma(X, \mathscr{F})$ for the global sections of a sheaf $\mathscr{F}$ on X. In particular, $\Gamma(\operatorname{Spec} R, \widetilde{M}) \cong M$ and $\Gamma(X, \mathcal{O}_X) = \mathcal{O}_X(X)$.

Definition 5.1.7: Quasi-coherent Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. An $\mathcal{O}_X$-module $\mathscr{F}$ is *quasi-coherent* if X can be covered by affine open subsets $U_i = \operatorname{Spec} R_i$ such that for each i there exists an R_i-module M_i with

$$\mathscr{F}\big|_{U_i} \;\cong\; \widetilde{M}_i.$$

The full subcategory of quasi-coherent sheaves on X is denoted $\mathbf{QCoh}(X)$.

For the finitely generated analogue, a subtlety arises: the kernel of a map between finitely generated modules need not be finitely generated. For instance, the surjection $k[x_1, x_2, \dots] \to k[x_1, x_2, \dots]$ sending each x_i to 0 has kernel $(x_1, x_2, \dots)$, which is not finitely generated. (Images and cokernels of maps between finitely generated modules *are* finitely generated, so this is specifically a kernel problem.) The correct notion must therefore be stronger than finite generation:

Definition 5.1.8: Coherent Module

Let R be a ring and M an R-module. Then M is *coherent* if it is finitely generated and for every $n \geq 1$, every R-module homomorphism $R^n \to M$ has finitely generated kernel.
A ring R is called *coherent* if it is coherent as a module over itself.

Coherence is strictly stronger than finite presentation: the latter only requires *some* surjection $R^n \twoheadrightarrow M$ to have finitely generated kernel, whereas coherence demands this for *all* maps $R^n \to M$. Over a Noetherian ring, coherence, finite presentation, and finite generation are all equivalent.

Definition 5.1.9: Coherent Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. An $\mathcal{O}_X$-module $\mathscr{F}$ is *coherent* if X can be covered by affine open subsets $U_i = \operatorname{Spec} R_i$ such that for each i there exists a coherent R_i-module M_i with $\mathscr{F}|_{U_i} \cong \widetilde{M}_i$.

5.1.10 Distinction from Hartshorne Hartshorne defines "coherent sheaf" to mean "sheaf associated to a finitely generated module," but then immediately restricts to Noetherian schemes where the distinction vanishes. Our definition agrees with the one used by EGA and most modern references: coherence is an intrinsic condition on the sheaf (finitely generated with finitely generated kernels of all maps from free sheaves), which coincides with Hartshorne's definition precisely when X is locally Noetherian.

A key feature of quasi-coherence is that it is *affine-local*: checking it on one affine cover suffices.

Proposition 5.1.11: Quasi-coherence is Affine-local

Let $(X, \mathcal{O}_X)$ be a scheme and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is quasi-coherent if and only if for *every* affine open subset $U = \operatorname{Spec} R \subseteq X$, there is an isomorphism $\mathscr{F}|_U \cong \widetilde{M}$ for some R-module M.
If X is locally Noetherian, then $\mathscr{F}$ is coherent if and only if this holds with each M finitely generated (equivalently, coherent).

***Proof*:**
The "if" direction is immediate. For the "only if" direction, since the property is local and every affine scheme is covered by distinguished opens of any given affine cover, it suffices to prove the claim when $X = \operatorname{Spec} R$ is itself affine.

Let $M = \Gamma(\operatorname{Spec} R, \mathscr{F})$. We must show $\mathscr{F} \cong \widetilde{M}$. By hypothesis, there is an affine cover $\{D(g_i)\}$ of X with $\mathscr{F}|_{D(g_i)} \cong \widetilde{M_i}$ for some R_{g_i}-module M_i. Taking global sections over $D(g_i)$ gives $\mathscr{F}(D(g_i)) \cong M_i$. But $\mathscr{F}(D(g_i))$ is also obtained from M by localization at g_i: the restriction map $M = \mathscr{F}(X) \to \mathscr{F}(D(g_i))$ factors through M_{g_i}, and the quasi-coherence assumption on overlaps $D(g_i g_j)$ ensures this identification is consistent. Hence $M_i \cong M_{g_i}$, and by corollary 1.3.4 the natural map $\widetilde{M} \to \mathscr{F}$ is an isomorphism on a basis, hence an isomorphism of sheaves.

For the coherent case, if X is locally Noetherian then each R_{g_i} is Noetherian, so M_{g_i} is finitely generated if and only if it is coherent. The result follows by the same argument with the additional finite generation condition.

The affine-local property immediately yields the scheme-theoretic analogue of corollary 3.1.5:

Corollary 5.1.12: Equivalence of Categories: Module Version

If $X = \operatorname{Spec} R$, the functor $M \mapsto \widetilde{M}$ is an equivalence of categories:

$$\mathbf{QCoh}(\operatorname{Spec} R) \underset{\widetilde{(-)}}{\overset{\Gamma}{\rightleftarrows}} R\text{-Mod}$$

with inverse $\mathscr{F} \mapsto \Gamma(\operatorname{Spec} R, \mathscr{F})$. If R is Noetherian, this restricts to an equivalence between $\mathbf{Coh}(\operatorname{Spec} R)$ and finitely generated R-modules.
More generally, for any $\mathcal{O}_X$-module $\mathscr{F}$ on X and any R-module M, there is a natural adjunction isomorphism:

$$\operatorname{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathscr{F}) \;\cong\; \operatorname{Hom}_R(M, \Gamma(X, \mathscr{F})).$$

***Proof*:**
By proposition 5.1.11, every quasi-coherent sheaf on $\operatorname{Spec} R$ is of the form $\widetilde{M}$. Since $\Gamma(\operatorname{Spec} R, \widetilde{M}) \cong M$ and the tilde construction is functorial and exact, the two functors are inverse equivalences. The adjunction follows from the universal property of localization: a map $\widetilde{M} \to \mathscr{F}$ is determined by its value on global sections $M \to \Gamma(X, \mathscr{F})$, and conversely any such module map extends uniquely to a sheaf map by localization.

Example 5.1.13: Quasi-Coherent Sheaves

1. *The structure sheaf.* The structure sheaf $\mathcal{O}_X$ is quasi-coherent (even coherent): on each affine open $\operatorname{Spec} R$, it is $\widetilde{R}$. At each stalk, the $\mathcal{O}_{X,\mathfrak{p}}$-module $\mathcal{O}_{X,\mathfrak{p}} \cong R_\mathfrak{p}$ is free of rank one.

2. *Sheaves supported on a subvariety.* Let $X = \operatorname{Spec} k[x,y]$ and let $f \in k[x,y]$ be irreducible. The sheaf $\widetilde{k[x,y]/(f)}$ has stalks

$$\widetilde{k[x,y]/(f)}_\mathfrak{p} = \begin{cases} (k[x,y]/(f))_\mathfrak{p} = k[x,y]_\mathfrak{p}/(f)_\mathfrak{p} & \text{if } f \in \mathfrak{p}, \\ 0 & \text{if } f \notin \mathfrak{p}. \end{cases}$$

Geometrically, this is a "line bundle on $V(f)$, extended by zero to all of X." More generally, if $Y = V(\mathfrak{a}) \subseteq X$ is a closed subscheme defined by an ideal $\mathfrak{a} \subseteq R$, then the pushforward $\iota_* \mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-module isomorphic to $\widetilde{R/\mathfrak{a}}$.

3. *Torsion modules over* $\operatorname{Spec} \mathbb{Z}$. Let $X = \operatorname{Spec} \mathbb{Z}$ and $M = \mathbb{Z}/12\mathbb{Z}$. The stalks of $\widetilde{M}$ are:

$$\widetilde{M}_{(2)} \cong \mathbb{Z}/4\mathbb{Z}, \qquad \widetilde{M}_{(3)} \cong \mathbb{Z}/3\mathbb{Z}, \qquad \widetilde{M}_{(p)} = 0 \ \text{ for } p \neq 2,3, \qquad \widetilde{M}_{(0)} = 0.$$

The sheaf is supported on the two closed points (2) and (3), with "fibre dimensions" reflecting the prime factorization $12 = 2^2 \cdot 3$. More generally, if M is any torsion R-module, then $\widetilde{M}_{(0)} = 0$: torsion modules are not "spread" across the generic point.

4. *Eigenvalues and Jordan blocks.* Let $\alpha \in \operatorname{End}_\mathbb{C}(V)$ be an endomorphism of an n-dimensional $\mathbb{C}$-vector space, and view V as a $\mathbb{C}[x]$-module via $p(x) \cdot v = p(\alpha)(v)$. The associated sheaf $\widetilde{V}$ on $\mathbb{A}^1_\mathbb{C} = \operatorname{Spec} \mathbb{C}[x]$ is a quasi-coherent sheaf supported on the eigenvalues of α.

At a closed point λ, the stalk $\widetilde{V}_{(\lambda)} = V_{(x-\lambda)}$ is nonzero precisely when λ is an eigenvalue: if $\alpha - \lambda I$ is invertible, every element of V can be divided by $(x-\lambda)$, so the localization vanishes. If λ is an eigenvalue, the generalized eigenspace decomposition gives

$$\widetilde{V}_{(\lambda)} \cong \bigoplus_{j=1}^{r} \frac{\mathbb{C}[x]}{(x-\lambda)^{m_j}},$$

where $m_1, \dots, m_r$ are the sizes of the Jordan blocks for λ. The sheaf $\widetilde{V}$ thus encodes the full Jordan normal form of α: the support is the spectrum of eigenvalues, and the stalk structure records the Jordan block sizes. One can make this even cleaner by descending to $\operatorname{Spec}(\mathbb{C}[x]/m_\alpha(x))$, where m_α is the minimal polynomial.

5. *Extension by zero is not quasi-coherent.* Let $U \subseteq X$ be an open subscheme of an integral scheme with inclusion $\iota : U \hookrightarrow X$. The sheaf $\iota_! \mathcal{O}_U$ (extension by zero) is an $\mathcal{O}_X$-module but is typically *not* quasi-coherent. Indeed, if $V = \operatorname{Spec} R$ is an affine open not contained in U but with $U \cap V \neq \emptyset$, then $\iota_! \mathcal{O}_U|_V$ has no nonzero global sections (since elements must vanish outside U) but is not the zero sheaf on V, and hence cannot equal $\widetilde{M}$ for any R-module M.

6. *Restriction vs. pushforward.* If Y is a closed subscheme of X, the restriction $\mathcal{O}_X|_Y$ is typically *not* an $\mathcal{O}_Y$-module. For example, let $X = \operatorname{Spec} k[x,y]$ and $Y = \operatorname{Spec} k[x,y]/(y^2)$. In $\mathcal{O}_Y$, the function y is nilpotent ($y^2 = 0$), but in $\mathcal{O}_X|_Y$ there is no such nilpotence — restriction treats Y as a topological subspace, not as a scheme. The correct construction is the pushforward $\iota_* \mathcal{O}_Y$, which *is* a quasi-coherent $\mathcal{O}_X$-module (isomorphic to $\widetilde{R/\mathfrak{a}}$ where $\mathfrak{a}$ is the ideal of Y).

7. *The Hedgehog Theorem, algebraically.* Take $R = \mathbb{R}[x,y,z]/(x^2+y^2+z^2-1)$ and let M be the submodule of R^3 consisting of triples (X,Y,Z) with $xX + yY + zZ = 0$: the module of "tangent vectors to the sphere." On each affine open, $\widetilde{M}$ is locally free of rank 2 (isomorphic to $\mathcal{O}_X^2$), but it is not globally free — its complexification and analytification would contradict the Hairy Ball Theorem.

8. *Free and locally free sheaves.* The simplest quasi-coherent sheaves are the *free* sheaves $\mathcal{O}_X^{\oplus I}$ and the *locally free* sheaves, which restrict to free sheaves on an open cover. Locally free sheaves correspond to vector bundles (theorem 5.1.27). The category of locally free sheaves is not abelian: the multiplication-by-t map $\mathcal{O}_{\mathbb{A}^1_{\mathbb{C}}} \xrightarrow{\cdot t} \mathcal{O}_{\mathbb{A}^1_{\mathbb{C}}}$ has kernel the skyscraper sheaf at the origin, which is not locally free. Quasi-coherent sheaves arise as the "abelian completion" — obtained by adjoining all kernels and cokernels of maps between locally free sheaves.

9. *Quillen–Suslin.* On affine space $\mathbb{A}^n_k = \operatorname{Spec} k[x_1, \dots, x_n]$, every locally free sheaf is in fact free. This is the geometric content of the Quillen–Suslin theorem (formerly Serre's conjecture): every projective module over a polynomial ring is free. Nontrivial quasi-coherent sheaves on affine space therefore arise only as kernels and cokernels of maps between free sheaves.

10. *Projective and flat modules.* Let $X = \operatorname{Spec} R$ and M a projective R-module. Since projective modules over local rings are free, each stalk $\widetilde{M}_\mathfrak{p} \cong M_\mathfrak{p}$ is a free $R_\mathfrak{p}$-module: projective modules give the closest algebraic analogue of nontrivial vector bundles. If instead M is flat, the localization $M_\mathfrak{p}$ remains flat at each point. Geometrically, flatness is the condition that fibres vary "continuously" (cf. section 6.8.1).

11. *Number-theoretic line bundles.* Let $R = \mathbb{Z}[\sqrt{-5}]$, which is not a UFD: $2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$. The ideal $J = (2, 1+\sqrt{-5})$ is not principal, so the sheaf $\widetilde{J}$ is a line bundle on $\operatorname{Spec} R$

that is not globally free. On the distinguished opens $D(2)$ and $D(3)$, the ideal J becomes principal:

$$J \cdot R[1/2] = R[1/2] \qquad \text{(since 2 is a unit)},$$
$$J \cdot R[1/3] = (1+\sqrt{-5}) \cdot R[1/3] \qquad \text{(since } 2 = (1+\sqrt{-5})(1-\sqrt{-5})/3\text{)}.$$

Hence $\widetilde{J}$ is locally free of rank 1 but not globally free — the failure of unique factorization is detected by the nontriviality of this line bundle. More generally, the Picard group $\mathrm{Pic}\,(\mathrm{Spec}\,\mathcal{O}_K)$ of a number ring is the ideal class group of K (see section 5.4).

12. *Line bundles on* $\mathbb{P}^n_k$. There is a natural line bundle on $\mathbb{P}^n_k$ whose fibre over a point p is the one-dimensional subspace of $\mathbb{A}^{n+1}_k$ that p represents. We shall see in section 5.3 that every line bundle on $\mathbb{P}^n_k$ is isomorphic to a tensor power of this bundle or its dual.

13. *Skyscraper at the generic point.* Let X be an integral Noetherian scheme with function field $K(X)$. The constant sheaf $\underline{K(X)}$ is a quasi-coherent $\mathcal{O}_X$-module (on each affine open $\mathrm{Spec}\,R$, it corresponds to the R-module $\mathrm{Frac}(R)$), but it is not coherent unless X is a point.

5.1.2 Properties of Quasi-Coherent Sheaves

The following collects the basic algebraic properties of the tilde construction.

Proposition 5.1.14: Properties of Sheaf Module

Let R be a ring, $X = \mathrm{Spec}\,R$, M an R-module, and $\varphi : R \to S$ a ring homomorphism with $f = \varphi^a : \mathrm{Spec}\,S \to \mathrm{Spec}\,R$. Then:

1. $\widetilde{M}_{\mathfrak{p}} \cong M_{\mathfrak{p}}$ for every $\mathfrak{p} \in \mathrm{Spec}\,R$.
2. $\Gamma(\mathrm{Spec}\,R, \widetilde{M}) \cong M$.
3. The functor $M \mapsto \widetilde{M}$ is exact and fully faithful from R-Mod to $\mathcal{O}_{\mathrm{Spec}\,R}$-Mod.
4. $\widetilde{M \otimes_R N} \cong \widetilde{M} \otimes_{\mathcal{O}_{\mathrm{Spec}\,R}} \widetilde{N}$ for any R-module N.
5. $\widetilde{\bigoplus_{i\in I} M_i} \cong \bigoplus_{i\in I} \widetilde{M_i}$ for any family $\{M_i\}_{i\in I}$ of R-modules.
6. If N is an S-module, then $f_*(\widetilde{N}) \cong \widetilde{({}_R N)}$, where ${}_R N$ denotes N viewed as an R-module via φ.
7. $f^*(\widetilde{M}) \cong \widetilde{M \otimes_R S}$.

***Proof*:**
Parts (1)–(2) follow from the definition: stalks are computed as colimits of localizations M_f over distinguished opens containing $\mathfrak{p}$, which yields $M_{\mathfrak{p}}$; global sections are $M_1 = M$. Part (3): full faithfulness follows from (2) and corollary 5.1.12; exactness follows from the exactness of localization. Part (4): the isomorphism $(M \otimes_R N)_f \cong M_f \otimes_{R_f} N_f$ on each distinguished open

gives the result on a basis, hence globally. Part (5) is similar using $(\bigoplus M_i)_f \cong \bigoplus (M_i)_f$. Parts (6)–(7) follow from the canonical identifications of pushforward and pullback with restriction and extension of scalars.

It is important for exactness in (6) that the source is an affine scheme. For non-affine schemes, f_* fails to be exact — this failure is precisely what sheaf cohomology measures (chapter 6).

The next result shows that sections of quasi-coherent sheaves on affine schemes behave like elements of modules: they can be "cleared of denominators" and extended from distinguished opens.

Lemma 5.1.15: Quasi-coherence and Extending From $D(f)$

Let $X = \operatorname{Spec} R$, $f \in R$, and $\mathscr{F}$ a quasi-coherent sheaf on X. Then:

1. If $s \in \Gamma(X, \mathscr{F})$ satisfies $s|_{D(f)} = 0$, then $f^n s = 0$ for some $n > 0$.
2. If $t \in \mathscr{F}(D(f))$, then $f^n t$ extends to a global section in $\Gamma(X, \mathscr{F})$ for some $n > 0$.

Part (2) should be thought of as "clearing denominators": the section $t \in \mathscr{F}(D(f))$ may involve f in the denominator, and multiplying by a sufficiently high power of f eliminates the denominator to produce a global section. Part (1) is the quasi-coherent analogue of analytic continuation, tempered by the possibility of nilpotent elements: a section vanishing on a distinguished open must be "almost zero" globally.

***Proof*:**
By proposition 5.1.11, $\mathscr{F} \cong \widetilde{M}$ for $M = \Gamma(X, \mathscr{F})$.

(1). The condition $s|_{D(f)} = 0$ means $s/1 = 0$ in M_f. By the definition of localization, $f^n s = 0$ in M for some $n > 0$.

(2). The section $t \in \mathscr{F}(D(f)) = M_f$ can be written as $t = m/f^n$ for some $m \in M$ and $n > 0$. Then $f^n t = m/1 \in M = \Gamma(X, \mathscr{F})$, which is the desired global section.

The global section functor on an affine scheme is exact when restricted to quasi-coherent sheaves — a crucial property that distinguishes affine schemes:

Proposition 5.1.16: Exactness of Global Sections for Quasi-Coherent Sheaves

Let $X = \operatorname{Spec} R$ be an affine scheme and $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ an exact sequence of $\mathcal{O}_X$-modules with $\mathscr{F}'$ quasi-coherent. Then the sequence of global sections

$$0 \longrightarrow \Gamma(X, \mathscr{F}') \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{F}'') \longrightarrow 0$$

is exact.

***Proof*:**
Left-exactness of Γ holds for all sheaves (proposition 1.2.16), so it suffices to prove surjectivity on the right.

Let $s \in \Gamma(X, \mathscr{F}'')$. Since $\mathscr{F} \to \mathscr{F}''$ is surjective as a sheaf map, for each $x \in X$ there exists a distinguished open $D(g_x) \ni x$ such that $s|_{D(g_x)}$ lifts to some $\hat{s}_x \in \mathscr{F}(D(g_x))$. By quasicompactness, finitely many such opens $D(g_1), \dots, D(g_r)$ cover X. Choose lifts $t_i \in \mathscr{F}(D(g_i))$ of $s|_{D(g_i)}$.

On overlaps $D(g_i g_j)$, the sections t_i and t_j both lift s, so their difference $t_i - t_j$ lies in $\mathscr{F}'(D(g_i g_j))$. Since $\mathscr{F}'$ is quasi-coherent, lemma 5.1.15(1) gives an integer n_{ij} with $g_i^{n_{ij}}(t_i - t_j) = 0$ on $D(g_j)$ (after appropriate further localization). Taking n larger than all n_{ij} and applying lemma 5.1.15(2) to replace each t_i by $g_i^n t_i$ (which now lifts $g_i^n s$), one obtains local sections that agree on overlaps and hence glue to a global section $\bar{t} \in \Gamma(X, \mathscr{F})$ lifting $g_i^n s$ for each i.

Finally, since $D(g_1), \dots, D(g_r)$ cover X, we have $(g_1^n, \dots, g_r^n) = R$ (by corollary 2.1.18). Write $1 = \sum_i a_i g_i^n$ and let $\bar{t}_i$ be the global lift of $g_i^n s$ constructed above. Then $t := \sum_i a_i \bar{t}_i \in \Gamma(X, \mathscr{F})$ maps to $\sum_i a_i g_i^n s = s$, proving surjectivity.

This exactness characterizes affine schemes: in chapter 6, we shall see that $H^i(X, \mathscr{F}) = 0$ for all $i > 0$ and all quasi-coherent $\mathscr{F}$ if and only if X is affine (theorem 6.2.9).

The following collects closure properties of quasi-coherent and coherent sheaves:

Proposition 5.1.17: Closure Properties of Quasi-Coherent Sheaves

Let X be a scheme.

1. The kernel, cokernel, and image of any morphism of quasi-coherent sheaves are quasi-coherent. Any extension of quasi-coherent sheaves by quasi-coherent sheaves is quasi-coherent. In particular, $\mathbf{QCoh}(X)$ is an abelian category.
2. If X is locally Noetherian, the same statements hold with "quasi-coherent" replaced by "coherent."

Let $f : X \to Y$ be a morphism of schemes. Then:

3. If $\mathscr{G}$ is a quasi-coherent $\mathcal{O}_Y$-module, then $f^*\mathscr{G}$ is a quasi-coherent $\mathcal{O}_X$-module. If additionally X and Y are locally Noetherian and $\mathscr{G}$ is coherent, then $f^*\mathscr{G}$ is coherent.
4. If $\mathscr{F}$ is a quasi-coherent $\mathcal{O}_X$-module and either X is Noetherian or f is quasi-compact and quasi-separated, then $f_*\mathscr{F}$ is a quasi-coherent $\mathcal{O}_Y$-module.

***Proof*:**
All claims are local on X, so we may assume $X = \operatorname{Spec} R$ is affine.

(1) Kernels, cokernels, images. Under the equivalence $\mathbf{QCoh}(\operatorname{Spec} R) \simeq R\text{-Mod}$, kernels, cokernels, and images of $\widetilde{M} \to \widetilde{N}$ correspond to $\widetilde{\ker}$, $\widetilde{\operatorname{coker}}$, and $\widetilde{\operatorname{im}}$ respectively, by exactness of the tilde functor.

Extensions. Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence with $\mathscr{F}'$ and $\mathscr{F}''$ quasi-coherent. Set $M' = \Gamma(X, \mathscr{F}')$, $M = \Gamma(X, \mathscr{F})$, and $M'' = \Gamma(X, \mathscr{F}'')$. By proposition 5.1.16 (applied with $\mathscr{F}'$ quasi-coherent), the sequence $0 \to M' \to M \to M'' \to 0$ is exact. Applying the exact functor

$\widetilde{(-)}$ gives a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \widetilde{M'} \arrow[r] \arrow[d, "\cong"] & \widetilde{M} \arrow[r] \arrow[d] & \widetilde{M''} \arrow[r] \arrow[d, "\cong"] & 0 \\ 0 \arrow[r] & \mathscr{F}' \arrow[r] & \mathscr{F} \arrow[r] & \mathscr{F}'' \arrow[r] & 0 \end{tikzcd}$$

where the outer vertical maps are isomorphisms (since $\mathscr{F}'$ and $\mathscr{F}''$ are quasi-coherent). By the five lemma, the middle map $\widetilde{M} \to \mathscr{F}$ is an isomorphism, so $\mathscr{F}$ is quasi-coherent.

(2) follows by adding finite generation hypotheses throughout.

(3) On affine opens, pullback corresponds to extension of scalars: $f^*(\widetilde{M}) \cong \widetilde{M \otimes_R S}$ by proposition 5.1.14(7).

(4) The key observation is that quasi-compactness and quasi-separatedness ensure $f^{-1}(V)$ is covered by finitely many affine opens whose pairwise intersections are again finitely covered by affine opens. This yields an exact sequence

$$0 \longrightarrow f_*\mathscr{F} \longrightarrow \bigoplus_i f_*(\mathscr{F}|_{U_i}) \longrightarrow \bigoplus_{i,j,k} f_*(\mathscr{F}|_{U_{ijk}})$$

where the second and third terms are quasi-coherent (since each U_i and U_{ijk} is affine), and $f_*\mathscr{F}$ is their kernel, hence quasi-coherent by (1).

Without the quasi-compactness or quasi-separatedness hypotheses, the pushforward can fail to preserve quasi-coherence:

Example 5.1.18: Pushforward Not Quasi-Coherent

Let $X = \coprod_{i\in\mathbb{N}} \operatorname{Spec}\mathbb{Z}$, $Y = \operatorname{Spec}\mathbb{Z}$, and $f : X \to Y$ the identity on each component. Take $\mathscr{F} = \mathcal{O}_X$. If $f_*\mathcal{O}_X$ were quasi-coherent, then the natural map

$$\Gamma(Y, f_*\mathcal{O}_X) \otimes_{\mathbb{Z}} \mathbb{Z}[\tfrac{1}{2}] \longrightarrow (f_*\mathcal{O}_X)(D(2))$$

would be an isomorphism. The left side is $(\prod_{i\in\mathbb{N}} \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}[\frac{1}{2}]$: every element has the form $(a_i/2^n)_{i\in\mathbb{N}}$ for a *fixed* exponent n. The right side is $\prod_{i\in\mathbb{N}} \mathbb{Z}[\frac{1}{2}]$, which contains elements $(b_i/2^{n_i})$ where n_i grows without bound. Such elements have no preimage, so the map is not surjective.

Even between Noetherian schemes, the pushforward of a coherent sheaf need not be coherent:

Example 5.1.19: Pushforward of Noetherian Not Coherent

The structure morphism $f : \operatorname{Spec} k[x] \to \operatorname{Spec} k$ gives $f_*\mathcal{O}_{\operatorname{Spec} k[x]} = k[x]$, which is not a finitely generated k-module. The issue is that f is not a finite morphism. Sufficient conditions for f_* to preserve coherence include f being finite, projective, or proper (see theorem 6.6.6).

For later use, we record the pushforward result from the proof of proposition 5.1.17(4) as a separate statement:

Proposition 5.1.20: Pushforward is Quasi-Coherent for Qcqs Morphisms

Let $f : X \to Y$ be a quasi-compact quasi-separated morphism of schemes. If $\mathscr{F}$ is a quasi-coherent $\mathcal{O}_X$-module, then $f_*\mathscr{F}$ is a quasi-coherent $\mathcal{O}_Y$-module.

***Proof*:**
It suffices to show that for each affine open $\operatorname{Spec} A \subseteq Y$ and each $g \in A$, the natural map

$$\Gamma(f^{-1}(\operatorname{Spec} A), \mathscr{F}) \otimes_A A_g \xrightarrow{\cong} \Gamma(f^{-1}(D(g)), \mathscr{F})$$

is an isomorphism. This can be verified using the exact sequence from the proof of proposition 5.1.17(4): the quasi-compact quasi-separated hypothesis ensures that $f^{-1}(\operatorname{Spec} A)$ admits a finite affine cover with affine pairwise intersections, reducing the claim to the affine case where it follows from the exactness of localization.

5.1.3 Locally Free Sheaves and Vector Bundles

An important class of quasi-coherent sheaves — historically the first studied — arises from vector bundles. In this subsection, we establish the equivalence between vector bundles and locally free sheaves, and explain how $\mathbf{QCoh}(X)$ can be viewed as the abelian completion of the category of vector bundles on X.

Recall from section 0.5 that a vector bundle is locally trivial: over each point x, there is an open neighborhood U and an isomorphism $\pi^{-1}(U) \cong U \times \mathbb{A}^r_k$ respecting projection and linear on fibres. The algebraic translation of "locally trivial" is "locally free."

Definition 5.1.21: Vector Bundle

Let X be a scheme over a field k. A *vector bundle of rank r* on X is a scheme E together with a morphism $\pi : E \to X$ and an open covering $\{U_i\}$ of X with isomorphisms

$$\varphi_i : \pi^{-1}(U_i) \xrightarrow{\sim} U_i \times \mathbb{A}^r_k$$

such that for every pair i, j, the composite $\varphi_j \circ \varphi_i^{-1}|_{U_i \cap U_j}$ is of the form $(\mathrm{id}, \varphi_{ij})$ where $\varphi_{ij} \in \mathrm{GL}_r(\mathcal{O}_X(U_i \cap U_j))$. A *section* of E over $U \subseteq X$ is a morphism $s : U \to E$ with $\pi \circ s = \mathrm{id}_U$.

The transition matrices φ_{ij} satisfy the cocycle conditions $\varphi_{ii} = I$ and $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$ on triple overlaps. These are precisely the data needed to glue a sheaf from local pieces.

Definition 5.1.22: Free and Locally Free $\mathcal{O}_X$-module

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is *free of rank r* if $\mathscr{F} \cong \mathcal{O}_X^r$. It is *locally free of rank r* if there exists an open cover $\{U_i\}$ of X with $\mathscr{F}|_{U_i} \cong \mathcal{O}_{U_i}^r$ for each i. If X is connected, the rank is globally constant.

To identify which quasi-coherent sheaves are locally free, we need to know that "freeness of the stalk" propagates to a neighborhood. In differential geometry, this propagation is immediate: a smooth map of constant rank at a point has the same rank nearby, because the rank of a matrix is detected by the nonvanishing of minors, which is an open condition. The algebraic analogue is provided by Nakayama's lemma. We now formulate a sheaf-theoretic version that will serve as our main tool.

First, some notation. For a quasi-coherent sheaf $\mathscr{F}$ on a scheme X and a point $x \in X$ with residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$, the *fibre* of $\mathscr{F}$ at x is the $\kappa(x)$-vector space

$$\mathscr{F}(x) \ := \ \mathscr{F}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x) \ = \ \mathscr{F}_x \,/\, \mathfrak{m}_x \mathscr{F}_x.$$

This is the "value" of the sheaf at the point: a vector bundle of rank r has fibres that are r-dimensional $\kappa(x)$-vector spaces, while a general coherent sheaf may have fibres of varying dimension.

Lemma 5.1.23: Geometric Nakayama Lemma

Let X be a scheme, $\mathscr{F}$ a quasi-coherent $\mathcal{O}_X$-module of finite type (i.e., locally finitely generated), and $x \in X$.

1. *(Vanishing.)* If $\mathscr{F}(x) = 0$, then there exists an open neighborhood $U \ni x$ with $\mathscr{F}|_U = 0$.
2. *(Generators lift.)* If sections $s_1, \dots, s_n \in \mathscr{F}(U)$ are defined on a neighborhood $U \ni x$ and their images $\overline{s}_1, \dots, \overline{s}_n$ generate the fibre $\mathscr{F}(x)$, then after shrinking U the induced map $\mathcal{O}_U^n \to \mathscr{F}|_U$ is surjective.
3. *(Semicontinuity of fibre dimension.)* The function
$$x \ \longmapsto \ \dim_{\kappa(x)} \mathscr{F}(x)$$
is upper semicontinuous: for each $n \geq 0$, the set $\{x \in X : \dim \mathscr{F}(x) \leq n\}$ is open.

***Proof*:**

All three statements are local, so we may assume $X = \operatorname{Spec} R$ is affine and $\mathscr{F} = \widetilde{M}$ for a finitely generated R-module $M = Rm_1 + \cdots + Rm_k$. Let $\mathfrak{p} \in \operatorname{Spec} R$ correspond to x.

(1) Vanishing. The hypothesis $\mathscr{F}(x) = 0$ translates to $M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p} = 0$, i.e., $M_\mathfrak{p} = \mathfrak{p}M_\mathfrak{p}$. By the classical Nakayama lemma (applied to the finitely generated $R_\mathfrak{p}$-module $M_\mathfrak{p}$), we conclude $M_\mathfrak{p} = 0$.

Since M is finitely generated and each generator m_i satisfies $m_i/1 = 0$ in $M_\mathfrak{p}$, there exists $f_i \notin \mathfrak{p}$ with $f_i m_i = 0$ in M. Setting $f = f_1 \cdots f_k$ gives $f \notin \mathfrak{p}$ and $M_f = 0$, so $\mathscr{F}|_{D(f)} = \widetilde{M_f} = 0$.

(2) Generators lift. After shrinking U to a distinguished open and clearing denominators (using lemma 5.1.15), we may assume the s_i arise from elements $m_1, \dots, m_n \in M$. Define $\varphi : R^n \to M$ by sending the ith basis vector to m_i, and let $C = \operatorname{coker}(\varphi) = M/(Rm_1 + \cdots + Rm_n)$.

The hypothesis that $\overline{s}_1, \dots, \overline{s}_n$ generate $\mathscr{F}(x)$ translates to $C_\mathfrak{p}/\mathfrak{p}C_\mathfrak{p} = 0$, i.e., $C(x) = 0$. Since C is a quotient of the finitely generated module M, it is itself finitely generated. Applying part (1) to $\widetilde{C}$ yields $g \notin \mathfrak{p}$ with $C_g = 0$, meaning φ is surjective after localizing at g. Hence the map $\mathcal{O}_{D(g)}^n \to \mathscr{F}|_{D(g)}$ is surjective.

(3) Semicontinuity. Suppose $\dim_{\kappa(x)} \mathscr{F}(x) \leq n$. Choose n elements $s_1, \dots, s_n$ in $\mathscr{F}(U)$ (for some neighborhood $U \ni x$) whose images generate $\mathscr{F}(x)$. By (2), after shrinking U, the map $\mathcal{O}_U^n \to \mathscr{F}|_U$ is surjective. For any $y \in U$, applying $- \otimes_{\mathcal{O}_{X,y}} \kappa(y)$ to this surjection gives a

surjection $\kappa(y)^n \twoheadrightarrow \mathscr{F}(y)$, so $\dim_{\kappa(y)} \mathscr{F}(y) \leq n$. Hence $\{x \in X : \dim \mathscr{F}(x) \leq n\}$ is open.

The classical Nakayama lemma is a statement about a single local ring; the geometric upgrade converts it into a *neighborhood* statement by exploiting finite generation. The mechanism is simple but powerful: finitely many generators can be simultaneously controlled by inverting finitely many elements, i.e., by passing to a single distinguished open.

Part (3) should be compared with the situation for smooth vector bundles, which have *constant* fibre dimension. For coherent sheaves on schemes, constancy of fibre dimension is a strong condition — it is closely related to local freeness, as the next proposition makes precise.

Proposition 5.1.24: Locally Free if and only if Stalks are Free

Let X be a scheme and $\mathscr{F}$ a quasi-coherent sheaf of finite presentation *(*i.e., locally the cokernel of a map between finitely generated free modules*)*. Then $\mathscr{F}$ is locally free of rank r if and only if $\mathscr{F}_x$ is a free $\mathcal{O}_{X,x}$-module of rank r for every $x \in X$.
In particular, this applies to all coherent sheaves on locally Noetherian schemes, since over a Noetherian ring every finitely generated module is finitely presented.

The finite presentation hypothesis enters in a precise way: the geometric Nakayama lemma (lemma 5.1.23) requires finite type (= finitely generated) to show that generators of the fibre extend to generators of the sheaf nearby. This gives the surjectivity step. For the injectivity step, we must also apply the vanishing part of the lemma to the *kernel*, which requires the kernel to be finitely generated — and this is precisely what finite presentation guarantees.

***Proof*:**
($\Rightarrow$). If $\mathscr{F}|_U \cong \mathcal{O}_U^r$ for an open $U \ni x$, then $\mathscr{F}_x \cong \mathcal{O}_{X,x}^r$, since localization preserves free modules.

($\Leftarrow$). Suppose $\mathscr{F}_x \cong \mathcal{O}_{X,x}^r$ for all $x \in X$. Fix a point x and choose an affine neighborhood $\operatorname{Spec} R \ni x$ with $\mathscr{F}|_{\operatorname{Spec} R} \cong \widetilde{M}$, where M is a finitely presented R-module. Let $\mathfrak{p} \in \operatorname{Spec} R$ correspond to x, so that $M_\mathfrak{p} \cong R_\mathfrak{p}^r$ by hypothesis.

Step 1: Construct generators. Choose a basis $e_1, \ldots, e_r$ of the free $R_\mathfrak{p}$-module $M_\mathfrak{p}$. Write $e_i = m_i/g_i$ with $m_i \in M$ and $g_i \notin \mathfrak{p}$. Replacing $\operatorname{Spec} R$ by the distinguished open $D(g_1 \cdots g_r)$ (still an affine neighborhood of x on which $\mathscr{F}$ corresponds to a finitely presented module), we may assume $e_i = m_i \in M$.

The images $\overline{m}_1, \ldots, \overline{m}_r$ in the fibre $\mathscr{F}(x) = M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p} \cong \kappa(\mathfrak{p})^r$ form a basis.

Step 2: Surjectivity via geometric Nakayama. Define the map

$$\varphi : R^r \longrightarrow M, \qquad (a_1, \ldots, a_r) \longmapsto a_1 m_1 + \cdots + a_r m_r.$$

Since $\overline{m}_1, \ldots, \overline{m}_r$ generate $\mathscr{F}(x)$, lemma 5.1.23(2) gives a distinguished open $D(f) \ni x$ on which $\widetilde{\varphi} : \mathcal{O}_{D(f)}^r \to \mathscr{F}|_{D(f)}$ is surjective. Replacing $\operatorname{Spec} R$ by $D(f)$, we may assume φ is surjective.

Step 3: Injectivity via geometric Nakayama. Let $K = \ker(\varphi)$, so that $\widetilde{K}$ is a quasi-coherent sheaf on $\operatorname{Spec} R$ and we have the short exact sequence

$$0 \longrightarrow \widetilde{K} \longrightarrow \mathcal{O}_{\operatorname{Spec} R}^r \xrightarrow{\widetilde{\varphi}} \widetilde{M} \longrightarrow 0.$$

Since M is finitely presented, K is a finitely generated R-module[a]. At the stalk over x,

$$K_{\mathfrak{p}} \ = \ \ker(\varphi_{\mathfrak{p}} : R^r_{\mathfrak{p}} \to M_{\mathfrak{p}}).$$

But $\varphi_{\mathfrak{p}}$ sends $e_i \mapsto m_i$, which is a basis of $M_{\mathfrak{p}}$, so $\varphi_{\mathfrak{p}}$ is an isomorphism and $K_{\mathfrak{p}} = 0$. In particular, $\widetilde{K}(x) = K_{\mathfrak{p}}/\mathfrak{p}K_{\mathfrak{p}} = 0$. Since K is finitely generated, lemma 5.1.23(1) provides a distinguished open $D(g) \ni x$ with $K_g = 0$.

Conclusion. On the distinguished open $D(fg)$, the map φ is both surjective (by Step 2) and injective (by Step 3), hence an isomorphism $\mathcal{O}^r_{D(fg)} \xrightarrow{\sim} \mathscr{F}|_{D(fg)}$. Since x was arbitrary, $\mathscr{F}$ is locally free of rank r.

[a] If $R^s \xrightarrow{\alpha} R^t \to M \to 0$ is a finite presentation and $\varphi : R^r \twoheadrightarrow M$ is any surjection from a finitely generated free module, then Schanuel's lemma gives $K \oplus R^t \cong \ker(\alpha) \oplus R^r$; since $\ker(\alpha)$ is finitely generated by hypothesis, so is $K \oplus R^t$, hence K. Equivalently, being finitely presented is equivalent to *every* surjection from a finitely generated free module having finitely generated kernel.

The correspondence between vector bundles and locally free sheaves generalizes the classical fact that a vector bundle is determined by its transition functions. The key input from the scheme-theoretic side is the *relative spectrum* construction, which we develop in full generality in section 5.1.5 below. For the present theorem, we use it as follows: given a locally free sheaf $\mathcal{E}$ of rank r, the $\mathcal{O}_X$-algebra $\mathrm{Sym}(\mathcal{E}^\vee)$ is locally isomorphic to a polynomial ring in r variables, and $\underline{\mathrm{Spec}}_X(\mathrm{Sym}(\mathcal{E}^\vee))$ is the corresponding "total space" scheme over X.

Lemma 5.1.25: Coherent Sheaf Locally Free Criterion

Let $\mathcal{E}$ be a coherent sheaf on a locally Noetherian scheme X. Then:

1. A subsheaf of a locally free sheaf is locally free.
2. If $\mathcal{E}$ and $\mathcal{E}'$ are locally free of rank 1, then every nonzero morphism $\mathcal{E} \to \mathcal{E}'$ is injective.

***Proof*:**
Both statements are local and reduce to the corresponding facts about modules over local rings: a torsion-free module over a DVR is free, and a nonzero map between rank-1 free modules over a domain is injective.

The category of locally free sheaves is not abelian: quotients may fail to be locally free due to torsion. The following lemma shows how to "correct" this:

Lemma 5.1.26: Natural Quotient of Locally Free Sheaf

Let X be a Noetherian integral scheme, $\mathcal{E}$ a locally free sheaf of rank r, and $\mathcal{E}' \subseteq \mathcal{E}$ a locally free subsheaf of rank $r' \leq r$. Then there exists a locally free subsheaf $\overline{\mathcal{E}}' \subseteq \mathcal{E}$ of rank r' containing $\mathcal{E}'$ such that $\mathcal{E}/\overline{\mathcal{E}}'$ is locally free of rank $r - r'$.

***Proof*:**
Let $M = \mathcal{E}/\mathcal{E}'$ and $T \subseteq M$ its torsion subsheaf. Define $\overline{\mathcal{E}}'$ to be the kernel of the composition $\mathcal{E} \twoheadrightarrow M \twoheadrightarrow M/T$. The quotient $\mathcal{E}/\overline{\mathcal{E}}' \cong M/T$ is torsion-free and coherent, hence locally free by lemma 5.1.25(1). By the snake lemma applied to the diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{E}' \arrow[r] \arrow[d, hook] & \mathcal{E} \arrow[r] \arrow[d, equal] & M \arrow[r] \arrow[d, two heads] & 0 \\
0 \arrow[r] & \overline{\mathcal{E}}' \arrow[r] & \mathcal{E} \arrow[r] & M/T \arrow[r] & 0
\end{tikzcd}$$

we get $\overline{\mathcal{E}}'/\mathcal{E}' \cong T$, which is torsion, so $\overline{\mathcal{E}}'$ and $\mathcal{E}'$ have the same rank.

Theorem 5.1.27: Vector Bundle / Locally Free Sheaf Correspondence

Let X be a scheme over a field k. There is an equivalence of categories

$$\left\{ \begin{array}{c} \text{vector bundles} \\ \text{on } X \end{array} \right\} \underset{\underline{\mathrm{Spec}}_X \operatorname{Sym}(\mathscr{F}^\vee) \leftarrow\!\shortmid \mathscr{F}}{\overset{E \mapsto \mathcal{E}_E}{\rightleftarrows}} \left\{ \begin{array}{c} \text{locally free} \\ \text{sheaves on } X \end{array} \right\}$$

where morphisms on the left are bundle maps (linear on fibres) and on the right are $\mathcal{O}_X$-module homomorphisms.

***Proof*:**
From bundles to sheaves. Let $\pi : E \to X$ be a vector bundle of rank r. Define the *sheaf of sections* $\mathcal{E}_E$ by $\mathcal{E}_E(U) = \{s : U \to E \,|\, \pi \circ s = \mathrm{id}_U\}$, with $\mathcal{O}_X$-action $(f \cdot s)_p = f(p) \cdot s_p$ and addition defined fibre-wise. On a trivializing open U_i with $\pi^{-1}(U_i) \cong U_i \times \mathbb{A}^r_k$, every section decomposes uniquely as $s = f_1 e_1 + \cdots + f_r e_r$ where e_j are the coordinate sections and $f_j \in \mathcal{O}_X(U_i)$. The map $s \mapsto (f_1, \ldots, f_r)$ gives an $\mathcal{O}_X(U_i)$-module isomorphism $\mathcal{E}_E(U_i) \cong \mathcal{O}_X(U_i)^r$, so $\mathcal{E}_E$ is locally free of rank r.

From sheaves to bundles. Let $\mathcal{E}$ be a locally free sheaf of rank r. The total space is constructed via the *relative spectrum* (section 5.1.5):

$$E \;:=\; \underline{\mathrm{Spec}}_X(\operatorname{Sym}(\mathcal{E}^\vee)),$$

where $\mathcal{E}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X)$ is the dual sheaf and $\operatorname{Sym}(\mathcal{E}^\vee)$ is the symmetric algebra. On a trivializing open U_i where $\mathcal{E}|_{U_i} \cong \mathcal{O}_{U_i}^r$, we have $\operatorname{Sym}(\mathcal{E}^\vee)|_{U_i} \cong \mathcal{O}_X(U_i)[t_1, \ldots, t_r]$, and hence

$$\pi^{-1}(U_i) \;=\; \operatorname{Spec}\left(\mathcal{O}_X(U_i)[t_1, \ldots, t_r]\right) \;\cong\; U_i \times \mathbb{A}^r_k.$$

The transition functions of $\mathcal{E}$ give GL_r-valued cocycles on overlaps, furnishing E with the structure of a vector bundle. As a set,

$$E \;=\; \left\{(p, v) \;\middle|\; p \in X,\ v \in \mathcal{E}_p \otimes_{\mathcal{O}_{X,p}} \kappa(p)\right\},$$

which the reader should compare with the classical construction for curves where $E = \{(p, t) : p \in X,\ t \in \mathcal{E}_p/\mathfrak{m}_p\mathcal{E}_p\}$.

Functoriality. A bundle map $\varphi : E \to E'$ induces a sheaf morphism on sections by composition: $s \mapsto \varphi \circ s$. Conversely, a sheaf morphism $\mathcal{E} \to \mathcal{E}'$ induces a morphism of symmetric algebras (contravariantly on the duals), hence a bundle map via relative Spec. These operations are inverse to each other.

Definition 5.1.28: Line Bundle

A locally free $\mathcal{O}_X$-module of rank 1 is called a *line bundle* or *invertible sheaf*.

The term "invertible" is justified by the existence of a tensor-product inverse: if $\mathcal{L}$ is a line bundle, then $\mathcal{L}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$ satisfies $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \cong \mathcal{O}_X$. The group of isomorphism classes of line bundles under tensor product is the *Picard group* $\operatorname{Pic}(X)$, studied in depth in section 5.4.

5.1.29 K-Theory The monoid of isomorphism classes of locally free sheaves under direct sum can be group-completed to form the *Grothendieck group* $K_0(X)$ (see [55]). The tensor product makes $K_0(X)$ into a ring. Line bundles correspond to the units of this ring under the tensor product, and the inclusion $\operatorname{Pic}(X) \hookrightarrow K_0(X)^\times$ connects divisor theory to K-theory. For affine space, the Quillen–Suslin theorem ($K_0(\mathbb{A}^n_k) \cong \mathbb{Z}$) reflects the triviality of all vector bundles.

5.1.4 Ideal Sheaves and Closed Subschemes

Having developed the general theory of quasi-coherent sheaves, we return to closed subschemes. Recall from definition 3.1.22 that a closed immersion $\iota : Y \hookrightarrow X$ is a morphism for which $\iota^\# : \mathcal{O}_X \to \iota_*\mathcal{O}_Y$ is surjective. The kernel of this surjection is an $\mathcal{O}_X$-submodule of $\mathcal{O}_X$ — an *ideal sheaf* — and it encodes the scheme structure of Y inside X.

Definition 5.1.30: Ideal Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. An *ideal sheaf* on X is an $\mathcal{O}_X$-submodule $\mathcal{I} \subseteq \mathcal{O}_X$, i.e., a sheaf of ideals: for each open $U \subseteq X$, $\mathcal{I}(U)$ is an ideal of $\mathcal{O}_X(U)$, compatibly with restriction.

Lemma 5.1.31: Kernel and Ideal Sheaves

Let $\iota : Y \hookrightarrow X$ be a closed immersion. Then the kernel

$$\mathcal{I}_{Y/X} \ := \ \ker\bigl(\iota^\# : \mathcal{O}_X \to \iota_*\mathcal{O}_Y\bigr)$$

is an ideal sheaf on X, and there is a short exact sequence of $\mathcal{O}_X$-modules:

$$0 \longrightarrow \mathcal{I}_{Y/X} \longrightarrow \mathcal{O}_X \longrightarrow \iota_*\mathcal{O}_Y \longrightarrow 0.$$

***Proof*:**
As $\iota^\#$ is a surjective morphism of $\mathcal{O}_X$-modules, its kernel is an $\mathcal{O}_X$-submodule of $\mathcal{O}_X$, hence an ideal sheaf. The short exact sequence follows immediately.

Not every ideal sheaf arises from a closed immersion. The obstruction is quasi-coherence:

Example 5.1.32: Not All Ideal Sheaves Correspond To Closed Immersions

Let $X = \operatorname{Spec} k[x]_{(x)}$, which has two points: the closed point $\mathfrak{m} = (x)$ and the generic point $\eta = (0)$. Define an ideal sheaf $\mathcal{I}$ by

$$\mathcal{I}(X) = 0 \subseteq \mathcal{O}_X(X) = k[x]_{(x)}, \qquad \mathcal{I}(\{\eta\}) = k(x) = \mathcal{O}_X(\{\eta\}).$$

This is a valid $\mathcal{O}_X$-submodule, but it is not quasi-coherent: if $\mathcal{I} = \widetilde{I}$ for some ideal $I \subseteq k[x]_{(x)}$, then $\mathcal{I}(X) = I = 0$ forces $\mathcal{I}(\{\eta\}) = I_{(0)} = 0$, contradicting $\mathcal{I}(\{\eta\}) = k(x)$. By proposition 5.1.33 below, $\mathcal{I}$ cannot define a closed subscheme.

Proposition 5.1.33: Ideal Sheaves and Closed Immersions

Let X be a scheme and $\mathcal{I} \subseteq \mathcal{O}_X$ an ideal sheaf. Then $\mathcal{I}$ defines a closed immersion $Y \hookrightarrow X$ (with $\mathcal{O}_Y = \mathcal{O}_X/\mathcal{I}$) if and only if for every affine open $\operatorname{Spec} S \subseteq X$ and every $f \in S$, the restriction map induces an isomorphism $\mathcal{I}(\operatorname{Spec} S)_f \cong \mathcal{I}(D(f))$. This condition holds precisely when $\mathcal{I}$ is quasi-coherent.

***Proof*:**
($\Rightarrow$). If $\mathcal{I} = \mathcal{I}_{Y/X}$ for a closed immersion, then on each affine open $\operatorname{Spec} S$, the ideal sheaf restricts to $\widetilde{I}$ for $I = \mathcal{I}(\operatorname{Spec} S) \subseteq S$. The localization property $\widetilde{I}(D(f)) = I_f$ gives the required isomorphism.

($\Leftarrow$). The localization condition ensures that the quotient sheaves $\mathcal{O}_X(\operatorname{Spec} S)/\mathcal{I}(\operatorname{Spec} S)$ glue to form a well-defined quotient $\mathcal{O}_X/\mathcal{I}$, which defines the structure sheaf of the closed subscheme Y. The details are an application of the *qcqs lemma* (lemma 3.4.9).

We can now state the precise correspondence:

Proposition 5.1.34: Quasi-coherent Ideal Sheaves

Let X be a scheme. The assignment $Y \mapsto \mathcal{I}_{Y/X}$ defines a bijection

$$\{\text{closed subschemes of } X\} \xleftrightarrow{1:1} \{\text{quasi-coherent ideal sheaves on } X\}.$$

If X is locally Noetherian, then $\mathcal{I}_{Y/X}$ is coherent.

***Proof*:**
Given a closed immersion $\iota : Y \hookrightarrow X$, the ideal sheaf $\mathcal{I}_{Y/X}$ is quasi-coherent by proposition 5.1.33. Conversely, any quasi-coherent ideal sheaf $\mathcal{I}$ satisfies the localization condition of proposition 5.1.33 and hence defines a closed subscheme. Coherence in the locally Noetherian case follows because ideals of Noetherian rings are finitely generated.

Corollary 5.1.35: Ideal Sheaves On Affine Schemes

For an affine scheme $X = \operatorname{Spec} R$, there is a bijection between ideals $\mathfrak{a} \subseteq R$ and closed subschemes of X, given by $\mathfrak{a} \mapsto \operatorname{Spec}(R/\mathfrak{a})$.

With this correspondence in hand, we can define scheme-theoretic analogues of vanishing sets:

Definition 5.1.36: Vanishing Scheme

Let X be a scheme and $s \in \mathcal{O}_X(X)$ a global section. The *vanishing scheme* $V(s)$ is the closed subscheme corresponding to the quasi-coherent ideal sheaf generated by s: on each affine open $\operatorname{Spec} R \subseteq X$, it is defined by $\operatorname{Spec}(R/(s|_{\operatorname{Spec} R}))$. More generally, for a collection $S \subseteq \mathcal{O}_X(X)$, the vanishing scheme $V(S)$ corresponds to the ideal sheaf generated by S.

The advantage of vanishing *schemes* over vanishing *sets* is that they remember scheme-theoretic information such as intersection multiplicities.

Proposition 5.1.37: Properties of Vanishing Schemes

Let X be a scheme and $S_1, \ldots, S_n, \{S_i\}_{i \in \mathcal{I}}$ collections of global sections. Then:

1. $V(S_1) \cup \cdots \cup V(S_n) = V(S_1 \cap \cdots \cap S_n)$ (finite unions correspond to ideal intersections).
2. $\bigcap_{i \in \mathcal{I}} V(S_i) = V(\langle S_i | i \in \mathcal{I} \rangle)$ (intersections correspond to ideal sums).

***Proof*:**
The underlying topological statement follows from the corresponding properties of $V(-)$ and $I(-)$ (proposition 2.1.6). The scheme-theoretic content is that finite unions use ideal *intersections* (not products): we need $V(S) \cap V(S) = V(S)$, which holds for $S \cap S = S$ but would fail for $S \cdot S = S^2$. The key algebraic input is the inclusion $IJ \subseteq I \cap J \subseteq \sqrt{IJ}$.

Example 5.1.38: Vanishing Scheme Intersection

1. Consider $V(y - x^2) \cap V(y) \subseteq \mathbb{A}^2_k$. As a set, this is the single point $(0,0)$. As a scheme:

$$V(y - x^2) \cap V(y) \;=\; V(y - x^2,\, y) \;=\; \operatorname{Spec} \frac{k[x,y]}{(y - x^2,\, y)} \;\cong\; \operatorname{Spec} \frac{k[x]}{(x^2)}.$$

   The intersection point has *multiplicity* 2, reflecting the tangency between the parabola and the x-axis. This information is invisible to the underlying set but recorded by the scheme structure.

2. The intersection $V(y^2 - x^2) \cap V(y)$ gives an example where distributivity of unions and intersections fails at the scheme level. Since $y^2 - x^2 = (y - x)(y + x)$, we have $V(y^2 - x^2) = V(y - x) \cup V(y + x)$ (two lines through the origin). The scheme-theoretic computation shows

$$(V(y - x) \cup V(y + x)) \cap V(y) \;\neq\; (V(y - x) \cap V(y)) \cup (V(y + x) \cap V(y)),$$

   because the left side is $\operatorname{Spec} k[x]/(x^2)$ (a double point) while the right side is $\operatorname{Spec} k[x]/(x) \sqcup \operatorname{Spec} k[x]/(x) = \operatorname{Spec} k$ (a reduced point).

5.1.5 Relative Spectrum

In the proof of theorem 5.1.27, we constructed the total space of a vector bundle as $\underline{\mathrm{Spec}}_X(\mathrm{Sym}(\mathcal{E}^\vee))$. This is an instance of a general construction that associates a scheme over X to any quasi-coherent $\mathcal{O}_X$-algebra. Relative Spec is the scheme-theoretic analogue of the total space of a fibre bundle: it globalizes Spec from a single ring to a sheaf of rings over a base.

Definition 5.1.39: $\mathcal{O}_X$-algebra

Let $(X, \mathcal{O}_X)$ be a scheme. A *quasi-coherent $\mathcal{O}_X$-algebra* is a sheaf of commutative rings $\mathcal{A}$ on X that is simultaneously a quasi-coherent $\mathcal{O}_X$-module, with the ring multiplication $\mathcal{A}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{A}(U) \to \mathcal{A}(U)$ being $\mathcal{O}_X(U)$-bilinear for each open $U \subseteq X$.
Equivalently, $\mathcal{A}$ is a ring object in the category $\mathbf{QCoh}(X)$. On an affine open $\mathrm{Spec}\, R \subseteq X$, a quasi-coherent $\mathcal{O}_X$-algebra restricts to $\widetilde{A}$ for some R-algebra A.

Example 5.1.40: Quasi-Coherent $\mathcal{O}_X$-Algebras

1. The structure sheaf $\mathcal{O}_X$ is a quasi-coherent $\mathcal{O}_X$-algebra.
2. If $\mathcal{E}$ is a locally free sheaf of rank r, then $\mathrm{Sym}(\mathcal{E}) = \bigoplus_{n \geq 0} \mathrm{Sym}^n(\mathcal{E})$ is a quasi-coherent $\mathcal{O}_X$-algebra. On a trivializing open where $\mathcal{E} \cong \mathcal{O}_X^r$, we have $\mathrm{Sym}(\mathcal{E}) \cong \mathcal{O}_X[t_1, \ldots, t_r]$.
3. If $f : Y \to X$ is an affine morphism, then $f_*\mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-algebra.

The construction of relative Spec proceeds by gluing:

Theorem 5.1.41: Relative Spectrum

Let X be a scheme and $\mathcal{A}$ a quasi-coherent $\mathcal{O}_X$-algebra. There exists a scheme $\underline{\mathrm{Spec}}_X(\mathcal{A})$ together with an affine morphism $\pi : \underline{\mathrm{Spec}}_X(\mathcal{A}) \to X$ and an isomorphism of $\mathcal{O}_X$-algebras $\pi_*\mathcal{O}_{\underline{\mathrm{Spec}}_X(\mathcal{A})} \cong \mathcal{A}$, characterized by the following properties:

1. For each affine open $U = \mathrm{Spec}\, R \subseteq X$ with $\mathcal{A}|_U \cong \widetilde{A}$, there is a canonical isomorphism $\pi^{-1}(U) \cong \mathrm{Spec}\, A$.
2. As a set, $\underline{\mathrm{Spec}}_X(\mathcal{A}) = \{(x, \mathfrak{q}) : x \in X,\ \mathfrak{q} \in \mathrm{Spec}\,(\mathcal{A}_x)\}$.
3. The fibre over a point $x \in X$ with residue field $\kappa(x)$ is

$$\pi^{-1}(x) \;\cong\; \mathrm{Spec}\,(\mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)).$$

Furthermore, the construction gives an equivalence of categories:

$$\{\text{quasi-coherent } \mathcal{O}_X\text{-algebras}\} \;\simeq\; \{\text{affine morphisms } Y \to X\}^{\mathrm{op}}$$

with inverse $(\pi : Y \to X) \mapsto \pi_*\mathcal{O}_Y$.

***Proof*:**

Construction. Cover X by affine opens $U_i = \operatorname{Spec} R_i$ and write $\mathcal{A}|_{U_i} \cong \widetilde{A_i}$ for R_i-algebras A_i. On overlaps $U_i \cap U_j$, the quasi-coherence of $\mathcal{A}$ gives canonical isomorphisms between the restrictions of $\widetilde{A_i}$ and $\widetilde{A_j}$, hence isomorphisms between the corresponding spectra on the overlap. These isomorphisms satisfy the cocycle condition (being induced by identity maps of the sheaf $\mathcal{A}$), so by theorem 2.2.54 the schemes $\operatorname{Spec} A_i$ glue to a scheme $\underline{\operatorname{Spec}}_X(\mathcal{A})$ with a natural projection π to X.

Affine morphism. By construction, $\pi^{-1}(U_i) = \operatorname{Spec} A_i$ is affine for each i, so π is an affine morphism.

Equivalence. Given an affine morphism $\pi : Y \to X$, the pushforward $\pi_* \mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-algebra (by proposition 5.1.20, since affine morphisms are quasi-compact and separated). On each affine open $\operatorname{Spec} R \subseteq X$, we have $\pi^{-1}(\operatorname{Spec} R) = \operatorname{Spec} A$ and $\pi_* \mathcal{O}_Y|_{\operatorname{Spec} R} \cong \widetilde{A}$. The two operations are inverse by the equivalence $\mathbf{QCoh}(\operatorname{Spec} R) \simeq R\text{-Mod}$ applied to algebras.

Description of points and fibres. The set-theoretic description follows from the gluing construction: a point of $\underline{\operatorname{Spec}}_X(\mathcal{A})$ over $x \in U_i$ is a prime ideal of $(A_i)_{\mathfrak{p}}$ where $\mathfrak{p}$ corresponds to x. The fibre computation follows from $\operatorname{Spec}(A_i \otimes_{R_i} \kappa(\mathfrak{p})) \cong \operatorname{Spec}(A_i/\mathfrak{p}A_i)_{\mathrm{red}}$ after base change to the residue field.

The relative spectrum provides a unified perspective on several constructions we have already seen:

Example 5.1.42: Relative Spectrum

1. *Closed subschemes.* If $\mathcal{I} \subseteq \mathcal{O}_X$ is a quasi-coherent ideal sheaf, then $\underline{\operatorname{Spec}}_X(\mathcal{O}_X/\mathcal{I})$ is the closed subscheme defined by $\mathcal{I}$.

2. *Vector bundles.* If $\mathcal{E}$ is locally free of rank r, then $\underline{\operatorname{Spec}}_X(\operatorname{Sym}(\mathcal{E}^\vee))$ is the total space of the corresponding vector bundle, with fibres $\mathbb{A}^r$.

3. *Affine morphisms.* Every affine morphism $Y \to X$ arises as $\underline{\operatorname{Spec}}_X(\pi_* \mathcal{O}_Y) \to X$ for a unique quasi-coherent $\mathcal{O}_X$-algebra.

4. *Base change.* For any morphism $f : X' \to X$,

$$\underline{\operatorname{Spec}}_{X'}(f^* \mathcal{A}) \cong \underline{\operatorname{Spec}}_X(\mathcal{A}) \times_X X'.$$

The relative spectrum thus plays the same role for quasi-coherent algebras that Spec plays for rings: it is the universal "total space" construction, and it provides the bridge between the algebraic data (a sheaf of algebras) and the geometric data (a scheme over X).

5.1.6 Regular Embeddings and Local Complete Intersections

Relative Spec attaches a scheme over X to a sheaf of algebras, and the second item of example 5.1.42 identifies the total space of a vector bundle as the case of a symmetric algebra. One closed subscheme in particular deserves that treatment: a closed subscheme cut out, locally, by exactly as many equations as its codimension. Such subschemes behave like linear subspaces, and the sheaf of equations modulo its square turns out to be locally free, so relative Spec converts it into an honest vector bundle along the subscheme. This is the class of embeddings for which the geometry near Z is controlled by a bundle rather than by a general cone, and the adjunction and duality computations later in this book already assume it.

Definition 5.1.43: Regular Embedding

Let $i\colon Z \hookrightarrow X$ be a closed immersion of Noetherian schemes with quasi-coherent ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$ (definition 5.1.30, proposition 5.1.34). The immersion is a *regular embedding of codimension* d if every point of Z has an affine open neighbourhood $U = \operatorname{Spec} A$ in X for which the ideal $I = \mathcal{I}(U)$ is generated by a regular sequence $f_1, \dots, f_d$ of length d in A: that is, f_j is a nonzerodivisor on $A/(f_1, \dots, f_{j-1})$ for each j, and $A/I \neq 0$ ([15, Regular Sequence]).

The condition is local on Z and the codimension is required to be the same d at every point; a closed immersion that is regular of different codimensions on different components of Z is treated by working one component at a time.

Proposition 5.1.44: The Conormal Sheaf of a Regular Embedding

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension d with ideal sheaf $\mathcal{I}$. Then $\mathcal{I}/\mathcal{I}^2$, regarded as a sheaf of $\mathcal{O}_Z$-modules, is locally free of rank d. If $f_1, \dots, f_d$ generate $\mathcal{I}$ over an affine open U, their classes form a basis of $(\mathcal{I}/\mathcal{I}^2)|_{Z \cap U}$.

***Proof*:**
The statement is local, so take $U = \operatorname{Spec} A$ affine with $I = (f_1, \dots, f_d)$ generated by a regular sequence. The quotient I/I^2 is the degree-one piece of the associated graded ring $\bigoplus_{n \geq 0} I^n/I^{n+1}$, and for an ideal generated by a regular sequence that graded ring is the polynomial ring: [10, Associated Graded Ring of a Regular Sequence] gives an isomorphism of graded A/I-algebras

$$(A/I)[T_1, \dots, T_d] \xrightarrow{\sim} \bigoplus_{n \geq 0} I^n/I^{n+1}, \qquad T_j \longmapsto \overline{f_j}$$

Comparing degree-one pieces, I/I^2 is free over A/I on the classes of $f_1, \dots, f_d$. Since $\mathcal{I}/\mathcal{I}^2$ is annihilated by $\mathcal{I}$, it is a sheaf of $\mathcal{O}_Z$-modules, and these local descriptions exhibit it as locally free of rank d, as we sought to show.

Local freeness is exactly what relative Spec needs, so the equations of a regular embedding assemble into a vector bundle over Z.

Definition 5.1.45: Conormal Sheaf and Normal Bundle

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension d. The *conormal sheaf* is the locally free rank-d sheaf $\mathcal{I}/\mathcal{I}^2$ of proposition 5.1.44, and the *normal sheaf* is its dual $\mathcal{N}_{Z/X} = (\mathcal{I}/\mathcal{I}^2)^\vee$. The *normal bundle* of Z in X is the total space

$$N_Z X \;=\; \underline{\operatorname{Spec}}_Z\big(\operatorname{Sym}(\mathcal{I}/\mathcal{I}^2)\big)$$

a vector bundle of rank d over Z whose sheaf of sections is $\mathcal{N}_{Z/X}$, by example 5.1.42(2) applied to the locally free sheaf $\mathcal{N}_{Z/X}$.

Example 5.1.46: Regular and Irregular Embeddings

1. *Effective Cartier divisors.* A closed subscheme whose ideal sheaf is locally generated by a single nonzerodivisor is a regular embedding of codimension 1, a regular sequence of length one being exactly a nonzerodivisor. Its normal bundle is a line bundle.

2. *Linear subspaces.* For $\mathbb{P}^m \subseteq \mathbb{P}^n$ cut out by $x_{m+1}, \ldots, x_n$, those $n-m$ coordinates form a regular sequence in each standard affine chart meeting $\mathbb{P}^m$, so the inclusion is a regular embedding of codimension $n-m$.

3. *A point on a smooth curve.* If X is a smooth curve over a field (definition 2.7.11) and $P \in X$ a closed point, then $\mathcal{O}_{X,P}$ is a regular local ring of dimension one by proposition 2.7.12, so its maximal ideal is generated by one element, and $\{P\} \hookrightarrow X$ is a regular embedding of codimension 1.

4. *A failure.* Let $X = \operatorname{Spec} \bar{k}[x,y]/(xy)$ be the union of the two coordinate axes and $Z = \{P\}$ the origin, so $\dim X = 1$ and Z has codimension 1. The embedding is *not* regular. A regular embedding of codimension 1 would force the ideal $\mathfrak{m} = (x,y)$ to be locally generated by a single element, but $\mathfrak{m}^2 = (x^2, xy, y^2) = (x^2, y^2)$ because $xy = 0$, so $\mathfrak{m}/\mathfrak{m}^2$ has $\bar{k}$-dimension 2 and $\mathfrak{m}$ cannot be generated by fewer than two elements. Equally, no generator can be a nonzerodivisor: x kills y and y kills x. Singular points obstruct regularity of the embedding, which is why the constructions resting on a normal *bundle* require this hypothesis and the general case needs a cone instead.

Regular embeddings and smooth morphisms are the two classes of maps along which geometry is controlled by a bundle, and the morphisms built from them are the ones for which Riemann-Roch and the intersection-theoretic Gysin maps are available.

Two properties of regular embeddings are used by the intersection theory downstream and are proved here, where they belong, rather than assumed there.

Proposition 5.1.47: The Conormal Sequence of a Chain of Regular Embeddings

Let $i\colon Z \hookrightarrow Y$ and $j\colon Y \hookrightarrow P$ be regular embeddings of codimensions d and e. Then $j \circ i$ is a regular embedding of codimension $d+e$, and there is a short exact sequence of locally free sheaves on Z,

$$0 \longrightarrow N_Z Y \longrightarrow N_Z P \longrightarrow \big(N_Y P\big)\big|_Z \longrightarrow 0$$

which is locally split. Dually, $0 \to (\mathcal{I}_Y/\mathcal{I}_Y^2)|_Z \to \mathcal{I}_Z/\mathcal{I}_Z^2 \to \mathcal{I}_{Z/Y}/\mathcal{I}_{Z/Y}^2 \to 0$ is exact, where $\mathcal{I}_Y \subseteq \mathcal{I}_Z$ are the ideal sheaves in $\mathcal{O}_P$ and $\mathcal{I}_{Z/Y} = \mathcal{I}_Z/\mathcal{I}_Y$ is the ideal sheaf of Z in Y.

***Proof*:**
Work locally on P, on an affine open $\operatorname{Spec} A$ over which $\mathcal{I}_Y$ is generated by a regular sequence $g_1, \ldots, g_e$ and $\mathcal{I}_{Z/Y}$ by a regular sequence of length d in $A/(g)$; lift the latter to $h_1, \ldots, h_d \in A$. Then $\mathcal{I}_Z = (g_1, \ldots, g_e, h_1, \ldots, h_d)$: a section of $\mathcal{I}_Z$ reduces modulo $\mathcal{I}_Y$ to a combination of the $\bar{h}$'s, so differs from a combination of the h's by a section of $\mathcal{I}_Y$. The concatenation is a regular sequence, since the g's are one in A and the h's are one on $A/(g)$ by hypothesis. Hence $j \circ i$ is a regular embedding of codimension $d+e$.

For the sequence, write $J = \mathcal{I}_Y$ and $I = \mathcal{I}_Z$, so $J \subseteq I$. The ideal of Z in Y is $\bar{I} = I/J$, and

$$\bar{I}/\bar{I}^2 \;=\; (I/J)/\big((I^2+J)/J\big) \;=\; I/(I^2+J),$$

so $I/I^2 \to \bar{I}/\bar{I}^2$ is surjective with kernel $(I^2+J)/I^2 = J/(J\cap I^2)$. Moreover $J/J^2 \otimes_{\mathcal{O}_Y} \mathcal{O}_Z = J/(J^2+IJ) = J/IJ$, because $J \subseteq I$ forces $J^2 \subseteq IJ$, and $IJ \subseteq J \cap I^2$; so J/IJ surjects onto $J/(J\cap I^2)$. The composite

$$\big(J/J^2\big)\big|_Z \longrightarrow I/I^2 \longrightarrow \bar{I}/\bar{I}^2 \longrightarrow 0$$

is therefore exact in the middle and on the right, with no hypothesis beyond $J \subseteq I$.

Regularity supplies injectivity on the left and the splitting. In the local presentation above, $(J/J^2)|_Z$ is free on the classes of $g_1,\dots,g_e$ and I/I^2 is free on the classes of $g_1,\dots,g_e,h_1,\dots,h_d$, both by proposition 5.1.44, and the left-hand map sends the class of g_m to the class of g_m. It is therefore the inclusion of a free direct summand, split by the projection killing the h's. Dualizing a locally split short exact sequence of locally free sheaves gives the sequence of normal bundles in the statement, with the arrows reversed as displayed, and $(J/J^2)^\vee|_Z = (N_Y P)|_Z$ by definition 5.1.45.

Proposition 5.1.48: A Section of a Smooth Morphism Is a Regular Embedding

Let $p\colon P \to S$ be smooth of relative dimension e and let $s\colon S \to P$ be a section, so $p \circ s = \mathrm{id}_S$. Then s is a regular embedding of codimension e.

***Proof*:**
A section of a separated morphism is a closed immersion: s is the base change of the diagonal $P \to P \times_S P$ along $(\mathrm{id}_P, s\circ p)$, and the diagonal of a separated morphism is a closed immersion. So $s(S)$ is a closed subscheme; write $\mathcal{I}$ for its ideal sheaf.

The question is local, so take a point of $s(S)$ and a factorization as in definition 2.7.8: an open $U \subseteq P$ around it, an affine open $V = \operatorname{Spec} A \subseteq S$, and a closed immersion of U into $\operatorname{Spec} A[x_1,\dots,x_{e+r}]/(f_1,\dots,f_r)$ over V whose Jacobian has rank r. Shrinking further, the section is cut out inside U by the e functions $x_j - s^\#(x_j)$ for the e coordinates surviving after the r relations, since a point of U lies on $s(S)$ exactly when its coordinates agree with those of the section over the same point of V. These e functions generate $\mathcal{I}(U)$, and they form a regular sequence: modulo the first $j-1$ of them the ring is again of the same shape with one fewer free coordinate, and the next function is a coordinate minus a constant of the base, hence a nonzerodivisor there. So $\mathcal{I}$ is locally generated by a regular sequence of length e, as we sought to show.

Proposition 5.1.49: Regular Embeddings Pull Back Along Flat Morphisms

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension d and let $g\colon X' \to X$ be flat. Then the base change $i'\colon Z' = g^{-1}(Z) \hookrightarrow X'$ is a regular embedding of codimension d, and its conormal sheaf is the pullback: the canonical surjection $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}_{Z'}/\mathcal{I}_{Z'}^2$, for $h\colon Z' \to Z$ the induced morphism, is an isomorphism.

***Proof*:**
Base change of a closed immersion is a closed immersion, and the ideal sheaf of Z' is $\mathcal{I}\mathcal{O}_{X'}$. Work on an affine open $\operatorname{Spec} A \subseteq X$ over which $\mathcal{I}$ is generated by a regular sequence $f_1, \dots, f_d$, and let $\operatorname{Spec} A' \subseteq X'$ be an affine open mapping into it, so A' is flat over A. The images of $f_1, \dots, f_d$ generate $\mathcal{I}\mathcal{O}_{X'}$ on $\operatorname{Spec} A'$, and they remain a regular sequence: multiplication by f_j on $A/(f_1, \dots, f_{j-1})$ is injective by hypothesis, and tensoring that injection with the flat A-module A' keeps it injective, which says exactly that f_j is a nonzerodivisor on $A'/(f_1, \dots, f_{j-1})$. Flatness also gives $(\mathcal{I}/\mathcal{I}^2) \otimes_A A' \cong \mathcal{I}\mathcal{O}_{X'}/(\mathcal{I}\mathcal{O}_{X'})^2$, and this isomorphism is the canonical map, being induced by $f \mapsto f \otimes 1$ on generators; that is the asserted identification of conormal sheaves, completing the proof.

Definition 5.1.50: Local Complete Intersection Morphism

A morphism $f\colon X \to Y$ of Noetherian schemes is a *local complete intersection morphism*, or *l.c.i. morphism*, if every point of X has an open neighbourhood U for which $f|_U$ admits a factorization

$$U \xhookrightarrow{i} P \xrightarrow{p} Y$$

with i a regular embedding (definition 5.1.43) and p smooth (definition 2.7.8).

Proposition 5.1.51: Regular Embeddings and Smooth Morphisms Are l.c.i.

Every regular embedding is an l.c.i. morphism, and every smooth morphism is an l.c.i. morphism.

***Proof*:**
For a regular embedding $i\colon Z \hookrightarrow X$, take $U = Z$, $P = X$, and $p = \mathrm{id}_X$, which is smooth of relative dimension 0: the identity is covered by the factorization of definition 2.7.8 with $n = r = 0$, the empty Jacobian having rank 0.

For a smooth morphism $p\colon X \to Y$, take $U = X$, $P = X$, and $i = \mathrm{id}_X$. The identity is a closed immersion with zero ideal sheaf, which is generated by the empty regular sequence, so it is a regular embedding of codimension 0, completing the proof.

A single morphism can admit many such factorizations. Nothing in this book depends on comparing them: the definition above is the only thing used here, and it is independent of any choice because it only asserts that a factorization exists. The constructions that do depend on the choice, the Gysin maps and the Riemann-Roch correction terms, are cycle-level rather than scheme-level, and a reader who wants them will find the independence proved alongside them in *Intersection Theory* [16]. That pointer is a signpost, not a dependency.

5.2 Images of Scheme Morphisms

In the theory of sets, the image of a function is unambiguous: if $f : X \to Y$, then $f(X) = \{f(x) | x \in X\} \subseteq Y$, and it is automatically a subset. In algebraic geometry, things are not so clean. The image of a morphism of schemes need not be open, need not be closed, and need not even be locally closed—so it cannot, in general, carry a scheme structure coming from Y. This section develops two complementary responses to this problem:

1. *Chevalley's theorem*: the image is always *constructible*—a finite union of locally closed sets. This is the best purely topological statement one can make, and it has surprisingly powerful consequences for the geometry of fibers.
2. *The scheme-theoretic image*: when we insist on a scheme structure, we replace the set-theoretic image with the smallest closed subscheme through which the morphism factors. This is the "correct" notion of image in the category of schemes.

Both ideas require some commutative algebra—in particular, Grothendieck's Generic Freeness lemma, which says that finitely generated modules over a domain become free after inverting a single element. This is the key algebraic engine behind Chevalley's theorem.

5.2.1 Chevalley's Theorem

The reader who has worked primarily with affine morphisms may expect the image of a scheme morphism to be something reasonable—an open set, a closed set, or at worst a locally closed set. The following example shows that none of these hold in general.

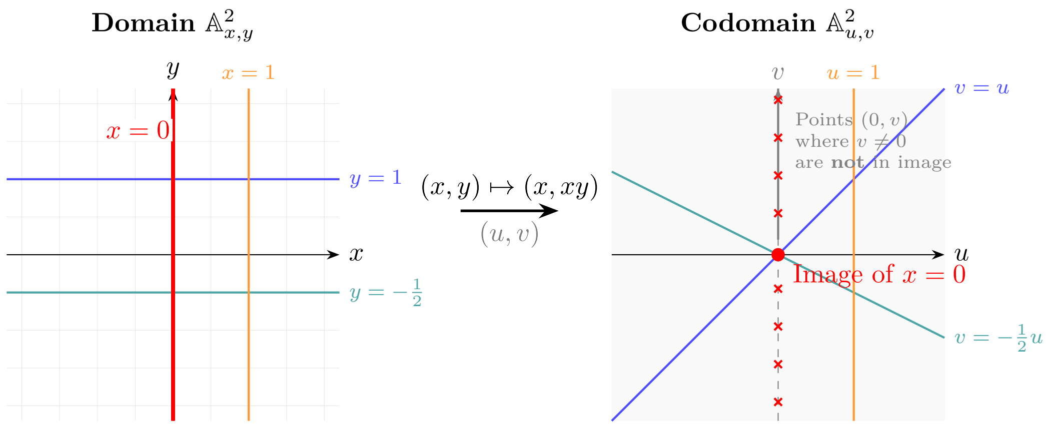
Example 5.2.1: Image That is Not Locally Closed

Let $f : \mathbb{A}^2_k \to \mathbb{A}^2_k$ be the morphism given by $(x, y) \mapsto (x, xy)$, corresponding to the ring map $k[u, v] \to k[x, y]$ with $u \mapsto x$, $v \mapsto xy$. Its image is:

$$\operatorname{im}(f) = \left(\mathbb{A}^2_k \setminus \{(u, v) \mid u = 0\}\right) \cup \{(0, 0)\}$$

In words: the entire plane with the v-axis removed, plus the origin put back in.

To see this, note that if $u = x \neq 0$, then $v = xy$ can be any element of k (by choosing $y = v/x$), so every point (u, v) with $u \neq 0$ is in the image. On the other hand, if $u = x = 0$, then $v = xy = 0$ regardless of y, so the only point on the v-axis that lies in the image is the origin.



This set is *not* locally closed. Recall that a locally closed subset is the intersection of an open set and a closed set. If $\operatorname{im}(f)$ were locally closed, its closure would be $\mathbb{A}^2_k$ (since $\operatorname{im}(f)$ is dense), and so $\operatorname{im}(f)$ would have to be open. But the complement $\mathbb{A}^2_k \setminus \operatorname{im}(f) = \{(0, v) \big| v \neq 0\}$ is not closed (its closure is the entire v-axis), so $\operatorname{im}(f)$ is not open either.

Because $\operatorname{im}(f)$ is not locally closed, it does not arise as the underlying set of any subscheme of $\mathbb{A}^2_k$.

However, the image in this example is not completely wild: it is the union of the open set $D(u)$ and the closed point $\{(0,0)\}$—a union of two locally closed sets. Chevalley's theorem says this is always the case.

Constructible Sets

To state Chevalley's theorem, we need the notion of a constructible set. The idea is simple: constructible sets are exactly the subsets you can build from open sets using complements and finite intersections. Equivalently (and more geometrically), they are finite unions of locally closed sets.

Definition 5.2.2: Constructible Set

Let X be a Noetherian topological space. A subset $S \subseteq X$ is *constructible* if it belongs to the smallest family of subsets containing all open sets and closed under:

1. complements
2. finite intersections

The first condition alone would give all open and all closed sets. Adding finite intersections then adds all locally closed sets (an open intersected with a closed set), and then all finite unions of locally closed sets (by De Morgan: a union of two constructible sets $S_1 \cup S_2 = (S_1^c \cap S_2^c)^c$ is constructible). In fact, one can show that these are *all* constructible sets:

Lemma 5.2.3: Constructible Sets as Locally Closed Sets

Let X be a Noetherian topological space. Then a subset $S \subseteq X$ is constructible if and only if it is a finite union of locally closed subsets. Consequently:

1. Finite unions and finite intersections of constructible sets are constructible.
2. If $f : X \to Y$ is continuous, then the preimage of a constructible set is constructible.

***Proof*:**
Every locally closed set is constructible (it is the intersection of an open and a closed set, both of which are in the family). Conversely, we must show that the collection of finite unions of locally closed sets is closed under complements and finite intersections.

Finite intersections. The intersection of two locally closed sets $(U_1 \cap Z_1) \cap (U_2 \cap Z_2) = (U_1 \cap U_2) \cap (Z_1 \cap Z_2)$ is locally closed. By distributing, the intersection of two finite unions of locally closed sets is again a finite union of locally closed sets.

Complements. It suffices to show the complement of a single locally closed set $U \cap Z$ is a finite union of locally closed sets. We have:

$$X \setminus (U \cap Z) = (X \setminus U) \cup (X \setminus Z)$$

The first is closed (hence locally closed) and the second is open (hence locally closed), so their union is a finite union of locally closed sets.

The two consequences are immediate: (1) follows from the definition, and (2) follows because continuous preimages preserve open sets, closed sets, intersections, and complements.

If the reader finds this reminiscent of the definition of measurable sets in measure theory, the analogy is apt: constructible sets form the "Boolean algebra" generated by open sets, just as measurable sets form the σ-algebra generated by open sets. The key difference is the finiteness constraint—no countable unions allowed.

For completeness, we record the definition given in EGA for the non-Noetherian case. A *retrocompact* open set $U \subseteq X$ is one where the inclusion $U \hookrightarrow X$ is quasi-compact. In the general definition, one replaces "open" by "retrocompact open" in definition 5.2.2. This ensures that constructible sets are described by "finite amounts of data" even when X is not Noetherian. For schemes of finite type over a Noetherian base—which includes essentially all schemes we care about—retrocompact open sets are the same as quasi-compact open sets, and the two definitions agree. We shall not need this generality.

5.2.4 Assumption For This Section We assume all topological spaces and schemes are Noetherian for the rest of this section, unless otherwise stated.

Generic Freeness

The algebraic engine behind Chevalley's theorem is Grothendieck's Generic Freeness lemma, which says that finitely generated modules over a domain become free after inverting a single element. Geometrically, this means that a coherent sheaf on an integral scheme is "generically" free—it is a vector bundle over a dense open subset.

Theorem 5.2.5: Generic Freeness (Grothendieck)

Let A be a Noetherian domain, S a finitely generated A-algebra, and M a finitely generated S-module. Then there exists a nonzero element $f \in A$ such that M_f is a free A_f-module.

Before giving the proof, let us unpack the geometric content. Taking $S = B$ and $M = B$ for a finitely generated A-algebra B: the theorem says that after shrinking $\operatorname{Spec} A$ to a distinguished open subset $D(f)$, the A_f-module B_f becomes free. In particular, $A_f \to B_f$ is faithfully flat (a nonzero free module is faithfully flat), and so the induced map $\operatorname{Spec} B_f \to \operatorname{Spec} A_f$ is *surjective*. This is the key input for Chevalley's theorem: a dominant finite type morphism from an integral scheme is surjective over a dense open.

***Proof*:**
We induct on the number of A-algebra generators of S. Write $S = A[t_1, \dots, t_n]$ (up to taking a quotient, which only makes M a module over a ring with fewer generators; explicitly, if $S = A[t_1, \dots, t_n]/I$ then we view M as an $A[t_1, \dots, t_n]$-module annihilated by I, and the result for $A[t_1, \dots, t_n]$ implies the result for S).

Base case: $n = 0$, so $S = A$. Then M is a finitely generated module over the Noetherian domain A. Let $K = \operatorname{Frac}(A)$, and let $r = \dim_K(M \otimes_A K)$. Choose elements $m_1, \dots, m_r \in M$

whose images form a K-basis of $M \otimes_A K$, and consider the A-linear map:

$$\varphi : A^r \longrightarrow M, \qquad e_i \longmapsto m_i.$$

After tensoring with K, this becomes an isomorphism. Both $\ker(\varphi)$ and $\operatorname{coker}(\varphi)$ are finitely generated A-modules that vanish after tensoring with K. For each generator of $\ker(\varphi)$ and $\operatorname{coker}(\varphi)$, there is a nonzero element of A annihilating it (since the generator maps to zero in the localization at (0)). Let f be the product of all such elements. Then $\ker(\varphi)_f = 0$ and $\operatorname{coker}(\varphi)_f = 0$, so $\varphi_f : A_f^r \xrightarrow{\sim} M_f$ is an isomorphism. Hence M_f is free of rank r.

Inductive step: $S = S_0[t]$ where $S_0 = A[t_1, \ldots, t_{n-1}]$. Let M be a finitely generated $S_0[t]$-module. We split M into its t-torsion and t-torsion-free parts.

Step 1: The t-torsion part. Let $T = \{m \in M : t^k m = 0 \text{ for some } k \geq 1\}$. Since M is Noetherian, T is finitely generated, and there exists $N \geq 1$ with $t^N T = 0$. Hence T is annihilated by t^N, so T is a module over $S_0[t]/(t^N)$. Now, $S_0[t]/(t^N)$ is a finitely generated S_0-module (with S_0-basis $1, t, t^2, \ldots, t^{N-1}$). Since T is finitely generated over $S_0[t]/(t^N)$, it is finitely generated over S_0. By the inductive hypothesis (applied with S_0 in place of S), there exists $f_1 \in A \setminus \{0\}$ such that T_{f_1} is a free A_{f_1}-module.

Step 2: The torsion-free part. The quotient $\overline{M} = M/T$ is a finitely generated $S_0[t]$-module on which t acts injectively. Let $K = \operatorname{Frac}(A)$. Then $\overline{M}_K = \overline{M} \otimes_A K$ is a finitely generated module over $K[t_1, \ldots, t_{n-1}, t]$, and since t acts injectively, $\overline{M}_K$ is a finitely generated *torsion-free* module over $K[t]$ (torsion-free with respect to t, viewing $\overline{M}_K$ as a $K[t]$-module via the other generators). Since $K[t]$ is a PID, torsion-free and finitely generated implies free:

$$\overline{M}_K \cong K[t]^s$$

for some $s \geq 0$[a]. Choose $\overline{m}_1, \ldots, \overline{m}_s \in \overline{M}$ whose images form a $K[t]$-basis of $\overline{M}_K$. The map $S_0[t]^s \to \overline{M}$ sending $e_i \mapsto \overline{m}_i$ becomes an isomorphism after tensoring with K. As in the base case, the kernel and cokernel are finitely generated and K-torsion, so there exists $f_2 \in A \setminus \{0\}$ with:

$$\overline{M}_{f_2} \cong A_{f_2}[t_1, \ldots, t_{n-1}, t]^s$$

The right side is a free A_{f_2}-module (of countably infinite rank when $s > 0$, with A_{f_2}-basis given by all monomials times the standard basis vectors).

Step 3: Reassembly. Taking $f = f_1 f_2$, we have that T_f is free over A_f (from Step 1) and $\overline{M}_f$ is free over A_f (from Step 2). The short exact sequence $0 \to T_f \to M_f \to \overline{M}_f \to 0$ splits, since $\overline{M}_f$ is free (hence projective). Therefore $M_f \cong T_f \oplus \overline{M}_f$ is free over A_f.

[a] Strictly speaking, $\overline{M}_K$ is a module over $K[t_1, \ldots, t_{n-1}][t]$, and we are using that it is torsion-free over $K[t_1, \ldots, t_{n-1}][t]$, which is a PID when we view it as a localization of the polynomial ring. The key point is that after localizing A to K, the t-torsion-free part becomes free over the resulting ring, which is a PID in the variable t.

The geometric consequence we need is immediate:

Corollary 5.2.6: Dominant Morphisms Hit a Dense Open

Let $f : X \to Y$ be a dominant morphism of finite type between Noetherian integral schemes. Then $f(X)$ contains a dense open subset of Y.

***Proof*:**
We may assume $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ are affine. Since f is dominant, the ring map $A \to B$ is injective (proposition 3.1.32), and B is a finitely generated A-algebra. By theorem 5.2.5 (with $S = B$ and $M = B$), there exists $a \in A \setminus \{0\}$ with B_a a free A_a-module. Since $B \neq 0$, we have $B_a \neq 0$ (localizing a nonzero module at a nonzero element of a domain gives a nonzero module). A nonzero free module is faithfully flat, so $A_a \to B_a$ is faithfully flat, and the induced map $\operatorname{Spec} B_a \to \operatorname{Spec} A_a$ is surjective. Thus $f(X) \supseteq D(a)$, a dense open subset of Y.

Chevalley's Theorem

We can now state and prove the main result. The proof is an application of Noetherian induction (theorem 0.3.3): the corollary above gives us control over a dense open subset, and the inductive hypothesis handles the closed complement.

Theorem 5.2.7: Chevalley's Theorem

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes. Then the image of every constructible subset of X is a constructible subset of Y. In particular, $f(X)$ is constructible.

***Proof*:**
Since a constructible set is a finite union of locally closed sets, and a finite union of constructible sets is constructible, it suffices to show that the image of a locally closed set is constructible. A locally closed subset of a Noetherian scheme is again a Noetherian scheme (with its induced reduced structure), and the restriction of a finite type morphism to a locally closed subscheme is still of finite type. Thus it suffices to show that $f(X)$ is constructible.

Step 1: Reduction to the affine case. The question is local on Y: if $Y = \bigcup_j V_j$ is an open cover, then $f(X)$ is constructible in Y if and only if $f(X) \cap V_j$ is constructible in V_j for each j. So we may assume $Y = \operatorname{Spec} A$ is affine. Since f is of finite type, X is quasi-compact, so we can cover X by finitely many affine opens $X = U_1 \cup \cdots \cup U_m$. Then $f(X) = f(U_1) \cup \cdots \cup f(U_m)$, and a finite union of constructible sets is constructible. So we may assume $X = \operatorname{Spec} B$ is also affine, with B a finitely generated A-algebra.

Step 2: Noetherian induction on Y. We prove by Noetherian induction on Y: the statement holds for Y, assuming it holds for every proper closed subset $Z \subsetneq Y$ (equipped with any scheme structure making it a Noetherian scheme). This is valid because Y is a Noetherian topological space (theorem 0.3.3).

Step 3: Decompose the base. Decompose $Y = \operatorname{Spec} A$ into its (finitely many) irreducible components $Y_1, \ldots, Y_r$, where $Y_i = V(\mathfrak{p}_i)$ for the minimal primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ of A (proposition 2.1.21). The image $f(X)$ is the union of the images $f(f^{-1}(Y_i)) \to Y_i$ for $i = 1, \ldots, r$. Each $f^{-1}(Y_i) \to Y_i$ is a morphism of finite type (base change preserves finite type). If $r > 1$, each Y_i is a proper closed subset of Y, and by the inductive hypothesis the image of $f^{-1}(Y_i) \to Y_i$ is constructible in Y_i. A constructible subset of a closed subspace is constructible in the ambient space (it is still a finite union of locally closed sets), so $f(X)$ is constructible. If $r = 1$, then Y is irreducible.

Step 4: The irreducible case. We may now assume $Y = \operatorname{Spec} A$ is irreducible. Replacing A by $A/\operatorname{Nil}(A)$ does not change the underlying topological space (proposition 2.1.10), so we may further assume A is a domain (Y is integral). Similarly, replacing B by $B/\ker(A \to B) \cdot B$ does not change the image of the map on spectra. After this, the morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ factors as $\operatorname{Spec} B \to \operatorname{Spec}(A/J) \hookrightarrow \operatorname{Spec} A$ for some ideal J. If $J \neq 0$, then $V(J) \subsetneq Y$ and we are done by the inductive hypothesis. If $J = 0$, then $A \hookrightarrow B$ and the morphism is dominant.

Step 5: The dominant case. We have A a Noetherian domain, B a finitely generated A-algebra with $A \hookrightarrow B$ injective, and $f : \operatorname{Spec} B \to \operatorname{Spec} A$ dominant. By corollary 5.2.6, there exists $a \in A \setminus \{0\}$ with $D(a) \subseteq f(X)$.

Consider the closed complement $Z = V(a) \subsetneq \operatorname{Spec} A$. Since $a \neq 0$ and A is a domain, Z is a proper closed subset. The preimage $f^{-1}(Z) = \operatorname{Spec}(B/aB)$ maps to $Z = \operatorname{Spec}(A/aA)$ by a morphism of finite type. By the Noetherian inductive hypothesis, the image of $f^{-1}(Z) \to Z$ is constructible in Z, hence constructible in Y. Therefore:

$$f(X) = D(a) \cup \big(\text{constructible subset of } V(a)\big)$$

is a union of two constructible sets, hence constructible.

Constructibility controls the image of a single morphism. Families ask for something transverse to that: a property visible on the generic fiber of $f\colon X \to S$ ought to persist on the fiber over every point of some nonempty open subset of the base. The pattern has appeared once already, in proposition 3.2.12, where an empty generic fiber was shown to force empty fibers over a whole neighborhood of η. Emptiness is the crudest property one can spread out this way. The next one up, and the one that carries most of the weight in practice, is density: given an open subscheme $U \subseteq X$, does density of U_η inside X_η propagate to the nearby fibers?

The question is not idle. Constructions on a family are routinely carried out on an open subset U, such as the locus where a chosen section does not vanish or the complement of a bad divisor, and what is then needed is that U still meets every part of X_s for most s. Density is exactly the condition that U_s meet every irreducible component of X_s, so the question is whether the components of a fiber can conspire, as s moves, to hide inside the closed complement $X \setminus U$. They can, but only over a proper closed subset of the base.

The engine is generic freeness again, applied in a shape slightly different from the one used for corollary 5.2.6: rather than tracking a whole image, it tracks a single equation, and it reports that the equation stays a nonzerodivisor on all nearby fibers.

Lemma 5.2.8: Nonzerodivisors Persist on Nearby Fibers

Let A be a Noetherian domain, let C be a finitely generated A-algebra, and let $c \in C$ be a nonzerodivisor. Write $g\colon \operatorname{Spec} C \to \operatorname{Spec} A$ for the induced morphism. Then there is a nonzero $a \in A$ such that for every $s \in D(a)$:

1. the image of c in $C \otimes_A \kappa(s)$ is a nonzerodivisor;
2. $D(c) \cap g^{-1}(s)$ is dense in $g^{-1}(s)$.

***Proof*:**
Step 1: A nonzerodivisor cuts out a dense distinguished open.
Let R be any ring and $r \in R$ a nonzerodivisor; we claim $D(r)$ is dense in $\operatorname{Spec} R$. The distinguished opens form a basis (corollary 2.1.13), so it suffices to check that $D(r) \cap D(h) = D(rh)$ is nonempty whenever $D(h)$ is. Recall that $D(x) = \emptyset$ says precisely that x lies in every prime of R, that

is, that x is nilpotent ([10]). So suppose $D(rh) = \emptyset$, say $(rh)^n = r^n h^n = 0$. A power of a nonzerodivisor is a nonzerodivisor, so $h^n = 0$ and $D(h) = \emptyset$. This proves the claim, and with it part (2) of the statement, granted part (1).

Step 2: Choosing a.
The quotient C/cC is a finitely generated module over the finitely generated A-algebra C. By theorem 5.2.5 there is a nonzero $a \in A$ such that $(C/cC)_a$ is a free A_a-module. Fix this a for the rest of the proof.

Step 3: The ideal cC_a is an A_a-module direct summand.
Localization at a is exact, so

$$0 \longrightarrow cC_a \longrightarrow C_a \longrightarrow (C/cC)_a \longrightarrow 0$$

is an exact sequence of A_a-modules. Its right-hand term is free, hence projective, so the sequence splits and cC_a is a direct summand of C_a as an A_a-module.

Moreover c remains a nonzerodivisor on C_a: if $c \cdot (x/a^m) = 0$ then $a^k cx = 0$ in C for some $k \geq 0$, so $c \cdot (a^k x) = 0$, so $a^k x = 0$, so $x/a^m = 0$. Multiplication by c is therefore an isomorphism of A_a-modules

$$\mu \colon C_a \xrightarrow{\sim} cC_a$$

Step 4: Base change to a fiber.
Fix $s \in D(a)$. Since a is invertible in $\kappa(s)$, the residue field $\kappa(s)$ is an A_a-algebra and $C \otimes_A \kappa(s) = C_a \otimes_{A_a} \kappa(s)$. Apply $- \otimes_{A_a} \kappa(s)$ to Step 3. The isomorphism μ becomes an isomorphism, and the inclusion $cC_a \hookrightarrow C_a$ is a split injection of A_a-modules, so it stays injective. Their composite is multiplication by the image of c, whence multiplication by c is injective on $C \otimes_A \kappa(s)$. That is part (1).

Step 5: Reading it on the fiber.
By proposition 3.2.10 the fiber $g^{-1}(s)$ is homeomorphic to $\operatorname{Spec}(C \otimes_A \kappa(s))$, and under this homeomorphism $D(c) \cap g^{-1}(s)$ corresponds to the distinguished open cut out by the image of c. Step 1 applied to $R = C \otimes_A \kappa(s)$ and r the image of c gives part (2), completing the proof.

Proposition 5.2.9: Fiberwise Density Spreads Out

Let S be a Noetherian integral scheme with generic point η, let $f \colon X \to S$ be a morphism of finite type, and let $U \subseteq X$ be an open subscheme. If U_η is dense in the generic fiber X_η, then there is a nonempty open $V \subseteq S$ such that U_s is dense in X_s for every $s \in V$.

***Proof*:**
Throughout, proposition 3.2.10 lets us identify the underlying space of the fiber X_s (definition 3.2.9) with the subspace $f^{-1}(s) \subseteq X$, and U_s with $U \cap f^{-1}(s)$; only the open subset underlying U matters. We record one fact used repeatedly: since $\overline{\{\eta\}} = S$ (definition 2.1.9), every nonempty open of S contains η, for otherwise it would be disjoint from a closed set containing η and hence from S. Consequently a finite intersection of nonempty opens of S is nonempty.

Step 1: Reduction to an affine base.
As S is Noetherian it has a cover by affine opens $\operatorname{Spec} A_i$ with A_i Noetherian (definition 2.4.1);

choose one, $S_0 = \operatorname{Spec} A$. Then $A = \mathcal{O}_S(S_0)$ is a domain by definition 2.4.18, and $\eta \in S_0$ with $\overline{\{\eta\}} \cap S_0 = S_0$, so η is the generic point of S_0, that is, the zero ideal of A. Replace S by S_0, X by $f^{-1}(S_0)$ and U by $U \cap f^{-1}(S_0)$. This alters neither the generic fiber nor any fiber over a point of S_0, the restricted morphism is still of finite type, and a nonempty open of S_0 is a nonempty open of S. So assume $S = \operatorname{Spec} A$ with A a Noetherian domain and $\eta = (0)$.

Step 2: Reduction to an affine total space.
Two observations about an open cover $T = W_1 \cup \cdots \cup W_m$ of a topological space and an open subset $O \subseteq T$. First, if O is dense in T then $O \cap W_\alpha$ is dense in W_α, since a nonempty open of W_α is a nonempty open of T. Second, if $O \cap W_\alpha$ is dense in W_α for every α then O is dense in T, since a nonempty open of T meets some W_α in a nonempty open of W_α.

Now f is quasicompact (definition 3.4.26) and S is affine, so X is quasicompact and admits a finite cover by affine opens $W_1, \dots, W_m$ with $\mathcal{O}_X(W_\alpha)$ a finitely generated A-algebra. The first observation, applied to the cover of X_η by the sets $X_\eta \cap W_\alpha$, says the hypothesis is inherited by each W_α. The second, applied to the cover of $f^{-1}(s)$ by the sets $f^{-1}(s) \cap W_\alpha$, says that if the conclusion holds for each W_α over a nonempty open $V_\alpha \subseteq S$, then it holds for X over $V = V_1 \cap \cdots \cap V_m$, which is nonempty by the opening remark. So assume $X = \operatorname{Spec} B$ with B a finitely generated A-algebra, and write $Z = X \setminus U = V(\mathfrak{b})$ for an ideal $\mathfrak{b} \subseteq B$.

Step 3: Components of X, and which of them dominate.
If $B = 0$ take $V = S$ and stop. Otherwise B is Noetherian by the Hilbert basis theorem, so X is a Noetherian topological space and proposition 2.1.21 writes $X = X_1 \cup \cdots \cup X_n$ with each X_i closed and irreducible and $X_i \not\subseteq X_j$ for $i \neq j$. Set $\mathfrak{p}_i = \bigcap_{\mathfrak{q} \in X_i} \mathfrak{q}$. Writing $X_i = V(\mathfrak{a}_i)$, we have $\mathfrak{a}_i \subseteq \mathfrak{p}_i$ and every $\mathfrak{q} \in X_i$ contains $\mathfrak{p}_i$, so $X_i = V(\mathfrak{p}_i)$. Each $\mathfrak{p}_i$ is prime: it is proper because $X_i \neq \emptyset$, and if $gh \in \mathfrak{p}_i$ then X_i is the union of the closed sets $X_i \cap V(g)$ and $X_i \cap V(h)$, so irreducibility puts X_i inside one of them, that is, $g \in \mathfrak{p}_i$ or $h \in \mathfrak{p}_i$. Finally $X_i \not\subseteq X_j$ reads $\mathfrak{p}_j \not\subseteq \mathfrak{p}_i$.

The morphism f sends $\mathfrak{q}$ to $\mathfrak{q} \cap A$. Call X_i *dominating* when $\mathfrak{p}_i \cap A = (0)$, and let I collect the dominating indices. For $i \notin I$ pick a nonzero $a_i \in \mathfrak{p}_i \cap A$; every $\mathfrak{q} \in V(\mathfrak{p}_i)$ contains a_i, so $f(\mathfrak{q}) \in V(a_i)$ and therefore $V(\mathfrak{p}_i) \cap f^{-1}(s) = \emptyset$ for $s \in D(a_i)$. Put $a' = \prod_{i \notin I} a_i$, an empty product being 1; it is nonzero because A is a domain. Then

$$f^{-1}(s) = \bigcup_{i \in I} \big(V(\mathfrak{p}_i) \cap f^{-1}(s)\big) \qquad \text{for all } s \in D(a')$$

Step 4: What the hypothesis says at η.
We claim $\mathfrak{b} \not\subseteq \mathfrak{p}_i$ for every $i \in I$. Suppose instead $\mathfrak{b} \subseteq \mathfrak{p}_i$ with $\mathfrak{p}_i \cap A = (0)$. Then $f(\mathfrak{p}_i) = (0) = \eta$, so $\mathfrak{p}_i \in f^{-1}(\eta)$. Consider

$$O = f^{-1}(\eta) \setminus \bigcup_{j \neq i} V(\mathfrak{p}_j)$$

an open subset of X_η. It contains $\mathfrak{p}_i$, since $\mathfrak{p}_i \in V(\mathfrak{p}_j)$ would say $\mathfrak{p}_j \subseteq \mathfrak{p}_i$, excluded in Step 3. Every prime of B lies in some X_j, so any point of O lies in $V(\mathfrak{p}_i)$; and $\mathfrak{b} \subseteq \mathfrak{p}_i$ gives $V(\mathfrak{p}_i) \subseteq V(\mathfrak{b}) = Z$. Thus O is a nonempty open subset of X_η disjoint from U, contradicting the density of U_η. For each $i \in I$ fix $b_i \in \mathfrak{b} \setminus \mathfrak{p}_i$.

Step 5: Spreading out one dominating component at a time.
Fix $i \in I$ and set $B_i = B/\mathfrak{p}_i$, a domain and a finitely generated A-algebra. The canonical homeomorphism $\operatorname{Spec} B_i \to V(\mathfrak{p}_i)$ carries the morphism $g_i \colon \operatorname{Spec} B_i \to \operatorname{Spec} A$ induced by $A \to B_i$ onto the restriction of f, and $A \to B_i$ is injective because $\mathfrak{p}_i \cap A = (0)$. The image β_i

of b_i in B_i is nonzero, hence a nonzerodivisor. By lemma 5.2.8 there is a nonzero $a_i' \in A$ such that $D(\beta_i) \cap g_i^{-1}(s)$ is dense in $g_i^{-1}(s)$ for every $s \in D(a_i')$. Under the homeomorphism, $g_i^{-1}(s)$ is $V(\mathfrak{p}_i) \cap f^{-1}(s)$ and $D(\beta_i)$ is the set of primes $\mathfrak{q} \supseteq \mathfrak{p}_i$ with $b_i \notin \mathfrak{q}$. Since $b_i \in \mathfrak{b}$, such a $\mathfrak{q}$ does not contain $\mathfrak{b}$ and so lies in U. A superset of a dense subset is dense, so

$$U \cap V(\mathfrak{p}_i) \cap f^{-1}(s) \quad \text{is dense in} \quad V(\mathfrak{p}_i) \cap f^{-1}(s) \qquad \text{for all } s \in D(a_i')$$

Step 6: Assembly.
Put $a = a' \prod_{i \in I} a_i'$, a nonzero element of the domain A, and let $V = D(a)$, a nonempty open of S. Fix $s \in V$ and abbreviate $C_i = V(\mathfrak{p}_i) \cap f^{-1}(s)$ for $i \in I$; these are closed in $f^{-1}(s)$ and, by Step 3, they cover it. Taking closures inside $f^{-1}(s)$, and using that the closure of a subset of the closed set C_i is the same computed in C_i or in $f^{-1}(s)$, Step 5 gives

$$\overline{U_s} \supseteq \bigcup_{i \in I} \overline{U_s \cap C_i} = \bigcup_{i \in I} C_i = f^{-1}(s)$$

So U_s is dense in X_s for every $s \in V$, as we sought to show.

The proof produces the open set explicitly. It is $D(a)$ for a a product of two kinds of factors: elements of A that kill the components of X lying over proper closed subsets of S, and, for each component that dominates, the element supplied by generic freeness for the quotient $B_i/\beta_i B_i$. Both kinds are computable in examples. Take $S = \operatorname{Spec} k[t]$, $X = \operatorname{Spec} k[t,x,y]/(xy - t)$ and $U = D(x)$. Here $B/xB = k[y]$ carried by the base as a t-torsion module, which becomes free, indeed zero, only after inverting t; and the answer $V = D(t)$ is correct, because the fiber over $t = 0$ is the union of the two axes $xy = 0$ while U misses the axis $x = 0$ entirely. Every other fiber is a single hyperbola on which x is invertible.

Integrality of S is used twice. Irreducibility supplies the membership relation exploited in Step 2, so that finitely many shrinkings of the base can be intersected without emptying it; and A being a domain is both what generic freeness demands and what makes $a \neq 0$ equivalent to $D(a) \neq \emptyset$. Over a base that is only Noetherian the statement must be applied to each irreducible component of S_{red} in turn.

The hypothesis cannot be relaxed from density of U_η to nonemptiness. With $S = \operatorname{Spec} k[t]$, $X = \operatorname{Spec} k[t,x,y]/(xy)$ and $U = D(x)$, the generic fiber has two components and U_η meets only one of them; every closed fiber looks the same, so no shrinking of the base helps. The content of the proposition is that this failure, once it is absent at η, is absent over a whole neighborhood.

One word on what this replaces. EGA IV_3, section 9, builds a general machine for questions of this shape: a scheme of finite presentation is written as a pullback from a Noetherian base of finite type over $\mathbb{Z}$, and a long list of properties is shown to ascend and descend along such limits, so that almost any reasonable condition holding at η holds over a neighborhood. That machine gives uniformity across many properties at once, at the cost of the limit formalism, which this book does not develop. The substitute used here is narrower and adequate for the families met later: generic freeness converts a statement about one module over the generic point into a statement over a distinguished open, and the constructible sets of definition 5.2.2 ensure that a condition assembled from finitely many such data is either empty or contains a nonempty open of an irreducible base. The price is that each property must be spread out by hand, as was done above for density.

The proof reveals more than the statement: it shows that the image of a dominant finite type morphism *contains* a dense open subset of the target. Let us record this and a few other consequences.

Corollary 5.2.10: Image Contains a Dense Open of its Closure

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes. Then $f(X)$ contains a dense open subset of $\overline{f(X)}$.

***Proof*:**
By theorem 5.2.7, $f(X)$ is constructible, hence a finite union of locally closed sets $f(X) = \bigsqcup_{i=1}^{n}(U_i \cap Z_i)$, where U_i is open and Z_i is closed. We may assume this decomposition is irredundant. Let $\overline{f(X)}$ be the closure of $f(X)$. Since $f(X)$ is dense in $\overline{f(X)}$, at least one of the locally closed pieces, say $U_1 \cap Z_1$, must be dense in some irreducible component of $\overline{f(X)}$. Then $U_1 \cap Z_1$ contains a dense open subset of that component. Applying this argument to each irreducible component gives a dense open subset of $\overline{f(X)}$ contained in $f(X)$.

Alternatively, one can argue directly: replace Y by $\overline{f(X)}$ with its reduced induced structure. Then $f : X \to \overline{f(X)}$ is dominant, and the result follows from Step 5 of the proof of theorem 5.2.7 (applied to each irreducible component).

Corollary 5.2.11: Open and Closed Morphisms via Constructibility

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes.

1. If f is an open map, then $f(X)$ is open. If f is a closed map (e.g., if f is proper by proposition 3.5.21), then $f(X)$ is closed.
2. More generally, $f(X)$ is open (resp. closed) if and only if it is stable under generization (resp. specialization).

***Proof*:**
Part (1) is immediate. For (2), a constructible set that is stable under generization is open (in a Noetherian space, this follows from the fact that the closure of a constructible set is determined by its specializations). The closed case is dual.

5.2.2 Application: Fiber Dimension

One of the most important applications of Chevalley's theorem is to the study of how fibers vary in a family. Given a morphism $f : X \to Y$, the *fiber* over a point $y \in Y$ is the scheme $X_y = X \times_Y \operatorname{Spec} \kappa(y)$ (definition 3.2.9). The dimension of this fiber can jump as y varies, but Chevalley's theorem constrains *how* it can jump.

Definition 5.2.12: Fiber Dimension

Let $f : X \to Y$ be a morphism of schemes and $x \in X$ a point with $y = f(x)$. The *fiber dimension of f at x* is:

$$\dim_x(f) = \dim_x(X_y)$$

where $X_y = f^{-1}(y)$ is the scheme-theoretic fiber over y, and $\dim_x$ denotes the local dimension at x (the maximum of the dimensions of the irreducible components of X_y containing x).

Theorem 5.2.13: Upper Semicontinuity of Fiber Dimension

Let $f : X \to Y$ be a morphism of finite type of Noetherian schemes. Then for each integer $n \geq 0$, the set:

$$\{x \in X : \dim_x(f) \geq n\}$$

is closed in X. In other words, the function $x \mapsto \dim_x(f)$ is upper semicontinuous.

***Proof*:**
The question is local, so we may assume $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ are affine with B a finitely generated A-algebra. For a point $x \in X$ corresponding to a prime $\mathfrak{q} \subseteq B$, with $y = f(x)$ corresponding to $\mathfrak{p} = \mathfrak{q} \cap A$, the fiber dimension at x is:

$$\dim_x(f) = \dim(B \otimes_A \kappa(\mathfrak{p}))_{\overline{\mathfrak{q}}}$$

where $\overline{\mathfrak{q}}$ is the image of $\mathfrak{q}$ in $B \otimes_A \kappa(\mathfrak{p})$. We use the following algebraic fact: for a finitely generated $\kappa(\mathfrak{p})$-algebra C and a prime $\mathfrak{q} \subseteq C$, we have $\dim C_{\mathfrak{q}} = \operatorname{trdeg}_{\kappa(\mathfrak{p})} \kappa(\mathfrak{q}) + \dim(C_{\mathfrak{q}}/\mathfrak{q}C_{\mathfrak{q}})$ by theorem 2.6.12. In particular, the fiber dimension is controlled by transcendence degrees.

We now use Chevalley's theorem. Consider the morphism $g : X \to X \times_Y X$ given by the diagonal, and its "relative" version. More directly: for each r, consider the closed subset $Z_r \subseteq X$ defined as follows. By working locally, we can express the fiber dimension condition using the rank of the Jacobian matrix of the defining equations. Alternatively, we argue as follows.

Consider the morphism $X \xrightarrow{f} Y$ and, for a given n, the subset $S_n = \{x \in X : \dim_x(f) \geq n\}$. To show S_n is closed, it suffices to show that for every irreducible closed subset $Z \subseteq X$ with $Z \cap S_n$ dense in Z, we have $Z \subseteq S_n$. This follows from a dimension-theoretic argument: the function $x \mapsto \dim_x(f)$ is constructible on X (as a consequence of Chevalley's theorem applied to appropriate projections), and a constructible function that achieves its minimum on a dense subset of an irreducible set achieves that minimum everywhere. The upper semicontinuity then follows from the fact that on each fiber, the local dimension is upper semicontinuous (theorem 2.6.12 combined with theorem 2.6.13).

A more self-contained argument proceeds by Noetherian induction on Y and uses the Generic Freeness lemma to control fibers over a dense open, much as in the proof of Chevalley's theorem. We leave the details to the reader as a guided exercise (cf. [65, Exercise 11.4.A]).

The upshot is that fibers can "jump up" in dimension on closed subsets, but not on open ones. Intuitively, the dimension of fibers can only increase as you specialize.

Example 5.2.14: Fiber Dimension in Practice

1. *Blowing up.* Let $f : \mathrm{Bl}_0 \mathbb{A}^2_k \to \mathbb{A}^2_k$ be the blow-up of the plane at the origin (see definition 7.2.26). The generic fiber (over any point $y \neq 0$) is a single point, so $\dim(f^{-1}(y)) = 0$. But the exceptional fiber $f^{-1}(0) \cong \mathbb{P}^1_k$ has dimension 1. The fiber dimension jumps from 0 to 1 at the origin—consistent with semicontinuity, since $\{0\}$ is closed.

2. *Projection of a surface.* Let $X = V(xz-y) \subseteq \mathbb{A}^3_k$ and $f : X \to \mathbb{A}^1_k$ the projection $(x, y, z) \mapsto x$. For $x = a \neq 0$, the fiber is $V(az - y) \cong \mathbb{A}^1_k$ (a line). For $x = 0$, the fiber is $V(y) \subseteq \mathbb{A}^2_k$, which is a copy of $\mathbb{A}^1_k$ parametrized by z. So the fiber dimension is 1 everywhere—no jumping. This is consistent with f being flat (the ring map $k[x] \to k[x, y, z]/(xz - y) \cong k[x, z]$ makes $k[x, z]$ a free $k[x]$-module).

3. *The $(x, y) \mapsto (x, xy)$ example revisited.* For the morphism $f : \mathbb{A}^2_k \to \mathbb{A}^2_k$ from example 5.2.1, the fiber over a point (a, b) with $a \neq 0$ is a single point $(a, b/a)$, so $\dim = 0$. The fiber over $(0, 0)$ is $V(x, xy) = V(x) = \mathbb{A}^1_k$ (the entire y-axis), so $\dim = 1$. The fiber over $(0, b)$ with $b \neq 0$ is $V(x, xy - b) = V(x, b) = \emptyset$. The locus where $\dim_x(f) \geq 1$ is $\{x = 0\} \subseteq \mathbb{A}^2_k$, which is indeed closed.

The examples above suggest a link between flatness and the constancy of fiber dimension. We know that flat morphisms between nice schemes are equidimensional (the fiber dimension does not jump). under the right algebraic hypotheses, the converse is true. This is the celebrated "Miracle Flatness" theorem.

Theorem 5.2.15: Miracle Flatness Theorem

Let $f : X \to Y$ be a morphism of finite type between locally Noetherian schemes. Suppose that:

1. Y is a regular scheme (e.g., a smooth variety),
2. X is Cohen–Macaulay.

Then f is flat if and only if all non-empty fibers X_y have the exact same "expected" dimension, namely $\dim X - \dim Y$. (In other words, f is flat $\iff$ f is equidimensional).

***Proof*:**
The question is entirely local, so we may assume $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ are affine, and we can localize at a point $x \in X$ and its image $y = f(x) \in Y$. This reduces the problem to a local homomorphism of local Noetherian rings $\varphi : (A, \mathfrak{m}) \to (B, \mathfrak{n})$.

By hypothesis, A is a regular local ring, B is a Cohen–Macaulay local ring, and the fiber dimension condition translates to the dimension formula:

$$\dim B = \dim A + \dim(B/\mathfrak{m}B).$$

We want to show that B is a flat A-module.

Since A is regular, its maximal ideal $\mathfrak{m}$ is generated by a regular sequence of parameters $a_1, \ldots, a_d$, where $d = \dim A$.

Because B is Cohen–Macaulay, it is universally catenary and the dimension formula tells us that quotienting B by the ideal $\mathfrak{m}B = (a_1, \ldots, a_d)B$ drops the dimension of B by exactly d. In a Cohen–Macaulay ring, any sequence of elements that drops the dimension by its length must be a regular sequence. Therefore, the images $\varphi(a_1), \ldots, \varphi(a_d)$ form a regular sequence in B.

We now invoke the *Local Criterion for Flatness* (see [15]): if $(A, \mathfrak{m}) \to (B, \mathfrak{n})$ is a local homomorphism, and an ideal $I \subseteq A$ is generated by a regular sequence that maps to a regular sequence in B, then B is flat over A. Applying this to $I = \mathfrak{m}$ completes the proof.

5.2.16 Remark: Why is it a "Miracle"? Checking flatness algebraically can be incredibly difficult, as it requires computing Tor groups or checking exactness for all modules. Checking that fibers have constant dimension is a purely topological/geometric task. The "miracle" is that geometry implies algebra here.
The hypotheses are absolutely necessary to prevent algebraic pathologies from hiding behind the geometry:

- If X is not Cohen–Macaulay, it might have embedded points (fuzz) that don't affect the topological dimension of the fibers but destroy algebraic flatness.
- If Y is not regular, the base might have singularities that force any flat family over it to also be highly singular, breaking the dimension count.

The interplay between fiber dimension and cohomology runs deep. In section 6.8.3 we shall prove a more refined semicontinuity result: for a projective morphism $f : X \to Y$ and a coherent sheaf $\mathscr{F}$ flat over Y, the function $y \mapsto h^i(X_y, \mathscr{F}_y)$ is upper semicontinuous. The Euler characteristic $y \mapsto \chi(X_y, \mathscr{F}_y)$ is locally constant. This is a far-reaching generalization whose proof uses a completely different technique (reducing to linear algebra on a complex of free modules), but the philosophical point is the same: cohomological invariants of fibers can only jump up on closed subsets.

5.2.3 The Scheme-Theoretic Image

Chevalley's theorem tells us that the set-theoretic image is constructible, but it says nothing about scheme structure. For many purposes—particularly when studying families or intersections—we need an image that lives in the category of schemes, not just the category of topological spaces. The set-theoretic image rarely has a natural scheme structure (as example 5.2.1 illustrates), so we do the next best thing: we take the smallest *closed* subscheme containing the image.

The idea is natural from the sheaf-theoretic perspective developed in this chapter. A regular function g on Y should "vanish on the image" if and only if g pulls back to zero on X, i.e., $\pi^\#(g) = 0$ in $\mathcal{O}_X$. The collection of all such g forms an ideal sheaf, and the corresponding closed subscheme is the scheme-theoretic image.

Definition 5.2.17: Scheme-Theoretic Image

Let $\pi : X \to Y$ be a morphism of schemes. We say the *image of* π *lies in a closed subscheme* $\iota : Z \hookrightarrow Y$ if the composition:

$$\mathcal{I}_{Z/Y} \longrightarrow \mathcal{O}_Y \xrightarrow{\pi^\#} \pi_*\mathcal{O}_X$$

is zero, where $\mathcal{I}_{Z/Y} = \ker(\mathcal{O}_Y \to \iota_*\mathcal{O}_Z)$ is the ideal sheaf of Z (definition 5.1.30). Equivalently, π factors through Z: every function on Y that vanishes on Z pulls back to zero on X.
The *scheme-theoretic image* of π is the closed subscheme of Y defined by the ideal sheaf:

$$\mathcal{I} = \ker\big(\mathcal{O}_Y \xrightarrow{\pi^\#} \pi_*\mathcal{O}_X\big)$$

That is, it is the smallest closed subscheme of Y through which π factors.

The scheme-theoretic image is well-defined: if the image lies in closed subschemes Z_1 and Z_2, it lies in their scheme-theoretic intersection $Z_1 \cap Z_2$ (the closed subscheme defined by $\mathcal{I}_{Z_1} + \mathcal{I}_{Z_2}$, see proposition 5.1.37). Hence the intersection of all closed subschemes containing the image exists and is the scheme-theoretic image.

Note that the scheme-theoretic image depends on the *scheme structures* of both X and Y, not just on the set-theoretic image. In particular, a non-reduced source can produce a non-reduced image, even when the set-theoretic image is a single point.

There is an important subtlety: the ideal sheaf $\mathcal{I} = \ker(\mathcal{O}_Y \to \pi_*\mathcal{O}_X)$ need not be quasi-coherent in general (the pushforward $\pi_*\mathcal{O}_X$ is not always quasi-coherent). However, by proposition 5.1.20, if π is quasi-compact and quasi-separated, then $\pi_*\mathcal{O}_X$ is quasi-coherent, and hence so is $\mathcal{I}$. In this case, the scheme-theoretic image is a honest closed subscheme (proposition 5.1.34).

The relationship between the scheme-theoretic image and the set-theoretic image is controlled by the quasi-compactness of π:

Proposition 5.2.18: Set-Theoretic Image and Scheme-Theoretic Image

Let $\pi : X \to Y$ be a quasi-compact morphism of schemes, and let Z be the scheme-theoretic image. Then the underlying set of Z is the closure of the set-theoretic image: $|Z| = \overline{\pi(X)}$.

***Proof*:**
Since π is quasi-compact, $\pi_*\mathcal{O}_X$ is quasi-coherent (proposition 5.1.20), and the question is local on Y. So assume $Y = \operatorname{Spec} A$ and $\pi_*\mathcal{O}_X(Y) = B$ (some A-algebra). The ideal sheaf $\mathcal{I}$ corresponds to $I = \ker(A \to B)$, and the scheme-theoretic image is $\operatorname{Spec}(A/I)$.

We must show $V(I) = \overline{\pi(X)}$. A prime $\mathfrak{p} \in \operatorname{Spec} A$ is in $V(I)$ if and only if $I \subseteq \mathfrak{p}$, i.e., every element of A that maps to zero in B lies in $\mathfrak{p}$. On the other hand, $\mathfrak{p} \in \overline{\pi(X)}$ if and only if $\mathfrak{p}$ is in the closure of the set-theoretic image.

If $\mathfrak{p} \in \pi(X)$, then $\mathfrak{p} = \pi(\mathfrak{q})$ for some $\mathfrak{q} \in X$, and the composition $A \to B \to \kappa(\mathfrak{q})$ sends I to zero, so $I \subseteq \mathfrak{p}$. Hence $\pi(X) \subseteq V(I)$. Since $V(I)$ is closed, $\overline{\pi(X)} \subseteq V(I)$.

For the reverse inclusion, suppose $\mathfrak{p} \supseteq I$. We must show $\mathfrak{p} \in \overline{\pi(X)}$, i.e., every open set containing $\mathfrak{p}$ meets $\pi(X)$. Let $D(a)$ be a basic open neighborhood of $\mathfrak{p}$, so $a \notin \mathfrak{p}$ and in particular $a \notin I$.

Then the image of a in B is nonzero, which means $\pi^{-1}(D(a)) \neq \emptyset$ (the function a does not vanish on X). Hence $D(a)$ meets $\pi(X)$.

Chevalley's theorem (theorem 5.2.7) says that the image of a morphism of finite type is constructible, and example 5.2.1 shows that constructible is the most one can ask for in general: images of morphisms of schemes are rarely closed, and rarely even locally closed. A group law changes the situation entirely. If the morphism is a homomorphism of group schemes (definition 4.2.1), then its image absorbs its own translates, and a constructible set that is stable under translation by its own points has no room left to be anything but closed. The classical form of this statement is proved for affine algebraic groups over an algebraically closed field of characteristic zero, where smoothness and generic smoothness are on hand; none of those three hypotheses is needed.

Proposition 5.2.19: The Image of a Homomorphism of Group Schemes Is Closed

Let k be a field and let $f\colon G \to H$ be a homomorphism of group schemes of finite type over k. Then:

1. the set-theoretic image $f(G)$ is a closed subset of H;
2. the scheme-theoretic image of f (definition 5.2.17) is a closed subgroup scheme of H whose underlying set is $f(G)$;
3. if k is perfect, the set $f(G)$ with its reduced induced structure is a closed subgroup scheme of H as well.

***Proof*:**
Both G and H are Noetherian, and f is a morphism of finite type.[a]

Step 1: Reduction to an algebraically closed field. Write $\pi_G\colon G_{\overline{k}} \to G$ and $\pi_H\colon H_{\overline{k}} \to H$ for the projections and $f' = f_{\overline{k}}$, a homomorphism of group schemes of finite type over $\overline{k}$. Every element of $\overline{k}$ is algebraic over k, so $\operatorname{Spec}\overline{k} \to \operatorname{Spec} k$ is an integral morphism; integrality is stable under base change, so π_H is integral and therefore a closed map by proposition 3.4.22. Both projections are surjective, the fiber over x being $\operatorname{Spec}(\kappa(x) \otimes_k \overline{k})$, the spectrum of a nonzero ring. From $\pi_H \circ f' = f \circ \pi_G$ and surjectivity of π_G we get $\pi_H(f'(G_{\overline{k}})) = f(\pi_G(G_{\overline{k}})) = f(G)$. So if $f'(G_{\overline{k}})$ is closed then so is its image $f(G)$ under the closed map π_H, and we may assume $k = \overline{k}$.

Step 2: Closed points. Set $E = f(G) \subseteq H$ and let $\Gamma \subseteq H(k)$ be the image of $G(k) \to H(k)$, a subgroup. Since H is of finite type over $k = \overline{k}$, the residue field at a closed point is a finite extension of k, hence k itself, so the closed points of H are exactly the elements of $H(k)$; moreover H is a Jacobson scheme (definition 3.7.4, by lemma 2.4.4), and the same holds for G. Two consequences will be used repeatedly. First, every nonempty constructible subset of H contains a closed point of H, because it contains a nonempty locally closed set $U \cap C$ and the closed points of C are dense in C; hence a constructible set all of whose closed points lie in a closed set C is contained in C. Second, $E \cap H(k) = \Gamma$: if $h \in H(k)$ lies in E, its fiber $G \times_{H,h} \operatorname{Spec} k$ is a nonempty k-scheme of finite type, so it has a closed point, and that point is a k-point mapping to h.

Step 3: Translations preserve the image. Let $\gamma = f(g)$ with $g \in G(k)$. On T-valued points the left translation λ_γ of example 4.3.2 satisfies $\lambda_\gamma(f(x)) = f(g)f(x) = f(gx)$, so $\lambda_\gamma \circ f = f \circ \lambda_g$ as morphisms of k-schemes. As λ_g is an automorphism of G, this gives $\lambda_\gamma(E) = f(\lambda_g(G)) = E$, and the same computation with right translations and with $\iota_H \circ f = f \circ \iota_G$ gives $\rho_\gamma(E) = E$ and $\iota_H(E) = E$.

Step 4: The closure. Let $Z = \overline{E}$, nonempty since the identity lies in E. Each of λ_γ, ρ_γ and ι_H is a homeomorphism of H carrying E onto E, hence carrying Z onto Z. By theorem 5.2.7 the set E is constructible, and its closed points are exactly the elements of Γ by Step 2, so $E \subseteq \overline{\Gamma}$ by the first consequence recorded there; therefore $Z = \overline{\Gamma}$.

Step 5: The closed points of Z act on Z. Let $z \in Z \cap H(k)$. Since $\lambda_\gamma(Z) = Z$ for $\gamma \in \Gamma$, the point γz lies in Z, so $\rho_z(\Gamma) \subseteq Z$; continuity and $Z = \overline{\Gamma}$ then give $\rho_z(Z) \subseteq Z$, and symmetrically $\lambda_z(Z) \subseteq Z$. Applying $\iota_H(Z) = Z$ shows $z^{-1} \in Z \cap H(k)$, whence $Z = \lambda_z(\lambda_{z^{-1}}(Z)) \subseteq \lambda_z(Z)$ and $\lambda_z(Z) = Z$, and likewise $\rho_z(Z) = Z$. In particular $Z \cap H(k)$ is a subgroup of $H(k)$.

Step 6: The covering argument. By corollary 5.2.10 there is a dense open $V \subseteq Z$ with $V \subseteq E$. Fix a closed point $z \in Z \cap H(k)$ and put $\theta = \lambda_z \circ \iota_H$, an automorphism of H over k carrying Z onto Z by Steps 4 and 5. Then $\theta(V)$ is a dense open subset of Z. In any topological space the intersection of two dense open subsets is dense open, so $V \cap \theta(V)$ is a nonempty open subset of Z and therefore contains a closed point v of Z, which is a closed point of H since Z is closed. Let $w = \theta^{-1}(v) \in V$; as θ is an automorphism over k, the point w is again k-rational. In the group $H(k)$ the relation $v = \theta(w) = zw^{-1}$ reads $z = vw$, and $v, w \in V \cap H(k) \subseteq E \cap H(k) = \Gamma$ by Step 2, so $z \in \Gamma \subseteq E$. Thus every closed point of Z lies in E. The difference $Z \setminus E$ is constructible and contains no closed point of H, so it is empty. Hence $E = Z$ is closed, which with Step 1 proves (1).

Step 7: The scheme-theoretic image. Return to an arbitrary field k and let $I \subseteq H$ be the scheme-theoretic image of f. Since f is quasi-compact, $|I| = \overline{f(G)} = f(G)$ by proposition 5.2.18 and part (1). The identity section $\epsilon_H = f \circ \epsilon_G$ factors through I by the defining property of I, and ι_H, being an automorphism of H with $\iota_H \circ f = f \circ \iota_G$ and ι_G an automorphism of G, carries I onto the scheme-theoretic image of $\iota_H \circ f$, which is I again. For multiplication, note that $\mathcal{O}_I \to f_*\mathcal{O}_G$ is injective, and that formation of the scheme-theoretic image of a quasi-compact separated morphism commutes with flat base change: on an affine open $\operatorname{Spec} A \subseteq H$ the image ideal is $\ker(A \to B)$ with $B = \Gamma(f^{-1}(\operatorname{Spec} A), \mathcal{O}_G)$ computed as the equalizer of a finite Čech complex, and tensoring by a flat A-algebra is exact. Base changing f along the flat projection $H \times_k G \to H$ shows that $I \times_k G$ is the scheme-theoretic image of $f \times \mathrm{id}_G$, and base changing along $I \times_k H \to H$ shows that $I \times_k I$ is the scheme-theoretic image of $\mathrm{id}_I \times f$. Pushforward is left exact, so the composite $G \times_k G \to I \times_k I$ again has scheme-theoretic image $I \times_k I$. Now $\mu_H \circ (f \times f) = f \circ \mu_G$ factors through I, so $G \times_k G$ factors through the closed subscheme $C = (I \times_k I) \times_{\mu_H, H} I$ of $I \times_k I$; the ideal sheaf of C is contained in the kernel of $\mathcal{O}_{I \times_k I} \to (f \times f)_*\mathcal{O}_{G \times_k G}$, which vanishes, so $C = I \times_k I$ and μ_H restricts to $I \times_k I \to I$. This proves (2).

Step 8: The reduced structure over a perfect field. Let Y be $f(G)$ with its reduced induced structure. Over a perfect field a reduced k-scheme is geometrically reduced, so $Y \times_k Y$ is reduced. Base change along $\operatorname{Spec} \overline{k} \to \operatorname{Spec} k$ carries $Y \times_k Y$ to $Y_{\overline{k}} \times_{\overline{k}} Y_{\overline{k}}$, whose underlying space is $Z \times Z$ on closed points; Step 5 sends such a pair to a closed point of Z, and the image of μ_H there is constructible, so it lies in Z by Step 2. Pushing forward along the surjection π_H as in Step 1 gives $\mu_H(Y \times_k Y) \subseteq f(G)$. A morphism from a reduced scheme whose set-theoretic image lies in a closed subset factors through the reduced induced structure on that subset, so μ_H restricts to $Y \times_k Y \to Y$, and the same applies to ι_H and ϵ_H, completing the proof.

[a] A group scheme of finite type over a field is separated: the identity is a k-rational point, hence a closed point with residue field k, so ϵ_H is a closed immersion, and the diagonal of H is its pullback along $(x, y) \mapsto xy^{-1}$. Since

G is quasi-compact and H is quasi-separated, f is quasi-compact, and it is locally of finite type because G is of finite type over k.

The group law entered only through translations, and the field only through the passage to $\overline{k}$, where closed points become $\overline{k}$-points and the Jacobson property lets a constructible set be tested on them. Nothing in the argument sees affineness, characteristic, reducedness or smoothness: G and H may be abelian varieties, k may have characteristic p, and either group may be infinitesimal, as μ_p and α_p are. What replaces the generic smoothness of the classical proof is corollary 5.2.10, which is available over any field and for any finite type morphism.

The gap between (2) and (3) is real. Over an imperfect field a product of reduced k-schemes can fail to be reduced, and a group scheme of finite type over such a field need not have its underlying reduced subscheme be a subgroup scheme; taking $f = \mathrm{id}_G$ for such a G turns it into a counterexample to (3). The scheme-theoretic image has no such defect, which is one more reason to treat it, rather than the reduced image, as the image of a morphism of schemes.

With (1) in hand the corestriction $G \to f(G)$ is a homomorphism of group schemes of finite type onto a closed subgroup scheme, and the kernel of definition 4.2.2 sits on the left of the resulting sequence. The caution about exactness stands: surjectivity on the right is faithful flatness of $G \to f(G)$, not surjectivity of $G(T) \to f(G)(T)$, and the two differ already for the n-th power map on $\mathbb{G}_m$ over $\mathbb{Q}$.

Without the quasi-compactness hypothesis, the scheme-theoretic image can be strictly larger than the closure of the set-theoretic image, as the pathological example in the following illustrates:

Example 5.2.20: Scheme-Theoretic Image

1. *Non-reduced source.* The morphism $\pi : \operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \to \mathbb{A}^1_k = \operatorname{Spec} k[x]$ given by $x \mapsto \epsilon$ has scheme-theoretic image $V(x^2) = \operatorname{Spec}(k[x]/(x^2))$: a polynomial $g(x)$ pulls back to zero precisely when $g(\epsilon) = 0$ in $k[\epsilon]/(\epsilon^2)$, which forces $g \in (x^2)$. The set-theoretic image is just the origin, but the scheme-theoretic image is a "fat point" of length 2, remembering the tangent direction.

   Compare this with the discussion of nilpotent thickening in example 2.2.8: the scheme-theoretic image captures the first-order infinitesimal data of the source.

2. *Reduced image from non-reduced source.* The morphism $\operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \to \mathbb{A}^1_k$ given by $x \mapsto 0$ has scheme-theoretic image $V(x) = \operatorname{Spec}(k)$: now $g(x)$ pulls back to $g(0)$, which vanishes precisely when $g \in (x)$. Although the source is non-reduced, the image is reduced. The difference from (1) is that the tangent direction is "crushed" by the map—the non-reduced structure of the source is not reflected in the image.

3. *Dense image = scheme-theoretic closure.* The open immersion $j : \mathbb{A}^1_k \setminus \{0\} = \operatorname{Spec} k[t, t^{-1}] \hookrightarrow \mathbb{A}^1_k = \operatorname{Spec} k[u]$ given by $u \mapsto t$ has scheme-theoretic image all of $\mathbb{A}^1_k$: a polynomial $g(u)$ that becomes zero in $k[t, t^{-1}]$ must be the zero polynomial. This is consistent with proposition 5.2.18: the set-theoretic image is $\mathbb{A}^1_k \setminus \{0\}$, whose closure is $\mathbb{A}^1_k$. In this sense, the scheme-theoretic image is the *scheme-theoretic closure* of the image.

4. *Pathological example (non-quasi-compact source).* Consider the morphism:

$$\pi : \coprod_{n \geq 1} \operatorname{Spec}\left(\frac{k[\epsilon_n]}{(\epsilon_n^n)}\right) \longrightarrow \mathbb{A}^1_k = \operatorname{Spec} k[x]$$

given by $x \mapsto \epsilon_n$ on the nth component. A polynomial $g(x)$ pulls back to zero on the nth component precisely when g vanishes to order $\geq n$ at the origin. Since this must hold for *all* n, the only such g is 0. The scheme-theoretic image is therefore all of $\mathbb{A}^1_k$, even though the set-theoretic image is just the origin.

This pathology arises because the source is not quasi-compact. For quasi-compact morphisms, the scheme-theoretic image has underlying set equal to the closure of the set-theoretic image (proposition 5.2.18).

The scheme-theoretic image has a clean universal property: it is the smallest closed subscheme $Z \hookrightarrow Y$ such that π factors through Z. This makes it functorial and well-behaved in diagram chases. For instance, if $\pi = \iota \circ g$ where $\iota : Z \hookrightarrow Y$ is a closed immersion, then the scheme-theoretic image of π is contained in Z (as a closed subscheme). This is the algebraic incarnation of the geometric intuition that "the image of a composition lies in the image of the outer map."

We close with a remark connecting the two threads of this section:

5.2.21 Set-Theoretic vs. Scheme-Theoretic: When Do They Agree? For a morphism $\pi : X \to Y$ of finite type between Noetherian schemes, the set-theoretic image $\pi(X)$ is constructible (Chevalley), and the scheme-theoretic image Z has underlying set $\overline{\pi(X)}$ (the closure). These agree—i.e., $|\text{scheme-theoretic image}| = \pi(X)$—precisely when the image is already closed. This happens, for instance, when π is proper (definition 3.5.20), or more generally when π is universally closed.
Even when the underlying sets agree, the scheme-theoretic image may carry non-reduced structure (as in example (1) above). The scheme-theoretic image is the "right" notion of image whenever one cares about scheme structure—for instance, when computing intersections, studying flatness, or constructing moduli spaces.

5.3 Q.C. Sheaves on Projective Schemes

On an affine scheme $\operatorname{Spec} R$, the tilde construction $M \mapsto \widetilde{M}$ establishes an equivalence between R-modules and quasi-coherent sheaves (corollary 5.1.12). For a projective scheme $X = \operatorname{Proj} S_\bullet$, the analogous construction starts with *graded* $S_\bullet$-modules and produces quasi-coherent sheaves on X. The main result of this section is that every quasi-coherent sheaf on $\operatorname{Proj} S_\bullet$ arises in this way (proposition 5.3.14), and that — when $S_\bullet$ is a polynomial ring — the graded ring itself is recovered as a direct sum of global sections of certain natural line bundles, the *twisting sheaves* (proposition 5.3.10).

5.3.1 The Tilde Construction on Proj

Recall from theorem 2.3.3 that the structure sheaf of $X = \operatorname{Proj} S_\bullet$ is defined on distinguished opens by $\mathcal{O}_X(D_+(f)) = (S_f)_0$: we localize at a homogeneous element f and then take the degree-zero part. The same recipe generalizes from the ring $S_\bullet$ to any graded module over it.

Definition 5.3.1: Sheaf Associated to M [on Projective Scheme]

Let $S_\bullet$ be a graded ring and $M_\bullet$ a graded $S_\bullet$-module. The *sheaf associated to* $M_\bullet$ on $X = \operatorname{Proj} S_\bullet$, denoted $\widetilde{M}$, is the $\mathcal{O}_X$-module defined on the basis of projective distinguished opens by

$$\widetilde{M}\big(D_+(f)\big) \;=\; \big(M[f^{-1}]\big)_0,$$

where $f \in S_\bullet$ is a nonzero homogeneous element and $(-)_0$ denotes the degree-zero component. The restriction maps are induced by further localization.

Let us unpack this. The localization $M[f^{-1}]$ inverts f in the graded sense: a typical element is m/f^k where $m \in M$ is homogeneous. This element has degree $\deg m - k\deg f$. Taking the degree-zero part selects those fractions m/f^k where $\deg m = k \deg f$. When $M = S_\bullet$ this recovers the structure sheaf $(S_f)_0 = \mathcal{O}_X(D_+(f))$; for a general module, the degree-zero localization $(M_f)_0$ is a module over the ring $(S_f)_0$.

Concretely, if $S_\bullet = R[x_0, \dots, x_r]$ is the standard polynomial ring graded by degree, then $D_+(x_i) \cong \operatorname{Spec}(S_{x_i})_0$ and

$$\widetilde{M}(D_+(x_i)) \;=\; (M_{x_i})_0,$$

which is a module over the affine coordinate ring $(S_{x_i})_0 \cong R[x_0/x_i, \dots, \widehat{x_i/x_i}, \dots, x_r/x_i]$.

Example 5.3.2: The Tilde Construction on $\mathbb{P}^1$

Let $S_\bullet = k[x,y]$ with the standard grading, so that $X = \mathbb{P}^1_k$. Take $M = S_\bullet/(x^2 - y^2)$, graded by the quotient grading, so that M has one generator in degree 0 and the relation $x^2 = y^2$ in degree 2.

On the chart $D_+(y) \cong \operatorname{Spec} k[t]$ (where $t = x/y$), the module of sections is

$$(M_y)_0 \;=\; \frac{k[x/y]}{((x/y)^2 - 1)} \;=\; \frac{k[t]}{(t^2-1)},$$

which is a $k[t]$-module. Similarly, on $D_+(x) \cong \operatorname{Spec} k[s]$ (where $s = y/x$):

$$(M_x)_0 \;=\; \frac{k[s]}{(1 - s^2)}.$$

Thus $\widetilde{M}$ restricts on each affine chart to the sheaf associated to a familiar quotient ring — the coordinate ring of two points $\{t = \pm 1\}$ on $\mathbb{A}^1$. The gluing of these local pieces recovers the sheaf on all of $\mathbb{P}^1$.

Proposition 5.3.3: Sheaf on Projective Scheme is Quasi-Coherent

Let $S_\bullet$ be a graded ring, $M_\bullet$ a graded $S_\bullet$-module, and $X = \operatorname{Proj} S_\bullet$. Then:

1. The stalk of $\widetilde{M}$ at a homogeneous prime $\mathfrak{p} \in X$ is $\widetilde{M}_\mathfrak{p} \cong M_{(\mathfrak{p})}$, the homogeneous localization of M at $\mathfrak{p}$.
2. $\widetilde{M}$ is a quasi-coherent $\mathcal{O}_X$-module. If $S_\bullet$ is Noetherian and $M_\bullet$ is finitely generated, then $\widetilde{M}$ is coherent.

***Proof*:**

On each basic open $D_+(f) \cong \operatorname{Spec}(S_f)_0$, the sheaf $\widetilde{M}$ restricts to the sheaf associated to the $(S_f)_0$-module $(M_f)_0$, which is quasi-coherent by the affine theory. The stalk computation follows from the same colimit argument as for the structure sheaf (theorem 2.3.3). For coherence, if $M_\bullet$ is finitely generated over a Noetherian ring $S_\bullet$, then each $(M_f)_0$ is finitely generated over the Noetherian ring $(S_f)_0$.

The structure sheaf $\mathcal{O}_X = \widetilde{S_\bullet}$ captures the degree-zero part of $S_\bullet$. To access sections of other degrees, we shift the grading:

Definition 5.3.4: Twisting Sheaf

Let $S_\bullet = \bigoplus_n S_n$ be a graded ring and $X = \operatorname{Proj} S_\bullet$. For each $n \in \mathbb{Z}$, define the *nth twisting sheaf* to be

$$\mathcal{O}_X(n) \;:=\; \widetilde{S(n)},$$

where $S(n)$ is the graded module with $S(n)_d = S_{n+d}$. The sheaf $\mathcal{O}_X(1)$ is called *the twisting sheaf* (or *Serre's twisting sheaf*) of X.

Note that $S(0) = S_\bullet$, so $\mathcal{O}_X(0) = \mathcal{O}_X$. Let us work out what the sections of $\mathcal{O}_X(n)$ look like concretely. On a distinguished open $D_+(f)$ for a homogeneous element f of degree e, the definition gives

$$\mathcal{O}_X(n)(D_+(f)) \;=\; \big(S(n)[f^{-1}]\big)_0 \;=\; \big(S[f^{-1}]\big)_n \;=\; (S_f)_n,$$

the elements of degree n in the localized ring S_f. For $n \neq 0$, this is no longer a ring, but it is a module over $\mathcal{O}_X(D_+(f)) = (S_f)_0$.

Let us see what this means explicitly on the standard projective space. Take $S_\bullet = k[x_0, \ldots, x_r]$ with the usual grading by total degree. On the chart $D_+(x_i) \cong \operatorname{Spec} k[x_0/x_i, \ldots, x_r/x_i]$, the ring $S_{x_i} = k[x_0, \ldots, x_r, x_i^{-1}]$ contains elements of every integer degree. An element of $(S_{x_i})_n$ is a sum of monomials in $x_0, \ldots, x_r, x_i^{-1}$ of total degree n. Since x_i is invertible, every such monomial can be written as x_i^n times a degree-zero monomial. More precisely, the map

$$(S_{x_i})_0 \xrightarrow[\cdot\, x_i^n]{\sim} (S_{x_i})_n, \qquad g \longmapsto x_i^n \cdot g, \tag{5.1}$$

is an isomorphism of $(S_{x_i})_0$-modules: multiplication by the unit x_i^n shifts the degree from 0 to n. This proves the following:

Proposition 5.3.5: $\mathcal{O}_X(n)$ is an Invertible Sheaf

Let $S_\bullet$ be a graded ring generated by S_1 as an S_0-algebra, and $X = \operatorname{Proj} S_\bullet$. Then $\mathcal{O}_X(n)$ is an invertible sheaf *(*line bundle*)* for each $n \in \mathbb{Z}$.

***Proof*:**

For each $f \in S_1$, multiplication by f^n defines an isomorphism $(S_f)_0 \xrightarrow{\sim} (S_f)_n$ of $(S_f)_0$-modules (as in (5.1)), giving $\mathcal{O}_X(n)|_{D_+(f)} \cong \mathcal{O}_{D_+(f)}$. Since $S_\bullet$ is generated by S_1, the opens $\{D_+(f) : f \in S_1\}$ cover X, so $\mathcal{O}_X(n)$ is locally free of rank 1.

Although $\mathcal{O}_X(n)$ is locally isomorphic to $\mathcal{O}_X$ on each chart, the sheaves $\mathcal{O}_X(n)$ for different n are *not* globally isomorphic — the local trivializations glue together differently, producing genuinely distinct line bundles. We shall now prove this by computing global sections. Before treating the general case, let us work through $\mathbb{P}^1$ in full detail.

Let $S_\bullet = k[x, y]$ and $X = \mathbb{P}^1_k = \operatorname{Proj} S_\bullet$, covered by the two charts

$$\begin{aligned} D_+(y) &\cong \operatorname{Spec} k[t], \quad t = x/y, \\ D_+(x) &\cong \operatorname{Spec} k[s], \quad s = y/x. \end{aligned}$$

On the overlap $D_+(x) \cap D_+(y) \cong \operatorname{Spec} k[t, t^{-1}]$, we have $s = 1/t$.

A section of $\mathcal{O}(n)$ over $D_+(y)$ is an element of $(S_y)_n$: a degree-n element of $k[x, y, y^{-1}]$. Any such element can be written uniquely as $y^n \cdot p(x/y) = y^n \cdot p(t)$ for a polynomial $p \in k[t]$. Similarly, a section over $D_+(x)$ is $x^n \cdot q(y/x) = x^n \cdot q(s)$ for a polynomial $q \in k[s]$. A global section must give a pair $(p(t), q(s))$ that *agree on the overlap.* On $D_+(x) \cap D_+(y)$, both expressions represent the same element of $(S_{xy})_n$:

$$y^n \cdot p(t) \;=\; x^n \cdot q(s) \;=\; (ty)^n \cdot q(1/t) \;=\; y^n \cdot t^n\, q(1/t).$$

Cancelling y^n (which is a unit on the overlap), the gluing condition is

$$p(t) \;=\; t^n \cdot q(1/t). \tag{5.2}$$

Let us now examine this condition for several values of n.

Case $n = 0$ (the structure sheaf). The condition becomes $p(t) = q(1/t)$. Writing $p(t) = \sum_{i \geq 0} a_i t^i$ and $q(s) = \sum_{j \geq 0} b_j s^j$, the right side is $\sum_j b_j t^{-j}$. For this to equal a polynomial in t (no negative powers), we need $b_j = 0$ for $j \geq 1$, so $q(s) = b_0$ is constant. Then $p(t) = b_0$ is also constant. Hence $\Gamma(\mathbb{P}^1, \mathcal{O}) = k$, recovering the familiar fact that the only global regular functions on $\mathbb{P}^1$ are constants (proposition 2.3.6).

Case $n = 1$ (the twisting sheaf). Now $p(t) = t \cdot q(1/t)$. Write $q(s) = b_0 + b_1 s + b_2 s^2 + \cdots$. Then

$$t \cdot q(1/t) \;=\; t\left(b_0 + \frac{b_1}{t} + \frac{b_2}{t^2} + \cdots\right) \;=\; b_0 t + b_1 + \frac{b_2}{t} + \cdots$$

For $p(t)$ to be a polynomial (no negative powers of t), we need $b_2 = b_3 = \cdots = 0$, giving $q(s) = b_0 + b_1 s$. Then $p(t) = b_1 + b_0 t$. The global section is thus parametrized by the pair $(b_0, b_1) \in k^2$, or equivalently by the degree-1 polynomial $b_0 x + b_1 y \in S_1$. Hence $\Gamma(\mathbb{P}^1, \mathcal{O}(1)) \cong S_1 \cong k^2$.

Case $n = 2$. By the same analysis with $p(t) = t^2 \cdot q(1/t)$, one finds $q(s) = b_0 + b_1 s + b_2 s^2$ and $p(t) = b_2 + b_1 t + b_0 t^2$. The global sections are parametrized by $(b_0, b_1, b_2) \in k^3$, corresponding to degree-2 polynomials $b_0 x^2 + b_1 xy + b_2 y^2 \in S_2$. So $\Gamma(\mathbb{P}^1, \mathcal{O}(2)) \cong S_2 \cong k^3$.

Case $n = -1$ *(negative twist).* Now $p(t) = t^{-1} \cdot q(1/t)$. Write $q(s) = b_0 + b_1 s + \cdots$. Then

$$t^{-1} \cdot q(1/t) \;=\; \frac{1}{t}\left(b_0 + \frac{b_1}{t} + \cdots\right) \;=\; \frac{b_0}{t} + \frac{b_1}{t^2} + \cdots$$

For this to be a polynomial in t, *every* coefficient must vanish: $b_0 = b_1 = \cdots = 0$. Hence $q = 0$ and $p = 0$: the only global section of $\mathcal{O}(-1)$ is zero. Geometrically, the transition function t^{-1} introduces a pole that no polynomial can cancel, so no nonzero section can be defined on both charts simultaneously.

General $n < 0$. The same argument applies: $p(t) = t^n q(1/t)$ for $n < 0$ forces all coefficients of q to vanish, so $\Gamma(\mathbb{P}^1, \mathcal{O}(n)) = 0$ for every $n < 0$. Note that these sheaves often be distinguished by having different local sections, where poles of different degrees can occur.

General $n \geq 0$. The gluing condition $p(t) = t^n q(1/t)$ is satisfied by polynomials q of degree $\leq n$, giving an $(n+1)$-dimensional space. Hence $\Gamma(\mathbb{P}^1, \mathcal{O}(n)) \cong S_n \cong k^{n+1}$ for $n \geq 0$.

The pattern generalizes to $\mathbb{P}^r_k$ with $S_\bullet = k[x_0, \ldots, x_r]$. The same argument — covering by the $r+1$ standard charts and imposing the gluing conditions — yields $\Gamma(\mathbb{P}^r_k, \mathcal{O}(n)) \cong S_n$ for $n \geq 0$, the vector space of homogeneous polynomials of degree n in $r+1$ variables, which has dimension $\binom{n+r}{r}$. For $n < 0$ and $r \geq 1$, the only global section is zero (the transition functions introduce poles that cannot be compensated on every chart).

The exceptional case is $\mathbb{P}^0_k = \operatorname{Proj} k[x_0] \cong \operatorname{Spec} k$: since there is only one chart, there are no nontrivial gluing conditions, and $\Gamma(\mathbb{P}^0_k, \mathcal{O}(n)) \cong k$ for all n.

Collecting these results:

$\dim_k \Gamma(\mathbb{P}^r_k, \mathcal{O}(n))$	$\mathbb{P}^0$	$\mathbb{P}^1$	$\mathbb{P}^2$	$\mathbb{P}^3$	$\mathbb{P}^4$
$n = -2$	1	0	0	0	0
$n = -1$	1	0	0	0	0
$n = 0$	1	1	1	1	1
$n = 1$	1	2	3	4	5
$n = 2$	1	3	6	10	15
general $n \geq 0$, $r \geq 1$	—	$\binom{n+1}{1}$	$\binom{n+2}{2}$	$\binom{n+3}{3}$	$\binom{n+4}{4}$

Since the spaces of global sections have different dimensions for different n (when $r \geq 1$), the line bundles $\mathcal{O}_X(n)$ are pairwise non-isomorphic. These dimensions will be computed systematically using sheaf cohomology in theorem 6.3.1, where we shall also determine $H^i(\mathbb{P}^r, \mathcal{O}(n))$ for $i > 0$: the vanishing of H^0 for negative twists is only the beginning of the story.

5.3.6 Why "Twisting"? The name comes from the transition functions. On the overlap $D_+(x) \cap D_+(y)$ of $\mathbb{P}^1_k$, the local trivializations (5.1) identify sections of $\mathcal{O}(n)$ with elements of the structure sheaf via multiplication by y^n (on the y-chart) and x^n (on the x-chart). The transition function relating these two trivializations is

$$\varphi_{xy} = \frac{x^n}{y^n} = t^n = s^{-n},$$

which is precisely the factor appearing in the gluing condition (5.2).
For $n = 0$, the transition function is 1: the bundle is trivial. For $n = 1$, the transition function is t: traversing the base $\mathbb{P}^1$ and returning to the starting chart, the fibre is multiplied by $t \cdot (1/t) = 1$ only after passing through *both* charts. Over $\mathbb{R}$, the total space of $\mathcal{O}(1)$ on $\mathbb{P}^1_{\mathbb{R}}$ is a Möbius-like band: the fibre must be "twisted" once to close up. For general n, the transition involves $|n|$ such twists (with opposite orientation for $n < 0$).
More precisely, the transition functions define an element of the Čech cohomology group $\check{H}^1(\mathbb{P}^1, \mathcal{O}_X^\times) \cong \operatorname{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$, and the integer n is exactly the degree of the line bundle. This connection between transition functions, line bundles, and Čech cohomology will be developed fully in section 5.4 and section 6.1.1.

Definition 5.3.7: n-Twisted Sheaf of $\mathscr{F}$

For any $\mathcal{O}_X$-module $\mathscr{F}$ on $X = \operatorname{Proj} S_\bullet$, the *$n$-twisted sheaf* is

$$\mathscr{F}(n) := \mathscr{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n).$$

Twisting by $\mathcal{O}_X(n)$ is a fundamental operation: it shifts the "degree of homogeneity" of sections. Since $\mathcal{O}_X(n)$ is locally free of rank 1, tensoring with it preserves quasi-coherence, coherence, and local freeness.

Proposition 5.3.8: Properties of Twisted Sheaves

Let $S_\bullet$ be a graded ring generated by S_1 as an S_0-algebra, and $X = \operatorname{Proj} S_\bullet$. Then:

1. $\mathcal{O}_X(n) \otimes_{\mathcal{O}_X} \mathcal{O}_X(m) \cong \mathcal{O}_X(n+m)$ for all $n, m \in \mathbb{Z}$. In particular, $\mathcal{O}_X(n)$ and $\mathcal{O}_X(-n)$ are tensor-product inverses, confirming that each $\mathcal{O}_X(n)$ is invertible.

2. For any graded $S_\bullet$-module $M_\bullet$, $\widetilde{M}(n) \cong \widetilde{M(n)}$.

3. Let $T_\bullet$ be another graded ring generated by T_1 over T_0, let $\varphi : S_\bullet \to T_\bullet$ be a graded homomorphism, and let $U \subseteq \operatorname{Proj} T_\bullet$ be the open locus where $f : U \to X$ is induced by φ. Then

$$f^*(\mathcal{O}_X(n)) \cong \mathcal{O}_{\operatorname{Proj} T}(n)\big|_U.$$

***Proof*:**
(1). On each $D_+(f)$ for $f \in S_1$, both sides restrict to $(S_f)_{n+m}$: the left side via the isomorphism $(S_f)_n \otimes_{(S_f)_0} (S_f)_m \xrightarrow{\sim} (S_f)_{n+m}$ given by multiplication in S_f, and the right side by definition. These local isomorphisms are compatible with restriction (further localization commutes with multiplication), hence glue to a global isomorphism by corollary 1.3.4.

(2). On $D_+(f)$, the left side gives $(M_f)_0 \otimes_{(S_f)_0} (S_f)_n \cong (M_f)_n$, and the right gives $(M(n)_f)_0 = (M_f)_n$.

(3). Reduce to the affine case: on $D_+(g) \subseteq \operatorname{Proj} T_\bullet$ mapping into $D_+(\varphi(g)) \subseteq X$, pullback corresponds to extension of scalars, and the grading shift commutes with this operation.

Having seen that $\Gamma(\mathbb{P}^r_k, \mathcal{O}(n)) \cong S_n$, it is natural to reassemble all degrees into a single graded object and ask whether we recover the entire graded ring $S_\bullet$.

Definition 5.3.9: Graded Module Associated to $\mathscr{F}$

Let $S_\bullet$ be a graded ring, $X = \operatorname{Proj} S_\bullet$, and $\mathscr{F}$ an $\mathcal{O}_X$-module. The *graded $S_\bullet$-module associated to $\mathscr{F}$* is

$$\Gamma_*(\mathscr{F}) := \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathscr{F}(n)),$$

with the $S_\bullet$-module structure induced by the multiplication maps $S_d \otimes \Gamma(X, \mathscr{F}(n)) \to \Gamma(X, \mathscr{F}(n+d))$ coming from proposition 5.3.8(1)–(2).

For polynomial rings, this recovers $S_\bullet$ exactly:

Proposition 5.3.10: Recovering the Graded Ring

Let R be a ring, $S_\bullet = R[x_0, \ldots, x_r]$, and $X = \operatorname{Proj} S_\bullet$. Then $\Gamma_*(\mathcal{O}_X) \cong S_\bullet$.

***Proof*:**
A global section $t \in \Gamma(X, \mathcal{O}_X(n))$ is a family of sections $t_i \in \mathcal{O}_X(n)(D_+(x_i)) = (S_{x_i})_n$ that agree on overlaps. The variables x_i are not zero divisors in $S_\bullet$, so the localization maps $S_\bullet \hookrightarrow S_{x_i} \hookrightarrow S_{x_0 \cdots x_r}$ are all injective, and these rings are subrings of the common localization $S' = S_{x_0 \cdots x_r}$. The agreement condition on overlaps means t lies in $\bigcap_{i=0}^r (S_{x_i})_n$ inside S'.

It remains to show $\bigcap_i S_{x_i} = S_\bullet$ inside S'. Every homogeneous element of S' can be written uniquely as $x_0^{a_0} \cdots x_r^{a_r} \cdot g(x_0, \ldots, x_r)$ with $a_j \in \mathbb{Z}$ and g not divisible by any x_i. Such an element lies in S_{x_i} if and only if $a_j \geq 0$ for all $j \neq i$. Belonging to *every* S_{x_i} requires $a_j \geq 0$ for all j (for each j, choose any $i \neq j$), which means the element lies in $S_\bullet$. Summing over degrees gives $\Gamma_*(\mathcal{O}_X) \cong S_\bullet$.

The reader should note two important features of this proof: the injectivity of the localization maps (which uses the fact that x_i are not zero divisors) and the combinatorial intersection argument. For a general graded ring $S_\bullet$ where the generators may be zero divisors, the conclusion can fail.

More fundamentally, the Γ_*–tilde correspondence is *not* an equivalence in either direction:

- The map $M_\bullet \mapsto \Gamma_*(\widetilde{M})$ need not recover M. For instance, if $M_0 = k$ and $M_n = 0$ for $n \neq 0$, then $\widetilde{M} = 0$ (there are no nonzero elements to localize in positive degrees), so $\Gamma_*(\widetilde{M}) = 0 \neq M$. The graded module M is concentrated in a single degree that is "invisible" to Proj.

- The map $M_\bullet \mapsto \widetilde{M}$ is not injective: the above example shows two different modules (M and 0) giving the same sheaf.

However, the other direction works perfectly: starting from a quasi-coherent sheaf and applying Γ_*, one can always recover the sheaf by applying the tilde construction. This is the key structural result of this section.

The proof requires a generalization of lemma 5.1.15 from distinguished opens of affine schemes to the projective setting. The role of the element $f \in R$ (which defines the distinguished open $D(f)$ in the affine case) is played by a global section $f \in \Gamma(X, \mathcal{L})$ of an invertible sheaf $\mathcal{L}$. The open set $X_f := \{x \in X : f_x \notin \mathfrak{m}_x\mathcal{L}_x\}$ is the locus where f "does not vanish" (cf. the evaluation of sections in definition 2.2.24); when $\mathcal{L} = \mathcal{O}_X$ this recovers the ordinary distinguished open, and when $\mathcal{L} = \mathcal{O}_X(1)$ on $\operatorname{Proj} S_\bullet$ this recovers $D_+(f)$.

Lemma 5.3.11: Extending Sections for Line Bundles

Let X be a scheme, $\mathcal{L}$ an invertible sheaf on X, $f \in \Gamma(X, \mathcal{L})$, and X_f the open set of points $x \in X$ where $f_x \notin \mathfrak{m}_x\mathcal{L}_x$. Let $\mathscr{F}$ be a quasi-coherent sheaf on X.

1. Suppose X is quasi-compact. If $s \in \Gamma(X, \mathscr{F})$ satisfies $s|_{X_f} = 0$, then $f^n s = 0$ in $\Gamma(X, \mathscr{F} \otimes \mathcal{L}^{\otimes n})$ for some $n > 0$.

2. Suppose additionally that X has a finite cover by affine opens $\{U_i\}$ with $\mathcal{L}|_{U_i}$ free and $U_i \cap U_j$ quasi-compact for all i, j. Then for any $t \in \Gamma(X_f, \mathscr{F})$, there exists $n > 0$ such that $f^n t \in \Gamma(X_f, \mathscr{F} \otimes \mathcal{L}^{\otimes n})$ extends to a global section of $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$.

The reader should think of this lemma as the projective analogue of lemma 5.1.15: part (1) says "sections that vanish on X_f are killed by a power of f," and part (2) says "sections defined only on X_f can be extended globally after multiplying by a power of f." The new feature is that f is a section of a line bundle rather than a global function, so "multiplying by f^n" lands in the twisted sheaf $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$ rather than in $\mathscr{F}$ itself. In the special case $\mathcal{L} = \mathcal{O}_X$, the section f is an ordinary global function, the open X_f is the distinguished open $D(f)$, and we recover lemma 5.1.15 exactly.

Before giving the proof, let us verify the hypotheses in the case we care about most.

Example 5.3.12: Hypotheses for Proj

Let $S_\bullet$ be a graded ring finitely generated by S_1 over S_0, $X = \operatorname{Proj} S_\bullet$, $\mathcal{L} = \mathcal{O}_X(1)$, and $f \in S_1$ a degree-1 element viewed as a section $f \in \Gamma(X, \mathcal{O}_X(1))$.

The open set X_f. A point $x \in X$ (corresponding to a homogeneous prime $\mathfrak{p}$) lies in X_f precisely when f_x generates the free $\mathcal{O}_{X,x}$-module $\mathcal{L}_x = \mathcal{O}_X(1)_x$, i.e., when f_x is a unit times the local generator. On the chart $D_+(g)$ for $g \in S_1$, the generator of $\mathcal{O}_X(1)|_{D_+(g)}$ is g (via (5.1)), and f_x is a unit times g precisely when $f/g \notin \mathfrak{p}$, i.e., when $\mathfrak{p} \in D_+(f)$. Hence $X_f = D_+(f)$.

The finite cover. Since $S_\bullet$ is generated by S_1, choose generators $f_0, \ldots, f_r \in S_1$; then $U_i = D_+(f_i)$ is affine (theorem 2.3.3), $\mathcal{L}|_{U_i}$ is free (by proposition 5.3.5), and $U_i \cap U_j = D_+(f_i f_j)$ is affine (hence quasi-compact).

Thus the hypotheses of both parts of lemma 5.3.11 are satisfied for any $f \in S_1$ on $\operatorname{Proj} S_\bullet$.

***Proof of lemma 5.3.11*:**
The strategy for both parts is to reduce to the affine case by trivializing $\mathcal{L}$ on a finite cover, applying lemma 5.1.15, and then patching the results.

(1) Annihilation. Cover X by finitely many affine opens $U_\alpha = \operatorname{Spec} A_\alpha$ (possible since X is quasi-compact) on which $\mathcal{L}$ is free; choose trivializations $\psi_\alpha : \mathcal{L}|_{U_\alpha} \xrightarrow{\sim} \mathcal{O}_{U_\alpha}$. Set $g_\alpha = \psi_\alpha(f|_{U_\alpha}) \in A_\alpha$, so that $X_f \cap U_\alpha = D(g_\alpha)$:

$$\begin{tikzcd} \mathcal{L}(U_\alpha) \arrow[r, "\psi_\alpha", "\sim"'] & \mathcal{O}_X(U_\alpha) = A_\alpha \\ f|_{U_\alpha} \arrow[r, mapsto] & g_\alpha \end{tikzcd}$$

Write $\mathscr{F}|_{U_\alpha} \cong \widetilde{M_\alpha}$ for an A_α-module M_α. The section $s|_{U_\alpha}$ corresponds to an element $s_\alpha \in M_\alpha$, and the hypothesis $s|_{X_f} = 0$ gives $s_\alpha|_{D(g_\alpha)} = 0$.

By the affine annihilation lemma (lemma 5.1.15(1)), $g_\alpha^{n_\alpha} s_\alpha = 0$ in M_α for some $n_\alpha > 0$. Using the trivialization $\psi_\alpha^{\otimes n} : \mathcal{L}^{\otimes n}|_{U_\alpha} \xrightarrow{\sim} \mathcal{O}_{U_\alpha}$, the element $g_\alpha^{n_\alpha} s_\alpha$ corresponds to $(f^{n_\alpha} s)|_{U_\alpha}$ intrinsically (independent of the choice of ψ_α). Taking $n = \max_\alpha n_\alpha$ gives $f^n s|_{U_\alpha} = 0$ for every α. Since the U_α cover X, the locality axiom gives $f^n s = 0$ globally.

(2) Extension. This is more involved: we must first extend the section locally, then correct the local extensions so that they glue.

Step 1: Local extensions. On each U_i of the given finite cover, the trivialization ψ_i converts $f|_{U_i}$ to $g_i \in A_i$ and gives $X_f \cap U_i = D(g_i)$. The section $t|_{D(g_i)} \in \mathscr{F}(D(g_i)) = (M_i)_{g_i}$. By the affine extension lemma (lemma 5.1.15(2)), there exists $n_i > 0$ such that $g_i^{n_i} \cdot t|_{D(g_i)}$ extends to an element $\hat{t}_i \in M_i = \mathscr{F}(U_i)$. Intrinsically, this means $f^{n_i} t|_{X_f \cap U_i}$ extends to a section of $\mathscr{F} \otimes \mathcal{L}^{\otimes n_i}$ over U_i. Taking $N = \max_i n_i$ (and multiplying by further powers of f where needed), we may assume each $\hat{t}_i$ is a section of $\mathscr{F} \otimes \mathcal{L}^{\otimes N}$ over U_i that restricts to $f^N t$ on $X_f \cap U_i$.

Step 2: Correcting overlaps. On $U_i \cap U_j$, both $\hat{t}_i$ and $\hat{t}_j$ restrict to $f^N t$ on $X_f \cap U_i \cap U_j$, so their difference $\delta_{ij} = \hat{t}_i - \hat{t}_j$ vanishes on $X_f \cap U_i \cap U_j$. Since $U_i \cap U_j$ is quasi-compact by hypothesis, part (1) applied to δ_{ij} on $U_i \cap U_j$ gives $m_{ij} > 0$ with $f^{m_{ij}} \delta_{ij} = 0$ on $U_i \cap U_j$. Set $m = \max_{i,j} m_{ij}$ (a finite maximum since the cover is finite).

Then the sections $f^m \hat{t}_i \in \Gamma(U_i, \mathscr{F} \otimes \mathcal{L}^{\otimes(N+m)})$ satisfy $f^m \hat{t}_i = f^m \hat{t}_j$ on every overlap $U_i \cap U_j$. By the gluability axiom, they glue to a global section $\bar{t} \in \Gamma(X, \mathscr{F} \otimes \mathcal{L}^{\otimes(N+m)})$. On X_f, this restricts to $f^{N+m} t$, so $\bar{t}$ is the desired extension with $n = N + m$.

5.3.13 The Mechanism of Extension The reader may find it helpful to compare the affine and projective extension results side by side:

Affine (lemma 5.1.15)	**Projective** (lemma 5.3.11)
$f \in R$ (ring element)	$f \in \Gamma(X, \mathcal{L})$ (section of line bundle)
Open: $D(f) \subseteq \operatorname{Spec} R$	Open: $X_f \subseteq X$
$t \in M_f$ (localization)	$t \in \mathscr{F}(X_f)$ (sections on X_f)
$f^n t \in M$ (cleared denominator)	$f^n t \in \Gamma(X, \mathscr{F} \otimes \mathcal{L}^{\otimes n})$ (twisted extension)

In the affine case, multiplying by f^n "clears the denominator" and lands back in M. In the projective case, the same multiplication lands in a *different* module — the global sections of the nth twist — because f is a section of $\mathcal{L}$, not of $\mathcal{O}_X$. The price we pay for working on a non-affine scheme is that extension requires passing to a twist; the benefit is that this works for *any* quasi-compact scheme with a suitable line bundle.

We now prove the key structural result: the tilde functor applied to $\Gamma_*(\mathscr{F})$ recovers $\mathscr{F}$.

Proposition 5.3.14: Partial Equivalence for Graded Rings

Let $S_\bullet$ be a graded ring, finitely generated by S_1 as an S_0-algebra. Let $X = \operatorname{Proj} S_\bullet$ and $\mathscr{F}$ a quasi-coherent sheaf on X. Then there is a natural isomorphism

$$\beta : \widetilde{\Gamma_*(\mathscr{F})} \xrightarrow{\sim} \mathscr{F}.$$

In particular, every quasi-coherent sheaf on $\operatorname{Proj} S_\bullet$ arises from a graded $S_\bullet$-module.

The hypothesis that $S_\bullet$ is generated by S_1 is essential: it ensures that X is covered by the affine opens $D_+(f)$ for $f \in S_1$, on which the twisting sheaf $\mathcal{O}_X(1)$ becomes free.

***Proof*:**
Step 1: Constructing β. We define β on the basis of distinguished opens $D_+(f)$ for $f \in S_+$ homogeneous. A section of $\widetilde{\Gamma_*(\mathscr{F})}$ over $D_+(f)$ is an element of the homogeneous localization $\Gamma_*(\mathscr{F})_{(f)}$: a fraction m/f^d where $m \in \Gamma(X, \mathscr{F}(d))$ is a global section of the dth twist and $\deg f \cdot d$ matches $\deg m$ in the graded module.

Now f^{-d} can be viewed as a section of $\mathcal{O}_X(-d)$ over $D_+(f)$: since $\mathcal{O}_X(d)|_{D_+(f)}$ is trivial, the element f^d is a generator, and f^{-d} is its inverse in the localized sections. Tensoring, the element $m \otimes f^{-d}$ is a section of $\mathscr{F}(d) \otimes \mathcal{O}_X(-d) \cong \mathscr{F}$ over $D_+(f)$. Define

$$\beta_{D_+(f)}\left(\frac{m}{f^d}\right) := m \otimes f^{-d} \in \mathscr{F}(D_+(f)).$$

Let us verify this is well-defined. If $m/f^d = m'/f^{d'}$ in $\Gamma_*(\mathscr{F})_{(f)}$, then $f^N(f^{d'}m - f^d m') = 0$ for some N, i.e., the sections $f^{d'}m$ and $f^d m'$ agree in $\mathscr{F}(d + d' + N \deg f)$ after multiplying by f^N. Restricting to $D_+(f)$ and dividing by $f^{d+d'+N}$ (which is a unit there), we get $m \otimes f^{-d} = m' \otimes f^{-d'}$ in $\mathscr{F}(D_+(f))$. Compatibility with restriction is immediate, so β is a well-defined morphism of sheaves.

Step 2: β is an isomorphism. It suffices to show that $\beta_{D_+(f)}$ is bijective for each $f \in S_1$. By example 5.3.12, the hypotheses of lemma 5.3.11 are satisfied with $\mathcal{L} = \mathcal{O}_X(1)$ and $X_f = D_+(f)$.

Surjectivity. Let $t \in \mathscr{F}(D_+(f))$. By lemma 5.3.11(2), there exists $n > 0$ such that $f^n t$ extends to a global section $s \in \Gamma(X, \mathscr{F} \otimes \mathcal{O}_X(n)) = \Gamma(X, \mathscr{F}(n))$. Then $s/f^n \in \Gamma_*(\mathscr{F})_{(f)}$ maps under β to $s \otimes f^{-n} = t$ on $D_+(f)$.

Injectivity. Suppose $m/f^d \in \Gamma_*(\mathscr{F})_{(f)}$ maps to zero, i.e., $m|_{D_+(f)} = 0$ in $\mathscr{F}(d)(D_+(f))$. By lemma 5.3.11(1) applied to the global section m of $\mathscr{F}(d)$ (which vanishes on $X_f = D_+(f)$), there exists $n > 0$ with $f^n m = 0$ in $\Gamma(X, \mathscr{F}(d+n))$. Hence $m/f^d = 0$ in $\Gamma_*(\mathscr{F})_{(f)}$.

Let us pause to summarize the relationship between graded modules and quasi-coherent sheaves on Proj. The two functors

$$\{\text{graded } S_\bullet\text{-modules}\} \underset{\Gamma_*(-)}{\overset{\widetilde{(-)}}{\rightleftarrows}} \mathbf{QCoh}(\operatorname{Proj} S_\bullet)$$

satisfy $\widetilde{\Gamma_*(\mathscr{F})} \cong \mathscr{F}$ for every quasi-coherent $\mathscr{F}$ (proposition 5.3.14), but $\Gamma_*(\widetilde{M})$ need not be isomorphic to M. In the language of category theory, Γ_* is a *right adjoint* that is faithful but not full, and the tilde functor is a *left adjoint* that is essentially surjective. The failure of the unit $M \to \Gamma_*(\widetilde{M})$ to be an isomorphism is measured by the *saturation* of M: two graded modules give the same sheaf if and only if they agree in sufficiently large degrees. This is analogous to the fact that Proj itself depends only on the "tail" of the graded ring (definition 2.3.1).

5.3.2 Characterizing Projective Subschemes

Combining the partial equivalence with the theory of ideal sheaves from section 5.1.4, we obtain concrete descriptions of closed subschemes of projective space and of projective schemes themselves.

Corollary 5.3.15: Characterizing Projective Subschemes

Let R be a ring. Then:

1. Every closed subscheme $Y \subseteq \mathbb{P}^r_R$ is determined by a homogeneous ideal $I \subseteq S_\bullet = R[x_0, \ldots, x_r]$: namely $Y \cong \operatorname{Proj}(S_\bullet/I)$.
2. A scheme Y over $\operatorname{Spec} R$ is projective if and only if $Y \cong \operatorname{Proj} S_\bullet$ for some graded ring $S_\bullet$ with $S_0 = R$ that is finitely generated by S_1 as an S_0-algebra.

***Proof*:**
(1). Let $\mathcal{I}_Y \subseteq \mathcal{O}_X$ be the ideal sheaf of Y in $X = \mathbb{P}^r_R$. Since the twisting functor is exact (tensor with a locally free sheaf preserves exact sequences) and Γ is left exact, the inclusion $\mathcal{I}_Y \hookrightarrow \mathcal{O}_X$ gives an inclusion of graded modules

$$\Gamma_*(\mathcal{I}_Y) \hookrightarrow \Gamma_*(\mathcal{O}_X) \cong S_\bullet,$$

where the isomorphism is proposition 5.3.10. Set $I = \Gamma_*(\mathcal{I}_Y)$, a homogeneous ideal of $S_\bullet$.

By proposition 5.3.14, $\mathcal{I}_Y \cong \widetilde{I}$, so Y is the closed subscheme of X defined by I, i.e., $Y \cong \operatorname{Proj}(S_\bullet/I)$. Moreover, I is the *largest* (or *saturated*) homogeneous ideal defining Y: by construction, it contains every homogeneous polynomial that vanishes on Y in every degree.

(2). By definition, Y is projective over $\operatorname{Spec} R$ if it admits a closed immersion into $\mathbb{P}^r_R$ for some r. By (1), any such Y is $\operatorname{Proj}(S_\bullet/I)$ for a homogeneous ideal $I \subseteq S_+$. Setting $S'_\bullet = S_\bullet/I$, we have $S'_0 = R$ (since $I \subseteq S_+$) and $S'_\bullet$ is finitely generated by S'_1 over S'_0.

Conversely, if $S_\bullet$ is finitely generated by $S_1 = R \cdot f_0 + \cdots + R \cdot f_r$ over $S_0 = R$, the surjection $R[x_0, \ldots, x_r] \twoheadrightarrow S_\bullet$ (sending $x_i \mapsto f_i$) induces a closed immersion $\operatorname{Proj} S_\bullet \hookrightarrow \mathbb{P}^r_R$.

Corollary 5.3.16: Characterizing Hypersurfaces in Projective Space

Let R be a UFD. Then every irreducible hypersurface $X \subseteq \mathbb{P}^n_R$ corresponds to a principal homogeneous ideal $(f) \subseteq R[x_0, \ldots, x_n]$.

***Proof*:**
By corollary 5.3.15, the hypersurface X corresponds to a homogeneous prime ideal $\mathfrak{p} \subseteq S_\bullet = R[x_0, \ldots, x_n]$ of height 1 (since X has codimension 1 in $\mathbb{P}^n_R$). Since R is a UFD, $S_\bullet$ is also a UFD (see [20]), and in a UFD every height-1 prime ideal is principal (exercise 5.3.1(5.3.1 (1))). Hence $\mathfrak{p} = (f)$ for some irreducible homogeneous polynomial f.

Definition 5.3.17: Degree of Projective Hypersurface

Let R be a UFD and $X \subseteq \mathbb{P}^n_R$ an irreducible hypersurface corresponding to $(f) \subseteq R[x_0, \ldots, x_n]$ via corollary 5.3.16. The *degree* of X is $\deg X := \deg f$.

This is well-defined because f is determined up to a unit in R, which does not affect the degree. When $R = k$ is an algebraically closed field, the degree of a hypersurface $V(f) \subseteq \mathbb{P}^n_k$ counts (with multiplicity) the number of intersection points with a generic line, in accordance with Bézout's theorem ([20]).

We conclude this section with the promised example showing that the tensor product of $\mathcal{O}_X$-modules cannot, in general, be computed section-wise.

Example 5.3.18: Tensor Products Require Sheafification

Let $X = \mathbb{P}^1_k$ for a field k. Consider $\mathscr{F} = \mathcal{O}_X(-1)$ and $\mathscr{G} = \mathcal{O}_X(1)$. By proposition 5.3.8(1),

$$\mathscr{F} \otimes_{\mathcal{O}_X} \mathscr{G} \;=\; \mathcal{O}_X(-1) \otimes \mathcal{O}_X(1) \;\cong\; \mathcal{O}_X(0) \;=\; \mathcal{O}_X,$$

so the global sections of the tensor product sheaf satisfy

$$(\mathscr{F} \otimes \mathscr{G})(\mathbb{P}^1_k) \;=\; \Gamma(\mathbb{P}^1_k, \mathcal{O}_X) \;\cong\; k.$$

On the other hand, the section-wise tensor product on global sections gives

$$\mathscr{F}(\mathbb{P}^1_k) \otimes_k \mathscr{G}(\mathbb{P}^1_k) \;=\; \underbrace{\Gamma(\mathbb{P}^1_k, \mathcal{O}(-1))}_{=0} \otimes_k \underbrace{\Gamma(\mathbb{P}^1_k, \mathcal{O}(1))}_{\cong k^2} \;=\; 0.$$

Hence $(\mathscr{F} \otimes \mathscr{G})(X) \cong k \ncong 0 = \mathscr{F}(X) \otimes \mathscr{G}(X)$.

The mechanism behind this is illuminating. Consider the presheaf $\mathbb{C}\mathrm{P} : U \mapsto \mathscr{F}(U) \otimes_{\mathcal{O}_X(U)} \mathscr{G}(U)$. On each chart $U_i = D_+(x_i)$, we have $\mathscr{F}(U_i) \cong \mathscr{G}(U_i) \cong \mathcal{O}_X(U_i)$ (both are free of rank 1), so $\mathbb{C}\mathrm{P}(U_i) \cong \mathcal{O}_X(U_i)$. But on global sections, $\mathbb{C}\mathrm{P}(X) = 0 \otimes k^2 = 0$. The presheaf $\mathbb{C}\mathrm{P}$ has "plenty of local sections" but "no global sections" — it violates the gluing axiom. Sheafification repairs this: the sheaf $\mathscr{F} \otimes \mathscr{G}$ is constructed by gluing the local sections of $\mathbb{C}\mathrm{P}$, and the global sections of the sheafification can be strictly larger than those of the presheaf.

This phenomenon is purely projective: on affine schemes, the presheaf tensor product is already a sheaf, so $\widetilde{M} \otimes \widetilde{N} \cong \widetilde{M \otimes N}$ and global sections commute with tensor products (proposition 5.1.14(4)).

Exercise 5.3.1

1. *(Krull's Principal Ideal Theorem.)* Let R be a Noetherian ring and $(x) \subsetneq R$ a proper principal ideal. Show that every minimal prime $\mathfrak{p}$ over (x) has height at most 1. *(Hint: localize at $\mathfrak{p}$ to reduce to a local ring with exactly one minimal prime over its principal ideal. Use the Artin–Rees lemma or Nakayama's lemma to show the chain of primes below $\mathfrak{p}$ has length at most 1.)*

Another example I came across is very interesting:

Example 5.3.19: Free Sheaves Need Not Be Projective Objects
Really cool! here

5.3.3 Maps to Projective Space

In example 3.1.14, we observed a fundamental asymmetry: a morphism $X \to \mathbb{A}^n_k$ is determined by n global functions $f_1, \ldots, f_n \in \mathcal{O}_X(X)$, but a morphism $X \to \mathbb{P}^n_k$ cannot be described by global functions alone — the homogeneous coordinates x_i on $\mathbb{P}^n$ are sections of the twisting sheaf $\mathcal{O}(1)$, not of the structure sheaf. We commented that defining a map to projective space requires a *line bundle* $\mathcal{L}$ on X together with global sections $s_0, \ldots, s_n \in \Gamma(X, \mathcal{L})$ that do not vanish simultaneously.

This subsection makes that observation precise. The main result (theorem 5.3.29) establishes a bijection between morphisms $X \to \mathbb{P}^n$ and equivalence classes of line bundles with generating global sections. Along the way, we develop two important tools: the notion of a *very ample sheaf* (which characterizes projective schemes), and a fundamental finite generation theorem for coherent sheaves on projective schemes (theorem 5.3.25).

Maps *from* projective space are better understood using cohomological methods (see chapter 6); theorem 3.5.28 shows that all proper schemes admit a surjective birational morphism from a projective scheme, making projective schemes the natural "compact" objects.

Twisting Sheaves on Relative Projective Space

The twisting sheaves of **??** were defined for $\operatorname{Proj} S_\bullet$ over the base $\operatorname{Spec} S_0$. For a general base scheme X, the projective space $\mathbb{P}^n_X$ is defined by base change, and the twisting sheaf is obtained by pulling back from the universal case:

Definition 5.3.20: Twisting Sheaf on $\mathbb{P}^n_X$

Let X be a scheme. The *twisting sheaf* $\mathcal{O}(1)$ on $\mathbb{P}^n_X = X \times_{\mathbb{Z}} \mathbb{P}^n_{\mathbb{Z}}$ is the pullback

$$\mathcal{O}_{\mathbb{P}^n_X}(1) \;:=\; g^*\big(\mathcal{O}_{\mathbb{P}^n_{\mathbb{Z}}}(1)\big),$$

where $g : \mathbb{P}^n_X \to \mathbb{P}^n_{\mathbb{Z}}$ is the projection. More generally, $\mathcal{O}_{\mathbb{P}^n_X}(d) := \mathcal{O}_{\mathbb{P}^n_X}(1)^{\otimes d}$ for $d \in \mathbb{Z}$.

By proposition 5.3.8(3), if $X = \operatorname{Spec} R$ and $\mathbb{P}^n_R = \operatorname{Proj} R[x_0, \dots, x_n]$, this recovers the twisting sheaf $\widetilde{S(1)}$ defined earlier. The base-change definition ensures that all properties established over $\mathbb{Z}$ (invertibility, non-isomorphism for different twists, etc.) propagate to $\mathbb{P}^n_X$ for any base X.

Definition 5.3.21: Very Ample Sheaf

Let X be a scheme over a base scheme Y, and $\mathcal{L}$ a line bundle on X. Then $\mathcal{L}$ is *very ample relative to* Y if there exists a closed immersion $\iota : X \hookrightarrow \mathbb{P}^n_Y$ (for some n) such that

$$\iota^*\big(\mathcal{O}_{\mathbb{P}^n_Y}(1)\big) \;\cong\; \mathcal{L}.$$

The condition says that $\mathcal{L}$ is the restriction of $\mathcal{O}(1)$ under an embedding into projective space. Very ample sheaves characterize projectivity:

Lemma 5.3.22: Characterizing Projective Schemes via Very Ample Sheaves

Let Y be a Noetherian scheme. A Y-scheme X is projective over Y if and only if X is proper over Y and possesses a very ample invertible sheaf relative to Y.

***Proof*:**
($\Rightarrow$). If X is projective over Y, then by definition there exists a closed immersion $\iota : X \hookrightarrow \mathbb{P}^n_Y$. The sheaf $\mathcal{L} = \iota^*(\mathcal{O}(1))$ is very ample, and X is proper by theorem 3.5.26.

($\Leftarrow$). Suppose X is proper over Y and $\mathcal{L}$ is very ample, so $\mathcal{L} \cong \iota^*(\mathcal{O}(1))$ for some locally closed immersion $\iota : X \to \mathbb{P}^n_Y$. Since X is proper over Y and $\mathbb{P}^n_Y$ is separated over Y, the morphism ι is proper (proposition 3.5.21). A proper locally closed immersion has closed image (proposition 3.4.22), so ι is in fact a closed immersion. Hence X is projective.

Global Generation

We need a notion expressing when a sheaf has "enough global sections" to control its local behaviour.

Definition 5.3.23: Generated by Global Sections

Let X be a scheme and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is *generated by global sections* if there exists a family of global sections $\{s_i\}_{i\in I} \subseteq \Gamma(X, \mathscr{F})$ whose images in the stalk $\mathscr{F}_x$ generate $\mathscr{F}_x$ as an $\mathcal{O}_{X,x}$-module for every $x \in X$.
Equivalently, $\mathscr{F}$ is generated by global sections if and only if it is a quotient of a free sheaf: the surjection $\bigoplus_{i\in I} \mathcal{O}_X \twoheadrightarrow \mathscr{F}$ sending the ith generator to s_i is surjective on stalks, hence surjective as a sheaf morphism.

Example 5.3.24: Generated by Global Sections

1. *Affine schemes.* Let $X = \operatorname{Spec} R$ and $\mathscr{F} = \widetilde{M}$. Choose any set of generators $\{m_i\}$ for M as an R-module. The corresponding global sections generate $\mathscr{F}$, since generation is checked on stalks (localizations), and generators of M remain generators after localization. Thus *every* quasi-coherent sheaf on an affine scheme is generated by global sections.

2. *The twisting sheaf.* Let $X = \operatorname{Proj} S_\bullet$ where $S_\bullet$ is generated by S_1 over S_0. The degree-1 elements $f_0, \ldots, f_r \in S_1$ define global sections of $\mathcal{O}_X(1)$ (they lie in $\Gamma(X, \mathcal{O}_X(1)) \supseteq S_1$). On $D_+(f_i)$, the section f_i generates the free module $\mathcal{O}_X(1)|_{D_+(f_i)} \cong \mathcal{O}_{D_+(f_i)}$. Since the $D_+(f_i)$ cover X, the sections $f_0, \ldots, f_r$ generate $\mathcal{O}_X(1)$.

3. *Negative twists.* The sheaf $\mathcal{O}_{\mathbb{P}^r_k}(-1)$ is *not* generated by global sections for $r \geq 1$: it has no nonzero global sections at all, as we computed in **??**.

The next theorem is one of the most important results about coherent sheaves on projective schemes: after twisting by a sufficiently large power of a very ample sheaf, any coherent sheaf becomes generated by *finitely many* global sections. This should be compared with the affine case, where global generation is automatic but says nothing about projective schemes.

Theorem 5.3.25: Global Section Generation of Twisted Sheaves

Let X be a projective scheme over a Noetherian ring R, let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X, and let $\mathscr{F}$ be a coherent $\mathcal{O}_X$-module. Then there exists an integer n_0 such that for all $n \geq n_0$, the twisted sheaf $\mathscr{F}(n)$ is generated by a finite number of global sections.

***Proof*:**
Reduction to $X = \mathbb{P}^r_R$. Let $\iota : X \hookrightarrow \mathbb{P}^r_R$ be a closed immersion with $\iota^*(\mathcal{O}(1)) = \mathcal{O}_X(1)$. The pushforward $\iota_*\mathscr{F}$ is coherent on $\mathbb{P}^r_R$ (closed immersions are finite, hence preserve coherence), and $\iota_*(\mathscr{F}(n)) \cong (\iota_*\mathscr{F})(n)$ by proposition 5.3.8(3). Moreover, $\mathscr{F}(n)$ is generated by global sections if and only if $\iota_*(\mathscr{F}(n))$ is (their global sections agree, and generation is checked on the support). We may therefore assume $X = \mathbb{P}^r_R = \operatorname{Proj} R[x_0, \ldots, x_r]$.

Local generators. Cover X by the standard affine charts $U_i = D_+(x_i)$ for $i = 0, \ldots, r$. Since $\mathscr{F}$ is coherent, on each U_i there is a finitely generated $(S_{x_i})_0$-module M_i with $\mathscr{F}|_{U_i} \cong \widetilde{M_i}$. Choose finite sets of generators $\{s_{i,1}, \ldots, s_{i,k_i}\} \subseteq M_i$.

Extending to global sections. Each $s_{i,j}$ is a section of $\mathscr{F}$ over the affine open $U_i = D_+(x_i)$. By lemma 5.3.11(2) (with $\mathcal{L} = \mathcal{O}_X(1)$ and $f = x_i$), there exists $n_{i,j} > 0$ such that $x_i^{n_{i,j}} s_{i,j}$ extends to a global section $t_{i,j} \in \Gamma(X, \mathscr{F}(n_{i,j}))$. Taking $n_0 = \max_{i,j} n_{i,j}$ (and multiplying by further powers of x_i where needed), we may assume all extensions live in $\Gamma(X, \mathscr{F}(n_0))$.

Generation. The twisted sheaf $\mathscr{F}(n_0)$ restricts on U_i to a module M_i', and the trivialization $\mathcal{O}_X(n_0)|_{U_i} \cong \mathcal{O}_{U_i}$ (via multiplication by $x_i^{n_0}$) gives an isomorphism $M_i \xrightarrow{\sim} M_i'$ sending $s_{i,j}$ to $x_i^{n_0} s_{i,j}$. Since $\{s_{i,j}\}_j$ generates M_i, the sections $\{x_i^{n_0} s_{i,j}\}_j$ generate M_i'. Hence the global sections $\{t_{i,j}\}_{i,j}$ generate $\mathscr{F}(n_0)$ on every chart U_i, and therefore everywhere. There are finitely many such sections (at most $\sum_{i=0}^r k_i$).

For $n > n_0$, the sheaf $\mathscr{F}(n) = \mathscr{F}(n_0)(n - n_0)$ is generated by products of the $t_{i,j}$ with degree-$(n - n_0)$ monomials in the x_i, which are again finitely many.

This result has two immediate and important consequences.

Corollary 5.3.26: Coherent Sheaves as Quotients of Twisted Free Sheaves

Let X be a projective scheme over a Noetherian ring R and $\mathscr{F}$ a coherent sheaf on X. Then $\mathscr{F}$ can be written as a quotient of a sheaf of the form $\bigoplus_{i=1}^m \mathcal{O}_X(n_i)$ for finitely many integers n_i.

***Proof*:**
By theorem 5.3.25, there exist $n \geq 0$ and a surjection $\mathcal{O}_X^m \twoheadrightarrow \mathscr{F}(n)$. Tensoring with $\mathcal{O}_X(-n)$ gives a surjection $\mathcal{O}_X(-n)^m \twoheadrightarrow \mathscr{F}$.

This should be viewed as a projective analogue of the fact that every finitely generated module over a ring is a quotient of a free module. On projective space, free modules are replaced by direct sums of line bundles. Iterating, one obtains a *free resolution* of $\mathscr{F}$ by direct sums of twisted structure sheaves — the projective analogue of a free resolution of a module. The Hilbert Syzygy Theorem asserts that such a resolution can always be chosen to have length at most $r + 1$ on $\mathbb{P}^r_R$; see section 6.4.

Finite Generation of Global Sections

On affine schemes, coherence of $\widetilde{M}$ corresponds to finite generation of M as an R-module. In particular, $\Gamma(\operatorname{Spec} R, \widetilde{M})$ is finitely generated. On projective schemes, global sections are *not* the same as the underlying module, so finite generation requires a separate argument. The following theorem provides it:

Theorem 5.3.27: Finite Generation of Coherent Sheaves over Projective Schemes

Let k be a field, R a finitely generated k-algebra, X a projective R-scheme, and $\mathscr{F}$ a coherent $\mathcal{O}_X$-module. Then $\Gamma(X, \mathscr{F})$ is a finitely generated R-module.
In particular, if $R = k$, then $\Gamma(X, \mathscr{F})$ is a finite-dimensional k-vector space.

The contrast with the affine case is stark: $\Gamma(\mathbb{A}^n_k, \mathcal{O}) = k[x_1, \ldots, x_n]$ is infinite-dimensional over k, reflecting the fact that $\mathbb{A}^n_k$ is not projective. The finiteness of global sections is a manifestation of the "compactness" of projective schemes.

***Proof*:**
By corollary 5.3.15, we may write $X = \operatorname{Proj} S_\bullet$ where $S_0 = R$ and $S_\bullet$ is finitely generated by S_1 as an S_0-algebra. Let $M_\bullet = \Gamma_*(\mathscr{F})$. By proposition 5.3.14, $\widetilde{M} \cong \mathscr{F}$. The proof proceeds in three steps: we first replace $M_\bullet$ by a finitely generated submodule, then reduce to a special case via filtration, and finally prove that special case using integral dependence.

Step 1: Replacing $M_\bullet$ by a finitely generated submodule. By theorem 5.3.25, there exists n_0 such that $\mathscr{F}(n)$ is generated by finitely many global sections for all $n \geq n_0$. Let $M'_\bullet \subseteq M_\bullet$ be the graded submodule generated by these finitely many sections (for a single n_0). Then $M'_\bullet$ is a finitely generated $S_\bullet$-module, the inclusion $M' \hookrightarrow M$ induces $\widetilde{M'} \hookrightarrow \widetilde{M} = \mathscr{F}$, and twisting by n_0 gives $\widetilde{M'}(n_0) \hookrightarrow \mathscr{F}(n_0)$. Since the generators of M' in degree n_0 are exactly the global sections generating $\mathscr{F}(n_0)$, this inclusion is an isomorphism in degree n_0, hence an isomorphism of sheaves (a map of quasi-coherent sheaves that is an isomorphism after twisting is an isomorphism). Twisting back gives $\widetilde{M'} \cong \mathscr{F}$.

We have shown: every coherent sheaf on $\operatorname{Proj} S_\bullet$ is the tilde of a finitely generated graded $S_\bullet$-module. It remains to show that if $M_\bullet$ is finitely generated, then $\Gamma(X, \widetilde{M}) = (M_\bullet)_0$ ... but this is *not* true in general; $\Gamma(X, \widetilde{M})$ can differ from M_0. Instead, we must show $\Gamma(X, \widetilde{M})$ is finitely generated over R directly.

Step 2: Reduction via filtration. Since $S_\bullet$ is a Noetherian ring and $M_\bullet$ is a finitely generated graded $S_\bullet$-module, there exists a finite filtration of graded submodules (see [10])

$$0 = M^{(0)} \subseteq M^{(1)} \subseteq \cdots \subseteq M^{(r)} = M_\bullet$$

such that each successive quotient $M^{(i)}/M^{(i-1)}$ is isomorphic to $(S_\bullet/\mathfrak{p}_i)(n_i)$ for some homogeneous prime $\mathfrak{p}_i$ and integer n_i. Applying the tilde construction and taking global sections, the short exact sequences $0 \to \widetilde{M^{(i-1)}} \to \widetilde{M^{(i)}} \to \widetilde{M^{(i)}/M^{(i-1)}} \to 0$ give left-exact sequences

$$0 \longrightarrow \Gamma(X, \widetilde{M^{(i-1)}}) \longrightarrow \Gamma(X, \widetilde{M^{(i)}}) \longrightarrow \Gamma(X, \widetilde{M^{(i)}/M^{(i-1)}}).$$

By induction, it suffices to show that each $\Gamma(X, \widetilde{S_\bullet/\mathfrak{p}}(n))$ is finitely generated over R.

Step 3: The integral domain case. We are thus reduced to proving the following: let $S_\bullet$ be a graded integral domain, finitely generated by $S_1 = Rx_0 + \cdots + Rx_r$ over $S_0 = R$ (a finitely generated k-algebra), and $X = \operatorname{Proj} S_\bullet$. Then $\Gamma(X, \mathcal{O}_X(n))$ is a finitely generated R-module for each $n \in \mathbb{Z}$.

Since $S_\bullet$ is a domain, multiplication by any x_i gives an injection $S(n) \hookrightarrow S(n+1)$, hence an injection $\Gamma(X, \mathcal{O}(n)) \hookrightarrow \Gamma(X, \mathcal{O}(n+1))$. It therefore suffices to prove finite generation for all $n \geq 0$.

Set $S'_\bullet = \bigoplus_{n \geq 0} \Gamma(X, \mathcal{O}_X(n)) = \Gamma_*(\mathcal{O}_X)$. This is a graded ring containing $S_\bullet$ (via the natural map $S_n \to \Gamma(X, \mathcal{O}(n))$) and contained in $\bigcap_i S_{x_i}$ (each global section restricts to a section on each $D_+(x_i)$).

$S'_\bullet$ is integral over $S_\bullet$. Let $s' \in S'_d$ for some $d \geq 0$. Since $s' \in S_{x_i}$ for each i, there exists N (independent of i, by taking a maximum) such that $x_i^N s' \in S_\bullet$. Since the monomials in $x_0, \ldots, x_r$ of degree N generate S_N, we may enlarge N so that $y \cdot s' \in S_\bullet$ for all $y \in S_{\geq N} := \bigoplus_{e \geq N} S_e$.

By induction, $y \cdot (s')^q \in S_{\geq N}$ for all $q \geq 1$ and all $y \in S_{\geq N}$. Taking $y = x_0^N$ gives $(s')^q \in x_0^{-N} \cdot S_\bullet$ for all $q \geq 1$. The module $x_0^{-N} \cdot S_\bullet$ is a finitely generated $S_\bullet$-submodule of the quotient field of $S_\bullet$. By the standard criterion for integral dependence (see [10]), s' is integral over $S_\bullet$.

Conclusion. Since $S_\bullet$ is a finitely generated k-algebra (hence a Nagata ring), its integral closure in the quotient field is a finitely generated $S_\bullet$-module ([10]). Since $S'_\bullet$ is contained in this integral closure, it is a finitely generated $S_\bullet$-module as well[a]. In particular, each graded piece $S'_n = \Gamma(X, \mathcal{O}_X(n))$ is a finitely generated S_0-module, i.e., a finitely generated R-module.

[a]More precisely, $S'_\bullet$ is a graded submodule of a finitely generated module over the Noetherian ring $S_\bullet$, hence itself finitely generated.

Corollary 5.3.28: Pushforward of Coherent Sheaf by Projective Morphism

Let $f : X \to Y$ be a projective morphism of schemes of finite type over a field k. Let $\mathscr{F}$ be a coherent sheaf on X. Then $f_*\mathscr{F}$ is coherent on Y.

Recall that this fails for non-projective morphisms: example 5.1.19 showed that the pushforward of $\mathcal{O}_{\mathbb{A}^1}$ along $\mathbb{A}^1 \to \operatorname{Spec} k$ is $k[x]$, which is not finitely generated over k.

***Proof*:**
The question is local on Y, so we may assume $Y = \operatorname{Spec} R$ where R is a finitely generated k-algebra. Since f is projective, X is a projective R-scheme. By proposition 5.1.20, $f_*\mathscr{F}$ is quasi-coherent, so $f_*\mathscr{F} \cong \widetilde{N}$ for $N = \Gamma(Y, f_*\mathscr{F}) = \Gamma(X, \mathscr{F})$. By theorem 5.3.27, N is a finitely generated R-module, so $f_*\mathscr{F}$ is coherent.

5.3.4 Characterizing Morphisms to Projective Space

With the notion of global generation in hand (definition 5.3.23), we can now make precise the correspondence between morphisms to projective space and line bundles with generating sections that was previewed in the introduction to this subsection.

Recall that for a line bundle $\mathcal{L}$ (locally free of rank 1), being generated by sections $s_0, \ldots, s_n$ has a particularly clean geometric interpretation: at each point $x \in X$, the fibre $\mathcal{L}(x)$ is a one-dimensional $\kappa(x)$-vector space, and the sections span this fibre if and only if at least one s_i is nonzero at x. Equivalently, the *base locus* $\{x \in X : s_0(x) = \cdots = s_n(x) = 0\}$ is empty. This is exactly the condition ensuring that the "ratio map" $x \mapsto [s_0(x) : \cdots : s_n(x)]$ is well-defined at every point.

Before stating the theorem, let us trace through the construction on a concrete case. Consider the Veronese map $\mathbb{P}^1_k \to \mathbb{P}^2_k$, classically given by $[s : t] \mapsto [s^2 : st : t^2]$. In our language:

- The line bundle is $\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(2)$ on $\mathbb{P}^1_k$.
- The global sections are the three degree-2 monomials: $s_0 = x_0^2$, $s_1 = x_0x_1$, $s_2 = x_1^2$ in $\Gamma(\mathbb{P}^1_k, \mathcal{O}(2)) \cong k[x_0, x_1]_2$.
- At each point of $\mathbb{P}^1_k$, at least one of x_0^2, x_0x_1, x_1^2 is nonzero, so these sections generate $\mathcal{O}(2)$.
- On the open set $X_0 = D_+(x_0)$ where $s_0 = x_0^2$ does not vanish, the ratios are $s_1/s_0 = x_1/x_0$ and $s_2/s_0 = (x_1/x_0)^2$ — these are the "affine coordinates" of the map, sending $D_+(x_0)$ into the affine chart $U_0 \subseteq \mathbb{P}^2_k$ by $t \mapsto (t, t^2)$.

The image is the conic $V(y_0y_2 - y_1^2) \subseteq \mathbb{P}^2_k$, and the map is a closed immersion. The general theorem below formalizes this pattern.

Theorem 5.3.29: Determining Maps To Projective Space

Let R be a ring and X an R-scheme. Then:

1. *(From morphisms to line bundles.)* If $\varphi : X \to \mathbb{P}^n_R$ is an R-morphism, then $\mathcal{L} := \varphi^*(\mathcal{O}(1))$ is an invertible sheaf on X, and the pullback sections $s_i := \varphi^*(x_i) \in \Gamma(X, \mathcal{L})$ for $i = 0, \ldots, n$ generate $\mathcal{L}$.

2. *(From line bundles to morphisms.)* If $\mathcal{L}$ is an invertible sheaf on X and $s_0, \ldots, s_n \in \Gamma(X, \mathcal{L})$ are global sections that generate $\mathcal{L}$, then there exists a unique R-morphism $\varphi : X \to \mathbb{P}^n_R$ such that $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$ and $s_i = \varphi^*(x_i)$ under this isomorphism.

In particular, there is a natural bijection:

$$\operatorname{Hom}_R(X, \mathbb{P}^n_R) \xleftrightarrow{1:1} \left\{ (\mathcal{L}, s_0, \ldots, s_n) \;\middle|\; \begin{array}{c} \mathcal{L} \text{ invertible on } X, \\ s_i \in \Gamma(X, \mathcal{L}) \text{ generating } \mathcal{L} \end{array} \right\} \Big/ \cong$$

where $(\mathcal{L}, s_0, \ldots, s_n) \cong (\mathcal{L}', s_0', \ldots, s_n')$ if there is an isomorphism $\alpha : \mathcal{L} \xrightarrow{\sim} \mathcal{L}'$ with $\alpha(s_i) = s_i'$ for all i.

***Proof*:**

Part (1): morphisms yield generating sections. Since pullback preserves locally free sheaves and their rank (proposition 5.3.8(3)), $\mathcal{L} = \varphi^*(\mathcal{O}(1))$ is an invertible sheaf on X. The homogeneous coordinates $x_0, \ldots, x_n$ generate $\mathcal{O}(1)$ on $\mathbb{P}^n_R$, and pullback preserves surjections, so the sections $s_i = \varphi^*(x_i)$ generate $\mathcal{L}$.

Part (2): generating sections yield a morphism. This is the substantial direction. We construct φ by defining it on an open cover and gluing.

Step 1: The open cover. For each $i = 0, \ldots, n$, define

$$X_i \;:=\; \{x \in X \mid (s_i)_x \notin \mathfrak{m}_x \mathcal{L}_x\},$$

the locus where s_i does not vanish. Since $\mathcal{L}$ is locally free of rank 1, the section s_i generates $\mathcal{L}_x$ as an $\mathcal{O}_{X,x}$-module at precisely the points where it maps to a nonzero element of the fibre. The generating hypothesis means that for each $x \in X$, some s_i does not vanish at x, so $X = X_0 \cup \cdots \cup X_n$.

Each X_i is open: on any affine open $\operatorname{Spec} A \subseteq X$ where $\mathcal{L}|_{\operatorname{Spec} A} \cong \mathcal{O}_{\operatorname{Spec} A}$, the section s_i corresponds to an element $a_i \in A$, and $X_i \cap \operatorname{Spec} A = D(a_i)$.

Step 2: Trivializing $\mathcal{L}$ on each X_i. On X_i, the section s_i generates $\mathcal{L}|_{X_i}$: for each $x \in X_i$, the element $(s_i)_x$ is a generator of the free rank-1 module $\mathcal{L}_x$. Therefore, the map

$$\psi_i : \mathcal{O}_{X_i} \xrightarrow{\sim} \mathcal{L}|_{X_i}, \qquad 1 \longmapsto s_i|_{X_i},$$

is an isomorphism of $\mathcal{O}_{X_i}$-modules. Intuitively, s_i provides a "local coordinate" for the line bundle on X_i.

Step 3: Constructing regular functions s_j/s_i. For each j, the section $s_j|_{X_i}$ is an element of $\mathcal{L}(X_i)$. Via the trivialization ψ_i, it corresponds to a unique regular function:

$$\frac{s_j}{s_i} \;:=\; \psi_i^{-1}(s_j|_{X_i}) \;\in\; \mathcal{O}_X(X_i).$$

At each stalk, this is the unique element $f \in \mathcal{O}_{X,x}$ satisfying $(s_j)_x = f \cdot (s_i)_x$ in $\mathcal{L}_x$. Such f exists and is unique because $\mathcal{L}_x$ is free of rank 1 with generator $(s_i)_x$.

In particular, $s_i/s_i = 1$, and on $X_i \cap X_j$ the ratios satisfy the multiplicative relation

$$\frac{s_k}{s_i} \;=\; \frac{s_k}{s_j} \cdot \frac{s_j}{s_i} \qquad \text{on } X_i \cap X_j, \tag{5.3}$$

since $(s_k)_x = (s_k/s_j)_x \cdot (s_j)_x = (s_k/s_j)_x \cdot (s_j/s_i)_x \cdot (s_i)_x$ and generators are unique.

Step 4: Local morphisms. The ith standard affine chart of $\mathbb{P}^n_R$ is $U_i = D_+(x_i) \cong \operatorname{Spec} R[y_0, \ldots, \hat{y}_i, \ldots, y_n]$, where $y_j = x_j/x_i$. Define a ring homomorphism

$$\varphi_i^\# : \; R[y_0, \ldots, \hat{y}_i, \ldots, y_n] \;\longrightarrow\; \mathcal{O}_X(X_i), \qquad y_j \;\longmapsto\; \frac{s_j}{s_i}.$$

This is well-defined: each s_j/s_i is a regular function on X_i by Step 3, and the map extends R-linearly (since X is an R-scheme). By theorem 3.1.4, this determines a morphism of schemes $\varphi_i : X_i \to U_i$.

Step 5: Gluing. We must show that $\varphi_i|_{X_i \cap X_j} = \varphi_j|_{X_i \cap X_j}$ as morphisms into $\mathbb{P}^n_R$. The transition functions on $\mathbb{P}^n_R$ between charts U_i and U_j are given by $y_k^{(j)} = y_k^{(i)}/y_j^{(i)}$, where $y_k^{(i)} = x_k/x_i$. Under φ_i^*, we have $y_k^{(i)} \mapsto s_k/s_i$. Hence

$$\varphi_i^*\left(\frac{y_k^{(i)}}{y_j^{(i)}}\right) \;=\; \frac{s_k/s_i}{s_j/s_i} \;=\; \frac{s_k}{s_j} \;=\; \varphi_j^*(y_k^{(j)}),$$

using (5.3). Since the local morphisms agree on overlaps, they glue by lemma 3.1.9 to a unique morphism $\varphi : X \to \mathbb{P}^n_R$.

Step 6: Identifying $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$. On U_i, the sheaf $\mathcal{O}(1)$ is trivial with generator x_i (recall from proposition 5.3.5 that multiplication by x_i trivializes $\mathcal{O}(1)$ on $D_+(x_i)$). The pullback $\varphi^*(\mathcal{O}(1))|_{X_i}$ is therefore trivial with generator $\varphi^*(x_i) = s_i$. But $\mathcal{L}|_{X_i}$ is also trivial with generator s_i (by ψ_i). These identifications are compatible on overlaps: the transition function for $\varphi^*(\mathcal{O}(1))$ between X_i and X_j is $\varphi^*(x_j/x_i) = s_j/s_i$, which is the same as the transition function for $\mathcal{L}$ (since $\psi_j^{-1} \circ \psi_i$ sends $1 \mapsto s_i/s_j$, i.e., the transition function of $\mathcal{L}$ between the ψ_i and ψ_j trivializations is s_i/s_j). Hence the local isomorphisms glue to a global isomorphism $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$ sending s_i to $\varphi^*(x_i)$.

Step 7: Uniqueness. Any R-morphism $\varphi' : X \to \mathbb{P}^n_R$ with $(\varphi')^*(\mathcal{O}(1)) \cong \mathcal{L}$ and $(\varphi')^*(x_i) = s_i$ must restrict on X_i to the morphism $X_i \to U_i$ induced by the ring map $y_j \mapsto s_j/s_i$ (since $(\varphi')^*(x_j/x_i) = s_j/s_i$ is forced). By lemma 3.1.9, φ' is uniquely determined by its restrictions to the cover $\{X_i\}$, so $\varphi' = \varphi$.

5.3.30 Comparison with the Affine Case The theorem establishes a clean parallel:

Target $\mathbb{A}^n_R$	**Target** $\mathbb{P}^n_R$
$\mathrm{Hom}_R(X, \mathbb{A}^n_R) \cong \mathcal{O}_X(X)^n$	$\mathrm{Hom}_R(X, \mathbb{P}^n_R) \cong \{(\mathcal{L}, s_0, \ldots, s_n)\}/\cong$
n global functions	line bundle + $(n+1)$ generating sections
always well-defined	need base locus to be empty

In the affine case, the structure sheaf $\mathcal{O}_X$ plays the role of the line bundle, and it is always generated by a single section (1), so no additional choice is needed. In the projective case, the choice of line bundle $\mathcal{L}$ is genuine extra data — different line bundles yield morphisms into *different* projective spaces (with different embeddings). The study of which line bundles on X are generated by global sections, and how the resulting morphisms to projective space behave, is a central theme of algebraic geometry; it is developed further in section 5.4.

Example 5.3.31: Maps Into Projective Space

1. *Linear automorphisms of* $\mathbb{P}^n_k$. Let $X = \mathbb{P}^n_k$, $\mathcal{L} = \mathcal{O}(1)$, and choose generating sections $s_i = \sum_{j=0}^n a_{ij} x_j$ for an invertible matrix $(a_{ij}) \in \mathrm{GL}_{n+1}(k)$. The resulting morphism $\varphi : \mathbb{P}^n_k \to \mathbb{P}^n_k$ sends $[x_0 : \cdots : x_n] \mapsto [s_0 : \cdots : s_n]$, which is a linear change of coordinates — an automorphism of $\mathbb{P}^n_k$. Conversely, any automorphism of $\mathbb{P}^n_k$ arising from $\mathcal{O}(1)$ with linear sections must be of this form. Rescaling all sections by a common scalar $\lambda \in k^*$ does not change the morphism (since $[\lambda s_0 : \cdots : \lambda s_n] = [s_0 : \cdots : s_n]$), so we obtain $\mathrm{PGL}_{n+1}(k) \subseteq \mathrm{Aut}(\mathbb{P}^n_k)$. In fact, equality holds: every automorphism of $\mathbb{P}^n_k$ is linear (this follows from $\mathrm{Pic}\,(\mathbb{P}^n_k) \cong \mathbb{Z}$, proved in corollary 5.4.57).

2. *The Veronese embedding.* Let $X = \mathbb{P}^n_k$, $\mathcal{L} = \mathcal{O}(d)$ for $d \geq 1$, and take the sections to be all $N + 1 = \binom{n+d}{d}$ monomials of degree d. These generate $\mathcal{O}(d)$, so the theorem gives a morphism $\nu_d : \mathbb{P}^n_k \to \mathbb{P}^N_k$. For $n = 1$, $d = 2$:
$$\nu_2 : \mathbb{P}^1_k \longrightarrow \mathbb{P}^2_k, \qquad [s : t] \longmapsto [s^2 : st : t^2].$$
The image is the conic $V(y_0 y_2 - y_1^2)$, and ν_2 is a closed immersion. More generally, ν_d is always a closed immersion (see definition 3.1.36); this is the *d-uple Veronese embedding.* A key consequence: every degree-d hypersurface in $\mathbb{P}^n$ becomes a *hyperplane section* of the Veronese image, reducing questions about higher-degree geometry to linear geometry in a larger projective space.

3. *Maps from affine schemes.* Let $X = \mathrm{Spec}\, A$ be affine. Every line bundle on an affine scheme is trivial: $\mathcal{L} \cong \mathcal{O}_X$. A trivialization $\psi : \mathcal{O}_X \xrightarrow{\sim} \mathcal{L}$ sends the generating sections s_i to elements $a_i = \psi^{-1}(s_i) \in A$. The generating condition $X = X_0 \cup \cdots \cup X_n$ translates to $D(a_0) \cup \cdots \cup D(a_n) = \mathrm{Spec}\, A$, i.e., $(a_0, \ldots, a_n) = A$. The morphism $\varphi : \mathrm{Spec}\, A \to \mathbb{P}^n_R$ is then given on $D(a_i)$ by the ring map $x_j/x_i \mapsto a_j/a_i \in A_{a_i}$.

   For a concrete case take $A = k[s,t]$ with $a_0 = s$, $a_1 = t$, $a_2 = 1$. Since $(s, t, 1) = A$, this defines a morphism $\mathbb{A}^2_k \to \mathbb{P}^2_k$. On $D(s)$: $y_1 \mapsto t/s$, $y_2 \mapsto 1/s$; on $D(1) = \mathrm{Spec}\, A$: $y_0 \mapsto s$, $y_1 \mapsto t$. This is the standard open embedding $\mathbb{A}^2_k \hookrightarrow \mathbb{P}^2_k$ sending $(s,t) \mapsto [s : t : 1]$.

4. *Projections.* Let $X = \mathbb{P}^n_k$ and choose a linear subspace $\Lambda \cong \mathbb{P}^{n-m-1}_k$ (the "centre of projection"). After a change of coordinates, $\Lambda = V(x_0, \ldots, x_m)$. The sections $x_0, \ldots, x_m \in$

$\Gamma(\mathbb{P}^n_k, \mathcal{O}(1))$ generate $\mathcal{O}(1)$ at every point *outside* Λ, but they all vanish on Λ. Thus, the theorem applies to $U = \mathbb{P}^n_k \setminus \Lambda$, giving a morphism $\pi : U \to \mathbb{P}^m_k$. This is the classical *projection from* Λ: the fibre $\pi^{-1}([a])$ is the $(n-m)$-plane spanned by Λ and the point $[a_0 : \cdots : a_m : 0 : \cdots : a_m : 0 : \cdots : 0]$ and Λ; geometrically, π sends a point $P \notin \Lambda$ to the unique point where the linear span $\langle P, \Lambda\rangle$ meets a complementary $\mathbb{P}^m$.

Note that π is *not* defined on Λ itself: the base locus of the sections $x_0, \ldots, x_m$ is exactly Λ. This illustrates the general principle that a collection of sections that fails to generate $\mathcal{L}$ everywhere still defines a morphism on the complement of the base locus — a *rational map* $X \dashrightarrow \mathbb{P}^m$ in the sense of section 3.6.

5. *The Segre embedding.* Let $X = \mathbb{P}^m_k \times_k \mathbb{P}^n_k$ with the line bundle $\mathcal{L} = \mathcal{O}(1,1) := \mathrm{pr}_1^*\mathcal{O}(1) \otimes \mathrm{pr}_2^*\mathcal{O}(1)$. The global sections $\{x_i \otimes y_j\}_{0\le i\le m,\, 0\le j\le n}$ generate $\mathcal{L}$ (at any point (P,Q), some x_i is nonzero at P and some y_j is nonzero at Q, so $x_i \otimes y_j$ is nonzero). The resulting morphism

$$\sigma : \mathbb{P}^m_k \times \mathbb{P}^n_k \longrightarrow \mathbb{P}^{(m+1)(n+1)-1}_k$$

is the *Segre embedding.* . It is a closed immersion, and its image is cut out by the 2×2 minors of the matrix (z_{ij}). The Segre embedding shows, in particular, that the product of projective schemes is again projective.

5.3.32 Base-Point-Free Linear Systems The data $(\mathcal{L}, s_0, \ldots, s_n)$ appearing in theorem 5.3.29 is closely related to the classical notion of a *linear system* on X. The vector subspace $V = \mathrm{span}_k\{s_0, \ldots, s_n\} \subseteq \Gamma(X, \mathcal{L})$ determines a linear system $|V|$, and the generating condition (empty base locus) says precisely that this linear system is *base-point free*. The morphism φ depends only on $\mathcal{L}$ and the subspace V (up to the choice of basis, which amounts to a linear automorphism of the target $\mathbb{P}^n$). Choosing a different basis of V gives a morphism that differs from φ by an element of PGL_{n+1}.
When $V = \Gamma(X, \mathcal{L})$ is the *complete* linear system and $\mathcal{L}$ is very ample, the resulting morphism is a closed immersion — this is essentially the definition of very ampleness (definition 5.3.21). Choosing a proper subspace $V \subsetneq \Gamma(X, \mathcal{L})$ gives a morphism to a *smaller* projective space, obtained by composing the complete embedding with a projection (as in example 5.3.31(4)).

5.3.5 Relative Proj and Projective Bundles

Relative Spec (theorem 5.1.41) globalized Spec: it turned a sheaf of algebras over a base into a scheme over that base, and recovered the total space of a vector bundle as $\underline{\mathrm{Spec}}_X(\mathrm{Sym}(\mathcal{E}^\vee))$. The same move applies to Proj, and it is the one this book has been using informally. A ruled surface is a family of projective lines over a curve; the blow-up of an ideal sheaf is Proj of the Rees algebra $\bigoplus_{n\ge 0} \mathcal{I}^n$; and the fibrewise projective space of a vector bundle is what turns a bundle into a variety one can do geometry on. All three are instances of a single construction, and this subsection gives it.

The input is a sheaf of algebras carrying a grading, since Proj sees only the graded structure.

Definition 5.3.33: Graded Quasi-Coherent $\mathcal{O}_X$-Algebra

Let X be a scheme. A *graded quasi-coherent $\mathcal{O}_X$-algebra* is a quasi-coherent $\mathcal{O}_X$-algebra $\mathcal{S}$ (definition 5.1.39) equipped with a decomposition

$$\mathcal{S} = \bigoplus_{n \geq 0} \mathcal{S}_n$$

into quasi-coherent $\mathcal{O}_X$-submodules, such that the multiplication carries $\mathcal{S}_m \otimes_{\mathcal{O}_X} \mathcal{S}_n$ into $\mathcal{S}_{m+n}$ and such that $\mathcal{S}_0 = \mathcal{O}_X$.
The algebra is *generated in degree one* if $\mathcal{S}_1$ is of finite type and the multiplication maps induce a surjection $\mathrm{Sym}^\bullet(\mathcal{S}_1) \twoheadrightarrow \mathcal{S}$.

The generation hypothesis is not decoration. On an affine open $U = \operatorname{Spec} A$ a graded quasi-coherent algebra restricts to $\widetilde{S}$ for a graded A-algebra S with $S_0 = A$, and it is exactly when S is generated by S_1 as an A-algebra, with S_1 finitely generated, that $\operatorname{Proj} S$ carries an invertible $\mathcal{O}(1)$ and is projective over $\operatorname{Spec} A$ (example 5.3.12, proposition 5.3.5). Everything below assumes it.

Example 5.3.34: Graded Algebras from Sheaves and Ideals

1. *Symmetric algebras.* For a quasi-coherent $\mathcal{E}$, the symmetric algebra $\mathrm{Sym}^\bullet \mathcal{E} = \bigoplus_{n \geq 0} \mathrm{Sym}^n \mathcal{E}$ of example 5.1.40 is graded, with $\mathrm{Sym}^0 \mathcal{E} = \mathcal{O}_X$ and $\mathrm{Sym}^1 \mathcal{E} = \mathcal{E}$. It is generated in degree one by construction, and $\mathcal{E}$ of finite type makes $\mathcal{S}_1$ so. On an open where $\mathcal{E} \cong \mathcal{O}_U^{r+1}$ it is the polynomial algebra $\mathcal{O}_U[x_0, \dots, x_r]$ with the usual grading.

2. *Rees algebras.* For a quasi-coherent ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$, the algebra $\bigoplus_{n \geq 0} \mathcal{I}^n$ is graded, with $\mathcal{I}^0 = \mathcal{O}_X$. It is generated in degree one because $\mathcal{I}^n$ is the n-fold product of $\mathcal{I}^1 = \mathcal{I}$. This is the algebra whose Proj is the blow-up (definition 7.2.26).

3. *A generator in the wrong degree.* Let $\mathcal{S} = \mathcal{O}_X[t]$ with t placed in degree two, so $\mathcal{S}_1 = 0$ and $\mathrm{Sym}^\bullet(\mathcal{S}_1) = \mathcal{O}_X \neq \mathcal{S}$: this is not generated in degree one. Its relative Proj is X itself, since on an affine $U = \operatorname{Spec} A$ the only standard open is $D_+(t) = \operatorname{Spec}(A[t]_t)_0 = \operatorname{Spec} A$. But every element of $A[t, t^{-1}]$ has even degree, so $(A[t]_t)_1 = 0$ and $\mathcal{O}(1)$ is the zero sheaf, not a line bundle. The hypothesis is exactly what makes the twisting sheaf invertible.

The construction now proceeds exactly as relative Spec did: build the answer over each affine open, and check that the pieces glue.

Theorem 5.3.35: Relative Proj

Let X be a scheme and $\mathcal{S}$ a graded quasi-coherent $\mathcal{O}_X$-algebra generated in degree one. There exists a scheme $\underline{\operatorname{Proj}}_X(\mathcal{S})$ with a morphism $\pi\colon \underline{\operatorname{Proj}}_X(\mathcal{S}) \to X$ and an invertible sheaf $\mathcal{O}_{\underline{\operatorname{Proj}}_X(\mathcal{S})}(1)$, characterized up to canonical isomorphism by:

1. for every affine open $U = \operatorname{Spec} A \subseteq X$ with $\mathcal{S}|_U \cong \widetilde{S}$, an isomorphism $\pi^{-1}(U) \cong \operatorname{Proj} S$ over U, carrying $\mathcal{O}_{\underline{\operatorname{Proj}}_X(\mathcal{S})}(1)|_{\pi^{-1}(U)}$ to $\mathcal{O}_{\operatorname{Proj} S}(1)$ (definition 5.3.4);
2. compatibility of these isomorphisms on overlaps.

The morphism π is projective over every affine open of X; if X is Noetherian, π is proper.

***Proof*:**
Step 1: the affine pieces and their overlaps. Cover X by affine opens $U_i = \operatorname{Spec} A_i$ with $\mathcal{S}|_{U_i} \cong \widetilde{S^{(i)}}$ for graded A_i-algebras $S^{(i)}$ generated in degree one. Set $P_i = \operatorname{Proj} S^{(i)}$ with its projection $\pi_i\colon P_i \to U_i$.

The gluing data comes from a single compatibility, which is where the generation hypothesis and quasi-coherence do their work. Let $f \in A_i$ and let $D(f) \subseteq U_i$ be the corresponding basic open. Quasi-coherence gives $\mathcal{S}|_{D(f)} \cong \widetilde{S^{(i)}_f}$, where $S^{(i)}_f = S^{(i)} \otimes_{A_i} (A_i)_f$ is graded by the grading of $S^{(i)}$, since f has degree zero. Localizing at a degree-zero element commutes with the Proj construction: a homogeneous prime of $S^{(i)}_f$ not containing the irrelevant ideal is the expansion of a unique such prime of $S^{(i)}$ meeting $\{1, f, f^2, \dots\}$ trivially, and the standard opens $D_+(g)$ for g homogeneous of positive degree correspond likewise. Hence

$$\pi_i^{-1}(D(f)) \;\cong\; \operatorname{Proj}\left(S^{(i)}_f\right)$$

as schemes over $D(f)$, compatibly with the degree-one parts and therefore with $\mathcal{O}(1)$.

Step 2: gluing. On an overlap $U_i \cap U_j$, cover by basic opens that are simultaneously basic in U_i and in U_j. Over such an open, Step 1 identifies both π_i^{-1} and π_j^{-1} with the Proj of the same localization of $\mathcal{S}$, because $\mathcal{S}$ is one sheaf and the two descriptions restrict to it. The resulting isomorphisms $\varphi_{ji}\colon \pi_i^{-1}(U_i \cap U_j) \to \pi_j^{-1}(U_i \cap U_j)$ are induced by the identity of $\mathcal{S}$, so φ_{ii} is the identity and $\varphi_{kj} \circ \varphi_{ji} = \varphi_{ki}$ on triple overlaps. The cocycle condition holds, and theorem 2.2.54 produces a scheme $\underline{\operatorname{Proj}}_X(\mathcal{S})$ with a morphism π to X restricting to each π_i. The sheaves $\mathcal{O}_{P_i}(1)$ glue by the same cocycle, since the identifications of Step 1 respect $\mathcal{O}(1)$; each is invertible by proposition 5.3.5, using generation in degree one, so the glued sheaf is invertible.

Step 3: properness. Over $U_i = \operatorname{Spec} A_i$, generation in degree one gives a surjection $A_i[x_0, \dots, x_r] \twoheadrightarrow S^{(i)}$ of graded rings, sending the variables to a finite generating set of the A_i-module $S^{(i)}_1$; it is surjective because $S^{(i)}$ is generated by $S^{(i)}_1$ as an A_i-algebra. So π_i is projective by corollary 5.3.15, and hence proper by theorem 3.5.26; separatedness is part of properness (definition 3.5.20). Properness, unlike projectivity, is local on the target, so for Noetherian X (the standing hypothesis of proposition 3.5.21) properness of every π_i gives properness of π, completing the proof.

Uniqueness. A scheme over X satisfying (1) and (2) is determined on each $\pi^{-1}(U_i)$ up to canonical isomorphism, and the compatibility in (2) makes these agree on overlaps, so any two such glue to canonically isomorphic schemes.

Relative Proj is the projective counterpart of relative Spec, and the contrast between them is the usual one. Relative Spec produces affine morphisms and loses nothing; relative Proj produces proper ones, and pays for properness by discarding the irrelevant ideal. What makes the projective version the more useful of the two here is precisely that properness: proper morphisms have pushforwards that preserve coherence (corollary 5.3.28), which is what makes the fibres of a family of projective spaces computable.

Two further properties are what make relative Proj usable downstream, and both are invisible in the construction as stated: it is compatible with arbitrary base change, and its twisting sheaf carries a universal quotient. The first says that a family of projective spaces restricts to a family over any base that maps in, which is the property every fibrewise argument silently uses.

Proposition 5.3.36: Relative Proj Commutes with Base Change

Let $\mathcal{S}$ be a graded quasi-coherent $\mathcal{O}_X$-algebra generated in degree one and let $h\colon Y \to X$ be *any* morphism of schemes. Then $h^*\mathcal{S}$ is a graded quasi-coherent $\mathcal{O}_Y$-algebra generated in degree one, and there is a canonical isomorphism

$$\underline{\operatorname{Proj}}_Y(h^*\mathcal{S}) \;\cong\; \underline{\operatorname{Proj}}_X(\mathcal{S}) \times_X Y$$

over Y, carrying $\mathcal{O}_{\underline{\operatorname{Proj}}_Y(h^*\mathcal{S})}(1)$ to the pullback of $\mathcal{O}_{\underline{\operatorname{Proj}}_X(\mathcal{S})}(1)$ along the projection. In particular, for $\mathcal{E}$ locally free of rank $r+1$ there is a canonical isomorphism $\mathbb{P}(h^*\mathcal{E}) \cong \mathbb{P}(\mathcal{E}) \times_X Y$ over Y, compatible with the twisting sheaves.

***Proof*:**
Both sides are local on Y and on X, so it suffices to treat $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} A'$ with $\mathcal{S}|_X \cong \widetilde{S}$ for a graded A-algebra S generated by S_1. Then $h^*\mathcal{S} \cong \widetilde{S'}$ with $S' = S \otimes_A A'$, graded by the grading of S (the factor A' sits in degree zero) and generated by $S'_1 = S_1 \otimes_A A'$, so the hypotheses transfer.

Since S is generated in degree one, the opens $D_+(f)$ for $f \in S_1$ cover $\operatorname{Proj} S$, and likewise the $D_+(f \otimes 1)$ cover $\operatorname{Proj} S'$. Localization commutes with the base change and preserves degrees, so

$$(S'_{f\otimes 1})_0 \;\cong\; (S_f)_0 \otimes_A A'$$

which says exactly that $D_+(f \otimes 1) \cong D_+(f) \times_{\operatorname{Spec} A} \operatorname{Spec} A'$. These isomorphisms are induced by the identity of S on each piece, so they agree on overlaps and glue to the asserted isomorphism. In degree one the same computation gives $(S'_{f\otimes 1})_1 \cong (S_f)_1 \otimes_A A'$, which is the compatibility of the twisting sheaves, completing the proof.

The case the rest of the book needs is the one where the graded algebra is a symmetric algebra.

Definition 5.3.37: Projective Bundle

Let X be a scheme and $\mathcal{E}$ a locally free $\mathcal{O}_X$-module of rank $r+1$. The *projective bundle* associated to $\mathcal{E}$ is

$$\mathbb{P}(\mathcal{E}) \;=\; \underline{\mathrm{Proj}}_X(\mathrm{Sym}^\bullet\mathcal{E})$$

with its projection $\pi\colon \mathbb{P}(\mathcal{E}) \to X$ and the invertible sheaf $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ supplied by theorem 5.3.35. The latter is the *tautological*, or *Serre*, twisting sheaf of the bundle.

A point of the fibre $\pi^{-1}(x)$ corresponds to a one-dimensional *quotient* of the fibre $\mathcal{E}(x) = \mathcal{E}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)$: this is the convention already fixed for ruled surfaces in section 7.2.5, and it is Grothendieck's. The opposite convention, in which $\mathbb{P}(\mathcal{E})$ is the bundle of one-dimensional subspaces, is equally standard and is used by other books in this series; the two differ by a dual. See box 7.2.36 for the consequences of confusing them.

Proposition 5.3.38: Local Triviality of a Projective Bundle

Let $\mathcal{E}$ be locally free of rank $r+1$ on X and let $U \subseteq X$ be an open over which $\mathcal{E}|_U \cong \mathcal{O}_U^{r+1}$. Then there is an isomorphism

$$\pi^{-1}(U) \;\cong\; \mathbb{P}^r_U$$

over U, carrying $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ to $\mathcal{O}_{\mathbb{P}^r_U}(1)$ (definition 5.3.20). In particular every fibre of π over a point $x \in X$ is $\mathbb{P}^r_{\kappa(x)}$.

***Proof*:**
A trivialization $\mathcal{E}|_U \cong \mathcal{O}_U^{r+1}$ induces an isomorphism of graded $\mathcal{O}_U$-algebras $\mathrm{Sym}^\bullet(\mathcal{E}|_U) \cong \mathcal{O}_U[x_0, \ldots, x_r]$, since the symmetric algebra of a free module of rank $r+1$ is the polynomial algebra on a basis and the isomorphism respects degrees. Relative Proj depends only on the graded algebra, and its formation commutes with restriction to an open by characterization (1) of theorem 5.3.35, so $\pi^{-1}(U) \cong \underline{\mathrm{Proj}}_U(\mathcal{O}_U[x_0, \ldots, x_r]) = \mathbb{P}^r_U$. The isomorphism of algebras is the identity in degree one, so it carries $\mathcal{O}(1)$ to $\mathcal{O}(1)$.

For the fibre statement, take U a trivializing neighbourhood of x and base change along $\mathrm{Spec}\,\kappa(x) \to U$; the fibre of $\mathbb{P}^r_U \to U$ over x is $\mathbb{P}^r_{\kappa(x)}$, as we sought to show.

Local triviality is what earns the name *bundle*: a projective bundle is not a proper morphism whose fibres happen to be projective spaces, but one that is Zariski-locally on the base a product. That is a genuinely stronger statement, and it is what makes the next proposition a one-line reduction.

The twisting sheaf is not a line bundle that happens to exist: it is a quotient of the pulled-back bundle, and that is what pins down its geometric meaning and, with it, the convention above.

Proposition 5.3.39: The Tautological Quotient

Let $\mathcal{E}$ be locally free of rank $r+1$ on X, with $\pi\colon \mathbb{P}(\mathcal{E}) \to X$. There is a canonical surjection of $\mathcal{O}_{\mathbb{P}(\mathcal{E})}$-modules

$$\pi^*\mathcal{E} \twoheadrightarrow \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$$

whose kernel is locally free of rank r. Over a point $x \in X$ it specializes to the one-dimensional quotient of $\mathcal{E}(x)$ named by the corresponding point of the fibre. Dually, $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(-1)$ is canonically a subsheaf of $\pi^*\mathcal{E}^\vee$, locally free of rank one.

***Proof*:**
The degree-one piece of $\mathrm{Sym}^\bullet\mathcal{E}$ is $\mathcal{E}$ itself, and for a graded ring S generated by S_1 the sheaf $\mathcal{O}_{\operatorname{Proj} S}(1)$ is generated by the images of S_1: on $D_+(f)$ with $f \in S_1$ the module $(S_f)_1$ is free of rank one on f (this is the computation behind proposition 5.3.5). The multiplication $S_1 \otimes (S_f)_0 \to (S_f)_1$ therefore gives a surjection, and these glue over the affine opens of X by theorem 5.3.35(2) to the asserted map $\pi^*\mathcal{E} \to \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$, surjective because surjectivity is checked on stalks and holds on each $D_+(f)$.

A surjection of locally free sheaves of ranks $r+1$ and 1 has locally free kernel of rank r: the sequence splits locally, since a surjection onto a locally free sheaf admits local sections. Restricting the surjection to the fibre over x gives a one-dimensional quotient of $\mathcal{E}(x)$, which is the quotient the point names by definition 5.3.37. Dualizing the surjection gives the injection $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(-1) \hookrightarrow \pi^*\mathcal{E}^\vee$ with locally free cokernel, as we sought to show.

Proposition 5.3.40: A Projective Bundle is Proper and Flat

Let $\mathcal{E}$ be locally free of rank $r+1$ on a scheme X. Then $\pi\colon \mathbb{P}(\mathcal{E}) \to X$ is flat with all fibres of dimension r, and it is proper when X is Noetherian.

***Proof*:**
Properness is theorem 5.3.35. For flatness, both flatness (definition 6.8.2) and the dimension of the fibres are local on the base, so by proposition 5.3.38 it suffices to treat $\mathbb{P}^r_A \to \operatorname{Spec} A$ for a ring A. That morphism is covered by the standard opens $D_+(x_i) = \operatorname{Spec} A[x_0/x_i, \dots, x_r/x_i]$, each a polynomial ring in r variables over A, hence free as an A-module and so flat over A. The local ring of $\mathbb{P}^r_A$ at any point is a localization of one of these, and a localization of a flat A-algebra is again flat over A, so every local ring of $\mathbb{P}^r_A$ is flat over A, which is flatness of the morphism. The fibre over a point of $\operatorname{Spec} A$ with residue field κ is $\mathbb{P}^r_\kappa$, of dimension r, completing the proof.

Example 5.3.41: Projective Bundles

1. *The trivial bundle.* For $\mathcal{E} = \mathcal{O}_X^{r+1}$, the definition gives $\mathbb{P}(\mathcal{E}) = \mathbb{P}^r_X$, recovering relative projective space and its twisting sheaf (definition 5.3.20). Relative Proj is thus a genuine generalization of the construction already in use.

2. *Ruled surfaces.* For $\mathcal{E}$ locally free of rank 2 on a smooth projective curve C, the surface $\mathbb{P}(\mathcal{E}) \to C$ is a ruled surface; this is the construction of definition 7.2.35, now available for bundles of every rank over every base.

3. *Twisting the bundle changes nothing.* For an invertible $\mathcal{L}$ there is a canonical isomorphism $\mathbb{P}(\mathcal{E} \otimes \mathcal{L}) \cong \mathbb{P}(\mathcal{E})$ over X, because tensoring by $\mathcal{L}$ induces a degree-preserving isomorphism $\mathrm{Sym}^n(\mathcal{E} \otimes \mathcal{L}) \cong (\mathrm{Sym}^n \mathcal{E}) \otimes \mathcal{L}^{\otimes n}$ for every n. Relative Proj is unchanged by twisting the degree-n piece by $\mathcal{L}^{\otimes n}$: over an open trivializing $\mathcal{L}$ the two graded algebras are identified compatibly with the grading, and those identifications glue. The twisting sheaves differ: the isomorphism carries $\mathcal{O}_{\mathbb{P}(\mathcal{E}\otimes\mathcal{L})}(1)$ to $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1) \otimes \pi^*\mathcal{L}$. A projective bundle therefore remembers less than the vector bundle it came from, which is exactly why ruled surfaces are classified by $\mathcal{E}$ only up to twist.

4. *The blow-up.* Taking $\mathcal{S} = \bigoplus_{n\geq 0} \mathcal{I}^n$ for $\mathcal{I}$ the ideal sheaf of a closed subscheme $Z \subseteq X$, rather than a symmetric algebra, gives $\underline{\mathrm{Proj}}_X(\mathcal{S}) = \mathrm{Bl}_Z X$ (definition 7.2.26); the exceptional divisor is the fibre of this Proj over Z. Blow-ups and projective bundles are the same construction applied to different graded algebras.

Relative Proj was built to package families of projective spaces, and it also settles a question left open much earlier. The normalization of a variety was produced as a *finite* morphism (theorem 3.4.35), and finiteness gives properness (proposition 3.5.22) but says nothing about projectivity. Every finiteness theorem for coherent sheaves proved so far asks for a projective scheme rather than a proper one (theorem 5.3.27, corollary 5.3.28), so a normalization sits outside their reach until the distance from finite to projective is measured. The construction of this subsection measures it: a finite morphism is a relative Proj.

The graded algebra to feed to theorem 5.3.35 is the one with $\mathcal{A} = f_*\mathcal{O}_Y$ placed in every positive degree. What matters is the choice in degree zero. Putting $\mathcal{A}$ there, as in the algebra $\mathcal{A}[t]$, violates the requirement $\mathcal{S}_0 = \mathcal{O}_X$ of definition 5.3.33 and would produce a Proj over Y rather than over X; keeping $\mathcal{O}_X$ in degree zero and $\mathcal{A}$ above it gives a graded algebra over the intended base, whose Proj has no points at infinity because the unit of $\mathcal{A}$ is a nowhere-vanishing section of degree one.

Proposition 5.3.42: Finite Morphisms Are Projective

Let X be a Noetherian scheme, let $f\colon Y \to X$ be a finite morphism, and set $\mathcal{A} = f_*\mathcal{O}_Y$. Let $\mathcal{S} = \mathcal{O}_X + t\,\mathcal{A}[t] \subseteq \mathcal{A}[t]$ be the graded $\mathcal{O}_X$-algebra with $\mathcal{S}_0 = \mathcal{O}_X$ and $\mathcal{S}_d = \mathcal{A}t^d$ for $d \geq 1$.

1. $\mathcal{A}$ is a coherent $\mathcal{O}_X$-algebra, $\mathcal{S}$ is generated in degree one, and there is an isomorphism
$$\underline{\mathrm{Proj}}_X(\mathcal{S}) \;\cong\; \underline{\mathrm{Spec}}_X(\mathcal{A}) \;\cong\; Y$$
over X. So f is a relative Proj, and it is projective over every affine open of X.

2. Suppose in addition that X is projective over a Noetherian ring R, with $\mathcal{O}_X(1)$ very ample relative to $\mathrm{Spec}\,R$. Then there are integers $m \geq 1$ and $n \geq 0$ and a closed immersion over X
$$\iota\colon Y \hookrightarrow \mathbb{P}\big(\mathcal{O}_X(-n)^{\oplus m}\big) \;\cong\; \mathbb{P}^{m-1}_X$$
so f is projective.

***Proof*:**
Step 1: the algebra $\mathcal{A}$. A finite morphism is affine: by definition 3.4.14 the preimage of an affine open $U = \mathrm{Spec}\,B \subseteq X$ is $\mathrm{Spec}\,A$ with A finite as a B-module. So $\mathcal{A} = f_*\mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-algebra (example 5.1.40(3)) with $\mathcal{A}|_U \cong \widetilde{A}$, and $Y \cong \underline{\mathrm{Spec}}_X(\mathcal{A})$ over X by example 5.1.42(3).

Each A is a finite module over the Noetherian ring B, so $\mathcal{A}$ is coherent by proposition 5.1.11 and definition 5.1.9.

Step 2: $\mathcal{S}$ is graded and generated in degree one. Over $U = \operatorname{Spec} B$ one has $\mathcal{S}|_U \cong \widetilde{S}$ for the graded ring $S = B \oplus \bigoplus_{d\geq 1} At^d \subseteq A[t]$ with $S_0 = B$, and these identifications commute with localization at elements of B because $\mathcal{A}$ is quasi-coherent and t is a formal variable. Hence $\mathcal{S}$ is a graded quasi-coherent $\mathcal{O}_X$-algebra in the sense of definition 5.3.33, its degree-zero part being $\mathcal{O}_X$. Its degree-one part $\mathcal{S}_1 = \mathcal{A}$ is coherent, hence of finite type, and the multiplication $\operatorname{Sym}^d(\mathcal{S}_1) \to \mathcal{S}_d$ is the d-fold product $\mathcal{A}^{\otimes d} \to \mathcal{A}$, which is surjective because $a \otimes 1 \otimes \cdots \otimes 1 \mapsto a$. So $\operatorname{Sym}^\bullet(\mathcal{S}_1) \twoheadrightarrow \mathcal{S}$ is surjective.

Step 3: the Proj *is the* Spec*.* Work over $U = \operatorname{Spec} B$ as above and put $t = 1 \cdot t \in S_1$. If $\mathfrak{p}$ is a homogeneous prime containing t and $x = at^d \in S_d$ with $d \geq 1$, then $x^2 = (a^2t^{2d-1}) \cdot t \in \mathfrak{p}$, since $2d-1 \geq 1$ makes a^2t^{2d-1} an element of S; primeness gives $x \in \mathfrak{p}$, so $\mathfrak{p}$ contains S_+ and is irrelevant. Hence $\operatorname{Proj} S = D_+(t)$. The map $A \to (S_t)_0$ sending $a \mapsto at/t$ is a homomorphism of B-algebras; it is surjective because a degree-zero element of S_t is s/t^d with $s = at^d \in S_d$, and injective because $at^{N+1} = 0$ in $A[t]$ forces $a = 0$. Therefore

$$\operatorname{Proj} S \;=\; D_+(t) \;=\; \operatorname{Spec}\,(S_t)_0 \;=\; \operatorname{Spec} A$$

over $\operatorname{Spec} B$. This isomorphism is induced by the unit of $\mathcal{A}$ and commutes with restriction to basic opens of $\operatorname{Spec} B$, so the local isomorphisms agree on overlaps and glue, by the characterizations in theorem 5.3.35(1),(2) and theorem 5.1.41(1), to an isomorphism $\underline{\operatorname{Proj}}_X(\mathcal{S}) \cong \underline{\operatorname{Spec}}_X(\mathcal{A}) \cong Y$ over X. Projectivity over each affine open is part of theorem 5.3.35, proving (1).

Step 4: a locally free sheaf mapping onto $\mathcal{A}$. Assume now the hypotheses of (2). Since $\mathcal{A}$ is coherent, theorem 5.3.25 gives $n \geq 0$ and $m \geq 1$ with $\mathcal{A}(n)$ generated by m global sections, that is (definition 5.3.23) a surjection $\mathcal{O}_X^{\oplus m} \twoheadrightarrow \mathcal{A}(n)$. Tensoring with the invertible sheaf $\mathcal{O}_X(-n)$ preserves surjectivity and undoes the twist (definition 5.3.7, proposition 5.3.8(1)), so there is a surjection $q\colon \mathcal{E} \twoheadrightarrow \mathcal{A} = \mathcal{S}_1$ with $\mathcal{E} = \mathcal{O}_X(-n)^{\oplus m}$ locally free of rank m. Composing $\operatorname{Sym}^\bullet(q)$ with the surjection of Step 2 gives a surjection of graded $\mathcal{O}_X$-algebras $\operatorname{Sym}^\bullet\mathcal{E} \twoheadrightarrow \mathcal{S}$.

Step 5: the closed immersion. Cover X by affine opens $U = \operatorname{Spec} B$ over which $\mathcal{O}_X(-n)$ is trivial. Over such a U one has $\operatorname{Sym}^\bullet\mathcal{E}|_U \cong B[x_1, \ldots, x_m]$ with its usual grading (example 5.3.34(1)) and $\pi^{-1}(U) \cong \mathbb{P}^{m-1}_U$ for the projection $\pi\colon \mathbb{P}(\mathcal{E}) \to X$ of definition 5.3.37, by proposition 5.3.38. Step 4 restricts to a graded surjection $B[x_1, \ldots, x_m] \twoheadrightarrow S$, which induces a closed immersion $\operatorname{Proj} S \hookrightarrow \mathbb{P}^{m-1}_B$ by corollary 5.3.15. All of these are induced by the single map of graded algebras, so they agree on overlaps and glue to a morphism $\iota\colon Y \to \mathbb{P}(\mathcal{E})$ over X; being a closed immersion can be checked on the members of an open cover of the target (proposition 3.1.24(1)), so ι is one. Finally $\mathcal{E} = \mathcal{O}_X^{\oplus m} \otimes \mathcal{O}_X(-n)$, so $\mathbb{P}(\mathcal{E}) \cong \mathbb{P}(\mathcal{O}_X^{\oplus m}) = \mathbb{P}^{m-1}_X$ over X by example 5.3.41(3),(1), as we sought to show.

Part (1) holds over any Noetherian base, the algebra $\mathcal{S}$ being manufactured from $\mathcal{A}$ alone. Projectivity of the base is spent in part (2) on one thing, namely producing a single locally free sheaf mapping onto $\mathcal{A}$ globally; over an arbitrary Noetherian X the local embeddings supplied by (1) have no reason to assemble into one closed immersion into a fixed $\mathbb{P}^{m-1}_X$, which is why the two parts are separated. The projective bundle carries the assembly: the target $\mathbb{P}(\mathcal{O}_X(-n)^{\oplus m})$ absorbs the twist that makes

$\mathcal{A}$ a quotient and then loses it again, since twisting a bundle by a line bundle does not change its projectivization.

Corollary 5.3.43: Normalization of a Projective Scheme Is Projective

Let k be a field, let X be an integral projective k-scheme, and let $\nu\colon \widetilde{X} \to X$ be its normalization (theorem 3.4.35). Then $\widetilde{X}$ is projective over k.

***Proof*:**

Step 1: $\widetilde{X}$ is finite over $\mathbb{P}^r_k$. Fix a closed immersion $j\colon X \hookrightarrow \mathbb{P}^r_k$, which exists by the definition of a projective k-scheme (corollary 5.3.15(2)). Each standard chart $\operatorname{Spec} S$ of $\mathbb{P}^r_k$ is the spectrum of a finitely generated k-algebra and $j^{-1}(\operatorname{Spec} S) = \operatorname{Spec}(S/I)$ for an ideal I (proposition 3.1.24(1)), so X is of finite type over k; being also integral, it is a legitimate input for theorem 3.4.35, which makes ν finite. The quotient S/I is generated by 1 as an S-module, so j is itself finite by definition 3.4.14. For an affine open $\operatorname{Spec} S \subseteq \mathbb{P}^r_k$ one gets $(j \circ \nu)^{-1}(\operatorname{Spec} S) = \nu^{-1}(\operatorname{Spec} S/I) = \operatorname{Spec} C$ with C finite over S/I and S/I finite over S, hence C finite over S. So $g = j \circ \nu$ is finite.

Step 2: apply the proposition over $\mathbb{P}^r_k$. The base $\mathbb{P}^r_k$ is Noetherian (lemma 2.4.7), is projective over the Noetherian ring k, and carries the very ample sheaf $\mathcal{O}(1)$ (definition 5.3.21, taking the identity as the closed immersion). By proposition 5.3.42(2) applied to g there is a closed immersion $\widetilde{X} \hookrightarrow \mathbb{P}^{m-1}_{\mathbb{P}^r_k}$ over $\mathbb{P}^r_k$ for some $m \geq 1$.

Step 3: down to k. Writing $\mathbb{P}^{m-1}_{\mathbb{P}^r_k} = \mathbb{P}(\mathcal{O}^{\oplus m})$ (example 5.3.41(1)) and applying proposition 5.3.36 to the structure morphism $h\colon \mathbb{P}^r_k \to \operatorname{Spec} k$ and the free sheaf $\mathcal{O}^{\oplus m}$ on $\operatorname{Spec} k$ gives

$$\mathbb{P}^{m-1}_{\mathbb{P}^r_k} \;\cong\; \mathbb{P}^{m-1}_k \times_k \mathbb{P}^r_k$$

The Segre embedding (example 5.3.31(5)) is a closed immersion of the right-hand side into $\mathbb{P}^M_k$ with $M = m(r+1) - 1$. A composite of closed immersions is a closed immersion (proposition 3.1.24(3)), so $\widetilde{X}$ admits a closed immersion into $\mathbb{P}^M_k$ and is projective over k, as we sought to show.

The route through a projective bundle is not decoration. The obvious candidate for an embedding, the pullback $\nu^*\mathcal{O}_X(1)$, does not serve: a very ample sheaf pulled back along a finite morphism need not be very ample[3], so lemma 5.3.22 cannot be invoked with that sheaf in hand. What the proposition produces instead is an embedding into a projective bundle over the base, and the Segre embedding flattens the result.

With corollary 5.3.43 the finiteness statements available for projective schemes, among them finite generation of global sections of a coherent sheaf (theorem 5.3.27) and coherence of pushforwards (corollary 5.3.28), apply verbatim to the normalization of a projective variety, which is what makes normalizing a harmless operation for every finiteness argument that follows.

[3]The degree-2 morphism from a curve of genus 2 to $\mathbb{P}^1_k$ pulls $\mathcal{O}(1)$ back to a line bundle of degree 2; were that bundle very ample it would embed the curve as a curve of degree 2 in a projective space, hence as a plane conic, which has genus 0.

5.4 Divisors: Geometric Representation of Line Bundles

Having established the correspondence between line bundles and maps to projective space (section 5.3.3), we now turn to the *geometry* of line bundles themselves. The central idea is simple: just as a regular function $f \in \mathcal{O}_X(X)$ determines a vanishing scheme $V(f)$ of codimension 1, a section of a line bundle determines a "generalized hypersurface" called a *divisor*. Since regular functions are sections of the trivial line bundle $\mathcal{O}_X$, divisors are a genuine generalization of zero sets.

Divisors measure how far the geometry of a scheme is from being controlled by global functions alone. There are two natural approaches to formalizing this idea:

1. *Weil divisors* are formal integer combinations of codimension-1 subvarieties — the "geometric" approach. They record *where* a section vanishes or has poles, and with what multiplicity.
2. *Cartier divisors* encode the same data "algebraically," as collections of local rational functions with prescribed vanishing behaviour, glued together by invertible transition functions.

On sufficiently nice schemes (integral, Noetherian, locally factorial), these two notions coincide. Both give rise to an abelian group — the *divisor class group* — that classifies line bundles up to isomorphism, and this group is isomorphic to the Picard group $\operatorname{Pic}(X)$ under mild hypotheses. If the reader has seen the divisor theory for Dedekind domains in [10], the present section generalizes that theory to arbitrary schemes and connects it to the geometry of line bundles[4].

The intuition for divisors is best developed by starting with examples from complex analysis (see [52, chapters 3 and 6]), where the two notions of divisor coincide and the key ideas can be seen concretely.

Divisors on $\mathbb{C}$. Let $X = \mathbb{C}$ and consider the free abelian group generated by the points of X: each element is a finite formal sum $D = \sum_i n_i[p_i]$ with $n_i \in \mathbb{Z}$ and $p_i \in \mathbb{C}$. Every such divisor determines a rational (meromorphic) function

$$f_D(z) \;=\; \prod_i (z - p_i)^{n_i},$$

which has a zero of order n_i at p_i when $n_i > 0$ and a pole of order $|n_i|$ when $n_i < 0$. Conversely, every nonzero rational function on $\mathbb{C}$ (i.e., every nonzero element of $\mathbb{C}(z)$) has finitely many zeros and poles, and hence determines a divisor. Such divisors are called *principal.* Since every divisor on $\mathbb{C}$ is principal, the quotient group is trivial:

$$\frac{\{\text{divisors on } \mathbb{C}\}}{\{\text{principal divisors}\}} \;=\; 0.$$

Divisors on $\mathbb{P}^1$. Now let $X = \mathbb{P}^1_{\mathbb{C}}$, the Riemann sphere. The divisor group is again the free abelian group on the points of X, but the principal divisors are more constrained: a meromorphic function on a compact Riemann surface must satisfy $\sum_{p \in X} \operatorname{ord}_p(f) = 0$ (the number of zeros equals the number of poles, counted with multiplicity)[5]. The *degree* map $D = \sum n_i[p_i] \mapsto \sum n_i$ therefore vanishes on principal divisors, and one can show it induces an isomorphism

$$\frac{\{\text{divisors on } \mathbb{P}^1\}}{\{\text{principal divisors}\}} \;\cong\; \mathbb{Z}.$$

[4]This connection is foundational for the *Riemann–Roch theorem* (theorem 7.1.19), one of the most important tools for understanding the geometry of curves and, more broadly, for classifying projective varieties.

[5]For instance, z^2 viewed on $\mathbb{P}^1$ has a double zero at 0 and a double pole at ∞, summing to zero.

The generator is the class of any single point $[p]$: two divisors are linearly equivalent if and only if they have the same total degree. Under the correspondence between divisors and line bundles, the divisor class of degree n corresponds to the twisting sheaf $\mathcal{O}_{\mathbb{P}^1}(n)$.

Divisors on an elliptic curve. Let $X = \mathbb{C}/\Lambda$ be an elliptic curve, where $\Lambda \subseteq \mathbb{C}$ is a lattice. As before, a meromorphic function on X has equal numbers of zeros and poles (counted with multiplicity), so principal divisors have degree zero. Hence the degree map again gives a surjection onto $\mathbb{Z}$, and the divisor class group fits into a short exact sequence

$$0 \longrightarrow \operatorname{Pic}^0(X) \longrightarrow \operatorname{Pic}(X) \xrightarrow{\deg} \mathbb{Z} \longrightarrow 0.$$

The degree-zero part $\operatorname{Pic}^0(X)$ is *nontrivial*: by Abel's theorem, a degree-zero divisor $D = \sum n_i[p_i]$ is principal if and only if $\sum n_i p_i = 0$ in the group law of X. This gives an isomorphism $\operatorname{Pic}^0(X) \cong X$ (as a group), recovering the classical fact that an elliptic curve is its own Jacobian. In particular, $\operatorname{Pic}(X)$ is *not* simply $\mathbb{Z}$ — the divisor class group detects genuinely new geometric information beyond the degree.

The progression from $\mathbb{C}$ (trivial divisor class group) to $\mathbb{P}^1$ (divisor class group $\cong \mathbb{Z}$) to elliptic curves (divisor class group $\cong \mathbb{Z} \oplus X$) illustrates a general principle: the more geometrically interesting a scheme is, the richer its divisor theory becomes. The remainder of this section develops the algebraic theory in full generality.

5.4.1 Weil Divisors

Weil divisors generalize the divisors on curves of [20] to higher-dimensional schemes; the classical theory, including linear equivalence and the Picard group of a curve, is developed there and is the case to keep in mind throughout. The key requirement is a well-defined notion of "order of vanishing" at each codimension-1 point. On a nonsingular curve, every local ring is a DVR, so the valuation provides this order directly. In higher dimensions, the codimension-1 points play the role of "prime divisors," and we need the local rings at these points to be DVRs — equivalently, regular of dimension 1.

Definition 5.4.1: Regular in Codimension 1

A scheme X is *regular in codimension* 1 if every local ring $\mathcal{O}_{X,\mathfrak{p}}$ of dimension 1 (i.e., at each codimension-1 point $\mathfrak{p}$) is a regular local ring.

The condition just defined is checked one point at a time, which makes it easy to state and useless as a hypothesis: no one verifies a condition at every codimension-1 point of a scheme by hand. What makes it usable is that a single global hypothesis implies it. Normality (definition 2.4.27) is that hypothesis, and it does more than supply the codimension-1 points: it pushes the whole non-regular locus down two steps in codimension, so that the theory of divisors developed below never meets a singular point along a prime divisor.

Proposition 5.4.2: Normal Schemes Are Regular Outside Codimension Two

Let X be a Noetherian normal integral scheme. Then:

1. for every point $\mathfrak{p} \in X$ with $\mathrm{kdim}\mathcal{O}_{X,\mathfrak{p}} = 1$, the local ring $\mathcal{O}_{X,\mathfrak{p}}$ is a discrete valuation ring; in particular X is regular in codimension 1 (definition 5.4.1);
2. every point at which X fails to be regular has codimension at least 2 (definition 2.6.5);
3. if in addition X is locally of finite type over a field, then $\mathrm{Sing}(X)$ (definition 2.7.15) is closed and each of its irreducible components has codimension at least 2 in X.

***Proof*:**
(1). Let $\mathfrak{p} \in X$ have $\mathrm{kdim}\mathcal{O}_{X,\mathfrak{p}} = 1$. Since X is integral, the local ring $\mathcal{O}_{X,\mathfrak{p}}$ is a domain, and since X is normal it is integrally closed in its fraction field (definition 2.4.27); it is Noetherian because X is. A Noetherian integrally closed local domain of dimension 1 is a discrete valuation ring [10]. Such a ring has principal maximal ideal $\mathfrak{m} = (\pi)$, so $\mathfrak{m}/\mathfrak{m}^2$ is generated by the class of π over the residue field and has dimension at most 1; the general inequality $\mathrm{kdim}R \leq \dim_{R/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2)$ recorded in definition 2.7.2 forces equality, so $\mathcal{O}_{X,\mathfrak{p}}$ is regular.
(2). By definition 2.6.5 the codimension of a point $\mathfrak{p}$ is $\mathrm{kdim}\mathcal{O}_{X,\mathfrak{p}}$, so the claim is that $\mathcal{O}_{X,\mathfrak{p}}$ is regular whenever its dimension is 0 or 1. Dimension 1 is part (1). For dimension 0: an integral scheme has a unique generic point η (lemma 2.4.20), the only point whose local ring has dimension 0, and $\mathcal{O}_{X,\eta}$ is the function field $K(X)$ (corollary 2.4.24). A field is a regular local ring, its maximal ideal being zero.
(3). Under the additional hypothesis, $\mathrm{Sing}(X)$ is closed by proposition 2.7.16. Let Z be an irreducible component of $\mathrm{Sing}(X)$. Being an irreducible closed subset of the integral scheme X, it has a unique generic point η_Z with $\overline{\{\eta_Z\}} = Z$ (lemma 2.4.20), and η_Z lies in the closed set $\mathrm{Sing}(X)$, so $\mathcal{O}_{X,\eta_Z}$ is not regular. Part (2) gives $\mathrm{codim}_X Z = \mathrm{kdim}\mathcal{O}_{X,\eta_Z} \geq 2$, completing the proof.

Part (1) is the statement the theory of Weil divisors runs on: it says that the order function of the next subsection is defined at every codimension-1 point of a normal Noetherian scheme, with no further hypothesis on the singularities. Parts (2) and (3) say where the singularities went. On a normal surface they can only be isolated points; on a normal threefold they occupy a curve or less. This is why normality, rather than regularity, is the standing hypothesis in divisor theory: it is weak enough to hold for the schemes that arise in practice, and strong enough that the codimension-1 geometry is untouched.

The hypothesis carried by part (3) is not decoration. Parts (1) and (2) are statements about individual local rings, and for a general Noetherian scheme the non-regular locus need not be a closed subset at all; a bound on the codimension of each of its points is then no bound on the codimension of its closure. Closedness is what converts the pointwise statement into a statement about a closed subset, and the finite-type hypothesis is one standard way to obtain it. Over a general Noetherian base one asks instead that the scheme be excellent, a condition designed to make loci of this kind closed; see [63].

The converse of part (1) fails. Regularity in codimension 1 is one half of Serre's criterion for normality, and by itself it permits schemes such as the union of two planes in $\mathbb{A}^4_k$ meeting at a point, whose local rings are regular at all codimension-1 points and whose local ring at the origin is not a domain.

Normality is the conjunction of that condition with a depth condition in higher codimension, and it is the conjunction that yields the extension property for rational functions recorded earlier.

Example 5.4.3: Regular in Codimension 1

1. Every regular scheme is regular in codimension 1, since all local rings are regular. In particular, every nonsingular curve satisfies this condition.
2. Every normal Noetherian scheme is regular in codimension 1. Indeed, if $\mathcal{O}_{X,\mathfrak{p}}$ is a one-dimensional integrally closed local domain, then it is a DVR (see [10]). This is the most common hypothesis encountered in practice.
3. The scheme $X = \operatorname{Spec} k[x,y]/(xy)$ (the union of two coordinate axes) is *not* regular in codimension 1: the local ring at the origin is not a domain, hence not regular. However, each irreducible component separately is regular.

When $\mathcal{O}_{X,\mathfrak{p}}$ is a DVR with valuation $\nu_{\mathfrak{p}}$, the integer $\nu_{\mathfrak{p}}(f)$ for a nonzero rational function f measures the order of vanishing (if positive) or the order of the pole (if negative) of f along the codimension-1 subscheme $\overline{\{\mathfrak{p}\}}$. This is the only datum we need to define Weil divisors.

5.4.4 Remark: Beyond the Regular Case If $\mathcal{O}_{X,\mathfrak{p}}$ is not regular, one can still define an "order of vanishing" using the *length* $\ell_{\mathcal{O}_{X,\mathfrak{p}}}(\mathcal{O}_{X,\mathfrak{p}}/(f))$ in place of the valuation. The key property $\operatorname{ord}(fg) = \operatorname{ord}(f) + \operatorname{ord}(g)$ is preserved, as one sees from the short exact sequence

$$0 \longrightarrow \mathcal{O}_{X,\mathfrak{p}}/(g) \xrightarrow{\cdot f} \mathcal{O}_{X,\mathfrak{p}}/(fg) \longrightarrow \mathcal{O}_{X,\mathfrak{p}}/(f) \longrightarrow 0,$$

where the left map is multiplication by f (which is injective when f is a non-zero-divisor) and additivity of length gives the result. This approach works on any Noetherian scheme, but the valuation-based definition is cleaner when available, and suffices for the applications in this text.

For narrative clarity, we impose the following standing hypotheses on the ambient scheme. The conditions are chosen to ensure that: the scheme has a function field to draw rational functions from (integral), restriction maps are injective so that rational functions are unambiguous (separated), the topology is well-behaved (Noetherian), and the order of vanishing is well-defined (regular in codimension 1).

Definition 5.4.5: Weil Divisor

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. A *prime divisor* on X is a closed integral subscheme of codimension 1. A *Weil divisor* is an element of the free abelian group generated by the prime divisors:

$$\operatorname{WDiv} X := \bigoplus_{\substack{Y \subseteq X \\ \operatorname{codim}(Y)=1}} \mathbb{Z} \cdot [Y].$$

A Weil divisor $D = \sum_i n_i [Y_i]$ is called *effective* if $n_i \geq 0$ for all i, written $D \geq 0$.

Let $Y \subseteq X$ be a prime divisor with generic point η_Y. Since Y has codimension 1 and X is regular in codimension 1, the local ring $\mathcal{O}_{X,\eta_Y}$ is a one-dimensional regular local ring, hence a DVR. Denote its valuation by ν_Y. Since X is separated, the prime divisor Y is uniquely determined by its generic point (cf. exercise **??**), and hence by its valuation ring $\mathcal{O}_{X,\eta_Y} \subseteq K(X)$.

Now let $K(X)$ denote the function field of X and let $f \in K(X)^*$ be a nonzero rational function. The integer $\nu_Y(f)$ records the behaviour of f along Y:

- $\nu_Y(f) > 0$: f vanishes along Y to order $\nu_Y(f)$ (a *zero* of order $\nu_Y(f)$).
- $\nu_Y(f) < 0$: f has a *pole* of order $|\nu_Y(f)|$ along Y.
- $\nu_Y(f) = 0$: f is regular and nonvanishing at the generic point of Y.

For the formal sum $\sum_Y \nu_Y(f)[Y]$ to be a well-defined Weil divisor, we need it to be a *finite* sum. This is guaranteed by the following lemma:

Lemma 5.4.6: Almost All Valuations Vanish

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1, and let $f \in K(X)^*$. Then $\nu_Y(f) = 0$ for all but finitely many prime divisors Y.

***Proof*:**
Since f is a rational function on an integral scheme, it is regular on some nonempty open subset; choose an affine open $U = \operatorname{Spec} R \subseteq X$ on which f is regular. The complement $C = X \setminus U$ is a proper closed subset. Since X is Noetherian, C is a finite union of irreducible closed subsets, each of codimension ≥ 1. In particular, C contains at most finitely many prime divisors of X.

It remains to show that only finitely many prime divisors of U have $\nu_Y(f) \neq 0$. On U, the function f is regular (i.e., $f \in R \subseteq K(X)$), so $\nu_Y(f) \geq 0$ for every prime divisor Y meeting U. Moreover, $\nu_Y(f) > 0$ if and only if f vanishes along $Y \cap U$, meaning $Y \cap U$ is contained in the closed subset $V(f) \subseteq U$. Since $f \neq 0$, this closed subset is proper, and a Noetherian space has only finitely many irreducible components of any given codimension. Hence only finitely many prime divisors Y of U satisfy $\nu_Y(f) > 0$.

Combining, $\nu_Y(f) \neq 0$ for at most finitely many prime divisors (those contained in C together with those in $V(f) \cap U$).

This ensures that the following formal sum is a well-defined Weil divisor:

Definition 5.4.7: Principal Divisor

Let X be as above and $f \in K(X)^*$ a nonzero rational function. The *principal divisor* (or *divisor of f*) is

$$\operatorname{div}(f) \;:=\; \sum_Y \nu_Y(f)\,[Y] \;\in\; \operatorname{WDiv} X,$$

where the sum runs over all prime divisors Y of X. The subgroup of principal divisors is

$$\operatorname{PDiv}(X) \;:=\; \{\operatorname{div}(f) \mid f \in K(X)^*\} \;\leq\; \operatorname{WDiv} X.$$

The assignment $f \mapsto \operatorname{div}(f)$ is a group homomorphism $K(X)^* \to \operatorname{WDiv} X$, since $\nu_Y(fg) = \nu_Y(f) + \nu_Y(g)$ for every prime divisor Y.

Note that $\operatorname{Div}(f/g) = \operatorname{Div} f - \operatorname{Div} g$ by the properties of discrete valuations, and hence this is even a group homomorphism from the multiplicative group $(\kappa(X))^*$ to the additive group $\operatorname{Div} X$!

Two Weil divisors D, D' are called *linearly equivalent*, written $D \sim D'$, if their difference is principal: $D - D' = \operatorname{div}(f)$ for some $f \in K(X)^*$. This is an equivalence relation (since principal divisors form a subgroup), and the quotient measures the "non-trivial" divisors on X.

Definition 5.4.8: Divisor Class Group

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. The *divisor class group* (or *Weil divisor class group*) of X is

$$\operatorname{Cl} X \;:=\; \frac{\operatorname{WDiv} X}{\operatorname{PDiv}(X)}.$$

Two divisors representing the same class in $\operatorname{Cl} X$ are said to be *linearly equivalent.*

A divisor is more than a bookkeeping device: it cuts out a space of rational functions, namely those whose poles it dominates. On a smooth projective curve that space is the one Riemann-Roch computes. It can be defined for a Weil divisor on any scheme regular in codimension 1, with no Cartier hypothesis, and the point of what follows is that on a projective scheme it is finite-dimensional even then.

Definition 5.4.9: The Sheaf of a Weil Divisor

Let X be a Noetherian, integral, separated scheme, regular in codimension 1, and let $D \in \operatorname{WDiv} X$. The *sheaf associated to D* is the subsheaf $\mathcal{O}_X(D)$ of the constant sheaf $K(X)$ given on an open $V \subseteq X$ by

$$\mathcal{O}_X(D)(V) \;=\; \{f \in K(X)^* \mid (\operatorname{div} f + D)|_V \geq 0\} \cup \{0\}$$

Its global sections are written $L(D) := \Gamma(X, \mathcal{O}_X(D))$, the *Riemann-Roch space* of D.

The condition is imposed only at codimension-1 points, since that is where $\operatorname{div} f$ lives, so $\mathcal{O}_X(D)$ sees nothing of what happens in codimension ≥ 2. That insensitivity is recorded in corollary 5.4.12 below and is what makes the sheaf useful on schemes that are singular in high codimension.

Proposition 5.4.10: The Sheaf of a Weil Divisor Is Coherent

In the situation of definition 5.4.9, assume in addition that X is *normal*. Then the sheaf $\mathcal{O}_X(D)$ is a coherent $\mathcal{O}_X$-module.

Normality is doing real work and cannot be relaxed to regularity in codimension 1. The proof clears denominators and then appeals to Algebraic Hartogs' Lemma (theorem 2.6.18) to conclude that a rational function with no poles in codimension 1 is regular, and that identity, $A = \bigcap_{\mathrm{height}(\mathfrak{p})=1} A_{\mathfrak{p}}$, is a property of normal domains. Regularity in codimension 1 alone gives the R_1 half of Serre's criterion and not the S_2 half, which is exactly what the intersection formula needs.

***Proof*:**
The defining condition is local and closed under addition and under multiplication by sections of $\mathcal{O}_X$, so $\mathcal{O}_X(D)$ is a sheaf of $\mathcal{O}_X$-modules; quasi-coherence is likewise local. Since X is Noetherian it suffices to show that $M := \mathcal{O}_X(D)(\operatorname{Spec} A)$ is a finite A-module for each affine open $\operatorname{Spec} A \subseteq X$ meeting the support of D, with A a Noetherian domain.

Write $D|_{\operatorname{Spec} A} = D_+ - D_-$ with $D_\pm$ effective. Choose a nonzero $a \in A$ vanishing on the support of D_+ to order at least that of D_+ at each of its finitely many codimension-1 points; such an a exists because D_+ has finitely many components in $\operatorname{Spec} A$ and A is a domain, so a product of suitable elements of the corresponding height-1 primes will do. Then for $f \in M$ the element af has $\operatorname{div}(af) \geq \operatorname{div}(a) - D_+ \geq 0$ at every codimension-1 point of $\operatorname{Spec} A$, so af lies in A by Algebraic Hartogs' Lemma (theorem 2.6.18), A being a normal domain and therefore the intersection of its height-one localizations. Hence $M \subseteq a^{-1}A$, a free A-module of rank one, and M is finite because A is Noetherian, as we sought to show.

Corollary 5.4.11: Riemann-Roch Spaces Are Finite-Dimensional

Let X be a normal integral projective scheme over a field k, and let $D \in \operatorname{WDiv} X$. Then $L(D)$ is a finite-dimensional k-vector space.

***Proof*:**
The sheaf $\mathcal{O}_X(D)$ is coherent by proposition 5.4.10, and $L(D) = H^0(X, \mathcal{O}_X(D))$, which is finite-dimensional over k by corollary 6.6.7, completing the proof.

The last statement is the one that makes divisor theory usable on a scheme whose singular locus is small but nonempty: passing to the regular locus loses nothing.

Corollary 5.4.12: Sections over a Big Open Subset

Let X be as in definition 5.4.9 and let $U \subseteq X$ be an open subset whose complement has codimension ≥ 2. Then restriction is an isomorphism

$$\Gamma\big(X, \mathcal{O}_X(D)\big) \xrightarrow{\sim} \Gamma\big(U, \mathcal{O}_X(D)|_U\big)$$

for every $D \in \operatorname{WDiv} X$. Consequently, if X is in addition normal and projective over a field k, then $\Gamma(U, \mathcal{O}_X(D)|_U)$ is a finite-dimensional k-vector space.

***Proof*:**
Both sides are subsets of $K(X)$, since U is a nonempty open of an integral scheme, and the map is restriction, hence injective. It is surjective because the defining condition of definition 5.4.9 is imposed only at codimension-1 points of X, and every such point lies in U: a codimension-1 point outside U would lie in $X \setminus U$, which has codimension ≥ 2. So a section over U already satisfies the condition defining a section over X. The final statement follows from corollary 5.4.11, completing the proof.

Before computing examples, we establish a geometric characterization of unique factorization domains in terms of divisor class groups as these characterize trivial divisor class groups of affine schemes. The underlying algebraic input is the classical fact that a Noetherian domain is a UFD if and only if every height-1 prime ideal is principal (cf. corollary 5.3.16). The following lemma provides the geometric bridge.

Lemma 5.4.13: Normal Domain: Intersection of Height-One Localizations

Let R be a Noetherian normal domain (i.e., an integrally closed Noetherian domain). Then

$$R = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} R \\ \operatorname{ht}(\mathfrak{p})=1}} R_{\mathfrak{p}},$$

where the intersection is taken inside $\operatorname{Frac}(R)$.

***Proof*:**
The inclusion $R \subseteq \bigcap_{\operatorname{ht}(\mathfrak{p})=1} R_{\mathfrak{p}}$ is immediate. For the reverse, suppose $x \in \operatorname{Frac}(R) \setminus R$. Since R is normal, x is not integral over R. We claim that $x \notin R_{\mathfrak{p}}$ for some height-1 prime $\mathfrak{p}$.

Write $x = a/b$ with $a, b \in R$ and $b \neq 0$. Since $x \notin R$, we have $a \notin (b)$. Consider the quotient ideal

$$(b : a) = \{r \in R \mid ra \in (b)\}$$

this is a proper ideal of R (since $1 \notin (b : a)$). Choose a minimal prime $\mathfrak{p}$ over (b). By Krull's principal ideal theorem (theorem 2.6.13), $\operatorname{ht}(\mathfrak{p}) = 1$ (since (b) is principal and R is a domain). Now $R_{\mathfrak{p}}$ is a one-dimensional normal local domain, hence a DVR. In $R_{\mathfrak{p}}$, the element b generates an $\mathfrak{m}_{\mathfrak{p}}$-primary ideal, so $\nu_{\mathfrak{p}}(b) \geq 1$. But $a \notin bR_{\mathfrak{p}}$ (since $\mathfrak{p}$ was chosen to contain $(b : a)$), which means $\nu_{\mathfrak{p}}(a) < \nu_{\mathfrak{p}}(b)$, and hence $\nu_{\mathfrak{p}}(x) = \nu_{\mathfrak{p}}(a) - \nu_{\mathfrak{p}}(b) < 0$. In particular, $x \notin R_{\mathfrak{p}}$.

This lemma says that on a normal scheme, a rational function that is regular at every codimension-1 point is in fact a regular function. Put geometrically: regular functions on a normal scheme are determined by their behaviour along prime divisors. This is the algebraic analogue of the Hartogs extension theorem in complex analysis (compare theorem 2.6.18), and it is the key input for the following characterization.

Proposition 5.4.14: Characterizing UFDs Geometrically

Let R be a Noetherian domain. Then R is a UFD if and only if $\operatorname{Spec} R$ is normal and $\operatorname{Cl}\operatorname{Spec} R = 0$.

***Proof*:**
($\Rightarrow$) A UFD is integrally closed (see [10]), so $\operatorname{Spec} R$ is normal. To show $\operatorname{Cl}\operatorname{Spec} R = 0$, it suffices to show that every prime divisor is principal. Let Y be a prime divisor with generic point η_Y corresponding to a height-1 prime $\mathfrak{p} \subseteq R$. Since R is a UFD, $\mathfrak{p}$ is principal: $\mathfrak{p} = (p)$ for some irreducible element $p \in R$. Then $\operatorname{div}(p) = [Y]$ (since $\nu_Y(p) = 1$ and $\nu_{Y'}(p) = 0$ for every other prime divisor Y', as p lies in no other height-1 prime of a UFD). Hence $[Y] = 0$ in $\operatorname{Cl}\operatorname{Spec} R$.

($\Leftarrow$) It suffices to show that every height-1 prime $\mathfrak{p}$ of R is principal (this characterizes Noetherian UFDs; cf. corollary 5.3.16). Let Y be the prime divisor corresponding to $\mathfrak{p}$. Since $\operatorname{Cl}\operatorname{Spec} R = 0$, the divisor $[Y]$ is principal: there exists $f \in K(X)^*$ with $\operatorname{div}(f) = [Y]$. This means:

- $\nu_Y(f) = 1$, so $f \in R_{\mathfrak{p}}$ and f generates $\mathfrak{p}R_{\mathfrak{p}}$;
- $\nu_{Y'}(f) = 0$ for every other prime divisor Y' (corresponding to every other height-1 prime $\mathfrak{q} \neq \mathfrak{p}$), so $f \in R_{\mathfrak{q}}^*$.

In particular, $\nu_{Y'}(f) \geq 0$ for all height-1 primes, so $f \in \bigcap_{\operatorname{ht}(\mathfrak{q})=1} R_{\mathfrak{q}} = R$ by lemma 5.4.13. Similarly, $\nu_{Y'}(f^{-1}) \geq 0$ for all $\mathfrak{q} \neq \mathfrak{p}$, so there is no obstruction from other primes. We claim $\mathfrak{p} = (f)$. Take any $g \in \mathfrak{p}$; then $\nu_Y(g) \geq 1$ and $\nu_{Y'}(g) \geq 0$ for all other prime divisors Y'. Hence

$$\nu_{Y'}(g/f) \;\geq\; 0$$

for *every* height-1 prime, and so $g/f \in R$ by lemma 5.4.13. This gives $g \in fR = (f)$, so $\mathfrak{p} \subseteq (f)$. Since $f \in \mathfrak{p}$ (as $\nu_Y(f) = 1 > 0$) and $\mathfrak{p}$ is prime, we conclude $\mathfrak{p} = (f)$.

Example 5.4.15: Divisor Class Groups

1. *Affine space.* The class group $\operatorname{Cl}\mathbb{A}^n_k = 0$, since $k[x_1, \ldots, x_n]$ is a UFD and proposition 5.4.14 applies. More generally, $\operatorname{Cl}\operatorname{Spec} R = 0$ for any UFD R.
2. *Dedekind domains.* Let R be a Dedekind domain (e.g., the ring of integers $\mathcal{O}_K$ of a number field K). The scheme $X = \operatorname{Spec} R$ is a normal Noetherian integral scheme of dimension 1, so it satisfies our standing hypotheses. The prime divisors are exactly the closed points of X, corresponding to the nonzero prime ideals of R. Every Weil divisor is thus a formal sum of nonzero primes — in other words, a fractional ideal written additively. Under this identification, the principal divisors correspond to principal fractional ideals, and the divisor class group $\operatorname{Cl} X$ is precisely the *ideal class group* $\operatorname{Cl}(R)$ from algebraic number theory [8] .

In particular:

- $\mathrm{Cl}\,\mathrm{Spec}\,\mathbb{Z} = 0$, since $\mathbb{Z}$ is a PID (equivalently, a UFD).
- $\mathrm{Cl}\,\mathrm{Spec}\,\mathbb{Z}[\sqrt{-5}] \cong \mathbb{Z}/2\mathbb{Z}$, the classical example of a non-trivial ideal class group.

This recovers the fundamental theorem of algebraic number theory: R is a UFD if and only if $\mathrm{Cl}\,(R) = 0$.

We now compute the divisor class group of projective space — the fundamental example for the rest of the text. Recall that every prime divisor of $\mathbb{P}^n_k$ is a hypersurface $V_+(f)$ for some irreducible homogeneous polynomial $f \in k[x_0, \dots, x_n]$ (unique up to scalar), and its *degree* is $\deg f$.

Proposition 5.4.16: Characterizing Divisors of Projective Space

Let k be a field, $X = \mathbb{P}^n_k$, and $H = V_+(x_0)$ the standard hyperplane. For a Weil divisor $D = \sum_i n_i[Y_i]$, define $\deg D = \sum_i n_i \deg Y_i$. Then:

1. For every $f \in K(X)^*$, one has $\deg(\mathrm{div}(f)) = 0$.
2. If D is a divisor of degree d, then $[D] = d[H]$ in $\mathrm{Cl}\,\mathbb{P}^n_k$.
3. The degree map induces an isomorphism $\deg\colon \mathrm{Cl}\,\mathbb{P}^n_k \xrightarrow{\sim} \mathbb{Z}$.

***Proof*:**
Part (1). A rational function on $\mathbb{P}^n_k$ is a ratio f/g of homogeneous polynomials of the same degree d. Factor $f = \prod_i f_i^{a_i}$ and $g = \prod_j g_j^{b_j}$ into irreducibles in the UFD $k[x_0, \dots, x_n]$. Then

$$\mathrm{div}(f/g) \;=\; \sum_i a_i\,[V_+(f_i)] \;-\; \sum_j b_j\,[V_+(g_j)],$$

so $\deg(\mathrm{div}(f/g)) = \sum_i a_i \deg f_i - \sum_j b_j \deg g_j = \deg f - \deg g = 0$.

Part (2). Let $D = \sum_i n_i[Y_i]$ with $\deg D = d$. Each prime divisor Y_i is cut out by an irreducible homogeneous polynomial f_i of degree $\deg Y_i$. The expression

$$r \;=\; \frac{\prod_i f_i^{n_i}}{x_0^d}$$

is a well-defined element of $K(X)^*$: after collecting positive and negative exponents, the numerator and denominator are homogeneous of the same total degree (since $\sum_i n_i \deg f_i = d$). Computing its divisor:

$$\mathrm{div}(r) \;=\; \sum_i n_i\,[Y_i] \;-\; d\,[H] \;=\; D - d[H].$$

Hence $[D] = d[H]$ in $\mathrm{Cl}\,\mathbb{P}^n_k$.

Part (3). By (1), the degree map descends to a well-defined homomorphism $\deg\colon \mathrm{Cl}\,\mathbb{P}^n_k \to \mathbb{Z}$. It is surjective (since $\deg[H] = 1$) and injective (since any D of degree 0 satisfies $[D] = 0 \cdot [H] = 0$ by (2)).

Under the correspondence between line bundles and Cartier divisors (proposition 5.4.54), the class $n[H] \in \operatorname{Cl} \mathbb{P}^n_k$ corresponds to the twisting sheaf $\mathcal{O}_{\mathbb{P}^n_k}(n)$. This recovers the classification of line bundles on projective space from corollary 5.4.57.

The next result is the primary computational tool for divisor class groups. It describes how $\operatorname{Cl} X$ changes when we pass to an open subset.

Before comparing class groups, it is worth isolating the bookkeeping fact on which the comparison rests. Passing from X to an open subset U can only destroy the prime divisors lying entirely in the complement; those that survive are recovered from their traces on U, and nothing appears on U that was not already visible on X. The statement is used often enough, and in enough places where the ambient scheme is chosen for convenience rather than fixed in advance, that it deserves a name of its own.

Lemma 5.4.17: Extension of Weil Divisors by Closure

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1, and let $U \subseteq X$ be a nonempty open subset.

1. Let $P(U)$ be the set of prime divisors of U, and let $P_U(X) := \{Y \mid Y \text{ a prime divisor of } X,\ Y \cap U \neq \emptyset\}$. Then $Y' \mapsto \overline{Y'}$, the closure taken in X, and $Y \mapsto Y \cap U$ are mutually inverse bijections between $P(U)$ and $P_U(X)$.

2. Extension by zero, defined on generators by $[Y'] \mapsto [\overline{Y'}]$, is a group homomorphism $\operatorname{WDiv} U \to \operatorname{WDiv} X$, and it is a section of the restriction homomorphism $\operatorname{WDiv} X \to \operatorname{WDiv} U$ that sends $[Y]$ to $[Y \cap U]$ when Y meets U and to 0 otherwise.

In particular the restriction homomorphism $\operatorname{WDiv} X \to \operatorname{WDiv} U$ is surjective.

***Proof*:**
First, U inherits the standing hypotheses of definition 5.4.5: open subschemes of Noetherian schemes are Noetherian, a nonempty open subscheme of an integral scheme is integral, separatedness is preserved by open immersions, and the local rings of U are local rings of X. So $\operatorname{WDiv} U$ is defined, and U is dense in the irreducible space X. Throughout, closed integral subschemes carry the reduced induced structure, and codimension is measured at the generic point.

Let Y' be a prime divisor of U, with generic point η, and put $Y := \overline{Y'}$. The closure of an irreducible subset is irreducible, so Y is a closed integral subscheme of X with generic point η, and $\mathcal{O}_{U,\eta} = \mathcal{O}_{X,\eta}$ because U is open, whence

$$\operatorname{codim}(Y, X) = \dim \mathcal{O}_{X,\eta} = \dim \mathcal{O}_{U,\eta} = \operatorname{codim}(Y', U) = 1$$

So Y is a prime divisor of X meeting U, as $\eta \in U$. Conversely, if Y is a prime divisor of X meeting U, then $Y \cap U$ is a nonempty open subset of the irreducible space Y, hence irreducible and dense in Y, and it is closed in U; it is therefore a closed integral subscheme of U with the same generic point as Y, and the displayed computation run backwards gives $\operatorname{codim}(Y \cap U, U) = 1$. The assignments are mutually inverse: $Y \cap U$ is dense in Y, so $\overline{Y \cap U} = Y$, and Y' is closed in U, so $\overline{Y'} \cap U = Y'$. This proves (1).

For (2), the prime divisors of U form a basis of the free abelian group $\operatorname{WDiv} U$, so $[Y'] \mapsto [\overline{Y'}]$

extends uniquely to a homomorphism. Restricting it sends $[\overline{Y'}]$ to $[\overline{Y'} \cap U] = [Y']$ by (1), so the composite is the identity on a basis and hence on all of $\mathrm{WDiv}\, U$. A homomorphism admitting a section is surjective, as we sought to show.

The lemma is what lets divisor computations be transported between a scheme and a large open subset of it. A divisor written down on U may be read as a divisor on X with no further choices, and the two descriptions of one prime divisor differ only by the passage between Y and $Y \cap U$. The case of interest is that in which $X \setminus U$ has codimension at least 2: then no prime divisor of X is disjoint from U, restriction and extension are mutually inverse isomorphisms $\mathrm{WDiv}\, X \cong \mathrm{WDiv}\, U$, and X may be replaced by its regular locus, or by any convenient open piece with small complement, without loss of divisor-theoretic information. This is the mechanism behind proposition 5.4.18(2) and corollary 5.4.12.

Proposition 5.4.18: Divisor Class Group and Closed Subsets

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1, and let $Z \subsetneq X$ be a proper closed subset with complement $U = X \setminus Z$.

1. There is a surjective group homomorphism $\mathrm{Cl}\, X \twoheadrightarrow \mathrm{Cl}\, U$ induced by restriction: $\sum n_i[Y_i] \mapsto \sum n_i[Y_i \cap U]$ (discarding components that do not meet U).
2. If $\mathrm{codim}(Z, X) \geq 2$, then this map is an isomorphism.
3. If Z is irreducible of codimension 1 (hence a prime divisor), there is an exact sequence

$$\mathbb{Z} \xrightarrow{1 \mapsto [Z]} \mathrm{Cl}\, X \longrightarrow \mathrm{Cl}\, U \longrightarrow 0.$$

Point (2) explains why the class group is insensitive to removing closed subsets of "small" codimension. It is the divisor-theoretic form of the algebraic Hartogs lemma (theorem 2.6.18): regular functions extend across codimension-2 subsets, and correspondingly, the divisor theory "doesn't see" such subsets.

***Proof*:**
Part (1). Since U is a dense open subset of the integral scheme X, we have $K(U) = K(X)$. Every prime divisor of U extends uniquely to a prime divisor of X (take the closure in X), so the restriction map $\mathrm{WDiv}\, X \to \mathrm{WDiv}\, U$ is surjective on generators and hence surjective. Moreover, a principal divisor $\mathrm{div}(f)$ on X restricts to $\mathrm{div}_U(f)$ on U (the valuations at prime divisors meeting U are unchanged), so the map descends to a surjection $\mathrm{Cl}\, X \twoheadrightarrow \mathrm{Cl}\, U$.

Part (2). If $\mathrm{codim}(Z, X) \geq 2$, then Z contains no codimension-1 points of X. Hence every prime divisor of X meets U, and the restriction $\mathrm{WDiv}\, X \to \mathrm{WDiv}\, U$ is an isomorphism (not a surjection). For injectivity on class groups: if $D \in \mathrm{WDiv}\, X$ becomes principal on U, say $D|_U = \mathrm{div}_U(f)$, then $D - \mathrm{div}_X(f)$ is a Weil divisor on X supported on Z. Since Z contains no prime divisors, this forces $D - \mathrm{div}_X(f) = 0$, so D is principal. This is the divisor-theoretic counterpart of the algebraic Hartogs lemma (theorem 2.6.18): regular functions extend across codimension-2 subsets, and correspondingly the divisor theory "does not see" such subsets.

Part (3). The kernel of $\mathrm{Cl}\, X \to \mathrm{Cl}\, U$ consists of classes representable by divisors supported entirely on Z. Since Z is irreducible of codimension 1, every such divisor is an integer multiple of $[Z]$, so the kernel is exactly the image of the map $\mathbb{Z} \to \mathrm{Cl}\, X$, $1 \mapsto [Z]$.

By strategically removing closed subsets and applying the exact sequence in part (3), one can often reduce the computation of $\mathrm{Cl}\, X$ to an affine (and hence often factorial) situation. Part (2) tells us that removing "small" closed subsets does not change the class group, which frequently allows us to simplify X without losing divisor-theoretic information.

Example 5.4.19: More Divisor Class Groups

1. *Punctured affine space.* Let $X = \mathbb{A}^n_k \setminus \{0\}$ for $n \geq 2$. Since $\mathrm{Cl}\, \mathbb{A}^n_k = 0$ and $\{0\}$ has codimension $n \geq 2$, the surjection $\mathrm{Cl}\, \mathbb{A}^n_k \twoheadrightarrow \mathrm{Cl}\, X$ is an isomorphism (no prime divisors are lost). Hence $\mathrm{Cl}\, X = 0$.

2. *Complements of hypersurfaces in $\mathbb{P}^2$.* Let $Y \subseteq \mathbb{P}^2_k$ be an irreducible curve of degree d. Then Y is a prime divisor, and proposition 5.4.18(3) gives
$$\mathbb{Z} \xrightarrow{1 \mapsto [Y]} \mathrm{Cl}\, \mathbb{P}^2_k \longrightarrow \mathrm{Cl}\, (\mathbb{P}^2_k \setminus Y) \longrightarrow 0.$$
Since $[Y] = d[H]$ in $\mathrm{Cl}\, \mathbb{P}^2_k \cong \mathbb{Z}$ by **??**, the image of $\mathbb{Z}$ is $d\mathbb{Z} \subseteq \mathbb{Z}$, and hence
$$\mathrm{Cl}\, (\mathbb{P}^2_k \setminus Y) \;\cong\; \mathbb{Z}/d\mathbb{Z}.$$
For instance, the complement of a conic ($d = 2$) has class group $\mathbb{Z}/2\mathbb{Z}$, while the complement of a line ($d = 1$) has trivial class group — geometrically, every codimension-1 subvariety of $\mathbb{P}^2 \setminus L$ is already a principal divisor. The d-torsion can be understood via Bézout's theorem ([20]): any line L meets Y in exactly d points (counted with multiplicity), so $d[L \cap (\mathbb{P}^2 \setminus Y)]$ is principal on the complement, but $[L \cap (\mathbb{P}^2 \setminus Y)]$ itself need not be.

3. *The quadric cone.* Let $\mathrm{char}\, k \neq 2$ and consider the quadric cone $X = \mathrm{Spec}\, R$ where $R = k[x, y, z]/(xy - z^2)$. The ring R is a normal domain[a], so X is normal, Noetherian, integral, and regular in codimension 1. However, R is *not* a UFD: the equation $xy = z^2$ exhibits a failure of unique factorization. By proposition 5.4.14, we expect $\mathrm{Cl}\, X \neq 0$.

   Let $Y = V(y, z) \subseteq X$ be the "ruling" of the cone (see fig. 5.1). Since $\mathfrak{p} = (y, z)$ is a height-1 prime of R, the subscheme Y is a prime divisor. We claim that $\mathrm{div}(y) = 2[Y]$. Indeed, in the local ring $\mathcal{O}_{X,\eta_Y} = R_{\mathfrak{p}}$, the relation $xy = z^2$ and the fact that $x \notin \mathfrak{p}$ (so x is a unit in $R_{\mathfrak{p}}$) give $y = z^2 x^{-1} \in (z)^2$. Since z generates the maximal ideal of the DVR $R_{\mathfrak{p}}$, the uniformizer is z and $\nu_Y(y) = 2$.

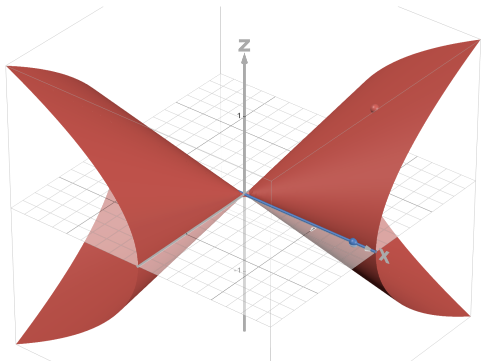


Figure 5.1: A ruling on the quadratic cone

Now apply the exact sequence to $Z = Y$:

$$\mathbb{Z} \xrightarrow{1 \mapsto [Y]} \operatorname{Cl} X \longrightarrow \operatorname{Cl}(X \setminus Y) \longrightarrow 0.$$

The complement is $X \setminus Y = \operatorname{Spec} R_y$ (inverting y). In R_y, the relation $xy = z^2$ allows us to eliminate $x = z^2 y^{-1}$, yielding $R_y \cong k[y, y^{-1}, z]$. This is a localization of the polynomial ring $k[y, z]$ (hence a UFD), so $\operatorname{Cl}(X \setminus Y) = 0$ by proposition 5.4.14. The exact sequence collapses to

$$\mathbb{Z} \longrightarrow \operatorname{Cl} X \longrightarrow 0,$$

so $\operatorname{Cl} X$ is cyclic, generated by $[Y]$. Since $\operatorname{div}(y) = 2[Y]$, we have $2[Y] = 0$ in $\operatorname{Cl} X$, giving $\operatorname{Cl} X \cong \mathbb{Z}/2\mathbb{Z}$ or $\operatorname{Cl} X = 0$.

To rule out the latter, we show that $\mathfrak{p} = (y, z)$ is not principal. Let $\mathfrak{m} = (x, y, z)$ be the maximal ideal at the vertex of the cone. The images $\overline{y}$ and $\overline{z}$ are linearly independent in the cotangent space $\mathfrak{m}/\mathfrak{m}^2 \cong k^3$ (the relation $xy - z^2 \in \mathfrak{m}^2$ imposes no constraint at the level of $\mathfrak{m}/\mathfrak{m}^2$). Hence $\mathfrak{p}$ requires at least two generators and cannot be principal. We conclude

$$\operatorname{Cl} X \ \cong\ \mathbb{Z}/2\mathbb{Z}.$$

Geometrically, a single ruling cannot be cut out by one equation, but the divisor $2[Y] = \operatorname{div}(y)$ *can*: the function y vanishes along Y with multiplicity 2.

4. *The smooth quadric surface.* Let $Q = V_+(xw - yz) \subseteq \mathbb{P}^3_k$. Under the Segre embedding (example 0.2.28), $Q \cong \mathbb{P}^1_k \times \mathbb{P}^1_k$, and Q carries two families of lines (rulings): "horizontal" lines $L_p = \{p\} \times \mathbb{P}^1$ and "vertical" lines $M_q = \mathbb{P}^1 \times \{q\}$ for $p, q \in \mathbb{P}^1$. Fix one line L_0 from the first family. Then $Q \setminus L_0 \cong \mathbb{A}^1 \times \mathbb{P}^1$ (removing one point from the first factor).

   Claim: $\operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1) \cong \mathbb{Z}$, generated by a vertical line. Removing a vertical section $M_0' = \mathbb{A}^1 \times \{q_0\}$ gives $\mathbb{A}^1 \times (\mathbb{P}^1 \setminus \{q_0\}) \cong \mathbb{A}^2$, which has $\operatorname{Cl} = 0$. The exact sequence

$$\mathbb{Z} \xrightarrow{1 \mapsto [M_0']} \operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1) \longrightarrow 0$$

   shows $\operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1)$ is cyclic, generated by $[M_0']$. To see that $[M_0']$ has infinite order, restrict to a fiber: the inclusion $\iota\colon \{s_0\} \times \mathbb{P}^1 \hookrightarrow \mathbb{A}^1 \times \mathbb{P}^1$ induces $\iota^*\colon \operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1) \to \operatorname{Cl}(\mathbb{P}^1) \cong \mathbb{Z}$ sending $[M_0'] \mapsto [\{q_0\}]$, which is a generator. Since ι^* is surjective, $[M_0']$ cannot be torsion.

   Returning to Q, the exact sequence for removing L_0 is

$$\mathbb{Z} \xrightarrow{1 \mapsto [L_0]} \operatorname{Cl} Q \longrightarrow \operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1) \cong \mathbb{Z} \longrightarrow 0.$$

   A vertical line M_0 on Q restricts to a generator of $\operatorname{Cl}(\mathbb{A}^1 \times \mathbb{P}^1)$, so the map $\mathbb{Z} \to \operatorname{Cl} Q$, $1 \mapsto [M_0]$ provides a splitting. Hence $\operatorname{Cl} Q \cong \mathbb{Z}[L_0] \oplus \mathbb{Z}[M_0]$, provided $[L_0]$ has infinite order. This follows by symmetry: swapping the two $\mathbb{P}^1$ factors exchanges horizontal and vertical rulings, so the same argument with M_0 removed shows $[L_0]$ has infinite order. Therefore

$$\operatorname{Cl} Q \ \cong\ \mathbb{Z} \oplus \mathbb{Z},$$

   generated by the two families of rulings. A hyperplane section of Q has class $[L_0] + [M_0]$[b]. In contrast to $\operatorname{Cl} \mathbb{P}^2_k \cong \mathbb{Z}$, the "degree" of a curve on Q is a *pair* of integers (a, b) recording how it meets each family of rulings — a first glimpse of how the Picard group can carry genuinely higher-rank information.

[a] A standard exercise: if $\operatorname{char} k \neq 2$ and $f \in k[x_1, \ldots, x_n]$ is square-free, then $k[x_1, \ldots, x_n, z]/(z^2 - f)$ is integrally closed.

[b] This can be checked directly: the hyperplane $V_+(x)$ meets Q in $V_+(x, yz) = V_+(x, y) \cup V_+(x, z)$, which is a horizontal ruling plus a vertical ruling.

The exact sequence in proposition 5.4.18 relates the class groups of a scheme and its open subsets. The following result provides a complementary tool, showing that the class group is insensitive to "thickening" in the affine-line direction.

Proposition 5.4.20: Class Group Stably Isomorphic

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. Then $X \times \mathbb{A}^1$ also satisfies all of these conditions, and the projection $\pi \colon X \times \mathbb{A}^1 \to X$ induces an isomorphism

$$\pi^* \colon \operatorname{Cl} X \xrightarrow{\sim} \operatorname{Cl}(X \times \mathbb{A}^1).$$

***Proof*:**
The scheme $X \times \mathbb{A}^1$ is Noetherian (locally $\operatorname{Spec} R[t]$, which is Noetherian by Hilbert's basis theorem), integral (with function field $K(X)(t)$), and separated (since separatedness is preserved under base change). It remains to verify regularity in codimension 1. The codimension-1 points of $X \times \mathbb{A}^1$ fall into two types:

- *Type 1:* the generic point of $Y \times \mathbb{A}^1$, where Y is a prime divisor of X. The local ring is $\mathcal{O}_{X,\eta_Y}[t]$ localized at $\mathfrak{m}_{\eta_Y} \cdot \mathcal{O}_{X,\eta_Y}[t]$. Since $\mathcal{O}_{X,\eta_Y}$ is a DVR (say with uniformizer u), this localization is a DVR with the same uniformizer u.
- *Type 2:* a point ξ whose image under π is the generic point of X. The local ring is $K(X)[t]$ localized at an irreducible polynomial, hence a DVR (since $K(X)[t]$ is a PID).

Thus $X \times \mathbb{A}^1$ is regular in codimension 1.

Definition of π^*. For a Weil divisor $D = \sum n_i[Y_i]$ on X, set $\pi^*(D) = \sum n_i[Y_i \times \mathbb{A}^1]$. Each $Y_i \times \mathbb{A}^1$ is a type 1 prime divisor of $X \times \mathbb{A}^1$. If $f \in K(X)^*$, then $\pi^*(\operatorname{div}_X(f)) = \operatorname{div}_{X\times\mathbb{A}^1}(f)$ (viewing f as an element of $K(X) \subseteq K(X)(t) = K(X \times \mathbb{A}^1)$), so π^* descends to a well-defined homomorphism $\pi^* \colon \operatorname{Cl} X \to \operatorname{Cl}(X \times \mathbb{A}^1)$.

Injectivity. Suppose $D \in \operatorname{WDiv} X$ and $\pi^*(D) = \operatorname{div}(h)$ for some $h \in K(X)(t)^*$. Since $\pi^*(D)$ involves only type 1 prime divisors, h must have $\nu_Z(h) = 0$ for every type 2 prime divisor Z. Write $h = g_1/g_2$ with $g_1, g_2 \in K(X)[t]$ coprime. Any irreducible factor of g_i of positive degree in t would contribute a nonzero valuation at a type 2 prime divisor, contradicting $\nu_Z(h) = 0$. Hence $g_1, g_2 \in K(X)^*$, so $h = g_1/g_2 \in K(X)^*$. It follows that $D = \operatorname{div}_X(h)$ is principal, and π^* is injective.

Surjectivity. It suffices to show that every type 2 prime divisor on $X \times \mathbb{A}^1$ is linearly equivalent to a $\mathbb{Z}$-linear combination of type 1 prime divisors (which lie in the image of π^*). Let $Z \subseteq X \times \mathbb{A}^1$ be a type 2 prime divisor. Localizing at the generic point of X gives a prime divisor of $\operatorname{Spec} K(X)[t]$. Since $K(X)[t]$ is a PID, this corresponds to a principal ideal (f) for some irreducible $f \in K(X)[t]$. Viewed as an element of $K(X)(t)^*$, the divisor $\operatorname{div}_{X\times\mathbb{A}^1}(f)$ involves Z with coefficient 1, plus possibly type 1 prime divisors arising from the denominators of the coefficients of f. Crucially, f cannot vanish along any *other* type 2 prime divisor $Z' \neq Z$: such a Z' would correspond to a different irreducible in $K(X)[t]$, and f is irreducible. Hence

$$\operatorname{div}(f) \;=\; [Z] + (\text{type 1 divisors}),$$

showing that $[Z]$ is linearly equivalent to a combination of type 1 divisors. Since type 1 divisors are in the image of π^*, the map π^* is surjective.

This result is the scheme-theoretic upgrade of the intuition that "adding a free variable doesn't change divisor theory." As immediate applications: $\mathrm{Cl}(\mathbb{A}_k^n) = 0$ for all n (by induction from $\mathrm{Cl}(\operatorname{Spec} k) = 0$), and $\mathrm{Cl}(X \times \mathbb{A}^n) \cong \mathrm{Cl}\, X$ for any X satisfying our hypotheses.

We now record several further applications that illustrate the interplay between the class group, unique factorization, and geometric structure.

Example 5.4.21: Applications of Stable Isomorphism

1. *Punctured affine space.* Let $X = \mathbb{A}_k^n \setminus \{0\}$ for $n \geq 2$. The complement $\{0\}$ has codimension $n \geq 2$ in $\mathbb{A}_k^n$, so proposition 5.4.18(2) gives $\mathrm{Cl}\, X \cong \mathrm{Cl}\, \mathbb{A}_k^n = 0$.
2. *Products with affine space.* The class group of the smooth quadric surface $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ is $\mathbb{Z} \oplus \mathbb{Z}$ (example 5.4.19). By proposition 5.4.20, $\mathrm{Cl}(\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{A}^n) \cong \mathbb{Z} \oplus \mathbb{Z}$ for all $n \geq 0$.
3. *Normality is necessary.* Let $C = \operatorname{Spec} k[x, y]/(y^2 - x^3)$ be the cuspidal cubic. The local ring at the origin is not integrally closed (y/x satisfies $(y/x)^2 = x$ but does not lie in the local ring), so C is not normal. The class group formalism as developed here does not directly apply; one must pass to the normalization $\widetilde{C} \cong \mathbb{A}_k^1$ (via $t \mapsto (t^2, t^3)$), which has $\mathrm{Cl} = 0$. The failure of normality at a single point introduces pathologies that the Cartier divisor formalism (definition 5.4.23) handles more gracefully.

5.4.2 Cartier Divisors

Weil divisors provide a geometric language for codimension-1 phenomena, but their definition requires strong hypotheses on the ambient scheme: integrality (for the function field), regularity in codimension 1 (for the valuations), and the Noetherian condition (for finiteness). On a general scheme — possibly reducible, non-reduced, or with non-regular local rings — none of these tools are available.

Cartier divisors provide an alternative that works on any scheme. The idea is to abandon the "global" perspective of Weil divisors (formal sums of codimension-1 subvarieties) in favour of a "local" one: a Cartier divisor is the data of rational functions on an open cover that are "locally principal," glued together by units on overlaps. Since principal divisors carry no interesting divisor-class information, the gluing data is precisely what the class group detects.

On sufficiently nice schemes (integral, Noetherian, locally factorial), the two theories agree and the distinction is a matter of perspective. The advantage of Cartier divisors, beyond their greater generality, is their direct connection to line bundles: we will see in proposition 5.4.54 that Cartier divisor classes are in natural bijection with isomorphism classes of invertible sheaves.

To define Cartier divisors on schemes that may not be integral, we need a replacement for the function field. The correct substitute is the *sheaf of total quotient rings*, which inverts all non-zero-divisors rather than all nonzero elements.

Definition 5.4.22: Sheaf of Total Quotient Rings

Let X be a scheme. For each affine open $U = \operatorname{Spec} R \subseteq X$, let $S_U \subseteq \Gamma(U, \mathcal{O}_X)$ denote the set of non-zero-divisors, and form the total quotient ring $S_U^{-1}\Gamma(U, \mathcal{O}_X)$. This assignment defines a presheaf on the affine opens of X; its sheafification is the *sheaf of total quotient rings*, denoted $\mathscr{K}_X$.
Write $\mathscr{K}_X^*$ for the subsheaf of abelian groups whose sections over an open U are the invertible elements of $\mathscr{K}_X(U)$, and $\mathcal{O}_X^*$ for the subsheaf of units of $\mathcal{O}_X$. The inclusion $\mathcal{O}_X^* \hookrightarrow \mathscr{K}_X^*$ makes $\mathcal{O}_X^*$ a subsheaf of abelian groups, and we may form the quotient sheaf $\mathscr{K}_X^*/\mathcal{O}_X^*$.

When X is integral, every nonzero element of an integral domain is a non-zero-divisor, so the total quotient ring of $\Gamma(U, \mathcal{O}_X)$ is simply its fraction field — which is $K(X)$ for every nonempty open U. Thus $\mathscr{K}_X$ is the constant sheaf associated to $K(X)$, and we recover the usual notion of rational function. In the non-integral case, $\mathscr{K}_X$ is genuinely more subtle: its sections are "rational functions" that may have poles but are not allowed to divide by zero-divisors.

Definition 5.4.23: Cartier Divisor

Let X be a scheme. A *Cartier divisor* on X is a global section of the quotient sheaf $\mathscr{K}_X^*/\mathcal{O}_X^*$:

$$\operatorname{CDiv} X \ := \ \Gamma(X,\ \mathscr{K}_X^*/\mathcal{O}_X^*).$$

This is an abelian group under the multiplication induced from $\mathscr{K}_X^*$. A Cartier divisor is *principal* if it lies in the image of the natural map

$$\Gamma(X, \mathscr{K}_X^*) \longrightarrow \Gamma(X, \mathscr{K}_X^*/\mathcal{O}_X^*) = \operatorname{CDiv} X.$$

Two Cartier divisors are *linearly equivalent* if they differ by a principal Cartier divisor.

Unwinding the definition of global sections of a quotient sheaf, a Cartier divisor can be represented by the following data: an open cover $\{U_i\}$ of X together with sections $f_i \in \Gamma(U_i, \mathscr{K}_X^*)$ such that on every overlap

$$\frac{f_i}{f_j} \ \in \ \Gamma(U_i \cap U_j,\ \mathcal{O}_X^*).$$

Two such collections $\{(U_i, f_i)\}$ and $\{(V_j, g_j)\}$ represent the same Cartier divisor if and only if, after passing to a common refinement, the ratios f_i/g_j are units wherever defined. The condition $f_i/f_j \in \mathcal{O}_X^*$ says that f_i and f_j agree up to an invertible regular function on the overlap — and since units contribute neither zeros nor poles, the "divisorial data" (zeros and poles of the f_i) glue to a well-defined global object.

A Cartier divisor is principal precisely when the cover can be taken to be $\{X\}$ itself: it is represented by a single global section $f \in \Gamma(X, \mathscr{K}_X^*)$.

5.4.24 Warning The group $\operatorname{CDiv} X = \Gamma(X, \mathscr{K}_X^*/\mathcal{O}_X^*)$ is *not* the same as the quotient $\Gamma(X, \mathscr{K}_X^*)/\Gamma(X, \mathcal{O}_X^*)$ in general. The former involves sections of the *quotient sheaf*, which may be represented by locally defined data that cannot be globalized. The discrepancy is measured by $H^1(X, \mathcal{O}_X^*)$, which we will identify with the Picard group in **??**.

In practice, the most frequently encountered Cartier divisors are *effective* — those with no poles:

Definition 5.4.25: Effective Cartier Divisor

A Cartier divisor $D \in \operatorname{CDiv} X$ is *effective* if it can be represented by data $\{(U_i, f_i)\}$ with each $f_i \in \Gamma(U_i, \mathcal{O}_X)$ (not in $\mathscr{K}_X^*$). Since $f_i \in \mathscr{K}_X^*$, each f_i is a non-zero-divisor in $\mathcal{O}_X(U_i)$.
An effective Cartier divisor determines a closed subscheme $D \hookrightarrow X$ whose ideal sheaf $\mathcal{I}_D \subseteq \mathcal{O}_X$ is locally generated by f_i on each U_i. This is a closed subscheme that is *locally principal*: on each U_i, it is cut out by a single equation.

An effective Cartier divisor is a closed subscheme cut out, locally, by one equation that is not a zero divisor. Both halves of that description are conditions on a single scheme, and neither survives the passage to a family on its own. If X lies over a base S and $D \subseteq X$ is locally principal, nothing prevents the local equation from degenerating over a special point of S: a non-zero-divisor on the total space may restrict to a zero divisor, or to 0, on one fiber. The subscheme D is then a divisor on X while D_s is not a divisor on X_s, and every construction that reads numerical data off the fibers collapses at that point of the base. The repair is a single condition on the morphism $D \to S$ rather than a list of conditions on the individual fibers.

Definition 5.4.26: Relative Effective Cartier Divisor

Let $f\colon X \to S$ be a morphism of schemes. A closed subscheme $D \subseteq X$ is an *effective Cartier divisor on X relative to S* (a *relative effective Cartier divisor on X/S*) if

1. the ideal sheaf $\mathcal{I}_D \subseteq \mathcal{O}_X$ is invertible, i.e. locally free of rank 1 (definition 5.1.28), so that D is an effective Cartier divisor on X; and
2. the composite $D \hookrightarrow X \xrightarrow{f} S$ is flat: for every $d \in D$, the local ring $\mathcal{O}_{D,d}$ is flat over $\mathcal{O}_{S,f(d)}$.

Both conditions are affine-local, and it pays to record what they say in coordinates. Let $\operatorname{Spec} A \subseteq S$ and $\operatorname{Spec} B \subseteq X$ be affine opens with $f(\operatorname{Spec} B) \subseteq \operatorname{Spec} A$, corresponding to a ring map $A \to B$, and shrink $\operatorname{Spec} B$ so that $\mathcal{I}_D$ is free there (definition 5.1.22). By the correspondence between closed subschemes and quasi-coherent ideal sheaves (proposition 5.1.34) and the module-sheaf dictionary (corollary 5.1.12), the restriction $\mathcal{I}_D|_{\operatorname{Spec} B}$ is $\widetilde{gB}$ for a single $g \in B$, and freeness of rank 1 says exactly that $B \to gB$, $1 \mapsto g$, is an isomorphism, i.e. $\operatorname{Ann}_B(g) = 0$. So

$$D \cap \operatorname{Spec} B \;=\; \operatorname{Spec}\left(B/gB\right)$$

with g a non-zero-divisor, and condition (2) says that B/gB is flat as an A-module.[6] Proposition 5.4.28 converts this into the condition one checks in practice: for every $s \in S$, the image $\overline{g}$ of g in the fiber ring $B \otimes_A \kappa(s)$ is again a non-zero-divisor. The converse implication is not formal; recovering flatness of B/gB from the fiberwise condition requires $X \to S$ itself to be flat, together with finiteness hypotheses and the local criterion of flatness (see [15]).

Both results below rest on one piece of commutative algebra: a flat quotient forces the defining equation to stay injective after every change of coefficients.

[6]Flatness of a module may be tested at primes, so the stalkwise condition of (2) is equivalent to flatness of B/gB over A for every such pair of affine opens; see [10].

Lemma 5.4.27: Non-Zero-Divisors and Flat Quotients

Let $A \to B$ be a ring homomorphism and let $g \in B$ satisfy $\operatorname{Ann}_B(g) = 0$. Assume $C := B/gB$ is flat as an A-module. Then for every A-module M the sequence of B-modules

$$0 \longrightarrow B \otimes_A M \xrightarrow{g \otimes \mathrm{id}} B \otimes_A M \longrightarrow C \otimes_A M \longrightarrow 0$$

is exact. In particular multiplication by g is injective on $B \otimes_A M$, and if M is an A-algebra, the image of g in the ring $B \otimes_A M$ is a non-zero-divisor.

***Proof*:**
Since $\operatorname{Ann}_B(g) = 0$, multiplication by g is injective on B, so $0 \to B \xrightarrow{g} B \to C \to 0$ is a short exact sequence of A-modules. Applying $- \otimes_A M$ and invoking the long exact sequence of $\operatorname{Tor}^A_\bullet(-, M)$ (see [15]) yields exactness of

$$\operatorname{Tor}^A_1(C, M) \longrightarrow B \otimes_A M \xrightarrow{g \otimes \mathrm{id}} B \otimes_A M \longrightarrow C \otimes_A M \longrightarrow 0$$

Flatness of C over A says precisely that $\operatorname{Tor}^A_1(C, -)$ vanishes, so the leftmost term is zero and multiplication by g on $B \otimes_A M$ is injective. When M carries an A-algebra structure, $B \otimes_A M$ is a ring and that injectivity is the assertion that the image of g is a non-zero-divisor, as we sought to show.

The first payoff is the statement that motivated the definition: the members of the family really are divisors.

Proposition 5.4.28: Relative Cartier Divisors and Fibers

Let $f\colon X \to S$ be a morphism of schemes and let $D \subseteq X$ be an effective Cartier divisor on X relative to S. For every $s \in S$, the fiber $D_s = D \times_S \operatorname{Spec} \kappa(s)$ is an effective Cartier divisor on the fiber $X_s = X \times_S \operatorname{Spec} \kappa(s)$ (definition 3.2.9).

***Proof*:**
Step 1: D_s is a closed subscheme of X_s. The morphism $D \to S$ factors through X, so associativity of fiber products (theorem 3.2.1) gives $D_s = D \times_S \operatorname{Spec} \kappa(s) = D \times_X X_s$. Thus $D_s \to X_s$ is the base change of the closed immersion $D \hookrightarrow X$ (definition 3.1.22) along $X_s \to X$, hence a closed immersion by proposition 3.2.5(2). It remains to show that $\mathcal{I}_{D_s}$ is invertible, which may be checked on an affine open cover of X_s.
Step 2: the local model. Let $x \in X_s$, and choose affine opens $\operatorname{Spec} A \subseteq S$ containing s and $\operatorname{Spec} B \subseteq X$ containing the image of x, with $f(\operatorname{Spec} B) \subseteq \operatorname{Spec} A$ and $\mathcal{I}_D$ free on $\operatorname{Spec} B$. As in the local description above, $D \cap \operatorname{Spec} B = \operatorname{Spec}(B/gB)$ with $\operatorname{Ann}_B(g) = 0$ and $C = B/gB$ flat over A. Put $\overline{B} = B \otimes_A \kappa(s)$ and let $\overline{g}$ be the image of g in $\overline{B}$. The open $\operatorname{Spec} \overline{B} = \operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} \kappa(s)$ of X_s contains x, and such opens cover X_s as $\operatorname{Spec} B$ ranges over the chosen cover of X. Right exactness of the tensor product gives

$$C \otimes_A \kappa(s) \;=\; (B/gB) \otimes_A \kappa(s) \;=\; \overline{B}/\overline{g}\,\overline{B}$$

so $D_s \cap \operatorname{Spec} \overline{B} = \operatorname{Spec}(\overline{B}/\overline{g}\,\overline{B})$, and the ideal of D_s on this open is $\overline{g}\,\overline{B}$.
Step 3: invertibility. Apply lemma 5.4.27 with the A-algebra $M = \kappa(s)$: the element $\overline{g}$ is a

non-zero-divisor in $\overline{B}$. Multiplication by $\overline{g}$ is then an isomorphism $\overline{B} \to \overline{g}\,\overline{B}$ of $\overline{B}$-modules, so $\mathcal{I}_{D_s}$ is free of rank 1 on $\operatorname{Spec} \overline{B}$ by corollary 5.1.12. These opens cover X_s, so $\mathcal{I}_{D_s}$ is invertible and D_s is an effective Cartier divisor on X_s, as we sought to show.

Proposition 5.4.28 runs the cheap implication: flatness of $\mathcal{O}_D$ over the base is a hypothesis about every A-module at once, and specializing it to $M = \kappa(s)$ costs nothing. The converse runs against the grain, since it must produce flatness, a statement quantified over all A-modules, out of information carried by one fiber at a time. Two extra inputs close the gap. The first is a source of flatness on the total space: the morphism f is assumed flat, so the ambient local ring contributes no Tor of its own. The second is a theorem reducing flatness to the vanishing of a single Tor against the residue field of the base.

The form of that theorem used below is the local criterion for flatness proved in *Homological Algebra* [15]: for a local homomorphism $(A, \mathfrak{m}) \to (B, \mathfrak{n})$ of Noetherian local rings and a finitely generated B-module M, the module M is flat over A if and only if $\operatorname{Tor}_1^A(M, A/\mathfrak{m}) = 0$. The same criterion, in the same local-homomorphism form, is recorded in *Commutative Algebra* [10], where its proof is referred back to the homological account. The Noetherian and finiteness hypotheses in the proposition are present for one reason: they are what the criterion asks for.

Proposition 5.4.29: Fiberwise Criterion for a Relative Cartier Divisor

Let S be a Noetherian scheme and let $f\colon X \to S$ be a morphism locally of finite presentation, which over a Noetherian base is the same as locally of finite type (definition 3.4.25). Assume X is flat over S, and let $D \subseteq X$ be a closed subscheme whose ideal sheaf $\mathcal{I}_D \subseteq \mathcal{O}_X$ is invertible. If for every $s \in S$ the fiber D_s is an effective Cartier divisor on the fiber X_s (definition 3.2.9), then D is an effective Cartier divisor on X relative to S (definition 5.4.26); that is, $D \to S$ is flat.

***Proof*:**
Step 1: reduction to stalks. Invertibility of $\mathcal{I}_D$ is a hypothesis, so only the flatness of $D \to S$ is at issue, and that condition is imposed one point at a time. Fix $d \in D$, put $s = f(d)$, and write

$$A = \mathcal{O}_{S,s}, \qquad B = \mathcal{O}_{X,d}, \qquad C = \mathcal{O}_{D,d}$$

Step 2: the hypotheses of the criterion. Since S is Noetherian, A is a Noetherian local ring. Choose affine opens $\operatorname{Spec} A_0 \subseteq S$ containing s and $\operatorname{Spec} B_0 \subseteq f^{-1}(\operatorname{Spec} A_0)$ containing d; then B_0 is a finitely generated algebra over the Noetherian ring A_0, hence Noetherian by the Hilbert basis theorem [10], and B is a localization of B_0, hence Noetherian local. The homomorphism $A \to B$ is local because $f(d) = s$, and it is flat because X is flat over S.
Step 3: the local equation. The sheaf $\mathcal{I}_D$ is invertible, so its stalk at d is free of rank 1 over B. By the local description following definition 5.4.26, $\mathcal{I}_{D,d} = gB$ for an element $g \in B$ with $\operatorname{Ann}_B(g) = 0$, and $C = B/gB$. In particular C is a finitely generated B-module and

$$0 \longrightarrow B \xrightarrow{g} B \longrightarrow C \longrightarrow 0$$

is a short exact sequence of A-modules.
Step 4: what the fiber hypothesis says. Write $\kappa = \kappa(s)$ and $\overline{B} = B \otimes_A \kappa$, and let $\overline{g}$ be the image of g. Localization commutes with the tensor product, so $\overline{B}$ is the local ring of X_s at d, and right

exactness gives $C \otimes_A \kappa = \overline{B}/\overline{g}\,\overline{B}$; hence $\overline{g}\,\overline{B}$ is the stalk of $\mathcal{I}_{D_s}$ at d. By hypothesis D_s is an effective Cartier divisor on X_s, so this ideal is free of rank 1 over $\overline{B}$. It is generated by $\overline{g}$, so composing the surjection $\overline{B} \to \overline{g}\,\overline{B}$, $1 \mapsto \overline{g}$, with an isomorphism $\overline{g}\,\overline{B} \cong \overline{B}$ exhibits a surjective $\overline{B}$-linear endomorphism of $\overline{B}$, which is multiplication by a unit and therefore injective. Thus $\mathrm{Ann}_{\overline{B}}(\overline{g}) = 0$.
Step 5: the Tor *computation.* Apply $\mathrm{Tor}_\bullet^A(-,\kappa)$ to the sequence of Step 3 and use $B \otimes_A \kappa = \overline{B}$:

$$\mathrm{Tor}_1^A(B,\kappa) \longrightarrow \mathrm{Tor}_1^A(C,\kappa) \longrightarrow \overline{B} \xrightarrow{\overline{g}} \overline{B}$$

The leftmost term vanishes because B is flat over A, and the rightmost map is injective by Step 4. Exactness therefore forces $\mathrm{Tor}_1^A(C,\kappa) = 0$. By Step 2 the local criterion for flatness of *Homological Algebra* [15] applies to the local homomorphism $A \to B$ and the finitely generated B-module C, and gives that $C = \mathcal{O}_{D,d}$ is flat over $A = \mathcal{O}_{S,s}$. As $d \in D$ was arbitrary, $D \to S$ is flat, which together with the invertibility of $\mathcal{I}_D$ is the definition, as we sought to show.

Each hypothesis is doing one job. Flatness of X over S kills $\mathrm{Tor}_1^A(B,\kappa)$, and without it the fiberwise condition says nothing about the total space; the Noetherian base and the finite presentation of f put the stalks inside the range of the local criterion; and the fiber hypothesis is used only through the injectivity of $\overline{g}$ on $\overline{B}$.

Flatness of X over S cannot be dropped. Take $S = \operatorname{Spec} k[t]$, $X = \operatorname{Spec} B$ with $B = k[t,x]/(t^2)$, and $D = V(x) \subseteq X$. The element $t \in B$ is nonzero and killed by t, so B is not flat over $k[t]$. The element $g = x$ satisfies $\mathrm{Ann}_B(x) = 0$, so $\mathcal{I}_D = xB$ is invertible. The fiber over a point $s \neq [(t)]$ is empty, and over $s = [(t)]$ one has $\overline{B} = k[x]$ with $\overline{g} = x$, so D_s is a reduced point on $\mathbb{A}^1_k$ and is an effective Cartier divisor there. Every fiber condition holds, yet $\mathcal{O}_D = k[t]/(t^2)$ has t-torsion and is not flat over $k[t]$, so D is not a relative effective Cartier divisor.

Nothing in that argument sees the geometry of X_s. Flatness of $\mathcal{O}_D$ over the base is converted, by one application of the vanishing of Tor_1, into the statement that the local equation stays a non-zero-divisor modulo $\mathfrak{m}_s$. The same mechanism gives stability under arbitrary base change.

Proposition 5.4.30: Base Change of a Relative Cartier Divisor

Let $f\colon X \to S$ be a morphism of schemes, let $D \subseteq X$ be an effective Cartier divisor on X relative to S, and let $S' \to S$ be an arbitrary morphism. Write $X' = X \times_S S'$ and $D' = D \times_S S'$, and let $p\colon X' \to X$ be the projection (definition 3.2.3). Then D' is an effective Cartier divisor on X' relative to S', and the canonical map $p^*\mathcal{I}_D \to \mathcal{O}_{X'}$ is injective with image $\mathcal{I}_{D'}$; in particular

$$\mathcal{I}_{D'} \;\cong\; p^*\mathcal{I}_D$$

***Proof*:**
Step 1: D' is a closed subscheme of X'. As in the previous proof, $D' = D \times_S S' = D \times_X X'$ by theorem 3.2.1, so $D' \to X'$ is the base change of $D \hookrightarrow X$ along p and is a closed immersion by proposition 3.2.5(2).
Step 2: the local model. Fix affine opens $\operatorname{Spec} A \subseteq S$, then $\operatorname{Spec} A' \subseteq S'$ mapping into $\operatorname{Spec} A$, and $\operatorname{Spec} B \subseteq X$ mapping into $\operatorname{Spec} A$ with $\mathcal{I}_D$ free on $\operatorname{Spec} B$, so that $D \cap \operatorname{Spec} B = \operatorname{Spec}(B/gB)$ with $\mathrm{Ann}_B(g) = 0$ and $C = B/gB$ flat over A. Set $B' = B \otimes_A A'$ and let g' be the image of g in B'. Then $\operatorname{Spec} B'$ is an open of X', such opens cover X', and $D' \cap \operatorname{Spec} B' = \operatorname{Spec}(C \otimes_A A')$.

Step 3: flatness over S'. Let $M' \hookrightarrow N'$ be an injection of A'-modules. The identification $(C \otimes_A A') \otimes_{A'} M' \cong C \otimes_A M'$ is functorial in M', and $M' \to N'$ is in particular an injection of A-modules, so flatness of C over A makes $C \otimes_A M' \to C \otimes_A N'$ injective. Hence $C \otimes_A A'$ is flat over A', and $D' \to S'$ is flat.

Step 4: the ideal sheaf on X'. Apply lemma 5.4.27 with $M = A'$ viewed as an A-module. Since $B \otimes_A A' = B'$ and $g \otimes \mathrm{id}$ is multiplication by g', the sequence

$$0 \longrightarrow B' \xrightarrow{g'} B' \longrightarrow C \otimes_A A' \longrightarrow 0$$

is exact. Exactness on the left says $\mathrm{Ann}_{B'}(g') = 0$; exactness on the right identifies $C \otimes_A A'$ with $B'/g'B'$. So the ideal of D' on $\operatorname{Spec} B'$ is $g'B'$, free of rank 1 over B'. As these opens cover X', the sheaf $\mathcal{I}_{D'}$ is invertible, and with Step 3 this proves D' is an effective Cartier divisor on X' relative to S'.

Step 5: comparison with the pullback. Pull the ideal sequence

$$0 \longrightarrow \mathcal{I}_D \longrightarrow \mathcal{O}_X \longrightarrow \iota_* \mathcal{O}_D \longrightarrow 0$$

of lemma 5.1.31, in which $\iota \colon D \hookrightarrow X$ is the closed immersion, back along p. The functor p^* is right exact, and on $\operatorname{Spec} B'$ it is $- \otimes_B B'$ by proposition 5.1.14(7). The isomorphism $B \to gB$, $1 \mapsto g$, gives $(gB) \otimes_B B' \cong B'$, under which the canonical map $p^* \mathcal{I}_D \to p^* \mathcal{O}_X = \mathcal{O}_{X'}$ becomes multiplication by g' on B'. Step 4 shows this map is injective with image $g'B' = \mathcal{I}_{D'}$, which is the assertion that the pulled-back ideal sequence stays exact on the left. Injectivity and the identification of the image are local statements about a globally defined map of sheaves, so they hold on all of X', completing the proof.

Proposition 5.4.28 is the case $S' = \operatorname{Spec} \kappa(s)$ of proposition 5.4.30, and the two proofs are one computation run with different coefficients. Note which half of the definition does the work. Invertibility transports for free, since the pullback of an invertible sheaf along any morphism is invertible. What is not free is that $p^* \mathcal{I}_D$ still *embeds* in $\mathcal{O}_{X'}$, so that it is the ideal sheaf of a closed subscheme rather than an abstract line bundle equipped with a map to $\mathcal{O}_{X'}$ that kills something. That injectivity is the vanishing of $\mathrm{Tor}_1^A(C, A')$, which is flatness of $\mathcal{O}_D$ over the base.

Without the flatness hypothesis the notion is unstable in families, in a way that is easy to exhibit. Take $S = \operatorname{Spec} k[t]$, $X = \operatorname{Spec} k[t, x, y]$, and $D = V(t) \subseteq X$. The element t is a non-zero-divisor in $k[t, x, y]$, so D is an effective Cartier divisor on X in the absolute sense. But $D \to S$ is the inclusion of a point, $\operatorname{Spec} k[x, y] \to \operatorname{Spec} k[t]$, which is not flat, and the fibers of D jump from empty over $t \neq 0$ to all of $X_0 = \mathbb{A}^2_k$ over $t = 0$. Over the origin the ideal sheaf of D_0 is zero, so D_0 is not a divisor on X_0 at all. Flatness of $D \to S$ rules this out by forcing the local equation to remain a non-zero-divisor after restriction to every fiber, and that is the one thing needed for the divisorial character of the family to be constant across the base.

The two propositions together say that the assignment

$$S' \longmapsto \{D' \subseteq X \times_S S' \mid D' \text{ relative effective Cartier on } (X \times_S S')/S'\}$$

is a contravariant functor on S-schemes. Representability of that functor produces the scheme of relative divisors and, through it, the relative Picard functor (see [17]); the same stability is what lets divisor classes be intersected in families (see [16]).

The effective Cartier divisors are precisely the closed subschemes whose ideal sheaf is an invertible $\mathcal{O}_X$-module (i.e., locally free of rank 1). This characterization — which requires no reference to $\mathscr{K}_X$ at all — is the starting point adopted by the Stacks project[7], and it has the advantage of working without any integrality or Noetherian hypotheses. We will return to this perspective in proposition 5.4.58.

When X satisfies the standing hypotheses of the previous subsection (Noetherian, integral, separated, regular in codimension 1), Cartier divisors can be compared directly with Weil divisors.

Proposition 5.4.31: Cartier Divisors to Weil Divisors

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. There is a group homomorphism

$$\mathrm{CDiv}\, X \longrightarrow \mathrm{WDiv}\, X$$

that sends principal Cartier divisors to principal Weil divisors.

***Proof*:**
Since X is integral, $\mathscr{K}_X$ is the constant sheaf $K(X)$, so a Cartier divisor is represented by data $\{(U_i, f_i)\}$ with $f_i \in K(X)^*$ and $f_i/f_j \in \mathcal{O}_X^*(U_i \cap U_j)$. For each prime divisor Y of X, choose an index i with $\eta_Y \in U_i$ (such an i exists since the U_i cover X) and define the coefficient of $[Y]$ to be $\nu_Y(f_i)$. This is independent of the choice: if $\eta_Y \in U_j$ as well, then f_i/f_j is a unit near η_Y, so $\nu_Y(f_i/f_j) = 0$ and $\nu_Y(f_i) = \nu_Y(f_j)$.

The resulting sum $\sum_Y \nu_Y(f_i)[Y]$ is finite: since X is quasi-compact (it is Noetherian), we may take the cover $\{U_i\}$ to be finite, and on each U_i the rational function f_i has nonzero valuation at only finitely many prime divisors (by the same argument as lemma 5.4.6).

If $\{(U_i, f_i)\}$ and $\{(U_i, f_i')\}$ represent Cartier divisors D and D' (after passing to a common refinement), then $D + D'$ is represented by $\{(U_i, f_i f_i')\}$, and $\nu_Y(f_i f_i') = \nu_Y(f_i) + \nu_Y(f_i')$, so the map is a homomorphism. A principal Cartier divisor is represented by a single $f \in K(X)^*$, which maps to $\mathrm{div}(f) \in \mathrm{WDiv}\, X$.

The key question is: when is this map an isomorphism? It is surjective precisely when every Weil divisor is *locally principal* — that is, when each prime divisor is locally cut out by a single equation. The natural hypothesis guaranteeing this is that every local ring is a UFD (so every height-1 prime is principal).

Proposition 5.4.32: Weil and Cartier Divisors Under Locally Factorial Hypotheses

Let X be an integral, separated, Noetherian scheme that is locally factorial (i.e., $\mathcal{O}_{X,x}$ is a UFD for all $x \in X$; cf. definition 2.4.29). Then the homomorphism of proposition 5.4.31 is an isomorphism

$$\mathrm{CDiv}\, X \xrightarrow{\sim} \mathrm{WDiv}\, X,$$

and principal Cartier divisors correspond to principal Weil divisors. In particular, there is an induced isomorphism of class groups.

[7] `https://stacks.math.columbia.edu`, Tag 01WR.

***Proof*:**
Since every UFD is integrally closed, X is normal, and hence regular in codimension 1 (example 5.4.3). Because X is integral, $\mathscr{K}_X$ is the constant sheaf $K(X)$.

Injectivity. Suppose a Cartier divisor D, represented by $\{(U_i, f_i)\}$, maps to the zero Weil divisor. Then $\nu_Y(f_i) = 0$ for every prime divisor Y meeting U_i. For each affine open $V = \operatorname{Spec} R \subseteq U_i$, the element $f_i \in K(X)^*$ satisfies $\nu_{\mathfrak{p}}(f_i) = 0$ for every height-1 prime $\mathfrak{p}$ of R. Since R is a UFD (hence normal), lemma 5.4.13 gives $f_i \in R^*$, so f_i is a unit on V. As this holds for every affine open in every U_i, we have $f_i \in \mathcal{O}_X^*(U_i)$ for all i, meaning $D = 0$ in $\operatorname{CDiv} X$.

Surjectivity. Let $D = \sum n_j[Y_j]$ be a Weil divisor. For each point $x \in X$, the prime divisors through x correspond to height-1 primes of $\mathcal{O}_{X,x}$. Since $\mathcal{O}_{X,x}$ is a UFD, each such prime is principal, so $\operatorname{Cl}(\operatorname{Spec} \mathcal{O}_{X,x}) = 0$ by proposition 5.4.14. Hence the restriction of D to $\operatorname{Spec} \mathcal{O}_{X,x}$ is principal: there exists $f_x \in K(X)^*$ with

$$\nu_Y(f_x) = \begin{cases} n_j & \text{if } Y = Y_j \text{ passes through } x, \\ 0 & \text{for all other prime divisors through } x. \end{cases}$$

Now, D and $\operatorname{div}(f_x)$ can differ only at prime divisors not passing through x. Both D and $\operatorname{div}(f_x)$ involve finitely many prime divisors (the latter by lemma 5.4.6), and those not passing through x form a closed subset $Z_x \subseteq X$ with $x \notin Z_x$. Set $U_x = X \setminus Z_x$; then $D|_{U_x} = \operatorname{div}(f_x)|_{U_x}$.

It remains to verify that the collection $\{(U_x, f_x)\}_{x \in X}$ defines a Cartier divisor, i.e., that $f_x/f_{x'} \in \mathcal{O}_X^*(U_x \cap U_{x'})$ for all x, x'. On $U_x \cap U_{x'}$, both f_x and $f_{x'}$ represent D, so $\operatorname{div}(f_x/f_{x'}) = 0$ on $U_x \cap U_{x'}$. For any affine open $V = \operatorname{Spec} R \subseteq U_x \cap U_{x'}$, the ring R is a normal domain (each stalk is a UFD, hence integrally closed). By lemma 5.4.13, $f_x/f_{x'} \in \bigcap_{\operatorname{ht}(\mathfrak{p})=1} R_{\mathfrak{p}} = R$, and the same argument applied to $f_{x'}/f_x$ gives the inverse. Hence $f_x/f_{x'} \in R^* = \mathcal{O}_X^*(V)$, and since this holds on every affine open in $U_x \cap U_{x'}$, we conclude $f_x/f_{x'} \in \mathcal{O}_X^*(U_x \cap U_{x'})$.

These two constructions are mutually inverse (one verifies directly that composing in either order yields the identity), and both send principal divisors to principal divisors.

The locally factorial hypothesis is satisfied by all regular schemes (since regular local rings are UFDs by the Auslander–Buchsbaum theorem; cf. proposition 2.7.20) and by all smooth varieties over a field. In particular, the Weil and Cartier theories agree on any nonsingular variety — the setting where divisors are most commonly used. On a normal but non-factorial scheme (such as the quadric cone of example 5.4.19), the map $\operatorname{CDiv} X \to \operatorname{WDiv} X$ is a proper inclusion: the ruling $[Y]$ is a Weil divisor that is *not* locally principal at the vertex, hence has no Cartier counterpart.

When X is not locally factorial, the map $\operatorname{CDiv} X \to \operatorname{WDiv} X$ identifies Cartier divisors with a distinguished subgroup of Weil divisors:

Definition 5.4.33: Locally Principal Weil Divisor

Let X be a Noetherian, integral, separated scheme that is regular in codimension 1. A Weil divisor $D \in \operatorname{WDiv} X$ is *locally principal* if for every $x \in X$, there exists an open neighbourhood U of x and $f \in K(X)^*$ such that $D|_U = \operatorname{div}(f)|_U$.

By construction, the image of $\operatorname{CDiv} X \to \operatorname{WDiv} X$ consists precisely of the locally principal Weil divisors. We therefore have a chain of inclusions

$$\mathrm{PDiv}(X) \subseteq \{\text{locally principal Weil divisors}\} \subseteq \mathrm{WDiv}\,X,$$

and the locally factorial hypothesis of proposition 5.4.32 collapses the second inclusion to an equality.

Definition 5.4.34: Cartier Divisor Class Group

Let X be a scheme. The *Cartier divisor class group* of X is

$$\mathrm{CaCl}\,X := \frac{\mathrm{CDiv}\,X}{\{\text{principal Cartier divisors}\}}.$$

When X is integral, separated, Noetherian, and locally factorial, proposition 5.4.32 gives an isomorphism $\mathrm{CaCl}\,X \cong \mathrm{Cl}\,X$.

Example 5.4.35: Cartier Divisors

1. *Projective space.* On $\mathbb{P}^n_k$, every local ring is a localization of a polynomial ring, hence a UFD. By proposition 5.4.32, every Weil divisor is Cartier, and $\mathrm{CaCl}\,\mathbb{P}^n_k \cong \mathrm{Cl}\,\mathbb{P}^n_k \cong \mathbb{Z}$. A hyperplane $H = V_+(x_0)$ is represented by the Cartier data $\{(U_i, x_0/x_i)\}_{i=0}^n$: on the standard affine chart $U_i = D_+(x_i) \cong \mathbb{A}^n$, the hyperplane is cut out by the single function $x_0/x_i \in \mathcal{O}_X(U_i)$.

2. *Smooth curves.* Every nonsingular curve is locally factorial (one-dimensional regular local rings are DVRs, hence UFDs). Thus Weil and Cartier divisors coincide, and a divisor is simply a formal sum of closed points — the classical notion of a divisor on a curve.

3. *Normal, non-factorial schemes.* On the quadric cone $X = \mathrm{Spec}\,k[x,y,z]/(xy - z^2)$ (example 5.4.19), the ruling $Y = V(y,z)$ is a Weil divisor that is *not* locally principal at the vertex (x,y,z): the prime ideal (y,z) is not principal in the local ring at that point. However, $2[Y] = \mathrm{div}(y)$ *is* Cartier. In this case, $\mathrm{CDiv}\,X \subsetneq \mathrm{WDiv}\,X$, and $\mathrm{CaCl}\,X = 0 \neq \mathbb{Z}/2\mathbb{Z} = \mathrm{Cl}\,X$.[a]

4. *Non-reduced schemes.* Let $X = \mathrm{Spec}\,k[x]/(x^2)$, the "double point." Since x is a zero-divisor, the element $x \in \mathcal{O}_X(X)$ does not define an effective Cartier divisor. The total quotient ring is $k[x]/(x^2)$ itself (no non-zero-divisors to invert beyond units), so $\mathscr{K}_X = \mathcal{O}_X$ and $\mathrm{CDiv}\,X = 0$. This illustrates the limitations of the Cartier formalism on highly non-reduced schemes.

[a] The Cartier class group is trivial because the only Cartier divisors on X are principal: any locally principal divisor is an even multiple of $[Y]$, hence already principal.

Example 5.4.36: Weil $\neq$ Cartier on Singular Curves

Example. Consider the nodal cubic $X = V(y^2 - x^3 - x^2) \subseteq \mathbb{P}^2_k$ (with $\mathrm{char} k \neq 2$), which has a node at the origin $P = (0,0)$. The local ring $\mathcal{O}_{X,P} \cong k[x,y]_{(x,y)}/(y^2 - x^2 - x^3)$ is *not* a UFD: in the completion $\widehat{\mathcal{O}}_{X,P} \cong k[\![u,v]\!]/(uv)$, the element $y^2 - x^2 = x^3$ factors into two branches that cannot be separated in $\mathcal{O}_{X,P}$ itself[a]. The Weil divisor $1 \cdot P$ (a single closed point, considered as a codimension-1 subvariety) is *not* Cartier at P: its ideal sheaf $\mathcal{I}_P$ is not locally principal near P, because the maximal ideal $\mathfrak{m}_P \subseteq \mathcal{O}_{X,P}$ is not principal (it requires two generators). Hence $\mathcal{I}_P$ is a coherent ideal sheaf, but *not* an invertible sheaf.

More concretely, on the normalization $\nu\colon \widetilde{X} \cong \mathbb{P}^1 \to X$, the preimage $\nu^{-1}(P)$ consists of two points $\{P_1, P_2\}$ (the two branches), and the Weil divisor $P_1 + P_2$ on $\mathbb{P}^1$ is Cartier—but it is the pullback of P only as a *cycle*, not as a Cartier divisor on X.

The upshot:

[a]See example 3.1.30 for the formal branch structure of a node.

5.4.3 Families of Weil Divisors and Birational Transforms

Two operations carry a Weil divisor from one normal variety to another, and they are easily confused. The first is specialization inside a family. A divisor $\mathbf{D}$ on a product $S \times X$ has a fiber $\mathbf{D}_s$ over each parameter $s \in S$, got by restricting $\mathbf{D}$ to the regular locus of the fiber, where it is Cartier and restriction is defined, and taking the closure. The second is transport along a birational map $g\colon X \dashrightarrow Y$. Every prime divisor of X carries a discrete valuation of the function field $K(X)$; the birational transform g_* sends it to the prime divisor of Y carrying the same valuation when there is one, and to zero when there is none.

These two operations disagree, and they disagree by exceptional terms. A birational map contracts some prime divisors and extracts others; a fiber of a family records the contracted ones, and a birational transform discards them. Example 5.4.48 computes the smallest instance, on the blow-up of $\mathbb{P}^2$ at a point: the fiber of a family of lines whose special member passes through the blown-up point acquires the exceptional curve as a component, while the birational transform of that same line does not. The general accounting of the discrepancy, a finite list of linear subspaces of the parameter space outside of which the two agree, is **??**; that is a forward reference, and nothing below depends on it.

The distinction has consequences. Chevalley's structure theorem, that a connected algebraic group over a perfect field is an extension of an abelian variety by an affine group, is proved in *Abelian Varieties* [6] by an argument that manufactures a projective model X on which a group G acts only rationally and then compares the divisors $\mathbf{D}_g$ as g varies. The 2002 proof of Conrad [22] treats the translate of a divisor by a group element as if the two operations above agreed, and a translation of a rationally acting group can contract a boundary divisor; the repair, following Kollár [40], needs the two notions kept apart from the start. That is what this subsection does.

Definition 5.4.37: Family of Weil Divisors

Let k be an algebraically closed field, let X be a normal variety over k (definition 2.4.27), and let S be a smooth variety over k (definition 2.7.11). Write $\pi\colon S \times_k X \to S$ for the first projection and $X^{\mathrm{reg}} \subseteq X$ for the complement of $\operatorname{Sing}(X)$ (definition 2.7.15).

A *family of Weil divisors on X parametrized by S* is a Weil divisor $\mathbf{D} \in \operatorname{WDiv}(S \times_k X)$ (definition 5.4.5) such that $\operatorname{Supp} \mathbf{D}$ contains no fiber $\pi^{-1}(s)$, $s \in S$. The family is *effective* if $\mathbf{D} \geq 0$.

For a closed point $s \in S$, identify the fiber $\pi^{-1}(s)$ with X. The restriction of $\mathbf{D}$ to $S \times X^{\mathrm{reg}}$ is a Cartier divisor (proposition 5.4.38(1)), so it restricts to a Cartier divisor on $\{s\} \times X^{\mathrm{reg}} = X^{\mathrm{reg}}$. The *fiber of $\mathbf{D}$ over s* is the Weil divisor

$$\mathbf{D}_s \in \operatorname{WDiv} X$$

obtained from that Cartier divisor by proposition 5.4.31 and then extension by closure from X^{reg} to X (lemma 5.4.17).

Three points about the definition deserve to be separated. The hypothesis on the support is what makes the restriction to a fiber legitimate: a local equation for $\mathbf{D}$ that vanished identically on $\pi^{-1}(s)$ would restrict to zero, not to a divisor. The passage through X^{reg} is forced, since a Weil divisor on a singular X need not be Cartier anywhere near a singular point and there is then nothing to restrict; on X^{reg} the local rings are regular, hence factorial (proposition 2.7.20), and proposition 5.4.32 identifies the two theories. Finally, nothing is lost by the detour: X is normal, so $\operatorname{Sing}(X)$ has codimension at least 2 (proposition 5.4.2(3)), and restriction $\operatorname{WDiv} X \to \operatorname{WDiv} X^{\mathrm{reg}}$ is then an isomorphism with inverse extension by closure (lemma 5.4.17 and proposition 5.4.18(2)).

Kollár flags a delicacy at exactly this point, warning that the identification of Weil divisors on X with Cartier divisors on X^{reg} "can be misleading for families of divisors" and referring to the theory of K-flatness for the general case. The warning is real, and it is worth saying precisely when it applies. It applies when the parameter space is only normal rather than smooth, and when the ambient family is not a product, so that the fibers X_s themselves vary: in either case Sing of the total space need not be π^{-1}(something), the regular locus of the total space need not meet a given fiber in the regular locus of that fiber, and the fiber-by-fiber recipe above can fail to be flat or even to be additive in the family. None of that happens here. The parameter space is smooth, the ambient family is the constant product $S \times X$, and the next proposition shows that under those two hypotheses the naive definition produces an effective Cartier divisor relative to S in the sense already established for this book, definition 5.4.26, so that its fibers are governed by proposition 5.4.28 and no further foundational theory is needed.

Proposition 5.4.38: A Family of Weil Divisors Is a Relative Cartier Divisor

Let k, X, S and π be as in definition 5.4.37, and put

$$Y \;:=\; S \times_k X^{\mathrm{reg}}, \qquad \pi_Y \;:=\; \pi|_Y : Y \longrightarrow S$$

1. $S \times_k X$ is a Noetherian, integral, separated scheme that is regular in codimension 1, so that $\mathrm{WDiv}\,(S \times_k X)$ is defined, and its non-regular locus lies in $S \times_k \mathrm{Sing}(X)$, of codimension at least 2. The open subscheme Y is regular, integral, separated and Noetherian; every Weil divisor on $S \times_k X$ restricts on Y to a Cartier divisor, and the restriction of an effective Weil divisor is an effective Cartier divisor.
2. If $\mathbf{D}$ is an effective family of Weil divisors on X parametrized by S, then the closed subscheme $D \subseteq Y$ cut out by the effective Cartier divisor $\mathbf{D}|_Y$ is an effective Cartier divisor on Y relative to S (definition 5.4.26).
3. For every closed point $s \in S$, the fiber D_s supplied by proposition 5.4.28 is an effective Cartier divisor on X^{reg}, and its extension by closure to X is $\mathbf{D}_s$.

***Proof*:**
Step 1: Y is regular. Since X is normal, X^{reg} is a regular scheme, locally of finite type over k. An algebraically closed field is perfect, so theorem 2.7.13 makes X^{reg} smooth over k. The scheme S is smooth over k by hypothesis, so $Y = S \times_k X^{\mathrm{reg}}$ is smooth over k by proposition 2.7.9(3), and hence regular by proposition 2.7.12. It is Noetherian and separated because S and X^{reg} are and both properties pass to a fiber product of finite type k-schemes. It is integral: k is algebraically closed, so the integral k-schemes S and X^{reg} are geometrically integral (definition 3.2.8), and a product of geometrically integral k-schemes is integral.

The same three arguments, with X in place of X^{reg}, make $S \times_k X$ Noetherian, integral and separated. Its non-regular locus is contained in $S \times_k \mathrm{Sing}(X)$, because Y is regular. Let T be an irreducible component of $\mathrm{Sing}(X)$, so $\dim T \leq \dim X - 2$ by proposition 5.4.2(3). Both $S \times_k T$ and $S \times_k X$ are integral by the argument just given, and transcendence degree is additive along a tensor product of finitely generated k-domains with domain tensor product, so their function fields have transcendence degrees $\dim S + \dim T$ and $\dim S + \dim X$ over k. Now theorem 2.6.11 computes both dimensions and theorem 2.6.12 gives

$$\mathrm{codim}\big(S \times_k T,\ S \times_k X\big) \;=\; \dim X - \dim T \;\geq\; 2$$

Every codimension-1 point of $S \times_k X$ therefore lies in Y and has a regular local ring, which is regularity in codimension 1.
Step 2: Weil divisors on Y are Cartier. Each local ring of Y is regular, hence a unique factorization domain by proposition 2.7.20, so Y is locally factorial (definition 2.4.29). By proposition 5.4.32 the homomorphism $\mathrm{CDiv}\,Y \to \mathrm{WDiv}\,Y$ of proposition 5.4.31 is an isomorphism. The restriction map $\mathrm{WDiv}\,(S \times_k X) \to \mathrm{WDiv}\,Y$ is defined because Y is a nonempty open subscheme of $S \times_k X$ (lemma 5.4.17); composing with the inverse isomorphism gives the asserted Cartier divisor.

For effectivity, let $\{(U_i, f_i)\}$ be Cartier data on Y whose associated Weil divisor is effective, so $\nu_\Gamma(f_i) \geq 0$ for every prime divisor Γ of Y meeting U_i. Fix an affine open $\mathrm{Spec}\, R \subseteq U_i$. A regular local ring is normal (proposition 2.7.19), so R is a Noetherian normal domain, and every height-1 prime of R is the generic point of a prime divisor of Y meeting U_i. Thus f_i lies in $R_{\mathfrak{p}}$

for every height-1 prime $\mathfrak{p}$, and lemma 5.4.13 puts $f_i \in R$. As such opens cover U_i, each f_i is regular, which is definition 5.4.25.

Step 3: the hypotheses of the fiberwise criterion. The projection $\pi_Y : Y = S \times_k X^{\mathrm{reg}} \to S$ is the base change of the smooth morphism $X^{\mathrm{reg}} \to \operatorname{Spec} k$ along $S \to \operatorname{Spec} k$, hence smooth by proposition 2.7.9(1) and flat by proposition 2.7.10. It is of finite type, and S is Noetherian, so π_Y is locally of finite presentation. The ideal sheaf $\mathcal{I}_D$ is invertible by Step 2 and definition 5.4.25. So proposition 5.4.29 applies to $D \subseteq Y$ once its fiberwise hypothesis is verified.

Step 4: the fiberwise hypothesis. Let $s \in S$ be any point, not necessarily closed, and write $\kappa = \kappa(s)$. The fiber $Y_s = X^{\mathrm{reg}} \times_k \kappa$ is smooth over κ by proposition 2.7.9(1) and therefore regular (proposition 2.7.12); it is integral because X^{reg} is geometrically integral over the algebraically closed field k, so that $X^{\mathrm{reg}} \times_k \kappa$ is integral. Take an affine open $V \subseteq Y$ on which $\mathcal{I}_D$ is free, say $\mathcal{I}_D|_V = gB$ with $B = \mathcal{O}_Y(V)$, and suppose $V_s \neq \emptyset$. If the image $\overline{g}$ of g in $\mathcal{O}_{Y_s}(V_s)$ were zero, then V_s would be contained in $\operatorname{Supp} \mathbf{D}$; since V_s is a nonempty open of the irreducible space Y_s it is dense there, and $\operatorname{Supp} \mathbf{D}$ is closed, so $Y_s \subseteq \operatorname{Supp} \mathbf{D}$. The fiber $\pi^{-1}(s)$ of $S \times_k X \to S$ is integral with Y_s as a nonempty open subscheme, so Y_s is dense in $\pi^{-1}(s)$ and $\pi^{-1}(s) \subseteq \operatorname{Supp} \mathbf{D}$, contradicting definition 5.4.37. Hence $\overline{g} \neq 0$, and $\overline{g}$ is a non-zero-divisor because Y_s is integral. So D_s is an effective Cartier divisor on Y_s for every $s \in S$, and proposition 5.4.29 gives (2).

Step 5: identification of the fiber. Let $s \in S$ be a closed point. Since k is algebraically closed, $\kappa(s) = k$ and $Y_s = X^{\mathrm{reg}}$. Unwinding Step 4 in this case, the local ideal of D_s on V_s is generated by the image of a local equation for $\mathbf{D}|_Y$, which is exactly the restriction of the Cartier divisor $\mathbf{D}|_Y$ to $\{s\} \times X^{\mathrm{reg}}$ used in definition 5.4.37. Its extension by closure is therefore $\mathbf{D}_s$, as we sought to show.

The second operation moves a divisor along a birational map rather than along a family. It is defined valuation by valuation, which is what makes it available with no hypothesis whatever on the map.

Definition 5.4.39: Birational Transform of a Weil Divisor

Let X and Y be Noetherian, integral, separated schemes that are regular in codimension 1 (definition 5.4.5), and let $g \colon X \dashrightarrow Y$ be a birational map (definition 3.6.2). Write $g^\sharp \colon K(Y) \xrightarrow{\sim} K(X)$ for the induced isomorphism of function fields (corollary 3.6.6).
Let $D \subseteq X$ be a prime divisor with generic point η_D, so that $\mathcal{O}_{X,\eta_D} \subseteq K(X)$ is a discrete valuation ring with fraction field $K(X)$. Set

$$g_*[D] \;:=\; \begin{cases} [D'] & \text{if a prime divisor } D' \subseteq Y \text{ satisfies } g^\sharp\big(\mathcal{O}_{Y,\eta_{D'}}\big) = \mathcal{O}_{X,\eta_D}, \\ 0 & \text{if no prime divisor of } Y \text{ satisfies this,} \end{cases}$$

and extend $\mathbb{Z}$-linearly to a group homomorphism

$$g_* \colon \operatorname{WDiv} X \longrightarrow \operatorname{WDiv} Y$$

called the *birational transform* (also the *proper* or *strict transform*). A prime divisor D with $g_*[D] = 0$ is *contracted* by g.

The definition asks nothing of g beyond being birational. It does not ask Y to be proper, it does not ask g to be a morphism, and it does not ask g to be an automorphism or even to have a codimension-2 indeterminacy locus. This is what makes it the right instrument for a group G acting on X only rationally: each $g \in G$ gives a birational self-map of X, and g_* is defined for every one of them, whether or not the action extends to a morphism anywhere near the boundary. The usual description, as the closure of the image of D under an isomorphism between dense opens, is recovered in lemma 5.4.40(1), which also settles that the recipe above is unambiguous.

Lemma 5.4.40: Functoriality of the Birational Transform

Let X, Y and W be Noetherian, integral, separated schemes that are regular in codimension 1, and let $g\colon X \dashrightarrow Y$ and $h\colon Y \dashrightarrow W$ be birational maps.

1. For each prime divisor $D \subsetneq X$ there is at most one prime divisor $D' \subsetneq Y$ with $g^\sharp(\mathcal{O}_{Y,\eta_{D'}}) = \mathcal{O}_{X,\eta_D}$, so g_* is well defined. If moreover g is represented by a morphism on an open neighbourhood of η_D, then
$$g_*[D] \;=\; \begin{cases} \big[\,\overline{\{g(\eta_D)\}}\,\big] & \text{if } \operatorname{codim}\big(\overline{\{g(\eta_D)\}}, Y\big) = 1,\\ 0 & \text{otherwise.}\end{cases}$$
2. The composite $h \circ g$ is a birational map and $(h\circ g)^\sharp = g^\sharp \circ h^\sharp$.
3. If the prime divisor $D \subsetneq X$ is not contracted by g, then
$$(h\circ g)_*[D] \;=\; h_*\big(g_*[D]\big)$$
Hence $h_* \circ g_*$ and $(h\circ g)_*$ agree on the subgroup of $\operatorname{WDiv} X$ generated by the prime divisors that g does not contract.
4. $h_* \circ g_* = (h\circ g)_*$ on all of $\operatorname{WDiv} X$ if and only if every prime divisor contracted by g is contracted by $h \circ g$. Without that condition the two differ (example 5.4.41).

***Proof*:**
Throughout, η_X and η_Y denote generic points, and K denotes $K(X)$, with $K(Y)$ and $K(W)$ identified with K by the relevant function-field isomorphisms whenever that is unambiguous.

(1) Uniqueness. Let $D_1', D_2' \subsetneq Y$ be prime divisors with $g^\sharp(\mathcal{O}_{Y,\eta_{D_i'}}) = \mathcal{O}_{X,\eta_D} =: R$. Then R is a discrete valuation ring with fraction field K, and for each i the canonical morphism $\operatorname{Spec}\mathcal{O}_{Y,\eta_{D_i'}} \to Y$ becomes, after the identification $g^\sharp$, a morphism $\varphi_i\colon \operatorname{Spec} R \to Y$ carrying the closed point to $\eta_{D_i'}$ and the generic point to η_Y. The two agree on $\operatorname{Spec} K$, both being the canonical morphism $\operatorname{Spec} K(Y) \to Y$, and Y is separated over $\operatorname{Spec}\mathbb{Z}$ with Y Noetherian, so theorem 3.5.14 allows at most one such extension over $\operatorname{Spec} R$. Hence $\varphi_1 = \varphi_2$ and $\eta_{D_1'} = \eta_{D_2'}$; a prime divisor is the closure of its generic point, so $D_1' = D_2'$.

(1) The description at a point of definition. Suppose g is represented by a morphism $g_0\colon U \to Y$ with $\eta_D \in U$, and put $y := g_0(\eta_D)$. Since g is dominant, $g_0(\eta_X) = \eta_Y$ (lemma 3.6.4) and the local homomorphism $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,\eta_D} = R$ induced by g_0 agrees with $g^\sharp$ on fraction fields; write $V := g^\sharp(\mathcal{O}_{Y,y}) \subseteq R$, a local subring of K with $\mathfrak{m}_V \subseteq \mathfrak{m}_R$.

Assume first $\operatorname{codim}(\overline{\{y\}}, Y) = 1$. Then $\mathcal{O}_{Y,y}$, hence V, is a discrete valuation ring with fraction field K, and $V \subseteq R$. Let $x \in R$, the case $x = 0$ being trivial, and suppose $x \notin V$. A valuation ring of K contains x or $1/x$, so $x \notin V$ forces $1/x \in \mathfrak{m}_V \subseteq \mathfrak{m}_R$, making $1/x$ a nonunit of R; but $x \in R$ and $1/x \in R$ make $1/x$ a unit of R, a contradiction. Therefore $R \subseteq V$, so $V = R$, and $\overline{\{y\}}$ is a prime divisor of Y realizing the defining condition: $g_*[D] = [\overline{\{y\}}]$.

Assume instead $\operatorname{codim}(\overline{\{y\}}, Y) \neq 1$, and suppose some prime divisor $D' \subseteq Y$ had $g^\sharp(\mathcal{O}_{Y,\eta_{D'}}) = R$. The composite $\operatorname{Spec} R = \operatorname{Spec} \mathcal{O}_{X,\eta_D} \to U \xrightarrow{g_0} Y$ and the morphism $\operatorname{Spec} R \to Y$ attached to D' as in the uniqueness argument are two extensions over $\operatorname{Spec} R$ of the canonical $\operatorname{Spec} K(Y) \to Y$, so theorem 3.5.14 identifies them. Comparing images of the closed point gives $y = \eta_{D'}$, whose closure has codimension 1, a contradiction. So $g_*[D] = 0$.

(2). Represent g by a dominant morphism $g_0 \colon U \to Y$ and h by a morphism $h_0 \colon V \to W$ with $V \subseteq Y$ dense open. By lemma 3.6.4, $g_0(\eta_X) = \eta_Y$, and $\eta_Y \in V$ because V is a nonempty open of the irreducible space Y. Hence $g_0^{-1}(V)$ is a nonempty open of U and $h_0 \circ g_0|_{g_0^{-1}(V)}$ represents a rational map $h \circ g$; it is dominant, since it carries η_X to $h_0(\eta_Y) = \eta_W$. The same construction applied to g^{-1} and h^{-1} produces a dominant rational inverse, so $h \circ g$ is birational. On function fields, the isomorphism induced by a dominant rational map is the map on local rings at the generic points, so the isomorphism induced by $h_0 \circ g_0$ is $g^\sharp \circ h^\sharp$.

(3). Let $g_*[D] = [D'] \neq 0$, so $g^\sharp(\mathcal{O}_{Y,\eta_{D'}}) = \mathcal{O}_{X,\eta_D}$. Two cases.

If $h_*[D'] = [D''] \neq 0$, then $h^\sharp(\mathcal{O}_{W,\eta_{D''}}) = \mathcal{O}_{Y,\eta_{D'}}$, and applying $g^\sharp$ and using (2),

$$(h \circ g)^\sharp(\mathcal{O}_{W,\eta_{D''}}) \;=\; g^\sharp\Big(h^\sharp(\mathcal{O}_{W,\eta_{D''}})\Big) \;=\; g^\sharp(\mathcal{O}_{Y,\eta_{D'}}) \;=\; \mathcal{O}_{X,\eta_D}$$

so $(h \circ g)_*[D] = [D''] = h_*(g_*[D])$.

If $h_*[D'] = 0$, then $h_*(g_*[D]) = 0$, and it must be shown that $(h \circ g)_*[D] = 0$ as well. Were it $[D''] \neq 0$, the displayed computation read backwards would give $h^\sharp(\mathcal{O}_{W,\eta_{D''}}) = \mathcal{O}_{Y,\eta_{D'}}$, since $g^\sharp$ is injective, contradicting $h_*[D'] = 0$.

(4). Both $h_* \circ g_*$ and $(h \circ g)_*$ are homomorphisms $\operatorname{WDiv} X \to \operatorname{WDiv} W$, and $\operatorname{WDiv} X$ is free on the prime divisors, so they are equal precisely when they agree on every prime divisor D. On a D not contracted by g they agree by (3). On a D contracted by g the left side is $h_*(0) = 0$, so agreement there says exactly $(h \circ g)_*[D] = 0$, which is the stated condition, completing the proof.

Part (3) is the positive statement in the form it is used: a prime divisor that survives the first map is transported by the composite exactly as the two maps transport it in turn, and this needs nothing of the second map. What fails is the other half. A birational map can destroy a prime divisor, and a later birational map can create it again, and the birational transform has no memory of what was destroyed. The plane provides the smallest example, and it is the mechanism the Chevalley argument has to control.

Example 5.4.41: The Quadratic Transformation of the Plane

Let k be a field and let

$$\sigma \colon \mathbb{P}^2_k \dashrightarrow \mathbb{P}^2_k, \qquad \sigma(x_0 : x_1 : x_2) \;=\; (x_1x_2 : x_0x_2 : x_0x_1)$$

The three forms x_1x_2, x_0x_2, x_0x_1 vanish simultaneously exactly when at least two coordinates vanish, so σ is a morphism on the complement of the three coordinate points $p_0 = (1 : 0 : 0)$, $p_1 = (0 : 1 : 0)$, $p_2 = (0 : 0 : 1)$. On the open set where all $x_i \neq 0$,

$$\sigma\big(\sigma(x_0 : x_1 : x_2)\big) \;=\; \big(x_0^2x_1x_2 \;:\; x_0x_1^2x_2 \;:\; x_0x_1x_2^2\big) \;=\; (x_0 : x_1 : x_2)$$

so $\sigma \circ \sigma$ and the identity agree on a dense open subset and are therefore the same rational map: σ is a birational involution of the plane.

Let $L_0 = V_+(x_0)$, a prime divisor of $\mathbb{P}^2_k$ with generic point η_{L_0}. A point of L_0 other than p_1 and p_2 has $x_1x_2 \neq 0$, and σ sends it to $(x_1x_2 : 0 : 0) = p_0$. Thus σ is a morphism near η_{L_0} and carries it to the closed point p_0, whose closure has codimension 2 in $\mathbb{P}^2_k$. By lemma 5.4.40(1),

$$\sigma_*[L_0] \;=\; 0$$

The identity map has $(\mathrm{id})_*[L_0] = [L_0]$ directly from definition 5.4.39. Consequently

$$\sigma_*\big(\sigma_*[L_0]\big) \;=\; \sigma_*(0) \;=\; 0 \qquad \text{while} \qquad (\sigma \circ \sigma)_*[L_0] \;=\; [L_0] \;\neq\; 0$$

so the birational transform is not functorial.

The geometry behind the computation is that σ contracts L_0 to the point p_0 and, run a second time, extracts L_0 back out of p_0: it is a blow-down followed by a blow-up, packaged as one self-inverse map of the plane. The example is stated with $\mathbb{P}^2$ and homogeneous coordinates only, so that nothing about blow-ups is needed to verify it.

Families of Weil divisors arise most often not one at a time but as the total space of a linear system, and the parameter space is then a projective space rather than an abstract smooth variety. Setting that up requires care about one thing only, which is the dimension convention.

Definition 5.4.42: Linear System of Weil Divisors

Let X be a normal, projective, geometrically integral scheme over a field k (definition 3.2.8), and let $D_0 \in \mathrm{WDiv}\, X$. Write $L(D_0) = \Gamma(X, \mathcal{O}_X(D_0))$ for the Riemann-Roch space (definition 5.4.9), a finite-dimensional k-vector space (corollary 5.4.11).
Let $W \subseteq L(D_0)$ be a k-linear subspace with $\dim_k W = n \geq 1$, and let $\mathbb{P}(W)$ denote the projective space of *lines* in W, so $\dim \mathbb{P}(W) = n-1$. For $0 \neq f \in W$ the Weil divisor $\mathrm{div}(f) + D_0$ is effective (definition 5.4.9) and linearly equivalent to D_0 (definition 5.4.8), and it depends only on the line $[f] \in \mathbb{P}(W)$, since $\mathrm{div}(cf) = \mathrm{div}(f)$ for $c \in k^*$. The *linear system spanned by* W is the resulting set of effective Weil divisors

$$|W| \;:=\; \{\mathrm{div}(f) + D_0 \mid 0 \neq f \in W\}$$

together with the bijection $\mathbb{P}(W) \to |W|$, $[f] \mapsto \mathrm{div}(f) + D_0$. Its members are written D_λ for $\lambda \in \mathbb{P}(W)$, and $[D_\lambda]$ denotes the corresponding point of $\mathbb{P}(W)$. The system is *complete* if $W = L(D_0)$, and a *pencil* if $n = 2$.
A prime divisor $F \subseteq X$ is a *fixed component* of $|W|$ if $F \subseteq \mathrm{Supp}\, D_\lambda$ for every λ; the system is *without fixed components* if it has none. The *base locus* is $\mathrm{Bs}\,|W| = \bigcap_\lambda \mathrm{Supp}\, D_\lambda$.

The map $\mathbb{P}(W) \to |W|$ is injective, and this is where geometric integrality earns its place: $\mathrm{div}(f)+D_0 = \mathrm{div}(f') + D_0$ says $\mathrm{div}(f/f') = 0$, so f/f' and its inverse are regular by lemma 5.4.13 applied on each affine open of the normal scheme X, whence $f/f' \in \Gamma(X, \mathcal{O}_X)^*$; and $\Gamma(X, \mathcal{O}_X) = k$ because X is projective, hence proper (theorem 3.5.26), and geometrically integral (theorem 3.5.27). Over a field that is not algebraically closed and an X that is integral but not geometrically so, $\Gamma(X, \mathcal{O}_X)$ can be a proper finite extension of k and the map collapses.

5.4.43 Convention: Vector Dimension and Projective Dimension Two dimensions are attached to a linear system and they differ by one. The space W has a *vector* dimension $\dim_k W = n$; the system $|W|$, as a parameter space, has *projective* dimension $\dim \mathbb{P}(W) = n-1$. The convention in force here is:

- $\dim |W| := \dim \mathbb{P}(W) = \dim_k W - 1$, so that a pencil has $\dim |W| = 1$;
- a Grassmannian of subsystems is *always* written with the *vector space* as its second argument. Thus $\mathrm{Grass}(r, W)$, the Grassmannian of *Classical Algebraic Geometry* [20], is the variety of r-dimensional linear *subspaces of the vector space* W, whose points correspond to subsystems $|W'| \subseteq |W|$ of projective dimension $r-1$.

The notation $\mathrm{Grass}(r, |W|)$ is never used, because it is ambiguous and the ambiguity is not harmless. In the literature the same symbol is read both ways, and reading it projectively when it was meant vectorially shifts every subsystem by one dimension. Kollár's Corollary 24 in [40] writes $\mathrm{Grass}(n-1, |H|)$ where the argument computes with subspaces of vector dimension $n-1$; under his own Definition 15 the symbol reads projectively, which would make the intersection of the corresponding members zero-dimensional rather than a curve. Anything imported from that source must be reread with the vector convention.

A linear system carries a divisor on $X \times \mathbb{P}(W)$ whose fiber over λ is D_λ. It is the object that turns statements about "a general member" into statements about a dense open subset of a parameter space, and it is a family of Weil divisors in the sense of definition 5.4.37.

Definition 5.4.44: Incidence Correspondence of a Linear System

Let X, D_0 and W be as in definition 5.4.42, and fix a basis $f_1, \dots, f_n$ of W, identifying $\mathbb{P}(W)$ with $\mathbb{P}^{n-1}_k$ carrying homogeneous coordinates $\lambda_1, \dots, \lambda_n$ so that λ corresponds to $f_\lambda := \sum_i \lambda_i f_i$. Choose an open cover $\{U_\alpha\}$ of X^{reg} and $h_\alpha \in K(X)^*$ with $D_0|_{U_\alpha} = \mathrm{div}(h_\alpha)|_{U_\alpha}$, which is possible by proposition 5.4.32 applied on X^{reg}.
The *incidence correspondence* of $|W|$ is the effective Cartier divisor $\mathbf{D}_{|W|}$ on $X^{\mathrm{reg}} \times_k \mathbb{P}(W)$ represented by the data

$$\left\{ \left(U_\alpha \times \{\lambda_j \neq 0\},\ G_{\alpha j} := h_\alpha \sum_{i=1}^{n} \frac{\lambda_i}{\lambda_j} f_i \right) \right\}_{\alpha,\, j}$$

together with the Weil divisor on $X \times_k \mathbb{P}(W)$ obtained from it by proposition 5.4.31 and extension by closure (lemma 5.4.17), for which the same symbol is used. The classical order of the factors is used here; interchanging them puts this in the situation of definition 5.4.37 with parameter space $S = \mathbb{P}(W)$.

Proposition 5.4.45: The Incidence Correspondence Is a Family with the Expected Fibers

In the situation of definition 5.4.44:

1. the data $\{(U_\alpha \times \{\lambda_j \neq 0\}, G_{\alpha j})\}$ define an effective Cartier divisor, independent of the choice of the (U_α, h_α);
2. $\operatorname{Supp} \mathbf{D}_{|W|}$ contains no fiber of the projection $q\colon X \times_k \mathbb{P}(W) \to \mathbb{P}(W)$;
3. if k is algebraically closed, $\mathbf{D}_{|W|}$ is an effective family of Weil divisors on X parametrized by $S = \mathbb{P}(W)$ in the sense of definition 5.4.37, and its fiber over $\lambda \in \mathbb{P}(W)$ is D_λ.

***Proof*:**
(1). Each f_i lies in $L(D_0)$, so $\operatorname{div}(f_i) + D_0 \geq 0$, and on U_α this reads $\operatorname{div}(h_\alpha f_i)|_{U_\alpha} \geq 0$. The scheme X^{reg} is regular, hence normal (proposition 2.7.19, proposition 2.4.28), so lemma 5.4.13 applied on each affine open of U_α gives $h_\alpha f_i \in \mathcal{O}_X(U_\alpha)$. The functions λ_i/λ_j are regular on $\{\lambda_j \neq 0\}$, so $G_{\alpha j}$ is a regular function on $U_\alpha \times \{\lambda_j \neq 0\}$.

It is nowhere the zero function. The elements $h_\alpha f_1, \ldots, h_\alpha f_n$ of $\mathcal{O}_X(U_\alpha)$ are linearly independent over k, because the f_i are and $h_\alpha \neq 0$; they therefore span a k-subspace E of dimension n. Restricting $G_{\alpha j}$ to the fiber over a point $\mu \in \{\lambda_j \neq 0\}$ of the second factor gives $\sum_i c_i \cdot (h_\alpha f_i \otimes 1)$ in $\mathcal{O}_X(U_\alpha) \otimes_k \kappa(\mu)$, with $c_i \in \kappa(\mu)$ the values of λ_i/λ_j at μ and $c_j = 1$. A field extension is a free, hence flat, module, so $E \otimes_k \kappa(\mu)$ injects into $\mathcal{O}_X(U_\alpha) \otimes_k \kappa(\mu)$ and the $h_\alpha f_i \otimes 1$ stay linearly independent over $\kappa(\mu)$. Hence the restriction is nonzero, so $G_{\alpha j} \neq 0$.

It is moreover a non-zero-divisor. Shrink U_α to an affine open $\operatorname{Spec} R \subseteq X^{\mathrm{reg}}$, with R a domain since X is integral. Then $\operatorname{Spec} R \times \{\lambda_j \neq 0\}$ is the spectrum of the polynomial ring $R[\lambda_1/\lambda_j, \ldots, \lambda_n/\lambda_j]$ in the $n-1$ variables other than $\lambda_j/\lambda_j = 1$, which is again a domain. A nonzero element of a domain is a non-zero-divisor, and such opens cover $U_\alpha \times \{\lambda_j \neq 0\}$.

On overlaps,

$$\frac{G_{\alpha j}}{G_{\beta j'}} = \frac{h_\alpha}{h_\beta} \cdot \frac{\lambda_{j'}}{\lambda_j}$$

The second factor is a unit on $\{\lambda_j \lambda_{j'} \neq 0\}$. For the first, $\operatorname{div}(h_\alpha/h_\beta) = 0$ on $U_\alpha \cap U_\beta$, so h_α/h_β and its inverse are regular there by lemma 5.4.13, making it a unit. This is definition 5.4.23 together with definition 5.4.25. A second choice (U'_γ, h'_γ) produces data whose ratios with the first are again of the form h_α/h'_γ, units by the same argument, so the Cartier divisor is unchanged.
(2). Fix $\mu \in \mathbb{P}(W)$ and choose j with $\lambda_j(\mu) \neq 0$. The computation in (1) shows that $G_{\alpha j}$ restricts to a nonzero function on $(U_\alpha \times \{\lambda_j \neq 0\}) \cap q^{-1}(\mu)$, a nonempty open subset of $q^{-1}(\mu)$. So that open subset is not contained in $\operatorname{Supp} \mathbf{D}_{|W|}$, and neither is $q^{-1}(\mu)$.
(3). The projective space $\mathbb{P}(W) = \mathbb{P}^{n-1}_k$ is covered by copies of $\mathbb{A}^{n-1}_k$, each smooth over k by definition 2.7.8 with $r = 0$, so $\mathbb{P}(W)$ is a smooth variety. With (2), the two conditions of definition 5.4.37 hold. For the fiber, let $\lambda \in \mathbb{P}(W)$ be a closed point and pick j with $\lambda_j \neq 0$. Restricting $G_{\alpha j}$ to $U_\alpha \times \{\lambda\}$ replaces each λ_i/λ_j by a scalar in k and gives $h_\alpha f_\lambda/\lambda_j$, whose divisor on U_α is

$$\operatorname{div}(h_\alpha)|_{U_\alpha} + \operatorname{div}(f_\lambda)|_{U_\alpha} = \big(D_0 + \operatorname{div}(f_\lambda)\big)\big|_{U_\alpha} = D_\lambda|_{U_\alpha}$$

because a nonzero scalar has zero divisor. These opens cover X^{reg}, so the Cartier fiber is $D_\lambda|_{X^{\mathrm{reg}}}$, whose extension by closure is D_λ, as we sought to show.

Two features of the incidence correspondence are worth separating from the proof. Its fiber over λ recovers the member D_λ exactly, with multiplicities, which is what licenses the interchange of "a property of a general member of $|W|$" with "a property of the fiber of $\mathbf{D}_{|W|}$ over a general point of $\mathbb{P}(W)$". And by proposition 5.4.45(3) together with proposition 5.4.38, it is an effective Cartier divisor on $X^{\mathrm{reg}} \times \mathbb{P}(W)$ relative to $\mathbb{P}(W)$, so its formation commutes with base change on the parameter space (proposition 5.4.30). Kollár's Definition 15 asserts in addition that $\mathbf{D}_{|W|}$ is irreducible when $|W|$ has no fixed components; that assertion is not proved there, is not used below, and is not claimed here.

Finally, the recurring technical need: a construction is performed on a convenient dense open subset of the total space, and the conclusion has to be transported back to the fibers. The following lemma is what does the transporting, and it is Kollár's Claim 13.1 with the hypothesis that his statement omits.

Lemma 5.4.46: Fibers of a Family Are Recovered from a Dense Open

Let k, X, S and π be as in definition 5.4.37, let $\mathbf{D}$ be an effective family of Weil divisors on X parametrized by S, and let $U \subseteq S \times_k X$ be an open subset that meets every irreducible component of $\operatorname{Supp} \mathbf{D}$. Then there is a dense open subset $S^\circ \subseteq S$ such that for every closed point $s \in S^\circ$, writing $U_s = U \cap \pi^{-1}(s) \subseteq X$:

1. no irreducible component of $\operatorname{Supp} \mathbf{D}_s$ is contained in $X \setminus U_s$, so $\operatorname{Supp} \mathbf{D}_s = \overline{\operatorname{Supp} \mathbf{D}_s \cap U_s}$;
2. $\mathbf{D}_s$ is the extension by closure (lemma 5.4.17) of its restriction to U_s.

The hypothesis on U cannot be dropped, even when $\operatorname{Supp} \mathbf{D}$ is irreducible (example 5.4.47).

***Proof*:**
Let $Z := \operatorname{Supp} \mathbf{D}$ carry its reduced closed subscheme structure, so $Z \to S$ is of finite type, and let η be the generic point of S. Since $S \times_k X$ is Noetherian, proposition 2.1.21 writes $Z = Z_1 \cup \cdots \cup Z_r$ with each Z_i closed and irreducible, say with generic point ζ_i.

Step 1: $U \cap Z_\eta$ is dense in the generic fiber Z_η. Fix i. If some point of Z_i maps to η, then $\overline{\{\eta\}} = S$ forces $\overline{\pi(Z_i)} = S$; and conversely if $\overline{\pi(Z_i)} = S$ then $\overline{\{\pi(\zeta_i)\}} = \overline{\pi(\overline{\{\zeta_i\}})} = S$, so $\pi(\zeta_i) = \eta$. Thus $(Z_i)_\eta$ is empty unless Z_i dominates S, in which case it contains ζ_i.

Let Z_i dominate S. By hypothesis $U \cap Z_i$ is a nonempty open subset of the irreducible space Z_i, so it contains ζ_i. Any nonempty open subset of $(Z_i)_\eta$ has the form $O \cap (Z_i)_\eta$ with $O \subseteq Z_i$ open and nonempty, hence contains ζ_i; so U meets it. Therefore $U \cap (Z_i)_\eta$ is dense in $(Z_i)_\eta$. Since $Z_\eta = \bigcup_i (Z_i)_\eta$ is a finite union of closed subsets and a nonempty open of Z_η meets one of them in a nonempty open of it, $U \cap Z_\eta$ is dense in Z_η.
Step 2: spreading out. The scheme S is Noetherian and integral and $Z \to S$ is of finite type, so proposition 5.2.9 applied to the open subscheme $U \cap Z \subseteq Z$ produces a nonempty, hence dense, open $S^\circ \subseteq S$ such that $U \cap Z_s$ is dense in Z_s for every $s \in S^\circ$.
Step 3: the components of $\operatorname{Supp} \mathbf{D}_s$ *are components of* Z_s. Let $s \in S^\circ$ be a closed point and identify $\pi^{-1}(s)$ with X. Because $\mathbf{D}$ is effective, a local equation for $\mathbf{D}$ on $S \times X^{\mathrm{reg}}$ cuts out Z there, so its restriction to X^{reg} cuts out $Z_s \cap X^{\mathrm{reg}}$ and

$$\operatorname{Supp} \mathbf{D}_s \;=\; \overline{Z_s \cap X^{\mathrm{reg}}} \;\subseteq\; Z_s$$

Let T be an irreducible component of $\operatorname{Supp} \mathbf{D}_s$. By lemma 2.6.1, $T \cap X^{\mathrm{reg}}$ is then an irreducible component of $Z_s \cap X^{\mathrm{reg}}$, which on X^{reg} is cut out by a single regular non-zero-divisor; so $T \cap X^{\mathrm{reg}}$ has codimension 1 in X^{reg} by corollary 2.6.14. Codimension is the dimension of the local ring at the generic point, and T and $T \cap X^{\mathrm{reg}}$ have the same generic point with the same local ring, so $\operatorname{codim}(T, X) = 1$ and $\dim T = \dim X - 1$ by theorem 2.6.12.

On the other hand $Z_s \neq X$ by definition 5.4.37 and X is irreducible, so every irreducible closed subset of Z_s has codimension at least 1 in X and hence dimension at most $\dim X - 1$, again by theorem 2.6.12. If $T \subseteq T'$ with T' irreducible and closed in Z_s and $T \neq T'$, the same theorem applied inside T' would give $\dim T \leq \dim T' - 1 \leq \dim X - 2$, which is false. So T is maximal among the irreducible closed subsets of Z_s, that is, an irreducible component of Z_s.

Step 4: conclusion. Let T be an irreducible component of $\operatorname{Supp} \mathbf{D}_s$, hence of Z_s by Step 3. The set $T \setminus \bigcup_{T' \neq T} T'$, the union running over the other components of Z_s, is a nonempty open subset of Z_s, so Step 2 makes U_s meet it. Thus $U_s \cap T$ is a nonempty open subset of the irreducible space T and is dense in it, which is (1).

For (2), restriction $\operatorname{WDiv} X \to \operatorname{WDiv} U_s$ followed by extension by closure sends $[T]$ to $[\overline{T \cap U_s}]$ when T meets U_s and to 0 otherwise (lemma 5.4.17). By (1) every component of $\operatorname{Supp} \mathbf{D}_s$ meets U_s and satisfies $\overline{T \cap U_s} = T$, so the composite fixes $\mathbf{D}_s$, completing the proof.

Example 5.4.47: A Dense Open Missing Every Fiber

Take k algebraically closed, $X = \mathbb{P}^1_k$, $S = \mathbb{A}^1_k$, and let $q \in \mathbb{P}^1_k$ be a closed point. Put

$$\mathbf{D} \;:=\; \big[\, S \times \{q\} \,\big] \;\in\; \operatorname{WDiv}(S \times_k \mathbb{P}^1_k)$$

which is a prime divisor, so $\mathbf{D}$ is effective with irreducible support. Its intersection with a fiber $\pi^{-1}(s)$ is the single point (s, q), so no fiber lies in its support and $\mathbf{D}$ is a family of Weil divisors. Since $\mathbb{P}^1_k$ is regular, $X^{\mathrm{reg}} = X$, and restricting a local equation for $\mathbf{D}$ to a fiber gives

$$\mathbf{D}_s \;=\; [q] \qquad \text{for every closed point } s \in S$$

Now let $U := (S \times_k \mathbb{P}^1_k) \setminus \operatorname{Supp} \mathbf{D}$, the complement of a proper closed subset of an irreducible scheme, hence a dense open subset. Then $U_s = \mathbb{P}^1_k \setminus \{q\}$, so $\operatorname{Supp} \mathbf{D}_s \cap U_s = \emptyset$ and the closure of $\mathbf{D}_s \cap U_s$ is 0, not $\mathbf{D}_s$. This happens for *every* $s \in S$, so no shrinking of the parameter space repairs it.

Density of U is therefore not enough, and irreducibility of $\mathbf{D}$ is not what is missing: what lemma 5.4.46 needs, and what fails here, is that U meet $\operatorname{Supp} \mathbf{D}$ at all.

Everything above can now be put on one surface. The example below exhibits the two operations of this subsection on the same divisor and measures their difference: it is exactly one exceptional curve. It also shows why the difference is not a bookkeeping artefact, since the two answers are not even linearly equivalent.

Example 5.4.48: Flat Limit and Birational Transform on a Blown-Up Plane

Let k be algebraically closed, let $p = (1 : 0 : 0) \in \mathbb{P}^2_k$, and let

$$Y \;:=\; \Big\{ \big((x_0 : x_1 : x_2),\, (u_1 : u_2)\big) \in \mathbb{P}^2_k \times \mathbb{P}^1_k \;\Big|\; x_1 u_2 = x_2 u_1 \Big\}$$

with $\pi \colon Y \to \mathbb{P}^2_k$ the first projection. This is the blow-up of the plane at p, presented so that only homogeneous coordinates are needed.

The surface. If $(x_1, x_2) \neq (0, 0)$ the equation forces $(u_1 : u_2) = (x_1 : x_2)$, so π restricts to an isomorphism over $\mathbb{P}^2_k \setminus \{p\}$, with inverse $(x_0 : x_1 : x_2) \mapsto ((x_0 : x_1 : x_2), (x_1 : x_2))$, and $E := \pi^{-1}(p) = \{p\} \times \mathbb{P}^1_k$. Over the chart $\{x_0 \neq 0\}$, put $z_1 = x_1/x_0$ and $z_2 = x_2/x_0$; the equation reads $z_1 u_2 = z_2 u_1$, so Y is covered there by the two charts

$$(z_1, w),\ w = u_2/u_1, \quad \pi(z_1, w) = (z_1, z_1 w), \quad E = \{z_1 = 0\}$$

$$(z_2, w'),\ w' = u_1/u_2, \quad \pi(z_2, w') = (z_2 w', z_2), \quad E = \{z_2 = 0\}$$

each isomorphic to $\mathbb{A}^2_k$. Together with $\pi^{-1}(\mathbb{P}^2_k \setminus \{p\})$ these cover Y, so Y is smooth over k, hence regular (proposition 2.7.12) and normal (proposition 2.7.19, proposition 2.4.28). It is irreducible: in the chart (z_1, w) the complement of E is $\{z_1 \neq 0\}$, dense in $\mathbb{A}^2_k$, and similarly in the other chart, so $Y = \overline{Y \setminus E}$ is the closure of an irreducible set.

The family. Let $S = \mathbb{A}^1_k$ with coordinate t and let $\mathbf{L}$ be the Cartier divisor on $S \times_k Y$ given by the data

$$\left\{ \left(S \times \pi^{-1}(\{x_i \neq 0\}),\ \pi^\sharp\left(\frac{x_2 - t x_0}{x_i}\right) \right) \right\}_{i=0,1,2}$$

Each entry is a regular function, and on an overlap the ratio of two of them is $\pi^\sharp(x_{i'}/x_i)$, a unit; so this is an effective Cartier divisor (definition 5.4.25). Its restriction to the fiber over any point of S is the pullback of a nonzero linear form, hence nonzero, so $\operatorname{Supp} \mathbf{L}$ contains no fiber and $\mathbf{L}$ is an effective family of Weil divisors on Y parametrized by S (definition 5.4.37; here $Y^{\mathrm{reg}} = Y$). Its members are the total transforms of the lines

$$L_t \;=\; V_+(x_2 - t x_0) \;\subseteq\; \mathbb{P}^2_k$$

and $p \in L_t$ if and only if $t = 0$.

The general fiber. Let $t \neq 0$. In the chart (z_1, w) the local equation is $z_1 w - t$, which is a unit at every point of $E = \{z_1 = 0\}$, and in the chart (z_2, w') it is $z_2 - t$, again a unit along $E = \{z_2 = 0\}$. So $\operatorname{Supp} \mathbf{L}_t$ misses E entirely and lies in $\pi^{-1}(\mathbb{P}^2_k \setminus \{p\})$, where π is an isomorphism. Hence

$$\mathbf{L}_t \;=\; \left(\pi^{-1}\right)_* [L_t] \qquad (t \neq 0)$$

The special fiber. Let $t = 0$, so the local equations become $\pi^\sharp(x_2/x_i)$. In the chart (z_1, w) this is $z_1 w$, whose divisor is $\{z_1 = 0\} + \{w = 0\}$, that is E plus the curve $\widetilde{L_0}$ on which $\pi(z_1, 0) = (z_1, 0)$ runs through $L_0 \cap \{x_0 \neq 0\}$. In the chart (z_2, w') it is z_2, whose divisor is E with multiplicity 1; and over $\mathbb{P}^2_k \setminus \{p\}$ it cuts out $L_0 \setminus \{p\}$. Assembling,

$$\mathbf{L}_0 \;=\; \widetilde{L_0} \;+\; E$$

where $\widetilde{L_0}$ is the closure in Y of $\pi^{-1}(L_0 \setminus \{p\})$.

The birational transform. The map $\pi^{-1} \colon \mathbb{P}^2_k \dashrightarrow Y$ is birational and is a morphism on $\mathbb{P}^2_k \setminus \{p\}$, which contains the generic point of L_0; that generic point goes to the generic point of $\widetilde{L_0}$, of codimension 1. By lemma 5.4.40(1),

$$\left(\pi^{-1}\right)_* [L_0] \;=\; \widetilde{L_0} \;=\; \mathbf{L}_0 - E$$

The fiber of the family and the birational transform of the same line differ by the exceptional curve.

The difference is not principal. Comparing the Cartier data for two values of t, the ratio of the local equations is the single global rational function $\pi^\sharp\big((x_2 - tx_0)/x_2\big)$, so

$$\mathbf{L}_t - \mathbf{L}_0 \;=\; \operatorname{div}_Y\Big(\pi^\sharp\Big(\frac{x_2 - tx_0}{x_2}\Big)\Big)$$

and all fibers of the family are linearly equivalent. On the other hand $[E] \neq 0$ in $\operatorname{Cl} Y$. Indeed, suppose $E = \operatorname{div}_Y(f)$ with $f \in K(Y)^* = K(\mathbb{P}^2_k)^*$. Every prime divisor of Y other than E meets $Y \setminus E$, and $Y \setminus E \cong \mathbb{P}^2_k \setminus \{p\}$, whose prime divisors correspond to those of $\mathbb{P}^2_k$ with the same local rings (lemma 5.4.17(1) applied twice). So $\operatorname{div}_{\mathbb{P}^2_k}(f) = 0$; on each standard affine chart of $\mathbb{P}^2_k$, which is the spectrum of a polynomial ring and so a normal domain, lemma 5.4.13 makes f and f^{-1} regular, whence $f \in \Gamma(\mathbb{P}^2_k, \mathcal{O})^* = k^*$ by proposition 2.3.6 and $\operatorname{div}_Y(f) = 0 \neq E$, a contradiction.

Consequently, although $L_0 \sim L_t$ on $\mathbb{P}^2_k$, the birational transforms satisfy

$$\big(\pi^{-1}\big)_*[L_0] \;=\; \mathbf{L}_0 - E \;\not\sim\; \mathbf{L}_t \;=\; \big(\pi^{-1}\big)_*[L_t]$$

so the birational transform does not preserve linear equivalence, while the fibers of the family do. No intersection numbers are used anywhere above: every claim is an identity between divisors or a principal-divisor computation.

The mechanism is now visible in its simplest form. Passing to the fiber of a family keeps whatever the local equation picks up along the exceptional locus, and passing to the birational transform throws it away. In the Chevalley argument the role of π is played by a translation L_g of a group acting only rationally on a projective model, the role of E by a component of the boundary that L_g contracts, and the role of t by the group element; the conclusion is that the two families of divisors that an argument might reasonably call "the translate of D by g" are different divisors, and differ by boundary terms that must be accounted for rather than assumed away. That accounting is **??**.

5.4.4 The Picard Group

The discussion of divisors on projective schemes has made it clear that there is a strong connection to the twisting sheaves $\mathcal{O}(d)$. We now make this connection precise: Cartier divisors and line bundles are two sides of the same coin, and the passage between them is the central bridge linking the geometry of divisors to the algebra of sheaves.

The reader familiar with algebraic number theory will recognize a close parallel. In a Dedekind domain R, fractional ideals are finitely generated R-submodules of $\operatorname{Frac}(R)$, and they form a group under multiplication. The ideal class group $\operatorname{Cl}(R)$ — fractional ideals modulo principal ones — measures the failure of unique factorization (cf. example 5.4.15). The Picard group is precisely the scheme-theoretic generalization of this construction: invertible sheaves ("line bundles") play the role of fractional ideals, the tensor product replaces ideal multiplication, and the Picard group classifies them up to isomorphism.

Proposition 5.4.49: Invertible Sheaves Form a Group

Let X be a ringed space and let $\mathcal{L}, \mathcal{L}'$ be invertible sheaves on X. Then:

1. The tensor product $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}'$ is invertible.
2. The *dual* $\mathcal{L}^\vee := \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$ is invertible, and $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \cong \mathcal{O}_X$.
3. The tensor product is associative and commutative (up to natural isomorphism) with unit $\mathcal{O}_X$.

Hence the set of isomorphism classes of invertible sheaves on X forms an abelian group under $\otimes$.

The tensor product of line bundles operates stalkwise: $(\mathcal{L} \otimes \mathcal{L}')_x \cong \mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{L}'_x$. Since each stalk is a free module of rank 1, this tensor product "multiplies the values" rather than increasing the rank — analogous to multiplying scalars rather than wedging differential forms. This is why the tensor product of two line bundles remains a line bundle.

***Proof*:**
For (1), choose trivialisations $\mathcal{L}|_U \cong \mathcal{O}_U$ and $\mathcal{L}'|_V \cong \mathcal{O}_V$. On $U \cap V$, $(\mathcal{L} \otimes \mathcal{L}')|_{U\cap V} \cong \mathcal{O}_{U\cap V} \otimes \mathcal{O}_{U\cap V} \cong \mathcal{O}_{U\cap V}$, so $\mathcal{L} \otimes \mathcal{L}'$ is locally free of rank 1.

For (2), on a trivialising open where $\mathcal{L}|_U \cong \mathcal{O}_U$, the dual restricts as $\mathcal{L}^\vee|_U \cong \mathcal{H}om_{\mathcal{O}_U}(\mathcal{O}_U, \mathcal{O}_U) \cong \mathcal{O}_U$, so $\mathcal{L}^\vee$ is invertible. The evaluation map

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \longrightarrow \mathcal{O}_X, \qquad s \otimes \varphi \longmapsto \varphi(s)$$

is an isomorphism on each trivialising open (where it is the multiplication $R \otimes_R R \to R$), hence a global isomorphism.

Part (3) follows from the corresponding properties of the tensor product of modules.

Definition 5.4.50: Picard Group

The *Picard group* of a ringed space X is the abelian group

$$\operatorname{Pic} X \ := \frac{\{\text{invertible sheaves on } X\}}{\text{isomorphism}}$$

with group operation $[\mathcal{L}] \cdot [\mathcal{L}'] = [\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}']$, identity element $[\mathcal{O}_X]$, and inverse $[\mathcal{L}]^{-1} = [\mathcal{L}^\vee]$.

We shall see in theorem 5.4.55 that for integral schemes, $\operatorname{Pic} X \cong H^1(X, \mathcal{O}_X^*)$, giving a cohomological interpretation.

Proposition 5.4.51: Functoriality of the Picard Group

Let $f\colon X \to Y$ be a morphism of ringed spaces. The pullback of $\mathcal{O}_Y$-modules induces a well-defined group homomorphism

$$f^*\colon \operatorname{Pic} Y \longrightarrow \operatorname{Pic} X, \qquad [\mathcal{L}] \longmapsto [f^*\mathcal{L}].$$

Consequently, Pic defines a contravariant functor from the category of ringed spaces to the category of abelian groups.

***Proof*:**
Recall that the pullback of an $\mathcal{O}_Y$-module $\mathcal{L}$ is given by $f^*\mathcal{L} = f^{-1}\mathcal{L} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$. We must first verify that $f^*\mathcal{L}$ is an invertible sheaf on X whenever $\mathcal{L}$ is an invertible sheaf on Y.

Choose an open cover $\{V_i\}$ of Y that trivialises $\mathcal{L}$, meaning $\mathcal{L}|_{V_i} \cong \mathcal{O}_{V_i}$. For each i, let $U_i = f^{-1}(V_i)$, which forms an open cover of X. Because the pullback operation commutes with restriction to open subsets, we have

$$(f^*\mathcal{L})|_{U_i} \;\cong\; (f|_{U_i})^*(\mathcal{L}|_{V_i}) \;\cong\; (f|_{U_i})^*\mathcal{O}_{V_i} \;\cong\; \mathcal{O}_{U_i}.$$

Thus, $f^*\mathcal{L}$ is locally free of rank 1, making it an invertible sheaf on X. This ensures the assignment $[\mathcal{L}] \mapsto [f^*\mathcal{L}]$ is well-defined.

To see that f^* is a group homomorphism, we rely on the fact that the pullback functor for modules commutes with the tensor product. For any two invertible sheaves $\mathcal{L}, \mathcal{L}'$ on Y, there is a canonical isomorphism

$$f^*(\mathcal{L} \otimes_{\mathcal{O}_Y} \mathcal{L}') \;\cong\; f^*\mathcal{L} \otimes_{\mathcal{O}_X} f^*\mathcal{L}'.$$

Therefore, $f^*([\mathcal{L}] \cdot [\mathcal{L}']) = f^*([\mathcal{L}]) \cdot f^*([\mathcal{L}'])$.

Finally, the canonical isomorphisms $(g \circ f)^* \cong f^* \circ g^*$ and $(\mathrm{id}_Y)^* \cong \mathrm{id}_{\operatorname{Pic} Y}$ follow directly from the general categorical properties of module pullbacks, confirming that Pic is a contravariant functor.

Definition 5.4.52: Line Bundle Associated to a Cartier Divisor

Let X be a scheme and $D \in \operatorname{CDiv} X$ a Cartier divisor represented by data $\{(U_i, f_i)\}$ with $f_i \in \Gamma(U_i, \mathscr{K}_X^*)$ and $f_i/f_j \in \mathcal{O}_X^*(U_i \cap U_j)$. The *invertible sheaf associated to* D is the sub-$\mathcal{O}_X$-module $\mathcal{O}_X(D) \subseteq \mathscr{K}_X$ defined by

$$\mathcal{O}_X(D)|_{U_i} \;:=\; f_i^{-1} \cdot \mathcal{O}_X|_{U_i} \;\subseteq\; \mathscr{K}_X|_{U_i}.$$

On the overlap $U_i \cap U_j$, the generators satisfy $f_i^{-1} = (f_j/f_i) \cdot f_j^{-1}$ with $f_j/f_i \in \mathcal{O}_X^*$, so the local descriptions glue. The trivialisation $\mathcal{O}_X(D)|_{U_i} \xrightarrow{\sim} \mathcal{O}_X|_{U_i}$ is given by $g \mapsto gf_i$.[a]

[a] Note that X need not be integral for this construction; the sheaf $\mathscr{K}_X$ of total quotient rings replaces the function field.

When X is integral, one can describe the sections of $\mathcal{O}_X(D)$ more concretely. Since $\mathscr{K}_X$ is the constant sheaf $K(X)$, the sections over an open U are

$$\mathcal{O}_X(D)(U) \;=\; \{g \in K(X)^* \mid \operatorname{div}(g) + D \geq 0 \text{ on } U\} \;\cup\; \{0\},$$

where $\operatorname{div}(g) + D \geq 0$ means the Weil divisor is effective on U. Intuitively, $\mathcal{O}_X(D)(U)$ consists of all rational functions whose poles are "bounded by D": the divisor D prescribes how many poles are allowed along each prime divisor, and g must compensate. Historically, this is precisely how divisors first arose on Riemann surfaces — as a bookkeeping device for the poles and zeros of meromorphic functions — and the space $\mathcal{O}_X(D)(U)$ is the classical *Riemann–Roch space* $L(D)$.

Example 5.4.53: Number-Theoretic Line Bundle

Let $R = \mathbb{Z}[\sqrt{-5}]$ and $X = \operatorname{Spec} R$. Since R is a Dedekind domain, X is a normal Noetherian integral scheme of dimension 1, and every fractional ideal of R defines an invertible sheaf via the tilde construction (definition 5.1.6).

Consider the prime ideal $\mathfrak{p} = (2, 1 + \sqrt{-5})$, which lies over 2 and satisfies $\mathfrak{p}^2 = (2)$ (cf. example 5.4.15). As a Weil divisor, $[V(\mathfrak{p})]$ corresponds to the single closed point $\mathfrak{p} \in X$. This Weil divisor is Cartier (since every local ring of a Dedekind domain is a DVR, hence a UFD): at the point $\mathfrak{p}$ itself, the uniformizer $1 + \sqrt{-5}$ generates the ideal locally, while at every other closed point $\mathfrak{q} \neq \mathfrak{p}$, the ideal $\mathfrak{p}$ is the full local ring and is generated by 1.

The invertible sheaf $\widetilde{\mathfrak{p}}$ associated to this fractional ideal is *not* isomorphic to $\mathcal{O}_X$, since $\mathfrak{p}$ is not principal. Its class $[\widetilde{\mathfrak{p}}] \in \operatorname{Pic} X \cong \operatorname{Cl}(R) \cong \mathbb{Z}/2\mathbb{Z}$ is the nontrivial element, and $\widetilde{\mathfrak{p}}^{\otimes 2} \cong \widetilde{\mathfrak{p}^2} = \widetilde{(2)} \cong \mathcal{O}_X$ (the last isomorphism since (2) is principal as a fractional ideal). This recovers the ideal class group of $\mathbb{Z}[\sqrt{-5}]$ as the Picard group.

Proposition 5.4.54: Correspondence Between Line Bundles and Cartier Divisors

Let X be a scheme. The assignment $D \mapsto \mathcal{O}_X(D)$ defines a group homomorphism $\operatorname{CDiv} X \to \operatorname{Pic} X$ satisfying:

1. $\mathcal{O}_X(D + D') \cong \mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')$ for all $D, D' \in \operatorname{CDiv} X$.
2. $\mathcal{O}_X(-D) \cong \mathcal{O}_X(D)^\vee$ for all $D \in \operatorname{CDiv} X$.
3. $\mathcal{O}_X(D) \cong \mathcal{O}_X$ if and only if D is a principal Cartier divisor.

In particular, the homomorphism descends to an injection $\operatorname{CaCl} X \hookrightarrow \operatorname{Pic} X$.

Part (1) pairs well with the stalkwise intuition for the tensor product: if D prescribes poles and zeros along certain prime divisors and D' prescribes its own, then $\mathcal{O}_X(D) \otimes \mathcal{O}_X(D')$ "combines" the two prescriptions, gaining the zeros and poles of both.

***Proof*:**
Part (1). If $D = \{(U_i, f_i)\}$ and $D' = \{(U_i, g_i)\}$ (after passing to a common refinement), then $D + D' = \{(U_i, f_i g_i)\}$. On U_i, the multiplication map in $\mathscr{K}_X$ gives

$$\mathcal{O}_X(D)|_{U_i} \otimes \mathcal{O}_X(D')|_{U_i} \;=\; f_i^{-1}\mathcal{O}_{U_i} \otimes g_i^{-1}\mathcal{O}_{U_i} \;\xrightarrow{\sim}\; (f_i g_i)^{-1}\mathcal{O}_{U_i} \;=\; \mathcal{O}_X(D + D')|_{U_i}.$$

These local isomorphisms are compatible on overlaps (the transition functions multiply), yielding a global isomorphism.

Part (2). The dual $\mathcal{O}_X(D)^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X(D), \mathcal{O}_X)$ is locally generated by f_i (the functional sending $f_i^{-1} \mapsto 1$), which is $\mathcal{O}_X(-D)|_{U_i}$.

Part (3). If $D = \operatorname{div}(f)$ is principal (represented by a single global $f \in \Gamma(X, \mathscr{K}_X^*)$), then $\mathcal{O}_X(D) = f^{-1}\mathcal{O}_X \cong \mathcal{O}_X$. Conversely, if $\varphi \colon \mathcal{O}_X(D) \xrightarrow{\sim} \mathcal{O}_X$, then φ sends the local generator f_i^{-1} to some $h_i \in \mathcal{O}_X^*(U_i)$. The element $f = f_i h_i^{-1} \in \mathscr{K}_X^*$ is independent of i (on $U_i \cap U_j$, the transition functions force $f_i h_i^{-1} = f_j h_j^{-1}$), and $D = \operatorname{div}(f)$.

For a general scheme X, the injection $\operatorname{CaCl} X \hookrightarrow \operatorname{Pic} X$ may be strict: there can exist invertible sheaves that do not embed as sub-$\mathcal{O}_X$-modules of $\mathscr{K}_X$. However, the map is surjective — and hence an isomorphism — whenever X is integral:

Theorem 5.4.55: Isomorphism of Cartier Class Group and Picard Group

Let X be an integral scheme. The map $D \mapsto [\mathcal{O}_X(D)]$ induces an isomorphism of abelian groups

$$\operatorname{CaCl} X \xrightarrow{\sim} \operatorname{Pic} X.$$

***Proof*:**

Injectivity follows from proposition 5.4.54(3). For surjectivity, let $\mathcal{L}$ be an invertible sheaf. Since X is integral, the constant sheaf $\mathscr{K}_X = K(X)$ satisfies $\mathcal{L} \otimes_{\mathcal{O}_X} \mathscr{K}_X \cong \mathscr{K}_X$ (both sides are locally free $\mathscr{K}_X$-modules of rank 1, and $\mathscr{K}_X$ has a unique such module up to isomorphism). The natural map $\mathcal{L} \hookrightarrow \mathcal{L} \otimes \mathscr{K}_X$ (sending $s \mapsto s \otimes 1$) is injective since X is integral. Composing with any isomorphism $\mathcal{L} \otimes \mathscr{K}_X \xrightarrow{\sim} \mathscr{K}_X$ embeds $\mathcal{L}$ as a sub-$\mathcal{O}_X$-module of $\mathscr{K}_X$. On a trivialising open U_i, this embedding sends the local generator of $\mathcal{L}|_{U_i}$ to some $f_i^{-1} \in K(X)^*$, and the collection $\{(U_i, f_i)\}$ defines a Cartier divisor D with $\mathcal{L} \cong \mathcal{O}_X(D)$. A different choice of isomorphism $\mathcal{L} \otimes \mathscr{K}_X \cong \mathscr{K}_X$ changes each f_i by a common factor $h \in K(X)^*$, replacing D by $D + \operatorname{div}(h)$.

Cohomological perspective. Consider the short exact sequence of abelian sheaves

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathscr{K}_X^* \longrightarrow \mathscr{K}_X^*/\mathcal{O}_X^* \longrightarrow 1.$$

Since X is integral, $\mathscr{K}_X^* = K(X)^*$ is a constant sheaf on the irreducible space X, hence flasque. The long exact sequence in cohomology gives

$$\Gamma(X, \mathscr{K}_X^*) \longrightarrow \underbrace{\Gamma(X, \mathscr{K}_X^*/\mathcal{O}_X^*)}_{\operatorname{CDiv} X} \xrightarrow{\delta} \underbrace{H^1(X, \mathcal{O}_X^*)}_{\operatorname{Pic} X} \longrightarrow \underbrace{H^1(X, \mathscr{K}_X^*)}_{=0}$$

where the vanishing follows from flasqueness (corollary 6.1.7). The connecting homomorphism δ sends a Cartier divisor $\{(U_i, f_i)\}$ to the Čech 1-cocycle $\{f_i/f_j\} \in \check{H}^1(\{U_i\}, \mathcal{O}_X^*)$, which classifies the line bundle $\mathcal{O}_X(D)$. Its cokernel is zero and its kernel is exactly the image of $\Gamma(X, \mathscr{K}_X^*)$ — the principal divisors. Hence $\operatorname{CaCl} X \cong \operatorname{Pic} X$.

The cohomological proof explains the warning following definition 5.4.23: the discrepancy between global sections of the quotient sheaf $\mathscr{K}_X^*/\mathcal{O}_X^*$ and the literal quotient of global sections is precisely the connecting homomorphism δ, and the group $H^1(X, \mathcal{O}_X^*)$ classifies line bundles.

Corollary 5.4.56: Divisor Class Group and Picard Group

Let X be an integral, separated, Noetherian, locally factorial scheme. Then there is a chain of isomorphisms

$$\operatorname{Cl} X \xrightarrow[\textit{proposition } 5.4.32]{\sim} \operatorname{CaCl} X \xrightarrow[\textit{theorem } 5.4.55]{\sim} \operatorname{Pic} X.$$

Corollary 5.4.57: Classifying Line Bundles on Projective Space

Let k be a field. Every invertible sheaf on $\mathbb{P}^n_k$ is isomorphic to $\mathcal{O}_{\mathbb{P}^n_k}(d)$ for a unique $d \in \mathbb{Z}$. Hence

$$\operatorname{Pic} \mathbb{P}^n_k \cong \mathbb{Z},$$

with generator $[\mathcal{O}_{\mathbb{P}^n_k}(1)]$ corresponding to the hyperplane class.

***Proof*:**
The scheme $\mathbb{P}^n_k$ is integral, separated, Noetherian, and regular (hence locally factorial by proposition 2.7.20). By corollary 5.4.56 and proposition 5.4.16, $\operatorname{Pic} \mathbb{P}^n_k \cong \operatorname{Cl} \mathbb{P}^n_k \cong \mathbb{Z}$, generated by the hyperplane class. Since the hyperplane divisor $[H]$ corresponds to $\mathcal{O}_{\mathbb{P}^n_k}(1)$ under the construction of definition 5.4.52, the class $d[H]$ corresponds to $\mathcal{O}_{\mathbb{P}^n_k}(d)$, confirming the earlier computation of proposition 5.3.5.

We close this subsection by recording the connection between effective Cartier divisors and line bundles equipped with sections — the algebro-geometric counterpart of the classical fact that the zero locus of a holomorphic section of a holomorphic line bundle is a divisor (proposition 0.6.34).

Proposition 5.4.58: Effective Cartier Divisors and Line Bundles

Let X be an integral scheme and D an effective Cartier divisor with associated closed subscheme $Y \hookrightarrow X$ (definition 5.4.25).

1. The ideal sheaf of Y satisfies $\mathcal{I}_Y \cong \mathcal{O}_X(-D)$.

2. There is a natural bijection

$$\{ \text{ effective Cartier divisors on } X \ \} \longleftrightarrow \left\{ \begin{array}{c} \text{pairs } (\mathcal{L}, s) \text{ with } \mathcal{L} \text{ invertible and} \\ 0 \neq s \in \Gamma(X, \mathcal{L}), \\ \text{up to isomorphism} \end{array} \right\}$$

Under this bijection, D corresponds to $(\mathcal{O}_X(D), s_D)$, where s_D is the "canonical section" — the element $1 \in K(X)$ viewed as a global section of $\mathcal{O}_X(D)$.

***Proof*:**
Part (1). Let D be represented by $\{(U_i, f_i)\}$ with $f_i \in \mathcal{O}_X(U_i)$ non-zero-divisors. Then $\mathcal{O}_X(-D)|_{U_i} = f_i \cdot \mathcal{O}_{U_i}$, which is precisely the ideal sheaf of Y on U_i (by definition 5.4.25, $\mathcal{I}_Y$ is locally generated by f_i). Since both sides glue by the same transition functions, we get $\mathcal{I}_Y \cong \mathcal{O}_X(-D)$ as subsheaves of $\mathcal{O}_X$.

Part (2). Given an effective $D = \{(U_i, f_i)\}$, the element $1 \in K(X)$ lies in $\mathcal{O}_X(D)(U_i) = f_i^{-1}\mathcal{O}_{U_i}$ (since $1 \cdot f_i = f_i \in \mathcal{O}_X(U_i)$), so $s_D := 1$ is a well-defined nonzero global section. Under the trivialisation $\mathcal{O}_X(D)|_{U_i} \cong \mathcal{O}_{U_i}$, the section s_D corresponds to f_i, so s_D vanishes precisely along Y.

Conversely, given $(\mathcal{L}, s)$ with $s \neq 0$, choose trivialisations $\psi_i \colon \mathcal{L}|_{U_i} \xrightarrow{\sim} \mathcal{O}_{U_i}$ and set $f_i = \psi_i(s|_{U_i})$. Since $s \neq 0$ and X is integral, each f_i is a non-zero-divisor. On overlaps, $f_i/f_j = \psi_i \circ \psi_j^{-1}(1) \in \mathcal{O}_X^*(U_i \cap U_j)$, so $\{(U_i, f_i)\}$ is an effective Cartier divisor. The two constructions are mutually inverse.

Part (1) provides a useful dictionary: the ideal sheaf of an effective Cartier divisor is the "negative" line bundle $\mathcal{O}_X(-D)$, and the short exact sequence of $\mathcal{O}_X$-modules

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y \longrightarrow 0$$

will be a fundamental tool in cohomological computations (section 6.3). Part (2) establishes that studying line bundles with sections is *equivalent* to studying effective divisors — a perspective that will be central to the theory of linear systems (remark 5.3.32) and the Riemann–Roch theorem (theorem 7.1.19).

5.5 Differentials

The reader arrives at this section already fluent in its geometry. On a smooth manifold M, the cotangent space T_p^*M (definition 0.5.11) and the differential forms assembled from it (definition 0.5.14) carry the infinitesimal geometry of M, and the differential of a function $df_p \in T_p^*M$ (definition 0.5.12) records its first-order behavior at p. The point of contact with algebra is a description of these objects that uses no analysis at all. Writing $\mathfrak{m}_p$ for the ideal of functions vanishing at p,

$$T_pM \;\cong\; (\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee, \qquad T_p^*M \;\cong\; \mathfrak{m}_p/\mathfrak{m}_p^2$$

([53]), and under the second identification the covector df_p is the class of $f - f(p)$ modulo $\mathfrak{m}_p^2$. No limit and no smooth structure enter the right-hand sides: only an ideal, its square, and a quotient. This is verbatim the Zariski cotangent space introduced earlier for varieties (definition 0.1.25), and that coincidence is why differentials belong to algebraic geometry at all.

The same description settles which of the two objects is primitive. The assignment $f \mapsto df = [\,f - f(p)\,]$ produces a covector from a function by a single quotient; the tangent vector, an element of the dual, is what one obtains *afterward*, by dualizing. In the smooth category the distinction is immaterial, since both spaces are finite-dimensional and either determines the other. But the construction that survives with no appeal to limits, and so transports to an arbitrary ring, is the cotangent one. Functoriality points the same way: functions pull back along a morphism $f \colon X \to Y$, so the cotangent construction, covariant in the ring of functions, transforms along with them, while tangent vectors push forward, in the opposite direction. Algebraic geometry therefore takes the cotangent side as basic. The *sheaf of differentials* $\Omega_{X/Y}$ (definition 5.5.14) is the object it builds first; the universal derivation $d \colon b \mapsto db$ is the Leibniz and chain rules promoted to a definition; and the tangent sheaf (definition 5.5.25) appears only afterward, as its dual.

The payoff is an exact algebraic stand-in for the cotangent bundle. Once $\Omega_{X/Y}$ is in hand, its fiber at a point is canonically $\mathfrak{m}_p/\mathfrak{m}_p^2 = T_p^*$ (proposition 5.5.10): the sheaf of differentials is the cotangent bundle glued together globally, with no coordinate charts and no partitions of unity. Being the cotangent object and not its dual is what lets it see singularities. The dimension of $\mathfrak{m}_p/\mathfrak{m}_p^2$ jumps above the dimension of X exactly at a singular point (definition 0.1.27), the sheaf $\Omega_{X/Y}$ fails to be locally free there (theorem 5.5.12), and this failure is the algebraic test for smoothness (proposition 5.5.22); the tangent sheaf, a dual that is well behaved only in the smooth case, is the wrong place to look for it. Further downstream, the top exterior power of $\Omega_{X/Y}$ is the canonical sheaf standing behind Serre duality (theorem 6.5.9) and Riemann–Roch (theorem 7.1.19), and $\Omega_{X/Y}$ itself governs deformation theory. These are the section's eventual destinations; the road to them begins in commutative algebra.

We begin with that algebra: the module of Kähler differentials $\Omega_{B/A}$ attached to a ring map $A \to B$, before globalizing to schemes. The reader who knows forms on manifolds will recognize the parallel at once, since the Leibniz rule and the chain rule are built directly into the definition.

5.5.1 Kähler Differentials

The module of relative differentials is pure commutative algebra and is developed in [18]: derivations, $\Omega_{B/A}$ and its universal property, the diagonal construction, base change and localization, both fundamental exact sequences, and finite generation. The statements are recalled here because the sheaf-theoretic constructions below rest on them, but the proofs are not repeated; what this book owns is the passage from the module to the sheaf $\Omega_{X/S}$ and everything after it.

Definition 5.5.1: A-Derivation

Let A be a commutative ring, B an A-algebra, and M a B-module. An A-*derivation* of B into M is a map $d\colon B \to M$ satisfying:

1. *(Additivity)* $d(b + b') = d(b) + d(b')$,
2. *(Leibniz rule)* $d(bb') = b\, d(b') + b'\, d(b)$,
3. *(Vanishing on A)* $d(a) = 0$ for all $a \in A$ (treating A as mapped into B)

The set of all A-derivations $B \to M$ is denoted $\mathrm{Der}_A(B, M)$ and carries a natural B-module structure via $(b \cdot d)(b') = b\, d(b')$.

Condition (3) is the algebraic analogue of “the derivative of a constant is zero”: the elements of A, viewed as “constants” in the A-algebra B, have trivial differential. Note that (3) is automatic when A is a prime ring or prime field (generated by a single element) since then $d(1) = d(1 \cdot 1) = 2\, d(1)$ forces $d(1) = 0$. Condition (3) forces the derivation to be A-linear since for $a \in A$:

$$d(ab) = d(a)b + ad(b) = 0 + ad(b) = ad(b)$$

Definition 5.5.2: Module of Relative Differential Forms

Let A be a commutative ring and B an A-algebra. A *module of (relative) Kähler differentials* for B over A is a B-module $\Omega_{B/A}$ equipped with an A-derivation $d\colon B \to \Omega_{B/A}$ satisfying the following universal property: for every B-module M and every A-derivation $d'\colon B \to M$, there exists a unique B-module homomorphism $\varphi\colon \Omega_{B/A} \to M$ such that $d' = \varphi \circ d$:

$$\begin{tikzcd} B \arrow[r, "d"] \arrow[dr, "d'"'] & \Omega_{B/A} \arrow[d, dashed, "\exists!\,\varphi"] \\ & M \end{tikzcd}$$

Equivalently, d induces a B-module isomorphism

$$\mathrm{Hom}_B(\Omega_{B/A}, M) \xrightarrow{\sim} \mathrm{Der}_A(B, M), \qquad \varphi \longmapsto \varphi \circ d.$$

The universal property determines $(\Omega_{B/A}, d)$ up to unique isomorphism, as with any object defined by a universal property. There are two general constructions to $\Omega_{B/A}$: an explicit generators-and-relations presentation, and a more geometric one via the diagonal ideal.

Construction 1: generators and relations. Let F be the free B-module on symbols $\{db : b \in B\}$, and let R be the submodule generated by the relations

$$d(b + b') - db - db', \qquad d(bb') - b\,db' - b'\,db, \qquad da \quad (a \in A).$$

Set $\Omega_{B/A} = F/R$ and define $d\colon B \to \Omega_{B/A}$ by $b \mapsto db$. By construction, d is an A-derivation, and the universal property is immediate: any A-derivation $d'\colon B \to M$ extends uniquely to a B-module map $F \to M$ (sending $db \mapsto d'(b)$), which kills R and hence factors through $\Omega_{B/A}$.

Construction 2: the diagonal ideal. The following construction has a geometric interpretation that globalizes directly to schemes.

Proposition 5.5.3: Differentials via the Diagonal

Let B be an A-algebra. Let $\mu\colon B \otimes_A B \to B$ be the multiplication map $b \otimes b' \mapsto bb'$, and let $I = \ker \mu$. View $B \otimes_A B$ as a B-module via left multiplication ($b \cdot (b_1 \otimes b_2) = bb_1 \otimes b_2$). Then:

1. I/I^2 inherits the structure of a B-module.
2. The map $d\colon B \to I/I^2$ defined by $d(b) = 1 \otimes b - b \otimes 1 \pmod{I^2}$ is an A-derivation.
3. The pair $(I/I^2, d)$ satisfies the universal property of $\Omega_{B/A}$.

***Proof*:**
This is proved in [18], where the diagonal construction is verified against the universal property. The verification is pure commutative algebra and is not repeated here.

The geometric content of Construction 2 is the following: if $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$, then $B \otimes_A B$ is the coordinate ring of the fiber product $X \times_Y X$, and the multiplication map $\mu\colon B \otimes_A B \to B$ corresponds to the diagonal morphism $\Delta\colon X \to X \times_Y X$. The ideal $I = \ker \mu$ is the ideal sheaf of the diagonal, and I/I^2 is the *conormal sheaf* of the diagonal embedding. Thus the module of differentials is the algebraic incarnation of "functions vanishing on the diagonal, modulo those vanishing to second order", precisely the data captured by first-order Taylor expansions. This interpretation globalizes immediately to the sheaf of differentials on a morphism of schemes (cf. definition 5.5.14).

Note that we can define the higher order equivalently (these are known as a jet bundle):

$$\mathcal{P}^n_{B/A} = \frac{B \otimes_A B}{I^{n+1}}$$

but we don't focus on these in this section.

Example 5.5.4: Relative Differentials

1. *Polynomial rings.* Let $B = A[x_1, \dots, x_n]$. The Leibniz rule and the chain rule force $\Omega_{B/A}$ to be generated by $dx_1, \dots, dx_n$. Conversely, the partial derivatives $\partial/\partial x_i \colon B \to B$ are A-derivations, and the universal property provides B-linear maps $\Omega_{B/A} \to B$ sending $dx_i \mapsto 1$ and $dx_j \mapsto 0$ for $j \neq i$. Hence $dx_1, \dots, dx_n$ are linearly independent, and

$$\Omega_{B/A} \;\cong\; \bigoplus_{i=1}^{n} B\,dx_i,$$

   a free B-module of rank n. The universal derivation is $d(f) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}\,dx_i$, exactly as in multivariable calculus.

2. *Quotient rings.* Let $B = A[x_1, \dots, x_n]/J$ where $J = (f_1, \dots, f_r)$. Using the two exact sequences below (propositions 5.5.5 and 5.5.6), one obtains

$$\Omega_{B/A} \;\cong\; \frac{\bigoplus_{i=1}^n B\,dx_i}{\sum_{j=1}^r B\,df_j},$$

   where $df_j = \sum_i \frac{\partial f_j}{\partial x_i}\,dx_i$. This is the algebraic analogue of computing the cotangent space of a submanifold by imposing the constraint equations.

   For instance, if $B = k[x,y]/(y^2 - x^3)$ (the cuspidal cubic), then $\Omega_{B/k} = B\,dx \oplus B\,dy \,/\, (2y\,dy - 3x^2\,dx)$. At the origin $(x,y) = (0,0)$, the relation $2y\,dy - 3x^2\,dx$ vanishes identically, so $\Omega_{B/k} \otimes_B k \cong k^2$, two-dimensional rather than the expected one. This detects the singularity (cf. example 2.7.5).

3. *Separable field extensions.* Let K/k be a finite field extension. Then $\Omega_{K/k} = 0$ if and only if K/k is separable. To see this, first consider a simple extension $K = k(\alpha) \cong k[x]/(f)$, where f is the minimal polynomial of α. By the quotient ring computation above, $\Omega_{K/k} \cong K\,dx/(f'(\alpha)\,dx)$. If K/k is separable, f has no multiple roots, meaning $f'(\alpha) \neq 0$. Since K is a field, $f'(\alpha)$ is a unit, forcing $dx = 0$ and hence $\Omega_{K/k} = 0$. Conversely, if K/k is inseparable (which only occurs if $\operatorname{char} k = p > 0$), there exists some $\alpha \in K$ whose minimal polynomial takes the form $f(x) = g(x^p)$. Its formal derivative vanishes identically ($f'(x) = 0$), so the relation $f'(\alpha)\,dx = 0$ imposes no constraint on dx. Thus $\Omega_{k(\alpha)/k} \neq 0$, and hence $\Omega_{K/k} \neq 0$.

   Geometrically, a separable field extension represents a discrete covering of an isolated points, meaning there is no continuous space to move and therefore a trivial cotangent space ($\Omega_{K/k} = 0$). In contrast, an inseparable extension contains "infinitesimal fuzz" where points have collided in positive characteristic, spawning a hidden, non-trivial differential direction even in a zero-dimensional space.

4. *Localization.* If $S \subseteq B$ is a multiplicative set, then $\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$. This follows from the universal property: an A-derivation $S^{-1}B \to M$ is determined by its restriction to B, using the quotient rule $d(b/s) = (s\,db - b\,ds)/s^2$.

The following two exact sequences are used to compute differentials. They are the algebraic counterparts of the cotangent exact sequence and the conormal exact sequence in differential geometry:

Proposition 5.5.5: First Exact Sequence (Base Change, or Fibrations)

Let $A \to B \to C$ be ring homomorphisms. Then there is a natural exact sequence of C-modules

$$\Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0.$$

The first map sends $db \otimes 1 \mapsto d(\varphi(b))$ (where $\varphi\colon B \to C$ is the structure map), and the second map sends $dc \mapsto dc$.

***Proof*:**
Proved in [18] by applying $\mathrm{Hom}_C(-, M)$ and checking exactness of the resulting sequence of derivation modules.

Note that $A \to B \to C$ is has no exactness conditions set on it: geometrically this has many interpretations (this can be a fibration, it can be closed immersions, etc). This generality is partly to build good tools to construct differentials, and partly because singularities often break exactness (as demonstrated in examples in chapter 3). Geometrically, if you take the total differential of a space ($\Omega_{C/A}$) and quotient out the differentials coming from the base space ($\Omega_{B/A}$), you are left exactly with the differentials of the fibers ($\Omega_{C/B}$). In terms of tangent vectors (the dual picture), the proposition says the tangent vectors of a fiber are exactly the tangent vectors of the total space that project to zero on the base.

Proposition 5.5.6: Second Exact Sequence (Conormal)

Let B be an A-algebra, $J \subseteq B$ an ideal, and $C = B/J$. There is a natural exact sequence of C-modules

$$J/J^2 \xrightarrow{\delta} \Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow 0,$$

where $\delta(\overline{f}) = df \otimes 1$ for $f \in J$.

***Proof*:**
Proved in [18], by the same $\mathrm{Hom}_C(-, M)$ argument as the first sequence, together with the observation that a derivation into a C-module descends to C exactly when it kills I.

Geometrically, this tells you how to link the intrinsic cotangent space of the subspace $\Omega_{C/A}$ to the ambient cotangent space $\Omega_{B/A} \otimes_B C$. In other word, to find the differentials of a subspace ($\Omega_{C/A}$), start with the differentials of the ambient space ($\Omega_{B/A}$) and kill the gradients of the constraint equations ($\delta(J/J^2)$).

Corollary 5.5.7: Finite Generation of Differentials

If B is a finitely generated A-algebra, then $\Omega_{B/A}$ is a finitely generated B-module.

***Proof*:**
Immediate from the construction of $\Omega_{B/A}$ by generators and relations; see [18].

The next two results relate differentials to base change and to field extensions, the latter being crucial for detecting smoothness.

Proposition 5.5.8: Base Change for Differentials

Let $A \to B$ and $A \to A'$ be ring homomorphisms, and set $B' = B \otimes_A A'$. Then

$$\Omega_{B'/A'} \;\cong\; \Omega_{B/A} \otimes_B B'.$$

***Proof*:**
Proved in [18].

Proposition 5.5.9: Differentials and Field Extensions

Let k be a field and K/k a finitely generated[a] field extension. Then $\Omega_{K/k}$ is a K-vector space and:

$$\dim_K \Omega_{K/k} \geq \operatorname{trdeg}(K/k)$$

with equality if and only if K is separably generated over k (i.e., K is a separable algebraic extension of a purely transcendental extension of k).

[a] be sure not to confuse with finite! example 5.5.4 was a special case with finite extensions

5.5.2 Differentials and Local Rings

The module of differentials detects regularity of local rings, providing the algebraic foundation for the Jacobian criterion.

Proposition 5.5.10: Differentials and Local Rings

Let $(B, \mathfrak{m})$ be a Noetherian local ring that is a localization of a finitely generated k-algebra (where k is a field), with residue field $\kappa = B/\mathfrak{m}$. Assume the residue field κ is a separable extension of k (e.g., k is perfect or algebraically closed). Then there is a natural isomorphism of κ-vector spaces

$$\Omega_{B/k} \otimes_B \kappa \;\cong\; \mathfrak{m}/\mathfrak{m}^2.$$

***Proof*:**
Apply the second exact sequence (proposition 5.5.6) to the surjection $B \twoheadrightarrow \kappa$ with kernel $\mathfrak{m}$:

$$\mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\delta} \Omega_{B/k} \otimes_B \kappa \longrightarrow \Omega_{\kappa/k} \longrightarrow 0.$$

If κ/k is separable (e.g., if k is perfect), then $\Omega_{\kappa/k} = 0$, and δ is surjective. Since δ sends $\overline{f} \mapsto df \otimes 1$ and both sides have the same dimension (namely $\dim_\kappa \mathfrak{m}/\mathfrak{m}^2$), the map δ is an isomorphism.

Lemma 5.5.11: Characterizing Freeness via Residue and Quotient Fields

Let $(R, \mathfrak{m})$ be a Noetherian local domain with fraction field K, and let M be a finitely generated R-module. Then M is free of rank r if and only if

$$\dim_{R/\mathfrak{m}}(M \otimes_R R/\mathfrak{m}) \;=\; r \;=\; \dim_K(M \otimes_R K).$$

***Proof*:**
By Nakayama's lemma, M is generated by r elements if and only if $\dim_{R/\mathfrak{m}}(M/\mathfrak{m}M) = r$. If additionally $\dim_K(M \otimes_R K) = r$, then a minimal generating set of M is K-linearly independent, hence R-linearly independent (since R is a domain), so $M \cong R^r$.

Theorem 5.5.12: Characterizing Regular Local Rings via Differentials

Let B be a localization of a finitely generated k-algebra at a maximal ideal, where k is a perfect field. Set $n = \dim B$. Then B is a regular local ring if and only if $\Omega_{B/k}$ is a free B-module of rank n.

***Proof*:**
By the Nullstellensatz, the residue field κ is a finite extension of k. Since k is perfect, κ/k is separable, so $\Omega_{\kappa/k} = 0$. Thus, by proposition 5.5.10, $\dim_\kappa(\Omega_{B/k} \otimes_B \kappa) = \dim_\kappa(\mathfrak{m}/\mathfrak{m}^2)$.

Let $K = \mathrm{Frac}(B)$; then $\Omega_{B/k} \otimes_B K \cong \Omega_{K/k}$. Because B is localized at a maximal ideal, we have $\mathrm{trdeg}(K/k) = \dim B = n$, so $\Omega_{K/k}$ has dimension n by proposition 5.5.9 (using that k is perfect).

If B is regular, then $\dim_\kappa(\mathfrak{m}/\mathfrak{m}^2) = n = \dim B$, so both the fibre and the generic fibre of $\Omega_{B/k}$ have the same dimension n. By lemma 5.5.11, $\Omega_{B/k}$ is free of rank n.

Conversely, if $\Omega_{B/k}$ is free of rank n, then $\dim_\kappa(\mathfrak{m}/\mathfrak{m}^2) = n = \dim B$, which is the definition of regularity.

We shall see by box 5.5.21 that the result for maximal points is sufficient.

5.5.13 Non-Closed Points and Arbitrary Primes If B is localized at a non-maximal prime P, it represents the local ring of a positive-dimensional subvariety (i.e., its generic point). The residue field $\kappa(P)$ is then a transcendental function field over k, meaning its differentials no longer vanish ($\Omega_{\kappa(P)/k} \neq 0$). Consequently, the exact sequence from proposition 5.5.10 does not yield a simple isomorphism.
If B is a regular local ring at such a prime, $\Omega_{B/k}$ is indeed still a free module, but its rank is $\dim B + \operatorname{trdeg}_k \kappa(P)$. Geometrically, $\dim B$ captures the codimension (the normal directions transverse to the subvariety), while the transcendence degree captures the tangent directions along the subvariety itself. Restricting to maximal ideals sets the transcendence degree to zero, isolating the local geometry of a single point.

5.5.3 The Sheaf of Differentials

The algebraic constructions of the previous subsection globalize to schemes via the diagonal ideal, following the geometric interpretation of proposition 5.5.3. Let $f\colon X \to Y$ be a morphism of schemes and $\Delta\colon X \to X \times_Y X$ the diagonal morphism. By proposition 3.5.1, Δ is a locally closed immersion: the image $\Delta(X)$ is isomorphic to a closed subscheme of an open subset $W \subseteq X \times_Y X$. (When f is separated, Δ is a closed immersion into all of $X \times_Y X$, and we may take $W = X \times_Y X$.)

Definition 5.5.14: Sheaf of Differentials

Let $f\colon X \to Y$ be a morphism of schemes and let $\mathscr{I}$ be the ideal sheaf of $\Delta(X)$ in the open subset $W \subseteq X \times_Y X$ described above. The *sheaf of relative differentials* (or *cotangent sheaf*) of X over Y is the quasi-coherent $\mathcal{O}_X$-module

$$\Omega_{X/Y} \;:=\; \Delta^*(\mathscr{I}/\mathscr{I}^2),$$

The *universal derivation* $d\colon \mathcal{O}_X \to \Omega_{X/Y}$ is defined locally by $d(b) = 1 \otimes b - b \otimes 1 \pmod{\mathscr{I}^2}$, exactly as in proposition 5.5.3.

The sheaf $\mathscr{I}/\mathscr{I}^2$ is naturally an $\mathcal{O}_{\Delta(X)}$-module, and the isomorphism $X \cong \Delta(X)$ gives it the structure of an $\mathcal{O}_X$-module. Since $\mathscr{I}$ is a quasi-coherent ideal sheaf (proposition 5.1.34), the quotient $\mathscr{I}/\mathscr{I}^2$ and its pullback are quasi-coherent (proposition 5.1.17). When Y is Noetherian and f is of finite type, $\Omega_{X/Y}$ is coherent.

To see the local affine structure, let $V = \operatorname{Spec} A \subseteq Y$ and $U = \operatorname{Spec} B \subseteq X$ be affine opens with $f(U) \subseteq V$. Then $U \times_V U = \operatorname{Spec}(B \otimes_A B) \subseteq X \times_Y X$, and the diagonal $\Delta(U)$ is the closed subscheme defined by the kernel $I = \ker(B \otimes_A B \xrightarrow{\mu} B)$. Thus $\mathscr{I}/\mathscr{I}^2$ restricted to $U \times_V U$ is $\widetilde{I/I^2}$, and we recover

$$\Omega_{X/Y}|_U \;\cong\; \widetilde{\Omega_{B/A}}.$$

Hence the sheaf of differentials can equivalently be constructed by gluing the modules $\widetilde{\Omega_{B/A}}$ over an affine cover. Similarly, the derivations $d\colon B \to \Omega_{B/A}$ glue to the global derivation $d\colon \mathcal{O}_X \to \Omega_{X/Y}$.

The three fundamental properties of Kähler differentials (base change, the cotangent sequence, and the conormal sequence) globalize immediately.

Proposition 5.5.15: Sheaf of Differentials: Base Change

Let $f\colon X \to Y$ be a morphism, $g\colon Y' \to Y$ another morphism, and $f'\colon X' = X \times_Y Y' \to Y'$ the base change of f. Let $g'\colon X' \to X$ denote the projection. Then

$$\Omega_{X'/Y'} \;\cong\; (g')^* \, \Omega_{X/Y}.$$

***Proof*:**
On affine patches, this is proposition 5.5.8. The local isomorphisms are compatible on overlaps (by the uniqueness in the universal property of differentials), so they glue to a global isomorphism.

Proposition 5.5.16: Cotangent Exact Sequence

Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes. There is a natural exact sequence of $\mathcal{O}_X$-modules

$$f^*\Omega_{Y/Z} \longrightarrow \Omega_{X/Z} \longrightarrow \Omega_{X/Y} \longrightarrow 0.$$

***Proof*:**
Follows from proposition 5.5.5 on affine patches and gluing.

Proposition 5.5.17: Conormal Exact Sequence

Let $f\colon X \to Y$ be a morphism and $\iota\colon Z \hookrightarrow X$ a closed immersion with ideal sheaf $\mathscr{I}_Z \subseteq \mathcal{O}_X$. There is a natural exact sequence of $\mathcal{O}_Z$-modules

$$\mathscr{I}_Z/\mathscr{I}_Z^2 \xrightarrow{\delta} \iota^*\Omega_{X/Y} \longrightarrow \Omega_{Z/Y} \longrightarrow 0,$$

where $\delta(\overline{f}) = df \otimes 1$ for local sections f of $\mathscr{I}_Z$. The sheaf $\mathscr{I}_Z/\mathscr{I}_Z^2$ is the *conormal sheaf* of Z in X.

***Proof*:**
Follows from proposition 5.5.6 on affine patches and gluing.

The next result is specific to projective space and will be indispensable for cohomological computations throughout the rest of the book.

Theorem 5.5.18: The Euler Sequence

Let A be a ring, $Y = \operatorname{Spec} A$, and $X = \mathbb{P}^n_A$. There is an exact sequence of sheaves on X:

$$0 \longrightarrow \Omega_{X/Y} \longrightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

In particular, $\Omega_{X/Y}$ is locally free of rank n.

Recall that $\mathcal{O}_X(-1) \cong \mathcal{O}_X(1)^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X(1), \mathcal{O}_X)$ (proposition 5.3.5), so sections of $\mathcal{O}_X(-1)$ are naturally linear functionals on $\mathcal{O}_X(1)$. The map $\mathcal{O}_X(-1)^{n+1} \to \mathcal{O}_X$ is an evaluation map:

in homogeneous coordinates, it sends an $(n+1)$-tuple of linear functionals to their pairing with the tautological coordinates $(x_0, \dots, x_n)$. The kernel of this evaluation (the "relations among the coordinates") is precisely the sheaf of differentials.

It is also instructive to duality this sequence to see that the tangent sheaf (defined rigorously soon) is the quotient of the free sheaf by a action of $\mathcal{O}_X$, just like how projective space is classically constructed:

$$O \to \mathcal{O}_X \to \mathcal{O}_X(1)^{\oplus(n+1)} \to \mathcal{T}_X \to O$$

***Proof*:**

Let $S = A[x_0, \dots, x_n]$ be the homogeneous coordinate ring of X, and let $E = S(-1)^{n+1}$ be the free graded S-module with basis $e_0, \dots, e_n$ in degree 1. Define a degree-0 homomorphism of graded S-modules $\varphi\colon E \to S$ by $e_i \mapsto x_i$, and let $M = \ker\varphi$. The exact sequence of graded modules

$$0 \longrightarrow M \longrightarrow E \xrightarrow{\varphi} S$$

gives rise to an exact sequence of sheaves on X:

$$0 \longrightarrow \widetilde{M} \longrightarrow \mathcal{O}_X(-1)^{n+1} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Note that φ is not surjective as a map of graded modules (it misses degree 0), but it is surjective in all degrees ≥ 1 and hence surjective on all stalks of the associated sheaves.

It remains to show that $\widetilde{M} \cong \Omega_{X/Y}$. We construct compatible isomorphisms on each standard affine chart and verify that they glue.

Step 1: local structure of M. On the chart $U_i = D_+(x_i) \cong \operatorname{Spec} A[x_0/x_i, \dots, x_n/x_i]$, localizing at x_i makes $\varphi_{x_i}\colon E_{x_i} \to S_{x_i}$ a surjection of free S_{x_i}-modules. The kernel M_{x_i} is free of rank n, generated by $\{x_i e_j - x_j e_i \mid j \neq i\}$. Passing to the associated sheaf (and extracting degree-0 elements), $\widetilde{M}|_{U_i}$ is free over $\mathcal{O}_{U_i}$ with basis

$$\left\{ \frac{1}{x_i}\, e_j - \frac{x_j}{x_i^2}\, e_i \right\}_{j \neq i}$$

(the factor $1/x_i$ ensures these have degree 0 in M_{x_i}).

Step 2: defining the isomorphism φ_i. On U_i, the sheaf $\Omega_{X/Y}|_{U_i}$ is the free $\mathcal{O}_{U_i}$-module with basis $\{d(x_j/x_i)\}_{j \neq i}$ (example 5.5.4). Define $\varphi_i\colon \Omega_{X/Y}|_{U_i} \to \widetilde{M}|_{U_i}$ by

$$\varphi_i\!\left(d\!\left(\frac{x_j}{x_i}\right)\right) \;=\; \frac{1}{x_i^2}(x_i\, e_j - x_j\, e_i).$$

Since both sides are free of rank n and φ_i sends a basis to a basis, this is an isomorphism.

Step 3: gluing. On the overlap $U_i \cap U_j$, the chain rule gives

$$d\!\left(\frac{x_k}{x_i}\right) \;=\; \frac{x_k}{x_j}\, d\!\left(\frac{x_j}{x_i}\right) \;+\; \frac{x_j}{x_i}\, d\!\left(\frac{x_k}{x_j}\right).$$

Applying φ_i to the left-hand side and φ_j to the right-hand side, both yield $\frac{1}{x_i x_j}(x_j\, e_k - x_k\, e_j)$. Hence $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$, and the local isomorphisms glue to a global isomorphism $\Omega_{X/Y} \cong \widetilde{M}$.

5.5.19 Relation To Canonical Sheaf Taking the top exterior power of the Euler sequence (the determinant bundle) and applying the formula:

$$\bigwedge^{\text{top}} B \;\cong\; (\bigwedge^{\text{top}} A) \otimes (\bigwedge^{\text{top}} C)$$

yields the important formula:

$$\omega_{\mathbb{P}^n_A/A} \;:=\; \bigwedge^n \Omega_{\mathbb{P}^n_A/A} \;\cong\; \mathcal{O}_{\mathbb{P}^n_A}(-n-1),$$

which will be central to Serre duality (section 6.5).

5.5.4 Smoothness and the Cotangent Sheaf

We now focus on varieties over an algebraically closed field, where the cotangent sheaf provides a clean characterization of nonsingularity.

Definition 5.5.20: Abstract Nonsingular Variety

Let X be an abstract variety over an algebraically closed field k. Then X is *nonsingular* (or *smooth over* k) if every local ring $\mathcal{O}_{X,x}$ is a regular local ring.

5.5.21 Non-Closed Points? The definition requires regularity at *all* points, not just closed ones. However, since the local ring at a non-closed point $\mathfrak{p}$ is a localization of a local ring at a closed point (namely, $\mathcal{O}_{X,\mathfrak{p}} \cong (\mathcal{O}_{X,x})_{\mathfrak{p}}$ for a suitable closed point $x \in \overline{\{\mathfrak{p}\}}$), and localization preserves regularity (see [15]), regularity at all closed points already implies regularity everywhere. Thus the two conditions are equivalent.

Proposition 5.5.22: Nonsingular Varieties and the Cotangent Sheaf

Let X be an irreducible separated scheme of finite type over an algebraically closed field k, and let $n = \dim X$. Then $\Omega_{X/k}$ is locally free of rank n if and only if X is a nonsingular variety.

***Proof*:**
The question is local, so we may work at a closed point $x \in X$ with local ring $B = \mathcal{O}_{X,x}$ and residue field $\kappa(x) = k$ (since k is algebraically closed). The stalk $(\Omega_{X/k})_x = \Omega_{B/k}$, and by proposition 5.1.24, $\Omega_{X/k}$ is locally free of rank n at x if and only if $\Omega_{B/k}$ is a free B-module of rank n. By theorem 5.5.12, this holds if and only if B is a regular local ring of dimension n. Combining over all closed points and invoking box 5.5.21 for the non-closed points gives the result.

This characterization yields a short proof that the smooth locus is open and dense:

Corollary 5.5.23: Nonsingular Points are Open and Dense

Let X be a variety over an algebraically closed field k. Then the set of nonsingular points of X is a nonempty open subset.

***Proof*:**
Let $n = \dim X$ and let $K = K(X)$ be the function field. Since k is algebraically closed, the extension K/k is separably generated, so $\dim_K \Omega_{K/k} = \operatorname{trdeg}(K/k) = n$ by proposition 5.5.9. Now, $\Omega_{K/k}$ is the stalk of $\Omega_{X/k}$ at the generic point η of X. Since $\Omega_{X/k}$ is coherent and its generic stalk is a free K-module of rank n, proposition 5.1.24 implies that $\Omega_{X/k}$ is locally free of rank n on some nonempty open subset U containing η that is not what that proposition says actually! I thought I proved spreading out there but didn't! I actually never touched on spreading out in this book; must figure that out.. By proposition 5.5.22, the points of U are nonsingular.

The conormal exact sequence imposes strong constraints on nonsingular closed subvarieties. The following theorem refines proposition 5.5.17 when both ambient and subvariety are nonsingular.

Theorem 5.5.24: Nonsingular Closed Subschemes and Differentials

Let X be a nonsingular variety over an algebraically closed field k, and let $Y \subseteq X$ be an irreducible closed subscheme defined by the ideal sheaf $\mathscr{I}_Y$. Then Y is nonsingular if and only if the following two conditions hold:

1. $\Omega_{Y/k}$ is locally free.
2. The conormal sequence of proposition 5.5.17 is exact on the left:

$$0 \longrightarrow \mathscr{I}_Y/\mathscr{I}_Y^2 \longrightarrow \Omega_{X/k} \otimes \mathcal{O}_Y \longrightarrow \Omega_{Y/k} \longrightarrow 0.$$

When Y is nonsingular, $\mathscr{I}_Y$ is locally generated by $r = \operatorname{codim}_X(Y)$ elements, and the conormal sheaf $\mathscr{I}_Y/\mathscr{I}_Y^2$ is locally free of rank r on Y.

***Proof*:**
($\Leftarrow$) Assume (1) and (2) hold, and let $q = \operatorname{rank}\Omega_{Y/k}$. Since $\Omega_{X/k}$ is locally free of rank $n = \dim X$ (proposition 5.5.22), the short exact sequence in (2) forces $\mathscr{I}_Y/\mathscr{I}_Y^2$ to be locally free of rank $n-q$. By Nakayama's lemma, $\mathscr{I}_Y$ can be locally generated by $n-q$ elements, so $\dim Y \geq n-(n-q) = q$ (see [10]). On the other hand, at any closed point $y \in Y$, proposition 5.5.10 gives $q = \dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2) \geq \dim \mathcal{O}_{Y,y} = \dim Y$. Hence $q = \dim Y$, so $\dim_k(\mathfrak{m}_y/\mathfrak{m}_y^2) = \dim Y$ and Y is nonsingular. Moreover, $n-q = \operatorname{codim}_X(Y)$.

($\Rightarrow$) Assume Y is nonsingular of dimension $q = \dim Y$. Then (1) is immediate from proposition 5.5.22. It remains to show that the conormal map $\delta\colon \mathscr{I}_Y/\mathscr{I}_Y^2 \to \Omega_{X/k} \otimes \mathcal{O}_Y$ is injective. Let $r = n-q = \operatorname{codim}_X(Y)$.

At a closed point $y \in Y$, the cokernel of δ is $\Omega_{Y/k}$, which is locally free of rank q. Hence $\ker(\Omega_{X/k} \otimes \mathcal{O}_Y \to \Omega_{Y/k})$ is locally free of rank r near y. Choose local sections $f_1, \dots, f_r \in \mathscr{I}_Y$ in a neighbourhood of y such that $df_1, \dots, df_r$ generate this kernel, and let $\mathscr{I}'$ be the ideal sheaf generated by $f_1, \dots, f_r$ with associated closed subscheme Y'.

By construction, the images $\delta(\overline{f_1}), \dots, \delta(\overline{f_r})$ span a free subsheaf of rank r in $\Omega_{X/k} \otimes \mathcal{O}_{Y'}$ near y. Applying the conormal sequence (proposition 5.5.17) to Y':

$$\mathscr{I}'/(\mathscr{I}')^2 \xrightarrow{\delta'} \Omega_{X/k} \otimes \mathcal{O}_{Y'} \longrightarrow \Omega_{Y'/k} \longrightarrow 0,$$

the map δ' is injective (its image is free of rank r), so by the argument of the forward direction, Y' is nonsingular of dimension q near y. Since $Y \subseteq Y'$ and both are integral of the same dimension, we must have $Y = Y'$ and $\mathscr{I}_Y = \mathscr{I}'$ locally near y. Hence δ is injective near every closed point, giving the global left-exactness.

5.5.5 Tangent Sheaf, Canonical Sheaf, and the Adjunction Formula

Definition 5.5.25: Tangent Sheaf and Canonical Sheaf

Let X be a nonsingular variety of dimension n over a field k. The *tangent sheaf* is

$$\mathcal{T}_X \;:=\; \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X),$$

a locally free sheaf of rank n. The *canonical sheaf* (or *dualizing sheaf*) is the invertible sheaf

$$\omega_X \;:=\; \textstyle\bigwedge^n \Omega_{X/k}.$$

For projective space, the Euler sequence (theorem 5.5.18) gives $\omega_{\mathbb{P}^n_k} \cong \mathcal{O}_{\mathbb{P}^n_k}(-n-1)$. Since this sheaf has no nonzero global sections (the degree is negative), the geometric genus of projective space is zero, a first indication that $\mathbb{P}^n$ carries no "interesting" top-degree differential forms.

In [51], we established a connection between the topology of a space and its analytical structure: the dimension of the space of globally defined holomorphic 1-forms on a compact Riemann surface of genus g is exactly g. This result is historically significant since it provides one of the first explicit algebraic characterizations of a geometric and topological invariant. Rather than simply counting the "number of holes" in a surface, this equivalence demonstrates that the macroscopic shape of the manifold dictates and constrains the algebraic structure of the differential forms that can exist upon it. In modern algebraic geometry, this perspective becomes the starting point: the geometric genus of a curve is formally defined exactly by this property, as the dimension of the space of global regular 1-forms. The abstraction allows for defining the genus of varieties over arbitrary fields: by replacing complex-analytic holomorphic forms with the purely algebraic machinery of Kähler differentials, we can formally measure the "topology" of spaces:

Definition 5.5.26: Geometric Genus

Let X be a nonsingular projective variety of dimension n over a field k. The *geometric genus* of X is

$$p_g(X) \;:=\; \dim_k \Gamma(X, \omega_X) \;=\; \dim_k H^0\big(X, \textstyle\bigwedge^n \Omega_{X/k}\big).$$

5.5.27 Foreshadowing: Arithmetic and Geometric Genus In [20], we introduced the Hilbert function and the *arithmetic genus* p_a. Serre duality (section 6.5) will show that $p_g = p_a$ for nonsingular projective *curves*, but these two invariants can differ in higher dimensions. The arithmetic genus is defined in terms of the Hilbert polynomial (definition 6.7.6), while the geometric genus measures global sections of the canonical sheaf, a more intrinsic quantity.

The following fundamental result shows that the geometric genus is a birational invariant: it cannot change under birational maps. This makes it a useful tool for showing that varieties are *not* rational.

Theorem 5.5.28: Birational Invariance of Geometric Genus

Let X and X' be nonsingular projective varieties over a field k. If X and X' are birational, then $p_g(X) = p_g(X')$.

***Proof*:**
Let $F\colon X \dashrightarrow X'$ be a birational map, and let $V \subseteq X$ be the largest open subset on which F is represented by a morphism $f\colon V \to X'$.

Step 1: the complement $X \setminus V$ has codimension ≥ 2. Let $P \in X$ be a point of codimension 1. Since X is nonsingular, $\mathcal{O}_{X,P}$ is a DVR. The birational map gives a morphism from the generic point η of X to X', i.e., a map $\operatorname{Spec} K(X) \to X'$. Since X' is projective (hence proper over k), the valuative criterion of properness (theorem 3.5.25) provides a unique extension $\operatorname{Spec} \mathcal{O}_{X,P} \to X'$, which extends to a morphism on some open neighbourhood of P. Hence $P \in V$, and every codimension-1 point lies in V.

Step 2: pullback of canonical forms. The cotangent sequence (proposition 5.5.16) gives $f^*\Omega_{X'/k} \to \Omega_{V/k}$. Since X and X' are birational, both cotangent sheaves are locally free of the same rank $n = \dim X$, and f is an isomorphism on a dense open $U \subseteq V$. Taking n-th exterior powers gives $f^*\omega_{X'} \to \omega_V$, which restricts to an isomorphism on U. This induces a map on global sections:

$$f^*\colon \Gamma(X', \omega_{X'}) \longrightarrow \Gamma(V, \omega_V).$$

A nonzero global section of an invertible sheaf on an integral scheme cannot vanish on a dense open subset (U is dense in V), so f^* is injective.

Step 3: extending sections across codimension ≥ 2. We claim that the restriction map $\Gamma(X, \omega_X) \to \Gamma(V, \omega_V)$ is an isomorphism. It suffices to check this on affine opens: for any affine open $W = \operatorname{Spec} R \subseteq X$ with $\omega_X|_W \cong \mathcal{O}_W$, the map $\Gamma(W, \mathcal{O}_W) \to \Gamma(W \cap V, \mathcal{O}_{W\cap V})$ is bijective. Since X is nonsingular, R is a normal domain, and $W \setminus (W \cap V) = W \cap (X \setminus V)$ has codimension ≥ 2 in W by Step 1. The bijectivity now follows from lemma 5.4.13: a regular function on $W \cap V$ that is bounded at all codimension-1 points extends uniquely to W.

Conclusion. Combining Steps 2 and 3, we have an injection $\Gamma(X', \omega_{X'}) \hookrightarrow \Gamma(X, \omega_X)$, so $p_g(X') \leq p_g(X)$. By symmetry (applying the same argument to the inverse birational map F^{-1}), we obtain $p_g(X) \leq p_g(X')$. Hence $p_g(X) = p_g(X')$.

The geometric genus is therefore a birational invariant: any variety with $p_g > 0$ cannot be rational (i.e., birational to $\mathbb{P}^n$), since $p_g(\mathbb{P}^n) = 0$. For example, a smooth hypersurface of degree $d \geq n+2$ in $\mathbb{P}^n$ has $p_g > 0$ and is therefore not rational. The converse fails in general: the geometric genus alone does not determine the birational equivalence class.

We close this section with the normal and conormal sheaves and the adjunction formula, which computes the canonical sheaf of a subvariety from that of the ambient variety.

Definition 5.5.29: Normal and Conormal Sheaf

Let $Y \subseteq X$ be a nonsingular closed subvariety of a nonsingular variety X over k, with ideal sheaf $\mathscr{I}_Y$. The *conormal sheaf* of Y in X is $\mathscr{I}_Y/\mathscr{I}_Y^2$, a locally free $\mathcal{O}_Y$-module of rank $r = \operatorname{codim}_X(Y)$ (theorem 5.5.24). The *normal sheaf* is its dual:

$$\mathcal{N}_{Y/X} \;:=\; \mathcal{H}om_{\mathcal{O}_Y}(\mathscr{I}_Y/\mathscr{I}_Y^2,\, \mathcal{O}_Y),$$

also locally free of rank r.

Dualizing the exact sequence of theorem 5.5.24 gives the familiar normal bundle sequence from differential geometry (definition 0.5.7):

$$0 \longrightarrow \mathcal{T}_Y \longrightarrow \mathcal{T}_X \otimes \mathcal{O}_Y \longrightarrow \mathcal{N}_{Y/X} \longrightarrow 0.$$

This is the algebraic incarnation of the splitting of the tangent bundle of X along Y into tangential and normal directions.

Proposition 5.5.30: Adjunction Formula

Let $Y \subseteq X$ be a nonsingular closed subvariety of codimension r in a nonsingular variety X over k. Then

$$\omega_Y \;\cong\; \omega_X \otimes \textstyle\bigwedge^r \mathcal{N}_{Y/X} \;\otimes\; \mathcal{O}_Y.$$

In the important special case $r = 1$ (i.e., Y is a divisor), let $\mathcal{L} = \mathcal{O}_X(Y)$ be the invertible sheaf associated to Y (definition 5.4.52). Then

$$\omega_Y \;\cong\; (\omega_X \otimes \mathcal{L})\big|_Y.$$

***Proof*:**
The short exact sequence from theorem 5.5.24,

$$0 \longrightarrow \mathscr{I}_Y/\mathscr{I}_Y^2 \longrightarrow \Omega_{X/k} \otimes \mathcal{O}_Y \longrightarrow \Omega_{Y/k} \longrightarrow 0,$$

involves locally free sheaves of ranks r, $n = \dim X$, and $q = \dim Y$ respectively, with $n = q + r$. Taking top exterior powers of a short exact sequence of locally free sheaves gives

$$\textstyle\bigwedge^n(\Omega_{X/k} \otimes \mathcal{O}_Y) \;\cong\; \bigwedge^r(\mathscr{I}_Y/\mathscr{I}_Y^2) \;\otimes\; \bigwedge^q \Omega_{Y/k}.$$

The left-hand side is $\omega_X|_Y$, and the right-hand side is $\bigwedge^r(\mathscr{I}_Y/\mathscr{I}_Y^2) \otimes \omega_Y$. Tensoring both sides with $\bigwedge^r \mathcal{N}_{Y/X} = (\bigwedge^r(\mathscr{I}_Y/\mathscr{I}_Y^2))^\vee$ yields

$$\omega_Y \;\cong\; \omega_X|_Y \otimes \textstyle\bigwedge^r \mathcal{N}_{Y/X}.$$

When $r = 1$, the conormal sheaf is $\mathscr{I}_Y/\mathscr{I}_Y^2 \cong \mathcal{L}^{-1}|_Y$ by proposition 5.4.58 (since $\mathscr{I}_Y \cong \mathcal{O}_X(-Y) = \mathcal{L}^{-1}$). Its dual is $\mathcal{N}_{Y/X} \cong \mathcal{L}|_Y$, so

$$\omega_Y \;\cong\; \omega_X|_Y \otimes \mathcal{L}|_Y \;=\; (\omega_X \otimes \mathcal{L})\big|_Y.$$

Example 5.5.31: Canonical Sheaves via the Adjunction Formula

1. *Hypersurfaces in $\mathbb{P}^n$.* Let $Y = V_+(f) \subseteq \mathbb{P}^n_k$ be a nonsingular hypersurface of degree d. Then $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n}(d)$ and $\omega_{\mathbb{P}^n} = \mathcal{O}_{\mathbb{P}^n}(-n-1)$, so the adjunction formula gives
$$\omega_Y \;\cong\; \mathcal{O}_{\mathbb{P}^n}(-n-1+d)\big|_Y \;=\; \mathcal{O}_Y(d-n-1).$$
For a nonsingular plane curve of degree d in $\mathbb{P}^2$, $\omega_Y \cong \mathcal{O}_Y(d-3)$, and the geometric genus is $p_g = \dim_k H^0(Y, \mathcal{O}_Y(d-3)) = \binom{d-1}{2}$. This recovers the classical genus formula $g = (d-1)(d-2)/2$ for plane curves.

2. *Complete intersections.* More generally, if $Y = V_+(f_1) \cap \cdots \cap V_+(f_r)$ is a nonsingular complete intersection of hypersurfaces of degrees $d_1, \ldots, d_r$ in $\mathbb{P}^n_k$, iterating the adjunction formula gives
$$\omega_Y \;\cong\; \mathcal{O}_Y(d_1 + \cdots + d_r - n - 1).$$
In particular, Y is a *Fano variety* (i.e., $\omega_Y^\vee$ is ample) precisely when $\sum d_i < n+1$.

3. *Non-rational varieties in every dimension.* A nonsingular hypersurface Y of degree $d \geq n+2$ in $\mathbb{P}^n_k$ satisfies $\omega_Y \cong \mathcal{O}_Y(d-n-1)$ with $d-n-1 \geq 1$, so $p_g(Y) \geq 1$. By theorem 5.5.28, Y is not rational. Since such hypersurfaces exist in every dimension $n-1 \geq 1$, this produces non-rational varieties in all dimensions, a consequence of the interplay between differentials, the adjunction formula, and birational invariance.

5.5.6 *Invariant Differentials on Group Schemes

We saw in example 6.5.8 that the canonical sheaf (and more generally, the cotangent sheaf $\Omega_{X/k}$) carries geometric information about a variety X. However, when the scheme in question possesses a group structure, the situation becomes especially rigid. Algebraic group schemes are "everywhere locally the same," and this global symmetry forces their sheaves of differentials to be uniform across the entire space.

Let G be a group scheme over a field k (for instance, an affine algebraic group or an abelian variety). For any T-valued point $g \in G(T)$, the group operation defines a *left translation* morphism:
$$\ell_g \colon G_T \longrightarrow G_T, \qquad x \longmapsto g \cdot x.$$
Because G is a group, ℓ_g is an isomorphism of schemes (with inverse $\ell_{g^{-1}}$). This allows us to "transport" local geometric data from the identity element $e \in G(k)$ to any other point.

Definition 5.5.32: Left-Invariant Differentials

Let G be a group scheme over k, and let $\mu\colon G \times_k G \to G$ be the multiplication morphism. A global differential form $\omega \in \Gamma(G, \Omega_{G/k})$ is *left-invariant* if
$$\mu^*\omega \;=\; 1 \otimes \omega \quad \text{in} \quad \Gamma(G \times_k G,\, \Omega_{(G\times_k G)/k}).$$
(In the functor of points perspective, this means $\ell_g^*\omega = \omega$ for all points g.)

Let $\mathfrak{g} = T_eG$ be the Lie algebra of G, which is the Zariski tangent space at the identity. Its dual vector space, $\mathfrak{g}^\vee = (\mathfrak{m}_e/\mathfrak{m}_e^2)$, is the cotangent space at e. Any left-invariant differential form is completely determined by its value at the identity, giving a natural k-linear map from the space of left-invariant differentials to $\mathfrak{g}^\vee$. In fact, this map is an isomorphism: every cotangent vector at the identity extends uniquely to a global left-invariant differential form via translation.

This process of spreading out the tangent space at the identity to the entire group scheme yields a profound structural result.

Theorem 5.5.33: Group Schemes are Parallelizable

Let G be a group scheme of finite type over a field k. Then the cotangent sheaf $\Omega_{G/k}$ is globally free. Specifically, evaluation at the identity induces an isomorphism of $\mathcal{O}_G$-modules:

$$\Omega_{G/k} \;\cong\; \mathcal{O}_G \otimes_k \mathfrak{g}^\vee.$$

***Proof*:**
The formal proof relies on the multiplication map $\mu\colon G \times_k G \to G$ and the first projection $p_1\colon G \times_k G \to G$. Consider the map $\Phi\colon G \times_k G \to G \times_k G$ defined by $(g, h) \mapsto (g, g \cdot h)$. This is an isomorphism of schemes over G (relative to the first projection), with inverse $(g, x) \mapsto (g, g^{-1} \cdot x)$.

The isomorphism Φ transforms the second projection $p_2\colon G \times_k G \to G$ into the multiplication map μ. Therefore, it must induce an isomorphism on the pullbacks of the cotangent sheaf:

$$\Phi^* p_2^* \Omega_{G/k} \;\cong\; \mu^* \Omega_{G/k}.$$

Restricting this isomorphism to the slice $G \times \{e\}$ (i.e., pulling back along $\mathrm{id}_G \times e\colon G \to G \times_k G$) trivializes the differential components from the base, leaving an isomorphism between $\Omega_{G/k}$ and the pullback of the fiber at e, which is precisely $\mathcal{O}_G \otimes_k \mathfrak{g}^\vee$.

Corollary 5.5.34: Triviality of the Canonical Sheaf for Groups

If G is a nonsingular group scheme over k of dimension n, then its tangent sheaf $\mathcal{T}_G$ and its canonical sheaf ω_G are both trivial:

$$\mathcal{T}_G \;\cong\; \mathcal{O}_G^{\oplus n}, \qquad \text{and} \qquad \omega_G \;\cong\; \mathcal{O}_G.$$

This corollary provides a massive constraint on which varieties can be endowed with a group structure. For example, among all projective curves, only those of geometric genus $p_g = 1$ (elliptic curves) have a trivial canonical sheaf. Thus, it is geometrically impossible to put a group scheme structure on the projective line $\mathbb{P}^1_k$ (where $\omega \cong \mathcal{O}(-2)$) or on a nonsingular curve of genus $g \geq 2$. Similarly, an abelian variety of dimension g must have $\omega_A \cong \mathcal{O}_A$, which immediately distinguishes it from most other projective varieties of the same dimension and restricts its intersection theory.

5.6 *Formal Schemes and Local Geometry

As we have seen, one feature which clearly distinguishes the theory of schemes from the older classical theory of varieties is the possibility of having nilpotent elements in the structure sheaf. In particular, if $Y \subseteq X$ is a closed subvariety defined by a sheaf of ideals $\mathcal{I}$, then for any $n \geq 1$ we can consider the closed subscheme Y_n defined by the n-th power $\mathcal{I}^n$. For $n \geq 2$, this is a scheme with nilpotent elements. It carries information about Y together with the infinitesimal properties of the embedding of Y in X.

We wish to construct an object which carries information about all the infinitesimal neighborhoods Y_n of Y simultaneously. Such an object would be "thicker" than any individual Y_n, but contained inside any actual open neighborhood of Y in X. We might naturally call this the formal neighborhood of Y in X.

To see why a new geometric construction is necessary to capture this, recall that the algebraic construction of the formal power series ring $k[[x]]$ is given by the limit of the truncated polynomial rings $k[x]/(x^n)$. We shall now explore the geometric consequences of this algebraic operation.

Topologically, the scheme $\operatorname{Spec}(k[x]/(x^n))$ consists of exactly one point, as the ring is Artinian and possesses a unique prime ideal (x). Geometrically, one views this as a "fat point" – a point equipped with infinitesimal nilpotents that remember $n-1$ orders of derivatives. Since $k[[x]]$ is the algebraic limit of these rings as $n \to \infty$, one might intuitively expect the geometric limit of a sequence of one-point spaces to remain a one-point space. However, passing to the infinite limit crosses a structural threshold: the limit of infinite nilpotents coalesces into a genuine coordinate variable, making $k[[x]]$ a domain of Krull dimension 1. Consequently, the affine scheme $\operatorname{Spec}(k[[x]])$ possesses *two* points. One should visualize $\operatorname{Spec}(k[[x]])$ as the "germ" of a curve, an unimaginably small neighborhood that has lost all global properties but is just large enough to have a center and an infinitesimal edge.

Formal Completions

The fact that the Spec functor transforms a limit of one-point spaces into a two-point space demonstrates that Spec does not commute nicely with infinite limits. To rectify this topological mismatch, Grothendieck introduced formal schemes. We shall follow Hartshorne and define this globally as a completion along a closed subscheme.

Definition 5.6.1: Formal Completion

Let X be a Noetherian scheme, and let Y be a closed subscheme defined by a sheaf of ideals $\mathcal{I}$. Then we define the *formal completion* of X along Y, denoted $(\hat{X}, \mathcal{O}_{\hat{X}})$, to be the following ringed space: we take the topological space Y, and on it the sheaf of rings $\mathcal{O}_{\hat{X}} = \varprojlim \mathcal{O}_X/\mathcal{I}^n$. Here we consider each $\mathcal{O}_X/\mathcal{I}^n$ as a sheaf of rings on Y, and make them into an inverse system in the natural way.

The reader should notice that the structure sheaf $\mathcal{O}_{\hat{X}}$ actually depends only on the closed subset Y, and not on the particular scheme structure on Y. For if $\mathcal{J}$ is another sheaf of ideals defining a closed subscheme structure on Y, then since X is a Noetherian scheme, there are integers m, n such that $\mathcal{I} \supseteq \mathcal{J}^m$ and $\mathcal{J} \supseteq \mathcal{I}^n$. Thus the inverse systems $(\mathcal{O}_X/\mathcal{I}^n)$ and $(\mathcal{O}_X/\mathcal{J}^m)$ are cofinal with each other, and hence have the same inverse limit.

Furthermore, one sees easily that the stalks of the sheaf $\mathcal{O}_{\hat{X}}$ are local rings, so in fact $(\hat{X}, \mathcal{O}_{\hat{X}})$ is a locally ringed space. If $U = \operatorname{Spec} A$ is an open affine subset of X, and if $I \subseteq A$ is the ideal $\Gamma(U, \mathcal{I})$, then evaluating the sections reveals that $\Gamma(\hat{X} \cap U, \mathcal{O}_{\hat{X}}) = \hat{A}$, the I-adic completion of A. Thus the process of completing X along Y is strictly analogous to the I-adic completion of a ring. However, one should note that the local rings of $\hat{X}$ are in general *not* complete, and their dimension (which is equal to $\dim X$) is *not* equal to the dimension of the underlying topological space Y.

We shall call a formal scheme obtained by completing a single Noetherian scheme X along a closed subscheme Y *algebraizable*.[8]

Example 5.6.2: Formal Completions

1. Take X to be a Noetherian scheme, and let $Y = X$. Then clearly $\hat{X} = X$. Thus the category of Noetherian formal schemes naturally includes all Noetherian schemes.
2. Take X to be a Noetherian scheme, and let $Y = \{P\}$ be a closed point. Then $\hat{X}$ is a one point space $\{P\}$ with the completion $\hat{\mathcal{O}}_P$ of the local ring at P as its structure sheaf. An $\hat{\mathcal{O}}_P$-module M, considered as a sheaf on $\hat{X}$, is coherent if and only if M is a finitely generated module. This directly resolves our earlier algebraic paradox: $\hat{X}$ is topologically a single point, capturing the geometric intuition of a true limit of fat points without accidentally spawning a generic point.

5.6.3 Comparing Local Geometries To summarize the distinctions between these deeply related objects for a point P on a curve:

1. $\operatorname{Spec}(k[x]/(x^n))$ is a single point representing a "fat point" of finite thickness.
2. $\hat{X}$ (the formal completion at P) is a single point representing the true geometric limit of fat points (infinite formal thickness).
3. $\operatorname{Spec}(k[[x]])$ is a two-point space representing a "formal disk" or germ (a center point plus a surrounding generic neighborhood).

Universal Smoothness and Singularities

The utility of formal schemes shines when studying the local properties of varieties. In classical algebraic geometry, curves can exhibit wildly different global shapes, such as a line $y = x$ or a parabola $y = x^2$. However, if one selects a smooth point on any such curve over a field k and takes the formal completion of the local ring at that point, the resulting ring is always isomorphic to $k[[t]]$, where t is a local parameter.

Because their local algebra is identical, their formal geometry is identical. Thus, $\operatorname{Spec}(k[[x]])$ does not represent *a* curve; it represents the fundamental atomic structure of *any* smooth curve at a smooth point. Under an infinite-power formal microscope, every smooth curve is indistinguishable from a straight line.

This universal behavior fails precisely when the geometry ceases to be smooth.

Example 5.6.4: Singularities and Formal Completions
If a curve possesses a singularity, its formal completion at that point acts as a perfect algebraic memory of the singularity, remaining distinct from $k[[x]]$.

1. *The Cusp:* Consider the curve cut out by $y^2 = x^3$, which has a sharp cusp at the origin. The

[8] It is apparently quite difficult to find non-algebraizable formal schemes, maybe add an exercise based on Hartshorne p.194 later?

formal completion of the local ring at the origin is

$$\frac{k[[x,y]]}{(y^2 - x^3)}$$

This ring is not isomorphic to $k[[t]]$. Even infinitesimally, the formal scheme remembers the sharp turn.

2. *The Node:* Consider a curve with a self-intersection at the origin, locally modeled by $xy = 0$. Its formal completion at the crossing is

$$\frac{k[[x,y]]}{(xy)}$$

This represents two distinct microscopic branches crossing each other. Topologically, the formal completion is still a single point, but the underlying sheaf of topological rings completely captures the crossing branches.

6

Cohomology of Schemes

Let X be a scheme with a sheaf $\mathscr{F}$. A natural question is: can we quantify the obstruction to gluing local sections into global sections? On an affine scheme, there is no obstruction: the global section functor is exact for quasi-coherent sheaves, and all local data glues. But on a projective variety, or on a scheme built by gluing affine pieces in a non-trivial way, the functor $\Gamma(X, -)$ fails to be exact, and its failure is measured by the *sheaf cohomology groups* $H^i(X, \mathscr{F}) = R^i\Gamma(X, \mathscr{F})$, the right derived functors of the global section functor. These groups are the central objects of study in this chapter.

The sheaves to which this machinery applies include the structure sheaf $\mathcal{O}_X$, the twisted sheaves $\mathcal{O}_X(n)$, the sheaves of differentials $\Omega_{X/S}$ and their exterior powers, and the pushforwards of sheaves along closed immersions. Each choice of sheaf extracts different geometric information: the cohomology of $\mathcal{O}_X$ governs the arithmetic genus, the cohomology of line bundles governs linear systems and the Riemann–Roch theorem, and the cohomology of the canonical sheaf ω_X encodes duality phenomena that constrain the topology of the variety.

Though the derived-functor definition is theoretically clean, it is computationally demanding: it requires an injective resolution, and injective sheaves are typically enormous and non-explicit. For practical computations, we introduce *Čech cohomology*, which replaces injective resolutions with the combinatorial data of an open cover. For quasi-coherent sheaves on separated Noetherian schemes with an affine cover, Čech cohomology and derived-functor cohomology coincide, and this is our main computational tool. Using it, we compute the cohomology of projective space $\mathbb{P}^n_R$ with coefficients in the twisted sheaves $\mathcal{O}(d)$ — the single most important explicit computation in the subject.

With these foundations, the chapter develops in four main directions.

Vanishing and finiteness. Two structural results govern the cohomology of schemes: the cohomology of quasi-coherent sheaves vanishes on affine schemes (section 6.2), and the cohomology of coherent sheaves on projective schemes is finite-dimensional (section 6.6). The first confirms the philosophy that cohomology measures gluing; the second ensures that the invariants we extract are "finitely complicated." Together, they justify defining the Euler characteristic $\chi(X, \mathscr{F})$, the Hilbert polynomial, and the arithmetic genus (section 6.7).

Duality. Serre duality (section 6.5) is the algebraic analogue of Poincaré duality: for a smooth projective variety X of dimension n over a field k, the cohomology groups satisfy $H^i(X, \mathscr{F}) \cong H^{n-i}(X, \mathscr{F}^\vee \otimes \omega_X)^*$ for locally free $\mathscr{F}$. The proof passes through the Ext groups and the local-to-global spectral sequence (section 6.4), which provide the natural framework for the duality pairing. As a consequence, we obtain the Riemann–Roch theorem for curves and the classical genus formulas.

Families and variation. The higher direct image sheaves $R^i f_* \mathscr{F}$ (section 6.6) relativize cohomology over a base scheme, packaging the cohomology of fibers into a single sheaf. For flat families over a Noetherian base, the semicontinuity theorem (section 6.8) shows that the fiber cohomology $h^i(X_y, \mathscr{F}_y)$ is upper semicontinuous in y, and Grauert's theorem guarantees that constancy of h^i implies local freeness of $R^i f_* \mathscr{F}$ and compatibility with base change. These results are the technical heart of moduli theory.

GAGA. For projective varieties over $\mathbb{C}$, Serre's GAGA theorems (section 6.9) identify the algebraic cohomology developed in this chapter with the analytic cohomology of the complex geometry: the analytification functor induces an equivalence $\mathbf{Coh}(X) \simeq \mathbf{Coh}(X^{\mathrm{an}})$ and isomorphisms $H^i(X, \mathscr{F}) \cong H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$. Through GAGA, the Hodge decomposition, the algebraic de Rham theorem, and Chow's theorem become consequences of the algebraic theory, closing the loop between the algebraic and analytic perspectives on cohomology.

Prerequisites. The reader should be comfortable with the theory of sheaves and schemes (chapters 1–3), quasi-coherent and coherent sheaves (chapter 5), and the basic properties of projective schemes and differentials (sections 2.3 and 5.5). For the GAGA section, familiarity with the complex geometry review (sections 0.6 through 6.9 of chapter 1) is assumed. The reader may also want to review [15] for the derived-functor approach to (co)homology theory.

6.1 Definition and Basic Properties

Recall that a category $\mathbf{C}$ is *pre-additive* (or enriched over $\mathbf{Ab}$) if for every $A, B \in \mathrm{Obj}(\mathbf{C})$, the set $\mathrm{Hom}_{\mathbf{C}}(A, B)$ carries the structure of an abelian group, and composition of morphisms is bilinear:

$$f \circ (g + h) = (f \circ g) + (f \circ h) \qquad \text{and} \qquad (f + g) \circ h = (f \circ h) + (g \circ h).$$

An *additive category* is a pre-additive category that admits all finite products and coproducts, and in which these coincide (this common construction is sometimes called the *biproduct*).

An abelian category is an additive category with the "right" conditions to do cohomology[1]. Concretely (cf. definition 0.4.29), a category $\mathbf{A}$ is *abelian* if:

1. for every $A, B \in \mathrm{Obj}(\mathbf{A})$, the set $\mathrm{Hom}_{\mathbf{A}}(A, B)$ is an abelian group and composition is bilinear;
2. finite products and coproducts exist and coincide;
3. every morphism $f \in \mathrm{Hom}_{\mathbf{A}}(A, B)$ has a kernel and a cokernel;
4. every monomorphism is the kernel of its cokernel, and every epimorphism is the cokernel of its kernel;

[1] Technically, one can do cohomology on exact categories that are not abelian, such as the category of vector bundles, but it is more natural to see this as a special generalization of the abelian setting.

5. defining $\operatorname{im}(f) = \ker(\operatorname{coker}(f))$, the first isomorphism theorem gives

$$\operatorname{im}(f) = \ker(\operatorname{coker}(f)) = \operatorname{coker}(\ker(f)),$$

so every morphism factors as an epimorphism followed by a monomorphism.

The prototypical abelian category is R-**Mod**; by the Freyd–Mitchell embedding theorem, any small abelian category $\mathbf{A}$ fully-faithfully embeds into R-**Mod** via an exact functor for an appropriate (not necessarily commutative) ring R. This is useful theoretically, but for our purposes it is more practical to think of the abelian category of sheaves $\mathbf{Sh}_{\mathbf{Ab}}(X)$ as its own creature. The "sheafification" of R-**Mod** gives $\mathcal{O}_X$-**Mod**. The important full subcategories $\mathbf{QCoh}(X)$ and $\mathbf{Coh}(X)$ are both abelian (by theorem 1.2.28), and $\mathbf{QCoh}(X)$ has enough injectives (as we shall see shortly). Though not critical at the moment, it should be noted that $\mathbf{QCoh}(X)$ is not closed under the internal Hom sheaf *Hom*, while $\mathbf{Coh}(X)$ is.

A functor between abelian categories $F : \mathbf{A} \to \mathbf{B}$ is *additive* if for every $X, Y \in \operatorname{Obj}(\mathbf{A})$, the induced map

$$\operatorname{Hom}_{\mathbf{A}}(X, Y) \longrightarrow \operatorname{Hom}_{\mathbf{B}}(FX, FY)$$

is a group homomorphism. An additive functor F is *left exact* if it preserves kernels, equivalently if a short exact sequence $0 \to X \to Y \to Z \to 0$ yields an exact sequence

$$0 \longrightarrow FX \longrightarrow FY \longrightarrow FZ,$$

and *right exact* if it preserves cokernels, equivalently if the same short exact sequence yields

$$FX \longrightarrow FY \longrightarrow FZ \longrightarrow 0.$$

If F is both left and right exact, it is called *exact*. The most common (and arguably the most important) left-exact functors are the covariant $\operatorname{Hom}(X, -)$ and the contravariant $\operatorname{Hom}(-, Y)$. The most important right-exact functor is $- \otimes Y$[2].

Given a left-exact functor $F : \mathbf{A} \to \mathbf{B}$, we can form its *right derived functors* $R^i F : \mathbf{A} \to \mathbf{B}$ for $i \geq 0$. Abstractly, one passes from $\mathbf{A}$ to its derived category $D(\mathbf{A})$ — constructed by taking the category of cochain complexes $\operatorname{Kom}(\mathbf{A})$, passing to the homotopy category $K(\mathbf{A})$, and then localizing at quasi-isomorphisms — and the total derived functor $\mathbf{R}F : D(\mathbf{A}) \to D(\mathbf{B})$ captures the right information to "fix" exactness. The derived category does not lend itself well to direct computation, but if $\mathbf{A}$ has enough injectives, then injective resolutions are universal up to homotopy, and the right derived functors $R^i F$ can be computed as follows: given an object $A \in \mathbf{A}$, choose an injective resolution $0 \to A \to I^0 \to I^1 \to \cdots$, apply F, and take cohomology:

$$R^i F(A) = h^i(F(I^\bullet)).$$

This is independent of the choice of injective resolution (up to canonical isomorphism). The details of all of these constructions can be found in [15]. A "classical" approach using δ-functors is briefly introduced at the end of that same chapter.

Recall that if R is a ring, then every R-module embeds into an injective module (see [15]). From this, we shall be able to conclude that the module categories over ringed spaces have enough injectives:

[2]The functor $X \otimes -$ is also right exact and is left adjoint to $\operatorname{Hom}(X, -)$; this adjunction is the tensor–Hom adjunction.

Proposition 6.1.1: Ringed Space Has Enough Injectives

Let $(X, \mathcal{O}_X)$ be a ringed space. Then the category $\mathcal{O}_X$-**Mod** of sheaves of $\mathcal{O}_X$-modules has enough injectives.

***Proof*:**
Let $\mathscr{F}$ be a sheaf of $\mathcal{O}_X$-modules. For each $x \in X$, the stalk $\mathscr{F}_x$ is an $\mathcal{O}_{X,x}$-module, so there exists an injective $\mathcal{O}_{X,x}$-module I_x together with an embedding $\mathscr{F}_x \hookrightarrow I_x$. For each point $x \in X$, let $j_x : \{x\} \hookrightarrow X$ denote the inclusion of the one-point space $\{x\}$, and equip $\{x\}$ with the $\mathcal{O}_{X,x}$-module I_x (viewed as a sheaf on a point). Define:

$$\mathscr{I} = \prod_{x \in X} (j_x)_*(I_x).$$

We claim that $\mathscr{I}$ is an injective $\mathcal{O}_X$-module and that $\mathscr{F}$ embeds into it.

The embedding. For each $x \in X$, the stalk of $(j_x)_*(I_x)$ at x is I_x, and the embedding $\mathscr{F}_x \hookrightarrow I_x$ extends (by the adjunction below) to a sheaf map $\mathscr{F} \to (j_x)_*(I_x)$. Taking the product over all x yields a morphism $\mathscr{F} \to \mathscr{I}$ that is injective on every stalk, hence injective as a morphism of sheaves (by theorem 1.2.27).

Injectivity of $\mathscr{I}$. For any $\mathcal{O}_X$-module $\mathscr{G}$, we have:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{G}, \mathscr{I}) \cong \prod_{x \in X} \mathrm{Hom}_{\mathcal{O}_X}\big(\mathscr{G}, (j_x)_*(I_x)\big) \cong \prod_{x \in X} \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathscr{G}_x, I_x).$$

The first isomorphism is the universal property of products. The second uses the fact that for the inclusion $j_x : \{x\} \hookrightarrow X$, the stalk functor $\mathscr{G} \mapsto \mathscr{G}_x$ is left adjoint to $(j_x)_*$[a].

Now, the functor $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is the product of functors $\mathrm{Hom}_{\mathcal{O}_{X,x}}((-)_x, I_x)$. Each factor is the composition of the stalk functor $\mathscr{G} \mapsto \mathscr{G}_x$, which is exact (by **??**), with the functor $\mathrm{Hom}_{\mathcal{O}_{X,x}}(-, I_x)$, which is exact because I_x is injective. A product of exact functors is exact, so $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is exact, making $\mathscr{I}$ injective.

[a] More precisely, a morphism $\mathscr{G} \to (j_x)_*(I_x)$ is determined by its value on stalks at x, since $(j_x)_*(I_x)$ is supported at $\overline{\{x\}}$ and its stalk at x is I_x. The formal adjunction is $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{G}, (j_x)_* M) \cong \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathscr{G}_x, M)$ for any $\mathcal{O}_{X,x}$-module M.

Corollary 6.1.2: Sheaves of Abelian Groups Has Enough Injectives

Let X be a topological space. Then the category of sheaves of abelian groups on X has enough injectives.

***Proof*:**
View X as the ringed space $(X, \underline{\mathbb{Z}})$, where $\underline{\mathbb{Z}}$ is the constant sheaf. Sheaves of abelian groups on X are precisely $\underline{\mathbb{Z}}$-modules, so the result follows from proposition 6.1.1.

Definition 6.1.3: Global Section Functor and Sheaf Cohomology

Let $(X, \mathcal{O}_X)$ be a ringed space. The *global section functor*

$$\Gamma(X, -) : \mathcal{O}_X\text{-}\mathbf{Mod} \longrightarrow \mathbf{Ab}$$

is left exact (by proposition 1.2.16) but not right exact in general (see Example 1.2.7). Since $\mathcal{O}_X$-**Mod** has enough injectives, we may form the right derived functors. The i-th right derived functor

$$H^i(X, -) = R^i\Gamma(X, -)$$

is called the i-th *sheaf cohomology functor*. For a sheaf $\mathscr{F}$, the abelian group $H^i(X, \mathscr{F})$ is the i-th *cohomology group* of $\mathscr{F}$. By general properties of derived functors, $H^0(X, \mathscr{F}) = \Gamma(X, \mathscr{F})$ and a short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of sheaves induces a long exact sequence:

$$0 \to H^0(X, \mathscr{F}') \to H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}'') \xrightarrow{\delta} H^1(X, \mathscr{F}') \to H^1(X, \mathscr{F}) \to \cdots$$

To compute these cohomology groups, we need resolutions. Recall that it suffices to resolve by *acyclic* objects — those $\mathscr{G}$ with $H^i(X, \mathscr{G}) = 0$ for $i > 0$ — rather than by injective objects (see [15]). We shall show that the following class of sheaves is acyclic:

Definition 6.1.4: Flasque Sheaf

Let X be a topological space. A sheaf $\mathscr{F}$ on X is called *flasque* (or *flabby*) if for every inclusion of open sets $U \subseteq V$, the restriction map $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ is surjective.

Proposition 6.1.5: Injective Sheaf Modules Are Flasque

Let $(X, \mathcal{O}_X)$ be a ringed space. Then any injective $\mathcal{O}_X$-module is flasque.

***Proof*:**
Let $\mathscr{I}$ be an injective $\mathcal{O}_X$-module, and let $U \subseteq V$ be open subsets of X. For any open set $W \subseteq X$, let $j_W : W \hookrightarrow X$ denote the inclusion and define $\mathcal{O}_W = (j_W)_!(\mathcal{O}_X|_W)$, the extension by zero of $\mathcal{O}_X|_W$ to all of X[a]. The inclusion $U \subseteq V$ induces a natural monomorphism of $\mathcal{O}_X$-modules $0 \to \mathcal{O}_U \to \mathcal{O}_V$. Since $\mathscr{I}$ is injective, the functor $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is exact, giving a surjection:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_V, \mathscr{I}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_U, \mathscr{I}) \longrightarrow 0.$$

But $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_W, \mathscr{I}) \cong \mathscr{I}(W)$ for any open W, since a morphism $\mathcal{O}_W \to \mathscr{I}$ is determined by the image of the identity section $1 \in \mathcal{O}_X(W)$. Hence the restriction $\mathscr{I}(V) \to \mathscr{I}(U)$ is surjective, so $\mathscr{I}$ is flasque.

[a] Recall from definition 1.3.11 that $(j_W)_!$ extends a sheaf by zero outside W. In particular, $(j_W)_!(\mathcal{O}_X|_W)(V') = \{s \in \mathcal{O}_X(V' \cap W) \,|\, \mathrm{supp}(s) \text{ is closed in } V'\}$ for open $V' \subseteq X$.

Before proving that flasque sheaves are acyclic, we need one key property: the global section functor is *exact* on short exact sequences of flasque sheaves. Let us record this precisely.

Lemma 6.1.6: Exactness of Global Sections on Flasque Sequences

Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be a short exact sequence of sheaves on a topological space X. If $\mathscr{F}'$ is flasque, then the sequence

$$0 \longrightarrow \Gamma(X, \mathscr{F}') \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{F}'') \longrightarrow 0$$

is exact. Furthermore, if both $\mathscr{F}'$ and $\mathscr{F}$ are flasque, then $\mathscr{F}''$ is also flasque.

***Proof*:**
Left exactness of $\Gamma(X, -)$ gives exactness at $\Gamma(X, \mathscr{F}')$ and $\Gamma(X, \mathscr{F})$. We must show surjectivity on the right. Let $s'' \in \Gamma(X, \mathscr{F}'')$. Consider the collection of pairs (U, s) where $U \subseteq X$ is open and $s \in \mathscr{F}(U)$ maps to $s''|_U$ under the surjection $\mathscr{F} \to \mathscr{F}''$. This collection is non-empty (by surjectivity on stalks, we can find such pairs locally) and is partially ordered by extension. By Zorn's lemma, there is a maximal such pair (U_0, s_0).

We claim $U_0 = X$. Suppose not; pick $x \in X \setminus U_0$. By surjectivity on stalks, there is an open $V \ni x$ and $t \in \mathscr{F}(V)$ mapping to $s''|_V$. On $U_0 \cap V$, the difference $s_0|_{U_0 \cap V} - t|_{U_0 \cap V}$ maps to zero in $\mathscr{F}''(U_0 \cap V)$, so it lies in $\mathscr{F}'(U_0 \cap V)$. Since $\mathscr{F}'$ is flasque, this section extends to some $\sigma \in \mathscr{F}'(V)$. Replacing t by $t + \sigma$, we may assume s_0 and t agree on $U_0 \cap V$, and they glue to a section on $U_0 \cup V$, contradicting maximality. Hence $U_0 = X$.

For the second claim, suppose $\mathscr{F}'$ and $\mathscr{F}$ are both flasque. Let $U \subseteq V$ be open sets and $s'' \in \mathscr{F}''(V)$. By the surjectivity just proved (applied to V), there exists $s \in \mathscr{F}(V)$ mapping to s''. Since $\mathscr{F}$ is flasque, the restriction $\mathscr{F}(V) \to \mathscr{F}(U)$ is surjective, so the image of $s|_U$ in $\mathscr{F}''(U)$ equals $s''|_U$. But we already have the section s defined on all of V, and any element of $\mathscr{F}''(U)$ can be lifted to $\mathscr{F}(V)$ and then restricted. More precisely, given any $t'' \in \mathscr{F}''(U)$, apply the surjectivity result to U to get a lift $t \in \mathscr{F}(U)$, then use the flasqueness of $\mathscr{F}$ to extend t to all of V, giving a preimage in $\mathscr{F}''(V)$. Hence $\mathscr{F}''$ is flasque.

Corollary 6.1.7: Trivial Cohomology For Flasque Sheaves

Let $\mathscr{F}$ be a flasque sheaf on a topological space X. Then

$$H^i(X, \mathscr{F}) = 0 \qquad \text{for all } i > 0.$$

***Proof*:**
Since $\mathcal{O}_X$-**Mod** has enough injectives, we can embed $\mathscr{F}$ into an injective sheaf $\mathscr{I}$ and form the short exact sequence:

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{I} \longrightarrow \mathscr{G} \longrightarrow 0$$

where $\mathscr{G} = \mathscr{I}/\mathscr{F}$. By proposition 6.1.5, the sheaf $\mathscr{I}$ is flasque. Since both $\mathscr{F}$ and $\mathscr{I}$ are flasque, the quotient $\mathscr{G}$ is also flasque by the lemma above.

Since $\mathscr{F}$ is flasque, the global section functor preserves the exactness of this sequence (again by the lemma above):

$$0 \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{I}) \longrightarrow \Gamma(X, \mathscr{G}) \longrightarrow 0.$$

Comparing with the long exact sequence in cohomology and using $H^0 = \Gamma$, we see that the connecting homomorphism $\delta : H^0(X, \mathscr{G}) \to H^1(X, \mathscr{F})$ is zero (since the map $\Gamma(X, \mathscr{I}) \to \Gamma(X, \mathscr{G})$

is already surjective). Hence $H^1(X, \mathscr{F}) = 0$.

For $i \geq 2$, the long exact sequence gives isomorphisms $H^i(X, \mathscr{F}) \cong H^{i-1}(X, \mathscr{G})$ because $\mathscr{I}$ is injective and therefore $H^j(X, \mathscr{I}) = 0$ for all $j > 0$. Since $\mathscr{G}$ is flasque, we may repeat the argument with $\mathscr{G}$ in place of $\mathscr{F}$, obtaining $H^{i-1}(X, \mathscr{G}) = 0$ by induction on i. Hence $H^i(X, \mathscr{F}) = 0$ for all $i > 0$.

Flasque sheaves are therefore acyclic for $\Gamma(X, -)$, and we may compute cohomology using flasque resolutions rather than injective ones. Let us see some examples:

Example 6.1.8: Flasque Sheaves

1. It was just shown that all injective $\mathcal{O}_X$-modules are flasque. The converse fails: flasque sheaves need not be injective.

2. *Constant sheaves on irreducible spaces.* Let X be an irreducible topological space and A an abelian group. Then the constant sheaf $\underline{A}$ is flasque. Indeed, for any non-empty open $U \subseteq V$, both U and V are connected (every open subset of an irreducible space is connected), so $\underline{A}(V) = A = \underline{A}(U)$ and the restriction map is the identity. This need not hold on reducible spaces: on $X = \{0, 1\}$ with the discrete topology, the constant sheaf $\underline{\mathbb{Z}}$ satisfies $\underline{\mathbb{Z}}(X) = \mathbb{Z} \times \mathbb{Z}$ and $\underline{\mathbb{Z}}(\{0\}) = \mathbb{Z}$, but the restriction is projection onto the first factor, which is surjective. However, if X is, say, the union of two disjoint open intervals in $\mathbb{R}$, then a section $(1, 0) \in \underline{\mathbb{Z}}(X)$ cannot be extended to a section over a connected open set containing both components while remaining locally constant. The key point is that irreducibility forces all opens to overlap, making extension trivial.

3. *The Godement resolution.* Let $\mathscr{F}$ be any sheaf of abelian groups on X. We can always construct a canonical flasque resolution called the *Godement resolution*. For each $x \in X$, let $\iota_x : \{x\} \hookrightarrow X$ denote the inclusion. Define:

$$C^0(\mathscr{F}) = \prod_{x \in X} (\iota_x)_*(\mathscr{F}_x),$$

so that $\Gamma(U, C^0(\mathscr{F})) = \prod_{x \in U} \mathscr{F}_x$ for every open $U \subseteq X$. This sheaf is flasque because a section over U is simply a choice of element in each stalk $\mathscr{F}_x$ for $x \in U$, with no compatibility condition — so restriction just forgets the stalks outside U, which is always surjective.

Note, however, that $C^0(\mathscr{F})$ need *not* be injective. For example, if $\mathscr{F} = \underline{\mathbb{Z}}$, then each stalk of $C^0(\mathscr{F})$ is $\mathbb{Z}$, which is not a divisible group, so $C^0(\mathscr{F})$ is not injective in the category of sheaves of abelian groups (where injective stalks must be divisible).

The natural map $\epsilon : \mathscr{F} \to C^0(\mathscr{F})$, sending a section $s \in \mathscr{F}(U)$ to the tuple of its germs $(s_x)_{x \in U}$, is injective by the locality axiom for sheaves. Let $\mathscr{Q}^0 = C^0(\mathscr{F})/\mathrm{im}(\epsilon)$. Applying the same construction to $\mathscr{Q}^0$ yields $C^1(\mathscr{F}) = C^0(\mathscr{Q}^0)$, and iterating:

$$0 \longrightarrow \mathscr{F} \xrightarrow{\epsilon} C^0(\mathscr{F}) \longrightarrow C^1(\mathscr{F}) \longrightarrow C^2(\mathscr{F}) \longrightarrow \cdots$$

This is a flasque resolution of $\mathscr{F}$, and one can compute $H^i(X, \mathscr{F})$ as the cohomology of the complex $\Gamma(X, C^\bullet(\mathscr{F}))$. The Godement resolution is completely canonical (no choices involved), but the resulting sheaves are typically enormous and *not* quasi-coherent.

4. *The divisor sheaf.* Let X be an irreducible normal Noetherian scheme with function field K. The sheaf of Weil divisors is:

$$\mathscr{D}iv \cong \bigoplus_{x \in X^{(1)}} (\iota_x)_*(\underline{\mathbb{Z}})$$

where $X^{(1)}$ denotes the set of codimension-1 points of X. The exact sequence of sheaves:

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathcal{K}^* \longrightarrow \mathscr{D}iv \longrightarrow 0$$

relates the invertible regular functions, the non-zero rational functions (here $\mathcal{K}^*$ is the constant sheaf associated to K^*, hence flasque by item (2) above), and the divisor sheaf. Since $\mathscr{D}iv$ is a direct sum of pushforwards of flasque sheaves, it is flasque. The long exact sequence in cohomology, together with the acyclicity of $\mathcal{K}^*$ and $\mathscr{D}iv$, gives:

$$\Gamma(X, \mathcal{K}^*) \longrightarrow \Gamma(X, \mathscr{D}iv) \longrightarrow H^1(X, \mathcal{O}_X^*) \longrightarrow 0$$

and hence an isomorphism:

$$H^1(X, \mathcal{O}_X^*) \cong \frac{\mathrm{WDiv}\,(X)}{\mathrm{PDiv}(X)} = \mathrm{Cl}\, X,$$

recovering the divisor class group from section 5.4 as cohomological data. This illustrates a recurring theme: the H^1 of a sheaf of units is a "class group" measuring the failure of local-to-global extension for invertible objects.

Acyclic does not imply flasque. Though not central to our development, it is worth recording the main class of acyclic-but-not-flasque sheaves, as it explains why sheaf cohomology in the smooth category is far simpler than in the algebraic category:

Example 6.1.9: Acyclic, But Not Injective Or Flasque

Let M be a smooth manifold and consider the sheaf C_M^∞ of smooth real-valued functions. This sheaf is *not* flasque: for instance, the function $1/x$ is a smooth section of $C_{\mathbb{R}}^\infty$ over the open set $(0, 1)$, but it cannot be extended to any smooth function on $(-1, 1)$. Nor is C_M^∞ injective in general.

However, C_M^∞ is acyclic. The key reason is the existence of *partitions of unity* subordinate to any open cover (see theorem 0.5.47): given a surjection of sheaves $\mathscr{F} \to \mathscr{G} \to 0$ and a global section $s \in \Gamma(X, \mathscr{G})$, we can lift s locally and then glue the local lifts using a partition of unity. More precisely, any sheaf of C_M^∞-modules admits partitions of unity, and such sheaves (called *fine sheaves*) are acyclic for Γ.

This is exactly the mechanism that makes de Rham cohomology computable on manifolds. The sheaves Ω_M^p of smooth differential p-forms are fine (being C_M^∞-modules), hence acyclic, and the de Rham complex $0 \to \underline{\mathbb{R}} \to \Omega^0 \to \Omega^1 \to \cdots$ is an acyclic resolution of the constant sheaf $\underline{\mathbb{R}}$. As recalled in theorem 0.5.23, this gives the isomorphism $H^i_{\mathrm{dR}}(M) \cong H^i(M, \underline{\mathbb{R}})$.

In algebraic geometry, no such luxury exists: the structure sheaf $\mathcal{O}_X$ of a scheme does not admit partitions of unity (the Zariski topology is too coarse, and algebraic functions are too rigid). This is why we must work harder — using flasque resolutions, Čech cohomology, or injective resolutions — to compute sheaf cohomology in the algebraic setting.

Let us link this to the module structure on cohomology. If $(X, \mathcal{O}_X)$ is a ringed space, then $\Gamma(X, \mathcal{O}_X)$ is a ring, and for any $\mathcal{O}_X$-module $\mathscr{F}$, the global sections $\Gamma(X, \mathscr{F})$ carry a natural $\Gamma(X, \mathcal{O}_X)$-module structure. This structure propagates to cohomology:

Corollary 6.1.10: Section and Hom Functor

Let $(X, \mathcal{O}_X)$ be a ringed space and let $R = \Gamma(X, \mathcal{O}_X)$. Then for any $\mathcal{O}_X$-module $\mathscr{F}$, the cohomology groups $H^i(X, \mathscr{F})$ carry a natural R-module structure. In particular:

1. If X is a scheme over a ring A (i.e., equipped with a structure morphism $X \to \operatorname{Spec} A$), then each $H^i(X, \mathscr{F})$ is naturally an A-module.
2. If X is a scheme over a field k, then each $H^i(X, \mathscr{F})$ is a k-vector space.

Moreover, for any $\mathcal{O}_X$-module $\mathscr{F}$ and any open subset $U \subseteq X$, there is a natural isomorphism:

$$H^i(X, \mathscr{F}) \cong R^i\Gamma(X, \mathscr{F})$$

and the connecting homomorphisms in the long exact sequence are R-module maps.

***Proof*:**
The global section functor $\Gamma(X, -)$ lands in R-Mod (not just **Ab**), and the category $\mathcal{O}_X$-**Mod** has enough injectives. The right derived functors of $\Gamma(X, -)$ computed in **Ab** and in R-Mod coincide, since the forgetful functor R-Mod $\to$ **Ab** is exact and preserves injectives. Hence the cohomology groups inherit the R-module structure. For a scheme over A, the structure morphism $X \to \operatorname{Spec} A$ gives a ring homomorphism $A \to \Gamma(X, \mathcal{O}_X) = R$, making every R-module an A-module by restriction of scalars.

6.1.1 Computation: Čech Cohomology

The derived-functor definition of sheaf cohomology is clean and canonical, but it is difficult to use in practice because injective (or even flasque) resolutions are typically non-explicit. Čech cohomology provides a more "hands-on" approach: it computes the obstruction to extending sections from an open cover $\mathcal{U}$ of a topological space X to larger open sets, using only the sections of the sheaf over finite intersections of the cover. When X is a Noetherian separated scheme, $\mathscr{F}$ is quasi-coherent, and the cover is affine, the Čech groups coincide with the derived-functor groups. This will be our main computational tool.

The following construction will be familiar to those who have seen group cohomology or simplicial cohomology in algebraic topology (see [51, chapter 4]).

Let X be a topological space, $\mathcal{U} = (U_i)_{i \in I}$ an open covering of X, and fix a well-ordering on the index set I[3]. For any finite set of indices $i_0 < \cdots < i_p$ in I, write:

$$U_{i_0,\ldots,i_p} = U_{i_0} \cap \cdots \cap U_{i_p}.$$

Let $\mathscr{F}$ be a sheaf of abelian groups on X. Define the *Čech cochain groups*:

$$C^p(\mathcal{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} \mathscr{F}(U_{i_0,\ldots,i_p}).$$

[3] Alternatively, one can work with the "alternating" convention: allow all tuples of indices, require that the cochain α is alternating (i.e. $\alpha_{\sigma(i_0),\ldots,\sigma(i_p)} = \operatorname{sgn}(\sigma) \cdot \alpha_{i_0,\ldots,i_p}$) and vanishes when any two indices coincide. The two approaches yield isomorphic cohomology groups.

An element $\alpha \in C^p(\mathcal{U}, \mathscr{F})$ is a collection of sections $\alpha_{i_0,\ldots,i_p} \in \mathscr{F}(U_{i_0,\ldots,i_p})$, one for each ordered $(p+1)$-tuple. The *coboundary map* $d : C^p(\mathcal{U}, \mathscr{F}) \to C^{p+1}(\mathcal{U}, \mathscr{F})$ is defined by:

$$(d\alpha)_{i_0,\ldots,i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k \, \alpha_{i_0,\ldots,\widehat{i_k},\ldots,i_{p+1}}\big|_{U_{i_0,\ldots,i_{p+1}}}$$

where $\widehat{i_k}$ means the index i_k is omitted. A standard computation shows $d^2 = 0$, so $(C^\bullet(\mathcal{U}, \mathscr{F}), d)$ is a cochain complex.

Definition 6.1.11: Čech Cohomology

Let X be a topological space, $\mathcal{U}$ an open covering of X, and $\mathscr{F}$ a sheaf of abelian groups on X. The *Čech cohomology groups* of $\mathscr{F}$ with respect to $\mathcal{U}$ are:

$$\check{H}^p(\mathcal{U}, \mathscr{F}) = h^p\big(C^\bullet(\mathcal{U}, \mathscr{F})\big) = \frac{\ker\big(d : C^p \to C^{p+1}\big)}{\operatorname{im}\big(d : C^{p-1} \to C^p\big)}.$$

An important warning: $\check{H}^p(\mathcal{U}, -)$ is *not* in general a δ-functor. That is, a short exact sequence of sheaves need not induce a long exact sequence in Čech cohomology:

Example 6.1.12: Čech Cohomology is Not a δ-Functor

Suppose $\mathcal{U}$ consists of the single open set X itself. Then every intersection $U_{i_0,\ldots,i_p}$ equals X, and there is only one index, so $C^p(\mathcal{U}, \mathscr{F}) = 0$ for $p > 0$ while $C^0(\mathcal{U}, \mathscr{F}) = \mathscr{F}(X) = \Gamma(X, \mathscr{F})$. Hence $\check{H}^0(\mathcal{U}, \mathscr{F}) = \Gamma(X, \mathscr{F})$ and $\check{H}^p(\mathcal{U}, \mathscr{F}) = 0$ for all $p > 0$.

Now let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be a short exact sequence of sheaves. If $\check{H}^*(\mathcal{U}, -)$ were a δ-functor, it would produce a long exact sequence, which in this case would read:

$$0 \longrightarrow \Gamma(X, \mathscr{F}') \longrightarrow \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X, \mathscr{F}'') \longrightarrow 0.$$

But $\Gamma(X, -)$ is not right exact in general — this is the whole point of sheaf cohomology! — so this need not be exact on the right. Hence $\check{H}^*$ cannot be a δ-functor for this covering.

This failure is not a defect of the theory but a feature of the choice of cover. As we shall see, choosing an appropriate cover (an affine cover for quasi-coherent sheaves on a separated scheme) restores the connection to derived-functor cohomology. Before proving this, let us see Čech cohomology in action:

Example 6.1.13: Čech Cohomology

1. *Differentials on $\mathbb{P}^1_k$.* Let $X = \mathbb{P}^1_k$, let $\mathscr{F} = \Omega_{X/k}$ be the sheaf of differentials, and let $\mathcal{U} = \{U, V\}$ be the standard affine cover with $U = D_+(x_0) \cong \operatorname{Spec} k[t]$ and $V = D_+(x_1) \cong \operatorname{Spec} k[u]$, where $t = x_1/x_0$ and $u = x_0/x_1 = 1/t$. The Čech complex has two terms:

$$C^0 = \Gamma(U, \Omega) \times \Gamma(V, \Omega), \qquad C^1 = \Gamma(U \cap V, \Omega).$$

The sections of the sheaf of differentials are:

$$\Gamma(U, \Omega) = k[t]\, dt, \qquad \Gamma(V, \Omega) = k[u]\, du, \qquad \Gamma(U \cap V, \Omega) = k[t, t^{-1}]\, dt.$$

On the overlap, the coordinate change gives $u = 1/t$, so $du = -t^{-2}\, dt$. The coboundary map

$d : C^0 \to C^1$ sends $(f(t)\,dt,\ g(u)\,du)$ to:

$$g(u)\,du\big|_{U\cap V} - f(t)\,dt\big|_{U\cap V} \;=\; \big(-t^{-2}g(1/t) - f(t)\big)\,dt.$$

Computing $\check{H}^0$: A cocycle $(f(t)\,dt,\ g(u)\,du) \in \ker d$ satisfies $f(t) = -t^{-2}g(1/t)$. The left side is a polynomial in t, while the right side is a polynomial in t^{-1} with no constant term (the t^{-2} factor shifts degrees). These can only agree if both are zero. Hence $\check{H}^0(\mathcal{U}, \Omega) = 0$, confirming that $\mathbb{P}^1_k$ has no nonzero global differential forms.

Computing $\check{H}^1$: The image of d consists of all Laurent polynomials of the form $\big(-t^{-2}g(1/t) - f(t)\big)\,dt$ where $f \in k[t]$ and $g \in k[u]$. Writing a general element of $k[t, t^{-1}]\,dt$ as $\sum_{n\in\mathbb{Z}} a_n t^n\,dt$, the image of d is spanned by all $t^n\,dt$ with $n \neq -1$: the terms $t^n\,dt$ for $n \geq 0$ come from $f(t)$, and the terms $t^n\,dt$ for $n \leq -2$ come from $g(u)$. The single "missing" basis element is $t^{-1}\,dt$. Therefore:

$$\check{H}^1(\mathcal{U}, \Omega) \cong k,$$

generated by the class of $t^{-1}\,dt$. This one-dimensional space reflects the genus of $\mathbb{P}^1$ being zero (recall that for a smooth projective curve of genus g, one has $\dim_k H^1(X, \Omega) = g$; here we are computing $H^1(X, \Omega)$ rather than $H^0(X, \Omega)$, which we just showed vanishes). We will revisit this when we discuss Serre duality.

2. *The circle.* Let $X = S^1$ with the Euclidean topology, and consider the constant sheaf $\underline{\mathbb{Z}}$. Take the cover $\mathcal{U} = \{U, V\}$ where U and V are two overlapping open arcs whose intersection $U \cap V = W_1 \coprod W_2$ consists of two disjoint small open intervals. Since U, V, W_1, and W_2 are all connected, the constant sheaf gives:

$$C^0 = \underline{\mathbb{Z}}(U) \times \underline{\mathbb{Z}}(V) = \mathbb{Z} \times \mathbb{Z}, \qquad C^1 = \underline{\mathbb{Z}}(U \cap V) = \underline{\mathbb{Z}}(W_1) \times \underline{\mathbb{Z}}(W_2) = \mathbb{Z} \times \mathbb{Z}.$$

The coboundary $d : C^0 \to C^1$ sends (a, b) to $(b - a,\ b - a)$, since on each connected component of $U \cap V$ the sections restrict to constants. Thus:

$$\ker d = \{(a, b) : b - a = 0\} \cong \mathbb{Z} \qquad \text{and} \qquad \mathrm{im} d = \{(c, c) : c \in \mathbb{Z}\} \cong \mathbb{Z}.$$

The cokernel $(\mathbb{Z} \times \mathbb{Z})/\{(c, c)\} \cong \mathbb{Z}$ (via $(m, n) \mapsto m - n$). Hence:

$$\check{H}^0(\mathcal{U}, \underline{\mathbb{Z}}) = \mathbb{Z}, \qquad \check{H}^1(\mathcal{U}, \underline{\mathbb{Z}}) = \mathbb{Z},$$

which agrees with the singular cohomology $H^*(S^1; \mathbb{Z})$. The class in $\check{H}^1$ is generated by the cocycle $(1, 0) \in \mathbb{Z} \times \mathbb{Z}$: the two components of $U \cap V$ see different "transition data," reflecting the non-trivial loop in S^1.

Let us now build towards the theorem that Čech cohomology computes derived-functor cohomology under appropriate hypotheses. The strategy is to show that the Čech complex arises from a resolution of the sheaf, and that this resolution can be compared to any injective resolution. The key condition is that each open set in the cover must be "cohomologically trivial" for the sheaf in question, so that the main obstruction comes from their combinations.

Lemma 6.1.14: 0th Čech Cohomology

Let X be a topological space, $\mathcal{U}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups on X. Then:
$$\check{H}^0(\mathcal{U},\mathscr{F}) = \Gamma(X,\mathscr{F}).$$

***Proof*:**
By definition, $\check{H}^0(\mathcal{U},\mathscr{F}) = \ker(d : C^0(\mathcal{U},\mathscr{F}) \to C^1(\mathcal{U},\mathscr{F}))$. An element $\alpha \in C^0$ is a collection $\{\alpha_i \in \mathscr{F}(U_i)\}_{i\in I}$. The coboundary condition $d\alpha = 0$ says that for each $i < j$,

$$(d\alpha)_{ij} = \alpha_j\big|_{U_i\cap U_j} - \alpha_i\big|_{U_i\cap U_j} = 0,$$

i.e. the sections α_i and α_j agree on the overlap $U_i \cap U_j$. Since $\mathscr{F}$ is a sheaf, the gluability axiom gives a unique global section $s \in \Gamma(X,\mathscr{F})$ with $s|_{U_i} = \alpha_i$ for all i. Hence $\ker d = \Gamma(X,\mathscr{F})$.

We next address the lack of a δ-functor by "sheafifying" the Čech construction. Let $X, \mathcal{U}$, and $\mathscr{F}$ be as above. For each $p \geq 0$, define the *sheaf of Čech p-cochains* by:

$$\mathscr{C}^p(\mathcal{U},\mathscr{F}) = \prod_{i_0<\cdots<i_p} (f_{i_0,\ldots,i_p})_*\big(\mathscr{F}\big|_{U_{i_0,\ldots,i_p}}\big)$$

where $f_{i_0,\ldots,i_p} : U_{i_0,\ldots,i_p} \hookrightarrow X$ is the inclusion and $(f_{i_0,\ldots,i_p})_*$ is the pushforward. The coboundary maps $d : \mathscr{C}^p \to \mathscr{C}^{p+1}$ are defined by the same alternating-sum formula as before. The key property of this construction is:
$$\Gamma\big(X, \mathscr{C}^p(\mathcal{U},\mathscr{F})\big) = C^p(\mathcal{U},\mathscr{F}),$$
so the Čech cochain groups are the global sections of the Čech sheaf complex.

Lemma 6.1.15: Čech Cohomology Resolution

Let X be a topological space, $\mathcal{U}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups. Then the augmented complex

$$0 \longrightarrow \mathscr{F} \xrightarrow{\epsilon} \mathscr{C}^0(\mathcal{U},\mathscr{F}) \xrightarrow{d} \mathscr{C}^1(\mathcal{U},\mathscr{F}) \xrightarrow{d} \mathscr{C}^2(\mathcal{U},\mathscr{F}) \longrightarrow \cdots$$

is exact. That is, $\mathscr{C}^\bullet(\mathcal{U},\mathscr{F})$ is a resolution of $\mathscr{F}$.

***Proof*:**
The augmentation $\epsilon : \mathscr{F} \to \mathscr{C}^0(\mathcal{U},\mathscr{F})$ is defined by taking the product of the natural restriction maps $\mathscr{F} \to (f_i)_*(\mathscr{F}|_{U_i})$ for each $i \in I$. Exactness at $\mathscr{C}^0$ (i.e. $\ker d^0 = \operatorname{im}\epsilon$) follows from the sheaf axioms for $\mathscr{F}$, exactly as in lemma 6.1.14.

For the exactness of the complex at $\mathscr{C}^p$ for $p \geq 1$, it suffices to check on stalks. Let $x \in X$ and choose an index $j \in I$ such that $x \in U_j$. For each $p \geq 0$, we define a *contracting homotopy* $k : \mathscr{C}^p(\mathcal{U},\mathscr{F})_x \to \mathscr{C}^{p-1}(\mathcal{U},\mathscr{F})_x$ as follows. A germ $\alpha_x \in \mathscr{C}^p(\mathcal{U},\mathscr{F})_x$ is represented by a section $\alpha \in \Gamma(W, \mathscr{C}^p(\mathcal{U},\mathscr{F}))$ over some open neighborhood W of x that we may shrink until $W \subseteq U_j$.

For any p-tuple $i_0 < \cdots < i_{p-1}$, set:

$$(k\alpha)_{i_0,\ldots,i_{p-1}} = \alpha_{j,i_0,\ldots,i_{p-1}}\big|_W.$$

This is well-defined because $W \subseteq U_j$ implies $W \cap U_{i_0,\ldots,i_{p-1}} = W \cap U_{j,i_0,\ldots,i_{p-1}}$, so the section $\alpha_{j,i_0,\ldots,i_{p-1}}$ is defined on W. Passing to germs at x gives the map k on stalks. A direct computation shows:

$$(dk + kd)(\alpha) = \alpha \qquad \text{for all } \alpha \in \mathscr{C}^p(\mathcal{U}, \mathscr{F})_x,\ p \geq 1.$$

Hence k is a chain homotopy between the identity and the zero map on the stalk complex $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})_x$ in degrees $p \geq 1$. It follows that the cohomology of this stalk complex vanishes in positive degrees, which is precisely the exactness of the resolution.

Lemma 6.1.16: Čech Cohomology and Flasque Sheaves

Let X be a topological space, $\mathcal{U}$ an open covering, and $\mathscr{F}$ a flasque sheaf of abelian groups on X. Then $\check{H}^p(\mathcal{U}, \mathscr{F}) = 0$ for all $p > 0$.

***Proof*:**
By lemma 6.1.15, the complex $0 \to \mathscr{F} \to \mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ is a resolution. Since $\mathscr{F}$ is flasque, the sheaves $\mathscr{C}^p(\mathcal{U}, \mathscr{F})$ are flasque for each $p \geq 0$: indeed, $\mathscr{F}|_{U_{i_0,\ldots,i_p}}$ is flasque (restriction of a flasque sheaf is flasque), the pushforward $(f_{i_0,\ldots,i_p})_*$ preserves flasque sheaves, and a product of flasque sheaves is flasque. Thus $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ is a flasque resolution of $\mathscr{F}$, and we may use it to compute derived-functor cohomology:

$$H^p(X, \mathscr{F}) = h^p\big(\Gamma(X, \mathscr{C}^\bullet(\mathcal{U}, \mathscr{F}))\big) = h^p\big(C^\bullet(\mathcal{U}, \mathscr{F})\big) = \check{H}^p(\mathcal{U}, \mathscr{F}).$$

But $\mathscr{F}$ is flasque, so $H^p(X, \mathscr{F}) = 0$ for $p > 0$ by corollary 6.1.7. Hence $\check{H}^p(\mathcal{U}, \mathscr{F}) = 0$ for $p > 0$.

Lemma 6.1.17: Čech Cohomology Comparison Map

Let X be a topological space and $\mathcal{U}$ an open covering. Then for each $p \geq 0$, there is a natural map, functorial in $\mathscr{F}$:

$$\check{H}^p(\mathcal{U}, \mathscr{F}) \longrightarrow H^p(X, \mathscr{F}).$$

***Proof*:**
Choose an injective resolution $0 \to \mathscr{F} \to \mathscr{I}^\bullet$. We also have the Čech resolution $0 \to \mathscr{F} \to \mathscr{C}^\bullet(\mathcal{U}, \mathscr{F})$ from lemma 6.1.15. By the general comparison theorem for resolutions (see [15]), there exists a morphism of complexes $\mathscr{C}^\bullet(\mathcal{U}, \mathscr{F}) \to \mathscr{I}^\bullet$ extending the identity on $\mathscr{F}$, unique up to homotopy. Applying $\Gamma(X, -)$ and taking cohomology gives the desired maps $\check{H}^p(\mathcal{U}, \mathscr{F}) \to H^p(X, \mathscr{F})$, and functoriality follows from the naturality of the comparison.

We are now ready for the main result. It is worth stating first in the generality the argument actually has, because the hypothesis that makes it work is not affineness or quasi-coherence but a vanishing condition, and consumers elsewhere in the series need the general form: a covering computes derived-functor cohomology as soon as the sheaf has no higher cohomology on the members of the covering and their intersections. Such a covering is called *acyclic* for the sheaf, and the result is Leray's theorem.

Theorem 6.1.18: Leray's Acyclic Covering Theorem

Let X be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups on X. Suppose the covering is *acyclic* for $\mathscr{F}$: for every $q > 0$ and every finite tuple $i_0, \dots, i_p$,

$$H^q\big(U_{i_0,\dots,i_p}, \mathscr{F}\big) = 0$$

Then the comparison maps of lemma 6.1.17 are isomorphisms

$$\check{H}^p(\mathcal{U}, \mathscr{F}) \xrightarrow{\sim} H^p(X, \mathscr{F}) \qquad \text{for all } p \geq 0$$

***Proof*:**
For $p = 0$ both sides are $\Gamma(X, \mathscr{F})$ by lemma 6.1.14. The rest is induction on p, by dimension shifting.

Setup. The category of abelian sheaves on X has enough injectives, so embed $\mathscr{F}$ into an injective sheaf $\mathscr{G}$, which is flasque by proposition 6.1.5, and let $\mathscr{Q}$ be the quotient:

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{Q} \longrightarrow 0$$

The hypothesis passes to $\mathscr{Q}$. Fix a tuple and write $U = U_{i_0,\dots,i_p}$. Since $\mathscr{G}$ is flasque, $H^q(U, \mathscr{G}) = 0$ for $q > 0$ by corollary 6.1.7, so the long exact sequence on U gives $H^q(U, \mathscr{Q}) \cong H^{q+1}(U, \mathscr{F})$ for $q \geq 1$, which vanishes by hypothesis. So $\mathcal{U}$ is acyclic for $\mathscr{Q}$ as well, and the induction has somewhere to go.

Exactness of the Čech complex. The same long exact sequence in degree zero, together with $H^1(U, \mathscr{F}) = 0$, gives exactness of

$$0 \longrightarrow \mathscr{F}(U) \longrightarrow \mathscr{G}(U) \longrightarrow \mathscr{Q}(U) \longrightarrow 0$$

for every intersection U. This is exactly where the hypothesis is spent: without it the right-hand map need not be surjective. Taking products over all tuples, the Čech cochain complexes form a short exact sequence

$$0 \longrightarrow C^\bullet(\mathcal{U}, \mathscr{F}) \longrightarrow C^\bullet(\mathcal{U}, \mathscr{G}) \longrightarrow C^\bullet(\mathcal{U}, \mathscr{Q}) \longrightarrow 0$$

Dimension shifting. Since $\mathscr{G}$ is flasque, $\check{H}^p(\mathcal{U}, \mathscr{G}) = 0$ for $p > 0$ by lemma 6.1.16 and $H^p(X, \mathscr{G}) = 0$ for $p > 0$ by corollary 6.1.7. The two long exact sequences, in Čech and in derived-functor cohomology, therefore read

$$0 \to \check{H}^0(\mathcal{U}, \mathscr{F}) \to \check{H}^0(\mathcal{U}, \mathscr{G}) \to \check{H}^0(\mathcal{U}, \mathscr{Q}) \to \check{H}^1(\mathcal{U}, \mathscr{F}) \to 0$$

$$0 \to H^0(X, \mathscr{F}) \to H^0(X, \mathscr{G}) \to H^0(X, \mathscr{Q}) \to H^1(X, \mathscr{F}) \to 0$$

with isomorphisms $\check{H}^p(\mathcal{U}, \mathscr{Q}) \cong \check{H}^{p+1}(\mathcal{U}, \mathscr{F})$ and $H^p(X, \mathscr{Q}) \cong H^{p+1}(X, \mathscr{F})$ for $p \geq 1$. The comparison maps of lemma 6.1.17 give a morphism between the two four-term sequences which is an isomorphism in degree zero on all three sheaves, so the five lemma makes $\check{H}^1(\mathcal{U}, \mathscr{F}) \to H^1(X, \mathscr{F})$ an isomorphism.

For $p \geq 1$, the inductive hypothesis applied to $\mathscr{Q}$ (legitimate, since $\mathcal{U}$ is acyclic for $\mathscr{Q}$) gives that $\check{H}^p(\mathcal{U}, \mathscr{Q}) \to H^p(X, \mathscr{Q})$ is an isomorphism, and the two dimension-shifting isomorphisms above are compatible with the comparison maps, so $\check{H}^{p+1}(\mathcal{U}, \mathscr{F}) \to H^{p+1}(X, \mathscr{F})$ is an isomorphism. This completes the induction, as we sought to show.

The scheme-theoretic case now follows by producing the acyclicity: on a separated scheme the intersections of affine opens are again affine, and a quasi-coherent sheaf has no higher cohomology on an affine. Separatedness is precisely what makes the intersections affine, which is where that hypothesis enters.

Theorem 6.1.19: Čech Cohomology Equivalence

Let X be a Noetherian separated scheme, $\mathcal{U}$ an open affine covering of X, and $\mathscr{F}$ a quasi-coherent sheaf on X. Then for all $p \geq 0$, the natural maps

$$\check{H}^p(\mathcal{U}, \mathscr{F}) \longrightarrow H^p(X, \mathscr{F})$$

are isomorphisms.

***Proof*:**
By theorem 6.1.18 it suffices to check that $\mathcal{U}$ is acyclic for $\mathscr{F}$. Fix a tuple $i_0, \ldots, i_p$. The intersection $U_{i_0,\ldots,i_p}$ is a finite intersection of affine opens in a separated scheme, hence affine by proposition 3.5.8. A quasi-coherent sheaf on a Noetherian affine scheme has vanishing higher cohomology by theorem 6.2.9, so $H^q(U_{i_0,\ldots,i_p}, \mathscr{F}) = 0$ for every $q > 0$. That is the hypothesis of Leray's theorem, and the conclusion is the claim, as we sought to show.

6.1.2 Refinement and the Degree-One Comparison

Everything so far has been relative to a fixed covering, and theorem 6.1.18 gives agreement with derived-functor cohomology only when a covering acyclic for the sheaf can be found. For many sheaves no such covering exists: a covering that trivializes a line bundle is not acyclic for $\mathcal{O}_X^\times$, so Leray says nothing about $\check{H}^1(X, \mathcal{O}_X^\times)$. What survives without any acyclicity hypothesis is a comparison in degree one, and reaching it means first letting the covering vary.

Definition 6.1.20: Refinement of a Covering

Let $\mathcal{U} = (U_i)_{i \in I}$ and $\mathcal{V} = (V_j)_{j \in J}$ be open coverings of X. Then $\mathcal{V}$ *refines* $\mathcal{U}$ if there is a map $\varepsilon : J \to I$ with $V_j \subseteq U_{\varepsilon(j)}$ for every j; such an ε is a *refinement function*. Write $\mathcal{V} \preceq \mathcal{U}$.

A refinement function transports cochains, and the point of the next lemma is that the transported map on cohomology does not depend on which refinement function was chosen. Without that, the coverings could not be assembled into a single invariant.

Lemma 6.1.21: Refinement Induces a Well-Defined Map

Let $\mathscr{F}$ be a sheaf of abelian groups on X and let $\mathcal{V} \preceq \mathcal{U}$.

1. A refinement function ε induces a morphism of complexes $\varepsilon^\bullet : C^\bullet(\mathcal{U}, \mathscr{F}) \to C^\bullet(\mathcal{V}, \mathscr{F})$, by
$$(\varepsilon^p \alpha)_{j_0,\ldots,j_p} = \alpha_{\varepsilon(j_0),\ldots,\varepsilon(j_p)}\big|_{V_{j_0,\ldots,j_p}}$$
2. Any two refinement functions ε, η induce chain homotopic morphisms. Hence the induced map $\check{H}^p(\mathcal{U}, \mathscr{F}) \to \check{H}^p(\mathcal{V}, \mathscr{F})$ is independent of the choice, and these maps are compatible with composition of refinements.

***Proof*:**
For (1), $\varepsilon^\bullet$ commutes with d because the coboundary formula is a signed sum over omitted indices and ε acts on indices, so omitting the k-th index before or after applying ε gives the same term; the restrictions agree since $V_{j_0,\ldots,j_p} \subseteq U_{\varepsilon(j_0),\ldots,\varepsilon(j_p)}$.
(2). Define $h^p : C^p(\mathcal{U}, \mathscr{F}) \to C^{p-1}(\mathcal{V}, \mathscr{F})$ by

$$(h^p\alpha)_{j_0,\ldots,j_{p-1}} = \sum_{k=0}^{p-1} (-1)^k \, \alpha_{\varepsilon(j_0),\ldots,\varepsilon(j_k),\eta(j_k),\ldots,\eta(j_{p-1})}\big|_{V_{j_0,\ldots,j_{p-1}}}$$

which makes sense because $V_{j_0,\ldots,j_{p-1}}$ is contained in every open set appearing on the right. The standard simplicial computation, identical to the one that shows two simplicial approximations are homotopic, gives $dh + hd = \eta^\bullet - \varepsilon^\bullet$. Hence the two morphisms agree on cohomology. Compatibility with composition is immediate, since a composite of refinement functions is one, completing the proof.

Definition 6.1.22: Čech Cohomology of a Space

Let $\mathscr{F}$ be a sheaf of abelian groups on X. The open coverings of X, ordered by refinement, form a filtered preorder: two coverings $\mathcal{U}$ and $\mathcal{U}'$ admit the common refinement $(U_i \cap U'_{i'})_{i,i'}$. The *Čech cohomology of* X is the filtered colimit

$$\check{H}^p(X, \mathscr{F}) = \varinjlim_{\mathcal{U}} \check{H}^p(\mathcal{U}, \mathscr{F})$$

over that preorder, the transition maps being those of lemma 6.1.21.

The comparison with derived-functor cohomology now holds in degrees zero and one for an arbitrary sheaf on an arbitrary space, with no acyclicity hypothesis anywhere. This is the form consumers need when Leray is unavailable.

Theorem 6.1.23: The Degree-One Comparison

Let X be a topological space and $\mathscr{F}$ a sheaf of abelian groups on X. Then

$$\check{H}^0(X, \mathscr{F}) \cong H^0(X, \mathscr{F}), \qquad \check{H}^1(X, \mathscr{F}) \cong H^1(X, \mathscr{F})$$

naturally in $\mathscr{F}$.

***Proof*:**
Degree zero is lemma 6.1.14 for each covering, and the transition maps are the identity on $\Gamma(X, \mathscr{F})$, so the colimit is $\Gamma(X, \mathscr{F}) = H^0(X, \mathscr{F})$.

Setup for degree one. The category of abelian sheaves on X has enough injectives; embed $\mathscr{F}$ into an injective $\mathscr{I}$, which is flasque by proposition 6.1.5, and let $\mathscr{Q} = \mathscr{I}/\mathscr{F}$. Since $H^1(X, \mathscr{I}) = 0$ by corollary 6.1.7, the long exact sequence gives

$$H^1(X, \mathscr{F}) \cong \operatorname{coker}\big(\Gamma(X, \mathscr{I}) \to \Gamma(X, \mathscr{Q})\big)$$

A class gives a cocycle. Let $\xi \in H^1(X, \mathscr{F})$ be represented by $t \in \Gamma(X, \mathscr{Q})$. Because $\mathscr{I} \to \mathscr{Q}$ is an epimorphism of sheaves, it is surjective on stalks, so there is an open covering $\mathcal{U} = (U_i)$ and sections $s_i \in \mathscr{I}(U_i)$ mapping to $t|_{U_i}$. On U_{ij} the difference $s_j - s_i$ maps to 0 in $\mathscr{Q}$, hence lies in $\mathscr{F}(U_{ij})$; set $c_{ij} = s_j - s_i$. Then c is a Čech 1-cocycle, since $c_{jk} - c_{ik} + c_{ij} = 0$ by inspection, and its class in $\check{H}^1(X, \mathscr{F})$ is independent of the choices: changing s_i by $a_i \in \mathscr{F}(U_i)$ changes c by the coboundary da, and passing to a refinement changes nothing by lemma 6.1.21.

The two constructions are mutually inverse. In the other direction, a cocycle $c \in \check{C}^1(\mathcal{U}, \mathscr{F})$ may be written $c_{ij} = s_j - s_i$ for some $s_i \in \mathscr{I}(U_i)$, because $\mathscr{I}$ is flasque and hence $\check{H}^1(\mathcal{U}, \mathscr{I}) = 0$ for every covering by theorem 6.1.18; the images of the s_i in $\mathscr{Q}(U_i)$ then agree on overlaps and glue to a global section t, whose class in the cokernel is the corresponding element of $H^1(X, \mathscr{F})$. Chasing the definitions shows the two assignments invert one another, and both are natural in $\mathscr{F}$ because every step is, completing the proof.

6.1.3 Soft Sheaves and the Paracompact Comparison

Theorem 6.1.23 is the best that can be said with no hypothesis on the space. On a paracompact space one can say everything: Čech cohomology agrees with derived-functor cohomology in every degree. The route runs through a class of sheaves that is to the topological setting what flasque sheaves are to the general one, and which was promised earlier when partitions of unity were contrasted with the rigidity of schemes.

Throughout this subsection X is a paracompact Hausdorff space. Recall that this means every open cover admits a locally finite open refinement; the consequence used below is that such a cover admits a shrinking, an open cover $\{V_i\}$ with $\overline{V_i} \subseteq U_i$.

Definition 6.1.24: Soft Sheaf

A sheaf $\mathscr{F}$ of abelian groups on a paracompact Hausdorff space X is *soft* if for every closed $K \subseteq X$ the restriction $\mathscr{F}(X) \to \mathscr{F}(K) := \varinjlim_{U \supseteq K} \mathscr{F}(U)$ is surjective: every section defined on a neighbourhood of a closed set extends to all of X.

A *fine* sheaf, in the sense of the partition-of-unity discussion in section 2.2.4, is one admitting partitions of unity subordinate to any locally finite cover. Fineness is the property one can check; softness is the property that makes cohomology vanish, and the first implies the second.

Proposition 6.1.25: Flasque and Fine Sheaves Are Soft

On a paracompact Hausdorff space, every flasque sheaf is soft, and every fine sheaf of modules over a sheaf of rings is soft.

***Proof*:**
A flasque sheaf restricts surjectively from X to any open set, hence a fortiori to the colimit over neighbourhoods of a closed set.

Let $\mathscr{F}$ be fine and let s be a section over an open $U \supseteq K$ with K closed. The two opens U and $X \setminus K$ cover X; take a locally finite refinement and a subordinate partition of unity, and let η be the sum of the members supported in U, so that η is an endomorphism of $\mathscr{F}$ with $\eta = \mathrm{id}$ on a neighbourhood of K and $\eta = 0$ off U. Then $\eta(s)$, extended by zero, is a global section restricting to s near K.

Lemma 6.1.26: Sections and Quotients of Soft Sheaves

Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be exact on X with $\mathscr{F}'$ soft. Then $\mathscr{F}(X) \to \mathscr{F}''(X)$ is surjective. If $\mathscr{F}$ is soft as well, then $\mathscr{F}''$ is soft.

***Proof*:**
Let $t \in \mathscr{F}''(X)$. Locally t lifts, so choose a locally finite open cover $\{U_i\}_{i \in I}$ with lifts $s_i \in \mathscr{F}(U_i)$, and a shrinking $\{V_i\}$ with $\overline{V_i} \subseteq U_i$. Order I and consider the set of pairs (J, s) where $J \subseteq I$ and s is a section of $\mathscr{F}$ over a neighbourhood of $\overline{V_J} := \bigcup_{i \in J} \overline{V_i}$ lifting t there; order these by extension. A chain has an upper bound by local finiteness, so Zorn gives a maximal (J, s). If some $k \notin J$, then on a neighbourhood of $\overline{V_J} \cap \overline{V_k}$ the difference $s - s_k$ lifts 0, hence comes from a section of $\mathscr{F}'$ there; $\mathscr{F}'$ is soft, so that section extends globally, and subtracting the extension from s_k makes the two agree near the overlap. They then glue to a lift over a neighbourhood of $\overline{V_{J \cup \{k\}}}$, contradicting maximality. So $J = I$ and t lifts globally.

For the second claim let $K \subseteq X$ be closed and t'' a section of $\mathscr{F}''$ near K. Softness of $\mathscr{F}$ is not yet applicable, because t'' need not lift near K; but the first part applied to the closed subspace K, which is paracompact Hausdorff and on which the restricted sequence is still exact with restricted $\mathscr{F}'$ still soft, lifts $t''|_K$ to a section of $\mathscr{F}$ over K. That lift extends to X by softness of $\mathscr{F}$, and its image in $\mathscr{F}''$ is a global section agreeing with t'' on K.

Theorem 6.1.27: Soft Sheaves Are Acyclic

Let $\mathscr{F}$ be a soft sheaf on a paracompact Hausdorff space X. Then $H^i(X, \mathscr{F}) = 0$ for all $i > 0$.

***Proof*:**
Embed $\mathscr{F}$ in a flasque sheaf, $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{Q} \to 0$, which is possible because there are enough injectives (corollary 6.1.2) and injectives are flasque (proposition 6.1.5). By proposition 6.1.25 the sheaf $\mathscr{G}$ is soft, so $\mathscr{Q}$ is soft by lemma 6.1.26.

The long exact sequence in cohomology, together with $H^i(X, \mathscr{G}) = 0$ for $i > 0$ (corollary 6.1.7), gives

$$\mathscr{G}(X) \to \mathscr{Q}(X) \to H^1(X, \mathscr{F}) \to 0, \qquad H^i(X, \mathscr{F}) \cong H^{i-1}(X, \mathscr{Q}) \text{ for } i \geq 2$$

The first map is surjective by lemma 6.1.26, so $H^1(X, \mathscr{F}) = 0$, and this holds for every soft sheaf. Applying it to the soft sheaf $\mathscr{Q}$ and inducting on i kills the higher groups as well.

The comparison theorem now follows, because the same argument applies verbatim on the Čech side.

Theorem 6.1.28: Čech Cohomology on a Paracompact Space

Let X be a paracompact Hausdorff space and $\mathscr{F}$ a sheaf of abelian groups on X. Then for every i

$$\check{H}^i(X, \mathscr{F}) \cong H^i(X, \mathscr{F})$$

naturally in $\mathscr{F}$.

***Proof*:**
Step 1: Čech cohomology vanishes on flasque sheaves. Let $\mathscr{G}$ be flasque and $\mathcal{U} = \{U_i\}$ any open cover. A Čech q-cocycle c with $q \geq 1$ is killed as follows: well-order I, and for each multi-index set $b_{i_0 \cdots i_{q-1}} = \sum_j \widetilde{c_{j\, i_0 \cdots i_{q-1}}}$, where the tilde denotes an extension by flasqueness of the section $c_{j\, i_0 \cdots i_{q-1}}$ from $U_j \cap U_{i_0} \cap \cdots$ to $U_{i_0} \cap \cdots$, summed over a locally finite refinement. The cocycle identity then gives $\delta b = c$ exactly as in the partition-of-unity computation. Hence $\check{H}^q(\mathcal{U}, \mathscr{G}) = 0$ for $q \geq 1$, and passing to the colimit over covers, $\check{H}^q(X, \mathscr{G}) = 0$.

Step 2: Čech cohomology has a long exact sequence. This is where paracompactness enters. For $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ exact, the sequence of Čech complexes is exact only up to refinement, since a section of $\mathscr{F}''$ lifts only locally; on a paracompact space every cover has a locally finite refinement and the colimit over covers is filtered, so the failure vanishes in the colimit and the long exact sequence exists.

Step 3: comparison. Both $\check{H}^\bullet(X, -)$ and $H^\bullet(X, -)$ are now δ-functors on sheaves of abelian groups over X; they agree in degree 0, both being $\Gamma(X, -)$; and both vanish in positive degrees on flasque sheaves, by Step 1 and corollary 6.1.7. Since every sheaf embeds in a flasque one, both are effaceable, hence universal, and a morphism of universal δ-functors agreeing in degree 0 is an isomorphism. The comparison map of theorem 6.1.23 is such a morphism, so it is an isomorphism in every degree.

This is the form consumers outside algebraic geometry need. A complex manifold is paracompact, its sheaves of smooth forms are fine, and theorem 6.1.27 together with proposition 6.1.25 is what makes a resolution by smooth forms compute sheaf cohomology.

6.2 Vanishing Theorems

The computation of the previous section used two key vanishing phenomena: cohomology vanishes above the dimension of the space, and cohomology of quasi-coherent sheaves vanishes on affine schemes. In this section we prove both results. These are the foundations on which virtually all further cohomological arguments rest.

The first result — the *dimension theorem* — says that if X is a Noetherian topological space of dimension n, then $H^i(X, \mathscr{F}) = 0$ for all $i > n$ and all sheaves of abelian groups $\mathscr{F}$. This is the scheme-theoretic analogue of the general principle that cohomology should not exceed the dimension of the underlying space[4].

The second result — the *affine vanishing theorem* — says that on an affine Noetherian scheme $X = \operatorname{Spec} R$, all quasi-coherent sheaves have trivial higher cohomology: $H^i(X, \mathscr{F}) = 0$ for $i > 0$. This confirms the philosophy that cohomology measures the obstruction to gluing: on an affine scheme, there is no gluing to be done. Conversely, we shall show that this vanishing property *characterizes* affine schemes among Noetherian schemes.

6.2.1 Dimension Theorem for Schemes

We begin with two lemmas about the behaviour of cohomology on Noetherian spaces. The Noetherian hypothesis enters through a single but crucial property: on Noetherian topological spaces, filtered colimits of sheaves commute with the global section functor. This fails in general.

Lemma 6.2.1: Noetherian Space and Flasque Sheaves

Let X be a Noetherian topological space. Then a filtered colimit of flasque sheaves on X is flasque.

***Proof*:**
Let $(\mathscr{F}_\alpha)$ be a filtered (directed) system of flasque sheaves on X, and let $U \subseteq V$ be open subsets. For each α, the restriction map $\mathscr{F}_\alpha(V) \to \mathscr{F}_\alpha(U)$ is surjective. Since the system is filtered, the colimit functor on abelian groups is exact, so:

$$\varinjlim_{\alpha} \mathscr{F}_\alpha(V) \longrightarrow \varinjlim_{\alpha} \mathscr{F}_\alpha(U)$$

is surjective. Now, the key point is that on a Noetherian topological space, every open subset is quasi-compact, so filtered colimits of sheaves commute with the section functor:

$$\big(\varinjlim_{\alpha} \mathscr{F}_\alpha\big)(U) \;\cong\; \varinjlim_{\alpha} \mathscr{F}_\alpha(U)$$

for every open U. This is *not* true in general: for a presheaf the colimit is computed section-wise, but for a sheaf one may need to sheafify. On a Noetherian space, however, every open cover

[4]Recall from [51] that the corresponding statement in singular cohomology is that $H^i(M; \mathbb{Z}) = 0$ for $i > \dim M$ when M is a manifold; in de Rham cohomology, it is the vanishing of the de Rham groups above the manifold dimension.

has a finite subcover, so any section of the sheafification already comes from some $\mathscr{F}_\alpha$, and sheafification is unnecessary. Combining the two displayed equations, the restriction map

$$\big(\varinjlim_\alpha \mathscr{F}_\alpha\big)(V) \longrightarrow \big(\varinjlim_\alpha \mathscr{F}_\alpha\big)(U)$$

is surjective, so $\varinjlim_\alpha \mathscr{F}_\alpha$ is flasque.

Lemma 6.2.2: Colimit and Cohomology Commuting

Let X be a Noetherian topological space and $(\mathscr{F}_\alpha)$ a filtered direct system of sheaves of abelian groups. Then for each $i \geq 0$ there is a natural isomorphism:

$$\varinjlim_\alpha H^i(X, \mathscr{F}_\alpha) \;\cong\; H^i(X, \varinjlim_\alpha \mathscr{F}_\alpha).$$

***Proof*:**
For each $\mathscr{F}_\alpha$, choose a flasque resolution $0 \to \mathscr{F}_\alpha \to \mathscr{I}_\alpha^\bullet$ functorially (for instance, using the Godement resolution from Example 6.1.8(3), which is canonical). The cohomology groups are then:

$$H^i(X, \mathscr{F}_\alpha) = h^i\big(\Gamma(X, \mathscr{I}_\alpha^\bullet)\big).$$

Since filtered colimits are exact in the category of abelian sheaves, taking the colimit of these resolutions yields an exact sequence:

$$0 \longrightarrow \varinjlim_\alpha \mathscr{F}_\alpha \longrightarrow \varinjlim_\alpha \mathscr{I}_\alpha^\bullet.$$

By lemma 6.2.1, since X is Noetherian, each term $\varinjlim_\alpha \mathscr{I}_\alpha^p$ is flasque. Hence $\varinjlim_\alpha \mathscr{I}_\alpha^\bullet$ is a flasque resolution of $\varinjlim_\alpha \mathscr{F}_\alpha$, and we may use it to compute cohomology:

$$H^i(X, \varinjlim_\alpha \mathscr{F}_\alpha) = h^i\big(\Gamma(X, \varinjlim_\alpha \mathscr{I}_\alpha^\bullet)\big).$$

On the other hand, filtered colimits of abelian groups commute with the cohomology functor h^i (since filtered colimits are exact):

$$\varinjlim_\alpha H^i(X, \mathscr{F}_\alpha) = \varinjlim_\alpha h^i\big(\Gamma(X, \mathscr{I}_\alpha^\bullet)\big) \cong h^i\big(\varinjlim_\alpha \Gamma(X, \mathscr{I}_\alpha^\bullet)\big).$$

It remains to show $\varinjlim_\alpha \Gamma(X, \mathscr{I}_\alpha^p) \cong \Gamma(X, \varinjlim_\alpha \mathscr{I}_\alpha^p)$ for each p. This holds because global sections commute with filtered colimits of sheaves on Noetherian spaces (the same property used in the proof of lemma 6.2.1). Combining these identifications completes the proof.

Corollary 6.2.3: Direct Sum and Cohomology Commuting

Let X be a Noetherian topological space and $(\mathscr{F}_\alpha)_{\alpha \in A}$ a family of sheaves of abelian groups. Then for each $i \geq 0$:

$$\bigoplus_{\alpha \in A} H^i(X, \mathscr{F}_\alpha) \cong H^i\Big(X, \bigoplus_{\alpha \in A} \mathscr{F}_\alpha\Big).$$

***Proof*:**
A direct sum is a filtered colimit of finite partial sums, and cohomology commutes with finite direct sums (since it is an additive functor). Hence the result follows from lemma 6.2.2.

The following result shows that cohomology computations on a closed subset can be performed on the ambient space by pushing forward:

Proposition 6.2.4: Cohomology of Closed Subsets

Let X be a topological space, $Y \subseteq X$ a closed subset, $j : Y \to X$ the inclusion, and $\mathscr{F}$ a sheaf of abelian groups on Y. Then for all $i \geq 0$:

$$H^i(Y, \mathscr{F}) \cong H^i(X, j_*\mathscr{F}).$$

***Proof*:**
Let $0 \to \mathscr{F} \to \mathscr{I}^\bullet$ be a flasque resolution of $\mathscr{F}$ on Y. Since the pushforward along a closed immersion preserves flasque sheaves — indeed, for $U \subseteq V$ open in X, the surjection $\mathscr{I}^p(V \cap Y) \twoheadrightarrow \mathscr{I}^p(U \cap Y)$ is precisely the restriction map for $j_*\mathscr{I}^p$ — the sequence $0 \to j_*\mathscr{F} \to j_*\mathscr{I}^\bullet$ is a flasque resolution of $j_*\mathscr{F}$ on X. Now:

$$H^i(X, j_*\mathscr{F}) = h^i\big(\Gamma(X, j_*\mathscr{I}^\bullet)\big) = h^i\big(\Gamma(Y, \mathscr{I}^\bullet)\big) = H^i(Y, \mathscr{F}),$$

where the middle equality uses $\Gamma(X, j_*\mathscr{G}) = \mathscr{G}(j^{-1}(X)) = \mathscr{G}(Y) = \Gamma(Y, \mathscr{G})$ for any sheaf $\mathscr{G}$ on Y.

By virtue of this result, we shall often write $\mathscr{F}$ in place of $j_*\mathscr{F}$ when Y is a closed subset of X, without risk of ambiguity: the cohomology groups are the same. With these tools, we can prove our first main vanishing theorem:

Theorem 6.2.5: Noetherian Space and Dimension

Let X be a Noetherian topological space of dimension n. Then for all sheaves of abelian groups $\mathscr{F}$ on X:

$$H^i(X, \mathscr{F}) = 0 \qquad \text{for all } i > n.$$

***Proof*:**
We shall need the following notation. For a closed subset $Y \subseteq X$ with inclusion $j : Y \hookrightarrow X$, write $\mathscr{F}_Y = j_*(j^{-1}\mathscr{F})$ for the "restriction of $\mathscr{F}$ to Y, pushed back to X." For an open subset $U \subseteq X$ with inclusion $i : U \hookrightarrow X$, write $\mathscr{F}_U = i_!(i^{-1}\mathscr{F})$, the extension by zero of $\mathscr{F}|_U$. If

$U = X \setminus Y$, checking on stalks gives a short exact sequence:

$$0 \longrightarrow \mathscr{F}_U \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}_Y \longrightarrow 0.$$

We prove the theorem by a double induction: on the dimension of X and on the number of irreducible components.

Step 1: Reduction to X irreducible. Suppose X is reducible. Let Y be an irreducible component of X and $U = X \setminus Y$. The short exact sequence above gives a long exact sequence:

$$\cdots \to H^i(X, \mathscr{F}_U) \to H^i(X, \mathscr{F}) \to H^i(X, \mathscr{F}_Y) \to \cdots$$

It suffices to show vanishing for $\mathscr{F}_Y$ and $\mathscr{F}_U$ separately. For $\mathscr{F}_Y$: since Y is closed and irreducible with $\dim Y \leq n$, proposition 6.2.4 gives $H^i(X, \mathscr{F}_Y) = H^i(Y, j^{-1}\mathscr{F})$, which vanishes for $i > n$ by the inductive hypothesis applied to the irreducible space Y. For $\mathscr{F}_U$: the support of $\mathscr{F}_U$ is contained in $\overline{U}$, which is a closed subset with strictly fewer irreducible components than X (we have removed the component Y). By proposition 6.2.4 applied to $\overline{U}$ and induction on the number of irreducible components, we get the required vanishing. Hence we may assume X is irreducible.

Step 2: Base case $\dim X = 0$. If X is irreducible of dimension 0, its only open subsets are $\emptyset$ and X itself (a proper closed irreducible subset would give $\dim X \geq 1$). Hence the global section functor $\Gamma(X, -) : \mathbf{Ab}(X) \to \mathbf{Ab}$ is an equivalence of categories (a sheaf on X is determined by a single abelian group). In particular, $\Gamma(X, -)$ is exact, so all its derived functors vanish: $H^i(X, \mathscr{F}) = 0$ for $i > 0$.

Step 3: gluing. To verify the maps glue on the overlap $U_i \cap U_j$, we must check that $\varphi_i = \varphi_j$ when applied to the local generators $d(x_k/x_i)$. By definition, φ_i yields $\frac{1}{x_i^2}(x_i\, e_k - x_k\, e_i)$. To evaluate φ_j, we first use the quotient rule to express the generator in the U_j basis:

$$d\left(\frac{x_k}{x_i}\right) = d\left(\frac{x_k/x_j}{x_i/x_j}\right) = \frac{x_j}{x_i}\, d\left(\frac{x_k}{x_j}\right) - \frac{x_k x_j}{x_i^2}\, d\left(\frac{x_i}{x_j}\right).$$

Applying the linear map φ_j to this expression yields:

$$\begin{aligned} \varphi_j\left(d\left(\frac{x_k}{x_i}\right)\right) &= \frac{x_j}{x_i}\left[\frac{1}{x_j^2}(x_j\, e_k - x_k\, e_j)\right] - \frac{x_k x_j}{x_i^2}\left[\frac{1}{x_j^2}(x_j\, e_i - x_i\, e_j)\right] \\ &= \left(\frac{1}{x_i}\, e_k - \frac{x_k}{x_i x_j}\, e_j\right) - \left(\frac{x_k}{x_i^2}\, e_i - \frac{x_k}{x_i x_j}\, e_j\right). \end{aligned}$$

The e_j terms perfectly cancel, leaving $\frac{1}{x_i}\, e_k - \frac{x_k}{x_i^2}\, e_i$, which is exactly $\frac{1}{x_i^2}(x_i\, e_k - x_k\, e_i)$. Hence $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$, and the local isomorphisms glue to a global isomorphism $\Omega_{X/Y} \cong \widetilde{M}$.

Step 4: Subsheaves of $\underline{\mathbb{Z}}_U$. Let $U \subseteq X$ be open and $\mathcal{K} \subseteq \underline{\mathbb{Z}}_U$ a subsheaf. If $\mathcal{K} = 0$, there is nothing to prove. Otherwise, for each $x \in U$, the stalk $\mathcal{K}_x$ is a subgroup of $\mathbb{Z}$, hence either 0 or $d_x\mathbb{Z}$ for some positive integer d_x. Let d be the smallest positive integer occurring among the stalks. There is a non-empty open subset $V \subseteq U$ on which $\mathcal{K}|_V \cong d \cdot \underline{\mathbb{Z}}|_V \cong \underline{\mathbb{Z}}_V$ (as a subsheaf of $\underline{\mathbb{Z}}|_V$). Thus $\mathcal{K}_V \cong \underline{\mathbb{Z}}_V$, and we have:

$$0 \longrightarrow \underline{\mathbb{Z}}_V \longrightarrow \mathcal{K} \longrightarrow \mathcal{K}/\underline{\mathbb{Z}}_V \longrightarrow 0.$$

The quotient $\mathcal{K}/\underline{\mathbb{Z}}_V$ is supported on the closure $\overline{U \setminus V}$, which is a proper closed subset of X. Since X is irreducible, $\dim \overline{U \setminus V} < n$. By proposition 6.2.4 and the inductive hypothesis on dimension, we get $H^i(X, \mathcal{K}/\underline{\mathbb{Z}}_V) = 0$ for $i \geq n$. From the long exact sequence, it suffices to prove vanishing for $\underline{\mathbb{Z}}_V$.

Step 5: Completing the proof for $\underline{\mathbb{Z}}_U$. It remains to show that for any open subset $U \subseteq X$, we have $H^i(X, \underline{\mathbb{Z}}_U) = 0$ for $i > n$. Let $Y = X \setminus U$ and consider the exact sequence:

$$0 \longrightarrow \underline{\mathbb{Z}}_U \longrightarrow \underline{\mathbb{Z}} \longrightarrow \underline{\mathbb{Z}}_Y \longrightarrow 0.$$

Since X is irreducible, the constant sheaf $\underline{\mathbb{Z}}$ on X is flasque (every non-empty open subset of an irreducible space is connected, so the constant sheaf has the same sections on every non-empty open; see Example 6.1.8(2)). Hence $H^i(X, \underline{\mathbb{Z}}) = 0$ for all $i > 0$. Furthermore, Y is a proper closed subset of the irreducible space X, so $\dim Y < n$. By proposition 6.2.4 and the inductive hypothesis on dimension, $H^i(X, \underline{\mathbb{Z}}_Y) = 0$ for $i \geq n$. The long exact sequence now gives $H^i(X, \underline{\mathbb{Z}}_U) = 0$ for $i > n$, completing the proof.

6.2.2 Vanishing Theorem for Affine Schemes

We now turn to the second fundamental vanishing result: on an affine Noetherian scheme $X = \operatorname{Spec} R$, all quasi-coherent sheaves have trivial higher cohomology. The key insight is that if I is an injective R-module, then the associated sheaf $\widetilde{I}$ on $\operatorname{Spec} R$ is flasque. This is not obvious: the sections of $\widetilde{I}$ on a general open set involve a limit over distinguished opens, and there is no a priori reason for restriction maps to be surjective. The proof requires the Noetherian hypothesis in an essential way, via the Artin–Rees lemma and Noetherian induction.

Let R be a Noetherian ring, M an R-module, and $\mathfrak{a} \subseteq R$ an ideal. The $\mathfrak{a}$-*torsion submodule* of M is:

$$\Gamma_{\mathfrak{a}}(M) = \left\{ m \in M \;\middle|\; \mathfrak{a}^k m = 0 \text{ for some } k > 0 \right\}.$$

Lemma 6.2.6: $\mathfrak{a}$-Torsion of an Injective Module is Injective

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, and I an injective R-module. Then the submodule $J = \Gamma_{\mathfrak{a}}(I)$ is injective.

***Proof*:**
By Baer's criterion, it suffices to show that for any ideal $\mathfrak{b} \subseteq R$ and any homomorphism $\varphi : \mathfrak{b} \to J$, there exists a homomorphism $\psi : R \to J$ extending φ.

Since R is Noetherian, $\mathfrak{b}$ is finitely generated, and every element of J is annihilated by some power of $\mathfrak{a}$. Hence there exists $n > 0$ such that $\mathfrak{a}^n \cdot \varphi(\mathfrak{b}) = 0$, i.e. $\varphi(\mathfrak{a}^n \mathfrak{b}) = 0$. By the Artin–Rees lemma (see [10]) applied to the inclusion $\mathfrak{b} \subseteq R$ and the $\mathfrak{a}$-adic filtration, there exists $n' \geq n$ such that $\mathfrak{a}^n \mathfrak{b} \supseteq \mathfrak{b} \cap \mathfrak{a}^{n'}$. Hence $\varphi(\mathfrak{b} \cap \mathfrak{a}^{n'}) = 0$, and φ factors through $\mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'})$.

Consider the commutative diagram:

$$\begin{tikzcd} R \arrow[r, two heads] & R/\mathfrak{a}^{n'} \arrow[dr, dashed, "\psi'"] & & \\ \mathfrak{b} \arrow[u, hook] \arrow[r, two heads] & \mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'}) \arrow[u, hook] \arrow[r] & J \arrow[r, hook] & I \end{tikzcd}$$

Since I is injective, the composed map $\mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'}) \to I$ extends to a map $\psi' : R/\mathfrak{a}^{n'} \to I$. The image of ψ' is annihilated by $\mathfrak{a}^{n'}$, so it is contained in $J = \Gamma_{\mathfrak{a}}(I)$. Composing with the natural projection $R \twoheadrightarrow R/\mathfrak{a}^{n'}$, we obtain the required extension $\psi : R \to J$.

Lemma 6.2.7: Localization of Injective Modules is Surjective

Let R be a Noetherian ring, I an injective R-module, and $f \in R$. Then the natural localization map $I \to I_f$ is surjective.

***Proof*:**
Let $\varphi : I \to I_f$ be the natural map and let $x \in I_f$. By definition of localization, there exist $y \in I$ and $n \geq 0$ such that $x = \varphi(y)/f^n$.

For each $i > 0$, let $\mathfrak{b}_i = \mathrm{Ann}_R(f^i)$. Since R is Noetherian, the ascending chain $\mathfrak{b}_1 \subseteq \mathfrak{b}_2 \subseteq \cdots$ stabilizes: there exists m such that $\mathfrak{b}_{m+j} = \mathfrak{b}_m$ for all $j \geq 0$. Define a homomorphism $\psi : (f^{n+m}) \to I$ by $f^{n+m} \mapsto f^m y$. This is well-defined: if $rf^{n+m} = 0$, then $r \in \mathfrak{b}_{n+m} = \mathfrak{b}_m$, so $rf^m = 0$, hence $rf^m y = 0$.

Since I is injective, Baer's criterion extends ψ to a map $\Psi : R \to I$. Let $z = \Psi(1)$. Then:

$$f^{n+m} z = f^{n+m}\Psi(1) = \Psi(f^{n+m}) = \psi(f^{n+m}) = f^m y,$$

so in I_f we have $\varphi(z) = z/1 = f^m y/f^{n+m} = y/f^n = x$. Hence φ is surjective.

Lemma 6.2.8: Associated Sheaf of Injective Module is Flasque

Let R be a Noetherian ring and I an injective R-module. Then the associated sheaf $\widetilde{I}$ on $X = \operatorname{Spec} R$ is flasque.

***Proof*:**
We proceed by Noetherian induction (cf. theorem 0.3.3) on the closed subset $Y = \mathrm{supp}(\widetilde{I})$. If Y is a single closed point, then $\widetilde{I}$ is a skyscraper sheaf, which is evidently flasque.

In the general case, it suffices to show that for any open set $U \subseteq X$, the restriction $\Gamma(X, \widetilde{I}) \to \Gamma(U, \widetilde{I})$ is surjective. If $Y \cap U = \emptyset$, then $\Gamma(U, \widetilde{I}) = 0$ and there is nothing to prove. Otherwise, choose $f \in R$ such that the distinguished open $D(f)$ satisfies $D(f) \subseteq U$ and $D(f) \cap Y \neq \emptyset$. Let $Z = X \setminus D(f)$, and for any open W let $\Gamma_Z(W, \widetilde{I})$ denote the sections of $\widetilde{I}$ over W with support in Z.

Given a section $s \in \Gamma(U, \widetilde{I})$, its restriction to $D(f)$ yields an element $s' \in \Gamma(D(f), \widetilde{I})$. By the identification $\Gamma(D(f), \widetilde{I}) \cong I_f$ (from proposition 2.1.12) and the surjectivity of $I \to I_f$ (lemma 6.2.7), there exists $t \in I = \Gamma(X, \widetilde{I})$ with $t|_{D(f)} = s'$. Let t' be the restriction of t to U. Then $s - t'$ restricts to zero on $D(f)$, so it has support in Z: $s - t' \in \Gamma_Z(U, \widetilde{I})$.

It remains to show that $\Gamma_Z(X, \widetilde{I}) \to \Gamma_Z(U, \widetilde{I})$ is surjective. Let $J = \Gamma_{\mathfrak{a}}(I)$ where $\mathfrak{a} = (f)$. Then $\Gamma_Z(X, \widetilde{I}) = J$ (the sections of $\widetilde{I}$ over all of X supported on $Z = V(f)$ are precisely the f-torsion elements of I). By lemma 6.2.6, J is an injective R-module. The support of $\widetilde{J}$ is contained in $Y \cap Z$, which is strictly contained in Y (since $D(f) \cap Y \neq \emptyset$, we have removed a non-empty open piece of Y). By the Noetherian induction hypothesis, $\widetilde{J}$ is flasque. Since $\Gamma(U, \widetilde{J}) = \Gamma_Z(U, \widetilde{I})$, the flasqueness of $\widetilde{J}$ gives the required surjectivity.

Theorem 6.2.9: Affine Noetherian Scheme: Trivial Cohomology

Let $X = \operatorname{Spec} R$ be the spectrum of a Noetherian ring R, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Then:
$$H^i(X, \mathscr{F}) = 0 \qquad \text{for all } i > 0.$$

***Proof*:**
Since $\mathscr{F}$ is quasi-coherent on the affine scheme $X = \operatorname{Spec} R$, there exists an R-module M with $\mathscr{F} \cong \widetilde{M}$ (by corollary 5.1.12). Choose an injective resolution $0 \to M \to I^\bullet$ in the category of R-modules. The associated-sheaf functor $\widetilde{(-)}$ is exact (by proposition 5.1.14), so:

$$0 \longrightarrow \widetilde{M} \longrightarrow \widetilde{I}^\bullet$$

is an exact sequence of $\mathcal{O}_X$-modules. By lemma 6.2.8, each $\widetilde{I}^p$ is flasque. Hence $\widetilde{I}^\bullet$ is a flasque resolution of $\mathscr{F} = \widetilde{M}$, and we may use it to compute cohomology:

$$H^i(X, \mathscr{F}) = h^i\big(\Gamma(X, \widetilde{I}^\bullet)\big) = h^i(I^\bullet).$$

But $I^\bullet$ is an injective resolution of M, so its cohomology is $h^0(I^\bullet) = M$ and $h^i(I^\bullet) = 0$ for $i > 0$. Hence:
$$H^0(X, \mathscr{F}) = M = \Gamma(X, \mathscr{F}) \qquad \text{and} \qquad H^i(X, \mathscr{F}) = 0 \text{ for } i > 0.$$

As a useful corollary, we can embed any quasi-coherent sheaf on a Noetherian scheme into a quasi-coherent flasque sheaf. This is the tool we used in theorem 6.1.19 to compare Čech and derived-functor cohomology:

Corollary 6.2.10: Injective Embedding of Sheaves On Noetherian Schemes

Let X be a Noetherian scheme and $\mathscr{F}$ a quasi-coherent sheaf on X. Then $\mathscr{F}$ can be embedded in a quasi-coherent flasque sheaf.

***Proof*:**
Since X is Noetherian, it is quasi-compact, so it has a finite affine open cover $X = \bigcup_{j=1}^r U_j$ with $U_j = \operatorname{Spec} R_j$. For each j, let $\mathscr{F}|_{U_j} \cong \widetilde{M_j}$ for the R_j-module $M_j = \Gamma(U_j, \mathscr{F})$. Embed each M_j

into an injective R_j-module I_j, and define:

$$\mathscr{G} = \bigoplus_{j=1}^{r} (\iota_j)_*(\widetilde{I_j})$$

where $\iota_j : U_j \hookrightarrow X$ is the inclusion. For each j, the embedding $\mathscr{F}|_{U_j} \hookrightarrow \widetilde{I_j}$ induces $\mathscr{F} \to (\iota_j)_*(\widetilde{I_j})$ (by adjunction of restriction and pushforward). Taking the direct sum gives an injective morphism $\mathscr{F} \hookrightarrow \mathscr{G}$.

Each $\widetilde{I_j}$ is flasque by lemma 6.2.8 and quasi-coherent on U_j. The pushforward $(\iota_j)_*(\widetilde{I_j})$ is flasque (the pushforward of a flasque sheaf along any continuous map is flasque) and quasi-coherent by proposition 5.1.20[a]. A finite direct sum of flasque quasi-coherent sheaves is flasque and quasi-coherent, so $\mathscr{G}$ has the required properties.

[a]This is where the Noetherian hypothesis enters: the pushforward of a quasi-coherent sheaf along an open immersion is quasi-coherent when the target is Noetherian (or more generally, when the morphism is quasi-compact and quasi-separated), but can fail for non-Noetherian schemes.

The vanishing theorem tells us that the higher cohomology of quasi-coherent sheaves detects non-affineness. the converse is also true:

Theorem 6.2.11: Characterizing Affine Schemes via Cohomology

Let X be a Noetherian scheme. Then the following are equivalent:

1. X is affine.
2. $H^i(X, \mathscr{F}) = 0$ for all $i > 0$ and all quasi-coherent sheaves $\mathscr{F}$ on X.
3. $H^1(X, \mathscr{I}) = 0$ for all coherent sheaves of ideals $\mathscr{I} \subseteq \mathcal{O}_X$.

***Proof*:**
(1) ⇒ (2): This is theorem 6.2.9.

(2) ⇒ (3): Every coherent sheaf of ideals is quasi-coherent, so this is immediate.

(3) ⇒ (1): We verify the hypotheses of corollary 3.1.15: we shall find elements $f_1, \dots, f_r \in R = \Gamma(X, \mathcal{O}_X)$ such that each X_{f_j} is affine and $(f_1, \dots, f_r) = R$.

Step A: Finding the f_j. Let $\mathfrak{p}$ be a closed point of X and let U be an affine open neighborhood of $\mathfrak{p}$. Let $Y = X \setminus U$. Then Y and $Y \cup \{\mathfrak{p}\}$ are closed subsets, and their ideal sheaves $\mathscr{I}_Y \supseteq \mathscr{I}_{Y \cup \{\mathfrak{p}\}}$ fit in a short exact sequence:

$$0 \longrightarrow \mathscr{I}_{Y \cup \{\mathfrak{p}\}} \longrightarrow \mathscr{I}_Y \longrightarrow \kappa(\mathfrak{p}) \longrightarrow 0,$$

where $\kappa(\mathfrak{p}) = \mathcal{O}_{X,\mathfrak{p}}/\mathfrak{m}_\mathfrak{p}$ is the skyscraper sheaf at $\mathfrak{p}$. The long exact sequence gives:

$$\Gamma(X, \mathscr{I}_Y) \longrightarrow \Gamma(X, \kappa(\mathfrak{p})) \longrightarrow H^1(X, \mathscr{I}_{Y \cup \{\mathfrak{p}\}}) = 0,$$

where the last group vanishes by hypothesis (3). Hence the map $\Gamma(X, \mathscr{I}_Y) \to \kappa(\mathfrak{p})$ is surjective, so there exists $f \in \Gamma(X, \mathscr{I}_Y) \subseteq R$ with $f_\mathfrak{p} \equiv 1 \pmod{\mathfrak{m}_\mathfrak{p}}$.

By construction, f vanishes on Y, so $\mathfrak{p} \in X_f \subseteq U$. Furthermore, $X_f = U_{\bar{f}}$ where $\bar{f}$ is the image of f in $\Gamma(U, \mathcal{O}_U)$, so X_f is a distinguished open of the affine scheme U, hence affine.

Since every closed point of X has an affine neighborhood of the form X_f with $f \in R$, and X is Noetherian (hence quasi-compact) with dense closed points (by lemma 2.4.4), we can cover X by finitely many such opens: $X = X_{f_1} \cup \cdots \cup X_{f_r}$.

Step B: Showing $(f_1, \ldots, f_r) = R$. The elements $f_1, \ldots, f_r \in R$ define a morphism of sheaves $\alpha : \mathcal{O}_X^r \to \mathcal{O}_X$ by $(a_1, \ldots, a_r) \mapsto \sum_j f_j a_j$. Since the X_{f_j} cover X, the map α is surjective on stalks, hence surjective. Let $\mathscr{F} = \ker \alpha$:

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathcal{O}_X^r \xrightarrow{\alpha} \mathcal{O}_X \longrightarrow 0.$$

We need to show $H^1(X, \mathscr{F}) = 0$, for then $\Gamma(X, \mathcal{O}_X^r) \to \Gamma(X, \mathcal{O}_X)$ is surjective, giving $1 \in \operatorname{im}(\alpha)$, i.e. $\sum a_j f_j = 1$ for some $a_j \in R$, proving $(f_1, \ldots, f_r) = R$.

Filter $\mathscr{F}$ by:

$$0 = \mathscr{F} \cap \mathcal{O}_X^0 \subseteq \mathscr{F} \cap \mathcal{O}_X^1 \subseteq \cdots \subseteq \mathscr{F} \cap \mathcal{O}_X^r = \mathscr{F}$$

where $\mathcal{O}_X^j$ denotes the first j summands of $\mathcal{O}_X^r$. Each successive quotient $(\mathscr{F} \cap \mathcal{O}_X^{j+1})/(\mathscr{F} \cap \mathcal{O}_X^j)$ embeds into $\mathcal{O}_X^{j+1}/\mathcal{O}_X^j \cong \mathcal{O}_X$, so it is a coherent sheaf of ideals. By hypothesis (3), its H^1 vanishes. Climbing up the filtration via the long exact sequence, we deduce $H^1(X, \mathscr{F}) = 0$.

Hence $(f_1, \ldots, f_r) = R$, and by corollary 3.1.15, X is affine.

6.3 Application: Sheaf Cohomology of Projective Spaces

Let us now compute the cohomology of twisted sheaves on projective space — the single most important explicit computation in the cohomology of schemes. Let R be a Noetherian ring, $S_\bullet = R[x_0, \ldots, x_n]$ with the standard grading, and $X = \operatorname{Proj} S_\bullet = \mathbb{P}_R^n$. Recall from definition 5.3.4 that the Serre twisting sheaf $\mathcal{O}_X(1)$ is the invertible sheaf whose sections on the basic open $D_+(x_i)$ are the degree-1 elements of $(S_{x_i})_0$. For any sheaf $\mathscr{F}$ of $\mathcal{O}_X$-modules, we write:

$$\Gamma_*(\mathscr{F}) = \bigoplus_{d \in \mathbb{Z}} \Gamma(X, \mathscr{F}(d)),$$

which carries a natural structure of graded $S_\bullet$-module (multiplication by x_i corresponds to the natural map $\mathscr{F}(d) \to \mathscr{F}(d+1)$).

Theorem 6.3.1: Twisted Sheaf Cohomology

Let R be a Noetherian ring and let $X = \mathbb{P}^n_R$, with $n \geq 1$. Then:

1. $H^0(X, \mathcal{O}_X(d)) \cong (S_\bullet)_d$ for all $d \geq 0$, and $H^0(X, \mathcal{O}_X(d)) = 0$ for $d < 0$.
2. $H^n(X, \mathcal{O}_X(-n-1)) \cong R$.
3. The natural multiplication map
$$H^0(X, \mathcal{O}_X(d)) \times H^n(X, \mathcal{O}_X(-d-n-1)) \longrightarrow H^n(X, \mathcal{O}_X(-n-1)) \cong R$$
is a perfect pairing of finitely generated free R-modules, for each $d \geq 0$.
4. $H^i(X, \mathcal{O}_X(d)) = 0$ for $0 < i < n$ and all $d \in \mathbb{Z}$.

In summary:
$$H^i\left(\mathbb{P}^n_R, \mathcal{O}_{\mathbb{P}^n_R}(d)\right) = \begin{cases} (R[x_0, \ldots, x_n])_d & i = 0,\ d \geq 0, \\ \left(\dfrac{1}{x_0 x_1 \cdots x_n} R[x_0^{-1}, \ldots, x_n^{-1}]\right)_d & i = n,\ d \leq -(n+1), \\ 0 & \text{otherwise.} \end{cases}$$

6.3.2 Remark: Sheaf vs. Singular Cohomology The reader should note that this result is fundamentally different from the singular cohomology of complex projective space. When $R = \mathbb{C}$ and $d = 0$, the theorem says $H^i(\mathbb{P}^n_\mathbb{C}, \mathcal{O}_{\mathbb{P}^n_\mathbb{C}}) = 0$ for all $i > 0$, whereas the singular cohomology $H^i_{\text{sing}}(\mathbb{C}\mathbb{P}^n; \mathbb{Z})$ is nontrivial: it equals $\mathbb{Z}$ in each even degree $0 \leq i \leq 2n$ and vanishes in odd degrees (see [51, section 4.1]). The discrepancy arises because sheaf cohomology with coefficients in $\mathcal{O}_X$ measures obstructions to extending *holomorphic* (or algebraic) sections, not topological ones. Singular cohomology would instead correspond to sheaf cohomology with coefficients in the constant sheaf $\underline{\mathbb{Z}}$, which we shall connect via GAGA in section 6.9.

***Proof*:**
We use Čech cohomology with respect to the standard affine cover. Let $\mathscr{F} = \bigoplus_{d \in \mathbb{Z}} \mathcal{O}_X(d)$. By corollary 6.2.3, cohomology commutes with direct sums on Noetherian spaces, so $H^i(X, \mathscr{F}) = \bigoplus_{d \in \mathbb{Z}} H^i(X, \mathcal{O}_X(d))$. Hence it suffices to compute the cohomology of $\mathscr{F}$ as a graded module and then read off each graded piece.

Let $\mathcal{U} = \{U_0, \ldots, U_n\}$ where $U_j = D_+(x_j)$ is the standard affine cover of $\operatorname{Proj} S_\bullet$. By theorem 6.1.19, the Čech cohomology $\check{H}^p(\mathcal{U}, \mathscr{F})$ computes the derived-functor cohomology $H^p(X, \mathscr{F})$. The sections of $\mathscr{F}$ on each intersection are:
$$\mathscr{F}(U_{i_0, \ldots, i_p}) = S_{x_{i_0} \cdots x_{i_p}}$$
(the localization of $S_\bullet$ at the product $x_{i_0} \cdots x_{i_p}$, retaining only degree-0 elements for each summand $\mathcal{O}_X(d)$, but over all d we obtain the full localization). The Čech complex is therefore:
$$C^\bullet(\mathcal{U}, \mathscr{F}) : \quad \prod_i S_{x_i} \xrightarrow{d^0} \prod_{i<j} S_{x_i x_j} \xrightarrow{d^1} \cdots \xrightarrow{d^{n-1}} S_{x_0 x_1 \cdots x_n}.$$

Proof of (1): Computing H^0. By lemma 6.1.14, $H^0(X, \mathscr{F}) = \Gamma(X, \mathscr{F})$. The global sections of $\mathscr{F}$ are the elements lying in all localizations simultaneously: $\Gamma(X, \mathscr{F}) = \bigcap_{i=0}^n S_{x_i} \cap S = S$ by proposition 2.3.6. The graded piece of degree d is $(S_\bullet)_d$, which is a free R-module with basis $\{x_0^{m_0} \cdots x_n^{m_n} \big| m_i \geq 0, \sum m_i = d\}$ when $d \geq 0$, and is zero when $d < 0$ (there are no monomials of negative degree in a polynomial ring). Hence $H^0(X, \mathcal{O}_X(d)) \cong (S_\bullet)_d$ for $d \geq 0$ and $H^0(X, \mathcal{O}_X(d)) = 0$ for $d < 0$.

Proof of (2): Computing H^n. The group $H^n(X, \mathscr{F})$ is the cokernel of the last differential:

$$d^{n-1} : \prod_{j=0}^n S_{x_0 \cdots \widehat{x_j} \cdots x_n} \longrightarrow S_{x_0 x_1 \cdots x_n}.$$

The target $S_{x_0 \cdots x_n}$ is a free R-module with basis consisting of all Laurent monomials $\{x_0^{l_0} \cdots x_n^{l_n} \big| l_i \in \mathbb{Z}\}$. The image of d^{n-1} is the R-submodule generated by those Laurent monomials $x_0^{l_0} \cdots x_n^{l_n}$ for which at least one $l_j \geq 0$: indeed, a monomial with $l_j \geq 0$ lies in $S_{x_0 \cdots \widehat{x_j} \cdots x_n}$, so it is in the image of the j-th summand. Conversely, every element of each summand has at least one non-negative exponent.

Therefore, $H^n(X, \mathscr{F})$ is the free R-module with basis

$$\{x_0^{l_0} \cdots x_n^{l_n} \big| l_i < 0 \text{ for all } i\},$$

graded by total degree $\sum l_i$. In degree $-(n+1)$, there is exactly one such monomial, namely $x_0^{-1} x_1^{-1} \cdots x_n^{-1}$. Hence:

$$H^n(X, \mathcal{O}_X(-n-1)) \cong R.$$

More generally, for $d \leq -(n+1)$, the group $H^n(X, \mathcal{O}_X(d))$ is a free R-module of rank $\binom{-d-1}{n}$, and $H^n(X, \mathcal{O}_X(d)) = 0$ for $d > -(n+1)$ (since there are no "all-negative" monomials of degree greater than $-(n+1)$).

Proof of (3): The perfect pairing. For $d \geq 0$, the module $H^0(X, \mathcal{O}_X(d))$ is free over R with basis $\{x_0^{m_0} \cdots x_n^{m_n} \big| m_i \geq 0, \sum m_i = d\}$, and $H^n(X, \mathcal{O}_X(-d-n-1))$ is free with basis $\{x_0^{l_0} \cdots x_n^{l_n} \big| l_i < 0, \sum l_i = -d-n-1\}$. The multiplication map

$$H^0(X, \mathcal{O}_X(d)) \times H^n(X, \mathcal{O}_X(-d-n-1)) \longrightarrow H^n(X, \mathcal{O}_X(-n-1)) \cong R$$

sends $(x_0^{m_0} \cdots x_n^{m_n}) \cdot (x_0^{l_0} \cdots x_n^{l_n}) = x_0^{m_0+l_0} \cdots x_n^{m_n+l_n}$, and this product is nonzero in H^n if and only if all exponents remain strictly negative, i.e. $m_j + l_j < 0$ for all j. In particular, the element $x_0^{-m_0-1} \cdots x_n^{-m_n-1}$ (which has total degree $-d-n-1$) pairs with $x_0^{m_0} \cdots x_n^{m_n}$ to give $x_0^{-1} \cdots x_n^{-1}$, the generator of $H^n(X, \mathcal{O}_X(-n-1)) \cong R$. Any other basis monomial in $H^n(X, \mathcal{O}_X(-d-n-1))$ pairs with $x_0^{m_0} \cdots x_n^{m_n}$ to produce a monomial with at least one non-negative exponent, hence zero in H^n.

This shows that the monomial bases are dual to each other under the pairing, so the pairing is perfect.

Proof of (4): Vanishing of intermediate cohomology. We proceed by induction on n. For $n = 1$, the range $0 < i < 1$ is empty, so there is nothing to prove.

Let $n > 1$. We first show that every element of $H^i(X, \mathscr{F})$, for $0 < i < n$, is annihilated by some power of x_n. Localizing the Čech complex $C^\bullet(\mathcal{U}, \mathscr{F})$ at x_n yields the Čech complex of $\mathscr{F}|_{U_n}$ on the affine scheme $U_n = D_+(x_n)$ with respect to the open cover $\{U_j \cap U_n \big| 0 \leq j \leq n\}$. Each set in this cover is a distinguished affine open of U_n, and U_n itself is one of them. Since U_n is affine and $\mathscr{F}|_{U_n}$ is quasi-coherent, $H^i(U_n, \mathscr{F}|_{U_n}) = 0$ for $i > 0$ by theorem 6.2.9. Since localization is exact, it follows that $H^i(X, \mathscr{F})_{x_n} = 0$ for $i > 0$. In other words, every element of $H^i(X, \mathscr{F})$ is annihilated by some power of x_n.

Next, we show that for $0 < i < n$, multiplication by x_n induces a bijection on $H^i(X, \mathscr{F})$, forcing this module to be zero. Consider the exact sequence of graded $S_\bullet$-modules:

$$0 \longrightarrow S_\bullet(-1) \xrightarrow{\cdot x_n} S_\bullet \longrightarrow S_\bullet/(x_n) \longrightarrow 0,$$

which gives the short exact sequence of sheaves on X:

$$0 \longrightarrow \mathcal{O}_X(-1) \xrightarrow{\cdot x_n} \mathcal{O}_X \longrightarrow \mathcal{O}_H \longrightarrow 0,$$

where $H = V_+(x_n) \cong \mathbb{P}^{n-1}_R$ is the hyperplane $x_n = 0$. Twisting by all $d \in \mathbb{Z}$ and taking the direct sum:

$$0 \longrightarrow \mathscr{F}(-1) \xrightarrow{\cdot x_n} \mathscr{F} \longrightarrow \mathscr{F}_H \longrightarrow 0,$$

where $\mathscr{F}_H = \bigoplus_{d\in\mathbb{Z}} \mathcal{O}_H(d)$. The long exact sequence in cohomology gives:

$$\cdots \longrightarrow H^i(X, \mathscr{F}(-1)) \xrightarrow{\cdot x_n} H^i(X, \mathscr{F}) \longrightarrow H^i(X, \mathscr{F}_H) \longrightarrow \cdots$$

As graded $S_\bullet$-modules, $H^i(X, \mathscr{F}(-1))$ is simply $H^i(X, \mathscr{F})$ shifted by one degree, and the connecting map is multiplication by x_n.

Now $H \cong \mathbb{P}^{n-1}_R$, and $H^i(X, \mathscr{F}_H) = H^i(H, \bigoplus_d \mathcal{O}_H(d))$ by proposition 6.2.4. Applying the inductive hypothesis to $\mathbb{P}^{n-1}_R$, we get $H^i(X, \mathscr{F}_H) = 0$ for $0 < i < n-1$.

For $0 < i < n-1$: The long exact sequence gives $H^i(X, \mathscr{F}(-1)) \xrightarrow{\sim} H^i(X, \mathscr{F})$, i.e. multiplication by x_n is bijective. Since every element is also killed by a power of x_n, we conclude $H^i(X, \mathscr{F}) = 0$.

For $i = n-1$ (when $n \geq 2$): We must show multiplication by x_n is still bijective on $H^{n-1}(X, \mathscr{F})$. The relevant portion of the long exact sequence is:

$$H^{n-2}(X, \mathscr{F}_H) \longrightarrow H^{n-1}(X, \mathscr{F}(-1)) \xrightarrow{\cdot x_n} H^{n-1}(X, \mathscr{F}) \longrightarrow H^{n-1}(X, \mathscr{F}_H) \xrightarrow{\partial} H^n(X, \mathscr{F}(-1)).$$

First, we establish injectivity on the left. If $n \geq 3$, then $H^{n-2}(X, \mathscr{F}_H) = 0$ by the inductive hypothesis (since $0 < n-2 < \dim H = n-1$). If $n = 2$, the term on the left is $H^0(X, \mathscr{F}_H)$; in this case, the preceding map in the sequence is $H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}_H)$, which is the natural surjection $S_\bullet \to S_\bullet/(x_n)$ by part (1). Exactness then forces the map $H^0(X, \mathscr{F}_H) \to H^1(X, \mathscr{F}(-1))$ to be the zero map. In either case, multiplication by x_n is injective on $H^{n-1}(X, \mathscr{F}(-1))$.

Next, we establish surjectivity on the right. The term $H^{n-1}(X, \mathscr{F}_H)$ is the top cohomology of $H \cong \mathbb{P}^{n-1}_R$, which by part (2) is a free R-module generated by all-negative Laurent monomials in the variables $x_0, \ldots, x_{n-1}$. The connecting homomorphism ∂ maps a basis element $x_0^{l_0} \cdots x_{n-1}^{l_{n-1}}$ to the element $x_0^{l_0} \cdots x_{n-1}^{l_{n-1}} x_n^{-1}$ in $H^n(X, \mathscr{F}(-1))$. This map is visibly injective. Because ∂ is injective, exactness forces the preceding map $H^{n-1}(X, \mathscr{F}) \to H^{n-1}(X, \mathscr{F}_H)$ to be the zero map.

Therefore, multiplication by x_n is surjective on $H^{n-1}(X, \mathscr{F})$. Being both injective and surjective, multiplication by x_n is bijective on $H^{n-1}(X, \mathscr{F})$, and the annihilation argument completes the proof.

Cohomology of the Twisted Sheaves of Differentials

The same two inputs compute the cohomology of every $\Omega^p_X(k)$ on projective space, not just of the line bundles. The mechanism is the Euler sequence (theorem 5.5.18): its exterior powers express each Ω^p in terms of sums of line bundles, and theorem 6.3.1 finishes each step. The vanishing statement below is the one consumers ask for, and it is sharp: the exceptions listed are genuinely nonzero.

Theorem 6.3.3: Bott Vanishing

Let k be a field, $X = \mathbb{P}^n_k$ with $n \geq 1$, and $0 \leq p \leq n$. Then

$$H^q\big(X,\ \Omega^p_X(m)\big) = 0$$

for every q and m except in the following three cases:

1. $q = 0$ and $m > p$;
2. $q = n$ and $m < p - n$;
3. $q = p$ and $m = 0$, where the group is one-dimensional over k.

In particular $H^q(X, \Omega^p_X) = 0$ for $q \neq p$ and $H^p(X, \Omega^p_X) \cong k$, and $H^q(X, \Omega^p_X(m)) = 0$ for all $0 < q < n$ whenever $m \neq 0$.

***Proof*:**
Write $N = \binom{n+1}{p}$. Taking p-th exterior powers in the Euler sequence

$$0 \longrightarrow \Omega_X \longrightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_X \longrightarrow 0$$

of theorem 5.5.18 gives, for each $p \geq 1$, a short exact sequence

$$0 \longrightarrow \Omega^p_X(m) \longrightarrow \mathcal{O}_X(m-p)^{\oplus N} \longrightarrow \Omega^{p-1}_X(m) \longrightarrow 0 \tag{6.1}$$

The exterior power of a short exact sequence of locally free sheaves with locally free quotient is exact in this form: locally the sequence splits, and for a split sequence $0 \to A \to A \oplus B \to B \to 0$ the filtration of $\Lambda^p(A \oplus B)$ by the number of factors from A has $\Lambda^p A$ as its bottom piece and $\Lambda^{p-1}A \otimes B$ as the next; here $B = \mathcal{O}_X$ has rank one, so the filtration has exactly two steps and (6.1) is what remains after twisting by $\mathcal{O}_X(m)$ and using $\Lambda^p(\mathcal{O}_X(-1)^{\oplus(n+1)}) \cong \mathcal{O}_X(-p)^{\oplus N}$. The identification is independent of the local splitting because the two extreme pieces of the filtration are canonical.

Now induct on p. For $p = 0$ the claim is theorem 6.3.1: the cohomology of $\mathcal{O}_X(m)$ vanishes except in degree 0 for $m \geq 0$ and degree n for $m \leq -n-1$, which is cases (1) and (2) with $p = 0$, and case (3) reads $H^0(X, \mathcal{O}_X) = k$.

Let $p \geq 1$ and assume the statement for $p - 1$. The long exact sequence of (6.1) reads

$$\cdots \to H^{q-1}\big(\Omega^{p-1}_X(m)\big) \to H^q\big(\Omega^p_X(m)\big) \to H^q\big(\mathcal{O}_X(m-p)\big)^{\oplus N} \to H^q\big(\Omega^{p-1}_X(m)\big) \to \cdots$$

so $H^q(\Omega^p_X(m))$ is squeezed between $H^{q-1}(\Omega^{p-1}_X(m))$ and $H^q(\mathcal{O}_X(m-p))^{\oplus N}$, and vanishes as soon as both do.

Middle degrees. Let $0 < q < n$. Then $H^q(\mathcal{O}_X(m-p)) = 0$ by theorem 6.3.1(4). By induction $H^{q-1}(\Omega_X^{p-1}(m))$ vanishes unless $q-1 = p-1$ and $m = 0$, that is unless $q = p$ and $m = 0$. So for $m \neq 0$ all middle-degree groups vanish, and for $m = 0$ the only possibly nonzero one is at $q = p$, where the sequence gives $H^p(\Omega_X^p) \cong H^{p-1}(\Omega_X^{p-1}) \cong k$ by induction, the neighbouring $\mathcal{O}_X(-p)$ terms vanishing since $0 < p < n$ forces H^{p-1} and H^p of $\mathcal{O}_X(-p)$ to be zero. (For $p = n$ the group $H^n(\mathcal{O}_X(-n))$ also vanishes, as $-n > -n-1$.)

Degree zero. $H^0(\Omega_X^p(m))$ injects into $H^0(\mathcal{O}_X(m-p))^{\oplus N}$, which is zero for $m < p$; so H^0 vanishes unless $m \geq p$. At $m = p$ the sequence identifies $H^0(\Omega_X^p(p))$ with the kernel of $k^{\oplus N} \to H^0(\Omega_X^{p-1}(p))$, and that map is injective, as one checks on the standard basis of Λ^p of the coordinate forms; hence the exceptional case is $m > p$, which is (1).

Degree n. Dually, $H^n(\Omega_X^p(m))$ receives $H^n(\mathcal{O}_X(m-p))^{\oplus N}$ and surjects onto nothing beyond it, and $H^n(\mathcal{O}_X(m-p)) = 0$ unless $m - p \leq -n-1$, that is unless $m \leq p - n - 1$, which is (2). Combined with the induction hypothesis for $H^{n-1}(\Omega_X^{p-1}(m))$, which vanishes for $n-1 \neq p-1$ or $m \neq 0$, this leaves exactly the stated cases.

The diagonal case $H^p(X, \Omega_X^p) \cong k$ is the algebraic form of a topological fact: it is generated by the p-th power of the hyperplane class, and it is what makes the Hodge diamond of $\mathbb{P}^n$ a single line of 1s down the diagonal. The vanishing in the middle degrees is what allows the cohomology of a hypersurface in $\mathbb{P}^{n+1}$ to be read off a graded ring, which is how Hodge theory computes the Hodge numbers of hypersurfaces.

6.4 Ext Groups and Sheaf Ext

The cohomology groups $H^i(X, \mathscr{F})$ are the derived functors of $\Gamma(X, -) = \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, -)$. From this perspective, cohomology measures the failure of the global-section functor to be exact. But $\operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, -)$ is a special case of $\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$ for a general $\mathcal{O}_X$-module $\mathscr{F}$; the derived functors of this more general functor are the *Ext groups*. These carry strictly more information than cohomology alone: while $H^i(X, \mathscr{G})$ measures the obstruction to extending global sections of $\mathscr{G}$, the group $\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ measures the obstruction to extending morphisms from $\mathscr{F}$ to $\mathscr{G}$. In Serre duality, Ext groups will serve as the natural "dual" objects to cohomology groups, and computing them on projective space will be the key step.

There are two kinds of Ext: the *global Ext* groups $\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$, which are abelian groups (or modules over the global sections ring), and the *sheaf Ext* $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$, which is itself a sheaf on X. These are related by a spectral sequence, and in the most important case — when $\mathscr{F}$ is locally free — this spectral sequence degenerates, reducing Ext computations to ordinary cohomology. This reduction is what makes the theory practical.

6.4.1 Global and Local Ext

Recall from definition 1.2.21 that for $\mathcal{O}_X$-modules $\mathscr{F}$ and $\mathscr{G}$ on a ringed space $(X, \mathcal{O}_X)$, the *sheaf Hom* is the sheaf:

$$\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})(U) = \operatorname{Hom}_{\mathcal{O}_U}(\mathscr{F}|_U, \mathscr{G}|_U),$$

and the *global Hom* is:

$$\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = \Gamma(X, \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})) = \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})(X).$$

Both $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$ and $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)$ are left-exact functors. Since the category $\mathcal{O}_X$-**Mod** has enough injectives (by proposition 6.1.1), we may form their right derived functors:

Definition 6.4.1: Ext Groups for Sheaves

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{F}, \mathscr{G}$ be $\mathcal{O}_X$-modules.

1. The *global Ext groups* are the right derived functors of $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)$:
$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = R^i\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, -)(\mathscr{G}).$$
Concretely, choose an injective resolution $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ and compute:
$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = h^i\big(\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)\big).$$
2. The *sheaf Ext* (or *local Ext*) functors are the right derived functors of $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)$:
$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = R^i\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)(\mathscr{G}).$$
Concretely, choose an injective resolution $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ and compute:
$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = h^i\big(\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)\big).$$
This is a sheaf of $\mathcal{O}_X$-modules on X.

By convention, $\mathrm{Ext}^0_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = \mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ and $\mathcal{E}xt^0_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$.

Note the asymmetry: we derive in the *second* variable. One can also derive in the first variable (resolving $\mathscr{F}$ by locally free sheaves), and a standard comparison argument shows this gives the same Ext groups[5]. Both Ext groups come equipped with the standard long exact sequences of derived functors:

[5]More precisely, if $\mathscr{F}$ admits a locally free resolution $\mathcal{E}^\bullet \to \mathscr{F} \to 0$ (e.g., if X is a smooth variety or if $\mathscr{F}$ is coherent on projective space), then $\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \cong h^i(\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathscr{G}))$. This is the "balancing" of Ext; see [15] for the proof in the module category, which carries over to sheaves.

Proposition 6.4.2: Long Exact Sequences for Ext

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{F}$ be an $\mathcal{O}_X$-module.

1. *(Covariant variable.)* A short exact sequence $0 \to \mathscr{G}' \to \mathscr{G} \to \mathscr{G}'' \to 0$ of $\mathcal{O}_X$-modules induces long exact sequences:
$$0 \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}') \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}'') \xrightarrow{\delta} \mathrm{Ext}^1(\mathscr{F}, \mathscr{G}') \to \cdots$$
and similarly for $\mathcal{E}xt$.

2. *(Contravariant variable.)* A short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of $\mathcal{O}_X$-modules induces a long exact sequence:
$$0 \to \mathrm{Hom}(\mathscr{F}'', \mathscr{G}) \to \mathrm{Hom}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}(\mathscr{F}', \mathscr{G}) \xrightarrow{\delta} \mathrm{Ext}^1(\mathscr{F}'', \mathscr{G}) \to \cdots$$
and similarly for $\mathcal{E}xt$.

***Proof*:**
Part (1) is the standard long exact sequence of a right derived functor applied to a short exact sequence. Part (2) follows from the corresponding result for modules (see [15]), adapted to the sheaf setting by computing on injective resolutions of $\mathscr{G}$ and using the left-exactness of $\mathrm{Hom}(-, \mathscr{G})$ in the first variable.

The next result shows that Ext can be computed from a locally free resolution of the first argument, which is often more convenient than resolving the second argument by injectives:

Proposition 6.4.3: Computing Ext via Locally Free Resolutions

Let X be a Noetherian scheme and let $\mathscr{F}, \mathscr{G}$ be $\mathcal{O}_X$-modules with $\mathscr{F}$ coherent. Suppose $\mathscr{F}$ admits a locally free resolution:
$$\cdots \to \mathcal{E}^{-1} \to \mathcal{E}^0 \to \mathscr{F} \to 0.$$
Then:
$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \cong h^i\big(\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathscr{G})\big) \qquad \text{and} \qquad \mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \cong h^i\big(\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}^\bullet, \mathscr{G})\big).$$
In particular, if $\mathscr{F}$ is locally free, then $\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = 0$ and $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = 0$ for all $i > 0$.

***Proof*:**
The first statement is the "balancing" of Ext. Since locally free sheaves are $\mathrm{Hom}(\mathscr{F}, -)$-acyclic (a morphism from a free module is determined by where the generators go, and this respects surjections), the locally free resolution computes the same derived functors as an injective resolution of $\mathscr{G}$. For the sheaf version, the same argument works on each open set. The vanishing for locally free $\mathscr{F}$ follows because the resolution $\mathcal{E}^\bullet = [\mathscr{F}]$ (concentrated in degree 0) is already a locally free resolution, so $h^i(\mathrm{Hom}(\mathcal{E}^\bullet, \mathscr{G})) = 0$ for $i > 0$.

The sheaf Ext is a "local" invariant, while the global Ext is obtained by "assembling" the local data via global sections. The following result makes this precise at the stalk level:

Proposition 6.4.4: Stalks of Sheaf Ext

Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathscr{F}$ and $\mathscr{G}$ be $\mathcal{O}_X$-modules, and $x \in X$. Then there is a natural isomorphism:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})_x \;\cong\; \operatorname{Ext}^i_{\mathcal{O}_{X,x}}(\mathscr{F}_x, \mathscr{G}_x).$$

In particular, if $\mathscr{F}$ is locally free, then $\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = 0$ for all $i > 0$ (since the stalks $\mathscr{F}_x$ are free $\mathcal{O}_{X,x}$-modules, and Ext over a local ring vanishes for free first arguments).

***Proof*:**
Choose an injective resolution $0 \to \mathscr{G} \to \mathscr{I}^\bullet$. Taking stalks at x gives a resolution $0 \to \mathscr{G}_x \to \mathscr{I}^\bullet_x$. The stalk of the sheaf Ext is:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})_x = h^i\big(\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)\big)_x = h^i\big(\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^\bullet)_x\big)$$

where the second equality uses the exactness of the stalk functor (by **??**). Now, $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, \mathscr{I}^p)_x = \operatorname{Hom}_{\mathcal{O}_{X,x}}(\mathscr{F}_x, \mathscr{I}^p_x)$ (the stalk of the sheaf Hom is the module Hom of the stalks). It remains to show that $\mathscr{I}^p_x$ is an injective $\mathcal{O}_{X,x}$-module. This follows from the next lemma.

Lemma 6.4.5: Injective Sheaves and Restriction

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{I}$ be an injective $\mathcal{O}_X$-module.

1. For any open set $U \subseteq X$, the restriction $\mathscr{I}|_U$ is an injective $\mathcal{O}_U$-module.
2. For any point $x \in X$, the stalk $\mathscr{I}_x$ is an injective $\mathcal{O}_{X,x}$-module.

***Proof*:**
(1) Let $j : U \hookrightarrow X$ denote the inclusion. The functor $j_! : \mathcal{O}_U\text{-}\mathbf{Mod} \to \mathcal{O}_X\text{-}\mathbf{Mod}$ (extension by zero) is exact and left adjoint to the restriction functor j^* (by proposition 1.3.12). Hence j^* preserves injectives: for any monomorphism $\alpha : \mathscr{F} \hookrightarrow \mathscr{G}$ of $\mathcal{O}_U$-modules, the map

$$\operatorname{Hom}_{\mathcal{O}_U}(\mathscr{G}, \mathscr{I}|_U) = \operatorname{Hom}_{\mathcal{O}_X}(j_!\mathscr{G}, \mathscr{I}) \longrightarrow \operatorname{Hom}_{\mathcal{O}_X}(j_!\mathscr{F}, \mathscr{I}) = \operatorname{Hom}_{\mathcal{O}_U}(\mathscr{F}, \mathscr{I}|_U)$$

is surjective because $j_!\alpha$ is a monomorphism (exactness of $j_!$) and $\mathscr{I}$ is injective.

(2) The stalk $\mathscr{I}_x = \varinjlim_{U \ni x} \mathscr{I}(U)$ can be computed as follows. Let $j_x : \{x\} \hookrightarrow X$ be the inclusion and consider the skyscraper functor $(j_x)_* : \mathcal{O}_{X,x}\text{-}\mathbf{Mod} \to \mathcal{O}_X\text{-}\mathbf{Mod}$. Its left adjoint is the stalk functor $\mathscr{G} \mapsto \mathscr{G}_x$. Since the stalk functor is exact, it preserves injectives by the same adjunction argument as in part (1).

As a consequence, sheaf Ext behaves well with respect to restricting to open subsets:

Proposition 6.4.6: Open Subsets and Ext Groups

Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathscr{F}$ and $\mathscr{G}$ be $\mathcal{O}_X$-modules, and $U \subseteq X$ an open subset. Then there is a natural isomorphism of sheaves on U:

$$\mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F},\mathscr{G})\big|_U \;\cong\; \mathcal{E}xt^i_{\mathcal{O}_U}(\mathscr{F}|_U,\mathscr{G}|_U).$$

***Proof*:**
Let $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ be an injective resolution on X. By lemma 6.4.5(1), the restricted sequence $0 \to \mathscr{G}|_U \to \mathscr{I}^\bullet|_U$ is an injective resolution on U. Then:

$$\mathcal{E}xt^i_{\mathcal{O}_U}(\mathscr{F}|_U,\mathscr{G}|_U) = h^i\big(\mathcal{H}om_{\mathcal{O}_U}(\mathscr{F}|_U,\mathscr{I}^\bullet|_U)\big) = h^i\big(\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F},\mathscr{I}^\bullet)|_U\big) = \mathcal{E}xt^i_{\mathcal{O}_X}(\mathscr{F},\mathscr{G})\big|_U,$$

where the middle equality uses the fact that $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F},\mathscr{I}^p)|_U = \mathcal{H}om_{\mathcal{O}_U}(\mathscr{F}|_U,\mathscr{I}^p|_U)$ (the sheaf Hom is computed open-set by open-set), and the last equality uses the exactness of restriction (which commutes with taking cohomology of a complex of sheaves).

6.4.2 The Local-to-Global Spectral Sequence

The global Ext and the sheaf Ext are related by the factorization:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F},-) = \Gamma(X,-) \circ \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F},-).$$

This is simply the statement that a global morphism $\mathscr{F} \to \mathscr{G}$ is a global section of the sheaf Hom. Both functors on the right are left exact, and this composition of functors is the standard setting for the Grothendieck spectral sequence[6]. For this spectral sequence to exist, we need $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F},-)$ to send injective objects to $\Gamma(X,-)$-acyclic objects. The following key lemma establishes this:

Lemma 6.4.7: Sheaf Hom of Injectives is Flasque

Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathscr{F}$ an $\mathcal{O}_X$-module, and $\mathscr{I}$ an injective $\mathcal{O}_X$-module. Then $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F},\mathscr{I})$ is a flasque sheaf.

***Proof*:**
Let $U \subseteq V$ be open subsets of X. We must show that the restriction map

$$\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F},\mathscr{I})(V) = \mathrm{Hom}_{\mathcal{O}_V}(\mathscr{F}|_V,\,\mathscr{I}|_V) \longrightarrow \mathrm{Hom}_{\mathcal{O}_U}(\mathscr{F}|_U,\,\mathscr{I}|_U) = \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F},\mathscr{I})(U)$$

is surjective. Let $\varphi : \mathscr{F}|_U \to \mathscr{I}|_U$ be a morphism of $\mathcal{O}_U$-modules. Let $j : U \hookrightarrow V$ be the inclusion. By the adjunction $j_! \dashv j^*$, the morphism φ corresponds to a morphism $j_!(\mathscr{F}|_U) \to \mathscr{I}|_V$. The natural map $j_!(\mathscr{F}|_U) \to \mathscr{F}|_V$ gives a monomorphism in the category of $\mathcal{O}_V$-modules. Since $\mathscr{I}|_V$ is injective (by lemma 6.4.5), the morphism $j_!(\mathscr{F}|_U) \to \mathscr{I}|_V$ extends to a morphism $\psi : \mathscr{F}|_V \to \mathscr{I}|_V$. By construction, $\psi|_U = \varphi$, so the restriction map is surjective.

[6]See *Homological Algebra* [15] for the construction of the Grothendieck spectral sequence and the conditions under which it is defined.

Since flasque sheaves are $\Gamma(X, -)$-acyclic (by corollary 6.1.7), the hypotheses of the Grothendieck spectral sequence are satisfied. We obtain:

Theorem 6.4.8: Local-to-Global Ext Spectral Sequence

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathscr{F}, \mathscr{G}$ be $\mathcal{O}_X$-modules. Then there is a convergent spectral sequence:

$$E_2^{p,q} = H^p\big(X,\, \mathcal{E}xt^q_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})\big) \implies \operatorname{Ext}^{p+q}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}).$$

This is natural in both $\mathscr{F}$ and $\mathscr{G}$.

***Proof*:**
This is the Grothendieck spectral sequence for the composite functor $\operatorname{Hom}_{\mathcal{O}_X}(\mathscr{F}, -) = \Gamma(X, -) \circ \mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)$. By lemma 6.4.7, the functor $\mathcal{H}om_{\mathcal{O}_X}(\mathscr{F}, -)$ sends injective $\mathcal{O}_X$-modules to flasque sheaves, which are $\Gamma(X, -)$-acyclic. The E_2-page of the Grothendieck spectral sequence is $R^p\Gamma \circ R^q\mathcal{H}om(\mathscr{F}, -) = H^p(X, \mathcal{E}xt^q(\mathscr{F}, -))$, and the abutment is $R^{p+q}(\Gamma \circ \mathcal{H}om(\mathscr{F}, -)) = \operatorname{Ext}^{p+q}(\mathscr{F}, -)$.

The spectral sequence simplifies dramatically in the most important case:

Corollary 6.4.9: Degeneration for Locally Free Sheaves

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathcal{L}$ be a locally free $\mathcal{O}_X$-module of finite rank, with dual $\mathcal{L}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$. Then for all $\mathcal{O}_X$-modules $\mathscr{G}$ and all $i \geq 0$:

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G}) \;\cong\; H^i(X,\, \mathcal{L}^\vee \otimes_{\mathcal{O}_X} \mathscr{G}).$$

In particular, setting $\mathcal{L} = \mathcal{O}_X$ recovers sheaf cohomology: $\operatorname{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_X, \mathscr{G}) \cong H^i(X, \mathscr{G})$.

***Proof*:**
By proposition 6.4.4, the stalk of $\mathcal{E}xt^q_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})$ at $x \in X$ is $\operatorname{Ext}^q_{\mathcal{O}_{X,x}}(\mathcal{L}_x, \mathscr{G}_x)$. Since $\mathcal{L}$ is locally free, $\mathcal{L}_x$ is a free $\mathcal{O}_{X,x}$-module, so $\operatorname{Ext}^q_{\mathcal{O}_{X,x}}(\mathcal{L}_x, \mathscr{G}_x) = 0$ for all $q > 0$. Hence $\mathcal{E}xt^q_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G}) = 0$ for $q > 0$ (a sheaf with all zero stalks is the zero sheaf).

The spectral sequence of theorem 6.4.8 collapses at the E_2-page: only the $q = 0$ row survives, giving:

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G}) \;\cong\; H^i\big(X,\, \mathcal{E}xt^0_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})\big) = H^i\big(X,\, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})\big).$$

Finally, for a locally free sheaf of finite rank, the natural map $\mathcal{L}^\vee \otimes_{\mathcal{O}_X} \mathscr{G} \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathscr{G})$ is an isomorphism (this can be checked on stalks, where it reduces to the standard isomorphism $M^\vee \otimes N \xrightarrow{\sim} \operatorname{Hom}_R(M, N)$ for a free module M of finite rank over a ring R). Hence $\operatorname{Ext}^i(\mathcal{L}, \mathscr{G}) \cong H^i(X, \mathcal{L}^\vee \otimes \mathscr{G})$.

6.4.10 Remark: Why Locally Free is the Key Case The reader should pause to appreciate how much corollary 6.4.9 simplifies matters. For a general coherent sheaf $\mathscr{F}$, computing $\operatorname{Ext}^i(\mathscr{F}, \mathscr{G})$ requires either an injective resolution of $\mathscr{G}$ (typically enormous and non-explicit) or the full spectral sequence. But for a locally free $\mathscr{F}$, the Ext groups reduce to ordinary sheaf cohomology, which we can compute using the Čech machinery of section 6.1.1.

This is not a marginal simplification: on a smooth projective variety, *every* coherent sheaf has a finite locally free resolution (since the local rings are regular, hence have finite global dimension). So Ext between any two coherent sheaves can, in principle, be computed by resolving the first argument by locally free sheaves and applying the corollary iteratively. On projective space, the situation is even better: every coherent sheaf has a resolution by direct sums of twisted sheaves $\mathcal{O}(d)$ (by corollary 5.3.26), and the cohomology of these is known by theorem 6.3.1.

One can also give a direct proof of corollary 6.4.9 without invoking spectral sequences. Since this argument is short and illuminating, we record it:

***Proof*:**
(Alternative proof of corollary 6.4.9)
Since $\mathcal{L}$ is locally free of finite rank, it is a projective object in the category of $\mathcal{O}_X$-modules on any open set where it is free, and the functor $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, -)$ is exact (checking on stalks: if $\mathcal{L}_x \cong \mathcal{O}_{X,x}^r$, then $\operatorname{Hom}_{\mathcal{O}_{X,x}}(\mathcal{O}_{X,x}^r, -) = (-)^r$ is exact). Hence $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, -)$ preserves injective resolutions up to exactness. Let $0 \to \mathscr{G} \to \mathscr{I}^\bullet$ be an injective resolution. Then:

$$0 \to \mathcal{H}om(\mathcal{L}, \mathscr{G}) \to \mathcal{H}om(\mathcal{L}, \mathscr{I}^\bullet)$$

is exact, and each $\mathcal{H}om(\mathcal{L}, \mathscr{I}^p)$ is flasque by lemma 6.4.7. Thus this is a flasque resolution of $\mathcal{H}om(\mathcal{L}, \mathscr{G}) \cong \mathcal{L}^\vee \otimes \mathscr{G}$, and:

$$\operatorname{Ext}^i(\mathcal{L}, \mathscr{G}) = h^i(\operatorname{Hom}(\mathcal{L}, \mathscr{I}^\bullet)) = h^i(\Gamma(X, \mathcal{H}om(\mathcal{L}, \mathscr{I}^\bullet))) = H^i(X, \mathcal{L}^\vee \otimes \mathscr{G}).$$

6.4.3 Computations on Projective Space

We now combine the degeneration result with the computation of twisted sheaf cohomology (theorem 6.3.1) to obtain explicit Ext groups on projective space. These calculations are the foundation for Serre duality.

Proposition 6.4.11: Ext of Twisted Sheaves on Projective Space

Let R be a Noetherian ring, $X = \mathbb{P}^n_R$ with $n \geq 1$, and $a, b \in \mathbb{Z}$. Then:

$$\operatorname{Ext}^i_{\mathcal{O}_X}\big(\mathcal{O}_X(a),\, \mathcal{O}_X(b)\big) \;\cong\; H^i\big(X,\, \mathcal{O}_X(b-a)\big),$$

and hence by theorem 6.3.1:

$$\operatorname{Ext}^i_{\mathcal{O}_X}\big(\mathcal{O}_X(a),\, \mathcal{O}_X(b)\big) \;\cong\; \begin{cases} \big(R[x_0, \ldots, x_n]\big)_{b-a} & i = 0,\; b-a \geq 0, \\ R^{\binom{a-b-1}{n}} & i = n,\; b-a \leq -(n+1), \\ 0 & \text{otherwise.} \end{cases}$$

In particular:

1. $\operatorname{Ext}^i(\mathcal{O}_X, \mathcal{O}_X(d)) = H^i(X, \mathcal{O}_X(d))$ for all i, d.
2. $\operatorname{Hom}(\mathcal{O}_X(a), \mathcal{O}_X(b)) = 0$ if $b < a$, and is a free R-module of rank $\binom{b-a+n}{n}$ if $b \geq a$.
3. $\operatorname{Ext}^n(\mathcal{O}_X(a), \mathcal{O}_X(b)) \neq 0$ only when $b - a \leq -(n+1)$, i.e. $a \geq b + n + 1$.

***Proof*:**
The line bundle $\mathcal{O}_X(a)$ is locally free of rank 1, with dual $\mathcal{O}_X(a)^\vee \cong \mathcal{O}_X(-a)$ (by proposition 5.3.8). Hence by corollary 6.4.9:

$$\operatorname{Ext}^i(\mathcal{O}_X(a), \mathcal{O}_X(b)) \cong H^i(X, \mathcal{O}_X(-a) \otimes \mathcal{O}_X(b)) = H^i(X, \mathcal{O}_X(b-a)),$$

where we used $\mathcal{O}_X(c) \otimes \mathcal{O}_X(d) \cong \mathcal{O}_X(c+d)$. The explicit values follow from theorem 6.3.1. The rank in the $i = n$ case is the number of "all-negative" monomials $x_0^{l_0} \cdots x_n^{l_n}$ of total degree $b - a$ with each $l_j < 0$, which by the substitution $l_j = -m_j - 1$ (so $m_j \geq 0$ and $\sum m_j = a - b - n - 1$) equals $\binom{a-b-1}{n}$.

Now let us see how to handle sheaves that are not themselves line bundles but can be resolved by them:

Example 6.4.12: Ext Computations on Projective Space

1. *Self-Ext of the structure sheaf.* Taking $a = b = 0$ in proposition 6.4.11:

$$\operatorname{Ext}^i(\mathcal{O}_X, \mathcal{O}_X) \cong H^i(X, \mathcal{O}_X) = \begin{cases} R & i = 0, \\ 0 & i > 0. \end{cases}$$

   This says that the structure sheaf has no non-trivial self-extensions, which is consistent with $\mathcal{O}_X$ being "rigid" as a line bundle on projective space.

2. *The dualizing line bundle.* We shall see in the next section that the *dualizing sheaf* of $\mathbb{P}^n_R$ is $\omega_X = \mathcal{O}_X(-n-1)$. Note the "dual" behaviour: by part (3) of proposition 6.4.11, the only

nonzero Ext between $\mathcal{O}_X(d)$ and ω_X is:

$$\operatorname{Ext}^n(\mathcal{O}_X(d), \omega_X) = H^n(X, \mathcal{O}_X(-n-1-d)) \cong \begin{cases} R^{\binom{d}{n}} & d \geq 0, \\ 0 & d < 0. \end{cases}$$

Meanwhile $H^0(X, \mathcal{O}_X(d)) \cong R^{\binom{d+n}{n}}$ for $d \geq 0$, and one can verify that $\binom{d+n}{n} = \binom{d}{n}$ fails in general — but the *pairing* between $H^0(X, \mathcal{O}_X(d))$ and $\operatorname{Ext}^n(\mathcal{O}_X(d), \omega_X)$ is nonetheless perfect (we will prove this via Serre duality in the next section). For $d = 0$:

$$\operatorname{Ext}^n(\mathcal{O}_X, \omega_X) = H^n(X, \omega_X) \cong R,$$

which will serve as the "trace" isomorphism $H^n(X, \omega_X) \xrightarrow{\sim} R$ in the duality theorem.

3. *Ext from the cotangent sheaf (using the Euler sequence).* Let k be a field and $X = \mathbb{P}^n_k$. The Euler sequence (by theorem 5.5.18) is:

$$0 \longrightarrow \Omega_{X/k} \longrightarrow \mathcal{O}_X(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Although $\Omega_{X/k}$ is locally free (of rank n), let us use this short exact sequence to compute $\operatorname{Ext}^i(\Omega_{X/k}, \mathcal{O}_X)$ via the long exact sequence in the *contravariant* variable (proposition 6.4.2(2)):

$$\cdots \to \operatorname{Ext}^i(\mathcal{O}_X, \mathcal{O}_X) \to \operatorname{Ext}^i(\mathcal{O}_X(-1)^{n+1}, \mathcal{O}_X) \to \operatorname{Ext}^i(\Omega_{X/k}, \mathcal{O}_X) \to \operatorname{Ext}^{i+1}(\mathcal{O}_X, \mathcal{O}_X) \to \cdots$$

By proposition 6.4.11:

$$\operatorname{Ext}^i(\mathcal{O}_X(-1)^{n+1}, \mathcal{O}_X) \cong H^i(X, \mathcal{O}_X(1))^{n+1},$$

which is $(R^{n+1})^{n+1}$ for $i = 0$ and zero for $0 < i < n$. Also, $\operatorname{Ext}^i(\mathcal{O}_X, \mathcal{O}_X) = R$ for $i = 0$ and zero for $i > 0$.

For $i = 0$, the long exact sequence gives:

$$0 \to \operatorname{Hom}(\mathcal{O}_X, \mathcal{O}_X) \to \operatorname{Hom}(\mathcal{O}_X(-1)^{n+1}, \mathcal{O}_X) \to \operatorname{Hom}(\Omega, \mathcal{O}_X) \to 0$$

so $\operatorname{Hom}(\Omega_{X/k}, \mathcal{O}_X) \cong k^{(n+1)^2}/k \cong k^{n^2+2n}$. On the other hand, by corollary 6.4.9, $\operatorname{Hom}(\Omega, \mathcal{O}_X) = H^0(X, \Omega^\vee) = H^0(X, \mathcal{T}_X)$, where $\mathcal{T}_X$ is the tangent sheaf. This is the space of global vector fields on $\mathbb{P}^n_k$, which has dimension $n^2 + 2n = \dim_k \mathfrak{gl}_{n+1}(k)$: the global vector fields on projective space are precisely those induced by the action of GL_{n+1}, modulo scalars.

4. *Ext between non-locally-free sheaves (sketch).* Let $X = \mathbb{P}^2_k$ and let $C \hookrightarrow X$ be a smooth plane curve of degree d, with ideal sheaf $\mathscr{I}_C$. The short exact sequence $0 \to \mathscr{I}_C \to \mathcal{O}_X \to \mathcal{O}_C \to 0$, together with the identification $\mathscr{I}_C \cong \mathcal{O}_X(-d)$ (by corollary 5.3.16), gives:

$$0 \to \mathcal{O}_X(-d) \to \mathcal{O}_X \to \mathcal{O}_C \to 0.$$

This is a locally free resolution of $\mathcal{O}_C$ as an $\mathcal{O}_X$-module (of length 1). Hence by proposition 6.4.3:

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathcal{O}_C, \mathscr{G}) \cong h^i\big(\operatorname{Hom}(\mathcal{O}_X, \mathscr{G}) \to \operatorname{Hom}(\mathcal{O}_X(-d), \mathscr{G})\big)$$

for any $\mathcal{O}_X$-module $\mathscr{G}$. The complex in question is $\mathscr{G}(X) \xrightarrow{\cdot f} \mathscr{G}(X)(d)$ where f is the defining equation of C, so:

$$\begin{aligned}
\operatorname{Ext}^0(\mathcal{O}_C, \mathscr{G}) &= \ker\big(\mathscr{G}(X) \to \mathscr{G}(X)(d)\big), \\
\operatorname{Ext}^1(\mathcal{O}_C, \mathscr{G}) &= \operatorname{coker}\big(\mathscr{G}(X) \to \mathscr{G}(X)(d)\big), \\
\operatorname{Ext}^i(\mathcal{O}_C, \mathscr{G}) &= 0 \quad \text{for } i \geq 2.
\end{aligned}$$

Taking $\mathscr{G} = \omega_X = \mathcal{O}_X(-3)$, we get $\operatorname{Ext}^1(\mathcal{O}_C, \omega_X) = \operatorname{coker}\big(H^0(X, \mathcal{O}(-3)) \to H^0(X, \mathcal{O}(d-3))\big)$. For $d \geq 3$, this is a k-vector space of dimension $\binom{d-1}{2} - 1$, which (as we will verify via Serre duality) equals $\dim_k H^0(C, \omega_C) = g$, the genus of the curve. This is not a coincidence: Serre duality will identify $\operatorname{Ext}^1(\mathcal{O}_C, \omega_X)$ with $H^0(C, \omega_C)^\vee$ via the adjunction between the closed embedding and the ambient duality.

Let us close this section by recording how the spectral sequence simplifies for coherent sheaves on projective schemes, even when the first argument is not locally free. This will be used in the proof of Serre duality:

Proposition 6.4.13: Finiteness of Ext on Projective Schemes

Let X be a projective scheme over a Noetherian ring R, and let $\mathscr{F}$ and $\mathscr{G}$ be coherent sheaves on X. Then:

1. Each $\mathcal{E}xt^q_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ is a coherent sheaf on X.
2. Each $\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G})$ is a finitely generated R-module.
3. $\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = 0$ for $i > \dim X$ (where $\dim X$ denotes the Krull dimension).

***Proof*:**
(1) This is a local statement: on any affine open $U = \operatorname{Spec} A$ of X, the sheaves $\mathscr{F}|_U$ and $\mathscr{G}|_U$ correspond to finitely generated A-modules M and N (by corollary 5.1.12), and the sheaf Ext on U is:

$$\mathcal{E}xt^q_{\mathcal{O}_U}(\widetilde{M}, \widetilde{N}) \cong \widetilde{\operatorname{Ext}^q_A(M, N)}$$

(this follows from proposition 6.4.6 and the standard computation of Ext over a Noetherian ring via a free resolution of the finitely generated module M; each $\operatorname{Ext}^q_A(M, N)$ is finitely generated). Hence $\mathcal{E}xt^q(\mathscr{F}, \mathscr{G})$ is coherent.

(2) The local-to-global spectral sequence (theorem 6.4.8) gives $H^p(X, \mathcal{E}xt^q(\mathscr{F}, \mathscr{G})) \Rightarrow \operatorname{Ext}^{p+q}(\mathscr{F}, \mathscr{G})$. Since $\mathcal{E}xt^q(\mathscr{F}, \mathscr{G})$ is coherent and X is projective over a Noetherian ring, each cohomology group $H^p(X, \mathcal{E}xt^q(\mathscr{F}, \mathscr{G}))$ is a finitely generated R-module[a]. Since the E_2-page consists of finitely generated R-modules and the spectral sequence converges, the abutment $\operatorname{Ext}^i(\mathscr{F}, \mathscr{G})$ is finitely generated.

(3) By theorem 6.2.5, $H^p(X, -) = 0$ for $p > \dim X$. On the other hand, $\mathcal{E}xt^q(\mathscr{F}, \mathscr{G}) = 0$ for q larger than the projective dimension of $\mathscr{F}$ (which is finite since the local rings of a projective scheme over a Noetherian ring are Noetherian, hence $\mathscr{F}$ admits locally free resolutions of finite length). The spectral sequence then gives $\operatorname{Ext}^i(\mathscr{F}, \mathscr{G}) = 0$ for i sufficiently large; more precisely, for $i > \dim X$ when $\mathscr{F}$ is locally free, and in general for $i > \dim X + \operatorname{pd}(\mathscr{F})$ where pd denotes

the projective dimension. When X is smooth (hence regular), the projective dimension of any coherent sheaf is at most $\dim X$, so $\mathrm{Ext}^i(\mathscr{F},\mathscr{G}) = 0$ for $i > 2\dim X$; but the spectral sequence gives the sharper bound $i > \dim X$ directly when the E_2-page is examined.

[a] This uses the finiteness theorem for cohomology of coherent sheaves on projective schemes, which we will prove in section 6.6 (see corollary 5.3.28). No circularity arises: the finiteness theorem depends on the twisted sheaf computation (theorem 6.3.1) and the "coherent sheaves are quotients of twisted free sheaves" result (corollary 5.3.26), both of which are already established.

6.4.14 Remark: What the Spectral Sequence Encodes The local-to-global spectral sequence should be thought of as expressing the passage from local to global information for morphisms between sheaves. On the E_2-page:

- The *bottom row* $E_2^{p,0} = H^p(X, \mathcal{H}om(\mathscr{F},\mathscr{G}))$ captures the "global extension classes that are locally trivial." When $\mathscr{F}$ is locally free, this is the only nonzero row, and the spectral sequence degenerates.
- The *left column* $E_2^{0,q} = \Gamma(X, \mathcal{E}xt^q(\mathscr{F},\mathscr{G}))$ captures the "local extension classes that are globally defined." These are the obstructions arising from the local algebra (e.g., the singularities of $\mathscr{F}$ or the non-regularity of the local rings).

The differentials in the spectral sequence measure the interaction between these two types of obstruction. For smooth projective varieties with locally free sheaves, all the interesting information is in the bottom row — this is why Serre duality can be stated purely in terms of cohomology groups, without mentioning Ext.

6.5 Serre Duality Theorem

In classical topology, Poincaré duality asserts that for a compact oriented manifold M of dimension n, the singular cohomology groups satisfy $H^i(M;\mathbb{R}) \cong H^{n-i}(M;\mathbb{R})^*$. The duality is mediated by integration: the cup product $H^i \times H^{n-i} \to H^n$, followed by integration over the fundamental class $H^n(M;\mathbb{R}) \xrightarrow{\sim} \mathbb{R}$, gives a perfect pairing. In Kähler geometry, this takes the refined form of Hodge duality (theorem 0.6.28): $H^{p,q}(X) \cong H^{n-p,n-q}(X)^*$, with the pairing mediated by the wedge product of (p,q)-forms followed by integration of the resulting top form.

Serre duality is the algebraic analogue. For a smooth projective variety X of dimension n over a field k, the role of the top differential form is played by the *canonical sheaf* $\omega_X = \det \Omega_{X/k}$, and the role of integration is played by a "trace" isomorphism $H^n(X,\omega_X) \xrightarrow{\sim} k$. The resulting duality

$$H^i(X,\mathscr{F}) \;\cong\; H^{n-i}(X,\, \mathscr{F}^\vee \otimes \omega_X)^*$$

for locally free $\mathscr{F}$ is the central structural result in the cohomology of algebraic varieties. It constrains the Hilbert polynomial, underpins the Riemann–Roch theorem, and governs the geometry of linear systems.

We first prove the duality theorem for projective space, where the "dualizing sheaf" is simply $\mathcal{O}(-n-1)$ and the computations are completely explicit. We then define the dualizing sheaf for general projective schemes and extend the theorem to smooth projective varieties.

6.5.1 Duality on Projective Space

The twisted sheaf computation (theorem 6.3.1) already contains the seed of duality: the perfect pairing of part (3) shows that $H^0(X, \mathcal{O}(d))$ and $H^n(X, \mathcal{O}(-d-n-1))$ are dual to each other. Our task is to recognize this as the $i = 0$ case of a more general duality, then extend to all i and to all coherent sheaves.

The first step is to abstract the key property of $\mathcal{O}(-n-1)$ that makes the pairing work:

Definition 6.5.1: Dualizing Sheaf

Let X be a proper scheme of dimension n over a field k. A *dualizing sheaf* for X is a coherent sheaf ω_X on X together with a k-linear map $t : H^n(X, \omega_X) \to k$, called the *trace map*, such that for every coherent sheaf $\mathscr{F}$ on X, the natural pairing:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \times H^n(X, \mathscr{F}) \xrightarrow{\text{apply } H^n} H^n(X, \omega_X) \xrightarrow{t} k$$

is a perfect pairing of finite-dimensional k-vector spaces. In other words, the induced map

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \xrightarrow{\sim} H^n(X, \mathscr{F})^*$$

is an isomorphism, where $(-)^*$ denotes the k-linear dual.
Equivalently, ω_X *represents* the functor $\mathscr{F} \mapsto H^n(X, \mathscr{F})^*$ on the category of coherent sheaves on X.

Proposition 6.5.2: Uniqueness of the Dualizing Sheaf

If a dualizing sheaf exists, it is unique up to unique isomorphism. That is, if (ω_X, t) and (ω'_X, t') are two dualizing sheaves for X, then there is a unique isomorphism $\varphi : \omega_X \xrightarrow{\sim} \omega'_X$ such that $t = t' \circ H^n(X, \varphi)$.

***Proof*:**
Both ω_X and ω'_X represent the same functor $\mathscr{F} \mapsto H^n(X, \mathscr{F})^*$. By the Yoneda lemma (theorem 0.4.22), any two representing objects are uniquely isomorphic via an isomorphism that intertwines the representing natural transformations.

Proposition 6.5.3: Dualizing Sheaf of Projective Space

Let k be a field and $X = \mathbb{P}^n_k$ with $n \geq 1$. The twisted sheaf $\omega_X = \mathcal{O}_X(-n-1)$ is a dualizing sheaf for X, with trace map $t : H^n(X, \mathcal{O}(-n-1)) \xrightarrow{\sim} k$ being the isomorphism that sends the class of $x_0^{-1} x_1^{-1} \cdots x_n^{-1}$ to $1 \in k$.

***Proof*:**
This is precisely the content of theorem 6.3.1(3) (with $R = k$). Indeed, that theorem establishes that for each $d \geq 0$, the multiplication pairing:

$$H^0(X, \mathcal{O}(d)) \times H^n(X, \mathcal{O}(-d-n-1)) \longrightarrow H^n(X, \mathcal{O}(-n-1)) \cong k$$

is a perfect pairing of finite-dimensional k-vector spaces. Since $\mathrm{Hom}(\mathcal{O}(d), \mathcal{O}(-n-1)) \cong H^0(X, \mathcal{O}(-d-n-1))$ (by corollary 6.4.9), this gives $\mathrm{Hom}(\mathcal{O}(d), \omega_X) \cong H^n(X, \mathcal{O}(d))^*$ for every d. When $d < 0$, both sides are zero (the left because there are no nonzero maps from $\mathcal{O}(d)$ to $\mathcal{O}(-n-1)$ when $d > -n-1$, the right by theorem 6.3.1(4)).

For a general coherent $\mathscr{F}$, one reduces to line bundles by choosing a presentation of $\mathscr{F}$ by direct sums of $\mathcal{O}(d)$'s (using corollary 5.3.26) and comparing the two sides via the five lemma. The argument is identical to Step 2 of the proof of theorem 6.5.5 below, so we defer the details.

The dualizing sheaf definition captures the "$i = 0$" case of duality: it says $\mathrm{Hom}(\mathscr{F}, \omega) = \mathrm{Ext}^0(\mathscr{F}, \omega) \cong H^n(X, \mathscr{F})^*$. The full Serre duality extends this to all Ext groups. To state it, we need the *Yoneda pairing*:

6.5.4 The Yoneda Pairing For $\mathcal{O}_X$-modules $\mathscr{F}, \mathscr{G}, \mathscr{H}$, the *Yoneda product* (see [15]) is a bilinear map:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \times \mathrm{Ext}^j_{\mathcal{O}_X}(\mathscr{H}, \mathscr{F}) \longrightarrow \mathrm{Ext}^{i+j}_{\mathcal{O}_X}(\mathscr{H}, \mathscr{G}).$$

This is defined by splicing extensions: an element of $\mathrm{Ext}^i(\mathscr{F}, \mathscr{G})$ is represented by an equivalence class of i-fold extensions $0 \to \mathscr{G} \to \mathcal{E}^{i-1} \to \cdots \to \mathcal{E}^0 \to \mathscr{F} \to 0$, and the Yoneda product splices two such sequences together.
Now, since $H^j(X, \mathscr{F}) = \mathrm{Ext}^j_{\mathcal{O}_X}(\mathcal{O}_X, \mathscr{F})$, the Yoneda product specializes to give a pairing:

$$\mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \times H^{n-i}(X, \mathscr{F}) \longrightarrow H^n(X, \omega_X) \xrightarrow{t} k.$$

This defines for each i a natural map:

$$\eta^i_{\mathscr{F}} : \mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \longrightarrow H^{n-i}(X, \mathscr{F})^*.$$

When $\mathscr{F}$ is locally free, $\mathrm{Ext}^i(\mathscr{F}, \omega_X) \cong H^i(X, \mathscr{F}^\vee \otimes \omega_X)$ by corollary 6.4.9, and the Yoneda pairing reduces to the cup product:

$$H^i(X, \mathscr{F}^\vee \otimes \omega_X) \times H^{n-i}(X, \mathscr{F}) \xrightarrow{\cup} H^n(X, \mathscr{F}^\vee \otimes \omega_X \otimes \mathscr{F}) \xrightarrow{\mathrm{ev}\otimes\mathrm{id}} H^n(X, \omega_X) \xrightarrow{t} k,$$

where $\mathrm{ev} : \mathscr{F}^\vee \otimes \mathscr{F} \to \mathcal{O}_X$ is the evaluation map. In the Čech framework, the cup product of cochains $\alpha \in C^p(\mathcal{U}, \mathscr{F}^\vee \otimes \omega_X)$ and $\beta \in C^q(\mathcal{U}, \mathscr{F})$ is defined by:

$$(\alpha \cup \beta)_{i_0,\ldots,i_{p+q}} = \alpha_{i_0,\ldots,i_p} \otimes \beta_{i_p,\ldots,i_{p+q}}|_{U_{i_0,\ldots,i_{p+q}}},$$

followed by the evaluation and trace maps.

Theorem 6.5.5: Duality for Projective Space

Let k be a field, $X = \mathbb{P}^n_k$ with $n \geq 1$, and $\omega_X = \mathcal{O}_X(-n-1)$. Then for every coherent sheaf $\mathscr{F}$ on X, the Yoneda pairing induces isomorphisms:

$$\eta^i_{\mathscr{F}} : \mathrm{Ext}^i_{\mathcal{O}_X}(\mathscr{F}, \omega_X) \xrightarrow{\sim} H^{n-i}(X, \mathscr{F})^* \qquad \text{for all } 0 \leq i \leq n.$$

Both sides vanish for $i > n$.

***Proof*:**
We verify the theorem in two stages: first for line bundles, then for all coherent sheaves.

Step 1: Verification for $\mathscr{F} = \mathcal{O}_X(d)$. By corollary 6.4.9, $\mathrm{Ext}^i(\mathcal{O}(d), \omega_X) \cong H^i(X, \mathcal{O}(-d-n-1))$. The Yoneda pairing (which reduces to the cup product for locally free sheaves) gives a bilinear map:

$$H^i(X, \mathcal{O}(-d-n-1)) \times H^{n-i}(X, \mathcal{O}(d)) \longrightarrow H^n(X, \mathcal{O}(-n-1)) \xrightarrow{t} k.$$

We must show this pairing is perfect for each i. By theorem 6.3.1(4), the intermediate cohomology vanishes: $H^j(X, \mathcal{O}(m)) = 0$ for $0 < j < n$ and all $m \in \mathbb{Z}$. So for $0 < i < n$, both $H^i(X, \mathcal{O}(-d-n-1))$ and $H^{n-i}(X, \mathcal{O}(d))$ are zero and the pairing is trivially perfect (both sides vanish).

For $i = 0$: the pairing is $H^0(X, \mathcal{O}(-d-n-1)) \times H^n(X, \mathcal{O}(d)) \to k$. If $-d-n-1 < 0$ (i.e. $d > -(n+1)$), then $H^0 = 0$ and $H^n = 0$, and the pairing is perfect. If $-d-n-1 \geq 0$ (i.e. $d \leq -(n+1)$), this is the pairing of theorem 6.3.1(3) (with the degree parameter set to $-d-n-1 \geq 0$), which is perfect.

For $i = n$: by symmetry, the pairing is $H^n(X, \mathcal{O}(-d-n-1)) \times H^0(X, \mathcal{O}(d)) \to k$. If $d < 0$, both sides vanish. If $d \geq 0$, this is again the perfect pairing of theorem 6.3.1(3) with the two factors interchanged (the pairing is symmetric up to the sign conventions in the cup product, which can be checked to be trivial in this case since all sheaves involved are line bundles).

Hence $\eta^i_{\mathcal{O}(d)}$ is an isomorphism for all i and all $d \in \mathbb{Z}$. By the additivity of Ext and cohomology in direct sums, the same holds for $\mathscr{F} = \bigoplus_j \mathcal{O}(d_j)$ (any finite direct sum of line bundles).

Step 2: Extension to all coherent sheaves. Let $\mathscr{F}$ be an arbitrary coherent sheaf on $X = \mathbb{P}^n_k$. Since X has dimension n and its local rings are regular of dimension $\leq n$ (the local ring at a point of codimension c is a regular local ring of dimension c), every coherent sheaf on X admits a finite locally free resolution of length at most n. More precisely, by corollary 5.3.26 and theorem 5.3.27, we can build a resolution:

$$0 \to \mathcal{E}_m \to \mathcal{E}_{m-1} \to \cdots \to \mathcal{E}_1 \to \mathcal{E}_0 \to \mathscr{F} \to 0$$

where each $\mathcal{E}_j$ is a finite direct sum of line bundles $\mathcal{O}(d)$ and $m \leq n$. We prove $\eta^i_{\mathscr{F}}$ is an isomorphism by induction on the length m of such a resolution.

Base case ($m = 0$): Here $\mathscr{F} = \mathcal{E}_0$ is a direct sum of line bundles, handled in Step 1.

Inductive step: Suppose $m \geq 1$. Break the resolution into the short exact sequence:

$$0 \to \mathcal{K} \to \mathcal{E}_0 \to \mathscr{F} \to 0,$$

where $\mathcal{K} = \ker(\mathcal{E}_0 \to \mathscr{F})$ is the first syzygy. Since the resolution of $\mathscr{F}$ restricts to a resolution $0 \to \mathcal{E}_m \to \cdots \to \mathcal{E}_1 \to \mathcal{K} \to 0$ of $\mathcal{K}$ of length $m-1$, the inductive hypothesis gives that $\eta^i_{\mathcal{K}}$ is an isomorphism for all i.

The short exact sequence $0 \to \mathcal{K} \to \mathcal{E}_0 \to \mathscr{F} \to 0$ induces two long exact sequences: one from $\mathrm{Ext}^*(-, \omega_X)$ (contravariant in the first argument) and one from $H^{n-*}(X, -)^*$ (the dual of the covariant cohomology sequence, which is exact since dualization is exact over a field). The naturality of the Yoneda pairing ensures that the maps η^i assemble into a morphism between these long exact sequences:

$$\begin{tikzcd}
\operatorname{Ext}^{i-1}(\mathcal{E}_0,\omega) \arrow[r] \arrow[d, "\eta^{i-1}_{\mathcal{E}_0}", "\sim"'] & \operatorname{Ext}^{i-1}(\mathcal{K},\omega) \arrow[r] \arrow[d, "\eta^{i-1}_{\mathcal{K}}", "\sim"'] & \operatorname{Ext}^{i}(\mathscr{F},\omega) \arrow[r] \arrow[d, "\eta^{i}_{\mathscr{F}}"] & \operatorname{Ext}^{i}(\mathcal{E}_0,\omega) \arrow[r] \arrow[d, "\eta^{i}_{\mathcal{E}_0}", "\sim"'] & \operatorname{Ext}^{i}(\mathcal{K},\omega) \arrow[d, "\eta^{i}_{\mathcal{K}}", "\sim"'] \\
H^{n-i+1}(\mathcal{E}_0)^* \arrow[r] & H^{n-i+1}(\mathcal{K})^* \arrow[r] & H^{n-i}(\mathscr{F})^* \arrow[r] & H^{n-i}(\mathcal{E}_0)^* \arrow[r] & H^{n-i}(\mathcal{K})^*
\end{tikzcd}$$

The four outer vertical maps are isomorphisms (the $\mathcal{E}_0$ maps by Step 1, the $\mathcal{K}$ maps by the inductive hypothesis). By the five lemma, the middle map $\eta^i_{\mathscr{F}}$ is also an isomorphism. This holds for every i, completing the induction.

Vanishing for $i > n$: Both sides vanish: $\operatorname{Ext}^i(\mathscr{F},\omega_X) = 0$ for $i > n$ by proposition 6.4.13(3), and $H^{n-i}(X,\mathscr{F}) = 0$ for $n - i < 0$ trivially.

Example 6.5.6: Serre Duality on Projective Space

1. *Structure sheaf.* For $\mathscr{F} = \mathcal{O}_X$ on $X = \mathbb{P}^n_k$:

$$\operatorname{Ext}^i(\mathcal{O}_X, \mathcal{O}(-n-1)) \cong H^i(X, \mathcal{O}(-n-1)) \cong H^{n-i}(X, \mathcal{O}_X)^*.$$

For $i = n$, this gives $H^n(X, \mathcal{O}(-n-1)) \cong H^0(X, \mathcal{O}_X)^* = k$. For $0 < i < n$, both sides are zero.

2. *Cotangent sheaf on $\mathbb{P}^2$.* Let $X = \mathbb{P}^2_k$ and $\mathscr{F} = \Omega_{X/k}$. Since Ω is locally free of rank 2, duality gives:

$$H^i(X, \Omega^\vee \otimes \mathcal{O}(-3)) \cong H^{2-i}(X, \Omega)^*.$$

Now $\Omega^\vee = \mathcal{T}_X$, the tangent sheaf, and $\mathcal{T}_X \cong \mathcal{T}_{\mathbb{P}^2}$. From the dual of the Euler sequence (theorem 5.5.18):

$$0 \to \mathcal{O}_X \to \mathcal{O}_X(1)^3 \to \mathcal{T}_X \to 0,$$

we get $\mathcal{T}_X \otimes \mathcal{O}(-3) \cong \mathcal{T}_X(-3)$, whose cohomology can be computed from the twisted Euler sequence. In particular:

$$H^0(X, \Omega) \cong H^2(X, \mathcal{T}_X(-3))^* = 0$$

(no nonzero global 1-forms on $\mathbb{P}^2$), confirming a result we already knew from Example 6.1.13(1) for $\mathbb{P}^1$.

3. *Skyscraper sheaves and residues.* Let $p \in \mathbb{P}^1_k$ be a closed point and let $\mathscr{F} = \kappa(p)$ be the skyscraper sheaf at p. Then $H^0(X, \kappa(p)) = k$ and $H^i(X, \kappa(p)) = 0$ for $i > 0$ (a skyscraper sheaf has no higher cohomology). Serre duality gives:

$$\operatorname{Ext}^1(\kappa(p), \mathcal{O}(-2)) \cong H^0(X, \kappa(p))^* = k, \qquad \operatorname{Hom}(\kappa(p), \mathcal{O}(-2)) = 0.$$

The non-trivial Ext group $\operatorname{Ext}^1(\kappa(p), \omega_X) \cong k$ has a concrete interpretation: the short exact sequence $0 \to \mathcal{O}(-2) \to \mathcal{O}(-1) \to \kappa(p) \to 0$ (where the first map is multiplication by the linear form vanishing at p, suitably twisted) represents a generator. This is the algebraic avatar of the *residue map* at p: the element of $\operatorname{Ext}^1(\kappa(p), \omega)$ corresponds to extracting the residue of a meromorphic differential form at p.

6.5.2 Duality for Projective Schemes

We now extend Serre duality from projective space to projective schemes. The key new ingredient is identifying the correct dualizing sheaf. For smooth varieties, this is the canonical sheaf; for singular schemes, the dualizing sheaf exists under a mild Cohen–Macaulay hypothesis but is more subtle to construct.

Recall from definition 5.5.25 that for a smooth variety X of dimension n over a field k, the *canonical sheaf* (or *canonical bundle*) is:

$$\omega_X = \det \Omega_{X/k} = \bigwedge^n \Omega_{X/k},$$

the top exterior power of the sheaf of differentials. This is an invertible sheaf. For $X = \mathbb{P}^n_k$, the computation of theorem 5.5.18 shows that $\omega_{\mathbb{P}^n} \cong \mathcal{O}_{\mathbb{P}^n}(-n-1)$, consistent with our earlier identification.

For a smooth projective variety X of dimension n embedded in $\mathbb{P}^N_k$ as a closed subscheme, the canonical sheaf is related to $\omega_{\mathbb{P}^N}$ by the *adjunction formula*. Recall from proposition 5.5.17 that if $X \hookrightarrow \mathbb{P}^N$ is a regular embedding of codimension $c = N - n$, there is an exact sequence:

$$\mathscr{I}_X/\mathscr{I}_X^2 \longrightarrow \Omega_{\mathbb{P}^N}\big|_X \longrightarrow \Omega_X \longrightarrow 0,$$

where $\mathscr{I}_X$ is the ideal sheaf of X. Taking top exterior powers (and using that the conormal sheaf $\mathcal{N}^\vee_{X/\mathbb{P}^N} = \mathscr{I}_X/\mathscr{I}_X^2$ is locally free of rank c for a regular embedding), one obtains:

Proposition 6.5.7: Adjunction Formula

Let X be a smooth closed subvariety of dimension n in $\mathbb{P}^N_k$, with codimension $c = N - n$. Then:

$$\omega_X \;\cong\; \omega_{\mathbb{P}^N}\big|_X \otimes \det\big(\mathcal{N}^\vee_{X/\mathbb{P}^N}\big)^{-1} \;\cong\; \mathcal{O}_{\mathbb{P}^N}(-N-1)\big|_X \otimes \det(\mathcal{N}_{X/\mathbb{P}^N}).$$

If X is a complete intersection, cut out by hypersurfaces of degrees $d_1, \dots, d_c$, then $\mathcal{N}_{X/\mathbb{P}^N} \cong \bigoplus_{j=1}^c \mathcal{O}_X(d_j)$ and hence:

$$\omega_X \cong \mathcal{O}_X\left(\sum_{j=1}^c d_j - N - 1\right).$$

***Proof*:**
The conormal–cotangent exact sequence for the regular embedding $i : X \hookrightarrow \mathbb{P}^N$ is:

$$0 \longrightarrow \mathcal{N}^\vee_{X/\mathbb{P}^N} \longrightarrow \Omega_{\mathbb{P}^N}\big|_X \longrightarrow \Omega_X \longrightarrow 0.$$

This is exact on the left because X is smooth (equivalently, the embedding is a regular immersion; see theorem 5.5.24). Taking the top exterior power of the middle term (which has rank N) and using the multiplicativity of determinants on short exact sequences of locally free sheaves:

$$\det(\Omega_{\mathbb{P}^N}|_X) \cong \det(\mathcal{N}^\vee_{X/\mathbb{P}^N}) \otimes \det(\Omega_X),$$

so $\omega_X = \det(\Omega_X) \cong \det(\Omega_{\mathbb{P}^N}|_X) \otimes \det(\mathcal{N}^\vee_{X/\mathbb{P}^N})^{-1} = \omega_{\mathbb{P}^N}|_X \otimes \det(\mathcal{N}_{X/\mathbb{P}^N})$.

For a complete intersection, the ideal sheaf $\mathscr{I}_X$ is generated by global sections $f_1, \dots, f_c$ of degrees $d_1, \dots, d_c$. The conormal sheaf is $\mathscr{I}_X/\mathscr{I}_X^2 \cong \bigoplus_{j=1}^c \mathcal{O}_X(-d_j)$ (the surjection $\bigoplus \mathcal{O}(-d_j) \twoheadrightarrow \mathscr{I}_X$

induces an isomorphism on the conormal level), so $\mathcal{N}_{X/\mathbb{P}^N} = (\mathscr{I}_X/\mathscr{I}_X^2)^\vee \cong \bigoplus \mathcal{O}_X(d_j)$. Hence $\det(\mathcal{N}_{X/\mathbb{P}^N}) = \mathcal{O}_X(\sum d_j)$ and $\omega_X \cong \mathcal{O}_X(-N-1+\sum d_j)$.

Example 6.5.8: Canonical Sheaves via Adjunction

1. *Smooth plane curve of degree d.* Let $C \subset \mathbb{P}^2_k$ be a smooth curve cut out by a polynomial of degree d. Then $\omega_C \cong \mathcal{O}_C(d-3)$. For $d=1$ (a line, genus 0): $\omega_C \cong \mathcal{O}_C(-2)$, so $H^0(C,\omega_C)=0$, confirming $g=0$. For $d=3$ (an elliptic curve, genus 1): $\omega_C \cong \mathcal{O}_C$, so $H^0(C,\omega_C)=k$, confirming $g=1$. For general $d \geq 1$, the genus is $g = \dim_k H^0(C,\omega_C) = \binom{d-1}{2}$ (by proposition 7.1.14).

2. *Smooth surface of degree d in $\mathbb{P}^3$.* Let $S \subset \mathbb{P}^3_k$ be a smooth surface of degree d. Then $\omega_S \cong \mathcal{O}_S(d-4)$. For $d=4$ (a K3 surface): $\omega_S \cong \mathcal{O}_S$, so S has trivial canonical bundle. For $d<4$, ω_S has negative degree (in an appropriate sense), and for $d>4$, the canonical bundle is ample.

3. *Hypersurface of degree d in $\mathbb{P}^n$.* More generally, a smooth hypersurface $X \subset \mathbb{P}^n_k$ of degree d has $\omega_X \cong \mathcal{O}_X(d-n-1)$. The canonical class is trivial (i.e. X is Calabi–Yau in dimension $n-1$) precisely when $d=n+1$.

We can now state the full Serre duality theorem. For the proof, we shall use the dualizing sheaf of projective space together with a closed embedding to transfer duality from $\mathbb{P}^N$ to X:

Theorem 6.5.9: Serre Duality for a Projective Scheme

Let X be a projective scheme of dimension n over a field k, and suppose X admits a dualizing sheaf ω_X (with trace map $t: H^n(X,\omega_X) \to k$). Then for every coherent sheaf $\mathscr{F}$ on X, the Yoneda pairing:

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F},\omega_X) \times H^{n-i}(X,\mathscr{F}) \longrightarrow H^n(X,\omega_X) \xrightarrow{t} k$$

induces isomorphisms:

$$\operatorname{Ext}^i_{\mathcal{O}_X}(\mathscr{F},\omega_X) \;\cong\; H^{n-i}(X,\mathscr{F})^* \qquad \text{for all } 0 \leq i \leq n.$$

Furthermore, a dualizing sheaf exists in each of the following cases:

1. X is a closed subscheme of $\mathbb{P}^N_k$ that is equidimensional and Cohen–Macaulay. In this case:
$$\omega_X \cong \mathcal{E}xt^c_{\mathcal{O}_{\mathbb{P}^N}}(\mathcal{O}_X,\omega_{\mathbb{P}^N}),$$
where $c = N-n$ is the codimension. In particular, if X is a smooth variety, then $\omega_X \cong \det \Omega_{X/k}$.

2. $X = \mathbb{P}^n_k$, with $\omega_X = \mathcal{O}(-n-1)$.

***Proof*:**
We give the proof for $X = \mathbb{P}^n_k$ in full (this is theorem 6.5.5, already established above) and outline the key steps for the general case.

Existence of the dualizing sheaf. For a closed subscheme $i : X \hookrightarrow \mathbb{P}^N_k$ of codimension $c = N - n$, define:

$$\omega_X = \mathcal{E}xt^c_{\mathcal{O}_{\mathbb{P}^N}}(i_*\mathcal{O}_X, \, \omega_{\mathbb{P}^N}).$$

This is a coherent sheaf on $\mathbb{P}^N$ supported on X, hence can be regarded as a coherent sheaf on X (via proposition 6.2.4). When X is Cohen–Macaulay, the local Ext vanishes in all degrees except c (since the local rings of a Cohen–Macaulay scheme of codimension c have depth exactly c, and for a module of depth c over a regular local ring, the local Ext vanishes below degree c). This ensures ω_X is concentrated in a single degree and behaves well.

The trace map $t : H^n(X, \omega_X) \to k$ is constructed from the trace map for $\mathbb{P}^N$ via the local-to-global spectral sequence (theorem 6.4.8): the isomorphism $\operatorname{Ext}^N(\mathcal{O}_X, \omega_{\mathbb{P}^N}) \cong H^N(\mathbb{P}^N, \omega_{\mathbb{P}^N})^*$ (from theorem 6.5.5 applied to $\mathbb{P}^N$) and the identification of $\operatorname{Ext}^N(\mathcal{O}_X, \omega_{\mathbb{P}^N})$ with $H^n(X, \omega_X)$ (via the spectral sequence, using the Cohen–Macaulay vanishing) combine to give the trace.

When X is smooth, the identification $\omega_X \cong \det \Omega_{X/k}$ follows from proposition 6.5.7 and the fact that $\mathcal{E}xt^c(\mathcal{O}_X, \omega_{\mathbb{P}^N}) \cong \det(\mathcal{N}_{X/\mathbb{P}^N}) \otimes \omega_{\mathbb{P}^N}|_X$ for a regular embedding of codimension c.

Duality for the general case. Given that the dualizing sheaf exists with the correct trace, the proof that $\eta^i_{\mathscr{F}}$ is an isomorphism for all coherent $\mathscr{F}$ follows the same pattern as theorem 6.5.5. One resolves $\mathscr{F}$ by sheaves for which duality is known (using corollary 5.3.26), then applies the five lemma. The key point is that on a projective scheme over k, the coherent sheaves $\mathcal{O}_X(d)$ for $d \gg 0$ satisfy duality (this can be reduced to duality on $\mathbb{P}^N$ via the closed embedding and the adjunction between i^* and i_*), and every coherent sheaf admits a finite resolution by such sheaves.

For smooth varieties, the Ext groups simplify and Serre duality takes its most elegant form:

Corollary 6.5.10: Serre Duality: Nonsingular Projective Variety

Let X be a nonsingular projective variety of dimension n over a field k, with canonical sheaf $\omega_X = \det \Omega_{X/k}$. Then for every locally free coherent sheaf $\mathscr{F}$ on X:

$$H^i(X, \mathscr{F}) \;\cong\; H^{n-i}(X,\, \mathscr{F}^\vee \otimes \omega_X)^* \qquad \text{for all } 0 \le i \le n.$$

In particular:

1. $H^n(X, \omega_X) \cong k$.
2. $H^i(X, \mathcal{O}_X) \cong H^{n-i}(X, \omega_X)^*$ for all i.
3. The *arithmetic genus* $p_a(X) = (-1)^n(\chi(\mathcal{O}_X) - 1)$ can be computed from the cohomology of the canonical sheaf via:

$$p_a(X) = (-1)^n \Big(\sum_{i=0}^{n} (-1)^i \dim_k H^i(X, \mathcal{O}_X) - 1 \Big) = \dim_k H^0(X, \omega_X) - \cdots$$

where Serre duality determines the "upper half" of the cohomology from the "lower half."

***Proof*:**
Since $\mathscr{F}$ is locally free, $\operatorname{Ext}^i(\mathscr{F}, \omega_X) \cong H^i(X, \mathscr{F}^\vee \otimes \omega_X)$ by corollary 6.4.9. Substituting into theorem 6.5.9:

$$H^i(X,\, \mathscr{F}^\vee \otimes \omega_X) \cong H^{n-i}(X, \mathscr{F})^*.$$

Replacing $\mathscr{F}$ by $\mathscr{F}^\vee \otimes \omega_X$ (which is also locally free) and using $(\mathscr{F}^\vee \otimes \omega_X)^\vee \otimes \omega_X \cong \mathscr{F}$:

$$H^i(X, \mathscr{F}) \cong H^{n-i}(X,\, \mathscr{F}^\vee \otimes \omega_X)^*.$$

Part (1) follows by setting $\mathscr{F} = \mathcal{O}_X$ and $i = 0$: $H^0(X, \mathcal{O}_X) = k$ (since X is a variety, hence integral), so $H^n(X, \omega_X) \cong H^0(X, \mathcal{O}_X)^* = k$. Part (2) is the special case $\mathscr{F} = \mathcal{O}_X$.

Example 6.5.11: Applications of Serre Duality

1. *Curves: Riemann–Roch.* Let C be a smooth projective curve of genus g over k. Serre duality gives $H^1(C, \mathcal{L}) \cong H^0(C, \mathcal{L}^\vee \otimes \omega_C)^*$ for any line bundle $\mathcal{L}$. If $\mathcal{L} = \mathcal{O}_C(D)$ for a divisor D, this becomes:

$$h^1(C, \mathcal{O}(D)) = h^0(C, \omega_C(-D)) = h^0(C, \mathcal{O}(K - D)),$$

where K is a canonical divisor (with $\mathcal{O}(K) \cong \omega_C$) and $h^i = \dim_k H^i$. Substituting into the Euler characteristic formula $\chi(\mathcal{O}(D)) = h^0(\mathcal{O}(D)) - h^1(\mathcal{O}(D))$ and using the fact that $\chi(\mathcal{O}(D)) = \deg D + 1 - g$ (which follows from the additivity of χ on short exact sequences), we recover the *Riemann–Roch theorem* for curves (cf. theorem 7.1.19):

$$h^0(C, \mathcal{O}(D)) - h^0(C, \mathcal{O}(K - D)) = \deg D + 1 - g.$$

In particular, setting $D = K$: $h^0(\omega_C) - h^0(\mathcal{O}_C) = \deg K + 1 - g$, giving $\deg K = 2g - 2$ (since $h^0(\omega_C) = g$ by definition 5.5.26 and $h^0(\mathcal{O}_C) = 1$).

2. *The Hodge numbers of a smooth surface.* Let S be a smooth projective surface over $\mathbb{C}$. Serre duality gives:

$$H^0(S, \mathcal{O}_S) \cong H^2(S, \omega_S)^*,$$
$$H^1(S, \mathcal{O}_S) \cong H^1(S, \omega_S)^*,$$
$$H^2(S, \mathcal{O}_S) \cong H^0(S, \omega_S)^*.$$

   Since $h^{p,q} = \dim H^q(X, \Omega^p)$ by the Dolbeault theorem (theorem 0.6.17) and GAGA, these translate to symmetries of the Hodge diamond: $h^{0,0} = h^{2,2}$, $h^{0,1} = h^{2,1}$, and $h^{0,2} = h^{2,0}$. This is the algebraic form of Hodge symmetry (theorem 0.6.28).

3. *Kodaira vanishing (preview).* Let X be a smooth projective variety of dimension n over $\mathbb{C}$ and let $\mathcal{L}$ be an ample line bundle. The Kodaira vanishing theorem (which requires analytic or characteristic-0 methods beyond our present scope) asserts:

$$H^i(X, \omega_X \otimes \mathcal{L}) = 0 \qquad \text{for } i > 0.$$

   By Serre duality, this is equivalent to:

$$H^{n-i}(X, \mathcal{L}^{-1}) = 0 \qquad \text{for } i > 0,$$

   i.e. $H^j(X, \mathcal{L}^{-1}) = 0$ for $j < n$. Thus Kodaira vanishing can be stated either as acyclicity of $\omega_X \otimes \mathcal{L}$ or as "low-degree vanishing" of $\mathcal{L}^{-1}$. Serre duality reduces one formulation to the other.

4. *Self-duality of abelian varieties.* Let A be an abelian variety of dimension g over k. Then $\Omega_{A/k}$ is trivial: $\Omega_{A/k} \cong \mathcal{O}_A^g$ (the tangent bundle of a group scheme is trivial, by translation). Hence $\omega_A = \det(\mathcal{O}_A^g) \cong \mathcal{O}_A$, and Serre duality gives:

$$H^i(A, \mathcal{O}_A) \cong H^{g-i}(A, \mathcal{O}_A)^*.$$

   In particular, $\dim_k H^i(A, \mathcal{O}_A) = \binom{g}{i}$ (which follows from the Künneth formula for abelian varieties, or from the degeneration of the Hodge-to-de Rham spectral sequence).

6.5.12 Remark: Beyond Smooth Varieties For singular projective schemes, Serre duality still holds under the Cohen–Macaulay hypothesis (theorem 6.5.9(1)), but the dualizing sheaf ω_X is no longer necessarily invertible. It is a coherent sheaf, and its local structure reflects the singularities of X. The theory of dualizing complexes (due to Grothendieck and Hartshorne) extends duality to arbitrary proper schemes, replacing the dualizing sheaf with a complex of sheaves $\omega_X^\bullet \in D^b(\mathbf{Coh}(X))$. In this generality, Serre duality becomes Grothendieck duality:

$$\mathbf{R}\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X^\bullet) \cong \mathbf{R}\Gamma(X, \mathscr{F})^*$$

in the derived category. For Cohen–Macaulay X, the dualizing complex is concentrated in a single degree and is the dualizing sheaf we have defined. For Gorenstein schemes (where ω_X is invertible), the theory is especially clean. We shall not pursue derived duality in these notes, but the reader should be aware that theorem 6.5.9 is a special case of this more general structure.

6.6 Higher Direct Images

The sheaf cohomology groups $H^i(X, \mathscr{F})$ encode global information about a single scheme X. But in practice, schemes arise in *families*: a morphism $f : X \to Y$ presents X as a family of fibers $X_y = f^{-1}(y)$ parameterized by points of Y, and we want to understand how the cohomology $H^i(X_y, \mathscr{F}|_{X_y})$ varies as y moves. The *higher direct image sheaves* $R^i f_* \mathscr{F}$ are sheaves on the base Y that package this information: they are the "relative" version of sheaf cohomology, and they reduce to ordinary cohomology when Y is a point.

The central result of this section is the *finiteness theorem*: if $f : X \to Y$ is a projective morphism of Noetherian schemes and $\mathscr{F}$ is a coherent sheaf on X, then each $R^i f_* \mathscr{F}$ is a coherent sheaf on Y. When $Y = \operatorname{Spec} k$ is a point, this says that the cohomology groups of coherent sheaves on projective varieties are finite-dimensional — a fact we have already used (via forward reference) in proposition 6.4.13 and implicitly throughout our Serre duality arguments.

6.6.1 Definition and First Properties

Recall from definition 1.3.1 that the *direct image* (or pushforward) of a sheaf $\mathscr{F}$ on X along a continuous map $f : X \to Y$ is the sheaf $f_* \mathscr{F}$ on Y defined by $f_* \mathscr{F}(V) = \mathscr{F}(f^{-1}(V))$ for open $V \subseteq Y$. When f is a morphism of ringed spaces and $\mathscr{F}$ is an $\mathcal{O}_X$-module, the pushforward $f_* \mathscr{F}$ is an $\mathcal{O}_Y$-module (via the sheaf map $f^\# : \mathcal{O}_Y \to f_* \mathcal{O}_X$). The functor $f_* : \mathcal{O}_X\textbf{-Mod} \to \mathcal{O}_Y\textbf{-Mod}$ is left exact but not right exact in general.

Definition 6.6.1: Higher Direct Image Sheaves

Let $f : X \to Y$ be a morphism of ringed spaces. The *higher direct image sheaves* are the right derived functors of f_*:

$$R^i f_* : \mathcal{O}_X\textbf{-Mod} \longrightarrow \mathcal{O}_Y\textbf{-Mod}, \qquad R^i f_*(\mathscr{F}) = h^i(f_* \mathscr{I}^\bullet),$$

where $0 \to \mathscr{F} \to \mathscr{I}^\bullet$ is an injective resolution of $\mathscr{F}$ in the category of $\mathcal{O}_X$-modules. By construction, $R^0 f_* = f_*$, and a short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of $\mathcal{O}_X$-modules induces a long exact sequence of $\mathcal{O}_Y$-modules:

$$0 \to f_* \mathscr{F}' \to f_* \mathscr{F} \to f_* \mathscr{F}'' \xrightarrow{\delta} R^1 f_* \mathscr{F}' \to R^1 f_* \mathscr{F} \to \cdots$$

When $Y = \{pt\}$ is a point, the morphism $f : X \to \{pt\}$ is the unique map to the terminal space, and $f_* = \Gamma(X, -)$. Hence $R^i f_* \mathscr{F} = H^i(X, \mathscr{F})$, recovering ordinary sheaf cohomology. This is the sense in which higher direct images "relativize" cohomology.

The following result is fundamental: it says that higher direct images can be computed "stalkwise" by computing cohomology on the preimages of open sets.

Proposition 6.6.2: Sheafification of Presheaf Direct Images

Let $f : X \to Y$ be a continuous map of topological spaces and let $\mathscr{F}$ be a sheaf of abelian groups on X. Then $R^i f_* \mathscr{F}$ is the sheafification of the presheaf on Y defined by:

$$V \longmapsto H^i(f^{-1}(V),\, \mathscr{F}|_{f^{-1}(V)})$$

for open $V \subseteq Y$. If this presheaf is already a sheaf (e.g., when Y has a basis of open sets V for which $f^{-1}(V)$ is "cohomologically well-behaved" in an appropriate sense), then no sheafification is needed.

***Proof*:**
Let $0 \to \mathscr{F} \to \mathscr{I}^\bullet$ be an injective resolution. For each open $V \subseteq Y$:

$$(f_*\mathscr{I}^p)(V) = \mathscr{I}^p(f^{-1}(V)).$$

The presheaf $V \mapsto h^i(\mathscr{I}^\bullet(f^{-1}(V)))$ computes $H^i(f^{-1}(V), \mathscr{F}|_{f^{-1}(V)})$ provided the restriction $\mathscr{I}^\bullet|_{f^{-1}(V)}$ is an injective (or at least acyclic) resolution of $\mathscr{F}|_{f^{-1}(V)}$. By lemma 6.4.5, the restriction of an injective sheaf to an open subset is injective, so this is indeed the case.

Now, the derived functors $R^i f_* \mathscr{F}$ are defined as the cohomology sheaves $h^i(f_*\mathscr{I}^\bullet)$. The presheaf $V \mapsto h^i((f_*\mathscr{I}^\bullet)(V))$ need not be a sheaf (taking cohomology does not commute with the sheaf condition in general), so $R^i f_* \mathscr{F}$ is the *sheafification* of this presheaf. Since $(f_*\mathscr{I}^\bullet)(V) = \mathscr{I}^\bullet(f^{-1}(V))$, the presheaf in question is $V \mapsto H^i(f^{-1}(V), \mathscr{F})$, as claimed.

Proposition 6.6.3: Stalks of Higher Direct Images

Let $f : X \to Y$ be a continuous map of topological spaces, $\mathscr{F}$ a sheaf of abelian groups on X, and $y \in Y$. Then there is a natural map:

$$(R^i f_* \mathscr{F})_y \longrightarrow H^i(X_y,\, \mathscr{F}|_{X_y}),$$

where $X_y = f^{-1}(y)$ is the (set-theoretic) fiber. This map is an isomorphism if y has a basis of open neighborhoods $\{V_\alpha\}$ such that the natural maps $\varinjlim_\alpha H^i(f^{-1}(V_\alpha), \mathscr{F}) \to H^i(X_y, \mathscr{F}|_{X_y})$ are isomorphisms.

***Proof*:**
By proposition 6.6.2, the stalk of $R^i f_* \mathscr{F}$ at y is:

$$(R^i f_* \mathscr{F})_y = \varinjlim_{V \ni y} H^i(f^{-1}(V),\, \mathscr{F}|_{f^{-1}(V)}).$$

The inclusion $X_y = f^{-1}(y) \hookrightarrow f^{-1}(V)$ for each open $V \ni y$ induces a restriction map $H^i(f^{-1}(V), \mathscr{F}) \to H^i(X_y, \mathscr{F}|_{X_y})$, and these are compatible as V varies, yielding the natural map on the colimit. The condition for this to be an isomorphism is precisely that cohomology "commutes with the limit" of shrinking V to $\{y\}$.

6.6.4 Remark: When Do Stalks Compute Fiber Cohomology? The natural map $(R^i f_* \mathscr{F})_y \to H^i(X_y, \mathscr{F}|_{X_y})$ is *not* an isomorphism in general, and the discrepancy is important. The stalk sees the cohomology of all "thickenings" of the fiber (i.e., the preimages of open neighborhoods of y), while the fiber cohomology sees only the fiber itself. These can differ when nearby fibers contribute non-trivially.
There are two important situations where the map is an isomorphism:

1. When f is proper and the topological spaces are "reasonable" (e.g., locally Noetherian schemes). In this case, the proper mapping theorem ensures that the cohomological information does not "leak" from nearby fibers.

2. When $Y = \operatorname{Spec} R$ for a local ring $(R, \mathfrak{m})$ and $y = [\mathfrak{m}]$ is the closed point. Then the only open set containing y is Y itself, so the stalk is simply $H^i(X, \mathscr{F})$ and the fiber is $X_y = X \times_Y \operatorname{Spec}(R/\mathfrak{m})$.

In the scheme-theoretic setting, the correct fiber to consider is the *scheme-theoretic fiber* $X_y = X \times_Y \operatorname{Spec} \kappa(y)$ (see definition 3.2.9), and the comparison between the stalk and the fiber cohomology involves the *formal functions theorem* (Grothendieck's theorem on formal functions), which we shall not develop in full here.

We now establish the key properties that make higher direct images useful in the scheme-theoretic setting:

Proposition 6.6.5: Quasi-Coherence of Higher Direct Images

Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of schemes, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Then each $R^i f_* \mathscr{F}$ is quasi-coherent on Y.

***Proof*:**
The question is local on Y, so we may assume $Y = \operatorname{Spec} R$ is affine. Since f is quasi-compact and quasi-separated, the preimage $X = f^{-1}(Y)$ is a quasi-compact quasi-separated scheme. Choose a finite affine open cover $\mathcal{U} = \{U_1, \ldots, U_r\}$ of X. Each intersection $U_{i_0,\ldots,i_p} = U_{i_0} \cap \cdots \cap U_{i_p}$ is quasi-compact (by the quasi-separated hypothesis), and $\mathscr{F}$ is quasi-coherent on each.

On the affine base, the Čech-to-derived spectral sequence (which generalizes the comparison of theorem 6.1.19 to the relative setting) shows that $R^i f_* \mathscr{F}$ can be computed by a Čech-type complex. More precisely, the presheaf $V \mapsto H^i(f^{-1}(V), \mathscr{F})$ is already a sheaf when Y is affine (this uses the quasi-compact quasi-separated hypothesis), and it is computed by the Čech complex associated to the cover $\mathcal{U}$. Each term of this complex is a product of quasi-coherent sheaves on Y (since the pushforward of a quasi-coherent sheaf along a quasi-compact quasi-separated morphism is quasi-coherent, by proposition 5.1.20), and the cohomology of a complex of quasi-coherent sheaves is quasi-coherent.

6.6.2 The Finiteness Theorem

The most important structural result about higher direct images is that they preserve coherence for projective morphisms. This is the "relative" version of the statement that cohomology groups of coherent sheaves on projective varieties are finite-dimensional, and it is the technical engine behind semicontinuity theorems, base change, and the construction of Hilbert schemes.

The strategy is to reduce to the computation of cohomology of twisted sheaves on projective space (theorem 6.3.1), which we already know completely. The key tools are:

1. Every coherent sheaf on a projective scheme is a quotient of a direct sum of twisted sheaves (corollary 5.3.26).
2. Cohomology of twisted sheaves on $\mathbb{P}^n_R$ gives finitely generated R-modules (theorem 6.3.1).
3. Descending induction on i, using the dimension bound $H^i = 0$ for $i > \dim X$ (theorem 6.2.5).

Theorem 6.6.6: Finiteness of Higher Direct Images

Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $\mathscr{F}$ be a coherent sheaf on X. Then:

1. Each $R^i f_* \mathscr{F}$ is a coherent sheaf on Y.
2. $R^i f_* \mathscr{F} = 0$ for $i > \dim X$.

***Proof*:**
Part (2) follows immediately from theorem 6.2.5: the stalks $(R^i f_* \mathscr{F})_y$ are subquotients of $H^i(f^{-1}(V), \mathscr{F})$ for open neighborhoods V of y, and $\dim f^{-1}(V) \leq \dim X$.

For part (1), the question is local on Y, so we may assume $Y = \operatorname{Spec} R$ is affine. Since f is projective, there is a closed immersion $i : X \hookrightarrow \mathbb{P}^N_R$ for some N, and f factors as $f = \pi \circ i$ where $\pi : \mathbb{P}^N_R \to \operatorname{Spec} R$ is the structure morphism. By proposition 6.2.4, $R^i f_* \mathscr{F} \cong R^i \pi_*(i_* \mathscr{F})$. Since i is a closed immersion, $i_* \mathscr{F}$ is coherent on $\mathbb{P}^N_R$ (by proposition 3.1.24). Hence it suffices to prove the theorem for $\mathbb{P}^N_R$ over $\operatorname{Spec} R$, i.e., to show:

Claim: For any coherent sheaf $\mathscr{G}$ on $\mathbb{P}^N_R$, the R-module $H^i(\mathbb{P}^N_R, \mathscr{G})$ is finitely generated.

We prove the claim by descending induction on i. For $i > N$, we have $H^i = 0$ by theorem 6.2.5, and there is nothing to prove.

For the inductive step, assume that $H^j(\mathbb{P}^N_R, \mathscr{H})$ is finitely generated for all $j > i$ and all coherent $\mathscr{H}$. By corollary 5.3.26, $\mathscr{G}$ admits a surjection from a direct sum of twisted sheaves:

$$\mathcal{E} = \bigoplus_{j=1}^{s} \mathcal{O}_{\mathbb{P}^N}(d_j) \twoheadrightarrow \mathscr{G} \to 0$$

for some integers $d_1, \ldots, d_s$. Let $\mathcal{K} = \ker(\mathcal{E} \twoheadrightarrow \mathscr{G})$, which is coherent (since $\mathbf{Coh}(X)$ is an abelian category). The short exact sequence $0 \to \mathcal{K} \to \mathcal{E} \to \mathscr{G} \to 0$ induces a long exact sequence:

$$H^i(\mathbb{P}^N, \mathcal{E}) \longrightarrow H^i(\mathbb{P}^N, \mathscr{G}) \longrightarrow H^{i+1}(\mathbb{P}^N, \mathcal{K}) \longrightarrow H^{i+1}(\mathbb{P}^N, \mathcal{E}).$$

By theorem 6.3.1, $H^i(\mathbb{P}^N, \mathcal{E}) = \bigoplus_j H^i(\mathbb{P}^N, \mathcal{O}(d_j))$ is a finitely generated R-module (it is even free). By the inductive hypothesis, $H^{i+1}(\mathbb{P}^N, \mathcal{K})$ is finitely generated. Hence $H^i(\mathbb{P}^N, \mathscr{G})$, which sits between two finitely generated R-modules in the exact sequence, is itself finitely generated (using the standard fact that a subquotient of a finitely generated module over a Noetherian ring is finitely generated).

The most commonly used special case is when the base is a field:

Corollary 6.6.7: Finite-Dimensionality of Cohomology

Let X be a projective scheme over a field k, and let $\mathscr{F}$ be a coherent sheaf on X. Then each cohomology group $H^i(X, \mathscr{F})$ is a finite-dimensional k-vector space.

***Proof*:**
Apply theorem 6.6.6 with $Y = \operatorname{Spec} k$. The higher direct images $R^i f_* \mathscr{F}$ are coherent sheaves on $\operatorname{Spec} k$, i.e., finite-dimensional k-vector spaces. Since $R^i f_* \mathscr{F} = H^i(X, \mathscr{F})$ when Y is a point, the result follows.

This result justifies the use of the notation $h^i(X, \mathscr{F}) = \dim_k H^i(X, \mathscr{F})$ for the *Betti numbers* of the sheaf $\mathscr{F}$, and underpins the entire theory of Hilbert polynomials and Euler characteristics. Let us also record the general version for the base that we referenced earlier:

Corollary 6.6.8: (Reminder) Pushforward of Coherent Sheaves by Projective Morphisms

Let $f : X \to Y$ be a projective morphism of Noetherian schemes. Then f_* sends coherent sheaves on X to coherent sheaves on Y, and more generally, each $R^i f_*$ sends $\mathbf{Coh}(X)$ to $\mathbf{Coh}(Y)$.

***Proof*:**
This is a restatement of theorem 6.6.6(1).

6.6.9 Remark: Properness vs. Projectivity The finiteness theorem as stated requires f to be *projective*. The natural level of generality is *proper*: Grothendieck proved that theorem 6.6.6 holds whenever f is a proper morphism of Noetherian schemes. The proof in the proper case is substantially harder: one cannot reduce to $\mathbb{P}^N$ via a closed immersion, and instead uses Chow's lemma (theorem 3.5.28) to approximate f by a projective morphism, together with a careful analysis of the resulting spectral sequence. Since every projective morphism is proper (theorem 3.5.26), our statement covers the cases most commonly encountered in practice.

6.6.3 Computation of Higher Direct Images

Higher direct images can be computed using the Čech machinery, provided the morphism is sufficiently well-behaved:

Proposition 6.6.10: Čech Computation of Higher Direct Images

Let $f : X \to Y$ be a separated morphism of Noetherian schemes, $\mathscr{F}$ a quasi-coherent sheaf on X, and $\mathcal{U} = \{U_1, \ldots, U_r\}$ a finite open affine cover of X. Then for each open affine $V = \operatorname{Spec} R \subseteq Y$, the higher direct images are computed by:

$$R^i f_*(\mathscr{F})(V) \;\cong\; \check{H}^i(\mathcal{U}|_{f^{-1}(V)}, \mathscr{F}|_{f^{-1}(V)}),$$

where $\mathcal{U}|_{f^{-1}(V)} = \{U_j \cap f^{-1}(V)\}$ is the restricted cover.

***Proof*:**
For affine $V \subseteq Y$, the preimage $f^{-1}(V)$ is a separated Noetherian scheme (separatedness is preserved under base change), and the sets $U_j \cap f^{-1}(V)$ form an affine cover of $f^{-1}(V)$ (as U_j is affine and $f^{-1}(V)$ is separated, the intersection of two affine opens in $f^{-1}(V)$ is affine by proposition 3.5.8). By theorem 6.1.19, the Čech cohomology with respect to this affine cover computes the derived-functor cohomology:

$$\check{H}^i(\mathcal{U}|_{f^{-1}(V)}, \mathscr{F}) \cong H^i(f^{-1}(V), \mathscr{F}).$$

Since the presheaf $V \mapsto H^i(f^{-1}(V), \mathscr{F})$ is already a sheaf on affine opens of Y (as noted in proposition 6.6.5), this computes $R^i f_* \mathscr{F}$ on the basis of affine opens.

Example 6.6.11: Higher Direct Images

1. *Structure morphism of projective space.* Let $\pi : \mathbb{P}^n_R \to \operatorname{Spec} R$ be the structure morphism. Then:
$$R^i \pi_*\big(\mathcal{O}_{\mathbb{P}^n}(d)\big) \;\cong\; \widetilde{H^i(\mathbb{P}^n_R, \mathcal{O}(d))},$$
which by theorem 6.3.1 is the quasi-coherent sheaf associated to the R-module $H^i(\mathbb{P}^n_R, \mathcal{O}(d))$. In particular, $\pi_*(\mathcal{O}_{\mathbb{P}^n}) \cong \widetilde{R} = \mathcal{O}_{\operatorname{Spec} R}$ and $R^i \pi_*(\mathcal{O}_{\mathbb{P}^n}) = 0$ for $i > 0$. More generally:
$$\pi_*(\mathcal{O}(d)) \cong \begin{cases} (\widetilde{R[x_0, \ldots, x_n]})_d & d \geq 0, \\ 0 & d < 0, \end{cases}$$
and all higher direct images vanish for $0 < i < n$.

2. *Blowing up a point.* Let $Y = \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$ and let $f : X = \mathrm{Bl}_0 Y \to Y$ be the blowup of the origin. This is a projective morphism with f an isomorphism over $Y \setminus \{0\}$ and fiber $f^{-1}(0) \cong \mathbb{P}^1_k$ (the exceptional divisor E). Then:
$$f_* \mathcal{O}_X \cong \mathcal{O}_Y \qquad \text{and} \qquad R^i f_* \mathcal{O}_X = 0 \text{ for } i > 0.$$
The first statement holds because f is birational and Y is normal (so $f_* \mathcal{O}_X = \mathcal{O}_Y$ by the algebraic Hartogs lemma, theorem 2.6.18). The vanishing of higher direct images reflects the fact that the fibers are either points (away from the origin) or $\mathbb{P}^1$ (at the origin), and $H^i(\mathbb{P}^1, \mathcal{O}) = 0$ for $i > 0$.

   More interesting is the behavior of the ideal sheaf of the exceptional divisor. Since $E \cong \mathbb{P}^1$ and $\mathcal{O}_X(-E)|_E \cong \mathcal{O}_{\mathbb{P}^1}(1)$ (the conormal bundle of the exceptional divisor in a blowup is the tautological bundle), we have $R^1 f_*(\mathcal{O}_X(-E)) = 0$ but $f_*(\mathcal{O}_X(-E)) \cong \mathfrak{m}_0$, the maximal ideal of the origin in $\mathcal{O}_Y$.

3. *Projection of a product.* Let C be a smooth projective curve of genus g over k, and consider the projection $f = \mathrm{pr}_2 : C \times_k \mathbb{P}^1_k \to \mathbb{P}^1_k$. Then for the structure sheaf:

$$f_*\mathcal{O}_{C\times\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}, \qquad R^1 f_*\mathcal{O}_{C\times\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}^g.$$

The first isomorphism follows from $H^0(C, \mathcal{O}_C) = k$. For the second: by the Künneth formula (or by proposition 6.6.10 with the product cover), the stalk of $R^1 f_*\mathcal{O}$ at a point $p \in \mathbb{P}^1$ is $H^1(C, \mathcal{O}_C) \cong k^g$, and the resulting sheaf is locally free of rank g. Since $\mathbb{P}^1$ has no non-trivial vector bundles of rank g with vanishing H^1 (by the classification of vector bundles on $\mathbb{P}^1$), we get $R^1 f_*\mathcal{O} \cong \mathcal{O}^g$.

This illustrates the general principle: the rank of $R^i f_*\mathscr{F}$ at a point y is *at least* $h^i(X_y, \mathscr{F}|_{X_y})$, with equality when the fiber dimension is constant (the precise statement is the semicontinuity theorem, which we treat in the next section).

4. *Failure for non-proper morphisms.* Let $j : \mathbb{A}^1_k \setminus \{0\} \hookrightarrow \mathbb{A}^1_k$ be the open immersion. Then $j_*\mathcal{O} = \mathcal{O}_{\mathbb{A}^1}[1/x]$ (rational functions regular away from the origin), which is *not* coherent as an $\mathcal{O}_{\mathbb{A}^1}$-module: it is quasi-coherent (corresponding to the $k[x]$-module $k[x, x^{-1}]$) but not finitely generated. Also, $R^1 j_*\mathcal{O} = 0$ (since $\mathbb{A}^1 \setminus \{0\}$ is affine, hence acyclic), but the pushforward already fails coherence. This shows that the properness (or at least projectivity) hypothesis in theorem 6.6.6 cannot be dropped.

To close this section, we record the connection between higher direct images and the Leray spectral sequence, which is the fundamental tool for computing cohomology of a space in terms of a morphism to a simpler space:

Theorem 6.6.12: Leray Spectral Sequence

Let $f : X \to Y$ be a morphism of ringed spaces and let $\mathscr{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then there is a convergent spectral sequence:

$$E_2^{p,q} = H^p(Y, R^q f_*\mathscr{F}) \implies H^{p+q}(X, \mathscr{F}).$$

This is natural in $\mathscr{F}$.

***Proof*:**
This is the Grothendieck spectral sequence for the composite functor $\Gamma(X, -) = \Gamma(Y, -) \circ f_*$. We need f_* to send injective $\mathcal{O}_X$-modules to $\Gamma(Y, -)$-acyclic $\mathcal{O}_Y$-modules. Indeed, if $\mathscr{I}$ is an injective $\mathcal{O}_X$-module, then $f_*\mathscr{I}$ is flasque: for $U \subseteq V$ open in Y, the restriction $f_*\mathscr{I}(V) = \mathscr{I}(f^{-1}(V)) \to \mathscr{I}(f^{-1}(U)) = f_*\mathscr{I}(U)$ is surjective because $\mathscr{I}$ is flasque (by proposition 6.1.5) and $f^{-1}(U) \subseteq f^{-1}(V)$. Hence $f_*\mathscr{I}$ is acyclic for $\Gamma(Y, -)$, and the Grothendieck spectral sequence applies.

Example 6.6.13: Applications of the Leray Spectral Sequence

1. *Affine morphisms.* If $f : X \to Y$ is an affine morphism of Noetherian schemes and $\mathscr{F}$ is quasi-coherent, then $R^q f_* \mathscr{F} = 0$ for all $q > 0$ (since the fibers are affine). The Leray spectral sequence degenerates at E_2, giving $H^p(X, \mathscr{F}) \cong H^p(Y, f_*\mathscr{F})$. This reduces the cohomology of X to the cohomology of Y with coefficients in the pushforward.

2. *Projective bundles.* Let $\mathcal{E}$ be a locally free sheaf of rank $r+1$ on a Noetherian scheme Y and let $X = \mathbb{P}(\mathcal{E}) \to Y$ be the associated projective bundle. The higher direct images of the twisted sheaves $\mathcal{O}_X(d)$ are computed by the relative version of theorem 6.3.1: $f_*\mathcal{O}_X(d) \cong \operatorname{Sym}^d \mathcal{E}$ for $d \geq 0$, and $R^i f_* \mathcal{O}_X(d) = 0$ for $0 < i < r$. The Leray spectral sequence then computes $H^*(X, \mathcal{O}_X(d))$ in terms of $H^*(Y, \operatorname{Sym}^d \mathcal{E})$.

3. *Recovering the projection formula.* For a proper morphism $f : X \to Y$ and a locally free sheaf $\mathcal{E}$ on Y, the *projection formula* gives:

$$R^i f_*(\mathscr{F} \otimes f^*\mathcal{E}) \;\cong\; (R^i f_* \mathscr{F}) \otimes \mathcal{E}.$$

   This follows from the fact that tensoring with a locally free sheaf is exact and commutes with the formation of higher direct images (since $f^*\mathcal{E}$ is locally free on X and the tensor product with a flat module commutes with cohomology). Combined with the Leray spectral sequence, the projection formula reduces many cohomological computations to known cases.

6.6.14 Remark: Finiteness Results in Context The finiteness theorem (theorem 6.6.6) is one of several finiteness results that govern the cohomology of schemes. Let us place it in context:

1. *Affine vanishing* (theorem 6.2.9): $H^i = 0$ for $i > 0$ on affine schemes with quasi-coherent coefficients. This is the strongest vanishing result, but it applies only to affine schemes.

2. *Dimension bound* (theorem 6.2.5): $H^i = 0$ for $i > \dim X$ on Noetherian spaces. This holds for *all* sheaves of abelian groups, not just quasi-coherent ones.

3. *Finiteness* (theorem 6.6.6): H^i is finitely generated for coherent sheaves on projective schemes over Noetherian rings. This does *not* say the groups vanish, only that they are "finitely complicated."

4. *Serre vanishing* (see theorem 5.3.25): for a coherent sheaf $\mathscr{F}$ on a projective scheme X, there exists d_0 such that $H^i(X, \mathscr{F}(d)) = 0$ for all $i > 0$ and all $d \geq d_0$. This says that twisting "enough" kills all higher cohomology.

Together, these results are what cohomological arguments rest on in algebraic geometry. The finiteness theorem sits at the center: it is what allows us to define numerical invariants (Euler characteristics, Hilbert polynomials, Hodge numbers) that can then be studied using the vanishing results.

6.7 Euler Characteristics and Hilbert Polynomials

With the finiteness theorem (theorem 6.6.6) in hand, we know that the cohomology groups $H^i(X, \mathscr{F})$ of a coherent sheaf on a projective scheme over a field are finite-dimensional vector spaces. This means we can *count*: we can form the alternating sum of their dimensions, obtaining the *Euler characteristic* $\chi(X, \mathscr{F})$. The Euler characteristic is a cruder invariant than the individual cohomology groups, but it has a decisive advantage: it is *additive* on short exact sequences, while the individual h^i are not. This additivity makes the Euler characteristic computable by purely algebraic means, without explicitly resolving the sheaf.

When X is a projective scheme and $\mathscr{F}$ is twisted by a line bundle $\mathcal{O}_X(d)$, the Euler characteristic $\chi(X, \mathscr{F}(d))$ is a polynomial in d: the *Hilbert polynomial*. This polynomial encodes fundamental geometric invariants of X — its dimension, degree, and arithmetic genus — and is the basis for the theory of Hilbert schemes and moduli spaces (cf. the discussion in section 3.3.6).

For curves, the interplay between the Euler characteristic and Serre duality yields the *Riemann–Roch theorem*, the single most important computational tool in the theory of algebraic curves.

6.7.1 The Euler Characteristic

Definition 6.7.1: Euler Characteristic

Let X be a projective scheme over a field k and let $\mathscr{F}$ be a coherent sheaf on X. The *Euler characteristic* of $\mathscr{F}$ is:

$$\chi(X, \mathscr{F}) = \sum_{i=0}^{\dim X} (-1)^i \dim_k H^i(X, \mathscr{F}) = \sum_{i \geq 0} (-1)^i h^i(X, \mathscr{F}),$$

where $h^i(X, \mathscr{F}) = \dim_k H^i(X, \mathscr{F})$. The sum is finite by theorem 6.2.5 and each term is finite by corollary 6.6.7.

The definition immediately connects to the one given in the curves chapter (**??**), which was stated for line bundles on curves. The key property that makes the Euler characteristic tractable is the following:

Proposition 6.7.2: Additivity of the Euler Characteristic

Let X be a projective scheme over a field k. If $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is a short exact sequence of coherent sheaves on X, then:

$$\chi(X, \mathscr{F}) = \chi(X, \mathscr{F}') + \chi(X, \mathscr{F}'').$$

***Proof*:**
The short exact sequence induces a long exact sequence in cohomology:

$$0 \to H^0(X, \mathscr{F}') \to H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}'') \to H^1(X, \mathscr{F}') \to \cdots \to H^n(X, \mathscr{F}'') \to 0,$$

where $n = \dim X$. This is a finite exact sequence of finite-dimensional k-vector spaces. The alternating sum of dimensions in any finite exact sequence is zero (this is a standard exercise in linear algebra: if $0 \to V_0 \to V_1 \to \cdots \to V_m \to 0$ is exact, then $\sum(-1)^j \dim V_j = 0$). Grouping the terms by sheaf:

$$\sum_{i=0}^{n}\left[(-1)^i h^i(\mathscr{F}') - (-1)^i h^i(\mathscr{F}) + (-1)^i h^i(\mathscr{F}'')\right] = 0,$$

which rearranges to $\chi(\mathscr{F}) = \chi(\mathscr{F}') + \chi(\mathscr{F}'')$.

6.7.3 Remark: Euler Characteristic as a Group Homomorphism The additivity property can be rephrased as follows. The Grothendieck group $K_0(X)$ of coherent sheaves on X is the free abelian group generated by isomorphism classes $[\mathscr{F}]$ of coherent sheaves, modulo the relation $[\mathscr{F}] = [\mathscr{F}'] + [\mathscr{F}'']$ for every short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$. The Euler characteristic defines a group homomorphism:

$$\chi : K_0(X) \longrightarrow \mathbb{Z}, \qquad [\mathscr{F}] \longmapsto \chi(X, \mathscr{F}).$$

This is the starting point for K-theory in algebraic geometry. Many invariants (the Hilbert polynomial, the Chern character, the Todd class) factor through $K_0(X)$, and their computability ultimately rests on the additivity of χ.

Example 6.7.4: Euler Characteristics

1. *Twisted sheaves on $\mathbb{P}^n_k$.* By theorem 6.3.1, the only nonzero cohomology groups of $\mathcal{O}_{\mathbb{P}^n}(d)$ are H^0 (when $d \geq 0$) and H^n (when $d \leq -(n+1)$). Hence:

$$\chi(\mathbb{P}^n_k, \mathcal{O}(d)) = \begin{cases} \dbinom{d+n}{n} & d \geq 0, \\ 0 & -(n+1) < d < 0, \\ (-1)^n \dbinom{-d-1}{n} & d \leq -(n+1). \end{cases}$$

   A fact (which we will prove in the next subsection) is that all three cases are given by a single polynomial in d:

$$\chi(\mathbb{P}^n_k, \mathcal{O}(d)) = \binom{d+n}{n} = \frac{(d+1)(d+2)\cdots(d+n)}{n!}$$

   for *all* $d \in \mathbb{Z}$. For instance, when $n = 1$: $\chi(\mathbb{P}^1, \mathcal{O}(d)) = d+1$. When $d = -1$: $\chi(\mathbb{P}^1, \mathcal{O}(-1)) = 0$, which is correct since both H^0 and H^1 vanish. When $d = -2$: $\chi(\mathbb{P}^1, \mathcal{O}(-2)) = -1$, reflecting $h^0 = 0$ and $h^1 = 1$.

2. *Structure sheaf of a curve.* Let C be a smooth projective curve of genus g over k. Then $h^0(C, \mathcal{O}_C) = 1$ (since C is integral and proper) and $h^1(C, \mathcal{O}_C) = g$ (this is one definition of the genus; see definition 7.1.13). Hence:

$$\chi(C, \mathcal{O}_C) = 1 - g.$$

By Serre duality (corollary 6.5.10), $h^1(C, \mathcal{O}_C) = h^0(C, \omega_C) = p_g$, the geometric genus (definition 5.5.26). So $\chi(C, \mathcal{O}_C) = 1 - p_g$, confirming the equivalence of the arithmetic and geometric genus for smooth curves (proposition 7.1.14).

3. *Structure sheaf of a surface.* Let S be a smooth projective surface over k. Then:

$$\chi(S, \mathcal{O}_S) = h^0(\mathcal{O}_S) - h^1(\mathcal{O}_S) + h^2(\mathcal{O}_S) = 1 - q + p_g,$$

where $q = h^1(\mathcal{O}_S)$ is the *irregularity* and $p_g = h^2(\mathcal{O}_S) = h^0(\omega_S)$ is the geometric genus (the last equality by Serre duality). The integer $\chi(\mathcal{O}_S) = 1 - q + p_g$ is a fundamental invariant in the classification of surfaces.

4. *Additivity in action.* Let $C \subset \mathbb{P}^2_k$ be a smooth plane curve of degree d. The short exact sequence $0 \to \mathcal{O}_{\mathbb{P}^2}(-d) \to \mathcal{O}_{\mathbb{P}^2} \to \mathcal{O}_C \to 0$ gives:

$$\chi(C, \mathcal{O}_C) = \chi(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}) - \chi(\mathbb{P}^2, \mathcal{O}(-d)) = 1 - \binom{-d+2}{2} = 1 - \frac{(d-1)(d-2)}{2}.$$

Hence $g = \frac{(d-1)(d-2)}{2}$, the classical genus formula for plane curves. For $d = 1$: $g = 0$ (a line). For $d = 3$: $g = 1$ (an elliptic curve). For $d = 4$: $g = 3$ (a quartic).

6.7.2 Hilbert Polynomials

The Euler characteristic of a twisted sheaf $\mathscr{F}(d)$ is a polynomial in d. This is the most important non-obvious property of the Euler characteristic, and it rests on two facts: Serre vanishing (which eventually kills all higher cohomology) and the additivity of χ (which allows induction via hyperplane sections).

Theorem 6.7.5: Hilbert Polynomial

Let X be a projective scheme over a field k, equipped with a very ample sheaf $\mathcal{O}_X(1)$, and let $\mathscr{F}$ be a coherent sheaf on X. Then there exists a polynomial $P_{\mathscr{F}} \in \mathbb{Q}[t]$, called the *Hilbert polynomial* of $\mathscr{F}$, such that:

$$P_{\mathscr{F}}(d) = \chi(X, \mathscr{F}(d)) \qquad \text{for all } d \in \mathbb{Z}.$$

The degree of $P_{\mathscr{F}}$ equals the dimension of the support of $\mathscr{F}$, and the leading coefficient is positive.

***Proof*:**
We proceed by induction on $r = \dim(\operatorname{supp}\mathscr{F})$.

Base case ($r = 0$): If $\operatorname{supp}\mathscr{F}$ is zero-dimensional, it consists of finitely many closed points (since X is Noetherian). On a zero-dimensional space, H^0 is the only nonzero cohomology group, and twisting by $\mathcal{O}_X(d)$ has no effect on a sheaf supported at finitely many points (since the stalk of $\mathcal{O}_X(d)$ at a closed point is isomorphic to $\mathcal{O}_{X,x}$, with the twist absorbed into the local isomorphism). Hence $\chi(X, \mathscr{F}(d)) = h^0(X, \mathscr{F}) = \dim_k \Gamma(X, \mathscr{F})$, a constant polynomial of degree 0.

Inductive step: Let $r \geq 1$. Choose a hyperplane section $H \in |\mathcal{O}_X(1)|$ that does not contain any associated point of $\mathscr{F}$ (this is possible because $\mathscr{F}$ has finitely many associated points and $|\mathcal{O}_X(1)|$ is base-point free by theorem 5.3.25). Then multiplication by the section $s \in H^0(X, \mathcal{O}_X(1))$ defining H gives an injective map $\mathscr{F}(-1) \xrightarrow{\cdot s} \mathscr{F}$. Let $\mathscr{G}$ be the cokernel:

$$0 \longrightarrow \mathscr{F}(-1) \xrightarrow{\cdot s} \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow 0.$$

Since s vanishes on H, the sheaf $\mathscr{G}$ is supported on $\operatorname{supp}(\mathscr{F}) \cap H$, which has dimension at most $r-1$ (the hyperplane cuts the dimension by at least one, since it avoids the associated points). Twisting the whole sequence by $\mathcal{O}_X(d)$:

$$0 \longrightarrow \mathscr{F}(d-1) \longrightarrow \mathscr{F}(d) \longrightarrow \mathscr{G}(d) \longrightarrow 0.$$

By the additivity of χ:

$$\chi(X, \mathscr{F}(d)) - \chi(X, \mathscr{F}(d-1)) = \chi(X, \mathscr{G}(d)).$$

By the inductive hypothesis, $\chi(X, \mathscr{G}(d))$ is a polynomial in d of degree at most $r-1$. The function $\Delta P(d) = P(d) - P(d-1)$ is the *difference operator*; if ΔP is a polynomial of degree $\leq r-1$, then P is a polynomial of degree $\leq r$[a]. Since $\chi(X, \mathscr{F}(d))$ is an integer for all d and agrees with a polynomial in d after taking differences, it is itself a polynomial in d (with rational coefficients), of degree at most r.

To show the degree is exactly r (and the leading coefficient is positive), we use the fact that for $d \gg 0$, Serre vanishing (theorem 5.3.25) gives $\chi(X, \mathscr{F}(d)) = h^0(X, \mathscr{F}(d))$, which is a polynomial in d of degree equal to $\dim(\operatorname{supp}\mathscr{F})$ with positive leading coefficient (this follows from the theory of Hilbert functions in commutative algebra).

[a] This is the discrete analogue of the fact that if $f'(x)$ is a polynomial of degree $r-1$, then $f(x)$ is a polynomial of degree r. Concretely: the binomial coefficients $\binom{d}{j}$ form a basis for $\mathbb{Q}[d]$, and $\Delta\binom{d}{j} = \binom{d}{j} - \binom{d-1}{j} = \binom{d-1}{j-1}$, so the difference operator lowers the degree by exactly one.

Definition 6.7.6: Hilbert Polynomial and Arithmetic Genus

Let X be a projective scheme of dimension n over a field k. The *Hilbert polynomial* of X (with respect to $\mathcal{O}_X(1)$) is $P_X = P_{\mathcal{O}_X}$. The *arithmetic genus* of X is:

$$p_a(X) = (-1)^n\big(\chi(X, \mathcal{O}_X) - 1\big) = (-1)^n(P_X(0) - 1).$$

For a curve ($n = 1$): $p_a = 1 - \chi(\mathcal{O}_C) = g$. For a surface ($n = 2$): $p_a = \chi(\mathcal{O}_S) - 1 = p_g - q$.

Example 6.7.7: Hilbert Polynomials

1. *Projective space.* By Example 6.7.4(1), $P_{\mathbb{P}^n}(d) = \binom{d+n}{n}$. This is a polynomial of degree n in d with leading coefficient $1/n!$. The arithmetic genus is $p_a(\mathbb{P}^n) = (-1)^n(\binom{n}{n} - 1) = 0$.
2. *Plane curve of degree d.* Let $C \subset \mathbb{P}^2_k$ be a curve of degree d. The short exact sequence

$0 \to \mathcal{O}_{\mathbb{P}^2}(-d) \to \mathcal{O}_{\mathbb{P}^2} \to \mathcal{O}_C \to 0$ gives, after twisting by $\mathcal{O}(m)$ and taking χ:

$$P_C(m) = \binom{m+2}{2} - \binom{m-d+2}{2} = dm - \frac{d(d-3)}{2}.$$

This is a polynomial of degree 1 (as expected, since $\dim C = 1$) with leading coefficient d, which is the *degree* of C. The arithmetic genus is $p_a = 1 - P_C(0) = \frac{(d-1)(d-2)}{2}$.

3. *Hilbert polynomial of a line bundle on a curve.* Let C be a smooth projective curve of genus g and let $\mathcal{L}$ be a line bundle of degree ℓ on C. By the Riemann–Roch theorem (which we prove below):
$$P_{\mathcal{L}}(d) = \chi(C, \mathcal{L}(d)) = \ell + d \cdot \deg(\mathcal{O}_C(1)) + 1 - g.$$
In particular, $\chi(C, \mathcal{L}) = \deg \mathcal{L} + 1 - g$, confirming **??**.

4. *Twisted ideal sheaf.* Let $Z \subset \mathbb{P}^n_k$ be a closed subscheme with ideal sheaf $\mathscr{I}_Z$. Then the short exact sequence $0 \to \mathscr{I}_Z \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_Z \to 0$ gives $P_{\mathscr{I}_Z}(d) = P_{\mathbb{P}^n}(d) - P_Z(d)$. The Hilbert polynomial of $\mathscr{I}_Z$ encodes the "difference" between the ambient space and the subscheme, and this is the key to defining Hilbert schemes: two subschemes are in the same Hilbert scheme component if and only if they have the same Hilbert polynomial.

The following result connects the Hilbert polynomial to the classical Hilbert function from commutative algebra. It was already stated in **??**; we can now give a cohomological proof:

Proposition 6.7.8: Hilbert Polynomial and Hilbert Function

Let $X \subseteq \mathbb{P}^N_k$ be a projective scheme and $\mathscr{F}$ a coherent sheaf on X. Then for $d \gg 0$:

$$P_{\mathscr{F}}(d) = h^0(X, \mathscr{F}(d)).$$

That is, the Hilbert polynomial agrees with the *Hilbert function* $d \mapsto h^0(X, \mathscr{F}(d))$ for all sufficiently large d.

***Proof*:**
By theorem 5.3.25 (Serre vanishing), there exists d_0 such that $H^i(X, \mathscr{F}(d)) = 0$ for all $i > 0$ and all $d \geq d_0$. For such d:

$$P_{\mathscr{F}}(d) = \chi(X, \mathscr{F}(d)) = h^0(X, \mathscr{F}(d)) - 0 + 0 - \cdots = h^0(X, \mathscr{F}(d)).$$

The degree and leading coefficient of the Hilbert polynomial have geometric meaning:

Proposition 6.7.9: Degree and Leading Coefficient

Let $X \subseteq \mathbb{P}^N_k$ be a projective scheme of dimension n and let $\mathscr{F}$ be a coherent sheaf with $\dim(\operatorname{supp}\mathscr{F}) = r$. Write:

$$P_{\mathscr{F}}(d) = \frac{a_r}{r!} d^r + \text{lower order terms}.$$

Then a_r is a positive integer. When $\mathscr{F} = \mathcal{O}_X$, the integer a_n is the *degree* of X in $\mathbb{P}^N$, i.e., the number of points in the intersection of X with a general linear subspace of complementary dimension (counted with multiplicity).

***Proof*:**
The positivity of a_r follows from the fact that $P_{\mathscr{F}}(d) = h^0(X, \mathscr{F}(d))$ for $d \gg 0$, and this grows like $d^r/r!$ (the Hilbert function of a module of dimension r over a polynomial ring has this growth rate; see theorem 2.6.11 for the connection between Hilbert functions and dimension). The integrality of a_r follows from the inductive proof of theorem 6.7.5: the difference $\Delta P_{\mathscr{F}}$ has leading coefficient $a_r/(r-1)!$, and by induction this integer is the degree of the support restricted to a hyperplane section.

The identification of a_n with the degree of X when $\mathscr{F} = \mathcal{O}_X$ follows from Bézout's theorem ([20]): the intersection of X with n general hyperplanes is a zero-dimensional scheme of length a_n.

6.7.10 Remark: Invariance of the Hilbert Polynomial The Hilbert polynomial depends on the choice of embedding (or more precisely, on the choice of very ample sheaf $\mathcal{O}_X(1)$). However, the *constant term* $P_X(0) = \chi(X, \mathcal{O}_X)$ is independent of the embedding: it depends only on the abstract scheme X, since it is defined purely in terms of the cohomology of the structure sheaf. Similarly, the arithmetic genus $p_a(X) = (-1)^n(\chi(\mathcal{O}_X) - 1)$ is an intrinsic invariant. In fact, by theorem 5.5.28, the geometric genus $p_g = h^0(X, \omega_X)$ is a birational invariant for smooth varieties, and Serre duality relates it to the "top" cohomology of $\mathcal{O}_X$.
The higher coefficients of P_X do depend on the embedding, but they carry geometric information about the embedding itself (degree, sectional genus, etc.).

6.7.3 Riemann–Roch for Curves

We now bring together Serre duality and the Euler characteristic to prove the Riemann–Roch theorem for smooth projective curves. This result was stated in theorem 7.1.19; we can now give a complete proof using the cohomological tools developed in this chapter.

Recall from definition 7.1.1 that a *curve* over a field k is a separated scheme of finite type over k of pure dimension 1. A *smooth projective curve* is a smooth, projective, integral scheme of dimension 1 over k.

For a smooth projective curve C over k, the cohomology groups have the following structure: H^0 and H^1 are the only potentially nonzero groups (since $\dim C = 1$), the canonical sheaf $\omega_C = \Omega_{C/k}$ is an invertible sheaf, and Serre duality (corollary 6.5.10) gives $H^1(C, \mathcal{L}) \cong H^0(C, \mathcal{L}^\vee \otimes \omega_C)^*$ for any line bundle $\mathcal{L}$.

We need the following fundamental relationship between line bundles and divisors, which connects the degree of a divisor to the Euler characteristic:

Lemma 6.7.11: Degree and Euler Characteristic

Let C be a smooth projective curve over a field k and let $\mathcal{L}$ be a line bundle on C. Then:

$$\chi(C, \mathcal{L}) = \deg \mathcal{L} + \chi(C, \mathcal{O}_C) = \deg \mathcal{L} + 1 - g,$$

where $\deg \mathcal{L}$ is the degree of $\mathcal{L}$ (i.e., the degree of the associated Cartier divisor class; see proposition 5.4.54).

***Proof*:**
By proposition 5.4.54, every line bundle on C is of the form $\mathcal{O}_C(D)$ for some divisor D. We prove the formula by induction on the "complexity" of D, reducing to the case of effective divisors supported at a single point.

Case 1: $D = p$ a single closed point. Let $\kappa(p)$ be the residue field and let $[\kappa(p) : k] = \delta$. The short exact sequence $0 \to \mathcal{O}_C \to \mathcal{O}_C(p) \to \kappa(p) \to 0$ (where the first map is the inclusion of regular functions into functions with at most a simple pole at p, and the quotient is the skyscraper sheaf at p) gives:

$$\chi(\mathcal{O}_C(p)) = \chi(\mathcal{O}_C) + \chi(\kappa(p)) = (1 - g) + \delta = \deg(p) + 1 - g,$$

since $\deg(p) = [\kappa(p) : k] = \delta$ and $\chi(\kappa(p)) = h^0(\kappa(p)) = \delta$.

Case 2: D is a general divisor. Write $D = D' + p$ for some closed point p and divisor D' with one fewer term. The short exact sequence $0 \to \mathcal{O}_C(D') \to \mathcal{O}_C(D) \to \kappa(p) \to 0$ gives:

$$\chi(\mathcal{O}_C(D)) = \chi(\mathcal{O}_C(D')) + \delta = (\deg D' + 1 - g) + \delta = \deg D + 1 - g,$$

where we used the inductive hypothesis and $\deg D = \deg D' + \deg p = \deg D' + \delta$. Since every divisor can be written as a difference of effective divisors, and we can reduce the negative part by the same argument applied to $\mathcal{O}_C(-p)$ (with the reversed exact sequence), the formula holds for all divisors.

Theorem 6.7.12: Riemann–Roch Theorem

Let C be a smooth projective curve of genus g over a field k, and let D be a divisor on C with associated line bundle $\mathcal{L} = \mathcal{O}_C(D)$. Let K be a canonical divisor (so $\mathcal{O}_C(K) \cong \omega_C$). Then:

$$h^0(C, \mathcal{O}_C(D)) - h^0(C, \mathcal{O}_C(K - D)) = \deg D + 1 - g.$$

Equivalently, using the notation $\ell(D) = h^0(C, \mathcal{O}_C(D))$:

$$\ell(D) - \ell(K - D) = \deg D + 1 - g.$$

***Proof*:**
By Serre duality (corollary 6.5.10):

$$h^1(C, \mathcal{O}_C(D)) = \dim_k H^1(C, \mathcal{O}_C(D)) = \dim_k H^0(C, \mathcal{O}_C(D)^\vee \otimes \omega_C)^* = h^0(C, \mathcal{O}_C(K-D)),$$

where we used $\mathcal{O}_C(D)^\vee \otimes \omega_C \cong \mathcal{O}_C(-D) \otimes \mathcal{O}_C(K) \cong \mathcal{O}_C(K-D)$. Now apply lemma 6.7.11:

$$h^0(C, \mathcal{O}_C(D)) - h^1(C, \mathcal{O}_C(D)) = \chi(C, \mathcal{O}_C(D)) = \deg D + 1 - g.$$

Substituting $h^1(C, \mathcal{O}_C(D)) = h^0(C, \mathcal{O}_C(K-D))$ gives the result.

Example 6.7.13: Riemann–Roch in Practice

1. *Degree of the canonical divisor.* Setting $D = K$ in Riemann–Roch: $\ell(K) - \ell(0) = \deg K + 1 - g$. Since $\ell(K) = h^0(\omega_C) = g$ and $\ell(0) = h^0(\mathcal{O}_C) = 1$:

$$\deg K = 2g - 2.$$

   For a smooth plane curve of degree d: $\omega_C \cong \mathcal{O}_C(d-3)$ (by proposition 6.5.7), so $\deg K = d(d-3)$. Setting $g = (d-1)(d-2)/2$, we verify: $2g - 2 = (d-1)(d-2) - 2 = d^2 - 3d = d(d-3)$.

2. *Very ample line bundles on curves.* A line bundle $\mathcal{L}$ on a smooth curve C of genus g is very ample (hence gives an embedding into projective space) if $\deg \mathcal{L} \geq 2g + 1$. To see this: by Riemann–Roch, $\ell(D) = \deg D + 1 - g$ whenever $\deg D > 2g - 2 = \deg K$ (since then $\deg(K - D) < 0$, forcing $\ell(K - D) = 0$). Furthermore, for $\deg D \geq 2g + 1$, the linear system $|D|$ separates points and tangent vectors (the standard argument uses Riemann–Roch for $D - p$ and $D - 2p$, which have degrees $\geq 2g - 1$ and $\geq 2g - 1$ respectively, and one checks the required surjectivity).

3. *Elliptic curves.* Let C be a smooth projective curve of genus 1 with a rational point $O \in C(k)$. By Riemann–Roch with $D = 3O$:

$$\ell(3O) = 3 + 1 - 1 = 3$$

   (since $\deg(K - 3O) = 2 \cdot 1 - 2 - 3 = -3 < 0$, so $\ell(K - 3O) = 0$). The three-dimensional space $H^0(C, \mathcal{O}(3O))$ gives a closed embedding $C \hookrightarrow \mathbb{P}^2$, and the image is a smooth plane cubic (cf. example **??**).

4. *Rational curves.* Let C have genus 0. Then $\chi(\mathcal{O}_C) = 1$ and for any line bundle $\mathcal{L}$ of degree $d \geq 0$:

$$h^0(C, \mathcal{L}) = d + 1$$

   (since $h^1 = h^0(\omega_C \otimes \mathcal{L}^{-1})$ and $\deg(\omega_C \otimes \mathcal{L}^{-1}) = -2 - d < 0$). In particular, $h^0(\mathcal{O}_C(1)) = 2$ gives a morphism $C \to \mathbb{P}^1$ of degree 1, so $C \cong \mathbb{P}^1$. Every smooth rational curve over k with a rational point is isomorphic to $\mathbb{P}^1_k$.

6.7.14 Remark: Beyond Curves The Riemann–Roch theorem for curves is the "$n = 1$" case of a family of results that express the Euler characteristic in terms of topological and geometric invariants. In higher dimensions, the *Hirzebruch–Riemann–Roch theorem* asserts that for a smooth projective variety X of dimension n and a locally free sheaf $\mathcal{E}$:

$$\chi(X, \mathcal{E}) = \int_X \mathrm{ch}(\mathcal{E}) \cdot \mathrm{td}(T_X),$$

where $\mathrm{ch}(\mathcal{E})$ is the Chern character of $\mathcal{E}$, $\mathrm{td}(T_X)$ is the Todd class of the tangent bundle, and the integral is the degree of the component in codimension n (i.e., the intersection with the fundamental class). For curves, the Chern character of a line bundle is $1 + c_1(\mathcal{L})$ and the Todd class of T_C is $1 + \frac{1}{2}c_1(T_C) = 1 + \frac{1}{2}(2 - 2g)$, so the formula reduces to $\chi(\mathcal{L}) = \deg \mathcal{L} + 1 - g$. For surfaces, it gives the *Noether formula*:

$$\chi(\mathcal{O}_S) = \frac{1}{12}(K_S^2 + e(S)),$$

where K_S^2 is the self-intersection of the canonical class and $e(S)$ is the topological Euler characteristic. These generalizations belong to the intersection-theoretic framework developed in *Intersection Theory* [16].

6.8 Flat Morphisms and Semicontinuity

In the previous sections, we studied the cohomology of a *single* scheme. In practice, however, schemes arise in *families*: a morphism $f : X \to Y$ presents X as a family of fibers X_y parameterized by points $y \in Y$, and the higher direct images $R^i f_* \mathscr{F}$ package the cohomology of these fibers into a sheaf on the base. A fundamental question is: how does the function $y \mapsto h^i(X_y, \mathscr{F}_y)$ behave as y varies?

The answer involves two intertwined ideas. The first is *flatness*: the condition that the fibers of f vary "continuously," without sudden jumps in dimension or embedded components. Flat families are the algebraic analogue of fiber bundles, and they are the correct notion of "continuously varying family" in algebraic geometry. The second idea is *semicontinuity*: the dimensions $h^i(X_y, \mathscr{F}_y)$ can jump *up* at special points but not down, and when they are constant, the higher direct images are locally free and commute with base change.

These results are essential for moduli theory. The semicontinuity theorem ensures that the loci where cohomological invariants jump form closed subsets, and Grauert's theorem guarantees that the "generic" behavior of cohomology is well-controlled. We shall use both in the discussion of Hilbert schemes (section 3.3.6) and in relating algebraic and analytic cohomology via GAGA (section 6.9).

6.8.1 Flat Families

Definition 6.8.1: Family of Schemes

Let X, B be schemes. Then a *Family of schemes* is given by a morphism $\pi : X \to B$. The scheme B is sometimes called the *parameter space*.

The notion of a family of schemes as it stands is much too general: consider any family $\pi : X \to B$, choose $b \in B$, and create a new family by taking the disjoint union of $X \setminus \pi^{-1}(b)$ and Y for some arbitrary scheme Y where Y is sent to b. From this, it can be deduced that there is no real properties that can be deduced, and further constrictions must be given.

Recall from commutative algebra (see [10]) that an R-module M is *flat* if the functor $M \otimes_R -$ is exact. Geometrically, flatness of a morphism means that the fibers do not "degenerate" in an uncontrolled way:

Definition 6.8.2: Flat Morphism

Let $f : X \to Y$ be a morphism of schemes.

1. An $\mathcal{O}_X$-module $\mathscr{F}$ is *flat over* Y (or f*-flat*) if for every point $x \in X$, the stalk $\mathscr{F}_x$ is a flat $\mathcal{O}_{Y,f(x)}$-module (via the local homomorphism $f_x^\# : \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$).
2. The morphism f itself is *flat* if $\mathcal{O}_X$ is flat over Y, i.e., if $\mathcal{O}_{X,x}$ is a flat $\mathcal{O}_{Y,f(x)}$-module for every $x \in X$.

Example 6.8.3: Flat and Non-Flat Morphisms

1. *Open immersions.* Any open immersion $j : U \hookrightarrow X$ is flat: the stalk map $\mathcal{O}_{X,x} \to \mathcal{O}_{U,x}$ is the identity for $x \in U$.
2. *Localization.* The morphism $\operatorname{Spec} R_f \to \operatorname{Spec} R$ induced by the localization $R \to R_f$ is flat, since localization is an exact functor.
3. *Finite free morphisms.* If $f : \operatorname{Spec} S \to \operatorname{Spec} R$ makes S a free R-module, then f is flat. More generally, if S is a finitely generated projective R-module, then f is flat.
4. *The family of conics.* Consider the morphism $f : V(xy - t) \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$, where the fiber over $t = a \neq 0$ is the smooth hyperbola $xy = a$, and the fiber over $t = 0$ is the union of coordinate axes $xy = 0$. The coordinate ring is $k[x, y, t]/(xy - t) \cong k[x, y]$, and the map to $k[t]$ sends $t \mapsto xy$. This is flat: $k[x, y]$ is free over $k[xy] = k[t]$ (with basis given by the monomials $x^i y^j$ with either $i = 0$ or $j = 0$). The fiber over $t = 0$ is reducible, but the family is still flat because the reducibility is "expected" — the total space is smooth.
5. *A non-flat family.* The blowup morphism $f : \mathrm{Bl}_0\mathbb{A}^2 \to \mathbb{A}^2$ is *not* flat: the fiber over the origin is $\mathbb{P}^1$ (one-dimensional), while the fiber over every other point is a single point (zero-dimensional). Flatness would require the fibers to have the same dimension (at least for an equidimensional source), so this jump is an obstruction. More precisely, the structure sheaf $\mathcal{O}_{\mathrm{Bl}_0\mathbb{A}^2}$ is not flat over $\mathcal{O}_{\mathbb{A}^2,0}$.
6. *Families of subschemes.* Let $X = V(x^2 - t) \subset \mathbb{A}^1_x \times \mathbb{A}^1_t$ and let $f : X \to \mathbb{A}^1_t$ be the projection. For $t \neq 0$, the fiber is two reduced points $x = \pm\sqrt{t}$; for $t = 0$, the fiber is the non-reduced scheme $\operatorname{Spec} k[x]/(x^2)$. This family *is* flat: $k[x, t]/(x^2 - t) \cong k[x]$ is free of rank 2 over $k[t]$ (with basis $\{1, x\}$). The non-reduced fiber at $t = 0$ is the "correct" flat limit of the two-point fibers: it remembers the multiplicity. This is exactly the phenomenon we encountered with intersection multiplicities in section 2.2.2.

The key property of flat families for cohomology is that they interact well with base change. Recall from definition 3.2.3 that for a morphism $f : X \to Y$ and a base change $g : Y' \to Y$, we form the fiber product:

$$\begin{tikzcd} X' = X \times_Y Y' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"'] & Y \end{tikzcd}$$

For any $\mathcal{O}_X$-module $\mathscr{F}$, the *pullback* $g'^*\mathscr{F}$ is a sheaf on X', and we can form both $R^i f'_*(g'^*\mathscr{F})$ (compute cohomology on the fibers of f') and $g^*(R^i f_*\mathscr{F})$ (pull back the higher direct image). These are both sheaves on Y', and there is a natural *base change map*:

$$\beta^i : \ g^*(R^i f_*\mathscr{F}) \longrightarrow R^i f'_*(g'^*\mathscr{F}).$$

This map is not an isomorphism in general. When it is, we say that *cohomology commutes with base change in degree i*.

6.8.2 The Flat Base Change Theorem

The following theorem says that cohomology always commutes with flat base change:

Theorem 6.8.4: Flat Base Change

Let $f : X \to Y$ be a separated morphism of finite type between Noetherian schemes, and let $\mathscr{F}$ be a quasi-coherent sheaf on X that is flat over Y. Let $g : Y' \to Y$ be a flat morphism, and form the Cartesian square:

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"'] & Y \end{tikzcd}$$

Then for all $i \geq 0$, the base change map is an isomorphism:

$$g^*(R^i f_*\mathscr{F}) \xrightarrow{\sim} R^i f'_*(g'^*\mathscr{F}).$$

***Proof*:**
The question is local on Y and Y', so we may assume both are affine: $Y = \operatorname{Spec} R$ and $Y' = \operatorname{Spec} R'$, with the flat map g corresponding to a flat ring homomorphism $\varphi : R \to R'$.

Step 1: Reduction to Čech cohomology. Since f is separated and of finite type, X admits a finite open affine cover $\mathcal{U} = \{U_1, \dots, U_r\}$. By proposition 6.6.10, the higher direct images are computed by the Čech complex: $R^i f_*\mathscr{F}$ is the quasi-coherent sheaf on Y associated to the R-module $h^i(C^\bullet(\mathcal{U}, \mathscr{F}))$, where:

$$C^p(\mathcal{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} \mathscr{F}(U_{i_0} \cap \cdots \cap U_{i_p}).$$

Each intersection $U_{i_0,\dots,i_p}$ is affine (by separatedness), and $\mathscr{F}$ is quasi-coherent, so $\mathscr{F}(U_{i_0,\dots,i_p})$ is an R-module.

Step 2: Base change of the Čech complex. The base change $X' = X \times_Y Y'$ has the affine cover $\mathcal{U}' = \{U_1', \dots, U_r'\}$ where $U_j' = U_j \times_Y Y' = \operatorname{Spec}(A_j \otimes_R R')$ for $U_j = \operatorname{Spec} A_j$. The Čech complex for $g'^*\mathscr{F}$ on X' is:

$$C^p(\mathcal{U}', g'^*\mathscr{F}) = \prod_{i_0<\cdots<i_p} (g'^*\mathscr{F})(U'_{i_0,\dots,i_p}) = \prod_{i_0<\cdots<i_p} \mathscr{F}(U_{i_0,\dots,i_p}) \otimes_R R'.$$

The last equality uses the fact that for an affine base change $\operatorname{Spec}(A \otimes_R R') \to \operatorname{Spec} A$, the pullback of a quasi-coherent sheaf $\widetilde{M}$ is $\widetilde{M \otimes_R R'}$. Hence:

$$C^\bullet(\mathcal{U}', g'^*\mathscr{F}) \cong C^\bullet(\mathcal{U}, \mathscr{F}) \otimes_R R'.$$

Step 3: Flatness and cohomology. Since R' is a flat R-module (by hypothesis), the functor $- \otimes_R R'$ is exact. Exact functors commute with the formation of cohomology (i.e., with taking kernels modulo images):

$$h^i(C^\bullet(\mathcal{U}, \mathscr{F}) \otimes_R R') \cong h^i(C^\bullet(\mathcal{U}, \mathscr{F})) \otimes_R R'.$$

The left side computes $R^i f'_*(g'^*\mathscr{F})(Y')$, while the right side computes $g^*(R^i f_*\mathscr{F})(Y') = (R^i f_*\mathscr{F})(Y) \otimes_R R'$. Hence the base change map is an isomorphism on global sections over affine opens, which suffices.

The most important special case is base change to a point:

Corollary 6.8.5: Cohomology of Fibers via Base Change

Under the hypotheses of theorem 6.8.4, let $y \in Y$ be a point with residue field $\kappa(y)$, and let $X_y = f^{-1}(y)$ be the scheme-theoretic fiber (definition 3.2.9). Then:

$$(R^i f_*\mathscr{F}) \otimes_{\mathcal{O}_{Y,y}} \kappa(y) \;\cong\; H^i(X_y, \mathscr{F}_y),$$

where $\mathscr{F}_y = \mathscr{F}|_{X_y}$ is the restriction to the fiber, provided that $\mathscr{F}$ is flat over Y.

***Proof*:**
Apply theorem 6.8.4 to the (flat) base change $g : \operatorname{Spec} \kappa(y) \to Y$. Then $X' = X_y$, and the left side is $g^*(R^i f_*\mathscr{F}) = (R^i f_*\mathscr{F}) \otimes \kappa(y)$, while the right side is $H^i(X_y, \mathscr{F}_y)$.

Example 6.8.6: Flat Base Change in Practice

1. *Constant families.* Let $f = \mathrm{pr}_2 : X \times_k Y \to Y$ be the projection of a product (with X projective over k). Then for any coherent sheaf $\mathscr{F}$ on X, pulled back to $X \times Y$ via the first projection, the sheaf $\mathscr{F}$ is flat over Y (trivially, since it does not depend on Y at all). The base change theorem gives:

$$R^i f_*(\mathrm{pr}_1^*\mathscr{F}) \cong H^i(X, \mathscr{F}) \otimes_k \mathcal{O}_Y,$$

i.e., the higher direct images are the constant (free) sheaves with fiber $H^i(X, \mathscr{F})$. This is the trivial case, but it illustrates the principle: for a constant family, the cohomology does not vary.

2. *Base change to the generic point.* Let $f : X \to Y$ be a projective morphism with Y integral and $\eta \in Y$ the generic point. The base change $\operatorname{Spec} K(Y) \to Y$ (where $K(Y)$ is the function field) is flat, so:
$$(R^i f_* \mathscr{F}) \otimes \kappa(\eta) \cong H^i(X_\eta, \mathscr{F}_\eta).$$
This says the *generic rank* of $R^i f_* \mathscr{F}$ equals the cohomology of the generic fiber. Combined with semicontinuity, this tells us that the generic fiber has the "smallest" cohomology, and special fibers can only have more.

3. *Flat base change for field extensions.* Let X be a projective scheme over a field k and let K/k be a field extension. The base change $X_K = X \times_k \operatorname{Spec} K \to X$ is flat, so:
$$H^i(X_K, \mathscr{F}_K) \cong H^i(X, \mathscr{F}) \otimes_k K.$$
In particular, $h^i(X_K, \mathscr{F}_K) = h^i(X, \mathscr{F})$: the Betti numbers do not change under field extension. This is essential for reducing problems over arbitrary fields to the algebraically closed case.

Proposition 6.8.7: Associated Points and Morphisms

Let $f : X \to Y$ be a flat morphism of locally Noetherian schemes. Then:
$$\operatorname{Ass}(X) = \bigcup_{y \in \operatorname{Ass}(Y)} \operatorname{Ass}(X_y)$$
where $X_y = f^{-1}(y)$ is the fiber over y. In particular, associated points of X map to associated points of Y (among the points in the image).

***Proof*:**
This follows from the behavior of associated primes under flat base change: if $A \to B$ is flat and $\mathfrak{q} \in \operatorname{Ass}_B(B)$, then $\mathfrak{q} \cap A \in \operatorname{Ass}_A(A)$.

6.8.3 Semicontinuity

The flat base change theorem tells us that when the base change is flat, cohomology commutes with base change. But what happens for non-flat base changes, such as the inclusion of a point $y \hookrightarrow Y$? In general, the base change map $(R^i f_* \mathscr{F}) \otimes \kappa(y) \to H^i(X_y, \mathscr{F}_y)$ is only a surjection, not an isomorphism, and the fiber cohomology can be *larger* than what the stalk of the higher direct image predicts. The semicontinuity theorem makes this precise: the function $y \mapsto h^i(X_y, \mathscr{F}_y)$ is *upper semicontinuous*, meaning it can jump up on closed subsets but not on open ones.

To state the result precisely, we need a piece of commutative algebra. Let R be a Noetherian ring and $C^\bullet$ a bounded complex of finitely generated free R-modules. For any R-module M, write $h^i(C^\bullet \otimes_R M) = h^i(C^\bullet; M)$. When R is a local ring and $M = R/\mathfrak{m}$ is the residue field, the function $i \mapsto h^i(C^\bullet; R/\mathfrak{m})$ measures the fiber cohomology. The following result governs how this varies:

Lemma 6.8.8: Semicontinuity on Complexes

Let R be a Noetherian local ring with residue field $\kappa = R/\mathfrak{m}$, and let $C^\bullet$ be a bounded complex of finitely generated free R-modules. Then:

1. The function $\mathfrak{p} \mapsto \dim_{\kappa(\mathfrak{p})} h^i(C^\bullet \otimes_R \kappa(\mathfrak{p}))$ is upper semicontinuous on $\operatorname{Spec} R$: for each n, the set $\{\mathfrak{p} \in \operatorname{Spec} R \,|\, \dim h^i(C^\bullet \otimes \kappa(\mathfrak{p})) \geq n\}$ is closed.
2. The Euler characteristic $\mathfrak{p} \mapsto \chi(C^\bullet \otimes \kappa(\mathfrak{p})) = \sum(-1)^i \dim h^i(C^\bullet \otimes \kappa(\mathfrak{p}))$ is *locally constant* (it equals $\sum(-1)^i \mathrm{rk} C^i$, which is independent of $\mathfrak{p}$).
3. If $h^i(C^\bullet \otimes \kappa(\mathfrak{p}))$ is constant as a function of $\mathfrak{p}$ (say, equal to r), and R is reduced, then $h^i(C^\bullet)$ is a free R-module of rank r.

***Proof*:**

(1) The complex $C^\bullet$ is a sequence of maps between free modules: $\cdots \to R^{a_{i-1}} \xrightarrow{d^{i-1}} R^{a_i} \xrightarrow{d^i} R^{a_{i+1}} \to \cdots$. After tensoring with $\kappa(\mathfrak{p})$, these become linear maps between vector spaces: $\kappa(\mathfrak{p})^{a_{i-1}} \to \kappa(\mathfrak{p})^{a_i} \to \kappa(\mathfrak{p})^{a_{i+1}}$. Now, $\dim h^i(C^\bullet \otimes \kappa(\mathfrak{p})) = a_i - \mathrm{rk}(d^{i-1} \otimes \kappa(\mathfrak{p})) - \mathrm{rk}(d^i \otimes \kappa(\mathfrak{p}))$. The rank of a matrix can only *drop* under specialization (the vanishing of minors is a closed condition), so $\mathrm{rk}(d^j \otimes \kappa(\mathfrak{p}))$ is lower semicontinuous in $\mathfrak{p}$. Hence $\dim h^i$ is upper semicontinuous.

(2) The Euler characteristic is $\sum(-1)^i \dim h^i(C^\bullet \otimes \kappa(\mathfrak{p})) = \sum(-1)^i a_i$ (the alternating sum of the ranks of the free modules), which is independent of $\mathfrak{p}$. This uses the rank-nullity theorem: the terms involving ranks of differentials cancel in the alternating sum.

(3) If $\dim h^i(C^\bullet \otimes \kappa(\mathfrak{p}))$ is constant, then by upper semicontinuity it achieves its maximum everywhere, so the ranks of the differentials d^{i-1} and d^i are constant. By Nakayama's lemma (applied to the quotient $h^i(C^\bullet) = \ker d^i / \mathrm{im} d^{i-1}$), this forces $h^i(C^\bullet)$ to be locally free over R. If R is reduced, locally free and finitely generated implies free of constant rank.

The geometric content of this algebraic lemma is the semicontinuity theorem:

Theorem 6.8.9: Semicontinuity of Cohomology

Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $\mathscr{F}$ be a coherent sheaf on X that is flat over Y. Then:

1. *(Upper semicontinuity.)* For each $i \geq 0$, the function
$$y \longmapsto h^i(X_y, \mathscr{F}_y)$$
is upper semicontinuous on Y: for every $n \geq 0$, the set $\{y \in Y \,|\, h^i(X_y, \mathscr{F}_y) \geq n\}$ is a closed subset of Y.
2. *(Euler characteristic is locally constant.)* The function
$$y \longmapsto \chi(X_y, \mathscr{F}_y) = \sum_{i \geq 0} (-1)^i h^i(X_y, \mathscr{F}_y)$$
is locally constant on Y.

***Proof*:**
The question is local on Y, so assume $Y = \operatorname{Spec} R$ is affine.

Constructing the representing complex. The key technical input is the existence of a *Čech complex computing fiber cohomology uniformly*. Choose a finite affine open cover $\mathcal{U}$ of X. By proposition 6.6.10, the cohomology $H^i(X, \mathscr{F})$ is computed by the Čech complex $C^\bullet(\mathcal{U}, \mathscr{F})$. Since $\mathscr{F}$ is coherent and each intersection in the cover is affine, the terms $C^p(\mathcal{U}, \mathscr{F})$ are finitely generated R-modules. Since $\mathscr{F}$ is flat over $Y = \operatorname{Spec} R$, and each intersection $U_{i_0,\dots,i_p}$ is affine, the R-modules $\mathscr{F}(U_{i_0,\dots,i_p})$ are flat over R.

One can replace the Čech complex by a quasi-isomorphic bounded complex $K^\bullet$ of finitely generated *free* R-modules (by taking a free resolution of the Čech complex, truncated appropriately — this is possible because R is Noetherian)[a]. This complex has the property that:

$$h^i(K^\bullet \otimes_R \kappa(y)) \cong H^i(X_y, \mathscr{F}_y)$$

for every $y \in Y$. Indeed, tensoring with $\kappa(y)$ corresponds to restricting to the fiber, and the flatness of $\mathscr{F}$ ensures that the Čech complex commutes with base change (as in the proof of theorem 6.8.4).

Both statements now follow from lemma 6.8.8: part (1) is the upper semicontinuity of $\dim h^i(K^\bullet \otimes \kappa(y))$, and part (2) is the constancy of the Euler characteristic.

[a] More precisely, one uses the fact that a bounded above complex of flat modules over a Noetherian ring is quasi-isomorphic to a bounded above complex of free modules. Since our Čech complex is bounded, $K^\bullet$ can be chosen bounded as well.

Example 6.8.10: Semicontinuity in Families

1. *Jumping h^1 for line bundles on curves.* Let $f : \mathcal{C} \to B$ be a flat family of smooth projective curves of genus g, and let $\mathcal{L}$ be a line bundle on $\mathcal{C}$ of relative degree d (i.e., $\deg \mathcal{L}|_{C_b} = d$ for each fiber C_b). By Riemann–Roch (theorem 6.7.12):

   $$h^0(C_b, \mathcal{L}_b) - h^1(C_b, \mathcal{L}_b) = d + 1 - g$$

   for every $b \in B$. Since the Euler characteristic is constant, any jump in h^0 is exactly compensated by a jump in h^1. By semicontinuity, h^0 is upper semicontinuous, so the locus where h^0 is "large" (and correspondingly h^1 is large) is closed.

   Concretely, consider a pencil of plane cubics (genus 1) and a degree-0 line bundle $\mathcal{L}$. Generically, $h^0 = 0$ and $h^1 = 0$ (by Riemann–Roch, since $d + 1 - g = 0$). But at special fibers where $\mathcal{L}$ happens to be trivial, $h^0 = 1$ and $h^1 = 1$. The locus of such fibers is a closed subset of the base.

2. *The dimension of global sections of $\mathcal{O}(d)$.* Let $f : X \to \operatorname{Spec} k$ be the structure morphism of a smooth projective variety, and consider the family $\mathcal{O}_X(d)$ as d varies. Though this is not literally a flat family over a connected base, the semicontinuity principle manifests as follows: for any short exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of coherent sheaves, the function h^0 satisfies $h^0(\mathscr{F}) \leq h^0(\mathscr{F}') + h^0(\mathscr{F}'')$, but equality need not hold. The "excess" is controlled by $H^1(\mathscr{F}')$, which is the obstruction to lifting global sections.

3. *Jumping h^0 for the structure sheaf of a degeneration.* Consider a flat family $f : X \to \mathbb{A}^1_k$ of surfaces where the generic fiber is a smooth surface with $h^0(\mathcal{O}) = 1$ and the special fiber over

0 is a reducible surface (e.g., a union of two components). If the special fiber has $h^0(\mathcal{O}) = 2$ (two connected components), then h^0 has jumped up at 0, and the locus $\{h^0 \geq 2\}$ is the closed point $\{0\}$, consistent with semicontinuity.

6.8.4 Grauert's Theorem and Cohomology and Base Change

The semicontinuity theorem says h^i can jump up. The natural follow-up question is: what happens when it does *not* jump? The answer is Grauert's theorem: constancy of h^i implies that the higher direct image is locally free and commutes with *all* base changes, not just flat ones.

Theorem 6.8.11: Grauert's Theorem (Cohomology and Base Change)

Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $\mathscr{F}$ be a coherent sheaf on X that is flat over Y. Let $y \in Y$ be a point, and suppose the natural base change map:

$$\beta^i(y) : \ (R^i f_* \mathscr{F}) \otimes \kappa(y) \longrightarrow H^i(X_y, \mathscr{F}_y)$$

is surjective. Then:

1. $\beta^i(y)$ is an isomorphism. That is, surjectivity implies bijectivity.
2. β^i is an isomorphism in an open neighborhood of y. In particular, $R^i f_* \mathscr{F}$ is locally free near y if and only if $h^i(X_y, \mathscr{F}_y)$ is locally constant near y.
3. If $\beta^i(y)$ is an isomorphism (equivalently, surjective), then $\beta^{i-1}(y)$ is surjective if and only if it is an isomorphism.

In particular, if $h^i(X_y, \mathscr{F}_y)$ is constant on Y and Y is reduced, then $R^i f_* \mathscr{F}$ is a locally free sheaf of rank $h^i(X_y, \mathscr{F}_y)$, and the formation of $R^i f_* \mathscr{F}$ commutes with arbitrary base change $g : Y' \to Y$.

***Proof*:**
As in the proof of theorem 6.8.9, we reduce to the local algebra by replacing the higher direct images with the cohomology of a bounded complex $K^\bullet$ of finitely generated free R-modules (where $R = \mathcal{O}_{Y,y}$).

(1) The base change map $\beta^i(y)$ corresponds to the natural map $h^i(K^\bullet) \otimes_R \kappa \to h^i(K^\bullet \otimes_R \kappa)$. Write $K^\bullet$ as $\cdots \to K^{i-1} \xrightarrow{d^{i-1}} K^i \xrightarrow{d^i} K^{i+1} \to \cdots$. The surjectivity of $\beta^i(y)$ means that every cohomology class in $h^i(K^\bullet \otimes \kappa)$ lifts to $h^i(K^\bullet)$. By Nakayama's lemma (applied to the finitely generated R-module $h^i(K^\bullet)$), the map $h^i(K^\bullet) \to h^i(K^\bullet) \otimes \kappa$ is surjective if and only if it is a presentation by a minimal set of generators. The surjectivity of β^i then forces the natural map to be an isomorphism by a careful diagram chase involving the snake lemma applied to the sequences defining $\ker d^i$ and $\mathrm{im} d^{i-1}$[a].

(2) Since $\beta^i(y)$ is an isomorphism at y, the semicontinuity of h^i implies $h^i(X_{y'}, \mathscr{F}_{y'}) \leq h^i(X_y, \mathscr{F}_y)$ in a neighborhood of y. On the other hand, the rank of the locally free sheaf $R^i f_* \mathscr{F}$ can only drop on a closed subset (or rather, its fiber dimension is lower semicontinuous). The two semicontinuity conditions force constancy in a neighborhood, and the argument of part (1) applies at each nearby point.

(3) This follows from the exact sequence relating h^{i-1} and h^i in the complex $K^\bullet$: the surjectivity of β^i controls the cokernel, which in turn governs the kernel at the next level.

The final statement (for constant h^i on reduced Y) follows by combining (1) and (2) with lemma 6.8.8(3): on a reduced base, a finitely generated module with constant fiber dimension is locally free.

[a] The detailed argument proceeds as follows. By surjectivity of β^i, one shows that the map $\ker d^i \to \ker(d^i \otimes \kappa)$ is surjective. Since K^i is free, the kernel $\ker d^i$ is a submodule of a free module, hence flat (over a local ring, submodules of free modules are flat if the ring is a principal ideal domain; in general, one uses the flatness of $\mathscr{F}$ to ensure this). The surjectivity then implies that $\mathrm{im} d^{i-1} \otimes \kappa \to \mathrm{im}(d^{i-1} \otimes \kappa)$ is also well-behaved, and the result follows.

Corollary 6.8.12: Cohomology and Base Change: Clean Form

Let $f : X \to Y$ be a projective morphism of Noetherian schemes with Y integral, and let $\mathscr{F}$ be coherent and f-flat. Suppose $h^i(X_y, \mathscr{F}_y)$ is constant for all $y \in Y$. Then:

1. $R^i f_* \mathscr{F}$ is locally free of rank $h^i(X_y, \mathscr{F}_y)$.
2. For any morphism $g : Y' \to Y$, the base change map $g^*(R^i f_* \mathscr{F}) \xrightarrow{\sim} R^i f'_*(g'^* \mathscr{F})$ is an isomorphism.
3. If additionally $h^{i-1}(X_y, \mathscr{F}_y)$ is also constant, then $R^{i-1} f_* \mathscr{F}$ is locally free and commutes with base change as well.

***Proof*:**
Part (1) follows from theorem 6.8.11 together with the observation that on an integral (hence reduced) scheme, a coherent sheaf of constant fiber dimension is locally free. Part (2) follows because, once $R^i f_* \mathscr{F}$ is locally free, the base change map is an isomorphism for any g (this is immediate from the flat base change theorem, since tensoring a locally free module with anything commutes with cohomology). Part (3) follows from theorem 6.8.11(3).

Example 6.8.13: Grauert's Theorem in Practice

1. *Line bundles on a constant family of curves.* Let C be a smooth projective curve of genus g over k, and consider the family $f : C \times \mathrm{Pic}^d(C) \to \mathrm{Pic}^d(C)$ with the universal line bundle $\mathcal{L}$ of degree d. If $d > 2g - 2$, then $h^1(C, \mathcal{L}_b) = 0$ for all $b \in \mathrm{Pic}^d(C)$ (by Serre duality and degree considerations). By Grauert's theorem, $R^1 f_* \mathcal{L} = 0$ and $f_* \mathcal{L}$ is a locally free sheaf of rank $h^0 = d + 1 - g$ on $\mathrm{Pic}^d(C)$. This locally free sheaf is the *Picard bundle*, a fundamental object in the study of Jacobians.
2. *Failure of base change without flatness.* Consider the blowup $f : \mathrm{Bl}_0 \mathbb{A}^2 \to \mathbb{A}^2$ from Example 6.6.11(2). The structure sheaf $\mathcal{O}_{\mathrm{Bl}_0 \mathbb{A}^2}$ is *not* flat over $\mathbb{A}^2$ (the fiber dimension jumps). Accordingly, $R^1 f_* \mathcal{O} = 0$ even though $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}) = 0$ for the exceptional fiber — the vanishing of R^1 does not "see" the fiber at the origin because the family is not flat there.
3. *When constancy fails: the Brill–Noether locus.* Let C be a general smooth curve of genus g and consider the family of degree-d line bundles. The *Brill–Noether locus* $W^r_d(C) = \{L \in \mathrm{Pic}^d(C) \,|\, h^0(C, L) \geq r + 1\}$ is a closed subvariety of $\mathrm{Pic}^d(C)$ by the semicontinuity theorem. The Brill–Noether theorem asserts that for a general curve of genus g, $W^r_d(C)$ has the

"expected dimension" $\rho = g - (r+1)(g-d+r)$ when $\rho \geq 0$, and is empty when $\rho < 0$. This is a result in the geometry of algebraic curves, and semicontinuity is the foundational tool that ensures W^r_d is an honest algebraic variety.

6.8.14 Remark: The Coherent Base Change Complex The key technical tool behind both the semicontinuity and Grauert theorems is the existence of a bounded complex $K^\bullet$ of finitely generated free R-modules representing the derived pushforward $\mathbf{R}f_*\mathscr{F}$ in the derived category of R-modules. Such a complex exists whenever f is projective and $\mathscr{F}$ is coherent and f-flat: one takes the Čech complex associated to a finite affine cover and replaces it by a free resolution.

In the language of derived categories (which we have not formally developed), the statement is: $\mathbf{R}f_*\mathscr{F} \in D^b(\mathrm{Mod}_R)$ is a *perfect complex* (quasi-isomorphic to a bounded complex of finitely generated free modules). The fiber cohomology is then $H^i(X_y, \mathscr{F}_y) = h^i(K^\bullet \otimes^{\mathbf{L}} \kappa(y))$, where $\otimes^{\mathbf{L}}$ is the derived tensor product (which equals the ordinary tensor product since $K^\bullet$ consists of free modules).

This perspective explains why the semicontinuity theorem is fundamentally a result in commutative algebra (about cohomology of complexes of free modules under base change) rather than algebraic geometry. The geometry enters only in constructing the complex $K^\bullet$ — and this construction uses the projectivity of f and the coherence of $\mathscr{F}$ in an essential way.

6.9 *GAGA

We developed two parallel cohomological machines: the algebraic cohomology of coherent sheaves on schemes (the subject of this chapter), and the analytic cohomology of coherent analytic sheaves on complex manifolds (via the Dolbeault complex, Hodge theory, and the analytic structure sheaf). Serre's GAGA theorems — *Géométrie Algébrique et Géométrie Analytique* [30] — form the bridge between these two worlds. They assert that for projective varieties over $\mathbb{C}$, the algebraic and analytic theories carry *exactly the same information*: the category of coherent algebraic sheaves is equivalent to the category of coherent analytic sheaves, and this equivalence preserves cohomology.

This is a surprising result. The algebraic world uses the Zariski topology (which is coarse: open sets are large and plentiful) and polynomial or rational functions, while the analytic world uses the classical (Euclidean) topology (which is fine: open sets can be arbitrarily small) and holomorphic functions. There are far more analytic open sets, far more analytic functions, and far more analytic sheaves than algebraic ones. Yet on projective varieties, these extra objects contribute nothing new to the cohomological theory. The rigidity of projective varieties forces the algebraic and analytic categories to coincide.

We state the GAGA theorems precisely, outline the key ideas of the proof, and develop the main consequences. This section draws heavily on the complex geometry review of chapter 1[7]; we now close the loop by proving that the algebraic cohomology developed in this chapter matches the analytic cohomology developed there.

[7]Specifically: sections 0.6 through the end of that chapter, where analytification (definition 6.9.1), analytic sheaves (**??**), and the GAGA statement (**??**) were introduced.

6.9.1 Analytification

We begin by recalling the analytification functor, which passes from algebraic geometry over $\mathbb{C}$ to complex-analytic geometry. The reader who has studied the complex geometry chapter in detail may skim this subsection; it is included for self-containedness.

Definition 6.9.1: Analytification

Let X be a scheme of finite type over $\mathbb{C}$. The *analytification* X^{an} is a complex analytic space with:

- The same underlying set as the $\mathbb{C}$-points $X(\mathbb{C})$.
- The classical (Euclidean) topology.
- The sheaf $\mathcal{O}_{X^{\mathrm{an}}}$ of holomorphic functions.

If X is smooth, then X^{an} is a complex manifold.

Let X be a scheme of finite type over $\mathbb{C}$. The set of $\mathbb{C}$-valued points $X(\mathbb{C}) = \mathrm{Hom}_{\mathbf{Sch}}(\operatorname{Spec}\mathbb{C}, X)$ carries a natural structure of complex-analytic space: locally, if $X = \operatorname{Spec}\mathbb{C}[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$, then $X(\mathbb{C}) = V(f_1, \ldots, f_r) \subseteq \mathbb{C}^n$ with the subspace topology from $\mathbb{C}^n$ and the sheaf of holomorphic functions. When X is smooth, $X(\mathbb{C})$ is a complex manifold (definition 0.6.1).

Definition 6.9.2: The Analytification Functor

Let X be a scheme of finite type over $\mathbb{C}$.

1. The *analytification* of X, denoted X^{an}, is the complex-analytic space $X(\mathbb{C})$ equipped with its natural analytic structure sheaf $\mathcal{O}_{X^{\mathrm{an}}}$ (cf. definition 6.9.1).
2. There is a natural continuous map of topological spaces $\varphi : X^{\mathrm{an}} \to X$ (the "identity on points," where X carries the Zariski topology and X^{an} carries the classical topology), and a morphism of sheaves of rings $\varphi^{\#} : \mathcal{O}_X \to \varphi_*\mathcal{O}_{X^{\mathrm{an}}}$ (sending algebraic regular functions to holomorphic functions).
3. For a coherent algebraic sheaf $\mathscr{F}$ on X, the *analytification* is the coherent analytic sheaf:
$$\mathscr{F}^{\mathrm{an}} = \varphi^*\mathscr{F} = \mathscr{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X^{\mathrm{an}}}$$
on X^{an}. This defines an exact functor:
$$(-)^{\mathrm{an}} : \mathbf{Coh}(X) \longrightarrow \mathbf{Coh}(X^{\mathrm{an}}).$$

The analytification functor has good formal properties: it is exact (since $\mathcal{O}_{X^{\mathrm{an}}}$ is flat over $\mathcal{O}_X$ — this is a non-trivial fact, reflecting the faithfully flat nature of the completion of a local ring), it preserves tensor products, duals, and exterior powers of locally free sheaves, and it sends line bundles to line bundles. Furthermore, for each $i \geq 0$, there is a natural *comparison map*:

$$\gamma^i : \ H^i(X, \mathscr{F}) \longrightarrow H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}}),$$

induced by the morphism of ringed spaces $\varphi : (X^{\mathrm{an}}, \mathcal{O}_{X^{\mathrm{an}}}) \to (X, \mathcal{O}_X)$. The GAGA theorems assert that, for projective X, this map is an isomorphism and the functor $(-)^{\mathrm{an}}$ is an equivalence.

6.9.2 Statement of the GAGA Theorems

Theorem 6.9.3: Serre's GAGA Theorems

Let X be a projective scheme over $\mathbb{C}$. Then:

1. *(Equivalence of categories.)* The analytification functor

$$(-)^{\mathrm{an}} : \mathbf{Coh}(X) \longrightarrow \mathbf{Coh}(X^{\mathrm{an}})$$

is an equivalence of categories. In particular:

   (a) Every coherent analytic sheaf on X^{an} is the analytification of a unique (up to isomorphism) coherent algebraic sheaf on X.

   (b) Every morphism of coherent analytic sheaves on X^{an} is the analytification of a unique morphism of coherent algebraic sheaves on X.

2. *(Comparison of cohomology.)* For every coherent algebraic sheaf $\mathscr{F}$ on X and every $i \geq 0$, the comparison map

$$\gamma^i : \ H^i(X, \mathscr{F}) \xrightarrow{\sim} H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$$

is an isomorphism of finite-dimensional $\mathbb{C}$-vector spaces.

3. *(Equivalence for subschemes.)* The analytification functor induces a bijection between closed algebraic subschemes of X and closed analytic subspaces of X^{an}.

Part (2) is typically proved first and is the most important for applications; parts (1) and (3) follow from it by formal arguments. Let us outline the proof of part (2):

***Proof*:**
(Proof sketch of theorem 6.9.3(2))
The proof proceeds in three stages.

Stage 1: The case $X = \mathbb{P}^n_{\mathbb{C}}$ and $\mathscr{F} = \mathcal{O}(d)$. The algebraic cohomology $H^i(\mathbb{P}^n, \mathcal{O}(d))$ was computed in theorem 6.3.1. The analytic cohomology $H^i((\mathbb{P}^n)^{\mathrm{an}}, \mathcal{O}^{\mathrm{an}}(d))$ can be computed using the Dolbeault resolution (**??**): since $\mathbb{CP}^n$ is a compact Kähler manifold (example 0.6.22), the Hodge decomposition (theorem 0.6.28) gives $H^i(\mathbb{CP}^n, \mathcal{O}^{\mathrm{an}}) \cong H^{0,i}_{\bar{\partial}}(\mathbb{CP}^n)$, and the Hodge diamond of $\mathbb{CP}^n$ (example 0.6.30) shows this vanishes for $i > 0$. For the twisted sheaf $\mathcal{O}(d)$, the Dolbeault cohomology can similarly be computed explicitly, and one verifies that the comparison map γ^i is an isomorphism by matching dimensions on both sides.

Stage 2: Extension to all coherent sheaves on $\mathbb{P}^n_{\mathbb{C}}$. Given a coherent sheaf $\mathscr{F}$ on $\mathbb{P}^n$, choose a resolution by direct sums of twisted sheaves (corollary 5.3.26):

$$\cdots \to \mathcal{E}_1 \to \mathcal{E}_0 \to \mathscr{F} \to 0,$$

where each $\mathcal{E}_j = \bigoplus \mathcal{O}(d_{j,l})$. Since the analytification functor is exact, the analytified resolution computes the analytic cohomology. The comparison maps for the $\mathcal{E}_j$ are isomorphisms by Stage 1, and a five-lemma argument (identical to Step 2 of the proof of theorem 6.5.5) yields the isomorphism for $\mathscr{F}$.

Stage 3: Extension to projective schemes. For a general projective scheme $i : X \hookrightarrow \mathbb{P}^N_{\mathbb{C}}$, the pushforward $i_*\mathscr{F}$ is a coherent algebraic sheaf on $\mathbb{P}^N$, and $(i_*\mathscr{F})^{\text{an}} = i_*^{\text{an}}(\mathscr{F}^{\text{an}})$ (the analytification of a pushforward along a closed embedding is the analytic pushforward of the analytification). By proposition 6.2.4, $H^i(X, \mathscr{F}) = H^i(\mathbb{P}^N, i_*\mathscr{F})$ and similarly for the analytic side. Stage 2 gives the isomorphism for $i_*\mathscr{F}$ on $\mathbb{P}^N$, hence for $\mathscr{F}$ on X.

***Proof*:**
(Proof sketch of theorem 6.9.3(1))

Full faithfulness. For coherent sheaves $\mathscr{F}, \mathscr{G}$ on X, the natural map $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) \to \mathrm{Hom}_{\mathcal{O}_{X^{\text{an}}}}(\mathscr{F}^{\text{an}}, \mathscr{G}^{\text{an}})$ is an isomorphism. Indeed, $\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \mathscr{G}) = H^0(X, \mathcal{H}om(\mathscr{F}, \mathscr{G}))$ and $\mathrm{Hom}_{\mathcal{O}_{X^{\text{an}}}}(\mathscr{F}^{\text{an}}, \mathscr{G}^{\text{an}}) = H^0(X^{\text{an}}, \mathcal{H}om(\mathscr{F}, \mathscr{G})^{\text{an}})$, and part (2) with $i = 0$ gives the isomorphism.

Essential surjectivity. Let $\mathscr{G}^{\text{an}}$ be a coherent analytic sheaf on X^{an}. One must produce a coherent algebraic sheaf $\mathscr{G}$ with $\mathscr{G}^{\text{an}} \cong (\mathscr{G})^{\text{an}}$. The argument uses a descending induction: twisting $\mathscr{G}^{\text{an}}$ by $\mathcal{O}^{\text{an}}(d)$ for $d \gg 0$, the analytic Serre vanishing theorem (which holds on compact complex manifolds) gives that $\mathscr{G}^{\text{an}}(d)$ is generated by its global sections. These global sections are analytic, but by part (2) applied to $\mathcal{O}_{X^{\text{an}}}$ and its twists, they can be identified with algebraic sections. One then builds $\mathscr{G}$ algebraically from these generators and relations. The details require a careful interplay between the algebraic and analytic Hilbert polynomial theories; see [30] for a full account.

6.9.4 Remark: The Projectivity Hypothesis is Essential The GAGA theorems fail dramatically for non-projective (and even non-proper) varieties. The simplest example is the affine line: on $\mathbb{A}^1_{\mathbb{C}}$, the algebraic structure sheaf has $\Gamma(\mathbb{A}^1, \mathcal{O}) = \mathbb{C}[x]$ (polynomial functions), while the analytic structure sheaf has $\Gamma(\mathbb{C}, \mathcal{O}^{\text{an}}) = \mathcal{O}(\mathbb{C})$ (entire holomorphic functions), which is an enormously larger ring (containing e^x, $\sin x$, etc.). Hence the global section comparison already fails.
More strikingly, there exist coherent analytic sheaves on $\mathbb{C}$ that have no algebraic counterpart: the ideal sheaf of the analytic subset $\{e^{2\pi i n} | n \in \mathbb{Z}\} = \{1\}$ in $\mathbb{C}^*$ is coherent analytic but not algebraic, since the zero set of any algebraic function on $\mathbb{C}^*$ is either finite or all of $\mathbb{C}^*$.
For *proper* but non-projective varieties, a version of GAGA still holds (due to Grothendieck): the analytification functor is an equivalence on coherent sheaves, and cohomology comparison holds. The key requirement is properness — which ensures compactness of the analytic space — not projectivity. However, our proof relied on the embedding into $\mathbb{P}^N$ and the explicit computation of twisted sheaf cohomology, so we have stated the theorem in the projective case.

6.9.3 Consequences

The GAGA theorems have far-reaching consequences that connect algebraic geometry, complex analysis, and topology. We develop the most important ones.

Corollary 6.9.5: Chow's Theorem

Every closed analytic subspace of $\mathbb{CP}^n$ is algebraic. That is, if $Z \subseteq \mathbb{CP}^n$ is a closed complex-analytic subset, then Z is the zero set of finitely many homogeneous polynomials.

***Proof*:**
A closed analytic subspace $Z \subseteq (\mathbb{P}^n_{\mathbb{C}})^{\mathrm{an}}$ is determined by its ideal sheaf $\mathscr{I}_Z^{\mathrm{an}}$, which is a coherent analytic sheaf on $(\mathbb{P}^n)^{\mathrm{an}}$. By theorem 6.9.3(1), there exists a coherent algebraic sheaf $\mathscr{I}$ on $\mathbb{P}^n_{\mathbb{C}}$ with $\mathscr{I}^{\mathrm{an}} \cong \mathscr{I}_Z^{\mathrm{an}}$. Since the analytification of an ideal sheaf is an ideal sheaf (the inclusion $\mathscr{I} \hookrightarrow \mathcal{O}_{\mathbb{P}^n}$ analytifies to $\mathscr{I}^{\mathrm{an}} \hookrightarrow \mathcal{O}_{\mathbb{P}^n}^{\mathrm{an}}$), the sheaf $\mathscr{I}$ is an ideal sheaf on $\mathbb{P}^n_{\mathbb{C}}$, defining a closed algebraic subscheme Z_{alg} with $Z_{\mathrm{alg}}^{\mathrm{an}} = Z$. By theorem 6.9.3(3), this correspondence is a bijection.

Both comparisons below rest on one analytic fact, which is used three times in this section and has so far been invoked without proof. It is the holomorphic counterpart of the Poincaré lemma: on a complex manifold a closed holomorphic form is locally exact, so the holomorphic de Rham complex resolves the constant sheaf. The local statement is several-variable complex analysis; the sheaf statement is one line from it.

Lemma 6.9.6: The Holomorphic Poincaré Lemma

Let $P \subseteq \mathbb{C}^n$ be an open polydisc and let ω be a holomorphic p-form on P with $d\omega = 0$. If $p \geq 1$ then $\omega = d\eta$ for some holomorphic $(p-1)$-form η on P; if $p = 0$ then ω is constant.

***Proof*:**
Write $\omega = \sum_{|I|=p} f_I \, dz_I$ with each f_I holomorphic on P, and expand each f_I in a power series converging normally on compact subsets of P ([52, the higher-dimensional power series expansion]). Since d acts coefficientwise on such a series and the homotopy operator below is continuous for normal convergence, it suffices to treat a single monomial term, and then to observe that the classical Poincaré homotopy

$$(h\omega)(z) = \sum_{|I|=p} \sum_{k=1}^{p} (-1)^{k-1} \Big(\int_0^1 t^{p-1} f_I(tz)\, dt \Big) z_{i_k} \, dz_{i_1} \wedge \cdots \widehat{dz_{i_k}} \cdots \wedge dz_{i_p}$$

produces a form that is again *holomorphic*, because $f_I(tz)$ is holomorphic in z for each fixed t and the integral in t is over a compact interval, so differentiation under the integral sign preserves $\partial/\partial \bar{z}_j = 0$. The identity $d(h\omega) + h(d\omega) = \omega$ is the same computation as in the smooth case, carried out on a star-shaped domain, and a polydisc is star-shaped about its centre. With $d\omega = 0$ this gives $\omega = d(h\omega)$. For $p = 0$ a closed holomorphic function has vanishing differential, hence is constant on the connected P.

Corollary 6.9.7: The Holomorphic de Rham Resolution

Let X be a complex manifold. Then

$$0 \longrightarrow \underline{\mathbb{C}}_X \longrightarrow \mathcal{O}_X^{\mathrm{an}} \longrightarrow \Omega_X^{1,\mathrm{an}} \longrightarrow \Omega_X^{2,\mathrm{an}} \longrightarrow \cdots$$

is a resolution of the constant sheaf $\underline{\mathbb{C}}_X$ by sheaves of holomorphic forms. Consequently $\mathbb{H}^i(X, \Omega_X^{\bullet,\mathrm{an}}) \cong H^i(X, \underline{\mathbb{C}}_X)$.

***Proof*:**
Exactness is a statement about stalks, and every point of a complex manifold has a polydisc neighbourhood, so lemma 6.9.6 gives exactness in positive degrees and identifies the kernel in degree zero with the locally constant functions. The displayed isomorphism is then the hypercohomology of a resolution.

Corollary 6.9.8: Algebraic de Rham Theorem

Let X be a smooth projective variety over $\mathbb{C}$. Then the algebraic de Rham cohomology

$$H^i_{\mathrm{dR}}(X/\mathbb{C}) = \mathbb{H}^i(X,\, \Omega^{\bullet}_{X/\mathbb{C}})$$

(the hypercohomology of the algebraic de Rham complex $0 \to \mathcal{O}_X \to \Omega^1_{X/\mathbb{C}} \to \Omega^2_{X/\mathbb{C}} \to \cdots$) is naturally isomorphic to the singular cohomology:

$$H^i_{\mathrm{dR}}(X/\mathbb{C}) \;\cong\; H^i_{\mathrm{sing}}(X(\mathbb{C}); \mathbb{C}).$$

***Proof*:**
By GAGA, the algebraic de Rham complex analytifies to the holomorphic de Rham complex on X^{an}, and hypercohomology is preserved:

$$\mathbb{H}^i(X,\, \Omega^{\bullet}_{X/\mathbb{C}}) \;\cong\; \mathbb{H}^i(X^{\mathrm{an}},\, \Omega_X^{\bullet,\mathrm{an}}).$$

Since X^{an} is a compact Kähler manifold (example 0.6.22), the holomorphic Poincaré lemma (lemma 6.9.6) gives an exact sequence $0 \to \underline{\mathbb{C}} \to \Omega^{0,\mathrm{an}} \to \Omega^{1,\mathrm{an}} \to \cdots$, i.e., the holomorphic de Rham complex is a resolution of the constant sheaf $\underline{\mathbb{C}}$. Hence:

$$\mathbb{H}^i(X^{\mathrm{an}}, \Omega^{\bullet,\mathrm{an}}) \cong H^i(X^{\mathrm{an}}, \underline{\mathbb{C}}).$$

Finally, by the de Rham theorem (theorem 0.5.23), sheaf cohomology with coefficients in $\underline{\mathbb{C}}$ is isomorphic to singular cohomology: $H^i(X^{\mathrm{an}}, \underline{\mathbb{C}}) \cong H^i_{\mathrm{sing}}(X(\mathbb{C}); \mathbb{C})$.

Corollary 6.9.8 used properness twice: GAGA needs it, and so does the identification of analytic hypercohomology with singular cohomology through a compact Kähler manifold. Neither survives on an affine variety, which is never proper of positive dimension. Yet the affine case holds, for a reason that has nothing to do with GAGA: on an affine scheme the algebraic side degenerates completely, so the whole content sits in the comparison with the analytic topology. This is Grothendieck's theorem, and the affine case is the one consumers meet, because the complement of a hypersurface in $\mathbb{P}^{n+1}$ is affine.

Theorem 6.9.9: Grothendieck's Comparison Theorem

Let U be a smooth affine variety over $\mathbb{C}$. Then the complex of *global algebraic* differential forms computes the complex cohomology of the associated analytic space:

$$H^i\big(\Gamma(U, \Omega^\bullet_{U/\mathbb{C}})\big) \;\cong\; H^i\big(U^{\mathrm{an}}; \mathbb{C}\big)$$

***Proof Key ideas*:**
The algebraic side degenerates, and this half is complete. Each $\Omega^p_{U/\mathbb{C}}$ is quasi-coherent on the affine U, so $H^q(U, \Omega^p_{U/\mathbb{C}}) = 0$ for every $q > 0$ by theorem 6.2.9. In the hypercohomology spectral sequence

$$E_1^{p,q} = H^q\big(U, \Omega^p_{U/\mathbb{C}}\big) \Longrightarrow \mathbb{H}^{p+q}\big(U, \Omega^\bullet_{U/\mathbb{C}}\big)$$

the entire $q > 0$ region vanishes, so the sequence degenerates and

$$\mathbb{H}^i\big(U, \Omega^\bullet_{U/\mathbb{C}}\big) \;\cong\; H^i\big(\Gamma(U, \Omega^\bullet_{U/\mathbb{C}})\big)$$

The left side of the theorem is therefore the algebraic de Rham hypercohomology, exactly the object of corollary 6.9.8. No properness is used, and nothing about U beyond affineness.

The comparison is the content, and is not proved here. What remains is $\mathbb{H}^i(U, \Omega^\bullet_{U/\mathbb{C}}) \cong H^i(U^{\mathrm{an}}; \mathbb{C})$. On the analytic side the holomorphic Poincaré lemma (corollary 6.9.7) still makes $\Omega^{\bullet,\mathrm{an}}$ a resolution of $\underline{\mathbb{C}}$, so the right side is analytic hypercohomology; but algebraic and analytic hypercohomology cannot be compared through GAGA, which needs properness, and the analytic sheaves $\Omega^{p,\mathrm{an}}$ are *not* acyclic on U^{an}, so the analytic side does not degenerate. The standard route chooses a smooth compactification $U \subseteq \overline{U}$ whose boundary is a normal-crossings divisor D, replaces $\Omega^\bullet$ by the complex of forms with logarithmic poles along D, and compares on $\overline{U}$, where properness is restored. That needs resolution of singularities and the logarithmic de Rham complex, neither developed in this book. The argument is [28, the comparison theorem for smooth affine varieties].

Corollary 6.9.10: Comparison of Euler Characteristics

Let X be a smooth projective variety of dimension n over $\mathbb{C}$ and let $\mathscr{F}$ be a coherent algebraic sheaf on X. Then:

$$\chi_{\mathrm{alg}}(X, \mathscr{F}) = \sum_{i=0}^{n} (-1)^i \dim_{\mathbb{C}} H^i(X, \mathscr{F}) = \sum_{i=0}^{n} (-1)^i \dim_{\mathbb{C}} H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}}) = \chi_{\mathrm{an}}(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}}).$$

In particular, the algebraic and analytic Hilbert polynomials coincide: the algebraic Hilbert polynomial $P_{\mathscr{F}}$ of theorem 6.7.5 equals the analytic one.

***Proof*:**
Immediate from theorem 6.9.3(2): the comparison isomorphism preserves dimensions.

Corollary 6.9.11: Hodge Decomposition via GAGA

Let X be a smooth projective variety of dimension n over $\mathbb{C}$. Then the singular cohomology of $X(\mathbb{C})$ admits a Hodge decomposition:

$$H^k_{\text{sing}}(X(\mathbb{C});\,\mathbb{C}) \;\cong\; \bigoplus_{p+q=k} H^q(X,\,\Omega^p_{X/\mathbb{C}}),$$

where $H^q(X,\Omega^p_{X/\mathbb{C}})$ is the algebraic sheaf cohomology of the p-th sheaf of algebraic differentials. The *Hodge numbers* are:

$$h^{p,q}(X) = \dim_{\mathbb{C}} H^q(X,\Omega^p_{X/\mathbb{C}}),$$

and they satisfy the symmetries $h^{p,q} = h^{q,p}$ (Hodge symmetry) and $h^{p,q} = h^{n-p,\,n-q}$ (Serre duality).

***Proof*:**
By GAGA, $H^q(X,\Omega^p_{X/\mathbb{C}}) \cong H^q(X^{\text{an}},(\Omega^p)^{\text{an}})$. By the Dolbeault theorem (theorem 0.6.17), $H^q(X^{\text{an}},(\Omega^p)^{\text{an}}) \cong H^{p,q}_{\bar\partial}(X^{\text{an}})$, the Dolbeault cohomology. Since X^{an} is a compact Kähler manifold, the Hodge decomposition theorem (theorem 0.6.28) gives:

$$H^k_{\text{dR}}(X^{\text{an}}) \cong \bigoplus_{p+q=k} H^{p,q}_{\bar\partial}(X^{\text{an}}),$$

and the de Rham theorem identifies the left side with $H^k_{\text{sing}}(X(\mathbb{C});\mathbb{C})$. Combining these isomorphisms gives the decomposition.

The symmetry $h^{p,q} = h^{q,p}$ is the Hodge symmetry from complex conjugation on a Kähler manifold (theorem 0.6.28). The symmetry $h^{p,q} = h^{n-p,\,n-q}$ follows from Serre duality (corollary 6.5.10): since $\omega_X = \Omega^n_{X/\mathbb{C}}$, we have $H^q(X,\Omega^p) \cong H^{n-q}(X,(\Omega^p)^\vee \otimes \Omega^n)^* \cong H^{n-q}(X,\Omega^{n-p})^*$, giving $h^{p,q} = h^{n-p,\,n-q}$.

Example 6.9.12: GAGA in Action

1. *Line bundles and the exponential sequence.* On a compact complex manifold M, the *exponential sequence* of analytic sheaves:

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{2\pi i} \mathcal{O}^{\text{an}}_M \xrightarrow{\exp} (\mathcal{O}^{\text{an}}_M)^* \longrightarrow 0$$

induces a long exact sequence in cohomology:

$$\cdots \to H^1(M,\mathcal{O}^{\text{an}}) \to H^1(M,(\mathcal{O}^{\text{an}})^*) \xrightarrow{c_1} H^2(M,\mathbb{Z}) \to H^2(M,\mathcal{O}^{\text{an}}) \to \cdots$$

The group $H^1(M,(\mathcal{O}^{\text{an}})^*)$ classifies analytic line bundles (it is the analytic Picard group), and $c_1 : \operatorname{Pic}^{\text{an}}(M) \to H^2(M,\mathbb{Z})$ is the first Chern class map (definition 0.6.35).

For a smooth projective variety X over $\mathbb{C}$, GAGA identifies the analytic Picard group with the algebraic one: $\operatorname{Pic}(X) \cong \operatorname{Pic}^{\text{an}}(X^{\text{an}})$. The exponential sequence is *not* algebraic (the exponential map exp is transcendental), but its cohomological consequences — particularly the Chern class map and the Lefschetz theorem on $(1,1)$-classes — can be stated and applied purely algebraically via GAGA.

2. *Holomorphic vs. algebraic vector bundles.* Let X be a smooth projective variety over $\mathbb{C}$. By GAGA, the categories of algebraic and holomorphic vector bundles on X are equivalent (via **??** and its analytic counterpart, vector bundles correspond to locally free sheaves in both settings). Hence every holomorphic vector bundle on X^{an} is algebraic. This fails spectacularly for non-projective varieties: on $\mathbb{C}^*$, the flat line bundle with monodromy $e^{2\pi i\alpha}$ for irrational α is holomorphic but not algebraic.

3. *Singular cohomology from algebraic data.* Let $X = \mathbb{P}^n_{\mathbb{C}}$. The algebraic sheaf cohomology groups are: $H^i(\mathbb{P}^n, \mathcal{O}) = \mathbb{C}$ for $i = 0$ and 0 for $i > 0$ (by theorem 6.3.1). By GAGA, the same holds analytically. The Hodge decomposition gives $H^k_{\mathrm{sing}}(\mathbb{CP}^n; \mathbb{C}) = \bigoplus_{p+q=k} H^q(\mathbb{P}^n, \Omega^p)$. The Hodge diamond of $\mathbb{CP}^n$ (example 0.6.30) has $h^{p,p} = 1$ for $0 \le p \le n$ and all other $h^{p,q} = 0$, so:
$$H^k_{\mathrm{sing}}(\mathbb{CP}^n; \mathbb{C}) = \begin{cases} \mathbb{C} & k \text{ even, } 0 \le k \le 2n, \\ 0 & k \text{ odd.} \end{cases}$$
This recovers the classical computation from algebraic topology (see [51, section 4.1]), but now derived entirely from algebraic sheaf cohomology via GAGA and Hodge theory.

4. *The Lefschetz theorem on hyperplane sections.* Let $X \subset \mathbb{P}^N_{\mathbb{C}}$ be a smooth projective variety of dimension n and let $H \subset \mathbb{P}^N$ be a hyperplane with $Y = X \cap H$ smooth. The Lefschetz hyperplane theorem asserts that the restriction map:
$$H^i_{\mathrm{sing}}(X(\mathbb{C}); \mathbb{Z}) \longrightarrow H^i_{\mathrm{sing}}(Y(\mathbb{C}); \mathbb{Z})$$
is an isomorphism for $i < n - 1$ and injective for $i = n - 1$. The proof uses Morse theory (or its algebraic substitute, the weak Lefschetz theorem) and is fundamentally analytic. However, via GAGA, the consequence for algebraic cohomology is immediate: the restriction maps $H^i(X, \Omega^p) \to H^i(Y, \Omega^p|_Y)$ are isomorphisms in the range dictated by Lefschetz. This has purely algebraic applications — for instance, it implies that the Picard group of a smooth hypersurface $X \subset \mathbb{P}^n$ with $n \ge 4$ is isomorphic to $\mathbb{Z}$, generated by $\mathcal{O}_X(1)$.

6.9.13 Remark: GAGA as a Principle Beyond the precise theorems stated above, GAGA functions as a *guiding principle* in algebraic geometry over $\mathbb{C}$: any "reasonable" statement about coherent sheaves and their cohomology on projective varieties that can be proved analytically also holds algebraically, and vice versa. This principle is invoked (often implicitly) throughout the subject:

- Kodaira vanishing ($H^i(X, \omega_X \otimes \mathcal{L}) = 0$ for $i > 0$ and $\mathcal{L}$ ample) is proved analytically using L^2-estimates for the $\bar{\partial}$-operator, but applies to algebraic cohomology via GAGA.
- The degeneration of the Hodge-to-de Rham spectral sequence for smooth projective varieties is proved using Kähler geometry (or, in characteristic p, by the Deligne–Illusie method), but the algebraic consequence — that the Hodge numbers $h^{p,q}$ are well-defined invariants of the algebraic variety — is what matters for classification.
- The construction of moduli spaces often proceeds by first constructing an analytic moduli space (using complex-analytic deformation theory) and then invoking GAGA to conclude that it is algebraic.

In this way, GAGA is not a comparison theorem but a bridge that allows free passage between the algebraic and analytic worlds, using whichever tools are most natural for the problem at hand.

6.9.4 Summary

We close with a summary of the cohomology that this chapter has developed, viewed through GAGA. Let X be a smooth projective variety of dimension n over $\mathbb{C}$, and let $\mathscr{F}$ be a coherent sheaf on X. The following table summarizes the relationships between the various cohomology theories:

Algebraic	**Analytic**	**Via**
$H^i(X, \mathscr{F})$	$H^i(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$	GAGA
$H^q(X, \Omega^p_{X/\mathbb{C}})$	$H^{p,q}_{\bar{\partial}}(X^{\mathrm{an}})$	GAGA + Dolbeault
$\mathbb{H}^k(X, \Omega^\bullet)$	$H^k_{\mathrm{sing}}(X(\mathbb{C}); \mathbb{C})$	Algebraic de Rham
$\chi(X, \mathscr{F})$	$\chi(X^{\mathrm{an}}, \mathscr{F}^{\mathrm{an}})$	GAGA
$\mathrm{Pic}\,(X)$	$\mathrm{Pic}^{\,\mathrm{an}}(X^{\mathrm{an}})$	GAGA
$\mathrm{Ext}^i(\mathscr{F}, \omega_X)$	$H^{n-i}(X, \mathscr{F})^*$	Serre duality
$h^{p,q} = h^{n-p,\,n-q}$	Hodge + Serre	GAGA + duality

The algebraic side (left column) uses only the Zariski topology, coherent sheaves, and the machinery of this chapter. The analytic side (middle column) uses the classical topology, Dolbeault cohomology, and Hodge theory from the complex geometry chapter. The right column indicates which bridge theorem connects them.

For non-projective or non-smooth varieties, the dictionary partially breaks down: GAGA requires properness, Hodge decomposition requires Kähler structure (or smoothness), and Serre duality requires Cohen — Macaulay (or at least the existence of a dualizing sheaf). The interplay between these hypotheses — determining which parts of the dictionary survive under which conditions — is a central theme of modern algebraic geometry.

7

Application: Curves, Surfaces, and a bit Beyond

It is time to reap what we have sown. Over the preceding chapters, we have built much machinery: schemes, coherent sheaves, divisors, differentials, and cohomology. The reader has persevered through the abstractions and now wish to see what benefits are attained from this generalization. This chapter is the answer.

We turn towards the concrete geometry of *curves* and *surfaces*—the one-dimensional and two-dimensional objects that have driven algebraic geometry since its inception. It is no exaggeration to say that most of the machinery of the previous chapters was *invented* to understand these objects. Riemann and his contemporaries studied algebraic curves through the lens of complex analysis; the Italians (Castelnuovo, Enriques, Severi) built an elaborate—if occasionally imprecise—theory of surfaces; and it was the desire to put these theories on firm footing that led Grothendieck to develop the scheme-theoretic framework we now command.

The philosophy of this chapter is simple: we take the abstract theorems we have proved and *use* them. Serre duality becomes the Riemann–Roch theorem. The sheaf of differentials becomes the tool that controls ramification. Euler characteristics become genus. Cohomology groups become spaces of functions and forms that we can count, embed with, and classify by. The payoff is enormous: by the end of this chapter, we will have a nearly complete understanding of the geometry of curves, and a sweeping overview of the classification of surfaces that has guided a century of research.

The chapter splits naturally into two halves. The first half (§7.1–§7.1.8) develops the theory of algebraic curves. Here the main components are the Riemann–Roch theorem, Hurwitz's formula, and the rich interplay between divisors and morphisms. The second half (§7.2–§7.2.8) turns to surfaces, where we encounter the intersection pairing, blow-ups, and the celebrated Enriques–Kodaira classification.

Throughout, we will plant signposts toward the later parts of this book. The study of families of curves leads naturally to moduli spaces and stacks (*Moduli, Deformation Theory, and Stacks* [17]). The birational geometry of surfaces—blow-ups, contractions, minimal models—is the prototype for the Minimal Model Program in arbitrary dimension (a separate volume). And the cohomological invariants that drive classification here find their core expression in étale cohomology and derived categories, each in its own volume. The reader should view this chapter not as a self-contained destination, but as the crossroads from which the highways of modern algebraic geometry radiate.

7.0.1 Conventions for this Chapter Unless otherwise stated, k denotes an algebraically closed field and all schemes are k-schemes. A *curve* will mean a complete nonsingular curve over k (see Definition 7.1.1 for the precise scheme-theoretic formulation). A *surface* will mean a smooth projective surface over k. When we say *point*, we mean *closed point*—the only other point of an integral curve is its generic point η, which we will mention explicitly when needed. We shall occasionally relax these hypotheses (particularly the algebraic closure of k) and will always say so.

7.1 Algebraic Curves

Algebraic curves are "simple" in the sense that their birational geometry is trivial—every birational map between smooth projective curves extends to an isomorphism—and yet "rich" because a single discrete birational-invariant, the *genus*, controls a vast web of geometric, arithmetic, and analytic phenomena.

There are three perspectives on algebraic curves that inform and enrich each other, and the reader should keep all three in mind as we proceed:

1. *The geometric perspective.* A curve is a one-dimensional, complete, nonsingular scheme over k. This is the language of this book, and the perspective from which we will prove our theorems. The nonsingularity assumption will be addressed in this section. The corresponding classical theory — curves as varieties, divisors, the genus, and Bézout — is [20], and this section is its continuation rather than its replacement.

2. *The analytic perspective.* Over $\mathbb{C}$, the category of smooth projective curves (with regular maps) is equivalent to the category of compact Riemann surfaces (with holomorphic maps) by Serre's GAGA theorems (theorem 6.9.3). A Riemann surface is a complex manifold, meaning it carries a rigid *complex structure*. If we forget this complex structure, we are left with a smooth, compact, oriented 2-manifold. The topological and smooth classification of such surfaces[1] tells us they are determined entirely by their genus—the "number of holes." This gives an immediate and powerful topological intuition for many algebraic results.

3. *The algebraic perspective.* By the equivalence between smooth projective curves over k and finitely generated field extensions K/k of transcendence degree 1 (proved classically in [20], where two nonsingular projective curves are shown to be isomorphic exactly when their function fields are), studying curves is equivalent to studying function fields. This is the perspective that connects most directly to number theory: the function field of a curve over $\mathbb{F}_q$ behaves much like a number field, and many of our theorems will have arithmetic counterparts. I'm

[1] For 2-dimensional manifolds, the topological and smooth categories are isomorphic; every topological surface admits a unique smooth structure.

thinking of actually adding this proof here, since it feels like too big a result to reference away. I'm okay with doubling the proof; being both here and in classical-alg-geo.

These three viewpoints formalize into equivalences of categories, provided we are careful with our morphisms. When $k = \mathbb{C}$, GAGA (theorem 6.9.3) provides a direct equivalence between the category of smooth projective curves and the category of compact Riemann surfaces. Meanwhile, the functor from smooth projective curves to function fields forms an anti-equivalence[2] of categories. Throughout this chapter, we will freely draw intuition from whichever perspective is most illuminating at any given moment.

7.1.1 Definition and First Properties

Let us begin by nailing down the scheme-theoretic definition. Recall that a scheme is *integral* if it is both reduced and irreducible (proposition 2.4.19), and *regular* if all of its local rings are regular local rings (definition 2.7.4).

Definition 7.1.1: Curve

Let k be a field. A *curve over* k is a Noetherian, integral, separated scheme X of dimension 1 and of finite type over k. We say:

1. X is *complete* (or *proper*) if the structure morphism $X \to \operatorname{Spec} k$ is proper.
2. X is *nonsingular* (or *smooth*, or *regular*) if all local rings $\mathcal{O}_{X,x}$ are regular, or equivalently (since $k = \overline{k}$), if the structure morphism is smooth (theorem 2.7.13).

Let $\mathbf{Crv}_k$ be the category of all curves over k, $\mathbf{Crv}_k^{\text{prop}}$ be the subcategory of proper curves, and $\mathbf{SmProjCrv}_k$ be the category of smooth projective (i.e., complete and nonsingular) curves. As $\mathbf{SmProjCrv}_k$ is the best-behaved and most frequently used category, it will serve as our default throughout much of the text.

A few remarks are in order. First, our definition allows curves that are not complete (such as the affine line $\mathbb{A}^1_k$) and curves that are singular (such as the nodal cubic $V(y^2 - x^3 - x^2) \subseteq \mathbb{P}^2_k$). In this section, however, our standing assumption is that curves are *complete and nonsingular* unless explicitly stated otherwise.

Second, the integrality assumption means that a curve has a unique generic point η with $\overline{\{\eta\}} = X$, and the function field $\kappa(\eta) = K(X)$ is a finitely generated field extension of k of transcendence degree 1 (theorem 2.6.11). Conversely, every such field extension arises as the function field of some smooth projective curve—this is the algebraic perspective mentioned above.

Third, for a complete nonsingular curve over $k = \overline{k}$, every closed point $x \in X$ has $\mathcal{O}_{X,x}$ a discrete valuation ring (DVR)[3]. This means the closed points of X are in bijection with the DVRs of $K(X)/k$, a fact that was the classical starting point for the theory of abstract curves.

Note that the definition mentioned properness instead of projectiveness. However, all proper

[2] More precisely, the category of smooth projective curves over k with *dominant* morphisms is anti-equivalent to the category of finitely generated field extensions of k of transcendence degree 1. See corollary 3.6.6 and proposition 3.6.8.

[3] Indeed, $\mathcal{O}_{X,x}$ is a regular local ring of dimension 1, and a regular local ring of dimension 1 is precisely a DVR. See proposition 2.7.3 and [10].

nonsingular curves *are* projective. We shall prove this in corollary 7.1.27 but it is worth mentioning early to set the right intuition for the reader: curves embed in projective space.

Our standing assumption throughout this chapter is that curves are complete and nonsingular. But singular curves arise inevitably—as fibers in degenerating families, as images of morphisms, and as boundary points of moduli spaces—so the reader should know how they connect to the smooth theory. If X is a complete integral curve that is *not* nonsingular, the local rings $\mathcal{O}_{X,x}$ at the singular points are no longer DVRs: the clean bijection between closed points and discrete valuations of $K(X)/k$ breaks down.

While arbitrary integral curves are classified only up to birational equivalence by their function fields, smooth proper curves are classified up to exact isomorphism. Any singularity can be systematically remedied using *normalization*, which provides a birational morphism from a smooth curve. For curves, normality and nonsingularity coincide: a Noetherian local domain of dimension 1 is a DVR if and only if it is integrally closed[4].

Proposition 7.1.2: Affine Curves and Unique Proper Curve

Let $X = \operatorname{Spec} R$ be a nonsingular curve. Then there is a unique proper nonsingular curve Y up to isomorphism where $K(X) = K(Y)$.

***Proof sketch*:**
As X is finite type and affine, $X \cong \operatorname{Spec} k[x_1, ..., x_n]/I$, hence it admits a closed immersion into $\mathbb{A}^n$, which in turn naturally embeds into $\mathbb{P}^n$. Let X be re-labeled to represent the image of this sequence of embeddings. Then its projective closure $\overline{X}$ is closed and proper. It may not be nonsingular, so we take its normalization:

$$\nu \colon \widetilde{X} \longrightarrow \overline{X}$$

where $\widetilde{X}$ is the unique nonsingular curve with $K(\widetilde{X}) = K(X)$, constructed by taking the integral closure of each affine coordinate ring of X in its fraction field (definition 2.4.27). Normalization is a finite map, and finite morphisms are proper and hence preserve closed sets, giving the desired proper nonsingular curve.

One should intuitively visualize ν as pulling apart branches at a node and smoothing out a cusp:

1. For the *nodal cubic* $y^2 = x^3 + x^2$, the normalization is $\mathbb{P}^1$, with two preimages of the node (one for each branch meeting at the origin).
2. For the *cuspidal cubic* $y^2 = x^3$, the normalization is again $\mathbb{P}^1$, with a single preimage of the cusp.

These represent the two most common elementary singularities: a node (where two distinct tangent directions cross) and a cusp (where the tangent directions coincide). While a curve can have much more complicated singularities (e.g., higher-order cusps, tacnodes, or multiple crossing branches), the geometric intuition of normalization as "pulling apart" branches and "smoothing" cusps remains valid in general.

[4]The "if" direction: an integrally closed (i.e. normal) Noetherian local domain of dimension 1 is a DVR; see [10]. The "only if" direction is proposition 2.7.19, since every DVR is a regular local ring of dimension 1.

In both cases above, a singular curve of arithmetic genus 1 turns out to be geometrically *rational* ($\widetilde{X} \cong \mathbb{P}^1$). The discrepancy between the arithmetic genus of X and the geometric genus of $\widetilde{X}$ is measured by the δ-*invariant* at the singular points; see the remark after proposition 7.1.14.

When X is complete, $\widetilde{X}$ is precisely the "complete nonsingular model" of the function field $K(X)$ from corollary 7.1.27—every integral complete curve has a smooth projective model, and that model is its normalization.

Proposition 7.1.2 compactifies an affine nonsingular curve, and it records only the function field of the result. Both restrictions bite. A curve produced by a construction rather than by a presentation, as a fiber of a family, as an orbit, or as a one-dimensional slice of a larger variety, is quasiprojective and rarely affine; and the arguments that use a completion need the original curve to sit inside the completed one as an open subscheme, not to be recovered from it up to birational equivalence. The next theorem removes both restrictions, at the cost of one honest correction: what the construction returns is a regular curve, and regularity is smoothness only over a perfect field.

Theorem 7.1.3: Projective Completion of a Smooth Curve

Let k be a field and let V be a quasiprojective curve over k (definition 7.1.1, definition 2.3.5) which is smooth over k. Then:

1. there exist a regular projective curve $\overline{V}$ over k, in particular integral and hence connected, and an open immersion $j\colon V \hookrightarrow \overline{V}$ with dense image; the map j identifies $K(\overline{V})$ with $K(V)$;
2. the pair $(\overline{V}, j)$ is unique up to unique isomorphism under V: for any second such pair $(\overline{V}', j')$ there is exactly one isomorphism $f\colon \overline{V} \to \overline{V}'$ with $f \circ j = j'$;
3. if k is perfect, then $\overline{V}$ is smooth over k.

***Proof*:**
Step 1: an integral projective curve containing V as a dense open. By definition 2.3.5 there are a projective k-scheme Y and an open immersion $V \subseteq Y$. Let $\overline{X} \subseteq Y$ be the closure of V, taken with its reduced induced structure. It is irreducible, being the closure of the irreducible V, and reduced, hence integral (proposition 2.4.19). A composite of closed immersions is a closed immersion (proposition 3.1.24(3)), so the closed subscheme $\overline{X}$ of Y admits a closed immersion into some $\mathbb{P}^n_k$ and is projective (corollary 5.3.15(2)), hence of finite type and proper over k (theorem 3.5.26). The subset V is open in $\overline{X}$, and the induced structure agrees with the given one, both being reduced with the same underlying space; so V is a dense open subscheme of $\overline{X}$ and $K(\overline{X}) = K(V)$. That field has transcendence degree 1 over k, so $\operatorname{kdim}\overline{X} = 1$ (theorem 2.6.11).

Step 2: normalize. Since $\overline{X}$ is integral of finite type over k, theorem 3.4.35 supplies a normalization $\nu\colon \widetilde{X} \to \overline{X}$: the scheme $\widetilde{X}$ is integral, normal and of finite type over k, the morphism ν is finite, surjective and birational, and ν is an isomorphism over the normal locus of $\overline{X}$. As $\overline{X}$ is projective, corollary 5.3.43 makes $\widetilde{X}$ projective over k, hence separated, Noetherian and proper. Birationality gives $K(\widetilde{X}) = K(V)$ and therefore $\operatorname{kdim}\widetilde{X} = 1$ (theorem 2.6.11).

Step 3: V survives the normalization. Smoothness implies regularity (proposition 2.7.12), and a regular local ring is integrally closed (proposition 2.7.19), so every local ring of V is normal.

Those local rings are the local rings of $\overline{X}$ at the points of V, so V lies in the normal locus of $\overline{X}$ and ν restricts to an isomorphism $\nu^{-1}(V) \to V$. Set $\overline{V} = \widetilde{\overline{X}}$ and let $j\colon V \to \overline{V}$ be the inverse of that isomorphism followed by the open immersion $\nu^{-1}(V) \subseteq \overline{V}$. Its image is a nonempty open subset of the irreducible scheme $\overline{V}$, hence dense, and j identifies $K(V)$ with $K(\overline{V})$.

Step 4: $\overline{V}$ is regular, and smooth over a perfect field. The scheme $\overline{V}$ is Noetherian, integral and normal, so by proposition 5.4.2(2) every point at which it fails to be regular has codimension at least 2. No point has codimension 2: for $x \in \overline{V}$ the closure $\overline{\{x\}}$ is an irreducible closed subset, so $\operatorname{codim}_{\overline{V}}\overline{\{x\}} \leq \operatorname{kdim}\overline{V} = 1$ by lemma 2.6.7, and this codimension is $\operatorname{kdim}\mathcal{O}_{\overline{V},x}$ by definition 2.6.5. Hence $\overline{V}$ is regular (definition 2.7.4) and is a projective curve over k in the sense of definition 7.1.1. If k is perfect, regularity and smoothness coincide for k-schemes locally of finite type (theorem 2.7.13), so $\overline{V}$ is then smooth over k. This proves (1) and (3).

Step 5: a morphism between two completions. Let $(\overline{V}', j')$ be a second pair as in (1). The composite $j' \circ j^{-1}$, defined on the dense open $j(V) \subseteq \overline{V}$, is a rational map $\varphi\colon \overline{V} \dashrightarrow \overline{V}'$. Its source is integral, normal (proposition 2.7.19) and of finite type over k, and its target is proper over k (theorem 3.5.26), so by theorem 4.7.4 the indeterminacy locus of φ has codimension at least 2 in $\overline{V}$. By the computation of Step 4 no point of $\overline{V}$ has codimension 2, so that locus is empty and φ is a morphism $f\colon \overline{V} \to \overline{V}'$ with $f \circ j = j'$.

Step 6: the morphism is the unique isomorphism under V. The source $\overline{V}$ is proper over k and the target $\overline{V}'$ is separated over k, so f is proper (proposition 3.5.21(5)) and therefore separated, of finite type and closed (definition 3.5.20). Its image is closed and contains the dense subset $j'(V)$, so f is surjective, and $f \circ j = j'$ forces f to induce the identity on $K(V)$, so f is birational. Each fiber is finite: a fiber is a closed subset of $\overline{V}$ other than $\overline{V}$, since f is surjective onto a space with more than one point, so each of its irreducible components Z satisfies $\operatorname{codim}_{\overline{V}}Z \geq 1$ and hence $\operatorname{kdim}Z = 0$ by lemma 2.6.7, making Z a closed point; and the components are finite in number because $\overline{V}$ is Noetherian. So f is quasifinite. Both schemes are integral and Noetherian and $\overline{V}'$ is normal, so corollary 3.4.33 makes f an open immersion, and a surjective open immersion is an isomorphism. Finally, two isomorphisms $\overline{V} \to \overline{V}'$ under V agree on the dense open $j(V)$ of the reduced scheme $\overline{V}$, and $\overline{V}'$ is projective over k and so separated, hence the two agree (proposition 3.5.10). This proves (2), completing the proof.

The theorem contains proposition 7.1.2: an affine nonsingular curve is quasiprojective, so it has a completion, and the uniqueness recorded here is the stronger one, pinning down the model together with its embedding rather than up to a shared function field.

Part (3) is not a convenience. Over an imperfect field a smooth curve can fail to have any smooth projective completion, so the general conclusion has to be regularity, and example 2.7.14 is the local mechanism. For a curve, let p be odd, let $k = \mathbb{F}_p(t)$ and let $C = \operatorname{Spec} k[x,y]/(y^2 - x^p + t)$, an integral affine curve. The Jacobian of $y^2 - x^p + t$ is $(0, 2y)$, so C fails to be smooth over k exactly along $y = 0$, which on C is the single closed point P cut out by y, with residue field $k(t^{1/p})$. Thus $V = C \setminus \{P\}$ is a smooth quasiprojective curve. But C is regular at P as well: the relation $y^2 = x^p - t$ makes the maximal ideal of $\mathcal{O}_{C,P}$ equal to (y), a principal ideal in a one-dimensional Noetherian local domain. Steps 5 and 6 applied to the regular curve C in place of $\overline{V}$ exhibit C as an open subscheme of the completion of V, which therefore contains P and is not smooth over k. No smooth projective curve contains V as a dense open.

Thus, we get the fundamental dictionary:

$$\left\{\begin{matrix}\text{Smooth Projective Curves}\\ \text{over } k\end{matrix}\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\begin{matrix}\text{Finitely Generated Field Extensions}\\ \text{over } k\\ \text{of transcendental degree 1}\end{matrix}\right\} \tag{7.1}$$

7.1.4 The Analytic Perspective Over $\mathbb{C}$, smooth projective curves correspond to compact Riemann surfaces via the analytification functor (definition 6.9.1). By GAGA (theorem 6.9.3), this functor establishes an equivalence of categories between smooth projective curves over $\mathbb{C}$ and compact Riemann surfaces. In particular, every compact Riemann surface admits the structure of a smooth projective variety—a nontrivial fact whose proof requires showing that every compact Riemann surface carries a nontrivial meromorphic function (see [30]). The topological classification of compact oriented surfaces tells us that such a surface is determined up to diffeomorphism by its genus $g \geq 0$, the "number of holes." We thus inherit a topological invariant for our algebraic curves that will turn out to equal both the arithmetic and geometric genus (Proposition 7.1.14). Recall also the Hodge diamond of a curve from example 0.6.31: the single interesting Hodge number is $h^{0,1} = h^{1,0} = g$.

Next, we record a key rigidity property of maps between complete curves. This should be compared with the situation in complex analysis, where a nonconstant holomorphic map between compact Riemann surfaces is necessarily surjective (and a branched cover).

Proposition 7.1.5: Image of Complete Nonsingular Curve is Closed

Let X be a complete nonsingular curve over k, and let Y be any curve over k. Suppose $f: X \to Y$ is a morphism. Then either:

1. $f(X)$ is a single closed point of Y, or
2. f is surjective, Y is necessarily complete, and f is a finite morphism.

***Proof*:**
Since X is proper over k and Y is separated over k, the image $f(X) \subseteq Y$ is closed (proposition 3.5.21). As X is irreducible, so is $f(X)$. A closed irreducible subset of the curve Y is either a single closed point or all of Y (the only irreducible closed subsets of dimension 1 of a curve are the curve itself and its closed points). In the latter case, f is surjective, and Y is proper since it is the image of a proper scheme under a morphism to $\operatorname{Spec} k$.

For finiteness: since f is dominant, it induces an inclusion of function fields $K(Y) \hookrightarrow K(X)$. Both are finitely generated of transcendence degree 1 over k, so $K(X)/K(Y)$ is a finite extension; let $n = [K(X) : K(Y)]$. To show f is finite, we check the condition affine-locally. Let $V = \operatorname{Spec} S \subseteq Y$ be an affine open and let $U = f^{-1}(V)$. Since X is a complete nonsingular curve, the ring of regular functions on U is the integral closure R of S in $K(X)$. As S is a Noetherian integrally closed domain of dimension 1 (a Dedekind domain), R is a finitely generated S-module[a], hence $f^{-1}(V) = \operatorname{Spec} R$ and $f|_U$ is finite.

[a] This is the "finiteness of integral closure" for Dedekind domains; see [10].

This proposition has a simple but consequence: the *only* morphisms between complete nonsingular curves are the trivial ones (constant maps) and the finite ones (branched covers). There is no notion of an "immersion" or "partial" map—every nonconstant morphism is surjective. This rigidity is one of the key features that makes the theory of curves so clean compared to higher-dimensional geometry.

As every nonconstant morphism between complete nonsingular curves is finite and surjective, there is a basic invariants attached to such morphisms:

Definition 7.1.6: Degree of Morphism of Curves

The *degree* of f is the degree of the finite field extension:

$$\deg f \;=\; [K(X) : K(Y)].$$

One should think of $\deg f$ as the "number of sheets" of the covering: over a "generic" point $y \in Y$, the fiber $f^{-1}(y)$ consists of exactly $\deg f$ points (counted with multiplicity)[5]. In the analytic picture over $\mathbb{C}$, a degree-n map between compact Riemann surfaces is precisely an n-sheeted branched cover.

Pullback of Divisors

Weil divisors and Cartier divisors coincide on nonsingular curves (proposition 5.4.31), so we work freely with $\mathrm{WDiv}\, X = \bigoplus_{x \in X_{\mathrm{cl}}} \mathbb{Z} \cdot x$ and the class group $\mathrm{Cl}\, X$. Furthermore, as $\dim X = 1$, all (Weil) divisors are finite collections of points and hence can be thought of as "recipes" for a rational function to satisfy, namely imposing the zeros and poles required of a rational function.

We now construct the *pullback* of divisors along a finite morphism $f\colon X \to Y$ between complete nonsingular curves. For a closed point $y \in Y$, choose a uniformizer $t \in \mathcal{O}_{Y,y}$ (a generator of the maximal ideal $\mathfrak{m}_y$, so that $\nu_y(t) = 1$). For each point $x \in f^{-1}(y)$, the map $f^{\#}\colon \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ sends t to an element of the maximal ideal $\mathfrak{m}_x$. Define the map $f^*\colon \mathrm{WDiv}\,(Y) \to \mathrm{WDiv}\,(X)$ on closed points as:

$$f^*(y) \;=\; \sum_{x \in f^{-1}(y)} \nu_x(f^{\#}(t)) \cdot x \;=\; \sum_{f(x)=y} e_x \cdot x$$

where $e_x = \nu_x(f^{\#}(t))$ is the *ramification index* at x (which we shall study in earnest in §7.1.4). This is a finite sum since f is a finite morphism, and it is independent of the choice of uniformizer t: any other uniformizer has the form $t' = ut$ for a unit $u \in \mathcal{O}_{Y,y}^*$, and $\nu_x(f^{\#}(u)) = 0$ since $f^{\#}(u)$ is a unit in $\mathcal{O}_{X,x}$.

Extending by $\mathbb{Z}$-linearity, we obtain a group homomorphism $f^*\colon \mathrm{WDiv}\, Y \to \mathrm{WDiv}\, X$. One checks that f^* sends principal divisors to principal divisors[6], and hence descends to:

$$f^*\colon \mathrm{Cl}\, Y \to \mathrm{Cl}\, X.$$

[5] More precisely, $\sum_{x \in f^{-1}(y)} e_x = \deg f$ where e_x is the ramification index at x; we shall return to this in §7.1.4.

[6] If $g \in K(Y)^*$, then $f^*(\mathrm{div}_Y g) = \mathrm{div}_X(g \circ f)$: the valuation of $g \circ f$ at x equals e_x times the valuation of g at $f(x)$, which is exactly what the formula gives.

Proposition 7.1.7: Finite Map and Degrees

Let $f\colon X \to Y$ be a finite morphism of complete nonsingular curves. For any divisor D on Y:

$$\deg f^*(D) \;=\; (\deg f)\cdot(\deg D).$$

***Proof*:**
By $\mathbb{Z}$-linearity, it suffices to prove $\deg f^*(y) = (\deg f)(\deg y)$ for a single closed point $y \in Y$. Let $V = \operatorname{Spec} S \subseteq Y$ be an affine open neighbourhood of y. Since Y is a nonsingular curve, S is a Dedekind domain. Let R be the integral closure of S in $K(X)$; then $f^{-1}(V) = \operatorname{Spec} R$ and R is a finitely generated S-module (as argued in proposition 7.1.5).

Localize at the maximal ideal $\mathfrak{m}_y \subseteq S$. Set $B = S_{\mathfrak{m}_y} = \mathcal{O}_{Y,y}$, which is a DVR with uniformizer t, and let $R' = R \otimes_S B$, the semi-local ring (R' has finitely many maximal ideals) obtained by localizing R at $\mathfrak{m}_y$. Since R is a finitely generated torsion-free S-module of rank $n = [K(X) : K(Y)] = \deg f$, the localization R' is a free B-module of rank n (torsion-free and finitely generated over a PID).

The maximal ideals of R' correspond to the points $x_1, \ldots, x_r \in X$ lying over y, with $(R')_{\mathfrak{m}_i} = \mathcal{O}_{X,x_i}$ for each i. Since the maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_r \subseteq R'$ are pairwise coprime, the Chinese Remainder Theorem gives an isomorphism of k-algebras:

$$R'/tR' \;\cong\; \bigoplus_{i=1}^{r} R'_{\mathfrak{m}_i}/tR'_{\mathfrak{m}_i} \;=\; \bigoplus_{i=1}^{r} \mathcal{O}_{X,x_i}/t\,\mathcal{O}_{X,x_i}.$$

Taking dimensions over k of both sides, and noting that R'/tR' is an algebra over the residue field $\kappa(y) = B/tB$, we obtain:

$$n[\kappa(y):k] = \dim_k R'/tR' = \sum_{i=1}^{r} \dim_k(\mathcal{O}_{X,x_i}/t\,\mathcal{O}_{X,x_i}) = \sum_{i=1}^{r} \nu_{x_i}(t)[\kappa(x_i):k] = \sum_{i=1}^{r} e_{x_i}\deg(x_i).$$

The left-hand side is $(\deg f)(\deg y)$, and the right-hand side is $\deg f^*(y)$ by definition, yielding the claim.

This result is the scheme-theoretic incarnation of a basic topological fact: a degree-n map of compact Riemann surfaces pulls back a point (counted with multiplicity) to n points (counted with multiplicity). The next corollary is one of the most important structural results about divisors on curves.

Corollary 7.1.8: Degree Map and Integers

Let X be a complete nonsingular curve. Then principal divisors have degree 0, and the degree map induces a surjective homomorphism:

$$\deg\colon \operatorname{Cl} X \twoheadrightarrow \mathbb{Z}.$$

***Proof*:**
Let $g \in K(X)^*$ with $g \notin k^*$ (if $g \in k^*$, then $\mathrm{div} g = 0$ and the result is trivial). The element g defines a finite morphism $\varphi \colon X \to \mathbb{P}^1_k$ as follows[a]: the inclusion $k(g) \hookrightarrow K(X)$ induces a finite morphism φ to $\mathbb{P}^1_k$ (finite by proposition 7.1.5). Now, $\mathrm{div} g = \varphi^*([0]) - \varphi^*([\infty])$ where $[0]$ and $[\infty]$ are the divisors of the points 0 and ∞ on $\mathbb{P}^1_k$. By proposition 7.1.7:

$$\deg(\mathrm{div} g) = \deg \varphi^*([0]) - \deg \varphi^*([\infty]) = \deg \varphi - \deg \varphi = 0.$$

Thus the degree map $\deg \colon \mathrm{WDiv}\, X \to \mathbb{Z}$ kills principal divisors and descends to $\mathrm{Cl}\, X$. Surjectivity is clear: any closed point $x \in X$ gives a divisor of degree 1 (assuming k is algebraically closed or X has a k-rational point).

[a]Concretely, $g \in K(X)$ gives a k-algebra inclusion $k(t) \hookrightarrow K(X)$ via $t \mapsto g$, which corresponds to a dominant rational map $X \dashrightarrow \mathbb{P}^1_k$. Since X is a smooth proper curve, this extends to a morphism by the valuative criterion (theorem 3.5.25).

The kernel of the degree map, $\mathrm{Cl}^0 X = \ker(\deg \colon \mathrm{Cl}\, X \to \mathbb{Z})$, is an extremely important invariant of the curve. We will see in §7.1.7 that for an elliptic curve E, this group is isomorphic to $E(k)$ itself—the set of closed points equipped with the elliptic curve group law. More generally, $\mathrm{Cl}^0 X$ is the group of k-points of an abelian variety of dimension g called the *Jacobian* $\mathrm{Jac}(X)$, whose construction and properties are central to the study of abelian varieties and the moduli spaces of curves[7].

[7]The Jacobian is the identity component of the Picard scheme $\mathrm{Pic}_{X/k}$, treated in *Moduli, Deformation Theory, and Stacks* [17]. When $k = \mathbb{C}$, it is a complex torus $\mathbb{C}^g/\Lambda$ for a lattice Λ of rank $2g$—the period lattice of the curve. For $g = 1$, the Jacobian is isomorphic to the curve itself, and this is the source of the group law on elliptic curves.

7.1.9 Arithmetic Motif: Covers of Spec $\mathbb{Z}$ Let us pause to observe the arithmetic counterpart of what we have just done. Consider the ring of Gaussian integers $\mathbb{Z}[i]$ and the natural inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$. This induces a finite morphism of schemes:

$$\pi\colon \operatorname{Spec} \mathbb{Z}[i] \longrightarrow \operatorname{Spec} \mathbb{Z}$$

of degree $[\mathbb{Q}(i) : \mathbb{Q}] = 2$. It is a "two-sheeted cover" of Spec $\mathbb{Z}$, and just as for curves, the fibers over closed points exhibit exactly three behaviours familiar from algebraic number theory:

1. **Split:** Over $p \equiv 1 \pmod 4$ (e.g. $p = 5$), we have $p = (2+i)(2-i)$, so the fiber $\pi^{-1}([(p)])$ consists of two points—the primes $(2+i)$ and $(2-i)$—each with ramification index 1.
2. **Inert:** Over $p \equiv 3 \pmod 4$ (e.g. $p = 3$), the ideal (3) remains prime in $\mathbb{Z}[i]$, so the fiber is a single point with residue field $\mathbb{F}_9$. Here the "ramification index" is 1 but the residue field degree is 2.
3. **Ramified:** Over $p = 2$, we have $(1+i)^2 = 2i$, so $2 = -i(1+i)^2$. The fiber is a single point $(1+i)$ with ramification index 2.

In all cases, the "sum of ramification indices times residue field degrees" over each point equals $\deg \pi = 2$, just as in proposition 7.1.7. This is the number-theoretic incarnation of the degree formula. When we reach Hurwitz's theorem (§7.1.4), the reader should think of it as the "genus formula" for branched covers of Spec $\mathbb{Z}$—where the genus of Spec $\mathbb{Z}$ is arguably 0, and the "ramification" is measured by the discriminant. See [58] for the full analogy.

7.1.2 The Riemann–Roch Theorem

If there is one theorem that captures the essence of the theory of algebraic curves, it is Riemann–Roch. It is the theorem that converts the abstract cohomological machinery we have built into concrete geometric information: it counts sections, produces embeddings, and constrains the topology of morphisms. Virtually every result in this chapter flows from it.

The idea, in a sentence, is this: for a divisor D on a curve X of genus g, the Riemann–Roch theorem computes the coherent Euler characteristic $\chi(\mathcal{L}(D)) = \dim_k H^0(X, \mathcal{L}(D)) - \dim_k H^1(X, \mathcal{L}(D))$ in terms of the degree of D and the genus g. Geometrically, one should think of this Euler characteristic as the "expected" number of independent global sections of $\mathcal{L}(D)$, adjusted by an obstruction term (H^1) dictated by the global geometry of the curve. Combining this with Serre duality, which relates $H^1(X, \mathcal{L}(D))$ to $H^0(X, \omega_X \otimes \mathcal{L}(D)^\vee)$, we obtain a formula that balances sections of $\mathcal{L}(D)$ against sections of its "Serre dual." This balance is what makes the theorem so powerful.

Divisors on Curves: Degree, Linear Systems, and $\ell(D)$

Let X be a complete nonsingular curve over $k = \overline{k}$. We briefly consolidate the divisor theory from §5.4 in the particularly clean form it takes on curves.

Since X is a regular scheme of dimension 1, every local ring $\mathcal{O}_{X,x}$ at a closed point is a DVR, hence a UFD. In particular, X is locally factorial, and the natural map from Cartier divisors to Weil divisors is an isomorphism (proposition 5.4.31 and corollary 5.4.56):

$$\operatorname{CaCl} X \;\cong\; \operatorname{Cl} X \;\cong\; \operatorname{Pic} X.$$

Note that for a singular curve, the natural map $\operatorname{CaCl} X \to \operatorname{Cl} X$ (equivalently, $\operatorname{Pic} X \to \operatorname{Cl} X$) is *injective* (more generally it is always injective for integral schemes that are regular in codimension 1, definition 5.4.1 and proposition 5.4.31), but it need not be *surjective*. A Weil divisor supported at a singular point may fail to be locally principal, and hence may not correspond to any line bundle at all. On singular curves, *Cartier divisors* (equivalently, line bundles, elements of $\operatorname{Pic} X$) remain the correct objects for Riemann–Roch and for defining morphisms to projective space. Weil divisors are a coarser invariant that may contain extra classes not represented by line bundles. The reader should keep this distinction in mind whenever the nonsingularity hypothesis is relaxed. In this section, we continue with the assumption that all curves are nonsingular.

We write divisors additively as $D = \sum_{x \in X_{\mathrm{cl}}} n_x \cdot x$ where all but finitely many n_x are zero. The *degree* of D is $\deg D = \sum n_x$. For each divisor D, we have the associated invertible sheaf $\mathcal{L}(D)$, with the key property that for D, D' divisors:

$$\mathcal{L}(D + D') \cong \mathcal{L}(D) \otimes \mathcal{L}(D'), \qquad \mathcal{L}(-D) \cong \mathcal{L}(D)^{\vee},$$
$$\mathcal{L}(D) \cong \mathcal{O}_X \iff D \sim 0 \quad \text{(linearly equivalent to zero)}.$$

Recall that D is *effective*, written $D \geq 0$, if all $n_x \geq 0$. A global section $s \in H^0(X, \mathcal{L}(D))$ corresponds to a rational function $f \in K(X)^*$ with $\operatorname{div}(f) + D \geq 0$, together with the zero section.

Definition 7.1.10: Complete Linear System

The *complete linear system* of D is the set of effective divisors linearly equivalent to D:

$$|D| \;=\; \{D' \geq 0 : D' \sim D\} \;=\; \{\operatorname{div}(f) + D : f \in K(X)^*,\ \operatorname{div}(f) + D \geq 0\}.$$

There is a natural bijection $|D| \cong \mathbb{P}\big(H^0(X, \mathcal{L}(D))\big)$: the effective divisors in the class of D are parametrized by the projectivization of the space of global sections of $\mathcal{L}(D)$, since two sections $s, s' \in H^0(X, \mathcal{L}(D)) \setminus \{0\}$ define the same effective divisor if and only if $s' = \lambda s$ for some $\lambda \in k^*$.

We introduce the classical notation:

$$\ell(D) \;:=\; \dim_k H^0(X, \mathcal{L}(D)).$$

Thus $|D|$ is a projective space of dimension $\ell(D) - 1$ (and $|D| = \emptyset$ precisely when $\ell(D) = 0$). The number $\ell(D)$ is finite by theorem 5.3.27[8]. A *linear system* is any linear subspace of $|D|$, and a linear system of dimension 1 is called a *pencil*.

The following lemma is elementary but crucial; it says that divisors with sections are "non-negative" in a precise sense.

[8] One can also see this from the finiteness of cohomology on projective schemes, corollary 6.6.7.

Lemma 7.1.11: Divisors and Degree

Let D be a divisor on a complete nonsingular curve X.

1. If $\ell(D) > 0$, then $\deg D \geq 0$.
2. If $\ell(D) > 0$ and $\deg D = 0$, then $D \sim 0$ (i.e. $\mathcal{L}(D) \cong \mathcal{O}_X$).

***Proof*:**
If $\ell(D) > 0$, then $|D| \neq \emptyset$, so $D \sim D'$ for some effective $D' \geq 0$. Since $\deg D = \deg D' = \sum n_x \geq 0$ (with $n_x \geq 0$ for all x), we get (1). For (2), if additionally $\deg D = 0$, then $D' = \sum n_x \cdot x$ with all $n_x \geq 0$ and $\sum n_x = 0$, forcing $n_x = 0$ for all x, i.e. $D' = 0$, so $D \sim 0$.

The Canonical Divisor and Genus

Recall from §5.5 that the sheaf of relative differentials $\Omega_{X/k}$ on a smooth k-scheme X of dimension n is a locally free sheaf of rank n (corollary 5.5.7). When X is a smooth curve, $n = 1$, so $\Omega_{X/k}$ is an *invertible sheaf*—it is a line bundle on X.

By the definition of the canonical sheaf (definition 5.5.25), $\omega_X = \bigwedge^{\dim X} \Omega_{X/k}$. For a curve, this is simply $\omega_X = \Omega_{X/k}$. On the other hand, the dualizing sheaf in the sense of Serre duality (definition 6.5.1) is also ω_X for a smooth projective variety. These two coincide for smooth curves, which is the reason Serre duality is so immediately applicable.

Definition 7.1.12: Canonical Divisor

Let X be a complete nonsingular curve. A *canonical divisor* K (or K_X) is any divisor in the linear equivalence class corresponding to $\omega_X = \Omega_{X/k}$, i.e. $\mathcal{L}(K) \cong \omega_X$.

The canonical divisor is defined only up to linear equivalence, but its degree and the dimension $\ell(K)$ are well-defined invariants of X. We now define the fundamental discrete invariant of a curve.

Definition 7.1.13: Genus of a Curve

Let X be a complete nonsingular curve over k. The following three numbers are defined:

1. The *arithmetic genus*: $p_a(X) = 1 - P_X(0)$, where P_X is the Hilbert polynomial of X with respect to any very ample sheaf (definition 6.7.6).
2. The *geometric genus*: $p_g(X) = \dim_k H^0(X, \omega_X)$ (definition 5.5.26).
3. The *cohomological genus*: $\dim_k H^1(X, \mathcal{O}_X)$.

As we prove below, these three numbers coincide, and their common value is called simply the *genus* of X, denoted g (or $g(X)$).

Proposition 7.1.14: Equivalence of Genus

Let X be a complete nonsingular curve over $k = \overline{k}$. Then:

$$p_a(X) \;=\; p_g(X) \;=\; \dim_k H^1(X, \mathcal{O}_X).$$

***Proof*:**
Step 1: $p_a = \dim_k H^1(X, \mathcal{O}_X)$. By definition, $p_a(X) = 1 - P_X(0)$ where $P_X(n) = \chi(\mathcal{O}_X(n))$ is the Hilbert polynomial (theorem 6.7.5 and definition 6.7.6). Setting $n = 0$:

$$P_X(0) = \chi(\mathcal{O}_X) = \dim_k H^0(X, \mathcal{O}_X) - \dim_k H^1(X, \mathcal{O}_X).$$

Since X is a projective integral scheme over $k = \overline{k}$, the global sections $H^0(X, \mathcal{O}_X) = k$ (theorem 3.5.27), so $P_X(0) = 1 - \dim_k H^1(X, \mathcal{O}_X)$, and $p_a = 1 - P_X(0) = \dim_k H^1(X, \mathcal{O}_X)$.

Note also that $H^i(X, \mathcal{O}_X) = 0$ for $i \geq 2$ since $\dim X = 1$ (theorem 6.2.5), so the Euler characteristic of $\mathcal{O}_X$ is just $1 - g$ where $g = \dim_k H^1(X, \mathcal{O}_X)$.

Step 2: $\dim_k H^1(X, \mathcal{O}_X) = p_g$. Serre duality for a smooth projective variety X of dimension n over k gives a perfect pairing (theorem 6.5.9):

$$H^i(X, \mathscr{F}) \times H^{n-i}(X, \omega_X \otimes \mathscr{F}^\vee) \to H^n(X, \omega_X) \cong k.$$

Taking $n = 1$, $i = 1$, $\mathscr{F} = \mathcal{O}_X$:

$$H^1(X, \mathcal{O}_X) \;\cong\; H^0(X, \omega_X)^\vee.$$

Therefore $\dim_k H^1(X, \mathcal{O}_X) = \dim_k H^0(X, \omega_X) = p_g(X)$.

From now on, we write $g = g(X)$ for the genus. Since $g = \dim_k H^1(X, \mathcal{O}_X) \geq 0$, the genus is a non-negative integer.

The following formula is presented here for readers who have seen the equivalent in complex geometry (ex. in [52]:

Proposition 7.1.15: Degree Formula for Plane Curves

Let $C \subseteq \mathbb{P}^2_k$ be a smooth projective curve of degree d. Then the genus of C is given by:

$$g(C) \;=\; \frac{(d-1)(d-2)}{2}.$$

Note that the assumption that $C \subseteq \mathbb{P}^2_k$ is restrictive but necessary: the twisted cubic will be an example of a curve that is degree 3 that naturally embeds in $\mathbb{P}^3_k$ but is rational. This result will generalize to an inequality for $\mathbb{P}^3_k$ (the universal embedding space for all curves) in theorem 7.1.63.

***Proof*:**
Since C is a closed subscheme of $\mathbb{P}^2$ defined by a single homogeneous polynomial of degree d, its ideal sheaf is $\mathcal{I}_C \cong \mathcal{O}_{\mathbb{P}^2}(-d)$. This gives the short exact sequence of sheaves on $\mathbb{P}^2$:

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^2} \longrightarrow \mathcal{O}_C \longrightarrow 0.$$

The Euler characteristic $\chi(\mathscr{F}) = \sum_i (-1)^i \dim_k H^i(X, \mathscr{F})$ is additive on short exact sequences (proposition 6.7.2), so we have:

$$\chi(\mathcal{O}_C) \;=\; \chi(\mathcal{O}_{\mathbb{P}^2}) - \chi(\mathcal{O}_{\mathbb{P}^2}(-d)).$$

From the cohomology of projective space (theorem 6.3.1), we know that the Hilbert polynomial of $\mathbb{P}^2$ is $\chi(\mathcal{O}_{\mathbb{P}^2}(n)) = \frac{(n+1)(n+2)}{2}$ for all $n \in \mathbb{Z}$. Evaluating this at $n = 0$ and $n = -d$ yields:

$$\begin{aligned} \chi(\mathcal{O}_{\mathbb{P}^2}) &= 1, \\ \chi(\mathcal{O}_{\mathbb{P}^2}(-d)) &= \frac{(-d+1)(-d+2)}{2} \;=\; \frac{(d-1)(d-2)}{2}. \end{aligned}$$

Therefore, $\chi(\mathcal{O}_C) = 1 - \frac{(d-1)(d-2)}{2}$.

On the other hand, since C is a complete nonsingular curve, proposition 7.1.14 establishes that the genus g equals the arithmetic genus $p_a(C) = 1 - \chi(\mathcal{O}_C)$. Thus:

$$1 - g \;=\; 1 - \frac{(d-1)(d-2)}{2} \qquad\Longrightarrow\qquad g \;=\; \frac{(d-1)(d-2)}{2}.$$

Example 7.1.16: Genus Computations

1. *The projective line.* $g(\mathbb{P}^1_k) = 0$. Indeed, $H^1(\mathbb{P}^1_k, \mathcal{O}_{\mathbb{P}^1}) = 0$ by the computation of cohomology of projective space (theorem 6.3.1).

2. *Smooth plane curves.* Let $C \subseteq \mathbb{P}^2_k$ be a smooth curve of degree d. By the degree formula (proposition 7.1.15):
$$g(C) \;=\; \frac{(d-1)(d-2)}{2}.$$
This gives $g = 0$ for $d = 1, 2$ (lines and conics—both rational), $g = 1$ for $d = 3$ (elliptic curves), $g = 3$ for $d = 4$ (the first non-hyperelliptic curves), and so on. Notice the rapid growth: the genus of a plane curve of degree d is quadratic in d.

   Note: This also allows us to verify the degree of the canonical divisor for plane curves before proving it generally. By the adjunction formula (proposition 6.5.7), the canonical sheaf is $\omega_C \cong (\omega_{\mathbb{P}^2} \otimes \mathcal{O}_{\mathbb{P}^2}(C))|_C \cong \mathcal{O}_C(d-3)$. The degree of restricting $\mathcal{O}(d-3)$ to a degree d curve is $d(d-3)$. Notice that $2g - 2 = 2\left(\frac{d^2-3d+2}{2}\right) - 2 = d(d-3)$, which perfectly predicts the general formula $\deg K_X = 2g - 2$.

3. *The degree of the canonical divisor.* By Riemann–Roch (which we will prove momentarily), one can show that for *any* smooth curve of genus g:
$$\deg K_X = 2g - 2.$$
For $\mathbb{P}^1$ this gives $\deg K = -2$, consistent with $\omega_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(-2)$ (from the Euler sequence theorem 5.5.18). For a smooth plane cubic, $\deg K = 0$, consistent with $\omega_C \cong \mathcal{O}_C$—a fact!

7.1.17 Forward Reference: The Moduli Space $\mathcal{M}_g$ We have just seen that smooth projective curves over k are classified up to isomorphism by a discrete invariant $g \geq 0$—but this classification is very coarse. For a fixed genus g, there is typically a *continuous family* of non-isomorphic curves.

How large is this family? By a deformation-theoretic argument (made precise in *Moduli, Deformation Theory, and Stacks* [17]), the "moduli space" parametrizing smooth curves of genus g has the following dimensions:

$$\dim \mathcal{M}_g = \begin{cases} 0 & \text{if } g = 0 \quad (\text{only } \mathbb{P}^1), \\ 1 & \text{if } g = 1 \quad (\text{the } j\text{-line}), \\ 3g - 3 & \text{if } g \geq 2. \end{cases}$$

Making this precise requires the theory of *stacks*: the space $\mathcal{M}_g$ is not a scheme (because curves have automorphisms), but a Deligne–Mumford stack. Its construction is one of the triumphs of modern algebraic geometry, developed in *Moduli, Deformation Theory, and Stacks* [17]. For now, the reader should simply note that the genus is the "first" invariant of a curve, but far from the "last"—within each genus lies a rich geometric world.

7.1.18 Warning: The Genera Diverge for Singular Curves The equality $p_a = p_g = \dim_k H^1(X, \mathcal{O}_X)$ is a luxury of nonsingularity. For a complete integral curve X that is singular, these invariants come apart in an important and controlled way.

The *geometric genus* $g(\widetilde{X})$ is the genus of the normalization—it is a birational invariant, blind to the singularities. The *arithmetic genus* $p_a(X) = 1-\chi(\mathcal{O}_X)$ depends on the scheme structure of X, singularities included, and is the invariant that remains constant in flat families (theorem 6.7.5). They are related by the δ*-invariant*: for each singular point $P \in X$, set

$$\delta_P \;=\; \dim_k\Big(\widetilde{\mathcal{O}}_{X,P} \,/\, \mathcal{O}_{X,P}\Big)$$

where $\widetilde{\mathcal{O}}_{X,P}$ denotes the integral closure of $\mathcal{O}_{X,P}$ in $K(X)$ (equivalently, the stalk $(\nu_*\mathcal{O}_{\widetilde{X}})_P$ of the pushforward of the structure sheaf along the normalization map ν). Then:

$$p_a(X) \;=\; g(\widetilde{X}) \;+\; \sum_{P \in X_{\mathrm{sing}}} \delta_P.$$

The δ-invariant measures the "severity" of a singularity: a node (two branches crossing transversally) has $\delta = 1$, and an ordinary cusp has $\delta = 1$ as well[a].

For example, both the nodal cubic $y^2 = x^3 + x^2$ and the cuspidal cubic $y^2 = x^3$ are degree-3 plane curves, so $p_a = \frac{(3-1)(3-2)}{2} = 1$ (example 7.1.16). Yet both have normalization $\mathbb{P}^1$, giving $g(\widetilde{X}) = 0$. The discrepancy $p_a - g(\widetilde{X}) = 1$ is accounted for by the single singular point with $\delta = 1$.

This distinction matters for moduli: when a smooth elliptic curve ($g = 1$) degenerates in a flat family to a nodal rational curve, the arithmetic genus $p_a = 1$ is preserved—as it must be, by flatness—but the geometric genus drops to 0. The "missing genus" is absorbed into the singularity. This phenomenon is the reason that the compactification $\overline{\mathcal{M}}_g$ parametrizes *stable* (i.e. nodal) curves rather than smooth ones; see box 7.1.66.

[a] More exotic singularities can have larger δ: for instance, the curve $y^2 = x^{2k+1}$ has a singularity with $\delta = k$ at the origin. The δ-invariant also equals the *colength* of $\mathcal{O}_{X,P}$ in its normalization, which in the analytic picture over $\mathbb{C}$ counts the number of "missing points" when X is compared to $\widetilde{X}$ as a topological space.

Statement and Proof of Riemann–Roch

Recall that Serre duality (theorem 6.5.9) gives, for any invertible sheaf $\mathcal{L}$ on a smooth projective curve X of genus g:

$$H^1(X, \mathcal{L}) \;\cong\; H^0(X, \omega_X \otimes \mathcal{L}^\vee)^\vee.$$

In divisor language, if $\mathcal{L} = \mathcal{L}(D)$, then $\omega_X \otimes \mathcal{L}(D)^\vee \cong \mathcal{L}(K - D)$, and hence:

$$\dim_k H^1(X, \mathcal{L}(D)) \;=\; \ell(K - D).$$

This identification transforms the Euler characteristic $\chi(\mathcal{L}(D)) = \ell(D) - \ell(K - D)$ into a quantity involving only dimensions of spaces of global sections.

Theorem 7.1.19: Riemann–Roch Theorem

Let X be a complete nonsingular curve of genus g over $k = \overline{k}$, and let K be a canonical divisor. For any divisor D on X:

$$\ell(D) - \ell(K - D) \;=\; \deg D + 1 - g.$$

Equivalently, in sheaf-cohomological language:

$$\chi(\mathcal{L}(D)) \;=\; \deg D + 1 - g.$$

Before giving the proof, let us appreciate what this theorem says. The right-hand side is *topological*: it depends only on the degree (an integer) and the genus (a topological invariant of the underlying Riemann surface, when $k = \mathbb{C}$). The left-hand side is *algebraic*: it involves the dimensions of spaces of rational functions with prescribed poles. The theorem says that these two kinds of data are linked by a single equation.

The term $\ell(K - D)$ is sometimes called the *index of speciality* of D and is denoted $i(D) := \ell(K - D)$. A divisor with $i(D) = 0$ is called *non-special*; for such divisors, Riemann–Roch simply says $\ell(D) = \deg D + 1 - g$. We shall see that "most" divisors of sufficiently large degree are non-special.

***Proof*:**
We prove the cohomological form $\chi(\mathcal{L}(D)) = \deg D + 1 - g$; the classical form follows immediately from Serre duality as explained above.

Base case: $D = 0$. We have $\chi(\mathcal{O}_X) = \dim_k H^0(X, \mathcal{O}_X) - \dim_k H^1(X, \mathcal{O}_X) = 1 - g$, since $H^0(X, \mathcal{O}_X) = k$ (theorem 3.5.27) and $g = \dim_k H^1(X, \mathcal{O}_X)$ (proposition 7.1.14). This equals $\deg 0 + 1 - g = 1 - g$.

Induction step. We show: the formula holds for D if and only if it holds for $D + P$, where P is any closed point of X.

Consider P as a closed subscheme of X, defined by the ideal sheaf $\mathcal{L}(-P)$ (a single effective Cartier divisor). The structure sheaf $\mathcal{O}_P$ is a skyscraper sheaf supported at P with stalk k, which we denote k_P. This gives the short exact sequence:

$$0 \to \mathcal{L}(-P) \to \mathcal{O}_X \to k_P \to 0.$$

Tensoring with the invertible sheaf $\mathcal{L}(D + P)$ (which is locally free of rank 1, hence flat, hence tensoring preserves exactness):

$$0 \to \mathcal{L}(-P) \otimes \mathcal{L}(D + P) \to \mathcal{L}(D + P) \to k_P \otimes \mathcal{L}(D + P) \to 0.$$

Now $\mathcal{L}(-P) \otimes \mathcal{L}(D + P) \cong \mathcal{L}(D)$, and $k_P \otimes \mathcal{L}(D + P) \cong k_P$ (tensoring a skyscraper sheaf with an invertible sheaf does not change its stalk—the invertible sheaf is locally isomorphic to $\mathcal{O}_X$ near P). So we have:

$$0 \to \mathcal{L}(D) \to \mathcal{L}(D + P) \to k_P \to 0. \tag{7.2}$$

Taking Euler characteristics and using additivity (proposition 6.7.2):

$$\chi(\mathcal{L}(D + P)) = \chi(\mathcal{L}(D)) + \chi(k_P).$$

Since k_P is a skyscraper sheaf supported at a single point, $H^0(X, k_P) = k$ and $H^i(X, k_P) = 0$ for $i > 0$, so $\chi(k_P) = 1$. Therefore:

$$\chi(\mathcal{L}(D+P)) \;=\; \chi(\mathcal{L}(D)) + 1.$$

On the other hand, $\deg(D+P) = \deg D + 1$. So the right-hand side of the Riemann–Roch formula also increases by 1 when we replace D by $D+P$. This proves the claim: Riemann–Roch holds for D if and only if it holds for $D+P$.

Thus, every divisor D can be reached from the zero divisor by finitely many steps of adding or subtracting a point[a]. By the induction step, the formula holds for D if and only if it holds for 0. Since we verified it for $D = 0$, it holds for all D, as we sought to show.

[a] Write $D = \sum n_i P_i = \sum_{n_i>0} n_i P_i - \sum_{n_j<0} |n_j| P_j$. Starting from 0, add the positive points and subtract the negative points one at a time.

The proof is simple—almost suspiciously so. The heavy lifting is hidden in two ingredients from earlier chapters: Serre duality (which required the machinery of Ext groups, the local-to-global spectral sequence, and residues) and the additivity of the Euler characteristic (which required the long exact sequence in cohomology). What the proof really shows is that the Euler characteristic $D \mapsto \chi(\mathcal{L}(D))$ is an additive function of divisors that agrees with $D \mapsto \deg D + 1 - g$ at $D = 0$ and has the same “slope.” Since both sides are “discrete-linear” in D, they must agree everywhere.

7.1.20 Riemann–Roch as an Index Theorem The Riemann–Roch theorem is the simplest instance of an *index theorem*. In mathematics, an index theorem equates an analytical index (which counts the solutions to a differential equation, minus obstructions) to a purely topological invariant. In the analytic world (over $\mathbb{C}$), $\chi(\mathcal{L}) = h^0 - h^1$ computes the Fredholm index of the $\bar{\partial}$-operator (i.e. Cauchy-Riemann equations as a differential equation) on a line bundle —an analytic result that Riemann–Roch equates to the topology of the curve ($\deg D + 1 - g$). This theme is taken up in vastly greater generality in *Intersection Theory* [16]:

1. The *Hirzebruch–Riemann–Roch theorem* computes $\chi(\mathscr{E})$ for any vector bundle $\mathscr{E}$ on a smooth projective variety X in terms of the Chern character $\mathrm{ch}(\mathscr{E})$ and the Todd class $\mathrm{td}(T_X)$: $\chi(\mathscr{E}) = \int_X \mathrm{ch}(\mathscr{E}) \cdot \mathrm{td}(T_X)$.
2. The *Grothendieck–Riemann–Roch theorem* (GRR) generalizes this to a relative setting: for a proper morphism $f\colon X \to Y$, it computes the Chern character of the higher direct images $R^i f_* \mathscr{E}$ in terms of $\mathrm{ch}(\mathscr{E})$ and $\mathrm{td}(T_{X/Y})$.

For a curve, the Chern character of a line bundle $\mathcal{L}(D)$ is simply $1 + \deg D$, and the Todd class of T_X is $1 + \frac{1}{2}c_1(T_X) = 1 + (1-g)$, and HRR reduces exactly to $\chi(\mathcal{L}(D)) = \deg D + 1 - g$. The reader who is intrigued by the Riemann–Roch theorem should look forward to GRR in *Intersection Theory* [16].

Example 7.1.21: Immediate Application – Elliptic Curves

Let $E \subseteq \mathbb{P}^2_k$ be a nonsingular cubic $y^2z = x^3 - xz^2$ with $\mathrm{char}(k) \neq 2$. This curve is not rational[a]. Let $P_0 = [0:1:0]$, the unique point at infinity, which is an inflection point: the tangent line $\{z = 0\}$ meets E with multiplicity 3, giving the divisor relation $3P_0 \sim$ (hyperplane section).

Consider the kernel $K = \ker(\deg\colon \mathrm{Cl}\, E \to \mathbb{Z})$. Define a map φ from the set of k-rational points

$E(k)$ to K by $\varphi(P) = [P - P_0]$. This map is *injective*: if $[P - P_0] = [Q - P_0]$, then $P \sim Q$. But by Riemann–Roch, $\ell(P) = \deg(P) + 1 - g = 1 + 1 - 1 = 1$. The complete linear system $|P|$ has dimension $\ell(P) - 1 = 0$, meaning it consists of exactly one effective divisor, namely P itself. Thus $P = Q$.

To see *surjectivity*, take any $D \in K$, so $\deg D = 0$. By the Riemann–Roch theorem, $\ell(D + P_0) = \deg(D + P_0) + 1 - g = 1 + 1 - 1 = 1$. Thus, there exists a unique effective divisor of degree 1—which must be a single k-rational point P—such that $D + P_0 \sim P$, meaning $D \sim P - P_0$.

Geometrically, if D is supported entirely on k-rational points, one can explicitly see this by writing $D = \sum n_i(P_i - P_0)$ and using the chord-tangent process: if a line through P and Q meets E in a third point R, then $P + Q + R \sim 3P_0$, so $[P - P_0] + [Q - P_0] = [T - P_0]$ where T is the third intersection of the line P_0R with E. This iteratively reduces any such sum to $D = [P - P_0]$ for some $P \in E(k)$.

Thus φ is a bijection, and the group structure on K transfers to a group structure on $E(k)$, with identity P_0. One checks that the addition and inversion maps $+\colon E \times E \to E$ and $[-1]\colon E \to E$ are morphisms of schemes, making E a *group scheme* over k (cf. example 4.1.12). We will develop this more thoroughly in §7.1.7.

More generally, if X is any complete nonsingular curve, then $K = \mathrm{Cl}^0 X$ is the group of k-points of an abelian variety called the *Jacobian variety* $\mathrm{Jac}(X)$, and $\mathrm{Cl}\, X$ sits in a short exact sequence:

$$0 \to \mathrm{Jac}(X)(k) \to \mathrm{Cl}\, X \xrightarrow{\deg} \mathbb{Z} \to 0.$$

For surfaces and higher-dimensional varieties, the divisor class group has a richer structure: the quotient $\mathrm{Cl}\, X/\mathrm{Cl}^0 X$ becomes the *Néron–Severi group* $\mathrm{NS}(X)$, which is a finitely generated abelian group (but generally of rank > 1). This generalization will be central to the theory of surfaces in §7.2.1.

[a] We saw this in [57, chapter 21.3]. Alternatively, we will see shortly that E has genus 1, while $\mathbb{P}^1_k$ has genus 0; since the genus is a birational invariant, they cannot be isomorphic.

A quick word should be mentioned for the singular case: the Riemann–Roch theorem extends to singular curves after replacing the sheaf of Kähler differentials $\Omega_{X/k}$ with the correct substitute. On a smooth curve, $\Omega_{X/k}$ is an invertible sheaf and simultaneously serves as the dualizing sheaf for Serre duality. On a singular curve X, both of these properties fail: $\Omega_{X/k}$ acquires *torsion* at the singular points (reflecting the fact that Kähler differentials detect the non-reducedness of the diagonal at singularities), and it is no longer locally free[9]. In particular, $\Omega_{X/k}$ cannot play the role of the dualizing sheaf in Serre duality.

The correct replacement is the *dualizing sheaf* ω_X° in the sense of Serre duality (definition 6.5.1), which for a complete integral curve over $k = \overline{k}$ can be described concretely as the sheaf of *regular differentials* in the sense of Rosenlicht: meromorphic differentials η on the normalization $\widetilde{X}$ satisfying a residue condition at each preimage of a singular point[10]. On a smooth curve, $\omega_X^\circ \cong \Omega_{X/k}$, so there

[9] For a concrete example, on the cuspidal cubic $X = \operatorname{Spec} k[x, y]/(y^2 - x^3)$, the module of differentials is $\Omega_{X/k} = (A\,dx \oplus A\,dy)/(3x^2\,dx - 2y\,dy)$. Consider the differential $\tau = 2x\,dy - 3y\,dx$. It is a non-zero element, but $y \cdot \tau = 2xy\,dy - 3y^2\,dx = x(3x^2\,dx) - 3x^3\,dx = 0$ in $\Omega_{X/k}$ (and similarly $x \cdot \tau = 0$). Thus τ generates a torsion submodule supported precisely at the cusp $(0, 0)$, which prevents $\Omega_{X/k}$ from being locally free.

[10] More precisely, if $\nu\colon \widetilde{X} \to X$ is the normalization and $P \in X$ is a singular point with preimages $Q_1, \dots, Q_r$, then a meromorphic 1-form η on $\widetilde{X}$ is a *regular differential* at P if $\sum_{i=1}^r \mathrm{Res}_{Q_i}(f \cdot \eta) = 0$ for every $f \in \mathcal{O}_{X,P}$. The sheaf ω_X° is the subsheaf of $\nu_*\omega_{\widetilde{X}}(\text{poles})$ consisting of such forms. See [33, Chapter IV.2] and [43, Chapter 8] for full details.

is nothing new in the nonsingular case. On a singular curve, ω_X° is still a rank-1 torsion-free sheaf (though not necessarily invertible), and the key miracle is that Serre duality (theorem 6.5.9) continues to hold:

$$H^1(X, \mathscr{F}) \;\cong\; \mathrm{Hom}_{\mathcal{O}_X}(\mathscr{F}, \omega_X^\circ)^\vee$$

for any coherent sheaf $\mathscr{F}$ on X.

With this dualizing sheaf in hand, Riemann–Roch for a *complete integral* (possibly singular) curve reads:

for any line bundle $\mathcal{L}$ on X,

$$\chi(\mathcal{L}) \;=\; \deg \mathcal{L} + 1 - p_a(X)$$

where $p_a(X) = 1 - \chi(\mathcal{O}_X)$ is the arithmetic genus*.*

The line bundle can be More generally any rank-1 torsion-free sheaf $\mathcal{L}$; on a singular curve, these play the role that invertible sheaves play on smooth curves. The proof is identical to the smooth case: additivity of the Euler characteristic plus induction on twisting by a point. What changes is the *interpretation*: the genus that appears is p_a, not the geometric genus $g(\widetilde{X})$, and the duality term $\ell(K - D)$ must be understood via ω_X° rather than $\Omega_{X/k}$.

Due to the cohomological machinery we built up, the Euler characteristic formula $\chi(\mathcal{L}) = \deg \mathcal{L} + 1 - p_a$ works *verbatim* on any complete integral curve, smooth or not. It is only when one wants to split $\chi(\mathcal{L})$ into $h^0 - h^1$ and apply Serre duality to control each piece separately that the distinction between ω_X° and $\Omega_{X/k}$ becomes essential.

7.1.3 Applications of Riemann–Roch

Proposition 7.1.22: Degree of the Canonical Divisor

Let X be a complete nonsingular curve of genus g. Then:

$$\deg K_X \;=\; 2g - 2, \qquad \ell(K_X) = g.$$

***Proof*:**
Apply Riemann–Roch with $D = K$:

$$\ell(K) - \ell(K - K) = \deg K + 1 - g.$$

Now $\ell(K - K) = \ell(0) = \dim_k H^0(X, \mathcal{O}_X) = 1$, and $\ell(K) = p_g = g$ by proposition 7.1.14. So $g - 1 = \deg K + 1 - g$, giving $\deg K = 2g - 2$.

This formula is extremely satisfying: over $\mathbb{C}$, the degree of K_X is $2g - 2$, which is also the topological Euler characteristic of the Riemann surface with the *opposite* sign (recall $\chi_{\text{top}} = 2 - 2g$). This is the Gauss–Bonnet theorem.

Proposition 7.1.23: Non-Speciality for Large Degree

Let D be a divisor on a curve X of genus g. If $\deg D > 2g-2$, then $\ell(K-D) = 0$, and Riemann–Roch gives:

$$\ell(D) \;=\; \deg D + 1 - g.$$

In particular, $\ell(D) > 0$ for any divisor of degree $\geq g$.

***Proof*:**
If $\ell(K-D) > 0$, then by lemma 7.1.11, $\deg(K-D) \geq 0$, i.e. $\deg D \leq \deg K = 2g-2$. Contrapositive: if $\deg D > 2g-2$, then $\ell(K-D) = 0$.

For the "in particular": if $\deg D \geq g$, then $\ell(D) = \deg D + 1 - g \geq 1$ when D is non-special. If $\deg D \leq 2g-2$, we cannot guarantee non-speciality, but the Riemann–Roch formula still gives $\ell(D) \geq \deg D + 1 - g$, which is ≥ 1 whenever $\deg D \geq g$ (even without non-speciality, since $\ell(K-D) \geq 0$).

One of the most important applications of Riemann–Roch is the construction of embeddings of curves into projective space. Recall (definitions 5.3.21 and 5.3.23) that an invertible sheaf $\mathcal{L}$ on X is:

1. *generated by global sections* (or *base-point-free*) if the natural map $H^0(X,\mathcal{L}) \otimes_k \mathcal{O}_X \to \mathcal{L}$ is surjective,
2. *very ample* if it defines a closed immersion $\varphi_{\mathcal{L}} \colon X \hookrightarrow \mathbb{P}^N_k$ where $N+1 = \dim_k H^0(X,\mathcal{L})$.

For a line bundle $\mathcal{L}(D)$ on a curve, base-point-freeness means that for every point $P \in X$, there exists a section $s \in H^0(X,\mathcal{L}(D))$ with $s(P) \neq 0$; equivalently, $\ell(D-P) = \ell(D) - 1$ for every P. Very ampleness is a stronger condition: it requires that for every pair of (possibly equal) points $P, Q \in X$, we have $\ell(D-P-Q) = \ell(D) - 2$, i.e. sections separate points and tangent vectors[11].

Proposition 7.1.24: Base-Point-Free Criterion for Curves

Let D be a divisor on a complete nonsingular curve X of genus g. If $\deg D \geq 2g$, then $\mathcal{L}(D)$ is base-point-free.

***Proof*:**
Let $P \in X$ be any closed point. We need $\ell(D) > \ell(D-P)$, i.e. there exists a section of $\mathcal{L}(D)$ not vanishing at P. By Riemann–Roch:

$$\ell(D) = \deg D + 1 - g + \ell(K-D),$$
$$\ell(D-P) = \deg D - 1 + 1 - g + \ell(K-D+P).$$

So $\ell(D) - \ell(D-P) = 1 + \ell(K-D) - \ell(K-D+P)$. It suffices to show $\ell(K-D) = \ell(K-D+P)$. Since $\deg D \geq 2g$, we have $\deg(K-D) = 2g-2-\deg D \leq -2 < 0$, so $\ell(K-D) = 0$ by lemma 7.1.11. Also $\deg(K-D+P) = -2+1 = -1$ if $\deg D = 2g$ (or even more negative if $\deg D > 2g$), so $\ell(K-D+P) = 0$ as well. Therefore $\ell(D) - \ell(D-P) = 1 > 0$.

[11] The condition at $P = Q$ is the tangent vector condition: sections of $\mathcal{L}(D)$ must separate tangent directions at P, which means $\ell(D) - \ell(D-2P) = 2$.

Using this, completeness and projectivity coincide for nonsingular curves.

Corollary 7.1.25: Very Ample Criterion for Curves

If $\deg D \geq 2g+1$, then $\mathcal{L}(D)$ is very ample, and $\varphi_{|D|}$ embeds X as a closed subvariety of $\mathbb{P}_k^{\ell(D)-1}$.

***Proof*:**
We must show that for any two closed points $P, Q \in X$ (possibly equal), $\ell(D) - \ell(D-P-Q) = 2$. By Riemann–Roch, this reduces to showing $\ell(K-D) = \ell(K-D+P+Q)$. Since $\deg D \geq 2g+1$, we have $\deg(K-D+P+Q) \leq (2g-2)-(2g+1)+2 = -1 < 0$, so both terms vanish by lemma 7.1.11.

The upshot is clean: *for smooth curves, "proper" and "projective" are synonymous.* We shall use these terms interchangeably from now on.

These bounds are sharp: on a hyperelliptic curve of genus g, the divisor $D = g \cdot g_2^1$ (where g_2^1 is the degree-2 linear system) has degree $2g$ and is base-point-free but not very ample. We will study this in detail in §7.1.6. Note that the equivalence of projective and proper is very much a *curve* phenomenon—it *fails* in higher dimensions[12].

With this application of Riemann–Roch, we can seamlessly link the geometric, algebraic, and scheme-theoretic models of a curve into a single equivalency.

Before assembling that equivalence, one of its implications has to be isolated and proved on its own. Properness is a condition on how X behaves under limits, and projectivity is the existence of a closed immersion into a projective space; the first says nothing about equations, while the second is an assertion that equations exist. For curves the two coincide, and the passage from the first to the second is the only step of the equivalence with any content.

Proposition 7.1.26: Complete Nonsingular Curves Are Projective

Let k be a field and let X be a complete nonsingular curve over k (definition 7.1.1). Then X is projective over k: there is a closed immersion $X \hookrightarrow \mathbb{P}_k^n$ for some n.

***Proof*:**
Step 1: a nonconstant function. The scheme X is integral of finite type over k with $\dim X = 1$, so its function field $K(X)$ has transcendence degree 1 over k (theorem 2.6.11). Choose $f \in K(X)$ transcendental over k. The function field is the local ring at the generic point, a filtered colimit of the rings of sections (corollary 2.4.24), so f is a regular function on some nonempty open $V \subseteq X$, and f defines a morphism $V \to \mathbb{A}_k^1 \subseteq \mathbb{P}_k^1$, hence a rational map $\varphi\colon X \dashrightarrow \mathbb{P}_k^1$.
Step 2: the rational map is a morphism. Every local ring of X is regular, hence integrally closed (proposition 2.7.19), so X is a normal integral k-scheme of finite type, and $\mathbb{P}_k^1$ is proper over k. By theorem 4.7.4 the locus of indeterminacy of φ has codimension at least 2 in X. No nonempty irreducible closed subset of X has codimension at least 2, since $\operatorname{codim} + \dim \leq \dim X = 1$

[12] There exist complete nonprojective surfaces, the simplest being Hironaka's example (**??**) of a smooth complete threefold that is not projective. The completeness-projectivity gap is what makes proper morphisms interesting as a separate notion from projective ones; see remark 6.6.9.

(lemma 2.6.7), so that locus is empty and φ is a morphism $\varphi\colon X \to \mathbb{P}^1_k$.
Step 3: the morphism is finite. The induced map on function fields sends the coordinate t to f, which is transcendental over k, so this map is injective and φ is dominant; in particular $\varphi(X)$ is not a single closed point. The target $\mathbb{P}^1_k$ is a curve over k in the sense of definition 7.1.1, so proposition 7.1.5 applies and gives that φ is surjective and finite.
Step 4: descending projectivity along a finite morphism. The morphism $\varphi\colon X \to \mathbb{P}^1_k$ is finite and $\mathbb{P}^1_k$ is projective over k, so proposition 5.3.42(2) gives a closed immersion of X into a projective bundle over $\mathbb{P}^1_k$, itself projective over k; composing, X is projective over k, as we sought to show.

This closes without importing anything: the descent of projectivity along a finite morphism is proposition 5.3.42, proved earlier in this chapter. The route taken here goes through a map to $\mathbb{P}^1_k$ and not through theorem 7.1.19, and the choice is forced. The numerical criterion of corollary 7.1.25 produces very ample divisors on a complete nonsingular curve in one line, and it would give projectivity immediately by lemma 5.3.22; but its proof passes through theorem 7.1.19, whose classical form is obtained from Serre duality, and Serre duality was established for projective schemes (theorem 6.5.9). Using that chain to prove projectivity would assume what is to be proved. A map to the projective line, by contrast, is available on any curve carrying a nonconstant rational function, which is every curve.

Two remarks on the scope of the statement. Nonsingularity is not needed: any proper k-scheme of dimension at most 1 is projective over k, singular or not, and the proof of the general case replaces the map to $\mathbb{P}^1_k$ by a direct construction of an ample sheaf; see [63]. What does fail in higher dimension is the conclusion itself, not the method: proper nonprojective surfaces and threefolds exist, and the mechanism behind them is that a proper scheme of dimension at least 2 can carry curves whose classes obstruct every candidate for an ample class. On a curve there is nothing for such an obstruction to live on.

Corollary 7.1.27: Equivalence of Curve Models

Let X be a nonsingular curve over $k = \overline{k}$, with function field $K = K(X)$. Then the following are equivalent:

1. X is projective.
2. X is proper (complete).
3. For every DVR R of K/k, the valuation ring R dominates $\mathcal{O}_{X,x}$ for some (necessarily unique) closed point $x \in X$.

***Proof*:**
(1) $\Rightarrow$ (2) is standard since projective morphisms are proper (theorem 3.5.26). The converse (2) $\Rightarrow$ (1) is exactly proposition 7.1.26 above.

The equivalence (2) $\Leftrightarrow$ (3) is simply the valuative criterion of properness (theorem 3.5.25) applied to the morphism $X \to \operatorname{Spec} k$. Given a DVR R of K/k, properness guarantees a unique lift $\operatorname{Spec} R \to X$ sending the closed point of $\operatorname{Spec} R$ to a closed point $x \in X$, which means R dominates $\mathcal{O}_{X,x}$. Conversely, since the local rings at closed points of a nonsingular curve are exactly the DVRs of K/k, the properness criterion is satisfied. In the language of classical algebraic geometry, X is the unique *complete nonsingular model* of the function field K.

Proposition 7.1.28: Genus Zero Means Rational

Let X be a complete nonsingular curve of genus $g = 0$ over $k = \overline{k}$. Then $X \cong \mathbb{P}^1_k$.

***Proof*:**
Let P be any closed point of X. By Riemann–Roch:

$$\ell(P) = \deg P + 1 - g + \ell(K - P) = 1 + 1 + \ell(K - P) \geq 2.$$

By proposition 7.1.22 $\deg K = -2$, so $\deg(K - P) = -3 < 0$, forcing $\ell(K - P) = 0$ by lemma 7.1.11, giving $\ell(P) = 2$ exactly.

Since $\ell(P) = 2$, the complete linear system $|P|$ is a 1-dimensional projective space, i.e. a pencil. The corresponding morphism $\varphi_{|P|} \colon X \to \mathbb{P}^1$ has degree $\deg P = 1$, hence it is an isomorphism (a degree-1 finite morphism of smooth curves is an isomorphism of function fields, hence an isomorphism of curves by corollary 7.1.27).

Thus $\mathbb{P}^1_k$ is the *unique* smooth projective curve of genus 0 over $\overline{k}$. Over non-algebraically-closed fields, the situation is far richer: a smooth projective curve of genus 0 over k is a *conic*—a degree-2 curve in $\mathbb{P}^2_k$—and it is isomorphic to $\mathbb{P}^1_k$ if and only if it has a k-rational point (i.e. a point with residue field k). Over $\mathbb{Q}$, for instance, the conic $x^2 + y^2 + z^2 = 0$ has genus 0 but is not isomorphic to $\mathbb{P}^1_{\mathbb{Q}}$ since it has no rational points.

Example 7.1.29: Embedding Low-Genus Curves

1. *Genus 0:* As shown in proposition 7.1.28, $X \cong \mathbb{P}^1$. Taking $D = n \cdot P$ for $n \geq 1$ gives $\ell(D) = n + 1$ and the morphism $\varphi_{|D|}$ maps $\mathbb{P}^1 \to \mathbb{P}^n$ as the rational normal curve of degree n (the Veronese embedding, example 3.1.39). For $n = 2$, this is a conic in $\mathbb{P}^2$.

2. *Genus 1:* Let E be a genus one (nonsingular) curve and $O \in E$ some chosen point. We have $\deg K = 0$[a] and $\omega_E \cong \mathcal{O}_E$. Taking $D = 3O$: $\ell(3O) = 3 + 1 - 1 = 3$ (non-special since $\deg(3O) = 3 > 0 = 2g - 2$), so $|3O|$ embeds E as a cubic in $\mathbb{P}^2$. This is the standard Weierstrass embedding, which we will discuss thoroughly in §7.1.7.

3. *Genus 2:* $\deg K = 2$ and $\ell(K) = g = 2$, so $|K|$ is a pencil (a linear system of dimension 1). The canonical map $\varphi_{|K|} \colon X \to \mathbb{P}^1$ is a degree-2 map. This shall tell us that every genus-2 curve is hyperelliptic! The first non-trivial embedding uses D with $\deg D \geq 5$; taking $D = 5P$ gives $\ell(D) = 4$, so $X \hookrightarrow \mathbb{P}^3$ as a curve of degree 5.

4. *Genus 3:* $\deg K = 4$ and $\ell(K) = 3$, so $|K|$ maps $X \to \mathbb{P}^2$. If X is non-hyperelliptic, this is an embedding: X is realized as a smooth quartic in $\mathbb{P}^2$ (of degree 4, which is consistent with the classical degree formula $g = \frac{(4-1)(4-2)}{2} = 3$). If X is hyperelliptic, $|K|$ factors through the $2:1$ map to $\mathbb{P}^1$ and maps X onto a conic in $\mathbb{P}^2$.

[a] Using the Gauss-Bonnet Theorem connection mentioned earlier, this say that such curves have *flat curvature*

The canonical linear system $|K|$ is the most natural linear system on a curve: it is defined intrinsically, without any choices. Let us study it in detail.

Theorem 7.1.30: The Canonical Map and Embedding

Let X be a complete nonsingular curve of genus $g \geq 2$. Then:

1. *Existence:* The canonical linear system $|K|$ is everywhere base-point-free. Consequently, it defines a morphism
$$\varphi_K : X \longrightarrow \mathbb{P}^{g-1}.$$
2. *The Generic Case (Embedding):* If φ_K is injective X, then φ_K is a closed immersion. In this case, X is embedded as a curve of degree $2g-2$ in $\mathbb{P}^{g-1}$. We call this the *canonical embedding* of X.
3. *The Hyperelliptic Case (Factoring):* If φ_K is *not* injective, φ_K factors as
$$X \xrightarrow{2:1} \mathbb{P}^1 \hookrightarrow \mathbb{P}^{g-1}$$
where the first map is called the *hyperelliptic double cover*, and the second is the Veronese embedding of degree $g-1$. Thus, φ_K maps X 2-to-1 onto a rational normal curve of degree $g-1$. Curves where φ_K is not injective are called *hyperelliptic* (studied footer in section 7.1.6).

***Proof*:**
(1) Geometrically, for $|K|$ to be base-point-free means that for every point $P \in X$, there is some global differential form that does not vanish at P. Algebraically, this means imposing a vanishing condition at P must strictly reduce the dimension of the linear system, so we must show $\ell(K) - \ell(K-P) = 1$.

By Riemann–Roch,
$$\ell(K-P) = \deg(K-P) + 1 - g + \ell(P) = (2g-3) + 1 - g + \ell(P) = g - 2 + \ell(P).$$

Because $g \geq 2$, the curve is not rational. Therefore, $\ell(P) = 1$ for *every* point P (if we had $\ell(P) \geq 2$, there would exist a non-constant function with only a simple pole, giving a degree-1 isomorphism $X \cong \mathbb{P}^1$, implying $g = 0$). Substituting $\ell(P) = 1$ back yields $\ell(K-P) = g-1$. Since $\ell(K) = g$, the dimension drops by exactly 1. The linear system has no base points, and the morphism φ_K cleanly maps X into $\mathbb{P}^{g-1}$.

(2) For φ_K to be a closed immersion, it must separate both points and tangent vectors. Algebraically, this requires that imposing *two* vanishing conditions drops the dimension of the canonical system by exactly 2. That is, we need $\ell(K) - \ell(K-P-Q) = 2$ for all pairs of points P, Q (where $P = Q$ corresponds to separating tangent vectors).

Using Riemann–Roch again on the divisor $P+Q$:
$$\ell(K-P-Q) = \deg(K-P-Q) + 1 - g + \ell(P+Q) = (2g-4) + 1 - g + \ell(P+Q) = g - 3 + \ell(P+Q).$$

Subtracting this from $\ell(K) = g$, we find the dimension drop is:
$$\ell(K) - \ell(K-P-Q) = g - (g - 3 + \ell(P+Q)) = 3 - \ell(P+Q).$$

For this drop to equal 2, we must have $\ell(P+Q) = 1$. This means the divisor $P+Q$ admits no extra global sections beyond the constant functions. Conversely, the condition $\ell(P+Q) \geq 2$ for some pair P, Q means there exists a non-constant function with at most two poles. This defines a degree-2 morphism $X \to \mathbb{P}^1$ (the linear system $|P+Q|$ is a g^1_2 pencil). Thus, φ_K successfully embeds X if and only if no such g^1_2 exists; in other words, if and only if X is not hyperelliptic.

(3) Suppose X is hyperelliptic, possessing a degree-2 map to $\mathbb{P}^1$ given by a linear system $g^1_2 = |P+Q|$. We want to show that the canonical system is entirely built out of this smaller system. Let us examine the linear system $|(g-1)(P+Q)|$.

Let x, y be a basis for $H^0(X, \mathcal{O}(P+Q))$. We can form g distinct monomials of degree $g-1$: $x^{g-1}, x^{g-2}y, \dots, y^{g-1}$. These are linearly independent sections of $\mathcal{O}((g-1)(P+Q))$, which immediately tells us $\ell((g-1)(P+Q)) \geq g$.

Now we apply Riemann–Roch to this multiple of the g^1_2:

$$\ell((g-1)(P+Q)) - \ell(K-(g-1)(P+Q)) = \deg((g-1)(P+Q)) + 1 - g = (2g-2) + 1 - g = g-1.$$

Because we proved $\ell((g-1)(P+Q)) \geq g$, the left side of the equation forces $\ell(K-(g-1)(P+Q)) \geq 1$. This implies there is an effective divisor linearly equivalent to $K-(g-1)(P+Q)$. However, note the degree:

$$\deg(K-(g-1)(P+Q)) = (2g-2) - (g-1)(2) = 0.$$

By lemma 7.1.11, an effective divisor of degree 0 must be the trivial divisor. Therefore, we have a strict linear equivalence: $K \sim (g-1)(P+Q)$.

Geometrically the punchline is that the canonical map φ_K is given by evaluating the sections of $|K|$. Since $K \sim (g-1)(P+Q)$, these sections are exactly the degree-$g-1$ monomials in the basis x, y of our g^1_2. Evaluating a map via all monomials of degree d is precisely the definition of the d-th Veronese embedding. Thus, the canonical map evaluates the g^1_2 map $X \to \mathbb{P}^1$, and then directly applies the $(g-1)$-fold Veronese embedding $\mathbb{P}^1 \hookrightarrow \mathbb{P}^{g-1}$.

The canonical embedding is one of the most elegant constructions in algebraic geometry. It embeds the curve "using its own differential forms"—no external choices are needed. This intrinsic nature makes the canonical embedding the right tool for studying the moduli of curves: it is through the canonical embedding that one proves, for instance, that $\mathcal{M}_g$ is of general type for $g \geq 24$ (a result of Harris–Mumford and Eisenbud–Harris).

One down-side of the canonical embedding is that the embedding space $\mathbb{P}^{g-1}$ can vary as the genus varies. We can actually embed all (complete nonsingular) curves into a single projective space:

Proposition 7.1.31: Every Curve Embeds in Projective 3-Space

Let X be a complete nonsingular curve over an algebraically closed field k. Then X is isomorphic to a closed subvariety of $\mathbb{P}^3_k$.

***Proof*:**
By corollary 7.1.25, we can choose a divisor D of degree $\deg D \geq 2g+1$, which guarantees that $\mathcal{L}(D)$ is very ample. This gives a closed immersion $\varphi_{|D|}\colon X \hookrightarrow \mathbb{P}^N_k$, where $N = \ell(D) - 1$. If $N \leq 3$, we can simply compose this map with a linear inclusion $\mathbb{P}^N_k \hookrightarrow \mathbb{P}^3_k$ and we are done.

Suppose $N > 3$. We wish to simplify the embedding by projecting X into a lower-dimensional space. Choose a point $O \in \mathbb{P}^N_k \setminus X$ and consider the linear projection away from O:

$$\pi_O \colon \mathbb{P}^N_k \setminus \{O\} \longrightarrow \mathbb{P}^{N-1}_k.$$

We must ensure that the restriction $\pi_O|_X$ remains a closed immersion. For this to hold, the projection must separate points and tangent vectors:

1. *Separating points:* π_O maps two distinct points $P, Q \in X$ to the same point in $\mathbb{P}^{N-1}_k$ if and only if the line $L_{P,Q}$ connecting them passes through the center of projection O.

2. *Separating tangent vectors:* π_O fails to be an immersion at a point $P \in X$ if and only if the tangent line $T_P X$ to X at P passes through O.

To avoid these failures, we must pick O such that it avoids all secant lines and tangent lines of X. Let $\mathrm{Sec}(X)$ be the *secant variety* of X, defined as the closure of the union of all secant lines $L_{P,Q}$ for $P, Q \in X$. Since a secant line is parameterized by pairs of points $(P, Q) \in X \times X$, we can bound the dimension of the secant variety geometrically:

$$\dim \mathrm{Sec}(X) \;\leq\; \dim(X \times X) + 1 \;=\; 1 + 1 + 1 \;=\; 3.$$

(Note that tangent lines appear naturally in this locus as limits of secant lines as $Q \to P$).

Because $N \geq 4$ and our base field k is algebraically closed (and thus infinite), the ambient space $\mathbb{P}^N_k$ strictly contains the 3-dimensional variety $\mathrm{Sec}(X)$. We can therefore find a generic point $O \in \mathbb{P}^N_k \setminus \mathrm{Sec}(X)$. Projecting from this O yields a new closed immersion $X \hookrightarrow \mathbb{P}^{N-1}_k$.

We may proceed inductively. As long as the dimension of the ambient projective space is strictly greater than 3, we can always find a point outside the secant variety and project down. The process terminates when the ambient dimension reaches 3, leaving us with a closed immersion $X \hookrightarrow \mathbb{P}^3_k$.

This result confirms that $\mathbb{P}^3_k$ is large enough to house *any* smooth algebraic curve. By contrast, not every curve can be embedded in $\mathbb{P}^2_k$ (a curve in $\mathbb{P}^2_k$ of degree d has a strictly determined genus $g = \frac{(d-1)(d-2)}{2}$ (a consequence of theorem 7.1.38), and even allowing singularities, the normalization of a plane curve cannot achieve arbitrary topological types). Thus, $\mathbb{P}^3_k$ is the natural "universal" target for general curve embeddings, analogous to how the Whitney embedding theorem guarantees that any 1-dimensional smooth manifold embeds in $\mathbb{R}^3$ (and in this case we can even take $\mathbb{R}^2$ requiring $n = 2$ due to the circle)

We conclude this section with Clifford's theorem, which controls the behavior of *special* divisors.

Theorem 7.1.32: Clifford's Theorem

Let D be an effective divisor on a complete nonsingular curve X of genus $g \geq 1$ with $0 \leq \deg D \leq 2g - 2$ and $\ell(K - D) > 0$ (i.e. D is *special*). Then:

$$\ell(D) - 1 \;\leq\; \frac{\deg D}{2}.$$

Equality holds if and only if $D = 0$, $D = K$, or X is hyperelliptic and D is a multiple of the g^1_2.

***Proof*:**
The key idea is to use the tensor product $\mathcal{L}(D) \otimes \mathcal{L}(K-D) \cong \mathcal{L}(K)$ and bound $\ell(D) \cdot \ell(K-D)$ against $\ell(K)$ using the multiplication map

$$\mu\colon H^0(X, \mathcal{L}(D)) \otimes H^0(X, \mathcal{L}(K-D)) \longrightarrow H^0(X, \mathcal{L}(K)).$$

The image of μ has dimension at least $\ell(D) + \ell(K-D) - 1$: this is because given nonzero subspaces $V \subseteq K(X)$ and $W \subseteq K(X)$ of dimensions a and b respectively, the product space $V \cdot W$ has dimension at least $a + b - 1$ (fix a nonzero $w_0 \in W$; then $V \cdot w_0$ has dimension a, and each additional basis element of W contributes at least one new dimension). Since the image lies in $H^0(X, \omega_X)$, which has dimension g:

$$\ell(D) + \ell(K-D) - 1 \ \leq \ g.$$

Now apply Riemann–Roch: $\ell(D) = \deg D + 1 - g + \ell(K-D)$, so $\ell(K-D) = \ell(D) - \deg D - 1 + g$. Substituting:

$$\ell(D) + (\ell(D) - \deg D - 1 + g) - 1 \leq g \quad \Longrightarrow \quad 2\ell(D) \leq \deg D + 2 \quad \Longrightarrow \quad \ell(D) - 1 \leq \frac{\deg D}{2}.$$

The characterization of equality is more delicate and we leave it as a guided exercise[a].

[a] When equality holds, the multiplication map μ must be surjective. This forces a very rigid structure on $\mathcal{L}(D)$: either it is trivial ($D = 0$), canonical ($D = K$), or the curve is hyperelliptic and D is a multiple of the g^1_2. A clean proof can be found in [33, §IV.5].

Clifford's theorem is the "other side" of Riemann–Roch: while Riemann–Roch gives the *exact* value of $\ell(D) - \ell(K-D)$, Clifford controls $\ell(D)$ itself when the divisor is special. Together, they provide a nearly complete understanding of the function $D \mapsto \ell(D)$ on $\mathrm{Div} X$.

7.1.33 Forward Reference: Petri's Theorem and Syzygies The canonical embedding of a non-hyperelliptic curve $X \hookrightarrow \mathbb{P}^{g-1}$ opens the door to a classical line of inquiry: what are the *equations* that define the image? Petri's theorem says that for "most" curves, the ideal of X in $\mathbb{P}^{g-1}$ is generated by quadrics[a]. More precisely, the homogeneous ideal is generated in degree 2 unless X is trigonal (admits a degree-3 map to $\mathbb{P}^1$) or is a smooth plane quintic ($g = 6$).
This leads to the study of *syzygies* of canonical curves, culminating in Green's conjecture (now a theorem, proved by Voisin for general curves): the Betti numbers of the minimal free resolution of the canonical ideal detect the *Clifford index* of the curve. This is a place where the theory of *derived categories* (a separate volume) meets classical curve theory, and provides some of the most important applications of homological algebra to concrete geometry.

[a] In algebraic geometry, a *quadric* refers to the geometric hypersurface cut out by a single polynomial equation of degree 2 (a *quadratic* polynomial). To say an ideal is "generated by quadrics" means it is generated by homogeneous polynomials of degree 2.

7.1.4 Hurwitz's Theorem and Ramification

We now turn to the second central theorem of the classical theory of curves. If Riemann–Roch relates the *intrinsic* invariants of a curve (genus, divisors, sections), then Hurwitz's theorem relates the *extrinsic* geometry of a finite morphism $f\colon X \to Y$ to the genera of X and Y.

The intuition is topological. Over $\mathbb{C}$, a finite morphism of smooth projective curves is a branched covering of compact Riemann surfaces. Away from finitely many *branch points*, the map is a genuine covering space with $n = \deg f$ sheets. At a branch point, some of the sheets "come together"—this is ramification. The Euler characteristic (equivalently, genus) of X is determined by the Euler characteristic of Y, the degree of f, and the amount of ramification. Hurwitz's formula makes this precise in the algebraic setting, working over an arbitrary algebraically closed field.

Ramification

Throughout this section, $f\colon X \to Y$ is a finite morphism of complete nonsingular curves over $k = \overline{k}$, with $n = \deg f = [K(X) : K(Y)]$.

Recall that every local ring $\mathcal{O}_{X,x}$ (resp. $\mathcal{O}_{Y,y}$) at a closed point of a smooth curve is a DVR with discrete valuation ν_x (resp. ν_y). For a point $x \in X$ with $y = f(x)$, let $t \in \mathcal{O}_{Y,y}$ be a uniformizer (i.e. $\nu_y(t) = 1$). The local map $f^\#\colon \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ sends t to an element of $\mathcal{O}_{X,x}$, and we defined the ramification index as $e_x = \nu_x(f^\#(t))$. Let us now give this a more careful treatment.

Definition 7.1.34: Ramification Index

Let $f\colon X \to Y$ be a finite morphism of complete nonsingular curves, $x \in X$ a closed point, and $y = f(x)$. The *ramification index* of f at x is:

$$e_x \;=\; \nu_x(f^\#(t))$$

where t is any uniformizer of $\mathcal{O}_{Y,y}$. This is independent of the choice of t. We say:

1. f is *unramified* at x if $e_x = 1$.
2. f is *ramified* at x if $e_x > 1$. The point $y = f(x)$ is then called a *branch point*.
3. If $\operatorname{char} k = p > 0$, the ramification is *tame* if $p \nmid e_x$, and *wild* if $p \big| e_x$.

The well-definedness is immediate: if $t' = ut$ for a unit $u \in \mathcal{O}_{Y,y}^*$, then $f^\#(u)$ is a unit in $\mathcal{O}_{X,x}$ (local maps send units to units), so $\nu_x(f^\#(t')) = \nu_x(f^\#(u)) + \nu_x(f^\#(t)) = 0 + e_x$. Geometrically, e_x measures the "order of contact" between the fiber $f^{-1}(y)$ and the point x: if s is a uniformizer of $\mathcal{O}_{X,x}$, then $f^\#(t) = u \cdot s^{e_x}$ for a unit u, so the map locally looks like $s \mapsto s^{e_x}$ up to units.

Example 7.1.35: Ramification in Practice

1. *The squaring map.* Consider $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ given by $[x : y] \mapsto [x^2 : y^2]$, which in the affine chart $y \neq 0$ is $t \mapsto t^2$. This has degree 2. At $t = 0$: the uniformizer of the target is t, and $f^\#(t) = t^2$, so $e_0 = 2$ (ramified). Similarly at $t = \infty$ (using the chart $x \neq 0$, the map is $s \mapsto s^2$ in the uniformizer $s = 1/t$), so $e_\infty = 2$. At every other point $t = a \neq 0, \infty$: the uniformizer of the target at a^2 is $t - a^2$, and $f^\#(t - a^2) = t^2 - a^2 = (t-a)(t+a)$. For the point $x = a$, $\nu_a((t-a)(t+a)) = 1$ (since $t + a$ is a unit at $t = a$ when $\operatorname{char} k \neq 2$), so $e_a = 1$ (unramified). The fiber over a^2 consists of two points $\{a, -a\}$, each with $e = 1$, confirming $\sum e_x = 2 = \deg f$.
2. *Wild ramification.* Let k have characteristic $p > 0$ and consider $f\colon \mathbb{A}^1 \to \mathbb{A}^1$ given by

$t \mapsto t^p - t$ (the Artin–Schreier map). At $t = 0$: $f^\#(s) = t^p - t$ where s is the uniformizer of the target at $f(0) = 0$. We compute $\nu_0(t^p - t) = 1$ (the lowest-order term is $-t$), so $e_0 = 1$—the map is unramified at the origin! In fact, the Artin–Schreier map is étale everywhere on $\mathbb{A}^1$, yet it is a nontrivial cover: the fiber over every point has p elements. This phenomenon—unramified covers that are not trivial—does not occur in the topological category (where $\pi_1(\mathbb{A}^1_{\mathbb{C}}) = 0$), and is a distinctive feature of positive characteristic.

The Ramification Divisor via Differentials

A key way to understand ramification is through the sheaf of differentials. The morphism $f\colon X \to Y$ induces, via pullback of Kähler differentials (proposition 5.5.16), a natural map of sheaves on X:

$$f^*\Omega_{Y/k} \longrightarrow \Omega_{X/k}. \tag{7.3}$$

Since both $\Omega_{Y/k}$ and $\Omega_{X/k}$ are invertible sheaves on smooth curves, this map is an injection of invertible sheaves[13] (of $\mathcal{O}_X$-modules), hence corresponds to an effective divisor: the cokernel is a torsion sheaf supported at finitely many points. Let us make this precise.

In terms of local coordinates: at a point $x \in X$ with $y = f(x)$, let s be a uniformizer of $\mathcal{O}_{X,x}$ and t a uniformizer of $\mathcal{O}_{Y,y}$. Then $f^\#(t) = u \cdot s^{e_x}$ for a unit $u \in \mathcal{O}_{X,x}^*$, and:

$$f^*(dt) \;=\; d(f^\#(t)) \;=\; d(u \cdot s^{e_x}) \;=\; (u' s^{e_x} + e_x u s^{e_x - 1})\, ds \;=\; s^{e_x-1}(u's + e_x u)\, ds.$$

The factor s^{e_x-1} vanishes to order $e_x - 1$ at x. The factor $(u's + e_x u)$ is a unit at x *provided* $e_x \neq 0$ in k, i.e. provided the ramification at x is tame (or we are in characteristic zero). In the case of tame ramification, the map (7.3) vanishes to order exactly $e_x - 1$ at x.

Definition 7.1.36: Ramification and Branch Divisors

The *ramification divisor* of f is:

$$R \;=\; \sum_{x \in X_{\mathrm{cl}}} r_x \cdot x$$

where r_x is defined by the relation $f^*\Omega_{Y/k} \cong \Omega_{X/k} \otimes \mathcal{L}(-R)$, or equivalently, $r_x = \nu_x(f^*(dt)/ds)$ for local uniformizers. The *branch divisor* is $B = f_*(R) \in \operatorname{WDiv} Y$.
When the ramification is tame (in particular, always in characteristic 0):

$$r_x = e_x - 1.$$

When the ramification is wild ($\operatorname{char} k = p > 0$ and $p \mid e_x$), we have $r_x \geq e_x$, and the precise value depends on higher-order ramification data.

The relation $f^*\Omega_{Y/k} \cong \Omega_{X/k}(-R)$ can be rewritten in terms of canonical divisors. Since $\Omega_{Y/k} \cong \mathcal{L}(K_Y)$ and $\Omega_{X/k} \cong \mathcal{L}(K_X)$, and $f^*\mathcal{L}(K_Y) \cong \mathcal{L}(f^*K_Y)$, we obtain the fundamental *divisor relation*:

$$K_X \;\sim\; f^*K_Y + R. \tag{7.4}$$

This is the key equation from which Hurwitz's formula follows by taking degrees.

[13] It is injective because f is dominant and separable (we are assuming $\operatorname{char} k = 0$ or that the extension $K(X)/K(Y)$ is separable; see Remark 7.1.37), so the induced map $K(Y) \otimes_{K(Y)} \Omega_{K(Y)/k} \to \Omega_{K(X)/k}$ on generic fibers is injective.

7.1.37 Remark: Inseparable Maps If the field extension $K(X)/K(Y)$ is inseparable (which can only happen in characteristic $p > 0$), the map $f^*\Omega_{Y/k} \to \Omega_{X/k}$ is the *zero map*, and the preceding discussion breaks down entirely. The simplest example is the Frobenius morphism $\mathrm{Frob}\colon X \to X^{(p)}$, which is purely inseparable of degree p and for which $\mathrm{Frob}^*(\Omega_{X^{(p)}/k}) = 0$. In this book, we shall always assume our finite morphisms of curves are *separable* (i.e. $K(X)/K(Y)$ is a separable field extension) when discussing Hurwitz's formula. The inseparable case requires a more delicate analysis using Hasse–Schmidt derivations or the algebraic trace form to define the different ideal, which goes beyond our scope here; see [58] for the number field analogue.

Hurwitz's Formula

Theorem 7.1.38: Hurwitz's Formula

Let $f\colon X \to Y$ be a separable finite morphism of complete nonsingular curves of genera g_X and g_Y respectively, with $n = \deg f$. Then:

$$2g_X - 2 \;=\; n(2g_Y - 2) \;+\; \deg R$$

where $R = \sum r_x \cdot x$ is the ramification divisor. In the tamely ramified case (in particular, whenever $\mathrm{char}k = 0$):

$$2g_X - 2 \;=\; n(2g_Y - 2) \;+\; \sum_{x \in X_{\mathrm{cl}}} (e_x - 1).$$

***Proof*:**
Taking degrees of both sides of (7.4):

$$\deg K_X = \deg(f^*K_Y) + \deg R.$$

By proposition 7.1.22, $\deg K_X = 2g_X - 2$ and $\deg K_Y = 2g_Y - 2$. By the degree formula for pullbacks (proposition 7.1.7), $\deg(f^*K_Y) = n \cdot \deg K_Y = n(2g_Y - 2)$. Therefore:

$$2g_X - 2 = n(2g_Y - 2) + \deg R.$$

The second formula follows since $r_x = e_x - 1$ in the tame case.

The proof is almost embarrassingly short—the hard work was establishing the relation $K_X \sim f^*K_Y + R$ in §7.1.4, which itself rests on the theory of Kähler differentials from §5.5. This is an illustration of the scheme-theoretic philosophy: the right definitions make the proofs almost automatic.

Let us also give an alternative argument via Euler characteristics that avoids explicit mention of the ramification divisor, as it gives additional insight and generalizes more readily.

***Proof Alternative proof via Euler characteristics*:**
Consider the short exact sequence on X:

$$0 \to f^*\Omega_{Y/k} \to \Omega_{X/k} \to \Omega_{X/Y} \to 0$$

coming from the cotangent exact sequence (proposition 5.5.16) applied to the composition $X \xrightarrow{f} Y \to \operatorname{Spec} k$. Taking Euler characteristics:

$$\chi(\Omega_{X/k}) = \chi(f^*\Omega_{Y/k}) + \chi(\Omega_{X/Y}).$$

Now $\Omega_{X/k} = \omega_X \cong \mathcal{L}(K_X)$, and by Riemann–Roch (theorem 7.1.19):

$$\chi(\omega_X) = \deg K_X + 1 - g_X = (2g_X - 2) + 1 - g_X = g_X - 1.$$

For the pullback: $f^*\Omega_{Y/k} \cong f^*\omega_Y \cong \mathcal{L}(f^*K_Y)$, and

$$\chi(f^*\omega_Y) = \deg(f^*K_Y) + 1 - g_X = n(2g_Y - 2) + 1 - g_X.$$

The relative differentials $\Omega_{X/Y}$ are a torsion sheaf on X (supported at the ramification locus), hence $H^1(X, \Omega_{X/Y}) = 0$ (a torsion sheaf on a curve has no higher cohomology), so $\chi(\Omega_{X/Y}) = \dim_k H^0(X, \Omega_{X/Y}) = \deg R$. Putting it all together:

$$g_X - 1 = n(2g_Y - 2) + 1 - g_X + \deg R$$

which rearranges to $2g_X - 2 = n(2g_Y - 2) + \deg R$.

Example 7.1.39: Hurwitz in Action

1. *Double covers of* $\mathbb{P}^1$. Let $f\colon X \to \mathbb{P}^1$ be a degree-2 separable map in characteristic $\neq 2$. Then $n = 2$, $g_Y = 0$, and all ramification is tame with $e_x \in \{1, 2\}$. Let b be the number of branch points in $\mathbb{P}^1$ (since $n = 2$, each corresponds to exactly one ramification point in X where $e_x = 2$). Hurwitz gives:

   $$2g_X - 2 = 2(0 - 2) + b = -4 + b, \qquad \text{so } g_X = \frac{b-2}{2}.$$

   Since g_X must be a non-negative integer, b must be even and $b \geq 2$. For $b = 2$: $g_X = 0$ (the map $[x : y] \mapsto [x^2 : y^2]$). For $b = 4$: $g_X = 1$ (elliptic curves). For $b = 6$: $g_X = 2$ (genus-2 curves, which are all hyperelliptic). More generally, a *hyperelliptic curve* of genus g is a double cover of $\mathbb{P}^1$ with $2g + 2$ branch points.

2. *Cyclic covers.* Let $f\colon X \to \mathbb{P}^1$ have degree n with total ramification ($e_x = n$) over exactly b points and no other ramification. Then:

   $$2g_X - 2 = n(-2) + b(n-1), \quad \text{so} \quad g_X = \frac{(n-1)(b-2)}{2}.$$

   For $n = 3$, $b = 4$: $g_X = 2$ (a genus-2 curve). For $n = 3$, $b = 5$: $g_X = 3$ (a trigonal curve of genus 3). For $n = 3$, $b = 3$: $g_X = 1$ (an elliptic curve with an order-3 automorphism).

3. *Genus bound.* If $g_Y \geq 1$ and $n \geq 2$, then $2g_X - 2 \geq n(2g_Y - 2) + 0$, so $g_X \geq ng_Y - n + 1 \geq g_Y$. In particular, there are no nonconstant maps from a curve of genus g to a curve of genus g' where $g' > g$. This is a "topological obstruction" to the existence of maps.

7.1.5 Applications of Hurwitz's Theorem

One of the most applications of Hurwitz's formula is the finiteness—and in fact an explicit bound on the size—of the automorphism group of a curve of genus ≥ 2.

Theorem 7.1.40: Hurwitz's Automorphism Bound

Let X be a complete nonsingular curve of genus $g \geq 2$ over k with $\operatorname{char} k = 0$ (or more generally, $\operatorname{char} k \nmid |\mathrm{Aut}(X)|$). Then:

$$|\mathrm{Aut}(X)| \;\leq\; 84(g-1).$$

This bound is sharp: the Klein quartic $x^3y + y^3z + z^3x = 0$ in $\mathbb{P}^2$ has genus 3 and $|\mathrm{Aut}| = 168 = 84 \cdot 2$, achieving equality.

***Proof Sketch of proof*:**
Let $G = \mathrm{Aut}(X)$, which acts on X faithfully. The quotient $Y = X/G$ is again a smooth projective curve[a], and the quotient map $\pi \colon X \to Y$ is a finite morphism of degree $|G|$.

Apply Hurwitz's formula to π:

$$2g - 2 = |G| \cdot (2g_Y - 2) + \sum_{x \in X_{\mathrm{cl}}} (e_x - 1).$$

Since G acts transitively on each fiber, the ramification data depends only on the branch points $y_1, \dots, y_s \in Y$ and their stabilizer orders $e_1, \dots, e_s$ (with $e_i \geq 2$). The sum becomes $|G| \cdot \sum_{i=1}^{s}(1 - 1/e_i)$, and we get:

$$\frac{2g-2}{|G|} = 2g_Y - 2 + \sum_{i=1}^{s} \left(1 - \frac{1}{e_i}\right). \tag{7.5}$$

The left side is positive (since $g \geq 2$), so the right side must be positive. Call the right side μ. Since each $1 - 1/e_i \geq 1/2$, we have $\mu \geq 2g_Y - 2 + s/2$. For $\mu > 0$, the minimum occurs at $g_Y = 0$, $s = 3$, $e_1 = e_2 = e_3 = 2$... but this gives $\mu = -2 + 3/2 < 0$. A careful case analysis shows the minimum positive value of μ is:

$$\mu_{\min} = \frac{1}{42}, \qquad \text{achieved at } g_Y = 0,\ s = 3,\ (e_1, e_2, e_3) = (2, 3, 7).$$

Therefore $|G| = (2g-2)/\mu \leq 42(2g-2) = 84(g-1)$.

[a] Smoothness of the quotient follows from the fact that stabilizer groups at points are cyclic (being finite subgroups of $\mathrm{Aut}(\mathcal{O}_{X,x}) \cong k^*$ acting on the tangent space) and from the theory of quotients by finite groups; see proposition 4.4.2.

The proof demonstrates the power of combinatorics constrained by Hurwitz's formula. The special role of the triple $(2,3,7)$ reflects the fact that the $(2,3,7)$-triangle group is the lattice in $\mathrm{PSL}_2(\mathbb{R})$ of smallest co-area—a key connection between algebraic curves, hyperbolic geometry, and number theory.

Example 7.1.41: Automorphism Groups in Low Genus

1. *Genus* 0*:* $\mathrm{Aut}(\mathbb{P}^1_k) = \mathrm{PGL}_2(k)$, which is infinite (and 3-dimensional as an algebraic group). Hurwitz's bound does not apply since $g < 2$.

2. *Genus* 1*:* If E is an elliptic curve with origin O, then $\mathrm{Aut}(E)$ (as a pointed curve, i.e. automorphisms fixing O) is finite but can be nontrivial. Over $\mathbb{C}$: generically $\mathrm{Aut}(E, O) \cong \mathbb{Z}/2\mathbb{Z}$ (the inversion $P \mapsto -P$), but for special j-invariants $j = 0$ (resp. $j = 1728$) the automorphism group is $\mathbb{Z}/6\mathbb{Z}$ (resp. $\mathbb{Z}/4\mathbb{Z}$), corresponding to the extra symmetries of the hexagonal (resp. square) lattice.

   However, as a group scheme, E also has the translation automorphisms $\tau_P : Q \mapsto Q + P$ for every $P \in E(k)$, so the *full* automorphism group $\mathrm{Aut}(E)$ is an extension of $\mathrm{Aut}(E, O)$ by $E(k)$, which is again infinite.

3. *Genus* 2*:* Every genus-2 curve is hyperelliptic, so it comes with an involution ι (the hyperelliptic involution, definition 7.1.46), giving $|\mathrm{Aut}(X)| \geq 2$. The Hurwitz bound gives $|\mathrm{Aut}(X)| \leq 84$. The actual maximum for $g = 2$ is achieved by the Bolza curve $y^2 = x^5 - x$, which has $|\mathrm{Aut}| = 48$ over $\mathbb{C}$ (so the theoretical bound of 84 is not tight for $g = 2$).

A finite morphism $f : X \to Y$ of smooth curves is *unramified* if $e_x = 1$ for all $x \in X$, or equivalently if the ramification divisor $R = 0$, or equivalently if $\Omega_{X/Y} = 0$. Since f is automatically flat (a finite morphism between smooth curves over a field is flat[14]), an unramified finite morphism is *étale*.

When $R = 0$, Hurwitz's formula simplifies dramatically:

$$2g_X - 2 = n(2g_Y - 2).$$

Corollary 7.1.42: Étale Covers and Genus

Let $f : X \to Y$ be a finite étale morphism of complete nonsingular curves of degree n.

1. If $g_Y = 0$: then $2g_X - 2 = -2n < 0$, which is impossible for $n \geq 2$ (since $g_X \geq 0$). So $\mathbb{P}^1$ has no nontrivial finite étale covers: $\mathbb{P}^1$ is *simply connected* in the algebraic sense.

2. If $g_Y = 1$: then $g_X = 1$ as well. Étale covers of elliptic curves are elliptic curves (for instance, the multiplication-by-n map $[n] : E \to E$ is étale when $\mathrm{char}\, k \nmid n$).

3. If $g_Y \geq 2$: then $g_X = n(g_Y - 1) + 1 > g_Y$. The cover has strictly larger genus.

***Proof*:**
Immediate from Hurwitz with $\deg R = 0$.

Part (1) is a purely algebraic result that mirrors the topological fact $\pi_1(\mathbb{P}^1_{\mathbb{C}}) = 0$ (the Riemann sphere is simply connected). Part (2) reflects $\pi_1(E_{\mathbb{C}}) \cong \mathbb{Z}^2$, which has many finite quotients.

[14] Flatness follows from the fact that a finite torsion-free module over a Dedekind domain is flat; or from the "miracle flatness" theorem in dimension 1.

7.1.43 Forward Reference: The Étale Fundamental Group The étale fundamental group $\pi_1^{\text{ét}}(X, \overline{x})$ is defined as the profinite group classifying finite étale covers of X (see box 3.1.19 for an earlier preview). Over $\mathbb{C}$, GAGA (theorem 6.9.3) implies that the étale fundamental group of a smooth projective variety is the *profinite completion* of the topological fundamental group:

$$\pi_1^{\text{ét}}(X_{\mathbb{C}}) \;\cong\; \widehat{\pi_1^{\text{top}}(X(\mathbb{C}))}.$$

For a curve of genus g, π_1^{top} is the surface group with $2g$ generators and one relation, and its profinite completion is an enormously rich object. Over fields of positive characteristic or arithmetic base schemes, the étale fundamental group captures Galois-theoretic information and is one of the central objects of arithmetic geometry. This story will be developed in a separate volume.

To connect Hurwitz's theorem to number theory, this theorem has one of the most important arithmetic counterparts. Let L/K be a finite extension of number fields with $[L : K] = n$, and consider the associated map of spectra of rings of integers:

$$\pi\colon \operatorname{Spec} \mathcal{O}_L \longrightarrow \operatorname{Spec} \mathcal{O}_K.$$

Although $\operatorname{Spec} \mathcal{O}_K$ is not a "curve" in our sense (it is a scheme of finite type over $\operatorname{Spec} \mathbb{Z}$, not over a field, and its "genus" is not defined in the usual way), the *structure* of the Hurwitz formula carries over perfectly. For each prime $\mathfrak{p}$ of $\mathcal{O}_K$, the fiber $\pi^{-1}(\mathfrak{p})$ consists of primes $\mathfrak{q}_1, \dots, \mathfrak{q}_r$ of $\mathcal{O}_L$ with ramification indices e_i and residue field degrees f_i satisfying the fundamental identity:

$$\sum_{i=1}^{r} e_i f_i \;=\; n.$$

The *different ideal* $\mathfrak{d}_{L/K}$ is the analogue of the ramification divisor R: it is an ideal of $\mathcal{O}_L$ that measures the failure of $\Omega_{\mathcal{O}_L/\mathcal{O}_K}$ to vanish, and its norm is the *discriminant ideal* $\Delta_{L/K}$, an ideal of $\mathcal{O}_K$ corresponding to the pushforward branch divisor $B = \pi_*(R)$.

The arithmetic analogue of the branch divisor equation is the *discriminant formula*:

$$\Delta_{L/K} \;=\; \prod_{\mathfrak{p}\text{ ramified}} \mathfrak{p}^{\sum_{i=1}^{r} f_i(e_i - 1 + \text{wild terms})}.$$

In the tamely ramified case, the exponent of $\mathfrak{q}_i$ in the different is $e_i - 1$. Taking the norm down to K multiplies this exponent by the residue field degree f_i, so the exponent at $\mathfrak{p}$ is exactly $\sum f_i(e_i - 1)$, perfectly paralleling the degree of the geometric branch divisor over a point. In the wildly ramified case, the exponent is larger, just as in the geometric setting[15].

This analogy is the starting point for one of the most fruitful ideas in modern mathematics: the *arithmetic–geometric dictionary*, in which number fields play the role of function fields of curves over finite fields, primes play the role of points, and the discriminant plays the role of the branch divisor. Many of the most important theorems and conjectures in number theory—including the Riemann hypothesis for curves over finite fields (proved by Weil), the Langlands program, and the Birch–Swinnerton-Dyer conjecture—can be motivated by pushing this analogy as far as it will go.

[15] Both formulas arise from the same algebraic fact about differentials in extensions of Dedekind domains. The "different" in number theory and the "ramification divisor" in geometry are the same mathematical object, viewed from different perspectives. See [58] for a comprehensive treatment.

7.1.6 Hyperelliptic Curves

In the previous section we saw that the canonical map $\varphi_K : X \to \mathbb{P}^{g-1}$ is either an embedding (when X is not hyperelliptic) or a 2-to-1 map onto a rational normal curve (when X is hyperelliptic). The hyperelliptic curves are thus the "exceptional case" in the canonical embedding theorem, and they deserve special attention. They are also the curves that arise most concretely in practice: every hyperelliptic curve admits an explicit equation of the form $y^2 = f(x)$, and much of the classical theory of curves was developed by studying them.

Definition and Characterization

Definition 7.1.44: Hyperelliptic Curve

A complete nonsingular curve X of genus $g \geq 2$ over $k = \overline{k}$ is *hyperelliptic* if there exists a finite morphism $\varphi: X \to \mathbb{P}^1$ of degree 2.

We require $g \geq 2$ in the definition to exclude $\mathbb{P}^1$ (genus 0, which trivially maps to itself by degree 2) and elliptic curves (genus 1, which always admit degree-2 maps to $\mathbb{P}^1$ but are better studied in their own right in §7.1.7).

The degree-2 map φ, when it exists, is unique up to automorphisms of $\mathbb{P}^1$:

Proposition 7.1.45: Characterizations of Hyperelliptic Curves

Let X be a complete nonsingular curve of genus $g \geq 2$ over $k = \overline{k}$. The following are equivalent:

1. X is hyperelliptic (there exists a degree-2 morphism $X \to \mathbb{P}^1$).
2. There exist two closed points $P, Q \in X$ (possibly equal) with $\ell(P+Q) \geq 2$ (i.e. X admits a g^1_2—a linear system of degree 2 and dimension 1).
3. The canonical linear system $|K_X|$ is *not* very ample.
4. The canonical map $\varphi_K : X \to \mathbb{P}^{g-1}$ is not a closed immersion.

Moreover, when these conditions hold, the g^1_2 is unique: any two degree-2 maps $X \to \mathbb{P}^1$ differ by an automorphism of $\mathbb{P}^1$.

***Proof*:**
(1) $\Rightarrow$ (2): A degree-2 morphism $\varphi: X \to \mathbb{P}^1$ corresponds to a base-point-free linear system of degree 2 and dimension 1. Taking $P + Q = \varphi^*(y)$ for some $y \in \mathbb{P}^1$, we get $\ell(P+Q) \geq 2$.

(2) $\Rightarrow$ (1): If $\ell(P+Q) \geq 2$, the linear system $|P+Q|$ is at least a pencil, giving a morphism $\varphi: X \to \mathbb{P}^1$ of degree $\deg(P+Q) = 2$.

(1) $\iff$ (3): This was shown in the proof of theorem 7.1.30: $|K|$ fails to be very ample precisely when there exist P, Q (possibly $P = Q$) with $\ell(P+Q) \geq 2$, which is exactly the existence of a g^1_2.

(3) $\iff$ (4): A line bundle is very ample if and only if the associated morphism is a closed immersion.

Uniqueness of the g_2^1: Suppose there are two such linear systems, giving two degree-2 morphisms $f, g\colon X \to \mathbb{P}^1$. If they are not equivalent by an automorphism of $\mathbb{P}^1$, the product map $(f, g)\colon X \to \mathbb{P}^1 \times \mathbb{P}^1$ has an image C which is a curve of bidegree (a, b). The composition of the map $X \to C$ with the projections gives f and g, which both have degree 2, so $a \leq 2$ and $b \leq 2$. Since f and g are distinct, C is not a graph, forcing $a = b = 2$ and the map $X \to C$ to be birational. But the arithmetic genus of a $(2, 2)$ curve on $\mathbb{P}^1 \times \mathbb{P}^1$ is $(2-1)(2-1) = 1$. This implies $g(X) \leq 1$, contradicting $g \geq 2$. Thus the g_2^1 is unique.

The degree-2 map $\varphi\colon X \to \mathbb{P}^1$ is a separable finite morphism (assuming $\operatorname{char} k \neq 2$; we will address characteristic 2 briefly at the end). Over the generic point of $\mathbb{P}^1$, the fiber consists of two distinct points, and so $K(X)/K(\mathbb{P}^1) = K(X)/k(t)$ is a separable quadratic extension. This extension has a unique nontrivial automorphism: the *hyperelliptic involution*.

Definition 7.1.46: Hyperelliptic Involution

Let X be a hyperelliptic curve with g_2^1 given by $\varphi\colon X \to \mathbb{P}^1$. The *hyperelliptic involution* is the unique nontrivial automorphism $\iota\colon X \to X$ such that $\varphi \circ \iota = \varphi$. Equivalently, ι is the nontrivial element of $\mathrm{Gal}(K(X)/k(t)) \cong \mathbb{Z}/2\mathbb{Z}$.

For each point $y \in \mathbb{P}^1$, the fiber $\varphi^{-1}(y)$ is either two distinct points $\{x, \iota(x)\}$ (when $e_x = 1$) or a single point fixed by ι (when $e_x = 2$, a ramification point). The fixed points of ι are precisely the ramification points of φ, which we call the *Weierstrass points* of X.

By Hurwitz's formula (theorem 7.1.38):

$$2g - 2 = 2(0-2) + \sum (e_x - 1) = -4 + b$$

where b is the number of ramification points. Therefore:

$$b = 2g + 2.$$

A hyperelliptic curve of genus g has exactly $2g + 2$ Weierstrass points. These are the fixed points of ι.

The hyperelliptic structure gives an explicit affine model. If $\operatorname{char} k \neq 2$, the quadratic extension $K(X)/k(t)$ is generated by an element y with $y^2 = f(t)$ for some polynomial $f \in k[t]$. Since X is nonsingular, f must have no repeated roots. The degree of f is determined by the genus:

Proposition 7.1.47: Weierstrass Models

Let X be a hyperelliptic curve of genus g over k with $\operatorname{char} k \neq 2$. Then X is birational to the affine curve:

$$y^2 = f(x), \qquad f \in k[x], \quad \deg f \in \{2g+1, 2g+2\}, \quad f \text{ squarefree.}$$

Conversely, the smooth projective model of such a curve has genus g and is hyperelliptic. The two cases are related by a change of variable at infinity:

1. $\deg f = 2g+1$: there is a single point at infinity, which is a Weierstrass point (a branch point of φ).
2. $\deg f = 2g+2$: there are two points at infinity, neither of which is a Weierstrass point (the map φ is unramified at infinity).

***Proof Sketch*:**
The function field of X is $k(x, y)$ with $y^2 = f(x)$, and the map φ is $(x, y) \mapsto x$. The ramification points of φ correspond to the roots of f (where $y = 0$) plus possibly the point(s) at infinity, depending on the parity of $\deg f$. In both cases, the total number of branch points is $2g+2$ (either $2g+1$ finite roots plus the point at infinity, or $2g+2$ finite roots). The genus computation follows from Hurwitz.

To see that f must be squarefree: if $f(x) = (x-a)^2 g(x)$, then at the point $x = a$, $y = 0$ in the affine model, we have $y^2 = (x-a)^2 g(x)$. The Jacobian criterion (proposition 2.7.6) shows this point is singular, contradicting nonsingularity of X.

Example 7.1.48: Hyperelliptic Curves in Practice

1. **Genus 2:** Every genus-2 curve is hyperelliptic (since $\deg K = 2$ and $\ell(K) = 2$, the canonical system is the g^1_2). A typical model: $y^2 = x^5 + ax^3 + bx + c$ with $\deg f = 5$.
2. **Genus 3, hyperelliptic:** $y^2 = f(x)$ with $\deg f = 7$ or 8, squarefree. The canonical map sends $X \xrightarrow{2:1} \mathbb{P}^1 \xrightarrow{\text{Veronese}} \mathbb{P}^2$, so $\varphi_K(X)$ is a conic in $\mathbb{P}^2$. A generic genus-3 curve is *not* hyperelliptic; it is a smooth quartic in $\mathbb{P}^2$.
3. **Over $\mathbb{F}_p$:** Hyperelliptic curves over finite fields are a major topic in computational number theory and cryptography, since their Jacobians provide groups suitable for discrete-logarithm-based protocols. The Hasse–Weil bound constrains $|X(\mathbb{F}_q)|$, and counting points on hyperelliptic curves can be done efficiently.

We established in theorem 7.1.30(3) that for a hyperelliptic curve, the canonical map factors through the g^1_2. Let us make this completely explicit.

If D_0 denotes the unique g^1_2 (the divisor class of $\varphi^*(\text{point})$), then $\deg D_0 = 2$ and $\ell(D_0) = 2$. The key claim is:

Proposition 7.1.49: Canonical Class of a Hyperelliptic Curve

$K_X \sim (g-1)D_0$. In particular, the canonical linear system is $|K_X| = |(g-1)D_0|$.

***Proof*:**
We have $\deg K_X = 2g-2 = (g-1)\cdot 2 = (g-1)\deg D_0$. It suffices to show $\ell((g-1)D_0) \geq g$. To see this, consider the Weierstrass model $y^2 = f(x)$. The g^1_2 is given by $\varphi = x\colon X \to \mathbb{P}^1$, and we can set $D_0 = \varphi^*(\infty)$. The space $H^0(X, \mathcal{L}((g-1)D_0))$ consists of rational functions with poles only at the point(s) above ∞, of order $\leq 2(g-1)$. This space clearly contains the g linearly independent functions $\{1, x, x^2, \ldots, x^{g-1}\}$ (since x^j has a pole of order $2j \leq 2(g-1)$ at infinity), giving $\ell((g-1)D_0) \geq g$.

Now we apply Riemann–Roch to the divisor $(g-1)D_0$:

$$\ell((g-1)D_0) - \ell(K_X - (g-1)D_0) = \deg((g-1)D_0) + 1 - g = 2(g-1) + 1 - g = g-1.$$

Since $\ell((g-1)D_0) \geq g$, this forces $\ell(K_X - (g-1)D_0) \geq 1$. But the degree of $K_X - (g-1)D_0$ is exactly $(2g-2) - 2(g-1) = 0$. By lemma 7.1.11, a degree-0 divisor with a global section must be principal, so $K_X - (g-1)D_0 \sim 0$, meaning $K_X \sim (g-1)D_0$.

The canonical map therefore factors as:

$$\begin{tikzcd} X \arrow[r, "\varphi", "2:1"'] \arrow[rr, bend left, "\varphi_K"] & \mathbb{P}^1 \arrow[r, hook, "\nu_{g-1}"] & \mathbb{P}^{g-1} \end{tikzcd}$$

where $\nu_{g-1}\colon \mathbb{P}^1 \hookrightarrow \mathbb{P}^{g-1}$ is the Veronese embedding of degree $g-1$ (definition 3.1.36), mapping $[s:t] \mapsto [s^{g-1} : s^{g-2}t : \cdots : t^{g-1}]$. The image $\nu_{g-1}(\mathbb{P}^1) \subseteq \mathbb{P}^{g-1}$ is the rational normal curve of degree $g-1$, and $\varphi_K(X)$ is this rational normal curve, with φ_K being 2-to-1 onto it.

7.1.7 Elliptic Curves

Elliptic curves are arguably the single most studied class of algebraic varieties. They sit at the crossroads of algebraic geometry, number theory, complex analysis, and even mathematical physics and cryptography. For us, they are the curves of genus 1—the first genus where non-trivial phenomena appear—and if there is at least one k-point, they come equipped with an inherent addition structure, i.e. a group law.

We saw in example **??** that the degree-0 Picard group of an elliptic curve is isomorphic to the curve itself, giving a group structure on its points. In example 4.1.12, we noted that this makes elliptic curves into group schemes. We now develop this theory systematically.

This section is a quick application of the theory of curves and of group-schemes. A proper in-depth coverage in the EYNTKA series can be found at [11].

Definition and Basic Properties

Definition 7.1.50: Elliptic Curve

An *elliptic curve* over k is a pair (E, O) where E is a complete nonsingular curve of genus 1 over k, and $O \in E(k)$ is some (nonempty) choice of k-rational point (usually called the *origin* or *identity*).

The requirement that a rational point O exist is essential: a genus-1 curve without a rational point is not an elliptic curve (over k). Over $k = \overline{k}$, every genus-1 curve has rational points (closed points with residue field k), but over non-algebraically-closed fields this can fail[16].

The first application of Riemann–Roch to elliptic curves gives the Weierstrass embedding:

Proposition 7.1.51: Elliptic Curves are Plane Cubics

Let (E, O) be an elliptic curve over k. Then the complete linear system $|3O|$ is very ample and embeds E as a smooth cubic curve in $\mathbb{P}^2_k$.

***Proof*:**
Since $g = 1$, we have $\deg K_E = 2g - 2 = 0$, so $\omega_E \cong \mathcal{O}_E$ (the canonical sheaf is trivial—a distinguishing feature of elliptic curves). By Riemann–Roch (theorem 7.1.19):

$$\begin{aligned}
\ell(O) &= 1 + 1 - 1 + \ell(K - O) = 1 + \ell(-O) = 1 \quad (\text{since } \deg(-O) = -1 < 0),\\
\ell(2O) &= 2 + 1 - 1 + \ell(K - 2O) = 2 + \ell(-2O) = 2,\\
\ell(3O) &= 3 + 1 - 1 + \ell(K - 3O) = 3 + \ell(-3O) = 3 = -3 < 0).
\end{aligned}$$

So $|3O|$ has dimension 2, defining a morphism $\varphi\colon E \to \mathbb{P}^2$. This morphism has degree $\deg(3O) = 3$. To check it is a closed immersion, we verify very ampleness: for any two points P, Q (possibly $P = Q$), we need $\ell(3O) - \ell(3O - P - Q) = 2$. Since $\ell(3O) = 3$, this means we must show $\ell(3O - P - Q) = 1$.

By Riemann–Roch applied to the divisor $D = 3O - P - Q$:

$$\ell(3O - P - Q) = \deg(3O - P - Q) + 1 - g + \ell(K - 3O + P + Q) = 1 + 0 + \ell(-3O + P + Q).$$

Because the degree of the error term is $\deg(-3O + P + Q) = -1 < 0$, it has no global sections, so $\ell(-3O + P + Q) = 0$. Therefore $\ell(3O - P - Q) = 1$ unconditionally for all pairs P, Q. This confirms $|3O|$ is very ample.

Let us make the embedding explicit. Choose a basis for $H^0(E, \mathcal{L}(nO))$ progressively:

- $H^0(E, \mathcal{L}(O)) = \langle 1 \rangle$ (the constant function, which has a pole of order ≤ 1 at O but is regular, so actually no pole).
- $H^0(E, \mathcal{L}(2O))$: contains 1 and a new function x with a pole of order exactly 2 at O (and no other poles).

[16] For example, the Selmer curve $3x^3 + 4y^3 + 5z^3 = 0$ in $\mathbb{P}^2_{\mathbb{Q}}$ has genus 1 but no $\mathbb{Q}$-rational point (this is a nontrivial arithmetic fact). It is a principal homogeneous space ("torsor") for the Jacobian elliptic curve with its natural origin.

- $H^0(E, \mathcal{L}(3O))$: contains $1, x$ and a new function y with a pole of order exactly 3 at O.

Now consider $H^0(E, \mathcal{L}(6O))$, which has dimension $\ell(6O) = 6$. The seven elements $\{1, x, y, x^2, xy, x^3, y^2\}$ all lie in this 6-dimensional space (check: each has a pole of order ≤ 6 at O), so there must be a linear relation among them. After rescaling, this relation takes the form:

$$y^2 + a_1xy + a_3y \ = \ x^3 + a_2x^2 + a_4x + a_6 \tag{7.6}$$

This is the *generalized Weierstrass equation*. It defines a cubic in $\mathbb{P}^2$ (homogenizing with a variable z: the point O maps to $[0 : 1 : 0]$).

Though this is the general form of the Weierstrass equation, in the right field by doing appropriate change of variables it can be often simplified. If $\mathrm{char}(k) \neq 2$, we can complete the square in y to eliminate the xy and y terms. We group the y terms on the left:

$$y^2 + (a_1x + a_3)y = x^3 + a_2x^2 + a_4x + a_6$$

Adding $\left(\frac{a_1x+a_3}{2}\right)^2$ to both sides, we get:

$$\left(y + \frac{a_1x + a_3}{2}\right)^2 = x^3 + a_2x^2 + a_4x + a_6 + \frac{(a_1x + a_3)^2}{4}$$

We make the substitution $y_1 = y + \frac{a_1x+a_3}{2}$. Expanding the right side and grouping by powers of x yields:

$$y_1^2 = x^3 + \left(a_2 + \frac{a_1^2}{4}\right)x^2 + \left(a_4 + \frac{a_1a_3}{2}\right)x + \left(a_6 + \frac{a_3^2}{4}\right)$$

For simplicity, let us denote these new coefficients as a_2', a_4', and a_6', so the equation becomes

$$y_1^2 = x^3 + a_2'x^2 + a_4'x + a_6'$$

Next, if $\mathrm{char}(k) \neq 3$, we can eliminate the x^2 term by completing the cube in x. We make the substitution $x = x_1 - \frac{a_2'}{3}$:

$$y_1^2 = \left(x_1 - \frac{a_2'}{3}\right)^3 + a_2'\left(x_1 - \frac{a_2'}{3}\right)^2 + a_4'\left(x_1 - \frac{a_2'}{3}\right) + a_6'$$

Expanding the cubic and quadratic terms gives:

$$y_1^2 = \left(x_1^3 - a_2'x_1^2 + \frac{a_2'^2}{3}x_1 - \frac{a_2'^3}{27}\right) + \left(a_2'x_1^2 - \frac{2a_2'^2}{3}x_1 + \frac{a_2'^3}{9}\right) + a_4'\left(x_1 - \frac{a_2'}{3}\right) + a_6'$$

Notice that the x_1^2 terms cancel out. Grouping the remaining coefficients for x_1 and the constants leaves us with an equation of the form:

$$y_1^2 = x_1^3 + Ax_1 + B$$

Dropping the subscripts, this transforms (7.6) into the *short Weierstrass form*:

$$y^2 = x^3 + Ax + B, \qquad \Delta = -16(4A^3 + 27B^2) \neq 0.$$

The condition $\Delta \neq 0$ is equivalent to the cubic having no repeated roots, which is equivalent to the curve being nonsingular. Over fields of characteristic 2 or 3, completing the square or the cube requires division by zero, so the short form is not available and one must work with (7.6).

The Group Law via Divisors

We now establish the group structure on $E(k)$ using the theory of divisors. The idea is to identify $E(k)$ with $\mathrm{Pic}^0(E)$ via $P \mapsto [P - O]$.

Theorem 7.1.52: The Group Law on an Elliptic Curve

Let (E, O) be an elliptic curve over $k = \overline{k}$. The map

$$\sigma\colon E(k) \to \mathrm{Pic}^0(E), \qquad P \mapsto \text{class of } [P] - [O],$$

is a bijection. Transporting the group structure from $\mathrm{Pic}^0(E)$ to $E(k)$, the curve E becomes an abelian group with identity O, where:

1. The addition law $P+Q = R$ is characterized by: $[P]+[Q]-[R] \sim [O]$, i.e. $P+Q+R' \sim 3O$ where R' is the "third intersection point" on the line through P and Q (and then R is the third intersection of the line through R' and O).
2. The inverse $-P$ is the third intersection point of the line connecting P and O with E.
3. The maps $+\colon E \times E \to E$ and $[-1]\colon E \to E$ are morphisms of k-schemes.

In particular, (E, O) is a group scheme over k (definition 4.1.1).

***Proof*:**
Bijectivity of σ: Injectivity: if $[P - O] = [Q - O]$ in $\mathrm{Pic}^0(E)$, then $P \sim Q$. This implies that P is an effective divisor in the complete linear system $|Q|$. But by Riemann–Roch, $\ell(Q) = 1+1-1+\ell(-Q) = 1$. Since the linear system $|Q|$ has dimension $\ell(Q)-1 = 0$, it consists of exactly one effective divisor, which is Q itself. Thus $P = Q$.

Surjectivity: let $D \in \mathrm{Pic}^0(E)$, i.e. D is a divisor of degree 0 modulo linear equivalence. Then $D + [O]$ has degree 1, and $\ell(D + O) = 1 + 1 - 1 + \ell(K - D - O) = 1 + \ell(-D - O)$. Since $\deg(-D - O) = -1 < 0$, $\ell(-D - O) = 0$, so $\ell(D + O) = 1$. This means $|D + O|$ contains a unique effective divisor, which must be a single point P (an effective divisor of degree 1 on a curve is a single point). Thus $D + [O] \sim [P]$, i.e. $D \sim [P] - [O] = \sigma(P)$.

The geometric addition law: The "chord-tangent" description in (1) follows from the fact that three collinear points P, Q, R' on a plane cubic satisfy $P+Q+R' \sim 3O$ (they are the intersection of E with a line L, and L is linearly equivalent to the line $z = 0$ that meets E in $3O$). From $P + Q + R' \sim 3O$, we get $\sigma(P) + \sigma(Q) = [P] + [Q] - 2[O] = [3O - R'] - 2[O] = [O] - [R']$, and we need $\sigma(R) = [O] - [R']$, i.e. $[R] - [O] = [O] - [R']$, giving $R + R' \sim 2O$, which means R is the third intersection of the line OR' with E.

Morphism property: One must check that the maps $(P, Q) \mapsto P+Q$ and $P \mapsto -P$ are morphisms of schemes, not set-theoretic maps. This can be verified by writing explicit rational formulas in the Weierstrass coordinates (a computation we omit), or more elegantly by using the functor of points (section 3.3 and definition 3.3.1): one shows that for any k-scheme T, the map $E(T) \times E(T) \to E(T)$ defined by the same divisor-theoretic construction is functorial in T, hence defines a morphism $E \times E \to E$ by Yoneda (theorem 0.4.22).

The group law on an elliptic curve is *commutative*: this follows from $\mathrm{Pic}^0(E)$ being an abelian group.

In the language of group schemes, an elliptic curve is a *commutative group scheme* of dimension 1, or equivalently an *abelian variety of dimension* 1.

A word on the singular case

What happens to the group law when an elliptic curve degenerates into a singular cubic? The group structure does not disappear—it *degenerates* from an abelian variety into a linear algebraic group. Note that elliptic curves are defined to be smooth, and hence we do not say "singular elliptic curves" but rather "singular cubic curves".

Let $C \subseteq \mathbb{P}^2_k$ be a cubic curve with a single singular point P_{sing} (and $\operatorname{char} k \neq 2, 3$). The smooth locus $C_{\text{reg}} = C \setminus \{P_{\text{sing}}\}$ is still a group variety under the chord-tangent law. The chord-tangent construction from theorem 7.1.52 still makes sense on C_{reg}: lines meeting C define effective divisors, and the same "third intersection point" recipe gives a group law on the smooth points. What fails is compactness—C_{reg} is not proper—so we lose the identification with $\operatorname{Pic}^0(C)$ and must work with the *generalized Jacobian* instead. The resulting group depends on the singularity type:

1. Node ($C\colon y^2 = x^3 + x^2$, say): the normalization is $\mathbb{P}^1$ with two preimages of the node glued together (**??**). Removing the singular point leaves $C_{\text{reg}} \cong \mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$—the *multiplicative group* (example 4.1.5). Concretely, the group law on the smooth locus is isomorphic to $(k^*, \cdot)$.

2. Cusp ($C\colon y^2 = x^3$): the normalization is $\mathbb{P}^1$ with a single preimage of the cusp, and removing the singular point gives $C_{\text{reg}} \cong \mathbb{G}_a = \operatorname{Spec} k[t]$—the *additive group* (example 4.1.4). The group law on the smooth locus is isomorphic to $(k, +)$.

Thus the degeneration of elliptic curves mirrors a degeneration of group structures:

$$\underbrace{E}_{\text{abelian variety}} \rightsquigarrow \underbrace{\mathbb{G}_m}_{\text{node}} \rightsquigarrow \underbrace{\mathbb{G}_a}_{\text{cusp}}.$$

This trichotomy has an interesting arithmetic incarnation. If E is an elliptic curve over a local field K (say $K = \mathbb{Q}_p$), the *reduction* of E modulo the maximal ideal gives a cubic curve $\overline{E}$ over the residue field $\mathbb{F}_p$. The reduction type of E is classified by the group structure on $\overline{E}_{\text{reg}}$:

1. *Good reduction:* $\overline{E}$ is smooth; $\overline{E}_{\text{reg}} = \overline{E}$ is an elliptic curve over $\mathbb{F}_p$.

2. *Multiplicative (semistable) reduction:* $\overline{E}$ has a node; $\overline{E}_{\text{reg}} \cong \mathbb{G}_m$. This is further divided into *split* ($\mathbb{G}_m$ over $\mathbb{F}_p$) and *non-split* (a twist of $\mathbb{G}_m$).

3. *Additive (unstable) reduction:* $\overline{E}$ has a cusp; $\overline{E}_{\text{reg}} \cong \mathbb{G}_a$.

The *Néron model* of E over $\mathbb{Z}_p$ makes this classification functorial, and the *conductor* of E measures the severity of the bad reduction—directly paralleling the discriminant and ramification story from section 7.1.4. This is the starting point for the strong interplay between the geometry of singular fibers and the arithmetic of L-functions in the Birch–Swinnerton-Dyer conjecture.

The j-Invariant

Two elliptic curves over k may look different in their Weierstrass equations but be isomorphic as curves. The j-invariant is the *complete* isomorphism invariant over $\overline{k}$.

Definition 7.1.53: The j-Invariant

Let E be an elliptic curve over k with char$k \neq 2, 3$, given in short Weierstrass form $y^2 = x^3 + Ax + B$. The j-*invariant* of E is:

$$j(E) = -1728 \cdot \frac{(4A)^3}{\Delta} = -1728 \cdot \frac{64A^3}{-16(4A^3 + 27B^2)} = 1728 \cdot \frac{4A^3}{4A^3 + 27B^2}.$$

The factor $1728 = 12^3$ ensures that the formula works well in all characteristics. The general definition via the a_i in (7.6) is more involved but exists in all characteristics.

Proposition 7.1.54: j-Invariant Classifies Elliptic Curves

Let $k = \overline{k}$.

1. Two elliptic curves E_1, E_2 over k are isomorphic (as curves with marked point) if and only if $j(E_1) = j(E_2)$.
2. For every $j_0 \in k$, there exists an elliptic curve E over k with $j(E) = j_0$.

In other words, the j-invariant defines a bijection between isomorphism classes of elliptic curves over $\overline{k}$ and the elements of k.

***Proof Sketch*:**
(1) An isomorphism of short Weierstrass curves $y^2 = x^3 + Ax + B$ and $y'^2 = x'^3 + A'x' + B'$ must have the form $x' = u^2 x$, $y' = u^3 y$ for some $u \in k^*$ (this follows from analyzing which changes of variable preserve the Weierstrass form). Under this substitution, $A' = u^4 A$ and $B' = u^6 B$, so $j(E') = 1728 \cdot 4(u^4A)^3/(4(u^4A)^3 + 27(u^6B)^2) = 1728 \cdot 4A^3u^{12}/(4A^3 + 27B^2)u^{12} = j(E)$.

Conversely, if $j(E) = j(E')$, the relation $4A^3/(4A^3 + 27B^2) = 4A'^3/(4A'^3 + 27B'^2)$ constrains A'/A and B'/B so that a suitable u exists.

(2) For $j_0 \neq 0, 1728$: take $A = 3j_0(1728 - j_0)$ and $B = 2j_0(1728 - j_0)^2$. For $j_0 = 0$: take $y^2 = x^3 + 1$. For $j_0 = 1728$: take $y^2 = x^3 + x$.

Over non-algebraically-closed fields, j is a necessary but not sufficient condition for isomorphism: two elliptic curves with the same j-invariant are *twists* of each other, classified by $H^1(\mathrm{Gal}(\overline{k}/k), \mathrm{Aut}(E))$. The classification of twists is an important topic in arithmetic geometry.

7.1.55 The j-Line as $\mathcal{M}_1$ The fact that elliptic curves over $\overline{k}$ are classified by a single invariant $j \in k$ means that the coarse moduli space of elliptic curves is the affine line: $\mathcal{M}_{1,1}^{\text{coarse}} \cong \mathbb{A}^1_k$. However, since elliptic curves have nontrivial automorphisms (at least the inversion $[-1]$, and more at $j = 0$ and $j = 1728$), the fine moduli space does not exist as a scheme—it is a Deligne–Mumford stack $\mathcal{M}_{1,1}$, whose coarse space is $\mathbb{A}^1$. This is the simplest nontrivial example of a moduli stack, and a perfect entry point for *Moduli, Deformation Theory, and Stacks* [17].

The Degree-Zero Picard Group and the Jacobian

We have shown that for an elliptic curve (E, O), the map $P \mapsto [P - O]$ is an isomorphism $E(k) \xrightarrow{\sim} \operatorname{Pic}^0(E)$. Let us place this in the broader context of the Jacobian and the Picard scheme.

For a general smooth projective curve X of genus g, the group $\operatorname{Pic}^0(X)$ is far richer than X itself. It is the group of k-points of an abelian variety $\operatorname{Jac}(X)$ of dimension g, called the *Jacobian* of X:

$$\operatorname{Jac}(X)(k) \;\cong\; \operatorname{Pic}^0(X).$$

For $g = 1$ with a marked point O, the Jacobian is isomorphic to (X, O) itself—this is the content of theorem 7.1.52. For $g \geq 2$, the Jacobian is a higher-dimensional abelian variety (an abelian variety of dimension g), and the *Abel–Jacobi map*

$$\alpha\colon X \hookrightarrow \operatorname{Jac}(X), \qquad P \mapsto [P - O] \quad \text{(for a choice of base point } O\text{)},$$

embeds X into its Jacobian as a curve. The Jacobian is the “abelian envelope” of the curve—it is the universal abelian variety through which morphisms from X to abelian varieties factor (the “Albanese” property).

7.1.56 Forward Reference: The Picard Scheme and Moduli The construction $X \mapsto \operatorname{Pic}^0(X)$ is not just a group—it is functorial, and it is representable. More precisely, the functor $T \mapsto \operatorname{Pic}^0(X_T/T)$ (families of degree-0 line bundles on fibers of X over a base T) is represented by a group scheme $\operatorname{Pic}^0_{X/k}$, the *Picard scheme*. For a smooth projective curve, $\operatorname{Pic}^0_{X/k}$ is an abelian variety—the Jacobian.

The existence and properties of the Picard scheme are among the crowning achievements of Grothendieck's foundations, and they play a central role in *Moduli, Deformation Theory, and Stacks* [17]: the Picard scheme is the starting point for constructing $\mathcal{M}_g$ (via the Torelli map $X \mapsto (\operatorname{Jac}(X), \Theta)$, where Θ is the theta divisor), and it provides the link between divisor theory and moduli theory.

Also: the Néron–Severi group $\operatorname{NS}(X) = \operatorname{Pic}(X)/\operatorname{Pic}^0(X)$, which is $\cong \mathbb{Z}$ for a curve (generated by the degree map), becomes a finitely generated group of potentially large rank for surfaces and higher-dimensional varieties. This will be essential in the surface theory of §7.2.1.

Isogenies and the Endomorphism Ring

A key feature of elliptic curves is that morphisms between them (respecting the group structure) are extremely well-behaved.

Definition 7.1.57: Isogeny

An *isogeny* $\varphi\colon E_1 \to E_2$ between elliptic curves is a morphism of schemes that sends O_1 to O_2 and is nonconstant (hence surjective and finite, by proposition 7.1.5).

Any isogeny is automatically a homomorphism of group schemes[17]: $\varphi(P+Q) = \varphi(P) + \varphi(Q)$. The most important isogeny is the multiplication-by-n map:

Proposition 7.1.58: Multiplication by n

For any integer $n \neq 0$, the map $[n]\colon E \to E$ defined by $P \mapsto \underbrace{P + P + \cdots + P}_{n}$ (and $[-n] = [n] \circ [-1]$) is a finite morphism of degree n^2.

***Proof Sketch*:**
That $[n]$ is a morphism follows from the group scheme structure. It is nonconstant for $n \neq 0$ (since E has points of infinite order when $\operatorname{char} k = 0$, or by a more careful argument in positive characteristic). For the degree: the kernel $E[n] = \ker([n])$ is a finite group scheme. When $\operatorname{char} k \nmid n$, it is étale of rank n^2, so $\deg[n] = n^2$. This can be proved by computing the degree of the divisor $[n]^*(O)$ using the addition formulas, or by using the Tate module argument below.

The *n-torsion subgroup* $E[n]$ is the kernel of $[n]$: as a group scheme, it is a finite group scheme of order (rank) n^2. Its structure depends on the characteristic:

Proposition 7.1.59: Structure of the n-Torsion

Let E be an elliptic curve over $k = \overline{k}$.

1. If $\operatorname{char} k \nmid n$ (including $\operatorname{char} k = 0$): $E[n](\overline{k}) \cong (\mathbb{Z}/n\mathbb{Z})^2$ as abstract groups (and $E[n]$ is an étale group scheme over k).
2. If $\operatorname{char} k = p$ and $n = p^m$: the group scheme $E[p^m]$ has rank p^{2m}, but its group of $\overline{k}$-points is either $E[p^m](\overline{k}) \cong \mathbb{Z}/p^m\mathbb{Z}$ (the *ordinary* case) or $E[p^m](\overline{k}) \cong \{0\}$ (the *supersingular* case), depending on the Hasse invariant of E. In the supersingular case, the group scheme is connected (supported entirely at the origin).

The *endomorphism ring* $\operatorname{End}(E) = \operatorname{Hom}(E, E)$ (where Hom means isogenies plus the zero map) is a ring under addition and composition. Its structure is constrained:

[17] This is a nontrivial "rigidity" result. It follows from the fact that the map $\varphi(P+Q) - \varphi(P) - \varphi(Q)$ is a morphism $E_1 \times E_1 \to E_2$ sending (O_1, O_1) to O_2; by the rigidity lemma for abelian varieties, such a map is constant, hence identically O_2.

Proposition 7.1.60: Endomorphism Ring of an Elliptic Curve

Let E be an elliptic curve over $k = \overline{k}$. Then $\mathrm{End}(E)$ is isomorphic to one of the following:

1. $\mathbb{Z}$ (the generic case).
2. An order in an imaginary quadratic field $\mathbb{Q}(\sqrt{-d})$ (the *complex multiplication* or *CM* case). Over $\mathbb{C}$, this occurs when the lattice $\Lambda \subseteq \mathbb{C}$ with $E \cong \mathbb{C}/\Lambda$ has extra symmetry.
3. A maximal order in a quaternion algebra over $\mathbb{Q}$ (only in characteristic $p > 0$; these are the *supersingular* elliptic curves).

7.1.61 Forward Reference: The Tate Module and ℓ-adic Representations Fix a prime $\ell \neq \mathrm{char} k$. The *$\ell$-adic Tate module* of E is the inverse limit:

$$T_\ell(E) \;=\; \varprojlim_m E[\ell^m](\overline{k}) \;\cong\; \mathbb{Z}_\ell^2.$$

This is a free $\mathbb{Z}_\ell$-module of rank 2 carrying a continuous action of $\mathrm{Gal}(\overline{k}/k)$. The resulting representation $\rho_\ell \colon \mathrm{Gal}(\overline{k}/k) \to \mathrm{GL}_2(\mathbb{Z}_\ell)$ is the simplest example of an *ℓ-adic Galois representation*. When $k = \mathbb{F}_q$ is a finite field, the Frobenius endomorphism $\mathrm{Frob}_q \in \mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q)$ acts on $T_\ell(E)$ with characteristic polynomial $t^2 - a_q t + q$, where $a_q = q + 1 - |E(\mathbb{F}_q)|$. The *Riemann hypothesis for elliptic curves over finite fields*—proved by Hasse—states that $|a_q| \leq 2\sqrt{q}$, or equivalently that the roots of $t^2 - a_q t + q$ have absolute value $\sqrt{q}$. This is the genus-1 case of the Weil conjectures, which will be a central topic of a separate volume.
For curves of higher genus, the Tate module is replaced by the ℓ-adic cohomology group $H^1_{\text{ét}}(X, \mathbb{Z}_\ell)$, a free $\mathbb{Z}_\ell$-module of rank $2g$. The Weil conjectures (proved by Deligne for varieties of arbitrary dimension) assert that the eigenvalues of Frobenius on $H^i_{\text{ét}}$ have absolute value $q^{i/2}$—a vast generalization of Hasse's bound.

We conclude with a brief but important observation about the arithmetic of elliptic curves.

Example 7.1.62: Elliptic Curves over Number Fields

If E is an elliptic curve over a number field K (e.g. $K = \mathbb{Q}$), the group $E(K)$ of K-rational points is a finitely generated abelian group by the *Mordell–Weil theorem*:

$$E(K) \;\cong\; \mathbb{Z}^r \oplus E(K)_{\text{tors}}$$

where $r \geq 0$ is the *Mordell–Weil rank* and $E(K)_{\text{tors}}$ is finite. Computing r is an extraordinarily difficult problem, connected to the *Birch–Swinnerton-Dyer conjecture* (one of the Millennium Prize Problems), which predicts that r equals the order of vanishing of the L-function $L(E, s)$ at $s = 1$.

7.1.8 Further Topics on Curves

We collect several additional results on curves that round out the theory and connect to later chapters. This section is more survey-like in character; the reader may skip it on a first pass and return as needed.

Castelnuovo's Bound and Space Curves

Our embedding theorems (corollary 7.1.25) show that any curve can be embedded in projective space, but they say nothing about which pairs (d, g) of degree and genus can occur for a curve in a given $\mathbb{P}^r$. This is an important question, especially for "space curves" (curves in $\mathbb{P}^3$).

Theorem 7.1.63: Castelnuovo's Bound

Let $X \subseteq \mathbb{P}^r$ be a nondegenerate curve of degree d (i.e. X is not contained in any hyperplane). Write $d - 1 = m(r-1) + \epsilon$ with $0 \leq \epsilon \leq r - 2$ (Euclidean division). Then:

$$g(X) \ \leq \ \pi(d, r) \ := \ \binom{m}{2}(r-1) + m\epsilon.$$

The bound $\pi(d, r)$ is called the *Castelnuovo bound*, and curves achieving equality are called *Castelnuovo curves* or *curves of maximal genus*. They exist for all admissible (d, r) and have been completely classified: they lie on surfaces of minimal degree in $\mathbb{P}^r$.

Example 7.1.64: Castelnuovo's Bound in Low Dimension

1. **Plane curves** $(r = 2)$**:** $d - 1 = m \cdot 1 + 0$, so $\pi(d, 2) = \binom{d-1}{2} = \frac{(d-1)(d-2)}{2}$. This recovers the genus formula for smooth plane curves—and shows that smooth plane curves are Castelnuovo curves.
2. **Space curves** $(r = 3)$**:** $d - 1 = 2m + \epsilon$ with $\epsilon \in \{0, 1\}$. For example, a degree-6 curve in $\mathbb{P}^3$: $d - 1 = 5 = 2 \cdot 2 + 1$, giving $\pi(6, 3) = \binom{2}{2} \cdot 2 + 2 \cdot 1 = 4$. So a nondegenerate curve of degree 6 in $\mathbb{P}^3$ has genus ≤ 4.

We omit the proof of Castelnuovo's bound, which uses the "uniform position principle" and careful Hilbert function estimates. The interested reader can find it in [33, § IV.6] or [64, § 21.7]. The bound is a key ingredient in the study of the Hilbert scheme of curves in $\mathbb{P}^r$, treated in *Moduli, Deformation Theory, and Stacks* [17].

Riemann–Roch in Families and Semicontinuity

One of the most powerful aspects of our cohomological machinery is that it works in *families*. Suppose $f : \mathcal{X} \to S$ is a flat proper family of curves (a flat proper morphism whose geometric fibers are smooth curves of genus g), and $\mathcal{L}$ is a line bundle on $\mathcal{X}$. For each point $s \in S$, we get a curve $X_s = f^{-1}(s)$ and a line bundle $\mathcal{L}_s = \mathcal{L}|_{X_s}$, and we may ask: how does $h^0(X_s, \mathcal{L}_s) = \ell(\mathcal{L}_s)$ vary with s?

The Euler characteristic behaves perfectly: by the flatness of f and the constancy of the Hilbert polynomial in flat families (theorem 6.7.5), $\chi(\mathcal{L}_s) = \deg \mathcal{L}_s + 1 - g$ is *constant* as s varies (since $\deg \mathcal{L}_s$ and g are locally constant in flat families). But the individual cohomology dimensions $h^i(X_s, \mathcal{L}_s)$ can jump.

The *semicontinuity theorem* (theorem 6.8.9) tells us that the jumping goes in a controlled direction:

1. The function $s \mapsto h^i(X_s, \mathcal{L}_s)$ is upper semicontinuous: it can only *jump up* on closed subsets of S.

2. Since χ is constant, if h^0 jumps up at some fiber then h^1 must also jump up to compensate.

Grauert's theorem (section 6.8.4) gives the converse: if $h^i(X_s, \mathcal{L}_s)$ is *constant* for all $s \in S$, then the higher direct image $R^i f_* \mathcal{L}$ is a locally free sheaf on S whose fiber at s is $H^i(X_s, \mathcal{L}_s)$, and moreover the base-change map is an isomorphism.

Example 7.1.65: Semicontinuity for Curves
Consider a flat family of smooth curves $\mathcal{X} \to S$ of genus $g = 3$ over a connected base S, equipped with a line bundle $\mathcal{L}$ of degree 3 on each fiber. Then $\chi(\mathcal{L}_s) = 3 + 1 - 3 = 1$ for all s. Generically, $h^0 = 1$ and $h^1 = 0$; but at special fibers where the divisor becomes "special" (i.e. $\ell(K - D) > 0$), both h^0 and h^1 jump up. The locus in S where this jump occurs is a *proper closed subset*—the "Brill–Noether locus." The study of such loci is one of the richest areas of modern curve theory.

7.1.66 Forward Reference: Deformation Theory of Curves The tangent space to the deformation functor of a smooth curve X of genus g is $H^1(X, T_X)$, where $T_X = \Omega^\vee_{X/k}$ is the tangent sheaf. By Serre duality, $H^1(X, T_X) \cong H^0(X, \omega_X^{\otimes 2})^\vee$, and by Riemann–Roch:

$$\dim H^0(X, \omega_X^{\otimes 2}) = \begin{cases} 0 & \text{if } g = 0, \\ 1 & \text{if } g = 1, \\ 3g - 3 & \text{if } g \geq 2. \end{cases}$$

This is the dimension of the moduli space $\mathcal{M}_g$! The obstructions to deformation lie in $H^2(X, T_X) = 0$ (since $\dim X = 1$), so the deformation space is always smooth. These ideas will be made precise using Schlessinger's criterion and cotangent complexes in *Moduli, Deformation Theory, and Stacks* [17].

7.1.67 Singular Curves in Moduli: Stable Curves and $\overline{\mathcal{M}}_g$ We have seen throughout this chapter that the theory of smooth curves is clean. But the moduli space $\mathcal{M}_g$ itself is *not* clean in one crucial respect: it is not compact (not proper as an algebraic space). Geometrically, a family of smooth curves can degenerate—smooth fibers can "pinch" and acquire singularities. The semicontinuity theorem (theorem 6.8.9) tells us that cohomological invariants jump in a controlled way, and the arithmetic genus is preserved by flatness (theorem 6.7.5), but the fibers themselves leave the category of smooth curves.

To compactify $\mathcal{M}_g$, we must enlarge the class of curves we parametrize. The *Deligne–Mumford compactification* $\overline{\mathcal{M}}_g$ adds as boundary points the *stable curves*: connected, complete curves X of arithmetic genus g whose only singularities are *nodes* (ordinary double points), subject to the stability condition $|\mathrm{Aut}(X)| < \infty$.

Why nodes and not cusps or worse? Several compelling reasons converge:

1. **Flatness and p_a.** A flat family whose smooth fibers have genus g can degenerate to a nodal curve of the same arithmetic genus: nodes contribute $\delta = 1$ to the arithmetic genus (box 7.1.18), so a rational curve with g nodes has $p_a = g$, matching the smooth fibers. By contrast, a cusp also has $\delta = 1$, but cuspidal singularities are "unstable"—they can be further deformed (perturbed into nodes or smoothed entirely), so they do not represent "maximally degenerate" limits.
2. **The dualizing sheaf.** On a nodal curve X of arithmetic genus $g \geq 2$, the dualizing sheaf ω_X° (**??**) is an ample line bundle if and only if the stability condition holds. This gives the compactification a clean moduli-theoretic meaning: stable curves are precisely the curves for which ω_X° provides a canonical polarization.
3. **The group-theoretic perspective.** Nodes give the mildest possible degeneration of the group structure: on a singular fiber of a degenerating family of elliptic curves, a node yields a $\mathbb{G}_m$-fiber (multiplicative reduction), while a cusp yields a $\mathbb{G}_a$-fiber (additive reduction) (**??**). The multiplicative case is "semistable" in the sense of geometric invariant theory, while the additive case is not.

The construction of $\overline{\mathcal{M}}_g$ as a proper Deligne–Mumford stack, and the proof that it is a projective variety (via the Knudsen–Mumford theorem), is one of the triumphs of *Moduli, Deformation Theory, and Stacks* [17]. For now, the reader should note the moral: *singular curves are not pathologies to be avoided, but essential objects that complete the moduli picture.* The theory of singular curves we have touched on in **??????** and box 7.1.18 provides exactly the foundations one needs to work at the boundary of $\overline{\mathcal{M}}_g$.

Curves over Non-Algebraically-Closed Fields

Throughout this chapter, we have assumed $k = \overline{k}$ for simplicity. Let us briefly indicate what changes over a general field k.

The main issues are:

1. **Rational points may not exist.** A smooth projective curve of genus 0 over k is isomorphic to a conic in $\mathbb{P}^2_k$, which is isomorphic to $\mathbb{P}^1_k$ if and only if it has a k-rational point (by the same Riemann–Roch argument as proposition 7.1.28, but $\ell(P) = 2$ requires a k-rational point P).

Over $\mathbb{Q}$, the existence of a rational point on a conic is determined by the Hasse–Minkowski theorem (local-global principle).

2. **Genus-1 curves without rational points.** A genus-1 curve without a k-rational point is not an elliptic curve (it has no origin for the group law). Such a curve is a *torsor* (principal homogeneous space) for its Jacobian, and its isomorphism class is an element of $H^1(k, E)$ (Galois cohomology with coefficients in the elliptic curve). The Shafarevich–Tate group $\mathrm{Sha}(E/k)$ classifies those torsors that are locally trivial at all completions of k.

3. **Regularity $\neq$ smoothness.** Over imperfect fields, a regular scheme need not be smooth (example 2.7.14). This complicates the theory of differentials and the adjunction formula.

Over finite fields $\mathbb{F}_q$, however, the theory is both tractable and extraordinarily rich:

Example 7.1.68: The Hasse–Weil Bound

Let X be a smooth projective curve of genus g over $\mathbb{F}_q$. The *Hasse–Weil bound* states:

$$\big||X(\mathbb{F}_q)| - (q+1)\big| \;\leq\; 2g\sqrt{q}.$$

For $g = 0$ ($X = \mathbb{P}^1$): $|X(\mathbb{F}_q)| = q + 1$ exactly. For $g = 1$ (elliptic curves): this is Hasse's theorem $|a_q| \leq 2\sqrt{q}$ from box 7.1.61. The general case is the *Riemann hypothesis for curves over finite fields*, proved by Weil in 1948. It is the statement that the eigenvalues of the Frobenius endomorphism on $H^1_{\text{ét}}(X, \mathbb{Q}_\ell)$ have absolute value $\sqrt{q}$, and its proof uses the theory of the Jacobian variety and intersection theory on $X \times X$. A full account will appear in a separate volume.

This concludes our treatment of curves. We have seen how the abstract cohomological machinery of the previous chapters—Riemann–Roch, Serre duality, and differentials—yields a complete picture: curves are classified by genus, their maps are controlled by Hurwitz, their embeddings by linear systems, and their families by semicontinuity. We now turn to the two-dimensional world, where the picture becomes vastly richer.

Part B: Algebraic Surfaces

7.2 Algebraic Surfaces

We now enter the world of surfaces, where the picture changes dramatically. On curves, the birational geometry was trivial: every birational map between smooth projective curves extends to an isomorphism. For surfaces, this fails spectacularly. One can *blow up* a point on a smooth surface to produce a new smooth surface that is birational but *not* isomorphic to the original. The theory of surfaces is, in large measure, the theory of understanding and controlling these blow-ups.

At the same time, surfaces are the first dimension where *intersection theory* becomes nontrivial: two curves on a surface meet in finitely many points, and the number of such points (counted with multiplicity) is a bilinear pairing on divisors. This intersection pairing is the engine that drives the entire theory, from the Riemann–Roch theorem for surfaces to the Enriques–Kodaira classification.

The reader who has absorbed the theory of curves will find that many of the same tools apply—divisors, linear systems, Serre duality, the canonical sheaf—but that the two-dimensional setting introduces genuinely new phenomena: self-intersection numbers, the Hodge index theorem, the Kodaira dimension, and the possibility of negative curves. These phenomena are not complications; they are the source of the extraordinary richness of surface theory.

Definition and First Examples

Definition 7.2.1: Surface

A *surface over* k is a Noetherian, integral, separated scheme X of dimension 2 and of finite type over k. We say:

1. X is *projective* if there exists a closed immersion $X \hookrightarrow \mathbb{P}^n_k$ for some n.
2. X is *nonsingular* (or *smooth*) if all local rings $\mathcal{O}_{X,x}$ are regular, or equivalently (over $k = \overline{k}$) if the structure morphism is smooth.

Unless otherwise stated, in this chapter a *surface* will mean a *smooth projective surface over* $k = \overline{k}$.

Unlike curves, where properness implies projectivity (corollary 7.1.27), there exist proper smooth surfaces that are not projective—Hironaka constructed such examples in dimension 3, and they exist in dimension 2 as well (though they are somewhat exotic). We restrict to projective surfaces to have an ample line bundle at our disposal.

Example 7.2.2: The Basic Surfaces

1. **The projective plane $\mathbb{P}^2$.** The simplest surface. It has $\operatorname{Pic}(\mathbb{P}^2) \cong \mathbb{Z}$, generated by the class of a line H. A curve of degree d on $\mathbb{P}^2$ has class dH. The canonical class is $K_{\mathbb{P}^2} = -3H$ (from the Euler sequence, theorem 5.5.18, applied to $\bigwedge^2$). Since K is anti-ample, $\mathbb{P}^2$ has "many" rational curves and is, in a sense we will make precise, as far from "general type" as possible.
2. **The product $\mathbb{P}^1 \times \mathbb{P}^1$.** This is the simplest *ruled surface*—a surface fibered over $\mathbb{P}^1$ with fibers isomorphic to $\mathbb{P}^1$. Its Picard group is $\operatorname{Pic}(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z}^2$, generated by the classes of the two rulings $F = \{p\} \times \mathbb{P}^1$ and $G = \mathbb{P}^1 \times \{q\}$. A divisor of *bidegree* (a, b) is one of class $aF + bG$. The canonical class is $K = -2F - 2G$. This surface is not isomorphic to $\mathbb{P}^2$ (their Picard groups differ!), but it is *birational* to $\mathbb{P}^2$: an explicit birational map is given by blowing up a point of $\mathbb{P}^2$ and then contracting the strict transform of a line through it.
3. **Smooth hypersurfaces in $\mathbb{P}^3$.** A smooth surface $S \subseteq \mathbb{P}^3$ of degree d has $\operatorname{Pic}(S) \cong \mathbb{Z}$ (generated by the hyperplane class H) for $d \geq 4$ by the Noether–Lefschetz theorem[a]. The canonical class is $K_S = (d-4)H$ by the adjunction formula (proposition 6.5.7). This yields a critical trichotomy:
   (a) $d \leq 3$: K_S is anti-ample (rational surfaces).

   (b) $d = 4$: $K_S = 0$ (K3 surfaces!).
   (c) $d \geq 5$: K_S is ample (surfaces of general type).

4. **Hirzebruch surfaces $\mathbb{F}_n$.** For each $n \geq 0$, the nth Hirzebruch surface is $\mathbb{F}_n = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-n))$, the projectivization of a rank-2 bundle on $\mathbb{P}^1$. It is a ruled surface with a special section C_0 satisfying $C_0^2 = -n$ and fibers F with $F^2 = 0$, $C_0 \cdot F = 1$. Notable cases: $\mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{F}_1 \cong \mathrm{Bl}_p \mathbb{P}^2$ (the blow-up of $\mathbb{P}^2$ at a point, which we will study in §7.2.4).

[a] The Noether–Lefschetz theorem says that for a *very general* smooth surface of degree ≥ 4 in $\mathbb{P}^3$, $\mathrm{Pic}\,(S) \cong \mathbb{Z}$. For *special* surfaces, the Picard group can be larger. The proof uses Hodge theory (specifically, the fact that $H^{1,1}(S) \cap H^2(S, \mathbb{Z}) = \mathbb{Z} \cdot H$ for a very general S).

7.2.3 Why Surfaces Are Harder Than Curves The central difference between curves and surfaces can be stated in one sentence: *on curves, every birational map between smooth projective models extends to an isomorphism; on surfaces, it does not.*

The reason is that on a smooth surface, one can *blow up* a closed point to produce a new smooth surface containing an *exceptional curve* $E \cong \mathbb{P}^1$ with $E^2 = -1$. This blow-up is birational (it is an isomorphism away from the point), but the new surface is not isomorphic to the old one. The theory of surfaces must therefore contend with a fundamental question that does not arise for curves: given a smooth projective surface X, is there a "simplest" or "minimal" birational model?

The answer—which is the content of the *minimal model theory* for surfaces—is yes: one can always contract (-1)-curves (by Castelnuovo's criterion, theorem 7.2.31) to arrive at a *minimal surface*. When the Kodaira dimension is ≥ 0, the minimal model is unique; when $\kappa = -\infty$, there are finitely many minimal models ($\mathbb{P}^2$ and the Hirzebruch surfaces $\mathbb{F}_n$ with $n \neq 1$).

In dimensions ≥ 3, the situation becomes far more complex: contracting curves can produce singularities, and one must allow *flips*—a story that is the subject of the Minimal Model Program (a separate volume).

Divisors and Linear Systems on Surfaces

Let X be a smooth projective surface over $k = \overline{k}$. Since X is regular, every local ring $\mathcal{O}_{X,x}$ is a regular local ring, hence a UFD (proposition 2.7.20). Therefore X is locally factorial (definition 2.4.29), and the natural map from Cartier divisors to Weil divisors is an isomorphism:

$$\mathrm{CaCl}\, X \;\cong\; \mathrm{Cl}\, X \;\cong\; \mathrm{Pic}\, X.$$

As with curves, we freely identify divisors with line bundles and use additive and multiplicative notation interchangeably.

Divisors on Surfaces

A *divisor* on X is a Weil divisor $D = \sum n_i C_i$, where the C_i are irreducible curves (codimension-1 subvarieties) on X and $n_i \in \mathbb{Z}$. Note the qualitative change from curves: on a curve, a divisor is a formal sum of *points*; on a surface, a divisor is a formal sum of *curves*. This is the shift in dimension that makes surface theory geometrically richer.

The associated invertible sheaf $\mathcal{L}(D)$ and the spaces of global sections $H^0(X, \mathcal{L}(D))$ are defined exactly as before. The complete linear system $|D|$ parametrizes effective divisors linearly equivalent to D, and is identified with $\mathbb{P}(H^0(X, \mathcal{L}(D)))$.

The Néron–Severi Group and the Picard Number

For a curve, the Picard group has a clean structure: $\operatorname{Pic}(X) \cong \operatorname{Pic}^0(X) \oplus \mathbb{Z}$, where Pic^0 is the Jacobian (connected, an abelian variety) and $\mathbb{Z}$ is the degree. For surfaces, the situation is more complex.

Definition 7.2.4: Néron–Severi Group and Picard Number

Let X be a smooth projective surface. The *Néron–Severi group* of X is:

$$\mathrm{NS}(X) \;=\; \operatorname{Pic}(X)/\operatorname{Pic}^0(X)$$

where $\operatorname{Pic}^0(X)$ is the group of line bundles that are *algebraically equivalent* to zero (equivalently, the group of k-points of the identity component of the Picard scheme $\operatorname{Pic}_{X/k}$). The *Picard number* (or *rank of the Néron–Severi group*) is $\rho(X) = \mathrm{rk}\mathrm{NS}(X)$.

By the *theorem of the base* (Néron–Severi theorem), $\mathrm{NS}(X)$ is a finitely generated abelian group. Its rank ρ is a fundamental invariant of X. Over $\mathbb{C}$, $\rho \leq h^{1,1}$ (the Hodge number), with equality holding only in special cases.

Example 7.2.5: Picard Groups of Basic Surfaces

1. $\operatorname{Pic}(\mathbb{P}^2) \cong \mathbb{Z}$, generated by the hyperplane class H. Here $\operatorname{Pic}^0 = 0$, $\mathrm{NS} \cong \mathbb{Z}$, $\rho = 1$.
2. $\operatorname{Pic}(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z}^2$, generated by the two rulings F and G. Again $\operatorname{Pic}^0 = 0$, $\mathrm{NS} \cong \mathbb{Z}^2$, $\rho = 2$.
3. For a smooth quartic surface $S \subseteq \mathbb{P}^3$ (a K3 surface): $\operatorname{Pic}^0(S) = 0$ (since $h^{0,1} = 0$), so $\operatorname{Pic}(S) = \mathrm{NS}(S)$. For a *very general* quartic, $\rho = 1$; but special quartics can have ρ as large as 20 (the maximum for K3 surfaces, achieved by the Fermat quartic $x^4 + y^4 + z^4 + w^4 = 0$ over $\overline{\mathbb{Q}}$, which has $\rho = 20$).
4. For an abelian surface A (a 2-dimensional abelian variety): $\operatorname{Pic}^0(A) \cong \widehat{A}$ (the dual abelian variety), which has dimension 2. The Néron–Severi group satisfies $1 \leq \rho \leq 4$. The group $\operatorname{Pic}^0(A)$ is "large"—it has the same dimension as A itself.

Ample, Nef, and Effective Divisors

On a curve, a divisor of positive degree is ample. On a surface, ampleness is more subtle. We will need several notions of "positivity" for divisors on surfaces:

1. A divisor D is *ample* if some positive multiple nD defines a closed immersion into projective space. By the Nakai–Moishezon criterion (which we will state in §7.2.1), D is ample if and only if $D^2 > 0$ and $D \cdot C > 0$ for every irreducible curve $C \subseteq X$.

2. A divisor D is *nef* ("numerically effective") if $D \cdot C \geq 0$ for every irreducible curve $C \subseteq X$.
3. A divisor D is *big* if $\ell(nD)$ grows like n^2 (quadratically, the maximum possible rate for a surface). Equivalently, the rational map $\varphi_{|nD|}$ is birational onto its image for $n \gg 0$.
4. A divisor D is *effective* if $D \geq 0$ (all coefficients non-negative).

These notions will be essential for the Enriques–Kodaira classification (§7.2.6). On curves, ample = positive degree and nef = non-negative degree, so all four notions collapse to conditions on the degree. The richer structure on surfaces is what makes their geometry so much more interesting.

7.2.6 The Cone of Curves and Duality The ampleness and nefness of divisors are best understood through the *cone of curves* $\mathrm{NE}(X) \subseteq \mathrm{NS}(X) \otimes \mathbb{R}$, defined as the convex cone generated by classes of irreducible curves. A divisor D is nef if and only if D pairs non-negatively with every element of $\mathrm{NE}(X)$, i.e. D lies in the *dual cone* $\mathrm{NE}(X)^\vee$ (the "nef cone"). The *Kleiman criterion* says that D is ample if and only if it lies in the interior of the nef cone.
In a separate volume, the *Cone Theorem* of Mori will describe the structure of $\mathrm{NE}(X)$: it is "rational polyhedral" on the K_X-negative side, generated by classes of rational curves. The extremal rays of this cone correspond to contractions of X, and the study of these contractions is the starting point of the Minimal Model Program.

7.2.1 Intersection Theory on Surfaces

We now develop the intersection pairing on a smooth projective surface—the single most important tool in the theory of surfaces. The idea is ancient and intuitive: two curves on a surface "should" meet in finitely many points, and the number of intersection points (counted with appropriate multiplicity) should depend only on the linear equivalence classes of the curves. Making this precise, and proving the resulting pairing has the properties one expects, is the content of this section.

This is the reader's first encounter with a genuine intersection product. *Intersection Theory* [16] develops intersection theory in full generality (Chow rings, Chern classes, the moving lemma), but the surface case is special enough to admit a self-contained treatment using only the cohomological tools we already have. The reader should view this section as both a preview of and a motivation for the general theory.

The Intersection Pairing

Let X be a smooth projective surface over $k = \overline{k}$. We wish to define a bilinear pairing:

$$\mathrm{Pic}\,(X) \times \mathrm{Pic}\,(X) \longrightarrow \mathbb{Z}, \qquad (D_1, D_2) \mapsto D_1 \cdot D_2.$$

There are several equivalent definitions; we begin with the one that follows most directly from the cohomology developed above.

Definition 7.2.7: Intersection Number (Cohomological)

Let X be a smooth projective surface and D_1, D_2 divisors on X. The *intersection number* of D_1 and D_2 is:

$$D_1 \cdot D_2 \;=\; \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-D_1)) - \chi(\mathcal{O}_X(-D_2)) + \chi(\mathcal{O}_X(-D_1 - D_2)).$$

This definition may seem opaque at first glance. The motivation comes from the inclusion-exclusion principle: if D_1 and D_2 are effective and meet transversally, then the "error" in the Euler characteristic when we subtract D_1 and D_2 separately versus subtracting their union should count exactly the points where they overlap. Let us verify that this formula gives the expected answer in simple cases.

Proposition 7.2.8: Properties of the Intersection Pairing

The intersection pairing has the following properties:

1. **Bilinearity:** $D_1 \cdot D_2$ is $\mathbb{Z}$-bilinear in D_1 and D_2.
2. **Symmetry:** $D_1 \cdot D_2 = D_2 \cdot D_1$.
3. **Dependence only on linear equivalence:** if $D_1 \sim D_1'$ and $D_2 \sim D_2'$, then $D_1 \cdot D_2 = D_1' \cdot D_2'$.
4. **Transversal effective divisors:** if C_1 and C_2 are effective divisors with no common component that meet transversally (i.e. their local equations generate the maximal ideal at each intersection point), then:
$$C_1 \cdot C_2 \;=\; |C_1 \cap C_2| \;=\; \#\{P \in X : P \in C_1 \cap C_2\}.$$
5. **General effective divisors:** if C_1 and C_2 are effective with no common component, then:
$$C_1 \cdot C_2 \;=\; \sum_{P \in C_1 \cap C_2} i_P(C_1, C_2)$$
where $i_P(C_1, C_2) = \dim_k \mathcal{O}_{X,P}/(f_1, f_2) \geq 1$ is the *local intersection multiplicity* at P (here f_1, f_2 are local equations for C_1 and C_2 at P).

***Proof*:**
(1)–(3): These follow formally from the properties of the Euler characteristic. For bilinearity, we use the additivity of χ on short exact sequences (proposition 6.7.2). The key computation is: for any three divisors A, B, C, we verify that $D_1 \cdot D_2$ as defined is linear in D_1 by replacing D_1 with $D_1 + E$ and expanding, using:

$$\chi(\mathcal{O}_X(-D_1 - E)) = \chi(\mathcal{O}_X(-D_1)) + \chi(\mathcal{O}_X(-E)) - \chi(\mathcal{O}_X) + (\text{cross terms}).$$

The cross terms assemble into $E \cdot D_2$, giving linearity. Symmetry is immediate from the definition. Dependence only on linear equivalence follows because $\chi(\mathcal{L})$ depends only on the isomorphism class of $\mathcal{L}$, hence only on the linear equivalence class of the corresponding divisor.

(4)–(5): These require a more hands-on argument. Consider the short exact sequence associated to an effective divisor C_2 on X:

$$0 \to \mathcal{O}_X(-C_2) \to \mathcal{O}_X \to \mathcal{O}_{C_2} \to 0. \tag{7.7}$$

Twisting by $\mathcal{O}_X(-C_1)$ (i.e. tensoring with $\mathcal{L}(-C_1)$, which is flat since it is locally free):

$$0 \to \mathcal{O}_X(-C_1 - C_2) \to \mathcal{O}_X(-C_1) \to \mathcal{O}_{C_2}(-C_1) \to 0$$

where $\mathcal{O}_{C_2}(-C_1) = \mathcal{O}_{C_2} \otimes \mathcal{L}(-C_1)$. Taking Euler characteristics and using (7.7) untwisted:

$$\chi(\mathcal{O}_X(-C_1 - C_2)) = \chi(\mathcal{O}_X(-C_1)) - \chi(\mathcal{O}_{C_2}(-C_1)),$$
$$\chi(\mathcal{O}_X(-C_2)) = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_{C_2}).$$

Substituting into the definition:

$$\begin{aligned} C_1 \cdot C_2 &= \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-C_1)) - \chi(\mathcal{O}_X(-C_2)) + \chi(\mathcal{O}_X(-C_1 - C_2)) \\ &= \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-C_1)) - [\chi(\mathcal{O}_X) - \chi(\mathcal{O}_{C_2})] + [\chi(\mathcal{O}_X(-C_1)) - \chi(\mathcal{O}_{C_2}(-C_1))] \\ &= \chi(\mathcal{O}_{C_2}) - \chi(\mathcal{O}_{C_2}(-C_1)). \end{aligned}$$

Now C_2 is a (possibly reducible, possibly singular) curve on X, and $\mathcal{L}(-C_1)|_{C_2}$ is a line bundle on C_2 whose degree on each irreducible component of C_2 is $-(C_1 \cdot C_{2,i})$. By the Riemann–Roch theorem on C_2 (generalized to possibly singular curves[a]):

$$\chi(\mathcal{O}_{C_2}) - \chi(\mathcal{O}_{C_2}(-C_1)) = \deg(\mathcal{L}(C_1)|_{C_2}).$$

When C_1 and C_2 have no common component, this degree equals $\sum_P i_P(C_1, C_2)$, the sum of local intersection multiplicities. When the intersection is transversal, each $i_P = 1$ and the sum is just $|C_1 \cap C_2|$.

[a] For a possibly singular or reducible curve, the quantity $\chi(\mathcal{L}) - \chi(\mathcal{O}) = \deg \mathcal{L}$ still holds by additivity and the definition of degree on each component.

The cohomological definition has the enormous advantage of being manifestly well-defined and bilinear. The price is that it is not obviously geometric—one must *prove* that it counts intersection points. Property (5) is the bridge: for effective divisors without common components, the intersection number literally counts points with multiplicity.

Example 7.2.9: Intersection Numbers on Basic Surfaces

1. **Lines in $\mathbb{P}^2$.** Let $L_1, L_2 \subseteq \mathbb{P}^2$ be two distinct lines. They are both in the class H (the hyperplane class). They meet in a single point, transversally, so $H \cdot H = 1$. More generally, curves of degrees d_1 and d_2 in $\mathbb{P}^2$ have intersection number $d_1 d_2$, recovering Bézout's theorem ([20]).
2. **Rulings on $\mathbb{P}^1 \times \mathbb{P}^1$.** Let $F = \{p\} \times \mathbb{P}^1$ and $G = \mathbb{P}^1 \times \{q\}$ be the two rulings. Then:
$$F \cdot G = 1, \qquad F \cdot F = 0, \qquad G \cdot G = 0.$$
The first is clear (two rulings from different families meet in one point). For $F^2 = 0$: two fibers $\{p\} \times \mathbb{P}^1$ and $\{p'\} \times \mathbb{P}^1$ with $p \neq p'$ are disjoint, so linearly equivalent effective divisors can be disjoint! This is a key difference from the curve case, where linearly equivalent effective divisors of positive degree always overlap.

3. **A curve of bidegree** (a, b) **on** $\mathbb{P}^1 \times \mathbb{P}^1$**.** If C has class $aF + bG$, then:

$$C \cdot F = b, \qquad C \cdot G = a, \qquad C^2 = 2ab.$$

For instance, the diagonal $\Delta \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ has bidegree $(1, 1)$, so $\Delta^2 = 2$.

Self-Intersection and the Normal Bundle

The most novel feature of intersection theory on surfaces—something that has no analogue for curves—is the *self-intersection number* $C^2 = C \cdot C$ of a curve C on X. This number can be positive, zero, or even negative. Understanding its geometric meaning is essential for everything that follows.

The key link is the *normal bundle*:

Proposition 7.2.10: Self-Intersection via the Normal Bundle

Let C be a smooth curve on a smooth surface X. Then:

$$C^2 \;=\; \deg \mathcal{N}_{C/X}$$

where $\mathcal{N}_{C/X} = (\mathcal{I}_C/\mathcal{I}_C^2)^\vee$ is the normal bundle (an invertible sheaf on C, since C has codimension 1 in X; see definition 5.5.29).

***Proof*:**
From the proof of proposition 7.2.8, we showed $C_1 \cdot C_2 = \deg(\mathcal{L}(C_1)|_{C_2})$ for effective divisors without common components. Setting $C_1 = C_2 = C$ is problematic since C certainly shares a component with itself. But by bilinearity and invariance under linear equivalence, we can define $C^2 = C \cdot C'$ where $C' \sim C$ is any divisor linearly equivalent to C with $C' \neq C$. Alternatively, we observe that $\mathcal{L}(C)|_C$ is exactly the normal bundle: the short exact sequence

$$0 \to \mathcal{I}_C/\mathcal{I}_C^2 \to \Omega_{X/k}|_C \to \Omega_{C/k} \to 0$$

(the conormal sequence, proposition 5.5.17) dualizes to relate the normal bundle $\mathcal{N}_{C/X} = (\mathcal{I}_C/\mathcal{I}_C^2)^\vee$ to the tangent sheaves. On the other hand, the ideal sheaf of C is $\mathcal{I}_C = \mathcal{L}(-C)$, so $\mathcal{I}_C/\mathcal{I}_C^2 \cong \mathcal{L}(-C)|_C$, and $\mathcal{N}_{C/X} \cong \mathcal{L}(C)|_C$. Therefore:

$$C^2 = \deg(\mathcal{L}(C)|_C) = \deg \mathcal{N}_{C/X}.$$

Geometrically, C^2 measures how C "moves" in X:

1. $C^2 > 0$: C can be deformed within X to a nearby curve that intersects C positively. Such curves are "ample-like." Example: lines in $\mathbb{P}^2$ have $L^2 = 1$.
2. $C^2 = 0$: C can be deformed to a disjoint curve (at least numerically). Such curves typically appear as fibers of fibrations. Example: fibers of $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$.
3. $C^2 < 0$: C is "rigid"—it *cannot* be deformed to a disjoint curve. In fact, C is the unique effective divisor in its numerical class. The prototypical example is the exceptional divisor E of a blow-up, with $E^2 = -1$.

Negative self-intersection is the source of much of the richness (and difficulty) of surface theory. It is a purely two-dimensional phenomenon: on a smooth curve, a "divisor" is a sum of points, and a point obviously has non-negative "self-intersection" (which is just its degree, always ≥ 0). The existence of curves with $C^2 < 0$ on surfaces is what makes birational geometry nontrivial.

Example 7.2.11: Self-Intersection Numbers

1. **Lines in $\mathbb{P}^2$:** $L^2 = 1$. More generally, a degree-d curve has $C^2 = d^2$. This is always positive.
2. **The exceptional curve of a blow-up:** if $\pi : \mathrm{wt}\, X \to X$ is the blow-up of a point and $E = \pi^{-1}(p)$ is the exceptional divisor, then $E^2 = -1$. We will prove this in §7.2.4.
3. **The negative section of $\mathbb{F}_n$:** the section $C_0 \subseteq \mathbb{F}_n$ with $C_0^2 = -n$ is rigid when $n > 0$: there is no other effective divisor in its numerical class. For $n = 0$, $C_0^2 = 0$ and C_0 moves in a pencil (the rulings).
4. **On abelian surfaces:** the self-intersection of any curve C on an abelian surface A is $C^2 \geq 0$ by the Hodge index theorem (see §7.2.3), with $C^2 = 0$ if and only if C is a fiber of a fibration $A \to E$ over an elliptic curve.

7.2.2 The Adjunction Formula on Surfaces

The adjunction formula expresses the canonical class of a curve on a surface in terms of the canonical class of the surface and the curve's self-intersection. It is the bridge between the *intrinsic* geometry of a curve (its genus) and its *extrinsic* geometry (how it sits in the surface).

Theorem 7.2.12: The Adjunction Formula

Let C be a smooth irreducible curve on a smooth projective surface X. Then:

$$2g(C) - 2 \;=\; C \cdot (C + K_X)$$

where $g(C)$ is the genus of C and K_X is the canonical divisor of X. Equivalently, $\omega_C \cong (\omega_X \otimes \mathcal{L}(C))|_C$, i.e. $K_C \sim (K_X + C)|_C$.

***Proof*:**
The conormal exact sequence (proposition 5.5.17) for the smooth closed embedding $\iota \colon C \hookrightarrow X$ gives:

$$0 \to \mathcal{N}^\vee_{C/X} \to \Omega_{X/k}|_C \to \Omega_{C/k} \to 0 \tag{7.8}$$

where $\mathcal{N}^\vee_{C/X} = \mathcal{I}_C/\mathcal{I}_C^2 \cong \mathcal{L}(-C)|_C$ is the conormal sheaf. Taking determinants (top exterior powers) of this short exact sequence of locally free sheaves on C:

$$\bigwedge^2(\Omega_{X/k}|_C) \;\cong\; \bigwedge^1(\mathcal{N}^\vee_{C/X}) \otimes \bigwedge^1(\Omega_{C/k}).$$

Now $\bigwedge^2 \Omega_{X/k} = \omega_X$ and $\bigwedge^1 \Omega_{C/k} = \omega_C$, so:

$$\omega_X|_C \;\cong\; \mathcal{L}(-C)|_C \otimes \omega_C.$$

Rearranging: $\omega_C \cong \omega_X|_C \otimes \mathcal{L}(C)|_C \cong (\omega_X \otimes \mathcal{L}(C))|_C$. In divisor language: $K_C \sim (K_X + C)|_C$. Taking degrees on C:

$$\deg K_C = \deg(K_X|_C) + \deg(\mathcal{L}(C)|_C) = K_X \cdot C + C^2 = C \cdot (K_X + C).$$

Since $\deg K_C = 2g(C) - 2$ (proposition 7.1.22), we obtain $2g(C) - 2 = C \cdot (C + K_X)$.

Note that this is the same adjunction formula we proved earlier (propositions 5.5.30 and 6.5.7) for smooth subvarieties of smooth varieties, specialized to the case of a curve on a surface. The novelty here is that we can express the result in terms of the intersection pairing, which makes it *numerical* and useful in practice.

Example 7.2.13: The Adjunction Formula in Action

1. **Smooth plane curves.** Let $C \subseteq \mathbb{P}^2$ be a smooth curve of degree d. Then $C \cdot C = d^2$ (Bézout) and $C \cdot K_{\mathbb{P}^2} = C \cdot (-3H) = -3d$. Adjunction gives:

$$2g - 2 = d^2 - 3d = d(d-3), \qquad g = \frac{(d-1)(d-2)}{2}.$$

   This is the genus formula we computed earlier, now recovered from the intersection pairing.

2. **Curves on $\mathbb{P}^1 \times \mathbb{P}^1$.** Let C be a smooth curve of bidegree (a, b) on $\mathbb{P}^1 \times \mathbb{P}^1$ (with $K = -2F - 2G$). Then $C^2 = 2ab$ and $C \cdot K = (aF + bG) \cdot (-2F - 2G) = -2b - 2a$, so:

$$2g - 2 = 2ab - 2a - 2b, \qquad g = (a-1)(b-1).$$

   For $(a, b) = (2, 2)$: $g = 1$ (an elliptic curve as a $(2, 2)$-curve). For $(a, b) = (3, 3)$: $g = 4$.

3. **Curves on K3 surfaces.** Let S be a K3 surface ($K_S = 0$) and $C \subseteq S$ a smooth curve. Then adjunction gives $2g(C) - 2 = C^2$, so $g(C) = C^2/2 + 1$. In particular, C^2 must be even, and $C^2 \geq -2$ with equality if and only if $g(C) = 0$, i.e. $C \cong \mathbb{P}^1$. Rational curves on K3 surfaces (with $C^2 = -2$) are called (-2)-*curves* or *nodal curves*, and they play a central role in the geometry of K3 surfaces.

4. **The exceptional divisor of a blow-up.** If $E \cong \mathbb{P}^1$ is the exceptional curve of a blow-up $\mathrm{Bl}_p X \to X$ of a smooth surface, then $g(E) = 0$, and adjunction gives $-2 = E^2 + K_{\mathrm{Bl}_p X} \cdot E$. We will show in §7.2.4 that $K_{\mathrm{Bl}_p X} = \pi^* K_X + E$, giving $K_{\mathrm{Bl}_p X} \cdot E = (\pi^* K_X) \cdot E + E^2 = 0 + E^2$, hence $-2 = 2E^2$, confirming $E^2 = -1$.

The adjunction formula has a useful extension to singular or reducible curves. If C is an effective divisor on X that is possibly singular or reducible (but still reduced), we define the *arithmetic genus* of C by $p_a(C) = 1 - \chi(\mathcal{O}_C)$, and the adjunction formula still holds in the form:

$$2p_a(C) - 2 \;=\; C \cdot (C + K_X). \tag{7.9}$$

This follows from the same proof: the short exact sequence $0 \to \mathcal{O}_X(-C) \to \mathcal{O}_X \to \mathcal{O}_C \to 0$ and the cohomological definition of the intersection number. When C is smooth and irreducible, $p_a(C) = g(C)$ and we recover the previous statement.

7.2.14 Forward Reference: Chow Rings and the General Intersection Product
The intersection pairing $\mathrm{Pic}\,(X) \times \mathrm{Pic}\,(X) \to \mathbb{Z}$ on a smooth projective surface is the simplest instance of the *intersection product* on the Chow ring:

$$A^p(X) \otimes A^q(X) \to A^{p+q}(X)$$

where $A^p(X)$ is the group of codimension-p algebraic cycles modulo rational equivalence (definition 8.1.4). For a surface, $A^0(X) = \mathbb{Z}$ (generated by the class of X), $A^1(X) = \mathrm{Pic}\,(X)$, and $A^2(X) = \mathbb{Z}$ (generated by the class of a point). The intersection pairing is then $A^1 \times A^1 \to A^2 \cong \mathbb{Z}$. *Intersection Theory* [16] defines the Chow ring for smooth varieties of any dimension, proves the *moving lemma* (which allows replacing cycles by transversal ones), defines Chern classes as elements of $A^*(X)$, and proves the Grothendieck–Riemann–Roch theorem. The intersection theory of surfaces, developed here "by hand," will be subsumed into this general framework.

We record one more fundamental result that constrains the intersection pairing: the Nakai–Moishezon criterion for ampleness.

Theorem 7.2.15: Nakai–Moishezon Criterion

Let D be a divisor on a smooth projective surface X. Then D is ample if and only if:

$$D^2 > 0 \qquad \text{and} \qquad D \cdot C > 0 \ \text{ for every irreducible curve } C \subseteq X.$$

***Proof Sketch*:**
The "only if" direction is easy: if D is ample, then $D \cdot C > 0$ for every curve C (since $D \cdot C = \deg(\mathcal{L}(D)|_C)$ and an ample line bundle restricted to a curve has positive degree), and $D^2 > 0$ by the same argument applied to general members of $|nD|$ for $n \gg 0$.

The "if" direction is harder and uses the Riemann–Roch theorem for surfaces (§7.2.3) and the Hodge index theorem (§7.2.3)[a]. We omit the details, referring to [33, § V.1].

[a] One shows: $D^2 > 0$ and $D \cdot H > 0$ for some ample H imply $\ell(nD) \sim n^2 D^2/2$ (quadratic growth), so $|nD|$ is a large linear system for $n \gg 0$. The condition $D \cdot C > 0$ for all curves ensures that the resulting map has no contracted curves, hence is finite, hence nD is very ample for $n \gg 0$.

The Nakai–Moishezon criterion is the surface version of the simple fact that a divisor on a curve is ample if and only if it has positive degree. In higher dimensions, the criterion generalizes: D is ample on an n-dimensional projective variety if and only if $D^{\dim V} \cdot V > 0$ for every subvariety $V \subseteq X$ of every dimension $1 \leq \dim V \leq n$. This is proved in a separate volume, using the machinery of *Intersection Theory* [16].

7.2.3 Riemann–Roch for Surfaces

The Riemann–Roch theorem for curves (theorem 7.1.19) computed $\chi(\mathcal{L}(D))$ in terms of $\deg D$ and the genus g. We now prove the analogous result for surfaces, where the formula involves the *intersection pairing* developed in the previous section. This is the first instance of a Riemann–Roch theorem in dimension greater than one, and it illustrates how the intersection product replaces the "degree" in higher dimensions.

The Riemann–Roch Theorem for Surfaces

Theorem 7.2.16: Riemann–Roch for Surfaces

Let X be a smooth projective surface over $k = \overline{k}$, and let D be a divisor on X. Then:

$$\chi(\mathcal{L}(D)) \;=\; \frac{1}{2}\, D \cdot (D - K_X) \;+\; \chi(\mathcal{O}_X).$$

Equivalently, expanding the intersection product:

$$\chi(\mathcal{L}(D)) \;=\; \frac{D^2 - D \cdot K_X}{2} + \chi(\mathcal{O}_X).$$

Before proving this, let us compare with the curve case. For a curve of genus g, Riemann–Roch says $\chi(\mathcal{L}(D)) = \deg D + 1 - g = \deg D + \chi(\mathcal{O}_X)$. The degree on a curve is a *linear* function of D. On a surface, the Euler characteristic is a *quadratic* function of D: it involves D^2 (the self-intersection) in addition to the linear term $D \cdot K_X$. This quadratic behavior is a hallmark of dimension 2.

***Proof*:**
The strategy is the same as for curves: verify the formula at $D = 0$, then show that both sides change by the same amount when we replace D by $D + C$ for an effective divisor C.

Base case: $D = 0$. We need $\chi(\mathcal{O}_X) = \frac{1}{2} \cdot 0 \cdot (0 - K_X) + \chi(\mathcal{O}_X) = \chi(\mathcal{O}_X)$.

Induction step. Let C be a smooth irreducible curve on X. We show: the formula holds for D if and only if it holds for $D + C$. Consider the standard short exact sequence:

$$0 \to \mathcal{O}_X(-C) \to \mathcal{O}_X \to \mathcal{O}_C \to 0.$$

Tensoring with $\mathcal{L}(D + C)$:

$$0 \to \mathcal{L}(D) \to \mathcal{L}(D + C) \to \mathcal{L}(D + C)|_C \to 0.$$

Taking Euler characteristics:

$$\chi(\mathcal{L}(D + C)) - \chi(\mathcal{L}(D)) \;=\; \chi(\mathcal{L}(D + C)|_C). \tag{7.10}$$

Now $\mathcal{L}(D + C)|_C$ is a line bundle on the curve C, and by Riemann–Roch on C (theorem 7.1.19):

$$\chi(\mathcal{L}(D + C)|_C) = \deg(\mathcal{L}(D + C)|_C) + 1 - g(C) = (D + C) \cdot C + 1 - g(C).$$

By the adjunction formula (theorem 7.2.12), $2g(C) - 2 = C \cdot (C + K_X)$, so $1 - g(C) = 1 - \frac{C^2 + C \cdot K_X + 2}{2} = -\frac{C^2 + C \cdot K_X}{2}$. Therefore:

$$\begin{aligned}
\chi(\mathcal{L}(D + C)|_C) &= D \cdot C + C^2 - \frac{C^2 + C \cdot K_X}{2} \\
&= D \cdot C + \frac{C^2 - C \cdot K_X}{2} \\
&= D \cdot C + \frac{1}{2} C \cdot (C - K_X).
\end{aligned}$$

On the other hand, the change in the right-hand side of the Riemann–Roch formula when D is replaced by $D+C$ is:

$$\begin{aligned}\frac{(D+C)^2-(D+C)\cdot K_X}{2}-\frac{D^2-D\cdot K_X}{2}\\ &=\frac{D^2+2D\cdot C+C^2-D\cdot K_X-C\cdot K_X-D^2+D\cdot K_X}{2}\\ &=D\cdot C+\frac{C^2-C\cdot K_X}{2}.\end{aligned}$$

The two sides match: $\chi(\mathcal{L}(D+C))-\chi(\mathcal{L}(D))=$ (change in RHS). Therefore the formula holds for D if and only if it holds for $D+C$.

Conclusion. Since we can reach any divisor D from 0 by adding and subtracting smooth irreducible curves[a], and the formula holds at $D=0$, it holds for all D.

[a]More precisely: any divisor D can be written as $D=C_1+\cdots+C_r-C_{r+1}-\cdots-C_s$ where each C_i is a smooth irreducible curve. To see this, use the fact that any very ample linear system on a smooth surface contains smooth members (by Bertini's theorem), so $D+nH$ is represented by a smooth curve for $n\gg 0$, and H itself can be represented by a smooth curve. Thus $D=(D+nH)-nH$, a difference of smooth effective divisors.

The proof has exactly the same structure as the proof for curves: the key inputs are the additivity of χ on short exact sequences and the Riemann–Roch theorem *on the divisor C itself* (one dimension down). This is the "inductive" nature of Riemann–Roch: the formula in dimension n reduces to the formula in dimension $n-1$ via the short exact sequence of a hypersurface section. This principle is what the Grothendieck–Riemann–Roch theorem (*Intersection Theory* [16]) systematizes.

Example 7.2.17: Riemann–Roch for Surfaces in Practice

1. $\mathbb{P}^2$**:** $K_{\mathbb{P}^2}=-3H$, $\chi(\mathcal{O}_{\mathbb{P}^2})=1$ (since $H^i(\mathbb{P}^2,\mathcal{O})=0$ for $i>0$ by theorem 6.3.1). For $D=dH$:

$$\chi(\mathcal{O}(d))=\frac{d^2-d(-3)}{2}+1=\frac{d^2+3d}{2}+1=\frac{d^2+3d+2}{2}=\binom{d+2}{2}.$$

This agrees with $\dim_k k[x,y,z]_d=\binom{d+2}{2}$ for $d\geq 0$—a reassuring check.

2. $\mathbb{P}^1\times\mathbb{P}^1$**:** $K=-2F-2G$, $\chi(\mathcal{O})=1$. For $D=aF+bG$:

$$\chi(\mathcal{O}(a,b))=\frac{2ab-(aF+bG)\cdot(-2F-2G)}{2}+1=\frac{2ab+2a+2b}{2}+1=(a+1)(b+1).$$

Check: $H^0(\mathbb{P}^1\times\mathbb{P}^1,\mathcal{O}(a,b))=H^0(\mathbb{P}^1,\mathcal{O}(a))\otimes H^0(\mathbb{P}^1,\mathcal{O}(b))$ by the Künneth formula, which has dimension $(a+1)(b+1)$ when $a,b\geq 0$, and higher cohomology vanishes.

3. **A smooth quartic in $\mathbb{P}^3$ (K3 surface):** $K_S=0$, so:

$$\chi(\mathcal{L}(D))=\frac{D^2}{2}+\chi(\mathcal{O}_S).$$

We will compute $\chi(\mathcal{O}_S)=2$ in the next subsection. Thus for an ample divisor D with $D^2=2d$: $\chi(\mathcal{L}(D))=d+2$.

The Arithmetic Genus

The *arithmetic genus* of the surface X is defined by analogy with curves:

$$p_a(X) \;=\; (-1)^{\dim X}(\chi(\mathcal{O}_X) - 1) \;=\; \chi(\mathcal{O}_X) - 1.$$

(The sign convention ensures $p_a \geq 0$ for surfaces of non-negative Kodaira dimension.) Thus $\chi(\mathcal{O}_X) = 1 + p_a$, and Riemann–Roch becomes:

$$\chi(\mathcal{L}(D)) = \frac{D \cdot (D - K)}{2} + 1 + p_a.$$

Setting $D = K_X$ in Riemann–Roch gives the useful identity:

$$\chi(\omega_X) = \frac{K_X^2 - K_X \cdot K_X}{2} + \chi(\mathcal{O}_X) = \chi(\mathcal{O}_X). \tag{7.11}$$

This is not surprising: by Serre duality (theorem 6.5.9), $H^i(X, \omega_X) \cong H^{2-i}(X, \mathcal{O}_X)^\vee$, so $\chi(\omega_X) = \sum(-1)^i h^i(\omega_X) = \sum(-1)^i h^{2-i}(\mathcal{O}_X) = \chi(\mathcal{O}_X)$.

Noether's Formula

The Riemann–Roch theorem for surfaces expresses $\chi(\mathcal{L}(D))$ in terms of intersection numbers and the invariant $\chi(\mathcal{O}_X)$. But what *is* $\chi(\mathcal{O}_X)$? Noether's formula gives the answer, expressing $\chi(\mathcal{O}_X)$ in terms of two fundamental numerical invariants of the surface.

Theorem 7.2.18: Noether's Formula

Let X be a smooth projective surface over $\mathbb{C}$. Then:

$$\chi(\mathcal{O}_X) \;=\; \frac{1}{12}\big(K_X^2 + \chi_{\text{top}}(X)\big)$$

where $\chi_{\text{top}}(X) = \sum_{i=0}^4 (-1)^i b_i(X)$ is the topological Euler characteristic (the alternating sum of Betti numbers of $X(\mathbb{C})$).

Over $\mathbb{C}$, we can use the Hodge decomposition (theorem 0.6.28) and GAGA (theorem 6.9.3) to relate algebraic and topological invariants. The Hodge numbers $h^{p,q} = \dim H^q(X, \Omega_X^p)$ satisfy $h^{p,q} = h^{q,p}$ (Hodge symmetry) and $b_k = \sum_{p+q=k} h^{p,q}$ (Hodge decomposition). For a surface:

$$\begin{aligned}
&b_0 = 1, \quad b_1 = 2h^{0,1}, \quad b_2 = h^{2,0} + h^{1,1} + h^{0,2} = 2h^{2,0} + h^{1,1},\\
&b_3 = 2h^{0,1}, \quad b_4 = 1.
\end{aligned}$$

So $\chi_{\text{top}} = 2 - 4h^{0,1} + 2h^{2,0} + h^{1,1}$.

On the other hand, $\chi(\mathcal{O}_X) = h^{0,0} - h^{0,1} + h^{0,2} = 1 - q + p_g$, where we introduce the standard notation:

Definition 7.2.19: Surface Invariants

For a smooth projective surface X over $\mathbb{C}$:

1. The *irregularity*: $q = q(X) = h^{0,1}(X) = \dim H^1(X, \mathcal{O}_X)$.
2. The *geometric genus*: $p_g = p_g(X) = h^{0,2}(X) = \dim H^0(X, \omega_X) = \dim H^2(X, \mathcal{O}_X)$ (by Serre duality).
3. The *nth plurigenus*: $P_n = P_n(X) = \dim H^0(X, \omega_X^{\otimes n})$ for $n \geq 1$, with $P_1 = p_g$.

***Proof Proof of Noether's formula (sketch)*:**
The full proof of Noether's formula requires either the Hirzebruch–Riemann–Roch theorem (which computes $\chi(\mathcal{O}_X)$ via Chern numbers) or a direct computation using the Hodge index theorem and the signature formula.

Via HRR: the Hirzebruch–Riemann–Roch theorem (proved in *Intersection Theory* [16]) states $\chi(\mathcal{O}_X) = \int_X \mathrm{td}(T_X)$, where td is the Todd class. For a surface:

$$\mathrm{td}(T_X) = 1 + \frac{1}{2}c_1(T_X) + \frac{1}{12}(c_1(T_X)^2 + c_2(T_X)).$$

Now $c_1(T_X) = -c_1(\Omega_X) = -K_X$ (in $A^1(X)$), so $c_1(T_X)^2 = K_X^2$, and $c_2(T_X) = \chi_{\mathrm{top}}(X)$ (the topological Euler characteristic, by the Gauss–Bonnet theorem or the Poincaré–Hopf theorem). Integrating the degree-2 part over X:

$$\chi(\mathcal{O}_X) = \frac{1}{12}(K_X^2 + \chi_{\mathrm{top}}(X)).$$

Example 7.2.20: Noether's Formula in Practice

1. **$\mathbb{P}^2$:** $K = -3H$, $K^2 = 9$, $\chi_{\mathrm{top}} = 3$ (Betti numbers $1, 0, 1, 0, 1$). Noether: $\chi(\mathcal{O}) = \frac{9+3}{12} = 1$.
2. **$\mathbb{P}^1 \times \mathbb{P}^1$:** $K = -2F - 2G$, $K^2 = 8$, $\chi_{\mathrm{top}} = 4$ ($b_0 = b_4 = 1$, $b_2 = 2$, $b_1 = b_3 = 0$). Noether: $\chi(\mathcal{O}) = \frac{8+4}{12} = 1$.
3. **A smooth quartic in $\mathbb{P}^3$ (K3 surface):** $K_S = 0$, so $K_S^2 = 0$. The topological Euler characteristic of a K3 surface is $\chi_{\mathrm{top}} = 24$ (Betti numbers $1, 0, 22, 0, 1$). Noether: $\chi(\mathcal{O}_S) = \frac{0+24}{12} = 2$. So $q = 0$ and $p_g = 1$ (since $\chi = 1 - q + p_g = 2$).
4. **An abelian surface A:** $K_A = 0$ (translation-invariant 2-form), $K^2 = 0$. The Betti numbers are $1, 4, 6, 4, 1$ (since $A(\mathbb{C}) \cong \mathbb{C}^2/\Lambda$ is a 4-torus), so $\chi_{\mathrm{top}} = 0$. Noether: $\chi(\mathcal{O}_A) = 0$. Indeed $q = 2$, $p_g = 1$, $\chi = 1 - 2 + 1 = 0$.

The Hodge Diamond of a Surface

Over $\mathbb{C}$, the Hodge numbers of a smooth projective surface are conveniently arranged in a *Hodge diamond* (recall definition 0.6.29):

$$\begin{array}{ccccc} & & h^{0,0} & & \\ & h^{1,0} & & h^{0,1} & \\ h^{2,0} & & h^{1,1} & & h^{0,2} \\ & h^{2,1} & & h^{1,2} & \\ & & h^{2,2} & & \end{array} \quad = \quad \begin{array}{ccccc} & & 1 & & \\ & q & & q & \\ p_g & & h^{1,1} & & p_g \\ & q & & q & \\ & & 1 & & \end{array}$$

The symmetries are: $h^{p,q} = h^{q,p}$ (Hodge symmetry) and $h^{p,q} = h^{2-p,2-q}$ (Serre duality). The diamond is determined by the three numbers $(q, p_g, h^{1,1})$, subject to the constraint $h^{1,1} \geq \rho$ (the Picard number) and $h^{1,1} \geq 1$ (since the class of an ample divisor is a nonzero $(1,1)$-class).

Example 7.2.21: Hodge Diamonds of Basic Surfaces

1. $\mathbb{P}^2$**:** $\begin{smallmatrix} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 1 & & 0 \\ & 0 & & 0 & \\ & & 1 & & \end{smallmatrix}$. Here $q = 0$, $p_g = 0$, $h^{1,1} = 1 = \rho$.

2. **K3 surface:** $\begin{smallmatrix} & & 1 & & \\ & 0 & & 0 & \\ 1 & & 20 & & 1 \\ & 0 & & 0 & \\ & & 1 & & \end{smallmatrix}$. Here $q = 0$, $p_g = 1$, $h^{1,1} = 20$, with $1 \leq \rho \leq 20$.

3. **Abelian surface:** $\begin{smallmatrix} & & 1 & & \\ & 2 & & 2 & \\ 1 & & 4 & & 1 \\ & 2 & & 2 & \\ & & 1 & & \end{smallmatrix}$. Here $q = 2$, $p_g = 1$, $h^{1,1} = 4$, with $1 \leq \rho \leq 4$.

4. **A smooth quintic surface in $\mathbb{P}^3$ (general type):** $K_S = H$, $K^2 = 5$, $\chi_{\text{top}} = 55$, $\chi(\mathcal{O}) = 5$. So $p_g = 4$ (from the exact sequence $0 \to \mathcal{O}(-5) \to \mathcal{O} \to \mathcal{O}_S \to 0$), $q = 0$, $h^{1,1} = 45$.

The Hodge Index Theorem

The intersection pairing $\text{Pic}\,(X) \times \text{Pic}\,(X) \to \mathbb{Z}$ is a symmetric bilinear form on the free part of $\text{NS}(X)$. The Hodge index theorem describes its *signature*—the number of positive and negative eigenvalues—and is one of the most important structural results in surface theory.

Theorem 7.2.22: Hodge Index Theorem

Let X be a smooth projective surface and let H be an ample divisor. Then the intersection form restricted to the orthogonal complement

$$H^{\perp} \;=\; \{D \in \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R} : D \cdot H = 0\}$$

is *negative definite*. Equivalently, the intersection form on $\text{NS}(X) \otimes \mathbb{R}$ has signature $(1, \rho - 1)$: exactly one positive eigenvalue and $\rho - 1$ negative eigenvalues.

The positive direction is spanned (approximately) by the ample class H itself: $H^2 > 0$ guarantees one positive eigenvalue. The theorem says that all other independent directions are negative. This is a very strong constraint on the geometry of divisors on a surface.

Proof Proof over $\mathbb{C}$**:**
Over $\mathbb{C}$, this follows from the Hodge–Riemann bilinear relations applied to the Kähler class $[\omega] \in H^{1,1}(X, \mathbb{R})$ (the class of the Kähler form associated to the ample divisor H; see definition 0.6.20 and proposition 0.6.21). The intersection pairing on $H^{1,1}(X, \mathbb{R})$ is given by the cup product:

$$(\alpha, \beta) \;=\; \int_X \alpha \wedge \beta.$$

The Hodge–Riemann relations (theorem 0.6.28) state that on the primitive cohomology $P^{1,1} = \{\alpha \in H^{1,1} : \alpha \wedge [\omega] = 0\}$ (those classes perpendicular to the Kähler class), the quadratic form $\alpha \mapsto -\int_X \alpha \wedge \alpha$ is positive definite. Restricting to algebraic classes ($\mathrm{NS}(X) \otimes \mathbb{R} \subseteq H^{1,1}(X, \mathbb{R})$) gives the result.

An algebraic proof (valid over any field) can be given using the Riemann–Roch theorem for surfaces and asymptotic estimates for $\ell(nD)$; see [33, § V.1]. The key idea is that if $D \cdot H = 0$ and $D^2 > 0$, then $\ell(nD)$ would grow quadratically, forcing $|nD|$ to be nonempty for large n. But $D \cdot H = 0$ with H ample would mean nD is trivial on the very ample system, a contradiction.

Corollary 7.2.23: Algebraic Index Inequality

Let C, D be divisors on a smooth projective surface with $C^2 > 0$. Then:

$$(C \cdot D)^2 \;\geq\; C^2 \cdot D^2$$

with equality if and only if D is numerically proportional to C (i.e. $D \equiv \frac{C \cdot D}{C^2} \cdot C$ in $\mathrm{NS}(X) \otimes \mathbb{Q}$).

***Proof*:**
This is the Cauchy–Schwarz inequality for the intersection form. Write $D = aC + D'$ where $a = (C \cdot D)/C^2 \in \mathbb{Q}$ and $D' \cdot C = 0$. Then $D^2 = a^2 C^2 + D'^2$ (since $D' \cdot C = 0$). By the Hodge index theorem, $D'^2 \leq 0$ (with equality iff $D' \equiv 0$). Therefore $D^2 \leq a^2 C^2 = (C \cdot D)^2 / C^2$, giving $(C \cdot D)^2 \geq C^2 \cdot D^2$.

The algebraic index inequality is used constantly in surface theory. For example, it immediately implies:

Corollary 7.2.24: Negative Self-Intersection Criterion

Let C be an irreducible curve on a smooth projective surface X with H ample.

1. If $C \cdot H = 0$, then $C^2 < 0$ (assuming C is nonzero in NS).
2. If $C^2 < 0$, then C is the unique effective divisor in its numerical class.

***Proof*:**
(1) Since $H^2 > 0$ and $C \cdot H = 0$ with $C \neq 0$, the Hodge index theorem gives $C^2 < 0$.

(2) If $C' \sim_{\text{num}} C$ is another effective divisor with $C' \neq C$, then $C' - C$ is a nonzero element of NS with $C'^2 = C^2 < 0$. But $C \cdot C' = C^2 < 0$, so C and C' "anti-intersect." However, C and C' are both effective and in the same numerical class, so $C \cdot C' = C^2$. If C and C' have no common component, then $C \cdot C' \geq 0$ (a sum of local intersection multiplicities), contradicting $C^2 < 0$. Therefore $C = C'$.

This corollary explains why curves with negative self-intersection are "rigid": they cannot be deformed to other curves in the same numerical class.

7.2.25 Forward Reference: The Hodge Conjecture and Higher-Dimensional Hodge Theory The Hodge index theorem is a consequence of the Hodge–Riemann bilinear relations, which hold on all Kähler manifolds. In higher dimensions, these relations control the signature of the intersection form on $H^{p,p}$, and the *Hard Lefschetz theorem* provides the mechanism.
A famous open problem is the *Hodge conjecture*: every class in $H^{p,p}(X, \mathbb{Q})$ on a smooth projective variety X is a $\mathbb{Q}$-linear combination of algebraic cycle classes. For surfaces, this is a theorem (the Lefschetz $(1,1)$-theorem: $\mathrm{NS}(X) \otimes \mathbb{Q} = H^{1,1}(X, \mathbb{Q}) \cap H^2(X, \mathbb{Q})$), but in higher dimensions it remains one of the Clay Millennium Problems. The interplay between Hodge theory and algebraic cycles will be explored further in a separate volume.

7.2.4 Blow-ups

Blow-ups are the fundamental birational operation in the theory of surfaces. They are the mechanism by which smooth surfaces can be modified—and the source of the key difference between curves and surfaces that we flagged in box 7.2.3. Every birational morphism between smooth surfaces factors as a sequence of blow-ups (by a theorem of Zariski), so understanding blow-ups is understanding birational geometry in dimension 2.

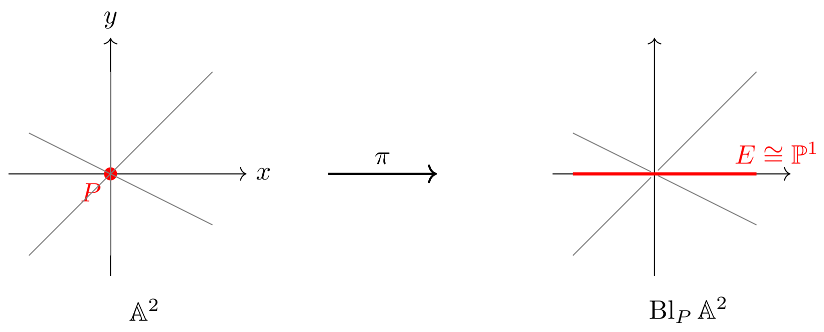
Blowing Up a Point: Geometric Motivation

Let us begin with the most geometric picture. Consider $\mathbb{A}^2_k$ with coordinates (x, y) and the origin $P = (0, 0)$. The *blow-up* of $\mathbb{A}^2$ at P replaces P by a copy of $\mathbb{P}^1$ parametrizing the *directions* through P. Every line through the origin has a well-defined slope $[x : y] \in \mathbb{P}^1$ (including the "vertical" line), and the blow-up "remembers" these directions even at the origin itself.

Concretely, define the blow-up as the closed subvariety of $\mathbb{A}^2 \times \mathbb{P}^1$:

$$\mathrm{Bl}_P \mathbb{A}^2 \;=\; \{((x, y), [s : t]) \in \mathbb{A}^2 \times \mathbb{P}^1 : xt = ys\} \;\subseteq\; \mathbb{A}^2 \times \mathbb{P}^1.$$

The projection $\pi \colon \mathrm{Bl}_P \mathbb{A}^2 \to \mathbb{A}^2$ onto the first factor is an isomorphism away from the origin: for $(x, y) \neq (0, 0)$, the ratio $[s : t] = [x : y]$ is uniquely determined. Over the origin, the fiber $\pi^{-1}(0, 0) = \{(0, 0)\} \times \mathbb{P}^1 \cong \mathbb{P}^1$: every direction is represented. This fiber is the *exceptional divisor E*.



The lines through the origin in $\mathbb{A}^2$ become curves in the blow-up that cross E at different points (corresponding to their slopes). The blow-up has "separated" the tangent directions at P. This is the reason blow-ups are used to *resolve singularities*: a node (two branches crossing at a point) can be resolved by blowing up and separating the two tangent directions (see example 3.1.30 for the formal algebraic version).

Blow-ups as Proj of Rees Algebras

We now give the scheme-theoretic definition, which works for any closed subscheme of any Noetherian scheme.

Definition 7.2.26: Blow-up

Let X be a Noetherian scheme and $\mathcal{I} \subseteq \mathcal{O}_X$ a coherent ideal sheaf defining a closed subscheme $Z \subseteq X$. The *blow-up of X along Z* (or *along $\mathcal{I}$*) is:

$$\mathrm{Bl}_Z X \;=\; \mathrm{Proj}\left(\bigoplus_{n\geq 0} \mathcal{I}^n\right)$$

where $\bigoplus_{n\geq 0} \mathcal{I}^n$ is the *Rees algebra* of $\mathcal{I}$—a graded $\mathcal{O}_X$-algebra whose degree-n piece is the nth power of the ideal sheaf (with $\mathcal{I}^0 = \mathcal{O}_X$). The natural map $\pi\colon \mathrm{Bl}_Z X \to X$ is called the *blow-up morphism*. The *exceptional divisor* is $E = \pi^{-1}(Z)_{\mathrm{red}}$ (set-theoretically, the preimage of Z).

The Proj construction here is the *relative Proj* over X (see the discussion following definition 2.3.4 and section 5.3): locally on an affine open $U = \operatorname{Spec} A$ of X with $\mathcal{I}|_U = \mathrm{wt}I$ for an ideal $I \subseteq A$, the blow-up restricts to $\mathrm{Proj} \bigoplus_{n\geq 0} I^n$, the classical blow-up algebra of I.

Proposition 7.2.27: Basic Properties of Blow-ups

Let $\pi\colon \mathrm{Bl}_Z X \to X$ be the blow-up of a Noetherian scheme X along a closed subscheme Z defined by $\mathcal{I}$.

1. π is a projective morphism (hence proper).
2. π is an isomorphism over $X \setminus Z$: the restriction $\pi^{-1}(X \setminus Z) \xrightarrow{\sim} X \setminus Z$ is an isomorphism. In particular, π is birational if Z has codimension ≥ 1.
3. The ideal sheaf $\mathcal{I} \cdot \mathcal{O}_{\mathrm{Bl}_Z X} = \pi^{-1}\mathcal{I} \cdot \mathcal{O}_{\mathrm{Bl}_Z X}$ is an *invertible* ideal sheaf on $\mathrm{Bl}_Z X$. It defines the exceptional divisor as a Cartier divisor: $E = V(\mathcal{I} \cdot \mathcal{O}_{\mathrm{Bl}_Z X})$.
4. **Universal property:** $\mathrm{Bl}_Z X$ is the *smallest* modification that makes $\mathcal{I}$ into a Cartier divisor. Precisely: if $f\colon Y \to X$ is any morphism such that $f^{-1}\mathcal{I} \cdot \mathcal{O}_Y$ is an invertible ideal sheaf, then f factors uniquely through π.

***Proof Sketch*:**
(1) Follows from the fact that Proj of a finitely generated graded $\mathcal{O}_X$-algebra is projective over X.

(2) Over the complement of Z, $\mathcal{I} = \mathcal{O}_X$, so $\mathcal{I}^n = \mathcal{O}_X$ for all n, and Proj of the polynomial algebra $\mathcal{O}_X[t]$ (in a single variable of degree 1) is just X itself.

(3) The tautological line bundle $\mathcal{O}_{\mathrm{Bl}}(1)$ on $\mathrm{Proj}\,(\bigoplus \mathcal{I}^n)$ satisfies $\mathcal{O}_{\mathrm{Bl}}(1) = \mathcal{I} \cdot \mathcal{O}_{\mathrm{Bl}}$ (the preimage ideal becomes principal).

(4) This is the universal property of the Proj of the Rees algebra, and can be found in [33, § IV] or [64, § 22].

Property (3) is the conceptual heart of the blow-up: the non-Cartier ideal $\mathcal{I}$ (which may be generated by more than one element) becomes Cartier on the blow-up. The exceptional divisor E is the scheme-theoretic vanishing locus of this newly-Cartier ideal.

Example 7.2.28: Blow-up of $\mathbb{A}^2$ at the Origin
Take $X = \mathbb{A}^2_k = \mathrm{Spec}\, k[x, y]$ and $Z = V(x, y)$ (the origin), with $\mathcal{I} = (x, y)$. The Rees algebra is:

$$\bigoplus_{n \geq 0} (x, y)^n \;\subseteq\; k[x, y][s, t]$$

where s, t are formal variables with $\deg s = \deg t = 1$ subject to the relation $xt = ys$ (from $x \cdot t - y \cdot s \in (x, y)^2$... more precisely, the Rees algebra is $k[x, y, s, t]/(xt - ys)$ graded by $\deg s = \deg t = 1$, $\deg x = \deg y = 0$). Then $\mathrm{Bl}_P\, \mathbb{A}^2 = \mathrm{Proj}\, k[x, y, s, t]/(xt - ys)$, which is exactly the incidence variety in $\mathbb{A}^2 \times \mathbb{P}^1$ from §7.2.4.

The exceptional divisor $E \cong \mathbb{P}^1$ is the fiber $\pi^{-1}(0, 0) = \mathrm{Proj}\, k[s, t]$. On the blow-up, the ideal (x, y) becomes the ideal sheaf $\mathcal{O}(-E)$, which is invertible (generated by x on the chart $s \neq 0$ and by y on the chart $t \neq 0$).

Intersection Theory on Blow-ups

We now specialize to the case that matters most for surface theory: the blow-up $\pi\colon \mathrm{wt}X = \mathrm{Bl}_P X \to X$ of a smooth projective surface X at a closed point P.

Theorem 7.2.29: Intersection Theory on the Blow-up

Let $\pi\colon \mathrm{wt}X \to X$ be the blow-up of a smooth projective surface at a point P, with exceptional divisor $E = \pi^{-1}(P) \cong \mathbb{P}^1$. Then:

1. $\mathrm{wt}X$ is a smooth projective surface.
2. $\mathrm{Pic}\,(\mathrm{wt}X) \cong \mathrm{Pic}\,(X) \oplus \mathbb{Z}\cdot E$. Every divisor on $\mathrm{wt}X$ can be written uniquely as $\pi^* D + mE$ for $D \in \mathrm{Pic}\,(X)$ and $m \in \mathbb{Z}$.
3. The intersection numbers on $\mathrm{wt}X$ are determined by:
$$E^2 = -1, \qquad (\pi^* D_1) \cdot (\pi^* D_2) = D_1 \cdot D_2, \qquad (\pi^* D) \cdot E = 0$$
for any $D, D_1, D_2 \in \mathrm{Pic}\,(X)$.
4. The canonical class satisfies:
$$K_{\mathrm{wt}X} \;=\; \pi^* K_X + E.$$
5. For an irreducible curve $C \subseteq X$ passing through P with multiplicity $m = \mathrm{mult}_P(C)$, the *strict transform* $\mathrm{wt}C \subseteq \mathrm{wt}X$ (the closure of $\pi^{-1}(C \setminus \{P\})$) satisfies:
$$\mathrm{wt}C \;=\; \pi^* C - mE, \qquad \mathrm{wt}C \cdot E = m, \qquad \mathrm{wt}C^2 = C^2 - m^2.$$

***Proof*:**
(1) Smoothness of $\mathrm{wt}X$ follows from the explicit local description: in local coordinates (x, y) centered at P, the blow-up is covered by two affine charts $U_s = \mathrm{Spec}\, k[x, t]$ (where $y = xt$) and $U_t = \mathrm{Spec}\, k[s, y]$ (where $x = sy$), both smooth. Projectivity follows from proposition 7.2.27(1).

(2) The restriction map $\mathrm{Pic}\,(\mathrm{wt}X) \to \mathrm{Pic}\,(X \setminus \{P\})$ is surjective (since π is an isomorphism over $X \setminus \{P\}$), with kernel generated by E. Since X is smooth of dimension 2 and P is a closed point (codimension 2), the restriction $\mathrm{Pic}\,(X) \to \mathrm{Pic}\,(X \setminus \{P\})$ is an isomorphism (removing a codimension-2 subset does not change the Picard group on a smooth variety; this follows from the exact sequence for divisor class groups, proposition 5.4.18). Combining: $\mathrm{Pic}\,(\mathrm{wt}X) \cong \mathrm{Pic}\,(X) \oplus \mathbb{Z} \cdot E$.

(3) The key computation is $E^2 = -1$. We use adjunction (theorem 7.2.12): $E \cong \mathbb{P}^1$ has $g(E) = 0$, so $2 \cdot 0 - 2 = E \cdot (E + K_{\mathrm{wt}X})$. We will show $K_{\mathrm{wt}X} \cdot E = -1$ independently (from (4) below), giving $-2 = E^2 + (-1)$, hence $E^2 = -1$.

For $(\pi^* D) \cdot E = 0$: the pullback $\pi^* D$ is trivial in a neighbourhood of E (since D is defined on X and π collapses E to a point, $\pi^* D$ restricted to E is trivial). More precisely, $\deg(\pi^* D|_E) = 0$ since $\pi^* D|_E = \pi|_E^*(D|_P)$ and D restricted to a point is trivial.

For $(\pi^* D_1) \cdot (\pi^* D_2) = D_1 \cdot D_2$: this follows from the projection formula for the proper morphism π.

(4) The canonical class formula $K_{\mathrm{wt}X} = \pi^* K_X + E$ is proved by comparing the sheaves of 2-forms. On the chart $U_s = \operatorname{Spec} k[x,t]$ with $y = xt$:

$$dx \wedge dy = dx \wedge d(xt) = dx \wedge (x\,dt + t\,dx) = x\,dx \wedge dt.$$

So the pullback of the 2-form $dx \wedge dy$ acquires a zero of order 1 along E (which is $\{x = 0\}$ in this chart). This means $\pi^*\omega_X \cong \omega_{\mathrm{wt}X} \otimes \mathcal{L}(-E)$, i.e. $\omega_{\mathrm{wt}X} \cong \pi^*\omega_X \otimes \mathcal{L}(E)$, giving $K_{\mathrm{wt}X} = \pi^* K_X + E$.

Now we can verify: $K_{\mathrm{wt}X} \cdot E = (\pi^* K_X) \cdot E + E^2 = 0 + E^2 = E^2$, and from adjunction $-2 = E^2 + E^2 = 2E^2$, giving $E^2 = -1$ as claimed.

(5) If C passes through P with multiplicity m, then the total transform $\pi^* C$ contains E with multiplicity m (the local equation of C at P starts with terms of degree $\geq m$, and pulling back introduces a factor of x^m or y^m along E). Thus $\pi^* C = \mathrm{wt}C + mE$, giving $\mathrm{wt}C = \pi^* C - mE$. The intersection numbers follow: $\mathrm{wt}C \cdot E = (\pi^* C - mE) \cdot E = 0 - m(-1) = m$, and $\mathrm{wt}C^2 = (\pi^* C - mE)^2 = C^2 - 2m \cdot 0 + m^2(-1) = C^2 - m^2$.

Example 7.2.30: Blow-ups in Practice

1. **$\mathrm{Bl}_P\,\mathbb{P}^2$.** Let H be the class of a line in $\mathbb{P}^2$. Then $\operatorname{Pic}(\mathrm{Bl}_P\,\mathbb{P}^2) = \mathbb{Z}H \oplus \mathbb{Z}E$ (identifying $\pi^* H$ with H by abuse). A line through P has strict transform $\mathrm{wt}L = H - E$ with $\mathrm{wt}L^2 = 1 - 1 = 0$ and $\mathrm{wt}L \cdot E = 1$. A conic through P has $\mathrm{wt}C = 2H - E$ with $\mathrm{wt}C^2 = 4 - 1 = 3$.

   The surface $\mathrm{Bl}_P\,\mathbb{P}^2$ is isomorphic to the Hirzebruch surface $\mathbb{F}_1$: the pencil of lines through P becomes a ruling of $\mathbb{F}_1$, and E is the negative section with $E^2 = -1$.

   The canonical class: $K_{\mathrm{Bl}_P\,\mathbb{P}^2} = \pi^*(-3H) + E = -3H + E$. One checks $K^2 = 9 - 1 = 8$ (consistent with Noether's formula: χ_{top} drops by 1 when we blow up, giving $\chi_{\mathrm{top}} = 4$, and $\frac{8+4}{12} = 1 = \chi(\mathcal{O})$).

2. **Resolving a node.** Consider the nodal cubic $C\colon y^2 = x^3 + x^2$ in $\mathbb{A}^2$. The singular point is the origin, where $\mathrm{mult}_0(C) = 2$. Blowing up the origin in $\mathbb{A}^2$: the strict transform $\mathrm{wt}C$ satisfies $\mathrm{wt}C \cdot E = 2$ (the two branches of the node meet E at two distinct points, corresponding to the two tangent directions $y = \pm x$). The strict transform is smooth: the blow-up has *resolved* the singularity.

3. **Resolving a cusp.** The cuspidal cubic $y^2 = x^3$ has $\mathrm{mult}_0 = 2$. A single blow-up gives a strict transform with $\mathrm{wt}C \cdot E = 2$ but $\mathrm{wt}C$ is *still singular* (it has an ordinary node). A second blow-up resolves the remaining singularity. In general, resolving a cusp requires more blow-ups than resolving a node—the "worse" the singularity, the more blow-ups needed.

Castelnuovo's Contractibility Criterion

We have seen that blowing up a point on a smooth surface introduces a (-1)-curve (a smooth rational curve E with $E^2 = -1$). Castelnuovo's criterion says that the converse holds: every (-1)-curve on a smooth surface arises from a blow-up and can be "blown down."

Theorem 7.2.31: Castelnuovo's Contractibility Criterion

Let X be a smooth projective surface and $E \subseteq X$ a smooth rational curve (i.e. $E \cong \mathbb{P}^1$) with $E^2 = -1$. Then there exists a smooth projective surface Y and a point $Q \in Y$ such that $X \cong \mathrm{Bl}_Q Y$, with E as the exceptional divisor. In other words, E can be *contracted*: there exists a morphism $\pi\colon X \to Y$ that maps E to the point Q and is an isomorphism on $X \setminus E$.

***Proof Sketch*:**
The idea is to construct the contraction morphism π using linear systems. Choose an ample divisor H on X. For $n \gg 0$, the linear system $|nH + (nH \cdot E) \cdot E|$ is a subsystem that separates points and tangent vectors away from E but maps E to a single point. More precisely, one constructs a line bundle $\mathcal{L}$ on X such that $\mathcal{L} \cdot E = 0$ (so E is contracted) and $\mathcal{L}$ is ample on $X \setminus E$ (so the rest of X is embedded). The resulting morphism $\varphi_{\mathcal{L}}\colon X \to Y$ contracts E to a point, and one verifies that Y is smooth at the image point (using the fact that $E^2 = -1$ and the local structure of the contraction).

A complete proof can be found in [33, § V.5]. The key technical ingredient is Grauert's criterion for contractibility, which states that a curve C on a surface can be contracted to a (possibly singular) point if and only if the intersection matrix $(C_i \cdot C_j)$ for the components of C is negative definite. For a single smooth rational curve, this is just $E^2 < 0$, and the smoothness of the resulting point requires $E^2 = -1$.

Castelnuovo's criterion has a clean restatement in terms of the adjunction formula:

Corollary 7.2.32: Detecting (-1)-Curves

Let E be an irreducible curve on a smooth projective surface X. Then E is a (-1)-curve (i.e. $E \cong \mathbb{P}^1$ with $E^2 = -1$) if and only if:

$$E^2 = -1 \qquad \text{and} \qquad K_X \cdot E = -1.$$

Equivalently (by adjunction): $E^2 = -1$ and $g(E) = 0$.

***Proof*:**
By adjunction, $2g(E) - 2 = E^2 + K_X \cdot E = -1 + K_X \cdot E$. If $K_X \cdot E = -1$, then $2g(E) - 2 = -2$, so $g(E) = 0$ and $E \cong \mathbb{P}^1$. Conversely, $g(E) = 0$ and $E^2 = -1$ gives $K_X \cdot E = -1$.

7.2.33 Forward Reference: Extremal Contractions and the MMP Castelnuovo's contractibility criterion is the surface case of the theory of *extremal contractions* in the Minimal Model Program (MMP). In the language of the MMP:

1. The *Cone Theorem* (Mori) identifies the extremal rays of the Mori cone $\mathrm{NE}(X)$: on a surface, the extremal rays on the K_X-negative side are generated by (-1)-curves.
2. Each extremal ray corresponds to a *contraction morphism*: for a (-1)-curve, this is exactly the blow-down of Castelnuovo's criterion.
3. The process of successively contracting (-1)-curves terminates (since ρ decreases by 1 at each step) and produces the *minimal model.*

In dimension ≥ 3, contracting an extremal ray can produce a variety with *terminal singularities*, and one must perform a *flip* to continue the MMP. The existence of flips (proved by Mori in dimension 3 and by Birkar–Cascini–Hacon–McKernan in general) is one of the most important results in modern algebraic geometry. The reader should think of Castelnuovo's criterion as the "toy model" for this vast theory.

We close this section by recording the key consequence for birational geometry:

Theorem 7.2.34: Factorization of Birational Morphisms

Let $f\colon X \to Y$ be a birational morphism between smooth projective surfaces. Then f is a composition of finitely many blow-ups at closed points:

$$X = X_n \xrightarrow{\pi_n} X_{n-1} \xrightarrow{\pi_{n-1}} \cdots \xrightarrow{\pi_1} X_0 = Y$$

where each π_i is the blow-up of X_{i-1} at a closed point.

***Proof Sketch*:**
If f is not an isomorphism, then there exists a point $Q \in Y$ over which f is not defined as an isomorphism. One shows that $f^{-1}(Q)$ contains a (-1)-curve E (this uses the fact that the exceptional locus of a birational morphism of smooth surfaces consists of curves with negative definite intersection matrix, and at least one component must be a (-1)-curve). By Castelnuovo's criterion, we can contract E to get $X \to X' \to Y$, where $X \to X'$ is a blow-down. By induction on $\rho(X) - \rho(Y)$ (which decreases by 1 at each step), the process terminates.

This factorization theorem is extremely powerful: it means that the birational geometry of smooth surfaces is completely controlled by blow-ups at points. In dimension ≥ 3, no such clean factorization exists in general—birational maps may involve flips, which are not morphisms in either direction. This is another instance of the simplicity of the two-dimensional theory.

7.2.5 Ruled Surfaces and Rational Surfaces

We now study two closely related classes of surfaces: the *ruled surfaces* (fibered over a curve with $\mathbb{P}^1$-fibers) and the *rational surfaces* (birational to $\mathbb{P}^2$). These are the surfaces of Kodaira dimension $\kappa = -\infty$—the "simplest" surfaces in the classification, yet geometrically very rich.

Ruled Surfaces

Definition 7.2.35: Ruled Surface

A *ruled surface* over a smooth projective curve C is a smooth projective surface S together with a surjective morphism $p\colon S \to C$ whose fibers are all isomorphic to $\mathbb{P}^1$.
More precisely, $S = \mathbb{P}(\mathscr{E})$ for some rank-2 locally free sheaf $\mathscr{E}$ on C, where $\mathbb{P}(\mathscr{E}) = \operatorname{Proj}(\operatorname{Sym}^\bullet \mathscr{E})$ is the projectivization—the surface whose fiber over $c \in C$ is $\mathbb{P}(\mathscr{E}_c) \cong \mathbb{P}^1$.

Here the projectivization $\mathbb{P}(\mathscr{E})$ parametrizes one-dimensional *quotients* of fibers (following Grothendieck's convention), so a surjection $\mathscr{E} \twoheadrightarrow \mathcal{L}$ for a line bundle $\mathcal{L}$ on C corresponds to a section $C \hookrightarrow \mathbb{P}(\mathscr{E})$.

7.2.36 Warning: The Other $\mathbb{P}(\mathscr{E})$ The quotient convention fixed here is Grothendieck's and is used throughout this book. The opposite convention, under which $\mathbb{P}(\mathscr{E})$ is the bundle of one-dimensional *subspaces* of $\mathscr{E}$, is equally standard and is the one Fulton uses; *Intersection Theory* [16] follows it, since its Segre and Chern classes are built that way. The two differ by a dual, $\mathbb{P}_{\text{quot}}(\mathscr{E}) = \mathbb{P}_{\text{sub}}(\mathscr{E}^\vee)$, so a formula carried between the two books must be dualized. The point matters wherever $\mathcal{O}_{\mathbb{P}(\mathscr{E})}(1)$ appears: the pushforward $f_*\mathcal{O}(d) \cong \operatorname{Sym}^d \mathscr{E}$ recorded above holds in this convention and acquires a dual in the other.

Two ruled surfaces $\mathbb{P}(\mathscr{E})$ and $\mathbb{P}(\mathscr{E}')$ over C are isomorphic (as surfaces over C) if and only if $\mathscr{E}' \cong \mathscr{E} \otimes \mathcal{L}$ for some line bundle $\mathcal{L}$ on C: twisting by a line bundle does not change the projectivization.

The Hirzebruch Surfaces

The most important examples are the ruled surfaces over $\mathbb{P}^1$. A rank-2 bundle on $\mathbb{P}^1$ splits as a direct sum of line bundles (by the Grothendieck–Birkhoff theorem), so $\mathscr{E} \cong \mathcal{O}(a) \oplus \mathcal{O}(b)$. Twisting by $\mathcal{O}(-b)$ gives $\mathscr{E} \otimes \mathcal{O}(-b) \cong \mathcal{O}(a-b) \oplus \mathcal{O}$, and since $a - b$ can be assumed ≥ 0 (by swapping if necessary), every ruled surface over $\mathbb{P}^1$ is isomorphic to one of the *Hirzebruch surfaces*:

Definition 7.2.37: Hirzebruch Surface

For $n \geq 0$, the nth *Hirzebruch surface* is:

$$\mathbb{F}_n \;=\; \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-n)).$$

Proposition 7.2.38: Invariants of $\mathbb{F}_n$

The Hirzebruch surface $\mathbb{F}_n$ has the following properties:

1. $\operatorname{Pic}(\mathbb{F}_n) \cong \mathbb{Z} \cdot C_0 \oplus \mathbb{Z} \cdot F$, where F is the class of a fiber of $p\colon \mathbb{F}_n \to \mathbb{P}^1$ and C_0 is the section corresponding to the surjection $\mathcal{O} \oplus \mathcal{O}(-n) \twoheadrightarrow \mathcal{O}$.
2. Intersection numbers: $F^2 = 0$, $C_0 \cdot F = 1$, $C_0^2 = -n$.
3. Canonical class: $K_{\mathbb{F}_n} = -2C_0 - (n+2)F$.
4. $\mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$ (with C_0 and F being the two rulings).
5. $\mathbb{F}_1 \cong \mathrm{Bl}_P\,\mathbb{P}^2$, with $C_0 = E$ (the exceptional divisor) and F the strict transform of a line through P.
6. For $n \geq 2$: C_0 is the unique irreducible curve with negative self-intersection on $\mathbb{F}_n$. In particular, $\mathbb{F}_n \ncong \mathbb{F}_m$ for $n \neq m$.

***Proof Sketch*:**
The Picard group and intersection numbers follow from the general theory of $\mathbb{P}(\mathscr{E})$-bundles and the tautological line bundle $\mathcal{O}_{\mathbb{P}(\mathscr{E})}(1)$; the section C_0 corresponds to $\mathcal{O}(1)$ and satisfies $C_0^2 = \deg(\mathscr{E}) \cdot (-1) + \deg(\text{surjection}) = -n$ by the self-intersection formula for sections of ruled surfaces.

For the canonical class: by the relative duality for a $\mathbb{P}^1$-bundle $p\colon S \to C$, $\omega_{S/C} \cong \mathcal{O}_S(-2) \otimes p^*(\det \mathscr{E})^\vee$, and then $\omega_S = \omega_{S/C} \otimes p^*\omega_C$. Working out for $\mathbb{F}_n$: $\det(\mathcal{O} \oplus \mathcal{O}(-n)) = \mathcal{O}(-n)$ and $\omega_{\mathbb{P}^1} = \mathcal{O}(-2)$, giving $K = -2C_0 + (n)F + (-2)F = -2C_0 - (n+2)F$.

The identification $\mathbb{F}_0 \cong \mathbb{P}^1 \times \mathbb{P}^1$ is clear. For $\mathbb{F}_1 \cong \mathrm{Bl}_P\,\mathbb{P}^2$: the pencil of lines through $P \in \mathbb{P}^2$ gives a morphism $\mathrm{Bl}_P\,\mathbb{P}^2 \to \mathbb{P}^1$ with $\mathbb{P}^1$-fibers and exceptional divisor as the (-1)-section, matching C_0 on $\mathbb{F}_1$.

For $n \geq 2$: if C is an irreducible curve on $\mathbb{F}_n$ different from C_0, write $C \sim aC_0 + bF$. Since C is effective and $C \neq C_0$, one shows $a \geq 0$ and (if $a > 0$) $b \geq an$, giving $C^2 = -a^2n + 2ab \geq a^2n \geq 0$. So C_0 is the only curve with negative self-intersection.

Example 7.2.39: Ruled Surfaces over Higher-Genus Curves
If C has genus $g \geq 1$, the ruled surfaces $\mathbb{P}(\mathscr{E}) \to C$ are much richer: rank-2 bundles on C no longer split. Their moduli form a variety of dimension $4g - 3$ (for stable bundles of fixed determinant). The geometry of these ruled surfaces reflects the geometry of vector bundles on C—a topic that connects to gauge theory, the Narasimhan–Seshadri theorem, and mathematical physics.

Rational Surfaces

Definition 7.2.40: Rational Surface

A smooth projective surface X is *rational* if it is birational to $\mathbb{P}^2$.

By theorem 7.2.34, every birational map from X to $\mathbb{P}^2$ factors through blow-ups and blow-downs. So every rational surface is obtained from $\mathbb{P}^2$ (or from some $\mathbb{F}_n$) by a sequence of blow-ups. The simplest examples are:

Example 7.2.41: Rational Surfaces in Practice

1. $\mathbb{P}^2$ **itself** and all $\mathbb{F}_n$ (including $\mathbb{P}^1 \times \mathbb{P}^1 = \mathbb{F}_0$).
2. **Del Pezzo surfaces.** The blow-up of $\mathbb{P}^2$ at $r \leq 8$ points in *general position* (no three collinear, no six on a conic, no eight on a singular cubic with one of them at the singular point). These have $K^2 = 9 - r$ and are Fano surfaces (i.e. $-K$ is ample) for $r \leq 8$.
3. **The cubic surface.** A smooth cubic surface $S \subseteq \mathbb{P}^3$ is isomorphic to $\mathrm{Bl}_{P_1,\ldots,P_6}\,\mathbb{P}^2$ (the blow-up of $\mathbb{P}^2$ at 6 points in general position). It has $K_S = -H$ (the hyperplane class), $K^2 = 3$, and $\rho = 7$. The 27 lines on S correspond to: the 6 exceptional divisors E_i, the 15 strict transforms of lines $\mathrm{wt}L_{ij}$ through pairs of blown-up points, and the 6 strict transforms of conics $\mathrm{wt}Q_i$ through 5 of the 6 points. Each line ℓ satisfies $\ell^2 = -1$ and $K \cdot \ell = -1$, so every line is a (-1)-curve. The intersection pattern of these 27 lines is governed by the Weyl group $W(E_6)$—a key connection to Lie theory.

The following classical theorem gives a cohomological characterization of rationality:

Theorem 7.2.42: Castelnuovo's Rationality Criterion

A smooth projective surface X over $\mathbb{C}$ is rational if and only if $q(X) = 0$ and $P_2(X) = 0$ (irregularity zero and second plurigenus zero).

The "only if" direction is clear: q and P_2 are birational invariants, and $\mathbb{P}^2$ has $q = P_2 = 0$. The "if" direction is a theorem whose proof uses the theory of linear systems on surfaces. We omit it, referring to [33, § V.6]. Castelnuovo's criterion is because it detects rationality from just two cohomological invariants.

7.2.6 The Enriques–Kodaira Classification

We arrive at the summit of the theory of surfaces: the classification of smooth projective surfaces by their *Kodaira dimension*. This classification, developed by the Italian school (Castelnuovo, Enriques) in the early 20th century and completed by Kodaira in the 1960s over $\mathbb{C}$, organizes the zoo of surfaces into a small number of families with sharply distinct geometric behavior. It is one of the most important achievements of algebraic geometry, and it serves as the model for the classification program in all dimensions.

Kodaira Dimension

The key invariant is the growth rate of the spaces of *pluricanonical forms*.

Definition 7.2.43: Kodaira Dimension

Let X be a smooth projective variety of dimension d over k. For each $n \geq 1$, the *nth plurigenus* is $P_n(X) = \dim_k H^0(X, \omega_X^{\otimes n})$. The *Kodaira dimension* $\kappa(X)$ is defined by the growth rate of P_n:

$$\kappa(X) = \begin{cases} -\infty & \text{if } P_n = 0 \text{ for all } n \geq 1, \\ \min\left\{k \geq 0 : \limsup\limits_{n\to\infty} \dfrac{P_n}{n^k} < \infty\right\} & \text{otherwise.} \end{cases}$$

Equivalently, $\kappa(X)$ is the dimension of the image of the *pluricanonical map* $\varphi_{|nK_X|} \colon X \dashrightarrow \mathbb{P}^{P_n - 1}$ for n sufficiently large and divisible (or $-\infty$ if all $P_n = 0$).

For a surface, $\kappa \in \{-\infty, 0, 1, 2\}$:

1. $\kappa = -\infty$: $P_n = 0$ for all n. The canonical class K_X is "anti-effective" in a strong sense.
2. $\kappa = 0$: P_n is bounded (either 0 or 1 for each n). The canonical class is "numerically trivial."
3. $\kappa = 1$: P_n grows linearly, i.e. $P_n \sim cn$ for some $c > 0$. The pluricanonical map gives a fibration onto a curve.
4. $\kappa = 2$: P_n grows quadratically, i.e. $P_n \sim cn^2$. The pluricanonical map is birational for $n \gg 0$. These are the *surfaces of general type.*

The Kodaira dimension is a birational invariant: if X and Y are birational smooth projective surfaces, then $\kappa(X) = \kappa(Y)$. This is because the plurigenera P_n are birational invariants (a consequence of the fact that the space of regular n-canonical forms is a birational invariant for smooth varieties).

Minimal Models

Definition 7.2.44: Minimal Surface

A smooth projective surface X is *minimal* if it contains no (-1)-curves (smooth rational curves E with $E^2 = -1$). Equivalently, X cannot be obtained as a blow-up of another smooth surface.

Theorem 7.2.45: Existence and Uniqueness of Minimal Models

1. **Existence:** Every smooth projective surface admits a birational morphism to a minimal surface, obtained by successively contracting (-1)-curves (theorem 7.2.31). The process terminates since the Picard number ρ drops by 1 at each step.
2. **Uniqueness for $\kappa \geq 0$:** If $\kappa(X) \geq 0$, the minimal model is unique up to isomorphism.
3. **Non-uniqueness for $\kappa = -\infty$:** If $\kappa(X) = -\infty$, the minimal models are $\mathbb{P}^2$ and the Hirzebruch surfaces $\mathbb{F}_n$ with $n \neq 1$ (note: $\mathbb{F}_1$ is *not* minimal since C_0 is a (-1)-curve).

***Proof Sketch*:**
Existence is clear from Castelnuovo's criterion and the fact that ρ is a non-negative integer.

Uniqueness for $\kappa \geq 0$: suppose X_1 and X_2 are two minimal models in the same birational equivalence class. The birational map $X_1 \dashrightarrow X_2$ can be resolved by a smooth surface Y dominating both:

$$\begin{tikzcd} & Y \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ X_1 \arrow[rr, dashed] & & X_2 \end{tikzcd}$$

By theorem 7.2.34, π_i is a composition of blow-ups. If π_1 is not an isomorphism, then Y contains a (-1)-curve E contracted by π_1. Since $\kappa \geq 0$, one can show that $K_Y \cdot E = -1$ forces π_2 to also contract E (otherwise $\pi_{2*}(E)$ would be a curve with $K_{X_2} \cdot \pi_{2*}(E) < 0$, contradicting $\kappa \geq 0$ via the "negativity of contraction" argument). By induction, π_1 and π_2 contract the same curves, so $X_1 \cong X_2$.

For $\kappa = -\infty$: a minimal rational surface is either $\mathbb{P}^2$ (if no ruling exists) or a Hirzebruch surface $\mathbb{F}_n$ with $n \neq 1$. The non-uniqueness arises because $\mathbb{P}^2$ and $\mathbb{F}_n$ (for various $n \neq 1$) are all minimal and birational.

Overview of the Classification

We now state the Enriques–Kodaira classification. The result organizes minimal surfaces into four classes according to κ, with $\kappa = 0$ further subdivided into four types. We work over $\mathbb{C}$ throughout.

$\kappa = -\infty$: Ruled and Rational Surfaces

A minimal surface with $\kappa = -\infty$ is either $\mathbb{P}^2$ or a ruled surface $\mathbb{P}(\mathscr{E})$ over a curve C of genus $g \geq 0$. The characterization is:

$$\kappa = -\infty \quad \Longleftrightarrow \quad P_{12} = 0 \quad \Longleftrightarrow \quad X \text{ is ruled or rational.}$$

The key invariant is the *base genus*: for a ruled surface $S \to C$, the base curve C has genus $g \geq 0$. When $g = 0$, the surface is rational (birational to $\mathbb{P}^2$).

$\kappa = 0$: Four Types

Minimal surfaces with $\kappa = 0$ fall into exactly four families, distinguished by their irregularity q and geometric genus p_g. In all cases, K_X is numerically trivial (i.e. $K_X \cdot C = 0$ for all curves C), and in fact $12K_X \sim 0$ (the canonical class has finite order dividing 12 in $\operatorname{Pic}(X)$).

1. **Abelian surfaces** ($q = 2$, $p_g = 1$, $K_X \sim 0$): These are 2-dimensional abelian varieties—quotients $\mathbb{C}^2/\Lambda$ for a lattice Λ of rank 4. They carry a group structure (addition of points) and have trivial canonical bundle. Their Hodge diamond has $h^{1,1} = 4$ and $\chi(\mathcal{O}) = 0$.
2. **K3 surfaces** ($q = 0$, $p_g = 1$, $K_X \sim 0$): These are simply connected surfaces with trivial canonical bundle. The prototype is a smooth quartic in $\mathbb{P}^3$. Their Hodge diamond has $h^{1,1} = 20$ and $\chi(\mathcal{O}) = 2$. K3 surfaces are among the most studied objects in algebraic geometry; we discuss them further below.

3. **Enriques surfaces** ($q = 0$, $p_g = 0$, $K_X \not\sim 0$ but $2K_X \sim 0$): These can be characterized as quotients of K3 surfaces by a fixed-point-free involution. They have $\chi(\mathcal{O}) = 1$ and $h^{1,1} = 10$.

4. **Bielliptic (hyperelliptic) surfaces** ($q = 1$, $p_g = 0$): These are quotients $(E_1 \times E_2)/G$ where E_1, E_2 are elliptic curves and G is a finite group acting freely. They fiber over both elliptic curves (after appropriate quotients) and have $\chi(\mathcal{O}) = 0$.

Example 7.2.46: K3 Surfaces: A Closer Look

K3 surfaces deserve special emphasis because of their extraordinary properties:

1. **Examples:** Smooth quartics in $\mathbb{P}^3$, smooth complete intersections of type $(2, 3)$ in $\mathbb{P}^4$ or $(2, 2, 2)$ in $\mathbb{P}^5$, double covers of $\mathbb{P}^2$ branched over a smooth sextic curve, and Kummer surfaces (resolutions of $A/\{\pm 1\}$ for an abelian surface A).

2. **Deformation equivalence:** All K3 surfaces are deformation-equivalent as complex manifolds. There is a single connected 20-dimensional moduli space of complex K3 surfaces (but not all are algebraic!). Algebraic K3 surfaces form a countable union of 19-dimensional families, parametrized by their Picard lattice.

3. **The period map:** The Hodge structure on $H^2(S, \mathbb{Z})$ determines S up to isomorphism (the global Torelli theorem for K3 surfaces, proved by Piatetski-Shapiro and Shafarevich). The lattice $H^2(S, \mathbb{Z}) \cong U^3 \oplus E_8(-1)^2$ is a fixed unimodular lattice of signature $(3, 19)$, and the moduli of K3 surfaces is governed by the period domain $\mathrm{SO}(3, 19)/\mathrm{SO}(2) \times \mathrm{SO}(1, 19)$.

4. **Rational curves:** K3 surfaces contain infinitely many rational curves (proved by Bogomolov–Hassett–Tschinkel and others). Each such curve C satisfies $C^2 = -2$ by adjunction ($K_S = 0$).

$\kappa = 1$: Elliptic Surfaces

A minimal surface with $\kappa = 1$ admits a *fibration* $f : X \to C$ over a curve C whose general fiber is a smooth curve of genus 1 (an elliptic curve). This fibration is the Iitaka fibration associated to the pluricanonical linear systems $|nK_X|$.

The *canonical bundle formula* describes K_X in terms of the fibration:

$$K_X \sim f^*(K_C + \mathcal{L}) + \sum_i (m_i - 1)F_i$$

where $\mathcal{L}$ is a line bundle on C of degree $\chi(\mathcal{O}_X)$, and the sum runs over the *multiple fibers* m_iF_i (fibers that are m_i times a reduced divisor). The invariants $\chi(\mathcal{O}_X)$, the genus of C, and the multiplicities m_i determine the surface.

Elliptic surfaces are ubiquitous: the "generic" surface of $\kappa = 1$ is an elliptic surface. Their study connects to the arithmetic of elliptic curves over function fields (via the Mordell–Weil group of the generic fiber), the theory of singular fibers (Kodaira's classification of singular fibers of elliptic fibrations), and string theory (where elliptic fibrations on Calabi–Yau manifolds play a central role).

$\kappa = 2$: Surfaces of General Type

A surface of *general type* is one with $\kappa = 2$, the maximal Kodaira dimension. For such surfaces, K_X is big: P_n grows like n^2, and the pluricanonical map $\varphi_{|nK_X|}$ is birational onto its image for n sufficiently large (in fact $n \geq 5$ always suffices by a theorem of Bombieri).

The numerical invariants of minimal surfaces of general type are constrained by two fundamental inequalities:

Theorem 7.2.47: Geography of Surfaces of General Type

Let X be a minimal surface of general type. Then:

1. **Bogomolov–Miyaoka–Yau inequality:** $K_X^2 \leq 3\chi(\mathcal{O}_X)$. Equality holds if and only if the universal cover of $X(\mathbb{C})$ is the unit ball in $\mathbb{C}^2$ ("ball quotients").
2. **Noether inequality:** $K_X^2 \geq 2\chi(\mathcal{O}_X) - 6$.
3. Both invariants are positive: $K_X^2 \geq 1$ and $\chi(\mathcal{O}_X) \geq 1$.

The allowed region in the (K^2, χ)-plane, bounded by the BMY inequality and the Noether line, is called the *geography* of surfaces of general type. Determining which lattice points (K^2, χ) in this region are realized by actual surfaces is a major open problem. The "boundary" surfaces (those on the Noether line or the BMY line) have very special geometry: Noether-line surfaces are fibered over curves with high-genus fibers, while BMY surfaces are ball quotients with connections to number theory.

Example 7.2.48: Surfaces of General Type

1. **Smooth hypersurfaces in $\mathbb{P}^3$ of degree $d \geq 5$:** $K_S = (d-4)H$, $K^2 = d(d-4)^2$, $\chi(\mathcal{O}) = \binom{d-1}{3}$. For $d = 5$: $K^2 = 5$, $\chi = 4$. Check: $5 \leq 3 \cdot 4 = 12$ (BMY) and $5 \geq 2 \cdot 4 - 6 = 2$ (Noether).
2. **Products of curves:** $X = C_1 \times C_2$ with $g(C_i) \geq 2$. Then $K_X = p_1^* K_{C_1} + p_2^* K_{C_2}$, $K^2 = 8(g_1 - 1)(g_2 - 1)$, $\chi(\mathcal{O}) = (g_1 - 1)(g_2 - 1)$. The BMY ratio is $K^2/\chi = 8$ (always < 9, approaching 9 for curves with many symmetries).
3. **The Godeaux surface:** the quotient of the Fermat quintic $\{x_0^5 + \cdots + x_3^5 = 0\} \subseteq \mathbb{P}^3$ by a free $\mathbb{Z}/5\mathbb{Z}$-action. It has $K^2 = 1$ and $\chi = 1$—the smallest possible invariants for a surface of general type.

7.2.7 The Classification Table

We summarize the Enriques–Kodaira classification in the following table. All surfaces are minimal and over $\mathbb{C}$.

κ	q	p_g	K^2	**Type**
$-\infty$	any	0	< 0	Ruled / Rational
0	2	1	0	Abelian surface
0	0	1	0	K3 surface
0	0	0	0	Enriques surface ($2K \sim 0$)
0	1	0	0	Bielliptic surface
1	any	any	0	Elliptic surface
2	any	≥ 1	> 0	General type

Several patterns are worth highlighting:

1. The canonical class K_X is "more positive" as κ increases: anti-ample for $\kappa = -\infty$, numerically trivial for $\kappa = 0$, semi-ample for $\kappa = 1$, and big for $\kappa = 2$. The Kodaira dimension measures "how positive K_X is."

2. The invariant K^2 is zero for $\kappa \leq 1$ (on minimal surfaces) and positive for $\kappa = 2$. This is why self-intersection of the canonical class is such a fundamental invariant.

3. The irregularity q and geometric genus p_g suffice to distinguish the four types of $\kappa = 0$ surfaces. For $\kappa = 1$ and $\kappa = 2$, more invariants are needed (the fibration structure and the geography, respectively).

7.2.49 Forward Reference: The Classification in Higher Dimensions and Positive Characteristic The Enriques–Kodaira classification of surfaces is the prototype for the *classification of algebraic varieties* in all dimensions. The organizing principle remains the same: classify by Kodaira dimension, with the Minimal Model Program providing the reduction to minimal models.

A separate volume will develop this program:

1. The *cone theorem* and *contraction theorem* generalize the theory of (-1)-curves to extremal rays of the Mori cone in any dimension.

2. *Flips* replace blow-downs when contractions produce singularities (this does not happen for surfaces, but is unavoidable in dimension ≥ 3).

3. The *abundance conjecture* (proved for surfaces, open in general) predicts that the pluricanonical map is always semi-ample for minimal varieties with $\kappa \geq 0$.

In positive characteristic, the classification has additional subtleties. For $\kappa = 0$: there exist *quasi-hyperelliptic surfaces* and *supersingular K3 surfaces* that have no analogue in characteristic zero. For general type: the BMY inequality $K^2 \leq 3\chi$ can fail in characteristic p, and Bombieri's theorem ($|5K|$ gives a birational map) requires modification. These topics remain active areas of research.

7.2.8 Further Topics on Surfaces

We conclude this chapter with several additional topics that complete the picture for surfaces and point toward future developments. As in §7.1.8, this section is more survey-like in character.

The Picard Scheme and Algebraic Equivalence

In the study of curves, we encountered the Jacobian variety $\mathrm{Jac}(X)$ representing the functor Pic^0. For surfaces, the corresponding object is the *Picard scheme* $\mathrm{Pic}_{X/k}$, a group scheme whose connected component $\mathrm{Pic}^0_{X/k}$ is an abelian variety of dimension $q = h^{0,1}(X)$ (the irregularity). The full Picard scheme has components indexed by degree (or, more precisely, by elements of the Néron–Severi group $\mathrm{NS}(X)$).

The various notions of equivalence for divisors on surfaces form a hierarchy:

1. **Linear equivalence** ($\sim$): $D_1 \sim D_2$ if $D_1 - D_2 = \mathrm{div}(f)$. This is classified by $\mathrm{Pic}(X)$.
2. **Algebraic equivalence** ($\sim_{\mathrm{alg}}$): $D_1 \sim_{\mathrm{alg}} D_2$ if there is a connected variety T and a divisor $\mathcal{D}$ on $X \times T$ such that $\mathcal{D}|_{X\times\{t_1\}} = D_1$ and $\mathcal{D}|_{X\times\{t_2\}} = D_2$ for $t_1, t_2 \in T$. The group $\mathrm{Pic}(X)/\sim_{\mathrm{alg}}$ is the Néron–Severi group $\mathrm{NS}(X)$.
3. **Numerical equivalence** ($\equiv$): $D_1 \equiv D_2$ if $D_1 \cdot C = D_2 \cdot C$ for all curves $C \subseteq X$. The group $\mathrm{Pic}(X)/\equiv$ is denoted $\mathrm{Num}(X)$.

The Néron–Severi theorem asserts that $\mathrm{NS}(X)$ is finitely generated, and a theorem of Matsusaka shows that $\mathrm{NS}(X)/\mathrm{torsion} \cong \mathrm{Num}(X)$ (numerical and algebraic equivalence coincide modulo torsion). For surfaces with $q = 0$ (such as $\mathbb{P}^2$, K3 surfaces, rational surfaces), $\mathrm{Pic}^0 = 0$ and all three notions coincide on $\mathrm{Pic}(X)_{\mathrm{free}}$.

Vanishing Theorems

Vanishing theorems for cohomology of line bundles are among the most powerful tools in algebraic geometry. We state the two most important ones; their proofs require techniques beyond our current scope (Hodge theory for Kodaira vanishing, delicate positivity arguments for Kawamata–Viehweg).

Theorem 7.2.50: Kodaira Vanishing

Let X be a smooth projective variety over $\mathbb{C}$ and $\mathcal{L}$ an ample line bundle on X. Then:

$$H^i(X, \omega_X \otimes \mathcal{L}) = 0 \qquad \text{for all } i > 0.$$

Equivalently (by Serre duality): $H^i(X, \mathcal{L}^{-1}) = 0$ for $i < \dim X$.

For surfaces, Kodaira vanishing says: if $\mathcal{L}$ is ample on X, then $H^1(X, \omega_X \otimes \mathcal{L}) = H^2(X, \omega_X \otimes \mathcal{L}) = 0$. Combined with Riemann–Roch for surfaces (theorem 7.2.16), this converts χ into h^0 for ample twists of ω_X:

$$h^0(X, \omega_X \otimes \mathcal{L}) = \chi(\omega_X \otimes \mathcal{L}) = \frac{(K+D)^2 - (K+D)\cdot K}{2} + \chi(\mathcal{O}_X)$$

where D is the divisor of $\mathcal{L}$.

The following generalization is the key vanishing theorem used in the Minimal Model Program:

Theorem 7.2.51: Kawamata–Viehweg Vanishing

Let X be a smooth projective variety over $\mathbb{C}$ and D a nef and big divisor on X. Then:

$$H^i(X, \omega_X \otimes \mathcal{L}(D)) = 0 \qquad \text{for all } i > 0.$$

This extends Kodaira vanishing from ample to nef-and-big divisors—a crucial relaxation because the canonical divisor of a minimal surface of general type is nef and big but typically not ample.

7.2.52 Forward Reference: Vanishing Theorems in the MMP Kawamata–Viehweg vanishing and its refinements (Nadel vanishing, Kollár vanishing) are the technical engine of the Minimal Model Program. In a separate volume, these vanishing theorems will be used to:

1. Prove the *basepoint-free theorem*: if D is a nef divisor on a minimal variety with $aD - K$ ample for some $a > 0$, then $|mD|$ is base-point-free for $m \gg 0$.
2. Prove the *rationality theorem*: the threshold a in the cone theorem is rational.
3. Establish the *non-vanishing theorem*: if $K + D$ is nef and big, then $|m(K + D)| \neq \emptyset$ for $m \gg 0$.

These are the "big three" consequences of vanishing that drive the MMP. The reader who wishes to understand why vanishing theorems are so central should keep in mind that they convert Euler characteristic computations (which are topological) into dimension computations (which are geometric), by killing the "error terms" H^i for $i > 0$.

Arithmetic Surfaces

We close this chapter by returning to the arithmetic motif that has accompanied us throughout. An *arithmetic surface* is a 2-dimensional scheme $\mathcal{X}$ that is integral, Noetherian, and equipped with a proper flat morphism

$$f\colon \mathcal{X} \to \operatorname{Spec} \mathcal{O}_K$$

where $\mathcal{O}_K$ is the ring of integers of a number field K. The generic fiber $\mathcal{X}_\eta = \mathcal{X} \times_{\mathcal{O}_K} K$ is a curve over K, and the closed fibers $\mathcal{X}_\mathfrak{p} = \mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_K/\mathfrak{p}$ are curves over finite fields $\mathbb{F}_q$.

Thus an arithmetic surface is a "family of curves parametrized by Spec $\mathcal{O}_K$": it interpolates between a curve over a number field (the generic fiber) and curves over finite fields (the special fibers). The generic fiber is the "algebraic curve" whose arithmetic we wish to study, and the special fibers record its "reduction modulo primes."

Example 7.2.53: Arithmetic Surfaces in Practice

1. **The projective line over $\mathbb{Z}$:** $\mathcal{X} = \mathbb{P}^1_\mathbb{Z}$, with $f\colon \mathbb{P}^1_\mathbb{Z} \to \operatorname{Spec} \mathbb{Z}$. The generic fiber is $\mathbb{P}^1_\mathbb{Q}$ and the fiber over (p) is $\mathbb{P}^1_{\mathbb{F}_p}$. This is the "trivial" arithmetic surface.
2. **An elliptic curve over $\mathbb{Z}$:** Let $E\colon y^2 = x^3 - x$ over $\mathbb{Q}$. The *Néron model* $\mathcal{E} \to \operatorname{Spec} \mathbb{Z}$ is an arithmetic surface whose generic fiber is E and whose fibers over primes p are elliptic curves

over $\mathbb{F}_p$ (for $p \neq 2$; the fiber over $p = 2$ is a singular curve, since E has bad reduction at 2).

The intersection theory on arithmetic surfaces is called *Arakelov theory*, named after Arakelov who developed it in the 1970s. The key idea is that Spec $\mathcal{O}_K$ is "missing" the archimedean places of K (the infinite primes), and Arakelov theory "compactifies" the arithmetic surface by adding data at infinity—specifically, a Hermitian metric on the line bundles at each archimedean place. This yields an intersection pairing

$$\mathrm{whDiv}(\mathcal{X}) \times \mathrm{whDiv}(\mathcal{X}) \to \mathbb{R}$$

(valued in $\mathbb{R}$, not $\mathbb{Z}$, because of the archimedean contributions) that extends the classical intersection pairing on surfaces. The arithmetic analogue of the adjunction formula, the Riemann–Roch theorem, and the Hodge index theorem all have Arakelov-theoretic versions, establishing a parallel between geometric and arithmetic intersection theory.

This is one of the most active frontiers of modern number theory. The Arakelov-theoretic Riemann–Roch theorem (Faltings, Gillet–Soulé) was used by Faltings in his proof of the Mordell conjecture (finiteness of rational points on curves of genus ≥ 2 over number fields), and Arakelov intersection theory continues to play a central role in the study of heights, Diophantine equations, and the arithmetic of moduli spaces.

This concludes our tour of curves and surfaces. We have traveled from the Riemann–Roch theorem for curves—the starting point of the entire theory—through Hurwitz's formula, the group law on elliptic curves, the intersection pairing on surfaces, blow-ups, and the Enriques–Kodaira classification. Along the way, we have seen the abstract machinery of schemes, cohomology, and duality transform into concrete geometric and arithmetic results.

The reader who has followed this chapter should be well-prepared for the next stages of the journey. The intersection theory of surfaces (§7.2.1) generalizes to the Chow ring and Chern classes of *Intersection Theory* [16]. The minimal model theory for surfaces (§7.2.6) becomes the full Minimal Model Program, treated in a separate volume. The moduli of curves and surfaces foreshadowed throughout becomes the theory of stacks and deformation theory of *Moduli, Deformation Theory, and Stacks* [17]. And the arithmetic themes woven into every section find their expression in étale cohomology and derived categories, each treated in a separate volume. The subject only grows from here.

7.3 A Glimpse Beyond: Higher-Dimensional Geometry

Throughout this chapter, we have built the theory of curves and surfaces around four interlocking themes: birational geometry (how varieties relate up to modification), intersection theory (the algebra of subvarieties meeting), classification (organizing varieties by discrete invariants), and cohomology (extracting information from sheaves). We have also noticed, repeatedly, that the passage from curves to surfaces introduced genuine complications—blow-ups, self-intersection, negative curves, the need for minimal models—that had no analogue in dimension one.

The reader may well wonder: what happens in dimension three and beyond? The short answer is that each of our four themes generalizes, but with essential new complications that required decades of effort and some of the biggest leaps mathematics of the 20th and 21st centuries to resolve. This section is a roadmap. We will state the main results and conjectures, explain the key new phenomena, and point to the parts of this book where the full theory will be developed. No proofs will be given; the goal is orientation, not rigor.

7.3.1 The Minimal Model Program

On curves, birational geometry is trivial: birational smooth projective curves are isomorphic (corollary 7.1.27). On surfaces, birational geometry is controlled by blow-ups and (-1)-curves: one successively contracts (-1)-curves to reach a minimal model, which is unique when $\kappa \geq 0$ (theorem 7.2.45). The entire process stays within the category of *smooth* varieties.

In dimension three and higher, this clean picture breaks. The fundamental problem is:

7.3.1 What Goes Wrong in Dimension ≥ 3 On a smooth threefold X, an *extremal contraction* (the analogue of contracting a (-1)-curve) can map a curve $C \subseteq X$ to a point. The image Y may be *singular*—unlike the surface case, where Castelnuovo's criterion guarantees smoothness of the contracted surface (theorem 7.2.31). Worse: when the contracted locus has codimension ≥ 2 (a "small contraction"), the image Y is not even $\mathbb{Q}$-factorial (Cartier divisors cannot be pulled back), and the standard intersection theory breaks down.
The solution is the *flip*: instead of accepting the singular image Y, one replaces the contracted locus by a different subvariety to obtain a new variety X^+ that is "less negative" in the K-direction. This surgery—called a *flip*—has no analogue in dimension 2.

The *Minimal Model Program* (MMP), developed by Mori, Kawamata, Kollár, Shokurov, and many others, systematizes this process. Here is the flowchart:

1. **Start** with a smooth projective variety X (or more generally, one with mild "terminal" singularities).
2. **Test:** Is K_X nef? If yes, X is a *minimal model*—stop.
3. **If not:** the Cone Theorem (Mori) guarantees the existence of a K_X*-negative extremal ray* $R \subseteq \overline{\mathrm{NE}}(X)$ and a corresponding *contraction morphism* $\varphi_R \colon X \to Y$.
4. **Three cases for the contraction:**
   (a) *Fiber type:* $\dim Y < \dim X$. Then X is a *Mori fiber space* (fibered over a lower-dimensional base with Fano fibers). This is the $\kappa = -\infty$ endpoint—stop.
   (b) *Divisorial:* The exceptional locus of φ_R has codimension 1. The image Y has terminal singularities, and $\rho(Y) = \rho(X) - 1$. Replace X by Y and go to step 2.
   (c) *Small (flipping):* The exceptional locus has codimension ≥ 2. Perform a *flip*: replace X by a new variety X^+ that agrees with X outside the exceptional locus but has "K-positive" curves where X had "K-negative" ones. Replace X by X^+ and go to step 2.
5. **Termination:** The process terminates after finitely many steps (since ρ decreases for divisorial contractions, and a separate argument handles flips).

The hard theorems, proved over the last four decades, are:

Theorem 7.3.2: Main Results of the MMP

1. **Cone Theorem** (Mori, Kawamata): The Mori cone $\overline{\mathrm{NE}}(X)$ is "rational polyhedral" on the K_X-negative side, generated by classes of rational curves.
2. **Existence of Flips** (Mori in dimension 3; Birkar–Cascini–Hacon–McKernan in general, 2010): Flips exist in all dimensions.
3. **Termination of Flips** (proved in dimension 3 by Shokurov; open in general, but the MMP terminates for varieties of general type by BCHM).
4. **Abundance Conjecture** (proved for surfaces and threefolds; open in general): If X is a minimal model (K_X nef), then K_X is *semi-ample*: some multiple $|mK_X|$ is base-point-free.

The BCHM theorem (2010) is one of the landmark results of 21st-century mathematics. It establishes the existence of minimal models for varieties of general type in all dimensions, resolving a conjecture that had been open for decades. A separate volume will develop this theory in full.

To appreciate the contrast with surfaces: on a surface, the MMP consists of at most $\rho - 1$ blow-downs, all within the smooth category, with no flips. The entire theory we developed in §7.2.4–§7.2.6 is the "trivial case" of the MMP—but it is the case that motivated the general program and provided the template.

Singularities in the MMP

The MMP forces us to work with singular varieties. The appropriate singularity classes form a hierarchy:

$$\text{smooth} \subsetneq \text{terminal} \subsetneq \text{canonical} \subsetneq \text{log terminal} \subsetneq \text{log canonical}.$$

Terminal singularities are the mildest: in dimension 2, they are smooth points (which is why the surface MMP stays smooth). In dimension 3, terminal singularities are isolated and classified (compound Du Val singularities). In higher dimensions, they are more complex but remain "mild" enough for intersection theory and vanishing theorems to work. The Kawamata–Viehweg vanishing theorem (theorem 7.2.51) extends to varieties with log terminal singularities, and this extension is what makes the MMP function.

7.3.2 Intersection Theory and Riemann–Roch in Higher Dimensions

On curves, the "intersection" of a divisor with the curve itself is just the degree. On surfaces, we developed a bilinear pairing $\mathrm{Pic}(X) \times \mathrm{Pic}(X) \to \mathbb{Z}$. In general, on a smooth projective variety X of dimension n, one needs a *multilinear* intersection product: n divisors $D_1, \ldots, D_n$ intersect in a number $D_1 \cdot D_2 \cdots D_n \in \mathbb{Z}$.

This is formalized by the *Chow ring*:

Definition 7.3.3: The Chow Ring (Preview)

For a smooth projective variety X of dimension n, the *Chow ring* is a graded commutative ring:

$$A^*(X) = \bigoplus_{p=0}^{n} A^p(X)$$

where $A^p(X)$ is the group of codimension-p algebraic cycles modulo rational equivalence (definition 8.1.4), with a ring structure given by the *intersection product*. The degree map $\deg\colon A^n(X) \to \mathbb{Z}$ (sending the class of a point to 1) completes the picture: the product $D_1 \cdots D_n \in A^n(X) \cong \mathbb{Z}$ gives the intersection number.

For surfaces, $A^0 = \mathbb{Z}$, $A^1 = \operatorname{Pic}(X)$, $A^2 = \mathbb{Z}$, and the intersection product is the pairing we defined in §7.2.1. In higher dimensions, the Chow ring is richer: A^p for $1 < p < n-1$ involves cycles of intermediate codimension (curves, surfaces, ... on X), and the intersection product is no longer just a bilinear pairing but a full ring structure.

The *moving lemma* (proved in *Intersection Theory* [16]) ensures that any two cycles can be moved to intersect properly, making the product well-defined. The proof is nontrivial and uses the geometry of the ambient projective space.

Chern Classes and Riemann–Roch

In dimension 1, the Riemann–Roch theorem involves the degree of a line bundle (a single number). In dimension 2, it involves D^2 and $D \cdot K$ (two numbers, assembled from intersection theory). The pattern becomes clear in the general *Hirzebruch–Riemann–Roch theorem*:

Theorem 7.3.4: Hirzebruch–Riemann–Roch (Preview)

Let X be a smooth projective variety of dimension n and $\mathscr{E}$ a vector bundle on X. Then:

$$\chi(\mathscr{E}) \;=\; \int_X \operatorname{ch}(\mathscr{E}) \cdot \operatorname{td}(T_X)$$

where $\operatorname{ch}(\mathscr{E}) = \operatorname{rk}(\mathscr{E}) + c_1(\mathscr{E}) + \frac{1}{2}(c_1^2 - 2c_2) + \cdots$ is the *Chern character* and $\operatorname{td}(T_X) = 1 + \frac{1}{2}c_1(T_X) + \frac{1}{12}(c_1^2 + c_2) + \cdots$ is the *Todd class*. The integral $\int_X$ means "take the degree-n component and apply $\deg\colon A^n(X) \to \mathbb{Z}$."

For a line bundle $\mathcal{L}$ on a curve: $\operatorname{ch}(\mathcal{L}) = 1 + c_1(\mathcal{L})$, $\operatorname{td}(T_X) = 1 + \frac{1}{2}c_1 = 1 + (1-g)$ (on the degree-1 part), recovering $\chi(\mathcal{L}) = \deg \mathcal{L} + 1 - g$. For a line bundle on a surface: one recovers $\chi(\mathcal{L}) = \frac{1}{2}D(D-K) + \chi(\mathcal{O}_X)$ (theorem 7.2.16). For a threefold, the formula involves D^3, $D^2 \cdot K$, $D \cdot K^2$, $D \cdot c_2(X)$, and $\chi(\mathcal{O}_X)$—the complexity grows, but the structure is always the same.

The generalization is the *Grothendieck–Riemann–Roch theorem* (GRR), which computes the behavior of the Chern character under proper pushforward:

$$\operatorname{ch}(f_!\mathscr{E}) \;=\; f_*(\operatorname{ch}(\mathscr{E}) \cdot \operatorname{td}(T_{X/Y}))$$

for a proper morphism $f\colon X \to Y$. This is the "master theorem" of which HRR and the classical RR for curves are special cases (taking $Y = \operatorname{Spec} k$). *Intersection Theory* [16] develops Chern classes, proves HRR, states GRR, and builds the intersection theory needed to make sense of the formulas.

7.3.3 The Geography of Higher-Dimensional Varieties

The Enriques–Kodaira classification organized surfaces by Kodaira dimension $\kappa \in \{-\infty, 0, 1, 2\}$. In dimension n, the Kodaira dimension takes values in $\{-\infty, 0, 1, \ldots, n\}$, and the classification by κ remains the organizing principle—but the "geography" within each class is far richer.

$\kappa = -\infty$: Fano Varieties

The higher-dimensional analogues of rational surfaces and Del Pezzo surfaces are the *Fano varieties*: smooth projective varieties X with $-K_X$ ample. These are the varieties with "the most rational curves."

A theorem constrains their complexity:

Theorem 7.3.5: Boundedness of Fano Varieties (BAB Conjecture)

Fano varieties of dimension n with at worst ϵ-log terminal singularities form a *bounded family*: they can all be parametrized by a finite number of algebraic families. In particular, there are only finitely many deformation types of smooth Fano varieties of each dimension.

This was conjectured by Alexeev and Borisov–Borisov and proved by Birkar (2019, for which he received the Fields Medal). In dimension 2, this is the elementary fact that Del Pezzo surfaces are blow-ups of $\mathbb{P}^2$ at ≤ 8 points. In dimension 3, the classification of smooth Fano threefolds (by Iskovskikh and Mori–Mukai) identifies 105 deformation families. In dimension 4 and beyond, the explicit classification is open, but boundedness is guaranteed.

$\kappa = 0$: Calabi–Yau Varieties and Holomorphic Symplectic Varieties

The $\kappa = 0$ case for surfaces gave us four types: abelian, K3, Enriques, and bielliptic. In higher dimensions, the "building blocks" of the $\kappa = 0$ world are conjectured to be three types (the *Beauville–Bogomolov decomposition*):

1. **Abelian varieties**: n-dimensional complex tori $\mathbb{C}^n/\Lambda$ with a polarization.
2. **Calabi–Yau varieties (strict sense)**: simply connected varieties with $K_X \sim 0$ and $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < n$. The prototype: a smooth quintic threefold in $\mathbb{P}^4$ (the "quintic Calabi–Yau," which has $h^{1,1} = 1$ and $h^{2,1} = 101$).
3. **Irreducible holomorphic symplectic (hyperkähler) varieties**: simply connected with $H^0(X, \Omega^2_X) = \mathbb{C} \cdot \sigma$ for a nondegenerate 2-form σ. The prototype: Hilbert schemes of points on K3 surfaces.

Calabi–Yau threefolds are central to *mirror symmetry*, a profound duality (originating in string theory) that exchanges $h^{1,1}$ and $h^{n-1,1}$ between "mirror pairs" of Calabi–Yau manifolds. Mirror symmetry has led to spectacular predictions in enumerative geometry (counting rational curves), many of which have been proved rigorously. The mathematical framework for mirror symmetry involves derived categories (a separate volume) and the theory of Gromov–Witten invariants.

$\kappa = \dim X$: Varieties of General Type

The higher-dimensional analogue of surfaces of general type consists of varieties where K_X is big. The BCHM theorem guarantees the existence of a minimal model, and the *canonical model* $X_{\mathrm{can}} = \operatorname{Proj} \bigoplus_n H^0(X, \omega_X^{\otimes n})$ is well-defined.

The moduli theory of varieties of general type is governed by the *KSBA compactification* (Kollár–Shepherd-Barron–Alexeev), which generalizes the Deligne–Mumford compactification $\overline{\mathcal{M}}_g$ of the moduli of curves. In dimension 2, the KSBA moduli space parametrizes "stable surfaces" (surfaces with semi-log-canonical singularities and ample canonical class), and its geometry is an active area of research. The boundedness of the moduli space (finiteness of components for fixed invariants) was established by Hacon–McKernan–Xu, building on the BCHM machinery.

7.3.4 Cohomological Frontiers

The cohomological tools we have used in this chapter—sheaf cohomology, Serre duality, the Euler characteristic—are just the beginning. We close with a brief survey of the cohomological theories that drive modern algebraic geometry in higher dimensions.

Étale Cohomology and the Weil Conjectures

Over $\mathbb{C}$, our cohomological invariants (Betti numbers, Hodge numbers) come from the topology of $X(\mathbb{C})$. Over finite fields $\mathbb{F}_q$, there is no underlying topological space in the classical sense, yet the Weil conjectures predict that the number of $\mathbb{F}_{q^n}$-points of a smooth projective variety satisfies properties: rationality of the zeta function, a functional equation, and the "Riemann hypothesis" (bounds on eigenvalues of Frobenius).

Grothendieck's invention of *étale cohomology* provides the ℓ-adic cohomology groups $H^i_{\text{ét}}(X, \mathbb{Q}_\ell)$ that play the role of singular cohomology over finite fields. The Weil conjectures, proved by Deligne in 1974 using étale cohomology, are:

Theorem 7.3.6: The Weil Conjectures (Deligne)

Let X be a smooth projective variety of dimension n over $\mathbb{F}_q$. Then:

1. The zeta function $Z(X,t) = \exp\left(\sum_{m=1}^{\infty} \frac{|X(\mathbb{F}_{q^m})|}{m} t^m\right)$ is a rational function of t.
2. $Z(X,t)$ satisfies a functional equation relating $Z(X,t)$ and $Z(X, q^{-n}t^{-1})$.
3. **Riemann Hypothesis:** The eigenvalues of Frobenius on $H^i_{\text{ét}}(X, \mathbb{Q}_\ell)$ have absolute value $q^{i/2}$.

For curves, this is Weil's theorem (the Hasse–Weil bound, example 7.1.68). For elliptic curves, the Riemann Hypothesis is Hasse's bound $|a_q| \le 2\sqrt{q}$ (box 7.1.61). The full Weil conjectures in all dimensions required a vast new cohomological apparatus, and their proof by Deligne is one of the crowning achievements of 20th century mathematics. A separate volume will develop étale cohomology and present Deligne's proof.

Hodge Theory in Higher Dimensions

Over $\mathbb{C}$, the Hodge decomposition (theorem 0.6.28) expresses $H^k(X, \mathbb{C})$ as a direct sum of $H^{p,q}$ spaces. For surfaces, we used the Hodge diamond and the Hodge index theorem (theorem 7.2.22) extensively. In higher dimensions, the Hodge–Riemann bilinear relations, the Hard Lefschetz theorem, and the Lefschetz $(1, 1)$-theorem all generalize, but the *Hodge conjecture* remains wide open:

Every rational (p, p)-class on a smooth projective variety over $\mathbb{C}$ is algebraic.

For $p = 1$, this is the Lefschetz $(1, 1)$-theorem (true, and used implicitly throughout our surface theory: it is the reason $\mathrm{NS}(X)$ captures all algebraic classes in $H^{1,1}$). For $p \geq 2$, the conjecture is one of the Clay Millennium Prize Problems. Its resolution would profoundly clarify the relationship between topology and algebraic geometry.

Derived Categories

Perhaps the most transformative development in algebraic geometry over the past thirty years is the realization that the *derived category* $D^b(\mathbf{Coh}(X))$—the bounded derived category of coherent sheaves—is a finer invariant of X than any collection of classical invariants (Hodge numbers, Chern numbers, etc.).

Two results illustrate this:

1. **Mukai's theorem:** An abelian variety A and its dual wh$A$ have equivalent derived categories: $D^b(\mathbf{Coh}(A)) \simeq D^b(\mathbf{Coh}(\mathrm{wh}A))$, even though A and wh$A$ need not be isomorphic. The equivalence is given by a *Fourier–Mukai transform*—a "categorified Fourier transform" whose kernel is the Poincaré bundle on $A \times \mathrm{wh}A$.
2. **Bondal–Orlov:** If X is a smooth projective variety with K_X ample or $-K_X$ ample (Fano or anti-Fano), then X is determined up to isomorphism by its derived category. But for $\kappa = 0$ (Calabi–Yau, abelian), derived equivalences can relate non-isomorphic varieties.

The derived category is also the natural home for *homological mirror symmetry* (Kontsevich, 1994): for a mirror pair $(X, X^\vee)$ of Calabi–Yau manifolds, the conjecture predicts an equivalence $D^b(\mathbf{Coh}(X)) \simeq D^\pi(\mathrm{Fuk}(X^\vee))$ between the derived category of coherent sheaves on X and the derived Fukaya category of $X^\vee$ (a symplectic invariant). This conjecture has been proved in several cases and remains one of the most active research frontiers at the intersection of algebraic geometry, symplectic geometry, and mathematical physics. A separate volume will develop the foundations of derived categories.

7.3.7 The Road Ahead The reader who has completed this chapter has seen, in miniature, the essential phenomena of algebraic geometry: the interplay of algebra and geometry (schemes), the power of cohomology (Riemann–Roch, Serre duality), the role of positivity (ampleness, nefness, the Kodaira dimension), and the richness of classification (genus for curves, the Enriques–Kodaira table for surfaces). Parts II and III of this book develop the first two of these themes to their full generality:

Part	Theme	Generalization of...
II	Intersection Theory	§7.2.1, §7.2.3
III	Moduli and Stacks	§7.1.7, §??

Three further directions leave this book for their own volumes in the series: birational geometry and the Minimal Model Program, étale cohomology, and derived categories.
The journey from curves to the frontier of modern algebraic geometry is a long one, but it is worth taking. Let us continue.

7.3.5 Hironaka's Example

Every example so far of a complete variety has been projective, and for curves section 7.1.6 showed the two notions coincide. In dimension three they do not. The example below is a smooth complete threefold carrying no ample divisor at all, and it is the reason proper morphisms are worth having as a notion separate from projective ones. The mechanism is simple enough to state in one line: blowing up two curves in the two possible orders gives two answers, and gluing them produces a variety in which an effective curve is algebraically equivalent to the negative of another.

Proposition 7.3.8: Hironaka's Example

Let X be a smooth projective threefold over an algebraically closed field, and let $C_1, C_2 \subseteq X$ be smooth curves meeting transversally at exactly two points P and Q, and nowhere else. Construct $\pi\colon M \to X$ by gluing two blow-ups:

- over $X \setminus \{Q\}$, blow up C_1 first, then the strict transform of C_2;
- over $X \setminus \{P\}$, blow up C_2 first, then the strict transform of C_1.

Over the overlap $X \setminus \{P, Q\}$ the two curves are disjoint, so the two orders give canonically isomorphic results and the pieces glue. Then M is a smooth complete threefold, and M is *not* projective.

***Proof*:**
Smoothness and completeness are inherited: blowing up a smooth subvariety of a smooth variety is smooth, and π is proper because it is proper on each of the two pieces and properness is local on the target.

For non-projectivity, work with 1-cycles modulo algebraic equivalence. Write ℓ for the class of the fibre of the exceptional divisor over a general point of C_1, and m for the class of the fibre over a general point of C_2.

The fibre over P. Near P the curve C_1 was blown up first, producing an exceptional $\mathbb{P}^1$ over P; the strict transform of C_2 meets that $\mathbb{P}^1$ in a single point, and blowing it up splits the fibre into two lines. So $\pi^{-1}(P)$ is a union $\ell_0 \cup \ell_1$ of two irreducible curves, and letting a general point of C_1 degenerate to P exhibits

$$\ell \sim \ell_0 + \ell_1$$

an algebraic equivalence. The second of the two components is the exceptional curve of the blow-up along C_2, so $\ell_1 = m$ and

$$\ell \sim \ell_0 + m$$

The fibre over Q. The construction is symmetric, the two curves exchanged, because over $X \setminus \{P\}$ it is C_2 that is blown up first. The same argument gives an irreducible curve m_0 in $\pi^{-1}(Q)$ with

$$m \sim m_0 + \ell$$

The relation. Substituting the second into the first,

$$\ell \sim \ell_0 + m_0 + \ell, \qquad \text{hence} \qquad \ell_0 + m_0 \sim 0$$

Now suppose M carried an ample divisor H. Intersection numbers with H are constant in an algebraic family, so $H \cdot (\ell_0 + m_0) = 0$. But ℓ_0 and m_0 are irreducible curves on M, so ampleness forces $H \cdot \ell_0 > 0$ and $H \cdot m_0 > 0$, whence $H \cdot (\ell_0 + m_0) > 0$. This contradiction shows no ample divisor exists, so M is not projective.

Two features are worth separating. The failure is not caused by singularities, since M is smooth, nor by non-completeness. It is caused by the blow-up construction being order-dependent, and the gluing is what converts that ambiguity into a numerical relation no ample class can satisfy. Note also what the example does *not* say: it is a statement about the absence of ample divisors. It has nothing to do with resolution of singularities, which is a different theorem of Hironaka's and is not proved in this book.

Part II

The Chow Group

Cycles are easy to write down and useless as they stand: the group of formal integer combinations of subvarieties is enormous, is not an invariant of anything, and admits no product. This part cuts it down to something that is. Rational equivalence identifies two cycles when one sweeps out to the other in a family over $\mathbb{P}^1$, and the quotient, the Chow group, turns out to carry exactly the structure the classical geometers were assuming when they deformed a configuration and trusted the count. Proper pushforward and flat pullback make it functorial, the localization sequence and homotopy invariance make it computable, and intersecting with a divisor gives the first operation that lowers dimension. That last operation, the first Chern class, is the seed of everything the rest of the book builds.

7.3.9 Convention: Standing Hypotheses Throughout the book a scheme is a *separated* scheme of finite type over a fixed algebraically closed field $\overline{k}$; such a scheme is called an *algebraic scheme*. A point is a closed point unless stated otherwise, a subvariety of a scheme is a closed integral subscheme, and a *variety* is a reduced and irreducible algebraic scheme, hence separated. Separatedness is folded in here rather than carried statement by statement because it is exactly the condition making the diagonal $\delta\colon X \to X \times X$ a closed immersion, and from chapter 12 onward every intersection product is built by pulling back along that diagonal. A non-separated example such as the line with a doubled origin is excluded by this convention, not by hypothesis.

7.3.10 Convention: $\mathbb{P}(E)$ Is the Bundle of Lines For a vector bundle E on X, this book writes $\mathbb{P}(E)$ for the bundle of one-dimensional *subspaces* of E: the fibre over x is the space of lines in E_x. This is Fulton's convention.

Algebraic Geometry [7] uses Grothendieck's opposite convention, under which $\mathbb{P}(\mathscr{E}) = \operatorname{Proj}(\operatorname{Sym}^\bullet \mathscr{E})$ parametrizes one-dimensional *quotients* of the fibres. The two are related by a dual: the bundle written $\mathbb{P}(E)$ here is the one written $\mathbb{P}(E^\vee)$ there, and a formula transported across the seam must be dualized. The Segre classes of Chapter 8 onward are defined by pushing forward powers of $c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$, so reading such a formula in the wrong convention silently computes the classes of $E^\vee$ instead of E.

8

Rational Equivalence

Recall from [20] that we had to define *projective plane curves* to be an equivalence relation on homogeneous polynomials of $k[x, y, z]$ up to a constant ($f \sim g$ if and only if $f = \alpha g$). This was specifically done keep track of multiplicity information. By the correspondence between homogeneous and affine curves, we can without loss of generality work with $[f(x, y)], [g(x, y)] \in k[x, y]$ and choose any representative; we shall simply call these *plane curves*. Given a choice of plane curves f, g let F, G denote the resulting variety. The intersection of two (affine or projective) plane curves F, G at P was defined as:

$$i(P, f \cdot g) = \dim_K \frac{\mathcal{O}\mathbb{A}^2_k, p}{(f, g)} = \dim_K \mathcal{O}_{Z,P}$$

where $Z = \mathcal{Z}(f, g)$. It was proven that:

1. (commutative on intersection): $i(P, F \cdot G) = i(P, G \cdot F)$
2. (linear on intersection): if $F_1 + F_2$ is the curve defined by $f_1 f_2$, then $i(P, (F_1 + F_2) \cdot G) = i(P, F_1 \cdot G) = i(P, F_2 \cdot G)$
3. (invariance of representation): if F' is given by $f + gh$ for some $h \in k[x, y]$:

$$i(P, F' \cdot G) = i(P, F \cdot G)$$

4. If F, G have have a common component through P (i.e. the dimension of the irreducible variety in $F \cap G$ containing p is greater than 0), then $i(P, F \cdot G) = \infty$. Otherwise, $i(P, F \cdot G) \in \mathbb{N}_{\geq 0}$.
5. If $P \notin F \cap G$:

$$i(P, F \cdot G) = 0$$

6. if $f(x, y) = x - a$ and $g(x, y) = y - b$ (more generally $\frac{\partial(f,g)}{\partial(x,y)} \neq 0$ at p) then:

$$i(P, F \cdot G) = 1$$

7. If F has no common component with G and H and P is a simple point of F:

$$i(P, G \cdot H) \geq \min i(P, F \cdot G), i(P, F \cdot H)$$

8.1 Cycles and Rational Equivalence

The classical intersection number recalled above attaches an integer to a pair of plane curves meeting at a point. Turning that local count into a theory valid on any variety and in every dimension calls for a home for “formal combinations of subvarieties” together with an equivalence relation flexible enough to make intersection numbers well defined. That home is the Chow group, the higher-dimensional descendant of the divisor class group of definition 5.4.8: where the divisor class group records codimension-one subvarieties modulo the divisors of rational functions, the Chow group records subvarieties of *every* dimension modulo the same mechanism.

This section builds the three objects in turn, the group of cycles $Z_k(X)$, the subgroup $\mathrm{Rat}_k(X)$ of cycles rationally equivalent to zero, and the quotient $A_k(X)$. Throughout the chapter X is an algebraic scheme (a scheme of finite type over the algebraically closed field $\overline{k}$), a *subvariety* means a closed integral subscheme, and a point means a closed point unless stated otherwise.

The construction rests on the order of vanishing of a rational function along a codimension-one subvariety, introduced for divisors in section 5.4. On a variety W a prime divisor is a codimension-one subvariety V, and the local ring $\mathcal{O}_{W,V}$ is a one-dimensional local domain with fraction field the function field $k(W)$. When W is regular in codimension one this ring is a discrete valuation ring and its valuation ν_V records orders of vanishing (definition 5.4.7); the length-theoretic definition below removes the regularity hypothesis, which matters as soon as W is an arbitrary subvariety of X.

Definition 8.1.1: Order Along a Subvariety

Let W be a variety and $V \subseteq W$ a codimension-one subvariety, and set $A = \mathcal{O}_{W,V}$, a one-dimensional local domain with fraction field $k(W)$. For nonzero $a \in A$ the quotient $A/(a)$ has finite length, and the *order of a along V* is

$$\mathrm{ord}_V(a) = \mathrm{length}_A\left(A/(a)\right)$$

Because $\mathrm{ord}_V(ab) = \mathrm{ord}_V(a) + \mathrm{ord}_V(b)$, this extends to a homomorphism $\mathrm{ord}_V : k(W)^* \to \mathbb{Z}$ by $\mathrm{ord}_V(a/b) = \mathrm{ord}_V(a) - \mathrm{ord}_V(b)$.

When $\mathcal{O}_{W,V}$ is a discrete valuation ring, in particular whenever W is normal, this length equals the valuation $\nu_V(a)$, so definition 8.1.1 agrees with the order function of section 5.4. For a fixed $r \in k(W)^*$ the integer $\mathrm{ord}_V(r)$ vanishes for all but finitely many V (lemma 5.4.6), so each sum $\sum_V \mathrm{ord}_V(r)[V]$ below is finite.

Definition 8.1.2: Cycle

Let X be an algebraic scheme. A *k-cycle* on X is a finite formal $\mathbb{Z}$-linear combination $\sum_i n_i[V_i]$ of k-dimensional subvarieties $V_i \subseteq X$ with $n_i \in \mathbb{Z}$. The k-cycles form the free abelian group

$$Z_k(X) = \bigoplus_{\dim V = k} \mathbb{Z}\,[V]$$

on the set of k-dimensional subvarieties of X, and the group of all cycles is $Z_*(X) = \bigoplus_{k\geq 0} Z_k(X)$.

A cycle remembers only which subvarieties occur and with what integer multiplicity; it detects neither embedded nor nonreduced structure, a point made precise by the equality $Z_k(X) = Z_k(X_{\mathrm{red}})$. Two cases sit at the extremes. A 0-cycle is a finite formal sum of closed points. At the top, if X has dimension n with n-dimensional irreducible components $X_1, \ldots, X_r$, then $Z_n(X)$ is free on $[X_1], \ldots, [X_r]$.

The *fundamental cycle* of X ranges over *all* irreducible components $X_1, \ldots, X_t$, of whatever dimension:

$$[X] = \sum_{i=1}^{t} \operatorname{length}_{\mathcal{O}_{X,X_i}}\left(\mathcal{O}_{X,X_i}\right)[X_i] \;\in\; Z_*(X)$$

weighting each component by the multiplicity with which X carries it, and reducing to $\sum_i [X_i]$ when X is reduced. When X is equidimensional the sum is concentrated in degree n; in general it is not, and it must not be truncated to the top-dimensional components. A scheme such as the union of a plane and a line meeting it in a point carries a fundamental cycle with a 2-dimensional and a 1-dimensional term, and discarding the line would make the Segre classes of chapter 10 disagree with the standard ones on exactly such a base.

The one construction that produces relations among cycles is the divisor of a rational function. Given a $(k+1)$-dimensional subvariety $W \subseteq X$ and a nonzero $r \in k(W)^*$, its *divisor* is the k-cycle

$$[\operatorname{div}(r)] = \sum_V \operatorname{ord}_V(r)\,[V] \in Z_k(X)$$

the sum running over the codimension-one subvarieties V of W, each of which is a k-dimensional subvariety of X. The sum is finite by lemma 5.4.6, and $[\operatorname{div}(rs)] = [\operatorname{div}(r)] + [\operatorname{div}(s)]$ because ord_V is a homomorphism.

Definition 8.1.3: Rational Equivalence

A k-cycle $\alpha \in Z_k(X)$ is *rationally equivalent to zero*, written $\alpha \sim 0$, if there are finitely many $(k+1)$-dimensional subvarieties $W_i \subseteq X$ and rational functions $r_i \in k(W_i)^*$ with

$$\alpha = \sum_i [\operatorname{div}(r_i)]$$

Two k-cycles are *rationally equivalent*, written $\alpha \sim \beta$, when $\alpha - \beta \sim 0$. The cycles rationally equivalent to zero form a subgroup $\operatorname{Rat}_k(X) \leq Z_k(X)$: it is closed under addition by definition and under negation because $[\operatorname{div}(r^{-1})] = -[\operatorname{div}(r)]$.

Geometrically, a subvariety W together with a rational function r presents $[\operatorname{div}(r)]$ as the difference between the zeros and the poles of r, that is, between the fibers over 0 and ∞ of the rational map $r\colon W \dashrightarrow \mathbb{P}^1$. Rational equivalence is thus the relation generated by sweeping one cycle to another through a family parametrized by the line. In the one codimension seen before, $k = n-1$ on an n-dimensional variety, a cycle is a Weil divisor and rational equivalence is exactly the linear equivalence of definition 5.4.8; the Chow group below carries $\operatorname{Cl} X$ to all dimensions at once.

Definition 8.1.4: Chow Group

The *k-th Chow group* of an algebraic scheme X is the quotient

$$A_k(X) = Z_k(X)/\operatorname{Rat}_k(X)$$

Its elements are *k-cycle classes*, and the class of a cycle α is written $[\alpha]$, or simply α when no confusion results. The *total Chow group* is

$$A_*(X) = \bigoplus_{k\geq 0} A_k(X) = Z_*(X)/\operatorname{Rat}_*(X), \qquad \operatorname{Rat}_*(X) = \bigoplus_{k\geq 0} \operatorname{Rat}_k(X)$$

Following the cohomological convention, much of the literature grades by codimension, writing $A^p(X) = A_{n-p}(X)$ or $CH^p(X)$ when X is equidimensional of dimension n; the two indexings carry the same information, and definition 8.1.4 is the promised general form of the construction previewed for surfaces in box 7.2.14. A cycle $\sum_i n_i[V_i]$ is *positive* (or *effective*) if it is nonzero with every $n_i \geq 0$, and a cycle class is positive if it has a positive representative. Positivity is the algebraic trace of an actual geometric subvariety, and deciding which classes are positive is among the central difficulties of the subject.

Passing to the quotient collapses exactly the ambiguity that a complete variety imposes on its points.

Example 8.1.5: Zero-Cycles on the Line

On $X = \mathbb{P}^1$ any two closed points are rationally equivalent. Write t for the affine coordinate and ∞ for the point at infinity. Distinct points $p, q \in \mathbb{A}^1 \subseteq \mathbb{P}^1$ are cut out by $t-p$ and $t-q$, and the rational function $r = (t-p)/(t-q)$ has

$$\operatorname{ord}_p(r) = 1, \qquad \operatorname{ord}_q(r) = -1, \qquad \operatorname{ord}_x(r) = 0 \text{ for } x \neq p, q$$

since r takes the finite nonzero value 1 at ∞. Hence $[\operatorname{div}(r)] = [p] - [q]$ and $[p] \sim [q]$; taking $r = t-p$ instead gives $[\operatorname{div}(t-p)] = [p] - [\infty]$, so $[\infty]$ lies in the same class. Every point of $\mathbb{P}^1$ therefore represents a single class h. A rational function on $\mathbb{P}^1$ has as many zeros as poles, so $\sum_V \operatorname{ord}_V(r) = 0$ for all r, and the total-coefficient map $\sum_i n_i[p_i] \mapsto \sum_i n_i$ descends to an isomorphism

$$A_0(\mathbb{P}^1) \xrightarrow{\sim} \mathbb{Z}, \qquad h \mapsto 1$$

The affine line behaves oppositely. On $X = \mathbb{A}^1$ the function $t-p$ has the single zero p and no pole, so $[\operatorname{div}(t-p)] = [p]$ and every point satisfies $[p] \sim 0$; thus $A_0(\mathbb{A}^1) = 0$. On an affine line a point can be swept away to nothing, whereas on the complete line it cannot, and the surviving invariant of a class is its degree.

The contrast in example 8.1.5 is why completeness recurs throughout the theory: the degree of a zero-cycle is an invariant of its rational equivalence class only on a complete scheme, a point taken up in section 8.2. The remaining first properties of Chow groups are collected next; each is either immediate or follows from the localization sequence established later in the chapter.

Example 8.1.6: First Properties of Chow Groups
Let X be an algebraic scheme.

1. *Reduction.* Since cycles see only subvarieties, $A_k(X) = A_k(X_{\mathrm{red}})$ for every k.
2. *Disjoint unions.* If $X = X' \sqcup X''$ is a disjoint union of open-and-closed subschemes, then $A_k(X) = A_k(X') \oplus A_k(X'')$, as every subvariety lies in one piece.
3. *Top-dimensional cycles.* If $\dim X = n$, there is no subvariety of dimension $n+1$, so $\mathrm{Rat}_n(X) = 0$ and $A_n(X) = Z_n(X)$ is free on the n-dimensional components of X. In particular the coefficient of such a component in a class $\alpha \in A_n(X)$ is well defined.
4. *Right-exactness.* For closed subschemes $X_1, X_2 \subseteq X$ there is an exact sequence
$$A_k(X_1 \cap X_2) \to A_k(X_1) \oplus A_k(X_2) \to A_k(X_1 \cup X_2) \to 0$$
with no left-exactness in general. This is a consequence of the localization sequence developed later in the chapter.

Exercise 8.1.1

1. Let X be the reduced scheme consisting of two distinct points. Compute $Z_k(X)$ and $A_k(X)$ for all k.
2. Show that $A_0(\mathbb{A}^n) = 0$ for every $n \geq 1$.
3. Prove from the definitions that $Z_k(X) = Z_k(X_{\mathrm{red}})$ and $\mathrm{Rat}_k(X) = \mathrm{Rat}_k(X_{\mathrm{red}})$, and deduce $A_k(X) = A_k(X_{\mathrm{red}})$.
4. Let X have dimension n with n-dimensional irreducible components $X_1, \ldots, X_r$. Show that $A_n(X) = \bigoplus_{i=1}^r \mathbb{Z}\,[X_i]$.
5. Let C be a smooth projective curve over $\overline{k}$. Show that $A_0(C) \cong \mathrm{Cl}\, C = \mathrm{Pic}\, C$, and that the degree gives a short exact sequence $0 \to \mathrm{Pic}^0 C \to A_0(C) \xrightarrow{\deg} \mathbb{Z} \to 0$.

8.2 Proper Pushforward and the Degree

Cycles and their classes change under morphisms, and the first of these functorialities is covariant: a proper morphism pushes cycles forward, respects rational equivalence, and on a complete scheme produces the degree of a zero-cycle.

A proper morphism $f\colon X \to Y$ carries closed sets to closed sets, so the image of a subvariety $V \subseteq X$ is again a subvariety $W = f(V) \subseteq Y$: the image of an irreducible set is irreducible, and properness (universal closedness) makes it closed. Without properness the image can fail to be closed, as for the projection of example 3.5.24, and pushforward is not defined. The inclusion of function fields $k(W) \hookrightarrow k(V)$ is a finite extension exactly when $\dim V = \dim W$, and its degree counts the points of

V over a general point of W.

Definition 8.2.1: Proper Pushforward

Let $f\colon X \to Y$ be a proper morphism and $V \subseteq X$ a subvariety with image $W = f(V)$. Set

$$\deg(V/W) = \begin{cases} [k(V) : k(W)] & \dim V = \dim W \\ 0 & \dim V > \dim W \end{cases}$$

(the extension $k(W) \hookrightarrow k(V)$ is finite in the first case), and put $f_*[V] = \deg(V/W)\,[W]$. Extending $\mathbb{Z}$-linearly gives the *pushforward* $f_*\colon Z_k(X) \to Z_k(Y)$.

Pushforward is covariant: for composable proper morphisms $(g \circ f)_* = g_* \circ f_*$, since degrees of function-field extensions multiply in towers. When $\dim V = \dim W$ the number $\deg(V/W)$ counts the sheets of the generically finite cover $V \to W$, matching the topological degree over $\mathbb{C}$. That f_* descends to Chow groups is the content of this section, and rests on how it acts on the divisor of a rational function.

Proposition 8.2.2: Pushforward of Divisor

Let $f : X \to Y$ be a proper surjective morphisms between varieties, and let $r \in k(X)^*$. Then:

1. if $\dim(Y) < \dim(X)$, then
$$f_*[\mathrm{div}(r)] = 0$$
2. if $\dim(Y) = \dim(X)$, then
$$f_*[\mathrm{div}(r)] = [\mathrm{div}(N(r))]$$
where $N(r)$ is the *norm* of r (computed as the determinant of the $R(Y)$-linear endomorphism of $R(X)$ given by multiplying by r).

***Proof*:**
The order function and the norm are multiplicative, so both sides of each identity are homomorphisms in r. Both parts reduce to two model computations; write $n = \dim X$.

Step 1: the line over a point. Let $Y = \operatorname{Spec} k$ and $X = \mathbb{P}^1$, with f the structure map. Every closed point of $\mathbb{P}^1$ has residue field $k = \overline{k}$, so $f_*[p] = [Y]$ for each p and $f_*[\mathrm{div}(r)] = \big(\sum_V \operatorname{ord}_V(r)\big)[Y]$. Writing $r = c\prod_j (t - a_j)^{m_j}$ in the affine coordinate t, the order of r at a_j is m_j and its order at ∞ is $-\sum_j m_j$ (in $s = 1/t$ one has $r = s^{-\sum_j m_j} u$ with u a unit at $s = 0$), so $\sum_V \operatorname{ord}_V(r) = 0$ and $f_*[\mathrm{div}(r)] = 0$. The same count shows that on any proper curve C over a field K a principal divisor has degree zero: normalize C and choose a finite map $\widetilde{C} \to \mathbb{P}^1_K$, then apply Step 2 to that map and this computation to $\mathbb{P}^1_K \to \operatorname{Spec} K$.

Step 2: finite morphisms of equal dimension. Suppose f is finite with $\dim Y = \dim X$, so that $K = k(Y) \subseteq L = k(X)$ is a finite extension. Fix a codimension-one subvariety $W \subseteq Y$ and let $A = \mathcal{O}_{Y,W}$, a one-dimensional local domain with fraction field K. Its integral closure B in L is a finite A-module whose maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_s$ correspond to the codimension-one subvarieties $V_i \subseteq X$ over W, with $B_{\mathfrak{m}_i} = \mathcal{O}_{X,V_i}$ and residue degree $[k(V_i) : k(W)] = [B/\mathfrak{m}_i : A/\mathfrak{m}]$. Only these V_i contribute to the coefficient of $[W]$ in $f_*[\mathrm{div}(r)]$, which is therefore $\sum_i [k(V_i) : k(W)] \operatorname{ord}_{V_i}(r)$, while the coefficient of $[W]$ in $[\mathrm{div}(N(r))]$ is $\mathrm{ord}_W(N(r))$. For $r \in B$ a nonzerodivisor these agree:

$$\mathrm{ord}_W(N(r)) = \mathrm{length}_A\big(A/N(r)A\big) = \mathrm{length}_A\big(B/rB\big) = \sum_i [k(V_i) : k(W)]\,\mathrm{ord}_{V_i}(r)$$

the middle equality is the determinant-length formula, since $N(r)$ is the determinant of multiplication by r on the finite A-module B, and the last is additivity of length across the semilocal decomposition of B, weighted by residue degrees (see [24, Appendix A] for this length bookkeeping). As $L = \mathrm{Frac}(B)$ and both sides are homomorphisms, the identity holds for all $r \in L^*$, which is part (2) for finite f.

Step 3: the general case. A proper surjective f with $\dim Y = \dim X$ is generically finite, and the coefficient of a codimension-one $W \subseteq Y$ in $f_*[\mathrm{div}(r)]$ is computed over the generic point of W, where f becomes finite; Step 2 gives (2). For (1), let $\dim Y < n$. Since $f_*[\mathrm{div}(r)] \in Z_{n-1}(Y)$ vanishes when $\dim Y < n-1$, only $\dim Y = n-1$ remains, and there $f_*[\mathrm{div}(r)] = c\,[Y]$ with c its coefficient over the generic point of Y. Base-changing to $K = k(Y)$ replaces f by a proper curve $C = X_K$ over the field K and r by a function in $k(C)^*$, and $c = \sum_{P \in C} \mathrm{ord}_P(r)\,[k(P) : K]$ is the degree of its principal divisor, which vanishes by Step 1. Hence $f_*[\mathrm{div}(r)] = 0$, completing the proof.

Theorem 8.2.3: Pushforward Preserves Rational Equivalence

Let $f \colon X \to Y$ be proper. If $\alpha \in Z_k(X)$ is rationally equivalent to zero, then so is $f_*\alpha$. Consequently f_* descends to a homomorphism

$$f_* \colon A_k(X) \to A_k(Y)$$

and $A_*(-)$ equipped with proper pushforward is a covariant functor.

***Proof*:**
It suffices to push forward a generator $\alpha = [\mathrm{div}(r)]$, where $r \in k(W)^*$ for some $(k+1)$-dimensional subvariety $W \subseteq X$. Replacing X by W and Y by $f(W)$ makes f proper and surjective, and proposition 8.2.2 identifies $f_*[\mathrm{div}(r)]$ with 0 (when $\dim f(W) < \dim W$) or with $[\mathrm{div}(N(r))]$ (when $\dim f(W) = \dim W$). Either value is a principal divisor, hence rationally equivalent to zero, so $f_*\alpha \sim 0$. Functoriality on classes then follows from $(g \circ f)_* = g_* \circ f_*$ on cycles, as we sought to show.

Definition 8.2.4: Degree of a Zero-Cycle

Let X be complete (proper over $\operatorname{Spec} k$), with structure morphism $\pi\colon X \to \operatorname{Spec} k$, and let $\alpha = \sum_P n_P[P]$ be a zero-cycle on X. The *degree* of α is

$$\deg(\alpha) = \int_X \alpha = \sum_P n_P\,[k(P) : k]$$

Since $\operatorname{Spec} k$ is a single point with no lower-dimensional subvarieties, $A_0(\operatorname{Spec} k) = Z_0(\operatorname{Spec} k) = \mathbb{Z}\,[\operatorname{Spec} k] \cong \mathbb{Z}$, and under this identification $\deg(\alpha) = \pi_*\alpha$. When the scheme is clear one writes $\int \alpha$ for $\int_X \alpha$.

By theorem 8.2.3 rationally equivalent zero-cycles have the same degree, so $\deg\colon A_0(X) \to \mathbb{Z}$ is well defined on a complete scheme. Extending by $\int_X \alpha = 0$ for $\alpha \in A_k(X)$ with $k > 0$ and using $\pi_X = \pi_Y \circ f$, any morphism $f\colon X \to Y$ of complete schemes satisfies

$$\int_X \alpha = \int_Y f_*\alpha$$

for every $\alpha \in A_*(X)$, a special case of the functoriality of pushforward.

8.2.5 Assuming Pushforward with subschemes Let $Y_1, \ldots, Y_r$ be closed subschemes of a scheme X. Let Y be a closed subscheme of X which contains all the Y_i. Given $\alpha_i \in A_*Y_i$, $i = 1, \ldots, r$, and $\beta \in A_*Y$, the notation "$\beta = \sum_{i=1}^r \alpha_i$ in A_*Y" stands in for the precise equation $\beta = \sum_{i=1}^r \varphi_{i*}(\alpha_i)$ where φ_i is the inclusion of Y_i in Y.

Example 8.2.6: Degrees, Bézout, and the Role of Properness

1. *Bézout's theorem.* Two curves $C, D \subseteq \mathbb{P}^2$ of degrees c, d with no common component meet in a zero-cycle $C \cdot D$ of degree cd. Pushing forward to a point, $\int_{\mathbb{P}^2}[C \cdot D] = cd$: the classical intersection number from the chapter opening is the degree of a zero-cycle, and its independence of the choice of C and D in their linear systems is rational equivalence together with theorem 8.2.3.

2. *Properness is necessary.* For the non-proper projection of example 3.5.24 the image of a subvariety can fail to be closed, so f_* is not even defined; and on $\mathbb{A}^1$ every point is rationally equivalent to zero (example 8.1.5), so no degree survives. The degree is an invariant of a rational equivalence class only on a complete scheme.

3. *A finite cover of curves.* A finite morphism $\varphi\colon C \to C'$ of smooth projective curves satisfies $\varphi_*[P] = [k(P) : k(\varphi(P))]\,[\varphi(P)]$. For a point $Q \in C'$ away from the branch locus, $\varphi^{-1}(Q)$ consists of $\deg\varphi = [k(C) : k(C')]$ reduced points, so $\varphi_*\big[\varphi^{-1}(Q)\big] = (\deg\varphi)\,[Q]$; since $\deg(\varphi_*\alpha) = \deg(\alpha)$ on the complete curves, the sheet count is exactly the degree recorded by φ.

Exercise 8.2.1

1. For the finite morphism $\varphi\colon \mathbb{P}^1 \to \mathbb{P}^1$, $[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$ (in characteristic $\neq 2$), compute $\varphi_*[p]$ for a closed point p and $\varphi_*[\varphi^{-1}(q)]$ for a closed point q. Verify that $\deg(\varphi_*\alpha) = \deg(\alpha)$ for every zero-cycle α, and explain where the degree $\deg\varphi = 2$ appears.
2. Let $\alpha = \sum_i n_i[P_i]$ be a zero-cycle on $\mathbb{P}^n$ over $\overline{k}$. Compute $\deg(\alpha)$ and prove it is invariant under rational equivalence.
3. Deduce from proposition 8.2.2 that a nonzero rational function on a smooth projective curve C has principal divisor of degree zero, and show by example that this fails on $\mathbb{A}^1$.
4. For the structure map $\pi\colon \mathbb{P}^1 \to \operatorname{Spec}\overline{k}$ and a rational function $r \in k(\mathbb{P}^1)^*$, verify $\int_{\mathbb{P}^1}[\operatorname{div}(r)] = \int_{\operatorname{Spec}\overline{k}} \pi_*[\operatorname{div}(r)]$, and say which hypothesis of definition 8.2.4 would fail were $\mathbb{P}^1$ replaced by $\mathbb{A}^1$.

8.3 Flat Pullback

Proper pushforward of section 8.2 is covariant: it moves cycles along a morphism in the direction of the arrow, lowering nothing and raising nothing in dimension. The second functoriality of the Chow group runs the other way. A morphism $f\colon X \to Y$ pulls a subvariety $V \subseteq Y$ back to its preimage $f^{-1}(V) \subseteq X$, and one would like to read that preimage as a cycle on X. For an arbitrary f this fails: the preimage can jump in dimension from fiber to fiber, can acquire embedded components, and carries no canonical multiplicities. The hypothesis that repairs every one of these defects at once is flatness. A flat morphism of relative dimension n has equidimensional fibers, so the preimage of a k-dimensional subvariety is pure of dimension $k+n$, and the scheme-theoretic preimage supplies the multiplicities with which its components must be counted. The resulting map $f^*\colon Z_k(Y) \to Z_{k+n}(X)$ raises dimension by n, is contravariant, and descends to Chow groups.

This section constructs f^*, records its three structural properties (it is a graded homomorphism, it composes contravariantly, and it commutes with proper pushforward across a fiber square), and proves that it respects rational equivalence. The two guiding examples are the ones every later computation rests on: restriction to an open subscheme, and the projection $X \times \mathbb{A}^n \to X$ of a trivial affine bundle.

Recall that a morphism $f\colon X \to Y$ of algebraic schemes is flat if each local ring $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. Flatness is the algebraic incarnation of the geometric demand that the fibers of f vary without collapsing or jumping, and its precise dimensional consequence is what makes pullback of cycles possible.

Definition 8.3.1: Flat Morphism of Relative Dimension n

A morphism $f\colon X \to Y$ of algebraic schemes is *flat of relative dimension* n if f is flat and, for every subvariety $W \subseteq Y$ and every irreducible component Z of $f^{-1}(W)$,

$$\dim Z = \dim W + n$$

Equivalently, f is flat and every nonempty fiber of f has pure dimension n.

The equivalence stated in the definition is the mechanism the whole construction turns on, so it is worth isolating. Flatness forces the fiber dimension to be locally constant, and it forces the preimage of a subvariety to inherit that constant as an additive shift; neither holds without it.

Proposition 8.3.2: Flat Preimages Are Equidimensional

Let $f\colon X \to Y$ be a flat morphism of algebraic schemes with Y a variety, and suppose every nonempty fiber of f has pure dimension n. Then for every subvariety $W \subseteq Y$ each irreducible component of $f^{-1}(W)$ has dimension $\dim W + n$. In particular $f^{-1}(W)$ is either empty or pure of dimension $\dim W + n$.

***Proof*:**
Write $g\colon f^{-1}(W) \to W$ for the restriction of f; it is the base change of f along the closed immersion $W \hookrightarrow Y$, hence flat, and its fibers are fibers of f, so each nonempty fiber of g has pure dimension n. Thus it suffices to prove the statement for $W = Y$: that a flat morphism $g\colon X' \to W$ onto a variety W, with all nonempty fibers of pure dimension n, has $\dim Z = \dim W + n$ for every irreducible component Z of X'.

Let $Z \subseteq X'$ be an irreducible component, with generic point η. Flatness satisfies going-down (the going-down property of flat local homomorphisms, [33, p. III.9]), so the minimal prime η of X' contracts to the minimal prime of W; that is, $g(\eta)$ is the generic point of W, and g restricts to a dominant morphism $g\big|_Z\colon Z \to W$ of varieties. Both are integral of finite type over $\overline{k}$, so their dimensions are transcendence degrees of function fields, and the dimension of a fiber of a dominant morphism of varieties can be read at a closed point.
The fiber-dimension theorem for the dominant morphism $g\big|_Z\colon Z \to W$ of varieties ([33, Exercise II.3.22]) provides a dense open $U \subseteq Z$ such that for every closed point $z \in U$, the fiber $(g\big|_Z)^{-1}(g(z))$ has dimension exactly $\dim Z - \dim W$ at z. The complement in Z of the other irreducible components of X' is also a dense open, so their intersection is nonempty and contains a closed point z; write $w = g(z) \in W$. Since z lies off the other components, X' coincides with Z near z, so the fiber $X'_w = g^{-1}(w)$ coincides with $(g\big|_Z)^{-1}(w)$ near z, and its dimension at z is $\dim Z - \dim W$. By hypothesis X'_w is nonempty of pure dimension n, so this local fiber dimension equals n. Therefore

$$n = \dim Z - \dim W$$

hence $\dim Z = \dim W + n$, as we sought to show.

The proposition is the geometric license behind the definition of flat pullback. If $V \subseteq Y$ is a k-dimensional subvariety and f is flat of relative dimension n, then $f^{-1}(V)$ is a closed subscheme of X, pure of dimension $k+n$, and one takes its associated cycle, weighting each component by the length that measures the multiplicity with which the scheme carries it. This is exactly the fundamental cycle of definition 8.1.2 applied to the scheme-theoretic preimage.

Definition 8.3.3: Flat Pullback

Let $f\colon X \to Y$ be a flat morphism of relative dimension n and $V \subseteq Y$ a k-dimensional subvariety. The scheme-theoretic preimage $f^{-1}(V) = X \times_Y V$ is a closed subscheme of X, pure of dimension $k+n$, with irreducible components $Z_1, \dots, Z_r$. The *flat pullback* of $[V]$ is the $(k+n)$-cycle

$$f^*[V] = \left[f^{-1}(V)\right] = \sum_{i=1}^{r} \operatorname{length}_{\mathcal{O}_{f^{-1}(V),Z_i}} \left(\mathcal{O}_{f^{-1}(V),Z_i}\right) [Z_i]$$

the length being taken in the local ring of the preimage scheme at the generic point of Z_i. Extending $\mathbb{Z}$-linearly gives the flat pullback

$$f^*\colon Z_k(Y) \to Z_{k+n}(X)$$

The multiplicities are the point of the construction. The preimage as a topological set records only the components Z_i; the scheme structure records how many times f "covers" each one, and it is these lengths that make f^* compatible with divisors and hence with rational equivalence. When f is an open immersion the preimage is reduced and every length is 1, so $f^*[V] = [V \cap X]$; this is the source of the restriction map computed in the next example. The relative dimension n is the amount by which f^* raises the dimension of a cycle, so f^* is graded of degree $+n$.

Example 8.3.4: Open Restriction and the Projection $X \times \mathbb{A}^n \to X$

Two flat morphisms recur throughout the theory, and their pullbacks can be read off the definition.

1. *Open immersion.* Let $j\colon U \hookrightarrow Y$ be the inclusion of an open subscheme. Such j is flat of relative dimension 0: it is a localization at the level of local rings, and its fibers are single reduced points (over $u \in U$) or empty (over $y \notin U$). For a k-dimensional subvariety $V \subseteq Y$ the preimage $j^{-1}(V) = V \cap U$ is the open subscheme of V, which is reduced. If $V \cap U \neq \emptyset$ it is a k-dimensional subvariety of U and $j^*[V] = [V \cap U]$; if $V \subseteq Y \setminus U$ the preimage is empty and $j^*[V] = 0$. Thus flat pullback along an open immersion is the *restriction of cycles* $Z_k(Y) \to Z_k(U)$ that forgets the part of a cycle supported off U, the map at the heart of the localization sequence of section 8.4.

2. *Trivial affine bundle.* Let $p\colon X \times \mathbb{A}^n \to X$ be the projection. It is flat, being a base change of the flat morphism $\mathbb{A}^n_{\overline{k}} \to \operatorname{Spec} \overline{k}$ along $X \to \operatorname{Spec} \overline{k}$, and every fiber is a copy of $\mathbb{A}^n$, pure of dimension n; so p is flat of relative dimension n. For a k-dimensional subvariety $V \subseteq X$ the preimage is $V \times \mathbb{A}^n$, which is integral of dimension $k+n$, so

$$p^*[V] = [V \times \mathbb{A}^n]$$

the cylinder over V. No multiplicity appears because $V \times \mathbb{A}^n$ is a variety. This is the computation that will make p^* an isomorphism on Chow groups (theorem 8.4.6).

Both examples exhibit the pattern the general theory generalizes: pullback restricts a cycle to an open part in the first case and thickens it uniformly along the fiber in the second. The structural properties below show that these behaviors are functorial.

The first property collects the ways in which f^* is well behaved as a map of cycle groups. It is a graded homomorphism raising dimension by the relative dimension, and it composes contravariantly, so that a tower of flat morphisms pulls back through the composite exactly as one expects.

Proposition 8.3.5: Basic Properties of Flat Pullback

Let $f\colon X \to Y$ be flat of relative dimension n.

1. *Homomorphism.* $f^*\colon Z_k(Y) \to Z_{k+n}(X)$ is a homomorphism of abelian groups for each k, and assembles to a graded homomorphism $f^*\colon Z_*(Y) \to Z_*(X)$ of degree n.

2. *Fundamental cycle.* $f^*[Y] = [X]$ for any algebraic scheme Y: flat pullback of the fundamental cycle of the base is the fundamental cycle of the total space.

3. *Composition.* If $g\colon Y \to Y'$ is flat of relative dimension m, then $g \circ f$ is flat of relative dimension $m + n$ and
$$(g \circ f)^* = f^* \circ g^*$$
as maps $Z_k(Y') \to Z_{k+m+n}(X)$.

4. *Identity.* $(\mathrm{id}_X)^* = \mathrm{id}$ on $Z_*(X)$.

***Proof*:**

1. *Homomorphism.* By construction f^* is defined on generators $[V]$ and extended $\mathbb{Z}$-linearly, so it is a homomorphism $Z_k(Y) \to Z_{k+n}(X)$ by fiat; proposition 8.3.2 is what guarantees the target group is $Z_{k+n}(X)$, since $f^{-1}(V)$ is pure of dimension $k + n$. Summing over k gives a degree-n graded homomorphism.

2. *Fundamental cycle.* Write $[Y] = \sum_i m_i[Y_i]$ over the irreducible components of Y, with $m_i = \mathrm{length}_{\mathcal{O}_{Y,Y_i}}(\mathcal{O}_{Y,Y_i})$. If Y is a variety this is the single generator $[Y]$ with multiplicity 1, and $f^{-1}(Y) = X$ gives $f^*[Y] = [X]$ at once.

   In general, $f^*[Y] = \sum_i m_i[f^{-1}(Y_i)]$ by definition of flat pullback, and since $Y = \bigcup_i Y_i$ we have $X = \bigcup_i f^{-1}(Y_i)$. The two sides are cycles, so it is enough to compare the coefficient of each subvariety of X, and the work is in matching up which subvarieties occur. Flatness enters through going-down: $\mathcal{O}_{Y,f(\eta)} \to \mathcal{O}_{X,\eta}$ is a flat local homomorphism for every $\eta \in X$, so every generization of $f(\eta)$ in Y lifts to a generization of η in X.

   Every component of X dominates a component of Y. Let X_j be a component of X, with generic point η_j, and put $y = f(\eta_j)$. A generization of y lifts to a generization of η_j; but η_j is a generic point of X, so its only generization in X is itself, and therefore y has no proper generization in Y. So y is the generic point of some component Y_i, and X_j is a component of $f^{-1}(Y_i)$. That i is unique: for $i' \neq i$ the generic point of Y_i does not lie in $Y_{i'}$, so $\eta_j \notin f^{-1}(Y_{i'})$ and only the i-th term of $\sum_i m_i[f^{-1}(Y_i)]$ contributes to the coefficient of $[X_j]$.

   Conversely, every component of every $f^{-1}(Y_i)$ is a component of X. Let W be one, with generic point η_W. Going-down applied at η_W lifts the generic point of Y_i to a generization of η_W inside $f^{-1}(Y_i)$, and η_W is maximal there, so W dominates Y_i, that is $f(\eta_W)$ is the generic point of Y_i. If W were not a component of X it would lie in some component $X_{j'}$, whose generic point generizes η_W and therefore maps to a generization of the generic point of Y_i; the latter being a generic point of Y, that forces $X_{j'}$ to dominate Y_i as well, so $\dim X_{j'} = \dim Y_i + n = \dim W$ by proposition 8.3.2, and $W = X_{j'}$. A component Y_i with $f^{-1}(Y_i) = \emptyset$ contributes to neither side.

Multiplicities. It remains to see that the multiplicity of X_j in $[X]$ equals m_i times its multiplicity in $[f^{-1}(Y_i)]$. This is a length computation: $\mathcal{O}_{Y,Y_i} \to \mathcal{O}_{X,X_j}$ is a flat local homomorphism of Noetherian local rings, and [10, Length Along a Flat Local Homomorphism] gives $\operatorname{length}(\mathcal{O}_{X,X_j}) = \operatorname{length}(\mathcal{O}_{Y,Y_i}) \cdot \operatorname{length}\big(\mathcal{O}_{X,X_j}/\mathfrak{m}_{Y_i}\mathcal{O}_{X,X_j}\big)$. Because Y_i is a subvariety and so reduced, $\mathcal{O}_{Y,Y_i}/\mathfrak{m}_{Y_i}$ is the function field of Y_i, whence $\mathcal{O}_{X,X_j}/\mathfrak{m}_{Y_i}\mathcal{O}_{X,X_j}$ is the local ring of $f^{-1}(Y_i)$ at X_j and the second factor is exactly the multiplicity of X_j in $[f^{-1}(Y_i)]$. Summing over i and j gives $f^*[Y] = [X]$, as we sought to show.

3. *Composition.* Composites of flat morphisms are flat, and relative dimensions add because fiber dimensions add along the tower: a fiber of $g \circ f$ over $y' \in Y'$ maps to the fiber $g^{-1}(y')$ of pure dimension m, with fibers of f of pure dimension n, so it has pure dimension $m+n$. Hence $g \circ f$ is flat of relative dimension $m+n$. For the cycle identity it suffices to check on a generator $[V]$ with $V \subseteq Y'$ a k-dimensional subvariety. Both sides are cycles supported on $(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V))$, so the claim is that the multiplicities match component by component. The scheme-theoretic preimage is computed by iterated fiber product,

$$(g \circ f)^{-1}(V) = X \times_{Y'} V = X \times_Y (Y \times_{Y'} V) = f^{-1}\big(g^{-1}(V)\big)$$

so the underlying scheme of $(g \circ f)^{-1}(V)$ equals the underlying scheme of $f^{-1}(g^{-1}(V))$. It remains to see that the length multiplicities agree, that is, that applying f^* to the cycle $g^*[V] = \sum_i \ell_i [W_i]$ (where W_i are the components of $g^{-1}(V)$ with lengths ℓ_i) reproduces the multiplicities of $(g \circ f)^{-1}(V)$. Fix a component Z of the common preimage, with generic point z lying over the generic point w of some W_i. Flatness of f makes $\mathcal{O}_{X,z}$ flat over $\mathcal{O}_{Y,w}$, and length is additive and multiplies through a flat local homomorphism along the fiber: the length of $\mathcal{O}_{(g\circ f)^{-1}(V),z}$ equals ℓ_i times the length of $\mathcal{O}_{f^{-1}(W_i),z}$. Summing z over the components of $f^{-1}(W_i)$ and i over the components of $g^{-1}(V)$ gives $(g \circ f)^*[V] = f^*(g^*[V])$, as required.

4. *Identity.* The identity map is flat of relative dimension 0 with $(\mathrm{id})^{-1}(V) = V$ reduced, so $(\mathrm{id}_X)^*[V] = [V]$ on generators, completing the proof.

Contravariant functoriality (part 3) is what will let a Chow-group computation be assembled from simpler flat maps, exactly as covariant functoriality of pushforward let degrees be computed in towers. The fundamental-cycle identity (part 2) is the seed of every computation in section 8.5: it identifies the generator of the top Chow group of the total space of a flat family in terms of the base.

The remaining two properties are the interaction of flat pullback with the machinery already built. The first is its compatibility with proper pushforward. Pullback and pushforward move cycles in opposite directions, and they do not commute for a single morphism; but across a fiber square, where one has a flat map and a proper map meeting at right angles, the two operations do commute. This is the cycle-level form of the projection formula, and it is the technical engine behind essentially every later compatibility.

Proposition 8.3.6: Compatibility of Flat Pullback and Proper Pushforward

Consider a fiber square of algebraic schemes

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

that is Cartesian, $X' = X \times_Y Y'$, with f proper and g flat of relative dimension n. Then f' is proper, g' is flat of relative dimension n, and

$$g^* \circ f_* = f'_* \circ g'^* \colon Z_k(X) \to Z_{k+n}(Y')$$

Consequently the identity holds on Chow groups.

***Proof*:**
Step 1: the maps in the base-changed square. Properness and flatness are stable under base change, so f' is proper and g' is flat; the fibers of g' over a point of X are fibers of g over its image in Y, so g' has the same pure relative dimension n as g. Both composites $g^* \circ f_*$ and $f'_* \circ g'^*$ therefore land in $Z_{k+n}(Y')$, and both are homomorphisms, so it suffices to check the identity on a generator $[V]$ with $V \subseteq X$ a k-dimensional subvariety.

Step 2: reduction to $X = V$. Replacing X by V, Y by $W = f(V)$, and forming the induced fiber squares, the four schemes become V, W, $V' = V \times_W W'$, $W' = W \times_Y Y'$, with $f\colon V \to W$ proper surjective and $g\colon W' \to W$ flat of relative dimension n. It suffices to prove

$$g^*(f_*[V]) = f'_*\big(g'^*[V]\big)$$

where now $f_*[V] = \deg(V/W)[W]$ by definition 8.2.1 and $g'^*[V] = [g'^{-1}(V)] = [V']$ (with its multiplicities).

Step 3: the two dimension regimes. There are two cases according to whether f drops dimension.

If $\dim V > \dim W$, then $\deg(V/W) = 0$, so the left side is $g^*(0) = 0$. On the right, $V' = V \times_W W'$ maps to W' under f' with fibers the fibers of $V \to W$, which are positive dimensional over the generic point; hence $\dim V' > \dim W'$ and every component of V' has image of strictly smaller dimension in W', so $f'_*[V'] = 0$ by the dimension clause of definition 8.2.1. Both sides vanish.

If $\dim V = \dim W$, write $d = \deg(V/W) = [k(V) : k(W)]$. The left side is $g^*(d\,[W]) = d\,[W']$, using $g^*[W] = [W'] = [g^{-1}(W)]$ (the base W is a variety and g is flat, so g^* of its fundamental cycle is the fundamental cycle of W' by proposition 8.3.5(2), which here is the total space $W' = g^{-1}(W)$). For the right side, $[V'] = g'^*[V]$ carries the pullback multiplicities flagged above, and $f'_*[V']$ must account for both those multiplicities and the residue-field degrees; flatness is what makes only the field degree survive. Fix a component W'_j of W', with generic point η_j. Localizing at η_j, the fiber of $f'\colon V' \to W'$ over η_j is Spec of $k(V) \otimes_{k(W)} k(W'_j)$: the generic fiber of $f\colon V \to W$ is $\operatorname{Spec} k(V)$, a $k(W)$-algebra of dimension d since V, W are varieties, and flat base change along $k(W) \hookrightarrow k(W'_j)$ preserves that dimension, so $\dim_{k(W'_j)}\big(k(V) \otimes_{k(W)} k(W'_j)\big) = d$. This dimension is exactly the total multiplicity that $f'_*[V']$ records over W'_j: summing the pullback multiplicity of each component of V' dominating W'_j against its residue degree over $k(W'_j)$ recovers the length d of the fiber algebra, because flatness of g (hence of g') forbids any length being created or lost in the base change. Thus $f'_*[V'] = \sum_j d\,[W'_j] = d\,[W']$, matching the left side. The two sides agree in both regimes, which establishes $g^* f_* = f'_* g'^*$ on cycles. Since each of g^*, f_*, f'_*, g'^* descends to Chow groups by theorem 8.2.3 and proposition 8.3.7 below, the identity descends as well, completing the proof.

The compatibility should be read as the statement that pushing a cycle forward and then restricting to a flat base change gives the same class as first restricting and then pushing forward within the base change. In the special case where g is an open immersion, it says that pushforward commutes with restriction to an open set, which is the reason the localization sequence of the next section is a sequence of A_*-functorial maps. In the case where f is the projection to a point, it recovers the invariance of the degree of a zero-cycle under a flat base change of the ground scheme.

Everything so far concerns cycles. To obtain a map of Chow groups, flat pullback must carry Rat_k into Rat_{k+n}. The mechanism is that flat pullback commutes with taking the divisor of a rational function, up to the multiplicities that flatness controls, so a cycle that is a sum of divisors pulls back to a sum of divisors.

Proposition 8.3.7: Flat Pullback Preserves Rational Equivalence

Let $f\colon X \to Y$ be flat of relative dimension n. If $\alpha \in Z_k(Y)$ is rationally equivalent to zero, then $f^*\alpha \in Z_{k+n}(X)$ is rationally equivalent to zero. Consequently f^* descends to a homomorphism

$$f^*\colon A_k(Y) \to A_{k+n}(X)$$

and $A_*(-)$ equipped with flat pullback is a contravariant functor on flat morphisms of algebraic schemes, raising dimension by the relative dimension.

***Proof*:**
It suffices to treat a generator of $\mathrm{Rat}_k(Y)$, namely $\alpha = [\mathrm{div}(r)]$ for a $(k+1)$-dimensional subvariety $W \subseteq Y$ and a nonzero $r \in k(W)^*$. The claim is that $f^*[\mathrm{div}(r)]$ is a sum of divisors of rational functions on subvarieties of X.

Step 1: reduction to the total space over W. The cycle $[\mathrm{div}(r)]$ is supported on W, and $f^*[\mathrm{div}(r)]$ is supported on $f^{-1}(W)$. Let $W' = f^{-1}(W) = X \times_Y W$ with its scheme-theoretic (possibly non-reduced) structure, and let $f_W\colon W' \to W$ be the restriction of f, which is flat of relative dimension n by base change. Because the components of $f^{-1}(W)$ are the components of W' and the pullback multiplicities are computed inside W', it is enough to prove that $f_W^*[\mathrm{div}(r)] \sim 0$ in $Z_{k+n}(W')$; the class then remains rationally equivalent to zero in X under the proper (closed immersion) pushforward along $W' \hookrightarrow X$, which preserves rational equivalence by theorem 8.2.3. Thus one may assume $Y = W$ is a variety and $r \in k(Y)^*$.

Step 2: the key identity on components. With Y a variety and $r \in k(Y)^*$, the divisor $[\mathrm{div}(r)] = \sum_V \mathrm{ord}_V(r)[V]$ runs over the codimension-one subvarieties $V \subseteq Y$. The assertion to prove is the flat-pullback divisor formula

$$f^*[\mathrm{div}(r)] = \left[\mathrm{div}(f^\# r)\right] \quad \text{on each component of } X \tag{8.1}$$

interpreted componentwise as follows. Let $X_1, \ldots, X_s$ be the irreducible components of X, each of dimension $n + \dim Y$ by proposition 8.3.2, with generic points dominating Y (flatness sends generic points to the generic point of the base). On each X_j the pullback $f^\# r$ of r is a nonzero rational function, since $r \neq 0$ and X_j dominates Y. The claim is

$$f^*[\mathrm{div}(r)] = \sum_{j=1}^{s} m_j \,[\mathrm{div}(f^{\#}r\big|_{X_j})]$$

where $m_j = \mathrm{length}_{\mathcal{O}_{X,X_j}}(\mathcal{O}_{X,X_j})$ is the multiplicity of the component X_j in the fundamental cycle $[X]$. Each term on the right is a divisor of a rational function on the subvariety $(X_j)_{\mathrm{red}}$, hence rationally equivalent to zero, so (8.1) gives $f^*[\mathrm{div}(r)] \sim 0$.

Step 3: verifying the identity. Fix a codimension-one subvariety $D \subseteq X_j$, a $(k+n)$-dimensional subvariety of X, and let $V = \overline{f(D)} \subseteq Y$. Two cases arise by the relative-dimension count of proposition 8.3.2 applied to $f\big|_{X_j}$.

If $\dim V = \dim Y$, then $V = Y$ and D dominates Y; the coefficient of $[D]$ in $[\mathrm{div}(f^{\#}r)]$ is $\mathrm{ord}_D(f^{\#}r)$, computed in the discrete-valuation-style local ring $\mathcal{O}_{X_j,D}$, while D does not appear in $f^*[\mathrm{div}(r)]$ because $[\mathrm{div}(r)]$ has no component equal to Y. But $\mathrm{ord}_D(f^{\#}r) = 0$. The point is that D dominates Y: its generic point ξ maps to the generic point of Y, at which r is a nonzero rational function, that is, a unit of $k(Y)$. Since $f^{\#}r$ is the pullback of r along the local homomorphism $\mathcal{O}_{Y,Y} = k(Y) \to \mathcal{O}_{X_j,D}$ induced on stalks at ξ, the image $f^{\#}r$ is a unit of $\mathcal{O}_{X_j,D}$, so $\mathrm{ord}_D(f^{\#}r) = 0$ and no such D contributes to either side. (A nonzero rational function need not be a unit along a codimension-one subvariety in general; here it is one precisely because D dominates the base and so sees r only at the generic point of Y.)

If $\dim V = \dim Y - 1$, then V is one of the codimension-one subvarieties appearing in $[\mathrm{div}(r)]$, and D is a component of $f^{-1}(V)$, lying on one or more of the components X_j. Two kinds of local ring enter the count, and keeping them distinct is the crux. Write

$$A = \mathcal{O}_{Y,V}, \qquad \widetilde{B} = \mathcal{O}_{X,D}$$

Here A is a one-dimensional local domain and $\mathrm{ord}_V(r) = \mathrm{length}_A(A/rA)$; the ring $\widetilde{B}$ is the local ring of the whole (possibly nonreduced) scheme X at the generic point of D, and it is flat over A because f is flat. The minimal primes of the one-dimensional ring $\widetilde{B}$ correspond to the components X_j passing through D; for such a component the quotient $B_j = \widetilde{B}/\mathfrak{p}_j = \mathcal{O}_{X_j,D}$ is the discrete-valuation-style local ring of the reduced component X_j at D, in which $\mathrm{ord}_D(f^{\#}r\big|_{X_j}) = \mathrm{length}_{B_j}(B_j/(f^{\#}r)B_j)$ is computed. When X is irreducible along D there is a single such j, but in general several components may meet along D.
The coefficient of $[D]$ in $f^*[\mathrm{div}(r)]$ is $\mathrm{ord}_V(r) \cdot e_D$, where

$$e_D = \mathrm{length}_{\mathcal{O}_{f^{-1}(V),D}}\big(\mathcal{O}_{f^{-1}(V),D}\big) = \mathrm{length}_{\widetilde{B}}\big(\widetilde{B}/\mathfrak{m}_A\widetilde{B}\big)$$

is the multiplicity of D in the pullback cycle $f^*[V]$, the second equality holding because $\mathcal{O}_{f^{-1}(V),D} = \widetilde{B} \otimes_A A/\mathfrak{m}_A = \widetilde{B}/\mathfrak{m}_A\widetilde{B}$. The coefficient of $[D]$ in $\sum_j m_j[\mathrm{div}(f^{\#}r\big|_{X_j})]$ is $\sum_{j\,:\,D \subseteq X_j} m_j \cdot \mathrm{ord}_D(f^{\#}r\big|_{X_j})$, the sum running over the components X_j that pass through D. That the two coefficients agree is two multiplicative facts chained together. First, flatness of $\widetilde{B}$ over A makes length multiply through the base change: filtering A/rA by its $\mathrm{ord}_V(r)$ composition factors $A/\mathfrak{m}_A$ and applying the exact functor $- \otimes_A \widetilde{B}$ gives

$$\mathrm{length}_{\widetilde{B}}\big(\widetilde{B}/r\widetilde{B}\big) = \mathrm{length}_A(A/rA) \cdot \mathrm{length}_{\widetilde{B}}\big(\widetilde{B}/\mathfrak{m}_A\widetilde{B}\big) = \mathrm{ord}_V(r) \cdot e_D$$

Second, order in the one-dimensional local ring $\widetilde{B}$ distributes over its minimal primes with the generic multiplicities as weights: each minimal prime $\mathfrak{p}_j$ has residue ring $B_j = \mathcal{O}_{X_j,D}$ and localization $\widetilde{B}_{\mathfrak{p}_j} = \mathcal{O}_{X,X_j}$ of length $m_j = \mathrm{length}_{\mathcal{O}_{X,X_j}}(\mathcal{O}_{X,X_j})$, so the additivity of order over the minimal primes of a one-dimensional local ring ([24, App. A.3]) gives, for the nonzerodivisor $f^{\#}r$,

$$\mathrm{length}_{\widetilde{B}}\big(\widetilde{B}/(f^{\#}r)\widetilde{B}\big) = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \mathrm{length}_{B_j}\big(B_j/(f^{\#}r)B_j\big) = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \mathrm{ord}_D(f^{\#}r\big|_{X_j})$$

Since $f^{\#}r$ is the image of r in $\widetilde{B}$, the left sides of the two displays coincide, whence $\mathrm{ord}_V(r)\cdot e_D = \sum_{j\,:\,D\subseteq X_j} m_j \cdot \mathrm{ord}_D(f^{\#}r\big|_{X_j})$: the two coefficients of $[D]$ match. When X is irreducible along D the sum has a single term and reduces to $e_D = m_j \cdot e_{D/V}$, where $e_{D/V} = \mathrm{length}_{B_j}(B_j/\mathfrak{m}_A B_j)$ is the ramification-type multiplicity of D over V computed in the reduced component ring B_j; the pullback multiplicity e_D over the full preimage scheme then exceeds it by exactly the factor m_j. Summing over the divisors D and, on the right, over the components X_j reproduces both sides of (8.1) term by term. Hence $f^*[\mathrm{div}(r)] = \sum_j m_j[\mathrm{div}(f^{\#}r\big|_{X_j})]$ is a sum of divisors of rational functions, so $f^*[\mathrm{div}(r)] \sim 0$.

Descent to Chow groups follows: $f^*(\mathrm{Rat}_k(Y)) \subseteq \mathrm{Rat}_{k+n}(X)$, so f^* induces $A_k(Y) \to A_{k+n}(X)$. Functoriality on classes is inherited from proposition 8.3.5(3),(4) on cycles, giving a contravariant functor as claimed, completing the proof.

The proposition completes the package: flat pullback is a well-defined operation on Chow groups, contravariant and dimension-raising, dual to the covariant dimension-preserving pushforward of section 8.2. The pair (f_*, f^*) makes A_* into a functor covariant for proper maps and contravariant for flat maps, with the compatibility of proposition 8.3.6 tying the two together across fiber squares. This is the structure the localization sequence and homotopy invariance of section 8.4 exploit, and it is why the affine-bundle projection $p\colon X \times \mathbb{A}^n \to X$, whose pullback was computed in example 8.3.4, will turn out to induce an isomorphism on Chow groups.

Example 8.3.8: Pullback to a Product and to an Open Set

The two flat morphisms of example 8.3.4 descend to Chow groups explicitly.

1. *Cylinder map.* For $p\colon X \times \mathbb{A}^1 \to X$ the induced map $p^*\colon A_k(X) \to A_{k+1}(X \times \mathbb{A}^1)$ sends the class of a subvariety V to the class of the cylinder $V \times \mathbb{A}^1$. On $X = \mathrm{Spec}\,\overline{k}$ this reads $A_0(\mathrm{Spec}\,\overline{k}) = \mathbb{Z} \to A_1(\mathbb{A}^1)$, sending the class of the point to $[\mathbb{A}^1]$, the fundamental class of the affine line, consistent with $p^*[Y] = [X]$ from proposition 8.3.5(2).

2. *Restriction to an open set.* For an open immersion $j\colon U \hookrightarrow Y$ the map $j^*\colon A_k(Y) \to A_k(U)$ restricts a class to U, forgetting components supported on the complement $Y \setminus U$. On $Y = \mathbb{P}^1$ with $U = \mathbb{A}^1$ this is the surjection $A_0(\mathbb{P}^1) = \mathbb{Z} \twoheadrightarrow A_0(\mathbb{A}^1) = 0$ from example 8.1.5: the class of a point on $\mathbb{P}^1$ restricts to a point of $\mathbb{A}^1$, which is rationally trivial. The kernel is generated by the class of the point at infinity, the part of $\mathbb{P}^1$ removed. This surjectivity with the removed piece generating the kernel is the prototype of the localization sequence.

Exercise 8.3.1

1. Let $f\colon X \to Y$ be flat of relative dimension 0 (for instance a finite flat morphism, or an open

immersion). Show directly from definition 8.3.3 that f^* maps $Z_k(Y) \to Z_k(X)$ for every k, preserving dimension, and that $f^*[V] = \sum_i e_i[Z_i]$ where Z_i are the components of $f^{-1}(V)$ and e_i their multiplicities in the fiber over the generic point of V.

2. Let $\pi\colon \mathbb{A}^{n+1} \setminus \{0\} \to \mathbb{P}^n$ be the tautological $\mathbb{G}_m$-bundle. It is flat of relative dimension 1. Compute $\pi^*[\mathbb{P}^k]$ for a linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$, identifying its class as a cycle on $\mathbb{A}^{n+1} \setminus \{0\}$.

3. Give an example of a flat morphism f for which $f^*\colon A_*(Y) \to A_*(X)$ is not injective, and one for which it is not surjective. (Hint: the projection $\mathbb{P}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$ and the open immersion $\mathbb{A}^1 \hookrightarrow \mathbb{P}^1$.)

4. Let $q\colon X \times \mathbb{A}^m \times \mathbb{A}^n \to X \times \mathbb{A}^m$ and $p\colon X \times \mathbb{A}^m \to X$ be the projections. Verify the composition law proposition 8.3.5(3) for $p \circ (\mathrm{id} \times \text{-})$ directly on the class of a subvariety $V \subseteq X$, confirming both routes give $[V \times \mathbb{A}^m \times \mathbb{A}^n]$.

5. Let $f\colon X \to Y$ be proper and $j\colon U \hookrightarrow Y$ an open immersion, with fiber square $X_U = f^{-1}(U) \to X$ over j. Using proposition 8.3.6, prove that $(f|_{X_U})_*(\alpha|_{X_U}) = (f_*\alpha)|_U$ in $A_*(U)$ for every $\alpha \in A_*(X)$, that is, proper pushforward commutes with restriction to an open subset.

6. Let $E \to X$ be a vector bundle of rank r over a variety X of dimension d, with projection π flat of relative dimension r. Compute $\pi^*[X]$ and identify the top Chow group $A_{d+r}(E)$, first showing that E is irreducible.

8.4 The Fundamental Sequences

The two functorialities already in hand, proper pushforward (section 8.2) and flat pullback (section 8.3), combine into the two structural results that make Chow groups computable. The first is a long-exact-sequence surrogate: cutting a scheme into a closed piece and its open complement relates the three Chow groups by a right-exact sequence, the algebraic analogue of the excision property of a homology theory. The second is homotopy invariance in its algebraic form: an affine bundle over X, and in particular the trivial bundle $X \times \mathbb{A}^n$, has the same Chow groups as X, shifted in degree by the fibre dimension. Neither result computes a single Chow group in isolation, but together they reduce the computation of $A_*(X)$ for the spaces of the next section to bookkeeping: stratify X by locally closed pieces on which the answer is known, and assemble the pieces through the sequence.

Both proofs rest on a single principle. A cycle on X is a formal combination of subvarieties; a subvariety either meets the open set U or is contained in the closed complement Z, and rational equivalence is generated by divisors of rational functions, which live on subvarieties and so obey the same dichotomy. The localization sequence is the exact translation of this observation into a statement about the three groups, and affine-bundle invariance is its first substantial payoff, extracted by Noetherian induction.

Throughout, $Z \subseteq X$ is a closed subscheme with open complement $U = X \setminus Z$, and

$$i\colon Z \hookrightarrow X, \qquad j\colon U \hookrightarrow X$$

denote the inclusions. The closed immersion i is proper, so it pushes cycles forward by definition 8.2.1: a subvariety $V \subseteq Z$ is also a subvariety of X, and $i_*[V] = [V]$ since $\deg(V/V) = 1$. The open immersion j is flat of relative dimension 0, so it pulls cycles back by definition 8.3.3: for a subvariety

$V \subseteq X$ the restriction is $j^*[V] = [V \cap U]$ when V meets U and $j^*[V] = 0$ when $V \subseteq Z$. These two operations are what the sequence is built from.

Theorem 8.4.1: Localization Sequence

Let X be an algebraic scheme, $Z \subseteq X$ a closed subscheme, and $U = X \setminus Z$ its open complement, with $i\colon Z \hookrightarrow X$ and $j\colon U \hookrightarrow X$ the inclusions. For every k the sequence

$$A_k(Z) \xrightarrow{i_*} A_k(X) \xrightarrow{j^*} A_k(U) \longrightarrow 0$$

is exact: j^* is surjective, and $\ker(j^*) = \operatorname{im}(i_*)$.

***Proof*:**
The map i_* is proper pushforward along the closed immersion (theorem 8.2.3) and j^* is flat pullback along the open immersion (proposition 8.3.7); both are defined on Chow groups, so the sequence is a sequence of homomorphisms. Exactness is checked at the two nonzero spots.

Step 1: surjectivity of j^*. A subvariety $V' \subseteq U$ is a k-dimensional closed integral subscheme of the open set U. Its closure $V = \overline{V'}$ in X is a k-dimensional subvariety of X with $V \cap U = V'$, since closure adds only points lying in the closed set Z and does not raise the dimension. Thus $j^*[V] = [V']$, and every generator of $Z_k(U)$ is hit by a cycle on X. Hence $j^*\colon Z_k(X) \to Z_k(U)$ is already surjective at the level of cycles, and a fortiori $j^*\colon A_k(X) \to A_k(U)$ is surjective.

Step 2: the inclusion $\operatorname{im}(i_*) \subseteq \ker(j^*)$. For a subvariety $V \subseteq Z$ the intersection $V \cap U$ is empty, so $j^* i_*[V] = j^*[V] = 0$. By linearity $j^* \circ i_* = 0$ on $Z_k(Z)$, hence on $A_k(Z)$, which gives $\operatorname{im}(i_*) \subseteq \ker(j^*)$.

Step 3: the inclusion $\ker(j^*) \subseteq \operatorname{im}(i_*)$. This is the substance of the theorem. Let $\alpha \in Z_k(X)$ be a cycle whose class in $A_k(X)$ lies in $\ker(j^*)$, so $j^*\alpha \sim 0$ on U; the goal is to modify α by a cycle rationally equivalent to zero on X until it is supported on Z, that is, until it lies in the image of $i_*\colon Z_k(Z) \to Z_k(X)$.

Split the cycle by whether its components meet U. Group the subvarieties occurring in α as those contained in Z and those meeting U, and write

$$\alpha = \alpha_Z + \beta, \qquad \alpha_Z \in Z_k(Z), \quad \beta = \sum_i n_i[V_i]$$

where each V_i meets U. Since $\alpha_Z = i_*\alpha_Z$ already lies in the image of i_*, it may be discarded, and it suffices to treat β. Applying j^* gives $j^*\alpha = j^*\beta = \sum_i n_i[V_i']$ with $V_i' = V_i \cap U$ the distinct k-dimensional subvarieties of U, and by hypothesis this cycle is rationally equivalent to zero on U:

$$\sum_i n_i[V_i'] = \sum_j [\operatorname{div}(r_j')] \quad \text{in } Z_k(U)$$

for finitely many $(k+1)$-dimensional subvarieties $W_j' \subseteq U$ and rational functions $r_j' \in k(W_j')^*$.

The idea is now to lift each W_j' and r_j' from U to X and compare divisors. Let $W_j = \overline{W_j'} \subseteq X$ be the closure, a $(k+1)$-dimensional subvariety of X with $W_j \cap U = W_j'$. Restriction of functions to the dense open $W_j' \subseteq W_j$ is an isomorphism of function fields $k(W_j) \xrightarrow{\sim} k(W_j')$, since a variety and a dense open subset have the same function field; let $r_j \in k(W_j)^*$ be the function corresponding to r_j'. Form the divisor $[\operatorname{div}(r_j)] \in Z_k(X)$. Its restriction to U is governed by the flat-pullback compatibility of the divisor construction: for a codimension-one subvariety $V \subseteq W_j$ meeting U, the local ring $\mathcal{O}_{W_j,V} = \mathcal{O}_{W_j',V\cap U}$ is unchanged by passing to the open set, so $\operatorname{ord}_V(r_j) = \operatorname{ord}_{V\cap U}(r_j')$, while a codimension-one subvariety of W_j contained in Z contributes a component of $[\operatorname{div}(r_j)]$ that lies in $Z_k(Z)$. Splitting the sum over these two kinds of V,

$$[\operatorname{div}(r_j)] = [\operatorname{div}(r_j')] + \gamma_j, \qquad \gamma_j \in Z_k(Z) \tag{8.2}$$

where the first term on the right is read inside $Z_k(X)$ via the identification $V \leftrightarrow \overline{V}$ of subvarieties of U with their closures, and γ_j collects the codimension-one subvarieties of W_j that lie in Z.

Step 3, conclusion. Sum (8.2) over j and use $\sum_j [\operatorname{div}(r_j')] = \beta$ as cycles on X under the closure identification (both sides are the cycle $\sum_i n_i[V_i]$, since $V_i = \overline{V_i'}$). This yields

$$\sum_j [\operatorname{div}(r_j)] = \beta + \sum_j \gamma_j$$

so that

$$\beta = \sum_j [\operatorname{div}(r_j)] - \sum_j \gamma_j$$

The first sum is a $\mathbb{Z}$-combination of divisors of rational functions on X, hence lies in $\operatorname{Rat}_k(X)$; the second sum $\gamma := \sum_j \gamma_j$ lies in $Z_k(Z)$. Therefore, in $A_k(X)$,

$$[\beta] = -[\gamma] = i_*(-[\gamma]) \in \operatorname{im}(i_*)$$

Adding back the discarded $\alpha_Z = i_*[\alpha_Z]$ gives $[\alpha] = i_*\big([\alpha_Z] - [\gamma]\big)$, exhibiting the class of α in the image of i_*. This proves $\ker(j^*) \subseteq \operatorname{im}(i_*)$, and with Steps 1 and 2 the sequence is exact, completing the proof.

The mechanism of the proof is worth isolating, because it recurs whenever Chow groups are computed by stratification. Surjectivity on the right is elementary: a subvariety of the open set extends to its closure, so no class on U is unreachable from X. The content is in the middle, and it is entirely a statement about lifting rational equivalences. A relation $j^*\alpha \sim 0$ on U is witnessed by finitely many rational functions on subvarieties of U; each such function extends across Z to a rational function on a subvariety of X (function fields do not see the boundary), and the extended divisor differs from the original only in components supported on Z. The failure of the sequence to be exact on the left, which is left as it stands, measures exactly how much rational equivalence is created by the extra room X provides over Z: a cycle on Z can become rationally equivalent to zero in X without being so in Z, because X contains $(k+1)$-dimensional subvarieties running out of Z into U. The affine line makes this concrete below.

8.4.2 Remark: No Left-Exactness The map $i_*\colon A_k(Z) \to A_k(X)$ need not be injective. Take $X = \mathbb{A}^1$, $Z = \{0\}$, and $k = 0$. Then $A_0(Z) = \mathbb{Z}$, generated by $[0]$, but $[0] \sim 0$ in $A_0(\mathbb{A}^1)$ by example 8.1.5, so $i_*[0] = 0$ and i_* is the zero map on a nonzero group. The kernel of i_* records cycles on Z that bound in X but not in Z, and there is no functorial description of it. This is why the localization sequence is only a three-term right-exact sequence, not a long exact sequence: Bloch's higher Chow groups supply the missing terms, but that theory is outside the scope of this book.

The localization sequence is stated for one closed subscheme, but its most frequent use is for a pair. Two closed subschemes X_1, X_2 of an ambient X can be compared through their union and intersection, and the resulting sequence is the Mayer-Vietoris right-exactness that was promised without proof in example 8.1.6. It is a companion to the localization sequence rather than a corollary of it: the derivation below runs directly at the level of cycles, using only that a subvariety of the union lies in one of the two pieces, so no appeal to theorem 8.4.1 is needed.

Corollary 8.4.3: Right-Exactness for Two Closed Subschemes

Let X_1 and X_2 be closed subschemes of an algebraic scheme, and set $X_1 \cup X_2$ and $X_1 \cap X_2$ (with reduced or any fixed scheme structure; Chow groups see only the reduced structure by example 8.1.6). For every k the sequence

$$A_k(X_1 \cap X_2) \xrightarrow{(\iota_{1*}, -\iota_{2*})} A_k(X_1) \oplus A_k(X_2) \xrightarrow{j_{1*}+j_{2*}} A_k(X_1 \cup X_2) \longrightarrow 0$$

is exact, where $\iota_m \colon X_1 \cap X_2 \hookrightarrow X_m$ and $j_m \colon X_m \hookrightarrow X_1 \cup X_2$ are the inclusions. There is no exactness at $A_k(X_1 \cap X_2)$ in general.

***Proof*:**
Write $Y = X_1 \cup X_2$. The argument is cleanest at the level of cycles, using the one structural fact that an irreducible closed subset of $Y = X_1 \cup X_2$ is contained in X_1 or in X_2. Consequently a subvariety $V \subseteq Y$ lies in X_1 or in X_2 (or in both, in which case $V \subseteq X_1 \cap X_2$), and because $Z_k(Y)$ is free on subvarieties, a cycle expressible both by subvarieties of X_1 and by subvarieties of X_2 has all its components in $X_1 \cap X_2$.

Step 1: surjectivity of $j_{1*} + j_{2*}$. Every subvariety of Y lies in X_1 or X_2, so at the level of cycles $Z_k(Y) = j_{1*}Z_k(X_1) + j_{2*}Z_k(X_2)$; the map $Z_k(X_1) \oplus Z_k(X_2) \to Z_k(Y)$, $(a_1, a_2) \mapsto a_1 + a_2$, is surjective. Passing to classes, $j_{1*} + j_{2*} \colon A_k(X_1) \oplus A_k(X_2) \to A_k(Y)$ is surjective.

Step 2: the composite is zero. For a subvariety $V \subseteq X_1 \cap X_2$ the two pushforwards $j_{1*}\iota_{1*}[V]$ and $j_{2*}\iota_{2*}[V]$ both equal $[V]$ in $Z_k(Y)$, since each inclusion sends $[V]$ to $[V]$. Hence $(j_{1*} + j_{2*}) \circ (\iota_{1*}, -\iota_{2*}) = j_{1*}\iota_{1*} - j_{2*}\iota_{2*} = 0$, giving $\operatorname{im}(\iota_{1*}, -\iota_{2*}) \subseteq \ker(j_{1*} + j_{2*})$.

Step 3: exactness in the middle. Let $(\eta_1, \eta_2) \in A_k(X_1) \oplus A_k(X_2)$ satisfy $j_{1*}\eta_1 + j_{2*}\eta_2 = 0$ in $A_k(Y)$. Represent η_1 and η_2 by cycles $a_1 \in Z_k(X_1)$ and $a_2 \in Z_k(X_2)$. The hypothesis says $a_1 + a_2 \sim 0$ in $Z_k(Y)$, so there are finitely many $(k+1)$-dimensional subvarieties $W_j \subseteq Y$ and $r_j \in k(W_j)^*$ with

$$a_1 + a_2 = \sum_j [\operatorname{div}(r_j)] \quad \text{in } Z_k(Y)$$

Each W_j lies in X_1 or in X_2, and the divisor $[\operatorname{div}(r_j)]$ of a rational function on W_j is a cycle supported on W_j, hence on X_1 or on X_2 accordingly. Collect the terms:

$$b_1 = \sum_{W_j \subseteq X_1} [\operatorname{div}(r_j)] \in \operatorname{Rat}_k(X_1), \qquad b_2 = \sum_{W_j \subseteq X_2,\, W_j \not\subseteq X_1} [\operatorname{div}(r_j)] \in \operatorname{Rat}_k(X_2)$$

where a W_j lying in both is assigned to the first group; then $\sum_j [\operatorname{div}(r_j)] = b_1 + b_2$ and so $a_1 + a_2 = b_1 + b_2$ in $Z_k(Y)$. Rearranging,

$$c := a_1 - b_1 = b_2 - a_2$$

The left expression $a_1 - b_1$ is a cycle supported on X_1, and the right expression $b_2 - a_2$ is a cycle supported on X_2; being the same cycle in the free group $Z_k(Y)$, it is supported on $X_1 \cap X_2$, that is $c \in Z_k(X_1 \cap X_2)$. Now compute classes. In $A_k(X_1)$, $b_1 \in \mathrm{Rat}_k(X_1)$ is zero, so $\eta_1 = [a_1] = [a_1 - b_1] = \iota_{1*}[c]$; in $A_k(X_2)$, $b_2 \in \mathrm{Rat}_k(X_2)$ is zero, so $\eta_2 = [a_2] = [a_2 - b_2] = -\iota_{2*}[c]$. Therefore

$$(\eta_1, \eta_2) = (\iota_{1*}[c], -\iota_{2*}[c]) = (\iota_{1*}, -\iota_{2*})\big([c]\big)$$

lies in the image of $(\iota_{1*}, -\iota_{2*})$. This gives $\ker(j_{1*} + j_{2*}) \subseteq \mathrm{im}(\iota_{1*}, -\iota_{2*})$, and with Step 2 exactness in the middle.

No left-exactness. The map $(\iota_{1*}, -\iota_{2*})$ need not be injective. Take X_1 and X_2 two distinct lines through the origin in $\mathbb{A}^2$, so $X_1 \cap X_2 = \{0\}$, in degree $k = 0$: then $A_0(X_1 \cap X_2) = \mathbb{Z}$ is generated by $[0]$, while $\iota_{1*}[0] = [0] \sim 0$ in $A_0(X_1) = A_0(\mathbb{A}^1)$ and likewise $\iota_{2*}[0] = 0$ in $A_0(X_2)$, so $(\iota_{1*}, -\iota_{2*})[0] = 0$ on a nonzero group. Thus there is no exactness at $A_k(X_1 \cap X_2)$, as we sought to show.

The corollary is the exact sequence recorded, without proof, as item (4) of example 8.1.6; the present derivation discharges that promissory note. Its shape is the Chow-group form of the Mayer-Vietoris sequence familiar from topology, truncated on the left for the same reason the localization sequence is: the intersection $X_1 \cap X_2$ can carry cycles that become rationally equivalent to zero once they are free to move in the larger $X_1 \cup X_2$. Where it is used most is in dévissage arguments, where a scheme is written as a union of simpler closed pieces and its Chow groups are assembled from those of the pieces and their overlaps.

A single application of the localization sequence recovers, from scratch, the computation of $A_0(\mathbb{A}^1)$ recorded in example 8.1.5, this time without touching a rational function by hand. The point is that $\mathbb{A}^1$ is $\mathbb{P}^1$ with a point removed, and the localization sequence turns the known $A_0(\mathbb{P}^1) = \mathbb{Z}$ into the answer for the open complement.

Example 8.4.4: Recomputing $A_0(\mathbb{A}^1)$ by Localization

Take $X = \mathbb{P}^1$, $Z = \{\infty\}$ the point at infinity, and $U = \mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$. The localization sequence in degree 0 reads

$$A_0(\{\infty\}) \xrightarrow{i_*} A_0(\mathbb{P}^1) \xrightarrow{j^*} A_0(\mathbb{A}^1) \longrightarrow 0$$

The left group is $A_0(\{\infty\}) = \mathbb{Z}$, generated by the class of the single point ∞, and $i_*[\infty] = [\infty] \in A_0(\mathbb{P}^1)$. By example 8.1.5 the group $A_0(\mathbb{P}^1) \cong \mathbb{Z}$ is generated by the class h of any point, and $[\infty] = h$ is a generator, so i_* is surjective. By exactness $\mathrm{im}(i_*) = \ker(j^*)$, and since i_* is onto, $\ker(j^*) = A_0(\mathbb{P}^1)$; the quotient by the whole group is trivial:

$$A_0(\mathbb{A}^1) = A_0(\mathbb{P}^1)/\ker(j^*) \cong \mathrm{coker}(i_*) = \mathbb{Z}/\mathbb{Z} = 0$$

Thus $A_0(\mathbb{A}^1) = 0$, in agreement with the direct computation. The single generator of $A_0(\mathbb{P}^1)$ is killed precisely because the point removed to form $\mathbb{A}^1$ was rationally equivalent to every remaining point; removing it collapses the whole group.

With the localization sequence available, the second structural result of the section follows by Noetherian induction. Homotopy invariance in algebraic geometry takes the form of affine-bundle invariance: pulling a cycle back along the projection of an affine bundle is an isomorphism on Chow groups. The motivating case is the trivial bundle $X \times \mathbb{A}^n \to X$, which asserts that appending an affine factor does not change the Chow group beyond the expected degree shift, the algebraic counterpart of the fact that $\mathbb{A}^n$ is contractible.

The result is stated for a general affine bundle, of which the trivial bundle is the leading case.

Definition 8.4.5: Affine Bundle of Relative Dimension n

An *affine bundle of relative dimension* n over X is a scheme $p\colon E \to X$ that is locally on X a product with affine space: there is an open cover $\{U_\lambda\}$ of X and isomorphisms $p^{-1}(U_\lambda) \cong U_\lambda \times \mathbb{A}^n$ over U_λ.

A vector bundle of rank n, with its transition maps in GL_n, is such a bundle, as is any torsor under a vector bundle; the trivial example is $X \times \mathbb{A}^n$ itself. In every case p is flat of relative dimension n, so flat pullback $p^*\colon A_k(X) \to A_{k+n}(E)$ is defined by definition 8.3.3.

Theorem 8.4.6: Affine-Bundle Invariance

Let $p\colon E \to X$ be an affine bundle of relative dimension n over an algebraic scheme X. Then the flat pullback

$$p^*\colon A_k(X) \xrightarrow{\sim} A_{k+n}(E)$$

is an isomorphism for every k. In particular, for the trivial bundle, $A_k(X) \xrightarrow{\sim} A_{k+n}(X \times \mathbb{A}^n)$.

***Proof*:**
The proof runs the argument directly at relative dimension n; the fibre dimension enters only through the trivial bundle, where a global tower is available. Note that a global tower of rank-one affine bundles is *not* claimed for a general bundle: the local identification $U\times\mathbb{A}^n = (U\times\mathbb{A}^{n-1})\times\mathbb{A}^1$ does not glue to a factorization $E \to X$ over X, since a general affine (or vector) bundle carries no global flag of subbundles. The trivial bundle $X \times \mathbb{A}^n$, being a literal product, *is* such a tower, $X \times \mathbb{A}^n = (X \times \mathbb{A}^{n-1}) \times \mathbb{A}^1$; the construction below invokes the tower only there. Fix n and prove that $p^*\colon A_k(X) \to A_{k+n}(E)$ is an isomorphism for all k, by Noetherian induction on the closed subschemes of X. For a closed subscheme $W \subseteq X$ write $E_W = p^{-1}(W)$ and $p_W\colon E_W \to W$ for the restricted bundle, again an affine bundle of relative dimension n.

Step 1: reductions. The scheme X is Noetherian, so its closed subschemes satisfy the descending chain condition and it suffices to prove: if p_W^* is an isomorphism for every proper closed subscheme $W \subsetneq X$, then p^* is an isomorphism for X. Chow groups depend only on the reduced structure (example 8.1.6), so take X reduced. If X is reducible, write $X = X_1 \cup X_2$ with X_1 an irreducible component and X_2 the union of the remaining components, both proper closed subschemes; the pullbacks $p^*_{X_1}, p^*_{X_2}, p^*_{X_1\cap X_2}$ are isomorphisms by hypothesis. The Mayer-Vietoris sequences of corollary 8.4.3 for $X = X_1 \cup X_2$ and for $E = E_{X_1} \cup E_{X_2}$ (whose intersection is $E_{X_1\cap X_2}$) form a commutative ladder linked by the vertical pullbacks; a diagram chase using the three outer isomorphisms shows p^* is an isomorphism for X. So assume X is a variety (reduced and irreducible), and let $\eta = \operatorname{Spec} k(X)$ be its generic point, with generic fibre $E_\eta \cong \mathbb{A}^n_{k(X)}$.

Step 2: surjectivity of p^*. It suffices to realize the class of a single $(k+n)$-dimensional subvariety $V \subseteq E$ as a pullback modulo the image of $A_{k+n}(E_Z)$ for some proper closed $Z \subsetneq X$. Let $W = \overline{p(V)} \subseteq X$. If $W \subsetneq X$ is proper, then $V \subseteq E_W$ is supported over $Z = W$, and $[V] \in \operatorname{im}\big(A_{k+n}(E_W) \to A_{k+n}(E)\big)$; by the inductive hypothesis p_W^* is onto, and compatibility of p^* with the closed immersion (the fibre square for p over $W \hookrightarrow X$ gives $p^*i_* = i_*p_W^*$ by proposition 8.3.6) writes $[V]$ as a pullback from X. If instead $W = X$, so V dominates X, restrict to the generic fibre. There $V_\eta \subseteq \mathbb{A}^n_{k(X)}$ is a closed subvariety, and its class is controlled by the trivial-bundle case of the theorem over the field $K = k(X)$: the projection $\mathbb{A}^n_K \to \operatorname{Spec} K$ has surjective flat pullback. This last fact is elementary and needs no base induction, being the trivial bundle over a point: writing the tower $\mathbb{A}^n_K = (\mathbb{A}^{n-1}_K) \times \mathbb{A}^1$ and inducting on n, the base case $A_*(\mathbb{A}^1_K)$ is generated over $A_*(\operatorname{Spec} K)$ by the pullback (every closed point of $\mathbb{A}^1_K$ is the divisor of its minimal polynomial, so $A_0(\mathbb{A}^1_K) = 0$, exactly as in example 8.1.5 with K in place of $\overline{k}$), and the tower propagates this up the coordinates. Thus $[V_\eta]$ is rationally equivalent on $\mathbb{A}^n_{k(X)}$ to a pullback $p^*_\eta(a_\eta)$ of a class a_η on $\operatorname{Spec} k(X)$. Spreading out the finitely many rational functions that realize this equivalence, together with a_η, to an open $U \subseteq X$ shows $[V]|_{E_U} = p^*_U(\alpha_U)$ for a class $\alpha_U \in A_k(U)$ and hence $[V] - p^*(\alpha_V)$ (with $\alpha_V \in A_k(X)$ lifting α_U via surjectivity of j^*, theorem 8.4.1) restricts to 0 on E_U, so its support lies over the proper closed $Z = X \setminus U$. In every case

$$[V] = p^*(\alpha_V) + \gamma_V \quad \text{in } A_{k+n}(E), \qquad \gamma_V \in \operatorname{im}\big(A_{k+n}(E_Z) \to A_{k+n}(E)\big)$$

for some proper closed $Z \subsetneq X$ and $\alpha_V \in A_k(X)$. By the inductive hypothesis p^*_Z is onto $A_*(E_Z)$, so $\gamma_V = i_* p^*_Z(\delta_V) = p^*(i_*\delta_V)$ for a class δ_V on Z, again by proposition 8.3.6. Hence $[V] = p^*(\alpha_V + i_*\delta_V)$ and p^* is surjective.

Surjectivity of p^* is now in hand for every X; injectivity is the remaining point, and it is not a formal consequence of the localization ladder, since the two rows are only right-exact and provide no left term to chase against. It is proved by exhibiting a left inverse to p^* for the trivial bundle and reducing the general bundle to that case.

Step 3: a left inverse for the trivial bundle. The construction is carried out for a single affine coordinate over an arbitrary base Y; the trivial bundle $X \times \mathbb{A}^n$ is a tower of n such single coordinates, and the rank-n left inverse is the composite of the n single-coordinate left inverses (Step 3d). Take $E = Y \times \mathbb{A}^1$, with coordinate t on the second factor and projection $p\colon Y \times \mathbb{A}^1 \to Y$, and define a homomorphism $\sigma\colon Z_{k+1}(Y \times \mathbb{A}^1) \to Z_k(Y)$ on a generator $[V]$, $V \subseteq Y \times \mathbb{A}^1$ a $(k+1)$-dimensional subvariety, by

$$\sigma[V] = \begin{cases} p_*\big[\operatorname{div}_V(t|_V)\big] & t|_V \in k(V)^* \text{ nonconstant} \\ 0 & t|_V \text{ constant} \end{cases}$$

extended additively; here "$t|_V$ constant" is read to include the degenerate case $t|_V \equiv 0$ (when $V \subseteq Y \times \{0\}$), which likewise contributes 0, since then $t|_V \notin k(V)^*$ and no divisor $\operatorname{div}_V(t|_V)$ is formed. When $t|_V$ is nonconstant, $[\operatorname{div}_V(t|_V)]$ is a k-cycle on V; the pushforward p_* of this particular cycle is well-defined even though $p|_V : V \to \overline{p(V)} \subseteq Y$ need not be proper. The point is that t is a regular function on $Y \times \mathbb{A}^1$, so its restriction $t|_V$ is regular on V and $\operatorname{div}_V(t|_V)$ is effective, with support contained in the zero locus $V \cap (Y \times \{0\}) \subseteq Y \times \{0\}$. On $Y \times \{0\}$ the projection p is the closed immersion identifying $Y \times \{0\}$ with Y, in particular proper, so $p_*[\operatorname{div}_V(t|_V)]$ is a well-defined k-cycle on Y by definition 8.2.1, computed component by component over the locus where $t|_V$ vanishes. Geometrically $\sigma[V]$ is the fibre of V over $t = 0$, projected to Y: the specialization of V to the zero section.

Step 3a: $\sigma \circ p^* = \mathrm{id}$. For a k-dimensional subvariety $W \subseteq Y$ the pullback is $p^*[W] = [W \times \mathbb{A}^1]$, and t restricts to the coordinate on $W \times \mathbb{A}^1$, which is nonconstant with $[\operatorname{div}_{W \times \mathbb{A}^1}(t)] = [W \times \{0\}]$ (the coordinate t has its single zero along $W \times \{0\}$ and no pole on $W \times \mathbb{A}^1$). Pushing forward, $p_*[W \times \{0\}] = [W]$, so $\sigma(p^*[W]) = [W]$. By additivity $\sigma \circ p^* = \mathrm{id}$ on $Z_k(Y)$, hence on $A_k(Y)$ once σ is known to descend.

Step 3b: σ respects rational equivalence. A generator of $\mathrm{Rat}_{k+1}(Y \times \mathbb{A}^1)$ is $[\mathrm{div}_S(g)]$ for a $(k+2)$-dimensional subvariety $S \subseteq Y \times \mathbb{A}^1$ and $g \in k(S)^*$, and unwinding the definitions gives $\sigma[\mathrm{div}_S(g)] = p_*\big(\sum_V \mathrm{ord}_V(g)\,[\mathrm{div}_V(t|_V)]\big)$, the outer sum over the codimension-one subvarieties $V \subseteq S$ on which $t|_V$ is nonconstant (a V with $t|_V$ constant, including $V \subseteq Y \times \{0\}$ where $t|_V \equiv 0$, drops out, contributing 0). This iterated divisor is symmetric in the two functions g and $t|_S$ modulo a principal divisor on S:

$$\sum_V \mathrm{ord}_V(g)\,[\mathrm{div}_V(t)] - \sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)] \in \mathrm{Rat}_k(S) \tag{8.3}$$

The relation (8.3) is the commutativity of two divisor operations on the $(k+2)$-dimensional S, equivalently the well-definedness of intersecting a cycle with a Cartier divisor; its proof requires the divisor-theoretic machinery of the next chapter, and is supplied there by theorem 9.3.2 (its surface case is exactly (8.3)), so no tool used here depends on it circularly. Granting (8.3), $\sigma[\mathrm{div}_S(g)]$ differs from $p_*\big(\sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)]\big)$ by the pushforward of a principal divisor, and $\sum_V \mathrm{ord}_V(t)\,[\mathrm{div}_V(g)]$ is itself a $\mathbb{Z}$-combination of principal divisors on the subvarieties V, so its pushforward lies in $\mathrm{Rat}_k(Y)$ by theorem 8.2.3. Either way $\sigma[\mathrm{div}_S(g)] \in \mathrm{Rat}_k(Y)$, so σ descends to

$$\sigma\colon A_{k+1}(Y \times \mathbb{A}^1) \to A_k(Y), \qquad \sigma \circ p^* = \mathrm{id}$$

In particular p^* is injective for the trivial rank-one bundle over any base Y, and being surjective by Step 2, it is an isomorphism there. The left inverse σ commutes with proper pushforward along closed immersions and with flat restriction to opens, since each of p_* and $\mathrm{div}(t)$ does; this compatibility is what carries the trivial-bundle result across the localization sequence in Step 3c.

Step 3d: the trivial rank-n bundle. For $E = X \times \mathbb{A}^n$ with coordinates $t_1, \ldots, t_n$, the tower

$$X \times \mathbb{A}^n \xrightarrow{q_n} X \times \mathbb{A}^{n-1} \xrightarrow{q_{n-1}} \cdots \xrightarrow{q_1} X$$

factors $p = q_1 \circ \cdots \circ q_n$ into single-coordinate trivial rank-one bundles $q_m\colon (X \times \mathbb{A}^{m-1}) \times \mathbb{A}^1 \to X \times \mathbb{A}^{m-1}$, so $p^* = q_n^* \circ \cdots \circ q_1^*$ by functoriality of flat pullback (proposition 8.3.5). Applying Step 3b with $Y = X \times \mathbb{A}^{m-1}$ gives a left inverse σ_m to each q_m^*; the composite $\sigma := \sigma_1 \circ \cdots \circ \sigma_n\colon A_{k+n}(X \times \mathbb{A}^n) \to A_k(X)$ satisfies $\sigma \circ p^* = \mathrm{id}$, so p^* is injective, and with Step 2 it is an isomorphism for the trivial bundle. The compatibility of each σ_m with proper pushforward along closed immersions and with flat restriction to opens passes to the composite, so σ has the same compatibility used in the next step.

Step 3c: injectivity for a general bundle. Let $p\colon E \to X$ be an affine bundle of relative dimension n over the variety X and $\alpha \in A_k(X)$ with $p^*\alpha = 0$. Choose a dense open $U \subseteq X$ over which E trivializes, $E_U \cong U \times \mathbb{A}^n$, and set $Z = X \setminus U$, a proper closed subscheme. Flat base change (proposition 8.3.5) gives $p_U^*(j^*\alpha) = j^*(p^*\alpha) = 0$, and p_U^* is injective by Step 3d applied to the trivial bundle over U; hence $j^*\alpha = 0$. By the localization sequence for $Z \subseteq X$ (theorem 8.4.1), $\alpha = i_*\beta$ for some $\beta \in A_k(Z)$. Pushing β into E through the fibre square over $Z \hookrightarrow X$ (proposition 8.3.6) gives $p^*\alpha = p^*i_*\beta = i_*(p_Z^*\beta) = 0$, so $p_Z^*\beta$ lies in the kernel of $i_*\colon A_{k+n}(E_Z) \to A_{k+n}(E)$. Over the total space E the left inverse of Step 3d, restricted to the trivializing locus and extended by $\sigma_Z := (p_Z^*)^{-1}$ over E_Z (the inductive isomorphism), assembles into a retraction $r_E\colon A_{k+n}(E) \to A_k(X)$ of p^*; the compatibility noted at the end of Step 3d makes the square with i_* commute, so that $r_E \circ i_* = i_* \circ \sigma_Z$ on $A_{k+n}(E_Z)$. Evaluating at $p_Z^*\beta$,

$$i_*\beta = i_*\big(\sigma_Z p_Z^*\beta\big) = r_E\big(i_* p_Z^*\beta\big) = r_E(0) = 0$$

so $\alpha = i_*\beta = 0$. Hence p^* is injective.

Steps 2 and 3 show p^* is an isomorphism for every X and every relative dimension n, completing the Noetherian induction; the tower entered only for the trivial bundle (Step 3d), where it is an actual global factorization, so no global tower was claimed for a general bundle, completing the proof.

The theorem is the algebraic-geometric form of a topological triviality: the projection $X \times \mathbb{A}^n \to X$ is a homotopy equivalence when $\mathbb{A}^n = \mathbb{C}^n$ is given the classical topology, so it induces isomorphisms on singular homology, and Chow groups behave the same way. The degree shift by n is the algebraic bookkeeping for the real dimension shift $2n$ on the topological side, since a k-cycle (complex dimension k) has real dimension $2k$ and A_{k+n} matches $H_{2(k+n)} = H_{2k+2n}$. The proof uses none of this: surjectivity of p^* needs only that every zero-cycle on an affine line over a field is rationally trivial ($A_0(\mathbb{A}^1_K) = 0$), propagated up the base by localization, while injectivity needs a left inverse, the specialization to the zero section. The one step left open, the divisor symmetry (8.3), is exactly the input that makes intersecting with a Cartier divisor well defined on rational equivalence, the payload of the next chapter; the isomorphism here is the first place that well-definedness is needed, and the forward reference is deliberate rather than a gap. It is discharged in theorem 9.3.2. The reason the argument needs Noetherian induction rather than a direct computation is the same reason the localization sequence is not left-exact: one cannot control what rational equivalences a cycle acquires on E without dismantling the base into pieces small enough that the bundle trivializes and the fibrewise vanishing takes over.

Exercise 8.4.1

1. Let $Z \subseteq X$ be closed with complement U. Show directly from theorem 8.4.1 that $A_k(U) \cong A_k(X)/\operatorname{im}\big(A_k(Z) \to A_k(X)\big)$, and use this to prove that if $A_k(Z) = 0$ then $j^*\colon A_k(X) \to A_k(U)$ is an isomorphism.

2. Let $X = \mathbb{A}^2$ and let $Z = \{0\}$ be the origin, so $U = \mathbb{A}^2 \setminus \{0\}$ is the punctured plane. Using the localization sequence together with the values $A_k(\mathbb{A}^2)$ (namely $A_2(\mathbb{A}^2) = \mathbb{Z}$ and $A_k(\mathbb{A}^2) = 0$ for $k < 2$, which you may take from section 8.5), compute $A_k(U)$ for all k.

3. Let X be a variety of dimension n and $U \subseteq X$ a nonempty open subset. Show that $A_n(X) \to A_n(U)$ is an isomorphism, and that both are $\mathbb{Z}$, generated by the fundamental class.

4. Compute $A_*(\mathbb{P}^n \times \mathbb{A}^m)$ in terms of $A_*(\mathbb{P}^n)$, assuming the value $A_k(\mathbb{P}^n) = \mathbb{Z}$ for $0 \le k \le n$ from section 8.5. State the answer as a graded group.

5. Let $p\colon E \to X$ be a vector bundle of rank r over an algebraic scheme X, with zero section $s\colon X \hookrightarrow E$. Affine-bundle invariance makes $p^*\colon A_k(X) \to A_{k+r}(E)$ an isomorphism. Explain why one cannot conclude $A_*(E) \cong A_*(X)$ by inverting p^* with a proper pushforward p_*, identifying the precise obstruction, and describe the degree by which p^* and the closed-immersion pushforward s_* each shift a cycle class.

6. Take X_1 and X_2 to be two distinct lines through the origin in $\mathbb{A}^2$, meeting only at $\{0\}$. Write out the sequence of corollary 8.4.3 in degree 0 and verify its exactness by hand, given that A_0 of a line is 0 and A_0 of a point is $\mathbb{Z}$.

8.5 First Computations

The machinery of the preceding sections is now assembled: cycles, rational equivalence, proper pushforward, flat pullback, and the two fundamental sequences. This section puts it to work on the spaces one meets first. We compute the Chow groups of affine space, of projective space, and of a smooth projective curve, and record the shape the answer takes for a projective bundle. These are the building blocks from which nearly every later computation is assembled, and they already display the two extremes of the theory: affine space, whose cycles almost all sweep away to nothing, and projective space, whose cycles are pinned down by a single integer in each dimension.

The pattern to watch is how the localization sequence and affine-bundle invariance from section 8.4 do the actual work. Affine space is an affine bundle over a point, so its Chow groups collapse; projective space is built from affine space by adding a hyperplane at infinity, so an induction along the localization sequence peels off one $\mathbb{Z}$ per dimension. Every computation below is an instance of one of these two moves.

Affine space is the first computation, and $\mathbb{A}^n$ retracts, in the sense of rational equivalence, all the way down to a point: every subvariety of dimension less than n can be swept off to zero, and only the fundamental class in top dimension survives. This is the higher-dimensional form of the observation $A_0(\mathbb{A}^1) = 0$ from example 8.1.5, and it follows in one line from affine-bundle invariance once one notices that $\mathbb{A}^n$ is an affine bundle over $\operatorname{Spec} \overline{k}$.

Proposition 8.5.1: Chow Groups of Affine Space

For every $n \geq 0$ the Chow groups of affine space are

$$A_k(\mathbb{A}^n) = \begin{cases} \mathbb{Z}\,[\mathbb{A}^n] & k = n \\ 0 & k \neq n \end{cases}$$

the group $A_n(\mathbb{A}^n)$ being free of rank one on the fundamental class $[\mathbb{A}^n]$.

***Proof*:**
The structure morphism $\pi\colon \mathbb{A}^n \to \operatorname{Spec} \overline{k}$ exhibits $\mathbb{A}^n$ as the trivial affine bundle of relative dimension n over a point. Affine-bundle invariance (theorem 8.4.6) states that flat pullback

$$\pi^*\colon A_k(\operatorname{Spec} \overline{k}) \xrightarrow{\sim} A_{k+n}(\mathbb{A}^n)$$

is an isomorphism for every k. The point $\operatorname{Spec} \overline{k}$ has a single subvariety, itself, of dimension 0, so $A_0(\operatorname{Spec} \overline{k}) = Z_0(\operatorname{Spec} \overline{k}) = \mathbb{Z}\,[\operatorname{Spec} \overline{k}]$ and $A_k(\operatorname{Spec} \overline{k}) = 0$ for $k \neq 0$. Transporting these through π^* gives $A_{k+n}(\mathbb{A}^n) = 0$ for $k \neq 0$, that is $A_j(\mathbb{A}^n) = 0$ for $j \neq n$, and $A_n(\mathbb{A}^n) = \pi^*\big(\mathbb{Z}\,[\operatorname{Spec} \overline{k}]\big)$. Flat pullback of the fundamental cycle of the base is the fundamental cycle of the total space (definition 8.3.3), so $\pi^*[\operatorname{Spec} \overline{k}] = [\mathbb{A}^n]$, and $A_n(\mathbb{A}^n) = \mathbb{Z}\,[\mathbb{A}^n]$ is free of rank one, completing the proof.

Affine space thus has no cycles in low dimension worth remembering, a fact that reads two ways. From below it says that any k-dimensional subvariety $V \subseteq \mathbb{A}^n$ with $k < n$ bounds: there is a family of cycles, parametrized by the line, sweeping $[V]$ to zero. From above it isolates the one invariant that does survive, the coefficient of the fundamental class, which for a top cycle records the multiplicity with which $\mathbb{A}^n$ appears in it. The vanishing in low dimension is what makes affine space so poor a stage for enumerative geometry, and simultaneously what makes it the natural chart in which to localize a computation on a larger space. This second role is exactly how projective space is handled next.

The contrast to keep in mind is the affine line. A single point of $\mathbb{A}^1$ is a 0-cycle in dimension $0 < 1$, so proposition 8.5.1 predicts $A_0(\mathbb{A}^1) = 0$, recovering the computation of example 8.1.5: the coordinate function $t - p$ has $[\operatorname{div}(t-p)] = [p]$, presenting the point as a principal divisor and hence as zero in the Chow group. The proposition is the assertion that this collapse happens in every dimension below the top, uniformly and for the same reason.

Projective space is the opposite. Nothing sweeps to zero for free, because a rational function on a subvariety of $\mathbb{P}^n$ has as many zeros as poles, and the surviving invariant in each dimension is an integer. The computation is an induction that peels $\mathbb{P}^n$ apart along a hyperplane: the closed subscheme $\mathbb{P}^{n-1} \subseteq \mathbb{P}^n$ at infinity, with open complement the affine chart $\mathbb{A}^n$. Affine space contributes nothing below the top by proposition 8.5.1, so the localization sequence forces every class of $\mathbb{P}^n$ in dimension $k < n$ to come from $\mathbb{P}^{n-1}$, and the induction runs.

The motivating principle is that projective space is *cellular*: it is stratified by the chain

$$\mathbb{P}^0 \subset \mathbb{P}^1 \subset \cdots \subset \mathbb{P}^{n-1} \subset \mathbb{P}^n$$

of linear subspaces, each obtained from the previous by adding an affine cell $\mathbb{A}^k = \mathbb{P}^k \setminus \mathbb{P}^{k-1}$. The Chow groups turn out to be free on the classes of these cells, one generator in each dimension. This is the algebraic form of the cell decomposition that computes the singular homology of $\mathbb{C}\mathrm{P}^n$ in [51], where the same chain of linear subspaces yields $H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = \mathbb{Z}$ and $H_{\mathrm{odd}} = 0$.

Theorem 8.5.2: Chow Groups of Projective Space

For every $n \geq 0$ and every $0 \leq k \leq n$, the Chow group $A_k(\mathbb{P}^n)$ is free abelian of rank one, generated by the class $[\mathbb{P}^k]$ of any linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$ of dimension k:

$$A_k(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^k], \qquad 0 \leq k \leq n$$

and $A_k(\mathbb{P}^n) = 0$ for $k > n$. Any two linear subspaces of the same dimension are rationally equivalent, so the generator does not depend on the choice.

***Proof*:**
The argument is an induction on n, run through projection from a point, which realizes the complement of a point in $\mathbb{P}^n$ as a line bundle over $\mathbb{P}^{n-1}$. Write h_k for the class $[\mathbb{P}^k]$ of a linear k-plane; the claim is that $A_k(\mathbb{P}^n) = \mathbb{Z}\,h_k$ is free of rank one for $0 \leq k \leq n$, and 0 otherwise. Groups $A_k(\mathbb{P}^n)$ with $k > n = \dim \mathbb{P}^n$ vanish because $\mathbb{P}^n$ has no subvariety of that dimension, so only $0 \leq k \leq n$ is at issue.

Step 1: base cases $n = 0$ and $k = 0$. For $n = 0$ the space $\mathbb{P}^0 = \operatorname{Spec} \overline{k}$ is a point, with $A_0(\mathbb{P}^0) = \mathbb{Z}\,[\mathbb{P}^0]$ and $A_k(\mathbb{P}^0) = 0$ for $k \neq 0$ by proposition 8.5.1, which is the claim. For general n and $k = 0$, every closed point of $\mathbb{P}^n$ is rationally equivalent to every other: two points p, q both lie on a line $\ell \cong \mathbb{P}^1 \subseteq \mathbb{P}^n$, and on ℓ they are rationally equivalent by example 8.1.5, so $[p] \sim [q]$ in $\mathbb{P}^n$. Thus $A_0(\mathbb{P}^n)$ is cyclic, generated by $h_0 = [p]$. Since $\mathbb{P}^n$ is complete, the degree map $\deg\colon A_0(\mathbb{P}^n) \to \mathbb{Z}$ of definition 8.2.4 is well defined and sends $h_0 = [p] \mapsto [k(p) : \overline{k}] = 1$, so h_0 has infinite order and $A_0(\mathbb{P}^n) = \mathbb{Z}\,h_0$ is free of rank one.

Step 2: projection from a point as a line bundle. Fix $n \geq 1$ and a closed point $P \in \mathbb{P}^n$. Choosing coordinates with $P = [0 : \cdots : 0 : 1]$, linear projection away from P is a morphism

$$q\colon \mathbb{P}^n \setminus \{P\} \to \mathbb{P}^{n-1}, \qquad [x_0 : \cdots : x_n] \mapsto [x_0 : \cdots : x_{n-1}]$$

whose fiber over a point $[a] \in \mathbb{P}^{n-1}$ is the line through P and $[a]$ with P removed, an affine line $\mathbb{A}^1$. This exhibits $\mathbb{P}^n \setminus \{P\}$ as a line bundle over $\mathbb{P}^{n-1}$, the total space of $\mathcal{O}_{\mathbb{P}^{n-1}}(1)$; in particular it is an affine bundle of relative dimension 1. Only the last clause is used below. Affine-bundle invariance (theorem 8.4.6) makes flat pullback

$$q^*\colon A_{k-1}(\mathbb{P}^{n-1}) \xrightarrow{\sim} A_k(\mathbb{P}^n \setminus \{P\}) \tag{8.4}$$

an isomorphism for every k.

Step 3: removing the point via localization. The point $\{P\}$ is a closed subscheme of $\mathbb{P}^n$ with open complement $\mathbb{P}^n \setminus \{P\}$. The localization sequence (theorem 8.4.1) for this pair reads

$$A_k(\{P\}) \xrightarrow{i_*} A_k(\mathbb{P}^n) \xrightarrow{j^*} A_k(\mathbb{P}^n \setminus \{P\}) \to 0 \tag{8.5}$$

exact at $A_k(\mathbb{P}^n)$ and $A_k(\mathbb{P}^n \setminus \{P\})$. The point contributes $A_k(\{P\}) = 0$ for every $k \geq 1$ and $A_0(\{P\}) = \mathbb{Z}\,[P]$. Hence for $k \geq 1$ the left term vanishes and j^* is an isomorphism, which combined with (8.4) gives, for $k \geq 1$,

$$A_k(\mathbb{P}^n) \xrightarrow[j^*]{\sim} A_k(\mathbb{P}^n \setminus \{P\}) \xleftarrow[q^*]{\sim} A_{k-1}(\mathbb{P}^{n-1}) \tag{8.6}$$

so $A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1})$ as abelian groups.

Step 4: the induction. Fix k with $1 \leq k \leq n$. Iterating (8.6) k times descends both indices by k:

$$A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1}) \cong \cdots \cong A_0(\mathbb{P}^{n-k})$$

and $A_0(\mathbb{P}^{n-k}) = \mathbb{Z}$ is free of rank one by Step 1. Together with the case $k = 0$ of Step 1, this proves $A_k(\mathbb{P}^n) \cong \mathbb{Z}$ is free of rank one for every $0 \leq k \leq n$. For $k > n$ the group vanishes as noted at the outset.

Step 5: the generator is the class of a linear subspace. It remains to identify the free generator with $h_k = [\mathbb{P}^k]$. Trace a linear $\mathbb{P}^k \subseteq \mathbb{P}^n$ through (8.6). Choose the point $P \in \mathbb{P}^k$; projection of the linear subspace $\mathbb{P}^k$ from the point $P \in \mathbb{P}^k$ is again linear, of one lower dimension, so $q(\mathbb{P}^k \setminus \{P\})$ is a linear $\mathbb{P}^{k-1} \subseteq \mathbb{P}^{n-1}$. Restricting the bundle $q\colon \mathbb{P}^n \setminus \{P\} \to \mathbb{P}^{n-1}$ of Step 2 over this $\mathbb{P}^{k-1}$ realizes $\mathbb{P}^k \setminus \{P\}$ as the affine bundle over $\mathbb{P}^{k-1}$; concretely $q^{-1}(\mathbb{P}^{k-1}) = \mathbb{P}^k \setminus \{P\}$ as a reduced irreducible subvariety, so flat pullback gives $q^*[\mathbb{P}^{k-1}] = [q^{-1}(\mathbb{P}^{k-1})] = [\mathbb{P}^k \setminus \{P\}]$ with multiplicity 1. Hence by construction of (8.6) the class $h_k = [\mathbb{P}^k]$ satisfies $j^* h_k = [\mathbb{P}^k \setminus \{P\}] = q^*[\mathbb{P}^{k-1}]$, so h_k corresponds to h_{k-1} under the isomorphism $A_k(\mathbb{P}^n) \cong A_{k-1}(\mathbb{P}^{n-1})$. Descending to $A_0(\mathbb{P}^{n-k})$, the class h_k maps to $h_0 = [\mathrm{pt}]$, which is the free generator by Step 1. Thus h_k is a generator of $A_k(\mathbb{P}^n) = \mathbb{Z}$. The top case $k = n$ is consistent with the fundamental-class computation $A_n(\mathbb{P}^n) = Z_n(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^n]$ of example 8.1.6, since $\mathbb{P}^n$ is irreducible and $h_n = [\mathbb{P}^n]$.

Step 6: independence of the linear subspace. Any two linear k-planes $\Lambda, \Lambda' \subseteq \mathbb{P}^n$ have classes differing by an integer multiple of the generator; the previous step assigns each of them the generator h_0 under the descent to $A_0(\mathbb{P}^{n-k})$, so $[\Lambda] = [\Lambda']$ in $A_k(\mathbb{P}^n)$. Concretely a projective linear transformation carrying Λ to Λ' sweeps out a rational equivalence between them, so the generator h_k is independent of the chosen k-plane, completing the induction and the proof.

The theorem says that the Chow groups of $\mathbb{P}^n$ take the simplest possible form: one copy of $\mathbb{Z}$ in each dimension from 0 to n, with a canonical generator, the class of a linear subspace. Writing the total group additively,

$$A_*(\mathbb{P}^n) = \bigoplus_{k=0}^{n} \mathbb{Z}\, h_k, \qquad h_k = [\mathbb{P}^k]$$

so $A_*(\mathbb{P}^n)$ is free of rank $n+1$. In codimension indexing the generator $h_k = [\mathbb{P}^k]$ has codimension $n-k$, and it is common to write $h = [\mathbb{P}^{n-1}]$ for the hyperplane class and record h_k as the intersection h^{n-k} of $n-k$ hyperplanes; the multiplicative structure that makes this literal is the intersection product developed in a later chapter. For now the additive statement is what matters: on projective space, a k-cycle carries exactly one integer of information, its degree, the coefficient of h_k.

This is the precise sense in which projective space is rigid where affine space is soft. A subvariety $V \subseteq \mathbb{P}^n$ of dimension k and degree d (its degree as a projective variety, the number of points in which it meets a general linear $\mathbb{P}^{n-k}$) has class $[V] = d\, h_k$, and no amount of rational equivalence can change d. The degree is a complete invariant of the class, and rational equivalence on $\mathbb{P}^n$ is, for irreducible cycles, exactly the relation "same dimension and same degree" (and for arbitrary cycles, "same degree in each dimension"). The comparison with singular homology is exact: under the cycle class map to $H_{2k}(\mathbb{C}\mathrm{P}^n)$ studied later, h_k maps to the generator, and the freeness of $A_k(\mathbb{P}^n)$ mirrors the freeness of $H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ ([51]).

The one-dimensional case deserves its own statement, both because curves are where the theory began and because on a curve the Chow group is an object the reader already knows under another name. On a smooth projective curve C the only cycles are 0-cycles and the fundamental class, and the 0-cycles modulo rational equivalence are precisely the divisor classes. This ties the general construction back to the divisor class group $\mathrm{Cl}\, C$ and the Picard group $\mathrm{Pic}\, C$ of section 5.4.

Proposition 8.5.3: Chow Groups of a Smooth Projective Curve

Let C be a smooth projective curve over $\bar{k}$. Then

$$A_1(C) = \mathbb{Z}\,[C], \qquad A_0(C) \cong \mathrm{Cl}\, C = \mathrm{Pic}\, C$$

and $A_k(C) = 0$ for $k \neq 0, 1$. Under the second isomorphism the degree of a zero-cycle corresponds to the degree of a divisor, giving a short exact sequence

$$0 \to \mathrm{Pic}^{\,0} C \to A_0(C) \xrightarrow{\deg} \mathbb{Z} \to 0$$

where $\mathrm{Pic}^{\,0} C$ is the group of divisor classes of degree zero.

***Proof*:**
Since $\dim C = 1$, the only nonzero groups are $A_0(C)$ and $A_1(C)$, and $A_k(C) = 0$ for $k \neq 0, 1$. The curve C is irreducible of dimension 1, so by example 8.1.6 its top group is $A_1(C) = Z_1(C) = \mathbb{Z}\,[C]$, free on the fundamental class.

The zero-cycles. A 0-cycle on C is a finite $\mathbb{Z}$-combination of closed points, so $Z_0(C)$ is the group $\mathrm{WDiv}\, C$ of Weil divisors on C. The subgroup $\mathrm{Rat}_0(C)$ of cycles rationally equivalent to zero is generated by the cycles $[\mathrm{div}(r)]$ for $r \in k(W)^*$ ranging over 1-dimensional subvarieties $W \subseteq C$; the only such W is C itself, so $\mathrm{Rat}_0(C)$ is generated by the principal divisors $[\mathrm{div}(r)]$ with $r \in k(C)^*$. Thus $\mathrm{Rat}_0(C)$ is exactly the group $\mathrm{Prin}\, C$ of principal divisors in the sense of definition 5.4.7, and

$$A_0(C) = Z_0(C)/\mathrm{Rat}_0(C) = \mathrm{WDiv}\,C/\,\mathrm{Prin}\,C = \mathrm{Cl}\,C$$

is the divisor class group of definition 5.4.8. Because C is smooth, the divisor-to-line-bundle correspondence of section 5.4 identifies $\mathrm{Cl}\,C$ with $\mathrm{Pic}\,C$.

The degree sequence. On a smooth projective curve linearly equivalent divisors have equal degree: if $D \sim D'$ then $\deg D = \deg D'$, since the Riemann-Roch theorem (theorem 7.1.19) gives $\chi(\mathcal{L}(D)) = \deg D + 1 - g$, a quantity depending only on the class of D, so $\deg D$ is determined by it. Hence deg descends to a homomorphism $\deg\colon A_0(C) \to \mathbb{Z}$, agreeing under the identification $A_0(C) = \mathrm{Cl}\,C$ with the degree of a divisor class. It is surjective because a single closed point has degree $[k(p) : \overline{k}] = 1$, and its kernel is the group $\mathrm{Pic}^0 C$ of degree-zero classes by definition. This gives the stated short exact sequence, completing the proof.

The proposition recovers, in the language of Chow groups, the entire divisor theory of a curve. The degree map splits the class group into a discrete part, the copy of $\mathbb{Z}$ recording degree, and a connected part $\mathrm{Pic}^0 C$, the Jacobian, carrying the continuous moduli of divisor classes of a fixed degree. For $C = \mathbb{P}^1$ the Jacobian is trivial and $A_0(\mathbb{P}^1) = \mathbb{Z}$, in agreement with example 8.1.5 and with the case $n = 1$ of theorem 8.5.2; for a curve of genus $g \geq 1$ the group $\mathrm{Pic}^0 C$ is a g-dimensional abelian variety and $A_0(C)$ is strictly larger than $\mathbb{Z}$, so the degree is no longer a complete invariant of a class. The Chow group of a curve is therefore the first place in the theory where an interesting continuous invariant appears, and it is the model on which the higher-dimensional theory is built.

The Chow group $A_0(C)$ also illustrates the failure of the total-degree map to see everything. On $\mathbb{P}^n$ the degree was a complete invariant of a k-cycle class; on a curve of positive genus it is not, because two points $p, q \in C$ need not be rationally equivalent even though $[p]$ and $[q]$ have the same degree. Their difference $[p] - [q]$ is a nonzero class in $\mathrm{Pic}^0 C$ precisely when p and q are not linearly equivalent, which for $g \geq 1$ is the generic situation. This is the geometric content of the statement that a curve of positive genus admits no nonconstant rational function of degree one.

The last computation of the section is stated but not proved, because its proof needs Segre classes, which are developed later. It generalizes the projective-space computation from the trivial bundle to an arbitrary vector bundle: the Chow groups of a projectivized vector bundle are a free module over the Chow groups of the base, on the powers of the tautological class. This is the projective-bundle formula, and it is the engine that will later define Chern classes.

Given a vector bundle E of rank $r+1$ on a scheme X, write $\mathbb{P}(E) \to X$ for the associated projective bundle, whose fiber over a point x is the projective space $\mathbb{P}(E_x) \cong \mathbb{P}^r$ of one-dimensional subspaces of the fiber E_x. Let $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1)) \in A^1(\mathbb{P}(E))$ be the first Chern class of the Serre-twist line bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ (the dual of the tautological subbundle, with the "one-dimensional subspaces" convention above), viewed as an operator on cycles. The projective-bundle formula states that the flat pullback π^* together with capping against powers of ξ makes $A_*(\mathbb{P}(E))$ a free module over $A_*(X)$ on the classes $1, \xi, \ldots, \xi^r$:

$$A_*(\mathbb{P}(E)) = \bigoplus_{i=0}^{r} \xi^i \cap \pi^* A_*(X)$$

Concretely, every class on $\mathbb{P}(E)$ is uniquely $\sum_{i=0}^{r} \xi^i \cap \pi^* \alpha_i$ with $\alpha_i \in A_*(X)$. Taking $X = \mathrm{Spec}\,\overline{k}$ and $E = \overline{k}^{\,n+1}$ recovers $\mathbb{P}(E) = \mathbb{P}^n$ and the free-module statement collapses to $A_*(\mathbb{P}^n) = \bigoplus_{i=0}^{n} \mathbb{Z}\,\xi^i$ of theorem 8.5.2, with ξ^i the class of a linear subspace of codimension i. The full statement, and the definition of the Chern class operators it rests on, are theorem 10.1.9 in section 10.1; the formula is recorded here because it is the natural common generalization of the two computations above, and because it is the shape every later Chow-group computation of a bundle takes.

Before the exercises, the three worked instances of the theory are collected side by side, so the common mechanism is visible in one place.

Example 8.5.4: The Chow Groups of $\mathbb{P}^1$, $\mathbb{P}^2$, and a Plane Curve

1. *The projective line.* For $n = 1$, theorem 8.5.2 gives $A_0(\mathbb{P}^1) = \mathbb{Z}\,[\mathrm{pt}]$ and $A_1(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1]$, so
$$A_*(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1] \oplus \mathbb{Z}\,[\mathrm{pt}]$$
The degree-0 part $A_0(\mathbb{P}^1) = \mathbb{Z}$ agrees with example 8.1.5, and with proposition 8.5.3 for the genus-zero curve $\mathbb{P}^1$, whose Jacobian $\mathrm{Pic}^0\mathbb{P}^1$ is trivial.

2. *The projective plane.* For $n = 2$,
$$A_*(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2] \oplus \mathbb{Z}\,[\mathbb{P}^1] \oplus \mathbb{Z}\,[\mathrm{pt}]$$
with $A_2(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2]$, $A_1(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^1]$ generated by the class of a line, and $A_0(\mathbb{P}^2) = \mathbb{Z}\,[\mathrm{pt}]$. A plane curve $D \subseteq \mathbb{P}^2$ of degree d has class $[D] = d\,[\mathbb{P}^1] = d\,h_1$ in $A_1(\mathbb{P}^2)$, since D is cut out by a form of degree d and hence is rationally equivalent to d lines (the pencil spanned by the form and ℓ^d for a linear form ℓ sweeps one to the other). Two plane curves of degrees c and d therefore have classes meeting, after the intersection product is available, in cd points, which is Bézout's theorem ([20]) read off from $h_1 \cdot h_1 = h_0$.

3. *A plane curve as a curve in its own right.* Let $C \subseteq \mathbb{P}^2$ be a smooth plane curve of degree d, so C is a smooth projective curve of genus $g = \binom{d-1}{2}$. Two computations of its Chow groups coexist. Viewed inside $\mathbb{P}^2$, the class $[C] = d\,h_1 \in A_1(\mathbb{P}^2)$ records only its degree. Viewed intrinsically, proposition 8.5.3 gives $A_0(C) \cong \mathrm{Cl}\,C = \mathrm{Pic}\,C$ with its degree sequence $0 \to \mathrm{Pic}^0 C \to A_0(C) \to \mathbb{Z} \to 0$, and for $d \geq 3$ the genus is positive, so $\mathrm{Pic}^0 C$ is nontrivial and $A_0(C)$ is strictly larger than $\mathbb{Z}$. The pushforward $A_0(C) \to A_0(\mathbb{P}^2) = \mathbb{Z}$ along the inclusion is the degree map, collapsing all of $\mathrm{Pic}^0 C$ to zero: the intrinsic Chow group of the curve sees the moduli that the ambient projective plane cannot.

Exercise 8.5.1

1. Compute $A_*(\mathbb{A}^2 \setminus \{p\})$, the Chow groups of the affine plane with one closed point removed. (Use the localization sequence for the closed point inside $\mathbb{A}^2$ together with proposition 8.5.1.)

2. Let $H \cong \mathbb{P}^{n-1}$ be a hyperplane in $\mathbb{P}^n$ and $U = \mathbb{P}^n \setminus H \cong \mathbb{A}^n$ its complement. Using the localization sequence, show directly that $j^*\colon A_k(\mathbb{P}^n) \to A_k(\mathbb{A}^n)$ is an isomorphism for $k = n$ and the zero map for $k < n$, and reconcile this with theorem 8.5.2.

3. Let $Q \subseteq \mathbb{P}^3$ be a smooth quadric surface. Assuming the isomorphism $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ and that $A_*(\mathbb{P}^1 \times \mathbb{P}^1)$ is free on the classes of a point, the two rulings, and the fundamental class, determine the rank of each $A_k(Q)$ and describe generators. Compare with $A_*(\mathbb{P}^2)$ and explain why $A_1(Q)$ has rank 2 rather than 1.

4. For $\mathbb{P}^m$ and $\mathbb{P}^n$, use theorem 8.5.2 and the projective-bundle formula (with $\mathbb{P}^m \times \mathbb{P}^n = \mathbb{P}(\mathcal{O}^{\oplus(m+1)})$ over $\mathbb{P}^n$) to show that $A_k(\mathbb{P}^m \times \mathbb{P}^n)$ is free of rank equal to the number of pairs

(i, j) with $0 \le i \le m$, $0 \le j \le n$, and $i + j = k$.

5. Let C be a smooth projective curve of genus 1 over $\overline{k}$. Show that $A_0(C)$ is not finitely generated, and identify the classes $[p] - [q]$ of differences of points inside $\operatorname{Pic}^0 C$. Contrast with $A_0(\mathbb{P}^1)$.

6. Let $X \subseteq \mathbb{A}^3$ be the affine cone $x^2 + y^2 = z^2$ over a conic, a surface singular at the origin. Using $A_k(X) = A_k(X_{\mathrm{red}})$ and the vanishing of $A_k(\mathbb{A}^n)$ in low degree as a guide, determine $A_2(X)$ and explain why the singular point does not contribute a class in $A_0(X)$ that survives rational equivalence.

9

Divisors, Line Bundles, and the First Chern Class

Chapter 8 built the Chow groups and their two functorialities, proper pushforward and flat pullback, but produced no way to multiply cycles: nothing so far intersects a k-cycle with anything to lower its dimension. The first and most basic such operation is intersection with a divisor. A Cartier divisor D on X, equivalently a line bundle together with a rational section, cuts each subvariety V not contained in its support down to a $(k-1)$-cycle $D \cdot [V]$, the divisor of the restricted section. This chapter makes that operation precise, proves it descends to rational equivalence, and establishes its one indispensable property, commutativity: intersecting with D and then with D' agrees with intersecting in the other order.

Commutativity is not a formality. It is the identity whose proof, resting on the vanishing degree of a principal divisor on a complete curve, supplies the well-definedness that the affine-bundle-invariance theorem of section 8.4 took on faith. Packaging intersection with a divisor as the *first Chern class* $c_1(L) \cap -$ of the associated line bundle turns these results into the seed of the entire theory of Chern classes, developed for bundles of higher rank in the chapter that follows.

9.1 Cartier Divisors, Line Bundles, and the Class Map

The Chow group of section 8.1 is graded by dimension, and every construction of sections 8.2 to 8.4 either preserves the grading (proper pushforward, flat pullback of the fundamental class) or shifts it by the fixed relative dimension of a flat map. What is still missing is an operation that *lowers* dimension by one, cutting a k-cycle down to a $(k-1)$-cycle by intersecting it with a hypersurface. The classical prototype is Bézout's theorem: a curve of degree d and a curve of degree e in $\mathbb{P}^2$ meet in de points. The intersection is a codimension-one operation, and the object one intersects with is a divisor. This section sets up the divisor side of that operation. It recalls, from *Algebraic Geometry* [7], the two faces of a divisor (the geometric Weil divisor and the sheaf-theoretic Cartier divisor) and the line bundle that packages the Cartier data, then constructs the first bridge from divisor theory to the Chow group: the class map $\operatorname{Pic}(X) \to A_{n-1}(X)$ sending a line bundle to the cycle of its rational sections. The intersection operation itself, and the proof that it is well defined on rational equivalence, occupy section 9.2; here the goal is to fix the input and record how it sits inside the Chow group of a fixed variety.

Throughout, X is an algebraic scheme over $\overline{k}$ and V ranges over subvarieties (closed integral subschemes); a point is a closed point, and $n = \dim X$.

Algebraic Geometry [7] developed two theories of divisors, and the reader who has that material in hand can move quickly here. A *Weil divisor* (definition 5.4.5) on a Noetherian integral separated scheme regular in codimension one is a finite formal integer combination $\sum_i n_i[Y_i]$ of prime divisors, where a prime divisor is a codimension-one subvariety. This is precisely a codimension-one algebraic cycle in the sense of section 8.1: when $\dim X = n$, the group $\operatorname{WDiv} X$ of Weil divisors is exactly the cycle group $Z_{n-1}(X)$. A *principal divisor* $\operatorname{div}(f) = \sum_Y \nu_Y(f)[Y]$ (definition 5.4.7) records the zeros and poles of a rational function $f \in K(X)^*$, weighted by the order ν_Y measured in the discrete valuation ring $\mathcal{O}_{X,\eta_Y}$; this is the same order-along-a-subvariety $\operatorname{ord}_Y(f)$ that defines rational equivalence in definition 8.1.1. The divisor class group $\operatorname{Cl} X = \operatorname{WDiv} X/\operatorname{PDiv}(X)$ of definition 5.4.8 is therefore literally the top Chow group $A_{n-1}(X)$ in the case where the only cycles rationally equivalent to zero come from functions on X itself, that is, on a variety.

A *Cartier divisor* (definition 5.4.23) encodes the same data locally and sheaf-theoretically: it is a global section of $\mathscr{K}_X^*/\mathcal{O}_X^*$, represented by an open cover $\{U_i\}$ with rational functions $f_i \in \Gamma(U_i, \mathscr{K}_X^*)$ whose ratios f_i/f_j are units on overlaps. A Cartier divisor is *principal* when it comes from a single global rational function; two Cartier divisors are linearly equivalent when they differ by a principal one. The gluing data $\{(U_i, f_i)\}$ builds a line bundle $\mathcal{O}_X(D)$, the sub-$\mathcal{O}_X$-module of $\mathscr{K}_X$ generated locally by f_i^{-1}, and this assignment is the heart of the theory: $D \mapsto \mathcal{O}_X(D)$ is a homomorphism $\operatorname{CDiv} X \to \operatorname{Pic} X$ that carries addition of divisors to tensor product of line bundles, sends $-D$ to the dual, and vanishes exactly on principal divisors. On an integral scheme it induces an isomorphism from the Cartier divisor class group $\operatorname{CaCl} X$ of definition 5.4.34 onto the whole Picard group (all of this is developed in section 5.4),

$$\operatorname{CaCl} X \xrightarrow{\sim} \operatorname{Pic} X$$

so a line bundle is the same thing as a Cartier divisor up to linear equivalence. This is the sense in which the section title lists “Cartier divisors” and “line bundles” as two names for one object.

The reader should keep both faces in view. The Cartier divisor is what one intersects with, because its local equations f_i give the equations to cut against; the line bundle $\mathcal{O}_X(D)$ is the coordinate-free invariant, the thing that survives when the choice of local equations is forgotten. The class map built below is precisely the passage from the line bundle back to a well-defined cycle class.

The two theories agree under a regularity hypothesis, and pinning down that hypothesis is what makes the class map land in the Chow group cleanly. The point is one of local principality: a Weil divisor is Cartier exactly when each of its prime divisors is locally cut out by a single equation. Regularity in codimension one is enough to send a Cartier divisor to a Weil divisor; local factoriality is what makes the map an isomorphism.

Proposition 9.1.1: Weil-Cartier Comparison

Let X be a variety of dimension n over $\overline{k}$.

1. If X is regular in codimension one (for instance X normal), there is a homomorphism
$$\operatorname{CDiv} X \longrightarrow \operatorname{WDiv} X = Z_{n-1}(X), \qquad D \longmapsto [D]$$
carrying principal Cartier divisors to principal Weil divisors. In local data $D = \{(U_i, f_i)\}$ the associated Weil cycle is $[D] = \sum_Y \nu_Y(f_i)[Y]$, the coefficient of a prime divisor Y being $\nu_Y(f_i)$ for any index i with $\eta_Y \in U_i$.

2. If X is locally factorial (every local ring $\mathcal{O}_{X,x}$ is a unique factorization domain; smooth varieties qualify), this map is an isomorphism $\operatorname{CDiv} X \xrightarrow{\sim} \operatorname{WDiv} X$ carrying principal divisors to principal divisors, hence an isomorphism of class groups
$$\operatorname{Pic} X \;\cong\; \operatorname{CaCl} X \;\cong\; \operatorname{Cl} X \;=\; A_{n-1}(X)$$

***Proof*:**
Both parts are proved in full in section 5.4 and only recalled here. The homomorphism of part (1) is proposition 5.4.31, and its upgrade to an isomorphism under local factoriality (part (2)) is proposition 5.4.32; the resulting chain of class-group isomorphisms $\operatorname{Pic} X \cong \operatorname{CaCl} X \cong \operatorname{Cl} X$ is corollary 5.4.56. Only the final equality $\operatorname{Cl} X = A_{n-1}(X)$ is new to this Part, and it is the identification made in the paragraph below: on a variety the codimension-one cycles modulo rational equivalence are exactly the Weil divisors modulo principal divisors, both quotients of $Z_{n-1}(X)$ by $\operatorname{PDiv}(X)$, as we sought to show.

The equality $\operatorname{Cl} X = A_{n-1}(X)$ deserves a moment's reflection, because it is easy to mistake for a definition. Rational equivalence on $Z_{n-1}(X)$ is generated by cycles $[\operatorname{div}(r)]$ where r is a rational function on a subvariety $W \subseteq X$ of dimension n. On a variety X, the only n-dimensional subvariety is X itself, so those generators are exactly the principal Weil divisors $\operatorname{div}(r)$ with $r \in K(X)^*$. Hence $A_{n-1}(X) = Z_{n-1}(X)/\operatorname{Rat}_{n-1}(X)$ and $\operatorname{Cl} X = \operatorname{WDiv} X/\operatorname{PDiv}(X)$ are the same quotient of the same group. Linear equivalence of divisors *is* rational equivalence of codimension-one cycles. This is the seam that will let the class map below produce Chow classes, not just divisor classes.

Example 9.1.2: The Quadric Cone: Cartier Strictly Inside Weil
The comparison map need not be surjective when factoriality fails. On the affine quadric cone $X = \operatorname{Spec} \overline{k}[x, y, z]/(xy - z^2)$, a normal surface singular at the vertex, example 5.4.35 computes that the ruling $Y = V(y, z)$ is a Weil divisor which is not Cartier at the vertex, while its double $2[Y] = \operatorname{div}(y)$ is, so
$$\operatorname{CaCl} X = 0 \subsetneq \operatorname{Cl} X = \mathbb{Z}/2\mathbb{Z} = A_1(X)$$
the final equality being the identification $\operatorname{Cl} X = A_{n-1}(X)$ of proposition 9.1.1. Normality supplies part (1), but the vertex obstructs local principality, so part (2) fails; on a smooth variety no such gap exists and Weil, Cartier, Picard, and A_{n-1} all coincide.

On a smooth variety the isomorphism $\operatorname{Pic} X \cong A_{n-1}(X)$ of proposition 9.1.1 already provides a bridge from line bundles to Chow classes. To intersect and to compute Chern classes on *arbitrary* algebraic schemes, one needs this bridge without any regularity hypothesis, and one needs it stated at the level of line bundles rather than Cartier divisors, so that the input is a coordinate-free invariant. The construction is direct: a line bundle L has rational sections, a rational section has a divisor of zeros and poles, and that divisor is a codimension-one cycle whose class in $A_{n-1}(X)$ turns out not to depend on the section chosen.

Fix a variety X of dimension n with function field $K(X)$, and a line bundle L on X. A *rational section* of L is a nonzero section of $L \otimes_{\mathcal{O}_X} K(X)$ over X, equivalently a section of L over some nonempty open subset. Since X is a variety, $L \otimes K(X)$ is a one-dimensional $K(X)$-vector space, so rational sections exist and any two differ by multiplication by an element of $K(X)^*$. On the open set where s is defined and nonvanishing, s trivializes L; along each prime divisor Y one may compare s with a local trivialization of L near η_Y and read off an integer $\operatorname{ord}_Y(s)$, the order of vanishing (or of the pole) of s along Y. Concretely, if t is a local generator of L near η_Y and $s = g \cdot t$ with $g \in K(X)^*$, set $\operatorname{ord}_Y(s) := \nu_Y(g)$; a different generator $t' = ut$ with $u \in \mathcal{O}^*_{X,\eta_Y}$ changes g to $u^{-1}g$ and leaves $\nu_Y(g)$ unchanged, so $\operatorname{ord}_Y(s)$ is well defined. The *divisor of* s is the Weil divisor

$$\operatorname{div}(s) \;=\; \sum_Y \operatorname{ord}_Y(s)\,[Y] \;\in\; Z_{n-1}(X)$$

finite because s is a regular nonvanishing section over some nonempty open $U \subseteq X$, where it trivializes L and so $\operatorname{ord}_Y(s) = 0$ for every prime divisor Y meeting U; the only possible contributors are the finitely many prime divisors in the proper closed complement $X \setminus U$. This is the same finiteness that makes the principal divisor $\operatorname{div}(f)$ of definition 5.4.7 a well-defined cycle. If $L = \mathcal{O}_X(D)$ for a Cartier divisor D and s is the canonical rational section (the constant $1 \in \mathscr{K}_X$, which against the local generator f_i^{-1} of $\mathcal{O}_X(D)|_{U_i}$ is written $1 = f_i \cdot f_i^{-1}$, giving $\operatorname{ord}_Y(s) = \nu_Y(f_i)$), then $\operatorname{div}(s)$ is exactly the Weil cycle $[D]$ of proposition 9.1.1.

Proposition 9.1.3: The Class Map

Let X be a variety of dimension n. For a line bundle L on X, choose any nonzero rational section s of L and set

$$c(L) \;:=\; [\operatorname{div}(s)] \;\in\; A_{n-1}(X)$$

Then $c(L)$ is independent of the choice of s, and the resulting assignment

$$c\colon \operatorname{Pic} X \longrightarrow A_{n-1}(X), \qquad L \longmapsto c(L)$$

is a group homomorphism. When X is locally factorial (in particular smooth), c is the isomorphism $\operatorname{Pic} X \cong A_{n-1}(X)$ of proposition 9.1.1; on a general variety c need be neither injective nor surjective.

***Proof*:**
Step 1: independence of the section. Let s, s' be two nonzero rational sections of L. Both lie in the one-dimensional $K(X)$-vector space $L \otimes K(X)$, so $s' = gs$ for a unique $g \in K(X)^*$. Along each prime divisor Y, choose a local generator t of L near η_Y and write $s = at$, $s' = a't$ with $a, a' \in K(X)^*$; then $a' = ga$ and

$$\operatorname{ord}_Y(s') = \nu_Y(a') = \nu_Y(g) + \nu_Y(a) = \nu_Y(g) + \operatorname{ord}_Y(s)$$

because ν_Y is a valuation. Summing over Y gives $\operatorname{div}(s') = \operatorname{div}(g) + \operatorname{div}(s)$ in $Z_{n-1}(X)$, where $\operatorname{div}(g) = \sum_Y \nu_Y(g)[Y]$ is the principal Weil divisor of $g \in K(X)^*$. Since $\operatorname{div}(g)$ is a principal divisor on the variety X, it is rationally equivalent to zero in $A_{n-1}(X)$ by the identification $\mathrm{Rat}_{n-1}(X) = \mathrm{PDiv}(X)$ recorded after proposition 9.1.1. Hence $[\mathrm{div}(s')] = [\mathrm{div}(s)]$ in $A_{n-1}(X)$, and $c(L)$ is well defined.

Step 2: homomorphism property. Let L, L' be line bundles with rational sections s, s'. Then $s \otimes s'$ is a nonzero rational section of $L \otimes L'$, and along each prime divisor Y, choosing local generators t of L and t' of L' near η_Y and writing $s = at$, $s' = a't'$, the product $s \otimes s' = (aa')(t \otimes t')$ is expressed against the local generator $t \otimes t'$ of $L \otimes L'$. Thus

$$\mathrm{ord}_Y(s \otimes s') = \nu_Y(aa') = \nu_Y(a) + \nu_Y(a') = \mathrm{ord}_Y(s) + \mathrm{ord}_Y(s')$$

so $\mathrm{div}(s \otimes s') = \mathrm{div}(s) + \mathrm{div}(s')$, and passing to classes gives $c(L \otimes L') = c(L) + c(L')$. The trivial bundle $\mathcal{O}_X$ has the rational section 1 with $\mathrm{div}(1) = 0$, so $c(\mathcal{O}_X) = 0$, and $c(L^{-1}) = -c(L)$ follows from $L \otimes L^{-1} = \mathcal{O}_X$. Therefore c is a homomorphism.

Step 3: comparison with the smooth case. When X is locally factorial, every line bundle is $\mathcal{O}_X(D)$ for a Cartier divisor D, and the canonical rational section of $\mathcal{O}_X(D)$ has $\mathrm{div}(s) = [D]$, the Weil cycle of D. Under $\mathrm{Pic}\, X \cong \mathrm{CaCl}\, X$ (section 5.4) and $\mathrm{CaCl}\, X \cong \mathrm{Cl}\, X = A_{n-1}(X)$ (proposition 9.1.1), the assignment $L = \mathcal{O}_X(D) \mapsto [D]$ is exactly that isomorphism, so c is an isomorphism. On a variety where these hypotheses fail, c can lose injectivity or surjectivity, as example 9.1.2 shows for the class-group comparison. This completes the proof.

The class map is the first of the two ways line bundles act on the Chow group. It records *where* a line bundle is, as a codimension-one class $c(L) \in A_{n-1}(X)$, but it does not yet say how L intersects cycles of other dimensions. That is the job of the operator $c_1(L) \cap -$ built in section 9.4, whose value on the fundamental class $[X]$ recovers $c(L)$: writing ahead, $c_1(L) \cap [X] = c(L)$. The class map is thus the top-dimensional case of the full first Chern class, and everything in this chapter can be read as promoting c from a map on line bundles to an operator on all of $A_*(X)$.

The value of stating c at the level of line bundles, rather than resting on the isomorphism $\mathrm{Pic} \cong A_{n-1}$, is that c makes sense with no regularity hypothesis on X: a rational section and its order along a prime divisor require only that X be a variety, and the group $A_{n-1}(X)$ is defined for every algebraic scheme. The intersection theory of section 9.2 is built for arbitrary X, and it needs c in this general form.

Example 9.1.4: The Class Map on Projective Space

Projective space is where every ingredient can be computed by hand. Since $\mathbb{P}^n$ has all local rings unique factorization domains, it is locally factorial, so $\mathrm{Pic}\, \mathbb{P}^n \cong \mathrm{Cl}\, \mathbb{P}^n \cong \mathbb{Z}$, generated by $\mathcal{O}_{\mathbb{P}^n}(1)$, and by theorem 8.5.2 the top Chow group is $A_{n-1}(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^{n-1}]$, generated by the class of a hyperplane. The class map identifies the two.

A rational section of $\mathcal{O}_{\mathbb{P}^n}(1)$ is a nonzero linear form. Take $s = x_0$, the section whose zero scheme is the hyperplane $H = V_+(x_0)$, whose Cartier data $\{(U_i, x_0/x_i)\}$ was recorded in example 5.4.35. On the standard chart $U_i = D_+(x_i) \cong \mathbb{A}^n$ the sheaf $\mathcal{O}_{\mathbb{P}^n}(1)$ is trivialized by x_i, and $s = x_0 = (x_0/x_i)\, x_i$, so s corresponds to the rational function $x_0/x_i \in K(\mathbb{P}^n)$ against the generator x_i. Along the prime divisor H, the generic point lies in charts U_i with $i \neq 0$, where x_0/x_i is a local equation cutting out H with order one, so $\mathrm{ord}_H(s) = 1$; along every other prime divisor x_0/x_i is a unit near the generic point, contributing zero. Hence

$$\mathrm{div}(x_0) = [H] = [\mathbb{P}^{n-1}], \qquad c\big(\mathcal{O}_{\mathbb{P}^n}(1)\big) = [\mathbb{P}^{n-1}] \in A_{n-1}(\mathbb{P}^n)$$

Choosing instead $s = x_1$ gives the hyperplane $V_+(x_1)$, a different divisor but the same class: $x_1 =$

$(x_1/x_0)\, x_0$ differs from x_0 by the rational function x_1/x_0, and $\mathrm{div}(x_1) - \mathrm{div}(x_0) = \mathrm{div}(x_1/x_0) \sim 0$, confirming the independence of proposition 9.1.3 in this case. More generally, $\mathcal{O}_{\mathbb{P}^n}(d)$ has rational sections given by degree-d forms; the section x_0^d has divisor $d[H]$, so $c(\mathcal{O}_{\mathbb{P}^n}(d)) = d\,[\mathbb{P}^{n-1}]$, and $c\colon \mathbb{Z} \to \mathbb{Z}$ is the identity. The generator $\mathcal{O}(1)$ of Pic maps to the generator [hyperplane] of A_{n-1}, and the class map is the announced isomorphism $\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}$.

There is one further refinement to record before intersecting, and it addresses a bookkeeping problem rather than a new geometric idea. When a Cartier divisor D is intersected with a subvariety V, one wants the resulting cycle to live on $V \cap |D|$, the part of V where D cuts, not just somewhere on V. Localizing the answer to the support is what makes the intersection product refined enough to carry information (for instance, an excess-intersection or a self-intersection formula) and what makes it functorial under the localization sequence of theorem 8.4.1. The device that carries this localized data is the *pseudo-divisor*: a line bundle, a chosen closed support, and a trivialization of the bundle away from that support.

Definition 9.1.5: Pseudo-Divisor

Let X be an algebraic scheme. A *pseudo-divisor* on X is a triple

$$\mathfrak{D} = (L, Z, s)$$

consisting of a line bundle L on X, a closed subset $Z \subseteq X$ (the *support*, written $|\mathfrak{D}|$), and a trivialization $s\colon \mathcal{O}_{X\setminus Z} \xrightarrow{\sim} L|_{X\setminus Z}$ of L over the complement of Z. The line bundle L is denoted $\mathcal{O}_X(\mathfrak{D})$.
A Cartier divisor D on X determines a pseudo-divisor $(\mathcal{O}_X(D),\, |D|,\, s_D)$ whose support is the support $|D|$ of D and whose trivialization s_D is the canonical rational section of $\mathcal{O}_X(D)$, which is a nonvanishing regular section away from $|D|$. Two Cartier divisors D, D' with the same support and the same associated line bundle give the same pseudo-divisor when their canonical sections agree off the support.

The three pieces of data are exactly what a divisor supplies for cutting. The line bundle L is the coordinate-free invariant whose class map value is $c(L)$; the support Z pins down where the cutting happens, so the product lands in $A_*(Z)$ rather than $A_*(X)$; and the trivialization s off Z is what lets one measure orders of vanishing, because on the locus where L is trivialized by s there is nothing to cut, and all the divisorial content is concentrated on Z. A Cartier divisor is the special case where $Z = |D|$ is as small as possible, but allowing Z to be larger is what gives the flexibility to restrict, pull back, and add pseudo-divisors while keeping supports under control. The next section takes a subvariety V not contained in the support and defines $\mathfrak{D} \cdot [V]$ as a class in $A_{k-1}(V \cap Z)$; when $V \subseteq Z$, the restricted line bundle $L|_V$ supplies the answer through its own class map. Here the pseudo-divisor is introduced only to fix the input; its use is the content of section 9.2.

Example 9.1.6: The Pseudo-Divisor of a Hyperplane

Let $X = \mathbb{P}^n$ and $H = V_+(x_0)$ the hyperplane at $x_0 = 0$. The associated pseudo-divisor is

$$\mathfrak{H} = \big(\mathcal{O}_{\mathbb{P}^n}(1),\ H,\ s\big)$$

with support the hyperplane H and trivialization s given by the section x_0 over the complement $\mathbb{P}^n \setminus H = D_+(x_0) \cong \mathbb{A}^n$, where x_0 is nonvanishing and so trivializes $\mathcal{O}(1)$. The line bundle $\mathcal{O}_{\mathbb{P}^n}(1)$ carries the class $c(\mathcal{O}(1)) = [H] \in A_{n-1}(\mathbb{P}^n)$ computed in example 9.1.4; the support restricts any intersection against $\mathfrak{H}$ to H; and the trivialization records that off H the hyperplane cuts nothing.

> Intersecting $\mathfrak{H}$ with a subvariety $V \not\subseteq H$ will, in section 9.2, produce the cycle $[V \cap H]$ with its multiplicities, supported on $V \cap H$, which is the algebraic incarnation of Bézout's degree count.

The class map and the pseudo-divisor together fix the two ways a line bundle enters intersection theory: as a class $c(L) \in A_{n-1}(X)$, which is its total content, and as a localized cutting datum (L, Z, s), which is how it acts. With the input settled, the operation $\mathfrak{D} \cdot -$ can be defined and shown well defined on rational equivalence, which is the subject of section 9.2.

Exercise 9.1.1

1. Let $X = \mathbb{P}^n$ and let $Q = V_+(q)$ be a smooth quadric hypersurface, cut out by a nondegenerate quadratic form q. Compute $c(\mathcal{O}_{\mathbb{P}^n}(Q))$, the class map value of the line bundle of Q, in $A_{n-1}(\mathbb{P}^n) = \mathbb{Z}\,[\mathbb{P}^{n-1}]$, and explain the answer in terms of the degree of Q.
2. Let X be a variety and $r \in K(X)^*$ a nonzero rational function. Show that the line bundle $\mathcal{O}_X(\operatorname{div}(r))$ associated to the principal Cartier divisor $\operatorname{div}(r)$ is trivial, and deduce that $c(\mathcal{O}_X(\operatorname{div}(r))) = 0$ directly from proposition 9.1.3. Which step of the class-map construction is this computing?
3. On the affine quadric cone $X = \operatorname{Spec} \overline{k}[x, y, z]/(xy - z^2)$ of example 9.1.2, the class map $c\colon \operatorname{Pic} X \to A_1(X)$ is a homomorphism $\operatorname{CaCl} X \to \operatorname{Cl} X$. Using the values $\operatorname{CaCl} X = 0$ and $\operatorname{Cl} X = \mathbb{Z}/2\mathbb{Z}$ from that example, describe c explicitly and state whether it is injective, surjective, or neither. What does this say about the necessity of the factoriality hypothesis in proposition 9.1.1(2)?
4. Pseudo-divisors add: given $\mathfrak{D} = (L, Z, s)$ and $\mathfrak{D}' = (L', Z', s')$, define $\mathfrak{D} + \mathfrak{D}' := (L \otimes L',\ Z \cup Z',\ s \otimes s')$. Verify that this is a well-defined pseudo-divisor, that $\mathcal{O}_X(\mathfrak{D} + \mathfrak{D}') = \mathcal{O}_X(\mathfrak{D}) \otimes \mathcal{O}_X(\mathfrak{D}')$, and that on the line-bundle level this addition is compatible with the homomorphism property of the class map in proposition 9.1.3.
5. Let L be a line bundle on a variety X, s a nonzero rational section, and Y a prime divisor. Show that the order $\operatorname{ord}_Y(s) := \nu_Y(a)$, defined by writing $s = a\,t$ against a local generator t of L near η_Y with $a \in K(X)^*$, does not depend on the choice of generator t. Then show that if s is a regular nonvanishing section of L near η_Y, then $\operatorname{ord}_Y(s) = 0$.

9.2 Intersecting with a Divisor

The class map of section 9.1 sends a line bundle to a cycle class in codimension one, but it says nothing about how that cycle meets the other cycles already living on X. This section supplies the missing operation. A Cartier divisor D acts on cycles by cutting each subvariety down along the zeros and poles of a rational section, an operation that lowers dimension by one and records where the cut takes place. The construction is the first instance of a general principle of the subject, that a line bundle acts on the Chow group as an operator, and it is the operation whose associativity and commutativity the rest of the chapter is built to establish.

The geometry is the one already visible in codimension one. A Cartier divisor is locally the divisor of a single rational function f_i on an open U_i, and the transition ratios f_i/f_j are units, so the zeros and poles of the local equations patch to a well-defined codimension-one cycle on X. To intersect D with a k-dimensional subvariety V, restrict this local data to V and read off the resulting $(k-1)$-cycle on V. Two features force themselves on the construction. First, the answer is supported on the intersection $V \cap |D|$, since away from the support of D the local equations are units and contribute nothing. Second, the recipe stumbles when V lies inside $|D|$, where the local equations $f_i|_V$ may vanish identically and $\operatorname{div}(f_i|_V)$ is undefined; the trivialization off $|D|$ is exactly what is lost. The definition below treats the two cases separately and unifies them through the restricted line bundle.

Definition 9.2.1: Intersection with a Cartier Divisor

Let D be a Cartier divisor on X with support $|D|$ and associated line bundle $\mathcal{O}_X(D)$, and let $V \subseteq X$ be a k-dimensional subvariety. The *intersection class* $D \cdot [V] \in A_{k-1}(V \cap |D|)$ is defined as follows.

1. If $V \not\subseteq |D|$, then the local equations of D restrict to a nonzero rational function s_V on V: choosing a representative $\{(U_i, f_i)\}$ of D, the ratios $f_i|_V$ agree up to units on overlaps and glue to a rational section s_V of $\mathcal{O}_X(D)|_V$ that is nonzero on the dense open $V \setminus |D|$. Its divisor is a $(k-1)$-cycle supported on $V \cap |D|$, and one sets
$$D \cdot [V] = \big[\operatorname{div}(s_V)\big] \in A_{k-1}(V \cap |D|)$$
2. If $V \subseteq |D|$, then no such nonzero section exists, and one instead uses the restricted line bundle. The class map of proposition 9.1.3 applied to $\mathcal{O}_X(D)|_V$ produces a class in $A_{k-1}(V)$; since $V \subseteq |D|$ gives $V \cap |D| = V$, one sets
$$D \cdot [V] = c\big(\mathcal{O}_X(D)|_V\big) \in A_{k-1}(V) = A_{k-1}(V \cap |D|)$$

Extending $\mathbb{Z}$-linearly over the subvarieties appearing in a cycle $\alpha = \sum_i n_i[V_i]$ gives the *intersection cycle* $D \cdot \alpha = \sum_i n_i\,(D \cdot [V_i])$, a class in $A_{k-1}(|D| \cap \operatorname{Supp}\alpha)$, where $\operatorname{Supp}\alpha = \bigcup_i V_i$.

Two conventions in the definition deserve unpacking, since both recur throughout the theory. The first is that the intersection class is recorded on the intersection $V \cap |D|$, not on all of X. This is a refinement one does not want to discard prematurely: the intersection of a curve with a divisor is a set of points on the curve, and remembering that the points lie in $V \cap |D|$ is exactly what makes localized intersection numbers and excess-intersection formulas possible later. When only the class in $A_{k-1}(X)$ is wanted, push forward along the inclusion $V \cap |D| \hookrightarrow X$, a proper map. The second convention is the treatment of $V \subseteq |D|$. Here the naive recipe fails because the local equations restrict to zero, so there is no rational section to take the divisor of; but $\mathcal{O}_X(D)|_V$ is still a line bundle on V, and its class under the class map $c\colon \operatorname{Pic}(V) \to A_{k-1}(V)$ of proposition 9.1.3 is a $(k-1)$-cycle class. Both clauses are the same construction, taking the class of the restricted bundle $\mathcal{O}_X(D)|_V$. When $V \not\subseteq |D|$ the bundle comes with a preferred rational section s_V and the class is computed as $[\operatorname{div}(s_V)]$; when $V \subseteq |D|$ that section degenerates and only the bundle survives. The packaging of the whole operation as the first Chern class $c_1(\mathcal{O}_X(D)) \cap -$ is the subject of section 9.4.

That the operation lowers dimension by exactly one is built into the definition: the divisor of a rational section on a k-dimensional variety is a codimension-one cycle, hence $(k-1)$-dimensional. Geometrically, $D \cdot [V]$ is the divisor cut out on V by the equation defining D, the higher-dimensional form of substituting a curve into the equation of a hypersurface and reading off the intersection points with multiplicity. The next example makes this concrete in the case a reader already knows from the classical degree of a plane curve.

Example 9.2.2: A Hyperplane Meeting a Plane Curve

Let $X = \mathbb{P}^2$ with homogeneous coordinates $[x_0 : x_1 : x_2]$, let $H = \{x_0 = 0\}$ be the line at infinity, a Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(H) = \mathcal{O}_{\mathbb{P}^2}(1)$, and let $V \subseteq \mathbb{P}^2$ be a smooth conic, say the closure of $\{x_1^2 = x_0x_2\}$, a subvariety of dimension 1 not contained in H. The unique intersection point turns out to be $p = [0 : 0 : 1]$, which lies in the chart $U_2 = \{x_2 \neq 0\}$; work there, with affine coordinates $u = x_0/x_2$ and $v = x_1/x_2$. On U_2 the conic is $V = \{u = v^2\}$ (from $x_1^2 = x_0x_2$), the divisor $H = \{x_0 = 0\}$ is $\{u = 0\}$, and against the local generator x_2 of $\mathcal{O}_{\mathbb{P}^2}(1)$ the section x_0 representing H is the function $x_0/x_2 = u$. Its restriction to V is $s_V = u|_V = v^2$, which vanishes exactly where V meets H: $u = 0$ forces $v = 0$, so p is the only intersection point, and there V is tangent to H. In the local coordinate v on V near p, the function $s_V = v^2$ has a double zero (the conic meets its tangent line to order two), so

$$H \cdot [V] = \big[\mathrm{div}(s_V)\big] = 2\,[p] \in A_0(V \cap H)$$

Pushing forward to $\mathbb{P}^2$ and taking degrees gives $\deg(H \cdot [V]) = 2$, the degree of the conic. The intersection class remembers more than the number 2: it records that both units of intersection are concentrated at the single point p, information the bare degree discards.

The example illustrates the two things the intersection class is meant to capture, the total count and its distribution, and shows how the classical intersection number of two plane curves reappears as the degree of $D \cdot [V]$. For a curve V of degree d and a line H in general position the same computation gives d distinct simple points, and $\deg(H \cdot [V]) = d$ in every case; this is the statement, once the well-definedness below is in hand, that intersecting with a hyperplane computes the degree. Before turning to that, the second worked example records the complementary phenomenon, what happens when V lies inside the support of D, so that the second clause of definition 9.2.1 is in force.

Example 9.2.3: A Divisor Restricted to a Component of Its Support

Let $X = \mathbb{P}^2$ again and let $D = L$ be a line, say $L = \{x_0 = 0\}$, so $|D| = L$ and $\mathcal{O}_{\mathbb{P}^2}(L) = \mathcal{O}_{\mathbb{P}^2}(1)$. Take $V = L$ itself, a 1-dimensional subvariety contained in $|D|$. The first clause does not apply, since x_0 restricts to zero on L; the second clause instructs one to compute the class of the restricted line bundle $\mathcal{O}_{\mathbb{P}^2}(1)|_L$. Now $L \cong \mathbb{P}^1$, and $\mathcal{O}_{\mathbb{P}^2}(1)|_L = \mathcal{O}_{\mathbb{P}^1}(1)$ by the adjunction of the hyperplane bundle to a linear subspace. The class map $\mathrm{Pic}\,(\mathbb{P}^1) = \mathbb{Z} \to A_0(\mathbb{P}^1) = \mathbb{Z}$ of proposition 9.1.3 carries $\mathcal{O}_{\mathbb{P}^1}(1)$ to the class of a single point $[q]$, so

$$L \cdot [L] = c\big(\mathcal{O}_{\mathbb{P}^2}(1)|_L\big) = [q] \in A_0(L)$$

the class of one point on L. This is the self-intersection $L \cdot L = 1$ of a line in the plane, the positive-degree form of the fact that two distinct lines meet in one point: moving L to a nearby line L' and intersecting $L \cdot [L'] = [L \cap L']$ recovers the same class, now visibly one transverse point. The second clause of the definition is what lets self-intersection make sense at all, since a divisor cannot literally be substituted into a variety it defines.

The two examples exhibit the operation on both sides of the containment condition and already hint at the compatibility the operation must have with rational equivalence. In example 9.2.3 the value $L \cdot [L]$ was computed by restricting the bundle, but it was also claimed to agree with $L \cdot [L']$ for a nearby line L', obtained by the first clause. Since $L \sim L'$ as divisors, these two computations must give the same class if the operation is to depend only on the rational-equivalence class of the cycle, and only then does the self-intersection number deserve its name. Establishing that agreement in general is the content of the section, and it is the property that will let the intersection class be defined on the Chow group rather than on cycles alone.

The stake is higher than notational convenience. Intersection with a divisor is meant to be an operator on $A_*(X)$, so that $D \cdot -$ carries $A_k(X)$ to $A_{k-1}(X)$ and can be iterated; without invariance under rational equivalence the operation lives only on cycles, where iteration is not controlled and no ring structure can form. The following proposition is the foundation of everything downstream, and it is one half of the identity (8.3) that the affine-bundle-invariance theorem of section 8.4 deferred to this chapter, the half asserting that $D \cdot -$ is well defined; the commutativity half is section 9.3.

Proposition 9.2.4: Rational Invariance of Divisor Intersection

Let D be a Cartier divisor on X. If $\alpha \in Z_k(X)$ is rationally equivalent to zero and no component of α lies in $|D|$, then $D \cdot \alpha \sim 0$ in $A_{k-1}(|D|)$. Consequently, for any class $a \in A_k(X)$ the intersection $D \cdot a \in A_{k-1}(|D|)$ is well defined by choosing a representative cycle none of whose components lies in $|D|$, and $D \cdot -$ descends to a homomorphism

$$D \cdot - \colon A_k(X) \to A_{k-1}(|D|)$$

independent of all choices.

The proof is the technical heart of the divisor calculus, and it is worth naming the obstacle before entering it. A cycle rationally equivalent to zero is a sum of divisors $[\mathrm{div}(r)]$ of rational functions r on $(k+1)$-dimensional subvarieties W. Intersecting with D replaces each such W-divisor by its trace under the section s_W of $\mathcal{O}_X(D)|_W$, and the claim is that this trace is again a boundary. On the surface W there are now two rational functions in play, the given r and the local equation of D, and the argument turns on the symmetry between them: the order of r measured along the zeros of D matches the order of the equation of D measured along the zeros of r, up to a principal divisor. That symmetry is a statement about a two-dimensional surface, and it is proved by pushing everything down to a complete curve, where a principal divisor has degree zero. Three inputs carry the argument: this degree-zero fact, proposition 8.2.2; the reciprocity lemma on the auxiliary curve that turns it into the two-function symmetry; and the localization sequence theorem 8.4.1 reducing the boundary strata to lower dimension. The degree-zero fact is why the argument could not be given in section 8.4 before the divisor machinery was in place.

***Proof*:**
By linearity it suffices to treat a single generator of $\mathrm{Rat}_k(X)$, so let $W \subseteq X$ be a $(k+1)$-dimensional subvariety and $r \in k(W)^*$, and set $\alpha = [\mathrm{div}(r)]$. The hypothesis that no component of α lies in $|D|$ is used only to make each $D \cdot [V]$ fall under the first clause of definition 9.2.1; the reduction below produces $D \cdot \alpha$ as the divisor of a rational function, and the containment cases are absorbed at the end.

Step 1: Reduction to the surface W. Replacing X by W and D by its restriction $D|_W$ changes neither side: the components V of $[\mathrm{div}(r)]$ are codimension-one subvarieties of W, the section s_V used to form $D \cdot [V]$ is the restriction to V of the section s_W of $\mathcal{O}_X(D)|_W$, and the inclusion $W \hookrightarrow X$ carries $A_{k-1}(|D|_W|) \to A_{k-1}(|D|)$ compatibly. It therefore suffices to prove the statement with $X = W$, so that D is a Cartier divisor on the $(k+1)$-dimensional variety W, $r \in k(W)^*$, and $D \cdot [\mathrm{div}(r)]$ is to be shown rationally equivalent to zero in $A_{k-1}(W)$. Restrict further to the dense open $W^\circ \subseteq W$ where D is principal, given by a single rational function $g \in k(W)^*$ with $\mathcal{O}_W(D)|_{W^\circ} = \mathcal{O}_{W^\circ} \cdot g^{-1}$; the codimension-one subvarieties of W meeting W°

are exactly those on which the computation is nontrivial, and those contained in the complement $W \setminus W^\circ$ contribute to both a principal divisor and the divisor of D and are handled by the localization sequence in Step 4. On W° the section s_V of clause (1) is $g|_V$, so

$$D \cdot [\operatorname{div}(r)] = \sum_V \operatorname{ord}_V(r) \left[\operatorname{div}_V(g|_V)\right]$$

the outer sum over the codimension-one subvarieties $V \subseteq W$ with r having a zero or pole along V, and the inner divisor formed on each such V.

Step 2: The two-function symmetry. The right-hand side is not visibly symmetric in r and g, but its symmetrization is the object that vanishes. Consider the two iterated sums

$$\Phi(r,g) = \sum_V \operatorname{ord}_V(r) \left[\operatorname{div}_V(g|_V)\right], \qquad \Phi(g,r) = \sum_V \operatorname{ord}_V(g) \left[\operatorname{div}_V(r|_V)\right]$$

both $(k-1)$-cycles on W, where in each the outer sum runs over codimension-one subvarieties $V \subseteq W$ and the inner divisor is formed on V of the restricted function. The claim of this step is the tame symmetry

$$\Phi(r,g) - \Phi(g,r) \sim 0 \quad \text{in } A_{k-1}(W)$$

Granting it, the proof is nearly complete: $\Phi(g,r) = \sum_V \operatorname{ord}_V(g) [\operatorname{div}_V(r|_V)]$ is a $\mathbb{Z}$-linear combination of divisors of the rational functions $r|_V$ on the k-dimensional subvarieties V, hence lies in $\operatorname{Rat}_{k-1}(W)$ by definition 8.1.3; combined with the symmetry, $\Phi(r,g) = D \cdot [\operatorname{div}(r)]$ is rationally equivalent to $\Phi(g,r) \sim 0$, which is the assertion.

Step 3a: Localizing the symmetry at a codimension-two locus. The symmetry is a statement whose content is concentrated in codimension two on W, so fix a codimension-two subvariety $Z \subseteq W$, a $(k-1)$-dimensional variety, and compare the coefficients of $[Z]$ on the two sides. A codimension-one $V \subseteq W$ contributes to the coefficient of $[Z]$ in $\Phi(r,g)$ only when $Z \subseteq V$, that is when Z is a prime divisor of V; write the coefficient of $[Z]$ in $\Phi(r,g)$ as

$$c_Z(r,g) = \sum_{Z \subseteq V} \operatorname{ord}_V(r) \operatorname{ord}_{Z,V}(g|_V)$$

the sum over codimension-one V with $Z \subseteq V \subseteq W$, and $\operatorname{ord}_{Z,V}$ the order along Z computed on V. The task is to show $c_Z(r,g) = c_Z(g,r)$ modulo the coefficient of $[Z]$ in a principal divisor, and this is a computation on the two-dimensional local geometry of W along Z. Passing to the local ring $\mathcal{O}_{W,Z}$, a two-dimensional local domain with fraction field $L = k(W)$ and residue field $K = k(Z)$, the codimension-one V containing Z correspond to the height-one primes $\mathfrak{p} \subseteq \mathcal{O}_{W,Z}$; each carries a discrete valuation $\operatorname{ord}_{\mathfrak{p}} = \operatorname{ord}_V$ on L, and the residue field $k(\mathfrak{p})$ of $\mathfrak{p}$ is the function field $k(V)$ of the divisor V. In these terms

$$c_Z(r,g) = \sum_{\mathfrak{p}} \operatorname{ord}_{\mathfrak{p}}(r) \operatorname{ord}_Z(g \bmod \mathfrak{p})$$

where $g \bmod \mathfrak{p}$ is the image of g in $k(\mathfrak{p}) = k(V)^*$ and ord_Z on $k(V)$ is the order along Z regarded as a prime divisor of V. The two factors play asymmetric roles: $\operatorname{ord}_{\mathfrak{p}}(r)$ is the transverse order of r along the entire divisor V, a single integer, while $\operatorname{ord}_Z(g \bmod \mathfrak{p})$ is the order of the residue $g|_V$ along the codimension-one subvariety $Z \subseteq V$. Swapping r and g interchanges these two roles rather than permuting equal factors, so $c_Z(r,g) = c_Z(g,r)$ is a genuine reciprocity between a transverse divisorial order and a residual order, not the commutativity of a product.

Step 3b: The exceptional curve and the tame symbol. The two integers in each term of $c_Z(r,g)$, the transverse order $\operatorname{ord}_{\mathfrak{p}}(r)$ and the residual order $\operatorname{ord}_Z(g \bmod \mathfrak{p})$, are the two ends of a single valuation datum, and completeness enters by resolving the two-dimensional geometry of $\mathcal{O}_{W,Z}$ into a curve. Choose a proper birational morphism $\pi\colon S \to \operatorname{Spec}\mathcal{O}_{W,Z}$ with S a regular two-dimensional scheme (a resolution of the localized surface, by normalization and blow-up), and let $E = \pi^{-1}(\mathfrak{m})$ be its exceptional fibre over the closed point, a complete curve over K whose integral components are $E_1, \dots, E_m$, each a complete curve over K with function field $k(E_i)$. The coefficient $c_Z(r,g)$ is invariant under π: since π is proper and birational, proposition 8.2.2 identifies the coefficient of $[Z]$ downstairs with the coefficient of the exceptional fibre upstairs, and among the prime divisors of S only the exceptional E_i map into Z, so the strict transforms of the V drop out and the coefficient is carried by $E_1, \dots, E_m$ alone. Along each exceptional component E_i the function r has a transverse order $a_i = \operatorname{ord}_{E_i}(r)$ and g a transverse order $b_i = \operatorname{ord}_{E_i}(g)$, both single integers, and once these orders are cleared r and g restrict to rational functions on E_i; their zeros and poles on E_i are exactly the points where the strict transforms $\widetilde{V}$ meet E_i. Attach to E_i the *tame symbol* of r and g, the residue element

$$\tau_i(r,g) = (-1)^{a_i b_i} \left(\frac{r^{\,b_i}}{g^{\,a_i}}\right)\Big|_{E_i} \;\in\; k(E_i)^*$$

which is a well-defined nonzero rational function on E_i because r^{b_i}/g^{a_i} has order zero along E_i. Reading off its order at a closed point $P \in E_i$ gives $\operatorname{ord}_P(\tau_i(r,g)) = b_i \operatorname{ord}_P(r|_{E_i}) - a_i \operatorname{ord}_P(g|_{E_i})$. The contribution of (i,P) to $c_Z(r,g)$ is the transverse order of r against the residual order of g, namely $a_i \operatorname{ord}_P(g|_{E_i})$, and its contribution to $c_Z(g,r)$ is $b_i \operatorname{ord}_P(r|_{E_i})$; their difference is thus $-\operatorname{ord}_P(\tau_i(r,g))$. Summing over S and using the π-invariance above, the two coefficients organize as

$$c_Z(r,g) - c_Z(g,r) = -\sum_{i=1}^{m} \sum_{P \in E_i} [k(P):K] \operatorname{ord}_P\big(\tau_i(r,g)\big)$$

the double sum over the components E_i of the exceptional curve and their closed points P. The pairing is asymmetric: the summand pairs the transverse order of one function along E_i against the residual order of the other at P, and swapping r and g replaces $\tau_i(r,g)$ by $\tau_i(r,g)^{-1}$ up to sign, so the difference is the sum of the point orders of the tame symbols, not a rearrangement of a symmetric product $\operatorname{ord}_P(r)\operatorname{ord}_P(g)$.

Step 3c: Assembling by degree zero. Each tame symbol $\tau_i(r,g)$ is a rational function on the complete curve E_i over K, so its divisor $\operatorname{div}_{E_i}(\tau_i(r,g))$ is a principal divisor on a complete curve. By proposition 8.2.2(1) applied to the proper structure map $E_i \to \operatorname{Spec} K$, its degree over K vanishes: $\deg_K \operatorname{div}_{E_i}(\tau_i(r,g)) = \sum_{P \in E_i} [k(P):K] \operatorname{ord}_P(\tau_i(r,g)) = 0$ for every i. Summing over the finitely many components,

$$c_Z(r,g) - c_Z(g,r) = -\sum_{i=1}^{m} \deg_K \operatorname{div}_{E_i}\big(\tau_i(r,g)\big) = 0$$

so $c_Z(r,g) = c_Z(g,r)$ for every codimension-two Z. This is the reciprocity that a naive term-by-term rearrangement would miss: it pairs the transverse order of one function against the residual order of the other, and its vanishing rests on completeness of each exceptional component E_i through the degree-zero property alone, with no external tame-symbol machinery imported. Consequently $\Phi(r,g) = \Phi(g,r)$ coefficient by coefficient on the locus where D is principal, any residual difference being a principal divisor supported off W°. This is the symmetry claimed in Step 2, as needed.

Step 4: The boundary components. It remains to account for the codimension-one subvarieties $V \subseteq W \setminus W^\circ$, where D was not represented by the single function g. Let $U = W^\circ$ and $Y = W \setminus W^\circ$ be its closed complement, of codimension at least one in W. The localization sequence (theorem 8.4.1) for $Y \subseteq W$,

$$A_{k-1}(Y) \to A_{k-1}(W) \to A_{k-1}(U) \to 0$$

is right exact, and Steps 1 through 3 established $D \cdot [\mathrm{div}(r)] \sim \Phi(g,r) \sim 0$ after restriction to U, that is, the class of $D \cdot [\mathrm{div}(r)]$ dies in $A_{k-1}(U)$. By exactness its class in $A_{k-1}(W)$ comes from $A_{k-1}(Y)$; but Y has dimension at most k, and the restriction $D|_Y$ is again a Cartier divisor, so the boundary class is computed by the same recipe on the lower-dimensional Y. To pin the inductive object down, decompose $Y = \bigcup_j Y_j$ into irreducible components, each of dimension at most k, with inclusions $\iota_j \colon Y_j \hookrightarrow W$. The class in $A_{k-1}(Y)$ produced by the boundary of the localization sequence is represented by $\sum_j (\iota_j)_*\big(D|_{Y_j} \cdot [\mathrm{div}(r_j)]\big)$, where $r_j := r|_{Y_j} \in k(Y_j)^*$ is the restriction of r to the generic point of Y_j, a nonzero rational function since the zero and pole components of r were split off in Steps 1 through 3, and $[\mathrm{div}(r_j)]$ is the principal cycle it cuts on the variety Y_j of dimension at most k. Each summand is thus $D|_{Y_j}$ intersected with a principal cycle on Y_j, the exact object to which the proposition applies one dimension lower. A Noetherian induction on $\dim W$ closes the argument: the base case $\dim W = 0$ is vacuous, and for the inductive step each summand $D|_{Y_j} \cdot [\mathrm{div}(r_j)]$ is rationally trivial on Y_j by the inductive hypothesis applied to the Cartier divisor $D|_{Y_j}$ and the principal cycle $[\mathrm{div}(r_j)]$ on the lower-dimensional Y_j; pushing forward along ι_j preserves rational equivalence, so the whole boundary class in $A_{k-1}(Y)$ vanishes. Hence $D \cdot [\mathrm{div}(r)] \sim 0$ in $A_{k-1}(W)$, and pushing forward along $W \hookrightarrow X$ gives $D \cdot [\mathrm{div}(r)] \sim 0$ in $A_{k-1}(|D|)$.

Step 5: Well-definedness of $D \cdot a$. Fix a class $a \in A_k(X)$. Every rational-equivalence class has a representative cycle none of whose finitely many components lies in $|D|$, since a component inside $|D|$ can be replaced by a rationally equivalent cycle whose components meet $|D|$ properly; on such a representative every $D \cdot [V]$ falls under the first clause of definition 9.2.1. Let $\alpha \sim \alpha'$ be two such representatives, so their difference lies in $\mathrm{Rat}_k(X)$, a finite $\mathbb{Z}$-combination of generators $[\mathrm{div}(r)]$ with $r \in k(W)^*$ on $(k+1)$-dimensional W. Steps 1 through 4 give $D \cdot [\mathrm{div}(r)] \sim 0$ in $A_{k-1}(|D|)$ for each generator, so summing gives $D \cdot (\alpha - \alpha') \sim 0$, and $D \cdot a := [D \cdot \alpha]$ is independent of the chosen representative. Additivity in a is immediate from additivity of $D \cdot -$ on cycles, so $D \cdot - \colon A_k(X) \to A_{k-1}(|D|)$ is a homomorphism, completing the proof.

The proposition is the license to speak of $D \cdot a$ for a class a, and with it the intersection with a divisor becomes an operator on the Chow group. Read backward, the argument explains why the well-definedness was out of reach in section 8.4: the entire content is the tame symmetry of Step 2, and that symmetry rests on the degree-zero property of a principal divisor on a complete curve, a fact about the divisor calculus that section did not yet have. The specialization map used there to invert affine-bundle pullback was built from exactly the operation $\mathrm{div}(t) \cdot -$ for the coordinate t, and its descent to Chow groups is this proposition applied to $D = \{t = 0\}$; the forward reference in that proof pointed here.

Two features of the argument recur and are worth isolating. The reduction of Step 1, replacing X by the $(k+1)$-dimensional W on which r lives, is the standard move that turns a statement about intersecting a principal cycle into a statement about a single surface with two functions on it, and it will reappear when commutativity is proved in section 9.3. The symmetry of Step 3, that pairing the zeros of r against g equals pairing the zeros of g against r, is the algebraic form of Weil reciprocity, here proved from the ground up rather than cited: the tame symbol is constructed by hand along each component of the exceptional curve, and its reciprocity is reduced to the vanishing degree of a principal divisor on that complete curve, so no external tame-symbol machinery is imported. The same reciprocity, promoted one dimension higher, is what makes $D \cdot D' = D' \cdot D$, and the two proofs share their engine.

The operator now available, $D \cdot - \colon A_k(X) \to A_{k-1}(|D|)$, invites the computation that motivated the whole construction. Intersecting repeatedly with a hyperplane on projective space should descend through the dimensions and terminate in a point, recovering the degree; the next example carries this out and previews the Chern-class computation of section 9.4.

Example 9.2.5: Iterated Hyperplane Sections on Projective Space

Let $X = \mathbb{P}^n$ and $H = \{x_0 = 0\}$ a hyperplane, so $\mathcal{O}_{\mathbb{P}^n}(H) = \mathcal{O}_{\mathbb{P}^n}(1)$ and $|H| \cong \mathbb{P}^{n-1}$. By theorem 8.5.2, $A_k(\mathbb{P}^n) = \mathbb{Z}\, h_k$ with h_k the class of a linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$. To compute $H \cdot h_k$, represent h_k by a linear $\mathbb{P}^k$ meeting H transversally, that is by an $L \cong \mathbb{P}^k$ not contained in H; then $L \cap H$ is a linear $\mathbb{P}^{k-1}$, and the restricted section x_0/x_j on L has its zero divisor equal to $L \cap H$ with multiplicity one. Hence

$$H \cdot h_k = [L \cap H] = h_{k-1} \in A_{k-1}(\mathbb{P}^{n-1})$$

pushed into $\mathbb{P}^n$, the class of a linear $\mathbb{P}^{k-1}$. Intersecting n times starting from the fundamental class $h_n = [\mathbb{P}^n]$,

$$\underbrace{H \cdot H \cdots H}_{k} \cdot [\mathbb{P}^n] = h_{n-k}$$

and in particular $H^n \cdot [\mathbb{P}^n] = h_0 = [\mathrm{pt}]$, a single reduced point of degree one. The iterated intersection walks down the tower of linear subspaces one step at a time, and its terminal value is the class of a point, whose degree 1 is the degree of $\mathbb{P}^n$ under the embedding $\mathcal{O}(1)$. The choice of a transversal representative L at each stage is legitimated precisely by proposition 9.2.4: the answer h_{k-1} does not depend on which $\mathbb{P}^k$ in the class h_k is used, so long as it meets H properly.

The computation is the prototype of every enumerative count the theory produces. Reinterpreted through the first Chern class in section 9.4, it reads $c_1(\mathcal{O}(1))^k \cap [\mathbb{P}^n] = h_{n-k}$, and its terminal instance $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\mathrm{pt}]$ is the statement that the top self- intersection of the hyperplane class on $\mathbb{P}^n$ is one, the algebraic form of the assertion that n general hyperplanes meet in a single point. That the intermediate steps are independent of the representatives chosen is the payoff of this section, and it is what turns a naive count of intersection points into an invariant.

Exercise 9.2.1

1. On $X = \mathbb{P}^2$ let $D = \{x_0 x_1 = 0\}$ be the union of two lines, a Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(D) = \mathcal{O}_{\mathbb{P}^2}(2)$, and let V be the conic $\{x_2^2 = x_0 x_1\}$. Compute $D \cdot [V] \in A_0(V \cap |D|)$ explicitly as a sum of points with multiplicities, and verify that its degree equals 4. Explain how the multiplicities distribute among the intersection points.

2. Let $V \subseteq \mathbb{P}^2$ be a smooth curve of degree d and H a line meeting V transversally in d distinct points. Using definition 9.2.1, show that $H \cdot [V]$ is the sum of those d points each with multiplicity one, and deduce $\deg(H \cdot [V]) = d$. Then explain why the value $\deg(H \cdot [V]) = d$ holds for every line H, transversal or not, without recomputing, citing proposition 9.2.4.

3. Let $D = \mathrm{div}(f)$ be a principal Cartier divisor on a variety X, so $\mathcal{O}_X(D) \cong \mathcal{O}_X$ is trivial. Show that $D \cdot a = 0$ in $A_{k-1}(X)$ for every class $a \in A_k(X)$. (Push $D \cdot a$ forward from $|D|$ into X;

the point is that a global trivialization of $\mathcal{O}_X(D)$ makes the restricted section extend to all of V.)

4. On $\mathbb{P}^3$ let H be a hyperplane and let V be a smooth cubic surface, regarded through its inclusion as a subvariety, with the induced very ample bundle $\mathcal{O}_V(1) = \mathcal{O}_{\mathbb{P}^3}(1)|_V$. Compute $H \cdot [V]$ as a class in $A_1(V \cap H)$, identify $V \cap H$ geometrically, and compute the degree of the further intersection $H \cdot (H \cdot [V])$.

5. Let $Q = \mathbb{P}^1 \times \mathbb{P}^1$ and let $D = f_1 = \{a\} \times \mathbb{P}^1$ be a ruling of the first kind, with $\mathcal{O}_Q(D) = \mathrm{pr}_1^*\mathcal{O}_{\mathbb{P}^1}(1)$. Compute $D \cdot [f_1]$ and $D \cdot [f_2]$, where $f_2 = \mathbb{P}^1 \times \{b\}$ is a ruling of the second kind, and interpret the results as the self-intersection f_1^2 and the cross-intersection $f_1 \cdot f_2$. Which case uses the second clause of definition 9.2.1?

6. Give an explicit example on $X = \mathbb{P}^1$ of a Cartier divisor D and two rationally equivalent zero-cycles $\alpha \sim \alpha'$ for which the naive formula would assign different answers if one ignored the requirement that components avoid $|D|$, and explain how definition 9.2.1 and proposition 9.2.4 together resolve the apparent discrepancy. (Here $k = 0$, so $D \cdot \alpha$ lands in $A_{-1} = 0$; use this to frame what the proposition is and is not asserting in the lowest dimension.)

9.3 Commutativity and the Projection Formula

Intersection with a divisor, made precise in section 9.2, is an operation $D \cdot -$ that lowers dimension by one and, by proposition 9.2.4, descends to rational equivalence. Nothing so far says these operations may be applied in either order. Given two Cartier divisors D and D' on X and a k-cycle α, one can form $D \cdot (D' \cdot \alpha)$ and $D' \cdot (D \cdot \alpha)$, each a class in $A_{k-2}(X)$; that they agree is the single algebraic fact on which the entire intersection calculus rests. Its proof does not follow from the definitions by formal manipulation, because the two orders of intersection produce different cycles before passing to rational equivalence. The content is that the difference is always a boundary, and the reason it is a boundary is the vanishing of the degree of a principal divisor on a complete curve, established back in proposition 8.2.2.

This is also the point at which a debt from chapter 8 comes due. The proof of affine-bundle invariance (theorem 8.4.6) deferred one identity, the divisor symmetry (8.3), on the grounds that the tools were not yet in place; that identity is exactly the commutativity proved here. This section discharges it in full and then draws the two functorial consequences, compatibility of $D \cdot -$ with proper pushforward and with flat pullback, that make the operation part of the same covariant/contravariant package as pushforward and pullback themselves.

The obstruction to commutativity lives on a surface, and the mechanism that removes it is a symmetry between two rational functions. Before the theorem, the mechanism deserves to be isolated as a lemma. Let S be a complete variety of dimension two and let $f, g \in k(S)^*$ be nonzero rational functions. Each has a divisor $[\mathrm{div}(f)] = \sum_W \mathrm{ord}_W(f)\,[W]$, the sum over the codimension-one subvarieties (the curves) $W \subseteq S$, and on each such curve W the other function g restricts, wherever it is neither zero nor a pole identically along W, to a rational function $g|_W$ with its own divisor of points. The two-function symmetry lemma compares the two zero-cycles obtained by cutting first with f and then with g, against the same two cuts in the opposite order.

The statement needs the order of one function measured against the divisor of another. For a curve $W \subseteq S$ with $\mathrm{ord}_W(g) = 0$, so that g is a unit at the generic point of W and restricts to a nonzero rational function $g|_W \in k(W)^*$, write $[\mathrm{div}_W(g)] = \sum_{P\in W} \mathrm{ord}_P(g|_W)\,[P]$ for the divisor of that restriction, a zero-cycle on W and hence on S. The lemma is a reciprocity between the iterated cuts $\sum_W \mathrm{ord}_W(f)\,[\mathrm{div}_W(g)]$ and $\sum_W \mathrm{ord}_W(g)\,[\mathrm{div}_W(f)]$.

Lemma 9.3.1: Two-Function Reciprocity on a Surface

Let S be a complete variety of dimension two and let $f, g \in k(S)^*$. Then

$$\sum_W \mathrm{ord}_W(f)\,[\mathrm{div}_W(g)] \;\sim\; \sum_W \mathrm{ord}_W(g)\,[\mathrm{div}_W(f)] \quad \text{in } A_0(S)$$

where each sum runs over the curves $W \subseteq S$ on which the relevant restriction is defined, and the divisors $[\mathrm{div}_W(g)]$, $[\mathrm{div}_W(f)]$ are the zero-cycles of the restricted functions. In fact both sides have the same degree, and their difference is a sum of principal divisors of rational functions on the curves $W \subseteq S$, hence rationally trivial.

The lemma is the two-dimensional heart of commutativity, and its proof is the one place where the degree-zero property of principal divisors on complete curves does the decisive work. The strategy is to reduce the equality of two zero-cycles to an equality of degrees along each curve, and then to recognize each such degree as the value at a point of the *tame symbol*, a bimultiplicative pairing on pairs of rational functions whose reciprocity across a complete curve is precisely the statement that a principal divisor has degree zero.

***Proof*:**
Step 1: the tame symbol at a curve. Fix a curve $W \subseteq S$ and let $A = \mathcal{O}_{S,W}$ be the local ring of S at the generic point of W. Because S is a variety and W has codimension one, A is a one-dimensional local domain with fraction field $k(S)$ and residue field $k(W)$; its normalization is a discrete valuation ring, and after replacing S by its normalization along W (which changes neither side of the asserted identity, both being computed by orders and residues that are birational invariants) one may assume A is itself a discrete valuation ring with valuation $v_W = \mathrm{ord}_W$ and uniformizer π. Define the *tame symbol* of f, g at W to be the residue class

$$\partial_W(f,g) \;=\; (-1)^{v_W(f)\,v_W(g)}\,\frac{f^{\,v_W(g)}}{g^{\,v_W(f)}}\Big|_W \;\in\; k(W)^*$$

the restriction to W of the indicated unit, which lies in A and is a unit there because its valuation

$$v_W\Big((-1)^{v_W(f)v_W(g)}\,f^{v_W(g)}g^{-v_W(f)}\Big) = v_W(g)\,v_W(f) - v_W(f)\,v_W(g) = 0$$

vanishes. The construction is bimultiplicative in f and g and antisymmetric, $\partial_W(g,f) = \partial_W(f,g)^{-1}$, both directly from the formula. The point of the sign is that this antisymmetry holds on the nose, not just up to squares.

Step 2: the tame symbol computes the difference of iterated divisors, pointwise. Fix a closed point $P \in S$ and a curve W through P. When $\mathrm{ord}_W(g) = 0$ the restriction $g|_W$ is a nonzero function on W and the coefficient of $[P]$ in $\mathrm{ord}_W(f)\,[\mathrm{div}_W(g)]$ is $\mathrm{ord}_W(f)\,\mathrm{ord}_P(g|_W)$; symmetrically for the other cut. Comparing the two iterated cuts at P amounts to comparing, over all curves $W \ni P$,

$$\sum_{W\ni P} \mathrm{ord}_W(f)\,\mathrm{ord}_P(g|_W) \quad \text{against} \quad \sum_{W\ni P} \mathrm{ord}_W(g)\,\mathrm{ord}_P(f|_W)$$

The tame symbol reconciles them, but the reconciliation is delicate at the curves where both functions have nonzero order, and getting it right is the whole content of the step. For a single curve W with $\operatorname{ord}_W(f) = a$ and $\operatorname{ord}_W(g) = b$, write $f = \pi^a u$ and $g = \pi^b w$ with u, w units at W. The tame symbol of *Step 1* then simplifies to

$$\partial_W(f, g) = (-1)^{ab}\, \bar{u}^{\,b}\, \bar{w}^{\,-a} \;\in\; k(W)^*$$

the powers of π cancelling, so its order at P is always

$$\operatorname{ord}_P\big(\partial_W(f, g)\big) = b \operatorname{ord}_P(\bar{u}) - a \operatorname{ord}_P(\bar{w}) \tag{9.1}$$

regardless of whether a or b vanishes. The iterated cuts, on the other hand, see only the restricted functions, and $f|_W = \bar{u}$ is defined precisely when $a = 0$ while $g|_W = \bar{w}$ is defined precisely when $b = 0$. Compare the two curve-by-curve.

If $a = 0$ and $b \neq 0$: then $\operatorname{ord}_W(g)\,[\operatorname{div}_W(f)]$ contributes $b \operatorname{ord}_P(\bar{u})$ at P while $\operatorname{ord}_W(f)\,[\operatorname{div}_W(g)] = 0$, so the iterated-cut difference is $b \operatorname{ord}_P(\bar{u})$, which is exactly (9.1) since $a = 0$ kills the second term. *If $a \neq 0$ and $b = 0$:* symmetrically the difference is $-a \operatorname{ord}_P(\bar{w})$, again (9.1). *If $a = b = 0$:* both restrictions are units, both sides vanish. In all three of these *non-mixed* cases,

$$\Big(\operatorname{ord}_W(g) \operatorname{ord}_P(f|_W) - \operatorname{ord}_W(f) \operatorname{ord}_P(g|_W)\Big) = \operatorname{ord}_P\big(\partial_W(f, g)\big)$$

the left side being the contribution of W to the iterated-cut difference at P, where a term whose restricted function is undefined is read as zero because its coefficient $\operatorname{ord}_W(g)$ or $\operatorname{ord}_W(f)$ then vanishes. The remaining case is the *mixed* one, $a \neq 0$ and $b \neq 0$: here *neither* $f|_W$ nor $g|_W$ is defined, so W contributes nothing to either iterated cut, yet $\partial_W(f, g) = (-1)^{ab}\bar{u}^{\,b}\bar{w}^{\,-a}$ is a genuine nonzero rational function on W whose divisor is generally nonzero. Thus on a mixed curve the iterated-cut difference and $\operatorname{ord}_P(\partial_W)$ do *not* agree; the tame symbol overcounts by exactly the mixed contribution. Summing the non-mixed agreement over the curves through P,

$$\Big(\sum_W \operatorname{ord}_W(g)\,[\operatorname{div}_W(f)] - \sum_W \operatorname{ord}_W(f)\,[\operatorname{div}_W(g)]\Big)_P = \sum_{\substack{W \ni P \\ \text{non-mixed}}} \operatorname{ord}_P\big(\partial_W(f, g)\big) \tag{9.2}$$

the sum on the right running only over curves $W \ni P$ on which at least one of f, g has order zero, and the left side denoting the coefficient of $[P]$ in the difference of the two iterated cuts. The mixed curves are deliberately excluded from the right side; they carry no iterated-cut contribution, and Step 3 will show their tame symbols are individually harmless.

Step 3: reciprocity along each curve kills the total. Fix a curve $W \subseteq S$. The tame symbol $\partial_W(f, g)$ is a nonzero rational function on the complete curve W (it is complete because S is complete and W is closed in S). Its divisor $[\operatorname{div}(\partial_W(f, g))] = \sum_P \operatorname{ord}_P(\partial_W(f, g))\,[P]$ is a principal divisor on W, so by proposition 8.2.2(1) applied to the structure map $W \to \operatorname{Spec} \overline{k}$, which is proper surjective with target of strictly smaller dimension, its pushforward vanishes: $\deg[\operatorname{div}(\partial_W(f, g))] = 0$. This holds for *every* curve W, mixed or not. Summing (9.2) over all closed points P and reorganizing the double sum by curves,

$$\deg\Big(\sum_W \operatorname{ord}_W(g)\,[\operatorname{div}_W(f)] - \sum_W \operatorname{ord}_W(f)\,[\operatorname{div}_W(g)]\Big) = \sum_{\text{non-mixed } W} \deg[\operatorname{div}(\partial_W(f, g))] = 0$$

each summand vanishing individually. Thus the difference of the two iterated zero-cycles has degree zero on S.

Step 4: from degree zero to rational equivalence. A zero-cycle of degree zero on the complete variety S need not be rationally equivalent to zero in general, but the difference here is more than a degree-zero cycle: (9.2), summed over all closed points P and reorganized by curves, exhibits it as a sum of principal divisors,

$$\sum_{W} \operatorname{ord}_W(g)\,[\operatorname{div}_W(f)] - \sum_{W} \operatorname{ord}_W(f)\,[\operatorname{div}_W(g)] = \sum_{\text{non-mixed } W} [\operatorname{div}(\partial_W(f,g))]$$

the right side running over the curves on which one of f, g has order zero, each term the divisor of the rational function $\partial_W(f,g)$ on the complete curve $W \subseteq S$. Each $[\operatorname{div}(\partial_W(f,g))]$ is by definition rationally equivalent to zero on W, hence on S under the proper pushforward along $W \hookrightarrow S$, which preserves rational equivalence by theorem 8.2.3. Therefore the difference is a finite sum of cycles each ~ 0, so it is itself ~ 0 in $A_0(S)$, which is the asserted reciprocity, as we sought to show.

The lemma is the algebraic incarnation of a classical reciprocity, but the mechanism it uses is elementary: on each curve W the tame symbol $\partial_W(f,g)$ is a single rational function, its divisor on the complete curve W is principal and so rationally trivial, and the difference of the two iterated cuts is a finite sum of such divisors. No appeal to Weil reciprocity or to the point-level tame symbol $\prod_P \partial_P(f,g) = 1$ as an external input is needed, and indeed none is available in this book; the entire force of the lemma comes from the single fact that a rational function on a complete curve has as many zeros as poles, counted with multiplicity and residue degree, applied one curve at a time. That fact is proposition 8.2.2(1), proved in chapter 8 by the model computation on $\mathbb{P}^1$. The reader should notice that the surface S never needed to be smooth; normalizing along each curve reduced every local computation to a discrete valuation ring, which is all the tame symbol requires.

With the reciprocity lemma in hand, commutativity is a matter of reducing the general assertion to the surface case the lemma governs. The reduction is entirely formal, following the pattern used throughout chapter 8: the operation $D \cdot -$ is a homomorphism, so it suffices to check the identity on the class of a single subvariety, and the intersection with D and D' is supported where the two divisors meet the subvariety, so one may replace the ambient scheme by the subvariety itself.

Theorem 9.3.2: Commutativity of Divisor Intersection

Let D and D' be Cartier divisors on an algebraic scheme X, and let $\alpha \in A_k(X)$. Then

$$D \cdot (D' \cdot \alpha) = D' \cdot (D \cdot \alpha) \quad \text{in } A_{k-2}(X)$$

More precisely, both sides are represented by well-defined classes in $A_{k-2}\big(|D| \cap |D'| \cap \operatorname{supp}\alpha\big)$, and they coincide there. This identity is the divisor symmetry (8.3) deferred in the proof of theorem 8.4.6.

***Proof*:**
Step 1: reduction to a variety. Both $D \cdot -$ and $D' \cdot -$ are homomorphisms $A_k \to A_{k-1}$ by definition 9.2.1 and proposition 9.2.4, so both iterated operations are homomorphisms $A_k(X) \to A_{k-2}(X)$. It therefore suffices to verify the identity on a generator $\alpha = [V]$, the class of a k-dimensional subvariety $V \subseteq X$. Restricting the line bundles $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ and their rational sections to V, the entire computation of $D \cdot (D' \cdot [V])$ and $D' \cdot (D \cdot [V])$ takes place

on V using the restricted divisors $D|_V$ and $D'|_V$. Replacing X by V, one is reduced to proving

$$D \cdot (D' \cdot [X]) = D' \cdot (D \cdot [X])$$

for X itself a k-dimensional variety and D, D' Cartier divisors on it, with the common value a class in $A_{k-2}(|D| \cap |D'|)$.

Step 2: choosing sections and unwinding the two orders. Let s and s' be the rational sections of $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ cutting out D and D', so that on the variety X,

$$D \cdot [X] = [\mathrm{div}(s)] = \sum_W \mathrm{ord}_W(s)\,[W], \qquad D' \cdot [X] = [\mathrm{div}(s')] = \sum_{W'} \mathrm{ord}_{W'}(s')\,[W']$$

the sums over the codimension-one subvarieties $W, W' \subseteq X$. To compute the double intersection one then intersects each $[W']$ with D and each $[W]$ with D'. On a codimension-one subvariety W not contained in $|D'|$, the section s' restricts to a rational section $s'|_W$ of $\mathcal{O}_X(D')|_W$, whose divisor is $[\mathrm{div}_W(s')]$, and by definition 9.2.1,

$$D' \cdot (D \cdot [X]) = \sum_W \mathrm{ord}_W(s)\,[\mathrm{div}_W(s')], \qquad D \cdot (D' \cdot [X]) = \sum_{W'} \mathrm{ord}_{W'}(s')\,[\mathrm{div}_{W'}(s)]$$

as cycles supported on $|D| \cap |D'|$. The two expressions are the two iterated cuts of the reciprocity lemma, with the roles of the two divisors interchanged.

Step 3: the difference as a sum of tame symbols. The difference

$$\Delta \;=\; \sum_{W'} \mathrm{ord}_{W'}(s')\,[\mathrm{div}_{W'}(s)] \;-\; \sum_W \mathrm{ord}_W(s)\,[\mathrm{div}_W(s')]$$

is a $(k-2)$-cycle supported on $|D| \cap |D'|$, and the claim is $\Delta \sim 0$ in $A_{k-2}(|D| \cap |D'|)$. The strategy is to write Δ as a $\mathbb{Z}$-combination of divisors of rational functions on $(k-1)$-dimensional subvarieties of X, using the tame symbol of lemma 9.3.1. For a $(k-1)$-dimensional subvariety $W \subseteq X$ on which one of the two sections has nonzero order, choose a common local trivialization of $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ near the generic point of W (a line bundle is locally trivial, and one open set trivializes both), turning s, s' into rational functions in the local ring $\mathcal{O}_{X,W}$, a two-dimensional local domain. The tame symbol

$$\partial_W(s, s') = (-1)^{\mathrm{ord}_W(s)\mathrm{ord}_W(s')} \left.\frac{s^{\mathrm{ord}_W(s')}}{s'^{\,\mathrm{ord}_W(s)}}\right|_W \;\in\; k(W)^*$$

is then a nonzero rational function on the $(k-1)$-dimensional variety W, defined exactly as in the lemma. Localizing X along W removes the trivialization ambiguity: two trivializations differ by a unit, which does not change $\partial_W(s, s')$ up to a rational function that is regular and nonvanishing at the generic point of W, so its divisor on W is well defined.

Step 4: applying reciprocity, one codimension-two point at a time. Fix a $(k-2)$-dimensional subvariety $Z \subseteq |D| \cap |D'|$ and compute the coefficient of $[Z]$ on both sides of the asserted identity $\Delta = \sum_{\text{non-mixed } W}[\mathrm{div}(\partial_W(s, s'))]$, the sum running over $(k-1)$-dimensional W on which one of s, s' has order zero. That coefficient is determined at the generic point ζ of Z, where $\mathcal{O}_{X,\zeta}$ has dimension two; the two-dimensional local situation there is governed by lemma 9.3.1, whose (9.2) reads, at ζ,

$$\Big(\sum_W \mathrm{ord}_W(s')\,[\mathrm{div}_W(s)] - \sum_W \mathrm{ord}_W(s)\,[\mathrm{div}_W(s')]\Big)_\zeta \;=\; \sum_{\substack{W \ni \zeta \\ \text{non-mixed}}} \mathrm{ord}_\zeta\big(\partial_W(s, s')\big)$$

the right sum over the $(k-1)$-dimensional subvarieties W through ζ on which one of s, s' has order zero, exactly as in the lemma; the mixed W, on which both sections have nonzero order, are excluded because they contribute to neither iterated cut. This is the coefficient of $[Z]$ in Δ set equal to the coefficient of $[Z]$ in $\sum_{\text{non-mixed } W}[\operatorname{div}(\partial_W(s,s'))]$. Since the two sides agree at every such Z, they agree as cycles:

$$\Delta = \sum_{\text{non-mixed } W} [\operatorname{div}(\partial_W(s,s'))]$$

Each term $[\operatorname{div}(\partial_W(s,s'))]$ is the divisor of a rational function on the $(k-1)$-dimensional subvariety W, hence rationally equivalent to zero on W, and its proper pushforward to $|D| \cap |D'|$ along $W \hookrightarrow |D| \cap |D'|$ preserves that equivalence by theorem 8.2.3. As a finite sum of cycles each ~ 0, therefore $\Delta \sim 0$, so

$$D \cdot (D' \cdot [X]) = D' \cdot (D \cdot [X]) \quad \text{in } A_{k-2}(|D| \cap |D'|)$$

Pushing forward to X along the inclusion, the identity holds in $A_{k-2}(X)$ for $\alpha = [X]$, and by Step 1 for every $\alpha \in A_k(X)$. The surface case $X = S$, in which the difference of the two orders is exhibited as a sum of principal cycles, is precisely the deferred divisor symmetry (8.3), with the two rational functions $s|_S$ and $s'|_S$ here playing the roles of g and the coordinate t there, completing the proof.

Commutativity is the property that makes the phrase "intersection number" unambiguous. Without it, the number attached to a chain of divisors would depend on the order in which they are imposed, and no well-defined product on cycles could exist. With it, one may write $D_1 \cdot D_2 \cdots D_r \cdot \alpha$ for an iterated intersection and speak of it without specifying an order, and the class of a complete intersection $D_1 \cap \cdots \cap D_r$ in A_* is symmetric in the defining divisors. The proof also settles the debt from chapter 8: the identity (8.3) that theorem 8.4.6 took on faith is the surface case here, so affine-bundle invariance is now unconditional. The tame symbol was the only tool required beyond the definitions, and its reciprocity was nothing more than the degree-zero property of principal divisors, propagated from curves to surfaces.

A first reading of the theorem should note what it does *not* require. The divisors need not be effective, the sections need not be regular, and X need not be smooth or even normal; the entire argument localizes to one-dimensional local rings, where a divisor is controlled by a single valuation. This is why intersection with a Cartier divisor is available on every algebraic scheme, in contrast to the intersection product of arbitrary cycles, which will demand the deformation-to-the-normal-cone machinery of a later chapter and works cleanly only on smooth varieties.

The second half of the section records how $D \cdot -$ interacts with the two functorialities of the Chow group. Both statements are what one would guess from the topological analogy, in which cap product with a cohomology class is natural for maps: pushforward absorbs the pullback of a divisor, and flat pullback commutes with intersection outright. The proofs run through the same reduction to a single subvariety and the same input, the pushforward of a principal divisor, that powered commutativity.

For a proper morphism $f\colon X \to Y$ and a Cartier divisor D on Y, the pullback f^*D is defined whenever no component of X maps into $|D|$, by pulling back the line bundle $\mathcal{O}_Y(D)$ and its rational section; it is the Cartier divisor $\operatorname{div}(f^\# s)$ of the pulled-back section. The projection formula compares intersecting upstairs and pushing down against pushing down and intersecting downstairs.

Proposition 9.3.3: Projection Formula for Divisors

Let D be a Cartier divisor on Y.

1. *Proper pushforward.* If $f\colon X \to Y$ is proper and $\alpha \in A_k(X)$ is a cycle no component of whose support maps into $|D|$, so that $f^*D \cdot \alpha$ is defined, then
$$f_*\big(f^*D \cdot \alpha\big) = D \cdot f_*\alpha \quad \text{in } A_{k-1}(Y)$$
and, more precisely, both sides are classes in $A_{k-1}\big(|D| \cap f(\mathrm{supp}\alpha)\big)$ and agree there, f being proper so that it carries $f^{-1}(|D|) \cap \mathrm{supp}\alpha$ onto $|D| \cap f(\mathrm{supp}\alpha)$.

2. *Flat pullback.* If $g\colon X' \to Y$ is flat of relative dimension n and $\beta \in A_k(Y)$, then
$$g^*\big(D \cdot \beta\big) = g^*D \cdot g^*\beta \quad \text{in } A_{k+n-1}(X')$$
where g^*D is the pullback Cartier divisor $\mathrm{div}(g^\# s)$; more precisely both sides are classes in $A_{k+n-1}\big(g^{-1}(|D| \cap \mathrm{supp}\beta)\big)$ and agree there.

***Proof*:**
Both statements are linear in α and β, so it suffices to verify each on the class of a subvariety, and both reduce, by the argument that opened the commutativity proof, to a computation with the divisor of a rational function that is settled by proposition 8.2.2 and proposition 8.3.6. The refined statements come for free: definition 9.2.1 records $D \cdot [V]$ on $V \cap |D|$ to begin with, and each identity below is an equality of cycles supported there, so no step passes to the ambient group. This refinement is what chapter 11 consumes.

1. *Proper pushforward.* Take $\alpha = [V]$ for a k-dimensional subvariety $V \subseteq X$, and set $W = f(V) \subseteq Y$, so $f_*[V] = \deg(V/W)\,[W]$ if $\dim W = k$ and $f_*[V] = 0$ if $\dim W < k$, by definition 8.2.1. Restricting to V and W, one may assume $X = V$, $Y = W$, and $f\colon V \to W$ proper surjective with V a variety. Let s be the rational section of $\mathcal{O}_Y(D)$ cutting out D; its pullback $f^\# s$ is a rational section of $f^*\mathcal{O}_Y(D)$ cutting out f^*D, and $f^*D \cdot [V] = [\mathrm{div}(f^\# s)]$ while $D \cdot [W] = [\mathrm{div}(s)]$, both by definition 9.2.1. There are two cases.

   If $\dim W < k$, then $f_*[V] = 0$, so the right side $D \cdot f_*[V]$ vanishes. On the left, $[\mathrm{div}(f^\# s)]$ is a $(k-1)$-cycle on V, and f pushes it into Y; but $f(V) = W$ has dimension $< k$, so $f_*[\mathrm{div}(f^\# s)]$ is a cycle of dimension $k-1$ supported on the $(<k)$-dimensional W, and proposition 8.2.2(1) applied to $f\colon V \to W$ (proper surjective, target of strictly smaller dimension) gives $f_*[\mathrm{div}(f^\# s)] = 0$. Both sides vanish.

   If $\dim W = k$, then $f\colon V \to W$ is proper surjective of equal dimension, generically finite of degree $d = \deg(V/W) = [k(V) : k(W)]$. Now $f^\# s$ is a rational section whose local expression, in a trivialization of $\mathcal{O}_Y(D)$ near a curve of W, is a rational function $r \in k(V)^*$ pulled back from $k(W)$, and proposition 8.2.2(2) gives $f_*[\mathrm{div}(f^\# s)] = [\mathrm{div}(N(f^\# s))]$ with N the norm along $k(V)/k(W)$. Because $f^\# s$ is pulled back from s, its norm is $N(f^\# s) = s^d$ up to the local trivialization, so $[\mathrm{div}(N(f^\# s))] = d\,[\mathrm{div}(s)] = d\,(D \cdot [W])$. Hence
$$f_*\big(f^*D \cdot [V]\big) = f_*[\mathrm{div}(f^\# s)] = d\,(D \cdot [W]) = D \cdot (d\,[W]) = D \cdot f_*[V]$$

as required.

2. *Flat pullback.* Take $\beta = [V]$ for a k-dimensional subvariety $V \subseteq Y$. Both sides are supported over $V \cap |D|$, so restricting to V one may assume $Y = V$ is a variety and D a Cartier divisor on it with cutting section s, so $D \cdot [V] = [\mathrm{div}(s)]$. Then $g^*(D \cdot [V]) = g^*[\mathrm{div}(s)]$, and by the flat-pullback divisor formula established in proposition 8.3.7, flat pullback commutes with taking the divisor of a rational function: $g^*[\mathrm{div}(s)] = [\mathrm{div}(g^\# s)]$ componentwise, with the multiplicities of the components of $g^{-1}(V)$ accounted by flatness. On the other side, g^*D is the Cartier divisor $\mathrm{div}(g^\# s)$ on X' by definition, and $g^*[V] = [g^{-1}(V)] = \sum_j m_j [X'_j]$ over the components X'_j of the preimage with their flat multiplicities m_j by proposition 8.3.5. Intersecting,

$$g^*D \cdot g^*[V] = g^*D \cdot \sum_j m_j [X'_j] = \sum_j m_j \,[\mathrm{div}((g^\# s)|_{X'_j})] = [\mathrm{div}(g^\# s)] = g^*(D \cdot [V])$$

the middle equality being definition 9.2.1 applied on each component and the last the same componentwise divisor formula. This is the claim; the compatibility of these classes with the base change is proposition 8.3.6, which guarantees the two routes land in the same class of $A_{k+n-1}(X')$. This establishes both parts, completing the proof.

The projection formula is the mechanism by which a computation on a resolved or covering space transfers to the base. In its pushforward form, intersecting the pullback of D upstairs and pushing the answer down is the same as pushing the cycle down and intersecting with D there, which is exactly what one does when computing an intersection number on Y by passing to a modification $X \to Y$ where the geometry is transparent. The scalar $d = \deg(V/W)$ in the equal-dimensional case is the reason degrees multiply under finite covers, and it is the algebraic source of formulas like $\deg(f^*D) = (\deg f)(\deg D)$ for a finite map of curves. In its flat form, the formula says intersection with a divisor is a local construction, insensitive to flat base change: pulling back the divisor and pulling back the cycle and then intersecting reproduces the pullback of the intersection. Together the two halves make $D \cdot -$ compatible with the covariant/contravariant package (f_*, g^*) of chapter 8, the bimodule structure that all of intersection theory is built to respect.

The abstractions above are best seen at work on the projective plane, where every class is a known multiple of a linear space and the intersection numbers can be read off by hand. The following example verifies commutativity concretely and recovers the intersection number of two lines.

Example 9.3.4: Two Lines on the Projective Plane

Let $X = \mathbb{P}^2$ over $\overline{k}$, with homogeneous coordinates $[x_0 : x_1 : x_2]$. By theorem 8.5.2, $A_2(\mathbb{P}^2) = \mathbb{Z}\,[\mathbb{P}^2]$, $A_1(\mathbb{P}^2) = \mathbb{Z}\,[L]$ generated by the class of a line L, and $A_0(\mathbb{P}^2) = \mathbb{Z}\,[P]$ generated by the class of a point. Take the two Cartier divisors $D = \{x_0 = 0\}$ and $D' = \{x_1 = 0\}$, each a line, with $\mathcal{O}_{\mathbb{P}^2}(D) \cong \mathcal{O}_{\mathbb{P}^2}(D') \cong \mathcal{O}(1)$ and cutting sections $s = x_0$, $s' = x_1$ (rational sections of $\mathcal{O}(1)$ in the trivialization on the chart $U_2 = \{x_2 \neq 0\}$, where the affine coordinates are $u = x_0/x_2$, $v = x_1/x_2$).

First order. Intersecting $[\mathbb{P}^2]$ with D gives $D \cdot [\mathbb{P}^2] = [\mathrm{div}(s)] = [D] = [L]$, the line $\{x_0 = 0\}$. Now intersect with D'. Since $D = \{x_0 = 0\}$ is not contained in $|D'| = \{x_1 = 0\}$, the section $s'|_D = x_1|_D$ is a rational section of $\mathcal{O}(1)|_D$; on the line $D \cong \mathbb{P}^1$ with coordinate $[x_1 : x_2]$ it vanishes to order one at the single point $x_1 = 0$, namely $[0 : 0 : 1]$. Thus

$$D' \cdot (D \cdot [\mathbb{P}^2]) = D' \cdot [L] = [\mathrm{div}_D(s')] = \big[\,[0 : 0 : 1]\,\big]$$

the class of the point where the two lines meet.

Reverse order. Intersecting $[\mathbb{P}^2]$ with D' first gives $D' \cdot [\mathbb{P}^2] = [\mathrm{div}(s')] = [D'] = [L]$, the line $\{x_1 = 0\}$. Intersecting with D, the section $s|_{D'} = x_0|_{D'}$ on $D' \cong \mathbb{P}^1$ with coordinate $[x_0 : x_2]$ vanishes to order one at $x_0 = 0$, again $[0 : 0 : 1]$, so

$$D \cdot (D' \cdot [\mathbb{P}^2]) = D \cdot [L] = [\mathrm{div}_{D'}(s)] = \big[\,[0 : 0 : 1]\,\big]$$

The two orders produce the same zero-cycle, the reduced point $[0 : 0 : 1]$, on the nose, which is theorem 9.3.2 made visible: here no rational equivalence was even needed, because the two lines are transverse and the tame symbol $\partial_W(x_0, x_1)$ has no zeros or poles. Taking degrees on the complete $\mathbb{P}^2$,

$$\deg\big(D \cdot D' \cdot [\mathbb{P}^2]\big) = 1$$

the intersection number of two distinct lines: they meet in exactly one point. This is the case $c = d = 1$ of Bézout's theorem ([20]), recovered from the divisor calculus.

Where reciprocity would enter. Replacing D' by D itself, one must intersect $[L]$ with the same line, which requires moving D within its linear system: since $\mathcal{O}(1)$ has many sections, choose instead $s'' = x_1$, whose divisor $\{x_1 = 0\}$ is linearly equivalent to $\{x_0 = 0\}$, and the previous computation gives $D \cdot D \cdot [\mathbb{P}^2] = \big[\,[0 : 0 : 1]\,\big]$ of degree 1, so the self-intersection $L^2 = 1$ on $\mathbb{P}^2$. The passage from $s = x_0$ to the linearly equivalent $s'' = x_1$ changes the cutting section by the rational function x_1/x_0, and that the answer is unchanged is precisely proposition 9.2.4, the well-definedness whose commutativity companion is proved in this section.

The example is the enumerative payoff in miniature. The number 1 is the degree of a zero-cycle, the two lines meet where their defining sections simultaneously vanish, and the independence of the answer from the order of intersection and from the choice of representative in each linear system is exactly what theorem 9.3.2 and proposition 9.2.4 guarantee. Iterating, r general hyperplanes in $\mathbb{P}^n$ intersect in $c_1(\mathcal{O}(1))^r \cap [\mathbb{P}^n] = [\mathbb{P}^{n-r}]$, and taking $r = n$ recovers a single reduced point of degree one, the statement that n general hyperplanes meet in one point. This is the computation the next section packages as the first Chern class.

Exercise 9.3.1

1. On $\mathbb{P}^2$, let $D = \{x_0 x_1 = 0\}$ be the union of two coordinate lines (a divisor in the class $2L$) and let $D' = \{x_2 = 0\}$ be a third line. Compute $D \cdot (D' \cdot [\mathbb{P}^2])$ and $D' \cdot (D \cdot [\mathbb{P}^2])$ as zero-cycles, verify they agree, and compute the degree of the product. Confirm the answer is $2 = \deg D \cdot \deg D'$, consistent with Bézout.

2. Let $S = \mathbb{P}^2$ with affine coordinates $u = x_0/x_2$, $v = x_1/x_2$ on the chart $x_2 \neq 0$. For the rational functions $f = u$ and $g = v$, compute the tame symbol $\partial_W(f, g)$ at each coordinate curve $W \in \{u = 0\}, \{v = 0\}$ and at the line at infinity, and verify directly that $\sum_W \mathrm{ord}_W(f)\,[\mathrm{div}_W(g)] = \sum_W \mathrm{ord}_W(g)\,[\mathrm{div}_W(f)]$ as zero-cycles.

3. Let $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ be the degree-two map $[x_0 : x_1] \mapsto [x_0^2 : x_1^2]$ (characteristic $\neq 2$) and let $D = [\infty]$ be the divisor of the point at infinity on the target. Using proposition 9.3.3(1), compute both $f_*(f^*D \cdot [\mathbb{P}^1])$ and $D \cdot f_*[\mathbb{P}^1]$, identify the scalar $d = \deg f$, and confirm they agree.

4. Let $g\colon \mathbb{A}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$ be the projection, flat of relative dimension one, and let $D = [0]$ be the origin on $\mathbb{P}^1$. Compute $g^*(D \cdot [\mathbb{P}^1])$ and $g^*D \cdot g^*[\mathbb{P}^1]$ and verify proposition 9.3.3(2) directly.

5. Deduce from theorem 9.3.2 that for any Cartier divisor D on a complete variety X of dimension

n, the iterated intersection $D^n \cdot [X] \in A_0(X)$ is symmetric under any reordering of the n copies of D (trivially, but state precisely why the number $\deg(D^n \cdot [X])$ is well defined), and use theorem 8.5.2 to compute $H^n \cdot [\mathbb{P}^n]$ for H a hyperplane.

6. Explain why the proof of theorem 9.3.2 does not require X to be normal, identifying the exact step where normalization is used and why it changes neither side of the identity. (One sentence on each.)

9.4 The First Chern Class of a Line Bundle

The three previous sections built a single operation and studied it thoroughly: a Cartier divisor D on X acts on cycles by $\alpha \mapsto D \cdot \alpha$, this action descends to rational equivalence (proposition 9.2.4), and two such actions commute (theorem 9.3.2). The divisor D, however, is not the most invariant object in sight. What acts on $A_*(X)$ is not D itself but its linear-equivalence class, since D and $D + \mathrm{div}(r)$ produce the same operator, and a linear-equivalence class of Cartier divisors is exactly a line bundle. The right name for the operator is therefore the *first Chern class* of the associated line bundle $\mathcal{O}_X(D)$, written $c_1(\mathcal{O}_X(D)) \cap -$.

The renaming records that the operator depends only on $L = \mathcal{O}_X(D)$, so it is defined for a line bundle whether or not a divisor is at hand; it makes the additive structure transparent, because tensoring line bundles adds their Chern classes; and it is the first instance of a construction that in the next chapter will attach to a vector bundle E of rank r a whole family of operators $c_1(E), \ldots, c_r(E)$, the higher Chern classes. Everything the first Chern class does is already contained in the divisor calculus of the preceding sections; this section repackages that calculus into its final, functorial form and reads off the consequences.

The starting point is that the operator $D \cdot -$ sees only the line bundle $\mathcal{O}_X(D)$. Two Cartier divisors with isomorphic associated bundles are linearly equivalent, that is, they differ by the divisor of a rational function, and a principal divisor intersects every subvariety in a principal divisor, hence acts as zero on $A_*(X)$; proposition 9.4.2 below records this and is the independence needed to base the definition on L rather than on D.

Definition 9.4.1: The First Chern Class of a Line Bundle

Let L be a line bundle on an algebraic scheme X, and suppose $L \cong \mathcal{O}_X(D)$ for some Cartier divisor D on X. The *first Chern class* of L is the operator on Chow groups

$$c_1(L) \cap -\colon A_k(X) \longrightarrow A_{k-1}(X), \qquad c_1(L) \cap \alpha := D \cdot \alpha$$

defined for every k, where $D \cdot \alpha$ is the intersection with the Cartier divisor D of definition 9.2.1. The class $c_1(L)$ is the operator itself; the notation $c_1(L) \cap \alpha$ reads "$c_1(L)$ capped with α".

The definition names D but must not depend on it: a line bundle admits many divisors of sections, and the Chern class has to be the same for all of them. The following proposition is the price of the definition, and its proof is short because the work was done in section 9.2.

Proposition 9.4.2: Independence of the Divisor

Let L be a line bundle on X and let D, D' be Cartier divisors with $\mathcal{O}_X(D) \cong L \cong \mathcal{O}_X(D')$. Then $D \cdot \alpha = D' \cdot \alpha$ in $A_{k-1}(X)$ for every $\alpha \in A_k(X)$. Consequently $c_1(L) \cap -$ of definition 9.4.1 is well defined, independent of the choice of D.

***Proof*:**
An isomorphism $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$ of line bundles is exactly a linear equivalence $D \sim D'$ of Cartier divisors: the tautological rational section of $\mathcal{O}_X(D)$ and that of $\mathcal{O}_X(D')$, compared through the isomorphism, differ by a global invertible section r of the sheaf of total quotient rings $\mathcal{K}_X^*$, so $D - D' = \operatorname{div}(r)$ is principal (on a variety $\mathcal{K}_X^* = k(X)^*$ and r is a rational function; on a general algebraic scheme r is a section of the total-quotient sheaf, which is all the definition of a principal Cartier divisor requires). It therefore suffices to show that a principal Cartier divisor $\operatorname{div}(r)$ acts as zero on rational equivalence classes.

Fix $\alpha \in A_k(X)$ and represent it by a cycle $\sum_i n_i[V_i]$. Since $\operatorname{div}(r) \cdot -$ is defined component by component and extended $\mathbb{Z}$-linearly (definition 9.2.1), it suffices to show $\operatorname{div}(r) \cdot [V] = 0$ in $A_{k-1}(X)$ for a single k-dimensional subvariety V. The associated bundle of the principal divisor $\operatorname{div}(r)$ is $\mathcal{O}_X$, and the rational section of $\mathcal{O}_X$ that represents $\operatorname{div}(r)$ is r itself; unwinding definition 9.2.1 for this section gives

$$\operatorname{div}(r) \cdot [V] = \big[\operatorname{div}\big(r|_V\big)\big] \in A_{k-1}(V)$$

where $r|_V \in k(V)^*$ is the restriction of r to V (a nonzero rational function when $V \not\subseteq |\operatorname{div}(r)|$, and otherwise computed through the restricted trivial bundle $\mathcal{O}_X|_V \cong \mathcal{O}_V$, whose section is again a rational function on V). In either case $\operatorname{div}(r|_V)$ is the divisor of a rational function on the variety V, hence a *principal* Weil divisor on V, which is rationally equivalent to zero by the very definition of rational equivalence (definition 8.1.3). Its class in $A_{k-1}(V)$, and so its image in $A_{k-1}(X)$, is therefore 0: intersecting a subvariety with a principal divisor produces a principal divisor, which dies in the Chow group. This is the same computation that drives the well-definedness of proposition 9.2.4, read now with the roles reversed, the divisor rather than the cycle being principal. Summing over the components of α gives $\operatorname{div}(r) \cdot \alpha = 0$, and so

$$D \cdot \alpha - D' \cdot \alpha = (D - D') \cdot \alpha = \operatorname{div}(r) \cdot \alpha = 0$$

giving $D \cdot \alpha = D' \cdot \alpha$, as we sought to show.

The proposition is the reason the notation $c_1(L)$ is legitimate at all. It shows the divisor served only to build the operator: once the operator exists, it is recorded by the line bundle alone. From here on a divisor is used to compute $c_1(L)$, never to define it, and the choice of divisor is a matter of convenience. On a variety, every line bundle admits a Cartier divisor, so the definition applies to all line bundles; on a general scheme one takes L to be of the form $\mathcal{O}_X(D)$, which covers every line bundle whose class lies in the image of the Cartier divisor class group. The definition and the properties below are stated for a general algebraic scheme, but every worked example and exercise that follows runs on a variety, where a rational section and its function field $k(V)$ are always at hand; the general-scheme statements specialize to these illustrations by taking X integral.

A first sanity check grounds the definition before its formal properties: the Chern class of the trivial bundle is the zero operator, and the two constructions available for a hyperplane bundle agree.

Example 9.4.3: The Trivial Bundle and the Hyperplane Bundle

1. *The trivial bundle.* For $L = \mathcal{O}_X$ one may take $D = 0$, the empty divisor. Then $c_1(\mathcal{O}_X) \cap \alpha = 0 \cdot \alpha = 0$ for every α, so $c_1(\mathcal{O}_X)$ is the zero operator. This matches the reading of c_1 as a measure of the twisting of L: a trivial bundle has no twisting to record.

2. *The hyperplane bundle on $\mathbb{P}^n$.* Let $L = \mathcal{O}_{\mathbb{P}^n}(1)$ and let $H \subseteq \mathbb{P}^n$ be a hyperplane, cut out by a linear form; then $\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1)$. The class map of proposition 9.1.3 sends $\mathcal{O}_{\mathbb{P}^n}(1) \mapsto [H] \in A_{n-1}(\mathbb{P}^n)$, the generator of theorem 8.5.2 in codimension one. Capping with $[\mathbb{P}^n]$ recovers this class,

$$c_1(\mathcal{O}_{\mathbb{P}^n}(1)) \cap [\mathbb{P}^n] = H \cdot [\mathbb{P}^n] = [H]$$

since $\mathbb{P}^n \nsubseteq H$ and the restricted section of $\mathcal{O}_{\mathbb{P}^n}(1)$ to $\mathbb{P}^n$ has divisor exactly H. The first Chern class of $\mathcal{O}(1)$, capped against the fundamental class, is the hyperplane class.

With the operator anchored, its formal properties follow. Each is a translation of a divisor identity already proved, and the point of collecting them is to see the first Chern class as a well-behaved endomorphism of the graded group $A_*(X)$: additive in α, commuting with its own kind, additive in L, compatible with pushforward and pullback, and eventually nilpotent. The statement below is the working summary of the theory of c_1, and every clause is used repeatedly in the chapters that follow.

Proposition 9.4.4: Properties of the First Chern Class

Let X be an algebraic scheme, L, L' line bundles on X, and $\alpha \in A_k(X)$.

1. *Homomorphism.* $c_1(L) \cap -\colon A_k(X) \to A_{k-1}(X)$ is a group homomorphism: $c_1(L) \cap (\alpha + \beta) = c_1(L) \cap \alpha + c_1(L) \cap \beta$.
2. *Commutativity.* For line bundles L, L',
$$c_1(L) \cap \big(c_1(L') \cap \alpha\big) = c_1(L') \cap \big(c_1(L) \cap \alpha\big)$$
in $A_{k-2}(X)$.
3. *Additivity.* $c_1(L \otimes L') = c_1(L) + c_1(L')$ as operators; that is, $c_1(L \otimes L') \cap \alpha = c_1(L) \cap \alpha + c_1(L') \cap \alpha$. In particular $c_1(L^\vee) = -c_1(L)$.
4. *Projection formula.* For a proper morphism $f\colon X' \to X$, a line bundle L on X, and $\alpha' \in A_k(X')$,
$$f_*\big(c_1(f^*L) \cap \alpha'\big) = c_1(L) \cap f_*\alpha'$$
and for a flat morphism $g\colon X'' \to X$ of relative dimension m and $\beta \in A_k(X)$,
$$c_1(g^*L) \cap g^*\beta = g^*\big(c_1(L) \cap \beta\big)$$
in $A_{k+m-1}(X'')$.
5. *Nilpotence.* If X is a variety of dimension n, then for any line bundle L,
$$c_1(L)^{n+1} \cap [X] = 0$$
where $c_1(L)^{n+1} \cap [X]$ means $c_1(L)$ capped $n+1$ times with the fundamental class $[X]$.

***Proof*:**
Choose Cartier divisors D, D' with $\mathcal{O}_X(D) \cong L$ and $\mathcal{O}_X(D') \cong L'$; by proposition 9.4.2 the choices do not affect any of the claims. Each part translates a divisor identity from section 9.2 or section 9.3.

1. *Homomorphism.* The intersection $D \cdot -$ is defined on cycles component by component and extended $\mathbb{Z}$-linearly (definition 9.2.1), so it is additive on $Z_k(X)$, and proposition 9.2.4 makes it descend to a well-defined additive map on $A_k(X)$. Hence $c_1(L) \cap - = D \cdot -$ is a homomorphism $A_k(X) \to A_{k-1}(X)$.
2. *Commutativity.* By definition $c_1(L) \cap (c_1(L') \cap \alpha) = D \cdot (D' \cdot \alpha)$ and $c_1(L') \cap (c_1(L) \cap \alpha) = D' \cdot (D \cdot \alpha)$. These are equal in $A_{k-2}(X)$ by theorem 9.3.2, the commutativity of intersection with two Cartier divisors.
3. *Additivity.* The associated-bundle construction of section 5.4 converts the sum of Cartier divisors into the tensor product of line bundles, $\mathcal{O}_X(D + D') \cong \mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')$, so $D + D'$ is a Cartier divisor with $\mathcal{O}_X(D + D') \cong L \otimes L'$. Using this divisor to compute the Chern class of $L \otimes L'$,
$$c_1(L \otimes L') \cap \alpha = (D + D') \cdot \alpha = D \cdot \alpha + D' \cdot \alpha = c_1(L) \cap \alpha + c_1(L') \cap \alpha$$

where the middle equality is the additivity of $- \cdot \alpha$ in the divisor, and it holds already at the level of cycles, on *every* subvariety V, with no repositioning of α required. Since $D \cdot -$ is defined component by component and extended $\mathbb{Z}$-linearly (definition 9.2.1), it suffices to check $(D+D')\cdot[V] = D\cdot[V]+D'\cdot[V]$ for a single k-dimensional subvariety V. Restricting the tensor-product isomorphism above to V gives $\mathcal{O}_X(D+D')|_V \cong \mathcal{O}_X(D)|_V \otimes_{\mathcal{O}_V} \mathcal{O}_X(D')|_V$, so a rational section s_V of $\mathcal{O}_X(D+D')|_V$ representing $(D+D')|_V$ is the product $s_V = s_V^D \cdot s_V^{D'}$ of representing rational sections of $\mathcal{O}_X(D)|_V$ and $\mathcal{O}_X(D')|_V$. By definition 9.2.1 the cap is the class of the divisor of the restricted section computed inside V, and the order function along a codimension-one subvariety W of V is additive on a product of rational sections, $\operatorname{ord}_W(s_V^D \cdot s_V^{D'}) = \operatorname{ord}_W(s_V^D) + \operatorname{ord}_W(s_V^{D'})$, since ord_W is a valuation. Summing over such W gives $[\operatorname{div}(s_V)] = [\operatorname{div}(s_V^D)] + [\operatorname{div}(s_V^{D'})]$ in $A_{k-1}(V)$, that is $(D+D')\cdot[V] = D\cdot[V]+D'\cdot[V]$; pushing forward to X and summing over the components of α gives the middle equality. This is a cycle-level identity holding for each V, so no moving lemma repositioning α off $|D|$ and $|D'|$ is needed, and none exists in general. Taking $L' = L^\vee$, so that $\mathcal{O}_X \cong L \otimes L^\vee$ and $c_1(\mathcal{O}_X) = 0$ by example 9.4.3, gives $c_1(L) + c_1(L^\vee) = 0$, hence $c_1(L^\vee) = -c_1(L)$.

4. *Projection formula.* The pullback along f of the Cartier divisor D, in the pseudo-divisor formalism of definition 9.1.5, is a pseudo-divisor f^*D with $f^*\mathcal{O}_X(D) \cong \mathcal{O}_{X'}(f^*D)$; so $c_1(f^*L) \cap \alpha' = f^*D \cdot \alpha'$. The proper case is the divisor projection formula $f_*(f^*D \cdot \alpha') = D \cdot f_*\alpha'$ of proposition 9.3.3, which rewritten with Chern-class notation is exactly $f_*(c_1(f^*L) \cap \alpha') = c_1(L) \cap f_*\alpha'$. The flat case is the compatibility $g^*(D \cdot \beta) = g^*D \cdot g^*\beta$ of the same proposition, and since $c_1(g^*L) \cap g^*\beta = g^*D \cdot g^*\beta$ this reads $c_1(g^*L) \cap g^*\beta = g^*(c_1(L) \cap \beta)$.

5. *Nilpotence.* Each cap with $c_1(L)$ lowers dimension by one, since $c_1(L) \cap -\colon A_j(X) \to A_{j-1}(X)$ by definition 9.4.1, so repeatedly capping $[X] \in A_n(X)$ gives

$$c_1(L)^{n+1} \cap [X] \in A_{n-(n+1)}(X) = A_{-1}(X)$$

and $A_j(X) = 0$ for $j < 0$, there being no subvarieties of negative dimension. Hence the class is zero.

Each clause has been reduced to a result of the preceding sections, completing the proof.

The well-definedness proved above lets $D \cdot -$ be evaluated on a representative chosen to avoid $|D|$, which is all the divisor calculus of this chapter needs. Later constructions cannot make that choice: in chapter 11 a cycle is intersected with the fibre at infinity of a degeneration, and *every* component of the relevant representative lies in that fibre. The following strengthening removes the avoidance hypothesis, and it is what makes those constructions legitimate.

Corollary 9.4.5: Divisor Intersection on a Closed Subscheme

Let D be a Cartier divisor on X and $T \subseteq X$ a closed subscheme, with no hypothesis relating T to $|D|$ and none making $D|_T$ a Cartier divisor on T. Then intersection with D descends to a homomorphism

$$D \cdot - \ : \ A_k(T) \longrightarrow A_{k-1}\big(T \cap |D|\big)$$

computed on any representative cycle, and compatible with the inclusion of T in X: for a cycle on T the two readings of definition 9.2.1, one in T and one in X, agree.

***Proof*:**
At the level of cycles there is nothing to prove: definition 9.2.1 computes $D \cdot [V]$ for a subvariety V from the restricted line bundle $\mathcal{O}_X(D)|_V$ and from whether $V \subseteq |D|$, and neither datum changes when V is regarded inside T rather than X. This is the reason the definition was framed through the pseudo-divisor $(\mathcal{O}_X(D), |D|, s_D)$ of definition 9.1.5: a pseudo-divisor restricts to any subscheme, whereas a Cartier divisor need not, its local equations being free to become zero divisors.

What needs an argument is that the operation kills cycles rationally equivalent to zero *on* T, and lands in the smaller group. Such a cycle is a sum of $\operatorname{div}(r)$ for $(k+1)$-dimensional subvarieties $W \subseteq T$ and $r \in k(W)^*$ (definition 8.1.3). The proof of proposition 9.4.6 treats exactly these generators, and in both of its cases the vanishing it produces is already recorded on W: in the case $W \subseteq |D|$ the class is computed in $A_{k-1}(W)$ before being pushed forward, and in the case $W \not\subseteq |D|$ the appeal to theorem 9.3.2 places the identity in $A_{k-1}(|D| \cap |\operatorname{div}(r)| \cap W)$. Both groups map to $A_{k-1}(T \cap |D|)$, so $D \cdot \operatorname{div}(r) = 0$ there, as we sought to show.

Proposition 9.4.6: Divisor Intersection on an Arbitrary Representative

Let D be a Cartier divisor on an algebraic scheme X and let $\alpha \in Z_k(X)$ be a cycle rationally equivalent to zero, with *no* hypothesis on its components. Then

$$D \cdot \alpha = 0 \quad \text{in } A_{k-1}(|D|)$$

Consequently $D \cdot -$ may be computed from any representative of a class in $A_k(X)$, including one all of whose components lie in $|D|$, and every representative gives the same class in $A_{k-1}(|D|)$.

***Proof*:**
Both sides are additive, so it suffices to treat $\alpha = [\operatorname{div}(r)]$ for a $(k+1)$-dimensional subvariety $W \subseteq X$ and a rational function $r \in k(W)^*$, these being the generators of the subgroup of cycles rationally equivalent to zero (definition 8.1.3). Write $\iota\colon W \hookrightarrow X$ for the inclusion.

Case 1: $W \subseteq |D|$. Every component of $\operatorname{div}(r)$ lies in W, hence in $|D|$, so clause (2) of definition 9.2.1 computes $D \cdot [\operatorname{div}(r)]$ from the restricted line bundle. By definition 9.4.1 that value is $c_1\big(\mathcal{O}_X(D)|_W\big) \cap [\operatorname{div}(r)]$, computed in $A_{k-1}(W)$. Now $[\operatorname{div}(r)] = 0$ in $A_k(W)$ by the definition of rational equivalence, and $c_1(\mathcal{O}_X(D)|_W) \cap -$ is a homomorphism on Chow *classes* (proposition 9.4.4(1)), so it sends 0 to 0. Pushing forward along the closed immersion $W \hookrightarrow |D|$ gives $D \cdot [\operatorname{div}(r)] = 0$ in $A_{k-1}(|D|)$.

Case 2: $W \not\subseteq |D|$. Then D restricts to a Cartier divisor $D|_W$ on W, and $[\operatorname{div}(r)] = \operatorname{div}(r) \cdot [W]$ exhibits α as the result of intersecting the fundamental class of W with the principal Cartier divisor $\operatorname{div}(r)$ on W. Commutativity of divisor intersection (theorem 9.3.2), applied on W to the pair $D|_W$ and $\operatorname{div}(r)$, gives

$$D|_W \cdot \big(\operatorname{div}(r) \cdot [W]\big) \;=\; \operatorname{div}(r) \cdot \big(D|_W \cdot [W]\big)$$

and that theorem records the identity in the refined group $A_{k-1}\big(|D| \cap |\operatorname{div}(r)| \cap W\big)$, so it may be read there. The right-hand side vanishes: the line bundle of a principal divisor is trivial, $\mathcal{O}_W(\operatorname{div}(r)) \cong \mathcal{O}_W$, so $\operatorname{div}(r) \cdot -$ is $c_1(\mathcal{O}_W) \cap -$, and c_1 of a trivial bundle is the zero operator by proposition 9.4.4(3) applied to $L = L' = \mathcal{O}_W$. Hence $D \cdot [\operatorname{div}(r)] = 0$ in $A_{k-1}(|D|)$, completing the proof.

Nothing in the argument is circular: theorem 9.3.2 is proved from the avoidance-hypothesis form proposition 9.2.4 alone, so it may be used here to remove that hypothesis.

Parts (1) through (4) say that c_1 is a homomorphism of graded groups compatible with all the structure already on A_*, and that $L \mapsto c_1(L)$ turns tensor product into addition, so $c_1 \colon \mathrm{Pic}\,(X) \to \mathrm{End}(A_*(X))$ is a homomorphism from the Picard group into the operators lowering degree by one. Part (2), the commutativity, is the fact that makes these operators generate a commutative subring of $\mathrm{End}(A_*(X))$; it is the discharged form of the divisor symmetry that theorem 8.4.6 in section 8.4 took on faith, now proved through theorem 9.3.2. Part (5), the nilpotence, is a dimension count: it says a single line bundle can only cut down the fundamental class n times before the result must vanish, and it is what makes the top self-intersection number $c_1(L)^n \cap [X]$, a zero-cycle, the natural degree-type invariant of L on a complete variety.

The nilpotence has a converse worth flagging: while $c_1(L)^{n+1} \cap [X]$ always vanishes, the intermediate power $c_1(L)^n \cap [X] \in A_0(X)$ need not, and its degree $\int_X c_1(L)^n$ is the numerical invariant of a line bundle on an n-dimensional complete variety. For $n = 1$ this is the degree of a line bundle on a curve; for a surface it is the self-intersection L^2. These invariants are the raw material of the intersection numbers computed throughout this book.

The first substantial computation is the self-intersection of $\mathcal{O}(1)$ on projective space, which reproduces the entire linear-subspace basis of theorem 8.5.2 from a single line bundle. It shows the first Chern class already generates the Chow ring of $\mathbb{P}^n$, and it is the model computation the higher Chern classes generalize.

Example 9.4.7: Powers of the Hyperplane Class on $\mathbb{P}^n$

Let $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$ act on $A_*(\mathbb{P}^n)$. By theorem 8.5.2 the Chow group $A_*(\mathbb{P}^n) = \bigoplus_{j=0}^n \mathbb{Z}\,[\mathbb{P}^j]$ is free on the classes of linear subspaces, with $[\mathbb{P}^j]$ the class of a j-plane. The claim is

$$c_1(\mathcal{O}_{\mathbb{P}^n}(1))^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}], \qquad 0 \le k \le n$$

so the k-th self-cap of the hyperplane bundle is the class of a linear subspace of codimension k.

The proof is induction on k. For $k = 0$ the left side is $[\mathbb{P}^n]$, the fundamental class, and $[\mathbb{P}^{n-0}] = [\mathbb{P}^n]$. Assume $c_1(\mathcal{O}(1))^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ for some $k < n$, so the class is represented by the subvariety $\mathbb{P}^{n-k} \subseteq \mathbb{P}^n$, a linear subspace of dimension $n-k$. Cap once more with H. There is a hyperplane H_0 with $\mathbb{P}^{n-k} \not\subseteq H_0$, because some linear form does not vanish identically on the nonempty $\mathbb{P}^{n-k}$ (equivalently, $\mathbb{P}^{n-k}$ is not contained in every hyperplane); its zero locus is such an H_0, and $H_0 \cap \mathbb{P}^{n-k}$ is then a hyperplane inside $\mathbb{P}^{n-k}$, of dimension $n-k-1$. Thus $\mathbb{P}^{n-k} \not\subseteq |H_0|$, and by definition 9.2.1 the cap $H \cdot [\mathbb{P}^{n-k}]$ is the divisor of the restricted linear form on $\mathbb{P}^{n-k}$, namely the hyperplane section $H_0 \cap \mathbb{P}^{n-k}$, a linear subspace $\mathbb{P}^{n-k-1}$ of dimension $n-k-1$:

$$c_1(\mathcal{O}(1))^{k+1} \cap [\mathbb{P}^n] = H \cdot [\mathbb{P}^{n-k}] = [H_0 \cap \mathbb{P}^{n-k}] = [\mathbb{P}^{n-k-1}]$$

completing the induction. In particular $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\mathbb{P}^0] = [\mathrm{pt}]$ is the class of a single point, and its degree is

$$\int_{\mathbb{P}^n} c_1(\mathcal{O}_{\mathbb{P}^n}(1))^n = 1$$

the statement that n general hyperplanes in $\mathbb{P}^n$ meet in a single reduced point. Capping once more, $c_1(\mathcal{O}(1))^{n+1} \cap [\mathbb{P}^n] = H \cdot [\mathrm{pt}] = 0$, in agreement with the nilpotence of proposition 9.4.4(5). Every generator $[\mathbb{P}^j]$ of $A_*(\mathbb{P}^n)$ is thus a power of $c_1(\mathcal{O}(1))$ applied to $[\mathbb{P}^n]$, so the first Chern class of the hyperplane bundle generates the Chow group of projective space as a module over the operators.

The computation is the first appearance of the ring structure that intersection theory is heading toward: on $\mathbb{P}^n$ the operators H^k multiply the way linear subspaces intersect, $[\mathbb{P}^a]\cdot[\mathbb{P}^b]=[\mathbb{P}^{a+b-n}]$ when $a+b\geq n$ and zero otherwise, and $\int_{\mathbb{P}^n} H^n = 1$ is the normalization. The Chow group $A_*(\mathbb{P}^n)$ is the truncated polynomial ring $\mathbb{Z}[H]/(H^{n+1})$ with $H = c_1(\mathcal{O}(1))$, and the nilpotence relation $H^{n+1}=0$ is exactly proposition 9.4.4(5). This ring will be built in general later in this Part, once the intersection product is available; here it is visible for projective space through the single line bundle $\mathcal{O}(1)$.

The second computation identifies c_1 on a curve with the classical degree, tying the abstract operator to the invariant every reader already knows. On a smooth projective curve the first Chern class is the degree map, and capping with $c_1(L)$ is subtraction of that degree from the point count.

Example 9.4.8: The First Chern Class on a Curve is the Degree

Let C be a smooth projective curve over $\overline{k}$ and L a line bundle on C. Since $\dim C = 1$, the only nonzero cap is on $A_1(C)=\mathbb{Z}\,[C]$, and it lands in $A_0(C)$. Write $L \cong \mathcal{O}_C(D)$ for a Cartier divisor $D=\sum_P n_P\,[P]$, which on a smooth curve is the same as a Weil divisor, a finite $\mathbb{Z}$-combination of closed points. Then

$$c_1(L)\cap[C] = D\cdot[C] = \sum_P n_P\,[P] \in A_0(C)$$

by definition 9.2.1, the section of $\mathcal{O}_C(D)$ having divisor D on the curve C itself. Pushing this zero-cycle to a point through $\pi\colon C\to \operatorname{Spec}\overline{k}$ and taking degrees (definition 8.2.4),

$$\int_C c_1(L)\cap[C] = \sum_P n_P\,[\overline{k}(P):\overline{k}] = \sum_P n_P = \deg D = \deg L$$

since every closed point of a variety over the algebraically closed field $\overline{k}$ has residue field $\overline{k}$. The number $\int_C c_1(L)$ is the degree of the line bundle. Under the identification $A_0(C)\cong \operatorname{Cl} C = \operatorname{Pic} C$ of proposition 8.5.3, capping with $c_1(L)$ against $[C]$ is the class map $L\mapsto[D]$, and its composite with the degree $A_0(C)\xrightarrow{\deg}\mathbb{Z}$ recovers the degree of the divisor class, fitting into the sequence $0\to\operatorname{Pic}^0 C\to A_0(C)\xrightarrow{\deg}\mathbb{Z}\to 0$ of proposition 8.5.3.

The two computations frame the whole theory. On projective space $c_1(\mathcal{O}(1))$ generates the Chow ring and its top power counts a single point; on a curve $c_1(L)$ is the degree, the invariant that governs Riemann-Roch through $\chi(L)=\deg L+1-g$ (theorem 7.1.19). Both come from one construction: the first Chern class turns a line bundle into an operator on cycles, and its powers are the numerical invariants that intersection theory computes. The chapter that follows generalizes the construction from line bundles to vector bundles of arbitrary rank: a rank-r bundle E carries Chern classes $c_1(E),\ldots,c_r(E)$, built so that c_1 of a line bundle is the operator of this section, and so that the total Chern class is multiplicative in short exact sequences. The machine that produces them, Segre classes of the projectivized bundle $\mathbb{P}(E)$, is the subject of the next chapter, and the divisor calculus of this chapter is the rank-one base case on which it is built.

Exercise 9.4.1

1. Compute $\int_{\mathbb{P}^1} c_1(\mathcal{O}_{\mathbb{P}^1}(d))$ for every integer d, both by the curve-degree formula of example 9.4.8 and directly from a divisor of a section of $\mathcal{O}_{\mathbb{P}^1}(d)$. Confirm that $c_1\colon \operatorname{Pic}(\mathbb{P}^1)\to A_0(\mathbb{P}^1)=\mathbb{Z}$ is an isomorphism.

2. Let S be a smooth projective surface and $L = \mathcal{O}_S(C)$ for a curve $C \subseteq S$. Express $c_1(L)^2 \cap [S] \in A_0(S)$ and its degree $\int_S c_1(L)^2$ in terms of the divisor C, and identify $\int_S c_1(L)^2$ with the self-intersection number $C \cdot C$ of the surface theory of section 7.2.1. Why does proposition 9.4.4(5) say $c_1(L)^3 \cap [S] = 0$?

3. Using proposition 9.4.4(3), show that $c_1 \colon \mathrm{Pic}\,(X) \to \mathrm{End}(A_*(X))$ is a group homomorphism from the Picard group to the additive group of degree-lowering operators. Deduce that $c_1(L^{\otimes m}) = m\, c_1(L)$ for every integer m, and that c_1 factors through the numerical class of L in the sense that linearly equivalent line bundles have equal first Chern class.

4. On $\mathbb{P}^1 \times \mathbb{P}^1$, let p_1, p_2 be the two projections and $L = p_1^*\mathcal{O}(a) \otimes p_2^*\mathcal{O}(b)$ the line bundle of bidegree (a, b). Using the projection formula proposition 9.4.4(4) and additivity, compute $\int_{\mathbb{P}^1 \times \mathbb{P}^1} c_1(L)^2$ in terms of a and b. (You may use $A_*(\mathbb{P}^1 \times \mathbb{P}^1)$ free on the classes of a point, the two rulings, and the fundamental class, with each ruling of self-intersection zero and the two rulings meeting in one point.)

5. Let $f \colon X \to Y$ be a proper surjective morphism of varieties with $\dim X = \dim Y = n$ and generic degree δ, and let L be a line bundle on Y. Show that $\int_X c_1(f^*L)^n = \delta \int_Y c_1(L)^n$. (Cap the projection formula n times and use $f_*[X] = \delta\,[Y]$.) Then explain why $c_1(f^*L)^{n+1} \cap [X] = 0$ holds regardless of δ.

Part III

Chern Classes and the Intersection Product

Everything so far has been rank one. The Chow group was cut out of the cycle group by rational equivalence, made functorial by proper pushforward and flat pullback, and then acted on by a single operator: intersecting with a Cartier divisor, which is to say capping with the first Chern class of a line bundle. That operator lowers dimension by one, commutes with itself, and satisfies a projection formula. It is also the only operator available, and a line bundle is the only bundle it sees.

This part removes both restrictions. A vector bundle of rank $r+1$ carries $r+1$ operators rather than one, and the projective bundle of its lines is the space on which they can be constructed: pushing powers of the tautological class down to the base produces the Segre classes, inverting the resulting series produces the Chern classes, and the geometry of the bundle forces the relations among them. With operators of every codimension in hand, the remaining question is whether two cycles can be multiplied rather than operated on. They can, and the mechanism is deformation to the normal cone, which replaces the classical manoeuvre of moving cycles into general position with a degeneration of the ambient embedding. The product it yields makes $A^*(X)$ a ring for smooth X, recovers Bézout's theorem, and subsumes the surface intersection pairing that *Algebraic Geometry* [7] built by hand.

10

Cones, Segre Classes, and Chern Classes of Vector Bundles

The construction of this chapter rests on one geometric observation. A vector bundle E on X has no interesting cycles of its own, being an affine bundle and so contributing nothing beyond $A_*(X)$ by theorem 8.4.6. Its projectivization $\mathbb{P}(E)$ does: it is proper over X, it carries the tautological line bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$, and the first Chern class of that line bundle is an operator of the kind Chapter 9 already supplies. Pushing its powers back down to X therefore produces operators on $A_*(X)$ out of nothing but the rank-one theory, and those operators are the Segre classes.

The chapter develops this in four movements. The Segre classes come first, along with the projective-bundle formula that computes $A_*(\mathbb{P}(E))$ completely and which was promised without proof in chapter 8. Chern classes follow by inverting the Segre series, and the splitting principle, made legitimate by the flag bundle, reduces their identities to the rank-one case already in hand. The Chern character and Todd class are then assembled as polynomial expressions in these operators, with the rational coefficients their denominators force. The chapter closes by extending Segre classes from vector bundles to cones, and so to arbitrary closed subschemes, which is the input chapter 11 needs.

10.1 Segre Classes and the Projective Bundle

A vector bundle E on X determines a projective bundle $\pi\colon \mathbb{P}(E) \to X$ carrying a tautological line bundle. Pushing powers of that line bundle's first Chern class down to X produces a family of operators on $A_*(X)$, the Segre classes, and the structural theorem behind them computes $A_*(\mathbb{P}(E))$ from $A_*(X)$ together with the powers of the tautological class. Both are built here, the second in full, discharging the formula recorded without proof in section 8.5.

The projective bundle itself is not built here. *Algebraic Geometry* [7] constructs Proj of a graded quasi-coherent algebra (definition 5.3.33) and its twisting sheaf (theorem 5.3.35), specializes the construction to a locally free sheaf (definition 5.3.37), and proves there that the resulting morphism is proper and flat (proposition 5.3.40) and locally trivial over a trivializing cover (proposition 5.3.38). One translation is needed at the point of import. What that volume writes $\mathbb{P}(\mathscr{E})$ parametrizes one-dimensional *quotients* of the fibres, while this book's $\mathbb{P}(E)$ parametrizes one-dimensional *subspaces* (box 7.3.10). Writing $\mathscr{E}$ for the sheaf of sections of E, the two are related by a dual:

$$\mathbb{P}(E) \;=\; \operatorname{Proj}\left(\operatorname{Sym}^\bullet \mathscr{E}^\vee\right)$$

so the bundle this book calls $\mathbb{P}(E)$ is the one [7] calls $\mathbb{P}(\mathscr{E}^\vee)$. Under that identification the twisting sheaf $\mathcal{O}(1)$ of the relative Proj is the universal rank-one quotient of $\pi^*\mathscr{E}^\vee$ (proposition 5.3.39); dualizing that surjection exhibits $\mathcal{O}(1)^\vee$ as the universal rank-one *subsheaf* of $\pi^*\mathscr{E}$, which is the tautological line. So $\mathcal{O}_{\mathbb{P}(E)}(1)$ is the dual of the tautological subbundle, which is what the subspace convention demands. Every formula below is read in that convention.

Throughout, E has rank $r+1$, so the fibre $\mathbb{P}(E_x)$ is a projective space of dimension r. All fibres of π are pure of dimension r, and π is flat, so π is flat of relative dimension r in the sense of definition 8.3.1; flat pullback $\pi^*\colon A_k(X) \to A_{k+r}(\mathbb{P}(E))$ is therefore defined by definition 8.3.3, and π being proper, pushforward $\pi_*\colon A_k(\mathbb{P}(E)) \to A_k(X)$ is defined by definition 8.2.1. These two maps and the divisor calculus of chapter 9 are all this section uses.

Definition 10.1.1: The Tautological Class of a Projective Bundle

Let E be a vector bundle of rank $r+1$ on an algebraic scheme X, let $\pi\colon \mathbb{P}(E) \to X$ be the associated bundle of lines in E, and let $\mathcal{O}_{\mathbb{P}(E)}(1)$ be the dual of the tautological subbundle. The *tautological class* of E is the first Chern class operator

$$\xi \;=\; c_1\left(\mathcal{O}_{\mathbb{P}(E)}(1)\right)$$

of definition 9.4.1, so that $\xi \cap -\colon A_k(\mathbb{P}(E)) \to A_{k-1}(\mathbb{P}(E))$ for every k. Its powers $\xi^i \cap -$ are the i-fold iterates of this operator, with $\xi^0 \cap -$ the identity.

The class ξ is an operator on the Chow groups of the total space, not an element of a ring: there is no product on $A_*(\mathbb{P}(E))$ until the intersection product of a later chapter supplies one. What makes ξ usable is that the operators $\xi \cap -$ commute among themselves and with every other first Chern class, by proposition 9.4.4(2), so an expression such as $\xi^i \cap \pi^*\alpha$ is unambiguous however it is bracketed.

Example 10.1.2: The Tautological Class on $\mathbb{P}^n$ and on a Line Bundle

Two extreme cases fix the meaning of ξ.

1. *A trivial bundle over a point.* Let $X = \operatorname{Spec} \overline{k}$ and $E = \overline{k}^{\,n+1}$, so $\mathbb{P}(E) = \mathbb{P}^n$ is the space of lines through the origin in $\overline{k}^{\,n+1}$ and $\mathcal{O}_{\mathbb{P}(E)}(1) = \mathcal{O}_{\mathbb{P}^n}(1)$ is the Serre twist. Here $\xi = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$, and example 9.4.7 computes its powers outright:
$$\xi^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}], \qquad 0 \le k \le n$$
with $\xi^{n+1} \cap [\mathbb{P}^n] = 0$. The tautological class of the trivial bundle over a point is the hyperplane class.

2. *A line bundle.* Let $E = L$ have rank one, so $r = 0$. A one-dimensional subspace of a one-dimensional fibre is the whole fibre, so $\mathbb{P}(L) = X$ and $\pi = \mathrm{id}$; in the Proj description, $\operatorname{Sym}^\bullet$

of an invertible sheaf has $\operatorname{Proj}(\operatorname{Sym}^\bullet \mathscr{E}^\vee) = X$ and $\mathcal{O}(1) = \mathscr{E}^\vee$, where $\mathscr{E}$ is the invertible sheaf of sections of L. The tautological subbundle is L itself, hence $\mathcal{O}_{\mathbb{P}(L)}(1) = L^\vee$ and, by proposition 9.4.4(3),

$$\xi = c_1(L^\vee) = -c_1(L)$$

The sign is forced by the convention: the tautological object is a *sub*bundle, and $\mathcal{O}(1)$ is its dual.

The tautological class lives on $\mathbb{P}(E)$, and the Chow groups one wants to compute with live on X. Transporting a class α on X up to $\mathbb{P}(E)$, cutting it with powers of ξ, and pushing the result back down produces an operator on $A_*(X)$ built from E alone. The bookkeeping is forced: π^* raises dimension by r, capping with ξ^m lowers it by m, and π_* preserves it, so the composite lowers dimension by $m - r$. Writing $m = r + i$ makes i the amount of dimension lost, and these composites are the Segre classes.

Definition 10.1.3: Segre Class Operators of a Vector Bundle

Let E be a vector bundle of rank $r + 1$ on an algebraic scheme X, with $\pi\colon \mathbb{P}(E) \to X$ and tautological class ξ. For an integer $i \geq -r$ the *i-th Segre class* of E is the operator defined on $\alpha \in A_k(X)$ by

$$s_i(E) \cap \alpha \;=\; \pi_*\big(\xi^{r+i} \cap \pi^*\alpha\big)$$

and $s_i(E) \cap \alpha = 0$ for $i < -r$. Each is a homomorphism $s_i(E) \cap -\colon A_k(X) \to A_{k-i}(X)$. The *total Segre class* of E is the sum

$$s(E) \;=\; \sum_{i \geq -r} s_i(E)$$

which is a finite sum of operators on any fixed class, since $s_i(E)$ lowers dimension by i and $A_j(X) = 0$ for $j < 0$.

That $s_i(E) \cap -$ is a homomorphism is inherited from its three constituents: π^* is additive by proposition 8.3.5(1), capping with ξ is additive by proposition 9.4.4(1), and π_* is additive by definition 8.2.1. Each of the three also descends to rational equivalence, by proposition 8.3.7, proposition 9.2.4 and theorem 8.2.3 respectively, so $s_i(E) \cap -$ is well defined on Chow groups without further argument. The content of the definition is not that these operators exist but that they are computable and that they encode E; the rank-one case already shows what they look like.

Example 10.1.4: The Segre Classes of a Line Bundle

Let L be a line bundle on X, so $r = 0$ and, by example 10.1.2(2), $\mathbb{P}(L) = X$ with $\pi = \mathrm{id}$ and $\xi = -c_1(L)$. The definition reads

$$s_i(L) \cap \alpha = \xi^{\,i} \cap \alpha = (-1)^i\, c_1(L)^i \cap \alpha, \qquad i \geq 0$$

there being no indices below $-r = 0$. The total Segre class is therefore the alternating series

$$s(L) = 1 - c_1(L) + c_1(L)^2 - c_1(L)^3 + \cdots$$

which on any fixed class terminates, since $c_1(L)^m \cap \alpha$ lies in $A_{k-m}(X)$ and this group vanishes once $m > k$. The series is the inverse of $1 + c_1(L)$: capping $s(L) \cap \alpha$ with $1 + c_1(L)$ telescopes, every term cancelling against its successor, and leaves α.

For a concrete instance take $X = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(1)$. Then $c_1(L)^i \cap [\mathbb{P}^n] = [\mathbb{P}^{n-i}]$ by example 9.4.7, so

$$s_i\big(\mathcal{O}_{\mathbb{P}^n}(1)\big) \cap [\mathbb{P}^n] = (-1)^i\,[\mathbb{P}^{n-i}], \qquad 0 \le i \le n$$

and $s_i(\mathcal{O}_{\mathbb{P}^n}(1)) \cap [\mathbb{P}^n] = 0$ for $i > n$. The Segre classes of a line bundle carry exactly the information in c_1, packaged with alternating signs.

The alternating signs in example 10.1.4 indicate that $s(E)$ is the wrong normalization for a class one wants to be additive on direct sums; the repair, carried out in section 10.2, is to invert the series. Before that, the operators need their formal properties: which of them vanish, which one is the identity, whether they commute, and how they behave under the two functorialities of the Chow group. All four are consequences of the corresponding properties of c_1 and of the flat-proper calculus of chapter 8.

Proposition 10.1.5: Basic Properties of Segre Classes

Let E be a vector bundle of rank $r + 1$ on an algebraic scheme X.

1. *Vanishing.* $s_i(E) \cap \alpha = 0$ for every $i < 0$ and every $\alpha \in A_*(X)$.
2. *Normalization.* $s_0(E) \cap \alpha = \alpha$ for every $\alpha \in A_*(X)$; that is, $s_0(E)$ is the identity operator. Consequently $s(E) = 1 + s_1(E) + s_2(E) + \cdots$.
3. *Commutativity.* If F is a second vector bundle on X, then for all i, j
$$s_i(E) \cap \big(s_j(F) \cap \alpha\big) = s_j(F) \cap \big(s_i(E) \cap \alpha\big)$$
4. *Functoriality.* If $f : X' \to X$ is proper and $\alpha' \in A_k(X')$, then
$$f_*\big(s_i(f^*E) \cap \alpha'\big) = s_i(E) \cap f_*\alpha'$$
and if $g : X'' \to X$ is flat of relative dimension m and $\beta \in A_k(X)$, then
$$s_i(g^*E) \cap g^*\beta = g^*\big(s_i(E) \cap \beta\big)$$

***Proof*:**
Formation of $\mathbb{P}(E)$ and of $\mathcal{O}_{\mathbb{P}(E)}(1)$ from $\mathrm{Sym}^\bullet \mathscr{E}^\vee$ commutes with base change along an arbitrary morphism (proposition 5.3.36), so for any morphism $h : Y \to X$ there is a fibre square

$$\begin{array}{ccc} \mathbb{P}(h^*E) & \xrightarrow{h'} & \mathbb{P}(E) \\ \downarrow{\scriptstyle \pi_Y} & & \downarrow{\scriptstyle \pi} \\ Y & \xrightarrow{h} & X \end{array}$$

with $\mathbb{P}(h^*E) = \mathbb{P}(E) \times_X Y$ and $\mathcal{O}_{\mathbb{P}(h^*E)}(1) = h'^*\mathcal{O}_{\mathbb{P}(E)}(1)$, hence $\xi_Y = h'^*\xi$ for the tautological classes. This square is used in every part.

1. *Vanishing.* By additivity it suffices to treat $\alpha = [V]$ for a k-dimensional subvariety $V \subseteq X$, with closed immersion $h : V \hookrightarrow X$. Since h is proper and π is flat, proposition 8.3.6 applied to the square above gives $\pi^* h_* = h'_* \pi_V^*$, so
$$\pi^*[V] = h'_*\,\pi_V^*[V]$$

and proposition 9.4.4(4) for the proper h', applied $r+i$ times with $\xi_V = h'^*\xi$, gives

$$\xi^{\,r+i} \cap \pi^*[V] = h'_*\big(\xi_V^{r+i} \cap \pi_V^*[V]\big)$$

Pushing forward by π and using $\pi \circ h' = h \circ \pi_V$ with functoriality of proper pushforward,

$$s_i(E) \cap [V] = h_*\, \pi_{V*}\big(\xi_V^{r+i} \cap \pi_V^*[V]\big) \tag{10.1}$$

The class inside h_* lies in $A_{k-i}(V)$, and for $i<0$ one has $k-i>k=\dim V$, so that group is zero: V has no subvariety of dimension exceeding its own. Hence $s_i(E)\cap[V]=0$.

2. *Normalization.* The inner class in (10.1) is $s_i(E|_V)\cap[V]$ computed on V, so that identity reduces the claim to the case $\alpha=[X]$ with X a variety of dimension k, where it reads $\pi_*(\xi^{\,r}\cap[\mathbb{P}(E)])=[X]$; note that $\pi^*[X]=[\mathbb{P}(E)]$ by proposition 8.3.5(2). The total space $\mathbb{P}(E)$ is a variety: it is covered by open sets isomorphic to $U_\lambda\times\mathbb{P}^r$ for a trivializing cover $\{U_\lambda\}$ of X (proposition 5.3.38), each of which is irreducible and reduced, and any two of them meet because their images in the irreducible X do. So $\dim\mathbb{P}(E)=k+r$ and $\pi_*(\xi^r\cap[\mathbb{P}(E)])$ lies in $A_k(X)=Z_k(X)=\mathbb{Z}\,[X]$ by example 8.1.6(3); write it as $m\,[X]$ with $m\in\mathbb{Z}$.

   The integer m is computed after restriction to a trivializing open. Choose a nonempty open $U\subseteq X$ with $E|_U$ trivial, and let $j\colon U\hookrightarrow X$ and $j'\colon \mathbb{P}(E|_U)\hookrightarrow\mathbb{P}(E)$ be the inclusions, both flat of relative dimension zero. Flat pullback commutes with π_* across the fibre square by proposition 8.3.6 and with capping by ξ by proposition 9.4.4(4), and $j'^*[\mathbb{P}(E)]=[\mathbb{P}(E|_U)]$ because restriction along an open immersion carries a fundamental cycle to a fundamental cycle (example 8.3.4(1)), so

$$j^*\big(m\,[X]\big)=\pi_{U*}\big(\xi_U^r\cap[\mathbb{P}(E|_U)]\big)$$

   Under a trivialization $\mathbb{P}(E|_U)\cong U\times\mathbb{P}^r$ the twisting sheaf becomes $\rho^*\mathcal{O}_{\mathbb{P}^r}(1)$ for the second projection $\rho\colon U\times\mathbb{P}^r\to\mathbb{P}^r$ (definition 5.3.20), and ρ is flat of relative dimension $\dim U$, so $\rho^*[\mathbb{P}^r]=[U\times\mathbb{P}^r]$ by proposition 8.3.5(2). Hence, by the flat case of proposition 9.4.4(4) and by example 9.4.7,

$$\xi_U^r\cap[U\times\mathbb{P}^r]=\rho^*\big(c_1(\mathcal{O}_{\mathbb{P}^r}(1))^r\cap[\mathbb{P}^r]\big)=\rho^*[\mathrm{pt}]=[U\times\{\mathrm{pt}\}]$$

   The projection π_U restricts to an isomorphism $U\times\{\mathrm{pt}\}\to U$, so $\pi_{U*}[U\times\{\mathrm{pt}\}]=[U]$ by definition 8.2.1. Thus $m\,[U]=[U]$ in $A_k(U)=\mathbb{Z}\,[U]$, which is free on the fundamental class by example 8.1.6(3), and therefore $m=1$. This gives $s_0(E)\cap[X]=[X]$, and with part (1) the total class has no terms in negative degree.

3. *Commutativity.* Let F have rank $f+1$, with $q\colon\mathbb{P}(F)\to X$ and tautological class η, and form the fibre product $W=\mathbb{P}(E)\times_X\mathbb{P}(F)$ with its two projections $q'\colon W\to\mathbb{P}(E)$ and $p'\colon W\to\mathbb{P}(F)$. Since q is proper and π is flat, q' is proper and p' is flat of relative dimension r; write $\rho=\pi\circ q'=q\circ p'\colon W\to X$, which is proper, and flat of relative dimension $r+f$ by proposition 8.3.5(3). Put $\tilde\xi=q'^*\xi$ and $\tilde\eta=p'^*\eta$. Then

$$\begin{aligned}
s_i(E)\cap\big(s_j(F)\cap\alpha\big)&=\pi_*\Big(\xi^{\,r+i}\cap\pi^*q_*\big(\eta^{\,f+j}\cap q^*\alpha\big)\Big)\\
&=\pi_*\Big(\xi^{\,r+i}\cap q'_*\,p'^*\big(\eta^{\,f+j}\cap q^*\alpha\big)\Big)\\
&=\pi_*\Big(\xi^{\,r+i}\cap q'_*\big(\tilde\eta^{\,f+j}\cap\rho^*\alpha\big)\Big)\\
&=\pi_*q'_*\Big(\tilde\xi^{\,r+i}\cap\tilde\eta^{\,f+j}\cap\rho^*\alpha\Big)=\rho_*\Big(\tilde\xi^{\,r+i}\cap\tilde\eta^{\,f+j}\cap\rho^*\alpha\Big)
\end{aligned}$$

the second line by proposition 8.3.6 for the fibre square defining W, the third by the flat case of proposition 9.4.4(4) together with $p'^*q^* = \rho^*$ (proposition 8.3.5(3)), and the fourth by the proper case of proposition 9.4.4(4) for q'. Running the same computation with the roles of E and F exchanged uses the same scheme W with its two projections interchanged, and expresses $s_j(F) \cap (s_i(E) \cap \alpha)$ as $\rho_*(\tilde{\eta}^{f+j} \cap \tilde{\xi}^{r+i} \cap \rho^*\alpha)$. The two expressions differ only in the order of two first Chern class cap operators, which commute by proposition 9.4.4(2), so they agree; this is the asserted commutativity.

4. *Functoriality.* For the proper case take $h = f$ in the fibre square above, so $f' : \mathbb{P}(f^*E) \to \mathbb{P}(E)$ is proper and $\pi_{X'}$ is proper and flat of relative dimension r. Then

$$f_*\big(s_i(f^*E) \cap \alpha'\big) = f_*\,\pi_{X'*}\big(\xi_{X'}^{r+i} \cap \pi_{X'}^*\alpha'\big) = \pi_*\,f'_*\big(f'^*\xi^{r+i} \cap \pi_{X'}^*\alpha'\big)$$

using $f \circ \pi_{X'} = \pi \circ f'$ and $\xi_{X'} = f'^*\xi$. The proper case of proposition 9.4.4(4) moves the cap outside, and proposition 8.3.6 for the same square gives $f'_*\pi_{X'}^* = \pi^* f_*$, so the right side is $\pi_*(\xi^{r+i} \cap \pi^* f_*\alpha') = s_i(E) \cap f_*\alpha'$.

For the flat case take $h = g$, so $g' : \mathbb{P}(g^*E) \to \mathbb{P}(E)$ is flat of relative dimension m. By proposition 8.3.5(3), $\pi_{X''}^* g^* = (g \circ \pi_{X''})^* = (\pi \circ g')^* = g'^*\pi^*$, and by the flat case of proposition 9.4.4(4) with $\xi_{X''} = g'^*\xi$,

$$\xi_{X''}^{r+i} \cap \pi_{X''}^* g^*\beta = g'^*\big(\xi^{r+i} \cap \pi^*\beta\big)$$

Applying $\pi_{X''*}$ and using $\pi_{X''*}g'^* = g^*\pi_*$, again proposition 8.3.6, gives $s_i(g^*E) \cap g^*\beta = g^*(s_i(E) \cap \beta)$, completing the proof.

Parts (1) and (2) say that the total Segre class is an operator of the form 1+(dimension-lowering terms), hence invertible: a series $1 + s_1 + s_2 + \cdots$ whose higher terms strictly lower dimension has a two-sided inverse of the same shape, computed by the usual recursion, and section 10.2 names that inverse the total Chern class. Part (3) is what makes any such formal manipulation legitimate, since inverting a series requires its terms to commute. Part (4) says the Segre classes are natural: they are computed in whichever space is most convenient and transported by the functorialities already available. The normalization in part (2) has an immediate structural consequence for π^* itself.

Corollary 10.1.6: Flat Pullback along a Projective Bundle is Split Injective

Let E be a vector bundle of rank $r+1$ on an algebraic scheme X, with $\pi : \mathbb{P}(E) \to X$ and tautological class ξ. Then for every k the map $\pi^* : A_k(X) \to A_{k+r}(\mathbb{P}(E))$ is injective, and $\beta \mapsto \pi_*(\xi^r \cap \beta)$ is a left inverse for it.

***Proof*:**
For $\alpha \in A_k(X)$ the composite is $\pi_*(\xi^r \cap \pi^*\alpha) = s_0(E) \cap \alpha = \alpha$ by definition 10.1.3 and proposition 10.1.5(2). A map admitting a left inverse is injective, as we sought to show.

No cycle class on the base is lost on passing to the projective bundle, and the loss can be undone by an explicit formula rather than an abstract argument. This is the mechanism that will make the flag bundle a legitimate device for proving identities among Chern classes rather than a heuristic; the iteration is carried out in section 10.2. A second consequence of proposition 10.1.5 is a computation, and it is the one that calibrates the whole normalization.

Example 10.1.7: The Segre Classes of a Trivial Bundle

Let $E = \mathcal{O}_X^{\oplus(r+1)}$ be trivial of rank $r+1$, so $\mathbb{P}(E) = X \times \mathbb{P}^r$ with π the first projection, ρ the second, and $\mathcal{O}_{\mathbb{P}(E)}(1) = \rho^*\mathcal{O}_{\mathbb{P}^r}(1)$, whence $\xi = c_1(\rho^*\mathcal{O}_{\mathbb{P}^r}(1))$. Then

$$s_0(E) = 1, \qquad s_i(E) = 0 \text{ for } i \neq 0$$

so $s(E) = 1$.

The case $i < 0$ is proposition 10.1.5(1) and the case $i = 0$ is proposition 10.1.5(2), so only $i \geq 1$ needs an argument. By (10.1) it suffices to take $X = V$ a variety of dimension k and $\alpha = [V]$. Then $\pi^*[V] = [V \times \mathbb{P}^r]$, and $[V \times \mathbb{P}^r] = \rho^*[\mathbb{P}^r]$ because ρ is flat of relative dimension k over the variety $\mathbb{P}^r$ (proposition 8.3.5(2)). The flat case of proposition 9.4.4(4) then moves the whole cap into the base:

$$\xi^{r+i} \cap \pi^*[V] = \rho^*\Big(c_1\big(\mathcal{O}_{\mathbb{P}^r}(1)\big)^{r+i} \cap [\mathbb{P}^r]\Big)$$

For $i \geq 1$ the exponent exceeds $r = \dim \mathbb{P}^r$, so the class in brackets vanishes by proposition 9.4.4(5), and therefore $s_i(E) \cap [V] = \pi_*(0) = 0$. A trivial bundle has trivial total Segre class, which is the normalization that makes $s(E)$, and with it the Chern classes built from it, an invariant of the twisting of E rather than of its rank.

The computation in example 10.1.7 used only the structure of $\mathbb{P}^r$ over a point, transported along a flat projection. The same transport, applied to a hyperplane rather than to the whole space, produces the identity that drives the main theorem: on a trivial projective bundle, the tautological class is the class of a sub-bundle of hyperplanes, so capping with ξ is the same as pushing forward from a smaller projective bundle.

Lemma 10.1.8: The Hyperplane Divisor on a Trivial Projective Bundle

Let X be an algebraic scheme, let $r \geq 1$, and let $P = X \times \mathbb{P}^r$ with projections $\pi\colon P \to X$ and $\rho\colon P \to \mathbb{P}^r$, and $\xi = c_1(\rho^*\mathcal{O}_{\mathbb{P}^r}(1))$. Fix a hyperplane $H \subseteq \mathbb{P}^r$, put $Q = X \times H$ with its closed immersion $\iota\colon Q \hookrightarrow P$, and let $\pi_Q\colon Q \to X$ be the projection. Then for every $\alpha \in A_k(X)$,

$$\xi \cap \pi^*\alpha = \iota_*\,\pi_Q^*\alpha \quad \text{in } A_{k+r-1}(P)$$

***Proof*:**
Since H is cut out by a linear form, $\mathcal{O}_{\mathbb{P}^r}(H) \cong \mathcal{O}_{\mathbb{P}^r}(1)$ by example 9.4.3(2). No component of P maps into H, so the local equations of H pull back to local equations of a Cartier divisor Q on P, and $\mathcal{O}_P(Q) \cong \rho^*\mathcal{O}_{\mathbb{P}^r}(1)$ because $\mathcal{O}(D)$ is formed from those same local equations. (This is the pullback of a Cartier divisor along an arbitrary morphism meeting the support properly, not the flat pullback of proposition 9.3.3, whose relative-dimension hypothesis ρ need not satisfy when X has components of different dimensions.) By definition 9.4.1 this identifies the operator: $\xi \cap \pi^*\alpha = Q \cdot \pi^*\alpha$.

Both sides of the asserted identity are additive homomorphisms $A_k(X) \to A_{k+r-1}(P)$, the left by proposition 8.3.7 and proposition 9.2.4, the right by proposition 8.3.7 and theorem 8.2.3, so it suffices to check them on the class of a k-dimensional subvariety $W \subseteq X$. The morphism π is flat of relative dimension r, being the base change of $\mathbb{P}^r \to \operatorname{Spec} \overline{k}$, so $\pi^*[W] = [W \times \mathbb{P}^r]$ by definition 8.3.3, the preimage $W \times \mathbb{P}^r$ being integral of dimension $k + r$ and hence carrying multiplicity one. This subvariety is not contained in $|Q| = X \times H$, so the first clause of definition 9.2.1 applies: writing s for the rational section of $\mathcal{O}_P(Q)$ cutting out Q, which is the pullback along ρ of the linear form defining H, its restriction to $W \times \mathbb{P}^r$ has divisor

$$\left[\operatorname{div}\left(s|_{W\times\mathbb{P}^r}\right)\right] = [W \times H]$$

with multiplicity one, since a linear form vanishes to order one along the hyperplane it defines. Hence $Q \cdot \pi^*[W] = [W \times H]$. On the other side π_Q is flat of relative dimension $r-1$, so $\pi_Q^*[W] = [W \times H]$ by the same reasoning, and ι_* carries this class to $[W \times H]$ in $Z_{k+r-1}(P)$, the closed immersion having degree one on each subvariety (definition 8.2.1). The two sides therefore agree on every generator, completing the proof.

The lemma converts a divisor operation into a pushforward, and pushforward is what the localization sequence controls. That is the whole strategy for the theorem below: peel $\mathbb{P}^r$ apart into a hyperplane and its affine complement, use theorem 8.4.6 on the complement, use the lemma to recognize the hyperplane contribution as one more power of ξ, and induct. The general bundle is then reduced to the trivial one by a second induction, over the closed subschemes of the base, since a bundle is trivial over some dense open set. Injectivity, by contrast, needs no induction at all: the Segre classes already invert the construction.

Theorem 10.1.9: The Projective Bundle Formula

Let E be a vector bundle of rank $r+1$ on an algebraic scheme X, with $\pi\colon \mathbb{P}(E) \to X$ and tautological class ξ. For every k the homomorphism

$$\Psi\colon \bigoplus_{i=0}^{r} A_{k-r+i}(X) \longrightarrow A_k\big(\mathbb{P}(E)\big), \qquad (\alpha_0,\dots,\alpha_r) \longmapsto \sum_{i=0}^{r} \xi^{\,i} \cap \pi^*\alpha_i$$

is an isomorphism. Every class $\beta \in A_k(\mathbb{P}(E))$ is thus uniquely of the form $\beta = \sum_{i=0}^{r} \xi^{\,i} \cap \pi^*\alpha_i$ with $\alpha_i \in A_{k-r+i}(X)$, and the components are recovered from β by the triangular system

$$\pi_*\big(\xi^{\,m} \cap \beta\big) = \alpha_{r-m} + \sum_{j=1}^{m} s_j(E) \cap \alpha_{r-m+j}, \qquad 0 \le m \le r \tag{10.2}$$

***Proof*:**
Each summand of Ψ lands in $A_k(\mathbb{P}(E))$: for $\alpha_i \in A_{k-r+i}(X)$ the class $\pi^*\alpha_i$ lies in $A_{k+i}(\mathbb{P}(E))$ and capping with ξ^i lowers dimension by i. The map is additive because π^* and $\xi \cap -$ are.

Step 1: the triangular system, and injectivity. Let $\beta = \sum_{i=0}^{r} \xi^i \cap \pi^*\alpha_i$ and fix $m \ge 0$. Then

$$\pi_*\big(\xi^{\,m} \cap \beta\big) = \sum_{i=0}^{r} \pi_*\big(\xi^{\,m+i} \cap \pi^*\alpha_i\big) = \sum_{i=0}^{r} s_{m+i-r}(E) \cap \alpha_i$$

directly from definition 10.1.3. By proposition 10.1.5(1) the terms with $m+i-r<0$ vanish, and by proposition 10.1.5(2) the term with $m+i-r=0$ is α_{r-m}; writing $j = m+i-r$ for the surviving terms, which satisfy $1 \le j \le m$ since $i \le r$, gives exactly (10.2).

Suppose now $\Psi(\alpha_0,\dots,\alpha_r) = 0$. Taking $m=0$ in (10.2) gives $\alpha_r = 0$. Assume inductively that $\alpha_r = \cdots = \alpha_{r-m+1} = 0$ for some $m \le r$; then every correction term in (10.2) involves one of these vanishing classes, so the identity reduces to $\alpha_{r-m} = 0$. Descending from $m=0$ to $m=r$

kills every component, so Ψ is injective. This argument used no hypothesis on X beyond those standing.

Step 2: surjectivity for a trivial bundle. Let $E = \mathcal{O}_X^{\oplus(r+1)}$, so $\mathbb{P}(E) = X \times \mathbb{P}^r$ and $\xi = c_1(\rho^*\mathcal{O}_{\mathbb{P}^r}(1))$ for the second projection ρ. The claim is proved by induction on r, uniformly in the base X.

For $r = 0$ one has $\mathbb{P}(E) = X$ with $\pi = \mathrm{id}$, so $\Psi(\alpha_0) = \alpha_0$ is the identity map of $A_k(X)$, which is surjective. Let $r \geq 1$ and assume the claim for $r-1$ over every base. Fix a hyperplane $H \subseteq \mathbb{P}^r$, put $Q = X \times H$ with closed immersion $\iota\colon Q \hookrightarrow P = X \times \mathbb{P}^r$, and let $j\colon X \times \mathbb{A}^r \hookrightarrow P$ be the inclusion of the open complement, $\mathbb{A}^r = \mathbb{P}^r \setminus H$. The localization sequence (theorem 8.4.1) reads

$$A_k(Q) \xrightarrow{\iota_*} A_k(P) \xrightarrow{j^*} A_k(X \times \mathbb{A}^r) \longrightarrow 0$$

Let $q = \pi \circ j\colon X \times \mathbb{A}^r \to X$ be the projection. By affine-bundle invariance (theorem 8.4.6) the pullback $q^*\colon A_{k-r}(X) \to A_k(X \times \mathbb{A}^r)$ is an isomorphism, and $q^* = j^*\pi^*$ by proposition 8.3.5(3). Given $\beta \in A_k(P)$, there is therefore $\alpha_0 \in A_{k-r}(X)$ with $j^*\beta = q^*\alpha_0 = j^*(\pi^*\alpha_0)$, whence $\beta - \pi^*\alpha_0$ lies in $\ker j^* = \operatorname{im} \iota_*$ by exactness: say

$$\beta = \pi^*\alpha_0 + \iota_*\gamma, \qquad \gamma \in A_k(Q)$$

Now $Q = X \times H \cong X \times \mathbb{P}^{r-1}$ is the trivial projective bundle of one lower fibre dimension, and its tautological class is $\xi_Q = \iota^*\xi$: choosing a second hyperplane $H' \neq H$, the restriction $\mathcal{O}_{\mathbb{P}^r}(1)|_H \cong \mathcal{O}_{\mathbb{P}^r}(H')|_H$ is the line bundle of the hyperplane $H' \cap H$ of H, which is $\mathcal{O}_{\mathbb{P}^{r-1}}(1)$ under the identification $H \cong \mathbb{P}^{r-1}$. By the inductive hypothesis,

$$\gamma = \sum_{i=0}^{r-1} \xi_Q^{\,i} \cap \pi_Q^*\alpha_i', \qquad \alpha_i' \in A_{k-(r-1)+i}(X)$$

Pushing forward, the proper case of proposition 9.4.4(4) applied i times with $\xi_Q = \iota^*\xi$ gives $\iota_*(\xi_Q^{\,i} \cap \pi_Q^*\alpha_i') = \xi^{\,i} \cap \iota_*\pi_Q^*\alpha_i'$, and lemma 10.1.8 identifies $\iota_*\pi_Q^*\alpha_i' = \xi \cap \pi^*\alpha_i'$. Hence

$$\iota_*\gamma = \sum_{i=0}^{r-1} \xi^{\,i+1} \cap \pi^*\alpha_i'$$

and setting $\alpha_{i+1} = \alpha_i' \in A_{k-r+(i+1)}(X)$ for $0 \leq i \leq r-1$ exhibits $\beta = \sum_{i=0}^{r} \xi^{\,i} \cap \pi^*\alpha_i$. This completes the induction on r, so Ψ is surjective for every trivial bundle.

Step 3: surjectivity in general. The scheme X is of finite type over $\overline{k}$, hence Noetherian, so its closed subschemes satisfy the descending chain condition and the following suffices: if $\Psi_{E|_W}$ is surjective in every degree for every proper closed subscheme $W \subsetneq X$, then Ψ_E is surjective. The empty scheme starts the induction.

First produce a dense trivializing open. The scheme X has finitely many irreducible components, with generic points $\eta_1, \ldots, \eta_s$. For each t choose an open $V_t \ni \eta_t$ over which E is trivial, and set $U_t = V_t \setminus \bigcup_{u \neq t} \overline{\{\eta_u\}}$, still an open neighbourhood of η_t since η_t lies on no other component. The U_t are pairwise disjoint: each U_u is contained in $\overline{\{\eta_u\}}$, because the components cover X, while U_t misses $\overline{\{\eta_u\}}$ for $u \neq t$. Their union $U = \bigsqcup_t U_t$ is an open set containing every generic point, hence dense, and $E|_U$ is trivial of rank $r+1$, being trivial on each of the disjoint pieces. If $U = X$ then E is trivial and Step 2 applies; otherwise let $Z = X \setminus U$ with its reduced structure, a proper closed subscheme.

Write $j\colon U \hookrightarrow X$ and $i_Z\colon Z \hookrightarrow X$ for the inclusions, and $j'\colon \mathbb{P}(E|_U) \hookrightarrow \mathbb{P}(E)$, $\iota\colon \mathbb{P}(E|_Z) \hookrightarrow \mathbb{P}(E)$ for the induced ones; these are the open and closed pieces of $\mathbb{P}(E)$, since forming $\mathbb{P}(E)$ commutes with base change. Let $\beta \in A_k(\mathbb{P}(E))$. Restricting to the trivializing part and applying Step 2 over U,

$$j'^*\beta = \sum_{i=0}^{r} \xi_U^{\,i} \cap \pi_U^* \alpha_i^U, \qquad \alpha_i^U \in A_{k-r+i}(U)$$

By theorem 8.4.1 the restriction $j^*\colon A_{k-r+i}(X) \to A_{k-r+i}(U)$ is surjective, so each α_i^U lifts to some $\alpha_i \in A_{k-r+i}(X)$. Flat pullback along j' commutes with π^*, since $\pi \circ j' = j \circ \pi_U$ gives $j'^*\pi^* = \pi_U^* j^*$ by proposition 8.3.5(3), and with capping by ξ by the flat case of proposition 9.4.4(4), using $\xi_U = j'^*\xi$. Hence $j'^*\big(\beta - \Psi_E(\alpha_0, \ldots, \alpha_r)\big) = 0$, and the localization sequence for the closed $\mathbb{P}(E|_Z) \subseteq \mathbb{P}(E)$ with open complement $\mathbb{P}(E|_U)$ produces $\delta \in A_k(\mathbb{P}(E|_Z))$ with

$$\beta - \Psi_E(\alpha_0, \ldots, \alpha_r) = \iota_*\delta$$

Since $Z \subsetneq X$ is a proper closed subscheme, the inductive hypothesis writes $\delta = \sum_{i=0}^{r} \xi_Z^{\,i} \cap \pi_Z^*\epsilon_i$ with $\epsilon_i \in A_{k-r+i}(Z)$. The proper case of proposition 9.4.4(4) with $\xi_Z = \iota^*\xi$, followed by proposition 8.3.6 for the fibre square over i_Z (whose horizontal map is proper and whose vertical map π is flat), gives

$$\iota_*\big(\xi_Z^{\,i} \cap \pi_Z^*\epsilon_i\big) = \xi^{\,i} \cap \iota_*\pi_Z^*\epsilon_i = \xi^{\,i} \cap \pi^*\big(i_{Z*}\epsilon_i\big)$$

Therefore $\beta = \Psi_E(\alpha_0 + i_{Z*}\epsilon_0, \ldots, \alpha_r + i_{Z*}\epsilon_r)$ lies in the image of Ψ_E. This closes the Noetherian induction, so Ψ is surjective for every vector bundle, and with Step 1 it is an isomorphism, completing the proof.

The theorem computes the Chow groups of a projective bundle from those of its base, and it does so with an explicit basis and an explicit inversion. The classes $1, \xi, \ldots, \xi^r$ behave like a basis of $A_*(\mathbb{P}(E))$ over $A_*(X)$, and it is standard to say that $A_*(\mathbb{P}(E))$ is a free $A_*(X)$-module on them; that phrasing anticipates a ring structure on $A_*(X)$ that does not exist yet, so the content here is the isomorphism Ψ of graded groups together with the fact that its components are the operators $\xi^i \cap \pi^*(-)$. The inversion (10.2) is what makes the statement usable in practice: to identify the components of a given class one pushes forward its caps with successive powers of ξ, and the Segre classes appear as the correction terms in a triangular system that can be solved from the top down. This is also the precise sense in which the Segre classes measure the failure of ξ to behave like the hyperplane class on a product, since a trivial bundle has all corrections zero by example 10.1.7.

Taking $X = \operatorname{Spec} \overline{k}$ and $E = \overline{k}^{\,n+1}$ recovers $A_k(\mathbb{P}^n) = \mathbb{Z}\left(\xi^{\,n-k} \cap [\mathbb{P}^n]\right)$, the statement of theorem 8.5.2: the group $A_j(\operatorname{Spec} \overline{k})$ is $\mathbb{Z}$ for $j = 0$ and zero otherwise, so only the index $i = n - k$ survives in the sum, and $\pi^*[\operatorname{Spec} \overline{k}] = [\mathbb{P}^n]$. The theorem is thus the exact generalization of the projective-space computation from a point to an arbitrary base, which is how it was announced in section 8.5, and its proof runs by reducing to that computation twice: once through example 9.4.7 in the normalization $s_0(E) = 1$, and once through affine-bundle invariance in Step 2. The corresponding statement in singular cohomology, that $H^*(\mathbb{P}(E))$ is free over $H^*(X)$ on the powers of the class of $\mathcal{O}(1)$, holds as well, but it enters the theory at a different point: in [51] the topological Chern classes are defined as pullbacks of the polynomial generators of $H^*(BU(n))$, and the projective-bundle statement is a consequence, proved once they are in hand. Here the dependence runs the other way, the formula above coming first and the Chern operators of section 10.2 being defined by inverting the total Segre class; and the argument owes nothing to topology, being valid over any algebraically closed field and on singular schemes.

Exercise 10.1.1

1. Let L be a line bundle on an algebraic scheme X. Using example 10.1.2(2), show that $s_i(L) \cap \alpha = (-1)^i c_1(L)^i \cap \alpha$ for every $i \geq 0$ and every $\alpha \in A_k(X)$, and deduce that the operators $s(L) \cap -$ and $(1 + c_1(L)) \cap -$ are mutually inverse on $A_k(X)$. Explain why the series terminates on any fixed class.

2. Let E have rank $r+1$ on X, with $\pi \colon \mathbb{P}(E) \to X$. Show that $\pi_* \pi^* \alpha = 0$ for every $\alpha \in A_*(X)$ when $r \geq 1$, while $\pi_* \pi^* = \mathrm{id}$ when $r = 0$. Show also that $\beta \mapsto \pi_*(\xi^r \cap \beta)$ is a left inverse of π^* for every r, and reconcile the two computations.

3. Take $X = \mathbb{P}^n$ and $E = \mathcal{O}_{\mathbb{P}^n}^{\oplus(m+1)}$, so that $\mathbb{P}(E) = \mathbb{P}^n \times \mathbb{P}^m$ with π the first projection. Use theorem 10.1.9 and theorem 8.5.2 to show that $A_k(\mathbb{P}^n \times \mathbb{P}^m)$ is free abelian of rank equal to the number of pairs (a, b) with $0 \leq a \leq m$, $0 \leq b \leq n$ and $a + b = k$, and identify a basis geometrically.

4. Let X be a complete variety of dimension n and E a vector bundle of rank $r+1$ on X. Show that
$$\int_X s_n(E) \cap [X] = \int_{\mathbb{P}(E)} \xi^{n+r} \cap [\mathbb{P}(E)]$$
and verify that both sides vanish for E trivial and $n \geq 1$.

5. Let E have rank $r+1$ on X and let $\alpha \in A_k(X)$. By theorem 10.1.9 there are unique classes $\beta_i \in A_{k-1-r+i}(X)$ with $\xi^{r+1} \cap \pi^* \alpha = \sum_{i=0}^r \xi^i \cap \pi^* \beta_i$. Determine β_r and β_{r-1} in terms of the Segre operators of E applied to α, and describe the recursion that determines the remaining β_i.

10.2 Chern Classes

The Segre operators of section 10.1 record everything a vector bundle contributes to the Chow group, but in a form nothing can be computed with. The list $s_0(E), s_1(E), s_2(E), \ldots$ does not terminate, so no finite collection of them is an invariant attached to E; already for a line bundle every $s_i(L)$ is nonzero as long as $c_1(L)^i$ is. Worse, the operators attached to a direct sum are not the product of those attached to the summands, so a bundle assembled from simpler pieces has Segre operators that cannot be read off the pieces. Inverting the total Segre operator repairs both defects at once, and the inverse is what the rest of the book uses.

The section establishes four properties of the resulting Chern operators: they vanish above the rank of the bundle, they are natural for flat pullback and satisfy a projection formula, they are multiplicative on short exact sequences, and in rank one they reduce to the operator $c_1(L) \cap -$ of section 9.4. The last two are proved by the splitting principle, which replaces X by a tower of projective bundles over it on which every bundle in sight is filtered by line bundles. That replacement is legitimate rather than heuristic because the projective-bundle formula makes π^* injective, so an identity verified upstairs descends.

Throughout, E is a vector bundle of rank $e \geq 1$ on an algebraic scheme X, the projection $\pi \colon \mathbb{P}(E) \to X$ has relative dimension $r = e - 1$, and $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$ is the tautological class of section 10.1, so that

$$s_i(E) \cap \alpha = \pi_*\big(\xi^{r+i} \cap \pi^* \alpha\big)$$

for $\alpha \in A_*(X)$, with $s_0(E)$ the identity and $s_i(E) = 0$ for $i < 0$. Every vector bundle named below is assumed of positive rank: the operators are manufactured from $\mathbb{P}(E)$, which is empty for the zero bundle, so neither $s(0)$ nor $c(0)$ is defined.

The first point to settle is that the inversion makes sense. A Segre operator lowers the dimension of a cycle class, and on a scheme of finite dimension an operator that lowers dimension by too much has nowhere to land; that is the whole reason the total operator is invertible.

Lemma 10.2.1: Dimension-Lowering Operators Vanish Above the Dimension

Let X be an algebraic scheme of dimension n. Call an operator on $A_*(X)$ *homogeneous of degree d* if it carries $A_k(X)$ into $A_{k-d}(X)$ for every k. Then every operator on $A_*(X)$ homogeneous of degree $d > n$ is the zero operator.

***Proof*:**
A k-dimensional cycle class satisfies $0 \le k \le n$, because X has no subvarieties of negative dimension and none of dimension exceeding n; hence $A_j(X) = 0$ for $j < 0$ and for $j > n$. An operator homogeneous of degree $d > n$ carries every $A_k(X)$ with $0 \le k \le n$ into $A_{k-d}(X)$ with $k - d < 0$, whose target vanishes, completing the proof.

Each $s_i(E)$ is homogeneous of degree i (section 10.1): π^* raises dimension by r and capping with ξ lowers it by one (definition 9.4.1). The lemma therefore applies to them and to every composite of them.

Lemma 10.2.2: Invertibility of the Total Segre Operator

Let X be an algebraic scheme of dimension n and E a vector bundle on X. For $\alpha \in A_k(X)$ only finitely many of the classes $s_i(E) \cap \alpha$ are nonzero, so

$$s(E) := \sum_{i \ge 0} s_i(E)$$

is a well-defined operator on $A_*(X)$. It is invertible: there is a unique operator μ on $A_*(X)$ with

$$\mu \circ s(E) = s(E) \circ \mu = \mathrm{id}$$

and it is given by the finite sum $\mu = \sum_{j=0}^{n} (-1)^j \nu^j$, where $\nu = s(E) - \mathrm{id} = \sum_{i \ge 1} s_i(E)$.

***Proof*:**
By lemma 10.2.1, $s_i(E) = 0$ for $i > n$, so the sum defining $s(E)$ has at most $n+1$ nonzero terms and is a well-defined operator.

The same lemma makes $\nu = \sum_{i \ge 1} s_i(E)$ nilpotent. Expanding ν^{n+1} produces a sum of composites $s_{i_1}(E) \circ \cdots \circ s_{i_{n+1}}(E)$ with every $i_t \ge 1$, each homogeneous of degree $i_1 + \cdots + i_{n+1} \ge n+1 > n$, hence each zero by lemma 10.2.1. Thus $\nu^{n+1} = 0$, and the finite sum $\mu = \sum_{j=0}^{n} (-1)^j \nu^j$ satisfies

$$\mu\,(\mathrm{id} + \nu) = (\mathrm{id} + \nu)\,\mu = \sum_{j=0}^{n} (-1)^j \nu^j + \sum_{j=0}^{n} (-1)^j \nu^{j+1} = \mathrm{id} + (-1)^n \nu^{n+1} = \mathrm{id}$$

the two sums telescoping. Since $s(E) = \mathrm{id} + \nu$, the operator μ is a two-sided inverse. Uniqueness is the usual argument for inverses in a ring: if μ' is another, then $\mu' = \mu' \circ (s(E) \circ \mu) = (\mu' \circ s(E)) \circ \mu = \mu$, completing the proof.

Nilpotence, not any positivity or finiteness of E, is what powers the inversion. The bundle may have arbitrary rank and X may be singular, reducible, or non-reduced; all that is used is that a cycle class cannot be cut down more times than the dimension of the ambient scheme allows. The inverse is therefore available on every algebraic scheme, which is the reason Chern classes in the Chow setting need no smoothness hypothesis.

Definition 10.2.3: Chern Classes of a Vector Bundle

Let E be a vector bundle on an algebraic scheme X. The *total Chern class* of E is the operator

$$c(E) := s(E)^{-1}$$

on $A_*(X)$ furnished by lemma 10.2.2, and the *i-th Chern class* $c_i(E)$ is its homogeneous component of degree i, an operator

$$c_i(E) \cap - \colon A_k(X) \longrightarrow A_{k-i}(X)$$

for every k. Equivalently, the $c_i(E)$ are determined by $c_0(E) = \mathrm{id}$ together with

$$\sum_{i+j=m} c_i(E) \circ s_j(E) = 0 \qquad \text{for every } m \geq 1$$

Unwinding the recursion $c_m(E) = -\sum_{j=1}^{m} c_{m-j}(E) \circ s_j(E)$ expresses each Chern class as an integer polynomial in the Segre classes:

$$\begin{aligned} &c_0(E) = \mathrm{id}, \qquad c_1(E) = -s_1(E), \qquad c_2(E) = s_1(E)^2 - s_2(E), \\ &c_3(E) = -s_1(E)^3 + 2\, s_1(E) s_2(E) - s_3(E) \end{aligned}$$

where a product of operators means composition. Each $c_m(E)$ is thus a $\mathbb{Z}$-combination of composites of $s_1(E), \ldots, s_m(E)$, a fact used below to transfer identities from Segre classes to Chern classes.

10.2.4 Note: Chern Classes Are Operators, Not Elements A Chern class here is an endomorphism of the graded group $A_*(X)$, not a member of it. There is no ring $A^*(X)$ yet: the intersection product that turns cycle classes into a ring for smooth X rests on the deformation construction of chapter 11, and until it is available the only multiplication in sight is composition of operators. Expressions such as $c_1(E)^2$ or $c(E)c(F)$ always mean composites, and an equation between them is an equation of maps $A_*(X) \to A_*(X)$, asserted for every class α. Once the product exists, capping against the fundamental class converts each operator into an element of $A^*(X)$ and the identities below become identities in a ring.

Every computation in this section runs through one imported object. The projective bundle carries a tautological line inside the pullback of E, and because this book and the scheme-theory volume use opposite conventions for $\mathbb{P}(-)$, that import has to be dualized in the open before it is used.

10.2.5 Note: The Tautological Subbundle and Its Quotient By box 7.3.10, the bundle of lines $\mathbb{P}(E)$ used here is $\operatorname{Proj}\left(\operatorname{Sym}^\bullet E^\vee\right)$, which is the projective bundle definition 5.3.37 of the *dual* sheaf. The universal surjection $\pi^*(E^\vee) \twoheadrightarrow \mathcal{O}_{\mathbb{P}(E)}(1)$ supplied there dualizes to an inclusion of vector bundles

$$\mathcal{O}_{\mathbb{P}(E)}(-1) \hookrightarrow \pi^* E$$

whose fibre over a point of $\mathbb{P}(E)$ lying above $x \in X$ and corresponding to a line $\ell \subseteq E_x$ is that line. Its cokernel Q is locally free of rank $\operatorname{rank} E - 1$, since the inclusion is a subbundle inclusion: over an open $U \subseteq X$ trivializing E the total space is $\mathbb{P}^r \times U$ and the inclusion is the tautological one on the first factor. Failing to dualize here replaces E by $E^\vee$ throughout and negates every odd Chern class, which is why the convention is restated at each import.

The definition inverts a series and so says nothing recognizable on its face. The rank-one case makes it concrete and settles the compatibility the notation presumes: the symbol c_1 has already been used in section 9.4 for the divisor operator of a line bundle, and the two meanings must agree.

Proposition 10.2.6: Chern Classes of a Line Bundle

Let L be a line bundle on an algebraic scheme X. Then

$$c(L) = \mathrm{id} + c_1(L)$$

where $c_1(L)$ on the right is the first Chern class operator of definition 9.4.1. In particular $c_1(L)$ in the sense of definition 10.2.3 is that operator, and $c_i(L) = 0$ for $i \geq 2$.

***Proof*:**
Both inputs are already computed in section 10.1: example 10.1.2(2) identifies $\mathbb{P}(L) = X$ with π the identity and $\mathcal{O}_{\mathbb{P}(L)}(1) = L^\vee$, whence $\xi = c_1(L^\vee) = -c_1(L)$, and example 10.1.4 evaluates the resulting Segre operators as $s_i(L) \cap \alpha = (-1)^i c_1(L)^i \cap \alpha$, so $s(L) = \sum_{i \geq 0} (-1)^i c_1(L)^i$. The operator $c_1(L)$ is nilpotent by the lemma 10.2.1, so this sum is the inverse of $\mathrm{id} + c_1(L)$: indeed

$$\left(\sum_{i \geq 0} (-1)^i c_1(L)^i\right)\left(\mathrm{id} + c_1(L)\right) = \mathrm{id}$$

the terms cancelling in pairs and the tail vanishing by nilpotence. Inverses being unique (lemma 10.2.2), $c(L) = s(L)^{-1} = \mathrm{id} + c_1(L)$. Reading off homogeneous components gives $c_1(L)$ in degree one and zero in every degree $i \geq 2$, as we sought to show.

The agreement is what licenses a single symbol for both constructions, and it fixes the sign convention of the whole theory: had $\mathbb{P}(E)$ been taken as the bundle of hyperplanes rather than of lines, the tautological class would be $c_1(L)$ instead of $-c_1(L)$ and the inversion would produce $c_1(L^\vee)$ where $c_1(L)$ is wanted. Box 7.3.10 therefore cannot be skipped. The rank-one case also shows that Chern classes terminate while Segre classes do not: $c_i(L) = 0$ beyond $i = 1$, while $s_i(L)$ is nonzero for every i with $c_1(L)^i \neq 0$.

Example 10.2.7: The Twisting Sheaves on Projective Space

Let $X = \mathbb{P}^n$ and $L = \mathcal{O}_{\mathbb{P}^n}(d)$ for an integer d. Write $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$, the hyperplane operator of example 9.4.7. Additivity of the first Chern class (proposition 9.4.4(3)) gives $c_1(\mathcal{O}_{\mathbb{P}^n}(d)) = d\,H$, so by proposition 10.2.6

$$c(\mathcal{O}_{\mathbb{P}^n}(d)) = \mathrm{id} + d\,H$$

Capping against the fundamental class and using $H^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ from example 9.4.7,

$$c(\mathcal{O}_{\mathbb{P}^n}(d)) \cap [\mathbb{P}^n] = [\mathbb{P}^n] + d\,[\mathbb{P}^{n-1}]$$

so the total Chern class of $\mathcal{O}(d)$ records the single integer d as the multiple of the hyperplane class it produces. The corresponding Segre operator is the full geometric series $s(\mathcal{O}_{\mathbb{P}^n}(d)) = \sum_{i\geq 0}(-d)^i H^i$, whose cap against $[\mathbb{P}^n]$ is $\sum_{i=0}^{n}(-d)^i[\mathbb{P}^{n-i}]$: every dimension carries a nonzero coefficient when $d \neq 0$, while the Chern class stops after one step. This is the terminating behaviour that theorem 10.2.9 establishes in general.

Termination in rank one came from an explicit computation. In general it comes from the projective-bundle formula, which pins down exactly one relation among the powers of ξ and identifies its coefficients as the Chern classes. The following criterion extracts from that formula the tool used twice below: a class on $\mathbb{P}(E)$ is detected by finitely many pushforwards.

Lemma 10.2.8: A Vanishing Criterion on the Projective Bundle

Let E be a vector bundle of rank $e = r+1$ on X, with $\pi\colon \mathbb{P}(E) \to X$ and $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$. A class $\beta \in A_*(\mathbb{P}(E))$ vanishes if and only if

$$\pi_*\big(\xi^m \cap \beta\big) = 0 \qquad \text{for } 0 \leq m \leq r$$

***Proof*:**

If $\beta = 0$ then every $\pi_*(\xi^m \cap \beta)$ vanishes, since capping and pushforward are homomorphisms.

Conversely, suppose the $r+1$ pushforwards vanish. By the projective-bundle formula (theorem 10.1.9) there are unique classes $\alpha_0, \ldots, \alpha_r \in A_*(X)$ with

$$\beta = \sum_{j=0}^{r} \xi^j \cap \pi^*\alpha_j$$

Capping with ξ^m and pushing forward, the definition of the Segre operators gives

$$\pi_*\big(\xi^m \cap \beta\big) = \sum_{j=0}^{r} \pi_*\big(\xi^{m+j} \cap \pi^*\alpha_j\big) = \sum_{j=0}^{r} s_{m+j-r}(E) \cap \alpha_j$$

Take $m = 0$. Every index $j < r$ contributes $s_{j-r}(E) = 0$, since $j - r < 0$, so the sum reduces to $s_0(E) \cap \alpha_r = \alpha_r$, and the hypothesis forces $\alpha_r = 0$. Proceed by descending induction: suppose $\alpha_r = \cdots = \alpha_{r-m+1} = 0$ for some $1 \leq m \leq r$. In the displayed sum for that m, the indices $j < r-m$ contribute zero because $m + j - r < 0$, the indices $j > r-m$ contribute zero because $\alpha_j = 0$ by the inductive hypothesis, and the single index $j = r-m$ contributes $s_0(E) \cap \alpha_{r-m} = \alpha_{r-m}$. The hypothesis forces $\alpha_{r-m} = 0$, and the induction runs down to $m = r$, giving $\alpha_0 = 0$. All coefficients vanish, so $\beta = 0$, completing the proof.

With a criterion for vanishing in hand, the relation defining the Chern classes can be produced. The statement below is the one that makes them computable: on $\mathbb{P}(E)$ the power ξ^e is not independent of the lower powers, and the coefficients expressing it in terms of them are exactly the Chern classes of E. Everything else in this section is extracted from it.

Theorem 10.2.9: The Chern Relation on the Projective Bundle

Let E be a vector bundle of rank $e = r+1$ on an algebraic scheme X, with $\pi\colon \mathbb{P}(E) \to X$ and $\xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$.

1. For every $\alpha \in A_*(X)$,

$$\sum_{i=0}^{e} \xi^{\,e-i} \cap \pi^*\big(c_i(E) \cap \alpha\big) = 0 \quad \text{in } A_*(\mathbb{P}(E))$$

2. $c_i(E) = 0$ for every $i > e$; that is, the Chern classes of E terminate at its rank.

3. The operators $c_0(E) = \mathrm{id}, c_1(E), \dots, c_e(E)$ are the unique operators on $A_*(X)$, homogeneous of degrees $0, 1, \dots, e$ respectively, whose degree-zero member is the identity and which satisfy the relation in (1).

***Proof*:**
Write $\gamma_\alpha = \sum_{i=0}^{e} \xi^{\,e-i} \cap \pi^*(c_i(E) \cap \alpha)$ for the class in question. Capping with $\xi^{\,m}$ and pushing forward, the definition of the Segre operators gives, for every $m \geq 0$,

$$\pi_*\big(\xi^{\,m} \cap \gamma_\alpha\big) = \sum_{i=0}^{e} \pi_*\Big(\xi^{\,m+e-i} \cap \pi^*\big(c_i(E) \cap \alpha\big)\Big) = \sum_{i=0}^{e} s_{m+1-i}(E) \cap \big(c_i(E) \cap \alpha\big) \tag{10.3}$$

using $m + e - i = r + (m+1-i)$, valid because $m + e - i \geq 0$ for $0 \leq i \leq e$ and $m \geq 0$, with the convention $s_j(E) = 0$ for $j < 0$ absorbing the terms where $i > m+1$.

1. Fix m with $0 \leq m \leq r$ and set $n = m+1$, so $1 \leq n \leq e$. In (10.3) the terms with $i > n$ vanish, so the right side is $\sum_{i=0}^{n} s_{n-i}(E) c_i(E) \cap \alpha$, which is the degree-$n$ homogeneous component of $s(E) \circ c(E) = \mathrm{id}$ applied to α. Since $n \geq 1$ that component is zero. Thus $\pi_*(\xi^{\,m} \cap \gamma_\alpha) = 0$ for $0 \leq m \leq r$, and lemma 10.2.8 gives $\gamma_\alpha = 0$.

2. By (1) the class γ_α vanishes, hence $\pi_*(\xi^{\,m} \cap \gamma_\alpha) = 0$ for *every* $m \geq 0$, not only for $m \leq r$. Reading (10.3) in that range and writing $n = m+1$, this says

$$\sum_{i=0}^{e} s_{n-i}(E) \circ c_i(E) = 0 \qquad \text{for every } n \geq 1$$

since α was arbitrary. Set $\widetilde{c} := \sum_{i=0}^{e} c_i(E)$, the truncation of the total Chern class at degree e. The displayed identities, together with $s_0(E) \circ c_0(E) = \mathrm{id}$ in degree zero, say precisely that $s(E) \circ \widetilde{c} = \mathrm{id}$. The inverse of $s(E)$ is two-sided and unique (lemma 10.2.2), so $\widetilde{c} = c(E)$, and comparing homogeneous components gives $c_i(E) = 0$ for $i > e$.

3. Let $c'_0 = \mathrm{id}, c'_1, \dots, c'_e$ be operators, homogeneous of the stated degrees, satisfying the

relation of (1) for every α. Subtracting the two relations, the terms with $i = 0$ cancel and

$$\sum_{i=1}^{e} \xi^{\,e-i} \cap \pi^*\Big(\big(c_i(E) - c_i'\big) \cap \alpha\Big) = 0$$

The exponents $e - i$ run over $0, 1, \ldots, e - 1 = r$, so this is a representation of the zero class in the form supplied by theorem 10.1.9. Uniqueness of that representation forces $(c_i(E) - c_i') \cap \alpha = 0$ for each i, and since α was arbitrary, $c_i' = c_i(E)$ for $1 \le i \le e$, completing the proof.

Part (2) is what gives a vector bundle a finite amount of Chern data: a rank-e bundle has exactly e potentially nonzero Chern classes, $c_1(E), \ldots, c_e(E)$, and the top one $c_e(E)$ carries the obstruction to finding a nowhere-vanishing section, as exercise 10.2.1 (3) below makes precise. Part (3) is the more useful of the three in practice. It says the Chern classes need not be computed by inverting a Segre series at all: any operators satisfying the relation on $\mathbb{P}(E)$ *are* the Chern classes. Every identity proved below is obtained by producing the relation from geometry and reading off its coefficients, and the Segre definition is never returned to.

The functoriality of the Segre operators transfers to the Chern classes for the same reason it held for them: the projective bundle of a pulled-back bundle is the pullback of the projective bundle. Two bundles on the same base are compared on the fibre product of their projective bundles, which is what makes their Chern classes commute.

Proposition 10.2.10: Properties of the Chern Class Operators

Let X be an algebraic scheme and E, F vector bundles on X.

1. *Commutativity.* For all $i, j \ge 0$ and $\alpha \in A_*(X)$,

$$c_i(E) \cap \big(c_j(F) \cap \alpha\big) = c_j(F) \cap \big(c_i(E) \cap \alpha\big)$$

2. *Projection formula.* If $f\colon X' \to X$ is proper and $\alpha' \in A_*(X')$, then

$$f_*\big(c_i(f^*E) \cap \alpha'\big) = c_i(E) \cap f_*\alpha'$$

3. *Flat pullback.* If $g\colon X'' \to X$ is flat of relative dimension m and $\alpha \in A_*(X)$, then

$$c_i(g^*E) \cap g^*\alpha = g^*\big(c_i(E) \cap \alpha\big)$$

***Proof*:**
Each $c_i(E)$ is a $\mathbb{Z}$-combination of composites of $s_1(E), \ldots, s_i(E)$ by the recursion following definition 10.2.3, so each of the three statements follows from the corresponding statement for Segre classes: expand the polynomials and apply the Segre statement to each factor of each monomial in turn. All three Segre statements are already proved in section 10.1.

1. *Commutativity.* proposition 10.1.5(3) gives $s_i(E) \cap (s_j(F) \cap \alpha) = s_j(F) \cap (s_i(E) \cap \alpha)$. Polynomials in commuting operators commute, so $c_i(E)$ and $c_j(F)$ commute as well.

2. *Proper pushforward.* This is the first identity of proposition 10.1.5(4), $f_*(s_i(f^*E) \cap \alpha') =$

$s_i(E) \cap f_*\alpha'$, together with the fact that f_* is additive, so the identity passes through any $\mathbb{Z}$-combination of composites.

3. *Flat pullback.* This is the second identity of proposition 10.1.5(4), $s_i(g^*E) \cap g^*\beta = g^*(s_i(E) \cap \beta)$, and the same additivity argument, completing the proof.

Commutativity is what allows the Chern classes of all bundles on X to be handled as though they were elements of a commutative ring, long before any such ring exists; every polynomial identity written below is an identity among commuting operators, and its meaning is unambiguous. The other two parts say that the Chern classes are attached to the bundle and not to the base: they survive restriction to an open set, pullback along a projective bundle, and pushforward along a proper map, which is exactly what a proof carried out after a base change needs in order to descend.

The identities that remain, multiplicativity on exact sequences chief among them, are proved by making E as simple as possible, and the simplest a bundle can be is filtered by line bundles. For such a bundle the relation of theorem 10.2.9 can be written down by hand, because the tautological line at a point of $\mathbb{P}(E)$ must have a nonzero image in one of the successive quotients. That observation is the whole proof of the next lemma, and through it of the Whitney formula.

Lemma 10.2.11: Chern Classes of a Filtered Bundle

Let E be a vector bundle of rank e on an algebraic scheme X, and suppose E admits a filtration by subbundles

$$0 = E_0 \subseteq E_1 \subseteq \cdots \subseteq E_e = E$$

with each quotient $L_i = E_i/E_{i-1}$ a line bundle. Then

$$c(E) = \prod_{i=1}^{e} \big(\mathrm{id} + c_1(L_i)\big)$$

as operators on $A_*(X)$; equivalently $c_m(E)$ is the m-th elementary symmetric expression

$$\sigma_m = \sum_{i_1 < \cdots < i_m} c_1(L_{i_1}) \cdots c_1(L_{i_m})$$

in the commuting operators $c_1(L_1), \dots, c_1(L_e)$, for $0 \leq m \leq e$.

***Proof*:**
Write $P = \mathbb{P}(E)$ with projection π and tautological class ξ, and for $1 \leq i \leq e$ let $P_i = \mathbb{P}(E_i) \subseteq P$ with projection π_i and inclusion $\iota_i \colon P_i \hookrightarrow P$; set $P_0 = \mathbb{P}(E_0) = \emptyset$, so $P_e = P$. Each P_i is a closed subscheme of P, because in a local trivialization of the filtration over $U \subseteq X$ the inclusion $P_i|_U \subseteq P|_U$ is the linear inclusion $\mathbb{P}^{i-1} \times U \subseteq \mathbb{P}^{e-1} \times U$, and these glue. Put

$$\theta_i = c_1\big(\mathcal{O}_P(1) \otimes \pi^* L_i\big) = \xi + c_1(\pi^* L_i)$$

the equality by additivity of the first Chern class (proposition 9.4.4(3)); the θ_i commute with each other by proposition 9.4.4(2).

Step 1: each θ_i pushes a class on P_i down onto P_{i-1}. The tautological subbundle of box 10.2.5 restricts along ι_i to the tautological subbundle of P_i, since a point of P_i is a line $\ell \subseteq (E_i)_x \subseteq E_x$ and the line attached to it is the same either way; thus $\iota_i^* \mathcal{O}_P(1) = \mathcal{O}_{P_i}(1)$ and $\mathcal{O}_{P_i}(-1) \subseteq$

$\pi_i^* E_i$. Composing that inclusion with the quotient map $\pi_i^* E_i \to \pi_i^* L_i$ gives a homomorphism $\mathcal{O}_{P_i}(-1) \to \pi_i^* L_i$, that is, a global section τ_i of the line bundle

$$M_i := \mathcal{O}_{P_i}(1) \otimes \pi_i^* L_i = \iota_i^*\big(\mathcal{O}_P(1) \otimes \pi^* L_i\big)$$

At a point of P_i above x corresponding to a line $\ell \subseteq (E_i)_x$, the section τ_i vanishes exactly when ℓ maps to zero in $(L_i)_x$, that is when $\ell \subseteq (E_{i-1})_x$. Its zero locus is therefore P_{i-1}. In a local trivialization as above, with homogeneous fibre coordinates $y_1, \dots, y_i$ chosen so that E_{i-1} is cut out by $y_i = 0$, the section τ_i is the linear form y_i, a nonzerodivisor; hence $\operatorname{div}(\tau_i)$ is an effective Cartier divisor on P_i with support P_{i-1} and with $\mathcal{O}_{P_i}(\operatorname{div}\tau_i) \cong M_i$.

Now let $\beta = \iota_{i*}\beta_i$ for some $\beta_i \in A_*(P_i)$. The map ι_i is a closed immersion, hence proper, so the projection formula (proposition 9.4.4(4)) gives

$$\theta_i \cap \iota_{i*}\beta_i = \iota_{i*}\big(c_1(M_i) \cap \beta_i\big)$$

using $\iota_i^*(\mathcal{O}_P(1) \otimes \pi^* L_i) = M_i$. By definition 9.4.1 the class $c_1(M_i) \cap \beta_i$ is $\operatorname{div}(\tau_i) \cdot \beta_i$, which by proposition 9.2.4 is a well-defined class in $A_*\big(|\operatorname{div}\tau_i|\big) = A_*(P_{i-1})$. Pushing forward along $P_{i-1} \hookrightarrow P_i \hookrightarrow P$, the class $\theta_i \cap \beta$ is the image of a class on P_{i-1}, which is the assertion of this step. For $i = 1$ the filtration gives $E_1 = L_1$, the section τ_1 is the identity of L_1 under $\mathcal{O}_{P_1}(-1) = L_1 = \pi_1^* L_1$, its divisor is empty, and $A_*(P_0) = A_*(\emptyset) = 0$, so θ_1 annihilates every class coming from P_1.

Step 2: the product of the θ_i is zero. Let $\beta \in A_*(P) = A_*(P_e)$. Applying Step 1 with $i = e$ exhibits $\theta_e \cap \beta$ as coming from $A_*(P_{e-1})$; applying it again with $i = e - 1$ exhibits $\theta_{e-1} \cap \theta_e \cap \beta$ as coming from $A_*(P_{e-2})$; after e steps the class $\theta_1 \cap \cdots \cap \theta_e \cap \beta$ comes from $A_*(P_0) = 0$ and so vanishes. The θ_i commute, so the product may be taken in any order:

$$\Big(\prod_{i=1}^{e} \theta_i\Big) \cap \beta = 0 \qquad \text{for every } \beta \in A_*(P)$$

Step 3: reading off the coefficients. Take $\beta = \pi^*\alpha$ for $\alpha \in A_*(X)$ and expand the product, which is legitimate because the factors commute:

$$0 = \Big(\prod_{i=1}^{e} \big(\xi + c_1(\pi^* L_i)\big)\Big) \cap \pi^*\alpha = \sum_{m=0}^{e} \xi^{\,e-m} \cap \Big(\sigma_m\big(c_1(\pi^* L_1), \dots, c_1(\pi^* L_e)\big) \cap \pi^*\alpha\Big)$$

The projection π is flat, so the flat compatibility of proposition 9.4.4(4) gives $c_1(\pi^* L) \cap \pi^*\alpha = \pi^*(c_1(L) \cap \alpha)$ for each L; iterating over the factors of each monomial,

$$\sigma_m\big(c_1(\pi^* L_1), \dots, c_1(\pi^* L_e)\big) \cap \pi^*\alpha = \pi^*\big(\sigma_m \cap \alpha\big)$$

with $\sigma_m = \sigma_m(c_1(L_1), \dots, c_1(L_e))$ formed on X. The displayed vanishing is therefore

$$\sum_{m=0}^{e} \xi^{\,e-m} \cap \pi^*\big(\sigma_m \cap \alpha\big) = 0$$

for every α, and $\sigma_0 = \mathrm{id}$. This is the relation of theorem 10.2.9(1) with σ_m in place of $c_m(E)$, so the uniqueness statement theorem 10.2.9(3) gives $c_m(E) = \sigma_m$ for $0 \le m \le e$, while $c_m(E) = 0$ for $m > e$ by part (2). Summing over m gives $c(E) = \prod_{i=1}^{e}(\mathrm{id} + c_1(L_i))$, as we sought to show.

The proof is geometric where it matters. A line inside E cannot map to zero in every quotient L_i of the filtration, and each factor θ_i cuts the support of a class down one step of the filtration, so after e factors nothing is left; the algebra afterwards only names the coefficients. Note also what the lemma does not require: no hypothesis on X, and no splitting of the filtration. A filtration with line-bundle quotients is enough, and an extension that does not split has the same Chern classes as the direct sum of its pieces.

Example 10.2.12: Split Bundles on Projective Space

1. *The trivial bundle.* The bundle $\mathcal{O}_X^{\oplus e}$ is filtered by the coordinate subbundles $\mathcal{O}_X^{\oplus i}$, with all quotients equal to $\mathcal{O}_X$. Each factor of lemma 10.2.11 is then $\mathrm{id} + c_1(\mathcal{O}_X) = \mathrm{id}$, since $c_1(\mathcal{O}_X) = 0$ by example 9.4.3, so
$$c\big(\mathcal{O}_X^{\oplus e}\big) = \mathrm{id}$$
and every trivial bundle has trivial total Chern class, of any rank and on any X.

2. *A sum of twists on* $\mathbb{P}^n$. Let $E = \mathcal{O}_{\mathbb{P}^n}(a) \oplus \mathcal{O}_{\mathbb{P}^n}(b)$ and write $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$. The summand $\mathcal{O}_{\mathbb{P}^n}(a)$ is a subbundle with quotient $\mathcal{O}_{\mathbb{P}^n}(b)$, so the lemma gives
$$c(E) = \big(\mathrm{id} + aH\big)\big(\mathrm{id} + bH\big) = \mathrm{id} + (a+b)H + ab\,H^2$$
and hence $c_1(E) = (a+b)H$ and $c_2(E) = ab\,H^2$. Capping with the fundamental class and using example 9.4.7,
$$c_1(E) \cap [\mathbb{P}^n] = (a+b)\,[\mathbb{P}^{n-1}], \qquad c_2(E) \cap [\mathbb{P}^n] = ab\,[\mathbb{P}^{n-2}]$$
for $n \geq 2$. On $\mathbb{P}^1$ the second class vanishes for a reason unrelated to the rank: H^2 lowers dimension by two on a one-dimensional scheme, so it is the zero operator by the degree count in lemma 10.2.2, and $c_2(E) = 0$ even though $\mathrm{rank} E = 2$.

Most bundles carry no filtration by line bundles. The projective bundle manufactures one, at the cost of replacing X by a larger scheme, and iterating the construction strips off one line bundle at a time until nothing is left. The scheme at the end of the tower is the following.

Definition 10.2.13: The Flag Bundle

Let E be a vector bundle of rank e on an algebraic scheme X. The *flag bundle* of E is the scheme $\mathrm{Fl}(E)$ over X defined by induction on e: for $e = 1$ it is X itself, and for $e > 1$ it is the flag bundle, formed over $\mathbb{P}(E)$, of the rank-$(e-1)$ quotient $Q = \pi^*E/\mathcal{O}_{\mathbb{P}(E)}(-1)$ of box 10.2.5. Its structure morphism $\mathrm{Fl}(E) \to X$ is the composite of the $e-1$ projective bundle projections in the tower, and its fibre over $x \in X$ is the variety of complete flags of subspaces of the fibre E_x. For a bundle of rank two the tower has one stage, so $\mathrm{Fl}(E) = \mathbb{P}(E)$.

The projective-bundle formula guarantees that nothing is lost in passing to the flag bundle: the pullback map on Chow groups is injective at each stage, so an identity of operators verified upstairs is an identity downstairs. That is the content of the splitting principle, and it is a theorem rather than a slogan precisely because of the injectivity.

Theorem 10.2.14: The Splitting Principle

Let X be an algebraic scheme and $E_1, \dots, E_t$ vector bundles on X. There is a scheme X' and a proper, flat morphism $f\colon X' \to X$ such that

1. $f^*\colon A_*(X) \to A_*(X')$ is injective, and
2. each f^*E_j admits a filtration by subbundles whose successive quotients are line bundles.

***Proof*:**
Suppose first that $t = 1$, and write $E = E_1$ of rank e; the morphism produced will be the structure morphism of the flag bundle $\mathrm{Fl}(E)$ of definition 10.2.13. Induct on e. If $e = 1$ the bundle is already a line bundle and $f = \mathrm{id}_X$ serves. Let $e > 1$ and let $\pi\colon \mathbb{P}(E) \to X$ be the projective bundle, proper and flat by proposition 5.3.40. The Flat pullback along it is injective by corollary 10.1.6, which also exhibits the left inverse $\beta \mapsto \pi_*(\xi^{e-1} \cap \beta)$. On $\mathbb{P}(E)$ the tautological subbundle of box 10.2.5 gives an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}(E)}(-1) \longrightarrow \pi^*E \longrightarrow Q \longrightarrow 0$$

with Q locally free of rank $e-1$. By the inductive hypothesis there is a proper flat $g\colon X' \to \mathbb{P}(E)$ with g^* injective and g^*Q filtered by subbundles with line-bundle quotients $N_2, \dots, N_e$. Set $f = \pi \circ g$, proper and flat as a composite of such, with $f^* = g^* \circ \pi^*$ injective as a composite of injections (proposition 8.3.5(3) identifies the composite pullback with the pullback of the composite). Pulling the displayed sequence back along g, which is exact because pullback of locally free sheaves is exact, gives a line subbundle $F_1 = g^*\mathcal{O}_{\mathbb{P}(E)}(-1) \subseteq f^*E$ with quotient g^*Q. Taking F_{j+1} to be the preimage in f^*E of the j-th step of the filtration of g^*Q produces

$$0 = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_e = f^*E$$

with F_1 of rank one and F_{j+1}/F_j isomorphic to the j-th quotient of the filtration of g^*Q, hence a line bundle. This proves the case $t = 1$.

For general t, apply the case $t = 1$ to E_1, obtaining $f_1\colon X_1 \to X$; then to $f_1^*E_2$ on X_1, obtaining $f_2\colon X_2 \to X_1$; and so on, and take $f = f_1 \circ \cdots \circ f_t$. Each stage is proper and flat with injective pullback, so the composite is too. A filtration by subbundles with line-bundle quotients pulls back to a filtration of the same shape, again because pullback is exact on locally free sheaves, so the filtration of $f_j^*E_j$ constructed at stage j survives all later stages. Hence every f^*E_j is filtered with line-bundle quotients, completing the proof.

Two features of the statement carry the weight. Injectivity of f^* is what turns the construction into a proof technique, since an identity of Chern operators is a family of identities of classes, and each descends. Flatness is what makes f^* available at all at this stage of the theory, there being no pullback for a general morphism until chapter 11. Both come from the projective-bundle formula, which is why that formula had to be proved before any of this.

The reduction the splitting principle enables is the multiplicativity of Chern classes on exact sequences. This is the property that makes them computable in practice: bundles are almost always presented by exact sequences, and the formula converts such a presentation directly into Chern data.

Theorem 10.2.15: The Whitney Sum Formula

Let
$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$
be an exact sequence of vector bundles on an algebraic scheme X. Then
$$c(E) = c(E')\, c(E'')$$
as operators on $A_*(X)$; that is, for every $m \geq 0$ and every $\alpha \in A_*(X)$,
$$c_m(E) \cap \alpha = \sum_{i+j=m} c_i(E') \cap \big(c_j(E'') \cap \alpha\big)$$

***Proof*:**
Let $e' = \mathrm{rank} E'$ and $e'' = \mathrm{rank} E''$, so $\mathrm{rank} E = e' + e''$.

Step 1: passing to a splitting cover. By theorem 10.2.14 applied to the pair E', E'' there is a proper flat $f\colon X' \to X$ with f^* injective, a filtration of f^*E' with line-bundle quotients $L'_1, \dots, L'_{e'}$, and a filtration of f^*E'' with line-bundle quotients $L''_1, \dots, L''_{e''}$. Pulling the given sequence back along f keeps it exact, pullback being exact on locally free sheaves, so
$$0 \longrightarrow f^*E' \longrightarrow f^*E \longrightarrow f^*E'' \longrightarrow 0$$
is exact on X'. Splice the two filtrations: take the filtration of f^*E' as the bottom e' steps of a filtration of f^*E, and above it take the preimages in f^*E of the steps of the filtration of f^*E''. The successive quotients are $L'_1, \dots, L'_{e'}$ followed by $L''_1, \dots, L''_{e''}$, all line bundles, so f^*E is filtered with line-bundle quotients.

Step 2: the identity upstairs. Lemma 10.2.11 applied to the three filtered bundles on X' gives
$$c(f^*E) = \prod_{i=1}^{e'} \big(\mathrm{id} + c_1(L'_i)\big) \cdot \prod_{j=1}^{e''} \big(\mathrm{id} + c_1(L''_j)\big) = c(f^*E')\, c(f^*E'')$$
the middle expression being the product over the quotients of the spliced filtration, regrouped using commutativity of first Chern class operators (proposition 9.4.4(2)). Comparing homogeneous components of degree m,
$$c_m(f^*E) = \sum_{i+j=m} c_i(f^*E')\, c_j(f^*E'')$$

Step 3: descending to X. Fix m and $\alpha \in A_*(X)$. Flat pullback naturality (proposition 10.2.10(3)) gives $f^*(c_m(E) \cap \alpha) = c_m(f^*E) \cap f^*\alpha$, and applying it twice,
$$f^*\Big(c_i(E') \cap \big(c_j(E'') \cap \alpha\big)\Big) = c_i(f^*E') \cap \big(c_j(f^*E'') \cap f^*\alpha\big)$$
Subtracting and using Step 2,
$$f^*\Big(c_m(E) \cap \alpha - \sum_{i+j=m} c_i(E') \cap \big(c_j(E'') \cap \alpha\big)\Big) = \Big(c_m(f^*E) - \sum_{i+j=m} c_i(f^*E')\, c_j(f^*E'')\Big) \cap f^*\alpha = 0$$
Since f^* is injective, the class inside vanishes. As α and m were arbitrary, the two operators agree, completing the proof.

The formula makes the total Chern class a homomorphism from short exact sequences to products, and this is what the Grothendieck group $K^\circ(X)$ of vector bundles will formalize when Riemann-Roch is taken up: the total Chern class is insensitive to the extension class, seeing only the sub and the quotient. Two immediate consequences deserve recording. First, $c(E)$ is invertible for every bundle E, with inverse $s(E)$: by definition 10.2.3 it is the inverse of $s(E)$, and that inverse is two-sided by lemma 10.2.2. So an exact sequence determines the Chern classes of any one of its three terms from the other two: $c(E'') = c(E)c(E')^{-1}$. Second, a trivial sub or quotient bundle contributes nothing, since $c(\mathcal{O}_X^{\oplus t}) = \mathrm{id}$ by example 10.2.12. Both are used repeatedly below.

Corollary 10.2.16: Chern Classes of a Dual Bundle

Let E be a vector bundle on an algebraic scheme X. Then

$$c_i\big(E^\vee\big) = (-1)^i\, c_i(E)$$

as operators on $A_*(X)$, for every $i \geq 0$.

***Proof*:**
Write $e = \mathrm{rank} E$ and let $f\colon X' \to X$ be as in theorem 10.2.14 for the single bundle E, with f^*E filtered by subbundles $0 = F_0 \subseteq \cdots \subseteq F_e = f^*E$ having line-bundle quotients $L_i = F_i/F_{i-1}$. Dualizing the filtration gives subbundles $(f^*E/F_i)^\vee \subseteq (f^*E)^\vee = f^*(E^\vee)$, increasing as i decreases, with successive quotients

$$\big(f^*E/F_{i-1}\big)^\vee/\big(f^*E/F_i\big)^\vee \cong \big(F_i/F_{i-1}\big)^\vee = L_i^\vee$$

each a line bundle, so $f^*(E^\vee)$ is filtered with line-bundle quotients $L_e^\vee, \ldots, L_1^\vee$. By lemma 10.2.11 and $c_1(L^\vee) = -c_1(L)$ (proposition 9.4.4(3)),

$$c\big(f^*(E^\vee)\big) = \prod_{i=1}^{e} \big(\mathrm{id} - c_1(L_i)\big)$$

whose degree-i component is $(-1)^i\sigma_i(c_1(L_1), \ldots, c_1(L_e)) = (-1)^i c_i(f^*E)$, again by lemma 10.2.11. Naturality (proposition 10.2.10(3)) rewrites both sides as pullbacks of operators on X, and injectivity of f^* gives $c_i(E^\vee) = (-1)^i c_i(E)$, as we sought to show.

The remaining ingredient of a working calculus is a supply of computed examples. The tangent bundle of projective space is the one every later computation leans on, and it is produced from the Euler sequence by a single application of the Whitney formula.

Example 10.2.17: The Tangent Bundle of Projective Space

The Euler sequence theorem 5.5.18 relates the sheaf of differentials of $\mathbb{P}^n$ to the twisting sheaves through the exact sequence

$$0 \longrightarrow \Omega_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow 0$$

Projective space is smooth, so $\Omega_{\mathbb{P}^n}$ is locally free of rank n and the sequence is a sequence of vector bundles; dualizing it, which preserves exactness, gives the sequence in the form used here,

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$

with $T_{\mathbb{P}^n} = \Omega^\vee_{\mathbb{P}^n}$ the tangent bundle. Write $H = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$. The Whitney formula (theorem 10.2.15) applied to this sequence reads

$$c\big(\mathcal{O}_{\mathbb{P}^n}\big)\, c\big(T_{\mathbb{P}^n}\big) = c\big(\mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)}\big)$$

The left factor is the identity by example 10.2.12(1), and the right side is $(\mathrm{id}+H)^{n+1}$, the direct sum of $n+1$ copies of $\mathcal{O}(1)$ being filtered with all quotients $\mathcal{O}_{\mathbb{P}^n}(1)$ and each factor contributing $\mathrm{id}+H$ by lemma 10.2.11. Hence

$$c\big(T_{\mathbb{P}^n}\big) = (\mathrm{id}+H)^{n+1}, \qquad c_i\big(T_{\mathbb{P}^n}\big) = \binom{n+1}{i} H^i$$

for $0 \le i \le n$, with $c_i(T_{\mathbb{P}^n}) = 0$ for $i > n$ in accordance with theorem 10.2.9(2). The binomial coefficient $\binom{n+1}{n+1} = 1$ in degree $n+1$ is not a contradiction: H^{n+1} is the zero operator on $A_*(\mathbb{P}^n)$ by the degree count of lemma 10.2.2, matching the nilpotence recorded in proposition 9.4.4(5).

Capping the top class against the fundamental class and using $H^n \cap [\mathbb{P}^n] = [\mathbb{P}^0]$ from example 9.4.7,

$$c_n\big(T_{\mathbb{P}^n}\big) \cap [\mathbb{P}^n] = (n+1)\,[\mathbb{P}^0], \qquad \int_{\mathbb{P}^n} c_n\big(T_{\mathbb{P}^n}\big) = n+1$$

the degree being taken with definition 8.2.4. For $n = 1$ this gives $\int_{\mathbb{P}^1} c_1(T_{\mathbb{P}^1}) = 2$, the degree of the tangent bundle of a rational curve, consistent with $\deg T_C = 2 - 2g$ for $g = 0$ through example 9.4.8. For $n = 2$ it gives $\int_{\mathbb{P}^2} c_2(T_{\mathbb{P}^2}) = 3$.

The integers $n+1$ produced by the top Chern class are the Euler characteristics of the underlying spaces, which is the first sign that $c_n(T_X)$ counts something. The general statement, that the degree of the top Chern class of the tangent bundle of a smooth complete variety is its topological Euler characteristic, needs the comparison between Chow groups and cohomology through the cycle class map and is not available yet; what is available is the computation itself, and it is already enough to run enumerative arguments once the intersection product exists.

10.2.18 Remark: The Topological Chern Classes A reader who has met Chern classes topologically, as the characteristic classes of a complex vector bundle in *Algebraic Topology* [51], will recognize the shape of this section: the same axioms, the same Whitney formula, the same splitting principle through a flag bundle. The two theories are not the same construction. The classes there live in singular cohomology and are built for complex bundles on topological spaces; the classes here live in the Chow group of an algebraic scheme over an arbitrary algebraically closed field, and the scheme is allowed to be singular, reducible, and non-reduced. Neither proof above uses the other theory, and none could: no topology is available over a field of positive characteristic. The two are compared, for a smooth complex variety, by the cycle class map built later in this book, which carries the algebraic classes to the topological ones.

Exercise 10.2.1

1. Using only definition 10.2.3, express $s_1(E)$, $s_2(E)$ and $s_3(E)$ in terms of $c_1(E), c_2(E), c_3(E)$, and check that the resulting formulas are obtained from the expressions for c_1, c_2, c_3 in terms of s_1, s_2, s_3 by exchanging the letters c and s. Explain in one sentence why that symmetry is forced by the definition.

2. Let $E = \mathcal{O}_{\mathbb{P}^n}(a) \oplus \mathcal{O}_{\mathbb{P}^n}(b)$ with $n \geq 2$. Compute $c(E)$, $c(E^\vee)$ and $c(E \otimes \mathcal{O}_{\mathbb{P}^n}(1))$ as operators on $A_*(\mathbb{P}^n)$, and compute $c_2(E) \cap [\mathbb{P}^n]$. Then take $n = 1$ and explain why $c_2(E) = 0$ there, identifying which of the two possible reasons (the rank of E or the dimension of the base) is responsible.

3. Let E be a vector bundle of rank $e \geq 2$ on an algebraic scheme X admitting a nowhere-vanishing global section τ, so that $\tau\colon \mathcal{O}_X \to E$ is an inclusion of a subbundle with locally free quotient Q of rank $e - 1 \geq 1$. Show that $c(E) = c(Q)$ and deduce $c_e(E) = 0$. Show separately that a line bundle with a nowhere-vanishing section is trivial, so that the conclusion holds in the excluded case $e = 1$ as well. Conversely, explain why the computation $\int_{\mathbb{P}^n} c_n(T_{\mathbb{P}^n}) = n+1 \neq 0$ of example 10.2.17 shows that $T_{\mathbb{P}^n}$ has no nowhere-vanishing global section.

4. Let E be a vector bundle of rank two on an algebraic scheme X and L a line bundle on X. Using theorem 10.2.14 and lemma 10.2.11, prove

$$c_1(E \otimes L) = c_1(E) + 2\, c_1(L), \qquad c_2(E \otimes L) = c_2(E) + c_1(E)\, c_1(L) + c_1(L)^2$$

as operators on $A_*(X)$. (You may use that tensoring a filtration by a line bundle again gives a filtration, with each quotient tensored by that line bundle.)

5. Compute $c(\Omega_{\mathbb{P}^n})$ as an operator on $A_*(\mathbb{P}^n)$ in closed form, and evaluate $\int_{\mathbb{P}^2} c_2(\Omega_{\mathbb{P}^2})$ and $\int_{\mathbb{P}^2} c_1(\Omega_{\mathbb{P}^2})^2$. Compare the first with $\int_{\mathbb{P}^2} c_2(T_{\mathbb{P}^2})$ and account for the agreement.

6. Let C be a smooth projective curve and E a vector bundle of rank two on C. Show that $c_2(E) = 0$ as an operator on $A_*(C)$, and that $c(E)$ is determined by the single class $c_1(E) \cap [C] \in A_0(C)$. Show further that if E sits in an exact sequence $0 \to L_1 \to E \to L_2 \to 0$ of bundles on C, then $\int_C c_1(E) = \deg L_1 + \deg L_2$.

10.3 The Chern Character and the Todd Class

The total Chern class of section 10.2 is multiplicative on short exact sequences: the Whitney formula turns an extension into a product. Riemann-Roch needs the opposite bookkeeping. It computes the Euler characteristic of a coherent sheaf, and Euler characteristics are *additive* on short exact sequences, so the characteristic class that Riemann-Roch pairs against must be additive too. This section builds that class, the Chern character, and alongside it the Todd class, the multiplicative correction factor the theorem attaches to the tangent bundle.

Both are universal power series in the Chern classes, and both differ from anything in section 10.2 in two ways. Their coefficients are rational, forced by denominators that no normalization removes, and they are infinite series rather than finite sums, which is legitimate only because a composite of more than $\dim X$ dimension-lowering operators is zero. Those two points come first, then the formal calculus of Chern roots that makes symmetric expressions in r formal symbols into operators, and then the two classes and the identities they satisfy.

The first point is a change of coefficients. Nothing in section 10.2 required it: Segre and Chern classes are operators on $A_*(X)$ with integer coefficients. The Chern character does require it, because its degree-two component is $\frac{1}{2}(c_1^2 - 2c_2)$, and no rescaling removes that $\frac{1}{2}$ while keeping additivity.

Definition 10.3.1: Chow Groups with Rational Coefficients

For an algebraic scheme X, set

$$A_k(X)_{\mathbb{Q}} := A_k(X) \otimes_{\mathbb{Z}} \mathbb{Q}, \qquad A_*(X)_{\mathbb{Q}} := \bigoplus_k A_k(X)_{\mathbb{Q}}$$

A class $\alpha \in A_k(X)$ is identified with $\alpha \otimes 1$, and every operator $A_k(X) \to A_{k-i}(X)$ extends uniquely to a $\mathbb{Q}$-linear operator $A_k(X)_{\mathbb{Q}} \to A_{k-i}(X)_{\mathbb{Q}}$. The degree map of definition 8.2.4 on a complete X is extended $\mathbb{Q}$-linearly and still written $\int_X : A_0(X)_{\mathbb{Q}} \to \mathbb{Q}$.

Since $\mathbb{Q}$ is flat over $\mathbb{Z}$, an injective map of Chow groups stays injective after this change of coefficients, so the injectivity statements of section 10.2 are available rationally without further comment. What is lost is torsion: the natural map $A_*(X) \to A_*(X)_{\mathbb{Q}}$ has kernel the torsion subgroup, so any torsion class becomes invisible. On $\mathbb{P}^n$ nothing is lost, since $A_*(\mathbb{P}^n) = \bigoplus_j \mathbb{Z}\,[\mathbb{P}^j]$ is free (theorem 8.5.2) and $A_*(\mathbb{P}^n)_{\mathbb{Q}} = \bigoplus_j \mathbb{Q}\,[\mathbb{P}^j]$; the class $\frac{3}{2}\,[\mathrm{pt}] \in A_0(\mathbb{P}^2)_{\mathbb{Q}}$, which will appear below as a component of $\mathrm{ch}(T_{\mathbb{P}^2})$, is a legitimate element of the rational Chow group and of nothing smaller.

The second point is that infinite series of operators are already finite sums. An operator lowering dimension can only be applied $\dim X$ times before the target group vanishes, which is the nilpotence recorded for a single line bundle in proposition 9.4.4(5), stated here for an arbitrary commuting family.

Lemma 10.3.2: Power Series in Dimension-Lowering Operators

Let X be an algebraic scheme of dimension n and let $\theta_1, \dots, \theta_m$ be pairwise commuting $\mathbb{Q}$-linear endomorphisms of $A_*(X)_{\mathbb{Q}}$ with $\theta_j\big(A_k(X)_{\mathbb{Q}}\big) \subseteq A_{k-d_j}(X)_{\mathbb{Q}}$ for integers $d_j \geq 1$. Write $\theta^\nu = \theta_1^{\nu_1} \cdots \theta_m^{\nu_m}$ and $|\nu| = \nu_1 + \cdots + \nu_m$.

1. $\theta^\nu = 0$ whenever $|\nu| > n$.
2. The assignment
$$\sum_\nu \lambda_\nu\, a^\nu \longmapsto \sum_{|\nu| \leq n} \lambda_\nu\, \theta^\nu$$
is a homomorphism of $\mathbb{Q}$-algebras $\mathbb{Q}[[a_1, \dots, a_m]] \to \mathrm{End}\big(A_*(X)_{\mathbb{Q}}\big)$ carrying a_j to θ_j.

***Proof*:**

1. A k-cycle is a combination of k-dimensional subvarieties, so $A_k(X)_{\mathbb{Q}} = 0$ unless $0 \leq k \leq n$. The operator θ^ν carries $A_k(X)_{\mathbb{Q}}$ into $A_{k-e}(X)_{\mathbb{Q}}$ where $e = \sum_j \nu_j d_j \geq |\nu|$. If $|\nu| > n$ then for every k with $A_k(X)_{\mathbb{Q}} \neq 0$ one has $k - e \leq n - |\nu| < 0$, so the target vanishes and $\theta^\nu = 0$.
2. Write T for the assignment. It is $\mathbb{Q}$-linear by construction, and $T(1) = \mathrm{id}$. For multiplicativity, let $\Phi = \sum_\mu \lambda_\mu a^\mu$ and $\Psi = \sum_\rho \kappa_\rho a^\rho$. The product $\Phi\Psi$ has coefficients $\sum_{\mu+\rho=\nu} \lambda_\mu \kappa_\rho$, so
$$T(\Phi\Psi) = \sum_{|\nu| \leq n} \Big(\sum_{\mu+\rho=\nu} \lambda_\mu \kappa_\rho \Big) \theta^\nu$$

On the other side, since the θ_j commute, $T(\Phi)T(\Psi) = \sum_{|\mu|\le n,\, |\rho|\le n} \lambda_\mu \kappa_\rho\, \theta^{\mu+\rho}$. Every term with $|\mu+\rho| > n$ vanishes by part (1), and every pair (μ,ρ) with $|\mu+\rho| \le n$ automatically satisfies $|\mu| \le n$ and $|\rho| \le n$, so the two expressions have the same nonzero terms, completing the proof.

The Chern classes $c_1(E), \ldots, c_r(E)$ of a bundle of rank r are exactly such a family: they commute with one another by section 10.2, ultimately because two divisor operators commute (theorem 9.3.2), and $c_i(E)$ lowers dimension by i. A formal power series in r variables therefore evaluates on them to a well-defined operator, given by a finite sum. This is the mechanism behind every definition below.

What remains is to say which power series. Both the Chern character and the Todd class are described most efficiently as symmetric expressions in r formal symbols $a_1, \ldots, a_r$, the Chern roots, which behave as though E were a direct sum of line bundles with those first Chern classes. The splitting principle of section 10.2 is what makes the symbols legitimate.

Definition 10.3.3: Chern Roots and Splitting Maps

Let E be a vector bundle of rank r on an algebraic scheme X. A *splitting map* for E is a flat proper morphism $f\colon X' \to X$ such that

1. $f^*\colon A_*(X) \to A_*(X')$ is injective, and
2. f^*E admits a filtration by subbundles $0 = E_0 \subset E_1 \subset \cdots \subset E_r = f^*E$ whose successive quotients $L_p = E_p/E_{p-1}$ are line bundles.

The operators $a_p := c_1(L_p)$ on $A_*(X')_{\mathbb{Q}}$, for $p = 1, \ldots, r$, are the *Chern roots* of E relative to f.

The splitting principle of section 10.2 produces a splitting map for any bundle: the flag bundle of E is a tower of projective bundles, proper and flat over X by proposition 5.3.40, and the projective-bundle formula of section 10.1 makes each stage of the tower inject on Chow groups. By flatness of $\mathbb{Q}$ over $\mathbb{Z}$, condition (1) also gives injectivity of $f^*\colon A_*(X)_{\mathbb{Q}} \to A_*(X')_{\mathbb{Q}}$. Writing $a_p = c_1(L_p)$ for the Chern roots, lemma 10.2.11 gives

$$c(f^*E) = \prod_{p=1}^{r} \big(\mathrm{id} + a_p\big), \qquad \text{that is} \qquad c_i(f^*E) = e_i(a_1, \ldots, a_r) \tag{10.4}$$

where e_i is the i-th elementary symmetric polynomial. The roots are not canonical, and they are not operators on X at all, only on X'. What descends to X is any symmetric expression in them, and the reason is the classical structure theorem for symmetric functions.

Any symmetric formal power series in the Chern roots is a power series in the elementary symmetric polynomials, hence a genuine operator: this is [18, Symmetric Power Series], the formal-power-series refinement of the classical fundamental theorem on symmetric functions proved there.

The two lemmas combine into the calculus this section runs on: a symmetric power series in r variables defines an operator attached to any rank-r bundle, and that operator may be computed as though the bundle were split.

Proposition 10.3.4: The Chern Root Calculus

Let E be a vector bundle of rank r on an algebraic scheme X and let $\Phi \in \mathbb{Q}[[a_1, \ldots, a_r]]$ be symmetric, with $\widetilde{\Phi}$ as in [18, Symmetric Power Series]. Define

$$\Phi(E) := \widetilde{\Phi}\big(c_1(E), \ldots, c_r(E)\big)$$

Then:

1. $\Phi(E)$ is a well-defined operator on $A_*(X)_{\mathbb{Q}}$, equal to a finite sum of composites of Chern classes of E, and it commutes with $\Psi(F)$ for every bundle F and symmetric Ψ.
2. For a flat morphism $g\colon X'' \to X$ and $\alpha \in A_*(X)_{\mathbb{Q}}$,
$$g^*\big(\Phi(E) \cap \alpha\big) = \Phi(g^*E) \cap g^*\alpha$$
3. If E itself admits a filtration by subbundles with line-bundle quotients $L_1, \ldots, L_r$, then $\Phi(E) = \Phi\big(c_1(L_1), \ldots, c_1(L_r)\big)$, the right-hand side being the evaluation of lemma 10.3.2.
4. If $f\colon X' \to X$ is a splitting map for E with Chern roots $a_1, \ldots, a_r$, then $f^*\big(\Phi(E) \cap \alpha\big) = \Phi(a_1, \ldots, a_r) \cap f^*\alpha$ for all $\alpha \in A_*(X)_{\mathbb{Q}}$.

***Proof*:**

1. The classes $c_1(E), \ldots, c_r(E)$ commute pairwise and $c_i(E)$ lowers dimension by i, so lemma 10.3.2 applies and $\widetilde{\Phi}$ evaluates to a well-defined operator, equal to the sum of the terms of $\widetilde{\Phi}$ of weight at most $\dim X$. Uniqueness of $\widetilde{\Phi}$ in [18, Symmetric Power Series] is what makes the value depend on Φ alone. Chern classes of any two bundles commute by section 10.2, so the images of the two evaluation homomorphisms commute.
2. Chern classes are natural under flat pullback, $c_i(g^*E) \cap g^*\beta = g^*(c_i(E) \cap \beta)$ by section 10.2, generalizing proposition 9.4.4(4). Applying this once for each factor of a monomial in the $c_i(E)$, and summing over the finitely many monomials of $\widetilde{\Phi}$ that survive, gives the claim.
3. By (10.4) applied to the filtration of E itself, $c_i(E) = e_i(a_1, \ldots, a_r)$ with $a_p = c_1(L_p)$. The evaluation homomorphism $\mathbb{Q}[[a_1, \ldots, a_r]] \to \mathrm{End}(A_*(X)_{\mathbb{Q}})$ of lemma 10.3.2 for the commuting family $a_1, \ldots, a_r$ is a ring homomorphism, so it may be applied to the identity $\Phi = \widetilde{\Phi}(e_1, \ldots, e_r)$ of [18, Symmetric Power Series] term by term, giving
$$\Phi(a_1, \ldots, a_r) = \widetilde{\Phi}\big(e_1(a), \ldots, e_r(a)\big) = \widetilde{\Phi}\big(c_1(E), \ldots, c_r(E)\big) = \Phi(E)$$
as operators.
4. Combining parts (2) and (3): the morphism f is flat, so $f^*(\Phi(E) \cap \alpha) = \Phi(f^*E) \cap f^*\alpha$, and f^*E carries a filtration with line-bundle quotients L_p, so $\Phi(f^*E) = \Phi(a_1, \ldots, a_r)$ by part (3). This completes the proof.

Part (4) is the working form. To prove an identity between operators built from bundles on X, pass to a splitting map, verify the corresponding identity of symmetric power series in the roots, and descend by injectivity of f^*. The roots never appear in a statement, only in a proof.

The first series to feed into this machine is the exponential.

Definition 10.3.5: The Chern Character

Let E be a vector bundle of rank r on an algebraic scheme X. The *Chern character* of E is the operator on $A_*(X)_{\mathbb{Q}}$ obtained from the symmetric power series

$$\operatorname{ch}(a_1,\dots,a_r)=\sum_{p=1}^{r}e^{a_p}=\sum_{p=1}^{r}\sum_{k\geq 0}\frac{a_p^k}{k!}$$

by proposition 10.3.4. It is written $\operatorname{ch}(E)$, and its component lowering dimension by k is written $\operatorname{ch}_k(E)$, so that $\operatorname{ch}(E)=\sum_{k\geq 0}\operatorname{ch}_k(E)$ with the sum finite by lemma 10.3.2.

The definition is stated through the roots for brevity, but no roots are used: what proposition 10.3.4 returns is a specific polynomial in $c_1(E),\dots,c_r(E)$ with rational coefficients, and the next proposition writes it out. In every positive degree that polynomial is the same for all ranks, once the convention $c_i(E)=0$ for $i>\operatorname{rank}E$ from section 10.2 is in force; the rank enters only in degree zero, where the constant term is the rank itself. That is what allows bundles of different ranks to be compared in a single identity: the positive-degree components match term by term and the ranks do their own bookkeeping in degree zero.

Proposition 10.3.6: Components of the Chern Character

Let E be a vector bundle of rank r on X, and write $c_i=c_i(E)$. Then $\operatorname{ch}_k(E)=\frac{1}{k!}\,p_k$, where $p_0=r$ and the operators p_k are determined by Newton's recursion

$$p_k-c_1p_{k-1}+c_2p_{k-2}-\cdots+(-1)^{k-1}c_{k-1}p_1+(-1)^k k\,c_k=0\qquad(k\geq 1)\tag{10.5}$$

In particular

$$\operatorname{ch}(E)=r+c_1+\tfrac{1}{2}\big(c_1^2-2c_2\big)+\tfrac{1}{6}\big(c_1^3-3c_1c_2+3c_3\big)+\cdots$$

so $\operatorname{ch}_0(E)=\operatorname{rank}E$ and $\operatorname{ch}_1(E)=c_1(E)$. For a line bundle L, $\operatorname{ch}(L)=e^{c_1(L)}$.

***Proof*:**
Work first with formal symbols. In $\mathbb{Q}[a_1,\dots,a_r][[t]]$ put $E(t)=\prod_{p=1}^{r}(1+a_pt)=\sum_{j\geq 0}e_jt^j$, where $e_j=e_j(a)$ and $e_j=0$ for $j>r$. Each factor $1+a_pt$ has constant term 1, hence is invertible, and the product rule gives

$$E'(t)=\sum_{p=1}^{r}a_p\prod_{q\neq p}(1+a_qt)=E(t)\sum_{p=1}^{r}\frac{a_p}{1+a_pt}$$

Expanding $\frac{a_p}{1+a_pt}=\sum_{k\geq 1}(-1)^{k-1}a_p^kt^{k-1}$ and summing over p turns this into

$$E'(t)=E(t)\sum_{k\geq 1}(-1)^{k-1}p_kt^{k-1},\qquad p_k:=\sum_{p=1}^{r}a_p^k$$

Comparing coefficients of t^{k-1} on both sides gives $ke_k=\sum_{m=1}^{k}(-1)^{m-1}e_{k-m}\,p_m$, which on multiplying by $(-1)^{k-1}$ and transposing is exactly (10.5) with e_i in place of c_i. Solving the recursion in low degrees, $p_1 = e_1$, then $p_2 = e_1 p_1 - 2e_2 = e_1^2 - 2e_2$, then $p_3 = e_1 p_2 - e_2 p_1 + 3e_3 = e_1^3 - 3e_1 e_2 + 3e_3$.

The power sums determine the Chern character, since interchanging the two finite-in-each-degree summations gives

$$\sum_{p=1}^{r} e^{a_p} = \sum_{k \geq 0} \frac{1}{k!} \sum_{p=1}^{r} a_p^k = r + \sum_{k \geq 1} \frac{p_k}{k!}$$

Each p_k is symmetric, and (10.5) exhibits it as an isobaric polynomial of weight k in $e_1, \ldots, e_k$; by the uniqueness in [18, Symmetric Power Series] that polynomial is the one proposition 10.3.4 substitutes into. Substituting $c_i = c_i(E)$ therefore gives $\mathrm{ch}_k(E) = \frac{1}{k!} p_k$ with p_k satisfying (10.5), and the displayed expansion follows from the three computed cases. Finally $\mathrm{ch}_0(E) = p_0 = r$ and $\mathrm{ch}_1(E) = p_1 = c_1$. For $r = 1$ the series is e^{a_1} and $e_1 = a_1$, so $\widetilde{\mathrm{ch}}(y_1) = e^{y_1}$ and $\mathrm{ch}(L) = e^{c_1(L)} = \sum_{k \geq 0} c_1(L)^k / k!$, a finite sum by lemma 10.3.2, as we sought to show.

The denominators do not cancel. Already the factor $\frac{1}{2}$ in $\mathrm{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2)$ survives on projective space, where $c_1^2 - 2c_2$ is an odd multiple of the point class for suitable bundles, so ch_2 is not the image of any operator on $A_*(X)$ with integer coefficients. The next example exhibits the smallest such case.

Example 10.3.7: The Chern Character of a Line Bundle on $\mathbb{P}^n$

Let $h = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$, so that $h^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ for $0 \leq k \leq n$ by example 9.4.7, and $h^{n+1} = 0$ as an operator by lemma 10.3.2. For the line bundle $\mathcal{O}_{\mathbb{P}^n}(d)$ one has $c_1(\mathcal{O}_{\mathbb{P}^n}(d)) = d\,h$ by proposition 9.4.4(3), so

$$\mathrm{ch}\big(\mathcal{O}_{\mathbb{P}^n}(d)\big) = e^{dh} = 1 + dh + \frac{d^2}{2} h^2 + \cdots + \frac{d^n}{n!} h^n$$

the series terminating because $h^{n+1} = 0$. Capping with the fundamental class on $\mathbb{P}^2$,

$$\mathrm{ch}\big(\mathcal{O}_{\mathbb{P}^2}(d)\big) \cap [\mathbb{P}^2] = [\mathbb{P}^2] + d\,[\mathbb{P}^1] + \frac{d^2}{2}\,[\mathrm{pt}]$$

For odd d the last coefficient is not an integer, so this class lies in $A_0(\mathbb{P}^2)_{\mathbb{Q}}$ and in no subgroup coming from $A_0(\mathbb{P}^2)$. Rational coefficients are not a convenience here; without them the Chern character of $\mathcal{O}_{\mathbb{P}^2}(1)$ does not exist.

The point of the exponential is the identity $e^{a+b} = e^a e^b$, which converts both operations on bundles into operations on operators. Both of the proofs below compare two bundles at once, so they need a single splitting map that works for several bundles simultaneously.

Theorem 10.3.8: Additivity and Multiplicativity of the Chern Character

Let X be an algebraic scheme and let E, F be vector bundles on X.

1. If $0 \to E' \to E \to E'' \to 0$ is an exact sequence of vector bundles on X, then
$$\mathrm{ch}(E) = \mathrm{ch}(E') + \mathrm{ch}(E'')$$
as operators on $A_*(X)_{\mathbb{Q}}$. In particular $\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F)$.
2. $\mathrm{ch}(E \otimes F) = \mathrm{ch}(E)\,\mathrm{ch}(F)$, the product on the right being composition of operators.

***Proof*:**
Write $r = \mathrm{rank}E$, $r' = \mathrm{rank}E'$, $r'' = \mathrm{rank}E''$ and $s = \mathrm{rank}F$; in part (1) the ranks satisfy $r = r' + r''$.

1. *Additivity.* By theorem 10.2.14 choose a flat proper $f\colon X' \to X$ with f^* injective that is a splitting map for both E' and E'', with Chern roots $a_1, \dots, a_{r'}$ and $b_1, \dots, b_{r''}$ respectively.

   The same f splits E. Pulling the exact sequence back gives $0 \to f^*E' \xrightarrow{\iota} f^*E \xrightarrow{\pi} f^*E'' \to 0$, again exact since pullback of vector bundles is exact. Let $0 = E'_0 \subset \cdots \subset E'_{r'} = f^*E'$ and $0 = E''_0 \subset \cdots \subset E''_{r''} = f^*E''$ be the two filtrations, with line-bundle quotients L_p and M_q, and define
$$H_p = \iota(E'_p) \quad (0 \le p \le r'), \qquad H_{r'+q} = \pi^{-1}\big(E''_q\big) \quad (1 \le q \le r'')$$
   This is a filtration of f^*E by subbundles: the first stretch is the filtration of the subbundle f^*E', the second consists of preimages of subbundles under a surjection of bundles. Its successive quotients are $L_1, \dots, L_{r'}$ followed by $H_{r'+q}/H_{r'+q-1} \cong E''_q/E''_{q-1} = M_q$, the isomorphism holding because π carries $H_{r'+q}$ onto E''_q with kernel $f^*E' = H_{r'}$ contained in $H_{r'+q-1}$. So f is a splitting map for E with Chern roots $a_1, \dots, a_{r'}, b_1, \dots, b_{r''}$.

   Now apply proposition 10.3.4(4) three times. For every $\alpha \in A_*(X)_{\mathbb{Q}}$,
$$f^*\big(\mathrm{ch}(E) \cap \alpha\big) = \Big(\sum_{p=1}^{r'} e^{a_p} + \sum_{q=1}^{r''} e^{b_q}\Big) \cap f^*\alpha = f^*\big(\mathrm{ch}(E') \cap \alpha\big) + f^*\big(\mathrm{ch}(E'') \cap \alpha\big)$$
   the first equality reading the roots of E off the spliced filtration and the second applying the proposition to E' and to E'' separately. Since f^* is injective on $A_*(X)_{\mathbb{Q}}$, this gives $\mathrm{ch}(E) \cap \alpha = \mathrm{ch}(E') \cap \alpha + \mathrm{ch}(E'') \cap \alpha$ for every α, which is the asserted identity of operators. The direct sum is the case of a split exact sequence.

2. *Multiplicativity.* First a statement about filtrations: if a bundle A has a filtration with line-bundle quotients $N_1, \dots, N_a$ and B one with quotients $P_1, \dots, P_b$, then $A \otimes B$ has a filtration with quotients the $N_i \otimes P_j$. Induct on a. For $a = 1$ the bundle $A = N_1$ is a line bundle, and tensoring the filtration B_j of B by it gives a filtration of $N_1 \otimes B$ with quotients $N_1 \otimes P_j$, since tensoring by a line bundle is exact and carries subbundles to subbundles. For $a > 1$, tensoring $0 \to A_{a-1} \to A \to N_a \to 0$ with the locally free B keeps it exact; the subbundle $A_{a-1} \otimes B$ has the required filtration by induction, and taking preimages of the filtration of $N_a \otimes B$ under the surjection $A \otimes B \to N_a \otimes B$ extends it, exactly as in part (1).

By theorem 10.2.14 choose a flat proper $g\colon X'' \to X$ with g^* injective that is a splitting map for both E and F, with Chern roots $w_1, \ldots, w_r$ and $u_1, \ldots, u_s$. Applying the previous paragraph to $g^*E \otimes g^*F = g^*(E \otimes F)$ produces a filtration whose quotients are the rs tensor products of line bundles from the two filtrations, and the first Chern class of such a tensor product is $w_p + u_j$ by proposition 9.4.4(3). So g is a splitting map for $E \otimes F$ with Chern roots the operators $w_p + u_j$. By proposition 10.3.4(4) and the formal identity $e^{w+u} = e^w e^u$,

$$g^*\big(\mathrm{ch}(E \otimes F) \cap \alpha\big) = \Big(\sum_{p,j} e^{w_p + u_j}\Big) \cap g^*\alpha = \Big(\sum_p e^{w_p}\Big)\Big(\sum_j e^{u_j}\Big) \cap g^*\alpha$$

and the right-hand side is $g^*\big(\mathrm{ch}(E)\,\mathrm{ch}(F) \cap \alpha\big)$, again by proposition 10.3.4(4) applied twice, the two operators commuting by proposition 10.3.4(1). Injectivity of g^* gives $\mathrm{ch}(E \otimes F) = \mathrm{ch}(E)\,\mathrm{ch}(F)$, completing the proof.

The theorem says that ch converts the two ways of building bundles into the two operations of a commutative ring: extensions become sums, tensor products become composites. That is the structure of the Grothendieck group of vector bundles, in which an exact sequence is declared to be a sum and the tensor product is the multiplication, and the Chern character is the map out of it that the Riemann-Roch theorem later in this book computes with. A reader who has met Chern classes topologically [51] will recognize the axioms driving the argument, the Whitney formula and naturality being the same there; the topological Chern character satisfies the same two identities. The construction here is independent of that one, valid over any algebraically closed field and on singular schemes, and landing in rational Chow groups rather than in cohomology.

Additivity comes at a price. The total Chern class is multiplicative, so ch cannot also be, and the class that Riemann-Roch attaches to the tangent bundle, correcting the Chern character of a bundle to its Euler characteristic, has to be multiplicative. The Todd class is that multiplicative partner. The specific power series below is not derivable from anything in this chapter; it is the one that makes Riemann-Roch true, and the first evidence for it is the computation at the end of this section, where the top Todd component of the tangent bundle of $\mathbb{P}^n$ integrates to $1 = \chi(\mathcal{O}_{\mathbb{P}^n})$.

Definition 10.3.9: The Todd Class

Write $Q(t) = \dfrac{t}{1-e^{-t}} \in \mathbb{Q}[[t]]$, which is a power series because $1 - e^{-t} = t\big(1 - \frac{t}{2} + \frac{t^2}{6} - \cdots\big)$ and the second factor has constant term 1, hence is invertible. For a vector bundle E of rank r on an algebraic scheme X, the *Todd class* of E is the operator on $A_*(X)_{\mathbb{Q}}$ obtained from the symmetric power series

$$\mathrm{td}(a_1, \ldots, a_r) = \prod_{p=1}^{r} \frac{a_p}{1 - e^{-a_p}}$$

by proposition 10.3.4, written $\mathrm{td}(E)$, with $\mathrm{td}_k(E)$ its component lowering dimension by k. When X is smooth, $\mathrm{td}(X)$ abbreviates $\mathrm{td}(T_X)$ for the tangent bundle T_X.

Proposition 10.3.10: Components of the Todd Class

Let E be a vector bundle on X and write $c_i = c_i(E)$. Then

$$\mathrm{td}(E) = 1 + \tfrac{1}{2}c_1 + \tfrac{1}{12}\big(c_1^2 + c_2\big) + \tfrac{1}{24}c_1c_2 + \cdots$$

In particular $\mathrm{td}_0(E) = 1$, so $\mathrm{td}(E)$ is an invertible operator on $A_*(X)_{\mathbb{Q}}$, and for a line bundle L,

$$\mathrm{td}(L) = Q\big(c_1(L)\big) = 1 + \tfrac{1}{2}c_1(L) + \tfrac{1}{12}c_1(L)^2 + \cdots$$

***Proof*:**
Expand Q first. From $1 - e^{-t} = t\big(1 - \frac{t}{2} + \frac{t^2}{6} - \frac{t^3}{24} + \cdots\big)$ one gets $Q(t) = (1-u)^{-1}$ with $u = \frac{t}{2} - \frac{t^2}{6} + \frac{t^3}{24} - \cdots$, so $Q(t) = 1 + u + u^2 + u^3 + \cdots$. Collecting terms,

$$Q(t) = 1 + \tfrac{t}{2} + \Big(\tfrac{1}{4} - \tfrac{1}{6}\Big)t^2 + \Big(\tfrac{1}{8} - \tfrac{1}{6} + \tfrac{1}{24}\Big)t^3 + \cdots = 1 + \tfrac{t}{2} + \tfrac{t^2}{12} + 0 \cdot t^3 + \cdots$$

Now write $q_1 = \frac{1}{2}$, $q_2 = \frac{1}{12}$, $q_3 = 0$ and expand $\prod_{p=1}^{r}\big(1 + q_1a_p + q_2a_p^2 + q_3a_p^3 + \cdots\big)$ in degrees at most three, using the power sums $p_k = \sum_p a_p^k$ of proposition 10.3.6. In degree one the only contribution is $q_1p_1 = \frac{1}{2}e_1$. In degree two the contributions are q_2p_2 from a single factor and $q_1^2e_2$ from two distinct factors, giving

$$\tfrac{1}{12}\big(e_1^2 - 2e_2\big) + \tfrac{1}{4}e_2 = \tfrac{1}{12}\big(e_1^2 + e_2\big)$$

In degree three the contributions are $q_3p_3 = 0$, then $q_1q_2\sum_{p\neq q} a_p^2a_q$ from two distinct factors, then $q_1^3e_3$ from three distinct factors. Since $\sum_{p\neq q} a_p^2a_q = p_1p_2 - p_3 = e_1e_2 - 3e_3$, the degree-three term is

$$\tfrac{1}{24}\big(e_1e_2 - 3e_3\big) + \tfrac{1}{8}e_3 = \tfrac{1}{24}e_1e_2$$

Substituting $e_i = c_i(E)$ through proposition 10.3.4 gives the stated expansion. Invertibility follows from $\mathrm{td}(E) = \mathrm{id} + N$ with $N = \sum_{k\geq 1}\mathrm{td}_k(E)$ a sum of dimension-lowering operators, so $N^{\dim X + 1} = 0$ by lemma 10.3.2 and $\mathrm{td}(E)^{-1} = \sum_{k\geq 0}(-N)^k$ is a finite sum. For $r = 1$ the series is $Q(a_1)$ with $e_1 = a_1$, so $\mathrm{td}(L) = Q(c_1(L))$, as we sought to show.

The first component $\frac{1}{2}c_1$ already forces rational coefficients, exactly as ch_2 did, and the degree-three coefficient of Q vanishes because $Q(t) - \frac{t}{2}$ is even, so Q has no odd-degree terms beyond the first. The next example computes the class in the smallest case, where a single Chern class controls everything.

Example 10.3.11: The Todd Class of a Line Bundle on $\mathbb{P}^2$
With $h = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$ as in example 10.3.7 and $h^3 = 0$, proposition 10.3.10 gives

$$\mathrm{td}\big(\mathcal{O}_{\mathbb{P}^2}(d)\big) = 1 + \frac{d}{2}h + \frac{d^2}{12}h^2$$

so that, capping with the fundamental class, $\mathrm{td}(\mathcal{O}_{\mathbb{P}^2}(d)) \cap [\mathbb{P}^2] = [\mathbb{P}^2] + \frac{d}{2}[\mathbb{P}^1] + \frac{d^2}{12}[\mathrm{pt}]$. For $d = 1$ the zero-dimensional component is $\frac{1}{12}[\mathrm{pt}]$, of degree $\frac{1}{12}$ under $\int_{\mathbb{P}^2}$. Individual Todd components carry denominators that no bundle removes; it is only the full expression appearing in Riemann-Roch, a Todd class multiplied by a Chern character, that is forced to have integral degree.

Proposition 10.3.12: Multiplicativity of the Todd Class

Let $0 \to E' \to E \to E'' \to 0$ be an exact sequence of vector bundles on an algebraic scheme X. Then
$$\operatorname{td}(E) = \operatorname{td}(E')\operatorname{td}(E'')$$
as operators on $A_*(X)_{\mathbb{Q}}$. In particular $\operatorname{td}(E \oplus F) = \operatorname{td}(E)\operatorname{td}(F)$, and $\operatorname{td}(E \oplus \mathcal{O}_X^{\oplus m}) = \operatorname{td}(E)$.

***Proof*:**
Choose, by theorem 10.2.14, a flat proper $f\colon X' \to X$ with f^* injective splitting both E' and E'', with roots $a_1, \ldots, a_{r'}$ and $b_1, \ldots, b_{r''}$. Part (1) of the proof of theorem 10.3.8 shows that f then splits E, with roots the union of the two lists. Applying proposition 10.3.4(4) three times, to E, to E' and to E'', and using the factorization of the defining product over the union of the two lists,

$$f^*\big(\operatorname{td}(E) \cap \alpha\big) = \Big(\prod_p Q(a_p) \prod_q Q(b_q)\Big) \cap f^*\alpha = f^*\big(\operatorname{td}(E')\operatorname{td}(E'') \cap \alpha\big)$$

for every $\alpha \in A_*(X)_{\mathbb{Q}}$, the two operators on the right commuting by proposition 10.3.4(1). Injectivity of f^* gives $\operatorname{td}(E) = \operatorname{td}(E')\operatorname{td}(E'')$. The direct sum is the split case, and a trivial bundle contributes $Q(0)^m = 1$ because $c_1(\mathcal{O}_X) = 0$ by example 9.4.3, completing the proof.

Additivity of ch and multiplicativity of td are what make both computable: any bundle presented by an exact sequence with known ends has both classes determined. The tangent bundle of projective space is presented in exactly that way, and the computation below is the one Riemann-Roch is checked against.

Example 10.3.13: The Tangent Bundle of $\mathbb{P}^n$

Example 10.2.17 has already imported the Euler sequence from theorem 5.5.18, dualized it to

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$

and read off $c(T_{\mathbb{P}^n}) = (\mathrm{id} + H)^{n+1}$. What is new here is the Chern character and Todd class the same sequence yields. Here $\mathbb{P}^n$ is projective n-space with its standard twisting sheaf, so the convention of box 7.3.10 does not intervene and no further dualization is needed. Write $h = c_1(\mathcal{O}_{\mathbb{P}^n}(1))$.

The Chern character. By theorem 10.3.8(1), $\operatorname{ch}(\mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)}) = \operatorname{ch}(\mathcal{O}_{\mathbb{P}^n}) + \operatorname{ch}(T_{\mathbb{P}^n})$, and the left side is $(n+1)e^h$ while $\operatorname{ch}(\mathcal{O}_{\mathbb{P}^n}) = 1$. Hence

$$\operatorname{ch}(T_{\mathbb{P}^n}) = (n+1)e^h - 1$$

The Todd class. By proposition 10.3.12 applied to the same sequence, $\operatorname{td}(\mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)}) = \operatorname{td}(\mathcal{O}_{\mathbb{P}^n})\operatorname{td}(T_{\mathbb{P}^n}) = \operatorname{td}(T_{\mathbb{P}^n})$, so
$$\operatorname{td}(T_{\mathbb{P}^n}) = Q(h)^{n+1}$$
with Q the series of definition 10.3.9. Both classes are computed from the single operator h, whose powers are known: $h^k \cap [\mathbb{P}^n] = [\mathbb{P}^{n-k}]$ and $h^{n+1} = 0$.

The case $n = 1$. Here $h^2 = 0$, so $\operatorname{ch}(T_{\mathbb{P}^1}) = 2(1+h) - 1 = 1 + 2h$ and $\operatorname{td}(T_{\mathbb{P}^1}) = (1 + \frac{h}{2})^2 = 1 + h$. The first agrees with the identification $T_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(2)$, whose Chern character is $e^{2h} = 1 + 2h$ by example 10.3.7. The dimension-lowering component of $\mathrm{td}(T_{\mathbb{P}^1})$ caps to $h \cap [\mathbb{P}^1] = [\mathrm{pt}]$, of degree $\int_{\mathbb{P}^1} [\mathrm{pt}] = 1$.

The case $n = 2$. Here $h^3 = 0$ and

$$\mathrm{ch}(T_{\mathbb{P}^2}) = 3\Big(1 + h + \tfrac{h^2}{2}\Big) - 1 = 2 + 3h + \tfrac{3}{2}h^2$$

which cross-checks against proposition 10.3.6: the Whitney formula applied to the Euler sequence gives $c(T_{\mathbb{P}^2}) = (1+h)^3 = 1 + 3h + 3h^2$, so $c_1 = 3h$ and $c_2 = 3h^2$, and then $\mathrm{ch}_0 = 2 = \mathrm{rank} T_{\mathbb{P}^2}$, $\mathrm{ch}_1 = c_1 = 3h$, and $\mathrm{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2) = \frac{1}{2}(9h^2 - 6h^2) = \frac{3}{2}h^2$. For the Todd class,

$$\mathrm{td}(T_{\mathbb{P}^2}) = \Big(1 + \tfrac{h}{2} + \tfrac{h^2}{12}\Big)^3 = 1 + \tfrac{3}{2}h + \Big(3 \cdot \tfrac{1}{12} + 3 \cdot \tfrac{1}{4}\Big)h^2 = 1 + \tfrac{3}{2}h + h^2$$

agreeing with $\mathrm{td}_2 = \frac{1}{12}(c_1^2 + c_2) = \frac{1}{12}(9+3)h^2 = h^2$ from proposition 10.3.10. Capping with the fundamental class,

$$\mathrm{ch}(T_{\mathbb{P}^2}) \cap [\mathbb{P}^2] = 2[\mathbb{P}^2] + 3[\mathbb{P}^1] + \tfrac{3}{2}[\mathrm{pt}], \qquad \mathrm{td}(T_{\mathbb{P}^2}) \cap [\mathbb{P}^2] = [\mathbb{P}^2] + \tfrac{3}{2}[\mathbb{P}^1] + [\mathrm{pt}]$$

in $A_*(\mathbb{P}^2)_{\mathbb{Q}}$.

The check. In both cases the zero-dimensional component of $\mathrm{td}(T_{\mathbb{P}^n}) \cap [\mathbb{P}^n]$ has degree 1 under $\int_{\mathbb{P}^n}$, and $\chi(\mathcal{O}_{\mathbb{P}^n}) = 1$ by theorem 6.3.1. The agreement of these two numbers is the simplest instance of the Riemann-Roch theorem proved later in this book, which computes, for a vector bundle E on a smooth projective X, the integer $\chi(E) = \int_X \mathrm{ch}(E)\,\mathrm{td}(X) \cap [X]$; and it is the reason $t/(1 - e^{-t})$, rather than some other multiplicative series, is the one attached to the tangent bundle.

10.3.14 Remark: Operators Now, Ring Elements Later Both classes have been defined as operators on $A_*(X)_{\mathbb{Q}}$, and every identity above is an identity of operators, composition serving as multiplication. This is the only option available so far, since $A_*(X)$ carries no product; chapter 11 and the chapter after it supply one for smooth X, at which point $A^*(X)_{\mathbb{Q}}$ becomes a graded ring, capping with the fundamental class turns each of these operators into a class, and $\mathrm{ch}(E)$ and $\mathrm{td}(E)$ are elements of that ring. No statement in this section changes when that happens: an identity of operators on $A_*(X)_{\mathbb{Q}}$ specializes to an identity of classes by applying it to $[X]$.

Exercise 10.3.1

1. Let $h = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$ and let $L = \mathcal{O}_{\mathbb{P}^2}(d)$. Compute $\mathrm{ch}(L) \cap [\mathbb{P}^2]$ and $\mathrm{td}(L) \cap [\mathbb{P}^2]$ in $A_*(\mathbb{P}^2)_{\mathbb{Q}}$, and compute the degree $\int_{\mathbb{P}^2}$ of the zero-dimensional component of $\mathrm{ch}(L)\,\mathrm{td}(T_{\mathbb{P}^2}) \cap [\mathbb{P}^2]$. Determine for which d this last number is an integer.
2. Show that $\mathrm{ch}_k(E^\vee) = (-1)^k \mathrm{ch}_k(E)$ for every vector bundle E and every k. (Dualize a filtration by subbundles and use proposition 9.4.4(3).) Deduce that $\mathrm{ch}(E \otimes E^\vee)$ has $\mathrm{ch}_1 = 0$ for every E.
3. Let E have rank r and let $\det E = \Lambda^r E$. Show that $c_1(\det E) = c_1(E)$ and hence $\mathrm{ch}(\det E) = e^{c_1(E)}$. (Use that the top exterior power of a filtered bundle is the tensor product of the line-bundle quotients.)

4. Compute $\mathrm{ch}(T_{\mathbb{P}^3})$ and $\mathrm{td}(T_{\mathbb{P}^3})$ as polynomials in $h = c_1(\mathcal{O}_{\mathbb{P}^3}(1))$, and verify that $\int_{\mathbb{P}^3}$ of the zero-dimensional component of $\mathrm{td}(T_{\mathbb{P}^3}) \cap [\mathbb{P}^3]$ equals 1. Check the degree-two and degree-three Todd components against proposition 10.3.10 using $c(T_{\mathbb{P}^3}) = (1+h)^4$.

5. Let S be a smooth projective surface and E a vector bundle of rank r on S. Write out the zero-dimensional component of $\mathrm{ch}(E)\,\mathrm{td}(T_S) \cap [S]$ in terms of $c_1(E), c_2(E), c_1(T_S), c_2(T_S)$, and specialize to $E = \mathcal{O}_S$ and to E a line bundle.

6. Show that $\mathrm{ch}(E \oplus \mathcal{O}_X^{\oplus m}) = \mathrm{ch}(E) + m$ and $\mathrm{td}(E \oplus \mathcal{O}_X^{\oplus m}) = \mathrm{td}(E)$, and explain why no bundle of positive rank can have $\mathrm{ch}(E) = 0$.

10.4 Segre Classes of Cones and Subschemes

The Segre operators of section 10.1 came out of one recipe: projectivize, cap with powers of the tautological class, push down. Local triviality was never used in that recipe. What it used was a graded sheaf of algebras on X whose relative Proj is proper over X and carries a canonical line bundle, and a vector bundle supplies one such algebra among many. This section runs the recipe on the general input, a *cone*, and then on the case that matters here: the normal cone of a closed subscheme $Z \subseteq X$. The resulting class $s(Z, X)$ is an invariant of the embedding defined with no smoothness, no regularity and no dimension hypothesis, and it is the object that chapter 11 will deform X onto.

The algebra that produces a cone is required to look like a polynomial algebra in the two respects that matter for Proj: it starts in degree zero at $\mathcal{O}_X$, and it is generated by its degree-one part, which is coherent.

Definition 10.4.1: Cone

Let X be an algebraic scheme. A *cone* over X is the relative spectrum

$$C = \mathrm{Spec}\left(S^\bullet\right) \xrightarrow{p} X$$

of a sheaf $S^\bullet = \bigoplus_{n \geq 0} S_n$ of graded quasi-coherent $\mathcal{O}_X$-algebras (definition 5.3.33) such that

1. $S_0 = \mathcal{O}_X$;
2. S_1 is coherent;
3. $S^\bullet$ is generated as an $\mathcal{O}_X$-algebra by S_1.

The relative spectrum is the construction of theorem 5.1.41, glued from the affine pieces $\mathrm{Spec}\,\Gamma(U, S^\bullet)$ over an affine open cover of X, exactly as the relative Proj of theorem 5.3.35 is. Writing $S_+ = \bigoplus_{n \geq 1} S_n$, the surjection $S^\bullet \to S^\bullet / S_+ = \mathcal{O}_X$ defines a closed immersion $X \hookrightarrow C$, the *vertex* of the cone.

Conditions (2) and (3) force every S_n to be coherent, so p is affine and of finite type; the grading is what distinguishes a cone from an arbitrary affine morphism, and it is the grading that supplies the vertex and, below, the compactification. A cone is not required to be flat over X, nor to have fibres of constant dimension, and both failures are the point: the cones that arise from geometry are singular, jump in dimension along strata, and carry exactly the information those defects encode.

Example 10.4.2: Bundles, the Zero Cone, and the Affine Cone Over a Projective Variety

1. *A vector bundle.* Let E be a vector bundle of rank e on X with sheaf of sections $\mathcal{E}$. The functions on the total space of E are the symmetric algebra on the dual, so $S^\bullet = \mathrm{Sym}^\bullet(\mathcal{E}^\vee)$ has $\mathrm{Spec}\,(S^\bullet) = E$, and conditions (1) to (3) hold with $S_1 = \mathcal{E}^\vee$ locally free of rank e. The vertex $X \hookrightarrow E$ is the zero section. The dual is exactly what makes $\mathrm{Proj}\,(\mathrm{Sym}^\bullet\mathcal{E}^\vee)$ the bundle of *lines* in E used throughout this book (box 7.3.10), and not the bundle of rank-one quotients of $\mathcal{E}$ that definition 5.3.37 writes as $\mathbb{P}(\mathcal{E})$.

2. *The zero cone.* Taking $S^\bullet = \mathcal{O}_X$, so that $S_+ = 0$, gives $C = X$ with its vertex the identity. The degenerate case earns its place: it is the normal cone of X in itself.

3. *An affine cone.* Let $Y = \mathrm{Proj}\,(R) \subseteq \mathbb{P}^n$ be a closed subscheme, where $R = \overline{k}[x_0, \ldots, x_n]/I$ for a homogeneous ideal I, and take $X = \mathrm{Spec}\,\overline{k}$. Then R itself satisfies (1) to (3), and $C = \mathrm{Spec}\,(R)$ is the affine cone over Y: the union of the lines of $\mathbb{A}^{n+1}$ through the origin lying over the points of Y, together with the origin, which is the vertex. For $I = 0$ this is $\mathbb{A}^{n+1}$ itself, the trivial bundle of rank $n+1$ over a point, recovering item (1).

A cone is affine over X, so p is proper only in the degenerate case where $S^\bullet$ is coherent as an $\mathcal{O}_X$-module and p is therefore finite; the zero cone of example 10.4.2(2), with $p = \mathrm{id}$, is such a case. The cones that carry geometry are not: a vector bundle of positive rank has fibres $\mathbb{A}^e$, and an affine scheme of positive dimension over a field is not proper. So p_* is unavailable in general, and pushing a cycle from C down to X is not an option. The remedy is the one projective space applies to affine space: compactify by adding a hyperplane at infinity. Adjoining one variable in degree one does this, and does it in a way that keeps the answer a relative Proj , so the canonical line bundle comes along for free.

Definition 10.4.3: Projective Cone and Projective Completion

Let $C = \mathrm{Spec}\,(S^\bullet)$ be a cone over X. Adjoin a variable z in degree one, giving the graded $\mathcal{O}_X$-algebra $S^\bullet[z]$ with $\big(S^\bullet[z]\big)_n = \bigoplus_{j=0}^n S_j\, z^{\,n-j}$, which again satisfies the conditions of definition 10.4.1. The *projective cone* of C and the *projective completion* of C are the relative Proj schemes

$$\mathbb{P}(C) = \mathrm{Proj}\,\big(S^\bullet\big), \qquad \mathbb{P}(C \oplus 1) = \mathrm{Proj}\,\big(S^\bullet[z]\big)$$

Both are proper over X (theorem 5.3.35). Write $q\colon \mathbb{P}(C \oplus 1) \to X$ for the structure morphism and $\mathcal{O}(1) = \mathcal{O}_{\mathbb{P}(C\oplus 1)}(1)$ for the canonical line bundle.

Three identifications justify the name, and each is read straight off the graded algebra. First, on the locus where z is invertible, $D_+(z) = \mathrm{Spec}\,\big((S^\bullet[z])_{(z)}\big)$, and the assignment $s/z^n \mapsto s$ for $s \in S_n$ is an isomorphism of $\mathcal{O}_X$-algebras $(S^\bullet[z])_{(z)} \cong S^\bullet$; so the open subscheme of $\mathbb{P}(C \oplus 1)$ where $z \neq 0$ is C itself. Second, z is a nonzerodivisor in $S^\bullet[z]$ and a global section of $\mathcal{O}(1)$, and its zero scheme is $\mathrm{Proj}\,\big(S^\bullet[z]/(z)\big) = \mathbb{P}(C)$; so $\mathbb{P}(C)$ is an effective Cartier divisor (definition 5.4.25) on $\mathbb{P}(C \oplus 1)$ with complement C, the hyperplane at infinity. Third, the surjection $S^\bullet[z] \to \mathcal{O}_X[z]$ killing S_+ realizes $X = \mathrm{Proj}\,(\mathcal{O}_X[z])$ as a closed subscheme of $\mathbb{P}(C \oplus 1)$ contained in the open piece C, where it is the vertex of definition 10.4.1.

So $\mathbb{P}(C \oplus 1)$ is C with $\mathbb{P}(C)$ glued on at infinity, and it is proper over X whereas C, outside the degenerate case just noted, is not. That properness is why it is introduced: it makes q_* available (definition 8.2.1 and theorem 8.2.3), and with $\mathcal{O}(1)$ in hand the construction of section 10.1 can be run verbatim.

The model case is the one the terminology is borrowed from. Take $X = \operatorname{Spec} \overline{k}$ and $S^\bullet = \overline{k}[t_1, \ldots, t_n]$, so that $C = \mathbb{A}^n$ is the trivial bundle of rank n over a point. Then $S^\bullet[z] = \overline{k}[t_1, \ldots, t_n, z]$ and $\mathbb{P}(C \oplus 1) = \mathbb{P}^n$, with $\mathbb{P}(C) = \mathbb{P}^{n-1}$ the hyperplane $z = 0$ at infinity, the open complement $D_+(z)$ the standard affine chart $\mathbb{A}^n$, and the vertex the origin of that chart. Compactifying a cone is compactifying affine space by a hyperplane, carried out fibrewise over X and allowing the fibres to be as singular as they like.

Definition 10.4.4: Segre Class of a Cone

Let C be a cone over an algebraic scheme X, with projective completion $q\colon \mathbb{P}(C \oplus 1) \to X$ and canonical line bundle $\mathcal{O}(1)$. The *Segre class* of C is the cycle class

$$s(C) \;=\; q_*\Big(\sum_{i \geq 0} c_1\big(\mathcal{O}(1)\big)^i \cap \big[\mathbb{P}(C \oplus 1)\big]\Big) \;\in\; A_*(X)$$

where $[\mathbb{P}(C \oplus 1)]$ is the fundamental cycle of definition 8.1.2 and $c_1(\mathcal{O}(1)) \cap -$ is the operator of definition 9.4.1. Write $s(C)_k$ for the component of $s(C)$ in $A_k(X)$.

The sum is finite. Each cap lowers dimension by one, so the term indexed by i lies in $A_{d-i}\big(\mathbb{P}(C \oplus 1)\big)$ for $d = \dim \mathbb{P}(C \oplus 1)$, and this group vanishes once $i > d$ because there are no cycles of negative dimension. Two features of the definition deserve emphasis. The Segre class of a cone is a *class*, not an operator: the capping has already happened, against the fundamental cycle of the compactification. And it is not homogeneous; $s(C)$ has components in several dimensions at once, and the interesting content is usually spread across them.

For the model cone $C = \mathbb{A}^n$ over $X = \operatorname{Spec} \overline{k}$ the class can be read off. The completion is $\mathbb{P}^n$, and $q_*\big(c_1(\mathcal{O}(1))^i \cap [\mathbb{P}^n]\big)$ lies in $A_{n-i}(\operatorname{Spec} \overline{k})$, which vanishes unless $i = n$. For $i = n$, example 9.4.7 gives $c_1(\mathcal{O}(1))^n \cap [\mathbb{P}^n] = [\mathrm{pt}]$, whose pushforward to the point is the fundamental class. Hence $s(\mathbb{A}^n) = [\operatorname{Spec} \overline{k}]$: the trivial bundle has trivial Segre class, as it must.

Before the definition can be trusted it has to be checked against the case it generalizes. A vector bundle is a cone, and its Segre class as a cone had better be its total Segre operator evaluated on the fundamental class of the base. The bundle has to have positive rank for that comparison to mean anything: the operators $s_i(E)$ of section 10.1 are manufactured from $\mathbb{P}(E)$, which is empty for the zero bundle, so neither $s(E)$ nor $c(E)$ is defined there. The rank-zero cone is the zero cone of example 10.4.2(2), whose Segre class is computed directly below.

Proposition 10.4.5: The Segre Class of a Vector Bundle as a Cone

Let E be a vector bundle of rank $e \geq 1$ on an algebraic scheme X, and let C be its total space regarded as a cone (example 10.4.2(1)). Then

$$s(C) \;=\; s(E) \cap [X]$$

where $s(E) = \sum_{j \geq 0} s_j(E)$ is the total Segre operator of section 10.1.

***Proof*:**
Step 1: the projective completion is a projective bundle. With $S^\bullet = \operatorname{Sym}^\bullet(\mathcal{E}^\vee)$ one has $S^\bullet[z] = \operatorname{Sym}^\bullet\big(\mathcal{E}^\vee \oplus \mathcal{O}_X\big) = \operatorname{Sym}^\bullet\big((\mathcal{E} \oplus \mathcal{O}_X)^\vee\big)$, the variable z being the coordinate dual to the added trivial summand. Hence $\mathbb{P}(C \oplus 1) = \mathbb{P}(E \oplus 1)$ is the bundle of lines in the rank-$(e+1)$ bundle $E \oplus 1$, its fibres are copies of $\mathbb{P}^e$, and its canonical bundle $\mathcal{O}(1)$ is the one whose first Chern class is the tautological class of section 10.1. In particular q is proper and flat of relative dimension e (proposition 5.3.40).

Step 2: the fundamental cycle is a flat pullback. Write $n = \dim X$ and let $X_1, \ldots, X_r$ be the n-dimensional irreducible components of X, so that the fundamental cycle is $[X] = \sum_\lambda m_\lambda [X_\lambda]$ with m_λ the length of $\mathcal{O}_{X,X_\lambda}$. Each preimage $q^{-1}(X_\lambda)$ is a projective bundle over the variety X_λ, hence integral, so $q^*[X_\lambda] = [q^{-1}(X_\lambda)]$ by proposition 8.3.5(2); and the length multiplicity of $q^{-1}(X_\lambda)$ in $[\mathbb{P}(E \oplus 1)]$ is again m_λ, because the local homomorphism $\mathcal{O}_{X,X_\lambda} \to \mathcal{O}_{\mathbb{P}(E\oplus 1),\, q^{-1}(X_\lambda)}$ is flat with fibre the function field of a projective space, so the two lengths agree. Summing over λ,

$$q^*[X] = \big[\mathbb{P}(E \oplus 1)\big]$$

Step 3: the terms with $i \geq e$. Definition 10.1.3 defines the Segre operators of a bundle F of rank f by $s_j(F) \cap \alpha = \pi_*\big(\xi^{\,f-1+j} \cap \pi^*\alpha\big)$, where $\pi \colon \mathbb{P}(F) \to X$ has relative dimension $f-1$ and $\xi = c_1(\mathcal{O}_{\mathbb{P}(F)}(1))$. Applied to $F = E \oplus 1$, of rank $e+1$, this reads

$$s_j(E \oplus 1) \cap \alpha = q_*\big(c_1(\mathcal{O}(1))^{\,e+j} \cap q^*\alpha\big), \qquad j \geq 0$$

so by Step 2 the terms of definition 10.4.4 indexed by $i = e + j$ with $j \geq 0$ sum to $\sum_{j\geq 0} s_j(E \oplus 1) \cap [X] = s(E \oplus 1) \cap [X]$.

Step 4: the terms with $i < e$ vanish. Flat pullback raises dimension by e and each cap lowers it by one, so $q_*\big(c_1(\mathcal{O}(1))^i \cap q^*[X]\big)$ lies in $A_{n+e-i}(X)$. For $i < e$ this index exceeds $n = \dim X$, and X carries no subvariety of dimension greater than its own, so the group is zero and every such term vanishes.

Step 5: dropping the trivial summand. This is the only step that uses $e \geq 1$: both E and $E \oplus 1$ then have positive rank, so both have invertible total Segre operators (lemma 10.2.2) and Chern operators to compare (definition 10.2.3). The split exact sequence $0 \to E \to E \oplus 1 \to \mathcal{O}_X \to 0$ and the Whitney formula (theorem 10.2.15) give $c(E \oplus 1) = c(E)\, c(\mathcal{O}_X)$. A line bundle has no Chern classes above the rank (proposition 10.2.6), so $c(\mathcal{O}_X) = \mathrm{id} + c_1(\mathcal{O}_X)$, and $c_1(\mathcal{O}_X) = 0$ by example 9.4.3; hence $c(E \oplus 1) = c(E)$ and, inverting, $s(E \oplus 1) = s(E)$. Combining with Steps 3 and 4,

$$s(C) = s(E \oplus 1) \cap [X] = s(E) \cap [X]$$

as we sought to show.

The proposition licenses the notation: for the cone of a vector bundle the class $s(C)$ of definition 10.4.4 and the operator $s(E)$ of section 10.1 determine one another, and writing s for both loses nothing. What the proposition also shows is where the new content will sit. On a bundle, the compactification $\mathbb{P}(E \oplus 1)$ is flat over X with fibres of constant dimension, the Segre operators exist, and $s(C)$ is a formal consequence of them. On a general cone none of that is available, and $s(C)$ does not come from any operator; it is a class computed once, by pushing forward from a compactification that may be singular and may jump in dimension over X.

The cone this was built for measures how a closed subscheme sits inside an ambient scheme. When

Z is a smooth subvariety of a smooth variety, the correct linear model is the normal bundle, and its rank is the codimension. When either smoothness hypothesis fails there is no bundle, but the ideal sheaf still has an associated graded algebra, and that algebra is always a cone.

Definition 10.4.6: Normal Cone of a Closed Subscheme

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme, with ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$. The graded sheaf of algebras $\bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$ has degree-zero part $\mathcal{O}_X/\mathcal{I} = \mathcal{O}_Z$, coherent degree-one part $\mathcal{I}/\mathcal{I}^2$, and is generated in degree one, so it satisfies the conditions of definition 10.4.1. The *normal cone* of Z in X is the cone over Z

$$C_Z X \;=\; \operatorname{Spec}\Big(\bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}\Big) \;\longrightarrow\; Z$$

On an affine open $U = \operatorname{Spec} A$ of X with $Z \cap U = \operatorname{Spec} A/I$, this is $\operatorname{Spec}\big(\bigoplus_{n\geq 0} I^n/I^{n+1}\big)$.

The normal cone is intrinsic to the pair (Z, X) and depends on the scheme structure of Z, not just on its underlying set: the ideal $\mathcal{I}$ is the input, and thickening Z changes it. When Z is a single reduced point P of a variety X, the normal cone is the classical tangent cone $\operatorname{Spec}\big(\operatorname{gr}_{\mathfrak{m}} \mathcal{O}_{X,P}\big)$, the cone of limiting directions of branches of X through P, and it is where the local singularity of X at P becomes visible.

Example 10.4.7: The Tangent Cone at a Node

Assume $\operatorname{char}\overline{k} \neq 2$ and let $X = \operatorname{Spec}\overline{k}[x,y]/(y^2 - x^2 - x^3) \subseteq \mathbb{A}^2$ be the affine nodal cubic, with P the origin and $\mathfrak{m} = (x, y)$ its maximal ideal.[a] The associated graded ring of $\overline{k}[x,y]$ for the filtration by powers of (x,y) is again $\overline{k}[x,y]$, a domain, so initial forms multiply: the initial form of gh is the product of those of g and h. Consequently the initial forms of the elements of the principal ideal $(y^2 - x^2 - x^3)$ are generated by the initial form of its generator, namely $y^2 - x^2$. Hence

$$\bigoplus_{n\geq 0} \mathfrak{m}^n/\mathfrak{m}^{n+1} \;\cong\; \overline{k}[x,y]/(y^2 - x^2)$$

and $C_P X = \operatorname{Spec}\overline{k}[x,y]/(y^2 - x^2)$ is the union of the two lines $y = x$ and $y = -x$: precisely the two branches of the node, straightened to their tangent directions. The normal cone is one-dimensional, matching $\dim X = 1$, but it is reducible where X is irreducible, and it is not a vector bundle over the point P (a rank-one bundle over a point is a single line). The failure is exactly the singularity.

[a] The hypothesis on the characteristic cannot be dropped. In characteristic 2 the substitution $u = y + x$ turns the equation into $u^2 = x^3$, so the curve is a cuspidal cubic with a single branch, and $y^2 - x^2 = (y + x)^2$ makes the ring computed below the coordinate ring of a single non-reduced double line, which is irreducible; the contrast drawn at the end of the example then fails.

The normal cone is a cone over Z, so it has a Segre class, and that class is what records how Z sits inside X when no bundle is available to record it.

Definition 10.4.8: Segre Class of a Closed Subscheme

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme. The *Segre class of Z in X* is the Segre class of the normal cone,

$$s(Z, X) \;=\; s\big(C_Z X\big) \;\in\; A_*(Z)$$

Two points about the target group. The class lives on Z, not on X: like the intersection classes of definition 9.2.1, it is recorded where the geometry happens, and one pushes forward along $Z \hookrightarrow X$ only when the cruder class in $A_*(X)$ is wanted. And no hypothesis whatsoever was imposed on the pair (Z, X): X may be singular, reducible, or nonreduced, Z may be any closed subscheme including X itself, and $s(Z, X)$ is defined regardless. That generality is what makes the class useful and also what makes it hard to compute; the two computations below are the cases where it can be read off.

The extreme case is immediate. Taking $Z = X$ makes $\mathcal{I} = 0$, so the normal cone is the zero cone of example 10.4.2(2), with projective completion $\operatorname{Proj}(\mathcal{O}_X[z]) = X$ and with $\mathcal{O}(1)$ free on the generator z, hence trivial. Every term of definition 10.4.4 beyond the first therefore vanishes, since $c_1(\mathcal{O}_X) = 0$ by example 9.4.3, and $s(X, X) = [X]$. A subscheme sitting inside itself has no normal directions, and its Segre class is its own fundamental cycle.

10.4.9 Foreshadowing: Why the Normal Cone The normal cone is not introduced here for its own sake. Chapter 11 constructs a family over $\mathbb{P}^1$ whose general fibre is the embedding $Z \hookrightarrow X$ and whose special fibre is the embedding of Z as the vertex of $C_Z X$; intersecting with Z inside X is then replaced by intersecting with the vertex inside the cone, where the answer can be read off. When $C_Z X$ is a vector bundle that replacement produces the Gysin map of a regular embedding; when it is not, the Segre class $s(Z, X)$ is what survives, and it is the term that appears in the excess intersection formula.

The first computable case is the one where the normal cone degenerates to a bundle. The condition on the ideal is the classical one: it should be generated, locally, by the right number of equations imposing independent conditions.

Regular embeddings and their normal bundles are scheme theory, and they belong to, and are defined in, *Algebraic Geometry*: a closed immersion whose ideal sheaf is locally generated by a regular sequence of length d is a *regular embedding of codimension d* (definition 5.1.43), its conormal sheaf $\mathcal{I}/\mathcal{I}^2$ is then locally free of rank d (proposition 5.1.44), and its *normal bundle* is the rank-d vector bundle

$$N_Z X \;=\; \underline{\operatorname{Spec}}_Z\Big(\operatorname{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)\Big) \;\longrightarrow\; Z$$

with sheaf of sections $(\mathcal{I}/\mathcal{I}^2)^\vee$ (definition 5.1.45). What this book adds is the cycle-level consequence below.

The definition is local on Z, and the number d is forced: the rank of $\mathcal{I}/\mathcal{I}^2$ is a local invariant, so a regular embedding has a well-defined codimension on each connected component of Z. The two extremes are familiar. The origin in $\mathbb{A}^n$, whose ideal $(x_1, \ldots, x_n)$ is a regular sequence, is a regular embedding of codimension n with trivial normal bundle of rank n; an effective Cartier divisor, whose ideal is locally generated by a single nonzerodivisor, is the codimension-one case, with normal bundle a line bundle. What makes the notion useful here is that the whole associated graded algebra, not its degree-one part, becomes a symmetric algebra.

Lemma 10.4.10: The Associated Graded Algebra of a Regular Sequence

Let A be a ring and let $a_1, \ldots, a_d$ be a regular sequence generating an ideal $I \subseteq A$. Then the homomorphism of graded A/I-algebras

$$(A/I)[T_1, \ldots, T_d] \longrightarrow \bigoplus_{n \geq 0} I^n/I^{n+1}, \qquad T_j \longmapsto a_j \bmod I^2$$

is an isomorphism. In particular I/I^2 is a free A/I-module of rank d on the classes of $a_1, \ldots, a_d$, and the canonical surjection $\mathrm{Sym}^\bullet_{A/I}(I/I^2) \to \bigoplus_{n \geq 0} I^n/I^{n+1}$ is an isomorphism.

***Proof*:**
This is [10, Associated Graded Ring of a Regular Sequence], proved there by induction on the length of the sequence. It is pure commutative algebra and is not re-derived here.

Proposition 10.4.11: Segre Class of a Regular Embedding

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$, with normal bundle $N = N_Z X$. Then the normal cone is the normal bundle,

$$C_Z X \;=\; N_Z X$$

and the Segre class of Z in X is

$$s(Z, X) \;=\; s(N) \cap [Z] \;=\; c(N)^{-1} \cap [Z]$$

where $s(N)$ and $c(N)$ are the total Segre and Chern operators of sections 10.1 and 10.2.

***Proof*:**
Work on an affine open $U = \operatorname{Spec} A$ of X on which $I = \mathcal{I}(U)$ is generated by a regular sequence $a_1, \ldots, a_d$. There is always a canonical surjection of graded A/I-algebras

$$\mathrm{Sym}^\bullet_{A/I}(I/I^2) \longrightarrow \bigoplus_{n \geq 0} I^n/I^{n+1}$$

sending a product of degree-one classes to the class of the corresponding product in I^n; it is surjective because the target is generated in degree one. Lemma 10.4.10 says that it is injective as well, and that I/I^2 is free of rank d on the classes of the a_j, so it is an isomorphism over U.

The surjection above is canonical, so the local isomorphisms are compatible on overlaps and glue to an isomorphism of graded $\mathcal{O}_Z$-algebras $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2) \xrightarrow{\sim} \bigoplus_{n \geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$. Taking relative spectra and comparing definition 10.4.6 with definition 5.1.43 gives $C_Z X = N_Z X$.

The normal cone is therefore the cone of a vector bundle of rank $d \geq 1$ over Z, so proposition 10.4.5 applies and gives $s(Z, X) = s(C_Z X) = s(N) \cap [Z]$. Finally $s(N) = c(N)^{-1}$, the Chern operators of definition 10.2.3 being defined by inverting the Segre series, completing the proof.

The normal cone is functorial in the subscheme, and for a regular embedding it sits inside a pulled-back normal bundle. Both facts are needed by chapter 11, and both are statements about cones as such, so they belong here rather than there.

Lemma 10.4.12: The Normal Cone of a Closed Subscheme

Let $Z, V \subseteq X$ be closed subschemes of an algebraic scheme and let $W = V \cap Z$ be their scheme-theoretic intersection. Then W is the scheme-theoretic preimage of Z under $V \hookrightarrow X$, and there is a canonical closed immersion

$$C_W V \hookrightarrow C_Z X$$

lying over the inclusion $W \hookrightarrow Z$.

***Proof*:**
The ideal sheaf of W in V is $\mathcal{I}_W = \mathcal{I}\mathcal{O}_V$, which is exactly the statement that W is the scheme-theoretic preimage of Z under the closed immersion $V \hookrightarrow X$. Consequently $\mathcal{I}^n\mathcal{O}_V = \mathcal{I}_W^n$ for every n, so the homomorphism of graded $\mathcal{O}_W$-algebras

$$\Big(\bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}\Big) \otimes_{\mathcal{O}_Z} \mathcal{O}_W \longrightarrow \bigoplus_{n\geq 0} \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$$

is surjective in each degree, the image of $\mathcal{I}^n \otimes \mathcal{O}_V \to \mathcal{O}_V$ being $\mathcal{I}_W^n$. On an affine open this is a surjection of graded rings, so by theorem 5.1.41(1) it induces a closed immersion of relative spectra; the local closed immersions are induced by a globally defined homomorphism of sheaves of algebras, so they glue. This exhibits $C_W V$ as a closed subscheme of the relative spectrum of the left side, which is $C_Z X \times_Z W$, again by the affine description of the relative spectrum. Composing with the closed immersion $C_Z X \times_Z W \hookrightarrow C_Z X$, the base change of $W \hookrightarrow Z$, gives the asserted closed immersion, completing the proof.

Lemma 10.4.13: The Normal Cone of a Preimage Embeds in the Pulled-Back Normal Bundle

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with ideal sheaf $\mathcal{I}$ and normal bundle $N = N_Z X$ (definition 5.1.43). Let $g\colon X' \to X$ be a morphism of algebraic schemes, let $Z' = g^{-1}(Z)$ be the scheme-theoretic preimage, and let $h\colon Z' \to Z$ be the induced morphism. Write $N' = h^*N$, a vector bundle of rank d on Z'. Then there is a canonical closed immersion of cones over Z'

$$j\colon C_{Z'}X' \hookrightarrow N'$$

Moreover:

1. j is an isomorphism whenever $Z' \hookrightarrow X'$ is itself a regular embedding of codimension d;
2. for $g = \mathrm{id}_X$ it is the identification $C_Z X = N$ of proposition 10.4.11.

***Proof*:**
Write $\mathcal{I}' \subseteq \mathcal{O}_{X'}$ for the ideal sheaf of Z'.

Step 1: the conormal surjection. By definition of the scheme-theoretic preimage, $\mathcal{I}'$ is the image of the map $g^*\mathcal{I} \to \mathcal{O}_{X'}$; that is, $\mathcal{I}'$ is generated by the pullbacks of local sections of $\mathcal{I}$. Composing $g^*\mathcal{I} \to \mathcal{I}'$ with the quotient $\mathcal{I}' \to \mathcal{I}'/\mathcal{I}'^2$ gives an $\mathcal{O}_{X'}$-linear surjection onto $\mathcal{I}'/\mathcal{I}'^2$. It kills the image of $g^*(\mathcal{I}^2)$, a product of two local sections of $\mathcal{I}$ pulling back to a product of two local sections of $\mathcal{I}'$. Since g^* is right exact, $g^*(\mathcal{I}/\mathcal{I}^2)$ is the cokernel of $g^*(\mathcal{I}^2) \to g^*\mathcal{I}$, so the surjection factors through it; and $g^*(\mathcal{I}/\mathcal{I}^2)$ is annihilated by $\mathcal{I}'$, hence is the $\mathcal{O}_{Z'}$-module $h^*(\mathcal{I}/\mathcal{I}^2)$. This gives a surjection of $\mathcal{O}_{Z'}$-modules

$$h^*\big(\mathcal{I}/\mathcal{I}^2\big) \twoheadrightarrow \mathcal{I}'/\mathcal{I}'^2$$

Step 2: the surjection of graded algebras. The symmetric algebra functor preserves surjections, so Step 1 gives a surjection $\mathrm{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \to \mathrm{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2)$ of graded $\mathcal{O}_{Z'}$-algebras. The graded algebra $\bigoplus_{n\geq 0} \mathcal{I}'^n/\mathcal{I}'^{n+1}$ of definition 10.4.6 is generated in degree one over $\mathcal{O}_{Z'}$, so the canonical map from $\mathrm{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2)$ onto it is surjective as well. Composing,

$$\mathrm{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \twoheadrightarrow \bigoplus_{n\geq 0} \mathcal{I}'^n/\mathcal{I}'^{n+1}$$

is a surjection of graded quasi-coherent $\mathcal{O}_{Z'}$-algebras.

Step 3: taking relative spectra. On an affine open $U \subseteq Z'$ the displayed surjection is a surjection of rings $\Gamma(U, \mathrm{Sym}^\bullet h^*(\mathcal{I}/\mathcal{I}^2)) \to \Gamma(U, \bigoplus \mathcal{I}'^n/\mathcal{I}'^{n+1})$, which induces a closed immersion of affine schemes in the opposite direction; these are compatible on overlaps because the surjection is a map of sheaves, so they glue to a closed immersion over Z' by theorem 5.1.41. The source of that closed immersion is $C_{Z'}X'$ by definition 10.4.6. Its target is, by definition 5.1.43 and example 10.4.2(1), the total space of the vector bundle on Z' whose sheaf of sections is $\big(h^*(\mathcal{I}/\mathcal{I}^2)\big)^\vee = h^*\big((\mathcal{I}/\mathcal{I}^2)^\vee\big)$, that is $N' = h^*N$; regularity of i enters here, and only here, through the local freeness of $\mathcal{I}/\mathcal{I}^2$ of rank d (lemma 10.4.10), which makes $h^*(\mathcal{I}/\mathcal{I}^2)$ locally free of rank d on Z'. The immersion is a morphism of cones because the algebra surjection is graded.

Step 4: part (1). Suppose $Z' \hookrightarrow X'$ is a regular embedding of codimension d. Then $\mathcal{I}'/\mathcal{I}'^2$ is locally free of rank d and the canonical map $\mathrm{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2) \to \bigoplus_n \mathcal{I}'^n/\mathcal{I}'^{n+1}$ is an isomorphism, both by lemma 10.4.10 applied to X'. So j is the morphism induced by the surjection $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ of locally free $\mathcal{O}_{Z'}$-modules of the same rank d. Such a surjection is an isomorphism: the target is locally free, hence projective, so the surjection splits and the source decomposes as the target plus the kernel; the kernel is then a direct summand of a locally free module of finite rank, hence itself locally free, and comparing ranks locally makes that rank zero, so the kernel vanishes. Therefore j is an isomorphism of cones.

Step 5: part (2). For $g = \mathrm{id}_X$ one has $\mathcal{I}' = \mathcal{I}$, $Z' = Z$, $h = \mathrm{id}_Z$, and the surjection of Step 2 is the canonical map $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2) \to \bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$. That map is the isomorphism exhibited in the proof of proposition 10.4.11, so j is the identification $C_Z X = N$ recorded there, completing the proof.

Both immersions above are induced by surjections of graded algebras out of $\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$, and when they are both available they agree. This is used whenever a normal cone is read inside a normal bundle.

Lemma 10.4.14: The Two Immersions Agree

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension d with normal bundle N, and let $V \subseteq X$ be a closed subscheme with $W = V \cap Z$. Then the composite

$$C_W V \;\hookrightarrow\; N|_W \;\hookrightarrow\; N$$

of the immersion j of lemma 10.4.13 (for $g\colon V \hookrightarrow X$) with the restriction of N to W coincides with the immersion $C_W V \hookrightarrow C_Z X = N$ of lemma 10.4.12.

***Proof*:**
Both immersions are the map on relative Spec induced by a surjection of graded $\mathcal{O}_W$-algebras onto $\bigoplus_n \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$, so it is enough to see that the two surjections coincide. The immersion of lemma 10.4.12 is induced by

$$\Big(\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}\Big)\otimes_{\mathcal{O}_Z}\mathcal{O}_W \;\twoheadrightarrow\; \bigoplus_n \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$$

and j is induced by $\operatorname{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \twoheadrightarrow \bigoplus_n \mathcal{I}_W^n/\mathcal{I}_W^{n+1}$. Because i is a regular embedding, proposition 10.4.11 identifies $\operatorname{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$ with $\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$ as graded $\mathcal{O}_Z$-algebras, and that identification is the identity in degree one. Pulling it back along h turns the second surjection into the first, since both are determined by their degree-one component, which in each case is the map $\mathcal{I}/\mathcal{I}^2\otimes\mathcal{O}_W \to \mathcal{I}_W/\mathcal{I}_W^2$ induced by $\mathcal{I}\mathcal{O}_V = \mathcal{I}_W$, as we sought to show.

For a regular embedding the Segre class carries no information beyond the normal bundle. The hypothesis $d \geq 1$ costs nothing: a regular embedding of codimension zero has zero ideal sheaf, so it is $Z = X$, and that case was settled directly above by the zero cone. The inverse $c(N)^{-1}$ is computed by the terminating geometric series $\mathrm{id} - c_1(N) + \big(c_1(N)^2 - c_2(N)\big) - \cdots$, terminating because every $c_j(N)$ with $j \geq 1$ lowers dimension and $A_k(Z) = 0$ for $k < 0$. The content of $s(Z, X)$ is therefore entirely in the irregular case: it is the invariant that continues to exist, and to enter the intersection formulas, when the ideal of Z needs more equations than the codimension allows.

The codimension-one case is the one already visible in chapter 9, and it recovers the tower of self-intersections of a divisor.

Example 10.4.15: An Effective Cartier Divisor and a Plane Conic
Let $D \subseteq X$ be an effective Cartier divisor (definition 5.4.25), so its ideal sheaf $\mathcal{I} = \mathcal{O}_X(-D)$ is invertible and locally generated by a single nonzerodivisor. This is a regular embedding of codimension one, with conormal sheaf $\mathcal{I}/\mathcal{I}^2 = \mathcal{O}_X(-D)|_D$ and normal bundle the line bundle $N_D X = \mathcal{O}_X(D)|_D$. A line bundle L has $c(L) = \mathrm{id} + c_1(L)$, so $c(L)^{-1} = \sum_{i\geq 0}(-1)^i c_1(L)^i$, and proposition 10.4.11 gives

$$s(D, X) \;=\; \sum_{i\geq 0}(-1)^i\, c_1\big(\mathcal{O}_X(D)|_D\big)^i \cap [D] \;=\; [D] - c_1\big(\mathcal{O}_X(D)|_D\big)\cap[D] + \cdots \tag{10.6}$$

the series terminating in dimension zero.
Take $X = \mathbb{P}^2$ and D a smooth conic, so $D \cong \mathbb{P}^1$ and $A_*(D) = \mathbb{Z}\,[D] \oplus \mathbb{Z}\,[\mathrm{pt}]$ by theorem 8.5.2; only the first two terms of (10.6) can be nonzero. The conic is cut out by a quadratic form, so $\mathcal{O}_{\mathbb{P}^2}(D) \cong \mathcal{O}_{\mathbb{P}^2}(2)$ (example 9.1.4). To evaluate the second term, push forward along the

closed immersion $\iota\colon D \hookrightarrow \mathbb{P}^2$ and use the projection formula proposition 9.4.4(4), additivity proposition 9.4.4(3) for $\mathcal{O}(2) = \mathcal{O}(1) \otimes \mathcal{O}(1)$, the class $\iota_*[D] = 2\,[\mathbb{P}^1]$ of a conic (example 8.5.4), and example 9.4.7:

$$\iota_*\Big(c_1\big(\mathcal{O}_{\mathbb{P}^2}(2)|_D\big) \cap [D]\Big) = c_1\big(\mathcal{O}_{\mathbb{P}^2}(2)\big) \cap \iota_*[D] = 2\,c_1\big(\mathcal{O}_{\mathbb{P}^2}(1)\big) \cap 2\,[\mathbb{P}^1] = 4\,[\mathrm{pt}]$$

Both $A_0(D)$ and $A_0(\mathbb{P}^2)$ are infinite cyclic on the class of a point and ι_* carries generator to generator, so $c_1(\mathcal{O}_{\mathbb{P}^2}(2)|_D) \cap [D] = 4\,[\mathrm{pt}]$ already in $A_0(D)$. Hence

$$s(D, \mathbb{P}^2) \;=\; [D] - 4\,[\mathrm{pt}]$$

The coefficient 4 is the self-intersection number of the conic, the degree $2 \cdot 2$ predicted by Bézout, and it enters the Segre class with a sign. For a divisor, then, $s(D, X)$ is a bookkeeping device for the alternating tower of self-intersections of D, and it contains exactly the information the line bundle $\mathcal{O}_X(D)|_D$ contains.

The irregular case is where the Segre class stops being a repackaging. The next computation is a closed subscheme whose normal cone is not a bundle, and the class detects the multiplicity of a singularity.

Example 10.4.16: The Vertex of the Quadric Cone

Let $X = \operatorname{Spec} \overline{k}[x, y, z]/(xy - z^2)$ be the affine quadric cone of example 9.1.2, a normal surface with an isolated singularity at the vertex, and let $Z = \{P\}$ be the reduced vertex, with ideal $\mathfrak{m} = (x, y, z)$.

The normal cone. Write $A = \overline{k}[x, y, z]/(xy - z^2)$. The defining polynomial is homogeneous, so A is a graded ring with A_n its degree-n piece and $\mathfrak{m}^n = \bigoplus_{m \geq n} A_m$; therefore $\mathfrak{m}^n/\mathfrak{m}^{n+1} = A_n$ and $\bigoplus_{n \geq 0} \mathfrak{m}^n/\mathfrak{m}^{n+1} = A$. The quadric cone is its own tangent cone at the vertex, and $C_P X = \operatorname{Spec} A = X$. It is not a vector bundle over the point P: a rank-r bundle over a point is $\mathbb{A}^r$, whereas X is singular at the origin. Equivalently, the embedding $\{P\} \hookrightarrow X$ is not regular, since $\mathfrak{m}$ needs three generators while $\dim X = 2$.

The projective completion. Adjoining w in degree one gives $A[w] = \overline{k}[x, y, z, w]/(xy - z^2)$, so

$$\mathbb{P}(C_P X \oplus 1) \;=\; \operatorname{Proj}\big(\overline{k}[x, y, z, w]/(xy - z^2)\big) \;=\; Q \subseteq \mathbb{P}^3$$

the quadric surface $\{xy = z^2\}$, which is integral because $xy - z^2$ is irreducible, and $\mathcal{O}(1) = \mathcal{O}_{\mathbb{P}^3}(1)|_Q$ by the surjection $\overline{k}[x, y, z, w] \to A[w]$.

The class. Since Z is a single point, $A_k(Z) = 0$ for $k \neq 0$, so only the term of definition 10.4.4 landing in A_0 survives, namely $i = \dim Q = 2$:

$$s(P, X) \;=\; \Big(\int_Q c_1\big(\mathcal{O}_{\mathbb{P}^3}(1)|_Q\big)^2 \cap [Q]\Big)\,[P]$$

Let $\iota\colon Q \hookrightarrow \mathbb{P}^3$ be the inclusion. The quadric is the divisor of the section $xy - z^2$ of $\mathcal{O}_{\mathbb{P}^3}(2)$, so $\iota_*[Q] = 2\,[\mathbb{P}^2]$ in $A_2(\mathbb{P}^3)$ by the class map (proposition 9.1.3 and example 9.1.4). By the projection formula proposition 9.4.4(4) and the fact that capping a linear-subspace class with $c_1(\mathcal{O}_{\mathbb{P}^3}(1))$ drops its dimension by one (example 9.4.7),

$$\iota_*\Big(c_1\big(\mathcal{O}_{\mathbb{P}^3}(1)|_Q\big)^2 \cap [Q]\Big) = c_1\big(\mathcal{O}_{\mathbb{P}^3}(1)\big)^2 \cap 2\,[\mathbb{P}^2] = 2\,[\mathrm{pt}]$$

Degrees are preserved by proper pushforward (definition 8.2.4), so the integral equals 2 and

$$s(P, X) \;=\; 2\,[P]$$

The contrast. Take instead the origin $P' \in \mathbb{A}^2$, where $\mathfrak{m} = (x, y)$ is a regular sequence and $\{P'\} \hookrightarrow \mathbb{A}^2$ is a regular embedding of codimension two. Proposition 10.4.11 gives $s(P', \mathbb{A}^2) = c(N)^{-1} \cap [P']$, and every $c_j(N) \cap [P']$ with $j \geq 1$ lies in $A_{-j}(P') = 0$, so $s(P', \mathbb{A}^2) = [P']$. At a smooth point of a surface the Segre class of the point is the point itself; at the vertex of the quadric cone it is twice the point. The integer it produces is the multiplicity of the singularity.

Exercise 10.4.1

1. Let X be an algebraic scheme and let $C = X$ be the zero cone of example 10.4.2(2), that is $S^\bullet = \mathcal{O}_X$. Show that $\mathbb{P}(C) = \emptyset$, that $\mathbb{P}(C \oplus 1) = X$ with q the identity, and that $\mathcal{O}(1)$ is the trivial bundle. Deduce $s(X, X) = [X]$, and then explain why proposition 10.4.11 cannot be used to reach the same answer, even though $X \hookrightarrow X$ is a regular embedding of codimension zero.
2. Let $Z = \mathbb{P}^1 \subseteq \mathbb{P}^3$ be a line, cut out by two of the homogeneous coordinates. Show that this is a regular embedding of codimension two with conormal sheaf $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, hence with normal bundle $N = \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(1)$, and compute $s(\mathbb{P}^1, \mathbb{P}^3)$.
3. Let $D \subseteq \mathbb{P}^2$ be a smooth plane curve of degree d. Show that $s(D, \mathbb{P}^2) = [D] - \beta$ for a class $\beta \in A_0(D)$ of degree d^2. Then explain why, for $d \geq 3$, one may *not* conclude $\beta = d^2\,[\mathrm{pt}]$, whereas for $d \leq 2$ one may.
4. Let $f \in \overline{k}[x, y, z]$ be a homogeneous polynomial of degree d defining a smooth plane curve $Y \subseteq \mathbb{P}^2$, let $X = \operatorname{Spec} \overline{k}[x, y, z]/(f)$ be the affine cone over Y, and let $P \in X$ be the vertex. Compute $s(P, X)$, generalizing example 10.4.16.
5. Let $X = \mathbb{A}^1$ and let $Z = \operatorname{Spec} \overline{k}[x]/(x^2)$ be the double point at the origin, with $Z_{\mathrm{red}} = \{P\}$ the reduced point. Compute both $s(Z, X)$ and $s(Z_{\mathrm{red}}, X)$, and say what the comparison shows about the dependence of the Segre class on the scheme structure of Z.

11

Deformation to the Normal Cone

Classical enumerative geometry intersected two cycles by moving them until they met properly and trusting that the count did not change. Making that rigorous requires a moving lemma, and moving lemmas are hard, restrictive, and unavailable on a singular ambient scheme. The construction of this chapter avoids the manoeuvre entirely: instead of moving the cycles, it degenerates the embedding.

Given a closed subscheme $Z \subseteq X$, there is a family over $\mathbb{P}^1$ whose fibre away from ∞ is the given embedding $Z \subseteq X$ and whose fibre at ∞ is the embedding of Z as the zero section of its normal cone (definition 10.4.6). Nothing has been perturbed; the ambient geometry has been deformed to a cone, where it is linear along the fibres and where intersecting is a matter of restricting to a zero section. Specializing a cycle to ∞ therefore converts an arbitrary intersection problem into one on a cone, and when the embedding is regular the cone is a vector bundle (definition 5.1.43) and the specialization can be inverted against theorem 8.4.6 to land back on Z.

That composite is the Gysin map, and it is the last piece of machinery the intersection product needs. This chapter builds the deformation space, proves that specializing is well defined on rational equivalence, and assembles the Gysin map for a regular embedding together with the refined version that chapter 12 will apply to the diagonal.

11.1 The Deformation to the Normal Cone

This section builds the family and establishes the three facts everything later rests on: it is flat over $\mathbb{P}^1$, its fibre away from ∞ is the given X, and its fibre at ∞ is the normal cone.

Throughout, $Z \subseteq X$ is a closed subscheme of an algebraic scheme, with ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$. Fix a coordinate t on $\mathbb{P}^1$, so that $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$ is the affine line with coordinate t, and write $U = \mathbb{P}^1 \setminus \{0\}$ for the complementary chart, an affine line with coordinate $u = t^{-1}$ vanishing exactly at ∞. Since X is of finite type over $\overline{k}$ (box 7.3.9), the product $X \times \mathbb{P}^1$ is Noetherian, so the blow-up construction of definition 7.2.26 applies to it.

The object to be built is a scheme over $\mathbb{P}^1$ whose fibre over every point of $\mathbb{A}^1$ is X and whose fibre over ∞ is $C_Z X$. The mechanism that produces such a family is visible in the local algebra. A function a in the ideal $\mathcal{I}$ vanishes on Z; dividing it by the parameter u rescales the normal directions to Z by the factor u^{-1}, and as u tends to 0 the rescaling magnifies an arbitrarily small neighbourhood of Z until only the behaviour of a to leading order in $\mathcal{I}$ survives. Adjoining all the functions au^{-1} with $a \in \mathcal{I}$ to $\mathcal{O}_X[u]$ therefore builds a family whose limit at $u = 0$ is manufactured from the successive quotients $\mathcal{I}^n/\mathcal{I}^{n+1}$, which is what $C_Z X$ is manufactured from (definition 10.4.6).

Adjoining $\mathcal{I}u^{-1}$ is exactly what makes the ideal $\mathcal{I} + (u)$ of $Z \times \{\infty\}$ principal, generated by u; and the universal solution to that demand is the blow-up (proposition 7.2.27, part (4)). The construction accordingly starts from a blow-up, and the rescaled algebra above is recovered as one of its charts.

Definition 11.1.1: The Deformation Space

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme, and let $\mathcal{J} \subseteq \mathcal{O}_{X\times\mathbb{P}^1}$ be the ideal sheaf of the closed subscheme $Z \times \{\infty\}$. The *deformation space* of Z in X is the blow-up

$$M_Z X \;=\; \mathrm{Bl}_{Z\times\{\infty\}}\big(X \times \mathbb{P}^1\big) \;=\; \mathrm{Proj}\Big(\bigoplus_{n\geq 0} \mathcal{J}^n\Big) \xrightarrow{\;p\;} X \times \mathbb{P}^1$$

of definition 7.2.26. Write $\pi\colon M_Z X \to \mathbb{P}^1$ for the composite of p with the projection $X \times \mathbb{P}^1 \to \mathbb{P}^1$.

Nothing has been achieved yet: $M_Z X$ is a blow-up like any other, and its fibre over ∞ is larger than the normal cone. The next lemma computes the part of $M_Z X$ lying over the chart U, splitting it into the piece that carries the cone and a piece that does not. The splitting is the reason the deformation is defined on an open subscheme of $M_Z X$ rather than on all of it.

Lemma 11.1.2: The Chart at Infinity

Let $\mathcal{R} \subseteq \mathcal{O}_X[u, u^{-1}]$ be the sheaf of $\mathcal{O}_X[u]$-subalgebras generated by $\mathcal{I}u^{-1}$, the *extended Rees algebra* of $\mathcal{I}$; grading by powers of u,

$$\mathcal{R} \;=\; \bigoplus_{m\geq 0} \mathcal{O}_X\, u^m \;\oplus\; \bigoplus_{n\geq 1} \mathcal{I}^n u^{-n} \tag{11.1}$$

Write $D_+(u) \subseteq p^{-1}(X \times U)$ for the open subscheme attached to the degree-one section u of $\bigoplus_n \mathcal{J}^n$. Then:

1. $D_+(u)$ is the relative spectrum $\mathrm{Spec}\,(\mathcal{R})$ over $X \times U$;
2. its closed complement $p^{-1}(X \times U) \setminus D_+(u)$ is canonically isomorphic to $\mathrm{Bl}_Z X$, carried by p into $X \times \{\infty\}$ by the blow-down morphism.

The closed subscheme $\mathrm{Bl}_Z X$ so obtained is closed in $M_Z X$.

***Proof*:**
Step 1: the ideal and its powers. Work on an affine open $\operatorname{Spec} A \subseteq X$ with $\mathcal{I}(\operatorname{Spec} A) = I$, and set $B = A[u]$, so that $\operatorname{Spec} A \times U = \operatorname{Spec} B$ and the ideal of $Z \times \{\infty\}$ there is $J = IB + uB$. Expanding the product of the two ideals gives

$$J^n \;=\; \sum_{j=0}^{n} I^j B \cdot u^{n-j}$$

Two consequences are read off this expansion. Every term with $j < n$ is divisible by u, so

$$J^n + uB \;=\; I^n B + uB \tag{11.2}$$

And for $n \geq 1$,

$$J^n \cap uB \;=\; u\, J^{n-1} \tag{11.3}$$

Indeed, write $x = \sum_{j=0}^{n} a_j u^{n-j}$ with $a_j \in I^j B$. The terms with $j < n$ all lie in uJ^{n-1}, since $I^j B\, u^{n-1-j} \subseteq J^{n-1}$. The remaining term $a_n \in I^n B$ is a polynomial in u with coefficients in I^n, so it lies in uB precisely when its constant coefficient vanishes, that is when $a_n \in uI^n B \subseteq uJ^{n-1}$. The reverse inclusion is immediate from $uJ^{n-1} \subseteq J^n$.

Step 2: the chart. Write $S = \bigoplus_{n \geq 0} J^n$, so that $p^{-1}(\operatorname{Spec} A \times U) = \operatorname{Proj}(S)$ by definition 11.1.1 and $D_+(u) = \operatorname{Spec}\left(S_{(u)}\right)$ is the affine chart of the relative Proj attached to the degree-one element $u \in J = S_1$ (theorem 5.3.35). Because u is a nonzerodivisor on B, the assignment $x/u^n \mapsto xu^{-n}$ is a well-defined injection of $S_{(u)}$ into $A[u, u^{-1}]$, and its image is $\bigcup_{n \geq 0} J^n u^{-n}$, which by Step 1 is the B-subalgebra R generated by Iu^{-1}. To see the grading (11.1), note $R = \sum_{n \geq 0} I^n B\, u^{-n}$, so the coefficient of u^m in R is $\sum_{n \geq 0,\, n+m \geq 0} I^n$, which is A for $m \geq 0$ and I^{-m} for $m < 0$. Hence $D_+(u) = \operatorname{Spec}(R)$ as a scheme over $\operatorname{Spec} B$, which is (1).

Step 3: the complement. Reduction modulo u gives a surjection of graded B-algebras $S \to \overline{S}$, where $\overline{S}_n = (J^n + uB)/uB$; by (11.2) this is the image of $I^n B$ in $B/uB = A$, namely I^n. So $\overline{S} = \bigoplus_{n \geq 0} I^n$ is the Rees algebra of I, its relative Proj is $\mathrm{Bl}_Z X$ over $\operatorname{Spec} A$ by definition 7.2.26, and the surjection exhibits it as a closed subscheme of $\operatorname{Proj}(S)$ lying over $V(u) = \operatorname{Spec} A \times \{\infty\}$, with structure morphism to $\operatorname{Spec} A$ the blow-down. Since u maps to 0 in $\overline{S}_1$, this closed subscheme is disjoint from $D_+(u)$.
For the reverse inclusion, the closed complement of $D_+(u)$ is $\operatorname{Proj}(S/uS)$, where uS is the homogeneous ideal generated by $u \in S_1$, whose degree-n part is uJ^{n-1} for $n \geq 1$ and 0 in degree 0. By (11.3) the ideal uJ^{n-1} is exactly the kernel of $J^n \to I^n$, so S/uS and $\overline{S}$ agree in every positive degree and differ only in degree 0, where $A[u] \to A$ kills u. In S/uS the degree-zero element u annihilates every positive degree, so for a homogeneous $f \in S_1$ the image of u in the localization $(S/uS)_{(f)}$ is $uf/f = 0$; hence $(S/uS)_{(f)} = \overline{S}_{(f)}$ for every such f, and the two relative Proj schemes coincide. The closed complement of $D_+(u)$ is therefore $\mathrm{Bl}_Z X$, which is (2).

Step 4: globalization. The algebra $\mathcal{R}$ and the surjection of Step 3 are defined without choices, so the local identifications agree on overlaps of an affine cover of X and glue. For the final assertion, $\mathrm{Bl}_Z X$ is closed in the open subscheme $p^{-1}(X \times U)$ and is contained in $p^{-1}(X \times \{\infty\})$, which is closed in $M_Z X$ and contained in $p^{-1}(X \times U)$; writing $\mathrm{Bl}_Z X = C \cap p^{-1}(X \times U)$ with C closed in $M_Z X$, one gets $\mathrm{Bl}_Z X = C \cap p^{-1}(X \times \{\infty\})$, an intersection of two closed subsets and hence closed in $M_Z X$, completing the proof.

The lemma is what makes the construction computable: over the chart at infinity the deformation is not a Proj at all but an affine scheme over $X \times U$, given by the single algebra (11.1), written R when $X = \operatorname{Spec} A$ is affine, and its reduction modulo u is the associated graded algebra of $\mathcal{I}$. The smallest instance already shows the mechanism.

Example 11.1.3: A Point on a Smooth Curve

Let X be a variety of dimension one and $P \in X$ a closed point at which X is smooth, and take $Z = \{P\}$ with its reduced structure. The local ring $\mathcal{O}_{X,P}$ is a discrete valuation ring; choosing a uniformizer s makes $\mathcal{I}$ locally generated by the single nonzerodivisor s, so $Z \hookrightarrow X$ is a regular embedding of codimension one (definition 5.1.43) and $\operatorname{gr}_{\mathfrak{m}_P}\mathcal{O}_{X,P} \cong \overline{k}[\overline{s}]$ is a polynomial ring in one variable. Hence $C_P X = \mathbb{A}^1$: the deformation replaces the curve by its tangent line at P and the point by the origin of that line.

The computation is explicit for $X = \mathbb{A}^1 = \operatorname{Spec} \overline{k}[x]$ and P the origin, where $I = (x)$. The algebra of (11.1) is $R = \overline{k}[x,u][xu^{-1}]$, and setting $v = xu^{-1}$ gives $x = uv$, so R is generated by u and v alone. These two are algebraically independent over $\overline{k}$: they lie in $\overline{k}(x,u)$, which has transcendence degree two, and they generate it, so they form a transcendence basis. Hence

$$R \;\cong\; \overline{k}[u,v], \qquad \operatorname{Spec}(R) = \mathbb{A}^2 \longrightarrow \mathbb{A}^1_u$$

is the trivial family with fibre $\mathbb{A}^1$. Over $u = c \neq 0$ the fibre is identified with X by $x = cv$, and over $u = 0$ it is $\operatorname{Spec}\big(R/uR\big) = \operatorname{Spec} \overline{k}[v]$, which is $C_P X$. The family is the rescaling $x \mapsto x/u$ of the line, and the closed subscheme $\{v = 0\}$ is the point P travelling along it.

The lemma isolates a piece of $M_Z X$ over ∞, namely $\operatorname{Bl}_Z X$, which the rescaling picture does not see and which is discarded.

Definition 11.1.4: The Deformation to the Normal Cone

With $Z \subseteq X$ and $M_Z X$ as in definition 11.1.1, and with $\operatorname{Bl}_Z X \subseteq M_Z X$ the closed subscheme of lemma 11.1.2(2), the *deformation to the normal cone* of Z in X is the open subscheme

$$M_Z^{\circ} X \;=\; M_Z X \setminus \operatorname{Bl}_Z X$$

together with the restriction of $\pi\colon M_Z X \to \mathbb{P}^1$, again written π, and the restriction of p, whose composite with the projection $X \times \mathbb{P}^1 \to X$ is written $\rho\colon M_Z^{\circ} X \to X$.

Everything the chapter needs is contained in the behaviour of π over the two charts, and lemma 11.1.2 has already computed one of them. Collecting that computation with the elementary behaviour over $\mathbb{A}^1$ gives the properties that section 11.2 and section 11.3 consume.

Theorem 11.1.5: The Fibres of the Deformation

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme, with $\pi\colon M^{\circ}_Z X \to \mathbb{P}^1$ as in definition 11.1.4.

1. Over $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$ the blow-down restricts to an isomorphism $\pi^{-1}(\mathbb{A}^1) \xrightarrow{\sim} X \times \mathbb{A}^1$ commuting with the projections to $\mathbb{A}^1$.
2. The fibre over ∞ is the normal cone, $\pi^{-1}(\infty) = C_Z X$ of definition 10.4.6, and it is an effective Cartier divisor on $M^{\circ}_Z X$, cut out by the function u.
3. π is flat.
4. There is a closed immersion $Z \times \mathbb{P}^1 \hookrightarrow M^{\circ}_Z X$ over $\mathbb{P}^1$ which over $\mathbb{A}^1$ is the inclusion $Z \times \mathbb{A}^1 \subseteq X \times \mathbb{A}^1$ of (1), and over ∞ is the vertex $Z \hookrightarrow C_Z X$ of definition 10.4.1.

***Proof*:**
Step 1: the chart over $\mathbb{A}^1$. The centre $Z \times \{\infty\}$ is disjoint from the open set $X \times \mathbb{A}^1$, so proposition 7.2.27(2) makes p an isomorphism $p^{-1}(X \times \mathbb{A}^1) \to X \times \mathbb{A}^1$. By lemma 11.1.2(2) the deleted subscheme $\mathrm{Bl}_Z X$ lies over $X \times \{\infty\}$, so it does not meet $p^{-1}(X \times \mathbb{A}^1)$, and therefore $\pi^{-1}(\mathbb{A}^1) = p^{-1}(X \times \mathbb{A}^1) \cong X \times \mathbb{A}^1$ compatibly with the projections to $\mathbb{A}^1$. This is (1).

Step 2: the fibre at infinity. Over the other chart, lemma 11.1.2 gives $\pi^{-1}(U) = \operatorname{Spec}(\mathcal{R})$ with $\mathcal{R}$ the extended Rees algebra, so the scheme-theoretic fibre over $\infty = \{u = 0\}$ is $\operatorname{Spec}\big(\mathcal{R}/u\mathcal{R}\big)$. Compute the quotient degree by degree in the grading (11.1). Writing $\mathcal{R}_m$ for the coefficient of u^m, so that $\mathcal{R}_m = \mathcal{O}_X$ for $m \geq 0$ and $\mathcal{R}_{-n} = \mathcal{I}^n$ for $n \geq 1$, the coefficient of u^m in $u\mathcal{R}$ is $\mathcal{R}_{m-1}$. Hence $(\mathcal{R}/u\mathcal{R})_m$ vanishes for $m \geq 1$, equals $\mathcal{O}_X/\mathcal{I} = \mathcal{O}_Z$ for $m = 0$, and equals $\mathcal{I}^n/\mathcal{I}^{n+1}$ for $m = -n$ with $n \geq 1$. Therefore

$$\mathcal{R}/u\mathcal{R} \;\cong\; \bigoplus_{n \geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$$

which is the graded algebra of definition 10.4.6, so $\pi^{-1}(\infty) = C_Z X$.

Step 3: the fibre at infinity is Cartier. The element u acts invertibly on $\mathcal{O}_X[u, u^{-1}]$ and hence injectively on the subsheaf $\mathcal{R}$, so u is a nonzerodivisor and $\pi^{-1}(\infty) \subseteq \pi^{-1}(U)$ is cut out by a single nonzerodivisor. On the complementary open $\pi^{-1}(\mathbb{A}^1)$ the fibre is empty, cut out by the unit 1. So $\pi^{-1}(\infty)$ is locally principal, generated at each point by a nonzerodivisor, which is an effective Cartier divisor (definition 5.4.25). With Step 2 this is (2).

Step 4: flatness. Flatness may be checked locally on source and target, so it suffices to treat the two charts separately. Over $\mathbb{A}^1$ the morphism is the projection $X \times \mathbb{A}^1 \to \mathbb{A}^1$ by Step 1, which is the base change of $X \to \operatorname{Spec} \overline{k}$ along $\mathbb{A}^1 \to \operatorname{Spec} \overline{k}$; every scheme is flat over a field and flatness is stable under base change, so this projection is flat. Over U the morphism is $\operatorname{Spec}(\mathcal{R}) \to U = \operatorname{Spec} \overline{k}[u]$. The ring $\overline{k}[u]$ is a principal ideal domain, over which a module is flat exactly when it is torsion-free. On an affine open $\operatorname{Spec} A \subseteq X$ one has $\mathcal{R}(\operatorname{Spec} A) = R \subseteq A[u, u^{-1}]$, and $A[u, u^{-1}]$ is torsion-free over $\overline{k}[u]$: given $f \in \overline{k}[u]$ nonzero of degree d with leading coefficient $c \in \overline{k}^*$, and a nonzero $x = \sum_{i \leq M} a_i u^i$ with $a_M \neq 0$, the coefficient of u^{M+d} in fx is $c\,a_M \neq 0$, so $fx \neq 0$. A submodule of a torsion-free module is torsion-free, so R is torsion-free and hence flat over $\overline{k}[u]$. This is (3).

Step 5: the constant subscheme. Let $\mathfrak{q} \subseteq \mathcal{R}$ be the ideal generated by $\mathcal{I}u^{-1}$; it contains $\mathcal{I}$, since

$\mathcal{I} = (\mathcal{I}u^{-1}) \cdot u$ and $u \in \mathcal{R}$. Its coefficient of u^m is $\mathcal{I} \cdot \mathcal{R}_{m+1}$, which for $m \geq 0$ is $\mathcal{I}$ and for $m = -n < 0$ is $\mathcal{I} \cdot \mathcal{I}^{n-1} = \mathcal{I}^n = \mathcal{R}_{-n}$. So

$$\mathcal{R}/\mathfrak{q} \;\cong\; \bigoplus_{m \geq 0} (\mathcal{O}_X/\mathcal{I})\, u^m \;=\; \mathcal{O}_Z[u]$$

and $\mathfrak{q}$ defines a closed immersion $Z \times U \hookrightarrow \operatorname{Spec}(\mathcal{R})$ over $X \times U$. Inverting u makes u^{-1} a unit, so $\mathfrak{q}$ generates $\mathcal{I}$ there and the immersion restricts over $U \setminus \{\infty\}$ to $Z \times (U \setminus \{\infty\}) \subseteq X \times (U \setminus \{\infty\})$; it therefore glues with the inclusion $Z \times \mathbb{A}^1 \subseteq X \times \mathbb{A}^1$ of Step 1 to a closed immersion $Z \times \mathbb{P}^1 \hookrightarrow M_Z^\circ X$ over $\mathbb{P}^1$. Modulo u the ideal $\mathfrak{q}$ becomes the ideal generated by the degree-one part $\mathcal{I}/\mathcal{I}^2$ of $\bigoplus_n \mathcal{I}^n/\mathcal{I}^{n+1}$, whose quotient is $\mathcal{O}_Z$; by definition 10.4.1 that is the vertex of $C_Z X$. So the immersion restricts over ∞ to the vertex, which is (4), completing the proof.

The theorem produces a single scheme over $\mathbb{P}^1$ carrying two embeddings at once. Away from ∞ the pair is the given $Z \subseteq X$, unchanged; at ∞ it is the vertex inside $C_Z X$. What moves in the family is the ambient scheme, not the subscheme: by (4) the subscheme is the constant family $Z \times \mathbb{P}^1$, and the ambient scheme degenerates around it until it becomes a cone with Z as its vertex.

The morphism $\rho\colon M_Z^\circ X \to X$ records how each fibre maps back to the original scheme. Over $\mathbb{A}^1$ it is the projection $X \times \mathbb{A}^1 \to X$ by Step 1 of the proof, and over ∞ it is the composite $C_Z X \to Z \hookrightarrow X$, since the structure map $\mathcal{O}_X \to \mathcal{R}$ becomes $\mathcal{O}_X \to \mathcal{O}_Z$ in degree zero after reducing modulo u. One property the family does not have is properness: its fibre over ∞ is affine over Z (definition 10.4.1), and an affine scheme of positive dimension over a field is not proper, so π is not a proper morphism as soon as $C_Z X$ has positive-dimensional fibres. Cycles will therefore be moved across the family by restriction to the Cartier divisor $\pi^{-1}(\infty)$ rather than by pushforward.

11.1.6 Remark: Why the Blow-Up Is Not Enough The subscheme deleted in definition 11.1.4 is not a technicality. Lemma 11.1.2 splits the part of $M_Z X$ lying over the chart U into $D_+(u)$ and $\mathrm{Bl}_Z X$, and the fibre over ∞ lies entirely in that part; its intersection with $D_+(u)$ is $C_Z X$ by Step 2 of the proof of theorem 11.1.5, while $\mathrm{Bl}_Z X$ maps into $X \times \{\infty\}$ and so lies in the fibre. The special fibre of the unmodified blow-up is therefore the disjoint union of $C_Z X$ and $\mathrm{Bl}_Z X$ as a set, and only the first piece records the normal directions along Z; the second is a copy of X modified along Z, carrying no information the construction is after. Deleting it costs the properness of the blow-down p over $X \times \mathbb{P}^1$ and gives a special fibre equal to $C_Z X$ on the nose.

The two extremes of the construction are the case where $C_Z X$ is a vector bundle and the case where it is not. The first is the case the Gysin map of section 11.3 will need, and codimension one is where it is easiest to see.

Example 11.1.7: An Effective Cartier Divisor and a Line in the Plane

Let $D \subseteq X$ be an effective Cartier divisor (definition 5.4.25). Its ideal sheaf is invertible and locally generated by one nonzerodivisor, so $D \hookrightarrow X$ is a regular embedding of codimension one with normal bundle $N_D X = \mathcal{O}_X(D)|_D$ (example 10.4.15), and $C_D X = N_D X$ by proposition 10.4.11. By theorem 11.1.5 the deformation therefore degenerates $D \subseteq X$ into D sitting inside the total space of a line bundle on itself, as the zero section, which is the vertex of that cone (example 10.4.2(1)). Take $X = \mathbb{P}^2$ and $D = L$ a line. Then $\mathcal{O}_{\mathbb{P}^2}(L) \cong \mathcal{O}_{\mathbb{P}^2}(1)$ by example 9.1.4, and its restriction to $L \cong \mathbb{P}^1$ is $\mathcal{O}_{\mathbb{P}^1}(1)$ (example 9.2.3), so $C_L \mathbb{P}^2$ is the total space of $\mathcal{O}_{\mathbb{P}^1}(1)$ over L. That total space is $\mathbb{P}^2$ with a point removed: linear projection away from a point $P \notin L$ exhibits $\mathbb{P}^2 \setminus \{P\}$ as a line bundle over L whose zero section is L itself, and the degree of that bundle is the self-intersection $L \cdot L = 1$ of example 9.2.3, so it is $\mathcal{O}_{\mathbb{P}^1}(1)$. (The proof of theorem 8.5.2 notes the projection but uses only that it is an affine bundle, which is all its own argument needs.) So the family degenerates the pair $L \subseteq \mathbb{P}^2$ into the pair $L \subseteq \mathbb{P}^2 \setminus \{P\}$, with L the zero section of a line bundle. On the special fibre, intersecting a cycle with L has become intersecting it with the zero section of a line bundle, and flat pullback along the projection of that bundle is an isomorphism on Chow groups (theorem 8.4.6).

When the embedding is not regular the cone is not a bundle, and the deformation exhibits the singularity of the pair rather than a linear model of it. The node is the case already computed in example 10.4.7, and the whole family can be written down.

Example 11.1.8: The Node and Its Tangent Cone

Assume $\operatorname{char}\overline{k} \neq 2$ and let $X = \operatorname{Spec} A$ with $A = \overline{k}[x,y]/(y^2 - x^2 - x^3)$ be the affine nodal cubic, P the origin, $Z = \{P\}$ and $I = (x,y)$. Set $v = xu^{-1}$ and $w = yu^{-1}$ in $A[u,u^{-1}]$, so that $x = uv$ and $y = uw$; the algebra $R = A[u][Iu^{-1}]$ of (11.1) is generated by u, v, w. Substituting into the defining relation gives $u^2w^2 = u^2v^2 + u^3v^3$, and u is a nonzerodivisor on the domain $A[u,u^{-1}]$, so

$$w^2 \;=\; v^2 + uv^3 \tag{11.4}$$

The relation is the only one. The ring A is free over $\overline{k}[x]$ on $1, y$, because its defining relation is monic of degree two in y; hence $A[u,u^{-1}]$ is free over $\overline{k}[x,u,u^{-1}]$ on $1, y$. Using (11.4) to reduce powers of w, the ring R is generated as a $\overline{k}[u,v]$-module by 1 and w. Those two are independent: if $f(u,v) + g(u,v)w = 0$, then, reading off the coefficients of the basis $1, y$, one gets $f(u, xu^{-1}) = 0$ and $g(u, xu^{-1}) = 0$ in $\overline{k}[x,u,u^{-1}]$, and u and xu^{-1} are algebraically independent over $\overline{k}$ (they generate $\overline{k}(x,u)$, of transcendence degree two), so $f = g = 0$. Therefore

$$R \;\cong\; \overline{k}[u,v,w]/\big(w^2 - v^2 - uv^3\big)$$

The fibres. Over $u = c \neq 0$ the fibre is $\operatorname{Spec}\overline{k}[v,w]/(w^2 - v^2 - cv^3)$, carried isomorphically onto X by $x = cv$, $y = cw$, as substituting confirms. Over $u = 0$ the fibre is $\operatorname{Spec}\overline{k}[v,w]/(w^2 - v^2)$, the union of the two lines $w = \pm v$: the tangent cone of example 10.4.7, in agreement with theorem 11.1.5(2). The family straightens the two branches of the node into their tangent lines while keeping the singular point fixed, and the cubic term uv^3, which distinguishes the nodal cubic from a pair of lines, is exactly the term that the rescaling drives to zero.

Exercise 11.1.1

1. Compute the deformation in the two degenerate cases. First take $Z = X$, so that $\mathcal{I} = 0$: identify $\mathcal{R}$, show that $M_X^\circ X = X \times \mathbb{P}^1$ with π the projection, and check that the fibre over ∞ is the zero cone of example 10.4.2(2). Then take $Z = \emptyset$, so that $\mathcal{I} = \mathcal{O}_X$: identify $\mathcal{R}$, show that $M_\emptyset^\circ X = X \times \mathbb{A}^1$, and reconcile the empty fibre over ∞ with definition 10.4.6.

2. Let $X = \mathbb{A}^n$ and let $Z = \{0\}$ be the origin, with $I = (x_1, \ldots, x_n)$. Show that the algebra of (11.1) is a polynomial ring $R \cong \overline{k}[u, v_1, \ldots, v_n]$ with $v_i = x_iu^{-1}$, so that $\pi^{-1}(U) \cong \mathbb{A}^{n+1} \to \mathbb{A}^1_u$ is the trivial family. Identify the isomorphism of the fibre over $u = c \neq 0$ with $\mathbb{A}^n$, identify $C_0\mathbb{A}^n$, and locate the closed subscheme $Z \times U$ of theorem 11.1.5(4) inside $\mathbb{A}^{n+1}$.

3. Let $X = \mathbb{A}^1 = \operatorname{Spec} \overline{k}[x]$ and let $Z = \operatorname{Spec} \overline{k}[x]/(x^2)$ be the double point at the origin, so $I = (x^2)$. Show that $\operatorname{Spec}(\mathcal{R})$ is the affine quadric cone of example 9.1.2, and identify the fibre over ∞. Comment on what the computation shows about the smoothness of the deformation space.
4. Let $D \subseteq \mathbb{P}^2$ be a smooth conic. Using example 10.4.15 and proposition 10.4.11, identify $C_D\mathbb{P}^2$ as the total space of a line bundle on $D \cong \mathbb{P}^1$, and determine which line bundle it is.
5. Let $X = \operatorname{Spec} \overline{k}[x, y]/(y^2 - x^3)$ be the cuspidal cubic and P the origin. Compute $C_P X$ and its fundamental cycle $[C_P X]$ in the sense of definition 8.1.2, and contrast the answer with the node of example 11.1.8.

11.2 Specialization of Cycles

A cycle on X is spread over $\mathbb{A}^1$, its closure is followed to the fibre at infinity, and what that closure cuts on the fibre is a cycle on the normal cone. The resulting operation $\sigma\colon A_k(X) \to A_k(C_Z X)$ is the specialization homomorphism. Three things have to be proved about it: that it is well defined on rational equivalence, that on a subvariety V it returns the normal cone of $V \cap Z$ in V, and that it is compatible with proper pushforward and flat pullback.

Throughout, $Z \subseteq X$ is a closed subscheme of an algebraic scheme, with ideal sheaf $\mathcal{I}$, and $\pi\colon M^\circ_Z X \to \mathbb{P}^1$ is the projection of definition 11.1.4, which by theorem 11.1.5(1) to (3) is flat with fibres

$$\pi^{-1}(\mathbb{A}^1) \;=\; X \times \mathbb{A}^1, \qquad \pi^{-1}(\infty) \;=\; C_Z X$$

Write $j\colon X \times \mathbb{A}^1 \hookrightarrow M^\circ_Z X$ for the open immersion of the first fibre, $i_\infty\colon C_Z X \hookrightarrow M^\circ_Z X$ for the closed immersion of the second, and $q\colon X \times \mathbb{A}^1 \to X$ for the projection. On $\mathbb{P}^1$ fix the coordinate u vanishing to order one at ∞, so that $\mathbb{P}^1 \setminus \{0\} = \operatorname{Spec} \overline{k}[u]$ and ∞ is the closed subscheme $\{u = 0\}$; on the complementary chart $\mathbb{A}^1$ the coordinate is $t = u^{-1}$.

The fibre at infinity is cut out by a single equation (theorem 11.1.5(2)), which is what makes definition 9.2.1 applicable. One further fact is needed before anything can be intersected: the line bundle carrying that equation is trivial along the fibre itself. It looks like an accident of the construction, and it is what will make the whole operation independent of choices.

Lemma 11.2.1: The Line Bundle of the Fibre at Infinity

Let D_∞ be the effective Cartier divisor of theorem 11.1.5(2), whose support is the fibre $\pi^{-1}(\infty) = C_Z X$. Then

$$\mathcal{O}_{M^\circ_Z X}(D_\infty) \;=\; \pi^*\mathcal{O}_{\mathbb{P}^1}(\infty), \qquad \mathcal{O}_{M^\circ_Z X}(D_\infty)\big|_{C_Z X} \;\cong\; \mathcal{O}_{C_Z X}$$

so the line bundle of D_∞ restricts to the trivial bundle on the fibre it cuts out.

***Proof*:**
On $\mathbb{P}^1$ the point ∞ is an effective Cartier divisor (definition 5.4.25): it is cut out by the nonzerodivisor u on $\operatorname{Spec} \overline{k}[u]$ and by the unit 1 on the complementary chart $\mathbb{A}^1$, and its line bundle is $\mathcal{O}_{\mathbb{P}^1}(\infty)$.

By theorem 11.1.5(2) the divisor D_∞ is cut out by $\pi^\# u$ over $\pi^{-1}(\mathbb{P}^1 \setminus \{0\})$, and over $\pi^{-1}(\mathbb{A}^1)$ it

is empty, cut out by the unit 1. Its local data $\{(\pi^{-1}(\mathbb{A}^1), 1), (\pi^{-1}(\mathbb{P}^1 \setminus \{0\}), \pi^\# u)\}$ are therefore the pullbacks along π of the local data just recorded for ∞ on $\mathbb{P}^1$, so $D_\infty = \pi^*(\infty)$ and the associated line bundle is $\pi^*\mathcal{O}_{\mathbb{P}^1}(\infty)$.
For the restriction, the composite $\pi \circ i_\infty \colon C_Z X \to \mathbb{P}^1$ factors through the closed subscheme $\{\infty\} = \operatorname{Spec} \overline{k}$, since i_∞ is the inclusion of the fibre over ∞, so

$$\mathcal{O}_{M_Z^\circ X}(D_\infty)\big|_{C_Z X} \;=\; i_\infty^* \pi^* \mathcal{O}_{\mathbb{P}^1}(\infty) \;=\; \big(\pi \circ i_\infty\big)^* \mathcal{O}_{\mathbb{P}^1}(\infty)$$

is the pullback of $\mathcal{O}_{\mathbb{P}^1}(\infty)|_{\{\infty\}}$ along the structure morphism $C_Z X \to \operatorname{Spec} \overline{k}$. A line bundle on the spectrum of a field is free of rank one, so that pullback is trivial, completing the proof.

The first consequence of the triviality is what the definition of σ needs: the operator $D_\infty \cdot -$ annihilates every class that already lives on the fibre at infinity. Geometrically, a cycle sitting inside the special fibre is not cut further by the equation of that fibre.

Lemma 11.2.2: Vanishing on Classes Supported at Infinity

For every $\gamma \in A_{k+1}(C_Z X)$,

$$D_\infty \cdot (i_\infty)_* \gamma \;=\; 0 \quad \text{in } A_k(C_Z X)$$

***Proof*:**
By proposition 9.2.4 the operator $D_\infty \cdot - \colon A_{k+1}(M_Z^\circ X) \to A_k(|D_\infty|) = A_k(C_Z X)$ is a homomorphism independent of all choices, so it may be evaluated on any cycle representing the class. Represent γ by a cycle $\sum_l n_l [T_l]$ with each $T_l \subseteq C_Z X$ a $(k+1)$-dimensional subvariety; then $(i_\infty)_* \gamma$ is represented by the same cycle, read on $M_Z^\circ X$, and every T_l is contained in $|D_\infty| = C_Z X$. The second clause of definition 9.2.1 is therefore in force on each term, and

$$D_\infty \cdot [T_l] \;=\; c\Big(\mathcal{O}_{M_Z^\circ X}(D_\infty)\big|_{T_l}\Big) \;\in\; A_k(T_l)$$

By lemma 11.2.1 the restricted bundle is trivial, and the class map of proposition 9.1.3 is a group homomorphism, so it sends the trivial bundle, the identity element of $\operatorname{Pic}(T_l)$, to zero. Hence $D_\infty \cdot [T_l] = 0$ for every l, and summing gives $D_\infty \cdot (i_\infty)_* \gamma = 0$, as we sought to show.

With these two facts the specialization map can be built. The recipe has three moves: thicken a class on X into a class on $X \times \mathbb{A}^1$, extend it across the family, and cut with the fibre at infinity. The middle move is the only one that involves a choice, since a class on the open part of $M_Z^\circ X$ has many extensions; the localization sequence measures the ambiguity exactly, and lemma 11.2.2 kills it.

Definition 11.2.3: Specialization of Cycles

Let $Z \subseteq X$ be a closed subscheme and $\alpha \in A_k(X)$. A *lift* of α is a class $\beta \in A_{k+1}(M_Z^\circ X)$ with

$$j^*\beta \;=\; q^*\alpha \quad \text{in } A_{k+1}\big(X \times \mathbb{A}^1\big)$$

The *closure lift* attached to a cycle $\sum_l n_l[V_l]$ representing α is the class of

$$\overline{\alpha \times \mathbb{A}^1} \;=\; \sum_l n_l\big[\overline{V_l \times \mathbb{A}^1}\big]$$

the closures being taken in $M_Z^\circ X$. The *specialization* of α is

$$\sigma(\alpha) \;=\; D_\infty \cdot \beta \;\in\; A_k\big(C_Z X\big)$$

for any lift β, the intersection being that of definition 9.2.1 with the Cartier divisor of lemma 11.2.1, whose support is $C_Z X$.

The definition presupposes two things, that a lift exists and that the answer does not depend on which one is chosen, and both come from the localization sequence for the closed fibre $C_Z X$ inside $M_Z^\circ X$. It also presupposes that the recipe respects rational equivalence on X, which is where flat pullback does its work.

Theorem 11.2.4: The Specialization Homomorphism

Let $Z \subseteq X$ be a closed subscheme of an algebraic scheme. Every class $\alpha \in A_k(X)$ admits a lift, and $D_\infty \cdot \beta$ is the same for all lifts β of α. The assignment of definition 11.2.3 is therefore a well-defined homomorphism

$$\sigma\colon A_k(X) \longrightarrow A_k\big(C_Z X\big)$$

for every k.

***Proof*:**
Step 1: existence of a lift. The fibre $C_Z X$ is closed in $M_Z^\circ X$ with open complement $X \times \mathbb{A}^1$, so theorem 8.4.1 gives an exact sequence

$$A_{k+1}\big(C_Z X\big) \xrightarrow{(i_\infty)_*} A_{k+1}\big(M_Z^\circ X\big) \xrightarrow{j^*} A_{k+1}\big(X \times \mathbb{A}^1\big) \longrightarrow 0 \tag{11.5}$$

Surjectivity of j^* produces a lift of $q^*\alpha$, so lifts exist.

Step 2: independence of the lift. Let β and β' both lift α. Then $j^*(\beta - \beta') = 0$, so by exactness of (11.5) there is $\gamma \in A_{k+1}(C_Z X)$ with $\beta - \beta' = (i_\infty)_*\gamma$. Since $D_\infty \cdot -$ is a homomorphism on $A_{k+1}(M_Z^\circ X)$ (proposition 9.2.4),

$$D_\infty \cdot \beta - D_\infty \cdot \beta' \;=\; D_\infty \cdot (i_\infty)_*\gamma \;=\; 0$$

by lemma 11.2.2. So the value depends only on α.

Step 3: additivity. Flat pullback q^* and open restriction j^* are homomorphisms (proposition 8.3.7), so the sum of a lift of α and a lift of α' is a lift of $\alpha + \alpha'$; and $D_\infty \cdot -$ is a homomorphism. Hence $\sigma(\alpha+\alpha') = \sigma(\alpha)+\sigma(\alpha')$, and σ is a homomorphism $A_k(X) \to A_k(C_Z X)$, completing the proof.

Nothing in the argument required α to be represented by any particular kind of cycle, and that is the point of routing the construction through Chow groups rather than through cycles: rational equivalence on X is absorbed by q^* before the family is entered, and the ambiguity created by the passage to $M_Z^\circ X$ is absorbed by lemma 11.2.2 after it. In practice, one lift is more useful than the others. Given a cycle $\sum_l n_l[V_l]$ representing α, the flat pullback $q^*[V_l] = [V_l \times \mathbb{A}^1]$ (example 8.3.4(2)) extends across the family by taking closures in $M_Z^\circ X$, and the closure lift of definition 11.2.3 restricts along the open immersion j to $q^*\alpha$ by example 8.3.4(1), so it is indeed a lift. Every one of its components dominates $\mathbb{P}^1$, hence lies in no fibre, and in particular none is contained in $|D_\infty|$; the first clause of definition 9.2.1 therefore governs every term. This is the representative used in every computation below.

Example 11.2.5: Specializing at a Point of the Projective Line

Let $X = \mathbb{P}^1$ and let $Z = \{P\}$ be a reduced closed point. The ideal $\mathcal{I}$ is generated at P by a local coordinate, so $\mathcal{I}/\mathcal{I}^2$ is one-dimensional over $\overline{k}$ and every $\mathcal{I}^n/\mathcal{I}^{n+1}$ is one-dimensional; the normal cone $C_P\mathbb{P}^1$ of definition 10.4.6 is the affine line $\mathbb{A}^1$ over the point P. The deformation space is concrete here: $X \times \mathbb{P}^1 = \mathbb{P}^1 \times \mathbb{P}^1$, the centre is the single point (P,∞), and $M_Z X$ is the blow-up of that surface at that point, with exceptional divisor a copy of $\mathbb{P}^1$. Removing $\mathrm{Bl}_P\mathbb{P}^1 = \mathbb{P}^1$, the strict transform of $\mathbb{P}^1 \times \{\infty\}$, leaves $M_Z^\circ X$, whose fibre over ∞ is the exceptional $\mathbb{P}^1$ with one point deleted, namely $\mathbb{A}^1 = C_P\mathbb{P}^1$.

Two computations are available, one in each dimension. In dimension one, $A_1(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1]$ by theorem 8.5.2, and the closure of $\mathbb{P}^1 \times \mathbb{A}^1$ is all of $M_Z^\circ X$, an integral surface, so the first clause of definition 9.2.1 gives $\sigma[\mathbb{P}^1] = [\mathrm{div}(\pi^\# u)]$. The zero scheme of $\pi^\# u$ is the fibre over ∞, which is the reduced curve $\mathbb{A}^1$, so by definition 8.1.1 the order of $\pi^\# u$ along it is the length of the local ring of a reduced curve at its generic point, namely one. Hence

$$\sigma\big[\mathbb{P}^1\big] \;=\; \big[\mathbb{A}^1\big] \;=\; \big[C_P\mathbb{P}^1\big] \;\in\; A_1\big(C_P\mathbb{P}^1\big) = \mathbb{Z}$$

and σ is an isomorphism in this dimension. In dimension zero the behaviour is the opposite. Let $Q \neq P$ be a closed point. The closure of $\{Q\} \times \mathbb{A}^1$ in $M_Z X$ is the strict transform of $\{Q\} \times \mathbb{P}^1$, which misses the centre (P,∞) entirely and therefore meets the fibre over ∞ in the single point (Q,∞); that point lies on the strict transform of $\mathbb{P}^1 \times \{\infty\}$, which was deleted. So the closure lift of $[Q]$ in $M_Z^\circ X$ does not meet $|D_\infty|$ at all, and definition 9.2.1 records the intersection class on the empty intersection:

$$\sigma[Q] \;=\; 0 \;\in\; A_0\big(C_P\mathbb{P}^1\big)$$

consistent with $A_0(\mathbb{A}^1) = 0$ (proposition 8.5.1). Since $A_0(\mathbb{P}^1) = \mathbb{Z}$ by theorem 8.5.2, specialization is not injective. A cycle disjoint from Z escapes to infinity as the family degenerates, and the operation records nothing of it.

The example computed σ on the class of a subvariety by taking the closure of its cylinder and reading the fibre at infinity, and the answer was in both cases a normal cone: the normal cone of Z in $\mathbb{P}^1$ in the first computation, and the empty cone $C_\emptyset\{Q\}$ in the second. That this always happens is the main theorem of the section, and it is what makes σ computable. Proving it requires knowing that the deformation space of a closed subscheme sits inside the deformation space of the ambient one, which is the content of the next two results.

The first is a statement about normal cones alone. If $V \subseteq X$ is a closed subscheme, the normal cone of $V \cap Z$ in V has an intrinsic map to the normal cone of Z in X, obtained by restricting the ideal $\mathcal{I}$ to V.

The second is the corresponding statement one level up, for the whole family rather than its special fibre. It is stated for an arbitrary morphism because the two compatibilities at the end of the section need the proper and the flat cases, while the theorem below needs the case of a closed immersion.

Lemma 11.2.6: Functoriality of the Deformation Space

Let $f\colon X' \to X$ be a morphism of algebraic schemes, $Z \subseteq X$ a closed subscheme, and $Z' = f^{-1}(Z)$ its scheme-theoretic preimage. Write $\pi'\colon M^\circ_{Z'}X' \to \mathbb{P}^1$ for the projection of the corresponding deformation space. Then there is a unique morphism $\widetilde{f}\colon M_{Z'}X' \to M_Z X$ over $f \times \mathrm{id}_{\mathbb{P}^1}$, and it satisfies

$$\widetilde{f}^{-1}\big(M^\circ_Z X\big) \;=\; M^\circ_{Z'}X'$$

so that it restricts to a morphism $\widetilde{f}\colon M^\circ_{Z'}X' \to M^\circ_Z X$. Moreover:

1. $\pi \circ \widetilde{f} = \pi'$; over $\mathbb{A}^1$ the morphism $\widetilde{f}$ restricts to $f \times \mathrm{id}_{\mathbb{A}^1}$, and over ∞ it restricts to the morphism of normal cones $C(f)\colon C_{Z'}X' \to C_Z X$ induced by the identity $\mathcal{I}\mathcal{O}_{X'} = \mathcal{I}'$.
2. If f is proper, then $\widetilde{f}$ and $C(f)$ are proper.
3. If f is flat of relative dimension n, then $\widetilde{f}$ and $C(f)$ are flat of relative dimension n, and the square formed by $\widetilde{f}$, $f \times \mathrm{id}_{\mathbb{P}^1}$ and the two blow-up morphisms is Cartesian.

***Proof*:**
Write $g = f \times \mathrm{id}_{\mathbb{P}^1}$, let $\mathcal{J} \subseteq \mathcal{O}_{X\times\mathbb{P}^1}$ be the ideal sheaf of $Z \times \{\infty\}$, and let $\mathcal{J}'$ be the ideal sheaf of $Z' \times \{\infty\}$ in $X' \times \mathbb{P}^1$.

Step 1: the ideals correspond. Scheme-theoretically $Z \times \{\infty\} = (Z \times \mathbb{P}^1) \cap (X \times \{\infty\})$, so $\mathcal{J}$ is the sum of the ideal sheaves of the two factors, and the same holds upstairs. The inverse image ideal of a sum is the sum of the inverse image ideals; the ideal of $Z \times \mathbb{P}^1$ pulls back to that of $Z' \times \mathbb{P}^1$ because $\mathcal{I}\mathcal{O}_{X'} = \mathcal{I}'$ by definition of the scheme-theoretic preimage, and the ideal of $X \times \{\infty\}$, generated by u, pulls back to the ideal of $X' \times \{\infty\}$, generated by the same u. Hence

$$\mathcal{J}\mathcal{O}_{X'\times\mathbb{P}^1} \;=\; \mathcal{J}'$$

Step 2: existence and uniqueness. On $M_{Z'}X'$ the ideal $\mathcal{J}'\mathcal{O}_{M_{Z'}X'}$ is invertible by proposition 7.2.27(3), and by Step 1 it is the inverse image ideal of $\mathcal{J}$ under the composite $M_{Z'}X' \to X' \times \mathbb{P}^1 \xrightarrow{g} X \times \mathbb{P}^1$. The universal property, proposition 7.2.27(4), therefore factors that composite uniquely through the blow-down $p\colon M_Z X \to X \times \mathbb{P}^1$, and the factorization is the asserted $\widetilde{f}$. Uniqueness of the factorization says precisely that $\widetilde{f}$ is the only morphism $M_{Z'}X' \to M_Z X$ lying over g. Since π is the second projection composed with p, and $\widetilde{f}$ lies over g, which is the identity in the second factor, $\pi \circ \widetilde{f} = \pi'$.

Step 3: the two open subschemes correspond. The assertion is local on X and on $\mathbb{P}^1$. Over $\mathbb{A}^1$ the blow-ups are isomorphisms, since the centres lie over ∞ (proposition 7.2.27(2)), and both deformation spaces contain the whole preimage of $\mathbb{A}^1$; there $\widetilde{f}^{-1}(X \times \mathbb{A}^1) = X' \times \mathbb{A}^1$ because $\pi \circ \widetilde{f} = \pi'$. Over $\mathbb{P}^1 \setminus \{0\} = \operatorname{Spec} \overline{k}[u]$, take an affine open $U = \operatorname{Spec} A$ of X with $\mathcal{I}|_U$ given by an

ideal $I \subseteq A$, and put $J = IA[u] + (u) \subseteq A[u]$, the ideal of $Z \times \{\infty\}$ over $U \times (\mathbb{P}^1 \setminus \{0\})$, so that by definition 7.2.26 the deformation space is there the relative Proj of the Rees algebra $\bigoplus_{n \geq 0} J^n$, in which u is an element of degree one. By lemma 11.1.2 the open subscheme $M_Z^\circ X = M_Z X \setminus \mathrm{Bl}_Z X$ is over $U \times (\mathbb{P}^1 \setminus \{0\})$ exactly the basic open $D_+(u)$ of that relative Proj, with coordinate ring the extended Rees algebra $A[u][I/u]$, its closed complement there being $\mathrm{Bl}_Z X$.
The same description applies upstairs over any affine open $U' = \operatorname{Spec} A'$ of $f^{-1}(U)$, with $I' = IA'$ and $J' = I'A'[u] + (u)$, giving $M_{Z'}^\circ X' = D_+(u)$ there. By Step 1, $J^n A'[u] = J'^n$ for every n, so the graded homomorphism $\bigoplus_n J^n \otimes_{A[u]} A'[u] \to \bigoplus_n J'^n$ is surjective in every degree; it therefore induces a morphism of relative Proj schemes defined on all of $M_{Z'}X'$, and that morphism lies over g; by the uniqueness of Step 2 it is $\widetilde{f}$. A morphism of Proj schemes induced by a graded homomorphism φ pulls the basic open $D_+(x)$ back to $D_+(\varphi(x))$, and $\varphi(u) = u$, so $\widetilde{f}^{-1}(D_+(u)) = D_+(u)$. Hence $\widetilde{f}^{-1}(M_Z^\circ X) = M_{Z'}^\circ X'$.

The three numbered assertions remain.

1. *The restrictions to the two fibres.* Over $\mathbb{A}^1$ both blow-ups are isomorphisms onto $X \times \mathbb{A}^1$ and $X' \times \mathbb{A}^1$, and $\widetilde{f}$ lies over g, so it restricts there to $f \times \mathrm{id}_{\mathbb{A}^1}$. Over ∞, the chart of Step 3 has coordinate ring $A[u][I/u]$, and the fibre over ∞ is cut out by u; on that chart the identification of the fibre with the normal cone given by theorem 11.1.5(2) reads

$$A[u][I/u]/(u) \;\cong\; \bigoplus_{n \geq 0} I^n/I^{n+1}$$

carrying the class of a/u, for $a \in I$, to the class of a in degree one. The morphism $\widetilde{f}$ is induced by $A[u] \to A'[u]$ and sends a/u to $f^\#(a)/u$, so on fibres it induces the homomorphism of graded algebras determined in degree one by $\bar{a} \mapsto \overline{f^\# a}$. That is the homomorphism defining $C(f)$.

2. *Properness.* Suppose f is proper. The blow-up $M_{Z'}X' \to X' \times \mathbb{P}^1$ is projective, hence proper, by proposition 7.2.27(1), and g is proper as a base change of f, so the composite $M_{Z'}X' \to X \times \mathbb{P}^1$ is proper. Factor $\widetilde{f}$ as

$$M_{Z'}X' \xrightarrow{(\mathrm{id}, \widetilde{f})} M_{Z'}X' \times_{X \times \mathbb{P}^1} M_Z X \xrightarrow{\mathrm{pr}_2} M_Z X$$

The first map is the graph of $\widetilde{f}$ over $X \times \mathbb{P}^1$, a closed immersion because $M_Z X \to X \times \mathbb{P}^1$ is separated, and the second is a base change of the proper morphism $M_{Z'}X' \to X \times \mathbb{P}^1$, hence proper. So $\widetilde{f}$ is proper. Restricting over the open subscheme $M_Z^\circ X$, whose preimage is $M_{Z'}^\circ X'$ by Step 3, preserves properness, and $C(f)$ is the base change of $\widetilde{f}$ along the closed immersion $C_Z X \hookrightarrow M_Z^\circ X$, hence proper.

3. *Flatness.* Suppose f is flat of relative dimension n; then so is g, being a base change of f. Flatness gives $\mathcal{J}^n \mathcal{O}_{X' \times \mathbb{P}^1} = g^* \mathcal{J}^n$ for every n, so the Rees algebra of $\mathcal{J}'$ is the pullback g^* of the Rees algebra of $\mathcal{J}$, and proposition 5.3.36 identifies

$$M_{Z'}X' \;\cong\; M_Z X \times_{X \times \mathbb{P}^1} \left(X' \times \mathbb{P}^1\right)$$

over $X' \times \mathbb{P}^1$. Under this identification the second projection is a morphism over g, hence equals $\widetilde{f}$ by the uniqueness of Step 2, and the square is Cartesian. So $\widetilde{f}$ is a base change of g, therefore flat with the same fibres, that is, flat of relative dimension n. The same holds after restricting to the two open subschemes, which correspond by Step 3, and $C(f)$ is again a base change of $\widetilde{f}$, so it too is flat of relative dimension n, completing the proof.

The computation of σ on a subvariety can now be carried out. The statement is the one the construction was designed to produce: specializing a subvariety returns the normal cone of its intersection with Z, which is the linearized model of how V meets Z.

Theorem 11.2.7: Specialization of a Subvariety

Let $Z \subseteq X$ be a closed subscheme, let $V \subseteq X$ be a k-dimensional subvariety, and let $W = V \cap Z$ be the scheme-theoretic intersection. Then

$$\sigma[V] \;=\; \big[C_W V\big] \;\in\; A_k\big(C_Z X\big)$$

the fundamental cycle of the normal cone $C_W V$, regarded as a closed subscheme of $C_Z X$ through lemma 10.4.12. In particular $C_W V$ is pure of dimension k.

***Proof*:**

Step 1: the closure lift. Let $\widetilde{V}$ be the closure of $V \times \mathbb{A}^1$ in $M_Z^\circ X$, with its reduced structure, a subvariety of dimension $k+1$. Since $V \times \mathbb{A}^1$ is closed in $X \times \mathbb{A}^1$, the open restriction of $[\widetilde{V}]$ is $j^*[\widetilde{V}] = [V \times \mathbb{A}^1] = q^*[V]$ by example 8.3.4, so $[\widetilde{V}]$ is a lift of $[V]$ and $\sigma[V] = D_\infty \cdot [\widetilde{V}]$.

Step 2: the smaller deformation space. Apply lemma 11.2.6 to the closed immersion $\iota\colon V \hookrightarrow X$, whose scheme-theoretic preimage of Z is W by lemma 10.4.12. It supplies a proper morphism

$$\widetilde{\iota}\colon M_W^\circ V \longrightarrow M_Z^\circ X$$

commuting with the projections to $\mathbb{P}^1$ and restricting over $\mathbb{A}^1$ to the closed immersion $V \times \mathbb{A}^1 \hookrightarrow X \times \mathbb{A}^1$. Over ∞ it restricts to the morphism $C(\iota)\colon C_W V \to C_Z X$, which is induced by the graded homomorphism attached to $\mathcal{I}\mathcal{O}_V = \mathcal{I}_W$ and is therefore the closed immersion of lemma 10.4.12.

The scheme $M_W V$ is integral. Indeed $V \times \mathbb{P}^1$ is a variety, and on an affine open $\operatorname{Spec} B \subseteq V \times \mathbb{P}^1$ the ideal $\mathfrak{b}$ of $W \times \{\infty\}$ is either the unit ideal, in which case the corresponding piece of the blow-up is $\operatorname{Spec} B$, or a nonzero proper ideal, in which case the charts of $\operatorname{Proj} \bigoplus_n \mathfrak{b}^n$ have coordinate rings $B[\mathfrak{b}/b]$ for $0 \neq b \in \mathfrak{b}$, subrings of the function field $k(V \times \mathbb{P}^1)$ containing B. Every chart is thus the spectrum of a domain with fraction field $k(V)(t)$, so $M_W V$ is reduced and irreducible, and so is its open subscheme $M_W^\circ V$. The latter has dimension $k+1$, since it contains $V \times \mathbb{A}^1$ as a nonempty, hence dense, open subscheme.

Step 3: the image and the degree. The morphism $\widetilde{\iota}$ is proper, hence closed, and $V \times \mathbb{A}^1$ is dense in $M_W^\circ V$, so the image is the closure of $\widetilde{\iota}(V \times \mathbb{A}^1) = V \times \mathbb{A}^1$, which is $\widetilde{V}$. Over $\mathbb{A}^1$ the morphism is an isomorphism onto $V \times \mathbb{A}^1$, which is dense in both $M_W^\circ V$ and $\widetilde{V}$, so the two schemes have the same function field and $\deg\big(M_W^\circ V/\widetilde{V}\big) = 1$. By definition 8.2.1,

$$\widetilde{\iota}_*\big[M_W^\circ V\big] \;=\; \big[\widetilde{V}\big]$$

Step 4: cutting the smaller family. The projection $\pi'\colon M_W^\circ V \to \mathbb{P}^1$ is flat by theorem 11.1.5(3), applied to the closed subscheme $W \subseteq V$. Each nonempty fibre is cut out locally by one equation, so every component has dimension at least k by Krull's principal ideal theorem, and each fibre is a proper closed subset of the integral $M_W^\circ V$, so every component has dimension at most k. Thus π' is flat of relative dimension k in the sense of definition 8.3.1, and its fibre over ∞, which is $C_W V$, is pure of dimension k.

Since $\pi \circ \widetilde{\iota} = \pi'$, the pullback Cartier divisor is $\widetilde{\iota}^* D_\infty = \pi'^*(\infty)$, cut out by $\pi'^\# u$; it is defined because $M^\circ_W V$ is dominant over $\mathbb{P}^1$, its restriction over $\mathbb{A}^1$ being the nonempty $V \times \mathbb{A}^1$, and so is not contained in $|D_\infty|$. The first clause of definition 9.2.1 therefore gives

$$\widetilde{\iota}^* D_\infty \cdot \big[M^\circ_W V\big] \;=\; \big[\operatorname{div}\big(\pi'^\# u\big)\big] \;=\; \sum_T \operatorname{ord}_T\big(\pi'^\# u\big)\,[T]$$

the sum over the codimension-one subvarieties T of $M^\circ_W V$. By definition 8.1.1 the coefficient $\operatorname{ord}_T(\pi'^\# u)$ is the length of $\mathcal{O}_{M^\circ_W V,T}/(\pi'^\# u)$, which is the local ring at T of the zero scheme of $\pi'^\# u$, that is, of the fibre $C_W V$. The coefficient vanishes unless $T \subseteq C_W V$, and the k-dimensional subvarieties of $C_W V$ are exactly its irreducible components, by the purity just established. Comparing with the fundamental cycle of definition 8.1.2,

$$\widetilde{\iota}^* D_\infty \cdot \big[M^\circ_W V\big] \;=\; \big[C_W V\big]$$

Step 5: descent to $C_Z X$. Combining the previous steps with the projection formula proposition 9.3.3(1) for the proper morphism $\widetilde{\iota}$,

$$\sigma[V] \;=\; D_\infty \cdot \widetilde{\iota}_*\big[M^\circ_W V\big] \;=\; \widetilde{\iota}_*\Big(\widetilde{\iota}^* D_\infty \cdot \big[M^\circ_W V\big]\Big) \;=\; \widetilde{\iota}_*\big[C_W V\big]$$

as classes in $A_k(C_Z X)$. By Step 2 the morphism $\widetilde{\iota}$ restricts over ∞ to the closed immersion of lemma 10.4.12, and a closed immersion carries a fundamental cycle to the fundamental cycle of its image, every degree in definition 8.2.1 being 1. Hence $\widetilde{\iota}_*[C_W V] = [C_W V]$ read inside $C_Z X$, as we sought to show.

The theorem is what turns σ from a construction into a computation. It also explains the two features of example 11.2.5: for $V = \mathbb{P}^1$ the intersection $V \cap Z$ is Z itself and the answer is $[C_P\mathbb{P}^1]$, while for $V = \{Q\}$ with $Q \notin Z$ the intersection is empty, the normal cone of the empty subscheme is empty, and the answer is 0. Specialization does not preserve the geometry of V, only its normal geometry along Z: the class $[C_W V]$ remembers how V meets Z and forgets everything about V away from Z. The multiplicities carry information: the fundamental cycle of definition 8.1.2 weights each component of $C_W V$ by the length of the local ring there, so a component along which the cone is non-reduced contributes with multiplicity greater than one. In the example below the cone is reduced and every multiplicity is one, the two components recording the two tangent directions of a node; the cuspidal cubic of exercise 11.2.1 (3) is the opposite case, a single tangent line doubled. And the purity statement is not an extra hypothesis but a by-product: the normal cone of any closed subscheme in a variety is pure of the dimension of that variety, because it is a fibre of a flat family whose other fibres are the variety itself.

Example 11.2.8: Specializing the Nodal Cubic to its Tangent Cone

Assume $\operatorname{char}\overline{k} \neq 2$, let $X = \operatorname{Spec}\overline{k}[x,y]/(y^2 - x^2 - x^3)$ be the affine nodal cubic, and let $Z = \{P\}$ be the reduced origin. The curve X is a variety of dimension one, so $A_1(X) = \mathbb{Z}\,[X]$ by example 8.1.6(3). Taking $V = X$ in theorem 11.2.7 gives $W = X \cap Z = Z$ and

$$\sigma[X] \;=\; \big[C_P X\big]$$

and example 10.4.7 computed the normal cone: $C_P X = \operatorname{Spec}\overline{k}[x,y]/(y^2 - x^2)$, the union of the two lines $L_+ = \{y = x\}$ and $L_- = \{y = -x\}$. The polynomial $y^2 - x^2 = (y-x)(y+x)$ is squarefree, so the cone is reduced and each component carries multiplicity one:

$$\sigma[X] \;=\; [L_+] + [L_-] \;\in\; A_1(C_P X)$$

which is free of rank two on the two lines, again by example 8.1.6(3). The specialization has replaced an irreducible curve by a reducible cone, one line for each branch of the node, and the two summands are exactly the two tangent directions along which X approaches P. The class of X in $A_1(X) = \mathbb{Z}$ carried no information about the singularity; its specialization, in a group of rank two, carries the branch structure.

What remains is to check that σ is compatible with the two functorialities of chapter 8. Both statements are forced by the shape of the construction: σ was built from the flat pullback q^*, the localization sequence, and the divisor operator, and each of the three commutes with proper pushforward and with flat pullback. The compatibilities are what allow σ, and hence the Gysin map of section 11.3, to be computed by moving to a cover or restricting to an open set.

Proposition 11.2.9: Compatibility with Proper Pushforward

Let $f\colon X' \to X$ be a proper morphism, $Z \subseteq X$ a closed subscheme, and $Z' = f^{-1}(Z)$. Write σ' for the specialization homomorphism of $Z' \subseteq X'$ and $C(f)\colon C_{Z'}X' \to C_Z X$ for the induced proper morphism of normal cones. Then for every $\alpha' \in A_k(X')$,

$$\sigma\big(f_*\alpha'\big) \;=\; C(f)_*\big(\sigma'(\alpha')\big) \quad \text{in } A_k\big(C_Z X\big)$$

***Proof*:**

Let $\widetilde{f}\colon M^\circ_{Z'}X' \to M^\circ_Z X$ be the proper morphism of lemma 11.2.6, and let β' be the closure lift of α', so that $j'^*\beta' = q'^*\alpha'$ and every component of β' dominates $\mathbb{P}^1$.

Step 1: the pushed-forward lift. Since $\pi \circ \widetilde{f} = \pi'$, the square with horizontal maps $\widetilde{f}$ and $f \times \mathrm{id}_{\mathbb{A}^1}$ and vertical maps the two open immersions j', j is Cartesian: the preimage $\widetilde{f}^{-1}(X \times \mathbb{A}^1)$ is $\pi'^{-1}(\mathbb{A}^1) = X' \times \mathbb{A}^1$, and $\widetilde{f}$ restricts there to $f \times \mathrm{id}_{\mathbb{A}^1}$ by lemma 11.2.6(1). As j is flat of relative dimension 0 and $\widetilde{f}$ is proper, proposition 8.3.6 gives

$$j^*\widetilde{f}_*\beta' \;=\; \big(f \times \mathrm{id}_{\mathbb{A}^1}\big)_* j'^*\beta' \;=\; \big(f \times \mathrm{id}_{\mathbb{A}^1}\big)_* q'^*\alpha'$$

The square with horizontal maps $f \times \mathrm{id}_{\mathbb{A}^1}$ and f and vertical maps q' and q is Cartesian as well, since $X' \times \mathbb{A}^1 = X' \times_X (X \times \mathbb{A}^1)$, so a second application of proposition 8.3.6, now with q flat of relative dimension 1 and f proper, gives $(f \times \mathrm{id}_{\mathbb{A}^1})_* q'^*\alpha' = q^* f_*\alpha'$. Hence $\widetilde{f}_*\beta'$ is a lift of $f_*\alpha'$.

Step 2: the projection formula. Every component of β' dominates $\mathbb{P}^1$, and $\pi \circ \widetilde{f} = \pi'$, so no component of the support of β' maps into $|D_\infty| = C_Z X$; the pullback divisor $\widetilde{f}^* D_\infty$ is therefore defined, and it equals $\pi'^*(\infty) = D'_\infty$. By proposition 9.3.3(1),

$$D_\infty \cdot \widetilde{f}_*\beta' \;=\; \widetilde{f}_*\big(D'_\infty \cdot \beta'\big)$$

as classes in $A_k(C_Z X)$. The left side is $\sigma(f_*\alpha')$ by Step 1 and theorem 11.2.4, and $D'_\infty \cdot \beta' = \sigma'(\alpha')$ is a class on $C_{Z'}X'$, on which $\widetilde{f}_*$ acts as $C(f)_*$ by lemma 11.2.6(1). Therefore $\sigma(f_*\alpha') = C(f)_*(\sigma'(\alpha'))$, completing the proof.

Proposition 11.2.10: Compatibility with Flat Pullback

Let $f\colon X' \to X$ be flat of relative dimension n, $Z \subseteq X$ a closed subscheme, and $Z' = f^{-1}(Z)$. Then the induced morphism $C(f)\colon C_{Z'}X' \to C_Z X$ is flat of relative dimension n, and for every $\alpha \in A_k(X)$,

$$\sigma'\big(f^*\alpha\big) \;=\; C(f)^*\big(\sigma(\alpha)\big) \quad \text{in } A_{k+n}\big(C_{Z'}X'\big)$$

***Proof*:**

Flatness of $C(f)$ is lemma 11.2.6(3), which also makes $\widetilde{f}\colon M^\circ_{Z'}X' \to M^\circ_Z X$ flat of relative dimension n with $\widetilde{f}^{-1}(C_Z X) = C_{Z'}X'$ and $\widetilde{f}^{-1}(X \times \mathbb{A}^1) = X' \times \mathbb{A}^1$.

Step 1: the pulled-back lift. Let β be the closure lift of α. Contravariant functoriality of flat pullback (proposition 8.3.5(3)) applied to the two equal composites $j \circ (f \times \mathrm{id}_{\mathbb{A}^1}) = \widetilde{f} \circ j'$ and $q \circ (f \times \mathrm{id}_{\mathbb{A}^1}) = f \circ q'$ gives

$$j'^*\widetilde{f}^*\beta \;=\; \big(f \times \mathrm{id}_{\mathbb{A}^1}\big)^* j^*\beta \;=\; \big(f \times \mathrm{id}_{\mathbb{A}^1}\big)^* q^*\alpha \;=\; q'^* f^*\alpha$$

so $\widetilde{f}^*\beta$ is a lift of $f^*\alpha$. Its components are components of flat preimages of the components of β, each of which dominates $\mathbb{P}^1$, so they dominate $\mathbb{P}^1$ as well and none lies in $|D'_\infty|$.

Step 2: the projection formula. The pullback Cartier divisor is $\widetilde{f}^*D_\infty = \pi'^*(\infty) = D'_\infty$, again because $\pi \circ \widetilde{f} = \pi'$. By proposition 9.3.3(2),

$$\widetilde{f}^*\big(D_\infty \cdot \beta\big) \;=\; D'_\infty \cdot \widetilde{f}^*\beta$$

as classes in $A_{k+n}(\widetilde{f}^{-1}(C_Z X)) \;=\; A_{k+n}(C_{Z'}X')$. The right side is $\sigma'(f^*\alpha)$ by Step 1 and theorem 11.2.4. On the left, $D_\infty \cdot \beta = \sigma(\alpha)$ is a class on $C_Z X$, and the restriction of $\widetilde{f}^*$ to classes on $C_Z X$ is the flat pullback along the base change $C(f)$, since flat pullback is computed on preimages and $\widetilde{f}^{-1}(C_Z X) = C_{Z'}X'$. Hence $\sigma'(f^*\alpha) = C(f)^*(\sigma(\alpha))$, as we sought to show.

The two propositions say that specialization is a natural transformation, covariant for proper morphisms and contravariant for flat ones, between the Chow groups of X and those of the normal cone. The hypothesis common to both, $Z' = f^{-1}(Z)$, is not a restriction imposed for convenience: the normal cone of Z' in X' maps to that of Z in X precisely because the ideal of Z pulls back to the ideal of Z', and without that identity there is no morphism $C(f)$ to state the compatibility with.

Exercise 11.2.1

1. Take $Z = X$. Exercise 11.1.1 (1) identifies the deformation space as $M^\circ_X X = X \times \mathbb{P}^1$ with special fibre $C_X X = X$. Deduce from theorem 11.2.7 that $\sigma\colon A_k(X) \to A_k(X)$ is the identity.
2. Let $X = \mathbb{P}^2$ and let $Z = L$ be a line, an effective Cartier divisor on X. Using example 10.4.15, identify $C_L\mathbb{P}^2$ with the total space of the line bundle $N = \mathcal{O}_{\mathbb{P}^2}(1)|_L$ on $L \cong \mathbb{P}^1$, and compute $A_k(C_L\mathbb{P}^2)$ for all k by theorem 8.4.6. Then compute $\sigma[\mathbb{P}^2]$, $\sigma[L']$ for a line $L' \neq L$, and $\sigma[Q]$ for a closed point Q, and say in which dimensions σ is an isomorphism.
3. Let $X = \operatorname{Spec}\overline{k}[x,y]/(y^2 - x^3)$ be the affine cuspidal cubic and $Z = \{P\}$ the reduced origin. Taking the cone $C_P X = \operatorname{Spec}\overline{k}[x,y]/(y^2)$ and its fundamental cycle $2\,[L]$ from

exercise 11.1.1 (5), compute $\sigma[X]$ as a cycle. Compare the answer with example 11.2.8 and say what distinguishes a cusp from a node in this computation. Does the answer depend on $\operatorname{char}\overline{k}$?

4. Let $\iota\colon Z \hookrightarrow C_Z X$ be the vertex of the normal cone (definition 10.4.1) and $i_Z\colon Z \hookrightarrow X$ the inclusion. Show that $\sigma \circ (i_Z)_* = \iota_*$ as homomorphisms $A_k(Z) \to A_k(C_Z X)$. (Hint: apply theorem 11.2.7 to a subvariety $V \subseteq Z$ and identify $C_V V$.)

5. Show that if $V \subseteq X$ is a subvariety with $V \cap Z = \emptyset$, then $\sigma[V] = 0$. Deduce that for $X = \mathbb{P}^n$ with $n \geq 1$ and $Z = \{P\}$ a closed point, σ vanishes on all of $A_0(\mathbb{P}^n)$, and check that this is consistent with a direct computation of $A_0(C_P\mathbb{P}^n)$.

11.3 The Gysin Map for a Regular Embedding

The specialization homomorphism of section 11.2 accepts an arbitrary closed subscheme, and its target is the Chow group of a cone that is usually not a bundle. This section takes the case in which it is. For a regular embedding the normal cone is the normal bundle (proposition 10.4.11), so specialization lands in $A_k(N)$, and flat pullback along $N \to Z$ is an isomorphism (theorem 8.4.6) that can be inverted. The composite lowers dimension by the codimension and lands on Z.

Two things have to be established beyond that one-line recipe. The first is that the recipe computes what a reader expects when the intersection is proper: the cycle $[V \cap Z]$, with the multiplicities the scheme structure supplies. The second is that the construction survives base change. For any morphism $X' \to X$ the normal cone of the preimage of Z still sits inside the pulled-back normal bundle, and the same recipe produces a homomorphism landing on that preimage rather than on Z. It is this refined form, not the plain one, that chapter 12 applies to the diagonal.

Both rest on the following comparison between the normal cone of a pullback and the pullback of the normal bundle. Regularity of i is what makes the comparison available: it is exactly the hypothesis under which the conormal sheaf is locally free, so that the symmetric algebra it generates is a bundle into which the associated graded algebra downstairs maps.

The lemma says that regularity of i propagates one step: it is not inherited by $Z' \hookrightarrow X'$, which can be badly irregular, but the failure is confined inside a bundle of the correct rank d. The cone $C_{Z'}X'$ may be singular, may be reducible, and may have components of several dimensions, and none of that prevents it from being a closed subscheme of a rank-d bundle on Z'. That containment is what lets the same recipe be applied over X' as over X, and the discrepancy between $C_{Z'}X'$ and all of N' is what the excess intersection formula of chapter 12 measures.

With $g = \mathrm{id}_X$ the lemma reduces to the case already in hand, and the construction can be stated.

Definition 11.3.1: The Gysin Map of a Regular Embedding

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 0$ with normal bundle $\pi\colon N \to Z$. For $d = 0$ the ideal sheaf is zero, i is an open and closed immersion, N is the zero bundle over Z, and the formula below reads as flat pullback along i; the construction is stated for $d \geq 1$ below and the degenerate case is recorded here so that statements about codimension-zero embeddings, which occur in chapter 12, have a definition to refer to. By proposition 10.4.11 the normal cone $C_Z X$ is N, so the specialization homomorphism of definition 11.2.3 is a homomorphism $\sigma\colon A_k(X) \to A_k(N)$ (theorem 11.2.4). The *Gysin map* of i is

$$i^! \;=\; (\pi^*)^{-1} \circ \sigma \;:\; A_k(X) \longrightarrow A_{k-d}(Z)$$

where $\pi^*\colon A_{k-d}(Z) \to A_k(N)$ is flat pullback along π.

The inverse in the definition exists because N is a vector bundle of rank d, hence an affine bundle of relative dimension d (definition 8.4.5), so π^* is an isomorphism by theorem 8.4.6. The codimension-zero case is excluded because it is empty of content: an ideal generated locally by the empty regular sequence is the zero ideal, so a regular embedding of codimension zero is $Z = X$.

The smallest case can be run through at once. Take $X = \mathbb{A}^1$ and $Z = \{0\}$, whose ideal is generated by the single nonzerodivisor t, so that i is a regular embedding of codimension one and N is a one-dimensional vector space over the point Z, with total space $\mathbb{A}^1$. By theorem 11.2.7, $\sigma[\mathbb{A}^1] = [C_Z\mathbb{A}^1] = [N]$, and $\pi^*[Z] = [N]$ by proposition 8.3.5(2), so $i^![\mathbb{A}^1] = [Z]$ in $A_0(Z) = \mathbb{Z}$: the origin with multiplicity one.

Proposition 11.3.2: The Gysin Map is a Homomorphism of Degree $-d$

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$. Then:

1. $i^!\colon A_k(X) \to A_{k-d}(Z)$ is a well-defined homomorphism of abelian groups for every k, and it is the unique map satisfying $\pi^* \circ i^! = \sigma$;
2. $i^! = 0$ on $A_k(X)$ for $k < d$.

***Proof*:**

1. A vector bundle of rank d is an affine bundle of relative dimension d in the sense of definition 8.4.5, a local trivialization of the bundle being in particular a local product decomposition with $\mathbb{A}^d$. Theorem 8.4.6 therefore makes $\pi^*\colon A_{k-d}(Z) \to A_k(N)$ an isomorphism for every k. The specialization homomorphism $\sigma\colon A_k(X) \to A_k(N)$ is a homomorphism defined on rational equivalence classes by theorem 11.2.4, and $C_Z X = N$ by proposition 10.4.11, so the composite $(\pi^*)^{-1} \circ \sigma$ is a homomorphism $A_k(X) \to A_{k-d}(Z)$. If $\tau\colon A_k(X) \to A_{k-d}(Z)$ satisfies $\pi^* \circ \tau = \sigma$, then $\pi^*\tau = \pi^* i^!$, and injectivity of π^* gives $\tau = i^!$.
2. For $k < d$ the target $A_{k-d}(Z)$ is a Chow group in negative dimension. No scheme has a subvariety of negative dimension, so $Z_{k-d}(Z) = 0$ and hence $A_{k-d}(Z) = 0$, forcing $i^! = 0$ there, as we sought to show.

The characterization $\pi^* \circ i^! = \sigma$ is how the map is used in practice: rather than inverting π^*, one produces a candidate class on Z and checks that its pullback to N is the specialization. The degree $-d$ matches the naive expectation that intersecting a k-dimensional cycle with a codimension-d subscheme leaves something of dimension $k-d$. What the construction adds to that expectation is that the answer is defined for every class, with no hypothesis on how Z and a representing cycle meet, and that it is a rational-equivalence invariant.

The first thing to verify is that the machinery reproduces the naive answer when the naive answer makes sense. The condition for that is exactly regularity of the induced embedding.

Proposition 11.3.3: The Gysin Map of a Regular Intersection

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$, let $V \subseteq X$ be a k-dimensional subvariety, and let $W = V \cap Z$ be the scheme-theoretic intersection, that is the preimage of Z under the inclusion $V \hookrightarrow X$. If $W \hookrightarrow V$ is a regular embedding of codimension d, then W is pure of dimension $k-d$ and

$$i^![V] \;=\; \iota_*[W] \;\in\; A_{k-d}(Z)$$

where $[W]$ is the fundamental cycle of definition 8.1.2 and $\iota\colon W \hookrightarrow Z$ is the inclusion.

***Proof*:**
Apply lemma 10.4.13 to the closed immersion $g\colon V \hookrightarrow X$, whose preimage of Z is W: it produces a closed immersion $j\colon C_W V \hookrightarrow N|_W$ of cones over W, and by part (1) of that lemma the hypothesis on $W \hookrightarrow V$ makes j an isomorphism. Thus $C_W V = N|_W$, the restriction of N to W, viewed as a closed subscheme of $C_Z X = N$.

By theorem 11.2.7, $\sigma[V] = [C_W V]$ in $A_k(N)$, the fundamental cycle of the normal cone of V along W read inside $C_Z X$. The previous paragraph identifies that cycle as $[N|_W]$. Now $\pi_W : N|_W \to W$ is flat of relative dimension d, being a vector bundle of rank d, so proposition 8.3.5(2) gives $\pi_W^*[W] = [N|_W]$; and flat pullback of a cycle is computed componentwise from scheme-theoretic preimages (definition 8.3.3), so the cycle $\pi_W^*[W]$ on $N|_W$ is the cycle $\pi^*\iota_*[W]$ on N. Hence

$$\pi^*\big(\iota_*[W]\big) = [N|_W] = \sigma[V]$$

and proposition 11.3.2(1) gives $i^![V] = \iota_*[W]$.

Finally, $\sigma[V]$ lies in $A_k(N)$, so $[N|_W]$ is a k-cycle; since π_W raises dimension by exactly d on each component, every irreducible component of W has dimension $k-d$, completing the proof.

The hypothesis is the scheme-theoretic form of "V meets Z properly and without excess tangency". It is not a condition on dimensions alone: a subvariety can meet Z in the expected dimension $k-d$ and still fail it, if the intersection is not cut out by d equations forming a regular sequence, and then $i^![V]$ carries multiplicities the set $V \cap Z$ does not display. When the hypothesis does hold, the Gysin map returns the fundamental cycle of the intersection scheme, so the multiplicities are the lengths that scheme records and nothing more.

Example 11.3.4: Linear Subspaces of Projective Space

Fix homogeneous coordinates $x_0, \dots, x_n$ on $\mathbb{P}^n$ and let $Z = \{x_0 = \cdots = x_{d-1} = 0\} \cong \mathbb{P}^{n-d}$ with $1 \le d \le n$. On each standard affine chart $\{x_r \neq 0\}$ meeting Z, the ideal of Z is generated by the d ratios $x_0/x_r, \dots, x_{d-1}/x_r$, which form a regular sequence in the polynomial ring of that chart: each is a nonzerodivisor modulo its predecessors, the quotients being polynomial rings in the remaining coordinates. So $i\colon Z \hookrightarrow \mathbb{P}^n$ is a regular embedding of codimension d and $i^!\colon A_m(\mathbb{P}^n) \to A_{m-d}(Z)$ is defined.

By theorem 8.5.2, $A_m(\mathbb{P}^n) = \mathbb{Z}\, h_m$ with h_m the class of a linear subspace $\mathbb{P}^m \subseteq \mathbb{P}^n$, and likewise $A_{m-d}(Z) = \mathbb{Z}\, h^Z_{m-d}$ for $Z \cong \mathbb{P}^{n-d}$. Take $m \ge d$ and represent h_m by $L = \{x_{m+1} = \cdots = x_n = 0\}$, the span of the coordinate points $e_0, \dots, e_m$. Then

$$L \cap Z \;=\; \{x_0 = \cdots = x_{d-1} = 0\} \cap L$$

is the span of $e_d, \dots, e_m$, a linear $\mathbb{P}^{m-d}$, and its ideal in $L \cong \mathbb{P}^m$ is generated by the restrictions of $x_0, \dots, x_{d-1}$, again a regular sequence of length d on every chart of L meeting it. So $L \cap Z \hookrightarrow L$ is a regular embedding of codimension d, and proposition 11.3.3 gives

$$i^!(h_m) \;=\; \big[\mathbb{P}^{m-d}\big] \;=\; h^Z_{m-d}$$

the intersection scheme being reduced and irreducible, so that its fundamental cycle carries multiplicity one. For $m < d$ the target $A_{m-d}(Z)$ vanishes and $i^!(h_m) = 0$ by proposition 11.3.2(2). The Gysin map of a linear subspace is therefore the shift $h_m \mapsto h^Z_{m-d}$, carrying generator to generator in every degree where the target is nonzero.

The computation is the expected one, and its value is that no general position was invoked to get it: the representative L was chosen for convenience, and proposition 11.3.2 guarantees that any other representative of h_m, including one contained in Z or tangent to it, returns the same class. That is the difference between this construction and a count of intersection points.

The codimension-one case can be compared with the operator built by hand in chapter 9, and the two agree wherever the first clause of definition 9.2.1 applies.

Proposition 11.3.5: Agreement with Intersection with a Divisor

Let D be an effective Cartier divisor on an algebraic scheme X with $D \neq X$, and let $i\colon D \hookrightarrow X$ be the inclusion, a regular embedding of codimension one with normal bundle $N = \mathcal{O}_X(D)|_D$ (example 10.4.15). Let $V \subseteq X$ be a k-dimensional subvariety with $V \not\subseteq |D|$, and let $W = V \cap D$ be the scheme-theoretic intersection, a closed subscheme with support $V \cap |D|$. Then W is pure of dimension $k-1$, the fundamental cycle of W and the intersection class of definition 9.2.1 agree,

$$[W] \;=\; D \cdot [V] \;\in\; A_{k-1}\big(V \cap |D|\big)$$

already as cycles and not as classes, and the Gysin class of V is their pushforward,

$$i^![V] \;=\; \iota_*\big(D \cdot [V]\big) \;\in\; A_{k-1}(D)$$

where $\iota\colon W \hookrightarrow D$ is the inclusion.

***Proof*:**
Recall that $W = V \cap D$ is the closed subscheme of V cut out by the ideal $\mathcal{I} \cdot \mathcal{O}_V$, where $\mathcal{I}$ is the ideal sheaf of D.

Step 1: the hypothesis of proposition 11.3.3 holds. Since D is an effective Cartier divisor, $\mathcal{I}$ is locally generated by a single nonzerodivisor $f \in \mathcal{O}_X(U)$. As V is integral and $V \not\subseteq |D|$, the restriction $f|_V$ is a nonzero element of the domain $\mathcal{O}_V(U \cap V)$, hence a nonzerodivisor, and it generates $\mathcal{I} \cdot \mathcal{O}_V$ there. So $W \hookrightarrow V$ is a regular embedding of codimension one, and proposition 11.3.3 gives $i^![V] = \iota_*[W]$ in $A_{k-1}(D)$, with W pure of dimension $k-1$.

Step 2: the fundamental cycle of W is the divisor of the section. The intersection class of definition 9.2.1 is computed by its first clause, V not being contained in $|D|$: the local equations of D restrict to a rational section s_V of $\mathcal{O}_X(D)|_V$, given on $U \cap V$ by $f|_V$, and $D \cdot [V] = [\operatorname{div}(s_V)] = \sum_T \operatorname{ord}_T(s_V)\,[T]$, the sum over codimension-one subvarieties $T \subseteq V$. Fix such a T and let $A = \mathcal{O}_{V,T}$, a one-dimensional local domain. The local ring of W at the generic point of T is $A/(f|_V)$, so the coefficient of $[T]$ in the fundamental cycle $[W]$ of definition 8.1.2 is $\operatorname{length}_A\big(A/(f|_V)\big)$, which is $\operatorname{ord}_T(s_V)$ by definition 8.1.1. The coefficient is zero exactly when $f|_V$ is a unit in A, that is when T is not a component of W, so the two cycles have the same components as well as the same multiplicities. Hence $[W] = [\operatorname{div}(s_V)]$ as cycles on $V \cap D$, so $[W] = D \cdot [V]$ in $A_{k-1}(V \cap |D|)$; pushing that identity forward along ι and combining it with Step 1 gives $i^![V] = \iota_*[W] = \iota_*\big(D \cdot [V]\big)$ in $A_{k-1}(D)$, completing the proof.

The proposition is the consistency check in codimension one: the operator $D \cdot -$, built in chapter 9 out of local equations and orders of vanishing, is the specialization construction in disguise. The restriction $V \not\subseteq |D|$ is exactly the first clause of definition 9.2.1; the second clause, where V lies inside the support and $D \cdot [V]$ is computed from the restricted line bundle, corresponds on the Gysin side to a cycle supported on the zero section of N, and evaluating it needs the self-intersection formula of chapter 12.

Everything so far records the answer on Z. The construction gives more: it records it on the part of Z that the cycle meets, and it does so for a cycle living on any scheme over X, not only on X itself. Lemma 10.4.13 is what makes this possible, since it supplies a bundle of rank d on the preimage of Z regardless of how singular that preimage is.

Definition 11.3.6: The Refined Gysin Homomorphism

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle N, and let $g\colon X' \to X$ be a morphism of algebraic schemes. Form the fibre square

$$\begin{array}{ccc} Z' & \xrightarrow{i'} & X' \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle g} \\ Z & \xrightarrow{i} & X \end{array}$$

with $Z' = g^{-1}(Z)$ and i' the induced closed immersion, and write $\pi'\colon N' = h^*N \to Z'$. Let $\sigma'\colon A_k(X') \to A_k(C_{Z'}X')$ be the specialization homomorphism of definition 11.2.3 for the pair (Z', X'), well defined by theorem 11.2.4, and let $j\colon C_{Z'}X' \hookrightarrow N'$ be the closed immersion of lemma 10.4.13. The *refined Gysin homomorphism* of i along g is

$$i^! \;=\; (\pi'^*)^{-1} \circ j_* \circ \sigma' \;:\; A_k(X') \longrightarrow A_{k-d}(Z')$$

where j_* is proper pushforward along the closed immersion j and $\pi'^*\colon A_{k-d}(Z') \to A_k(N')$ is the isomorphism of theorem 8.4.6.

The notation deliberately reuses $i^!$: the map depends on g, but the source and target groups record which g is meant, and no ambiguity arises in practice. Each of the three maps is defined for the reasons already in place. Specialization for (Z', X') needs no hypothesis on Z', which is why the construction tolerates a preimage that is singular or of the wrong dimension; j_* is defined because a closed immersion is proper, and proper pushforward preserves the dimension of a cycle (definition 8.2.1); and π'^* is invertible because N' is a bundle of rank d on Z', by definition 8.4.5 and theorem 8.4.6 as before. The only step that used regularity of i is lemma 10.4.13, and it used it to guarantee that the rank is d on the nose.

Proposition 11.3.7: The Refined Map Extends the Gysin Map

In the situation of definition 11.3.6 with $X' = X$ and $g = \mathrm{id}_X$, the refined Gysin homomorphism is the Gysin map $i^!$ of definition 11.3.1.

***Proof*:**
With $g = \mathrm{id}_X$ one has $Z' = Z$, $h = \mathrm{id}_Z$ and $N' = N$. By lemma 10.4.13(2) the closed immersion j is the identification $C_Z X = N$ of proposition 10.4.11, so j_* is the identity on $A_k(N)$ and $\sigma' = \sigma$. The composite $(\pi^*)^{-1} \circ \mathrm{id} \circ \sigma$ is definition 11.3.1, as claimed.

Example 11.3.8: Two Lines in the Plane

Let $X = \mathbb{P}^2$ and $Z = L = \{x_0 = 0\}$, an effective Cartier divisor and hence a regular embedding of codimension one with $N = \mathcal{O}_{\mathbb{P}^2}(1)|_L$ (example 10.4.15).
A transverse line. Take $g\colon X' = L' \hookrightarrow \mathbb{P}^2$ with $L' = \{x_1 = 0\}$. Then $Z' = g^{-1}(L)$ is cut out on L' by x_0, so $Z' = \{p\}$ with $p = [0 : 0 : 1]$, a reduced point, and $Z' \hookrightarrow L'$ is a regular embedding of codimension one. By lemma 10.4.13(1) the immersion j is an isomorphism, so $C_{Z'}X' = N' = N|_p$, a one-dimensional vector space over p. Specialization gives $\sigma'[L'] = [C_{Z'}L'] = [N']$ by theorem 11.2.7,

and $\pi'^*[p] = [N']$ by proposition 8.3.5(2), so

$$i^![L'] \;=\; [p] \;\in\; A_0(L \cap L') = \mathbb{Z}$$

The class is recorded on the single point $L \cap L'$, not on L, and since $\{p\}$ is the scheme-theoretic intersection $L' \cap L$, proposition 11.3.5 identifies it there with the intersection class $L \cdot [L'] \in A_0(L' \cap |L|)$. Pushing it forward along $\{p\} \hookrightarrow L$ returns the class of a point on L, which by the same proposition is the plain Gysin class of L' in $A_0(L)$.

The same line. Take instead $g = i$, so $X' = L$ and $Z' = L$. The normal cone of L in itself is the zero cone of example 10.4.2(2), so $C_{Z'}X' = L$ and j is the vertex of $N' = N = \mathcal{O}_{\mathbb{P}^2}(1)|_L$, that is the zero section $\zeta\colon L \to N$. Hence

$$i^![L] \;=\; (\pi^*)^{-1}\big(\zeta_*[L]\big) \;\in\; A_0(L)$$

Nothing in this section evaluates the right-hand side. What it must equal is known from chapter 9: example 9.2.3 computes $L \cdot [L] = [q]$, the class of one point, so the class of the zero section in $A_1(N)$ has to be $\pi^*[q]$. Identifying it is the self-intersection formula, proved in chapter 12.

The two halves of the example are the two regimes the refined map is built to handle uniformly. In the first, the preimage Z' has the expected dimension and the answer is a cycle on it. In the second, Z' is far too large, the cone degenerates onto the zero section, and the answer is carried entirely by the normal bundle. No hypothesis distinguishes the cases in definition 11.3.6; the difference shows up only in how much of N' the cone $C_{Z'}X'$ fills.

11.3.9 Foreshadowing: What the Gysin Map Is Not Yet The Gysin map is a homomorphism of Chow groups, not a product. Three things it is expected to satisfy are not proved here and are the business of chapter 12: that $i^!$ applied to the diagonal $X \hookrightarrow X \times X$ of a smooth variety turns $A_*(X)$ into a commutative ring; that $i^!$ of a class supported on Z is computed by capping with the top Chern class of N, which is the self-intersection formula and the missing step in example 11.3.8; and that two refined Gysin maps for two regular embeddings commute, which is what makes the resulting product associative. Everything in the present section is input to those proofs rather than a consequence of them.
The third item is settled before this chapter ends, in section 11.4; the first two belong to chapter 12.

The properties chapter 12 consumes are the two functorialities. They say that the refined Gysin map is a morphism of the theory over X: it commutes with pushing a cycle forward along a proper X-morphism and with pulling it back along a flat one, in each case through the induced morphism of preimages.

Theorem 11.3.10: Compatibility of the Refined Gysin Homomorphism

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$. Let $X' \to X$ and $X'' \to X$ be morphisms of algebraic schemes and $p\colon X'' \to X'$ a morphism over X; write $Z' \subseteq X'$ and $Z'' \subseteq X''$ for the preimages of Z and $q\colon Z'' \to Z'$ for the morphism induced by p.

1. *Proper pushforward.* If p is proper then so is q, and for every $\alpha \in A_k(X'')$,
$$i^!\big(p_*\alpha\big) \;=\; q_*\big(i^!\alpha\big) \quad \text{in } A_{k-d}(Z')$$
2. *Flat pullback.* If p is flat of relative dimension n then so is q, and for every $\beta \in A_k(X')$,
$$i^!\big(p^*\beta\big) \;=\; q^*\big(i^!\beta\big) \quad \text{in } A_{k+n-d}(Z'')$$

***Proof*:**
Write $C' = C_{Z'}X'$ and $C'' = C_{Z''}X''$, let $h'\colon Z' \to Z$ and $h''\colon Z'' \to Z$ be the structure morphisms, and set $N' = h'^*N$ and $N'' = h''^*N$ with projections π' and π''. Let $j'\colon C' \hookrightarrow N'$ and $j''\colon C'' \hookrightarrow N''$ be the closed immersions of lemma 10.4.13.

Since $X'' \to X$ factors through p, the preimage of Z in X'' is $Z'' = p^{-1}(Z')$, so q is the base change of p along $Z' \hookrightarrow X'$; properness and flatness of relative dimension n are both stable under base change, which proves the first assertion of each part. From $h'' = h' \circ q$ one gets $N'' = q^*N'$, so

$$\begin{tikzcd} N'' \arrow[r, "\bar q"] \arrow[d, "\pi''"'] & N' \arrow[d, "\pi'"] \\ Z'' \arrow[r, "q"] & Z' \end{tikzcd}$$

is a fibre square, called the *bundle square* below, with $\bar q$ the projection. The ideal sheaves satisfy $\mathcal{I}'\mathcal{O}_{X''} = \mathcal{I}''$ because $Z'' = p^{-1}(Z')$, so p induces a morphism of graded algebras and hence a morphism of cones $\bar p\colon C'' \to C'$ over q; and the *cone square*

$$\begin{tikzcd} C'' \arrow[r, "j''"] \arrow[d, "\bar p"'] & N'' \arrow[d, "\bar q"] \\ C' \arrow[r, "j'"] & N' \end{tikzcd}$$

commutes, since j' and j'' are induced by the canonical surjections of lemma 10.4.13 and those surjections are compatible with pullback along p, being built from the single conormal surjection of $\mathcal{I}$ on X. Concretely, both composites in the corresponding square of graded algebras are maps out of $\operatorname{Sym}^\bullet\big(h'^*(\mathcal{I}/\mathcal{I}^2)\big)$, hence are determined by their degree-one parts, and in degree one both send the class of a local section a of $\mathcal{I}$ to the class of its pullback in $\mathcal{I}''/\mathcal{I}''^2$, because $\mathcal{I}\mathcal{O}_{X''} = \mathcal{I}''$.

1. Let p be proper. Then $\bar q$ is proper, being the base change of q along π', and $\bar p$ is proper as well: it is the composite $C'' \hookrightarrow N'' \to N'$ of a closed immersion and a proper morphism, corestricted to the closed subscheme $C' \subseteq N'$. The compatibility of specialization with proper pushforward (proposition 11.2.9) gives
$$\sigma'\big(p_*\alpha\big) = \bar p_*\big(\sigma''\alpha\big)$$

Pushing forward along j' and using the cone square together with functoriality of proper pushforward (definition 8.2.1),

$$j'_*\sigma'\big(p_*\alpha\big) = j'_*\bar{p}_*\big(\sigma''\alpha\big) = \bar{q}_* j''_*\big(\sigma''\alpha\big)$$

It remains to move $\bar{q}_*$ past the inverse of π'^*. The bundle square has q proper and π' flat, so proposition 8.3.6 gives $\pi'^* q_* = \bar{q}_*\pi''^*$. Applying this to $\gamma = (\pi''^*)^{-1} j''_*\sigma''\alpha = i^!\alpha$,

$$\pi'^*\big(q_*(i^!\alpha)\big) = \bar{q}_*\big(\pi''^*\gamma\big) = \bar{q}_* j''_*\big(\sigma''\alpha\big) = j'_*\sigma'\big(p_*\alpha\big) = \pi'^*\big(i^!(p_*\alpha)\big)$$

Since π'^* is injective, $q_*(i^!\alpha) = i^!(p_*\alpha)$.

2. Let p be flat of relative dimension n. Flatness gives $\mathcal{I}'^m\mathcal{O}_{X''} = p^*\mathcal{I}'^m$ for every m and makes the formation of the quotients $\mathcal{I}'^m/\mathcal{I}'^{m+1}$ commute with pullback, so $C'' = C' \times_{Z'} Z''$ and the induced $\bar{p}\colon C'' \to C'$ is flat of relative dimension n. Consequently the cone square is a fibre square as well, both vertical maps being base change along $Z'' \to Z'$. The compatibility of specialization with flat pullback (proposition 11.2.10) gives

$$\sigma''\big(p^*\beta\big) = \bar{p}^*\big(\sigma'\beta\big)$$

Applying j''_* and using proposition 8.3.6 for the cone square, whose horizontal maps j', j'' are proper and whose vertical maps $\bar{p}, \bar{q}$ are flat,

$$j''_*\sigma''\big(p^*\beta\big) = j''_*\bar{p}^*\big(\sigma'\beta\big) = \bar{q}^* j'_*\big(\sigma'\beta\big)$$

Finally $\pi' \circ \bar{q} = q \circ \pi''$ is a composite of flat morphisms in two ways, so proposition 8.3.5(3) gives $\bar{q}^*\pi'^* = \pi''^* q^*$. Applying this to $\delta = (\pi'^*)^{-1} j'_*\sigma'\beta = i^!\beta$,

$$\pi''^*\big(q^*(i^!\beta)\big) = \bar{q}^*\big(\pi'^*\delta\big) = \bar{q}^* j'_*\big(\sigma'\beta\big) = j''_*\sigma''\big(p^*\beta\big) = \pi''^*\big(i^!(p^*\beta)\big)$$

and injectivity of π''^* gives $q^*(i^!\beta) = i^!(p^*\beta)$, completing the proof.

Part (1) is what makes the refined map a refinement rather than a separate construction: a class computed on a small piece of X and then pushed forward agrees with the class computed on X directly. Part (2) says that the map commutes with restriction to an open subscheme, that being flat pullback of relative dimension zero: $(i^!\alpha)|_{Z\cap U} = i^!(\alpha|_U)$. It does *not* say that $i^!\alpha$ can be reassembled from its restrictions to an open cover, and it could not: Chow groups are not a Zariski sheaf, and $A_0(\mathbb{P}^1) = \mathbb{Z}$ restricts to zero on each of the two standard charts (theorem 8.5.2, proposition 8.5.1). What the compatibility does give is the useful one-way statement of exercise 11.3.1 (4): a class vanishing on an open set containing Z has vanishing Gysin image. Both are also the form in which chapter 12 uses the map: the projection formula and the compatibility of the intersection product with proper pushforward are read off them.

The first corollary is the statement that the plain Gysin map of a subvariety is always the pushforward of a class living on the intersection.

Corollary 11.3.11: The Gysin Class of a Subvariety Lives on the Intersection

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $V \subseteq X$ be a k-dimensional subvariety, with inclusion $\iota_V\colon V \hookrightarrow X$ and $W = V \cap Z = \iota_V^{-1}(Z)$. Then

$$i^![V] \;=\; \iota_*\big(i_V^![V]\big)$$

where $i_V^!\colon A_k(V) \to A_{k-d}(W)$ is the refined Gysin homomorphism along ι_V and $\iota\colon W \hookrightarrow Z$ is the inclusion.

***Proof*:**
Apply theorem 11.3.10(1) with $X' = X$, $X'' = V$ and $p = \iota_V$, which is proper as a closed immersion. The preimage of Z in V is W and the induced morphism q is ι. By definition 8.2.1 the class $[V] \in A_k(V)$ has $p_*[V] = [V] \in A_k(X)$, the extension $k(V) \subseteq k(V)$ having degree one. On the left-hand side the refined Gysin homomorphism along id_X is the Gysin map of definition 11.3.1 by proposition 11.3.7, so the identity reads $i^![V] = \iota_*(i_V^![V])$, as we sought to show.

The class $i^![V] \in A_{k-d}(Z)$ discards information: it records a cycle on all of Z, whereas the geometry produced it on $V \cap Z$, which may be much smaller. The refined class $i_V^![V] \in A_{k-d}(V \cap Z)$ retains that localization, and it is the object every localized intersection number is a degree of. Proposition 11.3.3 is the special case in which the refined class is the fundamental cycle $[V \cap Z]$ itself, and example 11.3.8 exhibits both a case where it is and a case where it is not.

Exercise 11.3.1

1. Let $Z = \{x_2 = x_3 = 0\} \subseteq \mathbb{P}^3$, a line. Show that $Z \hookrightarrow \mathbb{P}^3$ is a regular embedding of codimension two, and compute $i^!\colon A_m(\mathbb{P}^3) \to A_{m-2}(Z)$ on the generator of each group, for $m = 0, 1, 2, 3$.

2. Let X be a variety of dimension n and let $D \subseteq X$ be an effective Cartier divisor with $D \neq X$, with inclusion i. Show that $i^![X] = [D]$, the fundamental cycle of the scheme D. Then take $X = \mathbb{P}^2$ and D a smooth plane cubic, and compute the image of $i^![\mathbb{P}^2]$ in $A_1(\mathbb{P}^2)$.

3. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $V \subseteq X$ be a k-dimensional subvariety with $V \cap Z = \emptyset$. Show that $i^![V] = 0$. Then let $Z = \{0\}$ be the origin in $X = \mathbb{A}^n$ with $n \geq 1$, verify that this is a regular embedding of codimension n, and determine $i^!$ on every group $A_k(\mathbb{A}^n)$.

4. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $u\colon U \hookrightarrow X$ be an open immersion. Show that the preimage of Z in U is $Z \cap U$ and that $i^!\big(u^*\alpha\big) = \big(i^!\alpha\big)\big|_{Z\cap U}$ for every $\alpha \in A_k(X)$. Deduce that if $Z \subseteq U$ and α restricts to zero on U, then $i^!\alpha = 0$.

5. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ and let $V \subseteq Z$ be a k-dimensional subvariety of Z. Show that $\sigma[V] = \zeta_*[V]$, where $\zeta\colon Z \to N$ is the zero section, and hence that $i^![V] = (\pi^*)^{-1}\zeta_*[V]$. Now take $X = \mathbb{A}^2$, $Z = \{y = 0\}$ and $V = Z$, and evaluate both $i^![Z]$ and $D \cdot [Z]$ of definition 9.2.1, checking that they agree.

11.4 Commutativity, Composition, and Sections

The Gysin map is built; the algebra it obeys is not yet available. Three identities are used downstream, and all three are properties of the construction of section 11.3 rather than of anything later, so they belong here.

They are not independent, and the dependency runs in a direction worth naming before starting. Commutativity of two Gysin maps looks the hardest and is in fact a consequence of the composition rule: two regular embeddings determine a single regular embedding of the product, that embedding factors through the two intermediate stages in two ways, and the composition rule applied to both factorizations equates the two orders. The work is therefore concentrated in composition, and the section is arranged accordingly. Composition in turn needs a commutativity statement of its own, but only the case of a *divisor*, and that case the construction hands over directly: specialization is intersection with a Cartier divisor, so two of them commute because divisors do.

Lemma 11.4.1: The Product of Two Regular Embeddings

Let $i\colon Z \hookrightarrow P$ and $i'\colon Z' \hookrightarrow P'$ be regular embeddings of codimensions d and d'. Then

$$i \times i' \;:\; Z \times Z' \;\hookrightarrow\; P \times P'$$

is a regular embedding of codimension $d + d'$, and it factors in two ways as a composite of regular embeddings,

$$Z \times Z' \xhookrightarrow{\mathrm{id}_Z \times i'} Z \times P' \xhookrightarrow{i \times \mathrm{id}_{P'}} P \times P', \qquad Z \times Z' \xhookrightarrow{i \times \mathrm{id}_{Z'}} P \times Z' \xhookrightarrow{\mathrm{id}_P \times i'} P \times P'$$

the four factors being regular of codimensions d', d, d and d' respectively.

***Proof*:**
Step 1: each factor is regular. The morphism $i \times \mathrm{id}_{P'}$ is the base change of i along the projection $\mathrm{pr}_1\colon P \times P' \to P$, which is flat, and the scheme-theoretic preimage of Z under that projection is $Z \times P'$. A regular embedding pulls back along a flat morphism to a regular embedding of the same codimension (proposition 5.1.49), so $i \times \mathrm{id}_{P'}$ is regular of codimension d. The same argument along pr_2 makes $\mathrm{id}_P \times i'$ and $\mathrm{id}_Z \times i'$ regular of codimension d', the latter as the base change of i' along $Z \times P' \to P'$; and $i \times \mathrm{id}_{Z'}$ is regular of codimension d, being the base change of i along the flat projection $P \times Z' \to P$. That accounts for all four factors named in the statement.

Step 2: the composite is regular. Work on affine opens $\operatorname{Spec} A \subseteq P$ and $\operatorname{Spec} A' \subseteq P'$ over which the ideals of Z and Z' are generated by regular sequences $f_1, \dots, f_d$ and $g_1, \dots, g_{d'}$. On $\operatorname{Spec}(A \otimes_{\overline{k}} A')$ the ideal of $Z \times Z'$ is generated by the images of the f's and the g's together, and that concatenation is a regular sequence. Indeed $A \to A \otimes_{\overline{k}} A'$ is flat, so the f's remain a regular sequence after it by the argument of Step 1; and modulo them the ring is $(A/(f)) \otimes_{\overline{k}} A'$, which is a flat base change of A', so the g's are regular there by the same argument with the two factors exchanged. Hence $i \times i'$ is a regular embedding of codimension $d + d'$.

The two factorizations are immediate from the definitions of the four morphisms, and Step 1 identified each factor as a regular embedding of the stated codimension, completing the proof.

The first identity says that the Gysin map of a section undoes the flat pullback along the morphism it is a section of. It is the only one of the three that can be proved directly from the proper-intersection computation already available.

Theorem 11.4.2: Gysin Maps of Sections

Let $\pi\colon P \to S$ be smooth of relative dimension $e \geq 1$ and let $s\colon S \to P$ be a section, so that $\pi \circ s = \mathrm{id}_S$. Then s is a regular embedding of codimension e and

$$s^! \circ \pi^* \;=\; \mathrm{id} \quad \text{on } A_k(S) \text{ for every } k$$

***Proof*:**

That s is a regular embedding of codimension e is proposition 5.1.48. Both $s^!$ and π^* are homomorphisms, so it suffices to prove the identity on the class of a k-dimensional subvariety $V \subseteq S$.

Step 1: the preimage and its components. The morphism π is flat of relative dimension e by proposition 2.7.10, so π^* is defined and $\pi^*[V] = [P_V]$ for $P_V = \pi^{-1}(V)$ the scheme-theoretic preimage (definition 8.3.3). The projection $\pi_V\colon P_V \to V$ is smooth of relative dimension e, being the base change of π along $V \hookrightarrow S$ (proposition 2.7.9(1)), hence flat by the same citation as above; since its base V is a variety, proposition 8.3.2 applies to π_V and P_V is pure of dimension $k+e$. Because $\pi \circ s = \mathrm{id}_S$, the section restricts to a section $s_V\colon V \to P_V$ of π_V, and indeed $s^{-1}(P_V) = (\pi \circ s)^{-1}(V) = V$ scheme-theoretically, so $s(S) \cap P_V = s_V(V)$.

Flatness of π_V makes every irreducible component of P_V dominate V, by the going-down argument already used in the proof of proposition 8.3.2. So the generic point of each component lies in the generic fibre, which is smooth over the field $k(V)$ and therefore regular (proposition 2.7.12); its local rings are consequently domains. Hence P_V is generically reduced along every component, so $[P_V] = \sum_j [V_j]$ with all multiplicities one (definition 8.1.2), and no two components share a generic point of the generic fibre. The subvariety $s_V(V)$ dominates V, so its generic point lies in the generic fibre and therefore in exactly one component, which is moreover the unique component containing $s_V(V)$; call it V_1, and let $V_2, \ldots, V_r$ be the others.

Step 2: the other components miss the section altogether. Let $j \geq 2$ and put $W_j = V_j \cap s(S) = V_j \cap s_V(V)$, using Step 1's identification of $s(S) \cap P_V$. Suppose $W_j \neq \emptyset$. Applying theorem 11.2.7 to the subvariety $V_j \subseteq P_V$, of dimension $k+e$, its normal cone $C_{W_j}V_j$ is pure of dimension $k+e$, and lemma 10.4.12 embeds it in $C_{s_V(V)}P_V$ over the inclusion $W_j \hookrightarrow s_V(V)$. Now $s_V(V) \hookrightarrow P_V$ is a section of the smooth morphism π_V, hence a regular embedding of codimension e (proposition 5.1.48), so that cone is the normal bundle N_V, of rank e over $s_V(V) \cong V$ (proposition 10.4.11). The part of N_V lying over W_j has dimension $\dim W_j + e$, so purity forces $\dim W_j = k$. But W_j is then a k-dimensional closed subscheme of the k-dimensional variety $s_V(V)$, hence all of it, so $V_j \supseteq s_V(V)$, contradicting the uniqueness of V_1 in Step 1. Therefore $W_j = \emptyset$ for every $j \geq 2$, and in particular $s^![V_j] = 0$, the Chow groups of the empty scheme vanishing.

Step 3: the remaining component. Step 2 has a consequence worth isolating, because the naive statement it replaces is false: an irreducible component of a smooth V-scheme need not itself be smooth over V, so one may not argue that $V_1 \to V$ is smooth because V_1 is the component through the section. What is true, and all that is needed, is that

$$U \;=\; P_V \setminus \bigcup_{j\geq 2} V_j$$

is an open subscheme of P_V containing $s_V(V)$, by Step 2, and $U \subseteq V_1$. Being a regular embedding of a given codimension is a local condition along the subscheme (definition 5.1.43), and $s_V(V) \hookrightarrow P_V$ is regular of codimension e as recorded in Step 2. Restricting to the open U and using that U is also open in V_1, the embedding $s_V(V) \hookrightarrow V_1$ is regular of codimension e.

The scheme-theoretic intersection $V_1 \cap s(S)$ is $s_V(V)$: this is a scheme-theoretic and not a set-theoretic claim, so it needs an argument rather than a comparison of points. By Step 1, $s^{-1}(P_V) = V$ scheme-theoretically; $s^{-1}(V_1)$ is a closed subscheme of V because the ideal sheaf of V_1 contains that of P_V; it has full support since $V_1 \supseteq s_V(V)$; and V is reduced, so $s^{-1}(V_1) = V$.

Proposition 11.3.3 therefore applies to the regular embedding s of codimension e and the $(k+e)$-dimensional subvariety V_1, and gives

$$s^![V_1] \;=\; \iota_*\big[s_V(V)\big] \;\in\; A_k\big(s(S)\big)$$

Identifying $s(S)$ with S through the isomorphism s, the right side is $[V]$.

Summing Steps 2 and 3 gives $s^!\big(\pi^*[V]\big) = s^![P_V] = [V]$, as we sought to show.

The remaining two identities concern a pair of Gysin maps rather than a single one, and both rest on one observation about definition 11.3.6 that is worth isolating first. The refined map is built from the specialization of the pair (Z', X'), the cone $C_{Z'}X'$, and the bundle N' into which that cone embeds. Nothing else about the ambient X enters. So an embedding may be enlarged, or crossed with a further factor, without changing the resulting homomorphism, provided the preimage and the normal bundle come along unchanged.

Lemma 11.4.3: The Refined Gysin Map Does Not See the Ambient Scheme

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with ideal sheaf $\mathcal{I}$, and let $\varphi\colon \widetilde{X} \to X$ be a morphism whose preimage $\widetilde{Z} = \varphi^{-1}(Z)$, with ideal sheaf $\widetilde{\mathcal{I}} = \mathcal{I}\mathcal{O}_{\widetilde{X}}$, is again a regular embedding $\widetilde{i}\colon \widetilde{Z} \hookrightarrow \widetilde{X}$ *of the same codimension* d. Then for every morphism $g\colon X' \to \widetilde{X}$, writing $Z' = g^{-1}(\widetilde{Z}) = (\varphi \circ g)^{-1}(Z)$,

$$\widetilde{i}^!_g \;=\; i^!_{\varphi\circ g} \;:\; A_k(X') \longrightarrow A_{k-d}(Z')$$

***Proof*:**
First, the hypothesis on codimensions already forces the conormal sheaf to be the pullback. The canonical map

$$\varphi_Z^*\big(\mathcal{I}/\mathcal{I}^2\big) \;\longrightarrow\; \widetilde{\mathcal{I}}/\widetilde{\mathcal{I}}^2, \qquad \varphi_Z\colon \widetilde{Z} \to Z,$$

is surjective, since $\widetilde{\mathcal{I}} = \mathcal{I}\mathcal{O}_{\widetilde{X}}$ is generated by the image of $\mathcal{I}$. Both sides are locally free of rank d (proposition 5.1.44), the source as the pullback of a rank-d bundle and the target because $\widetilde{i}$ is regular of codimension d. A surjection of locally free sheaves of equal rank is an isomorphism, its kernel being locally free of rank zero. So the canonical map is an isomorphism, and we may use it below.

The two closed subschemes $g^{-1}(\widetilde{Z})$ and $(\varphi \circ g)^{-1}(Z)$ of X' coincide because $\widetilde{\mathcal{I}} = \mathcal{I}\mathcal{O}_{\widetilde{X}}$ and pullback of ideals is transitive, so the target group is the same and both homomorphisms are the composite of definition 11.3.6 for one and the same pair (Z', X'). In particular they use the same specialization σ' and the same cone $C_{Z'}X'$, neither of which mentions the ambient scheme at all.

What must be compared is the bundle and the embedding of the cone in it. Write $h\colon Z' \to Z$ and $\widetilde{h}\colon Z' \to \widetilde{Z}$ for the structure morphisms, so $h = \varphi_Z \circ \widetilde{h}$. The hypothesis gives a canonical isomorphism $\widetilde{h}^*(\widetilde{\mathcal{I}}/\widetilde{\mathcal{I}}^2) \cong \widetilde{h}^*\varphi_Z^*(\mathcal{I}/\mathcal{I}^2) = h^*(\mathcal{I}/\mathcal{I}^2)$, that is $\widetilde{N}' \cong N'$ as bundles on Z', and hence π' is the same projection in both constructions.

It remains to see that the closed immersion j of lemma 10.4.13 is carried to the other by this isomorphism. That immersion is induced by a surjection of graded $\mathcal{O}_{Z'}$-algebras out of $\mathrm{Sym}^\bullet$ of the pullback conormal sheaf, and such a morphism is determined by its degree-one part. For i the degree-one part sends the class of a local section α of $\mathcal{I}$ to the class of $\alpha\mathcal{O}_{X'}$ in $\mathcal{I}'/\mathcal{I}'^2$; for $\widetilde{i}$ it sends the class of a local section of $\widetilde{\mathcal{I}}$ to the class of its extension to X'. The hypothesis isomorphism identifies the class of α with the class of $\alpha\mathcal{O}_{\widetilde{X}}$, and $(\alpha\mathcal{O}_{\widetilde{X}})\mathcal{O}_{X'} = \alpha\mathcal{O}_{X'}$ by transitivity of extension of ideals along $X' \to \widetilde{X} \to X$. So the two degree-one parts correspond, the two immersions j agree, and the three factors $(\pi'^*)^{-1}$, j_*, σ' agree one by one.

The hypothesis is satisfied whenever φ is flat, by proposition 5.1.49, and that is the only case used below: φ will be a projection off a product.

Three lemmas stand between here and the composition rule. The first is the only place a genuine commutativity statement is needed, and it is needed only for a divisor, where the construction delivers it directly: specialization *is* intersection with a Cartier divisor (definition 11.2.3), so two such intersections commute by theorem 9.3.2.

Lemma 11.4.4: The Gysin Map Commutes with Intersection by a Divisor

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$, let $g\colon X' \to X$ be a morphism with $Z' = g^{-1}(Z)$, and let D be a Cartier divisor on X'. Then for every $\alpha \in A_k(X')$,

$$i^!\big(D \cdot \alpha\big) \;=\; D \cdot \big(i^!\alpha\big) \quad \text{in } A_{k-d-1}(Z')$$

both sides being read through corollary 9.4.5, on X' for the left and on the closed subscheme Z' for the right.

***Proof*:**
Write $M = M^\circ_{Z'}X'$ for the deformation space of definition 11.1.4, with blow-down $\rho\colon M \to X'$, fibre $D_\infty = C_{Z'}X'$ over ∞, and $\kappa\colon C_{Z'}X' \hookrightarrow N'$, $\pi'\colon N' \to Z'$ as in definition 11.3.6, so that $i^! = (\pi'^*)^{-1} \circ \kappa_* \circ \sigma'$. Throughout, a Cartier divisor is intersected against classes on closed subschemes of the ambient through corollary 9.4.5; the relevant restrictions are in general *not* Cartier on those subschemes, and it is to avoid needing them to be that the corollary was stated.

Step 0: ρ^*D is a Cartier divisor on M. The blow-down is not flat, its fibre at infinity being a cone, so this needs an argument. Over $\mathbb{A}^1$ the map is the projection $X' \times \mathbb{A}^1 \to X'$, which is flat, and flat pullback of a Cartier divisor is Cartier. Over the chart at infinity, lemma 11.1.2 and (11.1) present $\mathcal{O}_M$ as the extended Rees algebra, a graded subsheaf of $\mathcal{O}_{X'}[u, u^{-1}]$. A local equation f of D is a non-zero divisor on $\mathcal{O}_{X'}$, hence on the free module $\mathcal{O}_{X'}[u, u^{-1}]$, hence on any subsheaf of it; so $\rho^{\#}f$ is a non-zero divisor and ρ^*D is Cartier.

Step 1: specialization. Let $\beta \in A_{k+1}(M)$ be a lift of α in the sense of definition 11.2.3, so that β restricts to $q^*\alpha$ on $X' \times \mathbb{A}^1$. Then $\rho^*D \cdot \beta$ is a lift of $D \cdot \alpha$: restriction along that open immersion is flat pullback of relative dimension zero, proposition 9.3.3(2) applies with no hypothesis on supports, and ρ^*D restricts to q^*D there by theorem 11.1.5(1). Hence, by theorem 11.2.4 and by theorem 9.3.2 applied on M to the Cartier divisors D_∞ and ρ^*D, the first by theorem 11.1.5(2) and the second by Step 0,

$$\sigma'\big(D \cdot \alpha\big) \;=\; D_\infty \cdot \big(\rho^*D \cdot \beta\big) \;=\; \rho^*D \cdot \big(D_\infty \cdot \beta\big) \;=\; \rho^*D \cdot \sigma'(\alpha)$$

an identity of classes on the cone, the refined form of theorem 9.3.2 placing both sides in the Chow group of $|D_\infty| \cap |\rho^*D| \cap \operatorname{supp}\beta$.

Step 2: pushforward along the cone. On N' put $D_{N'} = \pi'^*D$, the flat pullback along π' of D restricted to Z'; it is a Cartier divisor there, π' being flat. Its restriction to the cone and the restriction of ρ^*D agree, both being computed from the line bundle $\mathcal{O}_{X'}(D)$ pulled back to the cone with its canonical section. A closed immersion carries a cycle to itself, so corollary 9.4.5, applied to $C_{Z'}X' \subseteq N'$, gives

$$\kappa_*\big(\rho^*D \cdot \gamma\big) \;=\; D_{N'} \cdot \kappa_*\gamma$$

for every class γ on the cone, with no hypothesis on its support. This is the step at which the projection formula would otherwise be invoked, and proposition 9.3.3(1) could not be: its hypothesis is that no component of the support maps into $|D|$, which $\sigma'(\alpha)$ need not satisfy.

Step 3: the bundle inverse. Since π' is flat, proposition 9.3.3(2) gives $\pi'^*\big(D \cdot \delta\big) = D_{N'} \cdot \pi'^*\delta$ for every $\delta \in A_*(Z')$, again with no support hypothesis. Applying this to $\delta = i^!\alpha$ and using Steps 1 and 2,

$$\pi'^*\Big(D \cdot i^!\alpha\Big) = D_{N'} \cdot \pi'^*\big(i^!\alpha\big) = D_{N'} \cdot \kappa_*\sigma'(\alpha) = \kappa_*\sigma'\big(D \cdot \alpha\big) = \pi'^*\Big(i^!\big(D \cdot \alpha\big)\Big)$$

and π'^* is injective by theorem 8.4.6, which gives the identity.

The second lemma settles the composition rule when the outer embedding is the zero section of a vector bundle. This is the case in which the normal bundle of the composite genuinely splits, and it is where the conormal sequence of a chain does its work.

Lemma 11.4.5: Composition Through a Zero Section

Let E be a vector bundle of rank $e \geq 1$ on an algebraic scheme Y with zero section $\zeta\colon Y \hookrightarrow E$, and let $i\colon Z \hookrightarrow Y$ be a regular embedding of codimension $d \geq 1$. Let $g\colon S \to Y$ be a morphism, $E' = g^*E$ with projection $\pi'\colon E' \to S$, and let $\mathrm{pr}\colon E' \to E$ be the induced morphism. Then for every $\alpha \in A_k(E')$,

$$(\zeta \circ i)^!_{\mathrm{pr}}(\alpha) \;=\; i^!_g\Big(\zeta^!_{\mathrm{pr}}(\alpha)\Big) \quad \text{in } A_{k-d-e}\big(g^{-1}(Z)\big)$$

***Proof*:**
Write $S_Z = g^{-1}(Z)$. The preimage of Y under pr is the zero section of E', which we identify with S; the preimage of Z is S_Z. Both refined maps are therefore defined and land as stated.

That $\zeta^!_{\mathrm{pr}} = (\pi'^*)^{-1}$ is not merely the observation that the normal cone of a zero section is the bundle itself, which is the case $\mathcal{I} = 0$ of (11.6) below: that identifies the cone and the bundle and makes the immersion of lemma 10.4.13 an identity, but leaves $\zeta^!$ equal to $(\pi'^*)^{-1} \circ \sigma'$, and specialization is emphatically not the identity in general (example 11.2.5). What settles it is that the preimage of Y under pr is the zero section of E', a regular embedding of codimension e, so lemma 11.4.3 identifies $\zeta^!_{\mathrm{pr}}$ with the refined Gysin map of that zero section along id, which is its plain Gysin map (proposition 11.3.7); and $\pi'\colon E' \to S$ is smooth of relative dimension e with the zero section as a section, so theorem 11.4.2 gives $\zeta^! \circ \pi'^* = \mathrm{id}$, that is $\zeta^! = (\pi'^*)^{-1}$.

Step 1: the cone of the composite splits off the bundle. Let $\mathcal{I}$ be the ideal sheaf of S_Z in S and let $\mathcal{S}^\bullet = \mathrm{Sym}^\bullet(\mathcal{E}^\vee)$ present $E' = \mathrm{Spec}_S \mathcal{S}^\bullet$, so that $\mathcal{S}^0 = \mathcal{O}_S$ and $\mathcal{S}^1 = \mathcal{E}^\vee$. The ideal $\mathcal{J}$ of S_Z in E' is generated by $\mathcal{I}$ in degree zero together with all of $\mathcal{S}^1$: a function vanishes on the zero section over S_Z exactly when its degree-zero part lies in $\mathcal{I}$. Consequently

$$\mathcal{J}^n \;=\; \bigoplus_{q\le n} \mathcal{I}^{n-q}\mathcal{S}^q \;\oplus\; \bigoplus_{q>n} \mathcal{S}^q, \qquad \mathcal{J}^n/\mathcal{J}^{n+1} \;=\; \bigoplus_{p+q=n} \left(\mathcal{I}^p/\mathcal{I}^{p+1}\right) \otimes \mathcal{S}^q$$

so that the total graded algebra factors, $\bigoplus_n \mathcal{J}^n/\mathcal{J}^{n+1} = \left(\bigoplus_p \mathcal{I}^p/\mathcal{I}^{p+1}\right) \otimes_{\mathcal{O}_{S_Z}} \mathcal{S}^\bullet$. Taking Spec and using definition 10.4.6 this reads

$$C_{S_Z}E' \;\cong\; C_{S_Z}S \times_{S_Z} E'|_{S_Z} \tag{11.6}$$

Applied with $S = Y$ and $g = \mathrm{id}$ it identifies the normal bundle of the composite as $N_Z E = N_Z Y \oplus i^*E$, which is the splitting of proposition 5.1.47 for the chain $Z \subseteq Y \subseteq E$; the splitting is the one induced by the bundle projection.

Step 2: reduction. By theorem 8.4.6 every class on E' is $\pi'^*\beta$ for a unique $\beta \in A_{k-e}(S)$, and by linearity we may take $\beta = [V]$ for $V \subseteq S$ a subvariety. Both sides of the assertion commute with proper pushforward along $V \hookrightarrow S$ and its base change $V \times_S E' \hookrightarrow E'$, by theorem 11.3.10(1), so we may replace S by V. Then S is a variety, $\alpha = [E']$, and $\zeta^!\alpha = [S]$.

Step 3: comparison of fundamental cycles. Write $h\colon S_Z \to Z$ for the induced morphism, $r\colon h^*N_ZY \to S_Z$ and $q\colon h^*N_ZE \to h^*N_ZY$ for the projections, the latter using the splitting of Step 1, so that $r \circ q$ is the projection of h^*N_ZE. Both S and E' are varieties, so theorem 11.2.7 gives $\sigma[S] = [C_{S_Z}S]$ and $\sigma[E'] = [C_{S_Z}E']$. By (11.6) the second cone is the preimage of the first under q, so $q^*[C_{S_Z}S] = [C_{S_Z}E']$ by definition 8.3.3. Now definition 11.3.6 reads the two Gysin classes as

$$r^*\big(i^![S]\big) = [C_{S_Z}S], \qquad (r\circ q)^*\big((\zeta i)^![E']\big) = [C_{S_Z}E']$$

Combining the three displays, $(r\circ q)^*\big((\zeta i)^![E']\big) = q^*r^*\big(i^![S]\big) = (r\circ q)^*\big(i^![S]\big)$ by proposition 8.3.5(3), and $(r\circ q)^*$ is injective by theorem 8.4.6. Hence $(\zeta i)^![E'] = i^![S] = i^!\zeta^![E']$, which by Step 2 is the assertion in general.

The third lemma is the only input that is not about Gysin maps at all. It says that cutting a class on $X \times \mathbb{P}^1$ by a fibre gives an answer that does not depend on which fibre.

Lemma 11.4.6: Cutting by a Fibre of $X \times \mathbb{P}^1$ Is Independent of the Fibre

Let X be an algebraic scheme and for $t \in \mathbb{P}^1$ a closed point let D_t denote the Cartier divisor $X \times \{t\}$ on $X \times \mathbb{P}^1$. Then for every $\beta \in A_k(X \times \mathbb{P}^1)$ the class

$$D_t \cdot \beta \;\in\; A_{k-1}\big(X \times \{t\}\big) \;=\; A_{k-1}(X)$$

is the same for all t, the identification being the projection $X \times \{t\} \to X$.

***Proof*:**
Write $X \times \mathbb{P}^1 = \mathbb{P}(\mathcal{O}_X^{\oplus 2})$ with projection p and tautological class ξ; for the trivial bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ is pulled back from $\mathbb{P}^1$, and D_t is the divisor of the section of it cutting out t, so $\mathcal{O}(D_t) = \mathcal{O}_{\mathbb{P}(E)}(1)$ for every t. By theorem 10.1.9 with $r = 1$ there are unique classes $\alpha_0 \in A_{k-1}(X)$ and $\alpha_1 \in A_k(X)$ with

$$\beta \;=\; p^*\alpha_0 \;+\; \xi \cap p^*\alpha_1$$

It suffices to evaluate $D_t\cdot$ on each summand. For the first, take $\alpha_0 = [V]$ with $V \subseteq X$ a subvariety; then $p^*[V] = [V \times \mathbb{P}^1]$, which is not contained in $|D_t|$, so definition 9.2.1(1) computes $D_t \cdot p^*[V] = [V \times \{t\}]$, the divisor of the restricted section, and under $X \times \{t\} \cong X$ this is $[V]$. For the second, $D_t \cdot (\xi \cap p^*\alpha_1) = \xi^2 \cap p^*\alpha_1$ once pushed into $A_{k-2}(X \times \mathbb{P}^1)$, and $\xi^2 = 0$ by theorem 10.2.9, the bundle being trivial. To descend that vanishing to the refined class on $X \times \{t\}$, note that the first computation identifies the pushforward $\iota_{t*} \colon A_m(X) = A_m(X \times \{t\}) \to A_m(X \times \mathbb{P}^1)$ with $\gamma \mapsto \xi \cap p^*\gamma$, and that map is injective by the uniqueness clause of theorem 10.1.9, being the restriction of the isomorphism Ψ to the summand $\alpha_0 = 0$. Hence the refined class vanishes as well, and $D_t \cdot \beta = \alpha_0$ for every t, as we sought to show.

Theorem 11.4.7: Composition of Gysin Maps

Let $i \colon Z \hookrightarrow Y$ and $j \colon Y \hookrightarrow P$ be regular embeddings of codimensions $d \geq 1$ and $e \geq 1$. Then $j \circ i$ is a regular embedding of codimension $d + e$, and for every morphism $a \colon W \to P$, writing $W_Y = a^{-1}(Y)$, $W_Z = a^{-1}(Z)$ and $a' \colon W_Y \to Y$ for the induced morphism,

$$(j \circ i)^!_a \;=\; i^!_{a'} \circ j^!_a \;:\; A_k(W) \longrightarrow A_{k-d-e}(W_Z)$$

***Proof*:**
That $j \circ i$ is a regular embedding of codimension $d + e$ is the first assertion of proposition 5.1.47. Both sides commute with proper pushforward (theorem 11.3.10(1)), and $A_k(W)$ is generated by the classes of k-dimensional subvarieties, so it suffices to prove the identity for W a variety and the class $[W]$.

The two deformations. Let $M = M^\circ_Y P$ and $M' = M^\circ_{W_Y} W$ be the deformation spaces of definition 11.1.4, with projections to $\mathbb{P}^1$ written π and π'. The morphism a induces $\widetilde{a} \colon M' \to M$ over $\mathbb{P}^1$ by lemma 11.2.6, whose hypothesis $W_Y = a^{-1}(Y)$ holds by definition. Let

$$\chi \;:\; Z \times \mathbb{P}^1 \xhookrightarrow{i \times \mathrm{id}} Y \times \mathbb{P}^1 \hookrightarrow M$$

be the composite of $i \times \mathrm{id}$ with the closed immersion of theorem 11.1.5(4). Its first factor is a regular embedding of codimension d, being the base change of i along the flat projection

$Y \times \mathbb{P}^1 \to Y$ (proposition 5.1.49); its second is a regular embedding of codimension e. This last point needs the charts rather than the fibres: $\pi^{-1}(\mathbb{A}^1)$ is open, and there the immersion is $j \times \mathrm{id}$, but $\pi^{-1}(\infty)$ is closed, so regularity of the immersion into it says nothing about a neighbourhood. On the complementary chart $\pi^{-1}(\mathbb{P}^1 \setminus \{0\})$, lemma 11.1.2 presents the coordinate ring as an extended Rees algebra $\mathcal{R}$, in which the ideal of $Y \times \mathbb{P}^1$ is generated by $g_1/u, \ldots, g_e/u$ for $g_1, \ldots, g_e$ a local regular sequence cutting out Y in P. Here u is a non-zero divisor on $\mathcal{R}$ and $\mathcal{R}/u\mathcal{R} = \mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$, in which the images of the g_m/u are the coordinates and so form a regular sequence; hence $u, g_1/u, \ldots, g_e/u$ is a regular sequence, and permuting a regular sequence in a Noetherian local ring leaves it regular, so $g_1/u, \ldots, g_e/u$ is one. So χ is a regular embedding of codimension $d + e$, again by the chain result cited above, and $\widetilde{a}^{-1}(Z \times \mathbb{P}^1) = W_Z \times \mathbb{P}^1$.

Cutting by a fibre. For a closed point $t \in \mathbb{P}^1$ let D_t be the fibre $\pi'^{-1}(t)$ on M'. It is an effective Cartier divisor: for $t = \infty$ by theorem 11.1.5(2), and for $t \neq \infty$ because M' is a variety, as the proof of theorem 11.2.7 records, so that any nonzero local equation is a non-zero divisor. Theorem 11.1.5(1),(2) identify the fibres. Over $\mathbb{A}^1$ the fibre is $W \times \{t\}$, which is reduced, so definition 9.2.1(1) computes $D_t \cdot [M']$ as its fundamental cycle with multiplicity one; over ∞ the same intersection is by definition the specialization, which theorem 11.2.7 evaluates. So

$$D_t \cdot [M'] \;=\; [M'_t] \;=\; \begin{cases} [W] & t \neq \infty \\ [C_{W_Y} W] & t = \infty \end{cases} \tag{11.7}$$

Now apply $\chi^!$ along $\widetilde{a}$. The scheme M'_t maps to M_t, the inclusion $M_t \hookrightarrow M$ pulls $Z \times \mathbb{P}^1$ back to $Z \times \{t\}$, and the induced embedding is $j \circ i$ for $t \neq \infty$ and $\widetilde{\jmath} \circ i$ for $t = \infty$, where $\widetilde{\jmath}$ is the zero section of $N_Y P$; both are regular of codimension $d + e$. So lemma 11.4.3, applied with φ the inclusion of the fibre, gives

$$\chi^!\big(D_t \cdot [M']\big) \;=\; \begin{cases} (j \circ i)^!\,[W] & t \neq \infty \\ (\widetilde{\jmath} \circ i)^!\,[C_{W_Y} W] & t = \infty \end{cases} \tag{11.8}$$

The value at infinity. The cone $C_{W_Y} W$ is a closed subscheme of the pullback bundle $(a')^* N_Y P$ over W_Y (lemma 10.4.13). Lemma 11.4.5, stated for an arbitrary class on that bundle, therefore applies with $E = N_Y P$ and $S = W_Y$ to the pushforward of $[C_{W_Y} W]$; and theorem 11.3.10(1), for the closed immersion $C_{W_Y} W \hookrightarrow (a')^* N_Y P$, says that computing before or after that pushforward gives the same answer, the induced morphism on the preimages of Z being the identity of W_Z because the vertex of $C_{W_Y} W$ is W_Y, the cone algebra being generated in degree one. This gives

$$(\widetilde{\jmath} \circ i)^!\,[C_{W_Y} W] \;=\; i^!\Big(\widetilde{\jmath}^!\,[C_{W_Y} W]\Big) \;=\; i^!_{a'}\Big(j^!_a[W]\Big)$$

the last equality because $\widetilde{\jmath}^!$ is the inverse of the bundle pullback and $\sigma[W] = [C_{W_Y} W]$, which is exactly how definition 11.3.6 computes $j^!_a[W]$.

Independence of t. By lemma 11.4.4, applied to the regular embedding χ and the divisor D_t, whose restriction to $W_Z \times \mathbb{P}^1$ is the Cartier divisor $W_Z \times \{t\}$,

$$\chi^!\big(D_t \cdot [M']\big) \;=\; D_t \cdot \big(\chi^![M']\big)$$

and $\chi^![M']$ is a class on $W_Z \times \mathbb{P}^1$ not depending on t, so lemma 11.4.6 makes the right-hand side independent of t.

One point of bookkeeping remains. Lemma 11.4.4 records its identity in $A_*(W_Z \times \mathbb{P}^1)$, whereas (11.8) produces classes on $W_Z \times \{t\}$; the two are related by the pushforward ι_{t*} along the closed immersion $W_Z \times \{t\} \hookrightarrow W_Z \times \mathbb{P}^1$, through theorem 11.3.10(1) applied to $M'_t \hookrightarrow M'$. That costs nothing, because the proof of lemma 11.4.6 identifies ι_{t*} with $\gamma \mapsto \xi \cap p^*\gamma$, which is one and the same map for every t and is injective. Comparing the two lines of (11.8) at $t \neq \infty$ and at $t = \infty$, their images under ι_{t*} agree, hence so do the classes themselves, and $(j \circ i)^![W] = i^!_{a'}\big(j^!_a[W]\big)$, as we sought to show.

Commutativity now follows, and the route is the one announced at the start of the section: the two embeddings are made into a single embedding of a product, which factors in two ways.

Corollary 11.4.8: Commutativity of Gysin Maps

Let $i\colon Z \hookrightarrow P$ and $i'\colon Z' \hookrightarrow P'$ be regular embeddings of codimensions d and d', let W be an algebraic scheme, and let $a\colon W \to P$ and $b\colon W \to P'$ be morphisms. Put $W_1 = a^{-1}(Z)$, $W_2 = b^{-1}(Z')$ and $W_{12} = W_1 \cap W_2$. Then for every $\alpha \in A_k(W)$,

$$i^!_{a|W_2}\big(i'^!_b\alpha\big) \;=\; i'^!_{b|W_1}\big(i^!_a\alpha\big) \quad \text{in } A_{k-d-d'}(W_{12})$$

***Proof*:**
Let $c = (a,b)\colon W \to P \times P'$. Lemma 11.4.1 makes $i \times i'$ a regular embedding of codimension $d + d'$ and exhibits its two factorizations through $Z \times P'$ and through $P \times Z'$, all four factors being regular embeddings. The preimages under c are

$$c^{-1}(Z \times P') = W_1, \qquad c^{-1}(P \times Z') = W_2, \qquad c^{-1}(Z \times Z') = W_{12},$$

because the ideal of $Z \times P'$ is $\mathcal{I}_Z\mathcal{O}_{P\times P'}$ and $\mathrm{pr}_1 \circ c = a$, and symmetrically. The third follows from the first two: the ideal of $Z \times Z'$ is the sum $\mathcal{I}_Z\mathcal{O}_{P\times P'} + \mathcal{I}_{Z'}\mathcal{O}_{P\times P'}$, so $Z \times Z'$ is the scheme-theoretic intersection of $Z \times P'$ and $P \times Z'$, and extension of ideals along c carries a sum of ideals to the sum of the extensions.

Each of the four factors is the preimage of i or of i' along a projection off a product. Such a projection is the base change of a structure morphism to $\operatorname{Spec}\overline{k}$, hence flat, so lemma 11.4.3 applies to each factor and identifies its refined Gysin map with that of i or of i' along the corresponding composite. Explicitly, with $\varphi = \mathrm{pr}_1$ one gets $(i \times \mathrm{id}_{P'})^!_c = i^!_a$ and $(i \times \mathrm{id}_{Z'})^!_{c''} = i^!_{a|W_2}$ for $c''\colon W_2 \to P \times Z'$, and with $\varphi = \mathrm{pr}_2$ one gets $(\mathrm{id}_P \times i')^!_c = i'^!_b$ and $(\mathrm{id}_Z \times i')^!_{c'} = i'^!_{b|W_1}$ for $c'\colon W_1 \to Z \times P'$. Here c' and c'' are the morphisms induced by c on the preimages W_1 and W_2, so that $\mathrm{pr}_2 \circ c' = b|_{W_1}$ and $\mathrm{pr}_1 \circ c'' = a|_{W_2}$; they are exactly the morphisms that theorem 11.4.7 calls a', for the two factorizations respectively.

Apply theorem 11.4.7 to each factorization in turn, with c as the morphism to $P \times P'$. The first gives $(i \times i')^!_c = i'^!_{b|W_1} \circ i^!_a$ and the second gives $(i \times i')^!_c = i^!_{a|W_2} \circ i'^!_b$. The left sides agree, so the right sides do, which is the assertion.

Exercise 11.4.1

1. Theorem 11.4.2 assumes $e \geq 1$, and so does definition 11.3.6. Show that for $e = 0$ a section s of a smooth $\pi\colon P \to S$ of relative dimension zero is an open and closed immersion, so that the only sensible reading of $s^!$ is flat pullback along s, and that with that reading $s^! \circ \pi^* = \mathrm{id}$ follows from proposition 8.3.5(3) with no deformation at all.

2. In lemma 11.4.1, show that the normal bundle of $i \times i'$ is $\mathrm{pr}_1^* N \oplus \mathrm{pr}_2^* N'$, and that the two factorizations exhibit the two summands as the normal bundles of the two intermediate stages. Which of the two summands does lemma 11.4.3 match to N when applied with $\varphi = \mathrm{pr}_1$?

3. Let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension d and take $P' = P$ and $i' = i$, but keep two distinct morphisms $a, b\colon W \to P$. Write out what corollary 11.4.8 asserts, and identify the regular embedding of $P \times P$ to which lemma 11.4.1 applies. Then show that for $a = b$ the identity degenerates to a tautology, and explain why the corollary is in no case a statement about $i^!$ commuting with itself as an endomorphism.

4. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension d and $\varphi\colon U \hookrightarrow X$ an open immersion with $Z \subseteq U$. Check that φ satisfies the hypotheses of lemma 11.4.3, and deduce that the refined Gysin map of i may be computed in any open neighbourhood of Z.

5. In theorem 11.4.7 verify directly, from the definition of a regular embedding, that a local regular sequence for Z in Y lifted to P and appended to one for Y in P generates the ideal of Z in P, and that the concatenation is regular. Then write out the sequence $0 \to N_Z Y \to N_Z P \to N_Y P|_Z \to 0$ of proposition 5.1.47 for $Z = \{0\} \subseteq Y = \mathbb{A}^1 \subseteq P = \mathbb{A}^2$, identifying all three bundles and the splitting.

12

Intersection Products

The refined Gysin map of chapter 11 intersects a cycle with a regularly embedded subscheme. It is not yet a product: it takes a cycle on the ambient scheme and a regular embedding, not two cycles. The step that converts it into one is a piece of bookkeeping that costs nothing and gives everything.

For X smooth, the diagonal $\delta\colon X \hookrightarrow X \times X$ is a regular embedding, and two cycles α, β on X determine a single cycle $\alpha \times \beta$ on $X \times X$. Applying $\delta^!$ to it lands back on X and, as this chapter shows, the result is bilinear, commutative, associative, and has $[X]$ as a unit. That is a ring, and it is the Chow ring $A^*(X)$: the object the classical geometers were computing in when they counted, and which *Algebraic Geometry* [7] could construct only on a surface, by hand.

Nothing here needs a moving lemma. The cycles are never perturbed; the diagonal is what gets degenerated, and chapter 11 already did that work. The moving lemma is stated in its place, both because it is true and because seeing what it would have been used for is the clearest way to see why the deformation replaced it.

The chapter closes by paying two debts. The self-intersection formula computes i^*i_* as capping with the top Chern class, which recovers the $E^2 = -1$ of a blown-up surface as a special case of a general theorem; and the agreement theorem shows that on a smooth projective surface this product is the one built cohomologically upstream, so the surface theory that book developed by hand is subsumed rather than paralleled.

12.1 The Intersection Product

The first ingredient is a way of assembling a cycle on X and a cycle on Y into a single cycle on $X \times Y$. On cycles the rule is forced, $[V] \times [W] = [V \times W]$, and the content is that it survives rational equivalence. The mechanism that makes it survive is flat pullback along the two projections, which is already known to preserve rational equivalence (proposition 8.3.7), so the whole construction is arranged to be a composite of operations that do.

Lemma 12.1.1: Projections of a Product Are Flat

Let X and Y be algebraic schemes with Y pure of dimension l, and let $p\colon X \times Y \to X$ be the projection. Then p is flat of relative dimension l, and for every k-dimensional subvariety $V \subseteq X$ the scheme-theoretic preimage is $p^{-1}(V) = V \times Y$, so that

$$p^*[V] \;=\; [V \times Y]$$

the fundamental cycle of $V \times Y$, which is pure of dimension $k + l$.

***Proof*:**
The product is taken over $\operatorname{Spec} \overline{k}$, so p is the base change of the structure morphism $Y \to \operatorname{Spec} \overline{k}$ along $X \to \operatorname{Spec} \overline{k}$. Every module over a field is free, hence flat, so $Y \to \operatorname{Spec} \overline{k}$ is flat, and flatness is stable under base change; thus p is flat. The fibre of p over a point $x \in X$ is $Y \times_{\operatorname{Spec} \overline{k}} \operatorname{Spec} k(x)$, which is pure of dimension l, extension of the ground field preserving the dimension of a scheme of finite type. So p is flat of relative dimension l in the sense of definition 8.3.3.

The scheme-theoretic preimage of V is $p^{-1}(V) = (X \times Y) \times_X V = V \times Y$, and definition 8.3.3 computes $p^*[V]$ as the fundamental cycle of that preimage. For the dimension, restrict p to V: the induced morphism $V \times Y \to V$ is again flat with all fibres of pure dimension l, and its base V is a variety, so proposition 8.3.2 applies and every irreducible component of $V \times Y$ has dimension $k + l$, as we sought to show.

The lemma supplies both the cycle and the guarantee that it has the expected dimension, so the exterior product can be defined by the geometric formula and its bookkeeping checked afterwards.

Definition 12.1.2: The Exterior Product of Cycles

Let X and Y be algebraic schemes. For a k-dimensional subvariety $V \subseteq X$ and an l-dimensional subvariety $W \subseteq Y$, the product $V \times W$ is a closed subscheme of $X \times Y$, pure of dimension $k + l$ by lemma 12.1.1. The *exterior product* of $[V]$ and $[W]$ is its fundamental cycle (definition 8.1.2),

$$[V] \times [W] \;=\; [V \times W] \;\in\; Z_{k+l}(X \times Y)$$

Extending bilinearly gives the exterior product of cycles

$$\times \;:\; Z_k(X) \times Z_l(Y) \longrightarrow Z_{k+l}(X \times Y)$$

Example 12.1.3: The Two Rulings of a Quadric Surface

Take $X = Y = \mathbb{P}^1$. By theorem 8.5.2, $A_1(\mathbb{P}^1) = \mathbb{Z}\,[\mathbb{P}^1]$ and $A_0(\mathbb{P}^1) = \mathbb{Z}\,[\mathrm{pt}]$, the second generated by the class of any closed point. Fix closed points $p, q \in \mathbb{P}^1$. The four exterior products of generators are

$$[\mathbb{P}^1] \times [\mathbb{P}^1] = [\mathbb{P}^1 \times \mathbb{P}^1], \qquad [\mathbb{P}^1] \times [q] = [\mathbb{P}^1 \times \{q\}], \qquad [p] \times [\mathbb{P}^1] = [\{p\} \times \mathbb{P}^1], \qquad [p] \times [q] = [(p, q)]$$

each product of varieties being a variety here: $\mathbb{P}^1 \times \mathbb{P}^1$ is covered by four copies of $\mathbb{A}^2$, any two of

which meet, so it is irreducible, and it is reduced because those charts are.
The geometry is visible through the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$, $([a:b],[c:d]) \mapsto [ac:ad:bc:bd]$, whose image is the quadric surface $\{x_0x_3 = x_1x_2\}$. The middle two classes are the classes of the two families of lines on that quadric: $\mathbb{P}^1 \times \{q\}$ and $\{p\} \times \mathbb{P}^1$ each map to a line, and two members of the same family are disjoint while members of opposite families meet in one point. The first class is the fundamental class of the surface and the last is the class of a point.
Not every subvariety of a product is a product. The diagonal $\Delta = \{([a:b],[a:b])\} \subseteq \mathbb{P}^1 \times \mathbb{P}^1$ is a curve meeting both families, so it is neither of the form $V \times \mathbb{P}^1$ nor of the form $\mathbb{P}^1 \times W$; its class is nonetheless a sum of exterior products, $[\Delta] = [\mathbb{P}^1] \times [q] + [p] \times [\mathbb{P}^1]$, which exercise 12.1.1 (2) verifies by exhibiting a rational function.

Rational equivalence is not visible in definition 12.1.2, which is written on cycles. The route to it is to recognize the exterior product as a composite of a flat pullback and a proper pushforward, one factor at a time.

Proposition 12.1.4: The Exterior Product Descends to Rational Equivalence

Let X and Y be algebraic schemes.

1. *Flat descriptions.* Let $W \subseteq Y$ be an l-dimensional subvariety, let $p_W : X \times W \to X$ be the projection and $\iota_W : X \times W \hookrightarrow X \times Y$ the closed immersion. Then for every $\alpha \in Z_k(X)$,
$$\alpha \times [W] \;=\; (\iota_W)_*\big(p_W^*\alpha\big)$$
Symmetrically, if $V \subseteq X$ is a k-dimensional subvariety, $q_V : V \times Y \to Y$ the projection and $\iota_V : V \times Y \hookrightarrow X \times Y$ the closed immersion, then $[V] \times \beta = (\iota_V)_*(q_V^*\beta)$ for every $\beta \in Z_l(Y)$.

2. *Descent.* The exterior product carries $\mathrm{Rat}_k(X) \times Z_l(Y)$ and $Z_k(X) \times \mathrm{Rat}_l(Y)$ into $\mathrm{Rat}_{k+l}(X \times Y)$, so it induces a bilinear map
$$\times \;:\; A_k(X) \times A_l(Y) \longrightarrow A_{k+l}(X \times Y)$$

3. *Closed immersions.* If $f : X' \hookrightarrow X$ and $g : Y' \hookrightarrow Y$ are closed immersions of algebraic schemes, then $(f \times g)_*(\alpha \times \beta) = f_*\alpha \times g_*\beta$ for all $\alpha \in A_k(X')$ and $\beta \in A_l(Y')$.

***Proof*:**

1. Both sides are additive in α, so it suffices to take $\alpha = [V]$ for a k-dimensional subvariety $V \subseteq X$. The subscheme W is a variety, hence pure of dimension l, so lemma 12.1.1 applies to p_W and gives $p_W^*[V] = [V \times W]$, the fundamental cycle of $V \times W$ computed inside $X \times W$. Its components are the components of $V \times W$, each carrying the same length whether $V \times W$ is regarded in $X \times W$ or in $X \times Y$, and each maps isomorphically to itself under the closed immersion ι_W; so $(\iota_W)_*[V \times W] = [V \times W]$ by definition 8.2.1, the field extension in each degree being trivial. This is $[V] \times [W]$. The symmetric statement is the same argument with the roles of the two factors exchanged.

2. Let $\alpha \in \mathrm{Rat}_k(X)$ and $\beta \in Z_l(Y)$. Writing $\beta = \sum_j n_j[W_j]$ and using bilinearity, it is enough to show $\alpha \times [W] \in \mathrm{Rat}_{k+l}(X \times Y)$ for a single l-dimensional subvariety W. By

part (1) this class is $(\iota_W)_*(p_W^*\alpha)$. Flat pullback carries cycles rationally equivalent to zero to cycles rationally equivalent to zero (proposition 8.3.7), and so does proper pushforward along the closed immersion ι_W (theorem 8.2.3); hence $\alpha \times [W] \in \mathrm{Rat}_{k+l}(X \times Y)$. The symmetric half is identical, using the second formula of part (1) and additivity in the first variable.
For classes, let $\alpha \sim \alpha'$ in $Z_k(X)$ and $\beta \sim \beta'$ in $Z_l(Y)$. Then

$$\alpha \times \beta - \alpha' \times \beta' \;=\; (\alpha - \alpha') \times \beta \;+\; \alpha' \times (\beta - \beta')$$

and both terms on the right lie in $\mathrm{Rat}_{k+l}(X \times Y)$ by the previous paragraph. So the product of the classes is well defined, and it is bilinear because it is bilinear on cycles.

3. Both sides are bilinear, so it suffices to check on $\alpha = [V']$ and $\beta = [W']$ for subvarieties $V' \subseteq X'$ and $W' \subseteq Y'$. A closed immersion identifies V' with its image $f(V')$ and W' with $g(W')$, so $V' \times W'$ and $f(V') \times g(W')$ are the same closed subscheme of $X \times Y$; their fundamental cycles therefore agree, and each component of $V' \times W'$ maps isomorphically under $f \times g$. Hence $(f \times g)_*[V' \times W'] = [f(V') \times g(W')] = f_*[V'] \times g_*[W']$ by definition 8.2.1, completing the proof.

The proof is short because the exterior product was built out of the two functorialities of chapter 8 rather than defined by an independent construction, and that is the only reason it respects rational equivalence: nothing about a product of schemes is used beyond the flatness of its projections. Part (3) is the compatibility that lets a product computed on small closed subschemes be compared with the same product computed on the ambient scheme, and it is what the refined form of the intersection product below rests on.

What the exterior product does not do is produce a class on X. A class on $X \times X$ is returned to X by pulling back along the diagonal, and chapter 11 supplies the pullback provided the diagonal is a regular embedding. The diagonal morphism $\delta \colon X \to X \times X$ exists for every scheme, and it is a closed immersion exactly when X is separated, that being the definition of separatedness; every variety appearing from here on is separated, and the hypothesis is carried in the statements that use it.

The first step is to identify the conormal sheaf of the diagonal. It requires no hypothesis on X beyond separatedness, and it is what converts smoothness into the regularity of δ.

Lemma 12.1.5: The Conormal Sheaf of the Diagonal

Let X be an algebraic scheme, let $\delta \colon X \to X \times X$ be the diagonal and let $\mathcal{J} \subseteq \mathcal{O}_{X \times X}$ be the ideal sheaf of the closed subscheme $\delta(X)$. Regard $\mathcal{J}/\mathcal{J}^2$ as a sheaf of $\mathcal{O}_X$-modules through the isomorphism $\delta \colon X \to \delta(X)$. Then there is a canonical isomorphism

$$\mathcal{J}/\mathcal{J}^2 \;\cong\; \Omega_X$$

of $\mathcal{O}_X$-modules, under which the class of $1 \otimes a - a \otimes 1$ corresponds to da.

***Proof*:**
There is nothing to prove: this is how the sheaf of differentials is defined upstream. The cotangent sheaf is $\Omega_{X/Y} := \Delta^*(\mathscr{I}/\mathscr{I}^2)$ for $\mathscr{I}$ the ideal sheaf of the diagonal, with universal derivation $d(b) = 1 \otimes b - b \otimes 1$ modulo $\mathscr{I}^2$ (definition 5.5.14); taking $Y = \operatorname{Spec} \overline{k}$ and using that δ is a closed immersion gives the statement. The affine content, that I/I^2 with this d satisfies the

universal property of $\Omega_{B/A}$, is proposition 5.5.3, as we sought to show.

The identification uses no smoothness, and it is the reason the sheaf of differentials can be constructed from the diagonal in the first place: Ω_X is the conormal sheaf of δ, so it records the first infinitesimal neighbourhood of the diagonal and nothing more. What smoothness adds is that Ω_X is then locally free of the right rank, and local freeness of the conormal sheaf is the visible half of being a regular embedding. The other half, that the ideal is generated by a regular sequence and not by the right number of elements, is where the local rings of $X \times X$ enter.

Theorem 12.1.6: The Diagonal of a Smooth Variety Is a Regular Embedding

Let X be a separated smooth variety of dimension n over $\overline{k}$ and let $\delta\colon X \to X \times X$ be the diagonal. Then δ is a regular embedding of codimension n (definition 5.1.43), and its normal bundle is the tangent bundle of X,

$$N_\delta \;=\; T_X$$

***Proof*:**
Write $\Delta = \delta(X)$ and let $\mathcal{J}$ be its ideal sheaf.

Step 1: the diagonal is closed of codimension n. Separatedness of X says exactly that δ is a closed immersion, and δ is an isomorphism onto Δ, so Δ is a closed subscheme of $X \times X$ isomorphic to X and of dimension n. Smoothness is stable under products and dimensions add, so $X \times X$ is smooth of dimension $2n$ (proposition 2.7.9(3)); moreover $X \times X$ is pure of dimension $2n$ by lemma 12.1.1 applied with $V = X$. Its local rings are therefore regular, of dimension $2n$ at every closed point. The codimension of Δ in $X \times X$ is $2n - n = n$.

Step 2: the ideal is generated by n elements along the diagonal. Fix an affine open $U = \operatorname{Spec} A$ of X and keep the notation of lemma 12.1.5, so that $U \times U = \operatorname{Spec} B$ with $B = A \otimes_{\overline{k}} A$, and $J = \ker \mu$ is the ideal of $\delta(U)$. By lemma 12.1.5 there is an isomorphism $\Omega_A \cong J/J^2$ of A-modules, and smoothness of X makes Ω_A locally free of rank n.

The differentials da generate Ω_A, so after shrinking U around a chosen closed point there are $t_1, \ldots, t_n \in A$ whose differentials generate Ω_A: pick elements whose differentials span the fibre of Ω_A at the point, which has dimension n, and apply Nakayama together with coherence to spread the generation to a neighbourhood. The resulting surjection $A^n \to \Omega_A$ onto a locally free module of rank n is an isomorphism, since after localizing at any prime it becomes a surjective endomorphism of a finitely generated module over a commutative ring and every such endomorphism is injective. So $dt_1, \ldots, dt_n$ is a basis of Ω_A.

Transporting through lemma 12.1.5, the classes of

$$u_i \;=\; t_i \otimes 1 - 1 \otimes t_i, \qquad i = 1, \ldots, n$$

form a basis of J/J^2 over A. In particular $J = (u_1, \ldots, u_n) + J^2$. Let $z \in \delta(U)$ and put $R = \mathcal{O}_{X \times X, z}$ with maximal ideal $\mathfrak{m}$. Localizing the last equality gives $JR = (u_1, \ldots, u_n)R + J^2R$, and $J \subseteq \mathfrak{m}$ forces $J^2R \subseteq \mathfrak{m}\,JR$; so the finitely generated R-module $N = JR/(u_1, \ldots, u_n)R$ satisfies $N = \mathfrak{m}N$ and vanishes by Nakayama. Hence

$$JR \;=\; (u_1, \ldots, u_n)R \quad \text{for every } z \in \delta(U)$$

The coherent sheaf $\mathcal{J}/(u_1, \ldots, u_n)\mathcal{O}_{U \times U}$ therefore has vanishing stalk at every point of $\delta(U)$, so its support is a closed subset of $U \times U$ disjoint from $\delta(U)$, and on the open complement, an open neighbourhood of $\delta(U)$ in $X \times X$, the ideal sheaf of Δ is generated by $u_1, \ldots, u_n$.

Step 3: the generators form a regular sequence. Let $z \in \delta(U)$ be a closed point, corresponding to the closed point $x \in U$, and keep $R = \mathcal{O}_{X \times X, z}$. By Step 1 the ring R is regular local of dimension $2n$, and

$$R/JR = \mathcal{O}_{\Delta,z} = \mathcal{O}_{X,x}$$

is regular local of dimension n, again by smoothness of X. Regularity makes the embedding dimensions equal to the Krull dimensions, so $\dim_{\overline{k}} \mathfrak{m}/\mathfrak{m}^2 = 2n$ while the quotient $\mathfrak{m}/(\mathfrak{m}^2 + JR)$, which is the corresponding space for R/JR, has dimension n. The image of JR in $\mathfrak{m}/\mathfrak{m}^2$ is therefore n-dimensional, and it is spanned by the images of $u_1, \ldots, u_n$ by Step 2; those n images are consequently linearly independent in $\mathfrak{m}/\mathfrak{m}^2$, that is, $u_1, \ldots, u_n$ is part of a regular system of parameters of R. A regular system of parameters of a regular local ring is a regular sequence, and so is any part of one: extend $u_1, \ldots, u_n$ to a regular system of parameters $u_1, \ldots, u_{2n}$ of R and put $R_i = R/(u_1, \ldots, u_i)$, whose maximal ideal is generated by the images of $u_{i+1}, \ldots, u_{2n}$. Its embedding dimension is therefore at most $2n - i$, while its dimension is at least $2n - i$, since killing one element drops the dimension by at most one; the two agree, so R_i is regular local and hence a domain, and the image of u_{i+1} in it is nonzero, since otherwise $\mathfrak{m}R_i$ would be generated by $2n - i - 1$ elements. Each u_{i+1} is thus a nonzerodivisor on R_i, that is, $u_1, \ldots, u_n$ is a regular sequence in R.

This was proved at closed points, and it propagates to the rest of Δ by localization. If $z' \in \Delta$ is arbitrary, its closure meets the closed points of $X \times X$, since closed points are dense in a closed subset of a scheme of finite type over a field; choosing a closed point z in that closure, $\mathcal{O}_{X \times X, z'}$ is a localization of $\mathcal{O}_{X \times X, z}$ at a prime containing J. Localization is exact, so a nonzerodivisor stays a nonzerodivisor and a regular sequence stays one, the quotient remaining nonzero because $z' \in \Delta$. Thus every point of Δ has a neighbourhood in $X \times X$ on which $\mathcal{J}$ is generated by n sections whose images in the local rings along Δ form a regular sequence, which is the local condition of definition 5.1.43. Since Δ has codimension n by Step 1, δ is a regular embedding of codimension n.

Step 4: the normal bundle. The normal bundle of a regular embedding is the vector bundle whose sheaf of sections is the dual of the conormal sheaf (definition 5.1.45), and lemma 12.1.5 identifies that conormal sheaf $\mathcal{J}/\mathcal{J}^2$ with Ω_X. The dual of Ω_X is the tangent sheaf (definition 5.5.25), whose associated bundle is T_X. Hence $N_\delta = T_X$, completing the proof.

Two features of the statement are worth separating. The codimension is n, which is what makes $\delta^!$ lower dimension by exactly n and so makes the product of a k-cycle and an l-cycle a cycle of dimension $k + l - n$, the dimension a transverse intersection would have. The normal bundle is T_X, which is where the geometry of X enters the product: every formula below in which a Chern class of the tangent bundle appears is tracing back to this identification, and the self-intersection formula of section 12.3 is the first of them. Smoothness is used twice and cannot be weakened to either half alone: the local rings of $X \times X$ must be regular for Step 3, and Ω_X must be locally free for Step 2.

Example 12.1.7: The Diagonal of Affine Space

Let $X = \mathbb{A}^n = \operatorname{Spec} \overline{k}[t_1, \ldots, t_n]$. Then $X \times X = \operatorname{Spec} \overline{k}[t_1, \ldots, t_n, s_1, \ldots, s_n] = \mathbb{A}^{2n}$, and the ideal of the diagonal is $J = (t_1 - s_1, \ldots, t_n - s_n)$, so the generators of theorem 12.1.6 are visible globally: $u_i = t_i - s_i$. That they form a regular sequence needs no local algebra here, since $\overline{k}[t, s] = \overline{k}[t_1, \ldots, t_n, u_1, \ldots, u_n]$ is again a polynomial ring and each successive quotient

$\overline{k}[t,u]/(u_1,\ldots,u_i)$ is a polynomial ring, hence a domain, in which u_{i+1} is a nonzero element and therefore a nonzerodivisor. The conormal sheaf is free on the classes of the u_i, matching the basis $dt_1,\ldots,dt_n$ of $\Omega_{\mathbb{A}^n}$ under lemma 12.1.5, and N_δ is the trivial bundle of rank n, which is $T_{\mathbb{A}^n}$. The Chow groups are as small as possible: $A_k(\mathbb{A}^m) = 0$ for $k < m$ and $A_m(\mathbb{A}^m) = \mathbb{Z}\,[\mathbb{A}^m]$ by proposition 8.5.1. So the only product that can be nonzero is $[\mathbb{A}^n]\cdot[\mathbb{A}^n]$, and it is computed by the exterior product $[\mathbb{A}^n]\times[\mathbb{A}^n] = [\mathbb{A}^{2n}]$ followed by $\delta^!$. The intersection $\mathbb{A}^{2n}\cap\Delta$ is Δ itself, and $\Delta\hookrightarrow\mathbb{A}^{2n}$ is a regular embedding of codimension n, so proposition 11.3.3 gives $\delta^![\mathbb{A}^{2n}] = [\Delta]$, that is

$$[\mathbb{A}^n]\cdot[\mathbb{A}^n] \;=\; [\mathbb{A}^n]$$

the fundamental class reproducing itself.

The exterior product turns a pair of classes on X into one class on $X\times X$, and the diagonal is a regular embedding, so the Gysin map of definition 11.3.1 returns a class on X.

Definition 12.1.8: The Intersection Product

Let X be a separated smooth variety of dimension n over $\overline{k}$, with diagonal $\delta\colon X\hookrightarrow X\times X$, a regular embedding of codimension n by theorem 12.1.6. For $\alpha\in A_k(X)$ and $\beta\in A_l(X)$ the *intersection product* is

$$\alpha\cdot\beta \;=\; \delta^!\big(\alpha\times\beta\big) \;\in\; A_{k+l-n}(X)$$

where $\alpha\times\beta\in A_{k+l}(X\times X)$ is the exterior product of proposition 12.1.4 and $\delta^!\colon A_{k+l}(X\times X)\to A_{k+l-n}(X)$ is the Gysin map of definition 11.3.1.

The definition is the whole of the construction, and every property of the product is a property of one of its two inputs. It is defined for arbitrary classes, with no hypothesis on how representing cycles meet, because the Gysin map has none; and it depends only on the rational equivalence classes of the two factors, because both the exterior product and the Gysin map do. The dimension bookkeeping is the expected one: $k+l-n$ is the dimension of a transverse intersection of a k-dimensional and an l-dimensional subvariety of an n-dimensional variety, so in codimension the product adds. That grading, and the ring axioms it grades, are established in section 12.2; until then the product is exactly what definition 12.1.8 says it is.

Bilinearity and the vanishing below the expected dimension are inherited from the two inputs, and are recorded here for the computations that use them.

Proposition 12.1.9: Bilinearity and Vanishing in Low Dimension

Let X be a separated smooth variety of dimension n.

1. The product $A_k(X)\times A_l(X)\to A_{k+l-n}(X)$ is bilinear.
2. If $k+l<n$ then $\alpha\cdot\beta = 0$ for all $\alpha\in A_k(X)$ and $\beta\in A_l(X)$.

***Proof*:**

1. The exterior product is bilinear by proposition 12.1.4(2) and $\delta^!$ is a homomorphism of abelian groups by proposition 11.3.2(1), so the composite is bilinear in each variable.
2. The class $\alpha\times\beta$ lies in $A_{k+l}(X\times X)$ with $k+l<n$, and the codimension of δ is n, so

proposition 11.3.2(2) gives $\delta^!(\alpha \times \beta) = 0$. Equivalently the target group $A_{k+l-n}(X)$ is a Chow group in negative dimension and vanishes, as we sought to show.

The product of definition 12.1.8 records its answer on all of X, and that discards information the construction has available. If α is carried by a closed subscheme V and β by a closed subscheme W, the intersection ought to be carried by $V \cap W$, and the refined Gysin homomorphism of definition 11.3.6 delivers exactly that, because the preimage of $V \times W$ under the diagonal is the scheme-theoretic intersection. Indeed the ideal sheaf of $V \times W$ in $X \times X$ is generated by the pullbacks of $\mathcal{I}_V$ and $\mathcal{I}_W$ along the two projections, and composing either projection with δ is the identity, so that ideal pulls back to $\mathcal{I}_V + \mathcal{I}_W$, the ideal of $V \cap W$.

Definition 12.1.10: The Refined Intersection Product

Let X be a separated smooth variety of dimension n and let $V, W \subseteq X$ be closed subschemes, with $g \colon V \times W \to X \times X$ the closed immersion. Since $\delta^{-1}(V \times W) = V \cap W$, the refined Gysin homomorphism of definition 11.3.6 along g is a homomorphism

$$\delta^! \;:\; A_m(V \times W) \longrightarrow A_{m-n}(V \cap W)$$

For $\alpha \in A_k(V)$ and $\beta \in A_l(W)$ the *refined intersection product* is

$$\alpha \cdot_X \beta \;=\; \delta^!\big(\alpha \times \beta\big) \;\in\; A_{k+l-n}(V \cap W)$$

Proposition 12.1.11: The Refined Product Determines the Product

Let X be a separated smooth variety of dimension n, let $V, W \subseteq X$ be closed subschemes with inclusions i_V and i_W, and let $\iota \colon V \cap W \hookrightarrow X$ be the inclusion. For all $\alpha \in A_k(V)$ and $\beta \in A_l(W)$,

$$\iota_*\big(\alpha \cdot_X \beta\big) \;=\; \big(i_{V*}\alpha\big) \cdot \big(i_{W*}\beta\big) \;\in\; A_{k+l-n}(X)$$

***Proof*:**
Apply theorem 11.3.10(1) to the regular embedding δ with $X' = X \times X$, $X'' = V \times W$ and $p = g$ the closed immersion, which is proper. The preimage of $\delta(X)$ in X' is $\delta(X)$ itself and the preimage in X'' is $V \cap W$, so the induced morphism q of preimages is ι, read through the isomorphism $\delta \colon X \to \delta(X)$. The theorem gives

$$\delta^!\big(g_*\gamma\big) \;=\; \iota_*\big(\delta^!\gamma\big)$$

for $\gamma \in A_{k+l}(V \times W)$, where the left-hand $\delta^!$ is the refined map along $\mathrm{id}_{X \times X}$, which is the Gysin map of definition 11.3.1 by proposition 11.3.7. Take $\gamma = \alpha \times \beta$. By proposition 12.1.4(3) applied to the closed immersions i_V and i_W,

$$g_*(\alpha \times \beta) \;=\; i_{V*}\alpha \times i_{W*}\beta$$

so the left-hand side is $\delta^!\big(i_{V*}\alpha \times i_{W*}\beta\big)$, which is $(i_{V*}\alpha) \cdot (i_{W*}\beta)$ by definition 12.1.8, and the right-hand side is $\iota_*(\alpha \cdot_X \beta)$ by definition 12.1.10, as we sought to show.

The pattern is the one recorded in corollary 11.3.11 for a single subvariety: the class computed on the ambient scheme is the pushforward of a class computed on the intersection. The refined product is the finer object, and every localized intersection number in the classical sense is a degree taken of it; the product of definition 12.1.8 is its pushforward to X.

Example 12.1.12: Two Lines in the Projective Plane

Let $X = \mathbb{P}^2$, which is a smooth variety of dimension 2, and let $L = \{x_0 = 0\}$ and $L' = \{x_1 = 0\}$ be two distinct lines, meeting in the single reduced point $P = [0 : 0 : 1]$.

The exterior product. By definition 12.1.2, $[L] \times [L'] = [L \times L']$, and $L \times L' \cong \mathbb{P}^1 \times \mathbb{P}^1$ is a variety of dimension 2 by the argument of example 12.1.3, so the exterior product is the class of a single subvariety of $\mathbb{P}^2 \times \mathbb{P}^2$ with multiplicity one.

The Gysin pullback. The intersection of $L \times L'$ with the diagonal is $\delta^{-1}(L \times L') = L \cap L' = \{P\}$, a reduced point. In the affine chart of $L \times L'$ around (P, P), with coordinates y on $L \setminus \{x_2 = 0\}$ and z on $L' \setminus \{x_2 = 0\}$, that point is the origin of $\mathbb{A}^2$, cut out by the regular sequence y, z; so $\{P\} \hookrightarrow L \times L'$ is a regular embedding of codimension $2 = \operatorname{codim}(\delta)$. Proposition 11.3.3 therefore applies to δ and the subvariety $L \times L'$ and gives

$$[L] \cdot [L'] \;=\; \delta^{!}\big[L \times L'\big] \;=\; [P] \;\in\; A_0(\mathbb{P}^2)$$

The refined form of definition 12.1.10 records a class on $L \cap L' = \{P\}$, and that class is the generator of $A_0(\{P\}) = \mathbb{Z}$: by proposition 12.1.11 its image in $A_0(\mathbb{P}^2)$ is $[P]$, and the pushforward $A_0(\{P\}) \to A_0(\mathbb{P}^2)$ carries generator to generator.

The self-intersection. By theorem 8.5.2 the two lines have the same class, $[L] = [L'] = h_1$ in $A_1(\mathbb{P}^2)$. Since the product is defined on classes, the computation just made also evaluates

$$[L] \cdot [L] \;=\; h_1 \cdot h_1 \;=\; [P]$$

No cycle was perturbed to get this, and no representative of h_1 was required to meet L properly: the two factors were replaced by equal classes, which proposition 12.1.4(2) and proposition 11.3.2(1) permit. The refined product, by contrast, does depend on the representatives: computed on L against L it lives on $A_0(L)$ rather than on $A_0(\{P\})$, and evaluating it there is the business of section 12.3.

Exercise 12.1.1

1. Let $P = \operatorname{Spec} \overline{k}$ and let X be an algebraic scheme. Show that $A_0(P) = \mathbb{Z}[P]$, and that under the canonical isomorphism $X \times P \cong X$ the exterior product of proposition 12.1.4 sends $(\alpha, m[P])$ to $m\alpha$. Then show that the exterior product of cycles is associative: for algebraic schemes X, Y, T and cycles α, β, γ on them, $(\alpha \times \beta) \times \gamma = \alpha \times (\beta \times \gamma)$ under the identification $(X \times Y) \times T = X \times (Y \times T)$.

2. Let $S = \mathbb{P}^1 \times \mathbb{P}^1$ with coordinates $([a : b], [c : d])$, let $\Delta \subseteq S$ be the diagonal, and let $F = \mathbb{P}^1 \times \{[1 : 0]\}$ and $G = \{[0 : 1]\} \times \mathbb{P}^1$. Show that

$$[\Delta] \;=\; [\mathbb{P}^1] \times [\mathrm{pt}] + [\mathrm{pt}] \times [\mathbb{P}^1] \quad \text{in } A_1(S)$$

by computing the divisor of the rational function $f = (ad - bc)/(ad)$ on S chart by chart.

3. Let $X = \operatorname{Spec} \overline{k}[x, y]/(y^2 - x^3)$, a curve with a cusp at the origin p. Using lemma 12.1.5 and the presentation $\Omega_X = \big(\mathcal{O}_X\,dx \oplus \mathcal{O}_X\,dy\big)/\big(d(y^2 - x^3)\big)$, show that the conormal sheaf of the diagonal of X is not locally free of rank one, and deduce that $\delta\colon X \to X \times X$ is not a regular

embedding of codimension one. Where does the proof of theorem 12.1.6 break down for this X?

4. Let X be a separated smooth variety of dimension n and let $V, W \subseteq X$ be closed subschemes with $V \cap W = \emptyset$. Show that $[V] \cdot [W] = 0$ in $A_*(X)$, where $[V]$ and $[W]$ denote the images in $A_*(X)$ of the fundamental cycles. Give a second proof of the vanishing when $\dim V + \dim W < n$ that does not use disjointness.

5. Fix $0 \le a, b \le n$ with $a + b \ge n$, and let $L = \{x_{a+1} = \cdots = x_n = 0\}$ and $M = \{x_0 = \cdots = x_{n-b-1} = 0\}$ be coordinate subspaces of $\mathbb{P}^n$, so that $L \cong \mathbb{P}^a$, $M \cong \mathbb{P}^b$ and $L \cap M \cong \mathbb{P}^{a+b-n}$. Show that $L \cap M \hookrightarrow L \times M$ is a regular embedding of codimension n, and deduce from proposition 11.3.3 that $h_a \cdot h_b = h_{a+b-n}$ in $A_*(\mathbb{P}^n)$, where h_m is the class of a linear $\mathbb{P}^m$ as in example 11.3.4.

12.2 The Chow Ring

Section 12.1 produced a biadditive pairing on the Chow groups of a smooth variety. This section proves that the pairing obeys the three laws that make it a ring, and it indexes the groups by codimension so that the degrees add rather than subtract. The ring is then made contravariant: a morphism of smooth varieties pulls classes back, and so does a local complete intersection morphism between arbitrary algebraic schemes, which is the form Grothendieck-Riemann-Roch will need.

Two ingredients are imported rather than built. One is a pair of statements about regular embeddings that belong to scheme theory; the other is the commutativity of two Gysin maps, which chapter 11 flagged and did not prove. Both are isolated below, so that everything resting on them is visible at a glance.

12.2.1 Convention: Subscripts on Refined Gysin Maps When several refinements of one Gysin map occur in the same formula, the morphism along which the map is refined is written as a subscript: $i^!_g$ is the refined homomorphism of definition 11.3.6 for the morphism g, and an unadorned $i^!$ is the Gysin map of definition 11.3.1.

One reading has to be fixed first. Definition 11.3.1 is stated in codimension $d \ge 1$, whereas a regular embedding of codimension 0 is an open and closed immersion, its ideal sheaf vanishing locally. For such an $i\colon Z \hookrightarrow P$ both $i^!$ and its refinements $i^!_g$ mean flat pullback throughout this section, along i and along the induced immersion $g^{-1}(Z) \hookrightarrow P'$ for $g\colon P' \to P$; this is the reading definition 12.2.11 uses again. When the case arises from a diagonal or a graph the smooth variety in question has dimension 0, hence is $\operatorname{Spec} \overline{k}$: the embedding is then the identity under the canonical identification of $W \times \operatorname{Spec} \overline{k}$ with W, its flat pullback is the identity on Chow groups, and every statement below in which that happens holds by inspection. The arguments are written for the remaining case.

A cycle class on a smooth n-dimensional variety carries two equally good indices, its dimension and its codimension, and the product adds the second while subtracting the first from $2n$. A ring wants its grading to be additive, so the codimension index is the one to record.

Definition 12.2.2: The Chow Ring

Let X be a smooth variety of dimension n. For $0 \le p \le n$ write

$$A^p(X) \;=\; A_{n-p}(X)$$

and call $A^p(X)$ the group of cycle classes of *codimension* p. The graded abelian group

$$A^*(X) \;=\; \bigoplus_{p=0}^{n} A^p(X)$$

equipped with the intersection product $\alpha \cdot \beta = \delta^!(\alpha \times \beta)$ of section 12.1, is the *Chow ring* of X. That the product makes it a ring is theorem 12.2.7.

The two indices are interchangeable data, and both are used: a statement about the geometry of cycles is usually cleaner in dimension, a statement about the ring is cleaner in codimension. The class $[X]$ lies in $A^0(X)$, a divisor class lies in $A^1(X)$, and the class of a point on a complete X lies in $A^n(X)$.

Projective space is the first ring that can be written down, and it can be written down before the ring axioms are available, because each product in it is a single application of proposition 11.3.3.

Example 12.2.3: The Chow Ring of Projective Space

Let $X = \mathbb{P}^n$ with homogeneous coordinates $x_0, \ldots, x_n$. By theorem 8.5.2, $A_k(\mathbb{P}^n) = \mathbb{Z}\, h_k$ with h_k the class of a linear subspace $\mathbb{P}^k$, so $A^p(\mathbb{P}^n) = \mathbb{Z}\, h_{n-p}$ is infinite cyclic for $0 \le p \le n$ and zero otherwise.

Fix $p, q \ge 0$ with $p + q \le n$ and take the linear subspaces

$$L = \{x_0 = \cdots = x_{p-1} = 0\}, \qquad L' = \{x_p = \cdots = x_{p+q-1} = 0\}$$

of dimensions $n-p$ and $n-q$, so that $[L] = h_{n-p}$ and $[L'] = h_{n-q}$. Their product is $\delta^!\big([L] \times [L']\big) = \delta^![L \times L']$, and proposition 11.3.3 evaluates it once the scheme-theoretic intersection $W = (L \times L') \cap \Delta$ is known to be regularly embedded in $L \times L'$ with codimension n. Set-theoretically W consists of the pairs (z, z) with $z \in L \cap L'$, so $W \cong L \cap L' = \{x_0 = \cdots = x_{p+q-1} = 0\} \cong \mathbb{P}^{n-p-q}$.

Work on the chart $\{x_n \ne 0\}$ of each factor, which meets W because the coordinate point e_n lies in $L \cap L'$. Writing $u_i = x_i/x_n$ for the affine coordinates on the first factor and $v_i = x_i/x_n$ for those on the second, the chart of $L \times L'$ is the affine space with coordinate ring

$$\overline{k}\big[u_p, \ldots, u_{n-1};\; v_0, \ldots, v_{p-1},\; v_{p+q}, \ldots, v_{n-1}\big]$$

the coordinates $u_0, \ldots, u_{p-1}$ vanishing on L and $v_p, \ldots, v_{p+q-1}$ on L'. The diagonal is cut out on $\mathbb{P}^n \times \mathbb{P}^n$ in this chart by the n equations $u_i = v_i$ with $0 \le i \le n-1$, and restricting them to $L \times L'$ leaves

$$v_0,\; \ldots,\; v_{p-1}, \qquad u_p,\; \ldots,\; u_{p+q-1}, \qquad u_i - v_i \;\; (p+q \le i \le n-1)$$

a list of n linear forms in distinct variables. Each is a nonzerodivisor modulo its predecessors, every successive quotient being a polynomial ring, so the list is a regular sequence and $W \hookrightarrow L \times L'$ is a regular embedding of codimension n; the quotient is the polynomial ring on the remaining $n - p - q$ variables, so W is reduced and irreducible with fundamental cycle $[\mathbb{P}^{n-p-q}]$. The same computation runs on every chart meeting W. Proposition 11.3.3 therefore gives

$$h_{n-p} \cdot h_{n-q} \;=\; h_{n-p-q}$$

For $p+q > n$ the product lands in $A^{p+q}(\mathbb{P}^n) = A_{n-p-q}(\mathbb{P}^n) = 0$ and is zero, by proposition 11.3.2(2). Writing $h = h_{n-1}$ for the class of a hyperplane, induction on p gives $h^p = h_{n-p}$, and

$$A^*(\mathbb{P}^n) \;\cong\; \mathbb{Z}[h]/(h^{n+1})$$

once the ring axioms below license the power notation.

No general position argument entered the computation. The two subspaces were chosen in special position relative to the coordinates but in general position relative to each other, and proposition 11.3.2 guarantees that any other pair of representatives, including two equal ones, returns the same product. What the example does not show is that the answer depends on the two classes through a ring structure, and that is the content of the rest of the section.

The first import is scheme-theoretic. Both of its statements concern regular embeddings and smooth morphisms rather than cycles, so both belong to the scheme theory upstream, where they are now proved. Neither is re-derived here.

Lemma 12.2.4: Sections, Flat Base Change, and Graphs

1. Let $\pi\colon P \to S$ be a smooth morphism of relative dimension $e \geq 0$ and $s\colon S \to P$ a section of it. Then s is a regular embedding of codimension e with normal bundle $s^*T_{P/S}$.
2. Let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension d and $g\colon P' \to P$ a flat morphism. Then $g^{-1}(Z) \hookrightarrow P'$ is a regular embedding of codimension d, with normal bundle the pullback of the normal bundle of i.
3. Let X be a smooth variety of dimension $n \geq 0$, let V be an algebraic scheme, and let $f\colon V \to X$ be a morphism. Then the scheme-theoretic preimage of the diagonal under $f \times \mathrm{id}_X\colon V \times X \to X \times X$ is the graph Γ_f, the first projection restricts to an isomorphism $\Gamma_f \to V$, and $\gamma_f\colon V \to V \times X$ is a regular embedding of codimension n with normal bundle f^*T_X.

***Proof*:**
Parts (1) and (2) are upstream results and are cited rather than re-derived, both being statements about regular embeddings as such: part (1) is proposition 5.1.48 and part (2) is proposition 5.1.49. Part (3) follows from part (1). A T-valued point of $(f \times \mathrm{id}_X)^{-1}(\Delta)$ is a pair (t_V, t_X) of T-points of V and X with $f \circ t_V = t_X$, so the functor of points of the preimage is that of V, and the isomorphism is induced by the first projection, with inverse γ_f. The projection $\mathrm{pr}_1\colon V \times X \to V$ is the base change of $X \to \operatorname{Spec} \overline{k}$, hence smooth of relative dimension n, and γ_f is a section of it, so part (1) applies. Its relative tangent bundle is $\mathrm{pr}_2^* T_X$, whose pullback along γ_f is f^*T_X, completing the proof.

Part (3) extends the diagonal computation of section 12.1 from the identity morphism to an arbitrary one, and it is the reason the intersection product can be transported along a map. Taking $V = X$

and $f = \mathrm{id}_X$ returns the diagonal.

The second import is about Gysin maps. That two regular embeddings can be intersected against in either order is what makes the product associative, and that the Gysin map of a composite of two regular embeddings factors is what makes the pullback along a local complete intersection morphism well defined. Both are proved in section 11.4, as corollary 11.4.8 and theorem 11.4.7, and are used here without further comment; the section identity theorem 11.4.2 of the same section is the third.

This theorem and parts (1) and (2) of lemma 12.2.4 are the only results the arguments below use whose proofs are cited rather than given. On the theorem rest the associativity of the intersection product, the identity of theorem 12.2.10(1) between intersecting with the graph and pulling back, and with it the projection formula, the multiplicativity and functoriality of pullback, and the independence of the local complete intersection pullback from its factorization. On the lemma rests everything that passes through a graph: its part (3) is deduced from part (1), so the unit law, definition 12.2.8 and all four parts of theorem 12.2.10 rest on it, and Step 1 of theorem 12.2.12(1) uses parts (1) and (2) directly. Commutativity of the product itself and the grading rest on neither.

The last preparation is bookkeeping. A Gysin map applied to an exterior product in which one factor is inert should return that factor untouched.

Lemma 12.2.5: Gysin Maps and Exterior Products

Let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension $d \geq 1$, let W be an algebraic scheme, and let $\mathrm{pr}\colon P \times W \to P$ and $\mathrm{pr}'\colon W \times P \to P$ be the projections, whose scheme-theoretic preimages of Z are $Z \times W$ and $W \times Z$. Then for all $\xi \in A_k(P)$ and $\gamma \in A_l(W)$,

$$i^!_{\mathrm{pr}}(\xi \times \gamma) = \left(i^!\xi\right) \times \gamma, \qquad i^!_{\mathrm{pr}'}(\gamma \times \xi) = \gamma \times \left(i^!\xi\right)$$

***Proof*:**
Both sides are additive in γ, so it suffices to treat $\gamma = [W']$ for a subvariety $W' \subseteq W$ of dimension l, with inclusion $\iota\colon W' \hookrightarrow W$. Write $u = \mathrm{id}_P \times \iota\colon P \times W' \to P \times W$, a closed immersion and hence proper, and $\rho\colon P \times W' \to P$ for the projection. As W' is a variety of dimension l over $\overline{k}$, the structure morphism $W' \to \operatorname{Spec}\overline{k}$ is flat with l-dimensional fibre, so its base change ρ is flat of relative dimension l; and $\rho = \mathrm{pr} \circ u$, so u is a morphism over P.

Step 1: the exterior product as a pullback followed by a pushforward. For a subvariety $V \subseteq P$ the preimage $\rho^{-1}(V)$ is $V \times W'$, so $\rho^*[V] = [V \times W']$ by definition 8.3.3, and $u_*[V \times W'] = [V \times W']$ by definition 8.2.1, the immersion u carrying $V \times W'$ isomorphically onto its image. Hence $\xi \times [W'] = u_*\left(\rho^*\xi\right)$ for every ξ. The same computation on Z gives $q^*\eta = \eta \times [W']$ for the projection $q\colon Z \times W' \to Z$ and every $\eta \in A_*(Z)$.

Step 2: the two compatibilities. Theorem 11.3.10(2), applied with $X' = P$ and the identity, $X'' = P \times W'$ and ρ, and with ρ itself as the morphism over P, gives

$$i^!_\rho\left(\rho^*\xi\right) \;=\; q^*\left(i^!\xi\right) \;=\; \left(i^!\xi\right) \times [W']$$

where proposition 11.3.7 was used to read the refined map along the identity as $i^!$. Theorem 11.3.10(1), applied with $X' = P \times W$ and pr, $X'' = P \times W'$ and ρ, and with the proper morphism u over P, gives

$$i^!_{\mathrm{pr}}\left(u_*\eta\right) \;=\; q'_*\left(i^!_\rho\eta\right)$$

for the closed immersion $q'\colon Z \times W' \hookrightarrow Z \times W$ induced by u. Combining the two with Step 1, and using $q'_*\big(\zeta \times [W']\big) = \zeta \times [W']$ by definition 8.2.1 again, yields the first identity. Exchanging the two factors throughout gives the second, as we sought to show.

One comparison is used repeatedly below: when the preimage of Z is as well behaved as Z itself, the refined map is the plain one.

Proposition 12.2.6: Refined Gysin Maps with a Regular Preimage

Let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension $d \geq 1$ and $g\colon P' \to P$ a morphism of algebraic schemes such that $Z' = g^{-1}(Z)$ is regularly embedded in P' with codimension d. Then the refined Gysin homomorphism $i^!_g$ of definition 11.3.6 is the Gysin map $(i')^!$ of definition 11.3.1 for $i'\colon Z' \hookrightarrow P'$.

***Proof*:**
Write $C = C_{Z'}P'$ and $N' = h^*N$ as in definition 11.3.6, and let $j\colon C \hookrightarrow N'$ be the immersion of lemma 10.4.13. By part (1) of that lemma j is an isomorphism, and it is a morphism of cones over Z', so the projections satisfy $\pi_{N'} \circ j = \pi_C$. Since i' is a regular embedding, proposition 10.4.11 identifies C with the normal bundle of i', and definition 11.3.1 reads $(i')^! = (\pi_C^*)^{-1} \circ \sigma'$. Flat pullback along the isomorphism j is inverse to proper pushforward along it, each carrying a subvariety to its preimage or image with multiplicity one, so $j_*\pi_C^* = j_*j^*\pi_{N'}^* = \pi_{N'}^*$ by proposition 8.3.5(3). Hence $(\pi_{N'}^*)^{-1} \circ j_* = (\pi_C^*)^{-1}$, and composing with σ' turns definition 11.3.6 into definition 11.3.1, as we sought to show.

Theorem 12.2.7: The Chow Ring of a Smooth Variety

Let X be a smooth variety of dimension n, with diagonal $\delta\colon X \hookrightarrow X \times X$. Then, for all $\alpha, \beta, \gamma \in A^*(X)$:

1. the product is biadditive and $A^p(X) \cdot A^q(X) \subseteq A^{p+q}(X)$;
2. $\alpha \cdot \beta = \beta \cdot \alpha$;
3. $(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma)$;
4. $[X] \cdot \alpha = \alpha$.

Thus $A^*(X)$ is a commutative graded ring with unit $[X]$.

***Proof*:**

1. Biadditivity is proposition 12.1.9(1). For the degrees, let $\alpha \in A^p(X) = A_{n-p}(X)$ and $\beta \in A^q(X) = A_{n-q}(X)$. Their exterior product lies in $A_{2n-p-q}(X \times X)$, and δ is a regular embedding of codimension n (theorem 12.1.6), so $\delta^!$ lowers dimension by n by proposition 11.3.2(1). The product therefore lies in $A_{n-p-q}(X) = A^{p+q}(X)$.
2. Let $\tau\colon X \times X \to X \times X$ exchange the two factors. It carries a product of subvarieties $V \times W$ isomorphically onto $W \times V$, so $\tau_*(\alpha \times \beta) = \beta \times \alpha$ by definition 8.2.1, the exchange inducing an isomorphism of function fields. Since τ is an involution fixing the diagonal pointwise, $\tau^{-1}(\Delta) = \Delta$, which is regularly embedded in $X \times X$ with codimension n; proposition 12.2.6

therefore identifies $\delta^!_\tau$ with $\delta^!$. Applying theorem 11.3.10(1) with $X' = X \times X$ and the identity, $X'' = X \times X$ and τ, and with τ itself as the morphism over $X \times X$, which is proper because it is an isomorphism, and noting that the induced morphism of preimages is $\tau|_\Delta = \mathrm{id}_\Delta$,

$$\delta^!(\tau_*\xi) \;=\; \delta^!_\tau(\xi) \;=\; \delta^!(\xi)$$

for every $\xi \in A_*(X \times X)$. With $\xi = \alpha \times \beta$ this reads $\beta \cdot \alpha = \alpha \cdot \beta$.

3. Write $\Xi = \alpha \times \beta \times \gamma \in A_*(X \times X \times X)$, which is unambiguous because the exterior product is associative on cycles, and let $\mathrm{pr}_{12}, \mathrm{pr}_{23} \colon X \times X \times X \to X \times X$ be the two projections. Their preimages of the diagonal are $W_1 = \Delta \times X$ and $W_2 = X \times \Delta$, each identified with $X \times X$ by the two projections that are not constrained; under those identifications $\mathrm{pr}_{23}|_{W_1}$ and $\mathrm{pr}_{12}|_{W_2}$ are the identity of $X \times X$, so the refined maps along them are $\delta^!$ by proposition 11.3.7. By lemma 12.2.5, applied to $i = \delta$ with $W = X$ on each side in turn,

$$(\alpha \cdot \beta) \times \gamma = \delta^!_{\mathrm{pr}_{12}}(\Xi), \qquad \alpha \times (\beta \cdot \gamma) = \delta^!_{\mathrm{pr}_{23}}(\Xi)$$

Applying $\delta^!$ to each and using corollary 11.4.8 with $i = i' = \delta$, $W = X \times X \times X$ and the two projections,

$$(\alpha \cdot \beta) \cdot \gamma = \delta^!\big(\delta^!_{\mathrm{pr}_{12}}\Xi\big) = \delta^!\big(\delta^!_{\mathrm{pr}_{23}}\Xi\big) = \alpha \cdot (\beta \cdot \gamma)$$

both sides being computed on $W_{12} = (\Delta \times X) \cap (X \times \Delta)$, the small diagonal, which the first projection identifies with X.

4. By additivity it suffices to take $\alpha = [V]$ for a k-dimensional subvariety $V \subseteq X$, and by part (2) it suffices to evaluate $[V] \cdot [X] = \delta^![V \times X]$, the exterior product of the two fundamental cycles being the fundamental cycle of the variety $V \times X$ of dimension $k + n$ (definition 12.1.2). Apply lemma 12.2.4(3) to the inclusion $f \colon V \hookrightarrow X$: the scheme-theoretic intersection $(V \times X) \cap \Delta$, which is the preimage of Δ under $f \times \mathrm{id}_X$, is the graph Γ_f, isomorphic to V and regularly embedded in $V \times X$ with codimension n. Proposition 11.3.3 therefore gives $\delta^![V \times X] = \iota_*[\Gamma_f] = [V]$ in $A_k(\Delta) = A_k(X)$, where ι is the inclusion of the graph in the diagonal, which the projections identify with the inclusion of V in X, completing the proof.

The ring exists, and its two ends are already familiar: $A^0(X) = \mathbb{Z}[X]$ because a variety has only itself as a subvariety of top dimension, and $A^1(X) = A_{n-1}(X)$ is the group of divisor classes on which chapter 9 built the operator $D \cdot -$ by hand. The operators of section 10.2 can now be recorded as elements: for a vector bundle E on X the class $c_i(E) \cap [X]$ lies in $A^i(X)$, so Chern classes become members of a ring rather than endomorphisms of a graded group, and an identity of operators may be read as an identity in $A^*(X)$ by evaluating both sides on $[X]$.

The grading also explains an absence. On $\mathbb{P}^3$ two distinct lines meeting in a point have classes in A^2, so their product lies in $A^4(\mathbb{P}^3) = 0$: the ring records no trace of the point in which they meet. The computation is not defective. The intersection is not proper, one of the two lines can be moved off the other within its rational equivalence class, and the invariant answer to "how much do they meet" is zero.

A morphism $f \colon X \to Y$ of smooth varieties does not act on cycles in either direction in any naive way, but its graph is a regular embedding by lemma 12.2.4(3), and intersecting with the graph is a pullback.

Definition 12.2.8: Pullback Along a Morphism of Smooth Varieties

Let $f\colon X \to Y$ be a morphism of smooth varieties, of dimensions m and n, and let $\gamma_f\colon X \to X \times Y$ be its graph, a regular embedding of codimension n by lemma 12.2.4(3); for $n = 0$, where $Y = \operatorname{Spec} \overline{k}$ and γ_f is an isomorphism, $\gamma_f^!$ is flat pullback along it. The *pullback* along f is

$$f^*\colon A_l(Y) \longrightarrow A_{l+m-n}(X), \qquad f^*\beta \;=\; \gamma_f^!\big([X] \times \beta\big)$$

Equivalently $f^*\beta = \gamma_f^!\big(\mathrm{pr}_Y^*\beta\big)$, since $\mathrm{pr}_Y\colon X \times Y \to Y$ is flat of relative dimension m with $\mathrm{pr}_Y^*\beta = [X] \times \beta$.

In codimension the map preserves degree: $\beta \in A^q(Y)$ means $l = n - q$, and then $l + m - n = m - q$, so $f^*\beta \in A^q(X)$. The definition is the one the geometry suggests. To pull a cycle on Y back to X, place it on $X \times Y$, where it is inert in the first factor, and intersect with the graph, which is the locus where the second coordinate is the image of the first.

Example 12.2.9: Pullback to a Linear Subspace

Let $i\colon \mathbb{P}^m \hookrightarrow \mathbb{P}^n$ be the linear subspace $\{x_0 = \cdots = x_{n-m-1} = 0\}$ with $1 \le m < n$, and write h and h' for the hyperplane classes of $\mathbb{P}^n$ and $\mathbb{P}^m$. On the chart where $x_n \neq 0$ in both factors, with affine coordinates u_j on $\mathbb{P}^m$ for $n - m \le j \le n-1$ and v_j on $\mathbb{P}^n$ for $0 \le j \le n-1$, the graph $\Gamma_i = \{(z,z) \,\big|\, z \in \mathbb{P}^m\}$ is cut out of $\mathbb{P}^m \times \mathbb{P}^n$ by the n equations

$$v_j \quad (j < n-m), \qquad v_j - u_j \quad (j \ge n-m)$$

Fix $1 \le p \le m$ and represent h_{n-p} by $L = \{x_{n-m} = \cdots = x_{n-m+p-1} = 0\}$, a linear subspace of dimension $n-p$ meeting $\mathbb{P}^m$ in the linear subspace $\mathbb{P}^{m-p}$. Then $[\mathbb{P}^m] \times h_{n-p}$ is represented by the subvariety $\mathbb{P}^m \times L$, of dimension $m + n - p$, whose intersection with Γ_i is $\{(z,z) \,\big|\, z \in \mathbb{P}^{m-p}\}$. On the chart above, the coordinates $v_{n-m}, \dots, v_{n-m+p-1}$ vanish on L, so restricting the n equations of the graph to $\mathbb{P}^m \times L$ leaves

$$v_j \quad (j < n-m), \qquad u_j \quad (n-m \le j < n-m+p), \qquad v_j - u_j \quad (j \ge n-m+p)$$

again n coordinates and coordinate differences in distinct variables, hence a regular sequence exactly as in example 12.2.3. The intersection is therefore regularly embedded of codimension n, and reduced and irreducible, so proposition 11.3.3 gives

$$i^* h_{n-p} \;=\; \big[\mathbb{P}^{m-p}\big] \;=\; h'_{m-p}$$

For $p > m$ the target $A_{m-p}(\mathbb{P}^m)$ vanishes and the pullback is zero. The restriction of the class of a linear subspace is the class of a linear subspace of the same codimension, which is what a subspace meeting $\mathbb{P}^m$ properly should give; in the notation of example 12.2.3 this reads $i^*(h^p) = (h')^p$ for $p \le m$.

Theorem 12.2.10: Properties of Pullback

Let $f\colon X \to Y$ be a morphism of smooth varieties, of dimensions m and n.

1. For $\alpha \in A_*(X)$ and $\beta \in A_*(Y)$,
$$\gamma_f^!\big(\alpha \times \beta\big) \;=\; \alpha \cdot f^*\beta$$
and in particular $f^*[Y] = [X]$.
2. If f is proper then $f_*\big(\alpha \cdot f^*\beta\big) = f_*\alpha \cdot \beta$ for all $\alpha \in A_*(X)$ and $\beta \in A_*(Y)$.
3. $f^*\colon A^*(Y) \to A^*(X)$ is a homomorphism of graded rings.
4. $\mathrm{id}_X^* = \mathrm{id}$, and $(g \circ f)^* = f^* \circ g^*$ for a further morphism $g\colon Y \to Z$ of smooth varieties.

***Proof*:**
Throughout, $\Gamma_f \subseteq X \times Y$ is the graph and $\gamma_f\colon X \to X \times Y$ the regular embedding of codimension n onto it (lemma 12.2.4(3)).

1. By lemma 12.2.5, applied to γ_f with an inert factor X on the left,
$$\alpha \times f^*\beta \;=\; \alpha \times \gamma_f^!\big([X] \times \beta\big) \;=\; (\gamma_f)^!_{\mathrm{pr}_{23}}\big(\alpha \times [X] \times \beta\big)$$
where $\mathrm{pr}_{23}\colon X \times X \times Y \to X \times Y$ is the projection, whose preimage of Γ_f is $X \times \Gamma_f$, identified with $X \times X$ by the first two projections. Applying $\delta^!$ and invoking corollary 11.4.8 for the regular embeddings δ and γ_f with the projections pr_{12} and pr_{23},
$$\alpha \cdot f^*\beta \;=\; \delta^!\Big((\gamma_f)^!_{\mathrm{pr}_{23}}\big(\alpha \times [X] \times \beta\big)\Big) \;=\; (\gamma_f)^!_\rho\Big(\delta^!_{\mathrm{pr}_{12}}\big(\alpha \times [X] \times \beta\big)\Big)$$
On the left the outer map is refined along $\mathrm{pr}_{12}|_{X \times \Gamma_f}$, which is the identity of $X \times X$ under the identification above, so it is $\delta^!$ by proposition 11.3.7. On the right the preimage of Δ under pr_{12} is $\Delta \times Y$, identified with $X \times Y$, and ρ is the restriction of pr_{23} to it, which is the identity of $X \times Y$; so the outer map is $\gamma_f^!$, again by proposition 11.3.7. The inner map is evaluated by lemma 12.2.5 and the unit law theorem 12.2.7(4):
$$\delta^!_{\mathrm{pr}_{12}}\big((\alpha \times [X]) \times \beta\big) = \big(\alpha \cdot [X]\big) \times \beta = \alpha \times \beta$$
which gives the stated identity. Taking $\alpha = [X]$ and $\beta = [Y]$, the left side is $\gamma_f^![X \times Y]$; the variety $X \times Y$ meets Γ_f in Γ_f itself, a regular embedding of codimension n, so proposition 11.3.3 gives $\gamma_f^![X \times Y] = [\Gamma_f] = [X]$. Hence $[X] \cdot f^*[Y] = [X]$, and $f^*[Y] = [X]$ by the unit law.
2. Let f be proper. On cycles, $(f \times \mathrm{id}_Y)_*\big(\alpha \times \beta\big) = f_*\alpha \times \beta$: for subvarieties $V \subseteq X$ and $W \subseteq Y$ the image of $V \times W$ is $f(V) \times W$, and the field extension $k(f(V) \times W) \subseteq k(V \times W)$ is obtained from $k(f(V)) \subseteq k(V)$ by tensoring with $k(W)$ over $\overline{k}$ and passing to fraction fields, so the two degrees agree and definition 8.2.1 applies. Now $(f \times \mathrm{id}_Y)^{-1}(\Delta_Y) = \Gamma_f$ by lemma 12.2.4(3), and Γ_f is regularly embedded of codimension n, so proposition 12.2.6 identifies $(\delta_Y)^!_{f \times \mathrm{id}}$ with $\gamma_f^!$. Applying theorem 11.3.10(1) with $X' = Y \times Y$ and the identity,

$X'' = X \times Y$ and $f \times \mathrm{id}_Y$, and with $f \times \mathrm{id}_Y$ as the proper morphism over $Y \times Y$, whose induced morphism of preimages is f itself,

$$f_*\alpha \cdot \beta \;=\; \delta_Y^!\big((f \times \mathrm{id}_Y)_*(\alpha \times \beta)\big) \;=\; f_*\Big(\gamma_f^!\big(\alpha \times \beta\big)\Big) \;=\; f_*\big(\alpha \cdot f^*\beta\big)$$

the last step by part (1).

3. Degrees were checked after definition 12.2.8, additivity is clear from the definition, and $f^*[Y] = [X]$ is part (1). For multiplicativity, let $\beta, \beta' \in A_*(Y)$ and put $\Xi = [X] \times \beta \times \beta' \in A_*(X \times Y \times Y)$. By lemma 12.2.5,

$$[X] \times \big(\beta \cdot \beta'\big) = (\delta_Y)^!_{\mathrm{pr}_{23}}(\Xi), \qquad \big(f^*\beta\big) \times \beta' = (\gamma_f)^!_{\mathrm{pr}_{12}}(\Xi)$$

with $\mathrm{pr}_{23} \colon X \times Y \times Y \to Y \times Y$ and $\mathrm{pr}_{12} \colon X \times Y \times Y \to X \times Y$. Applying $\gamma_f^!$ to the first identity gives $f^*(\beta \cdot \beta')$, the map being refined along $\mathrm{pr}_{12}|_{X \times \Delta_Y}$, which is the identity of $X \times Y$. Corollary 11.4.8, for the regular embeddings γ_f and δ_Y with the projections pr_{12} and pr_{23}, exchanges the order:

$$f^*\big(\beta \cdot \beta'\big) = \gamma_f^!\Big((\delta_Y)^!_{\mathrm{pr}_{23}}\Xi\Big) = (\delta_Y)^!_{\psi}\Big((\gamma_f)^!_{\mathrm{pr}_{12}}\Xi\Big)$$

where $\psi \colon \Gamma_f \times Y \to Y \times Y$ is the restriction of pr_{23}. The first and third projections identify $\Gamma_f \times Y$ with $X \times Y$, and under that identification ψ is $f \times \mathrm{id}_Y$, whose preimage of Δ_Y is Γ_f; so $(\delta_Y)^!_\psi = \gamma_f^!$ by proposition 12.2.6. Hence

$$f^*\big(\beta \cdot \beta'\big) = \gamma_f^!\big(f^*\beta \times \beta'\big) = f^*\beta \cdot f^*\beta'$$

the last equality by part (1) with $\alpha = f^*\beta$.

4. For $f = \mathrm{id}_X$ the graph is the diagonal, and part (1) with $\alpha = [X]$ gives $\mathrm{id}^*\beta = [X] \cdot \beta = \beta$. For the composite, let $\zeta \in A_*(Z)$ and put $\Theta = [X \times Y] \times \zeta \in A_*(X \times Y \times Z)$. By lemma 12.2.5 and $[X \times Y] = [X] \times [Y]$,

$$[X] \times g^*\zeta = [X] \times \gamma_g^!\big([Y] \times \zeta\big) = (\gamma_g)^!_{\mathrm{pr}_{23}}(\Theta)$$

with $\mathrm{pr}_{23} \colon X \times Y \times Z \to Y \times Z$, so $f^*g^*\zeta = \gamma_f^!\big((\gamma_g)^!_{\mathrm{pr}_{23}}\Theta\big)$, the outer map being read on $X \times \Gamma_g \cong X \times Y$. Corollary 11.4.8, for the regular embeddings γ_f and γ_g with the projections pr_{12} and pr_{23}, exchanges the order, giving $(\gamma_g)^!_\rho\big((\gamma_f)^!_{\mathrm{pr}_{12}}\Theta\big)$ with $\rho \colon \Gamma_f \times Z \to Y \times Z$ the restriction of pr_{23}. By lemma 12.2.5 and part (1),

$$(\gamma_f)^!_{\mathrm{pr}_{12}}\Theta = \gamma_f^![X \times Y] \times \zeta = [X] \times \zeta$$

Identifying $\Gamma_f \times Z$ with $X \times Z$, the morphism ρ becomes $f \times \mathrm{id}_Z$, whose preimage of Γ_g is $\{(x,z) \big| z = g(f(x))\} = \Gamma_{g \circ f}$, regularly embedded of codimension $\dim Z$ by lemma 12.2.4(3). So proposition 12.2.6 identifies $(\gamma_g)^!_\rho$ with $\gamma^!_{g \circ f}$, and

$$f^*g^*\zeta \;=\; \gamma^!_{g\circ f}\big([X] \times \zeta\big) \;=\; (g \circ f)^*\zeta$$

completing the proof.

Part (1) says that intersecting with the graph is the same as pulling back and then multiplying, which is what makes part (2) a statement about a ring rather than about a Gysin map. The projection formula is the identity used most often downstream: it converts a computation on X into a computation on Y whenever a class on X arose by restriction, and it is how an intersection number computed on a subvariety is compared with one computed on the ambient variety.

Smoothness of the source is more than the construction needs. What was used about X is that its graph in $X \times Y$ is regularly embedded, and by lemma 12.2.4(3) that holds for any algebraic scheme mapping to a smooth variety. Dropping smoothness of the target as well requires a hypothesis on the morphism instead: it must factor as a regular embedding followed by a smooth morphism.

Definition 12.2.11: Gysin Pullback for a Local Complete Intersection Morphism

Throughout this definition f is assumed to admit a *global* factorization $f = p \circ i$ with i a regular embedding and p smooth. The upstream definition (definition 5.1.50) requires such a factorization only locally on the source, which is strictly weaker and does not suffice to fix a single i and p; the global hypothesis is Fulton's convention and is what the construction below needs. Let $f\colon X \to Y$ be a local complete intersection morphism of algebraic schemes (definition 5.1.50) which admits a factorization on all of X, and fix one: $f = p \circ i$ with $i\colon X \hookrightarrow P$ a regular embedding of codimension d and $p\colon P \to Y$ smooth of relative dimension e. The *Gysin pullback* of f is

$$f^! \;=\; i^! \circ p^* \;:\; A_k(Y) \longrightarrow A_{k+e-d}(X)$$

where p^* is flat pullback (definition 8.3.3) and $i^!$ is the Gysin map of definition 11.3.1, read for $d = 0$ as flat pullback along i, which is then an open and closed immersion. The integer $e - d$ is the *virtual relative dimension* of f.

The factorization on all of X is a hypothesis, not a consequence. The definition just cited asks only that every point of X have a neighbourhood on which f factors in this way, and local factorizations need not be glued into one on all of X; without one there is no p^* and no $i^!$ to compose, and $f^!$ is not defined. Wherever $f^!$ is written a factorization on all of X is therefore part of the data. The cases used below carry an evident one: a regular embedding and a smooth morphism are their own factorizations, and a morphism of smooth varieties is factored by its graph.

Theorem 12.2.12: The Gysin Pullback Is Well Defined

Let $f\colon X \to Y$ be a local complete intersection morphism admitting a factorization on all of X.

1. $f^!$ and the virtual relative dimension $e - d$ are independent of the chosen factorization.
2. If f is smooth of relative dimension e then $f^!$ is flat pullback, and if f is a regular embedding then $f^!$ is its Gysin map.
3. If X and Y are smooth varieties then every morphism $f\colon X \to Y$ meets the hypotheses above, its graph supplying a factorization on all of X, and $f^!$ is the pullback f^* of definition 12.2.8.

***Proof*:**

1. Let $f = p\circ i = p'\circ i'$ be two factorizations, with data (P, i, p, d, e) and (P', i', p', d', e'). Form $P'' = P \times_Y P'$ with projections $q\colon P'' \to P$ and $q'\colon P'' \to P'$, and set $p'' = p \circ q = p' \circ q'$. Being base changes of p' and p, the morphisms q and q' are smooth of relative dimensions e' and e, so p'' is smooth of relative dimension $e + e'$. It suffices to prove $i^! \circ p^* = (i'')^! \circ (p'')^*$ for $i'' = (i, i')\colon X \to P''$, since the same argument with the roles of the two factorizations exchanged gives $i'^! \circ p'^* = (i'')^! \circ (p'')^*$.

   Step 1: factoring i''. Write $W_1 = q^{-1}(X) = X \times_Y P'$, a closed subscheme of P'', and let $\tilde{i}\colon W_1 \hookrightarrow P''$ be its inclusion; since q is flat, lemma 12.2.4(2) makes $\tilde{i}$ a regular embedding of codimension d. The projection $\bar{q}\colon W_1 \to X$ is the base change of p' along f, hence smooth of relative dimension e', and $\gamma = (\mathrm{id}_X, i')\colon X \to W_1$ is a section of it, hence a regular embedding of codimension e' by lemma 12.2.4(1). As $i'' = \tilde{i} \circ \gamma$, theorem 11.4.7 gives $(i'')^! = \gamma^! \circ \tilde{i}^!$ and makes i'' a regular embedding of codimension $d + e'$.

   Step 2: the comparison. Let $\beta \in A_k(Y)$. Flat pullback is functorial by proposition 8.3.5(3), so $(p'')^*\beta = q^*\big(p^*\beta\big)$. Applying theorem 11.3.10(2) to the regular embedding i with $X' = P$ and the identity, $X'' = P''$ and q, and with the flat q as the morphism over P, and using proposition 12.2.6 to read the refined map along q as $\tilde{i}^!$,

   $$\tilde{i}^!\big((p'')^*\beta\big) \;=\; \bar{q}^*\big(i^! p^*\beta\big)$$

   Since γ is a section of the smooth morphism $\bar{q}$, theorem 11.4.2 gives $\gamma^! \circ \bar{q}^* = \mathrm{id}$, whence

   $$(i'')^!(p'')^*\beta \;=\; \gamma^!\bar{q}^*\big(i^! p^*\beta\big) \;=\; i^! p^*\beta$$

   Step 3: the degree. Step 1 exhibits i'' as a regular embedding of codimension $d + e'$, and the same step applied to the other factorization exhibits it as one of codimension $d' + e$. The codimension of a regular embedding is determined by the rank of its conormal sheaf, so $d + e' = d' + e$ and therefore $e - d = e' - d'$.

2. If f is smooth, take $P = X$ with $i = \mathrm{id}_X$, a regular embedding of codimension zero, and $p = f$; then $f^! = \mathrm{id}^* \circ f^* = f^*$. If f is a regular embedding, take $P = Y$ with $p = \mathrm{id}_Y$, smooth of relative dimension zero, and $i = f$; then $f^!$ is the Gysin map of f. Part (1) makes both readings legitimate.

3. Let X and Y be smooth of dimensions m and n. Factor f as

   $$X \xrightarrow{\gamma_f} X \times Y \xrightarrow{\mathrm{pr}_Y} Y$$

   The first map is a regular embedding of codimension n by lemma 12.2.4(3), and the second is smooth of relative dimension m, being the base change of $X \to \operatorname{Spec} \bar{k}$. So f is a local complete intersection morphism with a factorization on all of X, and by part (1) its Gysin pullback may be computed from this factorization: $f^!\beta = \gamma_f^!\big(\mathrm{pr}_Y^*\beta\big)$, which is definition 12.2.8, completing the proof.

The virtual relative dimension is the bookkeeping device the name suggests: for a smooth morphism it is the actual relative dimension, for a regular embedding it is minus the codimension, and for a general local complete intersection morphism it is the difference of the two, which need not be realized by any fibre. That generality is what Grothendieck-Riemann-Roch is stated in, the morphisms there being proper local complete intersection morphisms between possibly singular schemes.

One classical statement remains to be placed. The construction in this chapter never moved a cycle; the ambient embedding was degenerated instead. The classical construction did the opposite, and it rested on the following.

Theorem 12.2.13: The Moving Lemma

Let X be a smooth quasi-projective variety of dimension n, let α be a k-cycle on X and β an l-cycle on X. Then there is a k-cycle α' rationally equivalent to α such that α' and β intersect properly: every irreducible component of $|\alpha'| \cap |\beta|$ has dimension $k+l-n$. Moreover the cycle class obtained by counting the components of $|\alpha'| \cap |\beta|$ with their intersection multiplicities depends only on the classes of α and β.

No proof is given, and none is used. The lemma is stated because it is what the intersection product looked like before chapter 11: a product defined for cycles in general position and extended to all pairs by first moving one of them, the whole construction resting on the second assertion above, that the answer does not depend on how the cycle was moved. That second assertion is the difficult half, and it is what makes the moving lemma a theorem rather than a definition; the argument goes back to Chow, and a complete treatment is in Fulton [24].

Two costs were paid on that route and are not paid here. The lemma requires quasi-projectivity, so the classical product exists only on quasi-projective varieties, whereas theorem 12.2.7 holds on every smooth variety. And a product defined by moving is defined only up to the choice made, so every property of it, commutativity included, has to be proved compatible with all choices. The deformation construction makes no choice: $\delta^!$ is a homomorphism on classes from the outset, and the proofs above use only its functorialities.

Exercise 12.2.1

1. Let L and L' be distinct lines in $\mathbb{P}^3$ meeting in a point. Compute $[L] \cdot [L']$ in $A^*(\mathbb{P}^3)$ from the grading alone, and compute $[L] \cdot [H]$ for a plane H containing neither line. Then explain why the first answer is not a defect of the theory, by exhibiting a cycle rationally equivalent to $[L]$ whose support misses L'.

2. Let X be a smooth variety and let $V, W \subseteq X$ be subvarieties with $V \cap W = \emptyset$. Show that $[V] \cdot [W] = 0$ in $A^*(X)$, working from the definition of the product rather than from the grading.

3. Let $f\colon X \to Y$ be a proper morphism of smooth varieties of the same dimension n, with $\deg f$ the degree of definition 8.2.1. Show that $f_*\big(f^*\beta\big) = (\deg f)\,\beta$ for every $\beta \in A^*(Y)$, and deduce that f^* is injective on $A^*(Y) \otimes \mathbb{Q}$ when $\deg f \neq 0$.

4. Let $p\colon P \to Y$ be smooth of relative dimension e and let $i\colon Z \hookrightarrow P$ be a regular embedding of codimension $d \geq 1$. Show that $p \circ i$ is a local complete intersection morphism of virtual relative dimension $e-d$. Then take $Y = \operatorname{Spec} \overline{k}$, $P = \mathbb{P}^n$ and $Z = \mathbb{P}^{n-d}$ a linear subspace, and compute the image of the generator of $A_0(\operatorname{Spec} \overline{k})$ under $(p \circ i)^!$.

5. Let $i\colon \mathbb{P}^2 \hookrightarrow \mathbb{P}^3$ be a plane and let $D \subseteq \mathbb{P}^3$ be a surface of degree r not containing it, so that $[D] = r\,h$ in $A^1(\mathbb{P}^3)$ by example 8.5.4. Compute $i^*[D]$ in $A^1(\mathbb{P}^2)$, and verify the projection

formula $i_*\big(i^*[D]\big) = [D] \cdot [\mathbb{P}^2]$ by computing both sides in $A^*(\mathbb{P}^3)$.

12.3 Self-Intersection and Excess

The Gysin map returns the fundamental cycle of the intersection scheme when the intersection is as proper as it can be (proposition 11.3.3). At the opposite extreme, when the cycle being intersected already lies inside Z, nothing proved so far evaluates it at all: the second half of example 11.3.8 left exactly that computation open, and for the same reason proposition 11.3.5 reached only the first clause of definition 9.2.1. This section closes the gap. The answer is the top Chern class of the normal bundle, and the statement interpolating between the two extremes, in which the intersection has the wrong dimension by a prescribed amount, is the excess intersection formula.

Everything below is stated for the Gysin map of a regular embedding, so by section 12.1 it is at the same time a statement about the product: on a smooth variety the product is $\delta^!$ of an exterior product, and δ is a regular embedding like any other. The formulas are then spent on blow-ups, which is where a subvariety is forced to meet the locus it is intersected with in the worst possible way.

Throughout, $i\colon Z \hookrightarrow X$ is a regular embedding of codimension $d \geq 1$ of algebraic schemes, with ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$ and normal bundle $\pi\colon N \to Z$.

The reduction to a single computation is immediate from the construction. Apply definition 11.3.6 to the morphism $g = i$ itself: the preimage of Z in Z is Z, so the normal cone in question is $C_Z Z$, the zero cone of example 10.4.2(2), and the immersion of that cone into N is the vertex, that is the zero section. Specialization is therefore the identity, and the whole matter is the identification of the class of the zero section inside the Chow group of the total space of the bundle.

Lemma 12.3.1: The Class of the Zero Section

Let $\pi\colon N \to Z$ be a vector bundle of rank $d \geq 1$ on an algebraic scheme Z, with zero section $\zeta\colon Z \to N$. Then for every $\alpha \in A_k(Z)$,

$$\zeta_*\alpha \;=\; \pi^*\big(c_d(N) \cap \alpha\big) \quad \text{in } A_k(N)$$

equivalently $(\pi^*)^{-1}\zeta_*\alpha = c_d(N) \cap \alpha$, the top Chern class of N applied to α.

***Proof*:**
Write $P = \mathbb{P}(N \oplus 1)$ for the projective completion of definition 10.4.3, where $N \oplus 1 = N \oplus \mathcal{O}_Z$ has rank $d+1$, with projection $p\colon P \to Z$ and tautological class $\eta = c_1(\mathcal{O}_P(1))$. Let $u\colon N \hookrightarrow P$ be the open immersion, so that $p \circ u = \pi$, and let $\bar\zeta = u \circ \zeta\colon Z \to P$, a section of p and hence a closed immersion. Fix $\alpha \in A_k(Z)$ and put $\beta = \bar\zeta_*\alpha \in A_k(P)$.

Step 1: two trivializations. Writing $S^\bullet = \mathrm{Sym}^\bullet(N^\vee)$, the completion is $P = \mathrm{Proj}\,\big(S^\bullet[z]\big)$ with z the degree-one variable dual to the trivial summand, and N is the chart $D_+(z)$. On that chart the sheaf $\mathcal{O}_P(1)$ is free on the generator z, so

$$\mathcal{O}_P(1)\big|_N \;\cong\; \mathcal{O}_N$$

Since $\bar\zeta$ factors through N, it follows that $\bar\zeta^*\mathcal{O}_P(1) \cong \mathcal{O}_Z$ as well. Finally, the exact sequence $0 \to N \to N \oplus 1 \to \mathcal{O}_Z \to 0$ and theorem 10.2.15 give $c(N \oplus 1) = c(N)\,c(\mathcal{O}_Z) = c(N)$, the total Chern class of a trivial bundle being the identity (example 10.2.12(1)); in particular $c_i(N \oplus 1) = c_i(N)$ for every i, and this vanishes for $i > d$ by theorem 10.2.9(2).

Step 2: η annihilates β. The morphism $\bar{\zeta}$ is proper, being a closed immersion, so the projection formula (proposition 9.4.4(4)) and Step 1 give

$$\eta \cap \bar{\zeta}_* \alpha \;=\; \bar{\zeta}_*\Big(c_1\big(\bar{\zeta}^* \mathcal{O}_P(1)\big) \cap \alpha\Big) \;=\; \bar{\zeta}_*\big(c_1(\mathcal{O}_Z) \cap \alpha\big) \;=\; 0$$

the last equality because c_1 of a trivial line bundle is the zero operator (example 9.4.3).

Step 3: the top coefficient. The bundle $N \oplus 1$ has rank $d+1$, so theorem 10.1.9 writes β uniquely as

$$\beta \;=\; \sum_{j=0}^{d} \eta^j \cap p^* \alpha_j, \qquad \alpha_j \in A_{k-d+j}(Z) \tag{12.1}$$

By definition 10.1.3 applied to $N \oplus 1$, whose projective bundle has relative dimension d, one has $p_*\big(\eta^j \cap p^*\gamma\big) = s_{j-d}(N \oplus 1) \cap \gamma$, which vanishes for $j < d$ and is γ for $j = d$ by proposition 10.1.5(1),(2). Applying p_* to (12.1) therefore gives $p_*\beta = \alpha_d$. On the other hand $p \circ \bar{\zeta} = \mathrm{id}_Z$, so functoriality of proper pushforward (definition 8.2.1) gives $p_*\beta = \alpha$. Hence $\alpha_d = \alpha$.

Step 4: the remaining coefficients. Theorem 10.2.9(1) applied to $N \oplus 1$, together with the identification of its Chern classes in Step 1, reads

$$\sum_{i=0}^{d} \eta^{d+1-i} \cap p^*\big(c_i(N) \cap \gamma\big) \;=\; 0$$

for every $\gamma \in A_*(Z)$, so that $\eta^{d+1} \cap p^*\gamma = -\sum_{i=1}^{d} \eta^{d+1-i} \cap p^*(c_i(N) \cap \gamma)$. Capping (12.1) with η, using Step 2 on the left and this relation on the term $j = d$, and reindexing the resulting sum by $j = d - i$,

$$0 \;=\; \sum_{j=0}^{d-1} \eta^{j+1} \cap p^* \alpha_j \;-\; \sum_{j=0}^{d-1} \eta^{j+1} \cap p^*\big(c_{d-j}(N) \cap \alpha_d\big) \;=\; \sum_{m=1}^{d} \eta^m \cap p^*\big(\alpha_{m-1} - c_{d-m+1}(N) \cap \alpha_d\big)$$

The exponents occurring lie in $\{0, 1, \ldots, d\}$, so this is a representation of the zero class in the form of theorem 10.1.9, and uniqueness forces every coefficient to vanish: $\alpha_{m-1} = c_{d-m+1}(N) \cap \alpha_d$ for $1 \le m \le d$. Taking $m = 1$ and using Step 3,

$$\alpha_0 \;=\; c_d(N) \cap \alpha$$

Step 5: restriction to the bundle. The open immersion u is flat of relative dimension zero. The square with $\bar{\zeta}$ and ζ over u is a fibre square, $\bar{\zeta}$ factoring through N, so proposition 8.3.6 gives $u^*\beta = \zeta_*\alpha$. Applying u^* to (12.1), every term with $j \ge 1$ contains a factor $\eta \cap -$, whose pullback is $c_1(\mathcal{O}_P(1)|_N) \cap u^*(-) = 0$ by proposition 9.4.4(4) and Step 1; and the term $j = 0$ is $u^* p^* \alpha_0 = \pi^* \alpha_0$ by proposition 8.3.5(3). Hence $\zeta_*\alpha = \pi^*\alpha_0 = \pi^*(c_d(N) \cap \alpha)$, as we sought to show.

The zero section meets every fibre of N in exactly one point, and yet the class it carries is not the pullback of α but the pullback of α cut down by the top Chern class. On a trivial bundle of rank $d \ge 1$ the top Chern class vanishes and the zero section therefore carries the zero class, and the same holds whenever N admits a nowhere-vanishing section: such a section spans a trivial line subbundle with locally free quotient Q of rank $d-1$, so theorem 10.2.15 gives $c(N) = c(Q)$ and $c_d(N) = 0$, no bundle carrying a nonzero Chern class above its rank. Nonvanishing of $\zeta_*\alpha$ is therefore an obstruction to the existence of such a section and not a criterion for one. On $\mathbb{P}^1$ the bundle $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(1)$ has c_2 the zero operator, the groups $A_{k-2}(\mathbb{P}^1)$ vanishing for every $k \leq 1$, and yet $\mathcal{O}_{\mathbb{P}^1}(-1)$ has no nonzero section, so every section of the sum vanishes wherever its second coordinate does. Since π^* is an isomorphism (theorem 8.4.6), the identity is equivalent to the evaluation of $(\pi^*)^{-1}$ on classes supported on the zero section, and that is what the Gysin map of chapter 11 was waiting for.

Theorem 12.3.2: The Self-Intersection Formula

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle N. Then for every $\alpha \in A_k(Z)$:

1. the refined Gysin homomorphism of i along i itself is the operator $c_d(N) \cap -$, that is $i^!_i\alpha = c_d(N) \cap \alpha$ in $A_{k-d}(Z)$, in the notation of box 12.2.1;

2. the Gysin map of definition 11.3.1 satisfies

$$i^!(i_*\alpha) \;=\; c_d(N) \cap \alpha \quad \text{in } A_{k-d}(Z)$$

***Proof*:**

1. Apply definition 11.3.6 with $X' = Z$ and $g = i$. The ideal sheaf of Z in Z is zero, so the preimage $Z' = i^{-1}(Z)$ is Z itself, $h = \mathrm{id}_Z$ and $N' = h^*N = N$. The associated graded algebra of the zero ideal is $\mathcal{O}_Z$ concentrated in degree zero, so $C_{Z'}X' = C_Z Z$ is the zero cone of example 10.4.2(2), whose total space is Z; and the immersion $j\colon C_Z Z \hookrightarrow N$ of lemma 10.4.13 is induced by the surjection of graded algebras $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2) \to \mathcal{O}_Z$ killing every positive degree, which is the vertex of definition 10.4.1, that is the zero section ζ. For the specialization homomorphism $\sigma'\colon A_k(Z) \to A_k(C_Z Z) = A_k(Z)$ of the pair (Z, Z), theorem 11.2.7 gives $\sigma'[V] = [C_V V] = [V]$ for every k-dimensional subvariety $V \subseteq Z$, the preimage of Z in V being V. Such classes generate $A_k(Z)$ (definition 8.1.4), so σ' is the identity. Therefore

$$i^!_i\alpha \;=\; (\pi^*)^{-1} j_*\sigma'\alpha \;=\; (\pi^*)^{-1}\zeta_*\alpha \;=\; c_d(N) \cap \alpha$$

   the last equality by lemma 12.3.1.

2. Apply theorem 11.3.10(1) with $X' = X$, $X'' = Z$ and $p = i$, which is proper as a closed immersion. The preimage of Z in Z is Z and the induced morphism q is id_Z, so the identity reads $i^!(i_*\alpha) = i^!_i\alpha$, where the left-hand side is the refined map along id_X, which is the Gysin map of definition 11.3.1 by proposition 11.3.7, and the right-hand side is the refined map along i evaluated in part (1). Combining the two gives $i^!(i_*\alpha) = c_d(N) \cap \alpha$, completing the proof.

The theorem says what it means to intersect a subvariety with itself. Pushing α into X and pulling it back records no information about how Z sits inside X except through the normal bundle, and the whole of that information is the single operator $c_d(N)$. Two consequences are worth separating. The composite $i^!i_*$ is not the identity, and it is not even injective: on a subvariety with trivial normal bundle it is zero. And the sign of a self-intersection number, which has no counterpart in a naive count of intersection points, is the sign of a Chern number of N; this is why the exceptional curve of a blow-up can meet itself a negative number of times. Since the Gysin pullback of an l.c.i. morphism restricts to the Gysin map on a regular embedding (theorem 12.2.12(2)), part (2) is at the same time a statement about the pullback of definition 12.2.11: for the l.c.i. morphism i the composite $i^! \circ i_*$ is the operator $c_d(N) \cap -$.

In codimension one the theorem completes the comparison begun in proposition 11.3.5, which matched the Gysin map with the operator $D \cdot -$ of chapter 9 only for cycles not contained in the support of D.

Corollary 12.3.3: The Gysin Map on a Divisor Class

Let D be an effective Cartier divisor on an algebraic scheme X with $D \neq X$, and let $i\colon D \hookrightarrow X$ be the inclusion. Then for every $\alpha \in A_k(D)$,

$$i^!(i_*\alpha) \;=\; c_1\big(\mathcal{O}_X(D)|_D\big) \cap \alpha$$

In particular, for a k-dimensional subvariety $V \subseteq D$ the class $i^![V]$ is the image in $A_{k-1}(D)$ of the intersection class $D \cdot [V]$ computed by the second clause of definition 9.2.1.

***Proof*:**
The inclusion of an effective Cartier divisor is a regular embedding of codimension one with normal bundle $N = \mathcal{O}_X(D)|_D$ (example 10.4.15), and $c_1(N)$ is the top Chern class of a line bundle, so theorem 12.3.2(2) gives the displayed identity. For the second assertion, let $V \subseteq D$ be a k-dimensional subvariety with inclusion ι_V into D. Since V is contained in $|D|$, the second clause of definition 9.2.1 computes $D \cdot [V] = c_1(\mathcal{O}_X(D)|_V) \cap [V]$ in $A_{k-1}(V)$. Pushing that class forward along ι_V and applying the projection formula (proposition 9.4.4(4)) to the proper morphism ι_V, whose pullback of $\mathcal{O}_X(D)|_D$ is $\mathcal{O}_X(D)|_V$, gives $\iota_{V*}(D \cdot [V]) = c_1(\mathcal{O}_X(D)|_D) \cap [V]$, which is $i^!(i_*[V])$ by the first assertion, as we sought to show.

Example 12.3.4: A Line Meeting Itself in the Plane

Let $X = \mathbb{P}^2$ and $L = \{x_0 = 0\}$, an effective Cartier divisor with $\mathcal{O}_{\mathbb{P}^2}(L) \cong \mathcal{O}_{\mathbb{P}^2}(1)$ (example 9.1.4), so that $N = \mathcal{O}_{\mathbb{P}^2}(1)|_L \cong \mathcal{O}_{\mathbb{P}^1}(1)$ (example 11.1.7). The second half of example 11.3.8 produced the class $i_i^![L] = (\pi^*)^{-1}\zeta_*[L]$, the refined Gysin map along i applied to $[L]$, and could go no further. Theorem 12.3.2(1) evaluates it:

$$i_i^![L] \;=\; c_1\big(\mathcal{O}_{\mathbb{P}^1}(1)\big) \cap [L] \;=\; [\mathrm{pt}]$$

by example 9.4.7, a single point of L with multiplicity one. This is the value predicted there from example 9.2.3, where $L \cdot [L] = [q]$ was computed from the restricted line bundle, and corollary 12.3.3 identifies the two computations in general rather than in this instance.
For a smooth conic $C \subseteq \mathbb{P}^2$ the same argument gives $N = \mathcal{O}_{\mathbb{P}^2}(2)|_C$, and example 10.4.15 evaluated $c_1(\mathcal{O}_{\mathbb{P}^2}(2)|_C) \cap [C] = 4\,[\mathrm{pt}]$, so $i^!(i_*[C]) = 4\,[\mathrm{pt}]$. The self-intersection number of a plane curve of degree m is m^2, and the Gysin map records it on the curve itself rather than in $A_0(\mathbb{P}^2)$.

The examples so far are divisors, where the normal bundle is a line bundle and $c_d(N)$ is the operator of chapter 9. The construction that supplies higher codimension, and with it the geometry the formula was built for, is the blow-up: its exceptional divisor is a projectivized normal bundle, and it meets itself in every point.

Lemma 12.3.5: The Exceptional Divisor of a Blow-up Along a Regular Centre

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with $Z \neq \emptyset$, ideal sheaf $\mathcal{I}$ and normal bundle N. Let $f\colon \widetilde{X} = \mathrm{Bl}_Z X \to X$ be the blow-up (definition 7.2.26) and let $\widetilde{Z} = f^{-1}(Z)$ be its exceptional divisor, the closed subscheme cut out by $\mathcal{I}' = \mathcal{I}\mathcal{O}_{\widetilde{X}}$. Then:

1. $\widetilde{Z} = \mathbb{P}(N)$, with the restriction $g\colon \mathbb{P}(N) \to Z$ of f as its bundle projection;
2. $j\colon \widetilde{Z} \hookrightarrow \widetilde{X}$ is a regular embedding of codimension one whose conormal surjection $g^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is the universal surjection $g^*(N^\vee) \to \mathcal{O}_{\mathbb{P}(N)}(1)$ of box 10.2.5. In particular the normal bundle of j is $\mathcal{O}_{\mathbb{P}(N)}(-1)$, sitting inside g^*N as the tautological subbundle with quotient the universal quotient bundle Q of rank $d-1$.

***Proof*:**

Realize the Rees algebra as the subalgebra $S = \bigoplus_{n\geq 0} \mathcal{I}^n t^n$ of $\mathcal{O}_X[t]$, so that $\widetilde{X} = \mathrm{Proj}\,(S)$ over X by definition 7.2.26. The algebra S is generated in degree one, so the affine charts $D_+(at)$ attached to local sections a of $\mathcal{I}$ cover $\widetilde{X}$; on such a chart $\widetilde{X}$ is $\mathrm{Spec}\,(S_{(at)})$, whose elements are the fractions y/a^n with y a local section of $\mathcal{I}^n$.

1. The homogeneous ideal $\mathcal{I}S \subseteq S$ has degree-n part $\mathcal{I} \cdot \mathcal{I}^n t^n = \mathcal{I}^{n+1} t^n$, so
$$S/\mathcal{I}S \;=\; \bigoplus_{n\geq 0} \mathcal{I}^n/\mathcal{I}^{n+1}$$
which is $\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$ by lemma 10.4.10, glued over an affine cover as in the proof of proposition 10.4.11. Taking relative Proj of the surjection $S \to S/\mathcal{I}S$ exhibits $\mathrm{Proj}\left(\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)\right)$ as the closed subscheme of $\widetilde{X}$ cut out by the ideal generated by $\mathcal{I}$, that is by $\mathcal{I}'$, and its structure morphism to Z is the restriction of f. Since N has $(\mathcal{I}/\mathcal{I}^2)^\vee$ as its sheaf of sections (definition 5.1.45), that relative Proj is $\mathbb{P}(N)$ in the convention of box 7.3.10.
2. *The ideal is invertible.* On the chart $D_+(at)$ the ideal $\mathcal{I}'$ is generated by the image of a, since every local section y of $\mathcal{I}$ satisfies $y = a \cdot (y/a)$ with y/a in $S_{(at)}$. That generator is a nonzerodivisor: if $a \cdot (y/a^n) = 0$ in $S_{(at)}$ with y a local section of $\mathcal{I}^n$, then $(at)^m\, a\, y\, t^n = 0$ in $\mathcal{O}_X[t]$ for some $m \geq 0$, that is $a^{m+1} y = 0$, and the same relation forces $y/a^n = 0$ in $S_{(at)}$. Hence $\widetilde{Z}$ is an effective Cartier divisor and j is a regular embedding of codimension one (example 10.4.15), which also recovers proposition 7.2.27(3); its conormal sheaf is $\mathcal{I}'/\mathcal{I}'^2 = \mathcal{I}' \otimes \mathcal{O}_{\widetilde{Z}}$, invertible on $\widetilde{Z}$.
Which invertible sheaf. The ideal $\mathcal{I}'$ is generated by the image of $\mathcal{I}$, so the canonical map $g^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is surjective. Compare it on $D_+(at) \cap \widetilde{Z}$ with the universal surjection of $\mathbb{P}(N) = \mathrm{Proj}\,\mathrm{Sym}^\bullet(\mathcal{I}/\mathcal{I}^2)$. There $\mathcal{I}'/\mathcal{I}'^2$ is free on the class of a by the previous paragraph, while $\mathcal{O}_{\mathbb{P}(N)}(1)$ is free on the image $\bar{a}$ of the degree-one element at. A local section b of $\mathcal{I}$ satisfies $b = a \cdot (b/a)$ upstairs, and its class satisfies $\bar{b} = \bar{a} \cdot \overline{b/a}$ downstairs, the reduction $S \to S/\mathcal{I}S$ carrying b/a to $\overline{b/a}$. Under the two trivializations the two surjections therefore have the same coefficient on every local section, so they agree and $\mathcal{I}'/\mathcal{I}'^2 = \mathcal{O}_{\mathbb{P}(N)}(1)$.

Dualizing gives the inclusion $\mathcal{O}_{\mathbb{P}(N)}(-1) \hookrightarrow g^*N$, which is the tautological subbundle of box 10.2.5, whose quotient Q is locally free of rank $d-1$, completing the proof.

The lemma is the reason the blow-up is the standard source of examples for the self-intersection formula: the exceptional divisor is a divisor whose normal bundle is the tautological line bundle, with first Chern class $-\xi$ for the tautological class ξ of section 10.1. In the smallest case the resulting number was computed by hand upstream, and the general formula reproduces it.

Example 12.3.6: The Exceptional Curve of a Blown-up Surface

Let $X = \mathbb{P}^2$ and let $Z = \{P\}$ with $P = [0:0:1]$. On the affine chart $\{x_2 \neq 0\}$ the ideal of P is $(x_0/x_2, x_1/x_2)$, a regular sequence of length two in the polynomial ring of the chart, so $Z \hookrightarrow X$ is a regular embedding of codimension two whose normal bundle N is a rank-two vector bundle over a point, that is a two-dimensional vector space. Write $f\colon \widetilde{X} = \mathrm{Bl}_P\mathbb{P}^2 \to \mathbb{P}^2$ for the blow-up, $\widetilde{Z}$ for its exceptional divisor, written E in theorem 7.2.29, and $j\colon \widetilde{Z} \hookrightarrow \widetilde{X}$ for its inclusion. By lemma 12.3.5,

$$\widetilde{Z} \;=\; \mathbb{P}(N) \;\cong\; \mathbb{P}^1, \qquad N_{\widetilde{Z}}\widetilde{X} \;=\; \mathcal{O}_{\mathbb{P}^1}(-1)$$

The self-intersection formula now evaluates the intersection of $\widetilde{Z}$ with itself. By corollary 12.3.3 and additivity of the first Chern class (proposition 9.4.4(3)),

$$j^!\big(j_*[\widetilde{Z}]\big) \;=\; c_1\big(\mathcal{O}_{\mathbb{P}^1}(-1)\big) \cap [\mathbb{P}^1] \;=\; -\,c_1\big(\mathcal{O}_{\mathbb{P}^1}(1)\big) \cap [\mathbb{P}^1] \;=\; -[\mathrm{pt}]$$

using example 9.4.7. Its degree (definition 8.2.4) is -1, which is the self-intersection number $E^2 = -1$ recorded upstream. The same computation applies at any point of any smooth surface, differing only in the verification that the maximal ideal at that point is generated by a regular sequence of length two; what the general formula contributes is that the number -1 is not an accident of the plane but the degree of the tautological line bundle on the projectivized normal space.

The negative sign is where the Gysin map departs furthest from counting points. A cycle contained in Z has no intersection points with Z to count that are not already in it, and the number the formula returns is a Chern number of the normal bundle, with whatever sign that bundle has. The upstream proof of $E^2 = -1$ argued from the surface pairing and the geometry of the blow-down; the argument here uses no property of surfaces and no property of blow-ups beyond lemma 12.3.5.

Between the proper case, where the answer is the fundamental cycle of $V \cap Z$, and the self-intersection case, where it is a top Chern class, lies everything else: the preimage Z' has some dimension between the expected one and $\dim X'$, and the normal cone fills only part of the pulled-back normal bundle. When the preimage is itself regularly embedded the discrepancy is a vector bundle, and it is the only new input the general formula needs.

The comparison rests on the conormal sheaves. For a regular embedding of codimension d' the conormal sheaf $\mathcal{I}'/\mathcal{I}'^2$ is locally free of rank d' (proposition 5.1.44), and since $\mathcal{I}' = \mathcal{I}\mathcal{O}_{X'}$ is generated by the image of $\mathcal{I}$, the canonical map $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is a surjection of vector bundles. A surjection of vector bundles has locally free kernel, being locally split, so dualizing gives an inclusion of vector bundles with locally free cokernel; in particular $d' \leq d$.

Definition 12.3.7: The Excess Bundle

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle N and ideal sheaf $\mathcal{I}$, let $g\colon X' \to X$ be a morphism of algebraic schemes, and form the fibre square of definition 11.3.6, with $Z' = g^{-1}(Z)$, ideal sheaf $\mathcal{I}' = \mathcal{I}\mathcal{O}_{X'}$, and $h\colon Z' \to Z$. Suppose $i'\colon Z' \hookrightarrow X'$ is a regular embedding of codimension d' with normal bundle N'. The dual of the conormal surjection $h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ is an inclusion of vector bundles $N' \hookrightarrow h^*N$ on Z', and the *excess bundle* of the square is its cokernel

$$E \;=\; h^*N/N'$$

a vector bundle of rank $e = d - d'$ on Z'. When $e = 0$ the excess bundle is the zero bundle, and the operator $c_0(E)$ is then read as the identity, definition 10.2.3 being stated for bundles of positive rank.

The excess bundle measures how much of the normal directions to Z fail to be normal directions to Z' after base change. It vanishes exactly when $d' = d$, that is when the preimage has the expected codimension and the two normal bundles agree; at the other extreme, when $X' = Z$ and $g = i$, the subbundle N' is zero and the excess bundle is all of N. Its rank is the number of dimensions by which the intersection is too large.

The identity that converts the excess bundle into a correction term is the following, which contains lemma 12.3.1 as the case $N' = 0$ and is deduced from it.

Lemma 12.3.8: The Class of a Subbundle

Let $0 \to N' \xrightarrow{j} N \to E \to 0$ be an exact sequence of vector bundles on an algebraic scheme Z, with $\operatorname{rank} E = e$ and $\operatorname{rank} N' = d'$, and write $\pi\colon N \to Z$ and $\pi'\colon N' \to Z$ for the projections. Then for every $\gamma \in A_m(Z)$,

$$j_*\pi'^*\gamma \;=\; \pi^*\bigl(c_e(E) \cap \gamma\bigr) \quad \text{in } A_{m+d'}(N)$$

***Proof*:**
If $e = 0$ then j is an isomorphism of bundles and both sides are $\pi^*\gamma$, the operator $c_0(E)$ being the identity by the convention of definition 12.3.7. So assume $e \geq 1$, write $\pi_E\colon E \to Z$ for the projection and $\zeta_E\colon Z \to E$ for the zero section, and let $\rho\colon N \to E$ be the morphism of total spaces induced by the surjection $N \to E$.

Step 1: the square is a fibre square with ρ flat. Both assertions are local on Z. Over an affine open on which N', N and E are free, the surjection $N \to E$ splits, so there $N \cong N' \times E$ over the base with ρ the second projection, which is flat of relative dimension d', and the scheme-theoretic preimage $\rho^{-1}\bigl(\zeta_E(Z)\bigr)$ is N' embedded by j. Hence ρ is flat of relative dimension d' and

$$\begin{tikzcd} N' \arrow[r, "j"] \arrow[d, "\pi'"'] & N \arrow[d, "\rho"] \\ Z \arrow[r, "\zeta_E"] & E \end{tikzcd}$$

is a fibre square, with proper horizontal maps and flat vertical maps.

Step 2: transport the zero-section identity. Proposition 8.3.6 applied to that square gives $\rho^*\zeta_{E*}\gamma = j_*\pi'^*\gamma$. By lemma 12.3.1 applied to the bundle E, the left-hand side is $\rho^*\pi_E^*(c_e(E)\cap\gamma)$, and $\pi_E \circ \rho = \pi$, so proposition 8.3.5(3) rewrites it as $\pi^*(c_e(E)\cap\gamma)$. Combining the two displays gives the assertion, completing the proof.

The refined Gysin map is built from a specialization followed by a pushforward into the pulled-back normal bundle, and when the preimage is regularly embedded the specialization is already the Gysin map of the preimage; the pushforward is then the subbundle inclusion, and lemma 12.3.8 evaluates it.

Theorem 12.3.9: The Excess Intersection Formula

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with normal bundle N, let $g\colon X' \to X$ be a morphism of algebraic schemes, and suppose the preimage $Z' = g^{-1}(Z)$ is regularly embedded in X' of codimension $d' \geq 1$. Let E be the excess bundle of definition 12.3.7, of rank $e = d - d'$. Then for every $\alpha \in A_k(X')$,

$$i_g^!\alpha \;=\; c_e(E) \cap i'^!\alpha \quad \text{in } A_{k-d}(Z')$$

where $i_g^!$ is the refined Gysin homomorphism of i along g and $i'^!$ is the Gysin map of i', in the notation of box 12.2.1.

***Proof*:**
Write $\mathcal{I}$ and $\mathcal{I}'$ for the ideal sheaves of Z in X and of Z' in X', let $N' = N_{Z'}X'$ with projection π', and let $\varpi\colon h^*N \to Z'$ be the projection of the pulled-back normal bundle.

Step 1: the immersion of the cone is the subbundle inclusion. By lemma 10.4.13 the closed immersion $j\colon C_{Z'}X' \hookrightarrow h^*N$ is induced by the canonical surjection of graded $\mathcal{O}_{Z'}$-algebras

$$\operatorname{Sym}^\bullet\big(h^*(\mathcal{I}/\mathcal{I}^2)\big) \longrightarrow \bigoplus_{n\geq 0} \mathcal{I}'^n/\mathcal{I}'^{n+1}$$

Since i' is a regular embedding, lemma 10.4.10 identifies the target with $\operatorname{Sym}^\bullet(\mathcal{I}'/\mathcal{I}'^2)$ by the canonical surjection, which is the identity in degree one, and identifies $C_{Z'}X'$ with N' (proposition 10.4.11). A homomorphism of graded algebras out of a symmetric algebra is determined by its degree-one part, so the displayed map is $\operatorname{Sym}^\bullet(\varphi)$ for $\varphi\colon h^*(\mathcal{I}/\mathcal{I}^2) \to \mathcal{I}'/\mathcal{I}'^2$ the conormal surjection. Taking relative spectra, $j\colon N' \to h^*N$ is the inclusion of vector bundles dual to φ, that is the inclusion of definition 12.3.7, with cokernel E of rank e.

Step 2: evaluation. Let $\sigma'\colon A_k(X') \to A_k(C_{Z'}X') = A_k(N')$ be the specialization homomorphism of the pair (Z', X'). By definition 11.3.1 applied to i', the Gysin map is $i'^! = (\pi'^*)^{-1} \circ \sigma'$, so $\sigma'\alpha = \pi'^*(i'^!\alpha)$. Hence, using definition 11.3.6 and then lemma 12.3.8 with $\gamma = i'^!\alpha$,

$$\varpi^*\big(i_g^!\alpha\big) \;=\; j_*\sigma'\alpha \;=\; j_*\pi'^*\big(i'^!\alpha\big) \;=\; \varpi^*\Big(c_e(E)\cap i'^!\alpha\Big)$$

The pullback ϖ^* is injective, being an isomorphism by theorem 8.4.6, so $i_g^!\alpha = c_e(E)\cap i'^!\alpha$, as we sought to show.

The formula says that an intersection of the wrong dimension is computed by intersecting properly on the preimage and then correcting by the top Chern class of what was left over. When $e = 0$ the correction is the identity and the two Gysin maps agree, which is the regime of proposition 11.3.3:

the preimage has the expected codimension and the answer is read off it directly. When $e = d$, so that $d' = 0$ and $Z' = X'$, the hypothesis $d' \geq 1$ puts the case outside the statement, though only nominally: definition 12.2.11 reads the Gysin map of a codimension-zero embedding as flat pullback along it, which for $i' = \mathrm{id}_{X'}$ is the identity, so with that reading the uncorrected term is α itself, the correction is all of $c_d(h^*N)$, and for $g = i$ the formula returns theorem 12.3.2(1). The formula also explains the role of the normal cone in the general construction: without a regularity hypothesis on Z' there is no excess bundle, and the discrepancy between $C_{Z'}X'$ and h^*N is recorded by a Segre class instead.

Example 12.3.10: A Plane Through a Line in Three-Space

Let $X = \mathbb{P}^3$ with coordinates $x_0, \ldots, x_3$, let $Z = L = \{x_2 = x_3 = 0\}$ be a line and let $X' = H = \{x_3 = 0\}$ be a plane containing it, with $g\colon H \hookrightarrow \mathbb{P}^3$ the inclusion. By exercise 10.4.1 (2) the embedding i is regular of codimension two with $N = \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus 2}$. The expected dimension of $H \cap L$ is $\dim H - 2 = 0$, but the preimage is $Z' = g^{-1}(L) = L$, of dimension one.

The excess bundle. The line L is an effective Cartier divisor on $H \cong \mathbb{P}^2$ cut out by x_2, so i' is a regular embedding of codimension $d' = 1$ with $N' = \mathcal{O}_{\mathbb{P}^2}(1)|_L \cong \mathcal{O}_{\mathbb{P}^1}(1)$ (example 9.1.4, example 10.4.15). The excess bundle has rank $e = 2 - 1 = 1$, and the Whitney formula (theorem 10.2.15) applied to $0 \to N' \to N \to E \to 0$ gives, with $h = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$,

$$c(E) \;=\; c(N)\, c(N')^{-1} \;=\; (\mathrm{id} + h)^2 (\mathrm{id} + h)^{-1} \;=\; \mathrm{id} + h$$

so $c_1(E) = h$.

The class. The plane H is a variety not contained in $|L|$, and its scheme-theoretic intersection with L is L, so $i'^{!}[H] = [L]$ by proposition 11.3.5, which is the computation of exercise 11.3.1 (2). Theorem 12.3.9 then gives

$$i_g^{!}[H] \;=\; c_1(E) \cap [L] \;=\; h \cap [L] \;=\; [\mathrm{pt}]$$

in $A_0(L)$, of degree one. The uncorrected answer $[L]$ has the wrong dimension and the wrong degree; the correction cuts it down by one dimension and returns the intersection number a plane and a line in general position would have. What the refined map adds is that the class is recorded on L, the locus where the two meet, and not only as a number.

Applying the formula to a blow-up produces the statement that lets Chow groups be computed after blowing up. The square in question is the defining one for the exceptional divisor, and lemma 12.3.5 has already identified all of its data.

Theorem 12.3.11: The Blow-up Formula

Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ with $Z \neq \emptyset$ and normal bundle N, let $f\colon \widetilde{X} = \mathrm{Bl}_Z X \to X$ be the blow-up with exceptional divisor $j\colon \widetilde{Z} = \mathbb{P}(N) \hookrightarrow \widetilde{X}$ and $g\colon \widetilde{Z} \to Z$, and let Q be the universal quotient bundle of rank $d-1$ on $\mathbb{P}(N)$. Then for every $\beta \in A_k(\widetilde{X})$:

1. $i^!_f\beta = c_{d-1}(Q) \cap j^!\beta$ in $A_{k-d}(\widetilde{Z})$, where $i^!_f$ is the refined Gysin homomorphism of i along f and $j^!$ is the Gysin map of j;
2. $i^!(f_*\beta) = g_*\Big(c_{d-1}(Q) \cap j^!\beta\Big)$ in $A_{k-d}(Z)$, where the left-hand $i^!$ is the Gysin map of definition 11.3.1.

***Proof*:**

1. The exceptional divisor is the scheme-theoretic preimage $f^{-1}(Z)$, so the square formed by j, f, i and g is the square of definition 11.3.6 for the morphism f. By lemma 12.3.5(2) the preimage $\widetilde{Z}$ is regularly embedded in $\widetilde{X}$ of codimension $d' = 1$, and its conormal surjection, which is the one definition 12.3.7 dualizes, is there identified with the universal surjection; so the excess bundle of the square is the universal quotient Q, of rank $e = d-1$. Theorem 12.3.9 therefore gives $i^!_f\beta = c_{d-1}(Q) \cap j^!\beta$.
2. The blow-down f is proper (proposition 7.2.27(1)), so theorem 11.3.10(1) applies with $X' = X$, $X'' = \widetilde{X}$ and $p = f$. The preimage of Z in $\widetilde{X}$ is $\widetilde{Z}$ and the induced morphism on preimages is g, so $i^!(f_*\beta) = g_*(i^!_f\beta)$, the left-hand side being the Gysin map of i by proposition 11.3.7. Substituting part (1) completes the proof.

Part (1) computes an intersection with Z upstairs, where the geometry is a divisor and therefore elementary, at the cost of one Chern class of the universal quotient bundle. Part (2) is the form used in practice: it computes the Gysin image of any class pushed down from the blow-up, and it is the identity from which the Chow groups of $\widetilde{X}$ are assembled from those of X and of $\mathbb{P}(N)$. The smallest instance is a consistency check that can be run in both directions.

Example 12.3.12: The Fundamental Class of a Blow-up

Let X be a variety of dimension n and $Z \subseteq X$ a nonempty subvariety with $Z \neq X$, regularly embedded of codimension $d \geq 1$, and take $\beta = [\widetilde{X}]$. The Rees algebra is a graded subalgebra of $\mathcal{O}_X[t]$ and hence a sheaf of domains, so $\widetilde{X}$ is integral, that is a variety; and f restricts to an isomorphism over $X \setminus Z$ (proposition 7.2.27(2)), so the two function fields agree and $f_*[\widetilde{X}] = [X]$ by definition 8.2.1. Also $i^![X] = [Z]$ by proposition 11.3.3, the intersection of X with Z being Z itself, and $j^![\widetilde{X}] = [\widetilde{Z}]$ by proposition 11.3.5 applied to the subvariety $\widetilde{X}$ of itself. Substituting into theorem 12.3.11(2),

$$[Z] \;=\; g_*\Big(c_{d-1}(Q) \cap \big[\mathbb{P}(N)\big]\Big)$$

The identity can be checked directly, and the check is a use of the Segre operators rather than of anything proved in this section. The tautological sequence $0 \to \mathcal{O}_{\mathbb{P}(N)}(-1) \to g^*N \to Q \to 0$ and the Whitney formula (theorem 10.2.15) give $c(Q) = c(g^*N)\,(\mathrm{id} - \xi)^{-1}$ with $\xi = c_1(\mathcal{O}_{\mathbb{P}(N)}(1))$, the inverse being the terminating series $\sum_{m\geq 0}\xi^m$, so

$$c_{d-1}(Q) \;=\; \sum_{m=0}^{d-1} \xi^m \circ c_{d-1-m}\big(g^*N\big)$$

Capping with $[\mathbb{P}(N)] = g^*[Z]$ and pushing forward, naturality of the Chern operators for the flat morphism g (proposition 10.2.10(3)) and the definition of the Segre operators (definition 10.1.3), for which $\mathbb{P}(N) \to Z$ has relative dimension $d-1$, give

$$g_*\Big(\xi^m \cap g^*\big(c_{d-1-m}(N) \cap [Z]\big)\Big) \;=\; s_{m-d+1}(N) \cap \big(c_{d-1-m}(N) \cap [Z]\big)$$

for each m. Every term with $m < d-1$ vanishes, the Segre operator having negative index, and the term $m = d-1$ is $s_0(N) \cap [Z] = [Z]$; summing over m gives $g_*\big(c_{d-1}(Q) \cap [\mathbb{P}(N)]\big) = [Z]$, so the two computations agree.

Exercise 12.3.1

1. Let $Y \subseteq \mathbb{P}^2$ be a smooth plane curve of degree m, with inclusion i. Show that $i\colon Y \hookrightarrow \mathbb{P}^2$ is a regular embedding of codimension one, identify its normal bundle, and compute the degree of $i^!(i_*[Y]) \in A_0(Y)$. Then explain why the answer determines $i^!(i_*[Y])$ itself when $m \leq 2$ but not when $m \geq 3$.

2. Let $Z = \{x_0 = 0\} \cong \mathbb{P}^{n-1}$ be a hyperplane in $\mathbb{P}^n$ with $n \geq 1$, with inclusion i. Identify the normal bundle of i and compute $i^!(i_*\alpha)$ for α the generator h^Z_m of $A_m(Z)$ for each $0 \leq m \leq n-1$. Compare the answer with the Gysin map $i^!$ of example 11.3.4 in the case $d = 1$.

3. Let $i\colon Z \hookrightarrow X$ be a regular embedding of codimension $d \geq 1$ whose normal bundle is trivial, $N \cong \mathcal{O}_Z^{\oplus d}$. Show that $i^! \circ i_* = 0$ on $A_k(Z)$ for every k. Then let $X = \mathbb{P}^1 \times \mathbb{P}^1$, let $Z = \{q\} \times \mathbb{P}^1$ for a point $q \in \mathbb{P}^1$, and verify the hypothesis, deducing that the ruling has self-intersection zero.

4. In $\mathbb{P}^4$ with coordinates $x_0, \ldots, x_4$, let $Z = \{x_3 = x_4 = 0\}$ and $X' = \{x_2 = x_4 = 0\}$ be two planes, so that $Z \cap X' = \{x_2 = x_3 = x_4 = 0\}$ is a line L, and let $g\colon X' \hookrightarrow \mathbb{P}^4$ be the inclusion. Show that $Z \hookrightarrow \mathbb{P}^4$ is a regular embedding of codimension two with $N = \mathcal{O}_{\mathbb{P}^2}(1)^{\oplus 2}$, identify the preimage Z' of Z in X' and its normal bundle, compute the excess bundle, and evaluate $i^!_g[X']$. Interpret the degree of the answer.

5. Let $\widetilde{X} = \mathrm{Bl}_P\mathbb{P}^2$ as in example 12.3.6, with exceptional divisor $\widetilde{Z} \cong \mathbb{P}^1$ and the remaining notation that of theorem 12.3.11. Identify the universal quotient bundle Q on $\widetilde{Z}$ and compute $c_1(Q)$. Then verify theorem 12.3.11(2) for $\beta = [\widetilde{X}]$ by computing both sides, and compute $i^!\big(f_*j_*[\widetilde{Z}]\big)$.

12.4 Bézout and the Agreement Theorem

Two counts were recorded on credit. Example 8.2.6 asserted that two plane curves of degrees c and d meet in a zero-cycle of degree cd, and example 8.5.4 read that assertion off an identity $h_1 \cdot h_1 = h_0$

in a ring that did not then exist. Both are settled here, in the sharp form: not only the total count, but its distribution over the points of $C \cap D$, which is what an intersection multiplicity is.

Every enumerative count in projective space reduces to one computation, the product of the classes of two linear subspaces, because theorem 8.5.2 leaves nothing else in $A_*(\mathbb{P}^n)$ to compute, and that computation was carried out in example 12.2.3. Written in dimension rather than codimension, with $h_k = [\mathbb{P}^k] \in A_k(\mathbb{P}^n)$ the class of a linear subspace of dimension k and $h = h_{n-1}$ the hyperplane class, it reads

$$h_r \cdot h_s \;=\; \begin{cases} h_{r+s-n} & \text{if } r+s \geq n \\ 0 & \text{if } r+s < n \end{cases}$$

for $0 \leq r, s \leq n$, together with $h^p = h_{n-p}$ for $0 \leq p \leq n$, the vanishing $h^{n+1} = 0$, and the presentation

$$A^*(\mathbb{P}^n) \;\cong\; \mathbb{Z}[h]/(h^{n+1})$$

as a graded ring with h in codimension one. Nothing else about $\mathbb{P}^n$ is needed below.

The presentation converts every question about cycles on $\mathbb{P}^n$ into a question about polynomials in one variable. A class in $A^*(\mathbb{P}^n)$ is a polynomial in h of degree at most n; multiplying two of them and truncating above degree n is the whole intersection calculus of projective space. The truncation is geometric: $h^p h^q = 0$ for $p + q > n$ says that two subvarieties whose codimensions exceed the ambient dimension can be moved apart, and rational equivalence records that they have been.

The single integer attached to a subvariety by theorem 8.5.2 now deserves a name, since it is the only input Bézout's theorem takes.

Definition 12.4.1: Degree of a Subvariety of Projective Space

Let $V \subseteq \mathbb{P}^n$ be a closed subvariety of dimension k. Since $A_k(\mathbb{P}^n) = \mathbb{Z}\, h_k$ is free of rank one (theorem 8.5.2), there is a unique integer $\deg V$ with

$$[V] \;=\; (\deg V)\, h_k \;\in\; A_k(\mathbb{P}^n)$$

called the *degree* of V. The same definition applies to a closed subscheme pure of dimension k, using its fundamental cycle.

The definition is a tautology until the integer is computed, and the two computations that matter are the classical ones: a hypersurface has the degree of its equation, and the degree of anything is the number of points it cuts on a complementary linear space.

Proposition 12.4.2: Degrees of Hypersurfaces and Hyperplane Sections

Let $H \subseteq \mathbb{P}^n$ be a hyperplane, so that $\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1)$.

1. If $V \subseteq \mathbb{P}^n$ is a closed subvariety of dimension k, then $H^k \cdot [V] = (\deg V)\, h_0$ in $A_0(\mathbb{P}^n)$, so $\deg V = \int_{\mathbb{P}^n} \big(H^k \cdot [V]\big)$, the iterated divisor operator of definition 9.2.1 applied k times.

2. A linear subspace $\mathbb{P}^k \subseteq \mathbb{P}^n$ has degree 1.

3. If F is a nonzero homogeneous form of degree $d \geq 1$ on $\mathbb{P}^n$, the hypersurface $V(F)$ has degree d.

***Proof*:**

1. Example 9.2.5 computes $H\cdot h_j = h_{j-1}$ for $1 \leq j \leq n$. The operator $H\cdot -$ is a homomorphism of Chow groups (proposition 9.4.4(1)), so from $[V] = (\deg V)\, h_k$ one gets $H\cdot[V] = (\deg V)\, h_{k-1}$, and iterating k times gives $H^k\cdot[V] = (\deg V)\, h_0$. The class h_0 is that of a single closed point, of degree $[\overline{k} : \overline{k}] = 1$ by definition 8.2.4, whence $\int_{\mathbb{P}^n}(H^k\cdot[V]) = \deg V$.

2. By theorem 8.5.2 the class of a linear $\mathbb{P}^k$ is the generator h_k itself, so the coefficient is 1.

3. Write $Z = V(F)$ for the closed subscheme cut out by F, a hypersurface, and let D be the associated effective Cartier divisor, with $\mathcal{O}_{\mathbb{P}^n}(D) \cong \mathcal{O}_{\mathbb{P}^n}(d)$ because F is a section of $\mathcal{O}_{\mathbb{P}^n}(d)$. Applying proposition 11.3.5 with $V = \mathbb{P}^n$, which is legitimate since $\mathbb{P}^n$ is a variety not contained in $|D|$, the fundamental cycle of Z is $[Z] = D\cdot[\mathbb{P}^n]$. By definition 9.4.1 this is $c_1(\mathcal{O}_{\mathbb{P}^n}(d)) \cap [\mathbb{P}^n]$, and $\mathcal{O}_{\mathbb{P}^n}(d) \cong \mathcal{O}_{\mathbb{P}^n}(1)^{\otimes d}$, so additivity of the first Chern class (proposition 9.4.4(3)) gives

$$[Z] \;=\; d\left(c_1(\mathcal{O}_{\mathbb{P}^n}(1)) \cap [\mathbb{P}^n]\right) \;=\; d\, h_{n-1}$$

the last equality being example 9.4.3(2). Hence $\deg Z = d$, completing the proof.

Part (1) is the reason the word "degree" is the same word as in the classical theory: cutting a k-dimensional subvariety by k hyperplanes leaves a zero-cycle whose degree is the invariant, and by proposition 9.4.6 the hyperplanes need not be chosen in general position for the answer to come out right. Part (3) recovers the plane-curve count of example 8.5.4(2) as the case $n = 2$. Degrees of subvarieties that are not hypersurfaces are computed by pushing forward from a parametrization, as the next example illustrates.

Example 12.4.3: The Twisted Cubic

Let $\nu\colon \mathbb{P}^1 \to \mathbb{P}^3$ be the third Veronese embedding, $[s:t] \mapsto [s^3 : s^2t : st^2 : t^3]$, and let $C = \nu(\mathbb{P}^1)$ be the twisted cubic, a closed subvariety of $\mathbb{P}^3$ of dimension one. Since ν is a closed immersion, $\nu_*[\mathbb{P}^1] = [C]$ by definition 8.2.1, the extension of function fields being trivial.
Let $H = \{x_0 = 0\} \subseteq \mathbb{P}^3$. The pullback ν^*H is the Cartier divisor on $\mathbb{P}^1$ cut by s^3, that is $3\,[0:1]$, so $\nu^*H\cdot[\mathbb{P}^1] = 3\left[[0:1]\right]$ by definition 9.2.1. The projection formula for divisors (proposition 9.3.3(1)) applied to the proper morphism ν gives

$$H\cdot[C] \;=\; \nu_*\big(\nu^*H\cdot[\mathbb{P}^1]\big) \;=\; 3\,[P], \qquad P = \nu([0:1]) = [0:0:0:1]$$

a zero-cycle of degree 3. By proposition 12.4.2(1) with $k = 1$, $\deg C = 3$: the twisted cubic has degree three, all of it concentrated at the single point where H is tangent to C to order three. A general plane, by contrast, meets C in three distinct points, and proposition 9.4.6 is what makes the two computations agree without either being privileged.

With degrees in hand, Bézout's theorem is a restatement of example 12.2.3.

Theorem 12.4.4: Bézout's Theorem

Let $V, W \subseteq \mathbb{P}^n$ be closed subvarieties of dimensions r and s. If $r + s \geq n$, then

$$[V] \cdot [W] \;=\; (\deg V)(\deg W)\, h_{r+s-n} \;\in\; A_{r+s-n}(\mathbb{P}^n)$$

and if $r + s < n$ then $[V] \cdot [W] = 0$. In the boundary case $r + s = n$,

$$\int_{\mathbb{P}^n} \big([V] \cdot [W]\big) \;=\; (\deg V)(\deg W)$$

***Proof*:**
By definition 12.4.1 the classes are $[V] = (\deg V)\, h_r$ and $[W] = (\deg W)\, h_s$. The intersection product is $\mathbb{Z}$-bilinear (proposition 12.1.9(1)), so

$$[V] \cdot [W] \;=\; (\deg V)(\deg W)\, (h_r \cdot h_s)$$

and example 12.2.3 evaluates $h_r \cdot h_s$ as h_{r+s-n} when $r + s \geq n$ and as 0 otherwise. For $r + s = n$ the class h_0 is that of a single reduced closed point, of degree one by definition 8.2.4, and the degree map is a homomorphism, so $\int_{\mathbb{P}^n}([V] \cdot [W]) = (\deg V)(\deg W)$, as we sought to show.

The theorem is stated for classes, and in that form it costs nothing beyond the ring structure. What it does not yet say is where the count sits. The product $[V] \cdot [W]$ is defined as $\delta^!([V] \times [W])$, and the refined form of the Gysin map records that class on the preimage of the diagonal, that is on $V \cap W$ (corollary 11.3.11 and definition 12.1.10). Reading the refined class off requires knowing that $V \cap W$ sits inside $V \times W$ as a regular embedding, which is a hypothesis and not a theorem. The following lemma verifies it in the case where the verification is cheapest and most used: two curves on a smooth surface.

Lemma 12.4.5: Distinct Curves on a Smooth Surface Meet Regularly

Let S be a smooth variety of dimension two over $\overline{k}$ and let $C, D \subseteq S$ be distinct one-dimensional closed subvarieties. Then:

1. $C \cap D$ is a finite set of closed points, and for $P \in C \cap D$ with local equations $f, g \in \mathcal{O}_{S,P}$ of C and D, the local ring of the scheme $C \cap D$ at P is $\mathcal{O}_{S,P}/(f, g)$, of length equal to its dimension $\dim_{\overline{k}} \mathcal{O}_{S,P}/(f, g)$ as a $\overline{k}$-vector space;

2. the closed immersion $C \cap D \hookrightarrow C \times D$ induced by the diagonal is a regular embedding of codimension two;

3. consequently

$$[C] \cdot [D] \;=\; \sum_{P \in C \cap D} \dim_{\overline{k}} \big(\mathcal{O}_{S,P}/(f, g)\big)\, [P] \;\in\; A_0(S)$$

***Proof*:**
Since S is smooth its local rings are regular, hence factorial, so each of C and D is an effective Cartier divisor and the local equations f, g exist and are nonzerodivisors.

1. $C \cap D$ is a closed subscheme of the irreducible curve C not equal to C, because $D \neq C$ and both are irreducible of dimension one; a proper closed subscheme of a curve is finite. Scheme-theoretically $C \cap D$ is cut out on S by the ideal (f, g), so its local ring at P is $\mathcal{O}_{S,P}/(f, g)$. That ring is Artinian local with residue field $\overline{k}$, since $\overline{k}$ is algebraically closed and P is a closed point, so each factor of a composition series is a copy of $\overline{k}$ and its length equals its $\overline{k}$-dimension.

2. Fix $P \in C \cap D$ and write $\mathcal{O} = \mathcal{O}_{S,P}$, a regular local ring of dimension two with residue field $\overline{k}$, and $A = \mathcal{O}/(f)$, $B = \mathcal{O}/(g)$, the local rings of C and D at P. Let R be the local ring of $C \times D$ at (P, P), that is the localization of $A \otimes_{\overline{k}} B$ at the maximal ideal corresponding to (P, P).
Step 1: R is Cohen-Macaulay of dimension two. The product of two curves over $\overline{k}$ is a surface, so $\dim R = 2$. Because $\mathcal{O}$ is regular of dimension two and f is a nonzerodivisor, A has depth one, so $\mathfrak{m}_A$ contains a nonzerodivisor a; likewise $\mathfrak{m}_B$ contains a nonzerodivisor b. Now B is free as a $\overline{k}$-module, so $A \to A \otimes_{\overline{k}} B$ is flat and $a \otimes 1$ is a nonzerodivisor, with quotient $(A/aA) \otimes_{\overline{k}} B$; and A/aA is again free over $\overline{k}$, so $B \to (A/aA) \otimes_{\overline{k}} B$ is flat and $1 \otimes b$ is a nonzerodivisor on that quotient. Localizing at the maximal ideal, the pair $a \otimes 1$, $1 \otimes b$ is a regular sequence in R of length two, so $\operatorname{depth} R \geq 2 = \dim R$ and R is Cohen-Macaulay.
Step 2: the diagonal cuts R by a system of parameters. The diagonal $\delta\colon S \hookrightarrow S \times S$ is a regular embedding of codimension two (theorem 12.1.6), so its ideal is generated near (P, P) by two elements θ_1, θ_2. The scheme-theoretic preimage of $\delta(S)$ under $C \times D \hookrightarrow S \times S$ is $C \cap D$, whose ideal in R is therefore generated by the images of θ_1 and θ_2. By part (1) that quotient is Artinian, so the two images form a system of parameters of the two-dimensional local ring R. In a Cohen-Macaulay local ring every system of parameters is a regular sequence ([10, A System of Parameters in a Cohen-Macaulay Local Ring Is a Regular Sequence]), so $C \cap D \hookrightarrow C \times D$ is a regular embedding of codimension two.

3. The subvarieties C and D are integral over the algebraically closed field $\overline{k}$, so $C \times D$ is a subvariety of $S \times S$ of dimension two, with $[C] \times [D] = [C \times D]$ (definition 12.1.2). Part (2) supplies the hypothesis of proposition 11.3.3 for the regular embedding δ of codimension two and the subvariety $C \times D$, so

$$[C] \cdot [D] \;=\; \delta^![C \times D] \;=\; [C \cap D]$$

the fundamental cycle of the scheme $C \cap D$ read on S through δ. By definition 8.1.2 that cycle is $\sum_P \operatorname{length}(\mathcal{O}_{C \cap D, P})\,[P]$, and part (1) identifies each length with $\dim_{\overline{k}} \mathcal{O}_{S,P}/(f, g)$, completing the proof.

The lemma is the bridge from a statement about classes to a statement about points. In $\mathbb{P}^2$ every curve is a hypersurface, so its degree is the degree of its equation by proposition 12.4.2(3); the lemma settles a pair of distinct integral curves, and additivity of the local count in each argument extends it to arbitrary curves without a common component. Bézout's theorem then acquires its classical form.

Corollary 12.4.6: Bézout's Theorem for Plane Curves

Let F and G be nonzero forms on $\mathbb{P}^2$ of degrees c and d with no common irreducible factor, and let $C = V(F)$ and $D = V(G)$ be the plane curves they cut out, so that C and D have no common component; neither is assumed irreducible or reduced. Then $C \cap D$ is finite and

$$\sum_{P \in C \cap D} \dim_{\overline{k}} \big(\mathcal{O}_{\mathbb{P}^2,P}/(f_P, g_P)\big) \;=\; cd$$

where f_P and g_P are the local equations of C and D at P, obtained by dehomogenizing F and G.

***Proof*:**
Factor $F = \prod_i F_i^{m_i}$ and $G = \prod_j G_j^{n_j}$ into powers of pairwise nonassociate irreducible forms; by hypothesis no F_i is an associate of any G_j. The curves $C_i = V(F_i)$ and $D_j = V(G_j)$ are one-dimensional closed subvarieties of $\mathbb{P}^2$, of degrees $\deg F_i$ and $\deg G_j$ by proposition 12.4.2(3), and $C_i \neq D_j$ for all i, j. For a form u and a point P write $u_P \in \mathcal{O}_{\mathbb{P}^2,P}$ for the dehomogenization of u at P, so that $f_P = F_P$ and $g_P = G_P$.

Step 1: the count for a pair of components. The plane $\mathbb{P}^2$ is a smooth variety of dimension two and $C_i \neq D_j$ are distinct one-dimensional closed subvarieties, so lemma 12.4.5 applies to the pair (C_i, D_j): the intersection $C_i \cap D_j$ is finite and $[C_i] \cdot [D_j] = \sum_P \dim_{\overline{k}}(\mathcal{O}_{\mathbb{P}^2,P}/(F_{i,P}, G_{j,P}))\,[P]$ in $A_0(\mathbb{P}^2)$. Each point is closed with residue field $\overline{k}$, so taking degrees (definition 8.2.4) and comparing with theorem 12.4.4 for $n = 2$, $r = s = 1$ gives

$$\sum_{P \in C_i \cap D_j} \dim_{\overline{k}} \big(\mathcal{O}_{\mathbb{P}^2,P}/(F_{i,P}, G_{j,P})\big) \;=\; (\deg F_i)(\deg G_j)$$

Since $C \cap D = \bigcup_{i,j} C_i \cap D_j$ as sets, $C \cap D$ is finite.

Step 2: the local count is additive in each argument. Fix P and write $\mathcal{O} = \mathcal{O}_{\mathbb{P}^2,P}$, a regular local ring of dimension two. Let u be a divisor of F and let v, w be divisors of G, or the same with the roles of F and G exchanged; then $V(u) \cap V(vw) \subseteq C \cap D$ is finite by Step 1, so $\mathcal{O}/(u_P, (vw)_P)$ and its two companions below are Artinian as in lemma 12.4.5(1), of finite $\overline{k}$-dimension. We may assume u_P and w_P lie in the maximal ideal, since otherwise one of the three quotients is zero and the additivity claimed below is trivial. Being Artinian, $\mathcal{O}/(u_P, w_P)$ exhibits u_P, w_P as a system of parameters of $\mathcal{O}$, hence as a regular sequence ([10, A System of Parameters in a Cohen-Macaulay Local Ring Is a Regular Sequence]); in particular w_P is a nonzerodivisor on $A = \mathcal{O}/(u_P)$. The sequence

$$0 \longrightarrow A/(v_P) \xrightarrow{w_P} A/(v_P w_P) \longrightarrow A/(w_P) \longrightarrow 0$$

is then exact: surjectivity on the right and exactness in the middle are immediate, and if $w_P h \in (v_P w_P)A$ then $w_P(h - v_P a) = 0$ in A for some a, so $h \in (v_P)A$ because w_P is a nonzerodivisor on A. Counting $\overline{k}$-dimensions gives

$$\dim_{\overline{k}} \mathcal{O}/(u_P, v_P w_P) \;=\; \dim_{\overline{k}} \mathcal{O}/(u_P, v_P) \;+\; \dim_{\overline{k}} \mathcal{O}/(u_P, w_P)$$

that is, additivity in the second argument, and the exchanged case gives additivity in the first.

Iterating over the two factorizations,

$$\dim_{\overline{k}}\left(\mathcal{O}_{\mathbb{P}^2,P}/(f_P,g_P)\right) \;=\; \sum_{i,j} m_i n_j \,\dim_{\overline{k}}\left(\mathcal{O}_{\mathbb{P}^2,P}/(F_{i,P},G_{j,P})\right)$$

Step 3: summation. Sum the last display over the finitely many points of $C\cap D$; a point outside $C_i\cap D_j$ contributes nothing to the (i,j) term, since there $F_{i,P}$ or $G_{j,P}$ is a unit, so Step 1 evaluates each term and

$$\sum_{P\in C\cap D}\dim_{\overline{k}}\left(\mathcal{O}_{\mathbb{P}^2,P}/(f_P,g_P)\right) \;=\; \sum_{i,j} m_i n_j(\deg F_i)(\deg G_j) \;=\; \Big(\sum_i m_i\deg F_i\Big)\Big(\sum_j n_j\deg G_j\Big) \;=\; cd$$

the last equality because $\deg F=\sum_i m_i\deg F_i$ and $\deg G=\sum_j n_j\deg G_j$, completing the proof.

This is the count promised in example 8.2.6 and the identity $h_1\cdot h_1=h_0$ anticipated in example 8.5.4(2), now with the local distribution attached: the integer $\dim_{\overline{k}}\mathcal{O}_{\mathbb{P}^2,P}/(f_P,g_P)$ says how many of the cd units of intersection are spent at P. Nothing in the argument assumed that the curves are irreducible, reduced, smooth, or that they meet transversally; only that they share no component, which is exactly what makes the intersection finite.

Example 12.4.7: A Cuspidal Cubic and Its Cuspidal Tangent

Let $C=\{y^2z=x^3\}\subseteq\mathbb{P}^2$, a cubic curve with a cusp at $P=[0:0:1]$, and let $D=\{y=0\}$, the tangent line to C at the cusp. The cubic is irreducible and is not the line, so the two share no component and corollary 12.4.6 predicts a total of $3\cdot 1=3$.

Set-theoretically, $y=0$ together with $y^2z=x^3$ forces $x^3=0$, hence $x=0$: the only intersection point is P. Working in the chart $z=1$ with $\mathcal{O}=\overline{k}[x,y]_{(x,y)}$, the local equations are $f=y^2-x^3$ and $g=y$, and

$$\mathcal{O}/(f,g) \;=\; \overline{k}[x,y]_{(x,y)}/(y^2-x^3,\,y) \;\cong\; \overline{k}[x]_{(x)}/(x^3)$$

of dimension 3 over $\overline{k}$, with basis $1,x,x^2$. So all three units of intersection sit at the cusp, and $[C]\cdot[D]=3\,[P]$. Compare example 9.2.2, where a conic met its tangent line in a single point of multiplicity two: in both cases the multiplicity is the order to which the equation of the line vanishes along the curve, and here the cusp raises it from two to three.

The integer attached to P in the example is not special to curves in a plane. Whenever two subvarieties of a smooth variety meet in the expected dimension, the product distributes over the components of the intersection with well-defined integer coefficients, and those coefficients are the intersection multiplicities.

Definition 12.4.8: Proper Intersection and Intersection Multiplicity

Let X be a smooth variety of pure dimension n and let $V, W \subseteq X$ be closed subvarieties of dimensions r and s. Say that V and W *meet properly* if every irreducible component of $V \cap W$ has dimension exactly $m = r + s - n$. In that case $V \cap W$ is pure of dimension m, so $A_m(V \cap W) = Z_m(V \cap W)$ is free on the classes of its irreducible components $Z_1, \ldots, Z_t$ (example 8.1.6(3)), and the refined intersection class of definition 12.1.10 may be written uniquely as

$$[V] \cdot [W] \;=\; \sum_{j=1}^{t} i(Z_j;\, V \cdot W;\, X)\,[Z_j] \;\in\; A_m(V \cap W)$$

The integer $i(Z_j;\, V \cdot W;\, X)$ is the *intersection multiplicity* of V and W along Z_j.

The definition would be empty without the refinement: the class $[V] \cdot [W]$ read in $A_m(X)$ carries no information about where the intersection happens, and it is the localization on $V \cap W$ supplied by the refined Gysin map (definition 11.3.6, corollary 11.3.11) that makes the coefficients meaningful. The proper-intersection hypothesis is what makes the coefficients unique, since A_m of an m-dimensional scheme is free on components; drop it and the class still exists but no longer decomposes canonically.

When the scheme-theoretic intersection is as good as its dimension suggests, the multiplicities are lengths and nothing else. Proposition 11.3.3 says that if $V \cap W \hookrightarrow V \times W$ is a regular embedding of codimension n, then $[V] \cdot [W] = [V \cap W]$ is the fundamental cycle, so $i(Z_j; V \cdot W; X)$ is the length of the local ring of $V \cap W$ at the generic point of Z_j. That is exactly what lemma 12.4.5 verified for curves on a surface, and it is the source of the numbers computed in example 12.4.7. In general the regularity hypothesis can fail, and then the multiplicity and the naive length part company. The correction is homological.

Theorem 12.4.9: Serre's Tor Formula

Let X be a smooth variety of pure dimension n and let $V, W \subseteq X$ be closed subvarieties meeting properly, with Z an irreducible component of $V \cap W$ and $\mathcal{O} = \mathcal{O}_{X,Z}$ the local ring of X at the generic point of Z. Then

$$i(Z;\, V \cdot W;\, X) \;=\; \sum_{j \geq 0} (-1)^j \operatorname{length}_{\mathcal{O}} \operatorname{Tor}_j^{\mathcal{O}}\big(\mathcal{O}/\mathcal{I}_V,\, \mathcal{O}/\mathcal{I}_W\big)$$

where $\mathcal{I}_V$ and $\mathcal{I}_W$ are the ideals of V and W in $\mathcal{O}$. The sum is finite, each term having finite length and all terms vanishing for $j > n$.

The proof is omitted: it runs entirely through homological algebra over regular local rings, in particular the vanishing, finite-length and additivity theorems for Tor that this book does not develop, and it is genuinely outside the scope of a treatment built on deformation to the normal cone. Serre proved the formula in *Algèbre locale, multiplicités*, and the comparison with the multiplicities of definition 12.4.8 is carried out in Fulton [24]. The formula matters here because it exhibits the multiplicity as a purely local, purely algebraic invariant of the two local rings, with no reference to the deformation that defined it.

The correction terms are invisible in the case the previous results have been using, and that is worth recording, since it says the classical count of plane curves needs no homological algebra at all.

Proposition 12.4.10: Vanishing of the Higher Terms for Divisors

Let X be a smooth variety of pure dimension n and let $C, D \subseteq X$ be closed subvarieties of dimension $n-1$ with $C \neq D$, so that C and D meet properly. For every irreducible component Z of $C \cap D$, with $\mathcal{O} = \mathcal{O}_{X,Z}$ and local equations $f, g \in \mathcal{O}$,

$$\mathrm{Tor}_j^{\mathcal{O}}\big(\mathcal{O}/(f),\, \mathcal{O}/(g)\big) = 0 \quad \text{for } j \geq 1$$

so Serre's formula reduces to $i(Z;\, C \cdot D;\, X) = \operatorname{length}_{\mathcal{O}} \mathcal{O}/(f, g)$.

***Proof*:**
Since X is smooth, $\mathcal{O}$ is a regular local ring and C, D are effective Cartier divisors near Z, so f and g are nonzerodivisors and

$$0 \longrightarrow \mathcal{O} \xrightarrow{f} \mathcal{O} \longrightarrow \mathcal{O}/(f) \longrightarrow 0$$

is a free resolution of $\mathcal{O}/(f)$ of length one. Hence $\mathrm{Tor}_j(\mathcal{O}/(f), \mathcal{O}/(g)) = 0$ for $j \geq 2$, while Tor_1 is the kernel of multiplication by f on $\mathcal{O}/(g)$. Because $\mathcal{O}$ is regular and g is a nonzerodivisor, $\mathcal{O}/(g)$ is Cohen-Macaulay, hence has no embedded associated primes ([10, A Cohen-Macaulay Ring Has No Embedded Associated Primes]), so its associated primes are the minimal primes of (g), which are the primes corresponding to the components of D passing through Z. As C and D share no component, f lies in none of them, so multiplication by f on $\mathcal{O}/(g)$ is injective and $\mathrm{Tor}_1 = 0$. The remaining term is $\mathrm{Tor}_0(\mathcal{O}/(f), \mathcal{O}/(g)) = \mathcal{O}/(f, g)$, and theorem 12.4.9 therefore reads $i(Z; C \cdot D; X) = \operatorname{length}_{\mathcal{O}} \mathcal{O}/(f, g)$, as we sought to show.

Everything so far has been internal to this book: a product, a ring, a count. The comparison that remains is external. *Algebraic Geometry* [7] builds a pairing on the divisors of a smooth projective surface without any of this machinery, setting $C \cdot D = \chi(\mathcal{O}_S) - \chi(\mathcal{O}_S(-C)) - \chi(\mathcal{O}_S(-D)) + \chi(\mathcal{O}_S(-C-D))$, and derives adjunction, the behaviour of self-intersection, and the classification of surfaces from it. Two constructions with nothing in common beyond the answer they are supposed to give must be shown to give it.

Theorem 12.4.11: The Agreement Theorem for Surfaces

Let S be a smooth projective surface over $\overline{k}$ and let C, D be Cartier divisors on S, with associated classes $[C] = c_1(\mathcal{O}_S(C)) \cap [S]$ and $[D] = c_1(\mathcal{O}_S(D)) \cap [S]$ in $A^1(S) = A_1(S)$. Then

$$\int_S ([C] \cdot [D]) \;=\; \int_S (C \cdot (D \cdot [S])) \;=\; C \cdot D$$

where the first is the intersection product of section 12.1, the second the iterated divisor operator of definition 9.2.1, and the third the cohomological intersection number definition 7.2.7.

***Proof*:**
Write $\langle C, D\rangle_1$, $\langle C, D\rangle_2$ and $\langle C, D\rangle_3$ for the three quantities.

Step 1: all three are $\mathbb{Z}$-bilinear and depend only on linear equivalence. For the first, $C \mapsto [C] = C \cdot [S]$ is additive in C by additivity of the first Chern class (proposition 9.4.4(3)) and depends only on $\mathcal{O}_S(C)$ by proposition 9.4.2; the product is $\mathbb{Z}$-bilinear (proposition 12.1.9(1)) and $\int_S$ is a homomorphism (definition 8.2.4). For the second, $D \mapsto D \cdot [S]$ is additive and linear-equivalence invariant for the same two reasons, and $C \cdot -$ is a homomorphism of Chow groups additive in C by proposition 9.4.4(1),(3). For the third, bilinearity and invariance under linear equivalence are proposition 7.2.8(1),(3).

Step 2: the three agree for distinct prime divisors. Let C and D be distinct one-dimensional closed subvarieties of S, with local equations f, g at a point P. Lemma 12.4.5(3) gives

$$\langle C, D\rangle_1 = \int_S \Big(\sum_{P \in C\cap D} \dim_{\overline{k}} \left(\mathcal{O}_{S,P}/(f,g)\right) [P]\Big) = \sum_{P\in C\cap D} \dim_{\overline{k}} \left(\mathcal{O}_{S,P}/(f,g)\right)$$

each P being a closed point with residue field $\overline{k}$. For the second quantity, $S \not\subseteq |D|$ and $D \not\subseteq |C|$, so the first clause of definition 9.2.1 applies twice: $D\cdot[S] = [\operatorname{div}(s_S)] = [D]$, and $C\cdot[D] = [\operatorname{div}(f|_D)] = \sum_P \operatorname{ord}_P(f|_D)\,[P]$. By definition 8.1.1 the coefficient is $\operatorname{ord}_P(f|_D) = \operatorname{length}_{\mathcal{O}_{D,P}}\left(\mathcal{O}_{D,P}/(f|_D)\right)$, and $\mathcal{O}_{D,P}/(f|_D) = \mathcal{O}_{S,P}/(f,g)$, whose length equals its $\overline{k}$-dimension. Hence $\langle C, D\rangle_2$ is the same sum. For the third, proposition 7.2.8(5) evaluates $C\cdot D$ on effective divisors with no common component as $\sum_P \dim_{\overline{k}} \mathcal{O}_{S,P}/(f,g)$, again the same sum. So the three agree on such a pair.

Step 3: reduction of the general case to Step 2. Fix a very ample divisor H on the projective surface S. By Serre's theorem on global generation after an ample twist (theorem 5.3.25) there is an $m \geq 1$ such that $\mathcal{O}_S(C+mH)$, $\mathcal{O}_S(D+mH)$ and $\mathcal{O}_S(mH)$ are all generated by their global sections. Set

$$A_1 = C + mH, \quad A_2 = mH, \quad B_1 = D + mH, \quad B_2 = mH$$

so that $C = A_1 - A_2$ and $D = B_1 - B_2$ as Cartier divisors, and by Step 1 it suffices to prove the agreement for each of the four pairs (A_i, B_j).

A globally generated line bundle has a nonzero global section, so each A_i is linearly equivalent to an effective divisor A_i'; let $\Gamma_1, \ldots, \Gamma_t$ be the finitely many prime components of A_1' and A_2' and choose a closed point $p_a \in \Gamma_a$ for each. For a fixed j, the sections of $\mathcal{O}_S(B_j)$ vanishing at p_a form a proper linear subspace of the finite-dimensional space $H^0(S, \mathcal{O}_S(B_j))$, since that bundle is globally generated; a vector space over the infinite field $\overline{k}$ is not a finite union of proper subspaces, so some section τ_j vanishes at no p_a. Its divisor $B_j' = \operatorname{div}(\tau_j)$ is effective, linearly equivalent to B_j, and contains none of the Γ_a.

Writing $A_i' = \sum_a m_a \Gamma_a$ and $B_j' = \sum_b n_b \Lambda_b$ as nonnegative combinations of prime divisors, no Γ_a equals any Λ_b, so bilinearity reduces each $\langle A_i', B_j'\rangle_\bullet$ to $\sum_{a,b} m_a n_b \langle \Gamma_a, \Lambda_b\rangle_\bullet$, and every term is covered by Step 2. Invariance under linear equivalence transfers the conclusion back from A_i', B_j' to A_i, B_j, and Step 1 assembles the four pairs into the general case, completing the proof.

The theorem is the seam. Every numerical statement about divisors on a smooth projective surface proved in *Algebraic Geometry* [7] out of Euler characteristics, the adjunction formula, the identification of C^2 with the degree of the normal bundle, the Hodge index theorem, is a statement about the product of this chapter, and conversely every general theorem proved here specializes to a surface fact that book had to obtain by hand. The surface theory is subsumed rather than paralleled, and section 12.3 already used that when it recovered $E^2 = -1$ from the self-intersection formula.

The middle term of the theorem is worth isolating. It says that on a smooth projective surface the ring product of two divisor classes computes the same integer as the iterated divisor operator built in chapter 9, so the operator calculus of that chapter, commutativity and the projection formula included, transfers verbatim to the ring. The two constructions were built from different material: one from local equations and orders of vanishing, the other from a degeneration of the diagonal. Nothing forced them to agree except that both were built to count.

Exercise 12.4.1

1. Let $V, W \subseteq \mathbb{P}^3$ be surfaces of degrees 2 and 3 with no common component. Compute $[V] \cdot [W]$ in $A_1(\mathbb{P}^3)$ and read off the degree of the curve $V \cap W$. Then let $V_1, V_2, V_3 \subseteq \mathbb{P}^3$ be surfaces of degrees d_1, d_2, d_3 and compute $\int_{\mathbb{P}^3} \big([V_1] \cdot [V_2] \cdot [V_3]\big)$.

2. Let $L \cong \mathbb{P}^r$ and $M \cong \mathbb{P}^s$ be linear subspaces of $\mathbb{P}^n$ with $r + s < n$. Show that $[L] \cdot [M] = 0$, and explain why this conclusion is *not* a consequence of the observation that L and M can be chosen disjoint. (Consider the case $L = M$.)

3. Keep $C = \{y^2 z = x^3\} \subseteq \mathbb{P}^2$ from example 12.4.7 and take instead $D' = \{x = 0\}$, the line joining the cusp $[0 : 0 : 1]$ to the inflection point $[0 : 1 : 0]$. Compute $C \cap D'$ and the multiplicity of $[C] \cdot [D']$ at each of its points, and verify corollary 12.4.6.

4. Let $Q = \mathbb{P}^1 \times \mathbb{P}^1$, with rulings $f_1 = \{a\} \times \mathbb{P}^1$ and $f_2 = \mathbb{P}^1 \times \{b\}$ as in exercise 9.2.1 (5), where $\mathcal{O}_Q(f_1) = \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^1}(1)$ and $\mathcal{O}_Q(f_2) = \mathrm{pr}_2^* \mathcal{O}_{\mathbb{P}^1}(1)$. Using theorem 12.4.11, write down the three numbers $\int_Q([f_1] \cdot [f_2])$, $\int_Q([f_1] \cdot [f_1])$ and $\int_Q([f_2] \cdot [f_2])$. Then show that the diagonal $\Delta \subseteq Q$ satisfies $\Delta \sim f_1 + f_2$ and compute $\int_Q([\Delta] \cdot [\Delta])$.

5. Let $L \subseteq \mathbb{P}^2$ be a line. Compute $\int_{\mathbb{P}^2}([L] \cdot [L])$ from example 12.2.3, and identify precisely which conclusion of lemma 12.4.5 fails when $C = D = L$. Explain how the intersection product still returns an answer, and reconcile it with example 9.2.3.

12.4.12 Reading the numbering A star on a number, as in §2.7⋆, marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream.

Introduction

A moduli problem is asking whether a class of geometric objects (or a class of isomorphism classes of geometric objects) that is paramaterized by some invariant (fixed genus, closed subscheme of $\mathbb{P}^n$ with a fixed Hilbert polynomial, vector bundles of a fixed rank, etc.) can be paramaterized by a geometric object (ex. a variety, a scheme, or other new objects defined in this textbook when schemes are insufficient). The parameterisation by the geometric object records how the objects vary: a family of objects over a base S should be precisely a morphism from S to the space, so that a curve in the space is a one-parameter family and the local structure at a point measures the ways the corresponding object can deform. When such a space exists, it converts questions about an entire class of objects into geometry on a single object.

The obstacle that shapes the whole subject is the presence of automorphisms. An object with a nontrivial automorphism cannot be separated from its symmetries by a space of points alone, and the "naïve" moduli space of such objects either fails to exist or discards the information the automorphisms carry. The book is organized around three responses to this difficulty, one for each of its parts. The first part formulates a moduli problem as a functor of points and asks when that functor is representable by a scheme; it builds the machinery: the Hilbert and Quot schemes and geometric invariant theory, that produces moduli spaces in the representable case. The second part linearizes the local question through deformation theory: the tangent space and the obstruction space of a moduli problem at an object are cohomology groups of that object, and they determine the dimension of the moduli space and whether it is smooth. The third part enlarges the notion of space to the algebraic stack, the geometric object that remembers automorphisms and represents the moduli problems that schemes cannot; Artin's criteria, fed by the deformation theory of the second part, certify when a moduli problem is such a stack. The final chapter applies these to the moduli of curves, where a single object carries a deformation theory, a coarse space built by invariant theory, and a stack structure coalescing automorphism information.

Prerequisites

The reader is assumed to be at home with schemes, quasi-coherent sheaves, and sheaf cohomology at the level of a first graduate course in algebraic geometry, together with a working knowledge of homological algebra. These prerequisites, and a few more specialized inputs, are cited as machinery where they are used rather than re-derived, with references to the companion volumes of this series: [7] for scheme theory and sheaf cohomology, [15] for derived functors and Ext, [19] for group schemes and their actions, and [13] for the Galois cohomology behind the arithmetic threads. Étale-cohomological facts about the Brauer group, needed only in the chapter on gerbes, are stated and attributed to the co-requisite [12]. A small number of theorems whose proofs lie beyond a first course, Artin's approximation theorem, the Keel-Mori theorem, and the stable reduction theorem, are stated and used in full with their proofs cited; everything else is proved.

The four movements of the book are largely cumulative and are best read in order, though a reader interested primarily in stacks could begin with the third part after absorbing the language of the first. What the reader should carry away is a single habit of thought: a moduli problem is a functor; when automorphisms obstruct its representability the remedy is to remember them, in a stack; and the local geometry of the resulting space is deformation theory, a computation in the cohomology of one object.

Part IV

Moduli Problems and Their Construction

Whether Moduli spaces exist is the theme of this Part. The idea, due to Grothendieck, is to record the moduli problem as a functor: to each scheme S one assigns the set of families over S, with pullback making the assignment contravariant, and a *moduli space* is then a geometric object (ex. for us this chapter almost always a scheme) that *represents* this functor (section 3.3). The first chapter develops this language, separates fine from coarse moduli spaces, and meets the obstruction, objects with automorphisms, that prevents many natural moduli functors from being representable at all.

The remaining chapters build the machinery that produces moduli spaces when they exist. The next constructs the Hilbert and Quot schemes, Grothendieck's representable functors, from which most moduli spaces are cut out as locally closed subschemes; the last develops geometric invariant theory, which forms quotients by group actions and so produces moduli spaces of objects taken up to isomorphism. Together they are the constructions that the deformation theory of the next Part refines and the stack-theoretic language of the final Part generalizes when schemes are not enough.

13

Moduli Problems and Representability via Schemes

A moduli problem asks for a space that parametrizes a class of geometric objects, but the phrase "the space of all such objects" conceals a real question: in what sense is that collection a scheme? This chapter answers it in the language Grothendieck introduced for the purpose. A moduli problem is encoded as a functor that assigns to each scheme S the set of families of objects over S; a moduli space, when one exists, is a scheme that *represents* this functor, and the universal family such a scheme carries makes the correspondence between its points and the objects being classified exact. section 13.1 sets up this dictionary and the notion of a fine moduli space, and section 13.2 carries it out in full for the Grassmannian, the fundamental example in which everything works as one would wish.

Representability is the *exception* rather than the rule: the obstruction is visible in the most classical moduli problem of all, that of elliptic curves. An object with nontrivial automorphisms admits families that are fiberwise constant yet globally "twisted", and no scheme can distinguish them from the trivial family. The best one can then hope for in terms of schemes is a *coarse moduli space*, which records the points but forgets how objects vary. Section 13.3 isolates this automorphism obstruction and works the j-line in detail, section 13.4 shows how adding rigidifying structure restores representability, and section 13.5 assembles the criteria that decide representability in general and open onto the deformation theory of the next Part.

13.1 Moduli Problems as Functors of Points

A first answer to constructing a moduli space is to take the *set* of isomorphism classes and look for a scheme structure on it. The difficulty is in determining how such a family varies. Sometimes, this can be eminently visible as in the case of elliptic curves paramaterized by the j-line, however over more complicated invariant (ex. Hilbert Polynomials of a closed-scheme) the moduli space is not so evident.

The remedy is to record, for every base scheme S, the set of families over S. This assignment is a functor, and the language for turning such a functor into geometry is the *functor of points* (section 3.3): a scheme is determined by the sets of maps into it from all other schemes, so a moduli problem is representable by scheme exactly when its functor of families agrees with the maps-into functor of some scheme. This section makes that precise. It defines the moduli functor, defines a fine moduli space as a scheme representing it, and extracts from the Yoneda lemma the object that makes representability useful in practice: a single universal family through which every other family is pulled back in exactly one way.

The starting datum of any moduli problem is a rule that says what a *family* over a base S is and how families pull back along a morphism of bases:

Definition 13.1.1: Moduli Problem (Functor)

A *moduli problem* is a contravariant functor $\mathcal{F}$

$$\mathcal{F} \colon \mathbf{Sch}^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

that to each scheme S (of finite type over the base field k) one assigns a set

$$\mathcal{F}(S) = \{\, \text{families of the objects over } S \,\}/\!\cong$$

where "family of objects" is the datum of the problem taken up to isomorphism of families[a]; a morphism $f : T \to S$ is assigned the *pullback* map

$$f^* \colon \mathcal{F}(S) \longrightarrow \mathcal{F}(T)$$

carrying a family over S to its restriction over T.

[a] in definition ref:HERE this is formalized as a fibered-category of groupoids

The functor $\mathcal{F}$ carries more information than the set of isomorphism classes, which is recovered as $\mathcal{F}(\operatorname{Spec} k)$: the extra data is the collection $\mathcal{F}(S)$ for positive dimensional S together with the pullback maps, and this is precisely the record of how objects vary. A family over S is a single object that deforms as one moves through S; pullback along $f \colon T \to S$ reindexes that variation by T. Contravariance is the reflection of a basic geometric fact that a family over S restricts to a family over any T mapping into S, so a map of bases $T \to S$ induces a map of families in the opposite direction. The whole subject is the study of when this functor is geometric, and how to build the geometry when it is not.

The definition is only as concrete as the phrase "family of the objects" is made to be, so it is worth fixing that phrase in the two cases that run through this chapter.

Example 13.1.2: Families of Subschemes and of Line Bundles

1. Fix a scheme X over k. The *moduli of closed subschemes of* X assigns to S the set

$$\mathcal{F}(S) = \{\, Z \subseteq X \times_k S \text{ closed, flat over } S \,\}$$

of closed subschemes of $X \times_k S$ that are flat over S. A morphism $f\colon T \to S$ pulls a family Z back to $f^*Z = Z \times_S T \subseteq X \times_k T$, the base change of Z along f; flatness is preserved under base change, so f^*Z is again a family over T. Over a point $\operatorname{Spec} k$ a family is a single closed subscheme of X, so $\mathcal{F}(\operatorname{Spec} k)$ is the set of closed subschemes of X. This is the functor whose representing object, when it exists, is the Hilbert scheme (section 3.3.6).

2. The *Picard problem* assigns to S the set

$$\mathcal{F}(S) = \{\text{ line bundles on } X \times_k S\}/\cong$$

of line bundles up to isomorphism, with $f\colon T \to S$ acting by pullback $(\mathrm{id}_X \times f)^*$ of line bundles. Over a point this is $\operatorname{Pic}(X)$, the group of line bundles on X; over a general S it records how a line bundle on X can be deformed as S varies.

The rest of the section asks the single question a moduli functor poses, namely whether it is the functor of points of a scheme. The best possible answer is that $\mathcal{F}$ is, on the nose, the functor $h_M = \operatorname{Hom}(-, M)$ of maps into a scheme M. When this happens, points of M are objects, maps into M are families, and the geometry of M is the geometry of the moduli problem. This is the notion of a *fine moduli space*.

Definition 13.1.3: Fine Moduli Space (Set-Valued)

A *fine moduli space* for the moduli functor $\mathcal{F}$ is a scheme M together with an isomorphism of functors

$$\eta\colon \mathcal{F} \xrightarrow{\sim} h_M = \operatorname{Hom}(-, M)$$

where $h_M\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ sends S to the set $\operatorname{Hom}(S, M)$ of morphisms $S \to M$. When such an M exists, $\mathcal{F}$ is said to be *representable* (definition 0.4.21), and M *represents* $\mathcal{F}$.

The isomorphism η is the content of the definition, not the scheme M alone: it consists of a bijection $\eta_S\colon \mathcal{F}(S) \to \operatorname{Hom}(S, M)$ for every S, compatible with pullback, so that a family over S is literally the same datum as a morphism $S \to M$. Taking $S = \operatorname{Spec} k$ recovers the underlying set-theoretic promise, that isomorphism classes of objects biject with k-points of M; but the definition asks for this in families, for all S at once, which is why it captures variation and the naive point-set does not. A fine moduli space, when it exists, is unique up to unique isomorphism, since a representing object of a functor is: if M and M' both represent $\mathcal{F}$, then $h_M \cong \mathcal{F} \cong h_{M'}$, and the Yoneda lemma turns this isomorphism of functors into a unique isomorphism $M \cong M'$ of schemes.

The word "isomorphism of functors" is doing real work and is easy to underestimate, so before extracting its main consequence it helps to see one case where the representing scheme and the isomorphism are both completely explicit.

Example 13.1.4: Projective Space as a Fine Moduli Space

Fix $n \geq 0$ and let $\mathcal{F}$ be the functor of *rank*-1 *locally free quotients of the trivial rank*-$(n+1)$ *bundle*: to a scheme S assign

$$\mathcal{F}(S) = \left\{ \mathcal{O}_S^{n+1} \twoheadrightarrow \mathcal{L} \ : \ \mathcal{L} \text{ a line bundle on } S \right\} \Big/ \cong$$

where two surjections $q\colon \mathcal{O}_S^{n+1} \twoheadrightarrow \mathcal{L}$ and $q'\colon \mathcal{O}_S^{n+1} \twoheadrightarrow \mathcal{L}'$ are identified if there is an isomorphism $\mathcal{L} \cong \mathcal{L}'$ carrying q to q'. A morphism $f\colon T \to S$ acts by pullback: f^*q is the surjection $\mathcal{O}_T^{n+1} = f^*\mathcal{O}_S^{n+1} \twoheadrightarrow f^*\mathcal{L}$, again a line-bundle quotient because pullback is right exact and preserves local freeness. Then $\mathbb{P}^n = \mathbb{P}^n_k$ represents $\mathcal{F}$, with universal quotient the tautological surjection

$$\mathcal{O}_{\mathbb{P}^n}^{n+1} \twoheadrightarrow \mathcal{O}_{\mathbb{P}^n}(1)$$

onto the twisting line bundle $\mathcal{O}_{\mathbb{P}^n}(1)$.

To verify the universal property directly, unwind what a T-point of $\mathbb{P}^n$ is. A morphism $g\colon T \to \mathbb{P}^n$ is the same as a line bundle $\mathcal{L} = g^*\mathcal{O}(1)$ on T together with $n+1$ global sections $s_0, \dots, s_n \in \Gamma(T, \mathcal{L})$ generating $\mathcal{L}$ at every point, modulo the choice of trivialization of $\mathcal{L}$; the sections are the pullbacks $s_i = g^*(x_i)$ of the homogeneous coordinates. Packaging the sections into a map is exactly the surjection

$$q_g\colon \mathcal{O}_T^{n+1} \longrightarrow \mathcal{L}, \qquad (a_0, \dots, a_n) \longmapsto \textstyle\sum_i a_i s_i$$

which is surjective precisely because the s_i generate $\mathcal{L}$ everywhere. This assignment $g \mapsto q_g$ is the pullback of the tautological quotient along g, since $\mathcal{O}(1)$ is generated by $x_0, \dots, x_n$ and $q_g = g^*\big(\mathcal{O}_{\mathbb{P}^n}^{n+1} \twoheadrightarrow \mathcal{O}_{\mathbb{P}^n}(1)\big)$. Conversely, a line-bundle quotient $q\colon \mathcal{O}_T^{n+1} \twoheadrightarrow \mathcal{L}$ furnishes generating sections $s_i = q(e_i)$ and hence a unique morphism $g\colon T \to \mathbb{P}^n$ with $q_g = q$, by the universal property of projective space as Proj. The two constructions are mutually inverse and compatible with further pullback, so

$$\mathcal{F}(T) \xrightarrow{\sim} \operatorname{Hom}(T, \mathbb{P}^n)$$

is a natural isomorphism, and $\mathbb{P}^n$ is a fine moduli space for rank-1 quotients of $\mathcal{O}^{n+1}$.

The example is the template for the whole chapter: a moduli functor is pinned to a scheme by exhibiting one distinguished family over that scheme and showing every other family comes from it by pullback in a unique way. Here the distinguished family is $\mathcal{O}_{\mathbb{P}^n}^{n+1} \twoheadrightarrow \mathcal{O}(1)$, and the phrase "comes from it by pullback in a unique way" is the statement $\mathcal{F} \cong h_{\mathbb{P}^n}$. That this pattern is not special to $\mathbb{P}^n$ but is forced by the definition of representability is the content of the next result.

Representability was defined as an isomorphism $\mathcal{F} \cong h_M$ of functors, which is a statement about all schemes S at once and so seems to require checking infinitely many bijections. The Yoneda lemma collapses this to a single piece of data: one family over M itself. The mechanism is that a natural transformation out of h_M is completely determined by where it sends the identity morphism id_M, and running this in reverse shows that an isomorphism $\mathcal{F} \cong h_M$ is the same as a family over M with a universal property. That family is what one constructs in practice, and what one produces when asked to "exhibit the moduli space".

Proposition 13.1.5: Universal Family

Let M be a scheme and $\mathcal{F}$ a moduli functor. Giving an isomorphism of functors $\eta\colon \mathcal{F} \xrightarrow{\sim} h_M$ is equivalent to giving a family $\mathcal{U} \in \mathcal{F}(M)$, the *universal family*, with the following universal property: for every scheme S and every family $\xi \in \mathcal{F}(S)$ there is a unique morphism $g\colon S \to M$ with $g^*\mathcal{U} = \xi$. In particular, $\mathcal{F}$ is representable by M if and only if such a universal family exists, and then $\mathcal{U}$ corresponds under η to the identity $\mathrm{id}_M \in h_M(M)$.

***Proof*:**
The argument is the Yoneda lemma (theorem 0.4.22) applied to the functor $\mathcal{F}$ and the representable functor h_M; the proof records what each direction of the equivalence produces so that the universal family and the isomorphism are explicit.

Step 1: From an isomorphism to a universal family. Suppose given a natural isomorphism

$\eta\colon \mathcal{F} \xrightarrow{\sim} h_M$. In particular $\eta_M\colon \mathcal{F}(M) \to h_M(M) = \mathrm{Hom}(M, M)$ is a bijection, so there is a unique family

$$\mathcal{U} := \eta_M^{-1}(\mathrm{id}_M) \in \mathcal{F}(M)$$

the preimage of the identity morphism. Fix a family $\xi \in \mathcal{F}(S)$ and set $g := \eta_S(\xi) \in \mathrm{Hom}(S, M)$. Naturality of η with respect to the morphism $g\colon S \to M$ is the commuting square

$$\begin{array}{ccc} \mathcal{F}(M) & \xrightarrow{\eta_M} & \mathrm{Hom}(M,M) \\ \big\downarrow{\scriptstyle g^*} & & \big\downarrow{\scriptstyle g^*} \\ \mathcal{F}(S) & \xrightarrow{\eta_S} & \mathrm{Hom}(S,M) \end{array}$$

where the right-hand g^* is precomposition with g. Chase the family $\mathcal{U} \in \mathcal{F}(M)$ in the top-left corner around the square. Across the top it maps to $\eta_M(\mathcal{U}) = \mathrm{id}_M$, and then down to $g^*(\mathrm{id}_M) = \mathrm{id}_M \circ g = g$; down the left it maps to $g^*\mathcal{U}$, and then across to $\eta_S(g^*\mathcal{U})$. Equality of the two routes yields $\eta_S(g^*\mathcal{U}) = g = \eta_S(\xi)$, and η_S is injective, so

$$g^*\mathcal{U} = \xi$$

This exhibits ξ as a pullback of $\mathcal{U}$. For uniqueness, suppose $g'\colon S \to M$ also satisfies $(g')^*\mathcal{U} = \xi$. Running the same naturality square for g' gives $\eta_S((g')^*\mathcal{U}) = g'$, hence $g' = \eta_S(\xi) = g$. Thus g is the unique morphism with $g^*\mathcal{U} = \xi$, and $\mathcal{U}$ has the claimed universal property.

Step 2: From a universal family to an isomorphism. Conversely, suppose given a family $\mathcal{U} \in \mathcal{F}(M)$ such that for every S and every $\xi \in \mathcal{F}(S)$ there is a unique $g\colon S \to M$ with $g^*\mathcal{U} = \xi$. Define, for each S, a map

$$\theta_S\colon \mathrm{Hom}(S, M) \longrightarrow \mathcal{F}(S), \qquad g \longmapsto g^*\mathcal{U}$$

which is a well-defined map of sets because pullback of families is defined. Naturality of $\theta = (\theta_S)_S$ is functoriality of pullback: for $f\colon T \to S$ and $g\colon S \to M$,

$$\theta_T(g \circ f) = (g \circ f)^*\mathcal{U} = f^*(g^*\mathcal{U}) = f^*\theta_S(g)$$

so θ is a natural transformation $h_M \to \mathcal{F}$. The universal property says exactly that each θ_S is a bijection: given $\xi \in \mathcal{F}(S)$, existence of a g with $g^*\mathcal{U} = \xi$ is surjectivity of θ_S, and uniqueness of that g is injectivity. A natural transformation that is bijective on every object is an isomorphism of functors, so $\theta\colon h_M \xrightarrow{\sim} \mathcal{F}$ is an isomorphism; its inverse $\eta = \theta^{-1}\colon \mathcal{F} \xrightarrow{\sim} h_M$ is the sought isomorphism.

Step 3: The two constructions are mutually inverse. It remains to check that starting from η, producing $\mathcal{U} = \eta_M^{-1}(\mathrm{id}_M)$, and then producing θ returns η^{-1}, and symmetrically. From Step 1, for $g \in \mathrm{Hom}(S, M)$ set $\xi = g^*\mathcal{U}$; the uniqueness there gives $\eta_S(g^*\mathcal{U}) = g$, that is $\eta_S \circ \theta_S = \mathrm{id}_{\mathrm{Hom}(S,M)}$, so $\theta = \eta^{-1}$. In particular $\mathcal{U} = \theta_M(\mathrm{id}_M) = \eta_M^{-1}(\mathrm{id}_M)$ corresponds to id_M under η, as claimed. The two passages are therefore inverse to one another, giving the asserted equivalence between isomorphisms $\mathcal{F} \cong h_M$ and universal families, completing the proof.

The proposition is the reason the functorial viewpoint is workable, not only correct. To prove a scheme M is a fine moduli space one need not construct a bijection $\mathcal{F}(S) \cong \mathrm{Hom}(S, M)$ for every S by hand; it suffices to write down one family $\mathcal{U}$ over M and verify that every family is a unique pullback of it. In the projective-space example (example 13.1.4) the universal family is the tautological quotient $\mathcal{O}^{n+1} \twoheadrightarrow \mathcal{O}(1)$, and the verification that every quotient over T is a unique pullback is precisely the universal property of $\mathbb{P}^n$ that was checked there; the proposition explains why that single verification is enough. The universal family also carries a concrete geometric meaning: its fiber over a point $m \in M$ is the object that m classifies, so $\mathcal{U}$ is a single geometric family whose fiber over each point of the moduli space is the corresponding object. This is the sense in which a fine moduli space "carries the objects on its back".

Representability is a strong demand, and the structural constraint it imposes is exactly where the interesting moduli problems fail it. A functor of the form h_M is not an arbitrary functor: it satisfies a gluing condition inherited from morphisms of schemes, and a moduli functor that violates this condition cannot be representable no matter how the objects are arranged.

The motivating paragraph for the warning is the following observation about morphisms. A morphism $S \to M$ is a local datum: to give it is to give morphisms $S_i \to M$ on the opens of a cover that agree on overlaps, because M is glued from its structure sheaf. Hence h_M is a *Zariski sheaf*: a compatible family of T-points on a cover glues to a unique T-point. Any representable moduli functor must inherit this, and the obstruction to it is the presence of automorphisms.

13.1.6 Warning: Automorphisms Obstruct Representability A representable functor h_M is necessarily a *Zariski sheaf*: for any scheme S with a Zariski open cover $\{S_i\}$, a collection of morphisms $g_i \colon S_i \to M$ with $g_i|_{S_i \cap S_j} = g_j|_{S_i \cap S_j}$ glues to a unique $g \colon S \to M$ restricting to each g_i. Consequently, if a moduli functor $\mathcal{F}$ fails the sheaf condition, it cannot be representable. Objects with nontrivial automorphisms manufacture exactly this failure. Suppose an object X_0 over k has a nontrivial automorphism. One can then build a family over a base S whose restriction to each member S_i of an open cover is the constant family $X_0 \times S_i$, so that the family is trivial locally on the cover, yet whose gluing over the overlaps is twisted by the automorphism so that the family is not globally the constant family $X_0 \times S$. Such a family has isomorphic fibers over every point of S but is not itself constant. Under any candidate isomorphism $\mathcal{F} \cong h_M$ its class in $\mathcal{F}(S)$ would have to glue, from the locally-constant data, to the class of the constant family, and the two are distinct elements of $\mathcal{F}(S)$: the sheaf condition is violated, and no fine moduli space exists.

This is the central obstruction of the chapter. It is the reason the moduli of elliptic curves has no fine moduli space and must be replaced by the coarse j-line (section 13.3), and the sheaf-theoretic diagnosis given here is sharpened into a working representability criterion in section 13.5. The class of examples alluded to, locally trivial but globally twisted families, is made fully explicit in example 13.3.4.

The warning identifies the fault line that organizes the entire book. When $\mathcal{F}$ is a sheaf and is locally representable, it is representable, and the Hilbert and Quot schemes of the next chapter provide the fundamental such functors. When $\mathcal{F}$ fails to be a sheaf because its objects have automorphisms, one has three responses, and the book pursues all of them in turn: settle for a coarse moduli space that co-represents $\mathcal{F}$ (section 13.3), rigidify the problem by adding structure to kill the automorphisms (section 13.4), or enlarge the category of geometric objects to stacks, in which automorphisms are remembered rather than fatal. The functor of points is the frame in which all three are stated.

Exercise 13.1.1

1. Work out the functor of points of $\mathbb{P}^1$ directly from example 13.1.4. Describe $\mathcal{F}(S)$ for the functor of rank-1 locally free quotients of $\mathcal{O}_S^2$, show that a T-point of $\mathbb{P}^1$ is the same as a pair of sections (s_0, s_1) of a line bundle $\mathcal{L}$ on T generating $\mathcal{L}$ at every point, taken modulo isomorphism of $\mathcal{L}$, and check that on $T = \operatorname{Spec} k$ this recovers the usual k-points $[a_0 : a_1]$ of

$\mathbb{P}^1$.

2. Let X_0 be an object over k with a nontrivial automorphism $\sigma\colon X_0 \to X_0$, and let $S = S_1 \cup S_2$ be a scheme with two opens whose intersection $S_{12} = S_1 \cap S_2$ is nonempty. Using σ, describe how to glue the constant families $X_0 \times S_1$ and $X_0 \times S_2$ over S_{12} into a family ξ over S that has X_0 as every fiber but need not be isomorphic to the constant family $X_0 \times S$. Explain, referring to box 13.1.6, why the existence of such a ξ shows that the set of isomorphism classes $\mathcal{F}(\operatorname{Spec} k)$ alone cannot determine a scheme structure representing $\mathcal{F}$.

3. Suppose $\mathcal{F}$ is represented by a scheme M with universal family $\mathcal{U}$, and let $\xi \in \mathcal{F}(S)$ be a family that happens to be the pullback $h^*\mathcal{U}$ of $\mathcal{U}$ along some morphism $h\colon S \to M$. Prove, using only proposition 13.1.5, that h is the *unique* morphism $S \to M$ classifying ξ, and deduce that the classifying map of the universal family $\mathcal{U}$ itself is id_M.

4. A scheme M is given as a fine moduli space for a functor $\mathcal{F}$ by an isomorphism $\eta\colon \mathcal{F} \xrightarrow{\sim} h_M$. Explain how to read off the universal family $\mathcal{U} \in \mathcal{F}(M)$ from η, and carry this out for $M = \mathbb{P}^n$ and the quotient functor of example 13.1.4: identify which element of $\mathcal{F}(\mathbb{P}^n)$ is $\eta_{\mathbb{P}^n}^{-1}(\mathrm{id}_{\mathbb{P}^n})$.

5. Define a functor $\mathcal{G}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ by $\mathcal{G}(S) = \{\text{invertible global functions on } S\} = \Gamma(S, \mathcal{O}_S)^\times$, with pullback the obvious restriction. Show that $\mathcal{G}$ is a Zariski sheaf, and prove that $\mathcal{G}$ is representable by the scheme $\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$ by exhibiting the universal family and verifying the universal property of proposition 13.1.5. Contrast this with the functor $\mathcal{G}'(S) = \Gamma(S, \mathcal{O}_S)^\times/(\text{constants } k^\times)$ and argue that $\mathcal{G}'$ fails the sheaf condition, hence is not representable.

13.2 Fine Moduli Spaces: the Grassmannian

The previous section made the abstract case: a moduli problem is a functor $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ (definition 13.1.1), a fine moduli space is a scheme M with $\mathcal{F} \cong h_M$ (definition 13.1.3), and representability is equivalent to the existence of a universal family (proposition 13.1.5). What that section could not yet supply is a moduli problem interesting enough to matter and simple enough to solve by hand from beginning to end. This section supplies it. The Grassmannian $\mathrm{Gr}(d, n)$, whose points are the d-dimensional quotient spaces of a fixed n-dimensional vector space, is the model fine moduli problem: the functor it represents is written down in one line, the representing scheme is built explicitly from affine charts, the universal family is exhibited concretely, and the whole space is shown to be projective through the Plücker embedding. Every later construction in this book, most immediately the Hilbert and Quot schemes of the following chapter, is cut out of a Grassmannian, so the effort spent here is repaid many times over.

The design choice that makes the construction work is to phrase a point of the Grassmannian not as a subspace but as a *quotient*. A d-dimensional subspace $W \subseteq k^n$ and the quotient $k^n \twoheadrightarrow k^n/W$ carry the same information, but the quotient behaves far better in families: a surjection of a trivial bundle onto a locally free sheaf is a functorial datum requiring no side condition, whereas “a locally free subsheaf” must carry the extra condition that the quotient be locally free too (so that the sub is a *subbundle*, not just a subsheaf). Phrasing points as quotients builds that condition in from the start. This is the same bookkeeping that made $\mathbb{P}^n$ the moduli space of rank-1 quotients of $\mathcal{O}^{n+1}$ in example 13.1.4, and the Grassmannian is the direct generalization from rank 1 to rank d.

The functor of points is stated for an arbitrary base scheme S: a family over S is a rank-d locally free quotient of the trivial rank-n sheaf $\mathcal{O}_S^n$, and the whole point of working with quotients is that pulling back a surjection of sheaves along a morphism $T \to S$ again gives a surjection, so the assignment is contravariant with no further hypotheses.

Definition 13.2.1: The Grassmannian Functor

Fix integers $0 \le d \le n$. The *Grassmannian functor* $\underline{\mathrm{Gr}}(d,n)\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ assigns to a scheme S the set

$$\underline{\mathrm{Gr}}(d,n)(S) = \left\{ \mathcal{O}_S^n \overset{q}{\twoheadrightarrow} \mathcal{Q} \;\middle|\; \mathcal{Q} \text{ locally free of rank } d \right\} \Big/ \cong$$

of surjections of $\mathcal{O}_S$-modules from the trivial sheaf $\mathcal{O}_S^n$ onto a locally free sheaf $\mathcal{Q}$ of rank d, where two surjections $q\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ and $q'\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}'$ are identified when there is an isomorphism $\varphi\colon \mathcal{Q} \xrightarrow{\sim} \mathcal{Q}'$ with $\varphi \circ q = q'$. For a morphism $f\colon T \to S$ the pullback map sends the class of q to the class of $f^*q\colon \mathcal{O}_T^n \twoheadrightarrow f^*\mathcal{Q}$.

Because f^* is right exact, f^*q is again surjective, and $f^*\mathcal{Q}$ is again locally free of rank d, so the pullback map is well defined and $\underline{\mathrm{Gr}}(d,n)$ is a functor. Two remarks fix the meaning of the equivalence relation. First, the identification by isomorphisms of $\mathcal{Q}$ is forced by the moduli philosophy of definition 13.1.1: a family is an equivalence class, and here the class of q remembers only the kernel $\ker q \subseteq \mathcal{O}_S^n$ together with the fact that the quotient is a bundle. Second, over a field, $\underline{\mathrm{Gr}}(d,n)(k)$ is exactly the set of d-dimensional quotient spaces of k^n, equivalently the set of $(n-d)$-dimensional subspaces $\ker q \subseteq k^n$, which is the classical Grassmannian of $(n-d)$-planes. The functor is the family-version of that set.

The subbundle picture is the same datum read backwards, and it is worth recording because half the literature (and half of this book) prefers it. Given a quotient $q\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ with $\mathcal{Q}$ locally free of rank d, the kernel $\mathcal{S} = \ker q$ is a locally free subsheaf of rank $n-d$ whose quotient is locally free, hence a *subbundle*; conversely a rank-$(n-d)$ subbundle $\mathcal{S} \hookrightarrow \mathcal{O}_S^n$ has locally free quotient of rank d. The two descriptions are interchanged by the short exact sequence

$$0 \to \mathcal{S} \to \mathcal{O}_S^n \xrightarrow{q} \mathcal{Q} \to 0 \tag{13.1}$$

and neither is more basic than the other. The convention here takes the quotient $\mathcal{Q}$ as primary and calls $\mathcal{S}$ the associated sub.

Definition 13.2.2: The Grassmannian

The *Grassmannian* $\mathrm{Gr}(d,n)$ is the scheme constructed in theorem 13.2.4 below, characterized by the property that it represents the functor $\underline{\mathrm{Gr}}(d,n)$ of definition 13.2.1.

Before proving representability, it helps to see the charts concretely in the smallest new case, so that the abstract cover in the proof reads as bookkeeping for a picture already understood.

Example 13.2.3: Charts on the Grassmannian of Lines in 3-Space

Take $d=2$, $n=3$, so $\mathrm{Gr}(2,3)$ parametrizes rank-2 quotients of $\mathcal{O}_S^3$, equivalently rank-1 subbundles (lines) of $\mathcal{O}_S^3$. Over a field a point is a line $\ell \subseteq k^3$ through the origin, spanned by a nonzero vector (a_0, a_1, a_2) up to scale, so $\mathrm{Gr}(2,3)(k)$ is exactly $\mathbb{P}^2$, the set of such lines. The three charts are the loci where $a_0 \neq 0$, $a_1 \neq 0$, $a_2 \neq 0$; on the first, scaling to $a_0 = 1$ leaves the two free coordinates $(a_1, a_2) \in \mathbb{A}^2$, and these are the affine coordinates on that chart. So $\mathrm{Gr}(2,3) \cong \mathbb{P}^2$, its dual realization as lines rather than planes. This is the case $n = 3$ of the identification $\mathrm{Gr}(n-1, n) = \mathbb{P}^{n-1}$, which is the duality of exercise 13.2.1 (6) applied to $\mathrm{Gr}(1, n) = \mathbb{P}^{n-1}$; the three charts here are indexed by the nonzero coordinate of the spanning line vector, which under that duality corresponds to the U_I charts of theorem 13.2.4 indexed by the two-subsets $I \subseteq \{1, 2, 3\}$.

The representability theorem is the technical heart of the section. Its proof is the archetype for every later representability argument: cover the functor by open subfunctors, each of which is manifestly representable by an affine space, then check that these charts glue to a scheme representing the whole. The charts are indexed by d-element subsets of the coordinates, and the chart U_I is the locus where a given family, restricted to those d coordinates, is already an isomorphism onto the quotient.

The one input this argument needs, beyond the definitions, is the criterion that a functor which is a Zariski sheaf and admits an open cover by representable subfunctors is itself representable. That criterion is developed in full generality later (see theorem 13.5.6); here the covering subfunctors are so explicit that the gluing can be exhibited by hand, and the proof below does exactly that rather than invoking the general statement. This keeps the argument self-contained and shows the mechanism the general theorem abstracts.

Theorem 13.2.4: Representability of the Grassmannian

For $0 \le d \le n$ there is a scheme $\mathrm{Gr}(d, n)$, smooth and projective over $\mathbb{Z}$ (hence over any base) of relative dimension $d(n-d)$, representing the functor $\underline{\mathrm{Gr}}(d, n)$ of definition 13.2.1. It is covered by $\binom{n}{d}$ affine charts, each isomorphic to $\mathbb{A}^{d(n-d)}$.

***Proof*:**
The construction proceeds by building the covering subfunctors, identifying each with an affine space, and gluing. Throughout, $[n] = \{1, \ldots, n\}$, and for a d-element subset $I \subseteq [n]$ the inclusion of the coordinates indexed by I is written $\iota_I : \mathcal{O}_S^d \to \mathcal{O}_S^n$.

Step 1: The covering subfunctors. For each d-element subset $I \subseteq [n]$ define a subfunctor $\underline{U}_I \subseteq \underline{\mathrm{Gr}}(d, n)$ by declaring, for a family $q : \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ over S, that $[q] \in \underline{U}_I(S)$ when the composite

$$\mathcal{O}_S^d \xrightarrow{\iota_I} \mathcal{O}_S^n \xrightarrow{q} \mathcal{Q}$$

is an isomorphism. This condition is invariant under the equivalence relation of definition 13.2.1: replacing q by $\varphi \circ q$ replaces the composite by $\varphi \circ (q \circ \iota_I)$, still an isomorphism. It is also stable under pullback, since f^* preserves isomorphisms, so $\underline{U}_I$ is a genuine subfunctor.

The condition defining $\underline{U}_I$ is open: the composite $q \circ \iota_I$ is a map between locally free sheaves of the same rank d, and the locus in S where such a map is an isomorphism is the non-vanishing locus of its determinant, an open subscheme. A family q over S therefore lies in $\underline{U}_I(T)$ after pullback along $f : T \to S$ if and only if f factors through this open subscheme, which is the statement that $\underline{U}_I \hookrightarrow \underline{\mathrm{Gr}}(d, n)$ is an open subfunctor.

Step 2: The charts cover. A family $q : \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ lies in $\underline{U}_I(S)$ for suitable I at least locally on S. Fix a point $s \in S$; the surjection q of a rank-n trivial bundle onto a rank-d bundle has, at the residue field $\kappa(s)$, a surjection $\kappa(s)^n \twoheadrightarrow \mathcal{Q} \otimes \kappa(s)$ of a rank-n space onto a rank-d space. Some d of the standard basis vectors of $\kappa(s)^n$ map to a basis of the d-dimensional target, that is, $q \circ \iota_I$ is an isomorphism at s for some I. By openness of the isomorphism locus (Step 1), $q \circ \iota_I$ is an isomorphism on a neighborhood of s. Thus the pullback of q to that neighborhood lies in $\underline{U}_I$, and the $\underline{U}_I$ cover $\underline{\mathrm{Gr}}(d, n)$ in the sense that every family is, locally on its base, in some chart.

Step 3: Each chart is an affine space. Fix I. Order the $n-d$ complementary coordinates as $j_1, \ldots, j_{n-d}$, the elements of $[n] \setminus I$. The claim is that $\underline{U}_I$ is represented by the affine space $\mathbb{A}^{d(n-d)}$ with coordinate ring $k[x_{ab}]$, where a ranges over I and b ranges over $[n] \setminus I$, that is, by the space of $d \times (n-d)$ matrices.

Given a family $[q] \in \underline{U}_I(S)$, the composite $\theta = q \circ \iota_I : \mathcal{O}_S^d \xrightarrow{\sim} \mathcal{Q}$ is an isomorphism, and $\theta^{-1} \circ q : \mathcal{O}_S^n \to \mathcal{O}_S^d$ is a surjection restricting to the identity on the coordinates I. Its remaining entries assemble into a $d \times (n-d)$ matrix $M(q)$ over $\mathcal{O}_S(S)$, whose (a, b) entry records the image in the a-th standard basis vector of $\mathcal{O}_S^d$ of the b-th complementary coordinate. Replacing q by $\varphi \circ q$ replaces θ by $\varphi \circ \theta$ and leaves $\theta^{-1} \circ q$ unchanged, so $M(q)$ depends only on the class $[q]$ and gives a well-defined map

$$\underline{U}_I(S) \longrightarrow \operatorname{Hom}(S, \mathbb{A}^{d(n-d)}) = \{d \times (n-d) \text{ matrices over } \mathcal{O}_S(S)\}$$

natural in S. Conversely, a matrix M over $\mathcal{O}_S(S)$ determines the surjection $q_M : \mathcal{O}_S^n \twoheadrightarrow \mathcal{O}_S^d$ that is the identity on the coordinates I and has entry M_{ab} on the complementary coordinate b; this q_M visibly lies in $\underline{U}_I(S)$, with $\theta = \mathrm{id}$, and $M(q_M) = M$. The two constructions are mutually inverse and natural, so $\underline{U}_I \cong h_{\mathbb{A}^{d(n-d)}}$. By the Yoneda direction of proposition 13.1.5, $\underline{U}_I$ is representable by $U_I := \mathbb{A}^{d(n-d)}$.

Step 4: Gluing. For two subsets I, J the overlap subfunctor $\underline{U}_I \cap \underline{U}_J$ is the locus in $U_I \cong \mathbb{A}^{d(n-d)}$ where, in addition, $q \circ \iota_J$ is an isomorphism. In the coordinates of U_I a family is the surjection q_M; the composite $q_M \circ \iota_J$ is the $d \times d$ submatrix of the full $d \times n$ matrix $(\,\mathrm{Id}_I | M\,)$ on the columns indexed by J, so $\underline{U}_I \cap \underline{U}_J$ is the principal open subset of U_I where the determinant of that submatrix is invertible, an open subscheme $U_{IJ} \subseteq U_I$. The transition isomorphism $U_{IJ} \xrightarrow{\sim} U_{JI}$ is read off by recomputing the normalized matrix relative to the coordinates J: it sends the matrix normalized on I to the matrix normalized on J, obtained by left-multiplying $(\,\mathrm{Id}_I | M\,)$ by the inverse of its J-submatrix and deleting the resulting identity block. These transition maps are morphisms of schemes (entries are rational functions with the relevant determinant inverted) and satisfy the cocycle condition on triple overlaps, because on $U_{IJ} \cap U_{IK}$ all three normalizations describe the *same* surjection q read in three coordinate frames, so passing $I \to J \to K$ agrees with $I \to K$.

By the gluing construction for schemes, the U_I glue along the U_{IJ} to a scheme $\mathrm{Gr}(d, n)$. The subfunctors $\underline{U}_I$ glue correspondingly to a functor which, by Steps 1 and 2, is $\underline{\mathrm{Gr}}(d, n)$, and each $\underline{U}_I$ is represented by the chart U_I; a family over an arbitrary S is one that restricts to a chart family on each piece of a cover of S, and the transition maps guarantee these glue to a unique morphism $S \to \mathrm{Gr}(d, n)$. Hence $\mathrm{Gr}(d, n)$ represents $\underline{\mathrm{Gr}}(d, n)$.

Step 5: Dimension, smoothness, projectivity. Each chart is $\mathbb{A}^{d(n-d)}$, so $\mathrm{Gr}(d, n)$ is smooth of dimension $d(n-d)$ and covered by $\binom{n}{d}$ such charts. The construction used only $\mathbb{Z}$-coefficients (the transition maps are integral rational functions), so $\mathrm{Gr}(d, n)$ is defined and smooth over $\mathbb{Z}$, hence over any base by base change. Projectivity is the content of proposition 13.2.7, which realizes $\mathrm{Gr}(d, n)$ as a closed subscheme of a projective space. This completes the proof.

The theorem is the first fine moduli space of the book realized from scratch. What the reader should take from it is less the statement than the shape of the argument, because it recurs: a moduli functor is representable when it can be covered by open subfunctors each of which is an affine space of parameters, and the parameters are the freedom left after normalizing a family into a standard form. Here the standard form is "the identity on a chosen set of d coordinates," the remaining freedom is a $d \times (n-d)$ matrix, and the transition between two normalizations is a determinant computation.

The Hilbert and Quot schemes of the next chapter are built by exhibiting them as closed subfunctors of Grassmannian functors, so representability there is inherited from the theorem just proved.

The case $d = 1$ recovers the running example of the previous section. A rank-1 locally free quotient of $\mathcal{O}_S^n$ is exactly the datum classified by $\mathbb{P}^{n-1}$, and unwinding the definitions gives $\mathrm{Gr}(1, n) = \mathbb{P}^{n-1}$, matching $\mathbb{P}^{n-1}$ as the moduli space of rank-1 quotients of $\mathcal{O}^n$ from example 13.1.4 (there stated for $\mathcal{O}^{n+1}$, giving $\mathbb{P}^n$; the index shift is the passage from $n + 1$ to n). The charts of the theorem specialize to the standard affine charts of projective space: the subset $I = \{i\}$ is the locus where the i-th coordinate of the quotient is nonzero, and the $1 \times (n - 1)$ matrix of remaining entries is exactly the tuple of affine coordinates on that chart. So the Grassmannian is the common generalization of projective space to higher-rank quotients, and $\mathbb{P}^{n-1}$ is its rank-1 case.

Representability produces, through proposition 13.1.5, a universal family on $\mathrm{Gr}(d, n)$: the family corresponding to the identity morphism $\mathrm{Gr}(d, n) \to \mathrm{Gr}(d, n)$. Naming it and its associated sub is worthwhile, because these two bundles are the source of nearly every natural construction on the Grassmannian, from its tangent bundle to the tautological classes computed against it in intersection theory.

Definition 13.2.5: Universal Quotient and Universal Sub

On $G = \mathrm{Gr}(d, n)$ the *universal quotient bundle* $\mathcal{Q}$ is the locally free sheaf of rank d appearing in the family $q_{\mathrm{univ}} \colon \mathcal{O}_G^n \twoheadrightarrow \mathcal{Q}$ that corresponds, under the isomorphism $\underline{\mathrm{Gr}}(d, n) \cong h_G$ of theorem 13.2.4, to the identity morphism $\mathrm{id}_G \in h_G(G)$. Its kernel $\mathcal{S} = \ker q_{\mathrm{univ}}$, a locally free sheaf of rank $n - d$, is the *universal subbundle* (or *tautological subbundle*). They sit in the universal short exact sequence

$$0 \to \mathcal{S} \to \mathcal{O}_G^n \xrightarrow{q_{\mathrm{univ}}} \mathcal{Q} \to 0$$

on $\mathrm{Gr}(d, n)$.

The universal property is the statement of proposition 13.1.5 specialized to this family: for every scheme S and every rank-d locally free quotient $q \colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}'$, there is a unique morphism $\varphi_q \colon S \to \mathrm{Gr}(d, n)$ with $\varphi_q^* q_{\mathrm{univ}} = q$ (up to the identification of definition 13.2.1), that is, $\varphi_q^* \mathcal{Q} \cong \mathcal{Q}'$ compatibly with the surjection from $\mathcal{O}_S^n$. Every family of d-dimensional quotients is a pullback of the one universal family, in exactly one way. This is what it means for $\mathrm{Gr}(d, n)$ to be a *fine* moduli space in the sense of definition 13.1.3: the universal bundle is a single geometric object over the moduli space from which all others are cut by pullback.

The tautological name comes from the field-point picture. Over the classical Grassmannian, the fiber of $\mathcal{S}$ at a point $[\ell]$ (a subspace $\ell \subseteq k^n$) is the subspace ℓ itself, and the fiber of $\mathcal{Q}$ is the quotient k^n/ℓ; the subbundle is "tautologically" the space its point names. The next example makes the universal bundle explicit in charts, staying within the promised distance of the definition.

Example 13.2.6: The Universal Sequence in a Chart

On the chart $U_I \cong \mathbb{A}^{d(n-d)}$ of theorem 13.2.4 the universal quotient is the free sheaf $\mathcal{O}_{U_I}^d$, with universal surjection $q_{\mathrm{univ}}|_{U_I} = (\,\mathrm{Id}_I \,|\, M\,) \colon \mathcal{O}_{U_I}^n \twoheadrightarrow \mathcal{O}_{U_I}^d$, where $M = (x_{ab})$ is the matrix of chart coordinates. The universal sub $\mathcal{S}|_{U_I}$ is the kernel, freely generated by the $n - d$ columns

$$s_b = e_b - \sum_{a \in I} x_{ab}\, e_a \qquad (b \in [n] \setminus I)$$

where $e_1, \ldots, e_n$ is the standard basis of $\mathcal{O}_{U_I}^n$: each s_b is annihilated by $(\,\mathrm{Id}_I \big| M\,)$ since its I-coordinates $-x_{ab}$ exactly cancel the contribution of its complementary coordinate e_b. Over the overlap of two charts these local frames differ by the transition matrices of Step 4 of the proof, which is the concrete content of $\mathcal{Q}$ and $\mathcal{S}$ being globally defined bundles rather than trivial ones.

The Grassmannian was built as a smooth scheme covered by affine charts, but its charts alone do not reveal that it is projective; the affine cover exhibits it as separated and of finite type, not as a closed subscheme of any $\mathbb{P}^N$. Projectivity comes from a single global construction: send a d-dimensional subspace to its top exterior power, a line in $\wedge^d k^n$, which is a point of the projective space $\mathbb{P}(\wedge^d k^n)$. In families this is the map classified by the line bundle $\wedge^{n-d}\mathcal{S}^\vee$ (or equivalently $\wedge^d \mathcal{Q}$), and the resulting morphism turns out to be a closed embedding whose image is cut out by explicit quadratic equations. This is the mechanism that makes the Grassmannian, and through it every Hilbert and Quot scheme, a projective variety.

The functorial version uses the quotient. A rank-d quotient $q\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ induces on top exterior powers a surjection $\wedge^d q\colon \wedge^d \mathcal{O}_S^n = \mathcal{O}_S^{\binom{n}{d}} \twoheadrightarrow \wedge^d \mathcal{Q}$ onto a line bundle, which is precisely the datum classifying a morphism $S \to \mathbb{P}(\wedge^d k^n) = \mathbb{P}^{\binom{n}{d}-1}$ (a rank-1 quotient of a trivial bundle, in the sense of example 13.1.4). This assignment is natural in S, hence a morphism of functors, hence a morphism of the representing schemes.

Proposition 13.2.7: The Plücker Embedding

The morphism

$$\mathrm{p}\colon \mathrm{Gr}(d,n) \longrightarrow \mathbb{P}\Big(\bigwedge\nolimits^d k^n\Big) = \mathbb{P}^{\binom{n}{d}-1}$$

induced by $q \mapsto \wedge^d q$ is a closed embedding. Its image is the zero locus of the quadratic *Plücker relations*: writing the homogeneous coordinates as $p_{i_1\cdots i_d}$ indexed by d-element subsets $i_1 < \cdots < i_d$ of $[n]$ (the coordinates on the decomposable vectors $e_{i_1} \wedge \cdots \wedge e_{i_d}$), the image is cut out by

$$\sum_{t=0}^{d} (-1)^t \, p_{i_1\cdots i_{d-1} j_t} \, p_{j_0 \cdots \widehat{j_t} \cdots j_d} = 0 \tag{13.2}$$

for all $i_1 < \cdots < i_{d-1}$ and $j_0 < \cdots < j_d$ in $[n]$, where $\widehat{j_t}$ denotes omission. Consequently $\mathrm{Gr}(d,n)$ is projective over $\mathbb{Z}$, hence over any base.

***Proof*:**
The argument has three parts: the map is a morphism (already established above), it is a locally closed embedding checked on charts, and its image is exactly the Plücker locus, which is closed. Work with the sub $\mathcal{S}$ of rank $n-d$; equivalently one may phrase everything through $\wedge^d\mathcal{Q}$, and the two give the same map since $\wedge^d \mathcal{Q} \cong \wedge^n \mathcal{O}_S^n \otimes (\wedge^{n-d}\mathcal{S})^\vee$ is the same line bundle up to the trivial twist $\wedge^n \mathcal{O}_S^n \cong \mathcal{O}_S$.

Step 1: Injectivity on points and the decomposable coordinates. On $\bar{k}$-points, p sends a subspace $W = \ker q \subseteq \bar{k}^n$ of dimension $n-d$ (equivalently the quotient of dimension d) to the line $\wedge^d(\bar{k}^n/W)$, whose Plücker coordinates $p_{i_1\cdots i_d}$ are the $d \times d$ minors of a $d \times n$ matrix representing q. A subspace is recovered from these minors: the coordinate $p_{i_1\cdots i_d}$ is nonzero exactly when $q \circ \iota_I$ is an isomorphism for $I = \{i_1 < \cdots < i_d\}$, that is, exactly on the chart U_I, and on that chart the ratios of minors recover the normalized matrix M of theorem 13.2.4. Hence distinct points of $\mathrm{Gr}(d,n)$ have distinct images, so p is injective on points; the same computation applies with S-valued families, giving injectivity of the map of functors.

Step 2: A closed embedding on each chart. Fix I and work over the standard affine chart $D_+(p_I) \subseteq \mathbb{P}^{\binom{n}{d}-1}$ where the coordinate p_I is nonzero. By Step 1, $\mathrm{p}^{-1}(D_+(p_I)) = U_I$. On U_I the family is $(\,\mathrm{Id}_I | M\,)$, and its $d \times d$ minor on columns J is a polynomial in the entries x_{ab} of M; normalizing so that $p_I = 1$, the minor on the columns I with one index a replaced by a complementary index b equals $\pm x_{ab}$ exactly. Thus the coordinate functions p_J/p_I on $D_+(p_I)$ pull back to polynomials in the x_{ab} that include, among them, all $d(n-d)$ coordinates $\pm x_{ab}$ themselves. The comorphism

$$k\big[\,p_J/p_I \big| J\,\big] \longrightarrow k[x_{ab}] = \mathcal{O}(U_I)$$

is therefore surjective (the generators x_{ab} are in the image), so $\mathrm{p}\,|_{U_I} : U_I \to D_+(p_I)$ is a closed immersion. A morphism that is a closed immersion over each member of an open cover of the target, with the preimages covering the source, is a locally closed immersion; here the images $\mathrm{p}(U_I) = \mathrm{p}(\mathrm{Gr}(d,n)) \cap D_+(p_I)$ cover $\mathrm{p}(\mathrm{Gr}(d,n))$, so p is an immersion onto a locally closed subscheme.

Step 3: The image is the closed Plücker locus. It remains to identify the image with the closed subscheme $Y \subseteq \mathbb{P}^{\binom{n}{d}-1}$ defined by (13.2), which will show the immersion is closed. The relations (13.2) are exactly the condition that a vector $\omega \in \wedge^d k^n$ be *decomposable*, $\omega = v_1 \wedge \cdots \wedge v_d$ for some $v_i \in k^n$: this is the classical decomposability criterion of multilinear algebra: a d-vector is decomposable if and only if its coordinates satisfy the quadratic Plücker relations. It enters here only to describe the underlying point set, since the scheme-theoretic identification of the image is carried out independently at the chart level below. Every point of $\mathrm{p}(\mathrm{Gr}(d,n))$ is decomposable by construction, so $\mathrm{p}(\mathrm{Gr}(d,n)) \subseteq Y$. Conversely, a decomposable line $[v_1 \wedge \cdots \wedge v_d]$ with the v_i linearly independent spans a d-dimensional subspace, hence a point of $\mathrm{Gr}(d,n)$ mapping to it, so $Y \subseteq \mathrm{p}(\mathrm{Gr}(d,n))$ on points. Scheme-theoretically, $Y \cap D_+(p_I)$ is defined by the relations (13.2) normalized at $p_I = 1$, and these solve to express every p_J/p_I as a polynomial in the $\pm x_{ab}$ minors, giving $Y \cap D_+(p_I) \cong \mathbb{A}^{d(n-d)} = U_I$; so the immersion $U_I \hookrightarrow D_+(p_I)$ has image exactly $Y \cap D_+(p_I)$. Therefore p is a closed immersion with image Y.

Since $\mathrm{Gr}(d,n)$ is thereby a closed subscheme of $\mathbb{P}^{\binom{n}{d}-1}$, it is projective over the base, completing the proof.

The proposition closes the account of $\mathrm{Gr}(d,n)$ as a fine moduli space: it is representable (theorem 13.2.4), it carries a universal family (definition 13.2.5), and it is projective (through the Plücker embedding just constructed). The three facts are not independent decorations. The Plücker embedding is itself the classifying map of the line bundle $\wedge^d \mathcal{Q}$ built from the universal quotient, so projectivity is read off the universal family. This is the pattern the book will exploit repeatedly: a moduli space is projective because a natural line bundle on it, assembled from the universal family, is very ample.

The Plücker relations (13.2) deserve a concrete instance, since for $d = 1$ or $d = n-1$ they are vacuous (the Grassmannian is a projective space and there is nothing to cut out) and only become substantive at $d = 2$, $n = 4$.

Example 13.2.8: The Plücker Quadric $\mathrm{Gr}(2,4)$

Take $d=2$, $n=4$: $\mathrm{Gr}(2,4)$ parametrizes rank-2 quotients of $\mathcal{O}_S^4$, equivalently rank-2 subbundles, equivalently (over a field) 2-planes through the origin in k^4, or lines in $\mathbb{P}^3$. Its dimension is $d(n-d) = 2\cdot 2 = 4$, and the Plücker space is $\mathbb{P}(\wedge^2 k^4) = \mathbb{P}^5$ with coordinates $p_{12}, p_{13}, p_{14}, p_{23}, p_{24}, p_{34}$.

The charts. There are $\binom{4}{2} = 6$ charts U_I, one for each pair $I \subseteq \{1,2,3,4\}$. On U_{12} a family is normalized to the 2×4 matrix

$$\begin{pmatrix} 1 & 0 & x_{13} & x_{14} \\ 0 & 1 & x_{23} & x_{24} \end{pmatrix}$$

whose four entries $x_{13}, x_{14}, x_{23}, x_{24}$ are the affine coordinates on $U_{12} \cong \mathbb{A}^4$; here the first two columns (indices $I = \{1,2\}$) are the identity block. The Plücker coordinates are the 2×2 minors: $p_{12} = 1$, $p_{13} = x_{23}$, $p_{14} = x_{24}$, $p_{23} = -x_{13}$, $p_{24} = -x_{14}$, and $p_{34} = x_{13}x_{24} - x_{14}x_{23}$. The chart U_{34} normalizes on the last two columns instead, and its coordinates are related to those of U_{12} by inverting the relevant minor, as in Step 4 of theorem 13.2.4.

The single relation. For $d=2$, $n=4$ the relations (13.2) reduce to one quadric. Here $d-1=1$, so there is a single i-index, and $d+1=3$, so there are three j-indices; the only choice up to the labeling is $i_1 = 1$ and $(j_0, j_1, j_2) = (2,3,4)$, and the sum over $t = 0,1,2$ gives $p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23}$:

$$p_{12}\,p_{34} - p_{13}\,p_{24} + p_{14}\,p_{23} = 0 \tag{13.3}$$

Substituting the chart values verifies it: $1\cdot(x_{13}x_{24} - x_{14}x_{23}) - x_{23}\cdot(-x_{14}) + x_{24}\cdot(-x_{13}) = x_{13}x_{24} - x_{14}x_{23} + x_{14}x_{23} - x_{13}x_{24} = 0$. Thus $\mathrm{Gr}(2,4)$ is the smooth quadric fourfold in $\mathbb{P}^5$ cut out by (13.3), the smallest Grassmannian that is not itself a projective space.

Exercise 13.2.1

1. Show directly from definition 13.2.1 that the assignment sending a rank-d quotient $q\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ to its kernel subbundle $\mathcal{S} = \ker q$ defines an isomorphism of functors between $\underline{\mathrm{Gr}}(d,n)$ and the functor $S \mapsto \{\text{rank-}(n-d) \text{ subbundles of } \mathcal{O}_S^n\}$. (A rank-$(n-d)$ subbundle means a subsheaf $\mathcal{S} \hookrightarrow \mathcal{O}_S^n$, locally a direct summand, with locally free quotient.)

2. Using the chart $U_I \cong \mathbb{A}^{d(n-d)}$ of theorem 13.2.4, compute the transition map $U_{12} \cap U_{13} \to U_{13}$ explicitly for $\mathrm{Gr}(2,3)$: write the coordinates on U_{13} as rational functions of the coordinates on U_{12}, and identify the open locus of U_{12} on which the transition is defined.

3. Verify that $\mathrm{Gr}(1,n) = \mathbb{P}^{n-1}$ by comparing functors of points: show that a rank-1 locally free quotient $\mathcal{O}_S^n \twoheadrightarrow \mathcal{L}$ is exactly the datum that classifies a morphism $S \to \mathbb{P}^{n-1}$, and match the affine charts of theorem 13.2.4 with the standard charts of $\mathbb{P}^{n-1}$. Relate this to example 13.1.4.

4. On $\mathrm{Gr}(2,4)$, write down the two-plane $W \subseteq k^4$ corresponding to the Plücker point $[p_{12} : p_{13} : p_{14} : p_{23} : p_{24} : p_{34}] = [1:0:0:0:0:0]$, and verify it satisfies (13.3). Which chart U_I does this point lie in, and what are its affine coordinates there? Then check whether the point $[1:0:0:0:0:1]$ satisfies (13.3), and interpret the answer geometrically.

5. Show that the universal quotient bundle $\mathcal{Q}$ on $\mathrm{Gr}(1,n) = \mathbb{P}^{n-1}$ is the line bundle $\mathcal{O}(1)$, and hence that the Plücker embedding of proposition 13.2.7 is the identity in this case. (Use example 13.1.4 to identify the universal rank-1 quotient.)

6. Prove that $\mathrm{Gr}(d,n)$ and $\mathrm{Gr}(n-d,n)$ are isomorphic as schemes, and describe the isomorphism

at the level of functors of points. (Consider the dual: a rank-d quotient of $\mathcal{O}_S^n$ has a rank-$(n-d)$ kernel; dualize the short exact sequence (13.1).)

13.3 Coarse Moduli Spaces and the Automorphism Obstruction

The Grassmannian of section 13.2 is the fortunate case: its moduli functor is representable, and the representing scheme carries a universal family from which every other family is pulled back. Most moduli problems are not so obliging. The functor one writes down at the start almost never has a fine moduli space, and the reason is structural rather than accidental. It is already visible in the oldest moduli problem of all, the classification of elliptic curves up to isomorphism, where the culprit can be named precisely: the objects being classified have nontrivial automorphisms.

This section isolates that obstruction and extracts from the wreckage the best geometric object that survives. When no scheme represents $\mathcal{F}$, one may still ask for a scheme M whose $\bar{k}$-points match the isomorphism classes of objects and which receives a map from $\mathcal{F}$ that is as close to an isomorphism as the automorphisms permit. Such an M is a *coarse* moduli space. It remembers which objects there are but forgets how families of them twist, and this forgetting is exactly what a scheme is forced to do in the presence of automorphisms. The elliptic-curve moduli problem is worked in full: its coarse space is the affine j-line $\mathbb{A}^1_j$, and the same automorphisms that produce the coarse space also produce explicit families with pairwise isomorphic fibers that are not isomorphic as families, which is a direct proof that no fine moduli space can exist.

The warning issued at the end of section 13.1 (box 13.1.6) was that a representable functor is a sheaf, so a moduli functor that fails the sheaf condition cannot be representable. The failure mechanism is worth stating in the abstract before meeting it in an example. Suppose $\mathcal{F}$ is a moduli functor in the sense of definition 13.1.1, and suppose an object x of the moduli problem has an automorphism $\alpha \neq \mathrm{id}$. Take the constant family $x \times S$ over a base S. Both id and $\alpha \times \mathrm{id}_S$ are automorphisms of this family, and they are distinct. If $\mathcal{F}$ were represented by a scheme M (definition 13.1.3), a family over S would be the same data as a morphism $S \to M$, and the automorphisms of the family would be the automorphisms of that morphism, of which there is exactly one, the identity. A family that admits two distinct self-isomorphisms therefore cannot correspond to a morphism into any scheme. The automorphism is the obstruction, and it does not go away by choosing a cleverer scheme.

There is a weaker demand a scheme can meet. Rather than asking M to represent $\mathcal{F}$, ask only that M receive a natural map from $\mathcal{F}$ that is initial among all such maps to schemes, and that on geometric points this map be a bijection onto the isomorphism classes. This is the notion of a coarse moduli space.

Definition 13.3.1: Coarse Moduli Space

Let $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ be a moduli functor over k (definition 13.1.1). A *coarse moduli space* for $\mathcal{F}$ is a scheme M over k together with a natural transformation

$$\eta\colon \mathcal{F} \to h_M$$

satisfying the following two conditions.

1. *Universality.* For every scheme N over k and every natural transformation $\theta\colon \mathcal{F} \to h_N$, there is a unique morphism of schemes $g\colon M \to N$ with $\theta = h_g \circ \eta$, where $h_g\colon h_M \to h_N$ is postcomposition with g.

2. *Bijection on geometric points.* The map

$$\eta(\operatorname{Spec}\bar{k})\colon \mathcal{F}(\operatorname{Spec}\bar{k}) \to h_M(\operatorname{Spec}\bar{k}) = M(\bar{k})$$

   is a bijection.

A moduli functor is said to *admit a coarse moduli space* when such a pair (M, η) exists.

Condition (1) is a universal property, so it pins down the pair (M, η) up to unique isomorphism when it exists: the coarse space is the closest scheme-valued functor h_M underneath $\mathcal{F}$, or in categorical language, M corepresents $\mathcal{F}$ (h_M is the universal object under $\mathcal{F}$ in the category of representable functors). Condition (2) supplies the geometric content that a bare universal property would not: without it, the constant map from $\mathcal{F}$ to a point would satisfy (1) vacuously in trivial cases, and one wants M to parametrize the isomorphism classes. Together the two conditions say that M is a scheme whose $\bar{k}$-points *are* the isomorphism classes of objects, compatibly with all families, and that among schemes carrying such a map M is the finest. What M does not do is carry a universal family: there is a map $\mathcal{F} \to h_M$ but in general no inverse, so a morphism $S \to M$ need not come from any single family over S. A coarse moduli space records the objects but not the moduli of families.

Proposition 13.3.2: A Fine Moduli Space is Coarse

Let $\mathcal{F}$ be a moduli functor over k, and suppose $\mathcal{F}$ is represented by a scheme M in the sense of definition 13.1.3, with a natural isomorphism $\Phi\colon \mathcal{F} \xrightarrow{\sim} h_M$. Then (M, Φ) is a coarse moduli space for $\mathcal{F}$.

***Proof*:**
Both defining conditions of definition 13.3.1 must be checked for the pair (M, Φ).

Step 1: Universality. Let N be any scheme over k and $\theta\colon \mathcal{F} \to h_N$ a natural transformation. Since Φ is a natural isomorphism it has an inverse $\Phi^{-1}\colon h_M \to \mathcal{F}$, and $\theta \circ \Phi^{-1}\colon h_M \to h_N$ is a natural transformation between representable functors. By the Yoneda lemma (theorem 0.4.22), natural transformations $h_M \to h_N$ correspond bijectively to morphisms $M \to N$, so there is a unique morphism $g\colon M \to N$ with $h_g = \theta \circ \Phi^{-1}$. Composing on the right with Φ gives $h_g \circ \Phi = \theta$, which is the required factorization. Uniqueness of g is immediate: any g' with $h_{g'} \circ \Phi = \theta$ satisfies $h_{g'} = \theta \circ \Phi^{-1} = h_g$, and Yoneda forces $g' = g$.

Step 2: Bijection on geometric points. The component of Φ at $\operatorname{Spec}\bar{k}$ is a bijection $\mathcal{F}(\operatorname{Spec}\bar{k}) \to$

$h_M(\operatorname{Spec} \bar{k})$ because Φ is a natural isomorphism, so every component is an isomorphism of sets. This is precisely condition (2).

Both conditions hold, so (M, Φ) is a coarse moduli space for $\mathcal{F}$, completing the proof.

The proposition says that fineness is strictly stronger than coarseness: a fine moduli space is automatically a coarse one, with the coarsening map η taken to be the representing isomorphism Φ itself. The converse fails, and the gap between the two notions is exactly the phenomenon this section exists to explain. A coarse space exists in many cases where a fine one does not, and when a fine space does exist the two coincide. This is why the search for a moduli space proceeds in two stages: first ask for the ideal object (a fine space), and when the automorphism obstruction blocks it, retreat to the coarse space, which measures how much was lost.

The loss is not abstract. It manifests as families that a fine moduli space would have to distinguish but a coarse one cannot, because the families have identical fibers everywhere yet are not isomorphic. Such families are the fingerprint of a nontrivial automorphism, and they deserve a name.

Definition 13.3.3: Isotrivial Family

Fix a class of objects with a notion of family, and let x_0 be a fixed object over k. A family $\pi\colon X \to S$ over a k-scheme S is *isotrivial* (with fiber type x_0) if

1. every geometric fiber $X_{\bar{s}}$, for $\bar{s}$ a geometric point of S, is isomorphic to x_0; yet
2. $X \to S$ is not isomorphic, as a family over S, to the constant family $x_0 \times S \to S$.

A family satisfying (1) alone is said to have *constant fibers*; the point of the definition is that (1) does not force triviality.

An isotrivial nontrivial family is the local-to-global gap made concrete. Pointwise it looks exactly like the constant family $x_0 \times S$, since every fiber is a copy of x_0, but globally the copies are stitched together with a twist that no fiberwise isomorphism can undo. A moduli functor sends both this family and the constant family to the *same* data over each geometric point (both assign x_0 to every point), so any natural map $\mathcal{F} \to h_M$ to a scheme sends the two families to the same morphism $S \to M$. If $\mathcal{F}$ were representable, families would be determined by their classifying morphism, and the two families would have to be isomorphic, contradicting nontriviality. The existence of a single isotrivial nontrivial family thus rules out a fine moduli space. The mechanism producing such families is always the same: an automorphism of x_0 used as gluing data.

Example 13.3.4: An Isotrivial Nontrivial Family of Elliptic Curves

Work over an algebraically closed field k with $\operatorname{char} k \neq 2$, and let E be an elliptic curve over k with j-invariant $j(E) \notin \{0, 1728\}$, so that $\operatorname{Aut}(E) = \{\pm 1\}$, generated by the inversion automorphism $\iota = [-1]\colon E \to E$, $P \mapsto -P$ (the automorphism group and the special values $0, 1728$ are recorded in [11]). The involution ι is nontrivial because E has points with $2P \neq 0$, on which $\iota(P) = -P \neq P$; indeed the fixed points of ι are exactly the 2-torsion $\{P \mid 2P = 0\}$, a proper closed subscheme, so $\iota \neq \mathrm{id}$ on the rest of E.

Let $S = \operatorname{Spec} R$ where $R = k[t, t^{-1}]$, so $S = \mathbb{G}_m$ is the punctured affine line, and consider its connected double cover

$$\varphi\colon \tilde{S} = \operatorname{Spec} k[u, u^{-1}] \to S, \qquad \varphi^{\#}(t) = u^2$$

which is an étale $\mathbb{Z}/2\mathbb{Z}$-torsor for $\operatorname{char} k \neq 2$; the nontrivial deck transformation is $\sigma\colon u \mapsto -u$. Form the constant family $E \times \tilde{S} \to \tilde{S}$ and glue it to itself over the two sheets of φ using the automorphism ι: concretely, take the quotient

$$X \;=\; \big(E \times \tilde{S}\big)/(\iota \times \sigma)$$

by the free $\mathbb{Z}/2\mathbb{Z}$-action that acts as ι on the fiber and as σ on the base. Since σ is free on $\tilde{S}$, the action is free, and the projection $E \times \tilde{S} \to \tilde{S}$ descends to a morphism $\pi\colon X \to \tilde{S}/\sigma = S$. This is the *quadratic twist* of E along the cover φ.

Every geometric fiber of π is isomorphic to E. Fix a geometric point $\bar{s}$ of S; its two preimages $\bar{u}, \sigma(\bar{u})$ in $\tilde{S}$ each carry the fiber E, and the gluing identifies $E \times \{\bar{u}\}$ with $E \times \{\sigma\bar{u}\}$ via ι. The fiber $X_{\bar{s}}$ is therefore a single copy of E, so condition (1) of definition 13.3.3 holds with fiber type $x_0 = E$. In particular every fiber has the same j-invariant $j(E)$.

The family $X \to S$ is not the constant family $E \times S$. Suppose it were, say via an S-isomorphism $\psi\colon E \times S \xrightarrow{\sim} X$. Pulling back along φ trivializes both sides over $\tilde{S}$, and ψ becomes an automorphism of $E \times \tilde{S}$ over $\tilde{S}$ commuting with the two descent data. The descent datum for the constant family is the identity gluing $\mathrm{id} \times \sigma$, while that for X is $\iota \times \sigma$; an isomorphism of the two families is a $\tilde{S}$-automorphism β of $E \times \tilde{S}$ with $\beta \circ (\mathrm{id} \times \sigma) = (\iota \times \sigma) \circ \beta$, that is, a family of automorphisms $\beta_u \in \mathrm{Aut}(E)$ with $\beta_{-u} = \iota \circ \beta_u$. Since $\mathrm{Aut}(E) = \{\pm 1\}$ is a constant (hence étale) group scheme and $\tilde{S} = \mathbb{G}_m$ is connected, the classifying morphism $u \mapsto \beta_u$ from $\tilde{S}$ to $\mathrm{Aut}(E)$ is constant, so β_u is a single automorphism independent of u, forcing $\beta = \iota \circ \beta$, hence $\iota = \mathrm{id}$. This contradicts $\iota \neq \mathrm{id}$, which holds for every elliptic curve. (The hypothesis $j(E) \neq 0, 1728$ is not needed for $\iota \neq \mathrm{id}$; it enters only to pin $\mathrm{Aut}(E)$ to exactly $\{\pm 1\}$, so that the classifying morphism lands in a constant group and the constancy argument applies cleanly.) So no such ψ exists, and condition (2) holds.

The family $X \to S$ is therefore isotrivial and nontrivial. Its very existence proves that the elliptic-curve moduli functor $\mathcal{M}_{1,1}$ (a moduli functor in the sense of definition 13.1.1) is not representable: a fine moduli space would send $X \to S$ and $E \times S \to S$, which have identical fibers everywhere, to the same classifying morphism $S \to M$, and would then be forced to declare the two families isomorphic. They are not.

The construction pinpoints where representability breaks. Every ingredient is standard, an elliptic curve, its inversion automorphism, and an étale double cover, and the twist is assembled from nothing more than the automorphism ι acting as gluing data over a nonsimply-connected base. The same recipe works whenever the objects of a moduli problem have a nontrivial automorphism and the base has a nontrivial torsor for the relevant automorphism group. This is the general shape of the automorphism obstruction: automorphisms of objects become gluing data for locally trivial but globally twisted families, and the moduli functor, which sees only fibers, cannot separate them.

Having seen that $\mathcal{M}_{1,1}$ is not representable, the task is to identify the coarse space that survives. Over $\bar{k}$ the isomorphism classes of elliptic curves are classified by a single number, the j-invariant, and the scheme whose points are those numbers is the affine line. This affine line, the *j-line*, is the coarse moduli space.

13.3.5 Note: the moduli functor $\mathcal{M}_{1,1}$ Throughout this section $\mathcal{M}_{1,1}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ denotes the moduli functor of elliptic curves in the sense of definition 13.1.1: for a k-scheme S, $\mathcal{M}_{1,1}(S)$ is the set of isomorphism classes of families $(E \to S, \sigma)$ with $E \to S$ smooth and proper of relative dimension 1, geometric fibers of genus 1, and σ a section. Pullback of families makes it contravariant. The j-invariant of a single elliptic curve over a field is developed in [11].

Theorem 13.3.6: The j-Line is a Coarse Moduli Space

Let k be an algebraically closed field. The affine line $\mathbb{A}^1_j = \operatorname{Spec} k[j]$, the j-*line*, is a coarse moduli space for the moduli functor $\mathcal{M}_{1,1}$ of elliptic curves, with the natural transformation

$$\eta\colon \mathcal{M}_{1,1} \to h_{\mathbb{A}^1_j}$$

sending a family $(E \to S) \in \mathcal{M}_{1,1}(S)$ to the morphism $S \to \mathbb{A}^1_j$ classifying its fiberwise j-invariant. The pair $(\mathbb{A}^1_j, \eta)$ satisfies both conditions of definition 13.3.1. However, $\mathcal{M}_{1,1}$ is not representable, so $\mathbb{A}^1_j$ is not a fine moduli space.

***Proof*:**
The j-invariant of a single elliptic curve over k takes every value of k and determines the isomorphism class over $\bar{k} = k$: two elliptic curves over an algebraically closed field are isomorphic if and only if they have the same j-invariant, and every element of k arises as a j-invariant ([11]). This standard input is used freely below.

Step 1: The natural transformation η is well defined. For a family $E \to S$ the assignment $s \mapsto j(E_s)$ is a morphism of schemes $j_E\colon S \to \mathbb{A}^1_j$. Concretely, working locally where $E \to S$ is given by a Weierstrass equation with coefficients in $\mathcal{O}_S(U)$ and discriminant a unit, the j-invariant is the universal rational expression in those coefficients, which is a regular function on U; these glue to a global morphism $j_E\colon S \to \mathbb{A}^1_j$ independent of the chosen Weierstrass presentation, since j is an isomorphism invariant. Setting $\eta_S(E \to S) = j_E$ gives a map $\mathcal{M}_{1,1}(S) \to h_{\mathbb{A}^1_j}(S)$, and naturality in S holds because forming a Weierstrass presentation and computing j commute with base change: for $f\colon T \to S$ one has $j_{f^*E} = j_E \circ f$, which is the equation $h_{\mathbb{A}^1_j}(f) \circ \eta_S = \eta_T \circ \mathcal{M}_{1,1}(f)$.

Step 2: Bijection on geometric points. A $\bar{k}$-point of $\mathbb{A}^1_j$ is an element of $k = \bar{k}$, and $\eta(\operatorname{Spec} \bar{k})$ sends an elliptic curve over $\bar{k}$ to its j-invariant. By the classification input, this map $\mathcal{M}_{1,1}(\operatorname{Spec} \bar{k}) \to \mathbb{A}^1_j(\bar{k}) = k$ is a bijection: it is surjective because every value of k is a j-invariant, and injective because equal j-invariants force an isomorphism over $\bar{k}$. This is condition (2) of definition 13.3.1.

Step 3: Universality. Let N be a scheme over k and $\theta\colon \mathcal{M}_{1,1} \to h_N$ a natural transformation; a morphism $g\colon \mathbb{A}^1_j \to N$ with $h_g \circ \eta = \theta$ must be produced, and shown unique. No hypothesis is placed on N: it need not be separated, so the argument is arranged so that every step rests on the naturality of θ with respect to *scheme* morphisms (restriction along open immersions and base change of families), never on the determination of a morphism into N by its values on closed points. The strategy is to realize g over an open cover of $\mathbb{A}^1_j$ by applying θ to tautological families, then to glue and to verify the factorization by functoriality.

Cover $\mathbb{A}^1_j$ by finitely many affine opens U_i each carrying a Weierstrass family $\mathcal{E}_{U_i} \to U_i$ whose fiberwise j-invariant is the restriction of the coordinate j. Such a cover exists: away from $j = 0$ and $j = 1728$ the family $y^2 = x^3 + a(j)x + b(j)$ with $a(j), b(j)$ regular and chosen so the fiberwise j-invariant equals the coordinate provides one open, and small opens around 0 and 1728 carrying suitable Weierstrass equations (with the two special j-values omitted from each of the other charts) complete the cover; on each U_i the class $\mathcal{E}_{U_i} \in \mathcal{M}_{1,1}(U_i)$ has $\eta_{U_i}(\mathcal{E}_{U_i}) = \iota_i$, the open immersion $\iota_i \colon U_i \hookrightarrow \mathbb{A}^1_j$, by Step 1. Applying θ gives a morphism $g_i := \theta_{U_i}(\mathcal{E}_{U_i}) \in h_N(U_i) = \mathrm{Hom}(U_i, N)$.

The g_i agree on overlaps as *morphisms of schemes*. Write $U_{ij} = U_i \cap U_j$ with open immersions $r_i \colon U_{ij} \hookrightarrow U_i$ and $r_j \colon U_{ij} \hookrightarrow U_j$. Both restricted families $r_i^*\mathcal{E}_{U_i}$ and $r_j^*\mathcal{E}_{U_j}$ over U_{ij} have fiberwise j-invariant the coordinate $j|_{U_{ij}}$, but they need not be isomorphic as families over U_{ij}: two Weierstrass families with equal j-invariant function are forms of one another, and can differ by a nontrivial quadratic twist recorded by a class in $H^1_{\text{ét}}(U_{ij}, \underline{\mathrm{Aut}}(\mathcal{E}))$ for the automorphism group scheme $\underline{\mathrm{Aut}}(\mathcal{E})$ of the fiber, which is exactly the twisting phenomenon that example 13.3.4 exhibits over $\mathbb{G}_m$ (reducedness of U_{ij} does not kill it, since $\mathbb{G}_m$ is reduced yet carries the nontrivial twist). The two families do become isomorphic after an étale cover, however: any two elliptic curves over a base with the same j-invariant are locally isomorphic in the étale topology, since a twist is split by the finite étale cover trivializing its class. Choose a surjective étale morphism $p \colon U'_{ij} \to U_{ij}$ over which $p^*r_i^*\mathcal{E}_{U_i}$ and $p^*r_j^*\mathcal{E}_{U_j}$ are isomorphic as families over U'_{ij}, hence equal in $\mathcal{M}_{1,1}(U'_{ij})$. Naturality of θ along $r_i \circ p$ and $r_j \circ p$ then gives, in $\mathrm{Hom}(U'_{ij}, N)$,

$$g_i \circ r_i \circ p = \theta_{U'_{ij}}(p^*r_i^*\mathcal{E}_{U_i}) = \theta_{U'_{ij}}(p^*r_j^*\mathcal{E}_{U_j}) = g_j \circ r_j \circ p$$

so the two morphisms $g_i \circ r_i$ and $g_j \circ r_j$ from U_{ij} to N agree after the base change p. Since p is faithfully flat of finite presentation, it is an epimorphism of schemes, and two morphisms into any scheme N that agree after such a base change are equal; hence $g_i \circ r_i = g_j \circ r_j$ in $\mathrm{Hom}(U_{ij}, N)$, an identity valid whether or not N is separated (the descent uses only that p is an fppf epimorphism, not any separatedness of N). The morphisms g_i therefore glue to a single morphism $g \colon \mathbb{A}^1_j \to N$ with $g \circ \iota_i = g_i$.

This g satisfies $h_g \circ \eta = \theta$. Both are natural transformations $\mathcal{M}_{1,1} \to h_N$, so it suffices to check that $(h_g \circ \eta)_S(E \to S) = \theta_S(E \to S)$ for every family $E \to S$; equivalently, that $g \circ j_E = \theta_S(E \to S)$ as morphisms $S \to N$, where $j_E = \eta_S(E \to S)$. The families $\{j_E^{-1}(U_i)\}$ cover S by open subschemes, so it suffices to verify equality after restriction to each $V_i := j_E^{-1}(U_i)$. On V_i the morphism j_E factors as $j_E|_{V_i} = \iota_i \circ f_i$ through a morphism $f_i \colon V_i \to U_i$, and $f_i^*\mathcal{E}_{U_i}$ is a Weierstrass family over V_i whose fiberwise j-invariant equals $j_E|_{V_i}$, the same as that of $E|_{V_i} \to V_i$. As on the overlaps, the two families $E|_{V_i}$ and $f_i^*\mathcal{E}_{U_i}$ need not be isomorphic over V_i (they may differ by a twist), but they become isomorphic after a surjective étale cover $q \colon V'_i \to V_i$, so they agree in $\mathcal{M}_{1,1}(V'_i)$. Naturality of θ along q and $f_i \circ q$ gives

$$\theta_{V'_i}(q^*E|_{V_i}) = \theta_{V'_i}(q^*f_i^*\mathcal{E}_{U_i}) = g_i \circ f_i \circ q = g \circ \iota_i \circ f_i \circ q = g \circ j_E|_{V_i} \circ q$$

so $\theta_{V_i}(E|_{V_i}) \circ q = (g \circ j_E|_{V_i}) \circ q$ as morphisms $V'_i \to N$; since q is a faithfully flat epimorphism these agree already over V_i, giving $\theta_{V_i}(E|_{V_i}) = g \circ j_E|_{V_i}$. Since these agree on the open cover $\{V_i\}$ of S, $\theta_S(E \to S) = g \circ j_E$ as required, so $h_g \circ \eta = \theta$.

Uniqueness of g is forced by the same functoriality, with no appeal to separatedness. If g' also satisfies $h_{g'} \circ \eta = \theta$, then applying both sides to the tautological family $\mathcal{E}_{U_i}$ over U_i gives $g' \circ \iota_i = g' \circ \eta_{U_i}(\mathcal{E}_{U_i}) = \theta_{U_i}(\mathcal{E}_{U_i}) = g_i = g \circ \iota_i$ as morphisms $U_i \to N$, that is, $g'|_{U_i} = g|_{U_i}$. Since $\{U_i\}$ is an open cover of $\mathbb{A}^1_j$ and morphisms of schemes into N form a Zariski sheaf (two morphisms agreeing on each member of an open cover of the source are equal, for any target N), $g' = g$. This establishes condition (1).

Step 4: $\mathcal{M}_{1,1}$ is not representable, so $\mathbb{A}^1_j$ is not fine. If $\mathcal{M}_{1,1}$ were representable by a scheme M, then by proposition 13.3.2 the pair (M, Φ) would be a coarse moduli space, and by the uniqueness of coarse spaces (the universal property of definition 13.3.1 determines M up to unique isomorphism) one would have $M \cong \mathbb{A}^1_j$ with Φ an isomorphism $\mathcal{M}_{1,1} \cong h_{\mathbb{A}^1_j}$. But example 13.3.4 exhibits, for $\operatorname{char} k \neq 2$, two families over $S = \mathbb{G}_m$, the constant family $E \times S$ and its quadratic twist $X \to S$, with the same fiberwise j-invariant, hence $\eta_S(E \times S) = \eta_S(X) = j_E$ the same morphism $S \to \mathbb{A}^1_j$. Under an isomorphism $\Phi \cong \eta$ these two families would be equal in $\mathcal{M}_{1,1}(S)$, that is, isomorphic as families over S, contradicting the nontriviality proved in example 13.3.4. Therefore $\mathcal{M}_{1,1}$ is not representable. When $\operatorname{char} k = 2$ the quadratic-twist construction of example 13.3.4 does not apply verbatim, since the double cover $u \mapsto u^2$ is inseparable; non-representability still holds, now witnessed by the extra automorphisms of the supersingular curve (whose j-invariant is $0 = 1728$ in characteristic 2), which produce a nontrivial twist by the same recipe over a base carrying an étale torsor for its larger automorphism group ([11]). In every characteristic, then, the coarse space $\mathbb{A}^1_j$ is not a fine moduli space, as we sought to show.

The theorem is the prototype every later chapter measures itself against. It exhibits both halves of the moduli story at once: a genuine geometric object, the j-line, that correctly parametrizes isomorphism classes, and a precise accounting of what that object cannot see, namely the twisting of families recorded in example 13.3.4. The coarse space is not a failure but a compromise forced by the automorphisms: it is the best scheme available, and proposition 13.3.2 guarantees it would upgrade to a fine space the instant the automorphisms were removed. The diagnosis also points at the cure. The obstruction is that $\mathcal{M}_{1,1}$ fails the sheaf condition (box 13.1.6) because étale-locally trivial families need not be globally trivial, and the twisting is governed by the automorphism group $\operatorname{Aut}(E)$. Two later strategies repair this: rigidify the objects until their automorphism groups become trivial, which recovers a fine moduli space for the rigidified problem, and enlarge the category of geometric objects to stacks, which remember automorphisms rather than quotienting them away. The first is taken up in section 13.4; the second is the subject of the descent and algebraic-stacks chapters later in the book.

Concretely, the coarse-versus-fine gap can be read off the automorphism groups directly. Over $\bar{k}$ with $\operatorname{char} k \notin \{2, 3\}$, a generic elliptic curve has $\operatorname{Aut}(E) = \{\pm 1\}$ of order 2, while the two special curves have larger groups: $j = 1728$ gives $\operatorname{Aut}(E) = \mathbb{Z}/4\mathbb{Z}$ and $j = 0$ gives $\operatorname{Aut}(E) = \mathbb{Z}/6\mathbb{Z}$ ([11]). The coarse space $\mathbb{A}^1_j$ is a smooth affine line, but the two special points $j = 0$ and $j = 1728$ are where the automorphism group jumps, and it is precisely at those points that a stack-theoretic refinement would record extra structure. The coarse space smooths over this jump; the fine-moduli-theoretic content lives in the automorphism groups the coarse space discards.

Exercise 13.3.1

1. Let (M, η) be a coarse moduli space for a moduli functor $\mathcal{F}$ in the sense of definition 13.3.1. Prove that (M, η) is unique up to unique isomorphism: if (M', η') is another coarse moduli space for $\mathcal{F}$, there is a unique isomorphism $g \colon M \to M'$ of schemes with $h_g \circ \eta = \eta'$.

2. Consider the terminal functor $\mathcal{F}(S) = \{*\}$ (a one-point set for every S). Show that the universality condition (1) of definition 13.3.1 already forces the coarse space to be the reduced point $M \cong \operatorname{Spec} k$ (so here condition (1) determines the geometry and condition (2) holds automatically). Then show that condition (2) alone does *not* pin down M: exhibit a scheme M' over k with a natural transformation $\eta' \colon \mathcal{F} \to h_{M'}$ for which condition (2) holds but

condition (1) fails, and explain in one sentence what universality contributes beyond a bijection on geometric points.

3. In the setup of example 13.3.4, suppose instead that E is an elliptic curve with $j(E) = 0$ over $\bar{k}$ with $\operatorname{char} k \notin \{2, 3\}$, so that $\operatorname{Aut}(E) = \mathbb{Z}/6\mathbb{Z}$. Using an order-3 automorphism ζ of E and a connected étale $\mathbb{Z}/3\mathbb{Z}$-cover $\tilde{S} \to S = \mathbb{G}_m$, construct an isotrivial nontrivial family whose fibers are all isomorphic to E. Verify conditions (1) and (2) of definition 13.3.3, and explain why the argument requires $\operatorname{char} k \neq 3$.

4. Let $\pi \colon X \to S$ be the quadratic twist of example 13.3.4, with $S = \mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$. Compute the classifying morphism $\eta_S(X) \colon S \to \mathbb{A}^1_j$ explicitly, and confirm that it equals $\eta_S(E \times S)$. Deduce that $\eta_S(X)$ and $\eta_S(E \times S)$ take the same value at each closed point of S, a single point of $\mathbb{A}^1_j$, and explain why this equality of classifying morphisms is exactly the statement that $\mathbb{A}^1_j$ cannot distinguish the two families.

5. Prove the converse-type statement implicit in definition 13.3.3: if a moduli functor $\mathcal{F}$ admits *no* isotrivial nontrivial family (that is, every family with constant fibers of a fixed type is isomorphic to the constant family), does it follow that $\mathcal{F}$ is representable? Either prove it or give a counterexample, identifying which of the two conditions for a fine moduli space can still fail.

6. Let $\mathcal{F}$ be a moduli functor with a coarse moduli space (M, η), and suppose $\mathcal{F}$ happens to have *no* nontrivial automorphisms (every object x has $\operatorname{Aut}(x) = \{\mathrm{id}\}$) but is nonetheless not representable by a scheme. Explain, using box 13.1.6, why the existence of a coarse moduli space (M, η) (constructed as in the j-line argument, by hand from an invariant) does not immediately produce a fine moduli space, and name (in prose) the kind of geometric object that can represent such an $\mathcal{F}$.

13.4 Rigidification and Level Structure

The previous section diagnosed a precise failure. The moduli functor of elliptic curves is not representable, and the obstruction is not a technicality of the construction but a feature of the objects themselves: every elliptic curve carries the automorphism $[-1]$, and this automorphism produces families that are locally trivial yet globally nontrivial, exactly the families a representable functor cannot support (theorem 13.3.6). The j-line $\mathbb{A}^1_j$ survives only as a coarse moduli space (definition 13.3.1), recording isomorphism classes on geometric points but carrying no universal family.

There are two responses to this obstruction, and this book pursues both. The deep response, developed in Part VI, keeps the objects as they are and enlarges the category of geometric spaces until the functor becomes representable by a stack. The present section pursues the elementary response: change the moduli problem by decorating each object with extra data, chosen so that no nontrivial automorphism fixes the data. An object rigidified in this way has no automorphisms as a decorated object, the mechanism of theorem 13.3.6 shuts off, and the decorated moduli functor becomes fine. The archetype of such decorating data is a *level structure* on an elliptic curve, a choice of basis for the N-torsion. Throughout, N is a positive integer invertible in k, so that the N-torsion is well behaved in every family under consideration.

The guiding principle is that automorphisms are the enemy of representability, so the cure is to add structure on which the automorphisms act freely enough to leave no fixed points but the trivial one. What "structure" means is best fixed by an explicit example before the general notion, and the N-torsion of an elliptic curve supplies it. Recall that for an elliptic curve E over a field in which N is invertible, the group of N-torsion points $E[N] = \{P \in E \big| NP = O\}$ is free of rank two over $\mathbb{Z}/N$, so $E[N] \cong (\mathbb{Z}/N)^2$ ([11]). A basis of $E[N]$ is the same as an isomorphism $(\mathbb{Z}/N)^2 \to E[N]$, and this is the datum to be added.

Definition 13.4.1: Level Structure

Let N be a positive integer invertible in k, and let E be an elliptic curve over a k-scheme S (a smooth proper morphism $E \to S$ of relative dimension one with geometrically connected genus one fibers, equipped with a section $O\colon S \to E$). The N-torsion $E[N] = \ker([N]\colon E \to E)$ is then a finite étale group scheme over S, locally on S isomorphic to the constant group $(\mathbb{Z}/N)^2$. A *level-N structure* on E is an isomorphism of S-group schemes

$$\alpha\colon (\mathbb{Z}/N)^2_S \xrightarrow{\sim} E[N]$$

where $(\mathbb{Z}/N)^2_S$ denotes the constant group scheme on S. Equivalently, over a field it is an ordered basis (P, Q) of $E[N]$ as a $\mathbb{Z}/N$-module, namely α sends the standard generators e_1, e_2 to P, Q.

This construction instantiates a general pattern. A class of geometric objects is *rigidified* by a choice of *rigidifying data*: extra structure attached to each object, functorial in the base, whose only automorphism compatible with a fixed object is the identity of that object. A level-N structure is the instance of this pattern for elliptic curves: it is a coordinate system on the N-torsion, and just as a choice of basis rigidifies a vector space by eliminating its linear automorphisms as ambient symmetries, a level structure is meant to eliminate the automorphisms of the elliptic curve. The role of the hypothesis N invertible in k is to keep $E[N]$ étale of the expected rank; in characteristic p dividing N the p-part of the torsion collapses (an ordinary curve has $E[p] \cong \mathbb{Z}/p$ rather than $(\mathbb{Z}/p)^2$), and the definition would attach the wrong amount of data. The second paragraph abstracts the pattern: the useful feature of a level structure is not that it involves torsion points but that a nontrivial automorphism of E must move it, and any decorating datum with that property will serve the same rigidifying purpose.

The level-N structures assemble into a moduli functor exactly as bare elliptic curves did, now with the decorating datum carried along.

Definition 13.4.2: The Level-N Moduli Functor

Fix N invertible in k. The *level-N moduli functor* $\mathcal{F}_N\colon \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ assigns to a k-scheme S the set

$$\mathcal{F}_N(S) = \left\{ (E, \alpha) \;\middle|\; \begin{array}{l} E \to S \text{ an elliptic curve,} \\ \alpha\colon (\mathbb{Z}/N)^2_S \xrightarrow{\sim} E[N] \text{ a level-}N\text{ structure} \end{array} \right\} \Big/ \cong$$

of isomorphism classes of pairs, where an isomorphism $(E, \alpha) \to (E', \alpha')$ is an isomorphism $\varphi\colon E \to E'$ of elliptic curves over S carrying α to α', meaning $\varphi[N] \circ \alpha = \alpha'$ on N-torsion. Pullback along $f\colon T \to S$ sends (E, α) to $(f^*E, f^*\alpha)$, making $\mathcal{F}_N$ contravariant.

The condition $\varphi[N] \circ \alpha = \alpha'$ is what distinguishes the level-N problem from the bare one. Two elliptic curves with level structure are identified only by an isomorphism that respects the chosen bases of their N-torsion, so the set $\mathcal{F}_N(S)$ remembers strictly more than the underlying moduli functor $\mathcal{F}$ of elliptic curves: forgetting α defines a natural transformation $\mathcal{F}_N \to \mathcal{F}$ that is generally many to one, since a single curve E admits many level structures. What has been gave by this extra bookkeeping is that the isomorphisms in $\mathcal{F}_N$ are far more rigid, and the next result measures precisely how rigid.

The obstruction to representing $\mathcal{F}$ was that the automorphism group of an object was nontrivial, which forced a family to be gluable from its restrictions in more than one way (box 13.1.6). The decisive property of a level structure is that it collapses the automorphism group of a decorated object to the trivial group, provided N is at least 3. The following lemma isolates this fact; it is the engine of the representability theorem.

Lemma 13.4.3: Level Structures Kill Automorphisms

Let $N \geq 3$ be invertible in k, let E be an elliptic curve over an algebraically closed field, and let α be a level-N structure on E. If $\varphi \in \mathrm{Aut}(E)$ is an automorphism of E as an elliptic curve fixing α, that is $\varphi[N] \circ \alpha = \alpha$, then $\varphi = \mathrm{id}_E$.

***Proof*:**
Write $\varphi[N]$ for the induced automorphism of $E[N]$. The condition $\varphi[N] \circ \alpha = \alpha$ says that $\varphi[N]$ is the identity on $E[N]$, because α is an isomorphism and so right-cancellable. The proof reduces to showing that an automorphism of E acting trivially on $E[N]$ is the identity, for $N \geq 3$.

Step 1: The automorphism group and its faithful action. An automorphism of E as an elliptic curve fixes the origin O, hence is a group automorphism of E ([11]). Every such φ restricts to a $\mathbb{Z}/N$-linear automorphism of the N-torsion, giving a group homomorphism

$$\rho_N \colon \mathrm{Aut}(E) \to \mathrm{Aut}_{\mathbb{Z}/N}(E[N]) \cong \mathrm{GL}_2(\mathbb{Z}/N)$$

the last isomorphism induced by any basis of $E[N]$. The claim is that ρ_N is injective for $N \geq 3$.

Step 2: Injectivity via the degree bound. Let $\varphi \in \ker \rho_N$, so φ acts as the identity on $E[N]$. The endomorphism $\psi = \varphi - \mathrm{id}$ of E then kills $E[N]$, so $E[N] \subseteq \ker \psi$. If $\psi \neq 0$, then ψ is an isogeny (a nonzero endomorphism of an elliptic curve is surjective with finite kernel). Since N is invertible in k, the group $E[N] \cong (\mathbb{Z}/N)^2$ consists of N^2 distinct étale (geometric) points, all lying in $\ker \psi$; the number of geometric points of the kernel is the separable degree $\deg_s \psi$, so $\deg_s \psi \geq N^2$. As $\deg \psi \geq \deg_s \psi$ always, this forces $\deg \psi \geq N^2$.

Now bound $\deg \psi$ from above. The endomorphism ring $\mathrm{End}(E)$ is either $\mathbb{Z}$ or an order in an imaginary quadratic field or, in characteristic p, an order in a quaternion algebra; in every case the degree is the norm form, a positive definite quadratic form, and φ is an automorphism, hence a unit, so $\deg \varphi = 1$. The triangle inequality for the norm on $\mathrm{End}(E)$ gives

$$\sqrt{\deg \psi} = \|\varphi - \mathrm{id}\| \leq \|\varphi\| + \|\mathrm{id}\| = \sqrt{\deg \varphi} + \sqrt{\deg(\mathrm{id})} = 1 + 1 = 2$$

so $\deg \psi \leq 4$. Combining $\deg \psi \geq N^2$ with $\deg \psi \leq 4$ yields $N^2 \leq 4$, that is $N \leq 2$, contrary to $N \geq 3$. Therefore $\psi = 0$, that is $\varphi = \mathrm{id}_E$, which shows $\ker \rho_N$ is trivial and ρ_N is injective.

Step 3: Conclusion. The automorphism φ of the hypothesis lies in $\ker \rho_N$ by the reduction in the opening paragraph, and $\ker \rho_N$ is trivial by Step 2. Hence $\varphi = \mathrm{id}_E$, completing the proof.

The numerical heart of the argument is the failure of $[-1]$ to fix a level-N structure once $N \geq 3$. The automorphism $[-1]$, present on every elliptic curve, acts on $E[N]$ as multiplication by -1; this is the identity on $E[N]$ precisely when $2P = O$ for all $P \in E[N]$, that is precisely when $N \leq 2$. So for $N = 1$ or $N = 2$ the automorphism $[-1]$ fixes every level structure and the rigidification fails to remove it, whereas for $N \geq 3$ it moves each level structure, and with the last universal automorphism broken the object becomes rigid. The endomorphism norm inequality in Step 2 is what upgrades this observation from $[-1]$ to *all* automorphisms simultaneously, including the extra automorphisms at $j = 0$ and $j = 1728$ ([11]). This is the exact sense in which a level-N structure "rigidifies" an elliptic curve.

The representability theorem that the lemma powers can now be stated.

Theorem 13.4.4: Representability of Level-N Moduli

Let $N \geq 3$ be invertible in k. The level-N moduli functor $\mathcal{F}_N$ of definition 13.4.2 is representable by a smooth affine k-scheme $Y(N)$: there is an isomorphism of functors $\mathcal{F}_N \cong h_{Y(N)}$, so $Y(N)$ is a fine moduli space (definition 13.1.3) for elliptic curves with level-N structure. In particular $Y(N)$ carries a universal elliptic curve with level-N structure $(\mathcal{E}, \alpha^{\mathrm{univ}})$ through which every family is a unique pullback.

***Proof*:**
The proof has two parts: the removal of the automorphism obstruction, which is the conceptual content and where lemma 13.4.3 enters, and the actual construction of $Y(N)$, which proceeds by cutting the representing scheme out of a space that is already known to be fine.

Step 1: Rigidity of objects. Fix a pair (E, α) over an algebraically closed field. An automorphism of (E, α) in the sense of definition 13.4.2 is an automorphism φ of E with $\varphi[N] \circ \alpha = \alpha$, and lemma 13.4.3 shows the only such φ is id_E. Thus every object of the level-N problem has trivial automorphism group. This is exactly the hypothesis whose failure obstructed representability of the bare functor $\mathcal{F}$ (theorem 13.3.6): with no nontrivial automorphisms there is no room to build a locally trivial but globally nontrivial family, so the obstruction of box 13.1.6 is removed.

Step 2: A rigid family is determined by its restrictions. Make Step 1 quantitative. Let (E, α) and (E', α') be two families over S that become isomorphic over each member of a Zariski (indeed étale) cover $\{S_i \to S\}$. On each overlap S_{ij} the two given isomorphisms φ_i, φ_j differ by an automorphism of the pair $(E|_{S_{ij}}, \alpha|_{S_{ij}})$. The scheme of automorphisms of a rigidified elliptic curve is unramified over the base (an automorphism is rigid, being determined by finitely many torsion conditions), and by Step 1 it has trivial geometric fibers; an unramified group scheme with trivial fibers is the trivial group, so the automorphism in question is the identity and $\varphi_i = \varphi_j$ on S_{ij}. The local isomorphisms therefore agree on overlaps and glue to a global isomorphism $(E,\alpha) \cong (E',\alpha')$. So a family with level structure is determined up to unique isomorphism by its restrictions to a cover, which is the sheaf property that a representable functor must have and that $\mathcal{F}$ lacked.

Step 3: Construction of the representing scheme. It remains to exhibit a scheme representing $\mathcal{F}_N$. Every elliptic curve is a plane cubic in Weierstrass form after a choice of origin and invariant differential, and the coefficients of a Weierstrass equation define a scheme mapping to the affine space of coefficients; the substitutions relating two Weierstrass equations of the same curve form the action of a solvable group G, and the representing scheme is the quotient by this action, which exists as a scheme precisely because the added data makes the action free (no residual automorphisms remain to fix a point). Concretely, fix a Weierstrass family

$$\mathcal{W}\colon \quad y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

over the affine space $\mathbb{A}^5 = \operatorname{Spec} k[a_1,a_2,a_3,a_4,a_6]$, and let $U \subseteq \mathbb{A}^5$ be the open locus where the discriminant Δ is invertible, so $\mathcal{W}|_U \to U$ is an elliptic curve. Adjoining a level-N structure means adjoining an ordered basis of the N-torsion. The N-torsion $\mathcal{W}[N] \to U$ is finite étale, and the functor on U-schemes assigning to $T \to U$ the set of bases of $(\mathcal{W}[N])_T$ is representable by a scheme $\widetilde{U} \to U$ finite étale (it is a torsor under the constant group $\mathrm{GL}_2(\mathbb{Z}/N)$ over the locus where N is invertible, cut out inside a product of copies of $\mathcal{W}[N]$ by the nondegeneracy conditions). The pair $(\mathcal{W}_{\widetilde{U}}, \alpha^{\mathrm{univ}})$ is then a family with level structure over $\widetilde{U}$, where α^{univ} is the tautological basis.

The redundancy in this presentation is the freedom to change Weierstrass coordinates. Two triples over the same base give the same elliptic curve exactly when they differ by a substitution

$$(x,y) \longmapsto (u^2x + r,\ u^3y + su^2x + t), \qquad u \in \mathbb{G}_m,\ r,s,t \in \mathbb{G}_a$$

and these substitutions form the solvable group $G = \mathbb{G}_m \ltimes \mathbb{G}_a^3$ of the opening sentence, of dimension 4, acting on the coefficient space $\mathbb{A}^5$ and lifting to an action on $\widetilde{U}$ that transports the tautological torsion basis along the coordinate change. Two points of $\widetilde{U}$ give isomorphic pairs (E,α) if and only if they lie in one G-orbit: a coordinate change matching both the curves and their level structures is exactly an element of G carrying one point to the other. By Step 1 such an element is unique when it exists (no automorphisms to spoil uniqueness), which is precisely the statement that G acts *freely* on $\widetilde{U}$ (a nonidentity element fixing a point would be a nontrivial automorphism of the corresponding rigidified pair). A free action of the smooth affine solvable group G on the smooth affine scheme $\widetilde{U}$ admits a geometric quotient

$$Y(N) = \widetilde{U}/G$$

which is again a smooth affine scheme; the quotient collapses the 4 redundant coordinate-change directions, so $Y(N)$ has dimension $\dim \widetilde{U} - \dim G = 5 - 4 = 1$, a curve, as the geometry demands. Its functor of points is by construction

$$h_{Y(N)}(T) = \{\, T \to \widetilde{U} \,\}/\!\sim\ = \{\text{elliptic curves over } T \text{ with level-}N \text{ structure}\}/\!\cong\ = \mathcal{F}_N(T)$$

naturally in T. The chain of equalities is exactly the assertion $\mathcal{F}_N \cong h_{Y(N)}$: the first equality is the definition of the quotient, and the middle equality holds because every pair (E,α) over T is

Zariski-locally Weierstrass with a chosen torsion basis, hence Zariski-locally a T-point of $\widetilde{U}$, and these local charts glue uniquely by Step 2. Smoothness of $Y(N)$ follows from smoothness of $\widetilde{U}$ over the smooth base U and the freeness of the G-action. Therefore $Y(N)$ is a fine moduli space for $\mathcal{F}_N$, with universal family the descent of $(\mathcal{W}_{\widetilde{U}}, \alpha^{\mathrm{univ}})$, as we sought to show.

The theorem is the payoff of the entire chapter in miniature: a moduli problem that failed to be representable has been repaired, not by changing the category of geometric spaces but by refining the objects until their automorphism groups vanish. Steps 1 and 2 are the conceptual core and are where lemma 13.4.3 is spent; they say that rigidity of objects implies rigidity of families, which is the sheaf-theoretic condition a representable functor requires. Step 3 then constructs $Y(N)$ concretely, and the construction is instructive: the representing scheme is a free quotient of an étale cover of the Weierstrass parameter space by the solvable group of coordinate changes, a pattern that recurs when GIT builds moduli spaces as quotients in chapter 15. The hypothesis $N \geq 3$ is not a convenience but the exact threshold from lemma 13.4.3: for $N \leq 2$ the universal automorphism $[-1]$ survives, the quotient in Step 3 is not free, and $\mathcal{F}_N$ is again only coarsely representable.

The fine moduli space produced by the theorem has a name and a classical life of its own, and it is worth seeing it as a concrete curve rather than an abstract representing object.

Example 13.4.5: The Modular Curve $Y(N)$

The scheme $Y(N)$ of theorem 13.4.4 is the (affine) *modular curve of full level N*. Over $\mathbb{C}$ it has a transcendental description that makes its geometry visible, though the description has one feature that is easy to miss: an elliptic curve over $\mathbb{C}$ is a complex torus $\mathbb{C}/(\mathbb{Z}+\mathbb{Z}\tau)$ with τ in the upper half-plane $\mathfrak{H}$, and its N-torsion is $\frac{1}{N}(\mathbb{Z}+\mathbb{Z}\tau)/(\mathbb{Z}+\mathbb{Z}\tau)$, with the standard basis $\{\frac{1}{N}, \frac{\tau}{N}\}$. A level-$N$ structure is a choice of ordered basis of this torsion, an *arbitrary* isomorphism $\alpha\colon (\mathbb{Z}/N)^2 \to E[N]$ as in definition 13.4.1, and here a discrete invariant enters. The Weil pairing e_N attaches to α the primitive N-th root of unity $\zeta_\alpha = e_N(\alpha(e_1), \alpha(e_2)) \in \mu_N$, and any isomorphism of pairs preserves ζ_α (the Weil pairing is functorial in the curve), so ζ_α is constant on each connected component of the moduli. There are $\varphi(N) = |(\mathbb{Z}/N)^\times|$ primitive N-th roots of unity, and the change of variables $g \in \mathrm{SL}_2(\mathbb{Z}/N)$ (determinant 1) fixes ζ_α while a general $g \in \mathrm{GL}_2(\mathbb{Z}/N)$ raises it to the $\det g$ power ($\zeta \mapsto \zeta^{\det g}$); the fixed-determinant (*symplectic*) level structures with a given ζ_α are exactly those parametrized by the principal congruence subgroup

$$\Gamma(N) = \ker\big(\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N)\big)$$

acting on $\mathfrak{H}$ by fractional linear transformations ([11]). Thus over $\mathbb{C}$ the modular curve is a disjoint union of one copy of $\Gamma(N)\backslash\mathfrak{H}$ for each of the $\varphi(N)$ values of the Weil-pairing invariant,

$$Y(N)(\mathbb{C}) = \coprod_{\zeta \in \mu_N^{\mathrm{prim}}} \Gamma(N)\backslash\mathfrak{H}$$

a smooth affine Riemann surface with $\varphi(N)$ connected components, whose points are exactly the isomorphism classes of elliptic curves with full level-N structure. (Fixing the value ζ_α once and for all, that is passing to symplectic level structures, is the standard way to obtain the connected modular curve $\Gamma(N)\backslash\mathfrak{H}$; the full GL_2-functor of definition 13.4.2 sees all $\varphi(N)$ components at once.) For $N \geq 3$ the group $\Gamma(N)$ acts freely on $\mathfrak{H}$ (it contains no elliptic elements and, crucially, not $-\mathrm{id}$, which is where $N \geq 3$ reappears), so each component is a manifold with no orbifold points, matching the smoothness asserted by the theorem.

The universal family over $Y(N)$ is the descended family whose fiber over the class of τ is the torus $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ with its tautological torsion basis; concretely it is the Weierstrass family $y^2 = 4x^3 - g_2(\tau)x - g_3(\tau)$ built from the Eisenstein series, carried down along $\mathfrak{H} \to \Gamma(N)\backslash\mathfrak{H}$ together with the marked basis of N-torsion. This is a genuine universal object: any family of elliptic curves with level-N structure over a base S is the pullback of $(\mathcal{E}, \alpha^{\mathrm{univ}})$ along the unique classifying map $S \to Y(N)$ furnished by theorem 13.4.4. For $N = 3$ one has $\varphi(3) = 2$, so $Y(3)$ has two connected components, each an affine curve of genus zero, a projective line with a few points removed; the removed points are the cusps, where the elliptic curve degenerates, and they will reappear as the boundary when $\mathcal{M}_{1,1}$ is compactified in the capstone.

The modular curve makes the abstract representability concrete: $Y(N)$ is an ordinary quasi-projective variety, its complex points are $\varphi(N)$ copies of a familiar quotient of the upper half-plane, and it carries a family that specializes to any elliptic-curve-with-level-structure one wishes to name. The appearance of $\Gamma(N)$ ties the algebraic construction to the classical theory of modular forms, where functions on $\Gamma(N)\backslash\mathfrak{H}$ are precisely modular forms of level N ([11]); the moduli-theoretic reading is that a modular form is a section of a line bundle on the fine moduli space $Y(N)$.

Rigidification produced a fine moduli space by enlarging the problem, but the original problem, the classification of elliptic curves without extra data, has not gone away. The relation between the two is exact and useful: the fine space $Y(N)$ maps down to the coarse space of the un-rigidified problem by forgetting the level structure, and the map is the quotient by the group that permutes level structures. This exhibits the j-line as a group quotient of a smooth curve, which is often the most tractable way to study it.

Proposition 13.4.6: Coarse Space as a Quotient of the Rigidified Space

Let $N \geq 3$ be invertible in k, with $k = \bar{k}$. The finite group $\mathrm{GL}_2(\mathbb{Z}/N)$ acts on the fine moduli space $Y(N)$ by changing the level structure: an element $g \in \mathrm{GL}_2(\mathbb{Z}/N)$ sends the class of (E, α) to the class of $(E, \alpha \circ g)$. Forgetting the level structure induces a morphism $Y(N) \to \mathbb{A}^1_j$ to the j-line that is invariant under this action and identifies the j-line with the quotient:

$$\mathbb{A}^1_j \;\cong\; Y(N)/\mathrm{GL}_2(\mathbb{Z}/N)$$

and this presents the coarse moduli space $\mathbb{A}^1_j$ of elliptic curves (theorem 13.3.6, definition 13.3.1) as the quotient of the fine rigidified space by the finite group acting on level structures.

***Proof*:**
Step 1: The action and its orbits. For $g \in \mathrm{GL}_2(\mathbb{Z}/N)$ and a level structure $\alpha\colon (\mathbb{Z}/N)^2 \to E[N]$, the composite $\alpha \circ g$ is again an isomorphism $(\mathbb{Z}/N)^2 \to E[N]$, hence another level structure on the same curve E; this defines a right action of $\mathrm{GL}_2(\mathbb{Z}/N)$ on the pairs (E, α) fixing E and permuting the decorating data. Since $\mathrm{GL}_2(\mathbb{Z}/N)$ acts simply transitively on the set of bases of $E[N] \cong (\mathbb{Z}/N)^2$, for a fixed E the orbit of (E, α) under $\mathrm{GL}_2(\mathbb{Z}/N)$ is the set of *all* pairs (E, α') with the same underlying curve. Passing to isomorphism classes, the $\mathrm{GL}_2(\mathbb{Z}/N)$-orbit of the point of $Y(N)$ representing (E, α) consists of exactly the classes of pairs whose curve is isomorphic to E.

Step 2: The forgetful map factors through the quotient. The assignment $(E, \alpha) \mapsto E$ is a natural transformation $\mathcal{F}_N \to \mathcal{F}$ of moduli functors, and post-composing with the coarse map $\mathcal{F} \to h_{\mathbb{A}^1_j}$ of theorem 13.3.6 gives a natural transformation $\mathcal{F}_N \to h_{\mathbb{A}^1_j}$, hence by Yoneda and representability of $\mathcal{F}_N$ a morphism of schemes $\pi\colon Y(N) \to \mathbb{A}^1_j$. Concretely π sends the class of (E, α) to $j(E)$. Since $j(E)$ depends only on E and not on α, the map π is constant on $\mathrm{GL}_2(\mathbb{Z}/N)$-orbits by Step 1, so it is $\mathrm{GL}_2(\mathbb{Z}/N)$-invariant and factors through the quotient scheme $Y(N)/\mathrm{GL}_2(\mathbb{Z}/N)$, which exists because $\mathrm{GL}_2(\mathbb{Z}/N)$ is finite acting on a quasi-projective scheme, inducing $\bar{\pi}\colon Y(N)/\mathrm{GL}_2(\mathbb{Z}/N) \to \mathbb{A}^1_j$.

Step 3: $\bar{\pi}$ is a bijection on points, hence an isomorphism. On $\bar{k}$-points, Step 1 shows the fibers of π are exactly the $\mathrm{GL}_2(\mathbb{Z}/N)$-orbits: two pairs (E, α), (E', α') map to the same j if and only if $E \cong E'$, if and only if they lie in one orbit. Therefore $\bar{\pi}$ is injective on $\bar{k}$-points. It is surjective because every $j_0 \in \mathbb{A}^1_j(\bar{k})$ is $j(E)$ for some elliptic curve E over $\bar{k}$ (theorem 13.3.6), and E admits a level-N structure over $\bar{k}$ (its N-torsion is $(\mathbb{Z}/N)^2$, so a basis exists), giving a point of $Y(N)$ over j_0. Thus $\bar{\pi}$ is a bijection on $\bar{k}$-points between normal varieties. Upgrading this to an isomorphism requires two inputs beyond the set-theoretic bijection, and both are verified here rather than asserted.

First, $\bar{\pi}$ is birational. The morphism $\pi\colon Y(N) \to \mathbb{A}^1_j$ is finite. To see this, compactify: adding the cusps to each component completes $Y(N)$ to a smooth projective curve $X(N)$, and completing the j-line adds the single point $j = \infty$ to $\mathbb{A}^1_j = \mathbb{P}^1 \setminus \{\infty\}$, giving $X(1) = \mathbb{P}^1$. The forgetful map extends to a nonconstant morphism $\bar{\pi}_c\colon X(N) \to X(1) = \mathbb{P}^1$ of smooth projective curves, and a nonconstant morphism of smooth projective curves is finite. All cusps of $X(N)$ lie over $j = \infty$: the cusps are by construction the points where the elliptic curve degenerates, which is exactly the fiber of $X(N) \to \mathbb{P}^1$ over ∞ ([11]). Restricting the finite morphism $\bar{\pi}_c$ over the cusp-free locus $\mathbb{A}^1_j = \mathbb{P}^1 \setminus \{\infty\}$ recovers $\pi\colon Y(N) \to \mathbb{A}^1_j$, which is therefore finite, being the base change of a finite morphism along the open immersion $\mathbb{A}^1_j \hookrightarrow \mathbb{P}^1$. Over a $\bar{k}$-point j_0 with $j_0 \neq 0, 1728$ the curve E has $\mathrm{Aut}(E) = \{\pm 1\}$ ([11]). By Step 1 the fiber $\pi^{-1}(j_0)$ is the set of isomorphism classes of level structures on E, that is the $|\mathrm{GL}_2(\mathbb{Z}/N)|$ bases of $E[N]$ modulo the action of $\mathrm{Aut}(E) = \{\pm 1\}$; since $\{\pm 1\}$ acts on bases through $\pm\mathrm{id} \in \mathrm{GL}_2(\mathbb{Z}/N)$ without fixed points (as $-\mathrm{id} \neq \mathrm{id}$ for $N \geq 3$), the fiber has exactly $|\mathrm{GL}_2(\mathbb{Z}/N)|/2$ points, each a reduced point because N is invertible in k and the level cover is étale there. This common value is both the degree of π over the ordinary locus and the size of the corresponding $\mathrm{GL}_2(\mathbb{Z}/N)$-orbit, by the orbit-stabilizer count $|\mathrm{GL}_2(\mathbb{Z}/N)|/|\{\pm 1\}|$ with stabilizer exactly the $\{\pm 1\}$ acting through $\mathrm{Aut}(E)$. Hence the generic fiber of $\bar{\pi} = \pi/\mathrm{GL}_2(\mathbb{Z}/N)$ is a single reduced point, so $\bar{\pi}$ is a finite morphism of degree one, that is birational.

Second, $Y(N)/\mathrm{GL}_2(\mathbb{Z}/N)$ is normal, being the quotient of the smooth (hence normal) $Y(N)$ by a finite group, and $\mathbb{A}^1_j$ is smooth. A finite birational morphism to a normal variety is an isomorphism (this is the form of Zariski's main theorem that does not require separability hypotheses beyond birationality, which was checked above and holds in every characteristic because N is invertible). Therefore $\bar{\pi}$ is an isomorphism $Y(N)/\mathrm{GL}_2(\mathbb{Z}/N) \cong \mathbb{A}^1_j$.

> That $\mathbb{A}^1_j$ is the coarse moduli space of elliptic curves is theorem 13.3.6, so the displayed isomorphism presents the coarse space as the finite-group quotient of the fine rigidified space, completing the proof.

The proposition is the reconciliation the section was aiming for. The un-rigidified moduli problem has no fine solution, but it has a coarse one, $\mathbb{A}^1_j$, and the coarse solution is recovered from the fine solution of the rigidified problem by dividing out the choices that the rigidification introduced. The group $\mathrm{GL}_2(\mathbb{Z}/N)$ measures exactly how much extra data a level structure carries, and quotienting by it erases that data. This is a template for the whole subject: a moduli problem with automorphisms is studied by rigidifying to a fine space and then quotienting by the finite group that acts on the rigidifying data, and the quotient is the coarse space. The template has a limitation the reader should carry forward. The quotient $Y(N)/\mathrm{GL}_2(\mathbb{Z}/N)$ remembers only orbits, so it forgets the stabilizers, the automorphism groups of the curves, exactly the information $\mathbb{A}^1_j$ was already blind to at $j = 0$ and $j = 1728$. The framework that keeps track of the stabilizers while still forming the quotient is the quotient *stack* $[Y(N)/\mathrm{GL}_2(\mathbb{Z}/N)]$, and building it is the task of Part VI.

Exercise 13.4.1

1. Show that for $N = 2$ the automorphism $[-1]$ fixes every level-2 structure on every elliptic curve, and conclude that lemma 13.4.3 fails for $N = 2$. Deduce that the level-2 moduli functor $\mathcal{F}_2$ is not representable by a fine moduli space over $\bar{k}$ (with $\operatorname{char} k \neq 2$), even though it carries more data than $\mathcal{F}$.
2. Let $N \geq 3$. Compute the order of $\mathrm{GL}_2(\mathbb{Z}/N)$ for N prime, and verify directly that $\mathrm{GL}_2(\mathbb{Z}/3)$ acts simply transitively on the set of ordered bases of $(\mathbb{Z}/3)^2$. Using proposition 13.4.6, interpret the degree of the forgetful map $Y(3) \to \mathbb{A}^1_j$ over a generic j (where $\mathrm{Aut}(E) = \{\pm 1\}$) and explain the discrepancy between $|\mathrm{GL}_2(\mathbb{Z}/3)|$ and this degree in terms of the residual action of $[-1]$.
3. Prove that the endomorphism-norm inequality used in Step 2 of lemma 13.4.3 is sharp for $N = 2$ by exhibiting, on an arbitrary elliptic curve, an endomorphism $\psi = \varphi - \mathrm{id}$ of degree 4 with φ an automorphism and $E[2] \subseteq \ker \psi$. (Suggestion: take $\varphi = [-1]$.)
4. A level structure is one instance of rigidifying data. For a smooth projective curve C of genus $g \geq 2$, an *n-canonical framing* is a choice of ordered basis of $H^0(C, \omega_C^{\otimes n})$ for $n \geq 3$. Explain, by analogy with definition 13.4.1 and lemma 13.4.3, why such a framing should rigidify the moduli problem of genus-g curves: identify the group that acts on the framings (playing the role of $\mathrm{GL}_2(\mathbb{Z}/N)$) and state what property of $\mathrm{Aut}(C)$ makes the framing kill automorphisms.
5. Show that over $\mathbb{C}$ each connected component $\Gamma(3)\backslash\mathfrak{H}$ of the modular curve $Y(3)$ of example 13.4.5 has genus zero, by computing the index $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(3)]$ and the number of cusps, then applying the Riemann-Hurwitz formula to the covering $\Gamma(3)\backslash\mathfrak{H} \to Y(1) = \mathbb{A}^1_j$. (You may use that $Y(1)(\mathbb{C}) = \mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$ has genus zero with one cusp and elliptic points of orders 2 and 3.) Then check that the full forgetful map $Y(3) \to Y(1)$, taken over both components, has degree 24, reconciling with part (b).

13.5 Criteria for Representability

The preceding sections have twice confronted the same fault line. A moduli problem is a functor $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ (definition 13.1.1), and the best possible outcome is a fine moduli space, a scheme M with $\mathcal{F} \cong h_M$ (definition 13.1.3). Yet functors that arise from geometry routinely fail to be representable, and the failure is not random. The warning issued in box 13.1.6 named the culprit: a functor of points is always a sheaf for the Zariski topology, so any moduli functor that is not such a sheaf cannot be representable. This section turns that observation into working tools. Two questions organize it. First, given a functor, what are the criteria that decide whether it is representable? Second, what first-order information does a functor carry even before that question is settled, and how is that information extracted?

The answer to the first question is a local-to-global principle: a functor representable in the Zariski topology and locally representable on an open cover is representable, exactly as a scheme is glued from affine charts. The answer to the second is the tangent space of a functor, computed by feeding the functor the dual numbers $k[\varepsilon]$. This tangent space is the first invariant of deformation theory, the thread the book picks up after this chapter: where this chapter asks whether a moduli space exists, the deformation theory that follows asks what it looks like near a point, and the tangent space of the functor is where those two investigations meet. The two tools of this section are complementary. The sheaf-and-cover criterion is a global test for existence; the tangent space is a pointwise linearization that survives even when no representing scheme does.

The first tool is the tangent space, and it comes from the dual numbers. A scheme X over k has, at each k-point x, a Zariski tangent space $T_xX = (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee$, and the classical characterization of tangent vectors identifies them with morphisms $\operatorname{Spec} k[\varepsilon] \to X$ sending the closed point to x. The dual numbers $k[\varepsilon] = k[\varepsilon]/(\varepsilon^2)$ are the algebraic model of an infinitesimally short arc: a k-point together with a tangent direction. A functor $\mathcal{F}$ of points has no underlying topological space and no local ring, so its tangent space cannot be defined through a cotangent module. What it does have is its value on $\operatorname{Spec} k[\varepsilon]$, and the same tangent vectors reappear there. The device is to compare $\mathcal{F}(k[\varepsilon])$ with $\mathcal{F}(k)$ along the map that sets ε to zero.

Throughout, for a k-algebra A write $\mathcal{F}(A)$ for $\mathcal{F}(\operatorname{Spec} A)$; this is the standard abuse that lets a contravariant functor on schemes be evaluated on rings covariantly. The reduction homomorphism $k[\varepsilon] \to k$, $\varepsilon \mapsto 0$, induces $\mathcal{F}(k[\varepsilon]) \to \mathcal{F}(k)$, and a choice of base point $x \in \mathcal{F}(k)$ singles out a fiber.

Definition 13.5.1: Tangent Space of a Functor

Let $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ be a functor and $x \in \mathcal{F}(k)$ a k-point. Let

$$\pi\colon \mathcal{F}(k[\varepsilon]) \longrightarrow \mathcal{F}(k)$$

be the map induced by the k-algebra homomorphism $k[\varepsilon] \to k$ sending $\varepsilon \mapsto 0$. The *tangent space of $\mathcal{F}$ at x* is the fiber

$$T_x\mathcal{F} \;=\; \pi^{-1}(x) \;=\; \{\xi \in \mathcal{F}(k[\varepsilon]) \mid \pi(\xi) = x\}$$

the set of lifts of x to a $k[\varepsilon]$-point of $\mathcal{F}$.

As stated, $T_x\mathcal{F}$ is only a pointed set: it always contains the *constant* lift, the image of x under the section $\mathcal{F}(k) \to \mathcal{F}(k[\varepsilon])$ induced by the structure map $k \hookrightarrow k[\varepsilon]$. For a functor of geometric origin it carries far more structure. When $\mathcal{F} = h_X$ is representable, the fiber $T_x\mathcal{F}$ is precisely the classical Zariski tangent space T_xX, a k-vector space; the identification is exactly the arc description of tangent vectors recalled above, and the full account of why the functorial definition recovers the geometric one is developed in definition 3.3.12. The vector-space structure does not require representability. It is enough that $\mathcal{F}$ be *reasonable* in the following precise sense.

Proposition 13.5.2: Vector Space Structure on the Tangent Space

Suppose $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ takes the fiber product of k-algebras

$$k[\varepsilon] \times_k k[\varepsilon] \;\cong\; k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2$$

to the corresponding fiber product of sets, that is, the natural map

$$\mathcal{F}\big(k[\varepsilon] \times_k k[\varepsilon]\big) \longrightarrow \mathcal{F}(k[\varepsilon]) \times_{\mathcal{F}(k)} \mathcal{F}(k[\varepsilon])$$

is a bijection. Then for every $x \in \mathcal{F}(k)$ the set $T_x\mathcal{F}$ carries a natural k-vector space structure whose zero is the constant lift.

***Proof*:**
Fix $x \in \mathcal{F}(k)$. The argument realizes addition and scalar multiplication as functorial consequences of ring maps out of $k[\varepsilon]$.

Step 1: Addition. The ring $B = k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2$ is the fiber product $k[\varepsilon] \times_k k[\varepsilon]$, with the two projections $p_i\colon B \to k[\varepsilon]$ given by $\varepsilon_i \mapsto \varepsilon$ and $\varepsilon_j \mapsto 0$ for $j \neq i$. By hypothesis

$$\mathcal{F}(B) \xrightarrow{\sim} T_x\mathcal{F} \times T_x\mathcal{F}$$

once one restricts to the fiber over x: an element of the right side is a pair (ξ_1, ξ_2) of lifts of x, and the isomorphism produces a unique $\zeta \in \mathcal{F}(B)$ restricting to ξ_i under $\mathcal{F}(p_i)$. The addition homomorphism

$$\sigma\colon B \to k[\varepsilon], \qquad \varepsilon_1, \varepsilon_2 \mapsto \varepsilon$$

is well defined because it carries the defining relations $(\varepsilon_1, \varepsilon_2)^2 = 0$ to $\varepsilon^2 = 0$, which holds in $k[\varepsilon]$, and it is a k-algebra map reducing to the identity modulo ε. Set $\xi_1 + \xi_2 := \mathcal{F}(\sigma)(\zeta)$. This lands in $T_x\mathcal{F}$ since σ covers $\varepsilon \mapsto 0$, and it is independent of choices because ζ was unique.

Step 2: Scalar multiplication. For $\lambda \in k$ the map $m_\lambda\colon k[\varepsilon] \to k[\varepsilon]$, $\varepsilon \mapsto \lambda\varepsilon$, is a k-algebra homomorphism reducing to the identity modulo ε. Define $\lambda \cdot \xi := \mathcal{F}(m_\lambda)(\xi)$ for $\xi \in T_x\mathcal{F}$. Since m_λ covers $\varepsilon \mapsto 0$, the result again lies in $T_x\mathcal{F}$.

Step 3: The axioms. The vector-space axioms are inherited from identities among the structure maps. Commutativity and associativity of $+$ follow from the symmetry of B under $\varepsilon_1 \leftrightarrow \varepsilon_2$ and from the associativity of the triple analogue $k[\varepsilon_1, \varepsilon_2, \varepsilon_3]/(\varepsilon_1, \varepsilon_2, \varepsilon_3)^2$, which is the iterated fiber product $(k[\varepsilon] \times_k k[\varepsilon]) \times_k k[\varepsilon]$ and so is respected by $\mathcal{F}$ through two applications of the fiber-product hypothesis. The zero element is the constant lift 0_x, the image of x under $k \hookrightarrow k[\varepsilon]$: composing σ with the inclusion of one factor recovers the other summand, so $\xi + 0_x = \xi$. The identity $m_\lambda \circ m_\mu = m_{\lambda\mu}$ gives $\lambda(\mu\xi) = (\lambda\mu)\xi$, and $m_1 = \mathrm{id}$ gives $1 \cdot \xi = \xi$. Distributivity $\lambda(\xi_1 + \xi_2) = \lambda\xi_1 + \lambda\xi_2$ holds because m_λ commutes with σ up to the symmetric substitution $\varepsilon_i \mapsto \lambda\varepsilon_i$, and $(\lambda + \mu)\xi = \lambda\xi + \mu\xi$ follows by applying $\mathcal{F}$ to the map $B \to k[\varepsilon]$, $\varepsilon_1 \mapsto \lambda\varepsilon$, $\varepsilon_2 \mapsto \mu\varepsilon$, and comparing with σ. All verifications are functorial images of ring-level identities, so no further choices intervene, completing the proof.

The hypothesis of proposition 13.5.2 is a functorial statement of a single geometric fact: two tangent vectors at a point can be added because a pair of tangent directions is the same datum as one map from the "doubled" dual numbers B. Every representable functor satisfies the hypothesis, since h_X turns the fiber-product presentation of B into a fiber product of Hom-sets; the deformation functors studied later satisfy it too, which is why their tangent spaces are vector spaces even when the functors are not representable. This is the sense in which the tangent space survives the failure of representability, and it is the reason first-order deformation theory can proceed by linear algebra.

Example 13.5.3: The Tangent Space to Projective Space at a Point

Take $\mathcal{F} = h_{\mathbb{P}^n}$, the functor represented by projective n-space over k, described in example 13.1.4 as the moduli of rank-one locally free quotients of $\mathcal{O}^{n+1}$. Fix a k-point $x = [a_0 : \cdots : a_n] \in \mathbb{P}^n(k)$, and after reordering assume $a_0 \neq 0$, so x lies in the standard chart $U_0 \cong \mathbb{A}^n$ with affine coordinates $t_i = x_i/x_0$ for $i = 1, \ldots, n$; in these coordinates x is the point $(a_1/a_0, \ldots, a_n/a_0)$, which after rescaling may be written $(c_1, \ldots, c_n)$. A $k[\varepsilon]$-point of $\mathbb{P}^n$ lifting x is a quotient $\mathcal{O}_{k[\varepsilon]}^{n+1} \twoheadrightarrow L$ with L locally free of rank one, reducing modulo ε to the given quotient at x. Because x lies in U_0, such a lift is the same as an A-point of $\mathbb{A}^n$ over $A = k[\varepsilon]$ reducing to x, that is, an n-tuple

$$\big(c_i + b_i\varepsilon\big)_{i=1}^n, \qquad b_i \in k$$

The reduction π discards the $b_i\varepsilon$ terms, sending this tuple to $(c_1, \ldots, c_n) = x$, so the fiber $T_x\mathbb{P}^n$ is the set of tuples $(b_1, \ldots, b_n) \in k^n$. Under the addition of proposition 13.5.2 the doubled dual numbers send $\varepsilon_1, \varepsilon_2 \mapsto \varepsilon$, which adds the two tuples coordinatewise, and m_λ scales them; thus $T_x\mathbb{P}^n \cong k^n$ as a k-vector space. This matches the classical Zariski tangent space $T_x\mathbb{P}^n$ of dimension n, computed here without ever writing down $\mathfrak{m}_x/\mathfrak{m}_x^2$.

The tangent space is a pointwise probe. To decide representability one needs a global constraint, and the sharpest one available at this stage is the sheaf condition. Box 13.1.6 flagged, informally, that a representable functor is a Zariski sheaf and that objects with automorphisms break this property. That warning is now upgraded to a proposition with a proof, and strengthened: representability forces the sheaf condition not only for the Zariski topology but for the far finer fppf topology, in which covers may be flat surjective rather than open.

Recall the sheaf condition in functorial form. A functor $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ is a *sheaf* for a class of covers if, for every cover $\{U_i \to S\}$ in the class, the sequence

$$\mathcal{F}(S) \longrightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \times_S U_j)$$

is an equalizer: an element of $\prod_i \mathcal{F}(U_i)$ that agrees on all overlaps $U_i \times_S U_j$ comes from a unique element of $\mathcal{F}(S)$. For the Zariski topology the $U_i \to S$ are open immersions covering S; for the fppf topology they are morphisms, flat and of finite presentation, jointly surjective.

Proposition 13.5.4: Representable Functors are fppf Sheaves

For every scheme X, the functor of points $h_X = \mathrm{Hom}(-, X)$ is a sheaf for the fppf topology on $\mathbf{Sch}$, and in particular for the Zariski topology. Consequently, any functor $\mathcal{F}$ that is not a Zariski sheaf is not representable.

***Proof*:**
The Zariski case is elementary and comes first; the fppf case is the content of faithfully flat descent for morphisms, which is imported from the descent chapter.

Step 1: The Zariski sheaf condition. Let $\{U_i \to S\}$ be a Zariski open cover, so the $U_i \subseteq S$ are open subschemes with $\bigcup_i U_i = S$ and $U_i \times_S U_j = U_i \cap U_j$. Given morphisms $f_i \colon U_i \to X$ agreeing on overlaps, $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$, a morphism of schemes is a continuous map of underlying spaces together with a map of structure sheaves, and both data are local on the source. The set-map $|S| \to |X|$ is defined by $s \mapsto f_i(s)$ for any i with $s \in U_i$; agreement on overlaps makes this independent of i, hence well defined and continuous since it is continuous on each open U_i. The sheaf map $\mathcal{O}_X \to f_*\mathcal{O}_S$ is assembled from the $\mathcal{O}_X \to (f_i)_*\mathcal{O}_{U_i}$ by the sheaf axiom on X, again using overlap agreement. The resulting $f \colon S \to X$ restricts to f_i on each U_i, and it is the unique such morphism because a morphism is determined by its restrictions to an open cover. This is exactly the equalizer condition.

Step 2: The fppf sheaf condition. Let $\{U_i \to S\}$ be an fppf cover. Refining, it suffices to treat a single faithfully flat morphism of finite presentation $u \colon U \to S$, since a disjoint union $\coprod_i U_i \to S$ of the covering maps is again faithfully flat and of finite presentation, and the equalizer over the family coincides with the equalizer for $\coprod_i U_i \to S$. The claim is then that

$$\mathrm{Hom}(S, X) \longrightarrow \mathrm{Hom}(U, X) \rightrightarrows \mathrm{Hom}(U \times_S U, X)$$

is an equalizer. A morphism $g \colon U \to X$ equalizing the two maps $U \times_S U \rightrightarrows U$ is a morphism satisfying $g \circ \mathrm{pr}_1 = g \circ \mathrm{pr}_2$ on $U \times_S U$, that is, a descent datum for the morphism g along the fppf cover u. Faithfully flat descent for morphisms to a scheme asserts that such a g descends uniquely to a morphism $S \to X$: the map on topological spaces descends because u is submersive (a quotient map, being faithfully flat and quasi-compact), and the map on structure sheaves descends because faithfully flat descent is effective for quasi-coherent modules, applied to $\mathcal{O}_X$-algebra structures. Faithfully flat descent for morphisms is the subject of a later chapter, where it is proved in full; invoking it here rather than reproving it avoids circularity, since the descent chapter presupposes the functor-of-points language established in this one. Granting that theorem, the displayed sequence is an equalizer.

Step 3: The contrapositive. If $\mathcal{F} \cong h_X$ for some scheme X, then $\mathcal{F}$ inherits the Zariski sheaf property of h_X by transport of the natural isomorphism. Hence a functor failing the Zariski sheaf condition cannot be isomorphic to any h_X, so it is not representable, as we sought to show.

The proposition is the precise form of the obstruction that recurs throughout the chapter. Box 13.1.6 phrased it informally; proposition 13.5.4 makes it a decision procedure. To show a functor is not representable, exhibit a Zariski cover $\{U_i \to S\}$ and local data $\{f_i\}$ that agree on overlaps but glue to *no* element of $\mathcal{F}(S)$, or glue to more than one. Objects with automorphisms produce exactly this failure: an automorphism lets one twist local families along overlaps so that they agree as isomorphism classes yet cannot be reconciled into a single global family. The j-line seen in theorem 13.3.6 is the prototype, and the worked example below isolates the mechanism in its barest form. The fppf strengthening matters for a reason that becomes central once stacks appear: moduli functors typically *are* sheaves for the fppf or étale topology even when they fail to be representable by a scheme, so the sheaf condition alone does not detect the failure, a point that returns when algebraic stacks are introduced later in the book. What separates a representable functor from a mere fppf sheaf is the second criterion, the local representability that the next result formalizes.

A scheme is not built all at once; it is glued from affine pieces along open subschemes. The functor of points should be constructible the same way, and it is: a functor that looks representable on an open cover and satisfies the Zariski sheaf condition is globally representable. This is the local-to-global criterion promised at the start, and it is the mechanism by which nearly every moduli space in the book is eventually constructed, the Grassmannian of theorem 13.2.4 among them. The statement requires the notion of an open subfunctor, the functorial counterpart of an open subscheme.

Definition 13.5.5: Open Subfunctor

Let $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ be a functor. A subfunctor $\mathcal{G} \subseteq \mathcal{F}$ (a functor with $\mathcal{G}(S) \subseteq \mathcal{F}(S)$ for all S, compatibly with restriction) is an *open subfunctor* if for every scheme T and every morphism $T \to \mathcal{F}$ (equivalently, by Yoneda, every $t \in \mathcal{F}(T)$), the fiber product $\mathcal{G} \times_{\mathcal{F}} T$ is representable by an open subscheme of T. A family of open subfunctors $\{\mathcal{G}_i \subseteq \mathcal{F}\}$ is an *open cover* if for every field K over k and every $x \in \mathcal{F}(K)$, there is some i with $x \in \mathcal{G}_i(K)$; equivalently, for every $t \in \mathcal{F}(T)$ the open subschemes $\mathcal{G}_i \times_{\mathcal{F}} T \subseteq T$ cover T.

The definition says that pulling an open subfunctor back along any T-point of $\mathcal{F}$ produces an open subscheme of T, and that these opens cover T as t ranges over the point. When $\mathcal{F} = h_X$ is representable, its open subfunctors are exactly the h_U for $U \subseteq X$ open, and an open cover of $\mathcal{F}$ is a Zariski open cover of X; the definition is the transcription of that situation into functorial language, made so that it can be applied before X is known to exist. With it, the criterion reads as follows.

Theorem 13.5.6: Zariski-Local Criterion for Representability

Let $\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ be a functor satisfying:

1. $\mathcal{F}$ is a sheaf for the Zariski topology; and
2. $\mathcal{F}$ admits an open cover $\{\mathcal{G}_i \subseteq \mathcal{F}\}_{i\in I}$ by open subfunctors, each of which is representable, say $\mathcal{G}_i \cong h_{X_i}$ for a scheme X_i.

Then $\mathcal{F}$ is representable: there is a scheme X with $\mathcal{F} \cong h_X$, obtained by gluing the X_i along the open subschemes cut out by their pairwise intersections $\mathcal{G}_i \cap \mathcal{G}_j$.

***Proof*:**
The proof reconstructs a scheme from the charts X_i and identifies its functor of points with $\mathcal{F}$. The two hypotheses do exactly complementary work: the open cover supplies the charts and their overlaps, and the sheaf condition supplies the unique global sections that make the gluing represent $\mathcal{F}$.

Step 1: The charts and their overlaps. For each i, fix an isomorphism $\mathcal{G}_i \cong h_{X_i}$; by Yoneda the identity of X_i is a distinguished element $u_i \in \mathcal{G}_i(X_i) \subseteq \mathcal{F}(X_i)$, the universal element of the chart. For a pair (i, j) consider the open subfunctor $\mathcal{G}_j \cap \mathcal{G}_i = \mathcal{G}_j \times_{\mathcal{F}} \mathcal{G}_i$. Pulling $\mathcal{G}_j$ back along the X_i-point $u_i \in \mathcal{F}(X_i)$ gives, by the open-subfunctor hypothesis applied to $\mathcal{G}_j$ and $T = X_i$, an open subscheme

$$X_{ij} \;:=\; \mathcal{G}_j \times_{\mathcal{F}} X_i \;\subseteq\; X_i$$

Concretely X_{ij} is the largest open of X_i over which the universal element u_i lands in $\mathcal{G}_j$. Symmetrically $X_{ji} \subseteq X_j$ is defined.

Step 2: The transition isomorphisms. Both $X_{ij} \subseteq X_i$ and $X_{ji} \subseteq X_j$ represent the same functor, namely $\mathcal{G}_i \cap \mathcal{G}_j$: indeed $h_{X_{ij}} = \mathcal{G}_j \times_{\mathcal{F}} \mathcal{G}_i = \mathcal{G}_i \cap \mathcal{G}_j$ by construction, and likewise $h_{X_{ji}} = \mathcal{G}_i \cap \mathcal{G}_j$. Yoneda turns the resulting isomorphism of functors into a unique isomorphism of schemes

$$\varphi_{ji} \colon X_{ij} \xrightarrow{\sim} X_{ji}$$

Because it is induced by the identity on the common functor $\mathcal{G}_i \cap \mathcal{G}_j$, one has $\varphi_{ii} = \mathrm{id}$ and $\varphi_{ij} = \varphi_{ji}^{-1}$.

Step 3: The cocycle condition. Gluing schemes requires the transition maps to satisfy the cocycle identity on triple overlaps. Restricting to the open subscheme $X_{ijk} \subseteq X_i$ representing $\mathcal{G}_i \cap \mathcal{G}_j \cap \mathcal{G}_k$, the isomorphisms φ_{ji}, φ_{kj}, φ_{ki} are all induced by the identity on the triple-intersection functor $\mathcal{G}_i \cap \mathcal{G}_j \cap \mathcal{G}_k$. Hence

$$\varphi_{ki} = \varphi_{kj} \circ \varphi_{ji} \quad \text{on} \quad X_{ijk}$$

since both sides are the unique isomorphism induced by that identity. The cocycle condition holds, so the gluing datum $\big(X_i, X_{ij}, \varphi_{ji}\big)$ is effective: there is a scheme X with open subschemes $\iota_i \colon X_i \hookrightarrow X$ covering X, such that $\iota_i(X_{ij}) = \iota_j(X_{ji})$ and $\iota_j \circ \varphi_{ji} = \iota_i$ on X_{ij}.

Step 4: A natural transformation $\mathcal{F} \to h_X$. The universal elements $u_i \in \mathcal{F}(X_i)$ are compatible on overlaps: on X_{ij} the element u_i restricts into $\mathcal{G}_j$ (that is what X_{ij} was), and under φ_{ji} it is carried to the restriction of u_j, because both are the universal element of $\mathcal{G}_i \cap \mathcal{G}_j$. Transporting along the ι_i, the sections $u_i \in \mathcal{F}(X_i)$ agree on the overlaps $\iota_i(X_{ij}) = \iota_j(X_{ji})$ inside X. Since $\mathcal{F}$ is a Zariski sheaf and $\{X_i \to X\}$ is a Zariski open cover of X, there is a *unique* $u \in \mathcal{F}(X)$ restricting to u_i on each X_i. By Yoneda, u is a natural transformation $h_X \to \mathcal{F}$, equivalently a morphism $\Phi \colon h_X \to \mathcal{F}$ of functors.

Step 5: Φ is an isomorphism. It suffices to check that $\Phi_T \colon h_X(T) \to \mathcal{F}(T)$ is a bijection for every scheme T; both sides are Zariski sheaves in T (the source by proposition 13.5.4, the target by hypothesis (1)), so bijectivity may be checked locally on T. Given $t \in \mathcal{F}(T)$, hypothesis (2) gives an open cover $\{T_i\}$ of T with $t|_{T_i} \in \mathcal{G}_i(T_i) \cong h_{X_i}(T_i)$; thus $t|_{T_i}$ corresponds to a morphism $g_i \colon T_i \to X_i \xrightarrow{\iota_i} X$, and by construction $\Phi(g_i) = t|_{T_i}$. On the overlap $T_i \cap T_j$ the two morphisms g_i, g_j land in $\iota_i(X_{ij}) = \iota_j(X_{ji})$ and are identified there by φ_{ji}, so $g_i|_{T_i \cap T_j} = g_j|_{T_i \cap T_j}$ as morphisms to X. Because h_X is a Zariski sheaf, the g_i glue to a unique $g \colon T \to X$ with $\Phi(g) = t$; this is surjectivity, with uniqueness of g giving injectivity. Hence Φ_T is a bijection for all T, so $\mathcal{F} \cong h_X$, completing the proof.

The theorem is what the entire representability argument rests on. It reduces the global question "is $\mathcal{F}$ representable?" to two local ones: does $\mathcal{F}$ glue in the Zariski topology, and can one cover it by pieces already known to be schemes? Both halves are indispensable. Dropping the sheaf condition, one could cover a functor by representable opens whose local charts refuse to glue, exactly the pathology that makes non-separated or non-representable objects; dropping local representability, a Zariski sheaf carries no charts to glue. The Grassmannian is representable precisely because its standard affine charts (example 13.2.8) are representable open subfunctors and the quotient-bundle functor is a Zariski sheaf, and theorem 13.2.4 is in essence this criterion applied to those charts. The same template constructs the Hilbert and Quot schemes in the next chapter, once suitable representable charts are produced. The criterion also clarifies where representability can fail: not in the gluing of charts, which is automatic once they exist, but in the sheaf condition, which is where automorphisms strike.

Proposition 13.5.4 converts non-representability into a concrete search: find a cover and local data that violate the sheaf condition. The following example carries out that search in the simplest nontrivial setting, the naive functor of line bundles up to isomorphism. Its failure is the same one that stops the j-line from being a fine moduli space in theorem 13.3.6, stripped of the geometry so that the mechanism is visible in a single cohomology group.

Example 13.5.7: The Naive Picard Functor is not a Zariski Sheaf

Work over k and consider the naive Picard functor

$$\mathcal{F}\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}, \qquad \mathcal{F}(S) = \operatorname{Pic}(S) = \{\text{line bundles } \mathcal{L} \text{ on } S\}/\cong$$

the set of isomorphism classes of invertible sheaves on S, with pullback along $f\colon T \to S$ given by f^*. This is a moduli functor in the sense of definition 13.1.1: its objects are line bundles and its morphisms are pullbacks. The claim is that $\mathcal{F}$ is *not* representable, and the obstruction is the passage to isomorphism classes, which forgets the automorphisms $\mathcal{L} \xrightarrow{c} \mathcal{L}$ ($c \in k^\times$) that every line bundle carries.

To see the sheaf condition fail, take $S = \mathbb{P}^1$ and the standard Zariski cover $S = U_0 \cup U_1$ by the two affine charts $U_0 = \operatorname{Spec} k[t]$, $U_1 = \operatorname{Spec} k[u]$ with $u = t^{-1}$, overlap $U_0 \cap U_1 = \operatorname{Spec} k[t, t^{-1}]$. On each chart every line bundle is trivial up to isomorphism, since $\operatorname{Pic}(\mathbb{A}^1) = 0$; and $\operatorname{Pic}(U_0 \cap U_1) = \operatorname{Pic}(\mathbb{G}_m) = 0$ as well. Hence

$$\mathcal{F}(U_0) = \mathcal{F}(U_1) = \mathcal{F}(U_0 \cap U_1) = \{*\}$$

are all singletons, so the fiber product $\mathcal{F}(U_0) \times_{\mathcal{F}(U_0 \cap U_1)} \mathcal{F}(U_1)$ is a single point: the trivial bundle on each chart is the unique local datum, and it is automatically compatible on the overlap.

The global set is not a point. Every line bundle on $\mathbb{P}^1$ is $\mathcal{O}_{\mathbb{P}^1}(n)$ for a unique $n \in \mathbb{Z}$, so $\mathcal{F}(\mathbb{P}^1) = \operatorname{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$, and each of these restricts to the trivial bundle on both charts. The restriction map

$$\mathcal{F}(\mathbb{P}^1) \longrightarrow \mathcal{F}(U_0) \times_{\mathcal{F}(U_0 \cap U_1)} \mathcal{F}(U_1)$$

is therefore $\mathbb{Z} \to \{*\}$, which is not injective. The equalizer condition fails in its *uniqueness* clause: the unique compatible local datum lifts to infinitely many global classes, one for each n, rather than to a single one. Concretely, gluing the two trivial charts by the transition unit $t^n \in k[t, t^{-1}]^\times$ produces $\mathcal{O}_{\mathbb{P}^1}(n)$, and all these gluings have identical restriction to the charts yet are pairwise non-isomorphic globally. Hence $\mathcal{F}$ is not a Zariski sheaf, and by proposition 13.5.4 it is not representable.

The mechanism is the one anticipated by box 13.1.6. Passing to isomorphism classes erases the transition data t^n, which is precisely the automorphism-valued gluing information $\mathcal{O}(U_0 \cap U_1)^\times$ that distinguishes the global bundles; the sheaf condition then cannot recover n from local data that all look trivial. This is the same failure that prevented the j-line from being a fine moduli space in theorem 13.3.6, where isotrivial families with nonconstant transition automorphisms restrict to identical local objects yet are globally distinct. The repair, developed in the stacks Part of the book, is to remember the automorphisms rather than quotient them away, replacing the set-valued functor $\mathcal{F}$ by a groupoid-valued one, that is, a stack. The correct Picard functor, defined as a sheafification and made representable by adding rigidifying data, is itself constructed later; the naive version here fails at the first step, the sheaf condition.

The example isolates the difference between a coarse and a fine moduli space in a single computation. The set $\mathcal{F}(S)$ discards the automorphisms of its objects by passing to isomorphism classes, and that quotient is exactly what destroys the sheaf condition: gluing data differing by transition automorphisms become indistinguishable when restricted to the charts, yet build different global objects. No scheme's functor of points has this defect, by proposition 13.5.4, which is why $\mathcal{F}$ cannot be representable no matter how the geometry is arranged. The tangent-space tool and the gluing criterion together frame the two directions the book now takes. The deformation theory to come linearizes the moduli problem at a point, computing the tangent space of definition 13.5.1 as a cohomology group and asking when deformations are unobstructed; the theory of algebraic stacks that follows it enlarges the target category so that functors like the one above become representable objects, algebraic stacks, whose functor of points is a groupoid keeping track of the automorphisms this example was unable to forget.

Exercise 13.5.1

1. Let $\mathcal{F} = h_{\mathbb{A}^1}$ be the functor represented by the affine line over k, so $\mathcal{F}(A) = A$ for a k-algebra A. Compute $\mathcal{F}(k[\varepsilon])$, the reduction map $\pi\colon \mathcal{F}(k[\varepsilon]) \to \mathcal{F}(k)$, and the tangent space $T_x\mathcal{F}$ at a point $x \in \mathbb{A}^1(k) = k$. Verify directly, using the doubled dual numbers $k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2$, that the addition of proposition 13.5.2 recovers ordinary addition on k, so $T_x\mathcal{F} \cong k$.

2. Let $\mathcal{F} = h_G$ for an affine algebraic group G over k, and take $x = e$ the identity. Show that $T_e\mathcal{F}$ with the vector-space structure of proposition 13.5.2 is the Lie algebra $\mathrm{Lie}(G)$ as a k-vector space. (Hint: a $k[\varepsilon]$-point reducing to e is an element of $G(k[\varepsilon])$ mapping to e under $\varepsilon \mapsto 0$; identify these with left-invariant derivations of $\mathcal{O}_{G,e}$.)

3. Prove that a filtered colimit of representable functors need not be representable, but that every open subfunctor of a representable functor h_X is of the form h_U for a unique open subscheme $U \subseteq X$. Deduce that in theorem 13.5.6 the charts X_i may be taken to be affine without loss of generality.

4. Fix a scheme X over k and define the *relative Picard functor* $\mathrm{Pic}_{X/k}\colon \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ by $\mathrm{Pic}_{X/k}(A) = \ker\big(\mathrm{Pic}\,(X \times_k \mathrm{Spec}\,A) \to \mathrm{Pic}\,(X)\big)$, the line bundles on $X_A := X \times_k \mathrm{Spec}\,A$ trivial along the reduction $X \hookrightarrow X_A$, up to isomorphism (this is a k-pointed functor: $\mathrm{Pic}_{X/k}(k) = \{[\mathcal{O}_X]\}$ is a single point, the class of the trivial bundle). Compute its tangent space $T_{[\mathcal{O}_X]}\mathrm{Pic}_{X/k}$ at that point using definition 13.5.1. Show that a $k[\varepsilon]$-point of $\mathrm{Pic}_{X/k}$ is a line bundle on $X[\varepsilon] = X \times_k \mathrm{Spec}\,k[\varepsilon]$ restricting to $\mathcal{O}_X$, and identify the resulting set with $H^1(X, \mathcal{O}_X)$. (Hint: line bundles are classified by H^1 of the units sheaf, and $1 + \varepsilon\mathcal{O}_X$ is a subgroup of $\mathcal{O}_{X[\varepsilon]}^\times$ isomorphic to the additive group $\mathcal{O}_X$.) Conclude that even though the naive functor $\mathcal{F} = \mathrm{Pic}$ of example 13.5.7 is not representable, the associated relative functor has a well-defined tangent space, a k-vector space, illustrating proposition 13.5.2.

5. Let $\mathcal{F}$ be a Zariski sheaf admitting a cover by two representable open subfunctors $\mathcal{G}_0 \cong h_{X_0}$, $\mathcal{G}_1 \cong h_{X_1}$ with $\mathcal{G}_0 \cap \mathcal{G}_1 \cong h_{X_{01}}$. Write out the scheme X produced by theorem 13.5.6 explicitly when $X_0 = X_1 = \mathbb{A}^1$ and $X_{01} = \mathbb{A}^1 \setminus \{0\} = \mathrm{Spec}\,k[t, t^{-1}]$, for the two gluings $t \mapsto t$ and $t \mapsto t^{-1}$. Identify the two resulting schemes and explain, via the sheaf condition, why one of them fails to be separated.

6. Let $n \geq 2$ be invertible in k and consider the presheaf $\mathcal{F}(S) = \mathcal{O}(S)^\times/(\mathcal{O}(S)^\times)^n$, global units modulo n-th powers. Show that $\mathcal{F}$ fails the fppf sheaf condition already over the single base $S = \mathrm{Spec}\,k$: for a suitable k, exhibit a nontrivial fppf cover $S' \to S$ on which a class becomes trivial though it is nonzero over S. Explain why the Zariski sheaf condition over $\mathrm{Spec}\,k$ imposes no such constraint (a point has no nontrivial Zariski cover), so $\mathcal{F}$ passes the Zariski test at this base while failing the fppf test. Conclude that the fppf half of proposition 13.5.4 is

stronger than its Zariski half and rests on descent rather than on open gluing.

14

The Hilbert and Quot Schemes

The previous chapter reduced a moduli problem to a functor and asked when that functor is representable; the Grassmannian was the model case in which the answer is yes. This chapter produces the two representable functors from which most moduli spaces in algebraic geometry are cut out: the Quot scheme, parametrizing quotients of a fixed coherent sheaf, and its special case the Hilbert scheme, parametrizing closed subschemes of a fixed projective scheme with a fixed Hilbert polynomial. Grothendieck's existence theorem, that these functors are representable by projective schemes, is the foundation on which Hilbert schemes, moduli of sheaves, and the quotient constructions of the next chapter all rest.

The engine is flatness, the algebraic form of a family varying continuously, and the Hilbert polynomial is the discrete invariant it holds constant. section 14.1 sets up the Quot and Hilbert functors on this basis; section 14.2 controls which sheaves can occur, through Castelnuovo–Mumford regularity, boundedness, and the flattening stratification; and section 14.3 assembles these into the representability theorem, realizing Quot and Hilb as closed subschemes of a Grassmannian. section 14.4 reads off the local structure, identifying the tangent space to the Hilbert scheme with the global sections of the normal sheaf; this is the first computation of deformation theory and the bridge to the next Part. section 14.5 closes with worked schemes: the Hilbert scheme of points, hypersurfaces, and the Grassmannian recovered as a Quot scheme.

14.1 Flat Families and Hilbert Polynomials

Before any Hilbert or Quot scheme can be built, the moduli problem it represents has to be pinned down, and that requires answering a question the first chapter deferred: what, precisely, is a family of subschemes, or of quotient sheaves? A family should be a collection of objects varying *continuously* over a base S, but "continuously" has to be given algebraic content, since over a scheme there is no metric to appeal to. The content is flatness. A family that is flat over S is one whose members do not jump in a way that changes their discrete invariants as the parameter moves, and the discrete invariant that flatness holds fixed is the Hilbert polynomial.

This section fixes the two notions the whole chapter runs on. It recalls the Hilbert polynomial of a coherent sheaf on projective space, proves that in a flat family the fiberwise Hilbert polynomial is locally constant on the base, and then uses that fact to define the Quot and Hilbert functors, the moduli functors whose representability is the theorem of section 14.3. Flatness is the reason those functors are sheaves in the first place, and the fixed Hilbert polynomial P is the label that carves the boundless functor of all quotients into the bounded, representable pieces $\underline{\mathrm{Quot}}^P$ that individual moduli spaces attach to. The chapter's guiding image, a flat family versus a non-flat one in which the Hilbert polynomial jumps, appears at the end of the section and makes the flatness condition tangible.

The first of the two notions is the Hilbert polynomial, a single element of $\mathbb{Q}[t]$ that packages the numerical invariants of a coherent sheaf on projective space. Fix once and for all a projective scheme $Y \subseteq \mathbb{P}^n_k$ over the base field k, with $\mathcal{O}_Y(1)$ the restriction of the twisting sheaf $\mathcal{O}_{\mathbb{P}^n}(1)$. For a coherent sheaf $\mathcal{F}$ on Y and an integer m, write $\mathcal{F}(m) = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_Y(m)$ for the m-th Serre twist. The one piece of upstream machinery needed here is that the cohomology of a coherent sheaf on a projective scheme is finite dimensional over k and vanishes in high degree after twisting; this is Serre's finiteness and vanishing theorem from [7], which the reader may take as given.

The Euler characteristic packages the cohomology of $\mathcal{F}$ into a single integer.

Definition 14.1.1: Hilbert Polynomial of a Coherent Sheaf

Let $Y \subseteq \mathbb{P}^n_k$ be projective and $\mathcal{F}$ a coherent sheaf on Y. Recall the *Euler characteristic*

$$\chi(\mathcal{F}) = \sum_{i \geq 0} (-1)^i \dim_k H^i(Y, \mathcal{F})$$

a finite sum, since $H^i(Y, \mathcal{F}) = 0$ for $i > \dim Y$ and each term is finite dimensional (section 6.7). The *Hilbert polynomial* of $\mathcal{F}$ is the function of the twisting parameter m given by

$$P_{\mathcal{F}}(m) = \chi\big(\mathcal{F}(m)\big)$$

which agrees with a polynomial in m of degree $\dim \mathrm{Supp}\mathcal{F}$ (theorem 6.7.5); for $\mathcal{F} = \mathcal{O}_Y$ it recovers the classical Hilbert polynomial of the homogeneous coordinate ring of Y (proposition 6.7.8).

The name "polynomial" is justified by a theorem of Snapper and Grothendieck: $P_{\mathcal{F}}$ agrees, for all integers m, with a polynomial in m with rational coefficients, of degree equal to $\dim \mathrm{Supp}\mathcal{F}$. This is proved in full in [7]; the argument is an induction on the dimension of the support using a hyperplane section, and it is the reason the Hilbert polynomial is an object of $\mathbb{Q}[t]$ rather than only a numerical function. Two features make $P_{\mathcal{F}}$ the right invariant for moduli. First, it is computed from cohomology, so it is a deformation invariant rather than a set-theoretic count. Second, for $m \gg 0$ the higher cohomology $H^i(Y, \mathcal{F}(m))$ vanishes for $i > 0$, so $P_{\mathcal{F}}(m) = \dim_k H^0(Y, \mathcal{F}(m))$ counts the twisted global sections, and this ties the polynomial to the classical Hilbert function of a graded module. The leading coefficients encode the geometry: for a subscheme the degree of P is the dimension, and the top coefficient is $(\deg Z)/(\dim Z)!$, so it records the degree of the subscheme in $\mathbb{P}^n$ up to the factorial normalization (the two coincide outright when $\dim Z = 1$, as in the plane-curve example below).

Example 14.1.2: The Hilbert Polynomial of a Plane Curve

Let $C \subseteq \mathbb{P}^2_k$ be a curve of degree d, the zero scheme of a single homogeneous polynomial of degree d, and let $\mathcal{O}_C$ be its structure sheaf. The Hilbert polynomial of $\mathcal{O}_C$ is computed from the ideal-sheaf sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^2} \longrightarrow \mathcal{O}_C \longrightarrow 0$$

the ideal sheaf $\mathcal{I}_C = \mathcal{O}_{\mathbb{P}^2}(-d)$ being generated by the degree-d equation. Twisting this exact sequence by $\mathcal{O}(m)$ gives $0 \to \mathcal{O}_{\mathbb{P}^2}(m-d) \to \mathcal{O}_{\mathbb{P}^2}(m) \to \mathcal{O}_C(m) \to 0$, and Euler characteristic is additive on short exact sequences, so

$$\chi(\mathcal{O}_C(m)) = \chi(\mathcal{O}_{\mathbb{P}^2}(m)) - \chi(\mathcal{O}_{\mathbb{P}^2}(m-d))$$

Using $\chi(\mathcal{O}_{\mathbb{P}^2}(m)) = \binom{m+2}{2}$, this gives

$$P_{\mathcal{O}_C}(m) = \binom{m+2}{2} - \binom{m-d+2}{2} = d\,m + \frac{-d^2+3d}{2} = d\,m + 1 - \frac{(d-1)(d-2)}{2}$$

The polynomial has degree 1, recording that C is a curve; its leading coefficient d is the degree of C; and its constant term $1-g$ with $g = \binom{d-1}{2}$ is 1 minus the arithmetic genus, the genus-degree formula read off the Hilbert polynomial. For a conic $d=2$ this reads $P(m) = 2m+1$, and for a cubic $d=3$ it reads $P(m) = 3m$, so that $\chi(\mathcal{O}_C) = P(0)$ equals 1 and 0 respectively, matching $g=0$ and $g=1$.

The example already shows the polynomial doing its job as a bookkeeper: a single element of $\mathbb{Q}[t]$ simultaneously records dimension, degree, and genus. In a moduli problem these are exactly the discrete data one wants held fixed while the geometry deforms, and the next result is what guarantees they *are* held fixed.

With the Hilbert polynomial in hand, the second notion is flatness, and the result binding the two together is that flatness holds the Hilbert polynomial constant along a family. A family of subschemes or sheaves parametrized by S is a single sheaf on $Y \times_k S$, whose restriction to the fiber $Y_s = Y \times_k \kappa(s)$ over a point $s \in S$ is the member of the family sitting over s. For the family to deserve the name, the members should vary continuously, and the algebraic form of continuity is that the sheaf on the total space $Y \times S$ be *flat* over S. Flatness is developed in full in [7]; the following refresher records the one statement this chapter leans on.

Definition 14.1.3: Flat Family over a Base

Let Y be a scheme over k, S a k-scheme, and $p\colon Y \times_k S \to S$ the projection. A coherent sheaf $\mathcal{G}$ on $Y \times_k S$ is a *flat family over* S if $\mathcal{G}$ is flat over S, meaning that for every point of $Y \times_k S$ the stalk $\mathcal{G}_x$ is a flat module over the local ring $\mathcal{O}_{S,p(x)}$. For $s \in S$ the *fiber* of the family is the coherent sheaf

$$\mathcal{G}_s = \mathcal{G} \otimes_{\mathcal{O}_S} \kappa(s)$$

on $Y_s = Y \times_k \kappa(s)$, the restriction of $\mathcal{G}$ to the fiber over s.

Flatness is not visible on any single fiber; it is a condition on how the fibers fit together as s moves. The archetype to keep in mind is that a family of divisors on $\mathbb{P}^1$ is flat exactly when the number of points, counted with multiplicity, is constant, whereas allowing a point to appear or disappear as the parameter passes through a special value destroys flatness. This is the discrete-invariant discipline flatness enforces, and the following proposition is its precise cohomological expression: the entire Hilbert polynomial, not just one of its coefficients, is locally constant along a flat family.

The proof rests on a single import, the theorem on cohomology and base change, which controls how the cohomology of a family interacts with restriction to a fiber. In the form needed here it says: for a flat family $\mathcal{G}$ over S that is proper over S, and for each degree i, there is a coherent sheaf on S computing H^i of the fibers, and after a Serre twist the higher cohomology of the fibers vanishes uniformly, so that the zeroth cohomology becomes locally free of the constant rank one expects. The statement and proof are in [7]; it is used in every constancy result in this chapter.

Proposition 14.1.4: Local Constancy of the Hilbert Polynomial

Let $Y \subseteq \mathbb{P}^n_k$ be projective, S a Noetherian k-scheme, and $\mathcal{G}$ a flat family over S of coherent sheaves on Y, that is, a coherent sheaf on $Y \times_k S$ flat over S with proper support over S. For $s \in S$ let $P_s \in \mathbb{Q}[t]$ be the Hilbert polynomial of the fiber $\mathcal{G}_s$ on Y_s, computed with respect to $\mathcal{O}_{Y_s}(1)$. Then the map

$$s \longmapsto P_s$$

is locally constant on S: every point of S has an open neighborhood on which P_s is a fixed polynomial. In particular, if S is connected, the fiberwise Hilbert polynomial is the same for every $s \in S$.

***Proof*:**
The claim is local on S, so it suffices to prove that the function $s \mapsto P_s(m)$ is locally constant on S for each fixed integer m, since a polynomial in $\mathbb{Q}[t]$ of bounded degree is determined by its values at enough integers and the local constancy of each value will then force local constancy of the polynomial. Fix m.

Step 1: Reduce to the vanishing of higher cohomology. Let $p\colon Y \times_k S \to S$ be the projection and $q\colon Y \times_k S \to Y$ the other one, so that $\mathcal{G}(m) := \mathcal{G} \otimes q^*\mathcal{O}_Y(m)$ is the m-twisted family, again flat over S with proper support, and its fiber over s is $\mathcal{G}_s(m)$. By definition

$$P_s(m) = \chi(\mathcal{G}_s(m)) = \sum_{i \geq 0} (-1)^i \dim_{\kappa(s)} H^i(Y_s, \mathcal{G}_s(m))$$

The strategy is to show each individual term $s \mapsto \dim_{\kappa(s)} H^i(Y_s, \mathcal{G}_s(m))$ is locally constant. This is false term by term for arbitrary m, since cohomology can jump on special fibers, but the alternating sum is stable, and the cleanest route to the alternating sum is to twist so hard that all but the zeroth term vanish and then descend.

Step 2: The Euler characteristic is locally constant. By the theorem on cohomology and base change ([7]) applied to the flat proper family $\mathcal{G}(m)$ over S, there is, locally on S, a bounded complex

$$K^\bullet\colon \quad 0 \to K^0 \to K^1 \to \cdots \to K^N \to 0$$

of finite locally free $\mathcal{O}_S$-modules whose cohomology computes the cohomology of the family universally, in the sense that for every base change $T \to S$ the cohomology of $K^\bullet \otimes_{\mathcal{O}_S} \mathcal{O}_T$ computes $H^\bullet$ of the pulled-back family. Specializing $T = \operatorname{Spec} \kappa(s)$, the complex $K^\bullet \otimes \kappa(s)$ computes $H^\bullet(Y_s, \mathcal{G}_s(m))$. Now the Euler characteristic of a bounded complex of finite-dimensional vector spaces equals the Euler characteristic of its cohomology, so

$$\sum_i (-1)^i \dim_{\kappa(s)} H^i(Y_s, \mathcal{G}_s(m)) = \sum_i (-1)^i \dim_{\kappa(s)} \left(K^i \otimes \kappa(s)\right)$$

Each K^i is locally free of some rank r_i over $\mathcal{O}_S$, so $\dim_{\kappa(s)}(K^i \otimes \kappa(s)) = r_i$ is constant on the open where $K^\bullet$ is defined and K^i has that rank. Hence

$$P_s(m) = \sum_i (-1)^i r_i$$

is constant on that open neighborhood of the chosen point. This is where flatness enters: without it the base-change complex $K^\bullet$ of finite locally free modules need not exist, and the Euler characteristic could jump.

Step 3: From one twist to the polynomial. Step 2 gives local constancy of $s \mapsto P_s(m)$ for the fixed m. Running the argument for every m shows that each coefficient of the polynomial, being a fixed $\mathbb{Q}$-linear combination of finitely many values $P_s(m)$, is locally constant, so the polynomial P_s itself is locally constant on S. On a connected S a locally constant function is constant, giving the final assertion, as we sought to show.

The proposition is the technical fact that makes the Hilbert polynomial a usable moduli label. It says that flatness and the Hilbert polynomial are two sides of one coin: a flat family cannot change its members' Hilbert polynomials, and conversely the Hilbert polynomial is exactly the invariant whose constancy flatness certifies. This is why the moduli functors below are indexed by a fixed polynomial P. Over a connected base every member of a flat family carries the same P, so demanding a fixed P is the same as asking for the connected component of the space of all flat families, and it is on such a component that a moduli space has any hope of being of finite type. The converse direction, that a family with locally constant Hilbert polynomial over a reduced base is automatically flat, is a standard criterion proved in [7]; it will be used silently when verifying flatness of explicit families.

With flatness and the Hilbert polynomial in hand, the moduli problems this chapter solves can be written down. The primary one classifies quotient sheaves of a fixed coherent sheaf. Fix a coherent sheaf $\mathcal{F}$ on the projective scheme Y; think of $\mathcal{F}$ as an ambient sheaf whose quotients are to be parametrized. A family of quotients over a base S is a quotient of the pullback of $\mathcal{F}$ to $Y \times S$, and the family is admissible when it is flat over S, so that its Hilbert polynomial is constant, and when that polynomial is a prescribed P.

For a k-scheme S write $Y_S = Y \times_k S$, with projection $p_S \colon Y_S \to Y$, and $\mathcal{F}_S = p_S^*\mathcal{F}$ for the pullback of $\mathcal{F}$ to Y_S. A morphism $g \colon T \to S$ induces $\mathrm{id}_Y \times g \colon Y_T \to Y_S$, and pulling back along it carries data over Y_S to data over Y_T; this is the mechanism of contravariance.

Definition 14.1.5: The Quot Functor

Let $Y \subseteq \mathbb{P}^n_k$ be projective, $\mathcal{F}$ a coherent sheaf on Y, and $P \in \mathbb{Q}[t]$ a polynomial. The *Quot functor*

$$\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} : \mathbf{Sch}_k^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

assigns to a k-scheme S the set of isomorphism classes of surjections of coherent sheaves on $Y_S = Y \times_k S$

$$\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$$

such that $\mathcal{Q}$ is flat over S and each fiber $\mathcal{Q}_s$ has Hilbert polynomial P on Y_s. Two quotients $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ and $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}'$ are identified when there is an isomorphism $\mathcal{Q} \cong \mathcal{Q}'$ commuting with the surjections from $\mathcal{F}_S$, that is, having the same kernel subsheaf of $\mathcal{F}_S$. To a morphism $g\colon T \to S$ the functor assigns the pullback

$$g^* : \underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}(S) \longrightarrow \underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}(T), \qquad \big(\mathcal{F}_S \twoheadrightarrow \mathcal{Q}\big) \longmapsto \big(\mathcal{F}_T \twoheadrightarrow (\mathrm{id}_Y \times g)^*\mathcal{Q}\big)$$

the surjection obtained by pulling $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ back along $\mathrm{id}_Y \times g$.

Two points make this a well-posed moduli functor rather than a formal gesture. First, pullback preserves the defining conditions: pulling back a surjection along $\mathrm{id}_Y \times g$ keeps it surjective, because $\otimes$ is right exact, and it preserves flatness over the base and hence the fiberwise Hilbert polynomial, since the fibers of the pulled-back family over points of T are the fibers of the original family over their images in S (proposition 14.1.4 guarantees the polynomial is the same P). So g^* lands in $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}(T)$, and functoriality $(g \circ h)^* = h^* \circ g^*$ is the standard compatibility of pullbacks. Second, quotients are recorded up to the coarsest sensible equivalence: identifying $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ with its isomorphic siblings is the same as recording only the kernel subsheaf $\mathcal{K} = \ker(\mathcal{F}_S \to \mathcal{Q}) \subseteq \mathcal{F}_S$, so a $\underline{\mathrm{Quot}}$-point is equally a flat family of *subsheaves* of $\mathcal{F}$ with the complementary Hilbert polynomial. The functor thus captures both quotients and sub-objects at once, which is why it subsumes so many moduli problems.

The single most important special case is the one where $\mathcal{F}$ is the structure sheaf of Y. A surjection $\mathcal{O}_{Y_S} \twoheadrightarrow \mathcal{Q}$ has for its kernel an ideal sheaf, so its quotient is the structure sheaf of a closed subscheme of Y_S; flatness over S makes this a flat family of closed subschemes. This is the moduli problem of closed subschemes with which the first chapter opened, now made precise.

Definition 14.1.6: The Hilbert Functor

Let $Y \subseteq \mathbb{P}^n_k$ be projective and $P \in \mathbb{Q}[t]$. The *Hilbert functor*

$$\underline{\mathrm{Hilb}}^P_Y : \mathbf{Sch}_k^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

is the special case $\mathcal{F} = \mathcal{O}_Y$ of the Quot functor. It assigns to a k-scheme S the set of closed subschemes $Z \subseteq Y_S = Y \times_k S$ that are flat over S and whose fibers $Z_s \subseteq Y_s$ have Hilbert polynomial P; equivalently, the set of surjections $\mathcal{O}_{Y_S} \twoheadrightarrow \mathcal{O}_Z$ that are flat over S with fiberwise Hilbert polynomial P. To $g : T \to S$ it assigns the pullback

$$Z \longmapsto (\mathrm{id}_Y \times g)^{-1}(Z) = Z \times_S T$$

the scheme-theoretic preimage, which is the closed subscheme of Y_T cut out by the pulled-back ideal sheaf.

The Hilbert functor is the moduli of closed subschemes of Y with fixed numerical type P: dimension, degree, and, for curves, genus, all read off P as in example 14.1.2. A $\underline{\mathrm{Hilb}}^P_Y$-point over S is a subscheme $Z \subseteq Y \times S$ whose fiber over each $s \in S$ is a subscheme of Y with the prescribed invariants, varying flatly with s. When $S = \operatorname{Spec} k$ this is a single closed subscheme $Z \subseteq Y$ with Hilbert polynomial P, so the k-points of the eventual Hilbert scheme are exactly the subschemes one wants to parametrize; the content of the next sections is that these k-points assemble into an actual projective scheme, with the flat families over general S recorded as its S-points. That this functor decomposes the "all subschemes" functor by discrete type is the reason a fixed P is imposed: the functor of all closed subschemes of Y, flat over the base, is the disjoint union $\coprod_P \underline{\mathrm{Hilb}}^P_Y$ over all Hilbert polynomials, and each piece is representable while the whole is only a countable coproduct.

Example 14.1.7: The Hilbert Functor of Points on the Line

Take $Y = \mathbb{P}^1_k$ and the constant polynomial $P = d$, a positive integer. A subscheme $Z \subseteq \mathbb{P}^1$ with Hilbert polynomial the constant d is a zero-dimensional subscheme of length d: since P has degree 0, $\dim Z = 0$, and $\chi(\mathcal{O}_Z(m)) = \dim_k H^0(Z, \mathcal{O}_Z) = d$ for all m because Z is affine, so Z is d points counted with multiplicity. Over S, a $\underline{\mathrm{Hilb}}^d_{\mathbb{P}^1}$-point is a closed subscheme $Z \subseteq \mathbb{P}^1 \times S$, flat over S, whose every fiber is length d. Working in the affine chart $\mathbb{A}^1 = \operatorname{Spec} k[x]$, such a Z over an affine base $S = \operatorname{Spec} A$ is cut out by a single monic polynomial

$$f(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0 \in A[x]$$

with $a_i \in A$, since flatness of $A[x]/(f)$ over A forces the generator to be monic of degree d so that $A[x]/(f)$ is free of rank d over A. The coefficients $(a_0, \ldots, a_{d-1})$ are d free parameters, so on this chart the functor is represented by affine space $\mathbb{A}^d = \operatorname{Spec} k[a_0, \ldots, a_{d-1}]$: the Hilbert scheme of d points on the line is $\mathbb{A}^d$ over the affine chart, the space of monic degree-d polynomials, and globally $\mathrm{Hilb}^d_{\mathbb{P}^1} \cong \mathbb{P}^d$, the space of binary forms of degree d up to scalar. A fiber over S jumps from d distinct points to a fat point exactly as the discriminant of f vanishes, and flatness is what keeps the length pinned at d throughout.

The example exhibits the entire program in miniature: a moduli functor of flat families with fixed Hilbert polynomial turns out to be represented by an explicit scheme, here $\mathbb{P}^d$, whose points are the objects being classified. The general theorem of section 14.3 promotes this from $\mathbb{P}^1$ and points to arbitrary Y and arbitrary P, and the intervening section 14.2 supplies the boundedness that makes the Grassmannian into which everything embeds finite dimensional. For the broader role of Hilbert schemes in moduli theory, see section 3.3.6.

The abstractions above are worth grounding in a case where flatness visibly succeeds and fails, so that the reader can see the Hilbert polynomial jump. The cleanest such case is a plane cubic acquiring an extra embedded point as a parameter degenerates.

Example 14.1.8: A Non-Flat Degeneration

Work over $S = \mathbb{A}^1_k = \operatorname{Spec} k[t]$ and inside $Y = \mathbb{P}^2_k$ with homogeneous coordinates $[x_0 : x_1 : x_2]$; pass to the affine chart $\mathbb{A}^2 = \operatorname{Spec} k[x,y]$ with $x = x_1/x_0$, $y = x_2/x_0$ to do the algebra. Consider the family of subschemes $Z_t \subseteq \mathbb{A}^2$ defined, for $t \in \mathbb{A}^1$, by the ideal

$$I_t = (xy,\ tx)$$

of $k[x,y]$. The total space is $Z \subseteq \mathbb{A}^2 \times \mathbb{A}^1 = \operatorname{Spec} k[x,y,t]$ cut out by $I = (xy, tx) \subseteq k[x,y,t]$, and the family is the projection to $S = \operatorname{Spec} k[t]$.

The general fiber. For $t \neq 0$ the element tx is a unit multiple of x, so $I_t = (xy, x) = (x)$, and $Z_t = \{x = 0\}$ is the line $x = 0$, a copy of $\mathbb{A}^1$ with coordinate y. Compactifying in $\mathbb{P}^2$, the closure is a line, with Hilbert polynomial $P_t(m) = m + 1$: degree 1, dimension 1, and $\chi(\mathcal{O}_{Z_t}) = 1$.

The special fiber. For $t = 0$ the ideal is $I_0 = (xy)$, so $Z_0 = \{xy = 0\}$ is the union of the two coordinate lines $x = 0$ and $y = 0$, the degree-2 plane curve cut out by the single degree-2 form xy. By example 14.1.2 with $d = 2$ its Hilbert polynomial on the projective closure is $P_0(m) = 2m + 1$: degree 2, dimension 1, and $\chi(\mathcal{O}_{Z_0}) = P_0(0) = 1$, so arithmetic genus $p_a = 1 - \chi = 0$. The Hilbert polynomial has jumped from $m + 1$ to the degree-2 polynomial $2m + 1$.

By proposition 14.1.4 the family $Z \to S$ cannot be flat, since $\mathbb{A}^1$ is connected yet the fiberwise Hilbert polynomial is not constant. The failure is visible in the algebra: in $k[x,y,t]$ the element x is annihilated modulo I by both y and t, so $\operatorname{Ann}(x) = (y,t) \bmod I$. Since $I = (xy, tx) = (x) \cap (y,t)$, the associated primes of $k[x,y,t]/I$ are (x) and (y,t), and the second one, (y,t), cuts out the horizontal line $\{y = 0, t = 0\}$ whose image in S is the single point $t = 0$. A component of the total space maps to a proper closed subset of the base, which is precisely the obstruction to flatness: the fiber over $t = 0$ acquires the extra line $y = 0$ that no nearby fiber sees. Concretely, the length of $\mathcal{O}_Z$ localized along $\{t = 0\}$ is not the flat limit of the nearby fibers; a whole extra line materializes at $t = 0$, and the Euler characteristic records it.

The flat repair. The flat limit of the lines $\{x = 0\}$ as $t \to 0$ is just the line $\{x = 0\}$ again: the flat family extending $Z_t = \{x = 0\}$ for $t \neq 0$ over the whole base is the constant family $\{x = 0\} \times \mathbb{A}^1$, cut out by the flat ideal $(x) \subseteq k[x,y,t]$. This constant family has $P_t(m) = m + 1$ for every t, in accordance with proposition 14.1.4, and it is the unique flat family agreeing with Z_t away from $t = 0$. The non-flat Z above differs from it only in the embedded jump at the origin, and the flat limit systematically discards exactly such jumps. This is the sense in which flatness selects the geometrically correct limit: it forbids the fiber from gaining or losing pieces as the parameter specializes.

The two families in the example are the picture to carry through the chapter. A flat family is one whose Hilbert polynomial is a constant of the motion; a non-flat family is one in which the polynomial jumps at special parameters, and the flat families are recovered by taking flat limits that suppress the jumps. The Hilbert and Quot functors are built from flat families precisely so that each of their S-points has a well-defined, constant Hilbert polynomial, and the fixed P in $\underline{\mathrm{Hilb}}^P_Y$ and $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$

is the value of that constant. The task of the rest of the chapter is to show that these functors, so constrained, are representable, and the first step is to prove that the flat families with a given P form a *bounded* collection, which is the subject of section 14.2.

Exercise 14.1.1

1. Let $S \subseteq \mathbb{P}^3_k$ be a smooth surface of degree d, the zero scheme of one homogeneous polynomial of degree d. Using the ideal-sheaf sequence $0 \to \mathcal{O}_{\mathbb{P}^3}(-d) \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_S \to 0$ and the values $\chi(\mathcal{O}_{\mathbb{P}^3}(m)) = \binom{m+3}{3}$, compute the Hilbert polynomial $P_{\mathcal{O}_S}(m)$. Read off from it the dimension and the degree of S, and evaluate $P_{\mathcal{O}_S}(0) = \chi(\mathcal{O}_S)$ for $d = 4$ (a quartic K3 surface), confirming that $\chi(\mathcal{O}_S) = 2$.

2. Consider the family over $S = \operatorname{Spec} k[t]$ given in $\mathbb{A}^2 = \operatorname{Spec} k[x, y]$ by the ideal $I_t = (y - tx^2)$ as t varies, a family of parabolas degenerating to the line $y = 0$. Show that the total-space ideal $(y - tx^2) \subseteq k[x, y, t]$ presents a family that *is* flat over $\operatorname{Spec} k[t]$, by exhibiting $k[x, y, t]/(y - tx^2)$ as a free $k[t]$-module, and confirm that every fiber is a copy of $\mathbb{A}^1$ over its residue field. Contrast this with example 14.1.8: why does this affine degeneration stay flat while that one does not? (Caution: taking the projective closure of each affine fiber separately does *not* give a flat family, since the smooth conic $\{yz = tx^2\}$ for $t \neq 0$ acquires an embedded point at infinity in its flat limit; the flatness here is a statement about the affine total space over $\operatorname{Spec} k[t]$.)

3. Let $\mathcal{G}$ be a flat family over a connected base S with fiberwise Hilbert polynomial P. Suppose $\deg P = 0$, so P is a constant d. Prove that every fiber $\mathcal{G}_s$ is a zero-dimensional sheaf of length d, and deduce that the function $s \mapsto \dim_{\kappa(s)} H^0(Y_s, \mathcal{G}_s)$ is constant equal to d (you may use that for a zero-dimensional sheaf all higher cohomology vanishes). Where in the argument is flatness used?

4. Fix $Y = \operatorname{Spec} k$ (the case $Y = \mathbb{P}^0_k$, a single point) and $\mathcal{F} = \mathcal{O}_Y^{\oplus r} = k^{\oplus r}$. Describe the S-points of $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ for $S = \operatorname{Spec} k$ in the case P is the constant $r' \leq r$. On a point every coherent sheaf is locally free, so Hilbert polynomial r' means a rank-r' quotient. Show that such a $\operatorname{Spec} k$-point is exactly a rank-r' quotient of $k^{\oplus r}$, and explain in one sentence how this recovers the Grassmannian $\mathrm{Gr}(r', r)$ of section 13.2 as $\underline{\mathrm{Quot}}$ over the point. (The full identification, including the passage from a point to $Y = \mathbb{P}^n$ where the analogous locally free quotients of $\mathcal{O}_{\mathbb{P}^n}^{\oplus r}$ have Hilbert polynomial $r'\binom{m+n}{n}$, is example 14.5.4.)

5. Prove that the Quot functor is a functor, that is, verify the two claims left implicit in definition 14.1.5: that pulling a surjection $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ back along $\mathrm{id}_Y \times g$ for $g\colon T \to S$ yields a surjection with $\mathcal{Q}$-pullback flat over T and fiberwise Hilbert polynomial still P; and that $(g \circ h)^* = h^* \circ g^*$. You may cite proposition 14.1.4 for the Hilbert-polynomial claim and the right-exactness of $\otimes$ for surjectivity.

6. Let $Y = \mathbb{P}^1_k$ and $P = d$ as in example 14.1.7. Give the flat family over $S = \mathbb{A}^1 = \operatorname{Spec} k[t]$ whose fiber over $t \neq 0$ is the d reduced points $\{0, 1, 2, \ldots, d-1\}$ scaled by t, namely $\{0, t, 2t, \ldots, (d-1)t\}$, and whose fiber over $t = 0$ is a single fat point of length d at the origin. Write the monic polynomial $f_t(x) \in k[t][x]$ cutting it out, verify flatness, and identify the limiting ideal at $t = 0$ as (x^d).

14.2 Boundedness and the Flattening Stratification

The Quot and Hilbert functors of section 14.1 package an entire universe of coherent quotients into a single functor of points, and the whole point of the chapter is to show that this functor is representable by a scheme. Representability by a scheme is a finiteness statement: a scheme locally of finite type is cut out, chart by chart, from affine space by finitely many equations, so its functor of points cannot see more than a bounded amount of data at once. The Quot functor, by contrast, is defined with no a priori bound at all. It ranges over *every* coherent quotient of $\mathcal{F}$ with a fixed Hilbert polynomial P, across every base scheme S, and nothing in the definition tells us that these quotients form anything like a set of bounded complexity. Before a construction can even begin, this must be brought under control.

The controlling principle is *boundedness*: the sheaves that can occur as fibers of a flat family with Hilbert polynomial P do not form an unlimited zoo, but a family that a single finite-dimensional object can capture. Concretely, all of them become generated by their global sections, with vanishing higher cohomology, after twisting by one and the same power $\mathcal{O}(m)$ of the hyperplane bundle, once m is large enough (depending only on P). Fixing such an m replaces each quotient by a linear-algebra datum, a quotient of the fixed vector space $H^0(\mathcal{F}(m))$, and this is exactly the datum a Grassmannian parametrizes. Boundedness is what lets a single Grassmannian catch every quotient with a given Hilbert polynomial simultaneously; this section proves it. The measure of "large enough" is Castelnuovo-Mumford regularity, developed first, and the same circle of ideas produces Grothendieck's flattening stratification, the tool that later carves the locus of flat quotients out of the Grassmannian as a locally closed subscheme.

Throughout, $\mathbb{P}^n = \mathbb{P}^n_k$ over a fixed base field k, and $\mathcal{O}(1)$ is the twisting sheaf. For a coherent sheaf $\mathcal{F}$ on $\mathbb{P}^n$ and $m \in \mathbb{Z}$ write $\mathcal{F}(m) = \mathcal{F} \otimes \mathcal{O}(m)$, and $H^i(\mathcal{F}) = H^i(\mathbb{P}^n, \mathcal{F})$ with $h^i(\mathcal{F}) = \dim_k H^i(\mathcal{F})$.

Regularity, the numerical invariant that measures "large enough", is where the section begins. Mumford isolated a single number of a coherent sheaf that simultaneously controls global generation, the vanishing of higher cohomology, and the surjectivity of multiplication maps on global sections. The definition looks asymmetric, tracking $H^i(\mathcal{F}(m-i))$ rather than $H^i(\mathcal{F}(m))$, but this shift is precisely what makes regularity propagate upward: if a sheaf is m-regular, it is m'-regular for every $m' \geq m$, so regularity has a well-defined threshold. That threshold is the numerical handle boundedness needs.

Definition 14.2.1: Castelnuovo-Mumford Regularity

Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n_k$ and let $m \in \mathbb{Z}$. The sheaf $\mathcal{F}$ is *m-regular* (in the sense of Castelnuovo and Mumford) if

$$H^i\big(\mathbb{P}^n, \mathcal{F}(m-i)\big) = 0 \qquad \text{for all } i \geq 1$$

The *Castelnuovo-Mumford regularity* of $\mathcal{F}$ is

$$\operatorname{reg}(\mathcal{F}) = \min\{\, m \in \mathbb{Z} \,\big|\, \mathcal{F} \text{ is } m\text{-regular} \,\}$$

when the set on the right is nonempty and bounded below, and $\operatorname{reg}(\mathcal{F}) = -\infty$ if $\mathcal{F}$ is m-regular for all m (that is, if $\mathcal{F}$ has finite length).

The single index i runs over $1, 2, \ldots, n$ because H^i of a coherent sheaf on $\mathbb{P}^n$ vanishes for $i > n$. Thus

m-regularity is a finite list of cohomological vanishings: $H^1(\mathcal{F}(m-1)) = 0$, $H^2(\mathcal{F}(m-2)) = 0$, and so on down to $H^n(\mathcal{F}(m-n)) = 0$. The definition asks each higher cohomology group to vanish after a twist that increases with the cohomological degree, and it is this staircase of twists that makes the regularity threshold well behaved. The following theorem records the three properties that make regularity the right invariant; it is the technical engine of the section.

Theorem 14.2.2: Mumford's Regularity Theorem

Let $\mathcal{F}$ be an m-regular coherent sheaf on $\mathbb{P}^n_k$. Then for every $\ell \geq 0$:

1. $\mathcal{F}$ is $(m+\ell)$-regular; in particular $H^i(\mathcal{F}(m+\ell-i)) = 0$ for all $i \geq 1$, and $H^i(\mathcal{F}(j)) = 0$ for all $i \geq 1$ and all $j \geq m-i$.
2. The twisted sheaf $\mathcal{F}(m+\ell)$ is generated by its global sections.
3. The multiplication map
$$H^0(\mathbb{P}^n, \mathcal{F}(m+\ell)) \otimes_k H^0(\mathbb{P}^n, \mathcal{O}(1)) \longrightarrow H^0(\mathbb{P}^n, \mathcal{F}(m+\ell+1))$$
is surjective.

***Proof*:**
The argument is an induction on $n = \dim \mathbb{P}^n$ using a general hyperplane, and the three statements are proved together because part (1) for $\mathbb{P}^{n-1}$ feeds parts (2) and (3) for $\mathbb{P}^n$. Assume first that the base field k is infinite; the general case is recovered at the end by a base extension that does not change any of the cohomology groups involved.

Step 1: Restriction to a hyperplane preserves regularity. Choose a hyperplane $H \subseteq \mathbb{P}^n$ whose defining section $s \in H^0(\mathcal{O}(1))$ is a non-zerodivisor on $\mathcal{F}$, meaning that the multiplication map $\mathcal{F}(-1) \xrightarrow{\cdot s} \mathcal{F}$ is injective. Such an H exists because k is infinite: the associated points of $\mathcal{F}$ are finite in number, and a general hyperplane avoids all of them, so s does not lie in any associated prime and multiplication by s has zero kernel. Let $\mathcal{F}_H = \mathcal{F}|_H = \mathcal{F} \otimes \mathcal{O}_H$ be the restriction, a coherent sheaf on $H \cong \mathbb{P}^{n-1}$. The short exact sequence

$$0 \longrightarrow \mathcal{F}(-1) \xrightarrow{\cdot s} \mathcal{F} \longrightarrow \mathcal{F}_H \longrightarrow 0 \tag{14.1}$$

twisted by $\mathcal{O}(m-i)$ gives the long exact cohomology sequence

$$H^i(\mathcal{F}(m-i)) \longrightarrow H^i(\mathcal{F}_H(m-i)) \longrightarrow H^{i+1}(\mathcal{F}(m-1-i))$$

For $i \geq 1$ the left group vanishes because $\mathcal{F}$ is m-regular, and the right group is $H^{i+1}(\mathcal{F}(m-(i+1)))$, which also vanishes because $\mathcal{F}$ is m-regular (the twist $m-1-i$ is exactly the one that m-regularity kills in degree $i+1$). Hence $H^i(\mathcal{F}_H(m-i)) = 0$ for all $i \geq 1$, so $\mathcal{F}_H$ is m-regular as a sheaf on $\mathbb{P}^{n-1}$.

Step 2: The vanishing $H^i(\mathcal{F}(m+\ell-i)) = 0$ for $\ell \geq 0$. Induct on n. When $n = 0$ every higher cohomology group vanishes trivially and there is nothing to prove. For $n \geq 1$, by Step 1 and the inductive hypothesis applied to $\mathcal{F}_H$ on $\mathbb{P}^{n-1}$, the sheaf $\mathcal{F}_H$ satisfies $H^i(\mathcal{F}_H(m+\ell-i)) = 0$ for all $i \geq 1$ and all $\ell \geq 0$. Twist (14.1) by $\mathcal{O}(m+\ell-i)$ and read off the piece

$$H^i(\mathcal{F}_H(m+\ell-i)) \longrightarrow H^{i+1}(\mathcal{F}(m+\ell-1-i)) \xrightarrow{\cdot s} H^{i+1}(\mathcal{F}(m+\ell-i))$$

Fix $i \geq 1$. The first group vanishes by the case just established for $\mathcal{F}_H$, so the multiplication map $\cdot s$ on $H^{i+1}(\mathcal{F}(m+\ell-1-i))$ is injective for every $\ell \geq 0$; equivalently, writing $j = m+\ell-1-i$, multiplication by s is injective on $H^{i+1}(\mathcal{F}(j))$ for all $j \geq m-1-i$. By Serre vanishing $H^{i+1}(\mathcal{F}(j)) = 0$ for $j \gg 0$. An injective endomorphism-up-to-twist of a sequence of groups that eventually vanishes forces the whole sequence to vanish: if $\cdot s\colon H^{i+1}(\mathcal{F}(j)) \hookrightarrow H^{i+1}(\mathcal{F}(j+1))$ is injective for all $j \geq m-1-i$ and the target vanishes for j large, then reading the injections backward, each $H^{i+1}(\mathcal{F}(j))$ injects into a vanishing group and so is itself zero. Reindexing by $i' = i+1 \geq 2$ and $j = m+\ell-i'$ gives $H^{i'}(\mathcal{F}(m+\ell-i')) = 0$ for all $i' \geq 2$ and $\ell \geq 0$. The remaining case $i' = 1$ is the vanishing $H^1(\mathcal{F}(m+\ell-1)) = 0$, and it is proved by a separate induction on ℓ that avoids any appeal to Step 3. The base case $\ell = 0$ is the vanishing $H^1(\mathcal{F}(m-1)) = 0$, which is part of the hypothesis that $\mathcal{F}$ is m-regular. For the inductive step, fix $\ell \geq 1$ and twist (14.1) by $\mathcal{O}(m+\ell-1)$, giving in cohomology the piece

$$H^1(\mathcal{F}(m+\ell-2)) \longrightarrow H^1(\mathcal{F}(m+\ell-1)) \longrightarrow H^1(\mathcal{F}_H(m+\ell-1))$$

The left group is $H^1(\mathcal{F}(m+(\ell-1)-1)) = 0$ by the inductive hypothesis on ℓ. The right group is $H^1(\mathcal{F}_H((m+\ell)-1)) = 0$: by Step 1 the sheaf $\mathcal{F}_H$ is m-regular on $\mathbb{P}^{n-1}$, so by the inductive hypothesis on n it is $(m+\ell)$-regular, and its degree-1 vanishing at level $m+\ell$ is exactly this group. Squeezed between two vanishing groups, $H^1(\mathcal{F}(m+\ell-1)) = 0$, closing the induction on ℓ. This uses only the vanishing of H^1 for $\mathcal{F}_H$ and for $\mathcal{F}$ at the previous level, and in particular no surjectivity of a multiplication map. Combining, $\mathcal{F}$ is $(m+\ell)$-regular for every $\ell \geq 0$, which is part (1); the auxiliary form $H^i(\mathcal{F}(j)) = 0$ for $j \geq m-i$ is the same statement reindexed.

Step 3: Surjectivity of multiplication, part (3). The claim is that

$$\mu_\ell\colon H^0(\mathcal{F}(m+\ell)) \otimes H^0(\mathcal{O}(1)) \longrightarrow H^0(\mathcal{F}(m+\ell+1))$$

is surjective for all $\ell \geq 0$. Restricting to the hyperplane H gives a commutative diagram whose rows come from (14.1). Consider

$$0 \to H^0(\mathcal{F}(m+\ell)) \to H^0(\mathcal{F}(m+\ell+1)) \to H^0(\mathcal{F}_H(m+\ell+1)) \to 0$$

which is exact on the right because $H^1(\mathcal{F}(m+\ell)) = 0$ by part (1). Pick $t \in H^0(\mathcal{F}(m+\ell+1))$. Its restriction $t|_H \in H^0(\mathcal{F}_H(m+\ell+1))$ lies in the image of the multiplication map for $\mathcal{F}_H$, which is surjective by the inductive hypothesis on $\mathbb{P}^{n-1}$ applied to the m-regular sheaf $\mathcal{F}_H$. So $t|_H = \sum_a \bar{u}_a \cdot \bar{v}_a$ with $\bar{u}_a \in H^0(\mathcal{F}_H(m+\ell))$ and $\bar{v}_a \in H^0(\mathcal{O}_H(1))$. Lift each $\bar{u}_a$ to $u_a \in H^0(\mathcal{F}(m+\ell))$ (possible since $H^0(\mathcal{F}(m+\ell)) \to H^0(\mathcal{F}_H(m+\ell))$ is surjective, again because $H^1(\mathcal{F}(m+\ell-1)) = 0$) and each $\bar{v}_a$ to $v_a \in H^0(\mathcal{O}(1))$. Then $t - \sum_a u_a v_a$ restricts to 0 on H, so it is divisible by s: it equals $s \cdot t'$ for a unique $t' \in H^0(\mathcal{F}(m+\ell))$, using the exactness of (14.1) on global sections. Writing $s = \bar{s} \in H^0(\mathcal{O}(1))$, we obtain $t = \sum_a u_a v_a + t' \cdot s$, which exhibits t in the image of μ_ℓ. The base case $n = 0$ is trivial: on a point $\mathcal{F}(m+\ell)$ and $\mathcal{F}(m+\ell+1)$ are the same finite-dimensional vector space, while $H^0(\mathcal{O}(1)) \cong k$ is one-dimensional and its nonzero elements act invertibly, so the multiplication map is surjective.

Step 4: Global generation, part (2). Fix $\ell \geq 0$. By part (3) applied repeatedly, the graded module $\bigoplus_{p \geq 0} H^0(\mathcal{F}(m+\ell+p))$ is generated in degree 0 over the homogeneous coordinate ring $\bigoplus_{p \geq 0} H^0(\mathcal{O}(p))$. Global generation of $\mathcal{F}(m+\ell)$ is a statement about stalks at closed points, and it follows once the evaluation map $H^0(\mathcal{F}(m+\ell)) \otimes \mathcal{O} \to \mathcal{F}(m+\ell)$ is surjective. This surjectivity is local, and at a closed point x the module $\bigoplus_p H^0(\mathcal{F}(m+\ell+p))$ being generated in degree 0 means precisely that sections of $\mathcal{F}(m+\ell)$ generate the stalk after inverting a linear form nonvanishing at x; since such a linear form is a unit at x, the sections already generate $\mathcal{F}(m+\ell)_x$ over $\mathcal{O}_x$ by Nakayama. Hence $\mathcal{F}(m+\ell)$ is globally generated.

Finally, for a base field k that is not infinite, pass to the algebraic closure $\bar{k}$. Cohomology of $\mathcal{F}$ commutes with the flat base change $\mathbb{P}^n_{\bar{k}} \to \mathbb{P}^n_k$, so $\mathcal{F}_{\bar{k}}$ is m-regular, the three conclusions hold for $\mathcal{F}_{\bar{k}}$, and each descends because the maps in (2) and (3) are defined over k and become surjective after a faithfully flat extension, hence were surjective already. This establishes all three parts, completing the proof.

The theorem says that a single number, the regularity, dominates the entire positive-twist behavior of $\mathcal{F}$: past the threshold every higher cohomology group is dead, every twist is globally generated, and the graded module of sections is generated in the lowest surviving degree. The asymmetry in the definition (twisting H^i by $m-i$) is what powers part (1), the propagation upward, and part (1) is what makes "m-regular for $m \gg 0$" a meaningful uniform statement across a family: to bound a whole family it suffices to bound its regularity by a single integer. That is the bridge to boundedness.

A first concrete instance shows the invariant is computable and grounds the definition immediately.

Example 14.2.3: The Regularity of the Ideal Sheaf of a Point

Let $Z = \{x\} \subseteq \mathbb{P}^n$ be a single reduced closed point, with ideal sheaf $\mathcal{I}_Z$ and structure sheaf $\mathcal{O}_Z = k(x)$, a skyscraper. The defining sequence is

$$0 \longrightarrow \mathcal{I}_Z \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_Z \longrightarrow 0$$

Twist by $\mathcal{O}(m-i)$ and take cohomology. The skyscraper $\mathcal{O}_Z$ has $H^0 = k$ and $H^i = 0$ for $i \geq 1$, and $\mathcal{O}_{\mathbb{P}^n}(j)$ has $H^i(\mathcal{O}(j)) = 0$ for $1 \leq i \leq n-1$ (all j), $H^n(\mathcal{O}(j)) = 0$ for $j \geq -n$, and $H^0(\mathcal{O}(j)) \neq 0$ for $j \geq 0$. For $\mathcal{I}_Z$ this yields, for $i \geq 1$ and the twist $m-i$: the group $H^i(\mathcal{I}_Z(m-i))$ sits between $H^{i-1}(\mathcal{O}_Z(m-i))$ and $H^i(\mathcal{O}(m-i))$. For $i \geq 2$ both neighbors vanish once $m - i \geq -n$, so $H^i(\mathcal{I}_Z(m-i)) = 0$ for all $m \geq 1$. For $i = 1$ the relevant sequence is

$$H^0(\mathcal{O}(m-1)) \longrightarrow H^0(\mathcal{O}_Z(m-1)) = k \longrightarrow H^1(\mathcal{I}_Z(m-1)) \longrightarrow H^1(\mathcal{O}(m-1)) = 0$$

The evaluation map $H^0(\mathcal{O}(m-1)) \to k(x)$ is surjective as soon as some degree-$(m-1)$ form is nonzero at x, that is, for $m - 1 \geq 0$. Hence $H^1(\mathcal{I}_Z(m-1)) = 0$ exactly for $m \geq 1$, and $H^1(\mathcal{I}_Z(-1)) = k \neq 0$. Therefore $\operatorname{reg}(\mathcal{I}_Z) = 1$: the ideal sheaf of a point is 1-regular but not 0-regular. Consistently, $\mathcal{I}_Z(1)$ is globally generated (the linear forms vanishing at x generate $\mathcal{I}_Z(1)$), while $\mathcal{I}_Z$ itself is not globally generated near x.

Regularity of the structure sheaf of a projective scheme translates directly into regularity of its ideal sheaf, and it is the ideal sheaf that the Hilbert functor sees.

Example 14.2.4: Regularity and the Hilbert Polynomial of a Twisted Cubic

Let $C \subseteq \mathbb{P}^3$ be the twisted cubic, the image of $\mathbb{P}^1 \hookrightarrow \mathbb{P}^3$, $[s:t] \mapsto [s^3 : s^2t : st^2 : t^3]$. Its ideal $\mathcal{I}_C$ is generated by the three quadrics $x_0x_2 - x_1^2$, $x_1x_3 - x_2^2$, $x_0x_3 - x_1x_2$, and $\mathcal{O}_C \cong \mathcal{O}_{\mathbb{P}^1}(3)$, so $H^0(\mathcal{O}_C(m)) = H^0(\mathcal{O}_{\mathbb{P}^1}(3m))$ has dimension $3m+1$ for $m \geq 0$, giving Hilbert polynomial $P_C(m) = 3m + 1$. The curve C is projectively normal (arithmetically Cohen-Macaulay), with minimal free resolution

$$0 \longrightarrow \mathcal{O}(-3)^{\oplus 2} \longrightarrow \mathcal{O}(-2)^{\oplus 3} \longrightarrow \mathcal{I}_C \longrightarrow 0$$

Reading regularity off the resolution, the generators sit in degree 2 and the syzygies in degree 3 with a linear shift, so $\mathcal{I}_C$ is 2-regular but not 1-regular: $\operatorname{reg}(\mathcal{I}_C) = 2$, and the quadric generators show $\mathcal{I}_C(2)$ is globally generated. From the defining sequence $0 \to \mathcal{I}_C \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_C \to 0$ and the 0-regularity of $\mathcal{O}_{\mathbb{P}^3}$, $\operatorname{reg}(\mathcal{O}_C) = \operatorname{reg}(\mathcal{I}_C) - 1 = 1$. The point to retain is that a curve of Hilbert polynomial $3m+1$ in $\mathbb{P}^3$ has ideal-sheaf regularity bounded by an explicit small integer determined by P_C. That is the boundedness the next theorem makes uniform.

With regularity in hand as the measure of "large enough", the family of sheaves that the Quot functor ranges over can now be brought under uniform control. The definition of the Quot functor imposes no size constraint on the quotients it collects, yet a scheme representing it must. Boundedness is the reconciliation: the set of all quotient sheaves that can appear, as fibers over any base and any point, with a fixed Hilbert polynomial P, is uniformly regular. A single integer m_0, depending only on P (and on the ambient $\mathcal{F}$ and $\mathbb{P}^n$), bounds the regularity of every ideal or quotient sheaf in sight. The proof produces an explicit bound by controlling the intermediate cohomology of a sheaf through the coefficients of its Hilbert polynomial.

Theorem 14.2.5: Boundedness of Sheaves with Fixed Hilbert Polynomial

Fix the ambient $\mathbb{P}^n_k$, a coherent sheaf $\mathcal{F}$ on $\mathbb{P}^n$ that is a quotient of $\bigoplus_{j=1}^N \mathcal{O}(a_j)$ for fixed integers a_j, and a polynomial $P \in \mathbb{Q}[t]$. There is an integer $m_0 = m_0(P, n, \mathcal{F})$, depending only on P, on n, and on the presentation of $\mathcal{F}$, such that every coherent quotient

$$\mathcal{F} \twoheadrightarrow \mathcal{Q}$$

whose kernel is a subsheaf $\mathcal{G} = \ker(\mathcal{F} \to \mathcal{Q})$ with Hilbert polynomial P is m_0-regular. In particular, the family of all such quotients $\mathcal{Q}$ (equivalently, of all subsheaves $\mathcal{G} \subseteq \mathcal{F}$ with Hilbert polynomial P) is *bounded*: there is a single m_0 with $\mathcal{G}$ and $\mathcal{Q}$ both m_0-regular for every member of the family, so each becomes globally generated with vanishing higher cohomology after the one twist $\mathcal{O}(m_0)$.

***Proof*:**
The heart of the matter is the case of a subsheaf of a fixed sheaf; the case of the ideal sheaves $\mathcal{I}_Z$ relevant to $\underline{\mathrm{Hilb}}^P_Y$ (definition 14.1.6) and of the kernels relevant to $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ (definition 14.1.5) both reduce to it, since an ideal sheaf is a subsheaf of $\mathcal{O}_Y \subseteq \mathcal{F}$. The argument is an induction on n producing a uniform regularity bound.

Step 1: Reduction to a bound on h^0 and h^1. A coherent sheaf $\mathcal{G}$ on $\mathbb{P}^n$ is m-regular once $H^i(\mathcal{G}(m-i)) = 0$ for all $i \geq 1$. For $i \geq 2$ these groups are controlled by restriction to hyperplanes and the inductive hypothesis, as in theorem 14.2.2; the delicate part is H^1, and the delicate input is a bound on $h^0(\mathcal{G}(j)) = \dim H^0(\mathcal{G}(j))$ in terms of P. The whole proof therefore reduces to producing, from the Hilbert polynomial P alone, an integer past which $H^1(\mathcal{G}(j))$ vanishes, uniformly over the family. Since $\chi(\mathcal{G}(j)) = P(j)$ for all j (the Euler characteristic is the Hilbert polynomial, independent of the member) and $\chi = h^0 - h^1 + \cdots$, a uniform upper bound on h^0 together with the Euler-characteristic identity pins down h^1.

Step 2: The uniform bound on global sections (Grothendieck's lemma). There is an integer-valued function $Q(t)$, depending only on n and the rank and degree data extractable from P, such that every subsheaf $\mathcal{G} \subseteq \mathcal{F}$ with $\mathcal{G}$ having Hilbert polynomial P satisfies

$$h^0(\mathcal{G}(j)) \leq Q(j) \qquad \text{for all } j \tag{14.2}$$

with Q independent of the particular $\mathcal{G}$. This is proved by induction on n. Restrict to a general hyperplane H; the sequence $0 \to \mathcal{G}(j-1) \to \mathcal{G}(j) \to \mathcal{G}_H(j) \to 0$ (using that a general H meets $\mathcal{G}$ in a subsheaf of $\mathcal{F}_H$) gives

$$h^0(\mathcal{G}(j)) \leq h^0(\mathcal{G}(j-1)) + h^0(\mathcal{G}_H(j))$$

On $H \cong \mathbb{P}^{n-1}$ the sheaf $\mathcal{G}_H$ is a subsheaf of the fixed sheaf $\mathcal{F}_H$, and its Hilbert polynomial is $P(t)-P(t-1)$, again fixed. By induction $h^0(\mathcal{G}_H(j)) \leq Q_H(j)$ for a function Q_H depending only on the data on $\mathbb{P}^{n-1}$. Summing the telescoping inequality from a value j_0 where $h^0(\mathcal{G}(j_0)) = 0$ (which exists uniformly: a subsheaf of $\mathcal{F}$ has no sections in sufficiently negative twist, and "sufficiently negative" is controlled by the fixed $\mathcal{F}$) yields the bound (14.2) with $Q(j) = \sum_{j_0 < p \leq j} Q_H(p)$, a function of P and n only. The base case $n = 0$ is immediate, where h^0 is the constant length read off from P.

Step 3: From the h^0 bound to a uniform vanishing of H^1. Combine (14.2) with the Euler-characteristic identity. On $\mathbb{P}^n$,

$$h^1(\mathcal{G}(j)) = h^0(\mathcal{G}(j)) - \chi(\mathcal{G}(j)) + \sum_{i\geq 2}(-1)^i h^i(\mathcal{G}(j)) = h^0(\mathcal{G}(j)) - P(j) + \sum_{i\geq 2}(-1)^i h^i(\mathcal{G}(j))$$

For $i \geq 2$ the groups $h^i(\mathcal{G}(j))$ vanish for $j \geq j_1$ where j_1 is a uniform bound coming from the inductive control of higher cohomology in Step 1. This threshold is uniform over the family: twisting the general-hyperplane sequence $0 \to \mathcal{G}(j) \to \mathcal{G}(j+1) \to \mathcal{G}_H(j+1) \to 0$ gives $H^i(\mathcal{G}_H(j+1)) \to H^{i+1}(\mathcal{G}(j)) \xrightarrow{\cdot s} H^{i+1}(\mathcal{G}(j+1))$, so for $i \geq 1$ the multiplication $\cdot s$ on $H^{i+1}(\mathcal{G}(j))$ is injective once $H^i(\mathcal{G}_H(j+1)) = 0$, and by the inductive hypothesis on $\mathbb{P}^{n-1}$ the sheaves $\mathcal{G}_H \subseteq \mathcal{F}_H$ are uniformly regular past a threshold depending only on $P(t) - P(t-1)$ and $n-1$; injectivity feeding into Serre vanishing then forces $h^{i+1}(\mathcal{G}(j)) = 0$ for all j past a threshold set by that same $\mathbb{P}^{n-1}$ datum, uniformly over the family. Reindexing gives the uniform j_1 for all $i \geq 2$. For such j,

$$h^1(\mathcal{G}(j)) = h^0(\mathcal{G}(j)) - P(j) \leq Q(j) - P(j)$$

The right-hand side is a fixed function of j that does not depend on the member $\mathcal{G}$, so $h^1(\mathcal{G}(j))$ is bounded above uniformly. This bound need not be zero on its own; the vanishing comes from combining it with the multiplication-by-s structure past a *uniform* threshold. Twisting the general hyperplane section $0 \to \mathcal{G}(j) \xrightarrow{\cdot s} \mathcal{G}(j+1) \to \mathcal{G}_H(j+1) \to 0$ and taking cohomology gives $H^1(\mathcal{G}(j)) \xrightarrow{\cdot s} H^1(\mathcal{G}(j+1)) \to H^1(\mathcal{G}_H(j+1))$; once $\mathcal{G}_H$ is $(j+1)$-regular on $\mathbb{P}^{n-1}$, the right group vanishes and the multiplication map is surjective. By the inductive control of regularity for the fixed family of subsheaves $\mathcal{G}_H \subseteq \mathcal{F}_H$ on $\mathbb{P}^{n-1}$, this surjectivity holds for all $j \geq j_2$ with j_2 a uniform bound depending only on P and n. Set $j_3 = \max(j_1, j_2)$, again uniform. Surjectivity of $\cdot s$ for all $j \geq j_3$ says that $H^1(\mathcal{G}(j+t))$ is the image of the fixed space $V := H^1(\mathcal{G}(j_3))$ under the iterated map s^t, that is, $H^1(\mathcal{G}(j_3+t)) = s^t V$ for every $t \geq 0$. Consequently the linear map $s^t \colon V \to H^1(\mathcal{G}(j_3+t))$ is surjective, so by rank-nullity $h^1(\mathcal{G}(j_3+t)) = \dim V - \dim\ker(s^t|_V)$. The kernels $\ker(s^t|_V)$ form an increasing chain of subspaces of the finite-dimensional space V, and such a chain stabilizes. Because $\dim V \leq Q(j_3) - P(j_3)$, the chain can strictly grow at most $\dim V$ times, so it has stabilized by $t = \dim V$; for all $t \geq \dim V$ the dimension $h^1(\mathcal{G}(j_3+t))$ is therefore constant. Serre vanishing forces $h^1(\mathcal{G}(j)) = 0$ for $j \gg 0$ for each individual $\mathcal{G}$, so that constant value is 0, and hence $h^1(\mathcal{G}(j_3+t)) = 0$ for all $t \geq \dim V$. Setting $B = Q(j_3) - P(j_3)$, this gives the uniform vanishing $H^1(\mathcal{G}(j)) = 0$ for all $j \geq j_3 + B$. The threshold $m_1 = j_3 + B$ is one and the same for every member, computed from Q, P, and n through the uniform bounds

j_1, j_2, and the a priori bound B; this is the sought uniform vanishing $H^1(\mathcal{G}(j)) = 0$ for all $j \geq m_1$. The surjectivity of $\cdot s$ is what pins $H^1(\mathcal{G}(j_3 + t))$ to a quotient of the single bounded space V, and the stabilizing-kernel count then forces the eventual 0 back to the uniform level m_1; without the a priori bound on $\dim V$ the surjective chain alone would place no ceiling on where the vanishing begins.

Step 4: Assembling the regularity bound. Set $m_0 = \max(m_1 + 1, j_1 + n)$ with j_1 as in Step 3, adjusted so that all of $H^i(\mathcal{G}(m_0 - i)) = 0$ hold: the H^1 vanishing is Step 3, and the H^i vanishings for $2 \leq i \leq n$ are the uniform higher-cohomology bounds. Every $\mathcal{G}$ in the family is then m_0-regular, and m_0 depends only on P, n, and $\mathcal{F}$. The quotient $\mathcal{Q} = \mathcal{F}/\mathcal{G}$ inherits a regularity bound from the regularity of $\mathcal{F}$ (fixed, hence some $m_{\mathcal{F}}$) and of $\mathcal{G}$ (the uniform m_0) through the long exact sequence of $0 \to \mathcal{G} \to \mathcal{F} \to \mathcal{Q} \to 0$: for $i \geq 1$, $H^i(\mathcal{Q}(m - i))$ is squeezed between $H^i(\mathcal{F}(m - i))$ and $H^{i+1}(\mathcal{G}(m - i))$, both of which vanish for $m \geq \max(m_{\mathcal{F}}, m_0)$. Enlarging m_0 to this maximum makes every $\mathcal{Q}$ in the family m_0-regular as well, as we sought to show.

Boundedness is the pivotal reduction of the chapter. It says that although the Quot functor is defined by an open-ended condition, the sheaves it ranges over are all controlled by one integer m_0, and past m_0 each quotient is reconstructed from its degree-m_0 global sections. After the twist by $\mathcal{O}(m_0)$, a flat quotient $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ over any base induces a quotient of the fixed vector space (or fixed free module) $H^0(\mathcal{F}(m_0))$, of fixed rank $P(m_0)$, and this is exactly a point of a Grassmannian. section 14.3 builds the Quot scheme by realizing the functor as a subfunctor of that Grassmannian; boundedness is what guarantees the Grassmannian is large enough to hold every quotient at once. Without it there would be no single finite-dimensional ambient space and no scheme to cut out.

14.2.6 Remark: Boundedness as finiteness of the moduli problem Boundedness is the sheaf-theoretic form of a finiteness principle that recurs throughout moduli theory. A moduli functor is representable by a scheme (or by a quasi-compact stack) only if the objects it classifies form a bounded family, that is, are parametrized by a scheme of finite type over k. When boundedness fails, as for the full functor of all coherent sheaves with no fixed Hilbert polynomial, the resulting moduli object is not a scheme and typically not even an algebraic space of finite type; it is an Artin stack that is not quasi-compact. Fixing the Hilbert polynomial P is the standard device that restores boundedness, which is why $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ carries P as a decoration and the Quot scheme is a disjoint union over P of quasi-projective pieces.

Boundedness caught every quotient inside one Grassmannian, but not every point of that Grassmannian gives a *flat* quotient with the right Hilbert polynomial, and flatness is exactly the condition in the definitions of $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ and $\underline{\mathrm{Hilb}}^P_Y$. The tool that recognizes where flatness holds is Grothendieck's flattening stratification. Given a coherent sheaf on a product $Y \times S$, it decomposes the base S into locally closed pieces on each of which the sheaf becomes flat, and on each piece the fibers have a single Hilbert polynomial. This turns the flatness-with-Hilbert-polynomial-P condition, which cuts out $\underline{\mathrm{Quot}}^P$ inside the Grassmannian, into a locally closed subscheme, hence a subscheme.

The stratification is governed by a universal property: it is the finest way of splitting S so that the sheaf becomes flat, and any base change that flattens the sheaf factors through it. That universal property is what lets the strata be defined once and for all, functorially in S.

Theorem 14.2.7: Grothendieck's Flattening Stratification

Let S be a Noetherian scheme, let $Y \subseteq \mathbb{P}^n_S$ be a closed subscheme projective over S, and let $\mathcal{F}$ be a coherent sheaf on Y (equivalently, on $\mathbb{P}^n_S$, extended by zero). Then there is a finite set of polynomials $P_1, \ldots, P_r \in \mathbb{Q}[t]$ and a decomposition of S into pairwise disjoint locally closed subschemes

$$S = \coprod_{a=1}^{r} S_a, \qquad S_a = S_{P_a}$$

indexed by the P_a, with the following properties:

1. Each $S_a \hookrightarrow S$ is a locally closed immersion, and $S = \bigcup_a S_a$ set-theoretically; the closure of S_a is contained in $\bigcup_{P_b \geq P_a} S_b$, where $P_b \geq P_a$ means $P_b(m) \geq P_a(m)$ for $m \gg 0$.
2. A point $s \in S$ lies in S_P if and only if the fiber $\mathcal{F}_s = \mathcal{F}|_{Y_s}$ on $Y_s = Y \times_S \operatorname{Spec} k(s)$ has Hilbert polynomial P.
3. The restriction $\mathcal{F}|_{Y \times_S S_a}$ is flat over S_a with all fibers of Hilbert polynomial P_a.
4. *(Universal property.)* A morphism $T \to S$ of Noetherian schemes has the property that the pullback $\mathcal{F}_T$ on $Y_T = Y \times_S T$ is flat over T with locally constant Hilbert polynomial if and only if $T \to S$ factors scheme-theoretically through the disjoint union $\coprod_a S_a \hookrightarrow S$.

***Proof*:**
The construction is local on S, so assume $S = \operatorname{Spec} A$ is affine and Noetherian. The tool is cohomology and base change together with the theorem that a coherent sheaf on a projective scheme over A is flat if and only if a certain graded module is locally free.

Step 1: Reduction to a graded module of finite presentation. Since $Y \subseteq \mathbb{P}^n_S$ is projective over $S = \operatorname{Spec} A$, replace $\mathcal{F}$ by the graded $A[x_0, \ldots, x_n]$-module $M = \bigoplus_{m \geq m_2} H^0(\mathbb{P}^n_S, \mathcal{F}(m))$, taking m_2 large enough (boundedness, theorem 14.2.5, applied to the fibers, or Serre vanishing over A) that for every $s \in S$ and every $m \geq m_2$,

$$H^i(\mathbb{P}^n_{k(s)}, \mathcal{F}_s(m)) = 0 \quad (i \geq 1), \qquad \dim_{k(s)} H^0(\mathbb{P}^n_{k(s)}, \mathcal{F}_s(m)) = P_s(m)$$

where P_s is the Hilbert polynomial of $\mathcal{F}_s$. By cohomology and base change ([7], cohomology-and-base-change over $\operatorname{Spec} A$), for $m \geq m_2$ the A-module $M_m = H^0(\mathbb{P}^n_S, \mathcal{F}(m))$ is finitely generated, and its formation commutes with base change: $M_m \otimes_A k(s) \cong H^0(\mathbb{P}^n_{k(s)}, \mathcal{F}_s(m))$. Thus $\dim_{k(s)}(M_m \otimes_A k(s)) = P_s(m)$ for all s and all $m \geq m_2$.

Step 2: Flatness as local freeness of the M_m. The sheaf $\mathcal{F}$ is flat over A in a neighborhood of s if and only if each M_m ($m \geq m_2$) is locally free of rank $P_s(m)$ near s; this is the graded-flatness criterion, that a coherent sheaf projective over an affine base is flat precisely when its sufficiently-positive-degree cohomology modules are locally free ([7], flatness criterion via H^0 of twists). For a single finitely generated A-module M_m, the function $s \mapsto \dim_{k(s)}(M_m \otimes k(s))$ is upper semicontinuous, and M_m is locally free of rank d exactly on the locus where this function equals d and stays constant, which is a locally closed subscheme of S defined by the vanishing and non-vanishing of appropriate Fitting ideals. Concretely, the d-th Fitting ideal $\mathrm{Fitt}_d(M_m)$ cuts out the closed locus where the rank is $\leq d$, and M_m is locally free of rank d on the locally closed subscheme

$$V(\mathrm{Fitt}_{d-1}(M_m)) \setminus V(\mathrm{Fitt}_d(M_m))$$

equipped with the scheme structure making M_m locally free there.

Step 3: Stratifying by the Hilbert polynomial. For each polynomial P, define $S_P \subseteq S$ to be the locus of points s with $P_s = P$, that is, with $\dim_{k(s)}(M_m \otimes k(s)) = P(m)$ for all $m \geq m_2$. Because $P_s(m) = \dim_{k(s)}(M_m \otimes k(s))$ and each such dimension function is upper semicontinuous with locally closed constancy loci, S_P is the intersection over $m \geq m_2$ of the locally closed loci $\{\dim(M_m \otimes k(s)) = P(m)\}$. By Noetherianity this intersection stabilizes after finitely many m, so S_P is locally closed, and it is nonempty for only finitely many P (the Hilbert polynomials occurring on the Noetherian S form a finite set, since S has finitely many generic points and P_s is constant on a stratification refining the irreducible components). Equip S_P with the Fitting-ideal scheme structure of Step 2, the one making every M_m locally free of rank $P(m)$; then $\mathcal{F}$ restricted to $\mathbb{P}^n \times_S S_P$ is flat over S_P with fibers of Hilbert polynomial P, giving parts (2) and (3). The finitely many nonempty S_P are pairwise disjoint by construction (a point has one Hilbert polynomial) and cover S, giving part (1); the closure relation follows because each fiber-dimension function $s \mapsto \dim_{k(s)}(M_m \otimes k(s)) = P_s(m)$ is upper semicontinuous, so $P_s(m)$ can only jump up under specialization for every $m \geq m_2$, forcing $P_s \geq P_a$ on the closure of S_a and hence $\overline{S_a} \subseteq \bigcup_{P_b \geq P_a} S_b$.

Step 4: The universal property. Let $g\colon T \to S$ be a morphism with T Noetherian. The pullback $\mathcal{F}_T$ is flat over T if and only if each g^*M_m is locally free over T of the correct rank, again by the graded-flatness criterion applied over T and the base-change isomorphism $g^*M_m \cong \bigoplus H^0(\mathbb{P}^n_T, \mathcal{F}_T(m))$ in the stabilized range. Since M_m becomes locally free of rank $P(m)$ precisely after restriction to S_P, and the Fitting-ideal description of local freeness is compatible with base change, g^*M_m is locally free of rank $P(m)$ if and only if g factors through the subscheme S_P cut out by the Fitting conditions. Requiring flatness with locally constant Hilbert polynomial is requiring this for all m simultaneously with a single P per connected component of T, which is exactly factorization of g through $\coprod_P S_P \hookrightarrow S$. This is the asserted universal property, completing the proof.

The flattening stratification says that flatness, though defined pointwise and delicate, is a locally closed condition on the base: the base splits into finitely many locally closed strata, indexed by Hilbert polynomials, over which the sheaf is flat, and any test scheme on which the sheaf flattens factors through this splitting. For the Quot scheme this is decisive. Inside the Grassmannian produced by boundedness sits the universal quotient; applying the flattening stratification to it isolates the single stratum with Hilbert polynomial P as a locally closed subscheme, and the universal property of the stratification is precisely the universal property that identifies this subscheme with the Quot functor. The stratification also explains, geometrically, why the Hilbert polynomial jumps up under specialization: a limit of subschemes can acquire embedded or lower-dimensional components, raising the Euler characteristic in each twist. The next example makes such a jump visible.

Example 14.2.8: The Flattening Stratification of a Jumping Rank

Consider the simplest family in which fiber dimension jumps. Let $S = \mathbb{A}^1 = \operatorname{Spec} k[s]$ and let $Y = \mathbb{P}^0_S = S$ itself, so $Y \times_S S = S$, and take the coherent sheaf

$$\mathcal{F} = \operatorname{coker}\bigl(\mathcal{O}_S \xrightarrow{\cdot s} \mathcal{O}_S\bigr) = \mathcal{O}_S/(s) = k[s]/(s)$$

the structure sheaf of the origin, viewed as a sheaf on S over S. The fiber of $\mathcal{F}$ at a point $a \in \mathbb{A}^1$ is

$$\mathcal{F}_a = \mathcal{F} \otimes_{k[s]} k(a) = \bigl(k[s]/(s)\bigr) \otimes_{k[s]} k[s]/(s-a) = k[s]/(s, s-a)$$

which is 0 when $a \neq 0$ (the ideal $(s, s-a)$ is the unit ideal) and is k when $a = 0$. So the fiber

Hilbert polynomial, here a constant since Y_s is a point, is

$$P_a = \begin{cases} 0 & a \neq 0 \\ 1 & a = 0 \end{cases}$$

The flattening stratification splits $S = \mathbb{A}^1$ into two strata: the open stratum $S_{P=0} = \mathbb{A}^1 \setminus \{0\}$, where $\mathcal{F}$ restricts to the zero sheaf, flat of fiber length 0; and the closed stratum $S_{P=1} = \{0\}$, where $\mathcal{F}$ restricts to k, flat over the point of fiber length 1. On S itself $\mathcal{F}$ is not flat: flatness over $k[s]$ would force $\mathcal{F}$ to be torsion-free, but $\mathcal{F} = k[s]/(s)$ is all torsion, and indeed $\dim_{k(a)} \mathcal{F}_a$ jumps from 0 to 1 at the origin, violating the constancy that flatness demands. The stratification repairs this by isolating the jump: the closed point where the fiber length rises becomes its own stratum. This models exactly what happens in the Quot scheme when a family of quotients degenerates and an embedded point appears in the special fiber, raising the Hilbert polynomial there.

A second example shows the stratification separating strata of different colength, the closed-subscheme situation of $\underline{\mathrm{Hilb}}^P$ where the Hilbert polynomial is the constant length of a zero-dimensional scheme.

Example 14.2.9: Stratifying a Family of Ideals by Colength

Let $S = \mathbb{A}^1 = \operatorname{Spec} k[s]$ and work in the affine line $\mathbb{A}^1 = \operatorname{Spec} k[x]$ over S, that is, in $\mathbb{A}^1_S = \operatorname{Spec} k[x][s]$; the family of closed subschemes is given by the ideal

$$I_s = (s\,x,\ x^2) \subseteq k[x]$$

with the parameter s ranging over S. The quotient sheaf is $\mathcal{F} = k[x][s]/(sx, x^2)$, viewed as a coherent sheaf on $\mathbb{A}^1_S$ flat-checked over $S = \operatorname{Spec} k[s]$. Compute the fibers over a point $a \in S$: setting $s = a$ gives the ideal $I_a = (ax, x^2) \subseteq k[x]$.

- For $a \neq 0$, the element ax is a times x with a a unit, so $I_a = (x, x^2) = (x)$, and the fiber $Z_a = V(x)$ is a single reduced point, of colength $\dim_k k[x]/(x) = 1$. Its (constant) Hilbert polynomial is $P_a = 1$.
- For $a = 0$, the ideal is $I_0 = (x^2)$, and the fiber $Z_0 = V(x^2)$ is a length-2 non-reduced point (a point with one tangent direction), of colength $\dim_k k[x]/(x^2) = 2$. Its Hilbert polynomial is $P_0 = 2$.

The colength jumps from 1 to 2 at the origin, so the family is not flat over S: flatness forces the fiber length to be locally constant, and here it is not. Concretely, $\mathcal{F} = k[x][s]/(sx, x^2)$ has s-torsion, since $x \cdot s = 0$ while $x \neq 0$ in $\mathcal{F}$, and a torsion module over the domain $k[s]$ is not flat. The flattening stratification of $S = \mathbb{A}^1$ therefore splits into two strata carrying different Hilbert polynomials: the open stratum $S_{P=1} = \mathbb{A}^1 \setminus \{0\}$, over which the family restricts to the flat family of single reduced points (colength 1); and the closed stratum $S_{P=2} = \{0\}$, over which the family is the length-2 fat point (colength 2). Each stratum carries a flat family with constant Hilbert polynomial, while the union does not; the closure relation $\overline{S_{P=1}} = \mathbb{A}^1 \supseteq S_{P=2}$ realizes the general principle that the Hilbert polynomial can only jump up under specialization ($1 \to 2$ as $a \to 0$). This is exactly the picture inside $\underline{\mathrm{Hilb}}^P$: the strata $S_{P=1}$ and $S_{P=2}$ belong to different Hilbert schemes Hilb^1 and Hilb^2, here the Hilbert schemes of 1 and 2 points of the affine line $\mathbb{A}^1 = \operatorname{Spec} k[x]$ (not the projective Hilb^P_Y of the rest of the section, since the ambient is the affine $\mathbb{A}^1_S$), and the flattening stratification is what separates a would-be family straddling both into its flat pieces.

The two engines of this section, boundedness and the flattening stratification, together reduce the construction of the Quot scheme to linear algebra plus a locally closed condition: boundedness embeds every quotient into a fixed Grassmannian, and the flattening stratification carves out the flat, Hilbert-polynomial-P locus as a locally closed subscheme of that Grassmannian. section 14.3 carries this out in full and identifies the resulting scheme with the representing object of $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ and, as the special case $\mathcal{F} = \mathcal{O}_Y$, of $\underline{\mathrm{Hilb}}^P_Y$.

Exercise 14.2.1

1. Let $\mathcal{F} = \mathcal{O}_{\mathbb{P}^n}$ be the structure sheaf of projective space. Using the cohomology of the line bundles $\mathcal{O}(j)$, show directly that $\mathcal{O}_{\mathbb{P}^n}$ is 0-regular, that is, $\mathrm{reg}(\mathcal{O}_{\mathbb{P}^n}) = 0$, and deduce that $\mathcal{O}(m)$ is globally generated with vanishing higher cohomology for every $m \geq 0$. Explain how this is consistent with theorem 14.2.2(1).
2. Prove the propagation property of regularity from the definition and theorem 14.2.2: if $\mathcal{F}$ is m-regular then $\mathcal{F}$ is m'-regular for every $m' \geq m$. Then give an example of a coherent sheaf $\mathcal{F}$ on $\mathbb{P}^1$ and an integer m such that $\mathcal{F}$ is m-regular but not $(m-1)$-regular, computing $\mathrm{reg}(\mathcal{F})$ explicitly.
3. Let $Z \subseteq \mathbb{P}^n$ be a hypersurface of degree d, defined by one homogeneous polynomial of degree d, with ideal sheaf $\mathcal{I}_Z \cong \mathcal{O}(-d)$. Compute the Hilbert polynomial of $\mathcal{O}_Z$, compute $\mathrm{reg}(\mathcal{I}_Z)$ and $\mathrm{reg}(\mathcal{O}_Z)$ using the exact sequence $0 \to \mathcal{O}(-d) \to \mathcal{O} \to \mathcal{O}_Z \to 0$, and confirm that the regularity is bounded by an explicit function of d (hence of the Hilbert polynomial), illustrating theorem 14.2.5.
4. Consider the family $\mathcal{F}$ on $\mathbb{P}^1_{\mathbb{A}^1}$ given, in the affine chart $\mathbb{A}^1 \times \mathbb{A}^1 = \operatorname{Spec} k[x][s]$, by the module $M = k[x][s]/(x, s)$ over $k[x][s]$, viewed as a coherent sheaf on $\mathbb{P}^1_S$ over $S = \operatorname{Spec} k[s]$. Determine the fibers $\mathcal{F}_s$ for $s \neq 0$ and $s = 0$, decide whether $\mathcal{F}$ is flat over S, and describe the flattening stratification of S explicitly, identifying each stratum and its Hilbert polynomial.
5. Explain, in the setup of theorem 14.2.5, why fixing the Hilbert polynomial P is necessary for boundedness by exhibiting an infinite family of subschemes of $\mathbb{P}^2$, one for each degree $d \geq 1$, whose ideal-sheaf regularities are unbounded. Then explain in one paragraph how boundedness is used in section 14.3 to reduce the Quot functor to a subfunctor of a single Grassmannian, referencing $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ (definition 14.1.5).

14.3 Grothendieck's Existence Theorem

The two preceding sections assembled the raw materials. Section 14.1 phrased the moduli problem: the Quot functor $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ (definition 14.1.5) sends a base S to the set of flat quotients of $\mathcal{F}_S$ with fiberwise Hilbert polynomial P, and the Hilbert functor $\underline{\mathrm{Hilb}}^P_Y$ (definition 14.1.6) is the special case $\mathcal{F} = \mathcal{O}_Y$, parametrizing flat families of closed subschemes. Section 14.2 supplied the two engines: boundedness (theorem 14.2.5), which says the family of quotients with fixed P becomes uniformly m-regular for m large, and the flattening stratification (theorem 14.2.7), which cuts out the locus in a base over which a coherent sheaf is flat with a prescribed Hilbert polynomial. This section fits the pieces together. The outcome is that both functors are representable by projective schemes, the theorem stated without proof in theorem 3.3.11 and proved here in full.

The strategy is one the reader has already seen in miniature. In section 13.2 the Grassmannian was built by exhibiting its functor as an explicit gluing of affine charts; here the target is not built from scratch but is cut out of a Grassmannian that already exists. Boundedness is what makes this possible: it collapses the infinite-dimensional datum of a quotient sheaf $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ into the finite-dimensional datum of a single surjection on degree-m global sections, and a surjection of a fixed vector space onto a quotient of fixed dimension is exactly a point of a Grassmannian. The flatness and Hilbert-polynomial conditions of the Quot functor then carve a closed subscheme out of that Grassmannian, and closed subschemes of projective schemes are projective. The whole construction is boundedness plus flattening, packaged through the representability of the Grassmannian from theorem 13.2.4.

Throughout, $Y \subseteq \mathbb{P}^n_k$ is a fixed projective scheme, $\mathcal{F}$ a fixed coherent sheaf on Y, and $P \in \mathbb{Q}[t]$ a fixed Hilbert polynomial; $\mathcal{O}_Y(1)$ is the very ample line bundle pulled back from $\mathbb{P}^n_k$, and for a sheaf $\mathcal{G}$ the twist $\mathcal{G}(m)$ denotes $\mathcal{G} \otimes \mathcal{O}_Y(m)$. For a base scheme S, $\mathcal{F}_S$ denotes the pullback of $\mathcal{F}$ to $Y \times_k S = Y_S$ along the projection $Y_S \to Y$, and $\mathcal{O}_{Y_S}(1)$ is the corresponding relatively very ample bundle. The first task is to record precisely how boundedness lets a quotient be recovered from finitely much data.

The mechanism is that for a fixed flat quotient, taking the degree-m piece of the pushforward is an equivalence between quotient sheaves and quotient vector spaces, once m is past the regularity bound. This is where the m-regularity of definition 14.2.1 does its work: it forces the higher cohomology of the relevant twists to vanish, so that the global-sections functor is exact on the sequences in play and commutes with base change. The following proposition isolates the statement.

Proposition 14.3.1: Recovery of a Quotient from Its Degree-m Sections

Let $Y \subseteq \mathbb{P}^n_k$ be projective, $\mathcal{F}$ coherent on Y, and P a Hilbert polynomial. There is an integer m_0, depending only on Y, $\mathcal{F}$, and P, with the following property. For every k-scheme S and every quotient $q\colon \mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ that is flat over S with fiberwise Hilbert polynomial P, and every $m \geq m_0$:

1. the pushforwards $\pi_{S*}\mathcal{F}_S(m)$ and $\pi_{S*}\mathcal{Q}(m)$ along $\pi_S\colon Y_S \to S$ are locally free of finite rank, the second of rank $P(m)$, and their formation commutes with arbitrary base change $T \to S$;

2. the induced map $H^0(q)(m)\colon \pi_{S*}\mathcal{F}_S(m) \twoheadrightarrow \pi_{S*}\mathcal{Q}(m)$ is surjective; and

3. the quotient q is recovered from $H^0(q)(m)$: the surjection $\mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ is the sheafification of the surjection of graded modules generated in degree m by $H^0(q)(m)$, so q and $H^0(q)(m)$ determine each other.

***Proof*:**
Step 1: The uniform regularity bound. By boundedness (theorem 14.2.5) there is an integer m_0 such that every quotient $\mathcal{F}_s \twoheadrightarrow \mathcal{Q}_s$ with Hilbert polynomial P, over any field-valued point s, together with its kernel $\mathcal{K}_s$, is m_0-regular. Enlarging m_0 if necessary, arrange also that $\mathcal{F}$ itself is m_0-regular. Fix $m \geq m_0$. For a field-valued point s of S, m-regularity of $\mathcal{Q}_s$ gives $H^i(Y_s, \mathcal{Q}_s(m)) = 0$ for all $i > 0$, and the same for $\mathcal{F}_s$ and for the kernel $\mathcal{K}_s = \ker(\mathcal{F}_s \twoheadrightarrow \mathcal{Q}_s)$.

Step 2: Local freeness and base change. The fiberwise vanishing $H^i(Y_s, \mathcal{Q}_s(m)) = 0$ for $i > 0$, holding at every point of S, is exactly the hypothesis of cohomology and base change ([7], cohomology-and-base-change for a flat family): the higher direct images $R^i\pi_{S*}\mathcal{Q}(m)$ vanish for

$i > 0$, the sheaf $\pi_{S*}\mathcal{Q}(m)$ is locally free, and its formation commutes with any base change $T \to S$. Its rank at s is $h^0(Y_s, \mathcal{Q}_s(m)) = \chi(\mathcal{Q}_s(m)) = P(m)$, the last equality because the higher cohomology vanishes and χ computes P. The same argument applied to $\mathcal{F}_S$ and to the kernel $\mathcal{K} = \ker q$ (which is flat over S because in $0 \to \mathcal{K} \to \mathcal{F}_S \to \mathcal{Q} \to 0$ both $\mathcal{F}_S$ and $\mathcal{Q}$ are S-flat, so $\mathcal{Q}$ flat gives $\mathrm{Tor}_1^{\mathcal{O}_S}(\mathcal{Q}, -) = 0$ and hence $\mathcal{K}$ flat) gives that $\pi_{S*}\mathcal{F}_S(m)$ and $\pi_{S*}\mathcal{K}(m)$ are locally free and compatible with base change, of ranks $\chi(\mathcal{F}_s(m))$ and $\chi(\mathcal{F}_s(m)) - P(m)$.

Step 3: Surjectivity on sections. Twist the short exact sequence $0 \to \mathcal{K} \to \mathcal{F}_S \to \mathcal{Q} \to 0$ by $\mathcal{O}_{Y_S}(m)$ and push forward along π_S. Because $R^1\pi_{S*}\mathcal{K}(m) = 0$ by Step 2, the resulting sequence

$$0 \to \pi_{S*}\mathcal{K}(m) \to \pi_{S*}\mathcal{F}_S(m) \xrightarrow{H^0(q)(m)} \pi_{S*}\mathcal{Q}(m) \to 0$$

is exact; in particular $H^0(q)(m)$ is surjective. This proves (1) and (2).

Step 4: Recovery. For m-regular sheaves the graded module $\bigoplus_{\ell \geq m} \pi_{S*}\mathcal{Q}(\ell)$ is generated in degree m over the section (homogeneous coordinate) ring $\bigoplus_{\ell \geq 0} \pi_{S*}\mathcal{O}_{Y_S}(\ell)$, a further consequence of m-regularity (definition 14.2.1): m-regular sheaves are globally generated in every degree $\geq m$, and the multiplication maps $\pi_{S*}\mathcal{Q}(\ell) \otimes \pi_{S*}\mathcal{O}_{Y_S}(1) \to \pi_{S*}\mathcal{Q}(\ell+1)$ are surjective for $\ell \geq m$. The sheaf $\mathcal{Q}$ is therefore reconstructed as the sheaf associated to this graded module, and the surjection q as the sheafification of the surjection on graded modules determined in degree m by $H^0(q)(m)$ together with the module structure of $\mathcal{F}_S$. Two quotients with the same map $H^0(q)(m)$ on degree-m sections have the same kernel submodule in degree m, hence generate the same graded submodule of $\bigoplus_{\ell \geq m} \pi_{S*}\mathcal{F}_S(\ell)$, and so sheafify to the same quotient q; thus q and $H^0(q)(m)$ determine each other, completing the proof.

This proposition is the technical crux, and its meaning is worth stating plainly before it is used. A flat quotient sheaf on Y_S is a priori an object of infinite type: it is a coherent sheaf, carrying data in every degree. Boundedness says that all this data is forced once the degree-m piece is fixed, for a single m that can be chosen uniformly across the entire moduli problem. So the passage $q \mapsto H^0(q)(m)$ trades a sheaf-theoretic object for a linear-algebra object: a surjection of the locally free sheaf $\pi_{S*}\mathcal{F}_S(m)$ onto the locally free sheaf $\pi_{S*}\mathcal{Q}(m)$ of rank $P(m)$. That surjection is a family of quotient vector spaces, which is the datum a Grassmannian classifies. The construction of the Quot scheme is now the observation that this trade is functorial and that the conditions defining the Quot functor become closed conditions on the Grassmannian side.

There is one subtlety the recovery hides, and it is exactly the one boundedness alone does not settle: not every surjection $\pi_{S*}\mathcal{F}_S(m) \twoheadrightarrow V$ onto a rank-$P(m)$ bundle comes from a flat quotient with the right Hilbert polynomial. A general such surjection sheafifies to *some* quotient of $\mathcal{F}_S$, but this quotient need be neither flat over S nor of fiberwise Hilbert polynomial P. Cutting out the surjections that do arise is the job of the flattening stratification, and it is why the Quot scheme is a proper closed subscheme of the Grassmannian rather than the whole of it.

Before the theorem, it helps to name the Grassmannian in play and the natural transformation into it. Fix $m \geq m_0$ and write $H = \pi_{S*}\mathcal{F}_S(m)$ for the base S; when $S = \operatorname{Spec} k$ this is the vector space $H^0(Y, \mathcal{F}(m))$, and for general S it is a locally free sheaf whose formation commutes with base change (proposition 14.3.1). A quotient $q \in \underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}(S)$ produces, by the proposition, a rank-$P(m)$ locally free quotient $H \twoheadrightarrow \pi_{S*}\mathcal{Q}(m)$. When $S = \operatorname{Spec} k$ this is a point of the Grassmannian $\mathrm{Gr}(P(m),\, h^0(Y, \mathcal{F}(m)))$ of $P(m)$-dimensional quotients of $H^0(Y, \mathcal{F}(m))$; over a general base it is an S-point of that same Grassmannian, by the functorial description of definition 13.2.1. The assignment $q \mapsto H^0(q)(m)$ is therefore a natural transformation of functors, and everything hinges on identifying its image.

Theorem 14.3.2: Representability of the Quot Functor

Let $Y \subseteq \mathbb{P}^n_k$ be projective, $\mathcal{F}$ a coherent sheaf on Y, and $P \in \mathbb{Q}[t]$ a Hilbert polynomial. The functor $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ of definition 14.1.5 is representable by a projective k-scheme $\mathrm{Quot}^P_{\mathcal{F}/Y}$. Concretely, for $m \gg 0$ the natural transformation $q \mapsto H^0(q)(m)$ embeds $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ as a closed subfunctor of the Grassmannian

$$\underline{\mathrm{Gr}}\big(P(m),\, h^0(Y, \mathcal{F}(m))\big)$$

of $P(m)$-dimensional quotients of $H^0(Y, \mathcal{F}(m))$, and $\mathrm{Quot}^P_{\mathcal{F}/Y}$ is the corresponding closed subscheme, hence projective.

***Proof*:**
Fix $m \geq m_0$ as in proposition 14.3.1, write $N = h^0(Y, \mathcal{F}(m))$ and $G = \mathrm{Gr}(P(m), N)$, the Grassmannian of $P(m)$-dimensional quotients of $H^0(Y, \mathcal{F}(m)) \cong k^N$, which is projective by theorem 13.2.4.

Step 1: The natural transformation to the Grassmannian. The assignment $\tau_S\colon q \mapsto \big(H \twoheadrightarrow \pi_{S*}\mathcal{Q}(m)\big)$, where $H = \pi_{S*}\mathcal{F}_S(m)$, is defined for every S by proposition 14.3.1. To land in $\underline{G} = \underline{\mathrm{Gr}}(P(m), N)$ one must identify H with the constant sheaf $\mathcal{O}_S^N$. Over the point $\operatorname{Spec} k$ the source is $H^0(Y, \mathcal{F}(m)) \cong k^N$; over general S, compatibility with base change (proposition 14.3.1(1)) gives $\pi_{S*}\mathcal{F}_S(m) \cong \mathcal{O}_S \otimes_k H^0(Y, \mathcal{F}(m)) = \mathcal{O}_S^N$ canonically, because $\mathcal{F}_S$ is pulled back from $\mathcal{F}$ and the pushforward commutes with the base change $S \to \operatorname{Spec} k$. Under this identification $\tau_S(q)$ is a rank-$P(m)$ locally free quotient of $\mathcal{O}_S^N$, that is, an element of $\underline{G}(S)$. Naturality in S is the base-change compatibility of proposition 14.3.1(1): for $f\colon T \to S$, $f^*\pi_{S*}\mathcal{Q}(m) = \pi_{T*}(f^*\mathcal{Q})(m)$, so τ commutes with pullback. Hence $\tau\colon \underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \to \underline{G}$ is a natural transformation.

Step 2: τ is injective, i.e., a monomorphism of functors. For a fixed S, if two quotients q, q' satisfy $\tau_S(q) = \tau_S(q')$ then they induce the same surjection $H^0(q)(m) = H^0(q')(m)$ on degree-m sections, so by the recovery statement proposition 14.3.1(3) they are equal as quotients of $\mathcal{F}_S$. Thus each τ_S is injective, and τ is a monomorphism.

Step 3: The image is a closed subfunctor via the universal quotient. It remains to characterize which S-points of G lie in the image of τ, and to show this is a closed condition. On G sits the universal quotient of definition 13.2.5, a surjection $\mathcal{O}_G^N \twoheadrightarrow \mathcal{V}$ with $\mathcal{V}$ locally free of rank $P(m)$. Identifying $\mathcal{O}_G^N = \mathcal{O}_G \otimes_k H^0(Y, \mathcal{F}(m))$ and pulling $\mathcal{F}$ back to $Y \times_k G = Y_G$, the composite

$$\mathcal{F}_G(m) \;\longleftarrow\; \pi_G^*\big(\mathcal{O}_G \otimes_k H^0(Y, \mathcal{F}(m))\big) \;\twoheadrightarrow\; \pi_G^*\mathcal{V}$$

is not yet a quotient of $\mathcal{F}_G$; rather, the universal surjection $\mathcal{O}_G^N \twoheadrightarrow \mathcal{V}$ determines a graded quotient of the section module $\bigoplus_{\ell \geq m} \pi_{G*}\mathcal{F}_G(\ell)$, and taking the associated sheaf (Proj-sheafification) of the graded module generated in degree m by $\mathcal{O}_G^N \twoheadrightarrow \mathcal{V}$ produces a universal coherent quotient

$$u_G\colon \mathcal{F}_G \twoheadrightarrow \widetilde{\mathcal{Q}}$$

on Y_G. This $\widetilde{\mathcal{Q}}$ is coherent but in general *not* flat over G: over a general point of G the surjection on sections need not sheafify to a flat quotient with Hilbert polynomial P. Apply the flattening stratification (theorem 14.2.7) to the coherent sheaf $\widetilde{\mathcal{Q}}$ on $Y_G \to G$: there is a locally closed stratification $\{G_\alpha\}$ of G such that the restriction of $\widetilde{\mathcal{Q}}$ to Y_{G_α} is flat over G_α with locally constant Hilbert polynomial. Let $G_P \subseteq G$ be the union of those strata on which the constant Hilbert polynomial equals P.

An S-point $g\colon S \to G$ lifts to an S-point of $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ under τ if and only if the pullback $g^*\widetilde{\mathcal{Q}}$ is flat over S with fiberwise Hilbert polynomial P: for then $g^*u_G\colon \mathcal{F}_S \twoheadrightarrow g^*\widetilde{\mathcal{Q}}$ is a Quot-family whose degree-m sections recover g, and conversely $\tau_S(q)$ pulls the universal quotient back to q by the recovery statement. By the universal property of the flattening stratification (theorem 14.2.7), $g^*\widetilde{\mathcal{Q}}$ is flat over S with Hilbert polynomial P if and only if g factors through G_P. Thus the image of τ_S is exactly $h_{G_P}(S) \subseteq h_G(S)$, naturally in S.

Step 4: G_P is closed, and representability. The flattening stratification (theorem 14.2.7) presents G_P as a *locally closed* subscheme of G: it is the union of those strata on which $\widetilde{\mathcal{Q}}$ is flat with constant Hilbert polynomial P, and as a finite union of locally closed sets it is constructible and of finite type over k. The content of this step is that G_P is in fact *closed*. A naive semicontinuity count does not deliver this. For a coherent sheaf $\mathcal{E}$ on G the fiber-dimension function $g \mapsto \dim_{\kappa(g)}(\mathcal{E} \otimes \kappa(g))$ is *upper* semicontinuous, so the individual loci $\{h^0(\widetilde{\mathcal{Q}}(\ell) \otimes \kappa(g)) \geq P(\ell)\}$ are closed; but their intersection is strictly larger than G_P, because a stratum $G_{P'}$ carrying a *larger* polynomial $P' > P$ also satisfies $h^0(\widetilde{\mathcal{Q}}(\ell) \otimes \kappa(g)) = P'(\ell) \geq P(\ell)$ for $\ell \gg 0$ and hence meets every such locus. There is no matching pointwise upper bound $h^0(\widetilde{\mathcal{Q}}(\ell) \otimes \kappa(g)) \leq P(\ell)$ available on all of G: at a point where $\widetilde{\mathcal{Q}}$ is not flat, $\otimes\,\kappa(g)$ does not commute with π_{G*}, and sheafification of the degree-m module can strictly *raise* the dimension of the fiber sections, so no such degree-m inequality controls h^0 at a general point. The closedness of G_P is instead a specialization statement, proved by the valuative criterion, and it is exactly here that boundedness does its final work.

Step 4a: closedness under specialization. Since G_P is constructible and G is Noetherian, G_P is closed if and only if it is stable under specialization: whenever a valuation ring dominates a point of G_P, its closed point again lies in G_P. Concretely ([7], valuative criterion for a constructible set on a Noetherian scheme), it suffices to test discrete valuation rings. Let R be a discrete valuation ring with fraction field K and residue field κ, and let $g\colon \operatorname{Spec} R \to G$ be a morphism whose generic point $g_K\colon \operatorname{Spec} K \to G$ factors through G_P; write $g_0\colon \operatorname{Spec} \kappa \to G$ for the closed point. The claim is that g_0 also factors through G_P.

Pull the universal quotient back along g to obtain a coherent quotient $g^*u_G\colon \mathcal{F}_R \twoheadrightarrow g^*\widetilde{\mathcal{Q}}$ on $Y_R = Y \times_k \operatorname{Spec} R$. Over the generic fiber this restricts to $\mathcal{F}_K \twoheadrightarrow (g^*\widetilde{\mathcal{Q}})_K$, which by hypothesis is flat over K with Hilbert polynomial P. Let $\mathcal{K}_R = \ker(g^*u_G)$ and form the *flat limit*: replace $\mathcal{K}_R$ by the saturation

$$\mathcal{K}_R^\flat := \left(\, j_*\,\mathcal{K}_K \,\right) \cap \mathcal{F}_R$$

where $j\colon Y_K \hookrightarrow Y_R$ is the generic-fiber inclusion, that is, the largest subsheaf of $\mathcal{F}_R$ agreeing with $\mathcal{K}_K$ over K and having no sections supported on the special fiber. Equivalently, $\mathcal{K}_R^\flat$ is the kernel of $\mathcal{F}_R \to j_*(\mathcal{F}_K/\mathcal{K}_K)$; the quotient $\mathcal{Q}_R := \mathcal{F}_R/\mathcal{K}_R^\flat$ is then R-flat, because it is a coherent sheaf on Y_R with no associated points on the special fiber, hence torsion-free and so flat over the discrete valuation ring R ([7], flatness over a Dedekind base is torsion-freeness). Its generic fiber is $\mathcal{F}_K/\mathcal{K}_K = (g^*\widetilde{\mathcal{Q}})_K$, with Hilbert polynomial P. Because $\mathcal{Q}_R$ is flat over the connected base $\operatorname{Spec} R$, its fiberwise Hilbert polynomial is constant (proposition 14.1.4), so the special fiber $(\mathcal{Q}_R)_0 = \mathcal{Q}_R \otimes_R \kappa$ is a flat quotient of $\mathcal{F}_\kappa$ with Hilbert polynomial exactly P. This is the flat limit of the generic Quot-family.

Step 4b: the flat limit is classified by g_0. It remains to see that this flat limit is the family that g_0 names, so that $g_0 \in G_P$. Both the flat limit $\mathcal{Q}_R$ and the original pullback $g^*\widetilde{\mathcal{Q}}$ agree over the generic fiber, so their degree-m section modules agree there; since $m \geq m_0$ is past the regularity bound and $\mathcal{V}$ pulls back to the locally free sheaf $g^*\mathcal{V}$ on $\operatorname{Spec} R$, the degree-m data of $\mathcal{Q}_R$ is a rank-$P(m)$ locally free quotient $g^*\mathcal{O}_G^N = \mathcal{O}_R^N \twoheadrightarrow \pi_{R*}\mathcal{Q}_R(m)$ of $\mathcal{O}_R^N$ extending the generic quotient $g_K^*(\mathcal{O}_G^N \twoheadrightarrow \mathcal{V})$. A locally free quotient of $\mathcal{O}_R^N$ over the discrete valuation ring R is determined by its generic fiber, since $\operatorname{Spec} R$ has $\operatorname{Spec} K$ as a dense open and the Grassmannian G is separated (theorem 13.2.4); the two R-points g and the one classifying $\pi_{R*}\mathcal{Q}_R(m)$ agree on $\operatorname{Spec} K$, hence agree, and in particular have the same closed point. So the closed point g_0 classifies precisely the degree-m data of the flat family $(\mathcal{Q}_R)_0$, and by the recovery statement proposition 14.3.1(3) this degree-m datum is $(\mathcal{Q}_R)_0$ itself. Thus $g_0^*\widetilde{\mathcal{Q}} = (\mathcal{Q}_R)_0$ is flat over κ with Hilbert polynomial P, so g_0 factors through G_P. This proves that G_P is stable under specialization, hence closed in G.

The specialization argument also settles the direction of the earlier semicontinuity, and shows the flat limit cannot escape to a larger polynomial: the flat limit carries polynomial P, not some $P' > P$, precisely because flatness over $\operatorname{Spec} R$ pins the fiberwise polynomial to its generic value P. The strata $G_{P'}$ with $P' > P$ that an unconstrained degeneration might have produced are exactly the non-flat sheafifications the flat limit replaces; they lie in $\overline{G_P}$ only as underlying points of G, never as flat Quot-families with the wrong polynomial. By Steps 2, 3, and 4a, τ restricts to an isomorphism of functors $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \cong h_{G_P}$ over the now-closed subscheme G_P, so $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ is representable by

$$\mathrm{Quot}^P_{\mathcal{F}/Y} := G_P$$

a closed subscheme of the projective scheme $G = \mathrm{Gr}(P(m), N)$. A closed subscheme of a projective k-scheme is projective, so $\mathrm{Quot}^P_{\mathcal{F}/Y}$ is projective over k, as we sought to show.

The theorem is the payoff of the entire chapter, and its architecture repays a second look. Every ingredient entered at a definite point. Boundedness (theorem 14.2.5) chose the degree m and guaranteed that a single Grassmannian, of the fixed finite dimension $\mathrm{Gr}(P(m), h^0(\mathcal{F}(m)))$, receives every family in the moduli problem at once. Cohomology and base change (proposition 14.3.1) turned the assignment $q \mapsto H^0(q)(m)$ into a natural transformation of functors, injective by the recovery of a sheaf from its sections. The flattening stratification (theorem 14.2.7) identified the image as a closed subscheme, isolating those points of the Grassmannian whose associated quotient is flat with Hilbert polynomial P. Representability of the Grassmannian (theorem 13.2.4) provided the ambient projective space in which the Quot scheme sits. The pattern, closed subfunctor of a Grassmannian cut out by a flatness condition, is the template for nearly every construction of a moduli space by hand, and it recurs in the geometric-invariant-theory quotients of chapter 15.

One point of the proof deserves emphasis because it is the source of most subtleties in practice. The universal object $\widetilde{\mathcal{Q}}$ on the full Grassmannian G is coherent but not flat; flatness is imposed, not automatic, and the flattening stratification is precisely the device that imposes it. This is the scheme-theoretic content of the slogan that "the Quot scheme parametrizes flat quotients": flatness is a closed condition, and the Quot scheme is where it holds. The reader should not expect the map to the Grassmannian to be surjective; its image is exactly the flat locus, which is a proper closed subset whenever the moduli problem is nontrivial.

The Hilbert scheme is the special case $\mathcal{F} = \mathcal{O}_Y$, and it requires no new argument. A quotient $\mathcal{O}_{Y_S} \twoheadrightarrow \mathcal{Q}$ is the same datum as a closed subscheme $Z \subseteq Y_S$ flat over S: the kernel is an ideal sheaf $\mathcal{I}_Z$, and $\mathcal{Q} = \mathcal{O}_Z$ is the structure sheaf of the subscheme it cuts out. Applying theorem 14.3.2

verbatim gives the following.

Corollary 14.3.3: Representability of the Hilbert Functor

Let $Y \subseteq \mathbb{P}^n_k$ be projective and $P \in \mathbb{Q}[t]$ a Hilbert polynomial. The Hilbert functor $\underline{\mathrm{Hilb}}^P_Y$ of definition 14.1.6 is representable by a projective k-scheme Hilb^P_Y. It is the closed subscheme $\mathrm{Quot}^P_{\mathcal{O}_Y/Y}$ of the Grassmannian $\mathrm{Gr}\big(P(m),\, h^0(Y, \mathcal{O}_Y(m))\big)$ for $m \gg 0$, and so is projective over k.

***Proof*:**
Take $\mathcal{F} = \mathcal{O}_Y$ in definition 14.1.5. A quotient $\mathcal{O}_{Y_S} \twoheadrightarrow \mathcal{Q}$ flat over S with fiberwise Hilbert polynomial P has kernel an ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_{Y_S}$; the closed subscheme $Z \subseteq Y_S$ it defines is flat over S (its structure sheaf $\mathcal{Q} = \mathcal{O}_Z$ is S-flat by hypothesis) with fiberwise Hilbert polynomial P, and conversely such a Z yields the quotient $\mathcal{O}_{Y_S} \twoheadrightarrow \mathcal{O}_Z$. This bijection is natural in S (pullback of a subscheme corresponds to pullback of its structure quotient), so $\underline{\mathrm{Hilb}}^P_Y = \underline{\mathrm{Quot}}^P_{\mathcal{O}_Y/Y}$ as functors. By theorem 14.3.2 the right-hand functor is represented by the projective scheme $\mathrm{Quot}^P_{\mathcal{O}_Y/Y}$, which is therefore also a representing object for $\underline{\mathrm{Hilb}}^P_Y$; set $\mathrm{Hilb}^P_Y := \mathrm{Quot}^P_{\mathcal{O}_Y/Y}$, completing the proof.

The corollary is the theorem stated and deferred upstream in theorem 3.3.11, now discharged: the Hilbert scheme exists and is projective. The identification of a quotient of $\mathcal{O}_Y$ with a closed subscheme is the reason the Hilbert functor is the geometrically most natural instance of the Quot functor, and it is why Hilb^P_Y, not the more general $\mathrm{Quot}^P_{\mathcal{F}/Y}$, is the object one meets first in practice. Every projective variety with Hilbert polynomial P, embedded in the fixed Y, is a point of Hilb^P_Y; the scheme organizes all such subschemes into a single projective parameter space, and its points can degenerate to non-reduced or reducible subschemes, which is exactly the extra generality flat families give over classical varieties.

The construction produced not only the representing scheme but, by the very meaning of representability, a universal family living over it. Representability says $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \cong h_{\mathrm{Quot}^P_{\mathcal{F}/Y}}$, and the identity morphism of $\mathrm{Quot}^P_{\mathcal{F}/Y}$ corresponds under this isomorphism to a distinguished quotient over the Quot scheme itself. Naming it is the last piece of the construction, because it is the object from which every family is obtained by pullback.

Definition 14.3.4: The Universal Quotient and Universal Subscheme

Let $\mathrm{Quot}^P_{\mathcal{F}/Y}$ represent $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ (theorem 14.3.2), and write $Q = \mathrm{Quot}^P_{\mathcal{F}/Y}$. The *universal quotient* is the surjection

$$u\colon \mathcal{F}_Q \twoheadrightarrow \mathcal{U}$$

of coherent sheaves on $Y_Q = Y \times_k Q$, flat over Q with fiberwise Hilbert polynomial P, that corresponds under the isomorphism $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \cong h_Q$ of theorem 14.3.2 to the identity morphism $\mathrm{id}_Q \in h_Q(Q)$. In the Hilbert case $\mathcal{F} = \mathcal{O}_Y$, the kernel of u is the ideal sheaf of a closed subscheme

$$\mathcal{Z} \subseteq Y \times_k \mathrm{Hilb}^P_Y$$

flat over Hilb^P_Y with fiberwise Hilbert polynomial P, called the *universal subscheme*, and $\mathcal{U} = \mathcal{O}_{\mathcal{Z}}$ is its structure sheaf.

The universal property is what earns the name, and it is worth stating explicitly because it is used constantly. For every k-scheme S and every Quot-family $q\colon \mathcal{F}_S \twoheadrightarrow \mathcal{Q}$ (flat over S, fiberwise Hilbert polynomial P), there is a unique morphism $\varphi_q\colon S \to \mathrm{Quot}^P_{\mathcal{F}/Y}$ such that q is the pullback of the universal quotient:

$$(\mathrm{id}_Y \times \varphi_q)^* u = q$$

as quotients of $\mathcal{F}_S$. Every family is a pullback of the single universal family, in exactly one way. In the Hilbert case this reads: every flat family $Z \subseteq Y_S$ of closed subschemes with Hilbert polynomial P is the pullback $Z = (\mathrm{id}_Y \times \varphi)^{-1}(\mathcal{Z})$ of the universal subscheme along a unique classifying morphism $\varphi\colon S \to \mathrm{Hilb}^P_Y$. This is precisely the statement that Hilb^P_Y is a *fine* moduli space in the sense of definition 13.1.3: the universal subscheme is a single geometric object over Hilb^P_Y from which all families are cut by pullback, exactly as the universal quotient bundle plays that role over the Grassmannian in definition 13.2.5.

The classifying morphism φ_q is not an abstraction; it can be computed. For a family q over S, φ_q is the morphism $S \to G_P \subseteq \mathrm{Gr}(P(m), N)$ classifying the rank-$P(m)$ quotient $\pi_{S*}\mathcal{F}_S(m) \twoheadrightarrow \pi_{S*}\mathcal{Q}(m)$ on degree-m sections, by the proof of theorem 14.3.2. So the universal property is the Grassmannian's universal property (theorem 13.2.4) restricted to the flat locus, and the effective content of "classify a subscheme" is "classify the $P(m)$-dimensional space of its degree-m equations". The following example makes the whole construction concrete for a family of two points on a line.

Example 14.3.5: The Hilbert Scheme of Two Points on $\mathbb{P}^1$

Take $Y = \mathbb{P}^1_k$ and $P(t) = 2$, the constant polynomial: a closed subscheme $Z \subseteq \mathbb{P}^1$ with $\chi(\mathcal{O}_Z(m)) = 2$ for all m is a length-2 subscheme, i.e. two distinct points or one double point. The claim is $\mathrm{Hilb}^2_{\mathbb{P}^1} \cong \mathbb{P}^2$, and the construction of theorem 14.3.2 exhibits the isomorphism.

A length-2 subscheme $Z \subseteq \mathbb{P}^1 = \mathrm{Proj}\, k[x_0, x_1]$ is cut out by a single homogeneous ideal, and its ideal is generated in degree 2 by one quadratic form $f = a_0 x_0^2 + a_1 x_0 x_1 + a_2 x_1^2$ (the vanishing of a nonzero binary quadric defines a length-2 subscheme, namely its two roots with multiplicity). Two forms define the same subscheme exactly when they are proportional, so the subschemes are the points $[a_0 : a_1 : a_2]$ of $\mathbb{P}^2 = \mathbb{P}(H^0(\mathbb{P}^1, \mathcal{O}(2)))$. This is the classifying data of the theorem made explicit: $H^0(\mathbb{P}^1, \mathcal{O}(2))$ is 3-dimensional, and a length-2 subscheme is the same as a 1-dimensional subspace of quadrics vanishing on it, equivalently a 2-dimensional quotient

$$H^0(\mathbb{P}^1, \mathcal{O}(2)) \twoheadrightarrow H^0(Z, \mathcal{O}_Z(2))$$

which is a point of $\mathrm{Gr}(2,3) \cong \mathbb{P}^2$ (a rank-2 quotient of a rank-3 space, dual to the line of equations). Here $m = 2$ already works: $\mathcal{O}_Z(2)$ has no higher cohomology and $h^0(\mathcal{O}_Z(2)) = 2 = P(2)$. The flattening stratification imposes no further condition, because for $Y = \mathbb{P}^1$ every 1-dimensional space of quadrics defines a flat length-2 family, so $\mathrm{Hilb}^2_{\mathbb{P}^1} = \mathrm{Gr}(2,3) = \mathbb{P}^2$ with no closed subfunctor to cut.

The universal subscheme $\mathcal{Z} \subseteq \mathbb{P}^1 \times \mathbb{P}^2$ is the incidence locus $\{([x],[a]) : a_0x_0^2 + a_1x_0x_1 + a_2x_1^2 = 0\}$, flat of degree 2 over $\mathbb{P}^2$. Its fiber over $[a] \in \mathbb{P}^2$ is the subscheme $Z_{[a]}$ cut by the quadric $[a]$: two distinct points when the discriminant $a_1^2 - 4a_0a_2 \neq 0$, and a single double point (a point with a tangent direction) on the discriminant conic $a_1^2 = 4a_0a_2$. The double points form a conic in $\mathbb{P}^2$, and the support map $Z \mapsto$ (support of Z) is the Hilbert-Chow morphism $\mathrm{Hilb}^2_{\mathbb{P}^1} = \mathbb{P}^2 \to \mathrm{Sym}^2 \mathbb{P}^1$, sending a length-2 subscheme to its support cycle; here $\mathrm{Sym}^2 \mathbb{P}^1$ is itself a $\mathbb{P}^2$, presented as the S_2-quotient $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathrm{Sym}^2 \mathbb{P}^1 = \mathbb{P}^2$ of the ordered pairs of points, so the Hilbert-Chow map is an isomorphism between these two copies of $\mathbb{P}^2$. This morphism will reappear in section 14.5. So the abstract construction returns, in this case, the classical parameter space of binary quadratic forms.

The example is the pattern in miniature: choose m (here $m = 2$), read the subscheme off its space of degree-m equations, land in a Grassmannian, and check the flatness condition (here vacuous). In nontrivial cases the flatness cut is a proper closed condition, but the shape of the argument does not change. The existence theorem, that the Hilbert and Quot functors are representable by projective schemes carrying universal families, is now established in full.

Exercise 14.3.1

1. Let $Y = \mathbb{P}^n_k$ and let P be the constant polynomial $P(t) = 1$, so $\underline{\mathrm{Hilb}}^1_{\mathbb{P}^n}$ parametrizes length-1 subschemes, i.e. reduced points. Using the construction of theorem 14.3.2, identify $\mathrm{Hilb}^1_{\mathbb{P}^n}$ explicitly as a familiar projective scheme, and describe the universal subscheme $\mathcal{Z} \subseteq \mathbb{P}^n \times \mathrm{Hilb}^1_{\mathbb{P}^n}$. (Hint: a length-1 subscheme is a k-point of $\mathbb{P}^n$; compare with example 14.3.5 and the case $\mathrm{Gr}(1, n+1)$.)

2. In the proof of theorem 14.3.2 the universal coherent quotient $\widetilde{\mathcal{Q}}$ on the full Grassmannian G is asserted to be coherent but not in general flat over G. Explain, using the example $Y = \mathbb{P}^1$, $\mathcal{F} = \mathcal{O}_{\mathbb{P}^1}$, $P(t) = 2$, why some surjection $H^0(\mathcal{O}(2)) \twoheadrightarrow V$ onto a 2-dimensional space might fail to sheafify to a flat length-2 quotient, or argue that in this specific example every such surjection does succeed and so $G_P = G$. Relate your answer to why $\mathrm{Hilb}^2_{\mathbb{P}^1} = \mathbb{P}^2$ with no closed condition imposed.

3. Prove that $\mathrm{Quot}^P_{\mathcal{F}/Y}$ is separated and of finite type over k directly from theorem 14.3.2, without invoking projectivity: deduce both from the fact that it is a closed subscheme of a Grassmannian, and state which property of the Grassmannian each inherits from.

4. Let $Y = \mathbb{P}^n_k$ and $P(t) = \binom{t+n}{n} - \binom{t+n-d}{n}$, the Hilbert polynomial of a degree-d hypersurface in $\mathbb{P}^n$. Show that the universal property of definition 14.3.4 identifies $\mathrm{Hilb}^P_{\mathbb{P}^n}$ with the projective space $\mathbb{P}(H^0(\mathbb{P}^n, \mathcal{O}(d)))$ of degree-d forms, and describe the classifying morphism $\varphi\colon S \to \mathrm{Hilb}^P_{\mathbb{P}^n}$ associated to a flat family of hypersurfaces over S. (You may cite the computation of P for a hypersurface; the content is the identification of the parameter space and the universal family.)

5. The natural transformation $\tau\colon \underline{\mathrm{Quot}}^P_{\mathcal{F}/Y} \to \underline{\mathrm{Gr}}(P(m), N)$ of theorem 14.3.2 depends on the choice of $m \geq m_0$. Show that the representing scheme $\mathrm{Quot}^P_{\mathcal{F}/Y}$ does not depend on this choice:

prove that for two admissible values m, m' the two closed subschemes $G_P \subseteq \mathrm{Gr}(P(m), N)$ and $G'_P \subseteq \mathrm{Gr}(P(m'), N')$ are canonically isomorphic, and explain why this is forced by representability rather than needing a separate verification.

6. Explain why the universal subscheme $\mathcal{Z} \subseteq Y \times \mathrm{Hilb}_Y^P$ of definition 14.3.4 being flat over Hilb_Y^P is not an extra hypothesis but a consequence of the construction, and identify where in the proof of theorem 14.3.2 this flatness is arranged. Contrast this with the universal coherent quotient $\widetilde{\mathcal{Q}}$ on the ambient Grassmannian, which is not flat.

14.4 The Tangent Space and Local Structure

The previous section produced the Hilbert scheme Hilb_Y^P as a projective scheme representing the functor $\underline{\mathrm{Hilb}}_Y^P$ (corollary 14.3.3), together with its universal subscheme $\mathcal{Z}$ (definition 14.3.4). Once a moduli problem is represented by a geometric object, the payoff is that all of the object's geometry can be read off functorially. The most immediate piece of geometry is the local structure at a point: the tangent space at a point $[Z]$, and, one degree deeper, whether the scheme is smooth there and of what dimension. This section carries out the first such computation, and it is the computation that opens the door to all of deformation theory.

The bridge is a single principle already available from the functor-of-points tangent space of definition 13.5.1: because Hilb_Y^P represents a functor, its tangent space at a k-point $[Z]$ is not a differential-geometric object to be constructed by hand but a set of maps out of the dual numbers, computed directly from the functor. A tangent vector at $[Z]$ is a family over $\operatorname{Spec} k[\varepsilon]$ whose special fiber is Z, that is, a way of pushing the subscheme Z infinitesimally inside Y. The content of the section is that these infinitesimal pushes are exactly the global sections of one sheaf on Z, the normal sheaf, and that the obstruction to thickening such a push to a genuine curve of deformations lives in the next cohomology group of the same sheaf. This dictionary between the geometry of Hilb_Y^P and the cohomology of $N_{Z/Y}$ is the prototype for the tangent-obstruction formalism that the deformation-theory Part of this book develops systematically.

The object that governs how Z sits inside Y to first order is the normal sheaf. For a closed subscheme it is built from the ideal sheaf, and the construction is a direct scheme-theoretic analogue of the normal bundle of a submanifold: the conormal sheaf $\mathcal{I}_Z/\mathcal{I}_Z^2$ records first-order functions vanishing on Z, and its dual records first-order directions transverse to Z.

Definition 14.4.1: Normal Sheaf of a Closed Subscheme

Let $Z \subseteq Y$ be a closed subscheme of a scheme Y, with ideal sheaf $\mathcal{I}_Z \subseteq \mathcal{O}_Y$. The *conormal sheaf* of Z in Y is the $\mathcal{O}_Z$-module

$$\mathcal{N}^\vee_{Z/Y} \;=\; \mathcal{I}_Z/\mathcal{I}_Z^2$$

where the $\mathcal{O}_Z = \mathcal{O}_Y/\mathcal{I}_Z$-module structure is the one inherited from the $\mathcal{O}_Y$-module structure on $\mathcal{I}_Z$ (multiplication by $\mathcal{I}_Z$ acts as zero on the quotient, so the action factors through $\mathcal{O}_Z$). The *normal sheaf* of Z in Y is its $\mathcal{O}_Z$-dual,

$$N_{Z/Y} \;=\; \mathcal{H}om_{\mathcal{O}_Z}\big(\mathcal{I}_Z/\mathcal{I}_Z^2,\, \mathcal{O}_Z\big)$$

a coherent $\mathcal{O}_Z$-module.

The conormal sheaf $\mathcal{I}_Z/\mathcal{I}_Z^2$ algebraizes the classical conormal bundle: a local defining function $f \in \mathcal{I}_Z$ has a well-defined class modulo $\mathcal{I}_Z^2$, and that class is precisely its first-order behaviour transverse to Z, its "differential in the normal directions." Dualizing turns cotangent data into tangent data, so a section of $N_{Z/Y}$ over an open set assigns to each local equation of Z a function on Z, subject to the Leibniz-type constraint of being $\mathcal{O}_Z$-linear. The interpretation to keep in mind throughout the section is that such a section is a recipe for perturbing each defining equation $f \mapsto f + \varepsilon\,\varphi(f)$ to first order, consistently across overlaps. When Z and Y are both smooth, the normal sheaf is locally free of rank $\dim Y - \dim Z$ and reduces to the normal bundle of differential geometry; in general it can fail to be locally free, and that failure is exactly where the local geometry of Hilb^P_Y becomes interesting.

Two computations make the definition concrete and will be reused below.

Example 14.4.2: The Normal Sheaf of a Hypersurface

Let Y be a scheme and let $Z = V(f) \subseteq Y$ be an effective Cartier divisor, the zero scheme of a single regular function f (or, locally, a nonzerodivisor) generating $\mathcal{I}_Z$. Then $\mathcal{I}_Z \cong \mathcal{O}_Y(-Z)$ is an invertible sheaf, and multiplication by f gives an isomorphism $\mathcal{O}_Z \xrightarrow{\sim} \mathcal{I}_Z/\mathcal{I}_Z^2$ sending $1 \mapsto \bar{f}$. Consequently

$$\mathcal{I}_Z/\mathcal{I}_Z^2 \;\cong\; \mathcal{O}_Y(-Z)\big|_Z, \qquad N_{Z/Y} \;\cong\; \mathcal{O}_Y(Z)\big|_Z$$

the restriction to Z of the line bundle $\mathcal{O}_Y(Z)$. For the concrete case $Y = \mathbb{P}^n$ and $Z \subseteq \mathbb{P}^n$ a hypersurface of degree d, the normal sheaf is $N_{Z/\mathbb{P}^n} \cong \mathcal{O}_{\mathbb{P}^n}(d)\big|_Z = \mathcal{O}_Z(d)$. A global section of $N_{Z/Y}$ is then a function on Z scaling the single equation f, matching the intuition that a hypersurface is deformed to first order by perturbing its one defining equation.

Example 14.4.3: The Normal Sheaf of a Reduced Point

Let Y be a smooth n-dimensional variety over k and let $Z = \{p\}$ be a single reduced k-point, so $\mathcal{O}_Z = k$. Choosing a regular system of parameters $t_1, \dots, t_n$ at p, the ideal $\mathcal{I}_Z = \mathfrak{m}_p$ satisfies $\mathfrak{m}_p/\mathfrak{m}_p^2 \cong k^{\oplus n}$, the Zariski cotangent space at p. Hence

$$N_{Z/Y} \;=\; \mathrm{Hom}_k\big(\mathfrak{m}_p/\mathfrak{m}_p^2,\, k\big) \;\cong\; T_pY$$

the Zariski tangent space of Y at p, an n-dimensional k-vector space. A global section is a single tangent vector at p, matching the intuition that one moves a point inside Y by choosing a direction to move it in.

With the normal sheaf in hand, the tangent space of the Hilbert scheme can be computed. Recall from definition 13.5.1 that for a functor $F\colon \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ with a distinguished point $\xi \in F(\operatorname{Spec} k)$, the tangent space is the fiber

$$T_\xi F \;=\; \left\{\, \eta \in F(\operatorname{Spec} k[\varepsilon]) \;\middle|\; \eta \mapsto \xi \text{ under } F(\operatorname{Spec} k[\varepsilon]) \to F(\operatorname{Spec} k) \,\right\}$$

where $k[\varepsilon] = k[t]/(t^2)$ is the ring of dual numbers and the map is induced by $k[\varepsilon] \to k$, $\varepsilon \mapsto 0$; proposition 13.5.2 equips this set with a canonical k-vector space structure. When $F = \underline{\mathrm{Hilb}}_Y^P$ is representable by Hilb_Y^P (corollary 14.3.3), the Yoneda lemma identifies $T_\xi F$ with the Zariski tangent space $T_{[Z]} \operatorname{Hilb}_Y^P = (\mathfrak{m}_{[Z]}/\mathfrak{m}_{[Z]}^2)^\vee$, so the functor computation *is* the scheme computation. Everything now reduces to unwinding what an element of $\underline{\mathrm{Hilb}}_Y^P(\operatorname{Spec} k[\varepsilon])$ restricting to $[Z]$ is.

Theorem 14.4.4: Tangent Space of the Hilbert Scheme

Let $Y \subseteq \mathbb{P}_k^n$ be a projective scheme and let $[Z] \in \operatorname{Hilb}_Y^P(k)$ be the point corresponding to a closed subscheme $Z \subseteq Y$ with Hilbert polynomial P. There is a canonical isomorphism of k-vector spaces

$$T_{[Z]} \operatorname{Hilb}_Y^P \;\cong\; H^0\big(Z,\, N_{Z/Y}\big)$$

between the Zariski tangent space of Hilb_Y^P at $[Z]$ and the global sections of the normal sheaf. The same statement holds for the Quot scheme with $N_{Z/Y}$ replaced by $\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{K}, \mathcal{Q})$, where $\mathcal{K}$ is the kernel of the universal quotient $\mathcal{F} \twoheadrightarrow \mathcal{Q}$.

***Proof*:**
By representability (corollary 14.3.3) and the Yoneda identification recalled above, it suffices to compute $\underline{\mathrm{Hilb}}_Y^P(\operatorname{Spec} k[\varepsilon])$ over $[Z]$ and exhibit a canonical bijection to $H^0(Z, N_{Z/Y})$ that respects the vector space structures.

Step 1: Identify a tangent vector with a flat first-order deformation. By definition of the Hilbert functor (definition 14.1.6), an element of $\underline{\mathrm{Hilb}}_Y^P(\operatorname{Spec} k[\varepsilon])$ is a closed subscheme

$$\widetilde{Z} \;\subseteq\; Y \times_k \operatorname{Spec} k[\varepsilon] \;=\; Y_\varepsilon$$

flat over $\operatorname{Spec} k[\varepsilon]$ with fiberwise Hilbert polynomial P. Restricting along the closed immersion $\operatorname{Spec} k \hookrightarrow \operatorname{Spec} k[\varepsilon]$ (cutting by $\varepsilon = 0$) recovers the special fiber; requiring the tangent vector to lie over $[Z]$ is requiring this special fiber to be Z. Thus a tangent vector is exactly a flat family $\widetilde{Z} \subseteq Y_\varepsilon$ restricting to $Z \subseteq Y$ over the closed point, that is, a *first-order deformation of Z inside Y*.

Step 2: Translate flatness into a splitting of ideals. The subscheme $\widetilde{Z}$ is determined by its ideal sheaf $\mathcal{I}_{\widetilde{Z}} \subseteq \mathcal{O}_{Y_\varepsilon}$. Because $\mathcal{O}_{Y_\varepsilon} = \mathcal{O}_Y \otimes_k k[\varepsilon] = \mathcal{O}_Y \oplus \varepsilon\mathcal{O}_Y$ as an $\mathcal{O}_Y$-module, and the special fiber of $\widetilde{Z}$ is Z, the multiplication-by-ε sequence

$$0 \longrightarrow \mathcal{O}_Y \xrightarrow{\;\cdot\,\varepsilon\;} \mathcal{O}_{Y_\varepsilon} \longrightarrow \mathcal{O}_Y \longrightarrow 0$$

is exact. The key point is that $\widetilde{Z}$ is flat over $k[\varepsilon]$ if and only if tensoring the structure sequence $0 \to \mathcal{I}_{\widetilde{Z}} \to \mathcal{O}_{Y_\varepsilon} \to \mathcal{O}_{\widetilde{Z}} \to 0$ with the residue field $k = k[\varepsilon]/(\varepsilon)$ keeps it exact on the left, that is, if and only if $\operatorname{Tor}_1^{k[\varepsilon]}(\mathcal{O}_{\widetilde{Z}}, k) = 0$. Unwinding the standard criterion for flatness over the dual numbers (a $k[\varepsilon]$-module M is flat if and only if the multiplication map $\varepsilon\colon M/\varepsilon M \to \varepsilon M$ is an isomorphism), flatness of $\mathcal{O}_{\widetilde{Z}}$ is equivalent to the requirement that

$$\varepsilon \cdot \mathcal{O}_{\widetilde{Z}} \;\cong\; \mathcal{O}_{\widetilde{Z}}/\varepsilon\,\mathcal{O}_{\widetilde{Z}} \;=\; \mathcal{O}_Z$$

that is, that reduction mod ε induces an isomorphism $\varepsilon\mathcal{O}_{\widetilde{Z}} \cong \mathcal{O}_Z$. Concretely: locally, if $\mathcal{I}_Z$ is generated by $f_1, \dots, f_r$, a flat lift is generated by elements $f_i + \varepsilon g_i$ with $g_i \in \mathcal{O}_Y$, and flatness forces the ambiguity in the g_i to be exactly by elements of $\mathcal{I}_Z$.

Step 3: Assemble the local data into a section of the normal sheaf. A flat first-order deformation, described locally by generators $f_i + \varepsilon g_i$ of $\mathcal{I}_{\widetilde{Z}}$, defines an $\mathcal{O}_Y$-linear map

$$\varphi\colon \mathcal{I}_Z \longrightarrow \mathcal{O}_Z, \qquad f_i \longmapsto \bar{g}_i$$

as follows. Given $f = \sum a_i f_i \in \mathcal{I}_Z$ with $a_i \in \mathcal{O}_Y$, the corresponding lift is $\sum a_i(f_i + \varepsilon g_i) = f + \varepsilon \sum a_i g_i$, so the "first-order part" is $\sum a_i g_i$, read modulo $\mathcal{I}_Z$ to land in $\mathcal{O}_Z$. Two facts must be checked, and both follow from flatness as pinned down in Step 2. First, φ is well defined: if $\sum a_i f_i = 0$ then the syzygy must be respected to first order, which is exactly the statement that $\sum a_i g_i \in \mathcal{I}_Z$, so φ does not depend on the presentation. Second, φ kills $\mathcal{I}_Z^2$: for $f, f' \in \mathcal{I}_Z$ the product ff' lifts to $(f + \varepsilon(\cdots))(f' + \varepsilon(\cdots)) = ff' + \varepsilon(f\cdot(\cdots) + f'\cdot(\cdots))$, whose first-order part lies in $\mathcal{I}_Z$ and so maps to 0 in $\mathcal{O}_Z$. Therefore φ descends to an $\mathcal{O}_Z$-linear map

$$\varphi\colon \mathcal{I}_Z/\mathcal{I}_Z^2 \longrightarrow \mathcal{O}_Z$$

that is, a global section $\varphi \in H^0(Z, N_{Z/Y})$. Geometrically φ is precisely the "perturb each equation $f \mapsto f + \varepsilon\,\varphi(f)$" recipe anticipated after definition 14.4.1.

Step 4: Invert the construction. Conversely, a global section $\varphi \in H^0(Z, N_{Z/Y}) = \mathrm{Hom}_{\mathcal{O}_Z}(\mathcal{I}_Z/\mathcal{I}_Z^2, \mathcal{O}_Z)$ produces a first-order deformation. Lift φ to an $\mathcal{O}_Y$-linear map $\mathcal{I}_Z \to \mathcal{O}_Y$ (choosing local representatives $g_i \in \mathcal{O}_Y$ of $\varphi(\bar{f}_i)$) and set, locally,

$$\mathcal{I}_{\widetilde{Z}} \;=\; \big(\, f_i + \varepsilon\, g_i \;:\; i = 1, \dots, r \,\big) \;\subseteq\; \mathcal{O}_{Y_\varepsilon}$$

That φ vanishes on $\mathcal{I}_Z^2$ guarantees these local ideals patch to a global ideal sheaf $\mathcal{I}_{\widetilde{Z}}$ independent of the lift of φ (a different lift changes each g_i by an element of $\mathcal{I}_Z$, which does not change the ideal $(f_i + \varepsilon g_i)$ since $\varepsilon\mathcal{I}_Z \subseteq \mathcal{I}_{\widetilde{Z}}$ already). The resulting $\widetilde{Z}$ is flat over $k[\varepsilon]$ by the criterion of Step 2 and restricts to Z over $\varepsilon = 0$. Since $\widetilde{Z} \to \operatorname{Spec} k[\varepsilon]$ is flat with special fiber of Hilbert polynomial P, the Hilbert polynomial is preserved (proposition 14.1.4), so $\widetilde{Z} \in \underline{\mathrm{Hilb}}_Y^P(\operatorname{Spec} k[\varepsilon])$. This is inverse to the construction of Step 3.

Step 5: Compatibility with the vector space structures. The zero section $\varphi = 0$ gives $\mathcal{I}_{\widetilde{Z}} = \mathcal{I}_Z \otimes_k k[\varepsilon]$, the trivial (constant) deformation, which is the base point of the tangent space. Under the $k[\varepsilon]$-module addition used to define the vector space structure in proposition 13.5.2, adding two deformations corresponds to adding the associated maps $\mathcal{I}_Z \to \mathcal{O}_Z$, and scaling $\varepsilon \mapsto c\varepsilon$ for $c \in k$ scales φ by c; both are immediate from the local formula $f_i \mapsto f_i + \varepsilon g_i$. Hence the bijection of Steps 3 and 4 is k-linear, giving the claimed isomorphism $T_{[Z]} \operatorname{Hilb}_Y^P \cong H^0(Z, N_{Z/Y})$.

For the Quot scheme, a point corresponds to a quotient $\mathcal{F} \twoheadrightarrow \mathcal{Q}$ with kernel $\mathcal{K}$, and the identical argument, replacing "ideal sheaf $\mathcal{I}_Z$" by "kernel subsheaf $\mathcal{K}$" and "$\mathcal{O}_Z$" by "$\mathcal{Q}$", identifies first-order deformations of the quotient with $\mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{K}, \mathcal{Q})$; the Hilbert case is the special case $\mathcal{F} = \mathcal{O}_Y$, $\mathcal{Q} = \mathcal{O}_Z$, $\mathcal{K} = \mathcal{I}_Z$, where $\mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Z, \mathcal{O}_Z) = \mathrm{Hom}_{\mathcal{O}_Z}(\mathcal{I}_Z/\mathcal{I}_Z^2, \mathcal{O}_Z) = H^0(N_{Z/Y})$ because any $\mathcal{O}_Y$-map to the $\mathcal{O}_Z$-module $\mathcal{O}_Z$ automatically kills $\mathcal{I}_Z \cdot \mathcal{I}_Z$, completing the proof.

The theorem is the first theorem of deformation theory in the book, and it should be read as a template rather than an isolated computation. Its shape recurs throughout the deformation-theory Part: a moduli functor is probed by maps out of $\operatorname{Spec} k[\varepsilon]$, an element of $F(\operatorname{Spec} k[\varepsilon])$ over a fixed point is unwound into linear-algebraic data on the object itself, and that data organizes into the global sections of a coherent sheaf (or, more generally, into a cohomology group). Here the sheaf is $N_{Z/Y}$ and the group is H^0; for deformations of an abstract smooth variety X the sheaf will be the tangent sheaf T_X and the group $H^1(X, T_X)$; for deformations of a coherent sheaf $\mathcal{F}$ it will be $\operatorname{Ext}^1(\mathcal{F}, \mathcal{F})$ ([15]). In the normal-sheaf computation the ambient Y pins everything down, so a first-order deformation is packaged by H^0 of a coherent sheaf alone, with no higher Ext term required.

Two features of the canonical isomorphism deserve emphasis. First, it is entirely local-to-global: whether a first-order deformation exists at all is a purely local question (the Hom sheaf $N_{Z/Y}$ is defined stalk by stalk), whereas the tangent space measures which local perturbations glue to a global one, and that gluing is exactly the passage from the sheaf to its space of global sections. Second, the isomorphism is canonical, so it transports functoriality: a morphism of Hilbert schemes induced by an inclusion $Y \hookrightarrow Y'$ corresponds, on tangent spaces, to the natural map $H^0(N_{Z/Y}) \to H^0(N_{Z/Y'})$ coming from the surjection $\mathcal{I}_{Z/Y'} \twoheadrightarrow \mathcal{I}_{Z/Y}$ of conormal data.

The tangent space measures first-order deformations, but not every first-order deformation extends to an actual curve of subschemes. A tangent vector is a family over $\operatorname{Spec} k[\varepsilon] = \operatorname{Spec} k[t]/(t^2)$; extending it means finding a compatible family over $\operatorname{Spec} k[t]/(t^3)$, then $\operatorname{Spec} k[t]/(t^4)$, and so on, ultimately over a smooth curve or a power series ring. The failure to extend from order n to order $n+1$ is measured by a class in a second cohomology group, the obstruction space. For the Hilbert scheme this space is $H^1(Z, N_{Z/Y})$, and its vanishing forces Hilb^P_Y to be as smooth as its tangent space predicts.

Proposition 14.4.5: The Obstruction Space and Smoothness of the Hilbert Scheme

Let $[Z] \in \operatorname{Hilb}^P_Y(k)$ with $Z \subseteq Y$ as above.

1. Fix $n \geq 2$ and let $\widetilde{Z}_n$ be a flat deformation of Z over $\operatorname{Spec} k[t]/(t^n)$. There is a canonical *obstruction class*
$$\operatorname{ob}(\widetilde{Z}_n) \;\in\; H^1\big(Z, N_{Z/Y}\big)$$
that vanishes if and only if $\widetilde{Z}_n$ extends to a flat deformation over $\operatorname{Spec} k[t]/(t^{n+1})$; when it vanishes, the set of such extensions is a torsor under $H^0(Z, N_{Z/Y})$.

2. If $H^1(Z, N_{Z/Y}) = 0$, then Hilb^P_Y is smooth at $[Z]$ of dimension
$$\dim_{[Z]} \operatorname{Hilb}^P_Y \;=\; h^0\big(Z, N_{Z/Y}\big) \;=\; \dim_k H^0(Z, N_{Z/Y})$$

The obstruction machinery in its full generality, including the precise construction of $\operatorname{ob}(\widetilde{Z}_n)$ as the class of a Čech-type cocycle recording the discrepancy between local lifts, belongs to the deformation theory developed in the next Part of this book; the general obstruction calculus and the proof that vanishing H^1 yields formal smoothness are given there in the setting of arbitrary deformation functors. The statement is recorded here because it is the payoff of the tangent-space computation for the reader's immediate use: it converts a cohomological vanishing into a concrete geometric conclusion about Hilb^P_Y. The mechanism to keep in mind is the "lift-and-compare" picture. Cover Z by opens on which the order-n deformation extends to order $n+1$ (each local piece extends because the local rings are smooth enough, or by the local triviality of the normal-sheaf data); on overlaps two such local extensions differ by a section of $N_{Z/Y}$; the resulting collection of differences is a 1-cocycle whose class in $H^1(Z, N_{Z/Y})$ is ob. The global extension exists precisely when this class is a coboundary, that is, when the local extensions can be adjusted to agree on overlaps. When H^1 vanishes there is no obstruction at any order, the deformation extends to all orders and, by projectivity and the formal criterion, Hilb_Y^P is smooth at $[Z]$; its dimension is then the dimension of the tangent space, $h^0(N_{Z/Y})$.

The proposition also explains why the Hilbert scheme is so often singular. There is no upper bound on $\dim H^1(N_{Z/Y})$ across all subschemes with a fixed Hilbert polynomial, and when it is nonzero the actual dimension of Hilb_Y^P at $[Z]$ can be strictly smaller than $h^0(N_{Z/Y})$, with the tangent space overcounting deformations that do not integrate. This is the algebraic origin of the pathologies of Hilbert schemes: Mumford's example (in his 1962 paper "Further pathologies in algebraic geometry", *Amer. J. Math.* **84**) of a component of the Hilbert scheme of space curves in $\mathbb{P}^3$ that is everywhere non-reduced, and Vakil's "Murphy's law" theorem (from "Murphy's law in algebraic geometry: Badly-behaved deformation spaces", *Invent. Math.* **164** (2006)) that Hilbert schemes realize arbitrarily bad singularity types, both trace to a large obstruction space $H^1(N_{Z/Y})$ obstructing the integration of tangent vectors. The clean cases below are exactly those where H^1 vanishes for dimensional reasons.

The computation becomes concrete when the normal sheaf splits or is a line bundle, so that its cohomology is computable by hand. The following worked case treats a complete intersection, where the conormal sequence is as simple as possible, and specializes to a finite set of points, the running touchstone of this chapter.

Example 14.4.6: Tangent Space of a Complete Intersection and of Points

Let $Y = \mathbb{P}^n_k$ and let $Z = V(f_1, \ldots, f_c) \subseteq \mathbb{P}^n$ be a smooth complete intersection of c hypersurfaces of degrees $d_1, \ldots, d_c$, so Z has codimension c and dimension $n - c$. The Koszul resolution shows that Z is regularly embedded, so the conormal sheaf is locally free and splits as a direct sum of the individual conormal line bundles:

$$\mathcal{I}_Z/\mathcal{I}_Z^2 \;\cong\; \bigoplus_{i=1}^{c} \mathcal{O}_Z(-d_i)$$

each summand generated by the class of f_i (example 14.4.2 applied to each equation). Dualizing,

$$N_{Z/\mathbb{P}^n} \;\cong\; \bigoplus_{i=1}^{c} \mathcal{O}_Z(d_i)$$

and the tangent space to the Hilbert scheme is the direct sum of the sections of these line bundles:

$$T_{[Z]} \mathrm{Hilb}^P_{\mathbb{P}^n} \;\cong\; H^0\big(Z, N_{Z/\mathbb{P}^n}\big) \;\cong\; \bigoplus_{i=1}^{c} H^0\big(Z, \mathcal{O}_Z(d_i)\big)$$

Each summand $H^0(Z, \mathcal{O}_Z(d_i))$ is the space of degree-d_i perturbations of the i-th equation, restricted to Z: a first-order deformation of a complete intersection is precisely a first-order wiggle of each defining polynomial, and $h^0(\mathcal{O}_Z(d_i))$ counts the independent such wiggles modulo those that do not move Z.

Two specializations make this quantitative.

1. *A single hypersurface.* Take $c = 1$, so $Z \subseteq \mathbb{P}^n$ is a smooth hypersurface of degree d. Then $N_{Z/\mathbb{P}^n} = \mathcal{O}_Z(d)$ and, from the exact sequence $0 \to \mathcal{O}_{\mathbb{P}^n}(d-d) = \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_{\mathbb{P}^n}(d) \to \mathcal{O}_Z(d) \to 0$

together with $H^1(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}) = 0$, one reads off

$$h^0(N_{Z/\mathbb{P}^n}) \;=\; h^0(\mathbb{P}^n, \mathcal{O}(d)) - 1 \;=\; \binom{n+d}{d} - 1$$

Moreover $H^1(Z, \mathcal{O}_Z(d)) = 0$ for a smooth hypersurface (again from the same sequence, since $H^1(\mathcal{O}_{\mathbb{P}^n}(d)) = H^2(\mathcal{O}_{\mathbb{P}^n}) = 0$), so proposition 14.4.5 gives that $\mathrm{Hilb}^P_{\mathbb{P}^n}$ is smooth at $[Z]$ of dimension $\binom{n+d}{d} - 1$. This is exactly the projective space $\mathbb{P}\big(H^0(\mathbb{P}^n, \mathcal{O}(d))\big) = \mathbb{P}^{\binom{n+d}{d}-1}$ of degree-d hypersurfaces, the identification developed in full in example 14.5.2.

2. *A finite set of points.* Take $Z = \{p_1, \ldots, p_r\} \subseteq \mathbb{P}^n$ a reduced set of r distinct k-points, with constant Hilbert polynomial $P = r$. By example 14.4.3, $N_{Z/\mathbb{P}^n} = \bigoplus_{j=1}^r T_{p_j}\mathbb{P}^n$ is a skyscraper sheaf, so

$$H^0(Z, N_{Z/\mathbb{P}^n}) \;=\; \bigoplus_{j=1}^{r} T_{p_j}\mathbb{P}^n, \qquad H^1(Z, N_{Z/\mathbb{P}^n}) = 0$$

the latter because Z is 0-dimensional and coherent cohomology of a sheaf on a 0-dimensional scheme vanishes above degree 0. Hence $\mathrm{Hilb}^r_{\mathbb{P}^n} = \mathrm{Hilb}^P_{\mathbb{P}^n}$ is smooth at $[Z]$ of dimension

$$h^0(N_{Z/\mathbb{P}^n}) \;=\; \sum_{j=1}^{r} \dim T_{p_j}\mathbb{P}^n \;=\; rn$$

This matches the expectation that r distinct points in $\mathbb{P}^n$ move in an rn-dimensional family, each point contributing its n tangent directions independently; the reduced-points locus is a smooth open subscheme of $\mathrm{Hilb}^r_{\mathbb{P}^n}$ of dimension rn. The subtler geometry, where points collide and the tangent space jumps, is the subject of example 14.5.1.

The two clean cases share a structural reason for their smoothness: in both, the obstruction group $H^1(N_{Z/Y})$ vanishes, so the Hilbert scheme is exactly as smooth as the tangent space predicts, and its dimension is computed by a single cohomology count. Vanishing H^1 thus reduces the local geometry of Hilb^P_Y at $[Z]$ to a single H^0 computation, with no correction from higher obstructions. This also isolates the precise input needed to construct many moduli spaces as smooth varieties: control of H^1 of a normal or endomorphism sheaf. The same control, transported to the tangent sheaf T_C of a curve, is what makes the moduli of curves smooth of dimension $3g-3$ in the deformation-theory Part and the capstone, and it is the normal-sheaf computation of theorem 14.4.4 that first exhibits the pattern.

Exercise 14.4.1

1. Let $Z = V(xy) \subseteq \mathbb{A}^2_k$ be the union of the two coordinate axes (the node at the origin, viewed as a subscheme of the plane), with $\mathcal{I}_Z = (xy)$. Compute the normal sheaf $N_{Z/\mathbb{A}^2}$ and describe $H^0(Z, N_{Z/\mathbb{A}^2})$ as a k-vector space. Interpret a first-order deformation of Z that does *not* come from moving the two lines rigidly, and explain in one sentence why the origin is the interesting point.

2. Let $Z \subseteq \mathbb{P}^3_k$ be a smooth conic (a degree-2 curve of arithmetic genus 0) lying in a plane $H \cong \mathbb{P}^2$. Compute $N_{Z/\mathbb{P}^3}$, its global sections $h^0(N_{Z/\mathbb{P}^3})$, and $h^1(N_{Z/\mathbb{P}^3})$, and conclude that the Hilbert scheme of conics in $\mathbb{P}^3$ is smooth at $[Z]$; identify its dimension. (Hint: use the normal-bundle sequence $0 \to N_{Z/H} \to N_{Z/\mathbb{P}^3} \to N_{H/\mathbb{P}^3}|_Z \to 0$, with $N_{Z/H} = \mathcal{O}_H(2)|_Z \cong \mathcal{O}_{\mathbb{P}^1}(4)$ from viewing

Z as a degree-2 divisor in H, and $N_{H/\mathbb{P}^3}|_Z = \mathcal{O}_H(1)|_Z \cong \mathcal{O}_{\mathbb{P}^1}(2)$.)

3. Show directly from theorem 14.4.4 that the tangent space to Hilb at the trivial subscheme $Z = Y$ (the whole ambient scheme, with $\mathcal{I}_Z = 0$) and at the empty subscheme $Z = \varnothing$ is zero in each case, and reconcile this with the fact that these are isolated reduced points of the corresponding Hilbert schemes.

4. Let $Z = \operatorname{Spec} k[x]/(x^2) \subseteq \mathbb{A}^1_k$ be the length-2 subscheme supported at the origin (a point with a tangent direction), with $\mathcal{I}_Z = (x^2) \subseteq k[x]$. Compute $\mathcal{I}_Z/\mathcal{I}_Z^2$, the normal sheaf $N_{Z/\mathbb{A}^1}$, and $h^0(N_{Z/\mathbb{A}^1})$. Compare your answer with the dimension of $\operatorname{Hilb}^2(\mathbb{A}^1)$ and explain what the two tangent directions parametrize.

5. Let $Z \subseteq \mathbb{P}^n_k$ be a smooth complete intersection of two quadrics ($c = 2$, $d_1 = d_2 = 2$) with $n \geq 3$. Using example 14.4.6, write down $N_{Z/\mathbb{P}^n}$ and compute $h^0(N_{Z/\mathbb{P}^n})$. For $n = 3$ (so Z is an elliptic curve, a smooth $(2,2)$ curve in $\mathbb{P}^3$), evaluate the dimension explicitly and comment on whether the smoothness criterion of proposition 14.4.5 applies.

14.5 Worked Hilbert Schemes

The previous sections built the Hilbert and Quot schemes in the abstract: $\underline{\operatorname{Quot}}^P_{\mathcal{F}/Y}$ is representable by a projective scheme (theorem 14.3.2), the Hilbert functor is the special case $\mathcal{F} = \mathcal{O}_Y$ (corollary 14.3.3), and the tangent space at a point $[Z]$ is $H^0(Z, N_{Z/Y})$ (theorem 14.4.4). What none of this yet supplies is the feel of these schemes as geometric objects one can hold in the hand. A representability theorem tells us Hilb^P_Y exists; it does not tell us that Hilb^n of a smooth surface is itself smooth, or that it maps to the symmetric product, or that the ideal (y, x^2) names a specific length-two subscheme sitting at the origin with a chosen tangent direction. This section closes that gap by working three families of examples all the way down to charts, ideals, and equations.

The three examples are chosen to exercise the whole chapter. The Hilbert scheme of points on a surface is where the local theory of theorem 14.4.4 does the most work: the normal-sheaf tangent space computes to exactly $2n$ everywhere, and the scheme is smooth despite parametrizing objects (fat points, tangent directions) that look singular. The Hilbert scheme of hypersurfaces is the opposite extreme, a case so rigid that the whole scheme is a single projective space read off from one Hilbert polynomial. The Grassmannian as a Quot scheme returns to section 13.2, showing that the fine moduli space built by hand there is a special case of the machine built here. Together they show the range of the theory, from a space as simple as $\mathbb{P}^N$ to one as intricate as the punctual Hilbert scheme.

The first of the three is the running touchstone of the book: the Hilbert scheme of n points on a surface. A "point" of this scheme is a length-n subscheme, and the subtlety is that length-n subschemes are not just unordered n-tuples of distinct points. When two points collide they can leave behind a tangent direction, and when more collide the bookkeeping of the limiting subscheme becomes intricate, yet the total space of all such subschemes is smooth. This is the phenomenon the example makes concrete.

Fix the Hilbert polynomial to be the constant $P = n$. A closed subscheme $Z \subseteq Y$ with constant Hilbert polynomial n is a zero-dimensional subscheme whose structure sheaf has length n over k, that is, $\dim_k H^0(Z, \mathcal{O}_Z) = n$. The Hilbert scheme $\operatorname{Hilb}^n(Y) := \operatorname{Hilb}^P_Y$ for this constant polynomial is the *Hilbert scheme of n points.* The following worked example states its smoothness and dimension for a smooth surface, gives the tangent-space argument that establishes them, constructs the Hilbert-Chow morphism, and carries the whole picture down to the explicit case $\operatorname{Hilb}^2(\mathbb{A}^2)$.

The governing fact is a theorem of Fogarty, who proved smoothness and irreducibility for a smooth surface over any algebraically closed field; the analogous statement fails in dimension three and higher, where Hilb^n acquires singular and even reducible components once n is large enough.[1] The tangent-space computation rests on the normal-sheaf identification of theorem 14.4.4, and the one input beyond the tools developed here, irreducibility of $\operatorname{Hilb}^n(Y)$, is cited to Fogarty rather than reproved.

Example 14.5.1: The Hilbert Scheme of Points on a Surface

Let Y be a smooth quasi-projective surface over k. The claim is that the Hilbert scheme of n points $\operatorname{Hilb}^n(Y)$ is smooth of dimension $2n$ over k and carries a projective *Hilbert-Chow morphism*

$$\pi\colon \operatorname{Hilb}^n(Y) \longrightarrow \operatorname{Sym}^n Y = Y^n/\mathfrak{S}_n$$

sending a length-n subscheme Z to its support counted with multiplicity, $\sum_p \operatorname{length}_p(\mathcal{O}_Z)\cdot[p]$, which is an isomorphism over the locus of n distinct points and is a resolution of singularities of $\operatorname{Sym}^n Y$.

Step 1: The tangent space at a reduced subscheme. Suppose first that $Z = \{p_1, \dots, p_n\}$ is n distinct reduced points of Y. By theorem 14.4.4 the tangent space is

$$T_{[Z]} \operatorname{Hilb}^n(Y) = H^0(Z, N_{Z/Y})$$

where $N_{Z/Y} = \mathcal{H}om(\mathcal{I}_Z/\mathcal{I}_Z^2, \mathcal{O}_Z)$ is the normal sheaf (definition 14.4.1). Since Z is a disjoint union of n reduced points on the smooth surface Y, at each p_i the ideal $\mathcal{I}_Z$ is the maximal ideal $\mathfrak{m}_{p_i}$, and $\mathfrak{m}_{p_i}/\mathfrak{m}_{p_i}^2$ is the two-dimensional cotangent space of the surface. Hence $N_{Z/Y}$ is a skyscraper sheaf whose stalk at each p_i is the two-dimensional tangent space $T_{p_i}Y$, and

$$H^0(Z, N_{Z/Y}) = \bigoplus_{i=1}^n T_{p_i} Y$$

has dimension $2n$. Each summand is the freedom to move one point in the two directions of the surface, so the reduced locus already has the expected tangent dimension $2n$.

Step 2: The local dimension is $2n$ everywhere. The locus $U \subseteq \operatorname{Hilb}^n(Y)$ of reduced subschemes (n distinct points) is open and nonempty, and it maps isomorphically to the open subset of $\operatorname{Sym}^n Y$ of distinct points; since $\operatorname{Sym}^n Y$ has dimension $2n$, so does U, and its tangent dimension is $2n$ by Step 1. The two inputs that promote this to smoothness of the whole scheme are irreducibility of $\operatorname{Hilb}^n(Y)$ (so U is dense and $\dim \operatorname{Hilb}^n(Y) = 2n$) and the tangent-dimension count of Step 3; irreducibility is the part beyond the tools here and is cited to Fogarty. Granting it, every point of $\operatorname{Hilb}^n(Y)$ is a limit of reduced configurations and $\dim \operatorname{Hilb}^n(Y) = 2n$.

Step 3: The tangent space has dimension exactly $2n$. Smoothness reduces to the equality $\dim_k H^0(Z, N_{Z/Y}) = 2n$ at every point $[Z]$: combined with $\dim_{[Z]} \operatorname{Hilb}^n(Y) = 2n$ from Step 2 and the general inequality $\dim_{[Z]} \operatorname{Hilb}^n(Y) \le \dim_k T_{[Z]} \operatorname{Hilb}^n(Y)$, agreement of the tangent

[1] Fogarty, *Algebraic families on an algebraic surface* (1968), established smoothness and irreducibility of $\operatorname{Hilb}^n(Y)$ for Y a smooth surface. In dimension ≥ 3 the scheme $\operatorname{Hilb}^n(\mathbb{A}^d)$ acquires singularities and, for n large enough, extra components beyond the closure of the reduced configurations; a tangent-space count at a sufficiently fat point already exceeds the expected dimension dn. The classical example is the length-8 non-smoothable zero-dimensional scheme of Iarrobino and Emsalem in $\mathbb{A}^4$, giving reducibility of $\operatorname{Hilb}^8(\mathbb{A}^4)$; for $\mathbb{A}^3$ irreducibility persists further and the smallest reducible case is larger and was settled only later. `% TODO: verify cite key` (Fogarty 1968, Amer. J. Math. 90).

dimension with the local dimension forces smoothness at $[Z]$. The tangent count is local: the normal sheaf $N_{Z/Y}$ is supported on the finite set underlying Z, so

$$\dim_k H^0(Z, N_{Z/Y}) = \sum_{p\in Z} \dim_k \operatorname{Hom}_{A_p}(I_p/I_p^2, A_p), \qquad A_p = \mathcal{O}_{Z,p} = \mathcal{O}_{Y,p}/I_p$$

and it suffices to show each local term equals $2\operatorname{length}_k(A_p)$. This is where the surface hypothesis enters: for a zero-dimensional subscheme of a smooth surface the local length identity $\dim_k \operatorname{Hom}_{A_p}(I_p/I_p^2, A_p) = 2\operatorname{length}_k(A_p)$ holds at every point, whether or not Z is a local complete intersection there, the factor two matching the codimension of Z in the surface. This identity is a nontrivial computation over the regular local ring $\mathcal{O}_{Y,p}$ of dimension two, not a formal consequence of the tangent-normal setup; it is Fogarty's computation, and the argument is cited to Fogarty rather than reproduced here (the same reference that supplies irreducibility). Summing over p and using $\sum_p \operatorname{length}_k(A_p) = n$ gives $\dim_k H^0(Z, N_{Z/Y}) = 2n$. With Step 2 this yields $\dim_k T_{[Z]} = 2n = \dim_{[Z]} \operatorname{Hilb}^n(Y)$ at every point, so $\operatorname{Hilb}^n(Y)$ is smooth of dimension $2n$.

Step 4: The Hilbert-Chow morphism. The universal subscheme $\mathcal{Z} \subseteq Y \times \operatorname{Hilb}^n(Y)$ (definition 14.3.4) is finite and flat of degree n over $\operatorname{Hilb}^n(Y)$. Its support, taken with multiplicities, defines for each $[Z]$ the effective zero-cycle $\sum_p \operatorname{length}_p(\mathcal{O}_Z)[p]$ on Y, a point of $\operatorname{Sym}^n Y$. That this assignment is a morphism of schemes, not just of sets, is the content of the construction of $\operatorname{Sym}^n Y$ as a scheme representing the functor of effective relative zero-cycles: the norm of the finite flat family $\mathcal{Z}$ produces a morphism $\pi\colon \operatorname{Hilb}^n(Y) \to \operatorname{Sym}^n Y$. The morphism π is itself projective, hence proper. This is a relative statement over the base $\operatorname{Sym}^n Y$ and holds for quasi-projective Y: it does not require $\operatorname{Hilb}^n(Y)$ to be projective, and indeed for the affine $Y = \mathbb{A}^2$ of Steps 5-6 the scheme $\operatorname{Hilb}^n(\mathbb{A}^2)$ is not projective while π still is, so the properness of π must be read off the morphism, not off an ambient projectivity of the source. It restricts to the identity on the open locus of n distinct points, where a reduced length-n subscheme and its unordered support carry the same information. Since $\operatorname{Hilb}^n(Y)$ is smooth of dimension $2n$ and $\operatorname{Sym}^n Y$ is normal of the same dimension with singularities exactly along the diagonals where points collide, π is a resolution of singularities.

Step 5: The case $\operatorname{Hilb}^2(\mathbb{A}^2)$ *explicitly.* Take $Y = \mathbb{A}^2 = \operatorname{Spec} k[x,y]$ and $n = 2$; a point is an ideal $I \subseteq k[x,y]$ with $\dim_k k[x,y]/I = 2$, and there are two geometric types. If Z is two distinct reduced points $p = (a_1, b_1)$, $q = (a_2, b_2)$, then $I = \mathcal{I}_p \cap \mathcal{I}_q$ and $k[x,y]/I \cong k \times k$ has length two, with tangent space $T_p\mathbb{A}^2 \oplus T_q\mathbb{A}^2$ of dimension $4 = 2 \cdot 2$, and this open locus of distinct pairs maps isomorphically to the corresponding locus of $\operatorname{Sym}^2 \mathbb{A}^2$. The subschemes supported at a single point, say the origin, are the interesting ones: a length-two ideal I with $k[x,y]/I$ supported at the origin must contain $\mathfrak{m}^2 = (x^2, xy, y^2)$ (a length-two quotient of the local ring is a quotient of $\mathcal{O}/\mathfrak{m}^2$) and have two-dimensional quotient spanned by 1 and one linear form, so

$$I_{[\lambda:\,\mu]} = (\mathfrak{m}^2,\ \mu x - \lambda y) = (x^2,\ xy,\ y^2,\ \mu x - \lambda y)$$

for a point $[\lambda\colon \mu] \in \mathbb{P}^1$: the relation $\mu x - \lambda y = 0$ leaves a two-dimensional quotient $k[x,y]/I_{[\lambda:\,\mu]}$ spanned by 1 and the class of any linear form not proportional to $\mu x - \lambda y$. When $\mu \neq 0$ the relation reads $x = (\lambda/\mu)y$, so y survives and $\{1, y\}$ is a basis; when $\lambda \neq 0$ it reads $y = (\mu/\lambda)x$, so x survives and $\{1, x\}$ is a basis. A concrete representative is $[\lambda\colon \mu] = [1\colon 0]$, giving

$$I_{[1:\,0]} = (y,\ x^2)$$

since $\mu x - \lambda y = -y$ and modulo y the ideal $\mathfrak{m}^2$ collapses to (x^2). The quotient $k[x,y]/(y, x^2) = k[x]/(x^2)$ has k-basis $\{1, x\}$, length two, supported at the origin with the tangent vector ∂_x singled out. A double point at the origin is therefore not a single subscheme but a $\mathbb{P}^1$ of them, one for each tangent direction $[\lambda\colon \mu]$.

Step 6: The blow-up of the diagonal. Assembling both types identifies $\mathrm{Hilb}^2(\mathbb{A}^2)$ globally. The symmetric product $\mathrm{Sym}^2\mathbb{A}^2 = \mathbb{A}^4/\mathfrak{S}_2$ is singular exactly along the diagonal $\Delta \cong \mathbb{A}^2$ where the two points coincide. Over a diagonal point the Hilbert-Chow morphism π has fiber the $\mathbb{P}^1$ of tangent directions just found, whereas over a distinct pair the fiber is a single point. A birational morphism from a smooth fourfold to $\mathrm{Sym}^2\mathbb{A}^2$ that replaces each diagonal point by a $\mathbb{P}^1$ of normal directions and is an isomorphism elsewhere is exactly the blow-up of $\mathrm{Sym}^2\mathbb{A}^2$ along its diagonal:

$$\mathrm{Hilb}^2(\mathbb{A}^2) \cong \mathrm{Bl}_\Delta\left(\mathrm{Sym}^2\mathbb{A}^2\right)$$

The exceptional divisor is the locus of nonreduced subschemes, a $\mathbb{P}^1$-bundle over the diagonal $\Delta \cong \mathbb{A}^2$, whose points are the ideals $I_{[\lambda\colon \mu]}$ translated to each point of $\mathbb{A}^2$. The Hilbert scheme of two points is the blow-up that resolves the diagonal of the symmetric square.

The example says the Hilbert scheme repairs the symmetric product. The symmetric product $\mathrm{Sym}^n Y$ is singular precisely along the loci where points come together, and its singularities record only the crude datum of a multiplicity. The Hilbert scheme sits above it, smooth, remembering the finer datum of the subscheme structure that a collision leaves behind, and the Hilbert-Chow morphism forgets that finer datum to recover the cycle. The $\mathrm{Hilb}^2(\mathbb{A}^2)$ computation exhibits exactly what extra information is carried over a diagonal point of $\mathrm{Sym}^2 Y$, namely a projective line of tangent directions.

The tangent-direction $\mathbb{P}^1$ over the origin is the fiber of the Hilbert-Chow morphism at the most singular point of Sym^2, and its appearance is forced by the algebra: a length-two subscheme concentrated at a point is exactly an ideal $I \supseteq \mathfrak{m}^2$ with one-dimensional $\mathfrak{m}/I$, and one-dimensional subspaces of the two-dimensional $\mathfrak{m}/\mathfrak{m}^2$ form a $\mathbb{P}^1$. The passage $(y, x^2) \leftrightarrow$ "origin with tangent ∂_x" is the dictionary to carry forward: an ideal is a subscheme, and the nonreduced structure encodes the limiting geometry of colliding points. Every later appearance of the Hilbert scheme of points, in deformation theory and in the moduli of sheaves, rests on this picture.

The second example swings to the opposite pole. Where the Hilbert scheme of points hides a smooth but intricate space behind a constant Hilbert polynomial, the Hilbert scheme of hypersurfaces of a fixed degree is a single projective space, transparent from the polynomial alone. This is the case where the general construction of theorem 14.3.2 collapses to the linear algebra of homogeneous forms.

A degree-d hypersurface in $\mathbb{P}^n$ is the zero locus $V(f)$ of a nonzero homogeneous polynomial f of degree d, taken up to scalar, so the naive parameter space is the projectivization of the space of such forms. Viewing this as a Hilbert scheme identifies the naive parameter space with a fine moduli space carrying a universal family, and it pins down which Hilbert polynomial to fix. The Hilbert polynomial is read off from the ideal sheaf sequence, and the identification of the scheme with $\mathbb{P}^N$ is a direct application of the representability theorem in a case where the ambient Grassmannian is already a projective space; the example records both.

Example 14.5.2: The Hilbert Scheme of Degree-d Hypersurfaces

Fix $n \geq 1$ and $d \geq 1$. The claim is that the Hilbert scheme of degree-d hypersurfaces in $\mathbb{P}^n$, taken for the Hilbert polynomial

$$P_d(t) = \binom{t+n}{n} - \binom{t-d+n}{n} \tag{14.3}$$

is the projective space

$$\mathrm{Hilb}^{P_d}_{\mathbb{P}^n} \cong \mathbb{P}\big(H^0(\mathbb{P}^n, \mathcal{O}(d))\big) = \mathbb{P}^N, \qquad N = \binom{n+d}{d} - 1$$

of dimension $\binom{n+d}{d} - 1$, with universal family the universal degree-d hypersurface in $\mathbb{P}^n \times \mathbb{P}^N$.

Step 1: The Hilbert polynomial. A degree-d hypersurface $X = V(f) \subseteq \mathbb{P}^n$ has ideal sheaf $\mathcal{I}_X = \mathcal{O}_{\mathbb{P}^n}(-d)$, since f generates the ideal in each chart and cutting by a single degree-d form twists $\mathcal{O}$ down by d. The ideal sheaf sequence

$$0 \to \mathcal{O}_{\mathbb{P}^n}(-d) \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_X \to 0$$

twisted by $\mathcal{O}(t)$ and passed to Euler characteristics gives

$$P_d(t) = \chi(\mathcal{O}_X(t)) = \chi(\mathcal{O}_{\mathbb{P}^n}(t)) - \chi(\mathcal{O}_{\mathbb{P}^n}(t-d)) = \binom{t+n}{n} - \binom{t-d+n}{n}$$

using $\chi(\mathcal{O}_{\mathbb{P}^n}(m)) = \binom{m+n}{n}$ for the projective space. This is (14.3), of degree $n-1$ in t with leading coefficient $d/(n-1)!$, recording that X is an $(n-1)$-dimensional subscheme of degree d.

Step 2: A subscheme with this polynomial is a hypersurface. Conversely, any closed subscheme $X \subseteq \mathbb{P}^n$ with Hilbert polynomial P_d is a degree-d hypersurface. Its ideal sheaf $\mathcal{I}_X$ has $\chi(\mathcal{I}_X(t)) = \binom{t+n}{n} - P_d(t) = \binom{t-d+n}{n} = \chi(\mathcal{O}_{\mathbb{P}^n}(t-d))$, and for $t \geq d$ the higher cohomology $H^{>0}(\mathbb{P}^n, \mathcal{I}_X(t))$ vanishes (Serre vanishing for the twist of a coherent sheaf on $\mathbb{P}^n$, [7]), so the Euler characteristic reduces to h^0: this counts the degree-t part of the saturated homogeneous ideal, forcing $h^0(\mathcal{I}_X(t)) = \binom{t-d+n}{n} = h^0(\mathcal{O}_{\mathbb{P}^n}(t-d))$. In degree $t = d$ this gives $h^0(\mathcal{I}_X(d)) = 1$, a single degree-$d$ form f in the ideal, and matching Hilbert functions in every degree $t \geq d$ shows the ideal is generated by f, so $\mathcal{I}_X = f \cdot \mathcal{O}_{\mathbb{P}^n}(-d) \cong \mathcal{O}_{\mathbb{P}^n}(-d)$. Thus $X = V(f)$ for a unique-up-to-scalar $f \in H^0(\mathbb{P}^n, \mathcal{O}(d))$.

Step 3: Identification with $\mathbb{P}^N$. Steps 1 and 2 show the closed points of $\mathrm{Hilb}^{P_d}_{\mathbb{P}^n}$ are exactly the nonzero degree-d forms up to scalar, that is, the points of $\mathbb{P}(H^0(\mathbb{P}^n, \mathcal{O}(d)))$. To upgrade this bijection of points to an isomorphism of schemes, compare universal families. Over $\mathbb{P}^N = \mathbb{P}(H^0(\mathcal{O}(d)))$ there is a tautological section: the universal form $F \in H^0(\mathbb{P}^n \times \mathbb{P}^N,\ \mathcal{O}_{\mathbb{P}^n}(d) \boxtimes \mathcal{O}_{\mathbb{P}^N}(1))$, whose vanishing locus $\mathcal{X} = V(F) \subseteq \mathbb{P}^n \times \mathbb{P}^N$ is flat over $\mathbb{P}^N$ with every fiber a degree-d hypersurface of Hilbert polynomial P_d. This is a family of subschemes of $\mathbb{P}^n$ with Hilbert polynomial P_d, hence by corollary 14.3.3 it is classified by a morphism $\Phi \colon \mathbb{P}^N \to \mathrm{Hilb}^{P_d}_{\mathbb{P}^n}$. Its inverse sends the universal subscheme over $\mathrm{Hilb}^{P_d}_{\mathbb{P}^n}$, a flat family of hypersurfaces, to the degree-d form cutting it out, a line bundle map $\mathcal{O}_{\mathrm{Hilb}} \to \pi_* \mathcal{O}_{\mathbb{P}^n}(d)$ classifying a morphism back to $\mathbb{P}^N$. The two are mutually inverse because a hypersurface and its defining form determine each other, so Φ is an isomorphism, and the dimension is $\dim H^0(\mathbb{P}^n, \mathcal{O}(d)) - 1 = \binom{n+d}{d} - 1$.

The example is the cleanest possible instance of the fine moduli philosophy. The scheme parametrizing hypersurfaces is not just a set of forms up to scalar, it is a projective space whose universal hypersurface pulls back to every family, and the whole construction of theorem 14.3.2 degenerates here to the linear algebra of the vector space $H^0(\mathbb{P}^n, \mathcal{O}(d))$. This is why classical algebraic geometry could speak of "the space of plane curves of degree d" long before Hilbert schemes: for hypersurfaces the space is a projective space, and no scheme theory beyond $\mathbb{P}^N$ is needed. The next example computes N in the smallest case and connects it to a classical count.

Example 14.5.3: Plane Curves of Degree d

Take $n = 2$, so $\mathbb{P}^2$ and hypersurfaces are plane curves. A degree-d plane curve is $V(f)$ for f a homogeneous form of degree d in three variables x_0, x_1, x_2, and

$$\dim_k H^0(\mathbb{P}^2, \mathcal{O}(d)) = \binom{d+2}{2} = \frac{(d+1)(d+2)}{2}$$

is the number of degree-d monomials in three variables. Hence $\mathrm{Hilb}^{P_d}_{\mathbb{P}^2} \cong \mathbb{P}^N$ with

$$N = \binom{d+2}{2} - 1 = \frac{(d+1)(d+2)}{2} - 1 = \frac{d(d+3)}{2}$$

For $d = 1$, lines in $\mathbb{P}^2$, this gives $N = 2$, so the space of lines is $\mathbb{P}^2$, the dual projective plane: a line in $\mathbb{P}^2$ is a hyperplane, hence a rank-1 quotient of $\mathcal{O}^3$, and the space of these is $\mathrm{Gr}(1, 3) = \mathbb{P}^2$ by the identification $\mathrm{Gr}(1, n) = \mathbb{P}^{n-1}$ of theorem 13.2.4. For $d = 2$, conics, $N = 5$: a conic $ax_0^2 + bx_1^2 + cx_2^2 + d'x_0x_1 + ex_0x_2 + fx_1x_2$ has six coefficients up to scalar, so conics form $\mathbb{P}^5$, the classical fact that five general points determine a conic (five linear conditions cut a point in $\mathbb{P}^5$). For $d = 3$, plane cubics, $N = 9$: nine points in general position determine a cubic, the count underlying the group law on an elliptic curve, and the running touchstone $\mathcal{M}_{1,1}$ of the first chapter lives inside this $\mathbb{P}^9$ as the locus of smooth cubics modulo projective equivalence.

The Hilbert polynomial P_d carries exactly the information the moduli problem needs: its degree records the dimension of the hypersurface, and its leading coefficient records the degree d. The dimension count $N = \binom{n+d}{d} - 1$ recovers the classical parameter counts for lines, conics, and cubics, and it is the first invariant one computes for any Hilbert scheme. The point where these hypersurface families connect to the harder theory is that $\mathbb{P}^N$ contains, as locally closed subsets, the loci of smooth, of singular, and of degenerate hypersurfaces, and cutting out the smooth locus and quotienting by PGL_{n+1} is exactly the moduli problem geometric invariant theory solves in the following chapter.

The third example returns to the beginning of the book. The first chapter built the Grassmannian $\mathrm{Gr}(d, n)$ from scratch as a fine moduli space of quotients (section 13.2, definition 13.2.1, theorem 13.2.4). The Quot scheme is a direct generalization of that construction, replacing quotients of a vector space by quotients of a coherent sheaf, so the Grassmannian ought to reappear as a Quot scheme. Confirming this reconciles the hand construction of section 13.2 with the general machine of section 14.3 and shows the two agree on their common ground.

The reconciliation is a matter of matching functors. The Grassmannian functor $\underline{\mathrm{Gr}}(d, n)$ assigns to S the rank-d locally free quotients of $\mathcal{O}_S^n$; the Quot functor $\underline{\mathrm{Quot}}^P_{\mathcal{F}/Y}$ assigns to S the flat quotients of $\mathcal{F}_S$ with fiberwise Hilbert polynomial P. Taking $Y = \operatorname{Spec} k$ a point, $\mathcal{F} = \mathcal{O}^n$ the trivial rank-n sheaf, and $P = d$ the constant polynomial identifies the two.

Example 14.5.4: The Grassmannian as a Quot Scheme

Let $Y = \operatorname{Spec} k$ be a point and $\mathcal{F} = \mathcal{O}^n = \mathcal{O}_Y^{\oplus n}$ the trivial rank-n sheaf, with $P = d$ the constant Hilbert polynomial. The claim is that the Quot functor $\underline{\mathrm{Quot}}^d_{\mathcal{O}^n/\operatorname{Spec} k}$ coincides with the Grassmannian functor $\underline{\mathrm{Gr}}(d, n)$ of definition 13.2.1, and hence

$$\mathrm{Quot}^d_{\mathcal{O}^n/\operatorname{Spec} k} \cong \mathrm{Gr}(d, n)$$

as fine moduli spaces, with the universal quotient $\mathcal{U}$ of definition 14.3.4 equal to the universal quotient bundle $\mathcal{Q}$ of definition 13.2.5.

Over the point $Y = \operatorname{Spec} k$ a scheme S over k has $Y_S = S$, and the pulled-back sheaf is $\mathcal{F}_S = \mathcal{O}_S^n$. A point of $\underline{\mathrm{Quot}}^d_{\mathcal{O}^n/\operatorname{Spec} k}(S)$ is a coherent quotient $q\colon \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q}$ with $\mathcal{Q}$ flat over S and every fiber of constant Hilbert polynomial d. A coherent sheaf on the zero-dimensional fiber $\operatorname{Spec} \kappa(s)$ with constant Hilbert polynomial d is a $\kappa(s)$-vector space of dimension d, so each fiber $\mathcal{Q} \otimes \kappa(s)$ has dimension d; a coherent sheaf on S that is flat with fibers of constant rank d is locally free of rank d. Thus

$$\underline{\mathrm{Quot}}^d_{\mathcal{O}^n/\operatorname{Spec} k}(S) = \left\{ \mathcal{O}_S^n \twoheadrightarrow \mathcal{Q} \;\middle|\; \mathcal{Q} \text{ locally free of rank } d \right\} / \cong$$

which is the definition of $\underline{\mathrm{Gr}}(d, n)(S)$ (definition 13.2.1) verbatim. The two functors are equal, and pullback is the same operation (f^* applied to the surjection) in both. By Yoneda, equal functors have isomorphic representing objects, so $\mathrm{Quot}^d_{\mathcal{O}^n/\operatorname{Spec} k} \cong \mathrm{Gr}(d, n)$. The universal quotient $\mathcal{U}$ over $\mathrm{Quot}^d_{\mathcal{O}^n/\operatorname{Spec} k}$ corresponds under the isomorphism of functors to the universal object over $\mathrm{Gr}(d, n)$, which is the universal quotient bundle $\mathcal{Q}$ of definition 13.2.5.

The example places the Grassmannian inside the theory rather than beside it. The hand construction of section 13.2, with its affine charts and Plücker embedding, and the general existence theorem of section 14.3, with its Grassmannian of degree-m pieces and closed-subfunctor cut, are two routes to the same scheme, and the general route specializes to the specific one when the ambient sheaf is $\mathcal{O}^n$ over a point. This is the reason the construction of theorem 14.3.2 embeds every Quot scheme in a Grassmannian: the Grassmannian is the universal case, the Quot scheme of a trivial sheaf over a point, and every other Quot scheme is cut out of one of these by the flatness and Hilbert-polynomial conditions. The chart-by-chart smoothness of theorem 13.2.4 thus becomes the local model against which the local structure of a general Quot scheme, governed by the tangent-space computation of theorem 14.4.4, is measured.

Taking $Y = \operatorname{Spec} k$ was the degenerate case; allowing Y to be positive-dimensional connects the Quot scheme to maps into the Grassmannian, but two subtleties intervene that are absent over a point. First, the constant polynomial $P = d$ is a trap: over a positive-dimensional Y it selects length-d torsion quotients, whose Quot scheme is a relative Hilbert scheme of points, not a Grassmannian. A rank-d locally free quotient of $\mathcal{O}_Y^n$ has Hilbert polynomial $d \cdot \chi(\mathcal{O}_Y(t))$, a nonconstant polynomial of degree $\dim Y$, so this is the polynomial to fix if one wants locally free quotients in the picture at all. Second, and less obvious, fixing that polynomial does not by itself force a quotient to be locally free, nor does it capture every map $Y \to \mathrm{Gr}(d, n)$: the locally free quotients are only an *open* subscheme of the Quot scheme, and the Hom-scheme into the Grassmannian breaks into components indexed by a discrete pullback invariant, of which the fixed polynomial selects just one. The next example works out exactly what survives of the naive identification.

Example 14.5.5: The Relative Grassmannian as a Quot Scheme

Let Y be a projective scheme over k with a fixed ample $\mathcal{O}_Y(1)$, and $\mathcal{F} = \mathcal{O}_Y^n$ the trivial rank-n sheaf. The Hilbert polynomial of a rank-d locally free quotient $\mathcal{O}_Y^n \twoheadrightarrow \mathcal{E}$ is $\chi(\mathcal{E}(t))$, and this depends on more than the rank: it is governed by the full numerical class (Chern character) of $\mathcal{E}$, not by d alone. Among these, the quotients with *trivial* discrete invariant, those pulled back from a constant map $Y \to \mathrm{Gr}(d, n)$ so that $\mathcal{E} \cong \mathcal{O}_Y^{\oplus d}$, have Hilbert polynomial

$$Q(t) = d \cdot \chi(\mathcal{O}_Y(t))$$

of degree $\dim Y$ in t; this is the polynomial to fix to see the trivial-invariant locally free quotients, whereas the constant $P = d$ selects length-d torsion quotients instead. Fixing Q does *not*, by itself, isolate even these, let alone the whole relative Grassmannian, and the example is a warning against the naive identification $\mathrm{Quot}^{Q}_{\mathcal{O}_Y^n/Y} = \underline{\mathrm{Hom}}(Y, \mathrm{Gr}(d,n))$, which is false on two counts.

The locally free quotients form only an open locus. A coherent quotient $\mathcal{O}_Y^n \twoheadrightarrow \mathcal{Q}$ with $\mathcal{Q}$ flat over $\operatorname{Spec} k$ and Hilbert polynomial Q need not be locally free of rank d: torsion, a rank drop along a divisor, or embedded points all leave $\chi(\mathcal{Q}(t)) = Q(t)$ unchanged while destroying local freeness. So the k-points of $\mathrm{Quot}^{Q}_{\mathcal{O}_Y^n/Y}$ include quotients that are not rank-d bundles, and the locally free quotients cut out an *open* subscheme

$$\mathrm{Quot}^{Q,\circ}_{\mathcal{O}_Y^n/Y} \subseteq \mathrm{Quot}^{Q}_{\mathcal{O}_Y^n/Y}$$

(openness of the locally free locus of a coherent sheaf on Y, [7]), not the whole Quot scheme.

The Hilbert polynomial pins one component of the Hom-scheme. A rank-d locally free quotient $\mathcal{O}_Y^n \twoheadrightarrow \mathcal{E}$ is, by the Grassmannian functor (definition 13.2.1), the same datum as a morphism $\varphi\colon Y \to \mathrm{Gr}(d,n)$, the quotient being $\mathcal{E} = \varphi^*\mathcal{Q}$ the pullback of the universal quotient bundle. But $\chi(\varphi^*\mathcal{Q}(t))$ depends on the class of φ: the Hom-scheme $\underline{\mathrm{Hom}}(Y, \mathrm{Gr}(d,n))$ is a disjoint union of pieces indexed by this discrete pullback invariant, and fixing $\chi = Q$ singles out one such piece. For $Y = \mathbb{P}^1$ and $\mathrm{Gr}(1,2) = \mathbb{P}^1$, a degree-$e$ map pulls the universal quotient back to $\mathcal{O}(e)$, with $\chi(\mathcal{O}(e)(t)) = t + e + 1$; the value $Q(t) = \chi(\mathcal{O}_{\mathbb{P}^1}(t)) = t + 1$ forces $e = 0$, selecting only the constant maps among all $\underline{\mathrm{Hom}}(\mathbb{P}^1, \mathbb{P}^1)$.

Both corrections combine into the true statement: the locally free locus of the Quot scheme, at the fixed Hilbert polynomial Q, is an open subscheme that recovers one component of the Hom-scheme,

$$\mathrm{Quot}^{Q,\circ}_{\mathcal{O}_Y^n/Y} \cong \underline{\mathrm{Hom}}(Y, \mathrm{Gr}(d,n))_Q$$

the subscript Q marking the component on which the pullback invariant matches Q. For Y reduced to a point every quotient of the fixed space k^n of dimension d is automatically locally free and there is a single component, so the open locus is everything and this reduces to $\mathrm{Gr}(d,n)$, recovering example 14.5.4. The new content over a positive-dimensional base is exactly what these two corrections express: the Quot scheme carries non-locally-free quotients its point-model does not see, and even its locally free locus is a component of a Hom-scheme rather than a bare product $\mathrm{Gr}(d,n) \times Y$; that product is the sub-locus of *constant* morphisms $Y \to \mathrm{Gr}(d,n)$, one component among many, not the whole Hom-scheme.

The three examples span the range of behaviour. At one end sits the Hilbert scheme of points, smooth of dimension $2n$ on a surface yet carrying the full subtlety of colliding points and tangent directions; at the other end sit the hypersurface Hilbert schemes, projective spaces read directly from a Hilbert polynomial; and threading them is the Grassmannian, the universal Quot scheme from which the general construction cuts every other. The reader who has followed the ideal (y, x^2) to a tangent direction, the binomial $\binom{n+d}{d} - 1$ to the dimension of a space of forms, and the trivial sheaf over a point back to section 13.2 has seen the whole chapter operate on the objects one meets.

Exercise 14.5.1

1. Compute the length of the subscheme $Z \subseteq \mathbb{A}^2$ cut out by the ideal $I = (x^2, xy, y^2)$, and decide whether $[Z] \in \mathrm{Hilb}^n(\mathbb{A}^2)$ for the appropriate n. Then, using the tangent-direction description in example 14.5.1, explain why (x^2, xy, y^2) does *not* lie in $\mathrm{Hilb}^2(\mathbb{A}^2)$, and identify which Hilb^n it does belong to.

2. For $\mathrm{Hilb}^2(\mathbb{A}^2)$, verify directly that the ideal $I_{[\lambda:\,\mu]} = (\mathfrak{m}^2,\ \mu x - \lambda y)$ has $\dim_k k[x,y]/I_{[\lambda:\,\mu]} = 2$ for every $[\lambda:\mu] \in \mathbb{P}^1$, and show that $I_{[\lambda:\,\mu]} = I_{[\lambda':\,\mu']}$ if and only if $[\lambda:\mu] = [\lambda':\mu']$. Conclude that the fiber of the Hilbert-Chow morphism over the doubled origin is a $\mathbb{P}^1$.

3. Using example 14.5.2, compute the dimension of the Hilbert scheme of degree-d surfaces in $\mathbb{P}^3$ (quadric surfaces $d = 2$, cubic surfaces $d = 3$). In each case write out $\binom{n+d}{d} - 1$ explicitly, and for quadrics identify the locus of singular quadrics inside $\mathbb{P}^N$ as the zero locus of a single determinant.

4. Prove that the Hilbert polynomial $P_d(t) = \binom{t+n}{n} - \binom{t-d+n}{n}$ of (14.3) has degree $n-1$ in t with leading coefficient $d/(n-1)!$, and interpret both facts geometrically for the hypersurface $X = V(f)$.

5. Deduce from example 14.5.4 that $\mathbb{P}^{n-1}$ is a Quot scheme: identify $\underline{\mathrm{Quot}}^1_{\mathcal{O}^n/\operatorname{Spec} k}$ with the functor $\underline{\mathrm{Gr}}(1,n)$ and hence with the functor of points of $\mathbb{P}^{n-1}$, matching the universal quotient with $\mathcal{O}(1)$. Relate this to example 13.1.4.

6. Using theorem 14.4.4 and example 14.5.1, compute the dimension of the tangent space to $\mathrm{Hilb}^3(\mathbb{A}^2)$ at the point given by the ideal (x, y^3), a single point of length three concentrated on the y-axis. Confirm the answer equals $2 \cdot 3 = 6$, consistent with smoothness. Then explain where the local argument of Step 3 of example 14.5.1 breaks down for a zero-dimensional subscheme of $\mathbb{A}^3$: contrast the two structural routes to controlling $\mathcal{I}_Z/\mathcal{I}_Z^2$, namely freeness of the conormal module (available exactly at local complete intersection points) and the length-two duality special to a regular local ring of dimension two, and say which route survives on a surface at every point and which is the one that fails once the ambient dimension is three.

15

Geometric Invariant Theory

The Hilbert and Quot schemes of chapter 14 construct moduli spaces of objects that carry an embedding: a closed subscheme of a fixed projective space, or a quotient of a fixed sheaf. Most moduli problems, however, ask for objects only up to isomorphism, with no embedding fixed, and the passage from embedded objects to abstract ones is a quotient by the group that moves the embedding, most often a projective linear group. Geometric invariant theory, Mumford's refinement of Hilbert's nineteenth-century invariant theory, is the machine that forms such quotients. Given a reductive group G acting on a projective scheme together with a choice of linearized ample line bundle, it isolates a locus of *semistable* points on which a good quotient exists and is again projective. This is the second construction engine of the Part: paired with the Hilbert scheme, it produces the moduli spaces of abstract objects, the moduli of curves among them, that the functor-of-points method alone leaves as unrepresentable stacks.

The chapter builds the theory in the order the constructions depend on one another. section 15.1 establishes the affine quotient $X/\!\!/G = \operatorname{Spec} k[X]^G$, which exists as a scheme because the invariant ring is finitely generated for reductive G (Hilbert's finiteness theorem); the reductive groups themselves, and their actions on affine varieties, are taken from [19]. section 15.2 introduces linearizations of line bundles, the data that globalizes the affine quotient, and section 15.3 uses a linearization to cut out the semistable and stable loci and to build the projective quotient $X^{\mathrm{ss}}/\!\!/G$. section 15.4 reduces the stability of a point to a finite computation through the Hilbert–Mumford numerical criterion, and section 15.5 turns the machine on moduli problems, realizing a coarse moduli space as a GIT quotient of a Hilbert scheme and sketching the construction of $\mathcal{M}_g$ that the capstone completes.

15.1 Group Actions and Quotients

The moduli problems of the previous chapters carried their objects already embedded: a point of a Hilbert scheme is a closed subscheme of a fixed $\mathbb{P}^n$, not an abstract variety, and two points that differ only by a projective change of coordinates parametrize the same abstract object. To pass from embedded objects to abstract ones, one must quotient by the group that moves the embedding, typically a projective linear group. The moduli space of abstract objects should be the space of orbits of this group acting on a Hilbert scheme. Geometric invariant theory is the machinery that makes "the space of orbits" into a scheme, and this section builds its foundation: the problem of quotienting a variety by an algebraic group action, and the affine solution to that problem.

The difficulty is that the naive orbit space, the topological quotient X/G with the quotient topology, is almost never a variety. Orbits can fail to be closed, and when the closure of one orbit meets another, any continuous G-invariant function takes the same value on both, so no scheme structure on the set of orbits can separate them. The remedy is to give up on parametrizing all orbits and instead build a scheme that parametrizes the *closed* orbits, identifying two points precisely when their orbit closures meet. This section makes that idea precise for an affine variety, where the construction reduces to a question in invariant theory: the quotient is the spectrum of the ring of invariant functions, and the theorem that makes it a scheme is the finite generation of that ring.

Throughout, k is algebraically closed, G is a reductive linear algebraic group over k, and X is a scheme of finite type over k carrying a (left) algebraic action $\sigma\colon G \times X \to X$. An action is *algebraic* when σ is a morphism of schemes; the basic theory of such actions, their orbits, and their stabilizers is developed in [19], and is used here freely.

Before constructing anything, the target of the construction must be pinned down. section 4.4 introduced the two clean notions of quotient a group action can have: a *categorical quotient*, universal among G-invariant morphisms out of X, and a *geometric quotient*, whose points are exactly the orbits and whose structure sheaf is the invariants of the pushforward. These are recalled in definition 4.4.1. The categorical quotient always exists once one has a reasonable ring of invariants but may collapse distinct orbits; the geometric quotient separates all orbits but exists only when every orbit is closed, which fails in the presence of the collapsing phenomenon described above. Geometric invariant theory works with an intermediate notion, tuned so that it exists in wide generality yet retains as much orbit information as the categorical quotient can carry.

That intermediate notion is the good quotient. It asks for a morphism that is affine-local (so it can be built by gluing spectra of invariant rings), whose structure sheaf is exactly the invariant functions (so it is a categorical quotient), and which separates any two disjoint closed invariant subsets (so on the locus of closed orbits it is as good as a geometric quotient). The last condition is the substantive one: it is what forces distinct closed orbits to land at distinct points, and it is exactly the property the topological quotient lacks.

Definition 15.1.1: Good Quotient

Let G act on a scheme X of finite type over k. A morphism $\pi\colon X \to Y$ to a scheme Y is a *good quotient* if:

1. π is G-invariant: $\pi \circ \sigma = \pi \circ \mathrm{pr}_2$ as morphisms $G \times X \to Y$ (equivalently, π is constant on orbits);
2. π is affine and surjective;
3. the natural map $\mathcal{O}_Y \to (\pi_*\mathcal{O}_X)^G$ is an isomorphism, where $(\pi_*\mathcal{O}_X)^G$ denotes the subsheaf of G-invariant sections of $\pi_*\mathcal{O}_X$;
4. π separates disjoint closed invariant subsets: if $W_1, W_2 \subseteq X$ are closed, G-invariant, and disjoint, then $\pi(W_1)$ and $\pi(W_2)$ are closed and disjoint in Y.

A good quotient is a *geometric quotient* if, in addition, the fibers of π are exactly the orbits: $\pi(x) = \pi(x')$ if and only if $Gx = Gx'$.

Conditions (1) and (3) together already make a good quotient a categorical quotient: any G-invariant morphism $f\colon X \to Z$ factors uniquely through π, because on each affine open $\operatorname{Spec} B \subseteq Y$ the map f corresponds to an invariant ring homomorphism, which by (3) lands in $(\pi_*\mathcal{O}_X)^G(\operatorname{Spec} B) = B$. Condition (2) is what lets the whole construction be checked on affine charts and glued, and it is why the entire theory reduces to the affine case treated below. Condition (4) is the separation property that distinguishes a good quotient from a bare categorical quotient: it guarantees that closed orbits, which are always disjoint from one another and closed, map to distinct points. A good quotient is therefore a geometric quotient precisely on the locus where every orbit is closed; off that locus, several orbits (an orbit and the smaller orbits in its closure) are glued to a single point, and this gluing is not a defect but the only way to obtain a scheme at all.

The affine construction rests on a single input from invariant theory: the ring of invariant functions is again a finitely generated k-algebra, so that its spectrum is a variety. This is Hilbert's finiteness theorem, one of the founding results of the subject. Its proof is an invariant-theory argument whose home in the series is [19], where reductive groups, their representation theory, and the Reynolds operator are developed; the statement is recorded here because it is the theorem that turns definition 15.1.4 into a scheme, and it is used throughout the chapter.

Theorem 15.1.2: Hilbert's Finiteness Theorem

Let G be a linearly reductive group over k acting rationally on a finitely generated k-algebra A by k-algebra automorphisms. Then the subalgebra of invariants

$$A^G = \{\, a \in A \,\middle|\, g \cdot a = a \text{ for all } g \in G \,\}$$

is a finitely generated k-algebra. The same conclusion holds for any reductive group G, without the linear-reductivity hypothesis.

The proof is cited to [19]; the statement above is a recollection of that result, not a claim to reprove it here. The one-paragraph idea is worth carrying, because it explains why reductivity is the operative hypothesis. When G is linearly reductive, every finite-dimensional representation splits as a direct sum of irreducibles, and the trivial isotypic component varies functorially with the representation.

Applied to A, viewed as an increasing union of finite-dimensional G-subspaces, this produces a G-equivariant projection $R\colon A \to A^G$, the *Reynolds operator*, which is a homomorphism of A^G-modules: $R(bc) = b\,R(c)$ whenever $b \in A^G$. From here Hilbert's argument runs as in the graded case of the Hilbert basis theorem. Let A^G_+ be the ideal of A generated by all homogeneous invariants of positive degree; by Noetherianity it is generated by finitely many invariants $f_1, \ldots, f_r$, and applying the Reynolds operator to a representation $f = \sum a_i f_i$ of an arbitrary invariant f, using that R is A^G-linear, shows by induction on degree that $f_1, \ldots, f_r$ generate A^G as a k-algebra. The Reynolds operator, and with it the projection onto invariants, is exactly what linear reductivity supplies; in positive characteristic linear reductivity can fail for a reductive group, and the finiteness is then recovered through Nagata's theorem, using the weaker *geometric reductivity* that Haboush's theorem establishes for all reductive groups. That general case is the second sentence of the statement and is likewise cited to [19]; it is what makes the construction below available for groups such as SL_n in every characteristic, not only in characteristic zero.

15.1.3 Warning: reductivity is necessary Finiteness fails without a reductivity hypothesis. Nagata's counterexample exhibits an action of a non-reductive group (a suitable additive group $\mathbb{G}_a^n$) on a polynomial ring whose invariant subalgebra is not finitely generated, disproving Hilbert's fourteenth problem in its original generality. The reductivity of G is therefore not a convenience of the proof but a condition the conclusion requires, which is why the quotient theory of this chapter is built for reductive groups.

With finiteness in hand, the affine quotient is immediate to define and carries all four good-quotient properties. Let $X = \operatorname{Spec} A$ be affine with $A = k[X]$ finitely generated, and let G act on X, inducing the dual rational action on A. The ring of invariants $A^G = k[X]^G$ is a finitely generated k-algebra by theorem 15.1.2, so its spectrum is again an affine variety of finite type over k, and the inclusion $A^G \hookrightarrow A$ induces a morphism of schemes.

Definition 15.1.4: Affine GIT Quotient

Let G be a reductive group acting on an affine scheme $X = \operatorname{Spec} A$ of finite type over k. The *affine GIT quotient* of X by G is the affine scheme

$$X/\!\!/G := \operatorname{Spec}\, A^G$$

together with the morphism $\pi\colon X \to X/\!\!/G$ induced by the inclusion $A^G \hookrightarrow A$.

The double-slash notation distinguishes this from a putative orbit space X/G; the two agree precisely when every orbit is closed. The following proposition collects the geometric content of the construction and is the reason the affine quotient deserves the name "quotient" at all.

Proposition 15.1.5: The affine GIT quotient is a good quotient

With notation as in definition 15.1.4, the morphism $\pi\colon X \to X/\!\!/G$ is a good quotient. In particular:

1. π is a categorical quotient of X by G;
2. π is surjective, and its geometric fibers each contain a unique closed orbit;
3. for closed points $x, x' \in X$, one has $\pi(x) = \pi(x')$ if and only if the orbit closures meet, $\overline{Gx} \cap \overline{Gx'} \neq \emptyset$.

***Proof*:**
The four conditions of definition 15.1.1 are verified in turn, and the three numbered consequences are read off from them.

Step 1: π is G-invariant, affine, and surjective. Invariance holds by construction: π is dual to the inclusion $A^G \hookrightarrow A$, whose image consists of functions constant on orbits, so $\pi \circ \sigma = \pi \circ \mathrm{pr}_2$. The morphism π is affine because both source and target are affine. For surjectivity it suffices to show that $\operatorname{Spec} A \to \operatorname{Spec} A^G$ hits every closed point, i.e. that for every maximal ideal $\mathfrak{m} \subset A^G$ the extension $\mathfrak{m}A$ is a proper ideal of A. If $\mathfrak{m}A = A$, write $1 = \sum_i a_i f_i$ with $a_i \in A$ and $f_i \in \mathfrak{m}$, and apply the Reynolds operator R; using its A^G-linearity and $R(1) = 1$ gives $1 = \sum_i R(a_i) f_i \in \mathfrak{m}$, since each $R(a_i) \in A^G$ and $f_i \in \mathfrak{m}$, contradicting that $\mathfrak{m}$ is proper. Hence $\mathfrak{m}A \neq A$, so $\mathfrak{m}$ is the contraction of a maximal ideal of A and lies in the image; the surjectivity of $\operatorname{Spec} A \to \operatorname{Spec} A^G$ for reductive G is established this way in [19]. Thus π is surjective.

Step 2: the structure sheaf condition. Since both schemes are affine, $\pi_*\mathcal{O}_X$ is the sheaf associated to A viewed as an A^G-module, and its G-invariant global sections are A^G by definition. On a basic open $D(f) \subseteq X/\!\!/G$ with $f \in A^G$, the invariant sections of $\mathcal{O}_X(\pi^{-1}D(f)) = A_f = A \otimes_{A^G} A^G_f$ are $(A_f)^G = (A^G)_f$, because inverting an invariant commutes with taking invariants (the Reynolds operator is A^G-linear, so localizing at $f \in A^G$ commutes with R). Hence $\mathcal{O}_{X/\!\!/G} \to (\pi_*\mathcal{O}_X)^G$ is an isomorphism.

Step 3: separation of disjoint closed invariant sets. Let $W_1, W_2 \subseteq X$ be closed, G-invariant, and disjoint, with vanishing ideals $I_1, I_2 \subseteq A$; disjointness means $I_1 + I_2 = A$. Both ideals are G-stable, so applying the Reynolds operator R to a relation $1 = a_1 + a_2$ with $a_i \in I_i$ gives $1 = R(1) = R(a_1) + R(a_2)$ with $R(a_i) \in I_i^G := I_i \cap A^G$, since R maps a G-stable ideal into its invariants. Therefore $I_1^G + I_2^G = A^G$, which says exactly that $V(I_1^G)$ and $V(I_2^G)$ are disjoint closed subsets of $X/\!\!/G$. As $\pi(W_i)$ is contained in $V(I_i^G)$ and is closed because π is a closed map on invariant sets by the same Reynolds argument applied to a single invariant ideal, the images $\pi(W_1)$ and $\pi(W_2)$ are closed and disjoint. This is condition (4), and it is the one place linear reductivity, through the Reynolds operator, does the essential work.

The four conditions establish that π is a good quotient.

Consequence (1). A good quotient is a categorical quotient: given any G-invariant morphism $f\colon X \to Z$, on each affine open $\operatorname{Spec} B \subseteq Z$ the pullback $f^\#\colon B \to A$ has image in A^G by invariance, hence factors through $A^G = \mathcal{O}_{X/\!\!/G}(X/\!\!/G)$, uniquely because $A^G \hookrightarrow A$ is injective. This is the universal property.

Consequence (3), and then (2). For the fiber statement, fix a closed point $y \in X/\!\!/G$ corresponding to a maximal ideal $\mathfrak{m} \subset A^G$. Two closed points $x, x' \in X$ satisfy $\pi(x) = \pi(x') = y$ if and only if every invariant $f \in A^G$ takes the same value on them, namely $f \pmod{\mathfrak{m}}$. Now the orbit closures $\overline{Gx}$ and $\overline{Gx'}$ are disjoint closed invariant sets when they do not meet, so by the separation just proved there is an invariant $f \in A^G$ with $f \equiv 0$ on $\overline{Gx}$ and $f \equiv 1$ on $\overline{Gx'}$, forcing $\pi(x) \neq \pi(x')$. Conversely if the closures meet, any invariant is constant on each closed set and on their common point, hence takes equal values at x and x', giving $\pi(x) = \pi(x')$. This proves (3). Finally each fiber $\pi^{-1}(y)$ is a nonempty closed G-invariant set, so it contains an orbit of minimal dimension, which is therefore closed; by (3) any two closed orbits in the fiber have meeting (hence equal) closures and so coincide, giving the unique closed orbit of (2). This completes the proof.

The proposition's content is worth stating in words. The affine quotient $X/\!\!/G$ does not parametrize all orbits; it parametrizes the *closed* orbits, one per point, and it glues a non-closed orbit to the unique closed orbit sitting in its closure. The invariant functions cannot tell a non-closed orbit apart from the smaller orbit it degenerates to, because a regular invariant is continuous and the two orbits share a limit point, so any quotient built from invariant functions must identify them. Two points map to the same place exactly when their orbit closures collide. This is the precise sense in which $X/\!\!/G$ is the best scheme-theoretic approximation to the orbit space, and the phenomenon it encodes, the collapse of non-closed orbits, is the feature to keep in view for the projective quotients of the following sections.

The following example is the smallest one in which the collapse is visible, and every subsequent GIT computation is a variation on it. It is worth working in complete detail.

Example 15.1.6: Invariant Rings for a Torus and for Binary Forms

1. *The hyperbola action.* Let $G = \mathbb{G}_m$ act on $X = \mathbb{A}^2 = \operatorname{Spec} k[x, y]$ by

$$t \cdot (x, y) = (tx,\, t^{-1}y)$$

so that on functions $t \cdot x = t^{-1}x$ and $t \cdot y = t\,y$ (the coordinate functions transform inversely to the points). A monomial $x^a y^b$ has weight $b - a$ under this action, so it is invariant if and only if $a = b$. Hence the ring of invariants is

$$k[x, y]^{\mathbb{G}_m} = k[xy]$$

the polynomial ring in the single invariant $u := xy$. The affine GIT quotient is therefore

$$\mathbb{A}^2/\!\!/\mathbb{G}_m = \operatorname{Spec} k[u] = \mathbb{A}^1$$

with $\pi(x, y) = xy$. The orbit structure explains the collapse. For $c \neq 0$ the level set $\{xy = c\}$ is a single orbit, closed in $\mathbb{A}^2$, mapping to the point $u = c$. Over $u = 0$ the fiber $\{xy = 0\}$ is the union of the two coordinate axes minus the origin, together with the origin itself, and it decomposes into *three* orbits: the punctured x-axis $\{(x, 0) : x \neq 0\}$, the punctured y-axis $\{(0, y) : y \neq 0\}$, and the fixed point $\{(0, 0)\}$. Only the last is closed; the closures of the two punctured axes each contain the origin. Consistent with proposition 15.1.5, all three orbits map to the single point $u = 0$, and they are identified there precisely because their closures meet at the origin. The two non-closed orbits are glued to the closed orbit in their common closure, and the quotient $\mathbb{A}^1$ sees only the closed orbits: one for each $c \neq 0$ and the origin for $c = 0$.

2. *Binary forms under* SL_2. Let $V_d = \operatorname{Sym}^d(k^2)^\vee$ be the space of binary forms of degree d, on which $G = \mathrm{SL}_2$ acts by linear substitution of variables; write $X = V_d$. For $d = 2$, a binary

quadratic $f = ax^2 + bxy + cy^2$ has the discriminant $\Delta = b^2 - 4ac$, and a direct check shows Δ is SL_2-invariant (the substitution matrices have determinant one). In fact $k[V_2]^{\mathrm{SL}_2} = k[\Delta]$, a polynomial ring in one generator, so

$$V_2/\!\!/\mathrm{SL}_2 = \operatorname{Spec} k[\Delta] = \mathbb{A}^1$$

The closed orbits are the nondegenerate conics $\{\Delta = c\}$ for $c \neq 0$; the fiber over $\Delta = 0$ consists of the perfect squares ℓ^2 (a single non-closed orbit) together with the zero form (the closed fixed point), and the two are identified because scaling ℓ^2 by t^2 via the diagonal one-parameter subgroup drives it to 0, so $0 \in \overline{\mathrm{SL}_2 \cdot \ell^2}$. This is the same collapse as in part (1), now realized inside the invariant theory of forms, and the discriminant is the invariant that both detects it and coordinatizes the quotient.

Both computations exhibit the same three-part anatomy. The invariant ring is finitely generated, as theorem 15.1.2 promises; its spectrum is an affine variety, here $\mathbb{A}^1$; and the fiber over the special point collects several orbits, exactly one of which is closed, with the non-closed orbits degenerating into it. The general point of the quotient corresponds to a closed orbit that is a full-dimensional level set of the invariants, while the special point corresponds to the most degenerate closed orbit, on which the invariants that cut out the generic orbits all vanish. This degeneration, an orbit whose closure meets a smaller orbit, is the affine case of *instability*, and isolating the points whose closed-orbit collapse can be avoided by a good choice of linearization is precisely the task of the next sections. For the moment the payoff is the construction itself: for a reductive group acting on an affine variety, the ring of invariants produces a scheme that is a categorical quotient, separating exactly the closed orbits.

Exercise 15.1.1

1. Let $\mathbb{G}_m$ act on $\mathbb{A}^n = \operatorname{Spec} k[x_1, \ldots, x_n]$ with weights $w_1, \ldots, w_n \in \mathbb{Z}$, so that $t \cdot x_i = t^{-w_i} x_i$ on functions. Describe the invariant ring $k[x_1, \ldots, x_n]^{\mathbb{G}_m}$ as the span of monomials of weight zero, and compute $\mathbb{A}^n/\!\!/\mathbb{G}_m$ explicitly when $(w_1, w_2, w_3) = (1, 1, -1)$.

2. Let $\mathbb{G}_m$ act on $\mathbb{A}^2$ by $t \cdot (x, y) = (tx, ty)$ (weight 1 on both coordinates). Compute $k[x, y]^{\mathbb{G}_m}$ and $\mathbb{A}^2/\!\!/\mathbb{G}_m$. Contrast the orbit picture with that of example 15.1.6(1): which orbits are closed, and which single orbit is the unique closed orbit in the fiber over the image of the origin?

3. Show directly from definition 15.1.1 that a good quotient $\pi \colon X \to Y$ is a categorical quotient. (You may assume X affine; the general case is glued from this one.)

4. Let $\pi \colon X \to X/\!\!/G$ be the affine GIT quotient of a reductive group action. Prove that for each closed point $y \in X/\!\!/G$ the fiber $\pi^{-1}(y)$ contains a *unique* closed orbit, and that this orbit lies in the closure of every orbit in the fiber. (Reduce to the separation property of definition 15.1.1(4).)

5. Give an example of a non-reductive group action for which the invariant ring is not finitely generated, or explain in one paragraph why theorem 15.1.2 cannot simply be extended to all linear algebraic groups, referencing the phenomenon of box 15.1.3.

6. The finite group $G = \mathbb{Z}/2$ acts on $\mathbb{A}^2 = \operatorname{Spec} k[x, y]$ (with $\operatorname{char} k \neq 2$) by $(x, y) \mapsto (-x, -y)$. Compute the invariant ring, identify $\mathbb{A}^2/\!\!/(\mathbb{Z}/2)$ as an affine variety in $\mathbb{A}^3$, and verify that this quotient is a *geometric* quotient (every orbit is closed).

15.2 Linearizations

The affine quotient of section 15.1 disposes of the affine case completely: when X is affine and G is reductive, the invariant ring $k[X]^G$ is a finitely generated k-algebra, and $X/\!\!/G = \operatorname{Spec} k[X]^G$ is a categorical quotient whose points are the closed orbits (definition 15.1.4). The moduli spaces this book is after are almost never affine. A moduli of curves, or of subschemes with a fixed Hilbert polynomial, is a quotient of a projective or quasi-projective parameter space, and Proj has no ring of global functions to take invariants of: the only global regular functions on a projective variety are the constants, so the recipe "take Spec of the invariant functions" returns a point and throws away all the geometry.

The way out is to reconstruct the projective variety from its homogeneous coordinate ring, which is not a ring of functions but a ring of *sections* of powers of an ample line bundle. If X is projective and L is ample, then $X = \operatorname{Proj} \bigoplus_{n\geq 0} H^0(X, L^{\otimes n})$, and one recovers the quotient by taking G-invariant sections in each degree and forming $\operatorname{Proj} \bigoplus_{n\geq 0} H^0(X, L^{\otimes n})^G$. For this to make sense G must act not only on X but on the sections of L, and there is no canonical way for a group acting on X to act on L: the action of $g \in G$ carries the fiber L_x to the fiber L_{gx}, but which isomorphism $L_x \to L_{gx}$ to use is extra data. That data, a compatible choice for every g and every x, is a *linearization* of L. It is the choice that glues the local affine quotients over the open sets $X_s = \{s \neq 0\}$ (for s an invariant section) into a single projective quotient, and different linearizations glue them differently. This section defines linearizations, organizes them into a group, and computes, in the one example every reader already knows, how changing the linearization changes the invariant sections and so the eventual quotient.

Throughout, X is a scheme of finite type over the algebraically closed field k, G is a linear algebraic group acting on X by a morphism $\sigma\colon G\times X \to X$, and L is a line bundle on X. Write $\mathrm{pr}_2\colon G\times X \to X$ for the second projection and, for $g \in G$, write $\sigma_g\colon X \to X$ for the automorphism $x \mapsto gx$.

Definition 15.2.1: G-Linearization of a Line Bundle

Let G act on X via $\sigma\colon G \times X \to X$, and let L be a line bundle on X with total space $\pi\colon \mathbb{L} \to X$. A *$G$-linearization* of L is an action $\widetilde{\sigma}\colon G \times \mathbb{L} \to \mathbb{L}$ of G on the total space $\mathbb{L}$ such that:

1. π is equivariant, that is $\pi(\widetilde{\sigma}(g, v)) = \sigma(g, \pi(v))$, so g carries the fiber L_x to the fiber L_{gx}; and
2. $\widetilde{\sigma}$ is *linear on fibers*: for each $g \in G$ and $x \in X$ the induced map $L_x \to L_{gx}$, $v \mapsto \widetilde{\sigma}(g, v)$, is a linear isomorphism of one-dimensional vector spaces.

A line bundle equipped with a G-linearization is called *G-linearized*. Equivalently, a G-linearization is an isomorphism of line bundles on $G \times X$

$$\Phi\colon \sigma^* L \xrightarrow{\sim} \mathrm{pr}_2^* L \tag{15.1}$$

satisfying the *cocycle condition* on $G\times G\times X$: writing $m\colon G\times G \to G$ for multiplication, the two ways of comparing $(m \times \mathrm{id}_X)^*$ and the iterated pullbacks agree, that is $\mathrm{pr}_{23}^*\Phi \circ (\mathrm{id}_G \times \sigma)^*\Phi = (m \times \mathrm{id}_X)^*\Phi$.

The two formulations say the same thing from opposite ends. The total-space description (15.1) in the form of $\widetilde{\sigma}$ is the geometric picture: g moves the line L_x over x to the line L_{gx} over gx by a definite linear map, and doing g then h agrees with doing hg in one step (this is what the cocycle condition encodes). The isomorphism Φ is the same data written on the level of sheaves, which is what one computes with: for a fixed g it restricts to isomorphisms $L_{gx} = (\sigma_g^* L)_x \to L_x$, and the cocycle condition is the associativity of the action. The linearity clause is not automatic and is the crux of the definition: an equivariant map of total spaces that scaled the fibers nonlinearly would not descend to an action on sections.

Proposition 15.2.2: The Induced Action on Sections

A G-linearization of L induces, for every $n \geq 0$, a linear representation of G on the space of global sections $H^0(X, L^{\otimes n})$, by the formula

$$(g \cdot s)(x) = \widetilde{\sigma}\big(g,\, s(g^{-1}x)\big) \tag{15.2}$$

where $\widetilde{\sigma}$ is the linearization of $L^{\otimes n}$ induced by that of L. The subspace of invariants $H^0(X, L^{\otimes n})^G$ consists of the sections s with $g \cdot s = s$ for all $g \in G$, equivalently the sections that are equivariant maps $X \to \mathbb{L}^{\otimes n}$. The representation and the formula (15.2) need nothing beyond the finite-type hypothesis on X; only the final clause uses projectivity, namely that when X is projective $\bigoplus_{n \geq 0} H^0(X, L^{\otimes n})^G$ is a graded subring of the section ring $\bigoplus_{n \geq 0} H^0(X, L^{\otimes n})$.

***Proof*:**
A linearization of L induces one on each tensor power $L^{\otimes n}$: the isomorphism Φ of (15.1) gives $\Phi^{\otimes n} \colon \sigma^* L^{\otimes n} \to \mathrm{pr}_2^* L^{\otimes n}$, and the cocycle condition for Φ passes to $\Phi^{\otimes n}$ because pullback and tensor product are functorial, so each $L^{\otimes n}$ is G-linearized. Fix n and abbreviate $M = L^{\otimes n}$.

Step 1: The formula defines a linear map. For $g \in G$ and a global section $s \in H^0(X, M)$ define $g \cdot s$ by (15.2). Concretely, s is a morphism $X \to \mathbb{M}$ splitting π; the map $g \cdot s$ sends x to $\widetilde{\sigma}(g, s(g^{-1}x))$, which lies in the fiber $M_{g \cdot g^{-1}x} = M_x$ by fiber-preservation, so $g \cdot s$ is again a section. It is regular because it is the composite of the regular maps $x \mapsto g^{-1}x$, then s, then $\widetilde{\sigma}(g, -)$. Linearity in s is the fiberwise linearity of $\widetilde{\sigma}$ (clause (2) of definition 15.2.1): for $a, b \in k$ and sections s, t,

$$\big(g \cdot (as + bt)\big)(x) = \widetilde{\sigma}\big(g, a\, s(g^{-1}x) + b\, t(g^{-1}x)\big) = a\,(g \cdot s)(x) + b\,(g \cdot t)(x)$$

since $\widetilde{\sigma}(g, -) \colon M_{g^{-1}x} \to M_x$ is linear.

Step 2: The action axioms. That $e \in G$ acts as the identity is the identity axiom for $\widetilde{\sigma}$. For $g, h \in G$,

$$\big(g \cdot (h \cdot s)\big)(x) = \widetilde{\sigma}\big(g, (h \cdot s)(g^{-1}x)\big) = \widetilde{\sigma}\big(g, \widetilde{\sigma}(h, s(h^{-1}g^{-1}x))\big) = \widetilde{\sigma}\big(gh, s((gh)^{-1}x)\big) = \big((gh) \cdot s\big)(x)$$

where the third equality is the action axiom $\widetilde{\sigma}(g, \widetilde{\sigma}(h, v)) = \widetilde{\sigma}(gh, v)$ for $\widetilde{\sigma}$ and $(gh)^{-1} = h^{-1}g^{-1}$. So $g \mapsto (s \mapsto g \cdot s)$ is a homomorphism $G \to \mathrm{GL}(H^0(X, M))$; that it is a morphism of algebraic groups, hence a rational representation, follows because (15.2) is given by the morphism σ and the linearization, both algebraic.

Step 3: Invariants form a ring. A section s is invariant iff $s(gx) = \widetilde{\sigma}(g, s(x))$ for all g, x, which is precisely the condition that $s \colon X \to \mathbb{M}$ be G-equivariant. If $s \in H^0(X, L^{\otimes m})^G$ and $t \in$

$H^0(X, L^{\otimes n})^G$ are invariant, their product $st \in H^0(X, L^{\otimes(m+n)})$ satisfies $g\cdot(st) = (g\cdot s)(g\cdot t) = st$, because the linearization on $L^{\otimes(m+n)}$ is the tensor product of those on $L^{\otimes m}$ and $L^{\otimes n}$; and $1 \in H^0(X, \mathcal{O}_X) = H^0(X, L^{\otimes 0})$ is invariant. Hence $\bigoplus_n H^0(X, L^{\otimes n})^G$ is a graded subring of the section ring, completing the proof.

The invariant subring $R(X, L)^G := \bigoplus_{n\geq 0} H^0(X, L^{\otimes n})^G$ is the object from which the projective quotient is built: the next section forms Proj of it, and theorem 15.1.2 of section 15.1 is what guarantees this ring is finitely generated so the Proj is a projective scheme. What proposition 15.2.2 isolates now is that the ring depends on the linearization, not on L alone: the same underlying line bundle carried by two different linearizations produces two different G-actions on the same vector spaces $H^0(X, L^{\otimes n})$, hence two different invariant subrings and two different quotients. The bookkeeping of linearizations is therefore not a formality but the input that selects the quotient. To handle it systematically one collects all linearizations of all line bundles into a single algebraic structure.

The set of linearized line bundles carries the operations one expects of line bundles: two linearizations tensor to a linearization of the tensor product, and the dual of a linearization linearizes the dual bundle. The trivial bundle $\mathcal{O}_X$ has the trivial linearization in which G acts on $\mathcal{O}_X = X \times \mathbb{A}^1$ only through X. These operations make the linearized line bundles into a group, refining the Picard group.

Proposition 15.2.3: The Equivariant Picard Group

Let G act on X. The isomorphism classes of G-linearized line bundles on X form an abelian group $\operatorname{Pic}^G(X)$ under tensor product, with identity the trivially linearized $\mathcal{O}_X$ and inverse given by the dual linearization on $L^\vee$. Forgetting the linearization is a homomorphism

$$\operatorname{Pic}^G(X) \longrightarrow \operatorname{Pic}(X) \tag{15.3}$$

whose kernel is the character group $\widehat{G} = \operatorname{Hom}(G, \mathbb{G}_m)$ of G whenever X is connected and complete with $H^0(X, \mathcal{O}_X) = k$. Moreover:

1. if G is connected and X is normal, then every line bundle is linearizable after passing to a power: for each $L \in \operatorname{Pic}(X)$ there is an integer $N \geq 1$ (depending on L) such that $L^{\otimes N}$ admits a G-linearization. Equivalently, the cokernel of (15.3) is a torsion group.

***Proof*:**
Step 1: The group structure. Given linearizations Φ_L of L and Φ_M of M in the cocycle form (15.1), the isomorphism $\Phi_L \otimes \Phi_M : \sigma^*(L \otimes M) \to \mathrm{pr}_2^*(L \otimes M)$ satisfies the cocycle condition because each factor does and the condition is multiplicative under tensor; this linearizes $L \otimes M$ and is well defined on isomorphism classes. Associativity and commutativity are inherited from tensor product of line bundles. The trivial bundle with $\Phi = \mathrm{id}$ is a two-sided identity. For an inverse, the dual isomorphism $(\Phi_L^\vee)^{-1} : \sigma^* L^\vee \to \mathrm{pr}_2^* L^\vee$ linearizes $L^\vee$, and $L \otimes L^\vee \cong \mathcal{O}_X$ carries the trivial linearization, so $[L^\vee] = [L]^{-1}$. Thus $\operatorname{Pic}^G(X)$ is an abelian group and (15.3) is a homomorphism, since the forgetful map discards Φ and respects tensor product.

Step 2: The kernel. An element of the kernel is a linearization Φ of the *trivial* class in $\operatorname{Pic}(X)$, that is a linearization of $\mathcal{O}_X$ up to isomorphism. A linearization of $\mathcal{O}_X$ is an isomorphism $\Phi : \sigma^*\mathcal{O}_X \to \mathrm{pr}_2^*\mathcal{O}_X$ of the trivial bundle on $G \times X$, hence multiplication by an invertible global function $u \in H^0(G \times X, \mathcal{O})^\times$. Since X is complete and connected with $H^0(X, \mathcal{O}_X) = k$, and G is affine, $H^0(G \times X, \mathcal{O})^\times = H^0(G, \mathcal{O}_G)^\times$: an invertible function on $G \times X$ is constant in the X direction, so $u = u(g)$ is a function of g alone. The cocycle condition on Φ becomes $u(gh) = u(g)u(h)$, that is $u\colon G \to \mathbb{G}_m$ is a homomorphism of algebraic groups, a character. Two such linearizations are isomorphic iff the characters coincide (an isomorphism of linearized trivial bundles is multiplication by a constant, which cancels). Hence the kernel is $\widehat{G} = \mathrm{Hom}(G, \mathbb{G}_m)$.

Step 3: Linearizability of a power. Part (1) is a theorem of Mumford, proved in his *Geometric Invariant Theory* (Chapter 1, §3): if G is a connected linear algebraic group acting on a normal variety X over k, then for every line bundle L there is $N \geq 1$ with $L^{\otimes N}$ G-linearizable. The proof compares $\sigma^* L$ and $\mathrm{pr}_2^* L$ on $G \times X$: their difference is measured by a class in $\mathrm{Pic}\,(G)$ (after restriction to slices), and the obstruction to constructing Φ lies in the finite group $\mathrm{Pic}\,(G)$ for G connected, finiteness of Pic of a connected linear algebraic group being itself a result of Mumford (*GIT* Chapter 1, §3), which is killed by passing to a suitable power $L^{\otimes N}$; the cocycle condition is then arranged using $\mathrm{Pic}\,(G \times G) = \mathrm{Pic}\,(G) \oplus \mathrm{Pic}\,(G)$ and connectedness. The full argument uses the structure of Pic of a connected algebraic group; it is cited rather than reproduced here because it draws on the Picard group of an algebraic group beyond this chapter's scope, and the statement is all that the constructions of this book require. This completes the proof.

The exact sequence packaged by proposition 15.2.3,

$$0 \longrightarrow \widehat{G} \longrightarrow \mathrm{Pic}^{\,G}(X) \longrightarrow \mathrm{Pic}\,(X)$$

(valid for X connected complete with only constant global functions) is the organizing principle. It says two things at once. First, whether a given line bundle can be linearized at all is a question about the image of the forgetful map, and part (1) answers it affirmatively up to a power for connected G and normal X, which is exactly the generality of moduli constructions: the ample bundle on a parameter space can always be linearized after replacing it by a power, and replacing L by $L^{\otimes N}$ does not change Proj of the section ring, so no geometry is lost. Second, when a linearization exists it is unique only up to the character group $\widehat{G}$: twisting a linearization by a character $\chi \in \widehat{G}$ produces another linearization of the same bundle, and this twisting is precisely the freedom that will move the semistable locus. For a semisimple group such as SL_{n+1} one has $\widehat{G} = 0$, so the linearization is unique and there is nothing to twist; for a group with characters, such as GL_{n+1} or a torus, the twist is a parameter that changes the invariants. This dichotomy is the whole content of the worked example.

To twist a linearization Φ of L by a character $\chi\colon G \to \mathbb{G}_m$ means to replace the action on the total space by $\widetilde{\sigma}_\chi(g, v) = \chi(g) \cdot \widetilde{\sigma}(g, v)$, scaling each fiber map by $\chi(g)$; equivalently, tensoring the linearized bundle by the trivial bundle carrying the linearization χ. On sections, by (15.2), this multiplies the action on $H^0(X, L^{\otimes n})$ by χ^n, so a section that transformed by a weight ψ under the χ-untwisted action transforms by $\psi + n\chi$ after the twist. Invariance is the vanishing of the total weight, so twisting shifts which sections are invariant. The following example computes this shift in the case of projective space and reads off its effect on the invariant sections.

Example 15.2.4: Linearizations of $\mathcal{O}_{\mathbb{P}^n}(1)$

Let $V = k^{n+1}$ and $\mathbb{P}^n = \mathbb{P}(V)$, and let $\mathrm{GL}(V) = \mathrm{GL}_{n+1}$ act on $\mathbb{P}^n$ in the standard way, $g \cdot [v] = [gv]$. This example carries out three computations: the canonical linearization of $\mathcal{O}(1)$, the resulting action on sections and its invariants, and the effect of twisting by a character.

The canonical linearization. The tautological line bundle $\mathcal{O}_{\mathbb{P}^n}(-1)$ has total space $\{([v], w) \in \mathbb{P}^n \times V \,|\, w \in kv\}$, the union of the lines parametrized by $\mathbb{P}^n$. The group $\mathrm{GL}(V)$ acts on $\mathbb{P}^n \times V$ by $g \cdot ([v], w) = ([gv], gw)$, and this preserves the incidence $w \in kv$ (if $w = \lambda v$ then $gw = \lambda\, gv$), so it restricts to an action on $\mathcal{O}(-1)$ that covers the action on $\mathbb{P}^n$ and is linear on fibers: over $[v]$ the fiber is the line kv, and g maps it isomorphically to $k(gv)$ by $w \mapsto gw$. This is a $\mathrm{GL}(V)$-linearization of $\mathcal{O}(-1)$ in the sense of definition 15.2.1, hence, dualizing (proposition 15.2.3), a canonical linearization of $\mathcal{O}(1) = \mathcal{O}(-1)^\vee$ and of every $\mathcal{O}(d) = \mathcal{O}(1)^{\otimes d}$.

The action on sections. Global sections of $\mathcal{O}(d)$ for $d \geq 0$ are the degree-d homogeneous polynomials on V,

$$H^0(\mathbb{P}^n, \mathcal{O}(d)) = \mathrm{Sym}^d(V^\vee)$$

and the induced action (15.2) is the natural action of $\mathrm{GL}(V)$ on $\mathrm{Sym}^d(V^\vee)$: a linear form $\ell \in V^\vee$ transforms by $(g \cdot \ell)(v) = \ell(g^{-1}v)$, and degree-$d$ forms by extension. To detect invariants it suffices to test the central torus $Z = \{t \cdot \mathrm{id}_V \big| t \in \mathbb{G}_m\} \subseteq \mathrm{GL}(V)$. On V the scalar $t \cdot \mathrm{id}$ acts by t, so on $V^\vee$ by t^{-1}, so on $\mathrm{Sym}^d(V^\vee)$ by t^{-d}: every degree-d form is an eigenvector of weight $-d$ for Z. For $d > 0$ this weight is nonzero, so

$$H^0(\mathbb{P}^n, \mathcal{O}(d))^{\mathrm{GL}_{n+1}} = 0 \qquad (d > 0)$$

already because no nonzero form is Z-invariant. The same computation restricted to $\mathrm{SL}(V) = \mathrm{SL}_{n+1}$ gives $H^0(\mathbb{P}^n, \mathcal{O}(d))^{\mathrm{SL}_{n+1}} = 0$ for $d > 0$: the representation $\mathrm{Sym}^d(V^\vee)$ is irreducible and nontrivial, so it has no nonzero invariants. Since SL_{n+1} acts transitively on $\mathbb{P}^n$, there is a single orbit and no invariant section can separate points; with this linearization no point of $\mathbb{P}^n$ is semistable, which foreshadows the definition of semistability in section 15.3.

Twisting by a character. A character of GL_{n+1} is a power of the determinant, $\chi = \det^a$ for $a \in \mathbb{Z}$ (this is $\widehat{\mathrm{GL}_{n+1}} \cong \mathbb{Z}$). Twisting the linearization of $\mathcal{O}(1)$ by χ multiplies the action on $H^0(\mathbb{P}^n, \mathcal{O}(d))$ by $\chi^d = \det^{ad}$ (proposition 15.2.2 applied to the twist). On the central torus Z the determinant is $\det(t \cdot \mathrm{id}) = t^{n+1}$, so $\det^{ad}$ contributes weight $ad(n+1)$, and the total Z-weight of a degree-d form after the twist is

$$-d + ad(n+1) = d\big(a(n+1) - 1\big)$$

which vanishes exactly when $a(n+1) = 1$. No integer a solves this for $n \geq 1$, so no twist by a GL_{n+1}-character produces invariant sections of $\mathcal{O}(d)$: the central weight can be shifted by multiples of $d(n+1)$ but never landed on 0. This is the arithmetic reason projective space itself has no GL_{n+1}-quotient. The twist does move the weight, which is the phenomenon that matters once the group has a positive-dimensional space of invariants to redistribute, as it does for the torus computation below.

A torus, where the twist changes semistability. Let $T = \mathbb{G}_m \subseteq \mathrm{GL}_{n+1}$ act on $\mathbb{P}^n$ diagonally with distinct weights $a_0 < a_1 < \cdots < a_n$, so $t \cdot [x_0 : \cdots : x_n] = [t^{a_0}x_0 : \cdots : t^{a_n}x_n]$, with the canonical linearization of $\mathcal{O}(1)$ restricted from GL_{n+1}. A monomial $x^{\mathbf{m}} = x_0^{m_0} \cdots x_n^{m_n}$ of degree $d = \sum m_i$ in $H^0(\mathbb{P}^n, \mathcal{O}(d))$ then has T-weight $-\sum_i m_i a_i$ under the untwisted linearization. Twisting by the character $t \mapsto t^r$ shifts every weight by rd, giving twisted weight $rd - \sum_i m_i a_i$. A monomial is invariant when this weight vanishes, that is when $\sum_i m_i a_i = rd$, the average weight equals r. The coordinate point $e_i = [0 : \cdots : 1 : \cdots : 0]$ is a T-fixed point, and the only degree-d monomial nonzero at it is x_i^d, whose twisted weight is $rd - da_i = d(r - a_i)$. That monomial is invariant, and e_i therefore semistable, at the single value $r = a_i$ and at no other: a fixed point of a $\mathbb{G}_m$-action has a one-point weight polytope, so 0 lies in its translated weight set only on the wall $r = a_i$. It is the generic points, those with several nonzero coordinates, whose status changes across the walls $r = a_i$: the semistable locus X^{ss} depends on r, and as r crosses a wall a different set of monomials becomes invariant. Two different twists give two different invariant subrings, hence two different GIT quotients of the same $\mathbb{P}^n$ under the same T-action. This is the concrete meaning of the twisting freedom recorded in proposition 15.2.3, and it is the mechanism the numerical criterion of section 15.4 will quantify.

The example demarcates the two regimes cleanly. For a semisimple group the character group vanishes, the linearization is rigid, and the action on projective space, being transitive, has no invariants and no quotient to speak of; the content of geometric invariant theory in that setting comes from acting on a larger parameter space with many orbits, not on $\mathbb{P}^n$ itself. For a group with characters, the linearization is a parameter, and the invariant sections, and with them the eventual quotient, vary with the chosen character. This is why the definition of semistability in the next section is stated relative to a fixed G-linearized ample line bundle: the choice of linearization is part of the data of the quotient, not a preliminary that can be suppressed. The affine quotient of definition 15.1.4 is the degenerate case where $L = \mathcal{O}_X$ carries the trivial linearization and Proj of the invariant ring is replaced by Spec of the degree-0 invariants; linearizations are what generalize that construction to the projective setting where the interesting moduli live.

Exercise 15.2.1

1. Let G act trivially on X (so $\sigma = \mathrm{pr}_2$). Show that a G-linearization of a line bundle L on X is the same datum as a homomorphism $G \to \mathbb{G}_m$, that is a character, together with the identity isomorphism on L; deduce that $\operatorname{Pic}^G(X) \cong \operatorname{Pic}(X) \times \widehat{G}$ for the trivial action when X is connected complete with $H^0(X, \mathcal{O}_X) = k$, and reconcile this with the kernel computation in proposition 15.2.3.

2. Verify the cocycle condition explicitly for the canonical linearization of $\mathcal{O}_{\mathbb{P}^n}(-1)$ built in example 15.2.4: writing the action on the total space as $g \cdot ([v], w) = ([gv], gw)$, check that the associated isomorphism $\Phi\colon \sigma^*\mathcal{O}(-1) \to \mathrm{pr}_2^*\mathcal{O}(-1)$ on $\mathrm{GL}_{n+1} \times \mathbb{P}^n$ satisfies the identity of (15.1). State where the linearity-on-fibers clause of definition 15.2.1 is used.

3. Let $T = \mathbb{G}_m$ act on $\mathbb{P}^1$ by $t \cdot [x_0 : x_1] = [x_0 : tx_1]$, with the canonical linearization of $\mathcal{O}(1)$. Compute the T-weights of the monomials $x_0^d, x_0^{d-1}x_1, \ldots, x_1^d$ in $H^0(\mathbb{P}^1, \mathcal{O}(d))$. For which twist $t \mapsto t^r$ does the space of invariant sections of $\mathcal{O}(d)$ become nonzero, and what monomials span it? Identify the two T-fixed points $[1 : 0]$ and $[0 : 1]$ as semistable or unstable for that twist.

4. Show that replacing a G-linearized ample line bundle L by its power $L^{\otimes N}$ (with the induced linearization) leaves the graded ring $\bigoplus_n H^0(X, L^{\otimes n})^G$ unchanged up to the Veronese regrading, and hence leaves Proj of the invariant ring unchanged. Explain why this makes the linearizability-of-a-power statement in proposition 15.2.3(1) sufficient for the constructions of this chapter.

5. Let G be a finite group acting on X. Show that $\operatorname{Pic}^G(X)$ is the group of G-equivariant line bundles and that the forgetful map $\operatorname{Pic}^G(X) \to \operatorname{Pic}(X)$ need not be injective by exhibiting, for $G = \mathbb{Z}/2$ acting on $X = \operatorname{Spec} k$ by the trivial action, a line bundle whose distinct linearizations are indexed by $\widehat{\mathbb{Z}/2} = \{\pm 1\}$ (assume $\operatorname{char} k \neq 2$). What does this say about the two linearizations of the trivial bundle?

6. Prove that if L and M are G-linearized and $s \in H^0(X, L)^G$, $t \in H^0(X, M)^G$ are invariant sections, then the open sets $X_s = \{s \neq 0\}$ and $X_t = \{t \neq 0\}$ are G-invariant, and the ring $H^0(X_s, \mathcal{O})^G$ is the degree-0 part of the localized invariant ring $\left(\bigoplus_n H^0(X, L^{\otimes n})^G\right)_{(s)}$. Explain in one sentence how this exhibits the projective quotient of the next section as glued from affine quotients (definition 15.1.4) over the X_s.

15.3 Stability and the Projective Quotient

The affine quotient $X/\!\!/G = \operatorname{Spec} k[X]^G$ of definition 15.1.4 solves the quotient problem for affine X, but it is silent about projective varieties, which carry no nonconstant global functions to be invariant. The way out is already visible in the construction of projective space itself: a projective variety is not built from a ring of functions but from a graded ring of *sections* of an ample line bundle, and Proj of that ring recovers the variety. A G-linearization (definition 15.2.1) endows each space of sections $H^0(X, L^{\otimes n})$ with a linear G-action, so the invariant sections $H^0(X, L^{\otimes n})^G$ form a graded subring, and its Proj is the natural candidate for the quotient.

Two things must be settled before this candidate is usable. First, not every point of X lies in the domain of this rational quotient map: a point where every invariant section vanishes has no invariant coordinate to record its image, and must be discarded. The points that survive, where enough invariant sections are nonzero to place the point in projective space, are the *semistable* points; among them, the ones whose orbits are already as large and as separated as the group allows are the *stable* points. Second, the graded ring of invariants must be finitely generated, so that its Proj is a genuine projective scheme; this is exactly Hilbert's finiteness theorem (theorem 15.1.2) applied to the section ring. This section defines the two loci, proves that their quotient is projective, and works out the picture for configurations of points on the line.

The first task is to define the two loci themselves. The definition selects points by the survival of invariant sections and then refines the selection by orbit geometry. Recall that a section $s \in H^0(X, L^{\otimes n})$ does not have a well-defined value in k at a point x, but the condition $s(x) \neq 0$ (that s does not vanish at x, equivalently that s generates the stalk $L_x^{\otimes n}$) is intrinsic. The nonvanishing locus $X_s = \{x \big| s(x) \neq 0\}$ is open, and if s is G-invariant it is G-stable and affine (it is the complement of the zero divisor of a section of an ample bundle). These invariant affine opens are the charts of the quotient.

Definition 15.3.1: Semistable, Stable, and Unstable Points

Let G be a reductive group acting on a scheme X of finite type over k, and let L be a G-linearized ample line bundle on X. A point $x \in X$ is:

1. *semistable* (with respect to L) if there is an integer $n > 0$ and an invariant section $s \in H^0(X, L^{\otimes n})^G$ with $s(x) \neq 0$;
2. *stable* if it is semistable, some such s can be chosen with X_s affine and the action of G on X_s closed (every orbit in X_s is closed in X_s), the stabilizer G_x is finite (equivalently $\dim Gx = \dim G$), and the orbit Gx is closed in X^{ss};
3. *unstable* if it is not semistable, that is, every invariant section of every positive power of L vanishes at x.

The *semistable locus* $X^{\mathrm{ss}} = X^{\mathrm{ss}}(L)$ is the set of semistable points, and the *stable locus* $X^{\mathrm{s}} = X^{\mathrm{s}}(L)$ the set of stable points.

The three conditions carve X into strata of decreasing pathology. An unstable point is invisible to invariant theory: no invariant survives to record it, so it cannot appear in any quotient built from invariant sections and is thrown away. A semistable point is recorded, but perhaps not faithfully, because invariant sections cannot always tell it apart from other points in the closure of its orbit. A stable point is recorded faithfully: its orbit is closed and its stabilizer finite, so its orbit is exactly one point of the quotient and the fiber over that point is a single orbit. The passage from semistable to stable is the passage from a *good* quotient, which may glue several orbits to a point, to a *geometric* quotient, whose points are exactly the orbits.

The loci are open. If x is semistable, witnessed by $s \in H^0(X, L^{\otimes n})^G$, then every point of the open set X_s is semistable by the same section, so $X^{\mathrm{ss}} = \bigcup_{n,s} X_s$ is a union of opens and hence open. The stable locus is open for a subtler reason, established in the proof of the main theorem below: the conditions of finite stabilizer and closed orbit are open conditions on the semistable locus. Both loci are visibly G-stable, since the defining conditions are phrased entirely in terms of invariant data.

15.3.2 Warning: Stability depends on the linearization The loci X^{ss} and X^{s} depend on L, not only on the underlying line bundle but on the chosen linearization. Two linearizations of the same ample bundle can differ by a character of G, and this shift can move a point across the boundary between stable, strictly semistable, and unstable. The dependence is not a defect but a feature: varying the linearization produces a family of birational models of the quotient, and the study of how the quotient changes as L crosses a wall is the subject of variation of GIT. For a fixed linearization, however, the loci are determined, and the ambiguity of definition 15.2.1 by a character is exactly what makes this variation possible.

A concrete instance grounds the definition before the theory begins. It also previews the extended computation of example 15.3.6.

Example 15.3.3: The Torus Acting on the Projective Line

Let $G = \mathbb{G}_m$ act on $X = \mathbb{P}^1$ by $t \cdot [x_0 : x_1] = [tx_0 : t^{-1}x_1]$, and linearize $L = \mathcal{O}_{\mathbb{P}^1}(2)$ by the natural action on $H^0(\mathbb{P}^1, \mathcal{O}(2))$, the space of binary quadratics in x_0, x_1. A monomial $x_0^a x_1^b$ with $a + b = 2$ is scaled by t^{a-b}, so the invariant sections in $H^0(\mathcal{O}(2))$ are the multiples of x_0x_1, and in $H^0(\mathcal{O}(2)^{\otimes n}) = H^0(\mathcal{O}(2n))$ the invariants are spanned by $(x_0x_1)^n$.

A point $[x_0 : x_1]$ is semistable iff $(x_0x_1)^n$ is nonzero there for some n, that is, iff $x_0x_1 \neq 0$. Thus the two fixed points $0 = [0 : 1]$ and $\infty = [1 : 0]$ are unstable, and every other point is semistable. The orbit of any $[x_0 : x_1]$ with $x_0x_1 \neq 0$ is $\{[tx_0 : t^{-1}x_1]\}$; in the coordinate $\xi = x_0/x_1$ on $\mathbb{P}^1 \setminus \{0, \infty\}$ the action is $\xi \mapsto t^2\xi$, and as t ranges over $\mathbb{G}_m$ the scalar t^2 ranges over all of $\mathbb{G}_m$, so the action is transitive on $X^{\mathrm{ss}} = \mathbb{P}^1 \setminus \{0, \infty\} \cong \mathbb{G}_m$. Each orbit is therefore all of X^{ss}, in particular closed in X^{ss}. The stabilizer is $\mu_2 = \{\pm 1\}$, since $t \cdot [x_0 : x_1] = [x_0 : x_1]$ forces $t = t^{-1}$ in projective coordinates, that is, $t^2 = 1$; this is finite, so the orbit has full dimension $\dim \mathbb{G}_m$ and every semistable point is stable. The invariant section ring is $R = \bigoplus_n H^0(\mathcal{O}(2n))^{\mathbb{G}_m} = k[w]$ with $w = x_0x_1$ a single generator in positive degree, so

$$X^{\mathrm{ss}} /\!\!/ G = X^{\mathrm{s}}/G = \operatorname{Proj} k[w] = \operatorname{Spec} k$$

is a single point. The only $\mathbb{G}_m$-invariant regular functions on X^{ss} are the constants, so there is no modulus: any two semistable points lie in one orbit, and the quotient records nothing beyond the single class.

The two unstable points are exactly the two toric fixed points, where the $\mathbb{G}_m$-action degenerates and no invariant survives. The stable locus is the open orbit-swept complement, a single orbit, and its geometric quotient is a point. This is the smallest nontrivial GIT quotient, and every feature of the general theory is already present in it: an unstable locus to discard, a stable locus with closed orbits, and a quotient recorded by the Proj of the invariant section ring, here a single point because the ring has only one generator in positive degree.

With the loci in hand, the main construction assembles them into the projective quotient itself. The mechanism is a globalization of the affine quotient: over each invariant affine chart X_s the good quotient is the affine GIT quotient of definition 15.1.4, and these affine quotients glue along overlaps to a good quotient of the whole semistable locus, whose total object is Proj of the invariant section ring. Finite generation of that ring, theorem 15.1.2, guarantees the Proj is a projective scheme.

Before stating the theorem, recall the two quotient notions it delivers. A *good quotient* $\pi\colon X^{\mathrm{ss}} \to Y$ (definition 15.1.1) is an affine, G-invariant, surjective map with $\mathcal{O}_Y = (\pi_*\mathcal{O}_{X^{\mathrm{ss}}})^G$ that separates disjoint closed invariant subsets; its fibers are unions of orbits, with exactly one closed orbit apiece. A *geometric quotient* is a good quotient whose fibers are single orbits. The theorem says the good quotient of the semistable locus is projective, and that it restricts to a geometric quotient on the stable locus.

Theorem 15.3.4: The Projective GIT Quotient

Let G be a reductive group acting on a projective scheme X over k, and let L be a G-linearized ample line bundle. Set

$$R = R(X,L)^G = \bigoplus_{n\geq 0} H^0(X, L^{\otimes n})^G$$

the graded ring of invariant sections. Then:

1. R is a finitely generated graded k-algebra, and the quotient
$$X^{\mathrm{ss}}/\!\!/G := \operatorname{Proj} R$$
is a projective scheme over k.

2. The inclusions of invariant affine charts induce a morphism $\pi\colon X^{\mathrm{ss}} \to X^{\mathrm{ss}}/\!\!/G$ that is a good quotient. In particular π is surjective, G-invariant, affine, and its fibers are in bijection with the closed G-orbits in X^{ss}; two semistable points have the same image iff the closures of their orbits meet in X^{ss}.

3. The stable locus X^{s} is open and G-stable, its image $Y^{\mathrm{s}} = \pi(X^{\mathrm{s}})$ is open in $X^{\mathrm{ss}}/\!\!/G$, and the restriction $\pi\colon X^{\mathrm{s}} \to Y^{\mathrm{s}} = X^{\mathrm{s}}/G$ is a geometric quotient. Its fibers are exactly the orbits of stable points.

***Proof*:**
The argument builds the quotient chart by chart, glues, and then identifies the fibers.

Step 1: Finite generation and the projective scheme. Since X is projective and L is ample, the full section ring $R(X,L) = \bigoplus_{n\geq 0} H^0(X, L^{\otimes n})$ is a finitely generated graded k-algebra with $\operatorname{Proj} R(X,L) = X$ (the standard fact that an ample bundle recovers its projective variety as Proj of its section ring). The G-action, made linear on each graded piece by the linearization of definition 15.2.1, makes $R(X,L)$ a graded k-algebra on which G acts by graded algebra automorphisms. Its invariant subring is the graded ring $R = R(X,L)^G$. By Hilbert's finiteness theorem (theorem 15.1.2), the invariant subalgebra of a finitely generated k-algebra under a reductive group is again finitely generated; applied to the graded algebra $R(X,L)$ this gives that R is a finitely generated graded k-algebra. Hence $\operatorname{Proj} R$ is a projective scheme over $\operatorname{Proj} R_0 = \operatorname{Spec} k$, proving (1). Replacing L by a power if necessary, one may assume R is generated in a single degree, which is the standing convenience for the Proj description; this changes neither the loci nor the quotient, since $H^0(X, L^{\otimes nm})^G$ has the same nonvanishing loci as its subring generated by $H^0(X, L^{\otimes m})^G$.

Step 2: The affine charts and the morphism π. For a homogeneous invariant $s \in R_n = H^0(X, L^{\otimes n})^G$, the nonvanishing locus $X_s \subseteq X$ is G-stable, open, and affine (the complement of the support of a section of an ample bundle is affine), and its ring of functions is the degree-zero part of the localization,

$$k[X_s] = (R(X,L)_s)_0 = \left\{ \frac{f}{s^m} \;\middle|\; f \in H^0(X, L^{\otimes nm}),\ m \geq 0 \right\}$$

On this affine open the G-action is an action of a reductive group on an affine scheme, so the affine GIT quotient of definition 15.1.4 applies: $X_s /\!\!/ G = \operatorname{Spec} k[X_s]^G$, and $k[X_s]^G = (R_s)_0$ is the degree-zero part of the localization of the *invariant* ring R at s. But $(R_s)_0$ is exactly the ring of functions on the basic open $D_+(s) \subseteq \operatorname{Proj} R$. Thus the affine quotient of the chart X_s is canonically the affine open $D_+(s)$ of the projective scheme $\operatorname{Proj} R$:

$$X_s /\!\!/ G = \operatorname{Spec}(R_s)_0 = D_+(s) \subseteq \operatorname{Proj} R$$

As s ranges over homogeneous invariants, the X_s cover X^{ss} (by the definition of semistability, every semistable point lies in some X_s) and the $D_+(s)$ cover $\operatorname{Proj} R$. The affine quotient maps $X_s \to D_+(s)$ agree on overlaps $X_s \cap X_{s'} = X_{ss'}$, because on the overlap both restrict to the affine quotient of $X_{ss'}$: localization at s' commutes with taking invariants (invariance is preserved under multiplication by the invariant s' and its inverse, both G-fixed), so $(R_{ss'})_0$ is the common ring of functions computed either way. The maps therefore glue to a single morphism

$$\pi \colon X^{\mathrm{ss}} \longrightarrow \operatorname{Proj} R = X^{\mathrm{ss}} /\!\!/ G$$

Step 3: π is a good quotient. Each of the properties defining a good quotient (definition 15.1.1) is affine-local on the target and holds on each chart $D_+(s)$ by the corresponding property of the affine GIT quotient $X_s \to X_s /\!\!/ G$ (which is a good quotient because G is reductive, the content of the affine theory around definition 15.1.4). Explicitly: π is affine, since $\pi^{-1}(D_+(s)) = X_s$ is affine over $D_+(s)$; π is G-invariant, since each chart map is; π is surjective, since each $X_s \to D_+(s)$ is; and $\mathcal{O}_Y = (\pi_* \mathcal{O}_{X^{\mathrm{ss}}})^G$, since on each chart this reads $k[X_s]^G = (R_s)_0$, which is the invariance statement of the affine quotient. The orbit-separation property, that π separates disjoint closed G-invariant subsets, is again checked chart by chart: on the affine X_s it is exactly the separation property of the affine GIT quotient $X_s \to X_s /\!\!/ G$ established for reductive G in the affine theory around definition 15.1.4 (invariant functions separate disjoint closed invariant sets). This is the affine good-quotient property, so π is a good quotient. The fiber description follows: a good quotient identifies two points iff their orbit closures meet, and each fiber contains a unique closed orbit, proving (2).

Step 4: The stable locus and the geometric quotient. It remains to show that on X^{s} the good quotient π becomes geometric, that is, its fibers are single orbits, and that X^{s} is open with open image. Let $x \in X^{\mathrm{s}}$, so Gx is closed in X^{ss} with finite stabilizer, and x lies in an invariant affine chart X_s on which every orbit is closed. Consider the fiber $\pi^{-1}(\pi(x))$. It is a closed invariant subset of the affine X_s containing the closed orbit Gx. If it contained a second orbit Gy, then Gy would also be closed in X_s (all orbits in X_s are closed), and Gx, Gy would be disjoint closed invariant sets in the same fiber, contradicting the orbit-separation property of Step 3, which forces the closed orbits in a single fiber to coincide. Hence $\pi^{-1}(\pi(x)) = Gx$ is a single orbit, so $\pi|_{X^{\mathrm{s}}}$ is a geometric quotient onto its image.

For openness, work chart by chart on the affine X_s and combine two facts. First, the finite-stabilizer condition is open: the function $x \mapsto \dim Gx$ is lower semicontinuous on X_s (orbit dimension can only jump down under specialization), so its maximal value $\dim G$ is attained on an open set, and $\dim G_x = \dim G - \dim Gx$, so $U_s = \{x \in X_s \,|\, \dim G_x = 0\} = \{x \,|\, \dim Gx = \dim G\}$ is open. Finite stabilizer does not by itself force a closed orbit, which is why the second condition must be imposed and its openness argued separately.

Second, the closed-orbit condition is open on U_s. Let $\pi_s \colon X_s \to X_s /\!\!/ G$ be the affine good quotient. For $x \in U_s$ every orbit through U_s has the maximal dimension $\dim G$, so if Gx is not closed in X_s its boundary $\overline{Gx} \setminus Gx$ is a nonempty G-stable closed set consisting of orbits of dimension strictly less than $\dim G$; each such boundary point therefore lies in the strictly-smaller-dimension locus $X_s \setminus U_s$. Consequently, for $x \in U_s$ the orbit Gx is closed in X_s if and only if the fiber $\pi_s^{-1}(\pi_s(x))$ meets U_s in Gx alone: the unique closed orbit in that fiber (which every good-quotient fiber contains) has dimension $\dim G$ and hence lies in U_s, and it equals Gx exactly when Gx is itself closed. The non-closed orbits in U_s are therefore precisely the points of U_s lying in the π_s-saturation of $X_s \setminus U_s$, that is, $U_s \cap \pi_s^{-1}\big(\pi_s(X_s \setminus U_s)\big)$: a maximal-dimension orbit fails to be closed exactly when its closure meets the lower-dimensional locus, which happens exactly when it shares a π_s-fiber with a point of $X_s \setminus U_s$. Since π_s is a good quotient it maps the closed G-stable set $X_s \setminus U_s$ to a closed set (good quotients send closed invariant sets to closed sets), so $\pi_s^{-1}(\pi_s(X_s \setminus U_s))$ is closed, and its complement intersected with U_s,

$$X_s^{\mathrm{s}} := U_s \setminus \pi_s^{-1}\big(\pi_s(X_s \setminus U_s)\big)$$

is open in X_s. This is exactly the set of $x \in X_s$ with finite stabilizer and closed orbit, so the closed-orbit condition is open on U_s. Taking the union over homogeneous invariants s, the stable locus $X^{\mathrm{s}} = \bigcup_s X_s^{\mathrm{s}}$ is open and G-stable. Its image $Y^{\mathrm{s}} = \pi(X^{\mathrm{s}})$ is open because π is a good quotient (open G-saturated sets have open image under a good quotient), and $X^{\mathrm{s}} = \pi^{-1}(Y^{\mathrm{s}})$ is G-saturated by construction. Thus $\pi \colon X^{\mathrm{s}} \to Y^{\mathrm{s}}$ is a geometric quotient X^{s}/G, completing the proof.

The theorem is the projective completion of the affine story. Where the affine quotient $X /\!\!/ G$ collapses the failure of orbit separation into a single scheme with no discarded points, the projective quotient must first discard the unstable locus, on which invariant theory is blind, and then perform the same collapse on what remains. Part (2) is the statement that on the semistable locus one can only build a good quotient: strictly semistable orbits, those that are not closed, get glued to the closed orbit in their closure, and the quotient cannot see the difference. Part (3) is the reward for the extra hypotheses defining stability: on the stable locus the orbits are already closed and separated, so nothing is glued, and the quotient is geometric. The moduli interpretation is immediate: if the objects being classified correspond to stable points and their automorphisms to the finite stabilizers, then X^{s}/G is a coarse moduli space, since its points are exactly the isomorphism classes.

15.3.5 Remark: Strictly semistable points and S-equivalence A point that is semistable but not stable is called *strictly semistable*. Its orbit is not closed in X^{ss}, so its closure contains a strictly smaller orbit, and π sends both to the same point. Two semistable points with the same image are called *S-equivalent*: they are identified in the good quotient precisely because their orbit closures meet. The points of $X^{\mathrm{ss}}/\!\!/G$ are thus in bijection not with orbits but with S-equivalence classes, equivalently with closed orbits in X^{ss}. This is why the semistable quotient is only a good quotient and not geometric: the price of including strictly semistable points is that distinct orbits can share an image. In the moduli of vector bundles, S-equivalence is exactly the relation identifying a bundle with the direct sum of the factors of its Jordan-Holder filtration.

The construction is tested on a concrete moduli problem: configurations of points on the line, that is, unordered configurations of n points on $\mathbb{P}^1$ taken up to the automorphisms of the line. This is the projective analogue of the affine example above and the setting in which the classical cross-ratio reappears as the first invariant.

The automorphism group of $\mathbb{P}^1$ is PGL_2, and n unordered points are the roots of a binary form of degree n, an element of $\mathbb{P}H^0(\mathbb{P}^1, \mathcal{O}(n)) = \mathbb{P}^n$ up to scalar. The action of PGL_2 on configurations lifts to the action of SL_2 on binary forms (the $2:1$ cover $\mathrm{SL}_2 \to \mathrm{PGL}_2$ has kernel $\pm I$, which acts on a degree-n form by the scalar $(-1)^n$ and hence trivially on $\mathbb{P}^n$ for every n), so the moduli problem is the GIT quotient of $\mathbb{P}^n = \mathbb{P}(\mathrm{Sym}^n(k^2)^\vee)$ by SL_2, linearized by $\mathcal{O}_{\mathbb{P}^n}(1)$ with its natural SL_2-action. Stability of a configuration is then decided by the invariant sections, which are the classical invariants of binary forms.

Example 15.3.6: n Points on $\mathbb{P}^1$ modulo PGL_2

Let SL_2 act on binary forms of degree n, that is, on $V = \mathrm{Sym}^n(k^2)^\vee$ with coordinates the coefficients of $F(x,y) = \sum_{i=0}^n a_i \binom{n}{i} x^{n-i} y^i$, and consider $X = \mathbb{P}(V) = \mathbb{P}^n$ with $L = \mathcal{O}_{\mathbb{P}^n}(1)$ linearized by the induced SL_2-action. A point of X is a configuration $D = \{p_1, \dots, p_n\}$ of n points on $\mathbb{P}^1$ (with multiplicity), the zero divisor of F. The stability of D is governed by the multiplicities of its points.

The stability criterion. A configuration D is

1. *stable* iff every point of $\mathbb{P}^1$ occurs in D with multiplicity strictly less than $n/2$;
2. *semistable* iff every point occurs with multiplicity at most $n/2$;
3. *unstable* iff some point occurs with multiplicity strictly greater than $n/2$.

The full derivation of this criterion from one-parameter subgroups is the content of the Hilbert-Mumford analysis of section 15.4; here the picture is read off directly from invariant sections in the small cases. The heuristic is that a form is unstable exactly when it can be pushed to 0 by the group, and a form with a root of multiplicity $> n/2$ can be scaled to concentrate all its mass at that root, sending every invariant to zero.

The case $n = 2$. A binary quadratic $F = a_0x^2 + 2a_1xy + a_2y^2$ has the discriminant $\Delta = a_0a_2 - a_1^2$ as an SL_2-invariant, and Δ spans the invariants in low degree. So D is semistable iff $\Delta \neq 0$, that is, iff the two roots are distinct, each of multiplicity 1, equal to the threshold $2/2 = 1$; since 1 is not strictly less than 1, no configuration of two points is stable, and the semistable ones are the pairs of distinct points. But any two ordered distinct points can be moved to $\{0, \infty\}$ by PGL_2, and the stabilizer of $\{0, \infty\}$ is positive-dimensional (the diagonal torus). The quotient $X^{\mathrm{ss}} /\!\!/ \mathrm{SL}_2$ is a single point: all pairs of distinct points are equivalent, and there is nothing to record. This reflects the direct fact that any pair of distinct points can be carried to $\{0, \infty\}$ by an element of PGL_2, so PGL_2 acts transitively on unordered pairs of distinct points.

The case $n = 3$. For a binary cubic the analogous invariant is again the discriminant, of degree 4 in the coefficients, nonzero iff the three roots are distinct. Here $n/2 = 3/2$, so a point may appear with multiplicity $1 < 3/2$ but a double point has multiplicity $2 > 3/2$ and is unstable. Since $3/2$ is not an integer, no multiplicity can equal the threshold, so semistable and stable coincide: D is stable = semistable iff its three points are distinct, and unstable as soon as any two collide. Any three distinct points can be moved to $\{0, 1, \infty\}$ by PGL_2: for an ordering of the three points the element carrying them to $(0, 1, \infty)$ in that order is unique, so the unordered set $\{0, 1, \infty\}$ is reached by the six elements corresponding to the six orderings. The stabilizer of the unordered set $\{0, 1, \infty\}$ is therefore the anharmonic group S_3 (order 6, generated by $z \mapsto 1 - z$ and $z \mapsto 1/z$), which is finite, so the quotient is again a single point. Three unordered points on $\mathbb{P}^1$ have no modulus.

The case $n = 4$ and the cross-ratio. Four points are the first configuration with a genuine modulus. Here $n/2 = 2$: D is stable iff all four points are distinct (each multiplicity $1 < 2$), strictly semistable iff its maximum multiplicity is exactly 2 (so all multiplicities are ≤ 2 but not all < 2), and unstable iff some point has multiplicity ≥ 3. The strictly semistable locus therefore comprises both the two-double-point configurations $(2, 2)$ and the one-double-point configurations $(2, 1, 1)$: for instance the quartic with roots $\{0, 0, 1, -1\}$, $F = x^4 - x^2y^2$, has $a_0 = 1$ and $a_2 = -1/6$, so $I = 3a_2^2 = 1/12 \neq 0$; not every invariant vanishes, and F is semistable but not stable. The ring of SL_2-invariants of binary quartics is generated by two invariants

$$I = a_0a_4 - 4a_1a_3 + 3a_2^2, \qquad J = a_0a_2a_4 - a_0a_3^2 - a_1^2a_4 + 2a_1a_2a_3 - a_2^3$$

of degrees 2 and 3, with the discriminant $\Delta = I^3 - 27J^2$ up to a scalar. The quotient is

$$X^{\mathrm{ss}} /\!\!/ \mathrm{SL}_2 = \operatorname{Proj} k[I, J] = \mathbb{P}(2, 3) \cong \mathbb{P}^1$$

the weighted projective line $\operatorname{Proj} k[I, J]$ with $\deg I = 2$, $\deg J = 3$, which is abstractly $\mathbb{P}^1$ with coordinate $j = I^3/\Delta$. Four ordered distinct points on $\mathbb{P}^1$ have a cross-ratio $\lambda \in \mathbb{P}^1 \setminus \{0, 1, \infty\}$; forgetting the ordering identifies the six values $\lambda, 1 - \lambda, 1/\lambda, \ldots$ under the anharmonic action of S_3, and the resulting invariant is exactly the classical

$$j(\lambda) = \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}$$

up to normalization. Thus the GIT quotient of four points on $\mathbb{P}^1$ is the j-line. Every strictly semistable quartic, whether of type $(2, 2)$ or $(2, 1, 1)$, is S-equivalent to the single closed strictly semistable orbit x^2y^2: the good quotient identifies all of them with that closed orbit, so the entire strictly semistable locus collapses to the one point of $X^{\mathrm{ss}} /\!\!/ \mathrm{SL}_2$ (the boundary point $j = \infty$, cross-ratio degenerating to $\{0, 1, \infty\}$) where stability fails.

The four-point computation is the first appearance of the j-invariant in this book from the GIT side, and it is the same j-line that arose as the coarse moduli space of elliptic curves in theorem 13.3.6, since an elliptic curve is a double cover of $\mathbb{P}^1$ branched at four points, and the cross-ratio of those four points determines the curve up to isomorphism. The stable locus (four distinct points) is a fine-enough quotient to be geometric; the strictly semistable locus (all configurations of maximum multiplicity exactly 2) contributes the single boundary point where distinct orbits are S-equivalent. The general pattern, that stability is a bound on how concentrated a configuration may be, is what the numerical criterion of the next section makes precise and computable for arbitrary n and arbitrary reductive G.

Exercise 15.3.1

1. Let $\mathbb{G}_m$ act on $\mathbb{A}^2$ by $t \cdot (x, y) = (tx, t^{-1}y)$, and extend to $X = \mathbb{P}^2$ (with coordinates $[x : y : z]$) by fixing the extra coordinate, $t \cdot [x : y : z] = [tx : t^{-1}y : z]$. Linearize $L = \mathcal{O}_{\mathbb{P}^2}(1)$ by this action on the coordinates. Compute the invariant sections in each degree, determine the semistable and stable loci, and identify $X^{\mathrm{ss}} /\!\!/ \mathbb{G}_m$ as a scheme.
2. Show directly from definition 15.3.1 that the unstable locus is closed and G-stable, and that the semistable locus is open and G-stable. Deduce that the boundary $X^{\mathrm{ss}} \setminus X^{\mathrm{s}}$ (the strictly semistable locus) is a closed subset of X^{ss}.
3. In the setting of example 15.3.6 with $n = 5$ (binary quintics under SL_2), determine the stability of the following configurations, using the multiplicity criterion: (i) five distinct points; (ii) one double point and three simple points; (iii) one triple point and two simple points; (iv) a point of multiplicity 4 and one simple point.
4. Let $\pi \colon X^{\mathrm{ss}} \to X^{\mathrm{ss}} /\!\!/ G$ be the good quotient of theorem 15.3.4. Prove that a G-invariant morphism $f \colon X^{\mathrm{ss}} \to Z$ to any scheme Z factors uniquely through π, that is, π is a categorical quotient. (Use the good-quotient property that $\mathcal{O}_Y = (\pi_* \mathcal{O}_{X^{\mathrm{ss}}})^G$ and reduce to the affine charts.)
5. Explain why the quotient of two points on $\mathbb{P}^1$ modulo PGL_2 (the case $n = 2$ of example 15.3.6) is a single point, while the quotient of four points is one-dimensional. Precisely: count the dimension of the expected geometric quotient as $\dim X^{\mathrm{s}} - \dim \mathrm{PGL}_2$ in each case, and reconcile the answer with the stability criterion.

15.4 The Hilbert-Mumford Numerical Criterion

The stability notions of the previous section are defined by the existence or non-existence of invariant sections: a point x is semistable when some G-invariant section of a power of the linearization L fails to vanish at x (definition 15.3.1). This is a clean definition, but it is a poor tool for deciding stability in practice. To certify that x is semistable one must exhibit an invariant section not vanishing at x; to certify that x is unstable one must show that *every* invariant section vanishes there. Both require understanding the full invariant ring $\bigoplus_{n \geq 0} H^0(X, L^{\otimes n})^G$, which is exactly the object whose computation is hard. For the action of SL_2 on binary forms of degree d, the invariant ring is a classical and intricate object whose generators were the subject of nineteenth-century invariant theory; reading stability off its structure directly is not feasible beyond small d.

The Hilbert-Mumford criterion resolves this by replacing the group G with its one-parameter subgroups. A one-parameter subgroup is a homomorphism $\lambda \colon \mathbb{G}_m \to G$, and it turns the abstract question of stability into a finite weight computation: for each λ one forms an integer $\mu(x, \lambda)$ measuring how the linearization degenerates as $\lambda(t) \cdot x$ is driven to its limit at $t \to 0$, and x is semistable if and only if $\mu(x, \lambda) \geq 0$ for every λ. The gain is decisive: whether x is unstable is decided by producing a single destabilizing λ, and $\mu(x, \lambda)$ is computed by diagonalizing the λ-action, a linear-algebra task. The criterion is what makes GIT a computational subject.

Fix a reductive group G acting on a projective scheme X over the algebraically closed field k, together with a G-linearized ample line bundle L (definition 15.2.1). The starting object is the simplest nontrivial family of subgroups of G.

Definition 15.4.1: One-Parameter Subgroup and the Numerical Function

A *one-parameter subgroup* (abbreviated *1-PS*) of G is a homomorphism of algebraic groups

$$\lambda \colon \mathbb{G}_m \longrightarrow G$$

It is *trivial* if it is the constant map to the identity. Fix a point $x \in X$ and consider the orbit map $t \mapsto \lambda(t) \cdot x$, a morphism $\mathbb{G}_m \to X$. Since X is projective, hence proper, this morphism extends uniquely to a morphism $\mathbb{A}^1 \to X$, and its value at $t = 0$ is a well-defined point

$$x_0 \;=\; \lim_{t \to 0} \lambda(t) \cdot x \;\in\; X$$

The point x_0 is fixed by λ: it satisfies $\lambda(s) \cdot x_0 = x_0$ for all $s \in \mathbb{G}_m$. Because x_0 is λ-fixed, $\mathbb{G}_m$ acts through λ on the one-dimensional fiber L_{x_0}, and every action of $\mathbb{G}_m$ on a one-dimensional vector space is scalar multiplication by a character $t \mapsto t^r$ for a unique integer r. The *numerical function* of the pair (x, λ) is

$$\mu(x, \lambda) \;=\; r$$

where $t \mapsto t^r$ is the character by which λ acts on L_{x_0}. Equivalently, $\mu(x, \lambda) = -r'$ where $t \mapsto t^{r'}$ is the character by which λ acts on the dual line $L_{x_0}^\vee = \mathcal{O}(-1)_{x_0}$, the fiber over x_0 of the tautological subbundle; the sign matches the weight formula below because the affine lift $\hat{x}$ lives in $\mathcal{O}(-1)$.

The sign convention is the one that makes the criterion read "semistable iff $\mu \geq 0$" rather than "$\mu \leq 0$"; what μ measures becomes visible in coordinates. Choose a λ-eigenbasis and think of L as embedding X in projective space, so that a point is represented by a nonzero vector $\hat{x}$ in the affine cone over X. Writing $\hat{x} = \sum_i \hat{x}_i$ in eigencoordinates with $\lambda(t)$ acting on the i-th coordinate by t^{w_i}, the limit $\lim_{t \to 0} \lambda(t) \cdot \hat{x}$ is governed by the *smallest* weight w_i appearing with $\hat{x}_i \neq 0$, and one computes

$$\mu(x, \lambda) \;=\; -\min\{\, w_i \,|\, \hat{x}_i \neq 0 \,\}$$

Thus $\mu(x, \lambda) > 0$ means the minimal weight is negative: some eigencoordinate of $\hat{x}$ is scaled up (its coordinate blown toward infinity) as $t \to 0$, which is precisely the direction in which the affine lift of the orbit runs away from the origin. A positive μ signals that the orbit resists being pulled into the vertex of the cone, and it is this resistance that stability records.

15.4.2 Note: Computing μ by weights In practice $\mu(x,\lambda)$ is never computed from the abstract fiber L_{x_0}. One fixes a G-equivariant closed embedding $X \hookrightarrow \mathbb{P}(V)$ with $L = \mathcal{O}_{\mathbb{P}(V)}(1)|_X$, where V is a finite-dimensional G-representation. Diagonalizing the $\mathbb{G}_m$-action on V through λ gives a weight decomposition $V = \bigoplus_{w\in\mathbb{Z}} V_w$, where $\lambda(t)$ acts on V_w by t^w. A lift $\hat{x} \in V \setminus \{0\}$ of x decomposes as $\hat{x} = \sum_w \hat{x}_w$, and then

$$\mu(x,\lambda) \;=\; -\min\{\, w \,|\, \hat{x}_w \neq 0 \,\}$$

The value is independent of the chosen lift $\hat{x}$ (rescaling $\hat{x}$ does not change which weights occur) and of the ample power of L used (passing to $L^{\otimes n}$ multiplies every weight by n, hence scales μ by n and preserves its sign).

The numerical function is additive and homogeneous in a way that lets one restrict attention to a manageable supply of one-parameter subgroups. The following elementary properties are used repeatedly.

Proposition 15.4.3: Elementary Properties of the Numerical Function

Let $x \in X$, let λ, λ' be one-parameter subgroups of G, and let $g \in G$. Then:

1. For every integer $n \geq 1$, the 1-PS λ^n defined by $t \mapsto \lambda(t^n)$ satisfies $\mu(x,\lambda^n) = n\,\mu(x,\lambda)$.
2. $\mu(g\cdot x,\, g\lambda g^{-1}) = \mu(x,\lambda)$, where $g\lambda g^{-1}$ is the 1-PS $t \mapsto g\lambda(t)g^{-1}$.
3. If x is itself λ-fixed, then $\mu(x,\lambda)$ is r where $t \mapsto t^r$ is the character of λ on L_x; in particular $\mu(x,\lambda) + \mu(x,\lambda^{-1}) = 0$ in this case, where $\lambda^{-1}(t) = \lambda(t^{-1})$.

***Proof*:**

1. Replacing t by t^n in the orbit map does not change the limit point: $\lim_{t\to 0} \lambda(t^n)\cdot x = x_0$ still, since $t^n \to 0$ as $t \to 0$. The character of λ^n on L_{x_0} is the n-th power $t \mapsto (t^n)^r = t^{nr}$ of the character $t \mapsto t^r$ of λ, so $\mu(x,\lambda^n) = nr = n\,\mu(x,\lambda)$.
2. The element g carries the orbit of x under λ to the orbit of $g\cdot x$ under $g\lambda g^{-1}$: indeed $(g\lambda g^{-1})(t)\cdot(g\cdot x) = g\cdot(\lambda(t)\cdot x)$. Passing to limits, the limit point of $g \cdot x$ under $g\lambda g^{-1}$ is $g\cdot x_0$. The G-linearization identifies the fiber $L_{g\cdot x_0}$ with L_{x_0} compatibly with the two $\mathbb{G}_m$-actions (this is exactly the content of a linearization, definition 15.2.1), so the two actions on the fibers have the same character, giving equal numerical functions.
3. If x is λ-fixed then $x_0 = x$, so the fiber in question is L_x and the first claim is the definition. For the second, λ^{-1} acts on L_x by $t \mapsto (t^{-1})^r = t^{-r}$, hence $\mu(x,\lambda^{-1}) = -r = -\mu(x,\lambda)$, and the two sum to zero.

This establishes all three properties, completing the proof.

The homogeneity in part (1) means the *sign* of $\mu(x,\lambda)$, which is all the criterion asks about, depends only on the ray spanned by λ in the lattice of one-parameter subgroups, not on the particular integral multiple chosen. Part (2) says the numerical function is a G-invariant notion once one moves the point and the subgroup together, so verifying stability may be carried out after replacing λ by any conjugate, which is what allows one to reduce to 1-PS landing in a fixed maximal torus. Part (3) is the source of the fixed-point sign flip that will force strictly positive μ at stable points.

The first concrete instance is the diagonal torus in SL_2, which will carry the binary-forms computation later in the section and already illustrates the entire mechanism.

Example 15.4.4: The Diagonal 1-PS in SL_2

Let $G = \mathrm{SL}_2$ act on $V = k^2$ in the standard way, with basis e_1, e_2, and let $\lambda_a \colon \mathbb{G}_m \to \mathrm{SL}_2$ be the diagonal 1-PS

$$\lambda_a(t) = \begin{pmatrix} t^a & 0 \\ 0 & t^{-a} \end{pmatrix}$$

for a positive integer a; the entries multiply to 1, so this indeed lands in SL_2. On V the weights are $+a$ on e_1 and $-a$ on e_2. Consider the induced action on the projective line $\mathbb{P}(V) = \mathbb{P}^1$, linearized by $L = \mathcal{O}_{\mathbb{P}^1}(1)$, whose global sections $H^0(\mathbb{P}^1, \mathcal{O}(1))$ are spanned by the two coordinate functions dual to e_1, e_2. Take the point $x = [1 : 0]$, represented by $\hat{x} = e_1$, of weight $+a$. The minimal weight occurring in $\hat{x}$ is $+a$, so

$$\mu([1 : 0], \lambda_a) = -a < 0$$

For $x = [0 : 1] = e_2$ the minimal weight is $-a$, giving $\mu([0 : 1], \lambda_a) = a > 0$. For a general point $x = [x_1 : x_2]$ with both $x_1, x_2 \neq 0$, the lift $\hat{x} = x_1 e_1 + x_2 e_2$ has minimal weight $-a$, so $\mu(x, \lambda_a) = a > 0$ as well. The single point $[1 : 0]$, whose lift e_1 sits at weight $+a$, is the only one on which λ_a takes a negative value. It is the *repelling* fixed point of the projective action: for any $x = [x_1 : x_2]$ with $x_2 \neq 0$ one has $\lambda_a(t) \cdot [x_1 : x_2] = [t^{2a} x_1 : x_2] \to [0 : 1]$ as $t \to 0$, so every other point flows away from $[1 : 0]$ and toward the attracting fixed point $[0 : 1]$. That $[1 : 0]$ is destabilized despite being repelling is exactly the point: its affine lift e_1, of positive weight, is driven to the vertex of the cone as $t \to 0$, and it is the behavior of the *lift*, not of the projective orbit, that μ records.

The computation in example 15.4.4 is the whole idea in miniature: the destabilizing behavior of a 1-PS is concentrated at a fixed point whose affine lift is driven to the vertex of the cone, and the sign of μ there detects whether that fixed point obstructs stability. It remains to prove that these signs, ranged over all λ, capture semistability and stability exactly.

The motivating observation is that an unstable point can always be recognized by a degeneration: if every invariant section vanishes at x, the affine lift of the orbit closure passes through the vertex of the affine cone, and a theorem of Hilbert and Mumford says this degeneration can be realized along a one-parameter subgroup. Conversely, if some 1-PS drives the lift to the vertex, that 1-PS visibly kills every invariant section at x. The criterion packages both directions.

Theorem 15.4.5: The Hilbert-Mumford Numerical Criterion

Let G be a reductive group acting on a projective scheme X over the algebraically closed field k, and let L be a G-linearized ample line bundle. For $x \in X$:

1. x is semistable if and only if $\mu(x, \lambda) \geq 0$ for every one-parameter subgroup λ of G.
2. x is stable if and only if $\mu(x, \lambda) > 0$ for every nontrivial one-parameter subgroup λ of G.

The two directions of the criterion cost very different amounts, and it helps to separate them before the proof. The *easy* direction of (1) is the implication "some $\mu(x, \lambda) < 0 \Rightarrow x$ unstable": a destabilizing 1-PS produces, by hand, a witness that every invariant section vanishes at x. This is elementary and is given in full below. The *hard* direction is its converse: an unstable point must admit a destabilizing 1-PS. This is the Hilbert-Mumford theorem proper, and it rests on a properness input (the statement that the vertex of the cone, if it lies in an orbit closure, is reached along a 1-PS). In the linearly reductive projective setting it is proved in full below; the one step that would otherwise require the semistable-reduction machinery of Mumford's *Geometric Invariant Theory* is isolated and cited there.

***Proof*:**

Fix a G-equivariant closed embedding $X \hookrightarrow \mathbb{P}(V)$ with $L = \mathcal{O}_{\mathbb{P}(V)}(1)|_X$ and V a finite-dimensional G-representation, which exists because L is ample and G-linearized (this is the standard realization of an ample linearization, definition 15.2.1; that some power of L affords such a projective embedding is Serre's theorem on ample line bundles, **??**). Replacing L by a power multiplies every μ by a positive integer and changes neither stability nor the sign conditions, so this loses nothing. Let $\hat{X} \subset V$ be the affine cone over X, and for $x \in X$ choose a nonzero lift $\hat{x} \in \hat{X}$. Semistability of x is equivalent to the vertex 0 *not* lying in the Zariski closure $\overline{G \cdot \hat{x}}$ of the orbit of $\hat{x}$ in V: this is the affine-cone reformulation of semistability from section 15.3, since an invariant section not vanishing at x is a G-invariant homogeneous function on $\hat{X}$ not vanishing at $\hat{x}$, and a nonzero invariant separates $\hat{x}$ from 0 exactly when $0 \notin \overline{G \cdot \hat{x}}$ (the good quotient separates disjoint closed invariant sets by the Reynolds operator, definition 15.1.1 and definition 15.1.4, which is what supplies the separating invariant). The proof translates each direction through this reformulation.

Step 1: The easy direction of (1). Suppose $\mu(x, \lambda) < 0$ for some 1-PS λ. Decompose $\hat{x} = \sum_w \hat{x}_w$ into weight components for λ. By the weight formula of box 15.4.2, $\mu(x, \lambda) = -\min\{w \big| \hat{x}_w \neq 0\} < 0$ means every weight w with $\hat{x}_w \neq 0$ satisfies $w > 0$. Then

$$\lambda(t) \cdot \hat{x} = \sum_{w>0} t^w \hat{x}_w \xrightarrow{t \to 0} 0$$

since each surviving power t^w has $w > 0$. Thus $0 = \lim_{t \to 0} \lambda(t) \cdot \hat{x}$ lies in $\overline{G \cdot \hat{x}}$, so x is not semistable. This proves that semistability forces $\mu(x, \lambda) \geq 0$ for all λ, which is the easy half of (1).

Step 2: The hard direction of (1). Suppose x is not semistable, so $0 \in \overline{G \cdot \hat{x}}$. The claim is that there is a 1-PS λ with $\lim_{t \to 0} \lambda(t) \cdot \hat{x} = 0$, for then, by the weight formula run in reverse, every weight of $\hat{x}$ under λ is strictly positive and $\mu(x, \lambda) = -\min\{w \big| \hat{x}_w \neq 0\} < 0$. That such a 1-PS exists is the Hilbert-Mumford theorem in the affine setting: if the closure of a G-orbit $G \cdot v$ in a representation V contains a point v_0 in a *closed* orbit, then v_0 can be reached from v along a one-parameter subgroup, that is, there is a 1-PS λ with $\lim_{t \to 0} \lambda(t) \cdot v = g \cdot v_0$ for some $g \in G$. Applied to $v = \hat{x}$ and the closed orbit $\{0\} = G \cdot 0$, this yields a 1-PS with $\lim_{t \to 0} \lambda(t) \cdot \hat{x} = 0$, as needed.

The existence of such a 1-PS is the technical heart of the theorem. For G linearly reductive it can be proved by the following reduction to a torus, which is complete except for one properness input. Since $0 \in \overline{G \cdot \hat{x}}$, there is a curve in the orbit approaching 0: concretely, a map from the spectrum of a discrete valuation ring R with fraction field K, sending the generic point into $G \cdot \hat{x}$ and the closed point to 0. Because G is reductive, the valuative criterion applied to the quotient map $G \to G/P$ (for a suitable parabolic) together with the structure of G over R lets one arrange this curve to factor through the normalizer of a maximal torus after a finite base change; the Cartan decomposition over the local field $G(K) = \coprod_\lambda G(R)\lambda(\pi)G(R)$, where π is a uniformizer of R and λ ranges over 1-PS of a fixed maximal torus, then writes the degenerating curve as $g_1\lambda(\pi)g_2$ with $g_i \in G(R)$. Specializing g_1, g_2 to the closed point and absorbing them into the base point and the conjugate of λ produces a 1-PS of G realizing the degeneration to 0. The single step that this book does not reprove is that the degenerating curve exists and factors through a torus after base change, that is, the properness reduction above; it uses the semistable-reduction machinery for reductive groups and is cited to Mumford's *Geometric Invariant Theory* [50] (Chapter 2), where it is proved in the generality needed. With the 1-PS in hand, $\mu(x, \lambda) < 0$ as computed above, so the contrapositive gives: $\mu(x, \lambda) \geq 0$ for all λ implies x semistable. Together with Step 1 this proves (1).

Step 3: The stable direction. Recall from definition 15.3.1 that x is stable when it is semistable, its stabilizer G_x is finite (equivalently $\dim G \cdot x = \dim G$), and its orbit $G \cdot x$ is closed in the semistable locus. In the affine-cone picture, stability of x is equivalent to the orbit $G \cdot \hat{x}$ being closed in $V \setminus \{0\}$ *and* of dimension $\dim G$ (the stabilizer of $\hat{x}$ being finite modulo the scalars that act on the cone). This is the standard translation from section 15.3.

Suppose first that x is stable, and let λ be a nontrivial 1-PS. Since x is semistable, $\mu(x, \lambda) \geq 0$ by (1). Suppose toward a contradiction that $\mu(x, \lambda) = 0$. Then the minimal weight of $\hat{x}$ under λ is 0, so $\hat{x}_0 := \lim_{t\to 0} \lambda(t) \cdot \hat{x} = \sum_{w=0} \hat{x}_w$ is a nonzero point of $\hat{X}$, and it is a λ-fixed point in the orbit closure $\overline{G \cdot \hat{x}}$. If $\hat{x}_0 \neq \hat{x}$ up to the orbit, that is, if $\hat{x}_0 \notin G \cdot \hat{x}$, then $\overline{G \cdot \hat{x}}$ strictly contains $G \cdot \hat{x}$, so $G \cdot \hat{x}$ is not closed in $V \setminus \{0\}$ (it accumulates on the nonzero point $\hat{x}_0$), contradicting stability. Hence $\hat{x}_0 \in G \cdot \hat{x}$, say $\hat{x}_0 = g \cdot \hat{x}$. Then $g^{-1}\lambda g$ fixes $\hat{x}$, so $g^{-1}\lambda(t)g$ lies in the stabilizer $G_{\hat{x}}$ for all t; but $G_{\hat{x}}$ is finite (stability), and a nontrivial 1-PS has infinite image, a contradiction. Therefore $\mu(x, \lambda) > 0$ for every nontrivial λ.

Conversely, suppose $\mu(x, \lambda) > 0$ for every nontrivial 1-PS λ. Then in particular $\mu(x, \lambda) \geq 0$ for all λ, so x is semistable by (1). It remains to show $G \cdot \hat{x}$ is closed in $V \setminus \{0\}$ and $\dim G \cdot x = \dim G$. If $G \cdot \hat{x}$ were not closed in $V \setminus \{0\}$, its closure would meet a strictly smaller orbit at some nonzero point $\hat{y}$; applying the Hilbert-Mumford theorem of Step 2 to the closed orbit $G \cdot \hat{y}$ inside $\overline{G \cdot \hat{x}}$ produces a nontrivial 1-PS λ with $\lim_{t\to 0} \lambda(t) \cdot \hat{x} = g \cdot \hat{y} \neq 0$. The limit being a nonzero point means the minimal weight of $\hat{x}$ under λ is 0, so $\mu(x, \lambda) = 0$, contradicting the hypothesis. Hence $G \cdot \hat{x}$ is closed. If instead $\dim G \cdot x < \dim G$, the stabilizer G_x is positive-dimensional; moreover, because $G \cdot \hat{x}$ was just shown closed, the stabilizer $G_{\hat{x}}$ of the lift is reductive by Matsushima's criterion (the stabilizer of a point with closed orbit in an affine G-variety is reductive), and G_x differs from $G_{\hat{x}}$ only by the scalars acting on the cone, hence is reductive as well. A positive-dimensional reductive group contains a nontrivial torus, hence a nontrivial 1-PS λ; that λ fixes x, so by part (3) of proposition 15.4.3, $\mu(x, \lambda) + \mu(x, \lambda^{-1}) = 0$, forcing one of the two values to be ≤ 0 and contradicting strict positivity. Hence $\dim G \cdot x = \dim G$ and x is stable. This proves (2), completing the proof.

The criterion converts an existence-of-invariants problem into a sign-of-weights computation. To test semistability one need only bound $\mu(x, \lambda)$ below by 0 across all 1-PS, and by part (2) of proposition 15.4.3 it suffices to check 1-PS in a single fixed maximal torus T, since every 1-PS is conjugate into T and μ is conjugation-invariant once the point is moved along. Inside T the 1-PS form a lattice $\mathbb{Z}^r$ (with $r = \dim T$), and $\mu(x, -)$ is a piecewise linear function on this lattice; deciding a sign across a lattice is a finite combinatorial problem. This is what reduces GIT stability to the linear algebra of weights.

15.4.6 Note: Why one maximal torus suffices The reduction to a single torus is the practical form of the criterion. Every 1-PS λ of G has image in some maximal torus, and any two maximal tori are conjugate; so for a fixed maximal torus T, every 1-PS is of the form $g\lambda' g^{-1}$ with λ' a 1-PS of T and $g \in G$. By proposition 15.4.3(2), $\mu(x, g\lambda' g^{-1}) = \mu(g^{-1} \cdot x, \lambda')$. Thus checking $\mu(x, \lambda) \geq 0$ for all λ is the same as checking $\mu(g \cdot x, \lambda') \geq 0$ for all 1-PS λ' of the fixed torus T and all $g \in G$, that is, over the orbit of x and one torus. When the orbit is understood (as it is for binary forms, where every form is an SL_2-translate of one with a controlled root pattern), this is a finite check.

The archetypal application, and the example that motivated Hilbert's original work, is the action of SL_2 on binary forms. A binary form of degree d is a homogeneous polynomial

$$f(x, y) = \sum_{i=0}^{d} a_i\, x^{d-i} y^i$$

of degree d in two variables; the space of such forms is the SL_2 representation $V_d = \mathrm{Sym}^d(k^2)^\vee$, of dimension $d + 1$, and SL_2 acts by substitution on the variables. The projectivization $\mathbb{P}(V_d) = \mathbb{P}^d$ is the space of binary forms up to scalar, and it is linearized by $\mathcal{O}_{\mathbb{P}^d}(1)$ with its natural SL_2-action. Since k is algebraically closed, a nonzero binary form factors into linear factors,

$$f(x, y) = c \prod_{j=1}^{d} (\beta_j x - \alpha_j y)$$

so f is determined up to scalar by its *roots* $[\alpha_j : \beta_j] \in \mathbb{P}^1$ counted with multiplicity; a form of degree d corresponds to an unordered d-tuple of points on $\mathbb{P}^1$. The stability question becomes a question about configurations of d points on the projective line, connecting directly to example 15.3.6.

Example 15.4.7: Stability of Binary Forms

A nonzero binary form f of degree d, viewed as a point of $\mathbb{P}(V_d)$, is

1. *semistable* if and only if every root of f has multiplicity at most $d/2$;
2. *stable* if and only if every root of f has multiplicity strictly less than $d/2$.

Setting up the weight computation. By box 15.4.6, it suffices to evaluate $\mu(f, \lambda)$ for 1-PS λ of the diagonal maximal torus $T \subset \mathrm{SL}_2$, after moving f by SL_2. Every nontrivial 1-PS of T is $\lambda_a(t) = \mathrm{diag}(t^a, t^{-a})$ for some integer $a \neq 0$, and by homogeneity (proposition 15.4.3(1)) the sign of μ is unchanged under $a \mapsto na$, so one may take $a = 1$ and its inverse; write $\lambda = \lambda_1$. A monomial basis of V_d is $\{x^{d-i} y^i\}_{i=0}^{d}$. The 1-PS $\lambda(t)$ sends $x \mapsto t^{-1}x$ and $y \mapsto t\,y$ acting on forms by substitution (the contragredient of the action on k^2), so it scales the monomial $x^{d-i}y^i$ by $t^{-(d-i)} t^i = t^{\,2i-d}$. The weight of the basis vector $x^{d-i}y^i$ is therefore

$$w_i \;=\; 2i - d$$

running through $-d$, $-d + 2, \ldots, d - 2$, d as i runs from 0 to d.

The weight formula for μ. Write $f = \sum_i a_i\, x^{d-i} y^i$. By box 15.4.2,

$$\mu(f, \lambda) \;=\; -\min\{\, w_i \big| a_i \neq 0 \,\} \;=\; -\min\{\, 2i - d \big| a_i \neq 0 \,\}$$

Let m be the multiplicity of the root $[1:0]$ of f, that is, the largest power of y dividing f (the point $[1:0]$ is the root $y=0$, cut out by y in the coordinates dual to e_1, e_2; here the monomial $x^{d-i}y^i$ is divisible by y^i). Then $a_i = 0$ for $i < m$ and $a_m \neq 0$, so the minimum index i with $a_i \neq 0$ is exactly m, giving

$$\mu(f, \lambda) \;=\; -(2m-d) \;=\; d - 2m$$

Thus the numerical function of the standard diagonal 1-PS reads off the multiplicity of the root at $[1:0]$: it is ≥ 0 if and only if $m \leq d/2$, and > 0 if and only if $m < d/2$.

From one 1-PS to all of them. A general 1-PS of SL_2, up to conjugacy and $a \mapsto na$, is $g\lambda g^{-1}$ for some $g \in \mathrm{SL}_2$, and $\mu(f, g\lambda g^{-1}) = \mu(g^{-1} \cdot f, \lambda)$ by proposition 15.4.3(2). The element g moves the root $[1:0]$ to an arbitrary point $p = g \cdot [1:0] \in \mathbb{P}^1$, and the multiplicity of p as a root of f equals the multiplicity of $[1:0]$ as a root of $g^{-1} \cdot f$. So as g ranges over SL_2,

$$\mu(f, g\lambda g^{-1}) \;=\; d - 2\, m_p$$

where $m_p = \mathrm{mult}_p(f)$ is the multiplicity of f at $p = g \cdot [1:0]$. Every point $p \in \mathbb{P}^1$ arises this way (the SL_2-action on $\mathbb{P}^1$ is transitive), and conversely the destabilizing direction detected by any 1-PS λ_a with $a < 0$ swaps the roles of $[1:0]$ and $[0:1]$, which is already covered by ranging over all p. Hence the full family of values of $\mu(f, -)$ over all nontrivial 1-PS is exactly

$$\{\, d - 2\, m_p \big| p \in \mathbb{P}^1 \,\}$$

Reading off stability. By theorem 15.4.5:

- f is semistable iff $d - 2m_p \geq 0$ for all p, that is, $m_p \leq d/2$ for every point p: no root has multiplicity exceeding $d/2$.
- f is stable iff $d - 2m_p > 0$ for all p and every nontrivial 1-PS, that is, $m_p < d/2$ for every point p: every root has multiplicity strictly below $d/2$.

Only the roots of f contribute a nonzero multiplicity, so the condition is vacuous away from the roots, and the two statements are exactly (1) and (2).

The computation makes the geometry transparent. Instability of a binary form is caused by a single root that is "too fat", concentrating more than half the degree; the destabilizing 1-PS is the diagonal torus that fixes that root and, running $t \to 0$, sweeps every other root away from it toward the antipodal fixed point, thinning the configuration everywhere except at the fat root. For $d = 2$ the threshold $d/2 = 1$ says a conic pair of points on $\mathbb{P}^1$ is semistable exactly when the two points are distinct (a double point has multiplicity $2 > 1$, unstable), and is never stable (a distinct pair has each multiplicity $1 = d/2$, so semistable but not stable), matching the elementary fact that SL_2 acts transitively on pairs of distinct points, so the quotient is a single point. For $d = 3$ the threshold $3/2$ says a binary cubic is stable exactly when its three roots are distinct and semistable under the same condition, since a multiplicity of 2 or 3 exceeds $3/2$; a nodal cubic form (two roots, one doubled) is unstable. For $d = 4$ the threshold is 2: a binary quartic is semistable when no root is worse than a double point and stable when all four roots are distinct, and the semistable-but-not-stable locus consists of forms with exactly one double root, which is the source of the strictly semistable boundary in the GIT quotient of four points on $\mathbb{P}^1$.

15.4.8 Remark: Reconciliation with points on $\mathbb{P}^1$ example 15.3.6 analyzed n (unordered) points on $\mathbb{P}^1$ modulo PGL_2 directly through invariant sections, and found a configuration stable when no point is repeated more than $n/2$ times (and semistable with equality allowed). A degree-d binary form is precisely an unordered d-tuple of points on $\mathbb{P}^1$ (its roots with multiplicity), and $\mathrm{PGL}_2 = \mathrm{SL}_2/\{\pm 1\}$ acts on both descriptions compatibly, since the center $\{\pm 1\}$ acts trivially on $\mathbb{P}(V_d)$ and on configurations. The multiplicity of a root is the number of times a point is repeated, so "no root of multiplicity $> d/2$" is "no point repeated more than $d/2$ times", and the two stability criteria coincide with $n = d$. The present derivation via theorem 15.4.5 recovers example 15.3.6 without computing a single invariant section: the weight of a diagonal 1-PS did all the work that a search through the invariant ring would otherwise require.

The reconciliation is the point of the criterion. In section 15.3 the stability of point configurations was extracted by exhibiting invariants, a route that becomes impractical as d grows; here the same answer falls out of a one-line weight count, valid for all d at once. This is the leverage the criterion provides: any GIT stability question with a computable weight decomposition, including the stability of pluricanonically embedded curves that underlies the construction of the moduli of curves, reduces to counting weights of one-parameter subgroups rather than searching an invariant ring.

Exercise 15.4.1

1. Let $\lambda\colon \mathbb{G}_m \to G$ be a 1-PS and $x \in X$ a point that is already fixed by the whole group G. Show that $\mu(x, \lambda)$ can be computed without taking any limit, and that $\mu(x, \lambda) + \mu(x, \lambda^{-1}) = 0$. Deduce that a G-fixed point is semistable if and only if it is a fixed point of trivial weight for every 1-PS, and is never stable when G has a nontrivial 1-PS.

2. Consider SL_2 acting on binary forms of degree $d = 6$. Using example 15.4.7, classify the following forms as stable, strictly semistable (semistable but not stable), or unstable: (a) $f = x^6 + y^6$; (b) $f = x^3y^3$; (c) $f = x^4y^2$; (d) $f = x^5y$. In each case name the destabilizing 1-PS when one exists.

3. Let $\mathbb{G}_m$ act on $V = k^3$ with weights $(2, -1, -1)$, so that $\lambda(t) = \mathrm{diag}(t^2, t^{-1}, t^{-1})$, and consider the induced action on $\mathbb{P}^2$ linearized by $\mathcal{O}(1)$. Compute $\mu(x, \lambda)$ for the points $[1 : 0 : 0]$, $[0 : 1 : 0]$, and $[1 : 1 : 1]$. Which of these points is destabilized by λ? Since $G = \mathbb{G}_m$ is a torus, every 1-PS is λ^n; using this, determine the unstable locus in $\mathbb{P}^2$ for this linearization.

4. Prove directly, without invoking the hard direction of theorem 15.4.5, that if $x \in X$ admits a nontrivial 1-PS λ fixing x with $\mu(x, \lambda) = 0$, then x is not stable. (This isolates the argument in Step 3 that a positive-dimensional stabilizer prevents stability.)

5. In the binary-forms computation, the weight of the monomial $x^{d-i}y^i$ under $\lambda(t) = \mathrm{diag}(t, t^{-1})$ was found to be $2i - d$. Verify this directly from the definition of the SL_2 action on $V_d = \mathrm{Sym}^d(k^2)^\vee$, being careful about the contragredient, and explain why replacing λ by λ^{-1} replaces the multiplicity m of the root at $[1 : 0]$ by the multiplicity of the root at $[0 : 1]$ in the formula $\mu(f, \lambda) = d - 2m$.

6. A binary form of degree d is called *polystable* if it is semistable and its SL_2-orbit is closed in the semistable locus. Using example 15.4.7 and the orbit-closure picture from the proof of theorem 15.4.5, show that a strictly semistable binary quartic ($d = 4$) with exactly one double root has a unique polystable form in the closure of its orbit, and identify it. What does this

say about the point of the GIT quotient to which every such form is sent?

15.5 Moduli via GIT

The two great engines for building moduli spaces have now both been assembled. The Hilbert scheme produces, for a fixed projective ambient space and a fixed Hilbert polynomial, a fine moduli space of *embedded* objects: it parametrizes closed subschemes together with the very embedding that pins them down inside $\mathbb{P}^N$ (corollary 14.3.3). Geometric invariant theory, developed across the preceding four sections, produces quotients by group actions, and in particular a projective quotient of the semistable locus. This section combines them. The auxiliary embedding that makes an object into a point of a Hilbert scheme is exactly the extra rigidity that the group PGL_{N+1} acts to forget, and the GIT quotient of the locus of embedded objects by that group is a scheme whose points are the objects taken up to isomorphism. When the numerical stability of GIT matches the geometric class of objects one wishes to parametrize, this quotient is a coarse moduli space.

The strategy is uniform across a wide range of moduli problems, and it is the historically decisive one: it is how Mumford first constructed the moduli space of curves $\mathcal{M}_g$, and the same template governs moduli of polarized varieties, of stable vector bundles, and of abelian varieties with level structure. The section states the recipe as a proposition, proves it, and then walks through the curves case in the detail that its importance warrants, deferring only the one deep stability input to the capstone where the full construction belongs.

The obstruction to fine representability, met already in the first chapter, is automorphisms: an object with a nontrivial automorphism cannot sit as a point of a fine moduli space, because a family with nonconstant isomorphism type but constant fiber can be built by twisting along the automorphisms. The j-line (theorem 13.3.6) is the paradigm. Its objects, elliptic curves, each carry the automorphism $[-1]$ (and the special curves $j = 0, 1728$ carry more), so no fine moduli space exists; yet a coarse moduli space, the affine j-line $\mathbb{A}^1$, does. GIT reproduces exactly this behaviour. On the stable locus the stabilizers are finite, which is precisely the algebraic statement of "finitely many automorphisms", and the geometric quotient there identifies orbits, that is, identifies embedded objects that differ only by a change of embedding, hence identifies isomorphic objects. The resulting scheme cannot in general be fine (the finite automorphisms survive as stabilizers), but it is coarse.

To state the recipe precisely, fix a moduli problem: a functor

$$F\colon \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$$

whose value $F(S)$ is the set of isomorphism classes of families over S of the objects one wishes to classify (smooth projective curves of genus g, say, or polarized varieties with a fixed Hilbert polynomial). The bridge to the Hilbert scheme is an *auxiliary embedding*: a functorial way to embed each object in a fixed projective space, canonical up to the choice of a basis of the linear system that produces it. This is what the projective linear group acts on.

Proposition 15.5.1: Coarse Moduli via a GIT Quotient

Let $F\colon \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ be a moduli problem for objects that admit a functorial projective embedding of a fixed type. Concretely, suppose there is a fixed projective space $\mathbb{P}^N$, a Hilbert polynomial P, and a locally closed subscheme

$$H \subseteq \mathrm{Hilb}^P(\mathbb{P}^N)$$

of the Hilbert scheme (corollary 14.3.3) whose points are exactly the images of the objects of F under their embeddings, such that:

1. the group PGL_{N+1} acts on $\mathrm{Hilb}^P(\mathbb{P}^N)$ by moving the embedding, and preserves H;
2. two points of H lie in the same PGL_{N+1}-orbit if and only if the objects they represent are isomorphic;
3. for a PGL_{N+1}-linearized ample line bundle on a PGL_{N+1}-invariant projective completion $\overline{H}$ of H, the stable locus $\overline{H}^{\mathrm{s}}$ (definition 15.3.1) equals H, and the stabilizer in PGL_{N+1} of each point of H is finite.

Then the geometric quotient

$$M := H/\mathrm{PGL}_{N+1} = \overline{H}^{\mathrm{s}}/\mathrm{PGL}_{N+1}$$

exists as a quasi-projective scheme (theorem 15.3.4) and is a coarse moduli space for F in the sense of definition 13.3.1.

***Proof*:**
The quotient M is produced by the projective GIT machinery, and the content of the proof is to check that it satisfies the two defining conditions of a coarse moduli space: it corepresents F on geometric points, and it is universal among schemes to which F maps.

Step 1: The quotient exists and is a geometric quotient. By hypothesis (3) the stable locus of $\overline{H}$ for the chosen linearization equals H, and every point of H has finite stabilizer. theorem 15.3.4 constructs the projective GIT quotient $\overline{H}^{\mathrm{ss}} /\!\!/ \mathrm{PGL}_{N+1}$ as a projective scheme and shows that on the stable locus the induced map

$$\overline{H}^{\mathrm{s}} \longrightarrow \overline{H}^{\mathrm{s}}/\mathrm{PGL}_{N+1}$$

is a geometric quotient. Since $\overline{H}^{\mathrm{s}} = H$, the quotient $M = H/\mathrm{PGL}_{N+1}$ exists, and because it is an open subscheme of the projective $\overline{H}^{\mathrm{ss}} /\!\!/ \mathrm{PGL}_{N+1}$ it is quasi-projective. Being a geometric quotient, its closed points are in bijection with the PGL_{N+1}-orbits of closed points of H.

Step 2: The geometric points of M are the isomorphism classes. By hypothesis (2), two closed points of H lie in one orbit exactly when the objects they carry are isomorphic. Combined with Step 1, the closed points of M are in natural bijection with PGL_{N+1}-orbits on H, hence with isomorphism classes of objects of F over k. Since $F(\operatorname{Spec} k)$ is by definition the set of isomorphism classes of objects over k, there is a canonical bijection

$$F(\operatorname{Spec} k) \xrightarrow{\sim} M(k) \tag{15.4}$$

which is the first requirement of definition 13.3.1.

Step 3: The natural transformation $F \to h_M$. A coarse moduli space must carry a morphism of functors $\eta\colon F \to h_M$ inducing (15.4) on k-points; construct it. Let $\xi \in F(S)$ be a family over a scheme S. The functorial embedding turns ξ into a family of embedded objects over S, that is, a closed subscheme $Z \subseteq \mathbb{P}^N_S$ flat over S with fibrewise Hilbert polynomial P, whose fibres land in H. By the universal property of the Hilbert scheme (corollary 14.3.3) this family is classified by a morphism

$$g_\xi \colon S \longrightarrow H$$

The composite $q \circ g_\xi \colon S \to M$ with the quotient map $q\colon H \to M$ depends on ξ only through its isomorphism class: an isomorphism $\xi \cong \xi'$ of families over S changes the embedding by an S-point of PGL_{N+1}, hence changes g_ξ to $g_{\xi'}$ by an element of the group along which q is invariant, so $q \circ g_\xi = q \circ g_{\xi'}$. Setting $\eta_S(\xi) := q \circ g_\xi$ gives a well-defined natural transformation $\eta\colon F \to h_M$. On k-points $\eta_{\operatorname{Spec} k}$ sends an object to the image in M of the point of H carrying its embedding, which is exactly the bijection (15.4).

Step 4: Universality. It remains to show M is initial among schemes receiving a natural transformation from F: given any scheme M' and any $\eta'\colon F \to h_{M'}$, there is a unique morphism $\psi\colon M \to M'$ with $\eta' = h_\psi \circ \eta$. Apply η' to the tautological family over H. The identity of H classifies a family $\zeta \in F(H)$ (the restriction to H of the universal embedded object), and $\eta'_H(\zeta)$ is a morphism $\varphi\colon H \to M'$. Because η' is natural and each object of F over a point is unchanged under the PGL_{N+1}-action on its embedding, φ is constant on PGL_{N+1}-orbits: for $t \in \mathrm{PGL}_{N+1}(S)$ the translated family $t \cdot \zeta_S$ is isomorphic to ζ_S as an element of $F(S)$, so $\varphi \circ (t\cdot) = \varphi$. A PGL_{N+1}-invariant morphism out of H factors uniquely through the geometric quotient $q\colon H \to M$ by the universal property of the categorical quotient underlying M (definition 4.4.1), giving a unique $\psi\colon M \to M'$ with $\varphi = \psi \circ q$. Chasing definitions, $h_\psi \circ \eta = \eta'$, and ψ is the only such morphism because q is an epimorphism of schemes and φ determines ψ on the dense image of q. This is the universal property of definition 13.3.1, completing the proof.

The proposition isolates what has to be checked to run the GIT construction for a given moduli problem, and it makes visible where the difficulty lies. Conditions (1) and (2) are formal: they say only that the embedding is canonical up to the group and that the group's orbits are exactly the isomorphism classes. These are usually easy, once the embedding is chosen well. Condition (3) is the entire mathematical burden, and it splits into two parts. The equality $\overline{H}^{\mathrm{s}} = H$ is a *stability theorem*, an assertion that the objects one wants to parametrize are precisely the GIT-stable ones for a suitable linearization; this is never formal and is proved case by case using the Hilbert-Mumford criterion (theorem 15.4.5). The finiteness of stabilizers is the algebraic incarnation of the objects having finite automorphism groups, and it is what forces M to be coarse rather than fine.

One should read the output correctly. The scheme M is a coarse moduli space, not a fine one, precisely because the stabilizers, though finite, are generally nontrivial. A point of M remembers an object up to isomorphism but forgets its automorphisms, and (15.4) is a bijection on points that does *not* upgrade to an isomorphism of functors $F \cong h_M$. The stack-theoretic repair of exactly this defect, which remembers the automorphism groups as the stabilizers of a quotient stack $[H/\mathrm{PGL}_{N+1}]$, is the subject of the descent-and-stacks part of the book; the present construction is the coarse space of that stack.

15.5.2 Why PGL and not GL The group acting is the projective linear group PGL_{N+1}, not GL_{N+1}, because two embeddings of the same object into $\mathbb{P}^N$ differ by a projective change of coordinates, and the scalars $\mathbb{G}_m \subseteq \mathrm{GL}_{N+1}$ act trivially on $\mathbb{P}^N$. Using GL_{N+1} would carry a redundant central $\mathbb{G}_m$ whose orbits are points, inflating stabilizers by a positive-dimensional torus and destroying the stability needed in proposition 15.5.1(3). Because PGL_{N+1} is reductive (it is the quotient of the reductive GL_{N+1} by a central torus), the GIT machinery of theorem 15.3.4 applies unchanged. Linearizations require a little care: an ample line bundle on $\mathrm{Hilb}^P(\mathbb{P}^N)$ is naturally GL_{N+1}-linearized, and one must check the central $\mathbb{G}_m$ acts trivially on the relevant power to descend the linearization to PGL_{N+1}, which is why stability is always stated for a specific power of the natural bundle.

Before the curves example, a small self-contained instance grounds the recipe and shows every hypothesis of proposition 15.5.1 met concretely.

Example 15.5.3: Configurations of Points on the Line

Take the moduli problem $F(S) = \{\text{families of } d \text{ distinct unordered points on } \mathbb{P}^1 \text{ over } S\}$ up to isomorphism, for $d \geq 3$. An unordered d-tuple of distinct points on $\mathbb{P}^1$ is the same as a degree-d divisor with no repeated point, hence a point of the Hilbert scheme $\mathrm{Hilb}^P(\mathbb{P}^1)$ with P the constant polynomial d; this Hilbert scheme is the symmetric product $\mathrm{Sym}^d\mathbb{P}^1 \cong \mathbb{P}^d$, and the locus H of *distinct* configurations is the complement of the discriminant, an open subscheme. The automorphism group of $\mathbb{P}^1$ is PGL_2, which acts on $\mathbb{P}^d$, and two configurations are projectively equivalent exactly when they are isomorphic as abstract d-pointed schemes, so conditions (1) and (2) of proposition 15.5.1 hold.

For condition (3), linearize $\mathcal{O}_{\mathbb{P}^d}(1)$ (equivalently, act on binary forms of degree d) and apply the stability calculation of example 15.3.6: a configuration is GIT-stable precisely when every point has multiplicity strictly less than $d/2$, which for *distinct* points means each multiplicity is 1, automatic once $d \geq 3$. Thus $\overline{H}^{\mathrm{s}}$ is exactly the locus of distinct configurations, and each such configuration has finite PGL_2-stabilizer (a finite group of Mobius transformations permuting the points). All three hypotheses hold, and proposition 15.5.1 yields the coarse moduli space

$$M_{0,d}^{\mathrm{un}} = H/\mathrm{PGL}_2$$

of d distinct unordered points on the line, a quasi-projective variety of dimension $d-3$. For $d = 3$ it is a point (any three distinct points are projectively equivalent to $0, 1, \infty$); for $d = 4$ it is one-dimensional and the invariant is the cross-ratio taken up to the symmetric-group action on the four points, recovering the classical fact that the moduli of four points on $\mathbb{P}^1$ is a j-line-like affine curve.

The reason GIT earns its place beside the Hilbert scheme is the moduli of curves. For a smooth projective curve C of genus $g \geq 2$ there is a canonical projective embedding built from the curve itself, with no choices beyond a basis, and this is exactly the functorial embedding that proposition 15.5.1 requires. The construction that follows is Mumford's, and it is the historical origin of geometric invariant theory: GIT was created to make this construction work.

The embedding comes from pluricanonical linear systems. For a smooth curve C of genus $g \geq 2$ the canonical bundle ω_C has degree $2g-2 > 0$, so for $n \geq 3$ the power $\omega_C^{\otimes n}$ is very ample of degree $n(2g-2)$, and by Riemann-Roch it has

$$N_n + 1 := h^0(C, \omega_C^{\otimes n}) = (2n-1)(g-1)$$

sections. A choice of basis of $H^0(C, \omega_C^{\otimes n})$ embeds C as a curve

$$C \hookrightarrow \mathbb{P}^{N_n}, \qquad N_n = (2n-1)(g-1) - 1$$

whose Hilbert polynomial $P(t) = n(2g-2)t - g + 1$ depends only on g and n. Different bases give embeddings differing by an element of PGL_{N_n+1}, and this is the entire ambiguity: the embedding is canonical up to the projective linear group. This is the setup of proposition 15.5.1, with $\mathbb{P}^N = \mathbb{P}^{N_n}$ and the moduli problem $F = \mathcal{M}_g$ of smooth projective genus-g curves.

Example 15.5.4: The GIT Construction of $\mathcal{M}_g$

Fix $g \geq 2$ and an integer $n \geq 3$ (in fact $n \geq 5$ suffices comfortably, and $n = 3$ works with more care). Set $P(t) = n(2g-2)t - g + 1$ and $N = (2n-1)(g-1) - 1$. The construction proceeds in four movements.

The Hilbert scheme of embedded curves. Inside $\mathrm{Hilb}^P(\mathbb{P}^N)$ consider the subset

$$H_g \;=\; \left\{\, [C \hookrightarrow \mathbb{P}^N] \;:\; C \text{ smooth of genus } g,\ \mathcal{O}_C(1) \cong \omega_C^{\otimes n} \,\right\}$$

of n-canonically embedded smooth genus-g curves. Smoothness is an open condition, and the requirement that the embedding line bundle $\mathcal{O}_C(1)$ be the n-th power of the canonical bundle (rather than some other bundle of the same degree) is a locally closed condition. Hence H_g is a locally closed subscheme of the Hilbert scheme, quasi-projective by corollary 14.3.3. Its points are exactly the smooth genus-g curves *together with* a projective frame for their n-canonical embedding.

The group and its orbits. The group $G = \mathrm{PGL}_{N+1}$ acts on $\mathrm{Hilb}^P(\mathbb{P}^N)$ and preserves H_g, since a projective change of coordinates carries an n-canonical embedding to another n-canonical embedding of the same curve. Two points of H_g lie in one G-orbit if and only if the underlying curves are isomorphic: an isomorphism of curves carries $\omega_C^{\otimes n}$ to $\omega_{C'}^{\otimes n}$ and hence intertwines the two embeddings by a projective transformation, and conversely an element of PGL taking one embedded curve to the other restricts to an abstract isomorphism. This is exactly conditions (1) and (2) of proposition 15.5.1.

Stability. The mathematical heart is that the n-canonically embedded smooth curves are the GIT-stable points, for a suitable linearization on a projective completion of H_g inside $\mathrm{Hilb}^P(\mathbb{P}^N)$. This is theorem 15.5.5 below. Granting it, condition (3) of proposition 15.5.1 holds: $\overline{H_g}^{\mathrm{s}} = H_g$, and each smooth curve of genus $g \geq 2$ has finite automorphism group (by Hurwitz, $|\mathrm{Aut}(C)| \leq 84(g-1)$), so all stabilizers are finite.

The quotient. proposition 15.5.1 now delivers the coarse moduli space

$$\mathcal{M}_g = H_g / \mathrm{PGL}_{N+1}$$

a quasi-projective variety of dimension $\dim H_g - \dim \mathrm{PGL}_{N+1}$. A count confirms the expected value: $\dim H_g = \dim \mathrm{PGL}_{N+1} + (3g-3)$, since the deformations of an n-canonically embedded curve split into the $(N+1)^2 - 1 = \dim \mathrm{PGL}_{N+1}$ directions moving the embedding and the $3g-3$ directions deforming the abstract curve (the deformation-theoretic count $\dim H^1(C, T_C) = 3g-3$ is the subject of the deformation-theory part of the book). Hence

$$\dim \mathcal{M}_g = 3g - 3$$

the classical dimension of the moduli of curves.

The one input taken on faith in example 15.5.4 is the stability theorem, and it is the reason GIT was invented. It is stated here in the form the construction uses.

Theorem 15.5.5: Asymptotic Stability of Pluricanonical Curves

Let $g \geq 2$ and let n be sufficiently large. A smooth projective curve of genus g, embedded in $\mathbb{P}^N$ by the complete linear system $|\omega_C^{\otimes n}|$, is a stable point of the Hilbert scheme $\mathrm{Hilb}^P(\mathbb{P}^N)$ for the natural SL_{N+1}-linearization, in the sense of definition 15.3.1. More generally, a Deligne-Mumford stable curve, embedded by $|\omega_C^{\otimes n}|$, is Hilbert stable (equivalently, Chow stable) for n large.

***Proof Key ideas*:**
The full proof is the technical core of Mumford's geometric invariant theory and occupies the capstone chapter on the moduli of curves; it is deferred there because it requires the compactified problem (stable curves and stable reduction) and a delicate Hilbert-Mumford computation that would be premature before the descent-and-stacks material and the deformation theory of nodes. The strategy is to test stability against every one-parameter subgroup via the numerical criterion (theorem 15.4.5). A 1-PS λ of SL_{N+1} acts on $\mathbb{P}^N$ by weights on the coordinates, and $\mu([C], \lambda)$ is computed from the weighted filtration of the homogeneous coordinate ring of C that λ induces. The estimate that $\mu > 0$ for all nontrivial λ, that is, stability, reduces to a statement about how the sections of $\omega_C^{\otimes n}$ distribute their vanishing across the curve: a destabilizing λ would concentrate too much of the linear system at a subscheme of C, and pluricanonical positivity forbids this once n is large. Mumford's key technical device is to control the leading term of μ using the asymptotic Riemann-Roch expansion of $h^0(C, \omega_C^{\otimes n}(-m\,D))$ for divisors D concentrated by λ, whence the name asymptotic stability. See the capstone construction and Mumford, *Geometric Invariant Theory*, for the complete argument.

The deferral is genuine: the stability theorem is long, its full proof belongs with the compactification and node deformations, and stating it here lets the recipe of proposition 15.5.1 be seen in action without the machinery. What the theorem gives is decisive. It says that the class of curves one wants to parametrize, the smooth ones (and, in the compactified problem, the Deligne-Mumford stable ones), coincides exactly with a GIT-stable locus. This is the non-formal matching demanded by proposition 15.5.1(3), the notion of *stability* in the moduli of curves, a nodal curve with finite automorphisms, is forced to agree with GIT stability, and this agreement of two a priori unrelated stabilities is what makes the construction succeed.

The genus-one case sits below this construction as its degenerate case. For $g = 1$ the canonical bundle ω_C has degree $2g - 2 = 0$, so it is not ample and the pluricanonical embedding of example 15.5.4 collapses: there is no n-canonical embedding, the automorphism group of an elliptic curve is positive-dimensional before rigidifying (translations), and the genus-one moduli problem must be handled with a marked point to obtain $\mathcal{M}_{1,1}$. The coarse space is then the j-line of theorem 13.3.6, constructed there directly through the j-invariant rather than through GIT. The two constructions agree in spirit: the j-invariant is the single invariant function separating isomorphism classes, exactly as the GIT quotient H_g/PGL is Proj of a ring of invariant functions. The j-line is the GIT quotient one would obtain from the moduli of 4 points on $\mathbb{P}^1$ (example 15.5.3 with $d = 4$, since an elliptic curve is a double cover of $\mathbb{P}^1$ branched at four points), and Mumford's theorem for $g \geq 2$ is the higher-genus generalization of the elementary invariant theory that produces j.

15.5.6 Historical Note: GIT and the Moduli of Curves Geometric invariant theory was developed by Mumford in the 1960s expressly to construct $\mathcal{M}_g$ as a quasi-projective variety, published in the first edition of *Geometric Invariant Theory* (1965). The stability of pluricanonical curves (theorem 15.5.5) is the technical result that made the construction possible, and its extension to the Deligne-Mumford stable curves gave the projective coarse space $\overline{M}_g$ of the compactified problem. The stack-theoretic viewpoint, in which $\mathcal{M}_g$ is the moduli stack whose coarse space is this GIT quotient, came later (Deligne-Mumford, 1969) and is the subject of the descent-and-stacks part of the book; historically the GIT construction came first and the stacks were introduced to explain what the coarse space was forgetting.

The GIT quotient is one of the two roads to a coarse moduli space, and comparing it with the direct construction of the j-line clarifies what each accomplishes. The direct construction (theorem 13.3.6) exhibits an explicit invariant and reads off the coarse space as its target; it works when a small ring of invariants can be written down by hand. The GIT construction is systematic: it parametrizes objects with an auxiliary embedding, forgets the embedding by a reductive group, and produces the coarse space as Proj of the full invariant ring, whose finite generation (theorem 15.1.2) guarantees a scheme without ever writing an invariant explicitly. For curves of genus $g \geq 2$, where no analogue of the j-invariant is available, GIT is the only elementary road to $\mathcal{M}_g$, and the descent-and-stacks part of the book will show what is lost along it and how stacks recover it.

Exercise 15.5.1

1. In example 15.5.4 the dimension of H_g was asserted to be $\dim \mathrm{PGL}_{N+1} + (3g-3)$, and $\dim \mathrm{PGL}_{N+1} = (N+1)^2 - 1$. For $g = 2$ and $n = 3$, compute N, $\dim \mathrm{PGL}_{N+1}$, and $\dim H_g$ explicitly, and verify that $\dim \mathcal{M}_g = 3g - 3$ comes out to the expected value.
2. Explain, using proposition 15.5.1, why the coarse moduli space $\mathcal{M}_g$ produced by the GIT construction cannot in general be a fine moduli space. Which specific hypothesis of the proposition is responsible, and what would have to be true of the objects for the space to be fine?
3. In example 15.5.3, take $d = 4$. Show directly that the coarse space $M^{\mathrm{un}}_{0,4}$ of four distinct unordered points on $\mathbb{P}^1$ is one-dimensional, and identify the invariant that coordinatizes it. Relate your answer to the j-line (theorem 13.3.6).
4. The genus-one canonical bundle has degree 0. Explain precisely why this makes the pluricanonical embedding of example 15.5.4 fail for $g = 1$, and why the genus-one moduli problem must be rigidified by a marked point before a GIT-style construction can be attempted. Which step of proposition 15.5.1 breaks without the marking?
5. theorem 15.5.5 asserts that n-canonically embedded smooth curves are GIT-*stable*, not just semistable. Using the stability-versus-semistability distinction of definition 15.3.1 and the role of stabilizers in proposition 15.5.1(3), explain what would go wrong in the construction of $\mathcal{M}_g$ if the curves were only semistable, and connect this to the finiteness of $\mathrm{Aut}(C)$.
6. The GIT construction produces $\mathcal{M}_g$ as a quotient H_g/PGL_{N+1} where H_g is a fine moduli space of embedded curves. Explain why H_g itself *is* fine (as a locally closed subscheme of a Hilbert scheme) while its quotient $\mathcal{M}_g$ is only coarse, and describe in one sentence the object of the descent-and-stacks part of the book that restores the lost information.

Part V

Deformation Theory

Part IV built moduli spaces; this Part studies them one point at a time. The question it answers is local: given an object of a moduli problem, in how many directions, and against what obstructions, can it be deformed? Deformation theory linearizes this question. The first-order deformations of an object form a vector space, and that vector space is a cohomology group attached to the object; the obstructions to extending a first-order deformation to higher order live in the next cohomology group. The local geometry of a moduli space, its tangent space and its dimension, its smoothness or singularity at a point, thus becomes a computation in the cohomology of a single object.

The two chapters develop this in increasing depth. chapter 16 treats first-order deformations, identifying the tangent space of a moduli functor with a concrete cohomology group in each of the main cases: the deformations of a subscheme with H^0 of its normal sheaf, of a smooth variety with H^1 of its tangent sheaf, of a coherent sheaf with a first Ext group; from these it reads off dimensions such as the $3g-3$ moduli of a curve of genus $g \geq 2$. The following chapter passes to the obstruction theory governing whether these first-order data integrate: it locates the obstructions in a second cohomology group, develops the naive cotangent complex that packages the tangent and obstruction spaces together, and proves Schlessinger's criteria, which say exactly when a deformation functor has a hull or is pro-representable.

Deformation theory is also the hinge between the two halves of the book. The automorphism obstruction that kept the moduli problems of Part IV from being representable reappears here as the group H^0 of infinitesimal automorphisms, now a cohomology group one can compute; and the tangent-obstruction formalism assembled in this Part is precisely the input that Artin's criteria will require, in the descent-and-stacks Part, to certify that a moduli problem is an algebraic stack.

16

First-Order Deformations

The tangent space to a moduli space measures the first-order deformations of the objects it parametrizes, and this chapter computes it. The method is the functor-of-points calculation already run for the Hilbert scheme in chapter 14: a tangent vector at a point $[x]$ is a morphism from the dual numbers $\operatorname{Spec} k[\varepsilon]$ carrying the closed point to $[x]$, which is the same datum as a deformation of the object x over $k[\varepsilon]$. What is new here is that the calculation applies to any moduli functor, whether or not it is representable, and that in each standard case the set of first-order deformations is naturally a cohomology group of the object being deformed.

section 16.1 sets up the general framework, the functor of Artin rings and its tangent space. The next three sections compute that tangent space in the three cases that recur throughout moduli theory: section 16.2 identifies the deformations of a closed subscheme with the global sections of its normal sheaf, recovering the Hilbert-scheme tangent space of chapter 14; section 16.3 classifies the deformations of a smooth variety by the first cohomology of its tangent sheaf, from which follow the rigidity of projective space and the $3g-3$ moduli of a curve; and section 16.4 treats deformations of a coherent sheaf and of a line bundle through Ext groups. section 16.5 closes the chapter by identifying the infinitesimal automorphisms of an object with the zeroth cohomology of the same sheaf, tying the automorphism obstruction of chapter 13 to the cohomological bookkeeping that the next chapter's obstruction theory extends.

16.1 Deformation Functors and the Dual Numbers

Part IV built moduli spaces globally: a moduli problem is a functor of points, and when it is representable there is a scheme whose points are the objects being classified. The natural next question is local. Given a specific object, say a smooth curve of genus g or a closed subscheme of $\mathbb{P}^n$, how does the moduli space look in an infinitesimal neighborhood of the corresponding point? The tangent space to a moduli space at a point is the linearization of the classification problem there, and its dimension bounds the number of independent directions in which the object can move. For the Hilbert scheme this tangent space was computed in theorem 14.4.4 as a space of global sections of the normal sheaf; the aim of this chapter is to see that every such computation is an instance of one uniform mechanism.

The mechanism is to probe the moduli problem with the smallest nontrivial thickening of a point, the spectrum of the dual numbers $k[\varepsilon]/(\varepsilon^2)$. A family over this base is an object over the point $\operatorname{Spec} k$ together with exactly one order of infinitesimal wiggle, and the set of such families is the tangent space. To make this precise and to make it apply to schemes, subschemes, and sheaves alike, we package the classification data as a functor on Artinian local k-algebras, evaluate it on the dual numbers, and identify the resulting set with a cohomology group. This section sets up the functor, the dual numbers, and the tangent space; sections 16.2 to 16.4 carry out the three identifications.

Throughout this chapter k is an algebraically closed field, and characteristic 0 is assumed wherever a cohomology-dimension count requires it (this is flagged where it is used). The base category on which deformation problems live is the category of Artinian local k-algebras.

Definition 16.1.1: The Category of Artinian Local k-Algebras

Let k be a field. The category $\mathbf{Art}_k$ has as objects the local Artinian k-algebras A whose residue field $A/\mathfrak{m}_A$ is k, where $\mathfrak{m}_A$ denotes the maximal ideal; a morphism $A \to B$ is a local homomorphism of k-algebras (equivalently, a k-algebra homomorphism, since any k-algebra map between such rings is automatically local). A *small extension* in $\mathbf{Art}_k$ is a surjection

$$\pi\colon A \twoheadrightarrow B$$

in $\mathbf{Art}_k$ whose kernel $I = \ker \pi$ satisfies $\mathfrak{m}_A \cdot I = 0$, that is, I is annihilated by the maximal ideal and so is a k-vector space.

An object of $\mathbf{Art}_k$ is a ring that is finite-dimensional as a k-vector space, local, and with residue field the ground field itself; geometrically, $\operatorname{Spec} A$ is a one-point scheme whose single point is thickened by nilpotents. The prototype is the ring of dual numbers introduced below, whose spectrum is a point carrying a single tangent direction. The requirement $A/\mathfrak{m}_A = k$ says that the fat point sits over the ground field with no residue-field extension, so that everything is measured relative to k; this is the algebraic counterpart of restricting attention to a single closed point of a k-scheme. The structure theory of such rings (that $\mathfrak{m}_A$ is nilpotent, that A is a finite-dimensional local k-algebra, and that A is complete) is developed in [10]; only the surface facts just recorded are needed here.

A small extension is the atomic step by which one fat point is built up from a smaller one. Because $\mathfrak{m}_A I = 0$, the kernel I is a module over $A/\mathfrak{m}_A = k$, hence a k-vector space, and the simplest small extensions have $I \cong k$ one-dimensional. Every surjection in $\mathbf{Art}_k$ factors as a composite of such one-dimensional small extensions, so a property that propagates along small extensions propagates along all of $\mathbf{Art}_k$. This is why small extensions are the correct inductive unit: lifting an object from B to A across a small extension is the elementary move whose iteration builds a deformation over an arbitrary Artinian base, and whose failure to be possible is exactly an obstruction (the subject of the next chapter). The smallest nontrivial small extension of all is the augmentation $k[\varepsilon] \twoheadrightarrow k$ killing ε, with kernel $(\varepsilon) \cong k$.

Definition 16.1.2: Deformation Functor

A *deformation functor* (or *functor of Artin rings*) is a covariant functor

$$D\colon \mathbf{Art}_k \longrightarrow \mathbf{Set}$$

such that $D(k)$ is a single point. The condition $D(k) = \{*\}$ is the *pointing*: it fixes the object being deformed. For A in $\mathbf{Art}_k$, an element of $D(A)$ is a *deformation over* A of the object encoded by that single point; the map $D(\pi)\colon D(A) \to D(k)$ induced by the augmentation $\pi\colon A \to k$ sends every deformation to the unique base point, expressing that a deformation over A restricts over the closed point to the original object.

The functor D records, for each fat point $\operatorname{Spec} A$, the set of ways the fixed object spreads out over that fat point. Covariance is forced by geometry: a k-algebra map $A \to A'$ corresponds to a morphism $\operatorname{Spec} A' \to \operatorname{Spec} A$ of fat points, and pulling a family back along this morphism produces a family over $\operatorname{Spec} A'$, giving a map $D(A) \to D(A')$. The pointing $D(k) = \{*\}$ is what distinguishes a deformation functor from an arbitrary moduli functor: one particular object has been singled out (the point of $D(k)$), and $D(A)$ collects only those families over $\operatorname{Spec} A$ that reduce to that object over the closed point. This is the local, infinitesimal restriction of the global moduli functor $F\colon \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ of Part IV: fixing a k-point $x_0 \in F(k)$ and restricting F to fat points supported at x_0 produces a deformation functor D, and D sees exactly the germ of F at x_0.

16.1.3 Warning: a deformation functor is not required to be representable A deformation functor D need not be representable, and indeed the interest lies precisely in the cases where it is not. When D is pro-representable by a complete local k-algebra R, the ring R is the completed local ring of the moduli space at the point, and $D(A) = \operatorname{Hom}_{k\text{-alg}}(R, A)$; the next chapter (Schlessinger's criteria) gives conditions under which this happens. Even when D is not pro-representable, the tangent space $D(k[\varepsilon])$ defined below is a well-behaved invariant. The general non-representability warned about for global moduli in box 13.1.6 has an infinitesimal counterpart here.

Probing D means evaluating it on the smallest Artinian algebra that carries a nonzero tangent direction. This is the ring of dual numbers, whose spectrum is a point with exactly one infinitesimal arrow attached.

Definition 16.1.4: Dual Numbers and First-Order Deformations

The *ring of dual numbers* over k is

$$k[\varepsilon] := k[t]/(t^2)$$

the quotient of the polynomial ring $k[t]$ by the square of the variable, with ε the image of t, so $\varepsilon^2 = 0$. It is an object of $\mathbf{Art}_k$: it is local with maximal ideal (ε), residue field $k[\varepsilon]/(\varepsilon) = k$, and dimension 2 as a k-vector space, with basis $\{1, \varepsilon\}$. A *first-order deformation* of the object classified by a deformation functor D is an element of $D(k[\varepsilon])$: a family over the fat point $\operatorname{Spec} k[\varepsilon]$, flat over $k[\varepsilon]$, whose restriction along the augmentation $k[\varepsilon] \to k$ ($\varepsilon \mapsto 0$) is the given object over $\operatorname{Spec} k$.

The geometric content of $\operatorname{Spec} k[\varepsilon]$ is a point with a single tangent vector: it is the length-two scheme supported at the origin of the affine line, the first infinitesimal neighborhood. A morphism $\operatorname{Spec} k[\varepsilon] \to Y$ into a k-scheme Y is the same as a k-point $y \in Y(k)$ together with a tangent vector to Y at y, because such a morphism is a local k-algebra map $\mathcal{O}_{Y,y} \to k[\varepsilon]$, which sends $\mathfrak{m}_y$ to (ε) and $\mathfrak{m}_y^2$ to $(\varepsilon)^2 = 0$, hence factors through $\mathfrak{m}_y/\mathfrak{m}_y^2$ and is therefore an element of $(\mathfrak{m}_y/\mathfrak{m}_y^2)^\vee = T_yY$. The word *first-order* names exactly this truncation: because $\varepsilon^2 = 0$, a first-order deformation records the linear term of a would-be family and discards all higher-order data. Flatness over $k[\varepsilon]$ is the condition that the family varies rather than degenerating; over the dual numbers it amounts to the concrete requirement that the family be, locally, a free $k[\varepsilon]$-module, which the following sections unwind case by case.

The set $D(k[\varepsilon])$ of first-order deformations is the tangent space to the deformation problem. To justify calling it a tangent space, one needs that under a mild condition it carries the structure of a k-vector space, not just that of a pointed set.

Definition 16.1.5: Tangent Space of a Deformation Functor

Let D be a deformation functor. Its *tangent space* is the set

$$t_D := D(k[\varepsilon])$$

of first-order deformations, pointed by the image of $D(k) = \{*\}$ under the inclusion $k \hookrightarrow k[\varepsilon]$ (the trivial, or product, deformation). Say that D satisfies the *fiber-product condition* (H_ε) if for every pair of maps $A' \to A \leftarrow A''$ in $\mathbf{Art}_k$ with $A'' \to A$ surjective, the natural map

$$D(A' \times_A A'') \longrightarrow D(A') \times_{D(A)} D(A'') \tag{16.1}$$

is a bijection whenever $A = k$. When (H_ε) holds, $t_D = D(k[\varepsilon])$ is a k-vector space.

The point marked in t_D is the trivial deformation: the augmentation $k \to k[\varepsilon]$ is a section of $k[\varepsilon] \to k$, and applying D to it embeds the single point $D(k)$ into $D(k[\varepsilon])$ as the deformation that does not move. Every tangent space has a distinguished zero, and this is it. The condition (H_ε) supplies the arithmetic. Its role is visible already for the specific fiber product

$$k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2 \;\cong\; k[\varepsilon] \times_k k[\varepsilon] \tag{16.2}$$

where the right side is the fiber product over the two augmentations $k[\varepsilon] \to k$. Granting (16.1) for this fiber product, D of the left side is $t_D \times t_D$, and the two k-algebra maps

$$k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2 \longrightarrow k[\varepsilon]$$

given by $\varepsilon_1, \varepsilon_2 \mapsto \varepsilon$ (addition) and by $\varepsilon \mapsto c\varepsilon$ for $c \in k$ (scaling) induce, after applying D, the addition $t_D \times t_D \to t_D$ and scalar multiplication $k \times t_D \to t_D$ making t_D a k-vector space. This is the deformation-theoretic incarnation of the functor tangent space from definition 13.5.1: there the tangent space of a functor F at a k-point was defined as $F(k[\varepsilon])$ probed against the same fiber product, and the vector-space structure was extracted by the same maps (16.2). The deformation functor D is that construction with the base point fixed once and for all, so t_D inherits its linear structure verbatim. The next result records this inheritance and gives the full argument.

Proposition 16.1.6: The Tangent Space is a k-Vector Space

Let D be a deformation functor satisfying the fiber-product condition (H_ε) of definition 16.1.5. Then $t_D = D(k[\varepsilon])$ carries a canonical structure of k-vector space, with zero the trivial deformation, for which every k-algebra map $k[\varepsilon] \to k[\varepsilon]$ over k induces the corresponding linear self-map of t_D. If moreover $D(A' \times_k A'') \to D(A') \times D(A'')$ is a bijection for all A', A'' (the strong form of (H_ε)), the construction is functorial in D.

***Proof*:**
Write $t = D(k[\varepsilon])$, pointed by $0 := D(k \hookrightarrow k[\varepsilon])(*)$, where $* \in D(k)$ is the unique element.

Step 1: Identify D of the double fiber product. Consider the ring $B := k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2$, with $\varepsilon_i\varepsilon_j = 0$ for all i, j. The two k-algebra maps $p_1, p_2 \colon B \to k[\varepsilon]$ sending $\varepsilon_1 \mapsto \varepsilon$, $\varepsilon_2 \mapsto 0$ and $\varepsilon_1 \mapsto 0$, $\varepsilon_2 \mapsto \varepsilon$ respectively exhibit B as the fiber product $k[\varepsilon] \times_k k[\varepsilon]$ over the two augmentations, as in (16.2): a k-algebra map into B is a pair of maps into $k[\varepsilon]$ agreeing modulo (ε), and modulo (ε) both land in k. Because $A = k$ here, condition (H_ε) applies and gives a bijection

$$D(B) \xrightarrow{\sim} D(k[\varepsilon]) \times_{D(k)} D(k[\varepsilon]) = t \times t$$

since $D(k)$ is a single point, so the fiber product over $D(k)$ is the full product.

Step 2: Define addition. The map $s \colon B \to k[\varepsilon]$ of k-algebras sending $\varepsilon_1 \mapsto \varepsilon$ and $\varepsilon_2 \mapsto \varepsilon$ is well defined: it carries the relations $(\varepsilon_1, \varepsilon_2)^2 = 0$ to $\varepsilon^2 = 0$, which holds in $k[\varepsilon]$, since each generator $\varepsilon_i\varepsilon_j$ maps to $\varepsilon^2 = 0$. Define addition on t as the composite

$$t \times t \xrightarrow{\sim} D(B) \xrightarrow{D(s)} D(k[\varepsilon]) = t$$

where the first arrow is the inverse of the bijection of Step 1. Concretely, a pair (v_1, v_2) of first-order deformations comes from a unique element of $D(B)$, and its image under $D(s)$ is declared to be $v_1 + v_2$.

Step 3: Define scalar multiplication. For $c \in k$, the k-algebra map $m_c \colon k[\varepsilon] \to k[\varepsilon]$ with $\varepsilon \mapsto c\varepsilon$ is well defined ($\varepsilon^2 \mapsto c^2\varepsilon^2 = 0$). Set $c \cdot v := D(m_c)(v)$ for $v \in t$. In particular $m_1 = \mathrm{id}$ gives $1 \cdot v = v$, and m_0 factors through k, so $0 \cdot v = 0$ is the trivial deformation.

Step 4: Verify the vector-space axioms. Commutativity of addition follows from the symmetry of B under $\varepsilon_1 \leftrightarrow \varepsilon_2$: the swap $\tau \colon B \to B$ interchanges p_1 and p_2, hence interchanges the two factors of $t \times t$, while $s \circ \tau = s$, so $v_1 + v_2 = v_2 + v_1$. Associativity uses the triple fiber product. Set $C := k[\varepsilon_1, \varepsilon_2, \varepsilon_3]/(\varepsilon_1, \varepsilon_2, \varepsilon_3)^2$; iterating (H_ε) over the presentation $C = (k[\varepsilon] \times_k k[\varepsilon]) \times_k k[\varepsilon]$ (each factor a small extension of k, so each application of (H_ε) is legitimate) gives $D(C) \cong t \times t \times t$, and the two ways of bracketing the sum are both computed by the single map $C \to k[\varepsilon]$, $\varepsilon_i \mapsto \varepsilon$, hence agree. The zero 0 is a two-sided identity because the element of $D(B)$ representing the pair $(v, 0)$ pulls back along $p_1 : \varepsilon_1 \mapsto \varepsilon$, $\varepsilon_2 \mapsto 0$ to v and along $s \colon \varepsilon_1, \varepsilon_2 \mapsto \varepsilon$ to $v + 0$, and s agrees with p_1 on this element because its second-coordinate deformation is trivial (the map s and the map p_1 differ only in the image of ε_2, whose associated deformation is 0); thus $v + 0 = v$. Distributivity follows from the compatibility of scaling with s: the identity $m_c \circ s = s \circ (m_c \times m_c)$ of k-algebra maps $B \to k[\varepsilon]$ (both send $\varepsilon_1, \varepsilon_2 \mapsto c\varepsilon$) gives $c(v_1 + v_2) = cv_1 + cv_2$, while $(c + c')v = cv + c'v$ is computed by the single map $\varepsilon \mapsto (c + c')\varepsilon = c\varepsilon + c'\varepsilon$ factoring through s applied to the pair $(c \cdot v, c' \cdot v)$, using that D of $\varepsilon \mapsto c\varepsilon$ and $\varepsilon \mapsto c'\varepsilon$ produce cv and $c'v$. Finally, additive inverses are automatic once addition, scaling, and distributivity hold: $v + (-1) \cdot v = (1 + (-1)) \cdot v = 0 \cdot v = 0$ by distributivity and $0 \cdot v = D(m_0)(v) = 0$ (as m_0 factors through k), so $-v = (-1) \cdot v = D(m_{-1})(v)$. All axioms hold, so t is a k-vector space.

Step 5: Functoriality. A natural transformation $\eta\colon D \to D'$ of deformation functors commutes with $D(-)$ applied to s and to each m_c, since these are induced by ring maps and η is natural; hence $\eta_{k[\varepsilon]}\colon t_D \to t_{D'}$ is k-linear. This uses only that η respects the fiber-product bijections, which holds because both D and D' satisfy the strong form of (H_ε) and η is a morphism of functors, completing the proof.

The proposition upgrades t_D from a pointed set to a vector space, and the upgrade is functorial in the deformation functor. This is what makes the tangent space computable: to find $\dim_k t_D$ one need only identify $D(k[\varepsilon])$ with some cohomology group and read off its dimension, and the sections that follow do exactly this in three cases. The condition (H_ε) is the first of Schlessinger's conditions, met by every deformation functor arising from a geometric moduli problem, because pulling families back along the two projections of a fiber product of fat points glues them uniquely; the full list of Schlessinger's conditions, controlling when D has a hull or is pro-representable, is the business of the next chapter. For the present chapter the only fact used is that the three deformation functors below satisfy (H_ε), which will be transparent from their explicit descriptions.

With the framework in place, the three running deformation problems of this chapter can be named. Each is a deformation functor in the sense of definition 16.1.2, and each has a tangent space that the corresponding later section identifies with a cohomology group.

Example 16.1.7: The Three Running Deformation Functors

Fix a smooth projective variety X over k, a closed subscheme $Z \subseteq X$, and a coherent sheaf $\mathcal{F}$ on X. The following are deformation functors, forming the three cases studied in sections 16.2 to 16.4.

1. *Deformations of a scheme.* $\mathrm{Def}_X\colon \mathbf{Art}_k \to \mathbf{Set}$ assigns to A the set of isomorphism classes of *flat families* $\mathcal{X} \to \operatorname{Spec} A$ together with an isomorphism $\mathcal{X} \times_{\operatorname{Spec} A} \operatorname{Spec} k \cong X$ of the closed fiber with X. Two such families are identified if there is an isomorphism over $\operatorname{Spec} A$ restricting to the identity on the closed fiber. Here $\mathrm{Def}_X(k) = \{X\}$ is the single point, the constant family. A first-order deformation is a flat family over $\operatorname{Spec} k[\varepsilon]$ restricting to X; section 16.3 shows $\mathrm{Def}_X(k[\varepsilon]) \cong H^1(X, T_X)$.

2. *Deformations of a closed subscheme.* $\mathrm{Def}_{Z/X}\colon \mathbf{Art}_k \to \mathbf{Set}$ holds the ambient X fixed and assigns to A the set of closed subschemes $\mathcal{Z} \subseteq X \times_k \operatorname{Spec} A$, flat over A, with closed fiber $\mathcal{Z} \times_{\operatorname{Spec} A} \operatorname{Spec} k = Z$. Here $\mathrm{Def}_{Z/X}(k) = \{Z\}$, and $\mathrm{Def}_{Z/X}(k[\varepsilon])$ is the tangent space to the Hilbert scheme at $[Z]$ computed in theorem 14.4.4; section 16.2 identifies it with $H^0(Z, N_{Z/X})$.

3. *Deformations of a coherent sheaf.* $\mathrm{Def}_{\mathcal{F}}\colon \mathbf{Art}_k \to \mathbf{Set}$ assigns to A the set of isomorphism classes of coherent sheaves $\mathcal{F}_A$ on $X \times_k \operatorname{Spec} A$, flat over A, with an isomorphism $\mathcal{F}_A|_{X \times \operatorname{Spec} k} \cong \mathcal{F}$. Here $\mathrm{Def}_{\mathcal{F}}(k) = \{\mathcal{F}\}$, and section 16.4 shows $\mathrm{Def}_{\mathcal{F}}(k[\varepsilon]) \cong \operatorname{Ext}^1_X(\mathcal{F}, \mathcal{F})$.

In each case the functor is covariant (pull back along $\operatorname{Spec} A' \to \operatorname{Spec} A$), the value on k is a single point (the object itself, as the constant family), and the fiber-product condition (H_ε) holds because flat families and flat subschemes and flat sheaves glue uniquely over a fiber product of fat points. So each tangent space is a k-vector space by proposition 16.1.6.

These three functors organize the chapter. Each isolates one kind of geometric object and asks how it deforms while its neighbors stay fixed: the ambient variety deforms in Def_X, a subscheme deforms inside a fixed ambient in $\mathrm{Def}_{Z/X}$, and a sheaf deforms on a fixed variety in $\mathrm{Def}_{\mathcal{F}}$. The pattern is uniform: in every case the tangent space is $D(k[\varepsilon])$, and in every case it turns out to be a cohomology group ($H^1(T_X)$, $H^0(N_{Z/X})$, $\mathrm{Ext}^1(\mathcal{F},\mathcal{F})$) whose dimension is the number of independent first-order deformations. The consistency check between $\mathrm{Def}_{Z/X}$ and the Hilbert scheme, namely that $\mathrm{Def}_{Z/X}(k[\varepsilon])$ recovers exactly the Zariski tangent space $T_{[Z]}\mathrm{Hilb}$ of theorem 14.4.4, is what ties the infinitesimal picture of this chapter to the global moduli spaces of Part IV. That check is carried out in section 16.2.

One reading of the trivial deformation deserves emphasis, because it recurs in every case and returns in section 16.5. The zero element of t_D is the constant family $X \times_k \operatorname{Spec} k[\varepsilon]$ (or $Z \times_k \operatorname{Spec} k[\varepsilon]$, or $\mathcal{F} \boxtimes k[\varepsilon]$), the deformation that does not move the object. A nonzero first-order deformation is a family that agrees with the constant one over the closed point but disagrees to first order in ε. Whether such a family is isomorphic to the constant one as a deformation is the question of whether its class in t_D vanishes, and the automorphisms of the constant deformation, which measure the redundancy in this identification, form their own cohomology group $H^0(T_X)$ studied at the end of the chapter.

Exercise 16.1.1

1. Show that $\mathbf{Art}_k$ has k itself as a terminal object and that the polynomial ring $k[t]$ is *not* an object of $\mathbf{Art}_k$. Then show that for any A in $\mathbf{Art}_k$ the maximal ideal $\mathfrak{m}_A$ is nilpotent, and deduce that A is a finite-dimensional k-vector space.
2. Let $\pi\colon A \twoheadrightarrow B$ be a surjection in $\mathbf{Art}_k$ with kernel I. Show that π factors as a finite composite of small extensions. (Filter I by the descending chain $I \supseteq \mathfrak{m}_A I \supseteq \mathfrak{m}_A^2 I \supseteq \cdots$ and observe each successive quotient is killed by $\mathfrak{m}_A$.)
3. Verify directly that a k-algebra homomorphism $\mathcal{O}_{Y,y} \to k[\varepsilon]$ inducing the residue field identity is the same datum as a tangent vector to the k-scheme Y at the k-point y, that is, an element of $(\mathfrak{m}_y/\mathfrak{m}_y^2)^\vee$. Conclude that $Y(k[\varepsilon]) \to Y(k)$ has fiber over y equal to the Zariski tangent space T_yY.
4. Prove the isomorphism $k[\varepsilon_1,\varepsilon_2]/(\varepsilon_1,\varepsilon_2)^2 \cong k[\varepsilon] \times_k k[\varepsilon]$ of (16.2) by exhibiting mutually inverse k-algebra maps, where the fiber product is taken over the two augmentations $k[\varepsilon] \to k$.
5. Let $D = h_R := \mathrm{Hom}_{k\text{-alg}}(R, -)$ be the functor of points of a complete local Noetherian k-algebra $(R, \mathfrak{m}_R)$ with residue field k, restricted to $\mathbf{Art}_k$. Show D is a deformation functor and compute its tangent space $t_D = D(k[\varepsilon])$ as a k-vector space, identifying it with $(\mathfrak{m}_R/\mathfrak{m}_R^2)^\vee$.
6. Give a deformation functor D that does not satisfy the fiber-product condition (H_ε), so that $D(k[\varepsilon])$ fails to be a vector space in the canonical way. (Consider a functor whose value on $k[\varepsilon]$ is a single point but which does not respect the fiber product (16.2), or a set-valued functor built to break additivity.)

16.2 Deformations of a Closed Subscheme

The functor of Artin rings from section 16.1 is a general device, but its first genuine payoff appears in the most concrete deformation problem of all: fixing an ambient variety X and asking how a closed subscheme $Z \subseteq X$ can be moved inside it. This is the infinitesimal version of the question the Hilbert scheme answers globally. In section 16.1 the deformation functor $\mathrm{Def}_{Z/X}$ was introduced as one of the running cases; here it is solved outright. The answer is a single cohomological invariant: first-order deformations of Z are the global sections of a single coherent sheaf on Z, the normal sheaf $N_{Z/X}$. Every deformation is a first-order motion of Z in the directions transverse to itself, and $N_{Z/X}$ is precisely the sheaf of such transverse directions.

The section proves this classification in full, then reads two consequences out of it. For a complete intersection the normal sheaf splits as a sum of line bundles, and the deformation space becomes a cohomology group one can write down explicitly; for a hypersurface or a finite set of points the count reproduces the tangent space to the Hilbert scheme computed in an earlier chapter, closing the circle between the functorial and the geometric pictures. Throughout, X is a smooth projective variety over an algebraically closed field k, and $Z \subseteq X$ is a closed subscheme with ideal sheaf $I_Z \subseteq \mathcal{O}_X$; smoothness hypotheses are stated where the normal sheaf is required to be locally free.

The object that governs the whole section is the normal sheaf, defined in an earlier chapter as the dual of the conormal sheaf I_Z/I_Z^2. Recall its statement so the classification below can be phrased against it.

Recall definition 14.4.1: for a closed subscheme $Z \subseteq X$ with ideal sheaf I_Z, the *conormal sheaf* is the $\mathcal{O}_Z$-module I_Z/I_Z^2, and the *normal sheaf* is its dual

$$N_{Z/X} \;=\; \mathcal{H}om_{\mathcal{O}_Z}(I_Z/I_Z^2,\, \mathcal{O}_Z)$$

When Z and X are both smooth, I_Z/I_Z^2 is locally free of rank equal to the codimension $c = \dim X - \dim Z$, and $N_{Z/X}$ is then a vector bundle of rank c on Z. In general $N_{Z/X}$ is only a coherent sheaf, but the classification below holds regardless: it is the sheaf $\mathcal{H}om_{\mathcal{O}_Z}(I_Z/I_Z^2, \mathcal{O}_Z)$, whose sections are the $\mathcal{O}_Z$-linear maps out of the conormal sheaf.

A homomorphism $I_Z/I_Z^2 \to \mathcal{O}_Z$ is the same datum as an $\mathcal{O}_X$-linear map $I_Z \to \mathcal{O}_Z$, because any such map kills I_Z^2 automatically: if $\varphi\colon I_Z \to \mathcal{O}_Z$ is $\mathcal{O}_X$-linear and $f, g \in I_Z$, then $\varphi(fg) = f\,\varphi(g)$, and $f \in I_Z$ maps to 0 in $\mathcal{O}_Z = \mathcal{O}_X/I_Z$, so $\varphi(fg) = 0$. Thus

$$H^0(Z, N_{Z/X}) \;=\; \mathrm{Hom}_{\mathcal{O}_Z}(I_Z/I_Z^2,\, \mathcal{O}_Z) \;=\; \mathrm{Hom}_{\mathcal{O}_X}(I_Z,\, \mathcal{O}_Z)$$

This identification is the mechanism behind the theorem: a global section of the normal sheaf is an $\mathcal{O}_X$-homomorphism from the ideal to the structure sheaf of Z, and it is precisely such a homomorphism that a first-order deformation produces.

The problem is to understand a flat family over the dual numbers $k[\varepsilon] = k[t]/(t^2)$ that restricts to Z over the closed point. Such a family is a closed subscheme of $X \times_k \operatorname{Spec} k[\varepsilon]$, flat over $\operatorname{Spec} k[\varepsilon]$, whose special fiber is Z (definition 16.1.4). Everything reduces to a statement about ideal sheaves inside $\mathcal{O}_X \otimes_k k[\varepsilon] = \mathcal{O}_X[\varepsilon]$.

Theorem 16.2.1: First-Order Deformations of a Closed Subscheme

Let X be a scheme over k and $Z \subseteq X$ a closed subscheme with ideal sheaf I_Z. The set of first-order deformations of Z inside X, that is, the set $\mathrm{Def}_{Z/X}(k[\varepsilon])$ of closed subschemes $\widetilde{Z} \subseteq X \times_k \operatorname{Spec} k[\varepsilon]$ that are flat over $\operatorname{Spec} k[\varepsilon]$ and restrict to Z over the closed point, is in natural bijection with

$$\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z) \;=\; H^0(Z, N_{Z/X})$$

the global sections of the normal sheaf. Under this bijection the trivial deformation $Z \times_k \operatorname{Spec} k[\varepsilon]$ corresponds to the zero section, and the bijection is k-linear, so $\mathrm{Def}_{Z/X}(k[\varepsilon])$ is a k-vector space with $H^0(Z, N_{Z/X})$ as its tangent space.

***Proof*:**

The argument is entirely local: a deformation is a subsheaf $\widetilde{I} \subseteq \mathcal{O}_X[\varepsilon]$, and flatness pins down the shape of $\widetilde{I}$ so tightly that it is determined by a single homomorphism out of I_Z. The proof establishes that shape, extracts the homomorphism, and checks that the construction is reversible.

Step 1: Flat deformations are extensions of I_Z by εI_Z. Write $A = k[\varepsilon]$ and $\mathcal{O}_X[\varepsilon] = \mathcal{O}_X \otimes_k A$, a sheaf of A-algebras on X fitting into the exact sequence

$$0 \longrightarrow \varepsilon\mathcal{O}_X \longrightarrow \mathcal{O}_X[\varepsilon] \xrightarrow{\varepsilon\mapsto 0} \mathcal{O}_X \longrightarrow 0 \tag{16.3}$$

in which multiplication by ε induces an isomorphism $\mathcal{O}_X \xrightarrow{\sim} \varepsilon\mathcal{O}_X$ of $\mathcal{O}_X$-modules. A deformation $\widetilde{Z}$ is given by its ideal sheaf $\widetilde{I} \subseteq \mathcal{O}_X[\varepsilon]$, and the condition that $\widetilde{Z}$ restrict to Z over the closed point is that the image of $\widetilde{I}$ under $\varepsilon \mapsto 0$ be exactly I_Z.

Flatness of $\widetilde{Z}$ over A is equivalent to flatness of $\widetilde{\mathcal{O}} := \mathcal{O}_X[\varepsilon]/\widetilde{I}$ over A. Since $A = k[\varepsilon]$ is a local Artinian ring with maximal ideal (ε) satisfying $\varepsilon^2 = 0$, the local criterion of flatness over the dual numbers takes an elementary form: an A-module M is flat over A if and only if the multiplication map

$$\varepsilon\colon M/\varepsilon M \longrightarrow \varepsilon M \tag{16.4}$$

is an isomorphism, equivalently $\mathrm{Tor}_1^A(M, k) = 0$. Applying this to $M = \widetilde{\mathcal{O}}$: writing $\widetilde{\mathcal{O}}/\varepsilon\widetilde{\mathcal{O}} = \mathcal{O}_Z$ (the special fiber), flatness says that $\varepsilon\widetilde{\mathcal{O}} \cong \mathcal{O}_Z$ via multiplication by ε, so $\widetilde{\mathcal{O}}$ sits in a short exact sequence

$$0 \longrightarrow \varepsilon\mathcal{O}_Z \longrightarrow \widetilde{\mathcal{O}} \longrightarrow \mathcal{O}_Z \longrightarrow 0 \tag{16.5}$$

where $\varepsilon\mathcal{O}_Z$ denotes the copy of $\mathcal{O}_Z$ carried by ε. Dually, on the level of ideals, flatness is equivalent to the requirement that

$$\widetilde{I} \cap \varepsilon\mathcal{O}_X[\varepsilon] \;=\; \varepsilon I_Z \tag{16.6}$$

that is, the part of $\widetilde{I}$ divisible by ε is precisely εI_Z and nothing more, which is the ideal-level form of the iso characterization (16.4).

Step 2: A flat deformation is a lift of the identity, hence a homomorphism $\varphi\colon I_Z \to \mathcal{O}_Z$. Fix a flat deformation with ideal $\widetilde{I}$ satisfying (16.6). Every element of $\widetilde{I}$ has the form $g_0 + \varepsilon g_1$ with $g_0, g_1 \in \mathcal{O}_X$, and reducing modulo ε sends $\widetilde{I}$ onto I_Z; so for $g_0 + \varepsilon g_1 \in \widetilde{I}$ the leading term satisfies $g_0 \in I_Z$. Given a section $f \in I_Z$, choose any lift $f + \varepsilon h \in \widetilde{I}$ of f (one exists because $\widetilde{I} \to I_Z$ is surjective). Two lifts of the same f differ by an element of $\widetilde{I} \cap \varepsilon\mathcal{O}_X[\varepsilon] = \varepsilon I_Z$, so the class of h in $\mathcal{O}_X/I_Z = \mathcal{O}_Z$ is independent of the choice of lift. Define

$$\varphi\colon I_Z \longrightarrow \mathcal{O}_Z, \qquad \varphi(f) \;=\; \overline{h} \;\text{ where } f + \varepsilon h \in \widetilde{I}$$

This is well defined by the previous sentence. It is additive because the sum of two lifts lifts the sum. It is $\mathcal{O}_X$-linear: for $a \in \mathcal{O}_X$ and $f \in I_Z$ with lift $f + \varepsilon h$, the element $a(f + \varepsilon h) = af + \varepsilon(ah)$ lies in $\widetilde{I}$ (an ideal), and it lifts af, so $\varphi(af) = \overline{ah} = \overline{a}\,\overline{h} = a\,\varphi(f)$ in $\mathcal{O}_Z$. Thus $\varphi \in \mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z)$.

Step 3: The homomorphism recovers the deformation, giving the inverse map. Conversely, let $\varphi \in \mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z)$ be given, and set

$$\widetilde{I}_\varphi \;=\; \left\{\, f + \varepsilon g \;:\; f \in I_Z,\; g \in \mathcal{O}_X,\; \overline{g} = \varphi(f) \text{ in } \mathcal{O}_Z \,\right\} \;\subseteq\; \mathcal{O}_X[\varepsilon] \tag{16.7}$$

Concretely $\widetilde{I}_\varphi$ consists of $f + \varepsilon g$ where $f \in I_Z$ and g reduces to $\varphi(f)$ modulo I_Z. This is an $\mathcal{O}_X[\varepsilon]$-submodule: for $a + \varepsilon b \in \mathcal{O}_X[\varepsilon]$ and $f + \varepsilon g \in \widetilde{I}_\varphi$,

$$(a + \varepsilon b)(f + \varepsilon g) \;=\; af + \varepsilon(ag + bf)$$

and $af \in I_Z$ with $\overline{ag + bf} = \overline{a}\,\varphi(f) + \overline{b} \cdot 0 = \varphi(af)$ (using $f \in I_Z$, so $\overline{bf} = 0$, and $\mathcal{O}_X$-linearity of φ), so the product lies in $\widetilde{I}_\varphi$. It satisfies (16.6): an element $f + \varepsilon g \in \widetilde{I}_\varphi$ is divisible by ε exactly when $f = 0$, and then $\overline{g} = \varphi(0) = 0$ forces $g \in I_Z$, so $\widetilde{I}_\varphi \cap \varepsilon\mathcal{O}_X[\varepsilon] = \varepsilon I_Z$. Hence $\widetilde{I}_\varphi$ defines a flat deformation. The homomorphism attached to $\widetilde{I}_\varphi$ by Step 2 is φ itself, since $f + \varepsilon g \in \widetilde{I}_\varphi$ has $\overline{g} = \varphi(f)$ by construction.

Step 4: The two constructions are mutually inverse and linear. Starting from a flat deformation $\widetilde{I}$, Step 2 produces φ, and Step 3 applied to φ returns the ideal $\{f + \varepsilon g : \overline{g} = \varphi(f)\}$. This equals $\widetilde{I}$: any $f + \varepsilon g \in \widetilde{I}$ has $\overline{g} = \varphi(f)$ by the definition of φ, and conversely if $\overline{g} = \varphi(f)$ then $f + \varepsilon g$ and the chosen lift $f + \varepsilon h$ of f differ by $\varepsilon(g - h)$ with $\overline{g - h} = 0$, so $g - h \in I_Z$ and $\varepsilon(g - h) \in \varepsilon I_Z \subseteq \widetilde{I}$, giving $f + \varepsilon g \in \widetilde{I}$. Thus the maps of Steps 2 and 3 are inverse bijections. The trivial deformation has $\widetilde{I} = I_Z \cdot \mathcal{O}_X[\varepsilon] = \{f + \varepsilon g : f, g \in I_Z\}$, for which every lift of $f \in I_Z$ can be taken with $h \in I_Z$, so $\varphi = 0$: the trivial deformation is the zero section. Finally, the bijection is k-linear because $\widetilde{I}_{\varphi_1+\varphi_2}$ and $\widetilde{I}_{c\varphi}$ are read off additively and by scaling from the defining condition $\overline{g} = \varphi(f)$ in (16.7). The identification $\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z) = H^0(Z, N_{Z/X})$ from the discussion preceding the theorem completes the proof.

The theorem is the local model of every tangent-space computation in deformation theory: a deformation of a subscheme is a first-order sliding of its defining equations, encoded by how far each generator of I_Z is allowed to move out of the ideal, measured modulo I_Z. The datum $\varphi(f) = \overline{h}$ records the ε-correction $f \rightsquigarrow f + \varepsilon h$ applied to the equation f, and the constraint that φ be $\mathcal{O}_X$-linear and factor through I_Z/I_Z^2 is exactly the constraint that these corrections assemble into a consistent flat family rather than an arbitrary perturbation of equations. A section of $N_{Z/X}$ is a choice of outward normal direction at every point of Z, and moving Z along it to first order is the deformation. When $H^0(Z, N_{Z/X}) = 0$ the subscheme is rigid inside X: it admits no nontrivial first-order motion, and the corresponding point of the Hilbert scheme is isolated to first order.

The bijection also explains the appearance of the same cohomology group on both sides of the moduli story. Globally, the Hilbert scheme Hilb_X parametrizes closed subschemes of X, and its tangent space at the point $[Z]$ was computed in an earlier chapter to be $H^0(Z, N_{Z/X})$ (theorem 14.4.4). The theorem above restates that computation in functor-of-points terms: a first-order deformation of Z is exactly a $k[\varepsilon]$-point of Hilb_X lying over $[Z]$, so the two tangent spaces must agree. What the Hilbert scheme sees as its Zariski tangent space, the deformation functor $\mathrm{Def}_{Z/X}$ sees as $\mathrm{Def}_{Z/X}(k[\varepsilon])$, and theorem 16.2.1 identifies both with the sections of the normal sheaf. This is the promised bridge between the geometric and the functorial construction of the tangent space; the remaining examples make the count explicit.

Example 16.2.2: Deformations of a Complete Intersection

Let X be a smooth projective variety and let $Z \subseteq X$ be a smooth complete intersection of codimension c, cut out by a regular sequence of sections $f_1, \dots, f_c$ of line bundles $\mathcal{L}_1, \dots, \mathcal{L}_c$ on X; that is, I_Z is generated by the f_i and the associated Koszul complex is a resolution. The conormal sheaf then splits as a direct sum of line bundles restricted to Z:

$$I_Z/I_Z^2 \;\cong\; \bigoplus_{i=1}^{c} \mathcal{L}_i^{-1}\big|_Z$$

with the class of f_i generating the i-th summand. Dualizing gives the normal sheaf as a direct sum of line bundles

$$N_{Z/X} \;\cong\; \bigoplus_{i=1}^{c} \mathcal{L}_i\big|_Z \tag{16.8}$$

so a first-order deformation is a c-tuple $(s_1, \dots, s_c)$ with $s_i \in H^0(Z, \mathcal{L}_i|_Z)$, and the deformation $f_i \rightsquigarrow f_i + \varepsilon\widetilde{s}_i$ perturbs the i-th equation by a lift $\widetilde{s}_i$ of s_i. The dimension of the space of first-order deformations is therefore

$$h^0(Z, N_{Z/X}) \;=\; \sum_{i=1}^{c} h^0\big(Z,\, \mathcal{L}_i|_Z\big)$$

which (16.8) reduces to a sum of ordinary cohomology computations.

Take the concrete case $X = \mathbb{P}^n$ and Z a complete intersection of hypersurfaces of degrees $d_1, \dots, d_c$, so $\mathcal{L}_i = \mathcal{O}_{\mathbb{P}^n}(d_i)$ and

$$N_{Z/X} \;\cong\; \bigoplus_{i=1}^{c} \mathcal{O}_Z(d_i)$$

The perturbations of the equation f_i that move Z are the degree-d_i forms restricted to Z; the forms lying in the ideal $(f_1, \dots, f_c)$ act trivially, since they only reparametrize the same subscheme. For a single smooth surface Z of degree d in $\mathbb{P}^3$ (the case $c = 1$),

$$h^0(Z, N_{Z/X}) \;=\; h^0\big(\mathbb{P}^3, \mathcal{O}(d)\big) - h^0\big(\mathbb{P}^3, \mathcal{O}\big) \;=\; \binom{d+3}{3} - 1$$

the count of degree-d forms modulo the one-dimensional span of f itself, using $H^0(\mathbb{P}^3, \mathcal{O}(d)) \twoheadrightarrow H^0(Z, \mathcal{O}_Z(d))$ (surjective because $H^1(\mathbb{P}^3, \mathcal{O}(d-d)) = H^1(\mathbb{P}^3, \mathcal{O}) = 0$) with kernel $k \cdot f$. This matches the dimension of the projective space of degree-d surfaces through the parametrized family, confirming that the Hilbert scheme of such surfaces is smooth of the expected dimension at $[Z]$.

The complete-intersection case is the one where the abstract normal sheaf becomes a concrete list of line bundles and the deformation count becomes a sum of dimensions of linear systems. More is true: a global complete intersection is *unobstructed*, so $[Z]$ is always a smooth point of the Hilbert scheme, and the first-order count $h^0(Z, N_{Z/X})$ is the actual local dimension there (a standard theorem; see Hartshorne's *Deformation Theory* Ch. 8 or Sernesi 4.6.6). The obstructions to extending a first-order deformation to higher order live in $H^1(Z, N_{Z/X})$, which for the splitting (16.8) is $\bigoplus_i H^1(Z, \mathcal{L}_i|_Z)$; but for a complete intersection every such obstruction vanishes even when this group is nonzero, since the deformations produced by perturbing the defining equations always integrate. Non-vanishing of $H^1(Z, N_{Z/X})$ only says where an obstruction *could* live, not that a genuine obstruction exists, and it is precisely the complete-intersection hypothesis that kills it. The singular and reducible Hilbert schemes of space curves discovered by Mumford arise instead from *non*-complete-intersection curves, where no such splitting is available; complete intersections are among the first examples where the Hilbert scheme is provably smooth, which is the subject the next chapter takes up when it treats obstruction theory in general.

The extreme special cases, a single hypersurface and a finite set of points, are worth isolating because they recover the Hilbert-scheme tangent space of theorem 14.4.4 in its most transparent form.

Example 16.2.3: Hypersurfaces and Points

1. *A degree-d hypersurface in $\mathbb{P}^n$.* Let $Z = V_+(f) \subseteq \mathbb{P}^n$ be a smooth hypersurface of degree d, so $c = 1$, $I_Z = \mathcal{O}_{\mathbb{P}^n}(-d) \cdot f$, and $N_{Z/X} = \mathcal{O}_Z(d)$ by (16.8). The first-order deformations are
$$H^0(Z, N_{Z/X}) \;=\; H^0(Z, \mathcal{O}_Z(d))$$
Restriction from $\mathbb{P}^n$ is surjective with kernel spanned by f, so
$$h^0(Z, N_{Z/X}) \;=\; \binom{n+d}{d} - 1$$
the number of degree-d monomials in $n+1$ variables minus one for the equation f itself. This is exactly $\dim \mathbb{P}\big(H^0(\mathbb{P}^n, \mathcal{O}(d))\big)$, the dimension of the projective space of degree-d hypersurfaces. The Hilbert scheme of degree-d hypersurfaces in $\mathbb{P}^n$ *is* this projective space, smooth of dimension $\binom{n+d}{d} - 1$, and theorem 16.2.1 recovers its tangent space at every point without ever leaving the functor of points. This is the promised recovery of theorem 14.4.4 from the deformation-theoretic side.

2. *A finite set of points.* Let $Z = \{p_1, \ldots, p_r\} \subseteq X$ be r distinct reduced points of a smooth variety X of dimension n. At each p_i the local ring is regular of dimension n, the maximal ideal is generated by a regular system of parameters, and the conormal space $\mathfrak{m}_{p_i}/\mathfrak{m}_{p_i}^2$ is the cotangent space, of dimension n. Hence the normal sheaf is a sum of skyscraper sheaves
$$N_{Z/X} \;=\; \bigoplus_{i=1}^{r} (i_{p_i})_* T_{p_i} X$$
where $i_{p_i} \colon \{p_i\} \hookrightarrow X$ and the stalk at p_i is the tangent space $T_{p_i}X = (\mathfrak{m}_{p_i}/\mathfrak{m}_{p_i}^2)^\vee$, of dimension n. Therefore
$$h^0(Z, N_{Z/X}) \;=\; \sum_{i=1}^{r} \dim_k T_{p_i} X \;=\; rn$$
A first-order deformation of the reduced scheme Z is a choice of tangent vector at each point, moving the r points independently to first order. This is the tangent space to the

Hilbert scheme of r points, $\mathrm{Hilb}^r(X)$, at the reduced configuration $[Z]$: since $\mathrm{Hilb}^r(X)$ has dimension rn near the locus of distinct points, the computation $h^0(Z, N_{Z/X}) = rn$ shows the Hilbert scheme is smooth of the expected dimension there, matching theorem 14.4.4.

3. *A nonreduced point.* The reduced case above conceals the subtlety that makes Hilbert schemes of points interesting. Take $X = \mathbb{A}^2$ and Z the length-2 subscheme defined by $I_Z = (x^2, y)$, supported at the origin. Here Z is not a complete intersection in the splitting sense of c regular parameters cutting a reduced point, but the theorem still applies: $\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z)$ classifies the deformations. Computing this Hom for $I_Z = (x^2, y)$ over $\mathcal{O}_Z = k[x, y]/(x^2, y)$ gives a 4-dimensional space, matching $\dim \mathrm{Hilb}^2(\mathbb{A}^2) = 2 \cdot 2 = 4$; the nonreduced point does not obstruct, and $\mathrm{Hilb}^2(\mathbb{A}^2)$ is smooth. That every $\mathrm{Hilb}^r(\mathbb{A}^2)$ (indeed every Hilb^r of a smooth surface) is smooth of dimension $2r$ is a theorem whose local input is exactly this Hom computation, extended to all finite colength ideals.

The point of these examples is that the single formula $H^0(Z, N_{Z/X})$ specializes to every count one would compute by hand: the dimension of a linear system of hypersurfaces, a tangent vector per point, or a Hom of ideals for a nonreduced scheme. The reduced-points case exhibits the tangent space as a direct sum of ambient tangent spaces, one per point, which is why the Hilbert scheme of distinct points looks locally like a symmetric product; the nonreduced case (part 3) is where the normal-sheaf description is needed, since there is no naive count of "directions to move" and the Hom of ideals is the only available handle. In every case the deformation-theoretic tangent space of theorem 16.2.1 agrees with the Zariski tangent space of the Hilbert scheme from theorem 14.4.4, as it must, since a first-order deformation of Z is precisely a $k[\varepsilon]$-point of Hilb_X over $[Z]$.

Exercise 16.2.1

1. Let $Z \subseteq X$ be a closed subscheme with ideal sheaf I_Z, and let $\varphi\colon I_Z \to \mathcal{O}_Z$ be an $\mathcal{O}_X$-linear map. Verify directly from the definition that φ kills I_Z^2, and conclude that $\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z) = \mathrm{Hom}_{\mathcal{O}_Z}(I_Z/I_Z^2, \mathcal{O}_Z)$. Explain in one sentence why this is the reason the normal sheaf, rather than $\mathcal{H}om_{\mathcal{O}_X}(I_Z, \mathcal{O}_X)$, is the correct object.
2. Let Z be a smooth plane curve of degree d in $\mathbb{P}^2$. Using example 16.2.3, compute $h^0(Z, N_{Z/\mathbb{P}^2})$ and compare it with the dimension of the projective space of degree-d plane curves. For which d is this space positive-dimensional, and what does $d = 1$ say?
3. Let $Z \subseteq \mathbb{P}^3$ be the smooth complete intersection of a quadric and a cubic (a curve of degree 6). Write down $N_{Z/\mathbb{P}^3}$ as a direct sum of line bundles on Z, and set up the computation of $h^0(Z, N_{Z/\mathbb{P}^3})$. You may leave the answer in terms of the degrees $\deg(\mathcal{O}_Z(2))$ and $\deg(\mathcal{O}_Z(3))$ and the genus of Z; identify the genus using the adjunction formula.
4. Consider $X = \mathbb{A}^2$ and the length-2 subscheme Z with $I_Z = (x^2, y)$ supported at the origin, as in example 16.2.3(3). Compute $\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z)$ explicitly by specifying where the generators x^2 and y may be sent, and verify that the dimension is 4. Which of your homomorphisms corresponds to sliding the doubled point along the x-axis?
5. Show that a smooth subvariety $Z \subseteq X$ with $H^0(Z, N_{Z/X}) = 0$ is rigid inside X: every first-order deformation is trivial. Give an example of such a Z (with X specified), and explain in one sentence how this is the subscheme analogue of the infinitesimal rigidity of $\mathbb{P}^n$ studied in the next section.
6. Let $Z \subseteq X$ be a complete intersection of hypersurfaces of degrees $d_1, \ldots, d_c$ in $X = \mathbb{P}^n$. Explain

why the degree-d_i forms lying in the ideal $(f_1, \ldots, f_c)$ contribute nothing to $H^0(Z, N_{Z/X})$, and relate this to the fact that $H^0(Z, N_{Z/X})$ is a quotient, not the whole space of equation-perturbations. What geometric operation on Z do the trivial perturbations correspond to?

16.3 Deformations of a Smooth Variety

The previous section deformed a closed subscheme $Z \subseteq X$ inside a fixed ambient space, and the answer, $H^0(Z, N_{Z/X})$, remembered that embedding: it measured the ways the equations cutting out Z could wiggle. The present section drops the ambient space entirely and asks a more intrinsic question. Given a smooth projective variety X, in how many ways can X itself be deformed as an *abstract* variety, with no reference to any surrounding $\mathbb{P}^N$? This is the deformation problem that governs moduli of varieties, and it is where the dimension of a moduli space first becomes visible as a cohomology group. The answer will be $H^1(X, T_X)$, the first cohomology of the tangent sheaf, and the mechanism producing it is the most transparent in deformation theory: a first-order deformation is a way of regluing X from its affine pieces, and the tangent sheaf records the infinitesimal ambiguity in that gluing.

Throughout this section k is algebraically closed of characteristic zero and X is a smooth projective variety over k, of dimension n. The tangent sheaf of X is the dual of the sheaf of Kähler differentials,

$$T_X = \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X)$$

a locally free sheaf of rank n (definition 5.5.25). Its sections are the algebraic vector fields on X; over an affine open $U = \operatorname{Spec} A$, the sections $T_X(U)$ are the k-derivations $\operatorname{Der}_k(A, A)$, the infinitesimal automorphisms of U. This is exactly the data that will glue the affine pieces of a deformation, so the identification of the tangent sheaf with derivations is the hinge of the whole argument.

The passage from the embedded problem of the previous section to the abstract problem here rests on one fact about smooth affine varieties: they cannot be deformed at all. This rigidity is what lets a global deformation be reconstructed purely from gluing data, and it is worth isolating before the main theorem.

Proposition 16.3.1: Rigidity of Smooth Affine Varieties

Let $U = \operatorname{Spec} A$ be a smooth affine variety over k. Then every first-order deformation of U over the dual numbers $k[\varepsilon]$ is trivial: any flat $k[\varepsilon]$-algebra $\tilde{A}$ with $\tilde{A} \otimes_{k[\varepsilon]} k \cong A$ is isomorphic, as a deformation, to $A \otimes_k k[\varepsilon] = A[\varepsilon]$.

***Proof*:**
Let $\tilde{A}$ be a flat $k[\varepsilon]$-algebra reducing to A modulo ε. Flatness over $k[\varepsilon]$ for a module is equivalent to the sequence

$$0 \to A \xrightarrow{\varepsilon} \tilde{A} \to A \to 0$$

being exact, where the outer terms are $\tilde{A}/\varepsilon\tilde{A} \cong A$ and the kernel of reduction is $\varepsilon\tilde{A} \cong A$; concretely, multiplication by ε identifies A with the ideal $\varepsilon\tilde{A} \subseteq \tilde{A}$, and $\tilde{A}$ is an extension of A by this square-zero ideal.

Step 1: Lift generators. Since A is smooth over k, it is in particular formally smooth. Present

$A = P/I$ with $P = k[x_1, \dots, x_m]$ a polynomial ring. Choose lifts $\tilde{x}_i \in \tilde{A}$ of the images of x_i in A, giving a $k[\varepsilon]$-algebra map $\tilde{P} = P[\varepsilon] \to \tilde{A}$. Because $\tilde{A}$ is generated over $k[\varepsilon]$ by the classes reducing to a generating set of A (the reduction is surjective and $\varepsilon\tilde{A}$ is generated by the same classes times ε), this map is surjective.

Step 2: Lift the relations using smoothness. A trivialization is a $k[\varepsilon]$-algebra isomorphism $\tilde{A} \cong A[\varepsilon]$ over A, that is, a ring-homomorphism section $s\colon A \to \tilde{A}$ of the reduction $\tilde{A} \to A$ splitting $0 \to A \to \tilde{A} \to A \to 0$. Producing s is a lifting problem against the square-zero surjection $\tilde{A} \to A$, whose kernel $\varepsilon\tilde{A} \cong A$ is a square-zero ideal. Formal smoothness of A over k is precisely the statement that such lifts exist: given the identity map $A \to A$ and the square-zero surjection $\tilde{A} \to A$, there is a k-algebra homomorphism $s\colon A \to \tilde{A}$ whose reduction modulo ε is the identity.

Step 3: Conclude triviality. Such an s is a ring-theoretic section of $\tilde{A} \to A$, so $\tilde{A} = s(A) \oplus \varepsilon\tilde{A}$ as a k-vector space, and the resulting map $A[\varepsilon] \to \tilde{A}$, $a + \varepsilon b \mapsto s(a) + \varepsilon s(b)$, is a $k[\varepsilon]$-algebra isomorphism restricting to the identity modulo ε. Hence $\tilde{A} \cong A[\varepsilon]$ as a deformation, completing the proof.

The content of proposition 16.3.1 is that all the significant information in a deformation of a global smooth variety is concentrated in how the affine charts are reglued, never in the charts themselves. Over each chart the deformation is trivial; the choices are made on the overlaps, and different trivializations on adjacent charts differ by an infinitesimal automorphism. Since infinitesimal automorphisms of an affine chart are its vector fields, the gluing data is a family of vector fields on overlaps, which is exactly the input to Čech cohomology of the tangent sheaf. This is the entire idea of the theorem; the proof below makes it precise.

Theorem 16.3.2: First-Order Deformations of a Smooth Variety

Let X be a smooth projective variety over an algebraically closed field k of characteristic zero. Then the set of isomorphism classes of first-order deformations of X over $k[\varepsilon]$ is in natural bijection with the cohomology group

$$H^1(X, T_X)$$

the trivial deformation $X \times_k \operatorname{Spec} k[\varepsilon]$ corresponding to the zero class. In particular the deformation functor of X has tangent space $H^1(X, T_X)$ in the sense of definition 16.1.5.

***Proof*:**

Fix a first-order deformation of X: a flat scheme $\mathcal{X}$ over $\operatorname{Spec} k[\varepsilon]$ together with an isomorphism of its special fiber $\mathcal{X} \times_{\operatorname{Spec} k[\varepsilon]} \operatorname{Spec} k$ with X. The strategy is to trivialize $\mathcal{X}$ over an affine cover of X, read off the discrepancy between trivializations on overlaps as a Čech 1-cochain with values in T_X, and check that the cocycle condition holds and that the class is independent of all choices.

Step 1: Trivialize on an affine cover. Choose a finite affine open cover $\mathfrak{U} = \{U_i\}_{i \in I}$ of X, with $U_i = \operatorname{Spec} A_i$. The deformation $\mathcal{X}$ restricts over each U_i to a flat deformation $\mathcal{X}_i = \mathcal{X}|_{U_i}$ of the smooth affine variety U_i. By proposition 16.3.1 each $\mathcal{X}_i$ is trivial: fix an isomorphism of deformations

$$\varphi_i\colon \mathcal{X}_i \xrightarrow{\sim} U_i \times_k \operatorname{Spec} k[\varepsilon]$$

restricting to the identity on the special fiber U_i. Each φ_i is a choice; the cocycle will measure how these choices fail to agree.

Step 2: Extract the transition automorphisms. On a double overlap $U_{ij} = U_i \cap U_j$, the two trivializations φ_i and φ_j both identify $\mathcal{X}|_{U_{ij}}$ with the trivial deformation $U_{ij} \times_k \operatorname{Spec} k[\varepsilon]$. Their comparison

$$\theta_{ij} = \varphi_i \circ \varphi_j^{-1} \colon U_{ij} \times_k \operatorname{Spec} k[\varepsilon] \xrightarrow{\sim} U_{ij} \times_k \operatorname{Spec} k[\varepsilon]$$

is an automorphism of the trivial deformation over U_{ij} restricting to the identity modulo ε. An automorphism of $U_{ij} \times_k \operatorname{Spec} k[\varepsilon]$ reducing to the identity is, on the coordinate ring $A_{ij}[\varepsilon]$, a $k[\varepsilon]$-algebra map

$$a + \varepsilon b \longmapsto a + \varepsilon\big(b + D_{ij}(a)\big)$$

for a uniquely determined k-linear map $D_{ij} \colon A_{ij} \to A_{ij}$. The requirement that this map be a ring homomorphism forces $D_{ij}(aa') = a\, D_{ij}(a') + a'\, D_{ij}(a)$, so D_{ij} is a k-derivation of A_{ij}: an element of $\operatorname{Der}_k(A_{ij}, A_{ij}) = T_X(U_{ij})$. Thus each overlap carries a vector field $D_{ij} \in T_X(U_{ij})$.

Step 3: Verify the cocycle condition. On a triple overlap $U_{ijk} = U_i \cap U_j \cap U_k$ the three comparisons compose to the identity, $\theta_{ij} \circ \theta_{jk} \circ \theta_{ki} = \mathrm{id}$, since $\theta_{ij}\theta_{jk} = (\varphi_i\varphi_j^{-1})(\varphi_j\varphi_k^{-1}) = \varphi_i\varphi_k^{-1} = \theta_{ik}$. Because each θ reduces to the identity modulo ε, composing two of them adds their derivations: writing out $\theta_{ij}\theta_{jk}$ on $a + \varepsilon b$ gives $a + \varepsilon(b + D_{jk}(a) + D_{ij}(a))$, the cross term $\varepsilon^2 D_{ij}(D_{jk}(a))$ vanishing. Hence the map $\theta \mapsto D$ turns composition into addition, and $\theta_{ij}\theta_{jk} = \theta_{ik}$ becomes

$$D_{ij} + D_{jk} = D_{ik}$$

on U_{ijk}. This is precisely the Čech 1-cocycle condition $D_{ij} - D_{ik} + D_{jk} = 0$ for the cover $\mathfrak{U}$ with values in T_X. The collection (D_{ij}) therefore defines a class

$$\delta(\mathcal{X}) \in \check{H}^1(\mathfrak{U}, T_X)$$

and, passing to the limit over refinements (or using that T_X is coherent on the projective scheme X, so Čech and derived cohomology agree), a class in $H^1(X, T_X)$.

Step 4: Independence of the trivializations. A different choice of trivializations $\varphi_i' = \psi_i \circ \varphi_i$ replaces each φ_i by composing with an automorphism ψ_i of the trivial deformation over U_i reducing to the identity, that is, with a vector field $E_i \in T_X(U_i)$. The new transition maps are $\theta_{ij}' = \psi_i\theta_{ij}\psi_j^{-1}$, and translating to derivations (again composition becomes addition) gives

$$D_{ij}' = D_{ij} + E_i - E_j$$

on U_{ij}. The difference $D' - D$ is the Čech coboundary of the 0-cochain (E_i), so $\delta(\mathcal{X})$ is independent of the chosen trivializations, hence a well-defined element of $H^1(X, T_X)$ attached to the deformation $\mathcal{X}$.

Step 5: The class determines the deformation. If two deformations $\mathcal{X}$ and $\mathcal{X}'$ have the same class in $H^1(X, T_X)$, then after passing to a common refinement of the two covers and choosing trivializations, their cocycles (D_{ij}) and (D_{ij}') differ by a coboundary $(E_i - E_j)$. Adjusting the trivializations of $\mathcal{X}'$ by the vector fields E_i as in Step 4 makes the two cocycles equal on every overlap. Equal transition data means the glued schemes are isomorphic as deformations: the isomorphisms $\varphi_i^{-1} \circ \varphi_i'$ over the U_i agree on overlaps (their discrepancy is exactly $D_{ij}' - D_{ij} = 0$) and so patch to a global isomorphism $\mathcal{X} \cong \mathcal{X}'$ of deformations. Thus δ is injective on isomorphism classes.

Step 6: Every class is realized. Conversely, given a Čech 1-cocycle (D_{ij}) with values in T_X for the cover $\mathfrak{U}$, form for each overlap the automorphism θ_{ij} of $U_{ij} \times_k \operatorname{Spec} k[\varepsilon]$ determined by D_{ij} as in Step 2. The cocycle condition $D_{ij} + D_{jk} = D_{ik}$ is exactly the compatibility $\theta_{ij}\theta_{jk} = \theta_{ik}$ needed to glue the trivial pieces $U_i \times_k \operatorname{Spec} k[\varepsilon]$ along the θ_{ij} into a scheme $\mathcal{X}$ flat over $\operatorname{Spec} k[\varepsilon]$ with special fiber X. Its class under δ is the given cocycle, so δ is surjective. The trivial deformation is glued by the identity transition maps, all $D_{ij} = 0$, and so corresponds to the zero class. Combining Steps 5 and 6, δ is a bijection between isomorphism classes of first-order deformations and $H^1(X, T_X)$, as we sought to show.

The bijection of theorem 16.3.2 is the prototype for every tangent-space computation in the theory of moduli of varieties, and it is worth saying precisely what makes it work. Two features of the smooth global case conspire. First, smoothness makes each affine chart rigid, so the local pieces contribute nothing and the deformation is entirely a gluing phenomenon. Second, the automorphisms doing the gluing reduce to the identity modulo ε, so their linearizations are vector fields and composition of automorphisms becomes addition of vector fields; this linearization is what turns the gluing bookkeeping into a linear cocycle condition, and hence into cohomology. The first cohomology H^1 appears, rather than H^0 or H^2, because the data lives on double overlaps (a 1-cochain) and the coboundaries that trivialize a deformation come from global reparametrizations on single charts (a 0-cochain). The two neighboring groups are not idle: $H^0(X, T_X)$ will reappear in section 16.5 as the infinitesimal automorphisms of X, and $H^2(X, T_X)$ houses the obstructions to lifting a first-order deformation to higher order, the subject of the next chapter.

For a curve the theorem already delivers the number the reader has been promised. Before verifying that, consider the simplest possible outcome: a variety with no first-order deformations at all.

Example 16.3.3: Infinitesimal Rigidity of Projective Space

Let $X = \mathbb{P}^n_k$ with $n \geq 1$. Then

$$H^1(\mathbb{P}^n, T_{\mathbb{P}^n}) = 0$$

so by theorem 16.3.2 projective space has no nontrivial first-order deformations: it is infinitesimally rigid.

The computation runs through the Euler sequence (theorem 5.5.18), whose dual is the short exact sequence of locally free sheaves

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$

Taking the long exact sequence in cohomology, the relevant segment is

$$H^1\big(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1)\big)^{\oplus(n+1)} \longrightarrow H^1(\mathbb{P}^n, T_{\mathbb{P}^n}) \longrightarrow H^2(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n})$$

The cohomology of line bundles on projective space is known: every twist $\mathcal{O}_{\mathbb{P}^n}(d)$ has cohomology concentrated in degrees 0 and n, so $H^i(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(d)) = 0$ whenever $0 < i < n$. For $n \geq 2$ this gives $H^1(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1)) = 0$, since the index 1 lies strictly between 0 and n. It gives $H^2(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}) = 0$ as well: for $n \geq 3$ the index 2 lies strictly between 0 and n, and for $n = 2$ the group $H^2(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2})$ is Serre dual to $H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-3)) = 0$. The middle term is squeezed between two zeros, so $H^1(\mathbb{P}^n, T_{\mathbb{P}^n}) = 0$ for $n \geq 2$. For $n = 1$ the tangent sheaf is $T_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(2)$, and $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2)) = 0$ because $\deg = 2 > 2g - 2 = -2$, so the vanishing holds there as well.

The vanishing matches geometric expectation: $\mathbb{P}^n$ is a homogeneous space, so rigid, and the moduli “space” of projective spaces of a fixed dimension is a single reduced point. Contrast this with the deformations of $\mathbb{P}^n$ as a *subvariety* of a larger projective space, which are plentiful; the abstract variety $\mathbb{P}^n$ moves not at all, even though its embeddings move freely.

The rigidity of $\mathbb{P}^n$ is the degenerate case. The first nontrivial count, and the historical origin of the phrase "moduli of curves," comes from applying the same theorem to a curve, where the dimension of $H^1(T_C)$ turns out to be a number Riemann already knew.

Example 16.3.4: Deformations of a Smooth Curve and the Number $3g-3$

Let C be a smooth projective curve of genus g over k. The tangent sheaf T_C is a line bundle, being the dual of the canonical bundle:

$$T_C = \Omega_{C/k}^\vee = \omega_C^{-1}$$

because on a curve $\Omega_{C/k} = \omega_C$ has rank one. By theorem 16.3.2 the first-order deformations of C are classified by $H^1(C, T_C) = H^1(C, \omega_C^{-1})$, and this dimension is computed by Riemann-Roch and Serre duality.

Serre duality. By Serre duality on the curve C (corollary 6.5.10), for any line bundle $\mathcal{L}$ one has $H^1(C, \mathcal{L}) \cong H^0(C, \omega_C \otimes \mathcal{L}^{-1})^*$. Applying this to $\mathcal{L} = \omega_C^{-1}$ gives

$$H^1(C, \omega_C^{-1}) \cong H^0(C, \omega_C^{\otimes 2})^*$$

so $\dim H^1(C, T_C) = h^0(C, \omega_C^{\otimes 2})$, the dimension of the space of *quadratic differentials* on C.

Riemann-Roch. The degree of the canonical bundle is $\deg \omega_C = 2g-2$ (theorem 7.1.19), so $\deg \omega_C^{\otimes 2} = 4g-4$. Riemann-Roch for the line bundle $\omega_C^{\otimes 2}$ reads

$$h^0(C, \omega_C^{\otimes 2}) - h^1(C, \omega_C^{\otimes 2}) = \deg \omega_C^{\otimes 2} + 1 - g = (4g-4) + 1 - g = 3g-3$$

It remains to show $h^1(C, \omega_C^{\otimes 2}) = 0$ for $g \geq 2$. By Serre duality again, $h^1(C, \omega_C^{\otimes 2}) = h^0(C, \omega_C^{-1})$, and ω_C^{-1} has degree $2-2g < 0$ once $g \geq 2$, so it has no nonzero global sections: a line bundle of negative degree on an integral curve has $h^0 = 0$. Hence for $g \geq 2$

$$\dim H^1(C, T_C) = h^0(C, \omega_C^{\otimes 2}) = 3g-3$$

Low genus. The two small genera behave differently and both are instructive.

- For $g = 0$, the curve is $C = \mathbb{P}^1$ and $T_{\mathbb{P}^1} = \mathcal{O}(2)$, so $H^1(\mathbb{P}^1, \mathcal{O}(2)) = 0$ by example 16.3.3. The rational curve is rigid: $\dim H^1(T_C) = 0$, and the value $3g-3 = -3$ is not the answer (the formula fails because $h^1(\omega^{\otimes 2}) \neq 0$ when the degree $4g-4 = -4$ of $\omega^{\otimes 2}$ is negative).
- For $g = 1$, the curve is an elliptic curve, $\omega_C \cong \mathcal{O}_C$ is trivial, so $T_C \cong \mathcal{O}_C$ and $\dim H^1(C, T_C) = h^1(C, \mathcal{O}_C) = g = 1$. There is a one-dimensional space of first-order deformations, matching the one-dimensional coarse moduli space of elliptic curves recorded by the j-line (theorem 13.3.6). Here $3g-3 = 0$ undercounts because the vanishing $h^1(\omega^{\otimes 2}) = 0$ used above requires $g \geq 2$; when $g = 1$ the line bundle $\omega_C^{\otimes 2} = \mathcal{O}_C$ has $h^1 = 1$, correcting the count from 0 up to 1.

Collecting the three cases,

$$\dim H^1(C, T_C) = \begin{cases} 0 & g = 0 \\ 1 & g = 1 \\ 3g-3 & g \geq 2 \end{cases}$$

For $g \geq 2$ this is exactly the dimension of the moduli space $\mathcal{M}_g$ of smooth curves of genus g, the quantity the capstone will construct as an algebraic stack; the deformation theory computes the answer long before the space itself is built.

The number $3g-3$ is the first place in this book where a moduli dimension falls out of a cohomology computation, and the interpretation of example 16.3.4 deserves to be underlined. The tangent space to the moduli of genus-g curves at a point $[C]$ is $H^1(C, T_C)$, and Serre duality recasts this as the space of quadratic differentials $H^0(C, \omega_C^{\otimes 2})$, a description that will resurface when the cotangent space to $\mathcal{M}_g$ is identified. The genus-one case is the cleanest illustration of the general principle that will organize Part VI: an elliptic curve has a one-dimensional family of first-order deformations, so its moduli is one-dimensional, and yet no fine moduli space exists because every elliptic curve carries the automorphism $[-1]$. The infinitesimal deformation count sees the correct dimension; the automorphisms, which will be diagnosed as $H^0(C, T_C)$ in section 16.5, are the obstruction to representability that forces the coarse space and, ultimately, the stack.

A word of caution keeps the bookkeeping straight across the three regimes.

16.3.5 Warning: The Formula $3g-3$ Requires $g \geq 2$ The equality $\dim H^1(C, T_C) = 3g-3$ holds only for $g \geq 2$. The Riemann-Roch computation always gives $h^0(\omega_C^{\otimes 2}) - h^1(\omega_C^{\otimes 2}) = 3g-3$, but this is the dimension of $H^1(T_C)$ only when the correction term $h^1(\omega_C^{\otimes 2}) = h^0(\omega_C^{-1})$ vanishes, which fails for $g \leq 1$. For $g = 0$ and $g = 1$ the correction term is nonzero and the deformation dimension is 0 and 1 respectively, not -3 and 0. The pattern is the general phenomenon that Riemann-Roch computes an Euler characteristic, and reading off a single cohomology dimension from it always requires a vanishing input for the complementary group.

Exercise 16.3.1

1. Let X be a smooth projective variety with $H^1(X, T_X) = 0$. Explain in one or two sentences, citing theorem 16.3.2, what this says about the deformations of X, and give one example of such an X of dimension 2.

2. Verify the vanishing $H^1(\mathbb{P}^n, T_{\mathbb{P}^n}) = 0$ for $n \geq 2$ in detail by writing out the full long exact sequence attached to the dual Euler sequence and citing the values of $H^i(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1))$ and $H^i(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n})$ used at each step.

3. Compute $\dim H^1(C, T_C)$ for a smooth projective curve of genus $g = 3$, and identify the space of quadratic differentials $H^0(C, \omega_C^{\otimes 2})$ whose dimension it equals.

4. In Step 2 of the proof of theorem 16.3.2, an automorphism of $U_{ij} \times_k \operatorname{Spec} k[\varepsilon]$ reducing to the identity was written on coordinate rings as $a + \varepsilon b \mapsto a + \varepsilon(b + D_{ij}(a))$. Show directly that the requirement that this map be a $k[\varepsilon]$-algebra homomorphism forces D_{ij} to be a k-derivation, and hence a section of T_X over U_{ij}.

5. Let A be a smooth projective surface with $H^1(A, \mathcal{O}_A) = g$ and trivial canonical bundle $\omega_A \cong \mathcal{O}_A$ (an abelian surface or a $K3$ surface). Compute $\dim H^1(A, T_A)$ in terms of the Hodge numbers of A, using $T_A \cong \Omega_A$ (which holds because ω_A is trivial and A is a surface) and Serre duality.

6. The proof of theorem 16.3.2 used the rigidity of smooth affine varieties (proposition 16.3.1). Explain what breaks in the argument if X is allowed to be singular, and identify which cohomology group would need to be replaced by a more sophisticated invariant to account for local (as opposed to gluing) deformations.

16.4 Deformations of Sheaves and Line Bundles

The previous sections computed the tangent space of a moduli problem whose objects were *subschemes* of a fixed ambient variety ($H^0(N_{Z/X})$, theorem 16.2.1) and whose objects were *varieties* themselves ($H^1(X, T_X)$, theorem 16.3.2). This section treats the third fundamental moduli problem of algebraic geometry: the moduli of *sheaves*. A coherent sheaf $\mathcal{F}$ on a fixed variety X is a piece of linear-algebraic data spread over X, and one wants a parameter space whose points are such sheaves. Chief among these problems is the moduli of line bundles, whose parameter space is the Picard scheme $\operatorname{Pic}(X)$ already met in section 16.3 through its tangent sheaf. The pattern established in section 16.2 and section 16.3 continues, but with a new cohomological home: the tangent space to a moduli of sheaves is an Ext group, and the degree-one self-extensions $\operatorname{Ext}^1_X(\mathcal{F}, \mathcal{F})$ are exactly the first-order deformations of $\mathcal{F}$.

The mechanism is the same square-zero extension that governed the earlier cases. A first-order deformation of $\mathcal{F}$ is a sheaf on the thickened space $X[\varepsilon] = X \times_k \operatorname{Spec} k[\varepsilon]$, flat over the dual numbers, restricting to $\mathcal{F}$ over the closed point. The whole content of this section is that flatness over $k[\varepsilon]$ turns such a thickened sheaf into a self-extension of $\mathcal{F}$ by $\mathcal{F}$, and conversely. Once that translation is in hand, the moduli of coherent sheaves inherits its tangent space from homological algebra, and the special case of line bundles collapses $\operatorname{Ext}^1(\mathcal{L}, \mathcal{L})$ to the single cohomology group $H^1(X, \mathcal{O}_X)$ that is the tangent space to $\operatorname{Pic}(X)$.

Throughout, k is algebraically closed of characteristic zero and X is a smooth projective k-variety unless stated otherwise; $k[\varepsilon] = k[\varepsilon]/(\varepsilon^2)$ denotes the dual numbers and $X[\varepsilon] = X \times_k \operatorname{Spec} k[\varepsilon]$ the trivial thickening of X over them. A first-order deformation is understood in the sense of definition 16.1.4.

Before stating the classification, a word on the homological tool that carries it. The group Ext^1 measures extensions, and this is precisely the structure a first-order deformation manufactures. The following refresher fixes notation and the one property used below; its proof belongs to homological algebra.

16.4.1 Note: The Ext^1 Refresher For coherent sheaves $\mathcal{F}, \mathcal{G}$ on a scheme X, the group $\operatorname{Ext}^i_X(\mathcal{F}, \mathcal{G})$ is the i-th right derived functor of $\operatorname{Hom}_X(\mathcal{F}, -)$ evaluated at $\mathcal{G}$, equivalently the i-th hyper-derived functor computed from an injective (or, for $\mathcal{F}$, locally free) resolution. The degree-one group $\operatorname{Ext}^1_X(\mathcal{F}, \mathcal{G})$ classifies *extensions* of $\mathcal{F}$ by $\mathcal{G}$: isomorphism classes of short exact sequences

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0$$

of coherent sheaves, where two extensions are identified when a commuting isomorphism of the middle terms fixing $\mathcal{G}$ and $\mathcal{F}$ exists. The zero class is the split extension $\mathcal{E} = \mathcal{G} \oplus \mathcal{F}$. Addition is the Baer sum, and this bijection is an isomorphism of k-vector spaces. For proofs of the derived-functor description, the extension interpretation, and the Baer sum, see [15].

The extension interpretation is the whole point: a first-order deformation produces a sheaf $\mathcal{E}$ on $X[\varepsilon]$ that, once its $k[\varepsilon]$-structure is unwound, is exactly an extension of $\mathcal{F}$ by $\mathcal{F}$, with the class 0 recording the trivial (product) deformation. The theorem below makes this precise and proves the correspondence is a bijection.

Theorem 16.4.2: First-Order Deformations of a Coherent Sheaf

Let X be a scheme over k and $\mathcal{F}$ a coherent sheaf on X. The set of isomorphism classes of first-order deformations of $\mathcal{F}$, that is, coherent sheaves $\mathcal{F}_\varepsilon$ on $X[\varepsilon]$ that are flat over $k[\varepsilon]$ together with an isomorphism $\mathcal{F}_\varepsilon \otimes_{k[\varepsilon]} k \cong \mathcal{F}$, is in natural bijection with

$$\operatorname{Ext}^1_X(\mathcal{F}, \mathcal{F})$$

The bijection is k-linear, and the zero class corresponds to the trivial deformation $\mathcal{F}_\varepsilon = \mathcal{F} \otimes_k k[\varepsilon]$.

***Proof*:**
Write $R = k[\varepsilon]$ and let $\pi\colon X[\varepsilon] \to X$ be the projection, so that $\mathcal{O}_{X[\varepsilon]} = \mathcal{O}_X \otimes_k R$ as a sheaf of R-algebras on the topological space of X (the closed immersion $X \hookrightarrow X[\varepsilon]$ is a homeomorphism on underlying spaces). Sheaves on $X[\varepsilon]$ are thus sheaves of $\mathcal{O}_X \otimes_k R$-modules on X, and the multiplication by ε is an $\mathcal{O}_X$-linear endomorphism whose square is zero.

Step 1: Flatness over the dual numbers is a square-zero condition. A module M over $R = k[\varepsilon]$ is flat if and only if it is free, since R is a local Artinian ring and finitely generated flat modules over a local ring are free; the same criterion sheafifies. The maximal ideal $(\varepsilon) \subseteq R$ satisfies $(\varepsilon) \cong k$ as an R-module, and the local criterion for flatness reduces to a single condition: an R-module M is flat over R if and only if the multiplication map

$$\varepsilon\colon M \otimes_R k \longrightarrow \varepsilon M$$

is an isomorphism, equivalently $\varepsilon M = \ker(\varepsilon\colon M \to M)$ and the induced map $M/\varepsilon M \to \varepsilon M$ is bijective (see [10] for the local flatness criterion over an Artinian base). Applied to the sheaf $\mathcal{F}_\varepsilon$ on $X[\varepsilon]$, this says: $\mathcal{F}_\varepsilon$ is flat over R if and only if multiplication by ε fits into a short exact sequence of $\mathcal{O}_X$-modules

$$0 \longrightarrow \mathcal{F}_\varepsilon \otimes_R k \xrightarrow{\varepsilon} \mathcal{F}_\varepsilon \longrightarrow \mathcal{F}_\varepsilon \otimes_R k \longrightarrow 0 \tag{16.9}$$

where the right map is reduction modulo ε. By hypothesis $\mathcal{F}_\varepsilon \otimes_R k \cong \mathcal{F}$, so (16.9) reads

$$0 \longrightarrow \mathcal{F} \xrightarrow{\varepsilon} \mathcal{F}_\varepsilon \xrightarrow{\rho} \mathcal{F} \longrightarrow 0 \tag{16.10}$$

a short exact sequence of $\mathcal{O}_X$-modules in which the sub and the quotient are both $\mathcal{F}$. This is an extension of $\mathcal{F}$ by $\mathcal{F}$ in the sense of box 16.4.1, and it determines a class in $\operatorname{Ext}^1_X(\mathcal{F}, \mathcal{F})$.

Step 2: The extension recovers the $\mathcal{O}_{X[\varepsilon]}$-module structure. Conversely, suppose given an extension (16.10) of $\mathcal{O}_X$-modules. Endow $\mathcal{F}_\varepsilon$ with an action of ε by declaring ε to act as the composite

$$\mathcal{F}_\varepsilon \xrightarrow{\rho} \mathcal{F} \xrightarrow{\text{(the inclusion)}} \mathcal{F}_\varepsilon$$

that is, $\varepsilon \cdot x = \iota(\rho(x))$ where $\iota\colon \mathcal{F} \hookrightarrow \mathcal{F}_\varepsilon$ is the given subobject inclusion. This action satisfies $\varepsilon^2 = 0$ because $\rho \circ \iota = 0$ (the sequence is exact at the middle), so it makes $\mathcal{F}_\varepsilon$ a sheaf of $\mathcal{O}_X \otimes_k R = \mathcal{O}_{X[\varepsilon]}$-modules. The exactness of (16.10) says exactly that $\varepsilon\colon \mathcal{F}_\varepsilon \to \mathcal{F}_\varepsilon$ has image equal to its kernel, both being $\iota(\mathcal{F})$, and that $\mathcal{F}_\varepsilon/\varepsilon\mathcal{F}_\varepsilon \cong \mathcal{F}$; by the criterion of Step 1 the resulting $\mathcal{O}_{X[\varepsilon]}$-module is flat over R and restricts to $\mathcal{F}$ over the closed point. The two constructions of Steps 1 and 2 are mutually inverse: passing from a flat $\mathcal{F}_\varepsilon$ to the sequence (16.10) and back returns the same ε-action, since that action was read off from the same inclusion and reduction maps.

Step 3: Isomorphism of deformations matches isomorphism of extensions. An isomorphism of first-order deformations is an $\mathcal{O}_{X[\varepsilon]}$-linear isomorphism $\varphi\colon \mathcal{F}_\varepsilon \to \mathcal{F}'_\varepsilon$ commuting with the identifications of the closed fibers with $\mathcal{F}$. Reducing φ modulo ε gives the identity on $\mathcal{F}$, and φ commutes with multiplication by ε; hence φ carries the sub $\iota(\mathcal{F})$ to $\iota'(\mathcal{F})$ by the identity and induces the identity on the quotient $\mathcal{F}$. This is precisely a morphism of extensions in the sense of box 16.4.1 inducing the identity on sub and quotient, that is, an equivalence of extensions. Conversely an equivalence of extensions is $\mathcal{O}_X$-linear and, commuting with the ε-action reconstructed in Step 2, is $\mathcal{O}_{X[\varepsilon]}$-linear. Isomorphism classes of flat deformations therefore correspond bijectively to equivalence classes of extensions, that is, to elements of $\mathrm{Ext}^1_X(\mathcal{F},\mathcal{F})$.

Step 4: Linearity and the trivial deformation. The trivial deformation $\mathcal{F}_\varepsilon = \mathcal{F}\otimes_k R = \mathcal{F}\oplus\varepsilon\mathcal{F}$ is the split extension $\mathcal{F}\oplus\mathcal{F}$, whose class is 0. That the bijection is k-linear follows because the Baer sum of extensions matches the additive structure on deformations induced by the functoriality of $-\otimes_R k$: scaling ε by $\lambda \in k^\times$ acts on (16.10) by scaling the sub-inclusion, reproducing scalar multiplication in $\mathrm{Ext}^1_X(\mathcal{F},\mathcal{F})$, and the Baer sum of two extensions computes the deformation glued from the two ε-actions. The correspondence is natural in X and in $\mathcal{F}$ because every step used only the flatness criterion and the abelian structure of $\mathcal{O}_X$-modules, both of which are functorial, completing the proof.

The theorem is the sheaf-theoretic member of the family $H^0(N_{Z/X})$, $H^1(T_X)$, $\mathrm{Ext}^1(\mathcal{F},\mathcal{F})$ of tangent-space formulas, and it subsumes the other two conceptually: a subscheme is recorded by its structure sheaf and a smooth variety's deformations are governed by $H^1(T_X) = \mathrm{Ext}^1_{\mathcal{O}_X}(\Omega_X,\mathcal{O}_X)$, so all three are instances of "Ext^1 of the relevant object with itself, or with a partner." The essential mechanism is (16.10): flatness over the dual numbers forces the middle sheaf $\mathcal{F}_\varepsilon$ to be an extension whose kernel and cokernel are both the sheaf being deformed, and the nilpotent ε is precisely the connecting datum that an extension class encodes. A deformation is trivial exactly when this extension splits, that is, when $\mathcal{F}_\varepsilon \cong \mathcal{F}\otimes_k k[\varepsilon]$; the failure to split, measured by the nonzero class in $\mathrm{Ext}^1_X(\mathcal{F},\mathcal{F})$, is the first-order geometry of the moduli of sheaves.

16.4.3 Remark: Automorphisms and Ext^0 The degree-zero group $\mathrm{Ext}^0_X(\mathcal{F},\mathcal{F}) = \mathrm{Hom}_X(\mathcal{F},\mathcal{F})$ is the sheaf-theoretic analogue of the infinitesimal automorphisms $H^0(X,T_X)$ met in section 16.3: it is the space of infinitesimal automorphisms of the trivial deformation. A stable sheaf has $\mathrm{Hom}_X(\mathcal{F},\mathcal{F}) = k$ (only scalars), so its moduli space is fine near $[\mathcal{F}]$; a strictly semistable or decomposable sheaf has larger $\mathrm{Hom}_X(\mathcal{F},\mathcal{F})$, and the extra endomorphisms are the infinitesimal counterpart of the automorphism obstruction to fine representability recorded in box 13.1.6. The full $T^0/T^1/T^2$ bookkeeping $(\mathrm{Ext}^0,\mathrm{Ext}^1,\mathrm{Ext}^2)$ parallels the tangent-obstruction picture, with Ext^0 the infinitesimal automorphisms treated in the next section (section 16.5) and the obstruction to lifting a first-order deformation living in $\mathrm{Ext}^2_X(\mathcal{F},\mathcal{F})$; that obstruction theory is developed in the next chapter.

The most important sheaves to deform are the line bundles, and here the Ext group collapses to a single cohomology group, recovering the tangent space to the Picard scheme. This is the payoff that ties the section back to $\mathrm{Pic}\,(X)$, whose points parametrize line bundles up to isomorphism and whose identity component $\mathrm{Pic}^0(X)$ is an abelian variety when X is smooth projective.

Proposition 16.4.4: Deformations of a Line Bundle and the Tangent Space to Pic

Let X be a smooth projective variety over k and $\mathcal{L}$ a line bundle on X. Then

$$\mathrm{Ext}^1_X(\mathcal{L},\mathcal{L}) \;\cong\; H^1(X,\mathcal{O}_X)$$

so the first-order deformations of $\mathcal{L}$ form the vector space $H^1(X,\mathcal{O}_X)$. This is the tangent space at $[\mathcal{L}]$ to the Picard scheme $\mathrm{Pic}\,(X)$; since $\mathrm{Pic}\,(X)$ is smooth in characteristic zero with identity component the abelian variety $\mathrm{Pic}^0(X)$, one has

$$T_{[\mathcal{L}]}\mathrm{Pic}\,(X) \;=\; T_0\mathrm{Pic}^0(X) \;\cong\; H^1(X,\mathcal{O}_X)$$

***Proof*:**
Step 1: Reduce $\mathrm{Ext}^1(\mathcal{L},\mathcal{L})$ *to* $H^1(\mathcal{O}_X)$. Because $\mathcal{L}$ is locally free of rank one, the sheaf-Hom $\mathcal{H}om_X(\mathcal{L},\mathcal{L})$ is the trivial line bundle $\mathcal{O}_X$: locally $\mathcal{L}\cong\mathcal{O}_X$ and every $\mathcal{O}_X$-linear endomorphism of $\mathcal{O}_X$ is multiplication by a section, so $\mathcal{H}om_X(\mathcal{L},\mathcal{L})\cong\mathcal{O}_X$ canonically, the isomorphism sending a local endomorphism to its value on 1. For a locally free sheaf $\mathcal{L}$ the local Ext sheaves vanish, $\mathcal{E}xt^i_X(\mathcal{L},\mathcal{L})=0$ for $i>0$, so the local-to-global spectral sequence

$$E_2^{p,q}=H^p\big(X,\mathcal{E}xt^q_X(\mathcal{L},\mathcal{L})\big) \implies \mathrm{Ext}^{p+q}_X(\mathcal{L},\mathcal{L})$$

degenerates to its bottom row, giving $\mathrm{Ext}^i_X(\mathcal{L},\mathcal{L})\cong H^i(X,\mathcal{H}om_X(\mathcal{L},\mathcal{L}))=H^i(X,\mathcal{O}_X)$ for all i (the local-to-global Ext spectral sequence is [15]). In degree one this is the claimed isomorphism

$$\mathrm{Ext}^1_X(\mathcal{L},\mathcal{L})\cong H^1(X,\mathcal{O}_X)$$

Equivalently, tensoring an extension of $\mathcal{L}$ by $\mathcal{L}$ with $\mathcal{L}^{-1}$ gives an extension of $\mathcal{O}_X$ by $\mathcal{O}_X$, so $\mathrm{Ext}^1_X(\mathcal{L},\mathcal{L})\cong\mathrm{Ext}^1_X(\mathcal{O}_X,\mathcal{O}_X)=H^1(X,\mathcal{O}_X)$; every line bundle deforms like the structure sheaf.

Step 2: Identify with the Picard tangent space. By theorem 16.4.2 the first-order deformations of $\mathcal{L}$ as a coherent sheaf are classified by $\mathrm{Ext}^1_X(\mathcal{L},\mathcal{L})$, and Step 1 identifies this with $H^1(X,\mathcal{O}_X)$. Every first-order deformation of a line bundle is again a line bundle, since flatness over $k[\varepsilon]$ preserves the property of being invertible (the restriction to the closed fiber is invertible, and invertibility is an open condition preserved under the square-zero thickening). Thus the deformation functor of $\mathcal{L}$ as a sheaf agrees with its deformation functor as a line bundle, which by definition of $\mathrm{Pic}\,(X)$ as the scheme representing the Picard functor is the tangent space

$$T_{[\mathcal{L}]}\mathrm{Pic}\,(X)=\mathrm{Pic}\,(X)\big(k[\varepsilon]\big)_{[\mathcal{L}]}\cong H^1(X,\mathcal{O}_X)$$

in the sense of the functorial tangent space definition 13.5.1. Finally, translation by $\mathcal{L}$ is an isomorphism of $\mathrm{Pic}\,(X)$ carrying the component of $[\mathcal{L}]$ to $\mathrm{Pic}^0(X)$ and $[\mathcal{L}]$ to the identity $[\mathcal{O}_X]$, so the tangent spaces agree and $T_{[\mathcal{L}]}\mathrm{Pic}\,(X)\cong T_0\mathrm{Pic}^0(X)\cong H^1(X,\mathcal{O}_X)$, as we sought to show.

The proposition is the source of the classical dimension count for the Picard variety. The tangent space to $\mathrm{Pic}\,(X)$ at any point is $H^1(X,\mathcal{O}_X)$, a cohomology group intrinsic to X and independent of the chosen line bundle, so $\mathrm{Pic}^0(X)$ is a smooth group of dimension $h^1(\mathcal{O}_X)$. Over $\mathbb{C}$ this is the abelian variety $H^1(X,\mathcal{O}_X)/H^1(X,\mathbb{Z})$, and the number $h^1(\mathcal{O}_X)$ is the *irregularity* $q(X)$; the proposition says the tangent space to the Picard scheme is the irregularity-dimensional space $H^1(X,\mathcal{O}_X)$. The reason line bundles are so much simpler to deform than general sheaves is Step 1: their local Ext sheaves vanish, so the deformation theory sees only the global cohomology of the trivial line bundle, never any local contribution.

The cleanest instance is a curve, where the Picard variety is the Jacobian and the dimension count reproduces the genus. The example computes the tangent space directly and matches it against the known dimension of $\operatorname{Pic}^0(C)$.

Example 16.4.5: Deformations of a Line Bundle on a Curve

Let C be a smooth projective curve of genus g over k, and let $\mathcal{L}$ be a line bundle on C. By proposition 16.4.4 the first-order deformations of $\mathcal{L}$ form the vector space $H^1(C, \mathcal{O}_C)$, so

$$\dim_k \operatorname{Ext}^1_C(\mathcal{L}, \mathcal{L}) = \dim_k H^1(C, \mathcal{O}_C) = h^1(\mathcal{O}_C)$$

The genus of a smooth projective curve is defined as $g = h^1(C, \mathcal{O}_C) = \dim_k H^1(C, \mathcal{O}_C)$, equivalently $g = h^0(C, \omega_C)$ by Serre duality (**??** for Serre duality and the genus of a curve). Therefore

$$\dim_k \operatorname{Ext}^1_C(\mathcal{L}, \mathcal{L}) = g$$

Every line bundle on C has a g-dimensional space of first-order deformations, and this dimension is the same for every $\mathcal{L}$.

This matches the geometry of the Picard variety. The degree-zero Picard scheme $\operatorname{Pic}^0(C)$ is the *Jacobian* $\operatorname{Jac}(C)$, an abelian variety of dimension g; over $\mathbb{C}$ it is the complex torus

$$\operatorname{Jac}(C) = H^1(C, \mathcal{O}_C)/H^1(C, \mathbb{Z}) = H^0(C, \omega_C)^\vee/H_1(C, \mathbb{Z})$$

a g-dimensional torus. Its tangent space at the origin is $H^1(C, \mathcal{O}_C)$, of dimension g, in agreement with the Ext-computation above:

$$\dim \operatorname{Jac}(C) = \dim T_0\operatorname{Jac}(C) = h^1(\mathcal{O}_C) = g$$

The deformation theory of a single line bundle on C has recovered the dimension of the entire moduli space of line bundles. For genus $g = 0$, $C = \mathbb{P}^1$ has $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}) = 0$, so every line bundle on $\mathbb{P}^1$ is infinitesimally rigid and $\operatorname{Pic}(\mathbb{P}^1) = \mathbb{Z}$ is discrete; for an elliptic curve $g = 1$, the Jacobian is one-dimensional and $\operatorname{Jac}(E) \cong E$ recovers the curve itself (see [11]). For $g \geq 2$ the g-dimensional Jacobian is the first higher-dimensional abelian variety attached to C.

The example closes the loop with section 16.3. There the tangent space to the moduli of curves was $H^1(C, T_C)$ of dimension $3g - 3$; here the tangent space to the moduli of *line bundles on a fixed curve* is $H^1(C, \mathcal{O}_C)$ of dimension g. The two are different moduli problems with different tangent spaces, both computed by the same square-zero mechanism, and both matching the dimensions of the geometric objects ($\mathcal{M}_g$ and $\operatorname{Jac}(C)$) they linearize. The Jacobian is the running touchstone for the abelian-variety chapters that follow this book, where $\operatorname{Jac}(C)$ and its higher analogues $\operatorname{Pic}^0(X)$ become the central objects.

Exercise 16.4.1

1. Let X be a scheme over k and $\mathcal{F}$ a coherent sheaf. Show directly, without invoking theorem 16.4.2, that the trivial deformation $\mathcal{F}_\varepsilon = \mathcal{F} \otimes_k k[\varepsilon]$ is flat over $k[\varepsilon]$ and that as an extension of $\mathcal{F}$ by $\mathcal{F}$ it is the split extension. Conclude that its class in $\operatorname{Ext}^1_X(\mathcal{F}, \mathcal{F})$ is zero.

2. Let X be a smooth projective surface with $h^1(\mathcal{O}_X) = 0$ (for instance $X = \mathbb{P}^2$ or a $K3$ surface). Compute the dimension of the space of first-order deformations of an arbitrary line bundle $\mathcal{L}$ on X, and describe $\operatorname{Pic}^0(X)$ set-theoretically. What does this say about how the line bundles of a fixed numerical class sit inside $\operatorname{Pic}(X)$?

3. Let $\mathcal{F}$ be a coherent sheaf on a smooth projective curve C of genus g, and suppose $\mathcal{F}$ is locally free of rank r (a vector bundle). Using $\operatorname{Ext}^1_C(\mathcal{F},\mathcal{F}) \cong H^1(C, \mathcal{E}nd(\mathcal{F}))$ and Riemann-Roch for the vector bundle $\mathcal{E}nd(\mathcal{F})$ of rank r^2 and degree 0, compute $\dim \operatorname{Ext}^1_C(\mathcal{F},\mathcal{F}) - \dim \operatorname{Hom}_C(\mathcal{F},\mathcal{F})$ and hence show $\dim \operatorname{Ext}^1_C(\mathcal{F},\mathcal{F}) = r^2(g-1) + \dim \operatorname{Hom}_C(\mathcal{F},\mathcal{F})$. For a stable bundle $\mathcal{F}$ (so $\operatorname{Hom}_C(\mathcal{F},\mathcal{F}) = k$), deduce the deformation dimension $r^2(g-1)+1$.

4. Let $\mathcal{L}$ be a line bundle on a smooth projective variety X. Prove the isomorphism $\operatorname{Ext}^1_X(\mathcal{L},\mathcal{L}) \cong \operatorname{Ext}^1_X(\mathcal{O}_X,\mathcal{O}_X)$ used in proposition 16.4.4 directly, by exhibiting the map $\mathcal{E} \mapsto \mathcal{E} \otimes \mathcal{L}^{-1}$ on extensions and checking it is a bijection with inverse $-\otimes\mathcal{L}$. Explain in one sentence why this is the deformation-theoretic statement that "all line bundles deform like $\mathcal{O}_X$."

5. An extension $0 \to \mathcal{F} \to \mathcal{F}_\varepsilon \to \mathcal{F} \to 0$ realizing a first-order deformation gives a sheaf $\mathcal{F}_\varepsilon$ on $X[\varepsilon]$. Suppose the class in $\operatorname{Ext}^1_X(\mathcal{F},\mathcal{F})$ is nonzero. Show that $\mathcal{F}_\varepsilon$ is nonetheless *not* of the form $\mathcal{G} \otimes_k k[\varepsilon]$ for any $\mathcal{O}_X$-module $\mathcal{G}$, and interpret this as the statement that the deformation is nontrivial. (Hint: compare the ε-action to that on a product.)

16.5 Automorphisms and Infinitesimal Automorphisms

The obstruction that kept the moduli functors of the opening chapter from being representable (box 13.1.6) was never the geometry of any single object; it was the presence of automorphisms. An object with a nontrivial automorphism can be glued to itself along that automorphism in a family, producing a nonconstant family all of whose fibers are isomorphic, and no scheme can separate points a family refuses to separate. This chapter has so far measured the first-order geometry of a deformation problem by a single cohomology group: $H^0(N_{Z/X})$ for a subscheme (theorem 16.2.1), $H^1(X, T_X)$ for a smooth variety (theorem 16.3.2), $\operatorname{Ext}^1_X(\mathcal{F},\mathcal{F})$ for a sheaf (theorem 16.4.2). This closing section measures the automorphisms themselves, and finds that they too are governed by the tangent sheaf, one cohomological degree below the deformations. The infinitesimal automorphisms of a smooth variety X are exactly $H^0(X, T_X)$.

Placing $H^0(X, T_X)$ next to $H^1(X, T_X)$ organizes the local structure of a deformation problem into a graded bookkeeping. The infinitesimal automorphisms sit in degree 0, the first-order deformations in degree 1, and the obstructions to extending a deformation, taken up in the next chapter, sit in degree 2. Writing these as T^0, T^1, and T^2 turns the qualitative dichotomy between fine and coarse moduli established in the opening chapter (theorem 13.3.6) into a computable invariant: a nonzero T^0 detects the automorphisms that force a coarse space in place of a fine one. This section develops T^0 and the T^0/T^1 bookkeeping; the T^2 layer is the subject of the next chapter.

Throughout, X is a smooth projective variety over an algebraically closed field k of characteristic 0, and $T_X = \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)$ its tangent sheaf. The tangent sheaf, its identification with k-linear derivations of $\mathcal{O}_X$, and the sheaf cohomology $H^i(X, T_X)$ are imported from **??** and are used without re-derivation; the content of this section is the deformation-theoretic meaning of $H^0(X, T_X)$.

Definition 16.5.1: Infinitesimal Automorphism

Let X be a smooth variety over k, and let $X_\varepsilon = X \times_k \operatorname{Spec} k[\varepsilon]$ be the trivial first-order deformation of X over the dual numbers $k[\varepsilon] = k[\varepsilon]/(\varepsilon^2)$ (definition 16.1.4). An *infinitesimal automorphism* of X is an automorphism

$$\varphi\colon X_\varepsilon \xrightarrow{\sim} X_\varepsilon$$

of X_ε over $\operatorname{Spec} k[\varepsilon]$ whose reduction modulo ε is the identity of X; that is, φ restricts to id_X on the closed fiber $X = X_\varepsilon \times_{\operatorname{Spec} k[\varepsilon]} \operatorname{Spec} k$. The set of infinitesimal automorphisms of X is denoted $T^0 = T^0(X)$.

An infinitesimal automorphism is a symmetry of the trivial deformation that is invisible at the level of the original space: it moves points only to first order in ε. The condition that φ reduce to the identity is what makes T^0 an infinitesimal object rather than the full automorphism group $\operatorname{Aut}(X)$; it isolates the part of a symmetry that a first-order neighborhood of the identity can see. The next result identifies this set with global vector fields, which simultaneously gives T^0 the structure of a k-vector space and identifies its dimension with a computable cohomological invariant.

Proposition 16.5.2: Infinitesimal Automorphisms as Global Vector Fields

Let X be a smooth variety over k. There is a canonical bijection

$$T^0(X) \xrightarrow{\sim} H^0(X, T_X)$$

sending an infinitesimal automorphism φ of X_ε to the global vector field D_φ measuring its first-order displacement. Under this bijection the composition of infinitesimal automorphisms corresponds to addition of vector fields, so $T^0(X)$ is a k-vector space and

$$\dim_k T^0(X) = h^0(X, T_X)$$

***Proof*:**
Write $A = k[\varepsilon]$, and let $p\colon X_\varepsilon \to X$ be the projection and $i\colon X \hookrightarrow X_\varepsilon$ the closed immersion of the fiber over $\varepsilon = 0$. An automorphism φ of X_ε over $\operatorname{Spec} A$ is the same datum as a k-algebra automorphism of the sheaf $\mathcal{O}_{X_\varepsilon} = \mathcal{O}_X \otimes_k A = \mathcal{O}_X \oplus \varepsilon\mathcal{O}_X$ that is A-linear and reduces modulo ε to a given automorphism of $\mathcal{O}_X$. The condition in definition 16.5.1 is that this reduction be the identity.

Step 1: A first-order automorphism is $\mathrm{id} + \varepsilon D$ *for a derivation* D. Let $\varphi^\sharp\colon \mathcal{O}_{X_\varepsilon} \to \mathcal{O}_{X_\varepsilon}$ be the comorphism. Because $\varphi^\sharp$ is A-linear and reduces to the identity modulo ε, on a local section $f \in \mathcal{O}_X$ it has the form

$$\varphi^\sharp(f) = f + \varepsilon\, D(f)$$

for a uniquely determined k-linear map $D\colon \mathcal{O}_X \to \mathcal{O}_X$; the value $\varphi^\sharp(\varepsilon) = \varepsilon$ is forced by A-linearity, so $\varphi^\sharp$ is determined by D. That $\varphi^\sharp$ is a ring homomorphism translates into a condition on D. Applying $\varphi^\sharp$ to a product fg and using $\varepsilon^2 = 0$,

$$\varphi^\sharp(fg) = fg + \varepsilon\, D(fg), \qquad \varphi^\sharp(f)\varphi^\sharp(g) = (f + \varepsilon D(f))(g + \varepsilon D(g)) = fg + \varepsilon\big(fD(g) + gD(f)\big)$$

Equating the two, $D(fg) = fD(g) + gD(f)$, so D is a k-linear derivation of $\mathcal{O}_X$. Conversely, for any derivation D the map $f \mapsto f + \varepsilon D(f)$, extended A-linearly with $\varepsilon \mapsto \varepsilon$, is a ring homomorphism reducing to the identity, and it is invertible with inverse $f \mapsto f - \varepsilon D(f)$ (again using $\varepsilon^2 = 0$). Thus $\varphi \mapsto D$ is a bijection between infinitesimal automorphisms of X_ε and k-linear derivations of $\mathcal{O}_X$.

Step 2: Derivations are global vector fields. By definition of the tangent sheaf, $T_X = \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)$, and the universal derivation $d\colon \mathcal{O}_X \to \Omega_{X/k}$ induces for every open $U \subseteq X$ a natural isomorphism between k-linear derivations $\mathcal{O}_X|_U \to \mathcal{O}_X|_U$ and $\mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)(U) = T_X(U)$, sending a derivation D to the $\mathcal{O}_U$-linear map $\omega \mapsto \langle \omega, D\rangle$ determined by $D = \langle d(-), D\rangle$. A derivation defined on all of X is a section over X, so the global k-linear derivations of $\mathcal{O}_X$ are exactly $H^0(X, T_X) = T_X(X)$. Set $D_\varphi \in H^0(X, T_X)$ equal to the section corresponding to the derivation D produced in Step 1. Steps 1 and 2 compose to the asserted bijection $T^0(X) \to H^0(X, T_X)$, $\varphi \mapsto D_\varphi$.

Step 3: Composition corresponds to addition. Let φ, ψ be infinitesimal automorphisms with associated derivations D, E. The composite $\psi \circ \varphi$ has comorphism $\varphi^\sharp \circ \psi^\sharp$, and on a section f,

$$\varphi^\sharp(\psi^\sharp(f)) = \varphi^\sharp(f + \varepsilon E(f)) = f + \varepsilon E(f) + \varepsilon D(f) = f + \varepsilon\,(D + E)(f)$$

where the cross term $\varepsilon D(E(f)) \cdot \varepsilon$ vanishes because it carries a factor $\varepsilon^2 = 0$. Hence the derivation of $\psi \circ \varphi$ is $D + E$, so the group operation of composition is carried to addition of global vector fields. The k-scaling $D \mapsto \lambda D$ likewise corresponds to the infinitesimal automorphism $f \mapsto f + \varepsilon\lambda D(f)$, so the bijection is one of k-vector spaces and $\dim_k T^0(X) = h^0(X, T_X)$, as we sought to show.

The proof exhibits the mechanism behind the whole T^i formalism in miniature. A first-order symmetry is a derivation because $\varepsilon^2 = 0$ kills every term beyond the linear one, so the nonlinear datum of an automorphism collapses to the linear datum of a vector field. That the group law becomes addition is why T^0 is a vector space and not just a group: the commutator of two infinitesimal automorphisms, which would record the Lie bracket $[D, E]$, lives in order ε^2 and is therefore invisible at first order. The Lie-algebra structure on $H^0(X, T_X)$ is real, but it belongs to the second-order theory; the first-order theory sees only the underlying vector space. This is the same reason the first-order deformations $H^1(X, T_X)$ of theorem 16.3.2 form a vector space rather than carrying the finer structure that second-order obstructions will impose in the next chapter.

With T^0 and T^1 both identified as cohomology of T_X, the tangent-obstruction bookkeeping of a smooth variety can be assembled. The following proposition records the assembly and, more importantly, ties T^0 back to the representability failure that opened the book.

Proposition 16.5.3: The T^0/T^1 Bookkeeping

Let X be a smooth projective variety over k. The first-order deformation theory of X is organized by the cohomology of the tangent sheaf:

1. $T^0(X) = H^0(X, T_X)$ is the space of infinitesimal automorphisms of X (proposition 16.5.2).
2. $T^1(X) = H^1(X, T_X)$ is the space of first-order deformations of X (theorem 16.3.2).

Both are finite-dimensional k-vector spaces. Moreover, $T^0(X) \neq 0$ if and only if X carries a nonzero global vector field, and in that case the automorphism group $\mathrm{Aut}(X)$ is positive-dimensional; the presence of such infinitesimal automorphisms is the local, first-order manifestation of the automorphism obstruction to fine representability recorded in box 13.1.6 and realized concretely by the j-line of theorem 13.3.6.

***Proof*:**
Statements (1) and (2) restate proposition 16.5.2 and theorem 16.3.2 respectively. Finite-dimensionality of $H^0(X, T_X)$ and $H^1(X, T_X)$ is the finiteness of sheaf cohomology of a coherent sheaf on a projective variety, imported from **??**.

For the last statement, $T^0(X) = H^0(X, T_X) \neq 0$ means precisely that X admits a nonzero global vector field D. Because X is a projective variety over k, its automorphism functor is represented by a group scheme $\underline{\mathrm{Aut}}(X)$ locally of finite type over k, whose tangent space at the identity is the Lie algebra $H^0(X, T_X)$; a nonzero global vector field is a nonzero tangent vector at the identity, so $\dim \underline{\mathrm{Aut}}(X) = \dim_k H^0(X, T_X) > 0$ and $\mathrm{Aut}(X)$ is positive-dimensional. The identification of $H^0(X, T_X)$ with $\mathrm{Lie}\, \underline{\mathrm{Aut}}(X)$ is exactly the case of proposition 16.5.2 in which the infinitesimal automorphisms are read as the $k[\varepsilon]$-points of the automorphism scheme lying over the identity, completing the proof.

The final clause is the payoff of the section. box 13.1.6 isolated the reason a moduli functor of objects-up-to-isomorphism can fail to be a sheaf, and hence fail to be representable by a scheme: an object with automorphisms admits families that are locally trivial but globally twisted, and the functor cannot see the twist. theorem 13.3.6 made this concrete for elliptic curves, where the extra automorphisms at $j = 0$ and $j = 1728$ force the moduli problem down to the coarse j-line. proposition 16.5.3 says that the same obstruction has an infinitesimal fingerprint: whenever X has a continuous family of automorphisms, $T^0(X) = H^0(X, T_X)$ is nonzero, and this nonvanishing is detectable by a first-order computation with the dual numbers, before any global family is written down. The deformation theory of the next chapter, extending the count to T^2, will diagnose obstructions the same way, turning the qualitative fine-versus-coarse dichotomy into arithmetic on three cohomology groups.

The two extremes of this dictionary are the most instructive: a space whose automorphisms form a positive-dimensional group, so $T^0 \neq 0$, and a space whose automorphism group is finite, so $T^0 = 0$. Projective space and high-genus curves furnish exactly these two poles.

Example 16.5.4: Infinitesimal Automorphisms of $\mathbb{P}^n$ and of Curves

1. *Projective space.* The Euler sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \longrightarrow T_{\mathbb{P}^n} \longrightarrow 0$$

imported from **??**, computes $H^0(\mathbb{P}^n, T_{\mathbb{P}^n})$. Taking the long exact sequence in cohomology and using $H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}) = k$, $H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1)) = k^{n+1}$, and $H^1(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}) = 0$,

$$0 \longrightarrow k \longrightarrow k^{(n+1)^2} \longrightarrow H^0(\mathbb{P}^n, T_{\mathbb{P}^n}) \longrightarrow 0$$

so $h^0(\mathbb{P}^n, T_{\mathbb{P}^n}) = (n+1)^2 - 1$. This is exactly $\dim \mathrm{PGL}_{n+1}$, and indeed

$$H^0(\mathbb{P}^n, T_{\mathbb{P}^n}) \cong \mathfrak{pgl}_{n+1} = \mathfrak{sl}_{n+1}$$

the Lie algebra of $\mathrm{PGL}_{n+1} = \mathrm{Aut}(\mathbb{P}^n)$: the global vector fields on $\mathbb{P}^n$ are the infinitesimal projective transformations, the traceless $(n+1)\times(n+1)$ matrices acting by the induced linear flow on homogeneous coordinates, with the scalar matrices (the kernel $\mathcal{O}_{\mathbb{P}^n}$ in the Euler sequence) acting trivially. Thus $T^0(\mathbb{P}^n) = \mathfrak{pgl}_{n+1} \neq 0$: projective space has a positive-dimensional automorphism group, and $\mathbb{P}^n$ is the paradigm of an object with many infinitesimal automorphisms. It is also infinitesimally rigid, $T^1(\mathbb{P}^n) = H^1(\mathbb{P}^n, T_{\mathbb{P}^n}) = 0$ (example 16.3.3), so its bookkeeping is $(T^0, T^1) = (\mathfrak{pgl}_{n+1}, 0)$: no deformations, but a large automorphism algebra.

2. *Curves of genus $g \geq 2$.* Let C be a smooth projective curve of genus g. On a curve the tangent sheaf is the dual of the canonical sheaf, $T_C = \Omega_C^\vee = \mathcal{O}_C(-K_C)$, a line bundle of degree $\deg T_C = 2 - 2g$. For $g \geq 2$ this degree is negative, and a line bundle of negative degree on a projective curve has no nonzero global sections, so

$$T^0(C) = H^0(C, T_C) = 0$$

A curve of genus at least 2 has *no* nonzero infinitesimal automorphisms. By proposition 16.5.3, $\mathrm{Aut}(C)$ is therefore 0-dimensional, hence a finite group (it is a closed subgroup scheme of finite type with trivial tangent space at the identity). This is consistent with the Hurwitz bound $|\mathrm{Aut}(C)| \leq 84(g-1)$, which asserts the same finiteness quantitatively. The deformation count sits one degree up: by Serre duality (imported from **??**) and Riemann-Roch,

$$T^1(C) = H^1(C, T_C) \cong H^0(C, \Omega_C^{\otimes 2})^\vee, \qquad \dim_k T^1(C) = 3g - 3$$

for $g \geq 2$ (example 16.3.4), matching the dimension of the moduli space of curves. The bookkeeping is $(T^0, T^1) = (0,\, 3g-3)$: a finite automorphism group and a $(3g-3)$-dimensional family of deformations.

3. *Genus* 0 *and* 1. For contrast, $g = 0$ gives $\mathbb{P}^1$, with $T^0 = H^0(\mathbb{P}^1, T_{\mathbb{P}^1}) \cong \mathfrak{pgl}_2$ of dimension 3 and $T^1 = 0$: the rational curve is rigid with a three-dimensional automorphism group. For $g = 1$, an elliptic curve has $T_C \cong \mathcal{O}_C$ (the canonical bundle is trivial), so $T^0 = H^0(C, \mathcal{O}_C) = k$ is one-dimensional, reflecting the translations, and $T^1 = H^1(C, \mathcal{O}_C) = k$ is one-dimensional, matching the one-dimensional j-line of theorem 13.3.6. The nonvanishing of T^0 at $g = 1$ is the infinitesimal trace of the automorphisms that forced a coarse rather than fine moduli space in theorem 13.3.6.

The two poles read off the same dictionary in opposite directions. For $\mathbb{P}^n$ the bookkeeping is $(T^0, T^1) = (\mathfrak{pgl}_{n+1}, 0)$, all symmetry and no deformation; for a curve of genus $g \geq 2$ it is $(0, 3g-3)$, no symmetry and a large deformation space. The transition happens exactly where $\deg T_X$ changes sign, which on a curve is the transition through genus 1. That T^0 vanishes precisely for $g \geq 2$ is the infinitesimal reason those curves have finite automorphism groups and, in the stack-theoretic framework of the book's third part, why $\mathcal{M}_g$ for $g \geq 2$ is a Deligne-Mumford stack with finite stabilizers rather than an Artin stack with positive-dimensional ones. The vanishing or nonvanishing of a single degree-0 cohomology group thus predicts, before any moduli space is constructed, whether the eventual moduli object will be a scheme, a Deligne-Mumford stack, or something with continuous automorphisms.

Exercise 16.5.1

1. Let X be a smooth projective variety with $H^0(X, T_X) = 0$. Show directly from definition 16.5.1 that the only automorphism of the trivial deformation X_ε over $\operatorname{Spec} k[\varepsilon]$ reducing to id_X is the identity, and explain in one sentence why this says X is "infinitesimally rigid as a symmetric object".

2. Compute T^0, T^1, and $\dim \operatorname{Aut}$ for each of the following, and state whether the automorphism group is finite or positive-dimensional:

    1. $\mathbb{P}^2$;
    2. a smooth plane quartic curve $C \subseteq \mathbb{P}^2$;
    3. an abelian surface A (so $T_A \cong \mathcal{O}_A^{\oplus 2}$).

3. Let C_1 and C_2 be smooth projective curves of genera $g_1, g_2 \geq 2$ and set $X = C_1 \times C_2$. Using $T_X \cong p_1^* T_{C_1} \oplus p_2^* T_{C_2}$ and the Künneth formula, show that $H^0(X, T_X) = 0$, and conclude that $\operatorname{Aut}(X)$ is finite.

4. Show that the Euler characteristic $\chi(T_X) = h^0(X, T_X) - h^1(X, T_X) + \cdots = \dim T^0 - \dim T^1 + \cdots$ is a single number combining the automorphism and deformation counts. For a smooth projective curve C of genus g, compute $\chi(T_C)$ from Riemann-Roch and check that it equals $\dim T^0 - \dim T^1$ in each of the cases $g = 0, 1$ and $g \geq 2$.

5. An elliptic curve E has $T^0(E) = H^0(E, \mathcal{O}_E) = k$. Interpret the nonzero global vector field on E as an infinitesimal automorphism explicitly, and explain how its nonvanishing is compatible with $\operatorname{Aut}(E, O)$ (automorphisms fixing the origin) being finite while $\operatorname{Aut}(E)$ as an abstract variety is positive-dimensional.

6. Suppose X is a smooth projective variety with $H^0(X, T_X) \neq 0$. Argue, using box 13.1.6, that the naive moduli functor parametrizing objects isomorphic to X cannot be represented by a scheme with X as an isolated reduced point, and relate this to the failure diagnosed for the j-line in theorem 13.3.6.

17

Obstructions and Pro-representability

First-order deformations, computed in chapter 16, are the linear approximation to a moduli problem; this chapter asks whether that approximation integrates to an actual family. A tangent vector to a moduli space is a deformation over the dual numbers, but an arc through the point is a deformation over a power-series ring, assembled one order at a time by lifting successively over the small extensions $k[t]/(t^{n+1}) \to k[t]/(t^n)$. At each stage the lift may be blocked, and the block is an element of a second cohomology group; when every such obstruction vanishes the moduli space is smooth at the point, and when they do not, the obstruction theory of this chapter measures the singularity.

section 17.1 locates the obstruction to extending a deformation of a smooth variety in $H^2(X, T_X)$ and defines what it means for a deformation functor to be unobstructed. section 17.2 packages the tangent and obstruction spaces into a single object, the naive cotangent complex, whose cohomology groups T^0, T^1, T^2 are the infinitesimal automorphisms, the first-order deformations, and the obstructions of a k-algebra. section 17.3 proves Schlessinger's criteria, the conditions on a functor of Artin rings under which it admits a hull, a formal versal deformation, or is pro-representable outright. section 17.4 closes Part V with computations: the unobstructedness of curves, abelian varieties, and K3 surfaces, an obstructed example, and the deformation of a node, whose single smoothing parameter is the local fact behind the stable-reduction theorem that the moduli of curves will require.

17.1 Extending Deformations and the Obstruction Space

The previous chapter linearized the deformation problem: a first-order deformation of a smooth projective variety X over the dual numbers $k[\varepsilon]$ is a class in $H^1(X, T_X)$, and this cohomology group is the tangent space to the moduli problem at $[X]$ (theorem 16.3.2). A tangent vector, though, is only the beginning of an arc. Knowing the direction in which X can be bent to first order says nothing about whether that bending can be continued to second order, to third order, and onward to a genuine one-parameter family; a tangent line to a plane curve need not be the start of any branch of the curve when the curve has a singularity at the point. The passage from a tangent vector to an actual deformation is exactly the passage from the linear approximation of the moduli space to the moduli space itself, and it is governed by a second cohomological invariant, one degree up from the tangent space.

This section isolates the mechanism. A deformation over an Artinian base is built by climbing a tower of *small extensions*, extending the deformation one infinitesimal step at a time; at each step the lift either exists or is blocked, and the block is measured by a single class in a *second* cohomology group. When that class vanishes the deformation extends, and the set of extensions is a torsor under the first cohomology group, the same group that measured first-order deformations. When the second cohomology group vanishes identically, every obstruction is forced to be zero and the moduli problem is smooth. This is the dichotomy that separates curves (always unobstructed) from surfaces (which can be obstructed), and it is the computational engine behind every smoothness statement about a moduli space in this book. Throughout, k is algebraically closed of characteristic zero, X is a smooth projective k-variety, and $T_X = \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X)$ is its tangent sheaf (definition 5.5.25).

The infinitesimal tower is built out of the smallest possible steps. Recall from section 16.1 that a *small extension* in $\mathbf{Art}_k$ is a surjection $A' \to A$ of local Artinian k-algebras with residue field k whose kernel (t) is a one-dimensional k-vector space annihilated by the maximal ideal, $\mathfrak{m}_{A'} \cdot t = 0$. Every surjection in $\mathbf{Art}_k$ factors as a finite composite of small extensions, so a deformation over an arbitrary Artinian base is assembled by lifting through a sequence of them. The atom of the whole theory is therefore a single small extension, and the atomic question is whether a deformation already built over A can be pushed one step further, over A'.

Definition 17.1.1: The Small-Extension Lifting Problem

Let $D\colon \mathbf{Art}_k \to \mathbf{Set}$ be a deformation functor and let

$$0 \to (t) \to A' \xrightarrow{\pi} A \to 0$$

be a small extension in $\mathbf{Art}_k$, with kernel $(t) \cong k$ satisfying $\mathfrak{m}_{A'} \cdot t = 0$. Given a deformation $\xi \in D(A)$, the *lifting problem* for ξ along π asks whether ξ lies in the image of the map $D(\pi)\colon D(A') \to D(A)$; a preimage $\xi' \in D(A')$ with $D(\pi)(\xi') = \xi$ is called a *lift* (or *extension*) of ξ over A'. The deformation ξ is *obstructed* along π if no lift exists, and *unobstructed* along π if a lift exists.

The definition is deliberately functorial and says nothing yet about how to decide the lifting problem; that is the work of the theorem below. The content of a deformation theory is a recipe that, to each pair (ξ, π), assigns a cohomology class whose vanishing is equivalent to the existence of a lift, and whose ambient group is a fixed invariant of X independent of ξ and π. For the deformation functor Def_X of a smooth projective variety that invariant is $H^2(X, T_X)$, and the assignment is constructed by re-gluing local lifts over an affine cover. The key structural fact, which makes the whole method work, is that the lifting problem is always solvable *locally*: over an affine open where X is smooth there is no obstruction, because a smooth affine scheme deforms trivially. The obstruction is entirely a global gluing phenomenon, and cohomology is the exact bookkeeping for gluing.

Before stating the theorem it is worth fixing what a deformation over an Artinian base concretely is, so that the phrase "re-glue over a cover" has an unambiguous meaning. A deformation of X over A is a flat scheme $\mathcal{X}_A \to \operatorname{Spec} A$ together with an identification of its special fiber $\mathcal{X}_A \times_{\operatorname{Spec} A} \operatorname{Spec} k$ with X. Flatness over A is the algebraic incarnation of "continuously varying," and over an Artinian base it forces the sheaf of functions on $\mathcal{X}_A$ to be, locally, a free A-module deforming $\mathcal{O}_X$. On an affine open $U = \operatorname{Spec} R$ of X a deformation over A is a flat A-algebra R_A with $R_A \otimes_A k \cong R$; smoothness of R over k is what guarantees such a flat lift exists and is unique up to isomorphism. These two facts, local existence and local uniqueness of lifts, are the input to the gluing argument.

Theorem 17.1.2: The Obstruction to Lifting a Deformation

Let X be a smooth projective variety over k, and let

$$0 \to (t) \to A' \xrightarrow{\pi} A \to 0$$

be a small extension in $\mathbf{Art}_k$ with kernel $(t) \cong k$. Let $\mathcal{X}_A \to \operatorname{Spec} A$ be a deformation of X over A. Then there is a canonically associated *obstruction class*

$$\operatorname{ob}(\mathcal{X}_A, \pi) \in H^2(X, T_X) \otimes_k (t)$$

with the following properties.

1. A lift of $\mathcal{X}_A$ to a deformation over A' exists if and only if $\operatorname{ob}(\mathcal{X}_A, \pi) = 0$.
2. When a lift exists, the set of isomorphism classes of lifts is a torsor (a principal homogeneous space) under the group $H^1(X, T_X) \otimes_k (t)$; in particular it is nonempty and any two lifts differ by a unique element of $H^1(X, T_X) \otimes_k (t)$.

In particular, if $H^2(X, T_X) = 0$ then every deformation of X lifts over every small extension.

***Proof*:**
Fix an affine open cover $\mathcal{U} = \{U_i\}$ of X by opens on which X is smooth, and small enough that each U_i, and each pairwise intersection $U_{ij} = U_i \cap U_j$ and triple intersection U_{ijk}, is affine. Write $\mathcal{X}_A|_{U_i}$ for the restriction of the deformation over A to U_i, which is a flat A-algebra deformation of $\mathcal{O}_X(U_i)$. Because a one-step lift and its ambiguity are governed by the same tensor factor (t) throughout, fix once and for all a generator t of the kernel, so that every class produced below is written as (a T_X-valued class) $\otimes\, t$; tensoring with (t) at the end restores canonicity and removes the choice.

Step 1: Local lifts exist. Each U_i is affine and smooth over k, so its ring $R_i = \mathcal{O}_X(U_i)$ is a smooth k-algebra. A smooth algebra is formally smooth: given the surjection $\pi\colon A' \to A$ and the flat A-algebra deformation $\mathcal{X}_A|_{U_i}$ of R_i, formal smoothness produces a flat A'-algebra $\mathcal{X}_i'$ lifting it, that is, a deformation of U_i over A' restricting to $\mathcal{X}_A|_{U_i}$ over A. Concretely, presenting $R_i = P_i/J_i$ over a polynomial ring P_i so that J_i is, locally on U_i, generated by a regular sequence (smoothness gives a local complete-intersection presentation) and lifting those local generators arbitrarily to A'-coefficients produces such an $\mathcal{X}_i'$; flatness over A' holds because the defining sequence stays regular under the nilpotent thickening. So on each chart a lift $\mathcal{X}_i'$ exists. The obstruction is not in producing the local pieces but in matching them.

Step 2: Local lifts are unique up to a T_X-valued isomorphism. On the overlap U_{ij} the two local lifts $\mathcal{X}_i'$ and $\mathcal{X}_j'$ both restrict to the same deformation $\mathcal{X}_A|_{U_{ij}}$ over A. Two lifts of the same object over A to the same small extension of A differ by an automorphism that is the identity over A; the group of such automorphisms of a smooth affine over the kernel (t) is exactly the module of k-derivations $\mathrm{Der}_k(R_{ij}, R_{ij}) \otimes_k (t)$, that is, the vector fields $T_X(U_{ij}) \otimes_k (t)$. Concretely, an isomorphism $\theta_{ij} \colon \mathcal{X}'_j|_{U_{ij}} \xrightarrow{\sim} \mathcal{X}'_i|_{U_{ij}}$ restricting to the identity over A acts on functions by $f \mapsto f + D_{ij}(\bar{f})\, t$ for a unique derivation $D_{ij} \in T_X(U_{ij})$, where $\bar{f}$ is the reduction of f modulo $\mathfrak{m}_{A'}$; the term $D_{ij}(\bar{f})t$ is well defined because $\mathfrak{m}_{A'}t = 0$, so only the residue $\bar{f}$ enters. Choose such an isomorphism θ_{ij} on each overlap. This is the gluing datum; whether it is a genuine gluing depends on whether it is consistent on triple overlaps.

Step 3: The cocycle. On a triple overlap U_{ijk} compare the two ways of passing from $\mathcal{X}'_k$ to $\mathcal{X}'_i$: directly via θ_{ik}, or around through $\mathcal{X}'_j$ via $\theta_{ij} \circ \theta_{jk}$. Both are isomorphisms $\mathcal{X}'_k|_{U_{ijk}} \xrightarrow{\sim} \mathcal{X}'_i|_{U_{ijk}}$ restricting to the identity over A, so their difference is again a T_X-valued datum: set

$$c_{ijk} \;=\; \theta_{ik}^{-1} \circ \theta_{ij} \circ \theta_{jk} \;\in\; T_X(U_{ijk})$$

using Step 2 to identify an automorphism of $\mathcal{X}'_k|_{U_{ijk}}$ over A with a vector field. Because each θ is additive in its derivation to first order in t (the products $\mathfrak{m}_{A'}t = 0$ kill all higher terms), the failure of associativity is exactly the additive combination

$$c_{ijk} \;=\; D_{jk} - D_{ik} + D_{ij}$$

in $T_X(U_{ijk})$. The collection (c_{ijk}) is a Čech 2-cochain for $\mathcal{U}$ with values in T_X. It is a *cocycle*: on a quadruple overlap U_{ijkl} the alternating sum $c_{jkl} - c_{ikl} + c_{ijl} - c_{ijk}$ vanishes, because it is the difference of two ways of reassociating the four-fold composite θ_{il}, and reassociation of composable isomorphisms is associative. Thus (c_{ijk}) defines a Čech cohomology class

$$\mathrm{ob}(\mathcal{X}_A, \pi) \;=\; [(c_{ijk})] \otimes t \;\in\; \check{H}^2(\mathcal{U}, T_X) \otimes_k (t) \;=\; H^2(X, T_X) \otimes_k (t)$$

where the last identification uses that $\mathcal{U}$ is an affine (hence Leray) cover for the coherent sheaf T_X, so Čech and derived-functor cohomology agree.

Step 4: The class is independent of the choices. Two choices enter: the local lifts $\mathcal{X}'_i$ and the overlap isomorphisms θ_{ij}. Changing θ_{ij} to another gluing isomorphism θ'_{ij} over A changes D_{ij} by a vector field $E_{ij} \in T_X(U_{ij})$, and replaces c_{ijk} by $c_{ijk} + (E_{jk} - E_{ik} + E_{ij})$, that is, by a Čech coboundary; the class is unchanged. Changing the local lift $\mathcal{X}'_i$ to another local lift $\mathcal{X}''_i$ over A' changes it by a vector field $F_i \in T_X(U_i)$ (Step 2, now over the affine U_i itself, where a lift is unique up to $T_X(U_i) \otimes (t)$), and adjusts D_{ij} by $F_j - F_i$; this again changes c_{ijk} by the coboundary of (F_i). In both cases the cohomology class is invariant. So $\mathrm{ob}(\mathcal{X}_A, \pi) \in H^2(X, T_X) \otimes_k (t)$ depends only on $\mathcal{X}_A$ and π.

Step 5: Vanishing is equivalent to liftability. If a global lift $\mathcal{X}_{A'}$ of $\mathcal{X}_A$ over A' exists, take $\mathcal{X}'_i = \mathcal{X}_{A'}|_{U_i}$ to be its restrictions; these already glue (they are restrictions of a single object), so one may take $\theta_{ij} = \mathrm{id}$, whence $c_{ijk} = 0$ and $\mathrm{ob}(\mathcal{X}_A, \pi) = 0$. Conversely, suppose $\mathrm{ob}(\mathcal{X}_A, \pi) = 0$. Then (c_{ijk}) is a Čech coboundary, so there are vector fields $(G_{ij}) \in T_X(U_{ij})$ with $c_{ijk} = G_{jk} - G_{ik} + G_{ij}$ on every triple overlap. Replacing each θ_{ij} by $\theta_{ij} \circ$ (the automorphism $-G_{ij}$) modifies D_{ij} to $D_{ij} - G_{ij}$ and hence replaces c_{ijk} by $c_{ijk} - (G_{jk} - G_{ik} + G_{ij}) = 0$. The corrected gluing data θ'_{ij} now satisfy $\theta'_{ik} = \theta'_{ij} \circ \theta'_{jk}$ on triple overlaps, so the local lifts $\mathcal{X}'_i$ glue, along the θ'_{ij}, to a scheme $\mathcal{X}_{A'}$ over A'; flatness is local, so $\mathcal{X}_{A'}$ is flat over A', and its restriction over A is $\mathcal{X}_A$ because each θ'_{ij} is the identity over A. Thus a lift exists, proving (1).

Step 6: The torsor structure. Suppose a lift exists; fix one, say $\mathcal{X}_{A'}$, presented by local pieces $\mathcal{X}'_i$ glued by θ_{ij} with $c_{ijk} = 0$. Any other lift is presented, over the same local pieces (replacing

$\mathcal{X}_i'$ by an isomorphic lift changes nothing), by gluing data θ_{ij}' differing from θ_{ij} by vector fields $\eta_{ij} \in T_X(U_{ij})$, subject to the cocycle condition $\eta_{jk} - \eta_{ik} + \eta_{ij} = 0$ that both glue. So (η_{ij}) is a Čech 1-cocycle, defining a class in $H^1(X, T_X)$; and (η_{ij}) a coboundary means precisely that the two lifts are isomorphic rel $\mathcal{X}_A$. Adding a class of $H^1(X, T_X) \otimes (t)$ to the gluing data sends one lift to another and acts simply transitively on isomorphism classes of lifts. This is the torsor structure of (2); when $H^1(X, T_X) = 0$ the lift, when it exists, is unique. This establishes both claims and, in the case $H^2(X, T_X) = 0$, the final assertion, completing the proof.

The theorem is the local structure theory of a moduli space packaged in two cohomology groups. The first, $H^1(X, T_X)$, is the tangent space: it measures the directions in which X can move to first order, and, by the torsor statement, the ambiguity in continuing a deformation once continuation is possible. The second, $H^2(X, T_X)$, is the *obstruction space*: it is the receptacle for the classes that block continuation. The dimension of the moduli space at $[X]$ is at most $\dim H^1(X, T_X)$, and equals it exactly when the obstructions all vanish; the gap between the tangent-space dimension and the actual dimension is cut out by the image of the obstruction map into $H^2(X, T_X)$. This is the precise sense in which H^1 is the linear approximation and H^2 measures the failure of the linear approximation to be exact. A reader who has met the analogous statement in another guise, the obstruction to extending a section or a class living one degree up in a long exact sequence, is seeing the same homological phenomenon: extending a structure over a thickening is a lifting problem, and lifting problems are controlled by the next cohomology group.

The proof also explains *why* the obstruction is a global object even though the lifting problem is posed globally from the start. Locally there is never any obstruction, because smooth affines are unobstructed (Step 1); the class $\mathrm{ob}(\mathcal{X}_A, \pi)$ measures only the failure of the local solutions to be compatible on overlaps, and that failure is a Čech 2-cocycle. This is the same mechanism by which a line bundle can be locally trivial yet globally nontrivial, one dimension up. The obstruction is a purely cohomological record of a gluing that cannot be carried out, not a local pathology.

One consequence deserves to be named, because it will organize every smoothness result in the sequel: unobstructedness is the property that turns the tangent space into the whole local picture.

Definition 17.1.3: Unobstructed Deformation Functor

A deformation functor $D\colon \mathbf{Art}_k \to \mathbf{Set}$ is *unobstructed*, or *formally smooth*, if for every small extension $A' \to A$ in $\mathbf{Art}_k$ the map $D(A') \to D(A)$ is surjective; that is, every deformation over A lifts over every small extension. Equivalently, a hull R of D (a complete local k-algebra pro-representing D up to the tangent space, in the sense of the next sections) is a formal power-series ring $R \cong k[\![x_1, \dots, x_n]\!]$ with $n = \dim_k D(k[\varepsilon])$, and the corresponding formal moduli space is smooth of dimension n at the point $[X]$.

The word "formally smooth" is chosen because the surjectivity condition on D is exactly the infinitesimal lifting criterion for smoothness of a scheme: a scheme M is smooth at a point m if and only if its functor of points lifts along every small extension near m. Applied to the local model of a moduli space, unobstructedness says that the moduli space, near $[X]$, looks formally like affine space of dimension $\dim H^1(X, T_X)$, with no equations cutting it down. When $D = \mathrm{Def}_X$ and $H^2(X, T_X) = 0$, theorem 17.1.2 shows Def_X is unobstructed and the moduli space is smooth of dimension $\dim H^1(X, T_X)$ at $[X]$; the two cohomology groups then give both the smoothness and the dimension in one stroke. The equivalence with a power-series hull is proved once Schlessinger's criteria are in place (see section 17.3); the definition records it now so that "unobstructed" and "power-series hull" are known to be the same condition wherever either appears.

The definition is only as useful as the vanishing statement $H^2(X, T_X) = 0$ is checkable. The following example is the whole payoff of the section: it makes the vanishing checkable for curves, deducing the smoothness of the moduli of curves from a one-line cohomological fact, and it exhibits the opposite phenomenon for surfaces, where $H^2(X, T_X)$ can be nonzero and a deformation can be obstructed.

Example 17.1.4: Curves are Unobstructed, Surfaces Need Not Be

1. *Curves.* Let C be a smooth projective curve of genus $g \geq 2$ over k. The tangent sheaf $T_C = \Omega_C^\vee$ is a line bundle, and C is a one-dimensional variety, so the coherent-sheaf cohomology $H^i(C, \mathcal{F})$ vanishes for all $i > 1$ and every coherent sheaf $\mathcal{F}$; this is Grothendieck's vanishing theorem for a Noetherian space of dimension 1. In particular
$$H^2(C, T_C) \;=\; 0$$
so by theorem 17.1.2 the deformation functor of C is unobstructed and the moduli space is smooth at $[C]$ of dimension $\dim H^1(C, T_C)$. The dimension is computed from Riemann-Roch. Since $T_C = \Omega_C^\vee$ has degree $-\deg \Omega_C = -(2g-2) = 2-2g < 0$, the space $H^0(C, T_C)$ of global vector fields vanishes for $g \geq 2$ (a nonzero global section of a negative-degree line bundle on a curve cannot exist), so C has no infinitesimal automorphisms in the sense of definition 16.5.1. Riemann-Roch on the curve (theorem 7.1.19) then gives
$$h^0(C, T_C) - h^1(C, T_C) \;=\; \deg T_C + 1 - g \;=\; (2-2g) + 1 - g \;=\; 3 - 3g$$
and with $h^0(C, T_C) = 0$ this yields $h^1(C, T_C) = 3g - 3$. Thus the tangent space to the moduli of curves at $[C]$ has dimension $3g - 3$, and since $H^2 = 0$ forces unobstructedness, $\mathcal{M}_g$ is smooth of dimension $3g - 3$ at every point coming from a curve with no automorphisms. The absence of obstructions is the reason the count $3g - 3$, first a tangent-space dimension, is the actual dimension of the moduli space and not just an upper bound.

2. *Surfaces.* For a smooth projective surface X the tangent sheaf T_X is a rank-2 bundle on a two-dimensional variety, and $H^2(X, T_X)$ need not vanish. By Serre duality on the surface (corollary 6.5.10),
$$H^2(X, T_X) \;\cong\; H^0(X, \Omega_X \otimes \omega_X)^\vee$$
where $\omega_X = \Omega_X^2$ is the canonical bundle, and the right-hand side is a space of global sections that can be nonzero. A surface of general type with ample canonical bundle can have $H^0(X, \Omega_X \otimes \omega_X) \neq 0$, hence $H^2(X, T_X) \neq 0$, so the receptacle for obstructions is nonzero and the vanishing-room argument that made curves unobstructed is no longer available. Nonzero $H^2(X, T_X)$ is only a *necessary* condition for an obstructed deformation, not a sufficient one: the obstruction map $\xi \mapsto \frac{1}{2}[\xi, \xi] \colon H^1(X, T_X) \to H^2(X, T_X)$ (exercise 17.1.1 (5)) can be identically zero even when its target does not vanish. This is precisely the phenomenon behind the unobstructedness of higher-dimensional Calabi-Yau manifolds (of dimension ≥ 3), where $H^2(X, T_X) \neq 0$ yet every obstruction still vanishes (the Bogomolov-Tian-Todorov theorem, section 17.4). An abelian surface exhibits the same nonzero target already in dimension two: its tangent bundle $T_X \cong \mathcal{O}_X^{\oplus 2}$ is trivial, so $H^2(X, T_X) \cong H^2(X, \mathcal{O}_X)^{\oplus 2}$ is two-dimensional, yet abelian surfaces are unobstructed.

Whether obstructions occur is therefore a question about the obstruction map, not about the dimension of its target, and it must be settled by a construction rather than by a vanishing count.

> Obstructed surfaces do exist. Kas and Horikawa exhibited surfaces of general type with obstructed deformations, whose moduli spaces are singular at the corresponding points, with local dimension there strictly less than $\dim H^1(X, T_X)$ (see [35] and [61] for the constructions and computations). The dichotomy with curves is dimensional at its root: on a curve, cohomology stops at degree 1 and there is no room for an obstruction to live, so unobstructedness is automatic; on a surface, degree 2 is available and an obstruction *can* be present, so smoothness of the moduli space is no longer free and must be established case by case.

The contrast in the example is the organizing principle of the rest of the chapter. For a curve, $H^2(C, T_C) = 0$ is automatic from dimension, so the moduli of curves is smooth and its dimension is read off from a single Riemann-Roch computation; this is the cohomological source of the number $3g - 3$ that reappears throughout the study of $\mathcal{M}_g$. For a surface, the obstruction space is a nontrivial invariant that can be nonzero, and deciding smoothness of the moduli space requires either computing $H^2(X, T_X)$ directly or invoking a structural reason for the obstructions to vanish. When $H^2(X, T_X) = 0$ outright, as for a K3 surface (where $\omega_X \cong \mathcal{O}_X$ forces $H^2(X, T_X) \cong H^0(X, \Omega_X)^\vee = 0$), theorem 17.1.2 already gives unobstructedness with no further input; the harder case is a higher-dimensional Calabi-Yau, where $H^2(X, T_X) \neq 0$ yet the obstruction map still vanishes, which is the content of the Bogomolov-Tian-Todorov theorem (section 17.4). The next section replaces the sheafy groups $H^i(X, T_X)$ by the algebraic cotangent functors $T^i(A/k, -)$, which compute the same obstruction theory for singular objects, where no tangent sheaf is available and the naive picture of "vector fields on overlaps" breaks down.

Exercise 17.1.1

1. Let X be a smooth projective variety with $H^2(X, T_X) = 0$, and let $A' \to A$ be an arbitrary surjection in $\mathbf{Art}_k$ (not necessarily small). Show that every deformation of X over A lifts to a deformation over A'. (Reduce to the small-extension case using that every surjection in $\mathbf{Art}_k$ factors as a finite composite of small extensions.)

2. Compute $H^1(\mathbb{P}^n, T_{\mathbb{P}^n})$ and $H^2(\mathbb{P}^n, T_{\mathbb{P}^n})$ for $n \geq 1$, and conclude that $\mathbb{P}^n$ is both rigid (no nontrivial first-order deformations) and unobstructed. (Use the Euler sequence $0 \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_{\mathbb{P}^n}(1)^{\oplus(n+1)} \to T_{\mathbb{P}^n} \to 0$, from theorem 5.5.18, and the cohomology of line bundles on $\mathbb{P}^n$.)

3. Let C be a smooth projective curve of genus g. Using example 17.1.4, determine $h^0(C, T_C)$ and $h^1(C, T_C)$ for $g = 0$ and $g = 1$, and interpret the results in terms of infinitesimal automorphisms and the dimension of the moduli of curves in those two genera.

4. In the setting of theorem 17.1.2, suppose $\mathcal{X}_A$ is a deformation of X over A that admits a lift over A', and suppose $H^1(X, T_X) = 0$. Show that the lift is unique up to isomorphism, and explain why the hypothesis $H^1 = 0$ says the deformation is rigid at every infinitesimal order.

5. Let $A = k[\varepsilon] = k[t]/(t^2)$ and $A' = k[t]/(t^3)$, so that $A' \to A$ is a small extension with kernel $(t^2) \cong k$. Interpret a lift of a first-order deformation $\xi \in H^1(X, T_X)$ over A to a deformation over A' as a "second-order deformation," and show directly from theorem 17.1.2 that the obstruction to the existence of such a lift is a class in $H^2(X, T_X)$ depending on ξ. Argue that this class is the quadratic term of the obstruction, so that unobstructedness at first order is exactly the vanishing of this quadratic obstruction for every ξ.

17.2 The Naive Cotangent Complex and the T^i Functors

The previous section measured deformations and obstructions of a *smooth* projective variety through the cohomology groups $H^i(X, T_X)$ of its tangent sheaf: first-order deformations sit in $H^1(X, T_X)$, obstructions in $H^2(X, T_X)$, and infinitesimal automorphisms in $H^0(X, T_X)$. That bookkeeping used smoothness at every turn, because the tangent sheaf $T_X = \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)$ only sees a variety's deformations correctly when the variety is smooth. Yet moduli problems force singular objects on us at once. The capstone stack $\overline{\mathcal{M}}_g$ parametrizes *nodal* curves; the deformation theory of a node, and of every other singularity that appears at the boundary of a moduli space, cannot be read off from $H^i(T_X)$ alone. What is needed is an invariant that specializes to $H^i(X, T_X)$ when X is smooth but continues to compute deformations and obstructions when X is singular.

That invariant is the cotangent complex, and its cohomology groups are the T^i functors of a k-algebra. This section constructs the version adequate for our purposes, the naive or Lichtenbaum-Schlessinger cotangent complex of a presentation $A = P/I$, and extracts from it the three functors T^0, T^1, T^2. The reader should read them as the algebraic, local counterparts of the geometric groups $H^0(T_X), H^1(T_X), H^2(T_X)$: T^0 collects infinitesimal automorphisms, T^1 classifies first-order deformations, and T^2 receives their obstructions. The section closes by reconciling the two languages: for a smooth affine $\operatorname{Spec} A$ the higher T^i vanish, and for any smooth X the local T^i sheafify to recover exactly the groups $H^i(X, T_X)$ of section 17.1.

The full theory of Illusie assigns to any morphism of schemes a cotangent *complex* $L_{A/k}$ in the derived category, whose homotopy groups compute deformations and obstructions in all degrees and whose construction requires simplicial resolutions. That machinery is out of scope for this book (BOOK.md §6): it belongs to derived algebraic geometry, and the naive two-term complex below suffices for every deformation problem treated here. The naive version agrees with the full cotangent complex in the degrees -1 and 0 that produce T^0, T^1, T^2, which is all the present chapter uses.

The construction begins with a presentation. Fix a finitely generated k-algebra A and write it as a quotient $A = P/I$ of a polynomial ring $P = k[x_1, \ldots, x_n]$ by an ideal I. Such a presentation always exists and is far from unique, but the functors extracted from it will turn out to depend only on A. Two modules over A organize the first-order geometry of the presentation. The first is the module of Kähler differentials $\Omega_{P/k}$, a free P-module on the symbols $dx_1, \ldots, dx_n$, restricted to A by base change to $\Omega_{P/k} \otimes_P A \cong A^n$. The second is the *conormal module* I/I^2, an A-module recording the equations cutting out $\operatorname{Spec} A$ inside affine space $\mathbb{A}^n = \operatorname{Spec} P$ to first order. They are linked by the universal derivation: the map sending a class $\bar{f} \in I/I^2$ to its total differential $df = \sum_i (\partial f/\partial x_i)\, dx_i$ lands in $\Omega_{P/k} \otimes_P A$ because differentiating a relation records how that relation constrains the tangent directions.

Definition 17.2.1: Naive Cotangent Complex

Let $A = P/I$ with $P = k[x_1, \dots, x_n]$ a polynomial ring and I an ideal. The *naive* (Lichtenbaum-Schlessinger) *cotangent complex* of A over k is the two-term complex of A-modules

$$L_{A/k}\colon \quad I/I^2 \xrightarrow{d} \Omega_{P/k} \otimes_P A$$

placed in cohomological degrees -1 and 0, where $d(\bar{f}) = df \otimes 1 = \sum_{i=1}^n \frac{\partial f}{\partial x_i}\, dx_i$.
For the second-order term, choose generators $f_1, \dots, f_r$ of I, so that I is the image of a surjection $P^r \twoheadrightarrow I$ from a free module, and let $\mathrm{Rel} \subseteq P^r$ be the module of relations (first syzygies) among the f_j. The submodule $\mathrm{Kos} \subseteq \mathrm{Rel}$ of *Koszul* (trivial) relations is generated by the elements $f_j e_k - f_k e_j$. The naive cotangent complex is extended to a three-term complex by the module of *nontrivial second syzygies*

$$L^{-2}_{A/k} \;:=\; (\mathrm{Rel} \,/\, \mathrm{Kos}) \otimes_P A$$

in cohomological degree -2, mapping to I/I^2 through the presentation of I.

The complex is short by design. Its degree-0 term is the restriction to A of the cotangent space of the ambient affine space; its degree-(-1) term is the conormal module, the equations; and its degree-(-2) term records the relations *among* the equations modulo the tautological Koszul ones. Reading the complex geometrically, $\Omega_{P/k} \otimes_P A$ is the cotangent bundle of $\mathbb{A}^n$ restricted to $\operatorname{Spec} A$, and d tells one which of its directions are killed by the defining equations. When $\operatorname{Spec} A$ is smooth of dimension d, the map d is injective with locally free cokernel $\Omega_{A/k}$ of rank d, and the whole complex collapses to $\Omega_{A/k}$ in degree 0; every correction term measures a failure of smoothness. The dependence on the presentation is genuine at the level of the complex but not at the level of the invariants below: the T^i are independent of all choices, a fact whose proof, a homotopy-invariance argument comparing two presentations through a common refinement, is the homological content this section imports and is carried out in full in [15].

The three functors of interest are the cohomology of this complex after dualizing into a coefficient module. Fixing an A-module M, apply $\mathrm{Hom}_A(-, M)$ to $L_{A/k}$ and read off cohomology. Concretely, the dualized complex is

$$\mathrm{Hom}_A(\Omega_{P/k} \otimes_P A, M) \xrightarrow{d^\vee} \mathrm{Hom}_A(I/I^2, M) \longrightarrow \mathrm{Hom}_A(L^{-2}_{A/k}, M)$$

sitting in cohomological degrees $0, 1, 2$, and the T^i are its cohomology in those degrees.

Definition 17.2.2: The T^i Functors

Let $A = P/I$ be a finitely generated k-algebra and M an A-module. The *cotangent functors* $T^i(A/k, M)$ for $i = 0, 1, 2$ are the cohomology groups of the complex $\mathrm{Hom}_A(L_{A/k}, M)$:

1. $T^0(A/k, M) = \ker\big(d^\vee : \mathrm{Hom}_A(\Omega_{P/k} \otimes_P A, M) \to \mathrm{Hom}_A(I/I^2, M)\big)$, the derivations that annihilate the equations;

2. $T^1(A/k, M) = \ker\big(\mathrm{Hom}_A(I/I^2, M) \to \mathrm{Hom}_A(L^{-2}_{A/k}, M)\big) \big/ \mathrm{im}(d^\vee)$;

3. $T^2(A/k, M) = \mathrm{coker}\big(\mathrm{Hom}_A(I/I^2, M) \to \mathrm{Hom}_A(L^{-2}_{A/k}, M)\big)$.

When $M = A$ one writes $T^i(A/k) := T^i(A/k, A)$.

The degree-zero functor admits a description that owes nothing to the presentation. A k-derivation of A into M is a k-linear map $\delta\colon A \to M$ satisfying the Leibniz rule $\delta(ab) = a\,\delta(b) + b\,\delta(a)$, and the collection of these is the module $\mathrm{Der}_k(A, M)$. A derivation of A is the same datum as a derivation of P that kills the ideal I, which is exactly an element of the kernel defining T^0. This gives the clean identification

$$T^0(A/k, M) = \mathrm{Der}_k(A, M) = \mathrm{Hom}_A(\Omega_{A/k}, M)$$

so that $T^0(A/k, A) = \mathrm{Der}_k(A, A)$ is the module of vector fields on $\mathrm{Spec}\, A$, the algebraic incarnation of $H^0(X, T_X)$.

The other two functors carry the deformation-theoretic content. A first-order deformation of the affine scheme $\mathrm{Spec}\, A$ over the dual numbers $k[\varepsilon]$ is a flat $k[\varepsilon]$-algebra $\tilde{A}$ reducing to A modulo ε, and such deformations up to equivalence are classified by $T^1(A/k, A)$. Equivalently, $T^1(A/k, M)$ classifies the square-zero extensions of A by M as k-algebras: the short exact sequences $0 \to M \to B \to A \to 0$ with $M^2 = 0$ in B, modulo isomorphism fixing the ends. The functor $T^2(A/k, A)$ is the home of obstructions: given a first-order deformation and a small extension of the base to lift it along, the obstruction to the lift is a class in $T^2(A/k, A)$ that vanishes exactly when a lift exists. These readings are the content of the next theorem, and they mirror term by term the smooth-case bookkeeping of section 17.1.

Before stating that theorem, one interpretive point deserves emphasis. The passage from the geometric groups $H^i(X, T_X)$ to the algebraic groups $T^i(A/k, A)$ is not a change of subject but a change of vantage. For a smooth variety the tangent sheaf already captures the deformation theory, and its cohomology is the answer. For a singular scheme the tangent sheaf is the wrong object, and the cotangent complex is the correction: T^1 and T^2 are what $H^1(T_X)$ and $H^2(T_X)$ *ought* to have been, computed through a resolution that remembers the equations and their relations rather than through a sheaf that has forgotten them.

Theorem 17.2.3: Interpretation of the Cotangent Functors

Let A be a finitely generated k-algebra.

1. $T^1(A/k, A)$ is in natural bijection with the set of first-order deformations of $\operatorname{Spec} A$ over $k[\varepsilon]$, up to equivalence; the trivial deformation corresponds to 0.
2. Given a first-order deformation of $\operatorname{Spec} A$ and a small extension $0 \to (t) \to A' \to k[\varepsilon] \to 0$ in $\mathbf{Art}_k$, there is a canonical obstruction class in $T^2(A/k, A) \otimes_k (t)$ whose vanishing is necessary and sufficient for a lift to exist; when a lift exists, the lifts form a torsor under $T^1(A/k, A) \otimes_k (t)$.
3. If A is smooth over k, then $T^i(A/k, M) = 0$ for all $i \geq 1$ and all M.
4. If X is a smooth k-scheme covered by affine opens $\operatorname{Spec} A_\alpha$, the local functors $T^i(A_\alpha/k, A_\alpha)$ sheafify (by (3), only the degree-zero sheaf survives, and it is the tangent sheaf T_X), and the local-to-global spectral sequence assembles them, converging to the geometric groups $H^i(X, T_X)$ of section 17.1. For X affine, or in low degree, this recovers $T^i(A/k, A) = H^i(X, T_X)$ directly.

***Proof*:**
The homological input, that $T^i(A/k, M)$ is independent of the presentation and computes the derived functors of derivations in degrees ≤ 2, is proved in [15]; the deformation-theoretic statements are deduced from the complex directly.

Step 1: First-order deformations and T^1. A first-order deformation of $\operatorname{Spec} A$ is a flat $k[\varepsilon]$-algebra $\tilde{A}$ with $\tilde{A} \otimes_{k[\varepsilon]} k = A$. Present it by lifting the presentation of A: since P is smooth, the surjection $P \to A$ lifts to a surjection $\tilde{P} := P[\varepsilon] \to \tilde{A}$, and flatness over $k[\varepsilon]$ forces $\tilde{A} = \tilde{P}/\tilde{I}$ where $\tilde{I}$ is generated by lifts $\tilde{f}_j = f_j + \varepsilon g_j$ of the generators f_j of I, with $g_j \in P$. Reducing modulo ε recovers A, so the data of the deformation is the tuple $(g_j) \in P^r$ modulo two indeterminacies. First, the g_j are only well-defined modulo I once reduced to A, and modulo the ambiguity in the lifts $\tilde{f}_j$; both amount to changing (g_j) by an element of the image of $d^\vee$, that is, by the derivation coming from an infinitesimal change of coordinates in $\tilde{P}$. Second, flatness requires that every relation $\sum_j p_j f_j = 0$ among the f_j lift to a relation $\sum_j \tilde{p}_j \tilde{f}_j = 0$, and the obstruction to lifting the relation is precisely the requirement that the induced functional on I/I^2 annihilate the second syzygies, that is, that (g_j) define a class in $\ker(\operatorname{Hom}_A(I/I^2, A) \to \operatorname{Hom}_A(L^{-2}_{A/k}, A))$. The two conditions together identify the equivalence classes of first-order deformations with

$$\frac{\ker\big(\operatorname{Hom}_A(I/I^2, A) \to \operatorname{Hom}_A(L^{-2}_{A/k}, A)\big)}{\operatorname{im}(d^\vee)} = T^1(A/k, A)$$

and the trivial deformation $\tilde{A} = A[\varepsilon]$, given by $g_j = 0$, maps to the zero class. This proves (1).

Step 2: Lifting over a small extension and T^2. Let $0 \to (t) \to A' \to k[\varepsilon] \to 0$ be a small extension, so $\mathfrak{m}_{A'} \cdot t = 0$ and $(t) \cong k$ as an A'-module through the residue field. A deformation $\tilde{A}$ over $k[\varepsilon]$ is presented as above; to lift it over A' is to choose lifts $\tilde{f}'_j \in P \otimes_k A'$ of the $\tilde{f}_j$ so that the relations among the $\tilde{f}_j$ continue to hold modulo t. Choose arbitrary set-theoretic lifts $\tilde{f}'_j$. For each relation $\sum_j \tilde{p}_j \tilde{f}_j = 0$ holding over $k[\varepsilon]$, the same combination $\sum_j \tilde{p}'_j \tilde{f}'_j$ of the lifted generators need not vanish, but it lies in $(t) \cdot (P \otimes_k A') = t \cdot P$ because it vanishes modulo t;

writing it as $t \cdot c_{(\mathrm{rel})}$ produces, as the relation ranges over the second syzygies, a well-defined element

$$\mathrm{ob} \in \mathrm{Hom}_A\big(L^{-2}_{A/k}, A\big) \otimes_k (t)$$

because the Koszul relations impose no condition (they lift tautologically) and the ambiguity in the set-theoretic lifts changes ob by the image of $\mathrm{Hom}_A(I/I^2, A) \otimes_k (t)$. Its class

$$\mathrm{ob}(\tilde{A}, A') \;\in\; T^2(A/k, A) \otimes_k (t)$$

is therefore canonical, and by construction the lifted generators can be adjusted to satisfy all relations exactly, that is, a lift $\tilde{A}'$ exists, if and only if ob vanishes. When it vanishes, two lifts differ by an element of $\mathrm{Hom}_A(I/I^2, A) \otimes_k (t)$ modulo those coming from coordinate changes, which is $T^1(A/k, A) \otimes_k (t)$; hence the lifts form a torsor under this group. This proves (2).

Step 3: Vanishing for smooth A. Suppose A is smooth over k of dimension d. By the Jacobian criterion the map $d\colon I/I^2 \to \Omega_{P/k} \otimes_P A$ is a split injection of A-modules with locally free cokernel $\Omega_{A/k}$ of rank d; the conormal sequence

$$0 \to I/I^2 \xrightarrow{d} \Omega_{P/k} \otimes_P A \to \Omega_{A/k} \to 0$$

is exact and split, and I is generated by a regular sequence, so its first syzygies are Koszul and the nontrivial second syzygies vanish: $L^{-2}_{A/k} = 0$. Applying $\mathrm{Hom}_A(-, M)$ to the split sequence keeps it exact, so $d^\vee$ is surjective, forcing $T^1(A/k, M) = 0$; and $L^{-2}_{A/k} = 0$ gives $T^2(A/k, M) = 0$. The same vanishing propagates to higher T^i in the full theory, but the two computed here are all the chapter requires. This proves (3).

Step 4: Sheafification and reconciliation. For X smooth, Step 3 gives $T^0(A_\alpha/k, A_\alpha) = \mathrm{Der}_k(A_\alpha, A_\alpha) = \Gamma(\operatorname{Spec} A_\alpha, T_X)$ and $T^{\geq 1}(A_\alpha/k, A_\alpha) = 0$ on each affine chart. The presheaf $U \mapsto T^0(\mathcal{O}_X(U)/k, \mathcal{O}_X(U))$ sheafifies to the tangent sheaf $T_X = \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)$, and the local-to-global passage from the affine-local groups to the global ones is governed by a spectral sequence whose E_2 page has the sheaf cohomology $H^p(X, \mathcal{T}^q_X)$ of the sheafified cotangent functors $\mathcal{T}^q_X$. Since $\mathcal{T}^0_X = T_X$ and $\mathcal{T}^q_X = 0$ for $q \geq 1$ by the local vanishing, the spectral sequence degenerates to the edge isomorphism

$$T^i(X/k, \mathcal{O}_X) \;\cong\; H^i(X, T_X)$$

identifying the global algebraic groups with the geometric groups of section 17.1. For X affine both sides are the derivations in degree 0 and vanish above, giving the claimed direct equality, which proves (4) and completes the proof.

The theorem answers the question left by the previous section. There the tangent and obstruction spaces of a smooth variety were the cohomology groups $H^1(T_X)$ and $H^2(T_X)$; here they are revealed as special cases of the cotangent functors T^1 and T^2, which continue to compute the same information for singular schemes on which the tangent sheaf fails. The notion of unobstructedness transfers verbatim: a deformation functor is unobstructed in the sense of definition 17.1.3 exactly when the relevant T^2 vanishes, so that every obstruction class in part (2) of theorem 17.2.3 is automatically zero and every lift exists. The class of schemes for which T^2 vanishes therefore deserves a name, and the next example identifies a large supply of them.

The most important such class is the local complete intersections, and the cleanest instance is a hypersurface. A hypersurface has a single defining equation, so its ideal is generated by a regular sequence of length one, its syzygies are all Koszul, and T^2 vanishes for the same reason it did in the smooth case: there are no nontrivial relations among the equations. What survives in T^1 is a finite-dimensional algebra built from the equation and its partial derivatives, the Jacobian or Tjurina algebra, which measures the singularity directly.

Example 17.2.4: The T^i of a Hypersurface

Let $A = P/(f)$ with $P = k[x_1, \dots, x_n]$ and f a nonzero nonunit, so $\operatorname{Spec} A$ is a hypersurface, and suppose $\operatorname{Spec} A$ has an isolated singularity (over a field of characteristic 0). The ideal $I = (f)$ is generated by the single regular element f, so the naive cotangent complex of definition 17.2.1 is short:

- $I/I^2 = (f)/(f^2) \cong A$, free of rank 1, generated by the class $\bar{f}$;
- $\Omega_{P/k} \otimes_P A \cong A^n$, free on $dx_1, \dots, dx_n$;
- the differential $d(\bar{f}) = \sum_{i=1}^n \partial_i f \, dx_i$, where $\partial_i f := \partial f / \partial x_i$;
- $L^{-2}_{A/k} = 0$, because I is generated by a regular sequence, so every relation among the (single) generator is Koszul.

Vanishing of T^2. Since $L^{-2}_{A/k} = 0$, the target $\operatorname{Hom}_A(L^{-2}_{A/k}, A) = 0$, and definition 17.2.2 gives

$$T^2(A/k, A) = \operatorname{coker}\big(\operatorname{Hom}_A(I/I^2, A) \to 0\big) = 0$$

The hypersurface is unobstructed. The same computation applies to any local complete intersection $A = P/(f_1, \dots, f_c)$ with $f_1, \dots, f_c$ a regular sequence: the first syzygies are Koszul, $L^{-2}_{A/k} = 0$, and $T^2(A/k, A) = 0$. Complete intersections are unobstructed, which is why they deform freely and why the boundary singularities appearing in moduli of curves, all of them nodes and hence hypersurface singularities, carry no obstruction to smoothing.

Computation of T^1. By definition 17.2.2, with $L^{-2}_{A/k} = 0$,

$$T^1(A/k, A) = \operatorname{coker}\big(d^\vee \colon \operatorname{Hom}_A(A^n, A) \to \operatorname{Hom}_A(A, A)\big)$$

Under the identifications $\operatorname{Hom}_A(A^n, A) \cong A^n$ and $\operatorname{Hom}_A(A, A) \cong A$, the dual map $d^\vee$ sends a functional $\varphi \colon A^n \to A$, given by $\varphi(dx_i) = a_i$, to the composite $\varphi \circ d$, whose value on the generator $\bar{f}$ is

$$(\varphi \circ d)(\bar{f}) = \varphi\Big(\sum_i \partial_i f \, dx_i\Big) = \sum_{i=1}^n a_i \, \partial_i f$$

so $d^\vee \colon A^n \to A$ is the map $(a_1, \dots, a_n) \mapsto \sum_i a_i \, \partial_i f$, whose image is the Jacobian ideal $(\partial_1 f, \dots, \partial_n f) \cdot A$. Therefore

$$T^1(A/k, A) = A\big/(\partial_1 f, \dots, \partial_n f) = \frac{P}{(f, \partial_1 f, \dots, \partial_n f)}$$

the *Jacobian* or *Tjurina algebra* of f. Because the singularity is isolated, the vanishing locus of f together with all its partials is the singular point alone, so this algebra is a finite-dimensional k-vector space; its dimension, the *Tjurina number* $\tau(f) = \dim_k T^1(A/k, A)$, counts the parameters in the versal deformation of the singularity.

The node. Take $n = 2$ and $f = xy$, the node $A = k[x, y]/(xy)$. Then $\partial_x f = y$, $\partial_y f = x$, and

$$T^1(A/k, A) = \frac{k[x, y]}{(xy, y, x)} = \frac{k[x, y]}{(x, y)} \cong k$$

so the node has a one-dimensional space of first-order deformations and $\tau = 1$. The single parameter is realized by the family $xy = t$, which smooths the node for $t \neq 0$; this is the local model that reappears in section 17.4 and, later, in the deformation theory of stable curves at the boundary of the moduli space.

The hypersurface computation isolates the mechanism behind unobstructedness. Obstructions live in T^2, and T^2 is built from the nontrivial relations among the defining equations; when the equations form a regular sequence there are no such relations, so T^2 vanishes and the scheme deforms without obstruction. The surviving invariant T^1, the Tjurina algebra, is a direct algebraic measurement of the singularity, and its dimension counts deformation parameters. This is the pattern the applications section exploits: to prove a moduli space smooth at a point, show that the T^2 of the object it parametrizes vanishes, and to compute its dimension there, compute the corresponding T^1.

Exercise 17.2.1

1. Let $A = k[x, y]/(y^2 - x^3)$ be the cuspidal cubic curve over a field of characteristic 0. Compute $T^1(A/k, A)$ and $T^2(A/k, A)$, and give the Tjurina number τ. Compare with the node of example 17.2.4: which singularity has more deformation parameters?

2. Show that a smooth affine k-scheme $\operatorname{Spec} A$ has $T^1(A/k, A) = 0$ directly from definition 17.2.2, and interpret the vanishing in terms of first-order deformations. Why does theorem 17.2.3(1) then say $\operatorname{Spec} A$ has no nontrivial first-order deformations?

3. Let $A = k[x, y, z]/(xy, xz, yz)$, the union of the three coordinate axes in $\mathbb{A}^3$. This is not a complete intersection: three equations cut out a one-dimensional scheme in three-space. Exhibit a nontrivial relation among the three generators that is not Koszul, and explain why this forces $L^{-2}_{A/k} \neq 0$, so that $T^2(A/k, A)$ can be nonzero. (You need not compute T^2; the point is to see the obstruction module come alive when the codimension exceeds the number of equations only by failing to be a complete intersection.)

4. Verify that $T^0(A/k, A) = \operatorname{Der}_k(A, A)$ is compatible with the two usages of section 17.1: for $A = k[x]$ (so $\operatorname{Spec} A = \mathbb{A}^1$ is smooth), compute $\operatorname{Der}_k(A, A)$ explicitly and confirm it agrees with the global vector fields $H^0(\mathbb{A}^1, T_{\mathbb{A}^1})$.

5. Let $A = P/(f)$ be a hypersurface as in example 17.2.4, and suppose $f = f_2 + f_3 + \cdots$ is the sum of its homogeneous pieces with $f_2 \neq 0$ a nondegenerate quadratic form (an ordinary double point, A_1 singularity, in n variables over characteristic 0). Compute $\dim_k T^1(A/k, A)$. *(Hint: after a linear change of coordinates the leading term is $x_1^2 + \cdots + x_n^2$; work out the Jacobian ideal of the pure quadric first and argue that the higher-order terms do not change the dimension.)*

17.3 Schlessinger's Criteria

The previous two sections assembled the local anatomy of a deformation problem: a tangent space of first-order deformations and an obstruction space receiving the failure to lift across a small extension. What is still missing is the global verdict. Given a deformation functor, does there exist a single formal object, living over a complete local ring, from which every deformation is induced? For a smooth projective variety with a fine moduli space, the answer is a completed local ring of that space, and the deformation functor is the functor of points of that ring restricted to Artinian algebras. But most deformation functors are not representable at all, precisely because the objects being deformed carry automorphisms, and automorphisms destroy the uniqueness that a universal family demands. This is the same obstruction that produced the j-line rather than a fine moduli space of elliptic curves in theorem 13.3.6.

Schlessinger's theorem cuts exactly along this fault line. It isolates a short list of conditions on a functor of Artin rings, phrased entirely in terms of how the functor behaves on fiber products, and it extracts two verdicts. The weaker three conditions guarantee a *hull*: a formal object that is versal, so every deformation is induced from it, and minimal, so its tangent space is as small as possible. The fourth condition upgrades versality to genuine pro-representability, producing a universal formal deformation that is unique with its inducing map. Everything a deformation theorist needs to know about whether a moduli problem has a well-behaved local model is encoded in these four conditions, and the proof of hull existence is a concrete order-by-order construction that this section carries out in full.

Throughout, k is an algebraically closed field, $\mathbf{Art}_k$ is the category of local Artinian k-algebras with residue field k, and $D\colon \mathbf{Art}_k \to \mathbf{Set}$ is a functor with $D(k)$ a single point. Recall from definition 16.1.5 that a *small extension* is a surjection $\pi\colon A' \to A$ in $\mathbf{Art}_k$ whose kernel (t) is a one-dimensional k-vector space annihilated by the maximal ideal $\mathfrak{m}_{A'}$, so that $\mathfrak{m}_{A'} \cdot t = 0$ and $t^2 = 0$.

The right notion of a solution to a deformation problem is not always a representing ring. When automorphisms are present, the best one can hope for is a formal object that induces all others but is not required to do so uniquely. To state this precisely, one works with $\widehat{\mathbf{Art}}_k$, the category of complete local Noetherian k-algebras with residue field k; for the theory of complete local rings, their construction as limits of Artinian quotients, and the Cohen structure theorem, see [10]. A complete local k-algebra R with maximal ideal $\mathfrak{m}_R$ defines a functor on $\mathbf{Art}_k$ by

$$h_R(A) = \mathrm{Hom}_{k\text{-alg}}(R, A) = \varinjlim_{n} \mathrm{Hom}_{k\text{-alg}}(R/\mathfrak{m}_R^n, A)$$

where the colimit is over n and identifies a homomorphism out of the complete ring R with a compatible system of homomorphisms out of its Artinian quotients, each of which factors through some $R/\mathfrak{m}_R^n$ because A is Artinian. The functor h_R is the object of study: it is the algebraic data determined by a formal deformation.

Definition 17.3.1: Hull of a Functor of Artin Rings

Let $D\colon \mathbf{Art}_k \to \mathbf{Set}$ be a functor with $D(k) = \{*\}$. A *formal element* of D is a complete local k-algebra $R \in \widehat{\mathbf{Art}}_k$ together with a compatible system $\xi = (\xi_n)$ of elements $\xi_n \in D(R/\mathfrak{m}_R^n)$, equivalently a morphism of functors $h_R \to D$. Such a ξ is *versal* if for every small extension $\pi\colon A' \to A$ in $\mathbf{Art}_k$ the map

$$h_R(A') \longrightarrow h_R(A) \times_{D(A)} D(A')$$

is surjective, that is, every deformation over A' that agrees over A with a deformation induced by ξ is itself induced by ξ. A versal element is a *hull* (or a *miniversal formal deformation*) if in addition the induced map on tangent spaces

$$d\xi\colon \mathrm{Hom}_{k\text{-alg}}(R, k[\varepsilon]) \longrightarrow D(k[\varepsilon])$$

is a bijection, so the formal object is as small as versality allows.

The versality condition says that ξ can be extended along any small extension whenever the obstruction to extending the underlying deformation vanishes: no deformation over A' compatible with ξ escapes being induced by ξ. The minimality condition pins down the tangent direction: a hull has exactly the tangent space of D and no spurious first-order directions. Versality alone does not force uniqueness of the map $h_R \to A'$ lifting a given map $h_R \to A$; it guarantees existence of such a lift but permits several. This is exactly what one must give up when the objects have automorphisms, and it is the precise sense in which a hull is weaker than a representing ring.

Definition 17.3.2: Pro-representable Functor

A functor $D\colon \mathbf{Art}_k \to \mathbf{Set}$ with $D(k) = \{*\}$ is *pro-representable* if there exists $R \in \widehat{\mathbf{Art}}_k$ and an isomorphism of functors

$$h_R \xrightarrow{\sim} D$$

on $\mathbf{Art}_k$. Equivalently, D admits a formal element (R, ξ) that is *universal*: for every small extension $\pi\colon A' \to A$ the map $h_R(A') \to h_R(A) \times_{D(A)} D(A')$ is a bijection, not only a surjection.

Pro-representability is the strongest possible outcome: the deformation functor is literally the functor of points of a formal parameter space R, and every deformation over an Artinian base is induced by a unique map out of R. The prefix "pro" records that R is a limit of the Artinian quotients through which the deformations factor, so D is representable in the pro-category of $\mathbf{Art}_k$ even when it is not representable by a single object of $\mathbf{Art}_k$. A hull is a pro-representable functor with the uniqueness relaxed to existence, and the entire content of Schlessinger's theorem is to say which functors reach each of these two levels.

Before quantifying the conditions, it is worth seeing the tangent space enter as a vector space rather than a bare set. The following is the mechanism by which the finiteness hypothesis (H3) below feeds into the construction.

Proposition 17.3.3: The Tangent Space as a Vector Space

Let $D\colon \mathbf{Art}_k \to \mathbf{Set}$ satisfy $D(k) = \{*\}$ and the condition that the natural map

$$D(A' \times_A A'') \longrightarrow D(A') \times_{D(A)} D(A'')$$

is a bijection whenever $A = k$ and $A' = A'' = k[\varepsilon]$. Then the set $t_D = D(k[\varepsilon])$ carries a canonical structure of a k-vector space, natural in D, for which the bijection above is an isomorphism of the underlying additive groups $D(k[\varepsilon] \times_k k[\varepsilon]) \cong t_D \oplus t_D$.

***Proof*:**
The ring $k[\varepsilon] \times_k k[\varepsilon]$ is isomorphic to $k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1^2, \varepsilon_1\varepsilon_2, \varepsilon_2^2)$, and under the stated bijection $D(k[\varepsilon] \times_k k[\varepsilon]) \cong t_D \times t_D$. *Addition.* The addition map on $k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1, \varepsilon_2)^2$ sending both ε_1 and ε_2 to ε is a k-algebra homomorphism $\sigma\colon k[\varepsilon] \times_k k[\varepsilon] \to k[\varepsilon]$, well-defined because it carries the relations $(\varepsilon_1, \varepsilon_2)^2 = 0$ to $\varepsilon^2 = 0$, which holds in $k[\varepsilon]$. Applying D and precomposing with the bijection defines a map $t_D \times t_D \to t_D$, which is the vector addition. *Scalar multiplication.* For $\lambda \in k$, the map $\varepsilon \mapsto \lambda\varepsilon$ is a k-algebra endomorphism of $k[\varepsilon]$, and D applied to it gives scalar multiplication by λ on t_D. The zero element is the image of $D(k) \to D(k[\varepsilon])$ induced by $k \to k[\varepsilon]$, which is a single point since $D(k) = \{*\}$. The vector-space axioms follow from functoriality applied to the corresponding identities among k-algebra homomorphisms of $k[\varepsilon]$ and $k[\varepsilon] \times_k k[\varepsilon]$; associativity and commutativity of addition, for instance, come from the associativity and symmetry of σ up to the automorphisms permuting the two copies of ε. Naturality in D is immediate since every ingredient is the image under D of a fixed homomorphism of test rings, completing the proof.

The proposition shows that once a deformation functor is even slightly compatible with fiber products, its set of first-order deformations is automatically a vector space, recovering the tangent space of definition 16.1.5 from an abstract categorical hypothesis rather than a hands-on cocycle description. This is the first payoff of phrasing everything through the fiber-product map, and it is the template for the conditions to come.

With the two target notions in place, everything reduces to a small list of axioms on a single natural map. Schlessinger's conditions all concern the map

$$\eta\colon D(A' \times_A A'') \longrightarrow D(A') \times_{D(A)} D(A'') \tag{17.1}$$

induced by the two projections $A' \times_A A'' \to A'$ and $A' \times_A A'' \to A''$, defined whenever $A' \to A$ and $A'' \to A$ are morphisms in $\mathbf{Art}_k$. The map η compares gluing on the ring side with gluing on the deformation side. It is always defined; the conditions ask how close it is to a bijection under progressively weaker hypotheses on the two morphisms. A functor of this shape, with $D(k) = \{*\}$, is often called a *deformation functor* or a *functor of Artin rings*, and the conditions below are its structural axioms.

Definition 17.3.4: Schlessinger's Conditions (H1)-(H4)

Let $D\colon \mathbf{Art}_k \to \mathbf{Set}$ be a functor with $D(k) = \{*\}$. Consider morphisms $A' \to A$ and $A'' \to A$ in $\mathbf{Art}_k$ and the associated map η of (17.1). The conditions are:

- *(H1)* η is surjective whenever $A'' \to A$ is a small extension.
- *(H2)* η is bijective when $A = k$ and $A'' = k[\varepsilon]$, so that $D(A' \times_k k[\varepsilon])$ maps isomorphically onto $D(A') \times D(k[\varepsilon])$.
- *(H3)* the tangent space $t_D = D(k[\varepsilon])$ is a finite-dimensional k-vector space.
- *(H4)* η is bijective whenever $A' = A''$ and $A' \to A$ is the *same* small extension, that is, whenever $A' \to A$ is a small extension and $A'' = A'$ with the identical map.

The four conditions are graded in strength and in what they control. Condition (H1) is a lifting axiom: it says two deformations that agree over A can be combined into a deformation over the fiber product $A' \times_A A''$ whenever one of the two sides is a small extension, so gluing along small extensions never fails for lack of an object. Condition (H2) makes η a bijection in the single case that governs the tangent space, and combined with (H1) it is exactly the hypothesis of proposition 17.3.3, so under (H1) and (H2) the tangent space t_D is a k-vector space. Condition (H3) makes that vector space finite-dimensional, which is what allows the construction to terminate in Noetherian rings at each stage. Condition (H4) is the uniqueness upgrade: it demands that η be an actual bijection, not just a surjection, in the self-fiber-product case that detects whether distinct liftings can be told apart, and it is precisely the condition that distinguishes a universal formal deformation from a versal one.

Before the theorem it helps to see how (H4) fails in the presence of automorphisms, since that failure is the entire reason pro-representability is the exception rather than the rule. Consider deforming an object E with a nontrivial automorphism group. Two liftings of a given deformation over A to the fiber product $A' \times_A A'$ can differ by the action of an automorphism that is the identity over A but nontrivial over A'; such an automorphism produces two different elements of $D(A' \times_A A')$ mapping to the same element of $D(A') \times_{D(A)} D(A')$, so η is not injective and (H4) fails. This is the same phenomenon that obstructs fine representability in theorem 13.3.6: the automorphisms of an elliptic curve are what force the j-line to be a coarse rather than fine space.

Everything now assembles into the central result. The theorem is stated for a functor D on $\mathbf{Art}_k$ with $D(k) = \{*\}$, and its two clauses give the two levels of solution introduced above.

Theorem 17.3.5: Schlessinger's Theorem

Let $D\colon \mathbf{Art}_k \to \mathbf{Set}$ be a functor with $D(k) = \{*\}$.

1. D admits a hull if and only if it satisfies conditions (H1), (H2), and (H3) of definition 17.3.4.
2. D is pro-representable if and only if it satisfies (H1), (H2), (H3), and (H4).

Moreover, when a hull exists it is unique up to (non-canonical) isomorphism, and its tangent space is canonically $t_D = D(k[\varepsilon])$.

The theorem is due to Schlessinger [60]. On the necessity side, if D has a hull (R, ξ), then h_R

satisfies all four conditions (part (a) of the exercises), and (H1) for D follows because the versal map $h_R \to D$ is smooth: given elements of $D(A')$ and $D(A'')$ agreeing over $D(A)$, versality lifts each into h_R, and (H1) for h_R then produces the required element of $D(A' \times_A A'')$. Finiteness of $\mathrm{Hom}(R, k[\varepsilon]) = (\mathfrak{m}_R/\mathfrak{m}_R^2)^\vee$ forces (H3). The full necessity of (H1) and (H2), whose (H2) half asserts *bijectivity* of η in the tangent case and so requires the minimality of the tangent map rather than versality alone, uses a separate minimality argument and is deferred to [60] and the exercises; the necessity is not needed for the hull-existence construction below. The substance is the converse for part (1): the construction of a hull from (H1)-(H3), carried out below in full. The upgrade to part (2) under (H4) is then a bookkeeping refinement of the same construction, tracking uniqueness at each stage.

***Proof*:**
Necessity of the conditions was discussed above and is deferred in part to [60]; a versal formal element makes the maps in definition 17.3.4 surjective and (H3) is the finiteness of the cotangent space of a Noetherian complete local ring, while the bijectivity in (H2) and (H4) uses the minimality of a hull. The content is the construction of a hull assuming (H1), (H2), and (H3); this occupies the remainder of the proof. Uniqueness of the hull and the upgrade to pro-representability under (H4) are addressed at the end.

Write $t = t_D = D(k[\varepsilon])$, a finite-dimensional k-vector space by (H1)-(H3) and proposition 17.3.3, and set $d = \dim_k t$. Let $S = k[[x_1, \dots, x_d]]$ be the power-series ring in d variables with maximal ideal $\mathfrak{n}$, so that $\mathrm{Hom}_{k\text{-alg}}(S, k[\varepsilon]) = (\mathfrak{n}/\mathfrak{n}^2)^\vee$ is d-dimensional and canonically dual to t. The strategy is to build the hull R as a quotient S/J for an ideal J obtained as the limit of a decreasing chain of ideals, constructing at each stage the smallest quotient of S through which a versal family extends.

Step 1: The tower of quotients. Fix a k-basis of t and let $x_1, \dots, x_d$ correspond to the dual basis, so that the tangent element of any quotient S/J with $J \subseteq \mathfrak{n}^2$ is identified with t itself. The hull will be produced as a limit $R = \varprojlim_q R_q$ of a tower

$$S/\mathfrak{n}^2 = R_1 \twoheadleftarrow R_2 \twoheadleftarrow R_3 \twoheadleftarrow \cdots$$

in which each R_q is a quotient S/J_q with $\mathfrak{n}^{q+1} \subseteq J_q \subseteq \mathfrak{n}^2$, so that R_q is Artinian, together with a compatible system of elements $\xi_q \in D(R_q)$. At each stage the ideal J_q is chosen as small as possible subject to ξ_{q-1} extending to R_q; smallness of J_q is what forces the resulting formal element to have the correct tangent space and to be versal.

Step 2: The base of the tower. Set $R_1 = S/\mathfrak{n}^2$. The tangent space of R_1 is $\mathrm{Hom}(R_1, k[\varepsilon]) = (\mathfrak{n}/\mathfrak{n}^2)^\vee \cong t$. Since $D(k[\varepsilon]) = t$ and R_1 has been built with tangent space canonically t, the identity of t corresponds under the duality to a distinguished element $\xi_1 \in D(R_1)$: concretely, $R_1 \cong k[\varepsilon] \times_k \cdots \times_k k[\varepsilon]$ (d factors), and by iterating (H2) the map $D(R_1) \to D(k[\varepsilon])^d = t^d$ is a bijection, so ξ_1 is the unique element mapping to the tuple of basis vectors. This ξ_1 induces the identity on tangent spaces by construction.

Step 3: The inductive extension. Suppose $R_q = S/J_q$ and $\xi_q \in D(R_q)$ have been constructed with $\mathfrak{n}^{q+1} \subseteq J_q \subseteq \mathfrak{n}^2$ and ξ_q inducing the fixed identification on tangent spaces. To pass to stage $q+1$, consider the collection of ideals

$$\mathcal{J} = \{J \subseteq S \mid \mathfrak{n}J_q \subseteq J \subseteq J_q \text{ and } \xi_q \text{ lifts to } D(S/J)\}$$

where ξ_q *lifts to* $D(S/J)$ means that $\xi_q \in D(S/J_q)$ lies in the image of $D(S/J) \to D(S/J_q)$ induced by the quotient $S/J \twoheadrightarrow S/J_q$. Because $\mathfrak{n}J_q \subseteq J \subseteq J_q$ and $J_q \supseteq \mathfrak{n}^{q+1}$, every such quotient

$S/J \to S/J_q$ is a composite of small extensions, so lifting is controlled by (H1). Here and below (H1), stated for a single small extension, is applied to a composite of small extensions by factoring the surjection into a chain of small extensions and lifting one stage at a time; each stage's kernel is a one-dimensional subspace of the total kernel, which is finite-dimensional and annihilated by $\mathfrak{n}$. The claim is that $\mathcal{J}$ has a smallest element J_{q+1}, and this smallest element, with any chosen lift ξ_{q+1}, gives the next stage.

Step 4: $\mathcal{J}$ is closed under intersection. Let $J', J'' \in \mathcal{J}$. To show $J' \cap J'' \in \mathcal{J}$, first reduce to $\mathfrak{n}J_q \subseteq J' \cap J''$, which is immediate, and $J' \cap J'' \subseteq J_q$, also immediate. It remains to show ξ_q lifts to $D(S/(J' \cap J''))$. There is a fiber-product presentation

$$S/(J' \cap J'') \cong S/J' \times_{S/(J'+J'')} S/J''$$

since the kernel of $S \to S/J' \times S/J''$ is exactly $J' \cap J''$ and the image is the fiber product over $S/(J' + J'')$. The projection $S/J'' \to S/(J' + J'')$ has kernel $(J' + J'')/J''$, which is annihilated by $\mathfrak{n}$ because $\mathfrak{n}(J' + J'') \subseteq \mathfrak{n}J_q \subseteq J' \cap J'' \subseteq J''$; a finitely generated module annihilated by $\mathfrak{n}$ is a finite-dimensional k-vector space, so this projection is a composite of small extensions, filtering the kernel by one-dimensional subspaces. By (H1) the map

$$D\big(S/(J' \cap J'')\big) \longrightarrow D(S/J') \times_{D(S/(J'+J''))} D(S/J'')$$

is surjective. Now ξ_q lifts to elements $\eta' \in D(S/J')$ and $\eta'' \in D(S/J'')$ by hypothesis, and both η' and η'' restrict to the same element of $D(S/(J' + J''))$, namely the further pushforward of ξ_q, because $S/(J' + J'')$ is a common quotient of S/J_q through which both restrictions factor. Thus (η', η'') is an element of the target fiber product, and by surjectivity it is the image of some $\zeta \in D(S/(J' \cap J''))$. Pushing ζ forward to $D(S/J_q)$ recovers ξ_q, so ξ_q lifts to $D(S/(J' \cap J''))$ and $J' \cap J'' \in \mathcal{J}$.

Step 5: Existence of a smallest element. Every $J \in \mathcal{J}$ satisfies $\mathfrak{n}J_q \subseteq J \subseteq J_q$, and $J_q/\mathfrak{n}J_q$ is a finite-dimensional k-vector space because S is Noetherian and J_q is finitely generated. The ideals in $\mathcal{J}$ correspond to k-subspaces of $J_q/\mathfrak{n}J_q$, and by Step 4 this collection of subspaces is closed under finite intersection. A collection of subspaces of a finite-dimensional vector space that is closed under intersection has a smallest member, namely the intersection of all of them, which is a finite intersection since the dimension is finite. Let J_{q+1} be this smallest ideal and set $R_{q+1} = S/J_{q+1}$. By definition ξ_q lifts to some $\xi_{q+1} \in D(R_{q+1})$; fix one such lift. This completes the inductive step, and the minimality of J_{q+1} is the "smallest quotient through which the versal family extends" that drives the construction.

Step 6: Passage to the limit. The ideals J_q form a decreasing chain with $\bigcap_q J_q =: J$, and setting $R = S/J = \varprojlim_q R_q$ produces a complete local Noetherian k-algebra; completeness holds because $\mathfrak{n}^{q+1} \subseteq J_q$ forces the $\mathfrak{m}_R$-adic topology to agree with the tower. The compatible system (ξ_q) defines a formal element $\xi = (\xi_q) \in \varprojlim_q D(R_q)$, that is, a morphism $h_R \to D$. Its tangent map is the identification of t with $D(k[\varepsilon])$ from Step 2, hence a bijection, so ξ satisfies the minimality requirement of definition 17.3.1 at once.

Step 7: Versality. It remains to show ξ is versal, meaning that for every small extension $\pi\colon A' \to A$ the map $h_R(A') \to h_R(A) \times_{D(A)} D(A')$ is surjective. Fix such a π and an element (u, ζ) of the target, where $u\colon R \to A$ is a k-algebra map and $\zeta \in D(A')$ restricts along π to $u_*\xi \in D(A)$. Since A is Artinian, u factors through some $R_q = S/J_q$, giving $\bar{u}\colon R_q \to A$ with $\bar{u}_*\xi_q = u_*\xi$. Choose a lift of $\bar{u}$ to a k-algebra homomorphism $v\colon S \to A'$: this is possible because S is a power-series ring, hence (formally) smooth, so any map to the quotient A lifts to the small extension A'. The composite $S \xrightarrow{v} A'$ need not kill J_{q+1}, but consider the ideal $J' = J_q \cap \ker\big(v \bmod (t)\big)$ inside S; more directly, form the fiber product

$$B = R_q \times_A A'$$

which sits in a small extension $B \to R_q$ with the same kernel (t) as π, since π is small and the fiber product with the surjection $\bar{u}$ preserves the kernel. By (H1) applied to the small extension $A' \to A$ and the map $R_q \to A$, the element $\xi_q \in D(R_q)$ and $\zeta \in D(A')$, agreeing over A, lift to an element $\beta \in D(B)$. Now $B = R_q \times_A A'$ is a quotient of S by an ideal J_B with $\mathfrak{n}J_q \subseteq J_B \subseteq J_q$: the surjection $S \to R_q \to A$ lifts to $S \to A'$ as above, giving $S \to B$, and its kernel J_B contains $\mathfrak{n}J_q$ because $B \to R_q$ is a small extension with $\mathfrak{m}_B \cdot (t) = 0$. The element β shows ξ_q lifts to $D(S/J_B)$, so $J_B \in \mathcal{J}$ and therefore $J_{q+1} \subseteq J_B$ by minimality. Hence $S \to B$ factors through $R_{q+1} = S/J_{q+1}$, yielding a map $w\colon R_{q+1} \to B \to A'$ whose composite $R_{q+1} \to A' \to A$ is $\bar{u}$ precomposed with $R_{q+1} \to R_q$, and pulling ξ_{q+1} back along w recovers (u, ζ) up to the lift chosen. Composing w with the projection $h_R \to h_{R_{q+1}}$ gives the required element of $h_R(A')$ mapping to (u, ζ). This establishes surjectivity, so ξ is versal, and (R, ξ) is a hull.

Step 8: Uniqueness of the hull. If (R, ξ) and (R', ξ') are two hulls, versality of each produces maps $h_{R'} \to h_R$ and $h_R \to h_{R'}$ compatible with the formal elements. The composite $h_R \to h_R$ induces the identity on tangent spaces (both hulls have tangent space t), and a map of complete local rings inducing an isomorphism on cotangent spaces $\mathfrak{m}/\mathfrak{m}^2$ is surjective by the complete Nakayama lemma; a surjective endomorphism of a Noetherian complete local ring is an isomorphism. Hence both composites are isomorphisms, and $R \cong R'$, though the isomorphism depends on the choices of lift and is not canonical. This proves clause (1).

Step 9: Pro-representability under (H4). Assume in addition (H4). The only place the construction used surjectivity rather than bijectivity of η was in the lifting steps; (H4) makes η bijective in the self-fiber-product case $A' = A''$, which is exactly the case that measures whether two lifts of a map $h_R \to A$ to $h_R \to A'$ can differ. Concretely, given $u\colon R \to A$ and a small extension $A' \to A$, two lifts $R \to A'$ of u differ by a k-derivation $R \to (t)$ that kills $\ker u$ and $\mathfrak{m}_R^2$, since (t) is annihilated by $\mathfrak{m}_{A'}$; hence the set of lifts of u, when nonempty, is a torsor under $\mathrm{Hom}(\mathfrak{m}_R/(\mathfrak{m}_R^2 + \ker u), (t))$, and the set of lifts of $u_*\xi$ in $D(A')$ is correspondingly a torsor under $t_D \otimes_k (t)$, which is $\cong t_D$ since (t) is one-dimensional, via the tangent-space action; (H4) forces the natural map between these two torsors, $h_R(A') \to h_R(A) \times_{D(A)} D(A')$, to be injective. Injectivity together with the surjectivity from Step 7 makes η for $h_R \to D$ a bijection on every small extension, so $h_R \to D$ is an isomorphism on all of $\mathbf{Art}_k$ by induction on the length of the Artinian ring, factoring any A as a tower of small extensions of k. Thus $D = h_R$ is pro-representable. Conversely, pro-representability makes every η a bijection, in particular (H4), completing the proof of clause (2) and the theorem, as we sought to show.

The proof follows a single template: build the parameter ring one infinitesimal order at a time, and at each order divide out by the smallest ideal that still allows the versal family to extend. Smallness is what makes the tangent space come out correctly, since a larger ideal would kill tangent directions that D possesses, and it is also what makes the hull unique. The single categorical input at every stage is condition (H1), which guarantees that deformations agreeing over a common base glue over the fiber product; (H2) and (H3) enter only to make the tangent space a finite-dimensional vector space, fixing the number of variables in the power-series ring. The passage from a hull to full pro-representability is entirely about injectivity, and injectivity is exactly what automorphisms destroy, which is why (H4) is the automorphism-free condition.

One structural remark clarifies the role of the completion. The hull R is a formal object: it lives in $\widehat{\mathbf{Art}}_k$, not in $\mathbf{Art}_k$, and represents D only in the pro-sense of a limit over the finite-length quotients. Whether this formal deformation *algebraizes* to an actual family over a ring of finite type, or over a Henselian ring, is a separate question answered by Artin's approximation and algebraization theorems; those provide the bridge from the formal local structure produced here to genuine algebraic moduli, and they are the subject of the algebraic-stacks development later in the book. For now the hull is the sharpest local invariant available, and Schlessinger's criteria are the precise test for its existence.

The conditions (H1)-(H3) are easy to verify for the deformation functors that arise in geometry, so the existence of a hull is the generic situation. Pro-representability is the exceptional case, gated entirely by (H4), and (H4) holds exactly when the objects being deformed have no infinitesimal automorphisms. The following example makes both statements concrete, tying the abstract criteria back to the cohomology groups of definition 16.1.5.

Example 17.3.6: Hulls and Universal Deformations in Geometry
Let X be a smooth projective variety over k and let $D = \mathrm{Def}_X$ be its deformation functor, sending $A \in \mathbf{Art}_k$ to the set of isomorphism classes of flat families $\mathcal{X} \to \operatorname{Spec} A$ with special fiber X.

1. Def_X *has a hull.* Conditions (H1) and (H2) follow from the gluing property of flat families: given flat deformations $\mathcal{X}' \to \operatorname{Spec} A'$ and $\mathcal{X}'' \to \operatorname{Spec} A''$ agreeing over $\operatorname{Spec} A$, with $A'' \to A$ a small extension, the flat family over $A' \times_A A''$ is obtained by gluing $\mathcal{X}'$ and $\mathcal{X}''$ along their common restriction, and flatness over the fiber product is checked on the local rings, where it reduces to the fact that $A' \times_A A''$ is the fiber product and a module flat over both factors and agreeing over A is flat over the fiber product (the standard flatness-descent lemma for a fiber square of rings, see [10]). This gives (H1); taking $A = k$, $A'' = k[\varepsilon]$ makes the gluing an equivalence, giving (H2). Condition (H3) is the finiteness of the tangent space, which for the smooth X under consideration is
$$t_{\mathrm{Def}_X} = D(k[\varepsilon]) = \operatorname{Ext}^1_X(\Omega_{X/k}, \mathcal{O}_X) = H^1(X, T_X)$$
the group of theorem 16.3.2, the second equality holding because $\Omega_{X/k}$ is locally free on the smooth X, so $\operatorname{Ext}^1_X(\Omega_{X/k}, \mathcal{O}_X) = H^1(X, \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)) = H^1(X, T_X)$. This is a finite-dimensional k-vector space because X is projective and $H^1(X, T_X)$ is coherent cohomology on a projective variety, hence finite over k. (For a singular scheme the first-order deformation space is instead $T^1_X = \operatorname{Ext}^1_X(\mathbf{L}_{X/k}, \mathcal{O}_X)$, which differs from $\operatorname{Ext}^1_X(\Omega_{X/k}, \mathcal{O}_X)$ once X is not lci; the smooth case sidesteps this correction.) By theorem 17.3.5(1), Def_X admits a hull. The hull is a complete local k-algebra whose tangent space is $H^1(X, T_X)$ and whose relations are cut out by the obstruction map into $H^2(X, T_X)$; when X is unobstructed the hull is the power-series ring $k[[H^1(X, T_X)]]$, by which is meant the completed symmetric algebra on $H^1(X, T_X)$, that is, the power-series ring on a chosen k-basis of $H^1(X, T_X)$.
2. *No infinitesimal automorphisms gives pro-representability.* The functor Def_X fails (H4) exactly when X carries infinitesimal automorphisms, that is, when $T^0 = H^0(X, T_X) \neq 0$: a nonzero global vector field integrates to a one-parameter family of automorphisms over the dual numbers, and this automorphism produces two distinct elements of $\mathrm{Def}_X(A' \times_A A')$ with the same image under η, exactly the failure of injectivity described after definition 17.3.4. When $H^0(X, T_X) = 0$ the object X is infinitesimally rigid as an automorphism problem, every lifting is unique, η is injective in the self-fiber-product case, and (H4) holds. By theorem 17.3.5(2), Def_X is then pro-representable: there is a complete local k-algebra R and a universal formal deformation $\mathcal{X}_{\mathrm{univ}} \to \operatorname{Spec} R$ from which every deformation of X over

an Artinian base is induced by a unique map. A curve C of genus $g \geq 2$ is the standard instance: $H^0(C, T_C) = 0$ because $\deg T_C = 2 - 2g < 0$, so Def_C is pro-representable with hull $k[[t_1, \ldots, t_{3g-3}]]$, since $\dim H^1(C, T_C) = 3g - 3$ and $H^2(C, T_C) = 0$ on a curve.

The two clauses of the example separate cleanly. The existence of a hull is a soft consequence of the geometry of flat families and the finiteness of coherent cohomology on a projective scheme, so essentially every geometric deformation functor clears (H1)-(H3). Whether that hull is universal is a question about automorphisms alone: $H^0(X, T_X)$ measures the infinitesimal automorphisms, and it vanishing is both necessary and sufficient for the hull to become a genuine universal formal deformation. The genus $g \geq 2$ curves are pro-representable and their hull is a smooth $(3g-3)$-dimensional formal disk, which is the local model of the moduli stack $\mathcal{M}_g$ and the reason $\mathcal{M}_g$ is smooth of dimension $3g - 3$. For an elliptic curve, by contrast, $H^0(E, T_E) = k \neq 0$, the functor is not pro-representable, and one is forced into the coarse space of theorem 13.3.6 or the stack-theoretic correction of Part VI.

17.3.7 Historical Note: Schlessinger's Functors of Artin Rings The criteria and the hull-existence theorem are the main results of Schlessinger's 1968 paper *Functors of Artin Rings* [60], which grew out of his 1964 Harvard thesis under John Tate. The framework recast deformation theory, previously a collection of cohomological computations, as the study of functors on $\mathbf{Art}_k$, and it made the hull the correct universal object in the presence of automorphisms, before the language of stacks made the automorphisms themselves into geometry. The four conditions have been the standard entry point to formal deformation theory ever since.

Exercise 17.3.1

1. Let $R \in \widehat{\mathbf{Art}}_k$ be a complete local k-algebra with residue field k. Show directly that the functor $h_R = \mathrm{Hom}_{k\text{-alg}}(R, -)$ satisfies conditions (H1), (H2), and (H4) of definition 17.3.4, and that it satisfies (H3) if and only if R is Noetherian. Conclude that a pro-representable functor satisfies all four conditions, recovering one direction of theorem 17.3.5(2).

2. Using proposition 17.3.3, show that for $R \in \widehat{\mathbf{Art}}_k$ Noetherian the tangent space of h_R is canonically $t_{h_R} = (\mathfrak{m}_R/\mathfrak{m}_R^2)^\vee$, the k-linear dual of the cotangent space. Deduce that if h_R is a hull of a functor D, then $\dim_k D(k[\varepsilon])$ equals the minimal number of generators of $\mathfrak{m}_R$.

3. Give an example of a functor $D\colon \mathbf{Art}_k \to \mathbf{Set}$ with $D(k) = \{*\}$ that satisfies (H1) and (H2) but not (H3), and explain in one sentence why the hull-existence construction of theorem 17.3.5 breaks down for it. (Suggestion: take the deformation functor of an infinite-dimensional object, or a functor whose tangent space is $k^{\oplus \mathbb{N}}$.)

4. Let X be a smooth projective variety with $H^0(X, T_X) = 0$ and $H^2(X, T_X) = 0$. Show that Def_X is pro-representable and that its hull is a power-series ring $k[[t_1, \ldots, t_n]]$ with $n = \dim_k H^1(X, T_X)$. Identify n for a smooth hypersurface of degree $d \geq 5$ in $\mathbb{P}^3$ (so that X is of general type and $H^0(T_X) = 0$), given the standard fact that for such a surface every deformation of X is realized by varying the defining degree-d form, so that $H^1(X, T_X)$ is computed by the space of degree-d forms modulo scaling and the projective linear group. State what geometric family this hull parametrizes, and explain why the same count *fails* to give $\dim_k H^1(X, T_X)$ for the quartic $d = 4$.

5. Using the failure of injectivity of η described after definition 17.3.4, explain precisely which

automorphisms obstruct (H4) for Def_X, defined as isomorphism classes of flat families with special fiber X. Show that the relevant automorphisms are the *infinitesimal* ones, measured by $H^0(X, T_X)$, and explain why a nontrivial *finite* automorphism group (for instance the hyperelliptic involution of a genus-2 curve C, for which $H^0(C, T_C) = 0$) does *not* obstruct (H4), so that Def_C is pro-representable. Where, then, do the finite automorphisms make themselves felt, and why does this concern the global moduli stack rather than the local functor Def_X?

17.4 Applications and Computations

The three sections before this one built an apparatus and left it idling: the obstruction space $H^2(X, T_X)$ and its algebraic avatar $T^2(A/k, A)$ (section 17.1, section 17.2), the criterion by which a deformation functor acquires a hull (section 17.3). This section puts the apparatus to work. The payoff of an obstruction theory is that it decides, for a given object, whether the deformations one can write down to first order extend to families over larger bases, and when they do, it pins down the local structure of the moduli space. Three questions organize the section: which classical objects are *unobstructed*, so that their moduli spaces are smooth; what the deformations of the simplest singular curve, the node, look like; and what an obstructed deformation, one that exists to first order but dies at second order, looks like when written out.

The first question is answered for the objects this book most cares about: curves, abelian varieties, and K3 surfaces are all unobstructed, and their moduli spaces are correspondingly smooth of the expected dimension. The second question produces the single number 1, the dimension of the space of smoothings of a node, and that number is the local fact behind the stable-reduction theorem that the moduli of curves in chapter 13's eventual sequel will need: a node deforms in exactly one way that separates its two branches, and this one-parameter smoothing $xy = t$ is the local model for how a nodal degeneration is resolved. The third question is settled by an example, the cone over a projective variety chosen so that a first-order deformation of the cone fails to lift, exhibiting $T^2 \neq 0$ in the flesh rather than as an abstract receptacle.

We begin with the unobstructedness results, since they are what make the moduli spaces of Part VI's capstone smooth.

Proposition 17.4.1: Unobstructedness of Curves, Abelian Varieties, and K3 Surfaces

Let k be an algebraically closed field of characteristic 0.

1. Every smooth projective curve C over k is unobstructed: $H^2(C, T_C) = 0$, so Def_C is formally smooth and its hull is a power-series ring in $h^1(C, T_C)$ variables. For $g = \mathrm{genus}(C) \geq 2$ this dimension is $3g - 3$, and $\mathcal{M}_g$ is smooth of dimension $3g - 3$.

2. Every abelian variety A of dimension g over k is unobstructed: $T_A \cong \mathcal{O}_A^{\oplus g}$ is trivial and $H^*(A, \mathcal{O}_A) \cong \bigwedge^* H^1(A, \mathcal{O}_A)$ is an exterior algebra, so $H^i(A, T_A) \cong \left(\bigwedge^i H^1(A, \mathcal{O}_A)\right) \otimes_k \mathfrak{a}$ with $\mathfrak{a} = T_{A,0}$; the deformation space $H^1(A, T_A)$ has dimension g^2, and the moduli space $\mathcal{A}_g$ of principally polarized abelian varieties is smooth of dimension $\binom{g+1}{2}$.

3. Every K3 surface S over k is unobstructed: $H^2(S, T_S) = 0$ (via the Calabi-Yau isomorphism $T_S \cong \Omega^1_S$ and $h^{1,0}(S) = 0$), so Def_S is formally smooth with hull a power-series ring in $h^1(S, T_S) = 20$ variables, and the moduli space of K3 surfaces is smooth of dimension 20 (respectively 19 for a fixed polarization). For a Calabi-Yau variety of dimension ≥ 3 the group $H^2(X, T_X)$ need not vanish, yet unobstructedness still holds: this is the Bogomolov-Tian-Todorov theorem, stated below with its proof cited.

***Proof*:**
The obstruction to lifting a deformation of a smooth projective X over a small extension lies in $H^2(X, T_X)$ by theorem 17.1.2, and unobstructedness in the sense of definition 17.1.3 follows as soon as this group vanishes; parts (1) and (3) proceed by exhibiting the vanishing, while part (2) has $H^2(A, T_A) \neq 0$ for $g \geq 2$ and requires a substitute argument. Throughout, the first-order deformation space is $H^1(X, T_X)$ by theorem 16.3.2, so once formal smoothness is established the hull is a power-series ring on $h^1(X, T_X)$ generators by definition 17.1.3.

Part (1): curves. A smooth projective curve C has dimension 1, so T_C is a coherent sheaf on a 1-dimensional scheme and $H^2(C, T_C) = 0$ by Grothendieck vanishing, since cohomology of a coherent sheaf vanishes above the dimension of the space (**??**). Thus C is unobstructed. The tangent space is $H^1(C, T_C)$ with $T_C = \omega_C^{-1}$ the anticanonical bundle, of degree $2 - 2g$; by Serre duality $H^1(C, T_C) \cong H^0(C, \omega_C \otimes T_C^{-1})^\vee = H^0(C, \omega_C^{\otimes 2})^\vee$, and $\omega_C^{\otimes 2}$ has degree $4g - 4 > 2g - 2$ for $g \geq 2$, so $H^1(C, \omega_C^{\otimes 2}) = 0$ and Riemann-Roch gives

$$h^1(C, T_C) = h^0(C, \omega_C^{\otimes 2}) = \deg(\omega_C^{\otimes 2}) - g + 1 = (4g - 4) - g + 1 = 3g - 3$$

Hence the hull of Def_C is $k[[t_1, \ldots, t_{3g-3}]]$ and, since the deformation theory of $\mathcal{M}_g$ at $[C]$ is governed by Def_C, the coarse moduli space is smooth of dimension $3g - 3$ at a point with no extra automorphisms.

Part (2): abelian varieties. An abelian variety is a smooth projective group scheme, and translations act transitively, so the tangent bundle is globally generated by the g translation-invariant vector fields forming a basis of the Lie algebra $\mathfrak{a} = T_{A,0}$. This trivializes T_A:

$$T_A \cong \mathcal{O}_A \otimes_k \mathfrak{a} \cong \mathcal{O}_A^{\oplus g}$$

Consequently $H^i(A, T_A) \cong H^i(A, \mathcal{O}_A) \otimes_k \mathfrak{a}$, and the cohomology of the structure sheaf of an abelian variety is the exterior algebra on $H^1(A, \mathcal{O}_A)$, a g-dimensional space: $H^i(A, \mathcal{O}_A) \cong$

$\bigwedge^i H^1(A, \mathcal{O}_A)$, so $h^i(A, \mathcal{O}_A) = \binom{g}{i}$. In particular $H^2(A, T_A) \cong \bigwedge^2 H^1(A, \mathcal{O}_A) \otimes \mathfrak{a}$ is nonzero for $g \geq 2$, so the vanishing route fails, yet unobstructedness holds for a different reason. In characteristic 0 the polarized deformations of A are realized by a single explicit family: varying the period lattice, equivalently the point of the Siegel upper half-space, produces a family of abelian varieties over a smooth base whose Kodaira-Spencer map at $[A]$ is an isomorphism onto the polarized first-order deformations. Every polarized first-order deformation is therefore tangent to an actual family over a smooth base and so extends to all orders, which is exactly unobstructedness. Concretely, the local moduli of A together with a principal polarization is the analytic germ of the symmetric space $\mathrm{Sp}(2g, \mathbb{R})/U(g)$, which is smooth of dimension $\binom{g+1}{2}$; algebraically, Def_A has tangent space $H^1(A, T_A) \cong \bigwedge^1 H^1(A, \mathcal{O}_A) \otimes \mathfrak{a}$ of dimension $g \cdot g = g^2$ (all deformations of the abelian variety forgetting the polarization), cut down to $\binom{g+1}{2}$ by the symmetry condition the polarization imposes, and the germ is smooth. The full argument that the obstruction map vanishes on abelian varieties, via the triviality of T_A and the Hodge-theoretic splitting, is [61, Ch. 3]; here the vanishing of the obstruction is the content of the classical construction of $\mathcal{A}_g$ as a quotient of Siegel space.

Part (3): K3 surfaces. A K3 surface S is a smooth projective surface with $\omega_S \cong \mathcal{O}_S$ and $H^1(S, \mathcal{O}_S) = 0$. Since $\omega_S \cong \mathcal{O}_S$ and $\dim S = 2$, the tangent bundle satisfies $T_S = (\Omega^1_S)^\vee \cong \Omega^1_S \otimes \omega_S^{-1} \cong \Omega^1_S$, using that on a surface $(\Omega^1_S)^\vee \cong \Omega^1_S \otimes (\det \Omega^1_S)^{-1} = \Omega^1_S \otimes \omega_S^{-1}$. Serre duality then gives

$$H^2(S, T_S) \cong H^0(S, T_S^\vee \otimes \omega_S)^\vee \cong H^0(S, \Omega^1_S)^\vee$$

and $h^0(S, \Omega^1_S) = h^{1,0}(S) = 0$ for a K3, so $H^2(S, T_S) = 0$. By theorem 17.1.2 the K3 is unobstructed, with hull $k[[t_1, \ldots, t_{20}]]$, where $h^1(S, T_S) = h^1(S, \Omega^1_S) = h^{1,1}(S) = 20$ by the same Calabi-Yau isomorphism and Hodge theory. For a Calabi-Yau variety X of dimension ≥ 3 the group $H^2(X, T_X)$ need not vanish, so the vanishing route fails and unobstructedness is the substantive Bogomolov-Tian-Todorov theorem, whose proof uses the Tian-Todorov lemma on the vanishing of a certain bracket in the differential-graded Lie algebra controlling deformations. That proof requires Hodge-theoretic input beyond the scope of this book, so it is cited rather than reproduced ([61, Thm. 14.10], [35, §6]), consistent with the deferral policy for results whose machinery lies outside the present development. In every case the moduli space is smooth of the stated dimension, completing the proof.

The three parts illustrate the two routes to unobstructedness that recur throughout the subject. The first route, taken by curves, is to kill the obstruction space outright: when $H^2(X, T_X) = 0$ there is nowhere for an obstruction to live, and formal smoothness is automatic. This is the cleanest situation and is exactly why $\mathcal{M}_g$ is smooth, a fact the moduli of curves relies on completely. The second route, taken by abelian varieties and by general Calabi-Yau manifolds, is subtler: the obstruction space is nonzero, yet the obstruction map into it is identically zero, so deformations lift despite having somewhere to fail. Abelian varieties see this through the triviality of the tangent bundle and the explicit construction via period lattices; Calabi-Yau manifolds see it through the Bogomolov-Tian-Todorov theorem. The lesson is that a nonzero H^2 is a warning, not a verdict: it flags where obstructions could appear, but only the vanishing of the actual obstruction cocycle, computed in theorem 17.1.2, decides whether the moduli space is smooth. For a K3 the warning is empty because $H^2(S, T_S)$ vanishes; for a Calabi-Yau threefold it is a nonzero group whose obstruction map is nonetheless zero, and the vanishing of that map is the content of Bogomolov-Tian-Todorov.

With smoothness in hand for the objects Part VI will compactify, the focus turns from smooth objects to singular ones, where the geometric $H^i(T_X)$ picture no longer suffices and the algebraic cotangent functors $T^i(A/k, A)$ of section 17.2 take over. The governing example is the node, both because it is the simplest singular point of a curve and because its deformation theory is the local input to the stable-reduction theorem. The computation is short enough to carry out in full from the naive cotangent complex.

Example 17.4.2: Deforming the Node

Let $A = k[x,y]/(xy)$, the coordinate ring of a node: two lines meeting transversally at the origin of $\mathbb{A}^2$, an A_1 (equivalently ordinary double point) singularity. Up to completion of the local ring, this is the local model of a nodal point on a curve. Its cotangent functors $T^i(A/k, A)$ are computed from the presentation $A = P/I$ with $P = k[x,y]$ and $I = (xy)$, a hypersurface, hence a complete intersection.

The naive cotangent complex. By definition 17.2.1 the two-term complex $L_{A/k}$ attached to this presentation is

$$I/I^2 \xrightarrow{d} \Omega_{P/k} \otimes_P A$$

placed in degrees 1 and 0. Here I/I^2 is free of rank 1 over A on the class of $f = xy$ (a single relation, no syzygies among a single generator, so the second-syzygy term vanishing gives $T^2 = 0$ immediately for a hypersurface, as recorded in example 17.2.4), and $\Omega_{P/k} \otimes_P A = A\,dx \oplus A\,dy$ is free of rank 2. The differential sends the generator $f = xy$ to

$$d(xy) = y\,dx + x\,dy \in A\,dx \oplus A\,dy$$

Computing T^0, T^1, T^2. The functors $T^i(A/k, A)$ are the cohomology of $\mathrm{Hom}_A(L_{A/k}, A)$ in the appropriate degrees, following definition 17.2.2 and theorem 17.2.3. Dualizing the two-term complex against A gives

$$\mathrm{Hom}_A(\Omega_{P/k} \otimes_P A,\ A) \xrightarrow{d^\vee} \mathrm{Hom}_A(I/I^2,\ A)$$

that is, $A^2 \xrightarrow{d^\vee} A$, where a pair $(a,b) \in A^2$ (representing the derivation $a\,\partial_x + b\,\partial_y$) maps to

$$d^\vee(a,b) = a \cdot y + b \cdot x \in A$$

the pairing of the derivation with the relation $d(xy) = y\,dx + x\,dy$. Then $T^0(A/k, A) = \ker d^\vee = \mathrm{Der}_k(A, A)$ is the module of vector fields, $T^1(A/k, A) = \mathrm{coker} d^\vee$ is the deformation space, and $T^2(A/k, A)$ is the obstruction space.

First, $T^2(A/k, A) = 0$: the node is a hypersurface, hence a complete intersection, hence l.c.i., and l.c.i. singularities have no second syzygies among a single defining equation, so they are unobstructed by example 17.2.4. There is no obstruction to deforming a node.

Next, $T^1(A/k, A) = \mathrm{coker}(d^\vee) = A/(x,y)$. The image of $d^\vee$ is the ideal generated by the two coordinates $d^\vee(1,0) = y$ and $d^\vee(0,1) = x$, so it is $(x,y) \subseteq A$, and the cokernel is

$$T^1(A/k, A) = A/(x,y) = k$$

a one-dimensional k-vector space. This is the Tjurina algebra $A/(f, \partial_x f, \partial_y f) = k[x,y]/(xy,\ y,\ x) = k$ of the isolated hypersurface singularity, computed as the Jacobian ring predicted by example 17.2.4. The node has exactly one first-order deformation up to isomorphism.

The versal deformation. A cocycle representing the generator of T^1 is the perturbation $f = xy \rightsquigarrow xy - t$ of the defining equation, since modulo (x,y) the constant -1 (the coefficient of t) spans $A/(x,y) = k$. Because $T^2 = 0$, this first-order deformation extends to all orders without obstruction, and because $\dim_k T^1 = 1$, one parameter suffices: the versal deformation of the node is

$$\operatorname{Spec} k[x, y, t]/(xy - t) \longrightarrow \operatorname{Spec} k[t]$$

with the node sitting over $t = 0$. For $t \neq 0$ the fiber is $\{xy = t\}$, a smooth conic (a hyperbola), while the fiber over $t = 0$ is the node itself. The single parameter t smooths the singularity: the two branches $x = 0$ and $y = 0$ that met at the origin are pulled apart into a single smooth curve as t moves off 0.

Reading the smoothing. The one-dimensionality of T^1 is the local input to the stable-reduction theorem in the moduli of curves. When a family of smooth curves degenerates to a singular limit, the limit can be taken to be a nodal curve, and each node contributes exactly this one-parameter smoothing $xy = t$: locally, the total space of a one-parameter degeneration to a nodal curve looks like $\{xy = t\}$ near each node, a surface smooth in the total space even though the special fiber is singular. The versal deformation $\{xy = t\}$ is therefore the local model for the boundary of the Deligne-Mumford compactification $\overline{\mathcal{M}}_g$: a node is a codimension-one degeneration, its single smoothing parameter is the normal direction to the boundary divisor, and the boundary of $\overline{\mathcal{M}}_g$ is where the nodes live. That the smoothing is unobstructed ($T^2 = 0$) is why the boundary is a divisor with normal crossings rather than a more singular locus, and why $\overline{\mathcal{M}}_g$ is smooth as a stack even where its coarse space is not. This connection is developed when the moduli of curves is constructed.

The node is the model unobstructed singularity, and its computation shows the mechanism by which a singular point deforms: the tangent space T^1 measures the independent smoothing directions, and the vanishing of T^2 guarantees each of them integrates to an actual family. Two features generalize. First, for any isolated hypersurface singularity $\{f = 0\}$, the same computation gives $T^1 = k[x_1, \ldots, x_n]/(f, \partial_1 f, \ldots, \partial_n f)$, the Tjurina algebra, whose dimension counts the smoothing parameters and whose vanishing detects smoothness; and $T^2 = 0$ always, so hypersurface singularities are always unobstructed. Second, the passage from the algebraic $T^i(A/k, A)$ to the geometric moduli is exactly the gluing of theorem 17.2.3: the local T^1 at each singular point assembles, together with the global $H^1(T)$ of the smooth part, into the deformation space of the whole curve, and this is what makes the boundary of $\overline{\mathcal{M}}_g$ a normal-crossings divisor whose components record which nodes are being smoothed.

Unobstructed examples are the well-behaved case; to see what the machinery is for, one needs a deformation that exists to first order but fails to extend, so that $T^2 \neq 0$ receives a nonzero obstruction. Such examples are less elementary than the node, because l.c.i. singularities are always unobstructed and one must leave the l.c.i. world to find $T^2 \neq 0$. The standard source is the affine cone over a projective variety embedded so that its second cotangent functor is nonzero, and the cleanest instance is the cone over a rational normal curve of high enough degree, or over a Segre-embedded product.

Example 17.4.3: An Obstructed Cone

Let $Y = \mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$ be the Segre embedding of the quadric surface, whose image is the smooth quadric $\{X_0X_3 - X_1X_2 = 0\} \subseteq \mathbb{P}^3$, and let A be the homogeneous coordinate ring of the corresponding *affine cone*

$$C(Y) = \operatorname{Spec} A, \qquad A = k[X_0, X_1, X_2, X_3]/(X_0X_3 - X_1X_2)$$

the affine quadric cone over Y, a three-dimensional hypersurface with an isolated singularity at the vertex. This particular cone is l.c.i. (it is a hypersurface), so it is unobstructed and would not serve; the point of naming it is to contrast it with the obstructed cone obtained by a non-complete-intersection embedding, taken up next.

Take instead $Y = \mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$ under the Segre embedding, whose homogeneous ideal is generated by the 2×2 minors of the 2×3 matrix (Z_{ij}) of coordinates, three quadrics

$$q_1 = Z_{00}Z_{11} - Z_{01}Z_{10}, \quad q_2 = Z_{00}Z_{12} - Z_{02}Z_{10}, \quad q_3 = Z_{01}Z_{12} - Z_{02}Z_{11}$$

cutting out a codimension-2 variety in $\mathbb{P}^5$, so the cone $C(Y) = \operatorname{Spec} A$ with $A = k[Z_{ij}]/(q_1, q_2, q_3)$ is a four-dimensional variety defined by three equations, hence *not* a complete intersection (three equations, codimension two). It is precisely the failure of the complete-intersection condition that lets T^2 be nonzero.

$T^2 \neq 0$. For this determinantal cone the second cotangent functor $T^2(A/k, A)$ is nonzero. The naive cotangent complex of definition 17.2.1 now has a nontrivial syzygy term. The three minors q_1, q_2, q_3 do not form a regular sequence (they cut out a codimension-2 locus with three equations), so there are syzygies among them beyond the trivial Koszul ones: the classical Hilbert-Burch resolution of a codimension-2 determinantal ideal presents (q_1, q_2, q_3) with two linear syzygies, one for each row of the matrix (Z_{ij}), and the departure of these from the Koszul syzygies is what feeds the second-syzygy term of $L_{A/k}$. The resulting computation of the second cotangent functor gives $T^2(A/k, A) \neq 0$ ([35, §3.1], and [61, Ch. 3] for deformations of determinantal schemes); this is the standard example of an obstructed singularity, cited here because the full determinantal T^2 computation is longer than the point it illustrates. By theorem 17.2.3 the cone can carry an obstructed deformation.

A first-order deformation that fails to lift. Perturb the three defining quadrics to first order,

$$q_i \rightsquigarrow q_i + \varepsilon\, g_i, \qquad \varepsilon^2 = 0$$

with g_i chosen quadrics. At first order the deformation is flat over $k[\varepsilon]$ exactly when the syzygies among the q_i lift modulo ε^2: each linear relation $\sum_i r_{ji}(Z)\, q_i = 0$ among the quadrics must extend to $\sum_i r_{ji}(Z)\,(q_i + \varepsilon g_i) = \varepsilon\, h_j$ with $h_j \in (q_1, q_2, q_3)$, and such a first-order lift can always be arranged, so every (g_1, g_2, g_3) satisfying this condition is a valid element of T^1. The obstruction appears one order up: lifting the first-order deformation to second order over $k[\varepsilon']/(\varepsilon'^3)$ requires quadratic corrections to the equations that keep the syzygies satisfied modulo ε'^3, and the class in $T^2(A/k, A)$ measures whether these corrections exist. For a class in T^1 pointing off the versal base, they do not: the second-order syzygy conditions cannot all be met simultaneously, and their failure lands in the nonzero group T^2. Concretely, the versal base space of $C(Y)$ is not smooth: it is a cone (a quadric cone in the relevant example) cut out by the obstruction equations in the T^1-directions, so a tangent direction pointing off the versal cone is a first-order deformation with a nonzero obstruction, and it does not lift. This exhibits the phenomenon the obstruction theory was built to detect: a class in T^1 that is not tangent to any actual family, whose obstruction in T^2 is nonzero.

The obstructed cone is where the abstract T^2 of section 17.2 becomes visible. The versal deformation of the node was a smooth base, Spec $k[t]$, one parameter for one first-order deformation, because $T^2 = 0$; the versal base of the obstructed cone is singular, a cone inside the affine space T^1 cut out by the obstruction equations valued in T^2. The dimension of T^1 still counts first-order deformations, but now not every first-order deformation integrates: the ones that do are exactly those tangent to the versal base, and the obstruction map $T^1 \to T^2$ (more precisely, the quadratic and higher terms of the Kuranishi map) records which do not. This is the general shape of a moduli space at an obstructed point: the tangent space is T^1, the ambient obstruction space is T^2, and the moduli space is locally the zero locus of the obstruction map inside T^1, smooth of dimension $\dim T^1$ exactly when $T^2 = 0$ or the obstruction map vanishes. The unobstructedness results of proposition 17.4.1 are precisely the statement that for curves, abelian varieties, and K3 surfaces this obstruction map vanishes, so their moduli spaces are smooth, while the cone shows what smoothness would fail to hold for a determinantal singularity carrying a nonzero obstruction.

Exercise 17.4.1

1. Let C be a smooth projective curve of genus g. Using proposition 17.4.1(1), compute $\dim_k H^1(C, T_C)$ for $g = 0$ and $g = 1$, and reconcile the answers with the geometry: explain why the $g = 0$ curve $\mathbb{P}^1$ is rigid (no moduli) and why the $g = 1$ answer is the 1-dimensional $\mathcal{M}_{1,1}$ deformation computed in chapter 13. (Recall $T_C = \omega_C^{-1}$ has degree $2 - 2g$, and treat the cases $g = 0, 1$ where the Riemann-Roch shortcut of the proposition's proof, valid for $g \geq 2$, does not directly apply.)

2. For the node $A = k[x, y]/(xy)$, verify by hand that the Tjurina algebra $A/(f, \partial_x f, \partial_y f)$ with $f = xy$ equals k, and check that the cusp $A' = k[x, y]/(y^2 - x^3)$ has $\dim_k T^1(A'/k, A') = 2$ by the same Jacobian-ring computation. Interpret the difference between the numbers 1 and 2 in terms of how many parameters are needed to smooth each singularity.

3. Write down the versal deformation $\operatorname{Spec} k[x, y, t]/(xy - t) \to \operatorname{Spec} k[t]$ of the node explicitly as a family, and prove that the total space $\operatorname{Spec} k[x, y, t]/(xy - t)$ is smooth (regular) at the origin even though the special fiber over $t = 0$ is singular there. Explain in one sentence why this is the reason $\overline{\mathcal{M}}_g$ can be smooth as a stack while parametrizing singular (nodal) curves.

4. Let S be a K3 surface. Using $\omega_S \cong \mathcal{O}_S$, $H^1(S, \mathcal{O}_S) = 0$, and Serre duality, verify all three of $h^0(S, T_S) = 0$, $h^1(S, T_S) = 20$, and $h^2(S, T_S) = 0$ from proposition 17.4.1(3). (You may use the Hodge numbers of a K3: $h^{0,0} = h^{2,0} = h^{0,2} = 1$, $h^{1,0} = h^{0,1} = 0$, $h^{1,1} = 20$.) Which of these three vanishings is the one that makes S unobstructed, and which makes Def_S pro-representable rather than only having a hull?

5. Let $A = k[x, y, z]/(xy, xz, yz)$, the coordinate ring of the three coordinate axes in $\mathbb{A}^3$ (a non-l.c.i. singularity: three equations, codimension two). Show that A is not a complete intersection by comparing the number of equations with the codimension, and explain why this makes it a candidate for $T^2(A/k, A) \neq 0$, in contrast to the node of example 17.4.2. (You need not compute T^2; identify the structural reason a non-complete-intersection can be obstructed.)

6. Let A be an abelian variety of dimension g. Using the triviality $T_A \cong \mathcal{O}_A^{\oplus g}$ and $H^i(A, \mathcal{O}_A) \cong \bigwedge^i H^1(A, \mathcal{O}_A)$ from proposition 17.4.1(2), compute $\dim_k H^i(A, T_A)$ for all i as a function of g, and identify $\dim_k H^1(A, T_A)$ as the dimension of the unpolarized deformation space. For $g = 2$, compare this with the dimension $\binom{g+1}{2}$ of $\mathcal{A}_g$ and explain the discrepancy in terms of the polarization constraint.

Part VI

Descent and Algebraic Stacks

The moduli problems of Part IV were functors, and the recurring verdict was that they are not representable by schemes. Part V diagnosed the obstruction one point at a time: the automorphisms of an object, visible as the infinitesimal automorphism group H^0, are what a scheme cannot record. This Part builds the geometry that can. It replaces the scheme, an object glued from affine pieces in the Zariski topology, with objects glued in finer topologies and remembering their automorphisms, and it shows that in this larger world the moduli problems finally acquire the spaces they were asking for.

The construction proceeds in three stages, and this Part devotes a chapter to each. chapter 18 develops descent theory: the recognition that the sheaf condition, and with it the notion of gluing, makes sense in any Grothendieck topology, and that quasi-coherent sheaves, morphisms, and (with a polarization) schemes descend along faithfully flat covers. The failure of a bare descent datum for schemes to be effective is the precise reason schemes are too rigid, and torsors and their classification by H^1 are the first sign of the automorphisms about to be geometrized. The following chapters pass to categories fibered in groupoids and to stacks, the bookkeeping for objects-with-automorphisms glued in a topology, and then to algebraic spaces and algebraic stacks, the geometric objects that represent the moduli functors schemes could not; Artin's criteria, fed by the deformation theory of Part V, certify when a moduli problem is such a stack. A final chapter treats gerbes and twisted sheaves, the 2-categorical layer above torsors.

The capstone is the moduli of curves. There $\mathcal{M}_g$ is the stack whose deformation theory is the $H^1(T_C)$ of Part V, whose coarse space is the geometric invariant theory quotient of Part IV, and whose Deligne–Mumford compactification $\overline{\mathcal{M}}_g$ by stable curves is proper and smooth as a stack even where its coarse space is singular. The three Parts of the book meet in that single object.

18

Descent Theory

A scheme is glued from affine pieces, and the glue is the Zariski topology: a morphism, a sheaf, or a subscheme is specified by giving it on an open cover together with the condition that the pieces agree on overlaps. The moduli problems of the earlier Parts strain against this. A family of curves can be locally trivial in a topology finer than the Zariski one and yet globally nontrivial; an object can have automorphisms, so that two local descriptions agree not on the nose but up to an isomorphism that must itself be tracked. To build the spaces these problems require, the first move is to enlarge the notion of an open cover, and then to ask which structures still glue.

That enlargement is Grothendieck's, and this chapter develops it. A Grothendieck topology axiomatizes the covering families for which gluing should work, and the étale, smooth, and faithfully flat topologies it produces are the ones in which moduli objects are locally trivial. The sheaf condition, rewritten as an equalizer over such a covering family, then makes sense on any site, and the central theorem, faithfully flat descent, says that quasi-coherent sheaves, morphisms, and, when a polarization is present, schemes are recovered from their restrictions to a flat cover together with gluing data on the overlaps. The point where this recovery fails, a descent datum for schemes with no descended polarization, is exactly the point where schemes must give way to the algebraic spaces and stacks of the chapters to follow. The chapter closes with torsors and their classification by the first cohomology of a site, the one-categorical first appearance of the automorphisms that stacks will geometrize and of the gerbes that Chapter 9 will build on them.

18.1 Grothendieck Topologies and Sites

Every construction in the first two Parts of this book rested on a single tacit assumption: that the geometric objects of a moduli problem glue in the Zariski topology. A family of curves over a base S was assembled from its restrictions to a Zariski-open cover; a coherent sheaf was recovered from its restrictions to open sets by the sheaf axiom. This assumption is where representability broke down.

The functor $\mathcal{M}_{1,1}$ of elliptic curves is not a Zariski sheaf in the sense that would let one build a fine moduli space: two families that agree Zariski-locally can differ globally, because an elliptic curve has automorphisms, and gluing along those automorphisms is invisible to the Zariski topology. The repair begins here. The Zariski topology is too coarse for descent because its covers are too large, too few isomorphisms are captured, and a morphism that is Zariski-locally an isomorphism is already an isomorphism. What Part VI needs is a notion of "cover" fine enough that faithfully flat maps count as covers, so that objects glued along them descend. That is exactly what a Grothendieck topology provides.

This section axiomatizes the word "cover." The insight, due to Grothendieck, is that gluing never used the points of a topological space, only the lattice of open sets and the relation "these opens cover that one." Abstracting that relation to an arbitrary category produces a *site*, and on a site one can speak of sheaves, cohomology, and descent with no topological space in sight. The étale, smooth, and fppf topologies on schemes are the payoff: covers there are surjective families of étale, smooth, or flat morphisms, and each strictly refines the Zariski topology. This chapter develops descent in that language; the present section builds the language itself, grounding each topology in a concrete cover as soon as the machinery permits.

The starting point is to isolate what a family of open inclusions $\{U_i \hookrightarrow X\}$ does in ordinary topology, stripped of any reference to points. Three closure properties do all the work in every gluing argument. A cover of X restricts to a cover of any open subset (a cover is stable under passing to a smaller open); covers compose (a cover of each member of a cover assembles to a cover of the whole); and the one-element family consisting of the identity, or any isomorphism, is a cover. In a general category the role of "restriction to a smaller open" is played by fiber product, and this substitution is the one nontrivial conceptual step. The following definition records these three axioms in the generality of an arbitrary category with fiber products, following [66] and the categorical foundations of [9].

Definition 18.1.1: Grothendieck Topology and Site

Let $\mathcal{C}$ be a category admitting fiber products. A *Grothendieck pretopology* on $\mathcal{C}$ assigns to each object X of $\mathcal{C}$ a collection of *covering families* $\{U_i \xrightarrow{u_i} X\}_{i \in I}$ of morphisms with target X, subject to three axioms.

1. *(Isomorphisms.)* If $u\colon U \to X$ is an isomorphism, then the one-element family $\{u\colon U \to X\}$ is a covering family.
2. *(Base change.)* If $\{U_i \to X\}_{i \in I}$ is a covering family and $V \to X$ is any morphism, then the fiber products $U_i \times_X V$ exist, and the family of projections $\{U_i \times_X V \to V\}_{i \in I}$ is a covering family of V.
3. *(Composition.)* If $\{U_i \to X\}_{i \in I}$ is a covering family and, for each i, $\{V_{ij} \to U_i\}_{j \in J_i}$ is a covering family, then the composite family $\{V_{ij} \to U_i \to X\}_{i \in I,\, j \in J_i}$ is a covering family of X.

A category $\mathcal{C}$ equipped with a Grothendieck pretopology is called a *site*.

The three axioms are precisely the three moves a gluing argument performs. Isomorphisms as covers guarantee that anything glued from a single copy of itself is unchanged, so the trivial cover trivializes; without it a sheaf could fail to satisfy $F(X) = F(X)$. Base change is the substitute for "restrict the cover to a smaller open": it says a cover pulls back to a cover, which is what lets one localize a descent problem. Composition says a cover of a cover is a cover, the property that makes "locally" a transitive relation, so that a statement true locally on a cover, and locally on that cover, is true locally. The axioms make no mention of surjectivity, injectivity, or any point-set notion; they are entirely about the arrows.

The word *pretopology* rather than *topology* flags a redundancy in the data. Two covering families of the same object can generate the same notion of "locally" even when they are not equal as families, just as in ordinary topology two different open covers of a space refine each other and so give the same sheaves. The invariant that a pretopology determines is not the set of covering families but the set of *sieves* they generate. A sieve on X is a collection S of morphisms into X closed under precomposition: if $(V \to X) \in S$ and $W \to V$ is any morphism, then the composite $(W \to X) \in S$. A covering family $\{U_i \to X\}$ generates the sieve of all morphisms $V \to X$ that factor through some u_i, and the object a Grothendieck topology assigns to X is a distinguished collection of *covering sieves* satisfying axioms equivalent to those above. The sieve formulation is cleaner for proving general theorems, since it quotients out the choice of indexing, but the pretopology formulation is what one computes with, and it is the one used throughout this book. Passing between the two is a bookkeeping exercise carried out in [66, Definition 2.24]; nothing below depends on which formulation is adopted, and "topology" will mean a pretopology given by explicit covering families.

The definition is abstract by necessity, but its content is already visible in the one topology the reader has used for years.

Example 18.1.2: The Zariski Site of a Scheme

Let X be a scheme and let $\mathcal{C} = \mathrm{Open}(X)$ be the category whose objects are the Zariski-open subsets of X and whose morphisms are the inclusions. Fiber products in this category are intersections: for opens $U, V \subseteq W$ the fiber product $U \times_W V$ is the intersection $U \cap V$. Declare a family of inclusions $\{U_i \hookrightarrow X\}$ to be a covering family exactly when $\bigcup_i U_i = X$ set-theoretically. The three axioms hold transparently.

1. The identity $X = X$ is a surjective one-element family, and any isomorphism of opens is an equality, so isomorphisms cover.
2. If $\bigcup_i U_i = X$ and $V \subseteq X$ is open, then $U_i \times_X V = U_i \cap V$ and $\bigcup_i (U_i \cap V) = V \cap \bigcup_i U_i = V$, so the pullback family covers V.
3. If $\bigcup_i U_i = X$ and $\bigcup_j V_{ij} = U_i$ for each i, then $\bigcup_{i,j} V_{ij} = \bigcup_i U_i = X$, so the composite family covers.

This is the site whose sheaves are exactly the sheaves on the topological space X: the abstract sheaf condition of section 18.2 specializes to the ordinary one. The Zariski topology is thus a Grothendieck topology on a category whose only morphisms are inclusions, and the leap of the next definition is to allow morphisms that are not inclusions.

The Zariski site is a Grothendieck topology in which the covering morphisms are open immersions and the underlying category is a poset. To capture descent along faithfully flat maps one enlarges the category to allow arbitrary scheme morphisms as members of a cover, and enlarges the supply of covering families to include surjective families of morphisms of a prescribed type. The prescription is the topology; changing the class of allowed morphisms changes the topology.

Fix a base scheme S and work in the category $\mathbf{Sch}/S$ of S-schemes. A topology on this category is specified by naming which surjective families count as covers, and there is a standard tower of four choices, each obtained by enlarging the class of covering morphisms from the last. The requirement common to all four is joint surjectivity: a family $\{U_i \to X\}$ covers only if the images of the U_i cover X as a set. What distinguishes the topologies is the type of morphism permitted. Recall from **??** that an étale morphism is a smooth morphism of relative dimension zero, equivalently a flat and unramified morphism locally of finite presentation; that a smooth morphism is flat, locally of finite presentation, with smooth fibers; and from **??** that a morphism is faithfully flat when it is flat and surjective. These are the four morphism classes that define the four topologies.

Definition 18.1.3: The Standard Topologies on Sch/S

Let τ be one of the labels *Zariski*, *étale*, *smooth*, or *fppf*. A family of morphisms $\{u_i : U_i \to X\}_{i \in I}$ of S-schemes is a *τ-covering family* if it is jointly surjective, that is $\bigcup_i u_i(U_i) = X$, and each u_i is a morphism of the corresponding type:

- *Zariski:* each u_i is an open immersion;
- *étale:* each u_i is étale;
- *smooth:* each u_i is smooth;
- *fppf:* each u_i is flat and locally of finite presentation (*fidèlement plat de présentation finie*).

Each choice satisfies the axioms of definition 18.1.1 and so makes $\mathbf{Sch}/S$ a site, written $(\mathbf{Sch}/S)_\tau$ and called the *big τ-site*. The *small étale site* $X_{\text{ét}}$ of a scheme X is the full subcategory of étale X-schemes $U \to X$, with the étale covering families; the small Zariski site is the site $\mathrm{Open}(X)$ of example 18.1.2.

The distinction between the big site and the small site is a matter of how much of the category one carries. The big τ-site $(\mathbf{Sch}/S)_\tau$ contains *all* S-schemes as objects and admits τ-covers between any of them; it is the natural home for a moduli functor, which is defined on all test schemes. The small étale site $X_{\text{ét}}$ retains only the schemes étale over a fixed X, and it is the natural home for the étale cohomology of X itself, because a sheaf on $X_{\text{ét}}$ is a local invariant of X that varies only in the étale direction. This book works predominantly with big sites, since the objects of a moduli problem are tested against arbitrary schemes; the small site returns when a fixed scheme's own cohomology is at issue. The fppf topology is the widest of the four that this book uses, and it is the one in which representable functors become sheaves, the reason it governs descent.

The four topologies are nested, and the nesting is the reason a hierarchy of descent statements exists. A cover in a coarser topology is automatically a cover in a finer one, so a sheaf for the finer topology is in particular a sheaf for the coarser, and a descent statement proved for the finer topology subsumes the coarser cases. Establishing the nesting is a matter of tracking the implications among the morphism classes: an open immersion is étale, an étale morphism is smooth, and a smooth morphism is flat and locally of finite presentation. The one non-formal input is that a smooth cover, though finer than an étale cover as a family, generates the *same* notion of sheaf on the small site, because smooth morphisms admit sections étale-locally.

Proposition 18.1.4: Comparison of the Standard Topologies

Let S be a scheme. Every Zariski-covering family is an étale-covering family, every étale-covering family is a smooth-covering family, and every smooth-covering family is an fppf-covering family. Consequently there are inclusions of topologies

$$\text{Zariski} \subseteq \text{étale} \subseteq \text{smooth} \subseteq \text{fppf}$$

in the sense that every τ-sheaf is a τ'-sheaf whenever τ is finer than τ'. Moreover, every smooth-covering family $\{U_i \to X\}$ admits, after further étale base change, a section: there is an étale cover $\{X_a \to X\}$ such that each $X_a \to X$ factors through some U_i. In particular the smooth and étale topologies define the same sheaves.

***Proof*:**
Step 1: The inclusions of morphism classes. Each inclusion of covering families follows from an implication between the defining properties of the morphisms, the joint-surjectivity requirement being identical across the four topologies.

An open immersion $u\colon U \to X$ is flat, since localization is flat, and unramified with trivial relative differentials, and locally of finite presentation; a morphism that is flat, unramified, and locally of finite presentation is étale by **??**. Hence every open immersion is étale, and a jointly surjective family of open immersions is a jointly surjective family of étale morphisms: every Zariski cover is an étale cover.

An étale morphism is by definition smooth of relative dimension zero, so in particular smooth. Hence a jointly surjective family of étale morphisms is a jointly surjective family of smooth morphisms: every étale cover is a smooth cover.

A smooth morphism is flat and locally of finite presentation by definition, these being two of the three conditions (the third being geometric regularity of the fibers). Hence a jointly surjective family of smooth morphisms is a jointly surjective family of flat, locally-finitely-presented morphisms: every smooth cover is an fppf cover.

Step 2: Finer topologies give more sheaves. Suppose τ is finer than τ', meaning every τ'-cover is a τ-cover, which is the content of Step 1 for adjacent pairs and follows by transitivity in general. If F is a τ-sheaf, then F satisfies the sheaf condition of section 18.2 for every τ-covering family, hence in particular for every τ'-covering family, since these form a subclass. Thus F is a τ'-sheaf. This is the asserted inclusion of topologies: finer topology, more covers to satisfy, fewer sheaves, and the sheaves for the finer topology are among those for the coarser.

Step 3: Smooth covers admit étale-local sections. Let $u\colon U \to X$ be a smooth surjective morphism. The structure theorem for smooth morphisms states that u is étale-locally on U a composite of an étale morphism with a projection from affine space: for each point of U there is an open neighborhood $U' \subseteq U$ and an étale morphism $U' \to \mathbb{A}^n_X$ over X. The projection $\mathbb{A}^n_X \to X$ has the zero section $X \to \mathbb{A}^n_X$, and pulling this section back along the étale map produces, over an étale cover of X, a section of u. The full statement is that a smooth surjection admits sections étale-locally on the target, proved via the local structure of smooth morphisms in **??**; see also [63, Tag 054L]. Concretely, there is an étale cover $\{X_a \to X\}$ and, for each a, a lift $X_a \to U$ of $X_a \to X$ through u. Applied to a smooth-covering family $\{U_i \to X\}$, choosing for each member such an étale refinement produces an étale cover of X factoring through the U_i, which is the claimed étale refinement.

Step 4: Smooth and étale sheaves coincide. A τ-sheaf is determined by its behavior on a cofinal system of covers. By Step 3 every smooth cover is refined by an étale cover, and by Step 1 every étale cover is a smooth cover; the two classes of covers are therefore mutually cofinal. A presheaf satisfies the sheaf condition for a cover if and only if it satisfies it for any refinement, since refinement composes covers (definition 18.1.1(3)) and the sheaf condition passes to refinements. Hence a presheaf is a smooth sheaf if and only if it is an étale sheaf, completing the proof.

The comparison proposition organizes Part VI's descent statements into a single ascending ladder. Descent along an fppf cover, once proved, descends along smooth, étale, and Zariski covers as special cases, because each of those is an fppf cover; and a functor that is an fppf sheaf is a fortiori a sheaf for the three coarser topologies. This is why faithfully flat descent, the subject of section 18.3, is stated and proved at the fppf level: it is the strongest statement, and the others follow. The coincidence of smooth and étale sheaves in the last clause is the reason algebraic stacks can be defined with either a smooth atlas or, in the Deligne-Mumford case, an étale atlas without changing the resulting sheaf theory, a point that returns when stacks are constructed in Chapter 7. On the small site of a fixed scheme the smooth and étale topologies are interchangeable; the distinction between them is felt only through the morphisms allowed in an atlas, not through the sheaves.

One further topology sits above the fppf topology and completes the tower, the fpqc topology, whose covers are the faithfully flat quasi-compact families. Descent holds in the fpqc topology as well, and it is the coarsest topology for which quasi-coherent descent is available, so it is the correct maximal setting for the descent theorems. There is a technical cost: the fpqc topology is not a small site in the naive sense, because the class of fpqc covers of a fixed scheme is not a set but a proper class, and the resulting sheaf theory requires a set-theoretic device (a universe, or a bound on the cardinality of covers) to be well-founded. This subtlety is inessential for the constructions of this book, all of which can be carried out at the fppf level, so the fpqc topology is invoked only when a descent statement is strictly stronger there, and its foundational treatment is deferred to [63, Tags 022A, 03NV]. The working hierarchy is

$$\text{Zariski} \;\subseteq\; \text{étale} \;\subseteq\; \text{smooth} \;\subseteq\; \text{fppf} \;\subseteq\; \text{fpqc}$$

with the last inclusion carrying the set-theoretic caveat just noted.

The definitions are now in place, and the promise of the section is to make them concrete. The following worked covers exhibit the difference between the Zariski and finer topologies in the smallest possible examples and introduce the cover that will power the theory of μ_n-torsors in section 18.5. Each is a family of morphisms; the question in each case is whether it is jointly surjective and of the required type.

Example 18.1.5: Three Model Covers

Work over a field k; write $\mathbb{A}^1 = \operatorname{Spec} k[x]$ and $\mathbb{G}_m = \operatorname{Spec} k[x, x^{-1}]$.

1. *A non-cover.* Consider the single-morphism family $\{\operatorname{Spec} k[x, x^{-1}] \to \operatorname{Spec} k[x]\}$, the inclusion of the punctured line $\mathbb{G}_m$ into $\mathbb{A}^1$ dual to the localization $k[x] \to k[x, x^{-1}]$. This morphism is an open immersion, hence étale, smooth, and flat of finite presentation. It is nonetheless *not* a covering family in any of the four topologies, because it is not jointly surjective: its image is the open set $x \neq 0$, missing the origin. Joint surjectivity is a genuine constraint, not an afterthought, and it is what fails here. Adjoining the origin, say via the second chart $\{\operatorname{Spec} k[x, (x-a)^{-1}] \to \operatorname{Spec} k[x]\}$ for a scalar $a \neq 0$, or more simply the identity chart at 0, restores a cover; a single localization does not.

2. *An fppf and étale cover of a point.* Let L/k be a finite field extension of degree n and consider

the one-morphism family $\{\operatorname{Spec} L \to \operatorname{Spec} k\}$ dual to the inclusion $k \hookrightarrow L$. A finite extension is a finite, flat (indeed free of rank n as a k-module) morphism of finite presentation, and it is surjective since $\operatorname{Spec} L$ and $\operatorname{Spec} k$ are both single points and the map hits it. Hence $\{\operatorname{Spec} L \to \operatorname{Spec} k\}$ is an *fppf cover of* $\operatorname{Spec} k$. If moreover L/k is separable, then $k \to L$ is unramified (the trace form is nondegenerate, equivalently $\Omega_{L/k} = 0$), so the morphism is étale and the family is an *étale cover*. This single-point cover is the model on which Galois descent runs: a field extension is a cover in the étale or fppf topology precisely when the Zariski topology, which sees $\operatorname{Spec} k$ as an irreducible point with no proper open cover, is blind. Over $k = \mathbb{R}$ the extension $\mathbb{C}/\mathbb{R}$ gives the étale cover $\{\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} \mathbb{R}\}$ underlying the descent of real forms.

3. *The Kummer cover.* Fix an integer $n \geq 1$ and consider the n-th power map on the multiplicative group,
$$\nu_n \colon \mathbb{G}_m \to \mathbb{G}_m, \qquad x \mapsto x^n$$
dual to the k-algebra map $k[t, t^{-1}] \to k[x, x^{-1}]$ sending $t \mapsto x^n$. As a module map, $k[x, x^{-1}]$ is free of rank n over $k[t, t^{-1}]$ with basis $1, x, \ldots, x^{n-1}$, so ν_n is finite, flat of rank n, and of finite presentation; it is surjective because every nonzero scalar has an n-th root in an extension, so on points it hits every point of the target. Thus $\{\nu_n \colon \mathbb{G}_m \to \mathbb{G}_m\}$ is an *fppf cover*. Its ramification is controlled by the derivative: Ω of the extension is generated by dx with relation $nx^{n-1}\, dx = dt \mapsto 0$, which vanishes identically iff n is invertible in k. Hence ν_n is *étale exactly when* $\operatorname{char} k \nmid n$, and in that case the family is an étale cover. The fiber of ν_n over the identity is $\operatorname{Spec} k[x, x^{-1}]/(x^n - 1) = \operatorname{Spec} k[\mu_n]$, the group scheme μ_n of n-th roots of unity, and ν_n is a μ_n-torsor (section 18.5): $\mathbb{G}_m$ acts on itself covering ν_n, freely and transitively on fibers over the fppf cover. This cover is the geometric source of the Kummer sequence and of every μ_n-torsor computation to come.

The three covers isolate the three phenomena that make the fppf topology the right setting for descent. The first shows that joint surjectivity cannot be dropped: a topology is not a class of "nice" morphisms but a class of *surjective* families of them, and forgetting surjectivity would make every open immersion a cover of its target and collapse the theory. The second shows that the étale and fppf topologies see structure the Zariski topology cannot: a field, Zariski-irreducible and uncoverable, acquires genuine covers by finite extensions, and this is the precise sense in which Galois theory is a descent theory, developed in section 18.4. The third exhibits the cover from which μ_n-torsors are built, and it already displays the characteristic subtlety of the fppf topology: the map ν_n is a perfectly good fppf cover in every characteristic, but it is étale only away from n, so a torsor theory that insisted on étale covers would lose the interesting phenomena in characteristic dividing n. The fppf topology retains them. These covers recur throughout Part VI as the concrete instances on which the abstract descent machinery is tested.

Exercise 18.1.1

1. Verify directly from definition 18.1.1 that the base-change axiom is consistent with the composition axiom in the following sense: if $\{U_i \to X\}$ is a covering family and $V \to X$ is any morphism, and if $\{W_{ij} \to U_i \times_X V\}$ is a covering family of each pullback, show that the composite $\{W_{ij} \to V\}$ is a covering family of V. Which two axioms did you use, and in what order?

2. Let k be a field. Show that the one-morphism family $\{\operatorname{Spec} k \xrightarrow{\mathrm{id}} \operatorname{Spec} k\}$ is a covering family of $\operatorname{Spec} k$ in every one of the four topologies of definition 18.1.3, and deduce that any presheaf

on $(\mathbf{Sch}/k)_\tau$ automatically satisfies the sheaf condition for this particular cover. What does this cover fail to tell you about the sheaf condition?

3. Consider the family $\{\operatorname{Spec} k[x, x^{-1}] \to \mathbb{A}^1,\ \operatorname{Spec} k[(x-1)^{-1}] \to \mathbb{A}^1\}$ of two open immersions into $\mathbb{A}^1 = \operatorname{Spec} k[x]$. Is it a Zariski cover? Is it an étale cover? Justify your answer using proposition 18.1.4, and identify the point of $\mathbb{A}^1$, if any, that is not in the image.

4. Let $L = k(\alpha)$ with $\alpha^p = a$ for a prime p equal to $\operatorname{char} k$ and $a \in k$ not a p-th power (a purely inseparable extension). Show that $\{\operatorname{Spec} L \to \operatorname{Spec} k\}$ is an fppf cover but *not* an étale cover. Contrast this with example 18.1.5(2), and explain in one sentence why the fppf topology is indispensable in positive characteristic.

5. For the Kummer cover $\nu_n \colon \mathbb{G}_m \to \mathbb{G}_m$ of example 18.1.5(3), compute the fiber product $\mathbb{G}_m \times_{\nu_n, \mathbb{G}_m, \nu_n} \mathbb{G}_m$ as an explicit affine scheme, and identify it with $\mathbb{G}_m \times \mu_n$. Interpret the two projections to $\mathbb{G}_m$ in terms of the group action; this fiber product is the object on which a descent datum for ν_n lives.

6. Explain why the sieve generated by a covering family $\{U_i \to X\}$ determines the sheaf condition, so that two covering families generating the same sieve impose the same condition on a presheaf. Illustrate with the two Zariski covers $\{U_1, U_2\}$ and $\{U_1, U_2, U_1 \cap U_2\}$ of a space $X = U_1 \cup U_2$: show they generate the same sieve.

18.2 Sheaves on a Site

The previous section built the covers. A Grothendieck topology on a category $\mathcal{C}$ selects, for each object, the families $\{U_i \to X\}$ that count as covers of it, and definition 18.1.3 supplied the four topologies this book uses on $(\mathbf{Sch}/S)$: Zariski, étale, smooth, and fppf. A covering family is machinery with a purpose, and the purpose is gluing. On an ordinary topological space the data that glues is a sheaf: an assignment of sets to open sets that is determined by its restrictions to any open cover, subject to a compatibility condition on overlaps. This section transports that notion to an arbitrary site, where overlaps become fiber products $U_i \times_X U_j$ and the compatibility condition becomes an equalizer diagram.

Three results organize the section. First, the sheaf condition itself, recast as an equalizer, together with the weaker separatedness condition that a presheaf may satisfy on the way to being a sheaf. Second, the sheafification functor, which forces any presheaf to become a sheaf in the universal way; it is the left adjoint to the inclusion of sheaves into presheaves, and the plus construction builds it explicitly, two applications sufficing. Third, and the payoff for moduli theory, Grothendieck's theorem that every representable functor $h_T = \operatorname{Hom}_S(-, T)$ is already a sheaf in the fppf topology. That theorem is why a moduli functor built out of representable pieces has a chance of being a sheaf, and its proof is faithfully flat descent of morphisms in disguise. The final example computes that $\mathbb{G}_m$, μ_n, and GL_n are fppf sheaves while a naive quotient presheaf is not, isolating the exact place where sheafification becomes unavoidable.

Throughout, $\mathcal{C}$ denotes a site: a category equipped with a Grothendieck topology, its covering families written $\{U_i \to X\}$. The reader should keep two examples in view, the small étale site $X_{\text{ét}}$ and the big site $(\mathbf{Sch}/S)_{\text{fppf}}$, and should read every abstract statement against them.

A presheaf is the raw assignment of data to objects, with no gluing required; it is a functor, nothing more. The sheaf condition is what turns local data into global data. On a space, a section over an open set U is recovered from its restrictions to the members of a cover of U, provided those restrictions agree on pairwise intersections. On a site the pairwise intersection $U_i \cap U_j$ has no meaning, but its correct replacement is the fiber product $U_i \times_X U_j$, and the correct formulation of "the restrictions agree" and "recovered from its restrictions" is a single equalizer diagram.

Definition 18.2.1: Presheaf and Sheaf on a Site

Let $\mathcal{C}$ be a site. A *presheaf* on $\mathcal{C}$ (of sets) is a functor $F\colon \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$; the image of a morphism $g\colon V \to U$ is the *restriction map* $F(g)\colon F(U) \to F(V)$, written $s \mapsto s|_V$. A presheaf of abelian groups is a functor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Ab}$.
A presheaf F is a *sheaf* if for every covering family $\{U_i \to X\}$ in $\mathcal{C}$ the diagram

$$F(X) \xrightarrow{e} \prod_i F(U_i) \underset{\mathrm{pr}_2^*}{\overset{\mathrm{pr}_1^*}{\rightrightarrows}} \prod_{i,j} F(U_i \times_X U_j) \tag{18.1}$$

is an equalizer of sets: e sends $s \in F(X)$ to the family $(s|_{U_i})_i$, the two parallel maps send $(s_i)_i$ to $(s_i|_{U_i \times_X U_j})_{i,j}$ and $(s_j|_{U_i \times_X U_j})_{i,j}$ respectively, and e identifies $F(X)$ with the subset of $\prod_i F(U_i)$ on which the two parallel maps agree.
A presheaf F is *separated* if for every covering family the map e in (18.1) is injective (equivalently, a section is determined by its restrictions to a cover, even if not every compatible family arises from one).

Unwinding the equalizer restores the two familiar sheaf axioms. That e is injective is *separatedness*: two global sections with the same restrictions to a cover are equal. That the image of e is exactly the equalizer subset is the *gluing* axiom: any family (s_i) of local sections that agrees on overlaps, meaning $s_i|_{U_i \times_X U_j} = s_j|_{U_i \times_X U_j}$ for all i, j, comes from a unique global section. A sheaf is a separated presheaf in which every compatible family glues; a separated presheaf need only enforce uniqueness, not existence. The single diagram (18.1) packages both axioms because an equalizer is by definition the universal solution to "equalize the two maps," and the universal solution being $F(X)$ itself is precisely the demand that global sections biject with compatible families.

The fiber products play the role intersections play on a space, and the difference is the entire content of descent theory. On a space, $U_i \cap U_j$ is the set-theoretic intersection, and a point of it is a point lying in both U_i and U_j. On a site with a cover $\{U_i \to X\}$ by, say, étale or flat maps, the map $U_i \to X$ need not be injective, so $U_i \times_X U_j$ can be strictly larger than any naive intersection; over a Galois cover $\operatorname{Spec} L \to \operatorname{Spec} k$ it is $\operatorname{Spec}(L \otimes_k L)$, which for L/k Galois of degree n is a product of n copies of L. The extra components are exactly what records the automorphisms, and they are why a sheaf on the étale or fppf site sees more than a Zariski sheaf does.

Example 18.2.2: The Structure Presheaf and a Constant Presheaf

Two presheaves on the big Zariski or étale site $(\mathbf{Sch}/S)_\tau$ illustrate the definition.

1. *The structure presheaf.* Send a scheme $T \to S$ to the ring $\mathcal{O}(T) = \Gamma(T, \mathcal{O}_T)$ of global functions, and a morphism to the induced pullback of functions. This is the functor $\mathbb{A}^1_S = \operatorname{Hom}_S(-, \mathbb{A}^1_S)$, the affine line viewed as a functor of points, since $\operatorname{Hom}_S(T, \mathbb{A}^1_S) = \mathcal{O}(T)$. It is a sheaf in every topology considered here, a fact that is the base case of theorem 18.2.4 below: global functions on T are recovered from their restrictions to any flat cover, and a compatible family of functions on the cover that agrees on the double overlaps glues to a function on T.

2. *A constant presheaf that is not a sheaf.* Fix a set A with at least two elements and send every nonempty scheme to A and the empty scheme to a point, with all restriction maps the identity. This presheaf is separated but fails gluing already in the Zariski topology: if $T = T_1 \sqcup T_2$ is a disjoint union of two nonempty schemes, the cover $\{T_1 \to T,\ T_2 \to T\}$ has $T_1 \times_T T_2 = \emptyset$, so *every* pair $(a_1, a_2) \in A \times A$ is compatible (the overlap condition is vacuous). The map $e\colon F(T) = A \to F(T_1) \times F(T_2) = A \times A$ is the diagonal $a \mapsto (a, a)$, which is injective, so F is separated; but its image is only the diagonal, so $F(T) = A$ cannot surject onto the equalizer $A \times A$, and gluing fails. The sheafification of this presheaf is the sheaf $T \mapsto A^{\pi_0(T)}$ of locally constant A-valued functions, recording one value of A per connected component; this is the constant sheaf $\underline{A}$.

The second item is the prototype of everything that follows: a presheaf can carry the right data on connected schemes yet fail to glue across a disjoint union, and the cure is to enlarge $F(T)$ so that it records a value per component. That enlargement is sheafification, and its universal character is what the next theorem establishes.

Not every presheaf is a sheaf, but every presheaf has a best sheaf approximation. The construction that produces it must be universal: any map from F to a sheaf should factor uniquely through the approximation, so that no information is lost and none is invented beyond what gluing forces. This universal property is an adjunction, the sheafification functor being left adjoint to the forgetful inclusion of sheaves into presheaves. The adjunction is not an abstract certificate alone; it is what lets one compute in the category of sheaves by lifting constructions from presheaves and sheafifying the result, and it is the reason the category of sheaves inherits limits and (after sheafification) colimits from presheaves. The explicit construction is the *plus construction*, which replaces sections by compatible families modulo refinement; applied once it separates, applied twice it sheafifies.

Theorem 18.2.3: Sheafification on a Site

Let $\mathcal{C}$ be a site and let $\mathrm{PSh}(\mathcal{C})$ and $\mathrm{Sh}(\mathcal{C})$ denote the categories of presheaves and sheaves of sets on $\mathcal{C}$. The inclusion $\iota\colon \mathrm{Sh}(\mathcal{C}) \hookrightarrow \mathrm{PSh}(\mathcal{C})$ admits a left adjoint

$$a\colon \mathrm{PSh}(\mathcal{C}) \longrightarrow \mathrm{Sh}(\mathcal{C}) \qquad \text{(sheafification)}$$

so that for every presheaf F and every sheaf G there is a natural bijection

$$\mathrm{Hom}_{\mathrm{Sh}(\mathcal{C})}(aF, G) \;\cong\; \mathrm{Hom}_{\mathrm{PSh}(\mathcal{C})}(F, \iota G) \tag{18.2}$$

The sheafification aF is constructed as F^{++}, where $(-)^+$ is the plus construction: F^+ is always separated, and F^+ is a sheaf whenever F is separated, so two applications yield a sheaf. There is a canonical map $\theta\colon F \to aF$, universal among maps from F to sheaves, and a preserves finite limits.

***Proof*:**
The plus construction and the two lemmas about it carry the whole argument; the adjunction (18.2) is then a formal consequence of the universal property of θ, which we record by citing the general adjoint-from-universal-arrows bookkeeping of [9].

Step 1: The plus construction. For an object $X \in \mathcal{C}$ and a covering family $\mathcal{U} = \{U_i \to X\}$, call a family $(s_i) \in \prod_i F(U_i)$ *compatible* if $s_i|_{U_i \times_X U_j} = s_j|_{U_i \times_X U_j}$ for all i, j, and write $\check{H}^0(\mathcal{U}, F) \subseteq \prod_i F(U_i)$ for the set of compatible families, which is the equalizer of the two parallel maps in

(18.1). If $\mathcal{V} = \{V_a \to X\}$ refines $\mathcal{U}$, meaning each $V_a \to X$ factors through some $U_{i(a)} \to X$, restriction along the refinement induces a map $\check{H}^0(\mathcal{U}, F) \to \check{H}^0(\mathcal{V}, F)$ that is independent of the chosen factorizations (two factorizations of $V_a \to X$ through U_i agree after pulling back the compatible-family condition to $V_a \times_X V_b$). The covering families of X form a filtered system under refinement, and one sets

$$F^+(X) \;:=\; \varinjlim_{\mathcal{U}} \check{H}^0(\mathcal{U}, F)$$

the colimit over that system. A morphism $g\colon Y \to X$ pulls a cover of X back to a cover of Y and so induces $F^+(g)\colon F^+(X) \to F^+(Y)$, making F^+ a presheaf. The map $F(X) \to \check{H}^0(\mathcal{U}, F)$ sending $s \mapsto (s|_{U_i})$ is compatible with refinement, so it induces a natural transformation $F \to F^+$.

Step 2: F^+ is separated. Let $\{U_i \to X\}$ be a cover and let $t, t' \in F^+(X)$ have equal restrictions to each U_i. Represent t, t' by compatible families over covers $\mathcal{U}_t, \mathcal{U}_{t'}$ of X; passing to a common refinement $\mathcal{V} = \{V_a \to X\}$ (the family of all $V_a \to X$ that factor through both a member of $\mathcal{U}_t$ and a member of $\mathcal{U}_{t'}$, a cover because covers are stable under the base-change and composition axioms of definition 18.1.3), represent both t and t' by compatible families $(s_a), (s'_a) \in \check{H}^0(\mathcal{V}, F)$. The hypothesis $t|_{U_i} = t'|_{U_i}$ means that after refining $\mathcal{V}$ further so each V_a factors through some U_i, the representatives agree: $s_a = s'_a$ for every a in this common refinement. Hence $t = t'$ in the colimit $F^+(X)$, so F^+ is separated. This uses only that any two covers have a common refinement, which the site axioms guarantee.

Step 3: F^+ is a sheaf when F is separated. Assume F separated. Separatedness of F^+ is Step 2, so it remains to glue. Let $\{U_i \to X\}$ be a cover and $(t_i) \in \prod_i F^+(U_i)$ a compatible family. Each t_i is represented by a compatible family over some cover $\{U_{ij} \to U_i\}$; the composite family $\{U_{ij} \to X\}$ is a cover of X by the composition axiom. Because F is separated, the representatives of t_i and $t_{i'}$ agree on the overlaps $U_{ij} \times_X U_{i'j'}$ (their images in F^+ agree by the compatibility of the t_i, and separatedness of F upgrades the F^+-agreement to equality of the underlying F-sections after a further common refinement). The resulting family over $\{U_{ij} \to X\}$ is therefore compatible and defines an element $t \in F^+(X)$ whose restriction to U_i is t_i; separatedness of F^+ makes t unique. Thus F^+ satisfies the sheaf condition.

Step 4: Sheafification and the universal property. For any presheaf F, apply the plus construction twice. By Step 2 the presheaf F^+ is separated; by Step 3, $F^{++} = (F^+)^+$ is a sheaf. Set $aF := F^{++}$ and let $\theta\colon F \to F^+ \to F^{++} = aF$ be the composite of the two canonical maps. Given any sheaf G and any morphism of presheaves $\varphi\colon F \to G$, the plus construction is functorial and $G^+ \cong G$ canonically because G is a sheaf (its compatible families glue uniquely, so the colimit defining $G^+(X)$ collapses to $G(X)$). Hence φ induces $F^{++} \to G^{++} \cong G$, giving a factorization $\varphi = \psi \circ \theta$ with $\psi\colon aF \to G$. Uniqueness of ψ follows because θ has dense image in the sense that any two maps $aF \to G$ agreeing after precomposition with θ agree on every compatible family, hence on aF, using separatedness of G. This is exactly the universal arrow from F to ι, so by the standard criterion ([9]) the assignment $F \mapsto aF$ is left adjoint to ι, establishing the natural bijection (18.2).

Step 5: Left exactness. Left exactness is checked objectwise, and finite limits of presheaves are computed objectwise, so it suffices to show that each $F \mapsto F^+(X)$ preserves finite limits. Fix a finite diagram of presheaves. For a cover $\mathcal{U}$ the assignment $F \mapsto \check{H}^0(\mathcal{U}, F)$ is a limit (an equalizer of two maps between products), hence commutes with the finite limit; it is not itself a *finite* limit when $\mathcal{U}$ has infinitely many members, but limits commute with limits, so this poses no obstruction. Passing to the colimit, $F^+(X) = \varinjlim_{\mathcal{U}} \check{H}^0(\mathcal{U}, F)$ is a filtered colimit by Step 2's common-refinement fact, and filtered colimits commute with finite limits in **Set**. Composing the two, $(-)^+$ preserves finite limits; iterating, $a = (-)^{++}$ preserves finite limits. Therefore a is exact on the left, completing the proof.

The adjunction is the working characterization of sheafification: aF is the initial sheaf receiving a map from F, so it adds exactly the gluing F was missing and nothing more. The plus construction makes this concrete by replacing each $F(X)$ with the colimit of compatible families over finer and finer covers, which is the operation of "allowing sections that are only locally defined." Left exactness of a matters downstream: it is why kernels of sheaf maps are computed presheaf-wise while cokernels require sheafification, and it is the technical fact that makes $\mathrm{Sh}(\mathcal{C})$ an abelian category when $\mathcal{C}$ carries abelian-group-valued sheaves, the setting in which the cohomology $H^i(X_\tau, -)$ of the final two sections lives. The étale and fppf topologies produce genuinely different sheafifications of the same presheaf, and the gap between them is where the arithmetic of a scheme resides.

The plus construction also settles a subtlety flagged by the disjoint-union example. The presheaf F of example 18.2.2(2) is separated (its map e is the injective diagonal), so by Step 3 a *single* plus already sheafifies it: F^+ is the sheaf $T \mapsto A^{\pi_0(T)}$ of locally constant functions, which records one value per connected component and glues across disjoint unions. Two applications are necessary only when the presheaf is not already separated: the first plus then enforces uniqueness and the second enforces existence, and neither can be skipped for a presheaf that fails both axioms. The general principle that $(-)^{++}$, not $(-)^+$, is what always yields a sheaf is thus not decorative, even though the separated example above needs only the one application.

The point of the whole apparatus, for moduli theory, is that the functors one wants to represent geometric objects should be sheaves. A moduli functor $F\colon (\mathbf{Sch}/S)^{\mathrm{op}} \to \mathbf{Set}$ that is not a sheaf cannot be represented by a scheme, because a scheme's functor of points *is* a sheaf. The next theorem is the source of that last fact, and it is Grothendieck's: the functor of points of any scheme is a sheaf not only in the Zariski topology, where it is elementary, but in the fppf topology, where it is the descent theorem of theorem 18.4.1 rephrased. It is the single most-used sheaf statement in the book, because it certifies that every representable piece of a moduli problem already satisfies the gluing every later construction needs.

Theorem 18.2.4: Representable Functors Are fppf Sheaves

Let S be a scheme and T an S-scheme. The representable presheaf

$$h_T = \mathrm{Hom}_S(-, T)\colon (\mathbf{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

is a sheaf in the fppf topology on $(\mathbf{Sch}/S)$. In particular it is a sheaf in the smaller étale and Zariski topologies. The same statement holds in the fpqc topology.

***Proof*:**
By the definition of the fppf topology (definition 18.1.3) it suffices to check the sheaf condition for a covering family, and because h_T carries finite disjoint unions to products, it suffices to treat a *single* surjective, flat, locally-finitely-presented morphism $U \to X$ of S-schemes; a general fppf cover $\{U_i \to X\}$ reduces to the single cover $U = \coprod_i U_i \to X$, which is again fppf. The fpqc strengthening is addressed at the end.

Step 1: Reduction to the equalizer for one cover. For the single cover $U \to X$, the sheaf condition

(18.1) reads: the sequence

$$h_T(X) \xrightarrow{e} h_T(U) \underset{\mathrm{pr}_2^*}{\overset{\mathrm{pr}_1^*}{\rightrightarrows}} h_T(U \times_X U) \tag{18.3}$$

is an equalizer of sets, where $\mathrm{pr}_1, \mathrm{pr}_2 \colon U \times_X U \to U$ are the two projections. Concretely, an element of $h_T(U)$ is an S-morphism $f \colon U \to T$, and the equalizer condition splits into two claims:

1. *Separatedness* (e injective): if $g, g' \colon X \to T$ satisfy $g \circ (U \to X) = g' \circ (U \to X)$, then $g = g'$.
2. *Gluing* (image of e is the equalizer): if $f \colon U \to T$ satisfies $f \circ \mathrm{pr}_1 = f \circ \mathrm{pr}_2$ as morphisms $U \times_X U \to T$, then f descends, meaning $f = g \circ (U \to X)$ for a unique S-morphism $g \colon X \to T$.

Step 2: Separatedness. Since $U \to X$ is fppf it is in particular faithfully flat and (being surjective) an epimorphism in the category of schemes for the purpose at hand: two morphisms $g, g' \colon X \to T$ agreeing after composition with a faithfully flat surjection are equal. This is the geometric form of the algebraic fact that a faithfully flat ring map $R \to R'$ is injective, so $g^\sharp, g'^\sharp \colon \mathcal{O}_T \to g_*\mathcal{O}_X$ are equal once they become equal after the injection $\mathcal{O}_X \hookrightarrow (U \to X)_*\mathcal{O}_U$; faithful flatness gives the injection ([10]). Hence e is injective and claim (1) holds. This separatedness is in fact already contained in the full descent equalizer invoked in Step 3, which yields injectivity and surjectivity of e onto the equalizer at once; Step 2 records the injective half separately because it has the transparent geometric reading above and isolates the one place faithful flatness enters directly.

Step 3: Gluing is descent of morphisms. Claim (2) is exactly the assertion that morphisms to T descend along the fppf cover $U \to X$. This is the content of theorem 18.4.1, proved in full in section 18.4: for S-schemes Y, Z the presheaf $R \mapsto \mathrm{Hom}_R(Y_R, Z_R)$ on the fppf site is a sheaf, equivalently an X-morphism is the same datum as a U-morphism whose two pullbacks to $U \times_X U$ coincide. Applying that theorem with $Y = X$ (as an X-scheme via the identity) and $Z = T \times_S X$ (so that $\mathrm{Hom}_X(X, Z) = \mathrm{Hom}_S(X, T) = h_T(X)$ and $\mathrm{Hom}_U(U, Z_U) = \mathrm{Hom}_S(U, T) = h_T(U)$), the descent equalizer for morphisms is precisely (18.3). A morphism $f \colon U \to T$ with $f \circ \mathrm{pr}_1 = f \circ \mathrm{pr}_2$ is a U-point of T whose two pullbacks to $U \times_X U$ agree, so it descends uniquely to an X-point $g \colon X \to T$ with $g \circ (U \to X) = f$. This establishes claim (2), and with Step 2 the sequence (18.3) is an equalizer. Therefore h_T is an fppf sheaf.

Step 4: The dependency and the fpqc case. The proof rests entirely on theorem 18.4.1: the gluing half of the sheaf condition for h_T *is* descent of morphisms, and no independent argument is needed. We record this dependency explicitly because the two theorems are stated in different sections and it is descent of morphisms, not representability, that supplies the mathematical work. The Zariski and étale cases follow because every Zariski or étale cover is an fppf cover (definition 18.1.3), so a sheaf for the finer fppf topology is a sheaf for the coarser ones. The fpqc strengthening replaces “locally of finite presentation” by “quasi-compact,” and its proof requires the same descent of morphisms along a faithfully flat quasi-compact cover together with a limit argument managing the loss of finite presentation; the set-theoretic subtleties of the fpqc topology (it is not a small site in the naive sense) place its full treatment beyond this book’s framework, and we cite [63] for the fpqc statement in complete generality, completing the proof.

Everything that follows rests on the theorem. Every moduli functor this book studies is assembled from representable functors h_T by limits, quotients, and gluing; theorem 18.2.4 certifies that the raw material is already fppf-local, so the only obstacle to a moduli functor being a sheaf is what the assembly does to it. When a quotient or a colimit destroys the sheaf property, the repair is sheafification (theorem 18.2.3), and the resulting sheaf is the first candidate for a representing object. This is the precise sense in which descent is the technical foundation of Part VI: to represent a moduli problem one first makes it a sheaf, and representable functors are where the sheaf property begins.

The reduction in the proof is worth isolating. Descent of morphisms (theorem 18.4.1) is proved by faithfully flat descent of the graph of the morphism, so theorem 18.2.4 is downstream of the module-level descent of section 18.3. The logical order is: modules descend, therefore morphisms descend, therefore representable functors are sheaves. Grothendieck's insight was that the representability question and the sheaf question are governed by the same faithfully flat descent, and this single flow is what makes the fppf topology, rather than the Zariski, the natural home of moduli.

The abstract theorems become concrete on the group schemes that recur throughout the book. The multiplicative group $\mathbb{G}_m$, the group of n-th roots of unity μ_n, and the general linear group GL_n are all representable, so theorem 18.2.4 makes each an fppf sheaf without further work; the value of computing them explicitly is to see the equalizer condition satisfied by hand and, by contrast, to watch a naive quotient presheaf fail it. The failure is not a pathology but the mechanism: quotients of sheaves of groups are the source of torsors and cohomology, and the gap between the presheaf quotient and its sheafification is exactly the first cohomology group that the next sections compute.

Example 18.2.5: $\mathbb{G}_m$, μ_n, GL_n, and a Quotient That Fails
Work over a base scheme S in the fppf topology on $(\mathbf{Sch}/S)$.

1. *The multiplicative group.* $\mathbb{G}_m = \operatorname{Spec}\mathbb{Z}[t,t^{-1}] \times_{\operatorname{Spec}\mathbb{Z}} S$ represents the functor $T \mapsto \mathcal{O}(T)^\times$, the units of the global function ring. It is representable, hence an fppf sheaf by theorem 18.2.4. Directly: for a cover $U \to X$, a unit $u \in \mathcal{O}(U)^\times$ with $\mathrm{pr}_1^* u = \mathrm{pr}_2^* u$ in $\mathcal{O}(U \times_X U)^\times$ descends to a unit on X because $\mathcal{O}(X)^\times$ is the equalizer of $\mathcal{O}(U)^\times \rightrightarrows \mathcal{O}(U \times_X U)^\times$, which is faithfully flat descent of the structure sheaf restricted to units.

2. *Roots of unity.* $\mu_n = \operatorname{Spec}\mathbb{Z}[t]/(t^n - 1) \times S$ represents $T \mapsto \{u \in \mathcal{O}(T)^\times \,|\, u^n = 1\}$, the n-th roots of unity in $\mathcal{O}(T)$. It is the kernel of the n-th power map $\mathbb{G}_m \xrightarrow{x \mapsto x^n} \mathbb{G}_m$, a representable subfunctor, hence an fppf sheaf. Kernels of maps of sheaves of groups are computed pointwise and remain sheaves, which confirms the equalizer condition without a separate check.

3. *The general linear group.* $\mathrm{GL}_n = \operatorname{Spec}\mathbb{Z}[x_{ij}, \det(x_{ij})^{-1}] \times S$ represents $T \mapsto \mathrm{GL}_n(\mathcal{O}(T))$, the invertible $n \times n$ matrices over $\mathcal{O}(T)$. Representable, hence an fppf sheaf; the case $n = 1$ recovers $\mathbb{G}_m$.

4. *A quotient presheaf that is not a sheaf.* Consider the presheaf Q on $(\mathbf{Sch}/S)_{\mathrm{fppf}}$ sending T to the naive quotient
$$Q(T) = \mathcal{O}(T)^\times / \big(\mathcal{O}(T)^\times\big)^n$$
the cokernel presheaf of $\mathbb{G}_m \xrightarrow{n} \mathbb{G}_m$, formed group-by-group. This Q is a presheaf of abelian groups but is *not* an fppf sheaf. Concretely, take $S = \operatorname{Spec} k$ with k a field containing a primitive n-th root of unity and $\operatorname{char} k \nmid n$, and take $X = \mathbb{G}_{m,k} = \operatorname{Spec} k[x, x^{-1}]$. The class of the coordinate $x \in \mathcal{O}(X)^\times$ is nonzero in $Q(X)$ because x is not an n-th power in $k[x, x^{-1}]$.

Pull back along the fppf cover $U = \operatorname{Spec} k[y, y^{-1}] \to X$, $x \mapsto y^n$ (the n-th power map, finite flat of degree n): on U the pullback of x is y^n, which *is* an n-th power, so the image of x in $Q(U)$ is 0. A nonzero section of $Q(X)$ restricting to 0 on a cover violates separatedness, so Q fails the sheaf condition of definition 18.2.1.

The last item is the entire moral of the section in miniature. The presheaf cokernel of $\mathbb{G}_m \xrightarrow{n} \mathbb{G}_m$ is not a sheaf because an element can become an n-th power fppf-locally without being one globally, which is the failure of separatedness observed on $x = y^n$. The sheafification of Q is the fppf sheaf $\mathbb{G}_m/\mathbb{G}_m^{[n]}$, and the discrepancy between an element being locally an n-th power and globally an n-th power is measured by a cohomology class: the connecting map in the long exact sequence of the Kummer sequence $1 \to \mu_n \to \mathbb{G}_m \xrightarrow{n} \mathbb{G}_m \to 1$ sends the class of x to a nontrivial element of $H^1(X_{\text{fppf}}, \mu_n)$, and that class is the μ_n-torsor $U = \operatorname{Spec} k[y, y^{-1}] \to X$ with $y^n = x$. The failure of the naive quotient to be a sheaf, the need for sheafification, and the appearance of a torsor are three faces of one phenomenon, and the sections that follow develop each face in turn: faithfully flat descent (section 18.3) proves the sheaf statements used here, and torsors and H^1 (section 18.5) identify the sheafification gap with a cohomology group.

A closing contrast sharpens the Zariski-versus-fppf distinction. Every functor above is already a Zariski sheaf, so passing to the fppf topology does not change whether $\mathbb{G}_m$, μ_n, or GL_n is a sheaf; it changes the *covers* against which the quotient Q is tested. Over the Zariski topology the cover $U \to X$ of item (4) is not available (the n-th power map is not a Zariski cover, being finite of degree n rather than an open immersion), so Q is closer to a sheaf Zariski-locally; it is precisely the finer fppf covers that detect the local n-th power and force the sheafification. This is why moduli problems are posed in the fppf or étale topology and not the Zariski: the finer topology sees the identifications, the automorphisms, and the local trivializations that the Zariski topology cannot.

Exercise 18.2.1

1. Let F be a presheaf of sets on a site $\mathcal{C}$. Show that F is separated in the sense of definition 18.2.1 if and only if the map $\theta\colon F \to F^+$ of the plus construction is injective on sections over every object, and that F is a sheaf if and only if θ is an isomorphism. Deduce that a sheaf G satisfies $G^+ \cong G$, the fact used in Step 4 of the proof of theorem 18.2.3.

2. Fix a finite Galois extension L/k with group $\Gamma = \mathrm{Gal}(L/k)$ and work on the small étale site of $\operatorname{Spec} k$. Compute the fiber product $\operatorname{Spec} L \times_{\operatorname{Spec} k} \operatorname{Spec} L$ as a scheme, and show that it is isomorphic to $\coprod_{\gamma \in \Gamma} \operatorname{Spec} L$, a disjoint union of $[L : k]$ copies of $\operatorname{Spec} L$ indexed by Γ. Explain, in one or two sentences, why this makes the sheaf condition (18.1) for the cover $\{\operatorname{Spec} L \to \operatorname{Spec} k\}$ into a condition involving the Galois action, and relate it to the extra components mentioned after definition 18.2.1.

3. Let $\{U_i \to X\}$ be a covering family in which one member $U_{i_0} \to X$ is an isomorphism. Show directly from definition 18.2.1 that every presheaf F satisfies the sheaf condition for this particular cover, so that a cover containing an isomorphism imposes no condition. (This is why the topology only needs to be tested on "genuine" covers.)

4. Give a presheaf on the big Zariski site of $\operatorname{Spec} \mathbb{Z}$ that is separated but not a sheaf. *(Hint: take a subpresheaf of a sheaf whose sections satisfy a global condition that is not local, for example the presheaf $T \mapsto \{f \in \mathcal{O}(T) \,|\, f$ is constant on $T\}$ of globally constant functions, and exhibit a cover on which local constants fail to glue to a global constant.)*

5. Using theorem 18.2.4, prove that the fiber product of two representable functors $h_T \times_{h_S} h_{T'}$ is a sheaf in the fppf topology, first by identifying it with a representable functor and second by a general argument that limits of sheaves are sheaves. Which of the two arguments also shows that an arbitrary (not necessarily finite) product of fppf sheaves is a sheaf?

6. Return to the quotient presheaf $Q(T) = \mathcal{O}(T)^\times/(\mathcal{O}(T)^\times)^n$ of example 18.2.5(4), and pin down exactly why the fppf cover $U = \operatorname{Spec} k[y, y^{-1}] \to X = \operatorname{Spec} k[x, x^{-1}]$, $x \mapsto y^n$, detects the failure of separatedness while no Zariski open cover of X can. First, for a Zariski open cover $\{U_i \hookrightarrow X\}$ of an integral scheme X, verify that the class of a unit $u \in \mathcal{O}(X)^\times$ restricting to 0 in every $Q(U_i)$ need not itself be 0 in $Q(X)$: the local n-th roots v_i with $u|_{U_i} = v_i^n$ satisfy $(v_i/v_j)^n = 1$ on $U_i \cap U_j$, so they define a Čech 1-cocycle valued in μ_n, and u is a global n-th power exactly when this cocycle is a coboundary, that is, when its class in $\check{H}^1(\{U_i\}, \mu_n)$ vanishes. Conclude that Zariski separatedness of Q is equivalent to the vanishing of $\check{H}^1_{\mathrm{Zar}}(X, \mu_n)$ and is therefore *not* automatic. Second, contrast this with the fppf cover above: show that the two pullbacks of y to $U \times_X U$ differ by the μ_n-action $y \mapsto \zeta y$, so y defines a nontrivial descent datum and does not descend to an n-th root of x on X; identify this descent datum with the μ_n-torsor of example 18.2.5. Pinpoint the step that would use Zariski covers being open immersions, and explain why it is unavailable for the finite-flat cover $x \mapsto y^n$.

18.3 Faithfully Flat Descent

The previous two sections built the language: a Grothendieck topology axiomatizes what it means to cover a scheme, and a sheaf on a site is a functor whose values glue along covers. What remains is to make gluing *do work*. The Zariski topology lets one build a sheaf on X from sheaves on an open cover $\{U_i\}$ together with agreement on the overlaps $U_i \cap U_j$. Descent theory is the assertion that the same reconstruction succeeds for the finer topologies of definition 18.1.3, where the "overlaps" are not intersections inside X but fiber products $U_i \times_X U_j$ that need not embed in X at all. The engine that makes this work is faithful flatness, and the section proves the foundational theorem of the whole chapter: a quasi-coherent sheaf on X is the same datum as a quasi-coherent sheaf on a faithfully flat cover $U \to X$ equipped with a compatible identification of its two pullbacks to $U \times_X U$.

This theorem is the reason Part VI can exist. Every later construction in the book depends on it. Grothendieck's theorem that representable functors are fppf sheaves (theorem 18.2.4) reduces to descent; the stack condition of the next chapter is descent for objects with automorphisms; and the very definition of an algebraic space or an algebraic stack presupposes that quasi-coherent sheaves, morphisms, and suitable schemes can be reconstructed from local data in the fppf or étale topology. The heart of the matter is a single exactness statement in commutative algebra, the exactness of the Amitsur complex of a faithfully flat ring map, which the section proves in full and then transports to schemes. The theorem is due to Grothendieck, who developed it in the FGA seminar [29]; the treatment here follows the modern account in [66].

Throughout, $U \to X$ denotes a single covering morphism (rather than a family $\{U_i \to X\}$; a family is recovered by taking $U = \coprod_i U_i$, and the two formulations are equivalent). The two projections $U \times_X U \to U$ are pr_1 and pr_2, and the three projections $U \times_X U \times_X U \to U \times_X U$ are $\mathrm{pr}_{12}, \mathrm{pr}_{13}, \mathrm{pr}_{23}$, where pr_{ij} forgets the factor not named. The first task is to say precisely what data on U should reconstruct a sheaf on X.

The naive hope is that a quasi-coherent sheaf on X is the same as a quasi-coherent sheaf on U. This is far too much to ask: pulling a sheaf back to U loses information, because $U \to X$ is not an isomorphism. What one gains on U must be paid for by a gluing datum recording how the sheaf on U is to be reassembled. Over the Zariski cover the gluing datum is the requirement that the restrictions to U agree on double overlaps; over a faithfully flat cover the double overlap becomes $U \times_X U$, and "agreement" becomes an isomorphism between the two pullbacks $\mathrm{pr}_1^* \mathcal{F}_U$ and $\mathrm{pr}_2^* \mathcal{F}_U$. Compatibility of this isomorphism on triple overlaps is the cocycle condition. This package is a descent datum.

Definition 18.3.1: Descent Datum for Quasi-Coherent Sheaves

Let $U \to X$ be a morphism of schemes. A *descent datum* for quasi-coherent sheaves on $U \to X$ is a pair $(\mathcal{F}, \varphi)$ consisting of a quasi-coherent sheaf $\mathcal{F}$ on U and an isomorphism of $\mathcal{O}_{U \times_X U}$-modules

$$\varphi\colon \mathrm{pr}_1^* \mathcal{F} \xrightarrow{\sim} \mathrm{pr}_2^* \mathcal{F}$$

on $U \times_X U$, subject to the *cocycle condition*: on $U \times_X U \times_X U$ the diagram of pullbacks

$$\mathrm{pr}_{13}^* \varphi = \mathrm{pr}_{23}^* \varphi \circ \mathrm{pr}_{12}^* \varphi$$

holds as an equality of isomorphisms $\mathrm{pr}_1^* \mathcal{F} \to \mathrm{pr}_3^* \mathcal{F}$, where $\mathrm{pr}_i\colon U \times_X U \times_X U \to U$ is the projection to the i-th factor and each $\mathrm{pr}_{ij}^* \varphi$ is interpreted through the canonical identifications $\mathrm{pr}_{ij}^* \mathrm{pr}_a^* \cong \mathrm{pr}_b^*$.

A *morphism of descent data* $(\mathcal{F}, \varphi) \to (\mathcal{G}, \psi)$ is a morphism $f\colon \mathcal{F} \to \mathcal{G}$ of $\mathcal{O}_U$-modules commuting with the gluing isomorphisms, that is, $\psi \circ \mathrm{pr}_1^* f = \mathrm{pr}_2^* f \circ \varphi$. Descent data on $U \to X$ and their morphisms form a category, written $\mathrm{Desc}(U \to X)$.

The cocycle condition is the precise sense in which the gluing is consistent. To read it, note that $\mathrm{pr}_{12}^* \varphi$ glues the first pullback to the second, $\mathrm{pr}_{23}^* \varphi$ glues the second to the third, and $\mathrm{pr}_{13}^* \varphi$ glues the first to the third directly; the condition demands that gluing first-to-second and then second-to-third agrees with gluing first-to-third in one step. Setting the second and third factors equal (pulling back along the diagonal $U \times_X U \to U \times_X U \times_X U$) and using the condition once forces φ to restrict to the identity on the diagonal $\Delta\colon U \to U \times_X U$, so a normalization $\Delta^* \varphi = \mathrm{id}_{\mathcal{F}}$ is automatic rather than an extra axiom. This is exactly the structure of transition functions for a vector bundle, promoted from a Zariski cover to an arbitrary cover: φ plays the role of the transition isomorphisms g_{ij}, and the cocycle condition is the familiar $g_{ik} = g_{jk} g_{ij}$.

There is a canonical source of descent data. Any quasi-coherent sheaf on X can be pulled back to U, and its two pullbacks to $U \times_X U$ are canonically isomorphic because both projections $U \times_X U \to U \to X$ agree after composition with $U \to X$. This produces the pullback functor into descent data, whose essential surjectivity, the statement that *every* descent datum arises this way, is the theorem the section is built to prove.

Definition 18.3.2: The Pullback Functor

Let $p\colon U \to X$ be a morphism. The *pullback functor*

$$p^*\colon \mathrm{QCoh}(X) \longrightarrow \mathrm{Desc}(U \to X)$$

sends a quasi-coherent sheaf $\mathcal{E}$ on X to the descent datum $(p^* \mathcal{E}, \varphi_{\mathrm{can}})$, where $\varphi_{\mathrm{can}}\colon \mathrm{pr}_1^* p^* \mathcal{E} \to \mathrm{pr}_2^* p^* \mathcal{E}$ is the canonical isomorphism arising from the equality $p \circ \mathrm{pr}_1 = p \circ \mathrm{pr}_2\colon U \times_X U \to X$. It sends a morphism $\mathcal{E} \to \mathcal{E}'$ to its pullback $p^* \mathcal{E} \to p^* \mathcal{E}'$, which automatically commutes with the canonical gluings.

The canonical isomorphism φ_{can} satisfies the cocycle condition because all three faces of the triple product agree after composition to X: on $U \times_X U \times_X U$ the three morphisms $\mathrm{pr}_1, \mathrm{pr}_2, \mathrm{pr}_3$ to U all become the same morphism to X, so the canonical identifications compose canonically. Thus $p^*\mathcal{E}$ is indeed a descent datum. The content of the theorem below is that, when p is faithfully flat and quasi-compact, this functor is an equivalence: no information is lost by passing to U with its gluing, and no descent datum is exotic.

Before the general theorem, the case that carries every idea is the affine one. If $X = \operatorname{Spec} B$ and $U = \operatorname{Spec} A$ with $B \to A$ a ring map, then $U \times_X U = \operatorname{Spec}(A \otimes_B A)$ and $U \times_X U \times_X U = \operatorname{Spec}(A \otimes_B A \otimes_B A)$. A quasi-coherent sheaf on U is an A-module N; the two pullbacks to $U \times_X U$ are $A \otimes_B A \otimes_A N = A \otimes_B N$ (via pr_1, tensoring on the left) and $N \otimes_A A \otimes_B A = N \otimes_B A$ (via pr_2, tensoring on the right); and a descent datum is an isomorphism of $(A \otimes_B A)$-modules $\varphi\colon A \otimes_B N \to N \otimes_B A$ satisfying the cocycle condition on $A \otimes_B A \otimes_B A$. The whole theorem is the statement that, for $B \to A$ faithfully flat, such an N with its φ comes from a unique B-module M by $N = A \otimes_B M$, with φ the tautological swap. The bridge to this statement is one exact sequence.

The algebraic core is the exactness of the Amitsur complex. A faithfully flat ring map $B \to A$ makes every B-module M recoverable from $A \otimes_B M$ together with a coequalizer datum, and the precise form of this recovery is that the augmented complex below is exact. Recall from [10] that a ring map $B \to A$ is *faithfully flat* if it is flat and the induced map $\operatorname{Spec} A \to \operatorname{Spec} B$ is surjective; equivalently, A is flat over B and $M \otimes_B A = 0$ implies $M = 0$ for every B-module M. The two facts imported from there are that a faithfully flat map is injective (so $B \hookrightarrow A$) and that a sequence of B-modules is exact if and only if it becomes exact after $- \otimes_B A$.

Lemma 18.3.3: Exactness of the Amitsur Complex

Let $B \to A$ be a faithfully flat ring map and M a B-module. Write $A^{\otimes n}$ for the n-fold tensor power $A \otimes_B \cdots \otimes_B A$. Then the augmented *Amitsur complex*

$$0 \longrightarrow M \xrightarrow{\eta} M \otimes_B A \xrightarrow{d^0} M \otimes_B A \otimes_B A \xrightarrow{d^1} M \otimes_B A^{\otimes 3} \longrightarrow \cdots$$

is exact, where $\eta(m) = m \otimes 1$ and $d^n = \sum_{i=0}^{n+1} (-1)^i \delta_i$ with δ_i inserting a factor 1 in the i-th slot of the A-factors. In particular the first three terms form an exact sequence

$$0 \longrightarrow M \xrightarrow{\eta} M \otimes_B A \xrightarrow{d^0} M \otimes_B A \otimes_B A$$

with $d^0(m \otimes a) = m \otimes 1 \otimes a - m \otimes a \otimes 1$.

***Proof*:**
The strategy is the standard reduction that gives the complex its power: a faithfully flat map is exact-detecting, so it suffices to prove exactness after applying $- \otimes_B A$, and after that base change the complex acquires a contracting homotopy for a purely formal reason, namely that $B \to A$ acquires a section.

Step 1: Reduction to the split case. By faithful flatness ([10]), a complex of B-modules is exact if and only if it stays exact after applying $- \otimes_B A$. Tensor the Amitsur complex with A over B. Since $A \otimes_B (M \otimes_B A^{\otimes n}) = M \otimes_B A^{\otimes(n+1)}$, the base-changed complex is the Amitsur complex of M regarded over A for the ring map $A \to A \otimes_B A =: A'$ given by $a \mapsto 1 \otimes a$, computed with the A-module $M \otimes_B A$. Concretely, after base change the map $B \to A$ is replaced by the map

$A \to A'$, and this map is *split*: the multiplication map $\mu\colon A' = A \otimes_B A \to A$, $a \otimes a' \mapsto aa'$, is a ring map that is a retraction, $\mu \circ (a \mapsto 1 \otimes a) = \mathrm{id}_A$. Thus it suffices to prove that the Amitsur complex of a ring map $A \to A'$ possessing an A-algebra retraction $\mu\colon A' \to A$ is exact for every A'-module (here $M \otimes_B A$).

Step 2: The contracting homotopy from a section. Rename the split situation: let $R \to S$ be a ring map with an R-algebra retraction $\sigma\colon S \to R$ (so σ restricts to the identity on R), and let L be a module over the base. The complex to kill is

$$0 \to L \xrightarrow{\eta} L \otimes_R S \xrightarrow{d^0} L \otimes_R S \otimes_R S \xrightarrow{d^1} \cdots$$

Define R-linear maps $h^n\colon L \otimes_R S^{\otimes(n+1)} \to L \otimes_R S^{\otimes n}$ by applying σ to the last tensor factor with an alternating sign,

$$h^n\big(\ell \otimes s_1 \otimes \cdots \otimes s_{n+1}\big) = (-1)^n\, \sigma(s_{n+1}) \cdot \big(\ell \otimes s_1 \otimes \cdots \otimes s_n\big)$$

with $h^0(\ell \otimes s) = \sigma(s) \cdot \ell$. The sign is what makes the signs in $d^n = \sum_i (-1)^i \delta_i$ telescope; without it the two terms of the homotopy relation would reinforce rather than cancel. A direct check on the face maps δ_i shows the cosimplicial identity

$$h^{n+1} \circ d^n + d^{n-1} \circ h^n = \mathrm{id}$$

in each degree $n \geq 0$, where in degree 0 the second term is replaced by $\eta \circ (\text{augmentation})$: for the first three terms, $h^0 \circ \eta = \mathrm{id}_L$ (because $\sigma(1) = 1$), and $h^1 \circ d^0 + \eta \circ h^0 = \mathrm{id}$ on $L \otimes_R S$. Verifying the middle identity explicitly, on a simple tensor $\ell \otimes s$, the degree-1 homotopy carries its sign $h^1(\ell' \otimes s_1' \otimes s_2') = -\sigma(s_2')(\ell' \otimes s_1')$, so

$$h^1(d^0(\ell \otimes s)) = h^1(\ell \otimes 1 \otimes s - \ell \otimes s \otimes 1) = -\sigma(s)(\ell \otimes 1) + \sigma(1)(\ell \otimes s) = \ell \otimes s - \ell \otimes \sigma(s)$$

while $\eta(h^0(\ell \otimes s)) = \eta(\sigma(s)\ell) = \sigma(s)\ell \otimes 1 = \ell \otimes \sigma(s)$ after moving the scalar $\sigma(s) \in R$ across the tensor product. The two therefore sum to $\ell \otimes s$, as required. A homotopy h with $hd + dh = \mathrm{id}$ forces every cohomology group to vanish, so the split complex is exact.

Step 3: Conclusion. The complex of Step 1 is exactly of the form treated in Step 2 (with $R = A$, $S = A' = A \otimes_B A$, $\sigma = \mu$, $L = M \otimes_B A$), hence exact. By the faithful-flatness reduction of Step 1, the original Amitsur complex over B is exact, which is what we sought to show.

The proof is worth pausing over, because its two-step shape, reduce to a split map by faithful flatness, then contract by hand, recurs everywhere in descent theory. Faithful flatness never lets one see the base B directly; it only guarantees that what is true after base change to A was already true over B. And after base change the situation trivializes completely, because the map $A \to A \otimes_B A$ always splits (via multiplication), so the complex acquires a formal contracting homotopy. This is the mechanism of splitting off a faithfully flat base. The exact sequence in low degrees is the one that matters for descent: M is the equalizer of the two maps $\eta_1, \eta_2\colon A \rightrightarrows A \otimes_B A$ applied to $M \otimes_B A$, precisely the sheaf condition for the cover $\operatorname{Spec} A \to \operatorname{Spec} B$.

Everything now assembles into the descent theorem. The module M is recovered from $M \otimes_B A$ as the equalizer of η_1 and η_2, and a descent datum supplies exactly the extra structure that identifies an A-module with $M \otimes_B A$ for a unique M.

Theorem 18.3.4: Faithfully Flat Descent for Quasi-Coherent Sheaves

Let $U \to X$ be a faithfully flat quasi-compact morphism of schemes. The pullback functor of definition 18.3.2

$$p^*: \mathrm{QCoh}(X) \longrightarrow \mathrm{Desc}(U \to X)$$

is an equivalence of categories. In the affine case $X = \operatorname{Spec} B$, $U = \operatorname{Spec} A$ with $B \to A$ faithfully flat, this says: the functor sending a B-module M to the A-module $A \otimes_B M$ with its canonical descent datum is an equivalence from B-modules to the category of A-modules equipped with a descent datum, with quasi-inverse $(N, \varphi) \mapsto M$, where

$$M = \left\{ n \in N \;\middle|\; \varphi(1 \otimes n) = n \otimes 1 \text{ in } N \otimes_B A \right\}$$

is the submodule of descent-invariant elements.

***Proof*:**
Both the source and target of p^* are local on X in the Zariski topology, and a faithfully flat quasi-compact morphism is, Zariski-locally on X, a faithfully flat morphism of affines (cover X by affine opens and, over each, cover the preimage by finitely many affine opens whose union is faithfully flat, using quasi-compactness). Descent data and quasi-coherent sheaves both glue along a Zariski cover of X, so it suffices to prove the theorem when $X = \operatorname{Spec} B$ and $U = \operatorname{Spec} A$ are affine with $B \to A$ faithfully flat. The affine statement is proved by constructing a quasi-inverse and checking the two composites are the identity.

Step 1: The descended module. Let (N, φ) be a descent datum, so N is an A-module and $\varphi: A \otimes_B N \xrightarrow{\sim} N \otimes_B A$ is an isomorphism of $(A \otimes_B A)$-modules satisfying the cocycle condition. Define

$$M = \ker\left(N \xrightarrow{\psi} N \otimes_B A\right), \qquad \psi(n) = \varphi(1 \otimes n) - n \otimes 1$$

the B-submodule of N on which the two structure maps agree. The claim is that the natural map $A \otimes_B M \to N$, $a \otimes m \mapsto am$, is an isomorphism of A-modules, and that under it φ becomes the canonical descent datum on $A \otimes_B M$.

Step 2: Comparison with the Amitsur equalizer. The cocycle condition makes φ compatible with the two ways of mapping N into $N \otimes_B A \otimes_B A$, which is precisely the statement that ψ fits into the Amitsur complex of $B \to A$ with coefficients twisted by φ. Concretely, transport the standard Amitsur complex of a B-module along φ: the isomorphism φ identifies the complex

$$0 \to M \to N \xrightarrow{\psi} N \otimes_B A \longrightarrow N \otimes_B A \otimes_B A$$

with the augmented Amitsur complex of the underlying B-module of N computed via the descent isomorphism, where the higher differentials are built from φ and its pullbacks and the cocycle condition is exactly what makes the composite of consecutive differentials vanish. Applying lemma 18.3.3 to the flat base change, the sequence

$$0 \to M \to N \xrightarrow{\psi} N \otimes_B A$$

is exact, so $M = \ker \psi$ is the equalizer.

Step 3: The comparison map is an isomorphism. Consider the map $\alpha: A \otimes_B M \to N$, $a \otimes m \mapsto am$. To prove α is an isomorphism it suffices, by faithful flatness ([10]), to prove that $A \otimes_B \alpha$ is an isomorphism of A-modules, since $- \otimes_B A$ reflects isomorphisms. Base-changing the exact sequence of Step 2 along the flat map $B \to A$ keeps it exact:

$$0 \to A \otimes_B M \to A \otimes_B N \xrightarrow{\mathrm{id}\otimes\psi} A \otimes_B N \otimes_B A$$

so $A \otimes_B M = \ker(\mathrm{id} \otimes \psi)$. On the other hand, the descent isomorphism $\varphi\colon A \otimes_B N \xrightarrow{\sim} N \otimes_B A$ carries this kernel isomorphically onto the kernel of the corresponding map out of $N \otimes_B A$, which by the cocycle condition is the equalizer computing N back again; unwinding the identifications, $A \otimes_B \alpha$ is carried to the identity of N. Hence $A \otimes_B \alpha$ is an isomorphism, so α is an isomorphism. This shows $N = A \otimes_B M$ with its canonical descent datum, so p^* is essentially surjective.

Step 4: Full faithfulness. For B-modules M, M' a morphism of descent data $p^*M \to p^*M'$ is an A-linear map $g\colon A \otimes_B M \to A \otimes_B M'$ commuting with the canonical gluing isomorphisms φ_{can}, that is, satisfying $\mathrm{pr}_2^* g \circ \varphi_{\mathrm{can}} = \varphi_{\mathrm{can}} \circ \mathrm{pr}_1^* g$ on $A \otimes_B A$. Under the canonical identifications this equation says that the two pullbacks $\mathrm{pr}_1^* g$ and $\mathrm{pr}_2^* g$ of g to $A \otimes_B A$ agree, that is, g is descent-invariant. The descent-invariant A-linear maps $A \otimes_B M \to A \otimes_B M'$ are then exactly the base changes $\mathrm{id}_A \otimes h$ of B-linear maps $h\colon M \to M'$. This is computed as an equalizer directly, avoiding any finite-presentation hypothesis on the modules. By Step 3 the target module M' is the equalizer of $\eta_1, \eta_2\colon A \otimes_B M' \rightrightarrows A \otimes_B A \otimes_B M'$; applying the left-exact functor $\mathrm{Hom}_B(M, -)$ to the resulting exact sequence $0 \to M' \to A \otimes_B M' \rightrightarrows A \otimes_B A \otimes_B M'$ presents $\mathrm{Hom}_B(M, M')$ as the equalizer

$$0 \to \mathrm{Hom}_B(M, M') \to \mathrm{Hom}_B(M, A \otimes_B M') \rightrightarrows \mathrm{Hom}_B(M, A \otimes_B A \otimes_B M')$$

whose middle and right terms are the A- and $(A \otimes_B A)$-linear map modules $\mathrm{Hom}_A(A \otimes_B M, A \otimes_B M')$ and $\mathrm{Hom}_{A\otimes_B A}(\cdots)$ by adjunction. Thus the invariant maps are $\mathrm{Hom}_B(M, M')$ outright, with no comparison of Hom with a tensor product needed. Hence every morphism of descent data descends uniquely to a B-linear map $M \to M'$, and p^* is fully faithful. An essentially surjective, fully faithful functor is an equivalence, completing the proof.

The theorem is the technical center of the chapter, and its meaning is worth restating in plain terms. A quasi-coherent sheaf on X can be handed over entirely by handing over its pullback to a faithfully flat cover U, together with the single isomorphism φ on $U \times_X U$ that says how the two pullbacks match. Nothing is lost. This is what makes it legitimate to *define* a sheaf on a base by working over a cover, the maneuver on which all of stack theory rests. The invariant submodule $M = \{n : \varphi(1 \otimes n) = n \otimes 1\}$ is the concrete recipe: it is the "G-invariants" of the descent, and in the Galois case below it will literally be a fixed-point set. The quasi-compactness hypothesis is what reduces the general statement to finitely many affines; for the coarser fpqc topology, where covers need not be quasi-compact, the same theorem holds but requires care with set-theoretic size, and the full fpqc statement is cited to [63].

Descent of quasi-coherent sheaves is the master case, and it immediately descends more than sheaves. An affine morphism $Y \to X$ is the same datum as a quasi-coherent sheaf of $\mathcal{O}_X$-algebras, namely $\mathcal{A} = f_*\mathcal{O}_Y$, with $Y = \underline{\mathrm{Spec}}_X \mathcal{A}$; and descent of sheaves upgrades to descent of sheaves of algebras with no new work, because the algebra structure is extra data that the sheaf equivalence carries along. The consequence is that a descent datum of affine schemes over U always glues to an affine scheme over X. This is the first *effectivity* result: not only sheaves but geometric objects, provided they are affine, can be reconstructed from a faithfully flat cover.

Theorem 18.3.5: Effectivity of Descent for Affine Morphisms

Let $U \to X$ be a faithfully flat quasi-compact morphism. The category of affine X-schemes is equivalent to the category of descent data of affine U-schemes, equivalently of quasi-coherent $\mathcal{O}_U$-algebras equipped with a descent datum compatible with the algebra structure. In the affine case $X = \operatorname{Spec} B$, $U = \operatorname{Spec} A$: a descent datum consisting of an A-algebra R together with an isomorphism of $(A \otimes_B A)$-algebras $\varphi\colon A \otimes_B R \xrightarrow{\sim} R \otimes_B A$ satisfying the cocycle condition descends to the B-algebra

$$R_0 = \left\{ r \in R \;\middle|\; \varphi(1 \otimes r) = r \otimes 1 \right\}$$

and $R = A \otimes_B R_0$ as A-algebras. Every descent datum of affine schemes is effective.

***Proof*:**
The strategy is to run the sheaf equivalence of theorem 18.3.4 and check that it respects algebra structures on both sides.

Step 1: Descend the underlying module. By theorem 18.3.4, the descent datum on the underlying quasi-coherent sheaf $\mathcal{A}_U = f_*\mathcal{O}_Y$ (in the affine case, the underlying A-module of R) descends uniquely to a quasi-coherent sheaf $\mathcal{A}$ on X (the B-module R_0), with $\mathcal{A}_U = p^*\mathcal{A}$ (that is, $R = A \otimes_B R_0$). The descended module R_0 is computed as the descent-invariant submodule, exactly as in the theorem.

Step 2: Descend the algebra structure. The multiplication and unit of R are morphisms of A-modules

$$\mu_R\colon R \otimes_A R \to R, \qquad u_R\colon A \to R$$

The hypothesis that φ is an isomorphism of $(A \otimes_B A)$-algebras says precisely that μ_R and u_R are morphisms of descent data, when $R \otimes_A R$ is given its induced descent datum from $\varphi \otimes \varphi$. Because p^* is an equivalence of categories (theorem 18.3.4) it is compatible with tensor products over the structure sheaf, so $(R \otimes_A R, \varphi \otimes \varphi)$ descends to $R_0 \otimes_B R_0$. A morphism of descent data descends uniquely to a morphism over X; applying this to μ_R and u_R produces B-linear maps

$$\mu_{R_0}\colon R_0 \otimes_B R_0 \to R_0, \qquad u_{R_0}\colon B \to R_0$$

whose pullback to A recovers μ_R and u_R.

Step 3: The descended structure is an algebra. The associativity, unit, and commutativity axioms for $(R_0, \mu_{R_0}, u_{R_0})$ are equalities of B-linear maps between finite tensor powers of R_0. Each such equality holds after applying the faithfully flat base change $- \otimes_B A$, where it becomes the corresponding axiom for R, which holds by hypothesis. By faithful flatness ([10]) a map of B-modules that vanishes after $- \otimes_B A$ already vanishes, so each axiom holds over B. Hence R_0 is a commutative B-algebra with $A \otimes_B R_0 = R$ as A-algebras.

Step 4: Equivalence. Conversely, any B-algebra R_0 yields the affine X-scheme $\operatorname{Spec} R_0$ whose pullback to U carries the canonical descent datum, and the two constructions are mutually inverse by theorem 18.3.4 applied to the underlying modules together with the uniqueness of descended morphisms in Steps 2 and 3. Passing from rings to sheaves of algebras and gluing over a Zariski cover of X, exactly as in the proof of theorem 18.3.4, gives the general statement for affine morphisms, completing the proof.

Effectivity for affine morphisms is the sharp boundary of what descends for free. The proof used nothing beyond the sheaf equivalence and the fact that an affine scheme is its algebra of functions; there was no geometry to lose. For non-affine schemes the situation changes: a descent datum of schemes along an fppf cover need not be effective, because the scheme structure can fail to glue even when the underlying sheaves do. The repair, requiring either a descending polarization or the passage to algebraic spaces, is the subject of the next section, and its failure is the reason Part VI must eventually leave schemes behind. For now, the affine case is enough to power the descent of morphisms and Grothendieck's sheaf theorem.

The example that grounds the entire theory, and that connects it to material the reader already knows, is descent along a field extension. Here the cover is $\operatorname{Spec} L \to \operatorname{Spec} k$ for a finite Galois extension L/k, which is faithfully flat (it is finite free of rank $[L:k]$, and surjective on spectra) and even étale when L/k is separable. The fiber product $\operatorname{Spec} L \times_{\operatorname{Spec} k} \operatorname{Spec} L$ is $\operatorname{Spec}(L \otimes_k L)$, and for L/k Galois with group $G = \operatorname{Gal}(L/k)$ there is the classical decomposition $L \otimes_k L \cong \prod_{g \in G} L$, sending $a \otimes b \mapsto (a \cdot g(b))_{g \in G}$. This turns the abstract descent datum into something completely concrete: a semilinear action of the Galois group.

Example 18.3.6: Descent Along a Finite Galois Extension

Let L/k be a finite Galois extension with group $G = \operatorname{Gal}(L/k)$, and consider the faithfully flat cover $\operatorname{Spec} L \to \operatorname{Spec} k$. A quasi-coherent sheaf on $\operatorname{Spec} L$ is an L-vector space N. Under the isomorphism $L \otimes_k L \cong \prod_{g \in G} L$, the two pullbacks of N become

$$L \otimes_k N \cong \prod_{g \in G} N, \qquad N \otimes_k L \cong \prod_{g \in G} N$$

and an isomorphism $\varphi\colon L \otimes_k N \to N \otimes_k L$ of $(L \otimes_k L)$-modules is the same as a family of k-linear isomorphisms $\rho_g\colon N \to N$, one for each $g \in G$, each g-*semilinear*: $\rho_g(\lambda n) = g(\lambda)\rho_g(n)$ for $\lambda \in L$, $n \in N$. The cocycle condition on $\operatorname{Spec}(L \otimes_k L \otimes_k L) \cong \prod_{g,h} \operatorname{Spec} L$ becomes the composition law $\rho_{gh} = \rho_g \circ \rho_h$. Thus:

> A descent datum on $\operatorname{Spec} L \to \operatorname{Spec} k$ is exactly a *semilinear G-action* on the L-vector space N.

Theorem 18.3.4 then reads: the functor $V \mapsto L \otimes_k V$ from k-vector spaces to L-vector spaces with semilinear G-action is an equivalence, with quasi-inverse $N \mapsto N^G = \{n \in N : \rho_g(n) = n \ \forall g\}$, the fixed-point space. An L-vector space with a compatible semilinear Galois action is thus the base change of a unique k-vector space, recovered as the G-invariants. This is the Galois-descent dictionary.

The example is the concrete face of theorem 18.3.4: the descent-invariant submodule $M = \{n : \varphi(1 \otimes n) = n \otimes 1\}$ of the theorem is here literally the fixed-point space N^G, because the condition $\varphi(1 \otimes n) = n \otimes 1$ unwinds, factor by factor over G, into $\rho_g(n) = n$ for every g. Faithfully flat descent thus contains classical Galois descent as the special case of the one-point étale cover $\operatorname{Spec} L \to \operatorname{Spec} k$. The same statement, run with a k-algebra in place of a vector space (invoking theorem 18.3.5), says that a k-algebra is the same as an L-algebra with semilinear G-action: forms of an algebra over k are classified by descent data over L.

The point at which this becomes a computational tool rather than a tautology is when one asks for the *forms* of a fixed object. Given a k-object V_0 (a vector space, an algebra, later a variety), a *twisted* *form* of V_0 is a k-object V that becomes isomorphic to V_0 after base change to L. Every twisted form gives a semilinear action on $L \otimes_k V_0$ that differs from the standard one by an automorphism $c_g \in \mathrm{Aut}_L(L \otimes_k V_0)$ for each g, and the cocycle condition $\rho_{gh} = \rho_g \rho_h$ forces the relation

$$c_{gh} = c_g \cdot {}^g c_h$$

This is exactly the condition for $(c_g)_{g \in G}$ to be a 1-*cocycle* of G with values in the automorphism group, and two forms are isomorphic precisely when their cocycles differ by a coboundary. Thus twisted forms of V_0 are classified by the pointed cohomology set $H^1(G, \mathrm{Aut}(L \otimes_k V_0))$. This is the first appearance of nonabelian Galois cohomology in the book; its formalism, the pointed sets $H^1(G, -)$ and their long exact sequences, is developed in [13] and returned to in the next section (section 18.4), where descent of schemes makes the twisted forms geometric.

Exercise 18.3.1

1. Let $B \to A$ be faithfully flat. Using only lemma 18.3.3 in low degrees, prove directly that $B \to A$ is injective, and that a sequence $0 \to M' \to M \to M'' \to 0$ of B-modules is exact if and only if it is exact after applying $- \otimes_B A$. (These are the two commutative- algebra inputs the section imports from [10]; the point of the exercise is an *alternative* derivation, showing they can equally be read off the Amitsur complex and so are of a piece with descent, rather than the route by which the lemma's own proof obtains them.)

2. Take $B = \mathbb{R}$ and $A = \mathbb{C}$, so $L = \mathbb{C}$, $k = \mathbb{R}$, and $G = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{1, \sigma\}$ with σ complex conjugation. Describe explicitly the semilinear action on $\mathbb{C}^n = \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^n$ coming from the standard descent datum, and verify that its fixed-point space is $\mathbb{R}^n$. Then exhibit a *different* semilinear action on $\mathbb{C}^2$ (a nonstandard ρ_σ) and compute its fixed-point space; confirm it is again a 2-dimensional $\mathbb{R}$-vector space, as theorem 18.3.4 predicts.

3. Verify the cocycle identity for the canonical descent datum of definition 18.3.2. That is, let $\mathcal{E}$ be a quasi-coherent sheaf on X, $p\colon U \to X$ a morphism, and φ_{can} the canonical isomorphism $\mathrm{pr}_1^* p^* \mathcal{E} \to \mathrm{pr}_2^* p^* \mathcal{E}$; show $\mathrm{pr}_{13}^* \varphi_{\mathrm{can}} = \mathrm{pr}_{23}^* \varphi_{\mathrm{can}} \circ \mathrm{pr}_{12}^* \varphi_{\mathrm{can}}$ on $U \times_X U \times_X U$.

4. Show that the cocycle condition forces the normalization $\Delta^* \varphi = \mathrm{id}_{\mathcal{F}}$, where $\Delta\colon U \to U \times_X U$ is the diagonal. (Hint: pull the cocycle identity back along a suitable map $U \times_X U \to U \times_X U \times_X U$ so that two of the three factors coincide, and use that φ is invertible.)

5. Let L/k be finite Galois with group G, and let A_0 be the L-algebra $A_0 = L \times L$ (two copies of L) with the semilinear G-action in which ρ_g acts on the factors both by g *and* by swapping them according to some homomorphism $\chi\colon G \to \mathbb{Z}/2$. Determine, in terms of χ, the descended k-algebra A_0^G, and identify it as a quadratic étale k-algebra. Which χ gives $k \times k$, and which gives a field?

6. Explain why the quasi-compactness hypothesis in theorem 18.3.4 is needed for the reduction to the affine case, and give an example of a faithfully flat but not quasi-compact morphism (e.g. an infinite disjoint union of open immersions covering X redundantly). What goes wrong with the finiteness argument in Step 1 of the proof if quasi-compactness is dropped?

18.4 Descent of Morphisms and of Schemes

The previous section proved that quasi-coherent sheaves descend along a faithfully flat cover: the category $\mathrm{QCoh}(X)$ is equivalent to the category of descent data on $U \to X$, and this equivalence is effective for affine morphisms (theorem 18.3.4, theorem 18.3.5). That result answers one half of what a moduli problem needs. To glue objects in the fppf topology one must descend not only the objects but the *morphisms* between them, and, one level up, the geometric spaces themselves. This section addresses both. The first result, that morphisms descend, is the exact input the chapter's sheaf theory rests on: it is what makes a representable functor an fppf sheaf, closing the gap left open in section 18.2. The second circle of results concerns properties of morphisms, and the third, effectivity of descent for schemes.

The third result records where descent breaks down. Descent data for schemes are *not* always effective: a compatible family of schemes over the pieces of a cover may fail to glue to a scheme over the base. Effectivity holds when a relatively ample line bundle descends alongside the schemes, which is the condition that keeps the descended object projective and hence a scheme. The failure of effectivity in general is not a technical nuisance. It is the precise reason the remainder of Part VI abandons schemes in favor of algebraic spaces and stacks: the category of schemes is too small to be closed under the gluing that moduli problems demand. Galois descent, the classical theory of forms of a variety over a field, is the étale-of-a-point special case of everything here, and it is where the abstract machinery first pays a concrete dividend.

Throughout, $U \to X$ is a faithfully flat cover, quasi-compact and locally of finite presentation where the fppf topology is invoked, with $U \times_X U$ carrying the two projections $\mathrm{pr}_1, \mathrm{pr}_2$ and $U \times_X U \times_X U$ the three projections $\mathrm{pr}_{12}, \mathrm{pr}_{13}, \mathrm{pr}_{23}$. For an X-scheme Y and a morphism $T \to X$ write $Y_T = Y \times_X T$ for the base change.

The starting point is the observation that a morphism between two X-schemes is determined by, and can be reconstructed from, its restriction to a cover, provided the restriction is compatible with the two ways of pulling back to the double overlap. This is the morphism analogue of the sheaf axiom, and its proof is a direct application of faithfully flat descent for sheaves to the sheaf of homomorphisms. The mechanism is worth stating cleanly before the theorem, because it recurs at every level of the descent hierarchy: a morphism is a section of a sheaf, and sections of a sheaf are exactly the data that satisfy the equalizer condition.

Theorem 18.4.1: Descent of Morphisms

Let X be a scheme and $U \to X$ a faithfully flat cover, quasi-compact and locally of finite presentation. Let Y, Z be X-schemes. Then the presheaf on the big fppf site $(\mathbf{Sch}/X)_{\mathrm{fppf}}$

$$\mathcal{F}\colon \quad (T \to X) \longmapsto \mathrm{Hom}_X(Y_T, Z_T)$$

is an fppf sheaf. Concretely, giving an X-morphism $g\colon Y \to Z$ is equivalent to giving a U-morphism $g_U\colon Y_U \to Z_U$ whose two pullbacks to $U \times_X U$ agree:

$$\mathrm{pr}_1^* g_U = \mathrm{pr}_2^* g_U \quad \text{in} \quad \mathrm{Hom}_{U\times_X U}\big(Y_{U\times_X U}, Z_{U\times_X U}\big)$$

That is, the sequence

$$\mathrm{Hom}_X(Y, Z) \longrightarrow \mathrm{Hom}_U(Y_U, Z_U) \rightrightarrows \mathrm{Hom}_{U\times_X U}\big(Y_{U\times_X U}, Z_{U\times_X U}\big)$$

is an equalizer.

***Proof*:**
The proof reduces the descent of morphisms to the descent of quasi-coherent sheaves already established, by exhibiting a morphism as a section of an affine scheme over the base and applying theorem 18.3.4. It is enough to prove the equalizer statement for the single cover $U \to X$; the general sheaf condition for the fppf topology follows because every fppf covering family is refined by such a single quasi-compact faithfully flat morphism, and the equalizer condition passes to the refinement.

Step 1: Injectivity of the restriction map. Suppose $g, g'\colon Y \to Z$ are two X-morphisms with $g_U = g'_U$ after base change to U; the assertion is that $g = g'$. Because $U \to X$ is faithfully flat, so is the base change $Y_U \to Y$; a morphism out of Y is determined by its composite with $Y_U \to Y$ precisely when the latter is an epimorphism of schemes, and a faithfully flat quasi-compact morphism is an epimorphism in the category of schemes. Concretely, g and g' agree if and only if they agree on the affine pieces, where equality of two ring maps $B \to A$ can be tested after the faithfully flat base change $A \to A'$, since $A \to A'$ is injective. Thus $g_U = g'_U$ forces $g = g'$, and the restriction map $\mathrm{Hom}_X(Y, Z) \to \mathrm{Hom}_U(Y_U, Z_U)$ is injective.

Step 2: The gluing datum defines a morphism, affine target case. Assume first that $Z \to X$ is affine, say $Z = \underline{\mathrm{Spec}}_X \mathcal{B}$ for a quasi-coherent $\mathcal{O}_X$-algebra $\mathcal{B}$. Then for any X-scheme W,

$$\mathrm{Hom}_X(W, Z) = \mathrm{Hom}_{\mathcal{O}_X\text{-alg}}\big(\mathcal{B}, (f_W)_*\mathcal{O}_W\big)$$

where $f_W\colon W \to X$ is the structure map, by the universal property of the relative $\underline{\mathrm{Spec}}$. Suppose given $g_U\colon Y_U \to Z_U$ with $\mathrm{pr}_1^* g_U = \mathrm{pr}_2^* g_U$. Translating through the adjunction, g_U is a map of $\mathcal{O}_U$-algebras $\mathcal{B}_U \to (f_{Y_U})_*\mathcal{O}_{Y_U}$, and the cocycle condition says exactly that this map of algebras is compatible with the two pullbacks to $U \times_X U$. The sheaf $\mathcal{B}$ and the pushforward $(f_Y)_*\mathcal{O}_Y$ are quasi-coherent on X (the pushforward because f_Y is quasi-compact and quasi-separated, which holds under the book's standing finite-type-over-k convention), and by theorem 18.3.4 the category of quasi-coherent sheaves on X is equivalent to the category of descent data on $U \to X$. A homomorphism of descent data is a homomorphism of the descended sheaves; hence the compatible algebra map $\mathcal{B}_U \to (f_{Y_U})_*\mathcal{O}_{Y_U}$ descends to a unique algebra map $\mathcal{B} \to (f_Y)_*\mathcal{O}_Y$ over X. This algebra map is the sought morphism $g\colon Y \to Z$, and it restricts to g_U by uniqueness in the descent equivalence. This proves the equalizer statement when $Z \to X$ is affine.

Step 3: Reduction of the general target to the affine case. For general Z, the point is that the morphism condition is local on the source Y and on the target Z, and locality is a Zariski (hence fppf) condition to which the sheaf property already proved applies. Cover Z by affine opens $Z = \bigcup_\alpha Z_\alpha$. A morphism $g_U : Y_U \to Z_U$ with matching pullbacks determines, over each preimage $g_U^{-1}(Z_{\alpha,U}) \subseteq Y_U$, a morphism into the affine Z_α. The preimages $g_U^{-1}(Z_{\alpha,U})$ are open in Y_U and, because the compatibility $\mathrm{pr}_1^* g_U = \mathrm{pr}_2^* g_U$ identifies them across the two pullbacks, they descend to open subschemes $Y_\alpha \subseteq Y$ with $(Y_\alpha)_U = g_U^{-1}(Z_{\alpha,U})$: an open subscheme of Y_U stable under the descent datum is the pullback of a unique open subscheme of Y, which is descent for open immersions, itself the affine-target case applied to the structure sheaf. On each Y_α the restricted datum $g_U|_{(Y_\alpha)_U} : (Y_\alpha)_U \to (Z_\alpha)_U$ has affine target, so Step 2 produces a morphism $g_\alpha : Y_\alpha \to Z_\alpha \hookrightarrow Z$. On overlaps $Y_\alpha \cap Y_\beta$ the two morphisms g_α, g_β agree, because they agree after the faithfully flat base change to U (both restrict to g_U) and Step 1 promotes that to equality on the base. The g_α therefore glue to a single X-morphism $g : Y \to Z$ restricting to g_U. This establishes the equalizer for arbitrary Z, and with it the sheaf property, completing the proof.

The theorem supplies what the chapter's sheaf theory needs. In section 18.2 the claim that a representable presheaf $h_T = \mathrm{Hom}_X(-, T)$ is an fppf sheaf (Grothendieck's theorem, theorem 18.2.4) was stated with its proof deferred to the descent of morphisms. That debt is now paid. Taking $Y = X$ and $Z = T$ fixed, so that for a test scheme $S \to X$ one has $Y_S = S$ and $Z_S = T_S$, the sheaf $\mathcal{F}$ of the theorem evaluates to $\mathcal{F}(S) = \mathrm{Hom}_X(S, T_S) = h_T(S)$; it is exactly the representable functor h_T. An fppf-local morphism into T that agrees on double overlaps glues to a genuine morphism into T, which is precisely the sheaf condition for h_T. Thus theorem 18.2.4 is a corollary of theorem 18.4.1, and the fppf topology is fine enough that representable functors remain sheaves in it. The strengthening to the fpqc topology, where the same conclusion holds for arbitrary faithfully flat quasi-compact covers without the finite-presentation hypothesis, requires a more delicate set-theoretic treatment of the site and is treated in full in [63]; the fppf statement above is what the moduli constructions of this book use.

A first concrete consequence is that morphisms between varieties over a field can be constructed after passing to a field extension, provided they are compatible with the two pullbacks. This is the mechanism behind the descent of forms in the Galois case below, and it deserves an explicit worked instance.

Example 18.4.2: Descending a Morphism from a Field Extension

Let k be a field, L/k a finite Galois extension with Galois group $\Gamma = \mathrm{Gal}(L/k)$, and let Y, Z be k-schemes. The cover is $\operatorname{Spec} L \to \operatorname{Spec} k$, and the double fiber product decomposes as a product over the Galois group,

$$\operatorname{Spec} L \times_{\operatorname{Spec} k} \operatorname{Spec} L \;\cong\; \coprod_{\sigma \in \Gamma} \operatorname{Spec} L$$

one copy for each $\sigma \in \Gamma$, the σ-th copy embedded by $(\mathrm{id}, \sigma) : L \otimes_k L \to L$. Under this identification the two projections $\mathrm{pr}_1, \mathrm{pr}_2$ send the σ-th copy to the source and target factors, and the compatibility condition $\mathrm{pr}_1^* g_U = \mathrm{pr}_2^* g_U$ of theorem 18.4.1 becomes the statement that the base-changed morphism $g_L : Y_L \to Z_L$ commutes with the semilinear Γ-action on both sides: for every $\sigma \in \Gamma$,

$$g_L \circ \sigma_Y = \sigma_Z \circ g_L$$

where σ_Y, σ_Z are the automorphisms of Y_L, Z_L induced by σ acting on L. Concretely, take

$Y = Z = \mathbb{A}^1_k = \operatorname{Spec} k[t]$ and $k = \mathbb{R}$, $L = \mathbb{C}$, $\Gamma = \{1, c\}$ with c complex conjugation. An L-morphism $g_L \colon \mathbb{A}^1_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{C}}$ is a polynomial $g_L(t) = \sum a_i t^i$ with $a_i \in \mathbb{C}$. The conjugation c acts on $\mathbb{A}^1_{\mathbb{C}}$ by $t \mapsto t$ on the coordinate and by $a_i \mapsto \bar{a}_i$ on the coefficients read as functions, so g_L is Γ-equivariant precisely when $\bar{a}_i = a_i$ for all i, that is, when every coefficient is real. The descended morphism is then the real polynomial map $g \colon \mathbb{A}^1_{\mathbb{R}} \to \mathbb{A}^1_{\mathbb{R}}$ with those coefficients. A polynomial with a complex coefficient, such as $g_L(t) = it$, fails equivariance and does not descend: it defines a morphism only over $\mathbb{C}$, not over $\mathbb{R}$.

Once objects and morphisms descend, the next question is which *properties* of a morphism can be detected after a faithfully flat base change. A property P of morphisms is called *fppf-local on the base* when, for every faithfully flat cover $U \to X$ locally of finite presentation, a morphism $f \colon Y \to X$ has property P if and only if the base change $f_U \colon Y_U \to U$ has property P. This is what one wants for constructions built by gluing: to check that a descended morphism is flat, or smooth, or proper, it suffices to check the property on the pieces of a cover, where the geometry is often simpler. The list of such properties is long and central to the theory, so the pattern of proof matters more than any single entry. The argument for flatness and for smoothness is the model; the rest follow the same faithfully flat descent of the defining local condition.

Theorem 18.4.3: Descent of Properties of Morphisms

Let $U \to X$ be a faithfully flat cover, quasi-compact and locally of finite presentation, and let $f \colon Y \to X$ be a morphism of schemes. For each property P in the list

$$\{\text{flat},\ \text{smooth},\ \text{étale},\ \text{proper},\ \text{finite},\ \text{an immersion},\ \text{surjective},\ \text{quasi-compact}\}$$

the morphism f has property P if and only if the base change $f_U \colon Y_U \to U$ has property P. In other words, each of these properties is fppf-local on the base.

***Proof*:**
Necessity in each case is immediate, since every property in the list is stable under arbitrary base change: if f has property P then so does f_U, being a base change of f. The content is sufficiency, that property P on f_U descends to f. The full table is verified property by property in [63], where each entry is reduced to a local statement in commutative algebra; the two representative cases are proved here in full, and the remaining entries follow the same pattern of descending a local condition along a faithfully flat ring map.

Step 1: Flatness descends. Flatness of a morphism is a condition on stalks: f is flat if and only if for every point $y \in Y$ the local ring $\mathcal{O}_{Y,y}$ is flat over $\mathcal{O}_{X,f(y)}$. Reduce to the affine situation $X = \operatorname{Spec} R$, $U = \operatorname{Spec} R'$ with $R \to R'$ faithfully flat, and Y affine over X near y, say $Y = \operatorname{Spec} A$ with A an R-algebra; flatness of f is flatness of A as an R-module, and flatness of f_U is flatness of $A' := A \otimes_R R'$ as an R'-module. Suppose A' is flat over R'. Let $0 \to M_1 \to M_2 \to M_3 \to 0$ be a short exact sequence of R-modules; applying $- \otimes_R A$ gives a complex whose exactness is to be shown. Base change to R': the complex $(- \otimes_R A) \otimes_R R'$ is

$$M_i \otimes_R A \otimes_R R' \;\cong\; (M_i \otimes_R R') \otimes_{R'} A'$$

by associativity of tensor product, and this is the sequence $M_i \otimes_R R'$ tensored over R' with the flat R'-module A'. Since $R \to R'$ is flat, the sequence $M_i \otimes_R R'$ is exact, and since A' is R'-flat, tensoring it with A' keeps it exact. Thus the complex $(M_\bullet \otimes_R A) \otimes_R R'$ is exact. But $R \to R'$ is *faithfully* flat, so a complex of R-modules is exact if and only if it becomes exact after $- \otimes_R R'$; therefore $M_\bullet \otimes_R A$ is itself exact. Hence A is flat over R, and f is flat.

Step 2: Smoothness descends. A morphism is smooth if and only if it is flat, locally of finite presentation, and has geometrically regular fibers of the expected dimension. Under flatness and local finite presentation, the geometric regularity of the fibers is encoded by the sheaf of relative differentials $\Omega_{Y/X}$ being locally free of rank equal to the fiber dimension: for a flat, finitely presented morphism the fiber over a point is regular of dimension n exactly when $\Omega_{Y/X}$ is locally free of rank n there. Thus smoothness is equivalent to flatness and local finite presentation together with local freeness of $\Omega_{Y/X}$ of the expected rank. Flatness of f descends by Step 1. Local finite presentation descends because $R \to R'$ is faithfully flat: an R-algebra A with $A \otimes_R R'$ of finite presentation over R' is of finite presentation over R, a faithfully flat descent statement for the finiteness of a presentation carried out in [63] and resting on [10]. It remains to descend the freeness and rank of $\Omega_{Y/X}$. Relative differentials commute with base change, so $\Omega_{Y_U/U}$ is the pullback of $\Omega_{Y/X}$ along the projection $Y_U \to Y$. If $\Omega_{Y_U/U}$ is locally free of rank n, then the pullback of $\Omega_{Y/X}$ along the faithfully flat morphism $Y_U \to Y$ is locally free of rank n; and local freeness of a coherent sheaf descends along a faithfully flat morphism, because a finitely presented module N over $\mathcal{O}_{Y,y}$ is free precisely when $N \otimes \mathcal{O}_{Y_U,y'}$ is free over the faithfully flat local extension, the rank being read off either side. Hence $\Omega_{Y/X}$ is locally free of rank n, and f is smooth.

Step 3: The remaining properties. Each of étale, finite, immersion, surjective, and quasi-compact is a combination of conditions already treated or a local condition of the same form. Étale is smooth of relative dimension 0, so it descends by Step 2 together with the descent of fiber dimension, the relative dimension at a point being unchanged under the faithfully flat base change $Y_U \to Y$ since fibers of f_U are base changes of fibers of f to field extensions. Surjectivity descends because $U \to X$ is surjective on points: f_U surjective forces f surjective by a diagram chase on underlying topological spaces, since $Y_U \to Y$ and $U \to X$ are surjective. Being an immersion, finite, or proper is checked by descending the corresponding property of the associated affine or the corresponding valuative or topological criterion, each reduced to a faithfully flat descent of a commutative-algebra condition in the manner of Steps 1 and 2; the full verifications, uniform in shape, are in [63]. This exhausts the list and completes the proof.

The descent of properties is what makes the atlas philosophy of Part VI work. An algebraic space or stack is presented by a cover from a scheme, and every geometric property one wants to attribute to the space, that it is smooth, that a morphism between two of them is proper, is defined by checking the property on the cover. theorem 18.4.3 is the guarantee that such definitions are well posed: the property read off the atlas does not depend on the atlas, because it descends. The same theorem is what lets the Hilbert and Quot schemes, and the GIT quotients built earlier in this book, be assembled by gluing over a cover while retaining flatness or properness. Flatness in particular is the most-used entry, because it is the technical hypothesis under which cohomology and base change behave, and its fppf-locality on the base is invoked silently throughout the moduli literature.

The final and most consequential result concerns descent of the schemes themselves, not their morphisms or their properties alone. A descent datum for schemes along a cover $U \to X$ is a scheme V over U together with an isomorphism $\varphi\colon \mathrm{pr}_1^* V \xrightarrow{\sim} \mathrm{pr}_2^* V$ over $U \times_X U$ satisfying the cocycle condition on the triple overlap. The datum is *effective* when it arises from base change of a single scheme over X: there exists a scheme $Y \to X$ and an isomorphism $Y_U \cong V$ over U carrying the canonical descent datum of Y_U to φ. For quasi-coherent sheaves, descent is always effective (theorem 18.3.4); for affine morphisms it is again always effective (theorem 18.3.5). For general schemes it is not. The obstruction is that gluing the affine pieces of V across the cover can produce an object that is a scheme locally over the cover but has no global scheme structure, because it cannot be covered by affine opens compatibly. The remedy, and the boundary of what schemes can do, is a polarization.

Theorem 18.4.4: Effective Descent for Polarized Schemes

Let $U \to X$ be a faithfully flat cover, quasi-compact and locally of finite presentation, and let (V, φ) be a descent datum for schemes along $U \to X$ with $V \to U$ quasi-projective. Suppose there is a line bundle $\mathcal{L}$ on V, relatively ample for $V \to U$, together with an isomorphism $\mathrm{pr}_1^* \mathcal{L} \xrightarrow{\sim} \mathrm{pr}_2^* \mathcal{L}$ over $\mathrm{pr}_1^* V \cong \mathrm{pr}_2^* V$ compatible with φ and satisfying the cocycle condition, that is, the polarization itself carries a descent datum. Then the descent datum (V, φ) is effective: there is a scheme $Y \to X$, quasi-projective, with $Y_U \cong V$ compatibly with φ, and $\mathcal{L}$ descends to a relatively ample line bundle on Y.

***Proof*:**
The strategy is to trade the scheme V for a quasi-coherent sheaf of graded algebras, descend the sheaf by theorem 18.3.4, and recover the scheme as a relative $\underline{\mathrm{Proj}}$. This converts an effectivity question for schemes, where descent can fail, into an effectivity question for quasi-coherent sheaves, where it never does.

Step 1: Reconstruct V from its polarization. Because $\mathcal{L}$ is relatively ample for the quasi-projective morphism $g\colon V \to U$, the scheme V is recovered from the graded $\mathcal{O}_U$-algebra of its sections,

$$\mathcal{S}_U \;:=\; \bigoplus_{n \geq 0} g_* \mathcal{L}^{\otimes n}$$

as the relative Proj, $V \cong \underline{\mathrm{Proj}}_U \mathcal{S}_U$, with $\mathcal{L}$ recovered as the relative $\mathcal{O}(1)$ (for $\mathcal{L}$ relatively ample rather than very ample one first replaces it by a Veronese power $\mathcal{L}^{\otimes d}$ generating $\mathcal{S}_U$ in degree one, which does not affect the descent argument since the polarization datum passes to the power). Each graded piece $g_* \mathcal{L}^{\otimes n}$ is a quasi-coherent $\mathcal{O}_U$-module, so $\mathcal{S}_U$ is a quasi-coherent sheaf of graded $\mathcal{O}_U$-algebras.

Step 2: The graded algebra carries a descent datum. The isomorphism $\mathrm{pr}_1^* \mathcal{L} \cong \mathrm{pr}_2^* \mathcal{L}$ over the double overlap, being compatible with φ and satisfying the cocycle condition, induces for each n an isomorphism of pushforwards

$$\mathrm{pr}_1^* \big(g_* \mathcal{L}^{\otimes n} \big) \;\xrightarrow{\sim}\; \mathrm{pr}_2^* \big(g_* \mathcal{L}^{\otimes n} \big)$$

compatible with the grading and the algebra structure, because pushforward commutes with the flat base changes $\mathrm{pr}_1, \mathrm{pr}_2$ (flat base change for quasi-coherent cohomology applies since pr_i are flat, being base changes of the flat $U \to X$). Assembling over n, the graded algebra $\mathcal{S}_U$ acquires a descent datum $\psi\colon\ \mathrm{pr}_1^* \mathcal{S}_U \xrightarrow{\sim} \mathrm{pr}_2^* \mathcal{S}_U$ of quasi-coherent sheaves of graded algebras, satisfying the cocycle condition because φ and the polarization datum do.

Step 3: Descend the algebra and take Proj. By theorem 18.3.4, applied to each graded piece and assembled, the descent datum $(\mathcal{S}_U, \psi)$ is effective as a quasi-coherent sheaf: there is a quasi-coherent sheaf of graded $\mathcal{O}_X$-algebras $\mathcal{S}$ on X with $\mathrm{pr}^* \mathcal{S} \cong \mathcal{S}_U$ compatibly with ψ, where $\mathrm{pr}\colon U \to X$. The algebra structure descends because it is a morphism of quasi-coherent sheaves $\mathcal{S}_U \otimes \mathcal{S}_U \to \mathcal{S}_U$ compatible with the descent data, and morphisms of quasi-coherent sheaves descend by the same theorem. Now set

$$Y \;:=\; \underline{\mathrm{Proj}}_X \mathcal{S}$$

Formation of relative Proj commutes with the flat base change $U \to X$, so $Y_U = \underline{\mathrm{Proj}}_U(\mathrm{pr}^* \mathcal{S}) = \underline{\mathrm{Proj}}_U \mathcal{S}_U \cong V$, and this isomorphism carries the canonical descent datum of Y_U to φ because it does so on the defining graded algebras. The relative $\mathcal{O}(1)$ on Y is a relatively ample line bundle descending $\mathcal{L}$. Thus $Y \to X$ is a quasi-projective scheme representing the descent datum, and $\mathcal{L}$ descends to a relatively ample bundle on it, as we sought to show.

The hypothesis that the polarization descends is exactly what cannot be dropped. Without a compatible relatively ample line bundle there is in general no scheme over X realizing the datum, and this failure is genuine, not an artifact of the proof. The standard obstruction is that the descended object, while locally a scheme, may be proper but non-projective, and a proper non-projective object has no ample line bundle to descend. The next result records the classical counterexample.

Theorem 18.4.5: Non-Effective Descent

There exist an algebraic space X, a faithfully flat (indeed étale) cover $U \to X$ by a scheme U, and a descent datum (V, φ) for schemes along $U \to X$ whose descended object over X would be proper, that is not effective in the category of schemes: no scheme $Y \to X$ has $Y_U \cong V$ compatibly with φ. The cover U is a Hironaka-type proper non-projective threefold carrying a free $\mathbb{Z}/2$-action, X is its free quotient, and $V = U$ with the tautological descent datum recording the action. The obstruction is the absence of a descending polarization, the free action exchanging two curves that no $(\mathbb{Z}/2)$-invariant ample bundle can distinguish, so the free quotient exists as an algebraic space but not as a scheme.

***Proof Key ideas*:**
The full construction, with the geometry of the proper non-projective threefold and the verification that the resulting descent datum admits no scheme representing it, is carried out in [66] and, in the setting of Grothendieck's original descent talks, in [29]; the argument uses the non-projective threefold geometry that is beyond the present chapter's scope, so the mechanism is described here and the detailed verification is cited. The idea is as follows. Hironaka constructed a smooth proper threefold W that is not projective, with a free action of a group of order two, $\mathbb{Z}/2$, whose two orbits of a certain curve are exchanged in a way no ample class can respect: the quotient $W/(\mathbb{Z}/2)$ exists as an algebraic space but not as a scheme, because a scheme quotient would require a $(\mathbb{Z}/2)$-invariant ample bundle, which does not exist.

Encoded as descent, take $U = W$ with the free $\mathbb{Z}/2$-action, let $X = W/(\mathbb{Z}/2)$ as an algebraic space so that $U \to X$ is an étale (hence fppf) cover with $U \times_X U = U \times \mathbb{Z}/2 = U \sqcup U$, and let $V = U$ with the tautological descent datum φ recording the $\mathbb{Z}/2$-action. Effectivity of this datum in schemes would produce a scheme Y with $Y_U \cong U$, that is, a scheme quotient $W/(\mathbb{Z}/2)$; its nonexistence as a scheme is exactly Hironaka's non-projectivity obstruction. Thus (V, φ) is a descent datum for schemes whose descended object over X would be proper that is not effective in the category of schemes, though it is effective in the larger category of algebraic spaces. This is the phenomenon claimed.

This is the fault line that Part VI is built to cross. Descent for quasi-coherent sheaves is always effective; descent for affine morphisms is always effective; but descent for schemes fails, and it fails for a structural reason, the possible absence of a polarization, not a repairable technical one. The moment a moduli problem forces a gluing of the second kind, a compatible family of proper objects with no globally coherent ample bundle, the glued object exists as a geometric space but not as a scheme. The response is to enlarge the category. Algebraic spaces, introduced in the chapter on algebraic spaces and stacks below, are precisely the objects obtained by allowing étale descent data for schemes to be effective by fiat, and algebraic stacks go one level further, allowing descent data with automorphisms. Hironaka's non-effective datum is a scheme-theoretic ghost that becomes a genuine algebraic space, and this is the first concrete evidence that the category of schemes is the wrong home for moduli. The counterexample is not a pathology to be avoided but the signpost pointing from schemes to spaces and stacks.

Descent theory has a classical ancestor that predates the language of sites and covers by decades: Galois descent, the theory of forms of an algebraic object over a field. When the cover is a Galois field extension $\operatorname{Spec} L \to \operatorname{Spec} k$, the machinery specializes exactly, and the abstract notion of a descent datum reduces to the concrete notion of a semilinear Galois action. The classification of forms by a nonabelian cohomology set $H^1(\mathrm{Gal}, \mathrm{Aut})$ is the one-point case of the torsor classification developed in the next section. Galois descent is where the theory of this chapter first meets objects a reader has seen before, and it supplies the running dictionary between cocycles and twisted forms.

Example 18.4.6: Galois Descent and Forms of a Variety

Let k be a field with separable closure k^{sep} and absolute Galois group $\Gamma = \mathrm{Gal}(k^{\mathrm{sep}}/k)$. Fix a k-scheme (or a k-vector space, or a k-algebra) X_0. A *form* of X_0 is a k-object X that becomes isomorphic to X_0 after base change to k^{sep}: an isomorphism $X_{k^{\mathrm{sep}}} \cong (X_0)_{k^{\mathrm{sep}}}$ exists, though no k-isomorphism $X \cong X_0$ need. Two forms are equivalent when they are k-isomorphic. The content of Galois descent is that the set of forms of X_0 up to equivalence is classified by nonabelian Galois cohomology.

The dictionary. By theorem 18.4.1 applied to the cover $\operatorname{Spec} k^{\mathrm{sep}} \to \operatorname{Spec} k$, and its extension to descent of schemes, a k-form of X_0 is the same datum as a descent datum on $(X_0)_{k^{\mathrm{sep}}}$, which by the Galois decomposition of the double fiber product is a continuous system of isomorphisms

$$c_\sigma \colon (X_0)_{k^{\mathrm{sep}}} \xrightarrow{\sim} {}^{\sigma}(X_0)_{k^{\mathrm{sep}}}, \qquad \sigma \in \Gamma$$

one for each element of the Galois group, satisfying the cocycle condition $c_{\sigma\tau} = {}^{\sigma}c_\tau \circ c_\sigma$. Composing with the tautological semilinear action lets one record the datum as a 1-cocycle $\sigma \mapsto a_\sigma \in \mathrm{Aut}\big((X_0)_{k^{\mathrm{sep}}}\big)$ valued in the automorphism group, and changing the chosen k^{sep}-isomorphism $X_{k^{\mathrm{sep}}} \cong (X_0)_{k^{\mathrm{sep}}}$ replaces a_σ by a coboundary. The forms of X_0 are therefore in bijection with the pointed cohomology set

$$\{k\text{-forms of } X_0\}/\cong \;\longleftrightarrow\; H^1\big(\Gamma, \mathrm{Aut}((X_0)_{k^{\mathrm{sep}}})\big)$$

the base point (trivial cocycle) corresponding to X_0 itself. The nonabelian H^1 here is a pointed set, not a group, and the formalism of cocycles, coboundaries, and twisting for a nonabelian coefficient group is developed in [13].

Worked form: the conics as forms of $\mathbb{P}^1$. Take $k = \mathbb{R}$, $k^{\mathrm{sep}} = \mathbb{C}$, $\Gamma = \{1, c\}$ with c complex conjugation, and $X_0 = \mathbb{P}^1_{\mathbb{R}}$. A smooth conic $C \subseteq \mathbb{P}^2_{\mathbb{R}}$ with no real point, such as the anisotropic conic

$$C\colon \quad x^2 + y^2 + z^2 = 0$$

becomes isomorphic to $\mathbb{P}^1_{\mathbb{C}}$ over $\mathbb{C}$ (every smooth conic over an algebraically closed field is a rational curve isomorphic to $\mathbb{P}^1$), so C is a form of $\mathbb{P}^1_{\mathbb{R}}$. It is not $\mathbb{R}$-isomorphic to $\mathbb{P}^1_{\mathbb{R}}$, because $\mathbb{P}^1_{\mathbb{R}}$ has real points and C does not. The automorphism group is $\mathrm{Aut}(\mathbb{P}^1_{\mathbb{C}}) = \mathrm{PGL}_2(\mathbb{C})$, and

$$H^1\big(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathrm{PGL}_2(\mathbb{C})\big)$$

has exactly two classes: the trivial class, giving $\mathbb{P}^1_{\mathbb{R}}$, and one nontrivial class, giving the anisotropic conic C. The two forms of $\mathbb{P}^1$ over $\mathbb{R}$ are the split line and the pointless conic, and the nontrivial cohomology class measures the failure of C to have a real point. Equivalently, quaternion algebras over $\mathbb{R}$ enter: the conic C is the Severi-Brauer variety of Hamilton's quaternions $\mathbb{H}$, and the two elements of H^1 correspond to the two $\mathbb{R}$-quaternion algebras $M_2(\mathbb{R})$ and $\mathbb{H}$, the split and division cases.

A second form: the two forms of $\mathbb{G}_m$. Over a field k the one-dimensional split torus $\mathbb{G}_m$ has automorphism group $\mathrm{Aut}(\mathbb{G}_{m,k^{\mathrm{sep}}}) = \{\pm 1\}$, generated by inversion $t \mapsto t^{-1}$. Its forms are classified by

$$H^1(\Gamma, \mathbb{Z}/2) = \mathrm{Hom}_{\mathrm{cont}}(\Gamma, \mathbb{Z}/2)$$

the continuous homomorphisms $\Gamma \to \mathbb{Z}/2$, which by Galois theory are the quadratic étale k-algebras: the trivial homomorphism gives the split torus $\mathbb{G}_m$, and a surjection $\Gamma \to \mathrm{Gal}(L/k) = \mathbb{Z}/2$ for a separable quadratic extension L/k gives the nonsplit norm-one torus

$$R^{(1)}_{L/k}\mathbb{G}_m \;=\; \ker\big(\mathrm{Nm}\colon R_{L/k}\mathbb{G}_m \to \mathbb{G}_m\big)$$

whose k-points are the norm-one elements of $L^\times$. For $k = \mathbb{R}$, $L = \mathbb{C}$, this nonsplit form is the circle group $\mathrm{SO}(2)$, the unit circle $\{z \in \mathbb{C} : |z| = 1\}$, which becomes $\mathbb{G}_m$ over $\mathbb{C}$ through $z \mapsto z$ but is anisotropic over $\mathbb{R}$. The two forms of $\mathbb{G}_m$ over $\mathbb{R}$, the multiplicative group and the circle, are the two elements of $H^1(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathbb{Z}/2)$, matching the two quadratic étale $\mathbb{R}$-algebras $\mathbb{R} \times \mathbb{R}$ and $\mathbb{C}$.

Galois descent shows the machinery of this chapter operating at its smallest nontrivial scale, a cover by a single point $\mathrm{Spec}\, L \to \mathrm{Spec}\, k$, and yields a classification a reader can compute by hand. The passage from the general descent datum to a 1-cocycle valued in an automorphism group is the same passage that, in the next section, turns a torsor into a class in $\check{H}^1$; the forms of X_0 classified by $H^1(\Gamma, \mathrm{Aut}(X_0))$ are the point-level instance of torsors classified by H^1 of a site. The automorphism group $\mathrm{Aut}(X_0)$ that appears here as the coefficient of the cohomology is the same automorphism obstruction that made $\mathcal{M}_{1,1}$ fail to be a fine moduli space in example 13.3.4: forms, torsors, and the automorphisms that obstruct fine representability are three faces of one phenomenon, and the sheaf-theoretic language of theorem 18.4.1 is what unifies them. The next section develops the torsor picture directly.

Exercise 18.4.1

1. Let $U \to X$ be a faithfully flat quasi-compact morphism and $g, g'\colon Y \to Z$ two X-morphisms of X-schemes. Using theorem 18.4.1, show directly that if $g_U = g'_U$ then $g = g'$, by reducing to the affine case and using that a faithfully flat ring map $A \to A'$ is injective. Explain where faithful flatness (as opposed to plain flatness) is used.
2. Let $k = \mathbb{R}$, $L = \mathbb{C}$, and consider the $\mathbb{C}$-morphism $g_L\colon \mathbb{A}^1_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{C}}$ given by $t \mapsto (2+3i)t^2 + 5t$. Does g_L descend to an $\mathbb{R}$-morphism $\mathbb{A}^1_{\mathbb{R}} \to \mathbb{A}^1_{\mathbb{R}}$? If not, exhibit the failure of the compatibility condition of theorem 18.4.1 explicitly. Then find the condition on the coefficients of a general $g_L(t) = \sum a_i t^i$ under which it descends.
3. Show that "being an open immersion" is fppf-local on the base, in the pattern of theorem 18.4.3:

for $U \to X$ faithfully flat and $f\colon Y \to X$, if $f_U\colon Y_U \to U$ is an open immersion then so is f. (*Hint: an open immersion is a flat monomorphism locally of finite presentation; descend flatness by the theorem, descend the monomorphism property using injectivity of* Hom*-restriction, and use that $U \to X$ is surjective on points to descend openness of the image.*)

4. Explain why theorem 18.4.4 does not contradict theorem 18.4.5. Specifically, identify which hypothesis of the polarized-descent theorem fails for Hironaka's example, and state in one sentence why the same descent datum *is* effective in the category of algebraic spaces.

5. Compute $H^1(\mathrm{Gal}(\mathbb{C}/\mathbb{R}), \mathrm{Aut})$ for the two forms of $\mathbb{G}_m$ over $\mathbb{R}$ from example 18.4.6, and identify the nonsplit form explicitly as the circle group. Then compute the forms of the two-dimensional split torus $\mathbb{G}_m^2$ over $\mathbb{R}$: what is $\mathrm{Aut}(\mathbb{G}_{m,\mathbb{C}}^2)$ as a Γ-module, and how many forms result? (*Hint:* $\mathrm{Aut}(\mathbb{G}_{m,\mathbb{C}}^2) = \mathrm{GL}_2(\mathbb{Z})$*; the forms are classified by conjugacy classes of order-dividing-2 elements* $\Gamma \to \mathrm{GL}_2(\mathbb{Z})$*, i.e. by* $\mathrm{GL}_2(\mathbb{Z})$*-conjugacy classes of involutions.*)

6. Let A/k be a finite-dimensional k-algebra, and let $A_0 = M_n(k)$ be the $n \times n$ matrix algebra. Explain why the forms of $M_n(k)$ over k, classified by $H^1(\Gamma, \mathrm{PGL}_n)$ (since $\mathrm{Aut}(M_n) = \mathrm{PGL}_n$ by the Skolem-Noether theorem), are exactly the central simple k-algebras of degree n, and connect this to the conic example: what is the form of $M_2(\mathbb{R})$ corresponding to the nontrivial class, and which conic is its Severi-Brauer variety?

18.5 Torsors and H^1

Faithfully flat descent, developed across the previous sections, answers the question of when an object defined locally on a cover glues to a global object. It does not by itself count the ways in which gluing can fail, or succeed nontrivially. That count is the content of the present section. The objects that measure it are torsors, and their classifying invariant is the first cohomology H^1 of the site. A torsor generalizes a familiar geometric object: a line bundle is a torsor under the multiplicative group, a rank-n vector bundle is a torsor under GL_n, and in each case the set of isomorphism classes is a first cohomology group. The section makes this dictionary precise, proves that isomorphism classes of G-torsors are classified by $\check{H}^1$ of the site, and identifies the concrete geometry behind three fundamental groups of coefficients: $\mathbb{G}_m$, μ_n, and GL_n.

The torsor is also the first place in the book where the automorphism obstruction of example 13.3.4 acquires a positive counterpart. In Part IV an object with nontrivial automorphisms resisted representation by a fine moduli space; the sheaf of trivializations of that obstruction is a torsor, and the failure of the moduli functor to be representable is measured by an H^1. Pushing the same idea one categorical level higher produces a gerbe, the two-dimensional analogue of a torsor, whose classifying invariant is H^2. The gerbe is named here and its full development deferred to Chapter 9; what this section supplies is the one-dimensional theory on which that development rests, together with the concrete operation, twisting a quasi-coherent sheaf by a torsor, that turns a cohomology class into geometry.

Throughout, τ denotes one of the topologies of definition 18.1.3 (Zariski, étale, smooth, or fppf), X a scheme over the base S, and G a sheaf of groups on the site $(\mathbf{Sch}/X)_\tau$ or on the small site X_τ. The multiplicative group $\mathbb{G}_m$, the group of n-th roots of unity μ_n, and the general linear group GL_n are the sheaves of groups $T \mapsto \mathbb{G}_m(T) = \Gamma(T, \mathcal{O}_T)^\times$, $T \mapsto \mu_n(T) = \{u \in \mathcal{O}_T(T)^\times \,|\, u^n = 1\}$, and $T \mapsto \mathrm{GL}_n(\mathcal{O}_T(T))$ respectively, each a representable sheaf in the fppf topology by theorem 18.2.4.

Definition 18.5.1: G-Torsor

Let G be a sheaf of groups on the site $(\mathbf{Sch}/X)_\tau$. A *G-torsor* (or *principal homogeneous space* under G) is a τ-sheaf P on X equipped with a left action $G \times P \to P$ such that

1. P is τ-locally nonempty: there is a covering family $\{U_i \to X\}$ with $P(U_i) \neq \varnothing$ for every i; and

2. the action is simply transitive: the map

$$G \times P \longrightarrow P \times P, \qquad (g, p) \longmapsto (g \cdot p,\, p)$$

   is an isomorphism of sheaves.

Equivalently, P is τ-locally isomorphic to G acting on itself by left translation: over each U_i of a covering family there is a $G|_{U_i}$-equivariant isomorphism $P|_{U_i} \cong G|_{U_i}$. A *morphism of torsors* $P \to P'$ is a G-equivariant morphism of sheaves; every such morphism is an isomorphism. The torsor P is *trivial* if it has a global section, that is, if $P(X) \neq \varnothing$.

The two formulations of the action condition are equivalent, and the equivalence is worth spelling out because it isolates what a torsor supplies over what a mere G-set supplies. Condition (2), that $(g, p) \mapsto (g \cdot p, p)$ is an isomorphism, says that once a point p of P is fixed, every other point is $g \cdot p$ for a unique g: the fiber of P is a copy of G, but with no distinguished element playing the role of the identity. A global section would supply such a distinguished element and thereby an equivariant isomorphism $P \cong G$; this is exactly why the existence of a global section is the same as triviality. The content of a nontrivial torsor is precisely that it looks like G from every local vantage but admits no global identification with G.

The claim that every morphism of torsors is an isomorphism is the first thing to verify, and it explains why the moduli of torsors is a *set* of isomorphism classes rather than a category with interesting morphisms.

Proposition 18.5.2: Torsor Morphisms Are Isomorphisms

Let P and P' be G-torsors on X in the topology τ. Any G-equivariant morphism of sheaves $f\colon P \to P'$ is an isomorphism. Consequently the category of G-torsors on X is a groupoid (every arrow is invertible), and the isomorphism classes of G-torsors form a set. The automorphism sheaf $\underline{\mathrm{Aut}}_G(P)$ of a torsor is the inner form ${}^P G$ of G, the twist of G by its conjugation action along P (the right-translation model of G glued by conjugation $h_j = g_{ij}\, h_i\, g_{ij}^{-1}$ along the transition cocycle); it recovers G exactly when P is trivial or G is abelian.

***Proof*:**
Being an isomorphism of sheaves is a τ-local property: a morphism of sheaves is an isomorphism if and only if it is one after restriction to each member of a covering family, since the sheaf condition recovers global sections from local ones. Choose a covering family $\{U_i \to X\}$ over which both P and P' trivialize, and refine so that a single family trivializes both. Over U_i, fix equivariant isomorphisms $P|_{U_i} \cong G|_{U_i}$ and $P'|_{U_i} \cong G|_{U_i}$; under these, the restriction $f|_{U_i}$ becomes a G-equivariant endomorphism of $G|_{U_i}$ acting on itself by left translation. Every

left-equivariant endomorphism of G is right translation by the fixed element $h_i = f(e) \in G(U_i)$, since $f(g) = f(g \cdot e) = g \cdot f(e) = gh_i$; this map is a bijection with inverse $g \mapsto gh_i^{-1}$. Hence $f|_{U_i}$ is an isomorphism for every i, and therefore f is an isomorphism. Every arrow of G-torsors is thus invertible, so the category is a groupoid; and since every torsor is glued from copies of a fixed local model G along a cover, the isomorphism classes form a set.

For the automorphism sheaf, apply the same computation with $P = P'$: over U_i an automorphism f restricts to right translation R_{h_i} by some $h_i \in G(U_i)$. The local models are glued along U_{ij} by the transition right translation $R_{g_{ij}}$, and for the restrictions of f to define a single global endomorphism they must commute with this gluing: $R_{h_i} \circ R_{g_{ij}} = R_{g_{ij}} \circ R_{h_j}$ on U_{ij}. Evaluating, $xg_{ij}h_i = xh_jg_{ij}$ for all x, so $g_{ij}h_i = h_jg_{ij}$, that is $h_j = g_{ij}\, h_i\, g_{ij}^{-1}$. The local elements h_i therefore do not agree on overlaps; they glue only after conjugation by the transition cocycle, so the family (h_i) is a section not of the constant sheaf G but of G twisted by the inner automorphisms $x \mapsto g_{ij}\, x\, g_{ij}^{-1}$. Sending f to this twisted section identifies $\underline{\mathrm{Aut}}_G(P)$ with G acting on itself by right translation, glued by inner automorphisms of the transition cocycle; this is the inner form of G cut out by P, and it recovers the constant G exactly when P is trivial. Thus $\underline{\mathrm{Aut}}_G(P)$ is a form of G, as we sought to show.

That every morphism of torsors is an isomorphism is the fact that keeps torsor theory one-dimensional. Torsors form a groupoid, not a category with genuinely different objects linked by non-invertible maps, so their classifying data are just the gluing isomorphisms on overlaps modulo the freedom to change local trivializations, with no higher coherence to track. The automorphism sheaf is a form of G, but it contributes no new isomorphism classes: it records the internal symmetry of a single torsor, not a way of building new ones. The moment the coefficient object acquires automorphisms whose own automorphisms matter, the classification jumps a categorical level, and the classifying invariant becomes H^2 rather than H^1; that is the gerbe, and it is the two-dimensional refinement that proposition 18.5.2 rules out for ordinary torsors.

Before the classification theorem, the mechanism by which a torsor produces a cohomology class must be made explicit, because the proof is nothing but that mechanism run forwards and backwards. A trivialized torsor over a cover is a copy of G over each cover-piece; the failure of the local trivializations to agree on overlaps is a family of elements of G indexed by pairs of cover-pieces, and the compatibility of these elements on triple overlaps is exactly the cocycle condition. This is the same descent datum of definition 18.3.1 in the special case where the fibered object is a torsor, and it is the reason the classification is a theorem about descent rather than a new construction.

Theorem 18.5.3: Classification of Torsors by H^1

Let G be a sheaf of groups on the site $(\mathbf{Sch}/X)_\tau$. There is a natural bijection

$$\{\text{isomorphism classes of } G\text{-torsors on } X \text{ in } \tau\} \xrightarrow{\sim} \check{H}^1(X_\tau, G)$$

between isomorphism classes of G-torsors and the first Čech cohomology of the site with coefficients in G, under which the trivial torsor corresponds to the distinguished (base-point) class. When G is nonabelian, both sides are pointed sets and the bijection is a bijection of pointed sets; when G is abelian, $\check{H}^1(X_\tau, G)$ is an abelian group, the bijection is a group isomorphism, and it agrees with the derived-functor cohomology $H^1(X_\tau, G)$.

***Proof*:**

Recall the Čech description of H^1. For a covering family $\mathcal{U} = \{U_i \to X\}$, write $U_{ij} = U_i \times_X U_j$ and $U_{ijk} = U_i \times_X U_j \times_X U_k$. A 1-*cocycle* on $\mathcal{U}$ with values in G is a family $(g_{ij}) \in \prod_{i,j} G(U_{ij})$ satisfying the cocycle condition

$$\mathrm{pr}_{13}^* g_{ik} = \mathrm{pr}_{12}^* g_{ij} \cdot \mathrm{pr}_{23}^* g_{jk} \quad \text{on } U_{ijk} \tag{18.4}$$

where pr_{ab} are the projections to the double overlaps. Two cocycles (g_{ij}) and (g'_{ij}) are *cohomologous* if there is a family $(h_i) \in \prod_i G(U_i)$ with $g'_{ij} = (\mathrm{pr}_1^* h_i)\, g_{ij}\, (\mathrm{pr}_2^* h_j)^{-1}$ on U_{ij}. The pointed set of cocycles modulo this equivalence is $\check{H}^1(\mathcal{U}, G)$, and $\check{H}^1(X_\tau, G)$ is the colimit over refinements of covering families. The base point is the class of the trivial cocycle $g_{ij} = e$.

Step 1: From a torsor to a cocycle. Let P be a G-torsor. Choose a covering family $\mathcal{U} = \{U_i \to X\}$ and trivializations, that is, equivariant isomorphisms $t_i \colon G|_{U_i} \xrightarrow{\sim} P|_{U_i}$; equivalently, by condition (2) of definition 18.5.1, sections $s_i = t_i(e) \in P(U_i)$, which exist by τ-local nonemptiness. On the overlap U_{ij} the two sections $\mathrm{pr}_1^* s_i$ and $\mathrm{pr}_2^* s_j$ are both points of $P(U_{ij})$, and by simple transitivity there is a unique $g_{ij} \in G(U_{ij})$ with

$$\mathrm{pr}_1^* s_i = g_{ij} \cdot \mathrm{pr}_2^* s_j$$

Comparing the three sections $\mathrm{pr}_1^* s_i$, $\mathrm{pr}_2^* s_j$, $\mathrm{pr}_3^* s_k$ over the triple overlap U_{ijk} and using uniqueness of the transition element gives, after pulling back along the appropriate projections, the identity (18.4): the composite of the transition from k to j with the transition from j to i is the transition from k to i. Thus (g_{ij}) is a 1-cocycle. A different choice of sections $s'_i = h_i \cdot s_i$ with $h_i \in G(U_i)$ replaces g_{ij} by the cohomologous cocycle $\mathrm{pr}_1^* h_i \cdot g_{ij} \cdot (\mathrm{pr}_2^* h_j)^{-1}$, and a refinement of the cover restricts the cocycle to the refinement. Hence the class $[(g_{ij})] \in \check{H}^1(X_\tau, G)$ is independent of all choices and depends only on the isomorphism class of P.

Step 2: From a cocycle to a torsor. Conversely, let (g_{ij}) be a 1-cocycle on $\mathcal{U}$. Build a sheaf P by descent: on each U_i take the trivial torsor $G|_{U_i}$, and glue $G|_{U_j}$ to $G|_{U_i}$ over U_{ij} by the isomorphism $\varphi_{ij} \colon \mathrm{pr}_2^*(G|_{U_j}) \to \mathrm{pr}_1^*(G|_{U_i})$ given by *right* translation by g_{ij}^{-1}, that is $x \mapsto x \cdot g_{ij}^{-1}$. Right translation is forced: the torsor G-action to be constructed is *left* translation, and only right translations commute with left translations, so only a right-translation gluing descends to a G-equivariant sheaf. Composing the transitions $k \to j \to i$ gives $\varphi_{ij} \circ \varphi_{jk} = R_{g_{ij}^{-1}} \circ R_{g_{jk}^{-1}}$, the map $x \mapsto x\, g_{jk}^{-1} g_{ij}^{-1} = x\,(g_{ij} g_{jk})^{-1}$; this equals $\varphi_{ik} = R_{g_{ik}^{-1}}$ precisely when $g_{ik} = g_{ij} g_{jk}$, so the descent cocycle condition of definition 18.3.1 for this gluing datum is equivalent to the 1-cocycle condition (18.4). Since G is a sheaf of groups rather than a quasi-coherent module, the relevant descent is that sheaves form a stack for τ: a gluing datum of τ-sheaves satisfying the cocycle condition is effective, and the local pieces glue to a τ-sheaf P on X (this is the sheaf-theoretic content underlying the faithfully flat descent of theorem 18.3.4, specialized to the sheaf of sets underlying G). The torsor G-action on P is the one induced by *left* translation on each $G|_{U_i}$: left translation commutes with the right translations φ_{ij} that glue the pieces, so it descends to a well-defined left action on P, and it is simply transitive because left translation is so on each $G|_{U_i}$. Thus P is a G-torsor trivialized over $\mathcal{U}$, with tautological section s_i on U_i the identity of the i-th piece $G|_{U_i}$. It remains to check that Step 1 applied to P returns the class $[(g_{ij})]$ with no inversion. Over U_{ij} the transition isomorphism φ_{ij} identifies the section $s_j = e$ of the j-th piece with the section $\varphi_{ij}(e) = e \cdot g_{ij}^{-1} = g_{ij}^{-1}$ of the i-th piece; call this common section $\mathrm{pr}_2^* s_j$. Applying left translation, the Step 1 recipe reads off the unique element $\gamma_{ij} \in G(U_{ij})$ with $\mathrm{pr}_1^* s_i = \gamma_{ij} \cdot \mathrm{pr}_2^* s_j$, that is $e = \gamma_{ij}\, g_{ij}^{-1}$, whence $\gamma_{ij} = g_{ij}$. Thus the recovered cocycle is (g_{ij}) itself, and no inversion arises. Hence Step 1 applied to this P returns the class $[(g_{ij})]$.

Step 3: The two constructions are mutually inverse. Step 1 sends a torsor to a class; Step 2 sends a cocycle to a torsor whose class is the given one, and cohomologous cocycles produce isomorphic torsors because a coboundary (h_i) is precisely a τ-local change of trivialization, which by proposition 18.5.2 induces an isomorphism of the glued torsors. Conversely a torsor P, trivialized over $\mathcal{U}$ with cocycle (g_{ij}), is isomorphic to the torsor glued from (g_{ij}) in Step 2 via the map that sends the local section s_i to the tautological section of the i-th piece. The two assignments are therefore mutually inverse bijections of pointed sets, and the trivial torsor, having a global section, has all transition elements equal to e after a suitable refinement, hence maps to the base-point class.

Step 4: The abelian case. When G is abelian, the cocycle condition (18.4) is additive and the set of cocycles is an abelian group under pointwise multiplication, with coboundaries a subgroup; thus $\check{H}^1(\mathcal{U}, G)$ and its colimit $\check{H}^1(X_\tau, G)$ are abelian groups. The group law transports to torsors as the *contracted product*: given torsors P, P' with cocycles (g_{ij}), (g'_{ij}), the torsor $P \wedge^G P' := (P \times P')/G$, quotient by the antidiagonal action $g \cdot (p, p') = (gp, g^{-1}p')$, has cocycle $(g_{ij}g'_{ij})$, the inverse of P is the torsor with cocycle (g_{ij}^{-1}), and the trivial torsor is the identity. That the Čech group $\check{H}^1$ agrees with the derived-functor cohomology $H^1(X_\tau, G)$ for abelian G is a standard comparison: $\check{H}^1 \to H^1$ is always an isomorphism in degree 1 on any site, since a class in either group is a torsor and the two descriptions of a torsor, gluing data versus a sheaf that is locally G, coincide. The derived statement, that $H^1(X_\tau, G) = R^1\Gamma(X, G)$ classifies the same torsors, is proved in [15]through the machinery of right-derived functors of global sections and holds for any abelian sheaf on a site. This gives the group isomorphism and completes the proof.

The classification is the organizing principle of the entire chapter turned into a single statement. Descent, in the form of theorem 18.3.4, manufactures a global object from local data glued by a cocycle; the classification recognizes that when the object being glued is a torsor, the gluing data are exactly a 1-cocycle, and the ambiguity in the gluing is exactly a coboundary. Every geometric object that is locally trivial in some topology, a line bundle, a vector bundle, a covering with structure group, a form of a fixed object, is a torsor under its automorphism sheaf, and its isomorphism classes are therefore an H^1. The theorem is the bridge that carries every such classification problem into cohomology, where exact sequences and long exact sequences become available. The point of Galois descent in example 18.4.6, that forms of an object over k are classified by H^1 of the Galois group with values in the automorphism group, is the special case of this theorem for the étale topology on $\operatorname{Spec} k$, where the site is the category of finite separable extensions and Čech H^1 is Galois cohomology; the point/field version is developed in full in [13].

The nonabelian case deserves a word, because the loss of a group structure on H^1 is not a defect but a feature of the geometry. When G is nonabelian, $\check{H}^1(X_\tau, G)$ is only a pointed set: there is no natural way to multiply two G-torsors, because the contracted product of Step 4 requires the antidiagonal action to be well-defined, which needs commutativity. What survives is the base point, the trivial torsor, and the distinguished role it plays in exact sequences of pointed sets. This is precisely the structure needed to classify forms and vector bundles, neither of which forms a group under any natural operation, and it is why GL_n- and PGL_n-torsors are handled with pointed-set cohomology rather than group cohomology.

The torsor is the one-dimensional case of a phenomenon that recurs one level up. To see the next level, consider not a single obstruction but the obstruction to trivializing an obstruction. When a moduli problem fails to be representable because its objects carry automorphisms, as in example 13.3.4, the sheaf recording local trivializations of the obstruction is a torsor, and its class in H^1 measures the failure. But the objects of the moduli problem themselves form a sheaf of categories, not a sheaf of sets, and the two-dimensional analogue of a torsor, a sheaf of categories that is locally equivalent to the classifying stack of G, carries a class in H^2. That object is a gerbe. The full theory of gerbes, their classification by H^2, and the Brauer group $H^2(X, \mathbb{G}_m)$ they compute, is the subject of Chapter 9; the present section supplies only its one-dimensional foundation and the name.

Definition 18.5.4: Twisting by a Torsor, and Gerbes

Let P be a G-torsor on X and let F be a quasi-coherent sheaf on X carrying a left action of G by $\mathcal{O}_X$-linear automorphisms (a G-equivariant sheaf). The *twist of F by P*, or *associated bundle*, is the quasi-coherent sheaf

$$P \times^G F := (P \times F)/G$$

the quotient of $P \times F$ by the diagonal action $g \cdot (p, v) = (g \cdot p,\ g \cdot v)$; it is τ-locally isomorphic to F and glued by the image of the transition cocycle (g_{ij}) of P under the action $G \to \mathrm{Aut}_{\mathcal{O}_X}(F)$. A *gerbe banded by G* (for G a sheaf of abelian groups) is the two-dimensional analogue: a stack of groupoids over X_τ that is τ-locally equivalent to the classifying stack BG and whose automorphism band is identified with G. Gerbes are classified by $H^2(X_\tau, G)$, generalizing the classification of torsors by H^1; their theory is developed in Chapter 9, and only their name is fixed here.

Twisting is the operation that turns a cohomology class into a new geometric object, and it is the concrete payoff of the classification theorem. Given a G-torsor P with class in H^1 and any representation of G on a sheaf F, the twist $P \times^G F$ is a sheaf that agrees with F locally but is reglued by the transition functions of P. The construction is functorial in F and multiplicative in P: the whole category of G-equivariant sheaves is transported by a single torsor, which is why a torsor is the correct notion of "structure group data". When $G = \mathbb{G}_m$ acts on $F = \mathcal{O}_X$ by scaling, the twist is a line bundle, and the following example makes this the first of three fundamental computations.

The passage from torsors to gerbes is the passage from H^1 to H^2, and the automorphism gerbe of a point of a moduli stack, the two-dimensional refinement of the automorphism obstruction of example 13.3.4, is what a stack sees where a scheme sees only a coarse point. This is the thread that ties example 13.3.4 and the j-line of theorem 13.3.6 to the gerbe chapter.

With twisting in hand, the three anchor examples identify the geometry behind the three coefficient sheaves that recur throughout algebraic geometry. Each is an instance of the classification theorem, and together they justify the claim that H^1 is where line bundles, roots of unity, and vector bundles live.

Example 18.5.5: $\mathbb{G}_m$-, μ_n-, and GL_n-Torsors

The three fundamental coefficient sheaves each classify a familiar class of geometric objects.

1. *$\mathbb{G}_m$-torsors are line bundles.* A line bundle L on X is a locally free $\mathcal{O}_X$-module of rank 1; its sheaf of local trivializations, the sheaf $\underline{\mathrm{Isom}}(\mathcal{O}_X, L)$ of nowhere-vanishing local generators, is a $\mathbb{G}_m$-torsor, since two local generators differ by a unit. Conversely, twisting $\mathcal{O}_X$ by a $\mathbb{G}_m$-torsor P via the scaling action produces a line bundle $P \times^{\mathbb{G}_m} \mathcal{O}_X$. By theorem 18.5.3 the isomorphism classes of line bundles are

$$\mathrm{Pic}\,(X) = H^1(X_{\mathrm{Zar}}, \mathbb{G}_m) = H^1(X_{\text{ét}}, \mathbb{G}_m)$$

the equality of the Zariski and étale computations being Hilbert's Theorem 90 ([13]): a $\mathbb{G}_m$-torsor that is étale-locally trivial is already Zariski-locally trivial, so no new line bundles appear on passing to the finer topology. The group law on $H^1(X, \mathbb{G}_m)$ is the tensor product of line bundles, and the abelian structure of Step 4 of theorem 18.5.3 is exactly $L \otimes L'$.

2. *μ_n-torsors and the Kummer sequence.* The n-th power map on units gives, in the fppf topology (and in the étale topology when n is invertible on X), a short exact sequence of sheaves of abelian groups

$$1 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \xrightarrow{(\cdot)^n} \mathbb{G}_m \longrightarrow 1 \tag{18.5}$$

the *Kummer sequence*, exact because every unit fppf-locally admits an n-th root (adjoin a root of $T^n - u$). Its long exact cohomology sequence reads

$$\mathcal{O}_X(X)^\times \xrightarrow{n} \mathcal{O}_X(X)^\times \to H^1(X, \mu_n) \to \operatorname{Pic}(X) \xrightarrow{n} \operatorname{Pic}(X)$$

which exhibits $H^1(X, \mu_n)$ as an extension

$$0 \to \mathcal{O}_X(X)^\times / (\mathcal{O}_X(X)^\times)^n \to H^1(X, \mu_n) \to \operatorname{Pic}(X)[n] \to 0$$

of the n-torsion $\operatorname{Pic}(X)[n]$ by the global units modulo n-th powers. A μ_n-torsor is thus the data of a line bundle together with a trivialization of its n-th power, the geometric object underlying cyclic covers and the Kummer theory of [13].

3. *GL_n-torsors are vector bundles.* A locally free sheaf E of rank n has, as its sheaf of local frames $\underline{\mathrm{Isom}}(\mathcal{O}_X^n, E)$, a GL_n-torsor, and twisting the standard representation $\mathcal{O}_X^n$ by a GL_n-torsor recovers a rank-n vector bundle. By theorem 18.5.3, isomorphism classes of rank-n vector bundles on X are the pointed set $H^1(X_{\mathrm{Zar}}, \mathrm{GL}_n) = H^1(X_{\text{ét}}, \mathrm{GL}_n)$, again with Zariski and étale agreeing by the generalized Hilbert 90 for GL_n ([13]). Since GL_n is nonabelian for $n \geq 2$, this H^1 is only a pointed set: vector bundles do not form a group, matching the general nonabelian phenomenon.

The three computations exhibit the entire range of the classification. For $\mathbb{G}_m$ the coefficient is abelian and H^1 is the Picard group, a genuine group under tensor product. For μ_n the same abelian formalism, run through the long exact sequence of the Kummer sequence (18.5), extracts the n-torsion of the Picard group and the units modulo n-th powers, a computation that recurs whenever a variety is studied through its n-torsion. For GL_n the coefficient is nonabelian and H^1 is a pointed set, the moduli of vector bundles up to isomorphism, with no group structure because there is no natural product of vector bundles of fixed rank. Each is an application of the single theorem theorem 18.5.3, and each identifies a cohomology group the reader has met before with a class of geometric objects.

The automorphism thread of Part IV now closes into this picture. In example 13.3.4 the elliptic curves with extra automorphisms obstructed the moduli functor $\mathcal{M}_{1,1}$ from being representable by a fine moduli space, and the coarse space, the j-line of theorem 13.3.6, forgot the automorphisms entirely. The torsor theory of this section explains why: over the j-line, the family of elliptic curves with a given j-invariant is not a single curve but a $\mathbb{G}_m$- or μ_n-torsor's worth of curves, twisted by the automorphism group, and the fibers over the special points $j = 0$ and $j = 1728$, where in characteristic $\neq 2, 3$ the automorphism group jumps to μ_6 and μ_4 respectively (in characteristics 2 and 3 these loci merge and the group can be larger), carry nontrivial H^1 with values in those groups. The correct object over the j-line is not a scheme but a stack whose points carry these automorphism groups, and the two-dimensional refinement of the obstruction, the automorphism gerbe of a point of $\mathcal{M}_{1,1}$, is the H^2 statement that Chapter 9 develops. The torsor and its H^1 are the first one-dimensional trace of the stack that Part VI is built to construct.

Exercise 18.5.1

1. Let P be a G-torsor on X in a topology τ. Prove that P is trivial if and only if it has a global section, and that a global section, when it exists, determines an equivariant isomorphism $P \cong G$; describe the set of all such isomorphisms and show it is a torsor under $G(X)$ acting by right translation.

2. Compute $H^1(\operatorname{Spec} k, \mathbb{G}_m)$ and $H^1(\operatorname{Spec} k, \mathrm{GL}_n)$ for a field k, and interpret each geometrically. (Hint: a line bundle on $\operatorname{Spec} k$ is a one-dimensional k-vector space; a rank-n vector bundle is an n-dimensional k-vector space. What is the classification up to isomorphism?)

3. Using the Kummer sequence (18.5), compute $H^1(\operatorname{Spec} k, \mu_n)$ for a field k with n invertible in k, and identify it with a familiar group from Kummer theory. Then compute $H^1(\mathbb{P}^1_k, \mu_n)$ using $\operatorname{Pic}(\mathbb{P}^1_k) = \mathbb{Z}$ and $\Gamma(\mathbb{P}^1_k, \mathcal{O})^\times = k^\times$.

4. Let L/k be a finite Galois extension with group Γ, and let V be a finite-dimensional k-vector space. Show that the twists of V by $\mathrm{GL}(V)$-torsors trivialized by $\operatorname{Spec} L \to \operatorname{Spec} k$ are classified by the Galois cohomology $H^1(\Gamma, \mathrm{GL}(V)(L))$, and deduce from example 18.4.6 that this pointed set is trivial: every such twist is isomorphic to V. (This is the vector-space case of Hilbert's Theorem 90; compare [13].)

5. Let P be a $\mathbb{G}_m$-torsor with class $[P] \in H^1(X, \mathbb{G}_m) = \operatorname{Pic}(X)$ corresponding to a line bundle L. Show that the twist $P \times^{\mathbb{G}_m} \mathcal{O}_X$ (scaling action) is isomorphic to L, and that the twist $P \times^{\mathbb{G}_m} \mathcal{O}_X$ by the action of weight d (where $\lambda \in \mathbb{G}_m$ acts by λ^d) is $L^{\otimes d}$. Conclude that twisting realizes the group law on $\operatorname{Pic}(X)$.

6. Explain why the classification theorem 18.5.3 produces a group when G is abelian but only a pointed set when G is nonabelian, by locating the exact step in the proof where commutativity is used. Then give an example of two GL_2-torsors (rank-2 vector bundles) on a curve for which no natural product exists, illustrating the absence of a group structure.

19

Fibered Categories and Stacks

Descent theory taught the objects of a moduli problem to glue in a Grothendieck topology, but it treated them as elements of a set: two local objects either agree on an overlap or they do not. The moduli problems that resisted representability throughout Part IV fail exactly this dichotomy. An elliptic curve has automorphisms, so two families that restrict to isomorphic curves over an overlap are identified not on the nose but through a chosen isomorphism, and different choices give different global objects. A structure that records objects together with the isomorphisms between them is not a sheaf of sets but a sheaf of groupoids, and the geometric incarnation of a sheaf of groupoids is a category fibered in groupoids.

This chapter builds the resulting language. The first section defines fibered categories and categories fibered in groupoids and proves the 2-Yoneda lemma, which embeds schemes among them and fixes the meaning of representability. The second imposes the descent condition from Chapter 6 one categorical level higher: a category fibered in groupoids whose morphisms and whose objects both glue in the topology is a stack, and the faithfully flat descent already proved becomes the statement that quasi-coherent sheaves and torsors form stacks, while stackification manufactures a stack from any fibered category. The third section constructs the first geometric stacks, the quotient stack $[X/G]$ and the classifying stack BG, whose points are torsors with an equivariant map; here $\mathcal{M}_{1,1}$ finally becomes an object in its own right, a quotient stack whose coarse space is the j-line and whose points carry the automorphisms the j-line forgot. The last section explains how geometry is imposed on a stack through representable morphisms and the diagonal, isolating the two conditions that Chapter 8 will promote to the definition of an algebraic stack.

19.1 Fibered Categories

Chapter 6 taught how data glue along a faithfully flat cover: a quasi-coherent sheaf on X is the same as a sheaf on a cover $U \to X$ with a gluing isomorphism satisfying the cocycle condition, and a torsor on X is classified by a first cohomology set. The data descended there were sets, sheaves, and morphisms between them. A moduli problem asks for something richer. The elliptic curves parametrized by $\mathcal{M}_{1,1}$ carry automorphisms, and any object with a nontrivial automorphism refuses to be recorded by a functor to sets: the value $F(T)$ of a moduli functor at a scheme T is a *set* of isomorphism classes, and the moment two families become isomorphic without being equal, the isomorphism, and with it the automorphisms, is discarded. This section builds the categorical vessel that keeps that information. Instead of a set-valued functor, a moduli problem is organized as a category $\mathcal{F}$ mapping to the category $\mathcal{C}$ of test schemes, whose fiber over each T is not a set but a groupoid of families and their isomorphisms.

The organizing structure is that of a *fibered category*: a functor $p\colon \mathcal{F} \to \mathcal{C}$ in which objects can be pulled back along morphisms of the base, coherently and essentially uniquely. When every fiber is a groupoid, this is a *category fibered in groupoids*, the correct home for a moduli problem with automorphisms. The section proves the 2-categorical Yoneda lemma, which shows that a scheme is faithfully recorded as such a fibered category and pins down what it means for a fibered category to be representable, and it grounds the abstraction in the three objects that thread the chapter: the fibered category of quasi-coherent sheaves, the classifying object BG of G-torsors, and $\mathcal{M}_{1,1}$. The 2-categorical language, groupoids, pseudofunctors, and the ordinary Yoneda lemma, is imported from [9]; the treatment of fibered categories follows the canonical reference [66].

Throughout, fix a base scheme S and a topology τ (default: the fppf topology, definition 18.1.3), and let $\mathcal{C} = (\mathbf{Sch}/S)_\tau$ be the category of S-schemes equipped with τ. A moduli problem is a rule assigning to each test scheme T a category of families over T, together with pullback along morphisms $T' \to T$. Recording this as a single category over $\mathcal{C}$ requires a mechanism for pullback that is intrinsic to the projection p, and that is what a cartesian arrow provides.

The definition of a cartesian arrow copies the universal property of a fiber product, stripped to the part that survives when there is no base change to speak of, only a projection p. An arrow $\varphi\colon x \to y$ in $\mathcal{F}$ lying over $f\colon U \to V$ in $\mathcal{C}$ deserves to be called the pullback of y along f if every arrow into y that factors through f on the base factors uniquely through φ upstairs.

Definition 19.1.1: Cartesian Arrow and Fibered Category

Let $p\colon \mathcal{F} \to \mathcal{C}$ be a functor. An arrow $\varphi\colon x \to y$ in $\mathcal{F}$ is *cartesian* (over $f = p(\varphi)\colon U \to V$) if for every arrow $\psi\colon z \to y$ in $\mathcal{F}$ and every factorization $p(\psi) = f \circ h$ of $p(\psi)$ through f in $\mathcal{C}$, there is a unique arrow $\theta\colon z \to x$ in $\mathcal{F}$ with $p(\theta) = h$ and $\varphi \circ \theta = \psi$. When such a cartesian φ exists, x is a *pullback* of y along f, written f^*y.
The functor p is a *fibered category* over $\mathcal{C}$ (and $\mathcal{C}$ is the *base*) if:

1. for every arrow $f\colon U \to V$ in $\mathcal{C}$ and every object y of $\mathcal{F}$ with $p(y) = V$, there is a cartesian arrow $\varphi\colon f^*y \to y$ over f; and
2. the composite of two cartesian arrows is cartesian.

For an object U of $\mathcal{C}$, the *fiber* $\mathcal{F}(U)$ is the subcategory of $\mathcal{F}$ whose objects are the x with $p(x) = U$ and whose arrows are those α with $p(\alpha) = \mathrm{id}_U$; such an α is called *vertical*. An object x with $p(x) = U$ is written x/U.

The universal property in the definition of cartesian is exactly the one that makes a pullback well-defined up to unique isomorphism. If both $\varphi\colon x \to y$ and $\varphi'\colon x' \to y$ are cartesian over the same f, applying the universal property of φ to φ' (with $h = \mathrm{id}_U$) and of φ' to φ produces mutually inverse vertical isomorphisms $x \cong x'$ over U. So the pullback f^*y is an object of the fiber $\mathcal{F}(U)$ determined up to a unique isomorphism compatible with the maps to y. This is the precise sense in which "pullback" makes sense in a fibered category even though no actual choice of pullback has been fixed: the fibered-category axioms guarantee existence, and the universal property guarantees uniqueness up to unique isomorphism, but a concrete pullback is a further choice.

Making that choice, once and for all, is a *cleavage*, and it turns the intrinsic pullback into a well-defined functor between fibers. A cleavage is the device that converts the geometry of $\mathcal{F}$ into the bookkeeping of a pseudofunctor.

Definition 19.1.2: Cleavage and Pullback Functors

A *cleavage* of a fibered category $p\colon \mathcal{F} \to \mathcal{C}$ is a choice, for each arrow $f\colon U \to V$ in $\mathcal{C}$ and each object y/V, of a cartesian arrow $\alpha_f(y)\colon f^*y \to y$ over f. Given a cleavage, each $f\colon U \to V$ induces a *pullback functor*

$$f^*\colon \mathcal{F}(V) \longrightarrow \mathcal{F}(U)$$

sending y/V to the chosen f^*y and sending a vertical arrow $\beta\colon y \to y'$ to the unique vertical arrow $f^*\beta\colon f^*y \to f^*y'$ with $\alpha_f(y') \circ f^*\beta = \beta \circ \alpha_f(y)$ (which exists and is unique because $\alpha_f(y')$ is cartesian). For composable $U \xrightarrow{f} V \xrightarrow{g} W$ there are canonical isomorphisms of functors

$$\epsilon_{f,g}\colon f^*g^* \xrightarrow{\sim} (g \circ f)^*, \qquad \mathrm{id}_U^* \cong \mathrm{id}_{\mathcal{F}(U)}$$

coming from the uniqueness of cartesian lifts, and these satisfy a compatibility (cocycle) condition for triple composites.

The isomorphisms $\epsilon_{f,g}$ are the reason a fibered category is *not* literally a contravariant functor from $\mathcal{C}$ to categories: pulling back along g and then along f agrees with pulling back along $g \circ f$ only up to the canonical isomorphism $\epsilon_{f,g}$, not on the nose, because the cleavage chose the two sides independently. The assignment $U \mapsto \mathcal{F}(U)$, $f \mapsto f^*$, $\epsilon_{f,g}$ is precisely the data of a *pseudofunctor* $\mathcal{C}^{\mathrm{op}} \to \mathbf{Cat}$, where the coherence condition on triple composites is the pentagon-type axiom for pseudofunctors (see [9] for the general notion, and [66] for the equivalence in this setting). The converse also holds: from a pseudofunctor one reconstructs a fibered category by the Grothendieck construction, taking objects to be pairs (U, x) with $x \in \mathcal{F}(U)$ and arrows $(U, x) \to (V, y)$ to be pairs (f, β) with $f\colon U \to V$ and $\beta\colon x \to f^*y$ vertical. The two viewpoints are equivalent, and the section moves freely between them: the fibered category $p\colon \mathcal{F} \to \mathcal{C}$ when the intrinsic geometry is cleaner, the pseudofunctor $U \mapsto \mathcal{F}(U)$ when an explicit rule for the fibers is at hand.

19.1.3 Convention: Fibered Categories versus Pseudofunctors A cleavage always exists (it is a choice of one cartesian arrow per (f, y), an application of choice at the level of the class of arrows), so every fibered category yields a pseudofunctor and conversely. The two formulations carry the same information, and no result below depends on a particular cleavage. When a moduli problem is presented by a rule "over T, the objects are such-and-such families", it is being described as a pseudofunctor; when it is presented as a category of all families at once with a projection to the base, it is being described as a fibered category. The word "pullback" means the intrinsic operation of definition 19.1.1, canonical up to unique isomorphism, whether or not a cleavage has been named.

A moduli problem with automorphisms lives in a fibered category of a special kind: one whose fibers are groupoids, so that every family has a well-defined group of automorphisms and every morphism of families is invertible. The two conditions, all arrows cartesian and all fibers groupoids, turn out to be the same.

Definition 19.1.4: Category Fibered in Groupoids

A fibered category $p\colon \mathcal{F} \to \mathcal{C}$ is a *category fibered in groupoids* (a *CFG*) if every arrow of $\mathcal{F}$ is cartesian. Equivalently (see proposition 19.1.5), $p\colon \mathcal{F} \to \mathcal{C}$ is a CFG if and only if:

1. for every arrow $f\colon U \to V$ in $\mathcal{C}$ and every y/V there is an arrow $x \to y$ over f; and
2. for every pair of arrows $\varphi\colon x \to z$, $\psi\colon y \to z$ in $\mathcal{F}$ and every arrow $h\colon p(x) \to p(y)$ in $\mathcal{C}$ with $p(\psi) \circ h = p(\varphi)$, there is a unique arrow $\theta\colon x \to y$ over h with $\psi \circ \theta = \varphi$.

Under a cleavage, the associated pseudofunctor takes values in the 2-category **Grpd** of groupoids: each fiber $\mathcal{F}(U)$ is a groupoid.

The equivalence of the "all arrows cartesian" description with the two-condition description, and with the requirement that every fiber be a groupoid, is worth proving in full, because it is the criterion one checks in practice: to verify that a rule $T \mapsto$ (a category of families) is a CFG one confirms that families pull back and that morphisms of families glue uniquely.

Proposition 19.1.5: Characterizations of a CFG

Let $p\colon \mathcal{F} \to \mathcal{C}$ be a functor. The following are equivalent:

1. p is a fibered category and every arrow of $\mathcal{F}$ is cartesian;
2. conditions (1) and (2) of definition 19.1.4 hold;
3. p is a fibered category and every fiber $\mathcal{F}(U)$ is a groupoid.

***Proof*:**
The argument runs (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1).

Step 1: (1) $\Rightarrow$ (2). Condition (2)(1) is the existence half of the fibered-category axiom, so it holds. For (2)(2), let $\varphi\colon x \to z$, $\psi\colon y \to z$ and $h\colon p(x) \to p(y)$ satisfy $p(\psi) \circ h = p(\varphi)$. Since ψ is cartesian (all arrows are, by hypothesis) and $p(\varphi) = p(\psi) \circ h$ factors $p(\varphi)$ through $p(\psi)$, the universal property of the cartesian arrow ψ yields a unique $\theta\colon x \to y$ with $p(\theta) = h$ and $\psi \circ \theta = \varphi$. This is exactly (2)(2).

Step 2: (2) $\Rightarrow$ (3). First, p is a fibered category. Existence of cartesian lifts: given $f\colon U \to V$ and y/V, condition (2)(1) supplies an arrow $\varphi\colon x \to y$ over f; this φ is cartesian, for given $\psi\colon z \to y$ with $p(\psi) = f \circ h$, apply (2)(2) to $\varphi\colon x \to y$, $\psi\colon z \to y$ and h (noting $p(\varphi) \circ h = f \circ h = p(\psi)$) to get the unique $\theta\colon z \to x$ over h with $\varphi \circ \theta = \psi$. Stability of cartesian arrows under composition follows directly from the universal property. Now each fiber is a groupoid: let $\alpha\colon x \to y$ be an arrow vertical over U (so $p(\alpha) = \mathrm{id}_U$). Apply condition (2)(2) to the pair $\varphi = \mathrm{id}_y\colon y \to y$, $\psi = \alpha\colon x \to y$ with target y and $h = \mathrm{id}_U$ (valid since $p(\mathrm{id}_y) \circ \mathrm{id}_U = \mathrm{id}_U = p(\alpha)$): this yields a unique arrow $\beta\colon y \to x$ over id_U with $\alpha \circ \beta = \mathrm{id}_y$, a vertical right inverse of α. Applying (2)(2) once more, this time to $\varphi = \alpha$, $\psi = \alpha$ with $h = \mathrm{id}_U$, shows that $\beta \circ \alpha = \mathrm{id}_x$: both $\beta \circ \alpha$ and id_x are vertical arrows $x \to x$ solving the same factorization $\alpha \circ (-) = \alpha$, and (2)(2) forces them equal. Hence β is a two-sided inverse and α is an isomorphism, so $\mathcal{F}(U)$ is a groupoid.

Step 3: (3) $\Rightarrow$ (1). Let $\varphi\colon x \to y$ be any arrow of $\mathcal{F}$, over $f = p(\varphi)\colon U \to V$. Since p is a fibered category there is a cartesian arrow $\varphi_0\colon x_0 \to y$ over f, and by its universal property (with $h = \mathrm{id}_U$, $\psi = \varphi$) there is a vertical $\theta\colon x \to x_0$ over id_U with $\varphi_0 \circ \theta = \varphi$. By hypothesis (3) the fiber $\mathcal{F}(U)$ is a groupoid, so θ is an isomorphism. An arrow equal to a cartesian arrow precomposed with a vertical isomorphism is again cartesian: the universal property transports across the isomorphism θ. Therefore φ is cartesian, and every arrow of $\mathcal{F}$ is cartesian, completing the proof.

The proposition is the working definition. A category over $\mathcal{C}$ is a CFG precisely when families pull back (condition (2)(1)) and morphisms of families are determined by their behavior on the base together with a compatible target (condition (2)(2)); the payoff is that every fiber is automatically a groupoid, so automorphisms are built in. In the pseudofunctor picture, a CFG is a pseudofunctor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Grpd}$, and this is the definitive form of a moduli problem: to each test scheme T, a groupoid of families over T; to each $T' \to T$, a pullback functor; and coherence isomorphisms for composites.

Morphisms between CFGs and the 2-morphisms between those are what make CFGs into a 2-category, the ambient world in which stacks live.

Definition 19.1.6: Morphisms and 2-Morphisms of CFGs

A *morphism* of CFGs $F\colon \mathcal{F} \to \mathcal{G}$ over $\mathcal{C}$ is a functor F with $p_{\mathcal{G}} \circ F = p_{\mathcal{F}}$ (so F carries $\mathcal{F}(U)$ into $\mathcal{G}(U)$ for each U). Because every arrow of a CFG is cartesian, such an F automatically preserves cartesian arrows. A 2-*morphism* $\eta\colon F \Rightarrow F'$ between two such morphisms is a natural transformation that is the identity on the base, meaning $p_{\mathcal{G}}(\eta_x) = \mathrm{id}_{p_{\mathcal{F}}(x)}$ for every object x of $\mathcal{F}$; equivalently, each component $\eta_x\colon F(x) \to F'(x)$ is a vertical arrow of $\mathcal{G}$. Since fibers are groupoids, every such η is invertible, so it is a natural isomorphism. CFGs over $\mathcal{C}$, their morphisms, and their 2-morphisms form a 2-category, in which the Hom between two objects is itself a groupoid. An *equivalence* of CFGs is an equivalence in this 2-category: an F admitting G with $F \circ G$ and $G \circ F$ 2-isomorphic to the identities.

Two CFGs are considered the same when they are equivalent, never when they are isomorphic: an equivalence of categories is the correct notion of sameness for groupoid-valued data, and demanding an isomorphism would be as unreasonable as demanding that two equivalent categories have equal objects. The groupoid $\underline{\mathrm{Hom}}_{\mathcal{C}}(\mathcal{F}, \mathcal{G})$ of morphisms and 2-morphisms is the internal Hom of the 2-category, and its appearance in the Yoneda statement below is what makes that statement a 2-categorical refinement of the ordinary one.

The first and most basic examples are the schemes themselves. A scheme X over S becomes a CFG in a canonical way, and the construction is the 2-categorical version of the functor of points section 3.3.

Definition 19.1.7: The CFG Represented by a Scheme

For an S-scheme X, the *representable CFG* $\underline{X}$ is the category whose objects are the S-morphisms $T \to X$ (for T ranging over $\mathcal{C}$) and whose arrows $(T' \xrightarrow{a} X) \to (T \xrightarrow{b} X)$ are the S-morphisms $g\colon T' \to T$ with $b \circ g = a$. The projection $p\colon \underline{X} \to \mathcal{C}$ sends $(T \to X)$ to T and g to g. A CFG $\mathcal{F}$ over $\mathcal{C}$ is *representable* (by a scheme) if there is an equivalence $\mathcal{F} \simeq \underline{X}$ for some S-scheme X.

The category $\underline{X}$ is fibered in groupoids because its fibers are discrete: the fiber $\underline{X}(T)$ is the *set* $\mathrm{Hom}_S(T, X)$ regarded as a groupoid with only identity arrows, since a vertical arrow is a $g = \mathrm{id}_T$ commuting with the structure maps, forcing $a = b$. Thus $\underline{X}$ has no nontrivial automorphisms anywhere; a scheme, viewed as a moduli problem, parametrizes objects with no automorphisms. This is the structural reason a moduli problem with automorphisms cannot be represented by a scheme, and the reason stacks are needed: representability by a scheme forces every fiber to be a set. The pseudofunctor associated with $\underline{X}$ is exactly the functor of points $h_X = \mathrm{Hom}_S(-, X)$, now landing in discrete groupoids rather than sets.

Representing X this way loses no information about X, and CFG-morphisms out of $\underline{X}$ are computed by evaluation. This is the content of the 2-Yoneda lemma, which lifts the ordinary Yoneda lemma of [9] to the groupoid-valued setting. The ordinary Yoneda lemma identifies natural transformations $h_X \Rightarrow F$ with elements of $F(X)$; the 2-categorical version identifies the *groupoid* of CFG-morphisms $\underline{X} \to \mathcal{F}$ with the fiber groupoid $\mathcal{F}(X)$, morphisms and 2-morphisms included.

Theorem 19.1.8: The 2-Yoneda Lemma

Let $\mathcal{F}$ be a CFG over $\mathcal{C}$ and X an object of $\mathcal{C}$. Evaluation at the object $\mathrm{id}_X \in \underline{X}(X)$ defines a functor

$$\mathrm{ev}\colon \underline{\mathrm{Hom}_{\mathcal{C}}}(\underline{X}, \mathcal{F}) \longrightarrow \mathcal{F}(X), \qquad F \longmapsto F(\mathrm{id}_X)$$

from the groupoid of morphisms $\underline{X} \to \mathcal{F}$ (and their 2-morphisms) to the fiber groupoid $\mathcal{F}(X)$, and this functor is an equivalence of groupoids. Consequently the assignment $X \mapsto \underline{X}$ embeds $\mathcal{C}$ 2-fully-faithfully into the 2-category of CFGs: for $X, Y \in \mathcal{C}$ the evaluation $\underline{\mathrm{Hom}_{\mathcal{C}}}(\underline{X}, \underline{Y}) \simeq \underline{Y}(X) = \mathrm{Hom}_S(X, Y)$ is an equivalence onto a discrete groupoid. A CFG is representable if and only if it is equivalent to $\underline{X}$ for some X.

***Proof*:**

A choice of cleavage on $\mathcal{F}$ is fixed throughout; for $g\colon T \to X$ and $\xi \in \mathcal{F}(X)$ write $g^*\xi \in \mathcal{F}(T)$ for the chosen pullback. The proof constructs a quasi-inverse to ev from this cleavage and checks that the two composites are 2-isomorphic to the identities.

Step 1: The quasi-inverse. Given $\xi \in \mathcal{F}(X)$, define a morphism $F_\xi\colon \underline{X} \to \mathcal{F}$ as follows. An object of $\underline{X}$ is a morphism $g\colon T \to X$; set $F_\xi(g) = g^*\xi \in \mathcal{F}(T)$. An arrow in $\underline{X}$ from $g'\colon T' \to X$ to $g\colon T \to X$ is a morphism $h\colon T' \to T$ with $g \circ h = g'$; the cleavage supplies canonical cartesian arrows $g^*\xi \to \xi$ and $(g')^*\xi \to \xi$ over g and $g' = g \circ h$, and the composite isomorphism ϵ of definition 19.1.2 identifies $(g')^*\xi = (g \circ h)^*\xi \cong h^*g^*\xi$. Define $F_\xi(h)\colon (g')^*\xi \to g^*\xi$ to be the cartesian arrow over h provided by the cleavage of $\mathcal{F}$ (the unique arrow over h lying above the identity of ξ). Functoriality of F_ξ up to the coherence isomorphisms ϵ makes F_ξ a morphism of CFGs; this is where the pseudofunctor coherence of definition 19.1.2 is used, and it is verified in [66]. By construction $p_{\mathcal{F}}(F_\xi(g)) = T = p_{\underline{X}}(g)$, so F_ξ is a morphism over $\mathcal{C}$.

Step 2: $\mathrm{ev} \circ (\xi \mapsto F_\xi)$ *is the identity.* Evaluating F_ξ at $\mathrm{id}_X \in \underline{X}(X)$ gives $F_\xi(\mathrm{id}_X) = \mathrm{id}_X^*\xi$, and the canonical isomorphism $\mathrm{id}_X^*\xi \cong \xi$ (the unit part of definition 19.1.2) is natural in ξ. Hence $\mathrm{ev}(F_\xi) \cong \xi$ naturally, so the composite $\mathrm{ev} \circ (\xi \mapsto F_\xi)$ is 2-isomorphic to $\mathrm{id}_{\mathcal{F}(X)}$.

Step 3: $(\xi \mapsto F_\xi) \circ \mathrm{ev}$ *is the identity.* Let $F\colon \underline{X} \to \mathcal{F}$ be a morphism of CFGs and put $\xi = F(\mathrm{id}_X) = \mathrm{ev}(F)$. A natural isomorphism $F \cong F_\xi$ must be produced. For $g\colon T \to X$, note that g is, as an arrow of $\underline{X}$, a morphism from $g\colon T \to X$ to $\mathrm{id}_X\colon X \to X$ (since $\mathrm{id}_X \circ g = g$). Write $\eta_g^{\mathrm{src}} := F(g\colon T \to X) \in \mathcal{F}(T)$ for the image of the *object* g, to keep it distinct from the image of the *arrow* g. Applying F to the arrow g gives an arrow $F(g)\colon \eta_g^{\mathrm{src}} \to F(\mathrm{id}_X) = \xi$ over g, and since every arrow of a CFG is cartesian, this $F(g)$ is a cartesian arrow over g. But $F_\xi(g) = g^*\xi$ is also equipped with a cartesian arrow to ξ over g, namely the cleavage arrow. Two cartesian arrows over the same g with the same target ξ differ by a unique vertical isomorphism (as shown following definition 19.1.1), yielding a unique isomorphism $\eta_g\colon \eta_g^{\mathrm{src}} \xrightarrow{\sim} F_\xi(g)$ in $\mathcal{F}(T)$ over id_T. The uniqueness makes $\eta = (\eta_g)_g$ natural in g: for $h\colon T' \to T$ with $g' = g \circ h$, both $\eta_{g'}$ and the mediating arrows are forced by the same universal property, so the naturality square commutes. Each η_g is vertical, so $\eta\colon F \Rightarrow F_\xi$ is a 2-morphism, and being a fiberwise isomorphism it is a 2-isomorphism. Hence $(\xi \mapsto F_\xi) \circ \mathrm{ev} \cong \mathrm{id}$, and ev is an equivalence.

Step 4: The embedding and representability. Taking $\mathcal{F} = \underline{Y}$ in the equivalence gives $\underline{\mathrm{Hom}}_{\mathcal{C}}(\underline{X}, \underline{Y}) \simeq \underline{Y}(X) = \mathrm{Hom}_S(X, Y)$, a discrete groupoid, so the 2-functor $X \mapsto \underline{X}$ induces equivalences on Hom-groupoids: it is a 2-fully-faithful embedding of $\mathcal{C}$ into CFGs. Finally, a CFG $\mathcal{F}$ is representable if and only if $\mathcal{F} \simeq \underline{X}$ for some X, which is the definition (definition 19.1.7); the equivalence just proved shows that when this holds the equivalence is recorded by an object $\xi \in \mathcal{F}(X)$ that is universal, the image of id_X, as we sought to show.

The lemma has two payoffs used constantly in the sequel. First, it lets one compute the groupoid of maps into a CFG by evaluation: a map $\underline{T} \to \mathcal{F}$ from a scheme is the same as an object $\xi \in \mathcal{F}(T)$, and a 2-morphism between two such maps is the same as an isomorphism between the corresponding objects. This is why the "T-points of $\mathcal{F}$" can be defined as the objects of $\mathcal{F}(T)$: the 2-Yoneda lemma guarantees that the categorical Hom recovers exactly this groupoid. Second, it makes "representable" a checkable condition: $\mathcal{F}$ is a scheme precisely when its fibers are discrete and glue to a representable functor of points, and the failure of representability for a moduli problem is detected in the fiber $\mathcal{F}(T)$ carrying nontrivial automorphisms. The embedding $\mathcal{C} \hookrightarrow \{\mathrm{CFGs}\}$ means that no generality is lost by working with CFGs from the outset: schemes sit inside as the CFGs with discrete fibers.

With the framework in place, the recurring objects of the chapter can be introduced. Each is a CFG; the difference among them is entirely in how much automorphism their fibers carry.

Example 19.1.9: Quasi-Coherent Sheaves, BG, and $\mathcal{M}_{1,1}$ as CFGs

Each of the following is a CFG over $\mathcal{C} = (\mathbf{Sch}/S)_\tau$; the automorphisms in the fiber are identified in each case.

1. *Quasi-coherent sheaves.* Let $\underline{\mathrm{QCoh}}$ have as objects the pairs $(T, \mathcal{E})$ with $T \in \mathcal{C}$ and $\mathcal{E}$ a quasi-coherent sheaf on T, and as arrows $(T', \mathcal{E}') \to (T, \mathcal{E})$ the pairs (f, φ) with $f\colon T' \to T$ and $\varphi\colon \mathcal{E}' \xrightarrow{\sim} f^*\mathcal{E}$ an isomorphism of $\mathcal{O}_{T'}$-modules. The projection sends $(T, \mathcal{E})$ to T. Pullback along $f\colon T' \to T$ is f^* of sheaves, and this is cartesian; every arrow is an isomorphism onto a pullback, so $\underline{\mathrm{QCoh}}$ is a CFG. The fiber over T is the groupoid $\mathrm{QCoh}(T)^\times$ of quasi-coherent sheaves on T with isomorphisms as arrows; the automorphisms of $(T, \mathcal{E})$ are $\mathrm{Aut}_{\mathcal{O}_T}(\mathcal{E})$, the sheaf automorphisms of $\mathcal{E}$. Restricting to locally free sheaves of rank n gives the CFG $\underline{\mathrm{Vect}}_n$, whose fiber over T is rank-n vector bundles and whose automorphism group at $(T, \mathcal{E})$ is $\mathrm{GL}(\mathcal{E}) = \mathrm{GL}_n(\Gamma(T, \mathcal{O}_T))$ when $\mathcal{E}$ is trivial.
2. *The classifying CFG BG.* Let G be a flat, finitely presented S-group scheme. Let BG have as objects the pairs (T, P) with $T \in \mathcal{C}$ and $P \to T$ a G-torsor for the topology τ (definition 18.5.1), and as arrows $(T', P') \to (T, P)$ the pairs (f, φ) with $f\colon T' \to T$ and $\varphi\colon P' \xrightarrow{\sim} P \times_T T' = f^*P$ an isomorphism of G-torsors over T'. Pullback of a torsor is a torsor, and every arrow is an isomorphism of torsors over a base change, so BG is a CFG. The fiber over T is the groupoid of G-torsors on T; its set of isomorphism classes is $H^1_\tau(T, G)$

(theorem 18.5.3). The automorphisms of (T, P) are the G-equivariant automorphisms of the torsor P, which for the trivial torsor $P = G \times_S T$ is the group $G(T)$ acting by translation. Thus BG has fibers with automorphism group G: it is the CFG that "remembers" the group.

3. *The moduli CFG of elliptic curves.* Let $\mathcal{M}_{1,1}$ have as objects the families $(T, E \to T)$ of elliptic curves, that is, smooth proper morphisms $E \to T$ with a section whose geometric fibers are genus-1 curves, and as arrows $(T', E') \to (T, E)$ the pairs (f, φ) with $f: T' \to T$ and $\varphi: E' \xrightarrow{\sim} E \times_T T'$ an isomorphism of elliptic curves over T' (respecting the sections). Pullback of a family along f is the fiber product $E \times_T T'$, which is again a family, and it is cartesian; every arrow is a fiberwise isomorphism, so $\mathcal{M}_{1,1}$ is a CFG. The fiber over T is the groupoid of elliptic curves over T; over an algebraically closed field of characteristic 0 (or of characteristic $\neq 2, 3$), the automorphisms of a single elliptic curve are $\{\pm 1\}$ generically, growing to a group of order 4 or 6 at $j = 1728$ or $j = 0$ (example 13.3.4). The presence of these automorphisms in every fiber is exactly why $\mathcal{M}_{1,1}$ is not representable by a scheme: by theorem 19.1.8 a representable CFG has discrete fibers.
4. *A scheme.* For an S-scheme X, the representable CFG $\underline{X}$ of definition 19.1.7 has fiber over T the discrete groupoid $\mathrm{Hom}_S(T, X)$: every automorphism is the identity. A scheme is the extreme case of a moduli problem, one with no automorphisms.

The four examples arrange themselves by the size of their automorphisms. A scheme $\underline{X}$ has trivial automorphisms and is representable; $\mathcal{M}_{1,1}$ carries the automorphisms $\{\pm 1\}$ and more at special j, and is not; BG has automorphism group G uniformly; and $\underline{\mathrm{QCoh}}$ has the full sheaf automorphism group in each fiber. In every case the fiber is a genuine groupoid, and the arrows of that groupoid, the isomorphisms discarded by a set-valued functor, are the data that a CFG keeps. That $\underline{\mathrm{QCoh}}$, BG, and $\mathcal{M}_{1,1}$ are not only CFGs but *stacks*, meaning their objects and morphisms glue in the topology τ, is the subject of section 19.2; that BG and $\mathcal{M}_{1,1}$ are quotient stacks built from group actions is the subject of section 19.3. For now the reader should hold onto the picture that a moduli problem with automorphisms is a CFG, and that the automorphisms live in the fibers where a functor of points would have flattened them to a set.

Exercise 19.1.1

1. Let $p: \mathcal{F} \to \mathcal{C}$ be a fibered category and $\varphi: x \to y$, $\varphi': x' \to y$ two cartesian arrows over the same $f: U \to V$. Prove directly from definition 19.1.1 that there is a unique vertical isomorphism $u: x \to x'$ over U with $\varphi' \circ u = \varphi$. (This is the "pullback unique up to unique isomorphism" statement.)
2. Show that an identity arrow $\mathrm{id}_y: y \to y$ is always cartesian, and deduce that in the representable CFG $\underline{X}$ of definition 19.1.7, every arrow is cartesian, so $\underline{X}$ is a CFG. Identify the fiber $\underline{X}(T)$ as a groupoid and confirm it is discrete.
3. Let G be an S-group scheme and BG the classifying CFG of example 19.1.9(2). Using the 2-Yoneda lemma (theorem 19.1.8) as a template, compute the groupoid $\underline{\mathrm{Hom}}_{\mathcal{C}}(\underline{T}, BG)$ of morphisms from a scheme T to BG, and show it is equivalent to the groupoid of G-torsors on T. Conclude that the isomorphism classes of maps $\underline{T} \to BG$ are classified by $H^1_\tau(T, G)$ (theorem 18.5.3).
4. A morphism of CFGs $F: \mathcal{F} \to \mathcal{G}$ over $\mathcal{C}$ is defined to be a functor with $p_{\mathcal{G}} \circ F = p_{\mathcal{F}}$. Prove that such an F automatically preserves cartesian arrows, so the requirement "preserves cartesian

arrows" is redundant for CFGs (though not for general fibered categories). Where does the argument use that all arrows of $\mathcal{F}$ are cartesian?

5. Give the pseudofunctor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Grpd}$ associated with $\underline{\mathrm{Vect}}_1$ (line bundles), specifying the value at each T, the pullback functors, and the automorphism group of an object. Explain why the isomorphism classes of objects in the fiber over T form the Picard group $\mathrm{Pic}(T)$, and identify the automorphism group of a line bundle $\mathcal{L}$ on T with $\Gamma(T, \mathcal{O}_T)^\times$.

6. Explain, using theorem 19.1.8, why a CFG whose fiber $\mathcal{F}(T)$ contains an object with a nontrivial automorphism for some T cannot be representable by a scheme. Apply this to $\mathcal{M}_{1,1}$ (example 19.1.9(3)) and to $B\mathbb{G}_m$ to conclude that neither is a scheme.

19.2 Prestacks and Stacks

The previous section built the container. A category fibered in groupoids over $\mathcal{C} = (\mathbf{Sch}/S)_\tau$ records objects with automorphisms varying over the base, exactly the shape of a moduli problem in which isomorphic objects must not be silently identified. What that section did *not* do is ask whether the objects and morphisms of a CFG can be reconstructed from local data. This is the same question the previous chapter answered for sheaves of sets and for quasi-coherent sheaves, now posed one categorical level up: given a covering family $\{U_i \to U\}$ and objects $x_i \in \mathcal{F}(U_i)$ together with isomorphisms of their restrictions on the overlaps, is there an object over U from which they all descend? A CFG for which the answer is always yes is a *stack*, and the passage from a fibered category to the descent condition is the step that turns a bookkeeping device into a geometric object.

The condition splits cleanly into two halves, and it is worth separating them from the start. Morphisms between objects are themselves a local datum: two objects over U are identified through their pullbacks to a cover, and an isomorphism between them should be constructible from compatible isomorphisms over the pieces. Objects are the harder datum: a family of objects on a cover, glued by isomorphisms on the overlaps satisfying a cocycle condition, should assemble into a single object over the base. A CFG in which morphisms glue is a *prestack*; a prestack in which objects also glue is a *stack*. The bulk of the work in verifying that a concrete moduli problem is a stack is already discharged by faithfully flat descent, and this section makes that reduction precise: $\underline{\mathrm{QCoh}}$, BG, and the CFG of G-torsors are stacks because theorem 18.3.4 and theorem 18.5.3 say exactly that their objects and morphisms descend. The section closes with stackification, the universal repair that forces a CFG to become a stack, and with the running example of a prestack that is not one.

Throughout, $\mathcal{F} \to \mathcal{C}$ is a CFG in the sense of definition 19.1.4, and τ is the fixed topology on $\mathcal{C}$ (fppf unless stated otherwise). For an object $U \in \mathcal{C}$ and two objects $x, y \in \mathcal{F}(U)$, the assignment sending an arrow $f\colon V \to U$ to the set $\mathrm{Isom}_{\mathcal{F}(V)}(f^*x, f^*y)$ of isomorphisms between the pullbacks in the fiber groupoid over V is a presheaf on the slice category $\mathcal{C}/U$: a morphism $g\colon W \to V$ over U acts by pulling back isomorphisms along g, using the pseudofunctor structure to identify $g^*(f^*x) \cong (fg)^*x$. This presheaf is the central object of the whole section, and it deserves a name.

Definition 19.2.1: The Isom Presheaf

Let $\mathcal{F} \to \mathcal{C}$ be a category fibered in groupoids, $U \in \mathcal{C}$, and $x, y \in \mathcal{F}(U)$. The *Isom presheaf* $\underline{\mathrm{Isom}}_U(x, y)$ is the presheaf of sets on the slice category $\mathcal{C}/U$ defined on objects by

$$\underline{\mathrm{Isom}}_U(x, y)(f : V \to U) = \mathrm{Isom}_{\mathcal{F}(V)}(f^*x, f^*y)$$

the set of isomorphisms in the fiber groupoid $\mathcal{F}(V)$ between the pullbacks f^*x and f^*y, and on a morphism $g : (W \to U) \to (V \to U)$ over U by the pullback map $\varphi \mapsto g^*\varphi$, formed through the canonical isomorphisms $g^*f^*x \cong (fg)^*x$ of the cleavage. The presheaf $\underline{\mathrm{Aut}}_U(x) = \underline{\mathrm{Isom}}_U(x, x)$ is the *Aut presheaf*; it takes values in groups.

The Isom presheaf is the fibered-category incarnation of the set of maps between two families. When $\mathcal{F} = \underline{\mathrm{QCoh}}$ and x, y are quasi-coherent sheaves on U, $\underline{\mathrm{Isom}}_U(x, y)(V \to U)$ is the set of $\mathcal{O}_V$-linear isomorphisms $x|_V \to y|_V$, so this presheaf is the sheaf-Hom of the previous chapter read as a presheaf on the site $\mathcal{C}/U$. When $\mathcal{F} = BG$ and x, y are G-torsors on U, $\underline{\mathrm{Isom}}_U(x, y)$ is the presheaf of torsor isomorphisms, which is itself a torsor under $\underline{\mathrm{Aut}}(x)$, the inner form ${}^xG$ of G (equal to G when G is abelian, as for $\mathbb{G}_m$ and μ_n). Whether these presheaves are sheaves is precisely the prestack condition, and the reader has already met the reason they are: a G-equivariant isomorphism of torsors, or an isomorphism of quasi-coherent sheaves, is a local datum that glues.

The two-tier definition that organizes the rest of the chapter can now be stated. The first tier asks that maps glue; the second asks that objects glue along a descent datum, and the descent datum is the CFG version of the gluing package of definition 18.3.1: instead of an isomorphism between the two pullbacks of a single sheaf, one has objects x_i over the pieces U_i of a cover and isomorphisms φ_{ij} over the double overlaps, subject to a cocycle condition on the triple overlaps.

Definition 19.2.2: Prestack and Stack

Let $(\mathcal{C}, \tau)$ be a site and $\mathcal{F} \to \mathcal{C}$ a category fibered in groupoids.
A *descent datum* for $\mathcal{F}$ on a covering family $\{f_i : U_i \to U\}$ of τ is a pair $(\{x_i\}, \{\varphi_{ij}\})$ consisting of an object $x_i \in \mathcal{F}(U_i)$ for each i and, writing $U_{ij} = U_i \times_U U_j$ with projections $\mathrm{pr}_i : U_{ij} \to U_i$ and $\mathrm{pr}_j : U_{ij} \to U_j$, an isomorphism

$$\varphi_{ij} : \mathrm{pr}_j^* x_j \xrightarrow{\sim} \mathrm{pr}_i^* x_i \quad \text{in } \mathcal{F}(U_{ij})$$

for each pair (i, j), subject to the *cocycle condition*: on the triple overlap $U_{ijk} = U_i \times_U U_j \times_U U_k$, with $\mathrm{pr}_{ab} : U_{ijk} \to U_{ab}$ the projections,

$$\mathrm{pr}_{ik}^* \varphi_{ik} = \mathrm{pr}_{ij}^* \varphi_{ij} \circ \mathrm{pr}_{jk}^* \varphi_{jk}$$

as isomorphisms $\mathrm{pr}_k^* x_k \to \mathrm{pr}_i^* x_i$ in $\mathcal{F}(U_{ijk})$, formed through the canonical cleavage identifications. A *morphism of descent data* $(\{x_i\}, \{\varphi_{ij}\}) \to (\{x_i'\}, \{\varphi_{ij}'\})$ is a family of isomorphisms $\psi_i : x_i \to x_i'$ in $\mathcal{F}(U_i)$ with $\varphi_{ij}' \circ \mathrm{pr}_j^* \psi_j = \mathrm{pr}_i^* \psi_i \circ \varphi_{ij}$ over each U_{ij}.
Every object $x \in \mathcal{F}(U)$ induces a *canonical* descent datum on $\{f_i : U_i \to U\}$: take $x_i = f_i^* x$ and let φ_{ij} be the canonical isomorphism $\mathrm{pr}_j^* f_j^* x \cong \mathrm{pr}_i^* f_i^* x$ arising from $f_i \circ \mathrm{pr}_i = f_j \circ \mathrm{pr}_j : U_{ij} \to U$.
A descent datum is *effective* if it is isomorphic to a canonical one.
The CFG $\mathcal{F}$ is a *prestack* (over $(\mathcal{C}, \tau)$) if for all $U \in \mathcal{C}$ and all $x, y \in \mathcal{F}(U)$ the Isom presheaf $\underline{\mathrm{Isom}}_U(x, y)$ of definition 19.2.1 is a τ-sheaf on $\mathcal{C}/U$. The CFG $\mathcal{F}$ is a *stack* if it is a prestack and, in addition, every descent datum on every covering family of τ is effective.

The definition packs two ideas that are worth prying apart. The prestack condition is descent for *morphisms*. Unwinding the sheaf axiom for $\underline{\mathrm{Isom}}_U(x, y)$ on a cover $\{U_i \to U\}$: an isomorphism $x \to y$ over U is the same as a family of isomorphisms $f_i^* x \to f_i^* y$ over the pieces that agree on the overlaps U_{ij}. The separated half of the sheaf axiom says a morphism over U is determined by its restrictions (no two distinct maps agree everywhere locally); the gluing half says compatible local maps come from a global one. A prestack is therefore a CFG in which Hom-sets are recovered from local data. The stack condition adds descent for *objects*: the objects x_i on the pieces, glued by the φ_{ij} satisfying the cocycle condition, assemble into a single x over U inducing them. The cocycle condition $\varphi_{ik} = \varphi_{ij} \circ \varphi_{jk}$ is the transitivity of the gluing, identical in form to the one in definition 18.3.1, and it is what guarantees the local objects can be consistently overlapped.

The two conditions are separate. Once $\mathcal{F}$ is a prestack, the effective descent data are unique up to unique isomorphism when they exist, because the gluing isomorphisms between two solutions are themselves controlled by the (now sheafy) Isom presheaves; the stack condition is then the pure *existence* statement that every descent datum has a solution. This is why one first checks that Isom is a sheaf and only then checks effectivity, and why the failure of a moduli problem to be a stack is always a failure of gluing objects, never a failure of gluing maps once the prestack condition holds.

Before the theorem, a first concrete instance grounds the descent datum in an object the reader already controls: a single quasi-coherent sheaf. This is the CFG $\underline{\mathrm{QCoh}}$ from example 19.1.9, whose fiber over T is the groupoid of quasi-coherent sheaves and their isomorphisms, and its stack condition is faithfully flat descent verbatim.

Example 19.2.3: The Descent Datum of a Quasi-Coherent Sheaf

Let $\mathcal{F} = \underline{\mathrm{QCoh}}$ and let $\{U_i \to U\}$ be a single-morphism cover $U' \to U$ (recover a family by $U' = \coprod_i U_i$). A descent datum for $\underline{\mathrm{QCoh}}$ on $U' \to U$ is a quasi-coherent sheaf $\mathcal{E}'$ on U' together with an isomorphism

$$\varphi\colon \mathrm{pr}_2^* \mathcal{E}' \xrightarrow{\sim} \mathrm{pr}_1^* \mathcal{E}' \quad \text{on } U' \times_U U'$$

satisfying the cocycle condition on $U' \times_U U' \times_U U'$. This is exactly the descent datum of definition 18.3.1, transcribed into the fibered-category language: the objects x_i are the single sheaf $\mathcal{E}'$ and the gluing isomorphism φ_{12} is the map φ. The prestack condition is that $\underline{\mathrm{Isom}}(\mathcal{E}, \mathcal{G})$ is an fppf sheaf for quasi-coherent $\mathcal{E}, \mathcal{G}$ on U, which holds because morphisms of quasi-coherent sheaves satisfy fppf descent: a section of $\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{G})$ over an object is recovered from its restrictions to any fppf cover. The effectivity of every descent datum is the essential surjectivity of the pullback functor in theorem 18.3.4. Thus every ingredient of "$\underline{\mathrm{QCoh}}$ is a stack" is already proved; the theorem below only records the packaging.

The example carries the content of the theorem in the affine, single-cover case; the theorem states the conclusion in general and adds the two group theoretic companions BG and the CFG of torsors, whose stack property is the torsor classification of theorem 18.5.3 read in the same way. The motivating point is that no new descent is needed: the entire content of this chapter's first stack examples was established in the previous chapter, and recognizing that is the payoff of separating the fibered category from its descent condition.

Theorem 19.2.4: The Stack of Quasi-Coherent Sheaves and Classifying Stacks

Over the site $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$:

1. The CFG $\underline{\mathrm{QCoh}}$, whose fiber over T is the groupoid of quasi-coherent $\mathcal{O}_T$-modules, is a stack. The same holds for the CFGs $\underline{\mathrm{Vect}}_n$ of rank-n vector bundles and $\underline{\mathrm{Coh}}$ of coherent sheaves.

2. For a flat, finitely presented, affine group scheme G over S, the CFG of G-torsors, whose fiber over T is the groupoid of G-torsors on T, is a stack. In particular the classifying CFG BG is a stack.

***Proof*:**

Each part is a prestack condition (Isom is a sheaf) followed by an effectivity condition (descent data glue), and both are the descent theorems of the previous chapter transcribed. The pseudofunctor coherence, the verification that the chosen cleavage makes the cocycle conditions strictly compatible with the associativity constraints of the 2-category of groupoids, is a formal bookkeeping check carried out in full in [66, Ch. 4]; it is cited rather than reproduced, as it is the same coherence for every fibered category and adds nothing to the geometry.

Step 1: $\underline{\mathrm{QCoh}}$ is a prestack. Fix U and quasi-coherent sheaves $\mathcal{E}, \mathcal{G}$ on U. The Isom presheaf $\underline{\mathrm{Isom}}_U(\mathcal{E}, \mathcal{G})$ is the presheaf of $\mathcal{O}$-linear isomorphisms, the subpresheaf of isomorphisms in $\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{G})$. The presheaf $\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{G})$ sends $V \to U$ to $\mathrm{Hom}_{\mathcal{O}_V}(\mathcal{E}|_V, \mathcal{G}|_V)$; that this is an fppf sheaf is faithfully flat descent for morphisms of quasi-coherent sheaves, which is the full-faithfulness half of theorem 18.3.4 applied to the sheaf $\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{G})$: a morphism $\mathcal{E}|_V \to \mathcal{G}|_V$ is determined by, and can be glued from, its restrictions to an fppf cover of V, because theorem 18.3.4 identifies $\mathrm{Hom}_{\mathcal{O}_V}(\mathcal{E}|_V, \mathcal{G}|_V)$ with the equalizer of the two pullbacks of Hom over the cover, which is exactly the sheaf condition for $\underline{\mathrm{Hom}}(\mathcal{E}, \mathcal{G})$. The subpresheaf of isomorphisms is a sheaf because being an isomorphism is fppf-local on the base (a morphism of quasi-coherent sheaves is an isomorphism iff it is so after faithfully flat base change, again by theorem 18.3.4). Hence $\underline{\mathrm{Isom}}_U(\mathcal{E}, \mathcal{G})$ is an fppf sheaf and $\underline{\mathrm{QCoh}}$ is a prestack.

Step 2: Descent data for $\underline{\mathrm{QCoh}}$ *are effective.* Effectivity is local on U, so it suffices to treat U affine, hence quasi-compact. A descent datum on a cover $\{U_i \to U\}$ is, by example 19.2.3, a quasi-coherent sheaf on $U' = \coprod_i U_i$ with a gluing isomorphism on $U' \times_U U'$ satisfying the cocycle condition. Because U is quasi-compact, the surjection $\coprod_i U_i \to U$ admits a finite subcover $U_{i_1}, \ldots, U_{i_r}$; replacing U' by the finite disjoint union $U'' = U_{i_1} \sqcup \cdots \sqcup U_{i_r}$ (and restricting the descent datum, which loses nothing since a descent datum for a cover restricts to one for any subcover realizing the same object) yields a morphism $U'' \to U$ that is faithfully flat and quasi-compact: a finite disjoint union of fppf morphisms is fppf, and finiteness of the union supplies the quasi-compactness that theorem 18.3.4 requires. That theorem then applies: the pullback functor $\mathrm{QCoh}(U) \to \mathrm{Desc}(U'' \to U)$ is an equivalence, hence essentially surjective, so every such descent datum is isomorphic to a canonical one $\mathrm{pr}^* \mathcal{E}$. This is exactly effectivity. Together with Step 1, $\underline{\mathrm{QCoh}}$ is a stack. The variants $\underline{\mathrm{Vect}}_n$ and $\underline{\mathrm{Coh}}$ are the full subcategories cut out by the fppf-local conditions "locally free of rank n" and "coherent"; both conditions descend along a faithfully flat quasi-compact cover ([10]), so the descended sheaf lies in the subcategory, and the same two steps apply.

Step 3: The CFG of G-torsors is a prestack. Fix U and two G-torsors P, Q on U. An arrow $V \to U$ pulls them back to torsors on V, and $\underline{\mathrm{Isom}}_U(P, Q)(V) = \mathrm{Isom}_G(P|_V, Q|_V)$ is the set of G-equivariant isomorphisms. This is a sheaf: a G-equivariant morphism of torsors is a morphism of the underlying fppf sheaves commuting with the G-action, and both the sheaf $\underline{\mathrm{Hom}}$ and the equivariance condition are fppf-local, so an equivariant isomorphism over V is determined by and glued from its restrictions to an fppf cover. Concretely $\underline{\mathrm{Isom}}_U(P, Q)$ is a torsor under the automorphism group scheme $\underline{\mathrm{Aut}}(P)$ when $P \cong Q$ locally; this is the inner form ${}^P G$, which equals G on the nose when G is abelian (as for the running anchors $\mathbb{G}_m$ and μ_n). Hence the CFG of torsors is a prestack.

Step 4: Descent data for G-torsors are effective. A descent datum on $\{U_i \to U\}$ is a torsor P_i on each U_i with gluing isomorphisms φ_{ij} over the overlaps satisfying the cocycle condition. Passing to $U' = \coprod_i U_i$, this is a G-torsor P' on U' with an isomorphism $\varphi\colon\ \mathrm{pr}_2^* P' \to \mathrm{pr}_1^* P'$ of G-torsors over $U' \times_U U'$ satisfying the cocycle condition. A G-torsor is, in particular, a quasi-affine (indeed affine, as G is affine) faithfully flat U'-scheme with G-action; its descent along the fppf cover $U' \to U$ is effective by faithfully flat descent for affine morphisms, the effectivity of affine descent that upgrades theorem 18.3.4 from quasi-coherent modules to their quasi-coherent sheaves of algebras ([66, Ch. 4]), and the descended scheme $P \to U$ inherits a G-action making it a torsor because the torsor axioms (the action map $G \times_S P \to P \times_U P$, $(g, p) \mapsto (gp, p)$, is an isomorphism and $P \to U$ is an fppf cover) are fppf-local on U and hold after pullback to U'. Thus every descent datum of torsors is effective, and the CFG of torsors, in particular BG, is a stack. The classification $H^1_{\mathrm{fppf}}(U, G) = \{G\text{-torsors on } U\}/\cong$ of theorem 18.5.3 is the set of isomorphism classes in the fiber $BG(U)$, so the stack BG is the groupoid-valued refinement of the cohomology functor $U \mapsto H^1_{\mathrm{fppf}}(U, G)$, completing the proof.

The theorem is the first payoff of the entire descent apparatus: the two principal geometric stacks of the book, $\underline{\mathrm{QCoh}}$ and BG, are stacks for a reason the reader has already proved. The Isom presheaf being a sheaf is descent for maps of sheaves or torsors, and the effectivity of descent data is descent for the objects themselves; there is nothing to check beyond re-reading theorem 18.3.4 and theorem 18.5.3

through the fibered-category dictionary. This is the sense in which a stack is descent theory for objects with automorphisms: the automorphisms are the reason a set-valued functor will not do, and descent is the reason the groupoid-valued replacement still glues. The classifying stack BG deserves particular emphasis. Its T-points are G-torsors on T, and its isomorphism classes over T are $H^1_{\text{fppf}}(T, G)$; BG is the object that remembers not only the cohomology set but the automorphisms $\underline{\mathrm{Aut}}(P) = G$ that made the moduli problem non-representable in the first place. The quotient $[X/G]$ of the next section will be the relative version, with $BG = [S/G]$ the case $X = S$.

The reader should hold in mind that both parts used the fppf topology because both descent theorems of the previous chapter were proved there, but the stack property of these two CFGs is not special to fppf. Both $\underline{\mathrm{QCoh}}$ and the CFG of G-torsors are stacks in every subcanonical topology, including Zariski: quasi-coherent sheaves glue along a Zariski cover by the classical gluing of sheaves, and G-torsors glue along a Zariski cover as schemes-with-action, so their descent data are effective there too. What changes with the topology is not effectivity but the *local structure* of the objects: a nontrivial G-torsor need not be Zariski-locally trivial (it becomes trivial only after passing to a finer fppf cover), so fppf descent sees a local triviality that Zariski descent cannot. The choice of topology is not incidental. A CFG is a stack *for a given topology*, and the same CFG can be a stack for one τ and fail for another; the example of such topology dependence is not $\underline{\mathrm{QCoh}}$ or BG but a CFG of *framed* objects, where the framing itself does not descend, as in the naive quotient constructed next. There the descent data whose gluing cocycle defines a nontrivial torsor are exactly the ineffective ones, so the CFG is a stack for a coarser topology in which those torsors have not yet appeared and fails for a finer one in which they have.

Not every CFG is a stack, and the ones that fail are the ones that need repair. Before constructing the repair, it helps to see the failure mechanism cleanly. A prestack can fail to be a stack because descent data fail to glue, and the simplest source of such failure is a moduli problem whose objects have been rigidified by extra structure that does not itself descend. The following is the running counterexample of the chapter, and it is the naive quotient that stackification will correct into $[X/G]$.

Example 19.2.5: The Naive Presheaf Quotient $\underline{X}/G$

Let G be a flat affine group scheme over S acting on an S-scheme X. Define a CFG $\mathcal{F} = [X/G]^{\text{pre}}$, the *naive presheaf quotient*, whose fiber over T is the groupoid whose objects are morphisms $T \to X$ and whose morphisms $u \to u'$ are elements $g \in G(T)$ with $g \cdot u = u'$. Equivalently $\mathcal{F}$ is the CFG associated to the action groupoid $[X(T)/G(T)]$ pointwise, before sheafification. This is a CFG: pullback along $S' \to S$ acts on maps to X and on group elements, and every arrow is invertible because $G(T)$ is a group.

$\mathcal{F}$ is a prestack. For $u, u' \in \mathcal{F}(T)$ the Isom presheaf sends $V \to T$ to $\{g \in G(V) : g \cdot u|_V = u'|_V\}$, a subsheaf of the representable sheaf $\underline{G}$ cut out by the equalizer condition $g \cdot u = u'$; since $\underline{G}$ is an fppf sheaf (it is representable) and the condition is fppf-local, $\underline{\mathrm{Isom}}_T(u, u')$ is an fppf sheaf. So morphisms glue.

$\mathcal{F}$ is not a stack. The objects of $\mathcal{F}(T)$ are maps $T \to X$, which record only the *trivial* torsor $G \times T \to T$ carrying its tautological equivariant map $(g, t) \mapsto g \cdot u(t)$ to X; a nontrivial torsor with an equivariant map to X is invisible to $\mathcal{F}$. A descent datum on a cover $\{U_i \to T\}$ consists of maps $u_i : U_i \to X$ and elements $g_{ij} \in G(U_{ij})$ with $g_{ij} \cdot u_j = u_i$ on U_{ij} satisfying the cocycle condition $g_{ik} = g_{ij} g_{jk}$. Such a datum glues to an object of $\mathcal{F}(T)$, a single map $T \to X$, exactly when the cocycle (g_{ij}) is a coboundary, that is, when the associated G-torsor is trivial. But (g_{ij}) is precisely the transition cocycle of a G-torsor on T (theorem 18.5.3), and there exist nontrivial such torsors on many T as soon as $H^1_{\text{fppf}}(T, G) \neq 0$. For such a datum there is no map $T \to X$ inducing it:

the descent datum encodes a nontrivial twisted family, which the naive quotient cannot represent. Hence $\mathcal{F}$ is a prestack that is not a stack, and the obstruction to gluing is measured by $H^1(T, G)$.

The failure is diagnostic. The naive presheaf quotient records only untwisted families, the ones built on the trivial torsor, so a descent datum whose gluing cocycle defines a nontrivial torsor has nothing to glue to. The information the CFG is missing is exactly a G-torsor: the correct quotient must allow objects to be G-torsors $P \to T$ with an equivariant map $P \to X$, not only maps $T \to X$. This is the definition of $[X/G]$ in the next section, and the example makes the definition inevitable rather than ad hoc: one adjoins to $\mathcal{F}$ precisely the twisted objects that its descent data demand. The general procedure for performing this repair, turning any CFG into the closest stack, is stackification.

Stackification is the reflection of the 2-category of CFGs onto its sub-2-category of stacks. The motivating demand is universal: given any CFG $\mathcal{F}$, one wants a stack $\mathcal{F}^{\mathrm{a}}$ and a morphism $\mathcal{F} \to \mathcal{F}^{\mathrm{a}}$ through which every morphism from $\mathcal{F}$ to a stack factors uniquely up to unique 2-isomorphism. The construction proceeds in the two steps the definition of a stack suggests: first force the Isom presheaves to become sheaves, producing a prestack; then adjoin the missing effective descent data, producing a stack. The example above shows both steps at work: $[X/G]^{\mathrm{pre}}$ is already a prestack, so the first step is vacuous, and the second step adjoins the twisted objects, producing $[X/G]$.

Theorem 19.2.6: Stackification

Let $(\mathcal{C}, \tau)$ be a site. The inclusion of the 2-category of stacks on $(\mathcal{C}, \tau)$ into the 2-category of categories fibered in groupoids over $\mathcal{C}$ admits a left 2-adjoint

$$(-)^{\mathrm{a}} : \{\text{CFGs}\} \longrightarrow \{\text{stacks}\}$$

the *stackification*. For each CFG $\mathcal{F}$ there is a morphism $\iota : \mathcal{F} \to \mathcal{F}^{\mathrm{a}}$ to a stack such that, for every stack $\mathcal{G}$, composition with ι induces an equivalence of groupoids

$$\underline{\mathrm{Hom}}(\mathcal{F}^{\mathrm{a}}, \mathcal{G}) \xrightarrow{\sim} \underline{\mathrm{Hom}}(\mathcal{F}, \mathcal{G})$$

Consequently $\mathcal{F}^{\mathrm{a}}$ is unique up to equivalence, and ι is an equivalence iff $\mathcal{F}$ is already a stack. The morphism ι is fully faithful on Isom sheaves and locally essentially surjective: every object of $\mathcal{F}^{\mathrm{a}}(U)$ is, on a cover of U, in the essential image of $\mathcal{F}$.

***Proof*:**
The construction is in two steps mirroring the two clauses of definition 19.2.2: sheafify morphisms to reach a prestack, then adjoin descent data to reach a stack. The set-theoretic size control (that the sheafifications and the glued objects form a set, not a proper class, so $\mathcal{F}^{\mathrm{a}}$ is a legitimate CFG) and the strict 2-categorical coherence of the resulting pseudofunctor are the bookkeeping carried out in full in [63, Tag 02ZM]; they are cited, and the argument below gives the construction and the universal property.

Step 1: The associated prestack. Define a new CFG $\mathcal{F}^{\mathrm{pre}}$ with the same objects as $\mathcal{F}$ over each U, but with morphism sheaves replaced by their sheafifications. Precisely, for $x, y \in \mathcal{F}(U)$ let $\underline{\mathrm{Isom}}_U(x, y)^{\#}$ be the τ-sheafification (theorem 18.2.3) of the Isom presheaf, and declare an isomorphism $x \to y$ in $\mathcal{F}^{\mathrm{pre}}(U)$ to be a global section of $\underline{\mathrm{Isom}}_U(x, y)^{\#}$. Composition is induced by the composition of Isom presheaves and passes to sheafifications because sheafification is a left-exact functor preserving finite products (theorem 18.2.3). The unit map $\underline{\mathrm{Isom}}_U(x, y) \to \underline{\mathrm{Isom}}_U(x, y)^{\#}$ gives a morphism of CFGs $\mathcal{F} \to \mathcal{F}^{\mathrm{pre}}$, the identity on objects. By construction the Isom presheaves of $\mathcal{F}^{\mathrm{pre}}$ are sheaves, so $\mathcal{F}^{\mathrm{pre}}$ is a prestack; and any morphism from $\mathcal{F}$ to a prestack $\mathcal{G}$ factors uniquely through $\mathcal{F}^{\mathrm{pre}}$, because a morphism to a prestack sends the Isom presheaves into sheaves and so factors through their sheafification by the universal property of theorem 18.2.3. This proves the claim of the theorem for the sub-2-category of prestacks.

Step 2: Adjoining effective descent data. Now let $\mathcal{F}$ be a prestack; construct a stack $\mathcal{F}^{\mathrm{a}}$ by adjoining the missing objects. Define $\mathcal{F}^{\mathrm{a}}(U)$ to be the groupoid whose objects are descent data on τ-covers of U: an object is a pair $(\{U_i \to U\}, (\{x_i\}, \{\varphi_{ij}\}))$ of a cover and a descent datum for $\mathcal{F}$ on it, and a morphism to $(\{V_a \to U\}, (\{y_a\}, \{\psi_{ab}\}))$ is, after passing to a common refinement of the two covers, a morphism of descent data as in definition 19.2.2. Two descent data related by refinement are declared isomorphic, so $\mathcal{F}^{\mathrm{a}}(U)$ is the (filtered) 2-colimit of the descent-data groupoids over all covers of U. Pullback along $V \to U$ pulls a cover of U back to a cover of V and a descent datum to a descent datum, making $\mathcal{F}^{\mathrm{a}} \to \mathcal{C}$ a CFG, and the tautological "one-element cover" embedding gives $\iota\colon \mathcal{F} \to \mathcal{F}^{\mathrm{a}}$, $x \mapsto (\{U \xrightarrow{\mathrm{id}} U\}, (x, \mathrm{id}))$.

Step 3: $\mathcal{F}^{\mathrm{a}}$ is a stack. That $\mathcal{F}^{\mathrm{a}}$ is a prestack follows because $\mathcal{F}$ is: an isomorphism of two descent data on a common refinement is a compatible family of isomorphisms of the underlying objects, whose gluing is governed by the (sheafy) Isom presheaves of $\mathcal{F}$, so $\underline{\mathrm{Isom}}$ for $\mathcal{F}^{\mathrm{a}}$ is a sheaf. For effectivity, a descent datum for $\mathcal{F}^{\mathrm{a}}$ on a cover $\{U_i \to U\}$ assigns to each U_i a descent datum for $\mathcal{F}$ on a cover of U_i, glued over the overlaps. Refining all these covers to a single cover $\{W_\alpha \to U\}$ of U (the composite of the two layers of covers, using that τ-covers compose), the data assemble into a single descent datum for $\mathcal{F}$ on $\{W_\alpha \to U\}$, which is by construction an object of $\mathcal{F}^{\mathrm{a}}(U)$ inducing the given one. Hence every descent datum for $\mathcal{F}^{\mathrm{a}}$ is effective, and $\mathcal{F}^{\mathrm{a}}$ is a stack.

Step 4: The universal property. Let $\mathcal{G}$ be a stack and $F\colon \mathcal{F} \to \mathcal{G}$ a morphism. Define $\widehat{F}\colon \mathcal{F}^{\mathrm{a}} \to \mathcal{G}$ on an object $(\{U_i \to U\}, (\{x_i\}, \{\varphi_{ij}\}))$ of $\mathcal{F}^{\mathrm{a}}(U)$ as follows: the images $F(x_i) \in \mathcal{G}(U_i)$ with the isomorphisms $F(\varphi_{ij})$ form a descent datum for $\mathcal{G}$ on $\{U_i \to U\}$, which is effective because $\mathcal{G}$ is a stack, so it glues to an object $z \in \mathcal{G}(U)$, well-defined up to unique isomorphism by the prestack property of $\mathcal{G}$. Set $\widehat{F}$ of the object to be z. This is functorial and satisfies $\widehat{F} \circ \iota \cong F$ because ι sends x to the trivial descent datum, which glues back to $F(x)$. Uniqueness up to unique 2-isomorphism: any $H\colon \mathcal{F}^{\mathrm{a}} \to \mathcal{G}$ with $H \circ \iota \cong F$ must send a descent datum to the gluing of $\{F(x_i)\}$, since H commutes with pullback and the descent datum is, locally on U, the image under ι of the x_i; the prestack property of $\mathcal{G}$ then makes the comparison 2-isomorphism $\widehat{F} \cong H$ unique. Therefore composition with ι induces the asserted equivalence $\underline{\mathrm{Hom}}(\mathcal{F}^{\mathrm{a}}, \mathcal{G}) \xrightarrow{\sim} \underline{\mathrm{Hom}}(\mathcal{F}, \mathcal{G})$, and $(-)^{\mathrm{a}}$ is left 2-adjoint to the inclusion.

Combining Steps 1 through 4, an arbitrary CFG is stackified by first applying Step 1 to reach a prestack and then Steps 2 and 3 to reach a stack, and the composite has the universal property of Step 4. That ι is an equivalence exactly when $\mathcal{F}$ is already a stack follows from the universal property applied to $\mathcal{G} = \mathcal{F}$, completing the proof.

Stackification is the free construction that makes descent data glue, and its universal property is what justifies calling $\mathcal{F}^{\mathrm{a}}$ "the" stack associated to $\mathcal{F}$: any construction on stacks applied to $\mathcal{F}^{\mathrm{a}}$ is determined by its restriction to $\mathcal{F}$, so passing to the stackification loses no mapping-out information. The two steps are worth holding separately in mind, because they diagnose two different pathologies. Step 1 repairs a CFG whose maps do not glue, replacing the Isom presheaves by their sheafifications; this is invisible when the objects already have sheafy automorphisms, as in $[X/G]^{\mathrm{pre}}$. Step 2 repairs a prestack whose objects do not glue, formally adjoining the missing twisted objects; this is the step that turns $[X/G]^{\mathrm{pre}}$ into $[X/G]$ by supplying the nontrivial-torsor families that example 19.2.5 found missing. The construction here is the concrete two-step one; the size-theoretic guarantee that the adjoined objects form a set and the strict coherence of the resulting pseudofunctor are the parts deferred to [63], because they are formal and identical for every site, contributing nothing to the geometry.

The contrast with $\underline{\mathrm{QCoh}}$ closes the loop. A stack is a CFG that needs no repair: $\iota\colon \mathcal{F} \to \mathcal{F}^{\mathrm{a}}$ is an equivalence, and $\underline{\mathrm{QCoh}}$ is such a CFG by theorem 19.2.4. The naive quotient $[X/G]^{\mathrm{pre}}$ is not, and its stackification is $[X/G]$; the discrepancy between them is measured by the nontrivial G-torsors, which is to say by $H^1(-, G)$, which is to say by exactly the automorphism data that made a set-valued moduli functor inadequate. Stackification is the machine that converts that inadequacy into a geometric object, and the quotient stack $[X/G]$ is the first geometric output of that machine.

Exercise 19.2.1

1. Let $\mathcal{F} \to \mathcal{C}$ be a CFG over a site $(\mathcal{C}, \tau)$. Show that $\mathcal{F}$ is a prestack if and only if, for every cover $\{U_i \to U\}$ and every pair $x, y \in \mathcal{F}(U)$, the map
$$\mathrm{Isom}_{\mathcal{F}(U)}(x, y) \longrightarrow \Big\{(\psi_i) : \psi_i \in \mathrm{Isom}_{\mathcal{F}(U_i)}(x|_{U_i}, y|_{U_i}),\ \psi_i|_{U_{ij}} = \psi_j|_{U_{ij}}\Big\}$$
is a bijection. Explain how injectivity is the "separated" half and surjectivity is the "gluing" half of the sheaf axiom for $\underline{\mathrm{Isom}}_U(x, y)$.

2. Prove that in any prestack, an effective descent datum, when it exists, is unique up to a unique isomorphism. (Hint: if $x, x' \in \mathcal{F}(U)$ both induce the descent datum $(\{x_i\}, \{\varphi_{ij}\})$, the local isomorphisms $x|_{U_i} \cong x_i \cong x'|_{U_i}$ are compatible on overlaps; glue them using the prestack condition.)

3. Consider the CFG $\underline{\mathrm{Vect}}_1$ of line bundles over $(\mathbf{Sch}/S)_{\mathrm{fppf}}$. Describe a descent datum on a cover $\{U_i \to U\}$ explicitly as a collection of line bundles L_i on U_i with transition isomorphisms φ_{ij}, and identify the cocycle condition with the usual 1-cocycle condition on transition functions. Conclude from theorem 19.2.4 that a line bundle on U is the same as a descent datum, and relate this to the identification $\mathrm{Pic}(U) = H^1_{\mathrm{fppf}}(U, \mathbb{G}_m)$.

4. Let $\mathcal{F} = \underline{\mathrm{Set}}$ be the CFG whose fiber over T is the groupoid of *constant* finite sets (sets with the trivial pullback maps), regarded over the small Zariski site of a fixed scheme X. Show that for X disconnected $\mathcal{F}$ is *not* even a prestack: exhibit a pair of objects whose Isom presheaf, the constant presheaf $\mathrm{Isom}(x, y)$, fails the gluing axiom on the cover by the connected components. Identify the associated prestack $\mathcal{F}^{\mathrm{pre}}$ (Isom presheaves sheafified) and the stackification $\mathcal{F}^{\mathrm{a}}$ (locally constant sheaves of finite sets). This is the "sheafify morphisms" step of stackification in its simplest form.

5. For the naive presheaf quotient $[X/G]^{\mathrm{pre}}$ of example 19.2.5, take $S = \operatorname{Spec} k$, $X = \operatorname{Spec} k$ a point, and $G = \mathbb{G}_m$. Identify $[X/G]^{\mathrm{pre}}$ concretely (its fiber over T has one object and automorphism group $\mathbb{G}_m(T)$), show it is a prestack but not a stack, and identify a nontrivial descent datum with a nontrivial line bundle. What is the stackification?

6. Show that the stack condition depends on the topology by exhibiting a single CFG that is a stack for one topology and fails to be a stack for a finer one. Take the naive presheaf quotient $\mathcal{F} = [\mathrm{pt}/\mu_n]^{\mathrm{pre}}$ of example 19.2.5 with $S = X = \operatorname{Spec} k$, n invertible in k, and μ_n in place of $\mathbb{G}_m$; its objects are *framed* (trivialized) μ_n-torsors and its descent data glue exactly when the gluing cocycle is a coboundary. Deduce that $\mathcal{F}$ is a τ-stack if and only if $H^1_\tau(T, \mu_n) = 0$ for every T. Then explain why this makes $\mathcal{F}$ a stack for a topology in which every μ_n-torsor over

the relevant base is trivial but not for the fppf topology, where nontrivial μ_n-torsors exist. Contrast this with the *full* CFG of μ_n-torsors, which (by theorem 19.2.4) is a stack for every subcanonical topology, and explain in one sentence why the framed CFG, and not the full torsor CFG, is where the topology dependence lives.

19.3 Quotient Stacks and Classifying Stacks

The previous two sections built the abstract apparatus: a moduli problem with automorphisms is a category fibered in groupoids (definition 19.1.4), and when its objects and morphisms glue in a topology it is a stack (definition 19.2.2). What has been missing is a supply of stacks one can hold in the hand. This section produces the first geometric ones. Given a group scheme G acting on a scheme X, there is a stack $[X/G]$, the *quotient stack*, whose points over a test scheme T are not orbits of T-points, but G-torsors on T carrying an equivariant map to X. The special case $X = \mathrm{pt}$ produces the *classifying stack* BG, whose T-points are precisely the G-torsors on T. These two constructions are what the theory is built from: every quotient stack is built from a torsor and an equivariant map, and BG is where the torsor classification of theorem 18.5.3 becomes a geometric object rather than a cohomology set.

The quotient stack repairs the central defect diagnosed in chapter 19's opening and in the $\mathcal{M}_{1,1}$ thread of Chapter 1: when G acts with stabilizers, the naive orbit space X/G forgets the automorphisms of the orbits and fails to represent the moduli problem. The stack $[X/G]$ keeps that information. Its coarse space, when one exists, is the good quotient $X/\!\!/G$ of geometric invariant theory (definition 15.1.1), and the two agree exactly where the action is free. The section closes by exhibiting $\mathcal{M}_{1,1}$ as a quotient stack in the Weierstrass presentation, recovering the j-line (theorem 13.3.6) as its coarse space and locating the generic μ_2 and special μ_4, μ_6 automorphisms of Chapter 1 inside the stack.

Throughout, $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$ and G is a flat, finitely presented, affine group scheme over S acting on an S-scheme X by a morphism $\sigma\colon G \times_S X \to X$. A G-torsor is as in definition 18.5.1: an S-scheme P with a G-action and a G-invariant map $P \to T$ that is fppf-locally on T isomorphic to the trivial torsor $G \times_S T$ with its translation action. The action being fixed, all torsors are for this same G.

The definition of $[X/G]$ is dictated by descent. If X/G were a fine moduli space, a T-point of it would be a map $T \to X/G$, that is, an orbit varying over T. Locally on T such an orbit is cut out by choosing a point of X in it, but the choice is ambiguous up to the G-action, and globally the ambiguities are recorded by a torsor. A T-point of the quotient stack is therefore the intrinsic version of "a G-orbit's worth of maps to X": a principal G-bundle $P \to T$ together with a G-equivariant map $P \to X$. When the torsor is trivial, $P = G \times_S T$, an equivariant map $G \times_S T \to X$ is the same as an ordinary map $T \to X$ (evaluate at the identity section), so the trivial-torsor points recover ordinary T-points of X; the general points twist this by a torsor.

Definition 19.3.1: The Quotient Stack $[X/G]$

Let G be a flat, finitely presented, affine group scheme over S acting on an S-scheme X. The *quotient stack* $[X/G]$ is the category fibered in groupoids over $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$ whose fiber over an S-scheme T is the groupoid $[X/G](T)$ with:

- *objects* the pairs $(P \xrightarrow{\pi} T,\ P \xrightarrow{f} X)$ where $\pi\colon P \to T$ is a G-torsor and $f\colon P \to X$ is a G-equivariant S-morphism (so $f(g \cdot p) = g \cdot f(p)$);
- *morphisms* $(P, f) \to (P', f')$ the isomorphisms of G-torsors $\alpha\colon P \xrightarrow{\sim} P'$ over T that commute with the maps to X, that is, $f' \circ \alpha = f$.

A morphism $h\colon T' \to T$ in $\mathcal{C}$ acts by pullback: the object (P, f) over T is sent to $(h^*P, f \circ \mathrm{pr}_P)$ over T', where $h^*P = P \times_T T'$ is the pulled-back torsor and $\mathrm{pr}_P\colon h^*P \to P$ the projection. This assignment makes $[X/G]$ fibered in groupoids, with the pullback square exhibiting each such h^*-arrow as cartesian.

The morphisms are worth dwelling on, because they are the entire reason $[X/G]$ is a stack and not a space. An object (P, f) over T has automorphisms: an automorphism is a torsor automorphism $\alpha\colon P \to P$ over T with $f \circ \alpha = f$. When $P = G \times_S T$ is trivial and f corresponds to a point $x\colon T \to X$, such an α is right-translation by a T-point g of G, and the condition $f \circ \alpha = f$ says g fixes x, that is, g lies in the stabilizer $\mathrm{Stab}_G(x)$. Thus the automorphism group of a point of $[X/G]$ is its stabilizer under the action. This is exactly the datum the orbit space X/G discards, and its retention is what lets $[X/G]$ see the difference between a free action (all stabilizers trivial) and one with fixed points.

The pullback of torsors is the cartesian structure, and it is functorial because fiber products are: $(h \circ h')^*P \cong h'^*(h^*P)$ canonically, so the pullbacks assemble into a pseudofunctor $\mathcal{C}^{\mathrm{op}} \to \mathbf{Grpd}$, $T \mapsto [X/G](T)$, in the sense of definition 19.1.4. That $[X/G]$ is fibered in groupoids is immediate from the definition of its morphisms: every arrow is a torsor isomorphism, hence invertible, so each fiber $[X/G](T)$ is a groupoid.

The first worked instance is the one that names the construction: taking X to be a single point produces a stack whose points over T are torsors with no further structure, since an equivariant map to a point is no constraint at all.

Definition 19.3.2: The Classifying Stack BG

Let G be a flat, finitely presented, affine group scheme over S. The *classifying stack* of G is the quotient stack

$$BG = [S/G] = [\mathrm{pt}/G]$$

of G acting (necessarily trivially) on the terminal object $S = \mathrm{pt}$. Concretely, the fiber $BG(T)$ is the groupoid of G-torsors on T: an object over T is a G-torsor $P \to T$, a morphism is an isomorphism of G-torsors over T, and a map $h\colon T' \to T$ acts by pullback of torsors. There is a distinguished object over S, the *trivial* torsor $G \to S$; the induced map $\mathrm{pt} = S \to BG$ it represents is the *universal torsor*.

The classifying stack is the geometric form of the torsor classification proved in theorem 18.5.3. There, G-torsors on T (up to isomorphism) were shown to be classified by the pointed cohomology set $H^1_{\mathrm{fppf}}(T, G)$, or $H^1_{\text{ét}}(T, G)$ for smooth G. The stack BG upgrades this classification from a *set* of isomorphism classes to a *groupoid* that remembers the isomorphisms themselves. The 2-Yoneda lemma (theorem 19.1.8) makes this precise: for any S-scheme T,

$$\underline{\mathrm{Hom}}_{\mathcal{C}}(\underline{T}, BG) \simeq BG(T) = \{G\text{-torsors on } T\}$$

an equivalence of groupoids. Passing to isomorphism classes on both sides recovers $\pi_0\big(\underline{\mathrm{Hom}}(\underline{T}, BG)\big) \cong H^1(T, G)$: the set of T-points of BG, up to 2-isomorphism, is the first cohomology of T with coefficients in G. The classifying stack is the object that represents the functor $T \mapsto H^1(T, G)$, refined to a groupoid so that the automorphisms $\underline{\mathrm{Aut}}(P) = G$ of the trivial torsor are visible; the automorphism group of the universal point $\mathrm{pt} \to BG$ is G itself. This is the sense in which BG deserves the name "classifying": it is the moduli stack of G-torsors, and a map into it is a torsor.

Before proving that these constructions are stacks, it is worth naming the universal torsor in the general case, because it is the structure that makes $[X/G]$ deserve to be called a quotient. Over X itself there is a tautological object of $[X/G]$: the trivial torsor $G \times_S X \to X$ equipped with the action map $\sigma\colon G \times_S X \to X$ as its equivariant map to X. This object is classified by a morphism $q\colon X \to [X/G]$, and the theorem below records that q is itself a G-torsor, the universal one from which every object of $[X/G]$ is pulled back.

Everything now rests on a single assertion, that $[X/G]$ satisfies descent, and that assertion is not new work: it is the descent already proved in Chapter 6, repackaged. A quotient-stack object over T is a torsor with an equivariant map, and both torsors and morphisms of torsors were shown to glue in the fppf topology. The stack axioms for $[X/G]$ are the sheaf condition for isomorphisms and the effectivity of descent for objects; the first is the statement that torsor isomorphisms glue, the second that torsor-with-map data glue, and both follow from theorems 18.3.4 and 18.5.3. The strict verification of the pseudofunctor coherence (that the chosen pullbacks compose associatively up to the coherence 2-isomorphisms) is bookkeeping, and is cited to [66].

Theorem 19.3.3: The Quotient Stack and Its Universal Torsor

Let G be a flat, finitely presented, affine group scheme over S acting on an S-scheme X. Then:

1. $[X/G]$ is a stack for the fppf topology on $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$.

2. The tautological object $(G \times_S X \to X,\ \sigma)$ defines a morphism $q\colon X \to [X/G]$, and q is a G-torsor: for every T and every object $(P, f) \in [X/G](T)$, the fiber product $X \times_{[X/G]} T$ is canonically isomorphic to P as a G-torsor over T. In particular q is the *universal G-torsor*: giving a G-torsor $P \to T$ together with a G-equivariant map $P \to X$ is the same as giving a morphism $T \to [X/G]$.

3. If a good quotient $X/\!\!/G$ in the sense of definition 15.1.1 exists, there is a canonical morphism
$$\pi\colon [X/G] \longrightarrow X/\!\!/G$$
which is universal among maps from $[X/G]$ to schemes, and which is an isomorphism precisely over the open locus of $X/\!\!/G$ above which G acts with trivial stabilizers.

***Proof*:**
The three parts are proved in turn, the first by reducing to Chapter 6 descent, the second by unwinding the 2-fiber product against the 2-Yoneda lemma, and the third by the universal property of the good quotient.

Step 1: $[X/G]$ is a stack. By definition 19.2.2, two conditions must hold: for objects $(P,f),(P',f') \in [X/G](T)$ the presheaf $\underline{\mathrm{Isom}}\big((P,f),(P',f')\big)$ on $\mathcal{C}/T$ is an fppf sheaf, and every descent datum on every fppf cover is effective.

For the isomorphism sheaf, fix a cover $\{T_i \to T\}$. An isomorphism $(P,f) \to (P',f')$ is a torsor isomorphism $\alpha\colon P \to P'$ over T with $f' \circ \alpha = f$. Torsor isomorphisms form an fppf sheaf: the functor $T' \mapsto \mathrm{Isom}_{T'}(P|_{T'}, P'|_{T'})$ is a torsor under the sheaf $\underline{\mathrm{Aut}}(P) \cong G$ (twisted), and torsors are fppf sheaves by theorem 18.5.3; equivalently, an isomorphism of torsors is a section of the fppf-sheaf $\underline{\mathrm{Isom}}(P,P')$, so compatible local isomorphisms α_i agreeing on overlaps glue to a global α. The compatibility with f, f' is a closed condition that glues because it holds on the members of the cover and X is a sheaf. Hence $\underline{\mathrm{Isom}}$ is a sheaf and $[X/G]$ is a prestack.

For effectivity, let $\{T_i \to T\}$ be an fppf cover and let a descent datum be given: objects $(P_i, f_i) \in [X/G](T_i)$ together with isomorphisms $\varphi_{ij}\colon (P_i,f_i)|_{T_{ij}} \to (P_j,f_j)|_{T_{ij}}$ over the double overlaps $T_{ij} = T_i \times_T T_j$, satisfying the cocycle condition on triple overlaps. Forgetting the maps f_i, the torsors P_i with their φ_{ij} form a descent datum for G-torsors, which is effective: G-torsors form a stack because they are locally trivial and torsor isomorphisms glue, so the P_i glue to a G-torsor $P \to T$ with $P|_{T_i} \cong P_i$. This is the content of theorem 18.5.3 read as descent (a torsor is an fppf-sheaf locally isomorphic to G, and such sheaves glue). The equivariant maps $f_i\colon P_i \to X$ agree on overlaps under the φ_{ij} by hypothesis, so they define compatible maps $P|_{T_i} \to X$; since X is a sheaf and $\{P|_{T_i} \to P\}$ is an fppf cover of P, these glue to a G-equivariant map $f\colon P \to X$ (equivariance is checked after pulling back to the cover, where it holds). The pair (P,f) is an object of $[X/G](T)$ restricting to (P_i,f_i), so the descent datum is effective. The coherence of the pseudofunctor structure is verified in [66]. Thus $[X/G]$ is a stack.

Step 2: q is the universal torsor. The tautological object $t_X = (G \times_S X \xrightarrow{\mathrm{pr}_X} X,\ \sigma)$ over X is a G-torsor (the trivial one on X) with equivariant map σ, hence an object of $[X/G](X)$; by 2-Yoneda (theorem 19.1.8) it corresponds to a morphism $q\colon X \to [X/G]$. To identify the fiber product, fix an object $(P,f) \in [X/G](T)$, corresponding to a morphism $\xi\colon T \to [X/G]$. By the universal property of the 2-fiber product, an object of $\big(X \times_{[X/G]} T\big)(T')$ over a T-scheme T' is a triple consisting of a T'-point $a\colon T' \to X$, the given point $\xi|_{T'}$, and a 2-isomorphism $q \circ a \cong \xi|_{T'}$ in $[X/G](T')$. Unwinding the definitions, $q \circ a$ is the trivial torsor $G \times_S T' \to T'$ with equivariant map $g \mapsto g \cdot a$, and a 2-isomorphism to $(P,f)|_{T'}$ is an isomorphism of torsors $G \times_S T' \xrightarrow{\sim} P|_{T'}$ commuting with the maps to X. Such an isomorphism is the choice of a section of $P|_{T'} \to T'$ (the image of the identity), and it determines a uniquely by $a = f \circ (\text{section})$. Therefore

$$\big(X \times_{[X/G]} T\big)(T') = \{\text{sections of } P|_{T'} \to T'\} = \mathrm{Hom}_T(T', P)$$

naturally in T', which says $X \times_{[X/G]} T \cong P$ as T-schemes, and the identification is G-equivariant. Hence the base change of q along any $\xi\colon T \to [X/G]$ is the torsor P; taking ξ to be the identity of $[X/G]$ (formally, working over the stack itself) exhibits q as a G-torsor over $[X/G]$. That every object of $[X/G]$ arises as a pullback of t_X is the same computation read backwards: an object (P,f) over T is $\xi^* t_X$ for the classifying map $\xi = \xi_{(P,f)}$, since its fiber product with q returns P with f. This is the universal property claimed.

Step 3: The map to the good quotient. Suppose $Y = X /\!\!/ G$ is a good quotient, so Y carries a G-invariant morphism $p\colon X \to Y$ that is universal among G-invariant maps to schemes and identifies Y with the ring of invariants locally (definition 15.1.1). To produce $\pi\colon [X/G] \to Y$ it suffices, by 2-Yoneda, to give for each object $(P, f) \in [X/G](T)$ a morphism $T \to Y$, functorially. The composite $p \circ f\colon P \to Y$ is G-invariant, because f is equivariant and p is invariant: $p(f(g \cdot x)) = p(g \cdot f(x)) = p(f(x))$. A G-invariant map out of a G-torsor $P \to T$ factors uniquely through T, since $P \to T$ is an fppf-local trivial torsor and a G-invariant map out of $G \times_S T$ factors through the projection to T; the factorization glues by descent (theorem 18.3.4). This produces $\bar{f}\colon T \to Y$ with $\bar{f} \circ \pi = p \circ f$, and the assignment $(P, f) \mapsto \bar{f}$ is functorial in T, hence defines $\pi\colon [X/G] \to Y$. The tautological object over X is sent to $p\colon X \to Y$, so $\pi \circ q = p$.

Universality: given any scheme Z and a morphism $[X/G] \to Z$, precompose with $q\colon X \to [X/G]$ to get a G-invariant map $X \to Z$ (invariant because $q \circ \sigma$ agrees with $q \circ \mathrm{pr}_X$ up to the tautological 2-isomorphism, so the two ways of mapping $G \times_S X$ to Z coincide), which factors uniquely through $Y = X /\!\!/ G$ by the universal property of the good quotient; the induced $Y \to Z$ composes with π to give back the original map, and uniqueness follows because q is an fppf cover and maps out of a stack are determined by their pullback along an fppf cover. Thus π is universal among maps from $[X/G]$ to schemes, that is, Y is the coarse space of $[X/G]$ in the categorical sense.

Finally, π is an isomorphism exactly over the free locus. On the open $U \subseteq Y$ over which G acts with trivial stabilizers, the action on $p^{-1}(U)$ is free with quotient a scheme (a good quotient of a free action is geometric, and $p^{-1}(U) \to U$ is then a G-torsor by definition 15.1.1), so $[p^{-1}(U)/G] \cong U$: a torsor $P \to T$ with equivariant map to $p^{-1}(U)$ is, since the action downstairs is free, the pullback of $p^{-1}(U) \to U$ along the induced $T \to U$, giving an equivalence $[X/G]|_U \simeq U$. Conversely, at a point of X with nontrivial stabilizer H, the fiber of π over its image contains a copy of BH (the automorphisms of that point form the group H, as computed after definition 19.3.1), which is not a scheme unless H is trivial, so π is not an isomorphism there. Hence π is an isomorphism precisely over the trivial-stabilizer locus, completing the proof.

The theorem is the structural payoff of the section, and its three parts say three complementary things. Part (1) is the promise kept: $[X/G]$ is a genuine stack, and its proof consumed no new descent, only the descent of Chapter 6 applied to torsors-with-a-map. Part (2) is the universal property that justifies the notation: $[X/G]$ is the target that turns a G-equivariant family into a map, exactly as a fine moduli space turns a family into a map to it, with the torsor absorbing the ambiguity that a scheme could not. The map $q\colon X \to [X/G]$ is a G-torsor, so X is a "cover" of the stack in the same sense that $\operatorname{Spec} L$ covers $\operatorname{Spec} k$ in Galois descent; this presentation $X \to [X/G]$ is what will make $[X/G]$ algebraic in Chapter 8, where an atlas is exactly such a smooth cover by a scheme. Part (3) situates the stack above its coarse space: the good quotient $X /\!\!/ G$ of geometric invariant theory (proposition 13.4.6) is the best scheme approximation to $[X/G]$, and the two coincide only where the action is free. Where stabilizers jump, the stack carries a residual BH that the coarse space cannot see. This is the precise mechanism by which the automorphism obstruction of Chapter 1 survives into the stack and is resolved there rather than lost.

One reading of part (3) deserves emphasis, because it reorganizes the whole of geometric invariant theory. GIT constructs $X /\!\!/ G$ by hand, through invariant sections of a linearized line bundle (definition 15.2.1), and the construction is delicate precisely because it must decide which orbits to keep (the semistable ones) and how to separate them. The quotient stack $[X^{\mathrm{ss}}/G]$ is the same data assembled without those choices: it exists for any action, and GIT is the statement that its coarse space is a projective scheme when the linearization is ample. The stack is the primary object and the projective GIT quotient is its coarse space, obtained by descending to schemes at the cost of forgetting stabilizers.

The construction acquires its full meaning through examples, and the richest cluster is the classifying stacks BG for the groups that appear in algebraic geometry. Each BG is a moduli stack of a familiar structure, because a G-torsor is a familiar structure in disguise: for $G = \mathrm{GL}_n$ a torsor is a rank-n vector bundle, for $G = \mathbb{G}_m$ a line bundle, for $G = \mu_n$ an n-torsion line bundle with a chosen trivialization of its n-th power. The classifying stacks are therefore the universal moduli of these bundles, and computing them concretely turns the abstract H^1-classification into geometry.

Example 19.3.4: Classifying Stacks of Line and Vector Bundles
Fix the base $S = \operatorname{Spec} k$ and the fppf topology.

1. *$B\mathbb{G}_m$ and line bundles.* A $\mathbb{G}_m$-torsor on T is the complement of the zero section of a line bundle, and the assignment (line bundle) $\mapsto$ (its associated $\mathbb{G}_m$-torsor of frames) is an equivalence between line bundles on T and $\mathbb{G}_m$-torsors on T. Hence

$$B\mathbb{G}_m(T) \simeq \{\text{line bundles on } T\}$$

as groupoids, with morphisms the bundle isomorphisms. The isomorphism classes are $\operatorname{Pic}(T) = H^1(T, \mathbb{G}_m)$, matching theorem 18.5.3. The automorphism group of a line bundle is $\mathbb{G}_m$ (multiplication by a unit), so every point of $B\mathbb{G}_m$ has automorphism group $\mathbb{G}_m$; the stack is a gerbe over a point in the sense to be developed in Chapter 9. The universal object is the tautological line bundle on $B\mathbb{G}_m$.

2. *BGL_n and vector bundles.* A GL_n-torsor on T is the frame bundle of a rank-n vector bundle, and again the associated-bundle construction gives an equivalence

$$BGL_n(T) \simeq \{\text{rank-}n \text{ vector bundles on } T\}$$

so BGL_n is the moduli stack of rank-n vector bundles, with a point's automorphism group the automorphisms GL_n of the trivial bundle. The determinant homomorphism $\det\colon \mathrm{GL}_n \to \mathbb{G}_m$ induces a morphism $BGL_n \to B\mathbb{G}_m$ sending a bundle to its determinant line bundle.

3. *$B\mu_n$ and torsion line bundles.* A μ_n-torsor on T is, by the Kummer sequence $1 \to \mu_n \to \mathbb{G}_m \xrightarrow{n} \mathbb{G}_m \to 1$, a line bundle L together with a trivialization of $L^{\otimes n}$. Thus $B\mu_n(T)$ is the groupoid of such pairs, and its isomorphism classes fit into

$$0 \to \mathcal{O}(T)^\times/(\mathcal{O}(T)^\times)^n \to H^1(T, \mu_n) \to \operatorname{Pic}(T)[n] \to 0$$

the Kummer sequence in cohomology. The automorphism group of the trivial μ_n-torsor is μ_n.

The three classifying stacks are the same construction viewed through three groups, and the pattern is uniform: BG is the moduli stack of the geometric objects that G-torsors encode, its isomorphism classes are $H^1(-, G)$, and the automorphism group of every point is G. The maps between them ($\det\colon BGL_n \to B\mathbb{G}_m$, or the Kummer inclusion $B\mu_n \to B\mathbb{G}_m$) are induced by homomorphisms of groups, which is the general principle that a group homomorphism $G \to H$ induces a morphism of classifying stacks $BG \to BH$ by pushing torsors forward. These stacks are not schemes, and not even algebraic spaces, precisely because their points have nontrivial automorphisms; they are the simplest stacks that are not spaces, and every quotient stack is built by mixing a BG of this kind into the geometry of X.

The example the book has been building toward is the moduli of elliptic curves. In Chapter 1 the functor $\mathcal{M}_{1,1}$ failed to be representable by a scheme because every elliptic curve has the automorphism $[-1]$, and special curves have more (example 13.3.4); the best a scheme could do was the coarse j-line (theorem 13.3.6). The quotient-stack construction resolves this: $\mathcal{M}_{1,1}$ is a quotient stack, and its coarse space is exactly the j-line, with the automorphisms sitting in the fibers.

Example 19.3.5: $\mathcal{M}_{1,1}$ as a Quotient Stack

Work over $S = \operatorname{Spec} k$ with char $k \neq 2, 3$. Every elliptic curve over a k-scheme T admits, fppf-locally on T, a Weierstrass equation

$$y^2 = x^3 + ax + b$$

with discriminant $\Delta = -16(4a^3 + 27b^2)$ invertible. Two such equations define isomorphic curves if and only if they are related by the change of variables $(x, y) \mapsto (\lambda^2 x, \lambda^3 y)$ for a unit $\lambda \in \mathbb{G}_m$, under which $(a, b) \mapsto (\lambda^4 a, \lambda^6 b)$. Hence the moduli stack is the quotient of the parameter space by this $\mathbb{G}_m$-action with weights $(4, 6)$:

$$\mathcal{M}_{1,1} \simeq \left[\left(\mathbb{A}^2_{a,b} \setminus \{\Delta = 0\}\right)/\mathbb{G}_m\right]$$

where $\lambda \cdot (a, b) = (\lambda^4 a, \lambda^6 b)$. An object over T is a $\mathbb{G}_m$-torsor $P \to T$ (a line bundle minus its zero section) together with a $\mathbb{G}_m$-equivariant map $P \to \mathbb{A}^2 \setminus \{\Delta = 0\}$; unwinding the weights, this is exactly a line bundle $\mathcal{L}$ on T (the "Hodge line bundle") with sections $a \in \mathcal{L}^{\otimes 4}$, $b \in \mathcal{L}^{\otimes 6}$ whose discriminant is nowhere zero, which is the twisted Weierstrass data of an elliptic curve over T.

The coarse space is the good quotient. After inverting Δ, the ring of invariants of the $\mathbb{G}_m$-action on $k[a, b]$ (weights $4, 6$) is generated by the single weight-0 element $j = 1728 \cdot \frac{4a^3}{4a^3 + 27b^2}$ (equivalently by a^3/Δ and b^2/Δ with one relation), and the good quotient of $\mathbb{A}^2 \setminus \{\Delta = 0\}$ is the affine j-line $\mathbb{A}^1_j$. By theorem 19.3.3(3) and proposition 13.4.6 the coarse space of $\mathcal{M}_{1,1}$ is therefore the j-line $\mathbb{A}^1_j = \operatorname{Spec} k[j]$, recovering theorem 13.3.6.

The automorphisms are the stabilizers. A point (a, b) has $\mathbb{G}_m$-stabilizer $\{\lambda : \lambda^4 a = a,\ \lambda^6 b = b\}$:

- generically (a, b both nonzero, so $j \neq 0, 1728$) the stabilizer is $\mu_2 = \{\pm 1\}$, matching the automorphism $[-1]$ of a generic elliptic curve;
- at $b = 0$ ($j = 1728$) the condition is $\lambda^4 = 1$, so the stabilizer is μ_4, matching the extra automorphism of the curve $y^2 = x^3 + ax$;
- at $a = 0$ ($j = 0$) the condition is $\lambda^6 = 1$, so the stabilizer is μ_6, matching the curve $y^2 = x^3 + b$.

These are precisely the automorphism groups recorded in example 13.3.4. Because μ_2 acts trivially on the whole parameter space (every (a, b) has ± 1 in its stabilizer), $\mathcal{M}_{1,1}$ carries a generic μ_2 in the automorphism group of every point: it is a μ_2-gerbe over its coarse space away from $j = 0, 1728$, and the residual gerbe of the stack point over $j = 1728$ (resp. $j = 0$) is $B\mu_4$ (resp. $B\mu_6$). The systematic study of these residual gerbes and of the gerbe structure of $\mathcal{M}_{1,1}$ over the j-line is the subject of Chapter 9.

The elliptic-curve example returns to Chapter 1. There, the automorphism $[-1]$ was an obstruction to fine representability, and the j-line was a lossy consolation prize that identified curves related by an isomorphism but forgot that each carries automorphisms. The quotient stack $\mathcal{M}_{1,1} = [(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ loses nothing: it is a fine moduli object in the 2-categorical sense, its T-points are exactly families of elliptic curves over T, and the automorphisms that obstructed a scheme now live visibly in the stabilizers. The j-line reappears as its coarse space, and the special automorphisms at $j = 0$ and $j = 1728$ are the points where the stabilizer jumps from μ_2 to μ_6 and μ_4. What was a defect of the theory is now a feature of the object: the stack is the moduli space, and the automorphisms are part of its geometry rather than an obstruction to its existence.

The construction is uniform across the running examples of the book. The Hilbert scheme of Chapter 2 was a fine moduli space because its objects (closed subschemes) have no automorphisms; there the stack $\underline{\mathrm{Hilb}}$ is representable, a scheme viewed as a stack with trivial stabilizers. The moduli of curves is the opposite extreme, where every object has automorphisms and only a stack will do. The quotient stack is the general mechanism spanning the two: it is representable exactly where the action is free, and it acquires nontrivial stabilizers exactly where the objects it parametrizes acquire automorphisms.

Exercise 19.3.1

1. Let G act on X, and let $(P, f) \in [X/G](T)$ be an object over T. Show that its automorphism group in the groupoid $[X/G](T)$ is the group of T-points of the stabilizer group scheme along f: when $P = G \times_S T$ is trivial and f corresponds to $x\colon T \to X$, this automorphism group is $\mathrm{Stab}_G(x)(T)$. Deduce that $[X/G] \to X/\!\!/G$ is an isomorphism near a point if and only if the stabilizer there is trivial.

2. Verify from definition 19.3.2 that a group homomorphism $\rho\colon G \to H$ of S-group schemes induces a morphism of stacks $B\rho\colon BG \to BH$, given on T-points by pushing a G-torsor P forward to the H-torsor $P \times^G H = (P \times_S H)/G$. Check that $B\rho$ sends the universal G-torsor to the pushforward of the trivial torsor, and that $B(\rho' \circ \rho) = B\rho' \circ B\rho$. Identify the determinant morphism $BGL_n \to B\mathbb{G}_m$ of example 19.3.4 as $B(\det)$.

3. Compute the ring of $\mathbb{G}_m$-invariants of $k[a, b]$ under the weight-$(4, 6)$ action $\lambda \cdot (a, b) = (\lambda^4 a, \lambda^6 b)$, and confirm that inverting $\Delta = -16(4a^3 + 27b^2)$ produces a ring whose spectrum is the affine j-line, with $j = 1728 \cdot \frac{4a^3}{4a^3+27b^2}$. (Compute the invariant monomials $a^i b^j$ with $4i + 6j = 0$, note there are none of positive degree, then pass to the localization at Δ where negative-weight invariants appear.) Explain in one sentence why this makes the coarse space of $\mathcal{M}_{1,1}$ the j-line.

4. Show that when G acts *freely* on X (trivial stabilizers everywhere) and a geometric quotient X/G exists as a scheme with $X \to X/G$ a G-torsor, the quotient stack $[X/G]$ is representable by the scheme X/G, that is, the canonical map $[X/G] \to X/G$ is an equivalence. (Use part (2) of theorem 19.3.3: a T-point of $[X/G]$ is a torsor with equivariant map to X, and freeness lets you descend it to a map $T \to X/G$.)

5. Let G be a finite group (constant group scheme) acting on a scheme X over k. Describe the objects and morphisms of $[X/G](T)$ concretely for $T = \operatorname{Spec} k$ with k algebraically closed: show that a k-point of $[X/G]$ is a point $x \in X(k)$ and that its automorphism group is the set-theoretic stabilizer $\mathrm{Stab}_G(x) \subseteq G$. How many isomorphism classes of k-points lie over a given G-orbit, and how does this compare with the single point that orbit contributes to X/G?

6. The naive presheaf quotient $(\underline{X}/G)^{\mathrm{pre}}$ assigns to T the set $X(T)/G(T)$ of orbits of T-points. Explain why the stackification of $\underline{X}/G$ (as a prestack of sets, viewed as a CFG with trivial automorphisms) is *not* in general $[X/G]$, and identify what extra data the passage to $[X/G]$

records that the set-theoretic sheafification of $X(T)/G(T)$ cannot. (Contrast the case of a free action, where they agree, with $BG = [\mathrm{pt}/G]$, where the naive quotient of $\mathrm{pt}(T)/G(T) = \mathrm{pt}$ is a point but BG is not.)

19.4 Representable Morphisms of Stacks

The previous sections built stacks as geometric objects in their own right: the quotient stack $[X/G]$ and the classifying stack BG are stacks in the fppf topology, and a scheme X sits among them through its representable stack $\underline{X}$. What is missing is a supply of *adjectives*. For an ordinary morphism of schemes one says it is flat, smooth, proper, or surjective, and these words carry the geometry. A morphism of stacks $f\colon \mathcal{F} \to \mathcal{G}$ has no underlying continuous map of topological spaces to test, so none of these adjectives has an immediate meaning. The device that repairs this, and that turns out to be the organizing principle for the whole geometric theory of stacks, is representability: a morphism f is understood by pulling it back along every map from a scheme and asking what happens over that scheme, where the classical adjectives *do* make sense.

This section isolates the one class of morphisms for which the transfer of geometric language is unproblematic, the representable morphisms, and shows how a property P of scheme morphisms is transported to them by base change. The payoff is the diagonal. A single morphism, the diagonal $\Delta_{\mathcal{F}}\colon \mathcal{F} \to \mathcal{F} \times \mathcal{F}$, packages all the automorphism data of a stack into the Isom-sheaves of section 19.2, and its representability is the first of the two conditions that will make a stack *algebraic* in the next chapter. Everything here is bookkeeping in the 2-category of stacks, but it is the bookkeeping without which no geometry on a stack can even be stated.

Throughout, $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$ and all stacks are stacks over $\mathcal{C}$; "scheme" means a scheme over S, identified with its representable stack $\underline{T}$ via the 2-Yoneda lemma (theorem 19.1.8). The 2-fiber product $\mathcal{F} \times_{\mathcal{G}} \mathcal{H}$ of stacks is the 2-categorical fiber product: its objects over T are triples (x, y, α) with $x \in \mathcal{F}(T)$, $y \in \mathcal{H}(T)$, and α an isomorphism in $\mathcal{G}(T)$ between the images of x and y. It is again a stack, and it enjoys the universal property of a fiber product up to canonical 2-isomorphism; the construction is imported from [9] and used freely.

The definition begins with the observation that some morphisms of stacks have schematic fibers. When they do, every geometric question about the morphism can be pushed down to a question about schemes.

Definition 19.4.1: Representable Morphism of Stacks

Let $f\colon \mathcal{F} \to \mathcal{G}$ be a morphism of stacks over $\mathcal{C}$. The morphism f is *representable by schemes* (or simply *representable*) if for every scheme T and every morphism $t\colon \underline{T} \to \mathcal{G}$ the 2-fiber product

$$\mathcal{F} \times_{\mathcal{G}} \underline{T}$$

is representable by a scheme, that is, equivalent as a stack to $\underline{T'}$ for some scheme T'. The morphism f is *representable by algebraic spaces* if, under the same hypotheses, $\mathcal{F} \times_{\mathcal{G}} \underline{T}$ is (equivalent to) an algebraic space.

The definition tests f against all schemes mapping to the target $\mathcal{G}$ and demands that each fiber over a scheme be a scheme. The word "fiber" is meant in the 2-categorical sense: the objects of $\mathcal{F} \times_{\mathcal{G}} \underline{T}$ over a T-scheme U are pairs consisting of an object $x \in \mathcal{F}(U)$ and an isomorphism in $\mathcal{G}(U)$ between

$f(x)$ and the image of $U \to T$. When this stack is a scheme, it has no nontrivial automorphisms in any fiber, so a representable morphism is exactly one whose fibers over schemes have trivialized their automorphisms. This is the precise sense in which representable morphisms are the "schematic" part of stack theory: they are the morphisms along which a scheme, pulled back, stays a scheme. The two versions of the definition differ only in what one is willing to accept as a fiber. Representability by algebraic spaces is the more flexible notion and is the one that survives in full generality, because algebraic spaces are closed under the quotient operations that produce stacks, whereas schemes are not (this is the lesson of the failure of effective descent for non-affine schemes, section 18.3); until algebraic spaces are constructed in the next chapter, the reader may keep the schematic version in mind, as every example below is representable by schemes.

A first sanity check: a morphism $\underline{X} \to \underline{Y}$ of representable stacks is representable, because for any $T \to \underline{Y}$ the fiber product $\underline{X} \times_{\underline{Y}} \underline{T}$ is $\underline{X \times_Y T}$, the representable stack of the scheme fiber product $X \times_Y T$, which exists as a scheme. So the notion extends the category of schemes without disturbing it. The content of the definition is entirely in the morphisms with stacky source or target, where it is a real restriction.

Example 19.4.2: A Morphism to BG from a Scheme

Let G be an affine group scheme over S and $BG = [S/G]$ the classifying stack (definition 19.3.2). Consider the structure morphism $\pi\colon BG \to \underline{S}$, and test its representability against a scheme $T \to \underline{S}$, that is, an S-scheme T. By definition of the 2-fiber product,

$$BG \times_{\underline{S}} \underline{T}$$

has, over a T-scheme U, the objects (P, θ) where P is a G-torsor on U (an object of $BG(U)$) and θ is an isomorphism in $\underline{S}(U)$ between $\pi(P)$ and the given point, which is no data at all since $\underline{S}(U)$ is discrete. Thus the fiber is again BG restricted to T-schemes, that is, $BG_T = [T/G]$, and this is a scheme only when it has no automorphisms, that is, only when G is trivial. For G nontrivial, $\pi\colon BG \to \underline{S}$ is *not* representable: the torsor P carries its automorphism group $\mathrm{Aut}(P) = G(U)$, which no scheme fiber can absorb. This is the prototype of a non-representable morphism, and it will be diagnosed precisely by the diagonal below: the automorphisms that obstruct representability of π are exactly the ones the diagonal of BG records.

The example already shows the pattern. Non-representability of a morphism of stacks is caused by automorphisms in the fibers, and the same automorphisms are what a stack carries at each of its points. Before turning them into a single object, the section records the reason representable morphisms matter: they are precisely the morphisms to which one can attach a geometric adjective.

Suppose $f\colon \mathcal{F} \to \mathcal{G}$ is representable and P is a property of morphisms of schemes. To say "f has P" one wants to say that every scheme-fiber of f, that is, every morphism $\mathcal{F} \times_{\mathcal{G}} \underline{T} \to \underline{T}$, has P. Two things must hold for this to be well-posed. First, the property must make sense on the fibers: since each $\mathcal{F} \times_{\mathcal{G}} \underline{T}$ is a scheme, the map to $\underline{T}$ is a morphism of schemes, so any P of scheme morphisms applies. Second, and this is the subtle point, the answer must not depend on *which* scheme $T \to \mathcal{G}$ one tests against. Two different maps $T \to \mathcal{G}$ and $T' \to \mathcal{G}$ can present overlapping information, and the fibers must give a consistent verdict. Consistency is guaranteed exactly when P descends, and this is where Chapter 6 enters.

Definition 19.4.3: Property of a Representable Morphism via Base Change

Let P be a property of morphisms of schemes that is *stable under base change* (if $g\colon Y \to Z$ has P, so does every base change $Y \times_Z Z' \to Z'$) and *fppf-local on the target* (a morphism $g\colon Y \to Z$ has P if and only if, for some fppf cover $\{Z_i \to Z\}$, each base change $Y \times_Z Z_i \to Z_i$ has P). Standard examples are: flat, smooth, étale, unramified, surjective, (locally) of finite presentation, proper, separated, quasi-compact, affine, and a closed (or open) immersion. A representable morphism $f\colon \mathcal{F} \to \mathcal{G}$ of stacks *has property* P if for every scheme T and every morphism $t\colon \underline{T} \to \mathcal{G}$ the base change

$$\mathcal{F} \times_{\mathcal{G}} \underline{T} \longrightarrow \underline{T}$$

of schemes has P.

The two hypotheses on P are exactly what make the definition consistent, and the mechanism is descent. The next proposition is where the whole apparatus of Chapter 6 pays off: it says that stability under base change plus fppf-locality is not only convenient but is the precise condition under which the base-change definition is independent of choices and can be checked on a single well-chosen test scheme rather than on all of them.

Proposition 19.4.4: Well-Definedness of Properties via Descent

Let P be a property of scheme morphisms that is stable under base change and fppf-local on the target, and let $f\colon \mathcal{F} \to \mathcal{G}$ be a representable morphism of stacks. Then:

1. If $\{U_i \to \underline{T}\}$ is any fppf cover refining a test $t\colon \underline{T} \to \mathcal{G}$, then $\mathcal{F} \times_{\mathcal{G}} \underline{T} \to \underline{T}$ has P if and only if each $\mathcal{F} \times_{\mathcal{G}} U_i \to U_i$ has P. In particular the verdict "f has P" can be checked fppf-locally on each test scheme.
2. If $\mathcal{G} = \underline{Y}$ is representable, so that f is a morphism of schemes $X \to Y$, then f has P in the sense of definition 19.4.3 if and only if $X \to Y$ has P as a morphism of schemes.
3. If $\mathcal{G}$ admits a representable morphism $\pi\colon \underline{U} \to \mathcal{G}$ from a scheme that is an fppf cover after base change to any test scheme (an *atlas*), then f has P if and only if the single base change $\mathcal{F} \times_{\mathcal{G}} \underline{U} \to \underline{U}$ has P.

***Proof*:**
The three statements share one engine, the fppf-locality of P on the target, applied to the descent of properties from Chapter 6 (theorem 18.4.3).

1. *Fppf-local check on a test scheme.* Fix $t\colon \underline{T} \to \mathcal{G}$ and write $Z = \mathcal{F} \times_{\mathcal{G}} \underline{T}$, a scheme by representability, with structure map $g\colon Z \to T$. For an fppf cover $\{U_i \to \underline{T}\}$, the compatibility of 2-fiber products gives canonical equivalences

$$\mathcal{F} \times_{\mathcal{G}} U_i \;\simeq\; Z \times_T U_i$$

because $U_i \to \underline{T} \to \mathcal{G}$ presents U_i as a T-scheme and fiber products compose. Hence $\mathcal{F} \times_{\mathcal{G}} U_i \to U_i$ is the base change $g_i\colon Z \times_T U_i \to U_i$ of g along $U_i \to T$. Since P is fppf-local on the target, g has P if and only if each g_i has P. This is the assertion.

2. *Agreement with the scheme-theoretic property.* Suppose $\mathcal{G} = \underline{Y}$ and $\mathcal{F} = \underline{X}$, so f is (the stack associated to) a morphism $h\colon X \to Y$ of schemes. For a test $t\colon \underline{T} \to \underline{Y}$, that is, a morphism $T \to Y$, the 2-fiber product is the representable stack of the scheme fiber product,
$$\underline{X} \times_{\underline{Y}} \underline{T} \;\simeq\; \underline{X \times_Y T}$$
so its structure map to $\underline{T}$ is h base-changed along $T \to Y$. If h has P, then every such base change has P because P is stable under base change, so f has P in the sense of definition 19.4.3. Conversely, taking the tautological test $t = \mathrm{id}_Y : \underline{Y} \to \underline{Y}$ (with $T = Y$) gives the base change $\underline{X} \times_{\underline{Y}} \underline{Y} \simeq \underline{X}$ with structure map h itself, so if f has P then h has P. The two notions coincide.

3. *Checking on an atlas.* Let $\pi\colon \underline{U} \to \mathcal{G}$ be representable and fppf-surjective, and suppose the base change $g_U : \mathcal{F} \times_{\mathcal{G}} \underline{U} \to \underline{U}$ has P; the claim is that $\mathcal{F} \times_{\mathcal{G}} \underline{T} \to \underline{T}$ has P for every test $t\colon \underline{T} \to \mathcal{G}$. Form the fiber product $V = \underline{U} \times_{\mathcal{G}} \underline{T}$, which is a scheme because π is representable, and let $q\colon V \to T$ be the projection. Then q is an fppf cover of T: it is a base change of the fppf cover π, and the three properties defining an fppf cover, flat, locally of finite presentation, and surjective, are each stable under base change. Two applications of the composition of fiber products give a canonical equivalence
$$\big(\mathcal{F} \times_{\mathcal{G}} \underline{T}\big) \times_T V \;\simeq\; \mathcal{F} \times_{\mathcal{G}} V \;\simeq\; \big(\mathcal{F} \times_{\mathcal{G}} \underline{U}\big) \times_U V$$
where the last step uses $V = \underline{U} \times_{\mathcal{G}} \underline{T}$ once more to view V as a U-scheme. Under this equivalence the base change of $\mathcal{F} \times_{\mathcal{G}} \underline{T} \to \underline{T}$ along the fppf cover $q\colon V \to T$ is identified with the base change of g_U along $V \to U$. Since g_U has P and P is stable under base change, this base change has P; since P is fppf-local on the target and q is an fppf cover, $\mathcal{F} \times_{\mathcal{G}} \underline{T} \to \underline{T}$ has P, as we sought to show. The converse is immediate: if f has P then in particular the base change along the test π has P, completing the proof.

The proposition is the reason the definition is usable in practice. Part (2) confirms that the base-change definition does not conflict with the classical one on schemes: a smooth morphism of schemes is smooth as a morphism of stacks, with no shift in meaning. Part (3) is the working tool. Testing a property against *all* schemes over $\mathcal{G}$ is impossible to carry out, but once $\mathcal{G}$ has an atlas, a single representable fppf-surjective $\underline{U} \to \mathcal{G}$, the property is decided by one base change $\mathcal{F} \times_{\mathcal{G}} \underline{U} \to \underline{U}$. This is precisely how "smooth", "proper", and the rest will be defined for morphisms between algebraic stacks in the next chapter: pull back along an atlas of the target and read the property off the resulting morphism of schemes. The transport of geometric adjectives to stacks is thus not an ad hoc extension of vocabulary but a theorem about descent.

Example 19.4.5: Smoothness of the Universal Torsor $X \to [X/G]$

Let G be a smooth affine group scheme over S acting on an S-scheme X, and let $q\colon X \to [X/G]$ be the projection to the quotient stack (definition 19.3.1). The morphism q is representable: for a test $t\colon \underline{T} \to [X/G]$, corresponding to a pair $(P \to T, P \to X)$ with P a G-torsor and $P \to X$ equivariant, the fiber product $\underline{X} \times_{[X/G]} \underline{T}$ is exactly the torsor P itself, which is a scheme (an fppf-locally trivial G-torsor over a scheme is a scheme, being fppf-locally $G \times T$; when G is affine, $P \to T$ is even affine, so P is a scheme). Thus every scheme-fiber of q is a G-torsor $P \to T$, and to decide whether q is smooth, flat, or surjective it suffices to decide these for $P \to T$. A G-torsor $P \to T$ is fppf-locally on T isomorphic to the projection $G \times T \to T$ (theorem 18.5.3), so it inherits every fppf-local property of the structure map $G \to S$. Because G is smooth over S, the projection $G \times T \to T$ is smooth and surjective, hence so is every G-torsor $P \to T$, and therefore, by proposition 19.4.4(1), the morphism $q\colon X \to [X/G]$ is smooth and surjective. In particular q is an atlas for $[X/G]$: a representable, smooth, surjective morphism from a scheme. This is the concrete meaning of the slogan that $[X/G]$ is "X modulo a smooth group": the quotient map is a smooth cover in the stack sense.

The example does more than illustrate the definition; it produces the first atlas. The map $X \to [X/G]$ is representable and smooth-surjective, which is exactly the data an algebraic stack will be required to possess. For $[X/G]$ this atlas is given by construction; the definition of a general algebraic stack in the next chapter axiomatizes the existence of such an atlas, and definition 19.4.3 is what lets one read the geometry of the stack off it.

There remains one representability question that no atlas resolves on its own, and it is the one that controls whether an atlas can be used at all. When is a morphism $\underline{T} \to \mathcal{F}$ from a *scheme* representable? Equivalently, when is the fiber product of two schemes over $\mathcal{F}$ again a scheme? The answer is encoded in a single morphism of $\mathcal{F}$, its diagonal, and this is the central result of the section.

The diagonal $\Delta_{\mathcal{F}}\colon \mathcal{F} \to \mathcal{F} \times \mathcal{F}$ is the morphism whose component in each factor is the identity; on objects over T it sends $x \in \mathcal{F}(T)$ to the pair (x, x). Its role is to detect isomorphisms. Given two objects $x, y \in \mathcal{F}(T)$, presented as a single morphism $(x, y)\colon \underline{T} \to \mathcal{F} \times \mathcal{F}$, the fiber product of $\Delta_{\mathcal{F}}$ against (x, y) measures how x and y can be identified. This measurement is exactly the Isom-sheaf of section 19.2, which is why the diagonal is the object that carries all automorphism information of $\mathcal{F}$.

Theorem 19.4.6: Representability of the Diagonal

Let $\mathcal{F}$ be a stack over $\mathcal{C}$ and $\Delta_{\mathcal{F}}\colon \mathcal{F} \to \mathcal{F} \times \mathcal{F}$ its diagonal. The following are equivalent.

1. The diagonal $\Delta_{\mathcal{F}}$ is representable (by schemes, respectively by algebraic spaces).
2. For every scheme T and all objects $x, y \in \mathcal{F}(T)$, the presheaf $\underline{\mathrm{Isom}}_T(x, y)$ on $(\mathbf{Sch}/T)$, sending $U \xrightarrow{u} T$ to the set of isomorphisms $u^*x \to u^*y$ in $\mathcal{F}(U)$, is representable by a T-scheme (respectively by an algebraic space over T).

When these hold, every morphism $\underline{T} \to \mathcal{F}$ from a scheme T is representable, and for any two morphisms $a\colon \underline{T_1} \to \mathcal{F}$, $b\colon \underline{T_2} \to \mathcal{F}$ from schemes the 2-fiber product $\underline{T_1} \times_{\mathcal{F}} \underline{T_2}$ is a scheme (respectively an algebraic space).

***Proof*:**
The proof turns on a single computation: the fiber product of two objects over $\mathcal{F}$ along the diagonal is the Isom-sheaf. Everything then follows by recognizing each stated fiber product as such an Isom-sheaf.

Step 1: The diagonal fiber product is the Isom-sheaf. Fix a scheme T and two objects $x, y \in \mathcal{F}(T)$, assembled into the morphism $(x, y)\colon \underline{T} \to \mathcal{F} \times \mathcal{F}$. Form the 2-fiber product

$$I = \underline{T} \times_{(\mathcal{F} \times \mathcal{F})} \mathcal{F}$$

taken against $\Delta_{\mathcal{F}}$. By the definition of the 2-fiber product, an object of I over a T-scheme $u\colon U \to T$ is a triple $\big(u,\ z \in \mathcal{F}(U),\ \alpha\big)$ where α is an isomorphism in $(\mathcal{F} \times \mathcal{F})(U)$ between (u^*x, u^*y) and $\Delta_{\mathcal{F}}(z) = (z, z)$. Such an α is a pair of isomorphisms $\alpha_1\colon u^*x \to z$ and $\alpha_2\colon u^*y \to z$ in $\mathcal{F}(U)$. The datum (z, α_1, α_2) is equivalent, up to unique isomorphism, to the single isomorphism $\beta = \alpha_2^{-1} \circ \alpha_1\colon u^*x \to u^*y$, obtained by taking $z = u^*y$ and $\alpha_2 = \mathrm{id}$. Hence the groupoid $I(U)$

is equivalent to the *set* of isomorphisms $u^*x \to u^*y$, with no nontrivial automorphisms, so I is (equivalent to) the presheaf $\underline{\mathrm{Isom}}_T(x,y)$. Because $\mathcal{F}$ is a stack, this presheaf is an fppf sheaf: it is precisely the Isom-sheaf whose sheaf property is the prestack half of the stack condition (definition 19.2.2). Thus

$$\underline{T} \times_{(\mathcal{F}\times\mathcal{F})} \mathcal{F} \;\simeq\; \underline{\mathrm{Isom}}_T(x,y)$$

as sheaves on $(\mathbf{Sch}/T)$.

Step 2: (1) ⇔ (2). By definition, $\Delta_{\mathcal{F}}$ is representable if and only if for every scheme T and every morphism $\underline{T} \to \mathcal{F}\times\mathcal{F}$ the fiber product $\underline{T}\times_{(\mathcal{F}\times\mathcal{F})}\mathcal{F}$ is a scheme (respectively an algebraic space). A morphism $\underline{T} \to \mathcal{F} \times \mathcal{F}$ is, by the universal property of the product, the same as a pair of objects $x, y \in \mathcal{F}(T)$. By Step 1 this fiber product is $\underline{\mathrm{Isom}}_T(x,y)$. Therefore $\Delta_{\mathcal{F}}$ is representable if and only if every $\underline{\mathrm{Isom}}_T(x,y)$ is representable by a scheme (respectively an algebraic space), which is precisely condition (2).

Step 3: Representability of morphisms from a scheme. Assume (1) holds and let $a\colon \underline{T_1} \to \mathcal{F}$ and $b\colon \underline{T_2} \to \mathcal{F}$ be two morphisms from schemes, corresponding by 2-Yoneda (theorem 19.1.8) to objects $x_1 \in \mathcal{F}(T_1)$ and $x_2 \in \mathcal{F}(T_2)$. Consider the 2-fiber product $\underline{T_1} \times_{\mathcal{F}} \underline{T_2}$. Its objects over a scheme U are triples $(U \to T_1,\ U \to T_2,\ \gamma)$ with γ an isomorphism in $\mathcal{F}(U)$ between the two pullbacks of x_1 and x_2. Writing $T = T_1 \times_S T_2$ and pulling x_1, x_2 back to objects $\tilde{x}_1, \tilde{x}_2 \in \mathcal{F}(T)$ along the two projections, a morphism $U \to T_1$ and a morphism $U \to T_2$ are the same as a single morphism $U \to T$, and the isomorphism γ is an element of $\underline{\mathrm{Isom}}_T(\tilde{x}_1,\tilde{x}_2)(U)$. Hence there is a Cartesian square

$$\underline{T_1} \times_{\mathcal{F}} \underline{T_2} \;\simeq\; \underline{\mathrm{Isom}}_T(\tilde{x}_1,\tilde{x}_2)$$

of sheaves on $(\mathbf{Sch}/T)$, and the right side is representable by a scheme (respectively an algebraic space) by (2). This is exactly the assertion that the 2-fiber product $\underline{T_1} \times_{\mathcal{F}} \underline{T_2}$ is a scheme (respectively an algebraic space) for every pair of morphisms from schemes. In particular, holding a fixed, every base change of $a\colon \underline{T_1} \to \mathcal{F}$ along a morphism $\underline{T_2} \to \mathcal{F}$ from a scheme is a scheme, which is precisely representability of a, completing the proof.

The equivalence is the technical heart of stack geometry, and its meaning is worth stating plainly. The diagonal of $\mathcal{F}$ is a single morphism, but it secretly is the whole family of Isom-sheaves $\underline{\mathrm{Isom}}_T(x,y)$ ranging over all schemes and all pairs of objects. Asking that the diagonal be representable is asking that isomorphisms between objects of $\mathcal{F}$ be parametrized by a scheme (or algebraic space): the automorphisms of $\mathcal{F}$ are themselves geometric. The second conclusion of the theorem is the payoff that makes atlases work. Once the diagonal is representable, any two maps from schemes to $\mathcal{F}$ have a schematic fiber product, so a scheme mapping to $\mathcal{F}$ is a representable morphism, and the notion of an atlas $\underline{U} \to \mathcal{F}$ acquires content: one can pull the atlas back against itself to get a scheme $\underline{U} \times_{\mathcal{F}} \underline{U}$, the "relation" that presents $\mathcal{F}$ as a quotient. This is why the definition of an algebraic stack, taken up in the next chapter, has exactly two clauses: the diagonal is representable, and there exists a smooth surjective representable morphism $\underline{U} \to \mathcal{F}$ from a scheme. The first clause is theorem 19.4.6; the second is the atlas of example 19.4.5. Representability of the diagonal is logically prior, because without it the atlas cannot even be tested for the properties (smooth, surjective) that its definition demands.

19.4.7 Note: The Two Diagonals and Separatedness The diagonal $\Delta_{\mathcal{F}}$ plays for stacks the role the scheme-theoretic diagonal plays for schemes, where the diagonal $\Delta_X : X \to X \times_S X$ being a closed immersion is separatedness. For a stack whose diagonal is representable, the adjectives of definition 19.4.3 apply to $\Delta_{\mathcal{F}}$: the stack $\mathcal{F}$ is called *separated* when $\Delta_{\mathcal{F}}$ is proper, *quasi-separated* when $\Delta_{\mathcal{F}}$ is quasi-compact, and the automorphism groups are unramified, that is, finite étale in the finite-type setting (a Deligne-Mumford condition met in the next chapter), exactly when $\Delta_{\mathcal{F}}$ is unramified. This is strictly stronger than finiteness of the automorphism groups: $B\mu_p$ in characteristic p has finite automorphism groups yet a ramified diagonal (example 19.4.8(4)), so it is not Deligne-Mumford. The single morphism $\Delta_{\mathcal{F}}$ thus simultaneously encodes the separation properties of $\mathcal{F}$ and the size of its automorphism groups, because both are properties of the Isom-sheaves it represents.

The theorem is now applied to the running stacks. In each case the diagonal is computed explicitly and its fibers are identified with stabilizer or automorphism groups, making concrete the claim that the diagonal records automorphisms.

Example 19.4.8: Diagonals of BG, $[X/G]$, $B\mathbb{G}_m$, and $B\mu_n$

Each computation identifies the diagonal Δ as a morphism of stacks and reads off its fibers.

1. *The classifying stack BG.* Let G be an affine group scheme over S. Over a scheme T the objects of BG are G-torsors, and for two torsors P, Q on T the sheaf $\underline{\mathrm{Isom}}_T(P,Q)$ is the sheaf of G-equivariant isomorphisms $P \to Q$. When $P = Q$ this is the sheaf of automorphisms of a torsor, which is representable: for the trivial torsor $P = G \times T$ one has $\underline{\mathrm{Aut}}(G \times T) \cong \underline{G}_T$, the group scheme G over T acting by right translation, and for a general P the automorphism sheaf is the inner form ${}^P G = P \times^G G$ (the twist of G by P under conjugation), again a scheme because G is affine and torsors are fppf-locally trivial. Hence $\underline{\mathrm{Isom}}_T(P,Q)$ is representable (it is a torsor under $\underline{\mathrm{Aut}}(P)$, empty or an $\underline{\mathrm{Aut}}(P)$-torsor according to whether P and Q are isomorphic), so by theorem 19.4.6 the diagonal $\Delta_{BG} : BG \to BG \times BG$ is representable. In fact the diagonal fiber over the point of $BG \times BG$ given by (P,P) is exactly $\underline{\mathrm{Aut}}(P)$; over the trivial torsor this is $\underline{G}$. One summarizes: the diagonal of BG "is" G. By contrast, the structure morphism $BG \to \underline{S}$ is representable only when G is trivial (example 19.4.2); the diagonal is representable for every affine G, which is the content that makes BG an algebraic stack while $BG \to \underline{S}$ remains non-representable.

2. *The quotient stack $[X/G]$.* Let G act on X. Over T an object of $[X/G]$ is a pair $(P \to T,\ f : P \to X)$ with P a G-torsor and f equivariant, and an isomorphism of two such objects is a torsor isomorphism commuting with the maps to X. Computing the diagonal $\Delta_{[X/G]}$ as in theorem 19.4.6, its fiber over a pair of objects is the sheaf of such compatible isomorphisms, which for the two objects coming from points $x_1, x_2 \in X(T)$ (that is, the trivial torsors with the equivariant maps determined by x_1, x_2) is the transporter
$$\mathrm{Transp}_G(x_1, x_2) = \{g \in G : g \cdot x_1 = x_2\}$$
a closed subscheme of $G \times T$ cut out by the equation $g \cdot x_1 = x_2$. In particular the diagonal is represented by the morphism
$$G \times X \longrightarrow X \times X, \qquad (g, x) \longmapsto (x,\ g \cdot x)$$
the "action-and-projection" morphism: its fiber over $(x_1, x_2) \in X \times X$ is exactly $\{g : g \cdot x_1 = x_2\}$, and over the diagonal (x, x) it is the *stabilizer* $\mathrm{Stab}_G(x) = \{g : g \cdot x = x\}$. Because

$G \times X \to X \times X$ is a morphism of schemes, it is representable, so theorem 19.4.6 makes $\Delta_{[X/G]}$ representable. The stabilizers are the automorphism groups of the points of $[X/G]$, exactly as expected from the moduli reading of a quotient stack. Setting $X = S$ recovers item (1): the action-and-projection morphism $G \times S \to S \times S$ collapses to $G \to S$ (both projections trivial), so Δ_{BG} is G.

3. *The classifying stack of line bundles* $B\mathbb{G}_m$. Take $G = \mathbb{G}_m$, so $B\mathbb{G}_m$ is the stack of line bundles: its fiber over T is the groupoid of invertible $\mathcal{O}_T$-modules and their isomorphisms. For a line bundle L on T the automorphism sheaf is $\underline{\mathrm{Aut}}(L) = \underline{\mathbb{G}_m}$, since an automorphism of an invertible module is multiplication by a unit, and this holds functorially in T: $\underline{\mathrm{Aut}}_T(L)(U) = \mathcal{O}_U(U)^\times = \mathbb{G}_m(U)$. Thus every point of $B\mathbb{G}_m$ has automorphism group $\mathbb{G}_m$, the diagonal fiber is $\underline{\mathbb{G}_m}$, and the diagonal
$$\Delta_{B\mathbb{G}_m} \colon B\mathbb{G}_m \longrightarrow B\mathbb{G}_m \times B\mathbb{G}_m$$
is representable, being $\mathbb{G}_m$ over each point. The Isom-sheaf $\underline{\mathrm{Isom}}_T(L, M)$ of two line bundles is the $\mathbb{G}_m$-torsor of trivializations of $L^{-1} \otimes M$, representable as the complement of the zero section in the total space of $L^{-1} \otimes M$; it is nonempty exactly when $L \cong M$. Every line bundle carries the scalar automorphisms $\mathbb{G}_m$, so no point of $B\mathbb{G}_m$ has trivial automorphisms, and $B\mathbb{G}_m \to \underline{S}$ is nowhere representable, consistently with item (1).

4. *The stack* $B\mu_n$. Take $G = \mu_n$, the group scheme of n-th roots of unity, $\mu_n = \mathrm{Spec}\big(\mathbb{Z}[t]/(t^n - 1)\big)$, an affine (finite, flat) group scheme over S. Then $B\mu_n$ is the stack of μ_n-torsors, and for a torsor P the automorphism sheaf is $\underline{\mathrm{Aut}}(P) \cong \underline{\mu_n}$ (finite flat of degree n), so the diagonal $\Delta_{B\mu_n}$ is represented by μ_n, a finite flat group scheme. Since μ_n is unramified over S exactly when n is invertible on S (equivalently μ_n is étale), the diagonal $\Delta_{B\mu_n}$ is unramified over such a base, so $B\mu_n$ has finite unramified automorphism groups and is a Deligne-Mumford stack there; when n is not invertible (for instance μ_p in characteristic p) the automorphisms are finite but not unramified, and $B\mu_n$ is an Artin stack that is not Deligne-Mumford. This is the smallest example in which the distinction between the two kinds of algebraic stack is visible on the diagonal alone.

In every case the fiber of the diagonal over a point is the automorphism group of that point: G for BG, the stabilizer $\mathrm{Stab}_G(x)$ for $[X/G]$, $\mathbb{G}_m$ for $B\mathbb{G}_m$, and μ_n for $B\mu_n$. The representability of the diagonal is the assertion that these automorphism groups vary in a scheme. This automorphism group at a point is what the next chapters will call the *residual gerbe* of the point, a copy of $B(\mathrm{Aut})$ sitting inside the stack over the point. The generic elliptic curve has automorphism group $\{\pm 1\}$, giving the residual gerbe $B\mu_2$; but at the two special j-invariants the automorphism group is larger (example 13.3.4). The curve $y^2 = x^3 + b$ with j-invariant 0 has the order-6 automorphism $(x, y) \mapsto (\zeta_3 x, -y)$, so $\mathrm{Aut} = \mathbb{Z}/6$ and the residual gerbe is $B(\mathbb{Z}/6)$; the curve $y^2 = x^3 + ax$ with j-invariant 1728 has the order-4 automorphism $(x, y) \mapsto (-x, iy)$, so $\mathrm{Aut} = \mathbb{Z}/4$ and the residual gerbe is $B(\mathbb{Z}/4)$. The gerbe structure is developed systematically when gerbes are introduced.

The four computations exhibit the mechanism promised at the start. A morphism of stacks is understood by its fibers over schemes; the fibers of the diagonal are the automorphism groups; and representability of the diagonal is the precise condition that these automorphism groups be geometric objects. The passage from BG (diagonal $= G$) to $[X/G]$ (diagonal $=$ action-and-projection, fibers $=$ stabilizers) to $B\mathbb{G}_m$ and $B\mu_n$ traces the full range of what the diagonal records, from a fixed group to a family of stabilizers to the arithmetic distinction between étale and non-étale automorphism schemes. With representable morphisms and the diagonal in hand, the geometry of an algebraic stack, smoothness, properness, dimension, can be defined by pulling back to an atlas, which is the task of the next chapter.

Exercise 19.4.1

1. Let $f\colon \mathcal{F} \to \mathcal{G}$ and $g\colon \mathcal{G} \to \mathcal{H}$ be morphisms of stacks. Show that if f and g are representable then so is the composite $g \circ f$, and that if $g \circ f$ is representable and the diagonal of $\mathcal{G}$ is representable then f is representable. (These are the "two out of three" stability properties that make representability behave like a class of geometric morphisms.)

2. Show that representability of morphisms is stable under base change: if $f\colon \mathcal{F} \to \mathcal{G}$ is representable and $\mathcal{G}' \to \mathcal{G}$ is any morphism of stacks, then the projection $\mathcal{F} \times_{\mathcal{G}} \mathcal{G}' \to \mathcal{G}'$ is representable.

3. Let G be a finite group (a constant finite group scheme) over a field k, acting on a k-scheme X. Compute the diagonal of $[X/G]$ explicitly as a morphism of schemes, and describe its fiber over a point $(x_1, x_2) \in X \times X$. When is $[X/G]$ a scheme? When does every point have trivial stabilizer, so that $[X/G]$ is representable by the ordinary quotient?

4. For the classifying stack $B\mathbb{G}_m$ over a field k, exhibit an atlas $\underline{U} \to B\mathbb{G}_m$ from a scheme U: find a representable, smooth, surjective morphism from a scheme. (Hint: the universal torsor $\mathrm{pt} \to B\mathbb{G}_m$ of definition 19.3.2, i.e. $\operatorname{Spec} k \to B\mathbb{G}_m$, classifies the trivial line bundle; verify it is a $\mathbb{G}_m$-torsor and hence smooth surjective. What is $\underline{U} \times_{B\mathbb{G}_m} \underline{U}$?)

5. Show that a stack $\mathcal{F}$ whose diagonal is representable and which admits an atlas $\pi\colon \underline{U} \to \mathcal{F}$ (a representable smooth surjective morphism from a scheme) has the property that $R = \underline{U} \times_{\mathcal{F}} \underline{U}$ is a scheme, and that the two projections $R \rightrightarrows \underline{U}$ are smooth. Interpret R as a groupoid in schemes presenting $\mathcal{F}$. (This is the "groupoid presentation" that the next chapter takes as an alternative definition of an algebraic stack.)

6. Determine for which base schemes S the diagonal of $B\mu_n$ is unramified, and deduce for which S the stack $B\mu_n$ is Deligne-Mumford. Contrast $B\mu_p$ with $B(\mathbb{Z}/p)$ (the constant group) in characteristic p: both have automorphism group of order p at every point, yet one is Deligne-Mumford and the other is not. Explain the discrepancy in terms of the diagonal.

20

Algebraic Spaces and Algebraic Stacks

Stacks solved the bookkeeping problem: a category fibered in groupoids that satisfies descent records objects with their automorphisms and glues them in a topology, and $\mathcal{M}_{1,1}$, BG, and $[X/G]$ are such objects. What stacks did not yet come with is geometry. A scheme carries a topology, a structure sheaf, a dimension, and a notion of smoothness; a bare stack carries none of these. The gap is closed by presenting a stack the way a scheme is presented by an affine cover: through an atlas, a surjection from a scheme that is well behaved enough to transport geometric notions from its source to the stack. This chapter carries out that construction and, with it, finally produces the geometric objects that represent the moduli problems the earlier Parts could not.

The construction is stratified by how singular the automorphisms are allowed to be. When a moduli problem has no automorphisms but still fails to be a scheme, the right object is an algebraic space, a sheaf that is étale-locally a scheme, and the first section builds these from étale equivalence relations. When the automorphisms are finite, the atlas can be taken étale and the object is a Deligne–Mumford stack; the second section defines these, characterizes them by the unramifiedness of their diagonal, and records the Keel–Mori theorem that extracts a coarse moduli space. When the automorphisms are positive dimensional, as for the stack of all coherent sheaves or of principal bundles, only a smooth atlas is available and the object is an Artin stack; the third section defines these and computes the dimension of a stack from a presentation.

The fourth section is the central one of the book. Artin's criteria answer the question the whole subject has been circling: given a moduli functor, when is it an algebraic stack? The answer is a checklist, and its central entries are precisely the deformation theory of Part V. A moduli problem is algebraic when it is a limit-preserving stack whose deformations are governed by a tangent space and an obstruction space satisfying Schlessinger's conditions and whose formal deformations algebraize. The criteria convert the local, infinitesimal data computed in Part V into the global existence of a smooth atlas, and applying them shows that the moduli of curves and the Hilbert and Quot functors are algebraic. The final section imposes the remaining geometry, quasi-coherent sheaves, the Picard group, cohomology, and the properties smooth, separated, and proper, on an algebraic stack through its atlas and its diagonal, closing with the fact that $\overline{\mathcal{M}}_g$ is smooth and proper as a stack even where its coarse space is singular, the statement the final chapter will prove.

20.1 Algebraic Spaces

The previous two chapters supplied two closely related repairs for the failure of a moduli functor to be a scheme. Descent theory recognized that the objects of a moduli problem glue in the étale or fppf topology, not the Zariski one (theorem 18.4.3), and the theory of fibered categories (chapter 19) built the categorical bookkeeping that keeps track of objects together with their automorphisms. This chapter turns those repairs into geometry. The first and gentlest step is to enlarge the category of schemes just far enough to include every quotient that ought to exist but does not: an *algebraic space* is a sheaf that looks like a scheme in the étale topology, and nothing more. It sits strictly between schemes and Deligne-Mumford stacks, and it is the object that finally represents moduli problems whose objects have no automorphisms but whose parameter space cannot be glued from affines in the Zariski topology.

The guiding image is the one that already governed Galois descent and the quotient stack. A scheme U mapping to an object X by an étale surjection presents X as a quotient, and the fibers of $U \times_X U \rightrightarrows U$ record which points of U have been identified. When X is itself a scheme, this is nothing new. The definition of an algebraic space is obtained by inverting the arrow of logic: rather than starting with X and forming the presentation, one starts with the presentation, an étale equivalence relation $R \rightrightarrows U$ on schemes, and *defines* X to be its quotient sheaf. The gain is that such a quotient sheaf need not be a scheme, and the objects that arise this way are exactly what is needed to run the moduli program one rung below the stacks of the coming sections.

Throughout, the base site is $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{et}}$, the category of S-schemes with the étale topology, and S is a fixed base scheme (often $\operatorname{Spec} k$ or a Noetherian scheme). A sheaf means an étale sheaf of sets on $\mathcal{C}$; a scheme U is identified with the sheaf it represents, $T \mapsto \operatorname{Hom}_S(T, U)$, which is an étale sheaf because étale descent is effective for morphisms to a scheme (theorem 18.4.3). All spaces below are quasi-separated unless noted.

The starting datum is an equivalence relation on a scheme, realized not set-theoretically but by an actual subscheme of the product whose two projections are étale. The étale hypothesis is what makes the quotient behave like a scheme locally; without it one would land in a wilder category still. The definition below packages the relation, its two structure maps, and the equivalence axioms into a single object.

Definition 20.1.1: Étale Equivalence Relation

Let U be an S-scheme. An *étale equivalence relation* on U is a scheme R together with a morphism

$$j = (s, t) \colon R \longrightarrow U \times_S U$$

such that:

1. j is a monomorphism (so R is, functorially, a relation: for every S-scheme T, the image of $R(T) \hookrightarrow U(T) \times U(T)$ is a subset);

2. that subset $R(T) \subseteq U(T) \times U(T)$ is an equivalence relation on $U(T)$, functorially in T (reflexive, symmetric, and transitive);

3. both projections $s = \mathrm{pr}_1 \circ j$ and $t = \mathrm{pr}_2 \circ j$ from R to U are étale morphisms in the sense of **??**.

The two maps are written as a parallel pair $s, t \colon R \rightrightarrows U$, the *source* and *target*. When the context demands the ambient monomorphism, the relation is written $R \hookrightarrow U \times_S U$.

The three conditions unwind to a picture that recurs throughout the chapter. Condition (1) says R is a subfunctor of $U \times_S U$, so a T-point of R is literally a pair of T-points of U that are declared equivalent; there is at most one way to be equivalent. Condition (2) is the reflexive, symmetric, transitive package, and it is what distinguishes an equivalence relation from a more general groupoid: reflexivity gives a diagonal section $e \colon U \to R$ with $j \circ e = \Delta_U$, symmetry an involution $\iota \colon R \to R$ swapping s and t, and transitivity a composition $R \times_{s,U,t} R \to R$. Condition (3), the étale hypothesis on both projections, is the geometric heart: the two ways of viewing R as living over U are both local isomorphisms, so passing between the "upstairs" scheme U and its quotient neither creates nor destroys infinitesimal information. It is exactly this that will let the quotient inherit the local structure of a scheme.

The elementary source of étale equivalence relations, and the one that shows the definition subsumes the classical case, is a free action of a finite (or étale) group scheme.

Example 20.1.2: Quotient by a Free Finite Group Action

Let a finite group G act freely on an S-scheme U, with the action morphism $a \colon G \times_S U \to U$, $(g, u) \mapsto g \cdot u$. Take $R = G \times_S U$ and

$$j = (s, t) = (\mathrm{pr}_U,\ a) \colon G \times_S U \longrightarrow U \times_S U$$

sending (g, u) to $(u, g \cdot u)$. On T-points, $R(T) = G \times U(T)$ maps to the pairs $(u, g \cdot u)$, whose image is the relation "$u \sim u'$ iff $u' = g \cdot u$ for some g": the orbit equivalence relation. It is reflexive ($g = e$), symmetric ($g \mapsto g^{-1}$), and transitive (multiply the group elements), so condition (2) holds. Freeness of the action is precisely the statement that j is a monomorphism: if $(u, g \cdot u) = (u, g' \cdot u)$ then $g \cdot u = g' \cdot u$ forces $g = g'$ because the stabilizer of u is trivial, so a T-point of the image determines (g, u) uniquely, giving condition (1). Finally G finite over S makes $G \times_S U \to U$ (the first projection) finite étale, so $s = \mathrm{pr}_U$ is finite étale. For $t = a$, introduce the shear isomorphism

$$\varphi \colon G \times_S U \xrightarrow{\sim} G \times_S U, \qquad (g, u) \longmapsto (g,\ g \cdot u)$$

whose inverse is $(g, u) \mapsto (g, g^{-1} \cdot u)$; then $t = a$ factors as $\mathrm{pr}_U \circ \varphi$, the composite of an isomorphism with the finite étale projection pr_U, so t is finite étale as well. Thus condition (3) holds and

> $R = G \times_S U \rightrightarrows U$ is an étale equivalence relation. Its quotient, constructed below, is the sheaf U/G.

The quotient of a scheme by such a relation is defined by a universal property in the category of étale sheaves, exactly as one would quotient a set by an equivalence relation. When the resulting sheaf happens to be a scheme (as it does, for instance, when U is quasi-projective and G acts freely) nothing new has been built. The point of the theory is that in general it is not, and the objects one gains deserve a name.

Definition 20.1.3: Algebraic Space

An *algebraic space* over S is an étale sheaf $X : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ on $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{et}}$ satisfying:

1. *(representable diagonal)* the diagonal morphism $\Delta_X : X \to X \times_S X$ is representable by schemes: for every S-scheme T and every morphism $T \to X \times_S X$, the fiber product $T \times_{X \times_S X} X$ is (representable by) a scheme;
2. *(étale atlas)* there is a scheme U and a morphism $p : U \to X$ that is surjective and étale (both properties are representable and hold after every base change $T \to X$, using (1)).

The morphism $p : U \to X$ is called an *étale atlas* or *étale presentation* of X.

Two features of this definition need immediate comment, since together they are the whole content. Condition (1), representability of the diagonal, is what makes the words "surjective" and "étale" in condition (2) meaningful: a property of a morphism $U \to X$ to a sheaf is defined by requiring it after base change to every scheme mapping to X, and the diagonal being representable is exactly what guarantees those base changes are schemes, so the property can be tested there. Condition (2) then asserts that X is étale-locally a scheme. The diagonal condition and the atlas condition are the two halves of algebraicity that will reappear, verbatim in spirit, in the definitions of Deligne-Mumford and Artin stacks in the next two sections, where a stack replaces a sheaf and "étale" is weakened to "smooth".

The definition by an atlas and the definition by a relation are two descriptions of the same objects, and moving between them is the basic fluency the chapter requires. Given an atlas $p : U \to X$, the fiber product

$$R = U \times_X U$$

carries its two projections $s, t : R \rightrightarrows U$, and because p is representable étale these projections are étale; the monomorphism $R \hookrightarrow U \times_S U$ (which lands in $U \times_S U$ because a point of R is a pair of points of U with the same image in X) is an equivalence relation because "$p(u) = p(u')$" is one. Thus every algebraic space with a chosen atlas produces an étale equivalence relation. Conversely, the quotient sheaf reconstructs the space.

20.1.4 Note: Algebraic Spaces as Quotients U/R Let $R \rightrightarrows U$ be an étale equivalence relation (definition 20.1.1). Its *quotient sheaf* U/R is the sheafification, in the étale topology, of the presheaf $T \mapsto U(T)/R(T)$ (orbits of T-points under the equivalence relation). Then U/R is an algebraic space, with $U \to U/R$ an étale atlas and $R = U \times_{U/R} U$; and every algebraic space arises this way from any of its atlases. Thus

$$\{\text{algebraic spaces}\} \longleftrightarrow \{\text{étale equivalence relations } R \rightrightarrows U\}\big/(\text{refinement of atlas})$$

is the working dictionary. The equivalence "étale-sheaf quotient of an étale equivalence relation = étale sheaf with representable diagonal and an étale atlas" is one of the foundational structural theorems; the direction that U/R has a representable diagonal (so that $U \to U/R$ is a bona fide étale surjection of the kind in definition 20.1.3) requires the machinery of [39] and is recorded in [63] (Tag 02WW and following). The easy direction, that an algebraic space with an atlas yields the relation $R = U \times_X U$, is the computation above.

The dictionary makes the first nontrivial fact free of charge: a scheme is an algebraic space, in the most economical way possible. This is the sanity check that the enlargement really is an enlargement and loses no old objects.

The claim that a scheme is an algebraic space is the base of every comparison that follows, and it is proved by exhibiting the identity as an atlas. The equivalence relation is then the trivial one, in which nothing is identified.

Theorem 20.1.5: Basic Properties of Algebraic Spaces

Let S be a base scheme and work over $\mathcal{C} = (\mathbf{Sch}/S)_{\text{et}}$. Then:

1. Every scheme is an algebraic space, presented by the identity atlas $U \xrightarrow{\text{id}} U$, whose associated étale equivalence relation is the diagonal $\Delta_U : U \hookrightarrow U \times_S U$.
2. The diagonal $\Delta_X : X \to X \times_S X$ of an algebraic space is representable, and X is quasi-separated (resp. separated) iff Δ_X is quasi-compact (resp. a closed immersion).
3. Algebraic spaces are closed under fiber products: if $X \to Z$ and $Y \to Z$ are morphisms of algebraic spaces, then $X \times_Z Y$ is an algebraic space.
4. A morphism $T \to X$ from a scheme T to an algebraic space X is representable: for every scheme T' and morphism $T' \to X$, the fiber product $T \times_X T'$ is a scheme.
5. Every quasi-separated algebraic space X contains a scheme as a dense open subspace; that is, there is a dense open $V \subseteq X$ which is representable by a scheme.

***Proof*:**
Parts (1) to (4) are short and are proved here; part (5) is one of the structural theorems whose proof is cited.

Part (1): a scheme is an algebraic space. Let U be an S-scheme, viewed as the sheaf $T \mapsto \mathrm{Hom}_S(T, U)$, which is an étale sheaf by effectivity of étale descent for morphisms to a scheme (theorem 18.4.3). Its diagonal $\Delta_U : U \to U \times_S U$ is a morphism of schemes, hence representable, and for a scheme mapping to $U \times_S U$ the fiber product $T \times_{U \times_S U} U$ is a scheme (it is the locus in T where the two maps $T \to U$ agree), so condition (1) of definition 20.1.3 holds. The identity $\mathrm{id}: U \to U$ is an isomorphism, hence surjective and étale, giving an atlas and condition (2). The associated relation is $R = U \times_U U = U$ with $j = \Delta_U$, the trivial equivalence relation identifying nothing. Thus U is an algebraic space.

Part (2): the diagonal. Representability of Δ_X is condition (1) of definition 20.1.3, so there is nothing to prove for the first clause. For the separation clauses, recall that quasi-compactness and being a closed immersion are representable properties of morphisms, so they may be tested after the base change of Δ_X along any scheme mapping to $X \times_S X$; a quasi-separated (resp. separated) algebraic space is defined precisely by requiring the representable morphism Δ_X to be quasi-compact (resp. a closed immersion), which is the stated equivalence.

Part (3): fiber products. Write $W := X \times_Z Y$ for the sheaf fiber product, which is an étale sheaf because sheaves are closed under limits. Two things must be checked: that Δ_W is representable, and that W has an étale atlas.

For the diagonal, fix a scheme T and a morphism $T \to W \times_S W$; the task is to show the base change $T \times_{W \times_S W} W$ (the fiber of Δ_W over T) is a scheme. The projection $W = X \times_Z Y \to X \times_S Y$ identifies $W \times_S W$ with a subsheaf of $(X \times_S X) \times_S (Y \times_S Y)$, and under this identification Δ_W is the restriction to W of the product $\Delta_X \times_S \Delta_Y : X \times_S Y \to (X \times_S X) \times_S (Y \times_S Y)$. Hence the base change of Δ_W along $T \to W \times_S W$ fits in a cartesian square with the base change of $\Delta_X \times_S \Delta_Y$ along the composite $T \to W \times_S W \to (X \times_S X) \times_S (Y \times_S Y)$:

$$T \times_{W \times_S W} W = T \times_{(X \times_S X) \times_S (Y \times_S Y)} (X \times_S Y)$$

The right-hand side is obtained by base-changing Δ_X (a scheme by (2)) along $T \to X \times_S X$ and then Δ_Y (a scheme by (2)) along the resulting scheme mapping to $Y \times_S Y$; each step yields a scheme because Δ_X and Δ_Y are representable. Thus $T \times_{W \times_S W} W$ is a scheme for every T, so Δ_W is representable.

For the atlas, choose étale atlases $a: U \to X$ and $b: V \to Y$. Pulling the sheaf $W = X \times_Z Y$ back along a on the X-side and along b on the Y-side replaces both X and Y by their atlas schemes, giving the sheaf

$$P = U \times_X (X \times_Z Y) \times_Y V = U \times_Z V,$$

the fiber product of the two scheme morphisms $U \to Z$ and $V \to Z$ over the algebraic space Z. This P is a scheme: since $U \to Z$ is a morphism from a scheme it is representable by (4), so its base change $U \times_Z V \to V$ along the scheme V is a scheme, whence $P = U \times_Z V$ is a scheme. The projection $P = U \times_Z V \to X \times_Z Y = W$ is the fiber product over Z of the two atlas maps a and b; it factors as the composite

$$U \times_Z V \xrightarrow{a \times_Z \mathrm{id}} X \times_Z V \xrightarrow{\mathrm{id} \times_Z b} X \times_Z Y = W,$$

each factor being a base change of an étale surjection (a, resp. b) along a morphism to X, resp. Y, hence itself étale and surjective; the composite is therefore étale and surjective. Thus $P \to W$ is an étale atlas by a scheme, and $W = X \times_Z Y$ has representable diagonal and an étale atlas, so it is an algebraic space, completing the proof of this part.

Part (4): a map from a scheme is representable. Let $T \to X$ be a morphism from a scheme, and let $T' \to X$ be any morphism from a scheme. Then $T \times_X T'$ is the fiber product of

$(T, T') \to X \times_S X$ against the diagonal Δ_X:

$$T \times_X T' = (T \times_S T') \times_{X \times_S X} X$$

and the right-hand side is a scheme because Δ_X is representable by condition (1) of definition 20.1.3 and $T \times_S T'$ is a scheme. Thus every map from a scheme is representable, which is what was to be shown.

Part (5): the dense open subscheme. Let X be a quasi-separated algebraic space with étale atlas $p \colon U \to X$. Over the generic points of U the étale map p is, after shrinking, an isomorphism onto its image, so X contains a scheme-theoretic dense open; the full statement (that a quasi-separated algebraic space has a dense open subscheme, and even generic separatedness) is one of the structural theorems whose proof uses Noetherian induction and the theory of étale-local structure of spaces. It is beyond the elementary framework here and is cited to [39] and [63] (Tag 06NH). This completes the proof of the parts proved in full and cites the remainder.

The four proved parts already organize the theory. Part (1) confirms that schemes sit inside algebraic spaces, part (2) records that the diagonal is the seat of separatedness, part (3) closes the category under the operation moduli theory uses constantly, and part (4) is what makes the notion workable: it says a scheme mapping to an algebraic space behaves like a scheme mapping to a scheme, so all the usual notions (open, closed, étale, smooth, flat, proper) transport to morphisms of algebraic spaces by testing on schemes. This last point is what makes algebraic spaces usable rather than only definable. Part (5), the dense open subscheme, is the precise sense in which an algebraic space is a scheme with only a small "non-scheme" locus: the pathology is always concentrated away from a dense open, so an algebraic space is birationally a scheme.

The representability of the diagonal deserves a second look, because it is the exact place where the stack theory of chapter 19 feeds in. There, a morphism of stacks was called representable when its base change to any scheme is a scheme (or algebraic space), and the representability of the diagonal (theorem 19.4.6) was identified as the condition encoding the $\underline{\mathrm{Isom}}$ sheaves. For an algebraic space the diagonal is representable by schemes rather than by algebraic spaces, and the $\underline{\mathrm{Isom}}$ sheaf of two objects is a subscheme of a product; this is the degenerate case of the stack story in which the automorphism sheaves are trivial. An algebraic space is thus exactly a stack whose objects have no nontrivial automorphisms, presented by an étale rather than a smooth atlas. The sections to come loosen each of these two constraints in turn.

The theory would be idle if every algebraic space were secretly a scheme. The example that follows is the one that justifies the whole enlargement: a free étale equivalence relation whose quotient is a genuine algebraic space and not a scheme, because the identification cannot be carried out inside any single affine chart.

Example 20.1.6: A Free Involution Quotient That Is Not a Scheme

Work over $S = \operatorname{Spec} k$ with k algebraically closed of characteristic $\neq 2$. A free finite group action on a *quasi-projective* scheme has a scheme quotient, because every orbit then lies in a common affine open (the exercises establish this). To produce a non-scheme algebraic space the atlas must therefore be non-quasi-projective, and the canonical source of such atlases is Hironaka's example of a smooth complete non-projective threefold.

Hironaka constructed a smooth complete (proper) threefold U over k, non-projective, carrying a free involution $\sigma \colon U \to U$ (an action of $\mathbb{Z}/2$ with no fixed points) whose orbit through a certain point is not contained in any affine open of U. Because σ is a free action of the finite étale group

$\mathbb{Z}/2$, it defines an étale equivalence relation $R = U \times (\mathbb{Z}/2) \rightrightarrows U$ in the sense of example 20.1.2: both projections $s, t\colon R \to U$ are finite étale (the constant group $\mathbb{Z}/2$ is finite étale over k in every characteristic, being $\operatorname{Spec}(k \times k)$; the hypothesis $\operatorname{char} k \neq 2$ is what makes the involution act freely), and freeness makes $(s, t)\colon R \hookrightarrow U \times_k U$ a monomorphism. The quotient sheaf

$$X = U/R = U/(\mathbb{Z}/2)$$

is therefore an algebraic space (definition 20.1.3), with étale atlas $U \to X$. It is a smooth proper algebraic space of dimension 3.

Why X is not a scheme. Suppose X were a scheme. Then the image $* \in X$ of the distinguished orbit would have an affine open neighborhood $\operatorname{Spec} A \subseteq X$, and its preimage $W \subseteq U$ under the étale double cover $U \to X$ would be a $\mathbb{Z}/2$-invariant open with $\operatorname{Spec} A = W/(\mathbb{Z}/2)$; since $\operatorname{Spec} A$ is affine and $\mathbb{Z}/2$ finite, the quotient map $W \to \operatorname{Spec} A$ would be a finite (hence affine) morphism, forcing W to be affine (an affine morphism over an affine scheme has affine total space), and W would contain the whole orbit of the distinguished point in a single affine open. This contradicts Hironaka's construction, in which that orbit lies in no affine open of U. Hence X has no affine open neighborhood of $*$ and is not a scheme. The verification that Hironaka's threefold has these properties, and that the resulting X is an algebraic space but not a scheme, is a standard structural computation carried out in [39] and [63] (the non-scheme algebraic space, Tag 02Z1 and the Hironaka example); its full details use resolution-of-singularities-era birational geometry beyond the present scope, so the construction of U itself is cited.

The obstruction is the one advertised in the outline of the theory: the distinguished orbit fails to be contained in a single affine open of the atlas, so the construction of a quotient scheme by patching affine invariant-ring quotients breaks down there, while the étale-sheaf quotient exists without difficulty. The involution is free and étale, so no stacky automorphism appears; X has trivial automorphism sheaves and is a bona fide space, one rung below the stacks of the next sections.

The example is worth holding onto because it isolates the single phenomenon that separates algebraic spaces from schemes. A scheme is glued from affines in the Zariski topology, and the gluing of a quotient by a group action requires each orbit to fit inside a stable affine chart. When some orbit refuses to fit, the Zariski gluing fails but the étale gluing does not, and the resulting object is an algebraic space with no scheme structure. Every non-scheme algebraic space is, at bottom, a variation on this theme: a free étale relation whose orbits are not uniformly affine.

This is exactly the shape of the failure encountered at the very start of the book, and naming the connection closes a thread carried since its beginning. There, $\mathcal{M}_{1,1}$ failed to be a fine moduli scheme because elliptic curves carry the automorphism $[-1]$, and the quotient presentation $[(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ (example 13.3.4 recorded the automorphisms; theorem 13.3.6 the j-line) exhibited it as a quotient. The distinction is that $\mathcal{M}_{1,1}$ has nontrivial automorphisms, so its correct home is a stack rather than a space. But the geometric mechanism, that a quotient by an étale (or smooth) presentation need not descend to a scheme, is identical: an algebraic space is the automorphism-free instance of that mechanism, a Deligne-Mumford stack the finite-automorphism instance.

Exercise 20.1.1

1. Let $R \rightrightarrows U$ be an étale equivalence relation with quotient sheaf $X = U/R$. Using theorem 20.1.5(4) and the description $R = U \times_X U$, prove that the source and target maps $s, t\colon R \to U$ are étale directly from the representability of the atlas $U \to X$, without invoking

the definition of R. (Base change $U \to X$ along $U \to X$.)

2. Show that a morphism $f\colon X \to Y$ of algebraic spaces has representable diagonal, i.e. $\Delta_f\colon X \to X \times_Y X$ is representable by schemes. Deduce that for any scheme T and morphism $T \to Y$, the fiber product $X \times_Y T$ is an algebraic space. (Use theorem 20.1.5(2),(3),(4).)

3. Consider the $\mathbb{Z}/2$-action $x \mapsto -x$ on $U = \mathbb{A}^1 = \operatorname{Spec} k[x]$ ($\operatorname{char} k \neq 2$), and let $R' \hookrightarrow \mathbb{A}^1 \times_k \mathbb{A}^1$ be the *closed* subscheme $\{(y-x)(y+x) = 0\}$, the full graph of the action. Show that R' is an equivalence relation but is *not* étale (the origin is a fixed point, so the projections ramify there), and that the quotient is nevertheless the scheme $\operatorname{Spec} k[x^2] \cong \mathbb{A}^1$. Explain in one sentence why this shows that a *non*-étale relation can still have a scheme quotient, so the étale hypothesis in definition 20.1.1 is a hypothesis about the presentation, not about the quotient.

4. Prove that an algebraic space X is a scheme if and only if it admits an étale atlas $U \to X$ that is a *Zariski* covering after a suitable choice, more precisely: X is a scheme iff there is an open subspace covering $\{X_i \hookrightarrow X\}$ by algebraic spaces each of which is a scheme. (One direction is trivial; for the other, glue the schemes X_i along $X_i \times_X X_j$, using that these fiber products are schemes by theorem 20.1.5(4).)

5. Let G be a finite group acting freely on a quasi-projective scheme U over k. Show that the quotient sheaf U/G is representable by a scheme. Explain why this is consistent with example 20.1.6: identify the single hypothesis of this exercise that Hironaka's atlas violates, and confirm that it is exactly the property whose failure lets the orbit escape every affine open. (Recall that on a quasi-projective scheme every finite set of points lies in a common affine open; apply this to each orbit.)

6. Explain the sense in which theorem 20.1.5(5) says an algebraic space is "birationally a scheme," and reconcile this with example 20.1.6: describe a dense open subscheme $V \subseteq X$ of Hironaka's quotient $X = U/(\mathbb{Z}/2)$ (take the image of a $\mathbb{Z}/2$-invariant affine open of U on which the action is free with a scheme quotient), and explain why the non-scheme locus is contained in the nowhere-dense closed complement.

20.2 Deligne-Mumford Stacks

The previous section built the first geometric objects beyond schemes: algebraic spaces, the étale-sheaf quotients of étale equivalence relations (definition 20.1.3). An algebraic space is a sheaf that is étale-locally a scheme, and it repairs the failures of representability that stem from gluing, not from automorphisms. The moduli problems this book exists to serve have a second, deeper source of non-representability: their objects carry automorphisms, and no sheaf, algebraic space or otherwise, can record them. Chapter 7 built the categorical object that does record them, the stack, and produced a supply of them as quotient stacks $[X/G]$ (definition 19.3.1) and classifying stacks BG (definition 19.3.2). This chapter imposes geometry on stacks. The present section treats the case where the automorphisms are as small as they can be while still being present: finite. A stack whose objects have finite, reduced automorphism groups admits an *étale* atlas, and this is exactly the class of *Deligne-Mumford stacks*. They are the stacks closest to schemes, and they are the stacks the moduli of curves lives in.

The organizing principle is a dictionary between the automorphisms of the objects and the geometry of the diagonal. A stack $\mathcal{X}$ has a diagonal $\Delta_{\mathcal{X}}\colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ whose fibers are the $\underline{\text{Isom}}$ sheaves of pairs of objects, and whose "fiber over the diagonal point" at an object x is the automorphism group scheme $\underline{\mathrm{Aut}}(x)$. Representability of the diagonal (Chapter 7, theorem 19.4.6) says these $\underline{\mathrm{Isom}}$ sheaves are algebraic spaces, which is half of what it means for a stack to be geometric. The other half is the atlas. The theorem of this section is that the atlas can be taken étale, rather than only smooth, exactly when the diagonal is unramified, which happens exactly when every automorphism group scheme is finite and reduced. The three conditions, an étale atlas, unramified diagonal, finite reduced automorphisms, are one condition wearing three faces, and Deligne-Mumford stacks are the stacks satisfying it.

Definition 20.2.1: Deligne-Mumford Stack

Let S be a base scheme and $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{et}}$ the big étale site. A stack $\mathcal{X}$ over $\mathcal{C}$ is a *Deligne-Mumford stack* (a *DM stack*) if:

1. the diagonal $\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ is representable (by algebraic spaces), quasi-compact, and separated; and
2. there exist a scheme U and a morphism $u \colon U \to \mathcal{X}$ that is *representable*, *étale*, and *surjective*.

Such a $u \colon U \to \mathcal{X}$ is called an *étale atlas* (or an *étale presentation*) of $\mathcal{X}$. Given an atlas, the fiber product $R = U \times_{\mathcal{X}} U$ is an algebraic space, both projections $s, t \colon R \to U$ are étale (being base changes of u), and the induced groupoid

$$R = U \times_{\mathcal{X}} U \rightrightarrows U$$

is an étale groupoid presenting $\mathcal{X}$ as its stack quotient $\mathcal{X} \simeq [U/R]$. When R is a scheme, this exhibits $\mathcal{X}$ as $[U/R]$ for an étale equivalence relation in the sense of definition 20.1.1, except that $R \hookrightarrow U \times_S U$ need not be a monomorphism (the groupoid $R \rightrightarrows U$ records automorphisms, so (s, t) can have positive-dimensional fibers only if automorphisms are positive dimensional, which the DM condition will forbid).

The definition is the stack-level analogue of the algebraic-space definition of the previous section, with two changes. First, the atlas is a map to a stack, so the requirement that it be *representable* (Chapter 7, definition 19.4.1) is not automatic and must be imposed: representability of u says that for every scheme T and every map $T \to \mathcal{X}$, the fiber product $U \times_{\mathcal{X}} T$ is a scheme (or algebraic space), which is what makes it meaningful to ask that u be étale, since "étale" is a property of morphisms of schemes and algebraic spaces. Second, the equivalence relation $R \rightrightarrows U$ is replaced by a groupoid, because two points of U over the same object of $\mathcal{X}$ can be identified in more than one way, once for each isomorphism between the objects they represent. The groupoid $R \rightrightarrows U$ is the geometric incarnation of the $\underline{\mathrm{Isom}}$ sheaves: $R = U \times_{\mathcal{X}} U$ parametrizes triples (a, b, φ) with $a, b \in U$ mapping to isomorphic objects and φ an isomorphism between them, so the fiber of $(s, t) \colon R \to U \times_S U$ over a diagonal point (a, a) is precisely $\underline{\mathrm{Aut}}$ of the object $u(a)$.

The requirement that the diagonal be representable is not redundant given the atlas; rather, the two together are what "geometric" means for a stack. A useful reformulation, which the following remark records for use throughout, is that representability of the diagonal is equivalent to the algebraicity of all $\underline{\mathrm{Isom}}$ sheaves.

20.2.2 Note: The Diagonal and the Isom Sheaves For a stack $\mathcal{X}$ over $\mathcal{C}$, the diagonal $\Delta_{\mathcal{X}}$ is representable by algebraic spaces if and only if, for every scheme T and every pair of objects $x, y \in \mathcal{X}(T)$, the sheaf

$$\underline{\mathrm{Isom}}_T(x, y) \colon (T' \to T) \longmapsto \mathrm{Isom}_{\mathcal{X}(T')}(x|_{T'}, y|_{T'})$$

on $\mathcal{C}/T$ is an algebraic space over T (Chapter 7, theorem 19.4.6). Taking $y = x$ gives the automorphism group algebraic space $\underline{\mathrm{Aut}}_T(x) = \underline{\mathrm{Isom}}_T(x, x)$, which is a group object over T; taking $T = \operatorname{Spec} K$ for a field gives the automorphism group scheme $\underline{\mathrm{Aut}}(x)$ of a single object. The diagonal being quasi-compact and separated translates into these $\underline{\mathrm{Isom}}$ sheaves being quasi-compact and separated over the base, which for the automorphism groups means, in particular, that they are separated group schemes.

The heart of the section is the equivalence that gives the class its name. Deligne and Mumford isolated the DM stacks in their proof of the irreducibility of the moduli of curves precisely because these are the stacks one can study with the étale-local methods of schemes, and the criterion for recognizing them is intrinsic: it reads off the automorphism groups, which are exactly the data a moduli problem hands to the geometer. The theorem below is the bridge between the two ways of looking at a stack, from above (an atlas one can take étale) and from within (objects whose automorphisms are finite).

Theorem 20.2.3: DM Stacks and the Unramified Diagonal

Let $\mathcal{X}$ be a stack over $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{et}}$ whose diagonal $\Delta_{\mathcal{X}}$ is representable, quasi-compact, and separated, and which admits a *smooth* representable surjection $V \to \mathcal{X}$ from a scheme (so $\mathcal{X}$ is an algebraic stack in the sense of the next section). The following are equivalent:

1. $\mathcal{X}$ is a Deligne-Mumford stack; that is, $\mathcal{X}$ admits an *étale* atlas $U \to \mathcal{X}$.
2. The diagonal $\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ is *unramified*.
3. For every scheme T and every object $x \in \mathcal{X}(T)$, the automorphism group algebraic space $\underline{\mathrm{Aut}}_T(x) \to T$ is *unramified*; equivalently, for every field K and every $x \in \mathcal{X}(K)$, the automorphism group scheme $\underline{\mathrm{Aut}}(x)$ is *finite and reduced* over K.

When S lies over $\mathbb{Q}$ (or more generally when the relevant automorphism groups have order prime to the residue characteristics), the reducedness in (3) is automatic, and the condition is simply that every $\underline{\mathrm{Aut}}(x)$ is *finite*.

***Proof*:**
The argument threads three reformulations of a single geometric fact: a morphism to a stack can be taken étale exactly when the smooth atlas one already has fails to be étale only along the automorphisms, and those automorphisms are the fibers of the diagonal.

Step 1: (2) ⇔ (3), the diagonal and the automorphisms. The diagonal $\Delta_{\mathcal{X}}$ is a representable morphism, so "unramified" is defined by base change: $\Delta_{\mathcal{X}}$ is unramified if and only if for every scheme T and every map $T \to \mathcal{X} \times_S \mathcal{X}$, the base change $T \times_{\mathcal{X} \times_S \mathcal{X}} \mathcal{X} \to T$ is an unramified morphism of algebraic spaces (Chapter 7, definition 19.4.3). A map $T \to \mathcal{X} \times_S \mathcal{X}$ is a pair of objects $x, y \in \mathcal{X}(T)$, and the base change is exactly the $\underline{\mathrm{Isom}}$ sheaf: $T \times_{\mathcal{X} \times_S \mathcal{X}} \mathcal{X} = \underline{\mathrm{Isom}}_T(x, y)$

by theorem 19.4.6. Hence $\Delta_{\mathcal{X}}$ is unramified if and only if every $\underline{\mathrm{Isom}}_T(x, y) \to T$ is unramified. Since unramifiedness is preserved and detected under base change, and the automorphism groups $\underline{\mathrm{Aut}}_T(x) = \underline{\mathrm{Isom}}_T(x, x)$ are the restrictions to the diagonal $x = y$, it is enough to test on the automorphism groups: the general $\underline{\mathrm{Isom}}_T(x, y)$ is either empty or, after passing to a cover trivializing the isomorphism, a torsor under $\underline{\mathrm{Aut}}_T(x)$, and a torsor under an unramified group space is unramified. This gives (2) ⇔ (3) in the form "every $\underline{\mathrm{Aut}}_T(x)$ is unramified over T."

For the pointwise reformulation, a group scheme G of finite type over a field K is unramified over K if and only if $\Omega_{G/K} = 0$, which by translation invariance holds if and only if the cotangent space at the identity vanishes, that is, if and only if G is étale over K. An étale K-group scheme of finite type is finite (over a field, étale plus finite type, hence quasi-compact, forces finite, since it becomes a finite disjoint union of copies of $\operatorname{Spec} \bar{K}$ after base change) and reduced (étale schemes over a field are reduced). Conversely a finite reduced group scheme over K is a finite étale K-scheme, hence unramified. When $\operatorname{char} K = 0$, Cartier's theorem makes every finite group scheme automatically reduced; when $|G|$ is prime to $\operatorname{char} K$, reducedness follows from the separate (and easier) fact that the order is invertible, so the group scheme is étale by order count. Either way "finite" alone suffices. This establishes the equivalence of (3)'s two formulations and its equivalence to (2).

Step 2: (1) ⇒ (2), an étale atlas forces an unramified diagonal. Suppose $u\colon U \to \mathcal{X}$ is an étale atlas. Form the fiber product $R = U \times_{\mathcal{X}} U$; by representability of u and of the diagonal, R is an algebraic space, and both projections $s, t\colon R \to U$ are base changes of u, hence étale. The diagonal of $\mathcal{X}$ fits into a cartesian square whose top row is the map $R \to U \times_S U$, in the sense that after base change along the étale surjection $U \times_S U \to \mathcal{X} \times_S \mathcal{X}$, the diagonal $\Delta_{\mathcal{X}}$ pulls back to $(s, t)\colon R \to U \times_S U$. Unramifiedness is étale-local on the target, since it is the vanishing of the relative cotangent sheaf and Ω commutes with the étale base change $U \times_S U \to \mathcal{X} \times_S \mathcal{X}$ (theorem 18.4.3 for the descent of étale-local properties along a cover). So $\Delta_{\mathcal{X}}$ is unramified if and only if $(s, t)\colon R \to U \times_S U$ is. Now $s\colon R \to U$ is étale, and the structure map $U \to S$ composed appropriately shows (s, t) factors as $R \xrightarrow{(s,t)} U \times_S U \xrightarrow{\mathrm{pr}_1} U$ with $\mathrm{pr}_1 \circ (s, t) = s$ étale. The morphism (s, t) is then unramified because $\Omega_{R/U \times_S U} = 0$: the surjection $\Omega_{R/U} \twoheadrightarrow \Omega_{R/U \times_S U}$ from the factorization has $\Omega_{R/U} = 0$ (as s is étale), so its quotient vanishes. Hence $\Delta_{\mathcal{X}}$ is unramified.

Step 3: (2) ⇒ (1), an unramified diagonal produces an étale atlas. This is the substance. Start from a smooth atlas $p\colon V \to \mathcal{X}$, which exists by hypothesis. The goal is to replace V by a scheme U mapping to $\mathcal{X}$ étale-ly and surjectively. The obstruction to p being étale is its positive relative dimension, and this dimension is accounted for entirely by the automorphisms: the fiber of p over an object x is the automorphism space $\underline{\mathrm{Aut}}(x)$ (up to torsor twisting), whose dimension is the relative dimension of p minus the "geometric" dimension. When the diagonal is unramified, Step 1 makes $\underline{\mathrm{Aut}}(x)$ unramified, hence zero-dimensional, so the automorphisms contribute nothing to the relative dimension and p is smooth of the "correct" dimension, with the excess directions absent.

To extract an étale atlas, work locally on V and slice. Fix a point $v \in V$ with image $x = p(v)$, and let $n = \dim_v(V/\mathcal{X})$ be the relative dimension of the smooth morphism p at v. Because $\Delta_{\mathcal{X}}$ is unramified, the diagonal $V \to V \times_{\mathcal{X}} V$ is an open immersion: indeed $V \times_{\mathcal{X}} V = V \times_{\mathcal{X}} V$ carries the two étale-locally-defined projections, and its diagonal $V \to V \times_{\mathcal{X}} V$ is a section of the unramified separated morphism $V \times_{\mathcal{X}} V \to V$, hence an open immersion (a section of an unramified separated morphism is an open immersion, being a monomorphism that is étale). Now choose a locally closed subscheme $W \hookrightarrow V$ through v that is transverse to the fibers of p

and of relative dimension 0 over $\mathcal{X}$: since p is smooth of relative dimension n at v, the fiber $p^{-1}(x)$ is smooth of dimension n near v in the appropriate sense, and one selects W by cutting V with n functions $f_1, \ldots, f_n$ vanishing at v whose differentials $df_1, \ldots, df_n$ restrict to a basis of the fiber cotangent space $\Omega_{p^{-1}(x)/\kappa(x)} \otimes \kappa(v)$. The restriction $p|_W : W \to \mathcal{X}$ is then unramified because its relative cotangent sheaf vanishes at v: the df_i span the fiber directions of p, so cutting them out kills $\Omega_{W/\mathcal{X}}$ at v. It is flat at v by a regular-sequence argument: $f_1, \ldots, f_n$ form a regular sequence on the smooth (hence Cohen-Macaulay) fibers of p near v, cutting the relative dimension from n down to 0 without dropping the fiber dimension by more than one at each step, so, checked étale-locally on the smooth atlas where the target is a scheme, miracle flatness applies to the Cohen-Macaulay source W over the regular target and $p|_W$ is flat at v. Unramified and flat at v makes $p|_W$ étale at v. Étaleness is an open condition, so after shrinking W around v the map $p|_W : W \to \mathcal{X}$ is étale.

Doing this at every point of $\mathcal{X}$ and taking the disjoint union of the resulting slices produces a scheme $U = \bigsqcup_\alpha W_\alpha$ with an étale map $u = \sqcup p|_{W_\alpha} : U \to \mathcal{X}$. This map is surjective because every point x of $\mathcal{X}$ is hit: x lifts to some $v \in V$ under the surjection p, and the slice W_α through v contains v, so u hits x. Thus $u : U \to \mathcal{X}$ is a representable, étale, surjective map from a scheme, that is, an étale atlas, and $\mathcal{X}$ is a Deligne-Mumford stack. This completes the cycle (1) $\Rightarrow$ (2) $\Rightarrow$ (1), and together with Step 1 the equivalence of all three conditions, as we sought to show.

The theorem is the recognition principle for the class. It says that to check a moduli stack is Deligne-Mumford, one never needs to construct an étale atlas by hand: it suffices to compute the automorphism group of a single object over each field and confirm it is finite (in characteristic 0) or finite and reduced (in general). This is a computation entirely internal to the moduli problem, made before any geometry is built, and it is why the DM condition is so usable. The deformation theory developed earlier is the tool that turns this into a smooth atlas: Schlessinger's hull (theorem 17.3.5) produces the formal neighborhood of a point, the tangent space T^1 (definition 17.2.2) measures its dimension, and $\underline{\mathrm{Aut}}$ finite means the extra dimensions that would force a smooth but non-étale atlas are absent, so the hull algebraizes to an étale slice. The passage from "finite automorphisms" to "étale atlas" in Step 3 is what this deformation-theoretic fact gives geometrically.

The other reason the DM class matters is that it is the largest class of stacks for which the classical scheme-theoretic invariants transport cleanly. Because an étale atlas is a local isomorphism for the étale topology, the étale-local geometry of a DM stack is the geometry of a scheme, and notions that are étale-local, smoothness, normality, the cotangent complex, coherent cohomology, descend from the atlas to the stack without correction. The one place a DM stack visibly departs from a scheme is at its points, where the automorphism group survives as a finite "residual gerbe," a copy of $B\underline{\mathrm{Aut}}(x)$ that the coarse space of the next theorem will collapse.

The construction of a coarse space is the tool that connects a DM stack to classical algebraic geometry: it produces the best possible algebraic space whose points are the isomorphism classes of objects, forgetting the automorphisms. For the moduli of elliptic curves this is the j-line (theorem 13.3.6); for the moduli of curves it is the classical M_g that predates the stack. The existence of the coarse space in general is the theorem of Keel and Mori, and it is the last general result of this section.

Theorem 20.2.4: Keel-Mori

Let S be a locally Noetherian scheme and $\mathcal{X}$ a Deligne-Mumford stack, separated and of finite type over S, with *finite inertia*: the inertia stack $I_{\mathcal{X}} = \mathcal{X} \times_{\Delta, \mathcal{X} \times_S \mathcal{X}, \Delta} \mathcal{X} \to \mathcal{X}$ (whose fiber over an object x is the automorphism group $\underline{\mathrm{Aut}}(x)$) is finite over $\mathcal{X}$. Then there exists a *coarse moduli space*: an algebraic space X and a morphism

$$\pi \colon \mathcal{X} \longrightarrow X$$

such that

1. π is *initial* among morphisms from $\mathcal{X}$ to algebraic spaces: every morphism $\mathcal{X} \to Y$ with Y an algebraic space factors uniquely through π;
2. π induces a *bijection* on geometric points: for every algebraically closed field K over S, the map $\pi \colon \mathcal{X}(K)/\cong \to X(K)$ from isomorphism classes of objects to K-points is a bijection; and
3. π is *proper* and *quasi-finite*, and the structure map $\mathcal{O}_X \to \pi_* \mathcal{O}_{\mathcal{X}}$ is an isomorphism.

Moreover X is of finite type and separated over S, its formation commutes with flat base change on X, and X is a *scheme* whenever $\mathcal{X}$ admits a quasi-projective coarse space, in particular whenever $\mathcal{X}$ is the quotient of a quasi-projective scheme by a finite group or, more generally, carries an ample line bundle descending to X.

***Proof Key ideas*:**
The full proof constructs X étale-locally by taking, on a suitable étale chart $\operatorname{Spec} A \to \mathcal{X}$ where the finite inertia acts by a finite group G on A, the invariant ring $\operatorname{Spec} A^G$, and then glues these local invariant quotients along the groupoid. The delicate part is that the local groups G vary and the gluing must be shown to be effective in the étale topology; the verification that the local quotients patch to a global algebraic space, and that π has the stated properties, uses the theory of finite flat groupoids and is carried out in full in [38] and, in the modern stack-theoretic language, in [59]. The argument is beyond the scope of the present treatment, which is why it is cited rather than reproduced; the universal property (1) and the bijection (2) are the facts this book uses and are used below without further comment. That X becomes a scheme under the quasi-projective hypotheses is the classical statement, going back to the GIT construction of coarse moduli (proposition 13.4.6), where the coarse space is cut out as a projective quotient and is manifestly a scheme.

The universal property is the precise sense in which X is the "best scheme approximation" to $\mathcal{X}$: it is the terminal object under $\mathcal{X}$ in the category of algebraic spaces, so any classical invariant one might compute by mapping $\mathcal{X}$ to a space factors through X. What π forgets is exactly the automorphisms: it records only the isomorphism class of an object, discarding its automorphism group, and the residual gerbe $B\underline{\mathrm{Aut}}(x)$ over a point of X is precisely the fiber of π that the coarse space cannot resolve. The condition $\mathcal{O}_X \cong \pi_* \mathcal{O}_{\mathcal{X}}$ records that π is a coarse space and not a mere continuous map: functions on $\mathcal{X}$ are the same as functions on X, so nothing is lost at the level of the structure sheaf, only at the level of automorphisms.

Two subtleties are worth flagging. First, the coarse space is an *algebraic space*, not always a scheme; the passage from the previous section's algebraic spaces to this section's stacks is what makes the non-scheme case unavoidable, since the coarse space of a DM stack can be a non-scheme algebraic space of exactly the shape produced in example 20.1.6. It becomes a scheme under the projective hypotheses that hold for the moduli of curves, which is why the classical M_g and $\overline{M}_g$ are quasi-projective schemes even though the general Keel-Mori coarse space need not be. Second, the finiteness of the inertia is essential and is exactly the DM (indeed the separated-DM) condition made quantitative: a stack with positive-dimensional automorphisms, such as $B\mathbb{G}_m$, has no coarse space in this sense, because collapsing a $\mathbb{G}_m$ to a point destroys too much, and the theory of the next section (Artin stacks) is where such stacks are handled.

The section's examples are the ones the whole book points toward. The moduli of elliptic curves, the moduli of curves of higher genus, and its Deligne-Mumford compactification are the canonical Deligne-Mumford stacks, and in each case the DM property is verified by the automorphism computation of theorem 20.2.3 rather than by constructing an atlas directly. The automorphisms that obstructed a fine moduli *scheme* in Chapter 1 (example 13.3.4) are precisely the finite groups that make these stacks Deligne-Mumford rather than schemes.

Example 20.2.5: The Moduli of Curves as DM Stacks

Work over a base S, and impose in each item the characteristic condition noted.

1. $\mathcal{M}_{1,1}$*, the moduli of elliptic curves.* Over $S = \operatorname{Spec} k$ with $\operatorname{char} k \neq 2, 3$, the stack $\mathcal{M}_{1,1}$ is the quotient stack $[(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ of example 19.3.5. The automorphism group of an elliptic curve E over an algebraically closed field is its group of automorphisms fixing the origin: generically $\mu_2 = \{\pm 1\}$ (the inversion $[-1]$), enlarging to μ_4 at $j = 1728$ and μ_6 at $j = 0$, as recorded in example 13.3.4. Each of these groups is finite, and each is reduced because $\operatorname{char} k \neq 2, 3$ keeps their orders prime to the characteristic, so the group schemes are étale by order count. By theorem 20.2.3, $\mathcal{M}_{1,1}$ is a Deligne-Mumford stack. Its coarse space is the j-line $\mathbb{A}^1_j = \operatorname{Spec} k[j]$ (theorem 13.3.6), recovered here as the Keel-Mori coarse space $\pi\colon \mathcal{M}_{1,1} \to \mathbb{A}^1_j$: the map π sends a curve to its j-invariant, is a bijection on geometric points (elliptic curves over $\bar{k}$ are classified up to isomorphism by j), and forgets the automorphism groups, which survive as the residual gerbes $B\mu_2$ generically, $B\mu_4$ over $j = 1728$, and $B\mu_6$ over $j = 0$.

2. $\mathcal{M}_g$*, the moduli of smooth curves of genus* $g \geq 2$*.* A smooth projective curve C of genus $g \geq 2$ over an algebraically closed field has a *finite* automorphism group: by the theorem of Hurwitz its order is bounded by $84(g-1)$, and in particular $\underline{\mathrm{Aut}}(C)$ is finite. Over a base where the relevant orders are prime to the residue characteristics (automatically over $\mathbb{Q}$), these groups are also reduced, so by theorem 20.2.3 the stack $\mathcal{M}_g$ is a Deligne-Mumford stack. Its Keel-Mori coarse space is the classical coarse moduli space M_g, a quasi-projective scheme by the GIT construction (proposition 13.4.6) that presents M_g as a projective quotient of a locus in a Hilbert scheme of pluricanonically embedded curves. The automorphisms are exactly what prevents M_g from being a fine moduli space: a curve with automorphisms cannot admit a universal family in a Zariski neighborhood, so no scheme carries the universal curve, and the stack is the object that does.

3. $\overline{\mathcal{M}}_g$*, the Deligne-Mumford compactification.* The compactification $\overline{\mathcal{M}}_g$ parametrizes *stable curves* of genus g: connected nodal projective curves C with at worst nodal singularities, arithmetic genus g, and finite automorphism group, the last condition being equivalent (for $g \geq 2$) to the ampleness of the dualizing sheaf ω_C, that is, to every smooth rational component meeting the rest of C in at least three points. Deligne and Mumford introduced this notion

in [23] precisely so that the compactified moduli space would remain a Deligne-Mumford stack: the stability condition is exactly the demand that $\underline{\mathrm{Aut}}(C)$ be finite, so theorem 20.2.3 applies verbatim and $\overline{\mathcal{M}}_g$ is a DM stack (over $\mathbb{Q}$, or with the usual characteristic caveats). Its coarse space is the projective scheme $\overline{M}_g$, and $\overline{\mathcal{M}}_g$ is smooth and proper as a stack even though $\overline{M}_g$ has quotient singularities, a point taken up in section 20.5 and in the capstone chapter.

In every case the DM property is a statement about the finiteness of automorphism groups, checked object by object, and the stack is the correct moduli object precisely because those finite automorphisms, which obstruct a fine moduli scheme, are recorded rather than discarded.

The three examples are one phenomenon at three levels of difficulty. The moduli of elliptic curves is the smallest case, where the whole stack is a single explicit quotient $[(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ and every automorphism group can be read off a Weierstrass equation. The moduli of smooth curves of higher genus is the case the theory was built for, where the finiteness of automorphisms is a nontrivial input (Hurwitz's bound) but the DM property still follows from a single application of the recognition theorem. The compactification is where the definition of the objects is engineered to preserve the DM property: stability is not a geometric afterthought but the exact condition that keeps automorphism groups finite, and this is why the Deligne-Mumford boundary is the right one. The coarse spaces (the j-line, M_g, and $\overline{M}_g$) are the classical objects, and the stacks are the refinements that see the automorphisms the coarse spaces forget.

The passage from finite to positive-dimensional automorphism groups is the subject of the next section. When a moduli problem has objects whose automorphism groups are positive-dimensional, such as the coherent sheaves whose automorphisms contain $\mathbb{G}_m$, no étale atlas exists and the DM condition fails; the atlas can then only be taken smooth, and the resulting class of stacks is the Artin (algebraic) stacks, of which Deligne-Mumford stacks are the special case of an unramified atlas.

Exercise 20.2.1

1. Let $\mathcal{X}$ be a stack over $(\mathbf{Sch}/S)_{\mathrm{et}}$ with an étale atlas $u\colon U \to \mathcal{X}$, and let $R = U \times_{\mathcal{X}} U$. Show that the two projections $s, t\colon R \to U$ are étale, and that the fiber of $(s,t)\colon R \to U \times_S U$ over a point (a,a) on the diagonal is the automorphism group scheme $\underline{\mathrm{Aut}}(u(a))$ of the object $u(a)$. Conclude that if $\mathcal{X}$ is DM then every such automorphism group is a finite étale scheme over the residue field.
2. Verify directly that a scheme X, viewed as a stack (a category fibered in sets), is a Deligne-Mumford stack. Identify its étale atlas, describe the groupoid $R \rightrightarrows U$, and confirm that every automorphism group is trivial, so the unramified-diagonal criterion of theorem 20.2.3 holds vacuously. What is its Keel-Mori coarse space?
3. For G a finite group (a finite constant group scheme) acting on a scheme X over a field k, show that the quotient stack $[X/G]$ is a Deligne-Mumford stack by exhibiting an étale atlas and computing the automorphism groups of its points. When is it separated with finite inertia, so that Keel-Mori applies, and what is the coarse space $[X/G] \to X/G$ then? Contrast this with $B\mathbb{G}_m$, and explain in one sentence why the latter is *not* a DM stack.
4. Let k be a field of characteristic $p > 0$ and consider $\mu_p = \operatorname{Spec} k[t]/(t^p - 1)$ as a group scheme over k. Show that μ_p is finite but *not* reduced (not étale) over k. Deduce that a stack having an object whose automorphism group scheme is μ_p fails the reducedness

condition of theorem 20.2.3(3) and so is *not* Deligne-Mumford in characteristic p, even though the automorphism group is finite. Relate this to the caveat "char $k \neq 2, 3$" in the $\mathcal{M}_{1,1}$ computation of example 20.2.5.

5. Using the Hurwitz bound $|\underline{\mathrm{Aut}}(C)| \leq 84(g-1)$ for a smooth projective curve C of genus $g \geq 2$ over $\bar{k}$ with char $\bar{k} = 0$, together with theorem 20.2.3, give the complete argument that $\mathcal{M}_g$ is a Deligne-Mumford stack. Which hypothesis of theorem 20.2.3 does the Hurwitz bound supply, and which hypotheses (representable diagonal, existence of a smooth atlas) are supplied elsewhere?

6. Let $\mathcal{X}$ be a separated DM stack of finite type over a Noetherian base with finite inertia, and let $\pi \colon \mathcal{X} \to X$ be its Keel-Mori coarse space. Show that a morphism $\mathcal{X} \to Y$ to a scheme Y factors uniquely through π, and deduce that two DM stacks with the same objects up to isomorphism (the same groupoid of geometric points) but different automorphism groups can have the *same* coarse space. Illustrate with $\mathcal{M}_{1,1}$ and its rigidification $\mathcal{M}_{1,1} /\!\!/ \mu_2$ (the μ_2-gerbe removed), which share the coarse j-line.

20.3 Algebraic Stacks

The previous section imposed geometry on a stack by demanding an *étale* atlas: a Deligne-Mumford stack (definition 20.2.1) is one presented by a scheme mapping étale-surjectively onto it, and the price of the étale hypothesis is that automorphism groups must be finite and reduced. That hypothesis is exactly what fails for the moduli problems this book has been circling around. A vector bundle has the automorphism group GL_n; a coherent sheaf has at least the scalars $\mathbb{G}_m$ inside its automorphisms; a principal G-bundle has the group G itself. These automorphism groups are positive dimensional, so no étale atlas can exist, and the moduli stacks of such objects lie outside the Deligne-Mumford world. The remedy is to relax the atlas from étale to *smooth.* A smooth morphism is submersive with smooth fibers of arbitrary dimension, so a smooth atlas can absorb the positive-dimensional automorphisms that an étale atlas cannot. The resulting objects are the *algebraic stacks* of Artin, and they are the widest class of geometric objects in this book: every Deligne-Mumford stack, every algebraic space, and every scheme is one, and the moduli of coherent sheaves and of principal bundles finally find their home here.

This section makes the definition, proves that the quotient stacks $[X/G]$ built in chapter 19 are algebraic (with $X \to [X/G]$ the smooth atlas), records how dimension and points are read off a presentation, and exhibits $\mathfrak{Coh}_X$ and $\mathfrak{Bun}_G$ as the motivating Artin, non-Deligne-Mumford examples. The passage from étale to smooth is the last enlargement of the geometric category the book needs; section 20.4 then gives the criterion that decides which stacks are algebraic, and the capstone chapter builds $\mathcal{M}_g$ inside this framework.

Throughout, the base site is $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$ (the fppf topology is the natural one for smooth atlases, since a smooth surjection is an fppf cover), and all stacks are quasi-separated and locally of finite presentation over a locally Noetherian base S unless stated otherwise. The two conditions that make a stack algebraic are a representable diagonal, encoding that its $\underline{\mathrm{Isom}}$ sheaves are geometric objects, and the existence of a smooth atlas.

Definition 20.3.1: Algebraic (Artin) Stack

Let $\mathcal{X}$ be a stack in groupoids over $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$. Then $\mathcal{X}$ is an *algebraic stack* (or *Artin stack*) if:

1. the diagonal $\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ is representable, quasi-compact, and separated (equivalently, for all schemes T and all objects $x, y \in \mathcal{X}(T)$, the sheaf $\underline{\mathrm{Isom}}_T(x, y)$ on $\mathcal{C}/T$ is an algebraic space, quasi-compact and separated over T); and

2. there exist a scheme U and a morphism $u \colon U \to \mathcal{X}$ that is representable, smooth, and surjective. Such a u is called a *smooth atlas* (or *presentation*) of $\mathcal{X}$.

A *Deligne-Mumford stack* (definition 20.2.1) is the special case in which the atlas u may be taken *étale*; the étale hypothesis is a strengthening of "smooth", so every Deligne-Mumford stack is algebraic.

The two clauses of the definition play complementary roles, and it is worth separating them. Clause (1), representability of the diagonal, is the condition that makes the notion of a *representable* morphism $u \colon U \to \mathcal{X}$ meaningful: by theorem 19.4.6, a morphism from a scheme T to $\mathcal{X}$ has representable fibers exactly when $\Delta_{\mathcal{X}}$ is representable, so without clause (1) one could not even ask that the atlas u in clause (2) be smooth (a property that is tested on the fibers $U \times_{\mathcal{X}} T$, which must be schemes or algebraic spaces for the test to make sense). Clause (1) is therefore logically prior: it is what lets the automorphisms of objects be geometric objects in their own right, and it is half of algebraicity all by itself. Clause (2) is the geometric input proper: it says $\mathcal{X}$ has "enough charts", a scheme U covering it smoothly, so that any question about $\mathcal{X}$ can be pulled back to a question about the scheme U and its self-intersection groupoid.

That self-intersection is the structure to keep in view. Given a smooth atlas $u \colon U \to \mathcal{X}$, form the fiber product

$$R = U \times_{\mathcal{X}} U$$

which, by clause (1), is an algebraic space (in fact a scheme when $\Delta_{\mathcal{X}}$ is representable by schemes), together with the two projections $s, t \colon R \rightrightarrows U$. Both projections are smooth, being base changes of the smooth u, and $R \rightrightarrows U$ is a groupoid in algebraic spaces: it has an identity $U \to R$ (the diagonal of u), an inverse $R \to R$ (swapping the two factors), and a composition $R \times_{s,U,t} R \to R$. The stack $\mathcal{X}$ is recovered as the stackification of this groupoid, exactly as $[X/G]$ was recovered from the groupoid $G \times_S X \rightrightarrows X$ in chapter 19. This is the precise sense in which an algebraic stack is "a scheme up to a smooth groupoid": the pair (U, R) is the presentation, and $\mathcal{X} = [U/R]$ is the quotient. The smooth-groupoid presentation is the working object for every computation on a stack, from dimension to quasi-coherent sheaves.

The definition would be inert without a supply of examples, and the richest supply is the one already constructed. Chapter 19 built the quotient stack $[X/G]$ for a group scheme G acting on a scheme X, and produced the morphism $q \colon X \to [X/G]$ classifying the tautological torsor. That morphism is the atlas: the theorem below shows q is smooth and surjective, so $[X/G]$ is algebraic whenever G is a smooth affine group scheme. This is the mechanism that makes BG, $B\mathbb{G}_m$, BGL_n, and $\mathcal{M}_{1,1}$ into geometric objects, and it is the source of essentially every algebraic stack met in practice, since a stack with a smooth atlas is, locally, a quotient stack.

Theorem 20.3.2: Quotient Stacks Are Algebraic

Let G be a smooth, separated group scheme of finite presentation over S acting on an S-scheme X. (Affineness of G is not needed for (1) and (2); it is used only where part (3) says so.) Then:

1. the quotient stack $[X/G]$ is an algebraic stack over $\mathcal{C} = (\mathbf{Sch}/S)_{\mathrm{fppf}}$;
2. the tautological morphism $q\colon X \to [X/G]$ of theorem 19.3.3 is a G-torsor, hence representable, smooth, and surjective; that is, q is a smooth atlas, and $R = X \times_{[X/G]} X \cong G \times_S X$ is the presentation groupoid;
3. in particular the classifying stack $BG = [\mathrm{pt}/G]$ is algebraic, with atlas $\mathrm{pt} = S \to BG$ and presentation groupoid $G \rightrightarrows \mathrm{pt}$; conversely, any algebraic stack with a smooth atlas $u\colon U \to \mathcal{X}$ is, Zariski-locally on U, of the form $[U'/R']$ for a smooth groupoid $R' \rightrightarrows U'$, and if moreover $\mathcal{X}$ has affine stabilizers (affine diagonal) it is étale-locally on the base a quotient stack $[V/\mathrm{GL}_N]$.

***Proof*:**
The three parts are proved in turn. Part (2) does the geometric work (identifying q and its base change R), part (1) then reads off algebraicity from the diagonal and the atlas, and part (3) specializes and reverses the construction.

Step 1: The atlas q is a G-torsor. By theorem 19.3.3(2), the tautological morphism $q\colon X \to [X/G]$ has the property that for every object $(P, f) \in [X/G](T)$, corresponding to a morphism $\xi\colon T \to [X/G]$, the fiber product satisfies

$$X \times_{[X/G],\xi} T \cong P$$

as G-torsors over T. In particular q is *representable*: its base change along any morphism $T \to [X/G]$ from a scheme is the scheme (indeed G-torsor) P. Moreover this base change $P \to T$ is a G-torsor, so it is faithfully flat and of finite presentation, and, because G is smooth over S, the torsor $P \to T$ is smooth (fppf-locally on T it is the projection $G \times_S T \to T$, which is smooth as a base change of the smooth $G \to S$, and smoothness is fppf-local on the base by theorem 18.4.3). A G-torsor is surjective (again fppf-locally it is $G \times_S T \to T$, which is surjective). Smoothness and surjectivity of q are properties of a representable morphism tested after base change to a scheme (definition 19.4.3), and they hold on every such base change, so q is smooth and surjective.

For the presentation groupoid, base-change q along itself. Taking $T = X$ and $\xi = q$ (which classifies the trivial torsor $G \times_S X \to X$ with equivariant map σ), the identity above gives

$$R = X \times_{[X/G]} X \cong G \times_S X$$

the trivial torsor pulled back, with the two projections $s, t\colon G \times_S X \rightrightarrows X$ being the action $\sigma\colon (g, x) \mapsto g \cdot x$ and the projection $\mathrm{pr}_X\colon (g, x) \mapsto x$. This is exactly the action groupoid of G on X.

Step 2: $[X/G]$ is algebraic. Two conditions must be checked: representability of the diagonal, and existence of a smooth atlas. The atlas is $q\colon X \to [X/G]$ from Step 1, which is a smooth surjection from a scheme. It remains to check that $\Delta_{[X/G]}$ is representable, quasi-compact, and separated.

The diagonal of $[X/G]$ is computed by the Isom sheaves. For a scheme T and two objects $(P,f),(P',f') \in [X/G](T)$, the sheaf $\underline{\mathrm{Isom}}_T((P,f),(P',f'))$ assigns to a T-scheme T' the set of G-torsor isomorphisms $P|_{T'} \to P'|_{T'}$ commuting with the maps to X. This sheaf is representable by a scheme affine over T: locally on T where both torsors trivialize, an isomorphism of trivial (left) G-torsors $G \times_S T' \to G \times_S T'$ is right-translation $g \mapsto g\gamma$ by a section $\gamma \in G(T')$ (this is the general form of an automorphism of the trivial torsor), and compatibility with the equivariant maps $g \mapsto g \cdot x$ and $g \mapsto g \cdot x'$ to X forces $g\gamma \cdot x' = g \cdot x$ for all g, that is, $\gamma \cdot x' = x$. Thus Isom is cut out inside the transporter

$$\mathrm{Transp}_G(x', x) = \{\gamma \in G : \gamma \cdot x' = x\}$$

a subscheme of $G \times_S T$ (the preimage of the diagonal under $(\gamma, t) \mapsto (\gamma \cdot x'(t), x(t))$), hence a scheme of finite presentation over T. Note that only representability is wanted here, so it is enough that $G \times_S T$ be a scheme; affineness of G would give an affine transporter, but the definition of an algebraic stack asks the diagonal to be representable, quasi-compact and separated, not affine. Separatedness and quasi-compactness of the transporter over T follow from the corresponding properties of $G \times_S T \to T$, which hold because $G \to S$ is separated by hypothesis and quasi-compact, being of finite presentation. Gluing these local descriptions along the trivializing cover (the identifications are torsor isomorphisms, which glue by descent, theorem 18.3.4), the sheaf Isom is representable by a scheme quasi-compact and separated over T. By theorem 19.4.6, representability of all these Isom sheaves is precisely representability of the diagonal $\Delta_{[X/G]}$, and the quasi-compactness and separatedness of the sheaves give the corresponding properties of Δ. Thus $\Delta_{[X/G]}$ is representable, quasi-compact, and separated, and with the smooth atlas q this establishes that $[X/G]$ is algebraic.

Step 3: BG, and the converse. Applying parts (1) and (2) with $X = S = \mathrm{pt}$ (the terminal object, with G acting trivially) gives that $BG = [\mathrm{pt}/G]$ is algebraic. Its atlas is $q\colon \mathrm{pt} = S \to BG$ (classifying the trivial torsor $G \to S$), and the presentation groupoid is $R = \mathrm{pt} \times_{BG} \mathrm{pt} \cong G \times_S \mathrm{pt} = G$, with both projections the structure map $G \to S$; that is, $G \rightrightarrows \mathrm{pt}$ with the multiplication as composition. This is the group G regarded as a one-object groupoid, whose stackification is BG.

For the converse, let $\mathcal{X}$ be algebraic with smooth atlas $u\colon U \to \mathcal{X}$ and groupoid $R = U \times_{\mathcal{X}} U \rightrightarrows U$. Working Zariski-locally on U, choose an open $U' \subseteq U$ and set $R' = R \times_{U \times U} (U' \times U')$; then $R' \rightrightarrows U'$ is a smooth groupoid in algebraic spaces and $\mathcal{X}|_{u(U')} \cong [U'/R']$ is its stackification, which is the assertion that $\mathcal{X}$ is locally a groupoid quotient. To upgrade this to a genuine quotient stack $[V/G]$ one needs R' to be an action groupoid; this holds étale-locally after a further construction. Where $\mathcal{X}$ has affine (or quasi-affine) diagonal and admits a locally free sheaf trivializing the atlas, the frame bundle of that sheaf is a GL_N-torsor $V \to \mathcal{X}$ from a scheme, and $\mathcal{X} \cong [V/\mathrm{GL}_N]$; the general local-quotient statement (that every algebraic stack with affine stabilizers is étale-locally a quotient stack $[V/\mathrm{GL}_N]$) is a structural theorem whose full proof uses the local structure of algebraic stacks and is beyond the present scope, cited to [59] and [63]. The smooth-groupoid presentation $\mathcal{X} = [U/R]$, on the other hand, is immediate from the atlas and holds for every algebraic stack, as we sought to show.

The theorem is what the whole chapter uses: it converts the categorical construction $[X/G]$ of chapter 19 into a geometric object with charts, and it does so by identifying the atlas $X \to [X/G]$ as a torsor, hence smooth. Two features deserve comment. First, the smoothness of the atlas is inherited directly from the smoothness of the group G: a G-torsor is smooth exactly when G is, which is why the theorem requires G smooth and why the positive-dimensional automorphism groups of coherent sheaves and bundles pose no obstruction to algebraicity, only to Deligne-Mumfordness. Second, the presentation groupoid $R \cong G \times_S X$ is the action groupoid, so the dimension of the stack is legible from the group: R is larger than $U = X$ by exactly $\dim G$, and the next definition turns this into the formula $\dim[X/G] = \dim X - \dim G$. The converse in part (3) is the reason quotient stacks are not a special case but the general picture: every algebraic stack with affine stabilizers is locally of the form $[V/\mathrm{GL}_N]$, so understanding quotient stacks is, locally, understanding all algebraic stacks with affine diagonal (the abelian-variety classifying stacks BA, whose stabilizers are proper, fall outside this local model).

Before the general examples, the smallest instances anchor the definition. The classifying stacks $B\mathbb{G}_m$ and BGL_n, built as groupoids in chapter 19, are now algebraic stacks by the theorem, and they are the simplest algebraic stacks that are *not* Deligne-Mumford.

Example 20.3.3: $B\mathbb{G}_m$ and BGL_n as Algebraic Stacks

Over $S = \operatorname{Spec} k$, the group $\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$ is smooth affine of dimension 1, so by theorem 20.3.2 the classifying stack $B\mathbb{G}_m$ is algebraic. Its atlas is the point $\mathrm{pt} \to B\mathbb{G}_m$ and its presentation groupoid is $\mathbb{G}_m \rightrightarrows \mathrm{pt}$. The atlas $\mathrm{pt} \to B\mathbb{G}_m$ is smooth (a $\mathbb{G}_m$-torsor, and $\mathbb{G}_m$ is smooth) and surjective, but it is not unramified: the automorphism group of the unique geometric point is $\mathbb{G}_m$ itself, which is positive dimensional, so the diagonal

$$\Delta\colon B\mathbb{G}_m \to B\mathbb{G}_m \times_k B\mathbb{G}_m$$

is not unramified, and $B\mathbb{G}_m$ is not a Deligne-Mumford stack (by theorem 20.2.3). No étale atlas exists: an étale morphism has discrete (finite) fibers, but a $\mathbb{G}_m$-torsor has one-dimensional fibers.

The same holds for BGL_n: the group GL_n is smooth affine of dimension n^2, so BGL_n is algebraic with atlas $\mathrm{pt} \to BGL_n$ and groupoid $GL_n \rightrightarrows \mathrm{pt}$, and the automorphism group of its point is GL_n, of dimension $n^2 > 0$, so BGL_n is not Deligne-Mumford. By contrast, for a *finite* group G (a smooth group scheme of dimension 0 in characteristic not dividing $|G|$) the atlas $\mathrm{pt} \to BG$ is étale, and BG is Deligne-Mumford: the distinction between Deligne-Mumford and Artin is exactly the distinction between $\dim G = 0$ and $\dim G > 0$ for the automorphism groups.

The example isolates the mechanism cleanly: BG is Deligne-Mumford precisely when G is finite, and Artin (but not Deligne-Mumford) when G is positive dimensional. The dimension of the automorphism group is the sole obstruction to the étale atlas, and the smooth atlas ignores it. This suggests that dimension is measured not by the atlas alone but by the atlas relative to the groupoid, and the following definition makes that precise.

Definition 20.3.4: Dimension and Points of an Algebraic Stack

Let $\mathcal{X}$ be an algebraic stack over S with smooth atlas $u\colon U \to \mathcal{X}$ and presentation groupoid $R = U \times_{\mathcal{X}} U \rightrightarrows U$.

1. The *dimension* of $\mathcal{X}$ at a point is

$$\dim_x \mathcal{X} = \dim_{\tilde{x}} U - \dim_{\tilde{x}}(u\text{-fiber})$$

where $\tilde{x} \in U$ lies over x and the u-fiber is the fiber of u through $\tilde{x}$; equivalently, since $s\colon R \to U$ is smooth of relative dimension equal to the fiber dimension of u,

$$\dim \mathcal{X} = \dim U - (\dim R - \dim U) = 2\dim U - \dim R$$

the last equality holding when U, R are equidimensional. For a quotient stack this specializes (see the remark below) to

$$\dim[X/G] = \dim X - \dim G$$

and in particular $\dim BG = -\dim G$.

2. The *underlying topological space* $|\mathcal{X}|$ is the set of isomorphism classes of points $x\colon \operatorname{Spec} K \to \mathcal{X}$ over fields K, where $x\colon \operatorname{Spec} K \to \mathcal{X}$ and $x'\colon \operatorname{Spec} K' \to \mathcal{X}$ are identified when there is a common field extension over which they become 2-isomorphic; equivalently $|\mathcal{X}| = |U|/|R|$ is the quotient of $|U|$ by the equivalence relation cut out by the image of $|R| \to |U| \times |U|$. It carries the quotient topology, for which $|u|\colon |U| \to |\mathcal{X}|$ is open.

3. The *residual gerbe* at a point $x \in |\mathcal{X}|$ with field of definition K is the reduced locally closed substack $\mathcal{G}_x \hookrightarrow \mathcal{X}$ through x; it is a gerbe over $\operatorname{Spec} K$ (banded by the automorphism group scheme $\underline{\operatorname{Aut}}(x)$), and it is the intrinsic remnant of the automorphisms at x. A point of $\mathcal{X}$ is thus not a bare point but a point together with its residual gerbe $B\underline{\operatorname{Aut}}(x)$.

Three points about the definition are worth drawing out. First, the dimension formula. The atlas u is smooth, and its fibers have the dimension of the automorphism groups of the objects being covered; for a quotient stack $[X/G]$ the atlas is $q\colon X \to [X/G]$ with fibers G-torsors, so the fiber dimension is $\dim G$, and $\dim[X/G] = \dim X - \dim G$. The subtraction is the point: a stack can have *negative* dimension, and BG does, with $\dim BG = -\dim G$. This is not a pathology but the correct bookkeeping: BG has a single point with a $\dim G$-dimensional automorphism group, and the automorphisms count against the dimension exactly as the parameters count for it. The Hilbert scheme, whose objects have no automorphisms, has atlas fibers of dimension 0 and so $\dim \mathrm{Hilb}$ equals the naive parameter count; the moduli stack $\mathcal{M}_g$, whose objects have finite automorphism groups (dimension 0), has $\dim \mathcal{M}_g = 3g-3$ equal to the deformation count, with the automorphisms invisible to dimension but visible to the stack structure.

Second, the topological space $|\mathcal{X}|$. It is the set of R-equivalence classes of points of the atlas, and it forgets everything about the automorphisms, retaining only which points are identified. For BG it is a single point; for $[X/G]$ it is the orbit space $|X|/|G|$ with its quotient topology; for $\mathcal{M}_{1,1}$ it is the underlying set of the j-line. The space $|\mathcal{X}|$ is where a stack looks like its coarse space, and it is the object on which one defines irreducibility, connectedness, and the dimension of an irreducible component.

Third, the residual gerbe. This is the datum that a point of a stack carries beyond its location in $|\mathcal{X}|$: at each point sits the classifying stack $B\underline{\mathrm{Aut}}(x)$ of its automorphism group, embedded in $\mathcal{X}$ as the residual gerbe. For $\mathcal{M}_{1,1}$ over a generic j-value the residual gerbe is $B\mu_2$ (the automorphism $[-1]$), and over $j = 0, 1728$ it jumps to $B\mu_6, B\mu_4$, recovering the picture of chapter 19. The residual gerbe is the local model for the stack near a point, and it is what the coarse space $|\mathcal{X}|$ discards. The systematic study of these gerbes is the subject of the next chapter.

Example 20.3.5: Dimensions of BG, $[X/G]$, and $\mathcal{M}_{1,1}$

The dimension formula is read off the presentation in each case.

1. $\dim B\mathbb{G}_m = -\dim \mathbb{G}_m = -1$. The atlas $\mathrm{pt} \to B\mathbb{G}_m$ has $\dim \mathrm{pt} = 0$ and the atlas fiber is a $\mathbb{G}_m$-torsor over pt, of dimension 1, so $\dim B\mathbb{G}_m = 0 - 1 = -1$. Similarly $\dim BGL_n = -n^2$ and $\dim BG = 0$ for a finite group G.

2. For $\mathcal{M}_{1,1} = [(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ with the weight-$(4, 6)$ action (example 19.3.5), the atlas is $X = \mathbb{A}^2 \setminus \{\Delta = 0\}$ of dimension 2 and $G = \mathbb{G}_m$ of dimension 1, so
$$\dim \mathcal{M}_{1,1} = \dim X - \dim \mathbb{G}_m = 2 - 1 = 1$$
matching the one-parameter family of elliptic curves recorded by the j-line. The single generic automorphism μ_2 is 0-dimensional and does not affect the dimension; it is finite, consistent with $\mathcal{M}_{1,1}$ being Deligne-Mumford of dimension 1.

3. For the affine plane modulo GL_2, $[\mathbb{A}^2/\mathrm{GL}_2]$ has $\dim = 2 - 4 = -2$, reflecting that the GL_2-action on $\mathbb{A}^2$ has a dense orbit (the nonzero vectors) with two-dimensional stabilizer, so the stack is concentrated at essentially one point with a large automorphism group.

The dimension examples show the formula behaving as bookkeeping should: parameters add, automorphisms subtract, and the balance is the dimension of the stack. With dimension and points in hand, the section reaches the examples that motivated the enlargement from étale to smooth atlases in the first place. These are the moduli stacks whose objects have positive-dimensional automorphism groups: the stack of coherent sheaves and the stack of principal bundles. They are the prototypical Artin, non-Deligne-Mumford stacks, and their construction is the reason algebraic stacks in Artin's sense exist.

The stack $\mathfrak{Coh}_X$ of coherent sheaves on a projective scheme X has, as its T-points, the flat families of coherent sheaves on $X \times T$ over T. A single coherent sheaf $\mathcal{F}$ has automorphism group $\underline{\mathrm{Aut}}(\mathcal{F}) = \mathrm{End}_{\mathcal{O}_X}(\mathcal{F})^\times$, the units of the endomorphism algebra, an open subscheme of the affine space $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{F})$, which always contains the scalars $\mathbb{G}_m$ (multiplication by a unit) and is positive dimensional. The presence of $\mathbb{G}_m$ in every automorphism group is the structural reason $\mathfrak{Coh}_X$ can never be Deligne-Mumford: its diagonal is not unramified. Yet it is algebraic, because it is covered by Quot schemes.

Example 20.3.6: $\mathfrak{Coh}_X$ and $\mathfrak{Bun}_G$: the Motivating Artin Stacks

Let X be a projective scheme over $S = \operatorname{Spec} k$ and let G be a smooth affine group scheme over k.

The stack of coherent sheaves. The stack $\mathfrak{Coh}_X$ has $\mathfrak{Coh}_X(T)$ the groupoid of T-flat coherent sheaves on $X_T = X \times_k T$, with morphisms the $\mathcal{O}_{X_T}$-isomorphisms. This is a stack for the fppf topology: quasi-coherent sheaves satisfy descent (theorem 18.3.4), and flatness and coherence are fppf-local. Its diagonal is representable: for two families $\mathcal{F}, \mathcal{G}$ over T, the sheaf $\underline{\mathrm{Isom}}_T(\mathcal{F}, \mathcal{G})$ is an open subscheme of the linear scheme $\underline{\mathrm{Hom}}_T(\mathcal{F}, \mathcal{G}) = \mathrm{Spec}_T \mathrm{Sym}\, \mathcal{H}om(\mathcal{F}, \mathcal{G})^\vee$ (the automorphisms being the invertible homomorphisms), hence affine of finite type over T; so Δ is representable, quasi-compact, and separated.

For the atlas, fix a numerical type (a Hilbert polynomial P) and an integer m. By boundedness (theorem 14.2.5), the family of coherent sheaves with Hilbert polynomial P is bounded, so for $m \gg 0$ every such sheaf $\mathcal{F}$ is m-regular, meaning $\mathcal{F}(m)$ is globally generated with $H^i(X, \mathcal{F}(m)) = 0$ for $i > 0$ and $h^0(X, \mathcal{F}(m)) = P(m)$. A choice of surjection $\mathcal{O}_X(-m)^{\oplus P(m)} \twoheadrightarrow \mathcal{F}$ is then a point of the Quot scheme $\mathrm{Quot}(\mathcal{O}_X(-m)^{\oplus P(m)}, P)$, and forgetting the choice of surjection gives a morphism

$$\mathrm{Quot}\big(\mathcal{O}_X(-m)^{\oplus P(m)}, P\big) \longrightarrow \mathfrak{Coh}_X^{\le P, m}$$

onto the open substack of m-regular sheaves with Hilbert polynomial P. This morphism is a $\mathrm{GL}_{P(m)}$-torsor (the fiber over $\mathcal{F}$ is the choice of basis of $H^0(X, \mathcal{F}(m))$, a $\mathrm{GL}_{P(m)}$-torsor), hence smooth and surjective; ranging over P and m, these Quot schemes cover $\mathfrak{Coh}_X$ by smooth morphisms from schemes. Thus $\mathfrak{Coh}_X$ is an algebraic stack, presented as

$$\mathfrak{Coh}_X^{\le P, m} \cong \big[\mathrm{Quot}(\mathcal{O}_X(-m)^{\oplus P(m)}, P) / \mathrm{GL}_{P(m)}\big]$$

a quotient stack on each numerical piece. Its automorphism groups contain $\mathbb{G}_m$, so it is Artin and not Deligne-Mumford. The full verification that these charts glue into a global algebraic stack is carried out via Artin's criteria (section 20.4); a complete treatment is in [41] and [59].

The stack of G-bundles. The stack $\mathfrak{Bun}_G$ has $\mathfrak{Bun}_G(T)$ the groupoid of G-torsors (principal G-bundles) on X_T, flat and of finite presentation over T, with morphisms the isomorphisms of G-torsors. When $G = GL_n$ this is the stack $\mathfrak{Bun}_{GL_n}$ of rank-n vector bundles on X. Its T-points have automorphism groups the global automorphisms of a G-bundle, which contain the center $Z(G) \supseteq \mathbb{G}_m$ (for $G = GL_n$) and are positive dimensional, so $\mathfrak{Bun}_G$ is Artin and not Deligne-Mumford. It is algebraic: for a bounded family of bundles one obtains a smooth atlas from a Quot-scheme parametrization of the bundle together with a rigidifying frame, presenting $\mathfrak{Bun}_G$ locally as a quotient stack $[V/\mathrm{GL}_N]$ exactly as above. The full proof is via Artin's criteria, cited to [41], [59], and [63].

These two stacks are the reason the Artin theory exists. The moduli of coherent sheaves and of principal bundles are among the central objects of algebraic geometry, and their automorphism groups are irreducibly positive dimensional: every coherent sheaf has its scalar automorphisms $\mathbb{G}_m$, every vector bundle its GL_n-worth of frames. There is no way to rigidify these away while retaining the moduli problem, because the scalars are intrinsic to the linear-algebraic structure of sheaves. The Deligne-Mumford framework, which requires finite automorphism groups, simply cannot accommodate them, and the smooth atlas of the Artin framework is the minimal enlargement that does. The presentation as a quotient of a Quot scheme by GL_N is the concrete form the abstract atlas takes, and it is the recurring shape of every algebraic stack in practice: a bounded family of objects, parametrized by a scheme (Quot or Hilbert), modulo the group of choices (GL_N) that the parametrization introduced.

The pattern that emerges across the section is uniform. An algebraic stack is a scheme up to a smooth groupoid; the smooth atlas is a scheme U covering it, the presentation groupoid $R \rightrightarrows U$ records the automorphisms and identifications, and every geometric question about the stack is answered by pulling back to U and checking compatibility over R. The quotient stack $[X/G]$ is the model case, and the local-quotient theorem says it is not only a case but the general picture. What separates the Artin world from the Deligne-Mumford world is a single number, the dimension of the automorphism groups: finite for Deligne-Mumford, positive dimensional for Artin, and the smooth atlas is the device that makes the positive-dimensional case geometric.

Exercise 20.3.1

1. Show directly from definition 20.3.1 that every scheme U, regarded as a stack via its functor of points $\underline{U}$, is an algebraic stack, and that every algebraic space is an algebraic stack. (For the scheme case: the identity $U \to \underline{U}$ is an atlas, and the diagonal of a scheme is a closed immersion, hence representable, quasi-compact, and separated. For an algebraic space X, use its étale atlas $U \to X$ from definition 20.1.3, noting étale implies smooth.) Conclude the inclusions: schemes $\subseteq$ algebraic spaces $\subseteq$ Deligne-Mumford stacks $\subseteq$ algebraic stacks.

2. Let G be a smooth affine group scheme over k acting on X. Using theorem 20.3.2, identify the presentation groupoid $R \rightrightarrows U$ of $[X/G]$ explicitly, and verify that the two projections $s, t\colon R \to U$ are both smooth of relative dimension $\dim G$. Deduce the dimension formula $\dim[X/G] = \dim X - \dim G$ from definition 20.3.4, and check it against $\dim BG = -\dim G$ and $\dim \mathcal{M}_{1,1} = 1$.

3. The stack $B\mathbb{G}_m$ over $\operatorname{Spec} k$ has a single geometric point with automorphism group $\mathbb{G}_m$. Show that its diagonal $\Delta\colon B\mathbb{G}_m \to B\mathbb{G}_m \times_k B\mathbb{G}_m$ is representable by computing the fiber product $\mathrm{pt} \times_{B\mathbb{G}_m \times B\mathbb{G}_m} \mathrm{pt}$ over the two maps $\mathrm{pt} \to B\mathbb{G}_m$ classifying the trivial torsor, and identify this fiber product with $\mathbb{G}_m$. Explain why the positive dimension of this $\underline{\mathrm{Isom}}$ scheme is exactly the failure of Δ to be unramified, so that $B\mathbb{G}_m$ is algebraic but not Deligne-Mumford (contrast with $B\mu_n$).

4. Let X be a projective scheme and fix a Hilbert polynomial P. Using boundedness (theorem 14.2.5), explain why the open substack $\mathfrak{Coh}_X^{\leq P,m}$ of m-regular sheaves with Hilbert polynomial P (for $m \gg 0$) is a quotient stack $[\mathrm{Quot}/\mathrm{GL}_{P(m)}]$ as in example 20.3.6. Identify the automorphism group of the point of $\mathfrak{Coh}_X$ corresponding to a stable sheaf $\mathcal{F}$ (one with $\mathrm{End}(\mathcal{F}) = k$), and confirm it is $\mathbb{G}_m$, so that even the "best-behaved" sheaves keep a positive-dimensional automorphism group.

5. Show that a group homomorphism $\rho\colon G \to H$ of smooth affine group schemes induces a morphism of algebraic stacks $B\rho\colon BG \to BH$, and compute the effect on dimensions and on atlases: the atlas $\mathrm{pt} \to BG$ maps to the atlas $\mathrm{pt} \to BH$, and $\dim BG - \dim BH = \dim H - \dim G = -\dim\ker\rho + \dim\operatorname{coker}$ (interpret suitably for the determinant $\det\colon GL_n \to \mathbb{G}_m$, where $\ker = SL_n$). Verify $\dim BGL_n = -n^2$ and $\dim B\mathbb{G}_m = -1$ are consistent with $\dim BSL_n = -(n^2-1)$ via the fibration $BSL_n \to BGL_n \to B\mathbb{G}_m$.

6. Explain why $[\mathbb{A}^1/\mathbb{G}_m]$ (with $\mathbb{G}_m$ acting by scaling) is an algebraic stack, compute its dimension, and describe its two points (the open orbit $\mathbb{A}^1 \setminus 0$ and the origin) together with their automorphism groups and residual gerbes. Show that $|[\mathbb{A}^1/\mathbb{G}_m]|$ is the two-point Sierpiński space (a generic point whose closure is the whole space, plus a closed point $B\mathbb{G}_m$), and contrast this with the coarse space, which is a single point. Why does this stack fail to be separated?

20.4 Artin's Representability Criteria

The whole book converges here. Part IV posed moduli problems as functors of points and learned that the best of them, the Hilbert and Quot schemes, are representable, while the most natural of them, the moduli of curves, are not. Part V opened the object being classified and read its infinitesimal structure: a first-order deformation is a class in a tangent cohomology group, and the obstruction to continuing it lives one degree up. Part VI built the geometric objects, algebraic spaces and algebraic stacks, capacious enough to represent moduli problems that no scheme could. What is still missing is the bridge. Given a moduli problem, one can compute its deformation theory by the methods of Chapter 5, but nothing so far turns that formal, order-by-order infinitesimal data into the statement that the moduli problem is an algebraic stack in the sense of definition 20.3.1. Artin's criteria are that bridge: a checklist, phrased entirely in terms of the functor and its deformation theory, whose satisfaction is equivalent to algebraicity.

The difficulty the criteria overcome is the gap between the formal and the algebraic. Schlessinger's theorem (theorem 17.3.5) produces, at each point of a moduli problem, a formal versal deformation living over a complete local ring, the hull of definition 17.3.1. A hull is a power series neighborhood of the point, but a power series ring is not a scheme of finite type, and a formal deformation is not a family over an actual scheme; it is the limit of families over Artinian thickenings. To build an atlas, a smooth surjection from a scheme, one must *algebraize* the hull: find an actual finite-type scheme, or at least a finitely presented one, whose completion at a point recovers the formal versal deformation, and which maps to the stack. This is the analytic heart of the theory, and it is supplied by Artin approximation, whose proof rests on Néron-Popescu desingularization. That proof is beyond the present scope and is cited; everything else, the precise statement of the criteria, the deformation-theoretic mechanism by which they assemble an atlas, and the verification of the criteria for $\mathcal{M}_g$ and for the Hilbert and Quot functors, is carried out in full.

The first task is to name the deformation-theoretic input to the criteria as a structure on the stack itself. Chapter 5 developed deformation and obstruction theory for a fixed variety or a fixed singular scheme, through the cotangent functors $T^i(A/k,-)$ (definition 17.2.2) and their interpretation (theorem 17.2.3). For a stack the same data must be attached uniformly to every point, and it must vary well enough that a formal versal deformation at one point remains versal nearby. The following packages this.

Definition 20.4.1: Deformation and Obstruction Theory for a Stack

Let $\mathcal{X}$ be a stack over $(\mathbf{Sch}/S)_{\text{fppf}}$ with S locally Noetherian, and let $x \in \mathcal{X}(k)$ be an object over a field k that is a residue field of S. A *deformation and obstruction theory* for $\mathcal{X}$ at x consists of:

1. a finite-dimensional k-vector space $T^1_{\mathcal{X}}(x)$, the *tangent space*, in natural bijection with the set of first-order deformations of x, that is, the lifts of x to an object over $k[\varepsilon]$ up to isomorphism, with the trivial deformation as zero;

2. for each small extension $0 \to (t) \to A' \to A \to 0$ in $\mathbf{Art}_k$ and each object $x_A \in \mathcal{X}(A)$ lifting x, an *obstruction space* $T^2_{\mathcal{X}}(x)$ (a finite-dimensional k-vector space, the same for all A) and a canonical *obstruction class*

$$\mathrm{ob}(x_A, A') \in T^2_{\mathcal{X}}(x) \otimes_k (t)$$

that vanishes if and only if x_A lifts to an object over A', and, when it vanishes, the set of such lifts is a torsor under $T^1_{\mathcal{X}}(x) \otimes_k (t)$.

The theory is said to satisfy *openness of versality* if, whenever a finite-type S-scheme U carries an object $\xi \in \mathcal{X}(U)$ whose induced formal deformation at a point $u \in U$ is versal (a hull for the deformation functor of $x = \xi(u)$ in the sense of definition 17.3.1), there is an open neighborhood $u \in V \subseteq U$ on which ξ is versal at every point.

The two vector spaces $T^1_{\mathcal{X}}(x)$ and $T^2_{\mathcal{X}}(x)$ are the abstraction of the tangent-obstruction pair of Chapter 5, lifted from a fixed object to a point of a stack. Their defining property is exactly the content of theorem 17.1.2 and theorem 17.2.3(2): a first-order deformation is a tangent vector, a lift across a small extension exists exactly when an obstruction class in the degree-two space vanishes, and the lifts, when they exist, form a torsor under the tangent space tensored with the kernel. What the definition adds is that these spaces are attached to *every* point of $\mathcal{X}$ and are required to be finite-dimensional, which is the input that makes Schlessinger's finiteness condition (H3) automatic.

Openness of versality is the subtler ingredient and deserves its own reading. A hull is a formal object: it controls the deformation functor at a single point, order by order. The condition demands that versality, once achieved formally at a point through an actual family over a finite-type base, is not an isolated accident but propagates to a Zariski neighborhood. Geometrically this is what makes an atlas possible. A single versal formal deformation gives a smooth map to $\mathcal{X}$ from a formal neighborhood; openness upgrades "formal neighborhood" to "Zariski neighborhood", so that a versal family over a scheme is a smooth map to $\mathcal{X}$ on an open set rather than only at one point. Without openness one could algebraize each hull separately and get a collection of pointwise-smooth maps that never assemble into a single smooth atlas; with it, each algebraized hull is smooth on a whole open, and these opens cover $\mathcal{X}$.

Before the theorem it is worth grounding the definition in the running touchstones, where T^1 and T^2 are already computed. For the Hilbert scheme, a point $[Z]$ has tangent space $T^1 = H^0(Z, N_{Z/Y})$ (theorem 14.4.4), and the obstruction lives in $T^2 = H^1(Z, N_{Z/Y})$; these are the deformation and obstruction spaces of the closed subscheme Z. For the moduli of curves, a point $[C]$ has tangent space $T^1 = H^1(C, T_C)$ and obstruction space $T^2 = H^2(C, T_C)$ in the sense of theorem 17.1.2; and because a curve is one-dimensional, $H^2(C, T_C) = 0$, so $\mathcal{M}_g$ is unobstructed and its deformation theory has vanishing obstruction space at every point. These are exactly the inputs the criteria will consume.

Example 20.4.2: The Deformation Theory of the Hilbert Functor
Let $Y \subseteq \mathbb{P}^n_k$ be projective and let $[Z] \in \underline{\mathrm{Hilb}}^P_Y(k)$ correspond to a closed subscheme $Z \subseteq Y$ with Hilbert polynomial P. The deformation and obstruction theory of $\underline{\mathrm{Hilb}}^P_Y$ at $[Z]$ is

$$T^1_{\underline{\mathrm{Hilb}}}([Z]) = H^0(Z, N_{Z/Y}), \qquad T^2_{\underline{\mathrm{Hilb}}}([Z]) = H^1(Z, N_{Z/Y})$$

the global sections and first cohomology of the normal sheaf. The tangent identification is theorem 14.4.4. For the obstruction, a first-order deformation $\widetilde{Z} \subseteq Y_\varepsilon$ is a section of $N_{Z/Y}$; to lift it across a small extension $A' \to A$ one perturbs the local equations of Z, and the mismatch on overlaps is a Čech 1-cocycle with values in $N_{Z/Y}$, whose class in $H^1(Z, N_{Z/Y})$ vanishes exactly when the local perturbations glue. Both groups are finite-dimensional because $N_{Z/Y}$ is coherent on the projective scheme Z. When Z is a smooth complete intersection in $Y = \mathbb{P}^n$, the normal sheaf $N_{Z/Y} = \bigoplus_i \mathcal{O}_Z(d_i)$ splits as a sum of positive line bundles (from example 14.4.2 applied to each equation), and for $\dim Z \geq 2$ the arithmetic Cohen-Macaulayness of a complete intersection gives $H^1(Z, \mathcal{O}_Z(d_i)) = 0$ (the intermediate cohomology of $\mathcal{O}_Z(m)$ vanishes for every m), whence $H^1(Z, N_{Z/Y}) = 0$: the complete intersection is unobstructed, and Hilb^P_Y is smooth at $[Z]$ of dimension $\sum_i h^0(\mathcal{O}_Z(d_i))$. This is the local structure of the Hilbert scheme, read entirely off the deformation theory.

The example shows the deformation theory of a moduli functor is not new work but a repackaging of the cohomology already computed in Chapters 2 and 5. This is the point of phrasing the criteria in terms of T^1 and T^2: whoever can compute the tangent and obstruction cohomology of the objects has, without further effort, the deformation and obstruction theory the criteria require.

With the deformation-theoretic input named, everything assembles into Artin's theorem. The statement is a list of five conditions, numbered (0) through (4) to match their logical roles: (0) and (1) are the "the functor is reasonable" conditions (a stack, and one of finite presentation), (2) is the representability of the diagonal, (3) is the deformation-theoretic input with effectivity, and (4) is openness of versality. The theorem asserts their conjunction is equivalent to algebraicity.

Theorem 20.4.3: Artin's Representability Criteria

Let S be a locally Noetherian scheme and let

$$\mathcal{X} \longrightarrow (\mathbf{Sch}/S)$$

be a category fibered in groupoids, with structure morphism $\mathcal{X} \to S$ locally of finite presentation. Then $\mathcal{X}$ is an algebraic stack in the sense of definition 20.3.1 if and only if the following conditions hold.

(0) *Stack.* $\mathcal{X}$ is a stack for the fppf topology (equivalently, since the objects are S-schemes, for the étale topology): isomorphisms form a sheaf and descent data are effective.

(1) *Limit-preserving.* $\mathcal{X}$ is locally of finite presentation as a fibered category: for every filtered colimit $A = \varinjlim A_i$ of $\mathcal{O}_S$-algebras the natural functor

$$\varinjlim_i \mathcal{X}(\operatorname{Spec} A_i) \longrightarrow \mathcal{X}(\operatorname{Spec} A)$$

is an equivalence of groupoids.

(2) *Representable diagonal.* The diagonal $\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$ is representable by algebraic spaces; equivalently the $\underline{\mathrm{Isom}}$ sheaves of any two objects over a base are algebraic spaces. On formal deformations this is the assertion that the automorphism-and-deformation functor of each object satisfies Schlessinger's conditions (H1)-(H4) of theorem 17.3.5, so that each has a hull.

(3) *Effective, coherent obstruction theory.* $\mathcal{X}$ admits a deformation and obstruction theory in the sense of definition 20.4.1, with finite-dimensional $T^1_{\mathcal{X}}(x)$ and $T^2_{\mathcal{X}}(x)$ depending in a coherent (constructible, base-change-compatible) way on x, and every formal object is *effective*: for each complete local Noetherian $\mathcal{O}_S$-algebra R with residue field k, the natural map

$$\mathcal{X}(\operatorname{Spec} R) \longrightarrow \varprojlim_n \mathcal{X}(\operatorname{Spec} R/\mathfrak{m}_R^{n+1})$$

is an equivalence, so a compatible system of objects over the Artinian quotients comes from an object over R itself.

(4) *Openness of versality.* The deformation and obstruction theory satisfies openness of versality: a formally versal deformation carried by a finite-type S-scheme is versal on an open neighborhood.

The theorem is due to Artin [2], building on the algebraization theorem of [1]; the modern formulation, with the coherence condition on the obstruction theory made precise, is that of Hall and Rydh [31], and a fully detailed treatment is in the Stacks Project [63]. Because the criteria characterize an intricate geometric property through purely functorial and infinitesimal data, they are the working tool by which every moduli stack in practice is shown to be algebraic: one never constructs an atlas by hand, one checks the list.

The full proof of the forward direction, that the criteria imply algebraicity, requires Artin approximation and Néron-Popescu desingularization, which are beyond the present scope. The deferral is warranted by the depth of the input: approximation is a theorem in commutative algebra about the density of algebraic solutions among formal ones over an excellent base, and its proof through the desingularization of a regular map of Noetherian rings is a long development in its own right (see [1], [31], [63], and for the commutative-algebra input [10]). What can and should be made precise is the mechanism by which the criteria assemble an atlas, since that mechanism is the entire conceptual content and is exactly what the verifications in the sequel exploit.

***Proof Key ideas*:**
The proof that Artin's criteria imply algebraicity constructs a smooth atlas point by point and then spreads it out; the analytic input, algebraization of the hull, is cited. The converse, that an algebraic stack satisfies the criteria, is comparatively direct.

Step 1: The criteria produce a hull at each point. Fix a point $x \in \mathcal{X}(k)$ over a residue field k of S. The deformation functor

$$D_x \colon \mathbf{Art}_k \longrightarrow \mathbf{Set}, \qquad D_x(A) = \{\text{objects of } \mathcal{X}(A) \text{ lifting } x\}/{\cong}$$

has $D_x(k) = \{*\}$. Condition (3) gives it a tangent space $T^1_{\mathcal{X}}(x)$, finite-dimensional, so Schlessinger's finiteness (H3) holds; condition (2) gives the fiber-product conditions (H1), (H2), (H4) through the representability of the diagonal (the $\underline{\text{Isom}}$ sheaves being algebraic spaces is what makes the automorphism-quotients form algebraic fiber products, which is the abstract form of Schlessinger's gluing axioms). By theorem 17.3.5, D_x has a hull: a complete local $\mathcal{O}_S$-algebra R_x with a formal versal deformation $\widehat{\xi}_x$, whose tangent space is $T^1_{\mathcal{X}}(x)$ and whose obstructions are captured by $T^2_{\mathcal{X}}(x)$. Concretely R_x is a quotient of the power series ring $\mathcal{O}_{S,s}[\![t_1, \dots, t_d]\!]$, $d = \dim T^1_{\mathcal{X}}(x)$, by an ideal generated by $\dim T^2_{\mathcal{X}}(x)$ obstruction equations.

Step 2: The hull is effective and algebraizes. The hull $\widehat{\xi}_x$ is a compatible system of objects over the Artinian quotients $R_x/\mathfrak{m}^{n+1}$. Effectivity (the second half of condition (3)) promotes this system to a genuine object $\widehat{\xi}_x \in \mathcal{X}(\operatorname{Spec} R_x)$ over the complete local ring itself. This is where the formal becomes almost-algebraic: an object over a complete local ring, not yet over a finite-type scheme. *Artin approximation* now closes the gap. Because $\mathcal{X}$ is limit-preserving (condition (1)) and $\mathcal{X} \to S$ is locally of finite presentation, an object over the completion R_x can be approximated to any finite order by an object over a finite-type $\mathcal{O}_S$-algebra: there exist a finite-type $\mathcal{O}_S$-algebra B, a point b with $\mathcal{O}_{U,b}$ Henselian and $\widehat{\mathcal{O}_{U,b}} \cong R_x$, and an object $\xi_x \in \mathcal{X}(U)$, $U = \operatorname{Spec} B$, whose induced formal deformation at b agrees with $\widehat{\xi}_x$. This is the step whose proof rests on Néron-Popescu desingularization and is cited to [1], [31], [63]. The map $\xi_x \colon U \to \mathcal{X}$ is, at b, formally smooth of relative dimension controlled by $T^1_{\mathcal{X}}(x)$: it is a formally versal family.

Step 3: Openness of versality spreads the atlas. The map $\xi_x \colon U \to \mathcal{X}$ is versal at the single point b. Condition (4), openness of versality, provides an open neighborhood $b \in V_x \subseteq U$ on which ξ_x is versal at every point. Versality at every point of V_x is precisely the statement that the induced morphism $V_x \to \mathcal{X}$ is smooth (formal smoothness plus finite presentation is smoothness): a formally versal family over a scheme, versal at all points, is a smooth map to the stack. This is the key transition, and it is what condition (4) exists to guarantee. Without it ξ_x would be smooth only at b, giving no open piece of an atlas.

Step 4: Assembling the global atlas. Carrying out Steps 1-3 at every point x produces smooth maps $V_x \to \mathcal{X}$ from finite-type schemes, one through each point of $\mathcal{X}$. Their disjoint union

$$U = \coprod_x V_x \longrightarrow \mathcal{X}$$

is smooth (smoothness is checked componentwise) and surjective (every point of $\mathcal{X}$ lies in the image of some V_x, by construction). This is a smooth atlas. The representability of the diagonal, condition (2), is the other half of definition 20.3.1; combined with the smooth surjective atlas, it certifies $\mathcal{X}$ as an algebraic stack. If moreover the automorphism groups are unramified, the atlas can be taken étale and $\mathcal{X}$ is Deligne-Mumford (theorem 20.2.3).

Step 5: The converse. Suppose $\mathcal{X}$ is already algebraic with a smooth atlas $p\colon U \to \mathcal{X}$. Condition (0) is part of the definition of a stack. Condition (2) is the representability of the diagonal, again part of the definition. For condition (1), limit-preservation, one uses that U and the groupoid $R = U \times_{\mathcal{X}} U \rightrightarrows U$ are of finite presentation over S (an atlas of an algebraic stack of finite presentation can be so chosen), and finite-presentation schemes commute with filtered colimits of rings by the standard limit formalism, which passes to the stack quotient. For condition (3), the deformation and obstruction theory is read off the atlas: the tangent space of $\mathcal{X}$ at $x = p(u)$ is the quotient of $T_u U$ by the image of the tangent space to the automorphism group (the fiber of $R \to U \times_S U$), and the obstruction theory descends from the smooth U, where it is the classical tangent-obstruction theory of theorem 17.1.2; effectivity holds because objects over a complete local ring are pulled back from the atlas through Grothendieck's existence theorem (section 14.3 and [7]). Condition (4), openness of versality, holds because smoothness of p is an open condition on U, so a versal family is versal on an open. Thus every algebraic stack satisfies the criteria, completing the proof.

The proof is the conceptual center of the chapter, and the division of labor in it is worth restating. Conditions (0) and (1) ensure the functor is manageable, a stack of finite presentation, so that the local-to-global and finite-order arguments have something to bite on. Condition (2) is one of the two defining data of an algebraic stack, the representable diagonal, and it feeds Schlessinger's theorem to produce a hull. Condition (3) supplies the deformation theory that gives the hull its shape and the effectivity that turns a formal system into an object over a complete ring. Artin approximation, the cited black box, algebraizes that object into a family over a finite-type scheme. Condition (4) then spreads the versality of that family from one point to an open, and the opens cover the stack. The single hardest step, and the only one deferred, is the algebraization; the rest is the deformation theory of Part V applied uniformly.

A reader who has followed Chapter 5 will recognize that conditions (2) and (3) are almost exactly Schlessinger's criteria promoted from a single deformation functor to a fibered category. Schlessinger asked when a functor of Artin rings has a hull; Artin asks the same of the deformation functor at every point, adds effectivity so the hull is an actual object, and adds openness so the pointwise hulls glue. The criteria are "Schlessinger with algebraization and spreading-out", and that is the precise sense in which they close the loop from Part V's formal deformation theory to Part VI's algebraic stacks.

The payoff is the theorem the book has been building toward since Chapter 1: the moduli of curves is an algebraic stack, and so, recovered as a special case, are the Hilbert and Quot functors. The verification is a matter of checking the five conditions, and every one of them is discharged by material already developed, modulo the single cited approximation step.

Theorem 20.4.4: $\mathcal{M}_g$ and the Hilbert and Quot Functors Are Algebraic

Fix $g \geq 2$ and work over $S = \operatorname{Spec} \mathbb{Z}$ (or over a field). Then:

1. the moduli stack $\mathcal{M}_g$ of smooth projective curves of genus g, and its Deligne-Mumford compactification $\overline{\mathcal{M}}_g$ of stable curves, are algebraic stacks, indeed Deligne-Mumford stacks, smooth of dimension $3g-3$ over S;

2. the Hilbert functor $\underline{\mathrm{Hilb}}_Y^P$ (definition 14.1.6) and the Quot functor $\underline{\mathrm{Quot}}_{\mathcal{F}/Y}^P$ of Chapter 2 are algebraic; being functors of sets (their objects have no nontrivial automorphisms), they are algebraic spaces, and in fact schemes.

***Proof*:**
The proof checks the five criteria of theorem 20.4.3 for $\mathcal{M}_g$; the Hilbert and Quot cases are the same argument with the automorphisms absent, and are treated in example 20.4.5. The one cited input is Artin approximation, invoked inside theorem 20.4.3 and not re-opened here.

Step 1: $\mathcal{M}_g$ is a stack of finite presentation (conditions (0) *and* (1)*).* That $\mathcal{M}_g$ is a stack for the fppf topology is descent for families of smooth curves: a curve over a base is a scheme proper and smooth with one-dimensional connected fibers of genus g, and such families descend along fppf covers because proper smooth morphisms and the numerical invariants defining them descend (theorem 18.4.3). This is condition (0). For condition (1), a smooth projective curve of genus g over $\operatorname{Spec} A$, with $A = \varinjlim A_i$ a filtered colimit, is a finitely presented object: it is cut out in some $\mathbb{P}_A^N$ by finitely many equations (a genus-g curve with $g \geq 2$ is tricanonically embedded by $\omega_C^{\otimes 3}$ in $\mathbb{P}^{5g-6}$, so pluricanonical embedding gives a uniform projective model), and a finitely presented scheme over a colimit ring descends to a finite stage A_i, uniquely up to a further stage. Hence $\varinjlim \mathcal{M}_g(\operatorname{Spec} A_i) \to \mathcal{M}_g(\operatorname{Spec} A)$ is an equivalence: $\mathcal{M}_g$ is limit-preserving. The uniform projective model is exactly the boundedness input, and it is theorem 14.2.5 applied to the tricanonical Hilbert polynomial $P(m) = (6m-1)(g-1)$ that guarantees all genus-g curves embed in one bounded family inside a single Hilbert scheme.

Step 2: The diagonal is representable (condition (2)*).* The diagonal of $\mathcal{M}_g$ encodes the $\underline{\mathrm{Isom}}$ sheaves: for two families of curves C, C' over a base T, the functor $\underline{\mathrm{Isom}}_T(C, C')$ assigns to $T' \to T$ the set of isomorphisms $C_{T'} \cong C'_{T'}$. For smooth projective curves this functor is representable by a scheme, quasi-projective and unramified over T: an isomorphism of curves is a point of the Hilbert scheme of the product $C \times_T C'$ (its graph), and the graphs of isomorphisms form a locally closed, unramified subscheme because a curve of genus $g \geq 2$ has only finitely many automorphisms (Hurwitz's bound $|\mathrm{Aut}(C)| \leq 84(g-1)$). Representability of $\underline{\mathrm{Isom}}$ is representability of the diagonal, and its unramifiedness is what will make $\mathcal{M}_g$ Deligne-Mumford rather than only Artin. On the formal side, the deformation-and-automorphism functor of a curve satisfies Schlessinger's (H1)-(H4) because the tangent space $H^1(C, T_C)$ is finite-dimensional and the automorphism scheme is finite and reduced, so theorem 17.3.5 applies and each curve has a hull.

Step 3: Deformation and obstruction theory, with effectivity (condition (3)*).* The deformation and obstruction theory of $\mathcal{M}_g$ at a curve $[C]$ is

$$T^1_{\mathcal{M}_g}([C]) = H^1(C, T_C), \qquad T^2_{\mathcal{M}_g}([C]) = H^2(C, T_C)$$

by theorem 17.1.2, with $T_C = \mathcal{O}_C(-K_C)$ the tangent sheaf. Both are finite-dimensional (coherent cohomology on a projective curve), so the coherence and finiteness parts of condition (3) hold; and because C is one-dimensional, $H^2(C, T_C) = 0$, so the obstruction space vanishes and $\mathcal{M}_g$ is unobstructed. By Riemann-Roch on the curve, $\dim H^1(C, T_C) = h^0(C, \Omega_C^{\otimes 2}) = 3g-3$ for $g \geq 2$ (the tangent sheaf has degree $2-2g$, and $h^0(T_C) = 0$ since $\deg T_C < 0$, while $h^1(T_C) = -\deg T_C + g - 1 = 3g-3$). Thus the tangent space to $\mathcal{M}_g$ has dimension $3g-3$ everywhere and the obstruction vanishes, which already forces smoothness of dimension $3g-3$ once algebraicity is known. Effectivity of formal deformations is Grothendieck's existence theorem (section 14.3, and [7]): a compatible system of curves over the Artinian quotients $R/\mathfrak{m}^{n+1}$ of a complete local ring R algebraizes to a formal family, and because the family is polarized (each curve carries the ample tricanonical bundle, whose sections give the projective embedding), Grothendieck's existence theorem promotes the formal family to an actual projective curve over $\operatorname{Spec} R$. Polarizability is what makes formal deformations of curves effective, and it is a feature of curves not shared by every deformation problem.

Step 4: Openness of versality (condition (4)*).* Openness of versality for $\mathcal{M}_g$ follows from the unobstructedness established in Step 3 together with the uppersemicontinuity of $H^1(C, T_C)$. Concretely, take a family of curves $\mathcal{C} \to U$ over a finite-type base with a point u at which the Kodaira-Spencer map $T_u U \to H^1(C_u, T_{C_u})$ is surjective (the condition for formal versality at u). Surjectivity of the Kodaira-Spencer map is an open condition on U: it is the non-vanishing of a minor of a map of coherent sheaves whose ranks are uppersemicontinuous, so the locus where the map is surjective is open. On that open the family is versal at every point, which is openness of versality. The vanishing of $H^2(C, T_C)$ is what makes formal versality equivalent to Kodaira-Spencer surjectivity in the first place, so the unobstructedness of Step 3 is again what the argument needs.

Step 5: Conclusion for $\mathcal{M}_g$ and its compactification. With all five criteria verified, theorem 20.4.3 certifies $\mathcal{M}_g$ as an algebraic stack. The diagonal is unramified (Step 2, finite reduced automorphism schemes), so by theorem 20.2.3 the atlas can be taken étale and $\mathcal{M}_g$ is Deligne-Mumford. Its smoothness and dimension $3g-3$ come from Step 3: the tangent space has constant dimension $3g-3$ and the obstruction vanishes, so the atlas is smooth of that dimension and $\mathcal{M}_g$ inherits smoothness. The same argument applies to $\overline{\mathcal{M}}_g$ with the single modification that stable curves have nodes, so the deformation theory at a nodal curve is governed by Ext groups rather than $H^i(T_C)$ (the tangent sheaf is replaced by the sheaf $\mathcal{H}om(\Omega_C, \mathcal{O}_C)$ and a local smoothing parameter is added at each node, computed in Chapter 5 as the T^1 of a node); the obstruction still vanishes because a nodal curve remains one-dimensional and its deformation-obstruction theory has no degree-two term. The finite automorphisms of a stable curve keep the diagonal unramified, so $\overline{\mathcal{M}}_g$ is Deligne-Mumford, smooth of dimension $3g-3$, and, as Chapter 10 will show, proper. This establishes part (1).

Step 6: The Hilbert and Quot functors. For $\underline{\mathrm{Hilb}}_Y^P$ and $\underline{\mathrm{Quot}}_{\mathcal{F}/Y}^P$ the same five checks succeed, and are simpler because the objects (closed subschemes, coherent quotients) have no nontrivial automorphisms, so the diagonal is trivial (an isomorphism of subschemes fixing the ambient Y is the identity), the $\underline{\mathrm{Isom}}$ sheaves are trivial, and the stack is a functor of sets. Conditions (0) and (1) are the sheaf property and finite presentation from Chapter 2; condition (2) holds trivially because the diagonal of a sheaf of sets is a monomorphism; condition (3) is the deformation theory $T^1 = H^0(N)$, $T^2 = H^1(N)$ of example 20.4.2, effective by Grothendieck's existence theorem; condition (4) is again openness of Kodaira-Spencer surjectivity. A stack that is a sheaf of sets with a trivial diagonal and a smooth atlas is an algebraic space, and a quasi-projective algebraic space with a trivial diagonal is a scheme, recovering the Grothendieck existence theorem of section 14.3 that Hilb^P_Y and $\mathrm{Quot}^P_{\mathcal{F}/Y}$ are projective schemes. This is part (2), completing the proof.

This theorem is the destination of the whole book, and it closes the loop that Part V opened. In Chapter 4 and Chapter 5 the deformation theory of curves was developed as a self-contained study of infinitesimal structure: $H^1(C, T_C)$ was the space of first-order deformations, $H^2(C, T_C)$ was where obstructions would live, and the vanishing $H^2 = 0$ on a curve made the moduli of curves unobstructed. At the time this was a statement about a formal deformation functor, with no geometric object it was the deformation theory *of*. Artin's criteria supply the missing object: they take exactly that deformation data, H^1 as tangent space, H^2 as obstruction space, $3g-3$ as the dimension, add the boundedness of theorem 14.2.5 to make the family finite-type and the effectivity of Grothendieck existence to make formal deformations algebraize, and conclude that a smooth Deligne-Mumford stack $\mathcal{M}_g$ of dimension $3g-3$ exists. The deformation theory was never idle: it was the local model of a global object that only the criteria could produce.

The single dependence on machinery beyond this book is worth locating precisely. Every condition except the algebraization inside Artin approximation was verified from Chapters 1 through 7: descent (Chapter 6) for the stack property, boundedness (Chapter 2) for finite presentation, Hilbert-scheme representability (Chapter 2) for the diagonal, the tangent-obstruction theory (Chapter 5) for the deformation theory, and Grothendieck existence (Chapter 2) for effectivity. Only the passage from an object over a complete local ring to an object over a finite-type scheme, the algebraization step, uses Néron-Popescu desingularization from outside. This is the boundary of a first course in the subject: the geometry and the deformation theory are internal, the approximation theorem is imported.

The abstract verification acquires its full meaning when carried out on a single functor where every group can be named. The Hilbert functor is the ideal test case: its objects have no automorphisms, so the diagonal degenerates and the criteria reduce to their sheaf-of-sets form, and one sees exactly how Artin's list recovers the concrete representability theorem of Chapter 2.

Example 20.4.5: Verifying the Criteria for the Hilbert Functor
Fix a projective scheme $Y \subseteq \mathbb{P}^n_k$ and a Hilbert polynomial P, and check the five conditions of theorem 20.4.3 for $\underline{\mathrm{Hilb}}^P_Y$ (definition 14.1.6), a functor of sets.

1. *Condition* (0)*, stack.* A functor of sets is a stack for the fppf topology exactly when it is an fppf sheaf. For $\underline{\mathrm{Hilb}}^P_Y$ this is fppf descent for flat families of closed subschemes: a subscheme $Z \subseteq Y_T$ flat over T with fiberwise Hilbert polynomial P is determined by its ideal sheaf, and quasi-coherent sheaves with descent data glue (theorem 18.3.4), the flatness and Hilbert polynomial descending because they are checked on fibers. So $\underline{\mathrm{Hilb}}^P_Y$ is an fppf sheaf.

2. *Condition* (1)*, limit-preserving.* A flat family of subschemes over $\operatorname{Spec} A$, $A = \varinjlim A_i$, is cut out by finitely many equations of bounded degree (the ideal is generated in degrees $\leq m_0$, where m_0 is the uniform regularity bound of theorem 14.2.5), and finitely many equations over a colimit ring descend to a finite stage. Hence $\varinjlim \underline{\mathrm{Hilb}}^P_Y(\operatorname{Spec} A_i) \to \underline{\mathrm{Hilb}}^P_Y(\operatorname{Spec} A)$ is a bijection: limit-preservation is boundedness plus finite-presentation descent. This is where theorem 14.2.5 does its work.

3. *Condition* (2)*, diagonal.* The objects are closed subschemes of a fixed Y; an isomorphism of two such subschemes over T compatible with the inclusion into Y_T is forced to be the identity, since both are the same closed subscheme. So the $\underline{\mathrm{Isom}}$ sheaf is trivial, the diagonal is a monomorphism, and it is representable trivially. This is the sense in which the Hilbert

functor has "no automorphisms": its diagonal is trivial, and this is precisely why it is representable by a scheme rather than only by a stack.

4. *Condition* (3)*, deformation theory.* At a point $[Z]$ the tangent space is $T^1 = H^0(Z, N_{Z/Y})$ (theorem 14.4.4) and the obstruction space is $T^2 = H^1(Z, N_{Z/Y})$ (example 20.4.2), both finite-dimensional. The Schlessinger conditions (H1)-(H4) of theorem 17.3.5 hold: the deformation functor of Z inside Y is $A \mapsto \{Z_A \subseteq Y_A \text{ flat}\}$, and it satisfies the fiber-product axioms because gluing flat subschemes over a fiber product of Artinian rings is unobstructed by the sheaf property. Effectivity is Grothendieck's existence theorem (section 14.3): a compatible system of subschemes $Z_n \subseteq Y_{R/\mathfrak{m}^{n+1}}$ algebraizes to a subscheme $Z \subseteq Y_R$ because the ambient Y is projective and the ideal sheaves form a compatible system with the required coherence.

5. *Condition* (4)*, openness of versality.* A family $Z \subseteq Y_U$ over a finite-type base is versal at a point u when the map $T_uU \to H^0(Z_u, N_{Z_u/Y})$ is surjective, an open condition by uppersemicontinuity of the ranks. On the versal open the family is a smooth (indeed, since the diagonal is trivial, étale-onto-image) map to the functor.

All five conditions hold, so $\underline{\mathrm{Hilb}}_Y^P$ is algebraic. Because its diagonal is trivial, it is an algebraic space; because it is bounded and carries an ample sheaf (it sits inside a Grassmannian, hence a projective space), it is a scheme. This recovers the Grothendieck existence theorem of section 14.3, that Hilb_Y^P is a projective scheme, now derived as a special case of Artin's criteria. The lesson is that the Hilbert scheme is the degenerate case of the theory, where trivial automorphisms collapse the stack to a scheme, and the criteria specialize to the concrete representability already proved by hand in Chapter 2.

The example makes visible what the criteria give and where. For the Hilbert functor, conditions (0) and (1) are descent and boundedness, condition (2) is vacuous because there are no automorphisms, condition (3) is the normal-sheaf cohomology $H^0(N)$ and $H^1(N)$, and condition (4) is the openness of a surjectivity. The single feature that separates the Hilbert functor from $\mathcal{M}_g$ is the diagonal: trivial for subschemes, unramified-but-nontrivial for curves. That difference is the whole difference between a scheme and a Deligne-Mumford stack, and Artin's criteria absorb it into a single condition on $\underline{\mathrm{Isom}}$, treating the representable and the stacky moduli problems by one uniform mechanism.

Exercise 20.4.1

1. Let $\mathcal{X}$ be an algebraic stack over a locally Noetherian S, of finite presentation, with a smooth atlas $U \to \mathcal{X}$. Show directly that $\mathcal{X}$ is limit-preserving (condition (1) of theorem 20.4.3), by reducing the claim for $\mathcal{X}$ to the corresponding claim for the finite-presentation schemes U and $R = U \times_{\mathcal{X}} U$ and using that a finitely presented scheme over $\varinjlim A_i$ descends to a finite stage. Explain in one sentence why finite presentation of the atlas is the hypothesis that makes this work.

2. For a smooth projective curve C of genus $g \geq 2$, use Riemann-Roch and Serre duality to compute $h^0(C, T_C)$ and $h^1(C, T_C)$, where $T_C = \mathcal{O}_C(-K_C)$ is the tangent sheaf. Conclude that $T^1_{\mathcal{M}_g}([C]) = H^1(C, T_C)$ has dimension $3g - 3$ and $T^2_{\mathcal{M}_g}([C]) = H^2(C, T_C) = 0$, and interpret the vanishing $h^0(T_C) = 0$ in terms of the automorphisms of C.

3. Explain why openness of versality (condition (4)) is not automatic from the other four conditions, by describing what could go wrong: give the geometric picture of a family that is

formally versal at one point u but not at nearby points, and explain why, without condition (4), the algebraized hulls of the proof of theorem 20.4.3 would fail to assemble into a smooth atlas. (You may argue heuristically; a rigorous counterexample is beyond the scope.)

4. Let $Z \subseteq \mathbb{P}^n$ be a smooth complete intersection of hypersurfaces of degrees $d_1, \ldots, d_c$ with $\dim Z \geq 2$. Using $N_{Z/\mathbb{P}^n} = \bigoplus_{i=1}^c \mathcal{O}_Z(d_i)$ (from example 14.4.2), compute the tangent and obstruction spaces $H^0(Z, N_{Z/\mathbb{P}^n})$ and $H^1(Z, N_{Z/\mathbb{P}^n})$ in the sense of example 20.4.2, and use that a complete intersection is arithmetically Cohen-Macaulay to deduce that the Hilbert scheme is smooth at $[Z]$. Then treat the plane curve $Z \subseteq \mathbb{P}^2$ of degree d (a hypersurface, $c = 1$, $\dim Z = 1$) directly: compute the dimension of Hilb at $[Z]$ explicitly and compare it with the dimension $\binom{d+2}{2} - 1$ of the space of degree-d plane curves.

5. Condition (3) of theorem 20.4.3 demands that formal deformations be *effective*. Explain, using Grothendieck's existence theorem (section 14.3), why effectivity holds for $\mathcal{M}_g$ (families of curves) but requires the presence of an ample line bundle (the tricanonical polarization). Contrast with the stack $\mathfrak{Coh}_X$ of coherent sheaves, where effectivity of formal deformations of a non-projective object is a hypothesis to be checked rather than free.

6. Reconcile the two ways this book has produced the Hilbert scheme: the direct construction of Chapter 2 (as a locally closed subscheme of a Grassmannian, via boundedness and the flattening stratification) and the derivation of example 20.4.5 (as a special case of Artin's criteria). Which inputs are shared between the two, and which is logically stronger? Explain in particular why the Chapter 2 construction is needed for the criteria to apply to $\mathcal{M}_g$ at all, so that the two approaches are not two proofs of one fact but stand in a dependency relation.

20.5 Geometry on a Stack

The previous four sections built the objects: an algebraic space is a sheaf étale-locally a scheme (definition 20.1.3), a Deligne-Mumford stack carries an étale atlas (definition 20.2.1), an Artin stack a smooth one (definition 20.3.1), and Artin's criteria (theorem 20.4.3) certify when a moduli problem lands among them. What remains is to do geometry on these objects. A scheme comes with a structure sheaf, quasi-coherent modules, line bundles and a Picard group, sheaf cohomology, and adjectives like smooth, normal, separated, and proper. This section transports each of these to an algebraic stack. The mechanism is uniform and was prefigured throughout the chapter: a property or a structure on a stack $\mathcal{X}$ is a property or structure on an atlas $U \to \mathcal{X}$ that is compatible with the groupoid $R = U \times_{\mathcal{X}} U \rightrightarrows U$ presenting $\mathcal{X}$. Descent (theorem 18.3.4, theorem 18.4.3) is exactly the guarantee that such atlas-level data glue to genuine data on the stack.

Two registers run in parallel. Quasi-coherent sheaves, the Picard group, and cohomology are *structures*: they are computed on the atlas and assembled by descent along $R \rightrightarrows U$. Adjectives split into two kinds. The atlas-local ones (smooth, normal, regular, locally of finite type) are read off any atlas because they are smooth-local on the source (definition 19.4.3); the separatedness-type ones (separated, proper) are conditions on the *diagonal* $\Delta_{\mathcal{X}}$, because separatedness of a space is a property of its diagonal, and a stack's diagonal encodes its $\underline{\mathrm{Isom}}$ sheaves. The section closes with the capstone the whole book has aimed at: $\overline{\mathcal{M}}_g$ is smooth and proper as a stack even though its coarse space $\overline{M}_g$ is singular. Smoothness is read on the atlas, properness from the valuative criterion, and neither survives the passage to the coarse space, which is why the stack is the right object.

Throughout, $\mathcal{X}$ is an algebraic stack over a base S with a presentation $R \rightrightarrows U$ by a groupoid of schemes, $s, t\colon R \to U$ the source and target maps (étale for a DM stack, smooth for an Artin stack), and $U \to \mathcal{X}$ the atlas. The two projections $\mathrm{pr}_1, \mathrm{pr}_2\colon R \to U$ of a groupoid presentation are s and t; the composition law is a map $m\colon R \times_{s,U,t} R \to R$.

Quasi-coherent sheaves are the first structure to transport, and the guiding principle is that a sheaf on $\mathcal{X}$ should be a sheaf on the atlas that does not depend on which point of the atlas one uses to see a given point of $\mathcal{X}$. Two points $u_1, u_2 \in U$ mapping to the same point of $\mathcal{X}$ are joined by a point of R, and asking a sheaf to be insensitive to the choice is asking for descent data along $R \rightrightarrows U$. This is the same descent that Chapter 6 proved for the fppf topology, now organized by a groupoid rather than a cover.

Definition 20.5.1: Quasi-Coherent Sheaves on a Stack

Let $\mathcal{X}$ be an algebraic stack with presentation $R \rightrightarrows U$, source and target $s, t\colon R \to U$. A *quasi-coherent sheaf* $\mathcal{F}$ on $\mathcal{X}$ is a quasi-coherent sheaf $\mathcal{F}_U$ on the atlas U together with an isomorphism of quasi-coherent sheaves on R

$$\varphi\colon s^*\mathcal{F}_U \xrightarrow{\sim} t^*\mathcal{F}_U$$

the *descent datum*, satisfying the cocycle condition $\mathrm{pr}_{13}^*\varphi = \mathrm{pr}_{23}^*\varphi \circ \mathrm{pr}_{12}^*\varphi$ on $R \times_{s,U,t} R$ and the normalization $e^*\varphi = \mathrm{id}$ along the identity section $e\colon U \to R$. A *morphism* $\mathcal{F} \to \mathcal{G}$ of quasi-coherent sheaves is a morphism $\mathcal{F}_U \to \mathcal{G}_U$ on U compatible with the descent data. The category of such is $\mathrm{QCoh}(\mathcal{X})$. The *structure sheaf* $\mathcal{O}_{\mathcal{X}}$ is $(\mathcal{O}_U, \mathrm{can})$, with the canonical identification $s^*\mathcal{O}_U \cong \mathcal{O}_R \cong t^*\mathcal{O}_U$.
Equivalently, $\mathrm{QCoh}(\mathcal{X})$ is the category of quasi-coherent sheaves on the *lisse-étale site* $\mathcal{X}_{\text{lis-ét}}$, whose objects are smooth morphisms $V \to \mathcal{X}$ from schemes and whose coverings are smooth surjections; a quasi-coherent sheaf assigns to each such $V \to \mathcal{X}$ a quasi-coherent sheaf on V, compatibly with smooth pullback.

The definition is descent made into a category, and its independence of the presentation is what makes it well posed. A different atlas $U' \to \mathcal{X}$ presents the same stack by a groupoid R', and the two presentations are related by a common refinement; the descent categories are equivalent because descent data glue and the equivalence is exactly theorem 18.3.4 applied to the two atlases against a common cover. The lisse-étale reformulation makes this coordinate-free: a quasi-coherent sheaf is a rule on *all* smooth schemes over $\mathcal{X}$ at once, and any single atlas recovers it by restriction. The structure sheaf $\mathcal{O}_{\mathcal{X}}$ is the descent of $\mathcal{O}_U$, and a quasi-coherent sheaf is a module over it in the same atlas-plus-descent sense.

The first example is the one that makes the definition concrete and that will recur: on a classifying stack the descent datum is a group action, so a quasi-coherent sheaf is a representation.

Example 20.5.2: $\mathrm{QCoh}(BG)$ as Representations

Take G an affine group scheme over k and $BG = [\mathrm{pt}/G]$ (definition 19.3.2). The atlas is $\mathrm{pt} = \operatorname{Spec} k \to BG$, the universal torsor, and its groupoid is $R = \mathrm{pt} \times_{BG} \mathrm{pt} = G$, with $s, t\colon G \to \mathrm{pt}$ both the structure map and composition $m\colon G \times G \to G$ the multiplication. A quasi-coherent sheaf on BG is thus a k-vector space V (a quasi-coherent sheaf on pt) with a descent isomorphism $\varphi\colon s^*V \to t^*V$ on G, that is, a k-linear map

$$\varphi\colon V \otimes_k \mathcal{O}_G \xrightarrow{\sim} V \otimes_k \mathcal{O}_G$$

of $\mathcal{O}_G$-modules, subject to the cocycle condition on $G \times G$ and the normalization at the identity.

Unwinding, φ is a comodule structure $V \to V \otimes_k \mathcal{O}_G$ over the Hopf algebra $\mathcal{O}_G$: the cocycle condition is coassociativity and the normalization is counitality. A comodule over $\mathcal{O}_G$ is precisely a *representation* of G (a linear action $G \times V \to V$), so

$$\mathrm{QCoh}(BG) \simeq \mathrm{Rep}(G)$$

the category of (algebraic) G-representations. For $G = \mathbb{G}_m$, with $\mathcal{O}_G = k[u, u^{-1}]$, a comodule is a $\mathbb{Z}$-grading $V = \bigoplus_{n\in\mathbb{Z}} V_n$ (the n-th weight space is where u acts by u^n), so $\mathrm{QCoh}(B\mathbb{G}_m)$ is the category of $\mathbb{Z}$-graded vector spaces. The structure sheaf $\mathcal{O}_{BG}$ is the trivial representation k.

The identification $\mathrm{QCoh}(BG) = \mathrm{Rep}(G)$ is the paradigm for all stacky geometry: a geometric construction on the stack is a representation-theoretic construction on the group, and conversely. A vector bundle on BG is a finite-dimensional representation; the structure sheaf is the trivial representation; tensor product of sheaves is tensor product of representations. For a quotient stack $[X/G]$ the same holds with X present: $\mathrm{QCoh}([X/G])$ is the category of G-equivariant quasi-coherent sheaves on X, a quasi-coherent sheaf $\mathcal{F}$ on X with an isomorphism $\sigma^*\mathcal{F} \cong \mathrm{pr}_X^*\mathcal{F}$ satisfying the cocycle condition, which is a G-linearization in the sense of the theory of Chapter 3. Equivariant geometry and stacky geometry are the same subject seen from two sides.

With sheaves in hand, line bundles and cohomology follow at once, since a line bundle is a quasi-coherent sheaf that is locally free of rank one and cohomology is the derived functor of global sections. The subtlety is that on a stack "global sections" means sections invariant under the groupoid, so $H^0(\mathcal{X}, \mathcal{F})$ is not $H^0(U, \mathcal{F}_U)$ but the R-invariant part, and the higher cohomology is the derived functor of this invariants functor. The two computational models, Čech on the groupoid and the lisse-étale site, agree, and each is useful.

Definition 20.5.3: Picard Group and Cohomology of a Stack

Let $\mathcal{X}$ be an algebraic stack with presentation $R \rightrightarrows U$.

1. The *Picard group* $\mathrm{Pic}\,(\mathcal{X})$ is the group of isomorphism classes of invertible sheaves (line bundles) on $\mathcal{X}$ under tensor product. By definition 20.5.1 a line bundle is a line bundle L on U with a descent datum $\varphi\colon s^*L \cong t^*L$ satisfying the cocycle condition, so

$$\mathrm{Pic}\,(\mathcal{X}) = \{(L, \varphi) : L \in \mathrm{Pic}\,(U),\ \varphi \text{ a groupoid descent datum}\}/\cong$$

2. For $\mathcal{F} \in \mathrm{QCoh}(\mathcal{X})$, the *cohomology* $H^i(\mathcal{X}, \mathcal{F})$ is the i-th right derived functor of the global sections functor $\Gamma(\mathcal{X}, -) = \mathrm{Hom}_{\mathrm{QCoh}(\mathcal{X})}(\mathcal{O}_{\mathcal{X}}, -)$, the functor of R-invariant sections. Equivalently it is the cohomology of $\mathcal{F}$ on the lisse-étale site $\mathcal{X}_{\text{lis-ét}}$.

When the nerve pieces U, R, $R \times_U R$, $\ldots$ are all affine (as they are for BG, $B\mu_n$, $B\mathbb{G}_m$, or any quotient by an affine group, so that the fiber products are affine), $H^i(\mathcal{X}, \mathcal{F})$ is computed by the *Čech complex of the groupoid*, the cochain complex

$$\Gamma(U, \mathcal{F}_U) \longrightarrow \Gamma(R, s^*\mathcal{F}_U) \longrightarrow \Gamma(R \times_{s,U,t} R, \ldots) \longrightarrow \cdots$$

whose p-th term is the global sections over the p-fold fiber product $R \times_U \cdots \times_U R$ (the degree-p piece of the nerve of the groupoid) and whose differential is the alternating sum of the simplicial face maps; its i-th cohomology is $H^i(\mathcal{X}, \mathcal{F})$.

The definition says cohomology on a stack is the cohomology of the *simplicial scheme* (the nerve of the groupoid), and this is what makes stacky cohomology computable and what connects it to classical constructions. For an affine scheme $\mathcal{X} = X$ presented by the trivial groupoid $X \rightrightarrows X$, the Čech complex collapses to $\Gamma(X, \mathcal{F})$ in degree zero and the definition returns ordinary sheaf cohomology ($H^{>0}$ vanishing on an affine); for a general scheme the derived-functor definition still returns ordinary sheaf cohomology, but the trivial-groupoid Čech complex sees only $\Gamma(X, \mathcal{F}) = H^0$ and one must resolve $\mathcal{F}$ (or pass to an affine atlas) to recover the higher cohomology. For a quotient stack the nerve is the bar construction of the action, and the cohomology is *equivariant* cohomology. For BG the nerve is the bar construction of G, and $H^i(BG, \mathcal{F})$ is the group cohomology (or, in the algebraic category, the derived-invariants $\mathrm{Ext}^i_{\mathrm{Rep}(G)}(k, V)$ of the representation $V = \mathcal{F}$) of G. The Picard group similarly becomes an equivariant Picard group, the group of G-linearized line bundles.

The computation that anchors this definition, and that will drive the examples, is the Picard group of $B\mathbb{G}_m$. It is the first place where a stack's Picard group differs from that of its coarse space (a point, with trivial Picard group), and the answer is dictated entirely by the representation theory of $\mathbb{G}_m$.

Proposition 20.5.4: The Picard Group of $B\mathbb{G}_m$

Over a field or local ring k, $\mathrm{Pic}\,(B\mathbb{G}_m) \cong \mathbb{Z}$ (over a general base ring the underlying rank-one modules contribute an extra factor $\mathrm{Pic}\,(k)$, giving $\mathrm{Pic}\,(B\mathbb{G}_m) \cong \mathrm{Pic}\,(k) \times \mathbb{Z}$). The isomorphism sends an integer n to the line bundle $\mathcal{O}(n)$ corresponding, under example 20.5.2, to the one-dimensional representation of $\mathbb{G}_m$ of weight n, and the weight is a complete invariant of a line bundle on $B\mathbb{G}_m$.

***Proof*:**
By example 20.5.2, $\mathrm{QCoh}(B\mathbb{G}_m)$ is the category of $\mathbb{Z}$-graded k-modules, with $\mathcal{F}$ corresponding to a $\mathbb{Z}$-graded module $V = \bigoplus_n V_n$. A line bundle is an invertible object of this category under the tensor product $(V \otimes W)_n = \bigoplus_{p+q=n} V_p \otimes W_q$. The invertible graded modules are exactly the modules that are free of rank one and concentrated in a single degree: if $V \otimes W \cong k$ (the unit, concentrated in degree 0), then comparing graded pieces forces V to be an invertible k-module concentrated in one degree n and W its inverse concentrated in degree $-n$. Over a general base k the underlying rank-one module contributes $\mathrm{Pic}\,(k)$; over a field or local ring it is free, and the only invariant is the degree $n \in \mathbb{Z}$. The tensor product of the degree-n line bundle with the degree-m one is concentrated in degree $n + m$, so $n \mapsto \mathcal{O}(n)$ is a group isomorphism $\mathbb{Z} \xrightarrow{\sim} \mathrm{Pic}\,(B\mathbb{G}_m)$, and the weight n determines the bundle, as we sought to show.

The proposition is the smallest instance of a stack seeing more than its coarse space. The coarse space of $B\mathbb{G}_m$ is a single point, whose Picard group is trivial, yet $\mathrm{Pic}\,(B\mathbb{G}_m) = \mathbb{Z}$: the stack carries a whole $\mathbb{Z}$ of line bundles, indexed by the weight of the $\mathbb{G}_m$-action, none of which descend to the point. These line bundles record the automorphism group $\mathbb{G}_m$ that every point of $B\mathbb{G}_m$ carries. The general pattern is $\mathrm{Pic}\,(BG) = \widehat{G} = \mathrm{Hom}(G, \mathbb{G}_m)$, the character group of G, because a line bundle on BG is a one-dimensional representation and a one-dimensional representation is a character. The same mechanism gives $\mathrm{Pic}\,(B\mu_n) = \mathbb{Z}/n$ and $\mathrm{Pic}\,(BGL_n) = \mathbb{Z}$ (via the determinant character), so the Picard group of a classifying stack is a direct read-out of the character theory of the group.

The last structure to transport is the vocabulary of adjectives. A scheme is smooth, normal, separated, or proper; each of these should have a meaning for a stack, and the meaning is governed by a single dichotomy. Some properties are *smooth-local on the source*: they hold for a scheme if and only if they hold after a smooth surjective base change, and such a property is read off any atlas. Others are properties of the *diagonal*, and separatedness is the prototype, since a scheme is separated exactly when its diagonal is a closed immersion. On a stack the diagonal carries the $\underline{\mathrm{Isom}}$ sheaves, so diagonal conditions are conditions on how objects are glued and how their automorphisms behave.

Definition 20.5.5: Geometric Properties of an Algebraic Stack

Let $\mathcal{X}$ be an algebraic stack with atlas $U \to \mathcal{X}$, presentation $R \rightrightarrows U$, and diagonal $\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times_S \mathcal{X}$.

1. *Atlas-local properties.* $\mathcal{X}$ is *smooth*, *regular*, *normal*, *reduced*, *locally of finite type*, or of a fixed *dimension* over S if the atlas U has the corresponding property. Each of these properties P is smooth-local on the source, so by definition 19.4.3 and theorem 18.4.3 the truth of P for U is independent of the atlas and defines P for $\mathcal{X}$.

2. *Diagonal conditions.* $\mathcal{X}$ is *quasi-separated* (resp. *separated*) if $\Delta_{\mathcal{X}}$ is quasi-compact and quasi-separated (resp. proper); equivalently $\mathcal{X}$ is separated when the $\underline{\mathrm{Isom}}$ sheaves are proper, so automorphisms and isomorphisms are parametrized by proper algebraic spaces. $\mathcal{X}$ is *Deligne-Mumford* (resp. *Artin*) according to whether $\Delta_{\mathcal{X}}$ is unramified (resp. representable), by theorem 20.2.3.

3. *Properness.* $\mathcal{X}$ is *proper* over S if it is separated, of finite type, and *universally closed*, the last verified by the *valuative criterion for stacks*: for every discrete valuation ring A with fraction field K, every 2-commutative square

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ \operatorname{Spec} A & \longrightarrow & S \end{array}$$

admits, after a finite extension of A, a lift $\operatorname{Spec} A \to \mathcal{X}$, and any two lifts agreeing over $\operatorname{Spec} K$ are related by a unique isomorphism (no extension needed for the uniqueness clause); existence of the lift after a finite extension gives universal closedness (hence properness together with separatedness and finite type), and uniqueness up to unique isomorphism gives separatedness.

The dichotomy in the definition is the organizing fact of stacky geometry, and the reason properties split is instructive. Smoothness, normality, and regularity are conditions on the *local rings* of a scheme, and smooth morphisms induce nice (regular, flat with regular fibers) maps on local rings, so these conditions transfer up and down a smooth atlas without loss. That is why they can be checked on any atlas. Separatedness and properness are conditions on *global* behavior, on how the object sits in a product with itself (the diagonal) and on limits of families (the valuative criterion), and these are exactly the places where a stack differs from a scheme because its points have automorphisms. The valuative criterion for a stack differs from the scheme version in one word: the lift is unique *up to unique isomorphism*, not literally unique, because the objects being lifted have automorphisms. This one modification is what will make $\overline{\mathcal{M}}_g$ proper while its coarse space is only proper after forgetting automorphisms.

The valuative criterion is abstract until one sees it as a completeness statement about families, so the grounding example is a family of curves degenerating over a punctured disk. The lift over $\operatorname{Spec} A$ is a limit of the family, and the criterion says the limit exists in the stack after a finite base change: this is stable reduction, the input to the capstone of the book.

Example 20.5.6: The Valuative Criterion as Stable Reduction

Let A be a discrete valuation ring with fraction field K and residue field κ, geometrically the local ring of a smooth curve at a point, $\operatorname{Spec} K$ the punctured disk and $\operatorname{Spec} \kappa$ the central point. A morphism $\operatorname{Spec} K \to \overline{\mathcal{M}}_g$ is a family of stable curves over the punctured disk, that is, a smooth (or stable) curve C_K of genus g over K. The valuative criterion for $\overline{\mathcal{M}}_g$ (definition 20.5.5(3)) asks whether this family extends to a stable curve over the whole disk $\operatorname{Spec} A$, filling in the central fiber. The *stable reduction theorem*, which Chapter 10 develops, asserts exactly this: after a finite extension $A \subseteq A'$ (a ramified base change, replacing the disk by a cover) the family C_K acquires a unique stable limit $C_{A'}$ over $\operatorname{Spec} A'$, a stable curve whose central fiber is a nodal degeneration of the generic one. Existence of the limit is universal closedness, so $\overline{\mathcal{M}}_g$ is universally closed; the finite extension is the "up to a finite extension of A" clause of the criterion, forced because the limiting curve may acquire automorphisms not present over K; and the uniqueness of the stable limit (up to unique isomorphism) is separatedness. Thus properness of $\overline{\mathcal{M}}_g$ is stable reduction, translated into the valuative criterion.

The example is the mechanism by which the geometry of the moduli stack is governed by the geometry of degenerating curves, and it explains why the boundary $\overline{\mathcal{M}}_g \setminus \mathcal{M}_g$ of nodal curves is exactly what properness requires: a family of smooth curves can degenerate, and for the moduli space to be proper the degenerate limits must be points of it, which forces the boundary to consist of nodal (stable) curves. The reason a finite extension of A is unavoidable is worth isolating: over K a curve may have a monodromy that only becomes trivial after a base change, and the stable limit lives over the cover; this is precisely the failure of the coarse space to be proper in the naive sense and the reason a stack, whose valuative criterion tolerates a finite extension, is the correct object.

All the pieces are now in place to compute, and the examples below run from the simplest classifying stacks to the capstone. They are chosen to display each of the three registers: the Picard group and cohomology of $B\mathbb{G}_m$ and $B\mu_n$ (structures via descent), the tangent and canonical bundles of a quotient stack (equivariant geometry), and the smoothness and properness of $\overline{\mathcal{M}}_g$ (adjectives via atlas and diagonal). The last is the statement the entire book has aimed at.

Example 20.5.7: Picard Groups, Cohomology, Tangent Bundles, and $\overline{\mathcal{M}}_g$

1. *The weight of a line bundle on $B\mathbb{G}_m$.* By proposition 20.5.4, $\operatorname{Pic}(B\mathbb{G}_m) = \mathbb{Z}$, the isomorphism sending a line bundle to its *weight*, the integer n such that the corresponding one-dimensional representation of $\mathbb{G}_m$ is $t \mapsto t^n$. Concretely, the tautological line bundle $\mathcal{O}(1)$ on $B\mathbb{G}_m$ (the universal one-dimensional representation, weight 1) generates, and $\mathcal{O}(n) = \mathcal{O}(1)^{\otimes n}$. A section of $\mathcal{O}(n)$ over $B\mathbb{G}_m$ is a $\mathbb{G}_m$-invariant element of the weight-n representation $k_{(n)}$, which is 0 for $n \neq 0$ and k for $n = 0$; hence $H^0(B\mathbb{G}_m, \mathcal{O}(n)) = 0$ for $n \neq 0$ and $= k$ for $n = 0$, so $\mathcal{O}(n)$ has no nonzero global sections unless it is trivial. This is the stacky version of the fact that a nontrivial character has no invariant vectors.

2. *Cohomology of $B\mu_n$.* Work over any field k. The group scheme μ_n is diagonalizable (the Cartier dual of $\mathbb{Z}/n$, of multiplicative type), so by the $\operatorname{QCoh}(BG) = \operatorname{Rep}(G)$ dictionary (example 20.5.2) its representations are exactly the $\mathbb{Z}/n$-graded k-vector spaces $V = \bigoplus_{m \in \mathbb{Z}/n} V_m$, and $H^i(B\mu_n, \mathcal{F}) = \operatorname{Ext}^i_{\operatorname{Rep}(\mu_n)}(k, V)$ for $V = \mathcal{F}$. The category of $\mathbb{Z}/n$-graded vector spaces is semisimple in every characteristic (a graded submodule is a direct summand, namely the sum of the graded pieces it meets), so μ_n is linearly reductive over any base, char $k|n$

included; all higher Ext vanish, and

$$H^i(B\mu_n, \mathcal{F}) = \begin{cases} V^{\mu_n} = V_0 & i = 0 \\ 0 & i > 0 \end{cases}$$

the invariants (the weight-0 part of the $\mathbb{Z}/n$-graded module V) in degree 0 and nothing above. In particular $H^i(B\mu_n, \mathcal{O}) = k$ for $i = 0$ and 0 for $i > 0$: the structure sheaf has the cohomology of a point, and this holds in *every* characteristic. The contrast is with the *constant* group $\underline{\mathbb{Z}/n}$, whose representation category obeys Maschke's theorem and is semisimple only when the order n is invertible in k; there the higher group cohomology $H^i(\underline{\mathbb{Z}/n}, k)$ is nonzero for $i > 0$ once $\operatorname{char} k \big| n$. When $\operatorname{char} k \nmid n$ the two group schemes are isomorphic and agree; when $\operatorname{char} k \big| n$ they differ, μ_n becoming infinitesimal (nonreduced) but remaining linearly reductive, while $\underline{\mathbb{Z}/n}$ stays étale but ceases to be linearly reductive. The higher cohomology that appears in char $\big| n$ is thus a feature of $B\underline{\mathbb{Z}/n}$, not of $B\mu_n$.

3. *The tangent and canonical bundle of* $[X/G]$. Let G be a smooth affine group scheme acting on a smooth variety X over k with finite stabilizers (so that the infinitesimal action is injective), and $q\colon X \to [X/G]$ the atlas. The tangent sheaf $T_{[X/G]}$ is defined by descent: it is the G-equivariant sheaf on X whose fiber is the quotient $T_X/\mathfrak{g}_X$, where $\mathfrak{g}_X \subseteq T_X$ is the image of the infinitesimal action $\mathfrak{g} \otimes_k \mathcal{O}_X \to T_X$ of the Lie algebra $\mathfrak{g} = \operatorname{Lie}(G)$ (the distribution tangent to the orbits, a subbundle since the stabilizers are finite). There is a short exact sequence of equivariant sheaves on X

$$0 \to \mathfrak{g} \otimes_k \mathcal{O}_X \to T_X \to q^* T_{[X/G]} \to 0$$

whose descent to $[X/G]$ computes the tangent bundle of the stack. Taking top exterior powers, the canonical bundle is

$$\omega_{[X/G]} = \det(T_{[X/G]})^{-1}, \qquad q^*\omega_{[X/G]} \cong \omega_X \otimes (\det \mathfrak{g})_X$$

as G-equivariant line bundles on X, where $\det \mathfrak{g}$ is the one-dimensional representation $\Lambda^{\mathrm{top}}\mathfrak{g}$ (the modular character). For $X = \mathrm{pt}$ this gives $\dim BG = -\dim G$ and $\omega_{BG} = \det \mathfrak{g}$, consistent with the general formula ($\omega_{\mathrm{pt}} = \mathcal{O}$, so $q^*\omega_{BG} = \det \mathfrak{g}$) and with the tangent-complex count: the tangent complex of BG is $\mathfrak{g}[1]$, concentrated in degree -1, so $\det(T_{BG}) = (\det \mathfrak{g})^{-1}$ and $\omega_{BG} = \det(T_{BG})^{-1} = \det \mathfrak{g}$. This recovers the negative dimension of BG (definition 20.3.4): the stack BG has dimension $-\dim G$ because that tangent complex sits in degree -1, the automorphisms subtracting dimension.

4. $\overline{\mathcal{M}}_g$ *is smooth and proper as a stack.* The capstone. Fix $g \geq 2$ over a base where the relevant characteristics are excluded as in example 20.2.5. The stack $\overline{\mathcal{M}}_g$ is *smooth* of dimension $3g - 3$ over the base: smoothness is an atlas-local property (definition 20.5.5(1)), and an atlas is the smooth locus of a Hilbert scheme of pluricanonically embedded stable curves (theorem 20.4.4), which is smooth because the deformation theory of a stable curve is unobstructed. A stable curve C is generically nodal, so Ω_C is not locally free and deformations are governed by $\operatorname{Ext}^i(\Omega_C, \mathcal{O}_C)$ rather than $H^i(C, T_C)$; the obstruction lives in $\operatorname{Ext}^2(\Omega_C, \mathcal{O}_C)$ and the tangent space is $\operatorname{Ext}^1(\Omega_C, \mathcal{O}_C)$ (these coincide with $H^i(C, T_C)$ only in the smooth locus, where Ω_C is locally free). The obstruction vanishes: a node is a hypersurface (local complete intersection) singularity, so Ω_C has projective dimension ≤ 1 locally and the local $\mathcal{E}xt$ sheaves $\mathcal{E}xt^j(\Omega_C, \mathcal{O}_C)$ vanish for $j \geq 2$, while $\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)$ is a skyscraper supported

at the nodes (the first-order smoothing of each node, Chapter 5 deformation theory of a node). The local-to-global spectral sequence $H^p(C, \mathcal{E}xt^q(\Omega_C, \mathcal{O}_C)) \Rightarrow \operatorname{Ext}^{p+q}(\Omega_C, \mathcal{O}_C)$ then has only the terms $H^p(C, \mathcal{H}om(\Omega_C, \mathcal{O}_C))$ $(q = 0)$ and $H^p(C, \mathcal{E}xt^1)$ $(q = 1)$; the $q = 0$ row vanishes above $p = 1$ because a curve has cohomological dimension one, and the $q = 1$ terms vanish for $p \geq 1$ because a skyscraper has no higher cohomology, so every contribution to $\operatorname{Ext}^2(\Omega_C, \mathcal{O}_C)$ is zero and the deformation theory is unobstructed. The tangent space $\operatorname{Ext}^1(\Omega_C, \mathcal{O}_C)$ has constant dimension $3g - 3$. The stack $\overline{\mathcal{M}}_g$ is *proper* over the base: it is separated (the diagonal is finite, since stable curves have finite automorphism groups, so the $\underline{\mathrm{Isom}}$ sheaves are finite hence proper, definition 20.5.5(2)), of finite type (boundedness of stable curves, theorem 14.2.5), and universally closed by the valuative criterion, which for $\overline{\mathcal{M}}_g$ is the stable reduction theorem (example 20.5.6). Thus $\overline{\mathcal{M}}_g$ is smooth and proper as a stack. Its coarse space $\overline{M}_g$ (theorem 20.2.4) is proper as a scheme but is *singular*: it has quotient singularities at the points corresponding to curves with automorphisms (the coarse space is locally $\overline{\mathcal{M}}_g$ modulo the automorphism group, and $\mathbb{A}^{3g-3}/\operatorname{Aut}(C)$ is generically a genuine quotient singularity when $\operatorname{Aut}(C) \neq 1$, the exception being the special case where the group acts by pseudo-reflections and the quotient stays smooth). Smoothness lives on the stack and is destroyed by passing to the coarse space, which is the precise sense in which $\overline{\mathcal{M}}_g$ is the right object and its coarse space loses information.

The four computations exhibit the whole apparatus of the section at work and, taken together, justify the stack-theoretic language the book has built. The Picard group and cohomology of $B\mathbb{G}_m$ and $B\mu_n$ are pure descent: they are the representation theory of the automorphism group, invisible on the coarse space. The tangent and canonical bundles of $[X/G]$ are equivariant geometry, the orbit distribution subtracting the dimension of the group and the modular character twisting the canonical bundle. The smoothness and properness of $\overline{\mathcal{M}}_g$ are the payoff: smoothness read on a Hilbert atlas and driven by the unobstructedness of stable curves, properness read from the diagonal (finite automorphisms) and the valuative criterion (stable reduction), and both features destroyed by the coarse space, whose quotient singularities are the fingerprint of the automorphisms the stack retains. The moduli of curves, worked in full in Chapter 10, is the object where all three registers of stacky geometry meet, and the geometry on a stack developed here is what makes it a smooth proper object rather than a singular scheme.

Exercise 20.5.1

1. Let G be an affine group scheme over k and $BG = [\mathrm{pt}/G]$. Using the identification $\operatorname{QCoh}(BG) \simeq \operatorname{Rep}(G)$ (example 20.5.2), show that $\operatorname{Pic}(BG) \cong \widehat{G} = \operatorname{Hom}(G, \mathbb{G}_m)$, the character group of G. Deduce $\operatorname{Pic}(B\mu_n) = \mathbb{Z}/n$ and $\operatorname{Pic}(BGL_n) = \mathbb{Z}$, and identify a generator of each.

2. Let G act on a scheme X over k. Prove directly from definition 20.5.1 that $\operatorname{QCoh}([X/G])$ is equivalent to the category of G-equivariant quasi-coherent sheaves on X (a sheaf $\mathcal{F}$ on X with an isomorphism $\sigma^*\mathcal{F} \cong \operatorname{pr}_X^* \mathcal{F}$ satisfying the cocycle condition), where $\sigma\colon G \times X \to X$ is the action. Identify the structure sheaf $\mathcal{O}_{[X/G]}$ and describe what a G-invariant global section of $\mathcal{O}_{[X/G]}$ is in terms of X and G.

3. Show that a $\mathbb{Z}$-graded k-vector space $V = \bigoplus_n V_n$, regarded as an object of $\operatorname{QCoh}(B\mathbb{G}_m)$ via example 20.5.2, has $H^i(B\mathbb{G}_m, V) = 0$ for all $i > 0$ and $H^0(B\mathbb{G}_m, V) = V_0$. (Use that $\mathbb{G}_m$ is linearly reductive over any base: its representation category is semisimple, so the invariants functor is exact.) Conclude that $H^i(B\mathbb{G}_m, \mathcal{O}(n)) = 0$ for all $i > 0$ and every n, and $= 0$ in

degree 0 unless $n = 0$.

4. For a discrete valuation ring A with fraction field K, state the valuative criterion (definition 20.5.5(3)) for a morphism of algebraic stacks $\mathcal{X} \to \mathcal{Y}$ to be *universally closed* and for it to be *separated*, being careful about the "up to unique isomorphism" and "after a finite extension of A" clauses. Explain, in one paragraph, why the scheme-theoretic valuative criterion (unique lift, no extension) is the special case where objects have no automorphisms, and why $\overline{\mathcal{M}}_g$ requires the stacky version.

5. Let G be a smooth affine group scheme of dimension d acting on a smooth n-dimensional variety X with finite stabilizers; assume in addition that the stabilizers are reduced (equivalently, that the diagonal is unramified, theorem 20.2.3), automatic in characteristic 0 but not in characteristic p (where e.g. μ_p-stabilizers are finite yet nonreduced and make $[X/G]$ Artin, not Deligne-Mumford), so that $[X/G]$ is Deligne-Mumford. Using the presentation with atlas $U = X$ and groupoid $R = G \times X$, compute $\dim[X/G] = \dim U - \dim R$ (definition 20.3.4), and confirm the answer $n - d$ agrees with the tangent-space count from the exact sequence $0 \to \mathfrak{g} \otimes \mathcal{O}_X \to T_X \to q^*T_{[X/G]} \to 0$ of example 20.5.7(3). Then specialize to $X = \mathrm{pt}$ to recover $\dim BG = -d$.

6. Explain why $\mathrm{Pic}\,(B\mathbb{G}_m) = \mathbb{Z}$ is not a contradiction with the fact that the coarse space of $B\mathbb{G}_m$ is a single point with trivial Picard group. Identify precisely which line bundles on the stack fail to descend to the coarse space, and relate this to the automorphism group $\mathbb{G}_m$ acting on the fibers. (Hint: a line bundle descends to the coarse space when $\mathbb{G}_m$ acts trivially on its fibers, that is, when its weight is 0.)

20.6⋆ Quotients of a Group Scheme by a Subgroup

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

The quotient stacks of section 20.3 were built from a group acting on a scheme, and nothing forced the two to be related. The case where they are, a group scheme acting on itself by translation, is the oldest quotient problem in the subject and the one with the cleanest answer: the stack collapses all the way to a scheme. Nothing later in this book uses the result, which is why this section carries a star, but it is the form in which the quotient G/H is consumed elsewhere in the series, and the machinery of chapter 20 is what makes the general case reachable at all.

The affine case is older and needs none of this. When G is an affine algebraic group over a field and $H \subseteq G$ is a closed subgroup, *Algebraic Groups and Representations* [19] constructs G/H by hand: Chevalley's theorem realizes H as the stabilizer of a line in a representation V, and G/H is then the orbit of that line in $\mathbb{P}(V)$, a quasi-projective variety. That argument is unavailable as soon as G is not affine, because a group with no nonconstant functions has no faithful representation to work in. What follows replaces the representation by descent.

Definition 20.6.1: The Quotient Sheaf G/H

Let k be a field, let G be a group scheme of finite type over k, and let $H \subseteq G$ be a closed subgroup scheme. The *quotient sheaf* G/H is the fppf sheafification of the presheaf

$$T \longmapsto G(T)/H(T)$$

on $(\mathbf{Sch}/k)_{\mathrm{fppf}}$, where $H(T)$ acts on $G(T)$ by right multiplication. The quotient map $\pi\colon G \to G/H$ is the sheafification of $g \mapsto gH$.

Sheafification is not a formality here. The presheaf quotient is almost never a sheaf: a T-point of G/H is a coset that exists only locally, and the obstruction to lifting it to a T-point of G is exactly the class of an H-torsor. That observation is the whole strategy, and it is why the torsor theory of section 18.5 is the right tool.

Proposition 20.6.2: Translation Is a Free Action

Let G be a group scheme over k and $H \subseteq G$ a closed subgroup scheme. Then

$$a\colon H \times_k G \longrightarrow G, \qquad (h, g) \longmapsto gh^{-1}$$

is a left action of H on G whose orbits are the left cosets gH, and it is *free*: the stabilizer group scheme of the action is trivial.

***Proof*:**
That a is a left action is the associativity of multiplication in G together with $(hh')^{-1} = h'^{-1}h^{-1}$, checked on T-points and hence true as a statement about morphisms by Yoneda.

For freeness, the stabilizer is the group scheme $\mathrm{Stab} = (H \times_k G) \times_{(a,\mathrm{pr}_G),\, G\times_k G,\, \Delta_G} G$, whose T-points are the pairs $(h, g) \in H(T) \times G(T)$ with $gh^{-1} = g$. Multiplying on the left by g^{-1} in the group $G(T)$ gives $h^{-1} = e$, hence $h = e$. So for every T the stabilizer has exactly the points (e, g), and $\mathrm{Stab} \cong G$ sits inside $H \times_k G$ as $\{e\} \times G$; the stabilizer of each point is the trivial group scheme, as we sought to show.

Freeness is what separates this situation from the quotient stacks of the previous section. There the stabilizers were the source of all stacky behavior, and example 20.3.3 showed $B\mathbb{G}_m$ failing to be a scheme precisely because a point has automorphisms. Here there are none, and the stack has no reason to remain a stack.

Theorem 20.6.3: The Quotient Is an Algebraic Space

Let k be a field, let G be a smooth separated group scheme of finite type over k, and let $H \subseteq G$ be a smooth closed subgroup scheme. Then G/H is an algebraic space over k, the map $\pi\colon G \to G/H$ is an H-torsor for the fppf topology, and π is faithfully flat and smooth.

***Proof*:**
Step 1: the quotient stack. The subgroup H is separated, being a closed subscheme of the separated G, and it is of finite presentation over k because it is of finite type over a field. It is smooth by hypothesis, so theorem 20.3.2 applies to the action of proposition 20.6.2 on $X = G$: the quotient stack $[G/H]$ is an algebraic stack, and the tautological morphism $q\colon G \to [G/H]$ is an H-torsor, hence representable, smooth and surjective. (This is the point at which the affineness hypothesis of the older form of that theorem would have bitten, since H is not assumed affine; the theorem is stated for a smooth separated group precisely so that this case is covered.)

Step 2: the stack is a sheaf. By theorem 20.3.2 the diagonal of $[G/H]$ is computed by the $\underline{\mathrm{Isom}}$ sheaves, and over a scheme T the sheaf $\underline{\mathrm{Isom}}_T((P,f),(P',f'))$ was identified with the transporter $\{\gamma \in H : \gamma \cdot x' = x\}$. By proposition 20.6.2 the action is free, so this transporter is either empty or a single point: if γ and γ' both carry x' to x then $\gamma^{-1}\gamma'$ stabilizes x' and is therefore the identity. Hence every $\underline{\mathrm{Isom}}$ sheaf is a subsheaf of the final sheaf, the diagonal of $[G/H]$ is a monomorphism, and no object of $[G/H](T)$ has a nontrivial automorphism. A stack all of whose fiber groupoids are setoids is the stack associated to a sheaf, and that sheaf is exactly G/H of definition 20.6.1: an object of $[G/H](T)$ is an H-torsor $P \to T$ with an equivariant map to G, which is the datum of a section of G/H over T, and isomorphism classes are what the sheafification records.

Step 3: from a smooth atlas to an étale one. Condition (1) of definition 20.1.3 holds: the diagonal of G/H is representable because it is a monomorphism whose base changes are the transporters of Step 2, which are schemes. For condition (2), Step 1 supplies the smooth surjection $\pi\colon G \to G/H$ from a scheme rather than an étale one. Step 3 of the proof of proposition 18.1.4 closes the gap: a smooth surjection admits sections étale-locally on the target, by the local structure of smooth morphisms (see also [63, Tag 054L]). Applying this to π produces an étale cover $\{V_a \to G/H\}$ together with lifts $V_a \to G$, and the disjoint union $U = \coprod_a V_a$ is a scheme with an étale surjection $U \to G/H$. So G/H has an étale atlas and is an algebraic space.

Step 4: the torsor statement. The morphism π is the atlas q of Step 1 read through the identification of Step 2, so it is an H-torsor, and a torsor is faithfully flat; it is smooth because H is, completing the proof.

An algebraic space is not yet a scheme, and example 20.1.6 shows the gap is real. What closes it for G/H is homogeneity: a group acts transitively on its own coset space, so any good behavior at one point can be translated everywhere. This is the same mechanism that made group varieties smooth as soon as they were smooth somewhere.

Theorem 20.6.4: The Quotient Is a Scheme

In the situation of theorem 20.6.3, the algebraic space G/H is a scheme, of finite type over k and of dimension $\dim G - \dim H$. If H is normal in G, then G/H is a group scheme and $\pi\colon G \to G/H$ is a homomorphism.

***Proof*:**
Step 1: quasi-separatedness. The diagonal of G/H is a monomorphism by Step 2 of theorem 20.6.3, and its base changes are the transporters, which are closed subschemes of $H \times_k G$ and hence quasi-compact. By theorem 20.1.5(2) the space G/H is quasi-separated.

Step 2: a dense open subscheme. By theorem 20.1.5(5) a quasi-separated algebraic space contains a dense open subspace that is representable by a scheme. Let $V \subseteq G/H$ be such an open subscheme; it is nonempty because G/H is.

Step 3: translating V over the whole space. The group G acts on G/H by left translation, and the action is transitive: on T-points the coset gH is carried to $g'gH$ by g', and any two cosets differ by such a g' fppf-locally. Each translate $g' \cdot V$ is again an open subscheme of G/H, being the image of the open V under an automorphism of G/H. Let $\bar{k}$ be an algebraic closure of k. Every point of $(G/H)(\bar{k})$ lies in some translate $g' \cdot V$, because transitivity moves a point of $V(\bar{k})$ onto it. The union W of the translates is therefore an open subset containing every closed point of G/H, and its complement is a closed subset of a space of finite type over k containing no closed point, hence is empty: a nonempty closed subset of such a space always contains a closed point. So the translates of V form an open cover of G/H by schemes.

A space covered by open subschemes is a scheme: the covering is by open subspaces each of which is representable, and the gluing data are the intersections, which are open subschemes of the members. Hence G/H is a scheme.

Step 4: finiteness and dimension. The morphism π is faithfully flat and of finite type, so G/H is of finite type over k; and π is smooth with fibers the cosets gH, each isomorphic to H, so $\dim G = \dim G/H + \dim H$.

Step 5: the normal case. If H is normal, the multiplication of G descends. On T-points, for $h, h' \in H(T)$,

$$(gh)(g'h') = gg'\big(g'^{-1}hg'\big)h'$$

and $g'^{-1}hg' \in H(T)$ by normality, so $(gh)(g'h')H = gg'H$: the composite $G \times_k G \to G \to G/H$ is constant on the $H \times H$-orbits and therefore factors through $G/H \times_k G/H$, by the universal property of the fppf sheaf quotient. The same computation gives the inverse and the identity, and the group axioms are inherited because they hold on G and π is an epimorphism of sheaves. Thus G/H is a group scheme and π a homomorphism, as we sought to show.

The result is what lets an arbitrary algebraic group be dismantled. Given a group scheme G over a field and a normal closed subgroup H, one may now form G/H and argue by induction on dimension, which is the engine behind the structure theory of algebraic groups: every such G is an extension of an abelian variety by a linear group. That theorem, Chevalley's, is the principal consumer of this section outside this book.

Exercise 20.6.1

1. Let G be an affine algebraic group over an algebraically closed field and $H \subseteq G$ a closed subgroup. Both theorem 20.6.4 and the construction cited from *Algebraic Groups and*

Representations [19] produce a quotient G/H. Explain why they agree, without comparing the constructions.

2. The classifying stack $B\mathbb{G}_m$ of example 20.3.3 is the quotient stack of a point by $\mathbb{G}_m$, and it is not an algebraic space. Identify which hypothesis of theorem 20.6.3 fails, and locate the exact step of its proof that breaks.

3. Let $G = \mathrm{GL}_2$ over a field k and let $H \subseteq G$ be the subgroup of upper-triangular matrices. Compute $\dim G/H$ using theorem 20.6.4, and identify the resulting scheme.

4. Give an example of a group scheme G over a field and a closed subgroup H that is not normal, for which G/H is a scheme but carries no group structure making π a homomorphism.

5. Show that $\pi\colon G \to G/H$ admits a section if and only if the H-torsor π is trivial, and deduce that a section exists whenever $H^1_{\mathrm{fppf}}(G/H, H)$ vanishes (theorem 18.5.3).

21

Gerbes and Twisted Sheaves

Torsors were the one-categorical trace of the automorphisms that stacks geometrize: a G-torsor is a sheaf locally isomorphic to G, and its class in H^1 measures the failure to be trivial. Raising this construction one categorical level produces a gerbe, a stack locally isomorphic to the classifying stack BG, whose obstruction to triviality lives in H^2. The gerbe was named at the end of the descent chapter and has since appeared unbidden at every turn: the residual gerbe $B\underline{\mathrm{Aut}}(x)$ is the local model of an algebraic stack at a point, and the extra automorphisms of a special elliptic curve make $\mathcal{M}_{1,1}$ carry a gerbe over its special points. This chapter studies gerbes in their own right, and in doing so closes Part VI by connecting the moduli theory of the book to the Brauer group and to the twisted sheaves that the Brauer group governs.

The development runs in four movements. The first defines a gerbe and its band and proves that gerbes banded by an abelian sheaf A are classified by $H^2(X, A)$, exactly one degree above the torsor classification of Chapter 6. The second specializes the band to $\mathbb{G}_m$ and finds the Brauer group: the class of a $\mathbb{G}_m$-gerbe lives in $H^2(X, \mathbb{G}_m)$, an Azumaya algebra produces such a gerbe through the boundary map of $1 \to \mathbb{G}_m \to \mathrm{GL}_n \to \mathrm{PGL}_n \to 1$, and the field case recovers the Brauer group of Galois cohomology. The third section defines twisted sheaves, the modules of a $\mathbb{G}_m$-gerbe, and shows that they form an abelian category in which the Brauer class is precisely the obstruction to the existence of a twisted line bundle. The fourth constructs their moduli, the twisted analogue of the moduli of sheaves, which is again an algebraic stack and which, because it can be a fine moduli space where the untwisted problem is only coarse, resolves questions such as the period-index problem that the untwisted theory cannot reach. A closing note records the topological counterpart of the whole picture, the Dixmier-Douady class and twisted K-theory, which belongs to other books.

21.1 Torsors, Banded Gerbes, and H^2

The torsor is a sheaf that is locally isomorphic to a group G and whose isomorphism classes are counted by H^1. This chapter climbs one categorical level. A torsor is a sheaf of sets locally modeled on G; its two-dimensional refinement is a stack of groupoids locally modeled on the classifying stack BG, and its isomorphism classes are counted not by H^1 but by H^2. That object is a gerbe. It was named at the end of Chapter 6 (definition 18.5.4), where the passage from H^1 to H^2 was flagged as the two-dimensional analogue of the passage from a sheaf to a torsor, and its full development promised here. This section delivers that development for the case that matters most: a gerbe banded by an abelian sheaf A is classified by $H^2_\tau(X, A)$, exactly one degree above the torsor theorem (theorem 18.5.3), and the proof is the two-cocycle mechanism run in place of the one-cocycle mechanism.

Two threads from earlier parts of the book meet here. The first is descent: a gerbe is a stack, and the objects of a stack glue along a cover with the same cocycle bookkeeping that governed descent in Chapter 6, one dimension up. The second is the automorphism obstruction of Part IV. When a moduli problem fails to be representable because its objects carry automorphisms, as for the elliptic curves of example 13.3.4, the correct object is not a scheme but a stack, and the local structure of that stack at a point with automorphism group G is precisely a gerbe banded by G. The residual gerbe of a stack point, built at the end of Chapter 8, is the H^2 counterpart of the j-line's coarse-space collapse, and it is the concrete geometric object this section is written to name. Throughout, τ is the étale or fppf topology, X a scheme over the base S, and the site is $(\mathbf{Sch}/X)_\tau$; for the abelian bands of interest the two topologies agree.

Definition 21.1.1: Gerbe

A *gerbe* over the site $(\mathbf{Sch}/X)_\tau$ is a stack in groupoids $\mathcal{G} \to (\mathbf{Sch}/X)_\tau$ that is

1. *locally nonempty*: every object U of the site admits a covering family $\{U' \to U\}$ with $\mathcal{G}(U') \neq \varnothing$; and
2. *locally connected*: for every U and any two objects $x, y \in \mathcal{G}(U)$, there is a covering family $\{U' \to U\}$ such that $x|_{U'} \cong y|_{U'}$ in $\mathcal{G}(U')$ for each U'.

A gerbe $\mathcal{G}$ is *neutral* (or *trivial*) if it has a global object, that is, if $\mathcal{G}(X) \neq \varnothing$. In that case, for any global object $x \in \mathcal{G}(X)$ there is an equivalence of stacks

$$\mathcal{G} \simeq B\underline{\mathrm{Aut}}(x)$$

with the classifying stack (definition 19.3.2) of the automorphism sheaf $\underline{\mathrm{Aut}}(x)$ of x.

The two axioms are the exact two-dimensional analogues of the two conditions defining a torsor (definition 18.5.1). Local nonemptiness is the direct lift of τ-local nonemptiness of a torsor: a torsor is a sheaf of sets that is locally inhabited, a gerbe is a stack of groupoids that is locally inhabited. Local connectedness is the lift of simple transitivity. A torsor is *simply* transitive, so its fibers, once inhabited, are a single G-orbit with trivial stabilizer, a set with one isomorphism class and no symmetry; a gerbe relaxes this to *transitivity only*, so its fibers, once inhabited, are a single isomorphism class of objects but each object carries an automorphism group. That surviving automorphism group is the new one-dimensional datum a gerbe carries over a torsor, and it is what the band will record. Neutrality plays the role that triviality played for torsors: a global object trivializes the gerbe just as a global section trivialized a torsor, and the trivialized gerbe is $B\underline{\mathrm{Aut}}(x)$, the stacky point with automorphisms $\underline{\mathrm{Aut}}(x)$, exactly as a trivialized torsor was G itself.

The equivalence $\mathcal{G} \simeq B\underline{\mathrm{Aut}}(x)$ deserves a word, since it is the anchor for every example below. The classifying stack $B\underline{\mathrm{Aut}}(x)$ of definition 19.3.2 is the stackification of the prestack whose only object over each U is the trivial $\underline{\mathrm{Aut}}(x)$-torsor, with automorphisms $\underline{\mathrm{Aut}}(x)(U)$; its objects over a general U are the $\underline{\mathrm{Aut}}(x)$-torsors on U. The equivalence sends a $\underline{\mathrm{Aut}}(x)$-torsor P over U to the object of $\mathcal{G}(U)$ obtained by twisting $x|_U$ by P, that is, the object locally isomorphic to x and reglued by the transition cocycle of P; conversely an object $y \in \mathcal{G}(U)$ gives the torsor $\underline{\mathrm{Isom}}(x|_U, y)$ of local isomorphisms $x \to y$, which is a torsor under $\underline{\mathrm{Aut}}(x)$ precisely because $\mathcal{G}$ is locally connected (so the Isom sheaf is locally nonempty) and its objects have automorphism group $\underline{\mathrm{Aut}}(x)$. A neutral gerbe is therefore no more and no less than the moduli of torsors under a fixed automorphism sheaf, packaged as a stack.

Before the band, the definition should be grounded in the object it was written to describe. The classifying stack of a group over a point is the first gerbe, and it is neutral by construction.

Example 21.1.2: BG as the Neutral G-Gerbe

Let G be a sheaf of groups on $(\mathbf{Sch}/X)_\tau$, and let BG be its classifying stack (definition 19.3.2): the stack whose objects over U are the G-torsors on U, with morphisms the torsor isomorphisms. Then BG is a gerbe over X, and it is neutral.

Local nonemptiness holds because the trivial torsor $G|_U$ is an object of $BG(U)$ for every U, so $BG(U)$ is never empty; the site already has a global object, namely the trivial torsor over X itself. Local connectedness holds because any two G-torsors P, P' on U become isomorphic on a common trivializing cover: each is locally isomorphic to G, so on a cover refining the trivializations of both, $P|_{U'} \cong G \cong P'|_{U'}$. Thus BG satisfies both gerbe axioms. It is neutral because it has the global object $G|_X$, the trivial torsor; and taking $x = G|_X$, the automorphism sheaf $\underline{\mathrm{Aut}}(G|_X)$ is G acting on itself by right translation (proposition 18.5.2), so the equivalence of definition 21.1.1 reads

$$BG \simeq B\underline{\mathrm{Aut}}(G|_X) = BG$$

tautologically. The gerbe BG is the model neutral gerbe: every neutral gerbe is of this form for $G = \underline{\mathrm{Aut}}(x)$ its automorphism band, and the content of the next definition is to attach that band to a gerbe that need not be neutral.

A gerbe that is not neutral has no global object to read an automorphism sheaf off of. Yet every object of $\mathcal{G}(U)$, over every U and every cover, has an automorphism sheaf, and because the gerbe is locally connected these local automorphism sheaves are all locally isomorphic. The band is the invariant that records this common local automorphism sheaf, glued across the gerbe. For a nonabelian automorphism sheaf the gluing is only well-defined up to inner automorphism, and the band is the corresponding object; for an abelian sheaf inner automorphisms are trivial and the band is a genuine identification. The abelian case is the one this book develops in full.

Definition 21.1.3: Band of a Gerbe, and A-Gerbes

Let $\mathcal{G}$ be a gerbe over $(\mathbf{Sch}/X)_\tau$. For each object U of the site and each $x \in \mathcal{G}(U)$, the automorphism sheaf $\underline{\mathrm{Aut}}(x)$ is a sheaf of groups on U. Local connectedness provides, for any two objects x, y over a common cover, an isomorphism $x|_{U'} \cong y|_{U'}$, which induces an isomorphism $\underline{\mathrm{Aut}}(x)|_{U'} \cong \underline{\mathrm{Aut}}(y)|_{U'}$ of automorphism sheaves. This isomorphism is canonical up to conjugation by $\underline{\mathrm{Aut}}(y)$, because the isomorphism $x \cong y$ is unique only up to composition with an automorphism. The *band* (or *lien*) of $\mathcal{G}$ is the resulting sheaf of groups on X, well-defined up to inner automorphism, obtained by gluing these local automorphism sheaves. Let A be a sheaf of *abelian* groups on X. A *banding of $\mathcal{G}$ by A* is a choice, for every object $x \in \mathcal{G}(U)$, of an isomorphism $A|_U \cong \underline{\mathrm{Aut}}(x)$, compatible with restriction and with the isomorphisms induced by morphisms of $\mathcal{G}$. Because A is abelian, inner automorphisms of $\underline{\mathrm{Aut}}(x)$ are trivial, so a banding is a *canonical* identification with no conjugation ambiguity. A gerbe equipped with a banding by an abelian sheaf A is an *A-gerbe*; when the band A is understood, "gerbe" means "A-gerbe".

The conjugation ambiguity is the whole reason the band is a subtle invariant for nonabelian coefficients and a trivial one for abelian coefficients. When $\underline{\mathrm{Aut}}(x)$ is nonabelian, the isomorphism $x \cong y$ that transports one automorphism sheaf to another is unique only up to an automorphism of the target, and composing with that automorphism conjugates the transported identification. The band is therefore not a sheaf of groups but a sheaf of groups modulo inner automorphisms, an object Giraud calls a *lien*. The full nonabelian theory, in which a gerbe is banded by a lien and the classification lives in a nonabelian H^2 that is a set with a torsor action rather than a group, is developed in Giraud's foundational treatise [26]; it is stated here and not developed. For an abelian A the ambiguity evaporates: conjugation by any element of an abelian group is the identity, so the transported identification is unique, the band is a genuine abelian sheaf, and a banding is a plain isomorphism $A \cong \underline{\mathrm{Aut}}$. The classification then lands in the ordinary abelian H^2, and that is the theorem this section proves in full.

The definition is grounded by the object that motivated Part VI: a point of an algebraic stack, whose automorphisms organize into a gerbe over its residue field.

Example 21.1.4: The Residual Gerbe of a Stack Point

Let $\mathcal{X}$ be an algebraic stack (definition 20.3.1) and let $x \in \mathcal{X}(k)$ be a point over a field k, with automorphism group scheme $G = \underline{\mathrm{Aut}}(x)$, a group scheme of finite type over k. The *residual gerbe* at x is the reduced locally closed substack $\mathcal{G}_x \hookrightarrow \mathcal{X}$ supported at the image point, and it is a gerbe over $\operatorname{Spec} k$ banded by G; when G is abelian it is a G-gerbe. Since $\operatorname{Spec} k$ has a single point and x is a global object of $\mathcal{G}_x$ over k, the residual gerbe is *neutral*, and by definition 21.1.1

$$\mathcal{G}_x \simeq BG = B\underline{\mathrm{Aut}}(x)$$

the classifying stack of the automorphism group over the residue field.

The running touchstone makes this concrete. Consider the moduli stack $\mathcal{M}_{1,1}$ of elliptic curves and its point $[E]$ at an elliptic curve E over $k = \bar{k}$ of characteristic $\neq 2, 3$. A generic elliptic curve has automorphism group $\mu_2 = \{\pm 1\}$, so the residual gerbe at a generic point is $B\mu_2$. At the two special j-invariants the automorphism group jumps: at $j = 0$ the curve $y^2 = x^3 + 1$ has $\underline{\mathrm{Aut}} = \mu_6$, and at $j = 1728$ the curve $y^2 = x^3 + x$ has $\underline{\mathrm{Aut}} = \mu_4$ (the values recorded in example 13.3.4). The residual gerbes at these points are therefore

$$\mathcal{G}_{[E],j=0} \simeq B\mu_6, \qquad \mathcal{G}_{[E],j=1728} \simeq B\mu_4$$

each a neutral abelian gerbe over $\operatorname{Spec} k$. This is the H^2 datum that the j-line of theorem 13.3.6 discards: the coarse space $\mathbb{A}^1_j$ records only the point j, forgetting the automorphism group entirely, whereas the stack $\mathcal{M}_{1,1}$ remembers it as a residual gerbe. The automorphism obstruction to representability, first seen in Part IV as the failure of a fine moduli space to exist, is now visible as a nonzero band on a residual gerbe.

The residual gerbes above are all neutral, because a point of a stack always carries the tautological global object x itself. The interesting gerbes, the ones that will realize the Brauer group in the next section, are the nonneutral ones, and the classification theorem is exactly the count of how nonneutral an A-gerbe can be. The count is a class in H^2, and the neutral gerbe is its base point, precisely as the trivial torsor was the base point of H^1.

The motivation for the classification is the same as for torsors, lifted one dimension. An A-gerbe is locally neutral by axiom: on a cover it acquires objects, and any two are locally isomorphic, so on a fine enough cover the gerbe looks like BA. What obstructs neutrality globally is the failure of these local objects to be chosen coherently, and that failure is measured by a Čech class. For a torsor the local objects were sections and their discrepancies were one-cochains; for a gerbe the local objects are objects of a groupoid, their pairwise discrepancies are isomorphisms, and the failure of those isomorphisms to compose coherently on triple overlaps is a two-cochain valued in the band. Its coherence on quadruple overlaps is the two-cocycle condition. This is the mechanism, and the theorem is its formalization.

Theorem 21.1.5: Classification of A-Gerbes by H^2

Let A be a sheaf of abelian groups on the site $(\mathbf{Sch}/X)_\tau$. There is a natural bijection

$$\{\text{equivalence classes of } A\text{-gerbes on } X \text{ in } \tau\} \xrightarrow{\sim} H^2_\tau(X, A)$$

between equivalence classes of gerbes banded by A and the second cohomology of the site with coefficients in A, under which the neutral gerbe BA corresponds to the zero class. The set of A-gerbes is an abelian group under the contracted product of gerbes, and the bijection is an isomorphism of abelian groups. For a nonabelian band the classification lives in Giraud's nonabelian H^2 and is a set with a distinguished class rather than a group; that generality is treated in [26].

***Proof*:**
The proof constructs the class of a gerbe as a Čech two-cocycle from local objects and their discrepancies, checks it is well-defined, and inverts the construction. It is the one-degree-up version of the proof of theorem 18.5.3, and the parallel is worth keeping in view: objects replace sections, isomorphisms of objects replace transition elements, and the coherence of those isomorphisms is one dimension higher. Recall the Čech description of H^2. For a covering family $\mathcal{U} = \{U_i \to X\}$, write $U_{ij} = U_i \times_X U_j$, and likewise U_{ijk}, U_{ijkl} for triple and quadruple overlaps. A 2-*cocycle* on $\mathcal{U}$ with values in A is a family $(\alpha_{ijk}) \in \prod_{i,j,k} A(U_{ijk})$ satisfying the cocycle condition

$$\alpha_{jkl} - \alpha_{ikl} + \alpha_{ijl} - \alpha_{ijk} = 0 \quad \text{on } U_{ijkl} \tag{21.1}$$

(the alternating sum over the four faces of the tetrahedron U_{ijkl}), and two cocycles are cohomologous if they differ by a coboundary $(\delta\beta)_{ijk} = \beta_{jk} - \beta_{ik} + \beta_{ij}$ of a 1-cochain $(\beta_{ij}) \in \prod_{i,j} A(U_{ij})$. The group of cocycles modulo coboundaries is $\check{H}^2(\mathcal{U}, A)$, and $H^2_\tau(X, A)$ is its colimit over refinements, which agrees with the derived-functor cohomology of A on the site by [15].

Step 1: From a gerbe to a two-cocycle. Let $\mathcal{G}$ be an A-gerbe. By local nonemptiness choose a covering family $\mathcal{U} = \{U_i \to X\}$ and objects $x_i \in \mathcal{G}(U_i)$, one over each cover piece. On a double overlap U_{ij} the restricted objects $x_i|_{U_{ij}}$ and $x_j|_{U_{ij}}$ are two objects of $\mathcal{G}(U_{ij})$; by local connectedness, after refining the cover so that each pair is already isomorphic over U_{ij}, choose an isomorphism

$$\varphi_{ij} : x_j|_{U_{ij}} \xrightarrow{\sim} x_i|_{U_{ij}}$$

the two-dimensional analogue of the transition element g_{ij} of a torsor. These isomorphisms need not compose coherently on triple overlaps. On U_{ijk} compare the two isomorphisms $x_k \to x_i$ obtained as $\varphi_{ij} \circ \varphi_{jk}$ and directly as φ_{ik}: both are isomorphisms $x_k|_{U_{ijk}} \to x_i|_{U_{ijk}}$, so they differ by a unique automorphism of x_i,

$$\varphi_{ij} \circ \varphi_{jk} = \alpha_{ijk} \cdot \varphi_{ik} \quad \text{on } U_{ijk} \tag{21.2}$$

where $\alpha_{ijk} \in \underline{\mathrm{Aut}}(x_i)(U_{ijk}) = A(U_{ijk})$, the last identification being the banding by A. The element α_{ijk} is the *discrepancy* of the transition isomorphisms on the triple overlap, and it is the exact two-dimensional analogue of the fact that for a torsor the transition elements composed on the nose (the one-cocycle condition $g_{ik} = g_{ij}g_{jk}$); here they compose only up to the automorphism α_{ijk}.

That (α_{ijk}) is a two-cocycle is the associativity of composition of the φ on the quadruple overlap U_{ijkl}. Compute the composite $x_l \to x_i$ in two ways. Bracketing as $(\varphi_{ij} \circ \varphi_{jk}) \circ \varphi_{kl}$ and applying (21.2) to the inner pair, then again, expresses it through φ_{il} and the automorphisms $\alpha_{ijk}, \alpha_{ikl}$; bracketing as $\varphi_{ij} \circ (\varphi_{jk} \circ \varphi_{kl})$ expresses it through φ_{il} and $\alpha_{jkl}, \alpha_{ijl}$. Since composition of isomorphisms is associative and A is abelian (so the automorphisms transported along φ_{ij} from $\underline{\mathrm{Aut}}(x_j)$ to $\underline{\mathrm{Aut}}(x_i)$ are unchanged, both being identified with A), equating the two bracketings gives

$$\alpha_{ijk} + \alpha_{ikl} = \alpha_{jkl} + \alpha_{ijl} \quad \text{on } U_{ijkl}$$

which is the cocycle condition (21.1). Thus (α_{ijk}) is a 2-cocycle on $\mathcal{U}$ with values in A.

Step 2: The class is independent of choices. Two kinds of choice were made: the objects x_i and the isomorphisms φ_{ij}. Changing the isomorphisms first, replace each φ_{ij} by $\varphi'_{ij} = \beta_{ij} \cdot \varphi_{ij}$ for some $\beta_{ij} \in \underline{\mathrm{Aut}}(x_i)(U_{ij}) = A(U_{ij})$. Substituting into (21.2) and using that A is abelian and central in each automorphism sheaf,

$$\varphi'_{ij} \circ \varphi'_{jk} = \beta_{ij}\, \varphi_{ij}\, \beta_{jk}\, \varphi_{jk} = (\beta_{ij} + \beta_{jk} + \alpha_{ijk})\, \varphi_{ik} = (\beta_{ij} + \beta_{jk} - \beta_{ik})\, \varphi'_{ik}$$

where the second equality moves β_{jk} past φ_{ij} using the banding (its image in $\underline{\mathrm{Aut}}(x_i)$ is the same $\beta_{jk} \in A$ because the band identifies all automorphism sheaves with A), and the third rewrites $\varphi_{ik} = \varphi'_{ik} - \beta_{ik}$ additively. Hence the new discrepancy is $\alpha'_{ijk} = \alpha_{ijk} + (\beta_{jk} - \beta_{ik} + \beta_{ij})$, differing from α_{ijk} by the coboundary $(\delta\beta)_{ijk}$: the class in $\check{H}^2(\mathcal{U}, A)$ is unchanged. Changing the objects x_i to isomorphic objects $x'_i \cong x_i$ over U_i transports the isomorphisms φ_{ij} along these isomorphisms and leaves the discrepancies invariant, again by centrality of A; and refining the cover restricts the cocycle to the refinement, changing nothing in the colimit. Thus $[(\alpha_{ijk})] \in H^2_\tau(X, A)$ depends only on the equivalence class of $\mathcal{G}$.

Step 3: From a two-cocycle to a gerbe. Conversely, let (α_{ijk}) be a two-cocycle on $\mathcal{U}$. Build a gerbe $\mathcal{G}_\alpha$ by descent for stacks (theorem 19.2.6) as follows. On each U_i take the neutral gerbe $BA|_{U_i}$, whose objects are A-torsors; the tautological object $\xi_i \in BA(U_i)$ is the trivial A-torsor.

Glue $BA|_{U_j}$ to $BA|_{U_i}$ over U_{ij} by the equivalence $\Phi_{ij}\colon BA|_{U_{ij}} \to BA|_{U_{ij}}$ that is the identity functor on objects and morphisms (both restrict to the same BA); the gluing carries ξ_j to ξ_i by the tautological isomorphism φ_{ij}. On a triple overlap the composite $\Phi_{ij} \circ \Phi_{jk}$ and Φ_{ik} agree as functors but their action on the tautological objects differs by the 2-cochain α_{ijk}: the natural transformation

$$\Phi_{ij} \circ \Phi_{jk} \Rightarrow \Phi_{ik}$$

is multiplication by $\alpha_{ijk} \in A(U_{ijk})$, acting through the band. The descent datum for a stack requires that these natural transformations satisfy the coherence pentagon on quadruple overlaps, and that coherence is exactly the two-cocycle condition (21.1): the alternating sum vanishing is the statement that the two ways of coherently associating four gluings agree. By theorem 19.2.6 the descent datum is effective, and the pieces glue to a stack $\mathcal{G}_\alpha$ over X.

The glued stack $\mathcal{G}_\alpha$ is a gerbe: it is locally nonempty because each $BA|_{U_i}$ has the object ξ_i, and locally connected because the objects of BA over any base, being A-torsors, are locally isomorphic. Its band is A, because the automorphism sheaf of the tautological object ξ_i is $\underline{\mathrm{Aut}}_{BA}(\text{trivial torsor}) = A$, glued by the banding. Thus $\mathcal{G}_\alpha$ is an A-gerbe. Applying Step 1 to $\mathcal{G}_\alpha$ with the objects ξ_i and the tautological isomorphisms φ_{ij} returns, by (21.2), the discrepancy α_{ijk}: the natural transformation $\Phi_{ij} \circ \Phi_{jk} \Rightarrow \Phi_{ik}$ was defined to be α_{ijk}. Hence the class of $\mathcal{G}_\alpha$ is $[(\alpha_{ijk})]$.

Step 4: The two constructions are mutually inverse. Step 1 sends a gerbe to a class; Step 3 sends a cocycle to a gerbe whose class is the given one. It remains to see that a gerbe $\mathcal{G}$ with class $[(\alpha_{ijk})]$ is equivalent to the gerbe $\mathcal{G}_\alpha$ built from that cocycle. The objects $x_i \in \mathcal{G}(U_i)$ and isomorphisms φ_{ij} chosen in Step 1 exhibit $\mathcal{G}$ as glued from the neutral gerbes $\mathcal{G}|_{U_i} \simeq B\underline{\mathrm{Aut}}(x_i) \simeq BA|_{U_i}$ along equivalences whose triple-overlap discrepancies are α_{ijk}, which is precisely the descent datum defining $\mathcal{G}_\alpha$. The equivalence sending x_i to the tautological object ξ_i is compatible with the gluing, hence extends to an equivalence $\mathcal{G} \simeq \mathcal{G}_\alpha$. Cohomologous cocycles give equivalent gerbes by Step 2 (a coboundary is a change of the isomorphisms φ_{ij}, which induces an equivalence of the glued gerbes), so the two assignments descend to mutually inverse bijections between equivalence classes of A-gerbes and $H^2_\tau(X, A)$. The neutral gerbe BA, having the global object $\xi =$ the trivial torsor, admits objects x_i all restricting from a single global object, so all transition isomorphisms can be chosen compatibly and every discrepancy α_{ijk} is trivial: the neutral gerbe maps to the zero class.

Step 5: The group structure. The abelian group structure on $H^2_\tau(X, A)$, addition of two-cocycles, transports to gerbes as the *contracted product*. Given A-gerbes $\mathcal{G}, \mathcal{G}'$ with cocycles $(\alpha_{ijk}), (\alpha'_{ijk})$, form the gerbe $\mathcal{G} \wedge^A \mathcal{G}'$ whose objects over U are pairs (x, x') of a local object of each, with automorphisms identified along the band by the codiagonal $A \times A \to A$ (adding the two automorphisms); its triple-overlap discrepancy is $\alpha_{ijk} + \alpha'_{ijk}$, so its class is the sum. The neutral gerbe BA is the identity, the gerbe with cocycle $(-\alpha_{ijk})$ is the inverse, and the bijection of Step 4 is a group isomorphism, completing the proof.

The theorem is the exact one-story ascent from the torsor classification, and the dictionary between the two is worth setting down, because every later computation reads it off. A torsor is trivialized by a global section; a gerbe is neutralized by a global object. The discrepancy of local sections on a double overlap is a transition element g_{ij}, a one-cochain; the discrepancy of local objects on a double overlap is an isomorphism φ_{ij}, and its own failure to compose is the two-cochain α_{ijk}. The one-cocycle condition $g_{ik} = g_{ij}g_{jk}$ was the associativity of transition on triple overlaps; the two-cocycle condition (21.1) is the associativity of transition one level up, the coherence of the isomorphisms on quadruple overlaps. The freedom to change local sections was a coboundary in degree one; the freedom to change local isomorphisms is a coboundary in degree two. Every structural feature of the torsor theorem reappears with its degree raised by one, which is the sense in which a gerbe is the two-dimensional torsor promised in Chapter 6.

The role of abelianness is the same as it was for torsors, and locating it explains what the nonabelian theory must repair. The proof used commutativity of A in exactly one place that mattered: transporting an automorphism along a transition isomorphism φ_{ij} changes it by conjugation, and for abelian A conjugation is trivial, so the band identifies all automorphism sheaves with one fixed A and the cochains live in a single abelian group. When the band is nonabelian this identification is ambiguous up to inner automorphism, the transported cochains no longer live in a fixed group, and the cocycle condition must be phrased for a lien rather than a sheaf of groups. The resulting nonabelian H^2 is not a group but a set on which a group of two-cocycles acts, with the neutral gerbe a distinguished point that need not be a zero; the full account, including the obstruction in H^3 to a band being realized by any gerbe at all, is Giraud's [26]. This book takes the abelian band as its object because the coefficient group that realizes the Brauer group in the next section, the multiplicative group $\mathbb{G}_m$, is abelian, and the entire $\mathbb{G}_m$-gerbe theory lives inside the theorem just proved.

The classification also completes the automorphism thread that has run since Part IV, and the completion is best seen on a nonneutral gerbe, since the residual gerbes of example 21.1.4 were all neutral. A nonzero class in H^2 produces a gerbe with no global object, and the next example builds one by hand from a nonzero two-cocycle, so that the base-point structure of the theorem is not left abstract.

Example 21.1.6: A Nonneutral Gerbe from a Two-Cocycle

Let $A = \mu_n$ and suppose X has a nonzero class $\alpha \in H^2_\tau(X, \mu_n)$, represented by a two-cocycle (α_{ijk}) on a cover $\mathcal{U} = \{U_i \to X\}$. By theorem 21.1.5 the gerbe $\mathcal{G}_\alpha$ built from (α_{ijk}) in Step 3 of the proof is a μ_n-gerbe with class $\alpha \neq 0$, hence *not* neutral: it has no global object, because a global object would trivialize the class. Concretely $\mathcal{G}_\alpha$ has, on each U_i, the object ξ_i, and these fail to glue to a global object precisely because the isomorphisms φ_{ij} gluing ξ_j to ξ_i do not compose coherently: their triple-overlap discrepancy is α_{ijk}, and if a global object existed one could choose the φ_{ij} to compose on the nose, forcing α_{ijk} to be a coboundary and $\alpha = 0$.

A source of such classes is at hand from the Kummer sequence of Chapter 6. The connecting map of the long exact cohomology sequence of $1 \to \mu_n \to \mathbb{G}_m \to \mathbb{G}_m \to 1$ (the sequence (18.5) of Chapter 6) in degree two,

$$\operatorname{Pic}(X) = H^1(X, \mathbb{G}_m) \xrightarrow{\delta} H^2(X, \mu_n)$$

sends a line bundle L to the class of the μ_n-gerbe of n-th roots of L: the gerbe $\mathcal{G}_L^{1/n}$ whose objects over U are pairs (M, ι) of a line bundle M on U and an isomorphism $\iota\colon M^{\otimes n} \cong L|_U$. This gerbe is locally nonempty because an n-th root of L exists locally (any line bundle is locally trivial, hence locally an n-th power), and locally connected because any two local n-th roots differ by a μ_n-torsor; its band is μ_n, acting on a root M by scaling. It is neutral exactly when L has a global n-th root, so its class $\delta[L] \in H^2(X, \mu_n)$ vanishes iff $L \in n \cdot \operatorname{Pic}(X)$. On a variety with a line bundle that is not an n-th power, for instance $L = \mathcal{O}(1)$ on $\mathbb{P}^m$ with $n \geq 2$ (so $L \notin n \operatorname{Pic}(\mathbb{P}^m) = n\mathbb{Z}$), the gerbe $\mathcal{G}_L^{1/n}$ is a nonneutral μ_n-gerbe, an explicit representative of a nonzero class in $H^2(\mathbb{P}^m, \mu_n)$. This is the first nonneutral gerbe of the book, and the mechanism, an obstruction to extracting a global root, is the template for the Brauer classes of the next section.

Exercise 21.1.1

1. Let $\mathcal{G}$ be a gerbe over $(\mathbf{Sch}/X)_\tau$. Prove directly from definition 21.1.1 that $\mathcal{G}$ is neutral if and only if it has a global object, and that a global object $x \in \mathcal{G}(X)$ determines an equivalence $\mathcal{G} \simeq B\underline{\mathrm{Aut}}(x)$. Identify precisely which axiom of a gerbe supplies local nonemptiness of the Isom sheaf $\underline{\mathrm{Isom}}(x|_U, y)$ and which supplies its simple transitivity, so that it is a torsor under $\underline{\mathrm{Aut}}(x)$.
2. Show that the residual gerbe $\mathcal{G}_x$ of a point $x \in \mathcal{X}(k)$ of an algebraic stack (example 21.1.4) is always neutral, and hence always of the form $B\underline{\mathrm{Aut}}(x)$ over $\operatorname{Spec} k$. Explain why this does not contradict the existence of nonneutral gerbes: what feature of $\operatorname{Spec} k$, as opposed to a higher-dimensional base, forces neutrality?
3. Carry out in full the verification that the discrepancy family (α_{ijk}) of (21.2) satisfies the two-cocycle condition (21.1), by expanding the composite $\varphi_{ij} \circ \varphi_{jk} \circ \varphi_{kl}$ on the quadruple overlap U_{ijkl} in the two bracketings and equating them. State explicitly where the abelianness of the band A is used.
4. Using the connecting map $\delta\colon \operatorname{Pic}(X) \to H^2(X, \mu_n)$ of example 21.1.6, prove that the μ_n-gerbe $\mathcal{G}_L^{1/n}$ of n-th roots of a line bundle L is neutral if and only if L admits a global n-th root, and hence that $\delta[L] = 0$ if and only if $L \in n \cdot \operatorname{Pic}(X)$. Deduce that $\mathcal{G}^{1/2}_{\mathcal{O}(1)}$ on $\mathbb{P}^2_k$ is a nonneutral μ_2-gerbe.
5. Prove that the contracted product $\mathcal{G} \wedge^A \mathcal{G}'$ of two A-gerbes (theorem 21.1.5, Step 5) has class $[\mathcal{G}] + [\mathcal{G}']$ by computing its triple-overlap discrepancy from those of $\mathcal{G}$ and $\mathcal{G}'$. Identify the neutral gerbe BA as the identity element and describe the inverse of an A-gerbe both as a gerbe and at the level of its class.
6. Explain, by locating the exact step in the proof of theorem 21.1.5 where commutativity of A is used, why the classification is a group isomorphism for an abelian band but only a bijection of pointed sets (with a distinguished class that need not be a zero) for a nonabelian band. Contrast this with the analogous phenomenon for torsors in theorem 18.5.3, and state in one sentence what new invariant (a lien, and an H^3-obstruction) the nonabelian theory of [26] must introduce that the abelian theory does not need.

21.2 $\mathbb{G}_m$-Gerbes and the Brauer Group

The previous section classified A-gerbes on X by $H^2_\tau(X, A)$ for an abelian band A, one cohomological degree above the torsors of Chapter 6. This section specializes the band to the multiplicative group $\mathbb{G}_m$, where the arithmetic content appears. The reason $\mathbb{G}_m$ is the band to single out is that $H^2_{\text{ét}}(X, \mathbb{G}_m)$ is not an abstract cohomology group but a classical invariant in disguise: over a field it is the Brauer group, the group of central simple algebras up to Morita equivalence, and over a general scheme it records the failure of Azumaya algebras and PGL_n-bundles to lift to GL_n. A $\mathbb{G}_m$-gerbe is thus a geometric avatar of a Brauer class, and the two-way dictionary between algebras and gerbes is what this section makes precise. The central construction is the boundary map of the short exact sequence $1 \to \mathbb{G}_m \to \mathrm{GL}_n \to \mathrm{PGL}_n \to 1$, which sends a PGL_n-torsor (equivalently a rank-n^2 Azumaya algebra) to the class in $H^2(X, \mathbb{G}_m)$ obstructing its lift to a vector bundle. This class is n-torsion, and the map it defines embeds the Azumaya Brauer group into its cohomological cousin. Gabber's theorem, that the embedding is a bijection onto the torsion subgroup, is stated and cited; the construction of the map and its torsion bound are carried out in full.

The band $\mathbb{G}_m$ deserves its own name at the level of gerbes because the entire theory of twisted sheaves in the next section is written for gerbes banded by it. Applying theorem 21.1.5 with $A = \mathbb{G}_m$ gives the classification for free; the definition records the terminology.

Definition 21.2.1: $\mathbb{G}_m$-Gerbe

Let X be a scheme and τ the étale (equivalently, for $\mathbb{G}_m$, the fppf) topology on $(\mathbf{Sch}/X)_\tau$. A $\mathbb{G}_m$-*gerbe* on X is a gerbe $\mathcal{G} \to (\mathbf{Sch}/X)_\tau$ banded by the multiplicative group $\mathbb{G}_m$ in the sense of definition 21.1.3: a gerbe together with a fixed identification $\mathbb{G}_m \cong \underline{\mathrm{Aut}}(x)$ of the automorphism sheaf of every local object x with $\mathbb{G}_m$, compatible with restriction. By theorem 21.1.5 the equivalence classes of $\mathbb{G}_m$-gerbes on X are in natural bijection with

$$H^2_{\text{ét}}(X, \mathbb{G}_m)$$

the neutral gerbe $B\mathbb{G}_{m,X}$ corresponding to the zero class. Given a class $\alpha \in H^2_{\text{ét}}(X, \mathbb{G}_m)$, write $\mathcal{G}_\alpha$ for a gerbe representing it.

The identification étale = fppf for $\mathbb{G}_m$-cohomology is Grothendieck's theorem that $H^i_{\text{ét}}(X, \mathbb{G}_m) = H^i_{\text{fppf}}(X, \mathbb{G}_m)$ for a smooth group scheme, so the choice of site does not matter and either may be used; this is why the definition brackets the two topologies together. The content of theorem 21.1.5 in this case is that a $\mathbb{G}_m$-gerbe is the same data as a class in a single cohomology group, and the neutral gerbe $B\mathbb{G}_{m,X}$, the classifying stack of the trivial $\mathbb{G}_m$-bundle, is the base point. A nonzero class is a gerbe with no global object, an obstruction to trivializing the automorphism-$\mathbb{G}_m$'s coherently across X.

The first example to have in hand is the automorphism gerbe of a stack point, which threads back to the $\mathcal{M}_{1,1}$ story of Chapter 1 and shows that $\mathbb{G}_m$-gerbes arise unbidden from the geometry already developed.

Example 21.2.2: The $\mathbb{G}_m$-Gerbe of a Stacky Point

Let k be a field and consider the classifying stack $B\mathbb{G}_{m,k} = [\operatorname{Spec} k/\mathbb{G}_m]$, the neutral $\mathbb{G}_m$-gerbe over $\operatorname{Spec} k$. Its single object over $\operatorname{Spec} k$ has automorphism group $\mathbb{G}_m(k) = k^\times$, the band, and it represents the zero class in $H^2_{\text{ét}}(\operatorname{Spec} k, \mathbb{G}_m)$. This is the trivial case: a point with a $\mathbb{G}_m$ of automorphisms, but with a global object, hence neutral.

A nonneutral example comes from an elliptic curve with extra automorphisms. Over the special points $j = 0$ and $j = 1728$ of the coarse space of $\mathcal{M}_{1,1}$ (theorem 13.3.6), the curve E has automorphism group larger than the generic $\{\pm 1\}$: cyclic of order 6 at $j = 0$ and of order 4 at $j = 1728$ (in characteristic 0, or away from $2, 3$; example 13.3.4). Over an algebraically closed field the residual gerbe at such a point is the classifying stack $B\mu_n$ (with $n = 6$ at $j = 0$, $n = 4$ at $j = 1728$, the order of $\underline{\mathrm{Aut}}(E)$), a μ_n-gerbe rather than a $\mathbb{G}_m$-gerbe: the automorphism group of the object bands the residual gerbe, and pushing forward along $\mu_n \hookrightarrow \mathbb{G}_m$ realizes it as a $\mathbb{G}_m$-gerbe whose class is the image of the μ_n-class. Over a field the class is torsion of order dividing n, and it records the extra symmetry of the special elliptic curve.

Over an algebraically closed k this residual gerbe is neutral: the curve E is a k-rational object, so it is the classifying stack $B\mu_n$ of a global object and its class is zero. A *nonneutral* example lives over a subfield k_0 over which the special curve E does *not* descend to a k_0-rational elliptic curve.

There the residual gerbe is a nontrivial *form* of $B\mu_n$, a μ_n-gerbe with no global object rather than the classifying stack of any k_0-rational curve, and its class in $H^2(\operatorname{Spec} k_0, \mu_n)$ can be nonzero. The obstruction is whether the object E itself descends to k_0, not whether its automorphisms are rational: a classifying stack $B\mu_n$ always has the trivial torsor as a global object and so is neutral regardless of $\mu_n(k_0)$. Only when E fails to descend does the gerbe acquire a nonzero class, recording the field of definition of the symmetric curve.

The example makes concrete the slogan that a $\mathbb{G}_m$-gerbe measures a failure of descent for objects with $\mathbb{G}_m$ of automorphisms. When the object exists, the gerbe is $B\mathbb{G}_m$ and the class is zero; when it exists only locally, the gluing discrepancies assemble into a 2-cocycle whose class is the obstruction. The same phenomenon over a field is the Brauer group, to which the discussion now turns.

The cohomological Brauer group is defined so that the field case reproduces the classical group-cohomological Brauer group of [13], and so that the torsion condition, invisible over a field where the whole group is torsion, becomes the operative restriction over a higher-dimensional base. Alongside it sits the Azumaya Brauer group, built from actual algebras rather than cohomology classes; the comparison of the two is the theorem of the section.

Definition 21.2.3: Brauer Group and Cohomological Brauer Group

Let X be a scheme. The *cohomological Brauer group* of X is the torsion subgroup

$$\operatorname{Br}'(X) = H^2_{\text{ét}}(X, \mathbb{G}_m)_{\text{tors}}$$

of the second étale cohomology group of $\mathbb{G}_m$. An *Azumaya algebra* on X is a sheaf $\mathcal{A}$ of $\mathcal{O}_X$-algebras that is, étale-locally on X, isomorphic to a matrix algebra $\operatorname{Mat}_n(\mathcal{O}_X)$ (equivalently, $\mathcal{A}$ is a locally free $\mathcal{O}_X$-algebra of finite rank whose fibers $\mathcal{A} \otimes \kappa(x)$ are central simple algebras over the residue fields $\kappa(x)$). Two Azumaya algebras $\mathcal{A}, \mathcal{B}$ are *Morita equivalent*, or *Brauer equivalent*, if there exist locally free sheaves $\mathcal{E}, \mathcal{F}$ of positive rank with

$$\mathcal{A} \otimes_{\mathcal{O}_X} \underline{\operatorname{End}}(\mathcal{E}) \cong \mathcal{B} \otimes_{\mathcal{O}_X} \underline{\operatorname{End}}(\mathcal{F})$$

The *(Azumaya) Brauer group* $\operatorname{Br}(X)$ is the set of Morita equivalence classes, a group under $\otimes_{\mathcal{O}_X}$ with identity the class of $\mathcal{O}_X$ and inverse the class of the opposite algebra $\mathcal{A}^{\text{op}}$. When $X = \operatorname{Spec} k$ is the spectrum of a field, $\operatorname{Br}(k)$ is the classical Brauer group of central simple k-algebras, and by [13]

$$\operatorname{Br}(\operatorname{Spec} k) = H^2\big(\operatorname{Gal}(k^{\text{sep}}/k), (k^{\text{sep}})^\times\big)$$

the Galois cohomology of the multiplicative group of a separable closure.

Two distinct groups now carry the name Brauer, and the distinction is the crux of the section. The Azumaya group $\operatorname{Br}(X)$ is built from algebras: étale-locally matrix algebras, glued by a PGL_n-cocycle since the automorphisms of Mat_n are PGL_n by the Skolem-Noether theorem. The cohomological group $\operatorname{Br}'(X)$ is a subgroup of a cohomology group, defined without reference to any algebra. Over a field the two coincide and both equal the Galois-cohomological group of [13], because over a field every class of $H^2(\operatorname{Gal}, (k^{\text{sep}})^\times)$ is torsion and is represented by a central simple algebra. Over a general scheme the coincidence is a theorem, Gabber's, and the torsion restriction in the definition of Br' is exactly what an Azumaya algebra can reach: a matrix-algebra bundle of rank n^2 produces an n-torsion class, so only torsion classes are candidates for algebras.

The Galois-cohomological description over a field is worth pausing on, since it grounds every field computation below. For k a field with absolute Galois group $G_k = \mathrm{Gal}(k^{\mathrm{sep}}/k)$, the group $\mathrm{Br}(k) = H^2(G_k, (k^{\mathrm{sep}})^\times)$ is built from the crossed-product construction: a 2-cocycle $c\colon G_k \times G_k \to (k^{\mathrm{sep}})^\times$ yields a central simple algebra $\bigoplus_{\sigma \in G} k^{\mathrm{sep}} u_\sigma$ with multiplication twisted by c, and every central simple algebra arises this way. This is the field model the gerbe picture generalizes: a $\mathbb{G}_m$-gerbe on X is the geometric object whose Čech 2-cocycle over an étale cover plays the role of the crossed-product cocycle over the field.

The bridge between the two Brauer groups is the boundary map of the defining sequence of PGL_n. This is the point where the gerbe interpretation does its work: the obstruction to lifting a PGL_n-torsor to a GL_n-torsor is a $\mathbb{G}_m$-gerbe, and its class in $H^2(X, \mathbb{G}_m)$ is the Brauer class of the corresponding Azumaya algebra. The following theorem constructs this map, bounds its image by n-torsion, and records that it embeds the Azumaya group into the cohomological one.

Theorem 21.2.4: Azumaya Algebras, PGL_n-Torsors, and $\mathbb{G}_m$-Gerbes

Let X be a scheme. Fix $n \geq 1$ and consider the short exact sequence of étale sheaves of groups

$$1 \longrightarrow \mathbb{G}_m \longrightarrow \mathrm{GL}_n \longrightarrow \mathrm{PGL}_n \longrightarrow 1 \tag{21.3}$$

where $\mathbb{G}_m$ is the center of GL_n (scalar matrices). Then:

1. The boundary map of nonabelian cohomology

$$\delta\colon H^1_{\text{ét}}(X, \mathrm{PGL}_n) \longrightarrow H^2_{\text{ét}}(X, \mathbb{G}_m)$$

assigns to a PGL_n-torsor P the class of the $\mathbb{G}_m$-gerbe $\mathcal{G}_P$ of local lifts of P to a GL_n-torsor. A rank-n^2 Azumaya algebra $\mathcal{A}$ corresponds to a PGL_n-torsor $P_{\mathcal{A}}$ (theorem 18.5.3 applied to the automorphism sheaf $\underline{\mathrm{Aut}}(\mathcal{A}) = \mathrm{PGL}_n$), and $\delta([P_{\mathcal{A}}]) = [\mathcal{A}] \in \mathrm{Br}(X)$ is its Brauer class.

2. The class $\delta([P]) \in H^2_{\text{ét}}(X, \mathbb{G}_m)$ is n-torsion.

3. The assignment $[\mathcal{A}] \mapsto \delta([P_{\mathcal{A}}])$ is a well-defined injective group homomorphism

$$\mathrm{Br}(X) \longleftrightarrow \mathrm{Br}'(X) = H^2_{\text{ét}}(X, \mathbb{G}_m)_{\mathrm{tors}}$$

from the Azumaya Brauer group to the cohomological Brauer group.

***Proof*:**
The three assertions are the construction of the boundary map, the torsion bound, and the injectivity of the induced homomorphism. They are proved in that order.

Step 1: The gerbe of local lifts and the boundary class. Let P be a PGL_n-torsor on X, classified by an étale-Čech 1-cocycle $\bar{g}_{ij} \in \mathrm{PGL}_n(U_{ij})$ on a cover $\{U_i \to X\}$ (theorem 18.5.3). Define a stack $\mathcal{G}_P$ over $(\mathbf{Sch}/X)_{\text{ét}}$ whose objects over $U \to X$ are the GL_n-torsors Q on U equipped with an isomorphism $Q/\mathbb{G}_m \cong P|_U$ of PGL_n-torsors, that is, the lifts of P across (21.3). Morphisms are isomorphisms of GL_n-torsors commuting with the identification of the associated PGL_n-torsor. Two facts make $\mathcal{G}_P$ a $\mathbb{G}_m$-gerbe.

First, $\mathcal{G}_P$ is locally nonempty: on each U_i the restriction $P|_{U_i}$ is trivial, so it lifts to the trivial

GL_n-torsor, giving an object of $\mathcal{G}_P(U_i)$. Second, $\mathcal{G}_P$ is locally connected with band $\mathbb{G}_m$: if Q, Q' are two lifts of $P|_U$, then étale-locally both are trivial, and any two lifts differ by an automorphism of P that lifts to GL_n, hence by an element of the kernel $\mathbb{G}_m$; concretely $Q' = Q \otimes L$ for a $\mathbb{G}_m$-torsor L, so any two objects are locally isomorphic and the automorphism sheaf of a lift is $\mathbb{G}_m$ (the scalars acting on Q that fix the induced PGL_n-torsor). Thus $\mathcal{G}_P$ is a $\mathbb{G}_m$-gerbe, and $\delta([P]) := [\mathcal{G}_P] \in H^2_{\text{ét}}(X, \mathbb{G}_m)$ is its class under theorem 21.1.5.

Explicitly, choose set-theoretic lifts $g_{ij} \in \mathrm{GL}_n(U_{ij})$ of the cocycle $\bar{g}_{ij}$. The failure of the g_{ij} to satisfy the cocycle condition is measured by

$$\alpha_{ijk} = g_{ij}\, g_{jk}\, g_{ki} \in \mathrm{GL}_n(U_{ijk}) \tag{21.4}$$

which, because $\bar{g}_{ij}\bar{g}_{jk}\bar{g}_{ki} = 1$ in PGL_n, lies in the kernel $\mathbb{G}_m(U_{ijk})$ (it is a scalar matrix). A direct expansion on U_{ijkl}, using associativity of matrix multiplication, gives the 2-cocycle identity $\alpha_{jkl}\,\alpha_{ijl}\,\alpha_{ijk}^{-1}\,\alpha_{ikl}^{-1} = 1$, so (α_{ijk}) is a Čech 2-cocycle with values in $\mathbb{G}_m$, and its class is $\delta([P])$. Changing the lifts g_{ij} by scalars $\lambda_{ij} \in \mathbb{G}_m(U_{ij})$ alters α_{ijk} by the coboundary of (λ_{ij}), so the class is independent of the choices. For the Azumaya algebra $\mathcal{A}$ of rank n^2, the sheaf $\underline{\mathrm{Aut}}_{\mathcal{O}_X\text{-alg}}(\mathcal{A})$ is PGL_n by the Skolem-Noether theorem, so $\mathcal{A}$ is classified by a PGL_n-torsor $P_{\mathcal{A}}$; the lifts of $P_{\mathcal{A}}$ to GL_n are precisely the locally free sheaves $\mathcal{E}$ with $\underline{\mathrm{End}}(\mathcal{E}) \cong \mathcal{A}$, and $\delta([P_{\mathcal{A}}]) = [\mathcal{A}]$ is the obstruction to the existence of such an $\mathcal{E}$ globally, which is the Brauer class.

Step 2: The torsion bound. The determinant gives a homomorphism $\det\colon \mathrm{GL}_n \to \mathbb{G}_m$ whose restriction to the central $\mathbb{G}_m$ of scalars is the n-th power map $\lambda \mapsto \lambda^n$. Apply det to the scalar-valued cochain α_{ijk} of (21.4): since α_{ijk} is the scalar matrix $\alpha_{ijk} \cdot \mathrm{id}$,

$$\det(\alpha_{ijk}) = \alpha_{ijk}^n$$

On the other hand $\det(g_{ij}) \in \mathbb{G}_m(U_{ij})$ are units, and applying the multiplicative det to (21.4) gives $\det(\alpha_{ijk}) = \det(g_{ij})\det(g_{jk})\det(g_{ki})$, which is the Čech coboundary $\partial(\det g)_{ijk}$ of the 1-cochain $(\det g_{ij})$. Therefore $\alpha_{ijk}^n = \partial(\det g)_{ijk}$ is a coboundary, so $n \cdot \delta([P]) = n \cdot [\alpha] = [\alpha^n] = 0$ in $H^2_{\text{ét}}(X, \mathbb{G}_m)$. Hence $\delta([P])$ is n-torsion.

Step 3: Well-definedness, homomorphism, and injectivity. First, $[\mathcal{A}] \mapsto \delta([P_{\mathcal{A}}])$ is well-defined on Morita classes: replacing $\mathcal{A}$ by $\mathcal{A} \otimes \underline{\mathrm{End}}(\mathcal{E})$ replaces the rank n by nm (with $m = \mathrm{rk}\,\mathcal{E}$) and the torsor $P_{\mathcal{A}}$ by the torsor of $\mathcal{A} \otimes \underline{\mathrm{End}}(\mathcal{E})$, but the associated gerbe of local lifts is unchanged up to equivalence, because $\underline{\mathrm{End}}(\mathcal{E})$ is a matrix algebra with a global lift $\mathcal{E}$, so it contributes the zero class and $\delta([P_{\mathcal{A} \otimes \underline{\mathrm{End}}(\mathcal{E})}]) = \delta([P_{\mathcal{A}}])$. Thus the map descends to $\mathrm{Br}(X)$.

It is a homomorphism because tensor product of Azumaya algebras corresponds to the Baer sum of the associated gerbes: if $\mathcal{A}$ has rank n^2 with cocycle α_{ijk} and $\mathcal{B}$ has rank m^2 with cocycle β_{ijk}, then $\mathcal{A} \otimes \mathcal{B}$ has rank $(nm)^2$, its structure sheaf glues by the Kronecker product $g_{ij} \otimes h_{ij}$, and the scalar discrepancy of $g_{ij} \otimes h_{ij}$ is $\alpha_{ijk}\,\beta_{ijk}$, so $\delta([\mathcal{A} \otimes \mathcal{B}]) = \delta([\mathcal{A}]) + \delta([\mathcal{B}])$ in additive notation. The identity $\mathcal{O}_X$ maps to the class of the neutral gerbe, namely 0.

For injectivity, suppose $\delta([\mathcal{A}]) = 0$, so the gerbe $\mathcal{G}_{P_{\mathcal{A}}}$ of local lifts is neutral and has a global object: a globally defined GL_n-torsor Q with $Q/\mathbb{G}_m \cong P_{\mathcal{A}}$. Equivalently there is a globally defined locally free sheaf $\mathcal{E}$ of rank n with $\underline{\mathrm{End}}(\mathcal{E}) \cong \mathcal{A}$. Then $\mathcal{A} = \underline{\mathrm{End}}(\mathcal{E}) = \mathcal{O}_X \otimes \underline{\mathrm{End}}(\mathcal{E})$ is Morita equivalent to $\mathcal{O}_X$, so $[\mathcal{A}] = 0$ in $\mathrm{Br}(X)$. Hence δ has trivial kernel and the induced map $\mathrm{Br}(X) \hookrightarrow \mathrm{Br}'(X)$ is injective, completing the proof.

The theorem constructs an injection $\mathrm{Br}(X) \hookrightarrow \mathrm{Br}'(X)$ and identifies the Azumaya Brauer group with a subgroup of the torsion of $H^2_{\text{ét}}(X, \mathbb{G}_m)$. The gerbe $\mathcal{G}_P$ is the geometric carrier of the obstruction: it is neutral exactly when the PGL_n-torsor lifts to a vector bundle, that is, when the Azumaya algebra is a matrix algebra, and its class is the precise cohomological measure of the failure to lift. The torsion bound is not an accident of the proof but the structural fact that an Azumaya algebra of rank n^2 can only see n-torsion: the determinant collapses the n-th power of the obstruction to a coboundary. This is why the cohomological Brauer group is defined as the *torsion* subgroup and not all of $H^2(X, \mathbb{G}_m)$; the nontorsion classes, which occur on non-separated or non-quasicompact schemes, can never come from an algebra.

Whether the injection is surjective onto the torsion is a substantially harder question, answered by Gabber.

21.2.5 Note: Gabber's Theorem ($\mathrm{Br} = \mathrm{Br}'$) Gabber's theorem states that for a quasi-compact scheme X admitting an ample line bundle (for instance any quasi-projective scheme over an affine base), the injection of theorem 21.2.4 is a bijection

$$\mathrm{Br}(X) = \mathrm{Br}'(X) = H^2_{\text{ét}}(X, \mathbb{G}_m)_{\text{tors}}$$

so every torsion class in $H^2_{\text{ét}}(X, \mathbb{G}_m)$ is the class of an Azumaya algebra. The proof is beyond the present scope: it constructs, from a torsion cohomology class, an Azumaya algebra representing it, using the theory of twisted sheaves developed in the next section together with a delicate argument producing a twisted vector bundle of the correct rank. The result and its proof are treated in [46] and the co-requisite [12]; the special case where X is the spectrum of a field is elementary and recovers the classical fact that every element of $\mathrm{Br}(k) = H^2(G_k, (k^{\text{sep}})^\times)$ is represented by a central simple algebra ([13]).

The surjectivity of $\mathrm{Br} \to \mathrm{Br}'$ has content: producing an algebra from a cohomology class reverses the natural direction. Gabber proved it for schemes carrying an ample line bundle, in particular quasi-projective ones (de Jong later gave a written proof); it can fail without such a hypothesis, where non-separated or otherwise pathological schemes carry torsion classes in Br' with no Azumaya representative, so $\mathrm{Br} \subsetneq \mathrm{Br}'$ in general. For the purposes of this book the injection is the working tool, and the distinction between Br and Br' can be elided precisely on the schemes where computations are done. The next section constructs the twisted sheaves that Gabber's proof requires, closing the loop from the algebra side back to the gerbe side.

With the correspondence established, the concrete content of the Brauer group is a series of computations over fields, where the group is fully torsion and coincides with the Galois cohomology of [13]. The quaternion algebra is the smallest nontrivial example and the model for everything: a μ_2-gerbe of order exactly 2, realized by an actual 4-dimensional division algebra.

Example 21.2.6: The Quaternion Algebra and Brauer Groups of Fields

The computations below are all over a field, where $\mathrm{Br}(k) = \mathrm{Br}'(k) = H^2(G_k, (k^{\text{sep}})^\times)$ is entirely torsion.

1. *The quaternion algebra as a gerbe of order* 2. Let k be a field of characteristic $\neq 2$ and $a, b \in k^\times$. The quaternion algebra

$$(a, b)_k = k\langle i, j\rangle/(i^2 = a,\ j^2 = b,\ ij = -ji)$$

is a 4-dimensional central simple k-algebra, split by the quadratic extension $k(\sqrt{a})$. Its class $[(a, b)_k] \in \mathrm{Br}(k)$ has order dividing 2: the opposite algebra $(a, b)_k^{\text{op}} \cong (a, b)_k$ (the map

$i \mapsto -i$, $j \mapsto -j$ is an anti-isomorphism to itself), so $[(a,b)_k]+[(a,b)_k] = [(a,b)_k \otimes (a,b)_k] = 0$ because $(a,b)_k \otimes (a,b)_k^{\mathrm{op}} \cong \underline{\mathrm{End}}((a,b)_k)$ is a matrix algebra ($16 = 4^2$-dimensional). Under theorem 21.2.4 it is a $\mathbb{G}_m$-gerbe realized as a μ_2-gerbe: the boundary of the PGL_2-torsor of $(a,b)_k$, whose Čech 2-cocycle is the sign cocycle valued in $\mu_2 \subset \mathbb{G}_m$. The class is nonzero (order exactly 2) precisely when $(a,b)_k$ is a division algebra, that is, when b is not a norm from $k(\sqrt{a})$; for example $(-1,-1)_{\mathbb{R}} = \mathbb{H}$, Hamilton's quaternions, is the nonzero element of $\mathrm{Br}(\mathbb{R})$.

2. $\mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2$. The absolute Galois group of $\mathbb{R}$ is $G_{\mathbb{R}} = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \mathbb{Z}/2$, generated by complex conjugation acting on $\mathbb{C}^\times$. Then

$$\mathrm{Br}(\mathbb{R}) = H^2(\mathbb{Z}/2, \mathbb{C}^\times) = \mathbb{R}^\times/N_{\mathbb{C}/\mathbb{R}}(\mathbb{C}^\times) = \mathbb{R}^\times/\mathbb{R}_{>0} = \{\pm 1\} \cong \mathbb{Z}/2$$

using the standard computation of the cohomology of a cyclic group as a norm quotient (the Herbrand quotient / Tate cohomology of $\mathbb{Z}/2$; see [13]): H^2 of a cyclic group $\langle\sigma\rangle$ acting on M is $M^\sigma/N(M)$, here $\mathbb{R}^\times$ modulo the norms $z\bar{z} = |z|^2 > 0$. The nontrivial class is $\mathbb{H}$, the quaternion algebra $(-1,-1)_{\mathbb{R}}$ of item (1).

3. $\mathrm{Br}(\mathbb{F}_q) = 0$. A finite field $\mathbb{F}_q$ has absolute Galois group $\widehat{\mathbb{Z}}$, procyclic, generated topologically by the Frobenius. Every central simple algebra over a finite field is a matrix algebra, since a finite division ring is a field by Wedderburn's little theorem, so

$$\mathrm{Br}(\mathbb{F}_q) = H^2(\widehat{\mathbb{Z}}, (\overline{\mathbb{F}}_q)^\times) = 0$$

the vanishing coming from the surjectivity of the norm map for the procyclic Galois group ([13]). Every Azumaya algebra over $\mathbb{F}_q$ splits, and every $\mathbb{G}_m$-gerbe over $\operatorname{Spec}\mathbb{F}_q$ is neutral.

4. *Local and global fields.* For a nonarchimedean local field K (a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q((t))$), local class field theory gives $\mathrm{Br}(K) = \mathbb{Q}/\mathbb{Z}$ via the invariant map, so every element has a representative division algebra of any prescribed order; for a global field K (a number field or a function field of a curve over $\mathbb{F}_q$), the fundamental exact sequence of global class field theory reads

$$0 \longrightarrow \mathrm{Br}(K) \longrightarrow \bigoplus_v \mathrm{Br}(K_v) \xrightarrow{\sum \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

summing over all places v, so a global Brauer class is a finite tuple of local invariants summing to zero. These are the central results of [13].

5. $\mathrm{Br}(\mathbb{P}^n_k) = \mathrm{Br}(k)$. For projective space over any field k, pullback along $\mathbb{P}^n_k \to \operatorname{Spec} k$ induces an isomorphism

$$\mathrm{Br}(k) \xrightarrow{\sim} \mathrm{Br}(\mathbb{P}^n_k)$$

so projective space carries no Brauer classes beyond those of the ground field. This is a computation in étale cohomology (using the projective bundle formula for $H^2(\mathbb{G}_m)$ and the vanishing of the relevant higher direct images), stated here and cited to [46] and the co-requisite [12]; it says that a Severi-Brauer variety over $\mathbb{P}^n_k$ contributes nothing new, and that the Brauer group is a birational invariant insensitive to adding projective-space factors.

The computations make the abstract correspondence tangible. Over $\mathbb{R}$ the Brauer group is a single $\mathbb{Z}/2$ carried by the quaternions, the unique division algebra; over a finite field it vanishes and every gerbe is neutral; over local and global fields it is the arithmetic heart of class field theory, the invariant map and the reciprocity sum being the organizing principles. That $\mathrm{Br}(\mathbb{P}^n_k) = \mathrm{Br}(k)$ shows the geometric Brauer group of a rational variety reduces to the arithmetic of the ground field, and it is the first instance of a computation where the base is a scheme rather than a point. The field cases are self-contained through the group cohomology of [13]; the higher-dimensional computations, where étale cohomology is unavoidable, are the province of [12], the co-requisite whose deep Brauer theory this book cites rather than develops. The next section builds the twisted sheaves that live on a $\mathbb{G}_m$-gerbe and turn the Brauer class into a category of geometric objects.

Exercise 21.2.1

1. Let P be a PGL_n-torsor on a scheme X with Čech 1-cocycle $\bar{g}_{ij}$, and let $g_{ij} \in \mathrm{GL}_n(U_{ij})$ be lifts. Verify in full the 2-cocycle identity for $\alpha_{ijk} = g_{ij}g_{jk}g_{ki}$ (equation (21.4)) on the quadruple overlaps U_{ijkl}, and confirm that changing the lifts g_{ij} by scalars $\lambda_{ij} \in \mathbb{G}_m(U_{ij})$ changes α_{ijk} by the coboundary of (λ_{ij}). Conclude that the class $\delta([P]) \in H^2(X, \mathbb{G}_m)$ is independent of the choice of lifts.

2. Give a second proof, distinct from the determinant argument of theorem 21.2.4, that $\delta([P])$ is n-torsion for a PGL_n-torsor P, by using instead the tensor-power structure: for a local rank-n lift $\mathcal{E}$ of P (so $\underline{\mathrm{End}}(\mathcal{E})$ is the Azumaya algebra of P), relate $n \cdot \delta([P])$ to the obstruction to gluing the determinant line bundles $\det \mathcal{E} = \wedge^n \mathcal{E}$, and explain why the argument via det is the cleaner one. (Hint: the class $n \cdot \delta([P])$ is the obstruction to gluing a rank-1 sheaf, whose torsor of lifts is a PGL_1-torsor and so visibly lifts.)

3. Compute $\mathrm{Br}(k)$ for k algebraically closed, and deduce that every Azumaya algebra over an algebraically closed field is a matrix algebra and every $\mathbb{G}_m$-gerbe over $\operatorname{Spec} k$ is neutral. Then explain, using theorem 21.2.4, why the residual μ_n-gerbe of the special point $j = 0$ of $\mathcal{M}_{1,1}$ (example 21.2.2) is neutral over $\bar{k}$ but may be nonneutral over a subfield.

4. Let $a, b \in k^\times$ with $\operatorname{char} k \neq 2$. Show that the quaternion algebra $(a, b)_k$ is split (isomorphic to $\mathrm{Mat}_2(k)$) if and only if the conic $ax^2 + by^2 = z^2$ has a k-rational point, if and only if b is a norm from $k(\sqrt{a})$. Deduce that $[(a, b)_k] = 0$ in $\mathrm{Br}(k)$ exactly in these cases, and compute $[(a, b)_k]$ for $k = \mathbb{R}$ in the four sign cases of $(a, b) \in \{\pm 1\}^2$.

5. Using the local-global exact sequence of [13] recalled in example 21.2.6(4), show that there is no division algebra over $\mathbb{Q}$ ramified at exactly one place, and construct explicitly a quaternion division algebra over $\mathbb{Q}$ ramified at exactly two places (say $\{2, \infty\}$ or $\{p, q\}$ for two odd primes). State the order of its class in $\mathrm{Br}(\mathbb{Q})$.

6. Explain why the cohomological Brauer group is defined as the *torsion* subgroup of $H^2(X, \mathbb{G}_m)$ rather than the whole group, by two routes: (i) from theorem 21.2.4, why no Azumaya algebra can produce a nontorsion class; and (ii) name the hypothesis in Gabber's theorem (box 21.2.5) under which $\mathrm{Br} = \mathrm{Br}'$ holds, and state what can go wrong without it. (You may cite the existence of nontorsion classes on non-quasi-projective schemes without constructing one.)

21.3 Twisted Sheaves

The previous two sections built the $\mathbb{G}_m$-gerbe $\mathcal{G}_\alpha$ attached to a class $\alpha \in H^2_{\text{ét}}(X, \mathbb{G}_m)$ (definition 21.2.1, theorem 21.1.5) and identified the torsion part of that cohomology group with the Brauer group (definition 21.2.3, theorem 21.2.4). So far the Brauer class has been an obstruction seen from the outside: a cohomology class, a gerbe that fails to be neutral, an Azumaya algebra that fails to be a matrix algebra. This section supplies the objects that the gerbe was built to carry, and through them the Brauer class acquires a concrete meaning. On an ordinary scheme one studies quasi-coherent sheaves; on a $\mathbb{G}_m$-gerbe the natural objects are the quasi-coherent sheaves on which the band $\mathbb{G}_m$ acts by scaling. These are the *twisted sheaves*. They form an abelian category $\mathrm{Tw}_\alpha(X)$ that reduces to $\mathrm{QCoh}(X)$ when $\alpha = 0$, and whose failure to contain a twisted line bundle is exactly the nonvanishing of α. The Brauer class, which measured the failure of the gerbe to be neutral in section 21.2, now measures the failure of a twisted line bundle to exist, and the equivalence with modules over an Azumaya algebra turns the whole theory into linear algebra over a division algebra.

A quasi-coherent sheaf on any algebraic stack was defined in Chapter 8 (definition 20.5.1): a compatible assignment of a quasi-coherent sheaf $\mathcal{F}_\xi$ on U to every object $\xi \in \mathcal{G}_\alpha(U)$, together with isomorphisms along pullbacks. On a gerbe there is extra structure. Every object ξ has automorphism group $\mathbb{G}_m$ acting on it, and this automorphism acts on the fiber $\mathcal{F}_\xi$ through some integer power of the scaling character. The integer is locally constant, so on a connected gerbe it is a single integer, the *weight*. Splitting $\mathrm{QCoh}(\mathcal{G}_\alpha)$ into weight components is the first structural fact, and the weight-one part is the category this section studies.

Definition 21.3.1: α-Twisted Sheaf

Let $\mathcal{G}_\alpha \to X$ be a $\mathbb{G}_m$-gerbe of class $\alpha \in H^2_{\text{ét}}(X, \mathbb{G}_m)$ (definition 21.2.1). A quasi-coherent sheaf $\mathcal{F}$ on $\mathcal{G}_\alpha$ (definition 20.5.1) has *weight* $w \in \mathbb{Z}$ if, for every object $\xi \in \mathcal{G}_\alpha(U)$, the tautological automorphism $t \in \mathbb{G}_m(U) = \underline{\mathrm{Aut}}(\xi)$ acts on the fiber $\mathcal{F}_\xi$ by the scalar t^w. An *α-twisted sheaf* is a quasi-coherent sheaf on $\mathcal{G}_\alpha$ of weight 1. The category of α-twisted sheaves is denoted $\mathrm{Tw}_\alpha(X)$ (equivalently $\mathrm{QCoh}(\mathcal{G}_\alpha, 1)$, the weight-one component of $\mathrm{QCoh}(\mathcal{G}_\alpha)$).
Concretely, choose an étale cover $\{U_i \to X\}$ trivializing α, with objects $x_i \in \mathcal{G}_\alpha(U_i)$ and gluing isomorphisms $\varphi_{ij} \colon x_j|_{U_{ij}} \xrightarrow{\sim} x_i|_{U_{ij}}$ whose failure of the cocycle condition is the chosen 2-cocycle representative $\alpha_{ijk} \in \mathbb{G}_m(U_{ijk})$ (theorem 21.1.5), so that

$$\varphi_{ij}\,\varphi_{jk} = \alpha_{ijk}\,\varphi_{ik} \tag{21.5}$$

Then an α-twisted sheaf is the same datum as a system $(\mathcal{E}_i, \psi_{ij})$ of quasi-coherent sheaves $\mathcal{E}_i$ on U_i and isomorphisms $\psi_{ij} \colon \mathcal{E}_j|_{U_{ij}} \xrightarrow{\sim} \mathcal{E}_i|_{U_{ij}}$ satisfying the *twisted cocycle condition*

$$\psi_{ij}\,\psi_{jk}\,\psi_{ki} = \alpha_{ijk} \cdot \mathrm{id}_{\mathcal{E}_i} \qquad \text{on } U_{ijk} \tag{21.6}$$

A morphism $(\mathcal{E}_i, \psi_{ij}) \to (\mathcal{E}'_i, \psi'_{ij})$ is a system of $\mathcal{O}_{U_i}$-linear maps $f_i \colon \mathcal{E}_i \to \mathcal{E}'_i$ commuting with the gluing, $f_i \psi_{ij} = \psi'_{ij} f_j$. When each $\mathcal{E}_i$ is locally free of finite rank r, the system is an *α-twisted vector bundle of rank r*, and $r = 1$ gives an *α-twisted line bundle.*

The two descriptions are the same object read two ways. The first is intrinsic: a sheaf on the gerbe, singled out by how the band acts. The second is the Čech translation, and it is the one used in computations. Its content is the single formula (21.6): the gluing isomorphisms ψ_{ij} do not satisfy the ordinary cocycle condition that would let them glue to a sheaf on X; they satisfy it only up to the scalar α_{ijk}. When $\alpha = 0$ one may choose the trivial cocycle $\alpha_{ijk} = 1$, and then (21.6) becomes the genuine cocycle condition and the $\mathcal{E}_i$ descend to an ordinary sheaf on X. For nonzero α they cannot descend, and the twisted sheaf lives only on the gerbe. This is the precise sense in which a twisted sheaf is “a sheaf that would exist on X if α vanished”.

The rank of an α-twisted vector bundle is well-defined, since the ψ_{ij} are isomorphisms and the twist by α_{ijk} is a scalar; here and throughout, "rank" of a twisted sheaf means the rank r of the underlying locally free $\mathcal{E}_i$ as in definition 21.3.1, which over a splitting cover is its dimension as a vector space, not the smaller rank it may carry as a module over the division algebra or Azumaya algebra representing α. The two senses differ by the degree of that algebra, and where the module rank is meant it is named as such. The higher numerical invariants require more care. A twisted Chern character can be defined once one fixes an auxiliary datum: choosing a $\mathbb{G}_m$-linearized line bundle $\mathcal{L}$ on the gerbe of weight 1 (equivalently a rank-one twisted sheaf, when one exists) untwists everything by tensoring, but the point of the theory is that such an $\mathcal{L}$ need not exist. The correct statement is that the rational Chern character of an α-twisted sheaf lives in a B-twisted Hodge-type group determined by a lift B of α; the precise formalism is developed in [4] and is the twisted analogue of the ordinary Chern character. For the arithmetic applications the rank alone already carries the decisive information, as the period-index discussion of the next section shows.

Example 21.3.2: Twisted Modules over a Point

Take $X = \operatorname{Spec} k$ for a field k and a $\mathbb{G}_m$-gerbe $\mathcal{G}_\alpha$ of class $\alpha \in H^2(\operatorname{Gal}(k^{\text{sep}}/k), (k^{\text{sep}})^\times) = \operatorname{Br}(k)$ (definition 21.2.3). Trivialize α over $\operatorname{Spec} k^{\text{sep}} \to \operatorname{Spec} k$, the limit of the finite étale covers $\operatorname{Spec} k' \to \operatorname{Spec} k$ over finite Galois k'/k, on which the 2-cocycle becomes a Galois 2-cocycle $\alpha_{\sigma,\tau} \in (k^{\text{sep}})^\times$. A finite-dimensional α-twisted vector space is then a k^{sep}-vector space V together with semilinear maps $\psi_\sigma \colon {}^\sigma V \to V$ for each $\sigma \in \operatorname{Gal}$, satisfying

$$\psi_\sigma \circ {}^\sigma\psi_\tau = \alpha_{\sigma,\tau} \cdot \psi_{\sigma\tau}$$

the finite-cover form of (21.6). This is exactly the data of a module over the crossed-product algebra $(k^{\text{sep}}/k, \alpha)$ built from the cocycle $\alpha_{\sigma,\tau}$, whose class in the Brauer group is α ([13, the crossed-product construction]). When $k = \mathbb{R}$ and α is the nontrivial class of order two, the crossed-product algebra is Hamilton's quaternions $\mathbb{H}$, and an α-twisted vector space over $\mathbb{R}$ is precisely a right $\mathbb{H}$-module, that is, a quaternionic vector space. Its rank as an $\mathbb{H}$-module measures how many copies of $\mathbb{H}$ it is built from, and the smallest nonzero one, $\mathbb{H}$ itself, is $\mathbb{H}$-module rank one but twisted rank two: its rank as a twisted sheaf (definition 21.3.1) is the dimension of the descent vector space over $\mathbb{C} = \mathbb{R}^{\text{sep}}$, which is two, the degree of $\mathbb{H}$. The Brauer obstruction has become the statement that the smallest twisted object is $\mathbb{H}$, not $\mathbb{R}$.

The example is the point-level model that the whole section generalizes. Over a point, an α-twisted sheaf is a module over the division algebra with Brauer class α, and the twisted rank of the smallest such module is the index of that division algebra. The passage from a field to a scheme replaces the division algebra by an Azumaya algebra and the module by a coherent $\mathcal{O}_X$-module structure, but the mechanism is identical: the Brauer class forces the twisted objects to be modules over an algebra that is not $\mathcal{O}_X$.

The first structural theorem organizes twisted sheaves into a category and records how the twist behaves under the sheaf operations. The category $\operatorname{Tw}_\alpha(X)$ ought to be abelian, since twisting is a global condition that kernels and cokernels respect; the tensor product of an α-twisted sheaf with a β-twisted sheaf ought to be $(\alpha+\beta)$-twisted, since the twisting scalars multiply; and the $\alpha = 0$ case ought to recover ordinary sheaves. The one substantive consequence is the last clause: a twisted *line* bundle exists if and only if the class vanishes. That statement is the geometric content of the Brauer group, and it is what elevates the twist from bookkeeping to obstruction.

Theorem 21.3.3: The Category of Twisted Sheaves

Let X be a scheme and $\alpha, \beta \in H^2_{\text{ét}}(X, \mathbb{G}_m)$.

1. The category $\mathrm{Tw}_\alpha(X)$ of α-twisted sheaves is abelian, and the full subcategory of coherent (or of locally free) α-twisted sheaves is closed under the operations that preserve those conditions.
2. Tensor product over $\mathcal{O}_X$ (computed componentwise on a common trivializing cover) sends an α-twisted sheaf and a β-twisted sheaf to an $(\alpha+\beta)$-twisted sheaf. In particular $\mathrm{Tw}_0(X) = \mathrm{QCoh}(X)$, and the internal hom $\mathcal{H}om_{\mathcal{O}_X}(-,-)$ sends an α-twisted sheaf and a β-twisted sheaf to a $(\beta-\alpha)$-twisted sheaf; the dual of an α-twisted sheaf is $(-\alpha)$-twisted.
3. There exists an α-twisted invertible sheaf if and only if $\alpha = 0$ in $H^2_{\text{ét}}(X, \mathbb{G}_m)$.

***Proof*:**

Fix an étale cover $\{U_i \to X\}$ on which α and β are represented by cocycles α_{ijk} and β_{ijk}, and describe twisted sheaves by the Čech data of definition 21.3.1.

Step 1: Abelianness. Let $f = (f_i)\colon (\mathcal{E}_i, \psi_{ij}) \to (\mathcal{E}'_i, \psi'_{ij})$ be a morphism of α-twisted sheaves. Form the kernel component by component, $\mathcal{K}_i = \ker(f_i\colon \mathcal{E}_i \to \mathcal{E}'_i)$. The gluing ψ_{ij} carries $\mathcal{E}_j$ isomorphically to $\mathcal{E}_i$ and, by the compatibility $f_i\psi_{ij} = \psi'_{ij} f_j$, carries $\ker f_j$ into $\ker f_i$; restricting ψ_{ij} therefore gives isomorphisms $\psi^K_{ij}\colon \mathcal{K}_j \to \mathcal{K}_i$. These inherit (21.6) from the ψ_{ij} because the identity is twisted by the same scalar α_{ijk} on the subsheaf as on the whole. Thus $(\mathcal{K}_i, \psi^K_{ij})$ is an α-twisted sheaf, and it is the kernel of f in $\mathrm{Tw}_\alpha(X)$: a morphism into $(\mathcal{E}_i, \psi_{ij})$ killed by f factors uniquely through it because it does so component by component and the factorizations are automatically compatible with the gluing. The cokernel is constructed identically with $\mathcal{C}_i = \operatorname{coker}(f_i)$, the restricted gluing again inheriting (21.6). The zero object is $(0, \mathrm{id})$ and biproducts are componentwise. Because kernels and cokernels are computed componentwise and $\mathrm{QCoh}(U_i)$ is abelian, every monomorphism is the kernel of its cokernel and every epimorphism the cokernel of its kernel, checked on each U_i; hence $\mathrm{Tw}_\alpha(X)$ is abelian. Coherence and local freeness are conditions on the $\mathcal{E}_i$ alone, preserved by the operations that preserve them on each U_i, which proves the closure statement.

Step 2: The tensor product adds classes. Let $(\mathcal{E}_i, \psi_{ij})$ be α-twisted and $(\mathcal{F}_i, \chi_{ij})$ be β-twisted on the common cover. Set $\mathcal{G}_i = \mathcal{E}_i \otimes_{\mathcal{O}_{U_i}} \mathcal{F}_i$ with gluing $\theta_{ij} = \psi_{ij} \otimes \chi_{ij}$. The twisted cocycle scalar of the tensor is the product of the two,

$$\theta_{ij}\theta_{jk}\theta_{ki} = (\psi_{ij}\psi_{jk}\psi_{ki}) \otimes (\chi_{ij}\chi_{jk}\chi_{ki}) = (\alpha_{ijk}\,\mathrm{id}) \otimes (\beta_{ijk}\,\mathrm{id}) = (\alpha_{ijk}\beta_{ijk})\,\mathrm{id}_{\mathcal{G}_i}$$

and since the cocycle of $\alpha+\beta$ is the product $\alpha_{ijk}\beta_{ijk}$ (the group law in H^2 is induced by multiplication in $\mathbb{G}_m$), the tensor product is $(\alpha+\beta)$-twisted. For $\alpha = \beta = 0$ the trivial cocycle makes (21.6) the ordinary cocycle condition, so a 0-twisted sheaf is descent data for an ordinary quasi-coherent sheaf on X, giving $\mathrm{Tw}_0(X) = \mathrm{QCoh}(X)$. For the internal hom, the gluing on $\mathcal{H}om(\mathcal{E}_i, \mathcal{F}_i)$ is $g \mapsto \chi_{ij} \circ g \circ \psi_{ij}^{-1}$, whose triple product carries the scalar $\beta_{ijk}\alpha_{ijk}^{-1}$ because the inverse gluing ψ^{-1} contributes α_{ijk}^{-1}; hence $\mathcal{H}om$ is $(\beta-\alpha)$-twisted, and the dual, the case $\mathcal{F} = \mathcal{O}_X$ (which is 0-twisted), is $(-\alpha)$-twisted.

Step 3: A twisted line bundle trivializes the class. Suppose $(\mathcal{L}_i, \lambda_{ij})$ is an α-twisted invertible sheaf:

each $\mathcal{L}_i$ is an invertible $\mathcal{O}_{U_i}$-module and $\lambda_{ij} \colon \mathcal{L}_j \to \mathcal{L}_i$ are isomorphisms with $\lambda_{ij}\lambda_{jk}\lambda_{ki} = \alpha_{ijk}\,\mathrm{id}$. Refining the cover if necessary, trivialize each $\mathcal{L}_i \cong \mathcal{O}_{U_i}$; then λ_{ij} becomes multiplication by a unit $\ell_{ij} \in \mathbb{G}_m(U_{ij})$, and the twisted cocycle condition reads

$$\ell_{ij}\,\ell_{jk}\,\ell_{ki} = \alpha_{ijk} \tag{21.7}$$

Equation (21.7) states exactly that α_{ijk} is the Čech coboundary $\delta(\ell)_{ijk}$ of the 1-cochain (ℓ_{ij}), so α is a coboundary and $[\alpha] = 0$ in $H^2_{\text{ét}}(X, \mathbb{G}_m)$.

Conversely, if $\alpha = 0$, choose a 1-cochain (ℓ_{ij}) with $\delta(\ell)_{ijk} = \alpha_{ijk}$, that is, with (21.7) holding. Set $\mathcal{L}_i = \mathcal{O}_{U_i}$ and $\lambda_{ij} = \ell_{ij}$; then $\lambda_{ij}\lambda_{jk}\lambda_{ki} = \alpha_{ijk}\,\mathrm{id}$, so $(\mathcal{L}_i, \lambda_{ij})$ is an α-twisted line bundle. Thus an α-twisted invertible sheaf exists if and only if α is a coboundary, that is, if and only if $\alpha = 0$, as required.

The three parts read as a dictionary between the cohomology of $\mathbb{G}_m$ and the linear algebra of twisted sheaves. Part (1) says the twist is a well-behaved decoration: it does not disturb the abelian structure, so all the homological constructions of ordinary sheaf theory transfer verbatim, and one may speak of exact sequences, extensions, and cohomology of twisted sheaves. Part (2) turns the group $H^2(X, \mathbb{G}_m)$ into a grading: the categories $\mathrm{Tw}_\alpha(X)$ are the graded pieces, the tensor product is the graded multiplication, and the degree-zero piece is ordinary sheaves. This is why the Brauer class is added, not multiplied, under tensor: the $\mathbb{G}_m$-weights compose additively in the exponent. Part (3) carries the geometric content. On an ordinary scheme, an invertible sheaf always exists, $\mathcal{O}_X$ itself. On the gerbe $\mathcal{G}_\alpha$, an α-twisted line bundle exists exactly when the gerbe is neutral, and its existence would furnish, by tensoring, an equivalence $\mathrm{Tw}_\alpha(X) \simeq \mathrm{QCoh}(X)$ that trivializes the whole theory. The Brauer class is the obstruction to that trivialization, seen now inside the category of sheaves rather than in the classification of gerbes.

The obstruction of part (3) is worth isolating in its cleanest form before building the Azumaya equivalence on top of it, because the two together explain why twisted sheaves of higher rank always exist even when line bundles do not.

Example 21.3.4: The Twisted Line Bundle Obstruction

Let $\mathcal{G}_\alpha \to X$ be a $\mathbb{G}_m$-gerbe. By theorem 21.3.3(3), an α-twisted line bundle exists exactly when $\alpha = 0$, and the same argument identifies twisted line bundles with a neutralization of the gerbe: a weight-one invertible sheaf on $\mathcal{G}_\alpha$ is a section of $\mathcal{G}_\alpha \to X$ up to the $\mathbb{G}_m$-band, so its existence is the statement that $\mathcal{G}_\alpha$ is neutral (theorem 21.1.5). The two obstructions, "the gerbe has a global object" and "there is a twisted line bundle", are the same vanishing $\alpha = 0$.

Two concrete instances make the obstruction visible.

1. Over $X = \operatorname{Spec}\mathbb{R}$ with $\alpha \in \mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2$ nontrivial, there is no α-twisted line bundle: a one-dimensional α-twisted vector space over $\mathbb{R}$ would be a one-dimensional right $\mathbb{H}$-module (example 21.3.2), and $\mathbb{H}$ has no such module, since $\mathbb{H}$ is a division algebra of dimension four and its smallest nonzero module is $\mathbb{H}$ itself, of rank one over $\mathbb{H}$ but real dimension four (hence sheaf-rank two, the $\mathbb{C}$-dimension of the descent vector space V in the twisted-sheaf datum of example 21.3.2, which is the degree of $\mathbb{H}$ and half its real dimension, not the real dimension four itself). There is thus an α-twisted sheaf of $\mathbb{H}$-module rank one (namely $\mathbb{H}$ as a module over itself), but no α-twisted *invertible* sheaf, because "invertible" means trivializing to $\mathcal{O}$ locally on X, which the twist forbids.

2. Over a smooth projective surface X with a nonzero Brauer class α arising from a conic bundle

or an elliptic fibration, the same phenomenon holds globally: there are α-twisted vector bundles of every rank divisible by the index of α, but none of rank one, and the smallest rank of an α-twisted vector bundle equals the index of the Brauer class. The nonexistence of the twisted line bundle is a curved-space refinement of the $\mathbb{R}$-computation, and it is what the moduli of twisted sheaves of the next section measures.

In both cases the twist obstructs rank one but not higher rank: the Brauer class kills the invertible object while permitting the vector bundle of rank equal to its index.

The example already contains the shape of the general theory. Whenever a twisted line bundle fails to exist, a twisted vector bundle of the index rank exists in its place, and its sheaf of endomorphisms is an Azumaya algebra representing α. This is the bridge to the second main result of the section, the equivalence between twisted sheaves and modules over an Azumaya algebra. It recasts the twisted category, which is defined on the auxiliary gerbe, as the category of modules over an algebra living on X itself, and it is the tool that makes twisted sheaves computable in classical terms.

The construction runs in both directions, and its input is a locally free α-twisted sheaf $\mathcal{V}$ of rank n. Such a $\mathcal{V}$ exists whenever α lies in $\mathrm{Br}(X)$: the tautological weight-one sheaf on the gerbe $\mathcal{G}_\alpha$ is a locally free α-twisted sheaf of rank n, living on the gerbe rather than on X, and it is this global rank-n twisted bundle that the construction takes as input. Given a locally free α-twisted sheaf $\mathcal{V}$ of rank n, its endomorphism sheaf $\mathcal{A} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{V},\mathcal{V})$ is 0-twisted by theorem 21.3.3(2), so it descends to an sheaf of $\mathcal{O}_X$-algebras on X; étale-locally $\mathcal{V}$ is free of rank n and $\mathcal{A}$ is the matrix algebra $M_n(\mathcal{O}_X)$, which is the defining property of an Azumaya algebra of degree n. In the other direction, tensoring with $\mathcal{V}$ turns a module over $\mathcal{A}$ into a twisted sheaf. These two functors are mutually inverse equivalences.

Definition 21.3.5: The Azumaya Module Equivalence

Let $\alpha \in \mathrm{Br}(X) \subseteq H^2_{\text{ét}}(X,\mathbb{G}_m)$ be represented by an Azumaya algebra $\mathcal{A}$ of degree n (definition 21.2.3, theorem 21.2.4), and let $\mathcal{V}$ be a locally free α-twisted sheaf of rank n with $\mathcal{A} \cong \mathcal{H}om_{\mathcal{O}_X}(\mathcal{V},\mathcal{V})$. The *Azumaya module equivalence* is the pair of functors

$$\mathrm{Tw}_\alpha(X) \longrightarrow \mathrm{Mod}(\mathcal{A}), \qquad \mathcal{E} \longmapsto \mathcal{H}om_{\mathcal{O}_X}(\mathcal{V},\mathcal{E})$$

$$\mathrm{Mod}(\mathcal{A}) \longrightarrow \mathrm{Tw}_\alpha(X), \qquad \mathcal{M} \longmapsto \mathcal{M} \otimes_{\mathcal{A}} \mathcal{V}$$

where $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{V},\mathcal{E})$ is a right $\mathcal{A} = \mathcal{H}om(\mathcal{V},\mathcal{V})$-module by precomposition ($g \cdot a = g \circ a$, so $(g \cdot a) \cdot b = g \cdot (ab)$), so that $\mathrm{Mod}(\mathcal{A})$ is the category of right $\mathcal{A}$-modules, and $\mathcal{M} \otimes_{\mathcal{A}} \mathcal{V}$ tensors such an $\mathcal{M}$ against the left $\mathcal{A}$-module $\mathcal{V}$, on which $\mathcal{A} = \mathcal{H}om(\mathcal{V},\mathcal{V})$ acts by evaluation.

Proposition 21.3.6: Twisted Sheaves are Azumaya Modules

The two functors of definition 21.3.5 are mutually inverse equivalences of categories $\mathrm{Tw}_\alpha(X) \simeq \mathrm{Mod}(\mathcal{A})$.

***Proof*:**
The argument establishes the equivalence first in the split case in full and then globally by descent.

Step 1: The split case. Suppose $\alpha = 0$, so $\mathcal{A} = M_n(\mathcal{O}_X)$ and $\mathcal{V} = \mathcal{O}_X^{\oplus n}$ is untwisted, with $\mathcal{A} = \mathcal{H}om(\mathcal{V}, \mathcal{V})$ the matrix algebra. The claim is that $\mathcal{E} \mapsto \mathcal{H}om(\mathcal{V}, \mathcal{E}) = \mathcal{E}^{\oplus n}$ and $\mathcal{M} \mapsto \mathcal{M} \otimes_{\mathcal{A}} \mathcal{V}$ are inverse equivalences between $\mathrm{QCoh}(X)$ and $\mathrm{Mod}(M_n(\mathcal{O}_X))$. This is Morita equivalence for the matrix algebra: the bimodule $\mathcal{V} = \mathcal{O}_X^{\oplus n}$ is a left $M_n(\mathcal{O}_X)$-module (by evaluation) and a right $\mathcal{O}_X$-module, and its dual $\mathcal{V}^\vee$ is the reverse bimodule, a right $M_n(\mathcal{O}_X)$-module and a left $\mathcal{O}_X$-module, with $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{V}, \mathcal{E}) = \mathcal{V}^\vee \otimes_{\mathcal{O}_X} \mathcal{E}$ carrying the right M_n-action. The canonical maps

$$\mathcal{V} \otimes_{\mathcal{O}_X} \mathcal{V}^\vee \xrightarrow{\sim} M_n(\mathcal{O}_X), \qquad \mathcal{V}^\vee \otimes_{M_n(\mathcal{O}_X)} \mathcal{V} \xrightarrow{\sim} \mathcal{O}_X$$

are isomorphisms of bimodules, the first because an $n \times n$ matrix is a sum of rank-one matrices $v \otimes w^\vee$, the second because the trace pairing $\mathcal{V}^\vee \otimes \mathcal{V} \to \mathcal{O}_X$ is M_n-balanced and surjective. From these two isomorphisms the composite $\mathcal{H}om(\mathcal{V}, \mathcal{E}) \otimes_{\mathcal{A}} \mathcal{V} = (\mathcal{V}^\vee \otimes_{\mathcal{O}_X} \mathcal{E}) \otimes_{M_n} \mathcal{V} = \mathcal{E} \otimes_{\mathcal{O}_X} (\mathcal{V}^\vee \otimes_{M_n} \mathcal{V}) = \mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{O}_X = \mathcal{E}$ recovers $\mathcal{E}$ naturally, and the reverse composite $\mathcal{H}om(\mathcal{V}, \mathcal{M} \otimes_{\mathcal{A}} \mathcal{V}) = \mathcal{V}^\vee \otimes_{\mathcal{O}_X} (\mathcal{M} \otimes_{M_n} \mathcal{V}) = \mathcal{M} \otimes_{M_n} (\mathcal{V} \otimes_{\mathcal{O}_X} \mathcal{V}^\vee) = \mathcal{M} \otimes_{M_n} M_n = \mathcal{M}$ recovers $\mathcal{M}$; both natural in the argument. The functors are therefore inverse equivalences in the split case, which is the classical Morita theorem for $M_n(\mathcal{O}_X)$ applied over the structure sheaf.

Step 2: The general case by descent. For arbitrary $\alpha \in \mathrm{Br}(X)$, choose an étale cover $p\colon U \to X$ on which $\alpha|_U = 0$, so that over U the algebra $\mathcal{A}|_U \cong M_n(\mathcal{O}_U)$ splits and $\mathcal{V}|_U$ is the untwisted $\mathcal{O}_U^{\oplus n}$; such a cover exists because α is represented by the Azumaya algebra $\mathcal{A}$, which étale-locally is a matrix algebra (theorem 21.2.4). Over U, Step 1 gives an equivalence $\mathrm{Tw}_0(U) = \mathrm{QCoh}(U) \simeq \mathrm{Mod}(\mathcal{A}|_U)$, and it is natural in U: the two functors $\mathcal{H}om(\mathcal{V}, -)$ and $- \otimes_{\mathcal{A}} \mathcal{V}$ commute with the pullback along $U \times_X U \rightrightarrows U$ because $\mathcal{H}om$ and $\otimes$ commute with flat base change. Twisted sheaves on X are, by definition 21.3.1, quasi-coherent sheaves on the gerbe, and a quasi-coherent sheaf on a stack is a sheaf on any étale atlas together with descent data (definition 20.5.1); likewise $\mathcal{A}$-modules on X satisfy étale descent because $\mathcal{A}$ is a quasi-coherent sheaf of algebras. The two categories $\mathrm{Tw}_\alpha(X)$ and $\mathrm{Mod}(\mathcal{A})$ are thus the descent categories of $\mathrm{QCoh}(U)$ and $\mathrm{Mod}(\mathcal{A}|_U)$ along p, and an equivalence over U compatible with the descent data descends to an equivalence over X. The compatibility is exactly the naturality just noted. Therefore $\mathcal{H}om(\mathcal{V}, -)$ and $\mathcal{V} \otimes_{\mathcal{A}} -$ are inverse equivalences $\mathrm{Tw}_\alpha(X) \simeq \mathrm{Mod}(\mathcal{A})$, completing the proof.

The equivalence is the computational payoff of the section. It says that studying α-twisted sheaves is the same as studying modules over a fixed Azumaya algebra $\mathcal{A}$ of class α, an object living on X with no reference to the gerbe. The gerbe was the conceptual home of the twist, the site where the band $\mathbb{G}_m$ acted by scaling; the Azumaya algebra is what it becomes on X, and the equivalence trades the extra dimension of the gerbe for the noncommutativity of $\mathcal{A}$. Over a point this recovers the statement of example 21.3.2 exactly: $\mathcal{A}$ is the division algebra D with $[D] = \alpha$, and $\mathrm{Mod}(\mathcal{A})$ is the category of D-modules, so a twisted vector space is a D-module. The twist has been converted into module theory over a division algebra, which is the form in which the arithmetic of the Brauer group is most transparent.

The equivalence also refines the twisted-line-bundle obstruction of theorem 21.3.3(3). An α-twisted line bundle corresponds under the equivalence to a right $\mathcal{A}$-module that is locally free of rank one over $\mathcal{O}_X$, which forces $\mathcal{A}$ to act on a rank-one module, hence $\mathcal{A} \cong \mathcal{O}_X$ and $\alpha = 0$. The obstruction and the equivalence are two views of the same fact: the Brauer class is nonzero precisely when the representing algebra is not the structure sheaf, and precisely when the twisted objects have no invertible member.

A twisted sheaf on a Severi-Brauer variety completes the picture and connects it back to the geometry that produced the Brauer class in the first place.

Example 21.3.7: Twisted Sheaves on a Severi-Brauer Variety
Let D be a central division algebra over a field k of degree n, with class $\alpha = [D] \in \mathrm{Br}(k)$, and let $P = \mathrm{SB}(D)$ be its Severi-Brauer variety, the form of $\mathbb{P}^{n-1}_k$ that becomes $\mathbb{P}^{n-1}$ over k^{sep} and whose functor of points sends a k-algebra R to the right ideals of $D \otimes_k R$ that are local direct summands of reduced dimension 1 (over $\bar{k}$ these are the right ideals of dimension n; for D a division algebra $\mathrm{SB}(D)(k) = \varnothing$, which is why the class is nonsplit). Over k^{sep} the twisted line bundle $\mathcal{O}(1)$ on $\mathbb{P}^{n-1}$ exists, but it does not descend to P, because its descent obstruction is exactly α: the Galois cocycle needed to glue $\mathcal{O}(1)$ over the twisting differs from the trivial one by $\alpha_{\sigma,\tau}$, so $\mathcal{O}(1)$ is a $p^*\alpha$-twisted line bundle on P rather than an ordinary one, where $p\colon P \to \operatorname{Spec} k$ is the structure map. What does descend is $\mathcal{O}(n) = \mathcal{O}(1)^{\otimes n}$, since $n\alpha = 0$ in $\mathrm{Br}(k)$ (the class is n-torsion, theorem 21.2.4), and more to the point the twisted sheaf $\mathcal{O}(1)$ on P is the geometric incarnation of the twisted objects over k: pushing it forward, $p_*\mathcal{O}(1)$ is an α-twisted vector space over k of twisted rank n, since $H^0(\mathbb{P}^{n-1}, \mathcal{O}(1))$ is n-dimensional over k^{sep}. Its endomorphism algebra recovers the division algebra, $\mathcal{H}om_k(p_*\mathcal{O}(1), p_*\mathcal{O}(1)) \cong D$ (definition 21.3.5 over the point), so $p_*\mathcal{O}(1)$ is the tautological twisted vector space whose associated Azumaya algebra is D itself, the point-level model of example 21.3.2 realized geometrically on P. The Severi-Brauer variety thus carries a canonical $p^*\alpha$-twisted line bundle whose existence on P, contrasted with its nonexistence on $\operatorname{Spec} k$, is the geometric statement that P splits α: pulling α back to P kills it, because on P the twisted line bundle $\mathcal{O}(1)$ finally exists.

The Severi-Brauer example returns to section 21.2. There the Brauer class was produced as the obstruction in a PGL_n-torsor, which is the same data as the Severi-Brauer variety. Here that variety reappears as the space on which the twisted line bundle exists after pullback, and the splitting of the Brauer class by P is read off the sudden existence of $\mathcal{O}(1)$ as an ordinary object upstairs. Twisted sheaves make the passage between the two descriptions of a Brauer class, the cohomological and the geometric, into a statement about where a line bundle does and does not exist.

Exercise 21.3.1

1. Let $\mathcal{G}_\alpha \to X$ be a $\mathbb{G}_m$-gerbe and let $\mathcal{F}$ be a quasi-coherent sheaf on $\mathcal{G}_\alpha$ of weight w (definition 21.3.1). Show that tensoring $\mathcal{F}$ with a weight-w' sheaf gives a weight-$(w + w')$ sheaf, and deduce that the weight-zero sheaves are exactly the pullbacks of quasi-coherent sheaves from X. Conclude that $\mathrm{QCoh}(\mathcal{G}_\alpha)$ decomposes as a direct sum $\bigoplus_{w\in\mathbb{Z}} \mathrm{QCoh}(\mathcal{G}_\alpha, w)$ over the weight, and identify $\mathrm{QCoh}(\mathcal{G}_\alpha, 1)$ with $\mathrm{Tw}_\alpha(X)$.

2. Verify directly from the Čech description that the tensor product of two α-twisted sheaves is 2α-twisted, and that if $2\alpha \neq 0$ there is still no 2α-twisted line bundle. Using this, show that the assignment $\alpha \mapsto \mathrm{Tw}_\alpha(X)$ upgrades $H^2_{\text{ét}}(X, \mathbb{G}_m)$ to a grading on the category of all twisted sheaves, and explain why the tensor unit lives in degree 0.

3. Over $X = \operatorname{Spec} \mathbb{R}$ with $\alpha \in \mathrm{Br}(\mathbb{R}) = \mathbb{Z}/2$ the nontrivial class, describe the abelian category $\mathrm{Tw}_\alpha(\mathbb{R})$ of coherent α-twisted sheaves explicitly as the category of finite-dimensional right $\mathbb{H}$-modules. What are the simple objects? What is the smallest twisted rank, and why is there no object of twisted rank one that is invertible in the sense of theorem 21.3.3(3)?

4. Let $\mathcal{A}$ be an Azumaya algebra of degree n on X with class α, and let $\mathcal{V}$ be a locally free α-twisted sheaf of rank n with $\mathcal{A} = \mathcal{H}om(\mathcal{V}, \mathcal{V})$. Using the Azumaya module equivalence (definition 21.3.5), show that the α-twisted sheaves that are locally free of rank n over $\mathcal{O}_X$ correspond to the $\mathcal{A}$-modules that are locally free of rank one over $\mathcal{A}$, and hence that a global

such twisted bundle exists if and only if $\mathcal{A}$ has a rank-one free module, that is, if and only if $\mathcal{A} \cong \mathcal{H}om(\mathcal{W}, \mathcal{W})$ splits off a twisted bundle $\mathcal{W}$. Relate this to the index of α.

5. Let $P = \mathrm{SB}(D) \to \operatorname{Spec} k$ be the Severi-Brauer variety of a degree-n division algebra D (example 21.3.7). Show that the pullback $p^*\colon \mathrm{Br}(k) \to \mathrm{Br}(P)$ kills $\alpha = [D]$, by exhibiting on P the $p^*\alpha$-twisted line bundle $\mathcal{O}(1)$ and invoking theorem 21.3.3(3). Explain why this does not contradict the nonexistence of an α-twisted line bundle over $\operatorname{Spec} k$, and interpret the statement "P splits α" in terms of twisted line bundles.

6. The internal hom $\mathcal{H}om(\mathcal{E}, \mathcal{F})$ of an α-twisted sheaf $\mathcal{E}$ and a β-twisted sheaf $\mathcal{F}$ is $(\beta - \alpha)$-twisted (theorem 21.3.3(2)). Deduce that $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{E})$ is 0-twisted for any α-twisted sheaf $\mathcal{E}$, hence descends to a genuine sheaf of $\mathcal{O}_X$-algebras on X, and that when $\mathcal{E}$ is locally free of rank n this algebra is Azumaya of degree n. Explain how this recovers the construction of the Azumaya algebra in definition 21.3.5 and why the class of that algebra is α rather than 0, despite $\mathcal{H}om(\mathcal{E}, \mathcal{E})$ itself being untwisted.

21.4 Moduli of Twisted Sheaves

This closing section reaches the payoff toward which the entire chapter has aimed. The previous three sections built the apparatus: a gerbe is the two-categorical object one level above a torsor, a $\mathbb{G}_m$-gerbe realizes a Brauer class geometrically, and an α-twisted sheaf is a quasi-coherent sheaf on the gerbe $\mathcal{G}_\alpha$ of class α on which the band acts by scalars. What has been missing is a moduli space. Chapter 2 constructed the Hilbert and Quot schemes as the fine moduli spaces of untwisted subschemes and quotients, and Chapter 8 assembled the whole framework into the statement that the stack $\mathfrak{Coh}$ of coherent sheaves on a projective X is algebraic. This section carries out the same program for twisted sheaves: it defines the stack $\mathcal{M}_\alpha$ of α-twisted sheaves, states that it is algebraic (the twisted analogue of Chapter 8, due to Lieblich), and then exhibits the reason the twisted theory earns its own development. The twisted moduli space is often a *fine* moduli space where the untwisted one is only coarse, and it converts the period-index problem for a Brauer class into a statement about the ranks of twisted sheaves.

The section is also where Part VI closes. The gerbes and twisted sheaves of this chapter are the last of the non-representable phenomena the book set out to organize, and their moduli are the last of the algebraic stacks it constructs. Once the twisted moduli stack is in hand, only the capstone remains, in which the moduli of curves assembles every strand of the book into a single object.

The definition of the twisted moduli stack copies the untwisted one word for word, with "coherent sheaf" replaced by "α-twisted sheaf". This is the right way to think about it: the twist is a decoration on the objects, not a change to the moduli formalism. A family of α-twisted sheaves over a base T is an α-twisted sheaf on the gerbe pulled back to X_T, flat over T, and the stack records these families and their isomorphisms exactly as $\mathfrak{Coh}$ records families of ordinary sheaves.

Definition 21.4.1: Moduli Stack of Twisted Sheaves

Let X be a projective scheme over a field k with a fixed ample line bundle $\mathcal{O}_X(1)$, and let $\alpha \in \mathrm{Br}(X)$ be a Brauer class with a chosen $\mathbb{G}_m$-gerbe $\mathcal{G}_\alpha \to X$ representing it (definition 21.2.1). The *moduli stack of α-twisted sheaves*, denoted $\mathcal{M}_\alpha$, is the stack over $(\mathbf{Sch}/k)$ whose fiber over a k-scheme T is the groupoid

$$\mathcal{M}_\alpha(T) = \left\{ \begin{array}{l} \mathcal{E} \text{ an } \alpha\text{-twisted sheaf on } X_T = X \times_k T, \\ \text{quasi-coherent of finite presentation and flat over } T \end{array} \right\}$$

with morphisms the isomorphisms of twisted sheaves. Here an α-twisted sheaf on X_T means a quasi-coherent sheaf on the pulled back gerbe $\mathcal{G}_\alpha \times_X X_T$ of weight 1 under the band (definition 21.3.1, definition 20.5.1); flatness over T is checked on the underlying sheaf of any local trivialization. For a polynomial $P \in \mathbb{Q}[m]$ the open and closed substack $\mathcal{M}_\alpha^P \subseteq \mathcal{M}_\alpha$ is the locus of families whose fibers have *twisted Hilbert polynomial* P, where the twisted Hilbert polynomial of an α-twisted sheaf $\mathcal{E}$ on X is $P_{\mathcal{E}}(m) = \chi(X, \mathcal{E}(m))$ computed from the twisted Chern character and the untwisted Todd class through the twisted Riemann-Roch formula.

The definition is the twisted mirror of the stack $\mathfrak{Coh}$ of Chapter 8, and every ingredient is one already in hand. The objects are the twisted sheaves of definition 21.3.1; the family condition is flatness over the base, unchanged from the untwisted case because flatness of a twisted sheaf is flatness of its local trivializations; and the numerical invariant that cuts $\mathcal{M}_\alpha$ into finite-type pieces is the twisted Hilbert polynomial, the twisted analogue of the invariant that stratified $\mathfrak{Coh}$. The dependence on α is the reason for the construction: $\mathcal{M}_0$ is the untwisted stack $\mathfrak{Coh}$, while for $\alpha \neq 0$ the objects of $\mathcal{M}_\alpha$ are never sheaves on X, because theorem 21.3.3 showed that an α-twisted invertible sheaf, hence any α-twisted sheaf that trivializes the twist, exists only when $\alpha = 0$. Different Brauer classes give different moduli problems, and this is what makes the twisted theory more than a relabeling.

One subtlety deserves a word. The twisted Hilbert polynomial requires a notion of $\chi(X, \mathcal{E}(m))$ for a twisted sheaf $\mathcal{E}$, and $\mathcal{E}$ is not a sheaf on X, so its cohomology must be taken on the gerbe. The gerbe $\mathcal{G}_\alpha \to X$ has X proper as its base, and coherent cohomology of a weight-1 sheaf on it is finite-dimensional because the weight-1 part descends, on a cover U trivializing α, to a coherent sheaf on the trivializing cover, whose cohomology is finite-dimensional by the finiteness of coherent cohomology on the proper scheme X (definition 20.5.1, and the cohomology of a stack developed in Chapter 8). The $\mathbb{G}_m$-gerbe itself has non-representable structure map with $\mathbb{G}_m$ inertia, so this finiteness is a property of the weight-1 descent, not of a properness of the gerbe. The Euler characteristic $\chi(X, \mathcal{E}(m)) = \sum_i (-1)^i \dim_k H^i(\mathcal{G}_\alpha, \mathcal{E}(m))$ is therefore a well-defined integer, and it is a numerical polynomial in m by the twisted Riemann-Roch theorem, exactly as in the untwisted case. The leading coefficient encodes the twisted rank and the lower coefficients the twisted Chern character.

A first concrete instance grounds the definition before the general theory. The cleanest case is $X = \operatorname{Spec} k$ a point, where a twist is a class in the field Brauer group and a twisted sheaf is a module over a division algebra.

Example 21.4.2: Twisted Sheaves over a Point

Let $X = \operatorname{Spec} k$ and let $\alpha \in \mathrm{Br}(k)$ be represented by a central division algebra D over k of degree d (so $\dim_k D = d^2$ and the period of α divides d). By definition 21.3.5 and the computation of example 21.3.4, the category of α-twisted sheaves on the point is the category of finitely generated right D-modules. Every such module is a direct sum of copies of D, so an α-twisted sheaf of rank rd

(as a k-vector space it has dimension rd^2) is $D^{\oplus r}$, unique up to isomorphism, with automorphism group $\mathrm{GL}_r(D)$. The moduli stack $\mathcal{M}_\alpha$ over $\operatorname{Spec} k$ therefore decomposes as a disjoint union

$$\mathcal{M}_\alpha = \coprod_{r\geq 0} B\mathrm{GL}_r(D)$$

of classifying stacks of the groups $\mathrm{GL}_r(D)$, one for each isomorphism class of D-module. Each component $B\mathrm{GL}_r(D)$ is a smooth algebraic stack over k of dimension $-\dim_k \mathrm{GL}_r(D) = -r^2d^2$, negative because the automorphisms outnumber the moduli: there is a single object and a positive-dimensional group fixing it. When $\alpha = 0$, $D = k$ and this recovers $\coprod_r B\mathrm{GL}_r$, the stack of finite-dimensional vector spaces, which is the point $X = \operatorname{Spec} k$ case of the untwisted $\mathfrak{Coh}$. The effect of the twist is to replace the structure group GL_r by the inner form $\mathrm{GL}_r(D)$ cut out by the division algebra, which is exactly the twisting the Brauer class encodes.

The point case already displays the two features that make the twisted moduli theory work. First, the twist acts by replacing the automorphism groups of the objects with inner forms: over the point, GL_r becomes $\mathrm{GL}_r(D)$, and in general the twisted sheaves are the modules over an Azumaya algebra (definition 21.3.5) whose fiberwise structure group is an inner form of GL_n. Second, the stack is still built from classifying stacks and quotient stacks, so the machinery of Chapters 7 and 8 applies without modification. The general theorem, that $\mathcal{M}_\alpha$ over a projective X is an algebraic stack, is the twisted counterpart of the algebraicity of $\mathfrak{Coh}$, and its proof runs along the same lines: verify Artin's criteria, using a twisted deformation theory and a twisted boundedness in place of the untwisted ones.

The algebraicity of $\mathcal{M}_\alpha$ is Lieblich's theorem. Its proof is the twisted analogue of the verification of Artin's criteria for $\mathfrak{Coh}$ carried out in Chapter 8, and rather than reproduce that long argument in twisted dress, the theorem is stated with its key ideas and the full proof is cited. The deferral is warranted because the substance is not new mathematics but the transfer of an argument already given in full: the twisted deformation theory, the twisted boundedness, and the verification of the criteria all mirror the untwisted case, and Lieblich's paper carries out the transfer with the care the general Brauer class requires.

Theorem 21.4.3: Algebraicity of the Twisted Moduli Stack

Let X be a projective scheme over a field k with a fixed ample sheaf, and let $\alpha \in \mathrm{Br}(X)$. Then the moduli stack $\mathcal{M}_\alpha$ of definition 21.4.1 is an algebraic stack in the sense of definition 20.3.1, locally of finite type over k. For a fixed twisted Hilbert polynomial P the substack $\mathcal{M}_\alpha^P$ is of finite type. If moreover X is a smooth projective surface and a stability condition is fixed, the open substack $\mathcal{M}_\alpha^{P,s}$ of stable α-twisted sheaves with invariants P admits a coarse moduli space that is a quasi-projective scheme, and the substack of semistable sheaves has a proper good moduli space.

***Proof Key ideas*:**
The full proof is Lieblich's [42]; it verifies Artin's criteria (theorem 20.4.3) for $\mathcal{M}_\alpha$ exactly as Chapter 8 verified them for $\mathfrak{Coh}$ and $\mathcal{M}_g$, with each untwisted input replaced by its twisted counterpart. The transfer of the algebraization step, which for stacks over a general base requires the theory of coherent cohomology on the gerbe over its proper base, is the technical content and is carried out there; the conceptual mechanism is the following.

Step 1: Twisted sheaves form a stack. That $\mathcal{M}_\alpha$ is a stack for the fppf topology (condition (0) of theorem 20.4.3) is descent for twisted sheaves. A twisted sheaf on X_T is a quasi-coherent sheaf on the gerbe $\mathcal{G}_\alpha \times_X X_T$, and quasi-coherent sheaves on a stack satisfy descent because they are defined by descent on the atlas (definition 20.5.1); the weight-1 condition under the band and the flatness over T are local and descend. So a compatible family of twisted sheaves over an fppf cover of T glues to a twisted sheaf over T, which is condition (0).

Step 2: Twisted deformation theory. The deformation and obstruction theory of $\mathcal{M}_\alpha$ at a twisted sheaf $\mathcal{E}$ is the twisted Ext theory: the tangent space is $\mathrm{Ext}^1_{\mathcal{G}_\alpha}(\mathcal{E}, \mathcal{E})$ and the obstruction space is $\mathrm{Ext}^2_{\mathcal{G}_\alpha}(\mathcal{E}, \mathcal{E})$, the self-extension groups computed in the abelian category $\mathrm{Tw}_\alpha(X)$ of theorem 21.3.3. These are finite-dimensional because $\mathcal{E}$ is coherent and its weight-1 cohomology descends to the proper base X, and they satisfy the tangent-torsor and obstruction-vanishing properties by the same homological argument that governs ordinary coherent sheaves, since $\mathrm{Tw}_\alpha(X)$ is an abelian category with enough of the structure of $\mathrm{QCoh}(X)$. This is the deformation-theoretic input to condition (3), and it is where the twist enters: the extensions are taken among twisted sheaves, so the infinitesimal structure is that of the twisted category, not the untwisted one.

Step 3: Twisted boundedness. The finite-type claim for $\mathcal{M}_\alpha^P$ (part of conditions (1) and (3)) rests on a twisted boundedness theorem: the α-twisted sheaves with fixed twisted Hilbert polynomial P, or fixed twisted Chern character, that are semistable form a bounded family. The proof mirrors the untwisted Castelnuovo-Mumford regularity bound of Chapter 2, applied on the gerbe. On an fppf cover trivializing α, a twisted sheaf becomes an ordinary sheaf on the cover with a GL_n-descent datum, and the untwisted boundedness on the cover, made equivariant, bounds the twisted sheaves upstairs. Boundedness is what makes $\mathcal{M}_\alpha^P$ finite-type rather than only locally of finite type, and it is the twisted analogue of the boundedness that cut $\mathfrak{Coh}$ into finite-type pieces.

Step 4: Effectivity and openness. Effectivity of formal twisted sheaves (the rest of condition (3)) is Grothendieck's existence theorem applied to the weight-1 descent: a compatible system of twisted sheaves over the Artinian quotients of a complete local ring algebraizes, because the base X is proper over $\operatorname{Spec} R$ and a twisted sheaf, carrying the ample sheaf pulled back from X, is a compatible system of coherent sheaves on the proper X_R to which the existence theorem applies. Openness of versality (condition (4)) is the upper-semicontinuity of Ext^1 in twisted families, identical to the untwisted argument. With all five criteria verified in their twisted forms, theorem 20.4.3 certifies $\mathcal{M}_\alpha$ as an algebraic stack.

Step 5: The coarse moduli space. For a surface with a fixed stability condition, the stable locus $\mathcal{M}_\alpha^{P,s}$ has automorphism groups reduced to the twisted scalars $\mathbb{G}_m$ (a stable twisted sheaf is twisted-simple), so it is a $\mathbb{G}_m$-gerbe over an algebraic space, and that space is a quasi-projective scheme by the twisted version of the Gieseker-Maruyama construction, built as a GIT quotient of a twisted Quot scheme. The semistable locus has a proper good moduli space by the twisted analogue of the Simpson-Langton-good-moduli-space theory, which is the good moduli space construction of Chapter 8 applied to the twisted stack. The details are in [42], completing the account of the key ideas.

The theorem places the twisted moduli stack on exactly the footing the book has built for every moduli problem: it is algebraic, cut into finite-type pieces by a numerical invariant, and equipped with a coarse space when a stability condition rigidifies the automorphisms. The proof is a checklist rerun of Artin's criteria, and the only new ingredients, the twisted Ext deformation theory and the twisted boundedness, are the twisted counterparts of the untwisted inputs from Chapters 2 and 5. This is the sense in which the twisted theory is not a separate subject but a decoration of the one already developed. What makes it worth developing separately is what the twisted moduli space can do that the untwisted one cannot, and that is the content of the next result.

The reason to construct $\mathcal{M}_\alpha$ rather than work with untwisted sheaves is that the twist can make a moduli problem *fine* where the untwisted version is only coarse, and that it converts an arithmetic question about the Brauer class, the period-index problem, into a geometric question about the twisted moduli space. Both phenomena are best seen on an example, and the canonical one is a Brauer class arising from an elliptic fibration, where the twisted moduli space of rank-one twisted sheaves is a new variety related to the original by the twist.

Example 21.4.4: Fine Moduli, Period, and Index

Three faces of the same construction show why the twisted moduli space earns its place.

1. *Fineness from the twist.* Let X be a smooth projective surface. The untwisted moduli space M of stable sheaves on X with fixed invariants is often only a coarse moduli space: a universal sheaf on $X \times M$ need not exist, because the stable sheaves have scalar automorphisms $\mathbb{G}_m$ that obstruct gluing a universal family, and the obstruction is a Brauer class $\alpha_M \in \mathrm{Br}(M)$ on the base M itself, the class of the $\mathbb{G}_m$-gerbe of Chapter 8's residual-gerbe type. Two distinct twists are in play and should be kept apart: a class $\alpha \in \mathrm{Br}(X)$ on X, which is what the stack $\mathcal{M}_\alpha$ of α-twisted sheaves is built from, and the class $\alpha_M \in \mathrm{Br}(M)$ on M, which is the obstruction to a universal family for the untwisted problem. The twist that repairs the coarse space is the second one, on M: there exists a $\mathrm{pr}_M^* \alpha_M$-twisted universal sheaf $\mathcal{U}$ on $X \times M$, and this $\mathcal{U}$ makes M a *fine* moduli space for α_M-twisted sheaves even though it was only coarse for untwisted ones. The obstruction α_M to a universal family, read as a Brauer class on M, is precisely the twist one absorbs to turn the coarse space fine.
2. *Period and index.* For $\alpha \in \mathrm{Br}(X)$ the *period* $\mathrm{per}(\alpha)$ is the order of α in $\mathrm{Br}(X)$, and the *index* $\mathrm{ind}(\alpha)$ is the greatest common divisor of the degrees of the Azumaya algebras representing α, equivalently the minimal rank of a locally free α-twisted sheaf. That $\mathrm{ind}(\alpha)$ is the minimal twisted rank is exactly definition 21.3.5: an Azumaya algebra of degree d representing α is $\underline{\mathrm{End}}$ of a locally free α-twisted sheaf of rank d, and conversely. So the period-index problem, the question of how large $\mathrm{ind}(\alpha)$ can be relative to $\mathrm{per}(\alpha)$, becomes the question of the minimal rank of a twisted sheaf, a question the moduli stack $\mathcal{M}_\alpha$ is designed to answer: the nonemptiness of $\mathcal{M}_\alpha^P$ for a P of small twisted rank is the existence of a twisted sheaf of that rank. The elementary bound $\mathrm{per}(\alpha) \mid \mathrm{ind}(\alpha)$ follows from definition 21.2.3 and the torsion bound of theorem 21.2.4: the class of a degree-d Azumaya algebra is d-torsion, so the period divides the index. The companion fact that period and index share the same prime factors is a further elementary consequence of the primary decomposition of the torsion Brauer group $\mathrm{Br}'(X)$ of definition 21.2.3 into its p-primary parts, obtained by applying the torsion bound of theorem 21.2.4 in each p-primary component rather than to α as a whole.
3. *A class from an elliptic fibration.* Let $\pi \colon S \to C$ be a relatively minimal elliptic surface over a smooth projective curve C, with a section absent so that π has no rational section but does have multisections. The relative Jacobian $J \to C$ is an elliptic surface with a section, and S is a torsor under J over C; the class of this torsor in the sheaf cohomology group $H^1(C, J)$, mapped into $\mathrm{Br}(J)$ through the Leray sequence of π, is a Brauer class $\alpha \in \mathrm{Br}(J)$ measuring the failure of S to have a section. A rank-one α-twisted sheaf on J is precisely the data that a genuine line bundle on S would provide if $S \to C$ had a section, and the twisted moduli space $\mathcal{M}_\alpha$ of rank-one α-twisted sheaves on J recovers S itself: $S \cong \mathcal{M}_\alpha^{P_1}$ for the twisted Hilbert polynomial P_1 of a rank-one twisted sheaf. The twisted moduli space of the

Jacobian reconstructs the original fibration, and the Brauer class is the exact obstruction that separates them. The order of α divides the multisection index of π, tying the period of the class to the geometry of the fibration.

The example is the justification for the chapter. In the first face, the twist converts a coarse moduli space into a fine one by absorbing the scalar automorphisms into a universal twisted sheaf; the obstruction to a universal family, which for untwisted sheaves is a genuine failure, becomes a Brauer class one can twist away. In the second, the period-index problem, a central question about how far a Brauer class is from being represented by a small algebra, is recast as a question about the minimal rank of a twisted sheaf, which the moduli stack is built to detect. In the third, the twisted moduli space of the Jacobian of an elliptic fibration reconstructs the original fibration, so the Brauer class is the precise geometric datum distinguishing a surface from its Jacobian. Each face is a problem the untwisted theory cannot even phrase cleanly, and each becomes a statement about $\mathcal{M}_\alpha$.

Two further points about the elliptic-fibration computation connect it to the book's running threads. The Brauer class α there is n-torsion where n is the multisection index, so it is a class of the kind theorem 21.2.4 produced from a PGL_n-torsor, and the twisted sheaves are the modules over the corresponding Azumaya algebra of definition 21.3.5. And the reconstruction $S \cong \mathcal{M}_\alpha^{P_1}$ is a twisted Fourier-Mukai phenomenon: the twisted derived category of J is equivalent to the untwisted derived category of S, an equivalence that lives one level below the moduli statement and is the twisted analogue of the classical Fourier-Mukai equivalence between an abelian variety and its dual. That derived-categorical statement is the subject of the closing box.

The Brauer class and its twisted sheaves have a topological counterpart that the book will not develop but that is too close to pass over in silence. The same cohomological class that measures a Brauer obstruction reappears, over the complex numbers or in a purely topological setting, as a degree-three cohomology class, and the twisted sheaves reappear as twisted vector bundles and twisted K-theory. The exponential sequence is the bridge, and the analogy is precise enough to state as a dictionary and stop, since developing it belongs to the topological and operator-algebraic volumes of the series.

21.4.5 Note: The Topological Bridge and Twisted Derived Categories Two lineages run parallel to the algebraic-geometric one of this chapter, and the correspondence with each is worth naming precisely.

The Dixmier-Douady class and twisted K-theory. On a topological space or a complex manifold X, the exponential sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^\times \to 0$ (or its continuous or smooth analogue) gives a boundary map $H^2(X, \mathcal{O}_X^\times) \to H^3(X, \mathbb{Z})$, and the image of a Brauer class $\alpha \in H^2(X, \mathbb{G}_m)$ under this map is its *topological* or *Dixmier-Douady class* in $H^3(X, \mathbb{Z})_{\mathrm{tors}}$. The dictionary is

$$\mathrm{Br}(X) = H^2(X, \mathbb{G}_m)_{\mathrm{tors}} \quad \longleftrightarrow \quad H^3(X, \mathbb{Z})_{\mathrm{tors}}$$

the algebraic Brauer group on the left, its topological counterpart on the right, matched by the exponential boundary. Where an α-twisted sheaf lives on the algebraic side, an α-twisted vector bundle lives on the topological side, and the Grothendieck group of twisted bundles is *twisted K-theory*, the receptacle for D-brane charges in the physics that motivated its study. The operator-algebraic incarnation, where the class classifies continuous-trace C^*-algebras and the twist is a bundle of compact operators, belongs to the C^*-algebra volume of the series (‘NateCStarAlgebras‘), and the homotopy-theoretic twisted K-theory to the algebraic-topology volume (‘NateAlgTop‘). The three theories, algebraic, topological, and operator-algebraic, are the same class H^3_{tors} read in three categories.

Twisted derived categories and Fourier-Mukai. On the derived side, a Brauer class α twists the derived category: $D^b(X, \alpha)$, the bounded derived category of α-twisted coherent sheaves, is a triangulated category depending on α, reducing to $D^b(X)$ when $\alpha = 0$. Twisted moduli spaces produce equivalences of these: for the elliptic fibration $S \to J$ of example 21.4.4, the reconstruction $S \cong \mathcal{M}_\alpha$ upgrades to a *twisted Fourier-Mukai equivalence* $D^b(J, \alpha) \simeq D^b(S)$, the twisted analogue of the classical Fourier-Mukai transform between an abelian variety and its dual. This is the theory of [4], and its systematic development, together with the derived and spectral algebraic geometry it lives in, belongs to a future volume on derived algebraic geometry. The point to carry forward is only the analogy: the Brauer class that twists the moduli problem also twists the derived category, and the twisted moduli space is where a twisted derived equivalence is realized geometrically.

The box closes the chapter by placing the algebraic theory of twisted sheaves among the several subjects that share it. The Brauer group is simultaneously an object of arithmetic (central simple algebras, cited to ‘NateGaloisCohom‘), of algebraic geometry (the $\mathbb{G}_m$-gerbes and twisted sheaves of this chapter), of topology (the Dixmier-Douady class and twisted K-theory), and of derived categories (twisted Fourier-Mukai). This chapter owns the second of these, borrows the first, and points to the third and fourth. With the twisted moduli stack constructed and its neighbors named, Part VI is complete: the framework of descent, stacks, gerbes, and their moduli is fully in place, and the capstone that follows will assemble it into the moduli of curves, the object toward which the entire book has been building.

Exercise 21.4.1

1. Let $X = \operatorname{Spec} k$ and let $\alpha \in \mathrm{Br}(k)$ be represented by a central division algebra D of degree d. Using example 21.4.2, describe the moduli stack $\mathcal{M}_\alpha$ of α-twisted sheaves completely: identify its connected components, the object and automorphism group of each, and the dimension of each component as an algebraic stack. Then explain why $\mathcal{M}_\alpha$ is not itself a scheme (its coarse space, a disjoint union of points $\operatorname{Spec} k$, discards all of the automorphism data), and

contrast this with the situation on a projective surface where a stability condition is imposed and the moduli space is a positive-dimensional scheme.

2. Show that $\mathcal{M}_0 = \mathfrak{Coh}(X)$: for the zero Brauer class, the moduli stack of 0-twisted sheaves is the untwisted stack of coherent sheaves of Chapter 8. Identify precisely which ingredient of definition 21.4.1 degenerates when $\alpha = 0$, and explain, using theorem 21.3.3, why for $\alpha \neq 0$ the objects of $\mathcal{M}_\alpha$ are never coherent sheaves on X.

3. The proof of theorem 21.4.3 identifies the tangent and obstruction spaces of $\mathcal{M}_\alpha$ at a twisted sheaf $\mathcal{E}$ as $\mathrm{Ext}^1_{\mathcal{G}_\alpha}(\mathcal{E}, \mathcal{E})$ and $\mathrm{Ext}^2_{\mathcal{G}_\alpha}(\mathcal{E}, \mathcal{E})$. Suppose X is a smooth projective surface and $\mathcal{E}$ is a stable α-twisted sheaf. Assuming the twisted analogue of Serre duality on the surface ($\mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \cong \mathrm{Hom}(\mathcal{E}, \mathcal{E} \otimes \omega_X)^\vee$) and that a stable twisted sheaf is twisted-simple ($\mathrm{Hom}(\mathcal{E}, \mathcal{E}) = k$), compute the expected dimension of the moduli *stack* $\mathcal{M}_\alpha^{P,s}$ near $[\mathcal{E}]$ as $\dim \mathrm{Ext}^1 - \dim \mathrm{Ext}^0$, deduce that the coarse moduli *space* has expected dimension $\dim \mathrm{Ext}^1$ (rigidifying the $\mathbb{G}_m$-gerbe raises the dimension by one), and explain in one sentence why the $\mathbb{G}_m$ of twisted scalars is what makes the coarse space a scheme rather than a stack.

4. Let $\alpha \in \mathrm{Br}(X)$ have period n, and recall from example 21.4.4 that the index $\mathrm{ind}(\alpha)$ is the minimal rank of a locally free α-twisted sheaf. Using the torsion bound of theorem 21.2.4 (the class of a degree-d Azumaya algebra is d-torsion) and definition 21.3.5, prove that $\mathrm{per}(\alpha) \,\big|\, \mathrm{ind}(\alpha)$. Then, phrasing the period-index problem as a question about the moduli stack, state what geometric fact about $\mathcal{M}_\alpha^P$ would establish the period-index bound $\mathrm{ind}(\alpha) \,\big|\, \mathrm{per}(\alpha)^r$ for a specific exponent r (you need not prove any such bound; state the moduli-theoretic reformulation).

5. For the elliptic fibration $\pi\colon S \to C$ of example 21.4.4, with relative Jacobian $J \to C$ and Brauer class $\alpha \in \mathrm{Br}(J)$, explain why a rank-one α-twisted sheaf on J plays the role that a line bundle would play if $S \to C$ had a section, and why the twisted moduli space $\mathcal{M}_\alpha$ of rank-one twisted sheaves on J recovers S. In particular, identify what the order of α measures about the fibration, and connect the reconstruction $S \cong \mathcal{M}_\alpha$ to the twisted Fourier-Mukai equivalence $D^b(J, \alpha) \simeq D^b(S)$ named in box 21.4.5.

6. The box box 21.4.5 records the correspondence $\mathrm{Br}(X) = H^2(X, \mathbb{G}_m)_{\mathrm{tors}} \leftrightarrow H^3(X, \mathbb{Z})_{\mathrm{tors}}$ via the exponential boundary map. Write down the long exact cohomology sequence of the exponential sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^\times \to 0$ around degree two, and use it to express the boundary map $H^2(X, \mathcal{O}_X^\times) \to H^3(X, \mathbb{Z})$ and to explain why its restriction to torsion is the map appearing in the dictionary. Then state, without proof, the topological analogue of an α-twisted sheaf and the topological invariant that plays the role of the twisted moduli space's Chern character.

22

Application: The Moduli of Curves

Part IV built moduli spaces as functors and asked when a scheme represents them; Part V linearized the question, computing the deformations and obstructions of a single object; Part VI replaced the scheme with the stack that finally represents the objects-with-automorphisms a moduli problem classifies. Each Part returned again and again to one example, the moduli of curves, and each left it in a more finished state: a functor whose non-representability was diagnosed by the automorphisms of an elliptic curve, then a tangent space of dimension $3g-3$ read off from the cohomology of the tangent sheaf, then an algebraic stack certified by Artin's criteria. This chapter assembles those threads into the theorem the whole book has been approaching.

The object is $\overline{\mathcal{M}}_g$, and its properties come from the three Parts in turn. Its smoothness and its dimension $3g-3$ are the unobstructed deformation theory of Part V. Its being a Deligne-Mumford stack, and the properness that makes it a good compactification, are the stack theory and the valuative criterion of Part VI, with stable reduction supplying the limits and the deformation theory of a node supplying the local model of the boundary. Its coarse moduli space, a projective variety singular where the stack is smooth, is the geometric invariant theory quotient of Part IV. The first section defines $\mathcal{M}_g$ and records that it is a smooth Deligne-Mumford stack; the second compactifies it by admitting stable curves; the third proves properness through stable reduction; and the fourth descends to the coarse space, revisits $\mathcal{M}_{1,1}$ and $\overline{\mathcal{M}}_{1,1}$ in full, and points toward the intersection theory of tautological classes that the next volume takes up. The moduli of curves is where deformation theory, geometric invariant theory, and stacks meet in a single object, and it is the natural place for this book to end.

22.1 The Moduli Stack $\mathcal{M}_g$

The book has built three machines, and this chapter runs them together on a single problem. Part IV asked when a moduli problem is representable and found that objects with automorphisms defeat fine representability, the j-line of the elliptic curves being the first casualty. Part V linearized a moduli problem: the tangent space to a moduli functor is a cohomology group, and for a smooth curve C that group is $H^1(C, T_C)$, of dimension $3g-3$. Part VI supplied the object that finally represents a moduli problem whose objects have automorphisms, the algebraic stack, together with its Deligne-Mumford refinement for objects with finite automorphism groups. This section assembles the three into the statement the whole book has pointed toward: for $g \geq 2$ the moduli of smooth genus-g curves is a smooth Deligne-Mumford stack of dimension $3g-3$. Every clause of that sentence is a theorem from an earlier chapter, and the work here is to see that the hypotheses of each apply to curves and to assemble the conclusions.

The starting point is the correct notion of a family. A single curve is a point of the moduli space; a family of curves over a base S is a map from S to it, and the moduli problem is the assignment $S \mapsto \{\text{families over } S\}$. For curves the right notion of family is forced by the demand that fibers be smooth curves and that the family vary algebraically, which is exactly a smooth proper morphism with the correct fibers. Throughout, k is an algebraically closed field, of characteristic 0 where a count of automorphisms requires it (this is flagged where it is used), and a curve means a smooth proper connected k-curve; $g \geq 2$ unless stated otherwise.

Definition 22.1.1: Family of Smooth Curves

A *family of smooth genus-g curves* over a scheme S is a morphism

$$\pi \colon \mathcal{C} \to S$$

that is smooth, proper, and whose geometric fibers are connected smooth curves of genus g. A morphism of families over a morphism $f \colon S' \to S$ of base schemes is a cartesian square

$$\begin{array}{ccc} \mathcal{C}' & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow \\ S' & \xrightarrow{f} & S \end{array}$$

exhibiting $\mathcal{C}'$ as the pullback $f^*\mathcal{C} = \mathcal{C} \times_S S'$; in particular for each morphism f the pullback $f^*\mathcal{C} \to S'$ is again a family of smooth genus-g curves. A *family of n-pointed smooth genus-g curves* is such a π together with n disjoint sections $\sigma_1, \ldots, \sigma_n \colon S \to \mathcal{C}$ of π, with pairwise disjoint images, whose values on each geometric fiber are n distinct points.
The *moduli stack of smooth genus-g curves* $\mathcal{M}_g$ is the category fibered in groupoids over $(\mathbf{Sch}/k)$ whose objects over S are the families $\pi \colon \mathcal{C} \to S$ and whose morphisms are the cartesian squares above; the fiber $\mathcal{M}_g(S)$ is the groupoid of families over S with isomorphisms of families. The *moduli stack of n-pointed curves* $\mathcal{M}_{g,n}$ is defined identically with n-pointed families in place of families.

The definition makes three demands, and each earns its place. Smoothness of π guarantees the fibers are smooth curves and that they vary without acquiring singularities. Properness rules out families that lose points off to infinity, so that a fiber is a complete curve of a well-defined genus rather than an affine piece of one. Connectedness of the fibers keeps the genus a single invariant of each fiber rather than a sum over components. That π^* of a family is a family is the pullback stability recorded in definition 22.1.1: smoothness, properness, and the fiber conditions are all stable under base change, so $\mathcal{M}_g$ really is a fibered category, its objects glued by the cartesian squares. The passage from a functor to a groupoid is the correction Part VI forced. A curve C has automorphisms, so two families can be isomorphic in more than one way; recording only isomorphism classes, as a set-valued functor does, discards the automorphisms and destroys the descent that makes the moduli problem geometric. The fibered category $\mathcal{M}_g$ keeps the isomorphisms, and the automorphisms of a curve reappear as the automorphisms of that curve viewed as an object of $\mathcal{M}_g(\operatorname{Spec} k)$.

The definition is abstract, so it should be grounded at once in the smallest family that is not a single curve: a one-parameter family whose fibers are elliptic curves, the case $g = 1$ that motivated the whole apparatus in Part IV.

Example 22.1.2: The Legendre Family

Fix k algebraically closed of characteristic $\neq 2$, let $S = \mathbb{A}^1_k \setminus \{0, 1\}$ with coordinate λ, and let $\mathcal{C} \subset \mathbb{P}^2_k \times S$ be the family of plane cubics

$$y^2 z = x(x - z)(x - \lambda z)$$

with $\pi\colon \mathcal{C} \to S$ the projection to S. For each $\lambda \in S(k)$ the fiber is the smooth plane cubic $y^2 = x(x - 1)(x - \lambda)$, a genus-1 curve, smooth precisely because $\lambda \neq 0, 1$ keeps the three branch points $0, 1, \lambda$ (and ∞) distinct. The morphism π is proper, being a closed subscheme of $\mathbb{P}^2 \times S$ projected to S, and it is smooth of relative dimension 1 because the fiberwise discriminant $\lambda^2(\lambda-1)^2$ is invertible on S; the section $z = 0$, $x = 0$ (the point at infinity) marks each fiber, so (π, σ) is a family of 1-pointed genus-1 curves and defines a map $S \to \mathcal{M}_{1,1}$.

The family exhibits both features that force a stack. First, the j-invariant of the fiber over λ is

$$j(\lambda) = 256\frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}$$

a degree-6 map $S \to \mathbb{A}^1_j$; two parameters λ, λ' give isomorphic fibers exactly when $j(\lambda) = j(\lambda')$, so the map to the coarse j-line (theorem 13.3.6) is six-to-one over a general j, recording the six values of λ permuted by the anharmonic group S_3 acting through $\lambda \mapsto 1 - \lambda, 1/\lambda, \ldots$. Second, every fiber carries the automorphism $y \mapsto -y$ fixing λ, the hyperelliptic involution, so the family admits a nontrivial automorphism over S itself. The coarse space forgets both the six-fold ambiguity and the involution; the stack $\mathcal{M}_{1,1}$ remembers them, and this is the running touchstone (example 13.3.4) that the general theory below organizes.

With the moduli stack defined, the central theorem can be assembled. Every adjective in "smooth Deligne-Mumford stack of dimension $3g - 3$" is a property of stacks made precise in Part VI, and each is verified by importing a result about curves. Algebraicity was proved in general in the algebraic-stacks chapter, by checking Artin's criteria; smoothness is the unobstructedness of curve deformations from Part V, together with the dimension count $\dim H^1(C, T_C) = 3g - 3$; the Deligne-Mumford property is the finiteness of $\operatorname{Aut}(C)$, which makes the diagonal unramified. The proof below is the assembly of these three imports, and its interest is entirely in seeing that each earlier theorem applies on the nose.

Theorem 22.1.3: $\mathcal{M}_g$ is a Smooth Deligne-Mumford Stack of Dimension $3g-3$

For $g \geq 2$, the moduli stack $\mathcal{M}_g$ is a smooth Deligne-Mumford stack over k, of dimension

$$\dim \mathcal{M}_g = 3g - 3$$

and $\mathcal{M}_{g,n}$ is a smooth Deligne-Mumford stack of dimension $3g-3+n$.

***Proof*:**
The three defining properties of the conclusion are established in turn: that $\mathcal{M}_g$ is an algebraic stack, that it is Deligne-Mumford, and that it is smooth of the stated dimension. Each step imports one theorem from an earlier chapter and checks its hypotheses for curves.

Step 1: $\mathcal{M}_g$ is an algebraic stack. That $\mathcal{M}_g$ is a stack in the fppf topology is descent for families of curves: a family $\pi\colon \mathcal{C} \to S$ is a scheme proper and smooth over S, and such schemes descend along fppf covers of S once a relative polarization is fixed, which the relative pluricanonical bundle $\omega_{\mathcal{C}/S}^{\otimes 3}$ supplies (it is relatively very ample for $g \geq 2$, since $\deg \omega_C^{\otimes 3} = 3(2g-2) = 6g-6 \geq 2g+1$ on each fiber). That $\mathcal{M}_g$ is moreover *algebraic* is theorem 20.4.4, where Artin's criteria were verified: $\mathcal{M}_g$ is a limit-preserving fppf stack with a deformation and obstruction theory ($H^1(C, T_C)$ and $H^2(C, T_C)$, from Part V) satisfying the Schlessinger-Artin conditions, and with effective formal versal deformations, so the criteria certify a smooth atlas $U \to \mathcal{M}_g$ from a scheme. Concretely the atlas is the locus in a Hilbert scheme of tricanonically embedded curves, on which the theorem of theorem 20.4.4 produces the smooth cover. Thus $\mathcal{M}_g$ is an algebraic stack.

Step 2: $\mathcal{M}_g$ is Deligne-Mumford. By theorem 20.2.3, an algebraic stack is Deligne-Mumford if and only if its diagonal is unramified, equivalently the automorphism group scheme $\underline{\mathrm{Aut}}(C)$ of every object is unramified over the base, equivalently finite and reduced for objects over a field. The object over $\operatorname{Spec} k$ is a curve C, and its automorphism group scheme is $\underline{\mathrm{Aut}}(C)$. For $g \geq 2$ this is a finite reduced group scheme: finiteness is definition 22.1.4 below (Hurwitz's bound $|\mathrm{Aut}(C)| \leq 84(g-1)$ makes the group finite), and reducedness holds because $H^0(C, T_C) = 0$ for $g \geq 2$, the tangent space to $\underline{\mathrm{Aut}}(C)$ at the identity being the space of infinitesimal automorphisms $H^0(C, T_C)$. That $H^0(C, T_C) = 0$ follows from $\deg T_C = \deg \omega_C^\vee = -(2g-2) < 0$: a global section of a line bundle of negative degree on a connected curve vanishes. Hence $\underline{\mathrm{Aut}}(C)$ is a finite reduced group scheme, i.e. a finite constant group scheme in characteristic 0, the diagonal of $\mathcal{M}_g$ is unramified, and $\mathcal{M}_g$ is Deligne-Mumford.

Step 3: $\mathcal{M}_g$ is smooth of dimension $3g-3$. Smoothness of a Deligne-Mumford stack is smoothness of a (equivalently, any) étale atlas, and by the deformation-theoretic criterion it is the unobstructedness of the deformation functor of every object together with a constant tangent-space dimension. The deformation functor of a curve C is unobstructed by proposition 17.4.1: the obstructions to lifting a first-order deformation lie in $H^2(C, T_C)$, which vanishes because a curve has cohomological dimension 1, so every first-order deformation extends and the functor is formally smooth. Its tangent space is the space of first-order deformations of C, computed in theorem 16.3.2 to be

$$T_{[C]}\mathcal{M}_g = H^1(C, T_C)$$

of dimension $\dim H^1(C, T_C) = 3g-3$ for $g \geq 2$. The dimension count is Riemann-Roch: $\chi(T_C) = \deg T_C + (1-g) = (2-2g) + (1-g) = 3-3g$, and since $h^0(T_C) = 0$ from Step 2, $h^1(T_C) = -\chi(T_C) = 3g-3$. A formally smooth Deligne-Mumford stack whose tangent space has constant dimension $3g-3$ is smooth of dimension $3g-3$, which is the claim.

For $\mathcal{M}_{g,n}$ the same three steps apply verbatim, with two changes. The automorphism group of an n-pointed curve $(C, p_1, \ldots, p_n)$ is the subgroup of $\mathrm{Aut}(C)$ fixing each p_i, still finite for $g \geq 2$, so the stack is again Deligne-Mumford. The tangent space gains n dimensions, one for the position of each marked point moving in the curve: the deformations of an n-pointed curve are classified by $H^1(C, T_C(-p_1 - \cdots - p_n))$, whose dimension is $3g - 3 + n$ by Riemann-Roch for $g \geq 2$. Hence $\mathcal{M}_{g,n}$ is smooth Deligne-Mumford of dimension $3g - 3 + n$, as required.

The theorem is the payoff of the whole book, and the way its proof splits across the three Parts is the point. Nothing in it is new; the entire content is that each hypothesis of an imported theorem holds for curves, and the assembly of the conclusions is the statement. The number $3g - 3$ deserves emphasis: it is the dimension of the space in which a curve can be deformed, and it appears first as $\dim H^1(C, T_C)$, a cohomological count, and second as the dimension of the moduli stack, a geometric one, the two identified by the deformation theory of Part V. That the two agree is the content of "the tangent space to the moduli space is the deformation space", the principle that made the local structure of moduli computable. For $g = 2$ this gives $\dim \mathcal{M}_2 = 3$, for $g = 3$ it gives 6, and in general the moduli of curves grows linearly in the genus, a fact with no analogue for the finite lists of low genus but which governs the geometry of $\mathcal{M}_g$ for all large g.

The Deligne-Mumford property is worth isolating, because it is exactly what fails for $g \leq 1$ and exactly what the finiteness of automorphisms gives. A Deligne-Mumford stack has an étale atlas, so it looks étale-locally like a scheme and its points have finite automorphism groups; an Artin stack with positive- dimensional automorphism groups, like $B\mathbb{G}_m$, does not. The line $\deg T_C < 0 \Rightarrow H^0(T_C) = 0$ that killed the infinitesimal automorphisms is the same computation that, run at $g = 1$ where $\deg T_C = 0$, returns $H^0(T_C) = k$, the one-dimensional space of translations, and at $g = 0$ where $\deg T_C = 2 > 0$ returns $H^0(T_C) = H^0(\mathbb{P}^1, \mathcal{O}(2))$, the three-dimensional Lie algebra of PGL_2. The genus condition $g \geq 2$ is precisely the condition that this Lie algebra vanish, that curves be rigid enough to have finite automorphism groups, and hence that their moduli be Deligne-Mumford rather than only Artin.

That finiteness of automorphism groups is what separates the well-behaved $\mathcal{M}_g$ for $g \geq 2$ from the Artin cases below deserves its own definition and a sharp bound. The automorphism group of a curve of genus at least 2 is not just finite but uniformly bounded in terms of the genus, a classical theorem of Hurwitz, and the bound quantifies the rigidity that makes the moduli Deligne-Mumford.

Definition 22.1.4: Automorphism Group of a Curve; Hurwitz's Bound

Let C be a smooth proper connected curve of genus g over $k = \bar{k}$. Its *automorphism group* $\mathrm{Aut}(C)$ is the group of k-automorphisms $C \to C$, equivalently the k-points of the automorphism group scheme $\underline{\mathrm{Aut}}(C)$. For $g \geq 2$ the group $\mathrm{Aut}(C)$ is finite, and in characteristic 0 it satisfies *Hurwitz's bound*

$$|\mathrm{Aut}(C)| \leq 84(g-1)$$

A curve is *hyperelliptic* if it admits a degree-2 map $C \to \mathbb{P}^1$; the corresponding involution $\iota \colon C \to C$ exchanging the two sheets, the *hyperelliptic involution*, is a central element of $\mathrm{Aut}(C)$ intrinsic to C, being the unique involution whose quotient is $\mathbb{P}^1$, and its fixed points are the $2g + 2$ Weierstrass points lying over the branch points.

The finiteness alone is what the moduli theory needs, and it is worth seeing why $g \geq 2$ forces it, since the mechanism is exactly the tangent-space computation of theorem 22.1.3. An algebraic group acting faithfully on a variety is finite as soon as its Lie algebra vanishes, and the Lie algebra of

$\underline{\mathrm{Aut}}(C)$ is $H^0(C, T_C)$, which is 0 for $g \geq 2$ because $\deg T_C = 2 - 2g < 0$. Thus $\underline{\mathrm{Aut}}(C)$ is a group scheme of dimension 0, and being of finite type it is finite. The Hurwitz bound $84(g-1)$ is the sharp refinement, and it is proved by a covering-space argument worth recording.

***Proof of Hurwitz's bound*:**
Work in characteristic 0 and let $G = \mathrm{Aut}(C)$, which is finite by the preceding paragraph. The quotient $C \to C/G = \bar{C}$ is a degree-$|G|$ map to a smooth curve $\bar{C}$ of some genus $\bar{g}$, ramified over finitely many points $q_1, \ldots, q_r$ of $\bar{C}$ with ramification indices $e_1, \ldots, e_r$. The Riemann-Hurwitz formula for the quotient map reads

$$2g - 2 = |G| \left(2\bar{g} - 2 + \sum_{i=1}^{r} \left(1 - \frac{1}{e_i}\right)\right)$$

the term $1 - 1/e_i$ being the contribution of each branch point, since the fiber over q_i has $|G|/e_i$ points each of ramification index e_i. Because $g \geq 2$ the left side is positive, so the bracketed quantity

$$\chi = 2\bar{g} - 2 + \sum_{i=1}^{r} \left(1 - \frac{1}{e_i}\right)$$

is a strictly positive rational number, and $|G| = (2g-2)/\chi$; maximizing $|G|$ is minimizing the positive χ. A finite case analysis on $(\bar{g}, r, e_1, \ldots, e_r)$ bounds χ below. If $\bar{g} \geq 2$ then $\chi \geq 2$. If $\bar{g} = 1$ then $\chi = \sum(1 - 1/e_i) \geq 1 - 1/2 = 1/2$ once $r \geq 1$ (and $\chi > 0$ forces $r \geq 1$). If $\bar{g} = 0$ then $\chi = -2 + \sum(1 - 1/e_i)$, and positivity forces $r \geq 3$; the minimum positive value of $-2 + \sum_{i=1}^{r}(1 - 1/e_i)$ over integers $e_i \geq 2$ is attained at $r = 3$ with $(e_1, e_2, e_3) = (2, 3, 7)$, giving

$$\chi = -2 + \left(1 - \tfrac{1}{2}\right) + \left(1 - \tfrac{1}{3}\right) + \left(1 - \tfrac{1}{7}\right) = -2 + \tfrac{1}{2} + \tfrac{2}{3} + \tfrac{6}{7} = \frac{1}{42}$$

since $\frac{1}{2} + \frac{2}{3} + \frac{6}{7} = \frac{21+28+36}{42} = \frac{85}{42}$ and $-2 + \frac{85}{42} = \frac{1}{42}$. Any other triple (e_1, e_2, e_3) of integers ≥ 2 with positive χ gives $\chi \geq 1/42$, as a direct check of the nearby triples $(2,3,8), (2,4,5), (3,3,4), \ldots$ confirms, each exceeding $1/42$. Hence $\chi \geq 1/42$ in all cases, and

$$|G| = \frac{2g-2}{\chi} \leq 42(2g-2) = 84(g-1)$$

which is the bound, as claimed.

The bound is sharp: the Klein quartic $x^3y + y^3z + z^3x = 0$ in $\mathbb{P}^2$ has genus 3 and automorphism group $\mathrm{PSL}_2(\mathbb{F}_7)$ of order $168 = 84 \cdot 2 = 84(3-1)$, realizing the $(2,3,7)$ triangle group that the extremal case picks out. The role of the bound in the moduli theory is qualitative, not quantitative: it certifies that $\mathrm{Aut}(C)$ is finite for every curve of genus ≥ 2, uniformly, which is precisely the input Step 2 of theorem 22.1.3 needed to conclude the diagonal is unramified and $\mathcal{M}_g$ is Deligne-Mumford. Where $\mathrm{Aut}(C)$ jumps, as at the Klein quartic or the hyperelliptic curves, the stack $\mathcal{M}_g$ acquires a point with larger automorphism group, a stacky point whose residual gerbe records the extra symmetry, exactly as the residual gerbe of $\mathcal{M}_{1,1}$ recorded the μ_4, μ_6 symmetry at the special j-invariants of example 13.3.4.

The hyperelliptic involution is the cleanest example of a curve automorphism, and it is worth seeing that hyperelliptic curves form a large family whose generic member has automorphism group exactly $\{1, \iota\}$, so that the involution is present on every hyperelliptic curve but is typically the whole automorphism group.

Example 22.1.5: The Hyperelliptic Involution and the Genus-2 Curve

Every curve of genus 2 is hyperelliptic. Riemann-Roch gives $h^0(C, \omega_C) = g = 2$, so the canonical map $C \to \mathbb{P}^1$ is a degree-2 map (the canonical bundle has degree $2g - 2 = 2$ and two sections), and this exhibits C as a double cover of $\mathbb{P}^1$. Explicitly, a genus-2 curve is

$$y^2 = f(x), \qquad f(x) = \prod_{i=1}^{6}(x - a_i)$$

a double cover of the x-line branched at the six roots $a_1, \ldots, a_6$ of a squarefree sextic (and at ∞ when $\deg f = 5$; six branch points on $\mathbb{P}^1$ in either normalization). The hyperelliptic involution is $\iota\colon (x, y) \mapsto (x, -y)$, exchanging the two sheets, with quotient the x-line $\mathbb{P}^1$ and six fixed points, the Weierstrass points lying over the branch points.

The moduli count matches theorem 22.1.3. A genus-2 curve is determined by its six branch points on $\mathbb{P}^1$ up to the action of PGL_2 on $\mathbb{P}^1$ (which is 3-dimensional) and the permutation S_6 of the points; so

$$\dim \mathcal{M}_2 = \dim\{6 \text{ points on } \mathbb{P}^1\} - \dim \mathrm{PGL}_2 = 6 - 3 = 3$$

exactly $3g - 3 = 3$ for $g = 2$. The generic such curve has automorphism group $\{1, \iota\} = \mathbb{Z}/2$, generated by the involution alone; special configurations of the six points (extra symmetry of the branch divisor under a subgroup of PGL_2) enlarge $\mathrm{Aut}(C)$, producing the stacky points where the automorphism group jumps. Since every genus-2 curve is hyperelliptic, the hyperelliptic locus is all of $\mathcal{M}_2$, of dimension 3; for $g \geq 3$ it is a proper substack, computed next.

The last low-genus phenomenon to record is the contrast between the moduli stacks for $g = 0, 1$ and the Deligne-Mumford stacks for $g \geq 2$, together with the dimensions and the hyperelliptic locus for small g. Assembling these in one place shows both why the theory bifurcates at $g = 2$ and how the number $3g - 3$ manifests concretely.

Example 22.1.6: The Moduli Stacks in Low Genus

The moduli stacks for small g exhibit the transition from Artin to Deligne-Mumford, and the dimension formula $3g - 3$ across the range.

1. *Genus* 0. The only smooth genus-0 curve over $k = \bar{k}$ is $\mathbb{P}^1$, with automorphism group $\mathrm{Aut}(\mathbb{P}^1) = \mathrm{PGL}_2$, a 3-dimensional algebraic group. A family of genus-0 curves is a $\mathbb{P}^1$-bundle, and the moduli stack is the classifying stack

$$\mathcal{M}_0 = B\mathrm{PGL}_2$$

an Artin stack of dimension -3, with a single point whose automorphism group is PGL_2. Its coarse space is a point: all $\mathbb{P}^1$'s are isomorphic. The dimension $3 \cdot 0 - 3 = -3$ is the negative of $\dim \mathrm{PGL}_2$, the signature of a point with a 3-dimensional automorphism group. Because PGL_2 is positive- dimensional, $\mathcal{M}_0$ is Artin but *not* Deligne-Mumford.

2. *Genus* 1. A smooth genus-1 curve has no distinguished point, and $\mathrm{Aut}(C)$ contains the 1-dimensional group of translations, so $\mathcal{M}_1$ (unpointed) is again not Deligne-Mumford. Rigidifying by a marked point kills the translations: $\mathcal{M}_{1,1}$, the moduli of elliptic curves, is a Deligne-Mumford stack of dimension $3 \cdot 1 - 3 + 1 = 1$, with generic automorphism

group $\mu_2 = \{\pm 1\}$ and special groups μ_4, μ_6 at $j = 1728, 0$. Its coarse space is the j-line $\mathbb{A}^1_j$ (theorem 13.3.6), one-dimensional, matching the stack dimension. This is the boundary case: $g = 1$ needs a marked point to become Deligne-Mumford, and the automorphism obstruction of example 13.3.4 is exactly the residual μ_2 that survives.

3. *Genus* 2. By example 22.1.5, $\dim \mathcal{M}_2 = 3 = 3 \cdot 2 - 3$, and $\mathcal{M}_2$ is a smooth Deligne-Mumford stack (theorem 22.1.3). Every genus-2 curve is hyperelliptic, so the hyperelliptic locus is all of $\mathcal{M}_2$.

4. *Genus* 3. Here $\dim \mathcal{M}_3 = 6 = 3 \cdot 3 - 3$. A generic genus-3 curve is a smooth plane quartic (canonical bundle very ample, embedding $C \hookrightarrow \mathbb{P}^2$ as a degree-4 curve), *not* hyperelliptic; the hyperelliptic curves form a proper substack.

The hyperelliptic locus $\mathcal{H}_g \subset \mathcal{M}_g$ of curves admitting a degree-2 map to $\mathbb{P}^1$ has dimension $2g - 1$ for all $g \geq 2$. A hyperelliptic curve is a double cover of $\mathbb{P}^1$ branched at $2g + 2$ points (Riemann-Hurwitz for a degree-2 cover, with one index-2 ramification point over each branch point: $2g - 2 = 2 \cdot (-2) + b \cdot (2 - 1) = -4 + b$ forces $b = 2g + 2$), and such a cover is determined by the branch divisor up to PGL_2 acting on $\mathbb{P}^1$:

$$\dim \mathcal{H}_g = (2g + 2) - \dim \mathrm{PGL}_2 = (2g + 2) - 3 = 2g - 1$$

For $g = 2$ this gives $2 \cdot 2 - 1 = 3 = \dim \mathcal{M}_2$, confirming that all genus-2 curves are hyperelliptic; for $g = 3$ it gives $2 \cdot 3 - 1 = 5 < 6 = \dim \mathcal{M}_3$, so the hyperelliptic locus is a divisor (codimension 1) in $\mathcal{M}_3$; and for large g the codimension $(3g - 3) - (2g - 1) = g - 2$ grows, so the generic curve is far from hyperelliptic.

The table across $g = 0, 1, 2, 3$ tells the whole story of why $g \geq 2$ is the regime of the theorem. At $g = 0$ and $g = 1$ the automorphism groups are positive-dimensional (all of PGL_2, or the translations), the moduli stacks are Artin and not Deligne-Mumford, and the coarse spaces collapse the moduli to a point or to the j-line. At $g \geq 2$ the automorphism groups are finite by Hurwitz, the stacks are Deligne-Mumford, and the dimension settles into the linear growth $3g - 3$ that governs all higher genus. The hyperelliptic locus interpolates: it is everything at $g = 2$, a divisor at $g = 3$, and a shrinking-codimension family thereafter, the last place where the double-cover description of Part IV's curves survives before the general curve becomes too generic to be so described. The stacky structure that example 13.3.4 first exhibited for $g = 1$ persists at every genus as the residual gerbes over the curves with extra automorphisms, and it is these that force $\mathcal{M}_g$ to be a stack rather than a variety even for $g \geq 2$, notwithstanding the finiteness of every individual automorphism group.

Exercise 22.1.1

1. Verify the dimension formula $\dim \mathcal{M}_{g,n} = 3g - 3 + n$ for $g \geq 2$ directly from Riemann-Roch, by computing $h^1(C, T_C(-p_1 - \cdots - p_n))$ for n distinct points $p_1, \ldots, p_n$ on a curve C of genus g. Confirm that h^0 of this twisted tangent sheaf vanishes, so that the dimension is $-\chi$, and recover the unpointed count at $n = 0$.

2. Show that a smooth genus-0 curve over $k = \bar{k}$ is isomorphic to $\mathbb{P}^1$, and that $\mathrm{Aut}(\mathbb{P}^1) = \mathrm{PGL}_2$ has Lie algebra $H^0(\mathbb{P}^1, T_{\mathbb{P}^1})$ of dimension 3. Conclude that $\mathcal{M}_0$ is not Deligne-Mumford, and identify the single object of $\mathcal{M}_0(\operatorname{Spec} k)$ and its automorphism group.

3. Using the Klein quartic $C\colon x^3y + y^3z + z^3x = 0$ in $\mathbb{P}^2$, verify that its genus is 3 (a smooth

plane curve of degree d has genus $\binom{d-1}{2}$) and that Hurwitz's bound $84(g-1)$ at $g=3$ equals 168. Identify which of the three cases $(\bar{g}, r, e_i)$ in the proof of Hurwitz's bound is realized by the quotient $C \to C/\mathrm{Aut}(C)$, given that $|\mathrm{Aut}(C)| = 168$.

4. Let $\mathcal{H}_g \subset \mathcal{M}_g$ be the hyperelliptic locus. Compute its dimension $2g-1$ from the double-cover description, and determine the codimension of $\mathcal{H}_g$ in $\mathcal{M}_g$ as a function of g. For which g is $\mathcal{H}_g$ a divisor, and for which is it all of $\mathcal{M}_g$?

5. Explain why the unpointed moduli stack $\mathcal{M}_1$ of genus-1 curves is not Deligne-Mumford, but $\mathcal{M}_{1,1}$ is. Compute $H^0(C, T_C)$ for a genus-1 curve C and interpret its dimension as the dimension of the automorphism group of C; explain how marking a point cuts this down to a finite group.

6. The Legendre family (example 22.1.2) gives a map $S = \mathbb{A}^1 \setminus \{0,1\} \to \mathcal{M}_{1,1}$. Explain why this map is neither injective nor a monomorphism of stacks: identify the finite fibers of the induced map to the coarse j-line and the automorphism over S that obstructs the family from being a monomorphism. What does this say about the difference between $\mathcal{M}_{1,1}$ and its coarse space?

22.2 Stable Curves and Deligne-Mumford Stability

The stack $\mathcal{M}_g$ of theorem 22.1.3 is smooth and Deligne-Mumford, but it is not proper: a family of smooth genus-g curves over a punctured disc need not extend to a family of smooth curves over the whole disc, because the limiting curve can degenerate. A curve moving in a one-parameter family can, in the limit, acquire a singularity, split into pieces, or send a subfamily off to infinity. To compactify $\mathcal{M}_g$ one must decide which singular curves to admit as limits. Admitting too few leaves $\mathcal{M}_g$ non-proper; admitting too many, such as arbitrary singular curves, produces a bloated space with no separation, since many different degenerations become indistinguishable. The correct class is the *stable curves*: curves with the mildest possible singularities, the nodes, subject to a rigidity condition that keeps their automorphism groups finite. This is the class Deligne and Mumford isolated, and the resulting compactification $\overline{\mathcal{M}}_g$ is the object toward which the chapter has been building.

This section develops the singular curves and the stability condition, and assembles the deformation-theoretic input to show that $\overline{\mathcal{M}}_g$ is a smooth Deligne-Mumford stack containing $\mathcal{M}_g$ as a dense open substack. The one property deferred is properness, which rests on the stable reduction theorem and is proved in section 22.3. Throughout, k is algebraically closed, of characteristic 0 where a count of automorphisms requires it, and $g \geq 2$ unless stated otherwise.

Definition 22.2.1: Nodal (Prestable) Curve

A *nodal curve* (or *prestable curve*) over k is a reduced, connected, proper k-scheme C of pure dimension 1 whose only singularities are *nodes*: points $p \in C$ at which the complete local ring is

$$\widehat{\mathcal{O}}_{C,p} \cong k[[x,y]]/(xy)$$

the local model of two smooth branches crossing transversally. Its *arithmetic genus* is

$$p_a(C) = \dim_k H^1(C, \mathcal{O}_C)$$

The *dual graph* Γ_C of C has one vertex for each irreducible component C_i, weighted by the geometric genus g_i of the normalization $\widetilde{C_i}$, and one edge for each node, joining the vertices of the two branches through that node (a self-edge, i.e. a loop, when both branches lie on the same component). The *dualizing sheaf* ω_C is the unique line bundle whose restriction to the smooth locus is the sheaf of Kähler differentials and which, near a node with branches $x = 0$ and $y = 0$, consists of the rational differentials η with at worst simple poles along each branch whose residues satisfy

$$\mathrm{Res}_{x=0}\eta + \mathrm{Res}_{y=0}\eta = 0$$

It is a line bundle of degree $\deg \omega_C = 2p_a(C) - 2$.

Three points in the definition carry the weight. The first is that a node is the mildest singularity a reduced curve can have: it is the singularity of two lines meeting at a point, and it has a single branch on each side rather than a cusp or worse. The second is that the arithmetic genus p_a, unlike the geometric genus, does not change when a smooth curve degenerates to a nodal one; it is the invariant that stays constant in a flat family, which is exactly why it, and not the geometric genus of the pieces, indexes the compactified moduli. The third is the dualizing sheaf ω_C. On a smooth curve ω_C is the canonical bundle Ω^1_C, but a nodal curve is singular, so Ω^1_C is no longer a line bundle; the residue condition at each node repairs this, producing a genuine line bundle ω_C that agrees with Ω^1_C away from the nodes and satisfies Serre duality and Riemann-Roch on the nodal curve. The degree formula $\deg \omega_C = 2p_a - 2$ is the nodal extension of $\deg \omega_C = 2g - 2$ for smooth curves, and it is what makes ω_C the right object to test for ampleness. The dualizing sheaf is developed in general in the megabook Part IV; here $\deg \omega_C = 2p_a - 2$ and the residue description are the facts used, and the reader may take them from **??** or verify them in the worked cases below.

To ground the definition, consider the simplest nodal curve, the one-nodal degeneration of an elliptic curve, and read off its dual graph, arithmetic genus, and dualizing sheaf directly.

Example 22.2.2: The Nodal Cubic and Its Dual Graph

Let $C \subset \mathbb{P}^2$ be the nodal cubic $y^2 z = x^3 + x^2 z$, the projective closure of the affine curve $y^2 = x^3 + x^2 = x^2(x+1)$. Near the origin $(x,y) = (0,0)$ the equation factors as $y^2 - x^2 = (y-x)(y+x)$ to leading order, so the two branches $y = x$ and $y = -x$ cross transversally: the origin is a node, and one checks that C is smooth elsewhere. The curve C is irreducible, so its dual graph Γ_C has a single vertex; the normalization $\widetilde{C} \to C$ separates the two branches at the node into two distinct points, and $\widetilde{C} \cong \mathbb{P}^1$ has geometric genus 0, so the vertex is weighted 0. The single node is a self-node on the one component, so Γ_C has one vertex of weight 0 and one loop.

The arithmetic genus is computed from the normalization sequence

$$0 \to \mathcal{O}_C \to \nu_* \mathcal{O}_{\widetilde{C}} \to k_p \to 0$$

where $\nu\colon \widetilde{C} \cong \mathbb{P}^1 \to C$ is the normalization and k_p is the skyscraper at the node measuring the two preimages glued to one point. Taking cohomology and using $H^1(\mathbb{P}^1, \mathcal{O}) = 0$ gives $H^1(C, \mathcal{O}_C) \cong k$, so $p_a(C) = 1$. The geometric genus of the normalization is 0; the node contributes the extra unit of arithmetic genus, so a node identifying two points of a genus-0 curve raises the arithmetic genus by one. The dualizing sheaf has degree $\deg \omega_C = 2p_a - 2 = 0$, and concretely a generating section is the differential $\omega = dx/y$ on the smooth locus, which on the normalization $\mathbb{P}^1$ has simple poles at the two points lying over the node with opposite residues ± 1, exactly the residue condition of definition 22.2.1. Thus $\omega_C \cong \mathcal{O}_C$ is trivial of degree 0, matching $2 \cdot 1 - 2 = 0$: the nodal cubic is the $j = \infty$ degeneration of the elliptic curves, and its dualizing sheaf is the limit of the trivial canonical bundle of a smooth genus-1 curve.

The nodal cubic has $\deg \omega_C = 0$, so ω_C is not ample: a genus-1 nodal curve is not stable on its own. Stability enters precisely to select, among nodal curves of arithmetic genus $g \geq 2$, those whose dualizing sheaf is ample, and the content of the definition is that this analytic condition on ω_C has two equivalent reformulations, one group-theoretic and one purely combinatorial in the dual graph. The equivalence is what makes stability both geometrically meaningful (ampleness of ω_C is what allows the pluricanonical embedding and coarse-space construction of section 22.4) and checkable by hand (the combinatorial condition is read off the dual graph).

Definition 22.2.3: Stable Curve

A *stable curve* of genus $g \geq 2$ is a nodal curve C of arithmetic genus $p_a(C) = g$ satisfying any one of the following equivalent conditions (their equivalence is theorem 22.2.4):

1. the dualizing sheaf ω_C is ample;
2. the automorphism group $\mathrm{Aut}(C)$ is finite;
3. *(combinatorial stability)* writing g_i for the geometric genus of the normalization $\widetilde{C_i}$ of each component, every component with $g_i = 0$ (normalization a smooth rational curve $\mathbb{P}^1$) meets the rest of C in at least 3 points (nodes on that component, counting a self-node twice), and every component with $g_i = 1$ meets the rest in at least 1 point; equivalently, every component satisfies $2g_i - 2 + k_i > 0$, where k_i is the number of those points.

A *semistable curve* weakens "≥ 3" to "≥ 2" for smooth rational components and imposes no condition on the higher-genus ones, equivalently requires ω_C nef rather than ample.

The combinatorial condition is the working criterion, so it deserves a translation; throughout it, "component" is read through its normalization, so that "genus" means the geometric genus g_i and a self-node of a component shows up as two of that component's attaching points. A smooth rational component is a copy of $\mathbb{P}^1$; on its own $\mathbb{P}^1$ has the 3-dimensional automorphism group PGL_2, so to rigidify it one must pin down at least three of its points, and the only points pinned down by the rest of the curve are the nodes where it attaches. A rational component meeting the rest in only one or two points therefore carries a positive-dimensional group of automorphisms fixing those attaching points, and this group acts on the whole curve, so $\mathrm{Aut}(C)$ is infinite. A genus-1 component (geometric genus $g_i = 1$) has the 1-dimensional translation group; one attaching point suffices to kill the translations, so genus-1 components need only meet the rest once. Every component of geometric genus ≥ 2 is already rigid with finite automorphisms and imposes no attaching condition. The unstable configurations are exactly the rational components with ≤ 2 special points and the genus-1 components with 0 special points, and stability forbids them.

The equivalence of the three conditions is the technical heart of the section, and it is proved by relating all three to the degree of ω_C on each component.

Theorem 22.2.4: Ampleness, Finite Automorphisms, and Combinatorial Stability

Let C be a nodal curve of arithmetic genus $g \geq 2$. The three conditions of definition 22.2.3 are equivalent:

$$\omega_C \text{ ample} \iff \mathrm{Aut}(C) \text{ finite} \iff C \text{ combinatorially stable}$$

***Proof*:**
The argument runs through the restriction of ω_C to each component. Fix an irreducible component C_i of C with normalization $\nu_i \colon \widetilde{C_i} \to C_i$ of geometric genus g_i, and let k_i be the number of points at which C_i meets the rest of C, that is the number of nodes lying on C_i, a self-node counted twice. The starting computation is the degree of ω_C on C_i.

Step 1: the degree of ω_C on each component. The adjunction description of the dualizing sheaf gives, on the normalization $\widetilde{C_i}$,

$$\nu_i^*(\omega_C|_{C_i}) \cong \omega_{\widetilde{C_i}}(D_i)$$

where D_i is the divisor of the k_i points of $\widetilde{C_i}$ lying over the nodes on C_i. Indeed a section of ω_C near a node is a differential with simple poles of opposite residue on the two branches, so pulling back to a single branch $\widetilde{C_i}$ allows a simple pole at each of the k_i marked points, which is exactly $\omega_{\widetilde{C_i}}(D_i)$. Taking degrees,

$$\deg\left(\omega_C|_{C_i}\right) = \deg \omega_{\widetilde{C_i}} + k_i = (2g_i - 2) + k_i$$

This is the local stability number of the component C_i. A line bundle on a proper curve is ample if and only if it has positive degree on every irreducible component, so

$$\omega_C \text{ ample} \iff (2g_i - 2) + k_i > 0 \text{ for every component } C_i \qquad (*)$$

which is the bridge to the other two conditions.

Step 2: ampleness $\iff$ combinatorial stability. Read the inequality $(2g_i - 2) + k_i > 0$ component by component. If $g_i \geq 2$ then $2g_i - 2 \geq 2 > 0$ and the inequality holds with no condition on k_i. If $g_i = 1$ then $2g_i - 2 = 0$, and the inequality reads $k_i > 0$, i.e. $k_i \geq 1$: the genus-1 component must meet the rest in at least one point. If $g_i = 0$ then $2g_i - 2 = -2$, and the inequality reads $k_i > 2$, i.e. $k_i \geq 3$: the rational component must meet the rest in at least three points. These are precisely the clauses of combinatorial stability in definition 22.2.3, so $(*)$ is the statement that ω_C is ample if and only if C is combinatorially stable.

Step 3: finite automorphisms $\Rightarrow$ combinatorial stability. Argue by contrapositive: an unstable component produces infinitely many automorphisms. Suppose some component C_i violates combinatorial stability. If C_i is rational ($g_i = 0$) with $k_i \leq 2$ special points, its normalization is $\mathbb{P}^1$ with at most two marked points; the subgroup of $\mathrm{PGL}_2 = \mathrm{Aut}(\mathbb{P}^1)$ fixing those ≤ 2 points is positive-dimensional (it is the full PGL_2 for $k_i \leq 1$ and the 1-dimensional torus $\mathbb{G}_m$ of automorphisms $z \mapsto \lambda z$ fixing $0, \infty$ for $k_i = 2$). Each such automorphism fixes the attaching nodes, so it extends by the identity on the rest of C to an automorphism of C; the resulting subgroup of $\mathrm{Aut}(C)$ is positive-dimensional, hence infinite. If C_i has arithmetic genus 1 with $k_i = 0$, it is a connected component of arithmetic genus 1 meeting nothing, so $C = C_i$ has arithmetic genus $1 < 2$, contradicting $g \geq 2$; thus the genus-1 clause is the statement that a genus-1 piece must attach, and its failure again yields the translations of the elliptic component as automorphisms fixing no attaching point. In every case $\mathrm{Aut}(C)$ is infinite, so finiteness of $\mathrm{Aut}(C)$ forces combinatorial stability.

Step 4: combinatorial stability $\Rightarrow$ finite automorphisms. Suppose C is combinatorially stable, so ω_C is ample by Steps 1 and 2. An automorphism $\sigma \in \mathrm{Aut}(C)$ preserves the canonical polarization ω_C, since ω_C is intrinsic to C, so $\mathrm{Aut}(C)$ is a group of automorphisms of the polarized variety (C, ω_C). For a projective variety with an ample line bundle the automorphism group preserving the polarization is a linear algebraic group of finite type, and its identity component acts trivially on the finite set of components and nodes, hence restricts to a subgroup of $\prod_i \mathrm{Aut}(\widetilde{C}_i, D_i)$, the automorphisms of each normalized component fixing its marked points. By ampleness each factor $(\widetilde{C}_i, D_i)$ has $\deg \omega_{\widetilde{C}_i}(D_i) = (2g_i - 2) + k_i > 0$, so $\deg T_{\widetilde{C}_i}(-D_i) = -((2g_i - 2) + k_i) < 0$ and hence $H^0(\widetilde{C}_i, T_{\widetilde{C}_i}(-D_i)) = 0$: a component with an ample marked canonical bundle has no infinitesimal automorphisms fixing its marked points. The Lie algebra of the identity component of $\mathrm{Aut}(C)$ therefore vanishes, so that identity component is trivial and $\mathrm{Aut}(C)$ is a finite group.

Steps 2, 3, and 4 close the cycle: ampleness $\iff$ combinatorial stability (Step 2), combinatorial stability $\Rightarrow$ finite automorphisms (Step 4), and finite automorphisms $\Rightarrow$ combinatorial stability (Step 3), so all three conditions are equivalent, as required.

The three faces of stability serve three purposes, and it is worth seeing which does what. Ampleness of ω_C is the geometric condition: it is what lets $\omega_C^{\otimes 3}$ embed a stable curve in projective space, so that stable curves become points of a Hilbert scheme and the coarse moduli space can be cut out by geometric invariant theory (section 22.4). Finiteness of $\mathrm{Aut}(C)$ is the stack condition: it is exactly what makes $\overline{\mathcal{M}}_g$ Deligne-Mumford rather than only Artin, the direct analogue of the finiteness of $\mathrm{Aut}(C)$ for smooth curves that made $\mathcal{M}_g$ Deligne-Mumford in theorem 22.1.3. Combinatorial stability is the computational condition: it is checked by drawing the dual graph and counting attaching points, and it is what one verifies in practice. The single inequality $(2g_i - 2) + k_i > 0$ per component ties all three together, and it is the degree of ω_C on that component, so the whole theory of stability is the demand that ω_C have positive degree everywhere.

With the definition in hand, the range of stable curves is best seen through examples, from the boundary case of genus 1 with a marked point up to the maximally degenerate genus-2 curves and the unstable curves that stability rejects.

Example 22.2.5: Stable and Unstable Curves in Low Genus

The following curves exhibit the stability condition across its range.

1. *The nodal cubic as a point of $\overline{\mathcal{M}}_{1,1}$.* The nodal cubic C of example 22.2.2 has arithmetic genus 1 and $\deg \omega_C = 0$, so ω_C is not ample and C is not stable as an unpointed curve. Adjoining one marked point p at a smooth point of C changes the count: the stability of a *pointed* curve tests $\omega_C(p) = \omega_C \otimes \mathcal{O}_C(p)$, of degree $0 + 1 = 1 > 0$, which is ample. Combinatorially, the single component is rational after normalization (geometric genus 0)

and carries the self-node (2 special points) plus the marked point p, for 3 special points, meeting the ≥ 3 threshold. So the pointed nodal cubic (C, p) is stable, and it is the point of $\overline{\mathcal{M}}_{1,1}$ lying over $j = \infty$, the degeneration adjoined to $\mathcal{M}_{1,1}$ to compactify it (section 22.4).

2. *A reducible stable genus-2 curve.* Take two smooth elliptic curves E_1, E_2 (genus 1 each) and glue them by identifying a point $q_1 \in E_1$ with a point $q_2 \in E_2$, forming a single node. The result $C = E_1 \cup_q E_2$ is nodal and connected; its dual graph is two vertices of weight 1 joined by one edge. Its arithmetic genus is $g_1+g_2+(\#\text{nodes}-\#\text{components}+1) = 1+1+(1-2+1) = 2$. Each component is genus 1 and meets the other in $k_i = 1$ point, so $(2 \cdot 1 - 2) + 1 = 1 > 0$ on each: combinatorial stability holds, and C is a stable genus-2 curve. Its automorphism group is finite: the automorphisms of each (E_i, q_i) fixing the node (generically the involution $\mathbb{Z}/2$, enlarged to $\mathbb{Z}/4$ or $\mathbb{Z}/6$ when E_i has $j = 1728$ or $j = 0$), together with the swap of the two components when $E_1 \cong E_2$ via an isomorphism carrying q_1 to q_2.

3. *An irreducible stable genus-2 curve with the maximal number of nodes.* A stable genus-g curve has at most $3g - 3$ nodes, attained on the maximally degenerate curves that are the deepest boundary points of $\overline{\mathcal{M}}_g$; for $g = 2$ the maximum is $3g - 3 = 3$. Realize it by an irreducible rational curve $\widetilde{C} \cong \mathbb{P}^1$ with three pairs of points glued to three nodes, so the dual graph is one vertex of weight 0 with three loops. The arithmetic genus is $g_i + \#\text{nodes} - \#\text{components} + 1 = 0 + 3 - 1 + 1 = 3$; to land at genus 2 take instead $\mathbb{P}^1$ with *two* pairs of points glued, giving one vertex of weight 0 with two loops and arithmetic genus $0 + 2 - 1 + 1 = 2$. The single rational component carries $k_i = 4$ special points (two nodes, each counted twice as a self-node), so $(2 \cdot 0 - 2) + 4 = 2 > 0$: the curve is stable. This two-noded irreducible curve and the one-node reducible curve of part (2) are two of the boundary strata of $\overline{\mathcal{M}}_2$; the three-noded stable genus-2 curves (two rational components meeting in three points, the theta graph, or a rational curve with a node attached to a nodal cubic, the dumbbell) are the maximally degenerate points, of codimension $3 = 3g - 3$, the dimension of $\overline{\mathcal{M}}_2$ itself.

4. *An unstable curve and its stabilization.* Let D be a smooth genus-2 curve and attach a smooth rational curve $R \cong \mathbb{P}^1$ to D at a single point r, forming a node. The curve $C = D \cup_r R$ is nodal of arithmetic genus $2 + 0 + (1 - 2 + 1) = 2$, but the rational *tail* R meets the rest in only $k_R = 1$ point, so $(2 \cdot 0 - 2) + 1 = -1 < 0$: on R the sheaf ω_C has negative degree, so ω_C is not ample and C is *unstable*. Concretely $R \cong \mathbb{P}^1$ carries the $\mathbb{G}_m$ of automorphisms fixing the node r and the point at infinity of R, an infinite group of automorphisms of C, so $\mathrm{Aut}(C)$ is infinite. The *stabilization* of C contracts the offending tail: collapse R to the point r, leaving the smooth genus-2 curve D, which is stable. Contracting a rational tail with one node, or a rational *bridge* with two nodes (a $\mathbb{P}^1$ meeting the rest in exactly two points), is the operation that turns any semistable curve into a stable one without changing the arithmetic genus, and it is the second half of the stable reduction procedure of section 22.3.

The examples exhibit the full range of stability: the boundary between stable and unstable is drawn component by component by the sign of $(2g_i - 2) + k_i$, the degree of ω_C there. The reducible genus-2 curve, the irreducible multinodal one, and the pointed nodal cubic are stable because every component clears the threshold; the tail curve is unstable because one component falls below it, and stabilization deletes exactly that component. The bound of $3g - 3$ nodes on a stable genus-g curve is not accidental: each node is a codimension-1 degeneration, and $3g - 3$ is the dimension of $\overline{\mathcal{M}}_g$, so a curve with $3g - 3$ nodes is a point of the deepest, 0-dimensional, boundary stratum. This is the combinatorial form of the dimension count $\dim \mathcal{M}_g = 3g - 3$ from theorem 22.1.3, and it foreshadows the boundary stratification of section 22.3.

The stable curves are now the objects of a moduli problem, and the deformation theory of Part V makes the resulting stack smooth. The key new input is the deformation theory of a node, computed in example 17.4.2: the node $k[x,y]/(xy)$ has a one-dimensional space of first-order deformations, spanned by the versal smoothing $xy = t$, which resolves the node into a smooth curve as t moves off 0. Each node thus contributes one deformation parameter that smooths it, and this parameter is transverse to the locus where the node persists, which is why the boundary is a divisor and why passing to stable curves does not destroy smoothness.

Theorem 22.2.6: $\overline{\mathcal{M}}_g$ is a Smooth Deligne-Mumford Stack

For $g \geq 2$, the moduli stack $\overline{\mathcal{M}}_g$ of stable genus-g curves is a smooth Deligne-Mumford stack over k of dimension

$$\dim \overline{\mathcal{M}}_g = 3g - 3$$

containing $\mathcal{M}_g$ as a dense open substack, with complement the boundary $\partial\overline{\mathcal{M}}_g = \overline{\mathcal{M}}_g \setminus \mathcal{M}_g$ of curves with at least one node. Its properness is established in section 22.3.

***Proof*:**
A *family of stable curves* over S is a proper flat morphism $\pi\colon \mathcal{C} \to S$ whose geometric fibers are stable genus-g curves, and $\overline{\mathcal{M}}_g$ is the category fibered in groupoids over $(\mathbf{Sch}/k)$ whose objects over S are such families and whose morphisms are the cartesian squares, exactly as for $\mathcal{M}_g$ in definition 22.1.1 with "smooth" relaxed to "at worst nodal." The proof establishes in turn that it is algebraic and Deligne-Mumford, that it is smooth of dimension $3g-3$, and that $\mathcal{M}_g$ is a dense open substack.

Step 1: algebraic and Deligne-Mumford. That $\overline{\mathcal{M}}_g$ is an algebraic stack follows as in theorem 22.1.3: families of stable curves descend along fppf covers once a relative polarization is fixed, and the relative pluricanonical bundle $\omega_{\mathcal{C}/S}^{\otimes 3}$ supplies one, since ω_C is ample of degree $2g-2$ on each stable fiber (Step 1 of theorem 22.2.4), so $\omega_C^{\otimes 3}$ has degree $6g-6 \geq 2g+1$ for $g \geq 2$; the degree bound is the smooth-curve heuristic, and on a nodal stable curve the very ampleness of $\omega_C^{\otimes 3}$ is the classical tricanonical embedding of stable curves, which embeds each stable curve in a fixed projective space. Artin's criteria are verified as for $\mathcal{M}_g$, with the deformation theory of stable curves in place of that of smooth curves. It is Deligne-Mumford by theorem 20.2.3: the diagonal is unramified because every stable curve has finite reduced automorphism group. Finiteness is definition 22.2.3(2), part of the definition of stability; reducedness holds because the tangent space to $\underline{\mathrm{Aut}}(C)$ at the identity is the space of infinitesimal automorphisms, which vanishes by the computation $H^0(\widetilde{C}_i, T_{\widetilde{C}_i}(-D_i)) = 0$ on each component from Step 4 of theorem 22.2.4. So $\underline{\mathrm{Aut}}(C)$ is finite and reduced, the diagonal is unramified, and $\overline{\mathcal{M}}_g$ is Deligne-Mumford.

Step 2: smooth of dimension $3g-3$. By the deformation-theoretic criterion, smoothness of the Deligne-Mumford stack is the unobstructedness of the deformation functor of every stable curve together with a constant tangent-space dimension. A stable curve C is a nodal curve, and its deformation theory splits into deformations that preserve the nodes and deformations that smooth them. The deformations preserving the singularities are governed, as for smooth curves, by the cohomology of the tangent complex, and the local contribution of each node is the versal smoothing (example 17.4.2): the node $k[x,y]/(xy)$ has $T^1 = k$, spanned by the deformation

$xy = t$. The full first-order deformation space of C fits in the local-to-global sequence

$$0 \to H^1(C, T_C) \to \mathrm{Def}^1_C \to H^0(C, \mathcal{T}^1_C) \to 0$$

where $\mathcal{T}^1_C$ is the sheaf of local first-order deformations, supported at the nodes with stalk k at each node (its global sections count the smoothing parameters, one per node), and $H^1(C, T_C)$ classifies the locally trivial deformations that move the components and the node positions while keeping the nodes. Curves are unobstructed by proposition 17.4.1: the obstructions lie in the degree-2 term Def^2_C, which vanishes because C is a curve, of cohomological dimension 1, so both the global term $H^2(C, T_C)$ and the higher local terms vanish. Hence every first-order deformation of C extends and the deformation functor is formally smooth. Its tangent space then has dimension

$$\dim \mathrm{Def}^1_C = \dim H^1(C, T_C) + \#\{\text{nodes}\}$$

from the local-to-global sequence, and the two terms recombine to a constant. Smoothing a node trades one node-preserving parameter for one smoothing parameter: near a boundary point with δ nodes, C is the limit of the smooth curves obtained by the δ versal smoothings, so $\dim H^1(C, T_C) = (3g-3) - \delta$ node-preserving directions and the δ smoothing directions add to $3g-3$, each smoothing parameter being transverse to the boundary divisor along which its node persists. The total is thus $3g-3$, the same as in the smooth case $\delta = 0$, and constant across the boundary. So $\overline{\mathcal{M}}_g$ is a formally smooth Deligne-Mumford stack with constant tangent dimension $3g-3$, hence smooth of dimension $3g-3$.

Step 3: $\mathcal{M}_g$ is a dense open substack. A smooth curve is a stable curve with no nodes, so $\mathcal{M}_g$ is the substack of $\overline{\mathcal{M}}_g$ where the fibers are smooth, which is the locus where the discriminant of π is invertible: an open condition, so $\mathcal{M}_g \hookrightarrow \overline{\mathcal{M}}_g$ is an open immersion. It is dense because the smoothing parameter of Step 2 exhibits every boundary point as a limit of smooth curves: given a stable curve C_0 with a node, the versal smoothing $xy = t$ deforms it to a smooth curve for $t \neq 0$, so a punctured neighborhood of $[C_0]$ in $\overline{\mathcal{M}}_g$ lies in $\mathcal{M}_g$. Thus $\mathcal{M}_g$ meets every open set and is dense, with complement the boundary $\partial\overline{\mathcal{M}}_g$ of curves carrying at least one node. This is the open immersion and the density, as required.

The theorem is the structural payoff of the compactification, and the way its proof reuses the earlier chapters is the point. Smoothness comes entirely from Part V: the unobstructedness of curve deformations (proposition 17.4.1) and the node's one-dimensional smoothing space (example 17.4.2) are the only inputs, and they combine so that $\dim = 3g-3$ holds uniformly across the boundary, with each node's smoothing parameter transverse to the boundary divisor it crosses. The Deligne-Mumford property comes from stability itself, the finiteness of $\mathrm{Aut}(C)$ being condition (2) of the definition, so that the compactified stack is Deligne-Mumford for the same reason $\mathcal{M}_g$ was. The one property not yet in hand is properness: nothing above shows that a family of stable curves over a punctured disc extends to the whole disc, and that is exactly the content of stable reduction, deferred to section 22.3.

The smoothness of $\overline{\mathcal{M}}_g$ as a stack should be held against the singularity of its coarse space. The coarse moduli space $\overline{M}_g$ (section 22.4) is singular in general, its singularities coming from the curves with extra automorphisms whose stabilizers act nontrivially on the deformation space; the stack $\overline{\mathcal{M}}_g$ resolves them by remembering the automorphism groups, and it is smooth precisely because the deformation space $\mathrm{Def}^1_C = k^{3g-3}$ is a smooth chart at every point, before the automorphism group is quotiented out. This is the sharpest form of the principle that a stack is the right object: the stack is smooth where its coarse space is singular, and the singularity of the coarse space comes from the automorphism groups the stack keeps and the coarse space forgets. It is the same phenomenon that made $\mathcal{M}_{1,1}$ a smooth stack with a stacky point at $j = 0, 1728$ while its coarse space, the smooth j-line, hid the μ_4, μ_6 symmetry (example 13.3.4); the boundary of $\overline{\mathcal{M}}_g$ extends the same picture to the nodal curves.

Exercise 22.2.1

1. Verify the degree formula $\deg \omega_C = 2p_a(C) - 2$ for the reducible stable genus-2 curve $C = E_1 \cup_q E_2$ of example 22.2.5(2), by computing $\deg(\omega_C|_{E_i})$ on each elliptic component from the adjunction formula $\deg(\omega_C|_{C_i}) = (2g_i - 2) + k_i$ and summing. Confirm that ω_C has positive degree on each component, so ω_C is ample.

2. Show that a stable curve of genus g has at most $3g - 3$ nodes. (Use the degree bound: each node contributes to the special-point count k_i on its components, and combinatorial stability forces $(2g_i - 2) + k_i \geq 1$; sum $\deg \omega_C = 2g - 2$ over components and bound the number of nodes.) For $g = 2$, exhibit the dual graph of a stable curve attaining the maximum $3g - 3 = 3$.

3. Let C be a nodal curve consisting of a smooth rational curve $R \cong \mathbb{P}^1$ meeting a smooth genus-g curve D (with $g \geq 2$) at exactly two points, forming two nodes (a rational *bridge*). Compute the arithmetic genus of C, decide whether C is stable, and describe its stabilization. Contrast with the rational *tail* of example 22.2.5(4).

4. Using the local-to-global sequence $0 \to H^1(C, T_C) \to \mathrm{Def}^1_C \to H^0(C, \mathcal{T}^1_C) \to 0$ from the proof of theorem 22.2.6, compute $\dim \mathrm{Def}^1_C$ for the one-node reducible curve $E_1 \cup_q E_2$ of example 22.2.5(2). Confirm the total is $3g - 3 = 3$, and identify how many of the three parameters are node-preserving and how many smooth the node.

5. Explain why the nodal cubic (example 22.2.2) is not a stable curve of genus 1 on its own, but becomes stable once a smooth marked point is added, by comparing $\deg \omega_C$ with $\deg \omega_C(p)$ and checking the combinatorial condition for the pointed curve. Why does the genus-1 case require a marked point where the genus-g ($g \geq 2$) case does not?

6. Show directly that the automorphism group of the rational-tail curve $C = D \cup_r R$ of example 22.2.5(4) is infinite, by exhibiting a one-parameter family of automorphisms of C supported on the tail R. Explain how this failure of finiteness matches the failure of ampleness of ω_C on R, illustrating the equivalence of theorem 22.2.4.

22.3 Stable Reduction and Properness

The two previous sections built $\overline{\mathcal{M}}_g$ and proved it a smooth Deligne–Mumford stack of dimension $3g - 3$ containing $\mathcal{M}_g$ as a dense open (theorem 22.2.6), with the properness clause of that theorem deferred to this section. What has not yet been shown is the property that makes the compactification worth the name: that $\overline{\mathcal{M}}_g$ is *proper*. Properness is the algebraic incarnation of compactness, and for a moduli space it has a concrete meaning. A one-parameter degeneration of smooth curves, a family over a punctured disc whose fibers are smooth genus-g curves, ought to have a well-defined limit inside the compactified moduli space, and that limit ought to be unique. This section proves exactly that. The existence of the limit is the *stable reduction theorem*, and its uniqueness is the separatedness of the stack; together they are the two halves of the valuative criterion for stacks (definition 20.5.5), and together they say $\overline{\mathcal{M}}_g$ is proper.

The section then reads off the geometry of the compactification. The complement $\partial\overline{\mathcal{M}}_g = \overline{\mathcal{M}}_g \setminus \mathcal{M}_g$, the locus of singular stable curves, is a divisor whose components are indexed by the ways a stable curve of genus g can degenerate along a single node. Its deeper strata are indexed by dual graphs, with codimension equal to the number of nodes. The local model of the whole picture is the node smoothing $xy = t$ of Part V (example 17.4.2): each node contributes one boundary-transverse coordinate, so the boundary is normal crossings in the stack. This is the payoff promised at the start of the chapter. As a stack $\overline{\mathcal{M}}_g$ is smooth and proper, its boundary a normal-crossings divisor, even though, as the next section shows, its coarse space $\overline{M}_g$ acquires singularities.

The central input is a theorem whose statement is elementary and whose proof draws on the semistable reduction of curves, a result at the boundary of what a first course develops. The setup is a family of smooth curves parametrized by a punctured disc, or algebraically over the punctured spectrum of a discrete valuation ring, together with the question of what happens as the parameter approaches the puncture. Without allowing singular curves the limit need not exist: a family of smooth plane cubics whose j-invariant runs off to infinity has no smooth limit, because the limiting curve is nodal. Deligne and Mumford's insight was that admitting stable curves, and permitting a finite base change to absorb monodromy, produces a limit that always exists and is unique.

Theorem 22.3.1: Stable Reduction

Let R be a discrete valuation ring with fraction field K and residue field k, and write $\eta = \operatorname{Spec} K$ for the generic point and $0 = \operatorname{Spec} k$ for the closed point of $S = \operatorname{Spec} R$. Let $C_\eta \to \eta$ be a smooth proper geometrically connected curve of genus $g \geq 2$ over K, or more generally a stable curve of genus g over K. Then there exists a finite extension $R \subseteq R'$ of discrete valuation rings, with fraction field K' a finite extension of K, such that the base-changed curve $C_{\eta'} = C_\eta \times_K K'$ extends to a family of *stable* curves

$$\mathcal{C} \longrightarrow S' = \operatorname{Spec} R'$$

whose generic fiber is $C_{\eta'}$ and whose special fiber $\mathcal{C}_0$ is a stable curve of genus g over the residue field k' of R'. The stable extension is unique: any two stable families over S' (after possibly enlarging R' further) with the same generic fiber are canonically isomorphic.

Geometrically the theorem says that a family of smooth curves over a punctured disc, however wildly it degenerates at the origin, can be completed to a family over the whole disc whose central fiber is a stable curve, provided one is willing to pass to a finite cover of the base disc first. The finite base change is not a defect of the method but a necessity: it is the algebraic reflection of the monodromy of the family around the puncture, and a multivalued limit becomes single-valued only after passing to a cover on which the monodromy is unipotent. The uniqueness clause is what pins down the limit and makes $\overline{\mathcal{M}}_g$ separated; it is the statement that a smooth family has at most one stable degeneration.

***Proof Key ideas*:**
The full proof rests on the semistable reduction theorem for curves, whose complete argument uses resolution of singularities of arithmetic surfaces and the structure theory of the minimal regular model, tools beyond the scope of this book. The complete proofs are in [23] (the original, via the reduction to the case of a curve with a level structure and the Néron model of its Jacobian) and in [32] (a modern exposition organized around the two steps below). The strategy is a two-step construction, and each step is instructive in itself.

Step 1: Semistable reduction. After a finite base change $R \subseteq R'$, the curve $C_{\eta'}$ extends to a flat proper family $\mathcal{X} \to S'$ whose total space $\mathcal{X}$ is a regular scheme and whose special fiber $\mathcal{X}_0$ is a *semistable* curve: a reduced nodal curve, but not necessarily one with ample dualizing sheaf. The base change is where monodromy is absorbed. The family C_η over the punctured disc carries a monodromy representation on the first cohomology of its fibers, and by the local monodromy theorem this monodromy is quasi-unipotent; passing to the finite cover $S' \to S$ on which it becomes unipotent is precisely what allows the total space to be resolved to a family with reduced nodal special fiber. Concretely one takes any flat proper extension of C_η over S and resolves the singularities of its total space, producing a regular arithmetic surface whose special fiber is a divisor with normal crossings. One defect can remain: the special fiber may be *nonreduced*, its components carrying multiplicities, with local model

$$x^a y^b = t$$

in $R[x, y]$ (a regular total space, since t is expressed by x, y) at a crossing of components of multiplicities a and b, whose special fiber $x^a y^b = 0$ is the node taken with those multiplicities. The finite base change is what clears the multiplicities. Passing to the cover $t = (t')^n$, where n is chosen so that the monodromy, quasi-unipotent by the local monodromy theorem, becomes unipotent, turns this local model into $x^a y^b = (t')^n$, a surface singularity of type A; taking its minimal resolution replaces the point by a chain of rational curves and yields a regular total space whose special fiber is now reduced and nodal, with each surviving node in the versal-smoothing model

$$xy = t'$$

the node from Part V (example 17.4.2). This is why the node is the local model of the entire compactification.

Step 2: Contraction to the stable model. The semistable special fiber $\mathcal{X}_0$ may fail to be stable in exactly one way: it may contain *unstable components*, smooth rational curves meeting the rest of $\mathcal{X}_0$ in only one or two points, on which the dualizing sheaf has non-positive degree (definition 22.2.3). These are the chains and tails of $\mathbb{P}^1$'s introduced by the resolution and base change. Each such component is contracted: a (-1)-curve or a chain of rational components is blown down within the family, replacing it by a smooth point or a node. The contraction is carried out on the whole family $\mathcal{X} \to S'$, not just the special fiber, and it changes nothing over the generic point η' (where the fiber is already smooth and has no such components). After finitely many contractions no unstable component remains, and the resulting special fiber has ample dualizing sheaf: it is a stable curve. On the contracted family $\mathcal{C} \to S'$ the relative dualizing sheaf $\omega_{\mathcal{C}/S'}$ is relatively ample, so $\mathcal{C} \to S'$ is a family of stable curves.

Uniqueness. Two stable families over S' with isomorphic generic fibers are isomorphic. The stable model is the relative canonical model of the family: it is Proj of the sheaf of algebras $\bigoplus_{n \geq 0} \pi_* \omega_{\mathcal{C}/S'}^{\otimes n}$, and this construction depends only on the family away from a codimension-two locus, hence only on the generic fiber together with the requirement of stability. Since the canonical model is unique, so is the stable limit. This is the clause that makes the limiting stable curve well-defined, completing the key ideas of the proof.

The theorem is the geometric engine of the compactification, and its structure deserves a closer look. The finite base change and the uniqueness clause are the two features that make the limit both exist and be unique, and they correspond precisely to the two halves of the valuative criterion below. That the intermediate semistable model is not unique, while the final stable model is, is the reason

$\overline{\mathcal{M}}_g$ rather than a moduli space of semistable curves is the correct compactification: only the stable condition rigidifies the limit. The reader should hold onto the node model $xy = t'$ of Step 1, since it is the shape every boundary point takes locally, and the transversality of its smoothing parameter t' is what makes the boundary a divisor rather than a higher-codimension locus.

Before the concrete degenerations, the passage from stable reduction to properness deserves to be spelled out, because it is the whole reason the theorem was proved. Properness for a Deligne–Mumford stack is checked, as for a scheme, by a valuative criterion: a map from the punctured spectrum of a valuation ring into the stack must extend across the puncture, uniquely up to unique isomorphism, after possibly a finite base change on the source. For a moduli stack a map from $\eta = \operatorname{Spec} K$ is exactly a family of objects over K, and the extension across $0 = \operatorname{Spec} k$ is exactly a limiting object. Stable reduction is this valuative criterion, read for the moduli of stable curves.

Theorem 22.3.2: $\overline{\mathcal{M}}_g$ is Proper and Separated

For $g \geq 2$ the Deligne–Mumford stack $\overline{\mathcal{M}}_g$ is proper and separated over the base field k. Consequently $\overline{\mathcal{M}}_g$ is a smooth proper Deligne–Mumford stack of dimension $3g - 3$, and $\mathcal{M}_g$ is an open (but not proper) substack.

***Proof*:**
Properness of a Deligne–Mumford stack of finite type over k is the conjunction of three properties: finite type, separatedness, and the existence half of the valuative criterion (universal closedness). These are the conditions packaged in the definition of a proper morphism of stacks (definition 20.5.5). Each is established from what the chapter has built.

Step 1: Finite type. The stack $\overline{\mathcal{M}}_g$ is of finite type over k. Every stable curve of genus g has ω_C ample (definition 22.2.3), and its tricanonical embedding by $\omega_C^{\otimes 3}$ realizes it as a curve of fixed degree $3(2g-2)$ and fixed Hilbert polynomial in a fixed projective space $\mathbb{P}^{5g-6}$. The stable curves therefore form a bounded family: they are parametrized by a locally closed subscheme of a single Hilbert scheme (definition 14.1.6), whose quotient by the projective linear group presents $\overline{\mathcal{M}}_g$. Boundedness of this Hilbert-scheme locus gives finite type. This is the same boundedness that will produce the coarse projective space in the next section.

Step 2: Separatedness (uniqueness of limits). Separatedness of a Deligne–Mumford stack is checked by the valuative criterion in its uniqueness form: for a discrete valuation ring R with fraction field K, any two extensions to $\operatorname{Spec} R$ of a given object over $\operatorname{Spec} K$ are canonically isomorphic. For $\overline{\mathcal{M}}_g$ an object over $\operatorname{Spec} R$ is a family of stable curves $\mathcal{C} \to \operatorname{Spec} R$, and an object over $\operatorname{Spec} K$ is its generic fiber. The uniqueness clause of theorem 22.3.1 is exactly this: two families of stable curves over $\operatorname{Spec} R$ with isomorphic generic fibers are isomorphic. Since the diagonal of a Deligne–Mumford stack is unramified and the automorphism groups of stable curves are finite (theorem 22.2.6), the isomorphism is unique, and $\overline{\mathcal{M}}_g$ is separated.

Step 3: Universal closedness (existence of limits). The existence half of the valuative criterion requires that *every* object over $\operatorname{Spec} K$ extend, after a finite base change $K \subseteq K'$, to an object over $\operatorname{Spec} R'$ for the corresponding valuation ring R'. An object over $\operatorname{Spec} K$ is a stable curve C_η over K; since a $\operatorname{Spec} K \to \overline{\mathcal{M}}_g$ can land in the boundary as well as in the open $\mathcal{M}_g$, the curve C_η may be smooth or already singular. Both are covered by theorem 22.3.1 in the generality stated there: a stable curve over K, smooth or nodal, acquires after a finite base change $R \subseteq R'$ a family of stable curves $\mathcal{C} \to \operatorname{Spec} R'$ with generic fiber $C_{\eta'}$.

When C_η is smooth this is the principal case of the theorem, the finite base change absorbing the monodromy of a smooth degeneration. When C_η is already a nodal stable curve there is no smooth-locus monodromy to absorb, but a base change may still be forced, now by monodromy permuting the components and nodes of C_η or acting on the Jacobians of the components; in either event the theorem supplies a stable limit. So every object over $\operatorname{Spec} K$ extends after the permitted finite base change. That base change is precisely the latitude the valuative criterion for stacks allows, and which distinguishes properness of a stack from properness of a scheme. Thus $\overline{\mathcal{M}}_g$ satisfies the existence half of the criterion and is universally closed.

Combining the three steps, $\overline{\mathcal{M}}_g$ is of finite type, separated, and universally closed over k, hence proper. With the smoothness and the open embedding of $\mathcal{M}_g$ from theorem 22.2.6, it is a smooth proper Deligne–Mumford stack of dimension $3g-3$, and $\mathcal{M}_g$ is open but omits the boundary, hence is not itself proper, as required.

This is the payoff the chapter was aimed at. The moduli of smooth curves $\mathcal{M}_g$ is not compact, for the reason any space of nondegenerate objects fails to be compact: degenerations escape it. Enlarging to $\overline{\mathcal{M}}_g$ by admitting stable curves plugs every such escape, and the plug is exactly a stable limit supplied by theorem 22.3.1. The finite base change in the existence step is where the stack framework proves its worth. A map into a scheme from a punctured disc extends without base change or not at all; a map into a Deligne–Mumford stack is allowed a finite cover of the source, and that latitude is exactly what a family with nontrivial monodromy requires. A properness that is invisible at the level of the coarse space, which sees only the underlying point set, is manifest at the level of the stack, which sees the automorphisms and the base changes they permit.

With properness in hand the geometry of the added locus can be described. The boundary $\partial\overline{\mathcal{M}}_g$ is the complement of the smooth locus, the closed substack parametrizing singular stable curves. Its structure is dictated by the possible degenerations: a generic point of the boundary parametrizes a stable curve with a single node, and there are exactly two ways a single node can arise. Either the node is non-disconnecting, so that the normalization of the curve is connected of genus $g-1$, or it is disconnecting, splitting the curve into two components of genera i and $g-i$. This dichotomy is the origin of the boundary divisors.

Definition 22.3.3: The Boundary and its Stratification

The *boundary* of $\overline{\mathcal{M}}_g$ is the reduced closed substack

$$\partial\overline{\mathcal{M}}_g = \overline{\mathcal{M}}_g \setminus \mathcal{M}_g$$

parametrizing stable curves that are singular. It is a divisor (pure codimension one), with irreducible components the *boundary divisors*:

1. Δ_0, the closure of the locus of irreducible stable curves with a single non-disconnecting node, whose normalization is a smooth connected curve of genus $g-1$; and
2. Δ_i for $1 \le i \le \lfloor g/2 \rfloor$, the closure of the locus of stable curves with a single disconnecting node splitting the curve into a component of genus i and a component of genus $g-i$ (with $\Delta_i = \Delta_{g-i}$, whence the range).

The whole boundary is stratified by *topological type*: to each stable curve C associate its *dual graph* Γ_C, the graph with one vertex of weight p_g for each irreducible component of geometric genus p_g and one edge for each node. The locally closed stratum $\mathcal{M}_\Gamma \subseteq \overline{\mathcal{M}}_g$ of curves with a fixed dual graph Γ has

$$\operatorname{codim} \mathcal{M}_\Gamma = \#\{\text{edges of } \Gamma\} = \#\{\text{nodes}\}$$

and its closure is the union of the strata whose graphs are obtained from Γ by smoothing edges. The boundary $\partial\overline{\mathcal{M}}_g$ is a *normal-crossings divisor* in the stack: at a curve with δ nodes, the δ smoothing parameters are independent local coordinates cutting out the δ branches of the boundary transversally.

The stratification is the combinatorial skeleton of the compactification, and the local deformation theory of Part V is what makes it precise. The tangent space to $\overline{\mathcal{M}}_g$ at a stable curve C with δ nodes decomposes: the deformations of C that preserve all δ nodes form a subspace of codimension δ, and the remaining δ directions each smooth one node independently, along the versal parameter t of the node model $xy = t$ (example 17.4.2). Each node thus contributes exactly one transverse coordinate to the boundary, which is why the number of nodes equals the codimension and why the δ boundary branches cross normally. The deepest strata, the curves with the maximal number $3g-3$ of nodes, are the 0-dimensional strata: totally degenerate stable curves whose components are all rational and whose dual graph is trivalent with $3g-3$ edges.

The dichotomy Δ_0 versus Δ_i is the first thing to compute, and it is entirely combinatorial. Smoothing the single node of a curve in Δ_0 raises the geometric genus from $g-1$ to g while keeping the curve connected, so Δ_0 has codimension one and its generic curve is irreducible. Smoothing the node of a curve in Δ_i merges the two components into one connected curve of genus $i + (g-i) = g$, again codimension one; here the generic curve is reducible with two components. The number of boundary divisors is therefore $1 + \lfloor g/2 \rfloor$: one Δ_0 and one Δ_i for each i from 1 to $\lfloor g/2 \rfloor$.

Example 22.3.4: The Boundary Divisors of $\overline{\mathcal{M}}_2$

Take $g = 2$, so $\dim \overline{\mathcal{M}}_2 = 3g-3 = 3$. The boundary divisors are indexed by Δ_0 and Δ_i for $1 \le i \le \lfloor 2/2 \rfloor = 1$, namely Δ_0 and Δ_1, two divisors of dimension 2.

1. Δ_0: an irreducible curve of arithmetic genus 2 with one non-disconnecting node, whose

normalization is a smooth genus-1 curve with two marked points (the two preimages of the node) glued. Its generic point is stable because the nodal curve has finite automorphism group: an automorphism restricts to one of the genus-1 normalization preserving the two-point preimage set, and only finitely many translations fix that pair, together with the involution swapping the two branches at the node. Smoothing the node yields a smooth genus-2 curve, so Δ_0 is a divisor.

2. Δ_1: two smooth genus-1 curves E_1, E_2 joined at a single disconnecting node (one point of each glued). This is stable exactly because each genus-1 component carries one node, meeting the rest in one point, which is the combinatorial threshold for a genus-1 component (definition 22.2.3). Its moduli within Δ_1 are the j-invariants of E_1 and E_2; the choice of gluing point on each component contributes nothing, since the translation automorphisms of an elliptic curve act transitively on its points, so the two j-invariants alone give $\dim \Delta_1 = 2$.

The dual graph of a generic Δ_0-curve is a single vertex of weight 1 with one loop; that of a generic Δ_1-curve is two vertices of weight 1 joined by one edge. The deepest stratum of $\overline{\mathcal{M}}_2$ is 0-dimensional, reached by curves with $3g - 3 = 3$ nodes: for instance two rational components ($\mathbb{P}^1$'s) joined at three nodes (the dual graph is the "theta graph" with two trivalent vertices and three edges), which is stable since each rational component meets the rest in three points. The codimensions line up: 3 nodes, codimension 3, a point in the 3-dimensional $\overline{\mathcal{M}}_2$.

The example makes the transversality concrete: at the two-node and three-node strata one literally counts nodes and reads off codimension. The remaining task is to see stable reduction in action on a family, so that the abstract limit of theorem 22.3.1 becomes a computation one can watch happen. The cleanest case is the one that motivated the whole story, a family of elliptic curves whose j-invariant degenerates. The genus is 1 rather than ≥ 2, so the relevant stack is $\overline{\mathcal{M}}_{1,1}$ with its marked point, but the mechanism, semistable reduction followed by contraction, is identical and lays bare the base change.

Example 22.3.5: Stable Reduction of a Degenerating Family

The Weierstrass family with $j \to \infty$. Over the punctured disc $\operatorname{Spec} R$ with $R = k[[t]]$, $K = k((t))$, take the Weierstrass family

$$E_\eta : \quad y^2 = x^3 + x^2 + t$$

over K. For $t \neq 0$ this is a smooth elliptic curve, with discriminant $-64t - 432t^2 = -64t(1 + \frac{27}{4}t)$, a unit times t on the nose, so $\operatorname{ord}_t \Delta = 1$ and the j-invariant satisfies $j(E_\eta) \sim c/t \to \infty$ as $t \to 0$. Setting $t = 0$ naively gives $y^2 = x^3 + x^2 = x^2(x+1)$, a nodal cubic with a node at the origin: the Weierstrass model already extends to a family whose special fiber is nodal, and the total space $y^2 = x^3 + x^2 + t$ is regular at that node (its equation has nonvanishing differential). The special fiber is the nodal cubic, an irreducible stable curve of arithmetic genus 1 (stable as a point of $\overline{\mathcal{M}}_{1,1}$, carrying the marked point at infinity). No base change is needed here because the monodromy of this degeneration is already unipotent, a single Dehn twist about the vanishing cycle; the limit is the nodal cubic, the point $j = \infty$ of the coarse space $\mathbb{P}^1 = \overline{M}_{1,1}$. This is the algebraic form of the Tate curve, whose analytic uniformization $E_\eta \cong \mathbb{G}_m / q^{\mathbb{Z}}$ with $q = q(t) \to 0$ degenerates to $\mathbb{G}_m$ with 0 and ∞ glued, again the nodal cubic; see [11] for the Tate parametrization and the q-expansion of j.

When a base change *is* forced, the mechanism appears in full. Take the family

$$E_\eta : \quad y^2 = x^3 + t$$

over $K = k((t))$ with $\operatorname{char} k \neq 2, 3$, whose discriminant is $-432t^2$ and whose j-invariant is identically 0: this is a family of curves each with extra automorphisms, degenerating with additive reduction. The naive limit $y^2 = x^3$ is a cuspidal cubic, which is *not* nodal, hence not semistable. Semistable reduction requires the base change $t = (t')^6$: over $R' = k[[t']]$ the substitution $x = (t')^2u$, $y = (t')^3v$ transforms the equation into $v^2 = u^3 + 1$, a smooth elliptic curve over k with good reduction. The finite base change of degree 6 is exactly the order of the automorphism group μ_6 that obstructed the naive limit; it absorbs the monodromy and produces a smooth (in particular stable) special fiber. This is the base change of theorem 22.3.1 made explicit.

A pencil of genus-2 curves. For $g = 2$, consider a pencil of smooth genus-2 curves specializing to a curve of the shape in example 22.3.4(2): two elliptic curves joined at a node. A genus-2 curve is hyperelliptic, $y^2 = f_6(x)$ with $\deg f_6 = 6$, so it is the double cover of $\mathbb{P}^1$ branched at the six roots of f_6. The disconnecting node of Δ_1 arises when the six branch points do not collide in a single pair but split into *two triples clustered over two distinct values* of x: the base $\mathbb{P}^1$ pinches into two $\mathbb{P}^1$'s joined at a node, and each carries three of the branch points *together with the node itself as a fourth.* A double cover of $\mathbb{P}^1$ must be branched over an even number of points (Riemann–Hurwitz), so three clustered branch points cannot bound a component on their own; the node is forced to be the fourth branch point, and the cover over each $\mathbb{P}^1$, branched at four points, is a genus-1 curve. Over $R = k[[t]]$ with $\operatorname{char} k \neq 2, 3$, take

$$y^2 = \big((x-1)^3 - t\big)\big((x+1)^3 - t\big)$$

For $t \neq 0$ the equation $(x-1)^3 = t$ has three distinct roots clustered near $x = 1$ and $(x+1)^3 = t$ three distinct roots near $x = -1$, six simple branch points in all, so the generic fiber is a smooth genus-2 curve. As $t \to 0$ the two triples collapse onto the distinct values $x = 1$ and $x = -1$, and the naive fiber $y^2 = (x-1)^3(x+1)^3$ is non-nodal at both points. This is where the base change of theorem 22.3.1 is forced, exactly as in the additive case above: after $t = (t')^6$ the substitution $x = 1 + (t')^2u$ turns the local factor $(x-1)^3 - t$ into $(t')^6(u^3 - 1)$, resolving the collision over $x = 1$ into a smooth elliptic tail $v^2 = (u^3 - 1)h(u)$, and symmetrically over $x = -1$. Each tail is elliptic precisely because the local cubic $v^2 = u^3 - 1$ has odd degree in u, hence is ramified at $u = \infty$, the point toward the neck, which supplies the fourth branch point counted above. The cover is therefore *ramified* over the neck of the base, not unramified. Because a single ramification point of a double cover has one preimage, the semistable model that separates the two tails carries a rational bridge over the neck of the base $\mathbb{P}^1$, a component meeting each tail in one point; this neck $\mathbb{P}^1$ is an unstable rational piece (two special points, non-positive dualizing degree), and contracting it, the contraction step of theorem 22.3.1, fuses the two tails at a single disconnecting node. The stable limit is therefore two genus-1 curves meeting at a node, a point of $\Delta_1 \subseteq \overline{\mathcal{M}}_2$. It is unique by theorem 22.3.1, and it lands in the boundary divisor Δ_1 described in example 22.3.4: the degeneration exhibits a boundary point of $\overline{\mathcal{M}}_2$ as the limit of smooth genus-2 curves.

The two elliptic examples show the two regimes cleanly: multiplicative degeneration ($j \to \infty$) needs no base change and lands on Δ_0-type boundary, the nodal curve; additive degeneration ($j = 0$ with a cusp) needs a base change, whose degree is the order of the extra automorphisms, and after it lands back in the interior with good reduction. The genus-2 pencil shows the disconnecting degeneration landing on Δ_1. Every stable degeneration one meets is a combination of these: a base change to kill monodromy, a semistable model, a contraction of unstable rational pieces, and a stable limit sitting in a boundary stratum indexed by the dual graph of the limit. This is the concrete content of the properness proved in theorem 22.3.2.

Exercise 22.3.1

1. State the valuative criterion for properness of a Deligne–Mumford stack (definition 20.5.5), separating its existence and uniqueness halves. Explain, in one sentence each, which half is supplied by the existence clause of theorem 22.3.1 and which by its uniqueness clause, and why the finite base change permitted in the criterion is essential for a stack but has no analogue in the criterion for a scheme.

2. For $g = 3$, list all boundary divisors of $\overline{\mathcal{M}}_3$ and give the dual graph and generic geometric type of each. What is $\dim \overline{\mathcal{M}}_3$, and what is the dimension of each boundary divisor?

3. Consider the family $E_\eta \colon y^2 = x^3 + t^2 x$ over $K = k((t))$ with $\operatorname{char} k \neq 2, 3$. Compute its j-invariant, determine whether the naive limit at $t = 0$ is semistable, and find the smallest base change $t = (t')^m$ after which the family acquires good (or at least semistable) reduction. Interpret m in terms of the automorphism group of the generic fiber.

4. Let C be a stable curve of genus g whose dual graph Γ has v vertices of geometric genera $p_1, \ldots, p_v$ and e edges. Prove the genus formula

$$g = \sum_{i=1}^{v} p_i + e - v + 1$$

   and use it to show that a totally degenerate stable curve (all components rational) lies in a 0-dimensional stratum of $\overline{\mathcal{M}}_g$, with dual graph a trivalent graph having $3g - 3$ edges and $2g - 2$ vertices.

5. Explain why $\mathcal{M}_g$ is *not* proper, by exhibiting a one-parameter family of smooth genus-g curves with no limit inside $\mathcal{M}_g$, and identify where its stable limit sits in $\overline{\mathcal{M}}_g$. Contrast this with the statement of theorem 22.3.2.

6. The boundary $\partial\overline{\mathcal{M}}_g$ is a normal-crossings divisor in the stack, yet its image in the coarse space $\overline{M}_g$ need not be. Using the node model $xy = t$ (example 17.4.2) and the fact that a stable curve with δ nodes has δ independent smoothing parameters, explain why the δ branches of the boundary cross transversally in the stack. Why does this transversality account for $\operatorname{codim} \mathcal{M}_\Gamma = \#\{\text{nodes}\}$?

22.4 The Coarse Space and Low Genus

The previous three sections built the stack: $\mathcal{M}_g$ is a smooth Deligne-Mumford stack of dimension $3g-3$ (theorem 22.1.3), its compactification $\overline{\mathcal{M}}_g$ by stable curves is smooth and proper (theorem 22.2.6, theorem 22.3.2), and the boundary $\partial\overline{\mathcal{M}}_g$ stratifies by the combinatorics of nodes (definition 22.3.3). The stack is the object that remembers everything, including the automorphism group of each curve, and it is the object on which the deformation theory of Part V and the descent theory of Part VI act cleanly. This closing section returns to the older question the whole book began with: is there a variety, a scheme of finite type over k, whose points are the isomorphism classes of stable curves? The answer is yes, but it is the coarse space, not the stack, and the passage from one to the other is where the automorphisms are paid for.

Two constructions of that coarse space run side by side, and this section presents both because each carries information the other does not. The Keel-Mori theorem (theorem 20.2.4) produces $\overline{M}_g$

abstractly, as an algebraic space receiving the stack, from the single hypothesis of finite inertia; it is fast and general but says nothing about projectivity. Geometric invariant theory produces $\overline{M}_g$ concretely, as a quotient of a Hilbert scheme of pluricanonically embedded curves by a projective linear group, and this construction sees the projective embedding directly. The section then spends the rest of its length making the abstract picture completely explicit in the one case where every number can be written down: $g = 1$, the moduli of elliptic curves, where the stack is $[(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$, the coarse space of $\overline{\mathcal{M}}_{1,1}$ is $\mathbb{P}^1$, and the generic automorphism μ_2 is visible in a fractional degree. A first look at the tautological classes λ, ψ, κ closes the book's technical development and points at the intersection theory that lies just beyond it.

The coarse moduli space is the best scheme-theoretic approximation to a stack: the universal target for maps that forget automorphisms. Its existence for $\mathcal{M}_g$ was promised in Part VI as an application of Keel-Mori, and its projectivity is the last classical fact this book owes the reader about the moduli of curves. The two are separate theorems with separate proofs, and it is worth stating them together so the division of labor is visible: existence and the algebraic-space structure are Keel-Mori in full; projectivity is geometric invariant theory, with the hard stability computation cited.

Theorem 22.4.1: The Coarse Space $\overline{M}_g$ is a Projective Variety

Let $g \geq 2$ and k algebraically closed. The Deligne-Mumford stack $\overline{\mathcal{M}}_g$ admits a coarse moduli space

$$\pi \colon \overline{\mathcal{M}}_g \longrightarrow \overline{M}_g$$

with the following properties.

1. *Existence and the space structure.* $\overline{M}_g$ exists as a proper algebraic space of finite type over k, and π is universal among morphisms from $\overline{\mathcal{M}}_g$ to algebraic spaces: any morphism $\overline{\mathcal{M}}_g \to Z$ to an algebraic space factors uniquely through π. The map π induces a bijection between the k-points of $\overline{M}_g$ and the isomorphism classes of stable genus-g curves over k.

2. *Projectivity.* $\overline{M}_g$ is a projective variety over k; the open subvariety $M_g \subset \overline{M}_g$, the coarse space of $\mathcal{M}_g$, is quasi-projective.

3. *Singularities.* $\overline{M}_g$ is normal, but singular in general: at a point $[C]$ whose curve has a nontrivial automorphism group $\mathrm{Aut}(C)$, the coarse space looks locally like the quotient of the smooth deformation space $H^1(C, T_C)$ by the finite group $\mathrm{Aut}(C)$, so $\overline{M}_g$ carries at worst finite-quotient singularities, while the stack $\overline{\mathcal{M}}_g$ is smooth everywhere, including at the very points where the coarse space fails to be.

***Proof*:**
The three parts are proved in order, the first in full from the Part VI theorem and the third by a direct local computation; the second cites the stability computation whose full form is beyond the scope here, as noted before the statement.

1. By theorem 22.2.6 the stack $\overline{\mathcal{M}}_g$ is a Deligne-Mumford stack, and by theorem 22.3.2 it is proper and separated over k; in particular it is of finite type with finite diagonal. A separated Deligne-Mumford stack of finite type over a field has finite inertia: the automorphism group scheme of a stable curve is finite (definition 22.2.3), which is exactly

the finite inertia hypothesis. The Keel-Mori theorem (theorem 20.2.4) applies verbatim to any such stack and produces a coarse moduli space $\pi\colon \overline{\mathcal{M}}_g \to \overline{M}_g$ that is an algebraic space, together with the universal property recorded in the statement and the bijection on geometric points. Properness of $\overline{M}_g$ follows from properness of $\overline{\mathcal{M}}_g$ because π is proper and surjective, and properness descends along such maps (definition 20.5.5); finite type is preserved for the same reason. This establishes (1) in full.

2. The coarse space $\overline{M}_g$ is constructed as a geometric invariant theory quotient, following the plan laid out in Part IV. Fix an integer $\nu \geq 3$. For a stable curve C the dualizing sheaf ω_C is ample (definition 22.2.3), and $\omega_C^{\otimes\nu}$ is very ample with

$$N_\nu + 1 := h^0(C, \omega_C^{\otimes\nu}) = (2\nu - 1)(g - 1)$$

for $\nu \geq 2$, by Riemann-Roch on the nodal curve ([7]) applied to the line bundle $\omega_C^{\otimes\nu}$ of degree $\nu(2g-2)$. The complete linear system $|\omega_C^{\otimes\nu}|$ embeds C as a curve of degree $\nu(2g-2)$ in $\mathbb{P}^{N_\nu}$, its *ν-canonical model*, with a fixed Hilbert polynomial $P(t) = \nu(2g-2)t - (g-1)$ independent of C. The locus of such embedded curves is a locally closed subscheme $H \subset \mathrm{Hilb}^P(\mathbb{P}^{N_\nu})$ of the Hilbert scheme of Ch. 2 (definition 14.1.6, section 14.3), and two points of H give isomorphic abstract curves precisely when they lie in the same orbit of $\mathrm{PGL}_{N_\nu+1}$ acting by change of projective coordinates. Thus

$$\overline{M}_g \;=\; H^{\mathrm{ss}} /\!\!/ \mathrm{PGL}_{N_\nu+1}$$

is a candidate description as a projective GIT quotient (theorem 15.3.4), once one knows the Hilbert points of ν-canonical stable curves are semistable for the natural linearization. That last input is the stability theorem for curves: for ν large the Hilbert point (equivalently, by the Hilbert-Mumford numerical criterion theorem 15.4.5, the asymptotic Chow point) of a ν-canonically embedded stable curve is Hilbert-stable, and conversely a semistable Hilbert point in H is the point of a stable curve, so on H the semistable and stable loci coincide and parametrize exactly the stable curves (theorem 15.5.5). The verification of asymptotic stability is a one-parameter-subgroup computation on the Hilbert point, carried out in full in [50] and [32]; the key idea is that a curve fails to be Hilbert-semistable exactly when it has a subcurve on which $\omega_C^{\otimes\nu}$ is too positive relative to the rest, which for $\nu \gg 0$ recovers the combinatorial stability condition of definition 22.2.3. Granting theorem 15.5.5, theorem 15.3.4 presents $\overline{M}_g$ as the projective quotient of the semistable locus, hence a projective variety; the stable locus, which contains M_g, has a geometric quotient that is the quasi-projective M_g. The two coarse spaces produced, by Keel-Mori and by GIT, agree by the uniqueness half of the coarse-space universal property in (1).

3. Normality of $\overline{M}_g$ follows from smoothness of the stack: $\overline{\mathcal{M}}_g$ is smooth (theorem 22.2.6), and the coarse space of a smooth Deligne-Mumford stack with finite inertia is normal, being étale-locally a quotient of a smooth scheme by a finite group. For the local structure at a point $[C]$, work in a versal deformation. The deformation theory of Part V gives a smooth versal base $\operatorname{Spec} k[[t_1, \ldots, t_{3g-3}]]$ for C, with tangent space $H^1(C, T_C)$ of dimension $3g-3$ and no obstructions (proposition 17.4.1), and the automorphism group $\mathrm{Aut}(C)$ acts on this base fixing the origin. Because $\overline{\mathcal{M}}_g$ is Deligne-Mumford, $\mathrm{Aut}(C)$ is finite (definition 22.2.3), and an étale neighborhood of $[C]$ in the stack is the quotient stack $[H^1(C, T_C)/\mathrm{Aut}(C)]$ (definition 19.3.1). Passing to coarse spaces, an étale neighborhood of $[C]$ in $\overline{M}_g$ is the quotient variety $H^1(C, T_C)/\mathrm{Aut}(C)$, a finite-quotient singularity. When $\mathrm{Aut}(C)$ acts freely away from the origin, as for a generic curve where $\mathrm{Aut}(C)$ is trivial, this quotient is smooth;

when the action has nontrivial stabilizers, the quotient acquires a genuine singularity, and the stack is exactly the object that resolves it by remembering the group. This is what is meant by saying the stack is smooth where the coarse space is singular, completing the proof.

The theorem is the reconciliation the book was built toward, and reading it against the narrative arc is worth the pause. Part IV posed the question of representability and diagnosed automorphisms as the obstruction to a fine moduli space; the coarse space is the concession one makes, a variety that represents the moduli problem up to the loss of the automorphism data. Part VI showed that the stack loses nothing, and the price of the stack is that it is not a scheme. The Keel-Mori theorem is the bridge between the two worlds: it says that whenever the automorphisms are finite, one can always descend from the stack to a scheme-like object, paying only the passage from smooth to singular. The projectivity, by contrast, is not a formal consequence of anything; it is the geometric input that ω_C is ample on a stable curve, run through the machine of geometric invariant theory. The reason stable curves were defined by ampleness of ω_C in definition 22.2.3, rather than by the combinatorial condition alone, is exactly so that this GIT construction would have a polarization to work with; the combinatorial condition and the ampleness condition were proved equivalent there precisely so that the intrinsic definition and the projective one would coincide.

The description of $\overline{M}_g$ as a GIT quotient is also the origin of the first invariants of the moduli space. The linearization used to form the quotient descends to an ample line bundle on $\overline{M}_g$, and its restriction to the boundary and its intersection numbers are the beginning of the intersection theory of the moduli space. Before turning to that, the entire construction is made completely explicit in genus one, where the coarse space is a familiar curve and every automorphism can be named.

The moduli of elliptic curves is the one case where the stack, the coarse space, the compactification, and every automorphism group can be written down by hand, and it has served as the running touchstone since Part IV. It is worth assembling the whole picture in one place, now that the language of stacks is available, because $\overline{\mathcal{M}}_{1,1}$ is the smallest nontrivial instance of every phenomenon in this chapter: a smooth proper Deligne-Mumford stack whose coarse space is a smooth projective curve, whose generic point carries a nontrivial automorphism, and whose boundary is a single point representing a nodal degeneration. The genus-one theory sits slightly outside the $g \geq 2$ conventions of this chapter, since an elliptic curve has $\dim H^1(T) = 1$ rather than $3g - 3 = 0$ and infinitely many automorphisms as a group scheme when unpointed; the pointed stack $\mathcal{M}_{1,1}$ is the object that behaves like the higher-genus $\mathcal{M}_g$.

Example 22.4.2: $\mathcal{M}_{1,1}$ and $\overline{\mathcal{M}}_{1,1}$ in Full

Work over $k = \bar{k}$ of characteristic $\neq 2, 3$. An elliptic curve over k has a Weierstrass model $y^2 = x^3 + ax + b$ with discriminant $\Delta = -16(4a^3 + 27b^2) \neq 0$, and two such models give isomorphic pointed curves precisely when they are related by $(a, b) \mapsto (u^4 a, u^6 b)$ for some $u \in \mathbb{G}_m$, the scaling $x \mapsto u^2 x$, $y \mapsto u^3 y$. The moduli stack of elliptic curves is therefore the quotient stack

$$\mathcal{M}_{1,1} \;\simeq\; \big[\,(\mathbb{A}^2_{a,b} \setminus \{\Delta = 0\})\,/\,\mathbb{G}_m\,\big]$$

with $\mathbb{G}_m$ acting by $u \cdot (a, b) = (u^4 a, u^6 b)$ (definition 19.3.1). This is the concrete realization of the elliptic curve as a family over its Weierstrass parameters, with the scaling ambiguity turned into a group action rather than divided out on the nose.

The coarse space is the j-line. The $\mathbb{G}_m$-invariant functions on $\mathbb{A}^2_{a,b} \setminus \{\Delta = 0\}$ are generated by the

j-invariant

$$j = 1728\,\frac{4a^3}{4a^3+27b^2}$$

a weight-zero rational function of (a, b), and the ring of invariants is exactly $k[j]$. The coarse space is Spec of this ring of invariants, the affine j-line

$$M_{1,1} = (\mathbb{A}^2 \setminus \{\Delta = 0\})/\!\!/\mathbb{G}_m \cong \mathbb{A}^1_j$$

recovering theorem 13.3.6: two elliptic curves are isomorphic if and only if they share the same j-invariant, and every value $j \in k$ is attained. The coarse map $\mathcal{M}_{1,1} \to \mathbb{A}^1_j$ is the universal map to a scheme, and it forgets exactly the automorphisms.

The compactification adds the nodal cubic at $j = \infty$. The stack $\overline{\mathcal{M}}_{1,1}$ of stable pointed genus-one curves adjoins to $\mathcal{M}_{1,1}$ the single isomorphism class of the nodal cubic, the curve $y^2 = x^3 + x^2$ (a rational curve with one node, marked at the smooth point at infinity), which is stable because its only pointed automorphisms form a finite group and ω_C is ample with the marking. As $j \to \infty$ a family of smooth elliptic curves degenerates to this nodal cubic, the boundary point of the moduli problem; the Tate-curve uniformization makes the degeneration explicit, presenting the nearby smooth curves as $\mathbb{G}_m/q^{\mathbb{Z}}$ with $q \to 0$ (see [11]). The coarse space of the compactified stack is therefore the projective j-line

$$\overline{M}_{1,1} = \mathbb{A}^1_j \cup \{j = \infty\} \cong \mathbb{P}^1$$

a smooth projective curve, in agreement with theorem 22.4.1(2): the coarse space is projective. In this genus the coarse space happens to be smooth, because the singularities of a coarse space are finite-quotient singularities of the deformation space, and here that space $H^1(E, T_E)$ is one-dimensional, on which any finite cyclic automorphism group acts by a character, and a one-dimensional linear quotient by a finite cyclic group is again a smooth line.

The stacky structure is the automorphism gerbe. The coarse space $\mathbb{P}^1$ forgets that every point carries automorphisms. A generic elliptic curve E has automorphism group $\mathrm{Aut}(E) = \mu_2 = \{\pm 1\}$, the inversion $P \mapsto -P$ fixing the marked origin, so the generic point of $\mathcal{M}_{1,1}$ is a μ_2-gerbe over its image in $\mathbb{A}^1_j$: the whole stack is generically a $B\mu_2$ living over the j-line, as recorded in example 13.3.4. At two special values the automorphism group jumps: at $j = 0$ the curve $y^2 = x^3 + 1$ has $\mathrm{Aut} = \mu_6$ (extra automorphisms from the sixth roots of unity acting by $x \mapsto \zeta^2 x$, $y \mapsto \zeta^3 y$), and at $j = 1728$ the curve $y^2 = x^3 + x$ has $\mathrm{Aut} = \mu_4$. These are the residual gerbes $B\mu_6$ and $B\mu_4$ of the two orbifold points. The coarse space $\mathbb{P}^1$ sees three ordinary points $j = 0, 1728, \infty$ where the stack sees $B\mu_6$, $B\mu_4$, and the nodal boundary; the passage to the coarse space discards exactly this gerbe structure, which is the automorphism obstruction of example 13.3.4 made geometric. The stack $\mathcal{M}_{1,1}$ remembers the μ_2, μ_4, μ_6; the scheme $\mathbb{A}^1_j$ does not.

The genus-one picture is the whole book in miniature: a moduli problem whose objects have automorphisms, whose stack is a smooth quotient $[(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$, whose coarse space is the classical j-line, and whose compactification is projective and glues in a nodal degeneration exactly as $\overline{\mathcal{M}}_g$ glues in stable curves. The one feature it does not exhibit, because its deformation space $H^1(E, T_E)$ is one-dimensional, is a singular coarse space; that first appears in genus three, where an automorphism can act on the base of dimension $3g - 3$ without pseudoreflections and so produce a genuine quotient singularity. Even in genus one the automorphisms leave a numerical trace: they force a fractional degree.

The intersection theory of $\overline{\mathcal{M}}_g$ is the subject that begins where this book ends, and it is organized by a distinguished collection of cohomology classes built from the universal curve. These are the tautological classes. They are defined here, and one of them is computed on $\overline{\mathcal{M}}_{1,1}$ to give the reader a concrete number, but their theory, the structure of the ring they generate and the theorems that compute their intersection numbers, is the content of a different volume and is only gestured at. The classes are built from three pieces of universal data: the Hodge bundle, the cotangent lines at the marked points, and their pushforwards.

Definition 22.4.3: Tautological Classes on $\overline{\mathcal{M}}_g$

Let $\pi\colon \overline{\mathcal{C}}_g \to \overline{\mathcal{M}}_g$ be the universal stable curve, the family whose fiber over a point $[C]$ is the curve C itself, with relative dualizing sheaf $\omega_\pi = \omega_{\overline{\mathcal{C}}_g/\overline{\mathcal{M}}_g}$.

1. The *Hodge bundle* is the rank-g vector bundle

$$\mathbb{E} \;=\; \pi_*\omega_\pi$$

on $\overline{\mathcal{M}}_g$, whose fiber over $[C]$ is the space $H^0(C, \omega_C)$ of holomorphic differentials, of dimension g. Its first Chern class is the *lambda class*

$$\lambda \;=\; \lambda_1 \;=\; c_1(\mathbb{E}) \;\in\; H^2(\overline{\mathcal{M}}_g)$$

and more generally $\lambda_i = c_i(\mathbb{E})$.

2. On the moduli of pointed curves $\overline{\mathcal{M}}_{g,n}$, with universal sections $\sigma_1, \dots, \sigma_n \colon \overline{\mathcal{M}}_{g,n} \to \overline{\mathcal{C}}_{g,n}$ marking the n points, the *psi-classes* are

$$\psi_i \;=\; c_1\big(\sigma_i^*\omega_\pi\big) \;\in\; H^2(\overline{\mathcal{M}}_{g,n})$$

the first Chern class of the cotangent line to the curve at the i-th marked point.

3. The *kappa classes* are the pushforwards

$$\kappa_a \;=\; \pi_*\big(\psi^{a+1}\big) \;\in\; H^{2a}(\overline{\mathcal{M}}_g)$$

where $\psi = c_1(\omega_\pi(\sum_i \sigma_i))$ is the relative cotangent class on the universal curve and π_* is integration over the fiber; on $\overline{\mathcal{M}}_g$ with no marked points the twist $\sum_i \sigma_i$ is empty, so $\psi = c_1(\omega_\pi)$ and κ_a reduces to Mumford's $\pi_*(c_1(\omega_\pi)^{a+1})$, while on $\overline{\mathcal{M}}_{g,n}$ the sections σ_i contribute the twist.

The subring of the cohomology (or Chow) ring generated by the λ_i, ψ_i, and κ_a, together with the classes of the boundary strata Δ_i, is the *tautological ring* of $\overline{\mathcal{M}}_g$.

The definition packages the natural bundles a family of curves carries. The Hodge bundle is the most basic: a family of curves has a family of spaces of differentials, and $\mathbb{E}$ is that family, its determinant $\det \mathbb{E}$ the natural line bundle whose Chern class λ measures how the differentials twist over the moduli space. The psi-classes record the same twisting for the cotangent line at a marked point, and the kappa classes are what one gets by integrating powers of the cotangent class along the fibers of the universal curve. The tautological ring is the subring these generate, and the central discovery of the subject, that intersection numbers of these classes satisfy recursions governed by an integrable hierarchy, is the Witten-Kontsevich theorem; it and the full structure of the tautological ring belong to intersection theory on moduli spaces, treated in [49] and pursued in the intersection-theory volume of this series. This book states the classes and stops, so that the reader who continues arrives with the definitions already in hand.

The one computation worth doing now is on $\overline{\mathcal{M}}_{1,1}$, where the Hodge bundle has rank one and the answer is a fraction, exhibiting the stacky structure of example 22.4.2 numerically. It is grounded in the following low-genus data, and stated with the caveat that the number is an orbifold degree.

Example 22.4.4: Low Genus: Dimensions, Boundary, and λ on $\overline{\mathcal{M}}_{1,1}$

The dimension $\dim \overline{\mathcal{M}}_g = 3g-3$ and the number of boundary divisors $\Delta_0, \Delta_1, \ldots, \Delta_{\lfloor g/2 \rfloor}$ of definition 22.3.3 for small g are as follows.

g	$\dim \overline{\mathcal{M}}_g = 3g-3$	$\#\{\Delta_i\}$	boundary divisors
2	3	2	$\Delta_0,\ \Delta_1$
3	6	2	$\Delta_0,\ \Delta_1$
4	9	3	$\Delta_0,\ \Delta_1,\ \Delta_2$
5	12	3	$\Delta_0,\ \Delta_1,\ \Delta_2$
6	15	4	$\Delta_0,\ \Delta_1,\ \Delta_2,\ \Delta_3$

In each row Δ_0 is the closure of the locus of irreducible curves with a single node (normalizing to a smooth curve of genus $g-1$ with two points glued), and Δ_i for $1 \le i \le \lfloor g/2 \rfloor$ is the closure of the locus of curves splitting into a genus-i and a genus-$(g-i)$ component meeting at one node. The count is $1 + \lfloor g/2 \rfloor$, so $g = 2$ has the two divisors Δ_0 and Δ_1: the irreducible one-nodal genus-two curves, and the pairs of elliptic curves joined at a point.

The boundary of $\overline{\mathcal{M}}_2$. A stable genus-two curve that is not smooth is one of two types. A point of Δ_1 is two elliptic curves E_1, E_2 joined transversally at a single node; its arithmetic genus is $1+1=2$ and it is stable because each component is a genus-one curve meeting the node (a genus-one component needs only one special point). A point of Δ_0 is an irreducible curve of geometric genus one with one node, the image of an elliptic curve with two points identified. These are the two codimension-one boundary strata; deeper strata, of higher codimension, glue in more nodes, and a point where the curve is a maximally degenerate stable curve with three nodes, all of whose components are rational, sits in codimension three, the deepest stratum in $\dim = 3$. There are two such configurations: the theta graph, two $\mathbb{P}^1$'s meeting at three points, and the dumbbell, two $\mathbb{P}^1$'s each carrying a self-node, joined by a bridge; both have first Betti number $b_1 = 2$ of the dual graph, which is what makes the arithmetic genus equal to 2. A genus-one component cannot appear in this stratum: it carries a one-dimensional modulus, its j-invariant, so any stratum containing one has dimension at least 1, hence codimension at most 2.

The degree of λ on $\overline{\mathcal{M}}_{1,1}$. On $\overline{\mathcal{M}}_{1,1}$ the Hodge bundle $\mathbb{E} = \pi_*\omega_\pi$ has rank one, its fiber over $[E]$ being the one-dimensional $H^0(E, \omega_E)$, and for a marked genus-one curve the cotangent line at the marked point coincides with the space of differentials, so $\lambda = \psi$ as classes on $\overline{\mathcal{M}}_{1,1}$. The degree of λ is computed by working on the coarse space $\mathbb{P}^1$ of example 22.4.2 and correcting for the generic automorphism. Over the Weierstrass presentation, a differential on $y^2 = x^3 + ax + b$ is a multiple of dx/y, which scales by u^{-1} under $(a,b) \mapsto (u^4 a, u^6 b)$; the Hodge line bundle is therefore the weight-one character of $\mathbb{G}_m$, and its degree over the compactified j-line is a single power of the boundary point. As a divisor class on the stack,

$$\deg_{\overline{\mathcal{M}}_{1,1}} \lambda \;=\; \frac{1}{24}$$

a fractional number, and the fraction records the automorphism orders. It is not a defect of the computation but the correct orbifold degree: the coarse space $\mathbb{P}^1$ has three special points $j = 0, 1728, \infty$ where the stack-automorphism groups are μ_6, μ_4, and, at the boundary node, μ_2 of order 2, and integrating a class over the stack divides the naive count by the order of the automorphism group at each point. The denominator 24 is exactly the number appearing in the theory of modular forms: the modular discriminant Δ, a modular form of weight 12, is a global section of $\mathbb{E}^{\otimes 12}$ over $\mathcal{M}_{1,1}$ with a simple zero at the boundary point $j = \infty$ (see [11]), so 12λ equals the class of the single boundary point, and dividing by the order 2 of that point's own stack-automorphism group μ_2 gives $\deg \lambda = \frac{1}{12} \cdot \frac{1}{2} = \frac{1}{24}$. Read against the coarse space, the moral is exact: a class that would have integer degree on a scheme has fractional degree on a stack, and the denominator is built from the automorphism orders that the coarse space theorem 22.4.1 threw away.

The fractional degree is the last word of the book's technical development, and it returns to Part IV. The automorphism obstruction was first seen as the failure of a fine moduli space to exist; it was then pinned to the nontrivial finite automorphism group $\mathrm{Aut}(C)$ that each stable curve carries, the source of the stackiness even when $H^0(C, T_C) = 0$ leaves no infinitesimal automorphisms; it was repaired by passing to a stack in Part VI; and it is now visible as a number, the denominator 24 that a scheme could never produce. The stack is the object on which every one of these statements is clean, and the coarse space is its image in the world of varieties. What lies beyond, the intersection theory of these tautological classes, the recursions among their integrals, and the moduli of higher-dimensional varieties for which the same three-act structure of representability, deformation, and stacks recurs, is the subject the closing chapter points the reader toward.

Exercise 22.4.1

1. Using theorem 22.4.1, explain precisely why $\overline{\mathcal{M}}_g$ admits a coarse moduli space that is an algebraic space but the stack itself is in general not an algebraic space. Which hypothesis of the Keel-Mori theorem (theorem 20.2.4) fails for a stack with a nonfinite inertia, and give an example from earlier in the book of a stack to which Keel-Mori does not apply.

2. Let C be a stable curve with automorphism group $\mathrm{Aut}(C)$ of order $m > 1$ acting on $H^1(C, T_C) \cong k^{3g-3}$. Describe the local structure of $\overline{M}_g$ near $[C]$ as a quotient singularity, and give a genus and a curve for which the quotient is smooth and one for which it is singular. Explain in one sentence why $\overline{\mathcal{M}}_g$ is smooth at $[C]$ regardless.

3. For $\mathcal{M}_{1,1} \simeq [(\mathbb{A}^2 \setminus \{\Delta = 0\})/\mathbb{G}_m]$ of example 22.4.2, verify that $j = 1728 \cdot 4a^3/(4a^3 + 27b^2)$ is $\mathbb{G}_m$-invariant under $(a, b) \mapsto (u^4 a, u^6 b)$, and confirm that it takes every value in k and that $j = 0$, $j = 1728$ correspond to the curves with $a = 0$, $b = 0$ respectively. Identify the automorphism group at each of these two points.

4. Compute $\dim \overline{\mathcal{M}}_g$ and the number of boundary divisors for $g = 2, 3, 6, 10$ using example 22.4.4. For $g = 2$, describe the two boundary divisors Δ_0 and Δ_1 as loci of stable curves, and state the codimension in $\overline{\mathcal{M}}_2$ of the deepest boundary stratum.

5. On $\overline{\mathcal{M}}_{g,n}$, let $\psi_i = c_1(\sigma_i^* \omega_\pi)$ be the psi-class of definition 22.4.3. Explain why on $\overline{\mathcal{M}}_{1,1}$ one has $\lambda = \psi$, where $\psi = \psi_1$ is the psi-class of the single marked point, and use this to state the value of $\int_{\overline{\mathcal{M}}_{1,1}} \psi$ as an orbifold degree. Why is the answer not an integer?

6. The tautological ring of definition 22.4.3 is generated by the λ_i, ψ_i, κ_a, and the boundary classes

Δ_i. Explain why the Hodge bundle $\mathbb{E} = \pi_* \omega_\pi$ has rank exactly g, and why $\lambda_i = c_i(\mathbb{E}) = 0$ for $i > g$. State, without proof, what further structure the classes λ_i satisfy that makes the tautological ring finitely described (naming the theorem you are pointing at).

23

What Next?

A book on moduli theory could continue almost without end, because the three tools it has assembled, the deformation theory that linearizes a classification problem, the geometric invariant theory that builds quotients, and the stacks that remember automorphisms, are the standard equipment of a large part of modern algebraic geometry. This closing chapter points to where they lead. It proves nothing; it is a map of the surrounding country, with the roads that leave from $\overline{\mathcal{M}}_g$ marked and the neighboring volumes of this series named where they carry the story on.

23.1 Intersection Theory on the Moduli of Curves

The tautological classes introduced at the end of the last chapter, the Hodge class λ and the ψ- and κ-classes, are the beginning of a subject in their own right. The Chow and cohomology rings of $\overline{\mathcal{M}}_{g,n}$ carry a distinguished subring, the tautological ring, generated by these classes and by the boundary strata, and the central problems ask for its structure and for the numbers obtained by integrating monomials in ψ and κ against the fundamental class. Those intersection numbers are governed by deep structures: the Witten conjecture, proved by Kontsevich, identifies their generating function with a solution of the Korteweg-de Vries hierarchy, and the ELSV formula ties them to Hurwitz numbers counting branched covers. The enumerative geometry of $\overline{\mathcal{M}}_g$ begun in Mumford's foundational essay [49] is developed systematically in the intersection-theory volume of this series ([16]), where the Grothendieck-Riemann-Roch computation of the classes λ_i and the structure of the tautological ring are carried out with the machinery of Chern classes and virtual fundamental classes.

23.2 Other Moduli Spaces

The moduli of curves is one of a family of spaces built by the same method. Principally polarized abelian varieties of dimension g form a Deligne-Mumford stack $\mathcal{A}_g$, smooth of dimension $\binom{g+1}{2}$, whose deformation theory was already glimpsed in the treatment of unobstructed objects; the Torelli map $\mathcal{M}_g \to \mathcal{A}_g$ sending a curve to its Jacobian ties the two together, and the study of its image is a subject with a long history. Level structures on abelian varieties lead to the Siegel modular varieties and, more generally, to the Shimura varieties at the confluence of Hodge theory, automorphic forms, and arithmetic; these are the subject of projected volumes on abelian and on Shimura varieties. In a different direction, the moduli of vector bundles and of principal G-bundles on a fixed curve, the stack $\mathfrak{Bun}_G$ met in the discussion of Artin stacks, is the geometric arena of the geometric Langlands program, and its cousins, moduli of Higgs bundles and of local systems, carry the non-abelian Hodge theory of Simpson. The twisted sheaves of the gerbes chapter reappear here: moduli of bundles are frequently only coarsely represented until one twists by a Brauer class, exactly as the period-index discussion foreshadowed.

23.3 Foundations Pushed Further

This book stopped its deformation theory at the naive cotangent complex and Schlessinger's criteria, enough for smooth objects and for the node, but the full theory reaches further. Illusie's cotangent complex $L_{X/S}$, a genuine object of the derived category rather than its first two terms, computes the deformations and obstructions of an arbitrary morphism and repairs the failures of the naive version at singular and non-l.c.i. points. Its natural home is derived algebraic geometry, where a scheme is replaced by a functor on simplicial or spectral rings and the tangent complex acquires higher homotopy; in that setting Artin's criteria become the representability theorem of Lurie, and the moduli functors of this book acquire derived enhancements whose classical truncations are the stacks constructed here. The categorical prerequisites, ∞-categories and the homotopy theory that supports them, belong to the category-theory volume ([9]) and to a projected volume on derived and spectral algebraic geometry, and they are the right language for the geometric Langlands correspondence and for the moduli of objects in a differential graded category.

23.4 Deformation Theory in Neighboring Subjects

The deformation theory of Part V is a local instrument that recurs wherever one asks how an object moves. The deformations of a subvariety, governed by the normal sheaf as in the Hilbert-scheme computation, control the geometry of curves on a surface and of complete intersections, the classical subject of Mumford's lectures and of Vistoli's work on the deformation theory of the Quot scheme ([66]). The deformations of a map, rather than of an object, are what the theory of rational curves is built on: the dimension estimate for a space of maps $\mathbb{P}^1 \to X$, together with the bend-and-break technique that produces rational curves by degenerating them, underlies Mori's proof of the cone theorem and the whole minimal model program, developed in Kollár's book on rational curves and in a projected volume on birational geometry. In each of these the pattern is the one established here: a tangent space and an obstruction space, each a cohomology group, that together linearize a moduli problem and predict the dimension of its solution.

23.5 Period Maps and Hodge Theory

A smooth projective family carries a variation of Hodge structure on its cohomology, and the period map records how the Hodge filtration moves as the fibers vary. The target of the period map is itself a moduli space, a quotient of a period domain by an arithmetic group, and the infinitesimal Torelli problem asks whether the period map is an immersion, a question answered by the same cohomological deformation theory used throughout this book. Hodge theory, the Hodge conjecture, and the algebraicity results of the theory of variations of Hodge structure are the subject of a projected volume on Hodge theory; the present book supplies the moduli-theoretic foundation on which the period map is defined.

23.6 A Short Reading List

The reader who wishes to go deeper has excellent guides. For the moduli of curves specifically, the book of Harris and Morrison [32] is the standard companion, and the original paper of Deligne and Mumford [23] remains the source of the stable-curve compactification and of stable reduction. For geometric invariant theory the reference is Mumford's own [50]. For descent and the foundations of stacks, the Grothendieck seminar as explained by Vistoli [66], the book of Laumon and Moret-Bailly [41], and the modern account of Olsson [59] are the standard treatments, with the Stacks Project [63] an encyclopedic resource. For deformation theory the books of Hartshorne [35] and Sernesi [61] carry the subject well past where this book leaves it. Each of these was cited in the chapters where its material was used, and together they chart the paths out of the country this book has surveyed.

The three ideas are worth restating one last time, because they are what the reader takes away. A moduli problem is a functor, and asking whether it is representable is asking for a space whose points are the objects classified. When automorphisms obstruct representability, the remedy is to remember them, and the object that does so is a stack. And the local structure of a moduli space, its smoothness and its dimension, is deformation theory, a computation in the cohomology of a single object. Every moduli space in the subjects above is built by holding these three ideas together, and the reader who has followed them here is equipped to meet the rest.

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